
M O D E L A N S W E R B A N K

RGUHS MD Community Medicine — Paper IV



H E A L T H S Y S T E M S , M A N A G E M E N T A N D I N T E R N A T I O N A L
H E A L T H

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143 topics in this volume · from a 688-topic bank · Textbook-grounded model answers · cited Recent Advances

G R O U N D E D I N

Park's Textbook of Preventive and Social Medicine, 28e — the spine for every topic
Gordis Epidemiology 7e — epidemiological method and study design
Handbook of Epidemiology (Ahrens & Pigeot) — advanced epidemiological method
Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e — a second PSM voice on Indian practice
Sunder Lal — Textbook of Community Medicine — Indian community medicine practice
Plotkin's Vaccines, 7ed — vaccines and immunisation
Nelson & Williams — Infectious Disease Epidemiology — infectious disease epidemiology
Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — public health and preventive medicine breadth
ICMR National Ethical Guidelines 2017 — research ethics in the Indian setting
Browner -- Designing Clinical Research (5e, 2022) — clinical research design
Routine Immunization I Manual For Medical Officers 2024 — current immunisation service delivery
Oxford Textbook of Public Health 7e — health systems and international health



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Health care systems and primary health care



Renewing health for all

Asked in: Paper I May 2019 — RGUHS MD Community Medicine

Definition — Health for All (HFA) is the social goal, resolved by the 30th World Health Assembly in May 1977, that all citizens of the world should by the year 2000 attain "a level of health that will permit them to lead a socially and economically productive life" (Park 28e, Ch 1, p 23). "Renewing Health for All" means the successive reaffirmation and re-specification of that unfinished goal after its 2000 AD deadline lapsed — through Health for All in the 21st Century (1998), the World Health Report 2008 on primary health care, the Sustainable Development Goals (2015) and the Astana Declaration (2018) — with **universal health coverage (UHC)** as its present operational expression. The goal was never abandoned; only its deadline, its targets and its delivery vehicle were rewritten.

1. The original Health for All movement and its monitoring framework

Milestone	Year	Operative content
30th World Health Assembly resolution	May 1977	Declared HFA by 2000 AD the main social target of governments and WHO (Park 28e, Ch 10, p 708)
Joint WHO–UNICEF conference, Alma-Ata (USSR), 6–12 September	1978	Governments of 134 countries declared existing inequality in health "politically, socially and economically unacceptable" and proclaimed primary health care (PHC) the way to achieve HFA; 8 essential elements, 4 principles — equitable distribution, community participation, intersectoral coordination, appropriate technology (Park 28e, Ch 1, p 23; Ch 10, pp 707–708)
Global Strategy for Health for All, 34th World Health Assembly	1981	Provided a global framework, flexible enough for all Member States to adapt; WHO set 12 global indicators , e.g. a minimum life expectancy of 60 years and a maximum infant mortality rate (IMR) of 50 per 1000 live births (Park 28e, Ch 10, p 709)
India's response — National Health Policy 1983	1983	Committed India to HFA; goals by 2000 AD were IMR from 125 (1978) to below 60, life expectancy 52→64 years, crude death rate 14→9, crude birth rate 33→21 per 1000, net reproduction rate of one, potable water to the whole rural population (Park 28e, Ch 10, p 709)

Four categories of HFA monitoring indicators (Park 28e, Ch 2, p 28): health policy indicators (political commitment, resource allocation, equity of distribution, community involvement); social and economic indicators related to health; indicators for the provision of health care (availability, accessibility, utilisation, quality); and health status indicators (low birth weight, nutritional status and psychosocial development of children, IMR, child mortality rate).

2. Why Health for All had to be renewed

- **The deadline passed with the goal unmet** — the subversion of the aims of Alma-Ata during the 1980s and the failure to achieve Health for All by 2000 left the agenda unfinished (Textbook of International Health, Ch 7, p 391).

- ▶ **Comprehensive PHC gave way to selective, vertical programmes** targeting single diseases, so services were organised around providers rather than around people (Harrison 22e, Ch 487, p 3863).
- ▶ **Widening disparities in health** between and within countries — the same driver that had produced Alma-Ata itself (Harrison 22e, Ch 487, pp 3905–3906).
- ▶ **Financial catastrophe from health care** — more than 100 million people are still driven into extreme poverty each year by health care costs, and most people can obtain services only if they can pay at the point of service (Harrison 22e, Ch 487, p 3863).
- ▶ **An epidemiological shift** — non-communicable diseases, mental health and injuries needed long-term, community-based chronic care that HFA-2000 targets never addressed (Harrison 22e, Ch 487, pp 3905–3906).
- ▶ **New and recurring threats** — pandemics and accelerated climate change exposed weak health systems as the obstacle to delivering even proven technologies (Harrison 22e, Ch 487, pp 3905–3906).
- ▶ **A new global development architecture** — the Millennium Development Goals (2000–2015) and then the 2030 Agenda required health targets restated in that language (Park 28e, Ch 1, p 24).

3. Health for All in the 21st Century (World Health Assembly, May 1998) — the first formal renewal

In May 1998 WHO adopted a resolution supporting a new global Health for All policy, **Health for All in the 21st Century**, succeeding the HFA by 2000 strategy launched in 1977. Its two aims were to build a shared vision by listing the 10 most important health issues, and to set 10 targets that motivate Member States to act and to set priorities for resource allocation; each country then sets its own targets within this framework (Mahajan & Gupta 4e, Ch 28, p 550).

Group	Global health target (Mahajan & Gupta 4e, Ch 28, pp 550–551)
Health outcome (4)	Health equity: by 2005, health equity indices in use within and between countries, assessed initially on child growth (stunting)
	Survival: by 2020, maternal mortality ratio < 100 per 100,000 live births, under-5 mortality < 45 per 1000 live births, life expectancy > 70 years
	Reverse the trends of five major pandemics: tuberculosis, HIV/AIDS, malaria, tobacco-related disease, and violence or trauma, by 2020
	Eradicate/eliminate: measles eradicated by 2020; lymphatic filariasis eliminated by 2020; Chagas disease transmission interrupted by 2010; leprosy eliminated by 2010; trachoma eliminated by 2020; vitamin A and iodine deficiency eliminated before 2020
Determinants of health (2)	Improve access to water, sanitation, food and shelter through intersectoral action, and manage major environmental risks, by 2020
	Measures to promote health — regulatory, economic, educational, organisational and community-based strategies that strengthen health-enhancing lifestyles, by 2020

Group	Global health target (Mahajan & Gupta 4e, Ch 28, pp 550–551)
Health policies and sustainable health systems (4)	Develop, implement and monitor national Health for All policies by 2005
	Access for all, throughout life, to comprehensive, essential, quality health care supported by essential public health functions, by 2010
	Establish global and national health information, surveillance and alert systems by 2010
	Make research-for-health policies and institutional mechanisms operational at global, regional and country level by 2010

4. Renewal after 2000 — from the Millennium Development Goals to Astana

Instrument (year)	What it renewed
Millennium Development Goals (2000)	Placed health at the heart of development; 3 of the 8 goals, 8 of the 18 targets and 18 of the 48 indicators were health related (Park 28e, Ch 1, p 24)
World Health Report 2008 — "Primary Health Care: Now More Than Ever"	Asserted the PHC approach was needed "now more than ever" and defined four reforms of PHC renewal — <i>universal coverage reforms</i> (prepaid pooled financing, remove user fees, reach the poorly served); <i>service delivery reforms</i> (people-centred, comprehensive, continuous care with an ongoing personal care team); <i>public policy reforms</i> (multisectoral action on social determinants, e.g. tobacco, transport, housing); <i>leadership reforms</i> (participatory, accountable governance of a mixed provider landscape) (Harrison 22e, Ch 487, pp 3862–3864)
2030 Agenda for Sustainable Development (2015)	17 goals; health centrally placed in SDG 3 ("Ensure healthy lives and promote well-being for all at all ages") with 13 targets; target 3.8 — universal health coverage including financial risk protection, access to quality essential services and to safe, effective, affordable essential medicines and vaccines — is the key to achieving all other health targets and a resilient health system (Park 28e, Ch 14, p 852)
Astana Declaration on Primary Health Care (2018)	Reinforced the importance of primary health care as the route to UHC, forty years after Alma-Ata (Oxford Textbook of Public Health 7e, Ch 47, p 1751)
World Health Assembly resolution (2019)	Reaffirmed the Alma-Ata principles and reframed PHC as three components — (i) primary care and essential public health functions as the core of integrated health services, (ii) empowered people and communities, (iii) multisectoral policy and action (Harrison 22e, Ch 487, p 3863)

Universal health coverage as the operational form of renewed HFA — coverage is extended along three axes: *breadth* (proportion of the population covered), *depth* (range of services covered) and *height* (proportion of costs covered, i.e. reducing out-of-pocket spending) (Harrison 22e, Ch 487, p 3863; Sunder Lal, Ch 17, p 740).

5. Renewing Health for All in India

National Health Policy (NHP) 2017 replaced NHP 2002 and 1983; its goals are the highest possible level of health and well-being for all at all ages, achievement of the SDGs by 2030, and progressive attainment of universal health coverage (Sunder Lal, Ch 16, p 697).

NHP 2017 objective	Target	Current Indian value (year)
Life expectancy at birth	70 years by 2025 (from 67.5)	— (Park 28e, Ch 23, p 1030)
Infant mortality rate	< 28 by 2019	26 (2022)
Neonatal mortality rate	16 by 2025	18 (2022)
Under-5 mortality rate	23 by 2025	29 (2022)
Maternal mortality ratio (per 100,000 live births)	100 by 2020	97 (2018–2020)
Total fertility rate	2.1 by 2025	2.0 (2024)
Stillbirth rate	Single digit by 2025	—
Government health expenditure	2.5% of GDP by 2025 (from 1.15%); state health spending > 8% of state budget by 2020	—
Households facing catastrophic health expenditure	Reduce by 25% by 2025	—

(Current values: Park 28e, Ch 9, p 659; targets: Park 28e, Ch 23, p 1030 and Sunder Lal, Ch 16, pp 697–698.)

Ayushman Bharat was launched in February 2018 on the recommendation of NHP 2017 to achieve UHC and meet the SDG commitment to "leave no one behind" (AFMC Red Book 2024, Ch 38, p 1209). It has two components: transformation of about 1.5 lakh sub-health centres and primary health centres into **Health and Wellness Centres (HWCs)**, and **Pradhan Mantri Jan Arogya Yojana (PM-JAY)** for secondary and tertiary hospitalisation. This is the explicit Indian move from selective to **comprehensive primary health care (CPHC)** along a life-course continuum, with "time to care" of not more than 30 minutes of travel; nearly 1.1 lakh sub-health centres had been converted by March 2022 (Sunder Lal, Ch 17, pp 739–746).

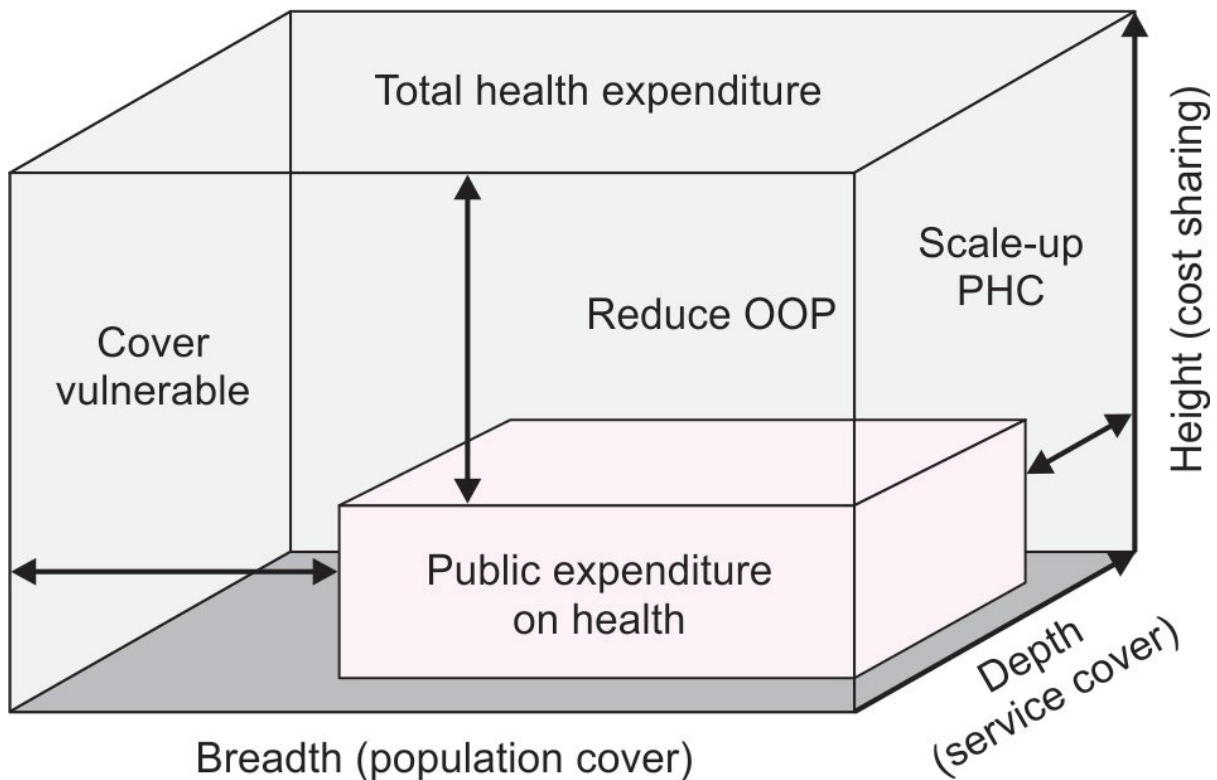
The 12 CPHC service packages at a Health and Wellness Centre (Sunder Lal, Ch 17, p 749):

- ▶ Care in pregnancy and childbirth
- ▶ Neonatal and infant health care services
- ▶ Childhood and adolescent health care services
- ▶ Family planning, contraceptive and other reproductive health care services
- ▶ Management of communicable diseases (national health programmes)
- ▶ General out-patient care for acute simple illnesses and minor ailments
- ▶ Screening, prevention, control and management of non-communicable diseases

- ▶ Care for common ophthalmic and ENT problems
- ▶ Basic oral health care
- ▶ Elderly and palliative health care services
- ▶ Emergency medical services including burns and trauma
- ▶ Screening and basic management of mental health ailments

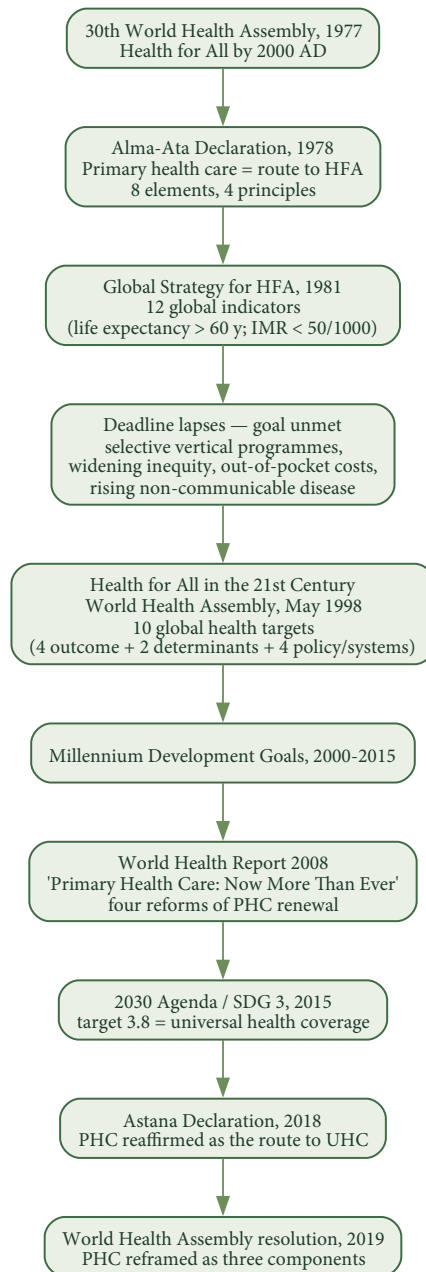
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — The universal coverage cube.** Labels to reproduce: outer box *Total health expenditure*; inner box *Public expenditure on health*; horizontal axis *Breadth (population cover)* with arrow "Cover vulnerable"; depth axis *Depth (service cover)* with arrow "Scale-up primary health care"; vertical axis *Height (cost sharing)* with arrow "Reduce out-of-pocket expenditure".

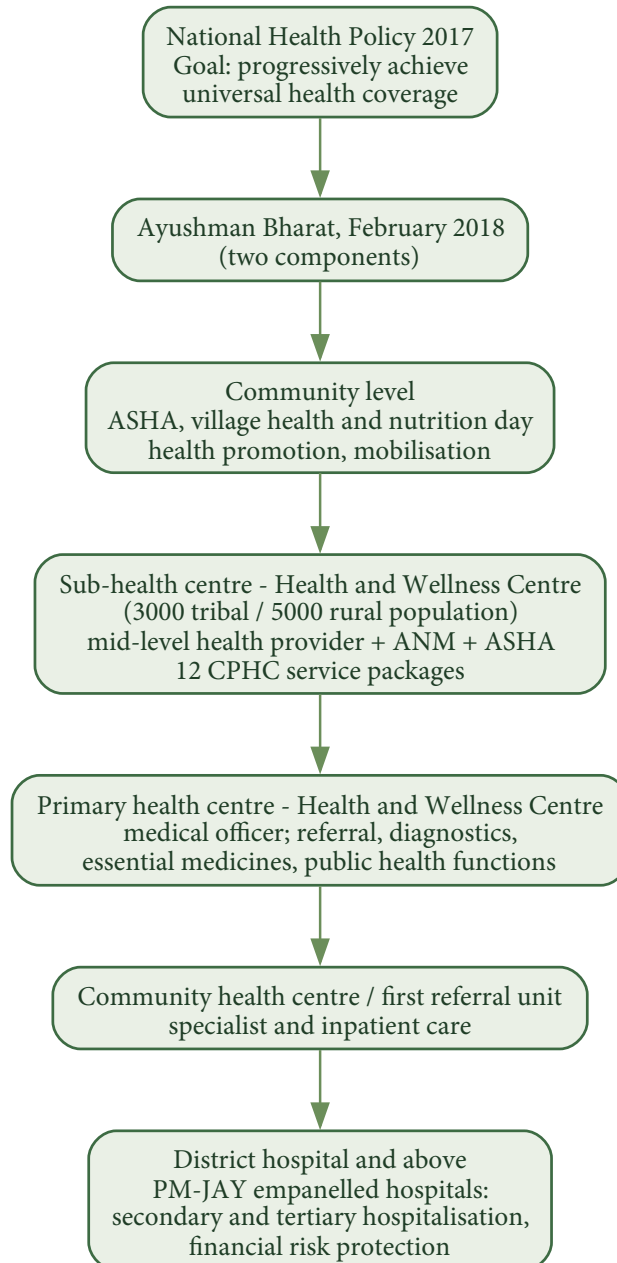


OOP: Out of pocket
PHC: Primary health care

Attaining universal coverage — breadth, depth and height (Sunderlal, p.740)



Renewing Health for All — the chain of renewal from Alma-Ata to universal health coverage



India's renewal of Health for All — comprehensive primary health care delivery under Ayushman Bharat

High-yield exam pointers

- **Dates to write:** HFA resolved 30th World Health Assembly **May 1977**; Alma-Ata **6–12 September 1978** (134 countries); Global Strategy **34th WHA, 1981**; HFA-21 adopted **May 1998**; World Health Report **2008**; SDGs **2015**; **Astana 2018**; WHA PHC resolution **2019**.
- **HFA-21 = 10 global targets = 4 + 2 + 4** (health outcome + determinants of health + health policies and sustainable health systems).
- **Four reforms of PHC renewal (WHR 2008):** universal coverage, service delivery (people-centred), public policy, leadership.
- **Three axes of UHC:** breadth (who is covered) — depth (which services) — height (what proportion of cost); SDG target **3.8**.

- ▶ **HFA-2000 global benchmarks:** life expectancy > 60 years, IMR < 50 per 1000 live births; 4 categories of HFA indicators.
- ▶ **Indian renewal chain:** NHP 1983 (HFA) → NHP 2002 → NHP 2017 (UHC) → Ayushman Bharat 2018 = HWC (12 CPHC packages) + PM-JAY.
- ▶ **Dated Indian figures to quote:** IMR 26 (2022), NMR 18 (2022), under-5 mortality 29 (2022), MMR 97 (2018–20), TFR 2.0 (2024), government health spending target 2.5% of GDP by 2025.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir (AAM)**, and Indian Public Health Standards 2022 now classify them as HWC–PHC (rural/urban) and HWC–sub health centre (rural/urban); write the current name first with the former name in brackets once. The indexed corpus carries the Astana Declaration 2018 only as a cited reference rather than as a full text, so the Astana content above is stated at the level the corpus supports.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW Ayushman Arogya Mandir portal, National Health Accounts 2021–22, Indian Public Health Standards 2022, World Malaria Report 2024, WHO Pandemic Agreement 2025

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandir (AAM)** and the vehicle has scaled: over 1.75 lakh AAM operationalised, 82% delivering all 12 expanded service packages by 30 November 2024 (up from 50% in March 2023), 1,16,978 community health officers in position, 1,35,343 Jan Arogya Samitis constituted for community oversight, and about 30.76 crore eSanjeevani teleconsultations delivered; the free-drug and diagnostic entitlement now stands at 106 medicines plus 14 tests at AAM–sub-health centre and 172 medicines plus 63 tests at AAM–primary health centre — the concrete machinery for reducing the "height" (cost-sharing) axis of the universal coverage cube [PMID 41816146].
- ▶ **Facility audits temper the dashboard.** In a Chhattisgarh block study only 58.8% of sub-centre AAMs had a community health officer posted (41.2% vacancy), none held the full complement of essential equipment and medicines, only 17.6% met the diagnostic standards of the operational guidelines, and teleconsultation was available in 70%; performance graded significantly better where a community health officer, laptop/webcam and teleconsultation were present ($p \leq 0.05$). The national community health officer deficit is about 11,111, alongside supply-chain and retention gaps [PMID 40821268; PMID 41816146].

Current national data

- ▶ **India's UHC Service Coverage Index rose from 57 (2015) to 63 (2021)**, above the South-East Asia regional average of 62, while out-of-pocket expenditure fell from 60.6% of total health expenditure in 2014–15 to 39.4% in 2021–22 (National Health Accounts); public-facility outpatient utilisation rose from 91.6 to 134.5 crore and inpatient from 4.3 to 6.3 crore between

2012–13 and 2021–22. Quote these four figures rather than the older textbook values [PMID 41816146].

- **Coverage and financial protection are diverging.** Bayesian projections for seven South Asian countries put India's Composite Coverage Index at 72% (2016) rising to 76% (2020) and projected past the SDG 80% service-coverage benchmark by 2030 — but catastrophic health expenditure in India is projected to reach 26.3% by 2030 (regional rise 7.2% in 2000 to 18.6% by 2030), with rural catastrophic expenditure exceeding urban by 6.0 percentage points, even as impoverishing expenditure falls from 3.5% to 2.6%. India's per-capita primary health care spending (US\$28.93) trails Bhutan (US\$66.00) and Sri Lanka (US\$59.52) [PMID 41192927].

Elimination, surveillance and disease status

- **Malaria has reversed direction.** Reported Indian cases fell to 161,753 in 2021 but rose to 176,522 in 2022 and 255,500 in 2024 — about a 58% increase since 2021 — with India accounting for over half of South-East Asia's malaria burden, mainly *Plasmodium falciparum* and *P. vivax* (World Malaria Report 2024); routine Indian surveys are urged to add sero-epidemiology alongside rapid diagnostic tests and microscopy, to verify elimination, detect low-density or dormant infections and measure intervention impact [PMID 41176154].
- **Kala-azar is at "not yet certified", not "eliminated".** The Kala-azar Elimination Programme — begun in 2005 by Bangladesh, India and Nepal with WHO — targets under 1 new case per 10,000 population at defined administrative levels in each country. Bangladesh first reached it in 2017 and WHO validated sustained elimination in 2023; India remains uncertified despite having maintained the required criteria for two consecutive years, and its case load was nearly five times Bangladesh's and more than twenty times Nepal's, leaving a potentially sizeable human reservoir. By 2023 cases had fallen 98.4% in India and regionwide to about 2% of pre-programme levels — but post-elimination surveillance in Bihar and Jharkhand found rK39 seroprevalence of 20–30% and *Leishmania*-infected sandfly pools (1.4% and 1.2%) despite indoor residual spraying. Lymphatic filariasis meanwhile remains endemic in 345 districts against a 2030 elimination goal [PMID 41856659; PMID 41324335; PMID 41951149].

Guideline and legislative updates

- **Indian Public Health Standards 2022 are now the benchmark for AAM infrastructure,** financed beyond National Health Mission grants through the PM–Ayushman Bharat Health Infrastructure Mission and XV Finance Commission health grants routed via Panchayati Raj institutions and urban local bodies; the stated next step is saturation of the remaining sub-centres and primary health centres followed by National Quality Assurance Standards certification, with SASHAKT as the training-monitoring platform and the Ayushman Bharat Health Account (ABHA) identifier creating longitudinal records [PMID 41816146].
- **Fund-flow rules changed.** The Single Nodal Agency public financial management reform for centrally sponsored schemes, including the National Health Mission, improved cash management, expenditure transparency and alignment of central releases with actual spending in Bihar and Odisha; but weak budget credibility, patchy digital connectivity and thin finance-staff capacity risk widening inequity in access to scheme funds, and compliance pressure on utilisation has outpaced attention to health outcomes [PMID 42480037].

Global and WHO developments

- ▶ **The WHO Pandemic Agreement was adopted at the World Health Assembly in May 2025**, concluding nearly three years of negotiation. Negotiated under Article 19 of the WHO Constitution, it represents a landmark commitment to strengthening pandemic preparedness and response globally. Article 12 requires legal contracts with manufacturers guaranteeing access to at least 20% of relevant pandemic-related products — half donated, the remainder at affordable prices, with distribution coordinated by WHO — operationalised through a Pathogen Access and Benefit-Sharing (PABS) annex to be finalised by April 2026; United States withdrawal from WHO and its refusal to adopt the Agreement constrain the treaty's global reach [PMID 41832575].
- ▶ **Business-as-usual will not deliver SDG target 3.8.** Modelling across 167 countries projects that only 34 countries reach a UHC Service Coverage Index of 80 or more by 2030 under current trends, versus 52 under primary-health-care expansion and 78 under accelerated equity-focused reform; critically, rising service coverage was positively associated with catastrophic health spending (coefficient 0.214), the "decoupling" of service coverage from financial protection that renewed Health for All must correct [PMID 42267267].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1 Man and Medicine: Towards Health for All; Ch 2–3 Concept of Health and Disease and indicators; Ch 9; Ch 10 Health Care of the Community; Ch 14 Sustainable Development Goals; Ch 23 Health Planning and Management); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 28, Health for All in the 21st Century and the 10 global health targets); Sunder Lal — Textbook of Community Medicine (Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community); Harrison's Principles of Internal Medicine 22e (Ch 487, Primary Care and Global Health); Oxford Textbook of Public Health 7e (Ch 47); Textbook of International Health (Ch 7); AFMC Red Book of Community Medicine 2024 (Ch 38, Ayushman Bharat). Named documents: Declaration of Alma-Ata 1978; Global Strategy for Health for All by the Year 2000 (1981); Health for All in the 21st Century (World Health Assembly, 1998); World Health Report 2008 — Primary Health Care: Now More Than Ever; 2030 Agenda for Sustainable Development (2015); Astana Declaration on Primary Health Care (2018); National Health Policy 2017 (Ministry of Health and Family Welfare, Government of India); Ayushman Bharat Comprehensive Primary Health Care through Health and Wellness Centres — Operational Guidelines 2018; Indian Public Health Standards 2022.

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Organization Pandemic Agreement (World Health Assembly, May 2025); National Health Accounts Estimates for India 2021–22; World Malaria Report 2024; Indian Public Health Standards 2022; PM–Ayushman Bharat Health Infrastructure Mission.



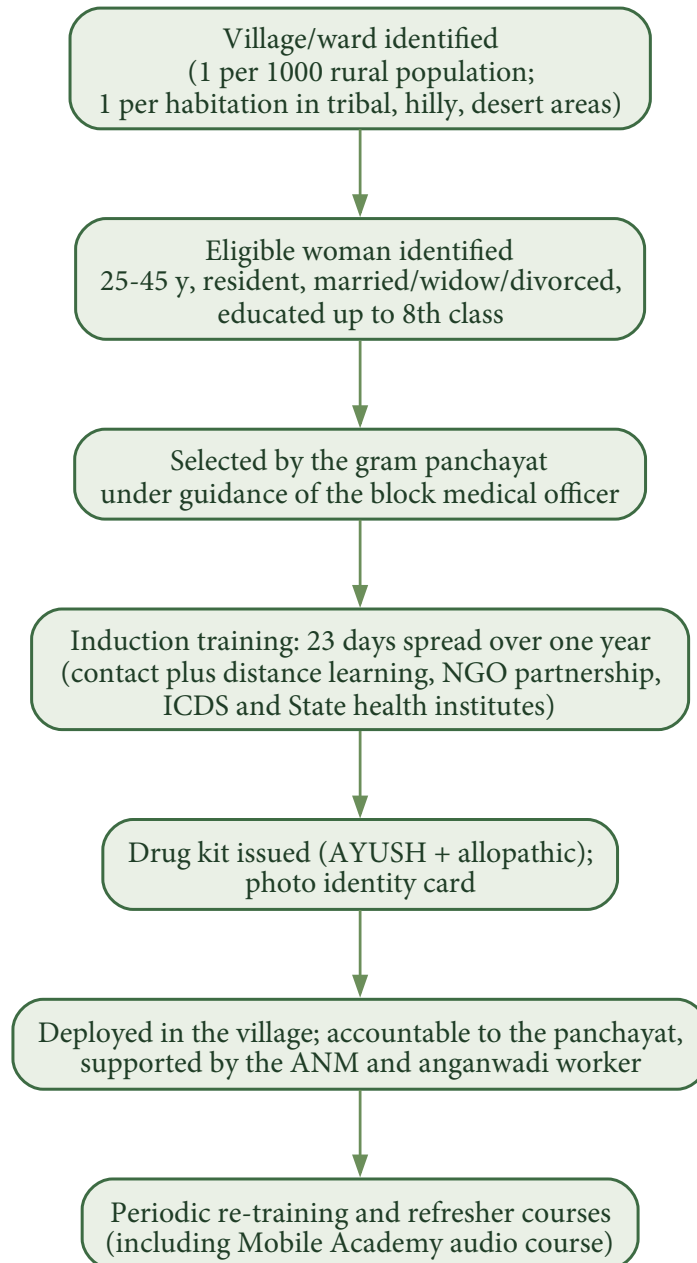
Accredited Social Health Activist (ASHA)

Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — The Accredited Social Health Activist (ASHA) is a trained female community health volunteer, resident in the village she serves, introduced as the first thrust area of the National Rural Health Mission (NRHM, 2005–2012) to act as the **interface between the community and the public health system** (Sunder Lal, Ch 17, p 738). She is not a salaried government employee but an honorary volunteer chosen by, and accountable to, the village panchayat, paid entirely through performance-based incentives (Jugal Kishore, Ch 1, p 134). Over 10.72 lakh ASHAs are currently in position and, with the anganwadi worker and the auxiliary nurse midwife (ANM), form the base of the primary health care pyramid in India (Sunder Lal, Ch 16, p 708).

Selection criteria and norms of coverage

Item	Norm
Sex and residence	Female, preferably a resident of the same village/ward she will serve (Park 28e, Ch 10, p 726)
Marital status	Married / widowed / divorced (Park 28e, Ch 7, p 539)
Age	25–45 years (Park 28e, Ch 7, p 539; Sunder Lal, Ch 17, p 738)
Education	Formal education up to 8th class ; relaxed only where no suitable candidate with this qualification is available (Park 28e, Ch 10, p 726; Jugal Kishore, Ch 1, p 166)
Personal attributes	Communication skills and leadership qualities; adequate representation from disadvantaged population groups so that those groups are served better (Park 28e, Ch 7, p 539)
Who selects	Chosen by the village panchayat, under the supervision and guidance of the block medical officer; accountable to the panchayat (Jugal Kishore, Ch 1, p 134)
Rural norm	One ASHA per 1000 population ; in tribal, hilly and desert areas relaxed to one ASHA per habitation (Park 28e, Ch 7, p 539; Ch 10, p 731)
Urban norm (NUHM)	One ASHA (urban link worker) per 1000–2500 urban poor population , covering about 200–500 households; states may instead entrust her responsibilities to the Mahila Arogya Samiti, in which case the incentives accrue to the Samiti (Park 28e, Ch 7, p 535)



Selection, training and deployment pathway of an ASHA

Training and the support structure

- ▶ **Induction training** — 23 days spread over one year, followed by **periodic re-training**; ASHAs are trained on a pedagogy of public health developed and monitored by a national experts group, implemented through community health resource organisations (Jugal Kishore, Ch 1, p 134).
- ▶ **Training models used** — contact-plus-distance learning, NGO/private partnership, ICDS training centres and State health institutes, and comprehensive women's health and empowerment models (Jugal Kishore, Ch 1, p 134).
- ▶ **Mobile Academy** — a free audio training course delivered on the ASHA's own mobile phone to refresh her knowledge base and improve her communication skills; an anytime-anywhere course, running in 16 States (Sunder Lal, Ch 17, p 739).

- ▶ **Resource persons** — the anganwadi worker and the ANM act as resource persons for her training; the ANM ensures she receives her performance compensation and travelling/daily allowance while attending training (Park 28e, Ch 10, p 727).
- ▶ **Financing** — the Government of India bears the cost of her training, her incentives and her medical kit; the remainder is funded from the State's financial envelope (Jugal Kishore, Ch 1, p 134).
- ▶ **Institutional platform** — she is a member of the **Village Health, Sanitation and Nutrition Committee (VHSNC)** along with the sarpanch, the female multipurpose health worker, the anganwadi worker, a school teacher and a self-help-group member; the committee holds an untied fund of Rs 10,000 per annum and prepares the village health action plan (Sunder Lal, Ch 16, p 697). By 2021, 5.55 lakh VHSNCs had been set up (Park 28e, Ch 7, p 539).

Roles and responsibilities

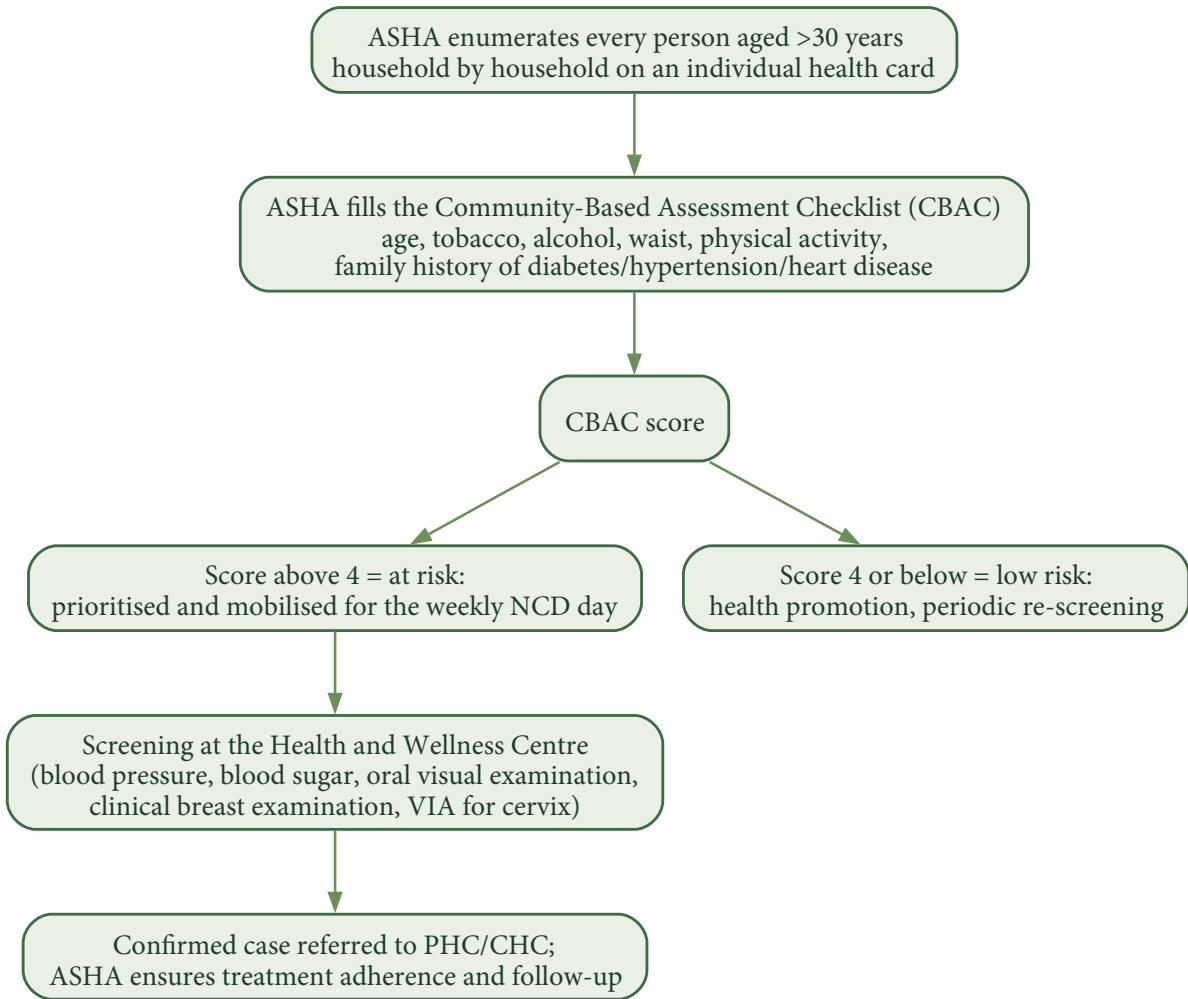
Domain	What the ASHA does
Awareness (health promotion)	Creates awareness on the determinants of health — nutrition, basic sanitation and hygienic practices, healthy living and working conditions, the health services that exist and the need to use them in time (Park 28e, Ch 10, pp 726–727)
Counselling	Counsels women on birth preparedness, importance of safe delivery, breastfeeding and complementary feeding, immunisation, contraception, prevention of reproductive tract and sexually transmitted infections, and care of the young child (Park 28e, Ch 10, p 727)
Mobilisation	Mobilises the community and facilitates access to services at the anganwadi centre, sub-centre and primary health centre — immunisation, antenatal and postnatal check-up, supplementary nutrition, sanitation (Park 28e, Ch 10, p 727)
Planning	Works with the VHSNC of the gram panchayat to develop a comprehensive village health plan (Park 28e, Ch 10, p 727)
Escort and referral	Arranges escort for, or accompanies, pregnant women and children needing treatment or admission to the nearest pre-identified facility — PHC, community health centre or first referral unit (Park 28e, Ch 10, p 727)
Curative care	Provides primary medical care for minor ailments — diarrhoea, fever, first aid for minor injuries; acts as a DOTS provider (directly observed treatment, short-course) under the national tuberculosis programme (Park 28e, Ch 10, p 727; Sunder Lal, Ch 17, p 738)
Depot holder	Holds and distributes oral rehydration therapy, iron and folic acid tablets, chloroquine, disposable delivery kits, oral pills and condoms; a drug kit with both AYUSH and allopathic formulations, decided by a Government of India expert/technical advisory group, is supplied to each ASHA and replenished through the anganwadi worker (Park 28e, Ch 10, p 727)
Vital events and surveillance	Records and reports all births and deaths, and any unusual health problem or disease outbreak, to the sub-centre/PHC (Park 28e, Ch 10, p 727)
Sanitation	Promotes construction of household toilets under the total sanitation campaign (Park 28e, Ch 10, p 727)

Domain	What the ASHA does
Disease-specific tasks	Participates in National Filaria Day; identifies skin patches with loss of sensation and brings them to the health worker's notice, and ensures leprosy patients take regular treatment (Sunder Lal, Ch 17, pp 738–739)
Records maintained	Separate survey and service registers — household survey, eligible couple, antenatal, births and deaths, immunisation and home-based newborn care registers (Sunder Lal, Ch 8, p 533)
Work pattern	Routine home visits of up to 2 hours a day, with more time given to priority households — those with a pregnant woman, a newborn, a child under 2 years, a malnourished child, an eligible couple aged 20–29 years, the poorest families, or a tuberculosis case (Sunder Lal, Ch 2, p 83)

Her role as a provider may be graded upward: States can train her further for newborn care and for management of a range of common ailments, particularly childhood illnesses (Park 28e, Ch 10, p 727).

Expanded role under the National Health Mission

Component	ASHA's specific task
Home-based newborn care (HBNC)	She is the main provider ; 6 home visits after an institutional delivery (days 3, 7, 14, 21, 28, 42) and 7 visits after a home delivery (days 1, 3, 7, 14, 21, 28, 42), the first within 6 hours of birth; after caesarean section with discharge on day 5–6, visits run from day 7 to day 42 (Sunder Lal, Ch 10, p 594; Park 28e, Ch 7, p 550)
HBNC content	Weights and records birth weight on the mother and child protection card, ensures early initiation of breastfeeding, delays bathing for 7 days, ensures birth-dose BCG, zero-dose OPV and hepatitis B, ensures birth registration, gives extra visits to low-birth-weight babies, identifies newborn illness for referral (Sunder Lal, Ch 10, p 594)
Home-based care for young child (HBYC)	Five further home contacts in the first week of the 4th, 7th, 10th, 13th and 16th month of the child's age (Sunder Lal, Ch 10, p 594)
Non-communicable disease screening	Enumerates all persons aged over 30 years household by household (about 37% of the population — roughly 1850 persons in 5000) and completes the Community-Based Assessment Checklist (CBAC) ; she is trained in oral visual examination, clinical breast examination and visual inspection with acetic acid (Sunder Lal, Ch 8, p 503)
Health and Wellness Centre team	At the upgraded sub-centre the team is ASHAs and ANMs led by a mid-level health provider (BAMS physician or BSc nurse); one ASHA per 1000 population is attached to the Health and Wellness Centre as part of the team (Sunder Lal, Ch 17, p 742; Park 28e, Ch 10, p 731)
Convergent action	Facilitates the VISHWAS campaign (village-based initiative to synergise health, water and sanitation) — eleven monthly themed campaigns led by the VHSNC, rolled out October 2017 (Sunder Lal, Ch 1, p 42)
Family planning	Along with trained dais she is the lynchpin of the family planning delivery system; a scheme uses her to deliver contraceptives at the beneficiary's doorstep (Park 28e, Ch 8, p 612)

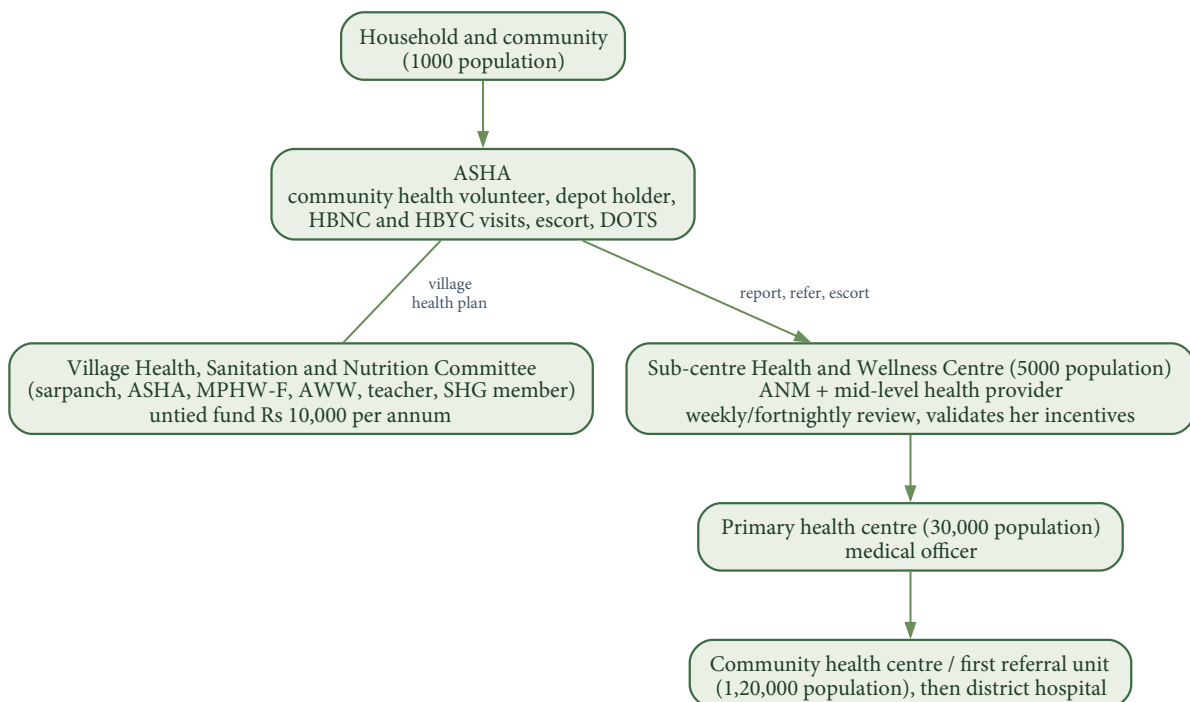


ASHA-initiated pathway for community-based screening of non-communicable disease

Incentives and payment

Head	Provision
Basis of payment	Honorary volunteer — no salary; performance-based incentive only, for promoting household toilets, universal immunisation, referral and escort services under reproductive and child health, and other programmes (Jugal Kishore, Ch 1, p 134)
Home-based newborn care	Rs 250 for the full set of postnatal home visits, paid on the 45th day after validation by the ANM as facilitator (Sunder Lal, Ch 10, p 594)
Low-birth-weight / SNCU follow-up	Additional incentive of Rs 50 per month for follow-up of low-birth-weight babies and newborns discharged from a sick newborn care unit, approved by the Mission Steering Group of the National Health Mission on 6 December 2013 (Park 28e, Ch 7, p 550)
Janani Suraksha Yojana	Paid in two instalments — 50% after the beneficiary is discharged, provided the ASHA accompanied her for delivery; the remaining 50% one month after delivery, once BCG has been given and she has assisted postnatal care and birth registration (Jugal Kishore, Ch 1, p 152)

Head	Provision
Urban (suggested monthly package)	Outreach sessions Rs 200; monthly Mahila Arogya Samiti meeting Rs 100; monthly meeting at the urban health centre Rs 200; health and nutrition day with the anganwadi worker Rs 100; community meeting Rs 50 each (upper limit Rs 200); baseline survey Rs 5 per household once a year; record maintenance Rs 50 per month; Rs 5 per fully immunised child (Jugal Kishore, Ch 1, p 168)
Programme-linked incentives	Built into national schemes — Janani Suraksha Yojana, tuberculosis, vector-borne disease and sterilisation programmes (Jugal Kishore, Ch 1, p 168)
Other supports	Photo identity card to raise self-esteem; ASHA Gruha (lodging) and help desks at facilities (Sunder Lal, Ch 17, p 758; Jugal Kishore, Ch 1, p 157)



ASHA's position in the village-to-district service, supervision and referral chain

Monitoring, evaluation and constraints

Aspect	Detail
Who supervises	The ANM holds a weekly/fortnightly meeting with her, reviews the fortnight's work, guides her on problems, informs her of outreach session dates, and teaches her the danger signs of pregnancy and labour (Park 28e, Ch 10, p 727)
Who monitors performance	The VHSNC monitors the services provided by the anganwadi worker, the ASHA and the sub-centre (Sunder Lal, Ch 17, p 744)
Convergence with ICDS	The anganwadi worker guides her in organising the monthly health day, is depot holder for replacement of consumed drugs, and updates the eligible-couple and infant lists with her help; she in turn mobilises pregnant and lactating women and infants for supplementary nutrition (Park 28e, Ch 10, p 727)

Aspect	Detail
Documented constraints	Irregular drug-kit replenishment; poor coordination between anganwadi worker, ANM and ASHA; services not reaching every household in some States; irregular and non-transparent payments; lack of effective institutional support structures; weak supervision (Jugal Kishore, Ch 1, p 157)
Attrition	20–30% of recruited ASHAs dropped out on learning that payment was performance-based and not permanent government service (Jugal Kishore, Ch 1, p 157)
Programme-overlap criticism	Trained dais under the earlier dai-training scheme (1,21,017 trained) were largely displaced when ASHAs took over the village role, a duplication of training resources (Jugal Kishore, Ch 1, p 157)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **home-based newborn care visit timeline** — a single horizontal day axis with two rows of marked visits:

- ▶ Axis labels: Day 1, Day 3, Day 7, Day 14, Day 21, Day 28, Day 42
- ▶ Row 1 label: **Institutional delivery** — 6 visits (mark days 3, 7, 14, 21, 28, 42)
- ▶ Row 2 label: **Home delivery** — 7 visits (mark days 1, 3, 7, 14, 21, 28, 42)
- ▶ Left-hand note: *First postnatal contact within 6 hours of birth*
- ▶ Right-hand note: *Incentive Rs 250, paid on day 45 after ANM validation*
- ▶ Below the axis: **Home-based care for young child** — 5 visits in the first week of months 4, 7, 10, 13 and 16

High-yield exam pointers

- ▶ **Selection quartet to write first** — female village resident, 25–45 years, married/widow/divorced, 8th class pass; **1 per 1000 population rural**, **1 per habitation** in tribal/hilly/desert areas, **1 per 1000–2500 urban poor**.
- ▶ **Training** — induction **23 days over one year** plus periodic re-training; refresher through the **Mobile Academy** mobile-phone audio course.
- ▶ **HBNC schedule (most-asked number)** — **6 visits** institutional (3, 7, 14, 21, 28, 42) and **7 visits** home delivery (1, 3, 7, 14, 21, 28, 42); **HBVC 5 visits** at months 4, 7, 10, 13, 16.
- ▶ **Money** — Rs 250 for HBNC on day 45 after ANM validation; Rs 50 per month for low-birth-weight/SNCU follow-up; Janani Suraksha Yojana incentive in two 50% instalments.
- ▶ **CBAC cut-off** — score **above 4** on the Community-Based Assessment Checklist means at risk of non-communicable disease and priority for the weekly NCD day.
- ▶ **She is an activist, not an employee** — honorary volunteer, chosen by and accountable to the gram panchayat, paid only by performance-based incentive; the Government of India funds her training, incentives and drug kit.
- ▶ **Numbers** — over **10.72 lakh** ASHAs in position; **5.55 lakh** VHSNCs set up by 2021, each with an untied fund of Rs 10,000 a year.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved textbooks give the count of "over 10.72 lakh ASHAs in position" without attaching a year, and Park 28e separately quotes 9.39 lakh ASHAs selected and supplied with drug kits in its family planning chapter — quote the figure with the book you are citing rather than as a dated national statistic; the Revised National Tuberculosis Control Programme is now the National Tuberculosis Elimination Programme, and Ayushman Bharat Health and Wellness Centres have been renamed Ayushman Arogya Mandirs (Park 28e, Ch 7, p 541), so use the current name first and the former name once in parentheses.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoHFW, PIB, National Health Mission, WHO

Programme and policy updates (India)

- ▶ **Educational qualification has been revised upward in practice.** Acting on recommendation 79 of the 15th Lok Sabha Committee on Empowerment of Women, the Ministry of Health and Family Welfare directed States to give due preference in ASHA selection to women qualified at least up to 10th standard wherever such candidates are available in good number; the 8th-class floor is retained so that tribal and marginalised habitations are not left without an ASHA, the rule applies only to fresh selections and not to serving ASHAs, and serving ASHAs are to be encouraged to complete matriculation and 12th pass so that they become eligible to train as ANMs — the current National Health Mission description of ASHA now carries the 10th-standard preference. Write "8th class as the minimum, 10th standard preferred where available" rather than 8th class alone. [MoHFW 2012-Aug]
- ▶ **Social security has widened beyond the insurance package.** About 37 lakh families of ASHAs, anganwadi workers and anganwadi helpers were brought under Ayushman Bharat PM-JAY (Rs 5 lakh per family per year) from February 2024, adding hospitalisation cover to the existing PMJJBY/PMSBY benefits. Under the Extended Pradhan Mantri Surakshit Matritva Abhiyan strategy the ASHA now earns a separate incentive for accompanying an identified high-risk pregnant woman for three extra visits over and above the PMSMA visit — 1.03 crore high-risk pregnancies have been identified so far. [PIB 2026-Jan]
- ▶ **ASHA-led task-shifting for chronic disease now has trial-grade support.** The IMPACT Diabetes cluster-randomised trial across 16 villages and 8 primary health centres in Guntur (Andhra Pradesh) and Rohtak (Haryana) had ASHAs screen 1,785 adults aged 30 years and over, confirm 418 with diabetes, and deliver nine months of tablet-based clinical-decision-support care under physician supervision: 21.8% versus 10.3% achieved a fall in HbA1c of 0.5% or more, primary-health-centre physician visits rose (85.0% versus 29.8%) and glucose-lowering medicine use rose (63.0% versus 43.1%). Mean HbA1c did not separate (8.6% versus 8.8%), and the authors flag her existing workload, the absence of any long-term support strategy and unreliable medicine supply as the limits on scale-up. [PMID 41120559]

Current national data

- **Home-based newborn care performance, 2025.** Between January and September 2025, 1.11 crore newborns (89%) received the scheduled ASHA visits nationally and 7.94 lakh of them (7.1%) were identified as sick and referred. Home-based care of the young child has been approved for 751 districts, including every aspirational district, in 2025–26; the ministry states the HBYC schedule as visits in the 3rd, 6th, 9th, 12th and 15th month — as does the published literature — rather than the first week of the 4th, 7th, 10th, 13th and 16th month that the textbook describes. Under the urban mission 93,554 ASHAs are in position with 80,566 Mahila Arogya Samitis formed, and 1,81,873 Ayushman Arogya Mandirs were operational on 30 November 2025. [PIB 2026-Jan] [PMID 42125563]
- **Knowledge and coverage remain the weak links.** A meta-analysis of 37 cross-sectional studies published between 2005 and 2022 put pooled adequate knowledge among ASHAs at 62% for maternal health, 69% for neonatal and child health, 62% for communicable disease and 73% for non-communicable disease; the determinants were level of education, years of experience, inadequate supervision and inadequate training. A mixed-methods study in a rural block of Khordha district, Odisha found 16.7% of infants received no home visit at all and only 13.3% received six or more; the barriers were absent refresher training (one ASHA's last training was five years earlier), non-functioning kit equipment such as the thermometer and watch, inability to count the respiratory rate, and filling the HBNC booklet at home after the visit. [PMID 39799364] [PMID 42125563]

Elimination, surveillance and disease status

- **Malaria — her incentive has more than doubled but her reach has not.** The incentive for ensuring complete treatment has been raised from Rs 75 to Rs 200 per confirmed malaria case; national annual parasite incidence was 0.18 in 2024, 160 districts reported zero malaria cases over 2022–24, India has exited the WHO high-burden-to-high-impact group, and indigenous transmission is targeted for interruption by 2027 under the National Framework for Malaria Elimination 2016–2030. Against that, an implementation-fidelity study in the PHC Ujina catchment of Nuh, Haryana found only 4 of 222 febrile under-five children (1.8%) were taken to the ASHA first, against a programme benchmark of 80%, and only 4.0% were tested for malaria within 24 hours — while all eight children she confirmed positive began treatment the same day and completed the full course, with follow-up in seven of the eight. Nearly three-quarters of mothers went first to a hakim or traditional healer. Quote the fidelity paradox — good protocol adherence, poor first-contact utilisation. [PIB 2026-Jan] [PMID 41356870]
- **Elimination target years have moved since the textbook edition.** All 633 endemic blocks reached the kala-azar target of under one case per 10,000 population at block level by end-2023 and were still sustaining it in October 2025; lymphatic filariasis elimination is targeted for 2027, ahead of the global 2030 goal, with 143 of 348 endemic districts having stopped mass drug administration after clearing transmission assessment survey 1; the Zero Measles-Rubella Campaign launched in April 2025 aims at MR elimination by 2026; and TB incidence fell 21% from 237 to 187 per lakh population between 2015 and 2024, with TB deaths down from 28 to 21 per lakh. [PIB 2026-Jan]

Guideline and legislative updates

- **Population-based NCD screening has become her largest single workload.** The programme is now the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD), renamed from NPCDCS, and screens five conditions — hypertension, diabetes, and oral, breast and cervical cancer — in everyone aged over 30. On the National NCD Portal, 74.97 crore persons aged 30 and above had been enumerated and 51.53 crore screened by 22 December 2025, the enumeration and CBAC step being ASHA work. [PIB 2026-Jan]
- **Payment and digital tooling are the two live reform fronts.** A qualitative study of the ASHA payment chain in Maharashtra and Assam traced its five steps — recording work, claim submission, verification, processing, disbursement — and found the first three still paper-based: an ASHA keeps more than ten registers, buys photocopies as proof of service costing about 5% of her monthly incentive, and travels for the multiple mandated signatures that make verification the rate-limiting step. Full digitisation of disbursement through the public financial management system improved timeliness, transparency and satisfaction over cash and cheque, but digital tools alone did not fix a district where managers de-prioritised the data entry. Separately, a digital health competency framework developed against the Ayushman Bharat Digital Mission rollout — which the authors argue is not yet matched by any standardised competency framework — and piloted with frontline health workers in two Uttar Pradesh districts scored 32 ASHAs at 39.9% (mean 11.97 of 30) on baseline digital competency, data collection being the weakest domain at 32.5%; relevant as U-WIN replaces the paper immunisation due-list nationwide. [PMID 39953777] [PMID 42266401]

Global and WHO developments

- **The cadre carries a WHO award and an unresolved status question.** India's roughly one million ASHAs were named one of six recipients of the WHO Director-General's Global Health Leaders Award at the opening of the 75th World Health Assembly in May 2022 — the only collective among the six — for linking rural communities to primary health care through the COVID-19 pandemic. The accompanying commentary sets the honour against the arithmetic: these women earn about US\$20–60 a month for a 6–7 hour day that stretched to 10–12 hours at the pandemic peak, and did surveillance, contact tracing and screening of returning migrants with minimal or no personal protective equipment while exposed to stigma, violence and social boycott. Its demands are recognition and respect from government and community, adequate training for each new role, protection from violence and infection, and appropriate and timely remuneration; its charge that seeking voluntary service from this cadre is unjust and unethical is the standard "activist or worker" argument to deploy in a critique question. [WHO 2022-May] [PMID 37384262]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India; Ch 8 — Demography and Family Planning; Ch 10 — Health Care of the Community); Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 2, Ch 8, Ch 10, Ch 16, Ch 17); Jugal Kishore — National Health Programmes of India (Ch 1, National Rural Health Mission and National Urban Health Mission sections). Coverage note: the corpus carries no separate captioned figure of the ASHA scheme, so the diagrams above are drawn/structural rather than reproduced plates, and it does not describe the ASHA facilitator or block-level ASHA support structures in enough detail to state them.

Recent-advances sources: 41120559 — Preliminary effectiveness and feasibility of ASHA-led mobile health intervention for diabetes care in Indian primary health care settings, *Scientific Reports* 2025, DOI [10.1038/s41598-025-20728-w](https://doi.org/10.1038/s41598-025-20728-w); 39799364 — Knowledge of Accredited Social Health Activists in India: a systematic review and meta analysis of evidence drawn from primary studies published between 2005 and 2022, *BMC Health Services Research* 2025, DOI [10.1186/s12913-024-12004-w](https://doi.org/10.1186/s12913-024-12004-w); 42125563 — Assessment of Home-based Newborn Care in a Rural Block of Odisha Using a Structured Framework: A Concurrent Mixed-method Study, *Indian Journal of Community Medicine* 2026, DOI [10.4103/ijcm.ijcm.666.24](https://doi.org/10.4103/ijcm.ijcm.666.24); 41356870 — Implementation Fidelity of Accredited Social Health Activist (ASHA) Workers in Malaria Case Management in a High-Burden District of North India, *Cureus* 2025, DOI [10.7759/cureus.96103](https://doi.org/10.7759/cureus.96103); 39953777 — Community health worker payment processes: a qualitative assessment of experiences in two Indian states, *Health Policy and Planning* 2025;40:483–95, DOI [10.1093/heapol/czaf010](https://doi.org/10.1093/heapol/czaf010); 42266401 — Strengthening digital competencies in India's health workforce: development and feasibility evaluation of a digital health competency framework for frontline healthcare workers in Uttar Pradesh, *Oxford Open Digital Health* 2026, DOI [10.1093/oodh/oqag009](https://doi.org/10.1093/oodh/oqag009); 37384262 — India's one million Accredited Social Health Activists (ASHA) win the Global Health Leaders award at the 75th World Health Assembly: Time to move beyond rhetoric to action?, *The Lancet Regional Health — Southeast Asia* 2022;3:100029, DOI [10.1016/j.lansea.2022.100029](https://doi.org/10.1016/j.lansea.2022.100029). Plus Ministry of Health and Family Welfare D.O. No. H-11016/1/2012-NRHM-II dated 22 August 2012 (educational qualification for ASHA selection) and the National Health Mission "About ASHA" page; Ministry of Health and Family Welfare, *Initiatives and Achievements* 2025 (PIB, January 2026); WHO Director-General's Global Health Leaders Award, 75th World Health Assembly, 22 May 2022.

♦ ♦ ♦

Appropriate technology in primary health care

Asked in: Paper IV 28-Jun-2024 — RGUHS MD Community Medicine

Definition — The Alma-Ata Conference (1978) defined primary health care as essential health care based on practical, **scientifically sound and socially acceptable methods and technology**, made universally accessible to individuals and families through their full participation and at a cost the community and country can afford (Park 28e, Ch 2, p 49; Oxford Textbook of Public Health 7e, Ch 16, p 630). **Appropriate technology** is one of its four operating principles and is defined by the World Health Organization (WHO) as *"technology that is scientifically sound, adaptable to local needs, and acceptable to those who apply it and those for whom it is used, and that can be maintained by the people themselves in keeping with the principle of self reliance with the resources the community and country can afford"* (Park 28e, Ch 10, p 720). "Appropriate" is a test of fit between a technology and its setting, not a synonym for cheap or second-rate.

Where the principle sits among the four principles of primary health care

Principle	Core idea	Indian illustration
Equitable distribution	Services shared equally by all irrespective of ability to pay; rich or poor, urban or rural (Park 28e, Ch 10, p 719)	Shifting the centre of gravity of the health budget from cities to the periphery
Community participation	Continuing involvement of people in planning, implementation and maintenance, with maximum reliance on local manpower, money and materials (Park 28e, Ch 10, p 720)	Accredited social health activist (ASHA), anganwadi worker, trained <i>dai</i> ; modelled on China's <i>bare-foot doctors</i>
Intersectoral coordination	Health cannot be delivered by the health sector alone — agriculture, animal husbandry, food, industry, education, housing, public works and communication are all involved (Park 28e, Ch 10, p 720)	Joint planning with other sectors to avoid duplication
Appropriate technology	Scientifically sound, locally adaptable, acceptable, community-maintainable and affordable methods and tools (Park 28e, Ch 10, p 720)	Oral rehydration solution, growth chart, standpipes, point-of-care tests

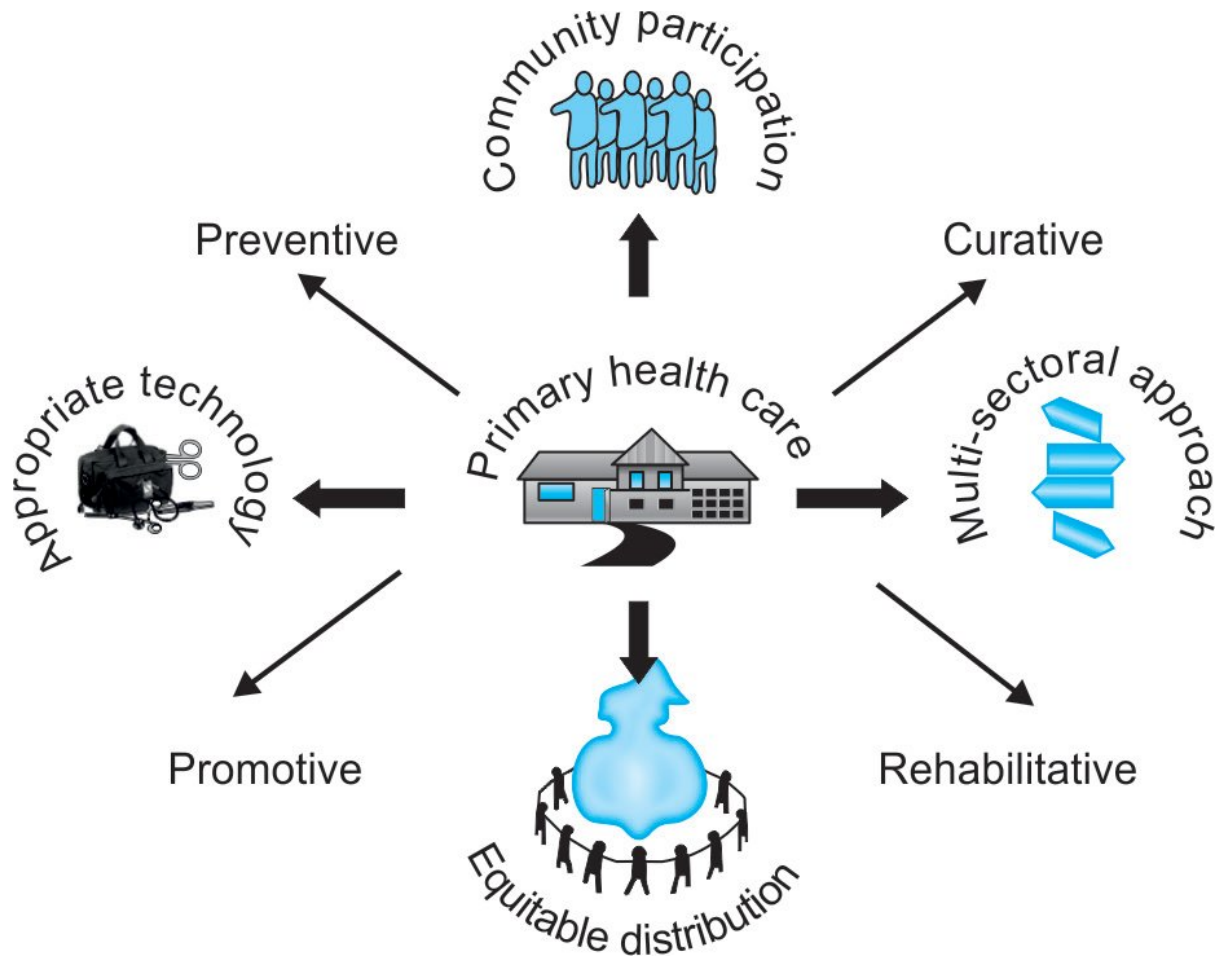


Fig 34.2 — Principles and components of primary health care (Suryakantha Environment Best, p.780)

The four are interdependent: technology the community cannot maintain defeats participation, and technology that captures the budget defeats equity.

Defining attributes — the tests a technology must pass

Attribute (from the WHO definition)	What it demands	Fails when
Scientifically sound	Proven efficacy and safety; not folk practice dressed up as low cost	Unvalidated remedies offered as "village-level" care
Adaptable to local needs	Matched to the local disease burden, climate, power supply and storage conditions	Analysers needing uninterrupted power at a sub-centre
Acceptable	To <i>both</i> those who apply it and those on whom it is used	Water-seal latrines given free by government in the 1960s went unused because the service approach was not based on the felt needs of the people (Park 28e, Ch 22, p 1016)
Maintainable by the people themselves	Self-reliance — local repair, local supply, no permanent external dependence	Equipment idle for want of a spare part or a service contract

Attribute (from the WHO definition)	What it demands	Fails when
Affordable	To the community <i>and</i> the country, counting recurrent as well as capital cost	Costly procedures adopted when cheaper valid alternatives exist

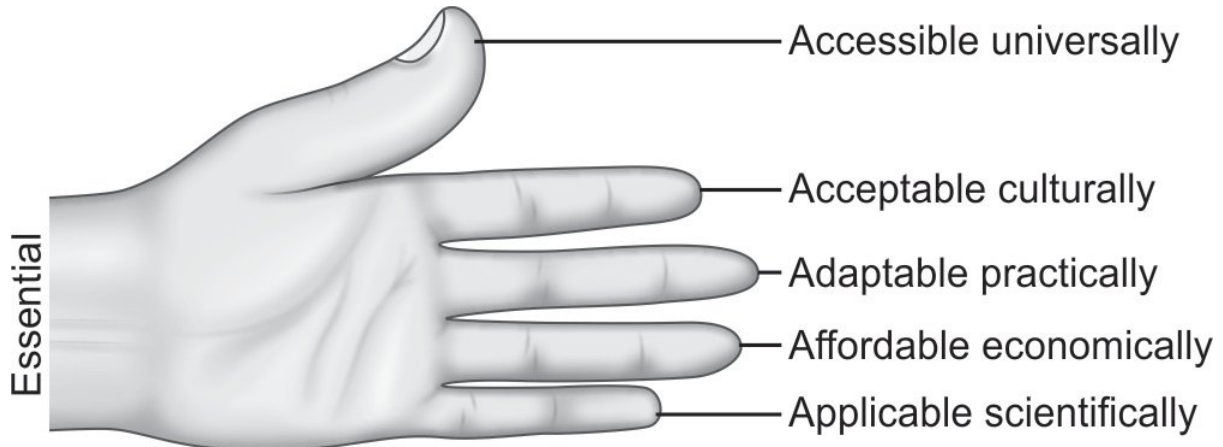
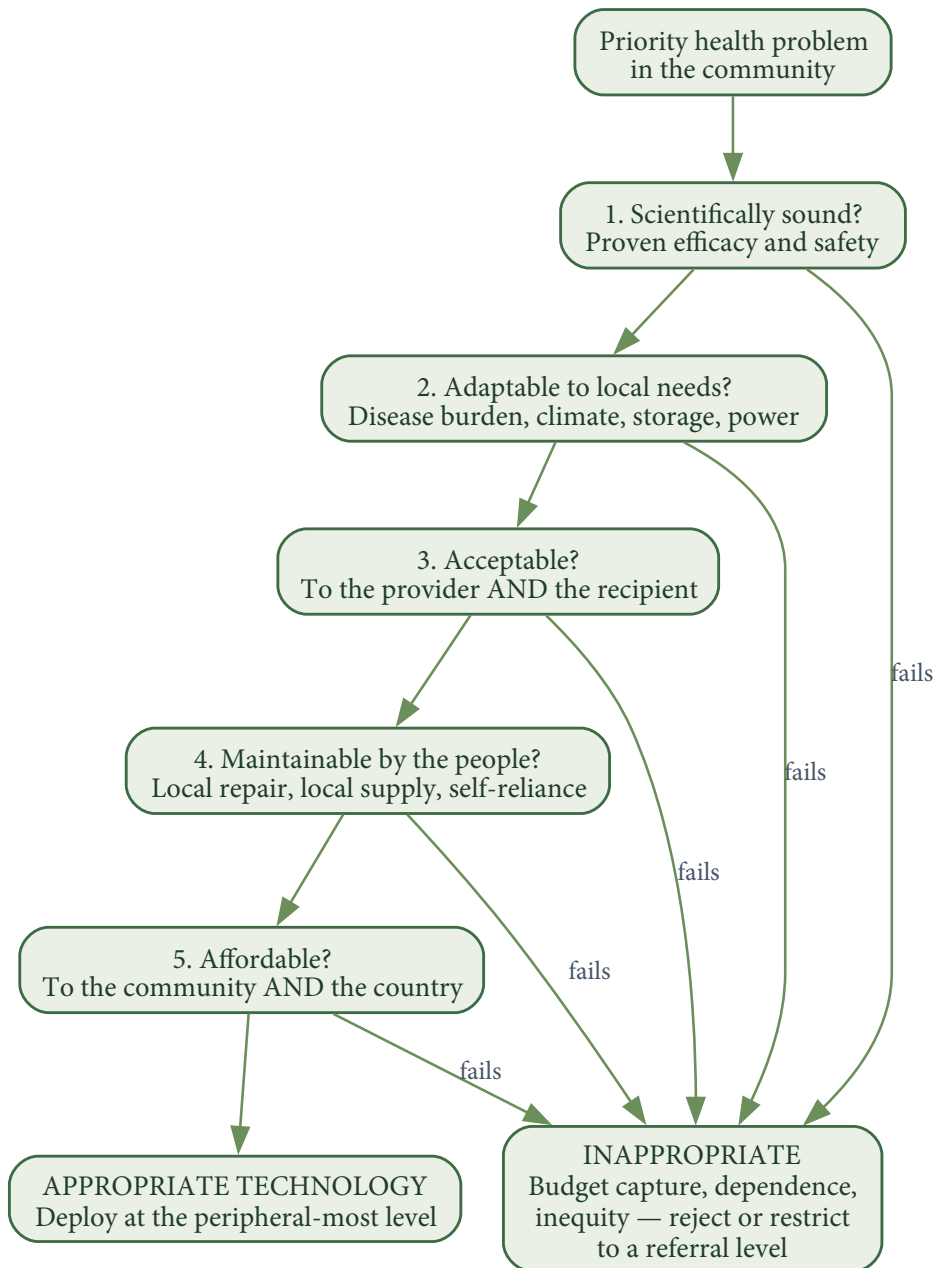


Fig 34.4 — Attributes of primary health care (Suryakantha Environment Best, p.781)

Why the principle matters — the argument against inappropriate technology

- **Budget capture.** Large, luxurious hospitals totally inappropriate to local needs absorb a major part of the national health budget and block improvement of general health services (Park 28e, Ch 10, pp 719–720); the hospital has been criticised as *an ivory tower of disease* that takes 50 to 80 per cent of the health budget (Park 28e, Ch 2, pp 67–68).
- **Opportunity cost at the point of care.** The same objection applies to costly equipment, procedures and techniques when cheaper, scientifically valid and acceptable ones exist — oral rehydration fluid instead of routine intravenous therapy, and socially acceptable standpipes, financially far more feasible than house-to-house water connections (Park 28e, Ch 10, pp 719–720).
- **Coverage over sophistication.** The large hospital serves only a small part of the population within a small radius and its services are mostly curative; such hospitals are "*more ivory towers of diseases than centres for the delivery of comprehensive health care services*" (Park 28e, Ch 10, p 721).
- **Sustainability.** Technology the community cannot maintain creates permanent dependence on external funds and expertise, contradicting the self-reliance clause of the WHO definition.
- **Equity.** Being deliverable at the peripheral-most level by a locally selected worker, appropriate technology reaches the rural and urban poor whom high technology misses.

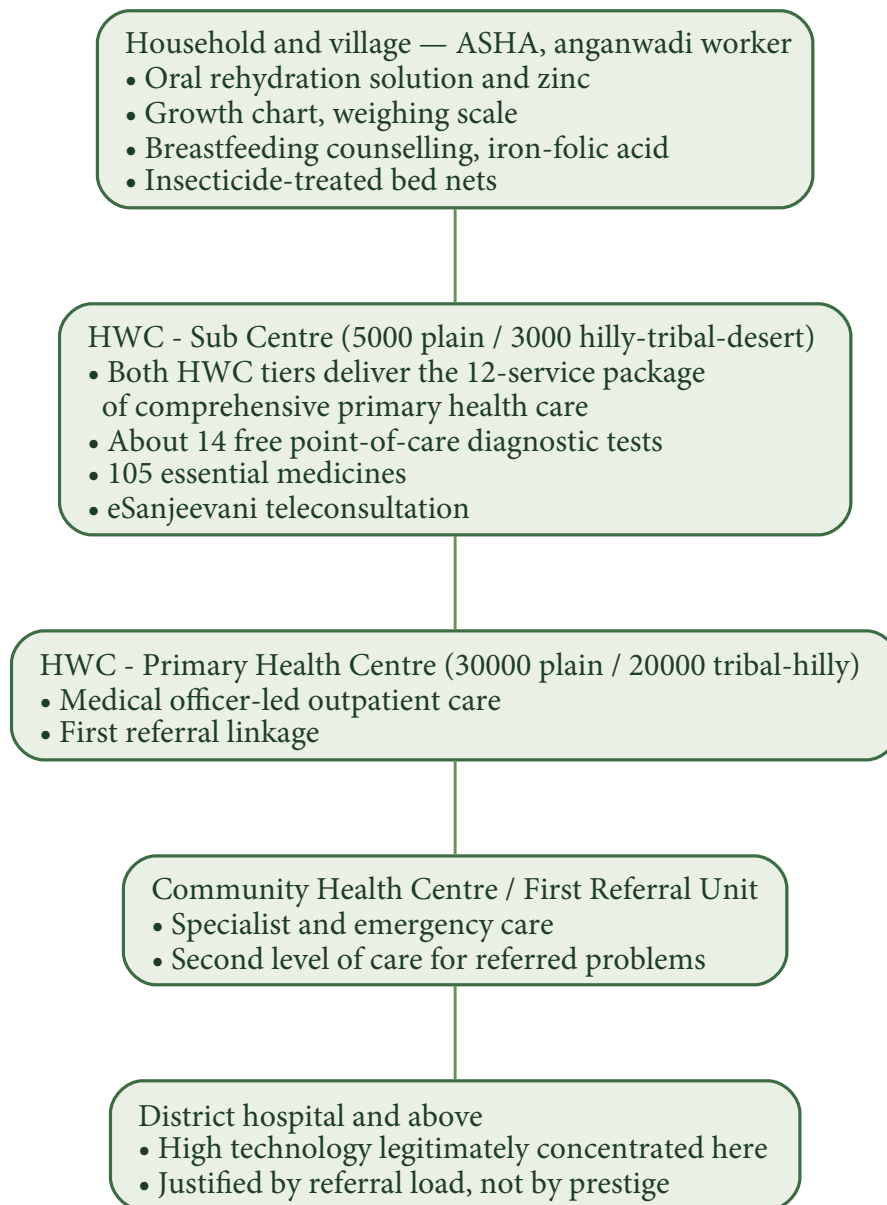


The five tests a technology must pass before it is "appropriate" for primary health care

Appropriate technology in Indian primary health care practice

Health problem	Appropriate technology	Delivery point / programme
Childhood diarrhoea	Oral rehydration solution (with zinc)	ASHA and anganwadi worker; O of UNICEF's GOBI strategy (Park 28e, Ch 25, pp 1058–1059)
Undernutrition	Growth chart with monthly weighing — a low-cost technology for reducing infant mortality (Park 28e, Ch 9, p 675)	Village health and nutrition day; G of GOBI
Infant feeding	Breastfeeding promotion — the single most effective measure for lowering infant mortality (Park 28e, Ch 9, p 675)	B of GOBI

Health problem	Appropriate technology	Delivery point / programme
Vaccine-preventable disease	Immunisation with a maintained cold chain	I of GOBI — measles, diphtheria, polio, pertussis, tetanus, tuberculosis (Park 28e, Ch 25, pp 1058–1059)
Anaemia, micronutrient deficiency	Iron–folic acid tablets, vitamin A, iodised salt	Food supplements — the FFF added to GOBI (Sunder Lal, Ch 17, p 741)
Undiagnosed common conditions	About 14 free point-of-care tests — haemoglobin, urine, blood sugar, malaria, dengue, HIV, syphilis, hepatitis B, sputum for tuberculosis, filaria, sickle cell, iodine in salt, water samples	Ayushman Arogya Mandir (formerly Ayushman Bharat Health and Wellness Centre, HWC) (Sunder Lal, Ch 17, p 748)
Access to medicines	Essential medicine list of 105 medicines per HWC; National List of Essential Medicines 2022	HWC; free drugs policy (Sunder Lal, Ch 17, p 748; Park 28e, Ch 24, p 1041)
Access to specialist opinion	eSanjeevani telemedicine — doctor-to-doctor from HWCs, and patient-to-doctor outpatient consultation	About 1.5 lakh HWCs connected (Sunder Lal, Ch 17, p 748)
Unsafe water	Standpipes, chlorination	Intersectoral, with local bodies (Park 28e, Ch 10, p 720)
Unsafe delivery, low service reach	Locally selected and locally trained village health guides, trained <i>dais</i> , ASHAs — "human technology" that overcomes cultural and communication barriers	Rural health scheme; ASHA one per 1000 population (Park 28e, Ch 10, pp 720, 726)



Appropriate technology matched to each level of the Indian health care chain

How appropriateness is judged and safeguarded

- ▶ **Explicit selection criteria for medicines** — a medicine enters the National List of Essential Medicines only if it is licensed in India, useful in a disease that is a public health problem in India, of proven efficacy and safety on valid scientific evidence, **cost-effective**, aligned with current treatment guidelines, **stable under Indian storage conditions**, judged on total treatment price rather than unit price, and assigned to the level of care — primary, secondary or tertiary — at which it is appropriate (Park 28e, Ch 24, p 1041).
- ▶ **Health technology assessment and cost-utility analysis** — the two techniques now used to allocate resources under scarcity, assessing effectiveness and cost-effectiveness respectively before a technology is adopted (Oxford Textbook of Public Health 7e, Ch 94, p 3528).
- ▶ **Facility standards** — Indian Public Health Standards 2022 set the benchmark of infrastructure, manpower and equipment for each level, defining services as "essential" (minimum assured) and "desirable" (Sunder Lal, Ch 17, p 748).

- ▶ **Generic and essential-medicines policy** — generic medicines sold near manufacturing cost; a national policy encouraging their use can save an estimated 60% of medicine costs (Oxford Textbook of Public Health 7e, Ch 107, pp 4046–4047).
- ▶ **Periodic reassessment** — appropriateness is not permanent. Affordable rapid diagnostics for multiple pathogens will shift sexually transmitted infection care from syndromic to causative treatment, replacing a technology appropriate in its time (Park 28e, Ch 5, p 409).

Critical comment — the limits of the principle

- ▶ **Appropriate is not inferior.** Primary health care is not inferior care; it rests on sound scientific methods, its hallmarks being universal availability, accessibility, acceptability and affordability (Sunder Lal, Ch 17, p 741). Reading "appropriate" as "cheap" risks a permanent two-tier system.
- ▶ **The selective primary health care detour.** GOBI-FFF and the child survival and safe motherhood programme used appropriate technologies to gain coverage quickly, but narrowed primary health care to a few vertical interventions (Sunder Lal, Ch 17, p 742). National Health Policy 2017 therefore shifted policy from selective care to assured comprehensive primary health care with referral linkages (Park 28e, Ch 10, p 728).
- ▶ **Technology alone does not deliver.** It fails without the other principles — supply chain, trained local workers, community demand, intersectoral support.
- ▶ **Digital equity.** Telemedicine extends specialist reach cheaply but needs connectivity and literacy; appropriateness must be re-tested in the poorest districts.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Four principles of primary health care** (embedded above) — a central "Primary health care" box with four arrows to *Equitable distribution*, *Community participation*, *Multi-sectoral approach*, *Appropriate technology*, and four diagonal arrows to the service dimensions *Preventive*, *Promotive*, *Curative*, *Rehabilitative*.
- ▶ **The five "A" attributes** (embedded above) — a hand with the five digits labelled *Accessible universally*, *Acceptable culturally*, *Adaptable practically*, *Affordable economically*, *Applicable scientifically*.

High-yield exam pointers

- ▶ **WHO definition (write verbatim):** technology that is *scientifically sound, adaptable to local needs, acceptable to those who apply it and those for whom it is used, and that can be maintained by the people themselves in keeping with the principle of self reliance with the resources the community and country can afford*.
- ▶ **Four principles mnemonic** — "ECIA": Equitable distribution, Community participation, Intersectoral coordination, Appropriate technology. Alma-Ata 1978; 8 essential components.
- ▶ **Five A's of primary health care:** Accessible universally, Acceptable culturally, Adaptable practically, Affordable economically, Applicable scientifically.
- ▶ **GOBI-FFF (UNICEF):** Growth charts, Oral rehydration, Breastfeeding, Immunisation + Female literacy, Family planning, Food supplements.

- **Park's two classic examples:** oral rehydration fluid in place of costly procedures; standpipes in place of house-to-house connections.
- **Current Indian anchors:** National List of Essential Medicines 2022 (first 1996; revised 2003, 2011, 2015, 2022); 105 essential medicines and about 14 point-of-care tests per Health and Wellness Centre; 12-service comprehensive primary health care package at both HWC tiers; eSanjeevani across about 1.5 lakh HWCs; Indian Public Health Standards 2022; National Health Policy 2017 shift from selective to comprehensive primary health care.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The WHO Astana Declaration on Primary Health Care (2018) renewed the Alma-Ata commitment, and the Ayushman Bharat Health and Wellness Centres have since been renamed Ayushman Arogya Mandir (Park 28e, Ch 7, pp 535, 541).

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NFHS 1993–2021, Comprehensive Annual Modular Survey 2022–23, ICMR, NP-NCD programme guidelines, WHO End TB Strategy, WHO/UNICEF Operational Framework for Primary Health Care

Programme and policy updates (India)

- **Ayushman Bharat Digital Mission (ABDM, launched September 2021)** is where the affordability and acceptability limbs of the WHO definition are now being tested. Across 537,278 outpatient visits over eight months at three government tertiary hospitals in Agra only 129,007 ABHA identities were generated (adoption 23.96%); digital health literacy among 425 surveyed attendees was only moderate (total DHLI mean 2.94 on a 1–4 scale), significantly lower in women (2.73 vs 3.16, $p < 0.001$) and falling with age (Spearman rho -0.230 , $p < 0.001$), and the dominant barriers were preference for in-person care (85.6%), cost of data plans (81.9%) and difficulty using health apps (65.6%), with 88% asking for training support [PMID 41787376].
- **Telemedicine has become a routine peripheral technology rather than a pilot.** In a hub-and-spoke evaluation of 78 health workers at two Karnataka primary health centres, 67% agreed it eased linkage to specialists at the hub and 64% that it cut patient travel cost and helped cover a larger population; but 59% felt it could breach patient confidentiality and 46% feared loss of health-worker jobs — the "acceptable to those who apply it" test failing before any technical one. The Telemedicine Practice Guidelines (2020) and ABDM supply the governing framework [PMID 41648971].

Current national data

- **Comprehensive Annual Modular Survey 2022–23 (NSS 79th round, 302,086 households).** Publicly funded health insurance (AB-PMJAY or state schemes) was associated with greater outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (aOR 1.168, 1.128–1.210) and with less catastrophic expenditure at the 20% threshold (outpatient aOR 0.869, 0.810–0.932; inpatient aOR 0.790, 0.737–0.846), but the benefit package still excludes outpatient care and out-of-pocket spending there persisted even among the insured — the remedies proposed being to fold outpatient care into the package and to strengthen primary care through Ayushman Arogya

Mandirs, which the authors call a cost-effective route that also cuts avoidable hospitalisation [PMID 42167946].

- **Breastfeeding — the "B" of GOBI — across five NFHS rounds (1993–2021):** exclusive breastfeeding for six months rose from 43.5% to 63.9% and early initiation from 9.6% to 41.2%, while mixed milk feeding worsened by about nine percentage points; 23 states and union territories have already met the 60% exclusive-breastfeeding Global Nutrition Target for 2030, five are on track and six are projected to miss it [PMID 42134478].

Elimination, surveillance and disease status

- **MiniDock MTB — a near-point-of-care molecular tuberculosis test purpose-built for the periphery.** In 1,380 participants aged 12 years and above enrolled at outpatient centres in seven high-burden countries including India, sensitivity was 85.7% (95% CI 80.4–90.0) on sputum and 79.6% (73.8–84.7) on tongue swabs, specificity above 97.5% for both; sensitivity closely matched sputum Xpert MTB/RIF Ultra (difference –2.8 percentage points, 95% CI –6.0 to 0.5) and beat smear microscopy by 24.3 points on sputum and 18.3 points on tongue swabs. The device is expected to cost under US\$180 with under US\$4 per test, is battery-operated and extraction-free, returns a visual result in 12–25 minutes, and scored a median System Usability Scale 75 (IQR 65–80) among 18 test-naïve staff working only from the instruction leaflet. This is the textbook definition of appropriate technology, and it fits WHO guidance that rapid molecular tests replace smear microscopy at every level, primary care included [PMID 42054680].
- **Lymphatic filariasis endgame surveillance is moving to molecular xenomonitoring —** detection of filarial DNA in mosquito vectors — because human microfilaraemia and antigenaemia surveys lose power at very low transmission. Current gravid-trap sampling is criticised as behaviourally biased towards zoophagic *Culex quinquefasciatus*; representative multi-physiological sampling with routine blood-meal profiling, stage-specific assays and harmonised analysis is proposed for post-mass-drug-administration surveillance [PMID 42033201].

Guideline and legislative updates

- **Programme renaming to carry into the exam:** NPCDCS is now the **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)**, delivered through Ayushman Arogya Mandirs, with ASHAs completing the Community-Based Assessment Checklist for every 1,000 rural (or 2,000 urban) people and thermal ablation or cryotherapy for eligible cervical lesions offered at the primary level. Screening coverage nonetheless remains below 2% — cervical 1.9%, breast 0.9% and oral 0.9% in women, oral 1.2% in men — the sharpest current demonstration that a cheap, scientifically valid technology is not yet "appropriate" until it actually reaches people [PMID 42285578].
- **Artificial-intelligence-assisted single-visit "screen-and-triage" at the primary health centre.** Among 871 women screened at three Indian PHCs with a portable digital cervicoscope, the AI classifier achieved 86.7% sensitivity, 92.0% specificity and 90.0% accuracy (AUC 0.894) against histopathology in the 40 biopsied women, while nurses given a month of structured training over about 100 screenings reached only 7.7% sensitivity on the same images. Reliance on remote experts added a 10–20 day delay before triage decisions reached the woman, and only 69 of 211 referred women (32.7%) attended colposcopy. The value of a single-visit tool lies in removing

that referral delay, not in the image itself — appropriateness judged by continuity of care, not by device performance [PMID 41867352].

Global and WHO developments

- ▶ **Effective primary health care coverage and child survival.** In 37,583 children (118,911 child-years) from Mozambique, Gabon and Ethiopia, rising PHC coverage — measured as services actually received, not facilities built — tracked falling under-five mortality in a dose–response manner, reaching a 58% reduction at consolidated high coverage (rate ratio 0.42; 95% CI 0.19–0.91). Microsimulation projected 373,108 preventable child deaths (95% uncertainty interval 313,152–444,244) between 2025 and 2035 if PHC services are dismantled as development assistance for health contracts [PMID 42566215].
- ▶ **Ten health-system "shifts" now define a PHC-oriented reform,** synthesised from the WHO/UNICEF Operational Framework for Primary Health Care across nine countries. The two most examinable are shift (iii), from hospital-centric resource allocation to strategic reallocations favouring primary care, and shift (iv), from physician-dominated care to multidisciplinary teams including non-physician providers. Specialist groups and hospital associations resisted the resource reallocation, and physicians resisted the shift to multidisciplinary teams — Park's "ivory tower of disease" objection restated as political economy. Leaders, administrators and bureaucrats resisted a different set of shifts: strategic purchasing, embedded learning and accountability, and expanded citizen participation [PMID 41955203].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 5 — Epidemiology of Communicable Diseases; Ch 7 — Health Programmes in India; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 — Health Care of the Community; Ch 22 — Communication for Health Education; Ch 24 — Essential Medicines and Counterfeit Medicines; Ch 25 — International Health); Sunder Lal — Textbook of Community Medicine (Ch 17 — Healthcare of the Community); Oxford Textbook of Public Health 7e (Ch 16 — How Access to Healthcare Affects Population Health; Ch 94; Ch 107); Suryakantha — Community Medicine with Recent Advances (Ch 34 — embedded Figs 34.2 and 34.4 only); Declaration of Alma-Ata 1978; National List of Essential Medicines, India, 2022; Indian Public Health Standards 2022; National Health Policy 2017.

Recent-advances sources: 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission: an analytical cross-sectional study among outpatient department attendees, BMC Health Services Research 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 41648971 — Telemedicine in rural settings: a cross-sectional study on its promoters and barriers, Indian Journal of Medical Research 2025, doi:[10.25259/IJMR_2513_2025](https://doi.org/10.25259/IJMR_2513_2025); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022–2023), BMJ Open 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 42134478 — Trends in breastfeeding indicators across Indian states and Union Territories (1993–2021): estimating progress towards the 2030 Global Nutrition Target, Preventive Medicine 2026, doi:[10.1016/j.ypmed.2026.108600](https://doi.org/10.1016/j.ypmed.2026.108600) (abstract-only); 42054680 — Pulmonary tuberculosis detection with MiniDock MTB using swab samples, New England Journal of Medicine 2026, doi:[10.1056/NEJMoa2509761](https://doi.org/10.1056/NEJMoa2509761); 42033201 — Strengthening lymphatic filariasis elimination programmes with molecular xenomonitoring: critical utility and unmet operational needs, Transactions of the Royal Society of Tropical Medicine and Hygiene 2026, doi:[10.1093/trstmh/trag041](https://doi.org/10.1093/trstmh/trag041) (abstract-only); 42285578 — Enhancing screening, early diagnosis and treatment initiation of oral, breast and cervical cancer in selected districts of India: an implementation research protocol, BMJ Open 2026, doi:[10.1136/bmjopen-2025-112428](https://doi.org/10.1136/bmjopen-2025-112428); 41867352 — Bringing cervical cancer screening closer to women: feasibility of artificial intelligence and remote assessment in primary health care, International Journal of Public Health 2026, doi:[10.3389/ijph.2026.1609094](https://doi.org/10.3389/ijph.2026.1609094); 42566215 — Primary health care coverage and child mortality in sub-Saharan Africa, JAMA Network Open 2026, doi:[10.1001/jamanetworkopen.2026.27655](https://doi.org/10.1001/jamanetworkopen.2026.27655); 41955203 — Primary health care-oriented reforms: the political economy dynamics underlying health-system shifts across nine countries, Health Policy and Planning 2026, doi:[10.1093/heapol/czag042](https://doi.org/10.1093/heapol/czag042).

Coverage note: the Park 28e pages carrying the appropriate-technology definition (Ch 10, p 720) and the service-approach example (Ch 22, p 1016) are optically degraded in the indexed scan; both passages were read in full and their wording confirmed against the identical text in the indexed Park 27e before quotation.



Bhore Committee 1946

Asked in: Paper IV 18-Nov-2019 — RGUHS MD Community Medicine

Definition — The **Bhore Committee** is the popular name of the **Health Survey and Development Committee**, appointed by the Government of India in 1943 under the chairmanship of **Sir Joseph Bhore**, which submitted its report in 1946 (Mahajan & Gupta 4e, Ch 26, p 485). It surveyed health conditions and health organisation in British India and recommended a complete remodelling of health services around an **integrated preventive-and-curative service** delivered through a network of **primary health centres (PHCs)**. Its report is the single most comprehensive health-planning document produced in India and became the blueprint for health planning after independence (Jugal Kishore — National Health Programmes of India, Health Planning; Park 28e, Ch 23, p 1031).

Constitution and terms of reference

Item	Detail
Official name	Health Survey and Development Committee (Govt. of India, 1946)
Popular name	Bhore Committee
Chairman	Sir Joseph Bhore
Year appointed / reported	Appointed 1943; report submitted 1946
Terms of reference	To survey the existing health conditions and health organisation in British India, and to make recommendations for their future development (Sathe — Health Management, Ch 18, p 192)
Method used	Situational analysis → identification of underlying causative conditions → objective setting → phased programme planning; the report is cited as a model of health-planning method (Sathe, Ch 18, p 196; Mahajan & Gupta 4e, Ch 26)
Status of report	Basis of most planning and health measures adopted by the national government after 1947 (Park 28e, Ch 23, p 1025)

Health situation documented by the committee

Indicator / problem	Value reported
Crude death rate (1937)	22.4 per 1,000 population
Infant mortality rate	162 per 1,000 live births
Maternal mortality ratio	About 20 per 1,000 births
Expectation of life at birth (1921–31)	26.91 years (males); 26.56 years (females)
Infant deaths as share of all deaths	24.3% in India versus 6.8% in England
Malaria	About 100 million cases and 2 million deaths every year
Other major killers	Cholera, plague, smallpox, dysentery and diarrhoea, respiratory disease — largely epidemic in behaviour

Indicator / problem	Value reported
Causes enumerated	Prevailing sanitary conditions, nutrition, inadequacy of the existing medical and preventive health organisation, general and health education, and the social background of ill-health

Source of the table: Sathe — *Health Management*, Ch 18, p 192; Jugal Kishore — *National Health Programmes of India, Health Planning*.

Guiding concepts the committee introduced

- **Comprehensive health care** — the term was first used by the Bhore Committee. It meant integrated preventive, curative and promotive services from "womb to tomb" for every individual residing in a defined geographic area, and it was to **replace the policy of providing more medical care alone** (Park 28e, Ch 10, p 719).
- **Criteria of comprehensive health care** laid down by the committee: (a) provide adequate preventive, curative and promotive services; (b) be as close to the beneficiaries as possible; (c) command the widest cooperation between the people, the service and the profession; (d) be available to all irrespective of ability to pay; (e) look after the vulnerable and weaker sections specifically; and (f) create and maintain a healthy environment in homes as well as workplaces (Park 27e, Ch 10).
- **Integration of preventive and curative services at all levels**, with a **single administrative officer** for the curative and preventive departments and a **unified health authority in every district** (Park 28e, Ch 23, pp 1034–1035).
- **Health centre as the unit of delivery** — following Lord Dawson's 1920 proposal in England and the 1931 call of the League of Nations Health Organization, the committee recommended health centres for integrated curative and preventive services; this is the origin of the Indian PHC (Park 28e, Ch 1, p 20).
- **The doctor as a "social physician"** — a basic doctor oriented to the community rather than to the hospital alone (Park plus, p 294).
- **Primary conditions for healthful living** must be secured alongside medical services: suitable housing; sanitary surroundings; safe drinking water; preventive and curative protection irrespective of ability to pay; qualitative and quantitative improvement in nutrition; elimination of unemployment; a living wage for the worker; improved agricultural and industrial production; improved rural communications; a vigorous and healthy community life; physical and mental recreation; and the living interest and practical cooperation of the people themselves (Sathe, Ch 18, p 192).

Core recommendation — a phased, three-tier integrated health service

The committee recommended an integrated service of **primary health units, secondary health units and district health units**, with ambulance, hospital, laboratory and specialist support, to be built through an immediate **short-term programme over 10 years** and a **long-term programme ("three million plan") over 30–40 years** (Mahajan & Gupta 4e, Ch 28; Red Book — Public Health and Preventive Medicine).

Feature	Short-term programme (10 years)	Long-term programme — "three million plan" (30–40 years)
Peripheral unit	One PHC per 40,000 population, with 30 beds	Primary health unit with a 75-bedded hospital per 10,000–20,000 population
PHC staffing	2 medical officers, 1 nurse, 4 public health nurses, 4 midwives, 4 trained <i>dais</i> , 2 sanitary inspectors, 2 health assistants, 1 pharmacist, 15 class IV employees	6 medical officers, 6 public health nurses, 2 sanitary inspectors, 2 health assistants and 11 ancillary staff
Intermediate unit	A secondary health centre for each subdivision, to support, coordinate and supervise the PHCs and act as referral centre	A 30-bedded hospital for every two primary health units; a secondary health unit with a 650-bedded hospital at taluk level (about 3 lakh population), covering roughly 30 primary health units
Apex unit	Strengthening of the existing district hospital	District hospital of 2,500 beds serving 10–30 lakh population
Linkage	—	All three tiers connected by telephone and ambulance
Manpower norms proposed	—	1 bed per 175 population; 1 doctor per 1,600; 1 nurse per 600

Sources: Mahajan & Gupta 4e, Ch 26, p 485 and Ch 28; Sunder Lal, Ch 17, p 748; Red Book — Public Health and Preventive Medicine; Jugal Kishore — National Health Programmes of India; Park plus, p 294.

Other recommendations

Domain	Recommendation
Medical education	Three months' training in preventive and social medicine for every medical graduate, to produce "social physicians"; internship training in public health; review of medical education and research (Park plus, p 294)
Health administration	The Director of Health Services should also be Secretary to the State Government; medical and public health organisations to be integrated at all levels ; a unified health authority in every district (Park 28e, Ch 23, pp 1034–1035)
Community participation	A village health committee in every village to secure active cooperation and support; a district health board of district health officials and public representatives (Jugal Kishore, Health Planning)
Financing	15% of government expenditure to be spent on health care
Drugs	No patents on pharmaceutical products
Disease control	A countrywide comprehensive malaria control programme , endorsed by the Planning Commission in 1951 (Jugal Kishore, Malaria)
Environment	Suitable housing, sanitary surroundings, safe water supply and elimination of unemployment to be ensured; with the Environmental Hygiene Committee this led to the National Water Supply and Sanitation Programme (Sathe, Ch 18, p 194)

Domain	Recommendation
Emphasis	Greater weight on preventive work than on curative work

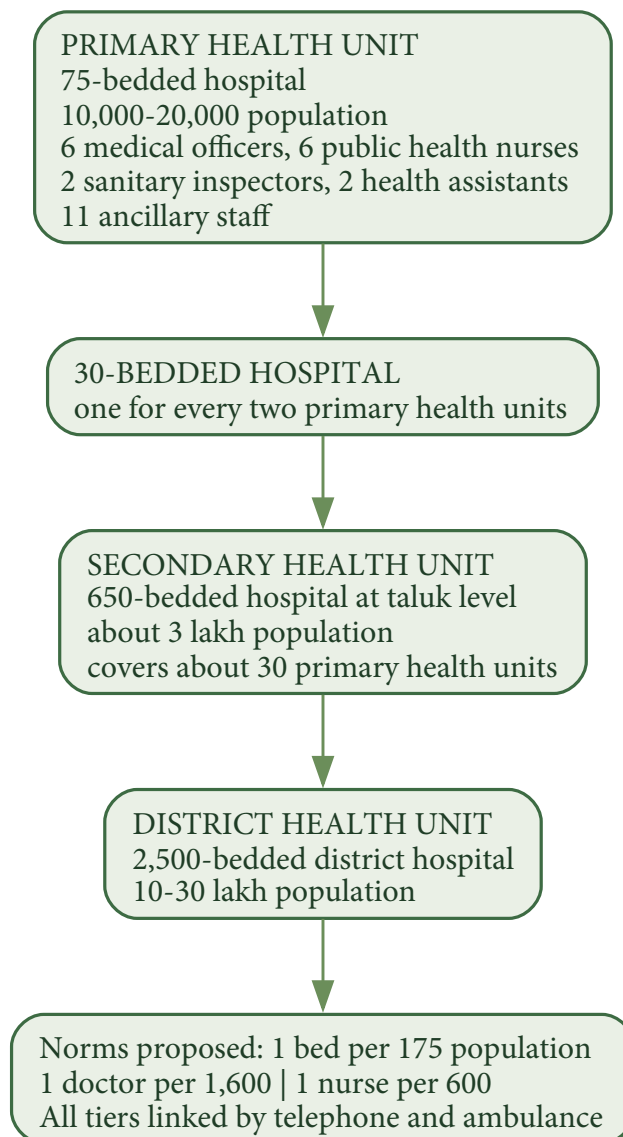
Implementation, impact and limitations

Recommendation	What actually happened
Network of PHCs	Accepted by the Central Council of Health at its first meeting, January 1953 ; a primary health unit was made an essential part of the Community Development Programme, one per Community Development Block, and PHCs were established from the First Five Year Plan (1951–56) (Mahajan & Gupta 4e, Ch 28; Suryakantha, Ch 38)
PHC staffing	Scaled down from two each of medical officers, lady health visitors and sanitary inspectors to one each , because of acute shortage of trained personnel (Mahajan & Gupta 4e, Ch 28)
Integration of preventive and curative services	West Bengal led by creating a post of Director of Health Services in 1947; integration is now the rule at State level (Park 28e, Ch 23; Mahajan & Gupta 4e, Ch 26)
Director of Health Services to be Secretary	Not implemented — the Health Secretary continues to come from the Indian Administrative Service (Park 28e, Ch 23, pp 1034–1035)
Comprehensive health care from womb to tomb	Became the basis of national health planning and, at Alma-Ata (1978) , was recognised as having all the hallmarks of primary health care , first proposed by the Bhore Committee (Park 27e, Ch 10; Red Book)
Population norm for PHC	Long-term norm of 10,000–20,000 never reached in the short term; the PHC today serves 30,000 in plain rural areas and 20,000 in tribal, hilly, desert and difficult areas , acts as referral unit for 6 sub-centres , and is being converted into a Health and Wellness Centre (Sunder Lal, Ch 17, p 748)

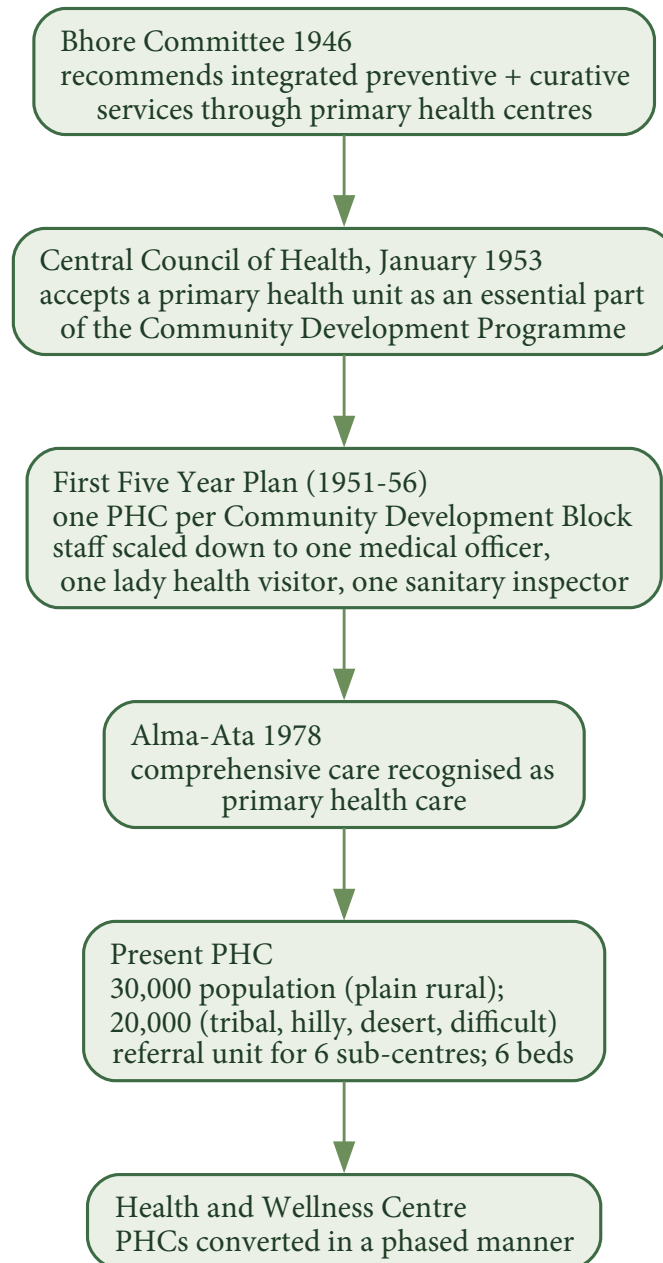
Limitations / criticisms (Park plus, p 294; Red Book — Public Health and Preventive Medicine):

- ▶ It **imposed a Western model** of health care, built entirely on the **allopathic system**, and left out the traditional and indigenous systems that dominated rural India's socio-cultural setting.
- ▶ It adopted the British "**welfare state**" model and ignored the Soviet model of health for all.
- ▶ The approach was **top-down and centralised**, not decentralised or community-driven.
- ▶ It gave importance to **vertical, single-disease programmes**, which later fragmented peripheral manpower until the multipurpose worker scheme corrected it.
- ▶ Implementation was **delayed for years** after 1947 by shortage of funds and trained personnel, so the bed and manpower ratios proposed were never achieved.

Diagrams



Bhore Committee long-term plan ("three million plan") — the three-tier integrated health service



From the Bhore Committee recommendation to today's primary health centre

High-yield exam pointers

- ▶ **Name and dates:** Health Survey and Development Committee, appointed 1943, reported 1946, chairman Sir Joseph Bhore — the first and most comprehensive Indian health-planning document.
- ▶ **Two programmes:** short-term = 10 years, PHC per 40,000, 30 beds, 2 medical officers; long-term = "three million plan", 30–40 years, 75-bedded primary health unit per 10,000–20,000.
- ▶ **Three tiers of the long-term plan:** 75-bed primary health unit → 650-bed secondary health unit (taluk) → 2,500-bed district hospital; norms 1 bed / 175, 1 doctor / 1,600, 1 nurse / 600.
- ▶ **Firsts to quote:** first use of the term "comprehensive health care", the "womb to tomb" phrase, the concept of the PHC in India, the "social physician", and 3 months of preventive and social medicine in medical training.

- ▶ **Administration:** integration of preventive and curative services at all levels; **Director of Health Services to be Secretary — never implemented;** 15% of government expenditure on health; no patents on pharmaceuticals.
- ▶ **1946 baseline figures:** crude death rate 22.4/1,000; IMR 162/1,000 live births; maternal mortality ratio about 20/1,000 births; life expectancy 26.9 years (male), 26.6 years (female).
- ▶ **Criticism mnemonic — "WAVE-T":** Western model, Allopathy-only, Vertical programmes, Elitist welfare-state model, Top-down (not decentralised).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the size of the long-term secondary health unit is quoted as a 650-bedded taluk hospital in Park plus and in Jugal Kishore's National Health Programmes of India, but as a 550-bedded unit covering about 30 primary health units in Mahajan & Gupta 4e — quote 650 beds, which is the figure the majority of Indian textbooks carry, and do not be surprised by the variant.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed

Programme and policy updates (India)

- ▶ **Ayushman Arogya Mandir — the current name of the Bhore-descended primary health centre.** The Ayushman Bharat Health and Wellness Centres, created in 2018 by upgrading sub-centres and PHCs (target 1.5 lakh facilities, each serving about 3,000–5,000 population), were renamed **Ayushman Arogya Mandir** by the Union Health Ministry in **November 2023**, with the tagline *Arogyam Paramam Dhanam* and an AYUSH-informed shift towards holistic wellness — counselling on lifestyle, yoga, mental health, nutrition and elderly care alongside the existing packages. The exercise carried ₹3,000 per facility and a ₹48 crore outlay, used regional-language signage, and required renamed facilities to upload photographs to the AB-HWC portal. Field observations flag the risk that it remains a name change without matching staffing and service change. Write this name, not "HWC", in current answers. [PMID 42257165]
- ▶ **Ayushman Bharat Digital Mission (ABDM, launched 2021)** now layers a digital spine over that same network: a 14-digit **ABHA** (Ayushman Bharat Health Account) number per citizen, a patient-controlled personal health record, Healthcare Professionals and Healthcare Facilities Registries, and the Unified Health Interface for bookings and teleconsultation. By March 2024, 568 million ABHA accounts and over 350 million linked health records, spanning about 230,000 facilities and 285,000 registered providers; eSanjeevani had delivered more than 270 million teleconsultations by August 2024. PM-JAY, the insurance pillar, covers ₹5 lakh per family per year for the poorest 50% of the population. [PMID 39414924]

Current national data

- ▶ **SRS 2023 — infant mortality rate 25 per 1,000 live births**, down from 40 per 1,000 in 2013. Set this against the committee's 1946 figure of 162 per 1,000 for the standard "what has changed since Bhore" question. The same paper also cites a 69% fall in India's infant mortality rate against a 55% global fall, and a 75% fall in under-five mortality against a 58% global fall, but attaches no period to those four figures — quote them as long-run child-survival gains, never as the 2013–

2023 change. Pneumonia is the second-ranked cause of infant death (16.2%) in the Registrar General's 2020–22 cause-of-death statistics. [PMID 42056953]

- **Citizens Survey 2022–23** — the Lancet Commission on a Citizen-Centred Health System for India interviewed 50,000 randomly selected households across 125 districts in 29 states and Union Territories from November 2022 to April 2023 (98% participation, 70% rural, 141 questions). Its validated questions were adapted from the NFHS, the National Sample Survey, CAHPS and the India Health Systems Reform Project, and it adds granular data those existing household surveys do not capture: provider preference across public and private sectors, perceived quality, perceived affordability and willingness to pay, digital health use, sources of health information, and the electoral prioritisation of health. Useful because UHC progress in India is otherwise judged on top-down metrics — insurance enrolment, utilisation, facility performance — which do not reach citizens' own experience. [PMID 41688486]

Elimination, surveillance and disease status

- **Kala-azar reached the elimination threshold in 2023** — fewer than 1 case per 10,000 population per year at **block** level, the sub-district implementation unit. Quote both the block-level definition and the year: the 2002 National Health Policy had set 2010, and the deadline slipped repeatedly before 2023 was met. By 2021 only 7 of 633 endemic blocks were still above threshold. Drivers included liposomal amphotericin B from 2014, synthetic pyrethroid indoor residual spraying from 2015–16, wage-loss compensation for patients, and KAMIS-based real-time surveillance. [PMID 40988627]
- **Lymphatic filariasis remains endemic in 345 districts**, with the declared national elimination target now **2030**; the operational load of mass drug administration, morbidity management and disability prevention, and post-MDA surveillance is argued to need public-private partnership of the kind already used in polio and TB. [PMID 41951149]
- **Malaria has moved from mass burden to residual focal transmission**. Against the committee's 1946 estimate of about 100 million cases and 2 million deaths a year, the current problem is last-mile: in Mizoram the parasite mix shifted from *P. falciparum* to *P. vivax*, which rose from 23% of cases in 2021 to 47% in 2024, with perennial rather than monsoon-limited transmission and 68% of Mamit district cases in 2025 — the pattern that makes elimination hardest. [PMID 42310537]

Guideline and legislative updates

- **District Residency Programme (National Medical Commission, from 2023)** — every postgraduate resident now serves 3 months in a district hospital, explicitly to strengthen primary health care and expose specialists-in-training to community-level problems. This is the closest modern statutory echo of Bhore's demand for 3 months of preventive and social medicine for every graduate and its "social physician"; commentary notes the fit is poor for pre- and para-clinical postgraduates. [PMID 42544003]
- **PM-JAY and financial protection**. Kerala's implementation (Karunya Arogya Suraksha Padhathi), which folded state and central schemes under a single purchaser and extended cashless cover to medicines from April 2019, cut net out-of-pocket health spending by about **31.7%** in a difference-in-differences analysis of 58,171 household observations across three southern states; the fall was concentrated in medicine spending and was largest among poorer households and those with chronic disease, while hospitalisation rates were unchanged. Benefit-

package design, not hospitalisation cover alone, is what delivers Bhore's "irrespective of ability to pay". [PMID 42284657]

Global and WHO developments

- **Alma-Ata (1978) → Astana (2018) → universal health coverage.** A realist review of 42 studies drawn from countries with the largest gains in the UHC Service Coverage Index between 2000 and 2021 found the usable evidence concentrated in India and China, and that primary-health-care reforms improved contact coverage and financial protection but fell short on reaching marginalised groups, on effective as opposed to merely contact coverage, and on service quality. Frontline workers, community organisations and multisectoral actors were the recurring active ingredient — the WHO operational framework's three components being integrated services, multisectoral policy and community engagement. [PMID 42293679]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1, Ch 10, Ch 23) and 27e (Ch 10); Mahajan & Gupta — Textbook of Preventive and Social Medicine, 4e (Ch 26 Health Planning, Administration and Management, p 485; Ch 28 Health Care of the Community); Sunder Lal — Textbook of Community Medicine (Ch 17, p 748); Jugal Kishore — National Health Programmes of India (Health Planning; Malaria); Sathe — Epidemiology and Management for Health Care for All (Ch 18, Development of Health Services, pp 192–196); Suryakantha — Community Medicine with Recent Advances (Ch 38, National Health Planning); Red Book — Public Health and Preventive Medicine; Park plus (p 294). Government of India (1946), *Report of the Health Survey and Development Committee*, Govt. of India Press, Shimla — the primary document, cited as reference 14 in Park 28e, Ch 23.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, *Journal of Family Medicine and Primary Care* 2026, [DOI](#); 39414924 — India's evolving digital health strategy, *npj Digital Medicine* 2024, [DOI](#); 42056953 — Seasonality of pneumonia mortality among infants in India from 2018–2020: a joinpoint analysis, *BMC Infectious Diseases* 2026, [DOI](#); 41688486 — The Citizens Survey 2022–23: a household-level dataset on Universal Health Coverage in India, *Scientific Data* 2026, [DOI](#); 40988627 — Integrating community efforts, innovative government interventions and data in the fight against kala-azar: lessons from Dumka, India, *International Health* 2026, [DOI](#); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, [DOI](#) (abstract-only); 42310537 — Rising burden of Plasmodium vivax in Mizoram: an emerging threat to malaria elimination in Northeast India, *BMC Infectious Diseases* 2026, [DOI](#) (abstract-only); 42544003 — Assessing the Impact of the District Residency Program on Pre- and Paraclinical Postgraduate Training in India, *Journal of the Association of Physicians of India* 2026, [DOI](#) (abstract-only); 42284657 — Does integrating pharmaceutical coverage into publicly financed health insurance reduce out-of-pocket spending? Difference-in-differences evidence from Kerala, India, *Social Science & Medicine* 2026, [DOI](#) (abstract-only); 42293679 — Understanding pathways from primary health care to universal health coverage outcomes: a realist review, *Frontiers in Public Health* 2026, [DOI](#).

Coverage note: Park 28e no longer carries the detailed Bhore Committee recommendations (its "Public Health in British India" section refers the reader to the 24th edition), so the short-term and long-term programme details above are grounded in the other indexed Indian textbooks named here rather than in the spine.



Challenges in implementing universal health coverage in India

Asked in: Paper IV 28-Jun-2024 — RGUHS MD Community Medicine

Definition — Universal health coverage (UHC) means that all people can access the promotive, preventive, curative, rehabilitative, long-term and palliative services they need, of assured quality, without incurring financial hardship; it is Sustainable Development Goal (SDG) target 3.8 (Park 28e, Ch 14, p 855; Oxford Textbook of Public Health 7e, Ch 107, pp 4044-4045). India's *High Level Expert Group* (HLEG) defined UHC as ensuring equitable access for every Indian citizen — regardless of income level, social status, caste or religion — to affordable, accountable and appropriate assured-quality health services (promotive, preventive, curative and rehabilitative) as well as services addressing the wider determinants of health, with the government as guarantor and enabler though not necessarily the only provider (Park 28e, Ch 23, p 1032). The Twelfth Five Year Plan set UHC as a key goal, to be reached in two parallel steps — a defined **Essential Health Package** financed by government and provided through the public health system, supplemented by contracted-in private providers (Park 28e, Ch 23, p 1032; Ch 7, p 540).

1. Financing challenges — the single most important constraint

Challenge	Grounded Indian evidence (with year)	Why it blocks UHC
Low total health spending	Total health spending 2018-19 $\approx$ 3.16% of gross domestic product (GDP) ; global evidence indicates a country must spend at least 5-6% of GDP , mostly public, before basic health needs are met (Sunder Lal, Ch 16, p 727)	Resource envelope too small for a guaranteed benefit package
Low public share	Government health spending 1.28% of GDP = 40.6% of total health spending ; it stagnated near 1.04% of GDP even at the peak of the National Rural Health Mission (Sunder Lal, Ch 16, p 727)	"Failure to attain minimum levels of public health expenditure remains the single most important constraint " (Sunder Lal, Ch 16, p 727)
Households finance the system	Household revenues including insurance are the single largest source, >60% of current health expenditure (National Health Accounts 2018-19; Sunder Lal, Ch 16, p 727)	Financing is regressive — payment at the point of illness, not by ability to pay
Out-of-pocket expenditure (OOPE)	India's OOPE is the highest in the world — 1.52% of GDP, Rs 2,155 per capita per year (Sunder Lal, Ch 16, pp 728-729)	Direct barrier to access and the engine of impoverishment
Catastrophic health expenditure (CHE)	Household health cost exceeding 10% of total monthly consumption expenditure, or 40% of non-food consumption expenditure (Sunder Lal, Ch 16, pp 728-729)	The internationally accepted marker of failed financial protection
Impoverishment	About 4.2%, i.e. nearly 50 million people, are pushed into poverty every year in India by health spending (Sunder Lal, Ch 16, pp 728-729)	Medical impoverishment defeats the core promise of UHC

Challenge	Grounded Indian evidence (with year)	Why it blocks UHC
Thin risk pooling	Private health insurance is only 6.6% of total health expenditure (Sunder Lal, Ch 16, p 729)	Prepayment and pooling remain too small to substitute for OOPE
Unmet policy target	National Health Policy (NHP) 2017 targets government health expenditure from 1.15% to 2.5% of GDP by 2025 , state health spending >8% of state budget by 2020 , and a 25% fall in households facing CHE by 2025 (Park 28e, Ch 23, p 1030)	The financing gap is a stated national commitment still unclosed

2. Service delivery and infrastructure challenges

Challenge	Grounded evidence	Consequence for coverage
Public facilities under-used	No more than 20-30% of people use the public primary health care system for curative care; indoor beds at primary health centres (PHC) and community health centres (CHC) are under-occupied (Sunder Lal, Ch 17, p 756)	Public capacity exists but does not convert into coverage
Referral chain broken	Most CHCs do not function as first referral units (FRU) ; the referral support system is weak; services are not integrated (Sunder Lal, Ch 17, p 756)	Continuity of care from sub-centre to district hospital fails
Commodity gaps	Equipment and medicines are in short supply at facilities (Sunder Lal, Ch 17, p 756)	Free-drug and free-diagnostic guarantee of NHP 2017 not realised
Rural deficit	66% of the population is rural ; setting up sub-centres, PHCs and CHCs with the manpower, infrastructure and finance required by Indian Public Health Standards (IPHS) and population norms "becomes an uphill task" (Sunder Lal, Ch 1, p 52)	Physical accessibility remains the limiting step
Urban and slum deficit	34% urban , nearly one-third of them in slums; yet >75% of dispensaries, 60% of hospitals and 80% of doctors are in urban areas, concentrated away from the slum poor (Sunder Lal, Ch 17, p 756; Ch 1, p 52)	High-density slums carry high tuberculosis and respiratory infection risk with least access
Tribal and hard-to-reach areas	Shortfall of 4,996 sub-centres (27% of requirement) in 11 states, 1,023 PHCs (40%) in 7 states, 209 CHCs (31%) in 10 states in tribal areas (Park 28e, Ch 13, p 850)	Nearly half the states have deficient tribal-area infrastructure
Weak community ownership	<i>Panchayati Raj</i> institutions are only marginally involved; community participation is lacking (Sunder Lal, Ch 17, p 756)	Removes the accountability and demand-generation arm of primary health care

3. Human resources for health

Challenge	Grounded evidence
Urban concentration of doctors	Up to 73.6% of doctors are in urban areas, where only 36% of the population lives (Park 28e, Ch 10, p 725)

Challenge	Grounded evidence
Causes of maldistribution	Absence of amenities in rural areas, lack of job satisfaction, professional isolation, lack of rural experience, and inability to adjust to rural life (Park 28e, Ch 10, p 725)
Density benchmark unmet	World Health Organization and World Bank set an "SDG index threshold" of 4.45 doctors, nurses and midwives per 1,000 population as the indicative minimum density (Oxford Textbook of Public Health 7e, Ch 107, p 4047)
Skill-mix distortion	The private sector holds 80% of doctors but only 25% of nurses (Sunder Lal, Ch 16, p 729)
Education and retention	Countries at every level of development struggle with education, deployment, retention and performance of the health workforce (Oxford 7e, Ch 107, p 4047)

4. A fragmented, largely unregulated mixed health system

Challenge	Grounded evidence
Private sector dominates curative care	Private sector holds 80% of doctors, 25% of nurses and 49% of beds , and is financed primarily by OOPE (Sunder Lal, Ch 16, p 729)
Public-private role split	Government is the dominant provider of preventive and promotive services to over 95% of the population , while curative utilisation flows to the private sector (Sunder Lal, Ch 16, p 729)
Not organised or regulated	"The private sector of health care services is not organized "; only some functions are regulated by statutory bodies, and services are available to those who can pay (Park 28e, Ch 10, p 746)
Quality of care	Poor-quality care is now a bigger barrier to reducing mortality than insufficient access ; governments must govern for quality through a shared vision, a quality strategy, strong regulation and continuous learning (Oxford 7e, Ch 21, pp 838-841)
Medicines and diagnostics	Essential medicines must be available at all times, in adequate amounts and assured quality, at a price the individual and community can afford; controlling substandard, spurious and counterfeit products needs a competent drug regulator with enforcement (Park 28e, Ch 24, pp 1041, 1053; Oxford 7e, Ch 3, pp 219-220)
Fragmentation	Limited complexity and fragmentation is desirable in a UHC financing architecture (Oxford 7e, Ch 21, p 841) — India's many parallel schemes work against this

5. Demand-side, demographic and epidemiological challenges

Challenge	Grounded evidence
Sheer population scale and diversity	India is vast, diverse and highly populated; reaching every region is itself a challenge because local beliefs, culture and practice shape each region's outlook (Oxford 7e, Ch 21, pp 841-842)
Double burden of disease	Non-communicable diseases (NCDs) were estimated to account for 63% of all deaths in 2016 , alongside a continuing communicable disease load (Park 28e, Ch 10, p 723)

Challenge	Grounded evidence
Ageing population	The 60+ population is nearly 12% of India's population , with life expectancy at 69.7 years ; their chronic and degenerative disease needs require expensive lifelong treatment — "a challenging task in universal health coverage" (Sunder Lal, Ch 1, p 53)
Rising cost of care	Cancer alone contributes an estimated 1.1 million new cases per year and about 3.9 million prevalent cases (Park 28e, Ch 10, p 723); the system must meet rising costs and rising public expectations (Park 28e, Ch 23, p 1032)
Equity gaps	Health resources are concentrated in favour of urban areas and the rural-urban divide persists (Sunder Lal, Ch 1, p 52); expenditure is maldistributed in favour of urban, upper and middle classes and biased against the rural poor (Sunder Lal, Ch 2, p 87)

6. Design gaps in the current UHC instruments

Instrument (current name)	What it provides	Residual challenge
Ayushman Bharat Health and Wellness Centres (AB-HWC), first opened 14 April 2018 at Bijapur, Chhattisgarh	1,72,148 centres operational as on 31 March 2024 , providing free essential drugs and diagnostic services (Park 28e, Ch 7, p 570), delivering an expanded range of 12 comprehensive primary health care services — preventive, promotive, curative, palliative and rehabilitative — free and close to the community (Sunder Lal, Ch 17, p 741)	Requires simultaneous strengthening of infrastructure, human resources, drugs, diagnostics, equipment, quality, monitoring and governance under revised IPHS 2022 (Park 28e, Ch 10, p 728)
Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana (AB PM-JAY), launched 23 September 2018 (subsumed the earlier Rashtriya Swasthya Bima Yojana)	Cover of Rs 5 lakh per family per year for 10.74 crore poor and vulnerable families (~50 crore people) for 1,685 procedures , cashless, portable, no cap on family size/age/gender, all pre-existing conditions covered (Park 28e, Ch 10, p 745; Ch 7, p 570)	Benefit is confined to secondary and tertiary hospitalisation with only 3 days pre- and 15 days post-hospitalisation follow-up — routine ambulatory care sits outside the package (Park 28e, Ch 10, p 745)
Targeting by Socio-Economic Caste Census (SECC)	Entitlement is based on the defined SECC database (Sunder Lal, Ch 17, pp 763-764)	An entitlement list based on a fixed database risks exclusion of the near-poor and the informal-sector non-poor
Senior-citizen expansion	All senior citizens aged 70 years and above get a distinct card, and those in families already covered get an additional top-up of Rs 5 lakh per year (Park 28e, Ch 7, p 571)	Adds a further liability that the same constrained public purse must fund

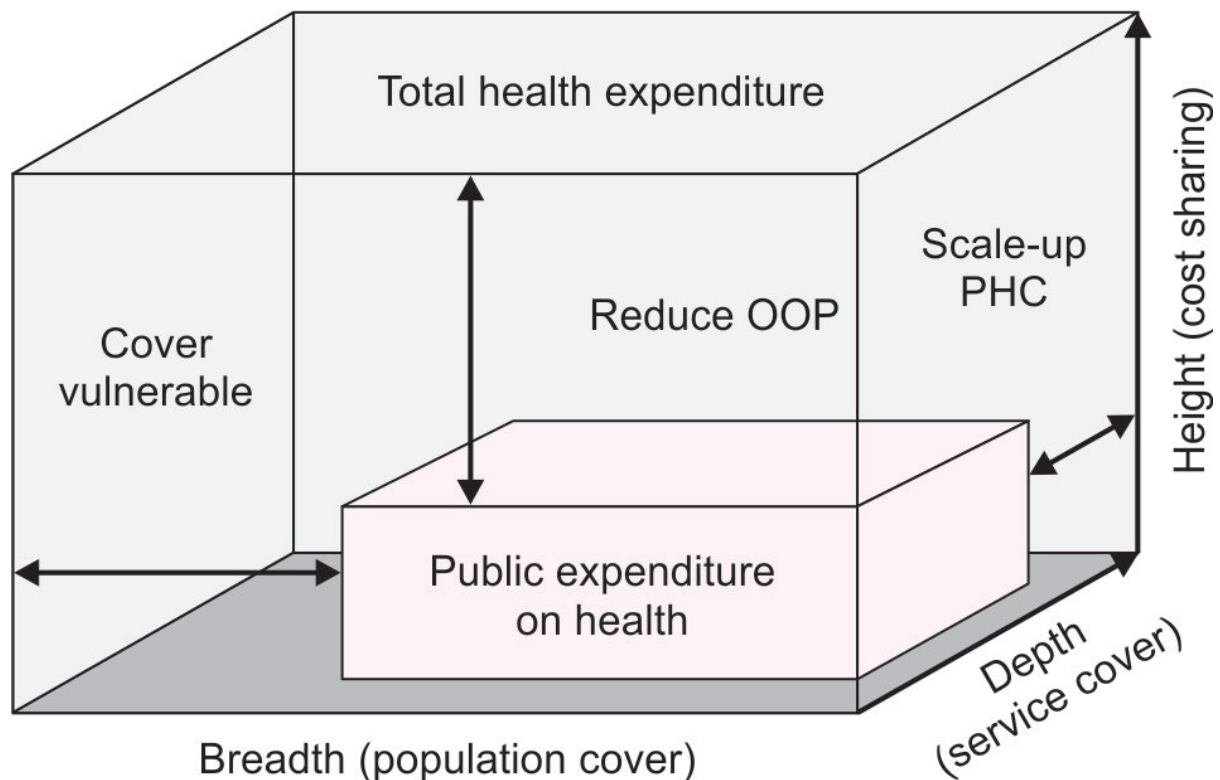
7. Monitoring and measurement challenges

UHC progress is tracked by two families of indicators — a **UHC index of service-coverage tracer indicators** (hospital access, health workforce density by cadre, access to medicines and vaccines, International Health Regulations capacities) and **financial protection indicators** (catastrophic and impoverishing out-of-pocket health spending) (Park 28e, Ch 14, p 858). India's stated systems targets

are a district-level electronic database and a strengthened surveillance system (Sunder Lal, Ch 16, p 698), and disaggregated data plus complete birth and death registration are themselves SDG monitoring requirements (Park 28e, Ch 14, p 858) — so incomplete registration and weak health information reporting are a challenge in their own right.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The UHC cube (attaining universal coverage)** — draw a large outer box labelled *Total health expenditure* with a small inner box labelled *Public expenditure on health*. Label the three axes: **Breadth (population cover)** with the arrow marked *Cover vulnerable*; **Depth (service cover)** with the arrow marked *Scale-up primary health care*; **Height (cost sharing)** with the arrow marked *Reduce out-of-pocket*. Add the key: OOP = out of pocket, PHC = primary health care.



OOP: Out of pocket
PHC: Primary health care

Fig 17.2 — Attaining universal coverage (Source: WHO, 2009) (Sunderlal, p.740)

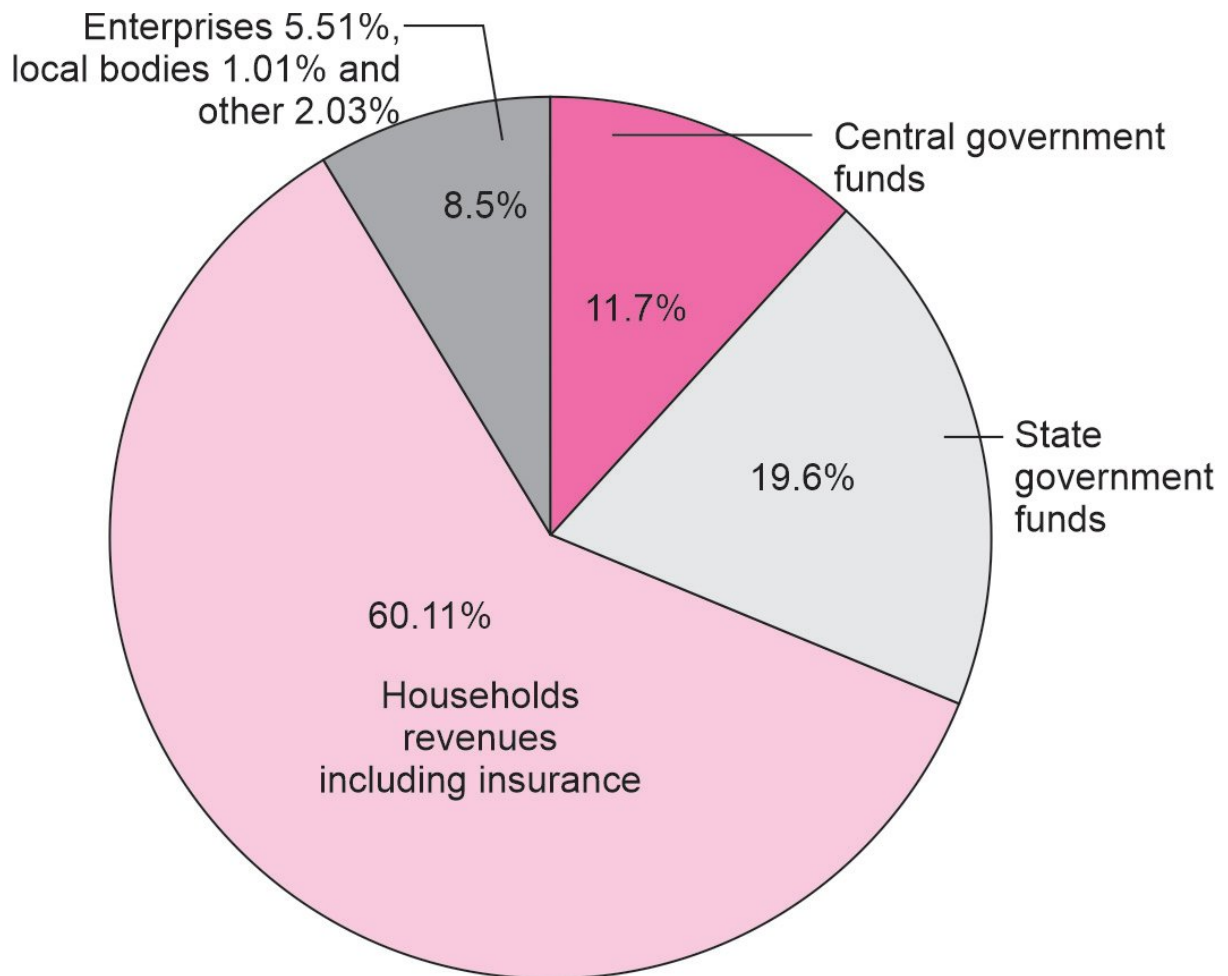
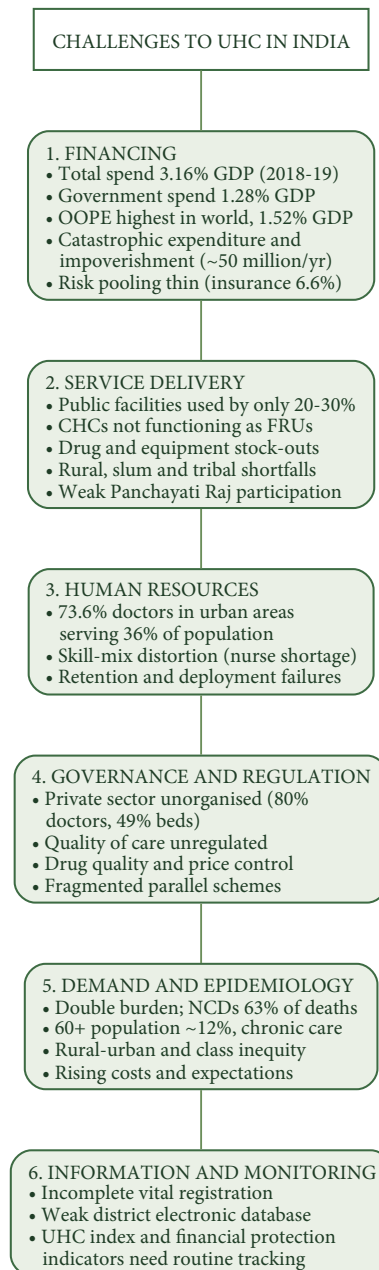
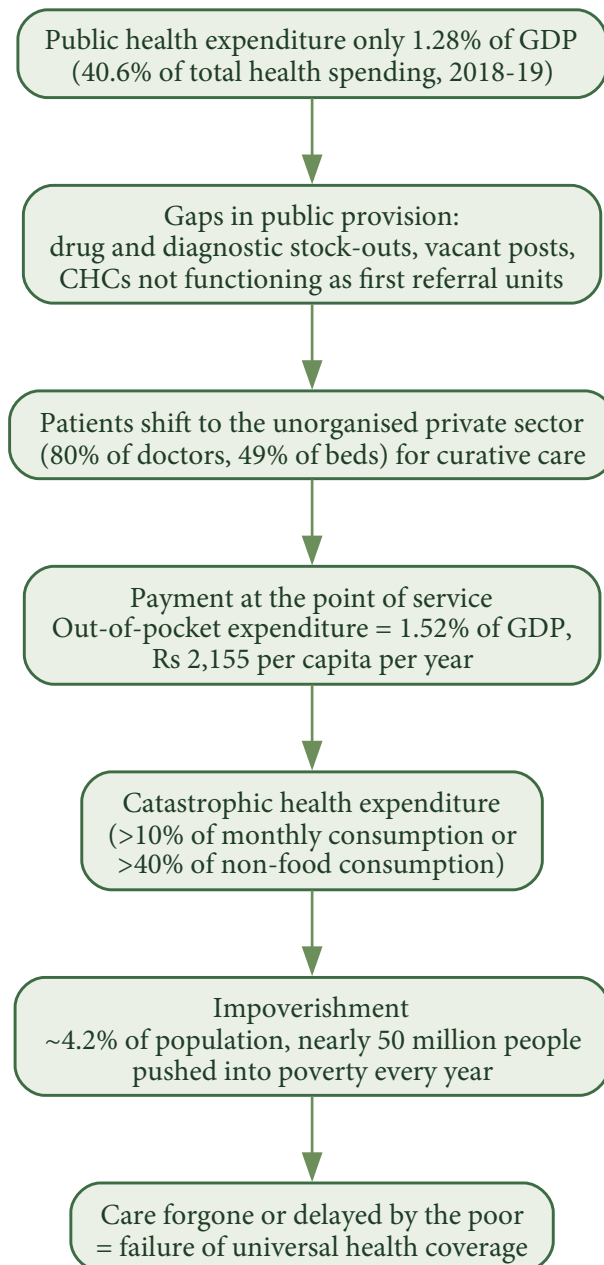


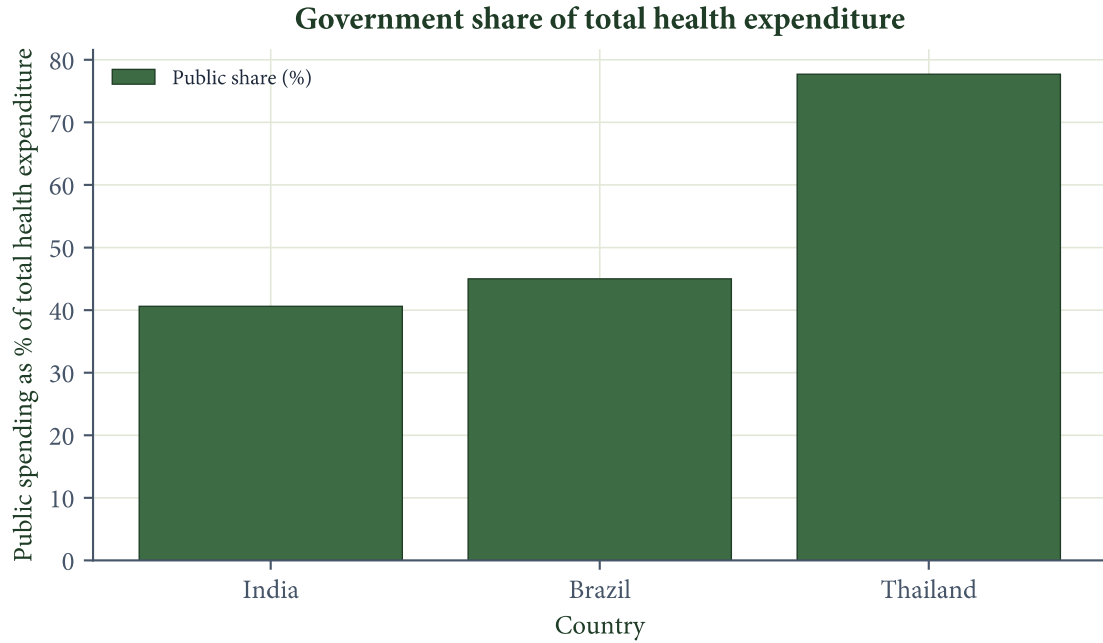
Fig 16.7 — Sources of health financing in India (NHA 2018-19) (Sunderlal, p.728)



Domains of challenge in implementing universal health coverage in India



How under-financing of the public system converts into medical impoverishment



India's public share of total health spending against two middle-income countries that have reached near-universal coverage (Sunder Lal, Textbook of Community Medicine, Ch 16, p 727).

High-yield exam pointers

- ▶ **Definition to write first:** SDG target 3.8 + the *High Level Expert Group* (HLEG) Indian definition — equitable access for all citizens regardless of income, social status, caste or religion, to affordable, accountable, appropriate, assured-quality services, with government as guarantor and enabler though not the only provider.
- ▶ **The three dimensions (UHC cube, WHO 2009)** — breadth = population covered, depth = services covered, height = proportion of cost covered; the corresponding Indian tasks are *cover the vulnerable, scale up primary health care, reduce out-of-pocket spending*.
- ▶ **The four financing numbers examiners look for:** total health spend 3.16% of GDP (2018-19); government spend 1.28% of GDP (40.6% of total); OOPE 1.52% of GDP / Rs 2,155 per capita; ~50 million (4.2%) impoverished per year.
- ▶ **Catastrophic health expenditure definition** — household health cost >10% of total monthly consumption expenditure or >40% of non-food consumption expenditure. Learn both denominators.
- ▶ **NHP 2017 financing targets** — government health expenditure 1.15% → 2.5% of GDP by 2025; state spending >8% of state budget by 2020; households facing catastrophic expenditure down 25% by 2025.
- ▶ **Human resource number** — 73.6% of doctors serve the 36% urban population; benchmark density 4.45 doctors, nurses and midwives per 1,000 population.
- ▶ **Ayushman Bharat two pillars** — AB-HWC (1,72,148 operational, 31 March 2024; 12 service packages) and AB PM-JAY (Rs 5 lakh per family per year, 10.74 crore families, 1,685 procedures, launched 23 September 2018), plus the 70+ senior-citizen top-up of Rs 5 lakh.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Ayushman Bharat Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir**, and the Indian Public Health Standards were revised in 2022 to set benchmarks explicitly aligned to universal health coverage and Sustainable Development Goal 3 (Park 28e, Ch 10, p 728). Park 28e quotes the National Health Policy 2017 baseline as 1.15% of gross domestic product while Sunder Lal quotes 1.35% for the same target; the spine figure of 1.15% has been used here. Coverage note: the retrieved corpus supports the financing, workforce, service-delivery and programme facts stated above, but carries no single consolidated "challenges to UHC in India" list, so the six domains are an organising framework built from separately retrieved and individually cited passages.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, National Health Accounts (India), NSS Comprehensive Annual Modular Survey 2022-23, WHO/World Bank UHC monitoring framework, National Health Policy 2017

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandirs, and PM-JAY is larger than the textbook figure** — the scheme is now described as Rs 5 lakh (about US\$6,000) per family per year for over 120 million poor households (12 crore), roughly 500 million people, above the 10.74 crore families quoted in the textbook, with no enrolment fee and automatic household-level eligibility; its design deliberately ties the hospitalisation package to the Ayushman Arogya Mandirs so that primary care, diagnostics and essential medicines sit alongside it, and linking Mandir outpatient services to scheme reimbursement is the stated route to closing the ambulatory-care gap that is the scheme's main design flaw [PMID 42167946].
- ▶ **Health technology assessment has been institutionalised inside the scheme** — a Health Financing and Technology Assessment (HeFTA) unit within the National Health Authority now supplies the evidence for including or excluding procedures from the health benefit package, for framing standard treatment guidelines and for package pricing, with reported cost savings to PM-JAY; the recommended next reform is value-based pricing integrated with strategic purchasing, which addresses the "who buys, at what price" gap the textbook does not cover [PMID 38715207].
- ▶ **The Ayushman Bharat Digital Mission is the newest UHC instrument, and adoption is its binding constraint** — at a government tertiary hospital in Agra only 129,007 of 537,278 outpatient visits over eight months (September 2024 to April 2025) generated an ABHA registration, an adoption rate of 23.96%; in the accompanying survey of 425 attendees across three government tertiary hospitals, 85.6% preferred in-person care, 81.9% cited the cost of a data plan, 88% wanted training or support, and 58.6% lacked adequate knowledge of the mission despite attending a participating facility [PMID 41787376]. Across large Indian platforms the same pattern recurs: Co-WIN's smartphone-dependent registration systematically excluded older, rural and non-smartphone-owning people until walk-in and offline routes were added late, and ABDM uptake has concentrated in urban tertiary hospitals with minimal penetration of

rural primary health centres — so a "digital-by-default" design that degrades non-digital pathways reproduces the very rural-urban and literacy divides it was meant to close [PMID 42215992].

Current national data

- ▶ **National Health Accounts now show a financing turn, so the textbook's 2018-19 numbers are superseded, and the residual burden has moved outpatient** — between 2013-14 and 2021-22 the government share of total health expenditure rose from 28.6% to 48% while the out-of-pocket share fell from 64.2% to 39.4%, so the state now finances more of India's health than households do, yet government health expenditure is still only about 1.9% of GDP against the National Health Policy 2017 target of 2.5% by 2025. What remains is no longer mainly a hospitalisation problem: catastrophic spending arises more often from ambulatory care than from hospitalisation (47.8% versus 43.1% of affected households), medicines alone are roughly two-thirds of out-of-pocket health spending, about 55 million Indians are still impoverished by health costs every year and around one household in six incurs catastrophic expenditure — all of it outside the PM-JAY package, with the Tamil Nadu Medical Services Corporation pooled-tender free-medicines model (running since 1995) cited as the proven domestic remedy [PMID 42548603].
- ▶ **Insurance buys access and protects the ward, but not the clinic — the residual failure is specifically outpatient** — in the Comprehensive Annual Modular Survey 2022-23 (NSS 79th round, 302,086 households) AB-PMJAY or state insurance was associated with higher outpatient use (adjusted OR 1.404, 95% CI 1.328-1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128-1.210), and with lower **inpatient** catastrophic expenditure at *both* thresholds — 10% (adjusted OR 0.721, 95% CI 0.676-0.769) and 20% (adjusted OR 0.790, 95% CI 0.737-0.846), both $p < 0.01$. **Outpatient** catastrophic expenditure fell only at the 20% threshold (adjusted OR 0.869, 95% CI 0.810-0.932) and showed no significant reduction at the 10% threshold (adjusted OR 1.048, 95% CI 0.985-1.114). So the scheme is not failing at low thresholds generally — it is failing at everyday ambulatory spending, which is exactly the gap the Arogya Mandir linkage is meant to fill [PMID 42167946].
- ▶ **A causal analysis of the same schemes finds no effect even on hospitalisation spending** — an instrumental-variable analysis of NSS rounds 60 (2004), 71 (2014) and 75 (2017-18) found no statistically significant relationship between publicly funded insurance enrolment and out-of-pocket expenditure per hospitalisation, and among insured households catastrophic expenditure at the 25% threshold reached 32.2% in private hospitals against 7.6% in public hospitals — the strongest published argument that insurance-led UHC cannot substitute for public provision, since the protection tracks *where* care is bought, not whether the patient is insured [PMID 41913189].

Elimination, surveillance and disease status

- ▶ **UHC can now be measured at district level, and the national picture is far worse than state averages suggest** — a district UHC index built for 687 of India's 707 districts from 24 tracer indicators across five domains (reproductive/maternal/newborn and child health, infectious disease, non-communicable disease, service capacity and access, financial risk protection) gave a median value of 44.0%, ranging from 26.4% in Baksa, Assam to 69.4% in the Nilgiris, Tamil

Nadu, with only 21.1% of districts above 50%; insurance coverage was 40.7%, 8.0% of the district population incurred catastrophic expenditure and 11.2% were impoverished, and the poorest wealth quintile sat at a median 36.9% against 43.9% for the district as a whole. Non-communicable disease coverage was the weakest domain at a median 9.8%, and 69.4% of the variation lay *between* states rather than within them [PMID 39219770].

- **The vertical disease programmes leak financial protection despite free drugs** — under the National Tuberculosis Elimination Programme, patients in a Papum Pare (Arunachal Pradesh) cohort of 320 incurred mean total costs of about Rs 41,481 from symptom onset to treatment completion, 28.44% borrowed money to pay, and only 55.78% of Nikshay-enrolled patients actually received the direct benefit transfer [PMID 41831939]. On the non-communicable side, a multicentre survey embedded in India's National Heart Failure Registry across 21 tertiary hospitals (n = 1,859) found out-of-pocket payments made up 92.6% (95% CI 92.5-92.8) of total health spending, with catastrophic spending in 37.7% and distress financing in 17.7% of households, and only 32.2% carrying any insurance — the authors attribute the residue to schemes that cover admission but not the outpatient visits and lifelong medicines that chronic disease actually costs [PMID 41836048].

Guideline and legislative updates

- **The catastrophic-expenditure definition itself is being revised, and the textbook cut-offs understate the burden** — an inverse rank-weighted index that sets income-sensitive, component-specific thresholds (rather than a single 10% of consumption or 40% of non-food consumption rule), applied to 2017-18 national data, identified twice as many households as catastrophic as the conventional uniform threshold did; moving from proportionate to income-sensitive thresholds raised the outpatient-medicines headcount from 12.2% to 20.7% and the inpatient-medicines headcount from 4.43% to 7.61%, while the transport headcount fell from 5.44% to 1.68% — confirming that drug spending, not travel, is what drives Indian catastrophic expenditure [PMID 41649954].

Global and WHO developments

- **Comparative work across 12 low- and middle-income countries names the exact structures India lacks** — stronger UHC performance on the WHO/World Bank service-coverage and financial-risk-protection indicators tracks with unified risk pooling rather than many parallel schemes, strategic purchasing, primary-care-based integrated delivery, a systemic rather than laissez-faire approach to the public/private mix, strategic investment in health infrastructure and workforce, and accountable governance [PMID 41401513].
- **The largest UHC success of the SDG era is now flagged as at risk, and it did not by itself build UHC** — global antiretroviral therapy coverage rose from 2% in 2000 to 77% in 2023 (Africa 0% to 82%, Eastern Mediterranean only 28%) and correlates strongly with reproductive, maternal, neonatal and child health coverage, but its correlation with the composite UHC index was weak and non-significant (about 0.19, $p > 0.05$), so a vertical programme can succeed without lifting the wider system. Growth has slowed since 2020 and global HIV financing peaked in 2017 and has since plateaued and declined; the lessons drawn for UHC are sustained political commitment, health-system strengthening, removal of financial barriers, community engagement and long-term financial sustainability [PMID 41462464].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 13 Tribal Health in India; Ch 14 Sustainable Development Goals; Ch 23 Health Planning and Management; Ch 24 Essential Medicines and Counterfeit Medicines); Sunder Lal — Textbook of Community Medicine (Ch 1 Concept of Health and Disease; Ch 2 Social and Behavioural Sciences; Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community); Oxford Textbook of Public Health 7e (Ch 21 Health policy in developing countries, incl. Case Study 3.5.3 Ayushman Bharat programme, India; Ch 103 Transformative learning for the future health workforce; Ch 107 Global health in the era of sustainable development). Named documents and data sources: National Health Policy 2017; Twelfth Five Year Plan and the High Level Expert Group on Universal Health Coverage; National Health Accounts 2018-19; Indian Public Health Standards 2022; Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana (2018) and Health and Wellness Centre progress as on 31 March 2024; WHO framework for attaining universal coverage (2009); Sustainable Development Goal target 3.8.

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health coverage: progress at risk, *Globalization and Health* 2025, [DOI](#).



Public health acts in Karnataka

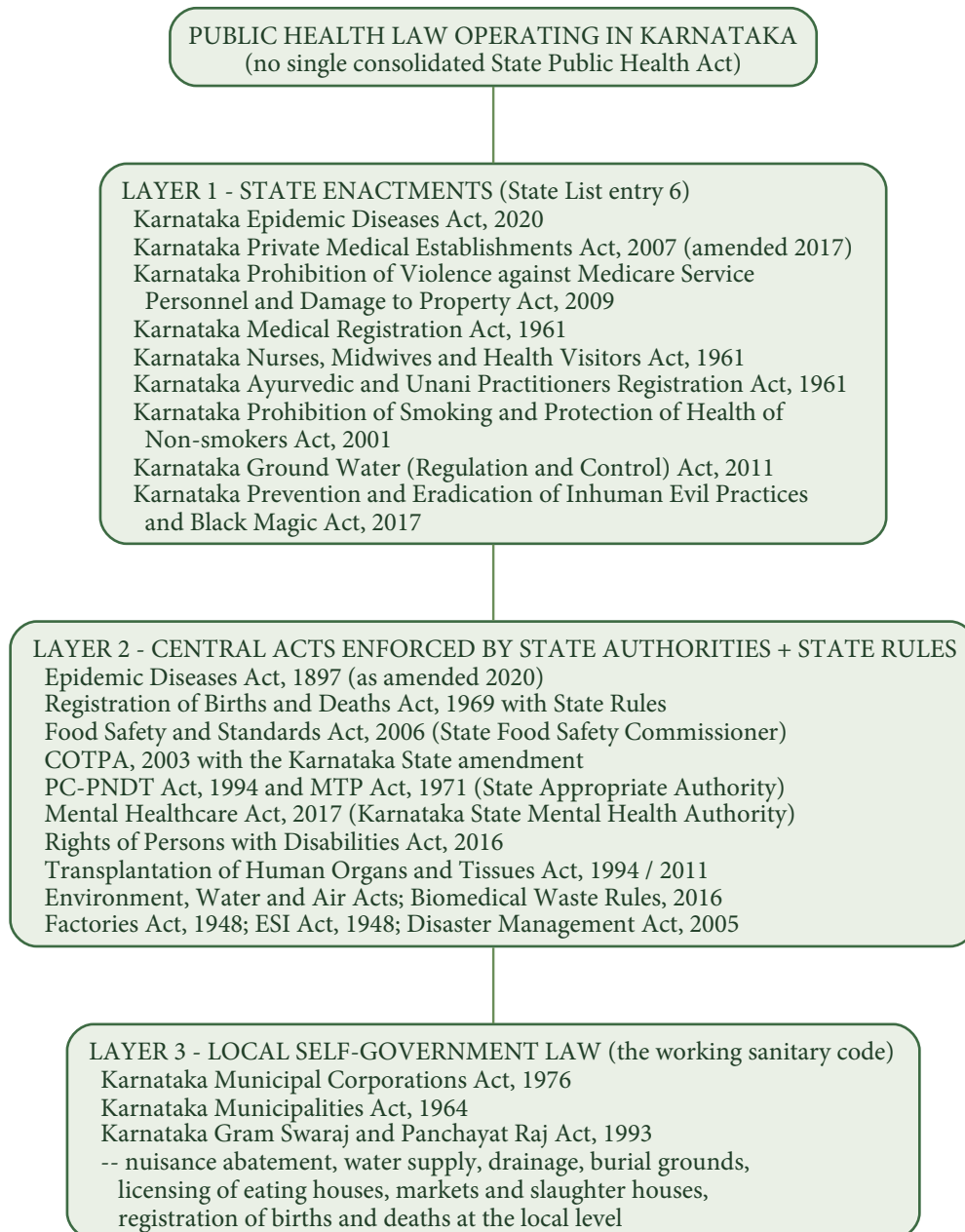
Asked in: Paper IV 08-Jan-2025 — RGUHS MD Community Medicine

Definition — Public health law is the study of the legal powers and duties of the State to assure the conditions for people to be healthy — to identify, prevent and ameliorate risks to health in the population — and of the limitations on the power of the State to constrain the autonomy, privacy, liberty or proprietary interests of the individual (Jugal Kishore, Ch 2, pp 833–834; Sathe & Sathe, Ch 9, p 107; Oxford Textbook of Public Health 7e, Ch 20, pp 774–775). Health is a **State subject** and the State is the ultimate authority responsible for all the health services operating within its jurisdiction (Park 28e, Ch 23, pp 1034–1035). Karnataka has **no single consolidated Public Health Act**; what operates is a three-layered patchwork — State enactments, central Acts run through State authorities and rules, and local self-government sanitary law.

Constitutional basis of public health legislation in the State

Constitutional provision	Effect in Karnataka
Articles 14, 15 and 21 (equality, non-discrimination, right to life)	The Constitution obliges Government to protect and fulfil the right to health as a fundamental right; the Supreme Court has repeatedly exhorted Government to give health rights legal recognition and the National Human Rights Commission has directed that a health law be enacted (Jugal Kishore, Ch 2, pp 837–838)
Article 47 (Directive Principle)	Duty of the State to raise the level of nutrition and the standard of living and to improve public health — the constitutional mandate expressly recited in modern health statutes (Jugal Kishore, Ch 2, pp 837–838)
Seventh Schedule — State List, entry 6 ("public health and sanitation; hospitals and dispensaries")	Only the Karnataka legislature can enact a general Public Health Act; the State is the ultimate authority for all health services within its jurisdiction (Park 28e, Ch 23, p 1034)
Seventh Schedule — Concurrent List, entries 29 and 30 (prevention of extension of infectious disease between States; vital statistics and registration of births and deaths)	Permits central Acts — Epidemic Diseases Act 1897, Registration of Births and Deaths Act 1969 — to operate in Karnataka alongside State law
Article 252(1)	A Union Act on a State-List subject binds a State only if that State adopts it by resolution ; Karnataka has not adopted the Clinical Establishments (Registration and Regulation) Act, 2010, which of its own force applies only to Arunachal Pradesh, Himachal Pradesh, Mizoram, Sikkim and the Union Territories (Jugal Kishore, Ch 2, pp 854–855)

Constitutional provision	Effect in Karnataka
Article 253	Enables legislation to give effect to international obligations — International Health Regulations 2005, WHO Framework Convention on Tobacco Control (Jugal Kishore, Ch 2, p 838)
Articles 243G and 243W (73rd and 74th Amendments, 1992)	Health, family welfare and education made a responsibility of village panchayats and nagarpalikas, so most day-to-day public health enforcement sits with local bodies under State municipal and panchayat law (Sunder Lal, Ch 9, p 553; Ch 2, pp 67–68)



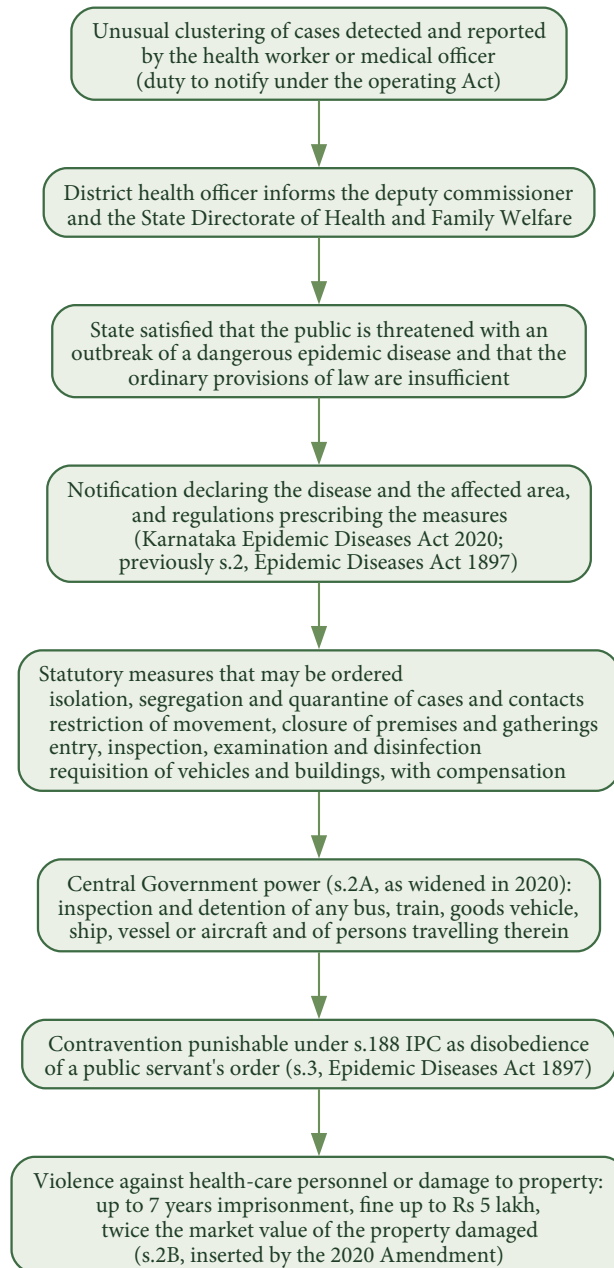
The three layers of public health law in force in Karnataka

Layer 1 — the State's own enactments

Domain	Act	Core provision and authority created
Epidemics and public health emergency	Karnataka Epidemic Diseases Act, 2020	Enacted during COVID-19 to give the State a standing statute for prevention and control of epidemic disease in place of ad-hoc reliance on the 1897 Act; empowers the State Government and district authorities to notify a disease and an affected area, prescribe containment regulations, and punish contravention
Private health sector	Karnataka Private Medical Establishments (KPME) Act, 2007, substantially amended in 2017	Compulsory registration of every private medical establishment of every system of medicine with a district-level registering authority; minimum standards; display of tariffs , a patients' charter of rights , and a district grievance-redressal mechanism empowered to penalise, suspend or cancel registration
Protection of health-care personnel	Karnataka Prohibition of Violence against Medicare Service Personnel and Damage to or Loss of Property in Medicare Service Institutions Act, 2009	Assault on doctors, nurses, students and paramedical staff, and damage to institutional property, made a cognizable and non-bailable offence with imprisonment and recovery of compensation for the property damaged
Registration of practitioners	Karnataka Medical Registration Act, 1961; Karnataka Nurses, Midwives and Health Visitors Act, 1961; Karnataka Ayurvedic and Unani Practitioners' Registration Act, 1961	Constitute the Karnataka Medical Council, the Karnataka Nursing Council and the State AYUSH registration board; maintain the State register and exercise disciplinary control, complementing the central council Acts (Jugal Kishore, Ch 2, pp 850, 854)
Tobacco and substance use	Karnataka Prohibition of Smoking and Protection of Health of Non-smokers Act, 2001 (now largely subsumed by COTPA 2003, which Karnataka has further amended by State amendment); Karnataka Excise Act, 1965	Smoke-free public places; the State amendment to COTPA raises the minimum legal age for sale of tobacco to 21 years and prohibits hookah bars and sale of loose cigarettes; excise law regulates availability of alcohol
Environment and water	Karnataka Ground Water (Regulation and Control of Development and Management) Act, 2011	Regulates extraction and protects drinking-water sources in notified over-exploited areas
Social determinants	Karnataka Prevention and Eradication of Inhuman Evil Practices and Black Magic Act, 2017; Karnataka Devadasis (Prohibition of Dedication) Act, 1982	Penalise harmful traditional practices that cause physical injury and delay care-seeking, and the dedication system linked to sexual exploitation and HIV risk
Service guarantee	Karnataka Guarantee of Services to Citizens Act, 2011 (Sakala)	Places notified health-department services under a statutory time limit with a compensatory cost payable on default

Layer 2 — central Acts working through State machinery

- ▶ **Epidemic Diseases Act, 1897** — under **section 2** the State may authorise any officer to take, and prescribe regulations for, such measures as it thinks necessary when the public is threatened with an outbreak of a dangerous epidemic disease, settling the expenses of compensation, travelling, temporary accommodation and segregation of infected persons, and requisitioning vehicles against compensation. Violation is punishable under **section 188 of the Indian Penal Code** — disobedience of a public servant's order (Jugal Kishore, Ch 2, pp 865–866).
- ▶ **Epidemic Diseases (Amendment) Act, 2020** — **widened section 2A**, which before 2020 covered only inspection of any **ship or vessel** leaving or arriving at a port (Jugal Kishore, Ch 2, p 866), so that the Central Government may now prescribe regulations for inspection of any **bus, train, goods vehicle, ship, vessel or aircraft** at any land port or aerodrome and for detention of it or of any person travelling therein; and **inserted section 2B**, prohibiting violence against health-care service personnel and damage to property — up to **7 years** imprisonment, fine up to **Rs 5 lakh**, **twice** the market value of property damaged, investigation within **30 days** and verdict within a year (AFMC Red Book 2024, Ch 39, pp 1238–1239).
- ▶ **Registration of Births and Deaths Act, 1969** — compulsory registration throughout the country and compilation of vital statistics in the States to ensure uniformity and comparability; Karnataka, with Tamil Nadu and Assam, had its *own* registration Act before it (Park 28e, Ch 21, p 996).
- ▶ **Disease notification** rests on this legal base — notification "is usually operative through certain legal Acts (e.g., Madras Public Health Act)", and a State with no specific Act merely invokes the Epidemic Diseases Act of 1897, extending it from year to year — Karnataka's position until 2020 (Park 28e, Ch 21, p 997).
- ▶ **Other central statutes administered by State authorities** — Food Safety and Standards Act 2006 (State Food Safety Commissioner); PC-PNDT Act 1994 and MTP Act 1971 (State Appropriate Authority); Mental Healthcare Act 2017 (Karnataka State Mental Health Authority); Rights of Persons with Disabilities Act 2016; Transplantation of Human Organs and Tissues Act 1994/2011; Environment (Protection) Act 1986 with the Biomedical Waste Management Rules 2016, enforced through the Karnataka State Pollution Control Board; Factories Act 1948, under which occupational diseases are notifiable, and the Employees' State Insurance Act 1948 (Jugal Kishore, Ch 2, pp 834–835; Sunder Lal, Ch 11, p 649).



Statutory response pathway when an epidemic disease threatens a district in Karnataka

Critical comment — strengths against deficiencies

Strength of the present arrangement	Deficiency the examiner expects you to name
Karnataka now has a dedicated State epidemic statute (2020) instead of extending a nineteenth-century Act year by year	The State still has no comprehensive Public Health Act defining a public health authority, its duties and a schedule of notifiable diseases — precisely the gap Park identifies for States that rely on the 1897 Act alone (Park 28e, Ch 21, p 997)
The KPME Act 2007/2017 regulates every system of medicine, mandates rate display and gives patients a statutory grievance forum	Karnataka has not adopted the Clinical Establishments Act 2010 under Article 252(1); the standing criticism is that the central Act ideally should apply in all States, and if States continue with their old Acts its purpose is defeated and uniform national minimum standards are lost (Jugal Kishore, Ch 2, p 857)

Strength of the present arrangement	Deficiency the examiner expects you to name
A dedicated Medicare Service Personnel Act (2009), reinforced since 2020 by section 2B of the central Act	Enforcement, not enactment, is the weak link — the four enforcement tools available to public health officials (permits, licences and registrations; administrative orders; civil penalties; injunctions) need an inspectorate the State does not have (Jugal Kishore, Ch 2, pp 835, 857)
Devolution under the 73rd and 74th Amendments places sanitation and vital registration with panchayats and municipalities closest to the people (Sunder Lal, Ch 9, p 553)	The operative sanitary law is scattered across the Acts of 1964, 1976 and 1993 with different officers and penalties; a State Public Health Act — as in Tamil Nadu — defines public health nuisance far more broadly, is more detailed and comprehensive than municipal and panchayat provisions, and applies uniformly across the whole State rather than only in municipal areas (Jugal Kishore, Ch 1, p 106)
Central Acts supply modern content for food safety, waste, tobacco, mental health and disability	Legacy and fragmentation — Karnataka was constituted in 1956 from five administrative units, and the inherited State public health statutes that continued in the transferred regions (the Madras Public Health Act in the Madras-Karnataka districts, the Bombay Public Health Act in the Bombay-Karnataka districts) were never consolidated into one State law; several State Acts still date from 1961–1976
The Sakala Act gives statutorily timed service delivery	Vital registration and notification remain unreliable in accuracy, timeliness and completeness , worsened by multiple registering agencies — health, panchayat, police and revenue (Park 28e, Ch 21, p 996)
Health remains a State subject, so Karnataka can legislate ahead of the Centre, as it has on tobacco age and on epidemics	There is no statutory right to health, no public health cadre in law and no financing guarantee ; the sanction is penal against the citizen rather than a duty on the State to provide services

Way forward

- **Enact a consolidated Karnataka Public Health Act** with a schedule of notifiable diseases, on the template of the Government of India's **Model Public Health Act (1955)** (Park 28e, Ch 11, p 798) — India's standing gap is a comprehensive Public Health Act covering epidemic and disaster preparedness, communicable and non-communicable disease surveillance, environmental sanitation, safe water supply and waste disposal (Jugal Kishore, Ch 2, p 835).
- **Constitute a statutory public health authority** on the model of the proposed **National Public Health Act 2022** — a 4-tier system of Public Health Authorities at **national, State, district and block (taluk)** levels covering disaster management, epidemics, lockdown, quarantine and bioterrorism, framed in the light of the International Health Regulations 2005 and the UN General Assembly Resolution on non-communicable diseases, 2011 (Sunder Lal, Ch 13, p 671; Ch 7, p 286).
- **Harmonise the private-sector law** — adopt the Clinical Establishments Act 2010 under Article 252(1), or align KPME standards and the district registering authority (district collector chairperson, district health officer convener) with the central template (Jugal Kishore, Ch 2, pp 855–857).

- ▶ **Unify the local-body sanitary provisions** into one schedule for corporations, municipalities and panchayats alike, defining public health nuisance broadly and giving health officers uniform powers of entry, inspection and direct action (Jugal Kishore, Ch 1, p 106).
- ▶ **Build enforcement capacity** — a trained public health cadre and health officers with legal training, since a public health law without an inspectorate is inoperative.
- ▶ **Add the missing subjects** — non-communicable disease risk factors, biomedical and electronic waste, climate and heat action, health data privacy, and a statutory duty of emergency medical care.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw a **three-tier pyramid of public health law in Karnataka**, one box per tier, widening downwards, with a side arrow marking the gap:

- ▶ Apex: Constitution of India
- ▶ Middle: State enactments of the Karnataka legislature
- ▶ Base: Central Acts with State rules and authorities, resting on local self-government sanitary law
- ▶ Side arrow label: *Gap — no consolidated Karnataka Public Health Act*

High-yield exam pointers

- ▶ **The one-line answer** — Karnataka has public health *Acts*, not a Public Health *Act*: a State layer, a central layer and a local-body layer, with no consolidating statute.
- ▶ **Constitutional anchors** — State List entry 6; Concurrent List entries 29 and 30; Articles 14/15/21 and 47; Article 252(1) for adopting a central Act; Article 253 for the International Health Regulations.
- ▶ **Four State Acts to name** — Karnataka Epidemic Diseases Act **2020**; Karnataka Private Medical Establishments Act **2007** (amended **2017**); Medicare Service Personnel Act **2009**; Karnataka Medical Registration Act **1961**.
- ▶ **Clinical Establishments Act 2010** applies of its own force only to Arunachal Pradesh, Himachal Pradesh, Mizoram, Sikkim and the Union Territories; every other State must adopt it by resolution under **Article 252(1)**.
- ▶ **2020 Amendment numbers** — section 2A **widened** (conveyances and travellers, not just ships); section 2B **inserted**: up to **7 years**, fine up to **Rs 5 lakh**, **twice** the market value of damaged property, investigation in **30 days**, verdict within a year.
- ▶ **Registration of Births and Deaths Act 1969** — Karnataka, Tamil Nadu and Assam had their own registration Acts before it.
- ▶ **The criticism to close with** — a State without a specific Act simply invokes the Epidemic Diseases Act of 1897 and extends it year to year; a **uniform Act similar to the Registration of Births and Deaths Act, 1969** is deemed necessary before India's notification system can improve (Park 28e, Ch 21, p 997).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The indexed corpus carries the constitutional framework, the central Acts and the general critique of Indian public health law but no Karnataka statute text, so these specifics are public-domain State legislation not corpus-verified and should be checked in the Karnataka Gazette before a figure is quoted — the Karnataka State amendment to the Cigarettes and Other Tobacco Products Act (21-year minimum age, hookah-bar and loose-cigarette bans), the 2017 amendment details of the Karnataka Private Medical Establishments Act, the section numbers and penalties of the Karnataka Epidemic Diseases Act 2020 and the Medicare Service Personnel Act 2009, and the Madras and Bombay Public Health Acts inherited in the transferred districts at the 1956 reorganisation. Park 28e prints the Tamil Nadu statute as the "Madras Public Health Act, 1930", whereas the enactment is of 1939.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, Gazette of India, Karnataka Legislature and Gazette, Kerala Gazette, MoHFW, NFHS-6, Sample Registration System, WHO

Programme and policy updates (India)

- **Kerala has now done what Karnataka has not** — the **Kerala Public Health Act, 2023** (Act 28 of 2023) consolidates the two inherited colonial statutes, the **Madras Public Health Act 1939** and the **Travancore–Cochin Public Health Act 1955**, into a single State law built on a One Health, preventive approach; it constitutes a **State Public Health Committee** chaired by the Health Minister (s.4) and a **District Public Health Committee** chaired by the District Panchayat President (s.5), and creates the statutory office of **Local Public Health Officer** (s.2(z)) with enforcement powers [Kerala Act 28 of 2023]. The Act was passed on **28 November 2023**, and the published critique is that it falters on the public health ethics and human rights frame — the Siracusa Principles, autonomy, privacy and confidentiality, transparency, accountability, rule of law and least harm are not given even token consideration, leaving a law open to abuse against already vulnerable communities; the follow-up exchange adds that State accountability should be written in explicitly and that the Act's position on migrant workers and tuberculosis patients runs against established public health principles [PMID 38755770] [PMID 39183604]. That safeguard gap is exactly what a future Karnataka Act must close. Rajasthan took the other route, becoming the first State to legislate a right to health (Right to Health Care Act 2022, Governor's assent 12 April 2023) [Rajasthan 2023-Apr].
- **Karnataka's own two moves, both 2025** — a draft **Karnataka Right to Health and Emergency Medical Services Bill, 2025** would create a State Health Authority and district/city authorities and promise free care with progressively lower out-of-pocket spending, but it ties "eligible patient" to National Food Security Act 2013 households, empanels private hospitals and outsources ambulance services, and caps penalties at Rs 25,000 — hence the criticism that it gatekeeps rather than confers a right [Karnataka 2025-draft]. Separately the **Karnataka Private Medical Establishments (Amendment) Bill, 2025**, passed on 19 December 2025, grants provisional registration within 10 days without inspection (6 months, renewable), makes permanent registration valid 5 years with a 10-day fast track for NABH/NQAS/QAI/QCI-

accredited facilities, brings **mental health establishments** under KPME for the first time, punishes employment of unqualified doctors or paramedical staff with up to 3 years and Rs 1 lakh, and reconstitutes the district Registration and Grievance Redressal Authority under the Deputy Commissioner with the District Health Officer as member-secretary and a woman member for grievances [Karnataka 2025-Dec]. The amendment answers a documented failure: more than a decade after KPME 2007, implementation in Bengaluru Urban was suboptimal — inadequate policy content needing clarity in certain provisions, lack of coordination between implementing agencies, workforce shortages, poor knowledge and competency among officials, insufficient State oversight of districts, corruption and political interference [PMID 34599123] — and the public health cadre that would staff any inspectorate is itself still under evolution nationally, its main challenge an inadequate workforce skilled in public health management [PMID 37884946].

Current national data

- ▶ **NFHS-6 (2023–24) fact sheets were released on 29 May 2026** (IIPS the nodal agency; about 6.79 lakh households across 715 districts) and now supersede NFHS-5 (2019–21) for every figure quoted in a public health answer: TFR steady at replacement **2.0**, contraceptive prevalence **69.1%** with unmet need down to **8.5%**, institutional births **90.6%**, skilled birth attendance **91.3%**, full immunisation **87.1%**, any antenatal care **95.9%**, and health insurance coverage **60.2%** against 41.0% in NFHS-5. Over the same period Ayushman Bharat Health and Wellness Centres have been **renamed Ayushman Arogya Mandir**, with more than **1.86 lakh** (1,86,298) operational as of **30 June 2026** [MoHFW/IIPS 2026-May].
- ▶ **Vital statistics still come from the Registrar General, not the survey** — SRS 2022–24 puts the **MMR at 87** per 100,000 live births (254 in 2004–06, a 65% fall), and SRS 2024 gives **IMR 24**, neonatal mortality **18** and under-five mortality **28** per 1,000 live births; the NFHS-6 fact sheets do not carry these. Quote the round with the figure — an undated rate is the commonest fault in this answer, and it is the Registration of Births and Deaths Act 1969 machinery that generates the SRS series in the first place [SRS 2022-24] [SRS 2024].

Elimination, surveillance and disease status

- ▶ **Kala-azar — target met, validation pending** — India reached the elimination threshold of fewer than 1 case per 10,000 population at block level in **all 633 endemic blocks by 2023**; reported cases fell from 34,803 in 2005 to 8,243 in 2015 and 524 in 2023, and the number of blocks still above threshold went 633 (2005) → 136 (2015) → 8 (2021) → 0 (2023), the last being in Pakur district, Jharkhand. India transitioned in December 2023 into a **consolidation phase**, which must sustain elimination for a **minimum of three consecutive years** before a dossier goes to WHO for validation — Bangladesh was validated first, in 2023 — with post-kala-azar dermal leishmaniasis, HIV–VL co-infection and undiagnosed cases the residual human reservoir [PMID 40828749].
- ▶ **Deadlines that have moved — which is precisely what the examiner is testing** — measles–rubella elimination has been **reset to 2026** after earlier target years lapsed, with a National Zero Measles–Rubella Elimination Campaign launched during World Immunization Week (24–30 April 2025) and a 73% fall in measles and 17% fall in rubella cases in 2024 against 2023; malaria elimination stands at **2030** and lymphatic filariasis mass drug administration at **2027** [MoHFW 2025-Apr]. On the regulatory side India now runs a National Action Plan on AMR with six State

action plans, stewardship programmes, laboratory-network strengthening and intersectoral coordination, but enforcement gaps, over-the-counter availability, limited awareness and fragmented data still hold it back — an argument for putting AMR duties in statute rather than in programme guidelines alone [PMID 41443506].

Guideline and legislative updates

- ▶ **The Karnataka tobacco amendment is now law, and the textbook description of it has changed** — the **Cigarettes and Other Tobacco Products (Karnataka Amendment) Act, 2024** received **Presidential assent on 23 May 2025** and was notified in the State Gazette at the end of that month. It raises the minimum age of sale from 18 to **21 years**, inserts **section 4A** banning hookah bars in any establishment including cafes, pubs and restaurants with **1–3 years' imprisonment and a fine of Rs 50,000–1 lakh** under the new section 21A, prohibits sale of loose or single cigarettes and any tobacco sale within **100 metres** of an educational institution, extends use in a public place to **spitting** of tobacco, and raises the spot fine from **Rs 200 to Rs 1,000**. Enforcement is the weakness to name: before the amendment, loose tobacco sale was widely prevalent (**94%**) across the four Karnataka cities surveyed, and among the 207 vendors and 204 smokers studied **92%** of vendors and **92%** of smokers were unaware of the section 7(2) ban [PMID 36925494]; and a 2026 survey of 189 public places in Udupi City Municipal Corporation found only **38%** compliant with section 4 (17.5% fully, 20.5% partially), **62%** non-compliant chiefly for absent signage (48%), and active smoking observed at **14%** of sites [PMID 41783619].
- ▶ **Two central Acts run through Karnataka's authorities have themselves moved on** — under the **MTP (Amendment) Act 2021** the contraceptive-failure ground now extends to **any woman and her partner**, not only a married woman and her husband, and runs up to **20 weeks** under section 3(2)(a) read with Explanation 1, while the **20–24 week window** in clause (b) is open only to the categories prescribed by **Rule 3B of the MTP Rules 2021** — survivors of rape or incest, minors, change of marital status during pregnancy, physical disability, mental illness, substantial foetal abnormality, and women in humanitarian or disaster settings [MTP (Amendment) Act 2021, s.3(2)(a) with Explanation 1] [MTP Rules 2021, Rule 3B]. Three consequences the examiner can ask for: the Act **directs State Governments to establish a Medical Board** — a gynaecologist, a paediatrician, a radiologist or sonologist and other members appointed by the State Government — to determine whether a pregnancy may be terminated **beyond 24 weeks** for substantial foetal abnormality, a live duty on the Karnataka Appropriate Authority; it opens abortion services to **unmarried women**; and the residual criticism is that the law still addresses only cisgender heterosexual women rather than all persons with a uterus, so non-binary, trans and queer people who can conceive fall outside it — rape and sexual assault as an eligibility ground reaching them only as far as sections 375 and 376 of the Indian Penal Code do [PMID 39790798]. The **Mental Healthcare Act 2017**, administered by the Karnataka State Mental Health Authority, is now read as a rehabilitation mandate as well as a rights charter: it mandates residential rehabilitation such as halfway homes and supported or sheltered accommodation, endorses community-based rehabilitation, and requires families and stakeholders on Mental Health Review Boards and State Mental Health Authorities [PMID 41616408].

Global and WHO developments

- ▶ **The WHO Pandemic Agreement was adopted by consensus on 20 May 2025** at the Seventy-eighth World Health Assembly — only the second legally binding health treaty in WHO's 77-year history, after the Framework Convention on Tobacco Control (2003). Its core is **Article 12**, a Pathogen Access and Benefit-Sharing system under which participating manufacturers make **20% of real-time production** of pandemic-related health products available to WHO, at least half as donation and the remainder at affordable prices. The operational modalities were deferred to a separate binding annex due at the Seventy-ninth World Health Assembly in May 2026; under **Article 31** the treaty opens for signature only after that annex is adopted, and **60 ratifications** are needed for entry into force under **Article 33** — so a failed annex negotiation would leave the 2025 adoption hollow [PMID 41035553]. For a State-law answer the hook is Article 253 of the Constitution, under which India legislates domestically to give effect to such instruments.
- ▶ **The International Health Regulations (2005) were amended in 2024** — the package was adopted by consensus at the **Seventy-seventh World Health Assembly on 1 June 2024** and introduces a definition of a "**pandemic emergency**" to trigger more effective international collaboration [PMID 40083009]; the new determination covers events at risk of becoming, or that have become, a pandemic, is treated as the highest level of global alert, and answers the criticism that the public health emergency of international concern (PHEIC) was a binary on-or-off call, giving two tiers where there was one [PMID 40022025]. Every State Party must now **establish or designate a National IHR Authority** — one entity, or two, alongside the long-standing National IHR Focal Point — with the broad function of coordinating implementation of the Regulations within its jurisdiction [PMID 40083009]; the amendments took effect on **19 September 2025**, and in federal or decentralised States that coordination extends to subnational entities and will hinge on adequate resources, personnel and a **legal mandate** [PMID 40953844]. Equity and money run through the package — increased obligations of collaboration and financial support during public health emergencies, and a new **Coordinating Financial Mechanism** to help States identify and access financing [PMID 40083009] — while the multisectoral and **subnational** emphasis of the amendments obliges countries to amend their legal and institutional frameworks [PMID 39402755]: the strongest external argument for a consolidated Karnataka Public Health Act that names a State public health authority and its district and taluk tiers.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 11 — Nutrition and Health, p 798, citing Govt. of India, 1955, Model Public Health Act, Ministry of Health; Ch 21 — Health Information and Basic Medical Statistics, pp 996–997; Ch 23 — Health Planning and Management, pp 1033–1035); Jugal Kishore — National Health Programmes of India (Ch 1 — Health Care Delivery System in India, p 106; Ch 2 — Legislations Related to Health, pp 833–838, 850, 854–857, 865–866); Sunder Lal — Textbook of Community Medicine (Ch 2, pp 67–68; Ch 7, p 286; Ch 9, p 553; Ch 11, p 649; Ch 13, p 671 — proposed National Public Health Act 2022); Sathe & Sathe — Epidemiology and Management for Health Care (Ch 9, p 107); Oxford Textbook of Public Health 7e (Ch 20 — Law and the public's health, pp 774–775); AFMC Red Book of Community Medicine 2024 (Ch 39 — Public Health Legislations, pp 1238–1239). Coverage note: no Karnataka-specific statute text is in the indexed corpus, so the State Act list is drawn from public-domain legislation while the constitutional framework, the central Acts and the critical comment are corpus-grounded.

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have PHEIC determinations changed since the COVID-19 pandemic? A document analysis of 101 WHO International Health Regulations Emergency Committees' statements, *BMC Public Health* 2025, doi:[10.1186/s12889-025-22056-0](https://doi.org/10.1186/s12889-025-22056-0); 39402755 — The amended International Health Regulations: implications and challenges for domestic legal frameworks, *International Journal of Health Planning and Management* 2025, doi:[10.1002/hpm.3853](https://doi.org/10.1002/hpm.3853) (abstract-only). Non-journal sources: Medical Termination of Pregnancy (Amendment) Act, 2021 — section 3(2)(a) with Explanation 1, and Medical Termination of Pregnancy (Amendment) Rules, 2021 — Rule 3B, Gazette of India; Kerala Public Health Act, 2023 (Act 28 of 2023), Kerala Gazette — sections 2(z), 4 and 5; Cigarettes and Other Tobacco Products (Karnataka Amendment) Act, 2024 — Presidential assent 23 May 2025, Karnataka Gazette May 2025; Karnataka Private Medical Establishments (Amendment) Bill, 2025 — passed 19 December 2025; draft Karnataka Right to Health and Emergency Medical Services Bill, 2025; Rajasthan Right to Health Care Act 2022 — assent 12 April 2023; Ministry of Health and Family Welfare and IIPS — NFHS-6 (2023–24) fact sheets, released 29 May 2026; Registrar General of India — SRS special bulletin on maternal mortality 2022–24 and SRS 2024; Ministry of Health and Family Welfare — National Zero Measles–Rubella Elimination Campaign, April 2025, and Ayushman Arogya Mandir status as on 30 June 2026; World Health Organization — Pandemic Agreement adopted 20 May 2025 (WHA78) and International Health Regulations amendments agreed 1 June 2024 (WHA77), in force 19 September 2025.



Community development programme

Asked in: Paper IV May 2019 — RGUHS MD Community Medicine

Definition — The United Nations defines **community development** as the process by which the efforts of the people themselves are united with those of governmental authorities to improve the economic, social and cultural conditions of communities, to integrate these communities into the life of the nation, and to enable them to contribute fully to national progress; it is a process designed to create conditions of economic and social progress for the whole community, with its active participation and the fullest possible reliance on the community's own initiative (Park 28e, Ch 23, p 1036). The **Community Development Programme (CDP)** is India's operational form of that idea — launched on **2 October 1952**, during the First Five Year Plan, for the all-round development of rural areas, and hailed as development "**of the people, for the people, by the people**" to exterminate "**the three ills of poverty, disease and illiteracy**" (Park 28e, Ch 23, p 1036).

Genesis, philosophy and place in Indian public health

Item	Detail
Launched	2 October 1952, during the First Five Year Plan (Park 28e, Ch 23, p 1036)
Rationale	To involve the rural population in its own welfare measures; at launch nearly 74 per cent of India's population lived in rural areas (Park 28e, Ch 23, p 1036)

Item	Detail
Guiding principle	Work from the people's own " felt needs "; self-help and the community's own initiative, with government aid as the supplement, not the substitute
Slogan and target	Development "of the people, for the people, by the people"; abolition of the three ills — poverty, disease, illiteracy
Historical standing	One of the two great human-development movements of the first half of the last century in India, the other being the provision of "basic health services" through primary health centres and subcentres (Park 28e, Ch 1, p 20)
Administrative innovation	Creation of the Community Development Block — since then the permanent unit of rural planning and development (Park 28e, Ch 23, p 1036)
Political counterpart	Panchayati Raj — democratic decentralisation, the three-tier structure recommended by the <i>Balwant Rai Mehta</i> Committee, 1957 (Sunder Lal, Ch 2, p 67)

Programme content — the multipurpose package

The CDP was envisaged as a **multipurpose programme**, and the examiner expects the list in Park's order (Park 28e, Ch 23, p 1036):

- ▶ **Improvement of agriculture** — the economic base of the village.
- ▶ **Improvement of communications** — village roads and links to markets and services.
- ▶ **Education.**
- ▶ **Health and sanitation** — delivered *through the establishment of primary health centres and subcentres*; this is the component of direct community-medicine interest.
- ▶ **Improvement of housing through self-help.**
- ▶ **Social welfare.**
- ▶ **Training in rural arts, crafts and industries** for local people.

Organisation and machinery

Panchayati Raj is a **three-tier structure of rural local self-government linking the village to the district**, and it is the machinery through which the CDP is delivered (Park 28e, Ch 23, pp 1035–1036).

Level	Body	Composition	Chief function in the CDP
Village	<i>Gram Sabha, Gram Panchayat, Nyaya Panchayat</i>	Gram Sabha = the entire adult population of the village, meeting at least twice a year ; it considers proposals for taxation, discusses the development programme and elects the Gram Panchayat, which has 9–15 elected members headed by a <i>Sarpanch/Sabhapati/Mukhiya</i> , with a Secretary; <i>Nyaya Panchayat</i> = the village court for speedy justice	Village planning and execution: agricultural production, rural industries, medical relief and maternal and child health , village roads and streets, tanks and sanitation ; the Panchayat's powers cover the entire field of civic administration including sanitation and public health (Park 28e, Ch 12, pp 841–842)

Level	Body	Composition	Chief function in the CDP
Block	<i>Panchayat Samiti / Janpada Panchayat</i>	All <i>Sarpanchas</i> of the village panchayats in the block; Members of the Legislative Assembly and Members of Parliament residing in the block; representatives of women, scheduled castes, scheduled tribes and cooperative societies; the Block Development Officer (BDO) is the ex-officio secretary	Execution of the community development programme in the block ; all government funds for Stage I and Stage II are channelled through it; the BDO and his staff give technical assistance and guidance to the village panchayats (Park 28e, Ch 23, p 1036)
District	<i>Zilla Parishad / Zilla Panchayat</i>	All heads of the Panchayat Samitis of the district; Members of Parliament and Members of the Legislative Assembly of the district; representatives of scheduled castes, scheduled tribes and women; two persons of experience in administration, public life or rural development; the Collector is a non-voting member ; total strength 40–70	Primarily a supervisory and coordinating body; functions and powers vary from state to state — in some states all development agencies, including health, are under its control (Park 28e, Ch 23, p 1036)

The Community Development Block (the operating unit) — about **100 villages** and a population of **80,000 to 1,20,000** (roughly one lakh); the area may or may not coincide with a tahsil; there are about **6,000 CD blocks** in the country, each headed by a **Block Development Officer** (Park 28e, Ch 23, pp 1035–1036).

The Village Level Worker (*Gram Sevak*) — the key grass-roots functionary, described by Park as "the key person responsible for transforming the economic and social life of the people". Each *gram sevak* is in charge of **10 villages** and attends to **5,000–6,000 people**, lives among them, probes their "**felt needs**", arouses interest in all-round family and village development, and functions as a **multipurpose worker and the link between the people and governmental agencies** (Park 28e, Ch 23, p 1037). He is the direct conceptual ancestor of today's multipurpose health worker.

Phasing and financing

Phase	Duration	Central assistance per block	Remark
Stage I	5 years of intensive development	Rs 12 lakh	Substantial central support; funds routed through the Panchayat Samiti
Stage II	The next 5 years	Rs 5 lakh	Tapering central support
Post-Stage II	After 10 years	Nil from the Centre	Financial arrangements become the responsibility of the State Governments ; the block continues as the permanent infrastructure for rural planning and development

(Park 28e, Ch 23, p 1036)

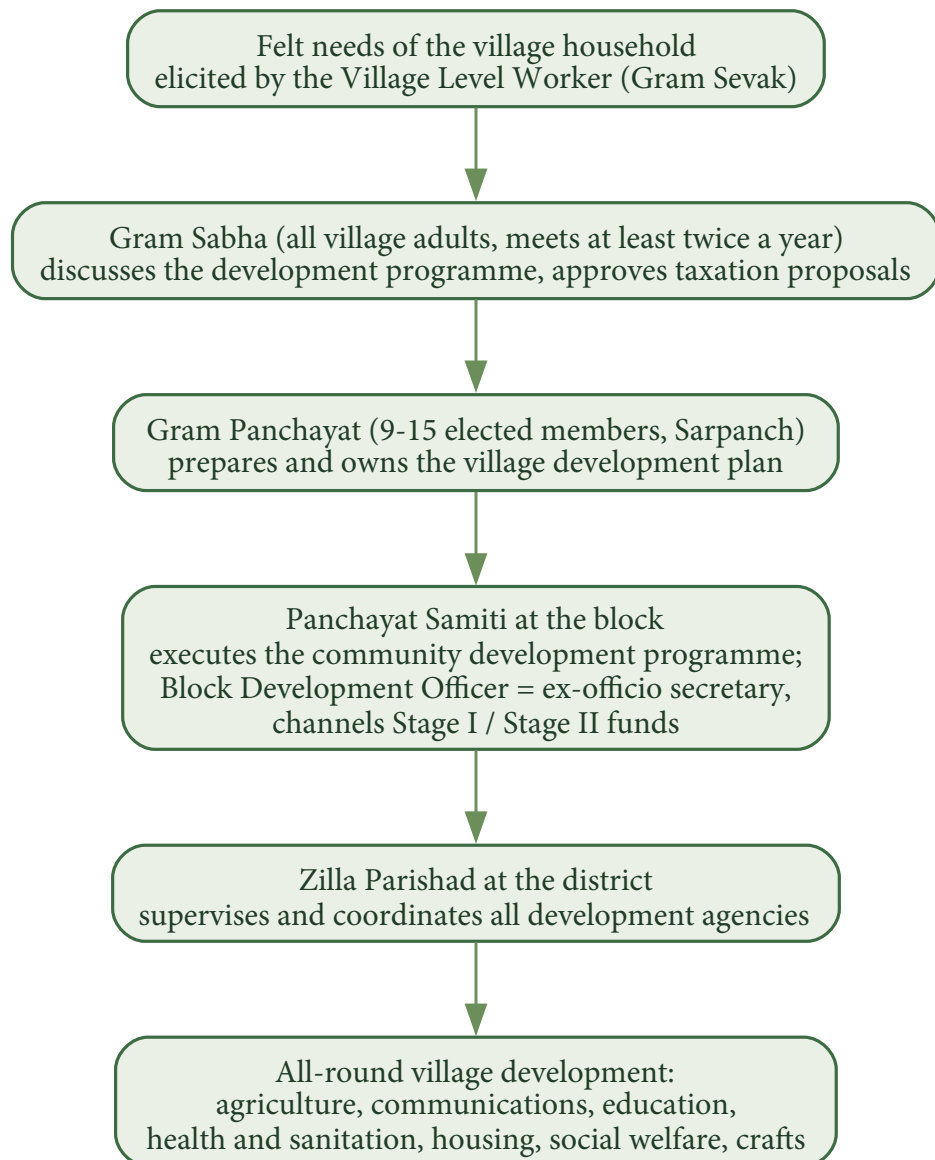
Evaluation — achievements, shortcomings and what followed

Achievements	Shortcomings
Created the CD block, which has survived as the permanent unit of rural planning, development and health administration	Did not bring about all-round improvement in rural areas, nor eliminate rural poverty and unemployment
Built the rural health infrastructure — one primary health centre with its subcentres per block of about one lakh population — described by Park as the lasting gain of the programme	The hope that the people would unite their efforts with the Government's to build a village community free of disparities in income and wealth was not realised
Gave India an early, world-leading experience of community involvement in health, later formalised as primary health care (Sunder Lal, Ch 17, pp 741–742)	The benefits did not reach the weaker sections of the community
Provided the administrative scaffolding on which Panchayati Raj and, later, decentralised health planning were built	It "tried to do too much too quickly with inadequate resources" and failed to survive as a movement — "a great opportunity lost" (Park 28e, Ch 1, p 20)

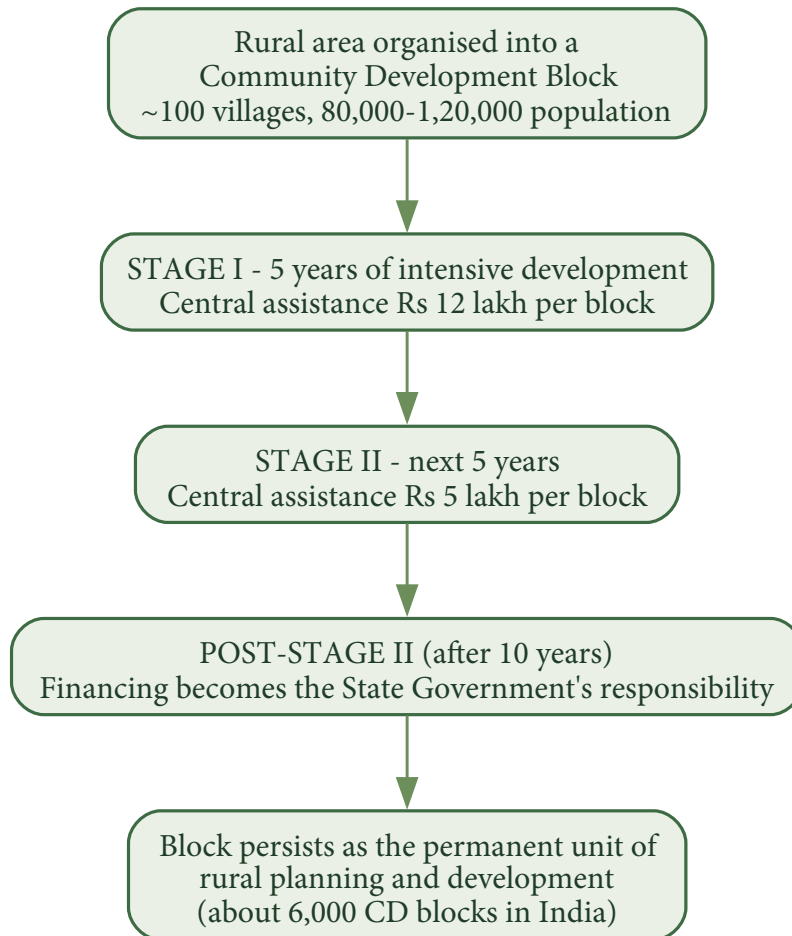
What followed — the search for effective anti-poverty strategy continued with the **Integrated Rural Development Programme (IRDP)**, launched in April 1978 to eliminate rural poverty and improve the quality of life of the rural poor, targeting agricultural labourers, small cultivators, village artisans and craftsmen with resources, skills, bank loans and subsidies, implemented through the **District Rural Development Agency (DRDA)**. In the Ninth Five Year Plan, Training of Rural Youth for Self-Employment (TRYSEM), Supply of Improved Toolkits to Rural Artisans (SITRA), Development of Women and Children in Rural Areas (DWCRA) and Ganga Kalyan Yojana (GKY) were merged with it, with a strategic shift from the individual-beneficiary approach to the formation of **self-help groups** built around locally viable activities (Park 28e, Ch 23, p 1037). The political consolidation came with the **73rd Constitutional Amendment Act, 1992** (passed December 1992, in force April 1993), which made Panchayati Raj Institutions statutory local self-governments responsible for **29 subjects including primary health care, education and family planning**, with **one-third of seats reserved for women** (Sunder Lal, Ch 2, pp 67–68; Ch 10, p 633).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

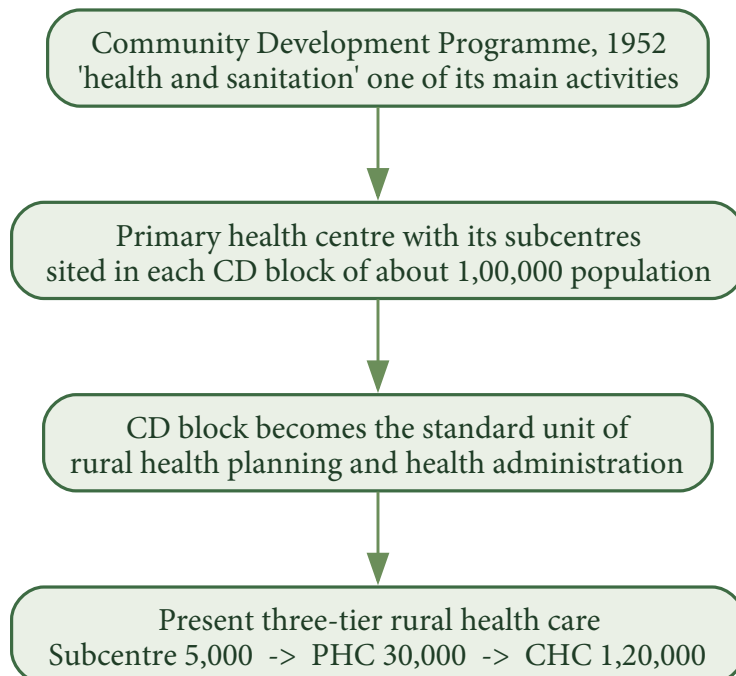
- **Drawn figure** — "Three-tier Panchayati Raj pyramid" (label from apex downwards):
Zilla Parishad / Zilla Panchayat — District — Collector is a non-voting member; Panchayat Samiti / Janpada Panchayat — Block (~100 villages, 80,000–1,20,000 population) — Block Development Officer, ex-officio secretary; Gram Panchayat (9–15 members, Sarpanch) with Gram Sabha and Nyaya Panchayat — Village — Village Level Worker (Gram Sevak), 10 villages. Mark the broad arrow up the left side as "**felt needs / people's participation**" and the arrow down the right side as "**funds, functions, functionaries**".



The community development process — how a village felt need becomes an executed block programme



Phasing and central financing of a Community Development Block



Health legacy of the Community Development Programme - from CD block to today's rural health care tiers

High-yield exam pointers

- **Dates and numbers:** launched 2 October 1952 (First Five Year Plan); block = ~100 villages, 80,000–1,20,000 population, ~6,000 blocks; Stage I = 5 years, Rs 12 lakh, Stage II = 5 years, Rs 5 lakh, then State funding.
- **Functionaries:** Block Development Officer heads the block and is *ex-officio secretary* of the Panchayat Samiti; Village Level Worker (Gram Sevak) covers 10 villages / 5,000–6,000 people.
- **Committee and Act:** *Balwant Rai Mehta* Committee 1957 recommended the three-tier structure; 73rd Constitutional Amendment Act 1992 (effective April 1993) — 29 subjects, one-third seats reserved for women.
- **Quotable phrases:** development "of the people, for the people, by the people"; abolition of "the three ills of poverty, disease and illiteracy"; the programme "tried to do too much too quickly with inadequate resources".
- **The health answer:** the one durable gain is the **primary health centre and subcentre network**, one PHC per CD block of ~1,00,000 population — write this even if the question looks purely administrative.
- **Failure line to write:** benefits did not reach the **weaker sections**, and rural poverty and unemployment were not eliminated.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's statement that "nearly 74 per cent" of India's population is rural belongs to the 1952 context of the programme; the same textbook's current figure is 640 per 1,000 population living in villages, in 6,64,235 villages, as of 2023 (Park 28e, Ch 12, p 831) — quote whichever you use with its context. Coverage note: the pre-1952 pilot projects sometimes cited for this topic are not described in the indexed corpus, so they are deliberately omitted here.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoHFW, National Health Mission, NFHS-5, Household Consumption Expenditure Survey 2022–23, Ministry of Panchayati Raj, WHO

Programme and policy updates (India)

- **The CDP's health component has been renamed — write the new name.** In November 2023 the Ayushman Bharat Health and Wellness Centres (the modern form of the "one PHC with its subcentres per block" that the CDP created) were rebranded **Ayushman Arogya Mandir**, with the tagline *Arogyam Paramam Dhanam* — health is the supreme wealth — and an explicit reorientation from illness care towards AYUSH-informed holistic wellness (physical, mental and social, with yoga, nutrition, lifestyle counselling and elderly care). The exercise was budgeted at ₹3,000 per facility, total outlay ₹48 crore, with facilities required to upload photographs to the AB-HWC portal for transparency and standardisation, and facility names written in the local state script with the tagline transliterated. The same authors caution that future cost-effectiveness remains questionable where facility and staffing changes were minor, and that the term "Mandir" raised religious and cultural inclusivity concerns, prompting responses in Ladakh, Nagaland and Meghalaya. [PMID 42257165]

- **Community ownership is now institutionalised as a named committee at every tier.** By 30 November 2024, 1,35,343 Jan Arogya Samitis had been constituted at Ayushman Arogya Mandirs, alongside Village Health Sanitation and Nutrition Committees (rural) and Mahila Arogya Samitis (urban) as the convergence platforms. The block survives as the unit of mass outreach: 4,189 Block Health Melas were held nationwide in April–May 2022 drawing over 40 lakh beneficiaries, later routinised through the Ayushman Bhav campaign into the monthly Ayushman Arogya Shivar. [PMID 41816146]
- **The self-help group has become the successor delivery platform for the CDP's "felt needs" approach.** In Bihar, health-and-nutrition layering on the JEEViKA SHG network (Bihar Rural Livelihoods Promotion Society, World Bank-supported) expanded from 101 blocks in 11 districts in 2016 to 349 blocks in 35 districts in 2018, while statewide SHG membership among recently delivered women rose from 32% to 47% between 2016 and 2021. Comparing 2021 with the 2016 baseline in the scale-up blocks, SHG-member mothers were more likely to report ≥ 3 antenatal visits (aOR 1.48), institutional delivery (aOR 1.71), dry cord care (aOR 2.64) and early initiation of breastfeeding (aOR 1.61), all $p < 0.0001$. [PMID 39882114]

Current national data

- **Panchayat performance now has a national metric, and it is low.** The Gram Panchayat Development Index (Ministry of Panchayati Raj, introduced 2018; six themes including health, nutrition and sanitation; maximum score 100, data from Mission Antyodaya) averaged only 39.4 for India in 2020 — Kerala 67.5 and Gujarat 59.7 at the top, Manipur 22.0 and Meghalaya 24.6 at the bottom. Rural multidimensional poverty from NFHS-5 (2019–21) was 0.110 for India (headcount 26.0%), and correlated inversely with panchayat performance ($r = -0.518$, $p < 0.01$); a 1% rise in local-government performance was associated with a 0.32% fall in rural MPI. National multidimensional poverty incidence fell from 55.8% (2006) to 16.8% (2021). [PMID 40007909]
- **Sanitation — the CDP's original "health and sanitation" head — is still unfinished, and still rural.** The Household Consumption Expenditure Survey 2022–23 (261,746 households, 36 states/UTs) found 12.5% of Indian households with no toilet at all — about 162.7 million people; prevalence was 17.4% rural against 1.6% urban, with 93% of the no-toilet households rural, and over 70% of the affected people in just six states (Bihar, Uttar Pradesh, Odisha, Rajasthan, Madhya Pradesh, Jharkhand). Rajasthan, which declared itself open-defecation free in 2018, still had almost 19% of residents without a toilet in 2023. [PMID 40536324]

Elimination, surveillance and disease status

- **The CD block is the statutory unit of kala-azar elimination.** Elimination as a public health problem is defined as an annual incidence of < 1 case per 10,000 population at block level (the block is the sub-district implementation unit); only 7 of 633 endemic blocks exceeded the threshold in 2021, and India reached the threshold in 2023 — a deadline first set for 2010 under the 2002 National Health Policy and revised repeatedly since. In Dumka, Jharkhand, cases fell from 158 (2019) to 43 (2023), with village-level governing bodies — Panchayati Raj Institution members, self-help groups under the state livelihood mission, ASHAs and Anganwadi workers — driving social mobilisation, and pucca housing built under the state Ambedkar Awas Yojana

(66 houses) alongside PMAY-G allotments used to modify the sandfly-favourable ecology. [PMID 40988627]

- ▶ **TB Mukht Panchayat is the largest formal re-use of the CDP's panchayat machinery for a disease-elimination goal.** Launched nationwide on **24 March 2023**, it leverages roughly **2,55,000 village local self-governments** to certify "TB-free" panchayats annually against NTEP verification indicators, with district teams including Panchayati Raj Institution officials doing the verification. Rajasthan's precursor **TB Mukht Gram Panchayat** (National Health Mission, **15 August 2022**) covered 1,412 gram panchayats/wards through Patient Support Groups of **10–12 members including the Sarpanch, Deputy Sarpanch, Panchayat Secretary, medical officers, TB champions, CHOs, ANMs, ASHAs and Anganwadi workers**, plus NiKshay Gram Sabhas held on Independence Day and Republic Day; five panchayats/wards were selected per TB unit and assessed in two stages on program delivery, active case finding, community awareness, community support, district capacity and TB incidence/prevalence/mortality — of 51 finally assessed, **29 achieved TB-free status and 13 were progressing**. [PMID 38756161] [PMID 40949527]

Guideline and legislative updates

- ▶ **Community Action for Health (CAH)**, the National Health Mission's formal community-participation strategy operating through Jan Arogya Samitis and VHSNCs, shows the same "benefits did not reach the weaker sections" failure mode the CDP was criticised for: in a 2021–22 comparative cross-sectional study of 73 facilities and 219 respondents in one aspirational district (Goalpara) and one non-aspirational district (Nagaon) of Assam, **94.1% of community members in the aspirational district and 53.6% in the non-aspirational district were unaware of CAH**, and the support needed for VHSNC member training and for funds to run the VHSNC differed significantly between the two districts (both $p = 0.010$). (Abstract-only; full text not in PMC.) [PMID 41045525]
- ▶ **Swachh Bharat Mission gives the strongest quantified evidence yet that a village-sanitation programme saves child lives.** SBM was launched on **2 October 2014** — the CDP's own anniversary date — explicitly as a *Jan Andolan* or people's movement, and built over 100 million household toilets between 2014 and 2020. In a quasi-experimental two-way fixed-effects analysis of **640 districts over 2011–2020**, districts where **> 30% of households received a toilet** under the mission had infant mortality lower by 5.3 and under-five mortality lower by **6.8 per 1,000 live births** (both $p < 0.05$), with a parallel gain in height-for-age z-score (+0.18, $p < 0.05$) above the same 30% threshold — corresponding to roughly **60,000–70,000 infant deaths averted annually**. [PMID 39223201]

Global and WHO developments

- ▶ **A "People" seventh building block has been proposed for the WHO health-systems framework.** The argument against the six blocks of the 2007 *Framework for Action* is that they privilege institutional capacity and omit community participation and the social determinants of behaviour; India's ASHAs, VHSNCs and panchayats are offered as the working proof, and the proposal aligns with WHO's Integrated People-Centred Health Services framework and the Global Strategy for Health Systems Resilience — the CDP's "of the people, by the people" premise restated as health-systems architecture. [PMID 41907566]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 23 (Health Planning and Management, pp 1031–1037), Ch 12 (Medicine and Social Sciences, pp 825–842), Ch 1 (Man and Medicine: Towards Health for All, p 20); Sunder Lal — Textbook of Community Medicine, Ch 2 (pp 67–69), Ch 10 (p 633), Ch 17 (Healthcare of the Community, pp 741–745).

Recent-advances sources: all verified against PubMed Central full text on 2026-08-16 except where marked. 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, *Journal of Family Medicine and Primary Care* 2026;15(3):1459–1460, doi:[10.4103/jfmpc.jfmpc.1966.24](https://doi.org/10.4103/jfmpc.jfmpc.1966.24) (PMC13241129); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026;15(1):16–25, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25) (PMC12975087); 39882114 — Effectiveness of a technical support program with women's self-help groups in catalyzing health and nutrition behaviors in Bihar — a multicomponent analysis, *Frontiers in Public Health* 2025;12:1389706, doi:[10.3389/fpubh.2024.1389706](https://doi.org/10.3389/fpubh.2024.1389706) (PMC11774890); 40007909 — Caste, local governance effectiveness, and multidimensional poverty in rural India: some evidences, *Frontiers in Sociology* 2025;10:1482825, doi:[10.3389/fsoc.2025.1482825](https://doi.org/10.3389/fsoc.2025.1482825) (PMC11850394); 40536324 — Prevalence and burden of no-toilet households in India: an analysis of 261,746 households in 36 states/Union Territories in 2022–2023, *Global Health Action* 2025;18(1):2511351, doi:[10.1080/16549716.2025.2511351](https://doi.org/10.1080/16549716.2025.2511351) (PMC12180355); 40988627 — Integrating community efforts, innovative government interventions and data in the fight against kala-azar: lessons from Dumka, India, *International Health* 2026;18(3):352–356, doi:[10.1093/inthealth/ihaf083](https://doi.org/10.1093/inthealth/ihaf083) (PMC13154818); 38756161 — How can TB Mukht Panchayat initiative contribute towards ending tuberculosis in India?, *The Lancet Regional Health — Southeast Asia* 2024;24:100376, doi:[10.1016/j.lansea.2024.100376](https://doi.org/10.1016/j.lansea.2024.100376) (PMC11096675); 40949527 — Assessing the TB Mukht Gram Panchayat Program in Rajasthan: a way toward tuberculosis elimination, *Indian Journal of Community Medicine* 2024;50(Suppl 1):S30–S34, doi:[10.4103/ijcm.ijcm.74.24](https://doi.org/10.4103/ijcm.ijcm.74.24) (PMC12430851); 41045525 — Status of Community Action for Health in selected districts of Assam, India: a quantitative study, *Indian Journal of Public Health* 2025;69(3):260–266, doi:[10.4103/ijph.ijph.1006.23](https://doi.org/10.4103/ijph.ijph.1006.23) (abstract-only — no PMC full text; bullet is abstract-consistent); 39223201 — Toilet construction under the Swachh Bharat Mission and infant mortality in India, *Scientific Reports* 2024;14(1):20340, doi:[10.1038/s41598-024-71268-8](https://doi.org/10.1038/s41598-024-71268-8) (PMC11369168); 41907566 — Strengthening the structure of public health: suggesting the incorporation of 'People' component as the 'seventh block', *Annals of Global Health* 2026;92(1):26, doi:[10.5334/aogh.5012](https://doi.org/10.5334/aogh.5012) (PMC13025245).

♦ ♦ ♦

Community participation in primary health care

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — Community participation (community involvement) is "the process by which individuals and families assume responsibility for their own health and welfare and for those of the community, and develop the capacity to contribute to their and the community's development" (World Health Organization/United Nations Children's Fund; Park 28e, Ch 1, pp 21-22; Sunder Lal, Ch 2, pp 68-69). It converts the health system from **health care for the people into health care by the people** — "placing people's health in people's hands" — and is one of the four principles on which the Alma-Ata (1978) primary health care strategy rests (Park 28e, Ch 2, p 34; Ch 10, p 720).

Place of community participation among the principles of primary health care

Principle (Alma-Ata, 1978)	Core requirement	How participation operates within it
Equitable distribution	Services shared equally by all, rich or poor, urban or rural, irrespective of ability to pay (Park 28e, Ch 10, p 719)	Local people identify who is being left out and press for outreach to them
Community participation	Continuing effort to secure meaningful involvement of the community in planning, implementation and maintenance of services, with maximum reliance on local manpower, money and materials (Park 28e, Ch 10, p 720)	The principle itself — universal coverage by primary health care cannot be achieved without the local community
Intersectoral coordination	Primary health care needs agriculture, animal husbandry, food, industry, education, housing, public works and communication besides the health sector (Park 28e, Ch 10, p 720)	Village-level committees are the point where sectors converge on one action plan
Appropriate technology	Technology that is scientifically sound, adaptable to local needs, acceptable to those who use it, and maintainable by the people themselves (Park 28e, Ch 10, p 720; Sunder Lal, Ch 17, p 741)	Acceptability and maintenance are decided by the community, not the provider

The definition of primary health care itself embeds participation: essential health care made universally accessible to individuals and families "through their full participation and at a cost that the community and the country can afford" (Park 28e, Ch 2, p 49). No standard pattern of participation can be prescribed, because economic, social, political and cultural conditions vary widely between and within communities — **flexibility of approach is essential** (Park 28e, Ch 2, p 34).

Forms of participation — what the community actually contributes

Park describes **three broad ways** in which a community can participate: by *providing* resources, by being *actively involved* in planning, management and evaluation, and by *joining in and using* the services, especially preventive and protective ones (Park 28e, Ch 2, p 34). Expanded across the programme cycle these become:

Domain	What the community does	Indian example
Resource contribution	Provides facilities, manpower, logistic support and possibly funds; generates its own resources for sub-centre and school buildings	Community land/labour for sub-centre and Anganwadi buildings (Sunder Lal, Ch 2, pp 68-69)
Needs assessment and priority setting	Determines its own health needs and priorities	Village Health Action Plan prepared by the Village Health, Sanitation and Nutrition Committee (Sunder Lal, Ch 16, p 697)
Implementation / service delivery	Organises and runs the programme with its own plus government resources	ASHA, Anganwadi worker, trained <i>dai</i> ; monthly Village Health, Nutrition and Sanitation Day (Park 28e, Ch 10, p 720)
Management and governance	Manages health services at village and facility level	Sub-centre managed by village <i>panchayat</i> ; <i>Rogi Kalyan Samiti</i> at PHC/CHC/district hospital (Sunder Lal, Ch 2, pp 68-69; Ch 17, pp 751-752)
Monitoring	Watches government-run services and feeds back	Community-based monitoring under the National Health Mission; committee verifies that the ANM and multipurpose worker visit on fixed days (Jugal Kishore, NHP, p 138)
Evaluation and accountability	Evaluates programmes and gives a "report card" to government	Review of every maternal and neonatal death in the village and its registration in the <i>panchayat</i> (Jugal Kishore, NHP, p 138)
Utilisation / self-care	Joins in and uses the services offered	Uptake of immunisation, antenatal care, contraception; self-care activities (Park 28e, Ch 2, p 34)

Engagement is a **continuum, not a yes/no state** — informing, consulting and active participation reflect progressively higher levels of involvement, and true empowerment means the community gains control over the decisions and actions that influence its health (Oxford Public Health 7e, Ch 100, pp 3829-3830).

Institutional mechanisms of community participation in India

Mechanism	Norm / coverage	Composition or selection	Untied fund	Principal function
Panchayati Raj Institutions	73rd and 74th Constitutional Amendment Acts, 1992 made health, family welfare and education a responsibility of village <i>panchayats</i> and <i>nagarpalikas</i> (Sunder Lal, Ch 9, p 553)	Elected local representatives	Administrative and financial powers devolved	Decentralised planning; PHCs proposed for transfer to elected local bodies (Sunder Lal, Ch 17, pp 751-752)

Mechanism	Norm / coverage	Composition or selection	Untied fund	Principal function
Village Health, Sanitation and Nutrition Committee (VHSNC)	One per village; 5.55 lakh set up by 2021 (Park 28e, Ch 7, p 539)	<i>Sarpanch</i> /PRI members, ASHA, auxiliary nurse midwife or multipurpose health worker (female), Anganwadi worker, school teacher, self-help group member, NGO (Sunder Lal, Ch 16, p 697; Ch 17, p 744)	Rs 10,000 per year	Village Health Action Plan; manages the village health fund; monitors ASHA, Anganwadi worker and sub-centre
Accredited Social Health Activist (ASHA)	One per 1000 population; relaxable to one per habitation in tribal, hilly and desert areas; urban ASHA one per 1000-2500 urban poor (200-500 households). Over 10.72 lakh working (Park 28e, Ch 7, p 539; Sunder Lal, Ch 17, pp 744, 758)	Resident woman of the village, married/widowed/divorced, 25-45 years, formal education up to 8th class, communication skills and leadership qualities; chosen by and accountable to the village <i>panchayat</i>	Performance-based incentives	Interface between the community and the public health system
Anganwadi worker	One per 400-800 population in plain areas, 300-800 in tribal and difficult areas (Sunder Lal, Ch 17, p 744)	Honorary village-based resident worker under Integrated Child Development Services, chosen by village people	—	Runs the Anganwadi centre; hosts the monthly health and nutrition day
Rogi Kalyan Samiti (patient welfare committee)	At PHC, CHC, sub-district and district hospital	Local representatives, officials and community members	PHC Rs 25,000/year untied + Rs 50,000/year maintenance; Rs 1 lakh/year to PHC- and CHC-level committees; Rs 5 lakh/year at district hospital (Jugal Kishore, NHP, pp 154-155; Sunder Lal, Ch 17, pp 751-752)	Community management and control of public hospitals; local quality improvement
Mahila Arogya Samiti (MAS)	Urban; 10-12 women for 50-100 slum households (250-500 population) (Sunder Lal, Ch 17, p 758)	Elects its own chairperson and treasurer; supported by urban ASHA	Rs 5,000 per year	Peer-educator group; community-based monitoring; referral linkage; manages a revolving fund

Mechanism	Norm / coverage	Composition or selection	Untied fund	Principal function
Mahila Swasthya Sangh (MSS)	79,512 working at village level (Sunder Lal, Ch 17, p 744)	Village women's health group	—	Women's collective for health promotion

Historical models to name. China's *barefoot doctors* were community participation on an unprecedented scale and directly influenced the Indian schemes; India's own **Village Health Guide scheme, launched 1977**, and the training of local *dais* used workers selected by the local community and trained locally, delivering care free of charge and overcoming cultural and communication barriers; ASHA and the Anganwadi worker are now regarded as an essential feature of primary health care in India (Park 28e, Ch 2, p 34; Ch 10, p 720).

Barriers to genuine participation

- ▶ **Social stratification** — traditional Indian society is cut across rigid religion and caste lines, and the role assigned to each caste group has been a serious obstacle to securing complete community participation (Park 28e, Ch 2, p 34).
- ▶ **Professional resistance** — the greatest resistance to the health guide's involvement in primary health care came from the **medical profession**, not from the lay public (Park 28e, Ch 2, p 34).
- ▶ **Tokenism** — participation "has become an aphorism that is still awaiting genuine realization in many countries" (Park 28e, Ch 2, p 34).
- ▶ **Elite capture** — those selected are often already confident, articulate, naturally identified leaders, while the most marginalised are rarely reached; top-down, provider-driven approaches remain the commonest (Oxford Public Health 7e, Ch 87, p 3306).
- ▶ **Committees that exist on paper** — the number of *Rogi Kalyan Samiti* meetings has fallen and needs strengthening; Panchayati Raj involvement in the Integrated Child Development Services is weak (Jugal Kishore, NHP, p 154; Sunder Lal, Ch 5, pp 229-230).
- ▶ **Loss of interest** — voluntary involvement dies when the community sees no concrete outcome, when meetings are full of jargon, and when no financial support is provided (Oxford Public Health 7e, Ch 100, pp 3830-3831).
- ▶ **No accountability loop** — representatives without a mechanism to link back to, and answer to, the wider community (Oxford Public Health 7e, Ch 100, p 3830).

Requirements for effective community participation

- ▶ The community must be seen as **part of the solution, not the problem** (Oxford Public Health 7e, Ch 100, p 3830).
- ▶ Community representatives need mechanisms to link to the wider community, and their input must be accountable to it; every strategy must deliberately include marginalised minority groups.
- ▶ **Issue selection by the community** — the community raises the issues that concern it, rather than receiving a pre-set agenda.
- ▶ Financial support, visible concrete outcomes and jargon-free meetings sustain participation.

- *Minkler's* five foundational principles of community organisation: **empowerment, community competence, participation, issue selection, and creating "critical consciousness"** (Oxford Public Health 7e, Ch 47, pp 1748-1749).
- Strong **political will** and legislation to make intersectoral coordination and devolution real, since participation without devolved authority and funds is consultation only (Park 28e, Ch 10, p 720).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Village Health, Sanitation and Nutrition Committee — spoke diagram.** Hub: *Village Health Action Plan + untied fund Rs 10,000/year*. Spokes labelled: *Sarpanch / Panchayati Raj member (chair), ASHA, Anganwadi worker, ANM or multipurpose health worker (female), school teacher, self-help group member, NGO representative*. Outputs arrow: *monthly Village Health, Nutrition and Sanitation Day → RMNCH+A services*.

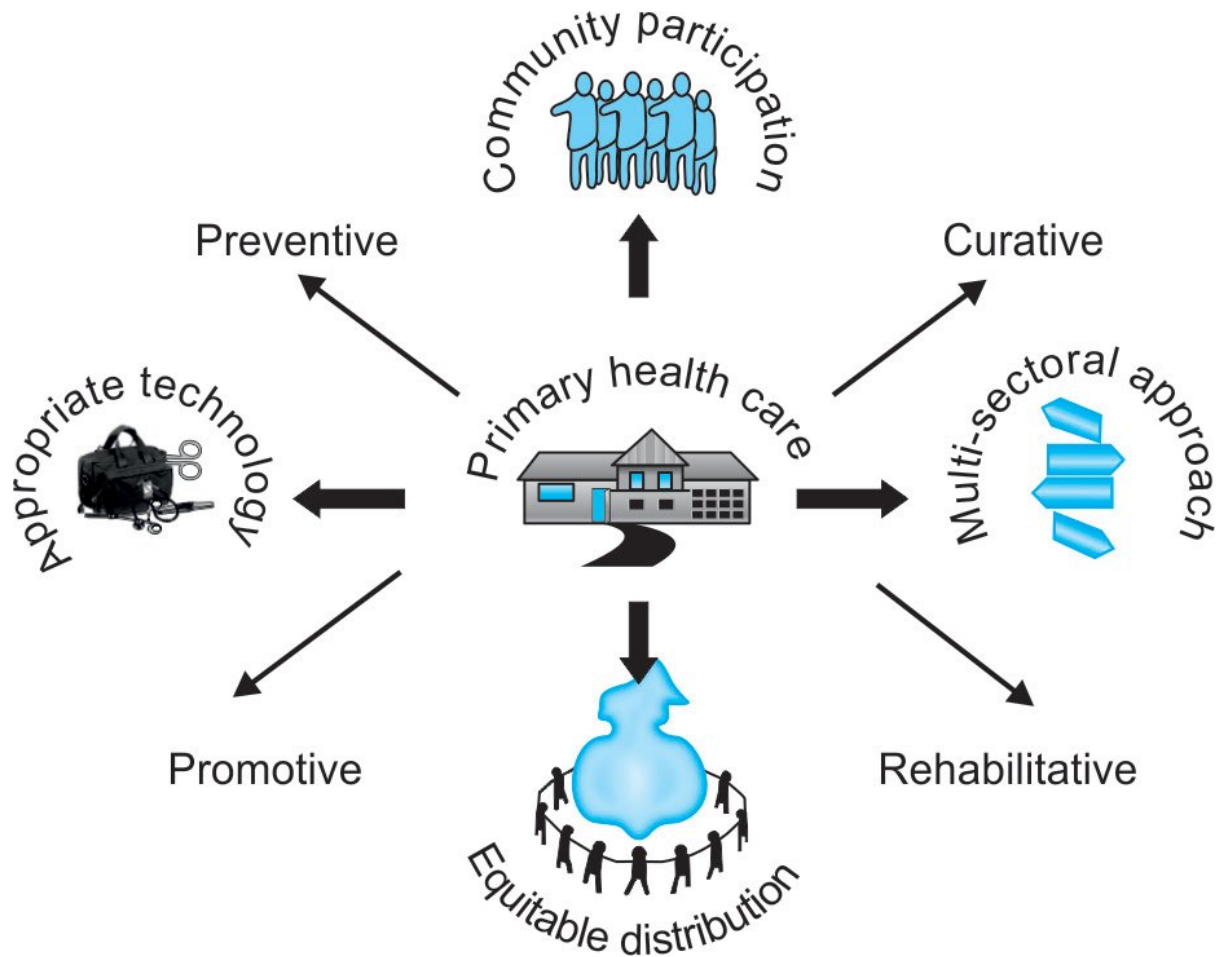
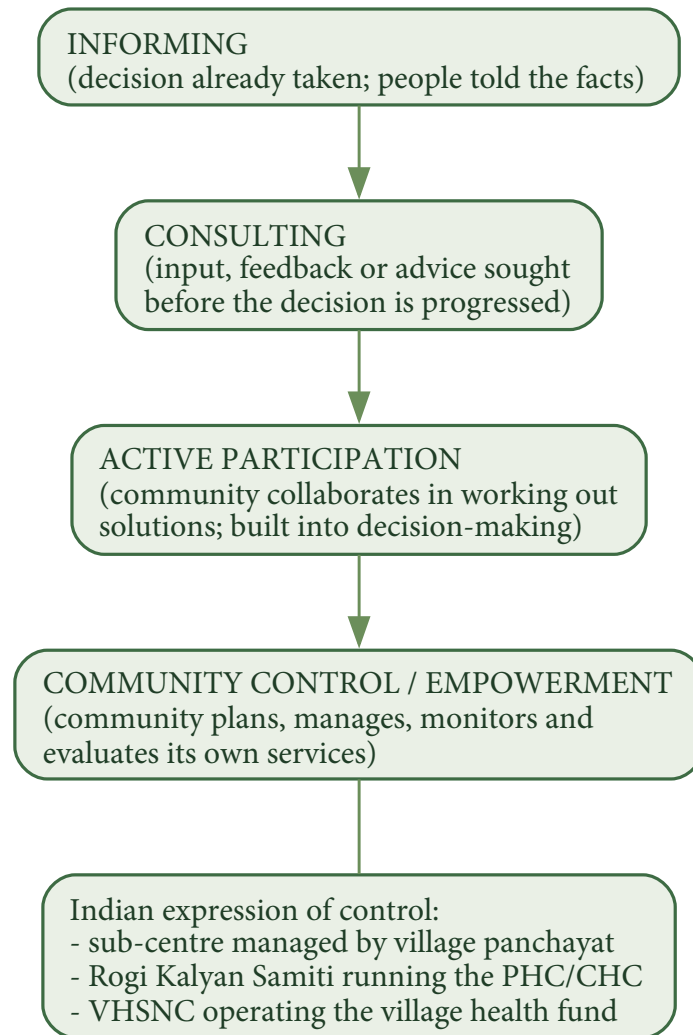
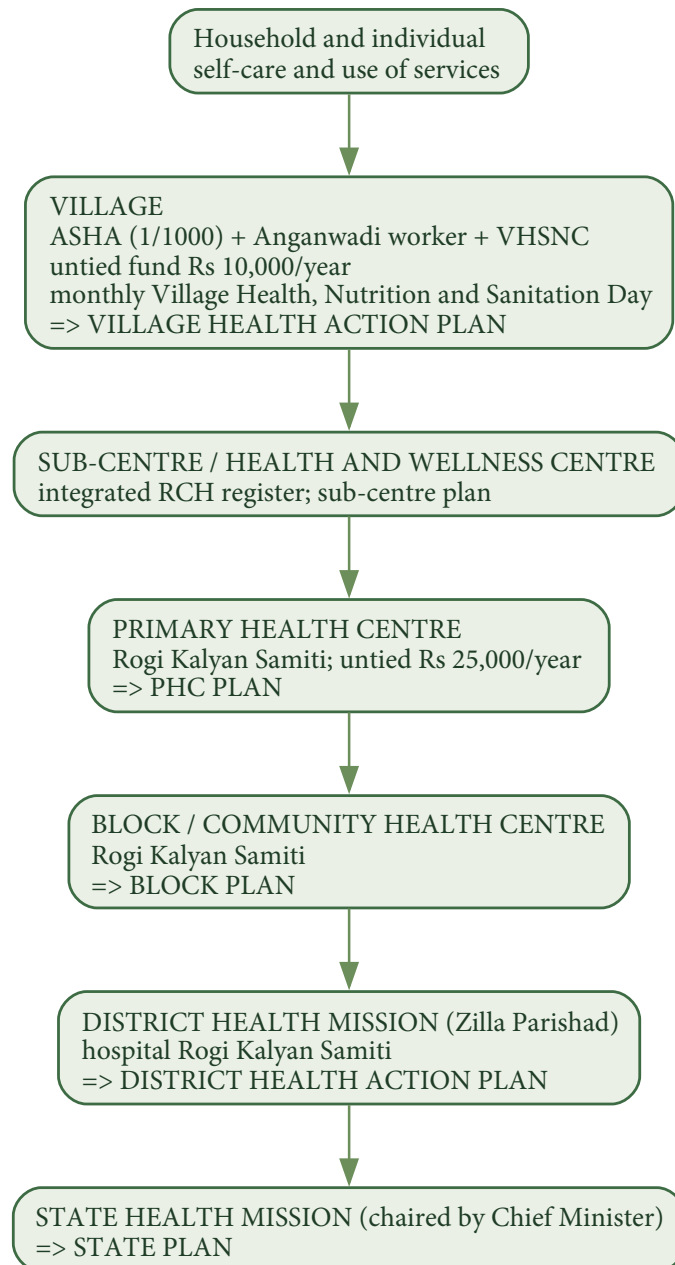


Fig 34.2 — Principles and components of primary health care (Suryakantha Environment Best, p.780)



Ladder of community involvement in primary health care — from being informed to community control



Community platforms and bottom-up health planning under the National Health Mission

High-yield exam pointers

- ▶ **Definition to quote verbatim** — WHO/UNICEF: "the process by which individuals and families assume responsibility for their own health and welfare and for those of the community, and develop the capacity to contribute to their and the community's development."
- ▶ **Four principles of primary health care (Alma-Ata 1978)** — equitable distribution, community participation, intersectoral coordination, appropriate technology. Slogan: *health care by the people*, "placing people's health in people's hands".
- ▶ **Park's three ways a community can participate** — provide resources; take part in planning, management and evaluation; join in and use the services.
- ▶ **Numbers** — ASHA 1 per 1000 population, woman 25-45 years, 8th class pass; VHSNC untied fund Rs 10,000/year, 5.55 lakh committees (2021); Mahila Arogya Samiti 10-12 women per 50-

100 urban slum households, Rs 5,000/year; Anganwadi worker 1 per 400-800 (plain) and 300-800 (tribal/difficult) population.

- **73rd and 74th Constitutional Amendment Acts, 1992** — health, family welfare and education devolved to *panchayats* and *nagarpalikas*.
- **Historical anchors** — barefoot doctors (China), Village Health Guide scheme (India, 1977), trained *dais*, ASHA under the National Health Mission (2005).
- **Barrier line that scores** — resistance to the village health guide came more from the medical profession than from the lay public.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the untied-fund and grant amounts above are the National Rural Health Mission-era norms as printed in the retrieved textbooks and are revised administratively from time to time; sub-centre and PHC-level Health and Wellness Centres have since been renamed **Ayushman Arogya Mandirs** (Park 28e, Ch 7, p 541), and the Village Health and Sanitation Committee was renamed the Village Health, **Sanitation and Nutrition** Committee when nutrition was added to its mandate (Park 28e, Ch 7, p 540).

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoHFW Ayushman Arogya Mandir portal data, National TB Elimination Programme (Ni-kshay), Central TB Division TB Mukh Panchayat indicators, WHO recommendation on participatory women's groups

Programme and policy updates (India)

- **The Health and Wellness Centre renaming, dated and costed** — the Union Health Ministry renamed every Ayushman Bharat Health and Wellness Centre an **Ayushman Arogya Mandir** in **November 2023** and gave it the tagline **Arogyam Paramam Dhanam** ("health is the supreme wealth"). The remit was widened from illness care to AYUSH-informed holistic wellness — lifestyle counselling, yoga, mental health support, nutrition and elderly care. Facilities are branded in the local state script at Rs 3,000 each, a national outlay of about Rs 48 crore, and must upload photographs to the programme portal. Critics note that with staffing and service packages largely unchanged the exercise risks being cosmetic, and that the word *Mandir* drew objections in Ladakh, Nagaland and Meghalaya. [PMID 42257165]
- **Jan Arogya Samiti (JAS) is the new facility-level community platform** created at each Ayushman Arogya Mandir to give people a formal role in planning and monitoring comprehensive primary health care — **1,35,343 constituted by 30 November 2024**. It sits alongside, not in place of, the VHSNC in rural areas and the Mahila Arogya Samiti in urban areas, and holds an **Arogya Sabha** twice a year, preceded by a **monthly village-level Samvad** (consultation). Supporting scale: over **1.75 lakh** Ayushman Arogya Mandirs operational, **1,16,978 Community Health Officers** in position, 82% of centres delivering all 12 expanded service packages by November 2024, and XV Finance Commission health grants now routed through Panchayati Raj Institutions and urban local bodies. [PMID 41816146]
- **TB Mukh Panchayat Initiative, launched 24 March 2023** by the Prime Minister at the One World TB Summit, Varanasi — a nationwide disease campaign built on the panchayat system of

local self-government, in contrast to earlier Indian community-engagement strategies in reproductive and child health and in HIV/AIDS, which were vertical. It leverages the country's **2.5 lakh (255,000) gram panchayats** on the principle of "healthy villages for a healthy nation". Panchayats write TB activities into the Panchayat Development Plan, review quarterly progress in *gram sabha* and Village Health and Sanitation Committee meetings, and submit an annual TB-free claim verified by a district team of health-department officials, Panchayati Raj Institution officers, representatives of doctors' organisations and public health professionals. [PMID 38756161; PMID 42558677]

Current national data

- ▶ **First evaluation of the Mahila Arogya Samiti** (realist evaluation, Chhattisgarh, 401 of the state's 3,700 MAS plus 20 focus groups with 159 members): **95.3% had acted on at least one of six domains and 79.8% on three or more** in the preceding three years — food security and nutrition 74.1%, sanitation 70.8%, social environment 64.1%, gender-based violence 60.8%, healthcare 59.1%, drinking water 56.4%; **97.3% had held a meeting in the previous month**. The examinable contrast: MAS women hold **complete autonomy over their agenda and their untied fund**, whereas in the VHSNC decision-making is shared with male panchayat representatives. Limits found — the Rs 5,000 annual fund is inadequate and arrives late, ward councillors are unresponsive, and because MAS has no formal statutory position authorities are not obliged to act on its feedback. [PMID 40349012]
- ▶ **Ni-kshay Mitra shows what voluntary community adoption actually delivers** — under Pradhan Mantri TB Mukh Bharat Abhiyaan (launched 2022, donors adopt patients for 6 months to 3 years), national Ni-kshay portal data for September 2022 to May 2023 recorded **82,205 registered donors but only 51% who actually delivered support**, and just **9.6% of 26.4 lakh notified patients documented as benefiting**. Where support did reach the patient, favourable treatment outcome was **95% versus 90%** (OR 2.2, 95% CI 2.10–2.31), and lack of support independently predicted an unfavourable outcome (Exp(B) 1.671). Individuals formed 66% of donors; corporates, institutions and NGOs converted registration into actual support most reliably. [PMID 41451044]
- ▶ **Meta-analysis of community health worker-led care in India** (13 studies, 9 pooled for blood pressure, $n = 7,114$) found ASHA-, peer-leader-, self-help-group- and medical-social-worker-delivered packages reduced **systolic blood pressure by 5.5 mm Hg (95% CI –9.0 to –1.9)**, rising to **7.5 mm Hg in those already hypertensive**, and BMI by 0.63 kg/m^2 ; there was no significant effect on diastolic pressure, fasting glucose, HbA1c or lipids, and GRADE certainty was low to very low. This is the quantitative case for task-shifting to the community cadre — real but modest. [PMID 42555695]

Elimination, surveillance and disease status

- ▶ **"TB-free village" is now an auditable status with six numeric indicators** set by the Central TB Division: ≥ 30 presumptive TB examinations per 1,000 population per year; **TB notification rate below 1 per 1,000** — the one ceiling among the six, so a panchayat fails by notifying *more* than this, not less; treatment success $\geq 85\%$; drug-susceptibility testing $\geq 60\%$; Ni-kshay Poshan Yojana instalment received by $\geq 50\%$; and nutritional support under the Abhiyaan to 100% of consenting patients. Certification is valid one year and is conferred on **World TB Day, 24 March**, with a

Mahatma Gandhi statuette. Field experience from Satna and Maihar (Madhya Pradesh) shows participation climbing from **8 panchayats in 2024 to 279 in 2026**, with claim success in Satna of 83.3%, 85.7% and 77.8% across the three rounds; the **TB notification rate was the commonest indicator to fail verification** (86% of disqualified panchayats in 2026; among those that failed, every block in both districts averaged above the ceiling, highest 1.87 ± 0.5 per 1,000 in Rampur Baghelan). [PMID 42558677]

- **The TB elimination deadline is the detail textbooks most often carry superseded** — India committed to ending TB by **2025, five years ahead of the global SDG/End TB target of 2030**. With incidence at 199 and mortality 23 per lakh population (2022) and persisting gaps in drug-resistant TB, biosafety, post-treatment follow-up and tribal coverage, commentators now argue the national target should be realigned to the global 2030 goal. Quote the 2025 commitment, but say that it is contested and that 2030 is the global benchmark. [PMID 40781735; PMID 42558677]

Guideline and legislative updates

- **Operational guidance for facility committees is outrunning their real authority** — an appraisal of Jan Arogya Samitis in a Punjab block found they did lead health-promotion activity and support service quality, but were **passive on monitoring services, supporting village-level committees and enabling social accountability**, largely from awareness gaps. Perceived **decision space, capacity and accountability were low to moderate**, and elected representatives reported significantly greater decision space and accountability than the service recipients they represent — the accountability loop the textbook identifies as missing, measured. The authors call for structured capacity-building for all committee members plus regular supportive supervision. [PMID 41074751]

Global and WHO developments

- **WHO recommends participatory learning and action cycles with women's groups to cut neonatal deaths**, on meta-analysis of seven trials showing a **33% fall in neonatal mortality and 49% fall in maternal mortality where over 30% of pregnant women had ever attended**. A six-trial cost reanalysis across India, Nepal, Bangladesh and Malawi put the average cost at **US\$203 per live birth** and **\$135–\$1,627 per neonatal life-year saved** — cost-effective on WHO income-based thresholds in every setting, though not in Nepal or one Malawi trial under the stricter supply-side threshold; staff costs were 65% of spend. National rural scale-up would take **1.2% of government health expenditure in India** (3.2–6.3% in Nepal, Bangladesh and Malawi) and could save about **42,800 neonatal lives a year in India**, 61,000 across the four countries. ASHA-facilitated groups were as effective as separately recruited facilitators, somewhat costlier. [PMID 33085753]

SOURCE & COVERAGE

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Health and wellness centre

Asked in: Paper IV 14-Jul-2023 — RGUHS MD Community Medicine

Definition — A Health and Wellness Centre (HWC), renamed **Ayushman Arogya Mandir (AAM)**, is an upgraded sub health centre or primary health centre that delivers an expanded package of **twelve comprehensive primary health care (CPHC) services** — preventive, promotive, curative, palliative and rehabilitative — free of cost, universally, to the whole population of a defined catchment area (Sunder Lal, Ch 17, p 741). It is the **first of the two pillars of Ayushman Bharat**, launched in 2018 on the National Health Policy 2017 vision of Health and Wellness Centres as the foundation of India's health system; the second pillar is Pradhan Mantri Jan Arogya Yojana (PM-JAY) for secondary and tertiary hospitalisation (Sunder Lal, Ch 17, p 763; Oxford Textbook of Public Health 7e, Ch 21, pp 841-842).

Concept: the paradigm shift it embodies

A. The key policy shifts of National Health Policy 2017 that the HWC operationalises

Domain	From	To
Clinical care	Stand-alone curative care	Preventive, promotive and rehabilitative approach for comprehensive wellness
Primary care	Selective care	Assured comprehensive primary health care, with linkage to referral hospitals for continuity
Drugs, diagnostics and emergency services in public facilities	User fees and cost recovery	Assured free drugs, diagnostics and emergency services to all
Secondary and tertiary care	Input-oriented, line-item budgeting	Output-based strategic purchasing from public and private hospitals (delivered as PM-JAY)
Infrastructure and human resources	Normative approach	Targeted approach to reach under-served areas
Urban health	Token interventions	On-scale assured interventions, with intersectoral action on urban determinants
National health programmes	Vertical, stand-alone	Integration with the health system for programme effectiveness
AYUSH	Stand-alone	Three-dimensional mainstreaming — co-location, standard medicines, yoga

(Seven key policy shifts — Sunder Lal, Ch 16, p 698; the clinical-care shift and the same list — Park 28e, Ch 10, p 728.)

B. What the concept adds on the ground

Element	What it means in practice
The gap being closed	A well-functioning primary health centre currently covers reproductive and child health plus a few disease-control programmes — together under 15% of the reasons people seek care; for the rest people use private providers or crowded district hospitals, and catastrophic expenditure pushes 63 million people below the poverty line each year (National Sample Survey 75th round) (Sunder Lal, Ch 17, pp 742, 749)
Life-cycle / continuum of care	Care organised across adolescence, pregnancy, birth, infancy, preschool, school, reproductive, adult and ageing phases, with two-way linkage from primary to secondary and tertiary levels (Sunder Lal, Ch 17, p 746)
Universality and equity	Universal, free and closer to the community — the entire catchment population , not a target group (Sunder Lal, Ch 17, p 741)
"Time to care"	Sited so that travel is no more than 30 minutes for primary care; centres established on geographical norms as well as population norms (Sunder Lal, Ch 17, pp 741-742)
Wellness, not only illness	"Eat RIGHT and SAFE ", physical activity, yoga and meditation sessions, tobacco cessation and no alcohol use are regular calendar activities at the centre (Sunder Lal, Ch 17, p 748)
Health-system strengthening	Standards set by the revised Indian Public Health Standards (IPHS) 2022 , aimed at universal health coverage and Sustainable Development Goal 3 (Park 28e, Ch 10, p 728)

Types, population norms, team and free entitlements (IPHS 2022)

Type of facility	Setting	Population norm	Team head and staffing	Assured free entitlements
HWC – Sub Health Centre (HWC-SHC)	Rural	1 per 5,000 plain; 3,000 hilly/tribal/desert	Community health officer (mid-level health provider) leading 1 male + 1 female multipurpose health worker (or two auxiliary nurse midwives), 5 ASHAs and support staff; 2 day-care beds with oxygen; outpatient standard 20 patients/day	105 essential medicines + about 14 point-of-care tests
Urban HWC (UHC)	Urban slum and vulnerable pockets	1 per 15,000–20,000	Medical officer, with a staff nurse/pharmacist, male multipurpose worker and one support staff; the auxiliary nurse midwife and ASHA are drawn from the linked HWC-UPHC / urban community health centre	105 essential medicines + about 14 point-of-care tests
HWC – Primary Health Centre (HWC-PHC)	Rural	1 per 30,000 plain; 20,000 tribal/hilly	Medical officer in charge, team as per IPHS norms; outpatient standard 75 patients/day	172 essential medicines + over 50 diagnostic services

Type of facility	Setting	Population norm	Team head and staffing	Assured free entitlements
HWC – Urban PHC (HWC-UPHC)	Urban, close to slums	1 per 50,000	Medical officer in charge; routine outpatient plus preventive and promotive care	172 essential medicines + over 50 diagnostic services
Multispecialty UPHC / Polyclinic	Urban	2.5–3 lakh, covering the catchment of 5–6 urban PHCs	6 specialists — medicine, obstetrics and gynaecology, paediatrics, ophthalmology, dermatology, psychiatry — with 2 medical officers, dentist and staff nurses; ambulatory/day-care outpatient only	Specialist outpatient care with point-of-care diagnostics for the six specialties

(Park 28e, Ch 10, pp 728, 731; Sunder Lal, Ch 17, Tables 17.2, 17.3, 17.5, pp 746, 748-750, 759; Ch 19, Table 19.1, p 779.) Note the contrast the examiner looks for: **the rural HWC-SHC health team is led by the community health officer, the urban HWC team by a medical officer** (Sunder Lal, Ch 17, p 746; Park 28e, Ch 10, p 731).

The twelve-service comprehensive primary health care package

#	Service package	Illustrative content at the centre
1	Care in pregnancy and childbirth	Early registration with identification number and Mother and Child Protection card, four antenatal checks, counselling on care and nutritional requirements
2	Neonatal and infant health care	Home-based newborn care (7 visits after home delivery, 6 after institutional delivery), identification of low-birth-weight/preterm/sick newborn with referral, breastfeeding and weaning support, growth monitoring
3	Childhood and adolescent health care including immunisation	Growth monitoring and infant and young child feeding, detection and follow-up of severe acute malnutrition, anaemia prevention and deworming, biannual school screening for the 4 Ds (defects at birth, deficiencies, diseases, developmental delay including disability), adolescent counselling
4	Family planning, contraceptive and other reproductive health services	Spacing and limiting methods, intrauterine contraceptive device insertion, counselling
5	Management of communicable diseases under national health programmes	Tuberculosis, leprosy, HIV and vector-borne disease services integrated at the centre
6	General out-patient care for acute simple illness and minor ailments	First-contact curative care with referral of complications
7	Screening, prevention, control and management of non-communicable diseases	Population-based screening for hypertension, diabetes and oral, breast and cervical cancer using the community-based assessment checklist; monthly follow-up for treatment adherence and lifestyle management

#	Service package	Illustrative content at the centre
8	Care for common ophthalmic and ENT problems	Screening for visual acuity, blindness, cataract and refractive error; awareness and referral
9	Basic oral health care	Mobilisation for oral cancer screening, topical fluoride application, tobacco-cessation counselling and referral
10	Elderly and palliative health care	Identification of high-risk elderly, home visits to home-bound and bedridden elderly, family support in palliative care, linkage to day-care centres
11	Emergency medical services including burns and trauma	First aid and stabilisation with referral transport
12	Screening and basic management of mental health ailments	Screening, suicide-risk assessment, counselling of patient and family, referral and follow-up

(Park 28e, Ch 10, pp 730-736; Sunder Lal, Ch 17, pp 749-750.) Every service is further graded as "essential" — the minimum assured service each facility must provide — or "desirable", to be achieved progressively (Sunder Lal, Ch 17, p 748).

Inputs to the centre

- **The mid-level health provider** — the key addition at the sub health centre is a **mid-level health provider (MLHP)** designated **community health officer (CHO)**: a BSc in community health, a GNM or BSc nurse, or an Ayurveda practitioner, trained and certified through a **six-month certificate programme in community health** run by IGNOU or state public health universities (Sunder Lal, Ch 17, p 742; Park 28e, Ch 7, p 570).
- **Community platform** — one ASHA per **1,000** population with over **10.72 lakh** ASHAs in service, and a village health, sanitation and nutrition committee holding untied money of **Rs 10,000 per annum** for the village health action plan; in urban slums the *Mahila Arogya Samiti* of **10–12 women covering 50–100 households of 250–500 population**, with an annual grant of **Rs 5,000** (Sunder Lal, Ch 17, pp 744, 758).
- **Flexible local money** — an untied grant of **Rs 10,000 per year at sub-centre level** for local health action, purchase of emergency medicines, repair of equipment and cleanliness, alongside performance-based ASHA incentives (Sunder Lal, Ch 16, p 704).
- **Infrastructure and identity** — about **Rs 7 lakh per centre** for repair and renovation of the existing sub-centre; every centre carries the prescribed Government of India branding and a **national identification number** (Sunder Lal, Ch 17, p 742).
- **Teleconsultation and diagnostics** — **eSanjeevani AB-HWC** gives doctor-to-doctor teleconsultation with specialists and super-specialists, all **1,50,000** centres being connected; the centre also serves as a hub-and-spoke platform for expanding the range of diagnostics (Sunder Lal, Ch 17, pp 748, 751).

Outputs, outcomes and impact

Level of the results chain	What is measured
Outputs (service utilisation)	Population enumeration and empanelment with a database of all families and individuals; individual health records and family health folders ; access to the expanded range of services. Utilisation — antenatals registered, antenatals given 4 antenatal checks and 180 iron-folic acid tablets, institutional deliveries, infants immunised, couples protected with contraceptives, persons screened for hypertension and diabetes, outpatient attendance and indoor admissions. Quality — National Quality Assurance Standards for HWC-SHC
Outcomes (coverage indicators)	Proportion of eligible couples using contraceptives; proportion of fully immunised infants and of completed immunisation; breastfeeding initiated within one hour; households using sanitary latrines and an improved drinking-water source. Also reduced out-of-pocket and catastrophic health expenditure, decongestion of secondary and tertiary facilities , and risk-factor mitigation
Impact	Reduction in disease incidence and prevalence , reduction in mortality , improved contraceptive rate, nutrition and reproductive and child health — gathered from routine monthly monitoring reports and periodic surveys (coverage evaluation surveys, National Family Health Survey)

(Sunder Lal, Ch 17, pp 742-743.)

Roll-out status

- The first Health and Wellness Centre was inaugurated on 14 April 2018 in Bijapur district, Chhattisgarh (Park 28e, Ch 7, p 570).
- All 1,50,000 sub health centres and all primary health centres were to be transformed; nearly 1.1 lakh sub health centres had been converted by **March 2022** (Sunder Lal, Ch 17, p 746). **1,72,148 centres were operational as on 31 March 2024** (Park 28e, Ch 7, p 570).
- The complementary pillar, AB-PM-JAY, launched **23 September 2018** subsuming the Rashtriya Swasthya Bima Yojana, gives cashless, paperless cover of **Rs 5 lakh per family per year** for secondary and tertiary hospitalisation to about **10.74 crore families (~50 crore people)** (Park 28e, Ch 7, p 570; Sunder Lal, Ch 17, p 763).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram — health system pyramid showing where the Health and Wellness Centre sits.** Labels, base upward: *Villages and urban wards; health volunteers (ASHA) 1:1000 → Sub-centre/HWC, health workers male and female and mid-level health provider (MLHP), 1:5,000 plains and 1:3,000 tribal-hilly → Primary health centres 1:30,000 plain and 1:20,000 tribal-hilly → Community health centres 1:120,000 plains and 1:80,000 tribal-hilly → Sub-district/taluka hospitals → District health organisations → State Ministry of Health and Family Welfare → National level, Ministry of Health and Family Welfare.* Mark the three tiers on the right as *Base, Middle level, Top level*.

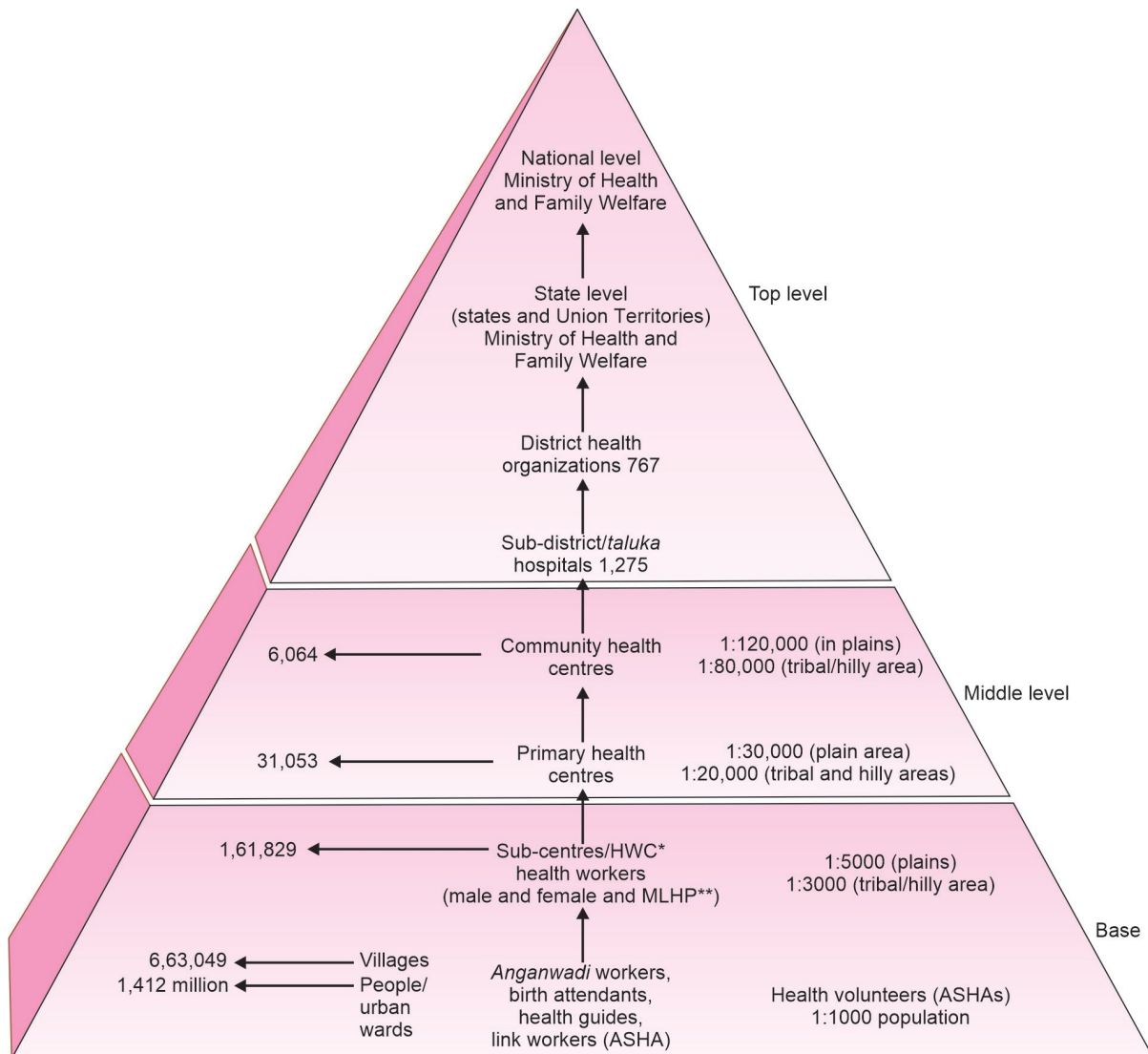
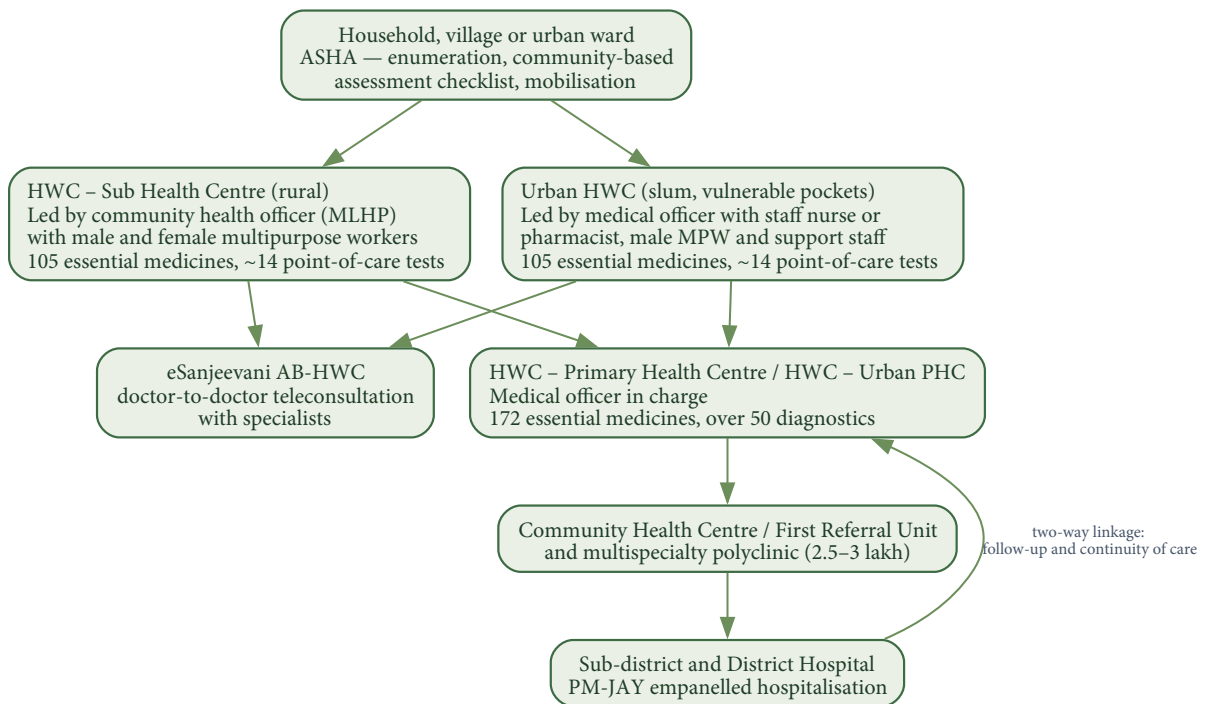
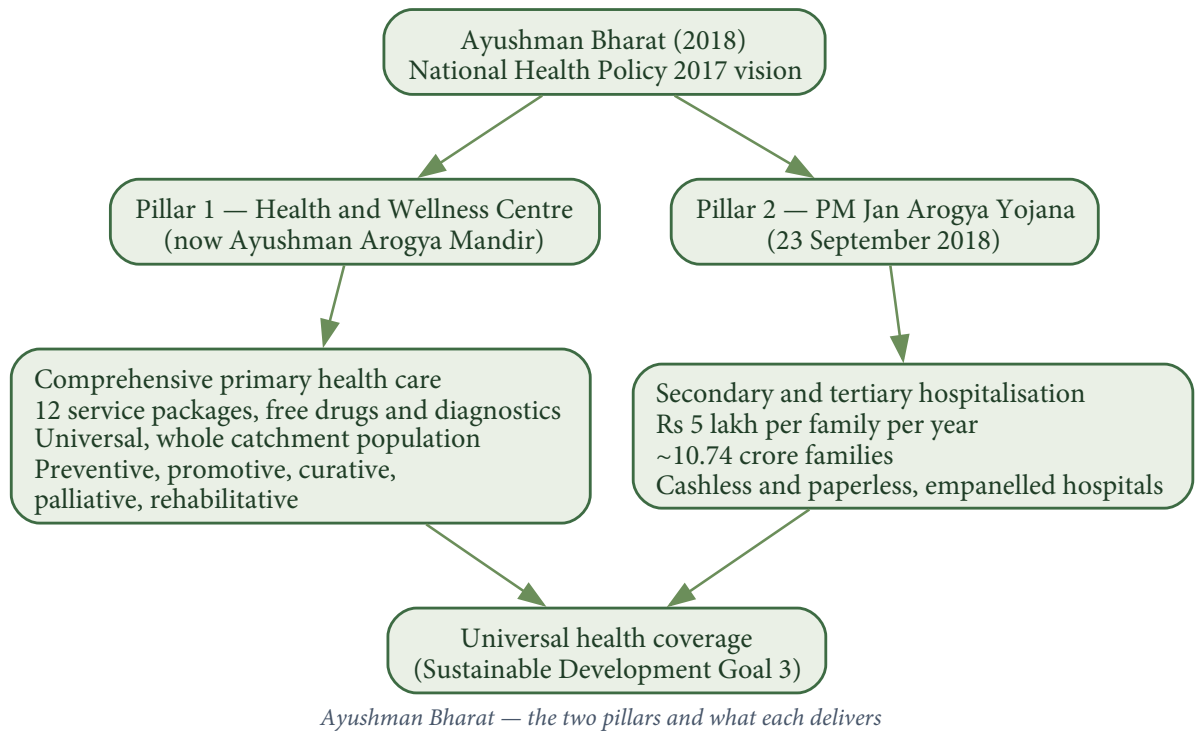


Fig 16.2 — Health system organization *HWC—Health and wellness centre, **MLHP—Mid (Sunderlal, p.709)



Service delivery and referral pathway through the Health and Wellness Centre

High-yield exam pointers

- ▶ HWC = Ayushman Arogya Mandir, launched 2018, Pillar 1 of Ayushman Bharat; policy parent National Health Policy 2017; standards IPHS 2022; goal universal health coverage.
- ▶ Seven key policy shifts of National Health Policy 2017 — primary care selective → comprehensive; secondary and tertiary input-budget → strategic purchasing; user fees → free drugs, diagnostics and emergency care; human resources normative → targeted; urban health

token → on-scale; national programmes vertical → integrated; AYUSH stand-alone → mainstreamed.

- ▶ **Twelve CPHC packages** in order — *pregnancy and childbirth; neonatal-infant; child-adolescent with immunisation; family planning and reproductive health; communicable disease programmes; out-patient minor ailments; non-communicable disease screening; eye and ENT; oral health; elderly and palliative; emergency and trauma; mental health.*
- ▶ **Norms and team head:** HWC-SHC 5,000/3,000 — **community health officer**; urban HWC 15,000–20,000 — **medical officer**; rural PHC 30,000/20,000 and urban PHC 50,000 — medical officer in charge; polyclinic 2.5–3 lakh (catchment of 5–6 urban PHCs); "time to care" 30 minutes.
- ▶ **Free entitlements by level:** HWC-SHC and urban HWC = 105 essential medicines + about 14 point-of-care tests; HWC-PHC and HWC-UPHC = 172 essential medicines + over 50 diagnostic services; eSanjeevani teleconsultation at both.
- ▶ **CHO (MLHP)** = BSc community health, GNM/BSc nurse or Ayurveda practitioner + **six-month certificate programme in community health** (IGNOU or state public health university); Rs 7 lakh per centre for renovation; sub-centre untied grant Rs 10,000 per year.
- ▶ **Dated numbers:** first centre 14 April 2018, Bijapur, Chhattisgarh; 1.1 lakh sub-centres converted by March 2022; 1,72,148 centres operational on 31 March 2024; PM-JAY Rs 5 lakh per family per year for 10.74 crore families.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Health and Wellness Centres have been officially renamed **Ayushman Arogya Mandir**, so give the current name first and add "erstwhile Health and Wellness Centre" once in parentheses; centre counts and PM-JAY beneficiary numbers are revised annually, so every figure must carry the date attached to it as above.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, MoHFW, PIB India, NFHS-6, Sample Registration System, NCVBDC, WHO

Programme and policy updates (India)

- ▶ **Scale, quality certification and a revised free-entitlement ladder.** More than 1,87,000 Ayushman Arogya Mandirs are now operational, with a cumulative footfall above 500 crore and over 46.7 crore eSanjeevani teleconsultations recorded since launch; public facilities certified under the National Quality Assurance Standards rose from 10 in 2016 to 64,671 as of May 2026. The entitlement ladder is now stated as 106 drugs and 14 diagnostics at sub health centre level, rising to 381 drugs and 134 diagnostics at district hospital — quote these with their date rather than the older 105-drug figure. [MoHFW 2026-Jun]
- ▶ **PM-JAY universalised for the elderly** — the second pillar's biggest eligibility change since 2018. The Ayushman Vay Vandana Card, launched 29 October 2024, extends the Rs 5 lakh cover to every citizen aged 70 years and above irrespective of income or socio-economic status;

seniors in families already covered under PM-JAY receive a separate annual top-up of up to Rs 5 lakh for themselves. [PIB 2024-Oct]

- **Implementation quality, not coverage, is now the live question.** A Plan-Do-Check-Act evaluation of 16 rural Lucknow centres found dedicated diagnostic space in only 6.25% of them and biomedical-waste facilities in 18.75%, yet households served by a centre reported higher institutional-delivery awareness (97.8% vs 87.0%, $p < 0.001$) and antenatal care (80.3% vs 65.9%, $p < 0.001$) than non-served households [PMID 42257150]; a six-state multicentric assessment found functioning NCD clinics in 66.7% of rural but only 33.3% of urban centres [PMID 42344085].

Current national data

- **NFHS-6 (2023-24) fact sheets were released on 29 May 2026** — 6.79 lakh households across 715 districts, with IIPS Mumbai as nodal agency. India figures against NFHS-5: health insurance coverage 41.0% → 60.2%, institutional births 88.6% → 90.6%, skilled birth attendance 89.4% → 91.3%, any antenatal care 92.6% → 95.9% and four or more visits 58.5% → 65.2%, full immunisation 83.8% → 87.1%, contraceptive prevalence 66.7% → 69.1% with unmet need down to 8.5%, stunting 35.5% → 29.3% and severe wasting 7.7% → 5.2%; total fertility rate stable at 2.0. [MoHFW 2026-May]
- **Mortality gains alongside an NCD counter-trend.** SRS 2022-24 puts the **maternal mortality ratio at 87 per 100,000 live births** (254 in 2004-06, a 65% fall) and SRS 2024 gives **IMR 24, neonatal mortality 18 and under-five mortality 28 per 1,000 live births**; U-WIN, the digital immunisation tracker, had **17.39 crore registered beneficiaries and 75.36 crore doses administered as on 2 June 2026**. Against this, NFHS-6 shows high blood sugar rising (**women 13.5 → 17.8%, men 15.6 → 20.9%**) and overweight or obesity rising (**women 24.0 → 30.7%, men 22.9 → 27.3%**), while cumulative population-based screening through the centres stands at about **42 crore for hypertension, 41.8 crore for diabetes, 32 crore oral, 33 crore breast and 7.4 crore cervical cancer**. [MoHFW 2026-Jun]

Elimination, surveillance and disease status

- **The centre has become the delivery platform for TB preventive treatment.** Across 208 districts in 23 states, **30,97,831 household contacts** of 8,51,974 people with pulmonary tuberculosis were identified; **91% were symptom-screened, 54% of those screened started preventive treatment and 84% completed it**, rising to **92% completion with the 3-month weekly rifapentine-isoniazid (3HP) regimen**. Decentralisation of services to Health and Wellness Centres with ASHA-supported adherence is credited for the coverage achieved. [PMID 42583442]
- **The elimination calendar every answer must date.** Lymphatic filariasis — elimination as a public health problem targeted for **2027**, with the twice-yearly rounds replaced from **February 2026** by a single annual mass drug administration across 124 districts in 12 endemic states, and Bihar clearing its first transmission assessment survey since MDA began in 2004. Measles and rubella — the WHO South-East Asia Region **reset the regional elimination target to 2026** in October 2024 after the 2023 target lapsed, and India is still classified endemic. Kala-azar is being held below one case per 10,000 at block level pending validation, and malaria elimination remains a **2030 goal**. [WHO 2024-Oct] [NCVBDC 2026-Feb]

Guideline and legislative updates

- ▶ **AYUSH mainstreaming has moved from policy statement into the standards themselves.** Over **12,500 AYUSH Health and Wellness Centres** established under the National AYUSH Mission now carry Ayushman Arogya Mandir branding; the Indian Public Health Standards have been revised to include AYUSH, an AYUSH essential-medicines list is stockable at Jan Aushadhi Kendras, and the Directorate General of Health Services hosts a dedicated AYUSH vertical — the three-dimensional mainstreaming promised in National Health Policy 2017 given administrative form. [PMID 41229518]
- ▶ **Task-shifting within the twelve packages now has accuracy and training evidence behind it.** Non-ophthalmologist-led diabetic retinopathy screening at a Punjab centre achieved **sensitivity 86.4% (95% CI 65.1-97.1) and specificity 94.3% (95% CI 88.5-97.7)** against specialist grading, with far fewer ungradable images than community artificial-intelligence screening (8.3% vs 38%) [PMID 41857844]; in-person mental-health training combined with tele-mentoring significantly raised knowledge-attitude-practice scores of community health officers in both Telangana and Tripura ($p < 0.01$), supporting service package 12 [PMID 41104324].

Global and WHO developments

- ▶ **Two new instruments now govern global health security.** The WHO Pandemic Agreement was adopted on 20 May 2025 at the Seventy-eighth World Health Assembly, and the **2024 amendments to the International Health Regulations (2005) entered into force on 19 September 2025**, introducing a "pandemic emergency" tier above the public health emergency of international concern and giving equity and One Health formal legal definition. [WHO 2025-May]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7, Health Programmes in India; Ch 10, Health Care of the Community); Sunder Lal — Textbook of Community Medicine (Ch 16, Health Planning and Management; Ch 17, Healthcare of the Community; Ch 19, Essential Medicines and Rational Use of Drugs); Oxford Textbook of Public Health 7e (Ch 21, Health policy in developing countries — case study on the Ayushman Bharat programme); National Health Policy 2017, Ministry of Health and Family Welfare; Ayushman Bharat — Comprehensive Primary Healthcare through Health and Wellness Centres, Operational Guidelines (NHSRC/National Health Mission); Revised Indian Public Health Standards (IPHS) 2022, Directorate General of Health Services, MoHFW; National Sample Survey 75th round.

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Consumer Protection Act

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — The Consumer Protection Act (CPA), enacted by Parliament in 1986 and recast as the Consumer Protection Act 2019, is comprehensive legislation for the better protection of the interests of consumers and for the promotion and enforcement of their rights. It provides a simple, speedy and inexpensive redressal of consumer grievances arising from defective goods, deficient services and unfair trade practices, outside the ordinary civil courts (Park 28e, Ch 12, p 837; Red Book — Public Health and Preventive Medicine, p 377).

Why the Act was needed, its objectives and the six consumer rights

Item	Content
Problem it solved	Under general law a professional must show the standard of care expected of that profession, but in India people did not go to civil courts freely — court fee is very heavy and a final verdict may take several years (Park 28e, Ch 12, p 837)
Remedy provided	A quick, efficacious and economic alternative: no court fee , the person can plead his own case without a lawyer, and the decision should be taken within 3 to 6 months (Park 28e, Ch 12, p 837; Suryakantha, Ch 52)
Objectives	Protect the consumer against unfair trade practices, sale of defective goods and deficient services, unsafe and hazardous goods, and overcharging; and establish redressal fora at different levels for speedy resolution (AFMC Red Book 2024, Ch 39, p 1235)
Right to safety	Protection against marketing of goods and services hazardous to life and property
Right to be informed	About quality, quantity, potency, purity, standard and price, so as to be protected against unfair trade practices (Jugal Kishore, Ch 2, pp 857–858)
Right to choose	Assured access, wherever possible, to a variety of goods and services at competitive prices
Right to be heard	Consumer interests to receive due consideration at appropriate fora
Right to seek redressal	Against unfair trade practice and unscrupulous exploitation
Right to consumer education	To acquire the knowledge and skill to be an informed consumer

Central and State Consumer Protection Councils exist to promote and protect these rights (Jugal Kishore, Ch 2, p 858).

Who is a "consumer" of medical care — Indian Medical Association vs V.P. Shantha (1995)

Health care entered the Act's ambit only because the Supreme Court held medical practitioners to be providers of "service", so a doctor is under the same obligation as any other service provider to compensate a patient for deficiency in that service. Applicability was disputed even among legal experts until the landmark three-judge judgment of **13 November 1995**, *Indian Medical Association vs V.P. Shantha and others*, which settled it (Mahajan & Gupta 4e, Ch 13, p 156); the judgment is reported as AIR 1996 SC 550 (Red Book — Public Health and Preventive Medicine, p 377).

Situation	Position under the Act
Doctor in independent private practice; private hospital or clinic charging the patient	Covered — patient is a consumer
Hospital that has both free and paying patients	Covered, including for the patient treated free there
Care paid for by a medical insurance policy, or by the employer as a condition of service	Covered — recipient is a consumer (Mahajan & Gupta 4e, Ch 13, p 156)
ESI hospitals (Employees' State Insurance)	Covered, following a Supreme Court judgment of May 2007
Armed Forces healthcare establishments	Covered — liable to penalty levied by the consumer redressal commission for deficiency in service, ranging from monetary compensation to suspension or removal of the erring medical officer's name from the medical register (AFMC Red Book 2024, Ch 39, pp 1235–1236)
Service rendered free of charge to every patient at that facility — charitable dispensary or hospital	Not "service"; a token amount taken only for registration does not alter this
Government dispensary, hospital or primary health centre where services are free to all	Outside the purview of the Act (Mahajan & Gupta 4e, Ch 13, p 156)
Medical officer employed by an employer to treat the employer — a "contract of personal service"	Excluded — master–servant relationship
Ordinary doctor–patient relationship — a "contract for personal service"	Not excluded; it falls squarely within "service"
Being under the disciplinary control of the Medical Council	Does not exclude the doctor from the Act

Medical negligence and the grounds on which a complaint lies

- **Negligence defined** — a person who holds himself out as ready to give medical advice and treatment impliedly undertakes that he possesses the skill and knowledge for the purpose; he then owes the patient a **duty of care in deciding what treatment to give** and a **duty of care in the administration of that treatment**. Breach of either duty gives the patient a right of action for negligence (Sunder Lal, Ch 1, p 57). Negligence in medical care is an offence under the Act.
- **Grounds of complaint** — an unfair or restrictive trade practice has been adopted; the goods bought suffer from a defect or the services availed suffer from a **deficiency in any respect**; or a price in excess of that fixed by law or displayed has been charged (Jugal Kishore, Ch 2, pp 857–858; Mahajan & Gupta 4e, Ch 13, p 155).
- **What the complaint must contain** — all the details of the case, an **expert certificate or opinion from a doctor of the concerned speciality** stating that the complaint is *prima facie* true and needs further investigation, and the compensation demanded (Park 28e, Ch 12, p 837).

- **Not a consumer** — a person buying goods for resale or for a large-scale profit-making activity, a lottery ticket holder, an insurance company, and a person receiving medical service under a contract of personal service (Jugal Kishore, Ch 2, p 858).
- **Consumer court versus Medical Council** — the two remedies are different and complementary. The National Medical Commission (which replaced the Medical Council of India) and the State Medical Councils have **only ethical jurisdiction**: they can cancel the doctor's registration temporarily or permanently but cannot punish him or award compensation. The consumer court, conversely, can **only award compensation** (Park 28e, Ch 12, p 837). The difference is illustrated by a Karnataka case in which the Medical Council let the doctor off with a warning while the Mangalore District Consumer Forum awarded Rs 3.5 lakh against the anaesthetist and Rs 1.5 lakh against the surgeon (Mahajan & Gupta 4e, Ch 13, p 155).

Three-tier redressal machinery

Level	Monetary limit of the compensation that the court may grant (as printed in Park 28e, Ch 12, pp 837–838)
District Consumer Disputes Redressal Commission (Park prints "District Consumer Court")	Up to Rs 20 lakh
State Consumer Disputes Redressal Commission	Rs 20 lakh to Rs 1 crore
National Consumer Disputes Redressal Commission, New Delhi	Above Rs 1 crore

Appellate route (the statutory hierarchy; not part of the Park passage above) — District Commission → State Commission → National Commission → **Supreme Court of India**.

Enforcement — failure to comply with an order of the District, State or National Commission is punishable with imprisonment of **not less than 1 month, extending to 3 years**, or a fine of **not less than Rs 2,000 extending to Rs 10,000**, or both (Jugal Kishore, Ch 2, p 858).

What the Consumer Protection Act 2019 added

New provision	Substance
Central Consumer Protection Authority (CCPA)	A regulator created to protect, promote and enforce consumer rights, alongside the renamed Consumer Disputes Redressal Commissions (AFMC Red Book 2024, Ch 39, p 1235)
Product liability	Manufacturer, seller and service provider made accountable for a defective product or service causing harm; the consumer may claim compensation for injury or damage
E-commerce and digital transactions	Explicit cover for online transactions, digital fraud and e-commerce grievances — relevant to online pharmacies and teleconsultation
Mediation	Mediation encouraged as an alternative dispute resolution route to expedite settlement

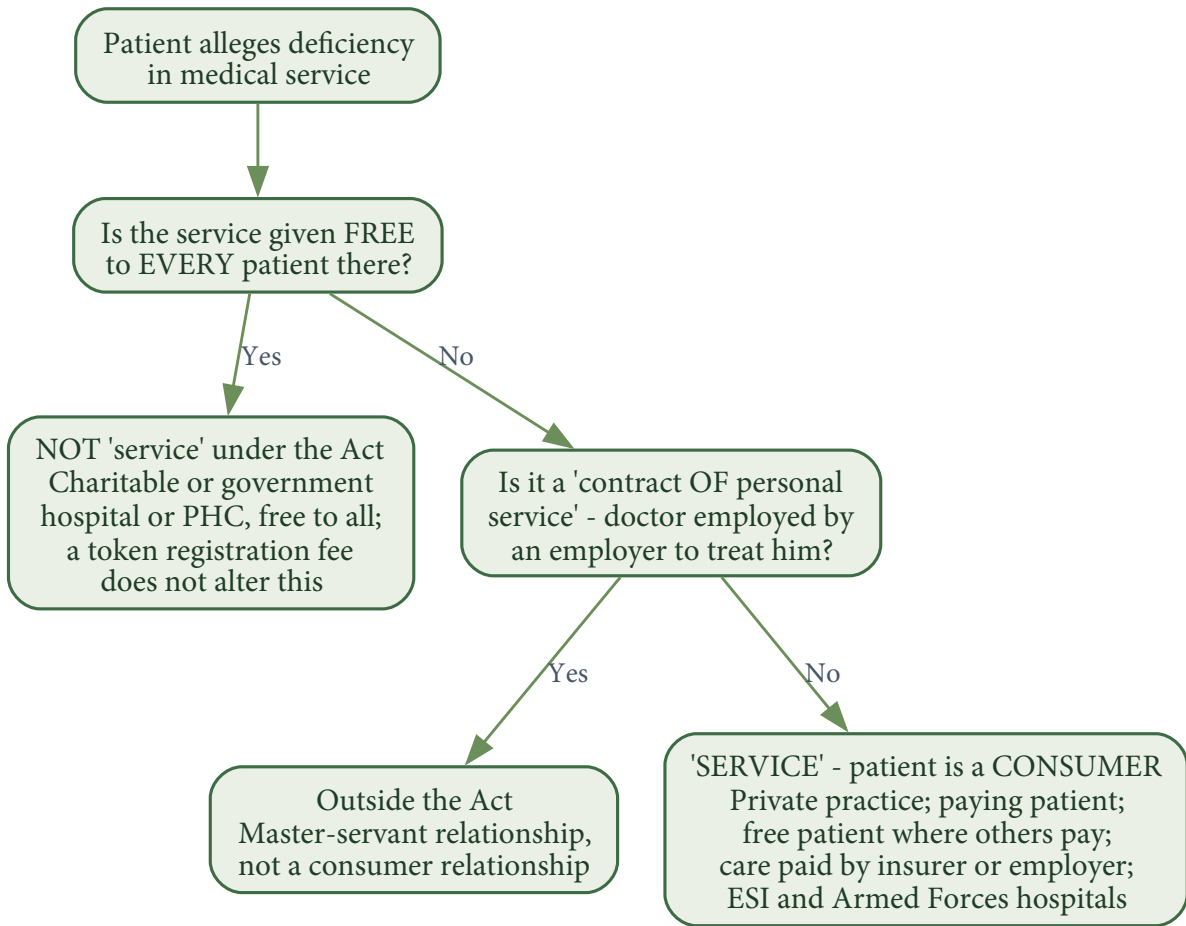
New provision	Substance
Misleading advertisements	Penalty up to Rs 10 lakh or imprisonment up to 2 years for the manufacturer or endorser; a subsequent offence up to Rs 50 lakh or imprisonment up to 5 years, with prohibition of the endorser (AFMC Red Book 2024, Ch 39, p 1235)
Pecuniary jurisdiction basis	Now determined by the value of the goods or services paid for, not by the compensation claimed

Implications for the medical profession and safeguards for the doctor

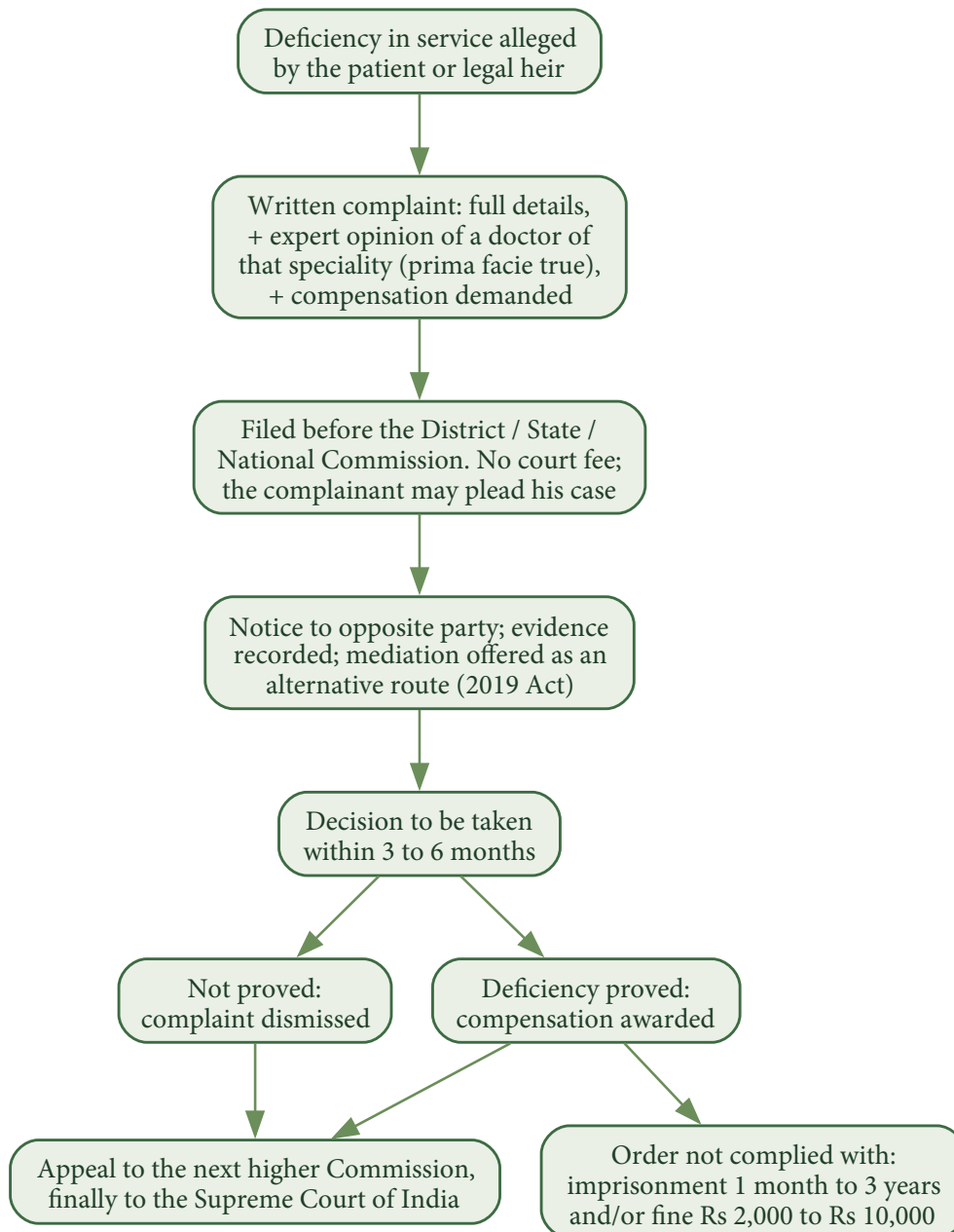
- **The Act raises quality** — legislation is one of the important methods of ensuring quality of health care, and bringing health services under the Act is regarded as a welcome move that safeguards the rights of the client and improves the quality of medical care (Jugal Kishore, Ch 2, p 858; Sunder Lal, Ch 1, p 57).
- **The profession's apprehension** — the Indian Medical Association objected that the judgment would make medical care costlier for the poor, and doctors fear frivolous allegations and loss of reputation. This criticism is held to be unwarranted: it rests merely on apprehension, free charitable and government facilities remain outside the Act, and **the same legal principles applied in civil courts are applied in the consumer fora** when deciding negligence (Mahajan & Gupta 4e, Ch 13, pp 156, 158; Red Book, p 395).
- **Practical safeguards** — be qualified and registered in the modern system of medicine; exercise the duty of care in deciding and in administering treatment; keep knowledge and skills updated; obtain **informed consent**; maintain complete medical records and the medical certificate register; display the required details; avoid unnecessary investigations and consultations; and observe all statutory laws and regulations (Sunder Lal, Ch 1, pp 55, 57).
- **Complementary legislation** — the **Clinical Establishments (Registration and Regulation) Act, 2010** provides for the registration and regulation of clinical establishments so as to prescribe minimum standards of facilities and services (Jugal Kishore, Ch 2, pp 854, 857). Its main features — grading of clinical establishments, standard treatment guidelines, protection of patient rights, standards of care and of prices/fees, negligence and unfair practice, and organ and tissue transplantation — work alongside the Consumer Protection Act in assuring the quality the consumer is entitled to (Sunder Lal, Ch 1, p 57).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **three-tier consumer redressal pyramid**, base to apex, with these exact labels: *District Consumer Disputes Redressal Commission* → *State Consumer Disputes Redressal Commission* → *National Consumer Disputes Redressal Commission, New Delhi* → *Supreme Court of India*; mark the left side "original jurisdiction by value" and the right side "appellate route".



Is a medical service a "service" under the Consumer Protection Act? (Indian Medical Association vs V.P. Shantha, 1995)



Course of a consumer complaint against a doctor or hospital, from filing to enforcement

High-yield exam pointers

- ▶ **Landmark case to name:** *Indian Medical Association vs V.P. Shantha*, Supreme Court, 13 November 1995 (AIR 1996 SC 550) — brought medical practice under "service".
- ▶ **Two exclusions only:** service free of charge to *everybody*, and a "contract of personal service". Everything else — a free patient in a hospital that charges others, ESI hospitals, Armed Forces establishments — is covered.
- ▶ **Six consumer rights** (mnemonic *S-I-C-H-R-E*): Safety, Information, Choice, Hearing, Redressal, Education.
- ▶ **Procedural triad:** no court fee, no lawyer needed, decision within **3 to 6 months**.

- **Compensation slabs to reproduce (Park 28e):** Rs 20 lakh / Rs 20 lakh–1 crore / above Rs 1 crore — then add that under the 2019 Act jurisdiction goes by the **value paid**, not the compensation claimed.
- **Consumer court awards compensation only;** the Medical Council/National Medical Commission can only cancel registration — quote the Karnataka case (Rs 3.5 lakh + Rs 1.5 lakh) to show the difference.
- **CPA 2019 additions:** Central Consumer Protection Authority, product liability, e-commerce cover, mediation, misleading-advertisement penalties (Rs 10 lakh/2 years, repeat Rs 50 lakh/5 years).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still prints the pre-2019 compensation limits and the older names — "District Consumer Court" and "Medical Council of India" — so reproduce those figures as the 1986 Act position while naming the Consumer Disputes Redressal Commissions and the National Medical Commission, which have since replaced them.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NCDRC, National Consumer Helpline, MoHFW/NMC Telemedicine Practice Guidelines 2020

Programme and policy updates (India)

- **The criminal-negligence track was rewritten on 1 July 2024** — the Bharatiya Nyaya Sanhita (BNS) replaced the Indian Penal Code, and **Section 106**, the successor to IPC 304A, raises death caused by a rash or negligent act to imprisonment up to **5 years**, but caps the punishment at **2 years** with fine where the act is done by a doctor performing a medical procedure. BNS Sections 18, 19, 26 and 30 (accident without criminal intent, good faith, act done with consent, act done for a person incapable of consenting) remain the practitioner's statutory defences [PMID 39439653]. The consumer-forum route is untouched: one and the same doctor may be proceeded against at the same time in civil court for damages, before the consumer commissions under the Consumer Protection Act 2019, under the criminal law, and before the professional regulator for misconduct — so the remedies run in parallel rather than one displacing another [PMID 40395703]. The same reform replaced the CrPC and the Indian Evidence Act with the Bharatiya Nagarik Suraksha Sanhita and the Bharatiya Sakshya Adhiniyam on the same date [PMID 42423515].
- **Filing is now digital** — consumer complaints are filed online through the **e-Jagriti** platform of the Ministry of Consumer Affairs, Food and Public Distribution, and the **National Consumer Helpline on toll-free 1915** takes grievances in 17 languages by phone, WhatsApp, SMS and the NCH/UMANG apps. Add this to Park's "no court fee, no lawyer needed" triad [NCDRC 2026] [NCH 2026].
- **Teleconsultation carries the same negligence exposure as a face-to-face consultation** — the practitioner must be registered with the NMC or a State Medical Council to teleconsult and must abide by the Telemedicine Practice Guidelines 2020 (NMC and MoHFW); misdiagnosis arising from poor video quality, an absent physical examination or missing investigations may still be

held negligent. The limitations of the teleconsultation should be documented and a telemedicine log kept, recording the time, the advice given and the documents reviewed [PMID 42088806].

Current national data

- ▶ A review of **32 reported Indian neurosurgical judgments** found negligence established in **40.6%**; **65.6%** of the NCDRC matters were appeals against State Commission orders; median compensation was **Rs 9.6 lakh** (mean Rs 19.7 lakh, range Rs 25,000 to Rs 10 crore); and lapses in the **postoperative** phase were the commonest basis of a finding of negligence (59.3%) [PMID 42088806].
- ▶ Of **26 Indian judgments** in which journal literature was produced as evidence, **62% were consumer-court cases** and the doctor succeeded in only **27%**; median compensation was **Rs 7.5 lakh**; published research was accepted as the *sole* evidence in 34.6% of cases, and the journal's impact factor had no bearing on the verdict (chi-square 0.16, $p = 0.68$). Courts have even preferred journal evidence over a medical board's opinion [PMID 40395703].
- ▶ **Quantum is escalating** — the NCDRC awarded **Rs 1 crore in August 2022** to the parents of a 6-year-old who died during squint surgery, against a background of Supreme Court awards of Rs 11 crore (2013) and Rs 1.7 crore (2015); commentators now urge a **no-fault liability system in the public sector with statutory caps on compensation**, and press the "6 Cs" of prevention — checklist, cost, consent, counselling, complications management and coverage by professional indemnity insurance [PMID 36872741] [PMID 40309544]. Where a postgraduate trainee is the alleged wrongdoer, the **supervising faculty member and the hospital** may be held liable alongside [PMID 39163069].

Guideline and legislative updates

- ▶ The single most examinable development — in *Bar of Indian Lawyers vs D.K. Gandhi* (Supreme Court, 2024) the Court held that **advocates are not liable for deficiency in service under the Consumer Protection Act**, reasoning that legal services lack a universal standard against which dereliction can be judged. The same bench expressly **doubted the 30-year-old V.P. Shantha ruling** that brought doctors within the Act and called for it to be reconsidered, reopening the question whether a doctor should be equated with any other service provider under the CPA 2019 — a debate sharpened by the growth of defensive medicine [PMID 41800776].
- ▶ **Pecuniary jurisdiction has moved** — the National Commission now entertains complaints **valued above Rs 2 crore**, so Park's "above Rs 1 crore" slab is superseded, and jurisdiction is fixed by the value of the goods or services paid for, not by the compensation claimed [NCDRC 2026].
- ▶ **Consumer protection is now a public-health instrument well beyond clinical care** — the CPA 2019 sits alongside the Food Safety and Standards Act 2006 and judicially expanded Article 21 to give an actionable right to safe, unadulterated food, and underpins SDGs 2, 3 and 12. A survey of **354 Indian youth aged 15–25** found statutory consumer-rights knowledge had **essentially zero correlation with actual dietary practice** (Kendall's tau -0.001), a quantified knowledge–action gap; the authors call for interpretative front-of-package labelling and for the CCPA to curb deceptive digital food-delivery advertising under the CPA 2019 [PMID 42549391].

Global and WHO developments

- **The standard of care is shifting internationally** — Indian courts still apply the **Bolam** test (adopted by the Supreme Court in *Jacob Mathew*), under which conformity with a responsible body of medical opinion defends the doctor, and mere deviation from standard practice is not negligence. English law has since moved to **Bolitho**, which requires that the supporting expert opinion also be *logically defensible*; if India adopts it, a doctor could be held negligent despite having followed expert guidance. Commentators also warn that the "reasonable doctor" yardstick is applied too narrowly when the working conditions that produced the alleged lapse are ignored [PMID 40395703] [PMID 36399918].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 12, Medicine and Social Sciences, pp 837–838); Sunder Lal — Textbook of Community Medicine (Ch 1, Concept of Health and Disease, Ethics in Medicine and Doctor–Patient Relationship, pp 55–57); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 13, Health and Law, pp 155–158); Jugal Kishore — National Health Programmes of India (Ch 2, Legislations Related to Health, pp 854–858); AFMC Red Book of Community Medicine 2024 (Ch 39, Public Health Legislations Related to Health, pp 1235–1236); Red Book — Public Health and Preventive Medicine (Consumer Protection Act as applicable to the medical profession, pp 377, 394–395); Suryakantha — Community Medicine with Recent Advances (Ch 52, Public Health Acts, p 939). Consumer Protection Act 1986; Consumer Protection Act 2019; Clinical Establishments (Registration and Regulation) Act 2010. **Coverage note:** the indexed corpus describes the 2019 Act's structure and penalties but does not state its revised pecuniary jurisdiction limits, its appeal time limits or the two-year limitation for filing, so those figures are deliberately not quoted.

Recent-advances sources: 39439653 — The New Bharatiya Nyaya Sanhita Laws: Progress or Pitfall for Doctors?, *Cureus* 2024, DOI [10.7759/cureus.69925](https://doi.org/10.7759/cureus.69925); 42423515 — Reform of criminal laws in India: potential impact on the criminal justice system and forensic investigation, *Medicine, Science and the Law* 2026, DOI [10.1177/00258024261465987](https://doi.org/10.1177/00258024261465987) (abstract-only); 42088806 — Medicolegal Aspects of Neurosurgery in India: A Narrative Review, *Cureus* 2026, DOI [10.7759/cureus.106418](https://doi.org/10.7759/cureus.106418); 40395703 — Can Research Articles Published in Medical Journals be Used as Expert Evidence in Medical Negligence Cases? A Cross-Sectional Retrospective Study of Indian Court Judgments, *Avicenna Journal of Medicine* 2025, DOI [10.1055/s-0045-1806761](https://doi.org/10.1055/s-0045-1806761); 36872741 — Alarming rise in consumer cases/medical malpractice claims with huge compensation, *Indian Journal of Ophthalmology* 2023, DOI [10.4103/IJO.IJO_2530_22](https://doi.org/10.4103/IJO.IJO_2530_22); 41800776 — Rethinking medical liability in India: Supreme Court's call for judicial review and ongoing uncertainty, *Indian Journal of Medical Ethics* 2025, DOI [10.20529/IJME.2025.058](https://doi.org/10.20529/IJME.2025.058) (abstract-only); 40309544 — Knowledge and awareness of dental professionals regarding Professional Indemnity Insurance (PII) in India: a systematic review, *Nigerian Medical Journal* 2025, DOI [10.71480/nmj.v66i1.588](https://doi.org/10.71480/nmj.v66i1.588); 39163069 — Medical Negligence by Postgraduate Students and Residents: The Liability Trail, *Journal of the Association of Physicians of India* 2024, DOI [10.59556/japi.72.0597](https://doi.org/10.59556/japi.72.0597) (abstract-only); 42549391 — Artificial intelligence in nutritional health: sustainable consumers' right to quality of life with special reference to youth, *Frontiers in Public Health* 2026, DOI [10.3389/fpubh.2026.1817284](https://doi.org/10.3389/fpubh.2026.1817284); 36399918 — The doctrine of "Reasonable Doctor" in medical negligence: need to be more reasonable, *Journal of Forensic and Legal Medicine* 2022, DOI [10.1016/j.jflm.2022.102451](https://doi.org/10.1016/j.jflm.2022.102451) (abstract-only). National Consumer Disputes Redressal Commission, ncdr.nic.in (accessed August 2026) — pecuniary jurisdiction above Rs 2 crore, online filing via e-Jagriti; National Consumer Helpline, consumerhelpline.gov.in (accessed August 2026) — 1915, 17 languages; Telemedicine Practice Guidelines 2020 (MoHFW and NMC).



First referral unit (FRU) in India

Asked in: Paper IV Nov 2017 — RGUHS MD Community Medicine

Definition — A first referral unit (FRU) is an existing health facility — a community health centre (CHC), sub-district hospital or district hospital — that can be *declared* fully operational only when it is equipped to provide **round-the-clock comprehensive emergency obstetric and newborn care (CEmONC)**, in addition to all **ten** stipulated minimum services (Sunder Lal, Ch 17, pp 754–755; Park 28e, Ch 7, p 543). It is the secondary-care node of the three-tier rural health system, positioned to receive the complicated case that the sub-centre and primary health centre (PHC) cannot manage. Its declared purpose is to prevent maternal and neonatal deaths (Sunder Lal, Ch 17, p 755).

Evolution and policy place of the FRU

Phase	What it did for the FRU
Child Survival and Safe Motherhood (CSSM) Programme, 1992–1997	Launched 20 August 1992 (<i>Sadbhavna Diwas</i>); phased over 5 years — begun in 51 districts in 1992–93 and expanded to all districts by 1996–97. In six very-high-maternal-mortality states — Assam, Bihar, Madhya Pradesh, Odisha, Rajasthan, Uttar Pradesh — additional inputs upgraded rural facilities for managing obstetric emergencies, and FRUs for emergency obstetric care were established in 219 districts over the 5-year period ; 1,748 referral units were identified nationally and supported with equipment kits E to P. CSSM thus laid the foundation of newborn care and FRUs at national level (Sunder Lal, Ch 10, p 572; Park 28e, Ch 7, p 542)
Reproductive and Child Health (RCH) Phase I, 1997–2005	Composite programme funded by the Government of India with World Bank and European Commission support; carried over all CSSM components and added reproductive tract infection/sexually transmitted infection (RTI/STI) care and adolescent reproductive and sexual health. Existing FRUs were strengthened with emergency obstetric kits, equipment kits and skilled manpower on contract. Funds were routed to <i>Panchayats</i> to assist poor people in obstetric emergencies, but state feedback showed no active involvement of Panchayats in running the scheme; incentives to health workers for round-the-clock services also failed in most states, working only in Andhra Pradesh and Tamil Nadu , where 3–4 staff nurses/auxiliary nurse midwives (ANMs) were posted at PHCs instead (Sunder Lal, Ch 10, p 572; Park 28e, Ch 7, pp 542–543)
RCH Phase II, from 1 April 2005	Operationalising FRUs at sub-district/CHC level for CEmONC declared a critical strategy ; it was decided that all FRUs be made operational for emergency and essential obstetric care in this phase (Park 28e, Ch 7, pp 543–544). Blood storage centres to be set up at FRUs and sub-district level per Government of India guidelines, for wider availability of safe blood for obstetric and trauma emergencies (Park 28e, Ch 7, p 524)
National Rural Health Mission (NRHM), 2005 → National Health Mission	Aim: operationalise all existing CHCs as 24-hour FRUs; anaesthetist and public-health programme manager on contract; Indian Public Health Standards (IPHS) adopted with ₹20 lakh per CHC for upgradation; <i>Rogi Kalyan Samiti</i> for accountability (Sunder Lal, Ch 17, pp 754–755)
National Health Policy 2017	"Proportion of specified type of facilities functioning as FRUs" made a monitorable policy indicator , measured against the norm of one FRU per 5,00,000 population (Park 28e, Ch 2, p 45)
IPHS 2022	CHCs formally classified as FRU-CHC and non-FRU CHC ; urban CHCs to function only as FRU-UCHCs (Sunder Lal, Ch 17, p 752)

Criteria for declaring a facility an FRU

(a) **Ten minimum services**, all available round the clock; three of them (starred) are the **critical determinants** of functionality (Park 28e, Ch 7, p 543; Sunder Lal, Ch 17, pp 754–755).

#	Service	Critical determinant?
1	24-hour delivery services — normal and assisted	—
2	Emergency obstetric care including surgical intervention (caesarean section)	Yes*
3	Newborn care — newborn stabilisation unit	Yes*
4	Emergency care of sick children	—
5	Full range of family planning services including laparoscopic sterilisation	—
6	Safe abortion services	—
7	Treatment of sexually transmitted and reproductive tract infections (STI/RTI)	—
8	Blood storage facility on a 24-hour basis	Yes*
9	Essential laboratory services	—
10	Referral (transport) services	—

(b) **Nine physical prerequisites** — to perform the full range of FRU function a facility must additionally have (Park 28e, Ch 7, p 543):

- ▶ Minimum **bed strength of 20–30** (relaxable initially to **10–12 beds** in difficult areas — the North-East states and underserved areas of Empowered Action Group states)
- ▶ A fully functional **operation theatre**
- ▶ A fully functional **labour room**
- ▶ An area earmarked and equipped for **newborn care** in the labour room and in the ward
- ▶ A **functional laboratory**
- ▶ **Blood storage facility**
- ▶ **24-hour water supply and electricity supply**
- ▶ Arrangements for **waste disposal**
- ▶ **Ambulance facility**

Access and coverage norms — 12–15% of all deliveries may need emergency obstetric services, and an FRU must therefore be reachable **within 2 hours of travel time** (Sunder Lal, Ch 10, p 586). Under IPHS 2022, an FRU-CHC carries **30 essential beds**, a functional operation theatre and blood storage unit, and must provide elective as well as emergency secondary-level surgery — going beyond obstetrics; a 50-bedded FRU-CHC additionally makes ophthalmic, orthopaedic and ENT services desirable (Sunder Lal, Ch 17, p 752).

Manpower prescribed — FRU-CHC versus non-FRU CHC (IPHS 2022, essential posts)

Post	Non-FRU CHC (30 beds)	FRU-CHC (30 beds)
Physician / family medicine specialist	1	1
Surgeon	–	1
Obstetrician and gynaecologist	–	1
Paediatrician	–	1
Anaesthesiologist	–	1
General duty medical officer (GDMO)	3	6
Staff nurses	29	46

(Sunder Lal, Ch 17, p 753, Table 17.8)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce this plate by hand and mark **every arrow in both directions**, to show two-way flow of patients, specialists, records and services between the levels.

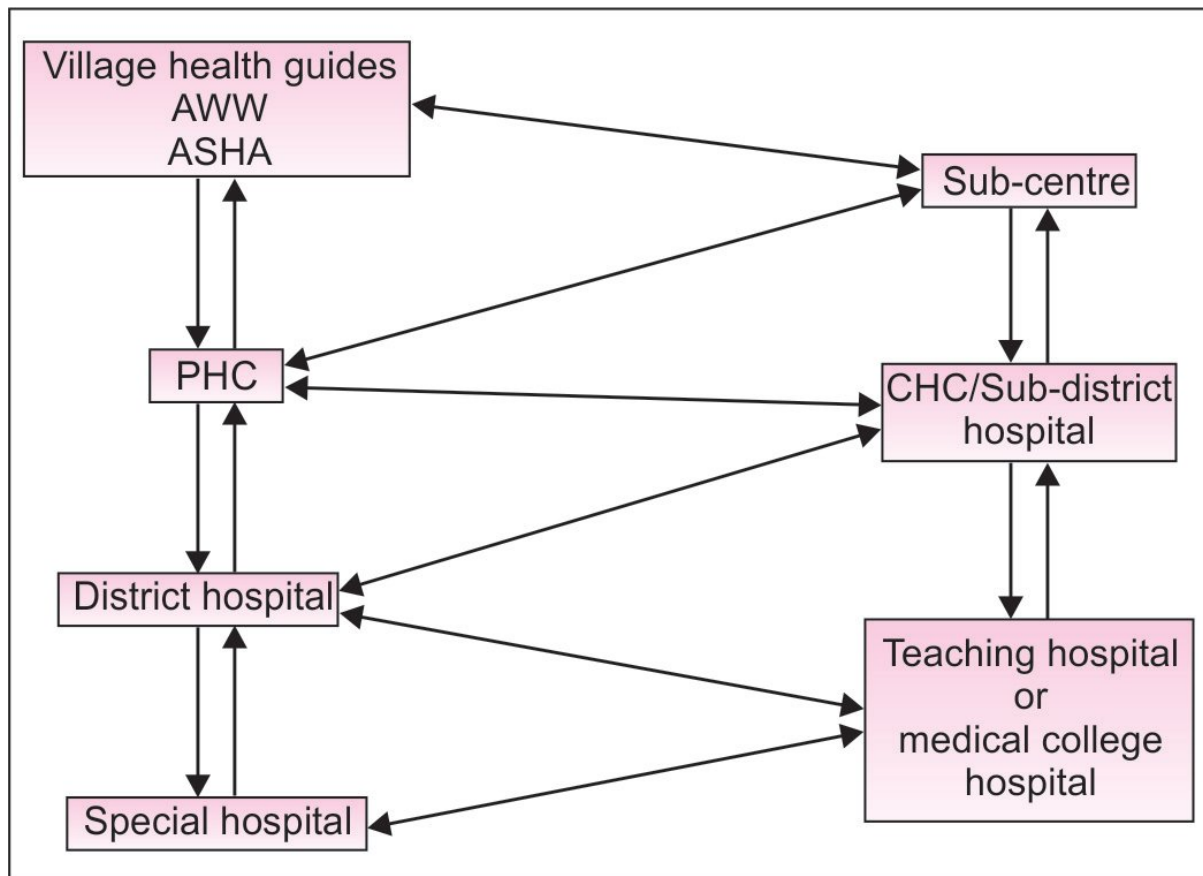
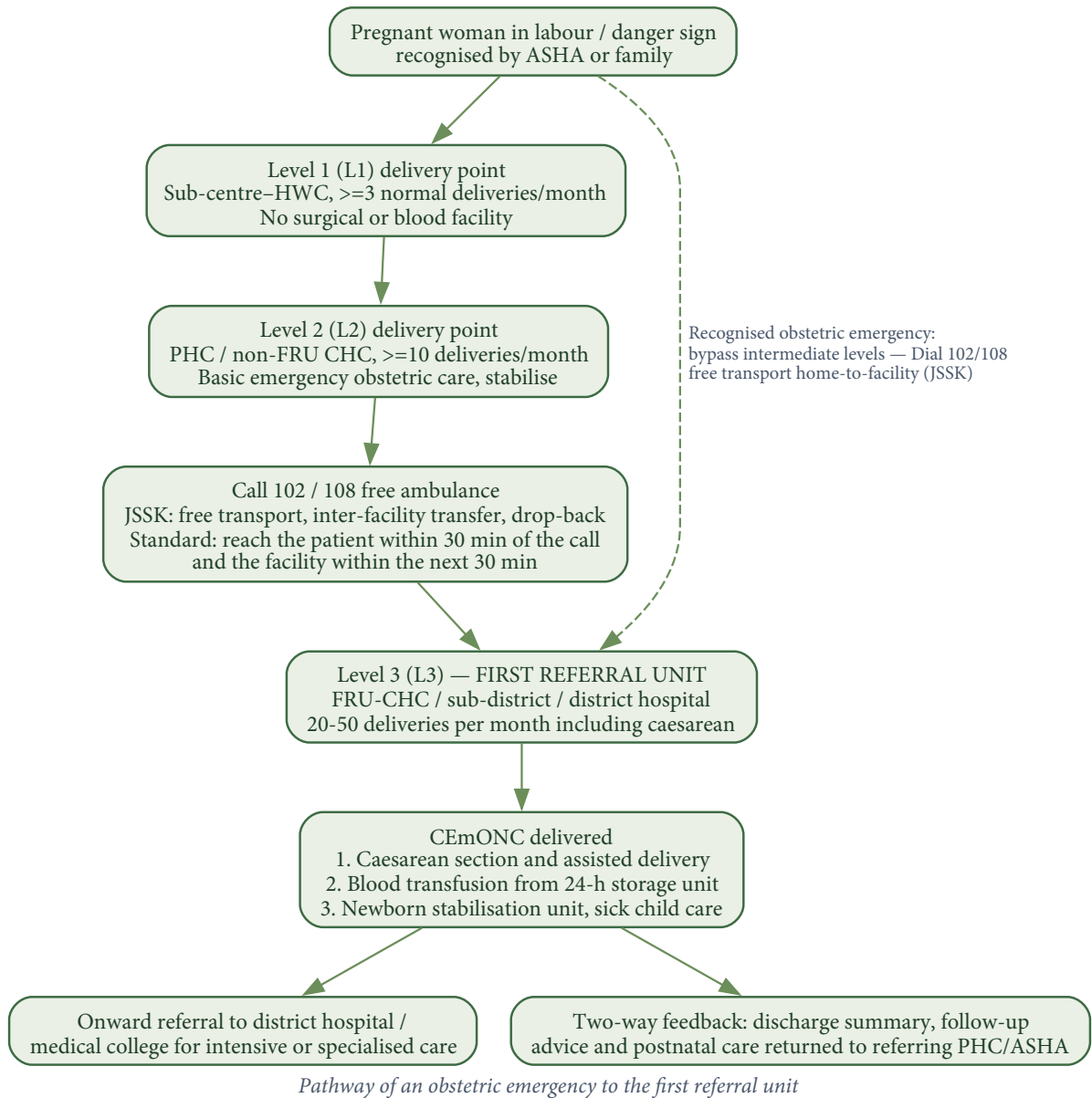


Fig 17.5 — Referral system (Sunderlal, p.762)



Critical appraisal — what works and what fails

Domain	Strength of the concept / achievement	Documented shortcoming
Concept	Correctly identifies CEmONC as the single most important intervention for saving women who develop complications (Sunder Lal, Ch 10, p 580)	Vertically driven by maternal-health targets; general surgical and paediatric emergency roles were added only in IPHS 2022
Coverage	6,064 CHCs functioning (5,480 rural, 584 urban) as on 31 March 2022 (Sunder Lal, Ch 17, p 752)	Of 3,559 designated FRUs only 1,685 (47%) were functional in 2022 (Sunder Lal, Ch 17, p 755)
Human resources	Multiskilling policy — 18-week life-saving anaesthetic skills and 16-week emergency obstetric (including caesarean) training for MBBS doctors (Sunder Lal, Ch 10, pp 580–581)	~80% shortfall of specialists at CHC level; where specialists are posted they are often not of the specified specialities, so most CHCs cannot be declared FRUs (Sunder Lal, Ch 10, p 580; Ch 17, p 753)

Domain	Strength of the concept / achievement	Documented shortcoming
Blood	Blood storage centres sanctioned at FRUs and sub-district level for remote areas where a full blood bank is not feasible (Park 28e, Ch 7, p 524)	A storage centre is only an outpost of the licensed bank network (590 blood component separation units, 34 model, 108 major and 591 district-level blood banks), so an FRU's blood supply is only as reliable as its parent district bank (Park 28e, Ch 7, p 524)
Referral linkage	National Ambulance Service — 11,879 Dial-108 and 10,716 Dial-102/104 vehicles operational as on June 2021, with free transport under Janani Shishu Suraksha Karyakram (Park 28e, Ch 7, pp 539–540)	Referral system is "poorly developed"; entry is a free-for-all — patients walk directly into secondary and tertiary care, so the FRU is bypassed upward and crowded from below (Sunder Lal, Ch 17, p 761)
Utilisation	Institutional births rose to 88.6% (Sunder Lal, Ch 10, p 580; NFHS-5, 2019–21)	Only 20–30% use public primary curative services; indoor beds at CHCs remain under-occupied against the IPHS planning assumption of 60% bed occupancy (Sunder Lal, Ch 17, pp 753, 756)
Quality	LaQshya covers district hospitals, sub-district hospitals, high-caseload CHCs, FRUs and medical colleges — 2,445 facilities identified (including 193 medical colleges) (Park 28e, Ch 7, pp 545–546)	Certification is the exception: only 441 labour rooms and 392 operation theatres state-certified , and 447 labour rooms and 365 operation theatres nationally certified till December 2021 — fewer than one in five identified facilities on either arm (Park 28e, Ch 7, pp 545–546)
Equity	Additional inputs targeted to high-maternal-mortality states from CSSM onward	Caesarean rate 14.3% in public versus 47.4% in private facilities (Sunder Lal, Ch 10, pp 580–581; NFHS-5, 2019–21) — the surgical load the FRU exists to carry is being absorbed by the private sector
Role clarity	Designed as the referral point for four PHCs serving 1,20,000 population (80,000 hilly/tribal)	The same secondary facility doubles as primary care for its own town — a dual role that dilutes its effectiveness (Sunder Lal, Ch 17, p 760)

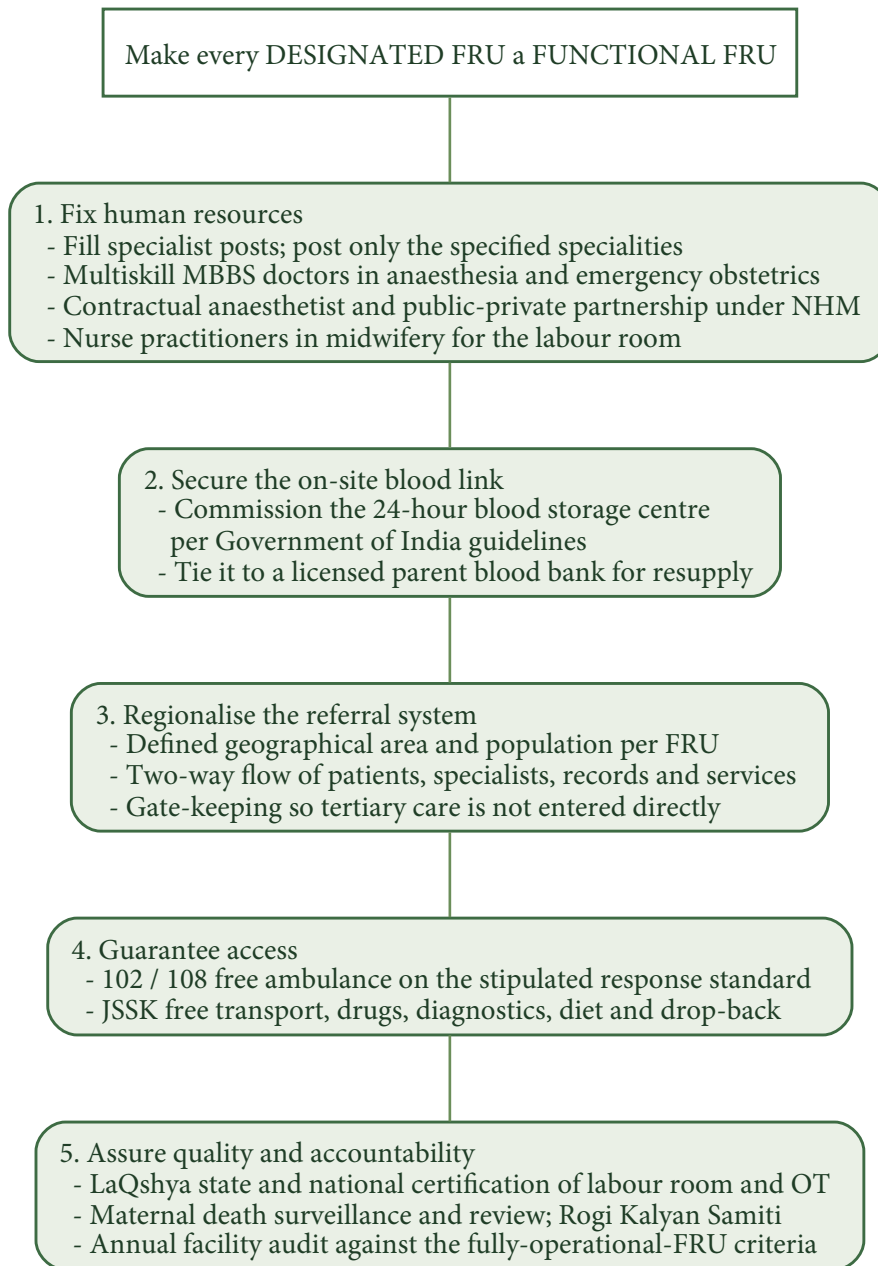
Evaluating an FRU — what a critical assessment must measure

- **Declaration versus function** — the single most useful audit is whether a *designated* FRU actually delivers each starred service in the criteria table above round the clock; failure of any one converts a designated FRU into a non-functional one.
- **Policy indicator with its operational cut-off** — proportion of specified facilities functioning as FRUs, with the **number of caesarean sections per year** as the operational proxy: a sub-district hospital or CHC must conduct a **minimum 60 caesarean sections per year (36 for hilly and North-Eastern states except Assam)** and a district hospital a **minimum 120 per year (72 for hilly and North-Eastern states except Assam)** (National Health Policy 2017 indicator 3.1.3a; Park 28e, Ch 2, pp 45–46).
- **Access indicators** — proportion of the catchment population within the stipulated travel time of an FRU, and availability of Dial 102/108 transport for that journey.

- ▶ **Utilisation indicators** — deliveries and caesarean sections conducted per month against the level-3 delivery-point benchmark, and bed occupancy against the IPHS planning assumption of 60%.
- ▶ **Quality indicators** — state and national LaQshya certification status of the labour room and maternity operation theatre, and findings of maternal death review and surveillance.
- ▶ **Impact indicators** — a community health centre's success is judged not by patients treated but by the fall in maternal mortality ratio, infant mortality rate, fertility rate and disease and disability rates in its assigned block population (Sunder Lal, Ch 16, pp 708–709).

Under IPHS 2022 the FRU's remit is being widened beyond maternity: every CHC at block headquarters is to be developed as a **Block Public Health Unit**, serving as the block's referral hub, disease-surveillance node and public-health planning platform (Sunder Lal, Ch 17, p 752). This is a welcome correction of the FRU's historically narrow, maternal-health-only framing, but it will succeed only if the specialist vacancies that already prevent the obstetric mandate from being met are first filled.

Strengthening the FRU — the way forward



Strategy to convert a designated FRU into a functional FRU

High-yield exam pointers

- ▶ **Three critical determinants of an FRU** (the examiner's favourite): **surgical intervention** (caesarean section) + **newborn care** + **blood storage**, all on a **24-hour** basis. Mnemonic: "Cut, Cot, Clot."
- ▶ **FRU = CEmONC centre**; **ten** minimum services plus **nine** physical prerequisites; sited at CHC, sub-district or district hospital; foundation laid by **CSSM 1992** (FRUs in **219** districts of six high-mortality states).
- ▶ **Norms to quote**: one FRU per **5,00,000** population (National Health Policy 2017 indicator 3.1.3a); accessible within **2 hours** travel time; **12–15%** of deliveries need emergency obstetric care.

- **The C-section operational cut-off** (exact, often asked): SDH/CHC ≥ 60 caesareans/year (36 for hilly and North-Eastern states except Assam); DH ≥ 120 /year (72 for the same states) (Park 28e, Ch 2, p 46).
- **The critique numbers: 3,559 designated FRUs, only 1,685 functional (2022); 80% specialist shortfall at CHC level; LaQshya** — 2,445 facilities identified, 447 labour rooms and 365 OTs nationally certified till Dec 2021.
- **Delivery-point benchmarks:** L1 ≥ 3 normal deliveries/month, L2 ≥ 10 /month with management of complications, L3 20–50/month including caesarean section (Park 28e, Ch 7, p 557); Sunder Lal gives the same scheme facility-wise — >3 at sub-centre, >10 at PHC, >20 at CHC, >50 at sub-district/district hospital (Ch 10, p 580).
- **Dated Indian figures:** maternal mortality ratio 97 per lakh live births in 2018–20, down from 167 in 2011–13 (Park 28e, Ch 9, pp 663–664); institutional births 88.6%, caesarean 14.3% public versus 47.4% private (NFHS-5, 2019–21).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Indian Public Health Standards were revised in 2022 and now split community health centres into first referral unit (FRU) and non-FRU types, so an answer written from the older "all CHCs to become FRUs" framing of the National Rural Health Mission should mention the 2022 reclassification; Park 28e states the delivery-point benchmark by level (L1/L2/L3) while Sunder Lal states the same scheme by facility, and the two are consistent rather than conflicting; the 47% functionality figure, the 80% specialist shortfall and the LaQshya certification counts are the most recent values carried in the retrieved textbooks and should be quoted with their year.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, National Family Health Survey-5, Rural Health Statistics, LaQshya guidelines, WHO EmONC monitoring framework

Programme and policy updates (India)

- **Health and Wellness Centres have been renamed Ayushman Arogya Mandir**, so the level-1 base of the referral chain now carries a new name but not new capacity — a block-level audit of 20 centres in Dhamtari district, Chhattisgarh, found a 41.2% community health officer vacancy, not one centre holding the full operational-guideline set of equipment and essential medicines, and only 17.6% of sub-centre Ayushman Arogya Mandirs meeting the prescribed diagnostic standard; cases that should be screened and stabilised at L1 therefore reach the FRU unfiltered [PMID 40821268].
- **Skilled-workforce projections put the specialist gap beyond 2030** — rural skilled health professional density (doctors, nurses, midwives) rose only from 7.55 per 10,000 in 2009 to 8.85 in 2019 and is projected to reach 11.46 by 2030, leaving a shortfall of 0.47–1.83 million; closing it would cost about ₹1.46–4.96 trillion, roughly 15% of government health spending. On this trajectory the very posts whose vacancy prevents a designated FRU from being declared functional are still unfilled at the Sustainable Development Goal deadline [PMID 42185866].

- **Tribal blocks remain the weakest node of the FRU network** — a state-level analysis of Rural Health Statistics from 2011 to 2021 found the tribal-area community health centre capacity index (functional ÷ required) essentially static, 0.79 to 0.78, while the overall tribal health-system capacity index fell from 0.95 to 0.79; roughly one CHC in five that the norms require — the very facility class designated as an FRU — has never been built [PMID 42192389].

Current national data

- **Bihar's 149 FRUs, April 2018 to March 2019, is the sharpest published audit of designation versus function:** 546,444 deliveries produced only 16,961 caesareans, a state-wide FRU caesarean rate of 3.1% (regional hospitals 0.3%, sub-district 1.2%, district 5.7%), and 17 designated FRUs conducted no delivery at all. Only 7% were LaQshya-certified, 26% had a blood bank or blood storage unit, 58% had an obstetrician, 38% an anaesthetist and 35% a doctor trained in emergency obstetric care; a functioning operation theatre was the single strongest correlate of caesarean rate (incidence rate ratio 21.0, 95% CI 7.9–55.8), ahead of obstetrician numbers (1.3) and emergency-obstetric-trained doctors (1.6) [PMID 37133240].
- **The public–private caesarean split now has an adjusted effect size** — NFHS-5 gives 13.95% in public against 47.18% in private facilities, with adjusted odds of caesarean in a private institution 4.11 (95% CI 3.98–4.24) times that in a public one, up from 3.70 in NFHS-4; in Bihar the odds ratio is 16.19 (95% CI 13.76–19.05) against a public-sector caesarean rate of 3.53%, which is below the floor at which caesarean section still averts maternal death [PMID 42406715].
- **Bypass of the secondary level is measurable at the receiving end** — an audit of 685 emergency obstetric admissions to a Puducherry tertiary hospital found 26.4% arrived as institutional referrals, most sent straight from primary health centres rather than through a secondary facility, and adverse fetal outcome was 1.88 (95% CI 1.21–2.95) times higher in referred than in self-referred women; the authors attribute the pattern to emergency obstetric services being unavailable at the secondary hospitals in between [PMID 37459016].

Elimination, surveillance and disease status

- **Maternal death surveillance has moved to the sub-district unit** — Karnataka's taluk-level review of deaths from 2017–18 to 2021–22 placed 39.82% within 48 hours of delivery and a further 20.95% at 2–7 days, identified nine persistent high-high hotspot taluks (Kalburgi, Yadgir and Chikkaballapura among them), and found primary postpartum haemorrhage (β 1.884) and sepsis (β 1.926) the strongest medical contributors — precisely the two emergencies an FRU's 24-hour theatre and blood storage exist to answer. Spatial clustering peaked in 2020–21 (Moran's I 0.191, $p = 0.001$), and antenatal care at sub-centre and sub-district hospital was protective [PMID 42386290].

Guideline and legislative updates

- **The LaQshya threshold is rarely met at first assessment** — a before-and-after evaluation at a tertiary referral hospital scored the labour room 48% and the maternity operation theatre 56% against the 70% certification cut-off, with quality management, infection control and emergency preparedness the failing domains while service provision scored near-full marks; targeted correction lifted partograph use from 0% to 78% and hand-hygiene compliance from 36% to 74%, showing the gap is in systems rather than in caseload [PMID 40978978].

Global and WHO developments

- ▶ **WHO's "availability of EmONC facilities" indicator failed validation in India** — across 282 birth facilities in Gonda and Meerut (Uttar Pradesh) and Krishnagiri and Thiruvallur (Tamil Nadu), not one of the 67 BEmONC-designated facilities had performed all seven basic signal functions in the preceding 90 days (17.9% partial), and only 3 of 13 CEmONC-designated facilities (23.1%) had performed all nine. Counting designated facilities instead of performed signal functions overstates coverage — the identical error as counting designated rather than functional FRUs [PMID 37708180].
- ▶ **The signal-function framework itself is under revision** — a review commissioned as part of that revision argues the 1997 emergency obstetric care list no longer matches the maternal and newborn landscape and must extend past clinical capability to health-systems and rights dimensions; a parallel multi-country validation found only 31% of Indian midwifery professionals performed every signal function they are authorised for, and assisted vaginal delivery — half of service 1 in the FRU's ten — is one Indian midwives are not authorised to perform at all [PMID 38820693], [PMID 37079621].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease: National Health Policy 2017 monitorable indicators and the FRU caesarean-section criteria; Ch 7 — Health Programmes in India: CSSM and RCH Phase I and II, FRU services and prerequisites, blood storage centres at FRUs, National Ambulance Service, LaQshya, delivery points; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics: maternal mortality ratio and NFHS-5 delivery-care indicators); Sunder Lal — Textbook of Community Medicine (Ch 10 — RMNCH+A: CSSM 1992–97 and the 219-district FRU upgradation, delivery points, multiskilling, access norms; Ch 16 — Health Planning and Management; Ch 17 — Healthcare of the Community, Indian Public Health Standards 2022 for CHC/FRU). Named documents: National Health Policy 2017; Indian Public Health Standards 2022; National Family Health Survey-5 (2019–21); Sample Registration System 2018–20; Janani Shishu Suraksha Karyakram; LaQshya guidelines.

Coverage note: the retrieved corpus supports the FRU definition, criteria, norms, staffing and functionality data fully; it does not carry a state-wise breakdown of functional FRUs, so none is quoted.

Recent-advances sources: 37079621 — Authorization of midwives to perform basic emergency obstetric and newborn care signal functions in Argentina, Ghana, and India: a multi-country validation study, PLoS One 2023, doi:[10.1371/journal.pone.0283029](https://doi.org/10.1371/journal.pone.0283029); 37133240 — What to scale first? A cross-sectional analysis of factors affecting cesarean delivery rates at first referral units in Bihar, India, Global Health Action 2023, doi:[10.1080/16549716.2023.2202465](https://doi.org/10.1080/16549716.2023.2202465); 37459016 — Outcome of emergency obstetric cases in relation to the referral pattern at a tertiary hospital in Puducherry, Indian Journal of Public Health 2023, doi:[10.4103/ijph.ijph_1124_22](https://doi.org/10.4103/ijph.ijph_1124_22) (abstract-only); 37708180 — Validating indicators for monitoring availability and geographic distribution of emergency obstetric and newborn care (EmONC) facilities, PLoS One 2023, doi:[10.1371/journal.pone.0287904](https://doi.org/10.1371/journal.pone.0287904); 38820693 — Evolving narratives on signal functions for monitoring maternal and newborn health services: a meta-narrative inspired review, Social Science & Medicine 2024, doi:[10.1016/j.socscimed.2024.116980](https://doi.org/10.1016/j.socscimed.2024.116980) (abstract-only); 40821268 — Assessment of the functioning of Health and Wellness Centers in Kurud block, Dhamtari district, Chhattisgarh, Cureus 2025, doi:[10.7759/cureus.88071](https://doi.org/10.7759/cureus.88071); 40978978 — Bridging gaps in maternal care: evaluating LaQshya implementation at a tertiary hospital, Cureus 2025, doi:[10.7759/cureus.90350](https://doi.org/10.7759/cureus.90350); 42185866 — Scaling-up skilled health professionals in the rural Indian public health system: projections for 2030, BMC Health Services Research 2026, doi:[10.1186/s12913-026-14550-x](https://doi.org/10.1186/s12913-026-14550-x); 42192389 — Public health system capacity of tribal and rural areas in India: a retrospective state-level analysis from 2011 to 2021, BMC Public Health 2026, doi:[10.1186/s12889-026-27725-2](https://doi.org/10.1186/s12889-026-27725-2); 42386290 — Spatial autocorrelation and regression approach for delineating maternal mortality and its associated factors in Karnataka, India, BMJ Open 2026, doi:[10.1136/bmjopen-2024-094191](https://doi.org/10.1136/bmjopen-2024-094191); 42406715 — Caesarean section epidemic in India: is private sector to blame? A multivariate logistic regression analysis of the National Family Health Survey, PLoS One 2026, doi:[10.1371/journal.pone.0352156](https://doi.org/10.1371/journal.pone.0352156). Named documents used through these:

National Family Health Survey-4 and -5; Rural Health Statistics 2009–2021; LaQshya guidelines; WHO EmONC signal-function monitoring framework.



Primary health care — role of medical officer

Asked in: Paper IV 22-Dec-2023 — RGUHS MD Community Medicine

Definition — The Declaration of Alma-Ata (1978) defines primary health care as *"essential health care based on practical, scientifically sound and socially acceptable methods and technology made universally accessible to individuals and families in the community through their full participation and at a cost that the community and the country can afford to maintain at every stage of their development in the spirit of self-reliance and self-determination"* (Park 28e, Ch 2, p 49). It is the **first level of contact** between the individual and the national health system, integrating promotive, preventive, curative and rehabilitative care as an integral part of the country's plan for socio-economic development. The Alma-Ata conference proclaimed it the means of attaining the World Health Organization (WHO) goal of **Health for All** (Park 28e, Ch 1, p 23).

Essential elements of primary health care (Alma-Ata, 1978)

Although the specific services vary between countries, the Declaration outlined **8 essential components** (Park 28e, Ch 10, pp 719-720).

#	Essential element	Illustrative Indian delivery point
1	Education concerning prevailing health problems and the methods of preventing and controlling them	Health education by accredited social health activist (ASHA), village health and nutrition day
2	Promotion of food supply and proper nutrition	Integrated Child Development Services, mid-day meal, Anemia Mukh Bharat
3	An adequate supply of safe water and basic sanitation	Jal Jeevan Mission, Swachh Bharat Mission
4	Maternal and child health care, including family planning	Reproductive and child health services at sub-centre and primary health centre (PHC)
5	Immunization against major infectious diseases	Universal Immunization Programme (UIP)
6	Prevention and control of locally endemic diseases	Vector-borne disease, tuberculosis and leprosy programmes
7	Appropriate treatment of common diseases and injuries	PHC outpatient department, first referral unit
8	Provision of essential drugs	Essential drug list, free drugs and diagnostics initiative

Mnemonic — "ELEMENTS": Education, Local endemic disease control, Expanded immunization, Maternal and child health with family planning, Essential drugs, Nutrition and food supply, Treatment of common ailments and injuries, Safe water and sanitation.

Principles (pillars) of primary health care

Principle	Meaning	Operational expression in India
Equitable distribution	Health services shared equally by all irrespective of ability to pay; rich or poor, urban or rural must have access. Corrects the "social injustice" of services concentrated in cities	Free care at government sub-centre/PHC; PHC per 20,000 in tribal and hilly areas against 30,000 in plains
Community participation	Continuing, meaningful involvement of individuals, families and the community in planning, implementation and maintenance of services, relying maximally on local manpower, money and material — "health by the people", "placing people's health in people's hands"	ASHA and trained <i>dais</i> ; village health, sanitation and nutrition committee; <i>Rogi Kalyan Samiti</i> ; <i>Panchayati Raj</i> institutions
Intersectoral coordination	Primary health care cannot be delivered by the health sector alone; it involves agriculture, animal husbandry, food, industry, education, housing, public works and communication. Requires joint planning to avoid duplication and strong political will	Convergence of health with women and child development, education, rural development and water and sanitation departments
Appropriate technology	Technology that is scientifically sound, adaptable to local needs, acceptable to those who use it, and maintainable by the people themselves with resources they can afford	Oral rehydration solution, hand pump, cold chain, rapid diagnostic tests, low-cost point-of-care devices

(Park 28e, Ch 10, pp 719-720)

Job responsibilities of the medical officer, primary health centre

The medical officer (MO) of a PHC is responsible for implementing **all** health and family welfare activities in the PHC area, both in an individual capacity and as overall in-charge of a defined population of **30,000 in plains and 20,000 in tribal/hilly areas** (Sunder Lal, Ch 16, pp 715-716; Park 28e, Ch 13, p 850). He or she is the **captain of the health team** — simultaneously a clinician and a manager. Where a second medical officer is sanctioned, that officer performs identical duties (Park 28e, Ch 10, pp 738-739).

A. Clinical (curative) work

- Organizes and performs duties necessary for **routine outpatient services**, and ensures emergency cases are attended to; devotes morning hours to the outpatient department and afternoons to supervising field work.
- **Screens cases needing specialized attention** and refers them to referral institutions; cooperates and coordinates with other institutions providing medical care in the area.
- Attends all calls from in-patients while on "on-call" duty.
- Practises and models care delivered with **respect, dignity, privacy and modesty**, and ensures **confidentiality** of patients.

- ▶ Performs any other duty a medical officer is expected to perform by virtue of the post.

B. Public health (preventive and promotive) work

- ▶ Arranges and guides delivery of health care at community level and at the PHC through health assistants, health workers and others.
- ▶ Ensures every team member is fully conversant with the national health and family welfare programmes for the area allotted to that functionary.
- ▶ Prepares **operational plans** to achieve the laid-down targets under the national programmes.
- ▶ Ensures effective implementation — education, motivation, service delivery and after-care — of the **Reproductive and Child Health Programme, Universal Immunization Programme, vector-borne disease, blindness control, non-communicable disease and mental health programmes**, control of leprosy, tuberculosis and sexually transmitted diseases, and Ayushman Bharat.
- ▶ Plans and implements the UIP to obtain maximum possible coverage, ensuring **proper vaccine storage, cold-chain equipment maintenance** and adequate supplies of vaccine and logistics.
- ▶ Ensures proper implementation of **Integrated Management of Neonatal and Childhood Illness (IMNCI)** guidelines.
- ▶ Institutionalizes **public health surveillance** and takes timely action on any outbreak or epidemic in the area.
- ▶ Visits schools in the PHC area at regular intervals and arranges **medical check-up and immunization**; organizes and conducts **tubectomy and vasectomy camps**; designs the tour programme to cover all basic health services including family planning.
- ▶ Is responsible for all **health education** activities in the area.

C. Administrative and managerial work

- ▶ **Supervises** the work of staff, scrutinizes their programmes and modifies them to suit work priorities; visits each sub-centre regularly on fixed days and hours to give guidance, supervision and leadership.
- ▶ Holds **monthly staff meetings** — spending one day each month at the PHC — to evaluate progress, discuss problems and decide corrective steps.
- ▶ Ensures maintenance of all **prescribed records and registers** at PHC level and issues certificates in the capacity of a medical officer; reports performance to the community health centre and district.
- ▶ Takes timely action on **legal and medico-legal matters**, Right to Information applications and court cases, and implements court orders expeditiously.
- ▶ Ensures appropriate **utilization of funds** under General Financial Rules, completes **audit** in advance and furnishes audit reports; disposes of obsolete/condemned items and vehicles per government orders.
- ▶ Undertakes **performance audit** of staff, arranges retraining where required, and settles staff grievances promptly.
- ▶ Facilitates, supervises, monitors and implements the provisions of all **health-sector Acts and Rules**.

D. Training, community mobilization and intersectoral work

- ▶ Organizes **training and continuing education** for PHC staff, ASHA, *anganwadi* workers and *dais*, under guidance of the district health authorities and the Health and Family Welfare Training Centres.
- ▶ Provides on-the-job, **skill-based** training by demonstration and return demonstration, training needs being identified during supervisory field visits (Sunder Lal, Ch 16, p 721).
- ▶ Monitors and guides **hospital/PHC committees, patient welfare societies (*Rogi Kalyan Samiti*) and village health and sanitation committees**.
- ▶ Ensures **inter-sectoral and inter-departmental coordination**, involving community leaders, *Panchayati Raj* institutions and social welfare agencies.
- ▶ Consults the community periodically for feedback and builds bottom-up **village and sub-centre action plans** (Sunder Lal, Ch 16, pp 707-708).

Summary of the managerial role: the success of a PHC depends largely on the team leadership the medical officer provides — he or she must be the **planner, promoter, director, supervisor, coordinator and evaluator** (Park 28e, Ch 10, pp 738-739).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Principles and components of primary health care.** Draw a central box "Primary health care" with eight arrows radiating out. Label the four **principles** (community participation, appropriate technology, equitable distribution, multi-sectoral/intersectoral approach) and the four **care components** (preventive, promotive, curative, rehabilitative).

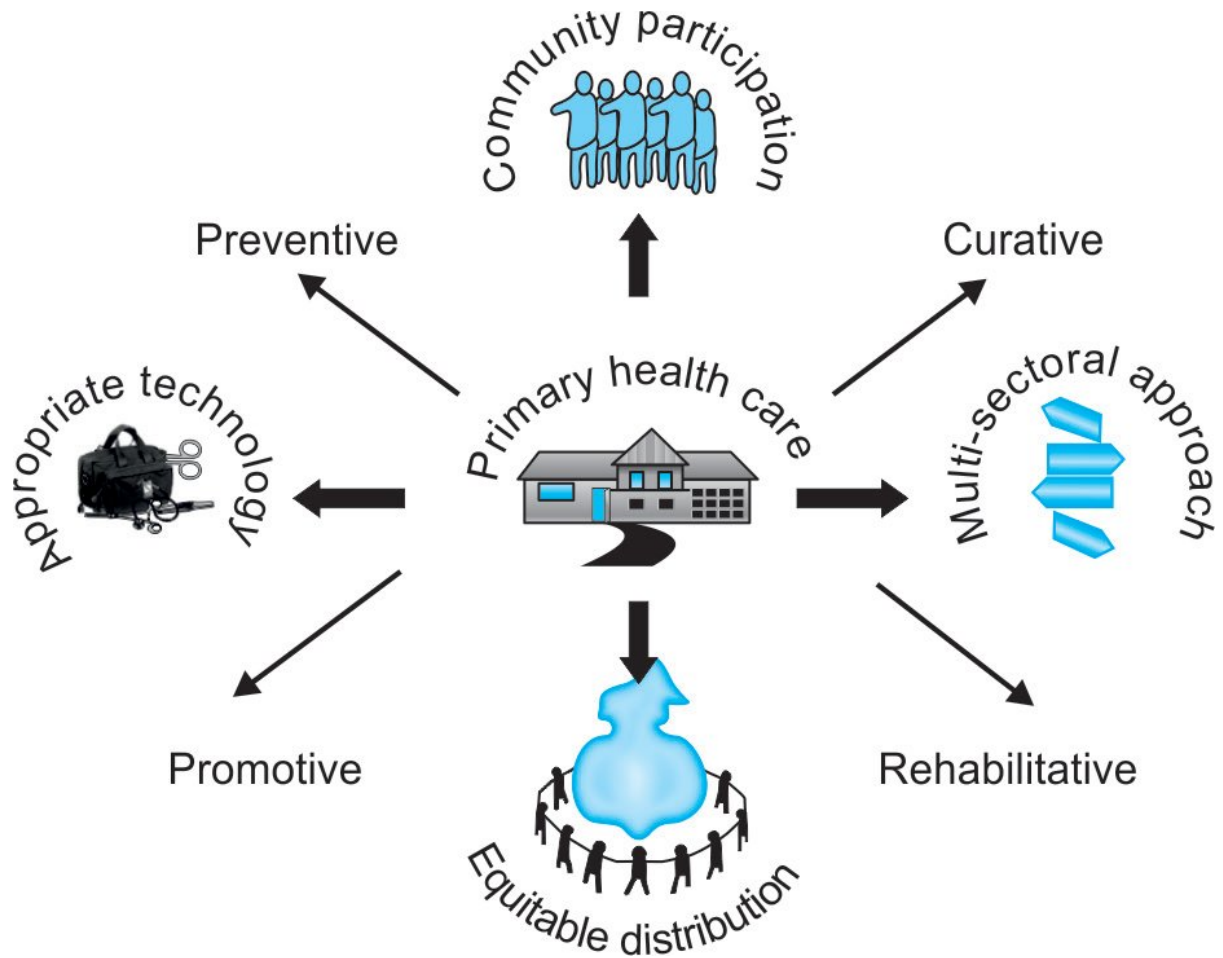
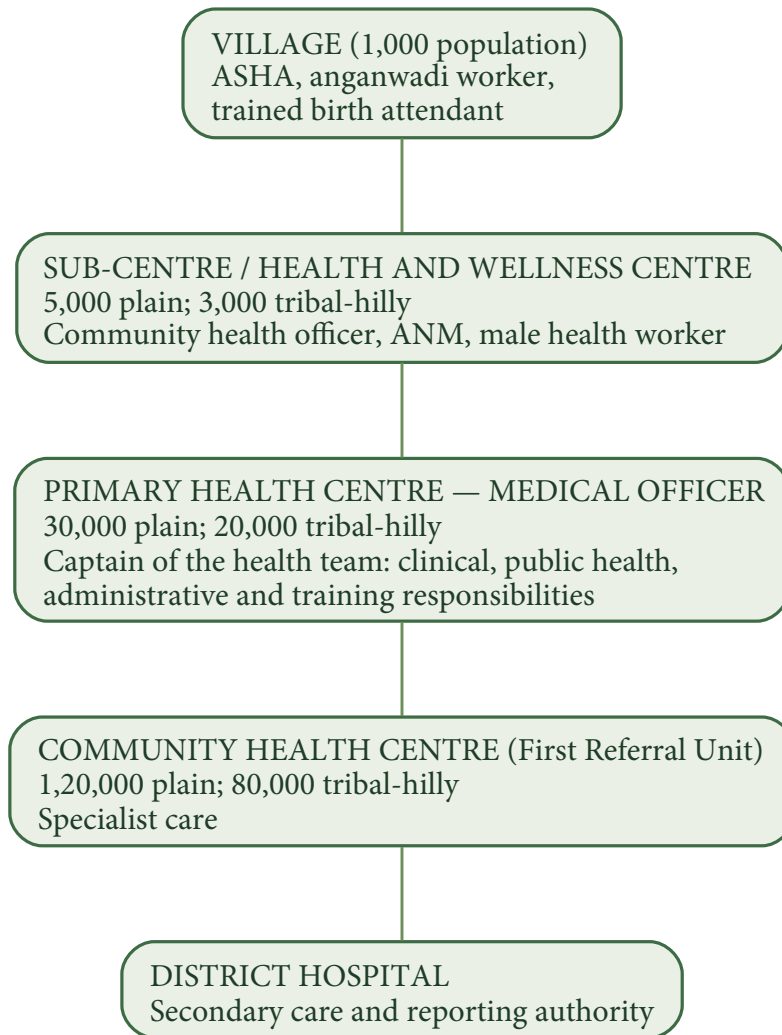
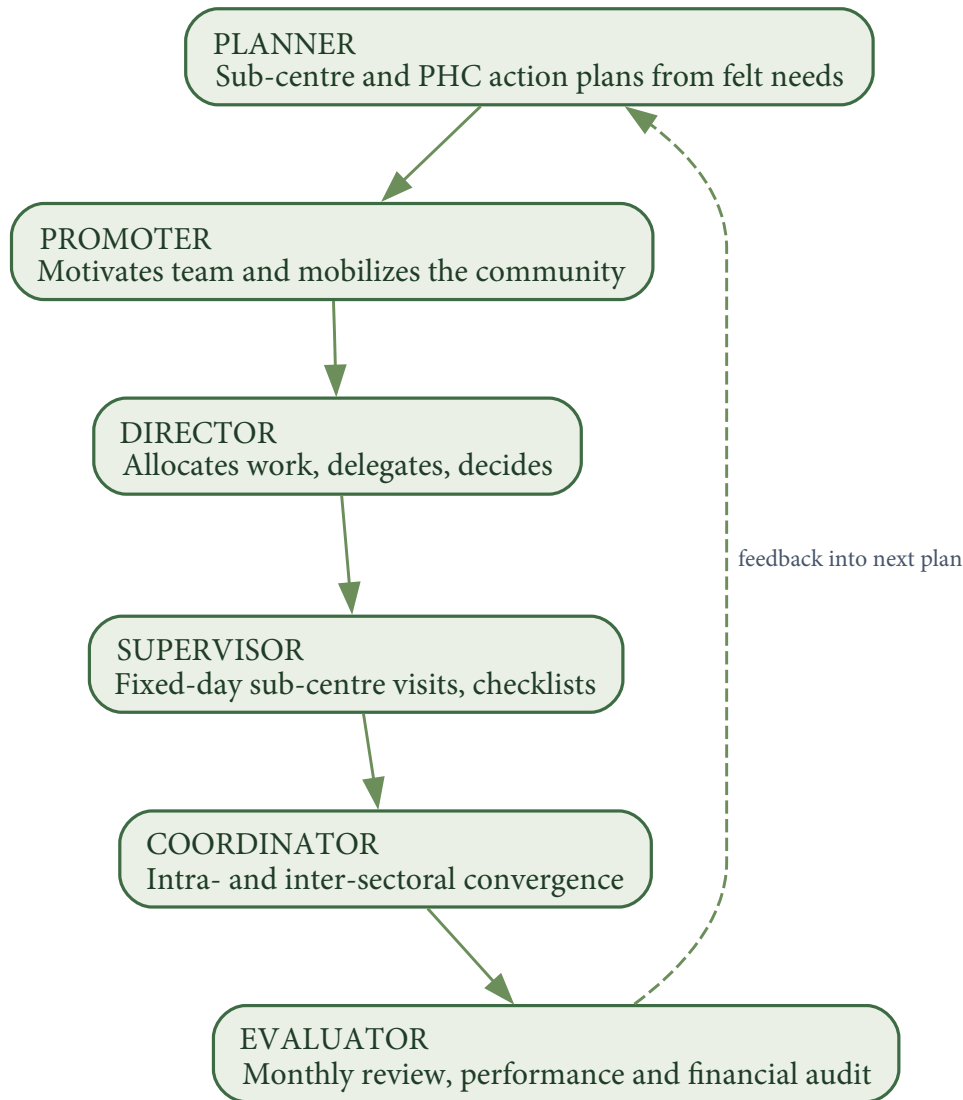


Fig 34.2 — Principles and components of primary health care (Suryakantha Environment Best, p.780)



Position of the medical officer, primary health centre, in the three-tier rural health care system — supervision flows downward, referral upward



The six managerial roles of the medical officer at a primary health centre

High-yield exam pointers

- ▶ **Alma-Ata 1978, USSR** — quote the definition verbatim (Park 28e, Ch 2, p 49).
- ▶ **8 elements** — mnemonic ELEMENTS; **4 principles** — equitable distribution, community participation, intersectoral coordination, appropriate technology.
- ▶ **Population norms** — sub-centre 5,000 (3,000 tribal/hilly), PHC 30,000 (20,000), community health centre 1,20,000 (80,000); ASHA 1 per 1,000 (Park 28e, Ch 10, pp 725, 731; Ch 13, p 850).
- ▶ **Sunder Lal's three heads** — clinical work, public health work, administrative work: use them as your sub-headings.
- ▶ **Six managerial roles** — planner, promoter, director, supervisor, coordinator, evaluator.
- ▶ **Anchor duties:** cold chain and UIP, IMNCI, school health, sterilization camps, fixed-day sub-centre visits, monthly staff meeting, records and certificates, medico-legal cases, General Financial Rules and audit, ASHA training, *Rogi Kalyan Samiti*.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Ayushman Bharat Health and Wellness Centres have been renamed **Ayushman Arogya Mandir** (Park 28e, Ch 7, pp 535, 541), and sub-centres and PHCs are being progressively upgraded to deliver the 12-service comprehensive primary health care package; the medical officer's job description was correspondingly redefined after the National Health Mission, National Health Policy 2017, Ayushman Bharat and Indian Public Health Standards (IPHS) 2022 (Sunder Lal, Ch 16, p 715).

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoHFW Ayushman Arogya Mandir portal, Indian Public Health Standards 2022, ICMR Standard Treatment Workflows, WHO Astana Declaration 2018

Programme and policy updates (India)

- ▶ **Health and Wellness Centres renamed Ayushman Arogya Mandir (AAM), with a redefined service package** — more than 1.75 lakh AAM operationalised nationally by upgrading existing sub-centres and PHCs; the old sub-centre/PHC menu limited to reproductive-child health and communicable disease has been replaced by **12 expanded service packages**, of which 82% of AAM were delivering all twelve as of 30 November 2024. Free medicines and diagnostics are notified level-wise — 106 drugs and 14 tests at AAM sub-health centre, 172 drugs and 63 tests at AAM-PHC/urban PHC (states may modify locally) [PMID 41816146].
- ▶ **A new mid-level cadre now leads the peripheral team, changing what the medical officer supervises** — the **Community Health Officer (CHO)/mid-level health provider** heads the AAM sub-health centre with clinical, public health and managerial roles of his or her own; **1,16,978 CHOs** were in position as of 30 November 2024 [PMID 41816146]. CHOs and ASHAs now screen the population with the **community-based assessment checklist (CBAC)** for communicable, non-communicable and mental disorders, with CHO-led follow-up linking diagnosis to continued treatment — so the PHC medical officer's role shifts further toward supervision, clinical referral anchoring and quality assurance [PMID 40534365].
- ▶ **AYUSH formally mainstreamed into primary care** — over **12,500 AYUSH Ayushman Arogya Mandirs** established under the National AYUSH Mission, and AYUSH services co-located in **7,011 PHCs, 3,051 CHCs and 460 district hospitals** (position as of 2024), besides more than 3,000 further National Health Mission facilities; AYUSH is also written into the updated Indian Public Health Standards, so cross-referral pathways and co-location supervision now sit within the medical officer's intersectoral coordination duties [PMID 41229518].

Current national data

- ▶ **Universal health coverage and financial protection** — India's UHC Service Coverage Index rose from **57 (2015) to 63 (2021)**, above the South-East Asia regional average of 62, and out-of-pocket expenditure fell from **60.6% (2014-15) to 39.4% (2021-22)** on National Health Accounts estimates — the headline numbers to quote when arguing that comprehensive primary health care advances Health for All [PMID 41816146].

- **Quality of care is weakest at the primary level, not the district level** — a six-state multicentric assessment using NP-NCD and IPHS checklists found district hospitals achieved the highest uninterrupted-treatment rate for hypertension (71.2%) and blood-pressure control in 45.3%, while urban health and wellness centres performed worst; facility type was significantly associated with clinical control ($p < 0.001$). The same analysis cites NFHS-5, where only 13% of hypertensive women and 9% of men had controlled blood pressure [PMID 42344085].
- **Immunisation coverage still has a quarter-sized gap** — meta-analysis of 45 Indian cross-sectional studies (>428,000 children, 2015-2025) found pooled **incomplete immunisation of 26% (95% CI 21-32%)** in under-fives despite the Universal Immunization Programme and Mission Indradhanush; the strongest determinant was absence of an immunisation card (OR 9.46, 95% CI 3.28-27.28), followed by non-institutional delivery (OR 3.59) and maternal illiteracy (OR 3.25) — all actionable by the PHC medical officer's UIP and outreach duties [PMID 42320383].

Elimination, surveillance and disease status

- **Quote the current elimination targets, not the superseded ones** — the **100-Day TB Elimination Campaign** (7 December 2024 to 17 March 2025) was launched to accelerate the national TB elimination goal; a difference-in-differences analysis in Maharashtra (53 campaign versus 27 no-campaign districts, NIKSHAY data) found the **relative change in notification rate was 10% higher in campaign districts (DID IRR 1.10, 95% CI 1.03-1.17, $p = 0.003$)** — read the direction carefully: notifications *fell* in both arms (10% decline where there was no campaign, 1% decline where there was), so the campaign blunted a decline rather than raising case-finding. The differential gain was confined to rural areas (DID IRR 1.21, 95% CI 1.11-1.32) and the public sector (DID IRR 1.17, 95% CI 1.06-1.29), with no significant effect in urban areas or private care [PMID 42457294]. **Lymphatic filariasis remains endemic in 345 districts** against a declared **2030** elimination goal, with mass drug administration, morbidity management and post-MDA surveillance still straining public health infrastructure [PMID 41951149].
- **Surveillance has gone digital but the laboratory backbone lags** — the Integrated Disease Surveillance Programme moved in **2018** from aggregate paper reporting to real-time case-based reporting on the **Integrated Health Information Platform (IHIP-IDSP)**; a qualitative dengue-surveillance study in western India nevertheless found that in the study state only **13% of sanctioned IDSP district public health laboratories were functional** — district public health laboratories worked in just three districts, and four of that state's 36 districts had no sentinel-hospital coverage. The same study documented parallel IHIP-IDSP and NVBDCP reporting streams and outbreak alerts generated only after formal confirmation by the District Surveillance Officer — the practical constraints on the medical officer's outbreak-detection duty [PMID 42245358].

Guideline and legislative updates

- **ICMR Standard Treatment Workflows (STWs) as the point-of-care guideline layer** — the compendium now covers **157 clinical conditions**; of 29 distinct antibiotics recommended, **41% fall in the WHO Access group, 55% Watch and about 4% Reserve, and 91% appear in the National List of Essential Medicines**, with availability mapped against **Indian Public Health Standards** by level of care. This gives the PHC medical officer a stewardship-aligned prescribing standard consistent with the essential-drugs element of Alma-Ata [PMID 42100260].

Global and WHO developments

- ▶ WHO's operational definition of primary health care has three components, not four principles — the Alma-Ata Declaration (1978) was renewed by the Astana Declaration (2018), and WHO now frames primary health care as a whole-of-society approach built on (i) primary care and essential public health functions as the backbone of integrated services, (ii) multisectoral policy and action, and (iii) empowerment of people and communities. Analysis of primary-health-care-oriented reforms across **nine countries** under WHO's PHC Implementation Solutions Initiative identified ten recurrent health-system shifts and showed such reforms are inherently redistributive — they reallocate resources and decision-making power, so political commitment, community engagement and durable institutional change determine whether they survive [PMID 41955203].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 1, Ch 2, Ch 7, Ch 10 — Health Care of the Community, Ch 13); Sunder Lal — Textbook of Community Medicine (Ch 16 — Health Planning and Management, Ch 17 — Healthcare of the Community); Suryakantha — Community Medicine with Recent Advances (figure, p 780); Declaration of Alma-Ata 1978; Indian Public Health Standards 2022.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, doi:[10.4103/jfmprc.jfmprc.478.25](https://doi.org/10.4103/jfmprc.jfmprc.478.25); 41229518 — India's journey in mainstreaming Ayush in primary health care: from tradition to integration, *Frontiers in Medicine* 2025, doi:[10.3389/fmed.2025.1629515](https://doi.org/10.3389/fmed.2025.1629515); 40534365 — Towards mental health equity: task shifting strategy to overcome barriers in primary healthcare services in India, *J Prev Med Public Health* 2025, doi:[10.3961/jpmpmh.25.144](https://doi.org/10.3961/jpmpmh.25.144); 42344085 — Assessment of hypertension and diabetes care in public health facilities across India: a multicentric cross-sectional study, *Indian J Community Med* 2026, doi:[10.4103/ijcm.ijcm.315.25](https://doi.org/10.4103/ijcm.ijcm.315.25); 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: a systematic review and meta-analysis (2015-2025), *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only); 42457294 — Effect of the '100 days TB elimination campaign' on TB notification in Maharashtra, *Indian J Tuberc* 2025, doi:[10.1016/j.ijtb.2025.08.014](https://doi.org/10.1016/j.ijtb.2025.08.014) (abstract-only); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42245358 — Informational challenges of dengue surveillance and response: a case study from India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1809584](https://doi.org/10.3389/fpubh.2026.1809584); 42100260 — Standard Treatment Workflows: scaling system-compatible approaches to rational antibiotic use, *Frontiers in Medicine* 2026, doi:[10.3389/fmed.2026.1789243](https://doi.org/10.3389/fmed.2026.1789243); 41955203 — Primary health care-oriented reforms: the political economy dynamics underlying health-system shifts across nine countries, *Health Policy and Planning* 2026, doi:[10.1093/heapol/czag042](https://doi.org/10.1093/heapol/czag042). Programme figures from the MoHFW Ayushman Arogya Mandir portal (position as of 30 November 2024); WHO Declaration of Astana 2018; Indian Public Health Standards 2022; ICMR Standard Treatment Workflows.



National Health Policy 2017

Asked in: Paper IV 13-May-2022 — RGUHS MD Community Medicine

Definition — The National Health Policy (NHP) 2017 is India's third national health policy, succeeding those of 1983 and 2002, framed in response to a disease burden shifting towards non-communicable disease, a rapidly growing private health-care industry, rising catastrophic health expenditure, and the adoption of the Sustainable Development Goals (Sunder Lal, Ch 16, p 697). Its primary aim is to inform, clarify, strengthen and prioritize the role of Government in shaping health systems in all dimensions — investment, organization of services, disease prevention and health promotion through cross-sectoral action, access to technology, human resources, medical pluralism, financial protection, and regulation (Park 28e, Ch 23, p 1030).

Goal and broad objectives

Goal (Sunder Lal, Ch 16, p 697)

- ▶ Attainment of the **highest possible level of "health and well-being for all at all ages"**, and achievement of the United Nations Sustainable Development Goals by 2030.
- ▶ **Progressively achieve universal health coverage (UHC)** — that is, achieve it in a graded, resource-linked manner rather than by a single declaration.

Broad objectives (Sunder Lal, Ch 16, p 697)

- ▶ Assuring **free comprehensive primary health-care services** covering reproductive, maternal, child and adolescent health, and the most prevalent communicable, non-communicable and occupational diseases.
- ▶ Improved access to **affordable, quality secondary and tertiary health care** through a combination of public and private providers.
- ▶ Achieve a **reduction in out-of-pocket and catastrophic health expenditure**.
- ▶ **Reinforce trust** in the public health-care system.
- ▶ **Align the growth of the private care sector with public health goals**.

The policy converts these into indicative, measurable targets under **three components** — (A) **health status and programme impact**, (B) **health systems performance**, and (C) **health system strengthening** — aligned to the Sustainable Development Goals (Park 28e, Ch 23, p 1030).

A. Specific goals — health status and programme impact

Domain	Specific quantitative goal	Target year
Life expectancy	Increase life expectancy at birth from 67.5 to 70 years	2025
Burden of disease	Establish regular tracking of the disability-adjusted life year (DALY) index by major disease category	2022
Fertility	Reduce total fertility rate (TFR) to 2.1 at national and sub-national level	2025
Child mortality	Reduce under-five mortality rate to 23 per 1,000 live births	2025

Domain	Specific quantitative goal	Target year
Infant mortality	Reduce infant mortality rate (IMR) to 28 per 1,000 live births	2019
Newborn/stillbirth	Reduce neonatal mortality rate to 16 and stillbirth rate to "single digit"	2025
Maternal mortality	Reduce maternal mortality ratio (MMR) to 100 per 100,000 live births	2020
HIV/AIDS	Achieve the global 90:90:90 target — 90% of people living with human immunodeficiency virus (HIV) know their status, 90% of those diagnosed on sustained antiretroviral therapy, 90% of those on therapy virally suppressed	2020
Leprosy, kala-azar, filariasis	Achieve and maintain elimination status — leprosy by 2018, kala-azar and lymphatic filariasis by 2017	2017–18
Tuberculosis	Achieve and maintain cure rate >85% in new sputum-positive patients , reduce incidence and reach elimination	2025
Blindness	Reduce prevalence of blindness to 0.25 per 1,000 and reduce disease burden by one-third of current level	2025
Non-communicable disease	Reduce premature mortality from cardiovascular disease, cancer, diabetes and chronic respiratory disease by 25%	2025

(Park 28e, Ch 23, p 1030; Park 28e, Ch 9, p 657; Sunder Lal, Ch 16, pp 697–698)

B. Specific goals — health systems performance

Sub-domain	Specific quantitative goal	Target year
Service utilization	Increase utilization of public health facilities by 50% from the current level of 30%	2025
Antenatal care	Sustain antenatal care coverage above 90%	2025
Delivery care	Skilled attendance at birth above 90%	2025
Immunization	>90% of newborns fully immunized by one year of age	2025
Family planning	Meet need of family planning above 90% at national and sub-national level	2025
Non-communicable disease control	80% of known hypertensive and diabetic individuals maintain "controlled disease status" at household level	2025
Tobacco (cross-sectoral)	Relative reduction in current tobacco use by 15% by 2020 and 30% by 2025	2020, 2025
Nutrition (cross-sectoral)	40% relative reduction in prevalence of stunting in under-five children	2025
Water and sanitation (cross-sectoral)	Access to safe water and sanitation to all (<i>Swachh Bharat Mission</i>)	2020

Sub-domain	Specific quantitative goal	Target year
Occupational health (cross-sectoral)	Reduce occupational injury by half from the current 334 per lakh agricultural workers	2020
Health behaviour (cross-sectoral)	National and state-level tracking of selected health behaviour	—

(Park 28e, Ch 23, p 1030; Sunder Lal, Ch 16, p 698)

C. Specific goals — health system strengthening

Sub-domain	Specific quantitative goal	Target year
Health finance	Increase government health expenditure from 1.15% to 2.5% of gross domestic product (GDP)	2025
Health finance	Increase state sector health spending to >8% of the state budget	2020
Financial protection	Decrease the proportion of households facing catastrophic health expenditure by 25% from current levels	2025
Human resource	Ensure doctors and paramedics as per Indian Public Health Standards (IPHS) norms in high-priority districts	2020
Community workforce	Increase community health volunteer-to-population ratio to IPHS norm in high-priority districts	2025
Infrastructure	Establish primary and secondary care facilities as per IPHS norms in high-priority districts	2025
Health information	Ensure a district-level electronic database and strengthen the surveillance system	—

(Park 28e, Ch 23, p 1030; Sunder Lal, Ch 16, p 698)

Strategies — policy thrust areas and the seven key policy shifts

1. **Raising public health expenditure** to 2.5% of GDP in a time-bound manner — the single most important financing commitment of the policy (Sunder Lal, Ch 16, p 698).

2. **Preventive and promotive health** — the policy adopts "Health in All" as a complement to "Health for All", addressing the social determinants of health through intersectoral coordination using a *whole-of-government* and *whole-of-society* approach, and names **seven priority areas** (Sunder Lal, Ch 16, p 698):

- ▶ *Swachh Bharat Abhiyan* — reduction of water-borne, excreta-related and vector-borne disease.
- ▶ Balanced healthy diets and regular exercise.
- ▶ Addressing tobacco, alcohol and substance abuse.
- ▶ *Yatri Suraksha* — preventing deaths from rail and road traffic accidents.
- ▶ *Nirbhay Nari* — action against gender violence.
- ▶ Reduced stress and improved safety at the workplace.
- ▶ Reduction of indoor and outdoor air pollution.

3. Seven key policy shifts in the organization of public health-care delivery (Sunder Lal, Ch 16, pp 698–699):

#	Area	Shift from	Shift to
i	Primary health care	Selective care	Assured comprehensive primary health care , with linkages to referral hospitals for continuity of care
ii	Secondary and tertiary care	Input-oriented, line-item budget financing	Output-based strategic purchasing from public and private hospitals
iii	Public hospitals	User fees and cost recovery	Assured free drugs, diagnostics and emergency services to all
iv	Infrastructure and human resources	Normative approach	Targeted approach to reach under-served areas
v	Urban health	Token interventions	On-scale assured interventions — primary care delivery and referral support for the urban poor
vi	National health programmes	Vertical stand-alone delivery	Integration with health systems for programme effectiveness and system efficiency
vii	AYUSH services	Stand-alone services	Three-dimensional mainstreaming — co-location of AYUSH, standard medicines, and yoga

4. **Strategy of care** — free primary health care provided directly by the **public sector**, supplemented by **strategic purchase** of secondary-care hospitalization and tertiary care from both public and non-government providers to fill critical gaps. This is the declared **main financing strategy**, with government acting as a **single payer** for district health system development (Sunder Lal, Ch 16, pp 699, 729).

Strategies — spectrum of services by level of care

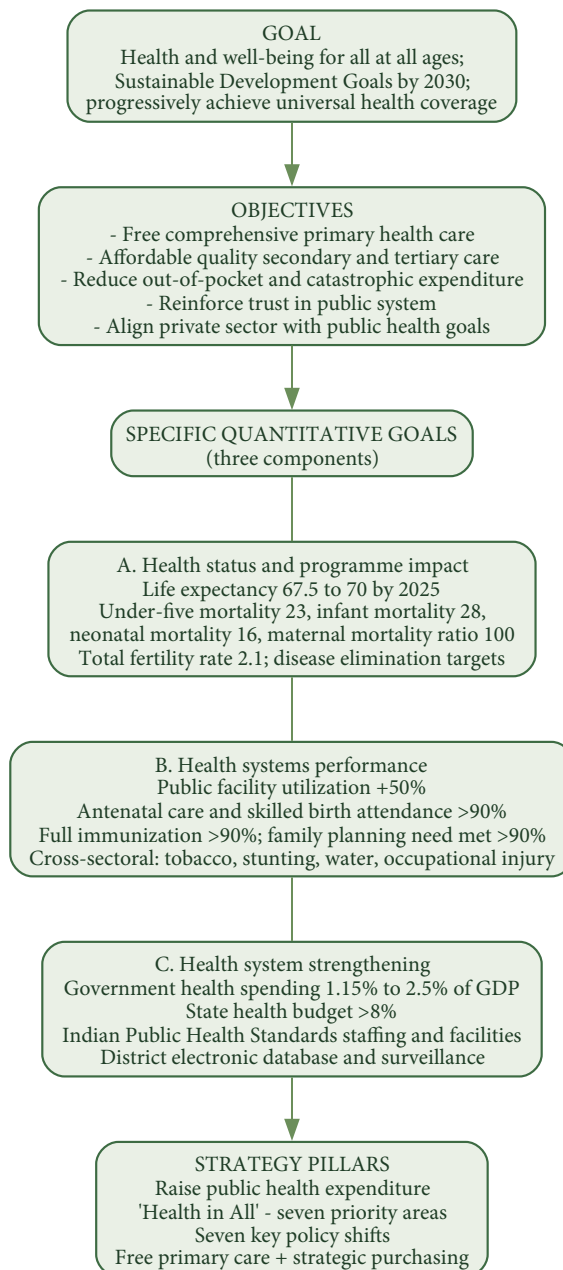
Beyond the level-wise package and its financing (second flowchart), the strategy specifies (Sunder Lal, Ch 16, pp 699–700):

- ▶ **Primary care** — special focus on the **tribal population, urban health and control of zoonotic diseases** alongside the Health and Wellness Centre package.
- ▶ **Reorienting public hospitals** — universal access to a progressively wider array of free drugs, diagnostics and emergency services.
- ▶ **Closing infrastructure and skill gaps** as per IPHS/national norms, following the 12th Five-Year Plan roadmap.
- ▶ **Urban health care** — urban health centres and urban community health centres for the urban poor and slums, with primacy to local bodies.
- ▶ **Human resources** — medical education integrated with service delivery, district hospitals converted into medical colleges, curricula revised, a cadre of **mid-level service providers** developed (including bridge courses for AYUSH graduates), and a **public health management cadre** created in every state.

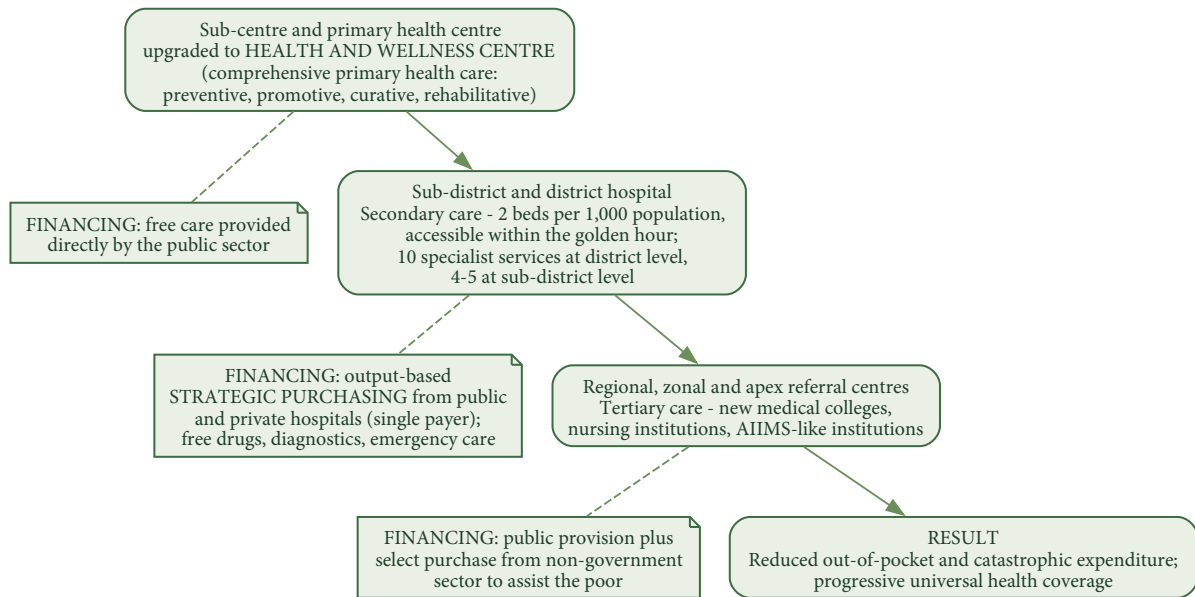
National health programmes and cross-cutting strategies (Sunder Lal, Ch 16, p 699)

- ▶ **Integration of national health programmes** — reproductive, maternal, newborn, child and adolescent health (RMNCH+A), communicable and non-communicable disease, mental health, trauma care and occupational health — with the general health system.
- ▶ **Universal immunization** as per the National Vaccine Policy 2011.
- ▶ **Integrated Disease Surveillance Programme** — networked laboratories backed by tertiary centres, with capacity to collect, analyse and respond to outbreaks.
- ▶ **Population stabilization** — improved static family planning services and **male participation in sterilization** raised from under 1% to 30%.
- ▶ **Women's health and gender mainstreaming** — care for reproductive morbidity and for women beyond the reproductive age (40+ years), gender-sensitive staff orientation, and free, dignified services for victims of gender violence.
- ▶ **Supportive supervision** in the more vulnerable districts.
- ▶ **Emergency care and disaster preparedness** — life-support ambulances and trauma centres, **one per 30 lakh urban and one per 10 lakh rural population.**

Diagrams



Architecture of the National Health Policy 2017 — goal, objectives, the three goal components and the strategy pillars



Service-delivery strategy of the National Health Policy 2017 — level of care and its mode of financing

High-yield exam pointers

- **Goal in one line:** "Health and well-being for all at all ages" + Sustainable Development Goals by 2030 + progressively achieve universal health coverage.
- **Mortality set:** under-five 23 (2025), infant mortality rate 28 (2019), neonatal 16 (2025), stillbirth single digit (2025), maternal mortality ratio 100 (2020), total fertility rate 2.1 (2025), life expectancy 67.5 → 70 (2025).
- **Financing numbers:** government health expenditure 1.15% → 2.5% of GDP by 2025; state health spending >8% of budget by 2020; catastrophic expenditure down 25% by 2025.
- **Three goal components:** health status and programme impact; health systems performance; health system strengthening.
- **Seven priority areas mnemonic — "SBD-TYN-WA":** Swachh Bharat, Balanced diet and exercise, Drug/tobacco/alcohol abuse, Yatri Suraksha, Nirbhay Nari, Workplace safety, Air pollution.
- **Signature phrases to quote:** "Health in All" complementing "Health for All"; *whole-of-government and whole-of-society approach*; **strategic purchasing** with government as **single payer**; **Health and Wellness Centre**; **golden hour** with 2 beds per 1,000.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Textbooks differ on three figures — Park 28e gives baseline government health expenditure as 1.15% of gross domestic product, the blindness target as 0.25 per 1,000 and the human immunodeficiency virus target as 90:90:90 (all matching the policy document), whereas Sunder Lal quotes 1.35%, 0.3% and the updated global 95:95:95 target; write the Park figures. The comprehensive-primary-care commitment has since been operationalised by converting sub-centres and primary health centres into Health and Wellness Centres, around which the revised Indian Public Health Standards 2022 are built.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, National Health Accounts estimates, Sample Registration System 2023, NFHS-5, National Sample Survey, WHO

Programme and policy updates (India)

- ▶ **Health and Wellness Centres now carry the name *Ayushman Bharat Arogya Mandir***, and Pradhan Mantri Jan Arogya Yojana (PM-JAY) is designed to align with them for primary care, diagnostics and essential medicines. In the Comprehensive Annual Modular Survey 2022–23 (302,086 households), publicly funded insurance was associated with higher outpatient use (adjusted odds ratio 1.404, 95% confidence interval 1.328–1.484), higher inpatient use (1.168, 1.128–1.210) and lower catastrophic inpatient expenditure at the 20% threshold (0.790, 0.737–0.846), while outpatient care remained the unclosed gap. Use the new name in the answer, and note that **strategic purchasing has delivered on hospitalization, not on outpatient care**. [PMID 42167946]
- ▶ **The insurance route to universal health coverage is now contested in the literature**. A study of three National Sample Survey rounds (60th 2004, 71st 2014, 75th 2017–18), with instrumental-variable regression on the 75th round to correct for self-selection into enrolment, found no statistically significant relationship between publicly funded insurance enrolment and out-of-pocket expenditure per hospitalization, and no significant difference in catastrophic expenditure between covered and uninsured households — a direct challenge to the policy's single-payer purchasing strategy. [PMID 41913189]

Current national data

- ▶ **Financing scoreboard against goal component C**. Between 2013–14 and 2021–22 the government share of total health expenditure rose from 28.6% to 48%, and the out-of-pocket share fell from 64.2% to 39.4%. However, **government health expenditure stands at about 1.9% of gross domestic product — short of the policy's 2.5%-by-2025 target, which has therefore been missed**. Catastrophic spending is in fact commoner for outpatient care than for hospitalization (47.8% against 43.1% of households), medicines alone account for roughly two-thirds of out-of-pocket spending, and health costs push about 55 million people into poverty each year — so the money must be aimed at free medicines and diagnostics, not only at insured admissions. [PMID 42548603]
- ▶ **Sample Registration System 2023 — infant mortality rate 25 per 1,000 live births**, down from 40 in 2013; pneumonia is the second leading cause of infant death, at 16.2%. The policy's target of

28 by 2019 has been surpassed; quote the 2023 figure when asked about progress rather than repeating the target. [PMID 42056953]

- ▶ **A district-level universal health coverage index** built from NFHS-5 (2019–21) and the 75th National Sample Survey round scored 687 districts on five tracer domains: median 43.9% (range 26.4–69.4), falling to 36.9% in the poorest wealth quintile, with about one in ten people facing catastrophic or impoverishing payments. The weakest domains were **non-communicable disease (median 9.8%) and service capacity/access (54.9%)** — precisely where the policy set its 80% hypertension/diabetes control goal and its Indian Public Health Standards staffing goal. [PMID 39219770]
- ▶ **Immunization remains short of the >90% goal.** A meta-analysis of 45 cross-sectional studies (over 428,000 participants, 2015–2025) put incomplete immunization among under-fives at 26% (95% confidence interval 21–32%); the strongest determinants were absence of an immunization card (odds ratio 9.46), non-institutional delivery (3.59) and maternal illiteracy (3.25). [PMID 42320383]

Elimination, surveillance and disease status

- ▶ **Tuberculosis — the 2025 elimination deadline drove a 100-Day TB Elimination Campaign** (7 December 2024 to 17 March 2025) built on intensified case-finding, early diagnosis, targeted screening and nutritional and financial support. TB notification fell by only 1% in the 53 campaign districts of Maharashtra versus a 10% fall in the 27 non-campaign districts, i.e. a 10% relative gain (difference-in-differences incidence rate ratio 1.10, 95% confidence interval 1.03–1.17, $p = 0.003$); **the campaign stabilised notifications rather than increasing them**, with the effect confined to rural areas (IRR 1.21) and the public sector (IRR 1.17) and absent in urban and private care. [PMID 42457294]
- ▶ **Lymphatic filariasis — the deadline has slipped and the textbook date is superseded.** The policy carried elimination by 2017; the disease is still endemic in **345 districts** and the declared national goal now stands at **2030**, sustained by mass drug administration, morbidity management and post-campaign surveillance, with public–private partnership proposed to close the operational gaps. Write 2030 as the current target and flag 2017 as the original policy date. [PMID 41951149]

Guideline and legislative updates

- ▶ **The "Health in All" nutrition priority now carries statutory backing.** Food Safety and Standards Authority of India regulations (2020) restrict high-fat, salt and sugar foods in and around schools; front-of-pack nutrition labelling is under development; and the Ayushman Bharat School Health Programme and PM POSHAN meal scheme carry the school component. Some 9–12% of Indian children and adolescents are now overweight or obese — a rising double burden set against the policy's 40% stunting-reduction goal. [PMID 41974918]

Global and WHO developments

- ▶ **Global progress towards eliminating viral hepatitis by 2030 is insufficient, and India carries the second-largest share of it.** WHO's 2022 assessment placed **35.3 million Indians** with chronic hepatitis B or C — 11.6% of the world total, behind only China — and worldwide hepatitis deaths (over 1.3 million) now exceed those from tuberculosis, making hepatitis a comparable public health threat. [PMID 41895313]

SOURCE & COVERAGE

- ▶ Park's Textbook of Preventive and Social Medicine, 28e — Ch 23 (Health Planning and Management), p 1030; Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), p 657.
- ▶ Sunder Lal — Textbook of Community Medicine — Ch 16 (Health Planning and Management), pp 697–700, 727–729; Ch 17 (Healthcare of the Community), p 744 (Indian Public Health Standards 2022).
- ▶ National Health Policy 2017, Ministry of Health and Family Welfare, Government of India (as reproduced in the above texts).

Recent-advances sources:

- ▶ PMID 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), *BMJ Open* 2026;16(5):e106382, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382).
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Primary health centre — IPHS and functions

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — Indian Public Health Standards (IPHS) are the uniform, benchmarked norms issued by the Directorate General of Health Services, Ministry of Health and Family Welfare, for every level of government health facility, stating what services must be delivered and what infrastructure, human resources, drugs, diagnostics and equipment are needed to deliver them. Each standard is graded as **essential** (the minimum acceptable functional standard, to be assured) or **desirable** (the achievable level states should aspire to). They were first issued in 2006 after the National Rural Health Mission, revised in 2012 and revised again in **2022** (Sunder Lal, Ch 17, pp 737, 744).

1. Objectives of IPHS and how they apply to a Primary Health Centre

Objective (IPHS)	What it fixes at the Primary Health Centre (PHC)
Define a uniform standard for high-quality services	One national benchmark, so a PHC in a hilly block is judged by the same yardstick as one in a plain block
Specify minimum assured (essential) and achievable (desirable) services	The 12-package comprehensive primary health care (CPHC) service list
Specify adequate workspace, human resources, drugs, diagnostics, equipment, administrative and logistic support	Staffing table, bed and oxygen norms, essential drug list, 50-plus diagnostics
Achieve and maintain acceptable quality of care	Standard treatment protocols; National Quality Assurance Standards (NQAS), Kayakalp and LaQshya certification
Facilitate monitoring and supervision	Facility survey, Health Management Information System reporting, National Identification Number on the Ayushman Bharat portal
Serve as a tool for governance, leadership and evaluation	Rogi Kalyan Samiti, Citizen's Charter, grievance redressal

(Sunder Lal, Ch 17, p 744; Park 28e, Ch 10, pp 728-729)

IPHS 2022 aligns the PHC with the key policy shifts of the **National Health Policy 2017** — curative to comprehensive care, selective to assured primary care, user fees to free drugs/diagnostics/emergency services, normative to "**time-to-care**" **targeted** infrastructure planning, token to on-scale urban health, and integration of national programmes with the health system — so as to advance Universal Health Coverage and Sustainable Development Goal 3 (Park 28e, Ch 10, p 728).

2. Classification and population norms of the PHC under IPHS 2022

All PHCs are being converted to **Health and Wellness Centres (HWCs)**, since renamed Ayushman Arogya Mandir (Park 28e, Ch 10, p 728).

Type	Population norm	Functional expectation
HWC-PHC (rural)	30,000 plain; 20,000 hilly, tribal, desert and difficult areas	States should aspire to make all PHCs 24 × 7; PHCs conducting deliveries prioritised for 24 × 7 status
HWC-UPHC (urban)	50,000, sited close to slums	Routine outpatient plus preventive-promotive care; those with indoor beds already conducting deliveries continue as 24 × 7
Multispeciality UPHC / Polyclinic (urban)	2.5–3 lakh (catchment of 5–6 urban PHCs)	Ambulatory specialist outpatient care only

(Sunder Lal, Ch 17, pp 748-749; Park 28e, Ch 10, p 731)

Norms are no longer rigid: siting must follow the "time to care" approach — accessible within 30 minutes of travel. One PHC is the referral unit for 6 sub-health centres and provides diagnostics, initiation of treatment, supervision, and administrative and financial control over them (Sunder Lal, Ch 17, p 748).

3. Infrastructure, beds and oxygen (IPHS 2022)

Facility	Beds (E = essential, D = desirable)	B-type oxygen cylinder (1500 L)	Oxygen concentrator (10 L)
PHC (rural)	6 (2 E + 4 D)	4	1
PHC (urban) and polyclinic	6 day-care (2 E + 4 D)	4	1
PHC / urban PHC, 24 × 7	10 (6 E + 4 D)	5	1

(Sunder Lal, Ch 17, p 749, Table 17.6)

Every bed must have an intravenous stand, bedside locker and attendant's stool; Ambu bag and oxygen must be functional and accessible. The PHC has an operation theatre complex, labour room and laboratory (Sunder Lal, Ch 17, pp 748-749).

4. Human resources and minimum performance standards

Human resource	Rural PHC (E / D)	24 × 7 PHC (E / D)
Medical Officer, MBBS	1 / 1	2 / 1
Medical Officer, AYUSH	– / 1	– / 1
Dentist	– / 1	– / 1
Staff nurse	2 / 1	7 / –
Pharmacist-cum-store keeper	1 / –	1 / 1
Laboratory technician / technologist	1 / –	2 / 1

Human resource	Rural PHC (E / D)	24 × 7 PHC (E / D)
Auxiliary nurse midwife (ANM, female health worker)	1 / –	1 / –
Male health worker / health assistant	1 / –	1 / –
Lady health visitor (female health assistant)	1 / –	1 / –
Cold chain / vaccine logistics assistant	– / 1	1 / –
Dresser; accountant (LDC-1); data entry operator	1 each / –	1 each / –
Cleaning staff	1 / –	4 / –

(Sunder Lal, Ch 17, p 749, Table 17.7). Desirable numbers are over and above the essential. The polyclinic additionally has 6 core specialists — medicine, obstetrics and gynaecology, paediatrics, ophthalmology, dermatology, psychiatry — on rotation for outpatient services only.

Minimum performance standards: Medical Officer 75 outpatients/day; dentist 20 outpatients/day; laboratory technician 100 tests/day; pharmacist 120 prescriptions dispensed/day; staff nurse 1 per 6 beds as per Indian Nursing Council norms. Outdoor services run 4 hours in the morning and 2 hours in the afternoon/evening, and the Medical Officer should spend at least 2 hours a day, twice a week, on field duty and monitoring (Sunder Lal, Ch 17, pp 749-750).

5. Assured service package

The **twelve comprehensive primary health care packages** at HWC-PHC are: (1) care in pregnancy and childbirth; (2) neonatal and infant health care; (3) childhood and adolescent health care; (4) family planning, contraception and other reproductive health care; (5) management of communicable diseases under national health programmes; (6) general outpatient care for acute simple illness and minor ailments; (7) screening, prevention, control and management of non-communicable diseases; (8) care for common ophthalmic and ear-nose-throat problems; (9) basic oral health care; (10) elderly and palliative care; (11) emergency medical services including burns and trauma; (12) screening and basic management of mental health ailments (Sunder Lal, Ch 17, p 749; Park 28e, Ch 10, p 731).

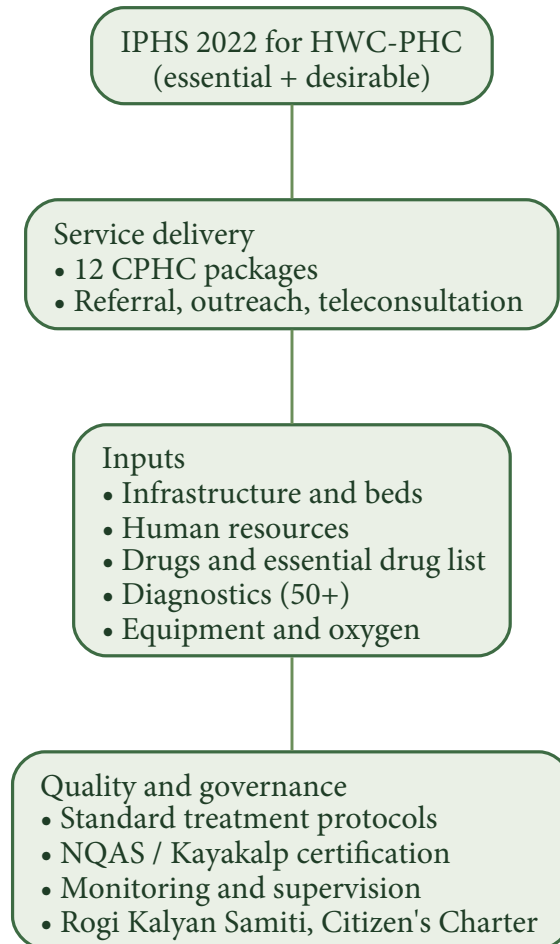
Alongside these, IPHS requires free essential drugs as per the essential drug list and **over 50 diagnostic services**, two-way referral with upward (CHC) and downward (outreach) linkages, outreach through Village and Urban Health and Nutrition Days, teleconsultation through **eSanjeevani**, and intersectoral collaboration with Women and Child Development, Education and local bodies. **Desirable services** at PHC include safe abortion by manual vacuum aspiration, school counselling, integrated counselling and testing centre services, issue of disability certificates, and blood cholesterol and electrocardiography (Sunder Lal, Ch 17, pp 749-750).

6. Challenges in achieving the standards in the state

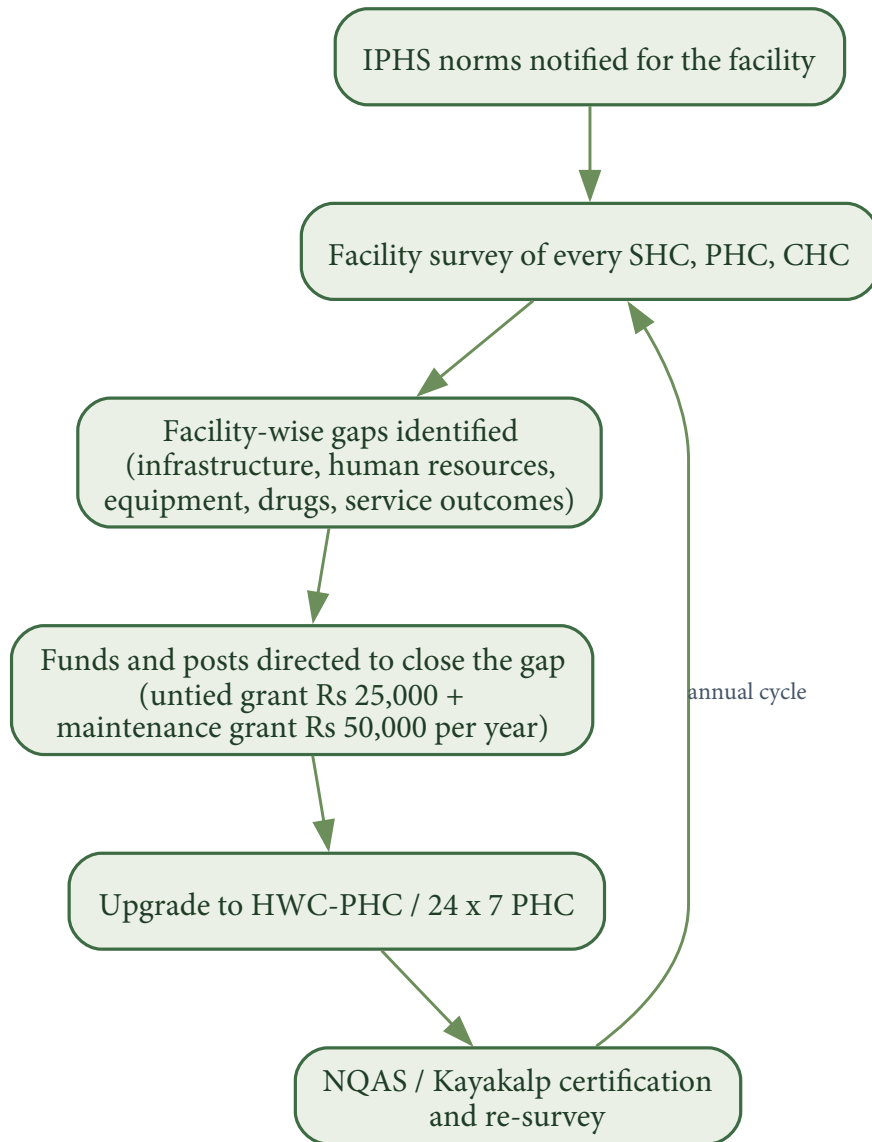
Domain	Challenge	Evidence base
Coverage and siting	Facilities distributed by rigid population norms rather than travel time; no functional PHC in many remote areas in dire need of care; in hilly/tribal blocks of deficient states the shortfall of PHCs reaches 40% of the required number	Sunder Lal, Ch 17, p 750; Park 28e, Ch 13, pp 850-851
Human resources	Single-doctor PHCs, unfilled specialist posts at the referral CHC (~80% shortfall of specialists nationally, so first referral units do not function), professional isolation, weak human-resource policy, poor working and living conditions, absenteeism	Sunder Lal, Ch 17, pp 753, 766; Park 28e, Ch 13, pp 850-851
Infrastructure	Majority of PHCs lack essential infrastructure and inputs; 94.5% run in government buildings, the rest in rented/unsuitable premises; labour room, operation theatre and oxygen support incomplete	Sunder Lal, Ch 17, p 750
Drugs and diagnostics	Essential drugs for common ailments unavailable in the majority of PHCs; stock-outs; the 50-plus diagnostics list not fully operational	Sunder Lal, Ch 17, pp 750, 766
Service output	Only about one-third of PHCs conduct deliveries , averaging 26 deliveries in three months — PHCs function suboptimally; services still largely limited to reproductive-child health and a few disease programmes, which is under 15% of what people seek care for	Sunder Lal, Ch 17, pp 749-750
Utilisation and trust	Over 70% of people use the private sector for outpatient care; only 45.7% of rural and 35% of urban hospitalisations occur in public hospitals (National Sample Survey, 1996–2018), driven by long waits, inconvenient hours, poor outreach and insensitivity to local needs	Sunder Lal, Ch 17, p 766
Financing	Out-of-pocket spending is 53.23% of current health expenditure; state funds only 19.63%; the Fifteenth Finance Commission target of 8% of the state budget on health, with 67% of health spending on primary care, is not met	Sunder Lal, Ch 16, pp 727-729
Governance and management	Weak planning, monitoring and supervision; untied and maintenance grants under-utilised; Rogi Kalyan Samitis and Panchayati Raj devolution largely nominal; slow conversion of PHCs to HWCs (16,635 of 24,935 rural PHCs converted as on 31 March 2021)	Sunder Lal, Ch 17, pp 750-751, 766
Quality certification	Few PHCs certified under NQAS/Kayakalp/LaQshya; standard treatment protocols and Citizen's Charter displayed but not audited	Park 28e, Ch 10, p 729

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram to label by hand — the two-way referral position of the HWC-PHC.** Draw a vertical ladder and label: *Community / ASHA and Anganwadi worker → HWC-Sub Health Centre (5,000 plain, 3,000 tribal; community health officer) → HWC-PHC (30,000 plain, 20,000 hilly; 6 sub-centres attached; Medical Officer) → Community Health Centre / First Referral Unit (specialists) → Sub-district and District Hospital.* Mark upward arrows "referral" and downward arrows "follow-up, supervision, supplies, diagnostics".



The eight standard-setting domains of IPHS 2022 for a Health and Wellness Centre - Primary Health Centre



How IPHS is operationalised - the facility survey and gap-closing cycle

High-yield exam pointers

- ▶ IPHS dates: 2006 → 2012 → 2022; every standard is either **essential** or **desirable**.
- ▶ Population norms: rural PHC 30,000 / 20,000 (hilly-tribal-desert), urban PHC 50,000, polyclinic 2.5–3 lakh; siting by 30-minute time-to-care; 1 PHC : 6 sub-centres.
- ▶ Beds: PHC 6 (2 E + 4 D); 24 × 7 PHC 10 (6 E + 4 D).
- ▶ Performance standards: Medical Officer 75 outpatients/day, dentist 20/day, laboratory technician 100 tests/day, pharmacist 120 dispensings/day, staff nurse 1 : 6 beds.
- ▶ The 12 CPHC packages — pregnancy/childbirth, neonatal-infant, child-adolescent, family planning, communicable disease programmes, general outpatient care, non-communicable disease, eye and ENT, oral health, elderly and palliative, emergency/burns/trauma, mental health.
- ▶ Challenge figures worth quoting with their year: 31,053 PHCs (24,935 rural + 6,118 urban) as on 31 March 2021; 94.5% in government buildings; ~1/3 conduct deliveries; ~80% specialist shortfall at CHC; out-of-pocket spend 53.23% of current health expenditure.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The question asks for challenges "in your state"; the indexed corpus carries only national and multi-state figures, so the challenge table above is written from national data (Rural Health Statistics as quoted in Sunder Lal, and the National Sample Survey 1996–2018) that applies to Karnataka — in the examination, quote your own state's Rural Health Statistics and Health Management Information System figures for PHC shortfall, Medical Officer vacancy and Ayushman Arogya Mandir conversion alongside these. Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir**, and the Ayushman Bharat Health Infrastructure Mission additionally funds block public health units and integrated public health laboratories that support the PHC.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW, PIB India, Indian Public Health Standards 2022, National Health Accounts 2022-23, Health Dynamics of India (Rural Health Statistics), WHO

Programme and policy updates (India)

- ▶ **"Health and Wellness Centre" is no longer the current name.** Ayushman Bharat-HWCs were renamed **Ayushman Arogya Mandir (AAM)**, tagline *Arogyam Parmam Dhanam*, with states directed to complete the rebranding by end-2023; **1,80,906 AAMs** (upgraded sub-centres and PHCs) were reported operational as on 31 October 2025. In a 2026 answer write **AAM-PHC** wherever the textbook writes HWC-PHC, and give the conversion figure with its date [MoHFW 2025].
- ▶ IPHS 2022 now carries an enforcement mechanism it lacked in 2006 and 2012 — an **open-source self-assessment toolkit and public dashboard (iphs.mohfw.gov.in) launched 28 June 2024**, on which every facility scores itself against the essential/desirable checklist; a facility counts as **IPHS-compliant only at a score of 80% or more**. As on 22 January 2025, **93% of facilities had been assessed** and 55% of those scored above 50%, against a state commitment of 50% of facilities compliant by 2025-26 [MoHFW 2025]. Quality certification has moved in parallel from token to scale — **50,373 public health facilities were NQAS-certified as on 31 December 2025**, 48,663 of them AAMs at the primary-care level and 1,710 secondary-care facilities, up from 6,506 in December 2023 and 22,786 in December 2024, against a stated objective of **at least 50% of public health facilities NQAS-certified by March 2026**; this is the direct answer to the textbook's "few PHCs certified under NQAS" challenge [PIB 2026-Jan].
- ▶ **PM-JAY was extended to every citizen aged 70 years and above** through the **Ayushman Vay Vandana card** from 29 October 2024, giving ₹5 lakh cover irrespective of income or socio-economic status — the elderly and palliative care package of the PHC now has a financing instrument attached at its referral end [PIB 2024-Oct].

Current national data

- ▶ **The financing figures in the body are superseded.** National Health Accounts estimates for 2022-23 put out-of-pocket expenditure at **43.4% of total health expenditure**, down from 64.2% in 2013-14; government health expenditure was **₹3.85 lakh crore (43.7% of total health**

expenditure; 1.43% of GDP) and government spending on primary care had risen to ₹1.4 lakh crore [MoHFW 2026]. Even so, a systematic review of Indian OOOPE estimates that about **90 million people** still cross the catastrophic threshold of more than 10% of household consumption spent on health [PMID 40552236].

- Facility counts worth quoting with their date: as on **31 March 2023** India had **1,69,615 sub-centres, 31,882 PHCs and 6,359 CHCs**; only **4,413 of the 21,964 specialists required at CHCs were in position — a 79.9% shortfall** (surgeons 83.3%, physicians 81.9%, paediatricians 80.5%, obstetrician-gynaecologists 74.2%), which is why first referral units do not function and the load falls back on the PHC [MoHFW 2024].
- An ICMR cross-sectional survey of 415 facilities in 19 districts of 7 states, fielded in two rounds in **September-December 2021 and February-June 2023**, measured readiness **for type 2 diabetes and hypertension services only** — taking IPHS as the assessment yardstick and scoring by the WHO service-readiness method, so it is not a general IPHS-compliance score. On that narrow measure the **PHC was the best-prepared peripheral facility (readiness 73%)**, ahead of sub-centres (61%) and CHCs (59%), with 64 of 100 surveyed PHCs already converted to HWCs; a facility counted as prepared only at 70% or more. The **medicines domain scored lowest at every level**: 45% of PHCs reported a chlorothiazide stock-out, 30% enalapril and 29% atenolol, stock-outs running a median of one to seven months, and the medicines, staff and treatment-guideline domains all fell below 70% at PHC level [PMID 40536370].

Elimination, surveillance and disease status

- **Tuberculosis: the 2025 national elimination deadline has passed unmet.** WHO's 2025 report records Indian incidence falling from 237 to 187 per lakh between 2015 and 2024 (a 21% decline, roughly twice the global pace) and mortality from 28 to 21 per lakh. The **100-day TB Mukh Bharat Abhiyaan (December 2024 to March 2025)** screened 12.97 crore vulnerable people in high-focus districts and detected 7.19 lakh cases, 2.85 lakh of them asymptomatic — vulnerability-based active case finding, not passive OPD detection, is now the operational model at PHC level [WHO 2025].
- **Kala-azar: India reached the elimination target — fewer than 1 case per 10,000 population at block level — in 2023**, with single-dose liposomal amphotericin B (introduced into the programme in 2014) the decisive tool: reported cases fell from 34,803 across 633 endemic PHC blocks in 2005 to 524 in 2023, and no block remained above the threshold. Sustaining it depends on the PHC detecting the two residual human reservoirs, **post kala-azar dermal leishmaniasis (in 5–15% of treated patients) and HIV-VL co-infection**, alongside undiminished indoor residual spraying and active case detection [PMID 40828749]. For malaria, India exited WHO's High Burden to High Impact grouping in 2024 and 160 districts across 23 states and union territories reported zero indigenous cases over 2022–24, against a 2030 elimination target [WHO 2025].

Guideline and legislative updates

- Two changes have rewritten PHC record-keeping. U-WIN, the Co-WIN-modelled electronic immunisation registry covering pregnant women and children up to 16 years against 12 vaccine-preventable diseases, with self-registration, SMS due-date reminders and QR-verifiable e-certificates, completed **nationwide rollout in November 2024** and replaces the paper due-list and

immunisation card. The **Registration of Births and Deaths (Amendment) Act 2023, in force from 1 October 2023**, makes registration digital through linked national and state databases and makes the birth certificate a single document for Aadhaar, school admission, passport, driving licence, marriage registration and voter enrolment — strengthening the civil registration function the PHC discharges [PIB 2024-Nov].

Global and WHO developments

- International health law has been rewritten twice in two years, and both instruments are examinable. The **International Health Regulations amendments adopted at the 77th World Health Assembly (1 June 2024) entered into force in September 2025**, creating a new highest alert tier — the **pandemic emergency**, declarable by the Director-General alongside a PHEIC — and adding equity provisions on access to health products (Article 13) and sustainable financing for developing countries (Article 44). The **WHO Pandemic Agreement was adopted by 124 governments at the 78th World Health Assembly on 20 May 2025** with 35 articles; it is only the second treaty negotiated under Article 19 of the WHO Constitution after the Framework Convention on Tobacco Control, and its Pathogen Access and Benefit-Sharing annex goes to the 2026 Assembly [PMID 40589223].

SOURCE & COVERAGE

Sunder Lal — Textbook of Community Medicine (Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community); Park's Textbook of Preventive and Social Medicine 28e (Ch 10 Health Care of the Community; Ch 13 Tribal Health in India); Revised Indian Public Health Standards (IPHS) 2022, Directorate General of Health Services, Ministry of Health and Family Welfare; National Health Policy 2017; Ayushman Bharat Comprehensive Primary Healthcare through Health and Wellness Centres — Operational Guidelines 2018; National Sample Survey 71st and 75th Rounds (2014, 2018). Coverage note: no Karnataka-specific facility or vacancy data exists in the indexed corpus, so no state-level figure has been stated.

Recent-advances sources: 40536370 — Preparedness of public & private health facilities for management of diabetes & hypertension in 19 districts in India, Indian Journal of Medical Research 2025, doi:[10.25259/IJMR.755.2024](https://doi.org/10.25259/IJMR.755.2024); 40552236 — Understanding out-of-pocket expenditure in India: a systematic review, Frontiers in Public Health 2025, doi:[10.3389/fpubh.2025.1594542](https://doi.org/10.3389/fpubh.2025.1594542); 40828749 — The story of elimination of visceral leishmaniasis (kala-azar) in India: challenges towards sustainment, PLoS Neglected Tropical Diseases 2025, doi:[10.1371/journal.pntd.0013321](https://doi.org/10.1371/journal.pntd.0013321); 40589223 — The 2025 World Health Assembly Pandemic Agreement and the 2024 Amendments to the International Health Regulations Combine for Pandemic Preparedness and Response, Medical Science Monitor 2025, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411). Also used: MoHFW update on Ayushman Arogya Mandir (operational figure as on 31 October 2025); MoHFW Indian Public Health Standards dashboard and assessment update (January 2025); PIB India — 50,000 NQAS certifications milestone (January 2026); PIB India — Ayushman Vay Vandana card launch (October 2024); PIB India — completion of nationwide U-WIN rollout (November 2024); MoHFW National Health Accounts Estimates for India 2022-23; MoHFW Health Dynamics of India (Infrastructure and Human Resources) 2022-23 (September 2024); WHO Global Tuberculosis Report 2025; Registration of Births and Deaths (Amendment) Act 2023 (in force 1 October 2023).



Health and environment problems in tribal population

Asked in: Paper III 23-Nov-2020 — RGUHS MD Community Medicine

Definition — **Scheduled Tribes (ST)** are those tribes or tribal communities notified under **Article 342** and defined in **Article 366(25)** of the Constitution of India; the list is State/Union Territory specific, so a community scheduled in one State need not be scheduled in another (Park 28e, Ch 13, p 848). Over **104 million** tribal people, spread across **705 tribes**, form **8.6 per cent** of India's population, nearly 90 per cent of them rural and mostly living in hilly and forested habitats (Park 28e, Ch 13, p 848). Being marginalised geographically, socio-economically and politically, they carry a **triple burden of disease** — persisting malnutrition and communicable disease, a rising non-communicable disease burden driven by environmental distress and changing lifestyles, and mental illness, especially addictions (Park 28e, Ch 13, p 848).

Magnitude and demographic profile

Parameter	Scheduled Tribes	All India	Source and year
Population	>104 million across 705 tribes = 8.6% of India's population ; males 52.5 million, females 52.0 million	1210 million (Census 2011)	Park 28e, Ch 13, p 848
Residence	~90% rural; hilly and forested terrain	—	Park 28e, Ch 13, p 848
Distribution	Concentrated in 10 States and 8 North-East States; Madhya Pradesh 15 million > Maharashtra 10 million > Odisha 9 million ≈ Rajasthan 9 million; highest concentration in the North-East	—	Park 28e, Ch 13, p 848
Sex ratio	990 (a tribal strength)	943	Census 2011 (Park 28e, Ch 13, p 848)
Child sex ratio (0–6 y)	957 (fell from 972 in 2001)	914	Census 2011 (Park 28e, Ch 13, p 848)
Literacy — persons	59.0% (47.1% in 2001)	73.0%	Census 2011 (Park 28e, Ch 13, p 848, Table 1)
Literacy — male / female	68.5% / 49.4%	80.9% / 64.6%	Census 2011 (Park 28e, Ch 13, p 848, Table 1)
Total fertility rate	2.09	1.99	NFHS-5, 2019-21 (Park 28e, Ch 13, p 849, Table 2)

Health problems

Maternal, child and nutritional deficit

Comparison of Scheduled Tribe with all-population indicators, NFHS-5 (2019-21), Park 28e, Ch 13, p 849, Table 2:

Indicator	Scheduled Tribes	All population
Infant mortality rate (per 1000 live births)	41.7	35.2
Neonatal mortality rate	28.8	24.9
Post-neonatal mortality rate	12.9	10.3
Under-5 mortality rate	50.3	41.9
Child mortality rate (1–4 years)	9.0	6.9
Infants fully immunised (%)	76.8	76.6
No vaccination at all (%)	4.1	3.6
BCG (%)	94.3	95.2
Measles (%)	86.7	87.9
Hepatitis B, 3rd dose (%)	81.2	88.6
Pregnant women receiving tetanus toxoid (%)	79.0	85.5
Under-5 children stunted (%)	40.9	35.5
Under-5 children wasted (%)	23.2	19.3
Under-5 children underweight (%)	39.5	32.1
Institutional delivery (%)	68.0	88.6
Deliveries by skilled health personnel (%)	71.5	89.4
Couples using any contraceptive (%)	64.4	66.7

Immunisation coverage of tribal children remains lower than that of the total population across almost every antigen — measles 86.7 against 87.9, BCG 94.3 against 95.2, Hepatitis B third dose 81.2 against 88.6, and tetanus toxoid in pregnancy 79.0 against 85.5 — even though the crude fully-immunised figure is close at 76.8 against 76.6 (Park 28e, Ch 13, p 848 and Table 2); Sunder Lal records that **only 26 per cent of tribal children receive all vaccines** (Sunder Lal, Ch 17, p 760). Early marriage, low body mass index and a high incidence of anaemia are the known drivers of high maternal mortality, and postnatal care is the weakest link — only about **37 per cent** of tribal women received any postnatal care within 48 hours of a home delivery (Park 28e, Ch 13, p 848).

Communicable diseases

Disease	Tribal burden
Tuberculosis	Estimated prevalence of pulmonary tuberculosis 703 per 100,000 in the tribal community against 256 per 100,000 in the rest of the country (Park 28e, Ch 13, p 849; Park 27e, Ch 13, p 825)
Malaria	Tribal people are about 8% of the national population but account for about 30% of all malaria cases, more than 60% of <i>Plasmodium falciparum</i> infections and as much as 50% of malaria mortality (Park 28e, Ch 13, p 849); drug-resistant falciparum malaria is the rule (Sunder Lal, Ch 17, p 760)

Disease	Tribal burden
Leprosy	Of 1,35,485 new cases detected in 2016-17, 25,474 (18.9%) were Scheduled Tribes (Park 28e, Ch 13, p 849)
Others	Yaws, diarrhoeal disease, skin infection, hepatitis, sexually transmitted infection and HIV (Sunder Lal, Ch 17, p 760; Park 28e, Ch 13, p 849)

Non-communicable, genetic, mental health and injury

Category	Tribal burden
Hypertension	One in four tribal adults is hypertensive; two of every three do not know the signs and symptoms; only 5% of men and 9% of women with hypertension knew their status (Park 28e, Ch 13, p 850)
Other NCDs	Rising cancer and diabetes from early epidemiological transition, rapid urbanisation, environmental distress and changing lifestyles (Park 28e, Ch 13, p 848)
Blindness	Shortage of ophthalmic manpower and services in tribal-predominant North-Eastern and hilly States (Park 28e, Ch 13, p 850)
Haemoglobinopathies	Sickle cell disease (anaemia + trait) and thalassaemia together 1–40% across different tribal communities, mostly the heterozygous form; sickle cell anaemia in 1 of every 86 births in central Indian tribal communities (Park 28e, Ch 13, p 850)
G6PD deficiency	0.7 to 15.6 per cent among 14 primitive tribal populations from four States with a high sickle-gene frequency (Park 28e, Ch 13, p 850)
Tobacco	72% of tribal men aged 15–54 years use tobacco, against 56% of non-tribal men (Park 28e, Ch 13, p 850)
Alcohol	About 50% of tribal men consume some form of alcohol; both addictions reduce productivity, deepen poverty, disrupt family harmony and generate law-and-order problems (Park 28e, Ch 13, p 850)
Injuries and violence	Snake, dog and scorpion bites are common because habitations are surrounded by forest; violence in conflict areas (Park 28e, Ch 13, p 850)

Why services do not reach — the health-system problem

- **Locational disadvantage** of sub-centres, primary health centres (PHC) and community health centres (CHC) relative to scattered hamlets, and **no transport for quick referral** of emergencies (Sunder Lal, Ch 17, p 760).
- **Infrastructure shortfall against the relaxed norms** — in 11 States a shortfall of **4,996 sub-centres (27%)**, in 7 States **1,023 PHCs (40%)** and in 10 States **209 CHCs (31%)**; in about half the States, tribal-area institutions are deficient by **27–40 per cent** (Park 28e, Ch 13, p 850).
- **Human resource vacuum** — vacant posts, no residential facilities, limited scope for professional growth, social and professional isolation, weak human-resource policy; compulsory rural service bonds exist in several States but are flouted and not enforced (Park 28e, Ch 13, pp 850–851; Park 27e, Ch 13, pp 826–827).
- **Cultural and communication mismatch** — information, education and communication (IEC) not tuned to tribal vocabulary, beliefs and practices; services not client-friendly in timing; **local traditional faith healers not involved** (Sunder Lal, Ch 17, p 760).

- **Weak convergence** — inadequate mobilisation of non-governmental organisations and poor integration with other health programmes (Sunder Lal, Ch 17, p 760).

Environment problems

Environmental problem	Health consequence
Inaccessible habitat — forest, hilly and riverbed terrain, dispersed and sparse habitation, poor road infrastructure	Locational disadvantage of sub-centres, PHCs and CHCs; no transport for emergency referral (Sunder Lal, Ch 2, p 68; Ch 17, p 760)
Vector-rich ecology — efficient vectors, triple insecticide resistance , innumerable breeding sites, plus outdoor sleeping habits, inadequate clothing, forest-based economy and high mobility	Sustained falciparum malaria transmission and mortality (Park 28e, Ch 7, p 498)
Forest interface with wildlife	Snake, dog and scorpion bites; zoonotic exposure (Park 28e, Ch 13, p 850)
Poor housing — ill-ventilated, ill-lit, overcrowded dwellings often shared with livestock	Tuberculosis, acute respiratory infection, diarrhoeal disease and vector-borne disease (Suryakantha, Ch 26, pp 721–722; Sunder Lal, Ch 3, p 108)
Indoor air pollution — unprocessed biomass fuel (firewood, charcoal, dung, crop residue) burnt on open fires in unventilated kitchens; more than 86.5% of rural households in India, China and sub-Saharan Africa use biomass fuel (Sunder Lal, Ch 3, p 121)	Acute and chronic respiratory disease, cardiovascular effects and raised lung-cancer risk in women and children, who are most exposed (Sunder Lal, Ch 3, p 113)
Unsafe water and open defecation	Soil pollution and water contamination; 70–80% of illnesses are related to water contamination and poor sanitation (Sunder Lal, Ch 3, pp 109–110)
Displacement and land alienation — dams, mining, deforestation, conservation reserves, urban clearance	Loss of livelihood and traditional food base, disruption of cultural identity, impoverishment and marginalisation (Park 28e, Ch 13, p 850; Oxford Textbook of Public Health 7e, Ch 92, p 3446; Ch 19, p 758)
Precarious livelihood — marginal farming and shifting cultivation, forest-based economy	Food insecurity underlying the stunting–wasting–anaemia complex (Sunder Lal, Ch 2, p 68)
Environmental distress with urbanisation and lifestyle change	Early epidemiological transition and non-communicable disease rise (Park 28e, Ch 13, p 848)

Measures taken — policy, infrastructure and programme adaptation

Measure	Content
Constitutional and planning framework	Scheduling under Article 342; Tribal Sub-Plan (TSP) , now the Scheduled Tribe Component (STC) , adopted in the 5th Five Year Plan for accelerated tribal development, run jointly by the Ministry of Tribal Affairs and the Ministry of Health and Family Welfare through tailored educational, infrastructural and livelihood schemes (Park 28e, Ch 13, p 850)

Measure	Content
Policy primacy	National Health Policy 1983, the 9th Plan, the National Population Policy 2000 and RCH-II all give primacy to tribal areas for primary health care (Sunder Lal, Ch 17, p 760)
Relaxed population norms	Sub-centre / Health and Wellness Centre 1 per 3,000 (vs 5,000 in plains); PHC 1 per 20,000 (vs 30,000); CHC 1 per 80,000 (Park 28e, Ch 13, p 850; Sunder Lal, Ch 17, p 760, Table 17.9)
Relaxed ASHA norm	One accredited social health activist (ASHA) per 1,000 population, relaxed to one ASHA per habitation in tribal and hilly areas, attached to the Health and Wellness Centre team (Park 28e, Ch 10, pp 726, 731)
Tribal service package under RCH-II	Assess unmet reproductive and child health need; improve accessibility, availability and acceptability; promote community participation and intersectoral coordination; promote and encourage the tribal system of medicine ; develop enough first referral units for obstetric emergencies (Suryakantha, Ch 43, p 887)
Tuberculosis elimination programme	Additional TB units and Designated Microscopy Centres in tribal/difficult areas; compensation for patient and attendant transport ; higher salary for contractual staff posted in tribal areas; enhanced vehicle maintenance and travel allowance; active case finding campaigns in tribal districts since 2015; Mobile TB Diagnostic Vans with X-ray and sputum microscopy — 35 vans in 17 districts of 5 States (Madhya Pradesh, Gujarat, Chhattisgarh, Rajasthan, Jharkhand) covering 17.65 million people (Park 28e, Ch 13, p 849; Park 27e, Ch 13, pp 825–826)
Vector borne disease control programme	Focused attention on tribal-dominated areas of the North-East, Andhra Pradesh, Chhattisgarh, Gujarat, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra and Odisha; Global Fund support to North-Eastern States and World Bank support to others for malaria; World Bank support for kala-azar elimination in Bihar, Jharkhand and West Bengal; 100% central assistance to North-Eastern States (Park 28e, Ch 13, pp 849–850)
Tribal Malaria Action Plan (TMAP)	Tribal-specific strategy for category 2 and 3 districts — hamlet-wise ASHAs instead of village-wise , engagement of anganwadi workers and local faith healers where an ASHA is not possible, mobile surveillance where routine services are absent, efficient supply-chain management, community mobilisation using traditional tools and practices (Park 28e, Ch 7, pp 498–499)
Leprosy eradication programme	State-wise ST-disaggregated data collected monthly; funds to non-governmental organisations working in tribal areas for IEC, deformity prevention and case follow-up; intensified IEC through rural media reaching remote and inaccessible populations (Park 28e, Ch 13, p 849)
Blindness and visual impairment programme (NPCBVI)	Centrally sponsored scheme, 60:40 in all States and 90:10 in North-Eastern States ; dedicated eye units in North-Eastern and other hilly States; contractual ophthalmic surgeons, assistants and eye-donation counsellors; multipurpose district mobile ophthalmic units for difficult areas (Park 28e, Ch 13, p 850)
Haemoglobinopathies	A sickle cell anaemia elimination programme has been launched (Sunder Lal, Ch 17, p 760)

Measures taken — environmental, intersectoral and the Expert Committee roadmap

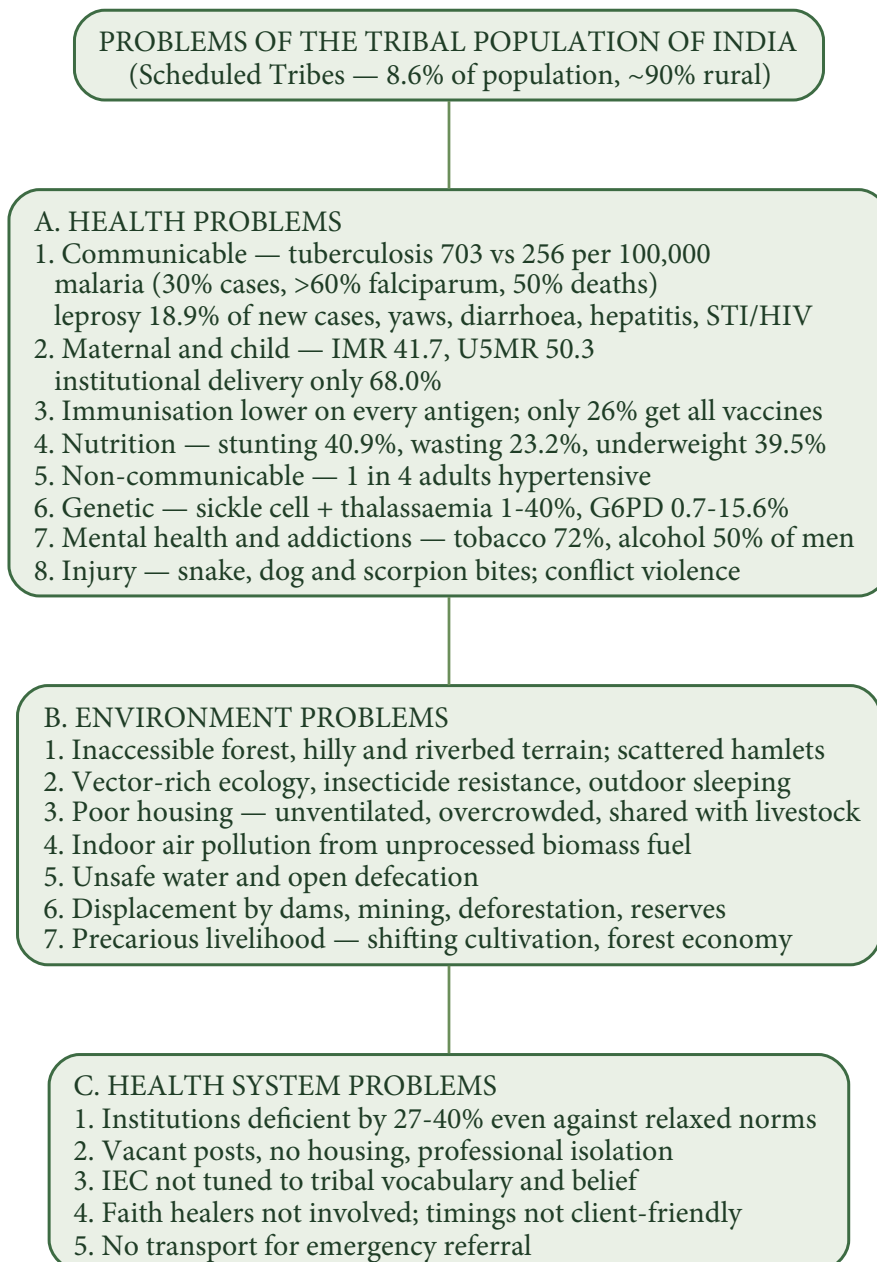
Environmental problem	Measure
Unsafe water	National Rural Drinking Water Programme, with the gram panchayat water and sanitation committee managing the completed scheme; the Jal Shakti Ministry created to deliver piped water to every household (Park 28e, Ch 7, p 573; Sunder Lal, Ch 8, pp 339–340)
Open defecation	Swachh Bharat Mission , launched 2 October 2014, with Swachh Bharat Mission–Gramin for solid and liquid waste management and open-defecation-free gram panchayats (Park 28e, Ch 7, p 573)
Indoor air pollution	Pradhan Mantri Ujjwala Yojana — free liquefied petroleum gas (LPG) connections and gas at subsidised rate to poor households (Sunder Lal, Ch 3, pp 113, 122–123); smokeless, fuel-efficient chullahs promoted by district rural development agencies and <i>Mahila Mandals</i> (Sunder Lal, Ch 3, p 113); improved stoves with ventilation and exhaust chimneys cut suspended particulate matter by up to 60% and carbon monoxide by up to 86% (Sunder Lal, Ch 3, p 121); solar cookers (Sunder Lal, Ch 3, p 113); National Clean Air Programme for ambient air (Sunder Lal, Ch 8, pp 339–340)
Undernutrition	Integrated Child Development Services, mid-day meal, National Food Security Act, POSHAN Abhiyaan (2018) and Mission Poshan 2.0 (2021), and Anaemia Mukta Bharat with its 6×6×6 strategy targeting a 3% annual fall in anaemia (Sunder Lal, Ch 5, p 218; Ch 8, p 339; Park 28e, Ch 7, p 546)
Housing and overcrowding	Improved ventilation and lighting, separation of livestock from living space, and healthy-housing education as part of environmental sanitation (Suryakantha, Ch 26, pp 721–722; Sunder Lal, Ch 3, p 108)

The Ministry of Health and Family Welfare and the Ministry of Tribal Affairs jointly constituted an **Expert Committee on Tribal Health under Dr Abhay Bang in October 2013**; its report, *Tribal Health in India: Bridging the Gap and a Roadmap for the Future*, recommends (Park 28e, Ch 13, p 851; Park 27e, Ch 13, p 827):

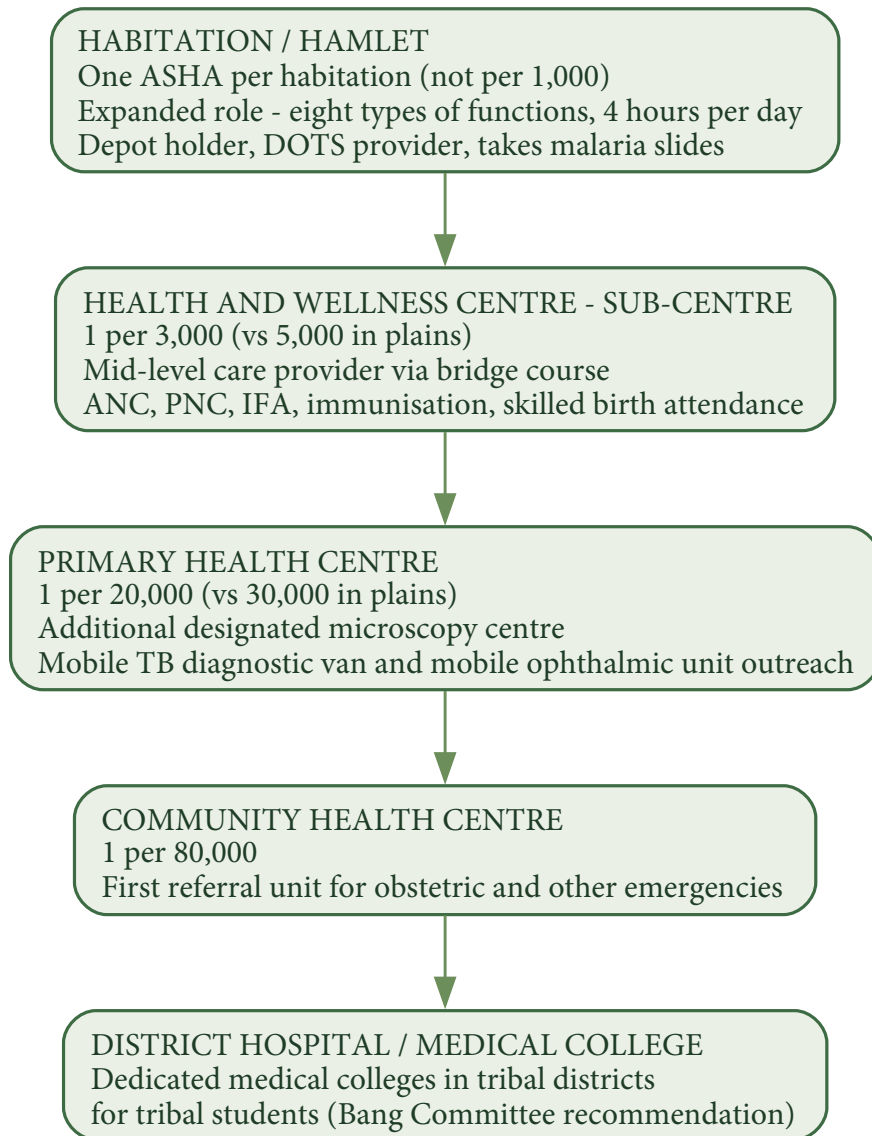
- ▶ The health care provider should, as far as possible, be a **local tribal** — the present workforce pattern is the opposite of this.
- ▶ The only sustainable way to a vibrant, responsive and accessible tribal health workforce is to **train and deploy local tribal people** in it.
- ▶ Place the **centre of gravity of the workforce close to the community**, not at the top with specialists and doctors.
- ▶ **Mid-level care providers** should be created through bridge courses and placed at the sub-centres.
- ▶ The **ASHA in tribal areas should have an expanded role**, with **eight types of functions** and a **total of 4 hours of work per day** expected of tribal ASHAs.
- ▶ To attract doctors to tribal areas, the **total salary needs to be substantially increased**.
- ▶ Create **dedicated medical colleges in tribal districts**, exclusively for tribal students from the scheduled areas.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

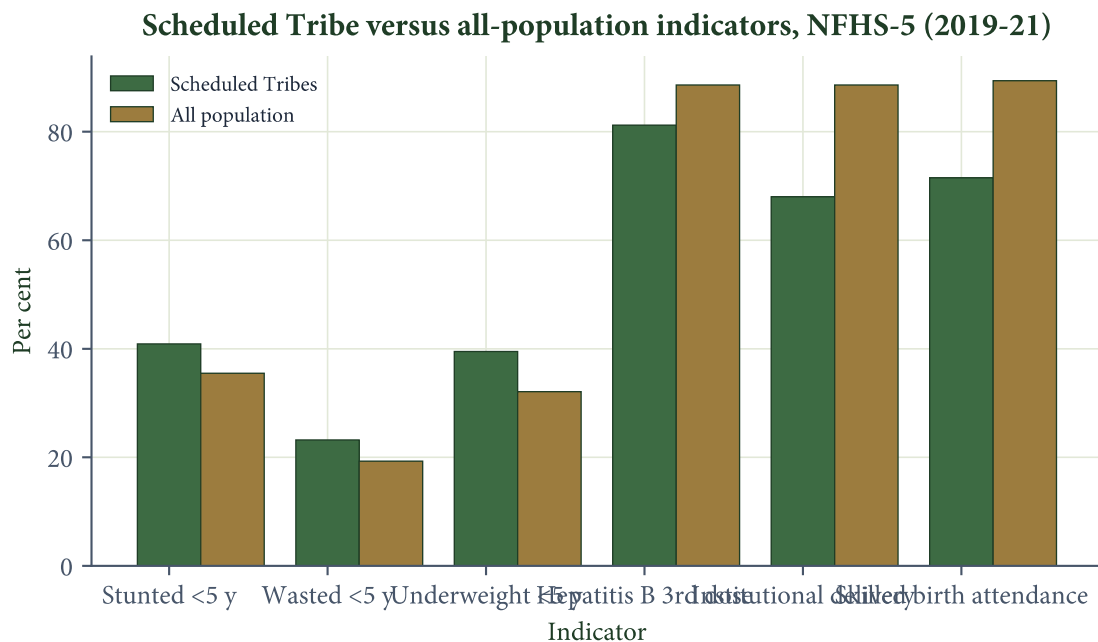
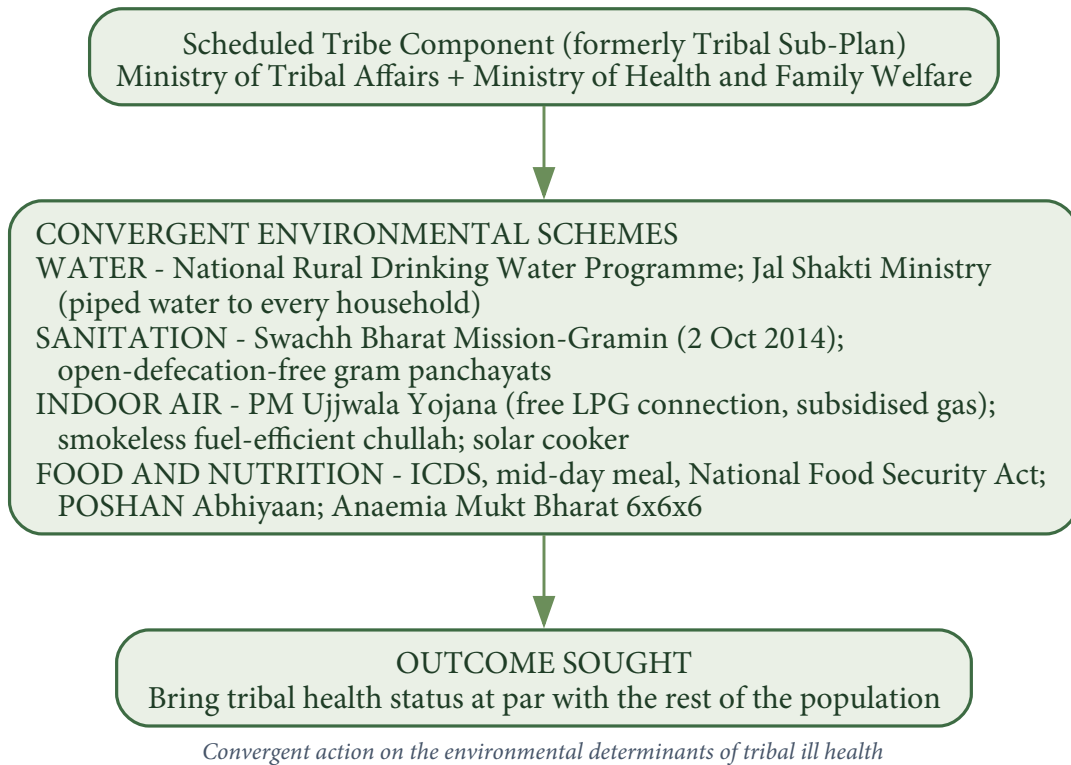
- **DRAWN figure** — "Triple burden of disease in the tribal community" (draw three concentric or overlapping bands with a central label). Central label: Tribal population — 104 million, 705 tribes, 8.6% of India. Band 1 label: Unfinished agenda — malnutrition, malaria, tuberculosis, maternal and child mortality. Band 2 label: Emerging — hypertension, diabetes, cancer (environmental distress, urbanisation, lifestyle change). Band 3 label: Third burden — mental illness and addictions (tobacco 72%, alcohol 50% of tribal men). Add a side box: Genetic overlay — sickle cell disease and thalassaemia 1–40%, G6PD deficiency 0.7–15.6%.



Classification of the common health and environment problems of the tribal population of India



Tribal-adapted health care delivery ladder with relaxed norms and programme add-ons



The widest Scheduled Tribe deficits are in institutional delivery and skilled birth attendance; nutrition and antigen-wise immunisation lag by roughly 4-7 percentage points. Source: NFHS-5 2019-21, reproduced in Park 28e, Ch 13, p 849, Table 2.

High-yield exam pointers

- **Constitution:** Scheduled Tribes notified under **Article 342**, defined in **Article 366(25)**; list is State/UT specific. **104 million, 705 tribes, 8.6%**, ~90% rural; Madhya Pradesh has the largest number, the North-East the highest concentration; sex ratio 990 vs 943.

- ▶ **Relaxed norms and the shortfall against them:** sub-centre 1:3,000, PHC 1:20,000, CHC 1:80,000, one ASHA per habitation; yet 4,996 sub-centres (27%), 1,023 PHCs (40%) and 209 CHCs (31%) short — a 27–40% deficit in about half the States.
- ▶ **Two must-quote disease numbers:** pulmonary tuberculosis 703 vs 256 per 100,000; tribal people are 8% of the population but carry 30% of malaria cases, >60% of falciparum and 50% of malaria deaths.
- ▶ **Genetics:** sickle cell disease + thalassaemia 1–40%; sickle cell anaemia 1 in 86 births in central India; G6PD deficiency 0.7–15.6%.
- ▶ **Addictions:** tobacco 72% of tribal men 15–54 y (vs 56% non-tribal); alcohol ~50%. Hypertension in 1 in 4 adults, but only 5% of men and 9% of women know their status.
- ▶ **The service gap in one line (NFHS-5 2019-21):** immunisation is lower on every antigen (Hepatitis B 3rd dose 81.2 vs 88.6, tetanus toxoid 79.0 vs 85.5), only 26% of tribal children get all vaccines, and institutional delivery 68.0 vs 88.6 is the widest gap.
- ▶ **Committee to name:** Dr Abhay Bang Expert Committee on Tribal Health, October 2013 — "Tribal Health in India: Bridging the Gap and a Roadmap for the Future"; core idea is a local tribal health workforce with the centre of gravity near the community, and a tribal ASHA with eight types of functions and 4 hours of work per day.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam — Park 28e Ch 13 carries three internal discrepancies between its running text and its Table 2, so quote the dated table values used above: the chapter text gives the Scheduled Tribe total fertility rate as 2.48 while Table 2 gives 2.09 against 1.99, the text gives the Scheduled Tribe infant mortality rate as 41.6 while Table 2 gives 41.7, and the text calls the 68.0 per cent tribal institutional delivery rate "much lower than the national average of 78.9 per cent" while Table 2 gives the all-population figure as 88.6; Sunder Lal (Ch 2, p 68; Ch 17, p 760) puts the tribal population at about 106 million and 8.2 per cent of the country, but the Census-based Park 28e figures of 104 million and 8.6 per cent are the ones to quote; the tuberculosis control programme is now the National Tuberculosis Elimination Programme (formerly RNTCP) and the vector borne disease programme is now the National Center for Vector Borne Diseases Control (formerly NVBDCP), so give the current name first if you use the abbreviation; and the retrieved corpus records only that "a sickle cell anaemia elimination programme has been launched" without naming the National Sickle Cell Anaemia Elimination Mission or its 2047 target, and carries no tribal-specific coverage data for the Swachh Bharat Mission, Jal Jeevan Mission or Pradhan Mantri Ujjwala Yojana, so those schemes are cited here as the general national environmental measures that apply to tribal areas rather than as tribal-specific interventions.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW, PIB India, ICMR, WHO World Malaria Report 2025, WHA76.16 and WHA79 (2026)

Programme and policy updates (India)

- ▶ **The sickle cell programme now has a name, a target year and a screening denominator.** The **National Sickle Cell Anaemia Elimination Mission** was announced in the Union Budget 2023-24 and launched on **1 July 2023 from Shahdol, Madhya Pradesh**, jointly by the Ministry of Health and Family Welfare and the Ministry of Tribal Affairs under the National Health Mission, aiming to **eliminate sickle cell disease as a public health problem by 2047** through universal screening of **7 crore people aged 0–40 years in 17 endemic States** in the first three years; the mission's own public dashboard — the screening denominator the field studies quote — had crossed **6 crore screened by mid-2025**, with roughly 2.15 lakh confirmed patients and 16.7 lakh carriers. A health technology assessment for a tribal cohort in six endemic States (Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha, Chhattisgarh) found the point-of-care tests (Hemotype SC, Sickle Scan, Sickle CERT, Gazelle) cost more per case screened than the solubility-plus-HPLC standard (US\$2.38–3.85 against US\$0.67) but detected **21,125–41,458 additional cases**; the National Health Mission then directed States to procure the kits by public tendering, which pushed prices below **US\$1.21 per test** and projects a **US\$37.86 million saving at full coverage**. Replaces the answer's bare line that "a sickle cell anaemia elimination programme has been launched". [PMID 42594133]; [PIB 2023-Jul]; [NSCAEM dashboard]
- ▶ **Two new tribal-development missions now carry the health, water and nutrition money.** PM-JANMAN (Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan), launched **15 November 2023** for Particularly Vulnerable Tribal Groups — ₹24,104 crore, about 22,000 habitations and 28 lakh people; and the **Dharti Aaba Janjatiya Gram Utkarsh Abhiyan** (Pradhan Mantri Janjatiya Unnat Gram Abhiyan), cleared by the Union Cabinet in **September 2024** — ₹79,156 crore, **17 line ministries, 63,843 tribal-majority villages and aspirational districts, over 5 crore tribal people**, running to March 2029. These are the current answer to "measures taken", alongside the Scheduled Tribe Component. [PIB 2023-Nov]; [PIB 2024-Sep]
- ▶ **Ayushman Bharat Health and Wellness Centres were renamed *Ayushman Arogya Mandir*** by a Ministry of Health and Family Welfare communication to States in **November 2023**, with rebranding to be completed by the end of that year; more than 1.63 lakh were operational by 15 December 2023, upgraded from sub-centres and primary health centres with explicit priority to rural and tribal areas. Write the relaxed tribal sub-centre norm of **1 per 3,000** under the new name and give the old one in brackets. [MoHFW 2023-Nov]
- ▶ **Tribal tuberculosis patients get their nutrition money late and without prioritisation.** Programme data on 6,568 adults notified in public facilities in Chhattisgarh (April–June 2024) showed **22 per cent severely and 13 per cent very severely undernourished**; **81 per cent** received the first Ni-kshay Poshan Yojana direct benefit transfer instalment but only **29 per cent within two months**, and only **21 per cent** received food baskets — with coverage the same across all undernutrition grades, so the most wasted were not prioritised. A ready criticism of the "measures taken" in a tribal State. [PMID 41042600]

Current national data

- ▶ **Quote the tribal tuberculosis excess with its survey round.** An analysis of **2,077,924 adults aged 15 years and above in NFHS-5 (2019-21)** gives point prevalence **416 per 100,000 (95% CI 383–452) in Scheduled Tribes against 277 (266–287) in non-Scheduled Tribes**, with tribal identity an independent risk factor after adjustment (**aOR 1.47, 95% CI 1.38–1.56**). The gradient is

geographic and constitutional: highest at **608 per 100,000** in non-Scheduled districts that are more than 60 per cent tribal, and lowest at **330** inside Fifth and Sixth Schedule Areas, suggesting the constitutional safeguards themselves buy some protection. Park's **703 against 256 per 100,000** remains quotable but must be dated separately; the 2025 systematic review of the tribal "missing millions" — 34 studies from 15 low- and lower-middle-income countries — takes as its background burden an earlier pooled meta-analytic prevalence of **7,030 per million** in Indian tribal populations, and finds the diagnosis-and-notification gap driven by illiteracy, stigma, geographical inaccessibility and cost on the community side and by thin diagnostic facilities, staff shortage and weak incentives on the system side. [PMID 41185051]; [PMID 41235544]

- **The environmental statements in this answer now have measured effect sizes, both from NFHS-5.** Indoor air pollution was associated with anaemia in 25,579 pregnant women (**crude OR 1.43, 95% CI 1.36–1.51**; adjusted OR 1.20, 1.13–1.27, attenuating to 1.07, 1.00–1.14 once wealth is added), and among the exposed, **63.1 per cent of Scheduled Tribe women were anaemic** against 57.3 per cent overall — the biomass-fuel link made numerical. Smokeless tobacco use during pregnancy or breastfeeding was **8.05 per cent in tribal women** against 2.62 per cent in Scheduled Castes, 1.85 per cent in other backward classes and 1.63 per cent in the residual "others" group, with Scheduled Tribe status carrying **AOR 2.20 (95% CI 1.95–2.48)**, and mass-media exposure failing to protect (AOR 1.49, 1.43–1.57). [PMID 41770749]; [PMID 40547460]

Elimination, surveillance and disease status

- **Malaria — the national picture improved but the tribal bottleneck moved to radical cure.** India exited the WHO High Burden to High Impact group in 2024 after roughly an 80 per cent fall in cases between 2015 and 2023, and now targets **zero indigenous cases by 2027 and elimination by 2030**. In Malkangiri and Koraput (Odisha), 269 confirmed *Plasmodium vivax* patients took chloroquine and artesunate–sulfadoxine–pyrimethamine with over 93 per cent adherence but only **58.7 per cent (95% CI 52.6–64.9) completed the 14-day primaquine course**, stopping mainly because symptoms settled (61.9%) or through forgetfulness (21.4%) — hence programme interest in shorter regimens or single-dose tafenoquine with G6PD testing, which intersects directly with the 0.7–15.6 per cent tribal G6PD deficiency in this answer. [WHO 2025]; [PMID 41505814]
- **Tuberculosis — the 2025 elimination deadline has lapsed and the debate is now about realigning to the global 2030 target.** Current commentary names Pradhan Mantri TB Mukh Bharat Abhiyan and Ni-kshay Poshan Yojana as the large initiatives while listing the absence of tribal-targeted strategies, multidrug resistance, weak biosafety and post-treatment relapse monitoring as unaddressed health-security gaps. Field performance supports the criticism: of **323 Saharia tribal patients in Madhya Pradesh** started on drug-resistant treatment (83 per cent rifampicin-resistant or multidrug-resistant), treatment success was **66.8 per cent against the programme target of over 75 per cent**, with 20.4 per cent lost to follow-up and 11.1 per cent deaths. [PMID 40781735]; [PMID 40337857]

Guideline and legislative updates

- **Sickle cell policy is moving from screening to comprehensive care, and the "mild Indian disease" teaching is now refuted.** The first community-based cohort with fortnightly home visits

for a year across five endemic tribal districts (Alluri Seetharama Raju, Chhotaudepur, Kandhamal, Mysuru, Udalguri) recorded 8,404 vaso-occlusive crisis episodes per 1,000 patient-years (95% CI 8,040–8,760), with 88.5 per cent of 252 patients having at least one crisis in the year, yet only 48.7 per cent of episodes treated at any health facility — the rest self-managed at home — and hydroxyurea taken intermittently by 92.1 per cent, regularly by only 2.4 per cent. This is the evidence base for adding acute pain management, hydroxyurea continuity, counselling and tertiary access to the mission's screening pillar. [PMID 42024909]

Global and WHO developments

- **Resolution WHA76.16 on the Health of Indigenous Peoples**, adopted at the Seventy-sixth World Health Assembly in May 2023, directs the Director-General to develop a **Global Plan of Action for the Health of Indigenous Peoples**, co-created with Indigenous Peoples, with a life-course approach emphasising reproductive, maternal and adolescent health. **Note the timeline, because it has slipped:** WHA76.16 originally mandated presentation to the **Seventy-ninth World Health Assembly in 2026**; Executive Board decision EB158(10) of February 2026 recommended deferral, and **WHA79 in May 2026 (agenda item 15.3) deferred the draft plan to the Eightieth World Health Assembly in May 2027**, routed through the **160th Executive Board of January 2027**, to allow further consultation — the **plan period remains 2027–2040**. This is the international counterpart to the Bang Committee roadmap and the natural closing line for a tribal-health answer. [WHO 2023-May]; [WHA79 2026-May]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 13 Tribal Health in India, pp 848–851 including Table 1 literacy and Table 2 comparison of maternal and child health indicators in Scheduled Tribes and all population 2019-21; Ch 7 Health Programmes in India, pp 498–499, 546, 573; Ch 10 Health Care of the Community, pp 726, 731); Park's Textbook of Preventive and Social Medicine 27e (Ch 13 Tribal Health in India, pp 824–827 — used for the clean text of the tuberculosis prevalence figures and the Expert Committee paragraph, which are garbled in the 28e scan); Sunder Lal — Textbook of Community Medicine (Ch 2, p 68; Ch 3 Environmental Health Problems, pp 108–113, 121, 122–123; Ch 5, p 218; Ch 8, pp 339–340; Ch 17 Healthcare of the Community, p 760 including Table 17.9); Suryakantha — Community Medicine with Recent Advances (Ch 26 Social Science, pp 721–722; Ch 43 National Health Programs, p 887); Oxford Textbook of Public Health 7e (Ch 19, p 758; Ch 92 Forced migrants and other displaced populations, p 3446); National Family Health Survey-5 (2019-21) and Census of India 2011 as cited within Park 28e Ch 13; Report of the Expert Committee on Tribal Health (Dr Abhay Bang), *Tribal Health in India: Bridging the Gap and a Roadmap for the Future*, Ministry of Health and Family Welfare and Ministry of Tribal Affairs.

Recent-advances sources: PMID 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, *PLoS One* 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); PMID 41185051 — Burden on the burdened: tuberculosis among Scheduled Tribes and non-Scheduled Tribes in constitutionally protected Scheduled and non-Scheduled areas of India, *Infectious Diseases of Poverty* 2025, doi:[10.1186/s40249-025-01375-9](https://doi.org/10.1186/s40249-025-01375-9); PMID 41235544 — Systematic review of barriers to and enablers of tuberculosis diagnosis, notification, and intervention for designing customised intervention package to minimise 'missing millions' in tribal communities of India, *Journal of Global Health* 2025, doi:[10.7189/jogh.15.04303](https://doi.org/10.7189/jogh.15.04303); PMID 42024909 — Epidemiology of vaso-occlusive crisis among sickle cell disease patients in India: a community-based multi-centric study, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_2879_2025](https://doi.org/10.25259/IJMR_2879_2025); PMID 41770749 — Exploring the association between indoor air pollution (IAP) and anaemia among pregnant women in India, *PLoS One* 2026, doi:[10.1371/journal.pone.0343845](https://doi.org/10.1371/journal.pone.0343845); PMID 40547460 — Prevalence and factors associated with tobacco use among pregnant and breastfeeding mothers in India: insights from the National Family Health Survey-5 (2019-21), *Frontiers in Public Health* 2025, doi:[10.3389/fpubh.2025.1495522](https://doi.org/10.3389/fpubh.2025.1495522); PMID 41505814 — Suboptimal primaquine adherence in *Plasmodium vivax* malaria: evidence from high-burden tribal districts in Odisha, *Journal of Infection and Public Health* 2025, doi:[10.1016/j.jiph.2025.103123](https://doi.org/10.1016/j.jiph.2025.103123) (abstract-only); PMID 41042600 — High burden of severe and very severe undernutrition among adults with tuberculosis: findings from a state wide operational research in Chhattisgarh, India, *Expert Review of Respiratory Medicine* 2025, doi:[10.1080/17476348.2025.2569845](https://doi.org/10.1080/17476348.2025.2569845) (abstract-only); PMID 40781735 — Enhancing India's health security efforts against tuberculosis: gaps and opportunities, *Health Security* 2025, doi:[10.1089/hs.2024.0109](https://doi.org/10.1089/hs.2024.0109) (abstract-only); PMID 40337857 — Drug-resistant tuberculosis treatment outcomes among Saharia: a marginalized community of Madhya Pradesh, India, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2025,

doi:[10.1093/trstmh/traf046](https://doi.org/10.1093/trstmh/traf046) (abstract-only). Press Information Bureau, India — 1 July 2023 (launch of the National Sickle Cell Anaemia Elimination Mission at Shahdol, Madhya Pradesh), 15 November 2023 (PM-JANMAN for Particularly Vulnerable Tribal Groups) and September 2024 (Dharti Aaba Janjatiya Gram Utkarsh Abhiyan / Pradhan Mantri Janjatiya Unnat Gram Abhiyan); Ministry of Health and Family Welfare, November 2023 — renaming of Ayushman Bharat Health and Wellness Centres as Ayushman Arogya Mandir; National Sickle Cell Anaemia Elimination Mission public dashboard (cumulative screening, confirmed cases and carriers, mid-2025 — the same public NSCAEM source cited for district selection in PMID 42024909); World Health Organization, *World Malaria Report 2025*; World Health Assembly resolution WHA76.16 on the Health of Indigenous Peoples, May 2023, together with the deferral trail for the Global Plan of Action for the Health of Indigenous Peoples — Executive Board decision EB158(10), February 2026, and the Seventy-ninth World Health Assembly, May 2026, agenda item 15.3, deferring the draft plan to the Eightieth World Health Assembly in May 2027 through the 160th Executive Board of January 2027, plan period still 2027–2040 (WHO Global Plan of Action for the Health of Indigenous Peoples page).



Health problems of tribal population

Asked in: Paper III 11-May-2022 — RGUHS MD Community Medicine

Definition — Scheduled Tribes (ST) are those tribes or tribal communities notified under **Article 342** of the Constitution of India, the term itself being defined in **Article 366(25)**; the list is state/union-territory specific, so a community scheduled in one state need not be scheduled in another. Over **104 million tribal people live in India, spread across 705 tribes, forming 8.6 per cent of the country's population** (Park 28e, Ch 13, p 848). Tribal communities are not homogeneous — they belong to more than **400 ethno-linguistic, religious and cultural groups** (Jugal Kishore, National Health Programmes of India, p 245) — and their marginalisation is geographic, socio-economic and political, which is why "the underprivileged, who need more, get less".

Population profile that shapes tribal health

Feature	Position (source year)
Distribution	Concentrated in 10 states and 8 North-Eastern states ; ~90 per cent live in rural , hilly and forested areas (Park 28e, Ch 13, p 848)
Largest numbers	Madhya Pradesh 15 million > Maharashtra 10 million > Odisha 9 million > Rajasthan 9 million; highest concentration is in the North-East (Park 28e, Ch 13, p 848)
Predominantly tribal states	6 states with >50 per cent ST population , and 9 more states where the majority of the ST population lives (Jugal Kishore, p 245)
Sex ratio	990 (Census 2011) vs national 943 — a tribal <i>strength</i> ; child sex ratio (0–6 y) fell 972 (2001) → 957 (2011) but stays above the general 914 (Park 28e, Ch 13, p 848)

Feature	Position (source year)
Literacy	ST 47.1 per cent (2001) → 59.0 per cent (2011); male 68.5, female 49.4 vs all-India 73.0 (Census 2011; Park 28e, Ch 13, p 848)
Livelihood/ecology	Marginal farming and shifting cultivation, forest-based economy, outdoor sleeping, high mobility, displacement (Sunder Lal, Ch 2, p 68; Park 28e, Ch 7, p 498)

Determinants of ill-health

Determinant	Health consequence
Poverty; macro- and micronutrient undernutrition	Low birth weight, low body mass index, anaemia, high maternal and child mortality
Poor environmental sanitation, hygiene and unsafe drinking water	Diarrhoeal disease, soil-transmitted helminths, hepatitis
Difficult terrain, sparse and dispersed habitations, no transport for referral	Delayed care, high case fatality, obstetric emergencies unmanaged
Locational disadvantage of sub-centres, primary health centres (PHC), community health centres (CHC); posts vacant	Low service utilisation
Cultural barriers, non-involvement of traditional faith healers, non-client-friendly services, inappropriate behaviour change communication	Low demand and poor acceptance
Weak human-resource policy, no incentive to serve tribal areas; poor monitoring and supervision	Chronic workforce shortage

(Jugal Kishore, National Health Programmes of India, p 246)

Health problems — the triple burden

Tribal communities carry a **triple burden**: unfinished malnutrition and communicable disease, a rising non-communicable disease load from early epidemiological transition, and a third burden of mental illness and addictions (Park 28e, Ch 13, p 848).

Category	Key tribal-specific burden
Communicable	Pulmonary tuberculosis prevalence 703 vs 256 per 100,000 in the rest of the country. Malaria — tribes are ~8 per cent of the population but account for ~30 per cent of all malaria cases, >60 per cent of <i>Plasmodium falciparum</i> and ~50 per cent of malaria mortality; <i>P. falciparum</i> predominates, with efficient vectors and triple insecticide resistance. Leprosy — of 1,35,485 new cases in 2016–17, 25,474 (18.90 per cent) were ST. Also sexually transmitted infections and HIV, skin infections, diarrhoea, hepatitis (Park 28e, Ch 13, pp 848–849; Ch 7, p 498)
Non-communicable	One in four tribal adults is hypertensive; only 5 per cent of men and 9 per cent of women knew their hypertensive status, and two of three did not know the symptoms. Blindness and visual impairment, especially in the tribal-predominant North-East (Park 28e, Ch 13, p 850)

Category	Key tribal-specific burden
Genetic (haemoglobinopathies)	Sickle cell disease (anaemia + trait) and thalassaemia together 1–40 per cent across tribal communities, mostly heterozygous; sickle cell anaemia 1 in 86 births in central Indian tribes. Glucose-6-phosphate dehydrogenase (G6PD) deficiency 0.7–15.6 per cent among 14 primitive tribal populations of four states (Park 28e, Ch 13, p 850)
Mental health and addictions	72 per cent of tribal men aged 15–54 years use tobacco vs 56 per cent non-tribal men; ~50 per cent of tribal men consume alcohol — driving NCDs, lost productivity, poverty, family disruption and law-and-order problems (Park 28e, Ch 13, p 850)
Injuries and violence	Snake, dog and scorpion bites are common as habitations abut forest; violence in conflict areas (Park 28e, Ch 13, p 850)
Maternal and reproductive	Early marriage, low body mass index, anaemia drive maternal mortality; only ~37 per cent of tribal women received any postnatal care within 48 hours of a home delivery (Park 28e, Ch 13, p 848)

Health-status gap against the general population

Indicator, NFHS-5 (2019–21)	Scheduled Tribe	All population
Infant mortality rate (per 1,000 live births)	41.7	35.2
Neonatal mortality rate	28.8	24.9
Post-neonatal mortality rate	12.9	10.3
Under-five mortality rate	50.3	41.9
Child mortality rate (1–4 y)	9.0	6.9
Under-five stunted (%)	40.9	35.5
Under-five wasted (%)	23.2	19.3
Under-five underweight (%)	39.5	32.1
Institutional delivery (%)	68.0	88.6
Delivery by skilled health personnel (%)	71.5	89.4
Total fertility rate	2.09	1.99
Couples using any contraceptive (%)	64.4	66.7
Children with no vaccination (%)	4.1	3.6
Hepatitis B third dose (%)	81.2	88.6
Pregnant women receiving tetanus toxoid (%)	79.0	85.5

(Park 28e, Ch 13, Table 2, p 849, from NFHS-5)

Health-system deficits

- **Relaxed norms exist but are unmet.** Tribal and hilly areas are entitled to **one health sub-centre per 3,000, one PHC per 20,000 and one CHC per 80,000 population** (Park 28e, Ch 13, p 850; Sunder Lal, Ch 17, Table 17.9, p 760).
- **Infrastructure shortfall** across 18 states and 3 union territories studied — **4,996 sub-centres short (27 per cent of requirement) in 11 states; 1,023 PHCs short (40 per cent) in 7 states; 209 CHCs short (31 per cent) in 10 states.** About half the states are deficient by 27–40 per cent (Park 28e, Ch 13, p 850).
- **Human-resource gap** from limited professional interaction and growth, social and professional isolation, weak human-resource policy, poor working conditions and limited social infrastructure. Compulsory rural-service bonds are flouted by most doctors and are not enforced (Park 28e, Ch 13, pp 850–851).

Promoting health in tribal populations

A. Objectives (Jugal Kishore, p 246): assess unmet needs; provide integrated, appropriate, quality services; ensure equitable access; stimulate demand; improve coverage and acceptability; promote community participation and intersectoral coordination; create employment for tribal people; promote tribal systems of medicine.

B. Programme-specific tribal packages under the National Health Mission

Programme	Tribal-specific action
Tuberculosis (NTEP, earlier RNTCP)	Additional TB units and designated microscopy centres in tribal/difficult areas; compensation for patient and attendant transport; higher salary for contractual staff posted in tribal areas; enhanced vehicle maintenance and travel allowance; TB health visitors. Mobile TB Diagnostic Vans with X-ray and sputum microscopy — 35 vans in 17 districts of 5 states (Madhya Pradesh, Gujarat, Chhattisgarh, Rajasthan, Jharkhand) covering ~17.65 million people. Campaign-mode active case finding in mapped tribal districts (Park 28e, Ch 13, pp 848–849)
Malaria and vector-borne disease	Tribal Malaria Action Plan (TMAP) ; hamlet-wise instead of village-wise ASHAs ; engagement of faith healers, local practitioners and anganwadi workers; mobile surveillance where routine services are absent; Global Fund support to the North-East and World Bank support elsewhere; 100 per cent central assistance to North-Eastern states (Park 28e, Ch 7, pp 498–499; Ch 13, pp 849–850)
Leprosy (NLEP)	Monthly ST-disaggregated case data; funds to non-governmental organisations for information-education-communication, deformity prevention and follow-up in tribal areas (Park 28e, Ch 13, p 849)
Blindness (NPCBVI)	Dedicated eye units in North-Eastern and hilly states; contractual ophthalmic surgeons, assistants and eye-donation counsellors; multipurpose district mobile ophthalmic units for difficult areas (Park 28e, Ch 13, p 850)
Haemoglobinopathies	National Sickle Cell Anaemia Elimination Mission — free screening and diagnostic services at all levels of health facility, with published national guidelines on haemoglobinopathies (Anemia Mukta Bharat Operational Guidelines, Ch 6, pp 84–85; Sunder Lal, Ch 17, p 760)

Programme	Tribal-specific action
Universal Immunization Programme	Inclusive microplanning with headcount survey of every tribal habitation including particularly vulnerable tribal groups; mapping migration season/place/duration; tagging vacant sub-centres to an auxiliary nurse midwife; strengthened alternate vaccine delivery and cold chain; U-WIN registration; outreach sessions at local bazaars, melas and festivals ; convergence with the Ministry of Tribal Affairs, tribal welfare, education, ICDS and panchayati raj institutions (Routine Immunization Manual for Medical Officers 2024, Ch 14, pp 314–315)
PM-JANMAN (Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan)	Under the Development Action Plan for Scheduled Tribes, FY 2023-24 to 2025-26, for 75 particularly vulnerable tribal group communities in 18 states and Andaman & Nicobar ; saturates households with housing, piped water, sanitation, roads, telecom, education, nutrition and a " Door Daraz Gaon Tak " mobile medical unit delivering immunization (Routine Immunization Manual 2024, Ch 14, p 316)

C. Recommendations of the Expert Committee on Tribal Health — constituted in **October 2013** jointly by the Ministry of Health and Family Welfare and the Ministry of Tribal Affairs under the chairmanship of **Dr Abhay Bang**, reporting as *Tribal Health in India: Bridging the Gap and a Roadmap for the Future*:

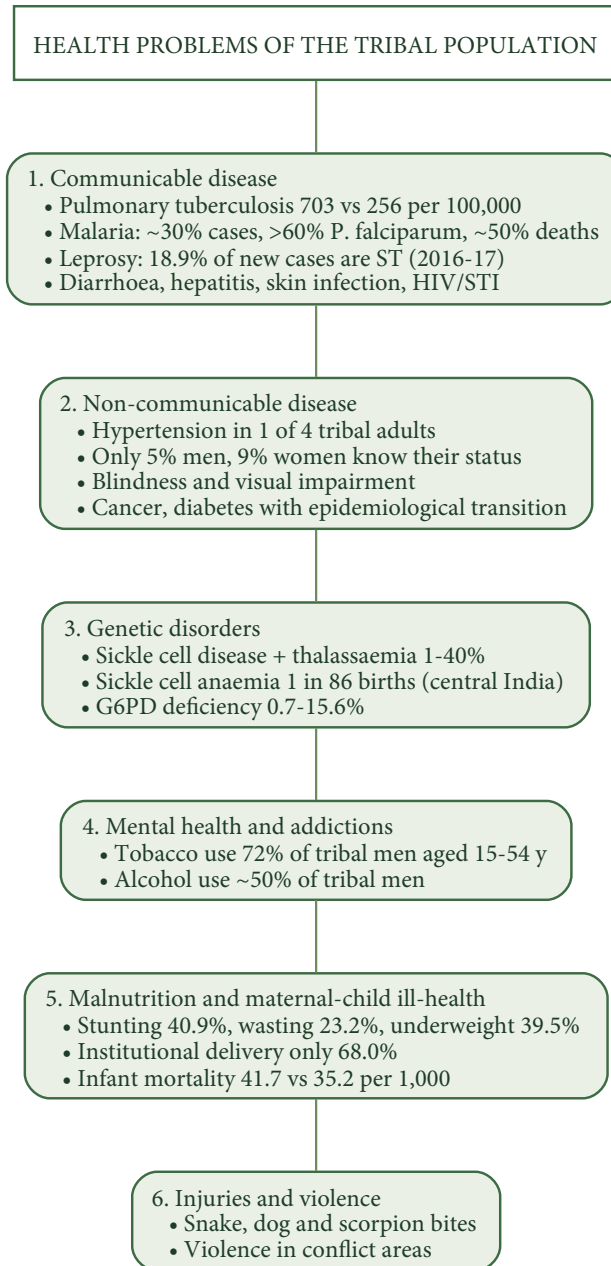
- ▶ The health care provider should, as far as possible, be a **local tribal**; the present workforce pattern is the opposite of this.
- ▶ The only sustainable route to a responsive, accessible tribal health workforce is to **train and deploy local tribal people** in it.
- ▶ Place the **centre of gravity of the workforce closer to the community**, not at the top with specialists and doctors.
- ▶ **Expand the role of the tribal ASHA** — expanded functions with a total of **4 hours of work per day**.
- ▶ Create **mid-level care providers through bridge courses**, posted at sub-centres.
- ▶ **Substantially increase total salary** to attract doctors to tribal areas.
- ▶ Create **dedicated medical colleges in tribal districts, exclusively for tribal students** from scheduled areas.

D. Community and cross-sectoral health promotion (Jugal Kishore, p 247): involve the community in planning, management and implementation; use community-based male and female workers as social mobilisers, educators and non-clinical providers; involve locally elected bodies and tribal boards; involve non-governmental organisations; promote tribal systems of medicine and tribal healers where appropriate. Strategy must be **multi-pronged, area-specific and need-based** — the North-Eastern states, with higher literacy and better economic status, need a different strategy from central India. Tribal strengths — strong community bonds, positive health values, faith in traditional healing, universal breastfeeding and a favourable sex ratio — are the entry points for promotion, not obstacles. Health systems must additionally be made **culturally safe**, with an indigenous health workforce, since concordance between provider and community improves communication, shared decision-making, trust and satisfaction, and reduces perceived racism (Oxford Textbook of Public Health 7e, Ch 89, pp

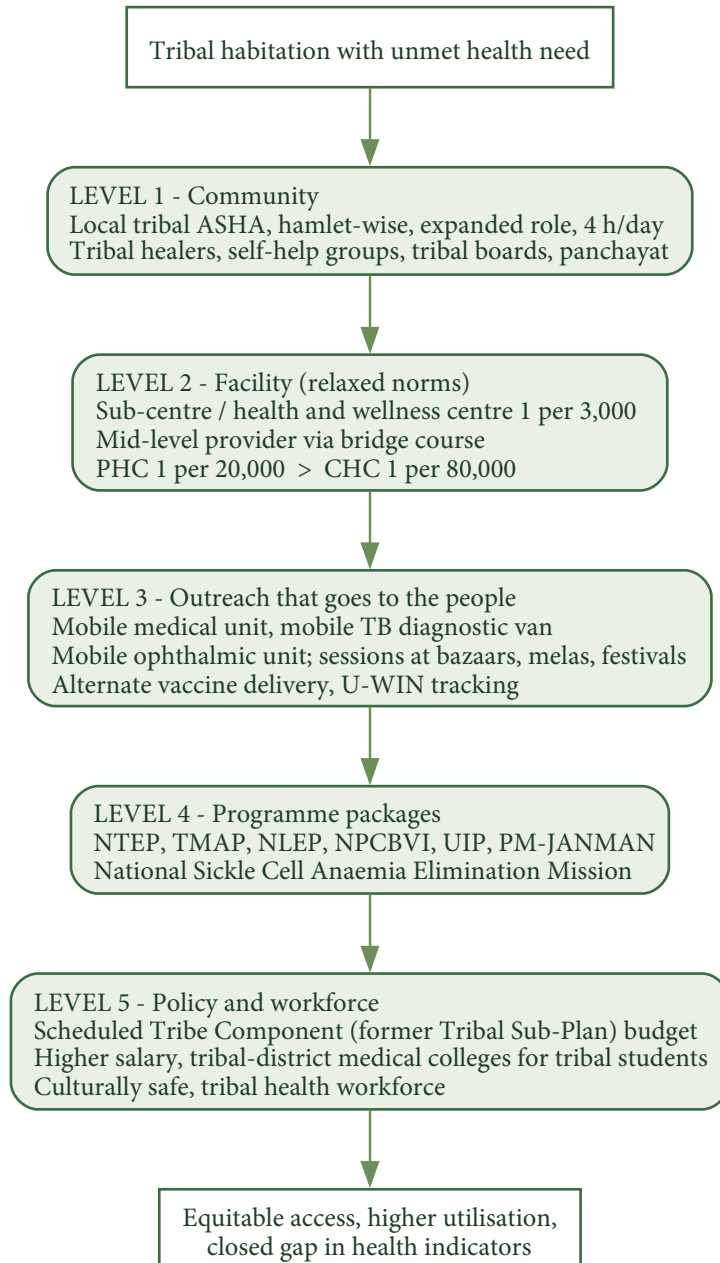
3347–3349). Structurally, the Tribal Sub-Plan, now the Scheduled Tribe Component, adopted in the Fifth Five Year Plan, remains the budgetary vehicle for this convergence (Park 28e, Ch 13, p 850).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

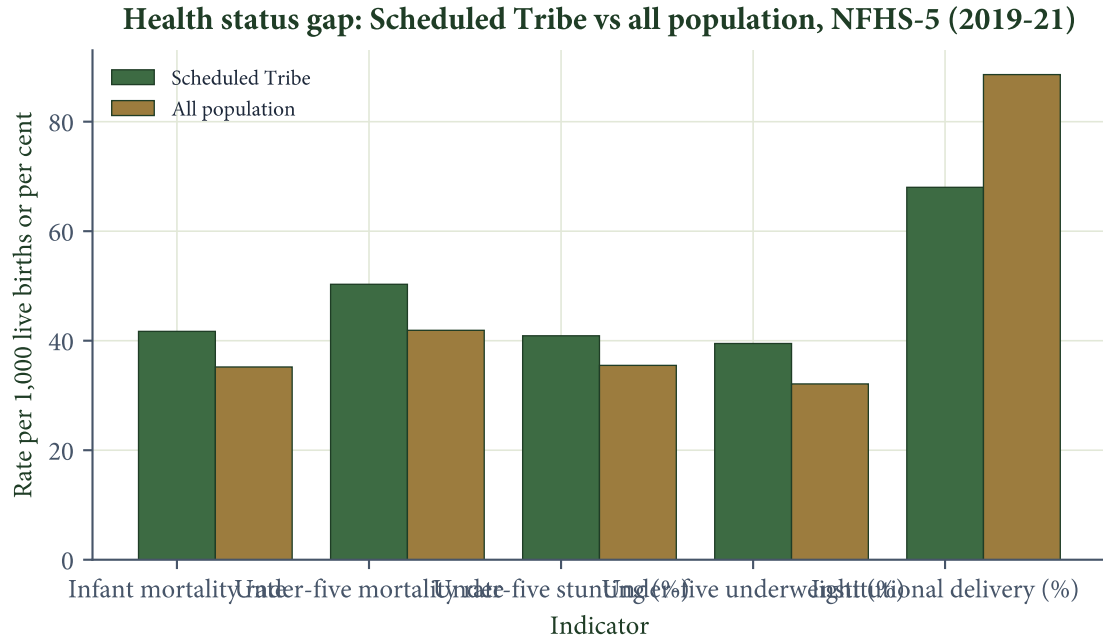
- ▶ **The triple burden of disease in the tribal community** — draw three overlapping circles labelled: (1) *Malnutrition and communicable disease* — tuberculosis, malaria, leprosy, diarrhoea; (2) *Non-communicable disease of early epidemiological transition* — hypertension, cancer, diabetes; (3) *Mental illness and addictions* — tobacco, alcohol; with a central overlap labelled "**Tribal health: unfinished agenda + new burden**", and a fourth outer ring labelled *Genetic disorders* — sickle cell, thalassaemia, G6PD deficiency.
- ▶ **Relaxed service-delivery pyramid for tribal areas** — a triangle with, from base upward: *Tribal ASHA (hamlet-wise) → Health sub-centre / health and wellness centre — 1 per 3,000 → PHC — 1 per 20,000 → CHC — 1 per 80,000 → District hospital*, with a side arrow labelled *Mobile medical unit / mobile TB diagnostic van / mobile ophthalmic unit*.



Classification of health problems in the tribal population of India



Strategy for promoting health in the tribal population



Source: NFHS-5 (2019-21), reproduced in Park's Textbook of Preventive and Social Medicine 28e, Ch 13, Table 2, p 849.

High-yield exam pointers

- **Constitutional anchor:** Article 342 notifies, Article 366(25) defines; list is state-specific. **104 million, 705 tribes, 8.6 per cent** of India's population; ~90 per cent rural.
- **Two tribal *strengths* to quote:** sex ratio **990** vs national 943 (Census 2011), and near-universal breastfeeding — examiners reward mentioning that tribal health is not uniformly worse.
- **The four killer numbers:** TB prevalence **703** vs **256 per 100,000**; malaria ~**30 per cent** of cases, >**60 per cent** of *P. falciparum*, ~**50 per cent** of deaths from 8 per cent of the population; sickle cell anaemia **1 in 86** births in central India; hypertension in **1 of every 4** tribal adults.
- **Relaxed norms mnemonic "3-20-80":** sub-centre 1/3,000, PHC 1/20,000, CHC 1/80,000 — with shortfalls of 27 per cent (sub-centres), 40 per cent (PHCs) and 31 per cent (CHCs).
- **Committee to name:** Expert Committee on Tribal Health, **Dr Abhay Bang**, constituted October 2013 by MoHFW + Ministry of Tribal Affairs — *Bridging the Gap and a Roadmap for the Future*; its one-line thesis is "**the tribal health worker should be a local tribal**".
- **Current programme names to write:** Tribal Malaria Action Plan, National Sickle Cell Anaemia Elimination Mission, PM-JANMAN for 75 particularly vulnerable tribal groups, Scheduled Tribe Component (former Tribal Sub-Plan).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's 28th edition gives the tribal total fertility rate as 2.48 in its demographic-profile text (from the Ministry of Tribal Affairs Annual Report 2017-18) but as 2.09 in its NFHS-5 comparison table — quote the NFHS-5 value of 2.09 with its year, since the examiner expects a dated survey figure; the Revised National Tuberculosis Control Programme referred to in the textbook is now the National Tuberculosis Elimination Programme, and the Tribal Sub-Plan is now the Scheduled Tribe Component.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW, National Health Mission, National Center for Vector Borne Diseases Control, WHO Immunization Agenda 2030

Programme and policy updates (India)

- **National Sickle Cell Anaemia Elimination Mission — the screening technology has now been settled by a formal health technology assessment.** The mission covers universal screening of about 7 crore people aged 0–40 years in 17 high-prevalence, largely tribal states, with the stated vision of ending sickle cell disease as a public health problem **before 2047**. A rapid assessment under Health Technology Assessment in India compared four point-of-care tests (Hemotype Sc, Sickle Scan, Sickle CERT, Gazelle) against the solubility test: point-of-care testing detected **21,125–41,458 additional disease or trait cases** and, unlike the solubility test, separates trait from disease at first contact. The National Health Mission then directed states to procure kits by public tender, which pushed prices **below US\$1.21 per test** and projects **US\$37.86 million saved at full coverage**; strip tests are advised for door-to-door community screening by ASHAs and Gazelle at facilities, with high-performance liquid chromatography reserved for confirmation. [PMID 42594133]
- **Ayushman Bharat Health and Wellness Centres now function as Ayushman Arogya Mandir — 1,86,900 centres operational nationally**, of which 1,34,694 are sub-health centres, 24,510 primary health centres and 12,257 AYUSH centres, delivering the expanded primary-care package including non-communicable disease screening, teleconsultation, mental health, and oral, eye and ENT care. Write the current name when describing the relaxed tribal sub-centre norm of one per 3,000 — the textbook term "health and wellness centre" is superseded. [MoHFW 2026-Aug]

Current national data

- **Sickle cell screening has crossed its national target, and the programme denominator is now the quotable figure: 7,28,27,076 persons screened, 20,50,442 sickle cell trait and 2,49,717 sickle cell disease detected**, with 5,29,760 confirmations pending — that is roughly **2.8 per cent trait and 0.34 per cent disease** across the screened, largely tribal population. Quote this dated programme figure alongside the textbook community range of 1–40 per cent. [MoHFW 2026-Aug]
- **Particularly vulnerable tribal group figures are the examinable sub-set of tribal data — 75 particularly vulnerable tribal groups among the 705 Scheduled Tribes**, Odisha alone holding 13. A 2024 review of Odisha's groups found **malaria prevalence 8.1 per cent** (25 per cent in Kutia

Kandha), sickle cell 3.3 per cent, beta-thalassaemia 0.3 per cent, G6PD deficiency 3.1 per cent, 33.8 per cent of children unvaccinated and group literacy of 37.4 per cent against Odisha's 73 per cent; the 2018 National Technical Report on Tribal Health had already flagged an absolute population decline in several of these groups. [PMID 39478584]

Elimination, surveillance and disease status

- ▶ **Malaria** — give the national position with its year. India recorded 0.26 million cases, 0.15 million *Plasmodium falciparum* and 86 deaths in 2024, the annual parasite incidence falling to 0.18 per 1,000 (from 3.29 in 1995). About 80 per cent of India's malaria is still confined to the roughly 20 per cent of the population living in tribal and forested areas, so the national decline and the tribal concentration must be stated together. [NCVBDC 2024] [PMID 40600283]
- ▶ **Bastar is the worked example of why elimination by 2030 is at risk in tribal blocks.** Chhattisgarh holds 2.29 per cent of India's population but contributes about 18 per cent of malaria cases and 40 per cent of malaria deaths. Surveillance across 2015–2023 showed the annual parasite incidence falling 25 per cent a year (IRR 0.75; 95% CI 0.71–0.79), yet incidence in the tribal southern Bastar division ran 12.5 times the central region (95% CI 3.7–43.0); the *P. falciparum* share rose to 87 per cent by 2023, test positivity climbed back to 5.7 per cent, and the annual blood examination rate stayed below the programme benchmark of 10 per cent throughout. Bastar, Dantewada, Bijapur and Narayanpur are projected to fall short of the zero-case target for 2030 despite the *Malaria Mukh Bastar* test-treat-track campaign running since 2020. [PMID 40600283]
- ▶ **Kala-azar has moved close to elimination but sits in the same forest-fringe districts — 449 cases and 9 deaths in 2024**, and 198 cases in January–June 2026, down from 9,241 cases in 2014, concentrated in Bihar, Jharkhand, West Bengal and Uttar Pradesh. [NCVBDC 2026]
- ▶ **Scrub typhus is an under-recognised tribal exposure and belongs in the communicable-disease list.** A community survey in nine districts dominated by particularly vulnerable tribal groups in Odisha (2022–23, n = 1,485) found IgG seroprevalence 48.95 per cent and IgM 18.19 per cent; independent risks were sitting on mud floors (aOR 2.00), childcare (aOR 1.63), forest produce collection (aOR 1.53) and large households (aOR 1.27) — an occupational-ecological pattern that fits the One Health and acute-febrile-illness agenda. [PMID 41192195]

Guideline and legislative updates

- ▶ **RATIONS (2023) converts food into a tuberculosis-preventive intervention and is the single most quotable Indian trial for this answer.** Cluster-randomised across 28 tuberculosis units of the National Tuberculosis Elimination Programme in four districts of Jharkhand, with about two-thirds of the 10,345 household contacts from Indigenous communities (Santhal, Ho, Munda, Oraon, Bhumij) and 34 per cent undernourished. Monthly rations to contacts (750 kcal, 23 g protein with micronutrients) cut incident tuberculosis by 39 per cent for all forms and 48 per cent for microbiologically confirmed disease (adjusted HR 0.59; 95% CI 0.42–0.83); 30 households (111 contacts) need supplementation to prevent one case. This is the evidence base for pairing nutrition support with tuberculosis case-finding in tribal districts. [PMID 37567200]

Global and WHO developments

- ▶ **Immunisation Agenda 2030 still counts India among the zero-dose concentrations.** Of an estimated 15.7 million zero-dose children globally in 2023, more than half lived in just eight

countries, India among them; forecasts to 2030 suggest only DTP3 would reach the 90 per cent global coverage target, and only under an optimistic scenario. WHO's Big Catch-Up is explicitly directed at marginalised and sub-national geographies — in India that means tribal habitations, migrant families and hamlets served only by outreach sessions. [PMID 40578370]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 13 Tribal Health in India, pp 848–851; Ch 7 Health Programmes in India, pp 498–499); Sunder Lal — Textbook of Community Medicine (Ch 2, p 68; Ch 17 Healthcare of the Community, p 760); Jugal Kishore — National Health Programmes of India (Tribal Health, pp 245–247); Routine Immunization I Manual for Medical Officers 2024 (Ch 14 Routine Immunization in Special Settings, pp 314–316); Anemia Mukta Bharat Operational Guidelines (Ch 6, pp 84–85); Oxford Textbook of Public Health 7e (Ch 89 The Health of Indigenous Peoples, pp 3337–3349). Named documents: Report of the Expert Committee on Tribal Health — *Tribal Health in India: Bridging the Gap and a Roadmap for the Future* (MoHFW and Ministry of Tribal Affairs); National Family Health Survey-5 (2019–21); Census of India 2011.

Recent-advances sources: 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, *PLoS One* 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 39478584 — Health status of particularly vulnerable tribal groups (PVTGs) of Odisha: a narrative review, *Journal of Health, Population and Nutrition* 2024, doi:[10.1186/s41043-024-00671-8](https://doi.org/10.1186/s41043-024-00671-8); 40600283 — Epidemiology of malaria in Chhattisgarh, India: a surveillance data analysis, 2015–2023, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2025, doi:[10.1093/trstmh/traf072](https://doi.org/10.1093/trstmh/traf072); 41192195 — Scrub typhus among particularly vulnerable tribal groups of Odisha, India, *Journal of Infection and Public Health* 2025, doi:[10.1016/j.jiph.2025.103026](https://doi.org/10.1016/j.jiph.2025.103026) (abstract-only); 37567200 — Nutritional supplementation to prevent tuberculosis incidence in household contacts of patients with pulmonary tuberculosis in India (RATIONS), *Lancet* 2023, doi:[10.1016/S0140-6736\(23\)01231-X](https://doi.org/10.1016/S0140-6736(23)01231-X) (abstract-only); 40578370 — Global, regional, and national trends in routine childhood vaccination coverage from 1980 to 2023 with forecasts to 2030 (Global Burden of Disease Study 2023), *Lancet* 2025, doi:[10.1016/S0140-6736\(25\)01037-2](https://doi.org/10.1016/S0140-6736(25)01037-2). Programme sources: National Sickle Cell Anaemia Elimination Mission screening dashboard and mission profile, MoHFW–National Health Mission (August 2026); Ayushman Arogya Mandir portal, MoHFW (August 2026); National Center for Vector Borne Diseases Control malaria situation table (2024) and kala-azar situation table (2024, with 2026 January–June provisional).



Indian Public Health Standards (IPHS) for sub-centres

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — Indian Public Health Standards (IPHS) are the benchmark standards laid down by the Ministry of Health and Family Welfare that specify, for each level of government health facility, the services to be assured and the infrastructure, human resources, drugs, diagnostics and equipment needed to deliver them. The sub-health centre (SHC) — now the Health and Wellness Centre–Sub Health Centre (HWC-SHC), rebranded *Ayushman Arogya Mandir* — is the most peripheral facility and the first point of contact between the primary health care system and the rural community (Park 28e, Ch 10, p 728). IPHS were first issued in 2006 after the launch of the National Rural Health Mission, revised in 2012, and revised again in 2022 (Sunder Lal, Ch 17, p 737).

Objectives and framework of IPHS 2022

- ▶ To define a **uniform standard** ensuring high-quality services.
- ▶ To specify the **minimum assured** ("essential") and **achievable** ("desirable") services at each level — SHC, PHC, CHC, sub-district and district facility.
- ▶ To specify adequate workspace, human resources, drugs, diagnostics, equipment and logistic support needed to deliver those services.
- ▶ To achieve and maintain acceptable **quality of care**.
- ▶ To facilitate **monitoring and supervision** of health facilities.
- ▶ To serve as a tool for **governance, leadership and evaluation**, by identifying facility-wise gaps through facility survey and directing funds to close them (Sunder Lal, Ch 17, pp 737, 744).

IPHS 2022 covers sub-district and district hospitals, community health centres, HWC-PHC (including multispecialty urban polyclinics) and HWC-SHC, defining for each the system-strengthening domains of infrastructure, human resources, drugs, diagnostics, equipment, quality improvement, monitoring and supervision, and governance and leadership — aligned to the policy shifts of the National Health Policy 2017, Universal Health Coverage and Sustainable Development Goal 3 (Park 28e, Ch 10, p 728).

IPHS 2022 standards for the Health and Wellness Centre–Sub Health Centre

Domain	IPHS 2022 standard for HWC-SHC (rural)
Population norm	1 SHC per 5,000 population in plain areas; 3,000 in hilly / tribal / desert areas. Norms are not rigid — "time to care" within 30 minutes of travel (Park 28e, Ch 10, p 731; Sunder Lal, Ch 17, p 742, 746)
Team leader	1 Community Health Officer (CHO) / mid-level health provider — BSc Community Health, or GNM/BSc nurse, or an Ayurveda practitioner, certified through a 6-month community-health certificate programme (IGNOU or state public health university) (Sunder Lal, Ch 17, p 742)
Multipurpose workers	2 — one female (auxiliary nurse midwife, ANM) plus one male multipurpose health worker, or two ANMs; support, sanitation and security services may be hired/outsourced (Sunder Lal, Ch 17, p 746, Table 17.3)

Domain	IPHS 2022 standard for HWC-SHC (rural)
Community-level worker	ASHA (accredited social health activist) @ 1 per 1,000 population (5 per SHC), or one per habitation in tribal/hilly/desert areas, attached to the HWC team (Park 28e, Ch 10, p 731; Sunder Lal, Ch 17, p 746)
Beds	2 day-care beds with oxygen support (Sunder Lal, Ch 17, p 746, Table 17.3)
Performance standard	Outpatient load 20 patients per day ; minimum 6 hours of routine sub-centre OPD clinic by the ANM per day , with the rest of the 5,000 population covered by outreach on a fixed work schedule known to the community and the supervisors (Sunder Lal, Ch 17, pp 745-746)
Drugs	Essential medicine list of 105 medicines , free of cost (Sunder Lal, Ch 17, p 748)
Diagnostics	About 14 point-of-care tests free — haemoglobin, urine, malaria, dengue, blood sugar, HIV, syphilis, hepatitis B, salt iodine, water testing, sputum for TB, filaria, sickle cell (Sunder Lal, Ch 17, p 748)
Teleconsultation	eSanjeevani AB-HWC (doctor-to-doctor teleconsultation, giving the SHC team access to specialists and superspecialists) and eSanjeevani OPD (patient-to-doctor); all 1.5 lakh HWCs are being connected (Sunder Lal, Ch 17, p 748)
Infrastructure and branding	₹7 lakh per HWC for repair/renovation of the existing sub-centre; renovated building, Government of India branding and a national identification number (Sunder Lal, Ch 17, pp 742, 746)
Financing	100% central assistance; Government of India bears ANM and lady health visitor (LHV) salaries, rent, contingency, drugs and equipment; male health worker's salary borne by the state; untied fund ₹10,000 per annum at sub-centre level (Sunder Lal, Ch 17, p 745; Ch 16, p 704)
Supervision	Health Assistant Female (LHV) and Health Assistant Male, one supervisor per 6 SHCs ; medical officer visits at least once a month (Sunder Lal, Ch 17, p 745)
Referral	Linked to the parent PHC/CHC for diagnostics, initiation of treatment, institutional delivery, medical termination of pregnancy and emergencies after first aid (Sunder Lal, Ch 17, p 745)

Assured service package at the sub-centre

Category	Services (IPHS 2022)
Essential (minimum assured)	Reproductive, maternal, newborn, child and adolescent health and family planning services; all national health programme services for communicable and non-communicable diseases with information–education–communication and behaviour-change communication; basic curative care for common illness, free of cost; free diagnostics and essential drugs; disease surveillance under the Integrated Disease Surveillance Programme with recognition of an outbreak at the incipient stage and first response; recording and reporting of births, deaths and causes of death (Sunder Lal, Ch 17, p 745)
Desirable	AYUSH treatment as per guidelines; disinfection of drinking-water sources and promotion of sanitary latrines (Sunder Lal, Ch 17, p 745)

Category	Services (IPHS 2022)
Twelve comprehensive primary health care packages	(1) Care in pregnancy and childbirth; (2) neonatal and infant care; (3) childhood and adolescent care including immunisation; (4) family planning, contraception and other reproductive health care; (5) management of communicable diseases — national health programmes; (6) management of common communicable diseases and outpatient care for acute simple illness and minor ailments; (7) screening, prevention, control and management of non-communicable diseases; (8) care for common ophthalmic and ENT problems; (9) basic oral health care; (10) elderly and palliative care; (11) emergency medical services including burns and trauma; (12) screening and basic management of mental health ailments (Park 28e, Ch 10, p 731)
Signature new tasks at SHC level	Population empanelment with universal screening of all aged 30 years and above for hypertension, diabetes and three common cancers — oral, breast and cervical (Park 28e, Ch 10, pp 734-735). Periodicity: every year for hypertension and diabetes, once in 5 years for the three cancers; ASHA first enumerates all persons over 30 and applies the Community-Based Assessment Checklist (CBAC) — 6 variables (age, family history of non-communicable disease, waist circumference, physical inactivity, tobacco use, alcohol use) — a score above 4 marking a person high-risk and prioritised for screening (Sunder Lal, Ch 16, p 703). Monthly follow-up of every confirmed diabetic and hypertensive for treatment adherence; wellness activities — yoga, physical activity, tobacco cessation; intersectoral collaboration for nutrition (Women and Child Development), school health (Education) and vector control (urban local bodies / panchayati raj institutions) (Sunder Lal, Ch 17, pp 747-748)

Diagrams

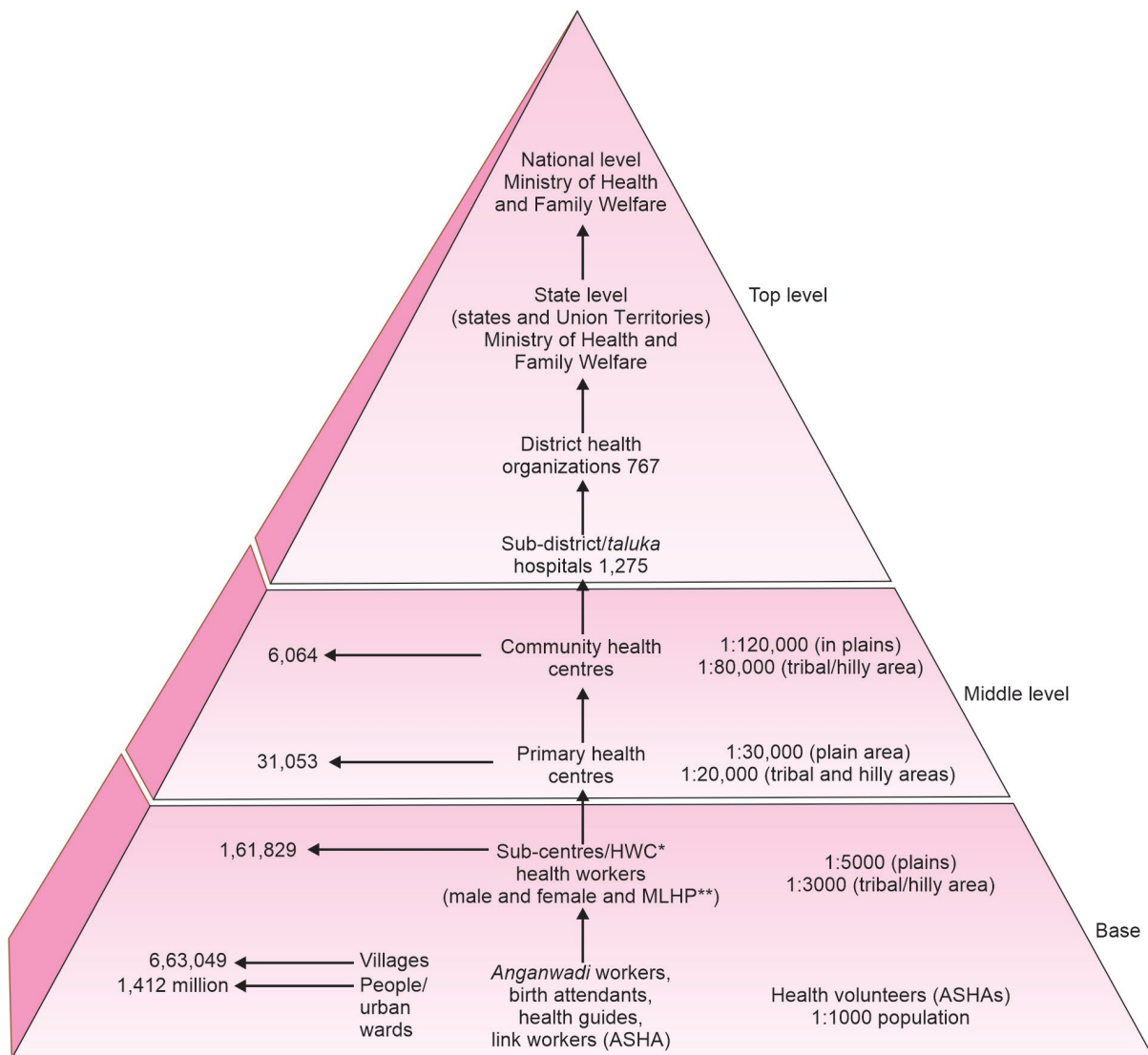
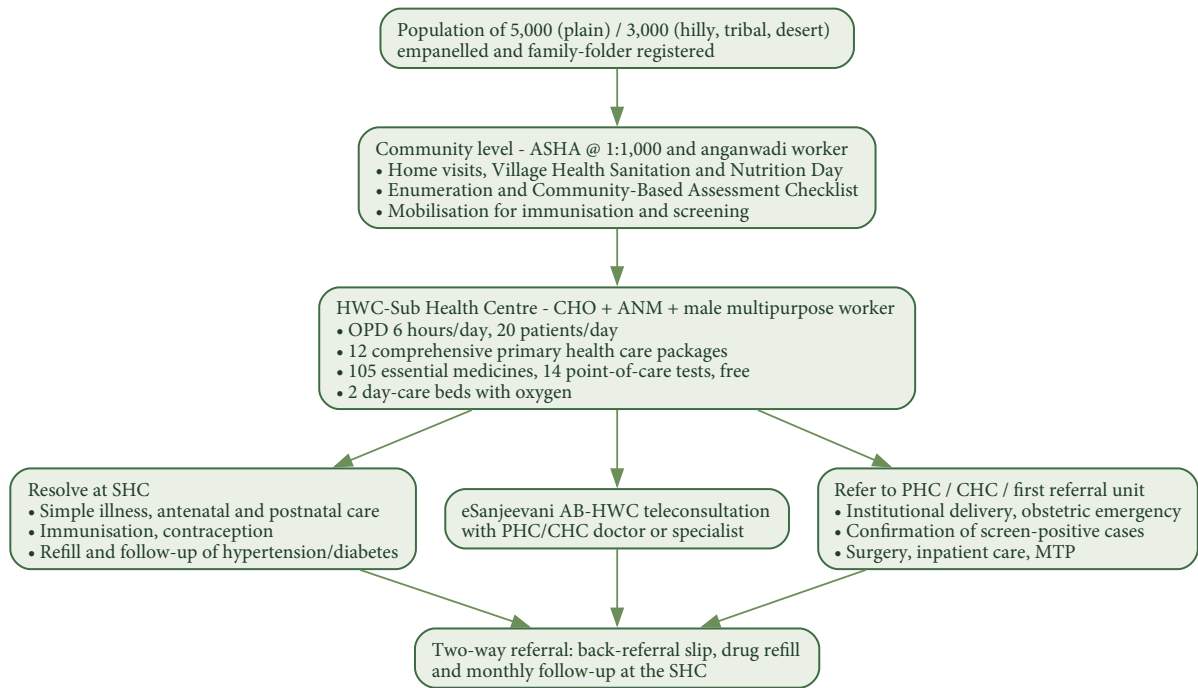
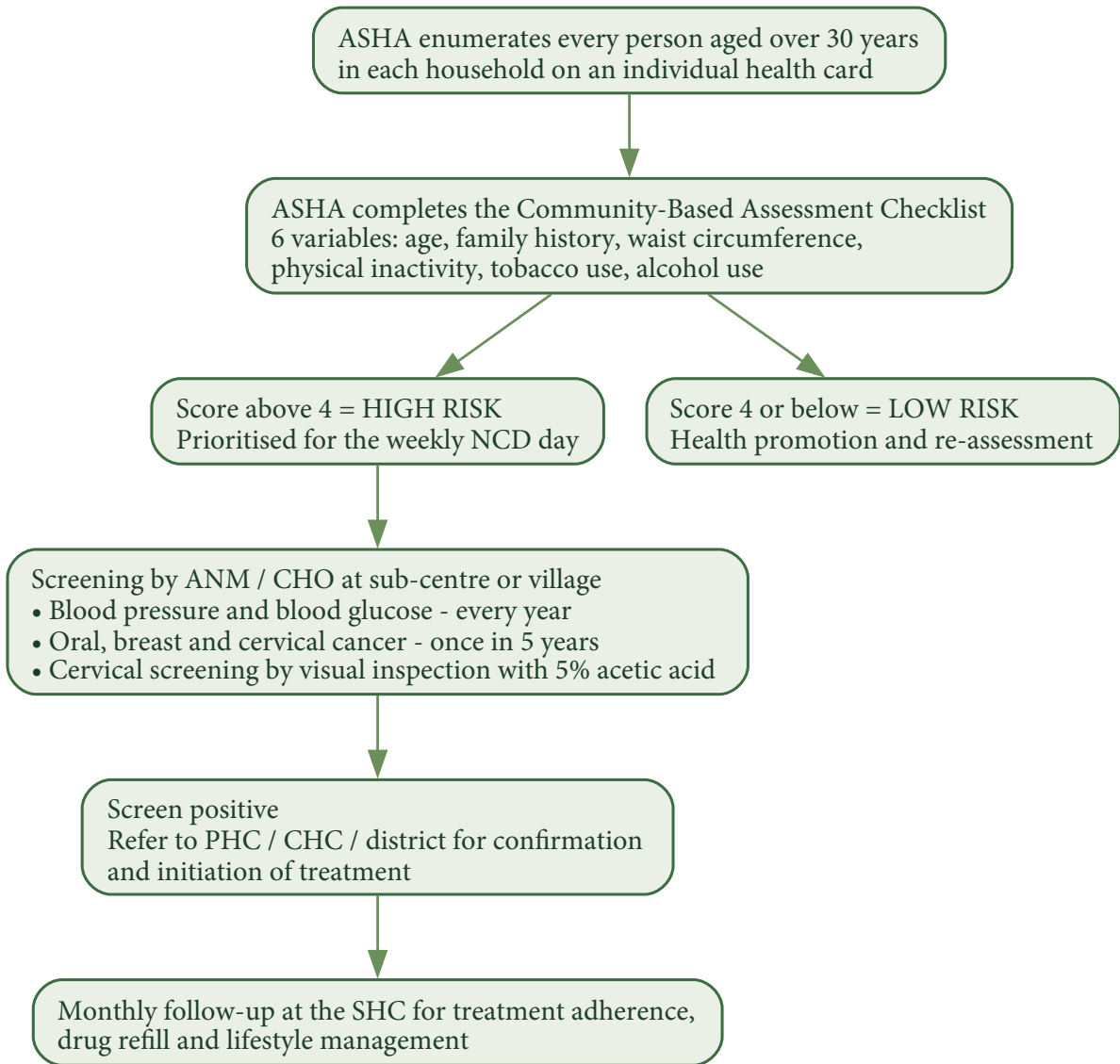


Fig 16.2 — Health system organization *HWC—Health and wellness centre, **MLHP—Mid (Sunderlal, p.709)



Service delivery and referral pathway of the Health and Wellness Centre-Sub Health Centre under IPHS 2022



Population-based non-communicable disease screening pathway at the HWC-Sub Health Centre

Recommendations to improve health care delivery by sub-centres

Over 70% of people use the private sector for outpatient care, and public hospitals take only 45.7% of rural and 35% of urban hospitalisation (National Sample Survey 75th round, 2018). The documented reasons — staff shortage, absenteeism, long waits, inconvenient clinic hours, poor outreach, weak planning and monitoring, poor quality, drug stock-outs, weak referral and supervision, and lack of community participation (Sunder Lal, Ch 17, p 766) — dictate the remedies below.

- **Close the human-resource gap, especially the male multipurpose worker.** India has 1,61,829 sub-centres (Sunder Lal, Ch 17, p 743), 1,57,935 of them functional on 31 March 2022 (Ch 17, p 745), served by 1,80,769 female but only 52,696 male health workers (Ch 4, p 146; Ch 16, p 708) — **nearly 63% of male health worker posts are vacant** (Ch 16, p 718). Fill every sanctioned post, use the National Health Mission provision for one additional contractual ANM where a male worker cannot be posted, and complete CHO placement at all HWC-SHCs.
- **Complete the transformation to Health and Wellness Centres.** Nearly 1.1 lakh of 1.5 lakh sub-health centres were transformed by March 2022 (Sunder Lal, Ch 17, p 746) — finish the rest with

renovated buildings, Government of India branding, national identification number and assured water, power and toilet.

- ▶ **Site facilities by the "time to care" approach, not by rigid population norms** — targeted rather than normative placement, to reach under-served, tribal and hard-to-reach areas (Park 28e, Ch 10, p 728; Sunder Lal, Ch 17, p 742).
- ▶ **Guarantee free drugs and diagnostics.** Keep buffer stock of the 105-medicine list and all 14 point-of-care tests; end stock-outs through pull-based indenting and logistics management.
- ▶ **Make the centre predictably available.** Enforce the fixed, publicly displayed work schedule and the annual sub-centre action plan built on a community needs assessment; hold the 6-hour OPD at hours convenient to the community (Sunder Lal, Ch 16, p 718; Ch 17, p 745).
- ▶ **Strengthen supportive supervision.** One supervisor per six sub-centres with checklist-based weekly visits, monthly medical officer visit, on-the-job continuing education, and assured supply of vaccines, drugs, contraceptives and equipment (Sunder Lal, Ch 17, p 745; Ch 16, p 717).
- ▶ **Institutionalise community participation and accountability.** Activate the Village Health Sanitation and Nutrition Committee (panchayati raj members, ANM, anganwadi worker, teacher, NGO, with ASHA as member secretary) and its ₹10,000 untied grant; hold the monthly Village Health and Nutrition Day; display sub-centre health information publicly for community-based monitoring (Sunder Lal, Ch 17, p 744; Ch 1, p 42).
- ▶ **Use flexible financing** — untied and annual maintenance grants at sub-centre level, plus the Pradhan Mantri Ayushman Bharat Health Infrastructure Mission (PM-ABHIM) and 15th Finance Commission health grants for upgradation (Sunder Lal, Ch 17, p 745).
- ▶ **Shift from selective to comprehensive primary health care.** Deliver all twelve packages, especially the non-communicable disease, elderly, palliative, oral and mental-health components previously absent at this level (Sunder Lal, Ch 17, pp 745-746).
- ▶ **Expand teleconsultation and two-way referral.** Connect every HWC-SHC to eSanjeevani, fix referral linkages with the parent PHC/CHC, and ensure back-referral with drug refill and monthly follow-up (Sunder Lal, Ch 17, p 748).
- ▶ **Embed quality assurance.** Apply the **National Quality Assurance Standards for HWC-SHC**, launched to ensure delivery of the 12 service packages (Sunder Lal, Ch 17, p 743), and the *Operational Guidelines for Improving Quality in Public Healthcare Facilities, 2021*; display a citizen's charter and conduct death audit (Sunder Lal, Ch 17, p 766).
- ▶ **Analyse service-delivery failures systematically** by system framework analysis, cause-effect (Ishikawa fishbone) diagrams, flowcharts and why-tree analysis, feeding findings back into replanning (Sunder Lal, Ch 17, pp 766-767).
- ▶ **Strengthen intersectoral convergence** with Women and Child Development for nutrition, Education for school health, and urban local bodies / panchayati raj institutions for vector control, water and sanitation (Sunder Lal, Ch 17, p 748).

High-yield exam pointers

- ▶ IPHS chronology 2006 → 2012 → 2022; 2022 edition covers SDH/DH, CHC, HWC-PHC, HWC-SHC; every service tagged **essential** or **desirable**.

- Population norms: SHC 5,000 / 3,000; PHC 30,000 / 20,000; CHC 1,20,000 / 80,000; ASHA 1:1,000; urban HWC 15,000–20,000; "time to care" ≤ 30 min.
- HWC-SHC team = 1 CHO + 1 ANM + 1 male multipurpose worker (or 2 ANMs) + 5 ASHAs; 2 day-care beds with oxygen; 20 patients/day; 6 h OPD by the ANM.
- 105 essential medicines, ≈14 free point-of-care tests, eSanjeevani AB-HWC (doctor-to-doctor) teleconsultation.
- Screening from age 30: hypertension and diabetes yearly, oral/breast/cervical cancer once in 5 years; CBAC has 6 variables, score >4 = high risk.
- Supervision 1 LHV/Health Assistant per 6 sub-centres (weekly visit); monthly medical officer visit; untied fund ₹10,000; renovation grant ₹7 lakh per HWC.
- Numbers to quote: >70% use private outpatient care, rural hospitalisation 45.7% public (NSS 75th round, 2018); 1,61,829 sub-centres, 1,80,769 female and 52,696 male health workers, ~63% male posts vacant.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Health and Wellness Centre has been renamed **Ayushman Arogya Mandir**, so either name may appear in the question. Coverage note: the retrieved corpus carries the IPHS 2022 population, staffing, drug, diagnostic and service norms in full, but not the room-by-room building layout or item-wise equipment list of the IPHS 2022 sub-centre volume, so those are not stated.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Indian Public Health Standards 2022, MoHFW Ayushman Arogya Mandir operational guidelines, ICMR, National TB Elimination Programme, WHO Global Strategy for cervical cancer elimination

Programme and policy updates (India)

- **First large-scale measurement of what a transformed sub-centre actually delivers.** A state-representative survey of 404 operational HWC-SHCs in Chhattisgarh (August–October 2023) found a mean of 358 outpatients per month, of whom 128 were non-communicable disease patients (81 hypertension, 40 diabetes). Measured against estimated population need, the HWC-SHC met 31% of the outpatient care need, 26% of hypertension care and 21% of diabetes care. Every centre had a CHO posted, 96% had at least one additional paramedical worker, and 99.2% had a functioning Jan Arogya Samiti; 58% did more than 20 teleconsultations a month but 22% did none. The weakest links were mental illness and chronic respiratory disease — clonazepam was stocked out in 91.8% and escitalopram in 94.8% of centres, so package 12 (mental health) exists on paper more than in the drug cupboard [PMID 39367295]
- **Community verdict on the SHC-to-HWC upgrade, and the national target.** Fourteen focus group discussions across Raipur, Kanker and Kondagaon found the single biggest perceived gain was the widened range of curative services, with hypertension and diabetes screening and follow-up named as the key value addition; medicines and point-of-care tests were felt to be reliably available. Services people still could not get at the centre were suturing of injuries, dental care,

mental health, emergency care, epilepsy, chronic respiratory disease and sickle cell disease. The paper records that the national target of operationalising 1,15,000 HWCs by upgrading sub-health centres was met by 2023 [PMID 38504174]

Current national data

- **Sub-centres and CHCs fall below the IPHS-benchmarked readiness threshold; PHCs clear it.** An ICMR-NCDIR cross-sectional survey of 415 facilities in 19 districts of 7 states (2021 and 2023), scored against IPHS 2012/2022 using the WHO Service Availability and Readiness Assessment method, set $\geq 70\%$ as the cut-off for calling a facility "prepared" for hypertension and diabetes care. Overall readiness was 61% for sub-centres and 59% for CHCs — both below that cut-off — while PHCs reached 73% and so cleared it; the study's conclusion is that PHCs were better prepared for diabetes and hypertension care than sub-centres, CHCs and sub-district hospitals. Medicines were the worst domain: of 105 sub-centres, stock-outs were reported for atenolol 68.5%, enalapril 66.7%, glimepiride 50.5%, amlodipine 44.8% and metformin 35.2%, with median stock-out lasting one to seven months. Only 52.3% of the sub-centres surveyed had actually been converted to HWCs, against 64% of PHCs — the transformation is not the same thing as the readiness [PMID 40536370]
- **Out-of-pocket spending, dated.** A 2025 systematic review puts out-of-pocket expenditure at 47.1% of total health expenditure in 2019-20, down from 69.4% in 2004-05, with roughly 90 million Indians crossing the catastrophic threshold (health spending above 10% of household consumption) and pharmacies accounting for about 43% of all out-of-pocket spending. This is the sharpest available justification for the 105-medicine free drug list and the free point-of-care tests at sub-centre level [PMID 40552236]

Elimination, surveillance and disease status

- **Tuberculosis — the deadline and the notification gap.** India's declared target under the National Tuberculosis Elimination Programme is elimination by 2025, five years ahead of the global SDG target of 2030; 2.59 million cases were notified in 2021. Because 60–70% of symptomatic patients first seek care in the private sector, under-notification remains the central problem, and NTEP now contracts Patient-Provider Support Agencies to bring private providers into diagnosis, notification and treatment support — the SHC's role shifts toward sputum collection, contact tracing and adherence follow-up rather than being the sole entry point [PMID 41941727]
- **Lymphatic filariasis is still endemic in 345 districts.** Despite mass drug administration since the early 2000s, the programme carries a continuing load of MDA, morbidity management and disability prevention, and post-MDA surveillance; public-private partnership is being proposed to sustain these toward the 2030 elimination goal. MDA drug distribution and lymphoedema morbidity registers sit squarely in the sub-centre team's workload [PMID 41951149]

Guideline and legislative updates

- **Task-shifting oral cancer screening to the CHO now has accuracy data.** A quasi-experimental study trained 46 CHOs at Ayushman Arogya Mandirs in Jodhpur for one day in oral visual examination; over 8 months they screened 209 people, and against ENT specialist confirmation their examination achieved sensitivity 83.87% (95% CI 70.92–96.82), specificity 98.31% (95% CI 96.42–100) and overall accuracy 96.17%. Mean knowledge score rose from 5.93 to 9.46 out

of 10 after training but fell back to 7.60 at two months ($P < 0.001$) — evidence that the IPHS oral-cancer screening mandate is deliverable at sub-centre level *provided* refresher training is built into the schedule [PMID 40341602]

- **The digital layer of IPHS 2022 is the weakest implemented part.** An assessment of **nine digital health interventions** at HWCs in Chandigarh found eVIN (86%) and the RCH portal (80%) effectively implemented, but eSanjeevani only "fairly" implemented at 59% and used by just 12.5% of 120 health workers, with 18.3% of clients aware of it; the HWC portal (38%) and the CPHC-NCD application (39%) were poorly implemented. **89.2% of workers said dual paper-plus-digital entry increased their workload** and 53% reported poor internet. Teleconsultation is an IPHS standard, not yet a routine [PMID 40854229]

Global and WHO developments

- **HPV vaccine enters the national programme.** A 2026 regional review records that **India launched a national HPV vaccination programme in 2026** — a voluntary, free, single-dose quadrivalent schedule for roughly 1.15–1.2 crore adolescent girls aged 14, delivered through schools and government facilities, following the earlier Punjab and Sikkim school-based programmes of 2016 (~97% coverage) and the licensing of the indigenous quadrivalent vaccine Cervavac. **Ten of the eleven WHO South-East Asia Region countries** have now introduced HPV vaccination nationally; the target framework remains the WHO 90–70–90 by 2030 cervical cancer elimination strategy. This adds an adolescent-girl cohort to the sub-centre's immunisation and outreach schedule [PMID 42518306]
- **What actually converts primary care into universal health coverage.** A realist review of 42 studies from the countries with the largest gains in the UHC service coverage index between 2000 and 2021 (India contributed 13 studies, the largest share) identified three working mechanisms: **a frontline health workforce that is locally positioned, adequately equipped, appropriately skilled and therefore well motivated; community formations empowered to provide navigation and address access barriers for care-seekers; and implementers, providers and community members aligned in viewing health as an entitlement extending beyond health services into nutrition, agriculture and transport** — precisely the CHO–ANM–male worker–ASHA team backed by the Jan Arogya Samiti and Village Health Sanitation and Nutrition Committee, plus the intersectoral convergence the IPHS asks of the sub-centre. The review's own caution is that PHC reforms still fell short in reaching marginalised groups, cutting financial barriers and improving service quality — equity gaps persisted even in the best-performing countries [PMID 42293679]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10, pp 728-735); Sunder Lal — Textbook of Community Medicine (Ch 17, pp 737, 742-748, 766-767; Ch 16, pp 703-704, 708, 717-718; Ch 4, p 146; Ch 1, p 42). Named documents cited therein: Indian Public Health Standards 2022 — Health and Wellness Centre–Sub Health Centre (Ministry of Health and Family Welfare); Ayushman Bharat Comprehensive Primary Healthcare through Health and Wellness Centres, Operational Guidelines; National Health Policy 2017; National Quality Assurance Standards for HWC-SHC; Operational Guidelines for Improving Quality in Public Healthcare Facilities, 2021; National Sample Survey 75th Round (2018); PM-ABHIM and 15th Finance Commission health grants.

Recent-advances sources: 39367295 — Performance of health and wellness centre in providing primary care services in Chhattisgarh, India, BMC Prim Care 2024, doi:[10.1186/s12875-024-02603-1](https://doi.org/10.1186/s12875-024-02603-1); 38504174 — How useful do communities find the health and wellness centres? A qualitative assessment of India's new policy for primary health care, BMC Prim Care 2024, doi:[10.1186/s12875-024-02343-2](https://doi.org/10.1186/s12875-024-02343-2); 40536370 — Preparedness of public & private health facilities for management of diabetes & hypertension in 19 districts in India, Indian J Med Res 2025, doi:[10.25259/IJMR_755_2024](https://doi.org/10.25259/IJMR_755_2024); 40552236 — Understanding out-of-pocket expenditure in India: a systematic review, Front Public Health 2025, doi:[10.3389/fpubh.2025.1594542](https://doi.org/10.3389/fpubh.2025.1594542); 41941727 — Impact and Economic Evaluation of the Patient-Provider Support Agency Model Under India's National Tuberculosis Elimination Program: Protocol for a Cohort Study, JMIR Res Protoc 2026, doi:[10.2196/76302](https://doi.org/10.2196/76302); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, Acta Trop 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 40341602 — Capacity building of community health officers for optimizing the screening and early diagnosis of oral cancer in Jodhpur, Rajasthan, India, Sci Rep 2025, doi:[10.1038/s41598-025-00225-w](https://doi.org/10.1038/s41598-025-00225-w); 40854229 — Implementation Status and Usability of Digital Health Interventions Among Health Care Workers and End Users at the Primary Health Care Level in Chandigarh, North India: Cross-Sectional Study, JMIR Form Res 2025, doi:[10.2196/69824](https://doi.org/10.2196/69824); 42518306 — HPV vaccination in the WHO South-East Asia Region: A scoping review of uptake, implementation challenges, and pathways to acceleration, Hum Vaccin Immunother 2026, doi:[10.1080/21645515.2026.2693359](https://doi.org/10.1080/21645515.2026.2693359); 42293679 — Understanding pathways from primary health care to universal health coverage outcomes: a realist review, Front Public Health 2026, doi:[10.3389/fpubh.2026.1808891](https://doi.org/10.3389/fpubh.2026.1808891).

♦ ♦ ♦

Health challenges of tribal population in India

Asked in: Paper III Nov 2018 — RGUHS MD Community Medicine

Definition — The term "Scheduled Tribes" (ST) first appeared in the Constitution of India; Article 366(25) defines them as those tribes or tribal communities, or parts of or groups within them, that are deemed under **Article 342** to be Scheduled Tribes for the purpose of the Constitution. The list is **state/union-territory specific** — a community notified as a scheduled tribe in one state need not be so in another. Over **104 million** tribal people live in India across **705 tribes**, accounting for **8.6 per cent** of the country's population, and have remained marginal geographically, socio-economically and politically, so that health and health care in tribal areas remain an unsolved problem (Park 28e, Ch 13, p 848).

Profile of the tribal population — the denominator

Parameter	Scheduled tribes	All-India comparison
Population	104 million across 705 tribes; 8.6% of national population (Census 2011)	—
Residence	Nearly 90% rural, in hilly and forested areas	—
Distribution	Largest numbers in Madhya Pradesh (15 million), Maharashtra (10 million), Odisha (9 million), Rajasthan (9 million); highest concentration in the 8 North-East states	Concentrated in 10 states + North-East
Sex ratio (2011)	990 — better than the general population	943
Child sex ratio, 0–6 y	957 (2011), fallen from 972 (2001)	914
Literacy (2011)	59.0% overall; male 68.5%, female 49.4%	73.0%; male 80.9%, female 64.6%
Life expectancy at birth	63.9 years	67 years
Total fertility rate (NFHS-5, 2019-21)	2.09	1.99

(Park 28e, Ch 13, pp 848–849, Tables 1 and 2; life expectancy from Park 27e, Ch 13, p 821.)

Challenge 1 — the triple burden of disease

Epidemiological transition is under way in tribal areas as in the rest of the country, so the tribal population carries a **triple burden**: malnutrition and communicable disease remain rampant, rapid urbanisation, environmental distress and changing lifestyles have raised non-communicable disease, and mental illness — especially addiction — is added on top (Park 28e, Ch 13, p 848).

Burden	Key facts to quote
1. Communicable disease — tuberculosis, malaria, leprosy, sexually transmitted infections, HIV/AIDS, skin infections, diarrhoeal disease, hepatitis	Tuberculosis: estimated prevalence of pulmonary tuberculosis 703 per 100,000 in the tribal community against 256 per 100,000 in the rest of the country. Malaria: tribal communities are ~8% of the population but account for ~30% of all malaria cases, >60% of <i>P. falciparum</i> infections and ~50% of malaria mortality; <i>P. falciparum</i> predominates and drug resistance is a problem. Leprosy: of 1,35,485 new cases detected in 2016-17, 25,474 (18.90%) were scheduled tribes (Park 28e, Ch 13, pp 849–850; Park 27e, Ch 13, p 825)
2. Non-communicable disease — early epidemiological transition	Hypertension in one of every four tribal adults; prevalence rises with age, tobacco, alcohol and sedentary living; two of three tribal adults did not know the signs and symptoms , and only 5% of men and 9% of women with hypertension knew their hypertensive status. Cancer and diabetes are rising. Blindness and visual impairment are a specific gap in the tribal-predominant North-East (Park 28e, Ch 13, p 850)
3. Mental health and addiction	72% of tribal men aged 15–54 years use tobacco versus 56% of non-tribal men; about 50% of tribal men consume alcohol — reducing productivity, increasing poverty, disrupting family harmony and generating law-and-order problems (Park 28e, Ch 13, pp 849–850)
Superadded: genetic (haemoglobinopathy) burden	Sickle cell disease (anaemia + trait) and thalassaemia together vary from 1% to 40% across tribal communities, mostly heterozygous; sickle cell anaemia occurs in 1 in 86 births among tribal communities of central India. G6PD deficiency 0.7–15.6% among 14 primitive tribal populations of four states (Park 28e, Ch 13, p 850)
Superadded: injury and violence	Tribal habitations abut forests — snake, dog and scorpion bites are common ; violence in conflict-affected areas (Park 28e, Ch 13, p 850)

Challenge 2 — the maternal, child and nutritional gap (NFHS-5, 2019-21)

Indicator	Scheduled tribe	All population
Infant mortality rate (per 1,000 live births)	41.7	35.2
Neonatal mortality rate	28.8	24.9
Post-neonatal mortality rate	12.9	10.3
Under-5 mortality rate	50.3	41.9
Child (1–4 y) mortality rate	9.0	6.9
Infants fully immunised (%)	76.8	76.6
No vaccination at all (%)	4.1	3.6
Hepatitis B, 3rd dose (%)	81.2	88.6
Pregnant women receiving tetanus toxoid (%)	79.0	85.5

Indicator	Scheduled tribe	All population
Under-5 children stunted (%)	40.9	35.5
Under-5 children wasted (%)	23.2	19.3
Under-5 children underweight (%)	39.5	32.1
Institutional delivery (%)	68.0	88.6
Deliveries by skilled health personnel (%)	71.5	89.4
Couples using any contraceptive (%)	64.4	66.7

(Park 28e, Ch 13, p 849, Table 2 — comparison of MCH indicators in ST and all population, 2019-21.)

No separate recent maternal mortality estimate exists for tribal women; **early marriage, early childbirth, low body mass index and high anaemia** are the known drivers. Postnatal coverage is the worst link — only about 37% of tribal women received any postnatal care within 48 hours of a home delivery (Park 27e, Ch 13, p 821).

Challenge 3 — why the delivery system itself fails in tribal areas

Domain	Barrier
Geographic / physical	Difficult terrain; sparse, dispersed population in forest and hilly regions; locational disadvantage of sub-centres, PHCs and CHCs ; no suitable transport for quick referral of emergencies; high mobility, forest-based economy, outdoor sleeping habits (Sunder Lal, Ch 17, p 760; Park 28e, Ch 7, p 498)
Infrastructure shortfall	Even against the relaxed tribal norms , in about half the states institutions are deficient by 27–40%: shortfall of 4,996 sub-centres (27%) in 11 states, 1,023 PHCs (40%) in 7 states, 209 CHCs (31%) in 10 states (Park 28e, Ch 13, pp 850–851)
Human resource	Vacant posts and no residential facility; limited scope for professional interaction or growth, a feeling of social and professional isolation , weak human-resource policies, poor working conditions, limited social infrastructure. Compulsory rural-service bonds are flouted by most doctors completing MBBS and are not enforced (Park 27e, Ch 13, p 823)
Cultural and behavioural	IEC not tuned to tribal vocabulary, beliefs and practices ; services not client-friendly in timings; traditional faith healers not involved ; culturally inappropriate or unsafe care even where care exists (Sunder Lal, Ch 17, p 760; Oxford 7e, Ch 89, pp 3343–3344)
Programme and management	Poor integration with other health programmes; inadequate mobilisation of non-governmental organisations (Sunder Lal, Ch 17, p 760)
Socio-economic	Poverty, illiteracy (ST female literacy 49.4%), displacement, out-of-pocket expenditure, patients pushed to multiple providers; persisting gap in human development indicators between STs and all categories despite the Scheduled Tribe Component (Park 28e, Ch 13, pp 849–850)

Modifications suggested in the public health care delivery system

The Ministry of Health and Family Welfare with the Ministry of Tribal Affairs constituted an **Expert Committee on Tribal Health in October 2013 under the chairmanship of Dr. Abhay Bang**, whose report "*Tribal Health in India — Bridging the Gap and a Roadmap for the Future*" is the reference

framework; its central principle is to **place the centre of gravity of the workforce not at the top (specialists and doctors) but closer to the communities** (Park 28e, Ch 13, p 851).

A. Re-design the workforce around local tribal people

- ▶ The health care provider should, **as far as possible, be a local tribal** — the present workforce pattern is the opposite of this. The only sustainable way to a vibrant, responsive and accessible tribal health workforce is to **train and deploy local tribal people**.
- ▶ Give the **ASHA in tribal areas an expanded role** — **eight types of function and a total of four hours of work per day**, with **hamlet-wise rather than village-wise ASHAs**.
- ▶ Create **mid-level care providers through bridge courses and place them at sub-centres**.
- ▶ **Substantially increase the total salary** of medical officers to attract doctors to tribal areas, over and above enhanced vehicle-maintenance and travel allowances.
- ▶ Create **dedicated medical colleges in tribal districts, exclusively for tribal students** in the scheduled areas — a "whole pathway" approach, since ethnic concordance between patient and provider improves communication, shared decision-making, trust and satisfaction, and reduces perceived racism (Park 28e, Ch 13, p 851; Oxford 7e, Ch 89, p 3349).

B. Bring the facility to the population, not the population to the facility

- ▶ **Close the 27–40% institutional shortfall** against the tribal norms of one sub-centre per 3,000, one PHC per 20,000 and one CHC per 80,000 population.
- ▶ Stop siting facilities on **rigid population norms** — use the "**time to care**" approach, **siting sub-centres and PHCs within 30 minutes of travel time** (Sunder Lal, Ch 17, p 748).
- ▶ Scale **doorstep diagnostics through mobile units: the Mobile TB Diagnostic Van (MTDV)** with X-ray and sputum-microscopy facilities — 35 vans across 17 districts of 5 states (Madhya Pradesh, Gujarat, Chhattisgarh, Rajasthan, Jharkhand), covering about **17.65 million people**; **multipurpose district mobile ophthalmic units** under the blindness programme; and **pre-scheduled, well-publicised mobile health services** in areas with civic disturbance (Park 28e, Ch 13, pp 849–850; Ch 7, p 499).
- ▶ Provide **residential quarters and referral transport**, without which posts stay vacant and emergencies stay unreferred.

C. Tune service content to tribal epidemiology

- ▶ **Active, campaign-mode case finding** — map tribal districts as a vulnerable population and conduct **door-to-door tuberculosis screening**; provide **additional TB units and Designated Microscopy Centres in tribal/difficult areas**, and **compensate patient and attendant transport**.
- ▶ **Universal newborn/community screening for sickle cell disease, thalassaemia and G6PD deficiency**, with counselling — justified by a 1–40% community prevalence and 1 in 86 births affected in central India.
- ▶ **Opportunistic hypertension screening at every contact**, since only 5% of men and 9% of women with hypertension know their status.
- ▶ Deliver malaria control as a tribal-specific package — the **Tribal Malaria Action Plan (TMAP)** — with early diagnosis and radical treatment, indoor residual spray, long-lasting insecticidal

nets, larval source management, and **regular, efficient supply-chain management** (Park 28e, Ch 7, pp 496–499).

D. Make the system culturally acceptable and community-owned

- ▶ Where an ASHA is not feasible, **train and equip anganwadi workers of ICDS, traditional faith healers, local health care providers, village headmen, Panchayati Raj members or school teachers** to diagnose and treat malaria, and **involve the forest department** in diagnosis, treatment and communication inside forest areas.
- ▶ Run **behaviour change communication using traditional tools and practices** and in the tribal vocabulary, with community mobilisation through **community volunteers** and **locally available, credible non-governmental organisations** (already funded under the leprosy programme for IEC, deformity prevention and follow-up).
- ▶ Shift clinic timings to be **client-friendly** for a population that works in forests and fields.

E. Governance, financing and convergence

- ▶ Use the **Scheduled Tribe Component (STC)**, formerly the **Tribal Sub-Plan** adopted in the Fifth Five Year Plan, to earmark funds, with **convergence between the Ministry of Health and Family Welfare and the Ministry of Tribal Affairs**.
- ▶ Continue **differential financing** — 100% central assistance to North-Eastern states, 90:10 central share for NE states under the blindness programme against 60:40 elsewhere, and externally aided support (Global Fund, World Bank) for malaria and kala-azar.
- ▶ Maintain **disaggregated ST reporting** in every programme — the leprosy programme already collects state-wise tribal data monthly — so that inequity is visible and correctable.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **DRAWN figure** — "The triple burden in tribal India" (draw three concentric or stacked bands). Outer band label: **Communicable disease and malnutrition** (tuberculosis 703/100,000; ~30% of India's malaria; stunting 40.9%). Middle band: **Non-communicable disease** (hypertension in 1 of 4 adults; cancer; diabetes; blindness). Inner band: **Mental illness and addiction** (tobacco 72% of men aged 15–54; alcohol ~50%). Add a side box: **Superadded** — genetic disorders (sickle cell, thalassaemia, G6PD deficiency) and animal-bite injury.

HEALTH CHALLENGES OF THE TRIBAL POPULATION
104 million, 705 tribes, 8.6% of India (Census 2011)

A. TRIPLE BURDEN OF DISEASE

1. Communicable — TB 703 vs 256 per 100,000; ~30% of malaria cases, >60% *P. falciparum*, ~50% malaria deaths; leprosy 18.9% of new cases
2. Non-communicable — hypertension in 1 of 4 adults; cancer; diabetes; blindness
3. Mental illness and addiction — tobacco 72% of men 15-54 y; alcohol ~50%

B. SUPERADDED BURDENS

- Genetic — sickle cell + thalassaemia 1-40%; sickle cell anaemia 1 in 86 births (central India); G6PD deficiency 0.7-15.6%
- Injury — snake, dog and scorpion bites; violence in conflict areas

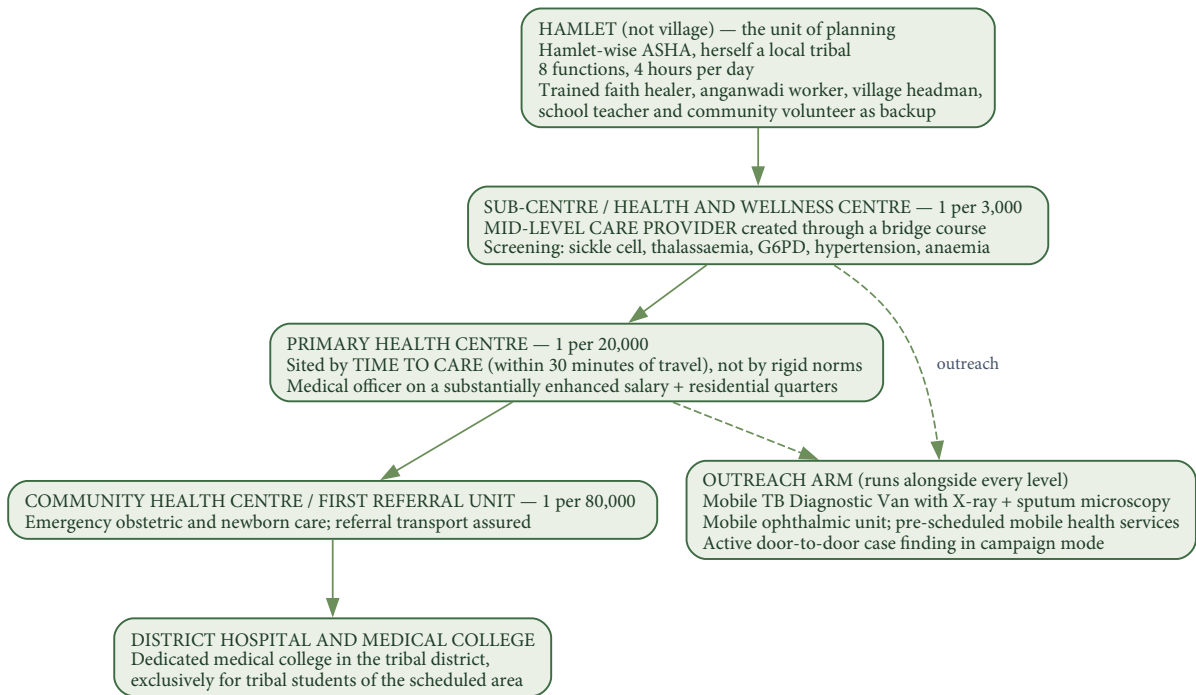
C. MATERNAL, CHILD AND NUTRITIONAL GAP (NFHS-5, 2019-21)

- IMR 41.7 vs 35.2; under-5 mortality 50.3 vs 41.9 per 1000 live births
- Stunting 40.9% vs 35.5%; wasting 23.2% vs 19.3%; underweight 39.5% vs 32.1%
- Institutional delivery 68.0% vs 88.6%; postnatal care within 48 h only ~37%

D. HEALTH SYSTEM AND SOCIAL BARRIERS

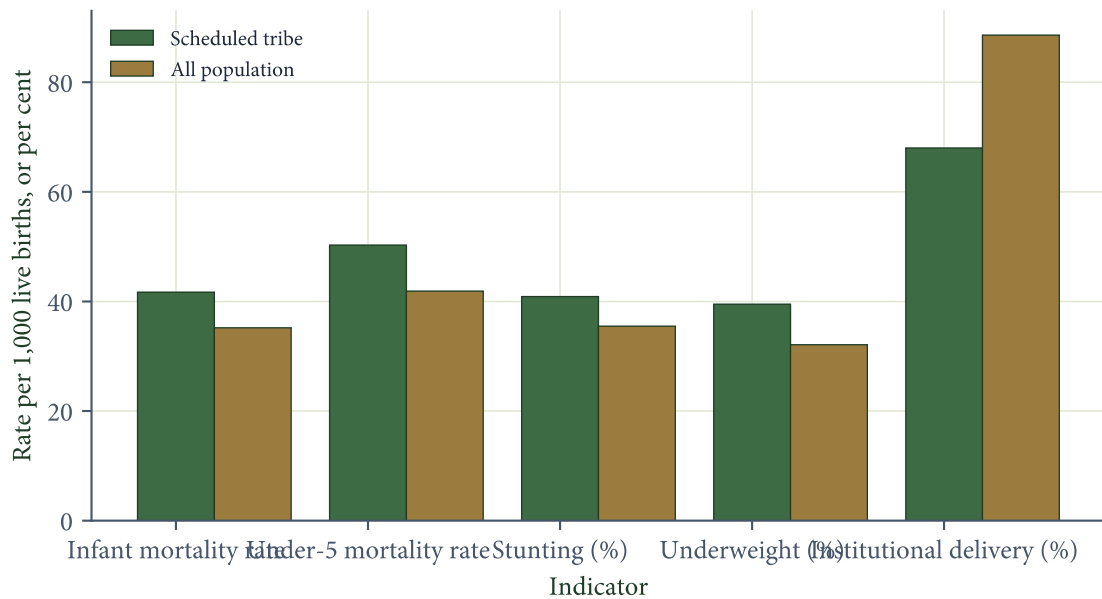
- Terrain, dispersed hamlets, locational disadvantage of HSC/PHC/CHC
- Shortfall of 4,996 sub-centres, 1,023 PHCs, 209 CHCs (27-40%)
- Vacant posts, professional isolation, bonds flouted
- IEC not in tribal vocabulary; faith healers excluded; timings unfriendly

Classification of the health challenges of the tribal population of India



Modified public health care delivery pathway for tribal areas, from hamlet to district, as recommended by the Expert Committee on Tribal Health

Scheduled tribe versus all-India health indicators, NFHS-5 (2019-21)



Comparison of selected maternal, child and nutritional indicators between the scheduled tribe and all-India populations. Source: NFHS-5 (2019-21), reproduced as Table 2 in Park's Textbook of Preventive and Social Medicine 28e, Ch 13, p 849.

High-yield exam pointers

- **Open with the constitutional definition:** Article 366(25) read with Article 342; the list is **state-specific**. Then the denominator — **104 million, 705 tribes, 8.6%** (Census 2011), **90% rural**.
- **The three numbers that carry the disease-burden mark:** tuberculosis **703 vs 256 per 100,000**; malaria — **8% of the population, ~30% of cases, >60% of *P. falciparum*, ~50% of deaths**; hypertension in **1 of 4 adults** with only **5% of men / 9% of women** aware of their status.

- **Relaxed tribal norms (always write all three):** sub-centre 1:3,000, PHC 1:20,000, CHC 1:80,000 — against 1:5,000 and 1:30,000 in the plains, and **one ASHA per habitation** in tribal, hilly and desert areas instead of one per 1,000 population; yet **shortfall of 4,996 sub-centres (27%), 1,023 PHCs (40%), 209 CHCs (31%).**
- **Name the committee:** Expert Committee on Tribal Health, MoHFW + Ministry of Tribal Affairs, October 2013, chaired by **Dr. Abhay Bang** — report *"Tribal Health in India: Bridging the Gap and a Roadmap for the Future"*.
- **Its one-line principle, worth a mark on its own:** *place the centre of gravity of the workforce not at the top — the specialists and doctors — but closer to the communities, and the provider should as far as possible be a local tribal.*
- **Four workforce asks:** expanded tribal ASHA (8 functions, 4 hours/day), mid-level provider via bridge course at the sub-centre, substantially higher medical-officer salary, dedicated medical colleges in tribal districts for tribal students.
- **Dated NFHS-5 gaps to quote:** IMR 41.7 vs 35.2, under-5 mortality 50.3 vs 41.9, stunting 40.9% vs 35.5%, institutional delivery 68.0% vs 88.6%.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam — the Tribal Sub-Plan is now called the **Scheduled Tribe Component**, the Revised National Tuberculosis Control Programme (RNTCP) named in the retrieved text is now the **National Tuberculosis Elimination Programme**, sub-centres and PHCs are being converted into **Ayushman Arogya Mandirs** (formerly Ayushman Bharat Health and Wellness Centres), and a **National Sickle Cell Anaemia Elimination Mission** aimed at tribal districts is referred to in the corpus only as "the sickle cell anaemia elimination programme has been launched" (Sunder Lal, Ch 17, p 760) without its targets, so quote it by name but do not attach numbers to it. Coverage note — the retrieved corpus supports the Expert Committee's workforce and infrastructure recommendations in detail but carries no tribal-specific maternal mortality ratio (Park explicitly states no separate recent estimate exists) and no separate tribal health index, so neither should be quoted.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, PIB India, MoHFW, NCVBDC, WHO

Programme and policy updates (India)

- **National Sickle Cell Anaemia Elimination Mission** — announced in the Union Budget for 2023-24, it sets **elimination of sickle cell disease by 2047** with universal screening of the **0–40 year** population of endemic, largely tribal districts, about seven crore people in all. The Prime Minister launched it from **Shahdol, Madhya Pradesh, in July 2023** [PIB 2023-Jul]. A health-technology assessment of point-of-care screening found the POC kits dearer per case screened than the existing solubility-plus-HPLC standard (US\$2.57–3.85 versus US\$0.67 per case screened; US\$157.92–230.73 versus US\$42.53 per case detected) but detecting an extra **21,125–41,458 cases**; on that evidence the **National Health Mission directed states to procure POC kits by public tender**, and competitive bidding with in-house manufacture brought the price below **US\$1.21 per test**, worth about US\$37.86 million in savings at full coverage. [PMID 42594133]

- ▶ **Two new tribal-specific missions have been added since the textbook edition.** PM-JANMAN (Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan), launched on Janjatiya Gaurav Divas, 15 November 2023 with an outlay of Rs 24,104 crore, targets the **75 Particularly Vulnerable Tribal Groups** with housing, piped water, anganwadi services and **mobile medical units**. **Dharti Aaba Janjatiya Gram Utkarsh Abhiyan**, launched 2 October 2024 with an outlay of Rs 79,156 crore, extends the same saturation approach through 25 interventions run by 17 ministries across **63,843 tribal villages in 549 districts**. Sub-centres and PHCs upgraded under Ayushman Bharat are now called **Ayushman Arogya Mandirs** (renamed November 2023) — write the current name in the exam. [PIB 2023-Nov]; [PIB 2024-Oct]; [MoHFW 2023-Nov]
- ▶ **The Tribal TB Initiative is the named tribal component of the National Tuberculosis Elimination Programme.** The Ministry of Tribal Affairs with the MoHFW released a **Joint Action Plan in 2020** and launched the initiative, under which a **100-day active case-finding drive across all 174 tribal districts** yielded roughly **10,000 additional TB cases**; a Tribal Action Plan has run since **2005**, and notified patients in tribal areas receive a one-time Rs 750 under the Tribal Support Scheme over and above the **Rs 500 per month** direct benefit transfer paid to every TB patient. The community-volunteer route the Expert Committee recommended is now being formally tested — **708 volunteers** from student and women's organisations across **189 villages** of Senapati district, Manipur were trained (2023–2025), of whom **339 were retained through the full training and assessment**; the predicted probability of sitting in the highest TB-knowledge band rose from **10% to 65%** (and in the lowest two bands fell from 61% to 8%), with improved intention to seek care within 1–2 weeks ($p = 0.0009$). [PMID 37202934]; [PMID 42599962]

Current national data

- ▶ **Mission screening yield to 31 July 2025:** over **6.07 crore people screened across 17 states**, about **2.15 lakh found to have sickle cell disease** and about **16.7 lakh carriers**, with 2.6 crore health cards issued, against a screening target of 7 crore. Roughly **95 per cent of cases sit in five states** — Odisha, Madhya Pradesh, Gujarat, Chhattisgarh and Maharashtra. This is the dated figure to quote alongside Park's "1 in 86 births". [PIB 2025-Jul]
- ▶ **Both headline burden ratios in the textbook have been revised upward by newer national work.** An ICMR study puts TB prevalence among tribal populations at **432 per 100,000** against **296 per 100,000** in the general population, with tribal communities contributing a **quarter of self-reported cases** and **9% of all notified patients** and **63% of the 170 tribal districts** exceeding the national average; the **Saharia**, a particularly vulnerable tribal group of Madhya Pradesh and Rajasthan, carry the country's heaviest burden at **1,504 per 100,000 against a national 217**. A district-wise NVBDCP survey attributes **47% of all malaria cases**, **70% of falciparum cases** and **46% of malaria deaths** to tribal communities forming about 8% of the population, with **over 90% of reported cases in eight states** — Odisha, Jharkhand, Chhattisgarh, Meghalaya, Arunachal Pradesh, Mizoram and the tribal districts of Madhya Pradesh and Maharashtra. [PMID 37202934]
- ▶ A community-based cohort followed fortnightly at home in five endemic districts found **223 of 252 patients (88.5%) had at least one vaso-occlusive crisis in a year** — **8,404 episodes per 1,000 patient-years** (95% CI 8,040–8,760), median four days each — yet only **48.7% of episodes were treated at any health facility**. The old teaching that Indian sickle cell disease runs mild does not survive this data [PMID 42024909]. Among tribal women with the disease, anaemia (**43% vs**

12.5%), pain crises (42% vs 11.5%) and transfusion during pregnancy (16% vs 1%; all $p < 0.001$) were commoner than in matched controls, and only 10% had used hydroxyurea periconceptionally [PMID 41834271].

Elimination, surveillance and disease status

- **Tuberculosis is now framed as an undernutrition problem in tribal India.** A meta-analysis put pooled malnutrition among tribal people with TB at 514 per 1,000 (95% CI 232–791) [PMID 39382499]. Programme data from Chhattisgarh on 6,397 adults notified under the National Tuberculosis Elimination Programme found 22% severely and 13% very severely undernourished, yet only 29% received the first nutrition cash transfer within two months and 21% received food baskets, with coverage no better in the worst-nourished — an equity failure inside the scheme itself [PMID 41042600].
- **Malaria radical cure is the weak link, not diagnosis.** In the high-burden tribal districts of Malkangiri and Koraput, adherence to chloroquine and artesunate-sulfadoxine-pyrimethamine exceeded 93%, but only 58.7% (95% CI 52.6–64.9) completed the 14-day primaquine course for *P. vivax*, most stopping once symptoms settled (61.9%) or from forgetfulness (21.4%); directly observed radical cure or single-dose tafenoquine with G6PD testing are the proposed fixes [PMID 41505814]. India's pledge is **elimination by 2027**, three years ahead of the Asia-Pacific regional goal of 2030 endorsed in November 2015 and carried in the National Framework for Malaria Elimination, and neither date is thought reachable without tribal-specific scale-up [PMID 37202934]. **Lymphatic filariasis meanwhile remains endemic in 345 districts** against a 2030 elimination goal [PMID 41951149]. **Kala-azar is close to elimination but transmission has not stopped.** The national year-wise record is 524 cases (4 deaths) in 2023 and 449 cases (9 deaths) in 2024, with 429 provisional cases in 2025 [NCVBDC]; and a confirmed indigenous, rK39-negative case diagnosed on bone-marrow amastigotes, with *Phlebotomus argentipes* caught near the patient's home in Maharajganj, eastern Uttar Pradesh, shows transmission occurring outside the classic Bihar–Jharkhand–West Bengal belt — the argument for sustaining surveillance as the target is approached [PMID 42440159].

Guideline and legislative updates

- **Universal Immunisation Programme — coverage is not the same as protection.** In tribal Phulbani, Odisha, first-dose measles coverage was 99.6% and second-dose 97.5%, yet only 55.1% of 1,185 children were seropositive (IgG above 250 IU/L) three months after the second dose, and India ranked fourth worldwide with 13,220 measles cases reported between August 2023 and January 2024 — a direct threat to the measles-rubella elimination target in tribal districts [PMID 42555691]. Of the newer UIP vaccines, the indigenous **Rotavac** gave an adjusted effectiveness of 54% (95% CI 45–62) against severe rotavirus gastroenteritis under routine programme conditions [PMID 41057654], while the indigenous quadrivalent HPV vaccine **Cervavac** (2023) still reaches under 1% of women, with a live policy push to bring HPV vaccination into the UIP and HPV nucleic-acid testing into the National Essential Diagnostics List [PMID 41645272].

Global and WHO developments

- The Seventy-sixth World Health Assembly adopted Resolution WHA76.16 on 30 May 2023, the first WHO resolution devoted to the health of Indigenous Peoples. It requests the Director-General to develop a **Global Plan of Action for the Health of Indigenous Peoples**, taking a life-

course approach and emphasising **reproductive, maternal and adolescent health**; the plan goes to the **Seventy-ninth World Health Assembly**, routed through the **158th session of the Executive Board**. This is the international frame in which the Expert Committee's recommendations should now be placed, alongside SDG target 3 and the disaggregated-reporting duty. [WHO 2023-May]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 13 Tribal Health in India, pp 848–851, including Table 1 literacy and Table 2 MCH indicators; Ch 7 Health Programmes in India, pp 496–500, malaria and the Tribal Malaria Action Plan); Park's Textbook of Preventive and Social Medicine 27e (Ch 13 Tribal Health in India, pp 821–823 — consulted only to read the same chapter's text where the 28e scan is illegible, for life expectancy, postnatal care, the tuberculosis prevalence figures and the Expert Committee recommendations); Sunder Lal — Textbook of Community Medicine (Ch 17 Healthcare of the Community, pp 746–760, primary health care in tribal areas and the relaxed tribal norms; Ch 2, p 68); Oxford Textbook of Public Health 7e (Ch 89 The health of Indigenous peoples, pp 3338, 3343–3344, 3349–3350); Report of the Expert Committee on Tribal Health — *Tribal Health in India: Bridging the Gap and a Roadmap for the Future*, Ministry of Health and Family Welfare and Ministry of Tribal Affairs (constituted October 2013); National Family Health Survey-5 (2019-21); Census of India 2011.

Recent-advances sources: 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, *PLoS One* 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 37202934 — Burden of tuberculosis and malaria among tribal populations and implications for disease elimination in India, *Indian Journal of Medical Research* 2023, doi:[10.4103/ijmr.ijmr_2128_22](https://doi.org/10.4103/ijmr.ijmr_2128_22); 42599962 — Tuberculosis knowledge, attitude and practice among volunteers: community-based intervention training outcome in tribal hard-to-reach areas in Manipur, *PLoS One* 2026, doi:[10.1371/journal.pone.0356469](https://doi.org/10.1371/journal.pone.0356469); 42024909 — Epidemiology of vaso-occlusive crisis among sickle cell disease patients in India: a community-based multi-centric study, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_2879_2025](https://doi.org/10.25259/IJMR_2879_2025); 41834271 — Reproductive health and pregnancy outcomes among women with sickle cell disease in tribal communities of India, *Pediatric Blood and Cancer* 2026, doi:[10.1002/1545-5017.70229](https://doi.org/10.1002/1545-5017.70229) (abstract-only); 39382499 — Malnutrition in tribal people living with tuberculosis in India: a systematic review and meta-analysis, *Indian Journal of Medical Research* 2024, doi:[10.25259/IJMR_2188_23](https://doi.org/10.25259/IJMR_2188_23); 41042600 — High burden of severe and very severe undernutrition among adults with tuberculosis: findings from a statewide operational research in Chhattisgarh, India, *Expert Review of Respiratory Medicine* 2025, doi:[10.1080/17476348.2025.2569845](https://doi.org/10.1080/17476348.2025.2569845) (abstract-only); 41505814 — Suboptimal primaquine adherence in *Plasmodium vivax* malaria: evidence from high-burden tribal districts in Odisha, *Journal of Infection and Public Health* 2025, doi:[10.1016/j.jiph.2025.103123](https://doi.org/10.1016/j.jiph.2025.103123) (abstract-only); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42440159 — rK39-negative visceral leishmaniasis with renal complications from eastern Uttar Pradesh, India, *Acta Parasitologica* 2026, doi:[10.1007/s11686-026-01338-6](https://doi.org/10.1007/s11686-026-01338-6) (abstract-only); 42555691 — Protective antibody response following two doses of measles vaccination and factors for non-response among children from tribal Odisha, India, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_3672_2025](https://doi.org/10.25259/IJMR_3672_2025); 41057654 — Impact of the indigenous rotavirus vaccine Rotavac in the Universal Immunization Program in India during 2016-2020, *Nature Medicine* 2025, doi:[10.1038/s41591-025-03998-9](https://doi.org/10.1038/s41591-025-03998-9); 41645272 — Cervical cancer elimination in India: repurposing

diagnostics, vaccination, and accelerating policy for the 2030 target, *Cancer* 2026, doi:10.1002/cncr.70292 (abstract-only). Press Information Bureau, Government of India — National Sickle Cell Anaemia Elimination Mission launch at Shahdol, Madhya Pradesh, July 2023; PM-JANMAN launch, November 2023; Dharti Aaba Janjatiya Gram Utkarsh Abhiyan launch, October 2024; National Sickle Cell Anaemia Elimination Mission screening milestone, July 2025. Ministry of Health and Family Welfare — renaming of Ayushman Bharat Health and Wellness Centres as Ayushman Arogya Mandir, November 2023. National Center for Vector Borne Diseases Control — Kala-azar situation in India, year-wise cases and deaths (2023, 2024, 2025 provisional). World Health Organization — Resolution WHA76.16 on the health of Indigenous Peoples, adopted 30 May 2023.



Health insurance in India

Asked in: Paper IV 22-Dec-2023 — RGUHS MD Community Medicine

Definition — Health insurance is a financing arrangement in which a **premium is paid in advance** — out of salary, income, or by the government or employer on the beneficiary's behalf — and agreed health services are received when illness occurs, so that the cost of sickness is met from a **pooled fund** instead of from the household's pocket at the moment of need (Sunder Lal, Ch 17, p 763). Its three defining features are **prepayment**, **pooling of risk** across many contributors, and **third-party payment** to the provider. In India it takes four forms: compulsory social health insurance, tax-financed government health assurance, voluntary private insurance and community-based insurance.

1. Why India needs health insurance — the financing backdrop

Financing indicator	Value (with year/source)
Total health expenditure	3.16% of gross domestic product (GDP), 2018-19 (Sunder Lal, Ch 16, p 727)
Government health expenditure	1.28% of GDP = 40.6% of total health spending (Sunder Lal, Ch 16, p 727)
Household revenues including insurance	60.11% of current health expenditure — the single largest source (National Health Accounts 2018-19; Sunder Lal, Ch 16, p 727)
Out-of-pocket expenditure (OOPE)	53.23% of current health expenditure; 1.52% of GDP; Rs 2,155 per capita per year — the highest in the world (Sunder Lal, Ch 16, pp 727-729)
Social security expenditure on health (premiums for CGHS, ESIS, ECHS, RSBY/PM-JAY)	9.6% of total health expenditure; social health insurance expenditure has risen from 4.2% to 16.2% over the past 14 years (Sunder Lal, Ch 16, p 729)
Private (voluntary) health insurance expenditure	6.6% of total health expenditure (Sunder Lal, Ch 16, p 729)

Financing indicator	Value (with year/source)
Catastrophic health expenditure (CHE)	Household health cost >10% of total monthly consumption expenditure OR >40% of non-food consumption expenditure; 17.3% of people crossed this between 2012-20, falling to 7% in 2022 (Sunder Lal, Ch 16, pp 728-729)
Impoverishment	~4.2%, nearly 50 million people, pushed into poverty every year by health spending (Sunder Lal, Ch 16, pp 728-729)
Policy commitment	National Health Policy (NHP) 2017 commits 2.5% of GDP to public health, with strategic purchasing — government as single payer buying secondary and tertiary care from public and private hospitals (Sunder Lal, Ch 16, p 729)

2. Principles and working vocabulary of health insurance

Term	Meaning
Risk pooling	Pre-paid funds of many are pooled so that the healthy and richer cross-subsidise the sick and poorer (Health Economics, Ch 12, p 188)
Actuarially fair premium	Premium set where expected payouts equal premiums paid plus administration: $P = p \times c + a$ (p = probability of the insured event, c = cost of the loss, a = administration cost) (Health Economics, Ch 11, pp 173-178)
Community rating vs experience rating	Community rating — everyone in the community pays the same premium irrespective of personal risk; experience rating — premium set by the group's claims experience or risk (e.g. age) (Health Economics, Ch 11, pp 173-174)
Cost sharing: deductible, co-payment, co-insurance	Deductible — fixed amount the insured pays before the insurer starts paying; co-payment — a specified amount per service; co-insurance — a percentage of the bill after deductibles and co-payments (Health Economics, Ch 11, pp 173-174)
Adverse selection	Only high-risk persons buy voluntary cover; premiums rise, healthy members drop out, and the market can collapse (Health Economics, Ch 11, pp 177-178)
Moral hazard	Insurance removes the price signal, so consumers over-use and providers over-supply services (Health Economics, Ch 11, pp 181-182)
Underwriting	Insurer reviews the application, decides cover offered and fixes the premium by the applicant's health status (Health Economics, Ch 11, pp 173-174)
Provider payment methods	Fee-for-service (encourages supplier-induced demand), capitation (a fixed sum per enrolled person per year, risks "cream skinning"), and package/case-based fixed rates as used by PM-JAY (Health Economics, Ch 10, pp 166-167)
The three dimensions of coverage	Breadth = proportion of population covered; depth = range of services covered; height = proportion of the cost covered (Health Economics, Ch 12, p 188)

3. Classification of health insurance in India

Type	Basis	Indian examples	Who pays the premium
Compulsory social health insurance (Bismarck model)	Statutory, membership compulsory, payroll deduction, run by public bodies, redistributive, earmarked funds (Health Economics, Ch 12, p 190)	Employees' State Insurance Scheme (ESIS) 1948; Central Government Health Scheme (CGHS) 1954; Ex-servicemen Contributory Health Scheme (ECHS) (Sunder Lal, Ch 16, p 729)	Employee + employer + government
Government-funded health assurance / protection (tax-financed, Beveridge-type)	Entitlement-based, non-contributory for the beneficiary, financed from general taxation	Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana (AB PM-JAY) 2018; erstwhile Rashtriya Swasthya Bima Yojana (RSBY) 2008; state health insurance schemes	Central + state government (60:40; 90:10 for North-Eastern states)
Voluntary / private health insurance	Voluntary enrolment, non-income-related premiums paid to a third party; individual, family-floater or group policies (Health Economics, Ch 11, pp 173-174)	Private insurer and corporate group policies	Individual or employer
Community-based health insurance	Voluntary, community-owned, targeted at the informal sector, small risk pool, flat premium per household, often an extension of micro-finance (Health Economics, Ch 12, pp 193-194)	Community and non-governmental organisation schemes	Community member, sometimes subsidised
Employer-provided/ organisation-based medical care (insurance-like but direct provision)	Care provided directly by the employing organisation	Railway health services; Armed Forces health services; corporate hospitals (Sunder Lal, Ch 17, p 743)	Employer

4. Compulsory social health insurance schemes

Feature	ESIS (ESI Act, 1948)	CGHS (1954)	ECHS
Beneficiary	Employee and family in non-seasonal factories/establishments employing 10 or more persons (threshold still 20 in Maharashtra and Chandigarh); extended by most states to shops, hotels, restaurants, cinemas, road motor transport and newspaper establishments; does not apply to mines and railways, which have separate Acts (Sunder Lal, Ch 11, p 650)	Central government employees and families, pensioners, Members and ex-Members of Parliament, sitting and retired judges, freedom fighters (free), accredited journalists, ex-Governors and ex-Vice-Presidents (Sunder Lal, Ch 17, p 764)	Ex-servicemen and authorised dependants — about 20 lakh ex-servicemen , ~ 10 million beneficiaries (Sunder Lal, Ch 17, p 764)
Wage/eligibility ceiling	Wages up to Rs 21,000 per month (raised from Rs 15,000 by ESIC notification 6.9.2016, effective January 2017) (Park 28e, Ch 18, p 971; Sunder Lal, Ch 11, p 650)	By pay level, no ceiling	Ex-servicemen status
Contribution	Total 4% of wages since July 2019 (reduced from 6.5%): employee 0.75%, employer 3.25% ; those with average daily wage up to Rs 176 are exempt from contributing but still get benefits (Sunder Lal, Ch 11, p 650)	Monthly contribution Rs 250 to Rs 1,000 ; pensioners may buy a "whole life" card by paying 10 years' subscription; per capita expenditure Rs 11,331 (Sunder Lal, Ch 17, pp 764-765)	Contributory
Service delivery	ESI dispensaries, ESI hospitals, reserved beds in other hospitals, panel practitioners; reimbursement where care is bought outside (Sunder Lal, Ch 17, p 763)	329 allopathic dispensaries (renamed wellness centres), 86 AYUSH dispensaries, 19 polyclinics, 73 laboratories, 74 dental clinics, 2 geriatric clinics, 4 hospitals; 1,403 empanelled hospitals and 272 diagnostic centres ; extended from Delhi to 72 cities (Sunder Lal, Ch 17, pp 764-765; Park 28e, Ch 10, p 746)	227 ECHS polyclinics (106 in military and 121 in non-military stations) plus empanelled hospitals, nursing homes and diagnostic centres (Sunder Lal, Ch 17, p 764)
Administration	Autonomous ESI Corporation — Union Minister for Labour is Chairman, Secretary, Ministry of Labour is Vice-Chairman (Park 28e, Ch 18, p 971)	Ministry of Health and Family Welfare	Ministry of Defence

Benefits under the ESI Act — seven statutory benefits (Park 28e, Ch 18, pp 971-973; Sunder Lal, Ch 11, p 652):

Benefit	Entitlement
1. Medical benefit (in kind)	Full medical care — outpatient, inpatient, emergency, specialist and super-specialist services — for the insured person and family, plus preventive, promotive and rehabilitative care and family planning services

Benefit	Entitlement
2. Sickness benefit	70% of daily wages for up to 91 days; includes extended sickness benefit and enhanced sickness benefit
3. Maternity benefit	Up to 26 weeks; confinement expenses raised from Rs 5,000 to Rs 7,500 per confinement
4. Disablement benefit	Temporary disablement — about 90% of wages for as long as the disablement lasts; permanent total disablement — life pension based on loss of earning capacity assessed by a medical board; permanent partial disablement — a proportionate life pension
5. Dependant's benefit	Pension at 90% of wages, shared among dependants in prescribed ratio, on death due to employment injury
6. Funeral expenses	Rs 15,000 in cash
7. Rehabilitation allowance	Plus vocational rehabilitation and old-age medical care
Additional — unemployment allowance	50% of wage for up to two years, for a person insured for two or more years who loses employment; medical care continues during this period

5. Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)

Element	Detail
Launch and lineage	Launched 23 September 2018; subsumed RSBY (which had given Rs 30,000 per family per year on a family-floater basis, maximum 5 members, to below-poverty-line families and 11 categories of unorganised workers, launched 2008 under the Ministry of Labour and Employment and transferred to the Ministry of Health and Family Welfare on 1.4.2015) (Park 28e, Ch 10, p 745)
Position in Ayushman Bharat	One of the two pillars — PM-JAY for secondary and tertiary hospitalisation, and Health and Wellness Centres for comprehensive primary health care (Sunder Lal, Ch 17, pp 741, 763)
Cover	Rs 5,00,000 per family per year for secondary and tertiary care hospitalisation (Park 28e, Ch 10, p 745)
Target group	10.74 crore (107 million) poor and vulnerable families — about 50 crore people — the largest government-funded health protection scheme in the world (Park 28e, Ch 10, p 745; Sunder Lal, Ch 17, p 763)
Eligibility	Entitlement-based: families in the defined Socio-Economic Caste Census (SECC) database and former RSBY beneficiaries; Aadhaar encouraged for identification but no one is denied for its absence; a woman is preferably recorded as head of family (Sunder Lal, Ch 17, pp 763-764)
Benefit design	Cashless and paperless at any empanelled public or private hospital; no cap on family size, age or gender; all pre-existing conditions covered; portable across the country; 1,685 procedures covering treatment, drugs, supplies, diagnostics and food; 3 days pre-hospitalisation and 15 days post-hospitalisation expenses (Park 28e, Ch 10, p 745)
Institutional structure	National Health Authority (NHA) at national level and State Health Agency/Authority (SHA) at state level, both autonomous (Sunder Lal, Ch 17, pp 763-764)

Element	Detail
Modes of implementation	(1) Trust/assurance model — SHA reimburses providers directly, without an insurance company; (2) Insurance model — SHA pays a premium per eligible family to an insurance company; (3) Mixed model — both together (Sunder Lal, Ch 17, p 764)
Funding	Centre:state = 60:40 for all states, 90:10 for North-Eastern states (Sunder Lal, Ch 17, p 763)
Safeguards	Pre-authorisation mandatory for all tertiary and selected secondary care packages; defined grievance redressal with safeguards against fraud and abuse by providers and users; <i>Ayushman Arogya Mitra</i> deployed at every empanelled hospital (Sunder Lal, Ch 17, pp 763-764)
Senior-citizen expansion	All eligible senior citizens aged 70 years and above get a new distinct card; those in families already covered get an additional top-up of up to Rs 5 lakh per year, not shared with family members under 70 ; those outside an existing PM-JAY family get Rs 5 lakh per year on a family basis; those already in CGHS/ECHS/Central Armed Police Force schemes may choose either; those holding private insurance or ESI cover remain eligible (Park 28e, Ch 7, pp 570-571)

6. Coverage achieved and residual constraints

Coverage indicator, NFHS-5 (2019-21)	Value
Households with at least one usual member covered by a health scheme or health insurance	41.0% (up from 29% in NFHS-4, 2015-16)
Rural vs urban households	42.4% rural, 38.1% urban
Women aged 15-49 covered	29.8%
Men aged 15-49 covered	33.3%
Highest / lowest states	Rajasthan 87.8%, Andhra Pradesh 80% ; lowest (<15%) in Andaman & Nicobar Islands and Jammu & Kashmir (13.8%)
Scheme mix among the insured	About 46% under a state health insurance scheme, 16% under RSBY ; only 3-7% under ESIS or CGHS

(NFHS-5 India Report, Ch 6, pp 483-484, 511-512)

Constraints, numbered:

- **Coverage is still incomplete** — coverage of health insurance in India remains "pathetically limited"; before PM-JAY, **86% of the rural and 82% of the urban population had no health expenditure support at all** (Sunder Lal, Ch 16, p 729).
- **The benefit is hospitalisation-only** — PM-JAY covers secondary and tertiary admission with just 3 days pre- and 15 days post-hospitalisation, so routine outpatient care, drugs and diagnostics, which drive most out-of-pocket spending, sit outside the package (Park 28e, Ch 10, p 745).
- **Regressive contribution base in ESIS** — only employees earning up to Rs 21,000 per month are covered while high-wage earners are excluded, described as "a classic case of the poor subsidising the poorest" and not necessarily the most equitable design (Sunder Lal, Ch 11, p 650).

- ▶ **Fragmented, parallel pools** — ESIS, CGHS, ECHS, PM-JAY, state schemes and private policies each run a separate risk pool; private insurance further fragments the pool by removing richer groups from it and adds inequity (Health Economics, Ch 12, pp 192-193).
- ▶ **Market failures in voluntary insurance** — adverse selection excludes the high-risk and the poor, moral hazard and diseconomies of scale raise costs, and administration alone takes **10-14% of total costs in unregulated markets** (Health Economics, Ch 11, pp 178-184).
- ▶ **Small, fragile community schemes** — community health insurance has small pools, limited cross-subsidy and difficulty enrolling the poorest (Health Economics, Ch 12, pp 193-194).
- ▶ **Exclusion error in entitlement targeting** — a fixed SECC-based list can miss the near-poor and the non-poor informal sector; PM-JAY is therefore proposed for extension to 80% of the population on a voluntary contributory basis (Sunder Lal, Ch 17, p 763).
- ▶ **Fraud and abuse risk** in a purchaser-provider system, which is why pre-authorisation and a grievance redressal mechanism are built in (Sunder Lal, Ch 17, pp 763-764).
- ▶ **An under-funded public purse** — insurance cannot substitute for public spending stuck at 1.28% of GDP against the NHP 2017 target of 2.5% (Sunder Lal, Ch 16, pp 727-729).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Sources of health financing in India (label exactly)** — a pie with four slices: *Households revenues including insurance 60.11%; State government funds 19.6%; Central government funds 11.7%; Enterprises 5.51%, local bodies 1.01% and other 2.03% (= 8.5%).* Add the caption "National Health Accounts, 2018-19".

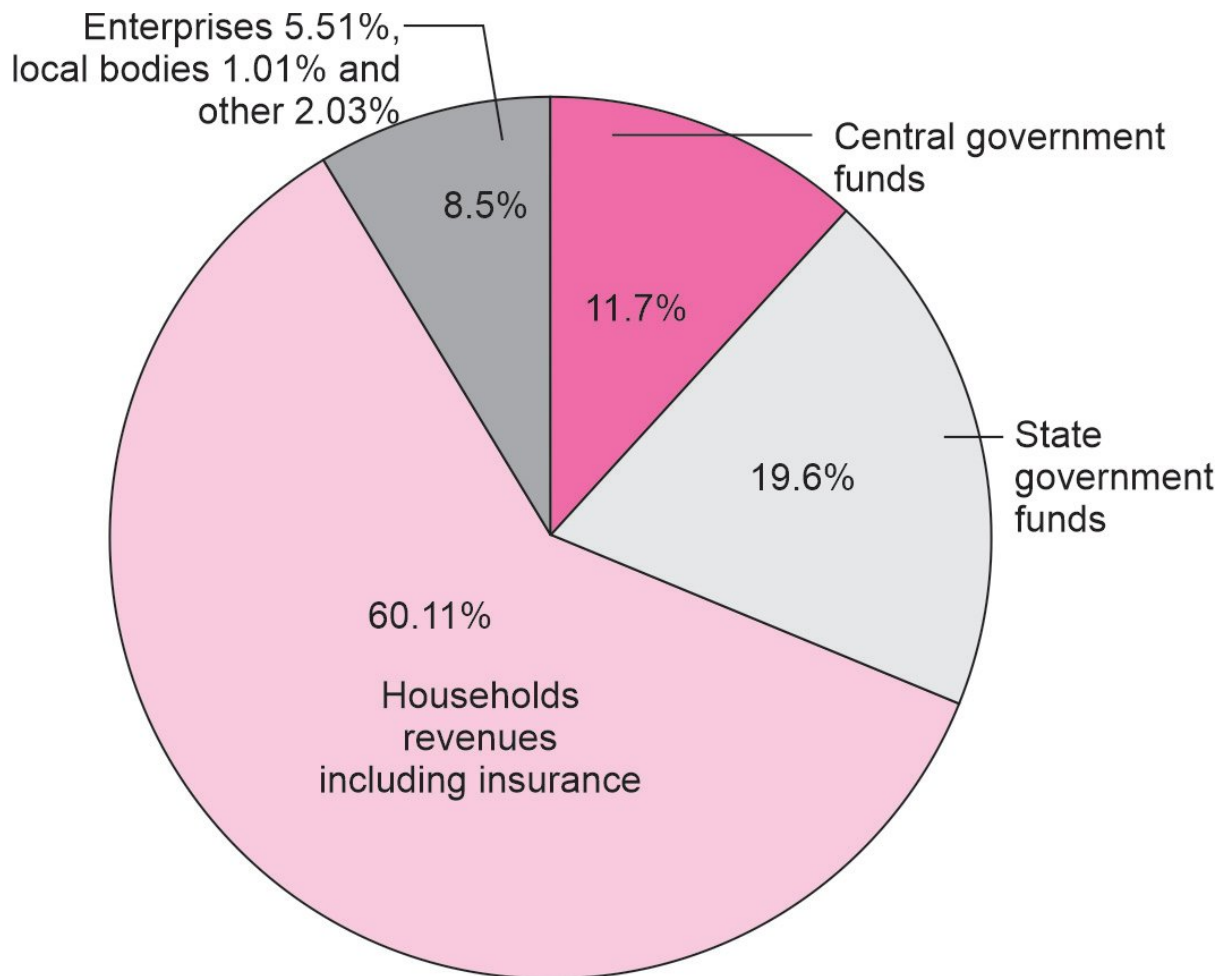
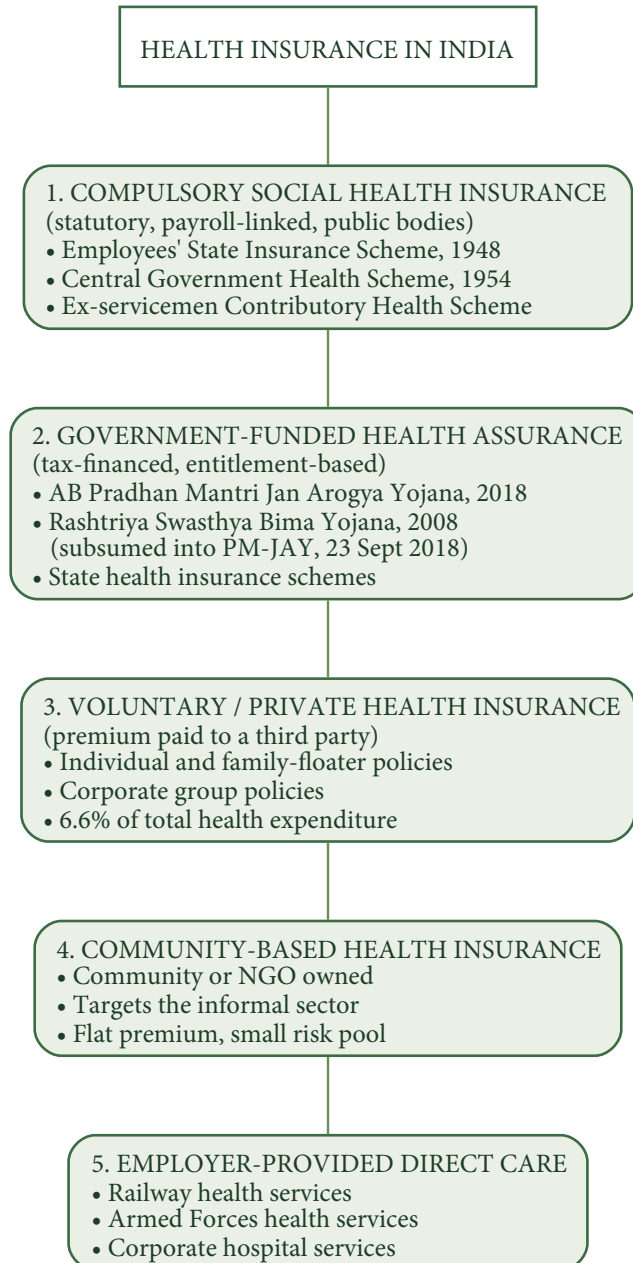
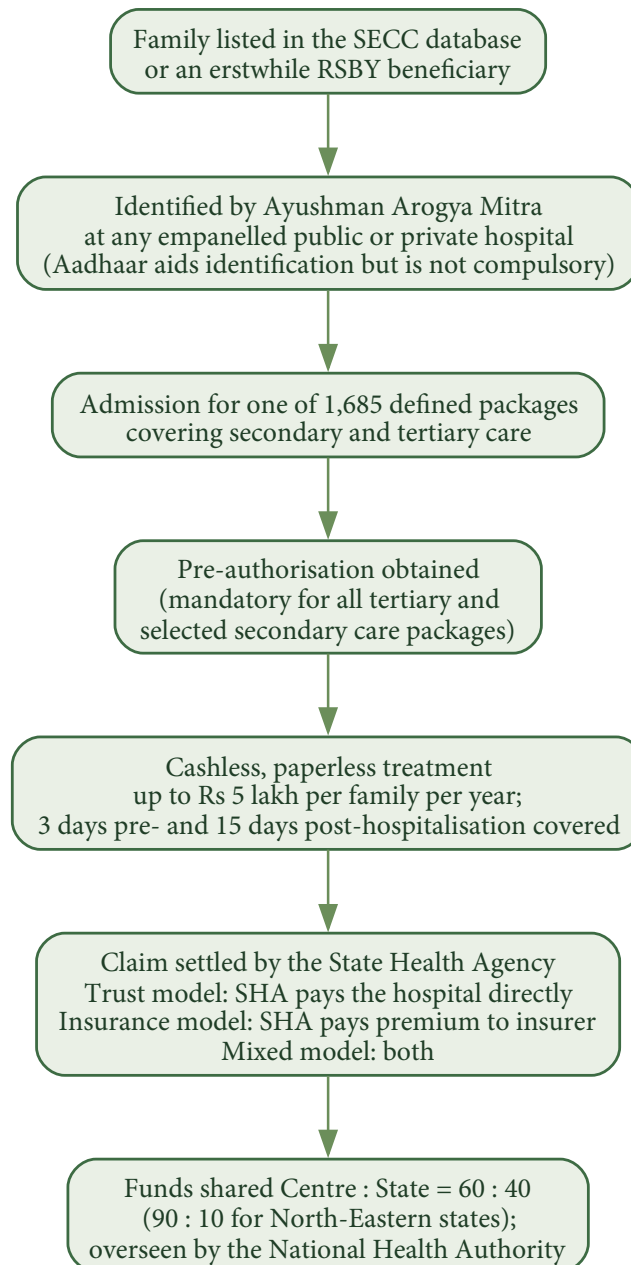


Fig 16.7 — Sources of health financing in India (NHA 2018-19) (Sunderlal, p.728)

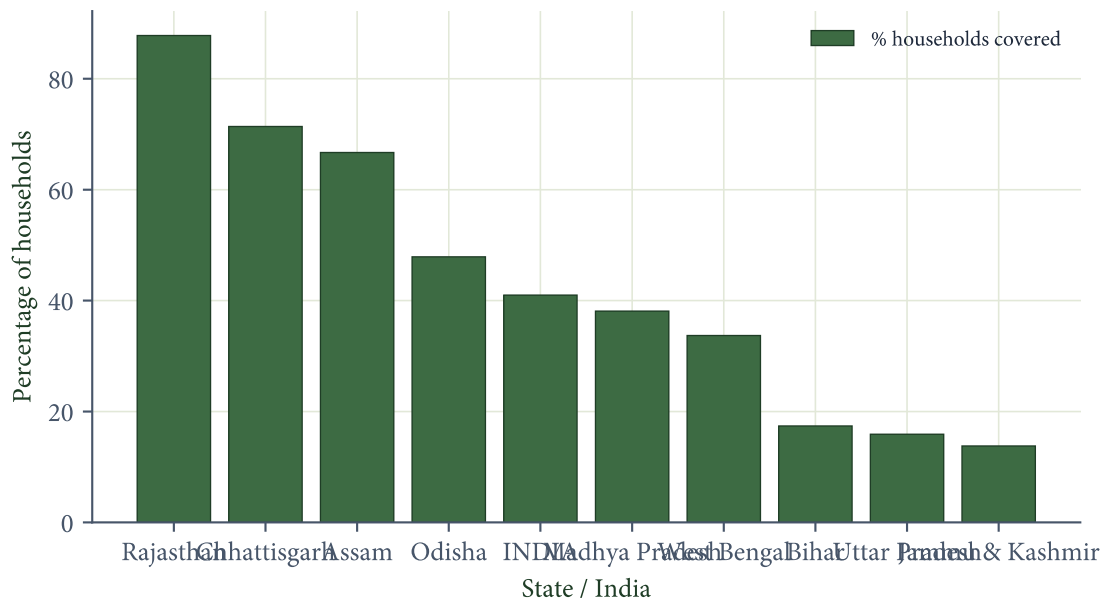


Classification of health insurance in India, with the schemes under each type



Beneficiary pathway and fund flow under Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana

Households with at least one member covered by health insurance or a health scheme



Wide inter-state variation in health insurance coverage, NFHS-5, 2019-21 (NFHS-5 India Report, Table 11.15).

High-yield exam pointers

- ▶ **Definition line to open with** — prepayment + risk pooling + third-party payment; "health insurance means you pay a premium out of your salary or income and get agreed-upon services" (Sunder Lal, Ch 17, p 763).
- ▶ **The four financing numbers** — total health spend 3.16% of GDP (2018-19); government 1.28% of GDP (40.6%); OOPe 53.23% of current health expenditure, 1.52% of GDP, Rs 2,155 per capita; private insurance 6.6% and social security 9.6% of total health expenditure.
- ▶ **ESIS numbers** — Act 1948; 10 or more employees; wage ceiling Rs 21,000/month; contribution 4% total = employee 0.75% + employer 3.25% (since July 2019); daily wage ≤ Rs 176 exempt; 7 benefits — medical, sickness (70% for 91 days), maternity (26 weeks), disablement (90%), dependant's (90%), funeral (Rs 15,000), rehabilitation.
- ▶ **PM-JAY numbers** — launched 23.09.2018; Rs 5 lakh/family/year; 10.74 crore families ≈ 50 crore people; 1,685 procedures; 3 days pre + 15 days post; 60:40 (90:10 NE); NHA and SHA; trust / insurance / mixed implementation models; 70 years and above get a distinct card with a Rs 5 lakh top-up.
- ▶ **Catastrophic health expenditure** — learn both denominators — >10% of total monthly consumption expenditure or >40% of non-food consumption expenditure; impoverishment ~4.2% (~50 million people) per year.
- ▶ **NFHS-5 (2019-21) coverage** — 41.0% of households (29% in NFHS-4), rural 42.4% > urban 38.1%, women 29.8%, men 33.3%.
- ▶ **Mnemonic for the classification** — "S-G-V-C-E": Social (ESIS/CGHS/ECHS) → Government assurance (PM-JAY) → Voluntary private → Community-based → Employer-provided.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park 28e states that PM-JAY covers **1,685 procedures**, while Sunder Lal quotes **1,592 procedures across 23 specialities** for the same scheme — the spine figure of 1,685 has been used here. Park 28e records a distinct PM-JAY card for people aged 70 years and above but does not print the card's brand name, so it has not been named. The Health and Wellness Centre pillar of Ayushman Bharat has since been renamed *Ayushman Arogya Mandir*. Coverage note: the retrieved corpus carries no consolidated "health insurance in India" chapter, so the classification framework above is assembled from separately retrieved and individually cited passages on health financing, social security legislation and Ayushman Bharat.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Accounts, NSS Comprehensive Annual Modular Survey 2022-23, NSS 75th round, NFHS-4 and NFHS-5, National Health Authority health benefit package, National Health Policy 2017

Programme and policy updates (India)

- ▶ The PM-JAY health benefit package has been re-issued and enlarged, so the "1,685 procedures" figure in the body is superseded — the HBP 2022 list carries over **1,949 procedures**, of which **1,284 (65.8%) are surgical or minimally invasive**, against 1,350 procedures (21.5% surgical) in HBP 1.0; cardiology and cardiothoracic packages rose from 163 in HBP 2.1 to 182, and cardiac claims alone absorb about **26% of total scheme expenditure**. The same review records the **90:10 central share as covering Jammu and Kashmir, Himachal Pradesh and Uttarakhand in addition to the North-Eastern states**, and puts PM-JAY's 2021-22 allocation at **about 8.65% of the national health budget**, comparable per head to countries running national surgical plans but far below Nigeria's [PMID 41393311].
- ▶ Health and Wellness Centres now function as *Ayushman Arogya Mandir*, and are the **designated route to close PM-JAY's outpatient gap** — the first national evaluation of AB-PMJAY four years after roll-out, using a nationally representative survey of **302,086 households**, concludes that publicly funded insurance protects inpatient care but leaves outpatient spending unprotected, and recommends extending benefits to outpatient services while strengthening primary care through Ayushman Arogya Mandirs and adapting implementation to state contexts [PMID 42167946].
- ▶ Health technology assessment has been institutionalised inside the purchaser itself — the Health Financing and Technology Assessment (HeFTA) unit created within the **National Health Authority** now generates the evidence used to include or exclude procedures from the health benefit package and to frame standard treatment guidelines, delivering documented cost savings to PM-JAY, with **value-based pricing** proposed as the next step in strategic purchasing [PMID 38715207].

Current national data

- ▶ **National Health Accounts have moved on by three rounds — quote 2021-22, not the 2018-19 figures in the table above.** Between 2013-14 and 2021-22 the **government share of total health**

expenditure rose from 28.6% to 48.0% while the out-of-pocket share fell from 64.2% to 39.4%, the two shares crossing over by the end of that period; total health expenditure is about 3.8% of GDP and government health expenditure about 1.9% of GDP, still short of the National Health Policy 2017 target of 2.5%. About 55 million people are pushed into poverty by health costs each year, and medicines account for roughly two-thirds of out-of-pocket spending — precisely the component PM-JAY does not buy; catastrophic spending was commoner for outpatient care (47.8%) than for hospitalisation (43.1%) [PMID 42548603].

- ▶ A survey newer than NFHS-5 now carries the coverage question — the Comprehensive Annual Modular Survey (NSS 79th round), 302,086 households, July 2022 to June 2023. Publicly funded health insurance was associated with greater use of outpatient care (adjusted OR 1.404, 95% CI 1.328-1.484) and inpatient care (adjusted OR 1.168, 95% CI 1.128-1.210). Inpatient catastrophic expenditure was significantly lower (adjusted OR 0.721, 95% CI 0.676-0.769 at the 10% threshold; 0.790, 95% CI 0.737-0.846 at 20%), but outpatient catastrophic expenditure showed no benefit at the 10% threshold (adjusted OR 1.048); among households that used inpatient care, mean annual inpatient out-of-pocket spending was Rs 37,572 per uninsured household against Rs 29,875 for PM-JAY or state-scheme households — still about a fifth of the latter's annual household expenditure [PMID 42167946].
- ▶ Coverage rose between NFHS-4 and NFHS-5 but the majority remain uncovered and the gradient is economic, not social — 59.01% of the sampled population still had no health cover, with concentration-index analysis showing a steep distributional gradient across economic strata and states while social and demographic determinants contributed comparatively little; the authors argue for a unified "One Nation, One Insurance" framework to end the fragmentation of parallel pools [PMID 40643314].
- ▶ National Sample Survey 75th round (2017-18) — cover on paper did not translate into protection. Despite scheme enrolment, 32.3%, 14.5% and 8.3% of households crossed the 10%, 25% and 40% catastrophic thresholds, differences from uninsured households being statistically non-significant, and instrumental-variable regression found no significant effect of enrolment on out-of-pocket expenditure per hospitalisation; private-hospital out-of-pocket spending rose over 2004 to 2017-18 while public-hospital spending fell [PMID 41913189].

Guideline and legislative updates

- ▶ Contract enforcement, not package price, is now the binding constraint — the case for regulating empanelled providers. Two state-representative household surveys in Chhattisgarh (2021 and 2022, about 15,000 individuals each), four years after roll-out and after an evidence-based increase in reimbursement prices, found enrolment unrelated to inpatient utilisation, patient satisfaction or length of stay, with catastrophic expenditure at the 10% threshold in 78.1% (2021) and 70.9% (2022) of enrolled patients treated in private hospitals; empanelled private hospitals continued to take cash co-payments while also claiming reimbursement ("double billing"), a practice their empanelment contracts explicitly prohibit [PMID 39135015].
- ▶ The evidential case for widening the benefit package to non-medical and indirect costs. A Haryana case-control study (772 respondents, six districts) found direct medical expenses 65% lower among PM-JAY beneficiaries (Rs 11,131 vs Rs 31,675) and catastrophic expenditure at the 30% threshold in 13.3% versus 45.9% of non-beneficiaries (OR 7.15, 95% CI 4.74-10.80), yet

beneficiaries incurred **19% higher non-medical expenses** (Rs 2,872 vs Rs 2,419) because travel, food and lodging lie outside the package; the authors recommend covering indirect costs, referral transport, assured drug availability at facilities and enforcement against balance billing, and note the scheme's own allocation rising nearly 29%, **from Rs 7,300 crore in financial year 2025 to Rs 9,406 crore in 2026** [PMID 40976675].

Global and WHO developments

- **South Asia is on course for the SDG service-coverage target but moving backwards on financial protection.** Bayesian projections put the regional composite coverage index at **81.5% by 2030 (from 53.5% in 2000)**, above the 80% universal-health-coverage benchmark, while **catastrophic health expenditure incidence more than doubles, from 7.2% to 18.6%**; India is projected among the worst affected at **26.3% by 2030**, although India's impoverishment from health spending falls modestly from **3.5% to 2.6%** — the twin-track message that service coverage and financial risk protection (SDG 3.8) must be tracked separately [PMID 41192927].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 18 Occupational Health); Sunder Lal — Textbook of Community Medicine (Ch 11 Occupational Health; Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community); Health Economics (Ch 10 Provider payments; Ch 11 Private health insurance; Ch 12 Achieving universal coverage); NFHS-5 India Report 2019-21 (Ch 6, Tables 11.13-11.15, Figure 11.2). Named documents and data sources: Employees' State Insurance Act 1948 and ESIC notification of 6.9.2016; Central Government Health Scheme 1954; Rashtriya Swasthya Bima Yojana 2008; Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana, launched 23 September 2018; National Health Policy 2017; National Health Accounts 2018-19; National Family Health Survey-5 (2019-21).

Recent-advances sources: 41393311 — Assessing Surgical Care under the Government-Funded Health Insurance Scheme-Pradhan Mantri Jan Arogya Yojana in India, *Annals of Global Health* 2025, doi:[10.5334/aogh.4908](https://doi.org/10.5334/aogh.4908); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 38715207 — Establishing a Health Technology Assessment Evidence Ecosystem in India's Pradhan Mantri Jan Arogya Yojana, *Health Systems and Reform* 2024, doi:[10.1080/23288604.2024.2327097](https://doi.org/10.1080/23288604.2024.2327097) (abstract-only); 42548603 — Can a bigger budget shrink the bill at the bedside? Public health financing as a route to lower out-of-pocket spending in India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1903360](https://doi.org/10.3389/fpubh.2026.1903360); 40643314 — Inequality in Health Insurance Coverage in a Pluralistic Health Insurance System: Evidence From India, *International Journal of Health Planning and Management* 2025, doi:[10.1002/hpm.70008](https://doi.org/10.1002/hpm.70008) (abstract-only); 41913189 — Do publicly funded health insurance schemes reduce out-of-pocket expenditure? Evidence from India, *BMC Health Services Research* 2026, doi:[10.1186/s12913-026-14456-8](https://doi.org/10.1186/s12913-026-14456-8); 39135015 — The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) after four years of implementation — is it making an impact on quality of inpatient care and financial protection in India?, *BMC Health Services Research* 2024, doi:[10.1186/s12913-024-11393-2](https://doi.org/10.1186/s12913-024-11393-2); 40976675 — Impact of India's publicly funded health insurance scheme on financial risk protection: a case-control study from Haryana state in India, *BMJ Open* 2025, doi:[10.1136/bmjopen-2024-093304](https://doi.org/10.1136/bmjopen-2024-093304); 41192927 — Progress towards universal health coverage in South Asia, 2000-2030: an examination of the twin elements of primary healthcare provision and financial protection, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-020052](https://doi.org/10.1136/bmjgh-2025-020052). National data and programme documents cited in the section: National Health Accounts estimates to 2021-22; NSS 79th round — Comprehensive Annual Modular Survey 2022-23; NSS 75th round 2017-18; NFHS-4 and NFHS-5; National Health Authority PM-JAY Health Benefit Package 2022; National Health Policy 2017.

♦ ♦ ♦

Public-private partnership in health care

Asked in: Paper IV 14-Jul-2023 — RGUHS MD Community Medicine

Definition — The World Health Organization (WHO) defines a **public-private partnership (PPP)** in the health sector as *the bringing together of a set of actors for the common goal of improving the health of populations, based on mutually agreed roles and principles* (Oxford Textbook of Public Health 7e, Ch 106, p 4021). Reich adds three defining elements: at least one private for-profit organisation working with at least one not-for-profit organisation; a **joint sharing of effort and of benefit** between the core partners; and a commitment to the **creation of social value** — improved health, especially for disadvantaged populations (Oxford 7e, Ch 51, p 1916). It is therefore a risk-and-benefit-sharing collaboration, not the mere purchase of a service.

1. Why partnership is needed — the Indian health-care mix

Driver	Evidence (with source)
Private sector dominates provision	Private sector holds 80% of doctors, 25% of nurses and 49% of beds and is the dominant provider of curative care, financed chiefly by out-of-pocket expenditure (Sunder Lal, Ch 16, p 729)
Most services already come from outside government	About 75% of health services are currently provided by the private sector (Park 28e, Ch 7, p 539; Suryakantha, Ch 36, p 797)
Most spending is non-governmental	The non-governmental health sector accounts for nearly 70% of health expenditure in India (Sunder Lal, Ch 17, p 739)
Government is the public-health backbone	Government sector is the dominant provider of preventive and promotive services to over 95% of the population (Sunder Lal, Ch 16, p 729)
Cost and impoverishment	Average medical expenditure per hospitalisation was ₹31,845 in private vs ₹4,450 in public facilities — about 7 times higher ; nearly 86% of rural and 82% of urban people had no health-expenditure support (Sunder Lal, Ch 16, p 729; Ch 17, p 763)
Universal coverage cannot be reached alone	Governments are encouraged to recognise that they cannot achieve universal health coverage (UHC) on their own; engagement may take the form of capacity development, innovation, outsourcing and consulting, with varying degrees of financial risk-sharing (Oxford 7e, Ch 21, pp 834-836)
Voluntary sector reaches the unreached	More than 7,000 voluntary agencies work in health-related activities, often in difficult, underserved and unserved areas (Sunder Lal, Ch 17, p 765)

2. Forms and models of partnership

Model	What it involves	Indian illustration
Contracting — the predominant model in India	Contract-in, contract-out, outsourcing, management of hospital facilities by leading non-governmental organisations (NGOs), hiring of staff, and contracted service delivery including family planning, medical termination of pregnancy and treatment of reproductive and sexually transmitted infections	Component (E) of the National Rural Health Mission — <i>People Public-Private Partnership for Public Health Goals</i> , pursued mainly through the Reproductive and Child Health (RCH-II) programme and by states with their own resources (Sunder Lal, Ch 17, p 739)

Model	What it involves	Indian illustration
Strategic purchasing / demand-side financing	Shift from input-oriented, line-item budgeting to output-based strategic purchasing from public and private hospitals, with government acting as a single payer ; the district health mission moves towards paying hospitals on the principle of " <i>money follows the patient</i> "	National Health Policy 2017 policy shift; district health mission reimbursement (Sunder Lal, Ch 16, pp 698, 729; Ch 17, p 761)
Accreditation and empanelment	Public funds purchase defined care packages from accredited public or private hospitals	Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana (AB PM-JAY); accredited private institutions under Janani Suraksha Yojana (Park 28e, Ch 10, p 745; Ch 7, p 544)
Social marketing and social franchising	Commercial marketing skills applied to health products and services through the 6 Ps — product, price, place, promotion, partnership and policy environment; franchising motivates private curative providers to deliver preventive services to weaker sections at affordable cost	Subsidised condoms, low-dose oral pills, oral rehydration salts and copper-T through private outlets, with sub-centres, primary health centres (PHCs), community health centres (CHCs) and Indian systems of medicine used as additional outlets (Sunder Lal, Ch 4, pp 160-162)
Partnership with professional bodies and industry	Professional associations and industry brought into programme delivery and information systems	Indian Medical Association and the Federation of Obstetric and Gynaecological Societies in family welfare; industry support for the management information system, transport and social marketing (Sunder Lal, Ch 17, pp 765-766; Ch 9, p 556)
Community-managed public institutions	Hospital management committee / <i>Rogi Kalyan Samiti</i> for community management of public hospitals; village health, sanitation and nutrition committees	National Rural Health Mission (Sunder Lal, Ch 17, p 761; Park 28e, Ch 7, p 539)
Product-development partnerships (global)	Not-for-profit and for-profit partners jointly develop products for diseases of poverty	International AIDS Vaccine Initiative — the first such partnership; Global Fund to Fight AIDS, Tuberculosis and Malaria (Oxford 7e, Ch 51, p 1916; Ch 106, p 4028)
Philanthropic and foundation support	Grants, donations and demonstration projects from private foundations	Rockefeller Foundation, Ford Foundation and CARE as international voluntary agencies in India; Gates Foundation and the GAVI Alliance globally (Park 28e, Ch 10, p 748; Oxford 7e, Ch 3, p 229)

3. Partnership in Indian national programmes and schemes

Programme / scheme	Form the partnership takes
National Tuberculosis Elimination Programme (NTEP)	The underlying principle is to extend public services to privately managed patients ; mandatory TB notification, the Nikshay database, Standards for TB Care in India and a ban on serodiagnostics are the tools used. Regulatory tools are limited, so partnership is preferred (Park 28e, Ch 7, pp 513-515). Incentive of ₹1,000 for notification of a TB case by the private sector, and ₹500 per month by direct benefit transfer to the patient for nutritional support (Sunder Lal, Ch 8, pp 364-365)
AB PM-JAY	Cashless, paperless hospitalisation at any empanelled hospital, public or private ; cover of ₹5 lakh per family per year for secondary and tertiary hospitalisation for 10.74 crore poor and vulnerable families (about 50 crore people) ; launched 23 September 2018 , subsuming Rashtriya Swasthya Bima Yojana; 1,685 procedures ; no cap on family size, age or gender; all pre-existing conditions covered; benefits portable across the country; 3 days pre- and 15 days post-hospitalisation cover ; <i>Ayushman Arogya Mitra</i> deployed at every hospital; Centre : State funding 60:40 (90:10 in North-Eastern states) (Park 28e, Ch 10, p 745; Sunder Lal, Ch 17, p 763)
National Programme for Control of Blindness	Eye camps organised by Lions Clubs, Rotary Clubs and NGOs; NGOs, voluntary organisations and private practitioners covered over 61% of the cataract backlog against 39% by the government sector (Sunder Lal, Ch 8, pp 517-518)
National Health Mission	Private-sector representation in the Mission, development of PPP guidelines for the health sector, and inclusion of NGOs in task groups (Suryakantha, Ch 36, p 797); PPP promoted for secondary and tertiary care under the district health mission (Sunder Lal, Ch 17, p 761)
Emergency and referral transport	Free ambulance services on toll-free Dial-108 (emergency, critical care and trauma) and Dial-102/104 (patient transport for pregnant women and children, linked to Janani Shishu Suraksha Karyakram entitlements); as on June 2021, 11,879 Dial-108 and 10,716 Dial-102/104 vehicles were operational, besides 5,859 empanelled vehicles (Park 28e, Ch 7, pp 539-540)
Janani Suraksha Yojana	Cash assistance for institutional delivery extended to accredited private health institutions (Park 28e, Ch 7, p 544)
HIV/AIDS and STI-RTI control	The National Strategic Plan for STI/RTI 2017-2024 delivers services in convergence with the National Health Mission and the private sector ; NGOs provide care and support to people living with HIV/AIDS (Sunder Lal, Ch 8, p 355; Suryakantha, Ch 44, pp 861-862)
Non-communicable disease control	Interventions to be built within the existing public sector health system plus feasible PPP models in all 640 districts (Suryakantha, Ch 22, p 543)
Indian Public Health Standards, CHC level	"Public-private partnership for laboratory services" is listed among the desirable services, and "to enhance public-private partnership" among the assured functions (Sunder Lal, Ch 17, pp 754-755)
Family welfare	Private practitioners given a satellite population of 5,000 for RCH services on a one-year contract at a time , with adequate compensation (Sunder Lal, Ch 9, p 556)

4. Role of the voluntary (non-governmental) sector

Park classifies the services rendered by voluntary health agencies under six heads (Park 28e, Ch 10, pp 746-747):

- ▶ **Supplementing the work of government agencies** — lending personnel, or contributing funds for special equipment, supplies or services, because government agencies operate under financial and statutory restrictions.
- ▶ **Pioneering** — exploring new ways and means of doing things; research is one such area.
- ▶ **Education** — health education, where the scope in India is unlimited and government agencies alone cannot cope.
- ▶ **Demonstration** — demonstrations and experimental projects, for example the Rockefeller Foundation's bore-hole latrine demonstration for hookworm control, whose modifications became part of India's environmental sanitation programme.
- ▶ **Guarding the work of government agencies** — acting as a critic and guide to government effort.
- ▶ **Advancing health legislation** — moulding public opinion and advancing health legislation for the benefit of the whole community.

5. Comparative strengths of the two sectors

	Public sector	Private sector
Advantages	Free diagnosis; free treatment; standardised regimen; referral and transfer system; supervision and monitoring; accountability for treatment outcome	Wide choice (more than 5 lakh practitioners); better access — convenient timings and shorter distances; personal attention and care; projected discounts; faith and perception of better care
Weaknesses	Under-utilisation, especially in favour of the poor; resource crunch for equipment, maintenance, consumables and infrastructure upgrading at district level	Delays in diagnosis; over-diagnosis (for example of TB, from over-dependence on X-rays); multiple non-standard regimens for inappropriate durations ; no mechanism to ensure a full course of treatment or to record treatment outcomes; high out-of-pocket cost

(Park 28e, Ch 7, pp 513-514; Sunder Lal, Ch 16, p 729; Ch 17, p 761)

6. Benefits, challenges and prerequisites for success

Benefits	Challenges and risks
Acceleration of innovation and research and development , and expansion of access to affordable health interventions (Oxford 7e, Ch 24, p 955)	Prioritisation and sustainability — gains may not survive beyond the funded period, and separately funded initiatives can distort national health priorities (Oxford 7e, Ch 106, pp 4032-4033)
Mobilises resources and ensures their efficient use, promotes improved delivery systems and develops localised models (Oxford 7e, Ch 51, p 1889)	Accountability and oversight are often lacking ; commercial considerations may limit public oversight during procurement, and contract provisions may block changes later needed to safeguard the public interest (Oxford 7e, Ch 106, p 4033)

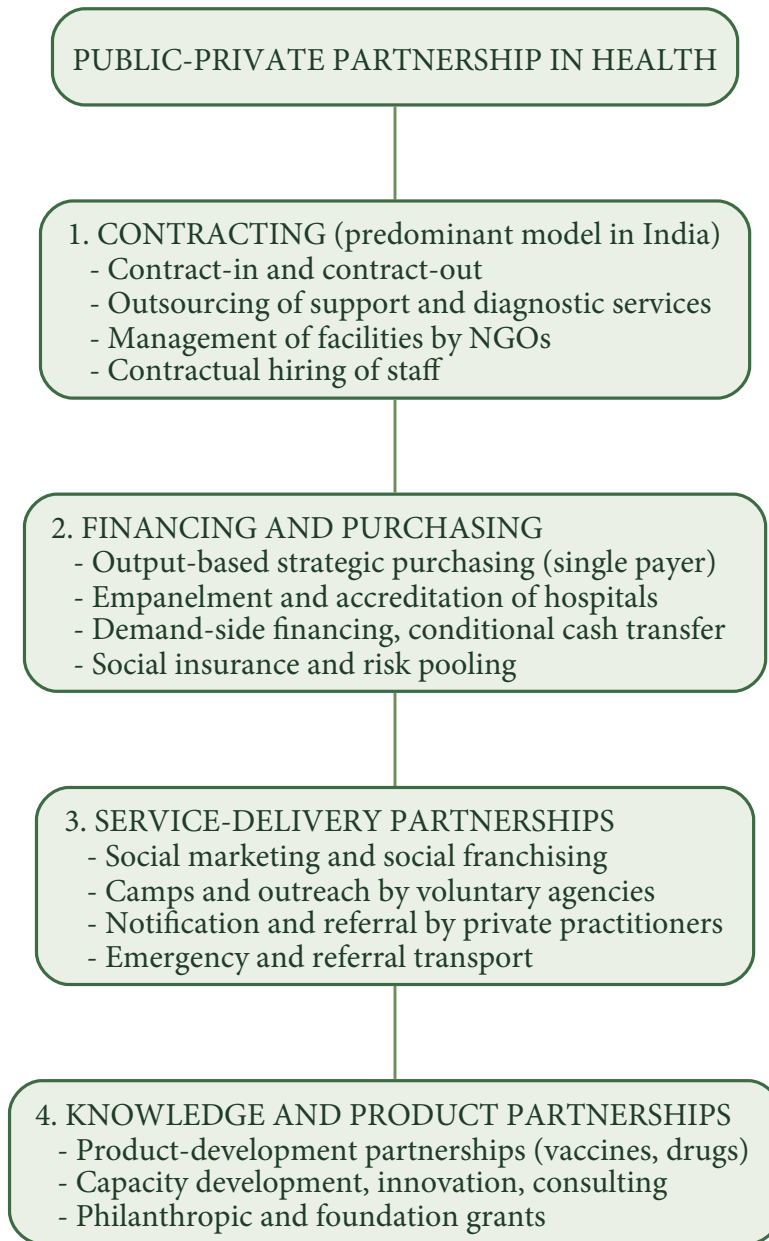
Benefits	Challenges and risks
Extends the reach of national programmes to where patients actually seek care	Conflicts of interest — profit-driven partners may not measure performance or success by unbiased means, and balancing commercial with public-health interest is difficult (Oxford 7e, Ch 24, p 955)
Provides financial risk protection when public money purchases private care	Cost — money borrowed from the private sector is often dearer than government borrowing and the cost is passed on to the public; the initial capital outlay may cancel out the efficiency gained (Oxford 7e, Ch 106, p 4033)
Brings in managerial and technical capacity that the public system lacks	Corruption, ethical and moral dilemmas, difficulty in assessing partnership compatibility and in evaluating partnership outcomes (Oxford 7e, Ch 106, pp 4032-4033)

Prerequisites for a successful partnership

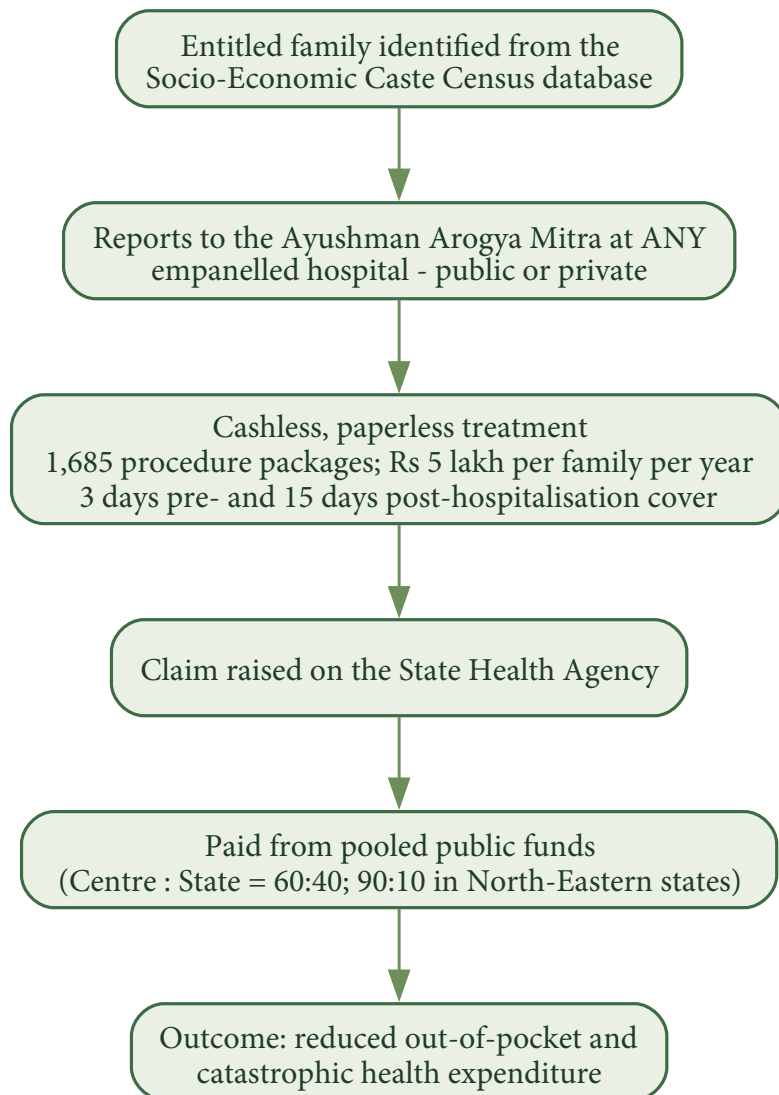
- ▶ **Regulatory frameworks and accreditation systems** to enable collaboration and foster partnership, since the private sector's needs differ from those of the public sector (Oxford 7e, Ch 21, pp 834-836).
- ▶ **Written guidelines or regulations governing partnerships, with mechanisms to audit compliance**, so that low- and middle-income countries reap the full benefit of these relationships (Oxford 7e, Ch 24, p 955).
- ▶ **Alignment of the growth of the private care sector with public health goals**, and improved access to affordable secondary and tertiary care through a **combination of public and private providers** — both explicit National Health Policy 2017 objectives (Sunder Lal, Ch 16, p 697).
- ▶ **Strategic purchasing** of secondary and tertiary care by government as a single payer, while free primary health care remains a public responsibility; National Health Policy 2017 commits 2.5% of gross domestic product to public health (Sunder Lal, Ch 16, p 729).
- ▶ **Regulation of the private sector including rural practitioners** to assure quality service at reasonable cost, mainstreaming of AYUSH, and effective risk-pooling and social insurance for the poor (Suryakantha, Ch 36, p 796).
- ▶ **Reinforcing trust in the public health-care system** and reducing out-of-pocket and catastrophic health expenditure as the measurable end-points of any partnership (Sunder Lal, Ch 16, p 697).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

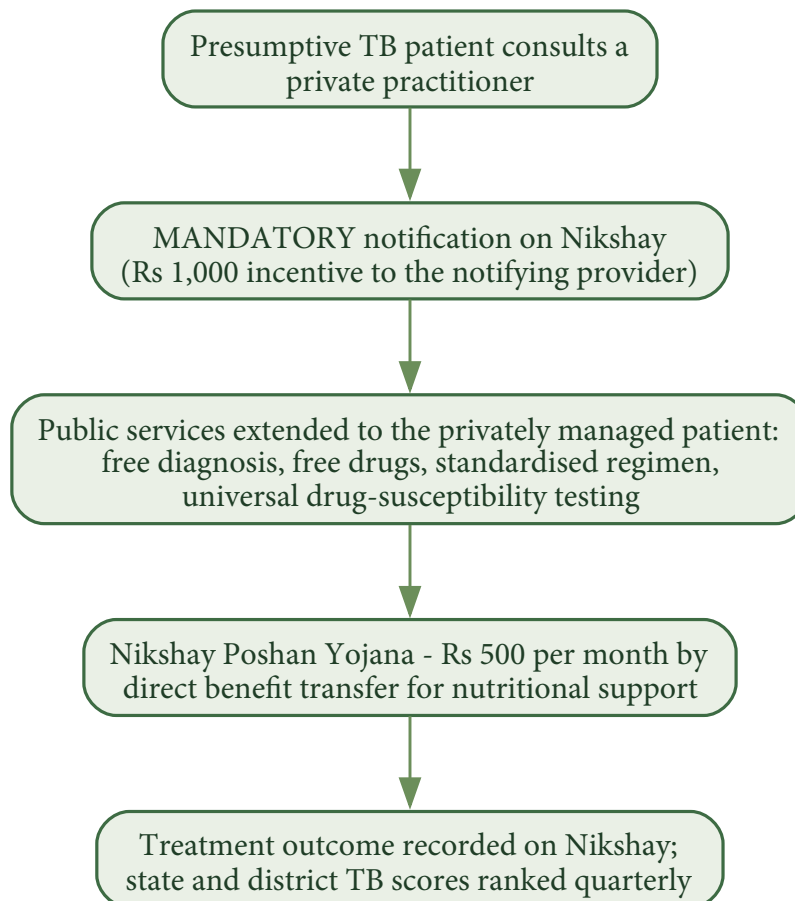
- ▶ **Labelled diagram** — the five sectors of health care in India, with the partnership interface marked (Park 28e, Ch 10, pp 726, 746-747). Draw five boxes feeding one central "Community" box, labelled exactly: **1. Public health sector** (primary health care, hospitals and dispensaries, health insurance schemes, other agencies) · **2. Private agencies** (private practitioners, nursing homes, corporate hospitals) · **3. Indigenous systems of medicine** (Ayurveda, Siddha, Unani, Homoeopathy) · **4. Voluntary health agencies** · **5. National health programmes**. Mark the public-private partnership interface as a double-headed arrow between sector 1 and sectors 2-4.



Modes of engaging the private and voluntary sector in health care



Public money purchasing private care - the AB PM-JAY pathway



Engaging the private provider in the National TB Elimination Programme

High-yield exam pointers

- ▶ **Definitions to reproduce:** WHO — "bringing together of a set of actors for the common goal of improving the health of populations based on mutually agreed roles and principles"; *Reich's* three elements — a for-profit plus a not-for-profit partner, joint sharing of effort and benefit, and creation of social value.
- ▶ **The four numbers that frame the answer:** private sector = **80% of doctors, 25% of nurses, 49% of beds; about 75%** of health services privately provided; **about 70%** of health expenditure non-governmental; government serves **over 95%** of the population for preventive and promotive care.
- ▶ **Contracting is the predominant PPP model in India** — contract-in, contract-out, outsourcing, under Component (E) of the National Rural Health Mission, "People Public-Private Partnership for Public Health Goals".
- ▶ **AB PM-JAY:** launched 23 September 2018, subsumed RSBY, ₹5 lakh per family per year, 10.74 crore families, 1,685 packages, 3 days pre- plus 15 days post-hospitalisation, funding 60:40 (90:10 in the North-East).
- ▶ **NTEP private engagement:** mandatory notification, Nikshay, ₹1,000 notification incentive, ₹500 per month patient direct benefit transfer, ban on serodiagnostics — "regulatory tools are limited and partnership is preferred".
- ▶ **Blindness programme is the classic Indian PPP success** — NGOs, voluntary agencies and private practitioners cleared **61%** of the cataract backlog against **39%** by government.

- **Park's six functions of voluntary health agencies:** Supplementing, Pioneering, Education, Demonstration, Guarding, Advancing legislation.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the programme figures quoted here are as printed in the cited editions (Park 28e, Sunder Lal, Suryakantha) — the AB PM-JAY beneficiary count, the ambulance fleet (June 2021) and the Nikshay Poshan Yojana nutritional benefit are revised periodically by the Ministry of Health and Family Welfare, so always quote a figure with its source and year and check the current MoHFW annual report before the examination.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 10 Health Care of the Community); Sunder Lal — Textbook of Community Medicine (Ch 4 Principles of Health Promotion and Education; Ch 8; Ch 9; Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community); Oxford Textbook of Public Health 7e (Ch 21 Health policy in developing countries; Ch 24; Ch 51 Governance and management of public health programmes; Ch 106 Private support of public health); Suryakantha — Community Medicine with Recent Advances (Ch 22, Ch 36 National Rural Health Mission, Ch 37 National Urban Health Mission, Ch 44); National Health Policy 2017; National Population Policy 2000; Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana (2018).



Role of NGOs in tribal health care

Asked in: Paper IV 03-Oct-2025 — RGUHS MD Community Medicine

Definition — A **voluntary health agency (non-governmental organisation, NGO)** is an organisation "administered by an autonomous board... collects funds for its support chiefly from private sources, and expends money... in conducting a programme directed primarily to furthering the public health" (*Gunn and Platt*, 1945; Park 28e, Ch 10, p 746). The sector is non-profit, supplements rather than replaces government services, and works with voluntary zeal in difficult, underserved and unserved areas (Sunder Lal, Ch 17, p 765).

Why tribal areas depend on the voluntary sector

Gap in tribal health care	Evidence (with source and year)
Size and dispersion	Over 104 million tribal people across 705 tribes = 8.6% of India's population (Report of the Expert Committee on Tribal Health, <i>Tribal Health in India: Bridging the Gap and a Roadmap for the Future</i> , MoHFW and Ministry of Tribal Affairs). Nearly 90% live in rural , mostly hilly and forested areas; largest numbers in Madhya Pradesh (15 million), Maharashtra (10 million), Odisha and Rajasthan (9 million each); scheduled-tribe sex ratio 990 (Census 2011) (Park 28e, Ch 13, p 848)

Gap in tribal health care	Evidence (with source and year)
Infrastructure shortfall despite relaxed norms	Norms already relaxed to 1 health sub-centre (HSC)/3,000, 1 primary health centre (PHC)/20,000, 1 community health centre (CHC)/80,000. On 'required versus shortfall' data for tribal areas of 18 states and 3 union territories (Expert Committee on Tribal Health policy brief), the deficient states showed a shortfall of 4,996 sub-centres = 27% (11 states), 1,023 PHCs = 40% (7 states) and 209 CHCs = 31% (10 states) (Park 28e, Ch 13, p 850)
Human resources	Vacant posts, limited scope for professional interaction or growth, a feeling of social and professional isolation, weak human-resource policy and poor working conditions; no transport for emergency referral (Park 28e, Ch 13, pp 850–851; Sunder Lal, Ch 17, p 760)
Tuberculosis	Estimated prevalence of pulmonary tuberculosis 703 against 256 per 100,000 in the rest of the country (Expert Committee on Tribal Health report; Park 28e, Ch 13, p 849)
Malaria	Tribal communities are about 8% of the national population but account for about 30% of all malaria cases, >60% of <i>P. falciparum</i> and ~50% of malaria mortality (Park 28e, Ch 13, p 849)
Maternal and child health gap	Scheduled-tribe infant mortality rate 41.6 , under-5 mortality rate 50.3 , neonatal mortality rate 28.8 per 1,000 live births; institutional delivery 68.0% vs 88.6% ; stunting 40.9% vs 35.5% ; only about 37% of tribal women received any postnatal care within 48 h of a home delivery (NFHS-5, 2019–21; Park 28e, Ch 13, pp 848–849)
Cultural and communication barriers	Information, education and communication (IEC) not tuned to tribal vocabulary, beliefs and practices; service timings not client-friendly; non-involvement of local traditional faith healers ; and explicitly, " inadequate mobilization of NGOs " (Sunder Lal, Ch 17, p 760)
Sector available to fill the gap	More than 7,000 voluntary agencies are involved in health-related activity in India, serving urban, rural and tribal communities (Sunder Lal, Ch 17, p 765)

Roles of NGOs — the six classical functions applied to tribal areas

Function (Park 28e, Ch 10, pp 746–747)	What it means in a tribal setting	Example (tribal where available)
1. Supplementing the work of government agencies	Government agencies cannot give complete service because they operate under financial and statutory restriction; NGOs lend personnel, or contribute funds for special equipment, supplies or services, and run facilities the state cannot staff	Management of the PHCs at Gumballi and Sugganahalli was contracted out by the Government of Karnataka to Karuna Trust in 1996 to serve the tribal community in the hilly areas (Jugal Kishore, Ch 1, pp 116–117)

Function (Park 28e, Ch 10, pp 746–747)	What it means in a tribal setting	Example (tribal where available)
2. Pioneering	Explore new ways of doing things, research included; when the effort bears fruit the government steps in and takes it over for larger numbers	Park's own instance — voluntary agencies first spearheaded the family planning movement in the face of much opposition , and the government later accepted family planning as national policy (Park 28e, Ch 10, p 746). Classical Indian model: the Comprehensive Rural Health Project, Jamkhed (<i>Arole</i> , in Newell, <i>Health by the People</i> , WHO 1975 — Reinke, Ch 13, ref 22)
3. Education	Health education and behaviour change in the local dialect, on a scale the government machinery cannot cope with unless supplemented by voluntary effort	Under the National Leprosy Eradication Programme funds are allotted to NGOs, who are encouraged to work in tribal areas for IEC, prevention of deformity and follow-up of cases, with intensified IEC through rural media in remote, inaccessible and tribal areas (Park 28e, Ch 13, p 849)
4. Demonstration	Demonstrations and experimental projects that prove a method works before the state scales it	SEARCH, Gadchiroli (Maharashtra) demonstrated home-based newborn care (HBNC) with an appreciable documented decline in infant mortality; the Government of India then approved national implementation of HBNC based on the Gadchiroli model (Sunder Lal, Ch 10, p 614). Park's classical instance is the bore-hole latrine demonstrated by the Rockefeller Foundation for hookworm, later part of the national environmental sanitation programme (Park 28e, Ch 10, p 746)
5. Guarding / guiding the work of government agencies	By setting a good example the voluntary agency can guide and constructively criticise the official agency	Institutionalised as part of the first pillar of the National Rural Health Mission — increasing participation and ownership by the community (decentralisation) — which includes a community monitoring system and "a greater space for NGO participation" alongside the ASHA programme, the village health, sanitation and nutrition committee, <i>Rogi Kalyan Samitis</i> and the district health society (Sunder Lal, Ch 17, p 737); District Blindness Control Societies are required to list voluntary agencies and actively involve them (Sunder Lal, Ch 8, p 518)
6. Advancing health legislation	Voluntary agencies "mobilise public opinion and advance legislation on health for the benefit of the whole community" (Park 28e, Ch 10, p 747)	Civil-society organisations shape policy and standards by advocating with stakeholders and the public, creating dialogue with policymakers and promoting the legalisation of health policies — on drug prices and access, breast-milk substitutes, tobacco control and the environment (Oxford Textbook of Public Health 7e, Ch 51, p 1890)

Named NGOs and what they deliver in tribal areas

Operational role	Named NGO / project and what it did
Managing peripheral facilities on contract	Karuna Trust, Karnataka — the Gumballi and Sugganahalli PHCs, contracted out in 1996 for the tribal hilly areas. A collaboration of the Government of Arunachal Pradesh, the Voluntary Health Association of India (VHAI) and Karuna Trust manages a significant number of PHCs in that predominantly tribal state (Jugal Kishore, Ch 1, pp 116–117)
Community-based providers in difficult terrain	SARTHI in 150 tribal villages of Gujarat and SEWA-Rural in Bharuch district, Gujarat successfully trained and deployed community-based providers and volunteers — in sparsely populated, difficult terrain a team of a man and a woman, preferably a married couple , works best (Jugal Kishore, Ch 1, p 247)
Taking over deficient facilities	Policy provides for mapping of NGOs in tribal areas , credible NGOs with a proven track record of clinical service provision being encouraged to take responsibility for managing reproductive and child health and other services in sub-centres, PHCs and CHCs where the public health system is deficient (Jugal Kishore, Ch 1, p 247)
Newborn and child survival at the doorstep	SEARCH, Gadchiroli — home-based newborn care, now delivered nationally by accredited social health activists (ASHAs) (Sunder Lal, Ch 10, p 614)
Outreach camps and mobile services	Lions Clubs, Rotary Clubs and many NGOs organised eye camps, and mobile eye clinics were added, to cover inaccessible rural, tribal and difficult areas ; NGOs, voluntary organisations and private practitioners cleared over 61% of the cataract backlog against 39% by the government sector, paid at a fixed rate per case. Dependence on camps has since fallen except in remote and tribal areas (Sunder Lal, Ch 8, pp 517–518). The All-India Blind Relief Society (1946) organises eye relief camps (Park 28e, Ch 10, p 747)
Disease-specific care and rehabilitation	Hind Kusht Nivaran Sangh (1950) — financial assistance to leprosy homes and clinics, training of medical workers and physiotherapists, field research, journal <i>Leprosy in India</i> . Tuberculosis Association of India (1939) — training and health education, and management of the New Delhi Tuberculosis Centre, the <i>Lady Linlithgow Sanatorium</i> at Kasauli, the <i>King Edward VII Sanatorium</i> at Dharampur and the Tuberculosis Hospital at Mehrauli (Park 28e, Ch 10, p 747)
Disaster relief and epidemic response	Indian Red Cross Society (established 1920, with branches all over India) — relief in earthquakes, floods and epidemics; milk, vitamins and medical supplies to hospitals and dispensaries; maternal and child welfare centres; family planning clinics; blood banks; first-aid and home-nursing training (Park 28e, Ch 10, p 747)
Sanitation and village development	Bharat Sevak Samaj (1952) — improvement of village sanitation, with branches in nearly all districts. Kasturba Memorial Fund (raised after Kasturba Gandhi's death in 1944) — welfare of village women through <i>gramsevikas</i> (Park 28e, Ch 10, p 747)
Technical resource and capacity building	Voluntary Health Association of India, Child In Need Institute (Kolkata), Family Planning Association of India (Mumbai), Gandhigram Institute of Rural Health and Family Welfare Trust (Dindigul) act as Regional Resource Centres supporting smaller NGOs (Jugal Kishore, Ch 1, p 259)

Where NGOs sit in the public–private partnership classification

Group (Jugal Kishore, Ch 1, pp 116–117)	Partner	Example
A	Government + the for-profit sector	Uttaranchal Mobile Hospital and Research Centre; contracting-out of IEC by the Gujarat State Malaria Control Society; contracting-in of cleaning and laundry; Arogya Raksha insurance scheme, Andhra Pradesh
B	Government + the non-profit sector — NGO work	Mother NGO / Service NGO schemes; Arpana Trust in Delhi's Molarbund colony; Karuna Trust managing the Gumballi and Sugganahalli PHCs (1996) for the tribal hilly areas ; the Arunachal Pradesh–VHAI–Karuna Trust collaboration
C	Government + an individual private service provider	Contractual additional auxiliary nurse midwife (ANM) in remote sub-centres; contractual staff nurses; private anaesthetist at Rs 1,000 per case; MTP-trained Safe Motherhood Consultant at Rs 500 per visit; the Haryana Urban RCH Model
D	Government + private sector and/or the non-profit sector and/or a private service provider and/or multilateral agencies — also NGO work	NGO distribution of medicated mosquito nets under the National Malaria Control Programme; District Blindness Control Societies (government, non-government and private); targeted interventions through the voluntary sector under the National AIDS Control Programme; NGO and private-practitioner expansion of DOTS under the tuberculosis programme

NGO/non-profit partnership therefore appears in **both group B and group D**; group A is for-profit and group C is individual private providers.

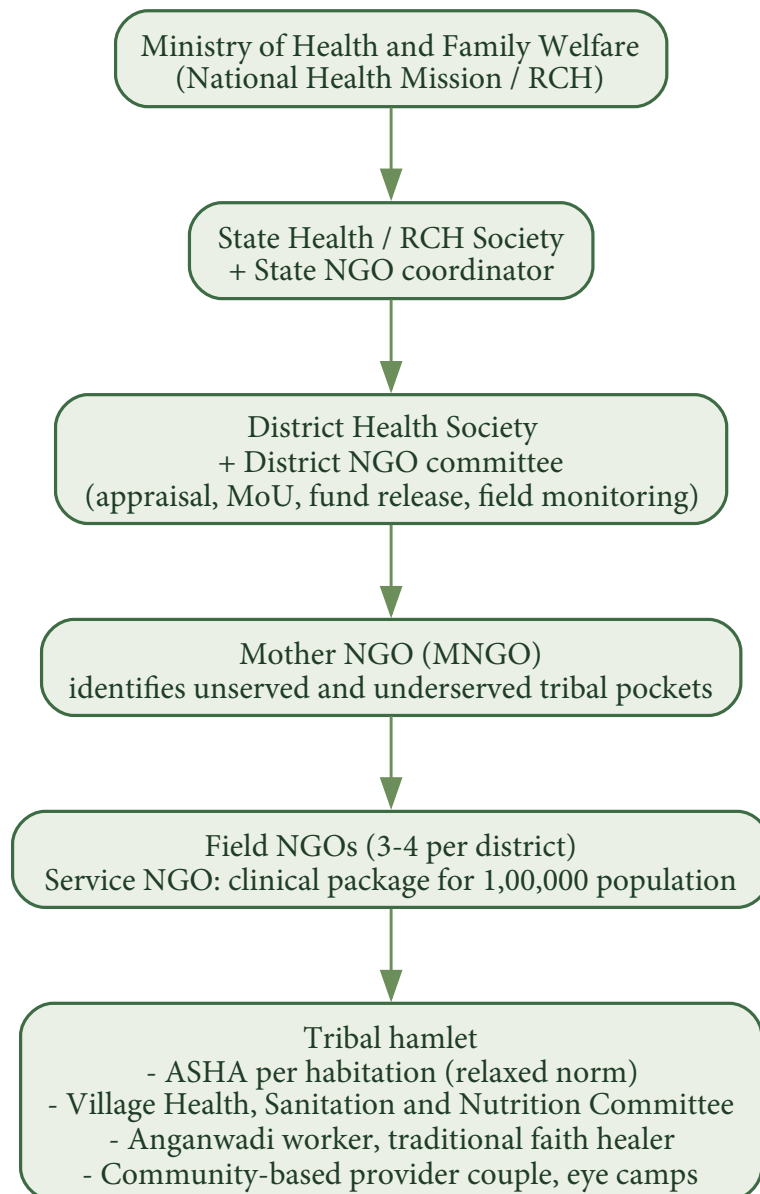
Formal government–NGO partnership mechanisms

- **Mother NGO (MNGO) scheme** — NGOs with substantial resources and proven competence are selected as MNGOs to improve reproductive and child health indicators **in underserved and unserved pockets** identified with district health officials; **306 MNGOs working in 414 districts** screen, select, guide and monitor smaller **field NGOs (FNGOs)**, each MNGO working with 3–4 FNGOs per district (Jugal Kishore, Ch 3, pp 989–990).
- **Service NGOs (SNGOs)** — clinical and specialised services (*dai* training, medical termination of pregnancy, male involvement) over a population of 100,000, in an area co-terminus with a CHC or block PHC (Jugal Kishore, Ch 3, pp 989–990).
- **Support and district machinery** — an Apex Resource Cell, Regional Resource Centres and best-practice centres, with **state NGO coordinators** who monitor implementation and facilitate NGO dialogue with the district health system (Jugal Kishore, Ch 1, p 259); the **district NGO committee** recommends MNGO proposals to the State Reproductive and Child Health Society, facilitates the memorandum of understanding, releases funds and conducts periodic field monitoring (Jugal Kishore, Ch 3, p 989).

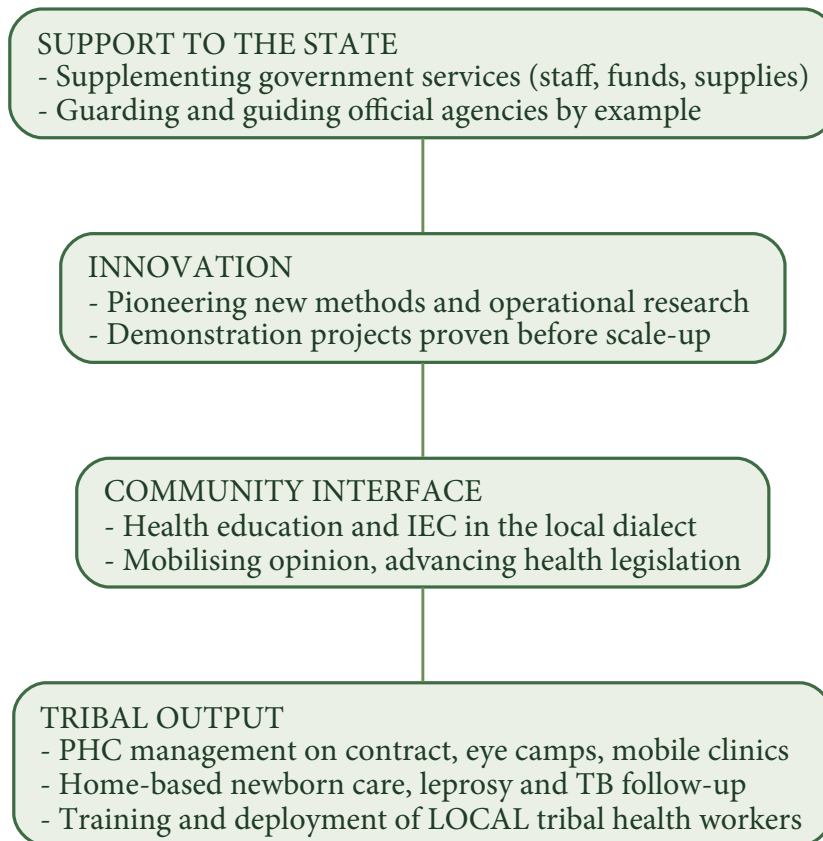
- **People public–private partnership, component (E) of the National Rural Health Mission (now the National Health Mission)** — contracting-in, contracting-out, outsourcing, **management of hospital facilities by leading NGOs**, hiring of staff, and service delivery including family planning, medical termination of pregnancy and treatment of sexually transmitted and reproductive tract infections (Sunder Lal, Ch 17, p 739), reinforcing the **National Population Policy, 2000** (Sunder Lal, Ch 17, p 765).
- **Programme-specific NGO windows for inaccessible areas** — the malaria programme's partnership schemes cover **outreach drug-distribution centres (DDCs) and fever treatment depots (FTDs)**, microscopy and treatment, and **insecticide-treated bed nets**, with partnership extended to faith-based and community-based organisations, *panchayati raj* institutions and **tribal-welfare departments** where accessibility is poor (Jugal Kishore, Ch 1, p 384).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Tribal health-care delivery with the NGO interface** — label: *tribal hamlet* → *ASHA (one per habitation) / anganwadi worker / traditional faith healer* → *sub-centre (1 per 3,000)* → *PHC (1 per 20,000)* → *CHC (1 per 80,000)* → *district hospital*; draw the **NGO/voluntary sector as a parallel side channel** feeding in at every level, arrows labelled *outreach eye camp, mobile eye clinic, contracted PHC, IEC and deformity prevention, home-based newborn care, training of local tribal workers*.



Channel through which a government-NGO partnership reaches a tribal hamlet



The six functions of a voluntary health agency and the tribal service they translate into

Limitations of the voluntary sector and how to strengthen it

- ▶ **Uneven distribution** — a wide inter-state differential exists in NGO coverage of villages; NGOs are unevenly spread across and within states and have a **limited area of operation** (Sunder Lal, Ch 17, p 765).
- ▶ **Under-mobilisation where most needed** — inadequate mobilisation of NGOs is itself listed among the causes of poor access in tribal areas, alongside difficult terrain and vacant posts (Sunder Lal, Ch 17, p 760).
- ▶ **Accountability** — grant-in-aid dependence requires pre-appraisal, appraisal and **independent evaluation of mother NGOs**, some national-level NGOs being entrusted with appraising others (Jugal Kishore, Ch 3, pp 989–990).
- ▶ **Align with the tribal workforce doctrine** — the **Expert Committee on Tribal Health** (October 2013, Ministry of Health and Family Welfare with the Ministry of Tribal Affairs, chaired by *Dr Abhay Bang*) held that the care provider should as far as possible be a **local tribal**, and that the centre of gravity of the workforce belongs **closer to the communities, not at the top**; it recommended an expanded ASHA role of 4 hours a day in tribal areas, mid-level providers through bridge courses at sub-centres and dedicated medical colleges in tribal districts (Park 28e, Ch 13, p 851). SEARCH, SARTHI and SEWA-Rural are the demonstrated vehicles for that train-and-deploy-locally model.

High-yield exam pointers

- ▶ **Definition to quote:** *Gunn and Platt* (1945) definition of a voluntary health agency; voluntary agencies are the "**motor trucks**" that penetrate the by-ways, official agencies the "**railway trunk lines**" that must run on tracks laid down by law (Park 28e, Ch 10, p 746) — the contrast that explains their value in tribal terrain.
- ▶ **Six functions, mnemonic S-P-E-D-G-A:** Supplementing government agencies, Pioneering, Education, Demonstration, Guarding/guiding government agencies, Advancing health legislation.
- ▶ **Tribal denominators:** 104 million tribal people, 705 tribes, 8.6% of the population (Census 2011 / Expert Committee on Tribal Health); relaxed norms **1 sub-centre/3,000, 1 PHC/20,000, 1 CHC/80,000**; ASHA norm relaxed to **one ASHA per habitation** in tribal, hilly and desert areas (Park 28e, Ch 7, p 539).
- ▶ **Three killer figures:** tuberculosis prevalence 703 vs 256 per 100,000; 8% of population but ~30% of malaria cases, >60% of *P. falciparum*, ~50% of malaria deaths; institutional delivery 68.0% vs 88.6% (NFHS-5, 2019–21).
- ▶ **Three named Indian NGO models to write:** SEARCH Gadchiroli → home-based newborn care adopted nationally; Karuna Trust → Gumballi and Sugganahalli PHCs contracted out in 1996 for tribal hilly areas; SARTHI (150 tribal villages, Gujarat) and SEWA-Rural (Bharuch) → community-based provider couples.
- ▶ **Partnership vocabulary:** Mother NGO – Field NGO – Service NGO; **306 MNGOs in 414 districts**; Apex Resource Cell and Regional Resource Centres; NGO work sits in **groups B and D** of the four-group public–private partnership classification (A = for-profit, C = individual private providers).
- ▶ **Committee to name:** Expert Committee on Tribal Health, *Dr Abhay Bang* (October 2013), MoHFW + Ministry of Tribal Affairs.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The corpus carries two counts for the Mother NGO scheme — 306 mother NGOs in 414 districts, and 102 mother NGOs with 800 field NGOs in 439 districts — so quote one and attribute it. The Jamkhed project appears in the indexed corpus only as a bibliographic reference, so name it but do not quote figures from it.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO Resolution 76.16 (2023) on the health of indigenous peoples, WHO Global Leprosy Strategy 2021–2030, National Strategic Plan for Malaria Elimination 2023–27

Programme and policy updates (India)

- ▶ **National Sickle Cell Anaemia Elimination Mission** — the Government of India has put substantial funding behind large-scale screening, treatment, counselling, education and the development of new technologies and gene therapies for sickle cell disease, and the programme's stated goal is elimination by 2047. The disease is concentrated in sickle-cell-endemic **tribal-**

dominated districts, which makes this the newest tribal-specific vertical programme. A multicentric community-mobilisation and information, education and communication (IEC) package across **six such districts** (9,837 people surveyed before the intervention, 9,696 after) raised the proportion who had heard of the disease with a difference-in-differences (DID) estimate of 55.5, and knowledge of its cause (DID 61.4), diagnosis (60.5), symptoms (55.5), treatment (57.7) and prevention (59.7) — the classical NGO functions of *education* and *demonstration* written into national programme design. [PMID 39644008] [PMID 40626777]

- **Health and Wellness Centres renamed *Ayushman Arogya Mandir*** (tag-line *Arogyam Parmam Dhanam*) — the single most examinable renaming since the textbook edition [PMID 41878109]. About **1.6 lakh** Ayushman Arogya Mandirs, **over 6,400** block-level non-communicable disease (NCD) clinics and **770** district NCD clinics together screen more than **8 crore adults a year** for hypertension and diabetes (2024 figures). The National Tuberculosis Elimination Programme (NTEP), the NCD programme and the National Mental Health Programme still run as separate vertical channels, so a tribal patient with tuberculosis plus diabetes plus depression is entered in three unlinked digital registers. [PMID 42397826]
- **MAHAN Trust, Melghat (Maharashtra)** — the working modern counterpart to SEARCH Gadchiroli, and a ready alternative example in the answer. An NGO serving about **3,00,000 people across 320 villages, over 80% of them tribal**, since **1997**, in a hilly, forested and hard-to-reach block. Village health workers chosen by the *gram sabha* deliver home-based child and adult care and community-based management of severe malnutrition in the local dialect, with documented falls in severe malnutrition and in mortality; health messages travel by song, folk play and drum-beating, and traditional faith healers are allowed to perform their rituals inside the hospital's intensive care unit. The authors frame tribal disease as a *quadruple* burden — malnutrition, infectious disease, non-communicable disease and mental illness, above all addiction. [PMID 40570231]

Current national data

- **Sickle cell disease in tribal districts, and tribal women's nutrition** — the Indian Sickle Cell Disease Registry, covering **324 patients at 12 PHCs across six tribal districts**, found **85.2% did not know their status before enrolment**; 83.6% were on hydroxyurea, 38.3% had been hospitalised and 12.0% transfused in the preceding year. On NFHS-5 data for **18,697 tribal women** of Bihar, West Bengal, Jharkhand and Odisha, **28.5% were underweight** and 7.6% overweight, underweight being commonest among rural women and in Odisha. [PMID 39024737] [PMID 38953825]
- **Nutrition support under NTEP, Chhattisgarh** — among 6,568 adults notified April–June 2024 in this tribal-preponderant state, **22% had severe and 13% very severe undernutrition**; **81%** received the first Ni-kshay Poshan Yojana instalment but only **29% within two months**, and just **21%** received food baskets, with coverage no better among the most undernourished. This is the inequity the Ni-kshay Mitra donor and NGO network exists to close. [PMID 41042600]

Elimination, surveillance and disease status

- **Kala-azar — elimination threshold reached, December 2023**. Endemic blocks still above **1 case per 10,000** fell from 633 (2005) to 136 (2015), 8 (2021) and **zero in 2023** (524 cases nationally that year; the last block to cross the line was in Pakur district, Jharkhand). India then entered a

consolidation phase of at least three consecutive years before it may submit a validation dossier to WHO. NGOs are named partners in the programme: **CARE India** did the community mobilisation and built the Kala-azar Management Information System for real-time reporting, **PATH** streamlined diagnostic and drug logistics and trained health personnel, and **DNDi** supported the treatment trials. Post-kala-azar dermal leishmaniasis (9,108 cases, 2014–23) and HIV–visceral leishmaniasis co-infection, which WHO put at 5–7% of Indian cases in 2022, remain the human reservoirs that could restart transmission. [PMID 40828749]

- **Malaria, tuberculosis and filariasis — the elimination dates the textbooks carry are the ones under revision.** Malaria cases fell roughly tenfold from **2.09 million (2001) to 0.2 million (2023)**, and India now targets **zero indigenous cases by 2027 with WHO certification by 2030** under the National Strategic Plan for Malaria Elimination 2023–27; Goa's annual parasite incidence reached zero in 2023, but imported cases in migrant construction labour persist (the Porvorim urban health centre alone reported 292 in 2023) [PMID 39953530]. For tuberculosis the **2025** elimination target set in 2018 is argued to need realignment with the global **2030** goal, with the **absence of targeted strategies for tribal populations** listed among the unaddressed gaps beside multidrug-resistant TB and relapse left unmonitored after treatment [PMID 40781735]. The tribal share of that gap now carries its own figure — a pooled prevalence of **7,030 per million** (the same order as the 703 per 100,000 in the body of this answer) against a national incidence of **196 per 100,000 in 2022**, some **2.8 million people**, a **fifth of them undiagnosed or unnotified**, and national goals of **44 per 100,000** incidence and **3 per 100,000** mortality by 2025; a systematic review of 34 studies (PROSPERO CRD42023439841) mapped tribal barriers and enablers onto the socio-ecological model and the implementation-research framework to design a customised package for the tribal *missing millions* [PMID 41235544]. The declared route into those blocks is the organised voluntary sector — **708 volunteers drawn from student and women's associations across 189 villages of Senapati district, Manipur** (98% of them scheduled tribe, one volunteer per 150 households), trained between **February 2023 and August 2025** to spot people with symptoms, collect and transport sputum and support patients through treatment; the share in the top knowledge band rose from **10% to 65%**, attitude scores improved modestly ($p = 0.0004$), but only 339 of the 708 stayed to the end — the retention problem that dogs volunteer schemes. India recorded an estimated **2.1 million new TB cases in 2024**, the world's highest, and the **National TB Prevalence Survey 2019–21** found only **36% of people with presumptive TB seek care** [PMID 42599962]. Lymphatic filariasis remains endemic in **345 districts** against a **2030** goal, and the non-profit and private sectors are described as a strategic but under-used resource for advocacy, social mobilisation, treatment refusal during mass drug administration (MDA), post-MDA surveillance, and finding and caring for patients. [PMID 41951149]

Guideline and legislative updates

- **Integration guidance for the Ayushman Arogya Mandir platform, and the coverage-is-not-access finding under the Pradhan Mantri Jan Arogya Yojana (PM-JAY).** MoHFW and the National Health Mission are urged to make **Ni-kshay, the NCD application and Tele-MANAS interoperable under the Ayushman Bharat Digital Mission within 12–18 months**, to make bidirectional TB–diabetes–common-mental-disorder screening mandatory at these centres, to have states refer 60–70% of TB patients for diabetes and 50–60% for depression within two years, and to begin shared-financing pilots in 2–3 years towards an institutionalised integrated care

package in five [PMID 42397826]. On the financing arm, among 320 enrolled PM-JAY beneficiaries in Jammu and Baramulla (27.2% scheduled tribe) only 48.4% had ever used the scheme; awareness of entitlement (odds ratio 1.253) and knowledge of the empanelled-hospital list (1.245) predicted use while poor road quality reduced it (0.761), and financial and documentation barriers were not independently significant. The recommended remedy is interpersonal, non-written IEC through ASHAs, anganwadi workers, panchayat and religious leaders — the voluntary-sector outreach function restated for an insurance-era scheme. [PMID 41878109]

Global and WHO developments

- **WHO Global Plan of Action for the Health of Indigenous Peoples** — under Resolution 76.16 (2023) WHO undertook to develop a global plan of action that builds in indigenous peoples' own understandings of and perspectives on health and wellbeing. Indigenous peoples are 6.2% of the world's population — probably an undercount, since some countries do not recognise them separately — and have shorter life expectancy at birth and poorer access to health care. This gives international standing to the Expert Committee on Tribal Health doctrine already in the answer — that the provider should as far as possible be a local tribal and the service culturally shaped. [PMID 40859176]
- **Leprosy — the elimination narrative as a hazard, and the NGO answer to it.** WHO's 1991 target (prevalence below 1 per 10,000) is widely misread as eradication, and the resulting loss of funding, political will and research investment is now flagged as the principal threat, alongside resistance mutations to rifampicin, dapsone and ofloxacin found in up to 8% of cases worldwide. Community engagement is the stated counter-strategy: under India's **National Leprosy Eradication Programme**, community members trained to recognise the signs, and affected people acting as advocates, in states such as Bihar increased reporting of suspected cases and reduced stigma, paralleling Brazil's community health agents and Nigeria's partnerships with religious leaders. The **Global Alliance to Eliminate Leprosy (1999)** paired WHO with the International Federation of Anti-Leprosy Associations (ILEP), the Nippon Foundation and the Novartis Foundation, and the **WHO Global Leprosy Strategy 2021–2030** targets zero transmission, zero disability and zero discrimination, with post-exposure prophylaxis and artificial-intelligence-assisted lesion recognition as the new tools. [PMID 41408601]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10, Ch 13, Ch 7); Sunder Lal — Textbook of Community Medicine (Ch 17, Ch 10, Ch 8); Jugal Kishore — National Health Programmes of India (Ch 1, Ch 3); Oxford Textbook of Public Health 7e (Ch 51); Reinke — Health Planning for Effective Management (Ch 13). Named documents: Report of the Expert Committee on Tribal Health, *Tribal Health in India: Bridging the Gap and a Roadmap for the Future* (MoHFW and Ministry of Tribal Affairs); National Family Health Survey-5 (2019–21); National Population Policy 2000; Census of India 2011.

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used: WHO Resolution 76.16 (2023) on the health of indigenous peoples; WHO Global Leprosy Strategy 2021–2030; National Strategic Plan for Malaria Elimination 2023–27 (India); National TB Prevalence Survey 2019–21 (India).



Bhore Committee recommendations

Asked in: Paper IV 17-Apr-2026 — RGUHS MD Community Medicine

Definition — The **Health Survey and Development Committee**, popularly the *Bhore Committee*, was appointed by the Government of India in 1943 under the chairmanship of **Sir Joseph Bhore** and submitted its report in 1946 (Mahajan & Gupta 4e, National Health Committees, p 485). Its terms of reference were to survey the existing health conditions and health organisation of British India and to make recommendations for future development. It remains the most comprehensive health policy and plan document ever prepared in India and became the blueprint for health planning in the independent country (Park 28e, Ch 23, p 1031; Jugal Kishore, Ch 1, p 52).

Situation documented by the Committee (the basis of its recommendations)

Indicator (British India)	Value recorded	Comparator
Crude death rate (1937)	22.4 per 1,000 population	—
Infant mortality rate (1937)	162 per 1,000 live births	—
Maternal mortality ratio	about 20 per 1,000 live births	—
Life expectancy at birth (1921–31)	26.9 years (male), 26.5 years (female)	—
Infant deaths as % of all deaths (1935–39)	24.3%	England and Wales (1938): 6.8%
Deaths under 10 years as % of all deaths	48.5%	England and Wales: 10.0%
Malaria	~100 million cases, ~2 million deaths per year	—
Tuberculosis mortality in cities	200–450 per 100,000	—
Population with safe drinking water / sanitary facilities	2% / 4.5%	—
Rural institutional cover	one institution for 105,626 people in 224 villages	—

(*Handbook of Epidemiology*, Ch 38, pp 1544–1545; *Sathe — Epidemiology and Management for Health Care, Development of Health Services*, p 192; *Jugal Kishore*, Ch 1, p 67)

The Committee attributed this state of affairs to insanitary conditions, poor nutrition, inadequacy of the existing medical and preventive organisation, illiteracy and lack of health education, and the social background of ill-health (Jugal Kishore, Ch 1, p 67).

Comprehensive health care and the conditions for healthful living

The term "comprehensive health care" was first used by this Committee — the provision of integrated preventive, curative and promotive services from "womb to tomb" to every individual residing in a defined geographic area, and it was to replace the earlier policy of providing medical care alone. Its stated criteria were:

Criterion	Requirement laid down by the Committee
(a)	Provide adequate preventive, curative and promotive health services
(b)	Be as close to the beneficiaries as possible
(c)	Command the widest cooperation between the people, the service and the profession
(d)	Be available to all irrespective of ability to pay
(e)	Look after specifically the vulnerable and weaker sections of the community
(f)	Create and maintain a healthy environment both in homes and in working places

(Park 28e, Ch 10, p 719)

The Committee further held that health is largely determined outside the health sector, and identified twelve primary conditions necessary for healthful living: suitable housing; sanitary surroundings; safe drinking water; preventive and curative health protection irrespective of ability to pay; qualitative and quantitative improvement in nutrition; elimination of unemployment; a living wage for the worker; improved agricultural and industrial production; improved communications, especially rural; a vigorous and healthy community life; physical and mental recreation; and the living interest and practical cooperation of the people themselves (Sathe, Development of Health Services, p 192).

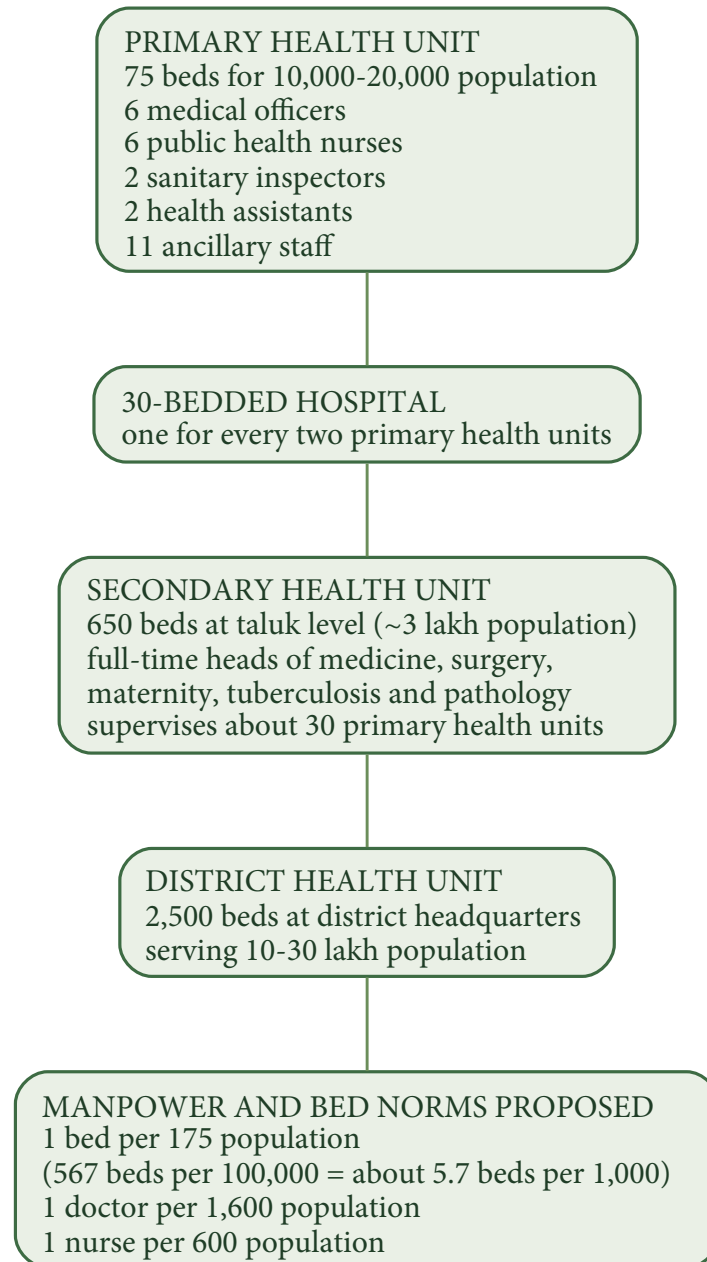
To build this service the Committee proposed a short-term programme over 10 years and a long-term programme spread over 30–40 years (Mahajan & Gupta 4e, Rural Primary Health Care, p 534).

Short-term programme (10 years)

Component	Recommendation
Primary health centre	One PHC per 40,000 population, with a 30-bedded hospital
PHC staff	2 medical officers, 1 nurse, 4 public health nurses, 4 midwives, 4 trained <i>dais</i> , 2 sanitary inspectors, 2 health assistants, 1 pharmacist, 15 class-IV employees
Secondary health centre	One secondary health unit per subdivision, to supervise, support and coordinate the PHCs
District level	Suitable strengthening of the existing district hospital

(*Red Book — Public Health and Preventive Medicine, Health Care System in India*, p 299; Mahajan & Gupta 4e, pp 485, 534)

Long-term programme — the "3 million plan"



Bhore Committee long-term plan ("3 million plan") — the proposed three-tier rural health structure, all tiers linked by telephone and ambulance

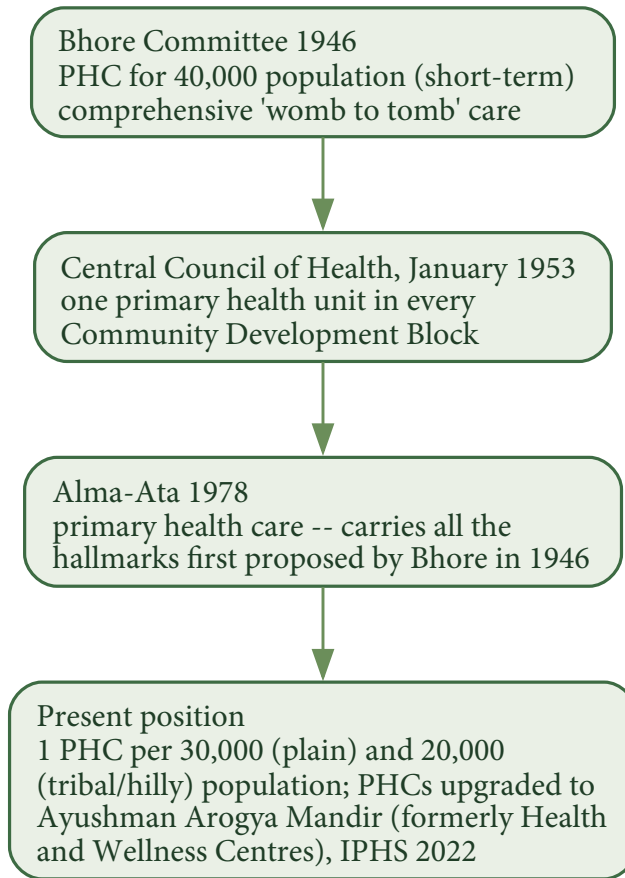
(Jugal Kishore, Ch 1, p 67; Park Plus, Ch 21, p 296)

Taken as a whole, and assuming a district population of 20 lakh and a primary health unit population of 15,000, the three-tier structure proposed above works out to about **7.5 beds per 1,000 population** — a separately derived aggregate, not a restatement of the 1-bed-per-175 norm (Mahajan & Gupta 4e, p 534).

Recommendations on administration, manpower, education and intersectoral action

- ▶ **Integration of the medical and public health departments** at every level, under a **single administrative officer** for the curative and preventive wings; the **Director of Health Services** should also be **Secretary** to the State Government; and a **unified health authority in each district** (Park 28e, Ch 23, pp 1034–1035).
- ▶ A **district health board** for every district, comprising district health officials and representatives of the public (Jugal Kishore, Ch 1, p 67).
- ▶ **Major changes in medical education** — a **three-month training in preventive and social medicine** for every undergraduate so as to produce "**social physicians**"; the Committee also reviewed the system of medical education and research as a whole (Park Plus, Ch 21, p 296; Red Book, p 299).
- ▶ A **village health committee** in every village, to secure the active cooperation and support of the people themselves (Jugal Kishore, Ch 1, p 67).
- ▶ **Greater emphasis on preventive work** than on curative medical relief (Jugal Kishore, Ch 1, p 67).
- ▶ **At least 15% of government expenditure to be spent on health care** (Jugal Kishore, Ch 1, p 67).
- ▶ **No patents on pharmaceutical products**, to keep essential medicines affordable (Jugal Kishore, Ch 1, p 67; Park Plus, Ch 21, p 296).
- ▶ **Intersectoral action on the conditions for healthful living listed above** — this recommendation, together with the report of the Environmental Hygiene Committee, led the Government of India to launch the **National Water Supply and Sanitation Programme** (Sathe, Development of Health Services, p 193).
- ▶ A **countrywide comprehensive programme against malaria**, endorsed by the Planning Commission in 1951 and launched as the **National Malaria Control Programme in April 1953** (Jugal Kishore, Ch 1, p 388).

Implementation status, legacy and criticisms



From the Bhore Committee to the present rural health infrastructure

Recommendation	Status
Network of primary health centres	Implemented — a PHC placed in every Community Development Block from 1953
Integration of preventive and curative services	Implemented — West Bengal led the process by creating the post of Director of Health Services in August 1947 ; completed by Maharashtra in May 1970
Comprehensive health care concept	Implemented in principle — became the basis of primary health care at Alma-Ata, 1978
Director of Health Services as Secretary to State Government	Not implemented — the Secretary continues to be an Indian Administrative Service officer
Bed norm of 1 per 175 population and the staffing norms	Not attained — PHC staffing was scaled down at the outset for want of personnel
15% of government expenditure on health	Not attained
Criticism 1	Model built exclusively on the allopathic system ; traditional health practices and indigenous systems prevalent in rural India were excluded
Criticism 2	Approach was not entirely decentralised but top-down — not community-led

Recommendation	Status
Criticism 3	Adopted the British "welfare state" model (ignoring the Soviet "health for all" model) and gave importance to vertical, single-disease programmes

(Red Book, p 299; Mahajan & Gupta 4e, p 534; Park 28e, Ch 10, p 719 and Ch 23, pp 1034–1035; Park Plus, Ch 21, pp 296–297; Sunder Lal, Ch 17, p 748)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Bhore Committee three-tier long-term structure (pyramid).** Labels, base upwards:
Primary health unit — 75 beds, 10,000–20,000 population; 30-bedded hospital for every two primary health units; Secondary health unit — 650 beds, taluk, ~3 lakh population; District health unit — 2,500 beds, 10–30 lakh population; side label linked by telephone and ambulance.
- ▶ **Short-term versus long-term programme (two-column box).** Left: *Short term, 10 years — PHC for 40,000, 30 beds, 2 medical officers*; Right: *Long term, 30–40 years, "3 million plan" — primary health unit for 10,000–20,000, 75 beds, 6 medical officers.*

High-yield exam pointers

- ▶ **Name and dates:** Health Survey and Development Committee, appointed 1943, chaired by Sir Joseph Bhore, reported 1946.
- ▶ **Two numbers that must appear:** short term PHC per 40,000 with 30 beds and 2 medical officers; long term 75-bedded primary health unit per 10,000–20,000 with 6 medical officers and 6 public health nurses.
- ▶ **"3 million plan" ladder:** 75-bed primary health unit → 30-bed hospital per two units → 650-bed secondary unit at taluk (3 lakh) → 2,500-bed district unit (10–30 lakh).
- ▶ **Norms:** 1 bed / 175 population (= 567 beds per 100,000), 1 doctor / 1,600, 1 nurse / 600; 15% of government expenditure on health.
- ▶ **Coined terms:** *comprehensive health care, "womb to tomb", social physician* (three-month training in preventive and social medicine).
- ▶ **1946 baseline to quote:** crude death rate 22.4/1,000, infant mortality rate 162/1,000 live births, life expectancy 26.9 years (male) / 26.5 years (female).
- ▶ **One-line verdict to close with:** the first and most comprehensive health-planning document for India, whose primary health centre and integrated preventive-plus-curative service remain the structural core of the system today.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park's 28th edition no longer carries the detailed committee-wise account (it refers the reader to the 24th edition), so the tier figures above are taken from Jugal Kishore, Mahajan & Gupta, the Red Book and Park Plus; Mahajan & Gupta gives the secondary health unit as 550 beds where Jugal Kishore and Park Plus give 650 beds, and the 650-bed figure is the one conventionally expected in RGUHS answers; internship training in public health is a present-day proposal listed under the current scenario in Park Plus, not a Bhore Committee recommendation, so it should not be attributed to the Committee.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Accounts, NFHS-5, NSS 75th round, Comprehensive Annual Modular Survey 2022–23, WHO

Programme and policy updates (India)

- ▶ Bhore's "primary health unit close to the people" now runs as a **nurse-led primary care team**: sub-centres upgraded under Ayushman Bharat comprehensive primary health care are staffed by a **Community Health Officer** (a nursing graduate with a six-month community health bridge course) working with the ANM, multipurpose worker and community health workers. A state-representative survey of **404 centres in Chhattisgarh (August–October 2023)**, each covering about **5,083 people**, found a mean of **358 outpatients per month including 128 NCD patients**, meeting **31% of the catchment population's outpatient need**, **26% for hypertension** and **21% for diabetes** [PMID 39367295].
- ▶ The **Jan Arogya Samiti** — the statutory community platform at each upgraded centre — had been constituted with monthly meetings in **99.2%** of those centres, making it the working descendant of the **village health committee** the Bhore Committee asked for in every village [PMID 39367295].
- ▶ A Punjab pilot of demand-side and supply-side strengthening at three model centres (2019–2021) showed, on difference-in-difference analysis against controls, monthly gains of **+265.7 people screened for NCDs**, **+21.3 newly diagnosed hypertensives and diabetics**, **+29.5 patients started on treatment**, **+102.2 outpatient attendances** and **+4.9 wellness sessions** — evidence that the Bhore-style package delivers only when supervision, training and community participation are added to infrastructure [PMID 38934822].
- ▶ Bhore's **single unified health authority** has re-emerged as a financing reform: the **Single Nodal Agency** system for centrally funded schemes, including the National Health Mission, was assessed in **Bihar and Odisha** (key-informant interviews February 2023 – February 2024). It improved cash management, expenditure transparency and alignment of central releases with actual spending, but weak budget credibility, patchy digital connectivity and thin finance staffing threaten to widen inequity in access to scheme funds [PMID 42480037].

Current national and state-level data

- ▶ **Comprehensive Annual Modular Survey 2022–23** (302,086 households — 173,096 rural, 128,990 urban — surveyed July 2022 to June 2023): **59.9%** of households had some health insurance and **49.7%** were covered by publicly funded schemes. Cover was associated with

greater outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128–1.210) and with less catastrophic expenditure at the 20% threshold (outpatient adjusted OR 0.869, 95% CI 0.810–0.932; inpatient adjusted OR 0.790, 95% CI 0.737–0.846), but **outpatient care remains the uncovered gap** [PMID 42167946].

- ▶ **The national financing figure to quote against Bhore's affordability criterion is the per-capita pair, not a state average:** National Health Accounts 2019–20 put per-capita total health expenditure at ₹4,864 against per-capita government health expenditure of ₹2,014 — that is, government met about 41% of the national total [PMID 39135015].
- ▶ **State-level financing, National Health Accounts 2019–20 linked to NFHS-5 across 21 major states** — these are **unweighted averages across states, not the national aggregate:** government spending averaged 39.44% of total health expenditure, out-of-pocket 51.2% and prepaid only 9.35%; per-capita health expenditure ranged from ₹1,588 in Bihar to ₹10,607 in Kerala, and NFHS-5 insurance coverage from 13.84% in Jammu and Kashmir to 87.84% in Rajasthan. On multilevel analysis, states whose dominant financing was government rather than out-of-pocket had significantly lower under-five mortality ($\beta -0.4793$) and neonatal mortality ($\beta -0.4868$) and better vaccination ($\beta +0.3941$) and WASH coverage ($\beta +0.3335$), all $p < 0.05$ — the modern evidence for Bhore's demand that health care be tax-financed [PMID 42218447].
- ▶ **NSS 75th round (2017–18, 113,823 households):** out-of-pocket payments pushed the poverty headcount from 12.26% to 13.51% overall (rural 17.71% → 19.05%), about 3.1 million households impoverished; the out-of-pocket share of total health expenditure fell from 64.2% (2013–14) to 48.2% (2018–19), yet government health spending remains near 2% of GDP — the clearest surviving failure of Bhore's demand that 15% of government expenditure go to health [PMID 41938958].

Elimination, surveillance and disease status

- ▶ **Kala-azar — target met, and the date matters:** India reached the elimination threshold as a public health problem in 2023, defined as under 1 case per 10,000 population at block level. The deadline had slipped repeatedly from the National Health Policy 2002 target of 2010. Jharkhand fell from 2,515 cases (2013) to 160 (2023), a 93.6% decline, and Dumka district crossed the threshold a year early, in 2022 [PMID 40988627].
- ▶ **Lymphatic filariasis remains the unfinished elimination:** despite a national programme running since the early 2000s, LF is still endemic in 345 districts, with the elimination goal now set at 2030; mass drug administration, morbidity management and post-MDA surveillance are straining programme capacity [PMID 41951149].

Guideline and legislative updates

- ▶ **Purchasing care from private hospitals has not replaced providing it** — the point on which Bhore's tax-funded public service model is being retested. A two-wave panel evaluation of AB-PMJAY in Chhattisgarh (2021 and 2022, about 15,000 individuals per wave, 1,627 hospitalisations) found roughly 90% of the population enrolled but no association with utilisation, patient satisfaction, length of stay, out-of-pocket expenditure or catastrophic expenditure, even after an evidence-based upward revision of package prices. Among enrolled patients admitted to private hospitals, 74.5% still crossed the 10% catastrophic-expenditure threshold, and mean out-of-pocket expenditure in private hospitals was about ten times that in

public hospitals, despite empanelment contracts expressly forbidding copayment [PMID 39135015].

Global and WHO developments

- The **Astana Declaration (2018)** restated primary health care as the route to universal health coverage, carrying forward the comprehensive, "womb to tomb" service Bhore described in 1946 and Alma-Ata adopted in 1978 [PMID 39367295]; against the **WHO benchmark of at least 5% of GDP** on health, India's roughly 2% shows how far the financing side of that model still lags [PMID 41938958].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10, Health Care of the Community, pp 719, 728; Ch 23, Health Planning and Management, pp 1031, 1034–1035); Mahajan & Gupta — Textbook of Preventive and Social Medicine, 4e (National Health Committees, p 485; Rural Primary Health Care, p 534); Jugal Kishore — National Health Programmes of India (Ch 1, pp 52, 67, 388); Red Book — Public Health and Preventive Medicine (Health Care System in India, p 299); Sathe & Sathe — Epidemiology and Management for Health Care (Development of Health Services, pp 192–193); Handbook of Epidemiology, Ahrens & Pigeot (Ch 38, pp 1544–1545); Sunder Lal — Textbook of Community Medicine (Ch 17, p 748); Park Plus (Ch 21, pp 296–297); Report of the Health Survey and Development Committee (Bhore Committee), Government of India, 1946.

Recent-advances sources: 39367295 — Performance of health and wellness centre in providing primary care services in Chhattisgarh, India, BMC Primary Care 2024, 10.1186/s12875-024-02603-1; 38934822 — Strengthening Comprehensive Primary Health Care through Ayushman Bharat: Role of Demand and Supply-side Interventions, Indian Journal of Public Health 2023, 10.4103/ijph.ijph_885_22 (abstract-only); 42480037 — Challenges to Achieving Universal Health Coverage Through Public Financial Management Reforms: Insights from the Implementation of Single Nodal Agency Reform in India, Health Systems and Reform 2026, 10.1080/23288604.2026.2693357 (abstract-only); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), BMJ Open 2026, 10.1136/bmjopen-2025-106382; 39135015 — The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) after four years of implementation, BMC Health Services Research 2024, 10.1186/s12913-024-11393-2; 42218447 — The role of health financing system and pluralism in achieving sustainable development goal health outcomes in India: a multilevel analysis, International Journal for Equity in Health 2026, 10.1186/s12939-026-02888-x; 41938958 — Impact of out-of-pocket health expenditure on poverty in India, Frontiers in Public Health 2026, 10.3389/fpubh.2026.1762886; 40988627 — Integrating community efforts, innovative government interventions and data in the fight against kala-azar: lessons from Dumka, India, International Health 2026, 10.1093/inthealth/ihaf083; 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, Acta Tropica 2026, 10.1016/j.actatropica.2026.108083 (abstract-only).



National Health Protection Mission

Asked in: Paper IV 17-Apr-2026 — RGUHS MD Community Medicine

Definition — The National Health Protection Mission (NHPM) is the health-financing arm of the Ayushman Bharat programme announced by the Government of India in February 2018 and launched on 23 September 2018 as **Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)**. It is a centrally sponsored, **entitlement-based**, fully tax-funded health assurance scheme providing cashless cover of **Rs 5,00,000 per family per year for secondary and tertiary care hospitalisation** to poor and vulnerable families identified from the Socio-Economic and Caste Census (SECC) 2011 (Park 28e, Ch 7, p 558; Park 28e, Ch 10, p 733). It is the second of the two pillars of Ayushman Bharat — the first being the **Ayushman Arogya Mandir (AAM)**, previously the Health and Wellness Centre, which delivers comprehensive primary health care (Park 28e, Ch 7, p 558). All pre-existing health protection or insurance schemes of the various Ministries were to be merged into the National Health Protection Mission, and it subsumed the erstwhile **Rashtriya Swasthya Bima Yojana (RSBY)** (Sunder Lal, Ch 17, p 764).

Why the Mission was needed — the case for financial protection

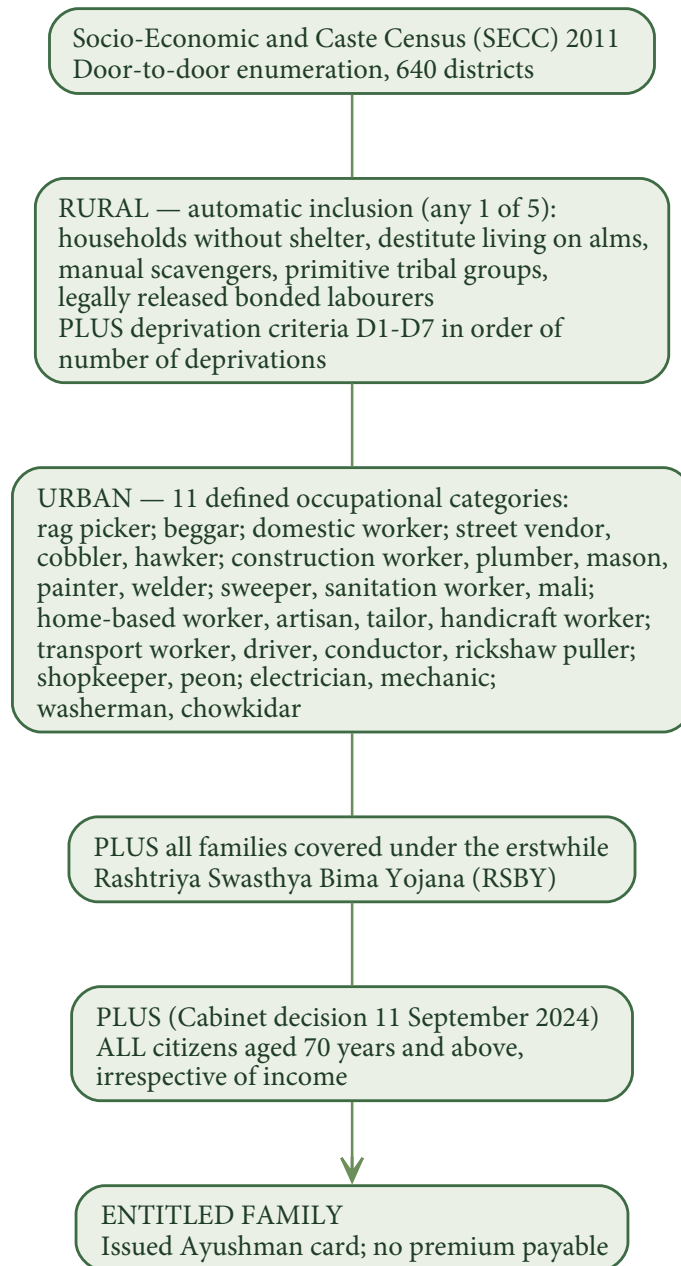
Problem	Evidence (with year)
Impoverishment by health costs	Out-of-pocket expenditure (OOPE) impoverishes an estimated 6 crore Indians every year ; PM-JAY is designed as financial risk protection against catastrophic expenditure (Park 28e, Ch 10, p 733)
Absence of any risk pooling	86% of the rural and 82% of the urban population were not covered by any health expenditure support scheme (Sunder Lal, Ch 17, p 763)
Low public financing	Total health spending 3.16% of GDP (2018–19) , of which government spending is only 1.28% of GDP ($\approx 40.6\%$ of total); failure to reach minimum public health expenditure is the "single most important constraint" (Sunder Lal, Ch 16, p 727)
Policy mandate	National Health Policy 2017 — progressively achieve universal health coverage (UHC) ; raise government health expenditure to 2.5% of GDP by 2025 (Sunder Lal, Ch 16, p 697 and Ch 10, p 577)
Predecessor scheme too shallow	RSBY (2008, Ministry of Labour; transferred to Ministry of Health and Family Welfare 01.04.2015) gave only Rs 30,000 per family per year , 1,516 packages, covering 2.75 crore families (Park 28e, Ch 10, p 733)
Global commitment	Sustainable Development Goal target 3.8 — UHC with financial protection, tracked by catastrophic and impoverishing out-of-pocket health spending (Park 28e, Ch 14, p 858)

Objectives and salient design features

- **Reduce catastrophic out-of-pocket expenditure** on hospitalisation and prevent medical impoverishment — the scheme is projected to prevent about **6 crore (60 million) Indians** from slipping below the poverty line annually (Sunder Lal, Ch 17, p 763).
- **Improve affordability, accessibility and quality** of secondary and tertiary care for the poor and vulnerable (Sunder Lal, Ch 17, p 763).

- ▶ **Entitlement-based, not enrolment-based** — no premium is paid by the beneficiary; every family appearing in the defined SECC database and in RSBY is entitled to claim. Aadhaar is encouraged for identification but **no person is denied benefit for want of an Aadhaar card** (Sunder Lal, Ch 17, pp 763-764).
- ▶ **Cashless and paperless** access at the point of service in any empanelled hospital — public or private — anywhere in India (**national portability**) (Sunder Lal, Ch 17, p 763).
- ▶ **No cap on family size, age or gender**; women are preferably recorded as head of the family so that the girl child and senior citizens are not excluded (Park 28e, Ch 10, p 733; Sunder Lal, Ch 17, p 763).
- ▶ **All pre-existing conditions covered from day one** (Park 28e, Ch 10, p 733).
- ▶ **Consolidation** — all ongoing health protection/insurance schemes across Ministries to be merged with the Mission (Sunder Lal, Ch 17, p 764).
- ▶ **Fraud and grievance control** — a defined public grievance redressal mechanism with safeguards against misuse by providers and users; **pre-authorisation is mandatory** for all tertiary care and selected secondary care packages (Sunder Lal, Ch 17, p 764).

Beneficiary identification



Entitlement pathway — who is covered under the National Health Protection Mission

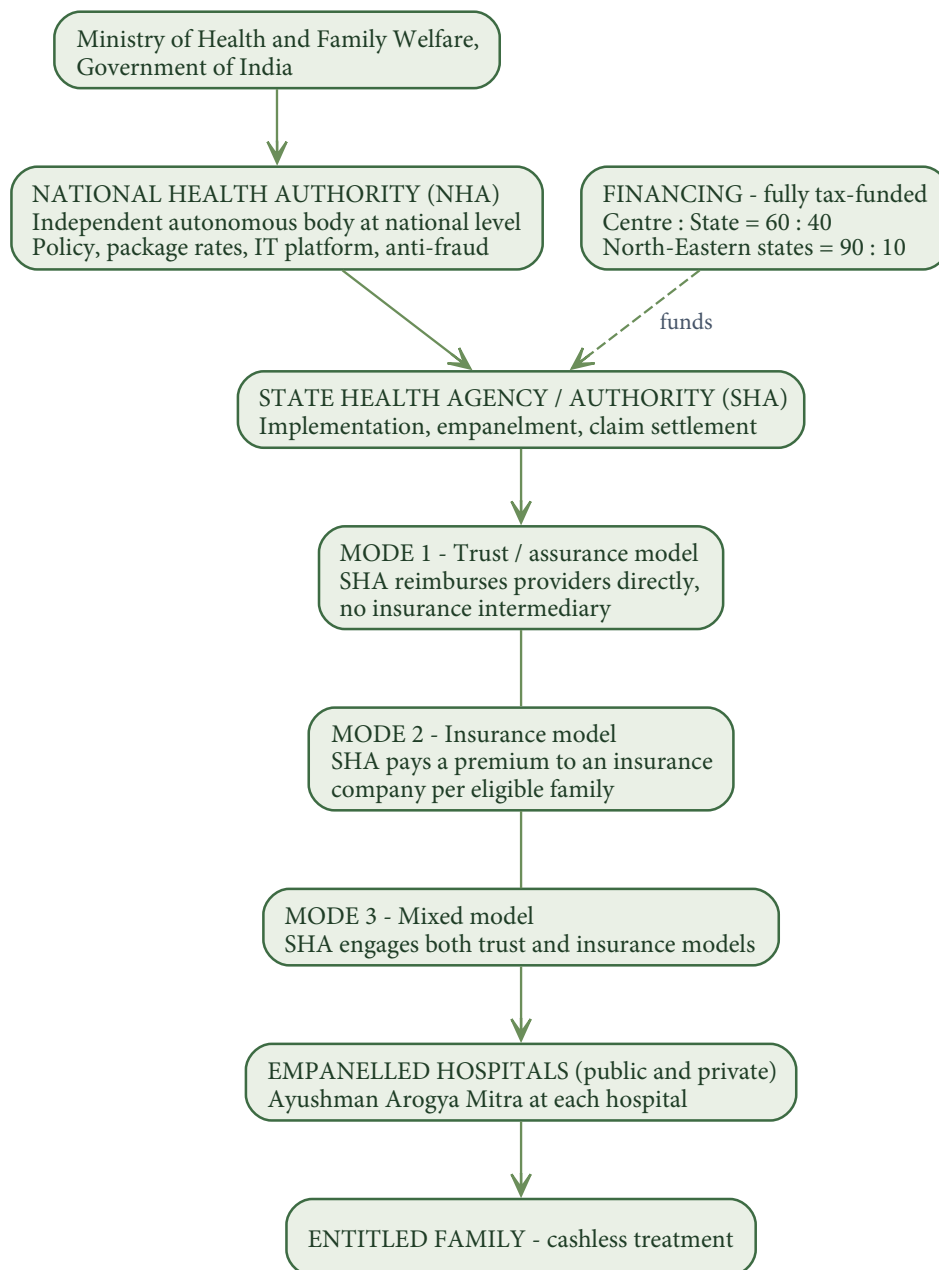
(Sunder Lal, Ch 2, pp 75-76; Sunder Lal, Ch 17, pp 763-764; Park 28e, Ch 7, pp 559-560)

Benefit package — what the entitlement buys

Component	Entitlement
Cover	Rs 5,00,000 per family per year, on a family floater basis, for secondary and tertiary care hospitalisation (Park 28e, Ch 10, p 733)
Families covered	10.74 crore poor and vulnerable families ≈ 50 crore people (Park 28e, Ch 10, p 733); Sunder Lal cites 107 million families / 500 million persons and >40% of the population (Sunder Lal, Ch 17, p 763)

Component	Entitlement
Packages	1,685 procedures covering treatment, food, drugs and supplies, and diagnostic services (Park 28e, Ch 10, p 733); the plan was launched with 1,592 procedures across 23 specialities at fixed package rates (Sunder Lal, Ch 17, p 764)
Continuum of care	3 days of pre-hospitalisation and 15 days of post-hospitalisation expenses, including drugs and diagnostics (Park 28e, Ch 10, p 733)
Mode of access	Cashless, paperless; portable across all empanelled public and private hospitals in India (Park 28e, Ch 10, p 733)
Exclusions from cap	No cap on family size, age or gender; all pre-existing conditions covered (Park 28e, Ch 10, p 733)
Facilitator	Ayushman Arogya Mitra deployed at every empanelled hospital to verify, assist and guide beneficiaries (Sunder Lal, Ch 17, p 763)

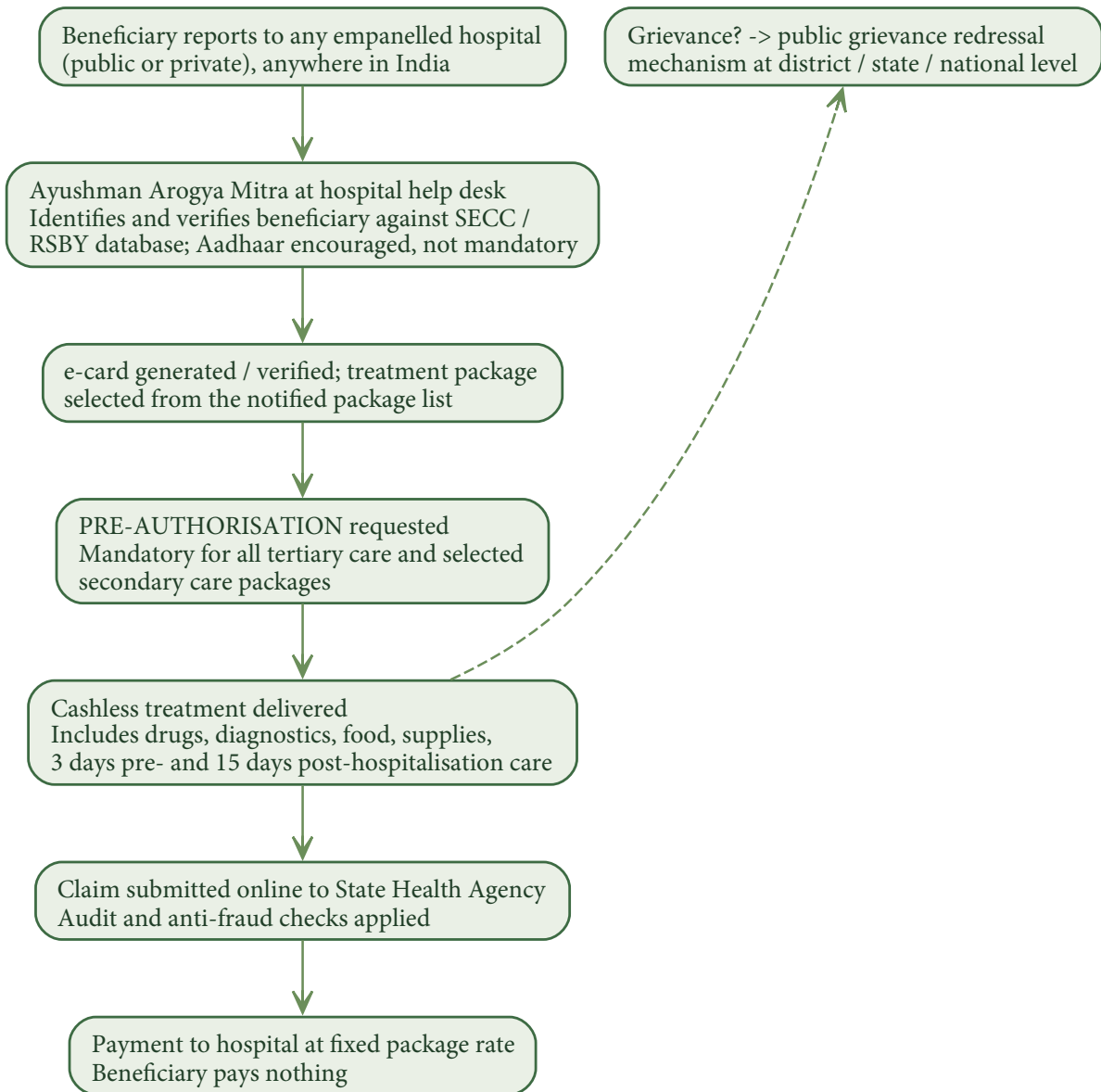
Institutional structure, implementation model and financing



Institutional structure and fund flow of the National Health Protection Mission

(Sunder Lal, Ch 17, p 764)

Service delivery — the cashless claim pathway



Cashless, paperless claim pathway at an empanelled hospital under PM-JAY

(Sunder Lal, Ch 17, pp 763-764)

Progress and achievements

Indicator	Reported achievement
Verified individual beneficiaries	Nearly 16 crore (Park 28e, Ch 10, p 733)
Hospitals empanelled	Over 21,800 (Park 28e, Ch 10, p 733)
Hospitalisations facilitated	Over 1.82 crore (Park 28e, Ch 10, p 733)
Out-of-pocket expenditure saved	Over Rs 22,500 crore (Park 28e, Ch 10, p 733)
Population under health protection	Rose to about 35% of the population after Ayushman Bharat, with the intention of covering all by 2025 (Sunder Lal, Ch 16, pp 729-730)

Indicator	Reported achievement
Planned expansion	To 80% of the population on a voluntary contribution basis (Sunder Lal, Ch 17, p 763)

Expansion to senior citizens (Union Cabinet, 11 September 2024) — all citizens aged 70 years and above are covered irrespective of income, benefiting about 4.5 crore families including 6 crore senior citizens with cover of up to Rs 5 lakh per family (Park 28e, Ch 7, pp 559-560):

Situation of the senior citizen	Entitlement
Already in a family covered by AB PM-JAY	An additional top-up of Rs 5 lakh per year , not to be shared with family members under 70 years
Not part of an existing PM-JAY family	Cover of up to Rs 5 lakh per year on a family basis
Already under CGHS, ECHS or CAPF schemes	Choice — continue with the existing scheme or opt for AB PM-JAY
Covered by private health insurance or ESI	Also eligible to benefit from AB PM-JAY
Card	A new, distinct card is issued to eligible senior citizens

The **Ayushman Bharat Digital Mission (September 2021)** complements the Mission by giving every citizen a digital health identity linked to longitudinal health records, with the National Health Authority as the implementing agency (Sunder Lal, Ch 17, p 763 and Ch 8, p 535).

Challenges and evaluation

- **Hospitalisation-only cover** — the package excludes outpatient care, where over 70% of illness episodes (67.5% rural, 73.8% urban) are treated in the private sector and where much of out-of-pocket expenditure on drugs and diagnostics arises (Sunder Lal, Ch 16, pp 729-730).
- **Provider moral hazard / supplier-induced demand** — third-party payment creates an incentive for unnecessary admissions and up-coding of packages; this is why pre-authorisation and claim audit are built in (Health Economics, Ch 8, pp 143-144).
- **Consumer moral hazard and adverse selection** — inherent to any insurance mechanism where the informed party holds more information than the payer; entitlement-based enrolment of whole SECC-defined groups partly protects against adverse selection (Health Economics, Ch 11, pp 177-178).
- **Exclusion error** — reliance on the SECC 2011 frame leaves out newly impoverished and migrant households.
- **Fragmentation of risk pools** across schemes reduces system efficiency; lower out-of-pocket payment offers better protection to the poor only when pooling is broad (Oxford Public Health 7e, Ch 21, p 833).
- **Prepayment does not abolish catastrophic expenditure** — copayments and services outside the package continue to generate it, and catastrophic spending does not disappear automatically with rising income (Oxford Public Health 7e, Ch 21, pp 833-834).

- **Supply-side constraint** — empanelment is skewed to urban private hospitals, so a cover on paper does not guarantee an available bed in poorly served districts; the parallel strengthening of Ayushman Arogya Mandirs is the intended counterweight.

Indicators for evaluating the Mission — proportion of entitled families holding an Ayushman card; hospitalisation rate per 1,000 entitled persons; claim rejection and turnaround time; proportion of claims in public versus private hospitals; average package cost; and the two SDG 3.8 financial-protection indicators, **catastrophic** and **impoverishing** out-of-pocket health expenditure (Park 28e, Ch 14, p 858).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The two pillars of Ayushman Bharat** — draw a horizontal base labelled *Ayushman Bharat* (announced February 2018) carrying two vertical pillars supporting a roof labelled *Universal Health Coverage — National Health Policy 2017*. Left pillar: *Pillar 1 — Ayushman Arogya Mandir (formerly Health and Wellness Centre): comprehensive primary health care; 1.5 lakh sub-centres and PHCs to be transformed; first centre 14 April 2018, Bijapur, Chhattisgarh; headed by Community Health Officer*. Right pillar: *Pillar 2 — National Health Protection Mission / PM-JAY: Rs 5 lakh per family per year; 10.74 crore families; secondary and tertiary hospitalisation; launched 23 September 2018*.
- **The health-financing triangle for PM-JAY** — draw a triangle with vertices labelled *Purchaser (National Health Authority / State Health Agency)*, *Provider (empanelled public and private hospitals)* and *Beneficiary (SECC-identified family)*. Label the arrows: purchaser → provider "package-rate payment after claim audit"; provider → beneficiary "cashless service"; beneficiary → purchaser "entitlement, no premium"; and a fourth arrow into the purchaser labelled "general taxation, Centre : State 60 : 40".

High-yield exam pointers

- **Dates:** announced February 2018 → launched **23 September 2018**; RSBY subsumed on the same date; senior-citizen expansion approved **11 September 2024**.
- **Numbers to write:** Rs 5 lakh per family per year; **10.74 crore** families ≈ **50 crore** people; **1,685** procedures (launched with 1,592 across 23 specialities); **3 days pre- + 15 days post-hospitalisation**; funding **60:40** (Centre:State), **90:10** for North-Eastern states.
- **Mnemonic for the design features** — "CAP FREE": Cashless and paperless; All pre-existing conditions covered from day one; Portable nationally; Family size, age and gender — no cap; Rs 5 lakh per family per year; Entitlement-based, no premium paid; Empanelled public *and* private hospitals.
- **Three modes of implementation:** **Trust/assurance, Insurance, Mixed** — always name all three.
- **Structure:** National Health Authority (national) → State Health Agency/Authority (state) → empanelled hospital → **Ayushman Arogya Mitra**.
- **Beneficiary identification:** SECC 2011 — rural 5 automatic-inclusion parameters + deprivation criteria D1–D7; urban **11 occupational categories**; plus all erstwhile RSBY families; plus all persons ≥70 years since 2024.

- **Context figures:** 86% rural / 82% urban had no health expenditure support; OOP expenditure impoverishes ~6 crore Indians a year; NHP 2017 target = government health expenditure 2.5% of GDP by 2025.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the scheme was announced in the 2018 Union Budget as the National Health Protection Mission (also called the National Health Protection Scheme) and was renamed **Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana** at launch, so write the current name first and the older name in parentheses. The two package counts differ because the Health Benefit Package has been revised across versions — Sunder Lal records the launch figure of 1,592 procedures across 23 specialities and Park 28e the later figure of 1,685 procedures; quote either with its source. The achievement figures (16 crore verified beneficiaries, 21,800 hospitals, 1.82 crore hospitalisations, Rs 22,500 crore saved) are quoted in Park 28e without an attached cut-off date, so state them as "as reported in Park 28e" rather than for a named year. The retrieved corpus carries no independent evaluation or National Health Accounts estimate of the Mission's effect on out-of-pocket expenditure, so none is claimed here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Accounts (India), NSS Comprehensive Annual Modular Survey 2022-23, National Health Authority policy briefs and dashboards, WHO-World Bank UHC Service Coverage Index

Programme and policy updates (India)

- **The Health Benefit Package (HBP) has been revised repeatedly and the current count supersedes both textbook figures** — HBP 1.0 carried 1,350 procedures, of which about 21.5% were surgical; across successive iterations (HBP 1.0 → 2.0 → 2.1) the most recent **HBP 2022 lists over 1,949 procedures, 1,284 (65.8%) of them surgical or minimally invasive**, with an *Unspecified Surgical Package* under which procedures costing less than Rs 1,00,000 may be booked directly and costlier ones only after scrutiny. Two further corrections to the standard answer: the **90:10 Centre:State share** covers not only the North-Eastern states but also **Jammu and Kashmir, Himachal Pradesh and Uttarakhand**, and as of 2021, 33 of the then 36 states/UTs had implemented PM-JAY — the three exceptions being **West Bengal, Odisha and New Delhi**. Always write the package count with its HBP version. [PMID 41393311]
- **Scale and budget line** — over **34.7 crore Ayushman cards** had been generated nationally by **August 2024**, and the central allocation to the scheme rose by about **28.8%, from Rs 7,300 crore (FY 2025) to Rs 9,406 crore (FY 2026)** — useful figures to quote against the older "16 crore verified beneficiaries" in Park 28e. [PMID 40976675]
- **Single Nodal Agency (SNA) public financial management reform** — assessed in Bihar and Odisha between February 2023 and February 2024, this restructuring of fund flow for centrally sponsored schemes, including the **National Health Mission**, delivered gains in cash management, expenditure transparency and alignment of central releases with state spending; weak budget credibility, patchy digital connectivity and thin finance staffing were identified as the risks that could widen inequity in access to scheme funds. [PMID 42480037]

Current national data

- **The financing crossover (National Health Accounts, 2013-14 to 2021-22)** — the government share of total health expenditure rose from **28.6% to 48.0%** while the out-of-pocket share fell from **64.2% to 39.4%**; total health expenditure is about **3.8% of GDP** and government health expenditure about **1.9% of GDP**, still short of the National Health Policy 2017 target of **2.5% by 2025**. Roughly **55 million Indians** are pushed into poverty by medical bills each year. These dated figures replace the 3.16%/1.28% of GDP (2018-19) numbers in the body. [PMID 42548603]
- **CAMS 2022-23 (NSS 79th round, 302,086 households)** — the first large national read-out four years after roll-out — 59.9% of households reported any insurance and **49.7% publicly funded health insurance**; PFHI cover was associated with higher outpatient use (adjusted OR **1.404**, 95% CI 1.328-1.484) and inpatient use (adjusted OR **1.168**, 95% CI 1.128-1.210). Mean inpatient out-of-pocket expenditure was **Rs 29,875** in PM-JAY/state-scheme households versus **Rs 37,572** in uninsured households, and inpatient catastrophic expenditure at the 10% threshold was **41.3% versus 54.2%**. [PMID 42167946]
- **Haryana case-control evaluation (772 hospitalised respondents, 2023)** — direct medical out-of-pocket expenditure was **65% lower** among beneficiaries (Rs 11,131 versus Rs 31,675) and catastrophic expenditure at a 30% threshold was **13.3% versus 45.9%** (OR 7.15, 95% CI 4.74-10.80). Beneficiaries nonetheless incurred **19% higher non-medical costs** (travel, food, lodging), which the package does not cover — the sharpest available evidence for the "hospitalisation-only cover" criticism. [PMID 40976675]

Elimination, surveillance and disease status

- **PM-JAY as a financial-protection arm of the National TB Elimination Programme** — under AB PM-JAY-Arogya Karnataka (July 2021 to June 2023), 9,173 confirmed TB patients were treated out of about 39.9 lakh approved pre-authorisations; **96.2% were below the poverty line** and the **public sector delivered 97.9% of TB care**, with roughly **Rs 10,000 of catastrophic cost averted per TB patient**. [PMID 40518205]

Guideline and legislative updates

- **Package pricing** — the long-standing objection is now quantified — at launch **42% of packages were reimbursed at about half the actual cost of care**, and successive HBP revisions have rationalised prices; costing work across 27 district hospitals in nine states shows a modelled 100-bed private district hospital funded **solely at PM-JAY rates reaches break-even in its fourth operational year** (second year if government provides free land). The Supreme Court has separately pressed the government to standardise private-hospital charges against publicly funded scheme rates. [PMID 40216706]
- **How catastrophic expenditure is counted is itself changing** — using NSS 75th round data (2017-18; 113,823 households), component-specific thresholds identify roughly **twice as many households in catastrophic expenditure as the conventional uniform 10% budget-share rule**. Read the outpatient-medicines headcount as a three-step ladder: **8.2% under the uniform 10% rule → 12.2% under a proportionate component-specific threshold → 20.7% under the income-sensitive inverse rank-weighted index (IRWI)**. The much-quoted 12.2% → 20.7% shift is therefore proportionate-to-IRWI, not uniform-to-IRWI. Relevant to how SDG 3.8 financial-protection indicators are reported. [PMID 41649954]

Global and WHO developments

- Realist review of countries with the largest gains in the WHO-World Bank UHC Service Coverage Index, 2000-2021 (42 studies, evidence concentrated in India and China) — primary-health-care reforms, which frequently ran on service expansion, benefit-package design and insurance or financing initiatives, did improve coverage and financial protection, but fell short on reaching marginalised groups, reducing financial barriers and improving service quality, with far more attention paid to contact coverage than to effective coverage. [PMID 42293679]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 — Health Programmes in India, Ayushman Bharat Programme, pp 558-560; Ch 10 — Health Care of the Community, Health Insurance, pp 733-734; Ch 14 — Sustainable Development Goals, p 858); Sunder Lal — Textbook of Community Medicine (Ch 2, pp 75-76 on SECC; Ch 16 — Health Planning and Management, pp 697, 727, 729-730; Ch 17 — Healthcare of the Community, pp 763-764; Ch 8, p 535 on the Ayushman Bharat Digital Mission); Oxford Textbook of Public Health 7e (Ch 21 — Health policy in developing countries, pp 831-834); Health Economics (Ch 8, pp 140-144; Ch 11, pp 177-182); National Health Policy 2017; Socio-Economic and Caste Census 2011; Union Cabinet decision of 11 September 2024 on senior-citizen coverage.

Recent-advances sources: 41393311 — Assessing Surgical Care under the Government-Funded Health Insurance Scheme — Pradhan Mantri Jan Arogya Yojana in India, *Annals of Global Health* 2025, doi:[10.5334/aogh.4908](https://doi.org/10.5334/aogh.4908); 40976675 — Impact of India's publicly funded health insurance scheme on financial risk protection: a case-control study from Haryana state in India, *BMJ Open* 2025, doi:[10.1136/bmjopen-2024-093304](https://doi.org/10.1136/bmjopen-2024-093304); 42480037 — Challenges to Achieving Universal Health Coverage Through Public Financial Management Reforms: Insights from the Implementation of Single Nodal Agency Reform in India, *Health Systems & Reform* 2026, doi:[10.1080/23288604.2026.2693357](https://doi.org/10.1080/23288604.2026.2693357) (abstract-only); 42548603 — Can a bigger budget shrink the bill at the bedside? Public health financing as a route to lower out-of-pocket spending in India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1903360](https://doi.org/10.3389/fpubh.2026.1903360); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 40518205 — Utilization of tuberculosis healthcare packages under Ayushman Bharat Pradhan Mantri Jan Arogya Yojana-Arogya Karnataka: A comprehensive socio-demographic analysis, *Indian Journal of Tuberculosis* 2025, doi:[10.1016/j.ijtb.2023.12.008](https://doi.org/10.1016/j.ijtb.2023.12.008) (abstract-only); 40216706 — Financial Viability of Private Hospitals Operating Under India's National Health Insurance Scheme Ayushman Bharat Pradhan Mantri-Jan Arogya Yojana (AB PM-JAY), *Applied Health Economics and Health Policy* 2025, doi:[10.1007/s40258-025-00966-9](https://doi.org/10.1007/s40258-025-00966-9); 41649954 — One size does not fit all: income-sensitive thresholds for catastrophic health expenditure, *Health Policy and Planning* 2026, doi:[10.1093/heapol/czag013](https://doi.org/10.1093/heapol/czag013); 42293679 — Understanding pathways from primary health care to universal health coverage outcomes: a realist review, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1808891](https://doi.org/10.3389/fpubh.2026.1808891). Also National Health Accounts estimates for India 2013-14 to 2021-22; NSS 79th round (Comprehensive Annual Modular Survey) 2022-23; NSS 75th round (Social Consumption: Health) 2017-18.

♦ ♦ ♦

Ayushman Bharat – Arogya Karnataka scheme

Asked in: Paper IV 19-Nov-2021 — RGUHS MD Community Medicine

Definition — Ayushman Bharat (AB) is the Government of India's flagship health initiative, announced in February 2018, resting on two pillars — the Ayushman Arogya Mandir (erstwhile Health and Wellness Centre) for comprehensive primary health care, and the Pradhan Mantri Jan Arogya Yojana (PM-JAY), a tax-funded cover for secondary and tertiary hospitalisation (Sunder Lal, Ch 17, p 763). Ayushman Bharat – Arogya Karnataka (AB-ArK) is the co-branded scheme created when Karnataka's own universal scheme *Arogya Karnataka* (launched 2 March 2018) was merged with PM-JAY and implemented in assurance mode from 30 October 2018, so that one card, one package list and one state trust serve the whole state population (Government of Karnataka, Health & Family Welfare Department, Order No. HFW 69 CGE 2018 dated 15 November 2018 — hereafter *GoK AB-ArK order 2018*).

Rationale — the financial-protection gap the scheme addresses

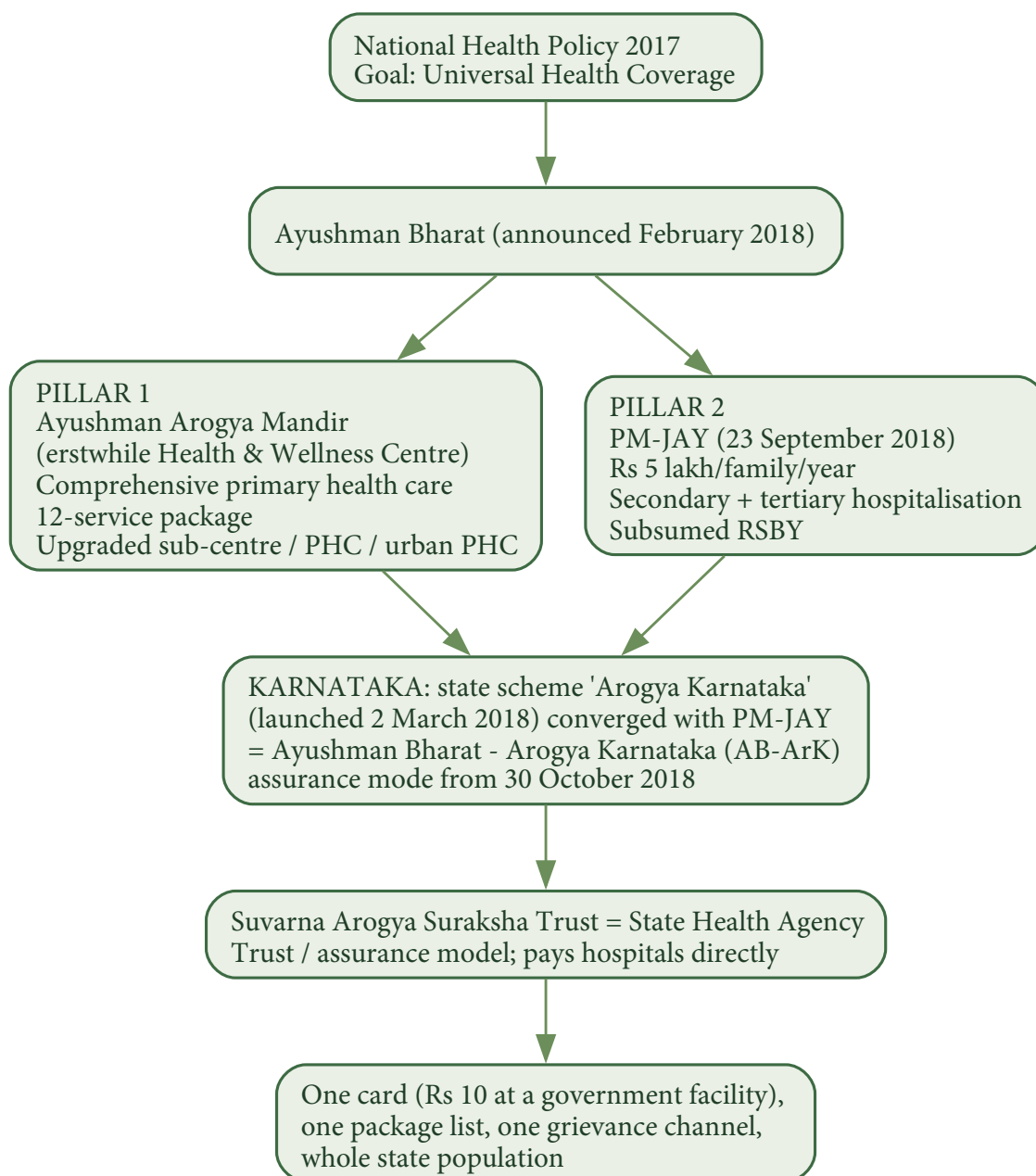
Ayushman Bharat was recommended by the National Health Policy 2017 to move India towards universal health coverage (UHC) and the Sustainable Development Goal commitment to "leave no one behind", replacing a sectoral, selective approach to service delivery with a continuum of care across primary, secondary and tertiary levels (AFMC Red Book 2024, Ch 38, p 1209). The gap it addresses is financial:

Problem indicator	Value (with year/source)
Out-of-pocket expenditure (OOPE) on health, India	Highest in the world; 1.52% of gross domestic product, ₹2,155 per capita per year (Sunder Lal, Ch 16, pp 728–729)
Population with no health-expenditure support before AB	86% rural, 82% urban (Sunder Lal, Ch 17, p 763)
Impoverishment from health spending	~4.2% of Indians (~50 million people) pushed into poverty every year (Sunder Lal, Ch 16, pp 728–729)
Catastrophic health expenditure — definition	Household health cost >10% of total monthly consumption expenditure, or >40% of non-food consumption expenditure (Sunder Lal, Ch 16, p 729)
Households facing catastrophic expenditure	17.3% (2012–20) falling to 7% (2022) (Sunder Lal, Ch 16, p 729)
Household with ≥1 member covered by any health scheme/insurance	41% (NFHS-5, 2019–21), up from 29% (NFHS-4); 46% of the insured are covered by a state scheme (NFHS-5 India Report, Ch 6, p 483)
Social security expenditure as share of total health expenditure	9.6% (Sunder Lal, Ch 16, p 729)

Karnataka entered this scheme with a fragmented legacy of overlapping state schemes — Yeshasvini (co-operative-society farmers, ₹210 annual contribution, 803 surgical procedures), Rashtriya Swasthya Bima Yojana (RSBY) and Vajpayee Arogyasri (below-poverty-line, ₹1.5 lakh family floater, 402 procedures plus 50 follow-up procedures) — each with its own card, package list and beneficiary register

(Park Plus, Ch 21, p 297; Jugal Kishore, Ch 1, pp 130–131). AB-ArK's first contribution to health was therefore **administrative**: collapsing these into a single entitlement.

Structure of Ayushman Bharat and its Karnataka convergence



Architecture of Ayushman Bharat and its convergence into Ayushman Bharat - Arogya Karnataka

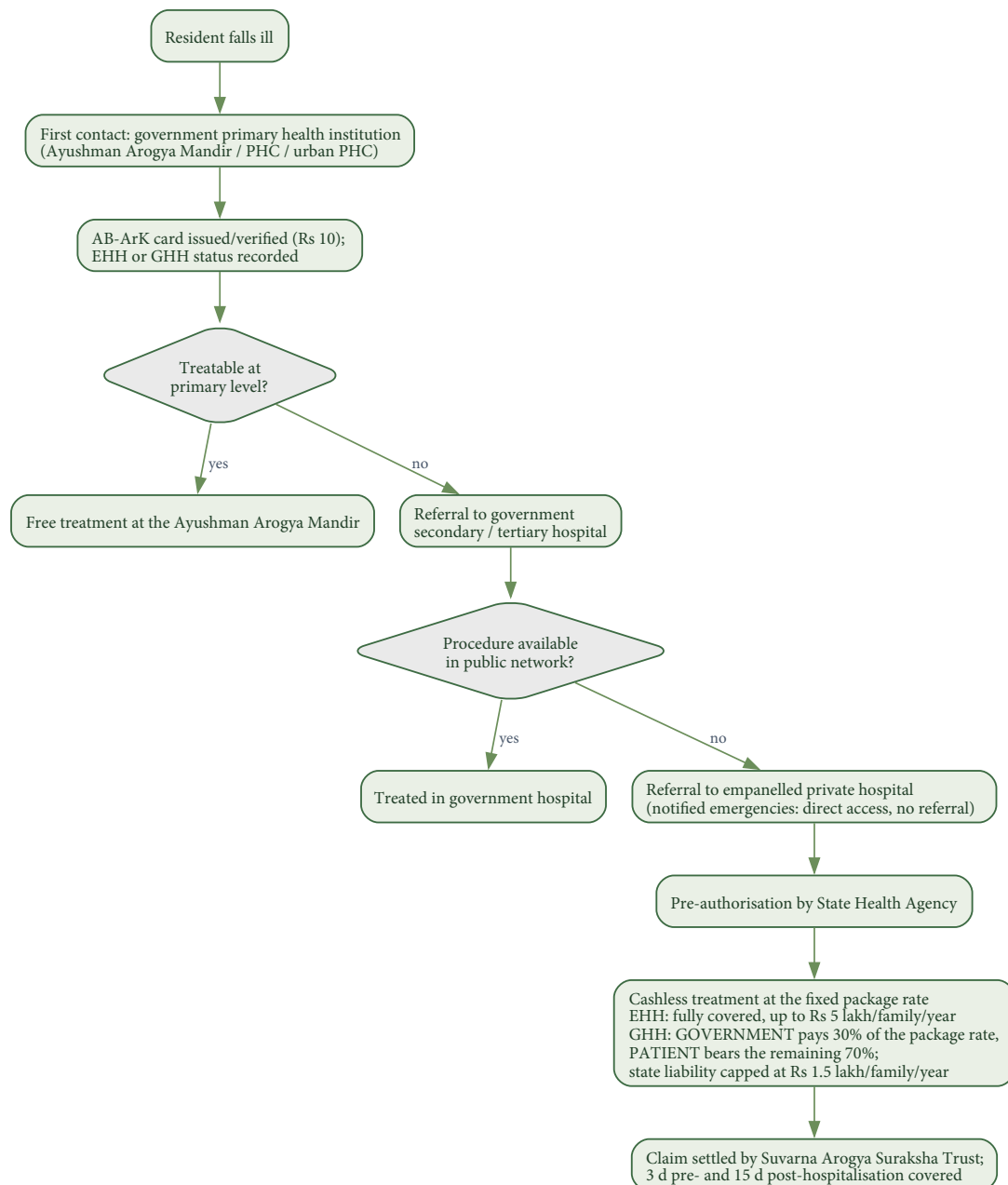
Feature	National PM-JAY	Ayushman Bharat – Arogya Karnataka
Target population	10.74 crore poor and vulnerable families (~50 crore people), the bottom 40%, identified from SECC 2011 deprivation and occupational criteria (AFMC Red Book 2024, Ch 38, p 1210)	Universal in intent — every resident of Karnataka, split into Eligible Households (EHH) holding a National Food Security Act 2013/below-poverty-line ration card or enrolled under RSBY, and General Households (GHH) without such a card (GoK AB-ArK order 2018)

Feature	National PM-JAY	Ayushman Bharat – Arogya Karnataka
Cover	₹5 lakh per family per year (Park 28e, Ch 10, p 745)	EHH: fully covered up to ₹5 lakh per family per year. GHH: the government pays 30% of the fixed package rate and the patient bears the remaining 70%, with state liability capped at ₹1.5 lakh per family per year (GoK AB-ArK order 2018)
Access	Cashless, paperless at any empanelled public or private hospital; portable across India (AFMC Red Book 2024, Ch 38, p 1210)	Referral-gated — complex secondary and tertiary procedures need referral from a government primary health institution; notified emergencies may go directly to an empanelled hospital without referral (GoK AB-ArK order 2018)
Family limit	No cap on family size, age or gender (AFMC Red Book 2024, Ch 38, p 1210)	Same; woman preferably recorded as head of family (Sunder Lal, Ch 17, pp 763–764)
Implementing body	National Health Authority (centre) and State Health Authority/Agency (state) (Sunder Lal, Ch 17, p 764)	Suvarna Arogya Suraksha Trust (SAST) acts as State Health Agency under the trust/assurance model , reimbursing hospitals directly (GoK AB-ArK order 2018)
Financing	Centre : state 60:40; 90:10 for north-eastern states (Sunder Lal, Ch 17, p 763)	Same 60:40 for the ~62 lakh SECC-2011/RSBY families; Karnataka funds 100% of the cost for all remaining resident families (GoK AB-ArK order 2018)
Legal basis for convergence	The scheme "is also implemented in convergence with the State schemes where the State governments have identified other beneficiaries based on similar parameter as of AB-PMJAY" (AFMC Red Book 2024, Ch 38, p 1210)	Karnataka used this clause to fold Yeshasvini, RSBY, Vajpayee Arogyasri and other state schemes into one entitlement (GoK AB-ArK order 2018)

Benefit package and delivery pathway

- **Package content** — services across **27 specialities** covering **1,949 procedures**, at fixed pre-agreed rates, including **3 days of pre-hospitalisation** and **15 days of post-hospitalisation** expenses (AFMC Red Book 2024, Ch 38, p 1210).
- **Package grading in Karnataka** — packages are tiered as *primary*, *simple secondary* (reserved for government hospitals to protect the public system), *complex secondary*, *tertiary* and *emergency* (GoK AB-ArK order 2018).
- **Beneficiary identification** — entitlement-based, not enrolment-based; the AB-ArK card is issued at a government primary health institution on payment of ₹10, and the **Ayushman Arogya Mitra** posted at every empanelled hospital helps the beneficiary claim (Sunder Lal, Ch 17, p 763; GoK AB-ArK order 2018).
- **Pre-authorisation** is mandatory for **all tertiary care** and **selected secondary care packages** (Sunder Lal, Ch 17, p 764).

- **Anti-fraud and grievance safeguards** — a well-defined complaint and public grievance redressal mechanism with robust safeguards against misuse, fraud or abuse by providers and users (Sunder Lal, Ch 17, p 764).



Patient pathway and gatekeeping under Ayushman Bharat - Arogya Karnataka

The primary-care pillar — Ayushman Arogya Mandir and its 12-service package

The twelve packages of comprehensive primary health care (CPHC) delivered at every Ayushman Arogya Mandir are (Park 28e, Ch 10, p 731; Sunder Lal, Ch 17, p 764):

- Care in pregnancy and childbirth
- Neonatal and infant health care services
- Childhood and adolescent health care services
- Family planning, contraceptive services and other reproductive health care services

- Management of communicable diseases, including national health programmes
- General outpatient care for acute simple illnesses and minor ailments
- Screening, prevention, control and management of non-communicable diseases
- Care for common ophthalmic and ear, nose and throat (ENT) problems
- Basic oral health care
- Elderly and palliative health care services
- Emergency medical services, including burns and trauma
- Screening and basic management of mental health ailments

Input	Norm
Population norm, rural sub-centre Ayushman Arogya Mandir	5,000 in plain areas; 3,000 in hilly/tribal/desert areas (Park 28e, Ch 10, p 731)
Population norm, urban Ayushman Arogya Mandir	One per 15,000–20,000 population, serving slums and other vulnerable pockets (Park 28e, Ch 10, p 731)
Access principle	"Time to care" — no more than 30 minutes of travel (Sunder Lal, Ch 17, p 742)
Team at the upgraded sub-centre	ASHAs (one per 1,000 population) and auxiliary nurse midwives (ANMs) led by a mid-level health provider / community health officer — a BAMS physician or BSc nurse (Park 28e, Ch 10, p 731; Sunder Lal, Ch 17, p 742)
Coverage achieved	Nearly 1.1 lakh sub-health centres transformed by March 2022, of a target of 1.5 lakh sub-centres and PHCs (Sunder Lal, Ch 17, pp 746, 764)
Additional platform role	Teleconsultation and an expanded range of diagnostics; over 1.17 lakh centres are telemedicine-enabled (Park 28e, Ch 10, p 731; Sunder Lal, Ch 8, pp 534–535)

How the scheme improves health

Dimension of universal health coverage	Mechanism in AB-Ark	Expected/observed gain
Breadth — population cover	Converging PM-JAY with Arogya Karnataka makes every resident eligible, not only the SECC-listed poor; nationally, states have expanded the base from 10.74 crore to 15.5 crore families (AFMC Red Book 2024, Ch 38, p 1210)	Household insurance cover rose from 29% (NFHS-4) to 41% (NFHS-5, 2019–21), and 46% of the insured are covered by a state scheme (NFHS-5, Ch 6, p 483)
Depth — service cover	The 12-package CPHC at Ayushman Arogya Mandir adds screening for non-communicable disease, mental health, oral, ENT, elderly and palliative care to the old selective package	Shifts detection and control of hypertension, diabetes and cancers to the sub-centre level, closer to home

Dimension of universal health coverage	Mechanism in AB-Ark	Expected/observed gain
Height — cost cover	Cashless, fixed-rate packages with pre- and post-hospitalisation cover remove the point-of-service payment for Eligible Households	Designed to prevent about 6 crore (60 million) people a year from slipping below the poverty line because of catastrophic health expenditure (Sunder Lal, Ch 17, p 763)
Continuity of care	Referral gatekeeping links the Ayushman Arogya Mandir to taluk, district and empanelled private hospitals in one continuum	Converts a fragmented, episodic system into a continuum-of-care model across the life cycle (AFMC Red Book 2024, Ch 38, p 1209)
Equity	No cap on family size, age or gender; woman preferred as head of family; simple-secondary packages reserved for government hospitals	Protects the girl child and the elderly, who were excluded by family-size caps in earlier schemes (Sunder Lal, Ch 17, pp 763–764)
Health-system strengthening	Ayushman Bharat Digital Mission (September 2021) gives every citizen a health account for longitudinal records and portability; Ayushman Bharat Health Infrastructure Mission (October 2021) equips all 733 districts for epidemic response (Sunder Lal, Ch 8, p 535; Ch 20, pp 786, 805–806)	Enables telemedicine, national portability and evidence-based purchasing

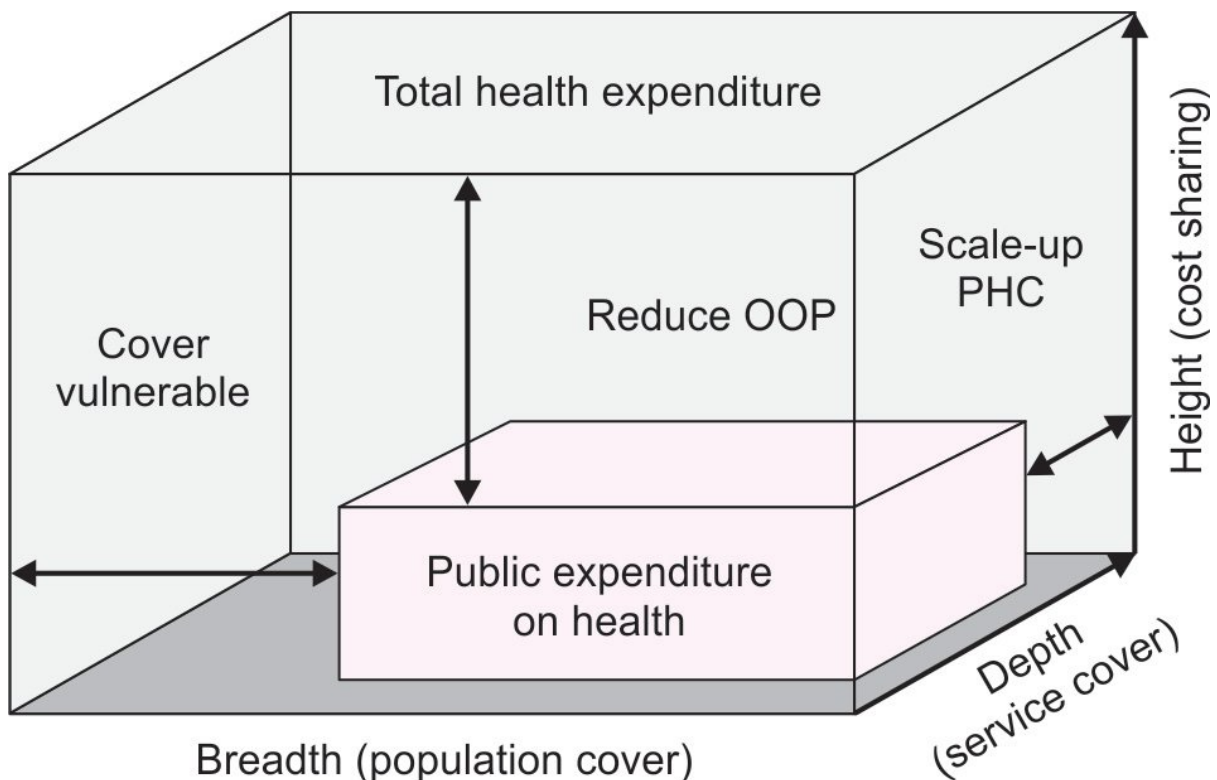
Limitations and measures that strengthen it

Limitation	Strengthening measure
Cover is for hospitalisation only — outpatient consultations, drugs and diagnostics, which dominate out-of-pocket expenditure, are outside PM-JAY	Free Drugs and Free Diagnostics Initiatives delivered through the Ayushman Arogya Mandir (Park 28e, Ch 10, p 729)
General Households still pay 70% of the package rate, so financial protection is partial for the non-ration-card population	Fixed, pre-negotiated package rates cap the exposure; state liability up to ₹1.5 lakh per family per year (GoK AB-Ark order 2018)
Awareness and card-holding fall short of entitlement	Ayushman Bhav campaign , launched 13 September 2023 , with <i>Ayushman Apke Dwar 3.0</i> for card saturation and health melas at Ayushman Arogya Mandirs (Park 28e, Ch 7, p 541)
Supply-side gaps — Karnataka runs 11 PHCs per CHC against a norm of 4 (Indian average 5), and 284 PHCs function without a doctor (Park Plus, Ch 21, p 296)	Indian Public Health Standards 2022 for each facility level, and the Ayushman Bharat Health Infrastructure Mission
Provider fraud, over-billing and unnecessary admissions	Mandatory pre-authorisation, fixed package rates, grievance redressal and anti-fraud safeguards (Sunder Lal, Ch 17, p 764)

Limitation	Strengthening measure
Elderly excluded when the family was not SECC-listed	A distinct card for people aged 70 years and above, irrespective of income (Park 28e, Ch 1, pp 6–7)
Public spending remains the binding constraint — OOPE stays the largest single source of health financing	Raise public health expenditure and use strategic purchasing, as in Thailand where public spending is 77.7% of total health expenditure (Sunder Lal, Ch 16, p 727)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **universal-health-coverage cube** and mark where AB-ArK acts on each axis. Exact labels: outer cube — **Total health expenditure**; inner block — **Public expenditure on health**; horizontal axis — **Breadth (population cover)** with the arrow labelled **Cover vulnerable**; depth axis — **Depth (service cover)** with the arrow labelled **Scale-up primary health care**; vertical axis — **Height (cost sharing)** with the arrow labelled **Reduce out-of-pocket expenditure**. Annotate: *Breadth* = Eligible + General Households; *Depth* = 12-package comprehensive primary health care plus the hospitalisation package list; *Height* = ₹5 lakh cashless cover for Eligible Households.



OOP: Out of pocket
PHC: Primary health care

Attaining universal coverage — the three dimensions of cover (Sunder Lal — Textbook of Community Medicine, Ch 17, p 740)

High-yield exam pointers

- ▶ **Four dates to write:** Ayushman Bharat announced **February 2018**; *Arogya Karnataka* launched **2 March 2018**; PM-JAY launched **23 September 2018**, subsuming RSBY; converged as **AB-ArK** in assurance mode from **30 October 2018**.
- ▶ **Karnataka two-tier rule (say who pays):** EHH — NFSA 2013/BPL ration card or RSBY-enrolled — fully covered up to **₹5 lakh/family/year**; GHH — no such card — the **government pays 30% of the fixed package rate and the patient bears the remaining 70%**, state liability capped at **₹1.5 lakh/family/year**; complex secondary and tertiary care needs a **referral from a government facility**, emergencies exempted.
- ▶ **Numbers to write:** ₹5 lakh/family/year; **10.74 crore families** ≈ **50 crore people** (bottom 40%), expanded to **15.5 crore**; **27 specialities, 1,949 procedures**; **3 days pre- and 15 days post-hospitalisation**; centre:state **60:40 (90:10 north-east)**.
- ▶ **Catastrophic health expenditure** = health cost >10% of total or >40% of non-food monthly household consumption expenditure.
- ▶ **Three implementation models:** trust/assurance (state health agency pays providers directly — Karnataka's model, through Suvarna Arogya Suraksha Trust), insurance (premium to an insurer), mixed.
- ▶ **Twelve CPHC packages** mnemonic — Pregnancy, Newborn, Child–adolescent, Family planning, Communicable/NHP, OPD, NCD, Eye–ENT, Oral, Elderly–palliative, Emergency, Mental health.
- ▶ **Dated Indian figures:** household insurance cover **41% (NFHS-5, 2019–21)** vs 29% (NFHS-4); OOPE **1.52% of GDP, ₹2,155 per capita**; catastrophic expenditure **7% (2022)**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The indexed textbook corpus documents Ayushman Bharat and PM-JAY fully but carries no passage on Arogya Karnataka, so every Karnataka-specific parameter above is cited to the Government of Karnataka Health and Family Welfare Department order for the converged scheme (Order No. HFW 69 CGE 2018 dated 15 November 2018) and the Department's published scheme notes, not to a textbook page. Park 28e prints **1,685 procedures** and Sunder Lal **1,592 procedures across 23 specialities**; the current Health Benefit Package figure of **1,949 procedures across 27 specialities** (AFMC Red Book 2024) is the one to write. The Health and Wellness Centre was formally renamed **Ayushman Arogya Mandir** in 2023, so write the current name first and the former name once in parentheses.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Accounts (India) 2021-22, NSSO 2021-22, WHO UHC Service Coverage Index, National Health Authority Health Benefit Package 2022, Suvarna Arogya Suraksha Trust

Programme and policy updates (India)

- ▶ **The Ayushman Arogya Mandir count and its service package have both moved past the textbook figures** — more than **1.75 lakh** centres are now operational on the Ministry of Health

and Family Welfare's Ayushman Arogya Mandir portal, against the 1.1 lakh transformed by March 2022 that the standard texts print. Rollout of the *complete* twelve-package set rose from **50% of centres in March 2023 to 82% by November 2024**, and 1,28,232 centres were tablet-enabled as on 30 November 2024. The Ayushman Bhav weekly health melas ran **17 September 2023 – 31 March 2024** and have since been converted into a routine **monthly Ayushman Arogya Shivar** — write the Shivar, not the mela, as the current activity. [PMID 41816146]

- ▶ **Always date the Health Benefit Package (HBP) round you quote** — PM-JAY opened with HBP 1.0 (1,394 packages across ~24 specialities); four revisions followed — HBP 2.0, 2.1, 2.2 and HBP 2022 (in force from April 2022) — taking the list to the current 1,949 procedures. Cost-based repricing halved the share of severely under-priced packages from 42% (HBP 1.0) to 20% (HBP 2.0). Analysis of 1.35 million claims from all 886 empanelled hospitals in Punjab (August 2019 – December 2022) showed each revision type raised utilisation, the gain being largest for packages that had been most under-priced (+1.84 claims per package per month), and similar in public and private hospitals. This is the modern answer to "how does a purchaser regulate provider behaviour" — strategic purchasing, not just a bigger cover. [PMID 39252993]
- ▶ **An evidence-based fix for the General Household 70% co-payment gap** — Kerala's *Karunya Arogya Suraksha Padhathi*, its PM-JAY implementation from April 2019, merged the state and national schemes under a **single purchaser** and extended cashless package cover to **day-care services and chronic-disease medicines**. A difference-in-differences analysis of 58,171 household observations (Kerala against Tamil Nadu and Karnataka, 2017–2020) found net household out-of-pocket spending fell by about 31.7%, concentrated almost entirely in medicine spending, with hospitalisation rates unchanged; gains were largest in poorer households and those with chronic disease. Benefit design that follows household spending beats hospitalisation-only expansion. [PMID 42284657]

Current national data

- ▶ **Dated financing figures that supersede the undated ones in the body** — out-of-pocket expenditure fell from **60.6% of total health expenditure in 2014-15 to 39.4% in 2021-22** (National Health Accounts estimates for India 2021-22), while government health spending rose from **1.13% to 1.35% of GDP** (NHA 2019-20). Public-facility outpatient contacts rose from **91.6 crore (2012-13) to 134.5 crore (2021-22)** and inpatient from **4.3 crore to 6.3 crore** (NSSO 2021-22). More than **34.7 crore Ayushman cards** had been created nationally as of **August 2024**, and the PM-JAY allocation was raised about 28.8%, **from ₹7,300 crore to ₹9,406 crore**, between the 2025 and 2026 financial years. [PMID 41816146; PMID 40976675]
- ▶ **Output of the primary-care pillar, cumulative to 30 November 2024** — **84.91 crore** people screened for hypertension, **74.24 crore** for diabetes, **48.98 crore** for oral cancer, **21.8 crore** women for breast cancer and **14.31 crore** for cervical cancer at Ayushman Arogya Mandirs; wellness sessions in 2024 stood at **6.7 times** the March 2021 level, reaching roughly 54 crore people. These are the numbers that show the "depth" axis of the coverage cube actually moving. [PMID 41816146]
- ▶ **Karnataka-specific figures worth writing verbatim** — Karnataka's total health expenditure in 2019-20 was ₹35,761 crore with out-of-pocket spending at 31.8% (₹11,368 crore), appreciably below the national share for the same year; more than **1.36 crore AB-PMJAY cards** had been

generated in the state by **19 March 2023**, with **35.88 lakh** beneficiaries treated at a cost of **₹2,982 crore**. Two retrospective series from a tertiary teaching hospital in coastal Karnataka (January–July 2023) quantify the protection: across 1,021 cardiac and 905 obstetric–gynaecological procedures, AB-PMJAY beneficiaries incurred **zero out-of-pocket expenditure**, uninsured patients bore **100%** (median ₹1,47,250 for angioplasty), and even privately insured patients paid **15–38%** of the bill. [PMID 39925993; PMID 39015849]

- **The impact evidence is genuinely split — say so, and say why** — repeat household surveys in Chhattisgarh (2021 and 2022, ~15,000 individuals) found enrolment made **no difference** to inpatient utilisation, length of stay, patient satisfaction, out-of-pocket spending or catastrophic expenditure; **70.9%** of enrolled patients using private hospitals still crossed the 10% catastrophic threshold in 2022, and mean out-of-pocket spending among the enrolled was ₹23,691 against a mean annual non-health household consumption of ₹97,365 [PMID 39135015]. A Haryana case-control study (n = 772) found the opposite — direct medical costs **65% lower** (₹11,131 vs ₹31,675) and catastrophic expenditure at the 30% threshold **13.3% vs 45.9%** (OR 7.15; 95% CI 4.74–10.80) — yet beneficiaries' **non-medical costs were 19% higher**, because travel, food, lodging and wage loss lie outside the package, and medicines plus diagnostics made up **81%** of what beneficiaries still paid in private hospitals [PMID 40976675]. The reconciling explanation is balance billing and uncovered indirect costs, not the size of the cover.

Elimination, surveillance and disease status

- **AB-ArK's own claims data now supply Karnataka-specific programme evidence** — analysis of the Suvarna Arogya Suraksha Trust pre-authorisation and claims database over 24 months (**July 2021 – June 2023**) covered **39,89,802 approved pre-authorisations**, of which 9,173 were confirmed tuberculosis admissions: **96.2%** below poverty line, **68.3%** male, **97.9%** treated in the public sector, and an average of roughly **₹10,000** of catastrophic cost averted per patient. This is the cleanest available demonstration that AB-ArK's financial protection feeds directly into the National TB Elimination Programme's elimination push. [PMID 40518205]

Guideline and legislative updates

- **The entitlement on paper is narrower than it looks, and narrower still in practice** — a documentary audit of the national scheme, **14 state health insurance schemes** and individual/group policies of four government and five private insurers found systematic exclusions by **age, gender and pre-existing condition**: congenital anomalies (including congenital heart disease and epilepsy) routinely excluded as "pre-existing", vaccination and growth-hormone therapy excluded in many policies, waiting periods up to **48 months**, and reproductive and gender-affirming services largely unfunded — an equity argument to deploy against the "no cap on family size, age or gender" claim [PMID 42450895]. Delivery falls short too: in an urban slum in Puducherry (2023, 273 households), **59%** were aware of the scheme and **65%** eligible, but only **49%** of eligible households had registered and **14%** had ever used the benefit, with cumbersome registration and limited empanelment cited [PMID 41805171].

Global and WHO developments

- **India's position on the WHO Universal Health Coverage Service Coverage Index** — the index rose from **57 in 2015 to 63 in 2021**, now marginally above the South-East Asia regional average of **62**, but still behind Thailand, Bhutan, Bangladesh and DPR Korea, each of which built its gain

on a statutory or plan-based primary-care platform (Thailand's National Health Security Act 2002 being the exemplar). Frame AB-ArK explicitly on the WHO UHC cube — population, service and cost cover — and against SDG target 3.8, which is the framing examiners expect for a "how does it improve health" stem. [PMID 41816146]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1, pp 6–7; Ch 7, p 541; Ch 10, pp 729–731, 745); Sunder Lal — Textbook of Community Medicine (Ch 8, pp 534–535; Ch 16, pp 727–729; Ch 17, pp 740–742, 746, 763–764; Ch 20, pp 786, 805–806); AFMC Red Book of Community Medicine 2024 (Ch 38, pp 1209–1210); Park Plus (Amrutha & Vijayalakshmi), Ch 21, pp 296–297; Jugal Kishore — National Health Programmes of India (Ch 1, pp 130–131); NFHS-5 India Report 2019–21 (Ch 6, p 483); National Health Policy 2017; Government of Karnataka, Health and Family Welfare Department — Ayushman Bharat – Arogya Karnataka Government Order No. HFW 69 CGE 2018 dated 15 November 2018, with the Department of Health and Family Welfare Services / Suvarna Arogya Suraksha Trust AB-ArK scheme notes.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 39252993 — Impact of health benefit package policy interventions on service utilisation under government-funded health insurance in Punjab, India (PM-JAY), *Lancet Regional Health Southeast Asia* 2024, doi:[10.1016/j.lansea.2024.100462](https://doi.org/10.1016/j.lansea.2024.100462); 42284657 — Does integrating pharmaceutical coverage into publicly financed health insurance reduce out-of-pocket spending? Difference-in-differences evidence from Kerala, India, *Social Science & Medicine* 2026, doi:[10.1016/j.socscimed.2026.119483](https://doi.org/10.1016/j.socscimed.2026.119483) (abstract-only); 40976675 — Impact of India's publicly funded health insurance scheme on financial risk protection: a case-control study from Haryana state in India, *BMJ Open* 2025, doi:[10.1136/bmjopen-2024-093304](https://doi.org/10.1136/bmjopen-2024-093304); 39135015 — The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana after four years of implementation — is it making an impact on quality of inpatient care and financial protection in India?, *BMC Health Services Research* 2024, doi:[10.1186/s12913-024-11393-2](https://doi.org/10.1186/s12913-024-11393-2); 39925993 — Out-of-pocket expenditure on selected surgeries in the obstetrics and gynaecology department for AB-PMJAY, private health insurance and uninsured patients in a tertiary care teaching hospital in Karnataka, *F1000Research* 2025, doi:[10.12688/f1000research.157203.2](https://doi.org/10.12688/f1000research.157203.2); 39015849 — Out-of-pocket expenditure for selected surgeries in the cardiology department for AB-PMJAY, private health insurance and uninsured patients in a tertiary care teaching hospital in Karnataka, India, *Cureus* 2024, doi:[10.7759/cureus.62444](https://doi.org/10.7759/cureus.62444); 40518205 — Utilization of tuberculosis healthcare packages under Ayushman Bharat Pradhan Mantri Jan Arogya Yojana–Arogya Karnataka: a comprehensive socio-demographic analysis, *Indian Journal of Tuberculosis* 2024, doi:[10.1016/j.ijtb.2023.12.008](https://doi.org/10.1016/j.ijtb.2023.12.008) (abstract-only); 42450895 — Inequities uncovered: examining exclusion in Indian health insurance policies, *Healthcare (Basel)* 2026, doi:[10.3390/healthcare14131885](https://doi.org/10.3390/healthcare14131885); 41805171 — From card to care: gaps in implementation of publicly funded health insurance among a vulnerable population in South India, *Journal of Public Health (Oxford)* 2026, doi:[10.1093/pubmed/fdag021](https://doi.org/10.1093/pubmed/fdag021) (abstract-only). National Health Accounts estimates for India 2021–22; NSSO 2021–22; WHO UHC Service Coverage Index 2021; National Health Authority Health Benefit Package 2022 (in force April 2022); Suvarna Arogya Suraksha Trust pre-authorisation and claims database, July 2021 – June 2023.

♦ ♦ ♦

Voluntary health agencies — HIV/AIDS control

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — A voluntary health agency is "an organisation that is administered by an autonomous board which holds meetings, collects funds for its support chiefly from private sources, and expends money, whether with or without paid workers, in conducting a programme directed primarily to furthering the public health by providing health services, health education, or by advancing research or legislation on health, or by a combination of these activities" (Park 28e, Ch 10, p 746). Because they are not bound by statutory and financial restrictions, they have been compared to **"motor trucks" that can penetrate the by-ways**, while official agencies are the **"railway trunk lines" that must run on tracks laid down by law** (Park 28e, Ch 10, p 746). More than **7,000 voluntary agencies** are engaged in health-related activity in India, working in underserved, unserved and difficult areas on a non-profit basis (Sunder Lal, Ch 17, p 765). In HIV/AIDS this sector is not an optional add-on: the National AIDS Control Programme (NACP) Phase-V (1 April 2021 – 31 March 2026) is explicitly built with "an integral vibrant community engagement" (Park 28e, Ch 7, p 517).

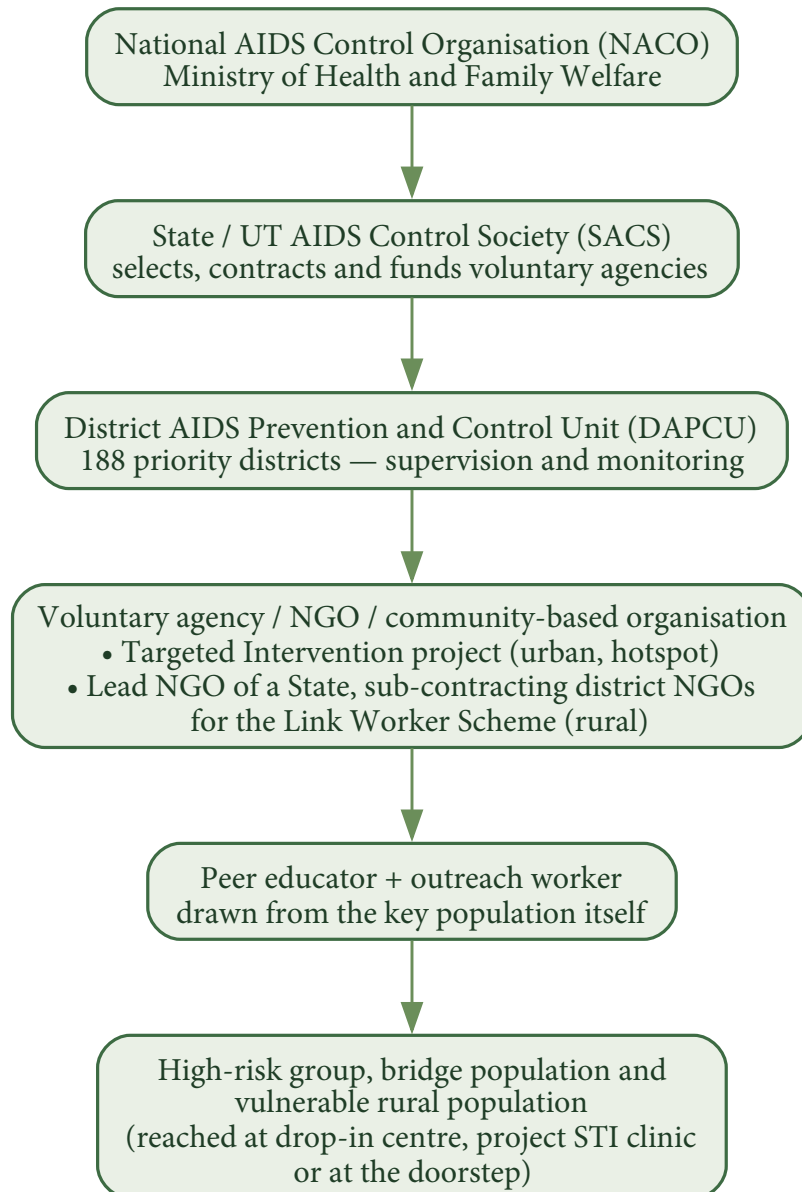
1. Why HIV/AIDS depends on the voluntary sector more than most programmes

- ▶ The **core high-risk groups (HRG)** — female sex workers (FSW), men who have sex with men (MSM), transgender persons/hijras, injecting drug users (IDU) — and the **bridge populations** (high-risk-behaviour migrants, long-distance truckers) are hidden, mobile and socially marginalised; a government clinic cannot enumerate or retain them (Park 28e, Ch 7, p 524).
- ▶ Prevention needs **>95% coverage** of these groups before incidence falls, which is only achievable by a peer-led outreach model, not by facility-based care (Sunder Lal, Ch 8, p 347).
- ▶ **Stigma and discrimination** deny people living with HIV (PLHIV) treatment, employment and schooling; a non-governmental intermediary is better placed to negotiate this than the state (Jugal Kishore, Ch 2, p 759).
- ▶ Knowledge remains poor and gendered — comprehensive knowledge of HIV/AIDS was **21.6% in adult women versus 30.7% in adult men** (NFHS-5) — so sustained interpersonal and community education is required (Sunder Lal, Ch 8, p 347).
- ▶ Government policy itself mandates the partnership. Promotion of public–private partnership to achieve public health goals is a declared departmental strategy, reinforcing the involvement of non-governmental organisations already spelt out in the National Population Policy 2000, and the National Rural Health Mission (now the National Health Mission) seeks to build community ownership of programmes through them (Sunder Lal, Ch 17, pp 765-766).

2. The six classical functions of a voluntary health agency, applied to HIV/AIDS

Function (Park 28e, Ch 10, pp 746-747)	Application in HIV/AIDS control in India
Supplementing government work — lending personnel, funds, equipment	Running Targeted Intervention (TI) projects, drop-in centres and project-based sexually transmitted infection (STI) clinics on behalf of State AIDS Control Societies (SACS)

Function (Park 28e, Ch 10, pp 746-747)	Application in HIV/AIDS control in India
Pioneering — exploring new ways of doing things, including research	The peer-educator model, drop-in centres, needle-syringe exchange and opioid substitution therapy (OST) entered the national programme after voluntary-sector demonstration; the family planning programme is Park's classical Indian example of such pioneering
Education	Behaviour change communication (BCC), mid-media and folk media, Adolescence Education Programme and Red Ribbon Clubs in schools and colleges (Sunder Lal, Ch 8, p 347)
Demonstration — pilot and experimental projects	Community-based HIV screening using finger-prick whole blood by lay/community workers before national scale-up (Park 28e, Ch 7, pp 520-521)
Guarding the work of government agencies	Community-led monitoring of ART centres, integrated counselling and testing centres (ICTC) and blood banks; ensuring confidentiality is not breached
Advancing health legislation and public opinion	Advocacy that led to the HIV and AIDS (Prevention and Control) Act, 2017 , a game-changer on which NACP Phase-V is built (Park 28e, Ch 7, p 517; Sunder Lal, Ch 8, p 350)

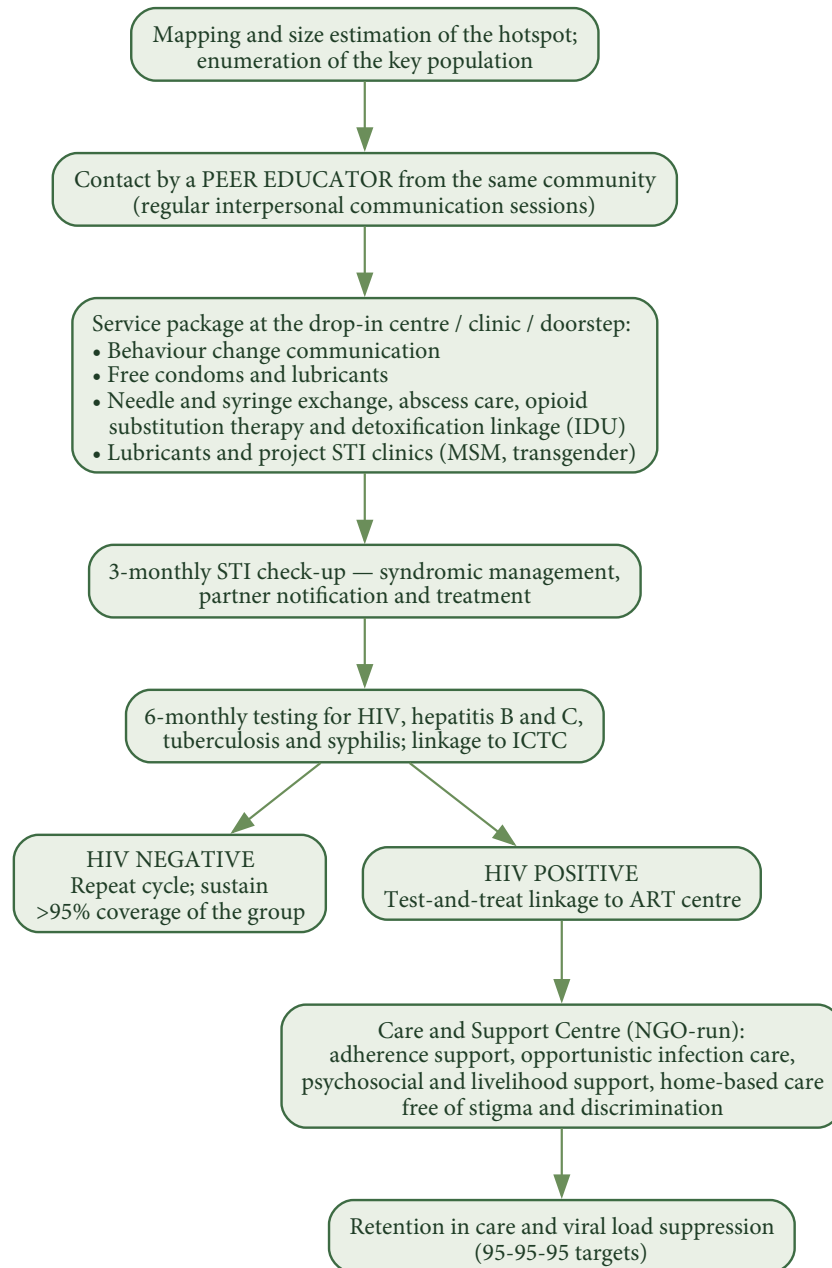


Where a voluntary agency plugs into the National AIDS Control Programme delivery chain

3. Voluntary-agency contribution to each NACP Phase-V service component

NACP component	What the voluntary agency actually does
Targeted Intervention (TI)	Implements the whole TI as an NGO/CBO peer-led outreach model; objective is to improve health-seeking behaviour of HRG and cut their risk of acquiring STI and HIV. As on September 2021, NACO supported TIs for FSW – 313, IDU – 200, migrants – 210, MSM – 94, truckers – 70, transgender/hijra – 40 (Park 28e, Ch 7, p 524)
Link Worker Scheme (LWS)	Community-based rural outreach in 144 high-priority districts (HIV prevalence $\geq 1\%$); a lead NGO per State sub-contracts district NGOs, link workers (one male, one female per cluster of five core villages) recruit village volunteers from elected leaders, self-help groups and youth groups (Sunder Lal, Ch 8, pp 348-349; Jugal Kishore, Ch 1, p 333)

NACP component	What the voluntary agency actually does
Basic services — ICTC and PPTCT	Refers and escorts clients to ICTC; conducts community-based screening, lay-worker screening, self-testing, index testing and partner notification ; helps achieve universal antenatal screening through the ICTC network (Sunder Lal, Ch 8, p 348)
STI/RTI management	Runs project-based STI clinics; 3-monthly medical check-up of HRG with syndromic management, partner notification and treatment; supplies counselling services at designated STI/RTI (<i>Suraksha</i>) clinics (Park 28e, Ch 7, p 525; Sunder Lal, Ch 8, pp 347-349)
Condom promotion	Free condom and lubricant distribution to HRG; social marketing through non-traditional outlets and female condom promotion (Sunder Lal, Ch 8, p 349)
Blood safety	Motivates and organises voluntary, non-remunerated blood donation camps; several State branches of the Indian Red Cross Society run blood banks (Park 28e, Ch 10, p 746; Sunder Lal, Ch 8, p 349)
Care, support and treatment (CST)	Staffs Care and Support Centres and community care centres linked to ART centres; delivers outreach, psychosocial support, nutrition and livelihood support, home-based care and adherence follow-up needed for near-100% ART compliance (Jugal Kishore, Ch 1, p 360; Sunder Lal, Ch 8, p 349)
Information, education and communication (IEC)	Mass media, mid-media/folk media, interpersonal communication, Red Ribbon Clubs, life-skills education for adolescents (Sunder Lal, Ch 8, p 347)
Mainstreaming, partnership and social protection	Links PLHIV to entitlements across the 18 key ministries and departments of the multisectoral response; runs <i>Mission Sampark</i> tracing of those who know their positive status but are not on ART (Sunder Lal, Ch 8, pp 346, 349)
Strategic information	Feeds routine service data from TI sites and drop-in centres into programme monitoring and sentinel surveillance (Sunder Lal, Ch 8, pp 349-350)



Targeted Intervention — the service pathway a voluntary agency delivers to one high-risk-group member

4. Named voluntary agencies in India and their HIV-relevant contribution

Agency (year founded)	Relevance to HIV/AIDS prevention and control
Indian Red Cross Society (1920, by an Act of the Indian Legislature)	Blood banks in State branches, first aid and home-nursing training through the St. John Ambulance Association, maternal and child welfare and family planning clinics — the natural partner for blood safety and voluntary blood donation (Park 28e, Ch 10, p 746; Ch 25, p 1061)

Agency (year founded)	Relevance to HIV/AIDS prevention and control
Voluntary Health Association of India (VHAI) (1970)	Non-profit federation of 27 State Voluntary Health Associations linking >4,500 health institutions; works by campaigns, policy research, advocacy, need-based training and parliamentary intervention — the advocacy and rights arm (Suryakantha, Ch 40, p 806)
Family Planning Association of India (FPAI) (1949, Mumbai)	Imparts education on family life, safe sex and prevention of sexually transmitted diseases ; runs clinics, mobile rural camps and training of doctors and paramedical workers; funds smaller NGOs (Suryakantha, Ch 40, p 806; Park 28e, Ch 10, p 747)
Tuberculosis Association of India (1939)	Training and research in tuberculosis, the commonest opportunistic infection and leading killer in PLHIV — supports HIV–TB collaborative activity (Suryakantha, Ch 40, p 806)
Central Social Welfare Board (1953)	Autonomous statutory channel that surveys, promotes and gives financial aid to voluntary welfare organisations — the funding conduit for small community-based organisations (Park 28e, Ch 10, p 747)
Professional bodies — Indian Medical Association, Trained Nurses Association of India, Federation of Obstetric and Gynaecological Societies	Set standards of professional education, publish journals, hold conferences, foster research; partner the government in reproductive health and prevention of parent-to-child transmission (Park 28e, Ch 10, p 748; Sunder Lal, Ch 17, pp 765-766)
International voluntary agencies — Rockefeller Foundation, Ford Foundation, CARE, Oxfam, Save the Children Fund, International Planned Parenthood Federation	Technical and financial support, demonstration projects and operational research (Park 28e, Ch 10, p 748; Ch 25, p 1061)

5. Level-of-prevention framework for the voluntary-sector role

Level	Voluntary-agency activity in HIV/AIDS
Primordial / health promotion	Life-skills and adolescence education, Red Ribbon Clubs, community norms against unsafe sex and needle sharing
Primary — specific protection	Condom and lubricant supply and social marketing, needle–syringe exchange, OST, voluntary blood donation, promotion of pre-exposure prophylaxis, community mobilisation for >95% HRG coverage
Secondary — early diagnosis and treatment	Community-based screening and self-testing, escorting to ICTC, 3-monthly STI check-up and 6-monthly HIV testing, index testing and partner notification, <i>Mission Sampark</i> tracing, PPTCT of pregnant women
Tertiary — disability limitation and rehabilitation	Care and Support Centres, ART adherence and retention support, opportunistic-infection and home-based care, nutrition, livelihood and social-protection linkage, legal aid against discrimination under the HIV and AIDS Act 2017

6. Strengths, limitations and safeguards for the partnership

The voluntary sector is a resource to be governed, not merely welcomed; its weaknesses are structural and each has a specific administrative remedy.

Strengths	Limitations	Safeguard needed
Community acceptance; access to hidden and criminalised populations; flexibility free of statutory restriction	Uneven inter-State distribution and limited area of operation of NGOs (Sunder Lal, Ch 17, p 765)	Map NGO coverage against burden; SACS to commission in gap districts
Low-cost peer-delivered outreach; innovation and demonstration capacity	Dependence on short-cycle donor funding threatens sustainability	Multi-year contracts and government budget lines
Advocacy voice independent of government	Variable technical quality and weak record-keeping	Standard TI operational guidelines, accreditation, supportive supervision by DAPCU
Trusted custodians of confidentiality	Risk of conflict of interest and weak accountability in public-private partnership (Oxford Textbook of Public Health 7e, Ch 106, pp 4032-4033)	Written memorandum of understanding, defined deliverables, independent evaluation and audit

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **NACP Phase-V service-component chart** (Park 28e, Ch 7, Fig 7, p 519) — draw a single parent box "National AIDS Control Programme" fanning to labelled boxes: *Targeted Intervention and Link Worker Scheme*; *Basic services (ICTC, PPTCT and HIV/TB)*; *STI/RTI management*; *Blood safety*; *Laboratory services*; *Care, support and treatment*; *Information, education and communication*; *Strategic information (monitoring, evaluation, surveillance, research and data analysis)*; *Administration and procurement*; *Finance*. Mark with an asterisk the components delivered largely through voluntary agencies (the first, second, third, sixth and seventh).
- **"Motor truck versus railway trunk line"** — draw a straight rail line labelled *Government agencies: run only on tracks laid by law* and a truck leaving it into by-lanes labelled *Voluntary agencies: reach the hotspot, the slum, the highway halt and the village*.

High-yield exam pointers

- Park's six functions: S-P-E-D-G-A — Supplementing, Pioneering, Education, Demonstration, Guarding, Advancing legislation.
- Core HRG = FSW, MSM, transgender/hijra, IDU; **bridge populations** = high-risk migrants and long-distance truckers.
- TI service rhythm: **STI check-up every 3 months, HIV/hepatitis B and C/TB/syphilis testing every 6 months**, coverage target >95%.

- ▶ NACP Phase-V (2021-26): reduce annual new HIV infections and AIDS-related deaths by **80% by 2025-26 from the 2010 baseline**; **95-95-95**; dual elimination of vertical transmission; elimination of HIV-related stigma.
- ▶ India's 2021 score card (Sunder Lal, Ch 8, Table 8.5): **2.4 million PLHIV**; 78% know status; 83% of those on ART; 85% virally suppressed; new infections down 46.3%, AIDS deaths down 76.5%.
- ▶ Legal peg: **HIV and AIDS (Prevention and Control) Act, 2017** — prohibits discrimination, State-appointed **ombudsman** hears violations.
- ▶ Structure to name: **NACO → SACS → DAPCU (188 priority districts) → TI NGO/CBO → peer educator**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the Targeted Intervention counts (September 2021) and the impact indicators (2021) are the latest figures carried in the retrieved textbook corpus; quote them with their year, as NACO publishes an updated India HIV Estimates report annually. The Link Worker Scheme staffing figures differ slightly between Sunder Lal (20 cluster link workers per district) and Jugal Kishore (two link workers per cluster of five core villages); state the structure, not the exact count.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO 2025 lenacapavir PrEP guideline, UNAIDS global estimates, NACO, DCGI, API 2026

Programme and policy updates (India)

- ▶ **Mitr ("friend") clinics — the current model answer for community-owned service delivery.** Three transgender-staffed single-window clinics in Hyderabad, Pune and Thane registered 5,226 clients between March 2021 and September 2024. Seventy per cent had **never been registered in the government Targeted Intervention programme**, and 75% came for gender-affirming care (laser hair removal 53%, hormone-therapy consultation 34%), which then acted as the entry point to HIV testing — seeking surgical referral was independently associated with being tested (aOR 1.51; 95% CI 1.02–2.25). Of 585 eligible interested clients, 576 (98%) started pre-exposure prophylaxis (PrEP). NACO's forward-looking white paper names the model and Telangana has funded its scale-up; the legal peg is the **Transgender Persons (Protection of Rights) Act, 2019**, which bars discrimination in health care and requires facilities to offer gender-affirming services [PMID 41059629].
- ▶ **Safe Zindagi — voluntary-sector HIV self-testing at national scale.** A web platform distributed a DCGI-approved oral-fluid self-test kit by courier or through pick-up points sited in private clinics, pharmacies and community-based organisations in Maharashtra and Telangana. Between June 2021 and September 2022, 5,324 clients ordered kits; 94% of recipients uploaded a result, 6% screened positive, and 72% of those completed confirmatory testing. Cite this as the newest form of Park's *demonstration* function [PMID 38861458].
- ▶ **Governance caution for the partnership.** An organisational ethnography of a Gates Foundation-funded Targeted Intervention NGO in western India (six months of participant observation, 17 interviews with NGO workers) found that target-driven, competitively contracted funding

produced good reported numbers through informal, personal and hierarchical arrangements, while frontline outreach workers were demotivated and sceptical of the reported success — concrete evidence for the written memorandum of understanding, defined deliverables and independent evaluation listed among the safeguards above [PMID 39233626].

Current national data

- ▶ **India's cascade on 2022–23 estimates supersedes the 2021 scorecard in the textbook:** approximately 2.5 million PLHIV (second largest burden globally), 79% aware of status, 86% of those on ART, 93% of those tested virally suppressed; adult general-population prevalence 0.21%, against key-population prevalence of 9.03% in people who inject drugs, 3.78% in transgender persons, 3.26% in MSM and 1.85% in female sex workers. Always quote the year with the figure [PMID 38861458].
- ▶ **HIV Sentinel Surveillance among primigravidae, seven southern States (rounds of 2015, 2017, 2019 and 2021; 182,066 records):** 470 new infections (0.26%), of which 74% were in women aged 15–24; incidence in young primigravidae fell from 0.32% (2015) to 0.20% (2021), but new infections rose again in 2021 in Andhra Pradesh, Karnataka and Tamil Nadu — evidence that antenatal sentinel surveillance still detects the general-population signal [PMID 40898796].

Elimination, surveillance and disease status

- ▶ **Community-led monitoring (CLM) has matured from advocacy into a surveillance instrument** — this is the modern, examinable form of Park's *guarding the work of government agencies*. The **Community-led Monitoring Project in Asia** was begun in 2021 by the Community Network for Empowerment (CoNE) in Manipur, India and Yayasan Peduli Hati Bangsa in Indonesia, together with the International Treatment Preparedness Coalition–Global and amfAR's TREAT Asia programme. The two community organisations **jointly monitor 12 public health facilities, from local district level up to referral hospitals, against availability, accessibility, acceptability and quality (AAAQ) indicators** drawn from national guidelines and policies. In April 2023 the project moved to a digital platform with a real-time public online dashboard — among the first in the Asia-Pacific region to publish CLM data live, and the only such project anywhere that also monitors hepatitis B and C alongside HIV. Digitisation transformed reach: **the two-country project covered 1,593 care recipients between April and December 2023, against 120 people reached by paper-based interviews and focus groups over January–December 2022;** among those in the digital round, 62% were in care for HIV, 34% for hepatitis C and 3% for hepatitis B. Quote the 12 facilities and the 1,593 as the combined India-and-Indonesia total, not as the Manipur arm's own figures [PMID 39360661].

Guideline and legislative updates

- ▶ **WHO recommended twice-yearly injectable lenacapavir as an additional PrEP option on 14 July 2025,** together with a simplified public-health approach to testing using HIV rapid tests. The operational point for this answer is that the simplified testing rule is what allows delivery through pharmacies, community clinics and tele-health rather than specialist centres — i.e. through exactly the voluntary-sector outlets that already run drop-in centres [WHO 2025-Jul].
- ▶ **Indian antiretroviral and prophylaxis position, 2025:** first-line ART is the fixed-dose combination tenofovir disoproxil fumarate/lamivudine/dolutegravir (TLD) under a Treat-All policy with routine viral-load monitoring; **DCGI has approved TDF/FTC for PrEP** and national

technical guidance exists, but programmatic rollout remains uneven. The NACO post-exposure prophylaxis document still lists a 28-day TDF + 3TC + EFV regimen (start as soon as possible, preferably within 2 hours, always within 72 hours) while institutions move to dolutegravir-based PEP in line with WHO's 2024 advice [PMID 42543939; API 2026]. PrEP is still **not available free of charge in the public sector**, which is precisely why the voluntary-sector clinics above have had to provide it themselves [PMID 41059629].

Global and WHO developments

- ▶ **UNAIDS global position (2025 data):** 41.0 million people living with HIV, 1.2 million new infections (42% below 2010, well short of the 80%-reduction trajectory), 570,000 AIDS-related deaths, 32.1 million on ART; global cascade **88–78–74**. Available resources were **US\$17.6 billion against the US\$21.9 billion needed annually**, with international funding down 18% and domestic sources now supplying about 60% [UNAIDS 2026].
- ▶ **The January 2025 United States foreign-aid freeze is the sharpest recent demonstration of donor dependence — the classic limitation of the voluntary sector.** A facility-level analysis of 31,628 PEPFAR-supported sites found that people on treatment held roughly steady (–1%), but HIV testing fell 18%, PrEP initiations fell 33%, and **new diagnoses fell 36% at community service sites versus 13% at continuously reporting facilities**; the supported workforce shrank by 62,541 (–24%). Community-delivered and key-population services are the first casualty when external funding stops, which is the argument for multi-year domestic budget lines [PMID 42507823].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India, National AIDS Control Programme; Ch 10 — Health Care of the Community, Voluntary Health Agencies; Ch 25 — International Health); Sunder Lal — Textbook of Community Medicine (Ch 8 — National AIDS Control Programme; Ch 17 — Healthcare of the Community); Jugal Kishore — National Health Programmes of India (NACP, Link Worker Scheme, community care and support); Suryakantha — Community Medicine with Recent Advances (Ch 40 — National Voluntary Health Agencies); Oxford Textbook of Public Health 7e (Ch 106 — Private support of public health). Programme documents named: National Strategic Plan for HIV/AIDS and STI 2017-2024; NACP Phase-V (2021-2026); HIV and AIDS (Prevention and Control) Act, 2017; NFHS-5.

Recent-advances sources: 41059629 — Laser hair removal to antiretrovirals: findings from a person-centred care model for transgender people in India, *Journal of the International AIDS Society* 2025, doi:[10.1002/jia2.70041](https://doi.org/10.1002/jia2.70041); 38861458 — HIV self-testing in India: implementation and qualitative evaluation of a web-based programme with virtual counsellor support, *Journal of the International AIDS Society* 2024, doi:[10.1002/jia2.26302](https://doi.org/10.1002/jia2.26302); 39233626 — Professionalisation experiences of a 'business-minded' HIV targeted intervention NGO in India: an organisational ethnography, *Global Public Health* 2024, doi:[10.1080/17441692.2024.2399674](https://doi.org/10.1080/17441692.2024.2399674) (abstract-only); 40898796 — New HIV infections among primigravida mothers in southern India: analysis from HIV Sentinel Surveillance (2015-2021), *Indian Journal of Public Health* 2025, doi:[10.4103/ijph.ijph_1094_24](https://doi.org/10.4103/ijph.ijph_1094_24) (abstract-only); 39360661 — Community-led monitoring of HIV and viral hepatitis services: lessons learned and impacts from India and Indonesia, *Journal of the International AIDS Society* 2024, doi:[10.1002/jia2.26373](https://doi.org/10.1002/jia2.26373) (full text PMC11447636; a two-country report — every facility count and reach figure in it is the combined India + Indonesia total, never the Manipur arm alone); 42543939 — Antiretroviral therapy in India (2025): drugs, indications, contraindications, mechanisms, and prophylaxis (PEP/PrEP), *Journal of the Association of Physicians of India* 2026, doi:[10.59556/japi.74.1559](https://doi.org/10.59556/japi.74.1559) (abstract-only); 42507823 — The impact of the United States foreign aid freeze on HIV service delivery in PEPFAR-supported countries: a facility-level analysis of 2024-2025 programme data, *Journal of the International AIDS Society* 2026, doi:[10.1002/jia2.70182](https://doi.org/10.1002/jia2.70182). Non-journal sources: WHO guidelines recommending twice-yearly injectable lenacapavir for HIV prevention, 14 July 2025; UNAIDS global HIV statistics fact sheet (2025 data); DCGI approval of TDF/FTC for pre-exposure prophylaxis and NACO national technical guidance, as summarised in the 2026 API review above.

♦ ♦ ♦

Health care delivery system in India — central level

Asked in: Paper IV 20-Jul-2020 — RGUHS MD Community Medicine

Definition — The health system in India has three main links: **Central, State and local or peripheral** (Park 28e, Ch 23, p 1033). At the Centre, the responsibility is **not direct service delivery** but **policy making, planning, guiding, assisting, evaluating and coordinating** the work of the State Health Ministries so that health services cover every part of the country and no State lags behind for want of them (Park 28e, Ch 23, p 1033). The official "organs" at the national level are the **Union Ministry of Health and Family Welfare (MoHFW)**, the **Directorate General of Health Services (DGHS)** and the **Central Council of Health and Family Welfare (CCHFW)**.

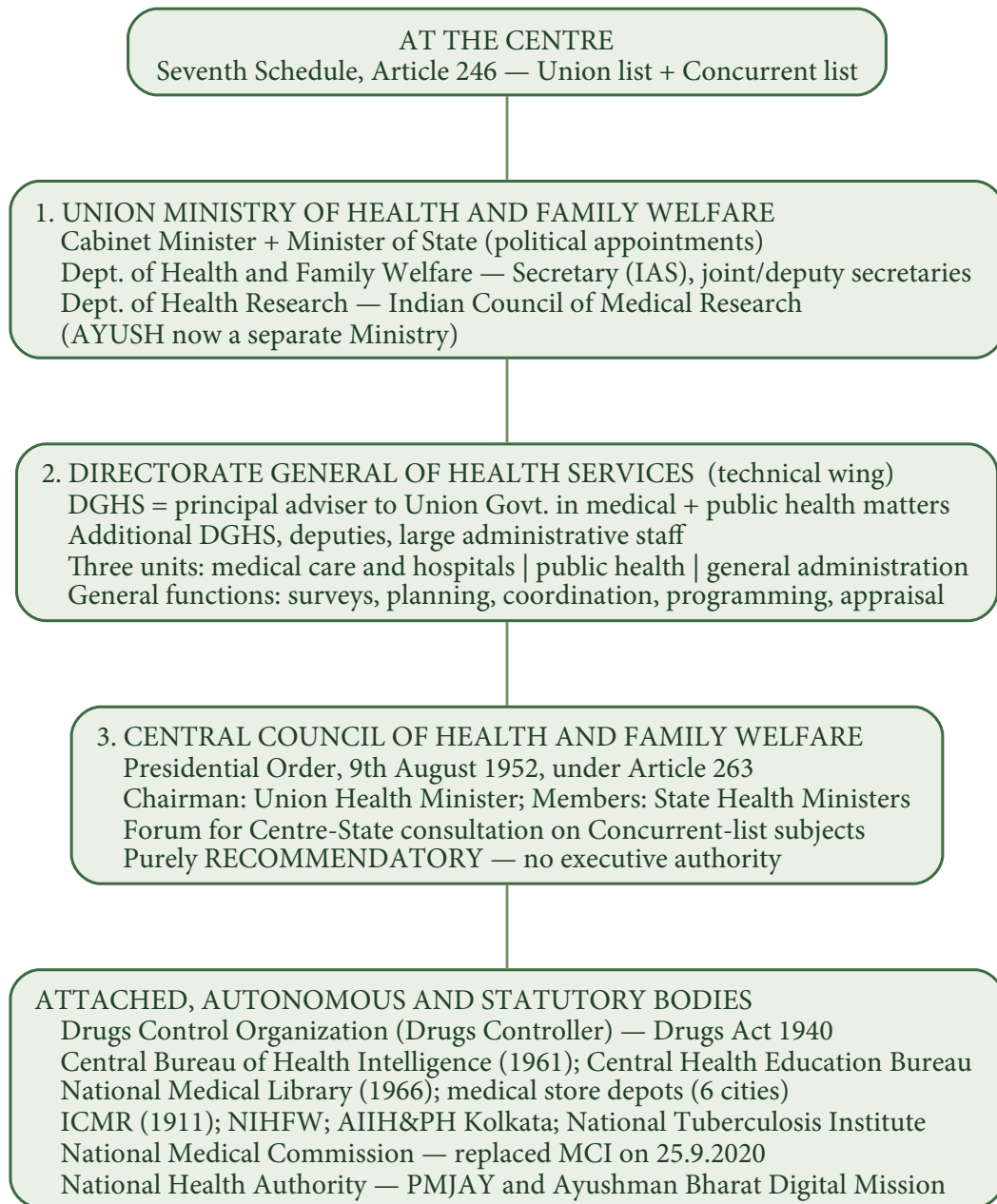
1. Constitutional basis of the Centre's role

The functions of the Union Health Ministry are set out in the **Seventh Schedule of Article 246** of the Constitution, under the Union list and the Concurrent list (Park 28e, Ch 23, p 1033).

List	Health subjects	Who legislates
Union list	(1) International health relations and administration of port quarantine (2) Administration of central institutes (All India Institute of Hygiene and Public Health Kolkata, National Centre for Disease Control Delhi) (3) Promotion of research through research centres (4) Regulation and development of the medical, pharmaceutical, dental and nursing professions (5) Establishment and maintenance of drug standards (6) Census, collection and publication of statistical data (7) Immigration and emigration (8) Regulation of labour in mines and oil fields (9) Coordination with States and other ministries for promotion of health	Centre alone
Concurrent list	(1) Prevention of extension of communicable disease from one unit to another (2) Prevention of adulteration of foodstuffs (3) Control of drugs and poisons (4) Vital statistics (5) Labour welfare (6) Ports other than major (7) Economic and social planning (8) Population control and family planning	Centre and States; State power restricted to the framework of central legislation
State list	Provision of medical care, preventive health services and pilgrimages within the State (position unchanged since the Government of India Act, 1935)	State — the ultimate authority for all health services in its jurisdiction

(Park 28e, Ch 23, pp 1033-1034)

2. The three official organs at the Centre



Official organs of the health care delivery system at the Central level

Functions of the Central Council of Health and Family Welfare (Park 28e, Ch 23, p 1034):

- ▶ To **consider and recommend broad outlines of policy** on health in all its aspects — remedial and preventive care, environmental hygiene, nutrition, health education, and promotion of facilities for training and research.
- ▶ To **make proposals for legislation** in fields relating to medical and public health matters, and to lay down the pattern of development for the country as a whole.
- ▶ To **recommend distribution of grants-in-aid** for health purposes to the States, and to review periodically the work accomplished through these grants.
- ▶ To **establish organisations** with appropriate functions for promoting and maintaining cooperation between Central and State health administrations.

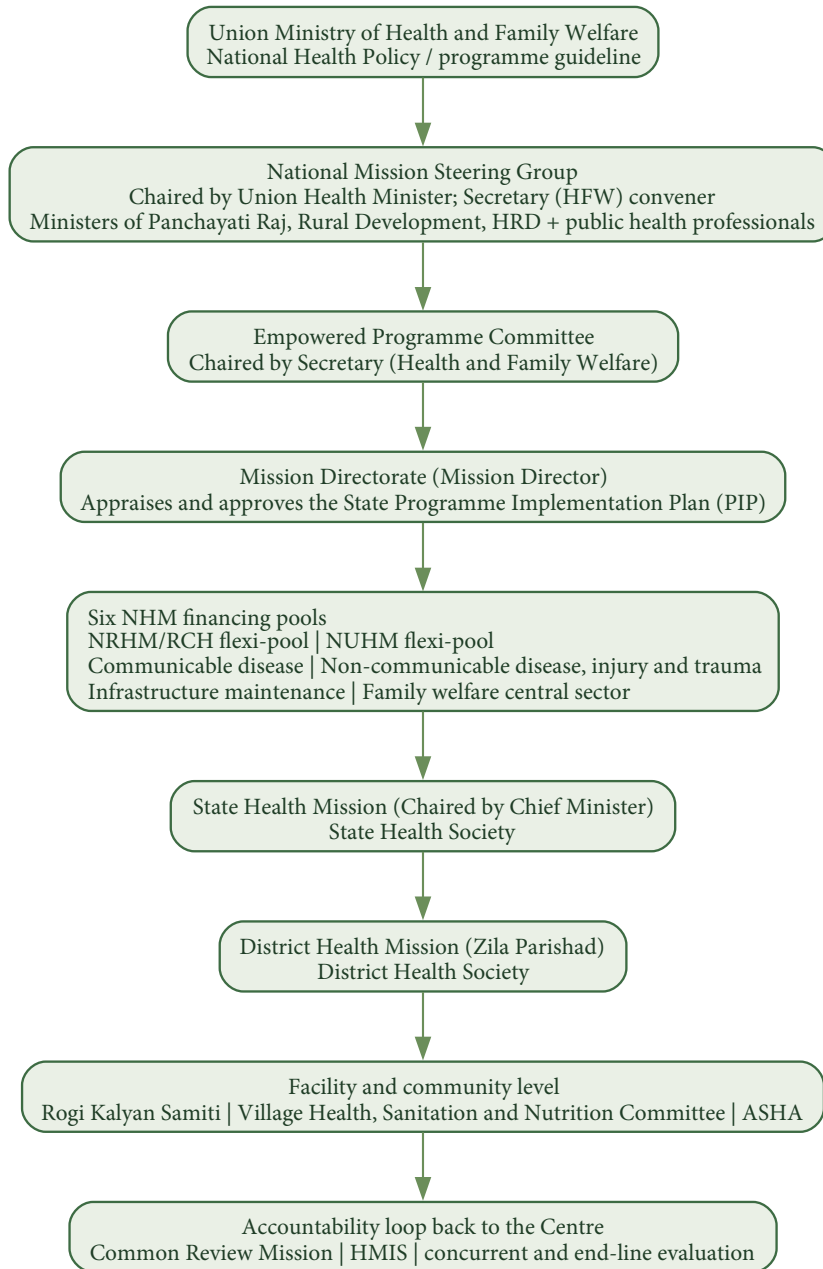
3. Specific functions of the Directorate General of Health Services

#	Function	Content (Park 28e, Ch 23, pp 1033-1034)
1	International health relations and quarantine	Major ports (Kolkata, Visakhapatnam, Chennai, Cochin, Mumbai, Kandla) and international airports (Mumbai-Santa Cruz, Kolkata-Dum Dum, Chennai-Meenambakkam, Tiruchirapalli, Delhi-Palam) directly controlled by DGHS; all assistance from international agencies routed through DGHS
2	Control of drug standards	Drugs Control Organization, headed by the Drugs Controller; lays down and enforces standards, controls manufacture and distribution; Drugs Act (1940) empowers the Centre to test imported drugs
3	Medical store depots	Mumbai, Chennai, Kolkata, Karnal, Guwahati, Hyderabad — supply civil medical requirements of Central and State Governments and handle foreign supplies
4	Post-graduate training	Administration of national institutes — AIIH&PH Kolkata, All India Institute of Mental Health Bengaluru, College of Nursing Delhi, National Tuberculosis Institute Bengaluru, National Institute of Communicable Diseases Delhi, Central Research Institute Kasauli, NIIHFW Delhi
5	Medical education	Direct charge of Lady Hardinge and Maulana Azad medical colleges and the colleges at Puducherry and Goa; guides and supports many others
6	Medical research	Through ICMR (founded 1911, New Delhi) — Cancer Research Centre, Tuberculosis Chemotherapy Centre Chennai, Virus Research Centre Pune, National Institute of Nutrition Hyderabad, Blood Group Reference Centre Mumbai; funds wholly from the Union Health Ministry budget
7	Central Government Health Scheme	Contributory comprehensive care for central government employees and dependants (see section 4)
8	National health programmes	Planning, guiding and coordinating all national programmes; these cost crores and "can hardly succeed without the help of the Central Government"
9	Central Health Education Bureau	Prepares education material for health awareness; trains categories of health workers in health education
10	Health intelligence	Central Bureau of Health Intelligence (est. 1961) — collection, compilation, analysis, evaluation and dissemination of health statistics; disseminates epidemic intelligence to States and international bodies; has an Epidemiological Unit, Health Economics Unit, National Morbidity Survey Unit and Manpower Cell
11	National Medical Library	Central Medical Library of DGHS, declared National Medical Library in 1966

4. Central-level agencies that deliver or finance care directly

Agency	Population served	Administrative home
Central Government Health Scheme (CGHS) , started 1954 in Delhi	Central government employees and dependants, MPs, judges, pensioners, freedom fighters, accredited journalists; extended to 72 cities; 12,50,515 card-holders and 36,69,224 beneficiaries (August 2020)	MoHFW / DGHS
Employees' State Insurance Scheme	Employees drawing wages not exceeding Rs 21,000/month, plus dependants	ESI Corporation — Union Minister for Labour is Chairman , Secretary (Labour) Vice-Chairman
Railway health services	Railway employees — 613 dispensaries and health centres, 125 hospitals, 13,702 beds	Ministry of Railways
Armed Forces / ECHS	Serving personnel; 227 ECHS polyclinics covering ~20 lakh ex-servicemen and ~10 million beneficiaries	Ministry of Defence
Ayushman Bharat — PMJAY (2018) , subsuming RSBY	10.7 crore poor and vulnerable families (~50 crore people), Rs 5 lakh/family/year for secondary and tertiary hospitalisation; Centre:State share 60:40 (90:10 for North-Eastern States)	National Health Authority, MoHFW

5. How central authority reaches the periphery



Route by which a central policy decision becomes a service at the periphery (National Health Mission)

6. Critical appraisal

Domain	Strength	Weakness
Governance / stewardship	Article 263 Council gives a standing Centre–State forum for the large number of Concurrent-list health subjects; the Centre confines itself to policy, planning, coordination and evaluation so no State lags behind	The Council is only recommendatory and the DGHS is only an adviser — health being a State subject, the State remains the ultimate authority, so central intent can stall at the State boundary

Domain	Strength	Weakness
Technical leadership	Concentrates functions no State can perform alone — port and airport quarantine, drug standards, ICMR-led research, CBHI health intelligence, national training institutes	The technical head is subordinate to a generalist Indian Administrative Service Secretary. The same technical-versus-administrative split persists downstream: the <i>Bhore</i> Committee (1946) recommended that the Director of Health Services also be Secretary to the State Government, and that recommendation has never been implemented
Financing	Central financing makes capital-intensive national programmes feasible; NHM's pooled flexi-financing against an appraised State PIP allows multi-year, State-specific planning; NHM cut out-of-pocket expenditure from 72% to 60%	Government health expenditure only 1.15% of GDP against the National Health Policy 2017 target of 2.5% by 2025 ; NHM review still found it at 1.45% of GDP. Out-of-pocket spending 62.4% (2014 estimate) against a NITI Aayog target of 50% by 2020. Activity-wise transfers arrive late, untied funds lie blocked, and most States lack separate finance and accounts managers
Programme design	A single central design gives national uniformity of case definitions, drugs and reporting, and eradicated smallpox, guinea-worm and poliomyelitis	Delivery has favoured a vertical approach — expensive and difficult to sustain ; where human resources, logistics and infrastructure are sub-critical, <i>all</i> national programmes do badly. NCDs (39.1%) and injuries (11.8%) now dominate the burden but central programme scope remains limited
Human resources	Central Health Service, national institutes and IPHS norms provide a common cadre and staffing standard	Acute shortage of skilled manpower and specialists persists; supervision is weak and HMIS data are not used for local action
Regulation of the private sector	National Medical Commission (replaced the Medical Council of India on 25.9.2020) and the Drugs Control Organization give central regulatory instruments	The private sector delivers ~80% of outpatient and ~60% of inpatient care yet regulatory standards are "not adequately defined and ineffectively enforced"
Fragmentation across ministries	Ayushman Bharat–PMJAY and the Ayushman Bharat Digital Mission under the National Health Authority create, for the first time, a single central purchaser and a single digital spine	Central health responsibility is split — ESI sits under the Ministry of Labour , railway and armed-forces health under their own ministries, and AYUSH is a separate Ministry — so no single central authority owns population health
Net effect on utilisation	Rs 5 lakh/family/year PMJAY cover for 10.7 crore families is the largest government-funded health protection scheme in the world	Public facilities are still bypassed: only 45.7% of rural and 35% of urban hospitalisations occur in public hospitals (NSSO, 1996–2018), reflecting shortage of personnel, absenteeism, long waits and poor outreach

Way forward (as set by National Health Policy 2017): raise government health spending to 2.5% of GDP by 2025 and State health spending to >8% of State budgets; cut households facing catastrophic health expenditure by 25% by 2025; ensure doctors and paramedics as per Indian Public Health Standards in high-priority districts; and shift the Centre's role from vertical programme manager to steward, purchaser and regulator.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Organogram of the health system at the Centre** — draw a vertical tree with these exact labels: *Seventh Schedule, Article 246* at the top; three boxes beneath it labelled **Union Ministry of Health and Family Welfare** (sub-labels: Cabinet Minister, Minister of State, Department of Health and Family Welfare, Department of Health Research), **Directorate General of Health Services** (sub-labels: Additional DGHS; medical care and hospitals; public health; general administration), and **Central Council of Health and Family Welfare** (sub-labels: Article 263, 9 August 1952; Chairman — Union Health Minister; Members — State Health Ministers). A fourth box below labelled **Attached and autonomous bodies**: Drugs Control Organization, Central Bureau of Health Intelligence, Central Health Education Bureau, National Medical Library, ICMR, NIHFW, National Medical Commission, National Health Authority.
- ▶ **Three links of the Indian health system** — a simple vertical chain labelled **Centre** → **State** → **District (local/peripheral)**, with the annotation "policy, planning, coordination" against the Centre and "ultimate authority for service delivery" against the State.

High-yield exam pointers

- ▶ **Three organs at the Centre** = MoHFW + DGHS + CCHFW. **Three links of the system** = Centre, State, local/peripheral.
- ▶ **Central Council of Health: Presidential Order, 9th August 1952, Article 263.** Chairman = Union Health Minister; members = State Health Ministers; four functions (policy outlines, legislation proposals, grants-in-aid distribution, Centre–State cooperation machinery).
- ▶ **Union Health Ministry functions = Seventh Schedule, Article 246** — 9 items on the Union list, 8 on the Concurrent list.
- ▶ **DGHS = 11 specific functions.** Remember the dated ones: **Central Bureau of Health Intelligence 1961, National Medical Library 1966, ICMR 1911, Drugs Act 1940, CGHS 1954, NMC replaced MCI 25.9.2020, Ayushman Bharat 2018.**
- ▶ **Numbers for the appraisal:** government health spend 1.15% of GDP → NHP 2017 target 2.5% by 2025; out-of-pocket 62.4% (2014) → NITI Aayog target 50%; NHM brought OOP 72% → 60%; private sector 80% outpatient / 60% inpatient; PMJAY Rs 5 lakh/family/year for 10.7 crore families, Centre:State 60:40 (90:10 North-East).
- ▶ **One-line critique to open with:** the Centre has the money, the technical institutions and the policy pen, but no executive authority over delivery — its instruments are grants-in-aid, guidelines and appraisal of State plans, never direct command.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park 28e still lists the Ministry's departments as "Department of Health" and "Department of Family Welfare", whereas these were merged into a single **Department of Health and Family Welfare** and a **Department of Health Research** was added, with AYUSH now a separate Ministry — write the current arrangement and mention the older split only in parentheses. Park 28e also carries two conflicting counts in the same chapter (29 States and 7 Union Territories on p 1033 versus 28 States and 9 Union Territories on p 1034); the later figure of 28 States and 9 Union Territories is the correct one.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, National Health Accounts Estimates for India 2021-22

Programme and policy updates (India)

- ▶ **Health and Wellness Centres have been renamed Ayushman Arogya Mandir (AAM)** — the single most examinable change to the Ayushman Bharat pillar described above. More than **1.75 lakh AAM** are operational; a new **Community Health Officer / mid-level health provider** cadre numbered **1,16,978 in position on 30 November 2024** (against a shortfall of about 11,111), and **82% of AAM had rolled out all 12 expanded service packages** by that date, up from 50% in March 2023. Free drugs and diagnostics were widened to **106 medicines and 14 tests at the AAM sub-health centre** and **172 medicines and 63 tests at the AAM primary health centre**, with facilities upgraded against IPHS 2022. [PMID 41816146]
- ▶ **Two new central financing channels now run alongside the NHM flexi-pools** for primary-care infrastructure — the **PM-Ayushman Bharat Health Infrastructure Mission (PM-ABHIM)** and **XV Finance Commission health grants routed through Panchayati Raj Institutions and Urban Local Bodies**. Community oversight has been institutionalised through **1,35,343 Jan Arogya Samitis**, and continuity of care through **eSanjeevani** teleconsultation and the **Ayushman Bharat Health Account (ABHA) digital health ID**. [PMID 41816146]
- ▶ **Single Nodal Agency (SNA) public financial management reform** has changed the route by which central NHM money reaches the States — **each centrally sponsored scheme now runs through a single designated State nodal account**, so releases and expenditure for that scheme are tracked in one place instead of across many implementing-agency accounts. Assessment in Bihar and Odisha found real gains in cash management, expenditure transparency and alignment of central releases with actual sub-national spending, but weak budget credibility, patchy digital connectivity and thin finance-staff capacity threaten to widen inter-State inequity, and the new pressure to demonstrate utilisation has crowded out attention to health outcomes. [PMID 42480037]

Current national data

- ▶ **Out-of-pocket expenditure has fallen to 39.4% of total health expenditure in 2021-22, from 60.6% in 2014-15** (National Health Accounts estimates for India) — this supersedes the 62.4% (2014) figure carried in the appraisal above and should be quoted with its year. Alongside it: **UHC Service Coverage Index 57 (2015) → 63 (2021)**, above the South-East Asia regional average

of 62; public-facility OPD attendance 91.6 crore (2012-13) → 134.5 crore (2021-22) and IPD 4.3 crore → 6.3 crore; and about 30.76 crore teleconsultations delivered by 30 November 2024. [PMID 41816146]

- ▶ **PM-JAY after four years showed no measurable financial protection in Chhattisgarh, the State with the most complete panel evaluation.** Two annual household waves (November 2021 and November 2022, ~15,000 individuals each) in this leading implementing State, with ~90% of the population enrolled, found enrolment unassociated with greater inpatient utilisation, lower out-of-pocket expenditure, shorter stay or higher patient satisfaction, even after reimbursement prices were revised upwards in January 2020. Among enrolled patients using **private** hospitals, **74.5% (95% CI 70.8-77.8)** still incurred catastrophic expenditure at the 10%-of-consumption threshold, and mean out-of-pocket expenditure in private hospitals was about **ten times** that in public hospitals — attributed to double-billing and ineffective enforcement of empanelment contracts. **The evidence is not one-sided:** the same paper notes an earlier six-State survey (2019-2020) in which patients who managed to use the scheme were **21% less likely** to incur catastrophic expenditure. Present the Chhattisgarh result as **State-level evidence**, not an all-India audit. [PMID 39135015]
- ▶ **A pan-India study of NSS Social Consumption (Health) rounds 60 (2004), 71 (2014) and 75 (2017-18) reached a similar verdict.** Descriptively, average out-of-pocket expenditure per hospitalisation **fell in public hospitals in every consumption quintile, rural and urban, while rising in private hospitals;** and in 2017-18 catastrophic expenditure at the 10% threshold affected **32.3% of insured households against 32.7% of the uninsured** — a statistically insignificant difference. On the 75th round, **instrumental-variable regression found no statistically significant relationship between publicly funded health insurance enrolment and out-of-pocket expenditure per hospitalisation,** and sub-group analysis showed **no significant impact in the bottom two quintiles, casual-labour households or rural areas.** The authors' inference is directly relevant to the Centre's stewardship role: insurance-based demand-side financing alone cannot deliver universal health coverage, and strengthening the public system carries the dual advantage of financial protection plus equity. [PMID 41913189]

Elimination, surveillance and disease status

- ▶ **Lymphatic filariasis remains endemic in 345 districts** against a national **elimination target of 2030.** Mass drug administration, morbidity management and post-MDA surveillance strain public health infrastructure, and the argument is being made for formal **public-private partnership** across the operational gaps — advocacy and social mobilisation, non-compliance with MDA, surveillance once MDA stops, finding and caring for lymphoedema patients, digital tools, operational research and vector control — on the polio-eradication and TB-programme model, where PPP has already worked in India. [PMID 41951149]

Guideline and legislative updates

- ▶ **Antimicrobial resistance is now framed as a health-systems and universal-health-coverage problem,** not a purely clinical one: India has a National Action Plan plus **six State action plans,** stewardship programmes, laboratory-network strengthening and One Health intersectoral coordination — but enforcement gaps, over-the-counter sale, pharmaceutical and hospital wastewater contamination and fragmented data remain the binding constraints; artificial-

intelligence-assisted surveillance is proposed to integrate the dispersed data streams. [PMID 41443506]

Global and WHO developments

- ▶ **The International Health Regulations (2005) were amended by resolution WHA77.17 on 1 June 2024, in force from 19 September 2025.** The package adds a new "pandemic emergency" tier above a PHEIC (wide geographical spread, exceeding health-system capacity, substantial socio-economic disruption, requiring coordinated international action), obliges every State Party to designate a **National IHR Authority**, and creates a **Coordinating Financial Mechanism** and a **States Parties Committee**. India's IHR obligations under the Union list therefore now have a named national authority and a new notification threshold. [WHO 2024-Jun]
- ▶ **The WHO Pandemic Agreement was adopted on 20 May 2025 at the Seventy-eighth World Health Assembly (124 in favour, none against, 11 abstentions).** Its core is a **Pathogen Access and Benefit-Sharing (PABS) system** under which participating manufacturers make **20% of real-time production** of vaccines, therapeutics and diagnostics available to WHO during a pandemic, with a Coordinating Financial Mechanism and a Global Supply Chain and Logistics Network alongside. **The PABS annex missed its May 2026 (WHA79) deadline and negotiations were extended to WHA80 in May 2027, or to an earlier special session of the Assembly;** the Agreement opens for signature only after the annex is adopted, and enters into force **30 days after the 60th ratification** — so it is **not yet in force**, and that is the examinable point. [WHO 2026-May]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 23 (Health Planning and Management, pp 1030, 1033-1034, 1040), Ch 7 (Health Programmes in India, pp 533-534, 541), Ch 10 (Health Care of the Community, pp 745-746), Ch 18 (Occupational Health, p 971); Sunder Lal — Textbook of Community Medicine, Ch 17 (Healthcare of the Community, pp 743, 746, 763-766); Jugal Kishore — National Health Programmes of India, Ch 1 (pp 86-87, 91, 99, 129-130, 138-139, 156, 159-160) and Ch 2 (pp 725, 741); Oxford Textbook of Public Health 7e, Ch 51 (Governance and management of public health programmes, pp 1877-1879, 1906-1907). Named documents: National Health Policy 2017; NITI Aayog Three Year Action Agenda 2017-18 to 2019-20; National Sample Survey Office rounds 1996-2018.

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Health care challenges in tribal populations

Asked in: Paper III 16-Nov-2019 — RGUHS MD Community Medicine

Definition — Scheduled Tribes (ST) are, under Article 366(25) of the Constitution, tribes or tribal communities deemed under Article 342 to be Scheduled Tribes; the list is state/union territory specific. Over 104 million tribal people across 705 tribes form 8.6 per cent of India's population and have remained marginal geographically, socio-economically and politically, so their health and health care remain an unsolved problem (Park 28e, Ch 13, p 848).

1. Magnitude and demographic profile — the setting of the problem

Parameter	Tribal (Scheduled Tribes)	Comparison	Source
Population	104 million; 705 tribes; 8.6% of India	Males 52.5 million, females 52.0 million	Park 28e, Ch 13, p 848
Residence	~90% rural, mostly hilly and forested areas	Concentrated in 10 states + 8 North-Eastern states	Park 28e, Ch 13, p 848
Largest numbers	Madhya Pradesh 15 million > Maharashtra 10 million > Odisha 9 million > Rajasthan 9 million	Highest concentration is in the North-East	Park 28e, Ch 13, p 848
Sex ratio (2011)	990 — better than the general population	National 943	Park 28e, Ch 13, p 848
Child sex ratio 0–6 y (2011)	957, fallen from 972 in 2001	General population 914	Park 28e, Ch 13, p 848
Literacy (Census 2011)	59.0% overall; males 68.5%, females 49.4%	All-India 73.0%; males 80.9%, females 64.6%	Park 28e, Ch 13, p 848, Table 1

This is a dispersed, hamlet-based population living where the terrain itself defeats a standard-issue health system.

2. A triple burden of disease — the need is greater, not merely different

Epidemiological transition is under way in tribal areas as elsewhere, so the tribal population now faces a **triple burden**: persisting malnutrition and communicable disease, rising non-communicable disease, and mental illness and addictions (Park 28e, Ch 13, pp 848–849).

Burden	Tribal-specific evidence
Tuberculosis	Estimated prevalence of pulmonary tuberculosis 703 per 100,000 versus 256 per 100,000 in the rest of the country — nearly three-fold (Park 28e, Ch 13, p 849)
Malaria	Tribal people are ~8% of the population but account for ~30% of all malaria cases, >60% of <i>P. falciparum</i> , and up to 50% of malaria deaths (Park 28e, Ch 13, p 849)
Leprosy	Of 1,35,485 new cases in 2016–17, 25,474 (18.9%) were Scheduled Tribes (Park 28e, Ch 13, p 849)
Other communicable	Skin infections, diarrhoeal diseases, hepatitis, yaws (Park 28e, Ch 13, p 849; Sunder Lal, Ch 17, p 760)

Burden	Tribal-specific evidence
Hypertension	One in four tribal adults is hypertensive; two of three do not know its signs or symptoms; only 5% of men and 9% of women knew their hypertensive status (Park 28e, Ch 13, p 850)
Genetic disorders	Sickle cell disease (trait + disease) and thalassaemia 1–40% across communities; sickle cell anaemia in 1 of 86 births in central Indian tribes; glucose-6-phosphate dehydrogenase (G6PD) deficiency 0.7–15.6% in 14 particularly vulnerable tribal groups (Park 28e, Ch 13, p 850)
Addictions	72% of tribal men aged 15–54 use tobacco versus 56% of non-tribal men; about 50% consume alcohol (Park 28e, Ch 13, pp 849–850)
Environmental/injury	Snake, dog and scorpion bites are common as habitations abut forests; violence in conflict areas (Park 28e, Ch 13, p 850)

Maternal and child health outcomes lag on almost every indicator (NFHS-5, 2019–21):

Indicator (NFHS-5, 2019–21)	Scheduled Tribes	All India
Infant mortality rate (per 1000 live births)	41.7	35.2
Neonatal mortality rate	28.8	24.9
Under-five mortality rate	50.3	41.9
Under-5 children stunted (%)	40.9	35.5
Under-5 children wasted (%)	23.2	19.3
Under-5 children underweight (%)	39.5	32.1
Institutional delivery (%)	68.0	88.6
Deliveries by skilled health personnel (%)	71.5	89.4
Pregnant women receiving tetanus toxoid (%)	79.0	85.5
Infants fully immunised (%)	76.8	76.6

Only about 37% of tribal women delivering at home receive postnatal care within 48 hours; early marriage, early childbearing, low body mass index and anaemia drive the high maternal mortality (Park 28e, Ch 13, p 848).

3. Supply-side challenges — infrastructure and human resources

Norms are already relaxed for tribal and hilly areas — one health sub-centre per 3,000, one primary health centre (PHC) per 20,000, one community health centre (CHC) per 80,000 — yet even these are unmet (Park 28e, Ch 13, p 850; Sunder Lal, Ch 17, p 760, Table 17.9).

- **Shortfall against the relaxed norm** (18 states, 3 union territories studied): 4,996 sub-centres short (27%) in 11 states; 1,023 PHCs short (40%) in seven states; 209 CHCs short (31%) in 10 states — in about half the states, deficient by 27–40 per cent (Park 28e, Ch 13, p 850).
- **Locational disadvantage** — existing sub-centres, PHCs and CHCs are sited where scattered hamlets cannot reach them (Sunder Lal, Ch 17, p 760).

- ▶ **Vacant posts and absent providers**, worsened by no residential facility at the centre (Sunder Lal, Ch 17, p 760).
- ▶ **Why the human-resource gap persists**: limited scope for professional interaction or growth, **social and professional isolation**, weak human-resource policies, poor working conditions and limited social infrastructure around health institutions (Park 28e, Ch 13, pp 850–851).
- ▶ **Failed corrective** — several states introduced a **compulsory rural-service bond** after the MBBS, but most doctors flout it and states cannot enforce it (Park 28e, Ch 13, p 851).
- ▶ **No functioning referral transport** for emergencies out of forest and hill terrain (Sunder Lal, Ch 17, p 760).

4. Demand-side challenges — access, culture and cost

- ▶ **Difficult terrain and sparse, dispersed settlement** in forest and hilly regions — the primary barrier to access and utilisation (Sunder Lal, Ch 17, p 760).
- ▶ **Cultural barriers** — customs, beliefs and practices profoundly shape health behaviour, and services are not client-friendly in **timing** or manner (Sunder Lal, Ch 2, p 68 and Ch 17, p 760).
- ▶ **Information, education and communication (IEC) not tuned to tribal vocabulary, beliefs and practices**, so messages do not land (Sunder Lal, Ch 17, p 760).
- ▶ **Non-involvement of local traditional faith healers**, the first point of contact for most tribal families (Sunder Lal, Ch 17, p 760).
- ▶ **Low female literacy, poverty and marginal-farmer/shifting-cultivation livelihoods** limit demand and the ability to bear indirect costs (Sunder Lal, Ch 2, p 68).
- ▶ **Economic burden** — travel, wage loss and being pushed between multiple providers generate heavy **out-of-pocket expenditure** (Park 28e, Ch 13, p 849; Sunder Lal, Ch 2, pp 84–85).

5. Programme, governance and information challenges

- ▶ **Displacement and traditional lifestyles** have made tribal development a planning problem since Independence; the Tribal Sub-Plan (TSP), now the **Scheduled Tribe Component (STC)**, was adopted in the Fifth Five Year Plan, yet the human-development gap between Scheduled Tribes and all categories persists (Park 28e, Ch 13, p 850).
- ▶ **Policy primacy without delivery** — the National Health Policy 1983, the Ninth Plan, the National Population Policy 2000 and Reproductive and Child Health Phase 2 all gave primacy to tribal areas, yet utilisation remains suboptimal (Sunder Lal, Ch 17, p 760).
- ▶ **Vertical disease-control programmes not integrated, and non-governmental organisations inadequately mobilised** (Sunder Lal, Ch 17, p 760).
- ▶ **Weak disaggregated data** — tribal-specific data are routinely collected for only some programmes (for example monthly state-wise Scheduled Tribe data under the National Leprosy Eradication Programme), handicapping planning elsewhere (Park 28e, Ch 13, p 849).

6. Responses to these challenges

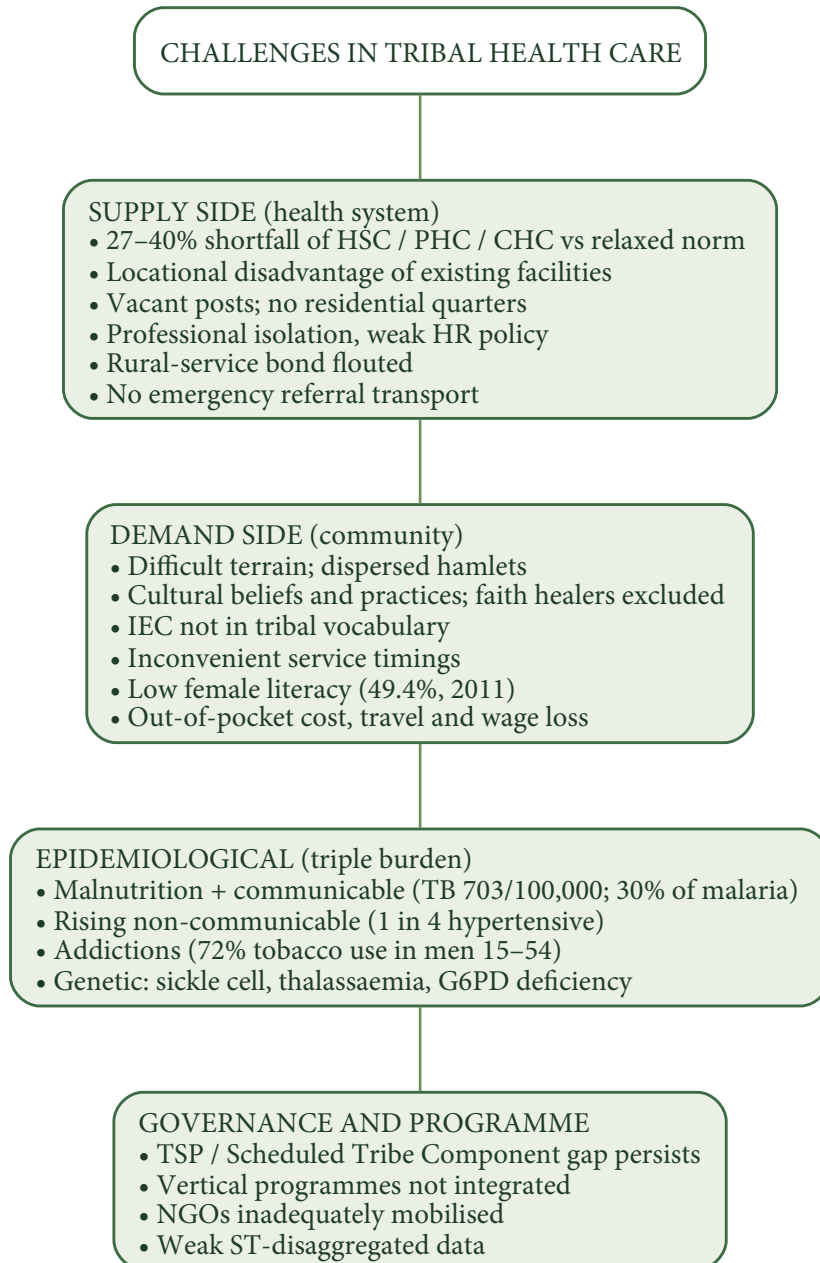
Response	Content
Tribal Malaria Action Plan (TMAP)	Tribal-specific malaria strategy for category 2 and 3 districts: hamlet-wise ASHAs instead of village-wise , engagement of anganwadi workers, faith healers and community volunteers, mobile surveillance where routine services are absent, intensive training of all cadres and supply-chain strengthening (Park 28e, Ch 7, pp 498–499)
Mobile TB Diagnostic Vans (MTDV)	35 vans with X-ray and sputum microscopy in 17 districts of 5 states (Madhya Pradesh, Gujarat, Chhattisgarh, Rajasthan, Jharkhand) covering ~17.65 million people, plus door-to-door active case finding campaigns from 2017 (Park 28e, Ch 13, p 849)
National Programme for Control of Blindness and Visual Impairment	Dedicated eye units in North-Eastern states, contractual ophthalmic manpower, multipurpose district mobile ophthalmic units for difficult areas (Park 28e, Ch 13, p 850)
Vector-borne disease control	Focused attention on tribal-dominated areas; Global Fund support to North-Eastern states and World Bank support elsewhere; 100% central assistance to the North-East (Park 28e, Ch 13, pp 849–850)

The **Expert Committee on Tribal Health** (constituted October 2013 jointly by the Ministry of Health and Family Welfare and the Ministry of Tribal Affairs, chaired by **Dr Abhay Bang**), in *Tribal Health in India: Bridging the Gap and a Roadmap for the Future*, recommended (Park 28e, Ch 13, p 851):

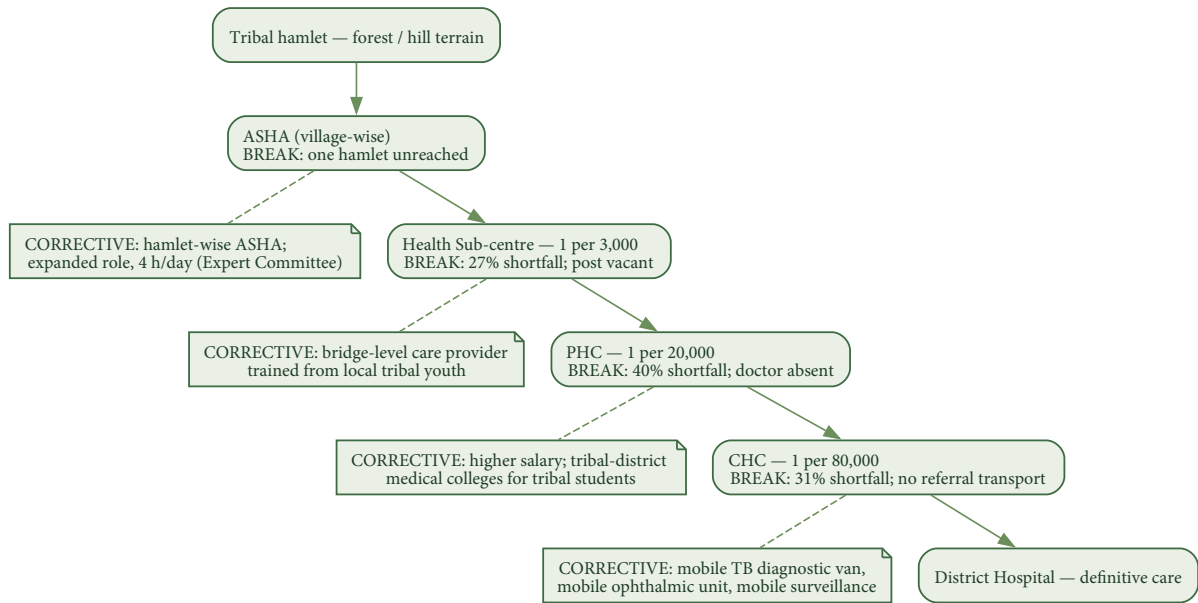
- ▶ The care provider should, as far as possible, be a **local tribal person** — the present workforce pattern is the opposite.
- ▶ **Train and deploy local tribal people** — the only sustainable route to a responsive tribal workforce.
- ▶ Shift the **centre of gravity of the workforce away from specialists and doctors, closer to the communities**.
- ▶ Create **bridge-level care providers** through dedicated courses and place them at sub-centres.
- ▶ Give the tribal ASHA an expanded role and functions, with **4 hours of work per day** expected.
- ▶ **Substantially increase total salary** to attract doctors to tribal areas.
- ▶ Create **dedicated medical colleges in tribal districts, exclusively for tribal students** of the scheduled areas.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

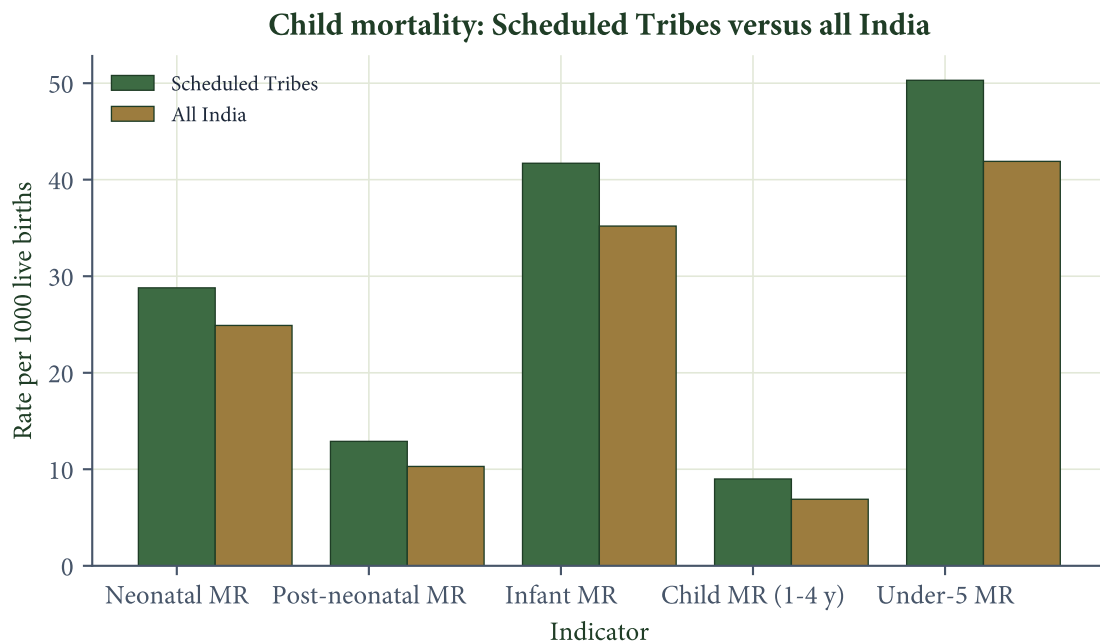
- ▶ **Labelled map to draw** — outline map of India, shade the **ten tribal-concentration states plus the eight North-Eastern states**; mark Madhya Pradesh (15 million), Maharashtra (10 million), Odisha (9 million) and Rajasthan (9 million) as the four largest tribal populations, and label the North-East as "**highest concentration**".



Challenges in providing health care to tribal populations — supply, demand and governance



Where the tribal patient's care pathway breaks, and the corrective placed at each step



Every child-mortality indicator is higher among Scheduled Tribes (NFHS-5, 2019-21; Park 28e, Ch 13, Table 2).

High-yield exam pointers

- ▶ **Constitutional anchor:** Article 366(25) defines, Article 342 notifies; list is state-specific. 104 million / 705 tribes / 8.6%, ~90% rural.
- ▶ **Relaxed norms "3-20-80":** sub-centre 1/3,000, PHC 1/20,000, CHC 1/80,000 — shortfall against them 27% / 40% / 31%.
- ▶ **Three killer comparisons:** tuberculosis 703 vs 256 per 100,000; 8% of the population but 30% of malaria cases, >60% of *P. falciparum*, 50% of malaria deaths; institutional delivery 68.0% vs 88.6% (NFHS-5, 2019-21).
- ▶ **Triple burden** = malnutrition and communicable disease + non-communicable disease + mental illness and addictions.

- ▶ **Committee to name:** Expert Committee on Tribal Health, Dr Abhay Bang, October 2013 — *Tribal Health in India: Bridging the Gap and a Roadmap for the Future.*
- ▶ **Its one-line message:** the care provider should be a **local tribal** — bridge-level providers at sub-centres, hamlet-wise ASHAs at 4 hours per day, tribal-district medical colleges for tribal students.
- ▶ **Planning instrument:** Tribal Sub-Plan (TSP), now **Scheduled Tribe Component (STC)**, adopted in the **Fifth Five Year Plan**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: Park 28e's running text on p 848 gives the all-India institutional delivery rate as 78.9 per cent and the Scheduled Tribe total fertility rate as 2.48, while its own Table 2 gives 88.6 per cent and 2.09 for the same National Family Health Survey-5 round; the table values have been used throughout.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Mission, MoHFW Health Technology Assessment in India, ICMR, NCVBDC, NFHS-5, WHO resolution WHA76.16

Programme and policy updates (India)

- ▶ **National Sickle Cell Anaemia Elimination Mission**, announced in the Union Budget 2023–24, targets elimination by 2047 through universal screening of the **0–40 year** population in endemic tribal districts. A health technology assessment for a 41.1-million tribal cohort in six endemic states (Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha, Chhattisgarh) found point-of-care tests cost more per case screened (US\$2.38–3.85) than solubility test with high-performance liquid chromatography confirmation (US\$0.67), but detected 21,125–41,458 additional cases; the **National Health Mission then directed states to procure kits by public tender**, which pulled the price below **US\$1.21 per test** and projects US\$37.86 million saved at full coverage [PMID 42594133].
- ▶ Nutrition support under the tuberculosis programme reaches tribal patients late and inequitably — in Chhattisgarh, **22% of notified adults had severe and 13% very severe undernutrition**; 81% received the first direct-benefit-transfer instalment but **only 29% within two months**, and food baskets reached 21% with no preference for the most undernourished [PMID 41042600].
- ▶ Tribal health is now framed as an inter-ministerial, Sustainable Development Goal-linked commitment — Ministry of Tribal Affairs, Tribal Pride Day (**Janjatiya Gaurav Divas**) and the International Day of the World's Indigenous Peoples — for a population restated at **106 million across 705 Scheduled Tribes**, with climate change, deforestation and political under-representation named as the newer barriers alongside the classical ones [PMID 38294024].

Current national data

- ▶ NFHS-5 (2019–21), 2.08 million adults aged 15+: tuberculosis point prevalence was **416 per 100,000 in Scheduled Tribes versus 277 in non-tribes**, and tribal identity stayed an independent risk factor after adjustment (aOR 1.47, 95% CI 1.38–1.56). Prevalence was **lowest (330) inside constitutionally protected Fifth/Sixth Schedule areas and highest (608) in non-Scheduled**

districts that are more than 60% tribal — the constitutional safeguard itself appears partly protective [PMID 41185051].

- Undernutrition in tribal women is improving but stays wealth-patterned: underweight among Scheduled Tribe women aged 15–49 fell from 31.7% (NFHS-4, 2015–16) to 25.5% (NFHS-5, 2019–21), the poorest quintile gaining most (–8.5 percentage points versus –3.7 in the richest) and the Erreygers concentration index narrowing from –0.177 to –0.134 [PMID 39849479].

Elimination, surveillance and disease status

- India's declared malaria targets are **zero indigenous cases by 2027 and elimination by 2030**; district-level mapping of National Centre for Vector Borne Diseases Control data for 2019–2023 across 700+ districts found Scheduled Tribe/Scheduled Caste and economically marginalised groups the most vulnerable, and a **post-pandemic geographical expansion of *P. vivax* into the North-East** [PMID 41326452].
- Radical cure is the weak link in tribal malaria — in Malkangiri and Koraput (Odisha) adherence exceeded 93% for 3-day chloroquine but **only 58.7% (95% CI 52.6–64.9) completed the 14-day primaquine course**, mostly stopping once symptoms settled; this is the operational argument for shorter regimens or single-dose tafenoquine with glucose-6-phosphate dehydrogenase testing [PMID 41505814].
- Drug-resistant tuberculosis outcomes in a particularly vulnerable tribal group (Saharia, Madhya Pradesh) reached **66.8% treatment success against the programme's >75% target**, with 20.4% lost to follow-up and 83% rifampicin-resistant or multidrug-resistant — active case finding alone does not close the loop [PMID 40337857].

Guideline and legislative updates

- Comprehensive sickle cell care now has to be written into maternal health services: an ICMR multi-state community study in five endemic tribal areas found women with sickle cell disease had far more antenatal anaemia (43% versus 12.5%), pain crises (42% versus 11.5%) and transfusion (16% versus 1%; all $p < 0.001$) than matched controls, yet **only 10% used hydroxyurea in the periconceptional-to-antenatal period** [PMID 41834271].

Global and WHO developments

- **World Health Assembly resolution WHA76.16 (2023)** commits WHO to a **Global Plan of Action for the Health of Indigenous Peoples**, requiring Indigenous concepts of health and wellbeing to be built into services — the global frame within which India's tribal-health obligations now sit; the supporting evidence base shows Indigenous populations carry about **twice the likelihood of multimorbidity** as non-Indigenous populations [PMID 40859176] [PMID 39067833].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 13 (Tribal Health in India), pp 848–851, and Ch 7 (Health Programmes in India), pp 498–499; Sunder Lal — Textbook of Community Medicine, Ch 17, p 760 and Ch 2, pp 68, 84–85; National Family Health Survey-5 (2019–21); Census of India 2011; Report of the Expert Committee on Tribal Health, Ministry of Health and Family Welfare and Ministry of Tribal Affairs.

Recent-advances sources: 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, PLoS One 2026, 10.1371/journal.pone.0355845; 41042600 — High burden of severe and very severe undernutrition among adults with tuberculosis: findings from a state wide operational research in Chhattisgarh, India, Expert Rev Respir Med 2025, 10.1080/17476348.2025.2569845 (abstract-only); 38294024 — Prioritizing the Health and Well-Being of the Indigenous Communities in India, Asia Pac J Public Health 2024, 10.1177/10105395231226298 (abstract-only); 41185051 — Burden on the burdened: tuberculosis among Scheduled Tribes and non-Scheduled Tribes in constitutionally protected Scheduled and non-Scheduled areas of India, Infect Dis Poverty 2025, 10.1186/s40249-025-01375-9; 39849479 — Wealth-based inequality in underweight among Scheduled Tribe women in India: a regional analysis, Int J Equity Health 2025, 10.1186/s12939-025-02392-8; 41326452 — Mapping malaria risk in India between 2019 and 2023: a tool for the public to track malaria, Sci Rep 2025, 10.1038/s41598-025-25671-4; 41505814 — Suboptimal primaquine adherence in Plasmodium vivax malaria: Evidence from high-burden tribal districts in Odisha, J Infect Public Health 2025, 10.1016/j.jiph.2025.103123 (abstract-only); 40337857 — Drug-resistant tuberculosis treatment outcomes among Saharia: a marginalized community of Madhya Pradesh, India, Trans R Soc Trop Med Hyg 2025, 10.1093/trstmh/traf046 (abstract-only); 41834271 — Reproductive Health and Pregnancy Outcomes Among Women With Sickle Cell Disease in Tribal Communities of India, Pediatr Blood Cancer 2026, 10.1002/1545-5017.70229 (abstract-only); 40859176 — Indigenous women's experiences, symptomology and understandings of menopause: a scoping review, BMC Womens Health 2025, 10.1186/s12905-025-03953-3; 39067833 — Multimorbidity among the Indigenous population: A systematic review and meta-analysis, Ann Epidemiol 2024, 10.1016/j.annepidem.2024.07.047 (abstract-only). Programme and policy dates used: National Sickle Cell Anaemia Elimination Mission (Union Budget 2023–24, elimination target 2047); National Health Mission directive on public procurement of point-of-care sickle cell tests; National Centre for Vector Borne Diseases Control district data 2019–2023 (zero indigenous malaria cases by 2027, elimination by 2030); NFHS-4 (2015–16) and NFHS-5 (2019–21); World Health Assembly resolution WHA76.16 (2023) — Global Plan of Action for the Health of Indigenous Peoples.

♦ ♦ ♦

Community health care delivery

Asked in: Paper III Oct 2017 — RGUHS MD Community Medicine

Definition — Health care is the multitude of services rendered to individuals, families and communities for the promotion, maintenance and restoration of health; because health is influenced by food, housing, sanitation, lifestyle and environment, "health care" implies far more than medical care (Park 28e, Ch 10, p 718). Health care delivery to a community is the organised provision of these services to a defined population through a **health care system** — the management sector that delivers the services and operates within the country's socio-economic and political framework (Park 28e, Ch 10, p 726). Since Alma-Ata (1978) the accepted vehicle is **primary health care**: "essential health care based on practical, scientifically sound and socially acceptable methods and technology, made universally accessible to individuals and families in the community through their full participation, and at a cost that the community and country can afford to maintain at every stage of their development in the spirit of self-reliance and self-determination" (Park 28e, Ch 2, p 49).

Principles of health care delivery

#	Cardinal principle	What it requires	Operational expression in India
1	Equitable distribution	Services shared equally by all irrespective of ability to pay; rich or poor, urban or rural must have access; the vulnerable and weaker sections specifically looked after (Park 28e, Ch 10, p 719)	Free drugs and free diagnostics initiative in public facilities; "time to care" siting norm of not more than 30 minutes travel to a primary facility (Sunder Lal, Ch 17, p 742)
2	Community participation	Continuing, meaningful involvement of individuals, families and communities in planning, implementation and maintenance of services, with maximum reliance on local manpower, money and materials — "health by the people", "placing people's health in people's hands" (Park 28e, Ch 10, p 720)	Accredited social health activist (ASHA), <i>Anganwadi</i> worker (AWW), village health sanitation and nutrition committee (VHSNC), <i>Rogi Kalyan Samiti</i> at the primary health centre (PHC), <i>Mahila Arogya Samiti</i> (MAS) in urban slums (Sunder Lal, Ch 17, pp 744, 751, 758)
3	Intersectoral coordination	Primary health care involves, besides the health sector, agriculture, animal husbandry, food, industry, education, housing, public works and communication; needs review of administrative systems, reallocation of resources, enabling legislation and joint planning to avoid duplication (Park 28e, Ch 10, p 720)	"Health in all policies" as complement to "health for all" — National Health Policy 2017 thrust area (Sunder Lal, Ch 16, p 698)
4	Appropriate technology	Technology that is scientifically sound, adaptable to local needs, acceptable to those who apply it and to those for whom it is used, and maintainable by the people themselves in keeping with self-reliance (Park 28e, Ch 10, p 720)	Growth monitoring, oral rehydration solution, breastfeeding, immunization, family planning, female literacy, vitamin A, iodized salt and iron-folic acid supplements — the UNICEF GOBI-FFF package (Sunder Lal, Ch 17, p 741)

Services built on these principles must in addition be comprehensive, accessible, acceptable, offer scope for community participation, and be available at a cost the community and country can afford (Park 28e, Ch 10, p 726).

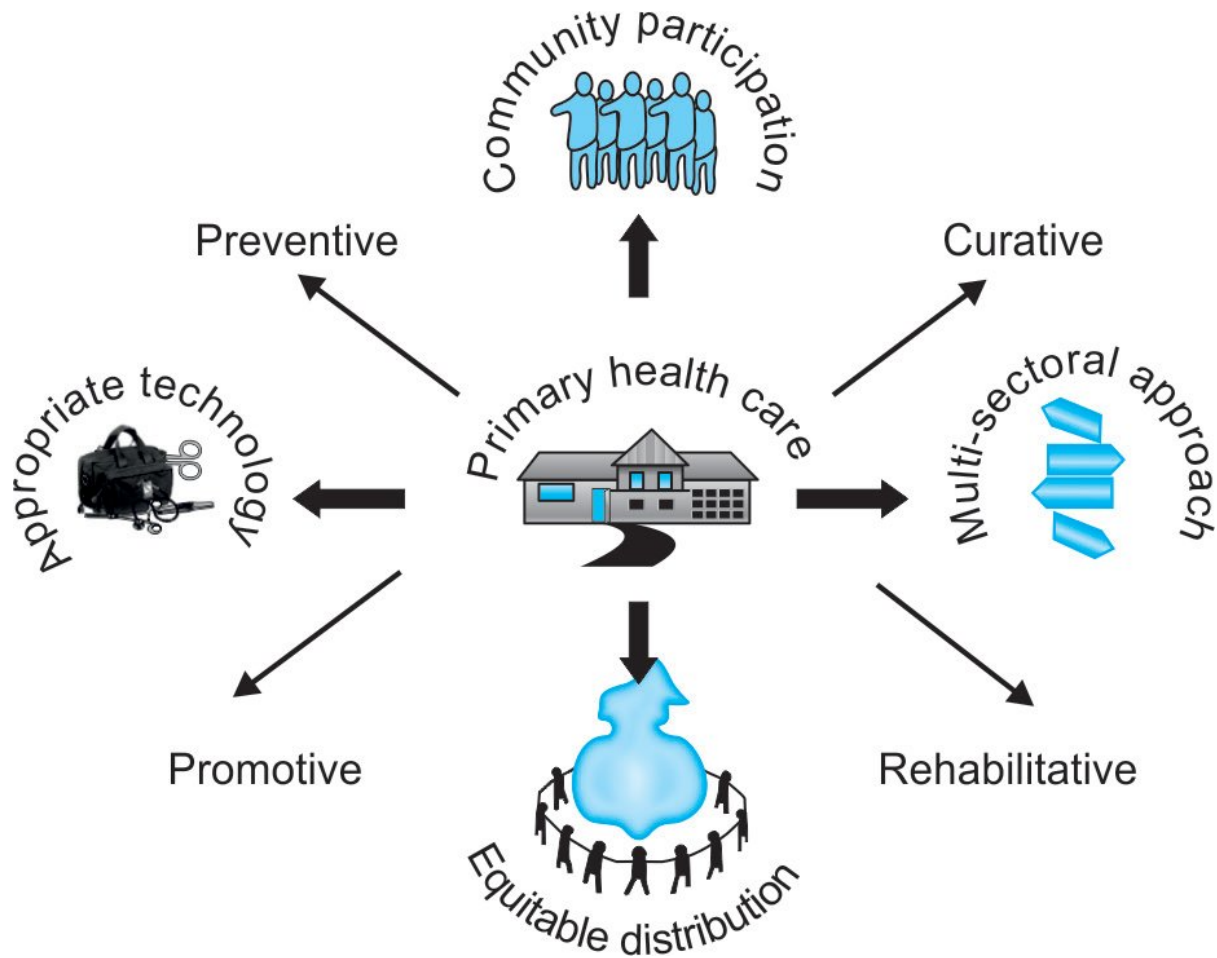


Fig 34.2 — Principles and components of primary health care (Suryakantha Environment Best, p.780)

Components — the essential content of care

A. The eight essential elements of primary health care (Alma-Ata Declaration, 1978) — Park 28e, Ch 10, p 719; Ch 2, pp 49-50; Sunder Lal, Ch 17, p 741:

#	Element	Chief Indian delivery vehicle
1	Education concerning prevailing health problems and the methods of preventing and controlling them	Information, education and communication activity by ASHA and health worker (female)
2	Promotion of food supply and proper nutrition	Integrated Child Development Services; village health, nutrition and sanitation day
3	An adequate supply of safe water and basic sanitation	Water supply and sanitation sector (intersectoral)
4	Maternal and child health care, including family planning	Reproductive, maternal, newborn, child health plus adolescent (RMNCH+A) services
5	Immunization against the major infectious diseases	Universal Immunization Programme

#	Element	Chief Indian delivery vehicle
6	Prevention and control of locally endemic diseases	National health programmes
7	Appropriate treatment of common diseases and injuries	Outpatient care at the primary facility
8	Provision of essential drugs	Essential drug list, free drugs initiative

B. The expanded comprehensive primary health care package — 12 services at the Ayushman Arogya Mandir (formerly Ayushman Bharat Health and Wellness Centre, launched 2018) — Park 28e, Ch 10, pp 730-731; Sunder Lal, Ch 17, pp 742, 749-750:

#	Service package	#	Service package
1	Care in pregnancy and childbirth	7	Screening, prevention, control and management of non-communicable diseases
2	Neonatal and infant health care services	8	Care for common ophthalmic and ENT problems
3	Childhood and adolescent health care services	9	Basic oral health care
4	Family planning, contraceptive and other reproductive health services	10	Elderly and palliative health care services
5	Management of communicable diseases — national health programmes	11	Emergency medical services including burns and trauma
6	Management of common communicable diseases and outpatient care for acute simple illnesses and minor ailments	12	Screening and basic management of mental health ailments

The centre additionally serves as a platform for **teleconsultation** and an expanded range of diagnostics (Park 28e, Ch 10, p 731).

C. The agencies that deliver — in India the system is represented by five major sectors, differing in health technology applied and in source of funds: the **public health sector**, the **private sector**, **indigenous systems of medicine**, **voluntary health agencies**, and the **national health programmes** (Park 28e, Ch 10, pp 726, 746). The private sector is unorganised, largely curative and available to those who can pay, yet supplies about 70% of outpatient and 60% of inpatient care (Sunder Lal, Ch 17, p 742).

Strategies of delivery

#	Strategy (WHO supportive activities for primary health care)	Indian operational form
1	Community involvement and participation	ASHA (1 per 1000 rural population; 1 per 1000–2500 urban poor), VHSNC, MAS
2	Intra- and inter-sectoral coordination	Convergence at the monthly village health, nutrition and sanitation day

#	Strategy (WHO supportive activities for primary health care)	Indian operational form
3	Provision of effective referral support	Community health centre as first referral unit; national ambulance service
4	Development and mobilization of resources	National Health Policy 2017 target — raise government health expenditure from 1.35% to 2.5% of GDP by 2025 and state health spending to over 8% of state budget
5	Managerial processes — policy, goals, objectives	National Health Policy 2017; decentralised participatory planning with the village as basic planning unit
6	Medical and health services research, including innovative approaches	Public–private partnership, contractual appointment, multi-skilling of doctors
7	Development and application of appropriate technology	GOBI-FFF package; teleconsultation at the Ayushman Arogya Mandir
8	Health manpower development	Mid-level health provider (community health officer); public health management cadre

(Sunder Lal, Ch 17, pp 738, 741, 744, 758; Ch 16, p 698; Park 28e, Ch 7, p 535)

Two structural strategies carry the rest: **standard-setting** through the Indian Public Health Standards, revised in 2022 to define an essential (minimum acceptable) and a desirable (aspirational) standard for every facility level, aligned to universal health coverage and Sustainable Development Goal 3 (Park 28e, Ch 10, p 728); and **financial protection** through the second pillar of Ayushman Bharat, Pradhan Mantri Jan Arogya Yojana, which covers secondary and tertiary hospitalisation for the poor — catastrophic health expenditure pushed 63 million people below the poverty line per annum (National Sample Survey 75th round, 2018) (Sunder Lal, Ch 17, pp 739-742).

Levels of care and population norms

Level	Facility	Population norm (plain / tribal-hilly-desert)	Key staff	Function
Community	Village — ASHA, AWW, trained <i>dai</i> , VHSNC	ASHA 1 : 1000; AWW 1 : 400–800 (300–800 tribal)	Volunteers, honorary workers	Mobilisation, home visits, escort, first contact
Primary	Ayushman Arogya Mandir – sub health centre	5000 / 3000	Health workers female and male, led by a mid-level health provider	Preventive, promotive and basic curative care
Primary	Ayushman Arogya Mandir – primary health centre	30,000 / 20,000	Medical officer	Comprehensive primary care, outreach, referral

Level	Facility	Population norm (plain / tribal-hilly-desert)	Key staff	Function
Secondary	Community health centre = first referral unit	1,20,000 / 80,000	4 specialists — surgeon, physician, obstetrician-gynaecologist, paediatrician (6 under Indian Public Health Standards, adding anaesthetist and eye surgeon); 30 indoor beds	Specialist and emergency obstetric care; referral for 4 primary health centres
Secondary	Sub-district and district hospital	District	District programme officers under the district health officer	Referral within one hour of travel; rapid response team
Tertiary	Medical college hospital, superspeciality institute	Regional / zonal / apex	Superspecialists	Complex diagnostic and therapeutic care, teaching

(Sunder Lal, Ch 17, pp 743-744, 751-752, 760-761; Park 28e, Ch 10, p 731; Ch 13, p 850; Ch 8, p 611)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Draw the **pyramid of levels of care** with three bands labelled from base upward: *Primary level (sub health centre, primary health centre); Secondary level (community health centre, taluka hospital, district hospital); Tertiary level (medical college hospital, apex hospital).*
- Draw the **four cardinal principles wheel**: a central box "Primary health care" with four arrows outward labelled *Equitable distribution, Community participation, Multi-sectoral (intersectoral) coordination, Appropriate technology*, and four diagonal arrows labelled *Preventive, Promotive, Curative, Rehabilitative*.

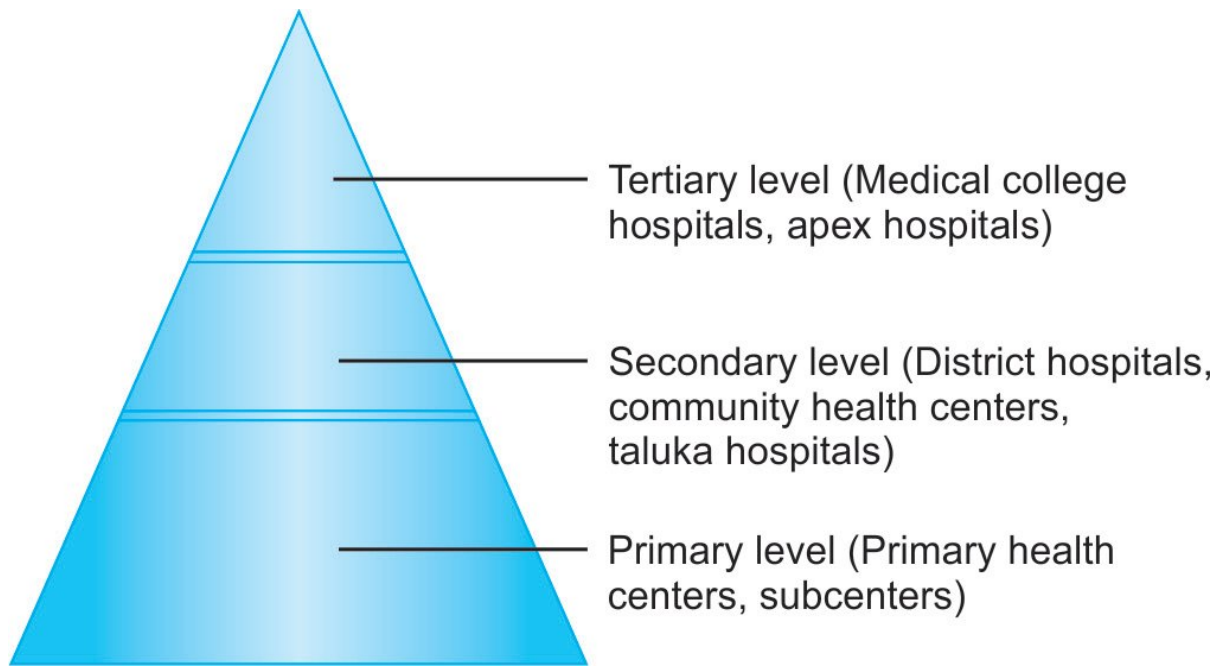
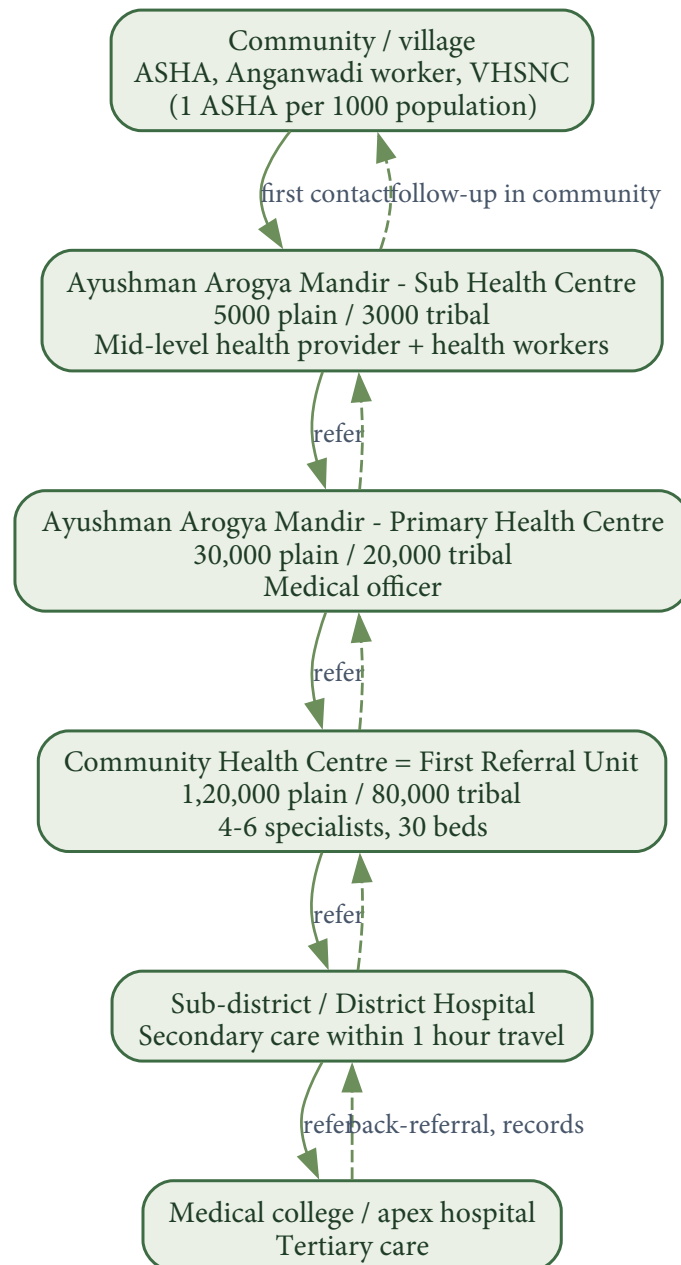
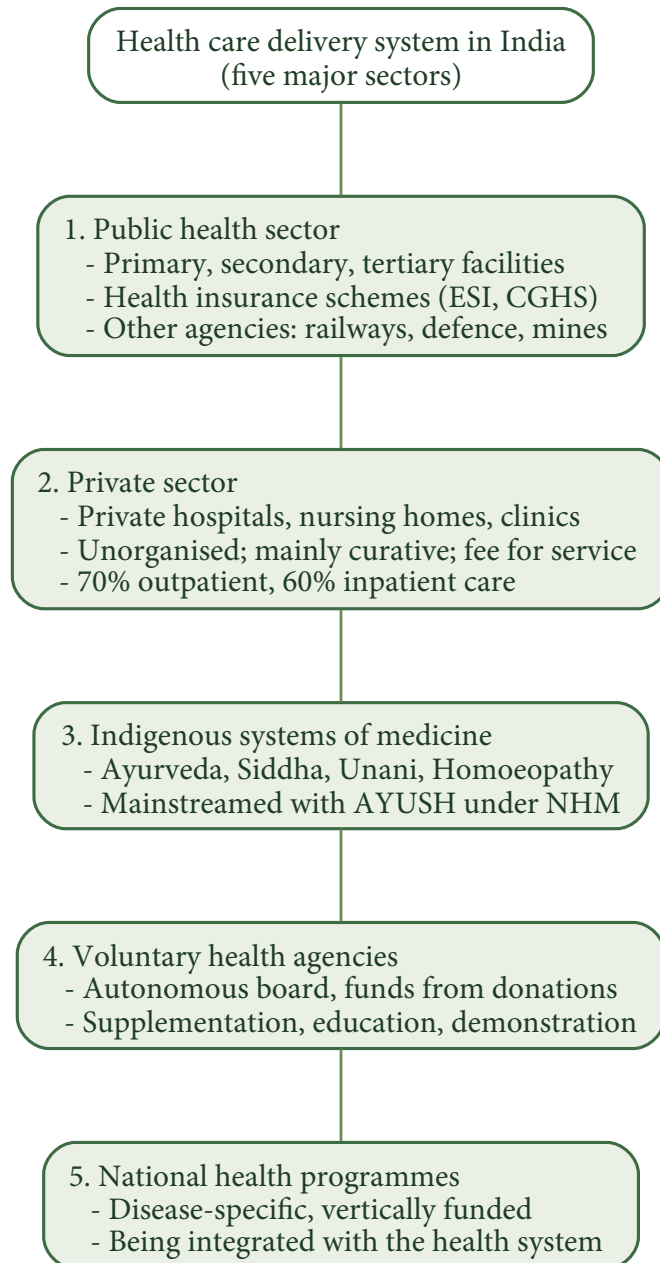


Fig 34.1 — Three-tier system of health care (Suryakantha Environment Best, p.778)



Two-way referral pathway for health care delivery to a community in India



The five sectors that deliver health care to the Indian community

High-yield exam pointers

- ▶ **Four principles (mnemonic ECIA)** — Equitable distribution, Community participation, Intersectoral coordination, Appropriate technology.
- ▶ **Eight elements of primary health care** — write the Alma-Ata list in order; the expanded package at the Ayushman Arogya Mandir is 12 services, not 8.
- ▶ **Population norms** — sub health centre 5000/3000; primary health centre 30,000/20,000; community health centre 1,20,000/80,000; ASHA 1:1000; community health centre serves as first referral unit for 4 primary health centres with 30 beds.
- ▶ **Current name first** — Ayushman Arogya Mandir (erstwhile Ayushman Bharat Health and Wellness Centre, launched 2018); Ayushman Bharat has two pillars — the Mandirs for

comprehensive primary care and Pradhan Mantri Jan Arogya Yojana for secondary/tertiary hospitalisation.

- **Dated figures** — Indian Public Health Standards revised 2022; about 1.1 lakh sub health centres converted to Health and Wellness Centres by March 2022; 5,481 community health centres functional (March 2021); National Health Policy 2017 target of 2.5% of GDP by 2025; catastrophic expenditure impoverishes 63 million per annum (National Sample Survey 75th round, 2018).
- **Alma-Ata definition** — reproduce verbatim; add "health by the people, placing people's health in people's hands".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Health and Wellness Centres were renamed **Ayushman Arogya Mandir** by the Government of India, so the current name should be written first with the former name in brackets once; Park 28e uses both names in different chapters, and older textbooks and question papers still use the earlier name.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, ICMR, National Health Authority, WHO Global Strategy for cervical cancer elimination (90–70–90), National Action Plan on AMR

Programme and policy updates (India)

- **Ayushman Arogya Mandir is the operative name, and the community health officer is the delivery point.** Current field studies describe the facilities as "Ayushman Arogya Mandir (erstwhile Health and Wellness Centers)" and place screening and prevention of non-communicable diseases including oral cancer, and basic oral health care, among the 12 comprehensive primary health care services the community health officer delivers; fifty trained community health officers in Jodhpur district, Rajasthan, screened 209 adults by oral visual examination over eight months (May–December 2023) and detected potentially malignant disorders in **14.84%** — leukoplakia 8.61%, oral submucous fibrosis 4.78%, erythroplakia 2.87% — each validated by an ENT specialist, with positives referred to the district hospital for biopsy [PMID 40320454]. Where the package was strengthened on both demand and supply sides in model centres in Punjab (2019–2021), difference-in-difference analysis against control blocks showed monthly gains of 265.7 people screened for non-communicable diseases, 21.3 newly diagnosed hypertensives and diabetics, 29.5 hypertensive and diabetic patients on treatment, 102.2 outpatient attendances and 4.9 wellness sessions [PMID 38934822].
- **National HPV vaccination programme (2026) — the newest national immunisation initiative.** India moved from sub-national pilots (school-based programmes in Punjab and Sikkim from 2016, reaching about 97% of target girls) to a nationwide voluntary, free, **single-dose** programme for adolescent girls aged 14 years, a target cohort of about 1.15–1.2 crore, delivered through government facilities and schools, with girls who turned 15 within three months of launch also eligible; the programme primarily used a quadrivalent vaccine (Gardasil 4), and India has also introduced its first indigenous quadrivalent vaccine, **Cervavac** [PMID 42518306]. Do **not** write that HPV vaccine has been folded into the Universal Immunization Programme — the 2026

AOGIN India policy statement still argues that integrating HPV vaccine *into* the Universal Immunization Programme is what is needed to reach the WHO target of 90% of girls vaccinated by age 15 [PMID 42169564].

- **Ayushman Bharat Digital Mission (launched 2021) — the "Scan and Share" service.** QR-code paperless outpatient registration linked to the 14-digit Ayushman Bharat Health Account: by October 2024 it was operating in 16,939 facilities across 35 of 36 states and union territories, had generated over 50 million tokens, and cut registration waiting time from about an hour to 2–5 minutes — although only 1.3% of tokens came from the private sector, which is the equity caveat to quote [PMID 39882498].

Current national data

- **Comprehensive Annual Modular Survey 2022–23 (National Sample Survey 79th round; over 302,000 households) is the current dated source on financial protection,** superseding the 75th round (2017–18) figure carried by the textbooks. Publicly funded health insurance, chiefly Pradhan Mantri Jan Arogya Yojana, was associated with greater outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128–1.210) and with less catastrophic expenditure at the 20% threshold (outpatient adjusted OR 0.869, 95% CI 0.810–0.932; inpatient 0.790, 95% CI 0.737–0.846). Outpatient care nonetheless remains the dominant source of household financial burden even among the insured — the argument for extending the benefit package to outpatient services anchored at the Ayushman Arogya Mandir [PMID 42167946].
- **Immunisation coverage gap, dated 2015–2025.** A meta-analysis of 45 Indian cross-sectional studies (over 428,000 participants) put incomplete immunisation among under-fives at a pooled 26% (95% CI 21–32), the strongest determinant being absence of an immunisation card (OR 9.46, 95% CI 3.28–27.28), followed by non-institutional delivery (OR 3.59) and maternal illiteracy (OR 3.25) [PMID 42320383]. A 15-year Delhi infectious-diseases hospital series corroborates the booster-dose gap — 2,131 culture-confirmed diphtheria cases, case-fatality 19.1%, and only 1.1% age-appropriately vaccinated [PMID 42555698].

Elimination, surveillance and disease status

- **Kala-azar — sustained-elimination (maintenance) phase under a renamed structure.** The centrally sponsored Kala-azar Control Programme (1990–91) evolved into the **National Kala-Azar Elimination Programme**, now coordinated by the **National Centre for Vector Borne Diseases Control** (the erstwhile National Vector Borne Disease Control Programme). The target, fixed by the 2005 India–Bangladesh–Nepal tripartite memorandum, is fewer than **one case per 10,000 population per year at block (sub-district) level** in India and Bangladesh and at district level in Nepal; the original National Health Policy 2002 deadline of 2010 was revised to 2015. All endemic blocks of Malda district, West Bengal, reached the target by 2016, yet total leishmaniasis notifications there fell from 289 (2015) to 28 (2021) and then rose to 61 in 2022 — the rise driven primarily by post-kala-azar dermal leishmaniasis while visceral cases kept falling (117 in 2015 to single figures in 2022). Post-kala-azar dermal leishmaniasis is the long-term **human reservoir** that threatens resurgence, so its detection and treatment is a core elimination activity [PMID 42591813]. India met the sub-district target across most endemic regions by late 2023 but must hold it for three consecutive years before elimination can be formally validated; multicentric

screening of 782 domestic animals across seven districts in five states meanwhile found *Leishmania donovani* infection in dogs, cattle, goats and cats, arguing for **One Health** surveillance of humans, vectors and reservoir hosts rather than a purely anthroponotic model [PMID 42206020].

- **Lymphatic filariasis — endgame, and the target year to quote is 2027.** The disease remains endemic in **345 districts**, and the programme is being asked to lean on public–private partnership for mass drug administration compliance, post-MDA surveillance, morbidity management and disability prevention, and vector control [PMID 41951149]. India carries about **55% of the global burden with roughly 487 million people at risk**; the national target is elimination by **2027** against the WHO global goal of **2030**, and an ICMR-led phase IIa placebo-controlled trial of **oxfendazole** (400 mg or 800 mg once daily for five days, primary endpoint absent adult-worm motility on ultrasound at 6 months) is testing the first true macrofilaricide, since present mass drug administration regimens clear microfilariae but not adult worms [PMID 42545617].

Guideline and legislative updates

- **The population norm alone is no longer accepted as the siting rule.** A spatial and cost analysis of Chamoli and Rudraprayag (Uttarakhand) using Indian Public Health Standards-aligned costs (capital Rs 1.85 crore per new primary health centre; recurring Rs 86 lakh per year, 2024 prices) found 23% of Chamoli's population beyond 15 km of any centre despite the 1 : 20,000 hilly-area norm. Strict compliance with the norm required 18 additional centres (Rs 275 crore; incremental cost-effectiveness ratio Rs 19,671 per additional person covered), whereas coverage-optimised placement of only 6 centres reached 96.0% coverage for Rs 158 crore (ratio Rs 8,373) — cite this when asked to criticise population norms as a planning instrument [PMID 42482081].
- **Antimicrobial resistance and equity legislation now sit inside the delivery framework, not beside it.** India has a National Action Plan on AMR plus six state action plans covering stewardship, laboratory networks, awareness and intersectoral coordination; the current recommendation is to embed AMR containment within universal health coverage and One Health surveillance rather than run it as a vertical programme [PMID 41443506]. Under the Transgender Persons (Protection of Rights) Act 2019 and the SMILE scheme (2022), at least one government hospital per state is to provide gender-affirming services, with a comprehensive health package delivered in conjunction with Pradhan Mantri Jan Arogya Yojana [PMID 42132281].

Global and WHO developments

- **Cervical cancer elimination and the single-dose shift.** The WHO Global Strategy (2020) sets the **90–70–90** targets — 90% of girls vaccinated, 70% of women screened, 90% of disease treated — and the 2022 WHO endorsement of a single-dose schedule is what made national rollout affordable in low- and middle-income settings. Ten of the 11 WHO South-East Asia Region countries have now introduced HPV vaccination nationally, with Bhutan, Sri Lanka and Thailand exceeding 90% coverage through school-based delivery; the four recurring barriers are policy and financing constraints, sociocultural resistance, weak health-system capacity, and vaccine hesitancy [PMID 42518306].

- **Primary health care remains the accepted route to universal health coverage, but the evidence is weakest exactly where the principles promise most.** A realist review of 42 studies drawn from the countries with the largest gains in the UHC Service Coverage Index between 2000 and 2021 (evidence concentrated in India and China) found that reforms improved contact coverage and financial protection through frontline workers, community organisations and multisectoral action, yet fell short in reaching marginalised groups, reducing financial barriers and improving service quality — attention went to contact coverage rather than effective coverage. The stated priorities are community engagement, regulation of the private sector, and multisectoral collaboration [PMID 42293679].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2, Ch 7, Ch 8, Ch 10, Ch 13); Sunder Lal — Textbook of Community Medicine (Ch 16, Ch 17); Suryakantha — Community Medicine with Recent Advances (figures, Ch 34); Oxford Textbook of Public Health 7e. Named documents relied on: Declaration of Alma-Ata 1978; Indian Public Health Standards (revised 2022); National Health Policy 2017; Ayushman Bharat Comprehensive Primary Health Care Operational Guidelines 2018; National Sample Survey 75th round (2018).

Recent-advances sources: 40320454 — Prevalence and determinants of oral potentially malignant disorders in western Rajasthan, *Sci Rep* 2025, doi:[10.1038/s41598-025-98096-8](https://doi.org/10.1038/s41598-025-98096-8); 38934822 — Strengthening comprehensive primary health care through Ayushman Bharat: role of demand and supply-side interventions, *Indian J Public Health* 2023, doi:[10.4103/ijph.ijph.885.22](https://doi.org/10.4103/ijph.ijph.885.22) (abstract-only); 42518306 — HPV vaccination in the WHO South-East Asia Region: a scoping review of uptake, implementation challenges and pathways to acceleration, *Hum Vaccin Immunother* 2026, doi:[10.1080/21645515.2026.2693359](https://doi.org/10.1080/21645515.2026.2693359); 42169564 — AOGIN India policy statement on the use of HPV vaccination for cervical cancer elimination, *Asian Pac J Cancer Prev* 2026, doi:[10.31557/APJCP.2026.27.5.1603](https://doi.org/10.31557/APJCP.2026.27.5.1603) (abstract-only); 39882498 — Scan and share to register outpatients, India, *Bull World Health Organ* 2025, doi:[10.2471/BLT.24.292828](https://doi.org/10.2471/BLT.24.292828); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from CAMS 2022–2023, *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: systematic review and meta-analysis (2015–2025), *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only); 42555698 — Persistent high diphtheria positivity in North India, 2007–2021, *Indian J Med Res* 2026, doi:[10.25259/IJMR.249.2026](https://doi.org/10.25259/IJMR.249.2026); 42591813 — Sustaining the elimination of visceral leishmaniasis: National Kala-Azar Elimination Programme in Malda district, West Bengal (2018–2022), *Cureus* 2026, doi:[10.7759/cureus.112553](https://doi.org/10.7759/cureus.112553); 42206020 — Insights into animal reservoir hosts in the sustainable phase of the National Kala-azar Elimination Programme in India, *One Health* 2026, doi:[10.1016/j.onehlt.2026.101444](https://doi.org/10.1016/j.onehlt.2026.101444); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Trop* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42545617 — A phase IIa, proof-of-concept, placebo-controlled randomised trial of oxfendazole as a macrofilaricidal drug in lymphatic filariasis, *Infect Dis Ther* 2026, doi:[10.1007/s40121-026-01419-z](https://doi.org/10.1007/s40121-026-01419-z) (abstract-only); 42482081 — Cost-effective primary health-centre allocation in Himalayan terrain: Chamoli and Rudraprayag districts, *Cost Eff Resour Alloc* 2026, doi:[10.1186/s12962-026-00778-1](https://doi.org/10.1186/s12962-026-00778-1) (abstract-only); 41443506 — Antimicrobial resistance in India: integrating the response into health systems for universal health coverage, *J Glob Antimicrob Resist* 2025, doi:[10.1016/j.jgar.2025.12.001](https://doi.org/10.1016/j.jgar.2025.12.001) (abstract-only); 42132281 — Breaking barriers: strategies for inclusive transgender healthcare in Indian hospitals, *Int J Health Plann Manage* 2026, doi:[10.1002/hpm.70087](https://doi.org/10.1002/hpm.70087) (abstract-only); 42293679 — Understanding pathways from primary health care to universal health coverage outcomes: a realist review, *Front Public Health* 2026, doi:[10.3389/fpubh.2026.1808891](https://doi.org/10.3389/fpubh.2026.1808891). Also relied on: WHO Global Strategy to accelerate the elimination of cervical cancer (2020, 90–70–90

targets); WHO single-dose HPV schedule recommendation (2022); National Action Plan on Antimicrobial Resistance (India) and the six state action plans.



Public health management cadre

Asked in: Paper IV 17-Apr-2026 — RGUHS MD Community Medicine

Definition — A public health management cadre is a **separate, professionally qualified service stream inside the state health department**, distinct from the clinical (medical services) and medical-education streams, whose members enter on a public health or allied qualification and are deployed exclusively to plan, deliver, regulate and evaluate **population-level** health services. India's National Health Policy 2017 directs that such a cadre be created in **all States**, with public health or related disciplines as the entry criterion and an appropriate career structure and recruitment policy (Jugal Kishore, National Health Programmes of India, Ch 2, p 745). Human resources for health are one of the core building blocks of a health system (Sunder Lal, Ch 17, p 766).



WHO health system building blocks — human resources as one of six (Oxford public health 7th ed, p.840)

Policy mandate for the cadre in India

Source	Recommendation	Citation
Bhore Committee, 1946	Integrate the medical and public health organisations at all levels under a single administrative officer, and set up a unified health authority in each district . Until then most States ran two departments — the Surgeon General/Inspector General of Civil Hospitals and the Director of Public Health	Park 28e, Ch 23, pp 1034–35
National Health Mission framework (component 9 — Public Health Management)	A specialised Public Health Cadre to infuse managerial expertise into health services; establishment and strengthening of state, district, city and block programme management units	Jugal Kishore, Ch 1, p 128

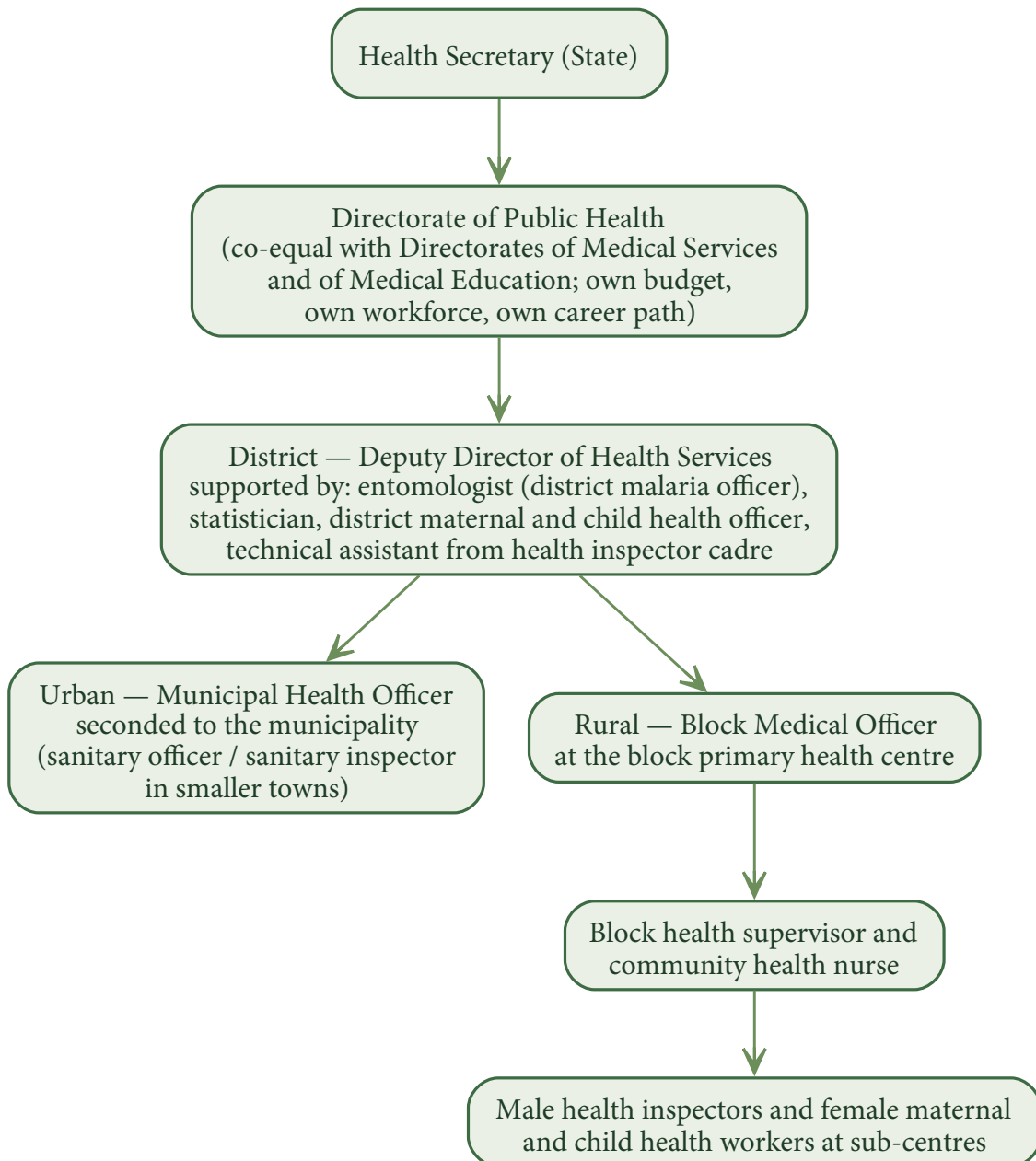
Source	Recommendation	Citation
High Level Expert Group on Universal Health Coverage	All-India and State-level Public Health Service Cadres, plus a specialised State-level Health Systems Management Cadre; a National Council for Human Resources in Health	Jugal Kishore, Ch 1, pp 88–89
National Health Policy 2017	Create a Public Health Management Cadre in all States, alongside a cadre of mid-level service providers and expanded allied technical skills	Jugal Kishore, Ch 2, p 745; Sunder Lal, Ch 16, pp 699–700
Indian Public Health Standards	A Public Health Manager is listed among the human resources for urban primary care facilities (urban PHC, polyclinic)	Park 28e, Ch 10, pp 731–32

Role of the cadre in health care

Mapped onto the World Health Organization's three core functions of a public health system — **assessment, policy formulation and assurance** (Oxford Textbook of Public Health 7e, Ch 100, p 3836) — and the Pan American Health Organization's eleven essential public health functions, EPHF 1–11 (PAHO 2002, in Oxford 7e, Ch 5, pp 266–67).

Core function	What the cadre does in practice	EPHF
Assessment and monitoring	Monitoring, evaluation and analysis of population health status; priority setting and resource allocation from that assessment	EPHF 1
Surveillance and risk control	Runs the surveillance units (integrated disease surveillance unit, plague surveillance unit); early detection of outbreaks, contact and source investigation through health inspectors	EPHF 2
Health promotion and social participation	Community awareness campaigns (vector breeding sources, diarrhoeal disease), health education, mobilising local bodies and panchayats	EPHF 3, 4
Policy, planning and institutional capacity	Prepares the district health action plan and the annual Epidemic Contingency Plan ; programme budgeting; runs state/district/city/block programme management units	EPHF 5
Regulation and enforcement	Uses public health law to detect and abate nuisances , assure food safety , monitor environmental hazards and civic hygiene (solid and liquid waste). Four enforcement tools: permits/licences/registrations, administrative orders, civil penalties, injunctions	EPHF 6
Equitable access and quality	Implements national health programmes, assures reach to the urban poor, tribal and left-wing-extremism-affected districts, and monitors service quality	EPHF 7, 9
Human resource development	Trains and supervises the peripheral workforce; in-service training in epidemiology and public health management	EPHF 8
Research	Operational and programme research to guide district action	EPHF 10
Emergency and disaster response	Mobilises people and material in floods, cyclones and mass-casualty events; water quality and chlorination monitoring, camp sanitation, fly and vector control, disposal of bodies with revenue department	EPHF 11

Core function	What the cadre does in practice	EPHF
Vital statistics	Collection of vital registration data (births and deaths) as a routine departmental function	EPHF 1
Intersectoral coordination	Reports mosquito nuisance, water contamination and pipeline breakage to the District Collector; sits on the quarterly Epidemic Coordination Committee the Collector chairs	EPHF 4, 5



Placement of the public health management cadre across levels of the state health system

Composition, entry and career structure

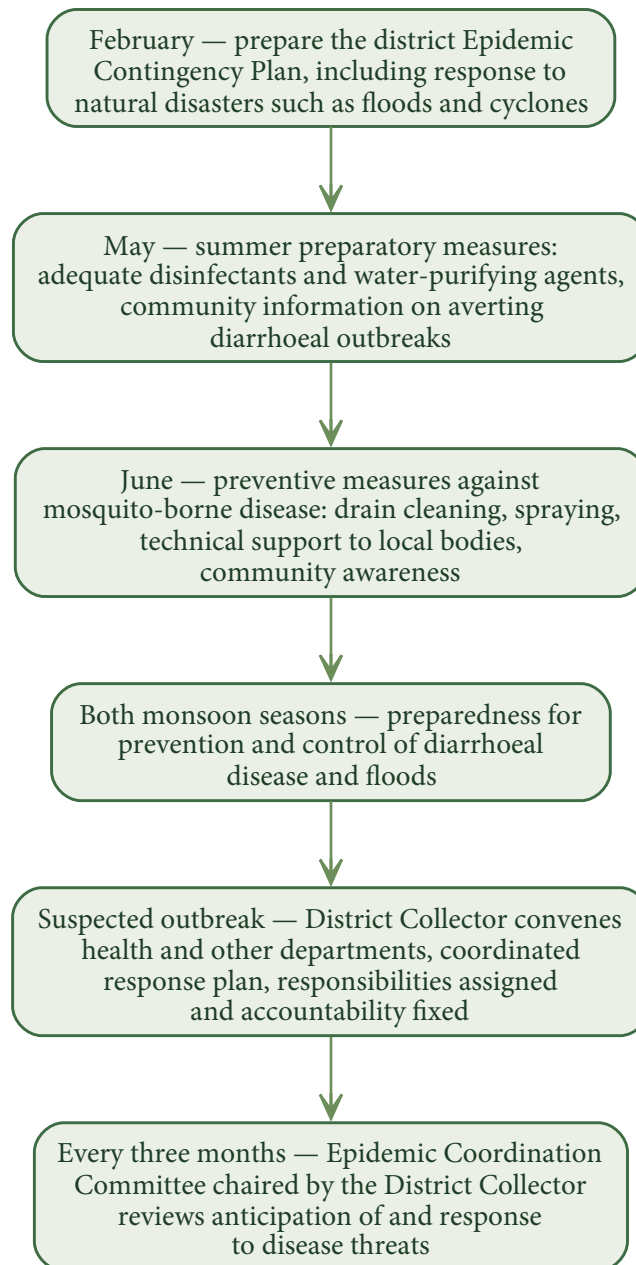
Element	Content
Who joins	Doctors with public health training form the major part ; also entrants from sociology, economics, anthropology, nursing, hospital management and communications who have since undergone public health management training (Jugal Kishore, Ch 2, pp 745–46)
Supporting technical staff	Entomologists, statisticians, maternal and child health officers — themselves experienced at district level before promotion to the directorate (Jugal Kishore, Ch 1, p 105)
Entry training	Three months of pre-placement training in public health ; a postgraduate diploma or degree in public health must be obtained within four years (Jugal Kishore, Ch 1, p 107)
Orientation	Towards an administrative and management role rather than a clinical role , and towards a population-wide rather than an individual-patient perspective (Jugal Kishore, Ch 1, p 107)
Career ladder	Municipal Health Officer (city or large town) → Deputy Director of Health Services (whole district) → Directorate of Public Health; with the option to teach and train future public health managers (Jugal Kishore, Ch 1, p 107)
Competency domains	Behaviours (principles, values, ethical practice); technical competence; context (strategic/policy, organisational, contractual, political); delivery (leadership, communication, programme management, prioritisation, management of resources) — United Kingdom skills and knowledge framework (Oxford 7e, Ch 100, pp 3837–38)
Capacity-building institutions	Public Health Foundation of India, set up as a public–private partnership and launched in 2006 , to establish and catalyse public health training institutes and an accreditation body; Indian Public Health Association (Jugal Kishore, Ch 3, p 991)

The Tamil Nadu model — the cadre in working practice

- ▶ **Three co-equal directorates** under the health secretary — Public Health, Medical Services, Medical Education — each with its **own dedicated budget, workforce and career path**, each able to rise to the same level within the department.
- ▶ The Directorate of Public Health's budget covers the entire public health workforce down to **sub-centres**, and all single-issue programmes **except tuberculosis and leprosy**.
- ▶ Dedicated technical strength — the State has **more than 100 entomologists** supporting vector-borne disease control.
- ▶ **Tamil Nadu Public Health Act** defines public health *nuisance* broadly (premises or animals kept in unhealthy conditions, stagnant water, ill-maintained drains, accumulation of refuse); health officers have **powers of entry and inspection** and may contain a nuisance by direct action. It applies **uniformly across the State**, unlike the health provisions of the Municipalities Act and Panchayati Raj Act.
- ▶ Municipal Health Officers are **seconded from the Directorate to the 37 largest municipalities**; the Deputy Director of Health Services holds residual responsibility for public health in all urban areas of the district.

- **Three staff nurse model** — a qualified staff nurse on each of three eight-hour shifts made primary health centres functional 24×7; Tamil Nadu was the **first State** to do this, and 30–35% of institutional deliveries are conducted at these PHCs.
- **2004 tsunami** — the World Health Organization recorded that the presence of a well-trained public health cadre enabled smooth mass mobilisation and deployment, resulting in the **complete avoidance of any kind of epidemic** (WHO 2006).

(All of section above: Jugal Kishore, *National Health Programmes of India*, Ch 1, pp 105–108.)



Annual public health work cycle organised by the district public health officer

Constraints and the way forward

Constraint	Consequence	Corrective measure
Post-Bhore integration dissolved separate public health directorates in most States	Most of the public health primary care system is inefficiently managed by clinical professionals ; wherever a public health cadre exists the results are much better, e.g. Tamil Nadu (Jugal Kishore, Ch 1, p 47)	Re-create a Directorate of Public Health with its own cadre and budget, as National Health Policy 2017 directs
Lack of incentives and career structure for public health professionals; public health training seen as an extension of clinical work	Poor recruitment and retention; low remuneration forces private practice alongside public health duty, so full effort is not given (Oxford 7e, Ch 3, pp 223–24; Ch 102, p 3909)	Defined entry qualification, faster promotion avenues, real administrative authority
Severe staff shortage in public health facilities — the source records over 85% of specialist doctor, 75% of doctor, 80% of laboratory technician, 53% of nursing and 52% of auxiliary nurse midwife posts vacant across States (Jugal Kishore, Ch 2, pp 734–35)	Programmes are delivered by a depleted and unsupervised periphery	Rational deployment of the already-available unemployed and under-employed professionals; simplified recruitment
No comprehensive National Public Health Act ; only scattered legislation	Epidemic and disaster preparedness, surveillance, sanitation and waste disposal lack a single enforceable legal base (Jugal Kishore, Ch 2, p 835)	A comprehensive public health act on the Tamil Nadu pattern, with periodic updating
Weak administrative control over municipal public health staff	Urban public health remains fragmented	Restore health authorities' administrative control over municipal public health staff
Fragmented training and non-integrated local-level training	Staff know only their own tasks and lack a holistic view of the public health system (Oxford 7e, Ch 102, p 3909)	In-service and distance-mode training in epidemiology and public health management under the National Health Mission

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — Public health cadre in the state health department (organogram). Labels: *Health Secretary; Directorate of Public Health; Directorate of Medical Services; Directorate of Medical Education; Deputy Director of Health Services (district); entomologist / district malaria officer; statistician; district maternal and child health officer; Municipal Health Officer; Block Medical Officer; block health supervisor; community health nurse; health inspector; sub-centre.*

Diagram 2 — Career ladder of the public health manager. Labels: *Medical graduate entry; 3 months pre-placement public health training; postgraduate diploma or degree in public health within 4 years; Municipal Health Officer; Deputy Director of Health Services; Directorate of Public Health; teaching/training arm.*

High-yield exam pointers

- ▶ **National Health Policy 2017** — "create a **Public Health Management Cadre in all States**"; entry on public health or related discipline; doctors with public health training form the major part, plus sociology, economics, anthropology, nursing, hospital management, communications.
- ▶ **Bhore Committee 1946** — integration of medical and public health services at all levels under a single administrative officer; a **unified health authority in each district**. This is the historical reason the separate cadre disappeared.
- ▶ **PAHO essential public health functions (2002): EPHF 1–11** — write at least 5 of the 11 when asked for the cadre's role; and the **WHO three core functions**: assessment, policy formulation, assurance.
- ▶ **Tamil Nadu career ladder**: Municipal Health Officer → Deputy Director of Health Services → Directorate; **3 months** pre-placement training + PG public health qualification **within 4 years**; >100 entomologists; MHOs seconded to **37** largest municipalities.
- ▶ **Tamil Nadu Public Health Act** — broad definition of *nuisance*; powers of entry, inspection and direct action; four enforcement tools — permits/licences/registrations, administrative orders, civil penalties, injunctions.
- ▶ **2004 tsunami** — trained public health cadre → **complete avoidance of any epidemic** (WHO 2006): the single best example to quote for the cadre's worth.
- ▶ **Indian Public Health Standards** list a **Public Health Manager** post for urban PHC and polyclinic; NHM mandates state, district, city and block **programme management units**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the National Health Policy 2017 mandate is a policy direction, and States have implemented it unevenly — only a few (Tamil Nadu being the long-standing example, with its separate Directorate of Public Health and the Tamil Nadu Public Health Act) run a genuine separate public health cadre, so answers should present Tamil Nadu as the working model rather than as the national norm. The vacancy percentages quoted above are given in the cited source without an attached year.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW Public Health Management Cadre guidance 2022, National Health Policy 2017, WHO essential public health functions, WHO South-East Asia Region UHC and health-workforce reports

Programme and policy updates (India)

- ▶ **The mandate is no longer just a policy line — there is now an operational design.** MoHFW released *Guidance for Implementation, Public Health Management Cadre* on **16 April 2022**, converting the National Health Policy 2017 direction into a structure. State health staff are reorganised into **four cadres — specialist, public health, health management and teaching**. The health management cadre runs national programmes and public health functions **from block level upwards** in two streams — roughly **70% postgraduates in public health and 30% MBAs** (human resources, procurement and supply chain, finance, operations, hospital and health management). Chief and Block Medical Officers are to be drawn from the public health cadre; serving general duty medical officers need **3–5 years of service plus a master's course in public health** for promotion within it; the existing **district health society** is named as the platform where the public health and clinical streams converge. [PMID 36438539]
- ▶ **Purpose and scale.** The cadre is framed as capacity for disease-burden estimation, planning of preventive health services and stronger public health surveillance, and as the delivery arm for the National Health Mission, PM-JAY and PMSSY. Because public health is a **State subject**, each State institutionalises its own customised version; the costing behind the policy estimated **33,236 posts** if the cadre ran from 2020 with roll-out to 2026. [PMID 37383994]
- ▶ **Uptake is four-speed, and this is the most examinable update.** The 13th Conference of the Central Council of Health and Family Welfare (**10–11 October 2019**) resolved that every State and UT would create the cadre by 2022, but reviews group States as **well established (Tamil Nadu, Maharashtra)**, some components in place (West Bengal, Rajasthan, Kerala), actively pursuing it (Odisha, Madhya Pradesh, Chhattisgarh) and still planning (Karnataka, Haryana, some north-eastern States). Even the names differ — Public Health Cadre in Odisha, Public Health Department in Maharashtra and West Bengal, Public Health Division in Gujarat. Only **Odisha** has restructured below the district, creating **Chief District Medical and Public Health Officer, District Public Health Officer and Block Public Health Officer** posts, with a **Special Secretary** heading public health at State level. [PMID 36438539; PMID 37884946]

Current national data

- **Situational analysis of four States** (Gujarat, West Bengal, Odisha, Maharashtra; 78 respondents interviewed April 2021 – January 2022): each has a separate public health hierarchy and Directorate, but **penetration below district level is very limited**, and **80–85% of the State health budget goes on salaries, pensions and maintaining human resources**, leaving 10–15% for everything else. Every State reported a shortage of formally trained professionals; about 50 informants blamed the **absence of permanent posts and of a career progression pathway**, pay disparity for contractual staff and lack of incentives. The High Level Expert Group put the block-to-State managerial requirement at **over 1.96 lakh posts**, with 20 regional centres for faculty development. [PMID 37884946]
- **The training pipeline is the binding constraint**. 84 universities across 20 States and 3 union territories run 116 public health programmes; MPH accounts for 73 (63%), of which 74% sit in private institutes, while 8 States and 5 UTs offer none at all. Master's-level public health programmes roughly **doubled between 2016 and 2021**, reaching **92 programmes across 76 institutions (21 medical, 55 non-medical)**, 47 of them (51%) started in that window. Admission criteria nevertheless stay narrow — non-medical institutions open their intake to a far wider range of disciplines than medical ones do — so the multidisciplinary intake the cadre policy assumes is not yet being produced. [PMID 36504927; PMID 35109861]

Elimination, surveillance and disease status

- **Why a standing cadre matters for programme continuity**. A Workload Indicators and Staffing Needs (WISN) study of the **National TB Elimination Programme**, run to WHO methodology across 18 districts in 7 States, found 73% of 372 respondents carrying COVID-19 duties on top of TB work, averaging **4.4 hours a day**; ten key screening and diagnosis cadres would have needed about **1.7 times a 40-hour week (mean 66.8 hours)** to cover both. ASHAs carried the highest workload ratio of all 28 cadres (WISN 2.15), peripheral health institutions ran a **36% staff shortage**, and CBNAAT/TrueNat platforms were switched to COVID-19 testing. The authors note the 2022 guidance gives the public health cadre an explicit mandate for **outbreak management in health facilities, disease control offices and educational settings**, and that **Tamil Nadu, Maharashtra, Chhattisgarh, West Bengal and Odisha had already trained and stationed such cadres**. [PMID 39113002]

Guideline and legislative updates

- **The essential public health functions list has moved on**. WHO's consolidated framework now runs to **12 essential public health functions**, and national training schemes are being mapped against it. A 2026 mapping of the ten ASEAN countries, with the United Kingdom as benchmark, found most ASEAN countries keep **separate training pathways for medical and non-medical professionals**, while the UK runs a **unified, competency-based scheme open to both**, with a master's and a portfolio route to specialist recognition; workforce plans existed in every country except Brunei and Vietnam. India is not part of that comparison, and its four-cadre design likewise keeps medical and non-medical entry streams separate — so the UK scheme is the contrast, not the parallel. For exams, cite PAHO's 2002 EPHF 1–11 as the historical list and the WHO 12 EPHFs as the current one. [PMID 41719597]

- **Tamil Nadu is now a named governance model in the literature, not merely a State example.** A multi-State qualitative study identified three contexts in which multisectoral working succeeded — Kerala's social-democratic context, Chhattisgarh's social-governance context, and **Tamil Nadu's public health governance context**, where a dedicated public health management cadre put administrative leadership in the hands of people trained in public health and management, so officials could engage other departments, collect data, diagnose problems, identify convergence areas and plan responsive action; the same cadre drives disease reporting and surveillance at primary care level on alerts from Panchayati Raj institutions. The recommended next steps are State-level convergence structures acting as **central management units** to supervise and take bottom-up feedback from local implementers, rationalised district committees with assigned deliverables, and **block convergence models** piloted through the Aspirational Blocks Programme with local self-governments. [PMID 38755604]

Global and WHO developments

- **Regional benchmarks worth quoting with their years.** In the WHO South-East Asia Region the UHC service coverage index rose from 47 in 2010 to 62 in 2021, against a 2030 minimum of 80, and the density of doctors, nurses and midwives (SDG indicator 3.c.1) reached **28.05 per 10,000 population, over 30% above 2014**. Nine of the eleven Member States now clear the 2006 threshold of 22.8 per 10,000 and three clear the 2016 Global Strategy threshold of 44.5. India contributed **294 new medical colleges since 2014**, roughly doubling MBBS capacity to over 100,000 seats. [PMID 38028168]
- **Global health governance is being pulled two ways at once.** Member States adopted a new **Pandemic Agreement expanding WHO's authority** at almost the same moment the United States — its largest funder and most powerful backer — moved to withdraw. The analysis reads the withdrawal as the end of an era of neoliberal internationalism rather than one administration's quirk, and argues that WHO's greatest risk is complacency and inaction by the remaining Member States. [PMID 41059528]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10 — Health Care of the Community; Ch 23 — Health Planning and Management); Jugal Kishore — National Health Programmes of India (Ch 1 — Health Care Delivery System in India, incl. the Tamil Nadu model; Ch 2 — National Policies Related to Health, incl. National Health Policy 2017; Ch 3 — professional bodies); Sunder Lal — Textbook of Community Medicine (Ch 16 — Health Planning and Management; Ch 17 — Healthcare of the Community); Oxford Textbook of Public Health 7e (Ch 1 — scope and concerns of public health; Ch 5 — essential public health functions, PAHO 2002; Ch 100 — the public health workforce; Ch 102 — training of public health professionals); National Health Policy 2017; National Health Mission framework; High Level Expert Group report on Universal Health Coverage; Indian Public Health Standards.

Recent-advances sources: 36438539 — Need for Proactive Role of IAPSM and other Public Health Professional Bodies in Roll out of Public Health Management Cadre in India, *Indian J Community Med* 2022, doi:[10.4103/ijcm.ijcm.712.22](https://doi.org/10.4103/ijcm.ijcm.712.22); 37383994 — India's public health management cadre policy, *Lancet Reg Health Southeast Asia* 2022, doi:[10.1016/j.lansea.2022.100053](https://doi.org/10.1016/j.lansea.2022.100053); 37884946 — Assessment of needs and gaps in public health cadre in India: a situational analysis, *BMC Health Serv Res* 2023, doi:[10.1186/s12913-023-10132-3](https://doi.org/10.1186/s12913-023-10132-3); 36504927 — The current state of public health education in India: a scoping review, *Front Public Health* 2022, doi:[10.3389/fpubh.2022.970617](https://doi.org/10.3389/fpubh.2022.970617); 35109861 — Transdisciplinarity of India's master's level public health programmes, *Hum Resour Health* 2022, doi:[10.1186/s12960-022-00713-4](https://doi.org/10.1186/s12960-022-00713-4); 39113002 — Assessing the impact of COVID-19 management on the workload of human resources working in India's National Tuberculosis Elimination Program, *BMC Health Serv Res* 2024, doi:[10.1186/s12913-024-11131-8](https://doi.org/10.1186/s12913-024-11131-8); 38755604 — Nurturing transformative local structures of multisectoral collaboration for primary health care, *BMC Health Serv Res* 2024, doi:[10.1186/s12913-024-11002-2](https://doi.org/10.1186/s12913-024-11002-2); 41719597 — Assessing public health training in Southeast Asia and the United Kingdom: a comparison of competency development for delivering the Essential Public Health Functions, *Public Health* 2026, doi:[10.1016/j.puhe.2026.106198](https://doi.org/10.1016/j.puhe.2026.106198) (abstract-only); 38028168 — Advancing Universal Health Coverage in the WHO South-East Asia Region with a focus on Human Resources for Health, *Lancet Reg Health Southeast Asia* 2023, doi:[10.1016/j.lansea.2023.100313](https://doi.org/10.1016/j.lansea.2023.100313); 41059528 — The World Health Organization and the Shifting US and Global Political Orders, *J Health Polit Policy Law* 2026, doi:[10.1215/03616878-12262672](https://doi.org/10.1215/03616878-12262672) (abstract-only). Plus MoHFW *Guidance for Implementation, Public Health Management Cadre* (16 April 2022) and National Health Policy 2017.



Role of ASHA in primary health care

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — The Accredited Social Health Activist (ASHA) is a trained female community health volunteer, resident of the village she serves, introduced under the National Rural Health Mission (NRHM, 2005) at the rate of one per 1000 rural population, who acts as the **interface between the community and the public health system** (Sunder Lal, Ch 17, p 738). She is chosen by and accountable to the village *panchayat*, is an honorary volunteer paid by performance-based incentive rather than salary, and is the operational instrument of the Alma-Ata principle of community participation in primary health care.

Profile, selection and status

Attribute	Norm
Population norm	One ASHA per 1000 population; relaxable in tribal, hilly and scattered habitations (Park 28e, Ch 10, p 726)
Sex and residence	Woman, preferably a resident of the same village; married, widowed or divorced (Park 28e, Ch 10, p 726)
Age	25–45 years (Sunder Lal, Ch 17, p 738)
Education	Formal education up to 8th class ; communication skill and leadership qualities (Park 28e, Ch 10, p 726)
Equity in selection	Adequate representation from disadvantaged population groups so that such groups are better served (Park 28e, Ch 10, p 726)
Selection authority	Selected by the village <i>panchayat</i> under the supervision and guidance of the block medical officer; accountable to the <i>panchayat</i> (Jugal Kishore, Ch 1, p 134)
Employment status	Honorary volunteer , not a government servant; performance-based compensation only (Sunder Lal, Ch 17, p 738)
Work schedule	Flexible — about 3–4 hours a day on 4–5 days a week (Sunder Lal, Ch 17, p 738)
Training	Induction training of 23 days spread over one year , then periodic re-training (Jugal Kishore, Ch 1, p 134); refreshed by <i>Mobile Academy</i> , a free audio course on her mobile phone (Sunder Lal, Ch 17, pp 738–739)
Numbers in position	Over 10.72 lakh ASHAs working (Sunder Lal, Ch 17, p 744)

Core roles of ASHA in primary health care

Role domain	What she actually does
1. Awareness and health education	Creates awareness on the determinants of health — nutrition, basic sanitation and hygienic practices, healthy living and working conditions — and on what services exist and why they must be used in time (Jugal Kishore, Ch 1, p 134)
2. Counselling	Counsels women on birth preparedness, importance of safe delivery, breastfeeding and complementary feeding, immunisation, contraception, prevention of reproductive tract and sexually transmitted infections, and care of the young child (Sunder Lal, Ch 17, p 738)

Role domain	What she actually does
3. Mobilisation and demand generation	Mobilises the community to access immunisation, antenatal care, postnatal care, family planning, supplementary nutrition and sanitation services at the <i>anganwadi</i> , sub-centre and primary health centre, chiefly on the monthly Village Health, Sanitation and Nutrition Day (Sunder Lal, Ch 17, p 738)
4. Escort and referral	Arranges escort and accompanies pregnant women and sick children to the nearest pre-identified facility — primary health centre, community health centre or first referral unit (Sunder Lal, Ch 17, p 738)
5. First-contact curative care	Provides primary medical care for minor ailments — diarrhoea, fever, first aid for minor injuries — and serves as a DOTS provider under the tuberculosis programme (Jugal Kishore, Ch 1, p 134)
6. Depot holder	Holds and distributes essential commodities in every habitation: oral rehydration salt, iron and folic acid tablets, disposable delivery kits, oral contraceptive pills and condoms; a drug kit is supplied to each ASHA (Sunder Lal, Ch 17, pp 738–739)
7. Vital events and surveillance	Records and reports all births and deaths in her village and notifies unusual health problems or disease outbreaks to the sub-centre / primary health centre (Sunder Lal, Ch 17, pp 738–739)
8. Sanitation	Promotes construction of household toilets under the total sanitation effort (Jugal Kishore, Ch 1, p 134)
9. Community-level planning	As member secretary of the Village Health, Sanitation and Nutrition Committee (VHSNC) she helps prepare and implement the village health action plan with the <i>anganwadi</i> worker, auxiliary nurse midwife (ANM) and self-help groups under <i>panchayat</i> leadership; the committee holds an untied fund of Rs 10,000 per annum (Sunder Lal, Ch 17, pp 738, 744)

Role in maternal, newborn and child health

Activity	Content and schedule
Antenatal	Mobilises every pregnant woman for the full antenatal package; the 8th–9th month visit is used to prepare the family for delivery (Sunder Lal, Ch 10, p 594)
Institutional delivery	Promotes and facilitates institutional delivery under Janani Suraksha Yojana ; her package is released after her postnatal visit and after the child has received BCG (Park 28e, Ch 7, p 544)
Home-Based Newborn Care (HBNC)	ASHA is the main provider . 6 visits after institutional delivery — days 3, 7, 14, 21, 28, 42; 7 visits after home delivery — days 1, 3, 7, 14, 21, 28, 42; first postnatal visit within 6 hours of birth (Sunder Lal, Ch 10, p 594)
HBNC incentive	Rs 250, paid on the 45th day after validation by the facilitator (ANM) (Sunder Lal, Ch 10, p 594)
Home-Based Care for Young Child (HBYC)	5 additional visits in the first week of the 4th, 7th, 10th, 13th and 16th month of age (Sunder Lal, Ch 10, p 594)
Immunisation	Brings beneficiaries to the <i>anganwadi</i> centre on immunisation and health-check days; supports the ANM's outreach session (Park 28e, Ch 10, p 727)

Activity	Content and schedule
Family planning	ASHAs, with trained <i>dais</i> , are described as the lynchpins of the family planning delivery system ; a scheme delivers contraceptives at the doorstep of beneficiaries through ASHA (Park 28e, Ch 8, p 612)

Role in national health programmes and NCD screening

Programme	ASHA's defined task
National Tuberculosis Elimination Programme	Identifies chest symptomatics, refers for sputum examination and acts as DOTS provider (Sunder Lal, Ch 17, p 738)
National Vector Borne Disease Control Programme — malaria	Performs passive case detection at community level using bivalent antigen-detecting rapid diagnostic tests ; treatment is begun within 24 hours of detection (Sunder Lal, Ch 8, p 376). The proportion of ASHAs equipped with rapid tests is itself a programme process indicator (Sunder Lal, Ch 8, p 386)
Lymphatic filariasis	Participates in National Filaria Day mass drug administration (Sunder Lal, Ch 17, p 738)
National Leprosy Eradication Programme	Identifies skin patch with loss of sensation , brings the case to the health worker, and ensures treatment regularity (Sunder Lal, Ch 17, pp 738–739)
NCD screening (population-based)	Enumerates every person aged 30 years and above household by household and completes the Community Based Assessment Checklist (CBAC) — age, family history of non-communicable disease, waist circumference, physical inactivity, tobacco use, alcohol use. A score above 4 marks the person high-risk , to be prioritised for the weekly NCD day, where the ANM screens for hypertension and diabetes (Sunder Lal, Ch 8, p 503 and Ch 16, p 703)
Common cancers	ASHAs and ANMs, supervised by the lady health visitor and medical officer, are trained in oral visual examination and clinical breast examination ; visual inspection with 5% acetic acid for cervical cancer is done at the health and wellness centre or above (Sunder Lal, Ch 8, p 503; Ch 16, p 703)

Position within the primary health care team

At the sub-centre, now the **Ayushman Arogya Mandir / Health and Wellness Centre**, the primary health care team is headed by the **Community Health Officer** — a BSc/GNM nurse or Ayurveda practitioner certified by a six-month community health programme — with the multipurpose workers and the ASHAs as team members (Park 28e, Ch 7, p 570). ASHA is therefore the lowest and most peripheral rung of the delivery system, and the only one who lives inside the community she serves.

Functionary	How the role meshes with ASHA's
Auxiliary nurse midwife (ANM)	Holds weekly/fortnightly meetings with ASHA; informs her of outreach session date, time and place; guides her on the eligible-couple register, on iron-folic-acid and tetanus toxoid, on oral pill dose schedule and side-effects, and on danger signs of pregnancy and labour for timely referral (Park 28e, Ch 10, p 727)
Anganwadi worker (AWW)	Acts as depot holder for the drug kit and replaces ASHA's consumed drugs; updates the list of eligible couples and infants with ASHA's help; guides her in organising the monthly health day (Park 28e, Ch 10, p 727)
ANM and AWW jointly	Act as resource persons for ASHA's training (Park 28e, Ch 10, p 727)

Functionary	How the role meshes with ASHA's
<i>Panchayat</i> and VHSNC	Select her, hold her accountable, and own the village health action plan she helps write (Jugal Kishore, Ch 1, p 134)

Urban ASHA under the National Urban Health Mission

Feature	Norm
Coverage	One urban ASHA per 1000–2500 urban poor population , about 200–500 households, preferably co-located at the slum <i>anganwadi</i> centre (Park 28e, Ch 7, p 535)
Five defined activities	Home visits up to 2 hours a day; attending the monthly Health, Sanitation and Nutrition Day; accompanying pregnant women and children to the urban health centre; holding the ward-level meeting; record keeping (Sunder Lal, Ch 17, p 758)
Community platform	Supports the <i>Mahila Arogya Samiti</i> , a peer group of 20–100 households doing community monitoring and referral (Jugal Kishore, Ch 1, p 167)
Recognition	Provided a photo identity card , which is observed to raise self-esteem; performance-based incentive fixed by the state (Sunder Lal, Ch 17, p 758)

How ASHA operationalises the principles of primary health care

Alma-Ata principle	ASHA's expression of it
Community participation	Selected by the community, accountable to the <i>panchayat</i> , member secretary of the VHSNC — "health by the people" made concrete (Park 28e, Ch 2, p 49; Ch 10, p 720)
Equity	Placed in every village, with deliberate representation of disadvantaged groups, taking the first level of care to the household (Park 28e, Ch 10, p 726)
Appropriate technology	Drug kit, oral rehydration salt, rapid diagnostic test, CBAC checklist — simple, cheap, locally usable tools (Sunder Lal, Ch 8, p 376)
Intersectoral coordination	Works jointly with the <i>anganwadi</i> (nutrition), sanitation (household toilets) and <i>panchayati raj</i> institutions (Jugal Kishore, Ch 1, p 134)
Universal, first-level access	One per 1000 population, resident in the village, available before travel to any facility is needed (Park 28e, Ch 10, p 726)

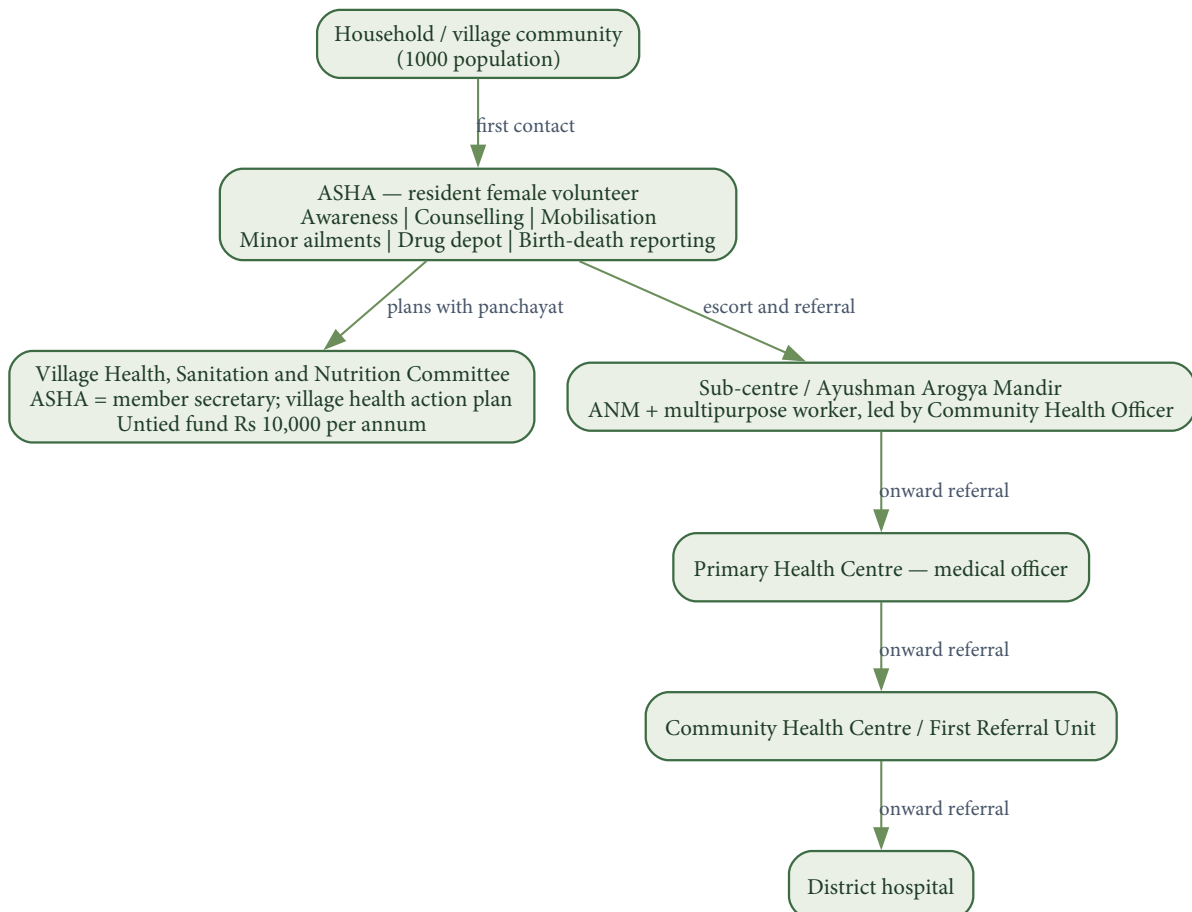
Constraints on role performance

Constraint	Consequence
Irregular and delayed incentive payments	Demoralisation; 20–30% dropout once volunteers learn payment is performance-linked and non-permanent (Jugal Kishore, Ch 1, p 157)
Drug kit replenishment failure	Depot-holder role collapses (Jugal Kishore, Ch 1, p 157)
Poor coordination between ASHA, ANM and <i>anganwadi</i> worker	Duplication and missed households (Jugal Kishore, Ch 1, p 157)

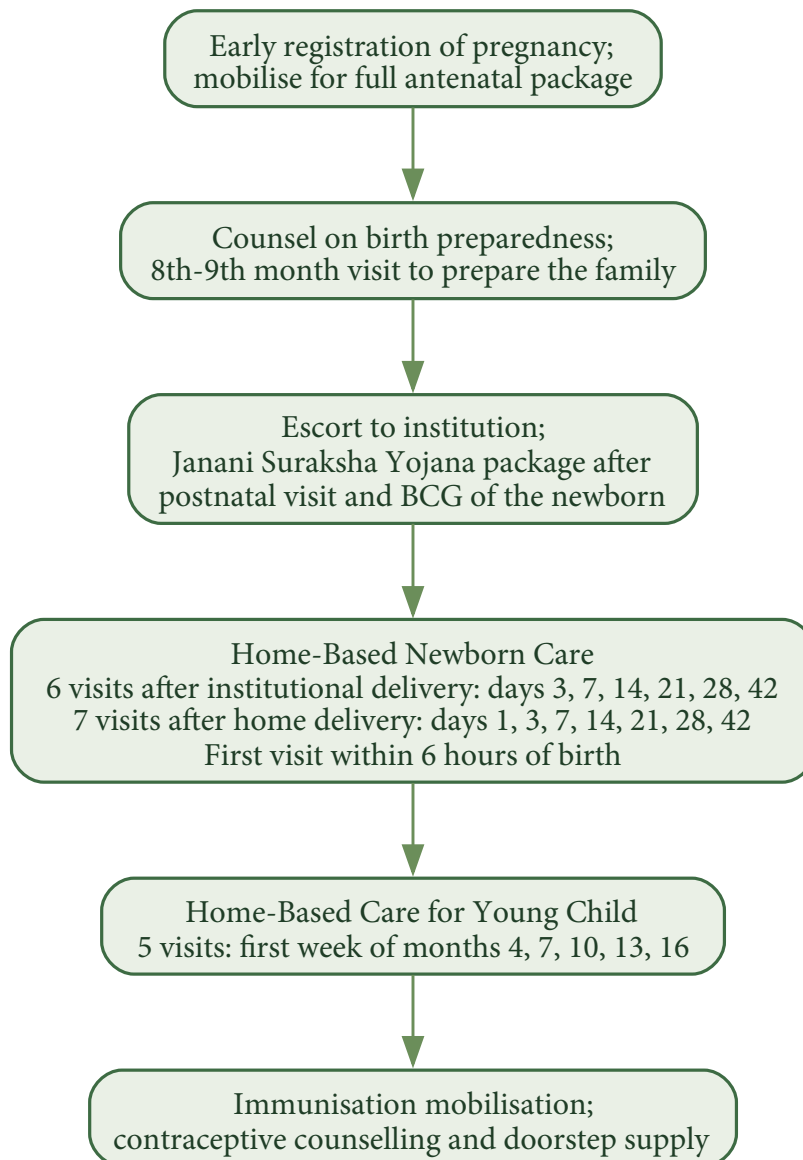
Constraint	Consequence
Weak institutional support structures and supervision	Skills not upgraded, problems unresolved (Jugal Kishore, Ch 1, p 157)
Incomplete household coverage in some states	Equity gain lost (Jugal Kishore, Ch 1, p 157)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

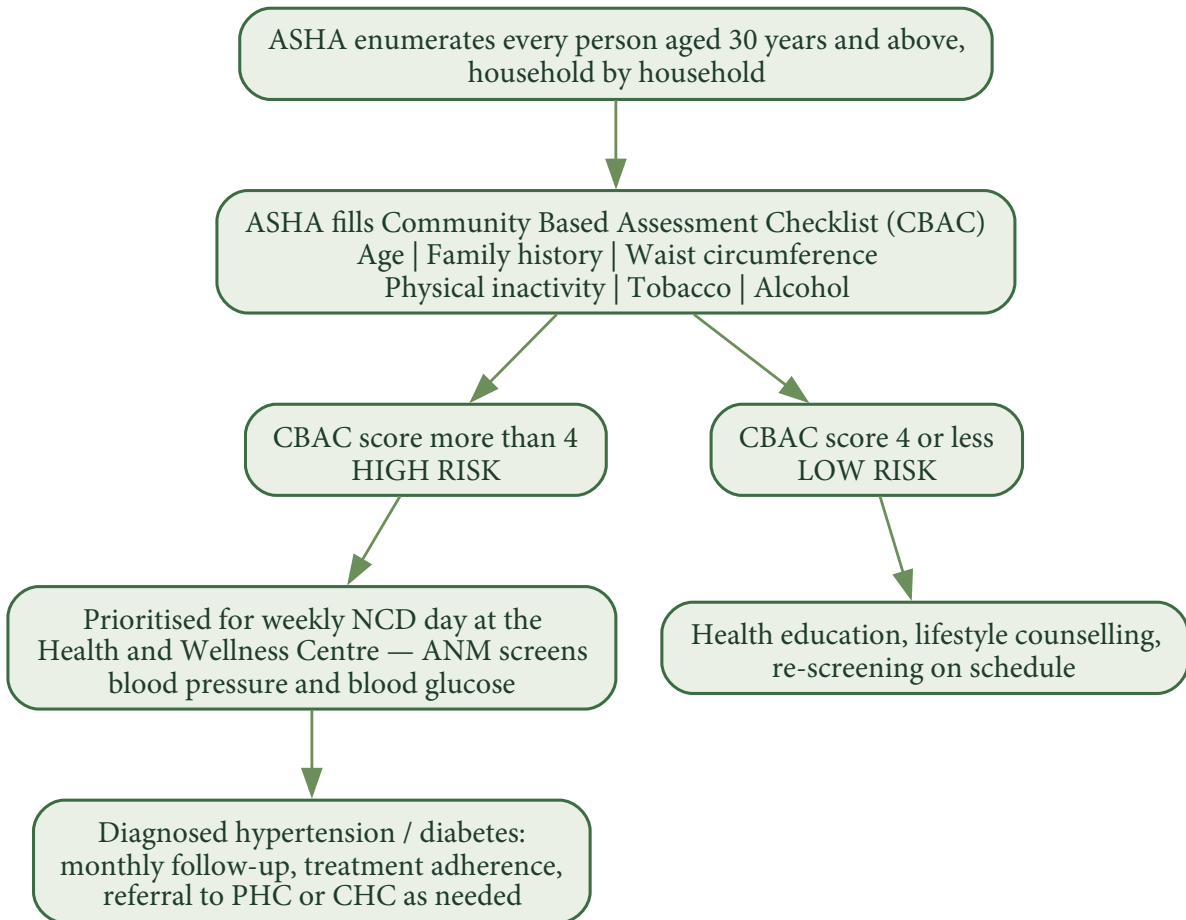
- **Labelled diagram — "ASHA's place in the rural health care pyramid."** Draw a pyramid with a village base and label, bottom upward: *Household / community* → *ASHA (1 per 1000)* and *Village Health, Sanitation and Nutrition Committee* → *Sub-centre, Ayushman Arogya Mandir — ANM, multipurpose worker, Community Health Officer* → *Primary health centre (medical officer)* → *Community health centre / First Referral Unit* → *District hospital*. Draw a two-headed arrow from ASHA to the community labelled "**interface: demand generation and accountability**" and a second arrow from ASHA to the sub-centre labelled "**referral and escort**".



ASHA as the first contact and referral link in the rural primary health care pathway



ASHA's role along the maternal, newborn and child health continuum



ASHA-initiated population-based screening for non-communicable disease at the Health and Wellness Centre

High-yield exam pointers

- ▶ **Norms to write first:** one ASHA per 1000 population (relaxed in tribal/hilly areas); woman aged 25–45 years, educated to 8th class, resident of the village, selected by the *panchayat*; honorary volunteer on performance-based incentive; induction training 23 days over one year.
- ▶ **Nine role headings (mnemonic "A-C-M-E-D-D-V-S-P"):** Awareness, Counselling, Mobilisation, Escort, Depot holder, Direct minor-ailment care and DOTS, Vital events and outbreak reporting, Sanitation, Planning through the VHSNC.
- ▶ **HBNC schedule is the commonest number asked:** 6 visits after institutional delivery (days 3, 7, 14, 21, 28, 42), 7 visits after home delivery (add day 1); incentive Rs 250 on the 45th day after ANM validation; HBYC 5 visits in months 4, 7, 10, 13, 16.
- ▶ **CBAC:** ASHA enumerates all persons ≥ 30 years; six variables; score >4 = high risk for the weekly NCD day.
- ▶ **Team integration:** *anganwadi* worker is the depot holder for her drug kit; ANM guides her on danger signs, oral pill schedule and outreach sessions; both act as her training resource persons.
- ▶ **Urban ASHA:** one per 1000–2500 urban poor (200–500 households), five defined activities, supports the *Mahila Arogya Samiti* of 20–100 households.
- ▶ **Structural anchor:** she is the community-level member of the sub-centre team at the **Ayushman Arogya Mandir**, headed by the Community Health Officer; 1,72,148 such centres were operational as on 31 March 2024 (Park 28e, Ch 7, p 570).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the sub-centre and primary health centre Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir**, and the figure of 10.72 lakh ASHAs quoted from Sunder Lal carries no year in the retrieved text, so quote it as "over 10 lakh" if the year is not certain; the Janani Suraksha Yojana ASHA package amount differs between rural and urban areas and between high- and low-performing states, so describe the payment trigger rather than a single rupee figure unless the current state rate is known.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, ICMR Antimicrobial Treatment Guidelines 2022

Programme and policy updates (India)

- ▶ **ASHA payment now runs as a formal five-step chain** — recording of work, claim submission, claim verification, claim processing, disbursement — mapped across Maharashtra and Assam through 53 interviews and 8 focus groups with 37 ASHAs, 46 supervisors and 34 managers. Disbursement is fully digital through the **Public Financial Management System (PFMS)**, which pays all ASHAs in an administrative area at the same time directly into her bank account; in Maharashtra these accounts were additionally linked to Aadhaar so that payment reaches the rightful beneficiary. This replaced cash and cheque and improved timeliness, transparency and satisfaction. The first three steps remain paper-based: an ASHA maintains more than ten registers, and the photocopies she must submit as proof of work cost her roughly 5% of her monthly incentive. Assam additionally runs the ASHA Payment and Performance Monitoring System (APPMS) for claim processing, whose reports are used for supervision, feedback and replacement of inactive ASHAs. [PMID 39953777]
- ▶ **Task-shifting to ASHA with a tablet-based clinical decision support system (IMPACT Diabetes)** — cluster randomised trial in 16 villages across 8 primary health centres of Guntur (Andhra Pradesh) and Rohtak (Haryana). ASHAs screened 1,785 adults aged 30 years and above and followed 418 with confirmed diabetes for nine months. More intervention participants achieved a $\geq 0.5\%$ fall in HbA1c (21.8% vs 10.3%, $p < 0.05$), physician visits at the primary health centre rose sharply (85.0% vs 29.8%) and glucose-lowering drug use rose (63.0% vs 43.1%). Mean HbA1c did not differ between arms, showing that ASHA-led support widens access but does not by itself substitute for intensification of treatment. [PMID 41120559]
- ▶ **HPV self-sampling has been brought inside the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)** — note the programme's current name — in a Sikkim pilot in which ASHAs delivered self-sampling kits to the doorstep, counselled women at Village Health and Nutrition Days and carried specimens to the district laboratory; more than 99% of women reported no difficulty or embarrassment with self-collection. The national policy in force since 2016 still specifies visual inspection with acetic acid every five years for women aged 30–65 years, so this is a screening-technology shift rather than a change of the written norm. ASHAs identified taxi fare for specimen transport and an unchanged honorarium

as the limiting factors; Sikkim has separately been vaccinating eligible girls against HPV in routine immunisation since 2016. [PMID 40341769]

Current national data

- ▶ **Cadre size and norm, with the year attached:** about 9.83 lakh ASHAs are in position, making this the largest community health worker programme in the world, and the one-per-1000 norm is an average rather than a reality — the actual ratio ranges from 285 population per ASHA in Arunachal Pradesh to 1,241 in Bihar. Typical performance-linked earnings are equivalent to about **US\$36–50 per month**. Current state practice commonly sets an educational bar higher than the classical 8th-class norm, relaxable according to need. NFHS-5 (2019–21) remains the current completed national round used for district-level planning, and the maternal mortality ratio contrast between the two states studied was 33 per 100,000 live births in Maharashtra against 195 in Assam. [PMID 39953777]
- ▶ **Pooled knowledge of ASHAs across 37 studies published 2005–2022:** maternal health 62%, neonatal and child health 69%, communicable disease 62%, non-communicable disease 73%. Knowledge of non-communicable disease now exceeds knowledge of communicable disease, attributed to the widening of her role since Ayushman Bharat in 2018 into NCD mobilisation, screening and follow-up, and further into oral health and eye care. A certification course for ASHAs has been launched by the Ministry of Health and Family Welfare with the National Institute of Open Schooling and the National Health Systems Resource Centre. Education level, years of experience, supervision and frequency of training were the determinants of knowledge. [PMID 39799364]
- ▶ **Community-health-worker-led NCD interventions in India, mostly outside the routine ASHA/CBAC pathway — systematic review of 13 studies (10 randomised, 3 pre-post), of which only 9 were meta-analysed:** systolic blood pressure was pooled from 9 studies ($n = 7,114$) and fell by 5.5 mmHg (95% CI -9.0 to -1.9), with a larger fall of 7.5 mmHg in the subgroup already hypertensive; body mass index was pooled from only 3 studies (828 intervention against 1,276 control) and fell by 0.63 kg/m² (95% CI -1.23 to -0.03). Diastolic pressure, fasting glucose, HbA1c and lipid profile showed no significant change; heterogeneity was high (I^2 79% for systolic pressure) and GRADE certainty was low to very low throughout. Read the cadre before quoting this in an ASHA answer: the interventions were delivered largely by **non-ASHA community workers** — peer leaders, self-help-group members, medical social workers, multipurpose workers and project community health workers — with only one included study (post-stroke home care in Ludhiana, 2023) actually ASHA-delivered. It is therefore evidence for community-level task-shifting in NCD care in general, and does not evaluate CBAC, the weekly NCD day or NP-NCD. [PMID 42555695]
- ▶ **Home-based newborn care coverage measured in the field rather than on paper:** lot quality assurance sampling of 434 mother–infant pairs across 31 sub-centres of Barrackpore-II block, West Bengal found **appropriate HBNC coverage of only 67.6%**, with **48.4% of sub-centre lots falling below the acceptable threshold**; a visit counted as appropriate only if it was on schedule within one day and included exclusive breastfeeding promotion, cord and eye care, temperature and weight measurement and hand hygiene. Irregular payment, low perceived credibility and weak coordination were the ASHA-side constraints. [PMID 41661029]

Elimination, surveillance and disease status

- **Tuberculosis** — the National TB Elimination Programme is shifting from directly observed therapy to digital adherence technologies, with the ASHA as the human backstop. India carries the largest burden globally at an estimated **2.8 million cases a year**, against the Government of India and Sustainable Development Goal elimination target of **2030**. **99DOTS** — the low-cost medication sleeve in which the patient makes a toll-free call to confirm each dose — has enrolled over **93,000 patients**, and Ni-kshay drives SMS reminders and direct benefit transfer. The reported constraints are uncharged phones, illiteracy among older patients and duplicate Ni-kshay identities; the fixes proposed by field staff were Aadhaar linkage of Ni-kshay IDs and a **dedicated NTEP ASHA at every centre** to raise the frequency of home visits. [PMID 40536374]

Guideline and legislative updates

- **Antimicrobial resistance is entering ASHA training under the national action plan**. A two-group interventional study of 73 ASHAs in Mysuru and Chamarajanagar during World Antimicrobial Awareness Week (18–24 November 2024) built games directly from the **ICMR antimicrobial treatment guidelines 2022**; allocation was by organisational affiliation rather than randomisation, though baseline scores were comparable. Post-test knowledge reached **86.1%** after gamified training against **54.5%** after conventional lectures ($p < 0.001$; effect size $r \approx 0.75$), with the largest gaps in distinguishing viral from bacterial respiratory illness (81% vs 40%) and gastrointestinal illness (84% vs 44%) and in recognising lower urinary tract infection (83% vs 50%). The stated policy implication is to embed such modules in routine ASHA training under the National Health Mission and the national action plan on AMR — which would add antimicrobial stewardship to her existing DOTS-provider role. [PMID 41188840]

Global and WHO developments

- **The community health worker cadre received its first formal global recognition at the 72nd World Health Assembly in 2019** as an instrument for advancing universal health coverage, and current global guidance holds that such workers should be appropriately and fairly compensated financially rather than left as unpaid volunteers — the international benchmark against which India's honorary, incentive-paid ASHA model is now judged, and the argument underlying the recurring demand for a fixed salaried component [PMID 39953777]. On the disease side, WHO's 2018 call to eliminate cervical cancer sets the frame for ASHA-delivered screening: GLOBOCAN 2022 counted 0.66 million new cervical cancers and 0.34 million deaths worldwide, of which India contributed 0.127 million cases and 0.079 million deaths [PMID 40341769].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 7 — Health Programmes in India; Ch 8 — Demography and Family Planning; Ch 10 — Health Care of the Community); Sunder Lal — Textbook of Community Medicine (Ch 8 — Epidemiology of Communicable and Non-Communicable Diseases; Ch 10 — RMNCH+A; Ch 16 — Health Planning and Management; Ch 17 — Healthcare of the Community); Jugal Kishore — National Health Programmes of India (Ch 1 — National Rural Health Mission and National Urban Health Mission); Alma-Ata Declaration on Primary Health Care, 1978 (as reproduced in Park 28e, Ch 1 and Ch 10). Coverage note: the Park 28e passages on ASHA selection and role integration are scanned with optical-character-recognition damage, so the role list and schedules above have been taken from the clean Sunder Lal and Jugal Kishore text and only cross-checked against Park.

Recent-advances sources: 39953777 — Community health worker payment processes: a qualitative assessment of experiences in two Indian states, *Health Policy and Planning* 2025, doi:[10.1093/heapol/czaf010](https://doi.org/10.1093/heapol/czaf010); 41120559 — Preliminary effectiveness and feasibility of ASHA-led mobile health intervention for diabetes care in Indian primary health care settings, *Scientific Reports* 2025, doi:[10.1038/s41598-025-20728-w](https://doi.org/10.1038/s41598-025-20728-w); 40341769 — A qualitative study on ASHA workers' perspective on HPV self-sampling in Sikkim India, *Scientific Reports* 2025, doi:[10.1038/s41598-025-87041-4](https://doi.org/10.1038/s41598-025-87041-4); 39799364 — Knowledge of Accredited Social Health Activists in India: a systematic review and meta analysis of evidence drawn from primary studies published between 2005 and 2022, *BMC Health Services Research* 2025, doi:[10.1186/s12913-024-12004-w](https://doi.org/10.1186/s12913-024-12004-w); 42555695 — Community health worker-led interventions for cardio-metabolic diseases in India: evidence from a systematic review and meta-analysis, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_3058_2025](https://doi.org/10.25259/IJMR_3058_2025); 41661029 — Appropriate home-based newborn care in a rural community in West Bengal, India using mixed-methods lot quality assurance sampling, *Journal of Tropical Pediatrics* 2026, doi:[10.1093/tropej/fmaf052](https://doi.org/10.1093/tropej/fmaf052) (abstract-only); 40536374 — A qualitative study on the barriers to tuberculosis treatment adherence using digital adherence technologies (DATs), *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR_1644_2024](https://doi.org/10.25259/IJMR_1644_2024); 41188840 — Innovative approaches to AMR education: evaluating the impact of gamification on empowering ASHA workers' knowledge and engagement in Mysuru, India, *BMC Medical Education* 2025, doi:[10.1186/s12909-025-08092-9](https://doi.org/10.1186/s12909-025-08092-9). ICMR Antimicrobial Treatment Guidelines 2022.



Voluntary health agencies — Indian Red Cross Society

Asked in: Paper IV 20-Jul-2020 — RGUHS MD Community Medicine

Definition — "A **voluntary health agency** may be defined as an organisation that is administered by an **autonomous board** which holds meetings, **collects funds for its support chiefly from private sources** and expends money, whether with or without paid workers, in conducting a programme directed primarily to furthering the public health by providing **health services or health education**, or by **advancing research or legislation for health**, or by a combination of these activities" (Gunn and Platt 1945, quoted in Park 28e, Ch 10, p 746). Voluntary health agencies have been compared to "**motor trucks**" which can penetrate the by-ways, and the official (government) agencies to "**Railway Trunk Lines**" which must run on tracks established by law (Park 28e, Ch 10, p 746). More than **7,000 voluntary agencies** are estimated to be involved in health-related activities in India (Sunder Lal, Ch 17, p 765).

Why the voluntary sector matters — its six classical functions

Function (Park's classification)	What the agency actually does
1. Supplementing the work of government agencies	Government agencies cannot give complete service because they work under financial and statutory restrictions; voluntary agencies lend personnel and contribute funds for special equipment, supplies or services (Park 28e, Ch 10, p 746)
2. Pioneering	Explore new ways of doing things, including research; when the effort succeeds the government takes it over for larger numbers — the family planning programme in India is the classical example of pioneering by voluntary agencies (Park 28e, Ch 10, p 746)
3. Education	Carry out extensive information, education and communication (IEC) work, which government agencies alone cannot cope with (Park 28e, Ch 10, p 746; Suryakantha, Ch 41, pp 823-824)
4. Demonstration	Demonstrations and experimental projects — the bore-hole latrine demonstrated by the Rockefeller Foundation to control hookworm in India became part of the national environmental sanitation programme (Park 28e, Ch 10, p 746)
5. Guarding the work of government agencies	By setting a good example they guide and criticise the work of government agencies (Park 28e, Ch 10, p 746)
6. Advancing health legislation	Mobilise public opinion and advance legislation on health matters for the benefit of the whole community (Park 28e, Ch 10, p 746)

Enumeration — voluntary health agencies working in India

#	Agency	Founded / headquarters	Principal field of work
1	Indian Red Cross Society (IRCS)	1920, by an Act of the Indian Legislature; New Delhi	Health promotion, disease prevention, mitigation of suffering; disaster relief, blood banks, first aid (Park 28e, Ch 10, p 747; Ch 25, p 1061)

#	Agency	Founded / headquarters	Principal field of work
2	Hind Kusht Nivaran Sangh (HKNS)	1950, New Delhi; precursor the Indian Council of the British Empire Leprosy Relief Association (BELRA), renamed LEPRO in 1950	Leprosy — grants to leprosy homes and clinics, health education, training of workers and physiotherapists, research, All-India Leprosy Workers Conferences, quarterly journal <i>Leprosy in India</i> (Park 28e, Ch 10, p 747)
3	Indian Council for Child Welfare (ICCW)	1952; affiliated to the International Union for Child Welfare	Child welfare through State and district councils — day-care services, <i>balwadis</i> , creches, holiday homes, recreation (Park 28e, Ch 10, p 747; Ch 9, p 699)
4	Tuberculosis Association of India (TAI)	1939, New Delhi; branches in all States	TB Seal campaign to raise funds, training of doctors, health visitors and social workers; runs the New Delhi Tuberculosis Centre, Lady Linlithgow Sanatorium (Kasauli), King Edward VII Sanatorium (Dharampur) and the Tuberculosis Hospital, Mehrauli (Park 28e, Ch 10, pp 746-747)
5	Bharat Sevak Samaj (BSS)	1952; non-political, non-official; branches in all States and nearly all districts	Helping people achieve health by their own actions and efforts ; improvement of village sanitation is a chief activity (Park 28e, Ch 10, p 747)
6	Central Social Welfare Board (CSWB)	August 1953, Government of India; autonomous, under the Ministry of Education	Surveys the needs of voluntary welfare organisations, promotes new ones, gives them financial aid; began " Family and Child Welfare Services " in rural areas in 1968; Industrial Cooperatives scheme for urban lower-middle-class women (Park 28e, Ch 10, p 747)
7	Kasturba Memorial Fund	1944, in commemoration of Kasturba Gandhi	Improving the lot of village women through <i>gramsevikas</i> ; funds welfare projects countrywide (Park 28e, Ch 10, p 747)
8	Family Planning Association of India (FPAI)	1949, Mumbai; branches all over the country	Pioneered family planning in India — grant-in-aid family planning clinics, medical termination of pregnancy and sterilisation services, mobile rural camps, training (Regional Training Centres at Hyderabad and Gwalior), <i>Parivar Pragati Pariyojana</i> (Park 28e, Ch 10, p 747; Suryakantha, Ch 41, p 824)
9	All India Women's Conference (AIWC)	1926; the only women's voluntary welfare organisation in the country	Maternal and child health (MCH) clinics, medical centres, adult and female literacy centres, milk centres, family planning clinics (Park 28e, Ch 10, p 747)
10	All-India Blind Relief Society	1946	Coordinates institutions working for the blind; organises eye relief camps and other relief measures (Park 28e, Ch 10, p 747)

#	Agency	Founded / headquarters	Principal field of work
11	Voluntary Health Association of India (VHAI)	1970; registered non-profit society	Federation of 27 State Voluntary Health Associations linking more than 4,500 health care institutions; policy research, advocacy, media and parliamentary intervention, training of trainers, health publications (Suryakantha, Ch 41, p 824)
12	Professional bodies	—	Indian Medical Association, All India Licentiate Association, All India Dental Association, Trained Nurses Association of India — annual conferences, journals, scientific sessions, research, standards of professional education, relief camps in calamities (Park 28e, Ch 10, p 747)
13	Other leprosy voluntary organisations (about 150)	—	Gandhi Memorial Leprosy Foundation (Sevagram, Wardha), German Leprosy Relief Association, Damien Foundation, Danish Save the Children Fund, JALMA (taken over by the Indian Council of Medical Research in 1975); federated as the National Leprosy Organization, 1965 (Park 28e, Ch 5, p 398)
14	Service clubs and other agencies	—	Rotary Clubs, Lions Clubs, Citizens Forum, Christian missionaries and private hospitals, HelpAge India, industrial establishments — eye camps, family welfare and care of the aged (Park 28e, Ch 8, p 616; Sunder Lal, Ch 17, p 765)
15	International voluntary agencies working in India	—	Rockefeller Foundation, Ford Foundation, Population Council, CARE (Cooperative for Assistance and Relief Everywhere), Oxfam, Save the Children Fund, International Planned Parenthood Federation, Aga Khan Foundation (Park 28e, Ch 10, p 747; Ch 25, p 1061; Suryakantha, Ch 43, pp 835-836)

Indian Red Cross Society — origin and objectives

The Red Cross is a **non-political, non-official international humanitarian organisation** devoted to the service of mankind in peace and war, founded by **Henry Dunant**, a Swiss businessman appalled by the neglect of the wounded at the **battle of Solferino (North Italy, 1859)**, and set out in his book *Un Souvenir de Solferino* (Park 28e, Ch 25, p 1060). The **First Geneva Convention (1864)** signed a treaty for the relief of the wounded and sick of armies in the field, bringing into being the **International Committee of the Red Cross (ICRC)**; in **1919** the **League of Red Cross Societies** was created with headquarters at Geneva to coordinate the national societies, which now number more than 90 (Park 28e, Ch 25, p 1060). Dunant received the **first Nobel Peace Prize in 1901**, and **World Red Cross Day is 8 May** (Suryakantha, Ch 43, p 836).

The Red Cross Society of India was established by an Act of the Indian Legislature in 1920 with three objectives (Park 28e, Ch 25, p 1061):

- ▶ Improvement of health
- ▶ Prevention of disease
- ▶ Mitigation of suffering

It works through a nationwide network of State and district branches (Park 28e, Ch 10, p 747).

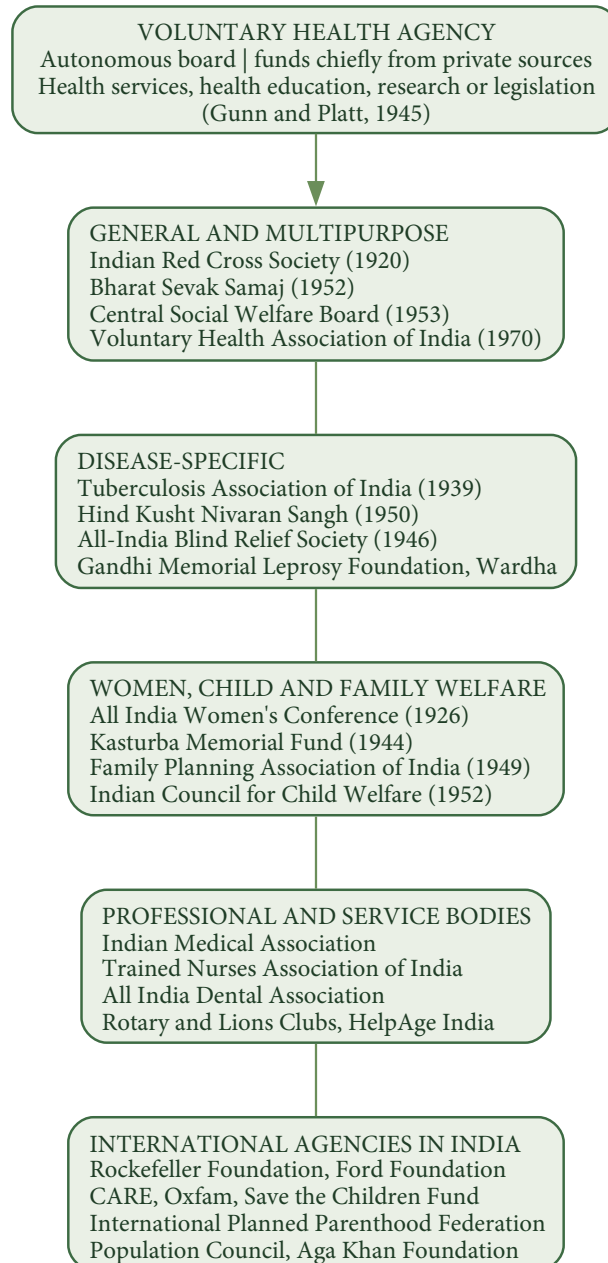
Functions of the Indian Red Cross Society

Function	Services rendered
(a) Relief work	When disaster strikes any part of the country — earthquakes, floods, drought, epidemics — the Society immediately mobilises all its resources and goes to the rescue of the affected people; relief is given as milk, medicines, food, vitamins, clothes and blankets (Park 28e, Ch 10, p 747; Suryakantha, Ch 43, p 836)
(b) Milk and medical supplies	A number of hospitals, dispensaries, maternity and child welfare centres, schools and orphanages receive assistance every year, consisting mainly of milk powder, medicines, vitamins and other supplies, and of equipment and drugs (Park 28e, Ch 10, p 747)
(c) Service to the armed forces and war veterans	Care of the sick and wounded of the armed forces is one of the primary obligations of the Red Cross. In peacetime it supplies military hospitals with newspapers, periodicals, musical instruments and other comfort goods, and it runs the well-equipped ' Red Cross Home ', Bengaluru, for permanently disabled ex-servicemen — the only one of its kind in India and the Far East (Park 28e, Ch 10, p 747; Ch 25, p 1061)
(d) Maternal and child welfare services	A large number of maternity and child welfare centres all over India are directly administered by, or affiliated to, the Red Cross. Its Bureau of Maternity and Child Welfare gives technical advice and financial aid for establishing model maternity and child welfare centres ; the Society has done pioneering work in the development of maternity and child health services (Park 28e, Ch 10, p 747; Ch 25, p 1061)
(e) Family planning	Several States run family planning clinics under the auspices of the Indian Red Cross , and the Society is listed among the well-known voluntary agencies of the family welfare programme (Park 28e, Ch 10, p 747; Ch 8, p 616)
(f) Blood bank and blood donation	Some State branches run blood banks and promote voluntary blood donation for those wounded in war or disaster (Park 28e, Ch 10, p 747; Suryakantha, Ch 43, p 836)
(g) First aid, home nursing and training	The St John Ambulance Association in India , which is part of the Red Cross, has trained several lakh men and women in first aid, home nursing and allied subjects (Park 28e, Ch 10, p 747)
(h) Health education	Conducts health education campaigns as part of its programmes to prevent human suffering (Park 28e, Ch 25, p 1061; Suryakantha, Ch 43, p 836)
(i) Junior Red Cross	One of the most active sections of the Society; gives lakhs of boys and girls the opportunity to take part in village uplift, first aid and anti-epidemic work , and builds up an international fraternity of youth promoting international friendliness, understanding and cooperation (Park 28e, Ch 25, p 1061; Suryakantha, Ch 43, p 836)

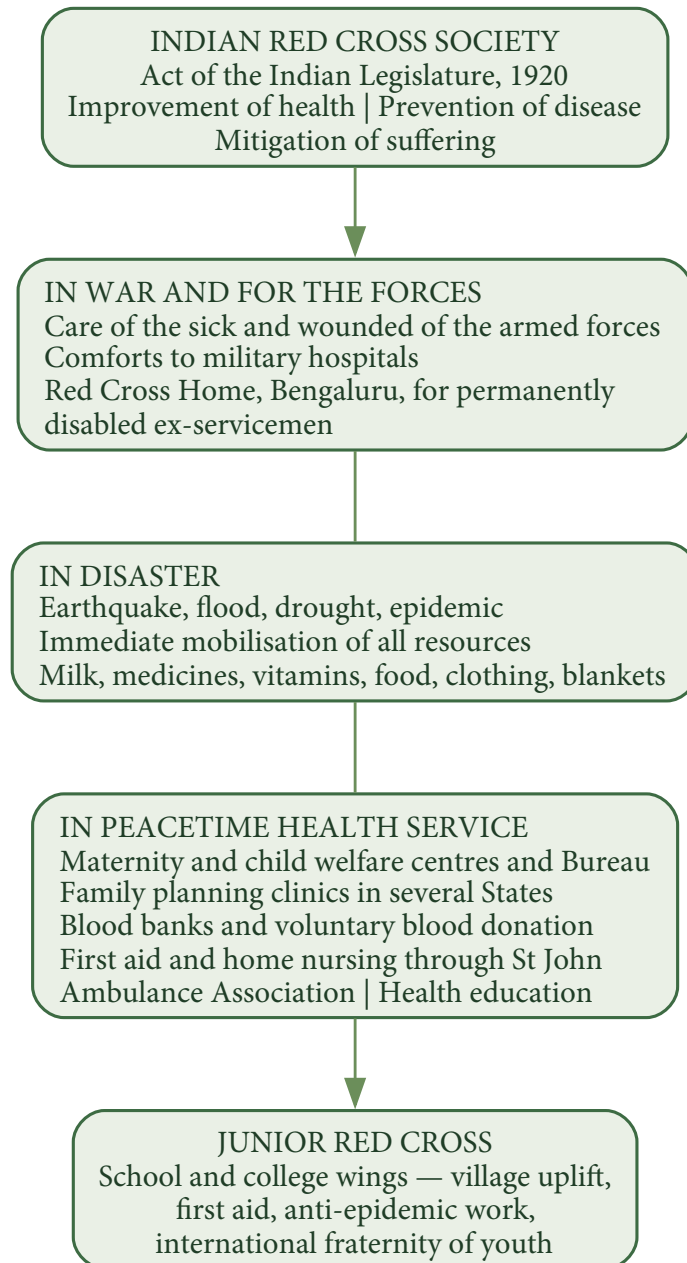
Taken together these fall into the Red Cross's standard programme areas: service to the armed forces, service to war veterans, disaster service, first aid and nursing, health education, and maternity and child welfare services (Park 28e, Ch 25, p 1061).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram — Indian Red Cross Society organogram and emblem.** Draw the red Greek cross on a white ground (the protective emblem proposed by Dunant) at the top; below it a vertical chain of four boxes labelled **National Headquarters, New Delhi → State / Union Territory branches → District branches → Sub-district units and Junior Red Cross (school and college wings)**; on the right of the chain draw three side-arrows labelled **Blood banks, St John Ambulance Association (first aid and home nursing) and Red Cross Home, Bengaluru (disabled ex-servicemen)**.



Voluntary health agencies working in India, grouped by field of work



Indian Red Cross Society (1920) — three objectives and the service streams that deliver them

High-yield exam pointers

- ▶ **Definition to write first:** Gunn and Platt (1945) — autonomous board, funds chiefly from private sources, programme of health services / health education / research or legislation for health (Park 28e, Ch 10, p 746). Follow it with the "motor trucks" versus "railway trunk lines" analogy.
- ▶ **Six functions of any voluntary health agency (mnemonic SPEDGA):** Supplementing government work, Pioneering, Education, Demonstration, Guarding government work, Advancing health legislation.
- ▶ **Dates that earn marks:** All India Women's Conference 1926; Tuberculosis Association of India 1939; Kasturba Memorial Fund 1944; All-India Blind Relief Society 1946; Family Planning Association of India 1949; Hind Kusht Nivaran Sangh 1950; Bharat Sevak Samaj and Indian

Council for Child Welfare 1952; Central Social Welfare Board 1953; Voluntary Health Association of India 1970.

- ▶ **Indian Red Cross triad:** improvement of health, prevention of disease, mitigation of suffering — Act of the Indian Legislature, 1920.
- ▶ **Red Cross milestones:** Henry Dunant, battle of Solferino 1859; *Un Souvenir de Solferino*; First Geneva Convention 1864 → ICRC; League of Red Cross Societies, Geneva, 1919; first Nobel Peace Prize 1901; World Red Cross Day 8 May.
- ▶ **Three names always quoted with the Indian Red Cross:** Red Cross Home, Bengaluru (disabled ex-servicemen), St John Ambulance Association (first aid and home nursing), Junior Red Cross.
- ▶ **Scale of the sector:** more than 7,000 voluntary agencies in health-related activities in India (Sunder Lal, Ch 17, p 765); the non-governmental health sector accounts for nearly 70% of health expenditure in India, which is why public–private partnership is a stated National Health Mission strategy (Sunder Lal, Ch 17, pp 739, 765-766).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The textbooks describe these bodies largely as they stood at their founding — the number of Indian Red Cross branches is not legible in our scanned copy of Park 28e (the 27e records "over 400 branches"), and the ministry attachments quoted, such as the Central Social Welfare Board under the Ministry of Education, are the textbook's wording rather than a current verification.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW, NTEP, National Blood Transfusion Council, WHO

Programme and policy updates (India)

- ▶ **Pradhan Mantri TB Mukh Bharat Abhiyaan (PMTBMBA), launched September 2022,** is the largest new voluntary-sector channel in Indian public health: a Ni-kshay Mitra — an individual, corporate, cooperative, institution, NGO, political party or elected representative — adopts consenting patients on TB treatment from a chosen health facility, block, urban ward, district or State and supplies nutritional, additional-nutrient, diagnostic and vocational support for 6 months up to 3 years, with community health workers doing the linkage and counselling; the linked Ni-kshay Poshan Yojana cash incentive has been doubled from ₹500 to ₹1,000 per month, and the TB Mukh Gram Panchayat initiative extends the same logic to local self-government. [PMID 41451044]; [PMID 40013645]
- ▶ **Ayushman Bharat Health and Wellness Centres were renamed "Ayushman Arogya Mandir" in November 2023,** with the tagline *Arogyam Paramam Dhanam* and an AYUSH-informed holistic-wellness mandate covering physical, mental and social wellness, yoga, lifestyle and nutrition counselling and elderly care; renaming was funded at ₹3,000 per facility (total outlay ₹48 crore) against the original target of converting 1.5 lakh sub-centres and PHCs, each serving 3,000–5,000 population. Use the new name in any primary-care answer — the textbook term "Health and Wellness Centre" is superseded. [PMID 42257165]

- **Community-Based Monitoring and Planning (CBMP)** — begun in 2007 under the National Rural Health Mission with voluntary agencies as nodal implementing partners across 324 primary health centres in 36 districts of nine States — is the textbook "guarding the work of government agencies" function institutionalised, through community scorecards, social audits and public hearings (*Jan Sunwais*); Maharashtra's programme grew from a 2013 pilot (13 districts, 615 villages) to about 1,000 villages in 17 districts by 2018. Under Ayushman Bharat (2018) the **Jan Arogya Samiti** at sub-centre and Ayushman Arogya Mandir level is now the participatory body, but its provider-centric composition de-emphasises civil-society facilitation and dilutes the mandate for scorecards, hearings and audits. [PMID 41536720]

Current national data

- **Blood transfusion services, e-RaktKosh national analytics to October 2025:** 4,484 licensed blood centres; annual collection rose from 12.6 million units in 2023 to 14.6 million units in 2024; voluntary non-remunerated donation reached 74.55% of collections in 2025 (the WHO benchmark being 100%); 72% of centres have blood component separation units and 234 are nucleic-acid-testing enabled. These are the current figures to quote alongside the Red Cross blood-bank and voluntary-blood-donation function. [PMID 42344650]
- **Ni-kshay portal national data, September 2022 to May 2023:** 82,205 Ni-kshay Mitra registered and 94% were linked to patients, but only 51% actually delivered support, and just 9.6% of 2.64 million notified patients received benefits. Support was associated with better outcomes — treatment success 95% with support versus 90% without (OR 2.2, 95% CI 2.10–2.31). Corporates, institutions and NGOs (n = 5,062) supported 101,603 patients and converted registration into actual support far better than individual donors; a Mandya district (Karnataka) evaluation of 255 patients found only 54.9% had ever been informed of the scheme. [PMID 41451044]; [PMID 40518206]

Elimination, surveillance and disease status

- **India was validated by WHO as having eliminated trachoma as a public health problem in 2024**, becoming the 20th validated country, against criteria of trachomatous inflammation-follicular prevalence <5% in children aged 1–9 years and trichiasis unknown to the health system <0.2% in adults aged ≥15 years, sustained without mass antibiotic treatment and backed by a health system able to manage incident trichiasis. Voluntary and corporate philanthropy was central — the **International Trachoma Initiative** antibiotic-donation programme underpinned the SAFE strategy — and post-validation impact and surveillance surveys are now the standing requirement. [PMID 39462916]
- **Lymphatic filariasis remains endemic in 345 districts** with the national elimination target now set at 2030; the ICMR-Vector Control Research Centre argues that public–private partnership, already proven in polio eradication and TB control, is still under-used in filariasis for advocacy, social mobilisation, treatment non-compliance during mass drug administration, post-MDA surveillance, patient identification, morbidity management and vector control. [PMID 41951149]

Guideline and legislative updates

- **The Foreign Contribution (Regulation) Amendment Act, 2020** has reshaped the funding base that the classical definition calls "chiefly private" — frequent licence renewals, tightened accounting and fresh restrictions on the use of foreign funds; Ministry of Home Affairs figures

cited in the literature record **16,754 organisations barred from receiving foreign contributions since 2014**. The compensating domestic stream is mandatory **corporate social responsibility spending under the Companies Act, 2013**, put at roughly **USD 225 million a year**, although it tends to cast voluntary agencies as implementing contractors rather than autonomous partners. [PMID 40777735]

- **Blood-service regulation has moved on from the textbook account: the National Blood Transfusion Council** is the apex policy body, professional (paid) donors were banned after the 1992 Common Cause public-interest litigation and the 1996 Supreme Court directive, the **National Blood Policy** followed in **2002**, oversight passed from NACO to the Directorate General of Health Services in **2021**, and **2025** brought revised **Voluntary Blood Donation Guidelines** plus a national External Quality Assessment Scheme; the 2026–2031 roadmap targets 100% voluntary donation, nationwide nucleic-acid testing and universal EQAS accreditation. [PMID 42344650]

Global and WHO developments

- **Civil society organisations remain the sustaining agent of adolescent family planning programmes** — a nine-country review spanning India, Bangladesh, Indonesia, Myanmar, Cameroon, Kenya, Madagascar, Malawi and Nigeria identified three pillars of sustainability: community participation and youth empowerment with grassroots mobilisation, multi-stakeholder government–CSO partnership with data-driven accountability, and cultural-religious adaptation that engages local gatekeepers. This is the contemporary statement of the pioneering role the Family Planning Association of India played in the 1950s. [PMID 41275306]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10 Health Care of the Community, pp 746-747; Ch 25 International Health, pp 1060-1061; Ch 8 Demography and Family Planning, p 616; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, p 699; Ch 5 Epidemiology of Communicable Diseases, p 398); Park's Textbook of Preventive and Social Medicine, 27e (Ch 10, pp 720-721, used to read the same passages where the 28e scan is garbled); Suryakantha — Community Medicine with Recent Advances (Ch 40/41 National Voluntary Health Agencies and Organizations, pp 823-825; Ch 42/43 Bilateral Agencies, p 836); Sunder Lal — Textbook of Community Medicine (Ch 17 Healthcare of the Community, pp 739, 765-766). The definition of a voluntary health agency is Gunn S.M. and Platt P.S. (1945), *Voluntary Health Programmes*, as quoted by Park.

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Public health legislations for women — PCPNDT Act

Asked in: Paper III 11-May-2022 — RGUHS MD Community Medicine

Definition — Public health legislation for women is the body of Indian statutory law that protects the survival, health, reproductive autonomy, dignity and working conditions of the girl child and the adult woman. Its constitutional footing lies in **Article 15** (special provisions for women and children), **Article 39** (equal pay for equal work), **Article 42** (just and humane conditions of work and maternity relief — the article translated into the Maternity Benefit Act, 1961) and **Article 51A** (renouncing practices derogatory to the dignity of women) (Jugal Kishore, Ch 2, p 781; Mahajan & Gupta 4e, Ch 13, pp 154-155). Legislation by itself cannot bring change: it must be **implemented to act as an engine of change**, and in a democratic system it can be effective only when people are literate and know and demand their rights, as also perform their duties — by itself a law lies buried in the statute book (Mahajan & Gupta 4e, Ch 13, pp 153-155).

Public health legislations for women in India — enumeration

A. Reproductive health and the girl child

Act (year)	Core public-health provision
Medical Termination of Pregnancy (MTP) Act, 1971 (amended 2002; amended Act and Rules 2021)	Termination up to 20 weeks on the opinion of one registered medical practitioner (RMP); >20 up to 24 weeks for prescribed categories of women on the opinion of not less than two RMPs . Grounds: risk to the woman's life or grave injury to her physical or mental health; substantial risk of serious foetal handicap; pregnancy from rape (the anguish is <i>presumed</i> to be grave mental injury); failure of contraception. Written consent of the guardian is needed if she is under 18 years or is a mentally ill person (Sunder Lal, Ch 9, pp 568-569)
Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994	Prohibits sex selection before and after conception and regulates prenatal diagnostic techniques — detailed below
Prohibition of Child Marriage Act, 2006 (PCMA) — repealed and replaced the <i>Child Marriage Restraint Act, 1929 (Sharda Act)</i>	Enforced with effect from 1 November 2007 ; prohibits child marriage rather than merely restraining it — makes child marriage an offence and provides punishment for those conducting, abetting, promoting or permitting a child marriage. Legal age at marriage remains 18 years for girls and 21 years for boys (raised from 15 and 18 by the 1978 amendment to the old Act). Prevalence of marriage below 18 years in India was 47%; child marriage is associated with high maternal and child mortality, malnutrition, anaemia and poor infant growth (Park 28e, Ch 9, p 698; Jugal Kishore, Ch 3, pp 907-908)
Protection of Children from Sexual Offences (POCSO) Act, 2012	Introduced by the Ministry of Women and Child Development ; protects the girl child (any person below 18 years) from penetrative and non-penetrative sexual assault, sexual harassment and pornography, and deems assault "aggravated" in specified circumstances (Park 28e, Ch 9, p 656)

B. Maternity entitlement and protection of the woman worker

Act (year)	Core public-health provision
Maternity Benefit Act, 1961; Maternity Benefit (Amendment) Act, 2017 (in force 1 April 2017)	Paid maternity leave of 26 weeks , of which up to 8 weeks may precede the expected date of confinement; 12 weeks (6 weeks pre- and 6 weeks post-delivery) after two children; crèche mandatory where more than 50 women workers are employed. Eligibility after completion of 80 days of service ; benefit allowed on submission of a medical certificate; the pregnant employee must not be given work that could adversely affect her health or that of the foetus (Park 28e, Ch 18, p 967; Jugal Kishore, Ch 3, p 923)
Employees' State Insurance (ESI) Act, 1948	Maternity benefit paid as a cash payment during maternity leave; free antenatal, natal and postnatal services to insured women. Under the ESI (General) Regulations, 1950 the claim falls due on the date of issue of the medical certificate for miscarriage, confinement, premature delivery or pregnancy-related illness (Park 28e, Ch 18, p 967; Jugal Kishore, Ch 3, p 923)
Factories Act, 1948	Section 66 — no woman to be employed between 7 p.m. and 6 a.m. ; Section 34 — prohibits carrying of excessive weights beyond the laid-down schedule; Section 79 — leave of 12 weeks; prohibits employment of women in hazardous occupations; separate toilets and washing facilities (Park 28e, Ch 18, p 967; Mahajan & Gupta 4e, Ch 9, p 110; Jugal Kishore, Ch 3, p 923)
Indian Mines Act, 1923 (and Mines Act, 1952)	Prohibits employment of women underground , which carries greater risk to health (Park 28e, Ch 18, p 967; Jugal Kishore, Ch 3, p 923)
Crèche and nursing-break statutes — Plantation Labour Act, 1951; Beedi and Cigar Workers (Conditions of Employment) Act, 1966; Contract Labour (Regulation and Abolition) Act, 1970	Crèche where 50 or more women are employed (Plantation Labour Act; Beedi and Cigar Workers Act) and where 20 or more women are employed (Contract Labour Act); working mothers given time off during duty hours for breastfeeding (Plantation Labour Act); under the Contract Labour Act a female employee (except midwives and nurses) is not to work beyond nine hours or between 7 p.m. and 6 a.m., relaxed by the Supreme Court in 2005 if security is ensured (Jugal Kishore, Ch 3, p 923)

C. Protection from violence, trafficking and exploitation

Act (year)	Core public-health provision
Immoral Traffic (Prevention) Act — enacted 1956 as the Suppression of Immoral Traffic in Women and Girls Act; amended and retitled 1986	Bans prostitution in its commercialised form; covers all persons, male or female, sexually exploited for commercial purposes (Park 28e, Ch 12, pp 839-840)
Dowry Prohibition Act, 1961; Dowry Prohibition (Amendment) Act, 1986	Minimum punishment for taking or abetting the taking of dowry raised to 5 years' imprisonment and a fine of Rs 15,000 ; every offence is non-bailable and non-compoundable ; dowry death (death of a woman by burns or bodily injury within 7 years of marriage, preceded by cruelty or harassment) carries not less than 7 years up to life imprisonment (Park 28e, Ch 12, p 840; Jugal Kishore, Ch 2, pp 888-889)

Act (year)	Core public-health provision
Indecent Representation of Women (Prohibition) Act, 1986	Prohibits indecent portrayal of women in advertisements, publications and other media (Sunder Lal, Ch 10, p 634)
Commission of Sati (Prevention) Act, 1987	Prevents the commission and glorification of <i>sati</i> (Sunder Lal, Ch 10, p 634)
Protection of Women from Domestic Violence Act, 2005	Covers physical, sexual, verbal, emotional and economic abuse, and unlawful dowry demands; extends to live-in relationships and to other women of the household including sisters and mothers. Gives a woman the right to reside in the matrimonial or shared household whether or not she has title to it. Sections 18-23 provide protection, residence, monetary, custody, compensation and interim orders ; Protection Officers are appointed in every district. A husband or live-in partner guilty of domestic violence can be imprisoned for one year and fined Rs 20,000 , and all offences under the Act are non-bailable (Jugal Kishore, Ch 2, pp 892-894)
Muslim Women (Protection of Rights on Marriage) Act, 2019 ("triple talaq" Act, July 2019)	Safeguards the marital and economic security of Muslim women (Sunder Lal, Ch 10, p 634)

D. Equal pay, workplace dignity and institutional machinery

Act (year)	Core public-health provision
Equal Remuneration Act, 1976	Payment of equal remuneration to male and female workers for the same or similar nature of work; also written into the Inter-State Migrant Workmen Act, 1979 (Jugal Kishore, Ch 2, pp 885-886; Ch 3, pp 923-924)
National Commission for Women Act, 1990	Constitutes the National Commission for Women, which (with the Department of Women and Child Development) reviews all laws, subordinate legislation, rules and administrative regulations to eliminate gender-discriminatory provisions , and reviews crimes against women regularly (Jugal Kishore, Ch 2, pp 780-781)
Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013	Mandates an Internal Complaints Committee in every workplace and a Local Complaints Committee in every district; gives statutory effect to the National Policy for Empowerment of Women 2001 commitment to prevent and punish sexual harassment at the place of work (policy commitment: Jugal Kishore, Ch 2, pp 780-781)

PC-PNDT Act, 1994 — genesis, scope and what it regulates

- **Why it was needed.** The sex ratio in India has been persistently adverse to women — **940 females per 1000 males (Census 2011)** — and the **child sex ratio (0-6 years) fell to an all-time low of 914 in Census 2011**, down 13 points from 927 in 2001 (rural 934 to 919; urban 906 to 902) (Park 28e, Ch 8, pp 582-583). Easy availability of sex-determination tests and abortion services acts as the catalyst for female foeticide.
- **Nomenclature and timeline.** Enacted as the **Prenatal Diagnostic Techniques (Regulation and Prevention of Misuse) Act, 1994**; brought into operation from **1 January 1996** (Park 28e, Ch 23, p 1039; Mahajan & Gupta 4e, Ch 13, p 157). The amended Act and Rules came into force from

February 2003, renaming it the **Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994** (Sunder Lal, Ch 9, p 567).

- **Object.** To prohibit sex selection before *and* after conception, and to regulate prenatal diagnostic techniques so that they are used only to detect genetic, metabolic, chromosomal or congenital abnormalities and sex-linked disorders — and not misused for sex determination leading to female foeticide (Sunder Lal, Ch 9, p 567).

Element regulated	Provision under the Act
Permitted indications for a prenatal diagnostic technique	(1) Chromosomal abnormalities (2) genetic metabolic disorders (3) haemoglobinopathies (4) sex-linked genetic diseases (5) congenital abnormalities (6) any other abnormality specified by the Central Supervisory Board (Sunder Lal, Ch 9, p 567)
Pre-conditions — at least one must be met	Pregnant woman above 35 years ; two or more spontaneous abortions or foetal losses; exposure to potentially teratogenic agents (drugs, radiation, infection, chemicals); family history in the woman or her spouse of mental retardation or physical deformity such as spasticity or other genetic disease; any other condition specified by the Board (Sunder Lal, Ch 9, pp 567-568)
Consent and counselling	All side effects and after-effects must be explained; written consent in the language she understands ; a copy of the consent must be given to the woman (Sunder Lal, Ch 9, p 568)
Absolute prohibitions	Communication of the sex of the foetus by words, signs or any other manner to the woman, her relatives or any other person; sex selection by any means before or after conception; advertisement relating to pre-conception or prenatal sex determination; seeking or encouraging the test by any person including the husband or a relative (Sunder Lal, Ch 9, p 568)
Who may set up a centre	A gynaecologist with experience of at least 20 procedures (chorionic villus aspiration or biopsy, amniocentesis, cordocentesis, foetoscopy, foetal skin or organ biopsy, foetal blood sampling) under supervision; or a sonologist, imaging specialist, radiologist or RMP with a postgraduate degree or diploma, or 6 months' training, or one year's experience in sonography or image scanning; or a medical geneticist (Sunder Lal, Ch 9, p 568)
Records	Every person conducting ultrasonography on a pregnant woman must keep complete records in the form prescribed under the Act; the Appropriate Authority has power to search and seize records (Sunder Lal, Ch 9, p 568; Mahajan & Gupta 4e, Ch 13, p 157)

PC-PNDT Act — machinery, offences and penalties

The Act specifies the administrative structures required for its implementation — the **Central Supervisory Board**, the **State Appropriate Authority and Advisory Committee**, the procedure for **registration** of establishments and the grounds for **suspension or cancellation** of registration, the offences and penalties, and the maintenance of records with power to search and seize (Mahajan & Gupta 4e, Ch 13, p 157). Enforcement is deliberately **decentralised to the district**: the **Civil Surgeon / Chief Medical Officer is the designated Appropriate Authority** who registers and recognises an ultrasonography or genetic centre, and no centre holding any machine capable of foetal sex determination may operate unless duly registered (Sunder Lal, Ch 9, p 568).

Offender	First conviction	Subsequent conviction
Medical geneticist, gynaecologist, RMP, owner or employee of a genetic counselling centre / laboratory / clinic who contravenes the Act, or who advertises sex determination or sex selection	Imprisonment up to 3 years + fine up to Rs 10,000	Imprisonment up to 5 years + fine up to Rs 50,000
Any person who seeks the aid of a centre or doctor for sex selection, or for a prenatal diagnostic technique on a pregnant woman (includes the husband and relatives)	Imprisonment up to 3 years + fine up to Rs 50,000	Imprisonment up to 5 years + fine up to Rs 1,00,000

Source for both rows: Sunder Lal, Ch 9, pp 568-569.

Public health impact so far. Despite the Act, the **sex ratio at birth for 2018-20 was 907** (rural 907, urban 910), with Kerala the highest among the bigger States (Park 28e, Ch 8, p 582); the NFHS-5 (2019-21) sex ratio at ages 0-6 years was **928** (NFHS-5 India Report, Ch 2). Implementation of women-related legislation in general remains **tardy** (Sunder Lal, Ch 10, p 634), which is why the Act is paired with the **Beti Bachao Beti Padhao** initiative of the Ministry of Women and Child Development in 100 gender-critical districts (Sunder Lal, Ch 10, p 634).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labelled diagram to sketch — gender violence through the life cycle (Jugal Kishore, Ch 2, p 891, after WHO 1996). Draw a horizontal life-course arrow with five labelled boxes hanging below it, labelled exactly: **Pre-birth** — female foeticide, battering during pregnancy, coerced pregnancy (rape); **Infancy** — female infanticide, malnutrition; **Girl below 5 years** — differential access to food and medical care, neglect of education, sexual and physical abuse, emotional violence; **Adolescent and adult woman** — early marriage, early and forced pregnancy, sexual and domestic violence, dowry harassment, honour killing, sexual harassment at workplace, acid burns, denial of entitlements; **Elderly** — desertion, neglect, medical neglect, widow abuse.

PUBLIC HEALTH LEGISLATIONS FOR WOMEN IN INDIA
Constitutional base: Articles 15, 39, 42 and 51A

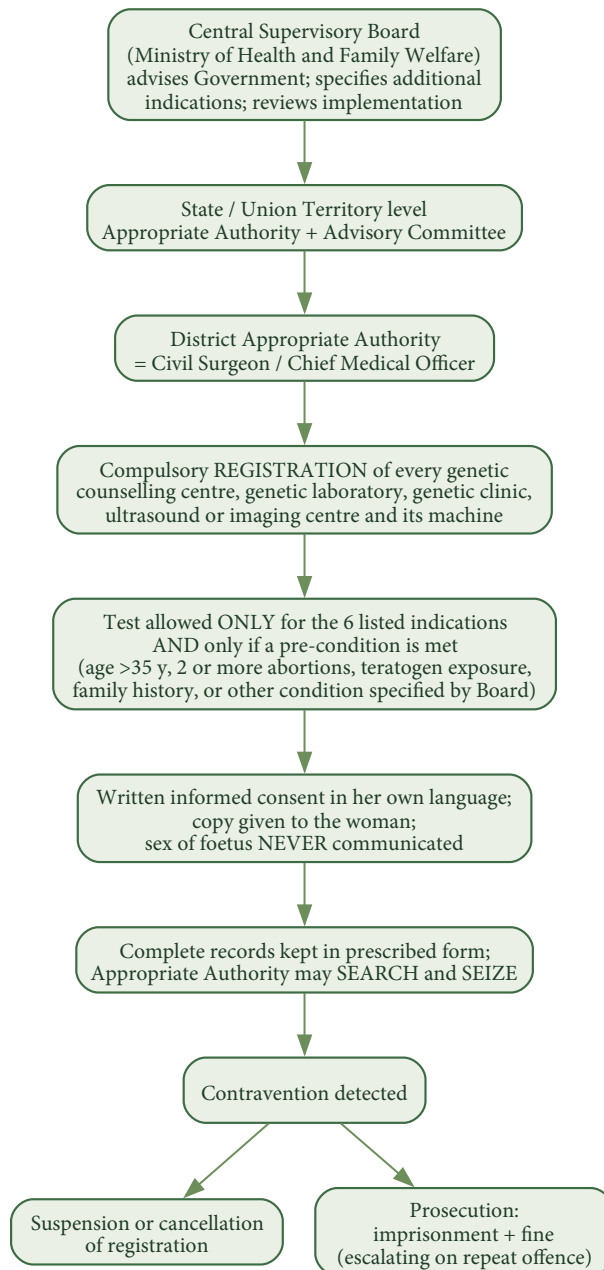
A. REPRODUCTIVE HEALTH AND THE GIRL CHILD
Medical Termination of Pregnancy Act, 1971 (amended 2021)
PC-PNDT Act, 1994 (amended 2003)
Prohibition of Child Marriage Act, 2006
(repealed Child Marriage Restraint Act, 1929)
Protection of Children from Sexual Offences Act, 2012

B. MATERNITY AND THE WOMAN WORKER
Maternity Benefit Act, 1961 and Amendment Act, 2017
Employees' State Insurance Act, 1948
Factories Act, 1948
Indian Mines Act, 1923 and Mines Act, 1952
Plantation Labour Act, 1951; Beedi and Cigar Workers Act, 1966
Contract Labour (Regulation and Abolition) Act, 1970

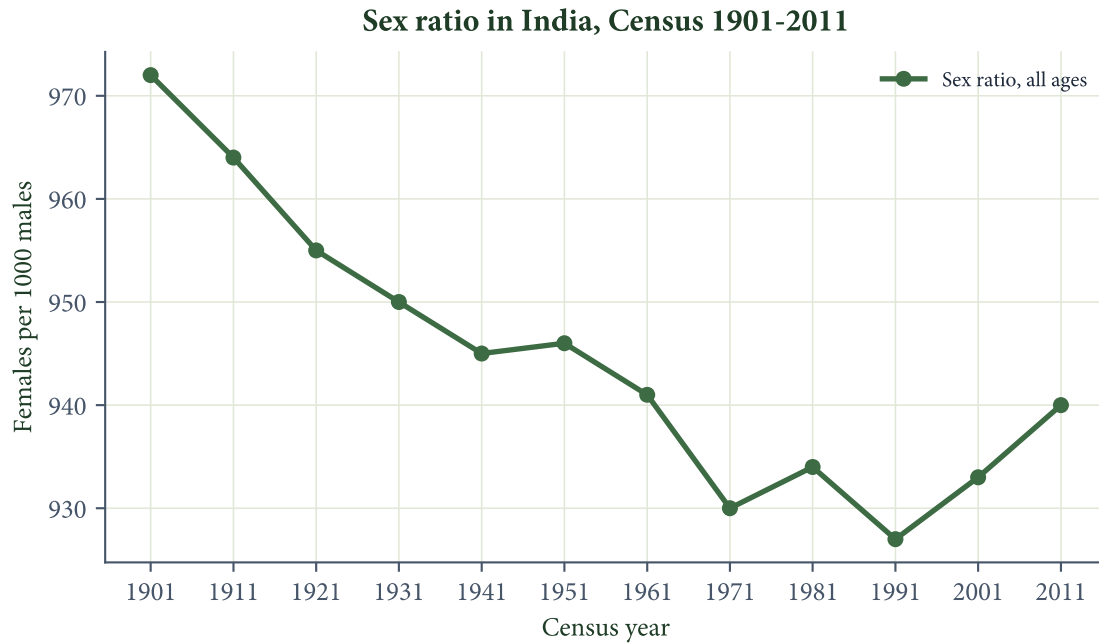
C. VIOLENCE, TRAFFICKING AND EXPLOITATION
Immoral Traffic (Prevention) Act, 1956 (amended 1986)
Dowry Prohibition Act, 1961 (amended 1986)
Indecent Representation of Women (Prohibition) Act, 1986
Commission of Sati (Prevention) Act, 1987
Protection of Women from Domestic Violence Act, 2005
Muslim Women (Protection of Rights on Marriage) Act, 2019

D. WORKPLACE DIGNITY, EQUAL PAY AND MACHINERY
Equal Remuneration Act, 1976
National Commission for Women Act, 1990
Sexual Harassment of Women at Workplace Act, 2013

Classification of public health legislations for women in India — four statutory groups



Implementation and enforcement pathway under the PC-PNDT Act, 1994



Sex ratio in India, Census 1901-2011 — the demographic justification for the PC-PNDT Act (Park 28e, Ch 8, p 582, Table 9)

High-yield exam pointers

- ▶ **Constitutional anchors:** Article 15 (special provisions for women and children), Article 39 (equal pay), Article 42 (maternity relief) → Maternity Benefit Act, 1961, Article 51A (dignity of women).
- ▶ **Child marriage — write the current statute first: Prohibition of Child Marriage Act, 2006** (repealed the Child Marriage Restraint Act, 1929, the *Sharda Act*), enforced **1 November 2007**; ages **18 years for girls, 21 years for boys**; punishes those conducting, abetting, promoting or permitting the marriage.
- ▶ **PC-PNDT dates:** enacted **1994**, in force **1 January 1996**, amended Act and Rules in force **February 2003** with renaming to *Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act*.
- ▶ **Six permitted indications** — chromosomal, genetic metabolic, haemoglobinopathies, sex-linked, congenital, other specified by the Central Supervisory Board. Pre-condition mnemonic: **age >35, ≥2 abortions, teratogen exposure, family history, Board-specified**.
- ▶ **Appropriate Authority = Civil Surgeon / Chief Medical Officer at district level** — registers the centre AND the machine. Penalties: doctor/centre — 3 years + Rs 10,000 first, 5 years + Rs 50,000 subsequent; person seeking the test — 3 years + Rs 50,000 first, 5 years + Rs 1,00,000 subsequent.
- ▶ **Crèche thresholds: 50 or more women** — Maternity Benefit Act (as in Park), Plantation Labour Act 1951, Beedi and Cigar Workers Act 1966; **20 or more women** — Contract Labour Act 1970.
- ▶ **Dated Indian figures:** sex ratio 940 (Census 2011); child sex ratio 0-6 years **914 (Census 2011)**, all-time low; sex ratio at birth **907 (SRS 2018-20)**; sex ratio 0-6 years **928 (NFHS-5, 2019-21)**; child marriage below 18 years **47% (Jugal Kishore)**.

- **MTP Act after the 2021 amendment:** ≤20 weeks on one RMP's opinion; >20 to 24 weeks for prescribed categories on two RMPs' opinion; maternity leave **26 weeks** under the Maternity Benefit (Amendment) Act, 2017.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written as answer content: the Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013 is named here from standard Indian public health teaching because the retrieved corpus records the National Policy for Empowerment of Women 2001 commitment to prevent and punish sexual harassment at work without naming the 2013 statute itself; every other Act, date, figure and penalty above is taken from a retrieved passage. Park's text states the crèche obligation as "where more than 50 women workers are employed", whereas the Maternity Benefit (Amendment) Act, 2017 as enacted fixes it at establishments having fifty or more employees. Section 79 of the Factories Act, 1948 is in fact the annual-leave-with-wages provision, and maternity leave for a woman worker is governed by the Maternity Benefit Act, although the retrieved textbook lists 12 weeks' leave under that section. Finally, by the Supreme Court ruling of October 2022, Rule 3B of the Medical Termination of Pregnancy Rules — which permits termination between 20 and 24 weeks on a change in marital status such as widowhood or divorce — was extended to unmarried women who experience a change in relationship status (Park 28e, Ch 8, p 605), so the phrase "married women" in the older contraceptive-failure ground is no longer restrictive.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Ministry of Women and Child Development (Mission Shakti), WHO abortion care guideline, NFHS-5, Census of India

Programme and policy updates (India)

- **Beti Bachao Beti Padhao is no longer a standalone scheme.** From 1 April 2022 the Ministry of Women and Child Development folded its women-centred schemes into a single umbrella, **Mission Shakti**, with two verticals. BBBP now sits in the **Sambal** (safety and security) vertical alongside the One Stop Centre, the Women Helpline and the newly added **Nari Adalat**. In an answer on women's legislation, name the parent mission and the vertical, not the old standalone scheme [MoWCD 2022].
- **The maternity and crèche schemes were similarly restructured.** The **Samarthya** (empowerment) vertical of Mission Shakti carries **PMMVY**, **Palna** (the erstwhile National Creche Scheme), **Sakhi Niwas** (working women's hostel) and **Shakti Sadan** — a merger of the earlier Ujjwala and Swadhar Greh homes. Mahila Shakti Kendra and Mahila Police Volunteers were discontinued in the same restructuring [MoWCD 2022].

Current national data

- **Anaemia in women rose between survey rounds, not fell.** A comparison of roughly 685,000 women in NFHS-4 (2015-16) with 690,000 in NFHS-5 (2019-21) found prevalence up from **51.4% to 56.2%**, highest in adolescents aged 15-19, and higher in rural, no-education, poorest-quintile and widowed women [PMID 41793337].

- **Spousal violence tracks shared attitudes, from NFHS-5 couple data (44,933 couples).** Risk of physical violence was highest when **both** partners justified wife-beating (AOR 2.02, 95% CI 1.98-2.07); where only the wife justified it the odds were higher (AOR 1.92, 1.88-1.96) than where only the husband did (AOR 1.20, 1.18-1.23). Where both partners had witnessed their father beating their mother, AOR was 3.44 (3.34-3.54) — a directly quotable justification for the Protection of Women from Domestic Violence Act, 2005 and for couple-focused counselling [PMID 42240301].
- **Child marriage is a socio-economic, not a religious, phenomenon.** Census 2011 with NFHS-5 spatial and multivariate analysis showed marriage under 18 years persisting across communities with a faster decline among Muslims over recent decades; education and wealth, not faith, were the dominant gradients — the evidence base for enforcing the Prohibition of Child Marriage Act, 2006 through schooling and cash transfers rather than prosecution alone [PMID 41797567].

Elimination, surveillance and disease status

- **Conditional sex ratio is the sharper surveillance tool than the crude sex ratio.** Across 18.4 million birth records (1999-2019), the Indian conditional sex ratio in 2016 was **866 girls per 1000 boys for a second birth after a first daughter and 769 for a third birth after two daughters**, against a natural range of 950-975; about **13.5 million girls were missing at birth in India between 1987 and 2016**. Skew is concentrated at higher birth orders after earlier daughters, which is where PC-PNDT enforcement and audit should be targeted [PMID 36165452].
- **The ban worked on sex ratio but had unintended child-health effects.** A quasi-experimental analysis of the prohibition on prenatal sex selection found the share of girls born rose, especially in families whose first child was a girl, but mortality outcomes worsened for both girls and boys in those same families and fertility rose — consistent with son-biased fertility stopping and greater competition for household resources. Enforcement removes the technique, not the preference [PMID 37904287].

Guideline and legislative updates

- **Maternity Benefit (Amendment) Act, 2017 — the gap is crèches and awareness, not leave.** An implementation audit of 300 working mothers in Mysuru found 82.7% availed leave and 98.8% of those took 26 weeks or more, but only **12.3% had a crèche**, 56.7% took nursing breaks, **9.7%** knew of the medical bonus and **10.3%** knew the legal provisions; uptake was 92.4% in government versus 75.6% in private employment. The deeper limitation is coverage — about **32.3% of salaried women and only 6.5% of all working women** are eligible at all, because the Act does not reach the informal sector [PMID 41454805].
- **Sexual Harassment of Women at Workplace Act, 2013 — compliance is measurable and incomplete.** A decade-long dataset of complaints disclosed by BSE100 companies (FY 2013-14 to 2023-24) documents both the statutory **90-day limit** for the Internal Committee to complete its inquiry and the recurring failure of companies either to constitute an Internal Committee or to make the mandated annual disclosure — the reason complaint counts are missing for several firm-years [PMID 40405307].

Global and WHO developments

- **WHO Abortion care guideline (2022; second edition 2025)** recommends removing medically unnecessary barriers — mandatory waiting periods, third-party authorisation and restrictions on

who may provide care — and supports **task-sharing** and **self-managed medical abortion up to 12 weeks**. Globally about **73 million induced abortions occur each year and roughly 45% are unsafe**. This is the international benchmark against which India's MTP Act, 1971 (amended 2021) — still a provider-opinion and indication-based law rather than a request-based one — is now judged [WHO 2025].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 12 Medicine and Social Sciences; Ch 18 Occupational Health; Ch 23 Health Planning and Management); Jugal Kishore — National Health Programmes of India (Ch 2 Legislations Related to Health, including "Legislation for Women Empowerment & Health"; Ch 3 Occupational Health Legislation and Legislation for Female Workers); Sunder Lal — Textbook of Community Medicine (Ch 9 Demography, Vital Statistics, Population Stabilization and Medical Genetics; Ch 10 RMNCH+A); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 13 Health and Law; Ch 9 Occupational Health); NFHS-5 India Report (2019-21); Sample Registration System sex ratio at birth 2018-20; Census of India 2011.

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Health team concept in health care delivery

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — A **health team** is "a group of persons who share a common health goal and common objectives, determined by community needs, towards the achievement of which each member of the team contributes in accordance with her/his competence and skills, and respecting the functions of the other" (World Health Organization, quoted in Park 28e, Ch 2, pp 48-49). The **health team concept** is the planned deployment of the right "mix" of professional, auxiliary and voluntary health personnel — each with a specific, recognised role, working under one leader — so that a defined population gets full health coverage. The composition is not fixed: the hospital team differs from the team that works in the community, and it changes with the level of care (Park 28e, Ch 2, p 48).

Why the health team concept was adopted

Basis	Content
Physician overload	The physician of today is overworked professionally, and many of his functions can be performed by auxiliaries given suitable training (Park 28e, Ch 2, p 48).
The auxiliary worker	Defined as one "who has less than full professional qualifications in a particular field and is supervised by a professional worker"; the World Health Organization no longer uses the term "paramedical" for the health professions allied to medicine (Park 28e, Ch 2, p 48).
Limits of any one person	"No one person can acquire all the necessary skills, or have enough time to do everything that must be done to satisfy the needs of even a small community" (Sunder Lal, Ch 16, p 715).
Division of labour, not mere variety	"The mere presence of a variety of health professionals is not sufficient to establish teamwork; it is the proper division and combination of their operations from which the benefits of divided labour will be derived" (Park 28e, Ch 2, p 49).
Full coverage	The health team approach aims to produce the right "mix" of health personnel for providing full health coverage of the entire population (Park 28e, Ch 2, p 49).
Cost	Around 70% or more of the primary health care budget is spent on manpower alone, so understanding and managing team behaviour is a core managerial task (Sunder Lal, Ch 16, p 711).
Task shifting	Redistribution of tasks to less specialised health care workers — used at scale in low- and middle-income settings to extend quality care affordably (Oxford Textbook of Public Health 7e, Ch 4, pp 247-248).

Essential features of an effective health team

- ▶ A **common goal and common objectives determined by community needs**, not by professional convenience (Park 28e, Ch 2, pp 48-49).
- ▶ Every member has a **specific and recognised role** and the freedom to exercise her/his particular skills, while respecting the functions of the others (Park 28e, Ch 2, p 48).
- ▶ A **written, current job description** for each member — most medical officers and health workers under-perform because job descriptions are absent or out of date (Sunder Lal, Ch 16, pp 715-

716). Under the National Health Mission, National Health Policy 2017, Ayushman Bharat and Indian Public Health Standards (IPHS) 2022, these job descriptions have been redefined.

- ▶ **The team must have a leader**, who evaluates the team adequately and knows the motivation of each member in order to stimulate and enhance their potential (Park 28e, Ch 2, p 49).
- ▶ **Auxiliaries are essential members**, not optional add-ons (Park 28e, Ch 2, p 49).
- ▶ **Supportive supervision and continuing education** — the health assistant (male and female) visits each sub-centre at least once a week on a fixed day; the auxiliary nurse midwife (ANM) holds monthly or fortnightly meetings with accredited social health activists (ASHAs) and anganwadi workers (AWWs) and acts as their resource person for training (Sunder Lal, Ch 16, pp 717-718).
- ▶ **Balanced leadership** with equal concern for the people and for the task, with the leadership style matched to the development level of the follower (Sunder Lal, Ch 16, p 712).
- ▶ **Intersectoral membership** — the team is not only health staff; it includes Panchayati Raj representatives, women and child development functionaries, teachers, self-help groups and non-governmental organisations (Sunder Lal, Ch 16, p 711).

Composition of the health team at each level of the Indian health care delivery system

Level (population norm)	Team leader	Other members	Illustrative team output
Village / community	ASHA as member secretary of the Village Health Sanitation and Nutrition Committee (VHSNC), under the village Panchayat	ASHA — one per 1,000 population, woman aged 25–45 years, educated up to 8th class, honorary volunteer working 3–4 hours a day on 4–5 days a week with performance-based compensation; AWW — one per 400–800 population in plain and 300–800 in tribal/difficult areas; trained <i>dai</i> (birth attendant) — one per village; VHSNC of Panchayati Raj members, ANM, AWW, ASHA, teacher and non-governmental organisations, with an untied fund of Rs 10,000 a year	Village health action plan; monthly Village Health, Nutrition and Sanitation Day (VHND) at the anganwadi with at least 4 hours of contact time between ANM, AWW, ASHA and beneficiaries (Sunder Lal, Ch 17, p 744; Ch 16, p 718)

Level (population norm)	Team leader	Other members	Illustrative team output
Urban slum	Urban ASHA (U-ASHA), linked to the urban primary health centre	U-ASHA — one per 1,000–2,500 urban poor population (about 200–500 households); <i>Mahila Arogya Samiti</i> (MAS) — a women's health committee of 10–12 members for 50–100 slum households (population 250–500), with an annual grant of Rs 5,000; ANM placed at the urban primary health centre for outreach sessions	Community-based peer education, demand generation and community-based monitoring; outreach health and nutrition days for slum dwellers (Sunder Lal, Ch 17, p 758; Park 28e, Ch 7, p 535)
Sub-centre — Ayushman Arogya Mandir (health and wellness centre) (5,000 plain; 3,000 tribal/hilly/desert)	Community health officer (CHO), the mid-level health provider — a BSc in community health, or a nurse (general nursing and midwifery) with BSc, or an Ayurveda practitioner, certified through a 6-month community-health certificate programme (IGNOU or a state public health university)	One male and one female multipurpose health worker (the female worker is the ANM); the ASHAs of the area; the male health worker visits each family once a fortnight	Comprehensive primary health care package of 12 services; population-based screening for hypertension, diabetes, obesity and the three common cancers — breast, cervical and oral (Sunder Lal, Ch 17, p 742; Ch 8, pp 529-530; Park 28e, Ch 10, p 731)
Primary health centre — health and wellness centre (rural 30,000 plain / 20,000 hilly and tribal; urban 50,000; polyclinic 2.5–3 lakh)	Medical officer (MBBS) — "the captain of the health team", who is planner, promoter, director, supervisor, coordinator and evaluator	Second medical officer, AYUSH medical officer and dental medical officer; staff nurses; pharmacist (one per 10,000); medical laboratory technician (one per 10,000); health assistant (male) and health assistant (female)/lady health visitor — one per 30,000 in plain and 20,000 in tribal/hilly areas, each supervising six sub-centres; ANM; cold-chain/vaccine logistics assistant; health educator/counsellor; public health manager; data entry operator; accountant; dresser	Universal Immunization Programme delivery and cold-chain maintenance; weekly maternal and child health clinic at each sub-centre; monthly staff meeting at the primary health centre to review progress (Park 28e, Ch 10, pp 725, 732, 738-739, 743)

Level (population norm)	Team leader	Other members	Illustrative team output
Community health centre / first referral unit (block; referral for four primary health centres; 30 beds)	Medical officer in charge of the community health centre	Presently four medical specialists — surgeon, physician, gynaecologist and paediatrician — with 21 paramedical and other staff; under IPHS, six specialists including an anaesthetist and an eye surgeon, supported by 24 paramedical and other staff with two nurse midwives; operation theatre, X-ray, labour room and laboratory	Emergency obstetric care, full range of family planning services including laparoscopic sterilisation and safe abortion; non-communicable disease clinic (Park 28e, Ch 8, pp 611-612)
District	Civil surgeon / chief medical officer, the leader of the district health organization	District programme officers and health programme managers for each national health programme; rapid response teams for outbreak investigation; under the multipurpose workers scheme an integrated set-up of a chief medical officer with three deputy chief medical officers (civil surgeon, district health officer, district family welfare officer), each in charge of one-third of the district	Secondary care within one hour of travel; supportive supervision of community health centres; outbreak response (Sunder Lal, Ch 17, p 761; Park 28e, Ch 23, p 1035)
Tertiary	Head of unit / department	Super-specialists, resident doctors, specialist nurses, technologists and allied health professionals in medical college hospitals, super-speciality institutions and large private hospitals	Complex diagnostic and therapeutic care; teaching and training for specialised staff; support to the primary and secondary levels (Sunder Lal, Ch 17, p 761; Park 28e, Ch 2, p 48)

Suitable examples of named multidisciplinary teams

Example	Composition	Levels spanned
Integrated Child Development Services team	AWW, ASHA and helper at village level; supervisor (<i>Mukhya Sevika</i>) one per 25 anganwadi workers at circle level; child development project officer one per one lakh population at block level; district programme officer at district level — working jointly with the female and male health workers, health supervisors and medical officers of the primary health centre	Village → circle → block → district (Sunder Lal, Ch 5, p 226)

Example	Composition	Levels spanned
District Mental Health Programme team	Five-member technical team — psychiatrist or trained medical officer as programme officer (mental health), clinical psychologist, psychiatric social worker, psychiatric/community nurse, record keeper — plus a two-member managerial team of a community nurse and an assistant	District hospital with liaison to primary health centres (Sunder Lal, Ch 15, pp 682-683)
Adolescent friendly health clinic team	Level 1 — multipurpose health workers at the sub-centre (walk-in services); Level 2 — weekly clinic at the primary health centre run by the medical officer, ANMs and a counsellor; Level 3 — daily clinic at community health centre / sub-district or district hospital / medical college by medical officers and specialists with counsellors	Sub-centre → primary health centre → secondary/tertiary (Sunder Lal, Ch 10, p 623)
Village Health, Nutrition and Sanitation Day team	ANM + AWW + ASHA + beneficiaries at the anganwadi centre once a month, monitored by the VHSNC	Village, supervised from the sub-centre (Sunder Lal, Ch 17, p 744)
Rapid response team	Multidisciplinary team maintained at district level for outbreak investigation and rapid response	District, deployed to the affected block/village (Sunder Lal, Ch 17, p 761)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw and label — "Health team pyramid of the Indian health care delivery system" (a triangle, four horizontal bands, labels written inside each band):

- ▶ Base band: *Community level* — ASHA (1 : 1,000), anganwadi worker (1 : 400–800), trained dai, VHSNC, Mahila Arogya Samiti
- ▶ Second band: *Sub-centre* — Ayushman Arogya Mandir (5,000 / 3,000) — community health officer (mid-level health provider), male and female multipurpose health workers
- ▶ Third band: *Primary health centre* (30,000 / 20,000) — medical officer as captain, health assistants (male and female), staff nurse, pharmacist, laboratory technician
- ▶ Apex band: *Community health centre and district* — specialists (surgeon, physician, obstetrician-gynaecologist, paediatrician, anaesthetist, eye surgeon); chief medical officer with district programme officers
- ▶ Left-side arrow pointing up labelled "managerial skills increase"; right-side arrow pointing down labelled "technical/service-delivery skills increase"

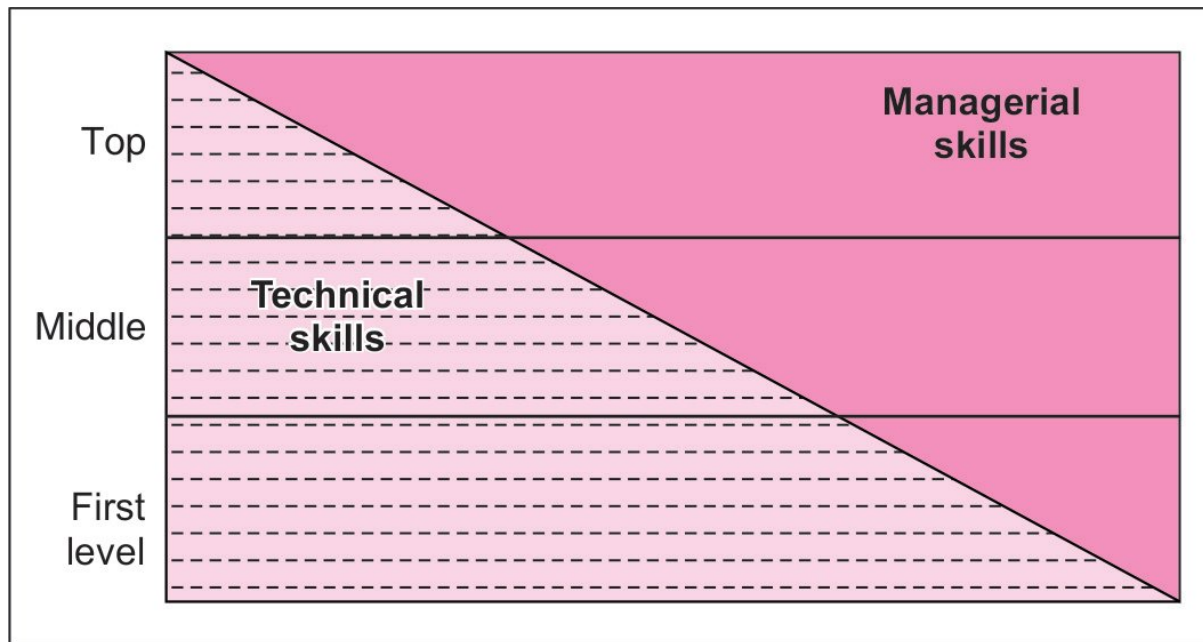
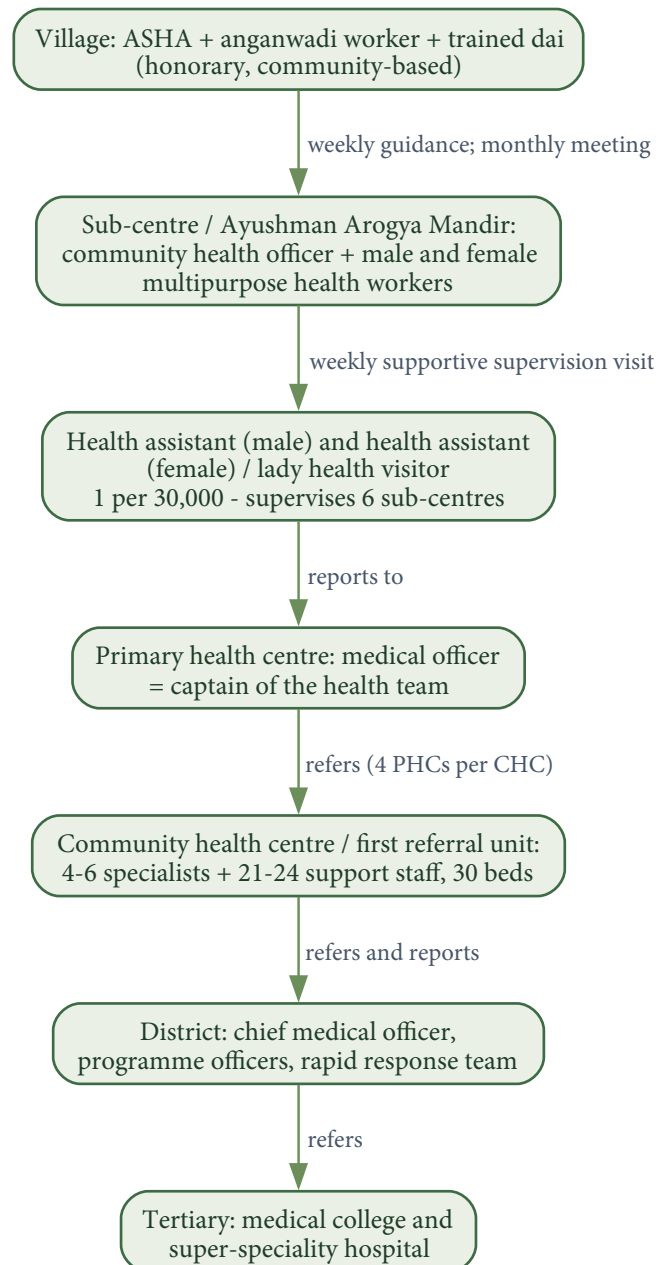
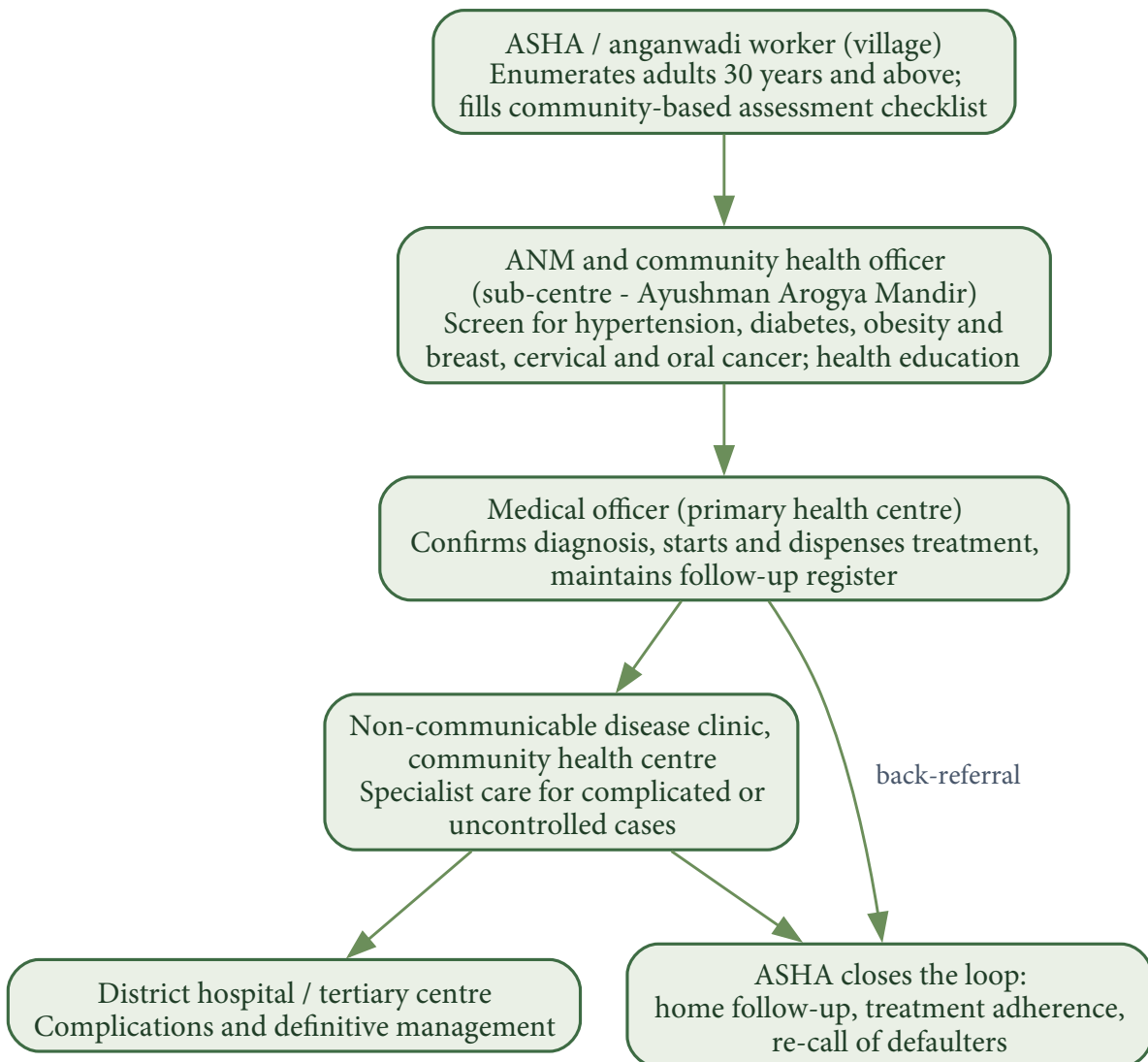


Fig 16.3 — Levels of organization and skills required (Sunderlal, p.710)



Supervision and referral chain that binds the health team across levels of care



Worked example - who in the health team does what in population-based screening for non-communicable disease

High-yield exam pointers

- ▶ **Definition to reproduce verbatim:** common health goal + common objectives + determined by community needs + each contributes per competence and skills + respecting the functions of the other.
- ▶ **Auxiliary worker** = less than full professional qualification in a field + supervised by a professional worker; "paramedical" is no longer used by the World Health Organization.
- ▶ **Population norms:** ASHA 1 : 1,000 · anganwadi worker 1 : 400–800 · sub-centre 5,000 (3,000 tribal/hilly/desert) · primary health centre 30,000 (20,000 hilly/tribal) · urban primary health centre 50,000 · health assistant 1 : 30,000 supervising 6 sub-centres · pharmacist and laboratory technician 1 : 10,000 each.
- ▶ **Mid-level health provider = community health officer** — BSc community health, or GNM nurse with BSc, or Ayurveda practitioner, plus a 6-month certificate in community health.
- ▶ **Community health centre staffing:** 4 specialists + 21 support staff at present; IPHS 6 specialists (adding anaesthetist and eye surgeon) + 24 support staff with 2 nurse midwives; 30 beds; referral for 4 primary health centres.

- **One-line quotable:** "The mere presence of a variety of health professionals is not sufficient to establish teamwork" — division and combination of operations is what yields the benefit.
- **Named team examples worth naming:** Integrated Child Development Services team, District Mental Health Programme five-member technical team, adolescent friendly health clinic three-level team, rapid response team, Village Health Nutrition and Sanitation Day team.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: health and wellness centres have been renamed **Ayushman Arogya Mandir**, so write the current name first and the former name in brackets once; the community health centre specialist numbers quoted here are the two figures Park gives side by side (the present four-specialist pattern and the six-specialist Indian Public Health Standards norm), and the Indian Public Health Standards were last revised in 2022.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW Ayushman Arogya Mandir portal, National Health Accounts 2021-22, WHO UHC Service Coverage Index, National Ayush Mission

Programme and policy updates (India)

- **The sub-centre and primary health centre team now works in an "Ayushman Arogya Mandir".** In November 2023 the Union Health Ministry renamed every Ayushman Bharat health and wellness centre as *Ayushman Arogya Mandir*, with the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth"), to reorient the team from illness care towards holistic wellness on AYUSH principles; renaming was funded at Rs 3,000 per facility (total outlay Rs 48 crore) with photographic verification uploaded to the portal. Post-renaming field visits still found adolescent privacy compromised at adolescent friendly health clinics, poor filing of biomedical waste and sterilisation records, and low uptake of the Ayushman Bharat Health Account — a name change by itself does not change team performance. [PMID 42257165]
- **The community health officer is now formally the team leader at the sub-centre, and the team now also earns as a team.** 1,16,978 community health officers were in position on 30 November 2024, leading the multipurpose workers and ASHAs with combined clinical, public health and managerial roles; 82% of Ayushman Arogya Mandirs had rolled out all 12 expanded service packages by that date. Performance-linked **team-based incentives**, paid on the collective performance of the primary health care team, have been introduced at Ayushman Arogya Mandir (there were previously no team-based incentives), alongside the individual activity-based incentives that continue for ASHAs; a national portal named SASHAKT now plans and monitors the quality of training on the expanded service packages for health care functionaries. A third community platform, the **Jan Arogya Samiti**, has been added alongside the Village Health Sanitation and Nutrition Committee (rural) and Mahila Arogya Samiti (urban) — 1,35,343 constituted by 30 November 2024, each holding an *Arogya Sabha* twice a year after the monthly village-level *Samvad*. [PMID 41816146] [PMID 39953777]
- **The AYUSH member of the team has a widened prescribing role.** The Union Cabinet approved 12,500 Ayush Ayushman Arogya Mandirs in March 2020 (outlay Rs 3,399.35 crore) and 11,771 were functional by 14 July 2024; Indian Public Health Standards for Ayush facilities have since

been published. Most examinally, Ayush doctors serving as community health officers — initially barred from prescribing Ayush medicines under the health and wellness centre guidelines — may now prescribe them, with supply routed through the National Ayush Mission. [PMID 40547778] [PMID 41229518]

Current national data

- ▶ **1,86,908 Ayushman Arogya Mandirs are functional** — 1,34,701 at sub-centre level, 24,510 primary health centre, 5,471 urban primary health centre, 9,968 urban health and wellness centre and 12,258 Ayush — against the original target of 1.5 lakh, which was met by December 2022. Always write the figure with its date. [MoHFW 2026-Aug]
- ▶ **What the reorganised team has bought, in national indicators:** UHC Service Coverage Index 57 (2015) rising to 63 (2021), above the South-East Asia regional average of 62; out-of-pocket expenditure falling from 60.6% (2014-15) to 39.4% (2021-22, National Health Accounts); public outpatient attendance rising from 91.6 crore (2012-13) to 134.5 crore (2021-22). [PMID 41816146]
- ▶ **Output and shortfall of the sub-centre team, measured.** A state-representative assessment of 404 health and wellness centres in Chhattisgarh found a mean of 358 outpatients treated per month including 128 with non-communicable disease, meeting 31% of the population's need for outpatient care, 26% for hypertension and 21% for diabetes [PMID 39367295]. A single-block facility survey of 20 centres found 41.2% of sub-centre teams with no community health officer in post (ASHAs present at 100% of centres, auxiliary nurse midwives at 94.1%), and high performance grading was separately associated with a community health officer being in post ($p < 0.001$) and with working teleconsultation ($p = 0.002$) [PMID 40821268].

Elimination, surveillance and disease status

- ▶ **Population-based screening done by the mid-level provider stands up to specialist validation.** Forty-six community health officers given one day of training in oral visual examination screened 209 attenders opportunistically at their Ayushman Arogya Mandirs over eight months; against remote ear-nose-throat specialist validation of the photographs their findings gave sensitivity 83.87% (95% CI 70.92-96.82), specificity 98.31% (95% CI 96.42-100) and overall accuracy 96.17% (95% CI 93.57-98.77) for oral potentially malignant disorders. [PMID 40341602]

Guideline and legislative updates

- ▶ **Task shifting now has randomised evidence from rural India.** In a three-hospital non-inferiority randomised controlled trial, 422 low-risk patients received spinal anaesthesia either from non-specialist medical officers trained through a structured curriculum or from consultant anaesthesiologists; failure occurred in 7 (3.0%) versus 5 (2.6%), a difference in success rate of 0.4% (95% CI 0.36-0.43; $p = 0.80$), with no significant difference in perioperative or postoperative complications. This is the evidence base for extending task sharing where specialists are unavailable. [PMID 39153752]
- ▶ **ASHA remuneration — process, not just rate.** A two-state study (53 in-depth interviews and 8 focus group discussions with ASHAs, supervisors and managers) mapped the five steps from recording work to disbursement; complete digitisation of disbursement through the public financial management system improved timeliness, transparency and worker satisfaction

compared with earlier cash and cheque payment, whereas digitising the claim-submission end drew mixed responses from workers and supervisors. [PMID 39953777]

Global and WHO developments

- **Task sharing with community health workers is World Health Organization-recommended practice for widening access, and is being tested well beyond its traditional scope.** A 2025 systematic review of six studies found community health worker-facilitated hearing aid provision (607 fittings) feasible and effective with high sustained device use, but training content and duration varied widely and only one study used World Health Organization-endorsed material — the review calls for standardised training, defined supervision models and culturally adapted outcome measures, which is the same "job description plus supervision" argument the health team concept rests on. [PMID 41238364]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 Concept of Health and Disease; Ch 7 Health Programmes in India; Ch 8 Demography and Family Planning; Ch 10 Health Care of the Community; Ch 23 Health Planning and Management); Sunder Lal — Textbook of Community Medicine (Ch 5 Nutrition; Ch 8 Epidemiology of Communicable and Non-Communicable Diseases; Ch 10 RMNCH+A; Ch 15 Mental Health; Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community); Oxford Textbook of Public Health 7e (Ch 4, task shifting); Indian Public Health Standards 2022 and National Health Mission / Ayushman Bharat operational guidance as cited within these texts.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of "Arogyam Paramam Dhanam" — a step towards holistic care, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmpc.jfmpc.1966.24](https://doi.org/10.4103/jfmpc.jfmpc.1966.24); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 40547778 — Policy mandates for Ayush in National Health Policy-2017: achievements, impediments, and future prospects, *Journal of Family Medicine and Primary Care* 2025, doi:[10.4103/jfmpc.jfmpc.1234.24](https://doi.org/10.4103/jfmpc.jfmpc.1234.24); 41229518 — India's journey in mainstreaming Ayush in primary health care: from tradition to integration, *Frontiers in Medicine* 2025, doi:[10.3389/fmed.2025.1629515](https://doi.org/10.3389/fmed.2025.1629515); 39367295 — Performance of health and wellness centre in providing primary care services in Chhattisgarh, India, *BMC Primary Care* 2024, doi:[10.1186/s12875-024-02603-1](https://doi.org/10.1186/s12875-024-02603-1); 40821268 — Assessment of the functioning of health and wellness centers in Kurud block, Dhamtari district, Chhattisgarh, *Cureus* 2025, doi:[10.7759/cureus.88071](https://doi.org/10.7759/cureus.88071); 40341602 — Capacity building of community health officers for optimizing the screening and early diagnosis of oral cancer in Jodhpur, Rajasthan, *Scientific Reports* 2025, doi:[10.1038/s41598-025-00225-w](https://doi.org/10.1038/s41598-025-00225-w); 39153752 — Task-sharing spinal anaesthesia care in three rural Indian hospitals: a non-inferiority randomised controlled clinical trial, *BMJ Global Health* 2024, doi:[10.1136/bmjgh-2023-014170](https://doi.org/10.1136/bmjgh-2023-014170); 39953777 — Community health worker payment processes: a qualitative assessment of experiences in two Indian states, *Health Policy and Planning* 2025, doi:[10.1093/heapol/czaf010](https://doi.org/10.1093/heapol/czaf010); 41238364 — Feasibility, efficacy and effectiveness of community health worker-facilitated hearing aid provision: a systematic review, *BMJ Open* 2025, doi:[10.1136/bmjopen-2025-108591](https://doi.org/10.1136/bmjopen-2025-108591). Facility counts from the Ministry of Health and Family Welfare Ayushman Arogya Mandir portal, August 2026.

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Social assistance and social insurance

Asked in: Paper I May 2017 — RGUHS MD Community Medicine

Definition — **Social security** is "security that society furnishes through appropriate organization, against certain risks to which its members are exposed"; in most countries the risks covered are **sickness, invalidity, maternity, old age and death**, and social security includes **social insurance and social assistance** (Park 28e, Ch 12, pp 845-846). The International Labour Organization (ILO) defines it as the protection society provides its members through a series of public measures against the economic and social distress that would otherwise be caused by substantial stoppage of earnings from sickness, maternity, injury, unemployment, old age and death (Park Plus, Ch 12). **Social assistance** is the *non-contributory*, tax-financed arm; **social insurance** is the *contributory*, pooled-premium arm.

Social assistance and social insurance compared

Feature	Social assistance	Social insurance
Working definition	A method of providing benefits, as of right, to persons usually of small means , in amounts sufficient to meet a minimum standard of living , out of the general revenues of the State (Park Plus, Ch 12)	A method of providing benefits, as a matter of right, to persons of small earnings , in amounts which combine the contributions of the beneficiary with subsidies from the employer and the State (Park Plus, Ch 12)
Contribution by beneficiary	Nil — non-contributory; this is the defining feature (Park Plus, Ch 12)	Compulsory periodic contribution , usually a payroll deduction, into a common pool from which benefits are paid (Park Plus, Ch 12)
Source of funds	General taxation / consolidated fund of the State	Tripartite pool — employee plus employer plus government subsidy
Basis of entitlement	Proven need — means test, below-poverty-line status, destitution	Record of contribution — claimable as a matter of right
Who is covered	Vulnerable groups: children, mothers, the aged, widows, orphans, the disabled, the unemployed (Park Plus, Ch 12; Suryakantha, Ch 17)	Organised-sector workers — suitable where the class to be covered is well organised, legally regulated and financially stable (Park Plus, Ch 12)
Nature of benefit	Flat, subsistence-level cash grant or benefit in kind (pension, food grain)	Earnings-related cash benefit plus medical care in kind
Direction of redistribution	Vertical — richer taxpayer to poorer beneficiary	Horizontal risk pooling — healthy to sick, young to old, within the insured pool (Oxford Textbook of Public Health 7e, Ch 21, pp 831-833)
Population reach	No limitation in principle; can reach hard-to-reach groups	Number of beneficiaries tied to the size of the formal sector ; informal workers need additional tax funding (Health Economics, Ch 12, pp 189-191)
Equity of financing	Generally progressive	Less progressive — encourages under-reporting of salaries; wage ceilings exclude high earners (Health Economics, Ch 12, Table 12.1)

Feature	Social assistance	Social insurance
International model	Tax-funded national service (Beveridge-type)	Bismarck model — run by a public agency, financed by compulsory payroll contributions split between employer and employee, non-profit, compulsory participation (Health Economics, Ch 12, pp 189-190)
Indian examples	National Social Assistance Programme, Annapurna, Ayushman Bharat–PMJAY, Integrated Child Development Services, Public Distribution System	Employees' State Insurance Scheme, Central Government Health Scheme, Ex-Servicemen Contributory Health Scheme

Social assistance in India

- **National Social Assistance Programme (NSAP)** — came into effect **15 August 1995**, a significant step towards fulfilling **Article 41** of the Directive Principles; it introduced a national policy for social assistance to the poor and a minimum national standard of assistance over and above what States provide (Jugal Kishore, National Health Programmes of India).
- **NSAP components** — Indira Gandhi National Old Age Pension Scheme (IGNOAPS), Indira Gandhi National Widow Pension Scheme (IGNWPS), Indira Gandhi National Disability Pension Scheme (IGNDPS), National Family Benefit Scheme (NFBS) and **Annapurna** (Jugal Kishore).
- **Annapurna** — **10 kg of rice or wheat per month, free of cost**, to destitute elderly persons who are eligible for but not receiving an old-age pension (Sunder Lal, Ch 5; Suryakantha, Ch 29).
- **Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana (AB-PMJAY)** — health cover of **Rs 5 lakh per family per year** for secondary and tertiary hospitalisation to about **10.74 crore poor and vulnerable families** identified from Socio-Economic Caste Census data; wholly tax-financed with **no premium from the beneficiary**, hence social assistance in character, and it subsumed the earlier Rashtriya Swasthya Bima Yojana (Park 28e, Ch 7; Sunder Lal, Ch 17, p 763).
- **Promotional and preventive assistance schemes** — food and nutrition security (Integrated Child Development Services, Public Distribution System), employment security, education security, and reservations and special educational institutions for the disabled, all drafted by the Centre and States and financed from general revenues (Park Plus, Ch 12).

Social insurance in India

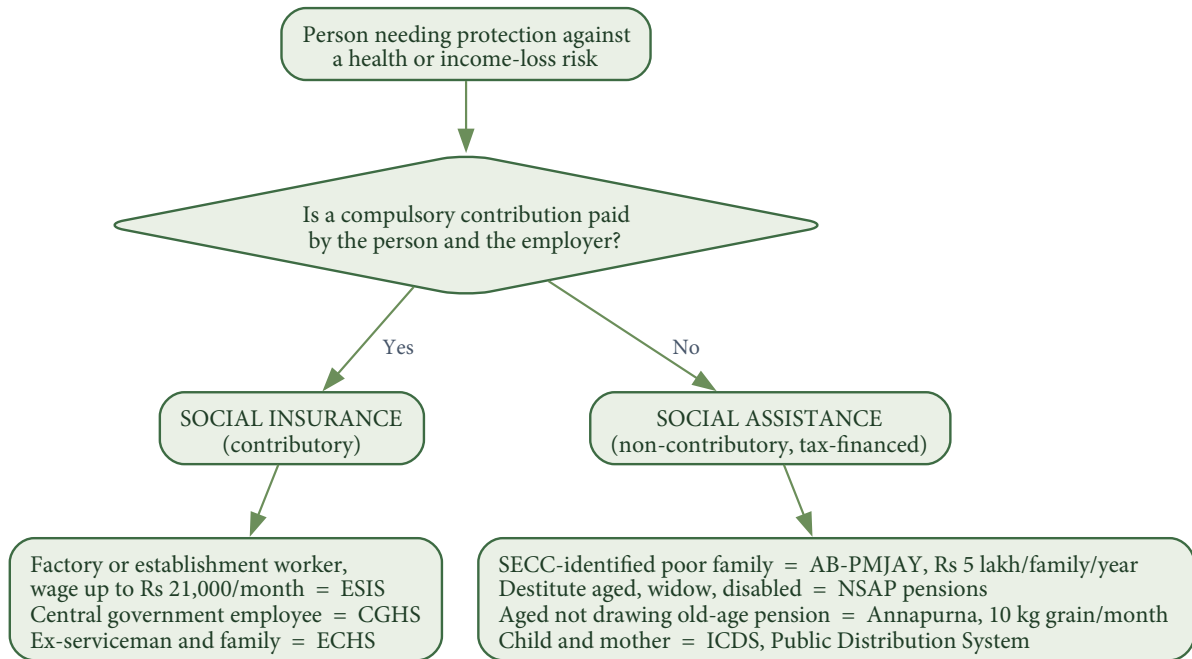
The compulsory social insurance schemes are the **Employees' State Insurance Scheme (ESIS)**, the **Central Government Health Scheme (CGHS)** and the **Ex-Servicemen Contributory Health Scheme (ECHS)**; social security expenditure on health accounts for about **9.6 per cent of total health expenditure**, against 6.6 per cent for private voluntary insurance (Sunder Lal, Ch 16, p 729).

Element	Employees' State Insurance Scheme
Statute	ESI Act 1948 — the first Indian law to introduce the principle of contribution by employer and employee (Park 28e, Ch 10, p 745)

Element	Employees' State Insurance Scheme
Applicability	All non-seasonal factories employing 10 or more persons ; extended by States to shops, hotels and restaurants, cinemas and theatres, road-motor transport and newspaper establishments (Park 28e, Ch 18; Sunder Lal, Ch 11, p 650)
Wage ceiling	Employees drawing wages up to Rs 21,000 per month (raised from Rs 15,000 by ESIC notification of 6.9.2016) (Park 28e, Ch 10 and Ch 18)
Contribution	Employer 3.25 per cent and employee 0.75 per cent of wages — total reduced from 6.5 to 4 per cent w.e.f. 1 July 2019 ; employees with a daily average wage up to Rs 176 are exempt from contribution but remain entitled to benefits (Park 28e, Ch 18; Sunder Lal, Ch 11, p 650)
State share	The State government's share of expenditure on medical care is one-eighth of the total cost , the balance borne by the ESI Corporation (Park 28e, Ch 18)
Benefits	(1) Medical (2) Sickness (3) Maternity (4) Disablement (5) Dependant's (6) Funeral expenses (7) Rehabilitation allowance (Park 28e, Ch 18)
Medical care delivery	Direct pattern — ESI service dispensaries and hospitals; indirect (panel) pattern — private practitioners appointed as "insurance medical practitioners" (Park 28e, Ch 18; Suryakantha, Ch 17)
Administration	Autonomous ESI Corporation — Union Minister for Labour as Chairman, Director General as chief executive, and a Medical Benefit Council headed by the Director General of Health Services (Park 28e, Ch 18)

CGHS — introduced in New Delhi in **1954** (previously the Contributory Health Service Scheme) to give comprehensive medical care to central government employees, "based on the principle of cooperative effort by the employee and the employer"; it provides out-patient care through dispensaries, drugs, laboratory and X-ray investigations, domiciliary visits, hospitalisation, specialist consultation, paediatric and immunisation services, antenatal-natal-postnatal care, emergency treatment, optical and dental aids and family welfare services, through allopathic and AYUSH wellness centres and empanelled hospitals (Park 28e, Ch 10, pp 745-746).

Other statutory social security measures for industrial workers — Workmen's Compensation Act 1923, Central Maternity Benefit Act 1961, Employees' Provident Fund and Miscellaneous Provisions Act 1952, Family Pension Scheme 1971 and Payment of Gratuity Act 1972; civil servants additionally have pension, gratuity, provident fund and family pension schemes (Park 28e, Ch 12, pp 845-846).



Routing an Indian beneficiary to a social insurance or a social assistance scheme

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — the wheel of social security.** Draw one circle with an outer rim divided into three arcs labelled **Social insurance**, **Social assistance** and **Social defence**, and spokes inside each arc. Label the social-insurance spokes: ESI benefits, Life insurance, Provident fund, Retirement benefit, Family pension, Plantation Act, Port and Dock Act, Mines Act. Label the social-assistance spokes: Widow pension, Assistance to leprosy patients, Family planning assistance, Geriatric care, Unemployment scheme, Immoral Traffic Act. Label the social-defence spokes: Prison reforms, Welfare of prisoners, Eradication of beggary, Juvenile delinquency, Anti-dowry Act, Criminals, Prostitution, Gambling, Suicides, Drug abuse, Alcoholism.
- **Diagram 2 — social security tree.** Reproduce the classification tree below with the exact box labels.

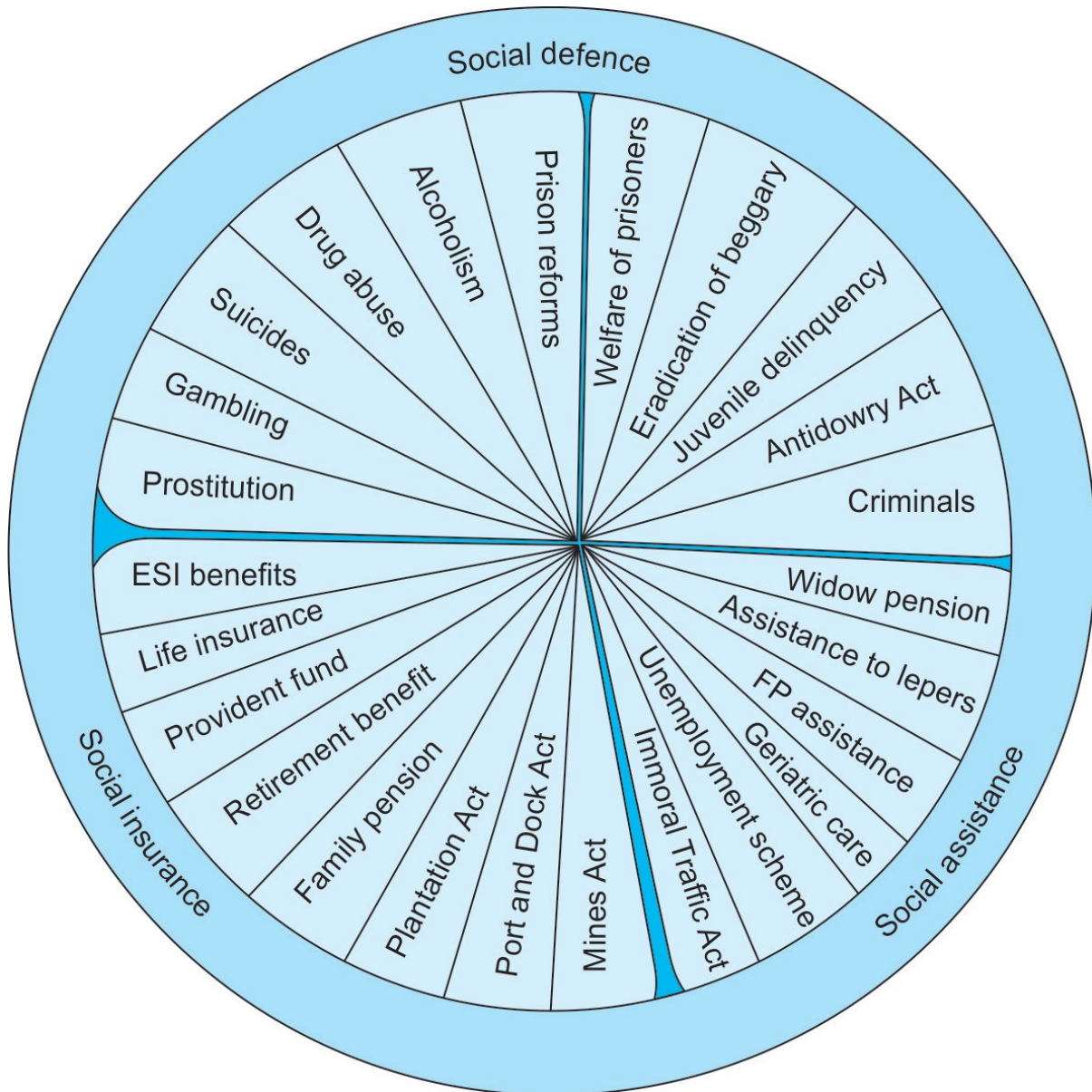
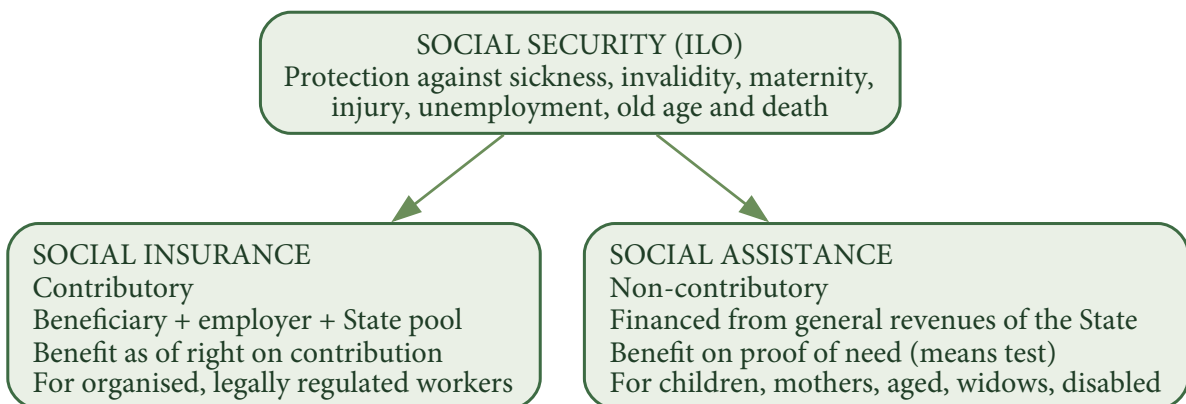


Fig 17.2 — Showing relation between social insurance, social assistance and social defence (Suryakantha Environment Best, p.256)



Social security and its two financing approaches

High-yield exam pointers

- ▶ **One-line discriminator:** social assistance = **non-contributory, tax-financed, need-based**; social insurance = **contributory, pooled-premium, claimable as of right**.
- ▶ **ESI arithmetic to reproduce:** Act 1948; 10 or more employees; wage ceiling Rs 21,000/month; employer 3.25 % + employee 0.75 % = 4 % (w.e.f. 1.7.2019); exempt below Rs 176/day; State bears **one-eighth** of medical-care cost; 7 **benefits**.
- ▶ **CGHS 1954, New Delhi** — ESIS and CGHS are the two well-organised health insurance schemes covering large groups of wage-earners in India (Park 28e, Ch 10).
- ▶ **NSAP — 15.8.1995, Article 41;** mnemonic for its five components **O-W-D-F-A:** Old age, Widow, Disability, Family benefit, Annapurna.
- ▶ **AB-PMJAY = Rs 5 lakh per family per year, about 10.74 crore families, SECC-based, no premium** — classify it as social *assistance*, not social insurance.
- ▶ **Three stages in the evolution of social security (ILO 1984):** paternalistic private charity and poor relief → compulsory-premium insurance schemes → prevention and universality (Park Plus, Ch 12).
- ▶ **Third arm in the figure — social defence:** the preventive, therapeutic and rehabilitative services protecting society from antisocial, criminal or deviant conduct (Park 28e, Ch 12).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Suryakantha states that the first social security Act was enacted in Germany in 1881 for industrial workers and that Sweden is the only country with its entire population under social security schemes; and Rashtriya Swasthya Bima Yojana, described as a health insurance scheme in earlier editions, has been subsumed by Ayushman Bharat-PMJAY, which is fully tax-financed and therefore social assistance in character.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO universal health coverage fact sheet 2025, ILO World Social Protection Report 2024-26, National Health Accounts India 2021-22, NSS Comprehensive Annual Modular Survey 2022-23

Programme and policy updates (India)

- ▶ **Health and Wellness Centres have been renamed Ayushman Arogya Mandir.** In November 2023 the Union Health Ministry renamed the Ayushman Bharat Health and Wellness Centres as **Ayushman Arogya Mandir (AAM)**, tagline *Arogyam Paramam Dhanam*, with facility names carried in the local script at Rs 3,000 per facility and a total outlay of about Rs 48 crore; more than 1.75 lakh AAMs are now operational [PMID 42257165] [PMID 41816146]. Write AAM rather than HWC — and note that this arm of Ayushman Bharat is wholly tax-financed, so like AB-PMJAY it belongs on the **social assistance** side of the table above.
- ▶ **Benefit-package design is now the working policy lever inside PM-JAY.** Analysis of 1.35 million claims from all 886 empanelled hospitals in Punjab (August 2019 to December 2022) found that moving from Health Benefit Package 1.0 to HBP 2.0 raised service utilisation, with the largest gains in packages that had been badly under-priced and were re-priced, and a similar effect in

government and private hospitals [PMID 39252993]. In Kerala, folding the state and national schemes into a single purchaser under the Karunya Arogya Suraksha Padhathi and extending cover to chronic-disease medicines cut net household out-of-pocket spending by about 31.7%, the fall being concentrated in medicines and largest in poorer households and those with chronic illness [PMID 42284657].

Current national data

- ▶ **National Health Accounts, India 2021-22 — quote the round with the figure.** Out-of-pocket expenditure has fallen to **39.4% of total health expenditure (2021-22)** from 60.6% in 2014-15, and India's WHO UHC service coverage index rose from **57 in 2015 to 63 in 2021**, above the South-East Asia regional average of 62 [PMID 41816146]. The intermediate points on the same series are 69.4% (2004-05) and 47.1% (2019-20), with total health expenditure at 3.83% of GDP in 2021-22 [PMID 40552236].
- ▶ **How the two arms actually divide the country (Comprehensive Annual Modular Survey 2022-23, NSS 79th round, 302,086 households).** 59.9% of households reported some health cover: **49.7% under publicly funded, non-contributory schemes (AB-PMJAY and state schemes), only 8.2% under ESIC, CGHS or employer social insurance**, and 2.0% private voluntary insurance [PMID 42167946]. The contributory arm remains the smallest — exactly what the formal-sector ceiling in the table above predicts.
- ▶ **The contributory arm still misses informal workers and the old.** In the Longitudinal Ageing Study in India, older informal workers paid less in absolute terms than formal-sector workers (INR 1,113 less for inpatient and INR 55 less for outpatient care) yet were significantly more likely to breach the 10%, 20% and 40% out-of-pocket thresholds [PMID 39096525]. In a Kerala panchayat survey of adults aged 65 and over, 82.3% drew a pension, predominantly **non-contributory social pensions**; a below-poverty-line ration card was the only significant predictor of receipt, the pension met health expenses for 65.3%, and pension amount explained 43.4% of the variation in monthly medicine spending [PMID 42257113].

Elimination, surveillance and disease status

- ▶ **Ni-kshay Poshan Yojana — social assistance built into an elimination programme.** The first national evaluation of this direct benefit transfer (INR 500 per month to every notified TB patient for nutritional support) covered 3,712,551 patients notified in nine states between 2018 and 2022: receipt of at least one instalment rose from **56.9% to 76.1%**, and the median wait for the first instalment halved from **200 days (2018) to 91 days (2022)**; non-receipt was highest among patients notified by the private sector (adjusted RR 2.10) [PMID 38273246].
- ▶ **Publicly funded insurance is being folded into the TB elimination programme.** Karnataka integrated AB-PMJAY-Arogya Karnataka with the National TB Elimination Programme, providing about US\$ 124.5 of cover per TB patient against a reported minimum uninsured hospitalisation cost of US\$ 159, in a setting where 19% of Indian TB patients face catastrophic costs and 89.15% of confirmed cases were treated in the government sector [PMID 38482155]. Under the Pradhan Mantri TB Mukh Bharat Abhiyaan, only 39,015 of 82,205 registered Ni-kshay Mitra community donors (51%) actually provided support and just 9.6% of notified patients were recorded as receiving it, but those who did had 95% treatment success against 90% without (OR 2.2, 95% CI 2.10 to 2.31, $p < 0.001$) [PMID 41451044].

Guideline and legislative updates

- **Whether tax-financed insurance actually delivers financial protection is now openly contested, and examiners expect both sides.** Two annual state-representative surveys in Chhattisgarh (about 15,000 individuals each, 2021 and 2022) found AB-PMJAY enrolment associated with neither higher inpatient utilisation nor lower out-of-pocket or catastrophic expenditure; among enrollees who used private hospitals, **78.1% (2021) and 70.9% (2022)** still incurred catastrophic expenditure at the 10% threshold [PMID 39135015]. An instrumental-variable analysis of NSS rounds 60, 71 and 75 likewise found no significant effect of scheme enrolment on out-of-pocket spending per hospitalisation, and no significant difference in catastrophic expenditure between covered and uninsured households [PMID 41913189]. Against this, the national CAMS 2022-23 analysis found publicly funded cover raised outpatient use (adjusted OR 1.404, 95% CI 1.328 to 1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128 to 1.210) and reduced inpatient catastrophic expenditure at the 20% threshold (adjusted OR 0.790, 95% CI 0.737 to 0.846), while leaving outpatient care as the residual burden — hence the standing recommendation to extend the benefit package to outpatient care and anchor it in the Ayushman Arogya Mandirs [PMID 42167946].

Global and WHO developments

- **WHO universal health coverage position (fact sheet of 5 December 2025).** About **4.6 billion people were not fully covered by essential health services in 2023**; in 2022 **2.1 billion people (26% of the world's population) faced financial hardship** from out-of-pocket health payments, of whom **1.6 billion were living in poverty or pushed deeper into it**. The UHC service coverage index rose from 54 to 71 between 2000 and 2023. The two monitoring indicators to name are **SDG 3.8.1 (service coverage index)** and **SDG 3.8.2 (proportion facing financial hardship)** [WHO 2025-Dec].
- **ILO World Social Protection Report 2024-26 (12 September 2024).** More than half the world's population is now covered by at least one social protection benefit — the ILO's own framing of the social security definition quoted at the top of this answer — but **3.8 billion people remain entirely unprotected**, and the report links that gap to climate vulnerability [ILO 2024-Sep].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7, Ch 10, Ch 12, Ch 18); Park Plus (Ch 12 — Medicine and Social Sciences); Suryakantha — Community Medicine with Recent Advances (Ch 17 Occupational Health including Fig 17.2, Ch 29 Preventive Geriatrics); Sunder Lal — Textbook of Community Medicine (Ch 5, Ch 11, Ch 16, Ch 17); Jugal Kishore — National Health Programmes of India (National Social Assistance Programme, 1995); Health Economics (Ch 12 — Achieving universal coverage, Table 12.1); Oxford Textbook of Public Health 7e (Ch 21). Coverage note: Park 28e itself carries only a two-line statement that social security includes social insurance and social assistance, so the operational definitions of the two terms and their point-by-point comparison are grounded on Park Plus and Suryakantha within the same indexed corpus.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness Centres as Ayushman Arogya Mandirs with a Tagline of 'Arogyam Paramam Dhanam' — A Step Towards Holistic Care, Journal of Family Medicine and Primary Care 2026, DOI [10.4103/jfmpc.jfmpc.1966.24](https://doi.org/10.4103/jfmpc.jfmpc.1966.24); 41816146 — Ayushman Arogya Mandir and India's Progress Toward Universal Health Coverage: A Narrative Review, Journal of Family Medicine and Primary Care 2026, DOI [10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 39252993 — Impact of Health Benefit Package Policy Interventions on Service Utilisation Under Government-Funded Health Insurance in Punjab, India, The Lancet Regional Health Southeast Asia 2024, DOI [10.1016/j.lansea.2024.100462](https://doi.org/10.1016/j.lansea.2024.100462); 42284657 — Does Integrating Pharmaceutical Coverage into Publicly Financed Health Insurance Reduce Out-of-Pocket Spending? Difference-in-Differences Evidence from Kerala, India, Social Science & Medicine 2026, DOI [10.1016/j.socscimed.2026.119483](https://doi.org/10.1016/j.socscimed.2026.119483) (abstract-only); 40552236 — Understanding Out-of-Pocket Expenditure in India: A Systematic Review, Frontiers in Public Health 2025, DOI [10.3389/fpubh.2025.1594542](https://doi.org/10.3389/fpubh.2025.1594542); 42167946 — Is Publicly Funded Health Insurance Associated with Healthcare Utilisation and Financial Protection in India? Evidence from a Cross-Sectional Household Survey (CAMS 2022-2023), BMJ Open 2026, DOI [10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 39096525 — Informal Employment and High Burden of Out-of-Pocket Healthcare Payments Among Older Workers: Evidence from the Longitudinal Ageing Study in India, Health Policy and Planning 2025, DOI [10.1093/heapol/czae074](https://doi.org/10.1093/heapol/czae074); 42257113 — Exploring Pension Access and Health-Seeking Behavior Among Older Adults in South Kerala: A Mixed-Methods Study, Journal of Family Medicine and Primary Care 2026, DOI [10.4103/jfmpc.jfmpc.624.25](https://doi.org/10.4103/jfmpc.jfmpc.624.25); 38273246 — Direct Benefit Transfer for Nutritional Support of Patients with TB in India: Analysis of National TB Program Data of 3.7 Million Patients, 2018-2022, BMC Public Health 2024, DOI [10.1186/s12889-024-17777-7](https://doi.org/10.1186/s12889-024-17777-7); 38482155 — Averting Catastrophic Tuberculosis Costs in an Indian State: Integration of Ayushman Bharat Arogya Karnataka with the National Tuberculosis Elimination Program, The Lancet Regional Health Southeast Asia 2024, DOI [10.1016/j.lansea.2023.100327](https://doi.org/10.1016/j.lansea.2023.100327); 41451044 — Assessing the Role of Pradhan Mantri TB Mukh Bharat Abhiyaan in India's Fight Against TB: A National-Level Data-Based Research Investigation, Indian Journal of Community Medicine 2025, DOI [10.4103/ijcm.ijcm.418.24](https://doi.org/10.4103/ijcm.ijcm.418.24); 39135015 — The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) After Four

Years of Implementation: Is It Making an Impact on Quality of Inpatient Care and Financial Protection in India?, BMC Health Services Research 2024, DOI [10.1186/s12913-024-11393-2](https://doi.org/10.1186/s12913-024-11393-2); 41913189 — Do Publicly Funded Health Insurance Schemes Reduce Out-of-Pocket Expenditure? Evidence from India, BMC Health Services Research 2026, DOI [10.1186/s12913-026-14456-8](https://doi.org/10.1186/s12913-026-14456-8). Also used: WHO universal health coverage fact sheet, 5 December 2025; ILO World Social Protection Report 2024-26, 12 September 2024.



Functions of public health laboratory

Asked in: Paper II 16-Jul-2020 — RGUHS MD Community Medicine

Definition — A public health laboratory is the laboratory arm of the health system whose client is the **population, not the individual patient**. Laboratory science is one of the constituent scientific disciplines of public health itself, alongside epidemiology, biostatistics, social sciences and demography (Park 28e, Ch 2, p 66). Its work answers the public-health functions of *investigating and diagnosing health problems* — *microbial, societal and chemical* and of *identifying, measuring, monitoring and anticipating community health needs through surveillance* (Oxford Textbook of Public Health 7e, Ch 1, pp 132-133).

Core functions

#	Function	What the laboratory actually does	Indian illustration
1	Diagnostic confirmation of the individual case	Converts a <i>suspected</i> or <i>probable</i> case into a confirmed case — "clinical diagnosis confirmed by an appropriate laboratory test" — so that case definitions used in surveillance are uniform	Free diagnostics assured at every level from sub-health centre upward under National Health Policy 2017 and Indian Public Health Standards (IPHS) 2022 (Jugal Kishore, p 490; Park 28e, Ch 10, p 728)
2	Support to disease surveillance	Laboratories are themselves reporting units in a surveillance system and are asked to collect and test specimens needed for laboratory confirmation and characterisation of notifiable conditions	"To improve laboratory support for disease surveillance" is a stated specific objective of the Integrated Disease Surveillance Programme (IDSP) , and upgradation of laboratories is one of its five project activities (Sunder Lal, Ch 7, p 306; Oxford 7e, Ch 42, p 1572)
3	Outbreak and epidemic investigation	Verifies the diagnosis at the outset, then tests clinical specimens (stool, blood, sputum, wound), food and water samples and other environmental samples chosen by the hypothesis about agent and source	Laboratory analysis is a formal step in the outbreak-investigation sequence; district surveillance units are charged with "coordinating activities of the public health laboratories, medical colleges, NGOs and private sector" (Nelson & Williams, Ch 5, pp 130-138; Jugal Kishore, pp 489-490)

#	Function	What the laboratory actually does	Indian illustration
4	Reference and specialised testing	Typing and sub-typing beyond the reach of a district laboratory — speciation, pulsed-field gel electrophoresis (PFGE) , plasmid analysis, molecular tests, culture and drug-susceptibility testing	National Institute of Virology (NIV) Pune and National Centre for Disease Control (NCDC) Delhi act as apex laboratories for confirmation during outbreak investigation, e.g. for Zika virus disease (Park 28e, Ch 5, pp 340-341)
5	Antimicrobial resistance (AMR) surveillance	Generates resistance data for common bacterial pathogens, defines trends, and maintains a national repository of bacterial strains and cultures to inform treatment guidelines	National Programme on Containment of AMR (launched 2013, NCDC the nodal agency) runs surveillance through a network of ten laboratories (Sunder Lal, Ch 19, p 783)
6	Sero-epidemiology and programme monitoring	Serological (immunity) surveys measure the proportion immune in a population — the only way herd immunity can be determined	Serological surveys are the stated method of determining herd immunity and hence of judging an immunisation programme (Park 28e, Ch 3, p 123)
7	Environmental monitoring — water	Sanitary survey, sampling, bacteriological, biological and chemical surveillance of drinking water; a surveillance programme must make provision for central and regional analytical laboratory services	See the water-testing table below (Park 28e, Ch 15, pp 877-881)
8	Food safety and detection of adulteration	Analyses statutory food samples and reports whether the article conforms to prescribed standards	137 food laboratories under state/union territory governments and local bodies, plus four Central Food Laboratories (Mysore, Kolkata, Ghaziabad, Pune) whose report is final and appellate; 13 National Reference Laboratories approved (Sunder Lal, Ch 5, p 219; Park 28e, Ch 11, pp 801-802)
9	Quality assurance of the laboratory network	External quality assessment (EQA), panel testing, on-site evaluation, random blinded rechecking, supply of standard reagents, accreditation to National Accreditation Board for Testing and Calibration Laboratories (NABL) standards	A state-level laboratory exists "for supporting the laboratory service network in the districts, for establishing external quality assurance, for ensuring the availability of quality reagents" (Jugal Kishore, p 493; p 289)
10	Training, research and human-resource development	Trains laboratory technicians and medical officers, standardises techniques, and undertakes applied research into the aetiology of diseases of local public-health importance, endemic or epidemic	Medical colleges are integrated into IDSP as reference laboratories, and for quality assurance, training, epidemic investigation and non-communicable disease surveillance (Jugal Kishore, pp 490-491)

#	Function	What the laboratory actually does	Indian illustration
11	Public health emergency preparedness	Provides tiered capacity to <i>rule out</i> , <i>rule in</i> and finally <i>confirm and type</i> a deliberate or unusual biological threat	The Laboratory Response Network model: Sentinel (level A) rules out, Reference (level B) rules in and does susceptibility testing, Regional (level C) gives confirmatory diagnosis and typing (Maxcy-Rosenau-Last 15e, Ch 79, pp 1318-1319)
12	Biosafety and safe disposal of its own waste	Segregates and pre-treats infectious laboratory waste before it leaves the premises	Microbiology, biotechnology and other clinical laboratory waste (cultures, stocks, specimens, live or attenuated vaccines) = Yellow category , pre-treated to sterilise on site per NACO/WHO guidance, then incinerated (Biomedical Waste Management Rules 2016, Schedule I; Park 28e, Ch 16, p 947)

The public health laboratory network in India — level by level

Level	Laboratory unit	Function
Sub-health centre / Ayushman Arogya Mandir	Point-of-care testing only	Free diagnostics and essential drugs; first recognition of an outbreak and first response (Sunder Lal, Ch 17, p 745)
Primary health centre	Basic laboratory	Routine microscopy and blood tests; medical officer holds responsibility for surveillance and laboratory services (Park 28e, Ch 7, p 494)
Block — community health centre	Block public health laboratory within the Block Public Health Unit (BPHU)	One of the BPHU's four functional areas (clinical service delivery, public health functions, block public health laboratory, health management information system unit); serves both clinical work and disease surveillance (Sunder Lal, Ch 17, p 752)
District	Integrated Public Health Laboratory (IPHL) ; district hospital microbiology laboratory	Diagnostic work and epidemiological investigation within the district; IPHLs are being set up in all districts under PM-ABHIM (Sunder Lal, Ch 20, pp 787, 805; Jugal Kishore, p 493)
State	State/intermediate reference laboratory attached to the State Epidemiology Centre	Supports the district network, runs external quality assurance, ensures quality reagents, investigates aetiology of endemic and epidemic disease (Jugal Kishore, p 493)
Regional / medical college	Referral laboratories linked to adjoining districts	IDSP referral laboratory network built on existing medical-college laboratories — begun in 9 states (Gujarat, Punjab, Rajasthan, Uttarakhand, Karnataka, Tamil Nadu, Maharashtra, Andhra Pradesh, West Bengal), with 23 further medical-college laboratories added (Jugal Kishore, pp 492-493)

Level	Laboratory unit	Function
National / apex	NCDC, NIV Pune, National AIDS Research Institute (NARI) Pune and programme-specific national reference laboratories	Final confirmation, typing, external quality assessment for the whole country, guideline-setting

Programme-specific tiering (examples to quote):

- **National Tuberculosis Elimination Programme** — National Reference Laboratories (6) → Intermediate Reference Laboratories at state level (34) → culture and drug-susceptibility testing laboratories in medical colleges and the private sector (61) → district and sub-district nucleic acid amplification test sites (6,496) → peripheral designated microscopy centres (24,573) (Park 28e, Ch 7, pp 510-511).
- **National AIDS Control Programme external quality assessment scheme** — apex laboratory NARI Pune (tier 1) → 13 National Reference Laboratories (tier 2) → 118 State Reference Laboratories (tier 3) → district level, i.e. all integrated counselling and testing centres and blood banks (tier 4) (Jugal Kishore, pp 341-342).
- **Dengue and chikungunya under NVBDCP** — 521 sentinel surveillance hospitals with laboratory support, backed by 14 Apex Referral Laboratories; diagnostic kits supplied through NIV Pune (Park 28e, Ch 7, p 502).

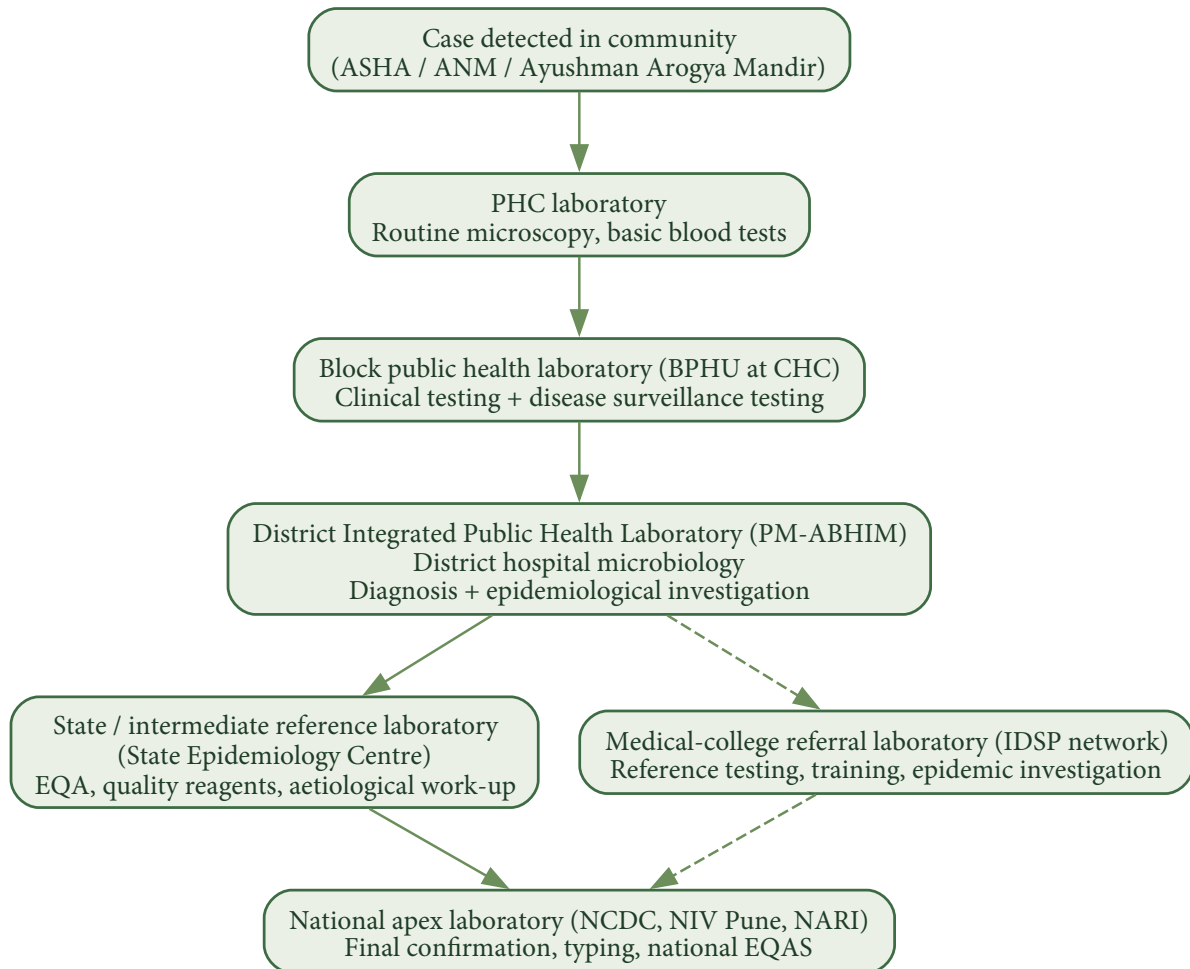
Environmental and food testing — what is actually done

Sample	Tests the public health laboratory performs	Standard / remark
Drinking water — bacteriological	Presumptive coliform test by the multiple tube method giving the most probable number (MPN) per 100 ml; confirmatory subculture in brilliant green bile broth; tests for faecal streptococci and <i>Clostridium perfringens</i> ; colony (plate) count	<i>Escherichia coli</i> must not be detectable in any 100 ml sample ; in large supplies, coliforms must be absent in 95% of samples taken through any 12-month period (Park 28e, Ch 15, pp 877, 881)
Drinking water — chemical	pH, colour, turbidity, chlorides, ammonia, chlorine demand and residual chlorine are the basic tests; also iron and toxic chemicals	Regular measurement of chlorine residual may in part replace bacteriological surveillance (Park 28e, Ch 15, p 881)
Drinking water — biological	Type and number of organisms present in the water	Complements the bacteriological result
Food	Analysis of statutory samples against prescribed standards under the Food Safety and Standards Act 2006 (in force 5 August 2011, Ministry of Health and Family Welfare)	Three samples are drawn — one to the designated laboratory, one left with the retailer, one kept with the food authority at district level (Sunder Lal, Ch 5, p 219)
Clinical chemistry for NCD programmes	Blood sugar, total cholesterol and lipid profile, blood urea, creatinine at community health centre and district hospital NCD clinics	Supports opportunistic screening for common cancers and NCD case management (Sunder Lal, Ch 8, p 506)

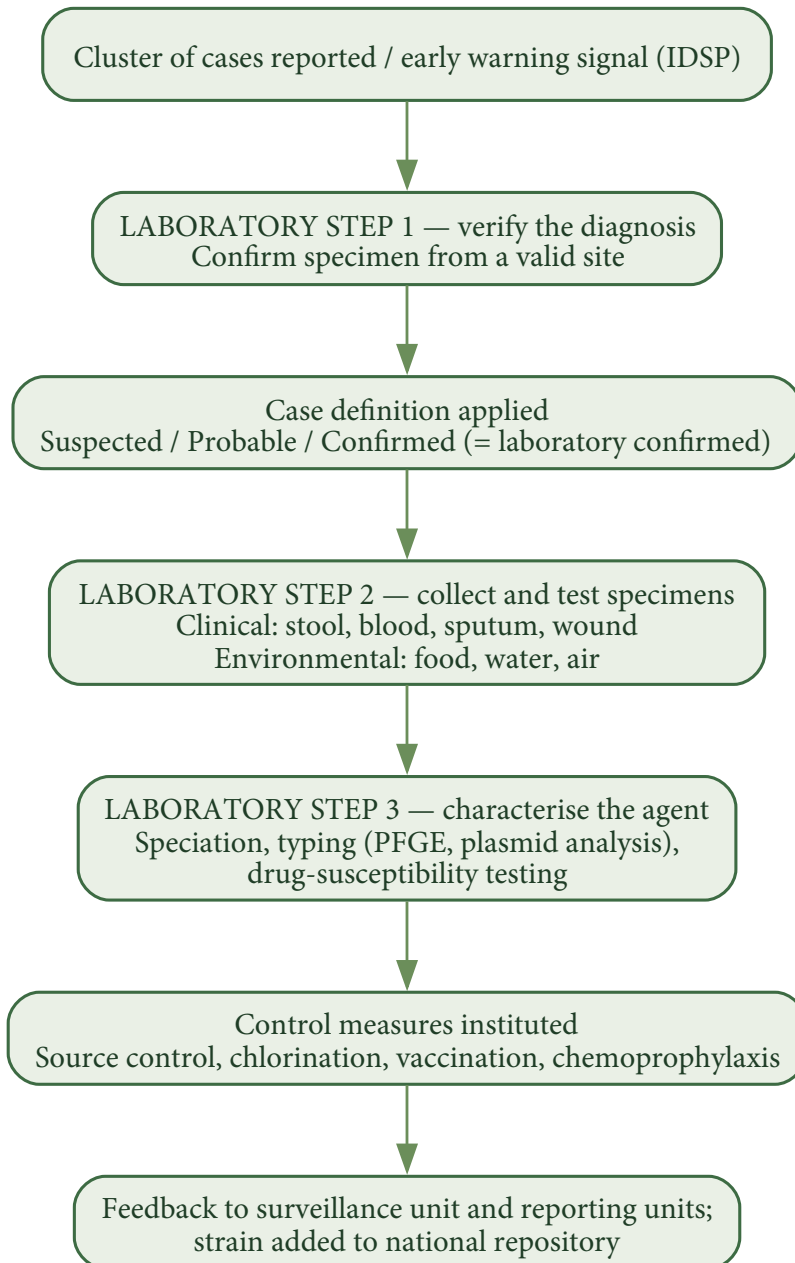
A **sanitary survey** — on-the-spot inspection of the whole water supply system by a qualified person — must accompany the sampling, because it is essential for adequate interpretation of the laboratory result (Park 28e, Ch 15, p 880).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Label list for a hand-drawn figure — "Elements of drinking-water surveillance": (1) Sanitary survey, (2) Sampling, (3) Bacteriological examination — presumptive coliform test, confirmatory test, faecal streptococci, *C. perfringens*, colony count, (4) Biological examination, (5) Chemical surveillance — residual chlorine.



Flow of a specimen through the tiered public health laboratory network in India



Role of the public health laboratory in outbreak detection and response

High-yield exam pointers

- ▶ **Definition line to open with:** laboratory sciences are one of the constituent scientific disciplines of public health (Park 28e, Ch 2, p 66) — the client is the population, not the patient.
- ▶ **Four functional areas of the Block Public Health Unit:** clinical service delivery, public health functions, **block public health laboratory**, health management information system unit — the laboratory serves both clinical care and disease surveillance.
- ▶ **PM-ABHIM laboratory components:** Integrated Public Health Laboratory in every district, 9 Biosafety Level-III laboratories, 4 new national institutes of virology, One Health national institute, strengthening of NCDC.

- ▶ **Water standard to quote:** *E. coli* must not be detectable in any 100 ml sample; coliforms absent in 95% of samples over any 12-month period; presumptive coliform test = multiple tube method giving MPN/100 ml.
- ▶ **Food:** 4 Central Food Laboratories (Mysore, Kolkata, Ghaziabad, Pune) — report is final; three samples drawn (laboratory / retailer / district food authority).
- ▶ **Tier mnemonic — "P-B-D-S-N":** PHC → Block → District (IPHL) → State reference → National apex; each higher tier adds reference testing, external quality assessment and training.
- ▶ **AMR:** National Programme on Containment of AMR, 2013, NCDC nodal, ten-laboratory surveillance network, national repository of strains.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the Integrated Public Health Laboratory in every district and the Block Public Health Unit laboratory are commitments under the Pradhan Mantri Ayushman Bharat Health Infrastructure Mission (PM-ABHIM) and are still being rolled out, so a textbook that predates 2021 will describe only the district hospital microbiology laboratory at that level. The retrieved corpus carries no single consolidated "functions of a public health laboratory" list for India, so the functions above are assembled from the laboratory content of the surveillance, water, food, tuberculosis, HIV and AMR chapters of the cited books; the tiered Laboratory Response Network (Sentinel-Reference-Regional) is a United States model quoted only as the standard illustration of emergency-preparedness tiering.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO consolidated TB diagnostics recommendations 2026, WHO Pandemic Agreement 2025, International Health Regulations (2024 amendments), ICMR

Programme and policy updates (India)

- ▶ **Four zonal National Institutes of Virology under PM-ABHIM** are being established with World Bank support, and a zonal viral-disease prioritisation exercise was run across the four zones in March 2024; 186 participants included the Virus Research and Diagnostic Laboratory (VRDL) network (26%) and IDSP public-health staff (28%), and dengue, influenza, measles, Japanese encephalitis and hepatitis A emerged as the top five priority viruses. Zonal priorities differed — Japanese encephalitis ranked second in the East zone but eighth in the North — so zonal research and outbreak-response priorities are being set zone-by-zone rather than uniformly [PMID 42522023].
- ▶ **Ayushman Bharat Health and Wellness Centres were renamed Ayushman Arogya Mandir in November 2023** with the tagline *Arogyam Paramam Dhanam*, at roughly ₹3,000 per facility and about ₹48 crore overall. Field assessments of the renamed centres report laboratory infrastructure gaps and poor upkeep of biomedical-waste logs and sterilisation records — the laboratory-side weak point at the first tier [PMID 42257165].
- ▶ **NTEP has moved to the Dual-phase Operational Strategy for Tuberculosis elimination (DOST)**, shifting from smear microscopy to upfront molecular testing; India now has over 9,000

nucleic acid amplification test sites covering all districts, and because these are multi-disease molecular platforms they can also cover many diagnostics on the National Essential Diagnostics List [PMID 42024898].

Current national data

- **IDSP weekly outbreak data, 2018–2023:** 583 of 6,948 reported outbreaks (8.3%) were zoonotic, led by Japanese encephalitis (29.5%), leptospirosis (18.7%), scrub typhus (13.9%) and rabies (10.6%); the north-east contributed 35.8%. Late reporting fell from 52.6% in 2019 to 5.2% in 2023, but follow-up reports were missing for 97.2% of zoonotic outbreaks — a laboratory-and-surveillance feedback loop that closes only at the front end [PMID 40492101].
- **Delhi NCR antimicrobial resistance surveillance network, 2020–2024:** 24 participating laboratories reported 10,689 typhoidal *Salmonella* isolates collated monthly on WHONET; multidrug resistance in *S. Typhi* fell to 1% by 2024 and ampicillin resistance fell from 9.6% to 4.1%, while ciprofloxacin non-susceptibility persisted at roughly 48–60% and reached 88–98% in *S. Paratyphi A*. Ceftriaxone resistance remained rare (3 isolates) [PMID 41643898].

Elimination, surveillance and disease status

- **Kala-azar has entered a post-elimination phase in which laboratory surveillance, not case-counting, carries the programme.** In endemic villages of Bihar and Jharkhand, rK39 seroprevalence was 20–30% and *Leishmania*-positive sand-fly pools were found despite indoor residual spraying (1.4% and 1.2% of pools by kDNA qPCR), showing that low-level transmission continues and that combined serological plus entomological testing is needed to sustain elimination [PMID 41324335].
- **Wastewater surveillance is now validated against clinical incidence for typhoid in an Indian city.** In Vellore (December 2022 to June 2023), multiplex qPCR for *ttr*, *staG* and *tvfB* detected *S. Typhi* in 80 of 1,446 wastewater samples (5.5%); each 10-fold rise in blood-culture-confirmed typhoid incidence carried 2.43 times higher odds of wastewater detection (95% CI 1.38–4.27), establishing sewage testing as a pre-vaccine-introduction monitoring tool [PMID 42419341].

Guideline and legislative updates

- **WHO issued its first recommendations on near-point-of-care NAATs and tongue-swab specimens for tuberculosis in February 2026.** Near-point-of-care NAAT on sputum was judged very accurate (sensitivity 85.4%, 95% CI 82.2–88.1; specificity 97.6%) and on tongue swabs accurate (sensitivity 76.4%, 95% CI 71.2–80.9; specificity 99.2%); the swab-optimised Truenat MTB Ultima (Molbio, Goa) is among the recommended low-complexity platforms, and the first-in-class card test is catalogued at USD 3.60 per test. Sputum remains the preferred specimen, and these tests still need reflex testing for rifampicin resistance [PMID 42024898].
- **Quality assurance of malaria rapid diagnostic tests is argued to belong among national laboratory functions,** resting on pre-market lot testing, post-market surveillance and competency-based training; ICMR and the Central Drugs Standard Control Organization have jointly issued a standardised quality-assurance protocol for these kits. Indian lot testing between 2007 and 2017 examined 6,056 lots — about 1.6 billion tests — of which 69 (1.1%) failed, and the review recommends that lot testing be made mandatory and carried out by authorised regulatory bodies: the malaria counterpart of the external quality assessment role the state laboratory plays for the district network [PMID 42165729].

Global and WHO developments

- The international legal framework the laboratory reports into has been rewritten twice in two years: amendments to the International Health Regulations (2005) were agreed at the 77th World Health Assembly on 1 June 2024 and entered into force in September 2025, and the WHO Pandemic Agreement was adopted at the 78th World Health Assembly on 20 May 2025, with equitable access to vaccines, diagnostics and therapeutics as a core pillar [PMID 40589223].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 3 Principles of Epidemiology; Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 11 Nutrition and Health; Ch 15 Environment and Health; Ch 16 Hospital Waste Management); Sunder Lal — Textbook of Community Medicine (Ch 5 Nutrition; Ch 7 Epidemiology; Ch 8 Communicable and Non-communicable Diseases; Ch 17 Healthcare of the Community; Ch 19 Essential Medicines; Ch 20 Recent Advances); Jugal Kishore — National Health Programmes of India (IDSP, RNTCP/NTEP, NACP chapters); Oxford Textbook of Public Health 7e (Ch 1 The scope and concerns of public health; Ch 42 Public health surveillance); Nelson & Williams — Infectious Disease Epidemiology (Ch 5 Outbreak Epidemiology); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 79 Disaster Preparedness and Response); Biomedical Waste Management Rules 2016 (Schedule I); Food Safety and Standards Act 2006; National Health Policy 2017; Indian Public Health Standards 2022.

Recent-advances sources: 42522023 — Strengthening India's pandemic preparedness with the four zonal institutes of virology: stakeholder consultations for outbreak response and collaborative research priorities, *Infectious Diseases of Poverty* 2026;15(1):84, doi:[10.1186/s40249-026-01484-z](https://doi.org/10.1186/s40249-026-01484-z); 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, *Journal of Family Medicine and Primary Care* 2026;15(3):1459-1460, doi:[10.4103/jfmprc.jfmprc_1966_24](https://doi.org/10.4103/jfmprc.jfmprc_1966_24); 42024898 — New tools and policies herald a major advance in tuberculosis diagnosis, *Indian Journal of Medical Research* 2026;163(3):265-268, doi:[10.25259/IJMR_923_2026](https://doi.org/10.25259/IJMR_923_2026); 40492101 — Zoonotic disease outbreaks reported under India's Integrated Disease Surveillance Programme, 2018-2023: a cross-sectional analysis of national surveillance data, *The Lancet Regional Health — Southeast Asia* 2025;37:100601, doi:[10.1016/j.lansea.2025.100601](https://doi.org/10.1016/j.lansea.2025.100601); 41643898 — Tracking antimicrobial resistance in typhoidal Salmonella, 2020-2024: output from an antimicrobial resistance surveillance network in New Delhi, India, *Indian Journal of Medical Microbiology* 2026;60:101074, doi:[10.1016/j.ijmm.2026.101074](https://doi.org/10.1016/j.ijmm.2026.101074) (abstract-only); 41324335 — High seropositivity to rK39 and capture of Leishmania-infected sand flies in Bihar and Jharkhand emphasize the need for post-elimination surveillance of visceral leishmaniasis in India, *The Journal of Infectious Diseases* 2026;233(3):e802-e811, doi:[10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543) (abstract-only); 42419341 — Integrating environmental and clinical surveillance for typhoid: insights from a prospective wastewater surveillance study in urban India, *The Lancet Microbe* 2026:101414, doi:[10.1016/j.lanmic.2026.101414](https://doi.org/10.1016/j.lanmic.2026.101414) (abstract-only); 42165729 — Quality assurance in rapid diagnostic tests for accurate diagnosis of malaria, *Indian Journal of Medical Research* 2026;163(4):461-468, doi:[10.25259/IJMR_2477_2025](https://doi.org/10.25259/IJMR_2477_2025); 40589223 — The 2025 World Health Assembly Pandemic Agreement and the 2024 amendments to the International Health Regulations combine for pandemic preparedness and response, *Medical Science Monitor* 2025;31:e950411, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411). Plus WHO recommendations on near-point-of-care NAATs and tongue-swab testing for tuberculosis (February 2026); amendments to the

International Health Regulations (2005) adopted 1 June 2024, in force September 2025; WHO Pandemic Agreement adopted 20 May 2025.



Health care package

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — A health care package is the explicitly defined and costed set of promotive, preventive, curative, rehabilitative and palliative services that a health system guarantees to a defined population at a defined level of care, financed publicly and delivered free or at negligible cost at the point of service. In India the operative package has two arms: the twelve-service **comprehensive primary health care** (CPHC) package delivered at **Ayushman Arogya Mandir (AAM)**, the renamed Ayushman Bharat Health and Wellness Centre (HWC), and the secondary and tertiary **health benefit package** purchased under **Ayushman Bharat–Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)** (Park 28e, Ch 7, pp 535, 570-571). Together they constitute India's operational route to universal health coverage (UHC).

Why a package is defined — principles and selection criteria

Principle	What it means	Application in India
Explicit entitlement	The citizen and the provider both know exactly which services are assured, and at which facility	Twelve CPHC packages defined facility-wise under Indian Public Health Standards (IPHS) 2022 (Park 28e, Ch 10, p 731)
Selection by value	Services chosen on value for money, disease burden and feasibility of implementation (<i>Disease Control Priorities</i> 3e)	Screening for hypertension, diabetes and oral, breast and cervical cancer added first among non-communicable diseases (Park 28e, Ch 7, p 570)
Comprehensive, not selective	Paradigm shift from a sectoral, selective set of vertical programmes to a comprehensive range of services	Before Ayushman Bharat, reproductive-child-health and disease-control programme services covered under 15% of the care people actually sought (Sunder Lal, Ch 17, p 749)
Life-cycle / continuum-of-care approach	Package follows the individual through adolescence, pregnancy, birth, infancy, preschool, school, reproductive, adult and ageing phases	Each of the twelve services carries promotive, preventive and curative components (Sunder Lal, Ch 17, pp 746, 764)
Free at the point of care	No user charge for the guaranteed services	Free essential drugs and diagnostics at every Ayushman Arogya Mandir (Park 28e, Ch 7, p 570)
Two-way referral linkage	Continuity between primary, secondary and tertiary levels	Systemic linkage plus a unique identity linking every family to a primary care facility (Sunder Lal, Ch 16, p 699)
Financial risk protection	Package protects against catastrophic out-of-pocket expenditure	AB PM-JAY cover of Rs 5,00,000 per family per year (Park 28e, Ch 10, p 745)

Principle	What it means	Application in India
Geographical accessibility	Package must be reachable	"Time to care" not more than 30 minutes of travel (Sunder Lal, Ch 17, p 742)

The twelve packages of comprehensive primary health care

Numbering as listed in Park 28e, Ch 10, p 731 and Sunder Lal, Ch 17, pp 749-750.

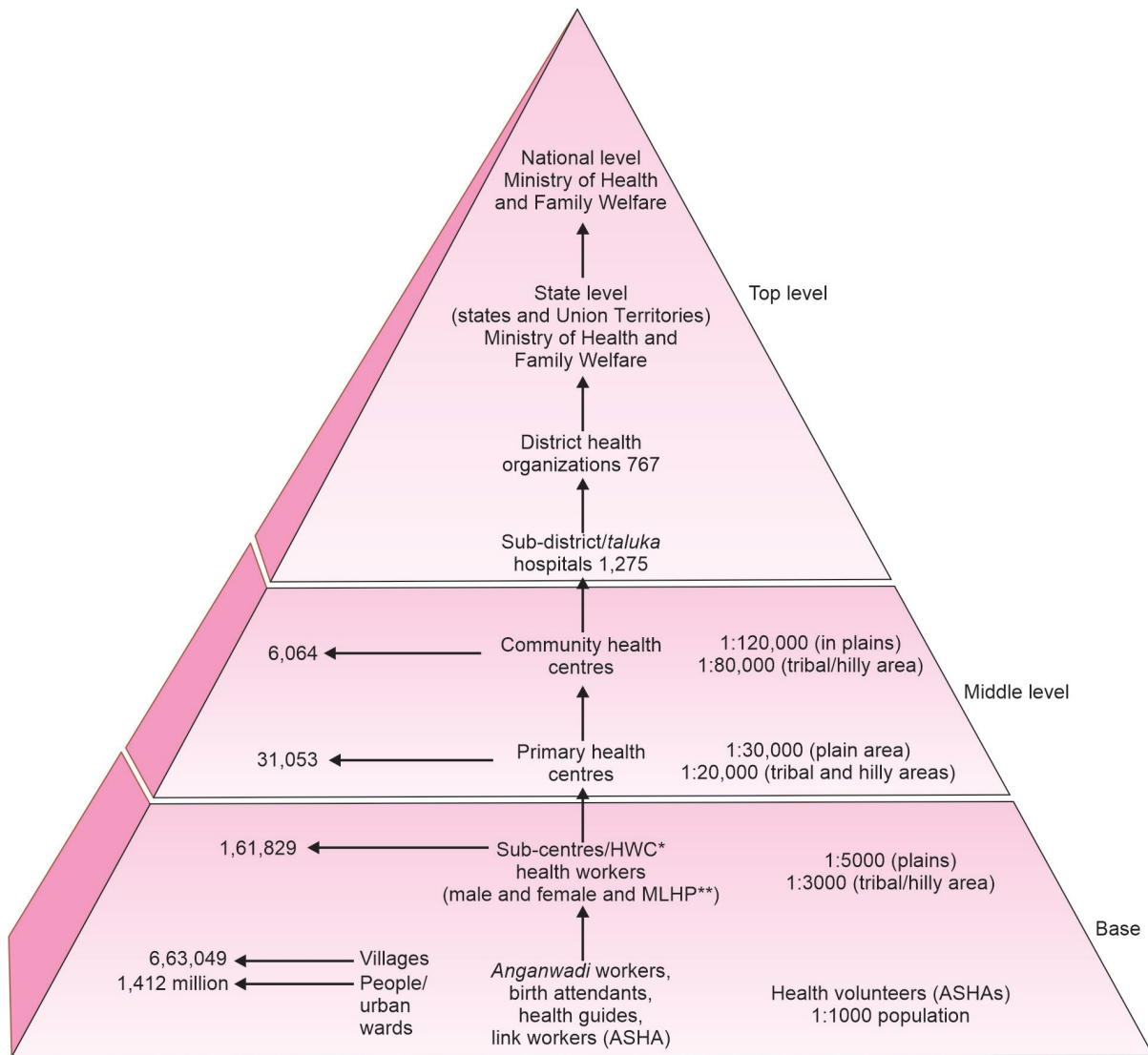
No.	Package	Illustrative content
1	Care in pregnancy and childbirth	Registration, antenatal check-ups, tetanus and adult diphtheria immunisation, iron-folic-acid and calcium supplementation, birth planning, institutional delivery, postnatal care
2	Neonatal and infant health care services	Home-based newborn care — 7 visits after home delivery and 6 after institutional delivery; identification of low-birth-weight, preterm and sick newborn; early breastfeeding counselling; growth monitoring; identification of birth asphyxia, sepsis and congenital anomaly with referral (Park 28e, Ch 10, p 733)
3	Childhood and adolescent health care services	Full immunisation, vitamin A, deworming, management of acute respiratory infection and diarrhoea, school health, Rashtriya Bal Swasthya Karyakram and adolescent health services
4	Family planning, contraceptive services and other reproductive health care services	Spacing and limiting methods, home delivery of contraceptives, menstrual hygiene, safe abortion referral, reproductive tract and sexually transmitted infection care
5	Management of communicable diseases — national health programmes	Tuberculosis, leprosy, vector-borne disease, HIV; case finding, treatment, adherence support and notification
6	Management of common communicable diseases and out-patient care for acute simple illness and minor ailments	General out-patient department care, first-contact curative care, essential drugs
7	Screening, prevention, control and management of non-communicable diseases	Population enumeration of those over 30 years by the accredited social health activist (ASHA), risk scoring on the community-based assessment checklist (CBAC) — score above 4 marks high risk; screening for hypertension, diabetes and oral, breast and cervical cancer; monthly follow-up for treatment adherence (Sunder Lal, Ch 8, pp 503, 529-530)
8	Care for common ophthalmic and ENT problems	Screening for visual acuity, cataract and refractive error; awareness and management of acute and chronic ear, nose and throat disease; referral for surgical care (Park 28e, Ch 10, pp 735-736)
9	Basic oral health care	Management of tooth abscess and caries, topical fluoride application, mobilisation for oral cancer screening, tobacco-cessation counselling and referral (Park 28e, Ch 10, p 736)

No.	Package	Illustrative content
10	Elderly and palliative health care services	Identification of high-risk elderly, home visits to home-bound and bedridden elderly, family support in palliative care, supportive appliances, linkage to day-care centres and hospices (Park 28e, Ch 10, pp 736-737)
11	Emergency medical services including burns and trauma	Community awareness, first aid for minor injury, fracture, animal bite and poisoning; triage on the airway–breathing–circulation–disability criteria and stabilisation before referral (Park 28e, Ch 10, p 737)
12	Screening and basic management of mental health ailments	Diagnosis and treatment of common mental illness, suicide risk assessment, emergency management of poisoning and seizures, counselling of patient and family, referral and follow-up (Park 28e, Ch 10, p 735)

Where the package is delivered — facility, norm and team

Facility	Population norm	Package delivered	Core team
Community / household	ASHA 1 per 1000 population, or 1 per habitation in tribal, hilly and desert areas	Home visits, village health sanitation and nutrition day, enumeration, CBAC screening, mobilisation	ASHA, anganwadi worker
AAM–Sub Health Centre (rural)	5000 plain; 3000 hilly, tribal and desert	All twelve packages at the primary level, open 6 hours a day, plus teleconsultation	Community health officer (CHO) / mid-level health provider 1, multipurpose health worker 1 male + 1 female, 5 ASHAs; 2 day-care beds with oxygen; performance standard 20 out-patients per day (Park 28e, Ch 10, p 731; Sunder Lal, Ch 17, Table 17.3)
Urban AAM (UHC)	1 per 15,000–20,000, chiefly slum and vulnerable populations	Same twelve packages, urban outreach	Medical officer, staff nurse/pharmacist, male multipurpose worker, support staff
AAM–PHC (rural)	30,000 plain; 20,000 hilly and tribal	Twelve packages plus medical officer out-patient care; delivery services at 24×7 PHC; 6 beds	Medical officer (MBBS), AYUSH medical officer, staff nurses
Urban PHC / Polyclinic	Urban PHC 50,000; polyclinic 2.5–3 lakh	Ambulatory specialist care in at least six specialties — medicine, obstetrics-gynaecology, paediatrics, ophthalmology, dermatology, psychiatry	Specialists on fixed-day rotation (Sunder Lal, Ch 17, pp 758-759)

Facility	Population norm	Package delivered	Core team
Community Health Centre / district hospital	CHC 1:1,20,000 plains, 1:80,000 tribal and hilly	Referral and secondary care package	Specialists



Health system organization showing the levels at which the package is delivered (Sunderlal, p.709)

Enablers that make the package deliverable

- **Human resource** — addition of the community health officer / mid-level health provider (a B.Sc. Nursing graduate or a BAMS physician) to the existing auxiliary nurse midwife (ANM) and ASHA team at the sub-centre (Sunder Lal, Ch 17, p 742).
- **Free essential medicines and diagnostics** at every centre, with expanded point-of-care testing (Park 28e, Ch 7, p 570).
- **Teleconsultation** linking the sub-centre team to the medical officer and specialists, with a robust information-technology system and electronic health records (Sunder Lal, Ch 17, p 764).

- **Population enumeration and empanelment** — family health folders and individual health records for every family in the catchment area, which convert the package into a traceable entitlement (Sunder Lal, Ch 17, pp 742-743).
- **Infrastructure and branding** — renovated buildings, national identity number, sufficient space, two day-care beds with oxygen support.
- **Capacity building, community mobilisation and intersectoral convergence**, with a functioning referral and transport support system.
- **Health promotion and wellness activities** — yoga sessions, meditation, open gyms, cyclathons, tobacco and alcohol cessation, "eat right and safe" messaging (Park 28e, Ch 7, p 570).

The secondary and tertiary arm — AB PM-JAY health benefit package

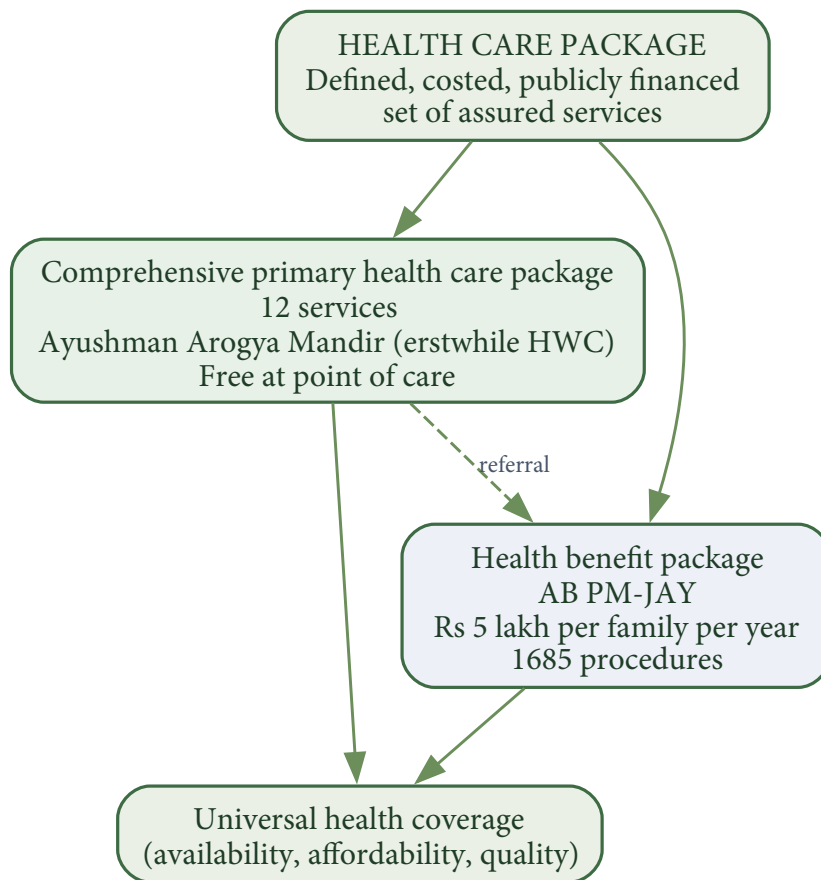
Feature	Provision
Cover	Rs 5,00,000 per family per year for secondary and tertiary care hospitalisation
Beneficiaries	About 10.74 crore poor and vulnerable families identified from the Socio-Economic Caste Census; entitlement-based, no cap on family size, age or gender
Benefit package	1685 procedures covering treatment, food, drugs and supplies, and diagnostics
Continuity	3 days of pre-hospitalisation and 15 days of post-hospitalisation follow-up care
Access	Cashless and paperless at any empanelled public or private hospital, portable across the country; all pre-existing conditions covered from day one
Extension	Senior citizens covered on a family basis, with a choice between AB PM-JAY and an existing scheme such as the Central Government Health Scheme or the Ex-Servicemen Contributory Health Scheme (Park 28e, Ch 7, p 571; Ch 10, p 745; Sunder Lal, Ch 17, p 763)

Monitoring and evaluation of the package

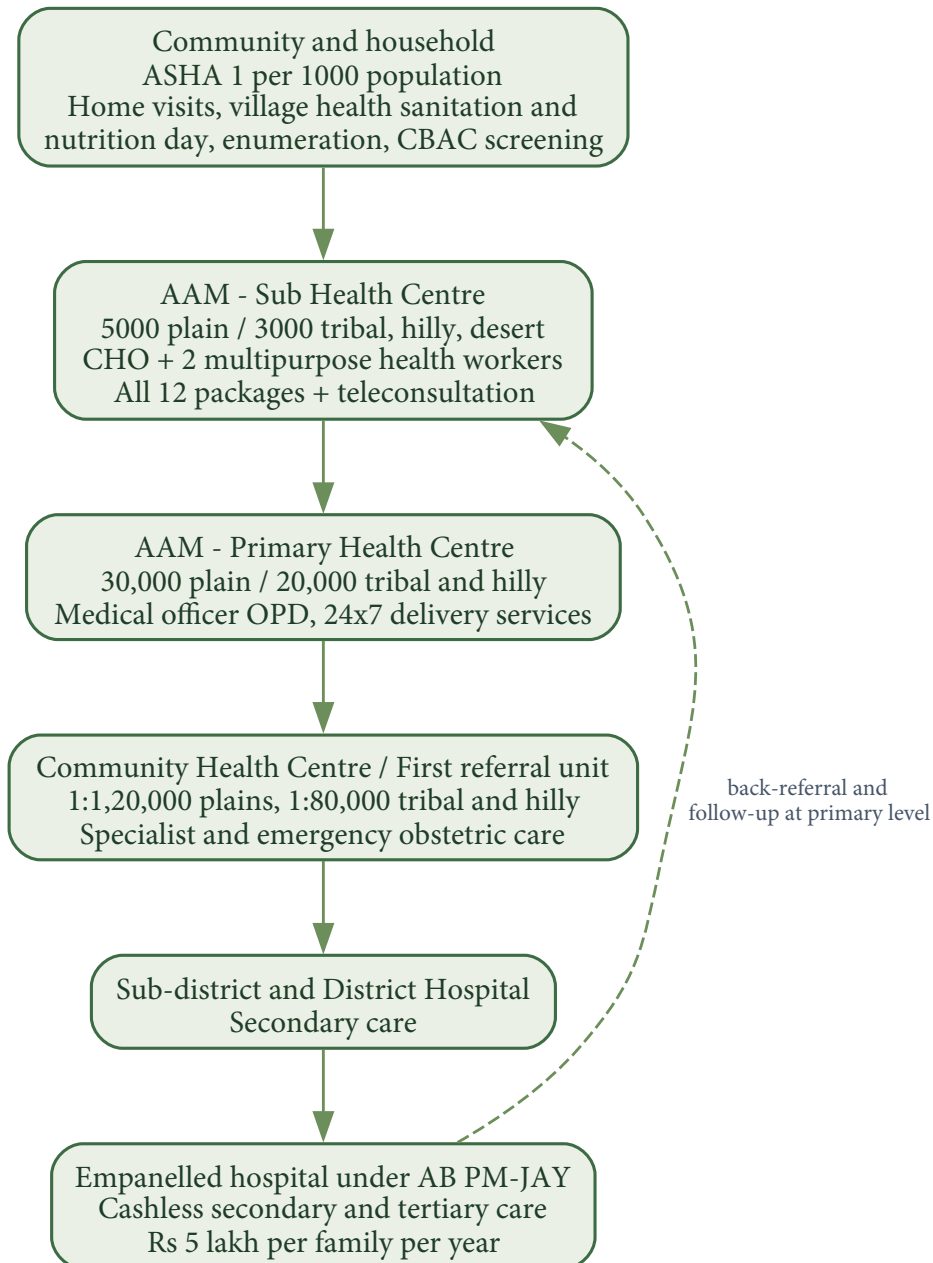
Level	Indicators
Input	Centres made functional (1,72,148 operational as on 31 March 2024, Park 28e, Ch 7, p 570); CHO in position; drug and diagnostic availability
Process	Population enumerated and empanelled; families with health folders; CBAC screening completed in those over 30 years; teleconsultations held
Output (service utilisation)	Antenatal registrations and those receiving 4 antenatal check-ups and 180 iron-folic-acid tablets; institutional deliveries; infants fully immunised; couples protected; persons screened for hypertension and diabetes (Sunder Lal, Ch 17, pp 742-743)
Outcome	Proportion of screened hypertensives and diabetics under control; out-patient footfall shifting from higher facilities to the primary level
Impact	Reduced out-of-pocket expenditure and catastrophic health expenditure; movement of the maternal mortality ratio, infant mortality rate and premature non-communicable disease mortality

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labelled diagram to sketch — "The two arms of India's health care package": draw two vertical columns rising to a common apex labelled *Universal health coverage*. Label the left column **Ayushman Arogya Mandir (erstwhile Health and Wellness Centre)** with the sub-labels *12-service comprehensive primary health care package, free drugs and diagnostics, 1,72,148 centres (31 March 2024)*. Label the right column **AB PM-JAY** with the sub-labels *Rs 5 lakh per family per year, 1685 procedures, 10.74 crore families, secondary and tertiary hospitalisation*. Draw a two-headed arrow between the columns labelled *two-way referral and continuity of care*.



Classification of India's health care package under Ayushman Bharat



Delivery pathway of the comprehensive primary health care package, community to empanelled hospital

High-yield exam pointers

- ▶ **Two components of Ayushman Bharat:** Ayushman Arogya Mandir (comprehensive primary care, 12 packages) + AB PM-JAY (Rs 5 lakh per family per year, secondary and tertiary hospitalisation).
- ▶ **Mnemonic for the 12 packages — "PIC-FaMily Needs Eye, Oral, Elder, Emergency, Mind":** Pregnancy and childbirth; Infant and neonatal; Childhood and adolescent; Family planning and reproductive; Communicable disease programmes; general OPD and Minor ailments; Non-communicable disease; Eye and ENT; Oral; Elderly and palliative; Emergency including burns and trauma; Mental health.
- ▶ **Population norms:** sub-centre AAM 5000 plain / 3000 tribal; urban AAM 15,000–20,000; PHC AAM 30,000 plain / 20,000 tribal; ASHA 1 per 1000 (IPHS 2022, Park 28e Ch 10 p 731).

- **Dated Indian figures:** first HWC at Bijapur, Chhattisgarh on 14 April 2018; 1,72,148 centres operational on 31 March 2024; 10.74 crore families and 1685 procedures under PM-JAY.
- **CBAC score above 4** = high risk for non-communicable disease, prioritised for the weekly NCD day; about 37% of the population is over 30 years (1850 in a population of 5000).
- **"Time to care" 30 minutes** is the accessibility standard for the primary care package.
- Package selection follows **disease burden, value for money and feasibility** (*Disease Control Priorities 3e*).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir**, and Park 28e uses both names interchangeably — write the current name first with the former in parentheses once. Park 28e itself lists the twelve packages in two slightly different orders in Chapter 7 and Chapter 10; the numbering used above follows Chapter 10 (p 731), which matches Sunder Lal, and examiners accept any order provided all twelve are named.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Accounts India, Comprehensive Annual Modular Survey 2022-23, NFHS-5, National Health Authority health benefit package 2022, Bulletin of the World Health Organization

Programme and policy updates (India)

- **The benefit package has grown well past the 1685 procedures the textbooks quote.** The first list (HBP 1.0) carried 1350 procedures, of which only about 21.5% were surgical; the HBP 2022 revision lists over 1949 procedures, 1284 of them (65.8%) surgical or minimally invasive [PMID 41393311]. A parallel account puts current cover at about 1970 secondary and tertiary procedures across 27 specialties. Always date the number you write. [PMID 37963613]
- **Package prices are now set from measured costs rather than negotiated estimates.** The Cost of Health Services in India study (63 hospitals, 14 states) found roughly 42% of HBP 1.0 procedures reimbursed at less than half their actual cost; applying that evidence halved the underpriced share, the 2022 revision raised rates for 832 packages, cut 122 and left 630 unchanged and brought in differential pricing by city tier, and the National Health Authority began a cost-surveillance pilot in 61 hospitals across five states in April 2022 to make costing routine [PMID 37963613]. Health technology assessment is now institutionalised inside the Authority as a Health Financing and Technology Assessment unit that governs inclusions and exclusions from the package [PMID 38715207].
- **A digital arm has been added alongside the two arms in the textbook.** The Ayushman Bharat Digital Mission (2021) runs on the 14-digit Ayushman Bharat Health Account plus health facility and health professional registries; its Scan-and-Share outpatient registration was live in 16,939 facilities across 35 of 36 states and union territories on 19 October 2024, cut registration waiting from about an hour to 2–5 minutes, and had linked 410 million health records [PMID 39882498].

Current national data

- ▶ **National Health Accounts, 2013-14 to 2021-22** — the government share of total health expenditure rose from 28.6% to 48.0% while the out-of-pocket share fell from 64.2% to 39.4%; government health spending nonetheless remains near 1.9% of GDP against the National Health Policy target of 2.5%. Medicines make up about two-thirds of out-of-pocket spending and catastrophic expenditure is commoner for outpatient care (47.8%) than for hospitalisation (43.1%) — precisely the ground the benefit package does not buy [PMID 42548603].
- ▶ **Comprehensive Annual Modular Survey 2022-23** (302,086 households) — 59.9% of households held some health cover, and PM-JAY is now cited at about 120 million households, roughly 500 million people, at Rs 5,00,000 a year. Publicly funded cover was associated with more outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128–1.210) and with less catastrophic expenditure at the 20% threshold (outpatient adjusted OR 0.869, 95% CI 0.810–0.932; inpatient adjusted OR 0.790, 95% CI 0.737–0.846) [PMID 42167946].
- ▶ **What an Ayushman Arogya Mandir actually delivers** — a state-representative assessment of 404 centres in Chhattisgarh recorded a mean 358 outpatients a month, 128 of them with non-communicable disease, meeting 31% of the population's outpatient need, 26% of the need for hypertension care and 21% for diabetes; psychotropic drugs were stocked out in over 90% of centres, the sharpest gap in the mental-health package [PMID 39367295].

Elimination, surveillance and disease status

- ▶ **Community health officers can carry the screening packages themselves.** After one day of training, 46 CHOs at Ayushman Arogya Mandir in Jodhpur detected oral potentially malignant disorders with 83.87% sensitivity, 98.31% specificity and 96.17% overall accuracy against remote ENT validation — worth citing because NFHS-5 records only 1.2% of men and 0.9% of women as ever screened for oral cancer [PMID 40341602].
- ▶ **Package delivery in tuberculosis** — under AB PM-JAY–Arogya Karnataka from July 2021 to June 2023, 9173 confirmed tuberculosis patients used TB-specific packages; 96.2% were below the poverty line and 97.9% were treated in the public sector, with roughly Rs 10,000 of catastrophic cost averted per patient [PMID 40518205].

Guideline and legislative updates

- ▶ **The financial-protection claim is genuinely contested, and an examiner rewards knowing both sides.** A Haryana case-control study (n=772) found direct medical out-of-pocket spending 65% lower among beneficiaries (Rs 11,131 versus Rs 31,675) and catastrophic expenditure at 13.3% versus 45.9% (OR 7.15; 95% CI 4.74–10.80) [PMID 40976675]. Against that, state-representative Chhattisgarh surveys in 2021 and 2022 found no effect on utilisation, patient satisfaction, length of stay or catastrophic expenditure, with mean out-of-pocket spending of Rs 47,052 in private against Rs 4,049 in public hospitals among the enrolled [PMID 39135015], and a national instrumental-variable analysis of successive National Sample Survey rounds found no significant association between enrolment and out-of-pocket spending per hospitalisation [PMID 41913189]. Non-medical costs — food, transport and wage loss — lie outside the package in every one of these studies.

Global and WHO developments

- ▶ **Primary care is the route to universal health coverage, but contact coverage is not effective coverage.** A realist review of 42 studies drawn from the countries with the largest gains in the WHO UHC Service Coverage Index between 2000 and 2021 found that benefit-package design and insurance financing dominated the reforms, that frontline workers and community organisations drove whatever worked, and that population coverage and financial protection improved while service quality and reach to marginalised groups lagged behind [PMID 42293679].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7, Health Programmes in India, pp 535, 541, 554, 570-571; Ch 10, Health Care of the Community, pp 728, 731-737, 745); Sunder Lal — Textbook of Community Medicine (Ch 8, pp 503, 529-530; Ch 16, pp 699, 706; Ch 17, pp 739-764); Oxford Textbook of Public Health 7e (Ch 4, p 247; Ch 21, p 843 — essential/minimum benefit package and universal health coverage). Named documents: Indian Public Health Standards 2022; Ayushman Bharat guidelines 2018; AB PM-JAY scheme documents as cited in the above texts.

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Hind Kusht Nivaran Sangh

Asked in: Paper IV 18-Nov-2019 — RGUHS MD Community Medicine

Definition — The Hind Kusht Nivaran Sangh (the Indian Leprosy Association) is a national **voluntary health agency** devoted exclusively to leprosy, founded in **1950** with its headquarters in **New Delhi**; its precursor was the Indian Council of the British Empire Leprosy Relief Association (BELRA), which was renamed LEPROA in 1950 (Park 28e, Ch 10, p 747). A voluntary health agency is an organisation administered by an **autonomous board**, which collects funds for its support chiefly from private sources and expends them — with or without paid workers — on a programme directed primarily at furthering the public health through health services, health education, research or health legislation (Park 28e, Ch 10, p 746). The Sangh has branches all over India and works in close cooperation with the Government and with other voluntary agencies.

Genesis and historical milestones

Year	Milestone	Significance for the Sangh
1874	Mission to Lepers (now The Leprosy Mission) founded by <i>Baily</i> at Chamba, Himachal Pradesh; headquarters later moved to Purulia, West Bengal	First organised voluntary anti-leprosy work in India (Park 28e, Ch 5, p 398)
1925	Indian Council of BELRA established	"Laid the foundation of organized leprosy work in India"; renamed Hind Kusht Nivaran Sangh in 1947 (Jugal Kishore — National Health Programmes of India, National Leprosy Eradication Program chapter)
1950	Hind Kusht Nivaran Sangh founded , headquarters New Delhi; the Indian Council of BELRA renamed LEPROA	Birth of an all-India Indian-led leprosy association (Park 28e, Ch 10, p 747)
1954	National Leprosy Control Programme initiated in the middle of 1954	The Sangh becomes the voluntary counterpart of an official programme (Park 28e, Ch 5, p 398)
1961	Anand Gram colony established by the Delhi branch, housing 400 leprosy-affected persons	Demonstration model of socio-economic rehabilitation (Jugal Kishore, National Health Programmes of India)
1965	National Leprosy Organization formed as a federating body	Common platform for the ~150 voluntary leprosy agencies to discuss problems and share experience; Park names the Sangh first in its list of the important agencies (Park 28e, Ch 5, p 398)
1981	Indian classification of leprosy proposed	Attributed to the Sangh by Mahajan & Gupta 4e, Ch 18, p 269
1983	Control programme converted into the National Leprosy Eradication Programme (NLEP)	Voluntary sector realigned to elimination and rehabilitation roles (Park 28e, Ch 5, p 398)

Aims and objectives

- ▶ To eradicate and control leprosy and to assist leprosy-affected persons in their social and economic rehabilitation.
- ▶ To act as a bridge between Government and non-governmental organisations engaged in leprosy elimination, strengthening collaboration and coordination for the common goal of prevention, treatment and rehabilitation of the cured person.
- ▶ To establish coordination among all the organisations working in the field of leprosy in Delhi, and to render them assistance whenever desired.
- ▶ To provide services for the treatment of leprosy and to conduct research on the social, psychological and economic conditions of patients.
- ▶ To educate patients, students and the public at large about all aspects of the disease and the plight of leprosy-affected persons — the core of stigma reduction.
- ▶ To establish institutions, clinics and hospitals for the control and treatment of leprosy.

(Objectives 1-6 as printed in Jugal Kishore — National Health Programmes of India; the list is the Sangh's own stated objectives, and objective 3 is worded for Delhi.)

Programme of work and contributions

Activity arm	What the Sangh actually does
Financial assistance	Grants-in-aid to various leprosy homes and clinics across the country (Park 28e, Ch 10, p 747)
Health education	Health education through publications and posters, aimed at early self-reporting and at reducing stigma (Park 28e, Ch 10, p 747)
Training	Training of medical workers and physiotherapists for leprosy service delivery (Park 28e, Ch 10, p 747)
Research	Research and field investigations, including the social, psychological and economic dimensions of the disease (Park 28e, Ch 10, p 747)
Professional forum	Organises the All-India Leprosy Workers Conferences (Park 28e, Ch 10, p 747)
Publication	Publishes <i>Leprosy in India</i> , a quarterly journal; also issues standard technical texts, e.g. <i>Job C.K. et al (1975)</i> , <i>Leprosy: Diagnosis and Management</i> , Hind Kusht Nivaran Sangh, New Delhi (Park 28e, Ch 5, reference 8)
Classification	The Indian classification of leprosy (1981) — a clinicobacterial classification widely used in India with five types: indeterminate, tuberculoid, borderline, lepromatous and pure neuritic. It adds pure neuritic leprosy to the four types of the Madrid classification (1953); Park 28e says the two systems are the most widely used in field programmes and "are not essentially different" (Park 28e, Ch 5, p 373; Mahajan & Gupta 4e, Ch 18, p 269)
Rehabilitation	Through Anand Gram (Delhi branch, 1961) — handloom cloth production, a physiotherapy and plastic surgery unit, poultry farming, cultivation of vegetables, a sewing centre and farming, so that no inmate needed to beg (Jugal Kishore)

The Sangh judged against the six functions of a voluntary health agency

Function (Park 28e, Ch 10, p 746)	How the Sangh discharges it
(a) Supplementing the work of government agencies	Lends personnel and funds to leprosy homes and clinics that government cannot reach under its financial and statutory restrictions
(b) Pioneering	Pioneered organised anti-leprosy work in India decades before the 1954 national programme; research is itself a form of pioneering
(c) Education	Publications, posters and public awareness work against stigma, supplementing what government agencies alone cannot cope with
(d) Demonstration	<i>Anand Gram</i> as a working demonstration project of socio-economic rehabilitation
(e) Guarding the work of government agencies	Acts as an informed critic and guide to the official programme by setting a good example of care
(f) Advancement of health legislation	Creates public opinion and advances legislation on health for the benefit of the whole community — here, for leprosy-affected persons

Park's analogy fixes the relationship exactly: voluntary agencies are the **"motor trucks"** that can penetrate the by-ways, while official agencies are the **"railway trunk lines"** that must run on tracks established by law (Park 28e, Ch 10, p 746).

Relationship with the National Leprosy Eradication Programme

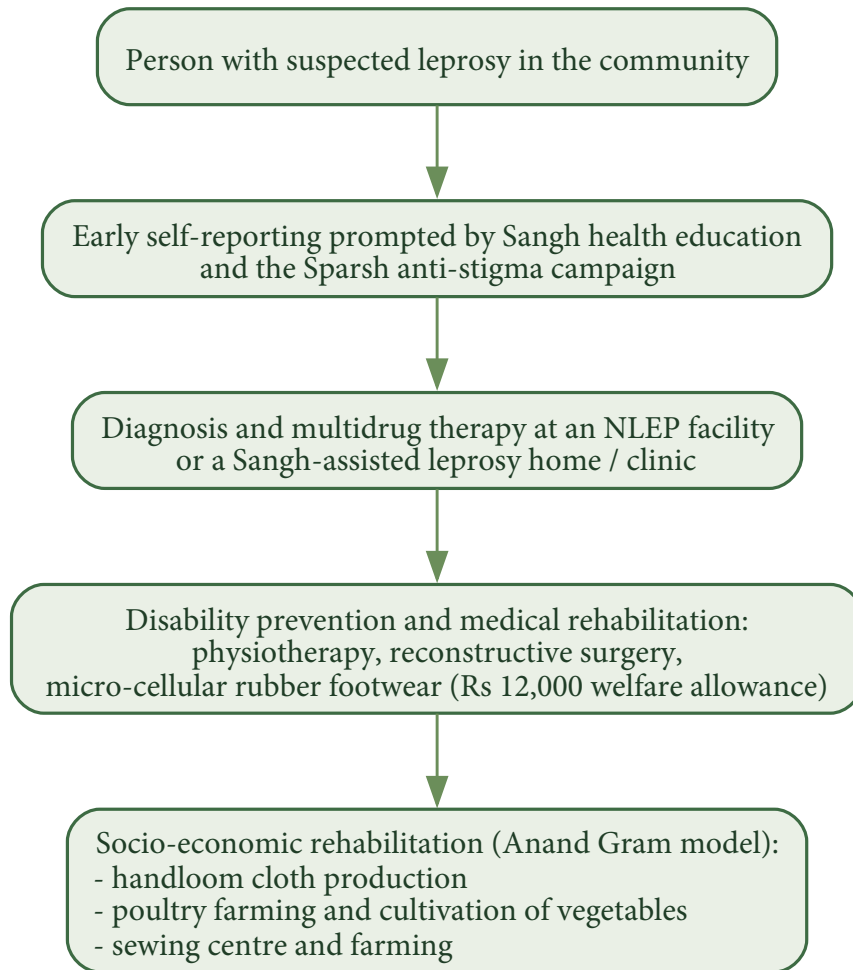
- **Complementary, not parallel** — leprosy services are fully integrated into the general health care system, so voluntary agencies now supplement rather than substitute for the primary health care network; 54 non-governmental organisations currently receive grant-in-aid from the Government of India under the **SET (survey, education and treatment) scheme**, serving remote and inaccessible areas, urban slums and industrial or labour populations (Park 28e, Ch 7).
- **Federated voice** — the Sangh works alongside the Gandhi Memorial Leprosy Foundation (Sevagram, Wardha), the German Leprosy Relief Association, the Damien Foundation, the Danish Save the Child Fund and the Indian Council of Medical Research's JALMA institute, Agra, under the National Leprosy Organization (Park 28e, Ch 5, p 398).
- **International partnership** — the International Federation of Anti-Leprosy Associations (ILEP), constituted by 10 agencies, supports NLEP with planning, monitoring and supervision, capacity building of general health care staff, information-education-communication (IEC), reconstructive surgery and socio-economic rehabilitation; 36 non-governmental organisations conducting reconstructive surgery are ILEP-supported (Park 28e, Ch 7).
- **Stigma reduction** — voluntary agencies partner the **Sparsh Leprosy Awareness Campaign**, launched in 2017 through Gram Sabhas and carried out in 60 per cent of India's villages, and the observance of **30 January as Anti-Leprosy Day** (Park 28e, Ch 7; Sunder Lal, Ch 8, p 371).
- **Rehabilitation partnership** — identified non-governmental institutions are formal support units for reconstructive surgery under disability prevention and medical rehabilitation (DPMR); 41

non-governmental organisations and 42 government medical colleges (83 centres) have been strengthened for this, and a welfare allowance of Rs 12,000 is paid to each patient undergoing reconstructive surgery (Park 28e, Ch 7).

- **Current policy hook** — Pillar 1 of the **National Strategic Plan and Roadmap for Leprosy 2023-2027** is *strengthen leadership, commitment and partnership*, and its action list explicitly includes strengthening partnership with existing partners and adding new partners and donors — the formal place of a body such as the Sangh in today's programme (Park 28e, Ch 7, citing Government of India 2023, Central Leprosy Division, Ministry of Health and Family Welfare).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labelled organogram — "Where the Hind Kusht Nivaran Sangh sits". Draw a central box, one box above it, one box below it, and a two-headed arrow to each side. Exact labels: centre — *Hind Kusht Nivaran Sangh, headquarters New Delhi (founded 1950; precursor Indian Council of BELRA)*; above — *Autonomous governing board; funds chiefly from private sources*; below — *Branches all over India (Delhi branch → Anand Gram colony, 1961)*; left two-headed arrow to — *Government of India: Central Leprosy Division, Ministry of Health and Family Welfare / National Leprosy Eradication Programme*; right two-headed arrow to — *Other voluntary agencies under the National Leprosy Organization (1965) and ILEP (10 agencies)*.



Service pathway of the Hind Kusht Nivaran Sangh, from case detection to socio-economic rehabilitation

High-yield exam pointers

- ▶ 1950, New Delhi — founding year and headquarters; precursor = Indian Council of BELRA (1925), itself renamed LEPRO in 1950.
- ▶ Indian classification of leprosy (1981): indeterminate, tuberculoid, borderline, lepromatous, pure neuritic — five types; the Madrid classification (1953) has the first four but no pure neuritic type.
- ▶ Journal = *Leprosy in India*, quarterly. Conference = All-India Leprosy Workers Conferences. Colony = Anand Gram (1961, Delhi branch, 400 persons).
- ▶ Six functions of any voluntary health agency: **Supplement, Pioneer, Educate, Demonstrate, Guard, Legislate** (mnemonic SPEDGL) — quote the "motor trucks vs railway trunk lines" analogy.
- ▶ Anti-leprosy dates to write: 1874 Mission to Lepers; 1954 National Leprosy Control Programme; 1965 National Leprosy Organization; 1983 NLEP; ~150 voluntary leprosy organisations in India.
- ▶ Current interface: ILEP (10 agencies), 54 NGOs on SET-scheme grant-in-aid, Sparsh campaign (2017, 60 per cent of villages), 30 January Anti-Leprosy Day, reconstructive-surgery welfare allowance Rs 12,000.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Textbooks disagree on the founding year — write **1950**, New Delhi, with the spine, since Jugal Kishore gives 1948 in its milestone table and 1949 in its text while Mahajan & Gupta 4e gives 1974. Write the Madrid classification as **1953**, because Park 28e's own reference for it is the International Leprosy Congress, Madrid (1953), whereas Mahajan & Gupta 4e prints 1959. Park 28e does not credit the Indian classification to the Sangh but to Chacko (1980), *A Manual of Leprosy*, The Leprosy Mission — only Mahajan & Gupta 4e attributes it to the Sangh. Park 28e names the Sangh's journal *Leprosy in India*, so do not equate it in the exam with the *Indian Journal of Leprosy*, which no retrieved passage says is the same publication.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, WHO Global Leprosy Strategy 2021-2030, National Strategic Plan and Roadmap for Leprosy 2023-2027 (Ministry of Health and Family Welfare)

Programme and policy updates (India)

- ▶ **India has pulled its deadline forward.** The National Strategic Plan and Roadmap for Leprosy 2023-2027 commits the country to **zero transmission by 2027** — three years ahead of the WHO and SDG 2030 timeline — under a triple goal of *zero transmission, zero disability and zero discrimination*. This is the plan under which the Sangh's partnership role (Pillar 1, leadership, commitment and partnership) is now assessed, so quote the 2027 date rather than 2030 [PMID 41054789; PMID 41878182].
- ▶ **The voluntary agency's job has shifted from substituting for government to supplying capability the state network lacks.** The Leprosy Mission Trust India ran molecular drug-resistance surveillance across 11 of its hospitals in eight states, and NLR India produced the first written peer-delivered guide to basic psychological support for leprosy and lymphatic filariasis (BPS-N), adapted from WHO Psychological First Aid, with significant falls in stigma and depression scores at three months. NLEP staff in three high-endemic West Bengal districts (Purulia, Birbhum, Uttar Dinajpur) name the gaps such partners must cover — stigma-driven concealment, wage loss and transport cost when a suspect is referred from camp to hospital, untraceable migrant patients, too few workers able to diagnose paucibacillary disease, and under-use of the **Nikusth** portal [PMID 41432848; PMID 39787071; PMID 41262280].

Current national data

- ▶ **India's NLEP indicators for 2024-25** — prevalence rate **0.57 per 10,000**, annual new case detection rate **7.00 per 100,000**, grade-2 disability rate **1.31 per million**. Elimination-level prevalence coexists with active transmission and late detection: a Telangana family cluster was missed across **three consecutive rounds** of the Leprosy Case Detection Campaign, the index case reaching grade-2 disability after a four-year diagnostic delay [PMID 41878182].
- ▶ **Dated burden figures to write instead of undated ones.** India reported **107,851 new cases in the 2023-24 reporting year**, **61.6 per cent of all new cases detected globally**; the global figure was **182,815 new cases in 2023**, with India, Brazil and Indonesia together carrying about 80 per cent and more than 95 per cent of cases clustered in 23 global priority countries. A handful of Indian

high-endemic districts account for roughly two-thirds of the world's new cases [PMID 41773049; PMID 40257140; PMID 40278757; PMID 41262280].

Elimination, surveillance and disease status

- **Antimicrobial resistance is now an Indian programme problem, not a laboratory curiosity.** Among 3,030 confirmed cases sampled across eight states (2021-2025), resistance was 7.8 per cent to rifampicin, 5.9 per cent to dapsone and 7.0 per cent to ofloxacin, with 2.1 per cent multidrug-resistant; rifampicin resistance reached 11.6 per cent in relapse and 13.8 per cent in defaulter cases, clustering around Prayagraj, Bokaro, Purulia and Champa. Primary resistance in newly diagnosed cases (2.4 per cent) means resistant strains are being transmitted, not only selected [PMID 41432848].
- **PEOPLE (2019-2023, Comoros and Madagascar, 109,436 enrolled)** tested how post-exposure prophylaxis should be delivered rather than whether it works: single double-dose rifampicin offered to everyone living within 100 m of an index case reduced incidence once baseline prevalence was accounted for (IRR 0.56, $p = 0.0030$), the individual protective effect was IRR 0.55 ($p = 0.0050$), and risk was two to four times higher within 75 m of an index patient. A companion fingerstick anti-PGL-I lateral-flow serosurvey of more than 17,000 contacts showed seroprevalence falling from 21.9-27.4 per cent to 11.6-22.8 per cent after two years of active case finding plus prophylaxis — a transmission indicator that does not wait years for new cases to surface [PMID 38762282; PMID 41329688].

Guideline and legislative updates

- **The preventive regimen and the treatment regimen have both settled.** WHO has recommended single-dose rifampicin as leprosy preventive treatment for contacts since 2018, and multidrug therapy is the uniform three-drug regimen (rifampicin, dapsone, clofazimine) for **6 months in paucibacillary and 12 months in multibacillary** disease. Reported chemoprophylaxis efficacy is 45-80 per cent; COLEP gave a 57 per cent fall in incidence, rising to 80 per cent (95% CI 50-92) in contacts who had also received childhood BCG [PMID 40257140; PMID 39609422; PMID 40278757].
- **Do not report every post-treatment event as "recurrence".** A 2026 Indian expert framework separates slowly regressing granuloma, immunological reaction, relapse, drug resistance and reinfection using slit-skin smear, histopathology and, where available, molecular viability assays, on the argument that one label conceals the cause and delays retreatment — relevant to any voluntary-sector clinic feeding data into national reporting [PMID 42119529].

Global and WHO developments

- **WHO's Towards zero leprosy global strategy 2021-2030** runs on four pillars: country-owned zero-leprosy roadmaps in all endemic countries; scaled-up prevention alongside integrated active case detection; management of complications and prevention of new disability; and combating stigma while ensuring human rights are respected. Pillars 1 and 4 are the formal openings for a voluntary agency of the Sangh's type. Verification is now proceeding country by country — WHO verified **Chile on 4 March 2026** as the first country in the Americas for elimination of leprosy [WHO 2021; WHO 2026-Mar].
- **Pipeline and outcome measures worth naming.** **LepVax**, the first leprosy-specific vaccine, was safe and immunogenic in phase 1; bedaquiline, telacebec and single-dose rifapentine are under

study for treatment and prophylaxis; and the WHO Skin NTDs app brings AI-assisted skin-lesion triage to frontline workers. Stigma and quality of life are now measured as programme outcomes — Indian evidence identifies disability, older age and female sex as the strongest determinants of poor quality of life, and shows stigma-driven mental-health harm differing between younger and older generations [PMID 40257140; PMID 41773049; PMID 42215552].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases, pp 373, 398 and chapter references; Ch 7 — Health Programmes in India, pp 494, 503-507; Ch 10 — Health Care of the Community, pp 746-747); Jugal Kishore — National Health Programmes of India (National Leprosy Eradication Program chapter and the voluntary-organisations section); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 18, p 269); Sunder Lal — Textbook of Community Medicine (Ch 8, p 371); National Strategic Plan and Roadmap for Leprosy 2023-2027, Central Leprosy Division, Ministry of Health and Family Welfare (2023), as summarised in Park 28e Ch 7.

Recent-advances sources: 41054789 — Leprosy in India: challenges, advances and the path to eradication, *Trans R Soc Trop Med Hyg* 2026, DOI [10.1093/trstmh/traf107](https://doi.org/10.1093/trstmh/traf107) (abstract-only); 41878182 — Grade 2 Disability in Leprosy in the Post-Elimination Era: A Case Report Highlighting Stigma and Programmatic Blind Spots, *Cureus* 2026, DOI [10.7759/cureus.104011](https://doi.org/10.7759/cureus.104011); 41432848 — Geographical trends of drug-resistant strains of *Mycobacterium leprae* in TLM hospitals: insights from a four-year study across eleven hospitals in eight Indian States, *Eur J Clin Microbiol Infect Dis* 2025, DOI [10.1007/s10096-025-05351-x](https://doi.org/10.1007/s10096-025-05351-x) (abstract-only); 39787071 — A new guide for basic psychological support for persons affected by neglected tropical diseases, *PLoS Negl Trop Dis* 2025, DOI [10.1371/journal.pntd.0011945](https://doi.org/10.1371/journal.pntd.0011945); 41262280 — Challenges in Implementation of National Leprosy Eradication Program: Cross-Sectional, Qualitative Study in Three High Endemic Districts of West Bengal, India, *Indian J Dermatol* 2025, DOI [10.4103/ijd.ijd_1204_23](https://doi.org/10.4103/ijd.ijd_1204_23); 41773049 — Quality of life in Indian people affected by leprosy? A systematic review from the country with the highest burden of leprosy cases, *Trans R Soc Trop Med Hyg* 2026, DOI [10.1093/trstmh/trag008](https://doi.org/10.1093/trstmh/trag008) (abstract-only); 40257140 — Leprosy research updates: Shaping the future of global health, *Indian J Med Res* 2025, DOI [10.25259/IJMR_195_2025](https://doi.org/10.25259/IJMR_195_2025); 40278757 — A Review of the Efficacy, Safety, and Feasibility of Rifamycin-Based Post-Exposure Chemoprophylaxis for Leprosy, *Trop Med Infect Dis* 2025, DOI [10.3390/tropicalmed10040084](https://doi.org/10.3390/tropicalmed10040084); 38762282 — Post-exposure prophylaxis in leprosy (PEOPLE): a cluster randomised trial, *Lancet Glob Health* 2024, DOI [10.1016/S2214-109X\(24\)00062-7](https://doi.org/10.1016/S2214-109X(24)00062-7) (abstract-only); 41329688 — Large-scale field application of a fingerstick blood test for *Mycobacterium leprae* infection, *PLOS Glob Public Health* 2025, DOI [10.1371/journal.pgph.0005270](https://doi.org/10.1371/journal.pgph.0005270); 39609422 — Leprosy, *Nat Rev Dis Primers* 2024, DOI [10.1038/s41572-024-00575-1](https://doi.org/10.1038/s41572-024-00575-1) (abstract-only); 42119529 — From Conundrums to Consensus: Addressing Gaps in the Management of Leprosy, *Am J Trop Med Hyg* 2026, DOI [10.4269/ajtmh.25-0687](https://doi.org/10.4269/ajtmh.25-0687) (abstract-only); 42215552 — Exploring generational differences in the impact of stigma on mental health among people affected by leprosy in rural India, *Sci Rep* 2026, DOI [10.1038/s41598-026-55020-y](https://doi.org/10.1038/s41598-026-55020-y). Plus WHO, *Towards zero leprosy: global leprosy (Hansen's disease) strategy 2021-2030* (2021) and the WHO leprosy fact sheet (Chile verified for elimination, 4 March 2026).



Indian Public Health Standards (IPHS) for PHCs

Asked in: Paper IV 14-Jul-2023 — RGUHS MD Community Medicine

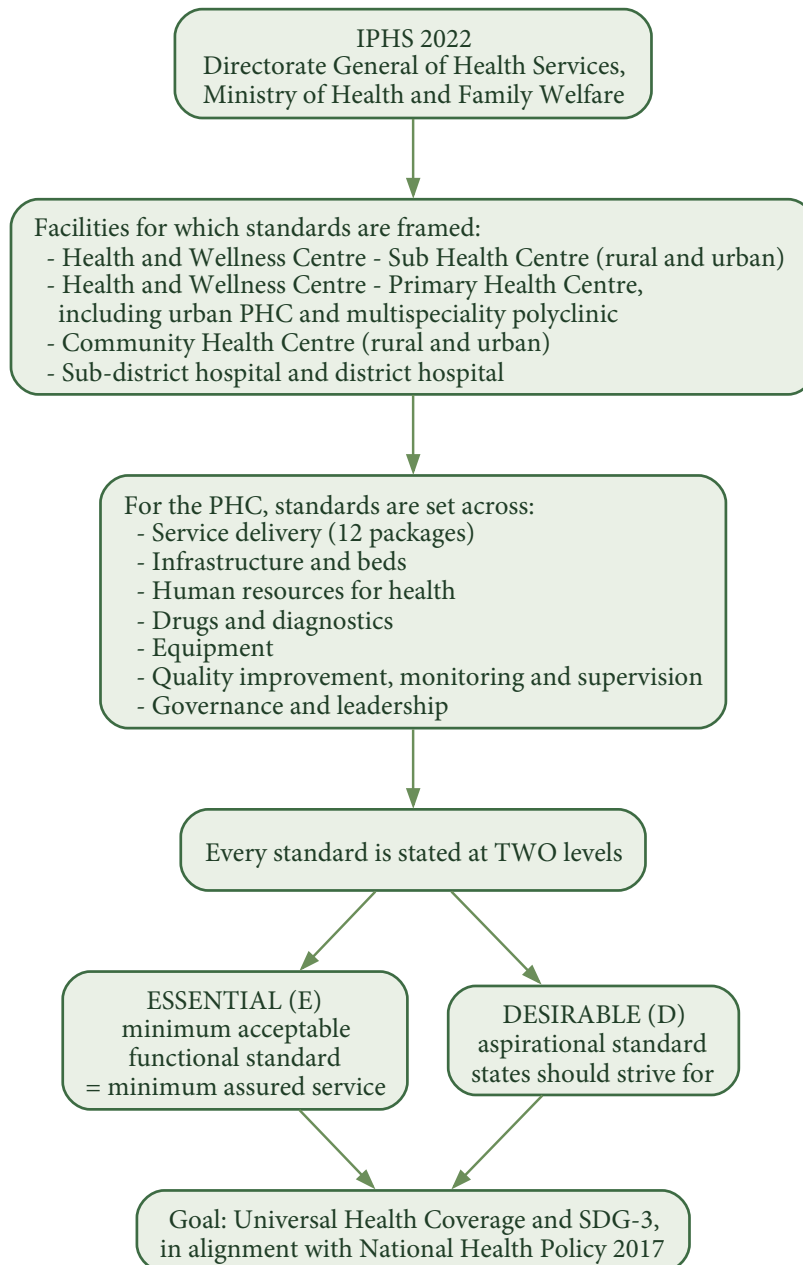
Definition — Indian Public Health Standards (IPHS) are the benchmark standards notified by the Directorate General of Health Services, Ministry of Health and Family Welfare, that lay down for each level of government health facility the **assured service package** and the **inputs required to deliver it** — workspace, human resources, drugs, diagnostics and equipment — together with the performance, monitoring and governance standards expected of it (Sunder Lal, Ch 17, p 744). For the primary health centre (PHC) they define what the citizen is entitled to receive and what the state is obliged to provide, and so convert the PHC from an ill-defined building into an auditable unit of care. First issued in 2006 after the launch of the National Rural Health Mission (NRHM), IPHS were revised in 2012 and again in 2022 (Sunder Lal, Ch 17, p 737).

Evolution, rationale and objectives

Revision	Context and what it added
IPHS 2006	Issued as a core strategy of NRHM (2005) for sub-centre, PHC, community health centre (CHC) and district hospital; used with the <i>facility survey</i> to identify facility-wise gaps and direct funding to closing them (Sunder Lal, Ch 17, p 737)
IPHS 2012	First revision of the same four levels of facility (Sunder Lal, Ch 17, p 737)
IPHS 2022	Rewritten around comprehensive primary health care — sub-centres and PHCs re-designated Health and Wellness Centres (HWC), urban HWCs, speciality urban PHCs (polyclinics) and universal screening for non-communicable disease; aligned to the seven policy shifts of the National Health Policy 2017 and to Universal Health Coverage and Sustainable Development Goal 3 (Sunder Lal, Ch 17, p 744; Park 28e, Ch 10, p 728)

Objectives of IPHS (Sunder Lal, Ch 17, p 744):

- ▶ To define a **uniform standard** to ensure high quality services.
- ▶ To specify the **minimum assured (essential)** and **achievable (desirable)** services at each level — sub-centre, PHC, CHC, sub-district and district facility.
- ▶ To specify adequate **workspace (buildings)**, **human resources for health**, **drugs**, **diagnostics**, **equipment**, and **administrative and logistical support** to deliver those services.
- ▶ To achieve and maintain an acceptable **quality of care**.
- ▶ To facilitate **monitoring and supervision** of health facilities.
- ▶ To serve as a tool for **governance, leadership and evaluation**.



IPHS 2022 - facilities covered and the essential/desirable ladder applied to each standard

Types of PHC and coverage norms under IPHS 2022

Item	Standard
Population covered	30,000 in plain rural area; 20,000 in tribal, hilly, desert and difficult areas (Sunder Lal, Ch 17, p 748)
Access norm	Not a rigid population norm — the "time to care" approach, within 30 minutes of travel time (Sunder Lal, Ch 17, p 748)
Referral position	One PHC is the referral unit for 6 sub-centres; it refers upward to the CHC/first referral unit (Sunder Lal, Ch 17, p 748)

Item	Standard
Categories (IPHS 2022)	(a) HWC-PHC (rural) — routine care with preventive and promotive interventions; (b) 24 × 7 HWC-PHC — prioritised for PHCs conducting deliveries, states should aspire to make all PHCs 24 × 7; (c) UPHC-HWC (urban) — routine outpatient care, since round-the-clock emergency and secondary care exist in cities; (d) Speciality UPHC / Polyclinic — adds fixed-day rotational outpatient care by 6 specialists: medicine, obstetrics and gynaecology, paediatrics, ophthalmology, dermatology and psychiatry (Sunder Lal, Ch 17, pp 748, 758–759)
Ownership and funding	Established and maintained by state governments under the Minimum Needs Programme and Basic Minimum Services Programme; primarily state funded (Sunder Lal, Ch 17, p 748)

Assured service package — the 12 packages of comprehensive primary health care at the HWC-PHC

- ▶ Care in pregnancy and childbirth.
- ▶ Neonatal and infant health care services.
- ▶ Childhood and adolescent health care services.
- ▶ Family planning, contraceptive services and other reproductive health care services.
- ▶ Management of communicable diseases — the national health programmes.
- ▶ General outpatient care for acute simple illnesses and minor ailments.
- ▶ Screening, prevention, control and management of non-communicable diseases.
- ▶ Care for common ophthalmic and ENT problems.
- ▶ Basic oral health care.
- ▶ Elderly and palliative health care services.
- ▶ Emergency medical services including burns and trauma.
- ▶ Screening and basic management of mental health ailments.

(Sunder Lal, Ch 17, pp 749–750)

Beyond these 12 packages the PHC also provides **referral support** for complicated cases sent up from the HWC-sub centres, and exercises diagnostic, supervisory, administrative and financial control over them; the complexity of care at PHC is a level above that at the sub-centre (Sunder Lal, Ch 17, p 750).

Human resources for health at a PHC as per IPHS 2022

Human resource	HWC-PHC (rural) — E	HWC-PHC (rural) — D	24 × 7 PHC — E	24 × 7 PHC — D
Medical officer (MBBS)	1	1	2	1
Medical officer (AYUSH)	–	1	–	1
Dentist	–	1	–	1
Staff nurse	2	1	7	–
Pharmacist and store keeper	1	–	1	1

Human resource	HWC-PHC (rural) — E	HWC-PHC (rural) — D	24 × 7 PHC — E	24 × 7 PHC — D
Medical laboratory technologist / laboratory technician	1	–	2	1
Optometrist / ophthalmic assistant / vision technician	–	1	–	1
Health worker (female), i.e. auxiliary nurse midwife (ANM)	1	–	1	–
Health worker / health assistant (male)	1	–	1	–
Health assistant (female) / lady health visitor	1	–	1	–
Health educator counsellor	–	1	–	1
Dental assistant	–	1	–	1
Cold chain / vaccine logistics assistant	–	1	1	–
Physiotherapist	–	–	–	1
Dresser	1	–	1	–
Lower division clerk / accountant	1	–	1	–
Data entry operator	1	–	1	–
Cleaning staff	1	–	4	–

E = essential, D = desirable; desirable posts are **over and above** the essential ones, and support staff are hired in-house only where the service is not outsourced (Sunder Lal, Ch 17, pp 749–750).

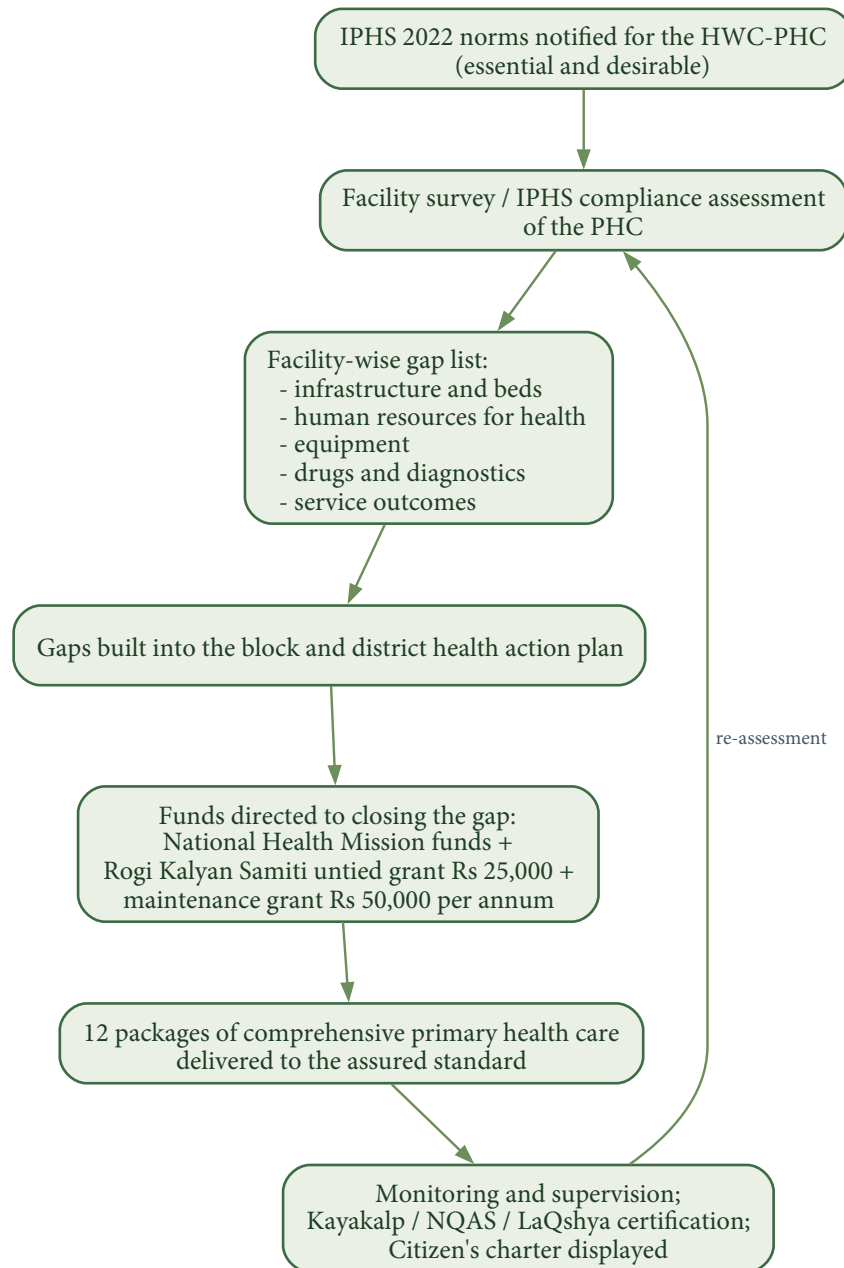
Infrastructure, drugs, diagnostics and minimum performance standards

Standard	Norm at PHC level
Beds and oxygen (Table 17.6)	Rural PHC — 6 beds (2 essential + 4 desirable), 4 B-type cylinders of 1500 L capacity, 1 oxygen concentrator of 10 L; urban PHC/polyclinic — 6 day-care beds (2 E + 4 D), same oxygen norm; 24 × 7 PHC/UPHC — 10 beds (6 E + 4 D) , 5 cylinders, 1 concentrator (Sunder Lal, Ch 17, pp 749–750)
Physical facilities	Operation theatre complex, labour room and laboratory; UPHCs that continue to conduct deliveries must have infrastructure as for a 24 × 7 PHC (Sunder Lal, Ch 17, pp 748, 758)
Drugs	172 essential medicines recommended for the HWC-PHC (Sunder Lal, Ch 17, p 750)
Diagnostics	Over 50 diagnostic services available at the HWC-PHC (Sunder Lal, Ch 17, p 750)

Standard	Norm at PHC level
Medical officer	Outpatient department — 75 patients per day (Sunder Lal, Ch 17, p 749)
Dentist	Outpatient department — 20 patients per day (Sunder Lal, Ch 17, p 749)
Laboratory technician	100 tests per day (Sunder Lal, Ch 17, p 749)
Pharmacist	120 prescriptions dispensed per day (Sunder Lal, Ch 17, p 749)
Staff nurse	1 per 6 beds, as per Indian Nursing Council norms (Sunder Lal, Ch 17, p 749)

Governance, community accountability and quality assurance

- ▶ **Rogi Kalyan Samiti / PHC management committee** at every PHC, to increase community control and participation (Sunder Lal, Ch 17, p 751).
- ▶ **Untied grant of ₹25,000** for local health planning and action and an **annual maintenance grant of ₹50,000** (Sunder Lal, Ch 17, pp 751–752).
- ▶ **Transfer of PHCs to elected panchayati raj institutions** for management and control (Sunder Lal, Ch 17, p 752).
- ▶ **Citizen's charter** displayed at the PHC — services offered, patients' rights and responsibilities, and the grievance redressal system; the charter also displays whether the facility is **Kayakalp / National Quality Assurance Standards (NQAS) / LaQshya certified** (Sunder Lal, Ch 17, p 752; Park 28e, Ch 10, pp 728–729).
- ▶ **Quality standards** — quality standards for urban PHCs were released in 2016 and have been superseded by the *Operational Guidelines for Improving Quality in Public Healthcare Facilities under IPHS 2022* (Sunder Lal, Ch 17, p 759).
- ▶ **Gap-closure cycle** — the facility survey identifies facility-wise gaps in infrastructure, human resources, equipment, drugs and supplies and in service outcomes, and National Health Mission funds are directed to closing them (Sunder Lal, Ch 17, p 737).



How IPHS is applied to a PHC - the facility-survey to gap-closure to certification cycle

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Position of the PHC in the rural health-care ladder under IPHS 2022** — draw a vertical ladder and label each rung with facility, population norm and staffing: *Village* (accredited social health activist, 1 per 1000; anganwadi worker) → *HWC-Sub Health Centre* (5000 plain / 3000 tribal; community health officer + 1 male and 1 female multipurpose worker + 5 ASHAs; 2 day-care beds) → *HWC-Primary Health Centre* (30,000 plain / 20,000 tribal, hilly, desert; medical officer MBBS; 6 beds; referral unit for 6 sub-centres) → *Community Health Centre / First Referral Unit* (1,20,000 plain / 80,000 tribal; 30 beds; specialists) → *Sub-district and District Hospital*. Mark the "time to care — within 30 minutes of travel" arrow alongside the sub-centre and PHC rungs.

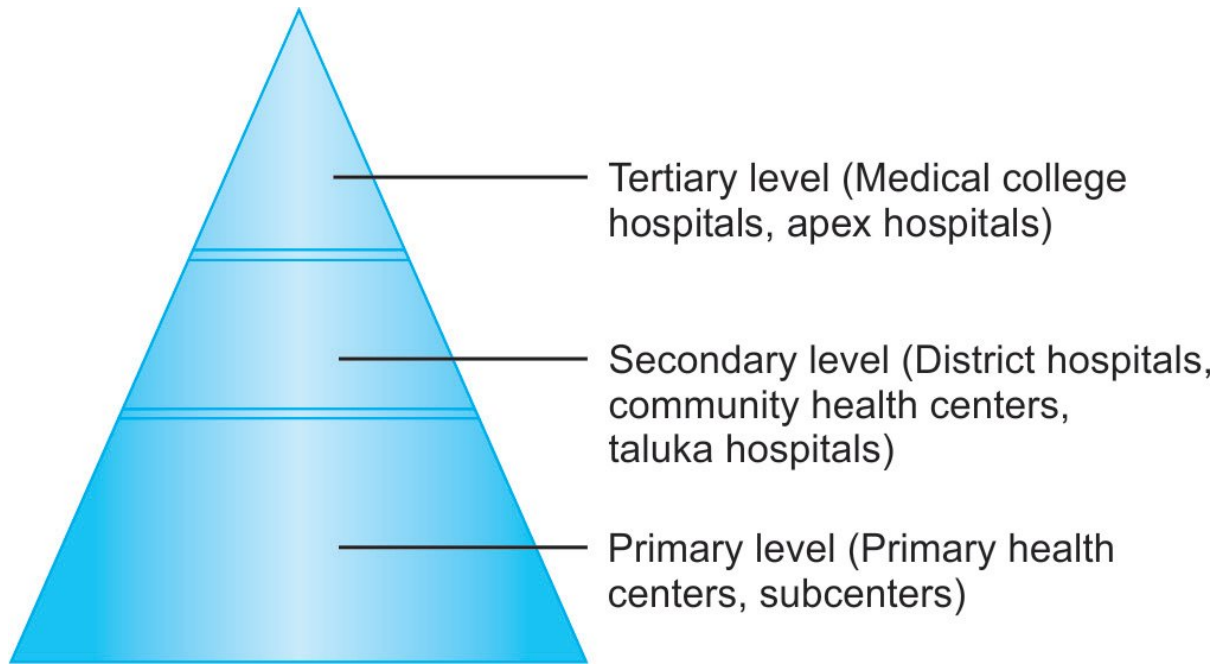


Fig 34.1 — Three-tier system of health care (Suryakantha Environment Best, p.778)

High-yield exam pointers

- ▶ **IPHS chronology:** 2006 (with NRHM) → revised 2012 → **revised 2022**; the 2022 revision covers HWC-SHC, HWC-PHC/UPHC/polyclinic, CHC and SDH/DH.
- ▶ Every IPHS norm is written twice — **E = essential (minimum assured)** and **D = desirable (aspirational)**; write both columns when asked for staffing or beds.
- ▶ **PHC population norm 30,000 plain / 20,000 tribal-hilly-desert-difficult**, "time to care" **30 minutes**, referral unit for **6 sub-centres, 6 beds** (10 beds if 24×7).
- ▶ **12 packages** of comprehensive primary health care — remember them as the life-course sequence: pregnancy → newborn/infant → child/adolescent → family planning → communicable disease → general outpatient → non-communicable disease → eye and ENT → oral → elderly and palliative → emergency/trauma → mental health.
- ▶ **172 essential medicines** and **50+ diagnostics** at the HWC-PHC.
- ▶ **Performance standards:** medical officer 75 outpatients/day, dentist 20/day, laboratory technician 100 tests/day, pharmacist 120 dispensings/day, staff nurse 1 per 6 beds.
- ▶ **Facility numbers (Rural Health Statistics):** 31,053 PHCs — 24,935 rural and 6,118 urban — functioning as on **31 March 2021**, of which 16,635 rural PHCs had been converted into HWC-PHCs; 94.5% run in government buildings (Sunder Lal, Ch 17, p 751).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir**, so a PHC-level unit is now called "Ayushman Arogya Mandir – Primary Health Centre" (Park 28e, Ch 10, p 728), and the facility counts quoted above are the Rural Health Statistics positions as on 31 March 2021 for PHCs and 31 March 2022 for sub-centres and CHCs.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, MoHFW Ayushman Arogya Mandir portal, NHSRC quality dashboard, ICMR, WHO

Programme and policy updates (India)

- ▶ **The facility has been renamed, and the answer should use the new name.** In November 2023 the Union Health Ministry re-designated all Ayushman Bharat Health and Wellness Centres as **Ayushman Arogya Mandir (AAM)** with the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth"), re-orienting them towards AYUSH-informed holistic wellness — physical, mental and social — with counselling on lifestyle, yoga, nutrition, mental health and elderly care added alongside curative work; the exercise was budgeted at ₹3,000 per facility (total outlay ₹48 crore), and every renamed centre had to upload photographs to the national portal for verification. Write it as a re-designation plus a wellness re-orientation, not as a fresh set of IPHS norms [PMID 42257165].
- ▶ **Scale, workforce and a second financing stream for closing IPHS gaps.** More than 1.75 lakh AAM have been operationalised nationally, led at sub-centre level by the Community Health Officer / mid-level health provider cadre — 1,16,978 in position as on 30 November 2024, against a residual shortfall of about 11,111. All 12 service packages had been rolled out in 50% of AAM by March 2023, rising to 82% by November 2024. Gaps against IPHS 2022 are now financed not only from National Health Mission grants but also through the **PM-Ayushman Bharat Health Infrastructure Mission (PM-ABHIM)** and **XV Finance Commission** health grants routed via panchayati raj institutions and urban local bodies, while community oversight has been formalised as **1,35,343 Jan Arogya Samitis** [PMID 41816146].
- ▶ **The drugs-and-diagnostics entitlement is now stated precisely.** 172 medicines and 63 **diagnostic tests** are notified at the AAM-PHC/UPHC, and 106 medicines with 14 tests at the AAM sub-health centre, states being free to modify locally — so the "over 50 diagnostics" in the body is now specified as 63 [PMID 41816146].
- ▶ **Teleconsultation is a routine PHC service standard, not an optional extra.** eSanjeevani, launched in November 2019, had crossed 163 million consultations across all 28 states and 8 union territories by September 2023, the provider-assisted AB-HWC hub-and-spoke model — community health officer at the spoke, doctor at a sub-district or district hub — accounting for over 93% of use; an average AAM conducts about 95 teleconsultations a month [PMID 41158553] [PMID 41816146]. Diabetes and hypertension have both entered the ten commonest AB-HWC diagnoses, but chiefly as **follow-up** of patients whose diagnosis was made in person, so teleconsultation is best described as sustaining chronic-disease treatment at a unit with no doctor on site rather than starting it [PMID 41158553].

Current national data

- **Facility counts that supersede the 31 March 2021 Rural Health Statistics figures quoted in the body.** As on 19 August 2026 the national portal reports **1,86,934 functional Ayushman Arogya Mandirs** — 1,34,718 sub-health centres, **24,512 PHCs**, 5,474 urban PHCs, 9,972 urban health and wellness centres and 12,258 AYUSH facilities [MoHFW 2026-Aug].
- **Universal health coverage and financial-protection indicators.** India's UHC Service Coverage Index rose from 57 (2015) to 63 (2021), above the South-East Asia regional average of 62; out-of-pocket expenditure fell from 60.6% (2014–15) to 39.4% (2021–22) on National Health Accounts estimates; outpatient use of public facilities rose from 91.6 to 134.5 crore and inpatient use from 4.3 to 6.3 crore between 2012–13 and 2021–22 [PMID 41816146].
- **IPHS 2022 is now the benchmark for national facility surveys — and PHCs outscore every other primary and intermediate level.** An ICMR-supported nine-district survey of 561 sub-centre-level AAM in the second half of 2024, using a checklist built from the IPHS 2022 guidelines, found 94% headed by a community health officer or mid-level health provider, a blood-pressure apparatus in 85% and a glucometer in 83.8%, and at least two essential drugs for each of hypertension and diabetes in about 78% [PMID 40991221]. A 415-facility assessment across 19 districts of seven states, scored by WHO's Service Availability and Readiness Assessment method against IPHS, gave **PHCs 73%** — above the 70% "prepared" threshold — versus 61% for sub-centres, 59% for CHCs and 57% for private primary care. State the claim as the authors do: PHCs beat sub-centres, CHCs and sub-district hospitals, while district hospitals, government and private medical colleges and private tertiary facilities all cleared 70%, the range topping out at 92% for private medical colleges [PMID 40536370].

Elimination, surveillance and disease status

- **Population-based NCD screening is now the platform's largest surveillance function, though coverage of need remains partial.** Cumulative screening at AAM to 30 November 2024: 84.91 crore for hypertension, 74.24 crore for diabetes, 48.98 crore for oral cancer, 21.8 crore women for breast cancer and 14.31 crore women for cervical cancer, each person tracked on the National NCD portal with an ABHA number to build a longitudinal record [PMID 41816146]. Set against this, a state-representative survey of 404 centres in Chhattisgarh found 358 outpatients a month per centre including 128 with NCDs, meeting 31% of the population's outpatient need, 26% for hypertension and 21% for diabetes, with mental illness and chronic respiratory disease the widest residual gaps [PMID 39367295].

Guideline and legislative updates

- **Quality certification has multiplied well beyond Kayakalp, NQAS and LaQshya.** MusQan, SaQushal, SaQsham, Swachh Swasth Sarvatra, Mera Aspataal, Quality Darpan, a dedicated NQAS for health and wellness centres (2020–21) and the AB-PMJAY Bronze certification now run in parallel, their checkpoints largely guided by IPHS and heavily overlapping with one another; facilities are therefore advised to climb a ladder — Kayakalp first, then SaQushal, then PM-JAY Bronze, then full NQAS [PMID 41280589]. National position to March 2026: 21,094 NQAS-certified facilities, 30,562 Kayakalp-incentivised facilities, 1,253 LaQshya-certified labour rooms and 920 maternity operation theatres [NHSRC 2026-Mar].

Global and WHO developments

- **Climate-resilient, low-carbon primary care is entering the standards themselves.** The National Action Plan on Climate Change and Human Health (2018) and the National Programme on Climate Change and Human Health (2019) set targets for climate–health surveillance and for green, climate-resilient health facilities, and sustainability criteria are being written into accreditation standards; India's health sector is reckoned the world's seventh-largest healthcare carbon emitter in absolute terms [PMID 41623435]. In a qualitative study of providers in the aspirational district of Yadgir, Karnataka, framed on WHO's Harmonized Health Facility Assessment, solarising PHCs was described as making vaccine cold storage reliable, deliveries safer and emergency care timelier — the practical link between IPHS infrastructure norms and climate resilience [PMID 42179708].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10); Sunder Lal — Textbook of Community Medicine (Ch 16, Ch 17, Ch 19); Suryakantha — Community Medicine with Recent Advances (Fig 34.1). Guideline documents cited within these: Revised Indian Public Health Standards Guidelines for Sub-centres, PHCs and CHCs, DGHS, Ministry of Health and Family Welfare, 2012 and 2022; Ayushman Bharat Comprehensive Primary Healthcare through Health and Wellness Centres — Operational Guidelines 2018; Operational Guidelines for Improving Quality in Public Healthcare Facilities, 2021; National Health Policy 2017; Rural Health Statistics 2021 and 2022.

Coverage note: the Park 28e passages for this section are optically degraded, so Park is cited only where its text was legible; the IPHS 2022 PHC standards are drawn from Sunder Lal Ch 17.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, J Family Med Prim Care 2026, doi:[10.4103/jfmpc.jfmpc_1966_24](https://doi.org/10.4103/jfmpc.jfmpc_1966_24); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, J Family Med Prim Care 2026, doi:[10.4103/jfmpc.jfmpc_478_25](https://doi.org/10.4103/jfmpc.jfmpc_478_25); 41158553 — Adoption and utilization of India's eSanjeevani national telemedicine service, Oxf Open Digit Health 2025, doi:[10.1093/oodh/oqaf025](https://doi.org/10.1093/oodh/oqaf025); 40991221 — Functionality of Ayushman Arogya Mandirs for hypertension and diabetes care in nine districts of India, Indian J Med Res 2025, doi:[10.25259/ijmr_1693_2025](https://doi.org/10.25259/ijmr_1693_2025); 40536370 — Preparedness of public and private health facilities for management of diabetes and hypertension in 19 districts in India, Indian J Med Res 2025, doi:[10.25259/ijmr_755_2024](https://doi.org/10.25259/ijmr_755_2024); 39367295 — Performance of health and wellness centre in providing primary care services in Chhattisgarh, India, BMC Prim Care 2024, doi:[10.1186/s12875-024-02603-1](https://doi.org/10.1186/s12875-024-02603-1); 41280589 — To gain insights and compare different quality certification programs for public health facilities in India, J Family Med Prim Care 2025, doi:[10.4103/jfmpc.jfmpc_1591_24](https://doi.org/10.4103/jfmpc.jfmpc_1591_24); 41623435 — Building India's low-carbon healthcare system: a comprehensive review of national policies, operational frameworks, and decarbonization strategies, Ann Glob Health 2026, doi:[10.5334/aogh.5096](https://doi.org/10.5334/aogh.5096); 42179708 — Powering care at the frontlines: healthcare providers' perspectives on solarised primary health centres in rural Karnataka, India, using the WHO HHFA framework, Front Med Technol 2026, doi:[10.3389/fmedt.2026.1800307](https://doi.org/10.3389/fmedt.2026.1800307). All nine re-grounded against PMC full text on 2026-08-19. Plus: MoHFW Ayushman Arogya Mandir portal (aam.mohfw.gov.in), national counts as on 19 August 2026; NHSRC quality assurance dashboard (qps.nhsrcindia.org), certified-facility counts to March 2026.

♦ ♦ ♦

Indian Red Cross Society

Asked in: Paper IV May 2019 — RGUHS MD Community Medicine

Definition — The Indian Red Cross Society (IRCS) is the Indian national society of the international Red Cross movement, established in 1920 under the **Indian Red Cross Society Act** and incorporated under **Parliament Act XV of 1920**, with the three statutory objectives of **improvement of health, prevention of disease and mitigation of suffering** (Jugal Kishore, Ch 3, p 989; Park 28e, Ch 25, p 1061). Its parent movement is a **non-political, non-official international humanitarian organisation** devoted to the service of mankind in peace and in war (Park 28e, Ch 25, pp 1060–1061). It is the largest of the world's voluntary agencies — described as the biggest relief organisation in the world, with two hundred million members and tens of thousands of employees (Suryakantha, Ch 43, p 836).

Origin and statutory identity

Year	Event	Significance
1859	Henry Dunant, a young Swiss businessman travelling through North Italy, is on the scene of the battle of Solferino and, appalled by the neglect of the thousands of wounded and dying, recruits volunteers from nearby villages; he gives his time, money and talent to the cause until he is bankrupt	Birth of the idea of impartial relief to the war-wounded (Park 28e, Ch 25, pp 1060–1061; Suryakantha, Ch 43, pp 835–836)
—	Dunant's book <i>Un Souvenir de Solferino</i> urges that voluntary national societies be founded "which in time of war would render aid to the wounded without distinction of nationality", with a protective emblem , trained workers , and services protected by international treaty	The three founding demands of the movement (Park 28e, Ch 25, p 1061)
1864	First Geneva Convention , Geneva — a treaty signed for the relief of the wounded and the sick of the armies in the field; the International Committee of the Red Cross (ICRC) thus comes into being	ICRC is an independent, neutral institution and the founder organisation of the Red Cross, with branches all over the world (Park 28e, Ch 25, p 1061)
1901	Dunant receives the first Nobel Prize for Peace	International recognition of the movement he founded (Suryakantha, Ch 43, p 836)
1919	League of Red Cross Societies created, headquarters at Geneva, to coordinate the work of the national societies — more than 90 in Park's count	Today the International Federation of Red Cross and Red Crescent Societies (IFRC) (Park 28e, Ch 25, p 1061; Ch 17, p 957)
1920	Red Cross Society of India established by an Act of the Indian Legislature — the Indian Red Cross Society Act , incorporated under Parliament Act XV of 1920	The Indian national society; the Act was last amended in 1992 and its rules framed in 1994 (Jugal Kishore, Ch 3, p 989; Park 28e, Ch 25, p 1061)
1936	Red Cross Society (Allocation of Property) Act, 1936 — the Indian Red Cross Society Act as previously administered by Burma (now Myanmar) terminated and the funds were transferred to the Society under this Act	Purpose of the Society restated in law: to collect money and gifts from the public for medical and other aid to the sick, wounded and war victims (Jugal Kishore, Ch 3, pp 968–969)

Governance, membership and reach

Element	Provision
President	The President of India is the President of the Society (Jugal Kishore, Ch 3, p 989)
Chairman	The Union Health Minister is the Chairman of the Society (Jugal Kishore, Ch 3, p 989)
Membership	The Society has 25–30 members (Jugal Kishore, Ch 3, p 969)
Managing Body	10–30 of those members form the Managing Body , in which the whole management is rested; it regulates and monitors finances, issues memberships and decides the inclusion of foreign representatives (Jugal Kishore, Ch 3, p 969)
Statutory allocation	The Act defines when and who may benefit from the Society and what percentage goes to the States and Union Territories (Jugal Kishore, Ch 3, p 969)
Branch network	35 State / Union Territory branches , with more than 700 district and sub-district branches (Jugal Kishore, Ch 3, p 989). Park's older, undated count is "a network of over 400 branches all over India" (Park 27e, Ch 10, p 725)
Guiding principles	The movement works from seven Fundamental Principles , through which it aims to influence the behaviour of all people (Jugal Kishore, Ch 3, p 969)
Character	Administered by an autonomous board and funded chiefly from private sources (Park 28e, Ch 10, p 746); the role of such voluntary organisations is complementary in nature and not an alternative to working through the government system (Jugal Kishore, Ch 3, p 989)

Core programme areas of the Society

Core programme area (Jugal Kishore, Ch 3, p 989)	Content
1. Promoting humanitarian principles and values	Encourages respect for other human beings and a willingness to work together to find solutions to problems (Jugal Kishore, Ch 3, pp 969, 989)
2. Disaster response	The largest portion of the Society's work — assistance to millions of people annually, from refugees to victims of natural disasters; covers emergency response units, humanitarian policy, and long-term rehabilitation (Jugal Kishore, Ch 3, p 969)
3. Disaster preparedness	Prompted by the sharp rise in natural disasters; makes Red Cross societies and communities aware of the risks they face , how to reduce vulnerability , and how to cope when disaster strikes (Jugal Kishore, Ch 3, p 969)
4. Health and care in the community	A cornerstone of humanitarian assistance and a large part of Red Cross spending, addressing those who die for want of even the most basic health services and elementary health education (Jugal Kishore, Ch 3, p 969)

- **Other major activities** — hospital services; blood bank; HIV/AIDS programmes; home for disabled servicemen; vocational training centres; tracing activities; maternity, child and family welfare; nursing; Junior Red Cross activities; preparedness and prevention of communicable and infectious diseases; and relief operations in **fire, railway and other accidents** and events (Jugal Kishore, Ch 3, p 989).

- **Capacity-building activities** — management and volunteer training; improving branch structures; planning; fund-raising; and gender equality (Jugal Kishore, Ch 3, p 989).

Activities of the Society as Park classifies them

Activity	What the Society does
(a) Relief work	When disaster strikes any part of the country — earthquakes, floods, drought, epidemics — the Society immediately mobilises all its resources and goes to the rescue of the affected people. Disaster services comprise distribution of milk, medicines, vitamin tablets, cod liver oil and hundreds of other items to famine-stricken people and to those hit by floods (Park 27e, Ch 10, p 725; Park 28e, Ch 25, p 1061)
(b) Milk and medical supplies	A number of hospitals, dispensaries, maternity and child welfare centres, schools and orphanages receive assistance every year, consisting mainly of milk powder, medicines, vitamins and other supplies (Park 28e, Ch 10, p 747)
(c) Armed forces	Care of the sick and the wounded among members of the forces is one of the primary obligations of the Red Cross. In peacetime the Society provides military hospitals with amenities such as newspapers, periodicals, musical instruments and other comfort goods, and runs a well-equipped hospital, the "Red Cross Home" at Bangalore (Bengaluru) — "the only one of its kind in India and the Far East" — for permanently disabled ex-servicemen , giving nursing and rehabilitative care (Park 28e, Ch 10, p 747; Ch 25, p 1061; Suryakantha, Ch 43, p 836)
(d) Maternal and child welfare services	A large number of maternity and child welfare centres all over India are either directly administered by, or affiliated to, the Red Cross; a bureau of Maternity and Child Welfare gives technical advice and financial aid for model maternity and child welfare centres. The Society is also one of the four important child welfare agencies of India (with the Indian Council for Child Welfare, the Central Social Welfare Board and the Kasturba Gandhi Memorial Trust), running day-care services (nursery schools, balwadis, creches for children of working mothers), holiday homes for children aged 12–16 years, and recreation facilities (Park 28e, Ch 10, p 747; Ch 9, p 699)
(e) Family planning	Several States run family planning clinics under the auspices of the Indian Red Cross (Park 28e, Ch 10, p 747)
(f) Blood bank and first aid	Some State branches have started blood banks , and maintaining blood banks and promoting blood donation for those wounded in wars or disasters is a stated activity. The St. John Ambulance Association in India , which is part of the Red Cross, has trained several lakh men and women in first aid, home nursing and allied subjects (Park 28e, Ch 10, p 747; Suryakantha, Ch 43, p 836)
Health education	The Indian Red Cross is named among the voluntary agencies engaged in health education work in India, and conducts health education campaigns; health education is also one of the movement's own programme areas (Park 28e, Ch 22, p 1024; Ch 25, p 1061; Suryakantha, Ch 43, p 836)

Services in war and disaster

The Society provides the following services in an emergency, whether war or disaster (Jugal Kishore, Ch 1, pp 654–655):

- **Food and other necessary supplies**
- **Medicines**

- ▶ Blood supply for emergency operations
- ▶ Psychological support
- ▶ Tracing (restoring contact between separated families)
- ▶ Warehousing
- ▶ Communication
- ▶ Other supportive services

Junior Red Cross and St. John Ambulance Association

- ▶ The **Junior Red Cross** is one of the **most active sections** of the Society; it gives **lakhs of boys and girls all over India** the opportunity to be associated with its activities (Park 28e, Ch 25, p 1061; Suryakantha, Ch 43, p 836).
- ▶ Its activities are **village uplift, first aid, anti-epidemic work**, and the building up of an **international fraternity of youth**, thereby promoting **international friendliness, understanding and cooperation** (Park 28e, Ch 25, p 1061).
- ▶ The **St. John Ambulance Association in India** is **part of the Red Cross** and is the Society's first-aid training arm; the Society also provides first-aid services to deserving cases through it (Park 28e, Ch 10, p 747; Suryakantha, Ch 43, p 836).

Place among the voluntary health agencies of India

A voluntary health agency is "an organisation that is administered by an **autonomous board** which holds meetings, collects funds for its support chiefly from **private sources** and expends money, whether with or without paid workers, in conducting a programme directed primarily to furthering the public health by providing **health services or health education**, or by advancing **research or legislation** for health, or by a combination of these activities" (Park 28e, Ch 10, p 746). More than **7,000 voluntary agencies** are estimated to be involved in health-related activities in India, health care being carried out by agencies such as the Red Cross, industrial establishments, Lion's Club and Helpage India, with successive national missions seeking public-private partnership through them (Sunder Lal, Ch 17, p 765). Park lists the Indian Red Cross Society **first** among the voluntary health agencies of India and classifies the service such agencies render under six functions.

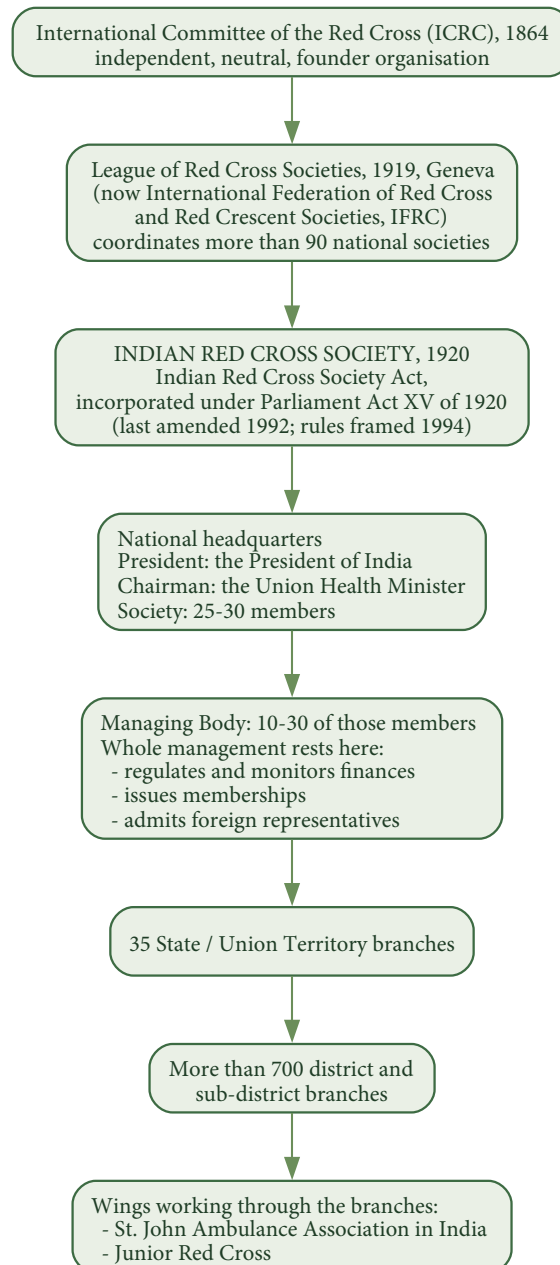
Function (Park 28e, Ch 10, p 746)	What Park means by it
(a) Supplementing the work of government agencies	Government agencies cannot provide complete service because they work under financial and statutory restrictions; voluntary agencies strengthen them by lending personnel or contributing funds for special equipment, supplies or services
(b) Pioneering	Voluntary agencies explore new ways of doing things, research being one form of pioneering; when the effort bears fruit the government takes the project over for larger numbers — Park's example is family planning, first spearheaded by voluntary agencies in the face of much opposition and later accepted as national policy
(c) Education	There is unlimited scope for health education in India, which government agencies cannot cope with unless supplemented by voluntary effort

Function (Park 28e, Ch 10, p 746)	What Park means by it
(d) Demonstration	Demonstrations and experimental projects advance the cause of public health — Park's example is the bore-hole latrine demonstrated by the Rockefeller Foundation for hookworm control, since an essential part of the environmental sanitation programme
(e) Guarding the work of government agencies	By setting a good example, voluntary agencies can guide and criticise the work of government agencies
(f) Advancing health legislation	Voluntary agencies mobilise public opinion and advance legislation on health matters for the benefit of the whole community — the Society's own footing in three statutes (1920, amended 1992 with rules in 1994; and the Allocation of Property Act 1936) is the Indian illustration

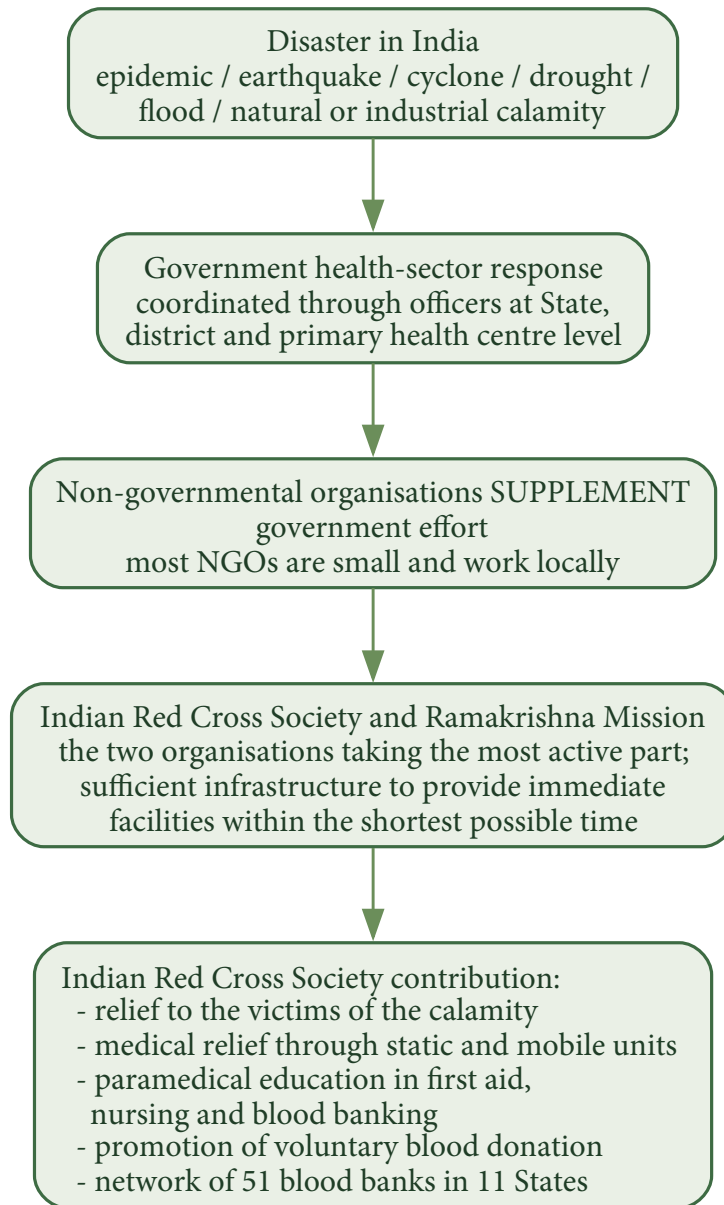
In maternity and child welfare specifically, the Society "has done pioneering work and has functioned as an **auxiliary of the country's health services**" (Park 28e, Ch 25, p 1061). Park's simile for the relationship is worth reproducing: voluntary health agencies are the "**motor trucks**" which can penetrate the by-ways, and the official agencies the "**railway trunk lines**" which must run on tracks established by law (Park 28e, Ch 10, p 746).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Lineage of the Red Cross movement to the Indian national society.** Labels to write: *Battle of Solferino, 1859 — Henry Dunant; Un Souvenir de Solferino — protective emblem, trained workers, protection by international treaty; First Geneva Convention 1864; International Committee of the Red Cross (ICRC) — independent, neutral, founder body; Dunant, first Nobel Prize for Peace, 1901; League of Red Cross Societies, 1919, Geneva (now IFRC) — more than 90 national societies; Indian Red Cross Society, 1920; World Red Cross Day — 8 May.*



Indian Red Cross Society — statutory identity, governance and branch structure



Where the Indian Red Cross Society sits in India's disaster response

High-yield exam pointers

- ▶ **Lineage in one line:** Solferino 1859 → *Un Souvenir de Solferino* → First Geneva Convention 1864 (ICRC) → Dunant, **first Nobel Peace Prize 1901** → League of Red Cross Societies 1919, Geneva (now IFRC) → **Indian Red Cross Society 1920. World Red Cross Day — 8 May every year.**
- ▶ **Statute and heads:** Indian Red Cross Society Act, incorporated under **Parliament Act XV of 1920**, amended 1992, rules 1994; **President of India = President, Union Health Minister = Chairman**; 25–30 members, of whom 10–30 form the **Managing Body**.
- ▶ **Three statutory objectives (H–P–M):** improvement of Health, Prevention of disease, Mitigation of suffering.
- ▶ **Reach:** 35 State/Union Territory branches and more than 700 district and sub-district branches (Jugal Kishore); Park's older undated count is "over 400 branches".
- ▶ **Four core programme areas:** humanitarian principles and values; disaster response (the largest share of work); disaster preparedness; health and care in the community.

- **Park's six activities (R–M–A–M–F–B):** Relief work; Milk and medical supplies; Armed forces; Maternal and child welfare; Family planning; Blood bank and first aid.
- **Two numbers and one institution:** 51 Red Cross blood banks in 11 States; more than 7,000 voluntary agencies in health work in India; the Red Cross Home, Bengaluru for permanently disabled ex-servicemen — "the only one of its kind in India and the Far East".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's counts of "over 400 branches" and "more than 90 national societies" are the textbook's own and carry no stated year, whereas Jugal Kishore gives the current structure of 35 State/Union Territory branches with more than 700 district and sub-district branches; the League of Red Cross Societies founded in 1919 is today named the International Federation of Red Cross and Red Crescent Societies (IFRC), the name Park itself uses in the disaster-management chapter.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, National Health Mission, Health Technology Assessment in India, National Blood Transfusion Council Standards 2022, Drugs and Cosmetics Rules, Disaster Management (Amendment) Act 2025, Indian Red Cross Society

Programme and policy updates (India)

- The Society's own current reach is **over 1,200 branches** across the country, organised in **four zones — East, West, North and South** — larger than either textbook count (Park's "over 400 branches" and Jugal Kishore's "more than 700 district and sub-district branches"). Operations quotable in an answer are relief for the **Wayanad landslide of 30 July 2024**, the **2025 Manipur flash floods** and the **Prayagraj Mahakumbh**, and a week-long **National Disaster Response Team (NDRT) training** run at National Headquarters from **11–18 July 2026** [IRCS 2026].
- The blood service in which the Society's blood banks sit is now large and tightly regulated: India has **more than 3,800 licensed blood centres** collecting about **14 million units a year**, most of them from **voluntary non-remunerated donors**. **Universal malaria screening of every donation is mandated by the Drugs and Cosmetics Rules**; the **National Blood Transfusion Council Standards, 2nd edition (2022)** go one step further and **recommend** — but do not mandate — a validated, sensitive antigen test for that screening [PMID 41906713].
- **National Sickle Cell Anaemia Elimination Mission** — announced in the **Union Budget 2023–24**, aiming to **eliminate sickle cell disease by 2047**, and targeting universal screening of about **7 crore tribal people aged 0–40 years nationally**. A rapid health technology assessment commissioned under Health Technology Assessment in India modelled the **4.11-crore tribal cohort aged 0–40 years of the six endemic States (Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha, Chhattisgarh)**: point-of-care kits cost more than the solubility-test standard of care (**US\$2.57–3.85 versus US\$0.67 per case screened**) but detected **21,125–41,458 additional cases**. On that evidence the **National Health Mission directed States to procure the kits by public tendering**, which drove prices below **US\$1.21 per test** and projects **US\$37.86 million saved at full coverage** [PMID 42594133]. It runs alongside the National Health Mission's **2016 haemoglobinopathy guidelines** and the **Thalassaemia Bal Sewa Yojana**, where a 68-centre national survey found leukodepleted red cells in **97% of centres** and prenatal diagnosis in **87%**,

yet monthly out-of-pocket costs of ₹500–16,000 despite free transfusion and chelation [PMID 41320182].

Current national data

- ▶ A 2025 nationwide online survey of India's licensed blood centres (439 analysable responses out of 4,153 centres) found 84.3% aware of vasovagal-reaction mitigation strategies but only 67.9% actually implementing them; fluid ingestion was the commonest physiological measure (94.8%), and of the 298 implementing centres only 44.3% had a dedicated donor counsellor — the case for a uniform national donor-care policy [PMID 41633805].
- ▶ Direct evidence on the **Junior Red Cross** wing: a pre–post study of 2,462 high-school JRC cadets found 73% had never attended a first-aid course; 80% could manage a cut injury correctly, but only 9.5% knew the first step for a seizure, 14.9% for burns and 34.6% for snakebite. Mean knowledge score rose from 3.91 to 6.65 and confidence improved significantly after a single 3-hour session taught by emergency physicians [PMID 40586649].

Elimination, surveillance and disease status

- ▶ **Malaria and the blood supply** — a national survey of 262 blood collection facilities (9,275,688 units, 2020–2024) found 97.7% screening every donation, 89.1% of them by antigen rapid diagnostic test; there were 1,231 confirmed positive donations (13.27 per 100,000 units), with the annual rate rising from 3.44 to 21.96 per 100,000 between 2020 and 2024. Only 3.1% of facilities reported transfusion-transmitted malaria, but external quality assessment ran in just 45.3%, lookback investigation in 33.6% and haemovigilance reporting in 49.6% — and endemic-area centres did less external quality assessment than non-endemic ones (28.7% versus 64.9%). The authors propose donor positivity as a **sentinel signal within India's 2030 malaria elimination goal** [PMID 41906713].
- ▶ **Hepatitis B in donors** — a meta-analysis of 40 Indian studies (January 2013 to October 2023) covering 2,222,736 donations gives a pooled HBsAg prevalence of 1.11% on the common-effect model and 0.95% (95% CI 0.79–1.16%) on the random-effects model the authors prefer — well below the 3–4.5% intermediate-endemicity figure the 2016 WHO fact sheet attributed to India, and consistent with the **National Viral Hepatitis Control Programme (2018)** goal of eliminating viral hepatitis as a public health problem by 2030. Prevalence is far higher in tribal areas (15.9%) than elsewhere (2.4%) [PMID 39513206].

Guideline and legislative updates

- ▶ **Haemovigilance now has two arms** — the Haemovigilance Programme of India (2012) for recipients and the **National Blood Donor Vigilance Programme** for donors, the latter having trialled the AABB donor-reaction **severity grading tool** across 50 Indian blood centres (overall agreement with the expert group 66.4%, Kendall's W 0.271); the programme's National Coordinating Centre concludes that the tool **may be implemented** in the programme — it has not yet been adopted [PMID 37533234]. A critical appraisal of 13 Indian knowledge-attitude-practice studies (1,684 participants) found awareness of transfusion reactions high but knowledge of the **reporting mechanism poor**, with lack of training, time pressure and fear of legal consequences the named barriers, and only one of the 13 studies reaching an acceptable methodological quality score [PMID 40825713].

- **Disaster Management (Amendment) Act, 2025** — cleared by the Rajya Sabha on 25 March and the Lok Sabha on 27 March 2025 — rewrites the statutory frame in which the Society's relief work sits: the NDMA and State Disaster Management Authorities themselves now prepare disaster management plans (previously the work of the national and State executive committees), national and State disaster databases become mandatory, Urban Disaster Management Authorities may be constituted for State capitals and cities with a municipal corporation, States may raise their own State Disaster Response Force, and the National Crisis Management Committee and High Level Committee gain statutory status [DM (Amendment) Act 2025-Mar].

Global and WHO developments

- **World Blood Donor Day, 14 June 2025** was observed under the slogan "*Give blood, give hope: together we save lives*", with objectives of raising awareness of the need for blood and plasma, recruiting and retaining regular voluntary donors, and securing government and international commitment to fund sustainable national blood programmes so that safe transfusion becomes universally available — the companion global date to the movement's own **World Red Cross Day on 8 May** [WHO 2025-Jun].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 10 (Health Care of the Community), pp 746–747; Ch 25 (International Health), pp 1060–1061; Ch 22 (Communication for Health Education), p 1024; Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), p 699; Ch 17 (Disaster Management), p 957. Park's Textbook of Preventive and Social Medicine, 27e — Ch 10, p 725 and Ch 25, pp 1035–1036, used only to read the identical passages where the 28e scan is optical-character-recognition damaged (the branch count and the verbatim activity list). Jugal Kishore — National Health Programmes of India, Ch 3, pp 968–969 and p 989; Ch 1, pp 654–655. Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 41 (Disaster Management), p 707. Suryakantha — Community Medicine with Recent Advances, Ch 43 (Bilateral Agencies), pp 835–836. Sunder Lal — Textbook of Community Medicine, Ch 17 (Healthcare of the Community), p 765. Coverage note: the corpus states that the movement works from seven Fundamental Principles but does not enumerate them, so they are not listed here.

Recent-advances sources: 41906713 — Transfusion-transmitted malaria in India: a national survey of epidemiology and testing practices, *Vox Sanguinis* 2026, doi:[10.1111/vox.70261](https://doi.org/10.1111/vox.70261); 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, *PLoS One* 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 39513206 — Burden of hepatitis B in asymptomatic blood donor population of India: a systematic review and meta-analysis, *Indian Journal of Medical Research* 2024, doi:[10.25259/ijmr.285_24](https://doi.org/10.25259/ijmr.285_24); 41633805 — Assessment of vasovagal reaction mitigation strategies in licensed blood centres in India (2025): a nationwide online survey, *Vox Sanguinis* 2026, doi:[10.1111/vox.70197](https://doi.org/10.1111/vox.70197) (abstract-only); 41320182 — Assessment of services for diagnosis, management and prevention of thalassaemia in India: the way forward, *Vox Sanguinis* 2025, doi:[10.1111/vox.70157](https://doi.org/10.1111/vox.70157) (abstract-only); 40586649 — Baseline knowledge and change in confidence after learning first aid from emergency physicians: a pre-post interventional study of 2462 high school Junior Red Cross cadets, *Indian Journal of Public Health* 2025, doi:[10.4103/ijph.ijph.1458_22](https://doi.org/10.4103/ijph.ijph.1458_22) (abstract-only); 37533234 — Knowledge assessment of a donor adverse reaction severity grading tool by the National Blood Donor Vigilance Programme of India, *Vox Sanguinis* 2023, doi:[10.1111/vox.13502](https://doi.org/10.1111/vox.13502) (abstract-only); 40825713 — Knowledge, attitude and practice studies on hemovigilance in India: a critical appraisal, *Transfusion Medicine* 2025, doi:[10.1111/tme.70011](https://doi.org/10.1111/tme.70011) (abstract-only). Also used: WHO World Blood Donor Day, 14 June 2025; Disaster Management (Amendment) Act 2025, March 2025; Indian Red Cross Society National Headquarters, 2026.

♦ ♦ ♦

Intersectoral coordination in primary health care

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — Intersectoral coordination is the planned working together of the health sector with every other sector whose activities determine health, so that their combined effect is greater than the sum of their separate effects (Sunder Lal, Ch 16, p 711). The Declaration of Alma-Ata (1978) states that "primary health care involves in addition to the health sector, all related sectors and aspects of national and community development, in particular agriculture, animal husbandry, food, industry, education, housing, public works, communication and other sectors" (Park 28e, Ch 10, p 720). It is one of the four founding principles of primary health care (PHC), alongside equitable distribution, community participation and appropriate technology (Park 28e, Ch 10, pp 719–720).

Why intersectoral coordination is indispensable to primary health care

- ▶ **The components of PHC cannot be provided by the health sector alone.** The eight essential elements declared at Alma-Ata belong to many inter-related sectors, not to one; health for all is therefore achievable only through inter- and intra-sectoral coordination (Sunder Lal, Ch 17, p 741).
- ▶ **Health is a development function.** The determinants of ill-health — the "causes of the causes" — lie in water and sanitation, nutrition, female literacy, early childhood development and women's empowerment, and need collective action across sectors (Sunder Lal, Ch 1, p 39).
- ▶ **Historical proof.** Sri Lanka, Costa Rica and Kerala achieved good health outcomes because effort in the health field was simultaneously reinforced by education, social welfare and land reform (Park 28e, Ch 2, p 35).
- ▶ **It is a stated obligation of governments.** Alma-Ata called on all governments to review their administrative systems, reallocate resources and introduce suitable legislation so that coordination can take place; this needs strong political will, and its operative element is planning *with* other sectors to avoid unnecessary duplication of activities (Park 28e, Ch 10, p 720).
- ▶ **It is one of WHO's eight supportive activities for primary health care** — the eight are community involvement and participation; intra- and inter-sectoral coordination; provision of effective referral support; development and mobilisation of resources; managerial processes (policy, goals and objectives); medical and health services research including innovative approaches; development and application of appropriate technology; and health manpower development (Sunder Lal, Ch 17, p 741).

Essential elements of primary health care and the sectors that must deliver them

Essential element of PHC (Alma-Ata)	Sector chiefly responsible besides health	Indian operational link
Education about prevailing health problems and their prevention	Education; information and broadcasting	School Health Programme, <i>Rashtriya Bal Swasthya Karyakram</i> , National Tobacco Control and National Oral Health Programmes run in convergence with the education department (Sunder Lal, Ch 16, p 711)

Essential element of PHC (Alma-Ata)	Sector chiefly responsible besides health	Indian operational link
Promotion of food supply and proper nutrition	Agriculture; women and child development; food and civil supplies	ICDS supplementary nutrition through the <i>Anganwadi</i> centre; mid-day meal scheme, now PM POSHAN (Park 28e, Ch 11, pp 803–804)
Adequate supply of safe water and basic sanitation	Public works; rural/urban development; drinking water and sanitation	<i>Swachh Bharat Abhiyan</i> — safe water and sanitation at the <i>Anganwadi</i> centre and in the village (Sunder Lal, Ch 16, p 711)
Maternal and child health care including family planning	Women and child development; panchayati raj	Joint home visits by <i>Anganwadi</i> worker, auxiliary nurse midwife (ANM) and accredited social health activist (ASHA) (Sunder Lal, Ch 16, p 711)
Immunisation against major infectious diseases	Panchayati raj; information and broadcasting; transport	Monthly Village Health, Nutrition and Sanitation Day (VHSND) at the <i>Anganwadi</i> centre (Sunder Lal, Ch 5, pp 229–230)
Prevention and control of locally endemic diseases	Animal husbandry; agriculture; municipal bodies	One Health — multiple sectors communicating and working together for better public health outcomes (WHO, cited in Sunder Lal, Ch 1, p 36)
Appropriate treatment of common diseases and injuries	Transport; local self-government	Equipped ambulances by population norm, with follow-up linked back to the primary health centre; formally listed collaborations with women and child development (nutrition), education (school health) and Jal Shakti (safe water and sanitation) (Sunder Lal, Ch 17, p 754)
Provision of essential drugs	Industry; pharmaceuticals	Supply of essential medicines at sub-centre, PHC and CHC under the National Health Mission (Sunder Lal, Ch 17, p 739)

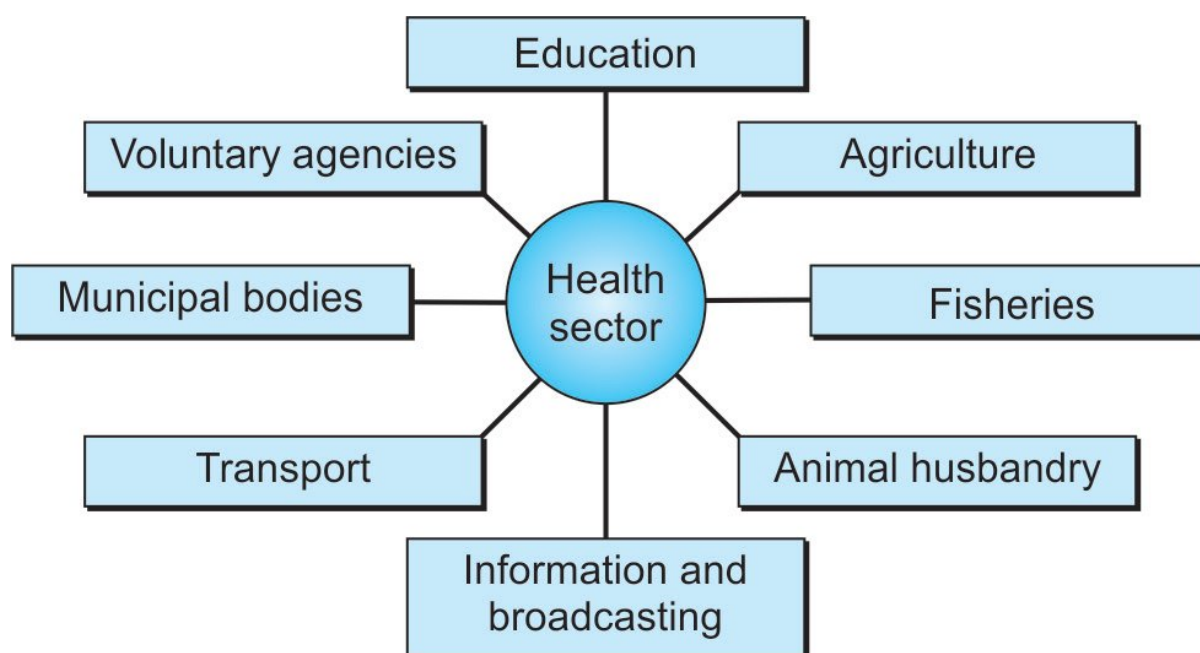


Fig 34.3 — Intersectoral co-ordination of primary health care (Suryakantha — *Community Medicine with Recent Advances*, Ch 35 (Chapter-34_Health Care), p 781)

Institutional mechanisms of coordination in India

Level	Coordinating body	Instrument / mode of coordination
National	Mission Steering Group of the National Health Mission, chaired by the Union Minister of Health and Family Welfare and including the Ministers of Panchayati Raj, Rural Development and Human Resource Development; Empowered Programme Committee chaired by the Secretary, Health (Jugal Kishore, Ch 1, pp 125, 138–139)	Policy direction; National Health Policy 2017's dedicated Health Impact Cell to analyse the health impact of existing and new policies of other sectors (Jugal Kishore, Ch 1, p 92)
National (population)	Coordination Cell proposed in the Planning Commission under the National Population Policy 2000 (Jugal Kishore, Ch 2, pp 754–755)	Intersectoral coordination for population and reproductive and child health goals
State	State Health Mission headed by the Chief Minister; State Health Society headed by the Chief Secretary (Jugal Kishore, Ch 1, p 126)	Merger of erstwhile separate programme societies into one society; sector-wide memoranda of understanding with other departments (Jugal Kishore, Ch 1, p 92)
District	District Health Mission under the Zila Parishad with the district health head as convener and all relevant departments, non-governmental organisations and private professionals represented; District Health Society headed by the District Collector (Jugal Kishore, Ch 1, pp 126, 138–139)	District Health Action Plan — the major instrument for convergence; facilitates coordination with women and child development, water supply and sanitation, especially during outbreaks (Jugal Kishore, Ch 1, p 126; Sunder Lal, Ch 17, p 739)
Block	Joint planning by the medical officer, child development project officer, block development officer, block education officer and the <i>Block Samiti</i> (Sunder Lal, Ch 16, p 711)	Block-level joint plan and joint review
Village	Village Health, Sanitation and Nutrition Committee (VHSNC) — panchayati raj members, ANM, <i>Anganwadi</i> worker, ASHA, schoolteacher and NGOs, with an untied grant of ₹10,000 per annum; over 5.55 lakh committees established (Sunder Lal, Ch 17, p 744)	Village Health Action Plan; monthly VHSND; the village is the first level at which programmes of all ministries converge (Sunder Lal, Ch 16, p 711)
Urban	District Health Society as the coordinating body under the National Urban Health Mission; <i>Mahila Arogya Samiti</i> in slums (Sunder Lal, Ch 17, pp 759–760)	Convergence with urban local bodies, education, ICDS, urban development and housing, water and sanitation
Facility	<i>Rogi Kalyan Samiti</i> at PHC and CHC, with a PHC untied grant of ₹25,000 and maintenance grant of ₹50,000 per annum (Sunder Lal, Ch 17, pp 751–752)	Community control and participation in facility management

The legal basis is the **73rd and 74th Constitutional Amendment Acts, 1992**, which made health, family welfare and education a responsibility of village *panchayats* and *nagarpalikas* and devolved administrative and financial powers to the three-tier panchayati raj structure of village, block and

district recommended by the *Balwant Rai Mehta* Committee, 1957 (Sunder Lal, Ch 9, p 553; Ch 2, p 67).

Convergence in practice — how it is actually done

- ▶ **Co-location of facilities** — siting the *Anganwadi* centre with the sub-centre or within the school; described as a best management practice (Sunder Lal, Ch 16, p 711).
- ▶ **Joint field work** — joint home visits and joint household surveys by the *Anganwadi* worker, ANM and ASHA (Sunder Lal, Ch 16, p 711).
- ▶ **A common shared event** — the VHSND held once a month on a fixed date at the *Anganwadi* centre by ASHA with the *Anganwadi* worker, ANM, self-help groups, community-based organisations and panchayati raj members, delivering immunisation, antenatal and postnatal check-ups and child and adolescent services (Sunder Lal, Ch 5, pp 229–230).
- ▶ **A single committee instead of many** — the National Rural Health Mission moved the separate village committees for drinking water, sanitation and ICDS towards one common village health and sanitation committee (Sunder Lal, Ch 17, p 739).
- ▶ **Hubs for health action** — the *Anganwadi* centre and the school system are the principal hubs at village level; the National Health Policy 2017 requires states to observe Village Health and Nutrition Days in complete convergence mode with *Anganwadi* centres as the hub for all health-related services (Sunder Lal, Ch 17, p 739; Jugal Kishore, Ch 1, p 92).
- ▶ **Institutional arrangement between ministries** — the Ministry of Health and Family Welfare to build arrangements with the Ministry of Women and Child Development so that convergent delivery of ICDS services becomes the norm (Jugal Kishore, Ch 1, p 92).
- ▶ **The logic in one line** — supplementary nutrition achieves little while childhood diarrhoea goes unchecked; only convergent action reduces morbidity, mortality and undernutrition and lifts coverage (Sunder Lal, Ch 16, p 711).

Global evolution of the idea

Milestone	Year	Contribution to intersectoral coordination
Declaration of Alma-Ata	1978	Named intersectoral coordination a principle of PHC and listed the partner sectors (Park 28e, Ch 10, p 720)
Ottawa Charter for Health Promotion	1986	Built on Alma-Ata and on the World Health Assembly debate on intersectoral action for health (Park 28e, Ch 2, p 49)
Adelaide Recommendations on Healthy Public Policy	1988	Shifted intersectoral action into healthy public policy (Oxford 7e, Ch 47, p 1766)
Healthy Cities movement (from the <i>Beyond Health Care</i> conference, Toronto 1984)	1984 onwards	City health plan built on intersectoral partnerships and community empowerment (Oxford 7e, Ch 99, pp 3770–3771)

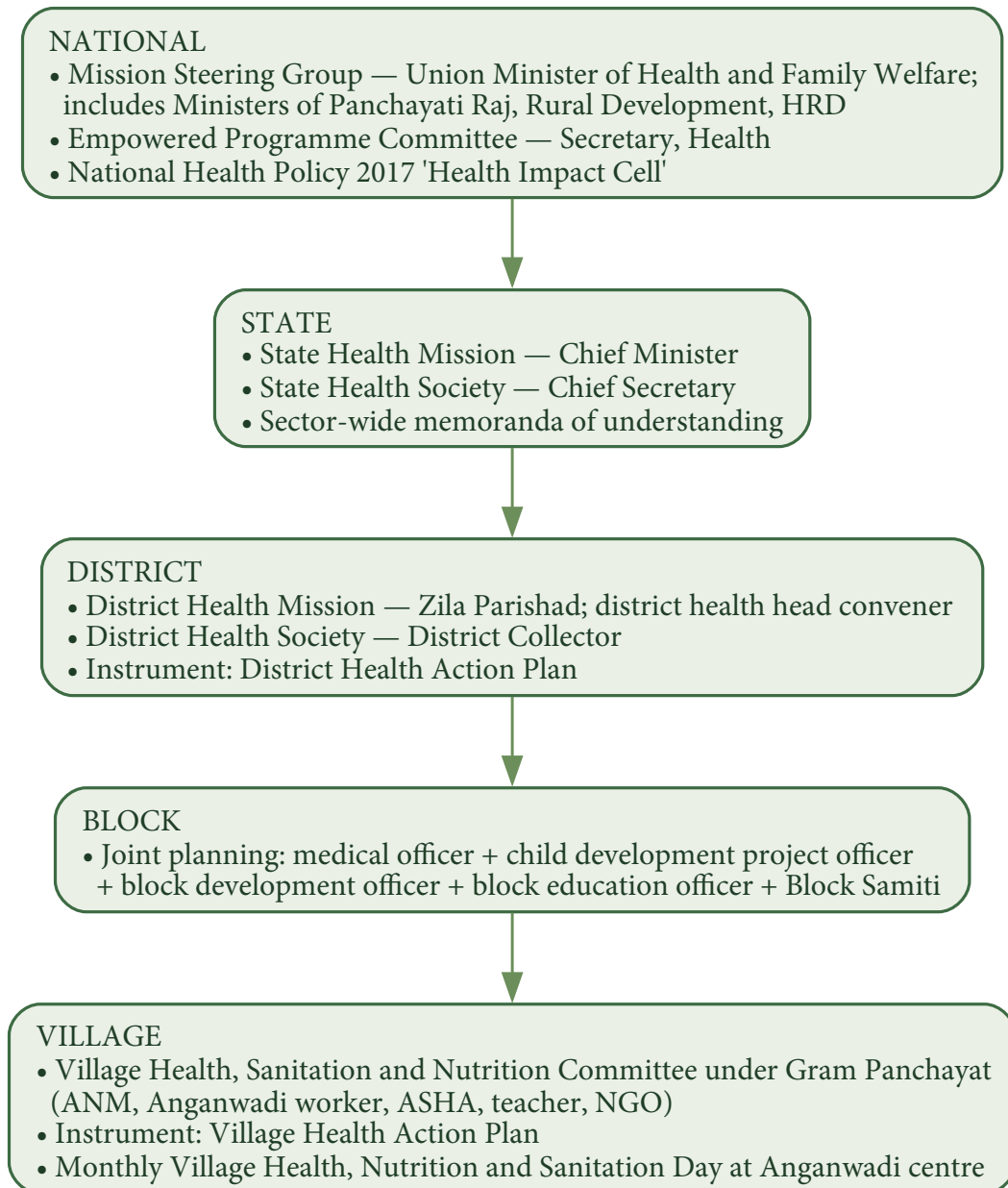
Milestone	Year	Contribution to intersectoral coordination
Adelaide Statement on Health in All Policies	2010	Shared governance for health; Health in All Policies defined by WHO as an approach across sectors that systematically takes account of the health implications of decisions, seeks synergies and avoids harmful health impacts, to improve population health and health equity (Oxford 7e, Ch 51, p 1900)
Sustainable Development Goals	2015	The 17 goals are interconnected and indivisible; Goal 3 is unachievable by the health sector alone (Oxford 7e, Ch 24, pp 955–956)
Astana Declaration on Primary Health Care	2018	Reinforced the importance of primary health care, which resists the separation of treatment from prevention and demands equity in power and resources (Oxford 7e, Ch 47, pp 1751–1752)

Constraints and measures to strengthen coordination

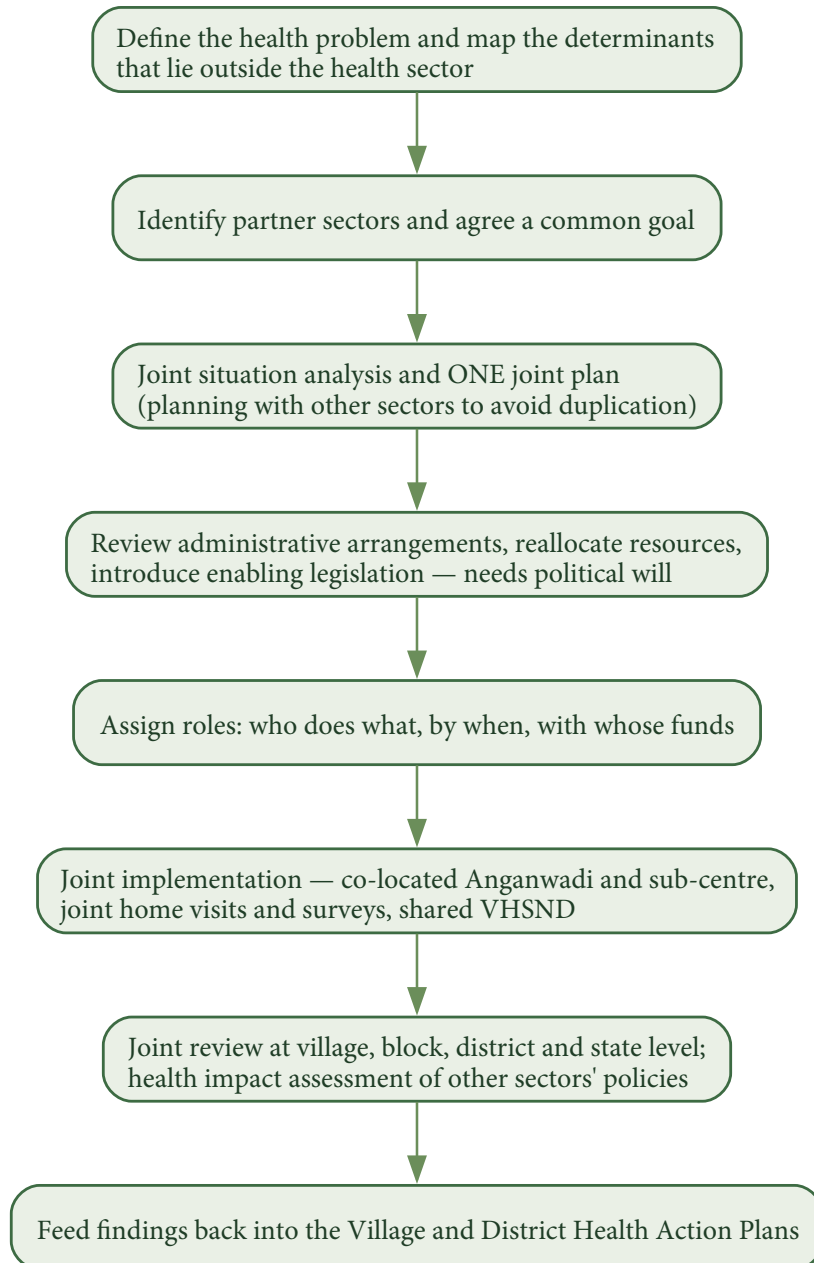
Constraint	Corrective measure
"Siloed" government departments — decision-making, funding and accountability are organised vertically within departments, discouraging horizontal collaboration (Oxford 7e, Ch 6, pp 314–315)	Review the administrative system and reallocate resources across sectors; enact enabling legislation as Alma-Ata directed (Park 28e, Ch 10, p 720)
Difficulty of sustaining both vertical (levels of government) and horizontal (cross-ministry) collaboration — essentially a governance and accountability problem (Oxford 7e, Ch 47, p 1766)	A single joint plan per level — District Health Action Plan and Village Health Action Plan with named roles (Sunder Lal, Ch 17, p 739)
Insufficient political commitment; marginalisation of the health department within government; limited capacity of public health to influence policymaking (Oxford 7e, Ch 51, p 1900)	High-level chairs for coordinating bodies — Chief Minister for the State Health Mission, Chief Secretary for the State Health Society, District Collector for the District Health Society (Jugal Kishore, Ch 1, p 126)
Duplication of activities between sectors and between vertical programmes (Park 28e, Ch 10, p 720; Jugal Kishore, Ch 2, pp 732–733)	Joint planning with other sectors; merger of separate programme societies into one state and one district health society (Jugal Kishore, Ch 1, p 622)
The health sector often lacks the expertise and capacity to measure the health impacts of other sectors' proposed projects or policies, so its ability to participate meaningfully in intersectoral work is limited (Oxford 7e, Ch 99, pp 3771–3772)	Health impact assessment, which helps institutionalise intersectoral processes in policymaking (Oxford 7e, Ch 99, p 3784); the National Health Policy 2017 Health Impact Cell (Jugal Kishore, Ch 1, p 92)
Weak participation of panchayati raj institutions and NGOs in practice — acknowledged as a weak component needing strengthening (Sunder Lal, Ch 5, pp 229–230)	Devolve funds, functions and functionaries to the village <i>panchayat</i> ; make ASHA member-secretary of a VHSNC that is a standing subcommittee of the <i>panchayat</i> (Sunder Lal, Ch 1, p 42; Ch 2, p 67)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Figure embedded above — *Intersectoral co-ordination of primary health care*: draw the health sector as a central circle, with eight boxes radiating from it, labelled exactly — Education, Agriculture, Fisheries, Animal husbandry, Information and broadcasting, Transport, Municipal bodies, Voluntary agencies.



Institutional levels of intersectoral coordination for primary health care in India



Steps in operationalising intersectoral coordination for a primary health care problem

High-yield exam pointers

- ▶ **Four principles of PHC (mnemonic ECIA)** — Equitable distribution, Community participation, Intersectoral coordination, Appropriate technology (Park 28e, Ch 10, pp 719–720).
- ▶ **Eight, not five** — intersectoral coordination is one of WHO's **eight** supportive activities for PHC; the count is a favourite trap (Sunder Lal, Ch 17, p 741).
- ▶ **The Alma-Ata sector list, verbatim** — agriculture, animal husbandry, food, industry, education, housing, public works, communication and other sectors.
- ▶ **Health for All, 30th World Health Assembly (1977)** — "attainment of a level of health that will enable every individual to lead a socially and economically productive life"; the Alma-Ata conference was held in September 1978, jointly organised by WHO and UNICEF.

- ▶ **Three-word test for the Alma-Ata obligation on governments** — review the administrative system, reallocate resources, introduce legislation; the operative element is joint planning to avoid duplication.
- ▶ **Indian coordination chairs** — State Health Mission: Chief Minister; State Health Society: Chief Secretary; District Health Mission: Zila Parishad chairperson; District Health Society: District Collector.
- ▶ **VHSNC numbers** — untied grant ₹10,000 per annum, over 5.55 lakh committees; PHC untied grant ₹25,000 and maintenance grant ₹50,000 per annum.
- ▶ **Health in All Policies** — the current name for intersectoral action; the idea arrived in three successive waves, intersectoral action first, then healthy public policy, then Health in All Policies.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the National Health Policy 2017 "Health Impact Cell" and the sector-wide memoranda of understanding are policy commitments as worded in the policy document; the retrieved corpus carries no dated national indicator of how well intersectoral coordination actually performs, so no coverage figure is quoted above.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, National Multisectoral Action Framework for TB-Free India 2019, National Action Plan for Rabies Elimination 2021, FSSAI 2020 school-food regulations, Astana Declaration 2018

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandirs**, and the same platform is being asked to carry tuberculosis, non-communicable disease and mental health care together — about 1.6 lakh Ayushman Arogya Mandirs, 770 district-level and over 6,400 block-level NCD clinics now screen more than eight crore adults a year for hypertension and diabetes, yet the National TB Elimination Programme, NP-NCD and the National Mental Health Programme still run as vertical silos with unlinked digital systems (Ni-kshay, the NCD application, Tele-MANAS); the convergence levers proposed are joint oversight committees, ABHA-linked longitudinal records under the Ayushman Bharat Digital Mission, and cross-training of frontline workers so a TB visit also screens for diabetes and depression [PMID 42397826].
- ▶ **TB Mukht Panchayat, launched 24 March 2023**, makes the village council rather than the health facility the unit of TB elimination: it works through roughly 2,55,000 village-level local self-governments and their VHSNC and Jan Arogya Samiti sub-committees, and operationalises the National Multisectoral Action Framework for TB-Free India (2019), which frames ending TB as a whole-of-society rather than a health-sector task; "TB-free" status is claimed yearly by the panchayat and verified by a district-level team of health department and panchayati raj institution officials together with doctors' organisations and public health professionals [PMID 38756161].
- ▶ **Kerala's Aardram mission** converted primary health centres into family health centres through decentralised governance — multi-sectoral training helped local governments engage with the health system as planning partners, and the decentralised structures then mobilised finance,

human resources and infrastructure modification with greater community participation; non-uniform commitment, sub-optimal engagement of *urban* local governments and unresolved sustainability and monitoring were the documented shortcomings. A usable modern Indian example to quote alongside the Alma-Ata obligation on governments [PMID 36949746].

Current national data

- ▶ **Money, not intent, is the binding constraint** — 2018 estimates put India's total per-capita expenditure on primary health care services at about US\$32, of which only US\$11 was government spending; health took 3.3% of government expenditure, and out-of-pocket payments formed 44–73% of total health expenditure across the five South Asian countries studied [PMID 39301269].
- ▶ **Adolescent services show what weak convergence costs:** a 2015–2025 scoping review of Rashtriya Kishor Swasthya Karyakram delivery found dedicated adolescent services in only 12.5% of assessed facilities and adolescent care at 6% of health and wellness centres, with poor intersectoral coordination listed alongside staff shortage, lack of privacy, provider bias against unmarried youth and weak referral systems among the barriers [PMID 42469808].

Elimination, surveillance and disease status

- ▶ **Rabies — National Action Plan for Rabies Elimination (2021), dog-mediated human rabies elimination by 2030 ("Zero by 30"):** Sikkim and Goa reached rabies-controlled status through One Health working, but nationally about 17.4 million animal bites and 20,565 human rabies deaths continue each year; the 2022–23 national survey found 71% of bite victims received at least three vaccine doses while only 43% completed the post-exposure schedule, and roughly one in five government facilities had no rabies immunoglobulin while about 30% had no vaccine stock [PMID 42494549].
- ▶ **Two separate coordination failures sit behind that burden.** Mass dog vaccination stays below the 70% coverage needed to interrupt transmission because vaccination and Animal Birth Control programmes are fragmented, episodic and largely dependent on non-governmental organisations instead of sustained government investment, with catch-neuter-vaccinate-release and oral dog vaccination implemented weakly and inconsistently. Separately, surveillance is poorly interoperable across the human-health, veterinary, municipal and wildlife sectors, so outbreaks are detected and answered late. The remedies proposed are a unified One Health digital platform with standardised case definitions and shared indicators for data sharing, and state and district One Health task forces carrying measurable cross-sectoral targets — canine vaccination coverage of at least 70%, post-exposure prophylaxis completed on time, fewer human rabies deaths — linked to performance-based financing [PMID 42494549].
- ▶ **Elimination deadlines have slipped and are being renegotiated** — India's 2025 TB elimination target, set in 2018 five years ahead of the global goal, is now argued to be unrealistic, with a call to realign to 2030 and strengthen multisectoral coordination alongside Pradhan Mantri TB Mukh Bharat Abhiyan and Ni-kshay Poshan Yojana [PMID 40781735]; lymphatic filariasis remains endemic in 345 districts against a 2030 elimination goal, for which public–private partnership is proposed as the under-used lever [PMID 41951149].

Guideline and legislative updates

- **Law can pull against the programme:** rabies control rests on the Prevention of Cruelty to Animals Act 1960 and the Animal Birth Control Rules 2001 (revised 2023), which mandate sterilisation and vaccination rather than culling, but August 2025 Supreme Court proceedings permitting forcible removal of street dogs in defined circumstances were perceived by many stakeholders as conflicting with them; continuing legal ambiguity, conflicting state legislation and recurrent culling campaigns divert effort from validated control measures — a live illustration of how legislative incoherence between sectors disables intersectoral action [PMID 42494549].
- **Food and school regulation as health policy:** India's building blocks against childhood obesity (9–12% of children and adolescents overweight or obese) are the FSSAI 2020 regulations restricting high-fat, salt and sugar foods in and around schools, the Ayushman Bharat School Health Programme, PM POSHAN meals and higher effective taxation of sugary drinks, with an interpretive front-of-pack nutrition labelling system still under development [PMID 41974918].

Global and WHO developments

- **Astana (2018) restated primary health care as three pillars — integrated services, multi-sectoral collaboration and community empowerment;** the 2024 Lancet Regional Health series on primary health care in South Asia turns this into nine cross-cutting action points, the first of which calls for multi-sectoral platforms that formulate unified primary health care policies with commonly defined goals, and notes that health ministries often lack the authority to fund or compel action outside their own portfolio — hence the call to site the coordinating platform inside the government's common planning and development machinery rather than inside the health department [PMID 39301269].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2, Ch 10 Health Care of the Community, Ch 11 Nutrition and Health); Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 2, Ch 5, Ch 9, Ch 16 Health Planning and Management, Ch 17 Healthcare of the Community); Jugal Kishore — National Health Programmes of India (National Health Mission structure; National Health Policy 2017; National Population Policy 2000); Oxford Textbook of Public Health 7e (Ch 6, Ch 24, Ch 47, Ch 51, Ch 99); Suryakantha — Community Medicine with Recent Advances (Ch 35, Fig 34.3, p 781). Named documents: Declaration of Alma-Ata 1978; Ottawa Charter 1986; Adelaide Recommendations 1988 and Adelaide Statement on Health in All Policies 2010; Sustainable Development Goals 2015; Astana Declaration on Primary Health Care 2018.

Recent-advances sources: 42397826 — Integrated tuberculosis-multimorbidity management in India: a SWOT analysis, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR.2790.2025](https://doi.org/10.25259/IJMR.2790.2025); 38756161 — How can TB Mukht Panchayat initiative contribute towards ending tuberculosis in India?, *The Lancet Regional Health Southeast Asia* 2024, doi:[10.1016/j.lansea.2024.100376](https://doi.org/10.1016/j.lansea.2024.100376); 36949746 — Re-engineering primary healthcare in Kerala, *Public Health Action* 2023, doi:[10.5588/pha.22.0033](https://doi.org/10.5588/pha.22.0033) (abstract-only); 39301269 — Reorienting health systems towards Primary Health Care in South Asia, *The Lancet Regional Health Southeast Asia* 2024, doi:[10.1016/j.lansea.2024.100466](https://doi.org/10.1016/j.lansea.2024.100466); 42469808 — Bridging the gap between policy and practice in adolescent health services: evidence from Indian primary public health centers, *BMC Health Services Research* 2026, doi:[10.1186/s12913-026-15085-x](https://doi.org/10.1186/s12913-026-15085-x) (abstract-only); 42494549 — One Health implementation for rabies elimination in India by 2030: progress, governance, challenges, and policy alignment, *One Health* 2026, doi:[10.1016/j.onehlt.2026.101520](https://doi.org/10.1016/j.onehlt.2026.101520); 40781735 — Enhancing India's health security efforts against *Mycobacterium tuberculosis*: gaps and opportunities, *Health Security* 2025, doi:[10.1089/hs.2024.0109](https://doi.org/10.1089/hs.2024.0109) (abstract-only); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 41974918 — Public health approaches to combating childhood obesity in India: a mini-review, *European Journal of Clinical Nutrition* 2026, doi:[10.1038/s41430-026-01742-1](https://doi.org/10.1038/s41430-026-01742-1) (abstract-only). Named documents cited through these: National Multisectoral Action Framework for TB-Free India 2019; National Action Plan for Rabies Elimination 2021; Prevention of Cruelty to Animals Act 1960 and Animal Birth Control Rules 2001 (revised 2023); FSSAI 2020 school-food regulations; Astana Declaration on Primary Health Care 2018.

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Job functions of Health Worker (Female)

Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — The health worker (female), better known as the **auxiliary nurse midwife (ANM)**, is the multipurpose grass-root health functionary posted at the sub-centre, the most peripheral contact point between the primary health care system and the community. Under the **Multipurpose Workers Scheme** one health worker (female) and one health worker (male) are posted at each sub-centre for 5,000 population (3,000 in tribal and hilly areas), and she limits her activities to 350–500 families (Park 28e, Ch 10, pp 738–739). She works under the supervision of the health assistant (female) or lady health visitor (LHV) and delivers the whole family-health and disease-control package at household level (Sunder Lal, Ch 16, pp 717–718).

Post, coverage and place in the health team

Item	Norm
Facility	Sub-health centre, now the Health and Wellness Centre — Sub Health Centre (Sunder Lal, Ch 17, p 746)
Population covered	5,000 in plain areas; 3,000 in hilly, tribal, desert and difficult areas (Sunder Lal, Ch 17, p 746)
Families she works among	350–500 families (Park 28e, Ch 10, p 738)
Staffing	Two ANMs (the second added on contract under the National Health Mission); one must stay at the sub-centre , and the 5,000 population is divided between them (Sunder Lal, Ch 17, p 745)
Immediate supervisor	Health assistant (female) / LHV — covers 30,000 population (20,000 tribal and hilly), six sub-centres, and visits each once a week on a fixed day (Sunder Lal, Ch 16, p 717)
Team below her	Accredited social health activist (ASHA), anganwadi worker (AWW), trained dai, female depot holder
Salary	Borne 100% by the Government of India (as for the LHV); the male health worker is paid by the state (Sunder Lal, Ch 17, p 745)

A. Maternal and child health

- ▶ Register pregnant women at the earliest contact and provide care throughout pregnancy; ensure every pregnant woman makes **at least four antenatal visits** as per the suggested schedule.
- ▶ Test urine for albumin and sugar, estimate haemoglobin, and refer all pregnant women to the primary health centre (PHC) for the **rapid plasma reagin (RPR) test for syphilis**.
- ▶ Refer abnormal pregnancy and cases with medical or gynaecological problems to the health assistant (female) or the PHC.
- ▶ Conduct deliveries in her area when called for, supervise deliveries conducted by dais, and refer difficult labour and newborns with abnormalities for institutional care, with follow-up after discharge.
- ▶ Under **Janani Suraksha Yojana**, identify beneficiaries and obtain approval of the competent authority before disbursement; submit the previous month's accounts by the **7th of each month**;

prepare the monthly work schedule at the mandatory meeting of accredited workers on the **3rd Friday of every month**.

- ▶ Carry out **name-based tracking of all pregnancies** for scheduled antenatal and postnatal services.
- ▶ Make post-natal home visits on **day 0, 3, 7 and 42** for home and sub-centre deliveries and on **day 3, 7 and 42** for institutional deliveries, advising on maternal care, newborn care and feeding.
- ▶ For a **low birth weight baby**, make a total of six post-natal visits from day 0 up to day 28 to screen for congenital abnormality and danger signs of sickness as per **integrated management of neonatal and childhood illness (IMNCI)** guidelines, with appropriate referral.
- ▶ Ensure initiation of breastfeeding within one hour of birth, exclusive breastfeeding, and timely complementary feeding at six months as per infant and young child feeding guidelines.
- ▶ Assess growth and development of infants and under-five children and make timely referral.
- ▶ Treat all cases of diarrhoea, acute respiratory infection (pneumonia) and other minor ailments, and refer severe dehydration, respiratory distress, severe acute malnutrition and other serious conditions as per IMNCI.
- ▶ Educate mothers individually and in groups, and assist the medical officer and health assistant (female) at antenatal and postnatal clinics held at the sub-centre.

B. Family planning, medical termination of pregnancy and nutrition

Domain	Job functions (Park 28e, Ch 10, pp 739–740)
Family planning	Maintain and continuously update the eligible couple and child register and use it to run the programme; distribute conventional and oral contraceptives; help acceptors reach services, accompanying them or arranging for the dai/ASHA to do so; follow up female acceptors, treat minor side-effects on the spot and refer the rest; establish female depot holders and keep them supplied
Community mobilisation for family welfare	Build rapport with acceptors, village leaders, ASHA and dais; identify women leaders and help the health assistant (female) train them; use mahila mandal meetings to educate women
Medical termination of pregnancy	Identify women needing termination and refer them to the nearest approved institution ; educate the community on the consequences of septic abortion and on where legal services are available
Nutrition	Identify malnutrition in children from birth to five years, treat and advise, refer severe cases to the PHC; distribute iron and folic acid tablets to pregnant women, nursing mothers and children up to five years; administer vitamin A solution as per guidelines

C. Immunisation and cold chain

- ▶ Immunise pregnant women with **tetanus toxoid**, and give BCG, pentavalent, DPT, oral polio vaccine, inactivated polio vaccine, hepatitis B and measles vaccines to all infants and children as per the national immunisation schedule.
- ▶ Prepare the work plan, estimate beneficiaries and logistics, and prepare the **due list** in coordination with the anganwadi worker and the ASHA mobiliser, ensuring turnout on session day.

- ▶ She is **responsible for cold chain maintenance** during both fixed and outreach sessions, and indents vaccine and logistics weekly on the basis of the due beneficiary list.
- ▶ Ensure injection safety and safe disposal, and manage immunisation waste as per Government of India / Central Pollution Control Board guidelines.
- ▶ Record, report and manage minor and serious **adverse events following immunisation (AEFI)**; submit monthly programme reports and weekly acute flaccid paralysis and measles surveillance reports; report serious AEFI and outbreaks **immediately**.
- ▶ Maintain the **tracking bag / tickler box** with updated counterfoils, track dropouts and left-outs, and display the work plan and village-wise dropout list at the sub-centre.

D. Disease control functions

Programme area	Her specific duties
Surveillance / notification	Notify the medical officer immediately of any abnormal rise in diarrhoea or dysentery, fever with rigors, fever with jaundice, acute flaccid paralysis in a child under 15 years, or fever with unconsciousness found on home visits; implement the Integrated Disease Surveillance Programme and report syndromes regularly (Sunder Lal, Ch 16, p 718)
Malaria	Take a blood smear from every fever case met on home visits and administer radical treatment if positive, informing the health worker (male)
Leprosy	Identify skin patches, especially with loss of sensation, and refer to the health worker (male) and medical officer
Tuberculosis, malaria, leprosy follow-up	Assist in maintaining records of patients under treatment, check treatment regularity and motivate defaulters
Diarrhoeal disease	Give oral rehydration solution to every case of diarrhoea, dysentery or vomiting
Blindness	Identify and refer all cases of blindness, including suspected cataract, to the medical officer
HIV, sexually transmitted and reproductive tract infections	Education, counselling, referral and follow-up
Lymphatic filariasis (endemic areas)	Identify and refer lymphoedema, elephantiasis and hydrocele; train patients in foot care and home-based management; identify and train drug distributors for mass drug administration of diethylcarbamazine plus albendazole on National Filaria Day
Non-communicable disease	Information, education and communication for prevention and early detection; community-based screening of persons over 30 years for hypertension, diabetes, obesity and the three common cancers (oral, breast, cervical), for which she is trained in oral visual examination, clinical breast examination and visual inspection with acetic acid , under the supervision of the LHV, medical officer and mid-level provider (Sunder Lal, Ch 8, pp 503, 529–530)
Care of the elderly	Domiciliary visits to elderly persons in her area under the National Programme for Health Care of the Elderly (Park 28e, Ch 7, p 564)

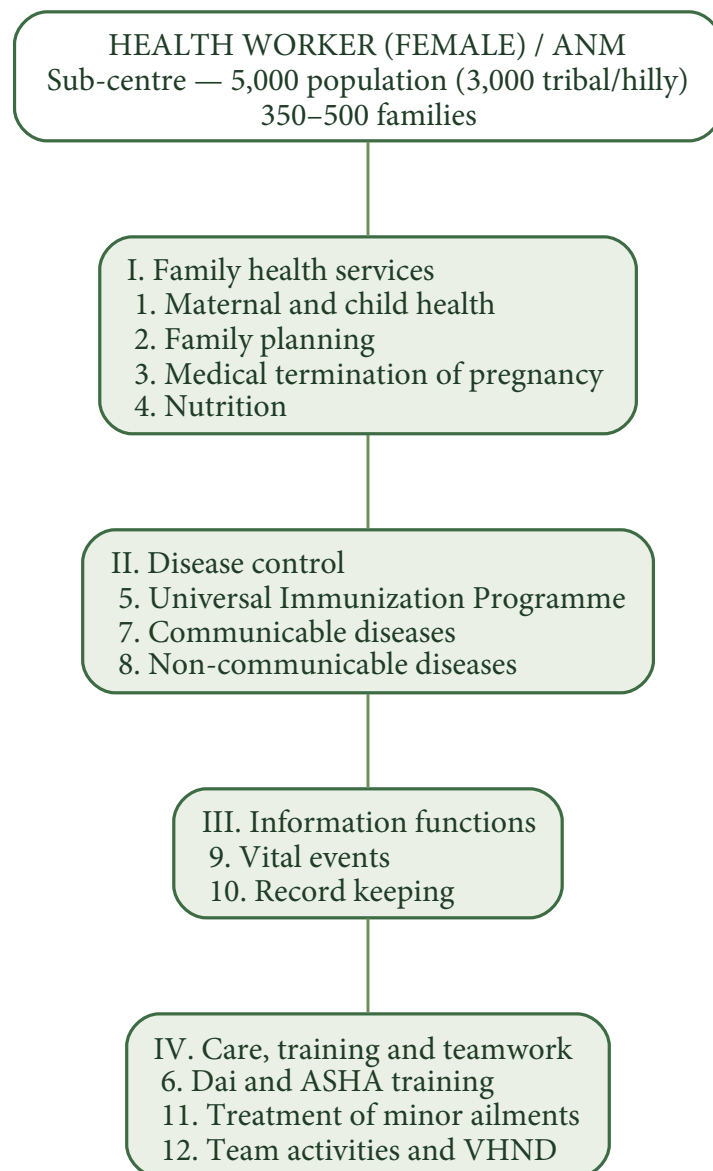
E. Vital events, records and reports

- ▶ Record and report all **births and deaths**, particularly maternal and infant deaths, to the health authority.
- ▶ Maintain registers of pregnant women at earliest contact, infants aged 0–1 year, children under five and adolescents, and **women aged 15–44 years**.
- ▶ Maintain prenatal, maternity and child-care records, and the eligible couple and child register.
- ▶ Maintain records of contraceptive distribution, intrauterine device insertions, cases sterilised, clinics held at the sub-centre, and supplies received and issued.
- ▶ Prepare and submit the prescribed weekly and monthly reports on time to the health assistant (female).

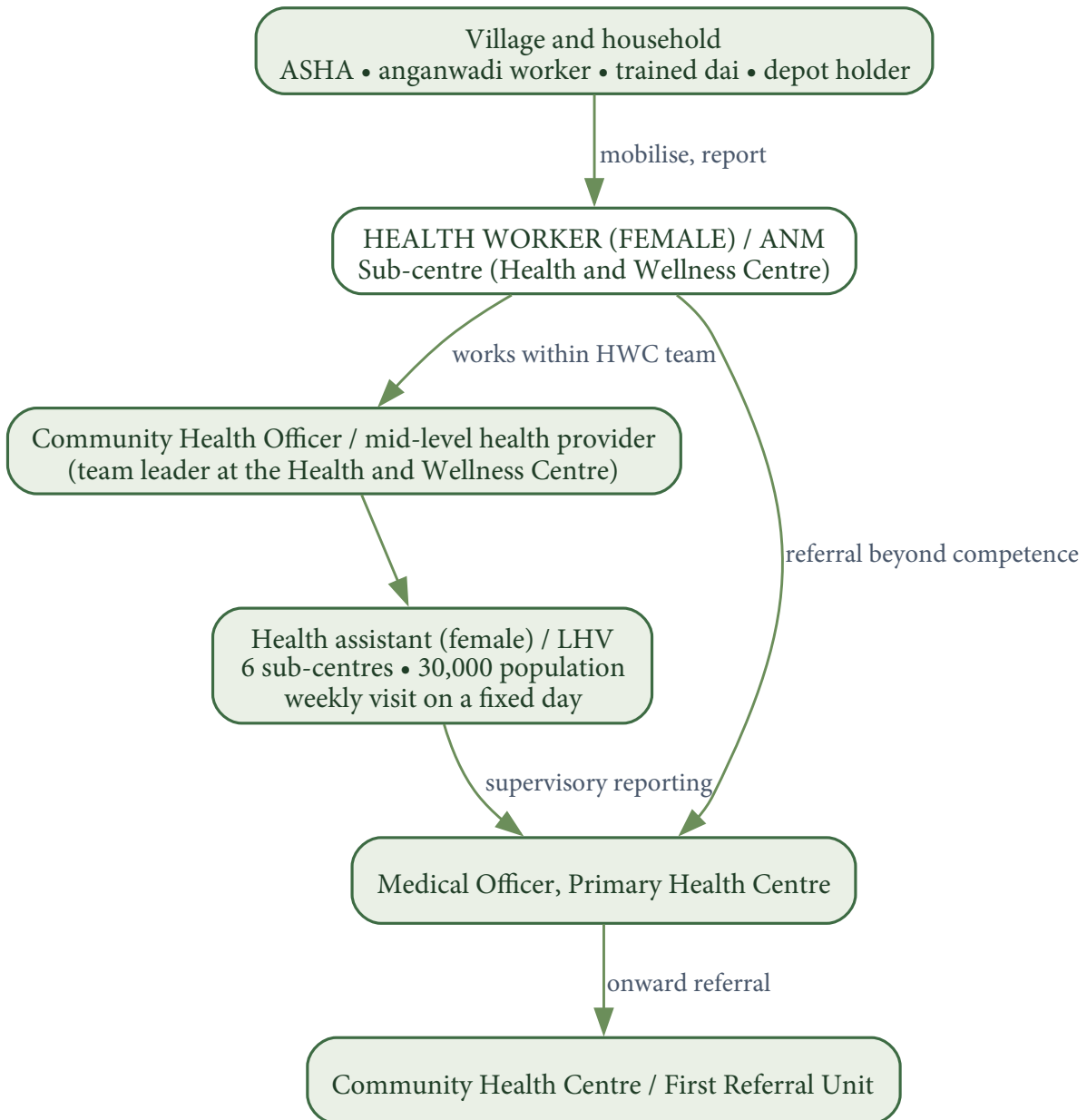
F. Primary medical care, training and team activities

- ▶ **Treatment of minor ailments** — treat minor ailments, give first aid for accidents and emergencies, refer beyond her competence to the PHC, community health centre or nearest hospital, and provide treatment under the **Indian System of Medicine** at local level.
- ▶ **Training** — list the dais in her area, involve them in family welfare, and help the LHV train them and the ASHAs.
- ▶ **Facilitating ASHA and anganwadi worker** — hold monthly or fortnightly meetings, act as resource person for their training, inform them of outreach session dates and venues, and use them to update eligible couple lists and mobilise beneficiaries.
- ▶ **Village Health and Nutrition Day** — organise it at the anganwadi centre **once a month with at least four hours of contact time** between the ANM, AWW, ASHA and beneficiaries.
- ▶ **Coordination** — attend staff meetings at the PHC or community development block, coordinate with the health worker (male), ASHAs and dais, and work with the panchayati raj institution and the Village Health, Sanitation and Nutrition Committee.
- ▶ **Planning** — undertake community needs assessment for workload assessment and prepare the **sub-centre action plan** with the panchayati raj institution, ASHAs and AWWs.
- ▶ **Sub-centre upkeep** — maintain cleanliness, dispose of medical waste as per guidelines, meet the health assistant (female) weekly for guidance, and participate in camps and campaigns.

Diagrams



The twelve job-function domains of the health worker (female) under the Multipurpose Workers Scheme



Supervision, reporting and referral chain of the health worker (female)

High-yield exam pointers

- **Norms to write first:** 1 ANM per 5,000 population (3,000 tribal/hilly), 350–500 families; supervised by LHV covering 6 sub-centres / 30,000 population, visiting each weekly on a fixed day.
- **Park's twelve headings** (write them as the skeleton): maternal and child health, family planning, medical termination of pregnancy, nutrition, immunisation, dai training, communicable diseases, non-communicable diseases, vital events, record keeping, treatment of minor ailments, team activities.
- **Her post-natal visit schedule:** day 0, 3, 7, 42 for home and sub-centre deliveries; day 3, 7, 42 for institutional deliveries; six visits up to day 28 for a low birth weight baby.
- **Janani Suraksha Yojana dates:** accounts of the previous month by the 7th; workers' meeting on the 3rd Friday of every month.

- ▶ **She owns the cold chain** at the sub-centre and the **eligible couple and child register** — two facts examiners look for.
- ▶ **Village Health and Nutrition Day**: monthly at the anganwadi centre, minimum **four hours** of contact time.
- ▶ **Registers**: pregnant women, infants 0–1 year, under-fives and adolescents, and **women 15–44 years**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — sub health centres have been converted to Health and Wellness Centres, now branded **Ayushman Arogya Mandir**, where the ANM works in a team led by a Community Health Officer (mid-level health provider), so a current answer should name this alongside the classical Multipurpose Workers Scheme job description, which remains the examiner's expected content.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, National Health Mission, MoHFW Immunization Division, WHO Immunization Agenda 2030, Gavi

Programme and policy updates (India)

- ▶ **The facility she staffs has been renamed.** Sub-health centres and PHCs upgraded under Ayushman Bharat comprehensive primary health care are now **Ayushman Arogya Mandir (AAM)**, the term that has replaced *Health and Wellness Centre*, and the service basket at these facilities widened from **6 packages to 12**; the ANM works within a primary-care team alongside the Community Health Officer (mid-level provider). A national micro-costing study with a time-and-motion component is measuring how **48 ANMs and 48 CHOs across 48 sub-centre AAMs** actually distribute working time across those 12 packages [PMID 40044205].
- ▶ **What a sub-centre now delivers, measured.** A state-representative assessment of **404 Health and Wellness Centres in Chhattisgarh** (August–October 2023) found a mean of **358 outpatients per centre per month**, of whom **128 were NCD patients** (93 collecting monthly medicines), and **58% of centres ran more than 20 teleconsultations a month** while 22% ran none. Acute illness was the largest share of clinical volume, followed by NCDs, communicable disease and reproductive-child health, with the ANM and community health workers carrying much of the outreach load [PMID 39367295].
- ▶ **Her registers have gone digital — and doubled.** ANMs now work the RCH portal, the ANMOL app, eVIN, the IDSP-IHIP S-form, CPHC-NCD and FP-LMIS. In a Chandigarh primary-care assessment, **89.2% (107/120) of health workers said parallel paper-and-digital entry had increased their workload**; ANMs were maintaining the same beneficiary in three places — RCH register, RCH portal and ANMOL app — and **46.3% spent over 90 minutes a day on RCH data entry alone**. eVIN and the RCH portal scored as effectively implemented (86% and 80%), whereas ANMOL was non-functional at every study centre [PMID 40854229].

Current national data

- **Coverage and programme scale.** Full immunisation coverage rose from **62% in 2015–16** to **76.4% in 2019–21**, still short of the **90% Immunization Agenda 2030 target**; the Universal Immunization Programme now protects against **12 vaccine-preventable diseases** and targets roughly **26 million infants and 29 million pregnant women each year** [PMID 42295252].
- **One child in four is still incompletely immunised.** A meta-analysis of 45 Indian cross-sectional studies (2015–2025, more than 428,000 participants) put pooled incomplete immunisation in under-fives at **26% (95% CI 21–32%)**. The strongest determinant was **absence of an immunisation card (OR 9.46; 95% CI 3.28–27.28)**, followed by non-institutional delivery (OR 3.59), maternal illiteracy (OR 3.25) and low socioeconomic status (OR 2.03) — the evidence case for her due list, tickler box and dropout tracking [PMID 42320383].

Elimination, surveillance and disease status

- **Kala-azar threshold met; filariasis target moved to 2030.** Elimination of kala-azar as a public health problem is defined as **fewer than 1 case per 10,000 population at block level**; the deadline was revised repeatedly and the country reached the threshold in **2023** (by 2021 only 7 of 633 endemic blocks were still above it) [PMID 40988627]. Lymphatic filariasis remains endemic in **345 districts**, the national elimination goal now stands at **2030**, and mass drug administration, morbidity management and post-MDA surveillance remain the operational load she supports [PMID 41951149].
- **Measles-rubella elimination is behind schedule.** India reported **13,220 measles cases between August 2023 and January 2024, ranking fourth worldwide**, and a tribal Odisha cohort with **99.6% MCV1 and 97.5% MCV2 coverage** still showed only **55.1% measles seropositivity** three months after the second dose — recorded coverage is not the same as protection [PMID 42555691]. The WHO South-East Asia Region target for rubella elimination is **December 2026**, and rubella IgM positivity of 7.2% in central India indicates continuing transmission [PMID 42159053].

Guideline and legislative updates

- **The polio schedule she administers has changed, and a hexavalent switch is being costed.** Fractional IPV is now given at **6 weeks, 14 weeks and 9 months from 2023**, replacing the earlier two fractional doses and adding an injection to those sessions. A costing by the national immunisation division found that a **hexavalent vaccine (pentavalent plus IPV)** would raise vaccine acquisition cost but save syringes, cold-chain space and staff time — about **24% of the savings coming from ANM session time for one fewer injection** — and would break even at a **50% fall in unit price**. Gavi opened hexavalent applications from **1 December 2023** [PMID 42295252].
- **Newer UIP vaccines are showing programme-level effect.** Ten-valent PCV (PNEUMOSIL) given in the **2+1 schedule** reduced vaccine-type nasopharyngeal carriage in rural Pune children by roughly a third against PCV-naïve controls (**adjusted OR 0.68; 95% CI 0.49–0.96**) [PMID 41442867]. Rotavirus vaccine — Rotavac from **26 March 2016**, Rotasiil added in 2018 and nationwide by 2019 — was followed by hospital rotavirus positivity falling from **37% before rollout to 17.5%**, with the steepest decline in infants [PMID 41832515].

Global and WHO developments

- **Task shifting now has an implementation framework.** WHO has asked countries to build regulatory and implementation frameworks before scaling task shifting and sharing — the mechanism by which screening, NCD follow-up and dispensing keep migrating to the ANM, CHO and ASHA. **SHIFT-SHARE 2.0**, refined through eight focus groups with 35 Indian stakeholders (November 2023 to September 2024), adds early risk identification, **workforce incentivisation as a core consideration**, and explicit exit or reversal pathways. Its cautionary case is Kerala, where prescribing shifted to public health nurses was later halted by court order — legal risk, not clinical capability, ended it [PMID 41855426].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 10 (Health Care of the Community), pp 738–744 (job description of members of the health team) and Ch 7 (Health Programmes in India), pp 550, 564. Sunder Lal — Textbook of Community Medicine, Ch 16 (Health Planning and Management), pp 717–718; Ch 17 (Healthcare of the Community), pp 742, 745–746; Ch 8, pp 503, 529–530; Ch 10, p 594. Indian Public Health Standards 2022 (as cited in the above texts).

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Major health problems in India

Asked in: Paper IV Jun 2017 — RGUHS MD Community Medicine

Definition — Assessment of the health status and the health problems of a population is the first requisite for any planned development of health services, and is also called **community diagnosis** (Park 28e, Ch 10, pp 721-722). Park groups the health problems of India under six heads — communicable disease, non-communicable disease, nutritional, environmental sanitation, medical care and population problems (Park 28e, Ch 10, p 723). India today carries a **triple burden**: unfinished communicable disease and undernutrition, a rapidly rising non-communicable disease and injury burden, and a health system that does not reach the periphery.

Health situation of India — the backdrop

Data required to define a community's health problems: (1) morbidity and mortality statistics; (2) demographic conditions; (3) environmental conditions bearing on health; (4) socio-economic factors; (5) cultural background, attitudes, beliefs and practices; (6) medical and health services available; (7) other services available. These problems are then **ranked by priority or urgency** for allocation of resources (Park 28e, Ch 10, pp 721-722).

Indicator	Current value (year)
Total population	1,450.93 million (2024)
Crude birth rate / crude death rate	16.0 / 7.0 per 1000 (2024); death rate was 21 in 1965
Annual growth rate; doubling time	0.89% (2024); 30 years
Population below 15 years / above 65 years	25.68% / 6.83% (2023)
Dependency ratio	48.73% (2020) — nearly one dependant per productive member
Rural population; density	64.0%; 492 per sq km (2024)
Life expectancy at birth (male / female)	71 / 74 years (mid-2024)
Infant mortality rate (IMR)	28 per 1000 live births (SRS 2020); rural 31, urban 19
Under-five mortality rate	32 (rural 36, urban 21) per 1000 live births (SRS 2020)
Neonatal / perinatal mortality rate	20 / 18 per 1000 (SRS 2020)
Maternal mortality ratio (MMR)	97 per lakh live births (SRS 2018-20), down from 167 (2011-13)

Interstate inequality is the striking feature — Kerala recorded the lowest IMR at 6 and Madhya Pradesh the highest at 43 per 1000 live births (2020). Mortality is highest in the 0-4 year age group, and 15-25 per cent of total deaths are still attributed to infectious and parasitic diseases (Park 28e, Ch 10, p 723).

1. Communicable disease problems

Communicable diseases continue to be a major problem; most of them can be prevented or treated with a minimum input of resources, which is the tragedy of their persistence (Park 28e, Ch 10, p 723).

Disease	Current Indian burden (year)
Malaria	0.17 million cases, of which 57.26% <i>P. falciparum</i> , and 83 deaths (2022); the falciparum share is rising, with resurgence in the North-Eastern states, Madhya Pradesh, Chhattisgarh, Jharkhand and Odisha
Tuberculosis	India accounts for about one-fifth of world incidence ; about 2.2 million develop tuberculosis a year, of whom about 0.62 million are new smear-positive, with about 0.24 million deaths a year; human immunodeficiency virus (HIV)-tuberculosis co-infection and drug-resistant tuberculosis have worsened its severity
Diarrhoeal diseases	About 5.59 million cases a year in children under 5 years; outbreaks continue because of poor environmental conditions
Acute respiratory infection (ARI)	19.39 million cases with 5,132 deaths (2022)
COVID-19	44.52 million cases reported up to September 2022
Leprosy	65,147 new cases (2020-21); child cases 9.49%, grade-2 deformity 4.14%, multibacillary 51.48%; prevalence rate about 0.68 per 10,000 — elimination achieved at national level , with wide district-to-district variation
Lymphatic filariasis	Endemic in 272 districts of 16 States and 4 Union Territories, population at risk over 670 million; in 2023, 345 endemic districts implemented mass drug administration (MDA) targeting about 746 million
HIV/AIDS	Epidemic stable ; about 2.5 million people living with HIV at the end of 2023
Others	Kala-azar, meningitis, viral hepatitis, Japanese encephalitis, dengue, enteric fever and helminthic infestations

2. Non-communicable disease (NCD) problems

India is in **rapid epidemiological transition**, with NCDs accounting for an estimated **63 per cent of all deaths in 2016**, driven by an ageing population and environmentally driven changes in behaviour (Park 28e, Ch 10, p 723). The shift in disease burden is quantified below.

Major disease group	% of total DALYs lost, 1990	% of total DALYs lost, 2016
Non-communicable diseases	30.5	55.4
Communicable, maternal, neonatal and nutritional diseases	60.9	32.7
Injuries	8.6	11.9

(DALY = disability-adjusted life year; ICMR-PHFI-IHME India State-Level Disease Burden Initiative 2017, quoted in Sunder Lal, Ch 8, p 498)

NCD	Magnitude in India
Cardiovascular disease; chronic respiratory disease	28% and 11% of all deaths respectively (2016)

NCD	Magnitude in India
Cancer	About 1.1 million new cases a year; nearly 3.9 million prevalent cases at any point of time
Tobacco	Tobacco-related cancers are about half of all cancers in men and 20% in women; about 1 million tobacco-related deaths a year
Blindness	More than 12 million blind; cataract 62.6% and refractive error 19.70% are the leading causes; intraocular lens implantation rose from <5% (1994) to 95% (2016-17)
Road traffic injury	449,002 crashes with 151,113 deaths and 451,361 injuries (Road Accident Report 2019)
Oral health	Neglected — most district hospitals have a dental surgeon post but lack equipment and maintenance

3. Nutritional problems

Nutritionally the Indian society is a **dual society** — a small well-fed group showing diseases of affluence alongside a very large undernourished group; the country therefore faces the **double burden of malnutrition** (Park 28e, Ch 10, pp 723-724; Sunder Lal, Ch 5, p 164).

Problem	Magnitude and features
Protein-energy malnutrition	Chiefly of the first years of life; about 80% of cases are mild and moderate , severe forms 1-2% of preschool children; marasmus is more frequent than kwashiorkor
Child undernutrition (NFHS-5, 2019-21)	Stunting 36% (rural 37%, urban 30%), wasting 19%, underweight 32%, overweight 3% among under-fives
Nutritional anaemia	India probably has the highest prevalence in women and children — 52.2% of pregnant women and 67.1% of children aged 6-59 months (rural 68.7%, urban 64.2%) are anaemic (NFHS-5); iron deficiency is by far the commonest cause; 19% of maternal deaths are attributed to anaemia
Low birth weight (<2500 g)	18.6% of Indian babies (Rapid Survey on Children 2014) against about 4% in some developed countries
Xerophthalmia (nutritional blindness)	About 0.04% of total blindness in India; keratomalacia is the major cause of nutritional blindness, usually at 1-3 years of age
Iodine deficiency disorders (IDD)	263 of 324 districts surveyed across 29 states and all Union Territories are endemic, i.e. prevalence above 10%; more than 71 million persons have goitre and other IDD — not confined to the sub-Himalayan "goitre belt"
Others	Lathyrism, endemic fluorosis in certain parts of the country, and widespread adulteration of foodstuffs

4. Environmental sanitation problems

- ▶ Perhaps the **most difficult problem to tackle** in this country, being multifaceted and multifactorial (Park 28e, Ch 10, p 724).
- ▶ The **twin problems** are lack of safe water in many parts of the country and lack of a sanitary way of excreta disposal.

- ▶ Diarrhoeal disease accounts for about **90 per cent** of the health problems related to water supply and sanitation; cholera, typhoid, dysentery, hepatitis A and E and poliomyelitis are the other waterborne diseases (Sunder Lal, Ch 7, p 278).
- ▶ Newer problems arise from population explosion, **urbanisation and industrialisation**, producing pollution hazards to human health in the air, in water and in the food chain.
- ▶ High death rate, high infant mortality rate, high sickness rate and poor standards of health in India are largely attributable to defective environmental sanitation (Park 28e, Ch 15, p 861).

5. Medical care problems

- ▶ The existing model is **hospital-based and disease-oriented**, and its benefits reach mainly the urban elite.
- ▶ Facilities are concentrated in urban areas — up to **73.6% of doctors** serve urban areas where only about 36% of the population lives, while nearly 64% live in rural areas that still rely heavily on indigenous practitioners.
- ▶ Maldistribution is attributed to absence of amenities in rural areas, lack of job satisfaction, professional isolation, lack of rural experience and inability to adjust to rural life.
- ▶ Service-side problems are overcrowding of hospitals, inadequate staffing and scarcity of certain essential drugs and medicines.
- ▶ **Resources are scarce** — India spends about 3 per cent of gross national product on health and family welfare development (Park 28e, Ch 10, p 725); cost-effectiveness and cost-benefit analysis are therefore used to allocate them.
- ▶ The corrective is the **primary health care approach**, which lays stress on equity, intersectoral coordination and community participation.

6. Population problems

- ▶ The population problem is one of the biggest facing the country, with inevitable consequences for employment, education, housing, health care, sanitation and the environment (Park 28e, Ch 10, p 724).
- ▶ The population crossed the **one billion** mark at the turn of the century and stood at 1,450.93 million in 2024.
- ▶ Its features are a large population base, high fertility in terms of both birth rate and family size, declining mortality, a **young population** (25.68% below 15 years in 2023) and a large illiterate segment — which explains why the birth rate has declined so slowly.
- ▶ **Population size and structure are the single most important factor** in health and manpower planning in India, and the situation calls for the two-child family norm.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram — Demographic transition in India (draw as a line graph):** X-axis *Years, 1901 to 2020*; Y-axis *Birth rate / death rate per 1000*; two labelled curves — **Birth rate** (49.2 in 1901 falling to 19.5 in 2020) and **Death rate** (42.6 in 1901 falling to 6.0 in 2020); label the widening space between the curves after 1921 as the "**demographic gap**", i.e. the population explosion.

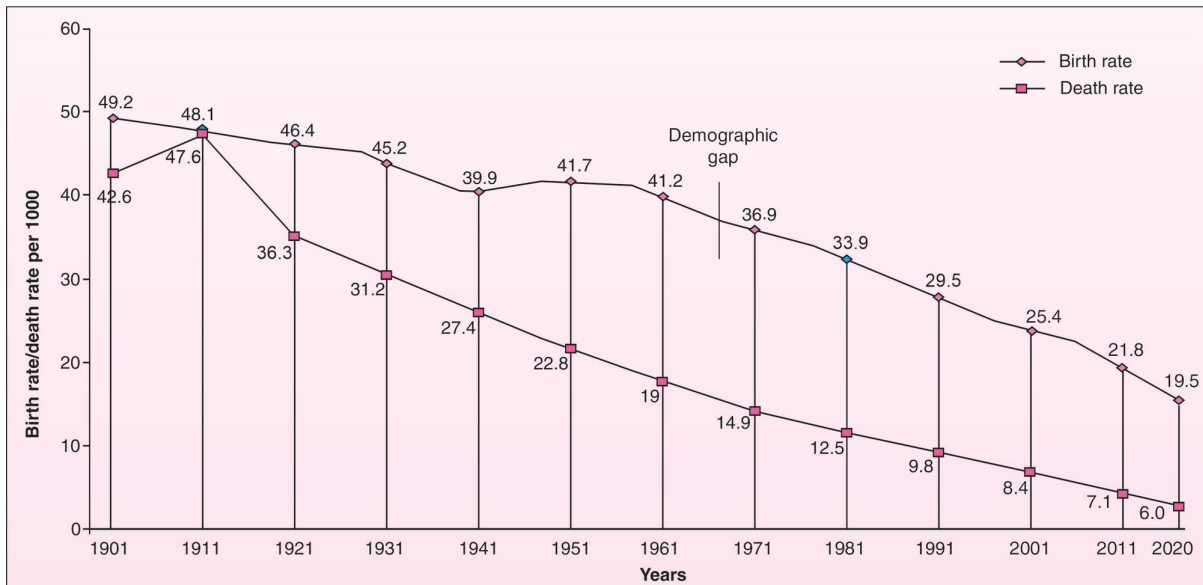
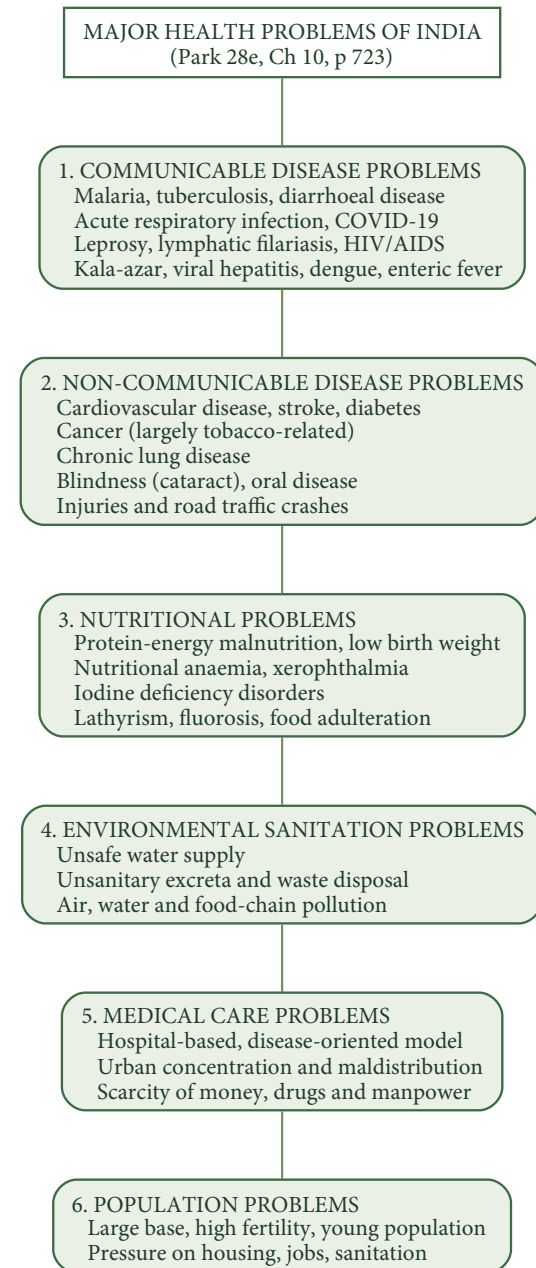
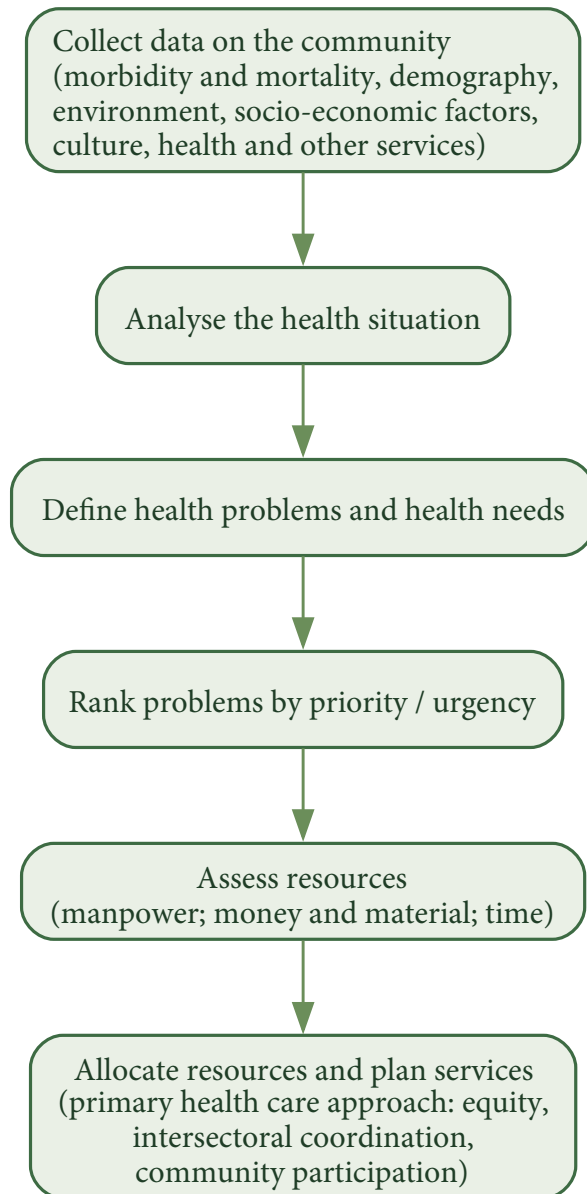


Fig 9.3 — Demographic transition in India (Sunderlal, p.541)



Park's classification of the major health problems of India



Community diagnosis - from data to priority-ranked health problems

High-yield exam pointers

- ▶ **Six heads (mnemonic "CNN-EMP")** — Communicable, Non-communicable, Nutritional, Environmental sanitation, Medical care, Population problems (Park 28e, Ch 10, p 723).
- ▶ Quote the **dated** figures: population 1,450.93 million (2024), crude birth rate 16.0 and crude death rate 7.0 (2024), IMR 28 and under-five mortality rate 32 (SRS 2020), MMR 97 per lakh live births (SRS 2018-20), life expectancy 71 years male and 74 years female (2024).
- ▶ **Epidemiological transition in one line:** NCD share of DALYs rose from 30.5% (1990) to 55.4% (2016), communicable-maternal-neonatal-nutritional fell from 60.9% to 32.7%, injuries rose from 8.6% to 11.9%.
- ▶ **NFHS-5 (2019-21) nutrition quartet:** stunting 36%, wasting 19%, underweight 32%, overweight 3%; anaemia 67.1% in children 6-59 months and 52.2% in pregnant women.

- ▶ India carries about **one-fifth of the world's tuberculosis incidence** (about 2.2 million cases and 0.24 million deaths a year); leprosy prevalence 0.68 per 10,000 means elimination has been achieved at national level.
- ▶ **Health manpower maldistribution:** 73.6% of doctors serve the roughly 36% urban population; health spending is about 3% of gross national product.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e itself quotes non-communicable disease mortality as 63 per cent of all deaths in Chapter 10 and 60 per cent in Chapter 7 for the same year (2016), and Sunder Lal quotes 62 per cent — any of these is acceptable provided the year is stated alongside.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO, MoHFW, NFHS-6 (2023-24), National Health Accounts 2021-22, ICMR-INDIAB 2023, NCVBDC, AOGIN India policy statement 2026

Programme and policy updates (India)

- ▶ **Health and Wellness Centres have been renamed Ayushman Arogya Mandir (AAM)** — in November 2023 the Union Health Ministry renamed the Ayushman Bharat HWCs as *Ayushman Arogya Mandirs*, tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth"), with an explicit shift towards AYUSH-informed holistic wellness — counselling on lifestyle, physical exercise and yoga, mental health support, nutrition and elderly care — rather than illness care alone; the exercise was budgeted at ₹3,000 per facility (total outlay about ₹48 crore) and renamed facilities must upload photographs to the AB-HWC portal. Substitute this name wherever an older text still says "HWC". [PMID 42257165]
- ▶ **What the renamed primary-care network has delivered** — across more than 1.75 lakh AAMs, India's UHC service coverage index rose from 57 (2015) to 63 (2021), out-of-pocket expenditure fell from 60.6% of total health expenditure (2014-15) to **39.4% (National Health Accounts 2021-22)**, and public-facility outpatient attendance rose from 91.6 crore (2012-13) to 134.5 crore (2021-22) with inpatient use from 4.3 to 6.3 crore; a new mid-level provider cadre now leads the sub-centre AAMs, 1,16,978 community health officers being in position by 30 November 2024. Workforce shortage, infrastructure and supply-chain gaps remain the limiting factors. [PMID 41816146]
- ▶ **PM-JAY has achieved coverage but not yet financial protection** — two state-representative household surveys in Chhattisgarh (about 15,000 individuals in each of the 2021 and 2022 waves) found enrolment under Ayushman Bharat PM-JAY was not associated with greater inpatient utilisation, better quality or lower catastrophic expenditure; among enrolled persons using private hospitals, catastrophic expenditure at the 10%-of-consumption threshold was 78.1% (2021) and 70.9% (2022), and contracted private hospitals went on overcharging patients — the mechanism identified being "double billing", taking cash from the patient while also claiming reimbursement for the same service. This is the sharpest current illustration of the "medical care problems" head. [PMID 39135015]

Current national data

- ▶ **NFHS-6 (2023-24) has superseded NFHS-5** — the fact sheets released by the Ministry of Health and Family Welfare show persisting child undernutrition, rising urban overweight, widening rural-urban and tribal inequity and falling exclusive breastfeeding, alongside real gains in immunisation coverage, antenatal care and reduction in childhood diarrhoea and pneumonia. Note the examinable caveat: this round carries **no mortality indicators**, so IMR, under-five mortality and MMR must still be quoted from the SRS with their year. [PMID 42554934]
- ▶ **ICMR-INDIAB reset the national NCD figures upward** — nationally representative survey of 113,043 adults aged 20 years and over across 31 states and union territories (fieldwork 2008-2020, published 2023): diabetes **11.4%** (95% CI 10.2-12.5), prediabetes 15.3%, hypertension 35.5%, generalised obesity 28.6%, abdominal obesity 39.5% and dyslipidaemia 81.2% — considerably higher than previous estimates. All of these except prediabetes were commoner in urban than in rural areas, and the diabetes epidemic is stabilising in the more developed states while still rising in most others. [PMID 37301218]

Elimination, surveillance and disease status

- ▶ **Trachoma eliminated as a public health problem (WHO-validated, 2024)** — India joined the list of 20 countries validated as trachoma-free; the WHO criteria are trachomatous trichiasis unknown to the health system below 0.2% in adults aged 15 years and over (roughly 1 case per 1000 total population), trachomatous inflammation-follicular below 5% in children aged 1-9 years sustained for at least two years without mass antibiotic treatment in each formerly endemic district, and a health system able to detect and treat incident cases with the resources to do so. The **SAFE strategy** — surgery for trichiasis, antibiotics, facial cleanliness and environmental improvement — delivered this, and impact and surveillance surveys must continue after validation to detect resurgence. [PMID 39462916]
- ▶ **Kala-azar reached the elimination threshold in December 2023** — reported incidence fell below **1 case per 10,000 population at block (subdistrict) level**; cases fell from 77,102 (1992) to 599 (2023) and deaths from 1,419 (1992) to 4 (2023), while South-East Asia's share of the world burden fell from 67% (2006) to 6% (2023). India has entered a **consolidation phase requiring at least three consecutive years** below the threshold before a dossier can be submitted to WHO for validation. Residual post-kala-azar dermal leishmaniasis (over 300 cases in 2023), kala-azar-HIV co-infection and bottlenecks in surveillance and vector control are the stated risks to sustainment; case-fatality was 3% in 2023 against the WHO roadmap target of 1%. [PMID 40037366]
- ▶ **Lymphatic filariasis — elimination now targeted for 2027 under the renamed NCVBDC** — the vector-borne disease programme is now the **National Centre for Vector Borne Diseases Control (NCVBDC)**, which targets LF elimination by 2027; indigenous cases are still reported from about 339 districts across 20 states and union territories, India carries roughly 40% of the global filariasis burden with about 650 million people at risk (update the textbook's "over 670 million"), and national mass drug administration coverage was about 69% in 2023 — many districts still short of the 65% effective coverage needed to interrupt transmission. A systematic review of 157 Indian studies (2001-2023) found 22,938 LF-positive persons among about 1.49 million examined, with diagnostic gaps and zoonotic dirofilariasis complicating the endgame. [PMID 41367637]

Guideline and legislative updates

- ▶ **HPV vaccination policy for cervical cancer elimination** — the AOGIN India 2026 policy statement recommends integrating HPV vaccine into the **Universal Immunization Programme** and endorses **single-dose** schedules on the strength of single-dose efficacy trial data, explicitly to move India towards the WHO 90-70-90 elimination targets (90% of adolescent girls vaccinated by age 15). [PMID 42169564]

Global and WHO developments

- ▶ **Tuberculosis — India carries about a quarter of the global burden and is not on trend for the 2025 target** — WHO figures relayed in a 2026 review put the global total at about **10.7 million** new TB cases and **1.2 million** deaths in 2024, TB again ranking as the leading infectious cause of death worldwide ahead of COVID-19, with the South-East Asia region supplying the largest share of cases (34%). India alone accounts for approximately **26%** of the global burden (25-26% in the review's own text), notified about **2.7 million** cases in 2023 and reported about **63,000** MDR/RR-TB cases the same year; national incidence fell from 237 to about 187 per 100,000 between 2015 and 2024, a 21% drop against a global average nearer 12%. COVID-19 disrupted diagnosis, notification and surveillance, and a fragmented private sector with persistent under-reporting remains a barrier, even as molecular diagnostics, the **Ni-kshay** digital surveillance platform, AI-assisted screening and patient-support schemes have strengthened the response — that review still writes of 2025 prospectively, as a target needing a more integrated and strategically prioritised approach. [PMID 42445255] How far off the pace India is has been quantified separately: incidence is falling only about **3% a year** against the roughly **15%** annual decline the 2025 goal demands, a widening gap against a national target set five years ahead of the global 2030 End TB and SDG deadline. [PMID 41275220] Update the textbook's "about one-fifth of world incidence" to roughly a quarter, and present 2025 as a target India is not on trend to meet rather than a deadline comfortably ahead.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10 Health Care of the Community, pp 721-725; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 650, 663-664; Ch 11 Nutrition and Health, p 782; Ch 15 Environment and Health, p 861; Ch 7 Health Programmes in India, p 571); Sunder Lal — Textbook of Community Medicine (Ch 5 Nutrition p 164; Ch 7 Epidemiology p 278; Ch 8 p 498; Fig 9.3 p 541); National Family Health Survey (NFHS-5) India Report 2019-21; Sample Registration System (SRS) 2020 and SRS Special Bulletin on Maternal Mortality 2018-20; ICMR-PHFI-IHME India State-Level Disease Burden Initiative 2017; Road Accident Report 2019; Rapid Survey on Children 2014.

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♦ ♦ ♦

Mudaliar Committee

Asked in: Paper IV Jun 2017 — RGUHS MD Community Medicine

Definition — The Mudaliar Committee is the **Health Survey and Planning Committee**, appointed by the Government of India under the chairmanship of **Dr A. Lakshmanaswami Mudaliar**. It was asked to evaluate the progress made in health since the *Bhore* Committee and during the first two Five Year Plans, to re-assess the country's health needs and resources, and to lay down guidelines for further national health planning (Park Plus, Ch 21, pp 296–297; Red Book — Public Health and Preventive Medicine, pp 299–300).

Identity, remit and place in health planning

Item	Detail
Official name	Health Survey and Planning Committee
Chairman	Dr A. Lakshmanaswami Mudaliar
Constituted / reported / published	Constituted 1959 ; report submitted 1961 ; report published 1962 as <i>Government of India (1962), Report of the Health Survey and Planning Committee</i> , Ministry of Health, New Delhi (Suryakantha, Ch 38, pp 809–810; Park 28e, Ch 23, p 1040)
Appointing authority	Government of India, Ministry of Health
Predecessor reviewed	<i>Bhore</i> Committee, 1946 — Health Survey and Development Committee (Mahajan & Gupta 4e, Ch 26, pp 500–501)
Terms of reference	(1) Evaluate progress made since the <i>Bhore</i> report and in the first two Five Year Plans; (2) review the first and second Five Year Plan health projects; (3) re-look at the health needs and resources of the country; (4) provide guidelines for further national health planning (Park Plus, Ch 21, pp 296–297; Jugal Kishore, p 68)
Planning period it shaped	Third and Fourth Five Year Plans — its recommendations formed the basis for establishing primary health centres (PHCs) in these two Plan periods (Suryakantha, Ch 38, pp 809–810)

Findings of the survey (the situation analysis)

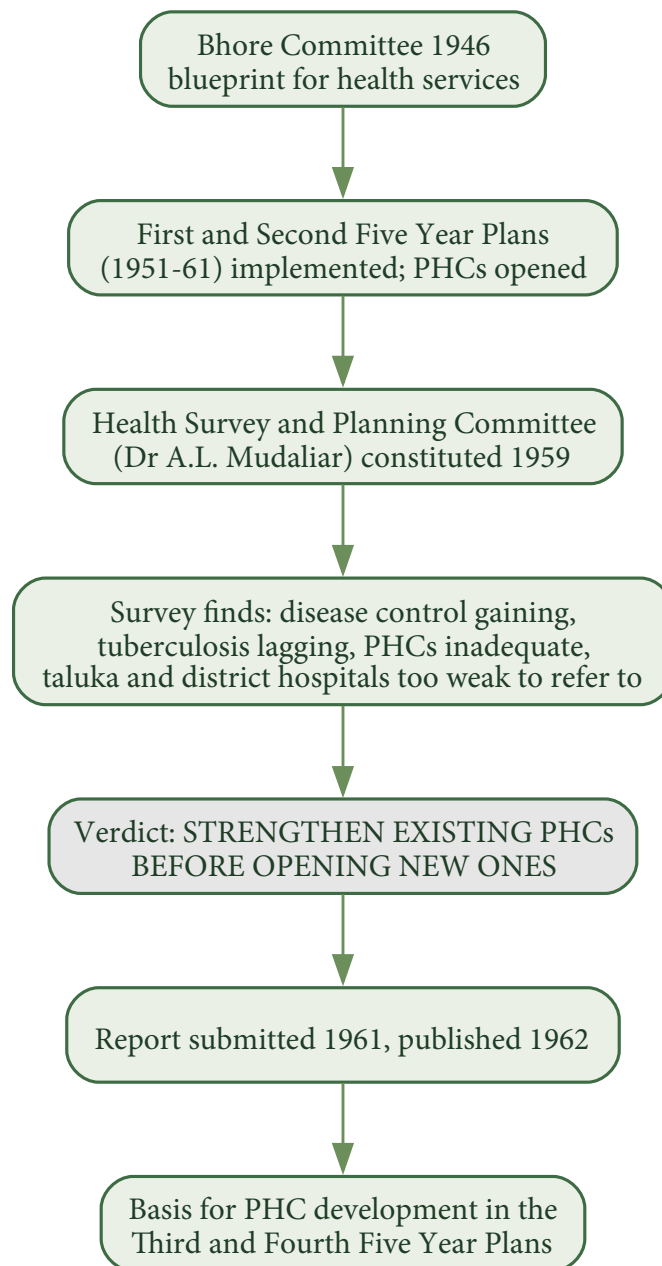
- **Communicable disease control had genuinely succeeded.** Malaria was reported to be under control; deaths from malaria, cholera and smallpox had been halved or sharply reduced, most of the effort of the 1950s having been aimed at communicable disease (Jugal Kishore, p 68; Park Plus, Ch 21, pp 296–297).
- **Overall morbidity and mortality had declined** — the death rate had fallen to **21.6 during 1956–61** and the expectation of life at birth had risen to **42 years** (Jugal Kishore, p 68).
- **Tuberculosis was the conspicuous failure** — the tuberculosis programme was reported to be lagging behind the other disease-control programmes (Jugal Kishore, p 68).
- **The primary health centre was the weak link.** The Committee expressed open dissatisfaction with the quality of care at PHCs — functioning was inadequate because of **under-staffing, lack of referral support and poor quality**, and the conditions in PHCs were found unsatisfactory (Park Plus, Ch 21, pp 296–297; Suryakantha, Ch 38, pp 809–810; Mahajan & Gupta 4e, Ch 26, pp 500–501).

- **PHCs could not serve any useful purpose** unless adequate facilities, resources and personnel were provided (Sathe & Sathe, Ch 18, p 205).
- **Plan outlay for health was inadequate** — the outlay for the health sector in the **Third Five Year Plan** did not match the country's needs (Sathe & Sathe, Ch 18, p 205).

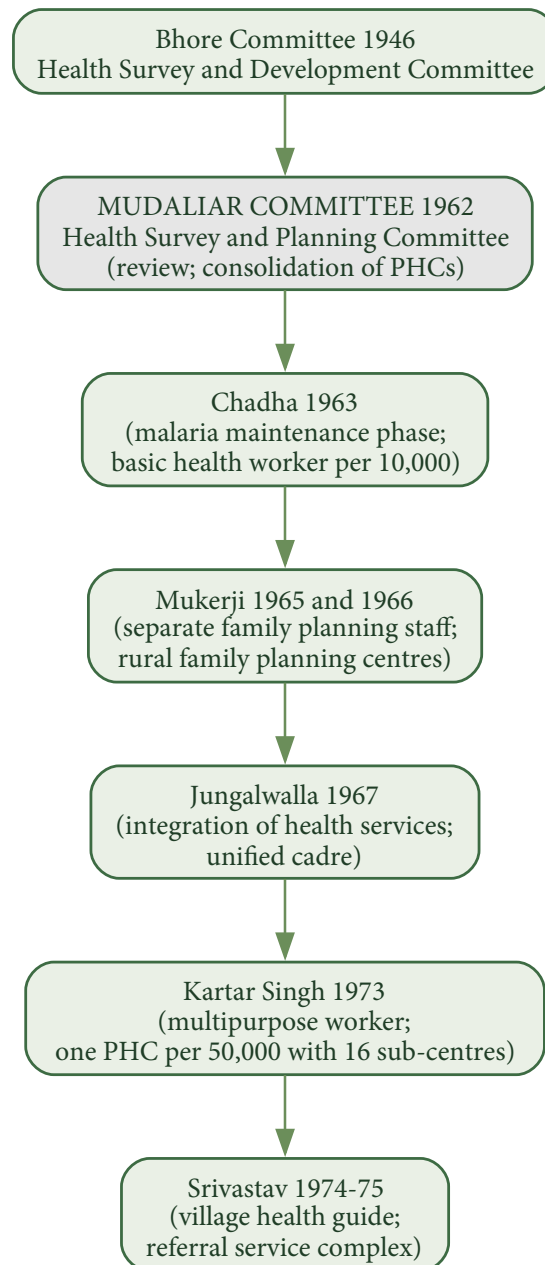
Major recommendations

Domain	Recommendation
Primary health care (the central recommendation)	Consolidate before expanding — strengthen the PHCs already established, and improve the quality of care they provide, <i>before</i> opening new ones (Suryakantha, Ch 38, pp 809–810; Mahajan & Gupta 4e, Ch 26, p 501)
Population norm for a PHC	A PHC should not be made to cater to more than 40,000 population (Mahajan & Gupta 4e, Ch 26, p 501)
Service content at the PHC	Curative, preventive and promotive services all to be provided at the PHC (Mahajan & Gupta 4e, Ch 26, p 501)
Referral chain	Strengthen sub-divisional (taluka) and district hospitals so that they can function effectively as referral centres for the PHCs; develop referral centres (Suryakantha, Ch 38, pp 809–810; Park Plus, Ch 21, p 297)
Outreach	Mobile service teams for rural areas (Park Plus, Ch 21, p 297)
Regional administration	A regional organisation in each State , between State headquarters and the district, headed by a Regional Deputy Director who supervises 2–3 District Medical Officers (Park Plus, Ch 21, p 297)
Health manpower cadre	Constitution of an All India Health Service on the pattern of the Indian Administrative Service (IAS) , to replace the erstwhile Indian Medical Service (Park Plus, Ch 21, p 297; Mahajan & Gupta 4e, Ch 26, p 501; Red Book — Public Health and Preventive Medicine, pp 299–300)
Doctor : population norm	Health manpower to be planned towards a doctor : population ratio of 1 : 3,500 (Mahajan & Gupta 4e, Ch 26, p 503)
Auxiliaries	Training of auxiliaries to help doctors (Sathe & Sathe, Ch 18, p 205)
Health financing	Levy of a small fee on those availing hospital services, except those who are really poor ; and a long-range health insurance policy for all citizens (Park Plus, Ch 21, p 297)
National programmes	Institute national programmes for smallpox, goitre, trachoma, tuberculosis and filariasis (Park Plus, Ch 21, p 297)
Quality of care	The 1961 report stressed encouragement of medical audit in India — the first official Indian push for it (Park Plus, Ch 21, p 305)
Central policy machinery	Make the Central Council of Health more effective (Jugal Kishore, p 68). This Council was set up by Presidential Order on 9 August 1952 under Article 263 of the Constitution, with the Union Health Minister as Chairman and the State Health Ministers as members (Park 28e, Ch 23, p 1034)

Diagrams



Why the Mudaliar Committee was set up and what came of it



Where the Mudaliar Committee sits in the chain of Indian health committees

Implementation and significance today

Recommendation	Position today
Strengthening of PHCs	Accepted and acted upon — the report was the basis for establishing PHCs through the Third and Fourth Five Year Plans (Suryakantha, Ch 38, pp 809–810)
PHC population norm of 40,000	Superseded twice — <i>Kartar Singh</i> (1973) proposed one PHC per 50,000 with 16 sub-centres (Red Book — Public Health and Preventive Medicine, pp 299–300); Indian Public Health Standards (IPHS) 2022 now set the rural PHC (Health and Wellness Centre – PHC) at 30,000 in plain areas and 20,000 in hilly and tribal areas , and the urban PHC at 50,000 (Park 28e, Ch 10, pp 731–732)

Recommendation	Position today
Regional organisation	Realised in modified form — a State Minister and Secretary of Health and Family Welfare with a State Directorate of Health Services, and a District Health Officer heading the district health office (Park Plus, Ch 21, p 297)
All India Health Service	Not created as such — a Central Health Service exists instead, providing medical manpower for the Directorate General of Health Services, the Central Government Health Scheme and allied departments (Jugal Kishore, pp 99–100); the Director General of Health Services cadre was created almost on par with the IAS (Park Plus, Ch 21, p 297)
Doctor : population 1 : 3,500	Achieved and surpassed — India has already met this norm; the ratio recorded is 1 : 2,222 (Mahajan & Gupta 4e, Ch 26, p 503), and 12,34,205 allopathic doctors registered up to 2019 gives 1 : 1,445 (Sunder Lal, Ch 17, p 766). The remaining problem is maldistribution — about 68% of doctors work in urban areas serving roughly 26% of the population (Mahajan & Gupta 4e, Ch 26, p 503)
Mobile service teams	Implemented but short — mobile medical units exist; Karnataka has 19, facing a tremendous shortfall (Park Plus, Ch 21, p 297)
Long-range health insurance for all citizens	Conceptually vindicated — pursued today through State and national schemes (Yeshasvini, Rashtriya Swasthya Bima Yojana, Vajpayee Arogyasri — Park Plus, Ch 21, p 297) and now Ayushman Bharat PM-JAY, with a 60:40 Centre–State funding share (90:10 for North-Eastern States) (Sunder Lal, Ch 17, p 763)
National programmes	Implemented (Park Plus, Ch 21, p 297)
Medical audit	Recommended in 1961; introduced in 1969 by the then Health Minister Dr Sushila Nayyar, but became operational only in 2007 , after the National Accreditation Board for Hospitals and Healthcare Providers (NABH) was established in 2005 (Park Plus, Ch 21, p 305)
What it did not address	The Committee's frame stopped at the PHC — no village-level worker was proposed. That came only later, with the multipurpose worker (<i>Kartar Singh</i> , 1973) and the village health guide drawn from the community itself (<i>Srivastav</i> , 1974–75, leading to the Rural Health Scheme of 1977) (Park Plus, Ch 21, pp 300–302; Suryakantha, Ch 38, p 811)

High-yield exam pointers

- ▶ **Name it right:** Mudaliar = Health Survey and **Planning** Committee; Bhore = Health Survey and **Development** Committee. Constituted 1959, reported 1961, published 1962.
- ▶ **Its one-line verdict:** *"Strengthen the existing primary health centres before opening new ones."*
- ▶ **Numbers to write:** PHC ≤ **40,000** population; doctor : population **1 : 3,500**; Regional Deputy Director over 2–3 District Medical Officers; death rate **21.6 (1956–61)**; life expectancy at birth **42 years**.
- ▶ **The five national programmes it named:** smallpox, goitre, trachoma, tuberculosis, filariasis (mnemonic **S-G-T-T-F**).
- ▶ **Two firsts:** the first recommendation for an **All India Health Service** on the IAS pattern, and the first call for a **long-range health insurance policy for all citizens** — the earliest Indian articulation of universal health coverage.

- **Current contrast to quote:** Mudaliar's 40,000 → Kartar Singh's 50,000 → IPHS 2022: 30,000 (plain) and 20,000 (hilly/tribal) per rural PHC.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Textbooks differ on the dates — Suryakantha, Sathe & Sathe, Jugal Kishore and the Red Book all record that the committee was **set up in 1959** and reported in 1961, whereas Mahajan & Gupta 4e (Ch 26, p 500) says it was "appointed by the Government in 1962", which is the year of publication; writing "constituted 1959, reported 1961, published 1962" covers all three. Jugal Kishore (p 68) prints the death rate as "21.6 per cent during the period 1956–61", which is a typographic slip, since a crude death rate is expressed per 1,000 population.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10 — Health Care of the Community, Indian Public Health Standards 2022 population norms, pp 731–732; Ch 23 — Health Planning and Management, Central Council of Health at p 1034 and the primary reference *Government of India (1962), Report of the Health Survey and Planning Committee*, Ministry of Health, New Delhi, at p 1040); Park Plus (Amrutha & Vijayalakshmi), Ch 21 — Health planning and management (pp 296–297 findings and recommendations, p 297 current scenario, p 305 medical auditing); Mahajan & Gupta — Textbook of Preventive and Social Medicine, 4e (Ch 26 — Health Planning, Administration and Management, pp 500–504); Suryakantha — Community Medicine with Recent Advances (Ch 38 — National Health Planning, pp 809–811); Jugal Kishore — National Health Programmes of India (p 68 Mudaliar Committee; pp 99–100 Central Health Service); Sathe & Sathe — Epidemiology and Management for Health Care (Ch 18 — Development of health services, p 205); Red Book — Public Health and Preventive Medicine (pp 299–300); Sunder Lal — Textbook of Community Medicine (Ch 17 — Healthcare of the Community, pp 763, 766).

Coverage note: Park 28e carries no dedicated Mudaliar Committee section — only the primary reference to the 1962 report (Ch 23, p 1040) and a passing citation in the environment chapter (Ch 15, pp 914–915) — so the findings and recommendations above come from the other indexed books, and they do not all carry the same list. The short common core (dissatisfaction with PHCs, strengthen existing PHCs before opening new ones, strengthen taluka and district hospitals as referral centres, an All India Health Service) is carried by Suryakantha, Mahajan & Gupta 4e and the Red Book; the six more detailed recommendations — the regional organisation under a Regional Deputy Director, mobile service teams, the small hospital fee sparing the really poor, the long-range health insurance policy, the five national programmes, and the encouragement of medical audit — are carried **only** by Park Plus (Ch 21, pp 296–297 and p 305) and are cited to it alone. The chairman's designation as Vice-Chancellor of the University of Madras is standard textbook attribution that the indexed corpus does not carry, and has therefore been left out.

♦ ♦ ♦

Recent rulings on prescription writing

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — A **prescription** is a written, signed and dated order from a registered medical practitioner (RMP) to a pharmacist, directing that a named medicine be dispensed to a named patient in a stated quantity and taken in a stated way. It is at once a therapeutic instruction, a legal document and a medical record that must be preserved and produced on demand. "Recent rulings" means the amendments, statutory notifications and practice guidelines issued since 2013 that changed **how** a prescription must be written, **what** may be written on it, and **how** the transaction must be recorded.

Standing legal framework the rulings modify

Instrument	What it binds the prescriber to
Indian Medical Council (Professional Conduct, Etiquette and Ethics) Regulations 2002 — compiled version amended up to 8 October 2016; the conduct code now sits under the National Medical Commission Act 2019	Display the registration number in the clinic and on every prescription and certificate; prescribe rationally, only generic names of drugs; maintain the patient's medical records for 3 years (Jugal Kishore, Ch 2, p 848)
Regulation 3.7.2	Write name and designation in full with registration particulars on the prescription letterhead; the appended Note directs that in a government hospital where the patient load is heavy the name of the prescribing doctor must be written below his/her signature (Sunder Lal, Ch 1, p 61)
Regulations 6.3 and 6.5	Every drug prescribed must explicitly state the proprietary formula as well as the generic name ; secret remedies whose composition the physician does not know may not be prescribed, dispensed or promoted (Sunder Lal, Ch 1, pp 61-62)
Regulation 7.2 (read with regulations 1.3 and 1.3.2)	Failure to maintain the records of indoor patients for 3 years , or refusal to provide them within 72 hours of a request by the patient or an authorised representative, is professional misconduct (Sunder Lal, Ch 1, p 62)
Regulation 7.8	Prescribing steroids or psychotropic drugs when there is no absolute medical indication , or selling Schedule 'H' and 'L' drugs and poisons to the public except to his patient , is gross professional misconduct (Sunder Lal, Ch 1, p 62)
Drugs and Cosmetics Act 1940 and Rules 1945	Quality, safety and efficacy of drugs and their import, manufacture, distribution and sale are regulated by the Central Drugs Standard Control Organisation (CDSCO) , headed by the Drugs Controller General (India) in the Directorate General of Health Services (Park 28e, Ch 24, p 1053). Section 26A empowers the Government to prohibit, by notification in the Official Gazette, the manufacture, sale or distribution of a drug considered dangerous (Jugal Kishore, Ch 2, pp 872-873) — the power used for the fixed-dose combination bans

Instrument	What it binds the prescriber to
Schedules H, H1 and X of the Drugs and Cosmetics Rules	Sale only against a written, signed, dated prescription naming the patient and the total quantity (standard Indian regulatory knowledge; not carried verbatim in the indexed corpus — see Note)
Narcotic Drugs and Psychotropic Substances Act 1985	Regulates narcotic drugs and psychotropic substances and gives effect to the international conventions, restricting their prescribing, supply and possession (Jugal Kishore, Ch 2, pp 869-870)
Drugs and Magic Remedies (Objectionable Advertisements) Act 1954	Prohibits advertisement of any drug or magic remedy claiming to diagnose, cure, mitigate, treat or prevent the diseases and conditions specified in its Schedule, and any advertisement making a false claim about a drug (Jugal Kishore, Ch 2, p 873)

The recent rulings themselves

Date	Ruling	Core provision	Effect on the prescriber
1 March 2014	Schedule H1 inserted into the Drugs and Cosmetics Rules (Rule 65)	A defined list of 46 molecules — third-generation cephalosporins, newer fluoroquinolones, carbapenems, anti-tuberculosis drugs and some habit-forming drugs — sold only against prescription; the chemist keeps a separate register recording the prescriber's name and address, the patient's name, the drug and the quantity, retained 3 years ; the pack carries a red-bordered warning box with a red Rx symbol	Ends over-the-counter sale of key antibiotics; the prescription becomes an auditable antimicrobial-resistance containment record
21 April 2017	Medical Council of India notification amending clause 1.5	"Every physician should prescribe drugs with generic names, legibly and preferably in CAPITAL letters , and shall ensure a rational prescription and use of drugs "	The single most-quoted ruling; breach is professional misconduct punishable under Chapter 8
March 2016 → September 2018	Ban on irrational fixed-dose combinations (FDCs) under Section 26A of the Drugs and Cosmetics Act	344 FDCs prohibited in 2016; after Supreme Court review (December 2017) and re-examination by the Drugs Technical Advisory Board, 328 FDCs prohibited in September 2018	Cough syrups, analgesic and antibiotic combinations struck off the prescribable list
Standing government instruction	Directions to all Central Government hospitals and Central Government Health Scheme (CGHS) dispensaries, and by State Governments to their physicians	Prescribe generic medicines as far as possible, dispensed through Jan Aushadhi generic drug stores	Links the prescription to the affordability agenda (Jugal Kishore, Ch 1, p 631)

Date	Ruling	Core provision	Effect on the prescriber
25 March 2020	Telemedicine Practice Guidelines, issued by the Board of Governors in supersession of the Medical Council of India and appended to the 2002 Regulations	Legalise teleprescription; classify medicines into List O, List A, List B and a Prohibited List; mandate a signed digital or scanned e-prescription to the patient	First ruling to define a valid prescription written without physical examination
2 August 2023	National Medical Commission (Registered Medical Practitioner Professional Conduct) Regulations 2023	Restated mandatory generic prescribing with graded penalties and directed prescribing from Jan Aushadhi and generic outlets	The generic-prescribing clause was placed in abeyance within weeks, so the operative rule remains the 2017 notification

Teleprescription categories under the 2020 guidelines

Category	What may be prescribed	Mode permitted
List O ("over the counter")	Medicines safe without direct supervision — e.g. paracetamol, oral rehydration salts, topical antifungals	Any mode — video, audio or text
List A	Relatively safe medicines with low abuse potential, at a first consultation that is by video, and re-prescribed at follow-up	Video for the first consult
List B	An add-on list for a follow-up consultation, for a patient already on treatment for the same condition	Any mode, follow-up only
Prohibited List	Drugs in Schedule X of the Drugs and Cosmetics Rules and narcotic and psychotropic substances under the NDPS Act 1985 — never by telemedicine	None

Delivery rules: the RMP must send a photograph, scan or digital copy of the **signed** prescription to the patient by email or messaging platform, may email it directly to a pharmacy only with the patient's **explicit consent**, and must not favour any particular pharmacy. India's public teleconsultation platform eSanjeevani runs in two variants — eSanjeevani Ayushman Bharat Health and Wellness Centre (doctor-to-doctor) and eSanjeevani OPD (patient-to-doctor) (Sunder Lal, Ch 17, p 748).

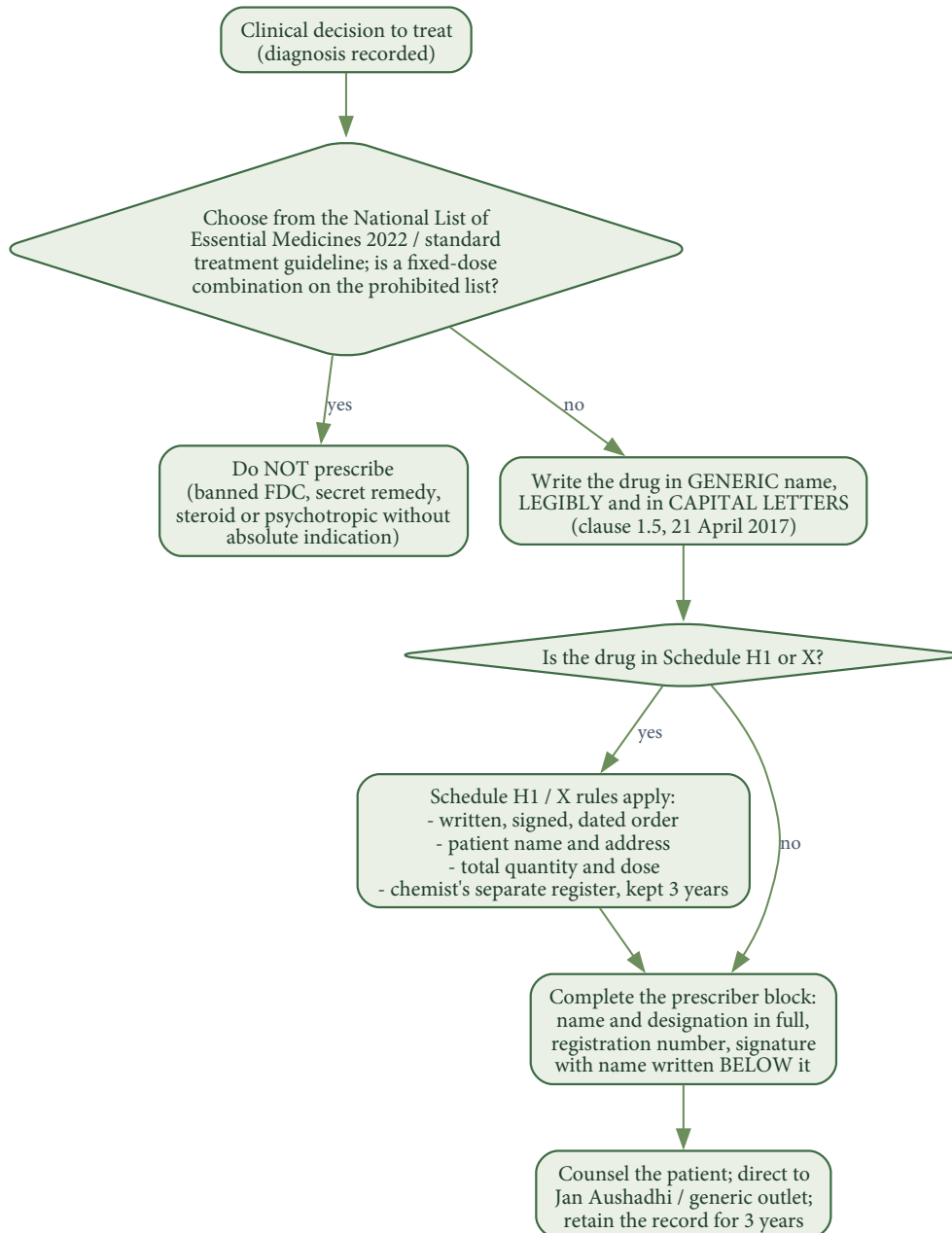
Public health rationale and enforcement

- **Antimicrobial resistance (AMR)** — India carries the world's highest burden. ICMR AMR surveillance data of 2016, generated as part of the WHO-coordinated Global Antimicrobial Resistance Surveillance System (GLASS), estimated that **50% or more of hospital antimicrobial use was inappropriate**; separately, a new Lancet study found that **unapproved antibiotic formulations constituted 47% in India**, with **azithromycin 500 mg** the most consumed antibiotic formulation and cephalosporins the most used class (Sunder Lal, Ch 19, p 783). Schedule H1 and the FDC bans are the direct regulatory answer; the **National Programme on Containment of Antimicrobial Resistance (launched 2013, National Centre for Disease Control the nodal agency)** promotes rational antibiotic prescribing (Sunder Lal, Ch 19, p 783).

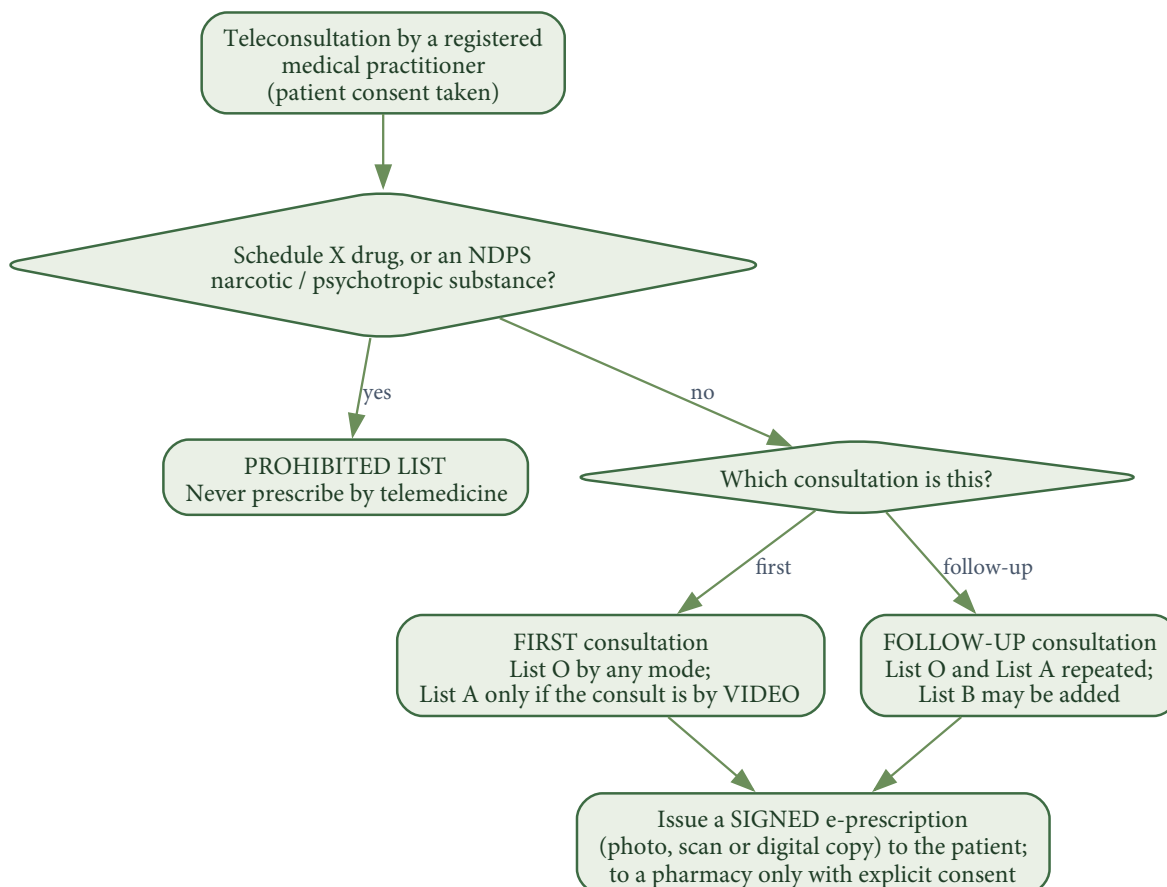
- ▶ **Out-of-pocket cost** — generic prescribing feeds the **Jan Aushadhi Kendras** (over 9000 outlets, essential medicines **50-90% cheaper** than the market) and **AMRIT** stores (affordable medicines and reliable implants for treatment) selling life-saving drugs at **half the market price** (Sunder Lal, Ch 19, pp 782-783).
- ▶ **Patient safety** — unsafe medication practices contribute about **134 million adverse events annually, resulting in around 2.6 million deaths**, the target of the **WHO Global Patient Safety Action Plan 2021-2030** (Sunder Lal, Ch 19, p 783).
- ▶ **Rational use of medicines (RUM)** — patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at the lowest cost to them and their community (**WHO 2004**) (Jugal Kishore, Ch 1, pp 628-629). Standard treatment guidelines and the **National List of Essential Medicines 2022** are the prescribing reference (Park 28e, Ch 24, p 1041).
- ▶ **Enforcement** — non-compliance is professional misconduct; the State Medical Council or the National Medical Commission may award such punishment as deemed necessary, or **direct removal of the practitioner's name from the register**, altogether or for a specified period, the deletion being widely publicised (Sunder Lal, Ch 1, p 64).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Labelled model prescription** — draw a single sheet and label: (1) prescriber block — name and designation in full, qualifications, speciality, **State Medical Council registration number**; (2) facility name, address and **date**; (3) patient identifiers — name, age, sex, weight, address, OPD number; (4) diagnosis; (5) **superscription** — the Rx symbol; (6) **inscription** — drug in **GENERIC name, CAPITAL letters**, strength and dosage form; (7) **subscription** — total quantity to be dispensed; (8) **signa** — dose, route, frequency, duration, relation to food; (9) non-drug advice and follow-up date; (10) "not to be repeated" instruction; (11) **signature with the prescriber's name written below it**; (12) Schedule H1 red-box warning and the chemist's separate register entry.



Writing a legally compliant prescription in India after the recent rulings



Decision path for prescribing in a teleconsultation (Telemedicine Practice Guidelines, 25 March 2020)

High-yield exam pointers

- **The one sentence to reproduce** — clause 1.5, Medical Council of India notification **21 April 2017**: drugs to be prescribed with **generic names, legibly, preferably in CAPITAL letters**, with rational prescription and use.
- **Dates mnemonic "14-17-18-20-23"** — Schedule H1 in force **2014**; generic and capital letters **2017**; 328 FDCs banned **2018**; Telemedicine Practice Guidelines **2020**; National Medical Commission conduct regulations **2023**.
- **Three "3s"** — indoor-patient records kept **3 years**; produced within **72 hours**; Schedule H1 chemist's register retained **3 years**.
- **Telemedicine lists** — **O** = over the counter (any mode), **A** = first consult by video, **B** = follow-up add-on, **Prohibited** = Schedule X and NDPS drugs.
- **Indian AMR figures (dated)** — 50% or more of hospital antimicrobial use inappropriate (ICMR AMR surveillance **2016**, under WHO GLASS); unapproved antibiotic formulations 47% of the Indian market and azithromycin 500 mg the most consumed formulation (**new Lancet study**).
- **Affordability link** — over 9000 Jan Aushadhi Kendras, medicines **50-90% cheaper**; AMRIT stores at half market price; prescribing reference is the **National List of Essential Medicines 2022**.
- **Penalty** — breach is professional misconduct; the council may award punishment or **remove the name from the medical register**, altogether or for a specified period.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the indexed corpus carries the parent 2002 conduct regulations and the Drugs and Cosmetics Act framework, but not the texts of the Schedule H1, clause 1.5, fixed-dose combination, telemedicine and 2023 notifications, which are written here as standard Indian regulatory knowledge.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, ICMR, National Health Authority — Ayushman Bharat Digital Mission, NMC Telemedicine Practice Guidelines, WHO AWaRe 2023

Programme and policy updates (India)

- **Jan Aushadhi now has hard evidence of financial protection** — a nine-state survey of 10,336 patients across 18 districts (2022-23) found medicine out-of-pocket spend among people buying only at Jan Aushadhi Kendras was 60.6-89.3% lower than at private pharmacies after adjustment, and 42% lower on propensity-matched analysis; catastrophic health expenditure ($\geq 40\%$ of non-food consumption) was 9.2% in outpatients and 35.4% in inpatients; uptake was held back by poor awareness, perceived stock shortages and generic-prescribing rates of only 18% in outpatients and 6% in inpatients, and the authors note that the Drugs Technical Advisory Board has proposed examining whether Jan Aushadhi pharmacists may substitute Schedule H, H1 and X prescriptions with generic equivalents [PMID 42152447]
- **Pradhan Mantri Bhartiya Jan Aushadhi Pariyojana scale-up and its limits** — Kendras rose from about 3,200 (2018) to over 9,000 (2022) against an official target of 10,500 by 2025, with a basket of more than 1,450 medicines; yet a larger number of brands correlated with *wider* price dispersion ($r = 0.79$), and although most Jan Aushadhi medicines met the WHO/HAI one-day-wage benchmark, chronic therapy still ate deep into it — latanoprost for glaucoma 0.55 and fluticasone propionate nasal spray 0.81 days' wage — leaving glaucoma treatment financially burdensome for low-income households; Uttar Pradesh and Bihar had the fewest Kendras per head despite the largest absolute numbers [PMID 41426670]
- **Ayushman Bharat Digital Mission "Scan and Share"** — the National Health Authority's QR-code outpatient registration, linked to the patient's Ayushman Bharat Health Account, ran in 16,939 facilities across 35 of 36 States and Union Territories by 19 October 2024, generating 56,801,213 tokens (only 1.3% in the private sector) and cutting registration waiting time from about an hour to 2-5 minutes — this is the digital record layer to which an e-prescription is now attached [PMID 39882498]

Current national data

- **Regulatory approval of a fixed-dose combination is no guarantee of rational prescribing** — the ICMR Rational Use of Medicines Centres Network audited 8,839 outpatient prescriptions from 13 tertiary centres and found a domperidone + proton-pump-inhibitor combination — a class the CDSCO approved during 2004-2008, and now a top seller at Jan Aushadhi Kendras — in 481 (5.44%); only 10.81% carried a nausea or vomiting indication, the dose was left out in 55% and the administration instruction in 86.7%, the duration ran beyond the recommended one week or was missing altogether in 75%, 10.81% went to patients over 60 despite the cardiac-safety age

caution, and the combination cost 1.8 times the proton-pump inhibitor alone; India still has no prescribing guideline for domperidone [PMID 40632092]

- **Generic prescribing is the weakest compliance point** — a single-centre audit of 2,104 inpatient prescriptions from six departments at SVS Medical College and Hospital, Mahabubnagar, Telangana (June-September 2024) found only 17.0% of drugs written by generic name, an average of 8.6 drugs per prescription, injectables in 83.4%, antibiotics in 66% of prescriptions, fixed-dose combinations in 26.1% and National List of Essential Medicines adherence of 59.3%; by AWaRe category 38.3% were Watch, 32.4% Access and 1.9% Reserve — the authors caution that one centre over four months cannot capture institutional, seasonal or regional prescribing differences [PMID 42583971]

Elimination, surveillance and disease status

- **Antimicrobial-use point prevalence survey in secondary care (April-September 2025)** — across 12 Indian secondary-care hospitals, 196 of 359 inpatients (54.6%) were on at least one antibiotic, 84.8% parenteral, 53.1% purely empirical, with culture sensitivity documented in only 28%; 74.2% of prescriptions were WHO Watch-group and only 22.1% Access, and stewardship programmes were largely absent — the surveillance denominator against which Schedule H1 is judged [PMID 42309479]
- **Stewardship infrastructure is the missing enforcement arm** — of five Indian academic medical centres surveyed (2017-2021), the three that responded all ran surgical-site-infection surveillance but none had an antimicrobial stewardship committee and only one had pharmacy informatics to track antimicrobial use; mean extended-spectrum beta-lactamase prevalence was 32.7% and carbapenem resistance 16.0% [PMID 40589099]

Guideline and legislative updates

- **Where the telemedicine rules actually sit, and their weak spot** — the Telemedicine Practice Guidelines of 25 March 2020 were added by the National Medical Commission as **Appendix-5** to the 2002 professional conduct regulations, and the Insurance Regulatory and Development Authority of India recognised teleconsultation for insurance claims on 11 June 2020; the consequent surge in consultations and in telemedicine apps and providers has outrun any mechanism to verify medical certificates issued remotely, which the authors trace to ambiguity in the guidelines [PMID 39817299]
- **Prescribing competence is being addressed as a national training gap** — the ICMR-funded Prescribing Skills Course, a 50-module massive open online course on the Government of India learning platform for interns and registered medical practitioners, ran cycles from 2020 to 2022 with completion of 19-31%; enrolment was far lower in regions with weaker health indices (rate ratio 0.35, 95% CI 0.32-0.38, $p < 0.001$) and practitioner scores rose from 60% to 73.3% ($p < 0.001$) in the third cycle [PMID 41821347]

Global and WHO developments

- **WHO 2023 AWaRe classification and AWaRe antibiotic book** — 41 essential antibiotics are set out as first- and second-choice options for more than 30 clinical infections in both primary care and hospital settings, and a further 257 antibiotics outside the Model List have been placed in an Access, Watch or Reserve group so that their use can be supervised; the stated purpose is to drive

revision of national essential-medicines lists and national prescribing guidelines [PMID 38342438]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 24, Essential Medicines and Counterfeit Medicines, pp 1041, 1053); Sunder Lal — Textbook of Community Medicine (Ch 1, Concept of Health and Disease, Ethics in Medicine and Doctor-Patient Relationship, pp 61-64; Ch 17, Healthcare of the Community, p 748; Ch 19, Essential Medicines and Rational Use of Drugs, pp 782-783); Jugal Kishore — National Health Programmes of India (Ch 1, Essential Medicines and Rational Use of Drugs, pp 628-629, 631; Ch 2, Legislations Related to Health, pp 848, 869-870, 872-873); Indian Medical Council (Professional Conduct, Etiquette and Ethics) Regulations 2002, amended up to 8 October 2016; Drugs and Cosmetics Act 1940 and Rules 1945; National Medical Commission Act 2019; National List of Essential Medicines 2022; Telemedicine Practice Guidelines 2020; WHO Global Patient Safety Action Plan 2021-2030.

Recent-advances sources: 42152447 — Do publicly supported generic pharmacies improve financial risk protection? Findings from a nationally representative cross-sectional survey in India, *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-113839](https://doi.org/10.1136/bmjopen-2025-113839); 41426670 — Medicine affordability and access in India: lessons from generic-branded price variation under the Jan Aushadhi Scheme, *Frontiers in Public Health* 2025, doi:[10.3389/fpubh.2025.1629835](https://doi.org/10.3389/fpubh.2025.1629835); 39882498 — Scan and share to register outpatients, India, *Bulletin of the World Health Organization* 2024, doi:[10.2471/BLT.24.292828](https://doi.org/10.2471/BLT.24.292828); 42583971 — Evaluation of prescribing pattern with special reference to antibiotics, using digital platform among inpatients in a tertiary care hospital, *Indian Journal of Pharmacology* 2026, doi:[10.4103/ijp.ijp_836_25](https://doi.org/10.4103/ijp.ijp_836_25); 40632092 — Prescribing trends of fixed-dose combination of domperidone and proton-pump inhibitor across tertiary care centers in India, is the combination rational?, *Indian Journal of Pharmacology* 2025, doi:[10.4103/ijp.ijp_744_23](https://doi.org/10.4103/ijp.ijp_744_23); 42309479 — Patterns of antibiotic use in Indian secondary care hospitals: results from a multicenter point prevalence survey, *International Journal of Infectious Diseases* 2026, doi:[10.1016/j.ijid.2026.108880](https://doi.org/10.1016/j.ijid.2026.108880) (abstract-only); 40589099 — Gaps and Opportunities in Antimicrobial Stewardship and Surgical Site Infection Surveillance Across India, *Surgical Infections* 2025, doi:[10.1089/sur.2025.043](https://doi.org/10.1089/sur.2025.043) (abstract-only); 39817299 — Unverified medical certifications surge amid telemedicine guideline ambiguities, *Indian Journal of Medical Ethics* 2024, doi:[10.20529/IJME.2024.054](https://doi.org/10.20529/IJME.2024.054) (abstract-only); 41821347 — Prescribing skills course: three years' experience and lessons learnt through India's massive open online course, *British Journal of Clinical Pharmacology* 2026, doi:[10.1002/bcp.70506](https://doi.org/10.1002/bcp.70506) (abstract-only); 38342438 — WHO's essential medicines and AWaRe: recommendations on first- and second-choice antibiotics for empiric treatment of clinical infections, *Clinical Microbiology and Infection* 2024, doi:[10.1016/j.cmi.2024.02.003](https://doi.org/10.1016/j.cmi.2024.02.003) (abstract-only). Plus Telemedicine Practice Guidelines, 25 March 2020 (Appendix-5 to the 2002 conduct regulations); IRDAI recognition of teleconsultation for insurance claims, 11 June 2020; National Health Authority Scan and Share service data to October 2024; WHO AWaRe classification 2023.

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Risk approach in health care

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — The **risk approach** is a strategy, developed and promoted by the World Health Organization (WHO), of identifying precisely the "**risk groups**" or "**target groups**" in a population by certain defined criteria and directing appropriate action to them first (Park 28e, Ch 2, p 57). It has been summed up as "**something for all, but more for those in need — in proportion to the need**". In essence it is a **managerial device for increasing the efficiency of health care services within the limits of existing resources** (Park 28e, Ch 2, p 57), based on the early detection of persons carrying high-risk factors (Park 28e, Ch 9, p 658).

Conceptual basis

Three linked ideas underlie the approach. First, where the disease "agent" is unidentified (coronary heart disease, cancer, peptic ulcer, mental illness), aetiology is discussed in terms of **risk factors** — either an attribute or exposure significantly associated with the development of a disease, or a determinant that can be modified by intervention (Park 28e, Ch 2, pp 56-57). Second, the **degree of risk is indirectly an expression of need**, so that a risk factor is a **proxy for need**, pointing to the need for promotive and preventive services (Park 28e, Ch 2, p 57). Third, since resources are scarce, identification of those at risk is imperative — it defines priorities and points to those most in need of attention (Park 28e, Ch 2, p 57).

Class of risk factor	Meaning	Example
Truly causative	The factor is itself a cause of the outcome; removing it removes that cause	Smoking for lung cancer (Park 28e, Ch 2, pp 56-57)
Merely contributory to the undesired outcome	Adds to the risk of the outcome without being its cause	Lack of physical exercise for coronary heart disease (Park 28e, Ch 2, pp 56-57)
Predictive only in a statistical sense	Marks the risk but has no causal role at all	Illiteracy for perinatal mortality (Park 28e, Ch 2, pp 56-57)
Modifiable	Amenable to intervention; useful in the care of the individual	Smoking, hypertension, elevated serum cholesterol, physical activity, obesity (Park 28e, Ch 2, p 57)
Unmodifiable (immutable)	Not subject to change; act as signals alerting health personnel to the possible outcome	Age, sex, race, family history, genetic factors (Park 28e, Ch 2, p 57)
Individual risk	Characterises the person	Age, sex, smoking, hypertension (Park 28e, Ch 2, p 57)
Collective / community risk	Characterises the family, group, community or environment	Presence of malaria, air pollution, substandard housing (Park 28e, Ch 2, p 57)

Risk factors are **suggestive, not proof of cause and effect** — their presence does not mean the disease will occur, nor does their absence guarantee freedom from it; the essential property is that they are **observable or identifiable before the event they predict**, and in the same individual they may act **additively or synergistically** (Park 28e, Ch 2, pp 56-57). The degree of risk is quantified

epidemiologically by **relative risk** (incidence among the exposed ÷ incidence among the non-exposed — a direct index of the strength of association) and by **attributable risk** and **population attributable risk**, which indicate how much disease could be prevented if the factor were removed (Park 28e, Ch 3, pp 93-98).

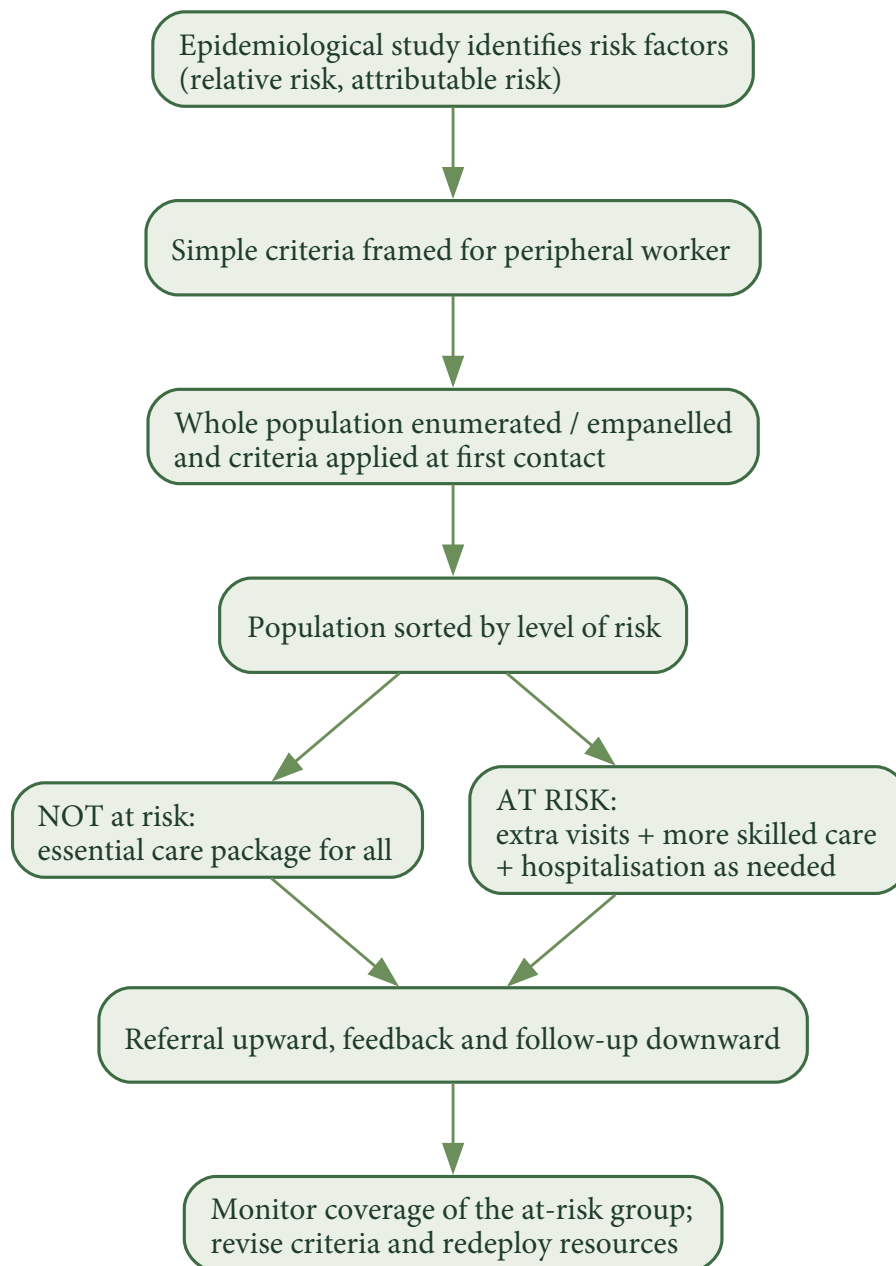
Defining "at-risk" groups (WHO guidelines)

Situation	Criteria used to define the risk group
a. Biological	Age group (infants with low birth weight, toddlers, elderly); sex (females in the reproductive age period); physiological state (pregnancy, cholesterol level, high blood pressure); genetic factors (family history of genetic disorders); other health conditions (disease, physical functioning, unhealthy behaviour)
b. Physical	Rural residence, urban slums; living conditions and overcrowding; environment — water supply, proximity to industries
c. Sociocultural	Social class; ethnic and cultural group; family disruption, education, housing; customs, habits and behaviour (smoking, lack of exercise, over-eating, drug addiction); access to health services; lifestyles and attitudes

(Park 28e, Ch 2, p 57, Table 9)

The classical risk groups so defined are **at-risk mothers**, **at-risk infants**, **at-risk families**, **the chronically ill**, **the handicapped** and **the elderly** (Park 28e, Ch 2, p 57).

Operational steps in a health service



Operational sequence of the risk approach in a health service

Four operational specifics the flowchart cannot carry, and which the examiner expects to be named:

- **Who applies the criteria** — the **auxiliary nurse midwife (ANM)** at the sub-centre, supported by the **accredited social health activist (ASHA)** and link worker who visit every house so that no eligible person is missed and the ANM keeps track of every case in her area (Park 28e, Ch 9, p 624). The criteria must therefore be simple enough to apply without laboratory support, because the approach depends on **maximum utilisation of all resources, including human resources not conventionally involved in such care** — traditional birth attendants, community health workers, women's groups (Park 28e, Ch 9, pp 625-626).

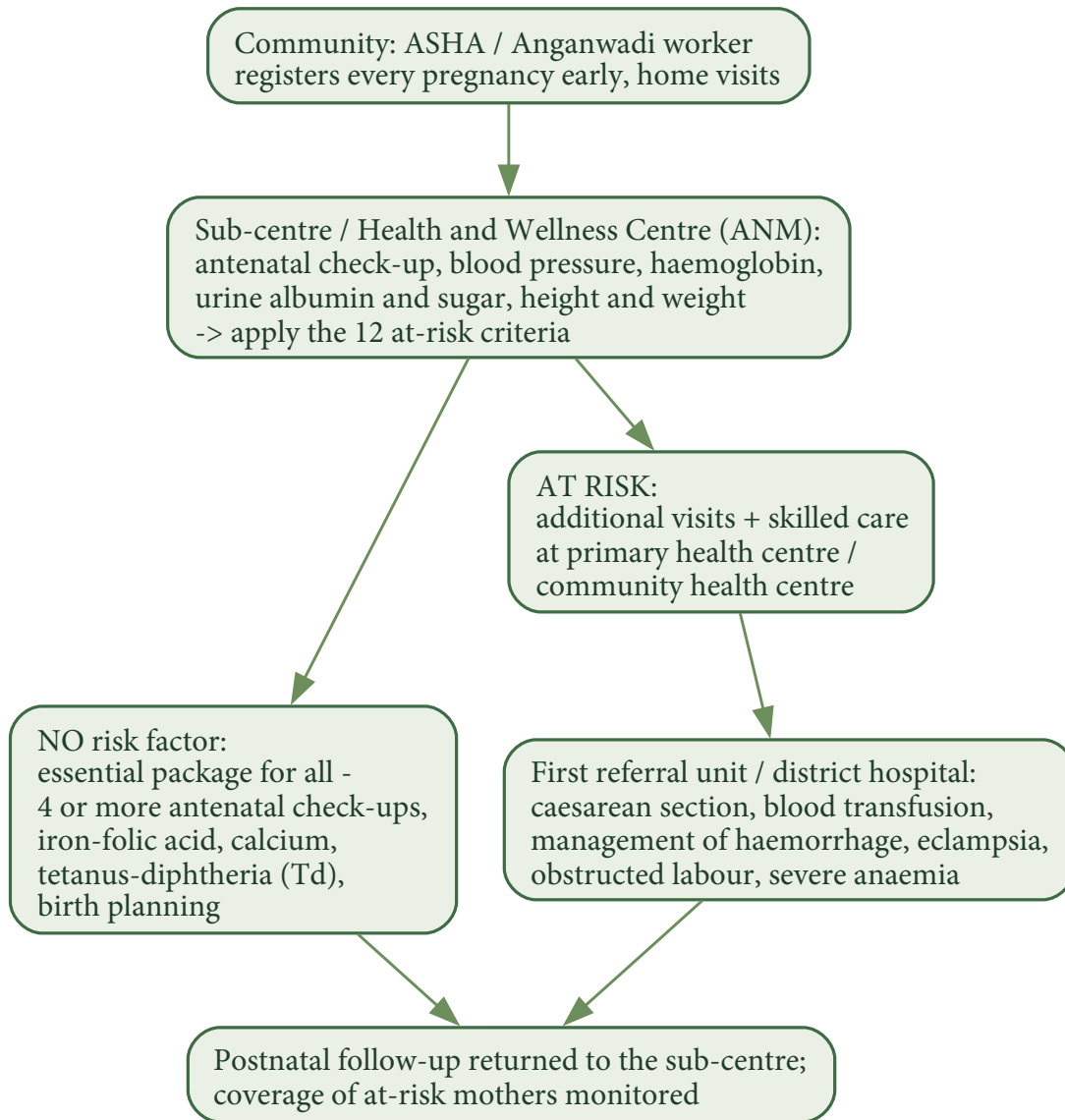
- **What record is marked** — every individual or family is enumerated and graded on the sub-centre register or antenatal register as at-risk or not, and a **line list** of the at-risk is maintained for follow-up (operational practice; the register-based tracking is described at Park 28e, Ch 9, p 624).
- **What care is differentiated** — essential care continues for **everyone**, while those with high-risk factors are given **additional and more skilled care including hospitalisation**, so that each person gets care appropriate to their need (Park 28e, Ch 9, p 658).
- **What indicator is monitored** — the stated purpose is improvement in both the **coverage and the quality** of care at all levels, particularly at the primary health care level (Park 28e, Ch 9, p 626); coverage of the identified at-risk group is therefore the working indicator, and the criteria are revised when the yield falls (operational practice).

Applications in health care

Field of application	At-risk group identified by	Action directed to them
Maternal health	12 antenatal criteria (below) applied at registration	Skilled care, institutional delivery, referral to the first referral unit (Park 28e, Ch 9, p 626)
Infant and child health	8 "at-risk infant" criteria (below)	Special intensive care, since these babies contribute most to perinatal, neonatal and infant mortality (Park 28e, Ch 9, p 635)
Non-communicable disease	Universal screening of all aged 30 years and above for hypertension, diabetes and three common cancers — oral, breast and cervical (Park 28e, Ch 10, p 735)	Referral by the ANM or staff nurse to the primary health centre, community health centre or district hospital for confirmation and treatment, with annual assessment (Park 28e, Ch 7, p 565)
HIV/AIDS	Sex workers, bridge populations practising high-risk sexual behaviour, attendees of sexually transmitted disease clinics	Targeted intervention under the National AIDS Control Programme (Sunder Lal, Ch 1, p 47)
Screening in general	Groups defined by epidemiological research (e.g. lower social groups for cancer cervix; family members of a diabetic, hypertensive or breast-cancer case)	High-risk (selective) screening raises the yield of new cases (Park 28e, Ch 4, p 165)
Environmental / occupational	Community-level risks — urban slums, overcrowding, proximity to industries, poor water supply	Area-based deployment of promotive and preventive services (Park 28e, Ch 2, p 57)

"At-risk" mothers — antenatal criteria (Park 28e, Ch 9, p 626)	"At-risk" infants (Park 28e, Ch 9, p 635)
1. Elderly primi (30 years and over)	1. Birth weight less than 2.5 kg
2. Short-statured primi (140 cm and below)	2. Twins
3. Malpresentations — breech, transverse lie	3. Birth order 5 and more
4. Antepartum haemorrhage, threatened abortion	4. Artificial feeding

"At-risk" mothers — antenatal criteria (Park 28e, Ch 9, p 626)	"At-risk" infants (Park 28e, Ch 9, p 635)
5. Pre-eclampsia and eclampsia	5. Weight below 70 per cent of expected weight (grade II and III malnutrition)
6. Anaemia	6. Failure to gain weight during three successive months
7. Twins, hydramnios	7. Children with protein-energy malnutrition and diarrhoea
8. Previous still-birth, intrauterine death, manual removal of placenta	8. Working mother / one parent
9. Elderly grandmultipara	
10. Prolonged pregnancy (14 days after expected date of delivery)	
11. Previous caesarean or instrumental delivery	
12. Pregnancy associated with general diseases	



Risk approach applied to antenatal care across the levels of care

Risk approach compared with the population approach

Feature	Risk (high-risk) approach	Population (mass) approach
Target	Individuals or groups at special risk	Whole population, irrespective of individual risk level
Identification	Requires detection by the optimum use of clinical and screening methods against defined criteria (Park 28e, Ch 2, p 60)	No screening programme needed to identify high-risk groups (Sunder Lal, Ch 1, p 47)
Aim	Bring preventive care to those at special risk (Park 28e, Ch 2, p 60)	Control the underlying causes in whole populations; shift the whole risk-factor distribution toward "biological normality" (Park 28e, Ch 6, p 442)

Feature	Risk (high-risk) approach	Population (mass) approach
Indian example	Targeted intervention for high-risk groups under the National AIDS Control Programme (Sunder Lal, Ch 1, p 47)	Regulating salt content and <i>trans</i> -fats in manufactured foods, control of tobacco use, promotion of physical activity, universal salt iodisation (Sunder Lal, Ch 1, p 47)
Motivation	High — the individual sees a personal reason to comply	Low — large community benefit but little apparent benefit to each participant (prevention paradox)
Impact on total disease burden	Small — deals with only the margin of the problem, and does not treat the underlying cause (Sunder Lal, Ch 1, p 47)	Large, sustainable and cost-effective (Sunder Lal, Ch 1, p 47)

The limiting principle is **Geoffrey Rose's axiom**: "*a large number of people exposed to a small risk may generate many more cases than a small number exposed to high risk*" — most events occur in the middle of the risk distribution, among people never judged high-risk (Sunder Lal, Ch 1, p 47; Oxford Textbook of Public Health 7e, Ch 56, pp 2110-2111). An estimate covering 22 new preventive interventions in the Netherlands, 1970-2010, attributed **three-quarters of the population health gain to the population strategy and one-quarter to the high-risk strategy** (Oxford Textbook of Public Health 7e, Ch 56, p 2111). The two are **complementary, not competing**: primordial prevention, the population strategy and the high-risk strategy should all be implemented together (Park 28e, Ch 2, p 60).

Advantages and limitations

Advantages	Limitations
Increases efficiency of services within existing resources; a genuine managerial tool (Park 28e, Ch 2, p 57)	Depends entirely on the power of the risk criteria to predict future disease
Improves both coverage and quality of care at all levels, especially at the primary health care level (Park 28e, Ch 9, p 626)	Requires continued and expensive detection , since the procedure usually involves the whole population and so resembles presymptomatic screening (Park 28e, Ch 2, p 57)
Ensures equity — care is proportional to need, and no one is denied essential care (Park 28e, Ch 9, p 658)	Does not treat the underlying cause of illness in the community, which the population approach does (Sunder Lal, Ch 1, p 47)
Uses all available human resources, including workers not conventionally involved in care (Park 28e, Ch 9, pp 625-626)	Risk labels are an operational convenience, not a description of the natural order — Rose's continuum, not a dichotomy (Sunder Lal, Ch 1, p 47)
Directs referral and hospitalisation to those who will benefit most (Park 28e, Ch 9, p 658)	If every risk factor is counted, over 90 per cent of India's pregnant women become "at risk" , so the approach loses focus; 50 per cent of maternal deaths occurred in women with no risk factor — hence every mother without exception needs professional care, and universal tracking is ensured under Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A) (Sunder Lal, Ch 10, p 578)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the risk approach pyramid — "something for all, more for those in need" — a triangle divided into three horizontal bands, with an arrow on the right showing increasing intensity of care upward and an arrow on the left showing decreasing population size upward. Label exactly:

- ▶ Base band: Whole population / all mothers and children — essential care package for all (community and sub-centre level)
- ▶ Middle band: "At-risk" group identified by defined criteria — additional visits and more skilled care (primary health centre / community health centre)
- ▶ Apex band: High-risk individuals — hospitalisation, specialist and referral care (first referral unit / district hospital)
- ▶ Right-hand arrow: Increasing intensity and skill of care
- ▶ Left-hand arrow: Decreasing number of people
- ▶ Caption under the figure: "Something for all, but more for those in need — in proportion to the need"

High-yield exam pointers

- ▶ Definition quote to write verbatim — "something for all, but more for those in need, in proportion to the need"; the risk approach is a **managerial device / managerial tool** for increasing efficiency **within the limits of existing resources**.
- ▶ A risk factor is a proxy for need — write this line; it links risk to service planning.
- ▶ Park's three-way split of risk factors — truly causative (smoking → lung cancer), merely contributory (lack of exercise → coronary heart disease), predictive only statistically (illiteracy → perinatal mortality); then modifiable vs unmodifiable, and individual vs collective.
- ▶ WHO Table of at-risk criteria — three headings only: **biological, physical, sociocultural**.
- ▶ Two lists must be reproduced: the **12 at-risk antenatal criteria** (start with elderly primi ≥ 30 years and short-statured primi ≤ 140 cm) and the **8 at-risk infant criteria** (start with birth weight < 2.5 kg).
- ▶ Rose's axiom and the prevention paradox — a large number at small risk generate more cases than a small number at high risk; quote the Netherlands estimate (three-quarters of the gain from the population strategy, one-quarter from the high-risk strategy).
- ▶ The killer criticism — with all risk factors counted, **>90 per cent** of Indian pregnant women are "at risk", and **50 per cent** of maternal deaths occur in women with no risk factor: hence India has moved to **universal tracking of every pregnancy** rather than risk-based selection alone.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still prints the older programme name "National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS)"; the programme is now the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD), and the community-level tool used to grade non-communicable disease risk before screening is the Community Based Assessment Checklist (CBAC), which Park 28e mentions only in passing (Ch 10, p 735).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e — Ch 2 (Concept of Health and Disease, pp 56-57, 60), Ch 3 (Principles of Epidemiology and Epidemiologic Methods, pp 93-98), Ch 4 (Screening for Disease, p 165), Ch 6 (Epidemiology of Chronic Non-communicable Diseases and Conditions, p 442), Ch 7 (Health Programmes in India, p 565), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 624, 625-626, 635, 658), Ch 10 (Health Care of the Community, p 735); Sunder Lal — Textbook of Community Medicine (Ch 1, pp 46-47; Ch 10, p 578); Oxford Textbook of Public Health 7e (Ch 56, pp 2110-2111 — prevention paradox and Rose's axiom; Netherlands estimate from Mackenbach et al. 2013 as cited there).



Role of ASHA in family welfare services in rural India

Asked in: Paper III Nov 2018 — RGUHS MD Community Medicine

Definition — The **accredited social health activist (ASHA)** is a trained female community health volunteer, resident of the village she serves, introduced under the National Rural Health Mission (now the National Health Mission) as the **interface between the community and the public health system** (Sunder Lal, Ch 17, p 738). In family welfare she is the first contact point for the eligible couple — counsellor, doorstep depot holder of spacing contraceptives, motivator and escort for clinical methods, and the recorder of vital events. Park calls the ASHA, the trained dai and the village health guide the **lynchpins of the family planning delivery system in India**, and records **9.39 lakh ASHAs selected and provided with drug kits** (Park 28e, Ch 8, p 612); Sunder Lal puts the current strength at **over 10.72 lakh** (Sunder Lal, Ch 17, p 744).

Selection, norms and terms of engagement

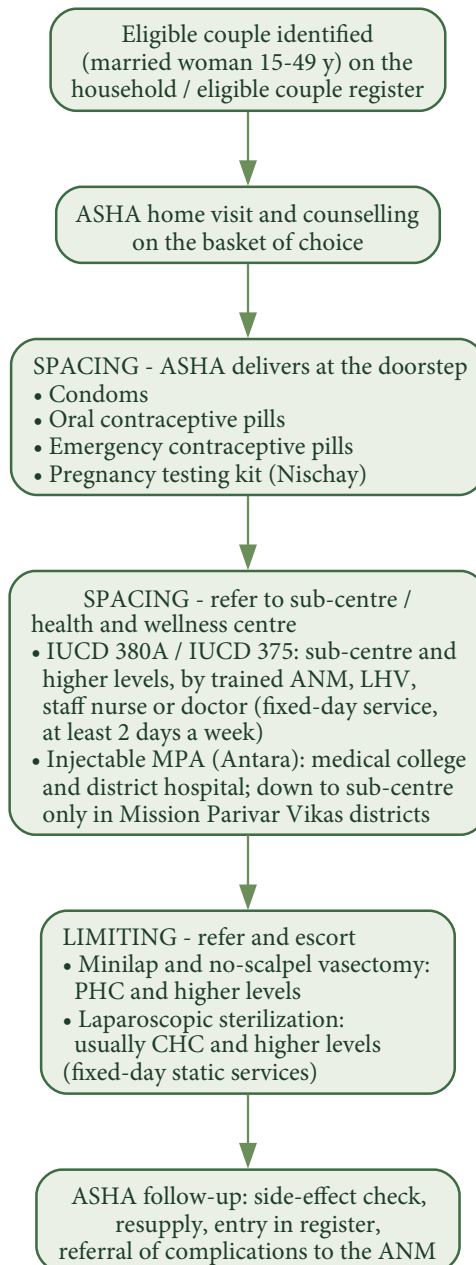
Item	Norm
Population norm	One ASHA per 1,000 rural population; in tribal, hilly and desert areas relaxed to one ASHA per habitation (Park 28e, Ch 10, pp 726, 731)
Residence and sex	Female, resident of the village — married, widowed or divorced (Park 28e, Ch 10, p 726)
Age	25–45 years (Park 28e, Ch 10, p 726)

Item	Norm
Education	Formal schooling up to 8th class (Park 28e, Ch 10, p 726; Sunder Lal, Ch 17, p 738)
Other criteria	Communication skill and leadership qualities; adequate representation from disadvantaged groups (Park 28e, Ch 10, p 726)
Selected by / accountable to	Chosen by and accountable to the village panchayat (Sunder Lal, Ch 17, p 738)
Work schedule	Flexible — 3–4 hours a day, 4–5 days a week (Sunder Lal, Ch 17, p 738)
Remuneration	Honorary volunteer on performance-based compensation , not a salaried worker (Sunder Lal, Ch 17, p 738)
Position in committee	Member secretary of the Village Health Sanitation and Nutrition Committee (VHSNC) , chaired by a woman panchayat member, holding an untied grant of ₹10,000 per year (Sunder Lal, Ch 1, p 42; Ch 17, p 738)

Family planning services delivered by ASHA

Service / scheme	What the ASHA does	Charge or incentive
Eligible couple listing	Helps the anganwadi worker (AWW) and auxiliary nurse midwife (ANM) list and annually update the register of eligible couples — all married couples with the wife aged 15–49 years, on average 190 per 1,000 population — and of children under one year (Sunder Lal, Ch 9, p 552; Park 28e, Ch 10, p 727)	Part of routine work
Behaviour change communication	Counsels on age at marriage, age at first pregnancy, gender equality, sexuality, adoption of contraception, spacing methods, sterilization and medical termination of pregnancy (Park 28e, Ch 7, p 535)	Part of routine work
Home Delivery of Contraceptives — she is a listed trained provider for oral contraceptive pills, condoms and emergency contraceptive pills (Park 28e, Ch 8, p 613, Table 24)	Delivers supplies at the doorstep ; began in 233 pilot districts of 17 states and is now operational across the whole country (Park 28e, Ch 8, p 612; Sunder Lal, Ch 10, p 626)	She charges the beneficiary Re 1 per pack of 3 condoms, Re 1 per cycle of oral pills, ₹2 per pack of one emergency contraceptive pill tablet (Sunder Lal, Ch 10, p 626)
Ensuring Spacing at Birth (ESB)	Counsels newly married couples to delay the first birth by 2 years after marriage and one-child couples to space the second birth by 3 years (Sunder Lal, Ch 10, p 626)	₹500 for ensuring spacing; ₹1,000 when the couple adopts a permanent limiting method up to two children only (Sunder Lal, Ch 10, pp 626–627; Park 28e, Ch 8, p 613)

Service / scheme	What the ASHA does	Charge or incentive
Pregnancy testing kits (<i>Nischay</i>)	Holds home-based kits that detect pregnancy as early as one week after the missed period , allowing early registration or safe, early medical termination of pregnancy (Sunder Lal, Ch 10, p 626)	Part of routine work
Depot holding	Depot holder for oral rehydration therapy, iron–folic acid tablets, disposable delivery kits, oral pills and condoms ; each ASHA is issued a drug kit (Sunder Lal, Ch 17, p 738; Park 28e, Ch 10, p 727)	Part of routine work
Motivation and escort for clinical methods	Mobilises acceptors for fixed-day static services at the correct level: IUCD 380A / IUCD 375 — sub-centre and higher levels , inserted by trained and certified ANM, lady health visitor, staff nurse or doctor; injectable medroxyprogesterone acetate (MPA, Antara) — medical colleges and district hospitals , extended down to all levels up to the sub-centre only in Mission Parivar Vikas (MPV) districts ; minilap and no-scalpel vasectomy — PHC and higher levels ; laparoscopic sterilization — usually CHC and higher levels (Park 28e, Ch 8, pp 613–614, Table 24)	Under the compensation scheme in public facilities the acceptor gets ₹2,000 vasectomy / ₹1,400 tubectomy and the ASHA ₹300 / ₹200 as motivator, in the 11 high-focus states (Park 28e, Ch 8, p 615, Table 25; Sunder Lal, Ch 10, p 627)
Mission Parivar Vikas	Intensified family planning counselling in 146 high-fertility districts (total fertility rate ≥ 3) of 7 high-focus states — Uttar Pradesh, Bihar, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand and Assam (Sunder Lal, Ch 10, p 627; Park 28e, Ch 8, p 613)	Higher amounts in MPV districts: acceptor ₹3,000 vasectomy, ₹2,000 tubectomy, ₹3,000 post-partum sterilization , with ₹400 / ₹300 / ₹400 to the ASHA (Park 28e, Ch 8, p 615, Table 25)

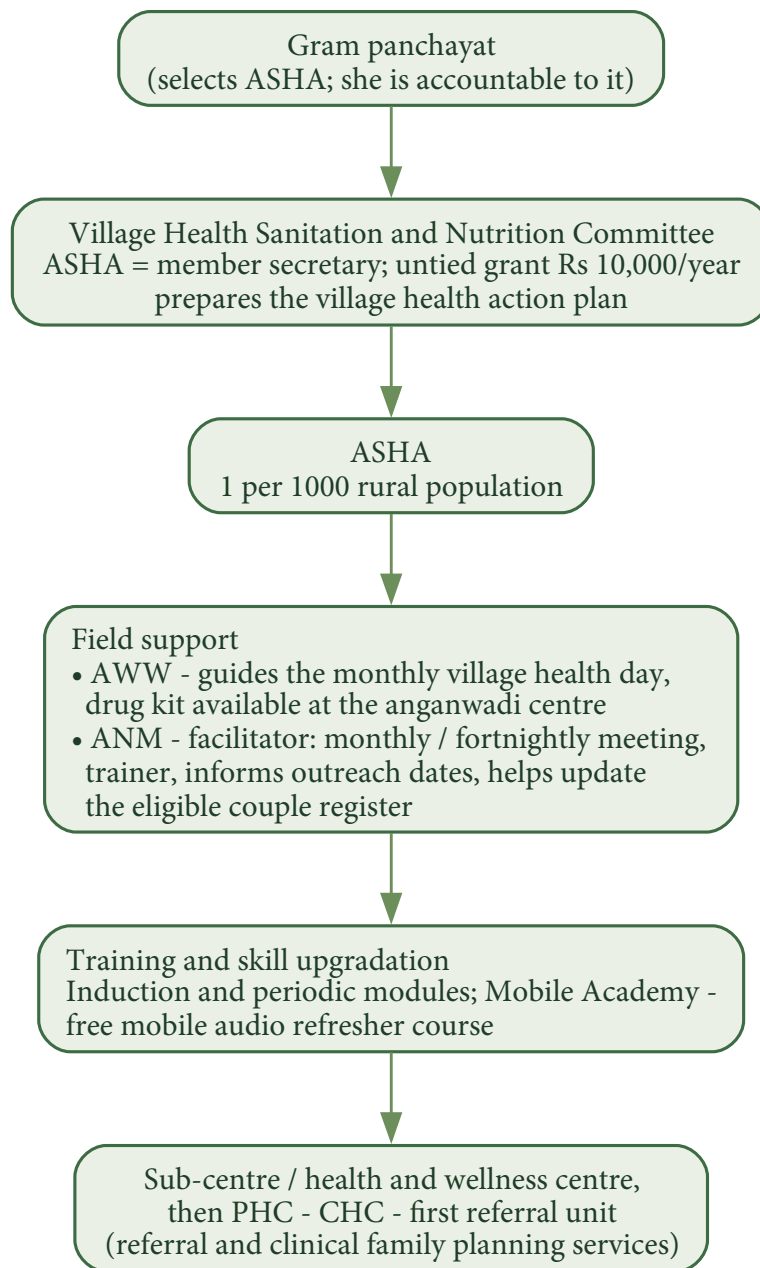


Family planning service pathway delivered through the ASHA in a village, by level of care

Role in the maternal, newborn and child components of family welfare

- ▶ **Early registration of pregnancy** and mobilisation of every pregnant woman for the full antenatal package; counselling on nutrition, danger signs, birth preparedness and clean, safe delivery (Park 28e, Ch 7, p 535).
- ▶ **Escort service** — accompanies pregnant women and sick children to the pre-identified primary health centre (PHC), community health centre (CHC) or first referral unit (Sunder Lal, Ch 17, p 738).
- ▶ **Janani Suraksha Yojana** — she is the effective link worker; her package is ₹600 in rural and ₹400 in urban areas, paid after her postnatal visit and once the child has received BCG, and it includes referral transport (not less than ₹250) and a delivery incentive (not less than ₹200) (Park 28e, Ch 7, p 544).

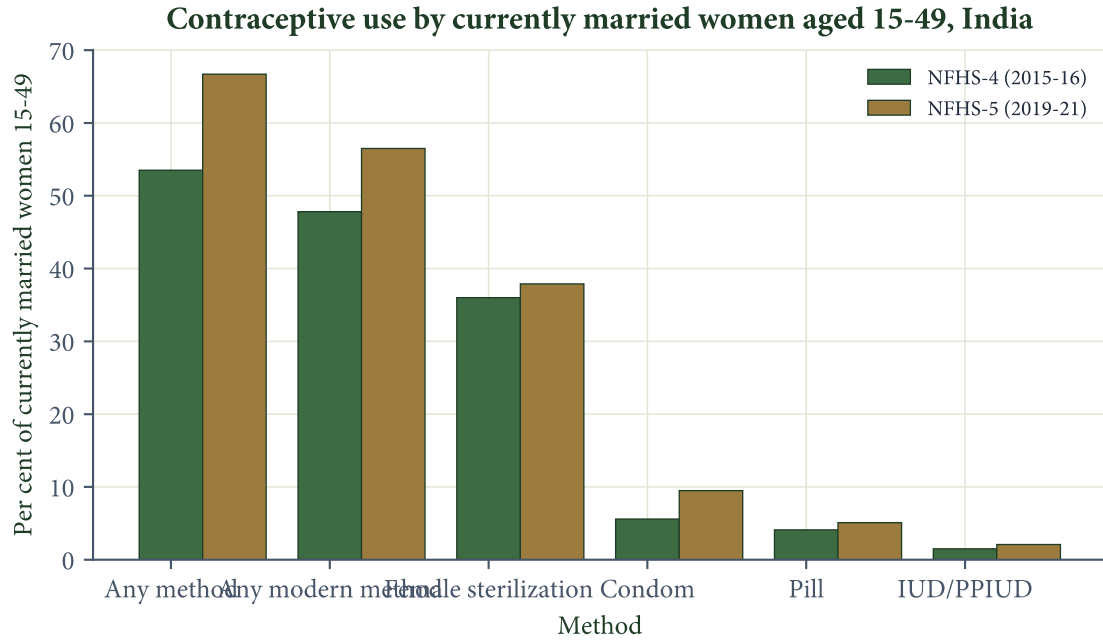
- **Home-based newborn care (HBNC)** — she is the principal provider: 6 visits after institutional delivery (days 3, 7, 14, 21, 28, 42) and 7 visits after home delivery (days 1, 3, 7, 14, 21, 28, 42), up to 42 days after delivery, with extra visits for low-birth-weight babies; incentive ₹250 paid on the 45th day after validation by the ANM (Sunder Lal, Ch 10, p 594; Ch 17, p 738).
- **Home-based care for the young child** — five further home contacts in the first week of the 4th, 7th, 10th, 13th and 16th month of age (Sunder Lal, Ch 17, p 738).
- **Immunisation and the Village Health, Nutrition and Sanitation Day** — brings beneficiaries to the anganwadi centre on the monthly day for immunisation, antenatal and postnatal check-up and supplementary nutrition (Sunder Lal, Ch 17, p 744).
- **Vital events and surveillance** — records and reports all **births and deaths** in her village and any unusual illness or disease outbreak to the sub-centre or PHC (Sunder Lal, Ch 17, p 738).



Support, supervision and accountability structure for the ASHA in family welfare service delivery

Evaluation — indicators against which her family welfare work is judged

Indicator	Definition / formula	Current Indian value
Contraceptive prevalence rate (any modern method)	Percentage of women aged 15–49 who are practising, or whose sexual partners are practising, any modern method of contraception	56.5%, NFHS-5 (2019–21) (Sunder Lal, Ch 10, p 627)
Method mix	Share of each method among current users	Female sterilization 37.9%, condom 9.5%, pill 5.1%, IUD/post-partum IUD 2.1%, injectable 0.6%, male sterilization 0.3% — NFHS-5 (2019–21) (Sunder Lal, Ch 10, p 627)
Couple protection rate (CPR)	Percentage of eligible couples effectively protected against childbirth by an approved method; weights 100% for sterilization and IUD, 95% for oral pills, 50% for condoms (Sunder Lal, Ch 10, pp 627–628)	Of 205 million eligible couples in 2011 only 44% were effectively protected; an effective CPR of more than 60% is required for a net reproduction rate (NRR) of 1, i.e. total fertility rate 2.1 (Sunder Lal, Ch 9, p 552; Park 28e, Ch 8, pp 591–592)
Effective coverage needed	Contraceptive coverage required to move the birth rate	About 75% of eligible couples must be given effective contraceptive coverage to bring the birth rate down significantly (Sunder Lal, Ch 9, p 552)
Unmet need for family planning	Percentage of currently married women who want to space or limit births but use no method	9% (spacing 4%, limiting 5%), down from 13% in NFHS-4 (NFHS-5 India Report 2019–21, pp 199–200)
Demand satisfied by modern methods	Modern-method use ÷ (unmet need + any-method use)	74.2%, up from 72% in NFHS-4 (Park 28e, Ch 8, pp 610–611)
Eligible couples per 1,000 population	Married couples with wife aged 15–49, listed and updated every year	About 190 per 1,000 (Sunder Lal, Ch 9, p 552); Park gives at least 150–180 per 1,000 , with about 20% of them aged 15–24 years (Park 28e, Ch 8, p 591)



Rise in the spacing methods the ASHA supplies at the doorstep. Source: Sunder Lal, Textbook of Community Medicine, Table 10.9 (NFHS-1 to NFHS-5).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The ASHA as the interface between the village and the public health system.** Draw a vertical ladder and label: Village households and eligible couples → ASHA (1 per 1,000 rural population; drug kit and contraceptive depot) → flanked on one side by Anganwadi worker (ICDS, 1 per 400–800 population; guides the monthly village health day) and on the other by ANM at sub-centre / health and wellness centre (facilitator, trainer and supervisor) → Primary health centre → Community health centre / first referral unit → District hospital. Add a side box Village Health Sanitation and Nutrition Committee — ASHA is member secretary, untied grant ₹10,000 per year and an arrow Gram panchayat → selects and holds the ASHA accountable.

Limitations of the ASHA in family welfare delivery

- **Volunteer status without a salary** — she is an honorary worker paid only performance-based compensation, so an activity that carries no incentive (counselling, register updating) competes poorly with one that does (Sunder Lal, Ch 17, p 738).
- **Skew toward limiting rather than spacing** — the method mix is still dominated by female sterilization (37.9%) with male sterilization at only 0.3%, and both books state that male participation through no-scalpel vasectomy and the spacing methods must be promoted vigorously (Sunder Lal, Ch 9, p 552; Ch 10, p 627). Sterilizing women who already have 3–4 children has little demographic impact (Sunder Lal, Ch 9, p 552).
- **Reporting inflation** — a paradox of high reported couple protection rate with a persistently high birth rate has been observed in some states, from over-reporting to complete targets (Sunder Lal, Ch 9, p 552).

- ▶ **Task load within a short working day** — beyond family welfare she provides directly observed treatment for tuberculosis, depot holding, immunisation mobilisation, leprosy case detection and filaria day work, all within 3–4 hours a day on 4–5 days a week (Sunder Lal, Ch 17, pp 738–739).
- ▶ **Dependence on the ANM and AWW** for training, outreach dates, supplies and validation of her claims, so weak facilitation directly reduces her output (Sunder Lal, Ch 16, p 718; Park 28e, Ch 10, p 727).

High-yield exam pointers

- ▶ **Selection formula:** resident woman, 25–45 years, 8th class pass, 1 per 1,000 rural population, chosen by and accountable to the **gram panchayat**; one per habitation in tribal, hilly and desert areas.
- ▶ **Doorstep contraceptive charges (memorise):** Re 1 / 3 condoms, Re 1 / cycle of oral pills, ₹2 / one emergency contraceptive pill tablet.
- ▶ **Service levels (Park Table 24):** oral pills, condoms and emergency pills — doorstep by ASHA; IUCD — sub-centre and above; injectable MPA (*Antara*) — medical college and district hospital, down to sub-centre only in MPV districts; minilap and no-scalpel vasectomy — PHC and above; laparoscopic sterilization — usually CHC and above.
- ▶ **ESB incentives:** ₹500 spacing, ₹1,000 for a permanent limiting method **up to two children only**. Mission Parivar Vikas: 146 districts, total fertility rate ≥ 3 , 7 high-focus states; MPV acceptor compensation ₹3,000 vasectomy / ₹2,000 tubectomy.
- ▶ **HBNC visit days** — "3-7-14-21-28-42" (6 visits, institutional); add day 1 for home delivery (7 visits); incentive ₹250 on the 45th day after ANM validation. **Home-based care for the young child:** months 4, 7, 10, 13, 16. JSY ASHA package ₹600 rural, ₹400 urban.
- ▶ **Numbers to quote:** NFHS-5 (2019–21) modern contraceptive prevalence rate 56.5%, unmet need 9%, demand satisfied by modern methods 74.2%; **effective couple protection rate** > 60% for NRR = 1, but only 44% of 205 million eligible couples (2011) were effectively protected.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the two textbooks differ on three numbers, so always quote the figure with its book — neither attaches a survey year to its ASHA head-count (Park 28e, 9.39 lakh selected and given drug kits; Sunder Lal, over 10.72 lakh currently working); Park defines the eligible couple as one whose wife is aged 15–45 and puts them at 150–180 per 1,000 population, while Sunder Lal and the National Family Health Survey use the 15–49 band and 190 per 1,000; and Sunder Lal's Ch 9 reverses the couple protection rate weights for IUD and oral pills, so use the Ch 10 set given above, which matches Park. The National Rural Health Mission is now a sub-mission of the National Health Mission and sub-centres and PHCs have been repositioned as health and wellness centres under Ayushman Bharat; ASHA incentive amounts are revised periodically by scheme and by state.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-5 (2019–21), WHO Selected Practice Recommendations for Contraceptive Use (4th edition)

Programme and policy updates (India)

- **The incentive design, not the ASHA herself, is the bottleneck in family planning.** Modelling of Uttar Pradesh payment data showed family planning is meant to be her single largest earning head — **42% of potential monthly earnings** — but yields only **3% of what she actually earns**, because payment turns on household behaviour she cannot control. Her modelled real earnings were **₹1,325 a month against ₹3,000** if she completed everything within her own control; most income instead came from routine work (36%) and antenatal care plus institutional delivery (31%), and 68% of ASHAs quoted the institutional-delivery incentive as **₹600** when it was **₹300** [PMID 36332076]. Interviews with 40 ASHAs across ten Uttar Pradesh districts identified the fixable levers as **supportive supervision by the ASHA Sangini, a stocked drug kit, and payment on time** — empty kits and delayed or partial payment were the main demotivators [PMID 41628200].
- **Injectable medroxyprogesterone acetate (*Antara*)** — rolled out in 2017 under Mission Parivar Vikas, **still barely used**. ASHA free-listing in rural and urban Nagpur ranked the barriers by salience score: **heavy menstrual bleeding 0.516, irregular menses 0.290, weight gain 0.195**, then fear built from other women's experiences; the service-side barriers were the compulsory 3-monthly repeat visit and the injection simply not being available at the PHC. Injectables were only **0.4% of the method mix in NFHS-5**, against IUD 2.7%, pills 4.4% and condoms 13.6% [PMID 40587115].
- **Health and Wellness Centres have been renamed Ayushman Arogya Mandir.** The sub-centre- and PHC-level centre to which the ASHA refers her eligible couples now carries this name; each is run by a **community health officer (CHO)** delivering the twelve comprehensive primary health care service packages — including non-communicable disease screening — alongside the ASHA and the ANM [PMID 40341602].

Current national data

- **Modern contraception is reaching the poor, but the spacing gap persists.** Across NFHS-1 to NFHS-5 (1992–93 to 2019–21) rural modern-method prevalence rose **33% → 56%** and demand satisfied **63% → 74%**; but the **modern reversible methods the ASHA actually supplies rose only 3% → 16% in rural areas**. Prevalence in the poorest quintile rose **27% → 49%** against **50% → 60%** in the richest, so inequality is narrowing yet still pro-rich — better-off women buy reversible methods privately [PMID 41220271].
- **The postpartum year is where her home visits and the contraceptive need actually coincide.** NFHS-5 reproductive-calendar data on 149,518 women: any-method use is **18.0% at one month postpartum, rising to 54.3% by twelve months**, still below the 66.7% for married women generally; women who are sexually active and using nothing rise from **1.5% at month 1 to 38.8% of all postpartum women by month 12** [PMID 40075525].
- **Meeting a health worker shifts the method — and the conversation, not the visit, is the active ingredient.** Among married non-pregnant, non-sterilised women in NFHS-5, **22.1% used a traditional method**, and traditional methods made up **44.1% of all reversible-method use**. Contact with a community health worker in the previous three months was associated with more modern reversible use (**adjusted RRR 1.21, 95% CI 1.17–1.25**) and less traditional-method use (**0.88, 95% CI 0.85–0.91**). But only **22.3% of women who met a worker discussed family**

planning at all, and where it was discussed contraceptive use was 49.2% versus 33.5% [PMID 41944354].

Elimination, surveillance and disease status

- Her remit has widened from family welfare into non-communicable disease and cancer screening. In the IMPACT Diabetes cluster-randomised trial (16 villages, 8 PHCs, two states) ASHAs screened 1,785 people and delivered nine months of decision-support-guided care, achieving a $\geq 0.5\%$ fall in HbA1c in 21.8% versus 10.3% of controls ($p < 0.016$), with PHC physician visits 85% versus 29.8% [PMID 41120559]. ASHA-delivered HPV self-sampling for cervical cancer screening in Sikkim — India's contribution to the WHO cervical cancer elimination call — was feasible and well accepted, limited mainly by low health literacy, sample transport and inadequate incentive [PMID 40341769].

Guideline and legislative updates

- Counselling is the intervention with the strongest recent Indian trial evidence. A cluster-randomised trial in Puducherry was stopped early at interim analysis: a single structured antenatal counselling session with audio-visual aid raised postpartum contraceptive acceptance at six weeks to 66.7% (78/117) versus 21% (25/119) on routine care [PMID 40257138]. A participatory learning and action cycle run through existing women's self-help groups in Madhepura, Bihar cut unmet need for limiting by 7.2 percentage points (95% CI -0.14 to 0.00) and raised contraceptive use by 5.8 points, though the effect on overall unmet need was not significant [PMID 40597148].
- Centchroman (ormeloxifene, *Chhaya*) — the weekly non-hormonal pill now has Indian multicentre data. 3,139 users across six medical college hospitals, followed 24 months: continuation 78.0% at 6 weeks, 47.3% at 6 months, 32.2% at 24 months, with 40.9% stopping within the first three months for menstrual disturbance (scanty flow in 60% by 18 months); failure rate 0.7–2.1%, satisfaction 75.2%, and no change in endometrial thickness, blood pressure or biochemistry. Pre-emptive counselling about scanty periods is what decides continuation [PMID 40186448].

Global and WHO developments

- WHO has rewritten the return-to-fertility counselling point for injectables. The Selected Practice Recommendations for Contraceptive Use, fourth edition, now directs providers to tell users of DMPA and norethisterone enanthate that ovulation may be delayed up to one year after the last injection, replacing the older "median 10 months" wording. The supporting review found median time to ovulation 212 days after subcutaneous DMPA 104 mg versus 183 days after intramuscular DMPA 150 mg (not significantly different), with 97.4% ovulating by 12 months; body mass index, age and region made no difference [PMID 41184158].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India; Ch 8 — Demography and Family Planning, Tables 24 and 25; Ch 10 — Health Care of the Community); Sunder Lal — Textbook of Community Medicine (Ch 1 — Concept of Health and Disease; Ch 9 — Demography and Vital Statistics; Ch 10 — RMNCH+A, Table 10.9; Ch 16 — Health Planning and Management; Ch 17 — Healthcare of the Community); National Family Health Survey-5 India Report, 2019–21 (chapter on Family Planning).

Recent-advances sources: 36332076 — Improving Community Health Worker Compensation: A Case Study From India Using Quantitative Projection Modeling and Incentive Design Principles, Glob Health Sci Pract 2022, doi:[10.9745/GHSP-D-21-00413](https://doi.org/10.9745/GHSP-D-21-00413); 41628200 — Factors affecting the motivation of community health workers: perspectives from Accredited Social Health Activists (ASHA) in Uttar Pradesh, India, PLoS One 2026, doi:[10.1371/journal.pone.0341811](https://doi.org/10.1371/journal.pone.0341811); 40587115 — Understanding the acceptability of depot medroxyprogesterone acetate (ANTARA) among women of reproductive age from the lens of grassroot level providers in rural and urban areas of Nagpur, India, Ann Afr Med 2025, doi:[10.4103/aam.aam_116_25](https://doi.org/10.4103/aam.aam_116_25); 40341602 — Capacity building of community health officers for optimizing the screening and early diagnosis of oral cancer in Jodhpur, Rajasthan, India, Sci Rep 2025, doi:[10.1038/s41598-025-00225-w](https://doi.org/10.1038/s41598-025-00225-w); 41220271 — Trends and patterns of inequality in modern contraceptive use in urban and rural India: are family planning programmes increasingly reaching the marginalized?, Health Policy Plan 2026, doi:[10.1093/heapol/czaf073](https://doi.org/10.1093/heapol/czaf073); 40075525 — The state of postpartum contraceptive use in India: descriptive lessons from nationally representative survey data, Reprod Health 2025, doi:[10.1186/s12978-025-01978-3](https://doi.org/10.1186/s12978-025-01978-3); 41944354 — Traditional or Modern Contraception? Association Between Health Worker Contact and Contraceptive Choice in India: Findings From NFHS 2019–2021, Stud Fam Plann 2026, doi:[10.1111/sifp.70052](https://doi.org/10.1111/sifp.70052); 41120559 — Preliminary effectiveness and feasibility of ASHA-led mobile health intervention for diabetes care in Indian primary health care settings, Sci Rep 2025, doi:[10.1038/s41598-025-20728-w](https://doi.org/10.1038/s41598-025-20728-w); 40341769 — A qualitative study on ASHA workers' perspective on HPV self-sampling in Sikkim India, Sci Rep 2025, doi:[10.1038/s41598-025-87041-4](https://doi.org/10.1038/s41598-025-87041-4); 40257138 — Effect of structured counselling on postpartum contraceptive acceptance: A cluster randomized controlled trial, Puducherry, Indian J Med Res 2025, doi:[10.25259/IJMR_887_2024](https://doi.org/10.25259/IJMR_887_2024); 40597148 — The impact of a participatory learning and action intervention on unmet need for contraception: a cluster-randomized controlled trial in rural Bihar, India, Reprod Health 2025, doi:[10.1186/s12978-025-02055-5](https://doi.org/10.1186/s12978-025-02055-5); 40186448 — Clinical outcomes among Centchroman users across six medical college hospitals in India, Int J Gynaecol Obstet 2025, doi:[10.1002/ijgo.70107](https://doi.org/10.1002/ijgo.70107) (abstract-only); 41184158 — Return to fertility after subcutaneous depot medroxyprogesterone acetate: a narrative review, BMJ Sex Reprod Health 2025, doi:[10.1136/bmjsexrh-2025-202839](https://doi.org/10.1136/bmjsexrh-2025-202839). Plus WHO Selected Practice Recommendations for Contraceptive Use, 4th edition.



Role of self-help groups in upliftment of community health

Asked in: Paper I 19-Nov-2020 — RGUHS MD Community Medicine

Definition — A **self-help group (SHG)** is a small group of the rural or urban poor who have voluntarily come together to improve the social and economic status of their members, chiefly by pooling regular small savings into a common fund (Park Plus, Ch 21, pp 314–315). In public health it is the operational unit of **community participation** — "a process by which individuals and families assume responsibility for their own health and welfare and for those of the community, and develop capacity to contribute to their and the community's development" (Sunder Lal, Ch 2, pp 68–69). Because health cannot be produced by the health sector alone, the SHG is the vehicle through which a community supplies resources, plans and monitors services, and generates demand for them.

Structure and formation of a self-help group

Feature	Specification
Composition	Homogeneous group of about 15–20 members drawn from the poor of one locality (Park Plus, Ch 21, pp 314–315)
Age of members	21–60 years (Park Plus, Ch 21, pp 314–315)
Family rule	Only one person per family may join a given SHG, so more families are covered (Park Plus, Ch 21, pp 314–315)
Sex composition	Normally either only men or only women; health-linked groups are almost always women's groups (Park Plus, Ch 21, pp 314–315)
Core discipline	Compulsory periodic meeting; every member saves a small amount regularly, learning prioritisation and financial discipline (Park Plus, Ch 21, pp 314–315)
Credit linkage	SHG–Bank Linkage Project, launched by NABARD in 1992, is the world's largest microfinance project (Park Plus, Ch 21, p 314)
Policy origin in India	Under the Integrated Rural Development Programme (IRDP), implemented in the Ninth Five Year Plan, the anti-poverty strategy shifted from an individual-beneficiary approach to formation of self-help groups built on locally viable activities and local occupational skills (Park 28e, Ch 23, p 1037)
Current umbrella	SGSY restructured as the National Rural Livelihoods Mission (NRLM): universal social mobilisation so that at least one member — preferably a woman — of every rural below-poverty-line household is in an SHG; federations of SHGs from village panchayat to district level; universal financial inclusion by linking SHGs to banks; placement-linked skill development (Jugal Kishore, p 693)

Why self-help groups matter for health — the participation basis

Framework	Content
Three ways a community can participate (Park 28e, Ch 2, p 34)	(1) Provide facilities, manpower, logistic support and possibly funds; (2) be actively involved in planning, management and evaluation; (3) join in and actually use the health services, especially preventive and protective measures

Framework	Content
Six types of community participation (Sunder Lal, Ch 2, pp 68–69)	(i) Manage village-level health services; (ii) generate own resources for buildings; (iii) determine own health needs and priorities; (iv) organise and implement programmes with own plus government resources; (v) monitor government programmes and give feedback; (vi) evaluate programmes and report back for corrective action
Health-sector rationale	Primary health care cannot be delivered by the health sector alone; the Alma-Ata Declaration requires all related development sectors and the full participation of the community (Park 28e, Ch 10, p 720)
Economic rationale	Women's control over economic resources is far less than men's, and women form the bulk of the landless poor; SHGs were formed specifically to correct this through group income generation (Sunder Lal, Ch 10, p 633)
Financial-risk rationale	Health improvement is more effective when tied to livelihood improvement, hence the need for micro-credit and savings mechanisms that let women save part of their income for family health needs (Jugal Kishore, p 694)

Roles of self-help groups in the upliftment of community health

Domain	Specific role of the SHG	Grounding
Financial risk protection	Pooled thrift meets urgent cash needs at the time of a health emergency, protecting the household from catastrophic out-of-pocket expenditure and indebtedness	Jugal Kishore, p 164; pp 694–695
Community risk pooling / insurance	Members pool an agreed amount (about ₹10–20 per family per month) in a bank account; the reserve meets emergency out-of-pocket costs, referral transport, health camps and premium financing for community health insurance	Jugal Kishore, pp 694–695; Suryakantha, Ch 36, pp 801–802
Income generation and the social determinants	Group activities, skill training, bank credit and marketing support raise household income, food security and schooling — acting on the upstream determinants of health	Jugal Kishore, pp 692–693; Sunder Lal, Ch 10, p 633
Maternal and child health	Under the National Population Policy 2000, a revolving fund is set up for income-generating activities by village-level SHGs, who also provide community-level health care services, and SHGs are used to organise and provide basic reproductive and child health services convergently with ICDS at the anganwadi centre	Mahajan & Gupta 4e, Ch 31, p 643; Sunder Lal, Ch 9, p 554
Service delivery platform	SHG members join the Village Health, Sanitation and Nutrition Day held monthly at the anganwadi centre, and the auxiliary nurse midwife (ANM) holds regular meetings with SHGs, panchayati raj institutions, teachers and NGOs	Sunder Lal, Ch 17, p 744; Ch 16, p 718

Domain	Specific role of the SHG	Grounding
Health promotion and behaviour change	Women's groups are the standard channel for interpersonal communication on breastfeeding, anaemia, safe pregnancy and delivery, sex-selective abortion and the sex ratio; a chosen subject is discussed monthly for 30–60 minutes	Sunder Lal, Ch 4, pp 146–147
Disease surveillance	SHGs, along with ASHAs, anganwadi workers and village panchayats, act as community informants reporting births, deaths, outbreaks and unusual events	Sunder Lal, Ch 7, p 302
Community monitoring and accountability	SHG volunteers are formal members of the Village Health, Sanitation and Nutrition Committee, which monitors the services provided by the anganwadi worker, ASHA and sub-centre	Sunder Lal, Ch 1, p 42; Ch 17, p 744
Environmental sanitation	Women's SHGs run municipal services directly — <i>Kudumbashree</i> SHGs served 15 of 74 wards of the Corporation of Cochin for source segregation and door-to-door solid waste collection	Solid Waste Management, Ch 2, pp 184–185
Mental health and disability	SHGs of parents of children with intellectual disability share and resolve problems, and characteristically grow into welfare associations, special schools and training centres — a recognised community-based rehabilitation measure	Suryakantha, Ch 30, p 763; Mahajan & Gupta 4e, Ch 34, p 662
Health of the aged	The Integrated Programme for Older Persons, for the first time, recognises formation of self-help groups and associations of older persons for advancement of their rights and use of their experience and services	Park 28e, Ch 9, pp 704–705
Empowerment / demand generation	Women-led thrift and self-help groups have been shown to empower members to demand improved health services (<i>Mahila Bachat Gat</i> , Maharashtra; urban health initiatives in Indore and Agra)	Jugal Kishore, p 164

Self-help-group-based community platforms in the Indian health system

Platform	Coverage and composition	Funds	Health functions
Village Health, Sanitation and Nutrition Committee (VHSNC)	Standing sub-committee of the gram panchayat; minimum 15 members, chairperson a <i>mahila panch</i> (scheduled caste preferred), member secretary ASHA ; includes NGO members and self-help group volunteers ; >5.55 lakh constituted by March 2021	Untied grant ₹10,000 per annum	Village health action plan; convergent action on social determinants; monitors ASHA, anganwadi worker and sub-centre; runs the monthly Village Health, Sanitation and Nutrition Day (Sunder Lal, Ch 1, p 42; Ch 17, p 744)

Platform	Coverage and composition	Funds	Health functions
Mahila Arogya Samiti (MAS) — urban slum	Women's health committee of 10–12 members for 50–100 urban slum households (population 250–500); elects chairperson and treasurer; supported by the urban ASHA; the National Urban Health Mission framework allows 5–20 members per samiti	₹5,000 per year per MAS; NUHM seed money ₹25 per household plus an annual performance grant of ₹25 per household	Community-based peer educator group ; interpersonal communication; community-based monitoring; linkage and referral; manages the revolving fund and the community risk pool (Sunder Lal, Ch 17, p 758; Park 28e, Ch 7, p 535; Suryakantha, Ch 36, pp 801–802; Jugal Kishore, pp 694–695)
Mahila Swasthya Sangh (MSS) / Mahila Mandal	Village-level women's health groups; 79,512 MSS working at village level	—	Enhance community participation in health and development at village level (Sunder Lal, Ch 17, p 744; Ch 2, pp 68–69)
Rogi Kalyan Samiti	Patient welfare committee at the primary health centre, community health centre and urban primary health centre	Institution-level untied funds	Community management of the facility; created together with MAS to ensure community participation under NUHM (Park 28e, Ch 7, p 535)
Thrift Credit Groups / DWCUA groups	Groups of at least 10 urban poor women under the Urban Self Employment Programme of SJSRY	Credit and subsidy	Employment and thrift for the urban poor, feeding into MAS-based health financing (Jugal Kishore, p 164)

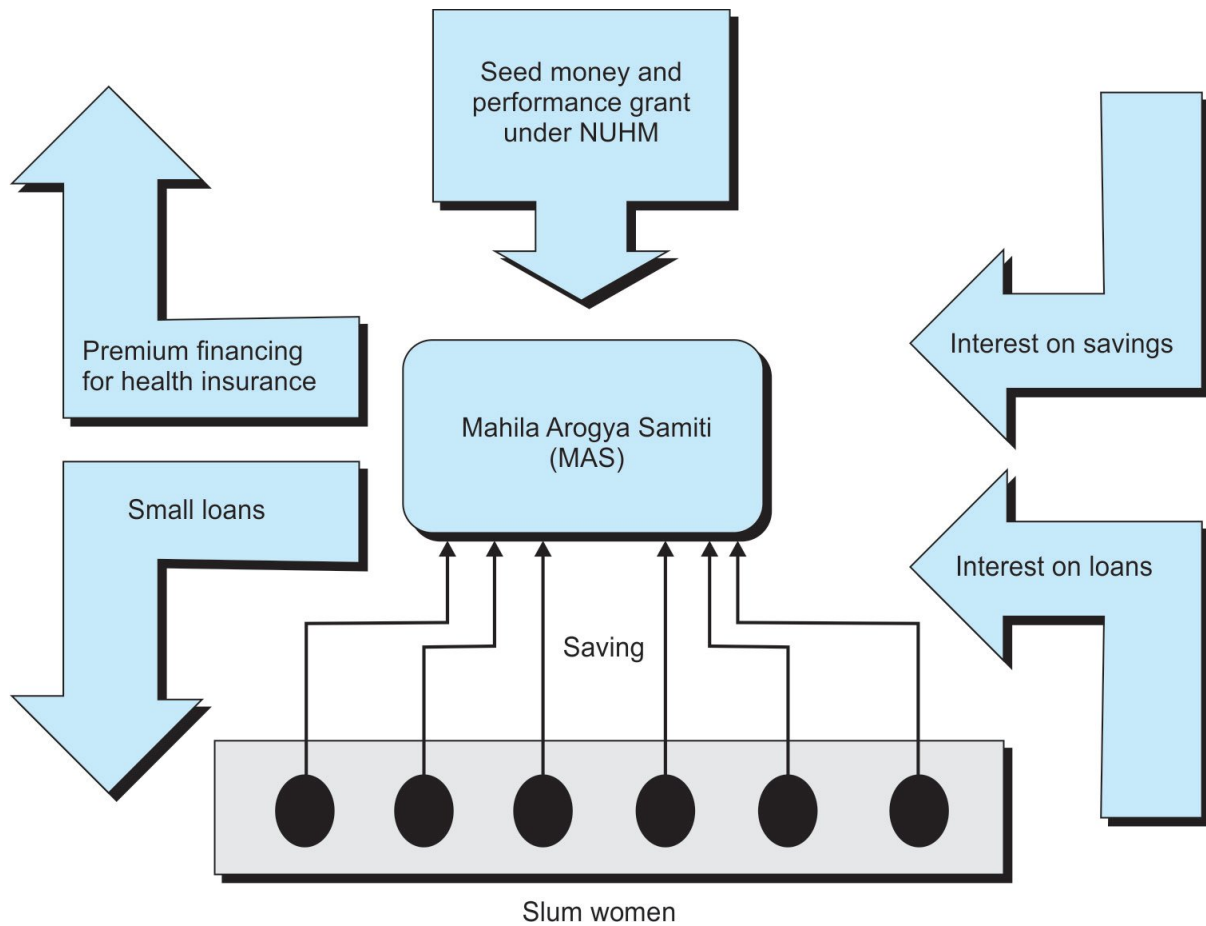
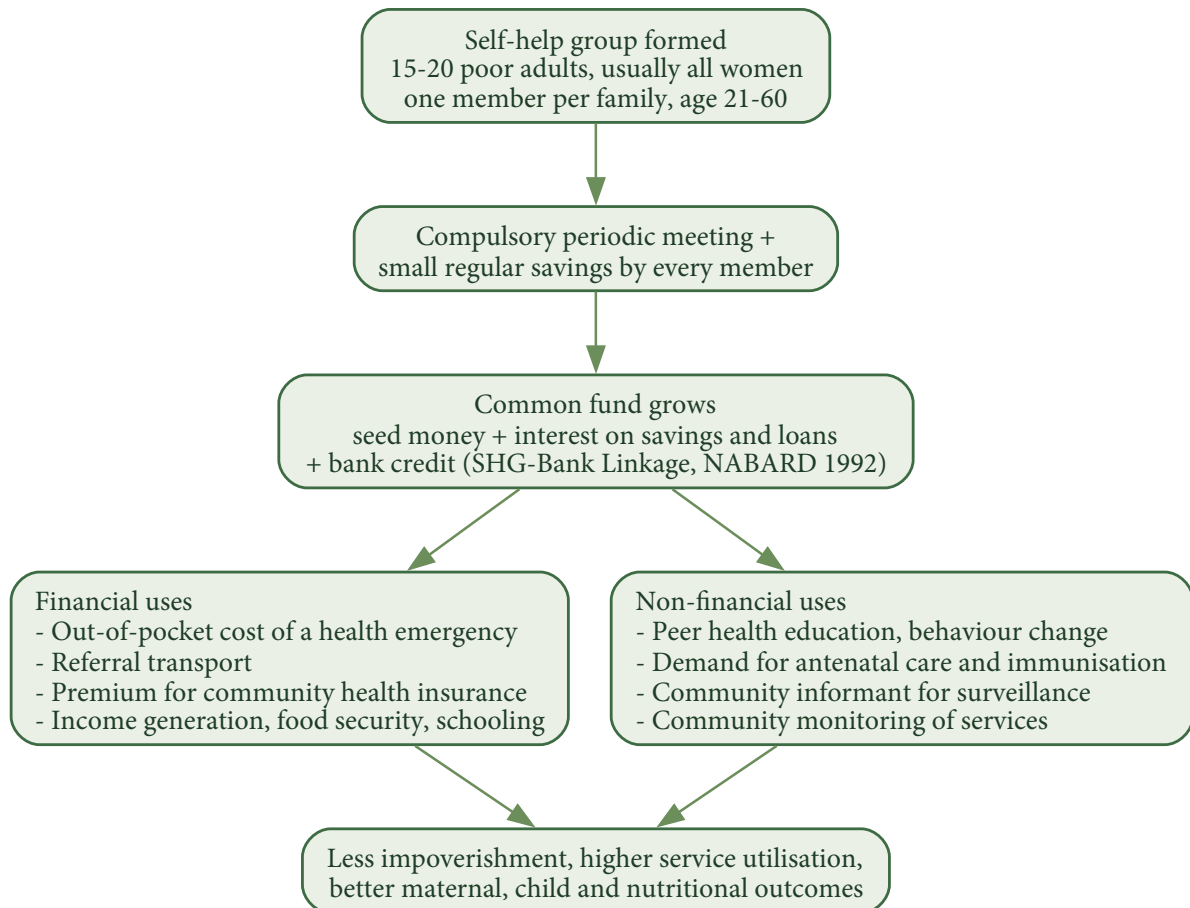
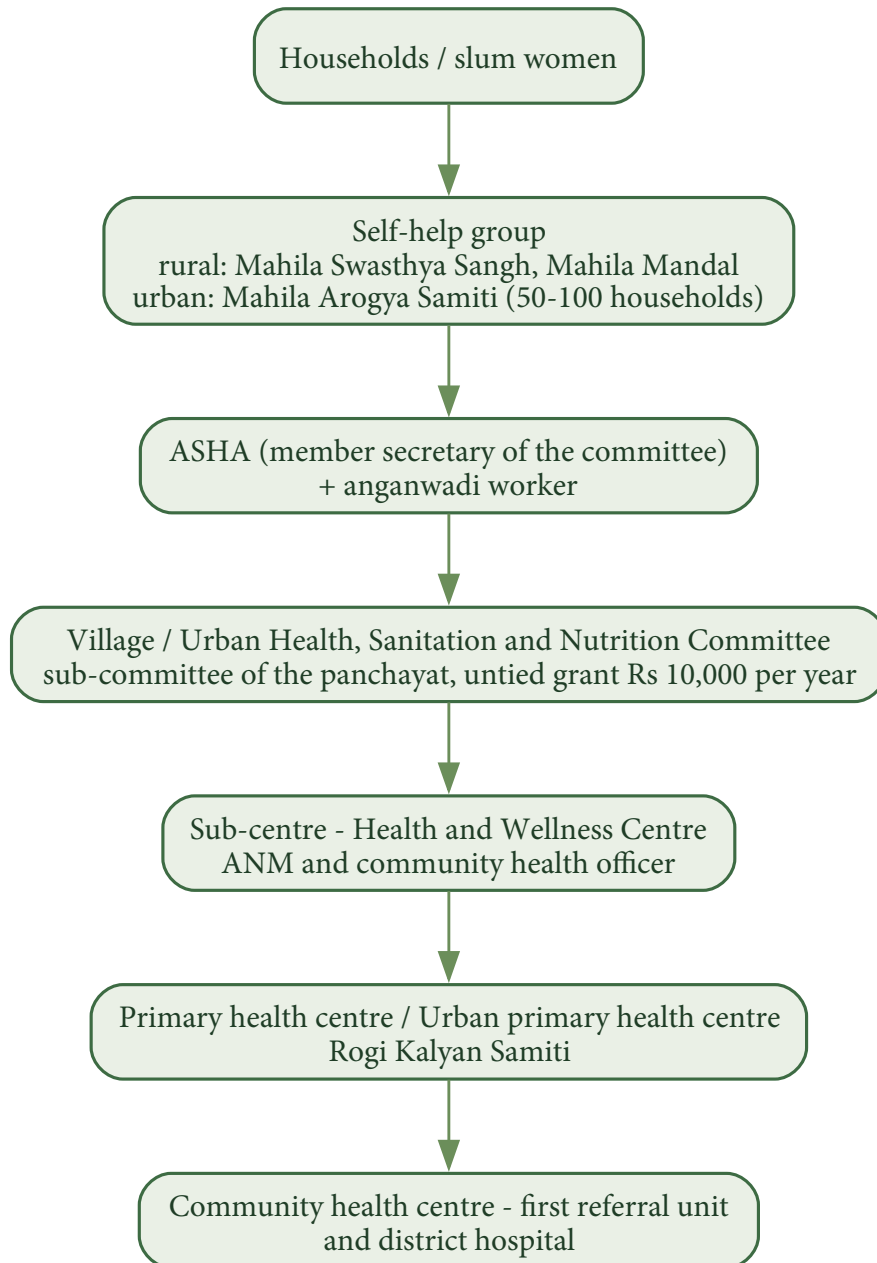


Fig 36.2 — Community risk pooling under NUHM (*Suryakantha Environment Best*, p.802)



How a self-help group converts collective thrift into better community health



Where the self-help group sits in the Indian health delivery chain

Constraints and how to strengthen self-help groups

- ▶ **Group meetings are seldom held or held badly** — workers are rarely trained to organise meaningful group meetings, the purpose is unclear and end results are obscure; the health manager must build worker capacity in group formation, group dynamics and group discussion (Sunder Lal, Ch 4, p 147).
- ▶ **Women's time is the limiting resource** — women are already busy, so only a meaningful, informative, interactive discussion of 30–60 minutes, at least monthly, will be attended (Sunder Lal, Ch 4, p 147).
- ▶ **Benefits do not reach the intended women** — 40% of benefits under SGSY were earmarked for women but in practice were not reaching them in that proportion (Sunder Lal, Ch 10, p 633).

- ▶ **Participation remains an aphorism** — community participation is still "awaiting genuine realisation" in many countries; no standard pattern can be prescribed because economic and social conditions vary widely (Park 28e, Ch 2, p 34).
- ▶ **Fragmentation in urban areas** — multiple agencies work in slums without coordination, so convergence with urban local bodies, education, ICDS, water and sanitation and housing is essential (Sunder Lal, Ch 17, pp 759–760).
- ▶ **Strengthening measures** — federate SHGs from panchayat to district level for voice and hand-holding support, ensure bank linkage for credit, provide skill and placement support (Jugal Kishore, p 693), give the group a defined statutory place (VHSNC membership, MAS) with an untied fund, and involve NGOs/community-based organisations as animators for social mobilisation and capacity building (Jugal Kishore, pp 692–693).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Community risk pooling through the Mahila Arogya Samiti** (redraw the plate above) — label: *Slum women* at the base; upward arrows marked *Saving* into a central box **Mahila Arogya Samiti (MAS)**; arrow in from above **Seed money and performance grant under NUHM**; arrows in from the right **Interest on savings** and **Interest on loans**; arrows out to the left **Premium financing for health insurance** and **Small loans**.
- ▶ **The self-help group as the base of the health delivery pyramid** — label from base upward: *Household* → *Self-help group / Mahila Arogya Samiti* → *ASHA and anganwadi worker* → *Village Health, Sanitation and Nutrition Committee* → *Sub-centre-Health and Wellness Centre* → *Primary health centre* → *Community health centre / first referral unit* → *District hospital*.

High-yield exam pointers

- ▶ SHG = 15–20 homogeneous poor members, age 21–60, one per family, single-sex, compulsory meeting, compulsory regular thrift.
- ▶ NABARD SHG–Bank Linkage Project, 1992 — the world's largest microfinance project; SGSY → NRLM (one woman per below-poverty-line household in an SHG, SHG federations up to district level, universal financial inclusion, skill placement).
- ▶ VHSNC: ≥15 members, chairperson *mahila panch*, **member secretary** ASHA, SHG volunteers as members, untied grant ₹10,000/year, >5.55 lakh committees (March 2021).
- ▶ MAS: 10–12 women, 50–100 urban slum households (population 250–500), ₹5,000 per year; NUHM seed money ₹25 per household; members pool ₹10–20 per family per month.
- ▶ 79,512 Mahila Swasthya Sangh and 10.72 lakh ASHAs at village level (Sunder Lal, Ch 17, p 744).
- ▶ Park's **three ways a community participates** — resources, planning/management/evaluation, and using the services; Sunder Lal's **six types** end in monitoring and evaluation.
- ▶ One-line clinchers: SHGs feature in NRLM, NUHM (MAS + Rogi Kalyan Samiti), National Population Policy 2000 revolving fund, Integrated Programme for Older Persons, and community-based rehabilitation in mental health.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the National Rural Livelihoods Mission and the urban livelihoods mission are now titled **Deendayal Antyodaya Yojana–NRLM and DAY-NULM**, and sub-centre/primary health centre Health and Wellness Centres have been renamed **Ayushman Arogya Mandirs** (Park 28e, Ch 7, p 541) — the retrieved corpus mostly uses the older names, and the membership size of the Mahila Arogya Samiti differs between the National Urban Health Mission framework document (5–20 members) and current textbook practice (10–12 members), so quote whichever you cite with its source. Coverage note: the indexed corpus has no chapter dedicated to self-help groups; this answer is assembled from the community-participation, National Urban Health Mission, poverty-alleviation and health-planning sections of the books listed below.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 Concept of Health and Disease; Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community; Ch 23 Health Planning and Management); Park Plus (Amrutha & Vijayalakshmi), Ch 21 Health Planning and Management; Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 2, Ch 4, Ch 7, Ch 10, Ch 16, Ch 17); Jugal Kishore — National Health Programmes of India (National Urban Health Mission and poverty-alleviation sections); Suryakantha — Community Medicine with Recent Advances (Ch 30 Mental Health; Ch 36 National Urban Health Mission); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 31 Family Planning and Population Policy; Ch 34 Mental Health); Solid Waste Management (Cochin source-segregation case study); National Population Policy 2000; National Urban Health Mission framework.



Role of voluntary health agencies in leprosy prevention and control

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — A voluntary health agency is "an organisation that is administered by an autonomous board which holds meetings, collects funds for its support chiefly from private sources, and expends money, whether with or without paid workers, in conducting a programme directed primarily to furthering the public health by providing health services, health education, or by advancing research or legislation for health, or by a combination of these activities" (Park 28e, Ch 10, p 746). Voluntary agencies have been compared to "**motor trucks**" which can penetrate the by-ways, and official agencies to "**railway trunk lines**" which must run on tracks established by law (Park 28e, Ch 10, p 746). In leprosy this distinction is decisive: the voluntary sector reached the disease in India eight decades before the State did, and it remains the formal partner of the national programme in exactly those areas the government machinery reaches least.

Why leprosy is the classical voluntary-sector disease in India

Feature of leprosy control	Consequence for the voluntary sector
Anti-leprosy work began in 1874, when the <i>Mission to Lepers</i> (now the Leprosy Mission) was founded by Baily at Chamba, Himachal Pradesh, headquarters later moving to Purulia, West Bengal; the official National Leprosy Control Programme was initiated only in the middle of 1954 (Park 28e, Ch 5, p 398)	Institutional care, custodial homes and case-holding were voluntary-sector creations that government later absorbed
About 150 voluntary organisations now work in the cause of leprosy in India; more than 7,000 voluntary agencies are involved in health-related activities overall (Park 28e, Ch 5, p 398; Sunder Lal, Ch 17, p 765)	A dense, disease-specific network already exists to be harnessed
Prevalence about 0.68 per 10,000 population; 65,147 new cases in 2020–21, of which child cases 9.49%, deformity grade II and above 4.14%, multibacillary 51.48% (Park 28e, Ch 10, p 723)	Elimination as a public-health problem achieved nationally, but residual transmission sits in pockets needing flexible, non-vertical outreach
India, Brazil and Indonesia together contributed 78.1% of new leprosy cases globally in 2022 (Park 28e, Ch 5, p 382)	India's residual burden is of global significance and attracts international voluntary funding
There are 612 self-settled leprosy colonies in the country where more than 50,000 leprosy-affected persons reside (Park 28e, Ch 7, p 503)	A stigmatised, geographically separate population that only paramedical workers and NGOs reach, weekly or fortnightly
Leprosy is an "iceberg" disease — the problem is one of discovering subclinical cases lying outside existing case-detection technology (Park 28e, Ch 5, p 389)	Demand generation and behaviour change, not drug supply, is the binding constraint — a voluntary-sector strength

Park's six functions of a voluntary health agency, applied to leprosy

Function (Park 28e, Ch 10, p 746)	How it is discharged in Indian leprosy work
1. Supplementing the work of government agencies — government cannot give complete service because it operates under financial and statutory restrictions; agencies help by lending personnel or contributing funds for special equipment, supplies or services	54 NGOs currently receive grant-in-aid from the Government of India under the Survey, Education and Treatment (SET) scheme (Park 28e, Ch 7, p 507)
2. Pioneering — exploring new ways of doing things, research being one form; when the effort bears fruit, government steps in and takes it over for larger numbers	Institutional leprosy care and ulcer care were pioneered by mission hospitals; the JALMA institute (Japanese Leprosy Mission for Asia) , Agra, was a voluntary body taken over by the Indian Council of Medical Research in 1975 and is today a tertiary-level referral institute of the programme (Park 28e, Ch 5, p 398; Ch 7, pp 503, 505)
3. Education — the scope for health education is unlimited and government alone cannot cope	Hind Kusht Nivaran Sangh conducts health education through publications and posters and publishes the quarterly <i>Leprosy in India</i> (Park 28e, Ch 10, p 747); NGOs deliver the intensive campaign themed " Towards Leprosy Free India " (Park 28e, Ch 7, p 503)
4. Demonstration — by putting up demonstrations and experimental projects, voluntary agencies have advanced the cause of public health; Park's own illustration is the bore-hole latrine demonstration by the Rockefeller Foundation against hookworm in India (Park 28e, Ch 10, p 746)	The leprosy parallel is the NGO-run reconstructive surgery unit, which demonstrated that correction of established deformity is feasible outside tertiary government hospitals (Park 28e, Ch 7, p 505)
5. Guarding the work of government agencies — by setting a good example and by criticising, voluntary agencies can guide the official agencies	Exercised formally through ILEP, which provides planning, monitoring and supervision of the programme and capacity building of general health-care staff (Park 28e, Ch 7, p 507)
6. Development of public opinion and advancing health legislation for the benefit of the whole community	Sustained advocacy against stigma, and for the repealing and amendment of all existing discriminatory laws on leprosy — an explicit Pillar-4 action of the current national plan (Park 28e, Ch 7, p 507)

Major voluntary agencies working on leprosy in India

Agency	Founded / headquarters	Contribution
The Leprosy Mission (formerly Mission to Lepers)	1874, Chamba → Purulia	Oldest anti-leprosy agency in India; an ILEP member agency (Park 28e, Ch 5, p 398; Ch 7, p 507)

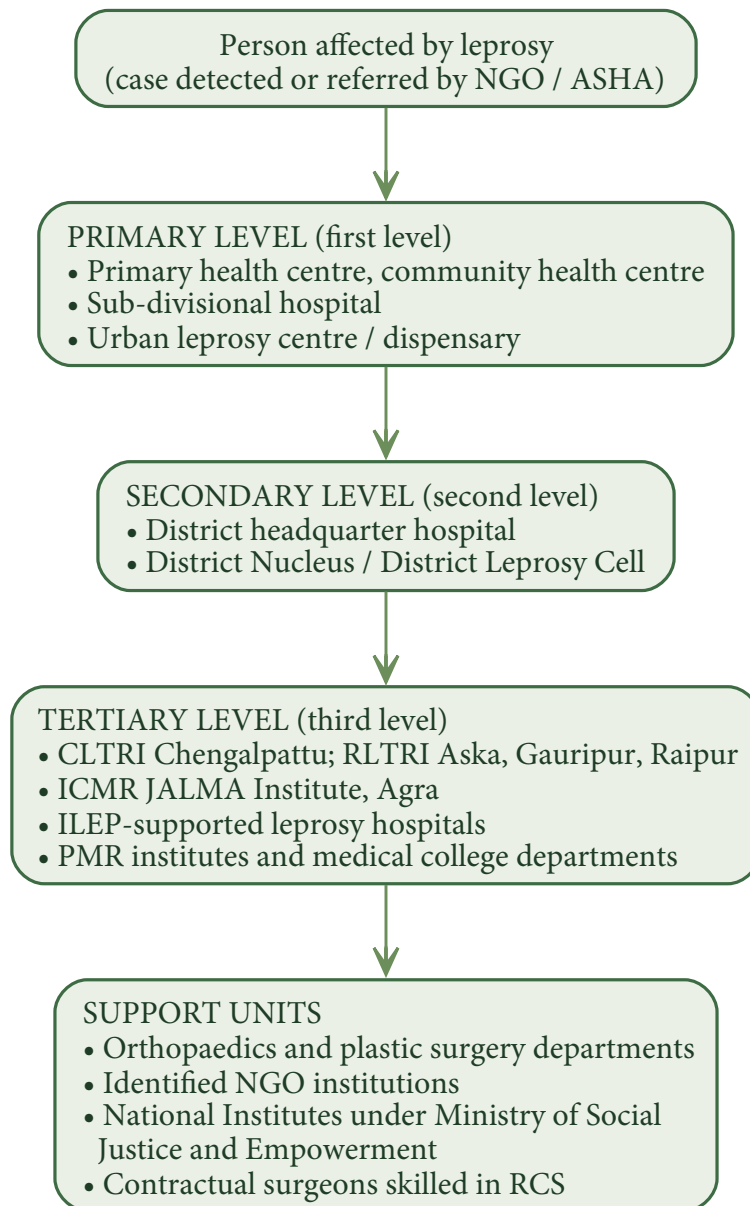
Agency	Founded / headquarters	Contribution
Hind Kusht Nivaran Sangh — precursor the Indian Council of the British Empire Leprosy Relief Association (BELRA), renamed LEPRO in 1950	1950, New Delhi	Financial assistance to leprosy homes and clinics; health education; training of medical workers and physiotherapists ; research and field investigations; All-India Leprosy Workers Conferences; publishes <i>Leprosy in India</i> ; branches all over India, works in close cooperation with government (Park 28e, Ch 10, p 747)
Gandhi Memorial Leprosy Foundation , Sevagram, Wardha; German Leprosy Relief Association ; Damien Foundation ; Danish Save the Child Fund	—	Named by Park among the important voluntary organisations that sprang up in the cause of leprosy in India (Park 28e, Ch 5, p 398)
National Leprosy Organization	1965 — a federation body	A common platform for voluntary agencies to discuss their problems and share their experiences (Park 28e, Ch 5, p 398)
International Federation of Anti-Leprosy Associations (ILEP) — 10 agencies in India: The Leprosy Mission, Damien Foundation of India Trust, Netherland Leprosy Relief, German Leprosy Relief Association, Lepra India, ALES, AIFO, Fontilles-India, AERF-India, American Leprosy Mission	Federation; active partner in NLEP	Planning, monitoring and supervision of the programme; capacity building of general health-care staff; IEC; reconstructive surgery services; socio-economic rehabilitation; supports 36 NGOs conducting reconstructive surgeries for disability correction (Park 28e, Ch 7, p 507)

Role within the National Leprosy Eradication Programme (NLEP)

The National Leprosy Control Programme was renamed the **National Leprosy Eradication Programme (NLEP)** in 1983 and implemented as a Centrally Sponsored Scheme (Park 28e, Ch 7, p 502). The voluntary sector enters it through defined channels:

- **SET (Survey, Education and Treatment) scheme** — the formal grant-in-aid route. Under it NGOs are involved in **disability prevention and ulcer care**, IEC, **referral of suspected cases**, **referral for reconstructive surgery (RCS)**, **research and rehabilitation**. NGO support is mainly required for **follow-up of cases under treatment in urban locations and difficult-to-reach areas** (Park 28e, Ch 7, p 505).
- **Geographical and social niches** — NGOs serve **remote and inaccessible areas, urban slums, industrial and labour populations and other marginalised population groups**; IEC, prevention of disability, case detection and referral, and follow-up for treatment completion are the important activities taken up by them (Park 28e, Ch 7, p 507).
- **Reconstructive surgery** — as on **January 2017 there were 115 recognised RCS centres, 61 Government and 54 NGO**. An incentive of Rs 12,000 is paid to every patient undergoing major reconstructive surgery, irrespective of financial status, by the district leprosy officer (Park 28e, Ch 7, p 505).

- ▶ **Self-settled colonies** — free care of ulcers, self-care training, counselling and **microcellular rubber (MCR) footwear** are provided to leprosy-affected persons residing in these colonies through paramedical workers and NGOs on a weekly or fortnightly basis (Park 28e, Ch 7, p 503).
- ▶ **Disability prevention and medical rehabilitation (DPMR)** — ILEP-supported leprosy hospitals are named tertiary-level care institutions and **identified NGO institutions** are named support units of the DPMR referral system (Park 28e, Ch 7, p 503).
- ▶ **Stigma reduction** — NGOs partner the **Sparsh leprosy awareness campaign**, launched in 2017 through Gram Sabhas and carried out with the help of Panchayat and Village Health and Sanitation Committee, covering 60% of villages across India, and the observance of **30 January as Anti-Leprosy Day** (Park 28e, Ch 7, p 505; Sunder Lal, Ch 8, p 371).



Three-tier DPMR referral pathway under NLEP, showing where voluntary agencies are placed (Park 28e, Ch 7, p 503)

Voluntary-sector contribution mapped to levels of prevention

HEALTH PROMOTION

- Sparsh leprosy awareness campaign via Gram Sabha
- Anti-Leprosy Day observance
- Advocacy to repeal discriminatory leprosy laws

PRIMARY PREVENTION

- Household contact survey and contact tracing
- Support to post-exposure prophylaxis of eligible contacts of the index case
- IEC to promote early self-reporting

SECONDARY PREVENTION

- Referral of suspected cases; skin camps
- Multidrug therapy delivery and case-holding
- Follow-up for treatment completion in urban and difficult-to-reach areas

TERTIARY PREVENTION (DPMR)

- Reaction and ulcer care, physiotherapy
- MCR footwear, aids, appliances, self-care kits
- Reconstructive surgery at NGO RCS centres
- Socio-economic rehabilitation

Voluntary health agency inputs across the levels of prevention in leprosy

Where the voluntary sector sits in the National Strategic Plan and Roadmap for Leprosy 2023–2027

Strategic pillar	Voluntary-sector contribution
Pillar 1 — Leadership and commitment	Explicit action: "strengthen partnership with existing partners and add new partners and donors" (Park 28e, Ch 7, p 506)
Pillar 2 — Accelerate case detection	Behaviour-change communication plan; NGO-run skin camps and referral of suspects (Park 28e, Ch 7, p 506)
Pillar 3 — Provide quality services	Coordination with medical colleges, private practitioners, NGO and private clinics/hospitals ; orientation of non-government stakeholders to follow NLEP guidelines (Park 28e, Ch 7, p 507)
Pillar 4 — Prevention of disease, disabilities, stigma, discrimination and violation of human rights	Counselling for self-care, monitoring availability and quality of assistive devices, post-treatment surveillance for 5 years, repealing and amendment of discriminatory laws (Park 28e, Ch 7, p 507)

Strategic pillar	Voluntary-sector contribution
Pillar 5 — Develop digital surveillance systems (Nikusth 2.0, integration of reporting on the Integrated Health Information Platform)	NGO case data fed into the national digital reporting stream (Park 28e, Ch 7, p 507)

The plan aims at interruption of transmission at district level, evidenced by **zero occurrence of new child cases for at least five consecutive years**; districts that achieve this then move on to elimination of leprosy as a disease, with **zero new cases reported for at least three consecutive years** (Park 28e, Ch 7, p 506).

Strengths and limitations of the voluntary sector

Strengths (Sunder Lal, Ch 17, pp 765–766)	Limitations
Comparative advantage of flexibility in procedures ; can penetrate the "by-ways" the statutory machinery cannot	Uneven distribution — a wide inter-state differential exists in the coverage of villages by NGOs, which have limited area of operation
Rapport building with communities ; at the cutting edge of programme implementation	Dependence on grant-in-aid and donor funds; sustainability risk as international leprosy funding contracts
Work in difficult, underserved and unserved areas with voluntary zeal, on a non-profit basis	Variable technical quality; needs orientation to NLEP guidelines (an explicit Pillar-3 action)
Assigned a supplementary or complementary role to government health-care delivery, aiding it in reaching the masses meaningfully	Risk of parallel vertical services if not integrated into general health care and district reporting

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **"Motor truck versus railway trunk line" schematic** (Park's analogy) — draw two arrows from a common *Community need in leprosy* box: a straight rail line labelled *Official agency — NLEP, running on tracks established by law, uniform coverage* and a branching road labelled *Voluntary agency — flexible, reaches the by-ways: self-settled colonies, urban slums, tribal and inaccessible areas*. Label the meeting point *SET scheme grant-in-aid / ILEP partnership*.
- **NGO–NLEP organogram** — boxes labelled: Central Leprosy Division (Directorate General of Health Services, Ministry of Health and Family Welfare) → District Leprosy Cell / District Nucleus → (branch 1) Primary Health Centre and Health and Wellness Centre, ASHA, ANM; (branch 2) SET-scheme NGO, ILEP-supported leprosy hospital, NGO RCS centre; both branches converge on *Person affected by leprosy*.

High-yield exam pointers

- **Definition to quote:** autonomous board, funds chiefly from private sources, programme of health services / health education / research or legislation for health.

- **Six functions (mnemonic S-P-E-D-G-A):** Supplementing, Pioneering, Education, Demonstration, Guarding the work of government agencies, Advancing public opinion and health legislation.
- **Dates:** Mission to Lepers 1874 (Baily, Chamba); Hind Kusht Nivaran Sangh 1950; National Leprosy Control Programme 1954; National Leprosy Organization 1965; JALMA to ICMR 1975; NLEP 1983; Sparsh campaign 2017.
- **Numbers:** ~150 leprosy NGOs; 54 NGOs on SET grant-in-aid; 115 RCS centres (61 government + 54 NGO, January 2017); 36 NGOs doing RCS under ILEP; 10 ILEP agencies in India; 612 self-settled colonies housing >50,000 affected persons; RCS incentive Rs 12,000.
- **Indian burden, dated:** prevalence 0.68 per 10,000; 65,147 new cases in 2020–21 (child 9.49%, deformity grade II and above 4.14%, multibacillary 51.48%).
- **Current plan:** National Strategic Plan and Roadmap for Leprosy 2023–2027 — five pillars, partnership named in Pillar 1; goal of zero new child cases for 5 years, then zero new cases for 3 years.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam. Park 28e itself carries two counts of reconstructive-surgery capacity — Ch 7 p 503 states that 41 NGOs and 42 Government Medical Colleges have been strengthened for reconstructive surgery, "totalling to 83 centres", while Ch 7 p 505 gives 115 recognised RCS centres (61 Government and 54 NGO) as on January 2017; either may be quoted provided the page is consistent. Park also gives the start of the National Leprosy Control Programme as the middle of 1954 in Ch 5 but as 1955 in Ch 7. The National Leprosy Eradication Programme now sits under the National Health Mission, and the state leprosy societies have been merged with the state health society under that mission. The programme figures above are the latest carried in the textbook; the annual National Leprosy Eradication Programme report updates them each year, so quote the year alongside every figure.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed (full text checked where available), WHO leprosy fact sheet, WHO Global Leprosy Strategy 2021–2030, India National Strategic Plan and Roadmap for Leprosy 2023–2027

Programme and policy updates (India)

- The National Strategic Plan for Leprosy 2023–2027 is now stated in its "**triple zero**" form — **zero transmission, zero disability, zero discrimination** [PMID 41878182] — and its goal is interruption of transmission by 2027, **three years ahead of Sustainable Development Goal 3.3**, the plan being explicitly aligned to the WHO global strategy 2021–2030 [PMID 38329161]. India's announcement of a national plan for zero transmission by 2027 is the headline commitment, made against surviving pockets of high prevalence and unknown transmission chains [PMID 41054789]. Because incubation is long, the transmission outcome will only be readable in 2037, and disability prevention, management and after-care will need attention for at least 20 years, to 2047 or beyond — the argument for keeping NGO rehabilitation capacity alive well past the target year [PMID 38329161].

- NLEP has implemented contact tracing with post-exposure prophylaxis since 2018, and its Leprosy Case Detection Campaign was initiated in 2017 — house-to-house surveys with ASHA-based surveillance, both new since the textbook's account of the SET scheme [PMID 40605889]. WHO guidelines now include screening the close contacts of a new patient together with **single-dose rifampicin PEP (SDR-PEP)**, which is the technical basis for NGO contact work; "close contacts" in trial usage means household members, family, neighbours and other social contacts [PMID 38378497]. Indian policy thinking pairs this chemoprophylaxis with **immunoprophylaxis using the indigenous MIP (Mw) vaccine**, which has demonstrable protective efficacy at five years and even up to 8–10 years, works in a cost-effective manner, and also protects against tuberculosis [PMID 38329161].

Current national data

- Quote the 2024–25 NLEP indicators in place of the older figures: prevalence **0.57 per 10,000**, annual new case detection rate **7.00 per 100,000**, and grade-2 disability rate **1.31 per million** [PMID 41878182].
- India recorded **107,851 new cases in 2022–23** and continues to account for the highest number of newly reported cases globally [PMID 42215552]; in 2022 India reported over **103,000** new cases against Brazil's **19,000** and Indonesia's **12,000**, while **150 countries reported fewer than 1,000 each** [PMID 39202679]. WHO reported **182,815 new cases globally in 2023**, **131,425** of them from South-East Asia, with Brazil, India and Indonesia together **78.1%**; of **7,218 new child cases**, **almost 77% (5,586) were Indian** — the sharpest available marker of recent transmission [PMID 39632641].

Elimination, surveillance and disease status

- **Grade-2 disability at diagnosis is the sentinel indicator of programme performance, and it is exactly where the voluntary sector is being asked to work.** A post-campaign study of 68 new cases in Visakhapatnam district found disability in **14%** (6% grade 1, 9% grade 2, against India's 2018–19 average of 7.7%), a median **eight-month** care-seeking delay, **10% child cases** and ulnar nerve involvement in **62%** [PMID 40605889].
- A Telangana family cluster of three cases was missed across **three consecutive rounds of the case detection campaign**, giving a four-year delay from symptom onset to multidrug therapy; the patient declined home visits for fear of stigma and disclosure. Psychosocial support and livelihood rehabilitation were facilitated with **The Lepra Society and local NGO partners**, and the report notes that self-help groups of affected persons can stay functional even after donor support is withdrawn. The recommended fixes are district-level grade-2 disability case audits and independent quality-assurance review of the detection campaigns [PMID 41878182].
- Antimicrobial resistance has become a surveillance priority: mutations conferring rifampicin (*rpoB*), dapsone (*folP1*) and ofloxacin (*gyrA*) resistance are detected in **up to 8% of cases globally**; around **two million people live with leprosy-related disability**; and premature complacency after the prevalence target has cost funding, political will and research investment. COVID-19 disruption alone cut case detection in the Americas by **35.9%** — 29,936 in 2019 to 19,195 in 2020 [PMID 41408601].

Guideline updates

- ▶ WHO recommends a **uniform MDT — the same rifampicin, clofazimine and dapsone combination for both types**, six months for paucibacillary and 12 months for multibacillary disease; the addition of clofazimine to the paucibacillary regimen initially emerged from **ICMR-JALMA, Agra**, itself a voluntary-sector institute taken over by ICMR [PMID 38329161] [WHO 2024].
- ▶ **PEOPLE (2024)** — cluster randomised across **16 villages in Madagascar and 48 in Comoros** with 109,436 people enrolled: single double-dose rifampicin (20 mg/kg) offered to anyone living within 100 m reduced leprosy incidence once baseline prevalence was controlled for (IRR 0.56, $p=0.0030$), individual-level protection was IRR 0.55 ($p=0.0050$), and risk was two to four times higher for those living within 75 m of an index patient — the evidence base for geographically targeted blanket prophylaxis rather than household-only PEP. Its author list includes an investigator from **Damien Foundation**, Brussels [PMID 38762282] — a name that also appears in Park's roll of leprosy voluntary agencies.

Global and WHO developments

- ▶ The **WHO Global Leprosy Strategy 2021–2030** calls for accelerated action towards **zero leprosy — zero disease, zero disability, zero stigma and discrimination** — and is part of the neglected tropical diseases road map 2021–2030, alongside the **Leprosy Elimination Framework**, a blueprint for countries to move beyond control to interruption of transmission [PMID 39632641]; India's national plan is aligned to it [PMID 38329161].
- ▶ The **Global Alliance to Eliminate Leprosy (GAEL, 1999)** brought WHO together with the Nippon Foundation, ILEP and the Novartis Foundation and secured free drug distribution and political commitment in underserved areas; the partnership was then undermined by disagreement over programme priorities, and a **2003 independent evaluation criticised WHO for sidelining its partners' concerns** — WHO's population-level strategies against the agencies' person-centred approach. A live illustration of Park's "guarding the work of government agencies" function [PMID 41408601].
- ▶ **Voluntary agencies are now the research engine of leprosy prevention.** The **PEP++ cluster-randomised trial** — the idea developed by NLR in 2016, implemented by NLR with NLR India, NHR Brasil, NLR Nepal and The Leprosy Mission International (TLMI) Bangladesh across India, Brazil, Bangladesh and Nepal — tests three doses of rifampicin plus clarithromycin at four-weekly intervals against SDR-PEP, with a target enrolment of **202,360 close contacts** and COLEP's 57% two-year protection as the benchmark [PMID 38378497]. **GLRA India** is running a storytelling-based stigma-reduction intervention among people affected by leprosy in **Sitapur, Uttar Pradesh** [PMID 42215552]. A 12-RCT systematic review and meta-analysis puts prophylactic leprosy vaccines at a pooled RR **0.61 (95% CI 0.41–0.91)** and therapeutic vaccination at a Ramu's score change of **–3.06 (95% CI –3.96 to –2.16)** [PMID 39632641].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases; Ch 7 — Health Programmes in India; Ch 10 — Health Care of the Community); Sunder Lal — Textbook of Community Medicine (Ch 8 — Epidemiology of Communicable and Non-Communicable Diseases and Related Disease-Specific National Health Programmes; Ch 17 — Healthcare of the Community); Government of India, National Strategic Plan and Roadmap for Leprosy 2023–2027, Central Leprosy Division, Ministry of Health and Family Welfare (January 2023), as reproduced in Park 28e.

Recent-advances sources: All items below are cited from PubMed. Full text was read via PubMed Central for every source except the two marked abstract-only, which have no PMC deposit and whose cited figures are taken from the PubMed abstract and author-affiliation record. 41054789 — Leprosy in India: challenges, advances and the path to eradication, *Trans R Soc Trop Med Hyg* 2026, doi:[10.1093/trstmh/traf107](https://doi.org/10.1093/trstmh/traf107) (abstract-only); 38329161 — Eradication of leprosy from India: reflections on past, present & future, *Indian J Med Res* 2024, doi:[10.4103/ijmr.ijmr.64.24](https://doi.org/10.4103/ijmr.ijmr.64.24) (PMC10954108); 41878182 — Grade 2 disability in leprosy in the post-elimination era, *Cureus* 2026, doi:[10.7759/cureus.104011](https://doi.org/10.7759/cureus.104011) (PMC13006927); 40605889 — Disabilities among newly reported leprosy cases in Visakhapatnam District, India, *Cureus* 2025, doi:[10.7759/cureus.85207](https://doi.org/10.7759/cureus.85207) (PMC12220206); 42215552 — Exploring generational differences in the impact of stigma on mental health among people affected by leprosy in rural India, *Sci Rep* 2026, doi:[10.1038/s41598-026-55020-y](https://doi.org/10.1038/s41598-026-55020-y) (PMC13454424); 39632641 — Efficacy of leprosy vaccines across the globe: systematic review and meta-analysis of randomized controlled trials, *Indian J Med Res* 2024, doi:[10.25259/IJMR.1289.2024](https://doi.org/10.25259/IJMR.1289.2024) (PMC11619031); 39202679 — Exploring active case detection approaches for leprosy diagnosis in varied endemic settings: a comprehensive scoping review, *Life (Basel)* 2024, doi:[10.3390/life14080937](https://doi.org/10.3390/life14080937) (PMC11355679); 41408601 — Leprosy elimination: are we winning the fight or losing focus?, *Ann Med* 2025, doi:[10.1080/07853890.2025.2604389](https://doi.org/10.1080/07853890.2025.2604389) (PMC12716464); 38762282 — Post-exposure prophylaxis in leprosy (PEOPLE): a cluster randomised trial, *Lancet Glob Health* 2024, doi:[10.1016/S2214-109X\(24\)00062-7](https://doi.org/10.1016/S2214-109X(24)00062-7) (abstract-only); 38378497 — The PEP++ study protocol: a cluster-randomised controlled trial on the effectiveness of an enhanced regimen of post-exposure prophylaxis, *BMC Infect Dis* 2024, doi:[10.1186/s12879-024-09125-2](https://doi.org/10.1186/s12879-024-09125-2) (PMC10877766). Plus WHO leprosy fact sheet (2024 data) and WHO Global Leprosy Strategy 2021–2030.



Rural Health Scheme 1977

Asked in: Paper IV May 2019 — RGUHS MD Community Medicine

Definition — The Rural Health Scheme is the community-participation reform of rural health care launched by the Government of India in 1977 on the principle of "placing people's health in people's hands", acting on the recommendations of the Shrivastav Committee of 1975 (Park 28e, Ch 10, p 726). It created a non-official, village-based health functionary — the **Community Health Worker**, later renamed **Village Health Guide (VHG)** — supported by the **trained dai**, and linked them to the existing sub-centre–primary health centre–community health centre chain (Mahajan & Gupta 4e, Ch 28, pp 552-553). It was a 100 per cent centrally sponsored scheme whose stated objective was to deliver primary health care at the doorstep of the rural population (Mahajan & Gupta 4e, Ch 31, p 640).

Genesis — the Shrivastav Committee and the 1977 launch

The **Shrivastav Committee** was constituted in 1974 as the "*Group on Medical Education and Support Manpower*" under Dr J.B. Shrivastav, Director General of Health Services, and reported in April 1975 (Suryakantha, Ch 39, p 811). Its two terms of reference were to reorient medical education to national needs and priorities, and to devise a curriculum for **health assistants** who would link medical officers with multipurpose health workers (Mahajan & Gupta 4e, Ch 26, pp 501-502).

#	Recommendation (Shrivastav Committee, 1975)	What it became
1	A cadre of semi-professional village-level health workers recruited from within the community itself (school teachers, postmasters, <i>gram sevaks</i> , social workers), to meet the community's primary health needs and act as the link between community and primary health centre (PHC)	Community Health Worker Scheme , launched 2 October 1977; later the Village Health Guide
2	Multipurpose health workers to be the link between the village worker and the PHC; the work of 2 male and 2 female health workers to be supervised by one male and one female health assistant , posted at the sub-centre and not at the PHC	Sub-centre staffing and supervisory pattern of the scheme
3	Develop a referral system from the PHC upward to hospitals at tehsil, district and regional level and to medical colleges	The tiered referral chain of the scheme
4	A medical and health education commission on the lines of the University Grants Commission	Not implemented; revived later by the Bajaj Committee (1986)

Acceptance of these recommendations in 1977 led directly to the launching of the Rural Health Scheme (Mahajan & Gupta 4e, Ch 26, pp 501-502). The scheme thus **preceded** the Alma-Ata Declaration of 1978, which followed close on the heels of these recommendations and set the goal of Health for All through the primary health care approach (Park 28e, Ch 10, p 726).

Components and organisational tiers

Component	Year	Functionary created	Coverage norm
Village Health Guide (originally Community Health Worker) scheme	2 October 1977	Village Health Guide	1 per village or per 1000 population; 500 in tribal, hilly or remote areas; 2 guides if village population exceeds 2000
Training of local <i>dais</i> (traditional birth attendants)	1978	Trained <i>dai</i>	1 per village or per 1000 population
Reorientation of Medical Education (ROME)	1977-78	Medical college faculty and students	3 PHCs attached to each medical college

Sources: Mahajan & Gupta 4e, Ch 28, p 553; Suryakantha, Ch 35, pp 782, 784.

The scheme rested on a **four-tier** service structure — village, sub-centre, PHC and community health centre (CHC) (Mahajan & Gupta 4e, Ch 28, pp 552-553):

Tier	Staffed by	Population norm
Village	Village Health Guide, trained <i>dai</i> , anganwadi worker	1000
Sub-centre	Health worker (male) + health worker (female)/auxiliary nurse midwife (ANM)	5000 in plains; 3000 in hilly/tribal/desert areas
Primary health centre	Medical officer and staff	20,000–30,000
Community health centre	Specialists	80,000–1,20,000

Source: Mahajan & Gupta 4e, Ch 28, p 553; sub-centre norms unchanged in current policy (Park 28e, Ch 10, p 731).

The Village Health Guide

Aspect	Provision
Eligibility	Permanent resident of the village; acceptable to all sections of the community; formal education up to at least 6th standard (relaxable if otherwise suitable); able and willing to spare 2–3 hours daily for community health work; preference to practising <i>dais</i>
Selection	By the Gram Panchayat ; reports to a Village Health Committee on all matters except technical ones, which go to the medical officer
Representation	Where two guides are appointed, one must be a woman and one from a Scheduled Caste/Tribe; by policy decision of 1986, male guides were to be replaced by female guides, a woman being more accessible and acceptable to the community
Training	3 months at the PHC, with a stipend of Rs 200 per month
Training content	Basic health concepts, treatment of minor ailments, first aid, environmental sanitation, family planning, personal hygiene and principles of health education

Aspect	Provision
On completion	A certificate, a manual (what to do and what not to do) and a kit of common medicines, modern and indigenous
Remuneration	Honorarium of Rs 50 per month; medicine stock replenished, annual supply worth Rs 600
Limits of practice	Cannot administer injections; cannot prescribe outside the supplied drug list; must not treat any case for more than 2 days without improvement

Source: Mahajan & Gupta 4e, Ch 28, p 553; training stipend and 1986 circular corroborated by Suryakantha, Ch 35, p 782.

Functions. She continued her own vocation and attended to health problems in her spare time, serving as the community's interface with the formal health services (Sathe & Sathe, Ch 19, p 212):

- ▶ Health education and motivating the community.
- ▶ Assisting immunisation and communicable disease control.
- ▶ Maternal and child health and family planning — informing eligible couples and supplying Nirodh (condoms) and oral pills (Park 28e, Ch 8, p 612).
- ▶ Creating awareness on environmental sanitation and personal hygiene.
- ▶ First aid and treatment of minor illnesses; referral of the rest.

The trained dai component

Almost all village deliveries were conducted at home, unattended by a doctor or trained functionary — the reason for the high maternal mortality and neonatal tetanus mortality that this component sought to reduce (Mahajan & Gupta 4e, Ch 28, pp 553-554).

Aspect	Provision
Target	One local <i>dai</i> trained per village or per 1000 population
Duration	1 month, with a stipend of Rs 300
Pattern	2 days a week at the PHC/sub-centre; 4 days a week in the field, accompanying the health worker (female)
Compulsory practice	At least 2 deliveries conducted under the supervision of the health worker (female)
Content	Reproductive anatomy, physiology of pregnancy and labour, recognition of danger signals, and the five cleans — clean hands, clean surface, clean blade, clean ligature, clean cord stump
On completion	A <i>dai's</i> (midwifery) kit; an incentive per delivery payable only if the case was registered at the sub-centre or PHC — a device to improve birth registration; she remained free to charge the community her own fee

Source: Mahajan & Gupta 4e, Ch 28, pp 553-554; five cleans from Suryakantha, Ch 35, p 782.

Reorientation of Medical Education (ROME)

The companion measure flowing from the same committee, introduced in 1977-78, with four objectives (Jugal Kishore, Ch 1, p 80):

- ▶ Introduce a **community bias** into undergraduate medical training, emphasising preventive and promotive services.
- ▶ Reorient **medical colleges** to become an integral part of the health care system instead of functioning in isolation.
- ▶ Reorient **faculty**, so that hospital-based, disease-oriented teaching was complemented by community-based, health-oriented teaching.
- ▶ Develop effective **referral linkages** between PHCs, district hospitals and medical colleges.

Three PHCs were attached to each medical college, with dormitories for house surgeons (Suryakantha, Ch 35, p 784). The first phase covered about **106 medical colleges** at a one-time grant-in-aid of about **Rs 16 lakh** each, but the objectives were not achieved to the desired extent (Jugal Kishore, Ch 1, p 80).

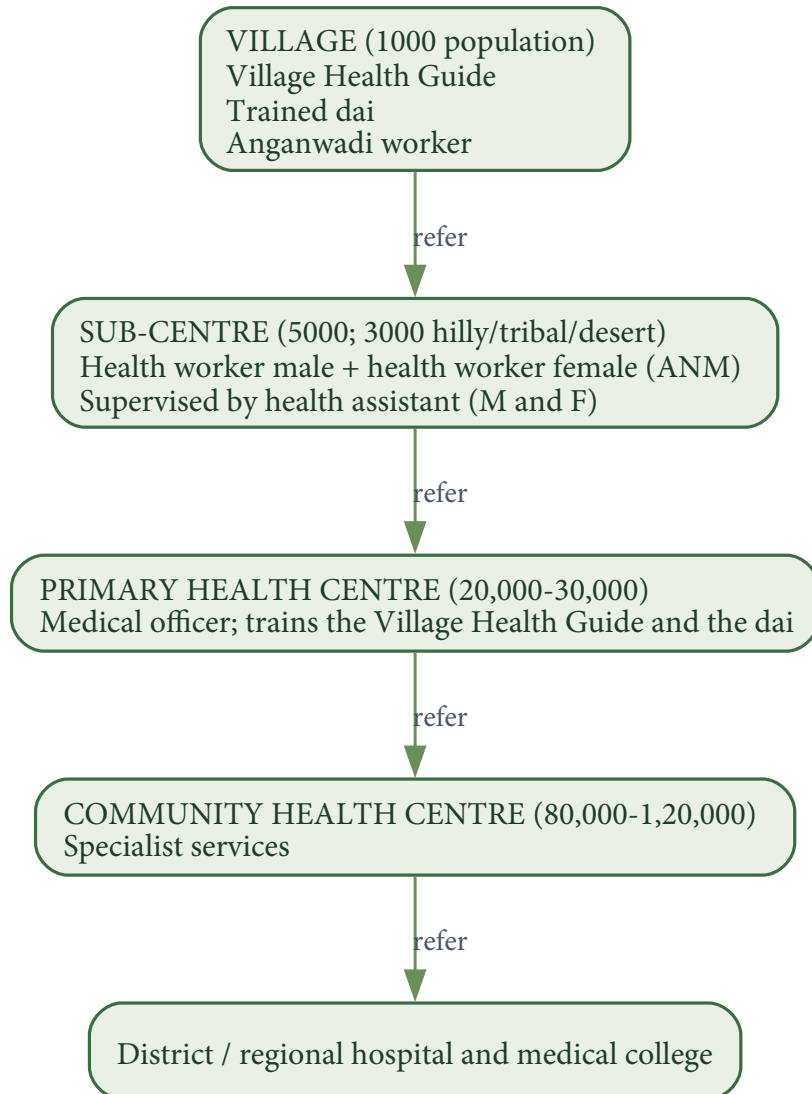
Evaluation and closure of the scheme

Domain	Finding
Nomenclature drift	Renamed Community Health Volunteer in 1979 to stress the voluntary character and remind workers they were not government servants; renamed Village Health Guide in 1981 (Sathe & Sathe, Ch 19, p 212)
Uneven uptake	State response was highly variable and central financial support was repeatedly changed; four states ran alternative schemes instead — Jammu & Kashmir (<i>Rehbar-e-Sehat</i>), Arunachal Pradesh (<i>Medics</i>), Tamil Nadu (<i>Mini Health Centres</i>) and Kerala (strengthening of PHCs) (Mahajan & Gupta 4e, Ch 28, p 553)
Worker grievances	Drugs not supplied; the Rs 50 honorarium too low and unpaid in some states; demands for identity cards, compassionate appointment and Class-IV government servant facilities (Sathe & Sathe, Ch 19, pp 212-213)
Performance	Successive evaluations gave "very disappointing results"; performance of health guides was unsatisfactory, and volunteers tended to get professionalised — drifting away from the community role they were created for (Sathe & Sathe, Ch 19, pp 212-213)
Professional resistance	The greatest resistance to the health guide's involvement in primary health care came from the medical profession rather than from the lay public (Park 28e, Ch 2, p 34)
Closure	The Umashankar Committee (July 1997) reviewed the scheme with the states and the Village Health Guide Association, and recommended discontinuation as it was not serving its purpose; the centrally sponsored scheme was discontinued from the Tenth Five Year Plan, i.e. 31 March 2003 (Sathe & Sathe, Ch 19, pp 212-213)

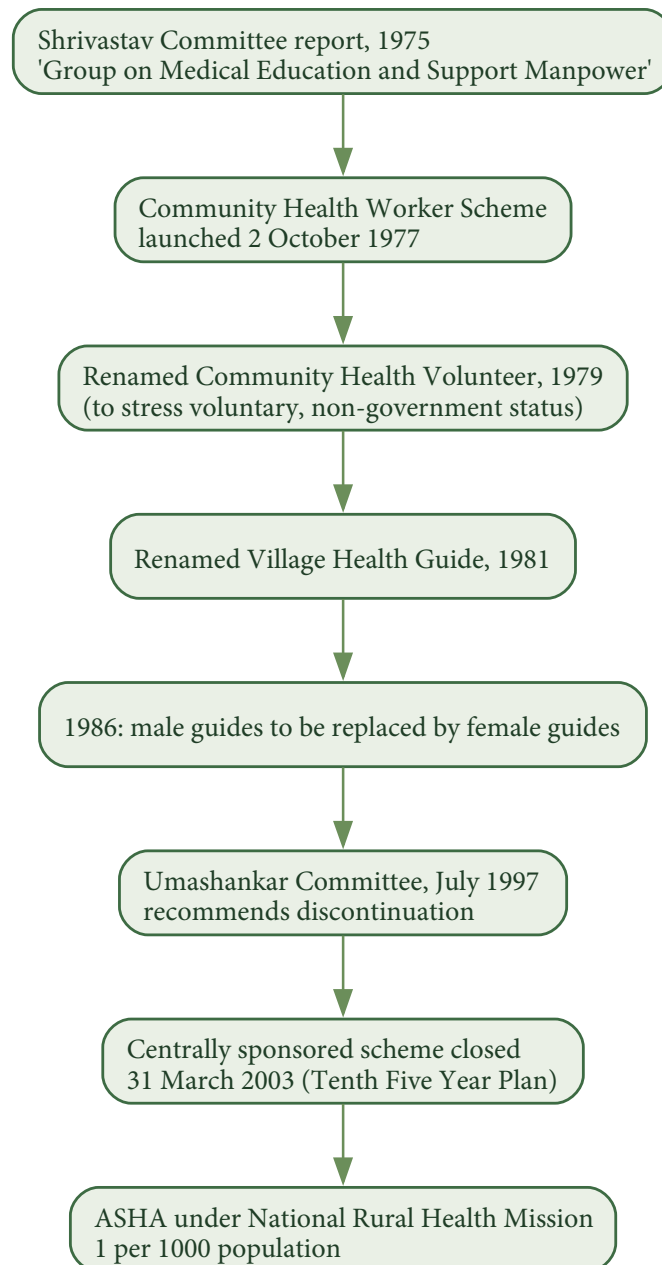
Legacy. The concept was revived, corrected and scaled as the **Accredited Social Health Activist (ASHA)** under the National Rural Health Mission. ASHA must be a resident of the village, a woman (married, widowed or divorced) preferably aged **25–45 years**, with formal education up to **eighth class**, one per 1000 population, relaxable to one per habitation in tribal, hilly and desert areas (Park 28e, Ch 10, p 726). The village-level schemes now in operation are the ASHA scheme, the Integrated Child Development Services scheme and training of local *dais* (Park 28e, Ch 10, p 726).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Four-tier structure of the Rural Health Scheme 1977** — draw a vertical ladder with the four rungs and label each with its functionary and population norm: *Village (Village Health Guide, trained dai, anganwadi worker — 1000); Sub-centre (health worker male + female/ANM — 5000, or 3000 hilly/tribal/desert); Primary health centre (medical officer — 20,000–30,000); Community health centre (specialists — 80,000–1,20,000)*; add an upward referral arrow to *District hospital* and a downward arrow marked *supportive supervision*.



Service and referral pathway created by the Rural Health Scheme, 1977



Evolution and closure of the village-level health worker, 1975 to 2005

High-yield exam pointers

- ▶ **Shrivastav Committee 1975** = "Group on Medical Education and Support Manpower"; set up 1974, reported April 1975; its acceptance in 1977 launched the Rural Health Scheme.
- ▶ Slogan: "**placing people's health in people's hands**"; launched **2 October 1977**; preceded Alma-Ata (1978).
- ▶ VHG arithmetic: **1 per 1000** population (500 tribal/hilly/remote) · **3 months** training · stipend **Rs 200/month** · honorarium **Rs 50/month** · drug kit **Rs 600/year** · minimum education **6th standard** · **2–3 hours** daily · no injections, no drugs outside the list, no case treated beyond **2 days** without improvement.
- ▶ **Dai** training: **1 month** · stipend **Rs 300** · 2 days PHC + 4 days field · minimum **2 supervised deliveries** · **five cleans** = clean hands, surface, blade, ligature, cord stump.

- ▶ Name trail to memorise: CHW (1977) → CHV (1979) → VHG (1981) → closed 31 March 2003 → ASHA (NRHM).
- ▶ Four alternative state schemes: J&K *Rehbar-e-Sehat*, Arunachal Pradesh *Medics*, Tamil Nadu *Mini Health Centres*, Kerala strengthening of PHCs.
- ▶ ROME 1977-78: community bias in undergraduate teaching; 3 PHCs per medical college; first phase 106 colleges, Rs 16 lakh grant each.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: Park's 28e still quotes "about 3.23 lakh health guides in position" (Ch 8, p 612) even though the centrally sponsored scheme closed on 31 March 2003, and the textbooks differ on small operational points — Park calls the structure three-tier where Mahajan & Gupta count four tiers by adding the village level, Suryakantha lists Karnataka as a fifth state with an alternative scheme, and the *dai's* per-delivery incentive is quoted as Rs 2 by Mahajan & Gupta and Rs 10 by Suryakantha; the tables above follow Mahajan & Gupta wherever they conflict.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO primary health care fact sheet 2025, National Medical Commission, Indian Public Health Standards 2022, NFHS-5, Sample Registration System, Health Management Information System, National Health Accounts

Programme and policy updates (India)

- ▶ The FACILITY has been renamed, not the village functionary — do not fold this into the CHW → CHV → VHG trail. In November 2023 the Union Health Ministry renamed the Ayushman Bharat Health and Wellness Centres — which are the upgraded **sub-centres and primary health centres** — as **Ayushman Arogya Mandir (AAM)**, with the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth") and an explicitly AYUSH-informed holistic wellness brief; the exercise was costed at Rs 3,000 per facility and about Rs 48 crore overall, and facilities must upload photographs of the renamed board to the portal. More than **1.75 lakh** AAMs are now operational. Each covers **3,000–5,000** population — the sub-centre tier, not the ~1,000-population village tier the Village Health Guide occupied, whose successor is still the ASHA. The examinable trail is therefore a *facility* one: HWC (2018) → **Ayushman Arogya Mandir (November 2023)**. [PMID 42257165] [PMID 41816146]
- ▶ The Community Health Officer is the institutional descendant of the Village Health Guide idea — but **salaried, professional and posted at the sub-centre**. The CHO or mid-level health provider is a nursing graduate with a further **six months** of community health nursing training who leads the sub-centre team of the auxiliary nurse midwife, multipurpose workers and ASHAs; **1,16,978** CHOs were in position as of 30 November 2024, against an estimated shortfall of about **11,111**, with retention in remote postings the chief difficulty. The sub-centre population norm is still quoted as **3,000–5,000**, the same band the 1977 scheme worked to. [PMID 41816146] [PMID 39367295]
- ▶ ASHA has become the world's largest community health worker programme, but has inherited the Village Health Guide's payment problem. About **9,83,032** ASHAs are in position; the norm

is one per 1,000 population, yet the real load ranges from 285 in Arunachal Pradesh to 1,241 in Bihar. ASHAs draw no salary and are paid task-linked incentives, and interviews with 40 ASHAs across ten Uttar Pradesh districts identify **delayed payment**, inadequate training and un replenished drug kits as the dominant demotivators — the same three grievances that discredited the Rs 50 honorarium and the unsupplied kit of the VHGs. [PMID 39953777] [PMID 41628200]

Current national data

- **What a sub-centre-level facility actually delivers now.** A state-representative assessment of 404 centres in Chhattisgarh found an average of **358 outpatients per month**, including **128 non-communicable disease patients**; the facilities met **31%** of the population's outpatient need, **26%** of hypertension care need and **21%** of diabetes care need, with the clearest gaps in mental illness and chronic respiratory disease. [PMID 39367295]
- **Headline national indicators to quote with their year.** India's UHC service coverage index rose from **57 (2015) to 63 (2021)**; out-of-pocket expenditure fell from **60.6% (2014–15) to 39.4% (2021–22)** on National Health Accounts estimates; sub-centres grew from **1,53,655 (Rural Health Statistics 2014–15) to 1,69,615** and primary health centres from **25,308 to 31,882 (Health Dynamics of India 2022–23)**. NFHS-5 unit data nonetheless show **16.3%** of women incurring distress financing for an institutional delivery, rising to **21.2%** in the poorest quintile. [PMID 41816146] [PMID 40482026]

Elimination, surveillance and disease status

- **Facility-based HMIS reporting is being explored as a complement to the Sample Registration System for mortality surveillance.** A Bayesian spatio-temporal analysis of SRS data (2014–2020) with NFHS-5 identified **institutional delivery and skilled birth attendance** as the two strongest health-system influencers reducing maternal mortality — health-system factors explaining the risk better than individual-level ones — with Bihar and Assam as spatial hotspots; those authors expressly caution that Health Management Information System data are of poor quality and should be used in scientific work only with care. Separately, the HMIS, covering about **2 lakh facilities**, recorded **4,23,408** excess rural facility deaths in eight states during the April–June 2021 Delta wave, a **270%** rise (95% CI 267–272) over the 2018–19 baseline — but those facilities are not fully representative of all rural deaths, and that study presents the HMIS as an opportunity to strengthen routine mortality surveillance rather than as an established substitute for the SRS. [PMID 38937489] [PMID 41748175]

Guideline and legislative updates

- **The Family Adoption Programme is the modern reincarnation of ROME.** Mandated by the National Medical Commission through its **March 2022** directive, it requires every MBBS student to adopt families in a village from the start of training, with the stated aim of improving the availability and accessibility of doctors in rural areas. A West Bengal study across 24 medical colleges found the recurring problems to be sustaining the adopted families, human resources and logistics, and the logbook format; a mixed-methods evaluation of 147 second-professional-year students and 575 community members reported improved confidence in patient care and communication in over **90%** of students. [NMC 2022-Mar] [PMID 40214316] [PMID 42330223]

- **The financing and service standards behind the current village tier — and the two committee layers that now sit under it.** Infrastructure is now upgraded against **Indian Public Health Standards 2022**, funded by the PM-Ayushman Bharat Health Infrastructure Mission and XV Finance Commission health grants routed through Panchayati Raj Institutions and urban local bodies. Free provision expanded from 9 drugs and 7 tests at the old sub-centre to **106 drugs and 14 diagnostic tests** at a sub-centre AAM (172 and 63 at a PHC-level AAM), and **82% of AAMs** had rolled out all **12 expanded service packages** by November 2024. Community structures are two-layered and coexist: the **Jan Arogya Samiti (JAS)** is the *facility-level* committee created at the AAM — **1,35,343** constituted, holding an Arogya Sabha twice yearly after monthly village *Samvad* — while the *community* platforms are the **Village Health, Sanitation and Nutrition Committee (VHSNC)** in rural areas and the Mahila Arogya Samiti in urban areas, states being asked to strengthen both alongside each other. It is the **VHSNC**, not the JAS, that stands in the line of the 1977 **Village Health Committee** to which the Village Health Guide reported. A Punjab appraisal found the JAS active in health promotion and service quality but passive in supporting village-level committees, in monitoring services and in social accountability, with only low-to-moderate decision space. [PMID 41816146] [PMID 41074751]

Global and WHO developments

- **The Alma-Ata line that this scheme anticipated has been formally renewed.** The **Declaration of Astana (2018)**, reiterated by World Health Assembly resolution 72/2, recommits member states to primary health care; WHO's operational framework names three components — integrated services, multisectoral policy, and **community engagement and empowerment**, the very principle behind "people's health in people's hands". WHO estimates that scaling up primary health care could save **60 million lives** and add **3.7 years** of life expectancy by 2030, delivering about 75% of the projected SDG health gains, at US\$200–328 billion a year. [WHO 2025-Dec]
- **The trained-dai component is supported by trial evidence of low certainty — say "some evidence of benefit", not "vindicated".** A meta-analysis of six interventional studies found that training traditional birth attendants reduced **perinatal mortality** (RR 0.69; 95% CI 0.61–0.78) and **7-day neonatal mortality** (RR 0.65; 95% CI 0.53–0.80), though not stillbirth (RR 0.70; 95% CI 0.39–1.26); the randomised subset ran the same way (perinatal 0.73, neonatal 0.70, stillbirth 0.81 and again not significant). The authors grade most of these outcomes **low to very low on GRADE** — two of the randomised trials were at high risk of bias and one raised some concerns, and heterogeneity for stillbirth was I^2 84%. The policy reading for the exam is that dai training helps where skilled attendance is unavailable, but is a bridge, not a substitute. [PMID 39264398]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2, Ch 8, Ch 10); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 26, Ch 28, Ch 31); Suryakantha — Community Medicine with Recent Advances (Ch 35, Ch 39); Sathe & Sathe — Epidemiology and Management for Health Care (Ch 19); Jugal Kishore — National Health Programmes of India (Ch 1); Report of the Group on Medical Education and Support Manpower (Shrivastav Committee), Government of India, 1975. Coverage note: the Park 28e scan of these pages is heavily degraded, so the operational detail (selection, training, kit, honorarium, closure) is grounded on Mahajan & Gupta 4e, Suryakantha and Sathe & Sathe rather than the spine.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of "Arogyam Paramam Dhanam" — a step towards holistic care, J Family Med Prim Care 2026, doi:[10.4103/jfmpc.jfmpc.1966.24](https://doi.org/10.4103/jfmpc.jfmpc.1966.24); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, J Family Med Prim Care 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 39367295 — Performance of health and wellness centre in providing primary care services in Chhattisgarh, India, BMC Prim Care 2024, doi:[10.1186/s12875-024-02603-1](https://doi.org/10.1186/s12875-024-02603-1); 39953777 — Community health worker payment processes: a qualitative assessment of experiences in two Indian states, Health Policy Plan 2025, doi:[10.1093/heapol/czaf010](https://doi.org/10.1093/heapol/czaf010); 41628200 — Factors affecting the motivation of community health workers: perspectives from Accredited Social Health Activists (ASHA) in Uttar Pradesh, India, PLoS One 2026, doi:[10.1371/journal.pone.0341811](https://doi.org/10.1371/journal.pone.0341811); 40482026 — Distress financing for institutional delivery in India: a regional analysis of economic inequality, coping mechanisms, and contributing factors, Int J Health Plann Manage 2025, doi:[10.1002/hpm.3950](https://doi.org/10.1002/hpm.3950) (abstract-only); 38937489 — Contributing factors for reduction in maternal mortality ratio in India, Sci Rep 2024, doi:[10.1038/s41598-024-65009-0](https://doi.org/10.1038/s41598-024-65009-0); 41748175 — Cause-specific excess mortality in rural India during the COVID-19 pandemic 2020-2023, BMJ Open 2026, doi:[10.1136/bmjopen-2024-097857](https://doi.org/10.1136/bmjopen-2024-097857); 40214316 — Family Adoption Program in MBBS curriculum: a qualitative study to assess the implementation challenges and possible solutions in the existing system, Indian J Public Health 2025, doi:[10.4103/ijph.ijph.453.23](https://doi.org/10.4103/ijph.ijph.453.23) (abstract-only); 42330223 — Enhancing collaboration, providing care, and fostering change: a 3-C approach to evaluate the impact of Family Adoption Program in undergraduate competency-based medical education curriculum, Indian J Public Health 2026, doi:[10.4103/ijph.ijph.850.24](https://doi.org/10.4103/ijph.ijph.850.24) (abstract-only); 41074751 — Exploring community participation through local health facility committees in Punjab: extent of decision space, capacity, and accountability, Indian J Public Health 2025, doi:[10.4103/ijph.ijph.421.24](https://doi.org/10.4103/ijph.ijph.421.24) (abstract-only); 39264398 — Maternal and child health training of traditional birth attendants and pregnancy outcomes: a systematic review and meta-analysis, J Epidemiol Glob Health 2024, doi:[10.1007/s44197-024-00300-x](https://doi.org/10.1007/s44197-024-00300-x). Non-journal sources: National Medical Commission Family Adoption Programme directive, March 2022; WHO primary health care fact sheet, December 2025.

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Salient features of Model Public Health Act 1955

Asked in: Paper IV Nov 2017 — RGUHS MD Community Medicine

Definition — The Model Public Health Act, 1955 is model (template) public health legislation issued by the Ministry of Health, Government of India, New Delhi (Park 28e, Ch 11, p 810; Ch 9, p 717). Being a *model*, it creates no rights or duties of itself: public health and sanitation are primarily a State responsibility, so it acquires legal force only where a State legislature enacts it. Its operative content, as reproduced in the standard texts, is three sets of **prescribed minimum standards** — introduced as "*the following minimum standards have been suggested for Restaurants and Eating Houses in India under the Model Public Health Act (1955)*" and as "*the following minimum standards for slaughter houses have been suggested under the Model Public Health Act (1955) in India*" (Park 28e, Ch 11, p 798; Park 27e, Ch 11, p 774), together with the minimum standards for **sanitation of the school and its environs**, which Park attributes jointly to three sources of which this Act is one (Park 28e, Ch 9, pp 683–684).

Why a model and not a central Act — the constitutional setting

Point	Content
Constitutional anchor	Health subjects are distributed under the Seventh Schedule of Article 246 of the Constitution of India; the functions of the Union Health Ministry are set out under (a) the Union list and (b) the Concurrent list (Park 28e, Ch 23, p 1033)
Union list — the Centre's own subjects	International health relations and administration of port quarantine ; administration of central institutes; immigration and emigration; regulation of labour in the working of mines and oil fields; coordination with States and with other ministries for promotion of health (Park 28e, Ch 23, p 1033)
Concurrent list — shared legislative power	The Centre and the States have simultaneous powers of legislation , the States' powers being restricted to the framework of such legislation as may be undertaken by the Centre. The list includes (1) prevention of extension of communicable diseases from one unit to another (2) prevention of adulteration of foodstuffs (3) control of drugs and poisons (4) vital statistics (5) labour welfare (6) ports other than major (7) economic and social planning (8) population control and family planning (Park 28e, Ch 23, p 1033)
Public health and sanitation sit with the States — hence a <i>model</i>	The Government of India Act, 1935 placed medical care, preventive health services and pilgrimages within the State in the "state" group; the position largely remained the same after the Constitution came into force in 1950, and the State is the ultimate authority responsible for all the health services operating within its jurisdiction (Park 28e, Ch 23, p 1034). Constitutionally, public health and sanitation in India are primarily the responsibility of the States (Jugal Kishore, Ch 1, pp 19–20), so the Centre could only commend a pattern for State legislatures to adopt
The legal landscape it was drafted into	Indian public health provisions were then scattered across subject-groups of different Acts: local authority (municipalities, municipal corporations, cantonments, zilla parishads), epidemic control , women and children , industrial workers , and miscellaneous such as the Prevention of Food Adulteration Act (Sathe & Sathe, Ch 9, p 107)

Salient features

Feature	Substance
A model, not an enactment	Issued by the Ministry of Health, Government of India, New Delhi in 1955 as a pattern for the States (Park 28e, Ch 11, p 810)
Route to enforceability	The model has no force of its own; it becomes operative only when a State legislature enacts it (Park 28e, Ch 23, pp 1033–1034)
Three documented domains	The provisions reproduced in the standard texts are three sets of prescribed minimum standards — restaurants and eating houses (ten numbered heads) and slaughter houses (eight numbered heads), both under food and environmental sanitation (Park 28e, Ch 11, p 798), and sanitation of the school and its environs (eleven heads), which Park attributes jointly to three sources of which the Model Public Health Act 1955 is one (Park 28e, Ch 9, pp 683–684, reference 122; Park 27e, Ch 9, pp 662–663, reference 120)
Standards are stated as <i>minimum</i> standards	Each list is introduced as the minimum standards suggested — a floor below which no establishment may fall, not an ideal specification (Park 28e, Ch 11, p 798; Park 27e, Ch 11, pp 774–775)
Written as verifiable numerical specifications	Floor areas in square feet, window and ventilator openings as a percentage of floor area, wall heights in feet, storage temperature in degrees Celsius, and frequency of refuse removal (Park 28e, Ch 11, p 798)
Whole-chain control for meat	For slaughter houses the standards run from ante-mortem and post-mortem examination of the animal , through condemnation of unfit meat and cold storage, to fly-proof covered transport of the dressed meat (Park 27e, Ch 11, p 774)

The prescribed minimum sanitary standards

Restaurants and eating houses — the ten heads

Head	Standard
(1) Location	Shall not be near any accumulation of filth or open drain, stable, manure pit and other sources of nuisances
(2) Floors	To be higher than the adjoining land, made with impervious material and easy to keep clean
(3) Rooms	(a) Rooms where meals are served not less than 100 sq. feet , providing accommodation for a maximum of 10 persons (b) Walls up to 3 feet smooth, corners rounded, impervious and easily washable (c) Lighting and ventilation — ample natural lighting aided by artificial lighting, with good circulation of air
(4) Kitchen	(a) Floor space minimum 60 sq. ft. (b) Window opening 25 per cent of floor area (c) Floor impervious, smooth, easy to keep clean and non-slippery (d) Doors and windows rat-proof, fly-proof and of the self-closing type (e) Ventilators 2 per cent of the floor area , in addition to smoke pipes
(5) Storage of cooked food	Separate room to be provided; for long storage, control of temperature is necessary
(6) Storage of uncooked foodstuffs	Perishable and non-perishable articles kept separately, in rat-proof and vermin-proof space; temperature control for perishable articles
(7) Furniture	Reasonably strong and easy to keep clean and dry

Head	Standard
(8) Disposal of refuse	Collected in covered, impervious bins and disposed of twice a day
(9) Water supply	An independent source, adequate, continuous and safe
(10) Washing facilities	To be provided; cleaning of utensils and crockery in hot water followed by disinfection

(Park 27e, Ch 11, pp 774–775; Park 28e, Ch 11, pp 798–799)

Slaughter houses — the eight heads

Head	Standard (Park)	Added detail (Suryakantha)
(1) Location	Preferably away from residential areas	Minimum 100 feet from the residential area
(2) Structure	Floors and walls up to 3 feet impervious and easy to clean	Floor above ground level, impervious, with a slope and channel; walls covered with tiles or glazed brick to a sufficient height
(3) Disposal of wastes	Blood, offal etc. not to be discharged into public sewers but collected separately	All refuse, blood, manure and garbage placed in vessels and removed as soon as possible
(4) Water supply	Independent, adequate and continuous	Abundant
(5) Examination of animals	Ante-mortem and post-mortem examination to be arranged; animals or meat found unfit for human consumption to be destroyed or denatured	Instruments used must be sterilised
(6) Storage of meat	Stored in fly-proof and rat-proof rooms; for overnight storage the room temperature shall be maintained below 5 °C	—
(7) Transportation of meat	Meat shall be transported in fly-proof covered vans	—
(8) Miscellaneous	Animals other than those to be slaughtered not allowed inside the shed	Dogs excluded; employees clean and in clean outer clothes; butchers healthy and personally hygienic

(Park 27e, Ch 11, p 774; Park 28e, Ch 11, p 798; Suryakantha, Ch 17, p 204)

Sanitation of the school and its environs — the eleven heads

Head	Prescribed minimum standard
(1) Location	Centrally situated with proper approach roads; at a fair distance from busy places and roads, cinema houses, factories, railway tracks and market places; premises properly fenced and kept free from all hazards

Head	Prescribed minimum standard
(2) Site	Suitable high land, not subject to inundation or dampness, and capable of being properly drained. The School Health Committee (1961) recommended 10 acres for higher elementary schools and 5 acres for primary schools, with one additional acre per 100 students ; in congested areas the nearest public park or playground to be made available
(3) Structure	Nursery and secondary schools single-storeyed as far as possible; exterior walls minimum 10 inches thick and heat-resistant
(4) Classroom	Verandahs attached; no classroom to hold more than 40 students ; per-capita space not less than 10 sq ft
(5) Furniture	Suited to the age group; single desks and chairs desirable; desks of the "minus" type; chairs with proper back-rests
(6) Doors and windows	Bottom sill 2 ft 6 in from floor level; combined door and window area at least 25 per cent of floor space ; windows on different walls for cross-ventilation; ventilators not less than 2 per cent of floor area
(7) Colour	Inside of the classroom white , periodically white-washed
(8) Lighting	Sufficient natural light, preferably from the left , never from the front
(9) Water supply	Independent source of safe potable water, continuous, distributed from taps
(10) Eating facilities	Vendors other than those approved by school authorities not allowed inside the premises; separate room for mid-day meals
(11) Lavatory	One urinal per 60 students and one latrine per 100 students , arranged separately for boys and girls

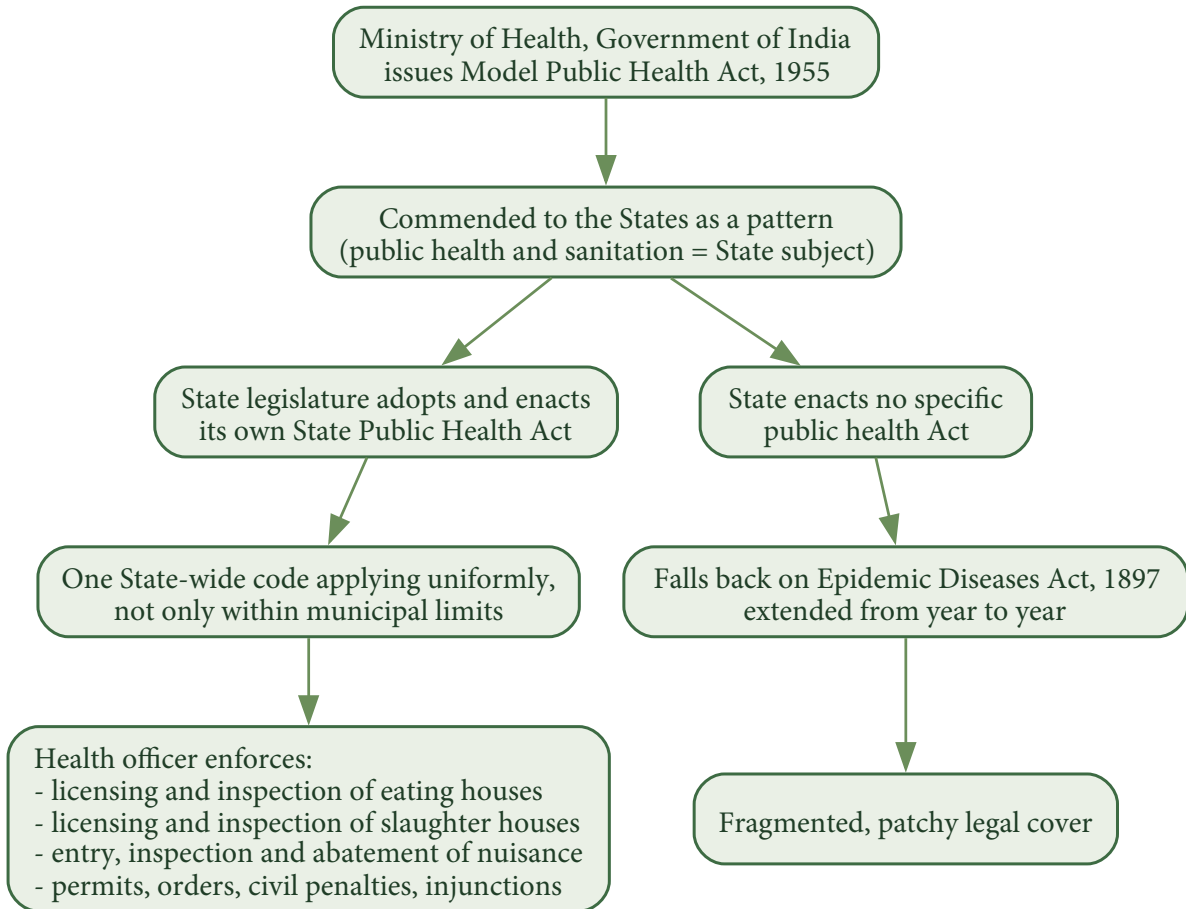
(Park 28e, Ch 9, pp 683–684; Park 27e, Ch 9, pp 662–663. Park introduces these as "the following minimum standards for sanitation of the school and its environs have been suggested in India" and attributes them to three references jointly, of which reference 122 in the 28e — 120 in the 27e — is Govt. of India (1955), Model Public Health Act.)

Objectives and enforcement of public health legislation — the frame the model sits in

Element	Content
Objectives of public health legislation (of all such laws, not of this Act specifically)	(a) protect and promote the health of their population (b) sustain the health policies and programmes (c) prevent ill health resulting from unsafe products and unsafe living conditions (d) fight new and re-emerging communicable diseases (e) support the development of health systems (f) combat continuing poverty, inequities in health and discrimination (AFMC Red Book 2024, Ch 39, p 1229)
Enforcement toolkit of a public health official (used by all public health legislation, not created by this Act)	Four primary enforcement tools — permits, licences and registrations; administrative orders; civil penalties; injunctions . Wherever these tools are lacking the legislation is ineffective (Jugal Kishore, Ch 2, p 835)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Labelled plan of a slaughter house as prescribed under the Model Public Health Act, 1955.** Labels to write: *minimum 100 feet from residential area; ante-mortem and post-mortem examination area; impervious floor above ground level, with slope and channel; walls impervious to 3 feet, tiled or glazed brick; independent, continuous water supply; separate collection of blood and offal — not into public sewers; fly-proof, rat-proof meat store below 5 °C; fly-proof covered van.*



How the Model Public Health Act 1955 was intended to become enforceable law in a State

Limitations

- **It was never self-executing.** Being a model, it created no rights or duties until a State legislature passed it, and the Centre had no power to compel adoption because under the Constitution the States are largely independent in matters relating to the delivery of health care (Park 28e, Ch 23, p 1033).
- **The uniformity it was meant to create never arrived.** Notification of disease continued to operate through isolated State Acts such as the *Madras Public Health Act, 1930*; some State Governments have **no specific Act at all** and simply invoke the *Epidemic Diseases Act, 1897*, "extending the same from year to year". Since legal provision is an essential pre-requisite for any notification system, the enactment of a **uniform Act similar to the Registration of Births and**

Deaths Act, 1969 is deemed necessary for any improvement in the notification system in India (Park 28e, Ch 21, p 997).

- ▶ **The corpus of Indian public health law stayed fragmented.** It suffers from antiquity, inconsistency, redundancy and ambiguity, rendering these laws ineffective or even counter-productive in advancing the population's health, and they often do not reflect contemporary scientific understanding of injury and disease or current legal norms for the protection of individual rights (Jugal Kishore, Ch 2, pp 835–836).
- ▶ **Weak enforcement machinery, and a risk of misuse.** Where the four enforcement tools are lacking the legislation is ineffective, and lack of awareness and lack of infrastructure are further reasons for failure (Jugal Kishore, Ch 2, pp 835–836); conversely, the wide health-officer powers a State public health Act confers can be misused — the **Goa Public Health Act Amendment of 1985 (Section 53.I.vii)** allowed the public health authorities and the police discretion to isolate people with HIV/AIDS and was repealed in 1996 (Jugal Kishore, Ch 1, p 372).
- ▶ **No comprehensive national public health law followed, and the policy record admits it.** India still lacks a comprehensive **National Public Health Act** handling preparedness for epidemics and disasters, surveillance of **both communicable and non-communicable** disease, environmental sanitation including safe water and waste disposal, and preventive plus curative services (Jugal Kishore, Ch 2, p 835); the **National Health Policy 1983, Para 17** had already called for urgent review of all existing legislation and a **unified, comprehensive legislation in the health field, enforceable all over the country** (Mahajan & Gupta 4e, Ch 26, p 520). [Recent advance] The 125-year-old Epidemic Diseases Act, 1897 is described as being replaced by a proposed **Public Health Act of 2022** covering epidemics, bioterrorism and disaster management, with a **four-tier structure of public health authorities at national, State, district and block levels** (Sunder Lal, Ch 7, p 286; Ch 13, p 671).

High-yield exam pointers

- ▶ **Ten eating-house heads, in Park's order** — L-F-R-K / C-U / F-R-W-W: Location, Floors, Rooms, Kitchen, Cooked-food storage, Uncooked-food storage, Furniture, Refuse disposal, Water supply, Washing facilities.
- ▶ **Eight slaughter-house heads** — L-S-D-W-E-S-T-M: Location, Structure, Disposal of wastes, Water supply, Examination of animals, Storage of meat, Transportation of meat, Miscellaneous.
- ▶ **The numbers to write:** 100 sq ft / 10 persons / 60 sq ft / 25 per cent / 2 per cent / 3 feet / twice a day / below 5 °C / 100 feet.
- ▶ **Attribution line to open the answer:** Model Public Health Act, 1955 — **Ministry of Health, Government of India, New Delhi**; a *model* only, operative in a State solely on State enactment (Seventh Schedule of Article 246; public health and sanitation a State responsibility).
- ▶ **Most-missed head:** *Examination of animals* — **ante-mortem AND post-mortem**, with unfit animals or meat **destroyed or denatured**; the only veterinary-inspection step in the whole Act.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the indexed corpus nowhere links **licensing or registration** of eating houses or slaughter houses to this Act; Park's date for the early State law, the *Madras Public Health Act, 1930***, should be reproduced exactly as printed; and the Public Health Act of 2022 is described by Sunder Lal as replacing the Epidemic Diseases Act, 1897, but the corpus does not confirm that it has been enacted, so write it as a proposal.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Kerala Public Health Act 2023, MoHFW Public Health Management Cadre guidelines 2022, ICMR, WHO International Health Regulations (2024 amendments), WHO Pandemic Agreement 2025

Programme and policy updates (India)

- **Kerala Public Health Act, 2023 — the model's route actually taken.** Kerala passed a comprehensive State public health Act on 28 November 2023, precisely the mechanism the 1955 model contemplated (a State legislature converting a template into binding law). A published ethics appraisal argues it gives little weight to the Siracusa Principles and to autonomy, privacy and confidentiality, transparency, accountability, rule of law and least-harm safeguards, disregards the COVID-19 lesson on guarding against State overreach, and could become a tool of abuse against already vulnerable communities — the same failure mode the corpus records for the repealed Goa amendment of 1985 [PMID 38755770]
- **Public Health Management Cadre (PHMC) — the enforcement machinery the model always lacked.** The Union health ministry released PHMC guidelines in 2022 setting out block-, district- and State-level cadre posts for medical, paramedical and non-medical professionals; the National Health Policy 2017 proposal was costed at about 33,236 posts if rolled out from 2020 to 2026, and the guidelines leave each State to write its own roadmap. A four-State situational analysis (Gujarat, West Bengal, Odisha, Maharashtra; 78 stakeholders) found every State has a dedicated public health cadre with a separate hierarchical structure, but a clearly demarcated Public Health Directorate only in Odisha and Gujarat — in West Bengal and Maharashtra the Directorate of Health Services oversees the work earmarked for the cadre. Nomenclature differs by State (Public Health Cadre in Odisha, Public Health Department in Maharashtra and West Bengal, public health Division in Gujarat), the workforce skilled in public health management is deficient in all four, and penetration below district level is limited — only Odisha reported a block-level public health officer post [PMID 37884946]

Current national data

- **The eating-house heads are still not enforced.** Of 102 active street food vending units observed in urban (Lawspet) and semi-urban (Villianur) Puducherry, **not one** displayed the licence or certification required under the Food Safety and Standards Act; only 29 units (28.3%) had dedicated handwashing facilities and 50 (49%) had dustbins, although all had separated raw from cooked food — a contemporary audit against the model Act's *water supply, washing facilities and disposal of refuse* heads [PMID 40257165]

- **School sanitation standards remain unmet in practice.** Among 566 students across 34 secondary schools (17 government, 17 non-government) in Jamnagar, Gujarat, only 51% reported access to a clean, functioning toilet and 37% used soap or ash for handwashing; good practice scores were reached by 28% overall, with upper socio-economic status the strongest correlate (AOR 2.5, 95% CI 1.5–4.3) and maternal secondary education next (AOR 1.8, 95% CI 1.1–3.0) [PMID 40369474]

Elimination, surveillance and disease status

- **IDSP is now IHIP-IDSP — but the notification problem Park flagged persists.** The Integrated Disease Surveillance Programme moved in 2018 from aggregate paper returns to a real-time, case-based Integrated Health Information Platform. Fieldwork of 57 interviews from November 2023 to March 2025 in two high-burden dengue districts of a western State found the district public health laboratories established under IHIP-IDSP functional in only three districts, four of the State's 36 districts without sentinel-hospital coverage, and dual overlapping reporting between IHIP-IDSP and the National Vector Borne Disease Control Programme, whose official count admits only ELISA-confirmed dengue [PMID 42245358]
- **ICMR-FoodNet — surveillance built around slaughter houses and food establishments.** India's first systematic foodborne-pathogen network sampling humans, food, animals and the environment together opened five centres in four North-Eastern States from 2020; phase II (January 2024) added seven centres, three of them veterinary, to cover all eight States. Its animal-husbandry module samples **slaughterhouses and butcher shops** — carcass intestinal contents, meat, milk and eggs, wash water, and swabs from butchers and food handlers — alongside market and hospital surveillance, and is aligned with the National Action Plan on antimicrobial resistance launched on 19 April 2017 [PMID 39432901]

Guideline and legislative updates

- **Slaughter-house waste disposal is now an antimicrobial-resistance question.** Of 309 samples from pig farms and slaughterhouses in West Bengal and Assam, 154 (49.8%) grew *E. coli* and 23 of these (14.9%) were ESBL producers, the highest yield being slaughterhouse effluent (26.1%); all carried *bla*CTX-M and 82.6% *bla*CMY, two effluent isolates carried *bla*NDM-1, and four imipenem-resistant isolates came only from slaughterhouse effluent. This gives modern force to the Act's rule that blood and offal must be collected separately and not discharged into public sewers [PMID 39282198]
- **The unfinished legislative agenda — occupational health.** Silicosis is described as a long-neglected public health emergency in India, with no law protecting health at work for over 90% of workers in the unorganised sector; the Employees' State Insurance Act 1948 reaches only the organised sector and the Employees' Compensation Act 1923 both, leaving most victims dependent on State assistance schemes — the same antiquity-and-fragmentation problem the 1955 model was meant to cure [PMID 41195485]

Global and WHO developments

- **IHR (2005) amendments and the WHO Pandemic Agreement.** The 77th World Health Assembly agreed a package of amendments to the International Health Regulations on 1 June 2024, entering into force in September 2025: through amended Articles 1 and 12 the Director-General may now declare a **pandemic emergency** alongside a PHEIC, a revised Article 13

requires the WHO Secretariat to facilitate access to health products, and a revised Article 44 adds a new coordinating financial mechanism for developing countries under the authority of the Health Assembly. On 20 May 2025 the 78th Assembly adopted the Pandemic Agreement with its 35 Articles — 124 governments in favour, and only the second legal agreement negotiated under Article 19 of the WHO Constitution after the Framework Convention on Tobacco Control — with the Pathogen Access and Benefit-Sharing annex (Article 31.2) to be considered at the 2026 Assembly, the Agreement taking effect in 2026 if adopted [PMID 40589223]

- **Rights safeguards in the IHR reform remain thin.** The 2024 amendments insert equity, solidarity, access-to-vaccines, financing and preparedness language, but are assessed as vague, leaving the most contentious provisions (notably travel restrictions) largely untouched. Accountability was strengthened only for monitoring the Regulations themselves: the new States Parties Committee for Implementation (Article 54 bis) is consultative, non-adversarial and non-punitive and meets only every two years, and the National IHR Authority each State must designate (Article 4) carries no duty to review the human-rights consequences of implementation — the international mirror of the domestic concern that broad health-officer powers can outrun individual-rights protection [PMID 39885739]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 9, p 717 reference 122; Ch 11 Nutrition and Health, pp 798–799 and p 810 reference 146; Ch 21, p 997; Ch 23 Health Planning and Management, pp 1033–1034), cross-checked against Park's Textbook of Preventive and Social Medicine, 27e (Ch 11, pp 774–775) for the sanitary-standards wording; Suryakantha — Community Medicine with Recent Advances (Ch 17, p 204); Jugal Kishore — National Health Programmes of India (Ch 1, pp 19–20, p 372; Ch 2, pp 834–836); Sathe & Sathe — Epidemiology and Management for Health Care (Ch 9 Organizational concepts, p 107); AFMC Red Book of Community Medicine 2024 (Ch 39, p 1229); Sunder Lal — Textbook of Community Medicine (Ch 7, p 286; Ch 13, p 671); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 26, p 520 — National Health Policy 1983, Para 17); Government of India (1955), *Model Public Health Act*, Ministry of Health, New Delhi. **Coverage note:** the primary text of the Act is not in the indexed corpus; what is textbook-documented is its three sets of prescribed minimum standards (eating houses, slaughter houses, school sanitation) and its place within Indian health legislation.

Recent-advances sources: 38755770 — Public health ethics and the Kerala Public Health Act, 2023, *Indian Journal of Medical Ethics* 2024, DOI [10.20529/IJME.2024.007](https://doi.org/10.20529/IJME.2024.007) (abstract-only); 37884946 — Assessment of needs and gaps in public health cadre in India: a situational analysis, *BMC Health Services Research* 2023, DOI [10.1186/s12913-023-10132-3](https://doi.org/10.1186/s12913-023-10132-3); 40257165 — Assessment of food safety and hygiene practices of street food vending units in urban and semi-urban areas of Puducherry, *National Medical Journal of India* 2024, DOI [10.25259/NMJI_793_2022](https://doi.org/10.25259/NMJI_793_2022) (abstract-only); 40369474 — Assessing and comparing knowledge, attitude and practices related to water, sanitation and hygiene among government and non-government school students in Gujarat, *BMC Public Health* 2025, DOI [10.1186/s12889-025-22937-4](https://doi.org/10.1186/s12889-025-22937-4); 42245358 — Informational challenges of dengue surveillance and response: a case study from India, *Frontiers in Public Health* 2026, DOI [10.3389/fpubh.2026.1809584](https://doi.org/10.3389/fpubh.2026.1809584); 39432901 — Surveillance of food and waterborne pathogens in North-East India: protocol for a laboratory-based sentinel surveillance study, *JMIR Research Protocols* 2024, DOI [10.2196/56469](https://doi.org/10.2196/56469); 39282198 — Isolation and characterization of extended-spectrum β -lactamase producing *E. coli* from pig farms and slaughterhouse, *Indian Journal of Microbiology* 2023, DOI [10.1007/s12088-023-01151-z](https://doi.org/10.1007/s12088-023-01151-z) (abstract-only); 41195485 — Silicosis: a public health emergency in India, *Indian Journal of Medical Ethics* 2025, DOI [10.20529/IJME.2025.051](https://doi.org/10.20529/IJME.2025.051) (abstract-only); 40589223 — Editorial: the 2025 World Health Assembly Pandemic Agreement and the 2024 amendments to the International Health Regulations combine for pandemic preparedness and response, *Medical Science Monitor* 2025, DOI [10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411); 39885739 — How did human rights fare in amendments to the International Health Regulations?, *Journal of Law, Medicine & Ethics* 2025, DOI [10.1017/jme.2024.172](https://doi.org/10.1017/jme.2024.172). Also used: Kerala Public Health Act 2023 (passed 28 November 2023); Ministry of Health and Family Welfare — Public Health Management Cadre guidelines, 2022; National Health Policy 2017 (PHMC proposal); National Action Plan on Antimicrobial Resistance, April 2017; WHO International Health Regulations (2005) amendments adopted by the 77th World Health Assembly, 1 June 2024, in force September 2025; WHO Pandemic

Agreement adopted by the 78th World Health Assembly, 20 May 2025.



Nurse as the pioneer of health

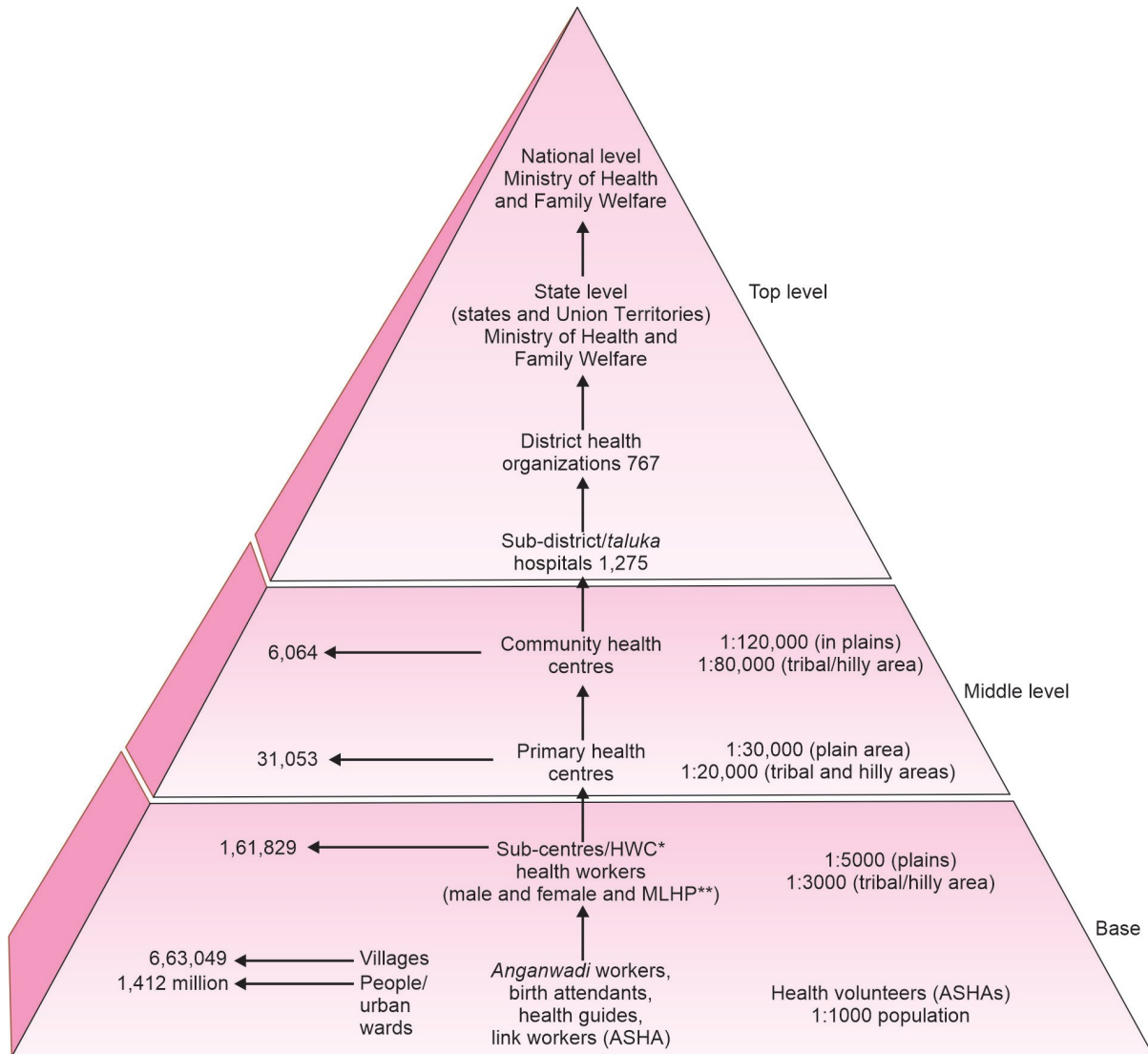
Asked in: Paper IV Jun 2017 — RGUHS MD Community Medicine

Definition — In Indian public health "the nurse" is not one person but a family of cadres — the auxiliary nurse midwife (ANM, designated health worker female), the lady health visitor (LHV, designated health assistant female), the staff nurse, the public health nurse, the nurse practitioner in midwifery (NPM) and the nurse-community health officer (CHO) at a Health and Wellness Centre. **The statement is largely correct and I agree with it**, because the nurse is the most numerous professional cadre, the first and often the only trained contact the community has, and the carrier of promotive and preventive care into the home. But she pioneers *within* a health team, and cannot be equated with the whole of it.

1. The nursing cadres of the Indian health system

Cadre	Basic training	Placement and population norm	Core public health role
Health worker female / ANM	1½ years (male multipurpose worker 1 year) (<i>Sunder Lal, Ch 16, p 721</i>)	Sub-centre; 1 per 5,000 in plain and 3,000 in tribal/hilly/desert areas; activities limited to 350–500 families (<i>Park 28e, Ch 10, pp 724, 727</i>)	The whole peripheral service package — maternal and child health, family planning, immunisation, nutrition, communicable disease, records
Health assistant female / LHV	Promotion cadre from ANM	1 per 30,000 (20,000 tribal/hilly); supervises 6 sub-centres, visits each once a week on a fixed day (<i>Park 28e, Ch 10, pp 724, 731</i>)	Supportive supervision, sub-centre maternal and child health and family welfare clinics, immunisation and cold-chain guidance, <i>dai</i> and ASHA training, school health
Staff nurse	General nursing and midwifery (GNM) or BSc nursing	Primary health centre (PHC) and community health centre (CHC); 1 per 6 beds (Indian Nursing Council norm) (<i>Sunder Lal, Ch 17, p 747</i>)	Institutional delivery, newborn care, labour-room and ward care, infection control

Cadre	Basic training	Placement and population norm	Core public health role
Community health officer (CHO) / mid-level health provider (MLHP)	GNM or BSc nurse (also BSc community health or Ayurveda practitioner) plus a 6-month certificate in community health through IGNOU or a state public health university (<i>Sunder Lal, Ch 17, p 742</i>)	Leads the Health and Wellness Centre–sub-centre team of ANMs and ASHAs	Delivers the 12-service comprehensive primary health care package, including screening for hypertension, diabetes and three common cancers
Nurse practitioner in midwifery (NPM)	18 months for GNM/BSc nurses with 2 years' experience conducting deliveries; curriculum built on the <i>Essential Competencies for Midwifery Practice</i> of the International Confederation of Midwives (<i>Park 28e, Ch 7, p 546</i>)	Labour rooms and midwifery-led care units	Respectful, woman-centred birth care that minimises unnecessary medical intervention
Public health nurse	Post-basic public health nursing	District and urban health services, school health services, child guidance clinics (<i>Park 28e, Ch 9</i>)	Home visiting, health education, school health, family follow-up



Health system organization — the nurse-staffed sub-centre and Health and Wellness Centre form the base of the pyramid (Sunderlal, p.709)

2. The case for agreement — the evidence

Argument	Grounded evidence
Numerically the dominant profession	Registered nurses and midwives 21,17,649 and ANMs 8,79,508 (registered up to 2019) against 12,34,205 allopathic doctors; nurse–population ratio 1:482 (Sunder Lal, Ch 17, p 766)
Denser than doctors even on international comparison	India has 17.3 nurses and midwives per 10,000 population against 7.3 doctors per 10,000 (2013–2021) (Park 28e, Ch 10, Table 3)
She is the first, and often the only, trained contact	1,57,935 sub-centres were functional as on 31 March 2022, each the first formal rung of primary health care and manned by health workers female and male, with a minimum of 6 hours of routine sub-centre outpatient clinic by the ANM per day (Sunder Lal, Ch 17, p 745)
In position at every tier	As on March 2021 — health workers (male and female) at sub-centres 2,39,911; staff nurses at PHCs 37,736 rural and 10,196 urban; nursing staff at CHCs 41,604 rural and 9,455 urban (Park 28e, Ch 10, Table 4)

Argument	Grounded evidence
Historical primacy in Indian health planning	The Bhore Committee (1946) long-term programme proposed a 75-bedded primary health unit for every 10,000–20,000 population staffed by 6 doctors and 6 public health nurses — parity with the physician in the very first national blueprint (<i>Park Plus, Ch 21, p 296</i>)
Her domain is the home, where health is actually made	<i>Florence Nightingale</i> : "the nurse can do more good in the home than in the hospital" (<i>Park 28e, Ch 22, p 1020</i>) and "the secret of national health lies in the homes of the people" (<i>Park 28e, Ch 12, p 829</i>); "home visiting is the backbone of all maternal and child health services" (<i>Park 28e, Ch 9, p 626</i>)
She now leads, and does not merely assist	At the Health and Wellness Centre–sub-centre the team of ASHAs and ANMs is led by a mid-level health provider who is typically a BSc or GNM nurse (<i>Sunder Lal, Ch 17, p 742</i>); nearly 1.1 lakh sub-health centres had been transformed into Ayushman Bharat Health and Wellness Centres by March 2022 (<i>Sunder Lal, Ch 17, p 746</i>)
The profession is expanding upward	The nurse practitioner in midwifery and the community health officer cadres convert the nurse from an auxiliary into an independent primary-care provider (<i>Park 28e, Ch 7, p 546</i>)
Global endorsement	World Health Day 2020 was themed the "Year of the Nurse and Midwife" (<i>Sunder Lal, Ch 18</i>)
Statutory standing	The Indian Nursing Council Act, 1947 regulates a uniform standard of training for nurses, midwives and health visitors (<i>Jugal Kishore — National Health Programmes of India, Ch 2</i>)

3. What the nurse actually delivers at the periphery

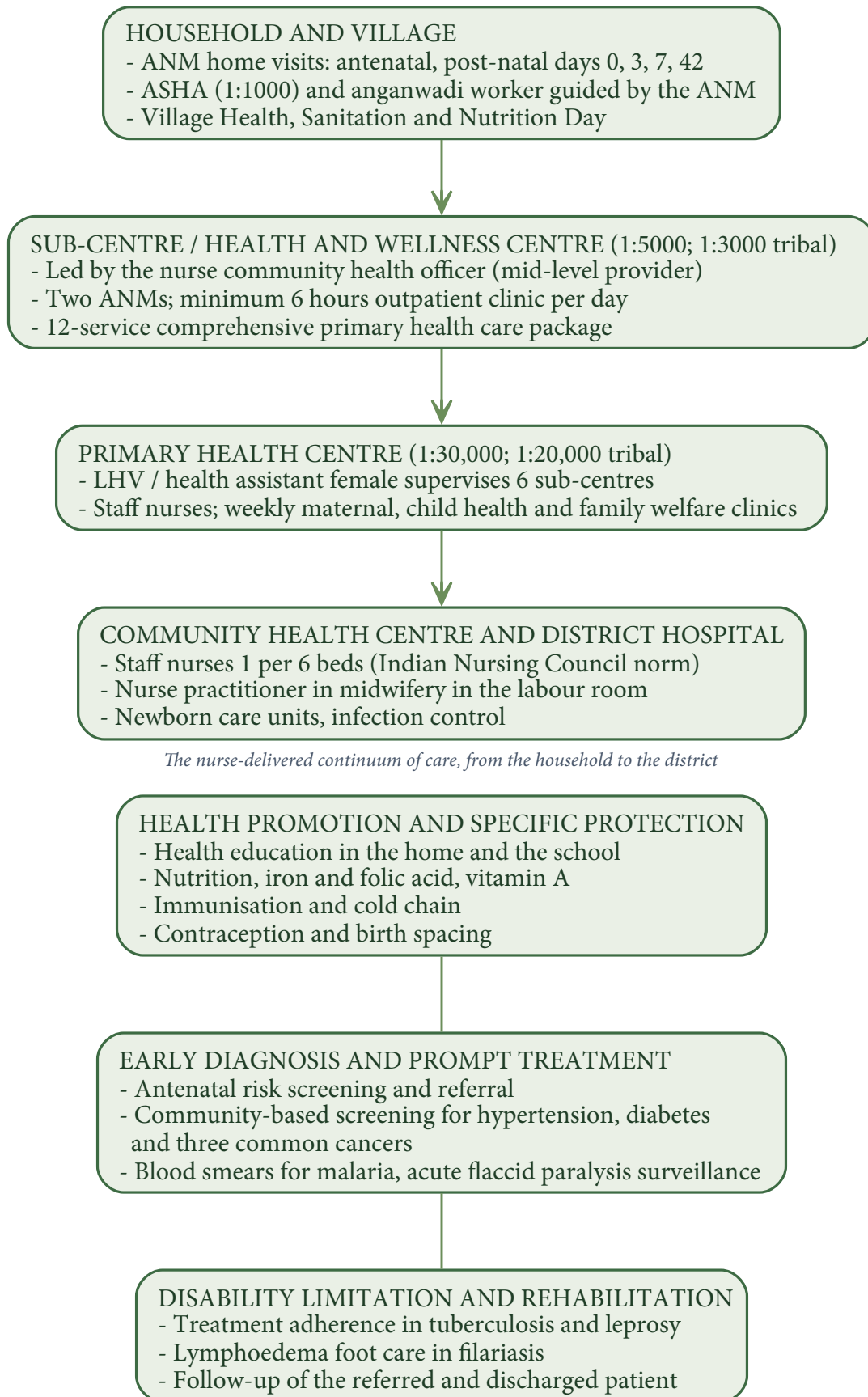
Job functions of the health worker female (ANM) — *Park 28e, Ch 10, pp 727–729*:

Domain	Specific tasks
Maternal health	Register the pregnancy at earliest contact; ensure at least 4 antenatal visits ; test urine for albumin and sugar and estimate haemoglobin; refer for RPR test for syphilis; conduct deliveries and supervise those conducted by <i>dais</i> ; refer difficult labour; complete formalities for Janani Suraksha Yojana disbursement; name-based tracking of every pregnancy
Newborn and child health	Post-natal home visits on days 0, 3, 7 and 42 for home and sub-centre deliveries and days 3, 7 and 42 for institutional deliveries; six post-natal visits for a low-birth-weight baby; early initiation of breastfeeding within 1 hour, exclusive breastfeeding and timely complementary feeding at 6 months; treat diarrhoea and pneumonia and refer per IMNCI guidelines
Family planning and medical termination of pregnancy	Maintain and update the eligible-couple register; distribute conventional and oral contraceptives; manage side-effects and refer; establish female depot holders; identify and refer women needing medical termination of pregnancy
Nutrition	Detect malnutrition from birth to 5 years; distribute iron and folic acid tablets to pregnant women, nursing mothers and young children; administer vitamin A
Immunisation (Universal Immunisation Programme)	Give tetanus toxoid, BCG, pentavalent, oral polio, inactivated polio, hepatitis B and measles vaccines; maintain the cold chain at fixed and outreach sessions ; ensure injection safety and safe waste disposal; record, report and manage adverse events following immunisation ; track dropouts and left-outs

Domain	Specific tasks
Communicable disease	Notify any abnormal increase in diarrhoea, fever with rigors, fever with jaundice or acute flaccid paralysis in a child under 15 years ; prepare blood smears for malaria; identify anaesthetic skin patches (leprosy); supervise treatment adherence in tuberculosis, malaria and leprosy and chase defaulters; counsel on HIV and sexually transmitted and reproductive tract infections; in filaria-endemic areas identify and train drug distributors for mass drug administration on National Filaria Day
Non-communicable disease	Information, education and communication for prevention and early detection
Vital statistics and records	Record and report births and deaths, particularly of mothers and infants; maintain eligible-couple, antenatal, maternity and child-care registers; submit weekly and monthly returns
Team building	Acts as guide and resource person for the ASHA (1 per 1,000 population) and the <i>anganwadi</i> worker; trains <i>dais</i> ; conducts the monthly Village Health, Sanitation and Nutrition Day

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Figure to reproduce by hand:** the health-system pyramid, labelled *Household and village* (ASHA 1:1,000) → *Sub-centre / Health and Wellness Centre*, 1:5,000 plain and 1:3,000 tribal — *ANM and nurse community health officer* → *Primary health centre* 1:30,000 → *Community health centre* 1:1,20,000 → *Sub-district hospital* → *District hospital* → *State* → *Ministry of Health and Family Welfare*, marking which tiers are nurse-staffed.



The nurse's contribution mapped to the levels of prevention

5. The counter-argument — why "pioneer" needs qualification

- ▶ **Health care is a team output, not a solo one.** "No one person can acquire all the necessary skills, or have enough time to do everything that must be done to satisfy the needs of even a small community" (*Sunder Lal, Ch 16, p 715*); at the primary health centre the **medical officer remains the leader of the health team**, accountable for planning, supervision, training and evaluation (*Sunder Lal, Ch 17, p 751*).
- ▶ **Numbers on paper are not numbers in the field.** The shortfall against sanctioned requirement is **49.2% for health assistant (female)/LHV** and **61.3% for health assistant (male)** (*Park Plus, Ch 21, pp 294–296*), and there is gross **maldistribution of health manpower between rural and urban areas** (*Park 28e, Ch 10, p 725*).
- ▶ **Density still lags comparable countries.** India's 17.3 nurses and midwives per 10,000 is well below Sri Lanka's 24.4 (*Park 28e, Ch 10, Table 3*), and the suggested national norm of **1 nurse per 5,000 population** is not uniformly met (*Park 28e, Ch 10, p 724*).
- ▶ **Training quality is the weak link.** Training schools give less community-based and more theoretical, hospital-based instruction, basic training faculty are poorly trained, and skill gaps persist in pelvic examination, intrauterine device insertion, abdominal examination and safe delivery (*Sunder Lal, Ch 16, p 721*).
- ▶ **Her scope has a ceiling.** Specialist care, surgery, anaesthesia and the definitive management of obstetric and medical complications belong to the referral tier; her strength is coverage and continuity, not definitive cure.
- ▶ **Health is determined largely outside the health sector.** Water, sanitation, housing, education, income and gender act on health irrespective of nursing effort, and the peripheral health team can address these social and environmental determinants only through coordination with other departments and local government (*Sunder Lal, Ch 16, pp 717–718*).

6. Verdict

I agree with the statement in its intended sense. Measured by numbers, by proximity to the household, by continuity of contact and by breadth of function, the nurse **pioneers** health — she reaches the family before the physician does and stays with it afterwards. The Bhore blueprint, the Nightingale tradition of the home visit and the present nurse-led Health and Wellness Centre all say the same thing. But "pioneer" means *first mover*, not *sole mover*: the nurse opens the ground, and the rest of the health team, a working referral system and intersectoral action must follow her onto it before health gains can be sustained.

High-yield exam pointers

- ▶ **Norms (Park 28e, Ch 10):** nurse **1:5,000**; health worker female and male **1:5,000** plain and **1:3,000** tribal/hilly; health assistant (LHV) **1:30,000** plain and **1:20,000** tribal, supervising 6 health workers; ASHA **1:1,000**; staff nurse **1 per 6 beds** (Indian Nursing Council).
- ▶ **Dated Indian figures:** nurses and midwives **17.3 per 10,000** and doctors **7.3 per 10,000** (2013–2021); registered nurses and midwives **21,17,649**, ANMs **8,79,508**, nurse–population ratio **1:482** (registered up to 2019); **1,57,935** sub-centres functional (31 March 2022); nearly **1.1 lakh** sub-centres converted to Health and Wellness Centres (March 2022).

- ▶ **ANM post-natal visit schedule:** days 0, 3, 7 and 42 for home or sub-centre delivery; days 3, 7 and 42 for institutional delivery; six visits for a low-birth-weight baby.
- ▶ **Cadre facts:** community health officer = **6-month** community health certificate on a GNM or BSc nursing base; nurse practitioner in midwifery = **18 months** for a nurse with **2 years'** delivery experience, on the International Confederation of Midwives competency framework.
- ▶ **Quotable lines:** Nightingale — "the nurse can do more good in the home than in the hospital" and "the secret of national health lies in the homes of the people"; Bhore Committee 1946 — **6 doctors and 6 public health nurses** per 75-bedded primary health unit; World Health Day 2020 = "Year of the Nurse and Midwife".
- ▶ **Answer structure for this stem:** cadres → arguments for → job functions → limitations → qualified verdict. Never write it as a one-sided eulogy; the examiner is testing whether "do you agree?" draws a balanced argument.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir**; the antenatal contact schedule in current World Health Organization guidance is eight contacts against the four antenatal visits still stated in the Indian job description quoted above; and the shortfall percentages for lady health visitor and health assistant posts are quoted from Park Plus, which does not state which Rural Health Statistics round they come from.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community; Ch 12 Medicine and Social Sciences; Ch 22 Communication for Health Education); Sunder Lal — Textbook of Community Medicine (Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community; Ch 18 International Health); Park Plus (Amrutha & Vijayalakshmi), Ch 21 Health Planning and Management; Jugal Kishore — National Health Programmes of India (Indian Nursing Council Act, 1947); Ayushman Bharat Comprehensive Primary Healthcare through Health and Wellness Centres, Operational Guidelines 2018; Indian Public Health Standards 2022. **Coverage note:** the corpus carries no dedicated chapter on nursing as a profession, so this answer is assembled from the health-manpower, job-description and health-care-delivery sections of these books.

♦ ♦ ♦

Urban health care — current challenges and solutions

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — Urban health care is the organised delivery of promotive, preventive, curative and rehabilitative services to people living in cities and towns, with a deliberate focus on the **urban poor** — those in listed and unlisted slums and other vulnerable groups. Its defining paradox is that cities hold the country's greatest concentration of doctors, hospitals and specialists yet the urban poor have the worst access to primary care — "**poverty in plenty**", the paradox the World Health Organization (WHO) marked by choosing "*Urban Health Matters*" as the World Health Day theme on 7 April 2010 (Sunder Lal, Ch 17, p 756). Urbanisation is now treated as a determinant of health in its own right, and of health inequity within and between urban populations (Oxford Textbook of Public Health 7e, Ch 6, pp 304–305).

Scale of the problem — the dated Indian numbers

Parameter	Value (with year)
Urban share of India's population	27.8% (2001) → 31.16% (Census 2011) → 34.0% (2021) ; India's pace of urbanisation is among the slowest in the world (Sunder Lal, Ch 9, p 543)
Absolute urban population	35.7 crore (2011) projected to 43.2 crore (2021) (Suryakantha, Ch 37, p 799)
Share of urban people living in slums	One-third or more of the urban population; up to half of the urban population by some estimates (Sunder Lal, Ch 1, p 52; Ch 2, p 68)
Growth of the slum population	About 7% per year (Suryakantha, Ch 37, p 799)
Urban housing stock	110.14 million of 330.84 million houses counted were urban (Census 2011); 48.8 million people were in slums in 1990, about 40% of them in million-plus cities (Park 28e, Ch 15, p 908)
Urban vital rates	Birth rate 16.1 , death rate 5.1 per 1,000 population, infant mortality rate 19 per 1,000 live births (SRS 2022) (Sunder Lal, Ch 2, p 68)
Global comparator	Nearly 1 billion people live in urban slums, informal settlements or as pavement dwellers, and have the worst health outcomes of all urban residents — frequently worse than the rural poor (Oxford 7e, Ch 58, pp 2151–2152)

The city average is the trap: aggregate urban indicators beat rural ones, but split the urban infant mortality rate into slum versus non-slum and the slum rate exceeds both the rural rate and the national average — documented in the slums of Delhi and Mumbai (Sunder Lal, Ch 10, pp 589–590).

Current challenges

Domain	Specific challenge
Demographic	Rapid, unplanned growth driven by migration , producing slums faster than services can follow; a large floating and temporary-migrant population that no register captures (Sunder Lal, Ch 17, p 756)
Physical and built environment	Overcrowding, poor housing, unsafe drinking water, absent sanitation and drainage, uncollected garbage, and air, water and soil pollution (Suryakantha, Ch 37, p 799; Sunder Lal, Ch 17, p 756)

Domain	Specific challenge
Triple burden of disease	<i>Communicable</i> : tuberculosis, acute respiratory infection, diarrhoea, vector-borne disease; <i>non-communicable</i> : obesity, hypertension, diabetes and cardiovascular disease driven by urban diet, tobacco, alcohol and inactivity; <i>undernutrition and stunting</i> persisting in poor children (Oxford 7e, Ch 4, pp 237–238)
Environmental exposures	Urban air pollution is linked to up to 1 million premature deaths a year , with over 90% of city air pollution in these countries from vehicle emissions; particulate matter causes cardiovascular death and asthma, and noise pollution causes hearing impairment, hypertension and ischaemic heart disease (Oxford 7e, Ch 58, pp 2159–2160)
Injuries	Pedestrian–motor vehicle conflict is a leading cause of urban injury, and rises further when new housing and commercial development increase pedestrian activity (Oxford 7e, Ch 58, p 2160)
No organised primary tier	Unlike the three-tier rural system (sub-centre → PHC → CHC), no comparable well-planned primary, secondary and tertiary structure was created for geographically defined slum areas, so primary care is simply absent in many pockets (Sunder Lal, Ch 17, p 756)
Fragmentation of providers	Municipal health services, state government hospitals, Employees' State Insurance, Central Government Health Scheme, railway and armed forces services, AYUSH, private practitioners and nursing homes, and non-governmental organisations all operate in the same city with little coordination (Sunder Lal, Ch 17, p 743)
Curative and private-sector bias	The private sector, largely unorganised, congregates in cities and provides mainly curative services available to those who can pay, pushing out-of-pocket expenditure up (Park 28e, Ch 10, p 746)
Access barriers for the poor	Poverty and ignorance, unfriendly treatment at government hospitals, and clinic hours that clash with daily-wage work (Suryakantha, Ch 37, p 799; Sunder Lal, Ch 9, p 543)
Legal invisibility	Residents of <i>unlisted</i> (non-notified) slums, the homeless, street children, rickshaw pullers, construction and brick-kiln workers, sex workers and temporary migrants fall outside enumeration and entitlement (Park 28e, Ch 7, p 535)
Information systems	Coverage of the urban population and reporting of work done is itself a major challenge; urban slum statistics have been called a "statistic of shame" (Sunder Lal, Ch 17, p 756; Ch 9, p 543)
Governance and political voice	Health is delivered by urban local bodies under the mayor/municipal commissioner while policy and funds sit with the state health department; the urban poor lack the political power to influence decisions that shape their living conditions (Sunder Lal, Ch 2, pp 67–68; Oxford 7e, Ch 58, pp 2151–2152)

Solutions I — restructuring urban primary health care (National Urban Health Mission)

The National Urban Health Mission (NUHM), launched in 2013 under the National Health Mission, is the organised national response. It covers all state capitals, district headquarters and **779 other cities/towns with a population of 50,000 and above (Census 2011)** in a phased manner; towns below 50,000 remain with the National Rural Health Mission (Park 28e, Ch 7, pp 535–536). Its goal is to improve the health of the urban poor through equitable access to quality care, a revamped public health

system, public–private partnership, community-based risk pooling and insurance, and active involvement of urban local bodies (Suryakantha, Ch 37, p 799).

Level	Population norm	Key content
Urban ASHA / link worker	1 per 1,000–2,500 urban poor, about 200–500 households; preferably co-located at the <i>anganwadi</i>	Home visits up to 2 h/day, monthly Health Sanitation and Nutrition Day, escorting pregnant women and children, ward meetings, record keeping; performance-based incentive from a revolving fund held by the ANM at the U-PHC (Park 28e, Ch 7, pp 535–536; Sunder Lal, Ch 17, p 758)
Mahila Arogya Samiti (MAS)	10–12 women per 50–100 households (250–500 population)	Peer-educator group: demand generation, community-based monitoring, referral linkage; annual grant of Rs 5,000; over 62,000 MAS formed (Sunder Lal, Ch 1, p 42; Ch 17, p 758)
ANM outreach	1 ANM per 10,000 population	Weekly outreach sessions in her area; immunisation and antenatal clinics at the U-PHC on other days (Park 28e, Ch 7, pp 535–536)
Urban Ayushman Arogya Mandir (urban health and wellness centre)	15,000–20,000 population	Comprehensive primary health care closer to slums; medical officer, staff nurse/pharmacist, male multipurpose worker, support staff; funded under the 15th Finance Commission health grants and PM-ABHIM; linked administratively to the nearest U-PHC (Park 28e, Ch 7, p 536; Ch 10, p 731)
Urban PHC (U-PHC)	30,000–50,000 (up to 75,000 in very dense areas; 5,000–10,000 for isolated slum clusters), sited within or ≤0.5 km of a slum, catering to about 25,000–30,000 slum people	OPD 12 noon to 8 pm, no inpatient care: consultation, basic laboratory diagnosis, drug and contraceptive dispensing, RCH services, counselling for communicable and non-communicable disease; staff — medical officer (+ part-time second MO), lady health visitor, pharmacist, 3 ANMs, public health manager, support staff and a monitoring and evaluation unit (Park 28e, Ch 7, p 536)
Urban CHC (U-CHC)	1 per 2.5 lakh population (5 lakh in metros), only in cities above 5 lakh	30–50 bedded (100 bedded in metros) in-patient referral facility for maternal complications, trauma, orthopaedics, dental, mental health, deafness, cancer and tobacco cessation (Park 28e, Ch 7, pp 536–537)
Referral tier	—	Urban local body maternity homes, state hospitals, medical colleges and empanelled private hospitals accredited as referral points (Sunder Lal, Ch 17, p 759)

3,350 U-PHCs have been upgraded to urban health and wellness centres, and after COVID-19 — which hit cities disproportionately — the declared paradigm shift is **universal comprehensive primary health care through urban health and wellness centres and polyclinics** (Sunder Lal, Ch 17, pp 746, 758). Revised Indian Public Health Standards (IPHS) 2022 set separate benchmarks for urban facilities — urban CHC, urban Ayushman Arogya Mandir (U-PHC and polyclinic) and urban health and wellness

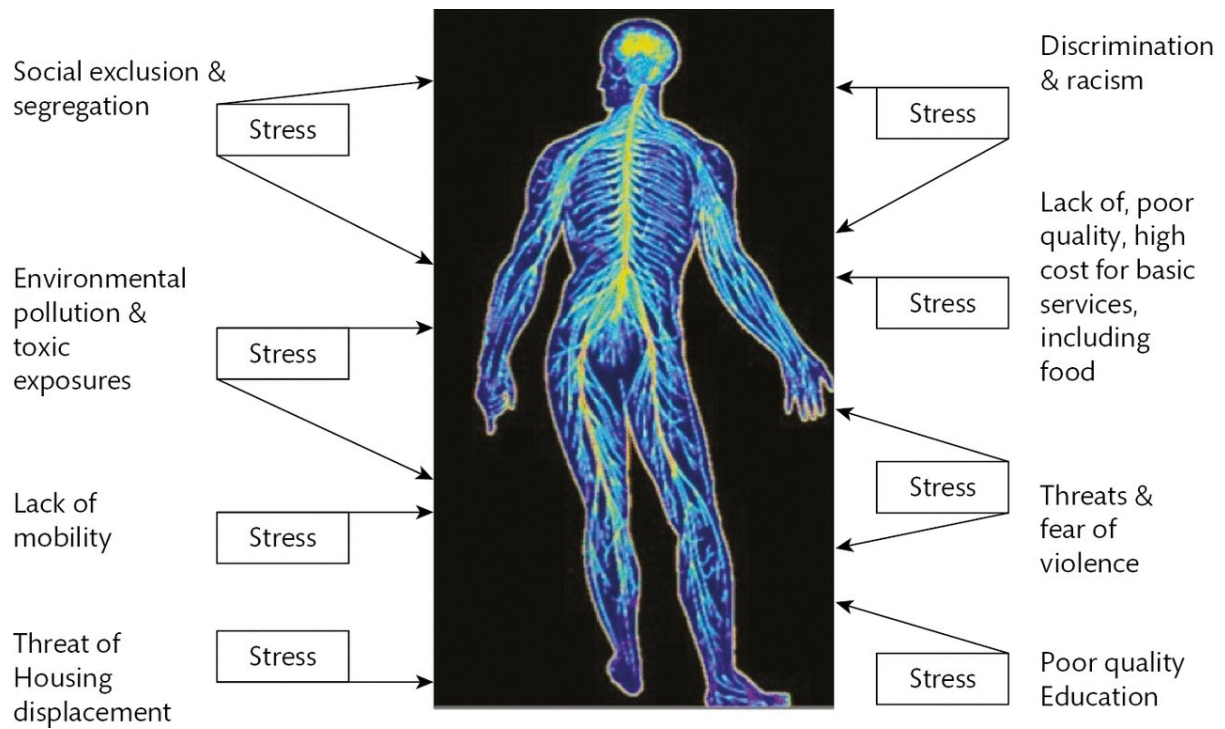
centre — covering infrastructure, human resources, drugs, diagnostics, equipment, quality, monitoring and governance (Park 28e, Ch 10, p 728).

Solutions II — beyond the health facility

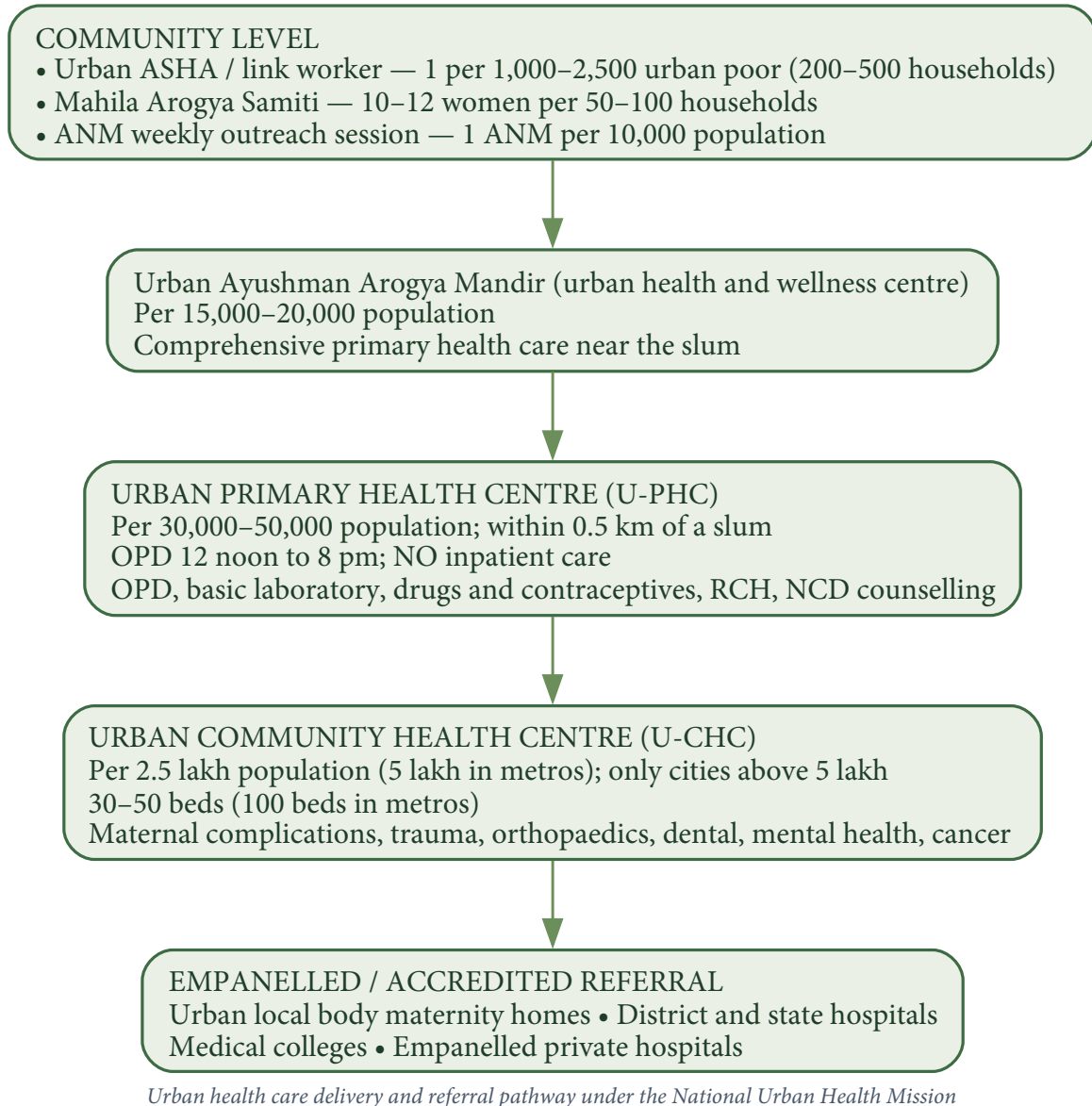
- ▶ **Convergence, because no single agency owns urban health** — with urban local bodies, education, ICDS, HIV/AIDS control, the national health programmes, the Ministries of Urban Development and Housing, urban renewal and slum-upgrading missions, basic services for the urban poor, and water and sanitation (Sunder Lal, Ch 17, pp 759–760).
- ▶ **Community participation as the delivery vehicle** — U-ASHA, Mahila Arogya Samitis and *Rogi Kalyan Samitis* give the community a role in planning and managing services, not merely receiving them (Sunder Lal, Ch 17, p 760).
- ▶ **Urban local body as the unit of planning**, with its own approved norms, the district health society as coordinating body, and geographic information system mapping of slums so unlisted settlements become visible (Sunder Lal, Ch 17, pp 759–760; Suryakantha, Ch 37, pp 799–800).
- ▶ **Co-location of vertical programmes** at the urban health centre — tuberculosis, integrated counselling and testing, AYUSH, disease surveillance and vector-borne disease control under one roof (Suryakantha, Ch 37, p 803).
- ▶ **Financial protection** — NUHM funded 60:40 centre to state (90:10 for north-eastern and special-category states), with subsidised risk pooling and insurance for slum families and self-financed premiums for the non-slum vulnerable (Sunder Lal, Ch 17, p 759; Suryakantha, Ch 37, pp 802–803).
- ▶ **Urban school health services** extended to government and private schools in or near slums within the U-PHC catchment area (Sunder Lal, Ch 17, p 759).
- ▶ **Three-stage monitoring** including community-based monitoring, a health management information system for reporting and feedback, and external evaluation (Suryakantha, Ch 37, p 803).
- ▶ **Healthy Cities approach** — the WHO movement begun after the *Beyond Health Care* conference (Toronto, 1984), re-linking city planning with public health through political commitment and a shared vision, wide community involvement rather than an expert-led agenda, a **city health plan** built on intersectoral partnership, and health in all urban policy (Oxford 7e, Ch 99, pp 3770–3771; Ch 58, p 2159).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **DRAWN figure** — "Urban stressors acting on the slum dweller." Draw a central human outline. Arrows in from the left labelled **Social exclusion and segregation; Environmental pollution and toxic exposures; Lack of mobility; Threat of housing displacement**. Arrows in from the right labelled **Discrimination; Lack of, poor quality or high cost of basic services including food; Threats and fear of violence; Poor quality education**. Mark every arrow "Stress". Caption: *Urban toxic stressors that adversely impact human health.*



Urban toxic stressors that adversely impact human health (Oxford Textbook of Public Health 7e, Ch 58, p 2165)



CHALLENGE: unplanned urbanisation, slum growth ~7%/year, no organised primary tier, fragmented providers, invisible urban poor

1. STRENGTHEN THE PRIMARY TIER

- Urban Ayushman Arogya Mandir per 15,000–20,000
- U-PHC per 30,000–50,000, evening OPD
- U-CHC per 2.5 lakh as first referral
- IPHS 2022 urban benchmarks

2. BUILD COMMUNITY INSTITUTIONS

- Urban ASHA / link worker
- Mahila Arogya Samiti (Rs 5,000/year)
- Rogi Kalyan Samiti
- Community-based monitoring

3. CONVERGE ACROSS SECTORS

- Urban local body as planning unit
- Water, sanitation, housing, slum upgrading
- ICDS, education, urban school health
- Co-locate vertical programmes at the U-PHC

4. PROTECT AND MEASURE

- NUHM 60:40 (90:10 special states) financing
- Risk pooling and insurance for slum families
- Slum mapping incl. unlisted settlements
- HMIS, three-stage monitoring, external evaluation

Four-pillar framework for tackling the current challenges of urban health care

High-yield exam pointers

- **Open with the paradox:** "*poverty in plenty*" — maximum providers, minimum access for the urban poor; WHO World Health Day 2010 theme "*Urban Health Matters*".
- **Dated numbers to write:** urban share 31.16% (Census 2011) → 34.0% (2021); slum population growing ~7%/year; urban birth rate 16.1, death rate 5.1, infant mortality rate 19 per 1,000 (SRS 2022); ~1 billion slum dwellers globally.
- **NUHM population norms are the marks-scoring block:** ASHA 1,000–2,500 · MAS 50–100 households · ANM 10,000 · urban health and wellness centre 15,000–20,000 · U-PHC 30,000–50,000 · U-CHC 2.5 lakh (5 lakh metro, 100 beds).

- ▶ **NUHM coverage rule:** all state capitals and district headquarters + 779 other cities/towns with population $\geq 50,000$ (Census 2011); below 50,000 $\rightarrow$ NRHM. Financing **60:40** (90:10 for north-eastern and special-category states).
- ▶ **Never write "urban indicators are better" without the split** — slum infant mortality exceeds rural infant mortality in Delhi and Mumbai.
- ▶ **Name the burden as triple:** communicable (tuberculosis, acute respiratory infection, diarrhoea, vector-borne) + non-communicable (obesity, hypertension, diabetes, cardiovascular disease) + undernutrition, on top of pollution and road traffic injury.
- ▶ **Close with Healthy Cities** — city health plan, political commitment, health in all urban policies.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam — health and wellness centres have been renamed **Ayushman Arogya Mandir**, so the urban primary facility appears in older texts as "urban health and wellness centre (U-HWC)" and in current documents as "urban Ayushman Arogya Mandir"; the historical predecessors of the present system were the 1,083 urban family welfare centres revamped into 871 urban health posts in 1983, and the sub-centre-equivalent urban tier below the U-PHC is a post-2021 addition funded through the 15th Finance Commission health grants and the Pradhan Mantri Ayushman Bharat Health Infrastructure Mission (PM-ABHIM). Coverage note — the retrieved corpus gives NUHM structure, norms and financing in full, but carries no post-2021 facility-count or NUHM performance evaluation data, so no coverage percentages are quoted here.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-4 and NFHS-5, CAMS 2022–23 (NSS 79th round), WHO Healthy Cities Programme, Lancet Global Health 2024 urban health Series

Programme and policy updates (India)

- ▶ **Ayushman Bharat Digital Mission (ABDM)** — **the renaming to memorise:** the National Digital Health Mission launched on 15 August 2020 was renamed the Ayushman Bharat Digital Mission on 27 September 2021, and its ABHA (Ayushman Bharat Health Account) digital health ID is the vehicle for nationwide electronic health records. In the 2020–21 pilot year, run only in the Union Territories (Chandigarh among them), digital health ID uptake among 419 adults at an urban primary health centre and its outreach screening camps was 78% (327), yet only 28.2% had even heard of electronic health records; the barriers were lack of awareness, low digital literacy and perceived lack of utility. [PMID 39382504]
- ▶ **PM-JAY's "card-to-care" gap in an urban slum:** in a 2023 community survey of 273 adults, one per household, in a coastal slum of Puducherry, 59% were aware of the scheme and 65% were eligible, but of the eligible households only 49% had registered and 14% had ever used the benefit — a difficult registration procedure and thin empanelment and treatment-package coverage were the reported obstacles. Entitlement on paper is not access. [PMID 41805171]
- ▶ **Surgical care as the neglected tier of slum health:** the case for treating "slum surgical health" as a field distinct from urban health. In an Ahmedabad slum pilot covering 12,730 people in 3,000

households over 6 months, 1.31% (167 people) had unmet surgical needs, and specially trained female community health workers who themselves live in the slum (*SATHIs*) moved 99 of them (59.3%) into met needs; the model has since been scaled to 350,000 slum dwellers in 70,000 homes. It is proposed alongside slum legalisation, sanitation and primary care, plus a baseline of the Lancet Commission on Global Surgery indicators under a National Surgical, Obstetric and Anaesthesia Plan. [PMID 39872131]

Current national data

- ▶ **CAMS 2022–23 (Comprehensive Annual Modular Survey, NSS 79th round) — 3,02,086 households, of which 1,28,990 urban:** publicly funded health insurance reached only 30% of urban households against 54.5% of rural, while urban inpatient out-of-pocket expenditure ran higher (Rs 42,970 vs Rs 30,688 rural). All-India — the models adjust for urban/rural residence rather than stratifying by it — insurance was associated with higher outpatient use (adjusted OR 1.404, 95% CI 1.328–1.484) and lower inpatient catastrophic expenditure at the 20% threshold (adjusted OR 0.790, 95% CI 0.737–0.846), but outpatient out-of-pocket spending stayed high even among insured households; the stated remedy is to extend the benefit package to outpatient care anchored on Ayushman Arogya Mandirs. [PMID 42167946]
- ▶ **NFHS-4 → NFHS-5 (2015–16 to 2019–21):** non-contributory public health insurance coverage rose only 8.0 percentage points in urban India (19.6% → 27.6%; 95% CI 7.3–8.7) against 11.7 pp in rural India, and the pre-existing pro-poor targeting in cities actually weakened ($p = 0.025$); in urban Uttar Pradesh the poorest two-fifths gained no more than everyone else, which the authors read as possible failure to target the urban poor. Urban PM-JAY eligibility rests on 11 occupational categories from SECC 2011 — categories the survey did not record, so urban targeting could not be assessed directly. [PMID 37640493]
- ▶ **Slum versus non-slum breastfeeding — NFHS-4 (2015–16), the seven states where slums were sampled, 933 slum vs 2,315 non-slum mother–child pairs:** early initiation of breastfeeding was numerically higher in slums (50.4%, 95% CI 47.2–53.7) than in urban non-slum areas (37.4%, 95% CI 35.4–39.4), but not statistically significant ($p = 0.768$). The one significant gap ran *against* the slum — exclusive breastfeeding 50.0% (95% CI 42.5–57.7) versus 55.8% (95% CI 51.5–60.0), $p = 0.004$ — with over 90% institutional delivery in both. Write the disaggregated split rather than the city average, and quote it as 2015–16 seven-state data, not a current national figure. [PMID 41950202]

Elimination, surveillance and disease status

- ▶ **Tuberculosis elimination needs urban community engagement, not only better diagnostics:** a four-week tele-education package (short videos, audio modules, quizzes, ASHA-led reinforcement) delivered to 112 households across an urban slum cluster and a tribal village network raised TB literacy and generated self-initiated referrals and higher screening uptake; the slum showed greater digital participation while the tribal site matched it through a blended offline model. [PMID 42362243]

Guideline and legislative updates

- ▶ **The Community Based Assessment Checklist (CBAC) underperformed on validation:** the frontline screening tool the Government of India introduced in 2016 for ASHAs and ANMs — under what was NPCDCS, renamed the National Programme for Non-Communicable Diseases

(NPNCD) in 2023 — was tested in a 2019–20 population-based screening in the urban primary health centre areas of Mysuru, on 32,686 adults aged 30 years and above with no known diabetes, hypertension or cancer. Area under the curve was 0.62 (95% CI 0.61–0.63) for diabetes risk and 0.61 (95% CI 0.60–0.61) for hypertension, and at the current cut-off score of more than 4 the sensitivity was 63.0% with specificity 54.4% (Youden's index 0.17 for diabetes, 0.15 for hypertension). The recommendation is that trained frontline workers perform a random capillary blood glucose and a blood-pressure reading alongside the checklist rather than rely on the score to triage. [PMID 42555696]

Global and WHO developments

- **Lancet Global Health 2024 Series on urban primary health care in South Asia reframes the problem:** geographical access to services inside cities is easy, but financial hardship from using care and poor quality of care are what defeat low-income residents; the prescription is effective and adequately resourced urban local bodies, increased public financing for health care, and service-delivery models designed for low-income urban communities, including structured government engagement with the private providers those communities already use. [PMID 39178875]
- **WHO Healthy Cities Programme now carries an Indian target:** the programme began in 1984, and in 2021 the Government of India recommended developing **500 Healthy Cities by 2030** in alignment with it; the accompanying analysis of ten global and international health-promotion conferences argues that the classical determinants-of-health models under-capture the newer legal, technological and commercial determinants and the spatial scales that now shape urban health. [PMID 41032407]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India — National Urban Health Mission, pp 535–537; Ch 10 Health Care of the Community, pp 728, 731, 746; Ch 15 Environment and Health, p 908); Sunder Lal — Textbook of Community Medicine (Ch 17 Healthcare of the Community, pp 743, 746, 756–760; Ch 1, pp 42, 52; Ch 2, pp 67–68; Ch 9, p 543; Ch 10, pp 589–590); Oxford Textbook of Public Health 7e (Ch 58 Urbanization and Health, pp 2151–2152, 2159–2160, 2165; Ch 6 Determinants of Health, pp 304–305; Ch 4, pp 237–238; Ch 99, pp 3770–3771); Suryakantha — Community Medicine with Recent Advances (Ch 37 National Urban Health Mission, pp 799–803); National Urban Health Mission 2013; Indian Public Health Standards 2022; Census of India 2011; Sample Registration System Bulletin 2022.

Recent-advances sources: 39382504 — Perception towards electronic health records & uptake of digital health IDs among the urban residents in northern India: a mixed methods study, *Indian J Med Res* 2024, doi:[10.25259/ijmr.664.23](https://doi.org/10.25259/ijmr.664.23); 41805171 — From card to care: gaps in implementation of publicly funded health insurance among a vulnerable population in South India, *J Public Health (Oxf)* 2026, doi:[10.1093/pubmed/fdag021](https://doi.org/10.1093/pubmed/fdag021) (abstract-only); 39872131 — Challenges to achieving surgical equity in slums, *Int J Public Health* 2025, doi:[10.3389/ijph.2024.1608098](https://doi.org/10.3389/ijph.2024.1608098); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022–2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 37640493 — Public health insurance coverage in India before and after PM-JAY: repeated cross-sectional analysis of nationally representative survey data, *BMJ Glob Health* 2023, doi:[10.1136/bmjgh-2023-012725](https://doi.org/10.1136/bmjgh-2023-012725); 41950202 — Prevalence and social determinants of breastfeeding practices in urban slums and urban non-slum areas in India: a comparative analysis, *PLoS One* 2026, doi:[10.1371/journal.pone.0323861](https://doi.org/10.1371/journal.pone.0323861); 42362243 — Use of tele-education for strengthening community participation in tuberculosis control, *Indian J Tuberc* 2026, doi:[10.1016/j.ijtb.2026.02.020](https://doi.org/10.1016/j.ijtb.2026.02.020) (abstract-only); 42555696 — Screening performance of community-based assessment checklist compared with point of care screening tests to identify high-risk individuals for diabetes and hypertension in urban Mysore, India, *Indian J Med Res* 2026, doi:[10.25259/IJMR.78.2026](https://doi.org/10.25259/IJMR.78.2026); 39178875 — Improving urban health through primary health care in south Asia, *Lancet Glob Health* 2024, doi:[10.1016/S2214-109X\(24\)00121-9](https://doi.org/10.1016/S2214-109X(24)00121-9) (abstract-only); 41032407 — Addressing global health challenges: a comprehensive framework for determinants of health, *Int J Inj Contr Saf Promot* 2025, doi:[10.1080/17457300.2025.2566336](https://doi.org/10.1080/17457300.2025.2566336) (abstract-only).



Public-private partnership – urban health centres

Asked in: Paper IV Nov 2016 — RGUHS MD Community Medicine

Definition — Urban health centres are the government primary-care facilities organising preventive, promotive and ambulatory curative care for the urban poor — the urban health and wellness centre (UHWC), the urban primary health centre (U-PHC), the specialist U-PHC/polyclinic and the urban community health centre (U-CHC) — under the **National Urban Health Mission (NUHM), 2013** (Sunder Lal, Ch 17, p 756). A **public-private partnership (PPP)** in health is the bringing together of a set of actors for the common goal of improving the health of populations based on mutually agreed roles and principles (World Health Organization definition, cited in Oxford Textbook of Public Health 7e, Ch 106, p 4021); *Reich* adds three points — at least one for-profit with at least one not-for-profit body, joint sharing of efforts and benefits, and commitment to creating social value for the disadvantaged (Oxford 7e, Ch 51, p 1916).

Why urban health centres were needed

Problem	Evidence
Migration drives unplanned slum growth; overcrowding, poor sanitation, unsafe drinking water, environmental pollution	Health indices of urban slum dwellers in some areas are worse than those of the rural population (Sunder Lal, Ch 17, p 756)
"Poverty in plenty" — abundant private practitioners, nursing homes, specialists and non-governmental organisations (NGOs) in cities, yet the urban poor have poor access	Sunder Lal, Ch 17, p 756
No parallel to the rural three-tier system — sub-centres were never established in urban areas	Sunder Lal, Ch 17, p 758
Urban share of India's population rising: 27.8% (2001) → 31.16% (2011) → 34.0% (2021)	Sunder Lal, Ch 9, p 543
Urban health recognised as a "thrust area" by National Health Policy 2002, National Population Policy 2000, Eleventh Five-Year Plan and Reproductive and Child Health (RCH)-2	Sunder Lal, Ch 17, p 756
National Health Policy 2017 shift — <i>urban health</i> : "from token interventions to on-scale assured interventions " for the urban poor	Park 28e, Ch 10, p 728

Evolution of urban health centres in India

Year	Development
1983	1,083 urban family welfare centres revamped into 871 urban health posts to cover the urban slum population (Sunder Lal, Ch 17, p 756)
Urban health post types	Type A <5,000; Type B 5,000–10,000; Type C 10,000–25,000; Type D 25,000–50,000. Beyond 50,000 the area is divided into sectors of 50,000. Types A, B, C are attached to a hospital for referral and supervision; Type D is attached for sterilisation, medical termination of pregnancy (MTP) and referral, and only Type D carries a post of medical officer (Park 28e, Ch 8, p 611)

Year	Development
2013	NUHM launched — affordable primary health care through U-PHCs, U-CHCs and outreach for the urban slum population (Sunder Lal, Ch 17, p 746)
2018 onwards	Under Ayushman Bharat, existing U-PHCs strengthened as Ayushman Arogya Mandirs (Park 28e, Ch 7, p 536); 3,350 of the U-PHCs upgraded to urban health and wellness centres (Sunder Lal, Ch 17, p 758)
Post-COVID-19	Paradigm shift to universal comprehensive primary health care through UHWCs and polyclinics; Urban Ayushman Arogya Mandirs (U-AAM) for 15,000–20,000 population approved under the 15th Finance Commission health grants and PM-ABHIM , linked to the nearest U-PHC for administrative, financial, reporting and supervisory purposes (Park 28e, Ch 7, p 536; Sunder Lal, Ch 17, p 746)

Coverage of NUHM: all State capitals, district headquarters and about **779 other cities/towns of population 50,000 and above (Census 2011)**, phased; the rest fall under the National Rural Health Mission. **Target groups:** urban poor in listed and unlisted slums, plus homeless people, street children, rickshaw pullers, construction and lime-kiln workers, sex workers and temporary migrants (Park 28e, Ch 7, pp 535–536).

Structure, population norms and beds

Facility	Population norm	Beds	Key features
UHCW / U-AAM	1 per 15,000–20,000 (no hilly/tribal norm)	Nil	First point of contact; comprehensive primary health care; team led by a medical officer (unlike the rural sub-centre, led by a community health officer) (Sunder Lal, Ch 17, p 746)
U-PHC (HWC-U-PHC)	30,000–50,000 (Park 28e, Ch 7, p 536); Sunder Lal, Table 17.5, p 749 states 50,000. Cities may fix 75,000 for very high-density areas, or 5,000–10,000 for isolated slum clusters	2 day-care beds essential, aspire to 4 ; no inpatient care (U-PHCs already conducting deliveries continue as 24 × 7 with 24 × 7 PHC infrastructure)	Sited preferably within a slum or within half a kilometre of one , serving a slum population of 25,000–30,000; OPD 12 noon to 8 pm (Sunder Lal, Ch 17, p 758)
Specialist U-PHC / polyclinic	2.5–3 lakh, covering the catchment of 5–6 U-PHCs (Sunder Lal, Ch 17, pp 749, 759)	2 essential + 4 desirable day-care beds	Day-care/ambulatory specialist care: fixed-day rotational multispecialty OPD in a minimum of six specialties — Medicine, Obstetrics & Gynaecology, Paediatrics, Ophthalmology, Dermatology, Psychiatry (states may start with 2–3 and expand); oral, physiotherapy and optometry services also planned (Sunder Lal, Ch 17, pp 758–759)

Facility	Population norm	Beds	Key features
U-CHC	2.5 lakh (non-metro); 5 lakh (metro); set up as a satellite hospital for every 4–5 U-PHCs, in cities of population above 5 lakh	30–50 beds (commonly written as 50-bedded); 100 beds in metros	Secondary-care referral centre — inpatient medical and surgical services, institutional deliveries; functions only as a first referral unit (FRU) (Sunder Lal, Ch 17, pp 752–753, 759; Park 28e, Ch 7, Fig 12, p 536)

Numbers achieved: recurrent cost support of Rs 20 lakh per year covers contractual staff of 6,118 urban PHCs, all being converted to health and wellness centres in phases; 350 U-CHCs set up so far (Sunder Lal, Ch 17, p 759).

Service package at urban health centres

The twelve packages envisaged under comprehensive primary health care (CPHC) delivered at the HWC-PHC/UHWC-PHC (Sunder Lal, Ch 17, pp 749–750):

#	Package	#	Package
1	Care in pregnancy and childbirth	7	Screening, prevention, control and management of non-communicable diseases
2	Neonatal and infant health care services	8	Care for common ophthalmic and ENT problems
3	Childhood and adolescent health care services	9	Basic oral health care
4	Family planning, contraceptive services and other reproductive health care services	10	Elderly and palliative health care services
5	Management of communicable diseases: national health programmes	11	Emergency medical services including burns and trauma
6	General out-patient care for acute simple illnesses and minor ailments	12	Screening and basic management of mental health ailments

Services provided at the U-PHC (Sunder Lal, Ch 17, pp 758–759) — the twelve CPHC packages, plus: outpatient consultation; services prescribed under RCH-II; national health programme services; collection and reporting of vital events and Integrated Disease Surveillance Programme (IDSP); referral; basic laboratory services; counselling; essential medicines and contraceptives; services for communicable and non-communicable diseases; social mobilisation and community-level activities. Tuberculosis, leprosy and non-communicable disease screening have been added (Park 28e, Ch 7, p 535). Post-COVID-19, **oxygen support by cylinder or concentrator** is an essential infrastructure requirement at U-PHCs (Sunder Lal, Ch 17, p 749, Table 17.6).

Staffing and standards: medical officer, staff nurse, lady health visitor, pharmacist, three auxiliary nurse midwives (ANMs) and a public health manager/mobilisation officer; ANMs report at the U-PHC and move out for outreach on designated days, holding immunisation and antenatal clinics there on other days (Park 28e, Ch 7, p 535, Fig 12 p 536). Minimum performance standards, Indian Public Health Standards (IPHS) 2022: medical officer **75 OPD/day**, dentist **20/day**, laboratory technician **100**

tests/day, pharmacist 120 prescriptions/day, staff nurse 1 per 6 beds; polyclinic specialist 60/day (psychiatrist 20–30) (Sunder Lal, Ch 17, pp 749, 759).

Community-level components:

Component	Norm and role
Urban ASHA (U-ASHA) / link worker	One per 1,000–2,500 urban poor population / 200–500 households; preferably co-located at the <i>anganwadi</i> ; home visits, monthly Health Sanitation and Nutrition Day, escorting pregnant women and children to the urban health centre, ward-level meetings, record keeping; performance-based incentive (Park 28e, Ch 7, p 535; Sunder Lal, Ch 17, p 758)
<i>Mahila Arogya Samiti</i> (MAS)	Women's health committee for 50–100 slum households (population 250–500); peer educator group, community-based monitoring, referral linkage; states may transfer ASHA responsibilities and incentives to the MAS (Park 28e, Ch 7, pp 535–536)
Urban health and nutrition day (UHND)	ANM outreach sessions delivering preventive, promotive and curative care to slum and vulnerable households (Sunder Lal, Ch 17, p 758)
<i>Rogi Kalyan Samiti</i>	Facility-level committee for planning and management; untied fund Rs 75,000–1,25,000 per U-PHC per year (Park 28e, Ch 7, p 535; Sunder Lal, Ch 17, p 759)

Governance and financing: urban local bodies (ULBs) are the **units of planning** and the district health society the coordinating body; very large municipal corporations receive funds directly from the state health society. Centre–state cost sharing is **60:40 (90:10 for North-Eastern and special-category states)**. Quality standards for U-PHCs (Ministry of Health and Family Welfare, 2016) are now superseded by the Operational Guidelines for quality under IPHS 2022 (Sunder Lal, Ch 17, pp 759–760).

Public-private partnership: rationale and models

Basis	Content
Rationale	The non-governmental health sector accounts for nearly 70% of health expenditure in India ; in the absence of an effective public system most households seek care from it (Sunder Lal, Ch 17, p 739)
Policy anchor	" People Public-Private Partnership (PPPP) for Public Health Goals " is Component (E) of the National Rural Health Mission; accreditation and PPP development is a core strategy (Sunder Lal, Ch 17, pp 738–739)
Predominant Indian model	Contracting — contract-in, contract-out, outsourcing, management of hospital facilities by leading NGOs, hiring staff, service delivery including family planning, MTP and treatment of sexually transmitted/reproductive tract infection (Sunder Lal, Ch 17, p 739)
Strategic purchasing	Government acts as single payer , purchasing care from private and public hospitals; National Health Policy 2017 seeks improved access to secondary and tertiary care "through a combination of public and private providers" (Sunder Lal, Ch 16, p 697)
Social marketing and social franchising	Commercial marketing techniques applied to increase outlets and improve product/service quality: condoms rose from 16 million pieces (1968–69) to 483.21 million (2017–18) and oral pills from 7.24 lakh cycles (1987–88) to 206.31 lakh cycles (2017–18) , each about one-third of national distribution (Sunder Lal, Ch 4, pp 161–162)

Basis	Content
Partnership with NGOs and professional bodies	Collaboration with NGOs decentralised to states under RCH-II; the Indian Medical Association and the Federation of Obstetricians and Gynaecologists are partners (Sunder Lal, Ch 17, pp 765–766)
Publicly funded health insurance	AB-PM-JAY (launched 23 September 2018, subsuming Rashtriya Swasthya Bima Yojana) — Rs 5 lakh per family per year for secondary and tertiary hospitalisation, 10.74 crore families (~50 crore people) , cashless and portable across empanelled public and private hospitals, 1,685 procedures , no cap on family size/age/gender, all pre-existing conditions covered, 3 days pre- and 15 days post-hospitalisation care (Park 28e, Ch 10, p 745)
Product-development PPP	Global form — the first was the International AIDS Vaccine Initiative for vaccine development (Oxford 7e, Ch 51, p 1916)

Public-private partnership applied to urban health centres

- **Referral empanelment** — ULB maternity homes, state government hospitals, medical colleges and private hospitals are empanelled/accredited as referral points for diabetes, trauma, orthopaedic, dental, mental health, critical illness, deafness control, cancer and tobacco cessation (Sunder Lal, Ch 17, p 759).
- **Contracting-out of U-PHC management** to NGOs and private providers — the predominant PPP model in India (Sunder Lal, Ch 17, p 739).
- **School health services** cover government and private schools in or near slums within the U-PHC catchment (Sunder Lal, Ch 17, p 759).
- **Programme partnership** — under the National Tuberculosis Elimination Programme the principle is to **extend public services to privately managed patients**, using Standards for TB Care in India, mandatory notification, *Nikshay* and the ban on serodiagnostics; regulatory tools are limited, so **partnership is preferred** (Park 28e, Ch 7, pp 514–515).
- **Convergence** with ULBs, education, Integrated Child Development Services, HIV/AIDS control, urban development and housing, water and sanitation, and women and child development — essential because multiple agencies operate in cities (Sunder Lal, Ch 17, pp 759–760).
- **Universal health coverage rationale** — governments cannot reach universal health coverage alone; engagement takes the form of capacity development, innovation, outsourcing and consulting, with varying degrees of financial risk sharing (Oxford 7e, Ch 21, pp 834–836).

Strengths and limitations of the two sectors

Reproduced from the public-versus-private table in Park 28e, Ch 7, p 514 (illustrated for tuberculosis).

	Public sector	Private sector
Advantages	Free diagnosis; free treatment; standardised regimen; referral and transfer system; supervision and monitoring; accountability of treatment outcome	Wide choice (>5 lakh practitioners); better access — convenient timings, shorter distances, personal attention and care, projected discounts; faith and perceptions of better care

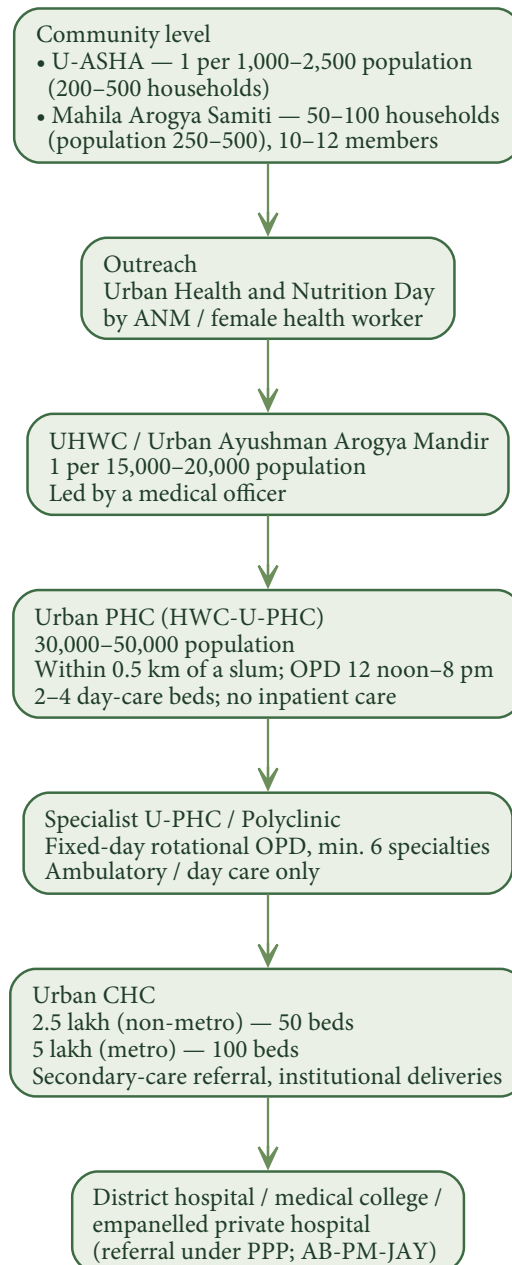
	Public sector	Private sector
Disadvantages	Staff's non-response to symptoms; delays between tests and receiving results; difficulty in transporting specimens; financial expenditure on travel, food, daily necessities and extra medicines; perceived low quality of services	Cost of clinical examination fees; cost of diagnostic tests; cost of drugs; irrational prescriptions; infrequent use of quality sputum tests for diagnosis of tuberculosis; no adherence-tracking mechanisms; fear of losing the patient if involved in the programme

Three further points frame the case for partnership:

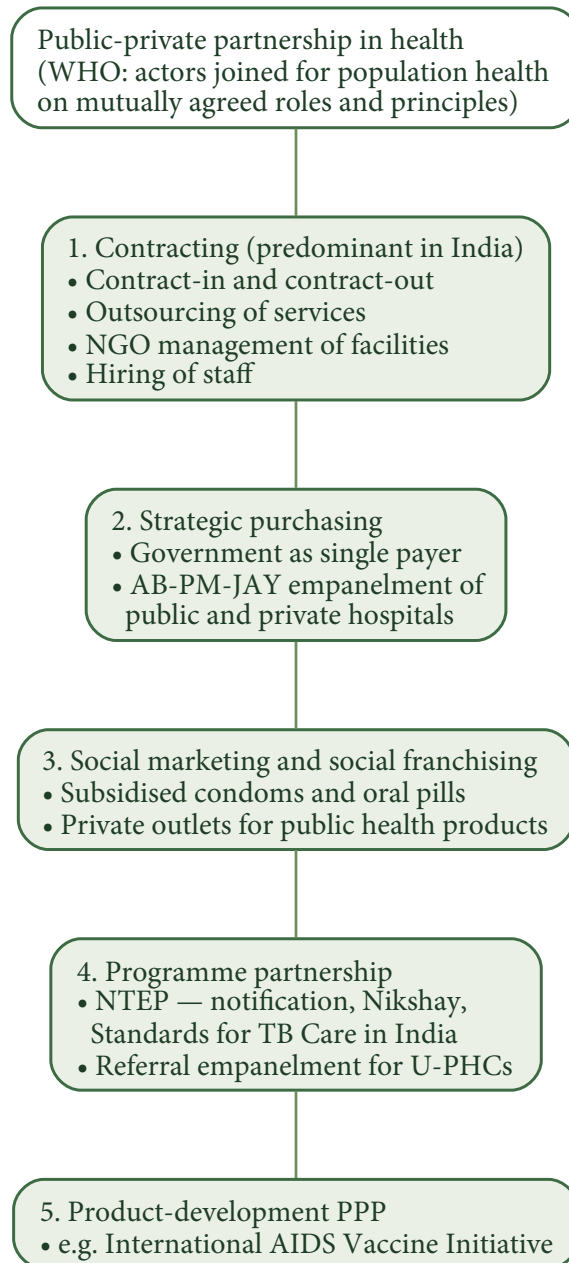
- **Quality concerns in the private sector** — delays in diagnosis, over-diagnosis from over-dependence on X-rays, multiple non-standard regimens for inappropriate durations, and no mechanism to ensure a full course of treatment or to record outcomes; "similar problems in varying degrees are encountered in other health sectors as well" (Park 28e, Ch 7, pp 513–514).
- **Narrow public service package** — even a well-functioning primary health centre presently provides services limited to reproductive, sexual and child health plus some national disease control programmes, together **less than 15% of all services** people seek; for the rest they resort to a private provider or a crowded district/medical college hospital (Sunder Lal, Ch 17, p 749). The twelve CPHC packages are the corrective.
- **Generic risks of PPP** (Oxford 7e, Ch 106, p 4033) — conflicts of interest; difficulty assessing partnership compatibility and evaluating outcomes; costlier private borrowing passed on to the public; limited transparency and government oversight during procurement; contract clauses that block later changes needed to safeguard the public interest. **Regulatory frameworks and accreditation systems are therefore prerequisites** for partnership (Oxford 7e, Ch 21, pp 834–836).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Labelled diagram to sketch:** *Urban health care facilities under NUHM* — draw an inverted pyramid with, from base upward: community (U-ASHA, *Mahila Arogya Samiti*), outreach (ANM, UHND), UHCW (15,000–20,000), U-PHC (30,000–50,000), polyclinic (2.5–3 lakh), U-CHC (2.5 lakh non-metro / 5 lakh metro), district hospital. Label the right margin with population norms and the left margin with the personnel at each level.



Tiers of urban primary health care delivery under NUHM, with population norms



Models of public-private partnership used in Indian health care

High-yield exam pointers

- **Population norms to write verbatim:** UHWC 15,000–20,000 → U-PHC 30,000–50,000 (75,000 in high-density; 5,000–10,000 for isolated slum clusters) → polyclinic 2.5–3 lakh (5–6 U-PHCs) → U-CHC 2.5 lakh non-metro (30–50 beds) / 5 lakh metro (100 beds), one for every 4–5 U-PHCs. No sub-centres exist in urban areas.
- **1983:** 1,083 urban family welfare centres → **871 urban health posts**; four types A (<5,000), B (5,000–10,000), C (10,000–25,000), D (25,000–50,000) — only Type D has a medical officer.
- **NUHM 2013** covers State capitals, district headquarters and ~779 towns of ≥50,000 (Census 2011); centre–state share **60:40** (90:10 for North-East and special-category states); **6,118 U-PHCs** (of which 3,350 upgraded to UHWCs) and **350 U-CHCs** established.

- ▶ **Twelve CPHC packages** — pregnancy/childbirth → neonatal/infant → childhood/adolescent → family planning → communicable disease → general OPD → non-communicable disease → eye and ENT → oral → elderly/palliative → emergency (burns and trauma) → mental health.
- ▶ **U-ASHA 1:1,000–2,500 population; MAS = 50–100 households; U-PHC OPD 12 noon–8 pm; medical officer 75 OPD/day, specialist 60/day.**
- ▶ **PPP one-liner:** private sector = ~70% of health expenditure in India; contracting is the predominant model; PPPP is **Component (E)** of NRHM.
- ▶ **AB-PM-JAY:** Rs 5 lakh/family/year, 10.74 crore families, 1,685 procedures, launched 23 September 2018.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the two spine texts differ on the U-PHC population norm — Park's Textbook of Preventive and Social Medicine 28e gives 30,000–50,000 while Sunder Lal's Table 17.5 states 50,000 — so quote the Park range and mention the variation in one line.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, National Health Accounts 2021–22, NSS Comprehensive Annual Modular Survey 2022–23

Programme and policy updates (India)

- ▶ **Health and wellness centres are now Ayushman Arogya Mandirs (AAM), tagline *Arogyam Parmam Dhanam*** — more than 1.75 lakh AAM operationalised nationwide (rural sub-centres and PHCs plus urban PHCs upgraded); urban infrastructure gap-filling to IPHS 2022 is financed through PM-ABHIM and XV Finance Commission grants routed via urban local bodies (rural through panchayati raj institutions); **1,16,978 community health officers** in position (30 Nov 2024); all twelve expanded packages rolled out at **82% of AAM by Nov 2024** (50% in March 2023); free drugs and diagnostics widened to **172 medicines and 63 tests at the AAM-PHC / urban PHC** (106 medicines, 14 tests at AAM sub-health centre); **1,35,343 Jan Arogya Samitis** constituted as the facility-level people's participation platform, holding *Arogya Sabha* twice yearly. Write "Ayushman Arogya Mandir / U-AAM", not "HWC", in 2026 answers [PMID 41816146].
- ▶ **eSanjeevani is now the teleconsultation arm of every urban health centre** — over 163 million consultations between Nov 2019 and Sep 2023 across all 28 states and 8 union territories, of which **more than 93% used the provider-assisted AB-HWC hub-and-spoke model** (community health officer at the AAM/U-PHC spoke connects to a doctor at a district or sub-district hospital hub, giving a digitally enabled referral pathway); infrastructure grew to **14,036 hubs and 108,610 spokes**; women were the majority of users and the 25–45 year group the largest; version 2.0 (March 2023) shifted use towards diabetes and hypertension follow-up [PMID 41158553].

Current national data

- ▶ **The "private sector = 70% of health expenditure" figure is superseded** — National Health Accounts show the **government share of total health expenditure rose from 28.6% (2013–14) to 48% (2021–22)** while the **out-of-pocket share fell from 64.2% to 39.4%**, the two lines crossing by

the end of the period; government health expenditure is still only ~1.9% of GDP against the National Health Policy 2017 target of 2.5%. Critically for urban primary care, catastrophic spending is commoner for outpatient (47.8%) than inpatient (43.1%) care, and medicines alone are about two-thirds of out-of-pocket spending — the burden sits precisely where U-PHC free drugs and diagnostics operate, not where PM-JAY operates [PMID 42548603].

- CAMS 2022–23 (NSS 79th round, 302,086 households) is the current national insurance dataset — 59.9% of households insured, 49.7% under publicly funded health insurance; insured households had higher outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient (adjusted OR 1.168, 95% CI 1.128–1.210) utilisation; mean inpatient out-of-pocket expenditure was Rs 29,875 under AB-PM-JAY/state schemes versus Rs 37,572 uninsured, and inpatient catastrophic expenditure at the 10% threshold 41.3% versus 54.2%. Outpatient catastrophic expenditure was *not* reduced (39.2% insured vs 38.1% uninsured) — the authors' remedy is to anchor outpatient cover in strengthened Ayushman Arogya Mandirs [PMID 42167946].

Elimination, surveillance and disease status

- Lymphatic filariasis — elimination target now 2030, still endemic in 345 districts; the argument is that mass drug administration, morbidity management and post-MDA surveillance strain public infrastructure alone, and that PPP (proven in polio eradication and TB control in India) is under-used for advocacy and social mobilisation, tackling MDA non-compliance, post-MDA surveillance, patient identification and lymphoedema care, digital innovation, operational research and vector control [PMID 41951149].
- Tuberculosis — the Kochi Declaration 2024, from the International Leadership Summit of Medical Associations, puts private-sector engagement at the centre of TB elimination because 60–80% of patients in high-burden countries first seek care from a private provider; it aligns with the WHO public–private mix approach and the UN High-Level Meeting targets for a TB-free world by 2030 — the modern citation for the NTEP "extend public services to privately managed patients" principle in the body [PMID 42396744].

Guideline and legislative updates

- PM-JAY's health benefit package has been revised well beyond the 1,685-procedure figure — HBP 1.0 listed 1,350 procedures (21.5% surgical); HBP 2022 lists over 1,949 procedures, of which 1,284 (65.8%) are surgical, and ~82% of empanelled hospitals provide surgical care; 26,137 hospitals empanelled (47% private, 53% public, roughly 2 per 100,000 population), centre–state split 60:40 (90:10 for North-Eastern states, J&K, Himachal Pradesh and Uttarakhand), with a reported annual gain of ~Rs 1.19 crore per district hospital. Equity gaps to quote as the PPP critique: only 7% of hospitals in Aspirational Districts are empanelled for cardiothoracic and vascular surgery versus 17% elsewhere, and men raised 23.6% more surgical claims than women once obstetric and gynaecological surgery is excluded [PMID 41393311].
- Enrolment is not access, and insurance alone does not cut out-of-pocket cost — a pan-India instrumental-variable analysis found no statistically significant reduction in out-of-pocket expenditure attributable to publicly funded health insurance, even in high-coverage states, with private-hospital admissions carrying higher expenditure and catastrophic-expenditure incidence despite cover; the authors recommend gate-keeping (private admission only on referral from a

public facility) and prioritising public hospitals [PMID 41913189]. Field data agree: of 320 verified PM-JAY beneficiaries in Jammu and Baramulla, only **48.4% had ever used the scheme**, with awareness of entitlement (OR 1.253) and knowledge of the empanelled-hospital list (OR 1.245) predicting use and poor road quality reducing it (OR 0.761), while financial and documentation barriers were not independent predictors [PMID 41878109].

Global and WHO developments

- **Urban primary health care in South Asia — the current framing for this exact question** — cities give easy *geographical* access to primary care, but financial hardship at the point of use and poor quality of care remain the binding constraints for low-income residents, and health and its social determinants vary sharply within and between South Asian cities. The prescribed response is multi-sectoral: **effective and adequately resourced urban local bodies**, increased public financing, and service-delivery models built for low-income urban communities that strengthen public provision, engage private providers where necessary, and work with the informal providers those communities already use [PMID 39178875].
- **Private-sector engagement in immunisation — 25 years of LMIC evidence** — the private share of childhood vaccination ranges from **3–4% to over 60%** (India 9–36% for routine vaccines, and 45–65% of *new* vaccines in urban for-profit settings); **zero-dose children cluster in urban-poor, remote and fragile settings** where non-state providers dominate. What works: formal public–private agreements with accreditation and licensing conditions, government supply of vaccines and cold-chain equipment, and inclusion of private providers in the national immunisation information system for coverage and AEFI reporting. The main risks are non-adherence to the national schedule and **financial exclusion through ancillary registration and service fees** even when the vaccine itself is free [PMID 39542515].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India, pp 513–515, 535–536; Ch 8 Demography and Family Planning, p 611; Ch 10 Health Care of the Community, pp 728, 745); Sunder Lal — Textbook of Community Medicine (Ch 4 pp 161–162; Ch 9 p 543; Ch 16 p 697; Ch 17 pp 738–739, 746, 749–753, 756, 758–760, 765–766); Oxford Textbook of Public Health 7e (Ch 21 pp 834–836; Ch 51 p 1916; Ch 106 pp 4021, 4033). Programme and guideline documents named: National Urban Health Mission 2013; National Health Policy 2017; Indian Public Health Standards 2022; Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana 2018; Census 2011.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: A narrative review, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmmpc.jfmmpc.478.25](https://doi.org/10.4103/jfmmpc.jfmmpc.478.25); 41158553 — Adoption and utilization of India's eSanjeevani national telemedicine service, *Oxford Open Digital Health* 2025, doi:[10.1093/oodh/oqaf025](https://doi.org/10.1093/oodh/oqaf025); 42548603 — Can a bigger budget shrink the bill at the bedside? Public health financing as a route to lower out-of-pocket spending in India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1903360](https://doi.org/10.3389/fpubh.2026.1903360); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022–2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42396744 — The Kochi Declaration 2024 — A catalyst for global TB elimination through private sector engagement, *International Journal of Tuberculosis and Lung Disease* 2025, doi:[10.5588/ijtld.25.0192](https://doi.org/10.5588/ijtld.25.0192) (abstract-only); 41393311 — Assessing Surgical Care under the Government-Funded Health Insurance Scheme — Pradhan Mantri Jan Arogya Yojana in India, *Annals of Global Health* 2025, doi:[10.5334/aogh.4908](https://doi.org/10.5334/aogh.4908); 41913189 — Do publicly funded health insurance schemes reduce out-of-pocket expenditure?: Evidence from India, *BMC Health Services Research* 2026, doi:[10.1186/s12913-026-14456-8](https://doi.org/10.1186/s12913-026-14456-8); 41878109 — Beyond coverage: why Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana struggles to deliver healthcare in Jammu and Kashmir, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1783458](https://doi.org/10.3389/fpubh.2026.1783458); 39178875 — Improving urban health through primary health care in south Asia, *Lancet Global Health* 2024, doi:[10.1016/S2214-109X\(24\)00121-9](https://doi.org/10.1016/S2214-109X(24)00121-9) (abstract-only); 39542515 — Private sector engagement for immunisation programmes: a pragmatic scoping review of 25 years of evidence on good practice in low-income and middle-income countries, *BMJ Global Health* 2024, doi:[10.1136/bmjgh-2023-014728](https://doi.org/10.1136/bmjgh-2023-014728).

♦ ♦ ♦

Urban social health activist

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — The Urban Social Health Activist (USHA) is the honorary female community health volunteer of the **National Urban Health Mission (NUHM)** — the urban counterpart of the rural Accredited Social Health Activist (ASHA) — selected from and resident in the slum she serves, and placed as the **first link between the urban poor community and the Urban Primary Health Centre (UPHC)** (Sunder Lal, Ch 17, p 758; Suryakantha, Ch 36, p 783). She is not a salaried service provider but a *demand generator*: she creates awareness, mobilises the community, escorts beneficiaries and holds a small drug kit, working for a performance-based incentive (Park 28e, Ch 7, pp 535–536).

Where USHA sits in the urban health delivery system

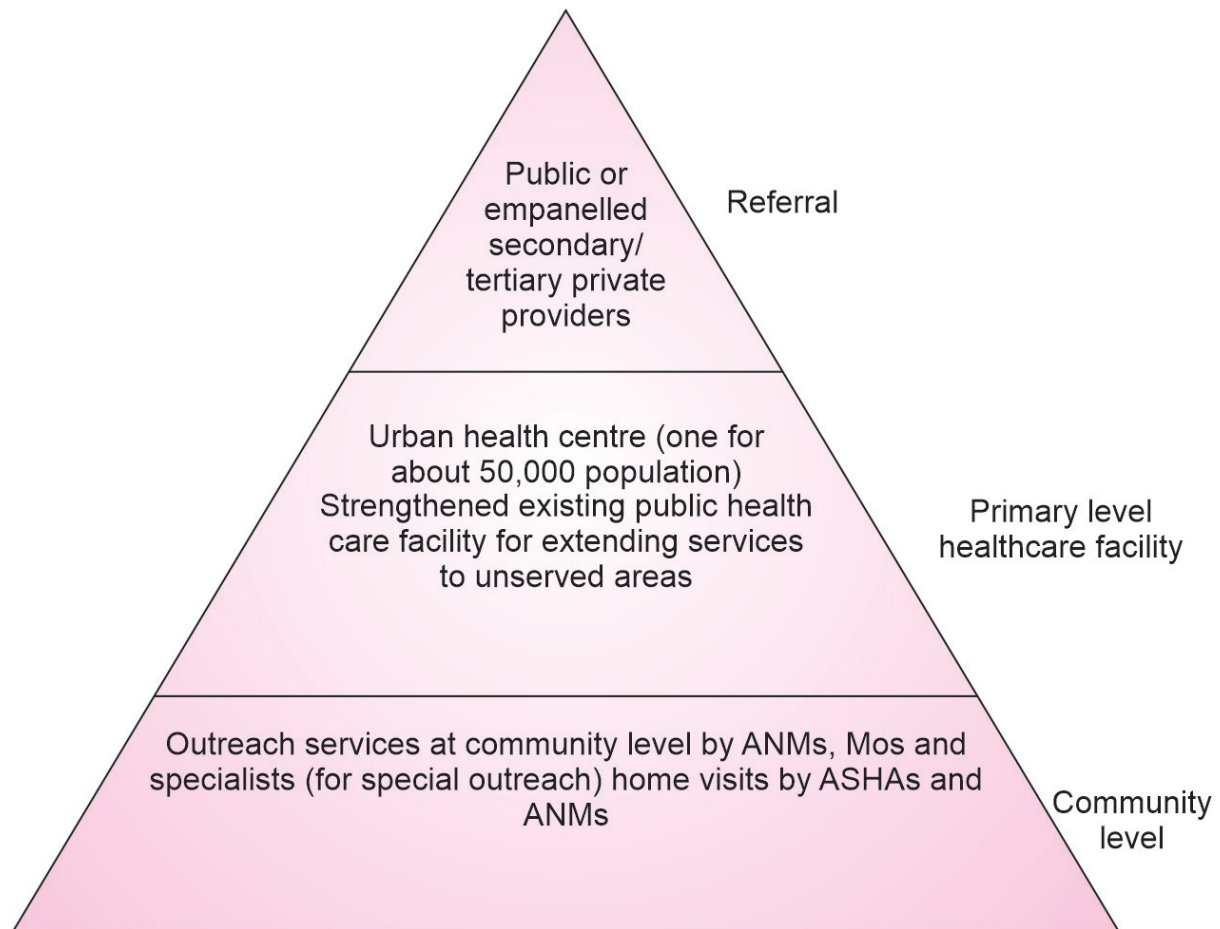


Fig 17.4 — Urban healthcare delivery model—various components, coverage of urban (Sunderlal, p.757)

Level	Unit	Population norm	Who mans it
Community — household	USHA (urban ASHA / Link Worker)	1 per 1,000–2,500 urban poor population, about 200–500 households; preferably co-located at the <i>Anganwadi</i> centre in the slum	The USHA herself (Park 28e, Ch 7, p 535; Sunder Lal, Ch 17, p 758)
Community — collective	<i>Mahila Arogya Samiti</i> (MAS)	50–100 households, 250–500 population; 10–12 women members	Elected chairperson and treasurer, supported by USHA (Park 28e, Ch 7, p 535; Sunder Lal, Ch 17, p 758)

Level	Unit	Population norm	Who mans it
Outreach	Auxiliary nurse midwife (ANM)	1 ANM per 10,000 population; 4–5 ANMs per U-PHC	ANM conducts the monthly Urban Health, Sanitation and Nutrition Day (Sunder Lal, Ch 17, p 758; Park 28e, Ch 7, p 536)
Primary facility	Urban Primary Health Centre (U-PHC)	~30,000–50,000 population; within or near a slum, half a kilometre radius; covers 25,000–30,000 slum people; OPD 12 noon to 8 pm, no in-patient care	Medical officer, lady health visitor, staff nurses, pharmacist/laboratory technician, ANMs, public health manager and support staff (Park 28e, Ch 7, p 536; Suryakantha, Ch 36, p 783)
Referral	Urban Community Health Centre (U-CHC)	1 per 2.5 lakh population, 30–50 bedded (50 bedded under Indian Public Health Standards 2022); metros 1 per 5 lakh with 100 beds; set up only in cities of above 5 lakh population, as a satellite hospital for every 4–5 U-PHCs	Specialist team providing in-patient medical and surgical services and institutional deliveries (Park 28e, Ch 7, p 536; Sunder Lal, Ch 17, pp 753, 759)

Note the structural point examiners look for: unlike rural areas, **no sub-centre exists in urban areas** — USHA and the outreach ANM *are* the sub-centre-equivalent, which is why the volunteer carries so much of the urban primary-care load (Sunder Lal, Ch 17, p 758). NUHM covers all state capitals, district headquarters and about **779 other cities/towns of 50,000 population and above (Census 2011)**, in a phased manner; towns below 50,000 remain with NRHM. Its target group is the urban poor in **listed and unlisted slums**, plus the homeless, street children, rickshaw pullers, construction-site workers, brick-kiln and lime-kiln workers, sex workers and temporary migrants (Park 28e, Ch 7, pp 535–536).

Selection criteria and terms of engagement

No separate urban selection norm is given in the texts; the USHA is selected on the **ASHA criteria under the National Health Mission**, the urban worker being described as "*similar to ASHA under NRHM*" (Sunder Lal, Ch 17, p 758).

Item	Norm
Sex and residence	A woman who "must be preferably resident of the village" she serves (Park 28e, Ch 7, p 539); in the urban set-up she is a "resident of same village/ward" — part of the community she works in (Sunder Lal, Ch 2, p 83)
Marital status	Married, widowed or divorced (Park 28e, Ch 7, p 539)
Age	25–45 years (Park 28e, Ch 7, p 539; Sunder Lal, Ch 17, p 738)
Education	Formal education up to 8th class (Park 28e, Ch 7, p 539; Sunder Lal, Ch 17, p 738)
Personal attributes	Communication skills and leadership qualities (Park 28e, Ch 7, p 539)
Equity in selection	Adequate representation from disadvantaged population groups, so that such groups are better served (Park 28e, Ch 7, p 539)
Coverage norm	1 per 1,000–2,500 urban poor population (the rural norm is 1 per 1,000) (Park 28e, Ch 7, p 535; Sunder Lal, Ch 2, p 83)

Item	Norm
Status	Honorary volunteer, not a salaried employee ; performance-based incentive fixed by the state (Sunder Lal, Ch 17, p 758)
Identity	Issued a photo identity card to boost her self-esteem and give her standing in the slum (Sunder Lal, Ch 17, p 758)
State flexibility	A state may engage a USHA or entrust her responsibilities to the MAS; in that case the incentives that would accrue to the USHA accrue to the MAS instead (Park 28e, Ch 7, p 535)

Roles and functions

A. Five core day-to-day activities (Sunder Lal, Ch 17, p 758):

- ▶ Home visits — up to 2 hours per day.
- ▶ Attend the monthly Health, Sanitation and Nutrition Day held by the ANM in her area.
- ▶ Accompany/escort pregnant women and children to the urban health centre.
- ▶ Hold the ward-level meeting.
- ▶ Record keeping.

B. Functions grouped by the role she plays

Role	What she actually does
Health activist / promoter	Active promoter of good health practices, enjoying community support; creates awareness on determinants of health — nutrition, basic sanitation, hygienic practices, healthy living and working conditions (Park 28e, Ch 7, p 535; Suryakantha, Ch 36, p 783)
Behaviour-change communicator	Interpersonal communication to generate demand for services and promote healthy practices in the slum (Sunder Lal, Ch 17, p 758)
Reproductive and child health counsellor	Awareness on RCH services, sexuality, gender equality, age at marriage and pregnancy; motivation for contraception, medical termination of pregnancy, sterilisation and spacing methods; early registration of pregnancy , pregnancy care, clean and safe delivery, nutritional care in pregnancy, identification of danger signs; counselling on immunisation, antenatal and postnatal care (Park 28e, Ch 7, p 535; Suryakantha, Ch 36, p 783)
Depot holder	Holds and distributes oral rehydration salts (ORS), iron and folic acid (IFA) tablets, chloroquine, oral contraceptive pills and condoms (Park 28e, Ch 7, p 535; Suryakantha, Ch 36, p 783)
Limited provider	Treats minor ailments with the drug kit provided (Suryakantha, Ch 36, p 783)
Link and escort	Facilitates access to services at the <i>Anganwadi</i> , U-PHC and urban local body; escorts/accompanies pregnant women and children needing treatment to the U-PHC or secondary/tertiary facility (Park 28e, Ch 7, pp 535–536)
Support to outreach	Identifies target beneficiaries and supports the ANM/MAS in conducting the regular monthly outreach session and in tracking service coverage (Park 28e, Ch 7, p 535; Suryakantha, Ch 36, p 783)

Role	What she actually does
Community organiser	Forms and promotes the Mahila Arogya Samiti in her community and helps it manage its revolving fund; encourages community participation in health activities (Park 28e, Ch 7, p 535; Sunder Lal, Ch 17, p 758; Suryakantha, Ch 36, p 783)
Disease-control mobiliser	Reinforces community action for immunisation and for prevention of water-borne and other communicable diseases — tuberculosis (DOTS), malaria, chikungunya, Japanese encephalitis (Park 28e, Ch 7, p 536)
Recorder and informant	Maintains records of vital events — births, deaths, immunisation, antenatal services in her locality — and reports any unusual health problem or disease outbreak in the slum to the ANM in charge of the area (Park 28e, Ch 7, p 536; Suryakantha, Ch 36, p 783)

Support, supervision, incentive and convergence

Element	Arrangement
Direct supervisor	ANM of the Primary Urban Health Centre; the elected urban local body (ULB) councillor also monitors her activities (Suryakantha, Ch 36, p 783)
Facilitators of her work	ANM guides and holds periodic meetings; <i>Anganwadi</i> worker acts as depot holder for replenishing her drug kit and guides her in organising the health day (Park 28e, Ch 10, p 727; Sunder Lal, Ch 5, p 229)
Payment	Performance-based incentive only , rate fixed by the state (Sunder Lal, Ch 17, p 758)
Payment mechanism	A revolving fund kept with the ANM at the U-PHC (in the PHC account), replenished from time to time (Park 28e, Ch 7, p 536)
Community platform she builds	MAS receives an annual grant of Rs 5,000 for its fortnightly/monthly meetings; capacity built through orientation workshops (Park 28e, Ch 7, p 535; Sunder Lal, Ch 17, p 758)
Outreach support to her area	ANM gets Rs 500 per month mobility support to run at least one routine outreach session per area per month (Sunder Lal, Ch 17, p 758)
Intersectoral convergence	With urban local bodies, ICDS, education, water and sanitation, housing/slum development, national health programmes and the Ministry of Urban Development (Sunder Lal, Ch 17, pp 759–760; Suryakantha, Ch 36, p 785)

Monitoring and evaluation of USHA performance

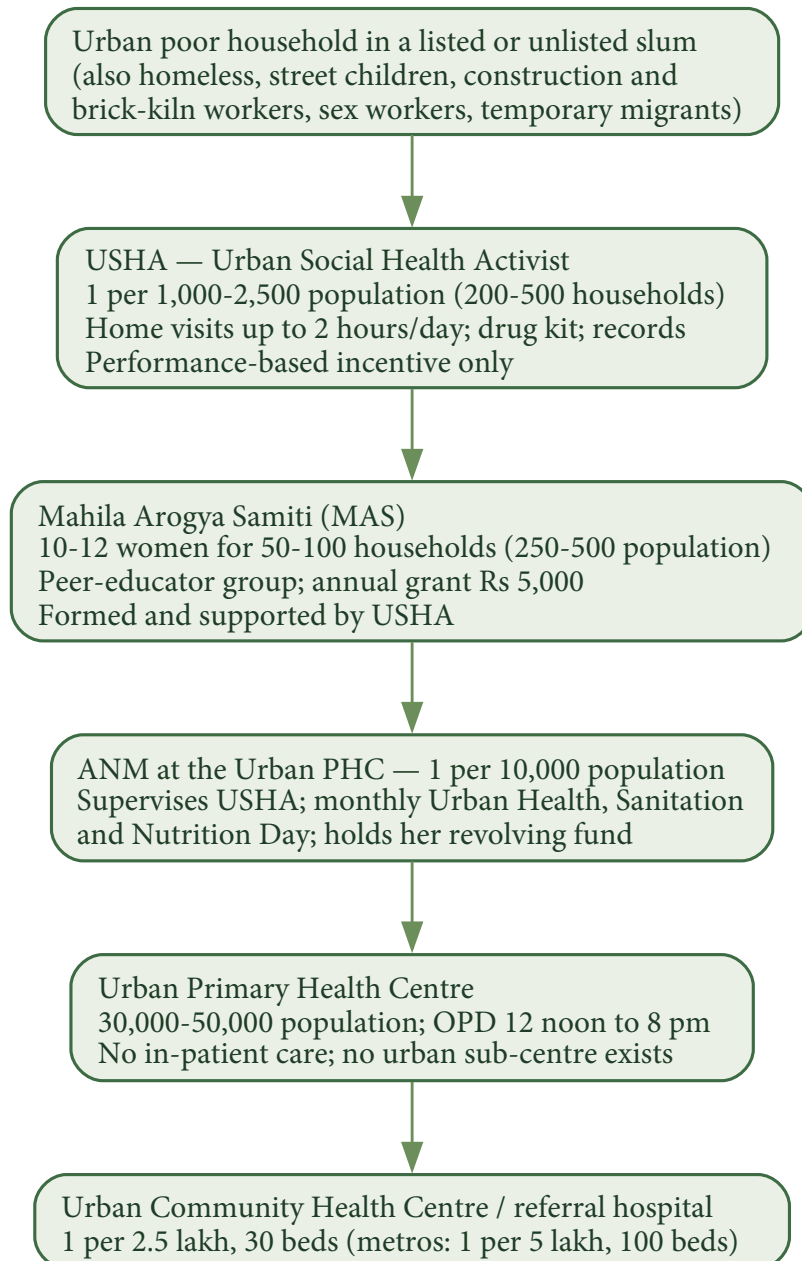
Park tabulates the NUHM city-level indicators as "**Process and input indicators in NUHM**" (Park 28e, Ch 7, Table 11), grouped under five sub-heads. Only **items 1–3, under *Community Processes*, are the USHA-specific indicators**; the rest are the downstream health-system and service indicators that her mobilisation feeds.

Sub-head in Table 11	Indicators	Bearing on USHA
Community Processes (items 1–3)	Number of <i>Mahila Arogya Samitis</i> formed; number of MAS members trained; number of ASHAs selected and trained	The direct measures of the USHA cadre — whether she exists, is trained, and has built her community collective
Health Systems (items 4–11)	ANMs recruited; special outreach health camps in slums/health-facility areas; Urban Health and Nutrition Days (UHNDs) in slums and vulnerable areas; U-PHCs and U-CHCs made operational; Rogi Kalyan Samitis created; OPD attendance at U-PHCs; deliveries in public health facilities	Sessions and attendance she mobilises the community to attend
RCH Services (items 12–16)	Antenatal care early registration in the first trimester; women with a first-trimester check-up; second dose of tetanus toxoid coverage; children fully immunised through public facilities; severely acute malnourished children identified and referred	Service uptake she is meant to generate by counselling and escort
Communicable Diseases (items 17–20)	Malaria cases detected on blood examination; tuberculosis cases identified as chest-symptomatic; suspected tuberculosis cases referred for sputum examination; multidrug-resistant tuberculosis cases on DOTS-plus	Her case-detection and referral function
Non-communicable Diseases (items 21–23)	Diabetes, cancer and hypertension cases screened in the city	Her role in the screening chain

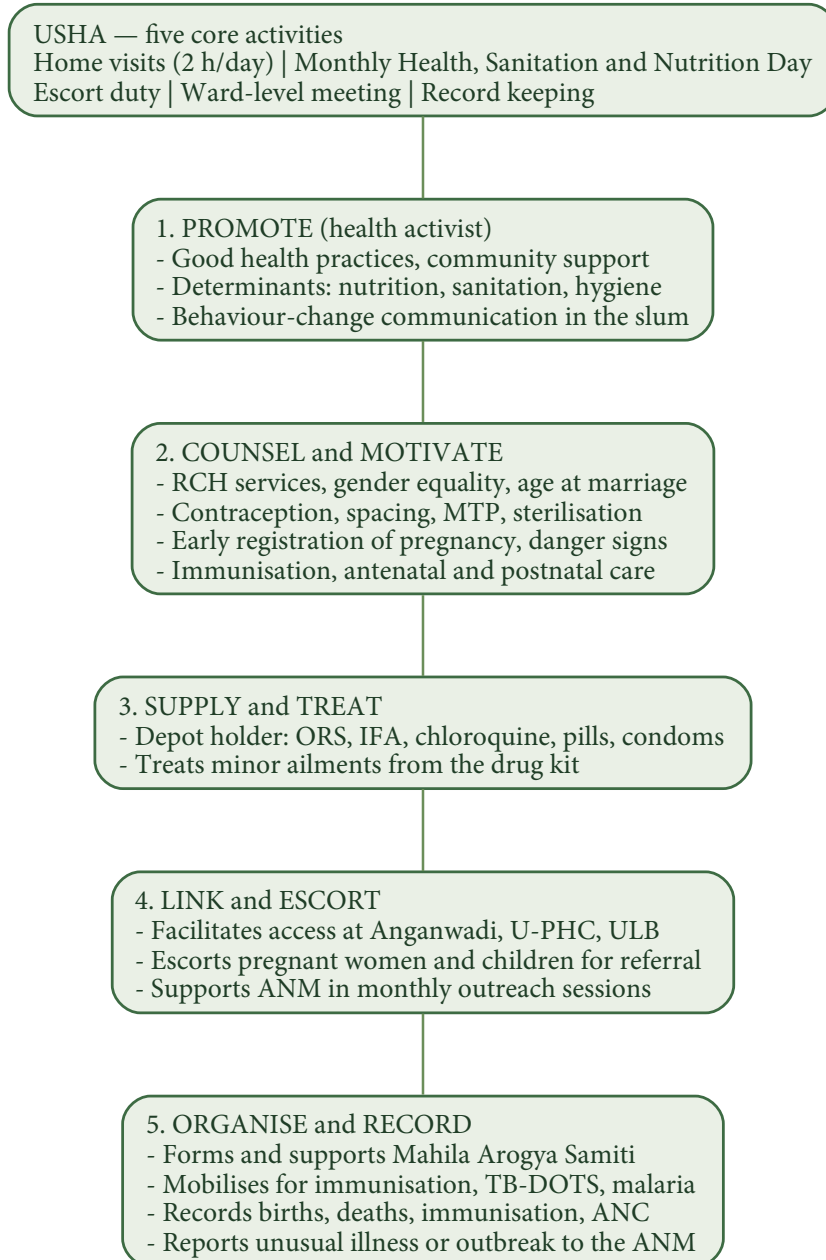
Baseline convention: for the newly created NUHM components (asterisked in the table, including MAS and ASHA indicators) **2013–14 is the baseline year and the value is zero**; for the remaining indicators the 2012–13 figure is the baseline (Park 28e, Ch 7, Table 11).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — "One USHA's beat in the slum" (draw an irregular outer boundary for the slum, filled with a grid of small squares for houses). Outer boundary label: **Listed or unlisted urban slum — 1,000–2,500 urban poor population**. Label across the grid of squares: **200–500 households — home visits up to 2 hours a day**. Shade four separate clusters of squares and label each cluster: *Mahila Arogya Samiti* — **50–100 households, 10–12 women**. Draw a small hut at the centre of the slum labelled *Anganwadi* centre — **USHA preferably co-located here; monthly Urban Health, Sanitation and Nutrition Day**. Draw an arrow leaving the slum boundary to a building outside it labelled **Urban Primary Health Centre** and label the arrow **escort and referral**.



Reporting line of the Urban Social Health Activist and her parallel community collective under the National Urban Health Mission



Functions of the Urban Social Health Activist grouped by the role she performs

High-yield exam pointers

- ▶ **Norms to write first:** 1 USHA per 1,000–2,500 urban poor population = 200–500 households; MAS = 50–100 households (250–500 population), 10–12 women, Rs 5,000/year; ANM = 1 per 10,000; U-PHC = 30,000–50,000; U-CHC = 1 per 2.5 lakh, 30–50 bedded (50 bedded under IPHS 2022), metros 1 per 5 lakh with 100 beds, only in cities above 5 lakh.
- ▶ **Selection (ASHA criteria, no separate urban norm):** woman resident of the ward/community she serves, 25–45 years, 8th class pass, married/widowed/divorced, communication and leadership skills.
- ▶ **Five activities mnemonic — "HEWRR":** Home visits 2 h/day, Escort, Ward meeting, Record keeping, plus attend the monthly Health, Sanitation and Nutrition Day.
- ▶ **Drug-kit contents to name:** ORS, IFA tablets, chloroquine, oral contraceptive pills, condoms.

- **Payment:** performance-based incentive from a **revolving fund held by the ANM in the U-PHC account** — she is a volunteer, not an employee; the state may divert her incentive to the MAS if it uses MAS instead of a USHA.
- **Programme anchor:** NUHM, a sub-mission of the National Health Mission; covers state capitals, district headquarters and about **779 towns of ≥50,000 population (Census 2011)**; her three Table 11 community-process indicators are **MAS formed, MAS members trained, ASHAs selected and trained**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam — the same functionary is named differently across sources: Park calls her the **ASHA or Link Worker (LW) in the urban poor settlement**, Sunder Lal calls her the **Urban Accredited Social Health Activist (U-ASHA)**, and Suryakantha and the NUHM framework use **Urban Social Health Activist (USHA)**, which is the term the RGUHS paper used; write "USHA (urban ASHA under NUHM)" once at the start and the naming is covered. The U-CHC bed norm is quoted differently in different places in the same corpus — the NUHM framework figure is 30–50 beds while the Indian Public Health Standards 2022 figure is a flat 50 beds (100 in metros), so write the range with the IPHS 2022 value attached. Under Ayushman Bharat the existing U-PHCs are being strengthened as **Urban Ayushman Arogya Mandirs / Urban Health and Wellness Centres, one per 15,000–20,000 population**, so newer literature may describe USHA as attached to a wellness centre rather than a U-PHC; the NUHM impact targets quoted in the textbooks were set for **2017** and are now dated, so quote them only with that year attached.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-4 and NFHS-5, NSS Comprehensive Annual Modular Survey 2022–23, MoHFW Ayushman Arogya Mandir programme data, Universal Immunization Programme

Programme and policy updates (India)

- **The U-PHC she reports to is now an Urban Ayushman Arogya Mandir.** Sub-centres and primary health centres in both rural and urban areas have been re-designated **Ayushman Arogya Mandir** (the erstwhile Health and Wellness Centre), with more than **1.75 lakh operational nationally** and 82% delivering the full 12-package expanded service list by 30 November 2024; **172 medicines and 63 diagnostic tests** are notified at the Ayushman Arogya Mandir Primary Health Centre / Urban PHC level, teams now draw a **collective performance-linked incentive** (there was none earlier), and urban infrastructure is additionally financed through **PM-ABHIM and Fifteenth Finance Commission grants routed via urban local bodies**. The *Mahila Arogya Samiti* remains the designated **urban** community platform (VHSNC being the rural counterpart), so the USHA's collective survives the rebranding intact. [PMID 41816146]
- **The Mahila Arogya Samiti has finally been evaluated.** A realist evaluation of **401 MAS across all 20 NUHM towns of Chhattisgarh** found **95.3%** had acted on at least one of six domains over three years and **79.8%** on three or more — food security and nutrition 74.1%, sanitation 70.8%, social environment 64.1%, gender-based violence 60.8%, healthcare 59.1%, drinking water 56.4%

— and 97.3% had held a meeting in the preceding month. Participatory selection of members, autonomy over the **Rs 5,000 untied fund**, annual training and facilitation by the community health worker were the enablers; the collectives could not, however, shift accountability above the neighbourhood level. Note the field norm here: 10 women, each representing about 20 households, so roughly 200 households per MAS. [PMID 40349012]

- **How her incentive actually reaches her — and why it stalls.** Fieldwork in Maharashtra and Assam mapped ASHA payment as a **five-step chain** (recording work → claim submission → verification → processing → disbursement); full digital disbursement through the **Public Financial Management System** improved timeliness and transparency over cash and cheque, while the still-manual earlier steps generate delays, deductions and out-of-pocket costs for the worker herself. India now has roughly **9.83 lakh ASHAs** — the largest such cadre in the world — earning about **USD 36–50 a month**, and the schooling criterion is described as **12 years, relaxable where candidates are unavailable**, above the 8th-class bar carried in the textbooks. Interviews with 40 ASHAs across ten Uttar Pradesh districts confirmed delayed payment, un replenished drug kits and thin training as the dominant demotivators, against a training norm of **23 days in year one and 12 days every year thereafter**. [PMID 39953777; PMID 41628200]
- **Her scope is being pushed from screening into treatment.** In the **IMPACT Diabetes** cluster-randomised trial across 16 village and peri-urban clusters served by eight primary health centres in two states, ASHAs screened **1,785 people**, identified **418 with diabetes** and delivered nine months of decision-support-guided care under physician supervision: **21.8% vs 10.3%** achieved a fall in glycated haemoglobin of $\geq 0.5\%$ ($p < 0.05$), with physician visits 85.0% vs 29.8% and medication adherence 63.0% vs 43.1%. [PMID 41120559]

Current national data

- **NUHM reach, and what the urban facilities actually look like.** Government management-information data for 2023 put NUHM coverage at about **275 million people in 1,213 cities and towns**, with **5,195 Urban PHCs operational by March 2023** — three-quarters carrying the minimum staffing and service package and only about **10% functioning 24×7**; **60–78%** of clinical and paramedical posts and **79–82%** of programme-management posts were filled against sanction. Pooled DLHS 2002–08 and NFHS-4 and NFHS-5 (2015–2021) data on roughly **203,000 urban births** show the richest–poorest gap in facility delivery more than halved from **57 to 21 percentage points**, antenatal care at public **Urban Health and Nutrition Days** in the poorest decile rising **11% → 23%**, and public-sector postnatal care in that decile **59% → 73%** — yet the poorest tenth still trails (facility delivery 77% vs 86% in the next decile). [PMID 42274885]
- **What the Ayushman card does and does not do for her community.** The **Comprehensive Annual Modular Survey 2022–23** (NSS 79th round), covering **302,086 households of which 128,990 urban**, found **49.7%** under a publicly funded scheme; coverage was associated with more outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128–1.210) and lower catastrophic expenditure at the 20% threshold (inpatient adjusted OR 0.790, 95% CI 0.737–0.846; outpatient 0.869, 95% CI 0.810–0.932) — but **outpatient out-of-pocket spending remains the residual burden**, which is precisely the argument for the primary-care platform the USHA feeds. [PMID 42167946]

Elimination, surveillance and disease status

- **One under-five in four is still incompletely immunised.** A meta-analysis of 45 Indian cross-sectional studies published 2015–2025, covering more than 428,000 participants, put pooled incomplete immunisation at 26% (95% CI 21–32%); the strongest determinants were **absence of an immunisation card** (OR 9.46, 95% CI 3.28–27.28), non-institutional delivery (OR 3.59), maternal illiteracy (OR 3.25) and low socioeconomic status (OR 2.03) — every one of them a USHA-facing task (card retention, escort to institutional delivery, due-list tracking) and the binding constraint on the Immunization Agenda 2030 targets. [PMID 42320383]
- **Community mobilisation for TB elimination is going digital.** A four-week tele-education package of videos, audio modules and quizzes **reinforced by ASHAs** reached 112 households across an urban slum cluster and a tribal village network; TB literacy rose at both sites and converted into **self-initiated referrals and higher screening uptake**, the urban slum showing greater digital participation while the tribal site matched it through a blended offline model. [PMID 42362243]

Guideline and legislative updates

- **HPV vaccination is moving into the Universal Immunisation Programme.** The indigenous quadrivalent vaccine **Cervavac** and its proposed introduction for **girls aged 9–14 years** is the next schedule addition the USHA will have to mobilise for alongside cervical screening; hesitancy, cold-chain and logistics, and low awareness are named as the limiting factors against the WHO cervical-cancer elimination targets. [PMID 42138645]

Global and WHO developments

- **Global evidence now backs community health workers prescribing, not merely referring — while the labour question stays open.** The ComBaCaL cluster-randomised trial, nested in a cohort spanning 103 rural Lesotho villages, had trained community health workers **initiate and titrate metformin, atorvastatin and aspirin** under tablet-based decision support: of 5,785 screened, 252 were diagnosed and 103 analysed, with 12-month glycated haemoglobin **6.5% vs 7.1%** for facility referral (adjusted mean difference –0.46%, 95% CI –1.14 to 0.22), better engagement in care and no safety difference. Set against it, ethnographic work in India argues that hierarchical task delegation, economic precarity and routine surveillance keep the ASHA's autonomy constrained while letting the system absorb ever more tasks without structural reform — the live policy question of whether the "honorary volunteer" designation should survive at all. [PMID 42174613; PMID 41952271]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India, pp 535–536, 538–539, Table 11 and Fig 12; Ch 10 Health Care of the Community, pp 726–727); Sunder Lal — Textbook of Community Medicine (Ch 2 Social and Behavioural Sciences, p 83; Ch 5 Nutrition, p 229; Ch 17 Healthcare of the Community, pp 738, 744, 752–753, 756–760); Suryakantha — Community Medicine with Recent Advances (Ch 36 National Urban Health Mission, pp 783–785). Programme and standards documents named by these texts: National Urban Health Mission Framework for Implementation, May 2013; Guidelines for Preparing NUHM Programme Implementation Plan 2013–14, Ministry of Health and Family Welfare; Indian Public Health Standards 2022. Coverage note: the corpus describes USHA's norms, selection, functions, supervision and incentive fully, but contains no separate USHA training-module schedule, so no training curriculum detail is claimed here.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 40349012 — Community participation through women's health collectives promoted by India's National Urban Health Mission: a realist evaluation in Chhattisgarh state, *International Journal for Equity in Health* 2025, doi:[10.1186/s12939-025-02498-z](https://doi.org/10.1186/s12939-025-02498-z); 39953777 — Community health worker payment processes: a qualitative assessment of experiences in two Indian states, *Health Policy and Planning* 2025, doi:[10.1093/heapol/czaf010](https://doi.org/10.1093/heapol/czaf010); 41628200 — Factors affecting the motivation of community health workers: perspectives from Accredited Social Health Activists (ASHA) in Uttar Pradesh, India, *PLOS ONE* 2026, doi:[10.1371/journal.pone.0341811](https://doi.org/10.1371/journal.pone.0341811); 41120559 — Preliminary effectiveness and feasibility of ASHA-led mobile health intervention for diabetes care in Indian primary health care settings, *Scientific Reports* 2025, doi:[10.1038/s41598-025-20728-w](https://doi.org/10.1038/s41598-025-20728-w); 42274885 — The role of public health services in reducing maternal and newborn health inequalities in urban India: a survey analysis of 200,000 births over two decades, *Journal of Urban Health* 2026, doi:[10.1007/s11524-026-01062-6](https://doi.org/10.1007/s11524-026-01062-6); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022–2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: a systematic review and meta-analysis (2015–2025), *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only); 42362243 — Use of tele-education for strengthening community participation in tuberculosis control, *Indian Journal of Tuberculosis* 2026, doi:[10.1016/j.ijtb.2026.02.020](https://doi.org/10.1016/j.ijtb.2026.02.020) (abstract-only); 42138645 — Human papillomavirus vaccination in India: paving the way to cervical cancer prevention, *Annals of African Medicine* 2026, doi:[10.4103/aam.aam.745.25](https://doi.org/10.4103/aam.aam.745.25) (abstract-only); 42174613 — Community health worker-led versus facility-based type 2 diabetes care in rural Lesotho: a cluster-randomized trial within the ComBaCaL cohort study, *BMC Medicine* 2026, doi:[10.1186/s12916-026-04943-4](https://doi.org/10.1186/s12916-026-04943-4); 41952271 — Power, surveillance, and the limits of resistance: the role of community health workers in India's health system, *Global Public Health* 2026, doi:[10.1080/17441692.2026.2657631](https://doi.org/10.1080/17441692.2026.2657631) (abstract-only).



Village level health functionaries

Asked in: Paper IV 14-Jul-2023 — RGUHS MD Community Medicine

Definition — Village level health functionaries are the locally resident, community-selected, honorary or part-time workers who deliver the first contact of primary health care inside the village itself, below the sub-centre, and act as the interface between the community and the public health system (Sunder Lal, Ch 17, p 744). They form the **base of India's three-tier rural health care organisation** and are the operational expression of the primary health care principle of community participation, greatly influenced by the Chinese *bare-foot doctor* experience (Park 28e, Ch 10, p 720). Park 28e lists three schemes in operation at the village level — the **ASHA scheme**, the **ICDS scheme** and the **training of local dais** (Park 28e, Ch 10, p 726).

Who the village level functionaries are

Functionary	Scheme / parent ministry	Population norm	Selection and training	Status and support
Accredited social health activist (ASHA)	National Health Mission, Ministry of Health and Family Welfare	1 per 1,000 population; relaxed to one per habitation in tribal, hilly and desert areas (Park 28e, Ch 10, p 726)	Woman resident of the village (married / widow / divorced), preferably 25–45 years , formal education up to 8th class , with communication skill and leadership qualities; selected by the gram panchayat under supervision of the block medical officer; induction training of 23 days spread over one year plus periodic re-training (Jugal Kishore, Ch 1, p 134)	Honorary volunteer on performance-based incentives ; works 3–4 hours a day on 4–5 days a week; issued a drug kit (Sunder Lal, Ch 17, p 738)
Anganwadi worker (AWW)	Integrated Child Development Services, Ministry of Women and Child Development	1 anganwadi centre per 400–800 population in plain areas and 300–800 in tribal / difficult areas; mini-anganwadi centre 150–400 (Park 28e, Ch 9, p 700; Sunder Lal, Ch 17, p 744)	Selected from the community she serves by the village people; trained 4 months in health, nutrition and child development (Park 28e, Ch 10, p 727)	Part-time honorary worker; honorarium Rs 1,500 per month as printed in Park 28e; assisted by an anganwadi helper
Trained dai (traditional birth attendant)	Dai training scheme begun 2001–02, later extended to all Empowered Action Group states	Norm one trained dai for each village ; national target one trained dai per 1,000 population (Park 28e, Ch 10, p 726; Ch 8, p 612)	The practising local dai is trained to conduct safe deliveries ; districts were chosen where the safe-delivery rate was below 30 per cent — initially 156 districts in 18 states and union territories (Park 28e, Ch 10, p 728)	Conducts home deliveries and also acts as family planning counsellor and motivator

Functionary	Scheme / parent ministry	Population norm	Selection and training	Status and support
Village health guide (health volunteer)	Village Health Guides' scheme, launched in 1977 (Park 27e, Ch 2)	1 per village or per 1,000 population, mostly women	Selected by the local community and trained locally, free of charge, so that they overcome cultural and communication barriers (Park 28e, Ch 10, p 720)	About 3.23 lakh health guides in position; the scheme now continues in some states only (Park 28e, Ch 8, p 612)
Mahila Swasthya Sangh (MSS)	Village-level women's health group	One per village or cluster	A women's collective rather than an individual worker	79,512 MSS working at village level (Sunder Lal, Ch 17, p 744)

Accredited social health activist — the principal village functionary

Her work is grouped under four heads (Sunder Lal, Ch 17, pp 738–739; Park 28e, Ch 10, pp 726–727):

- **Activist and counsellor** — creates awareness on the determinants of health, namely nutrition, basic sanitation and hygienic practice, healthy living and working conditions, and the need to use health and family welfare services in time; counsels women on birth preparedness, safe delivery, breastfeeding and complementary feeding, immunisation, contraception, prevention of reproductive tract and sexually transmitted infection, and care of the young child.
- **Link and mobiliser** — mobilises the community to use immunisation, antenatal check-up, postnatal check-up, family planning and ICDS services at the village, sub-centre and primary health centre, and **escorts or accompanies** pregnant women and children needing treatment or admission to the primary health centre, community health centre or first referral unit.
- **Care provider** — treats minor ailments such as diarrhoea, fever and minor injuries; is a **directly observed treatment, short-course (DOTS) provider** under the National Tuberculosis Elimination Programme; gives **home-based newborn care (HBNC)** through 6–7 home visits up to 42 days after delivery and **home-based care for the young child (HBYC)** through visits in the first week of the 4th, 7th, 10th, 13th and 16th month of age, for which she is paid **Rs 250** on the 45th day after validation by the auxiliary nurse midwife (Sunder Lal, Ch 10, p 594; Ch 17, p 738).
- **Depot holder** — keeps oral rehydration salts, iron and folic acid tablets, disposable delivery kits, oral contraceptive pills and condoms; the kit contains both AYUSH and allopathic formulations and is replenished through the anganwadi worker (Park 28e, Ch 10, p 727).
- **Recorder and surveillance worker** — records and reports all **births and deaths** in her village and any unusual health problem or disease outbreak to the sub-centre or primary health centre.
- **Disease-specific tasks** — participates in National Filaria Day, and identifies **skin patches with loss of sensation**, bringing them to the notice of the health worker and ensuring that leprosy patients take regular treatment.
- **Non-communicable disease screening** — enumerates every person **aged 30 years and above** and applies the **community-based assessment checklist (CBAC)**, which scores age, family history of non-communicable disease, waist circumference, physical inactivity, tobacco use and alcohol

use; a score above 4 marks the person as high risk and gives priority for screening (Sunder Lal, Ch 16, p 703; Ch 8, p 503).

- **Planner** — as member secretary of the village health sanitation and nutrition committee she helps prepare and implement the village health action plan, and promotes construction of household toilets.

Anganwadi worker and the ICDS service package

The anganwadi worker, along with the village health guide, is the community's primary link with health services for young children (Park 28e, Ch 10, p 727). She delivers an integrated package of six early-childhood services (Park 28e, Ch 9, p 700).

Service	Chief beneficiaries
Supplementary nutrition	Children under 6 years, pregnant and lactating women, adolescent girls
Immunisation	Children under 6 years, pregnant women
Health check-up, including maintenance of the growth chart	Children under 6 years, pregnant and nursing mothers
Medical referral services	Any beneficiary needing higher care
Nutrition and health education	Women 15–45 years
Non-formal pre-school education	Children 3–6 years

At the end of 2019 about 7,075 ICDS projects and 13.77 lakh anganwadi and mini-anganwadi centres were functional, with 305.09 lakh pre-school education beneficiaries and 836.25 lakh supplementary nutrition beneficiaries (Park 28e, Ch 9, p 701). Anganwadi work is supervised by the *Mukhyasevika*, with field supervision by the child development project officer (Park 28e, Ch 11, p 804).

The village-level platform, committee and supervision

- **Village health sanitation and nutrition committee (VHSNC)** — constituted in every village under the National Health Mission; its members are the *sarpanch*, the ASHA, the multipurpose health worker (female), the school teacher, the anganwadi worker and a self-help-group member, with panchayati raj institutions and non-governmental organisations represented (Sunder Lal, Ch 16, p 697; Ch 17, p 744).
- **Untied fund** — Rs 10,000 per annum to each village panchayat for local health action; over 5.55 lakh committees have been made functional and given the grant (Sunder Lal, Ch 16, p 697).
- **Functions of the committee** — prepares the **village health action plan**, which aggregates upward into the sub-centre, primary health centre, block and district plans; monitors the services given by the anganwadi worker, ASHA and sub-centre; ensures that the auxiliary nurse midwife and the multipurpose worker visit the village on fixed days; discusses and analyses every maternal and neonatal death and gets it registered in the panchayat; and manages the village health fund (Jugal Kishore, Ch 1, p 138; Sunder Lal, Ch 17, p 744).
- **Village health, nutrition and sanitation day (VHSND)** — held once a month on a fixed date at the anganwadi centre, organised by the ASHA with the anganwadi worker, auxiliary nurse

midwife, self-help groups and panchayati raj institutions, to deliver reproductive, maternal, newborn, child and adolescent health services and to act on the determinants of health (Sunder Lal, Ch 5, pp 229–230).

- ▶ **Supervision and integration** — the auxiliary nurse midwife holds weekly or fortnightly meetings with the ASHA and guides her on outreach sessions, the eligible-couple register, iron and folic acid, tetanus toxoid, oral pill schedules and the danger signs of pregnancy and labour; the anganwadi worker guides her in running the monthly health day and in replenishing her drug kit; both act as **resource persons for ASHA training** (Park 28e, Ch 10, p 727). The recognised convergence practices at village level are **co-location of anganwadi centres with sub-centres, joint home visits and joint household surveys** by the anganwadi worker, auxiliary nurse midwife and ASHA (Sunder Lal, Ch 16, p 711).
- ▶ **Records** — the ASHA maintains household survey, eligible couple, antenatal, birth and death, immunisation, home-based newborn care, daily diary, drug stock, VHSND and CBAC records, while the anganwadi worker maintains eleven registers (Sunder Lal, Ch 8, p 533). *Mobile Academy*, a free audio course delivered on her mobile phone, is used to refresh ASHA knowledge and communication skills (Sunder Lal, Ch 17, p 739).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **pyramid of the rural health care organisation**, labelling from the base upward:

- ▶ **Base — village (6,63,049 villages; 1,412 million people):** *anganwadi* workers, birth attendants, health guides, link workers (ASHA) — **health volunteers 1:1,000 population**
- ▶ **Sub-centre / health and wellness centre:** health workers male and female and mid-level health provider — **1:5,000 plains, 1:3,000 tribal or hilly**
- ▶ **Primary health centre:** **1:30,000 plain, 1:20,000 tribal and hilly**
- ▶ **Community health centre:** **1:1,20,000 plain, 1:80,000 tribal or hilly**
- ▶ **Sub-district / taluka hospital, then district health organisation, state, and Ministry of Health and Family Welfare**

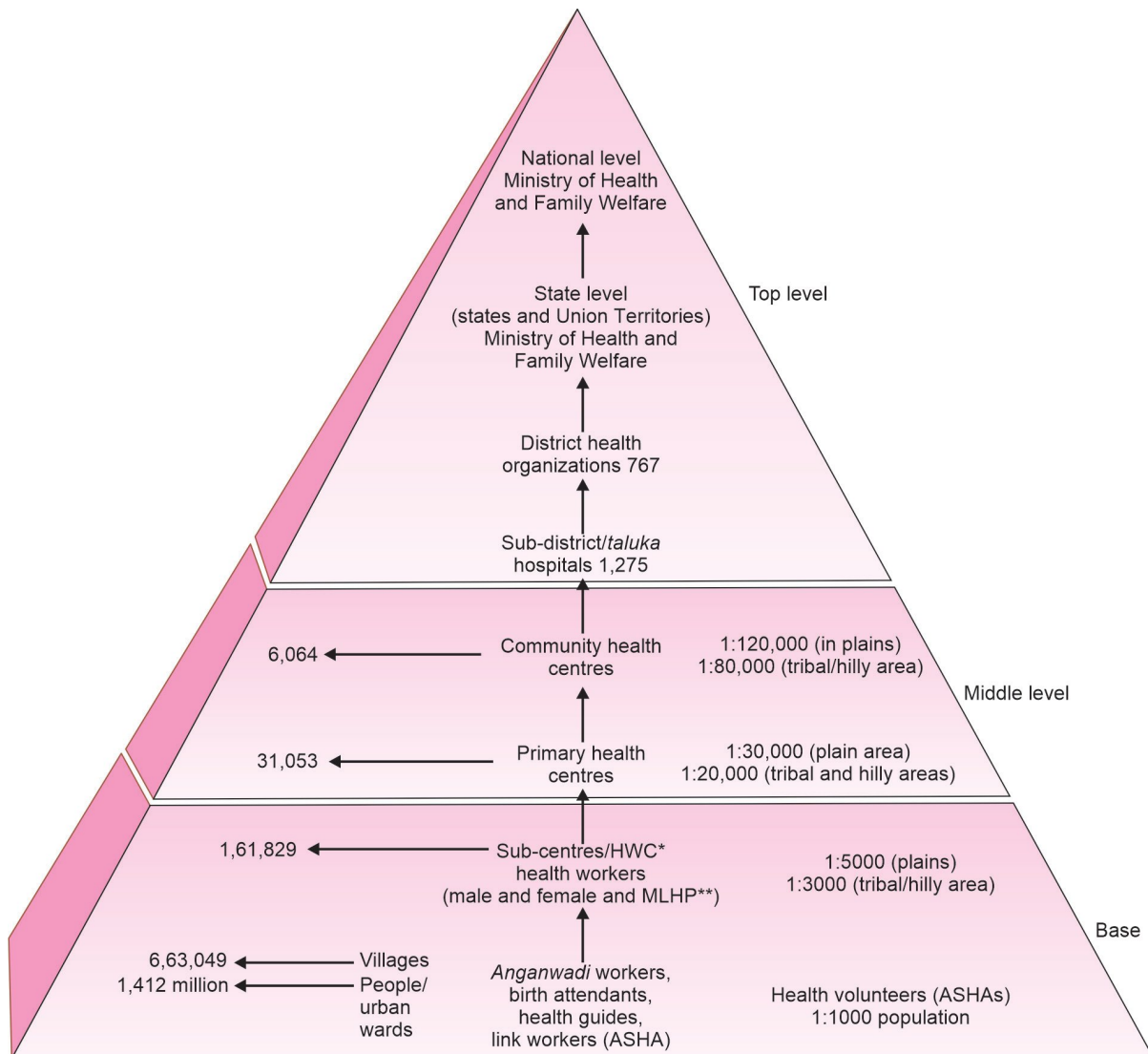
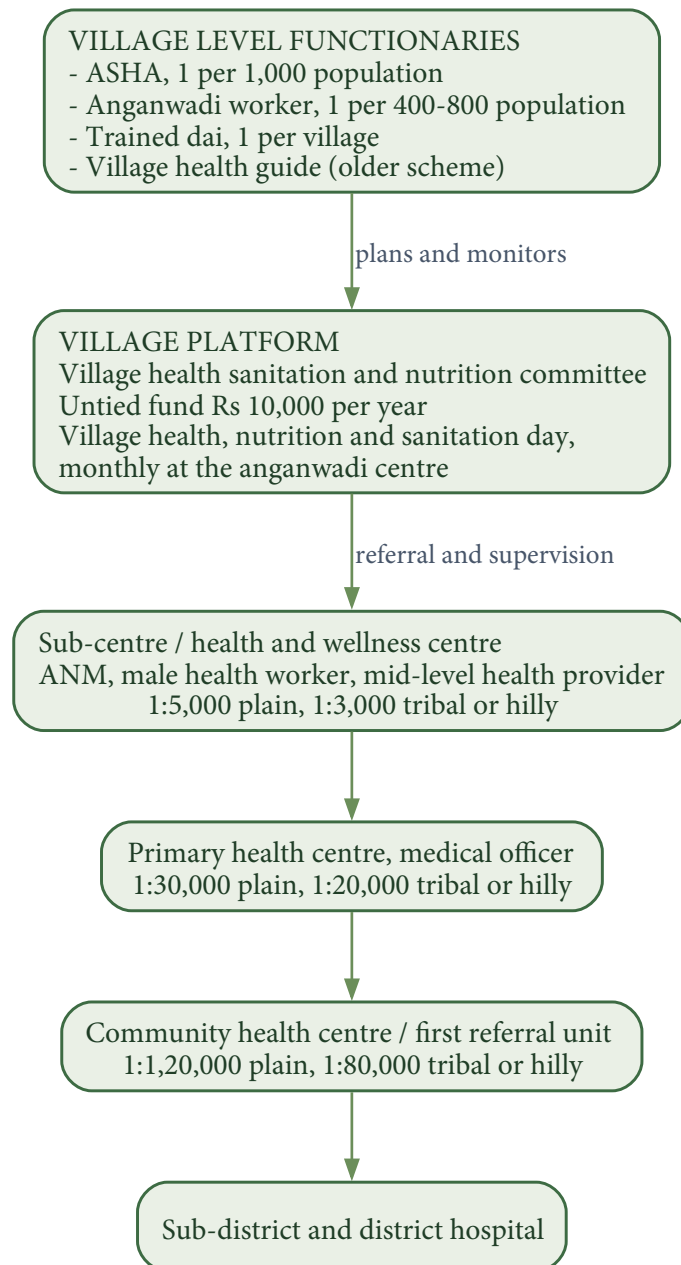
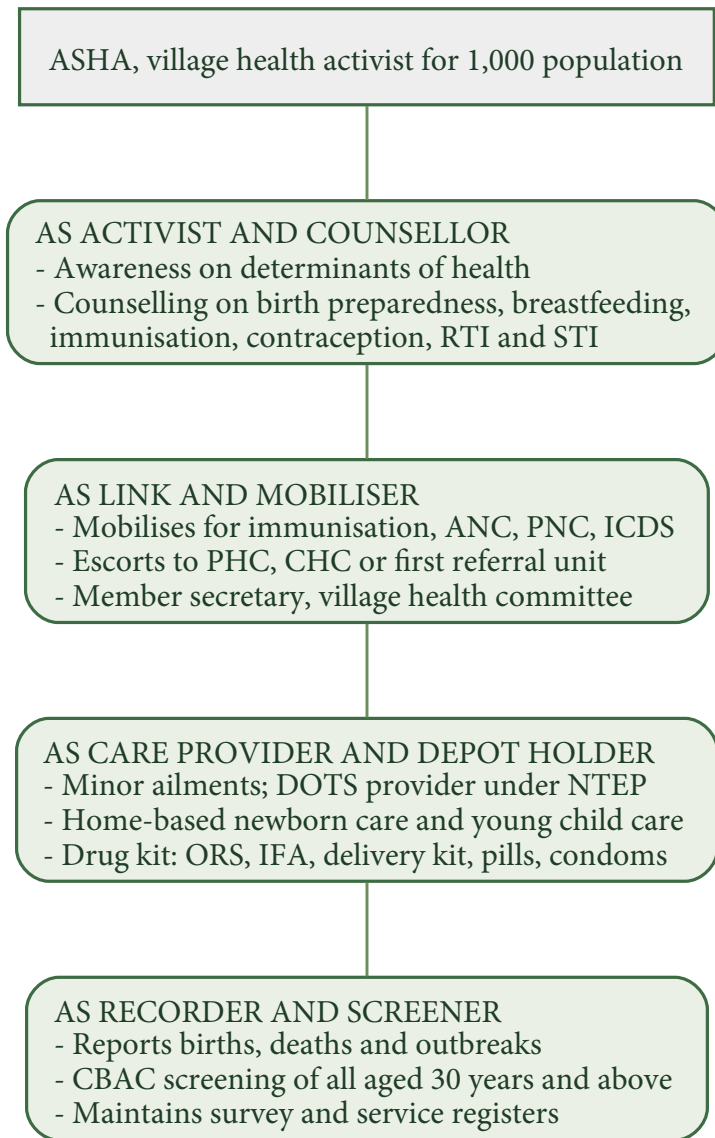


Fig 16.2 — Health system organization *HWC—Health and wellness centre, **MLHP—Mid (Sunderlal, p.709)



Village level functionaries, their platform and the referral chain above them



The role-set of the ASHA at village level

High-yield exam pointers

- ▶ **The village-level trio in Park 28e:** ASHA scheme, ICDS scheme, training of local dais (Ch 10, p 726).
- ▶ **Norms to write:** ASHA 1:1,000 (one per habitation in tribal, hilly and desert areas); anganwadi centre 400–800 plain and 300–800 tribal, mini-anganwadi 150–400; trained dai **one per village**; village health guide 1:1,000.
- ▶ **ASHA eligibility, mnemonic "R-W-25-45-8":** Resident of the village, Woman (married, widow or divorced), 25–45 years, class 8 schooling, chosen by the gram panchayat; **23 days of induction training over one year.**
- ▶ **ASHA payment to quote:** honorary volunteer on performance-based incentive; Rs 250 for the home-based newborn care visit set, paid on the 45th day after validation by the auxiliary nurse midwife.

- ▶ **VHSNC:** *sarpanch*, ASHA (member secretary), auxiliary nurse midwife, anganwadi worker, teacher, self-help-group member; **Rs 10,000 untied fund per year**; **5.55 lakh** committees functional; village health, nutrition and sanitation day once a month at the anganwadi centre.
- ▶ **CBAC:** six items — age, family history, waist circumference, physical inactivity, tobacco, alcohol; **score above 4 is high risk**; applied by the ASHA to everyone aged **30 years and above**.
- ▶ **ICDS scale (end of 2019):** 7,075 projects and **13.77 lakh** anganwadi and mini-anganwadi centres, delivering **six services**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: the worker counts and rates differ between the cited editions — Park 28e quotes 9.39 lakh ASHAs supplied with drug kits and an anganwadi worker honorarium of Rs 1,500 per month, whereas Sunder Lal states that over 10.72 lakh ASHAs are currently working, and honoraria and incentive rates are revised periodically by government order, so always quote the figure together with the book and year you are citing.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, NFHS-5, MoHFW Ayushman Arogya Mandir programme data, WHO

Programme and policy updates (India)

- ▶ **The facility just above the village functionary has been renamed.** In November 2023 the Ayushman Bharat Health and Wellness Centres were renamed and upgraded as **Ayushman Arogya Mandir (AAM)** with the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth"), reorienting the platform towards AYUSH-based holistic wellness — lifestyle and yoga counselling, mental health support, nutrition and elderly care — alongside the existing packages; the renaming carried a budget of Rs 3,000 per facility and a total outlay of about Rs 48 crore, and renamed facilities must upload photographs to the portal. In the exam write "sub-centre / Ayushman Arogya Mandir", not "health and wellness centre". [PMID 42257165]
- ▶ **Scale and staffing of the platform the ASHA now feeds into:** more than **1.75 lakh AAMs** operationalised and **1,16,978 community health officers** (mid-level health providers) in position as on 30 November 2024; 82 per cent of AAMs had rolled out all **12 expanded service packages** by November 2024 against 50 per cent in March 2023; **106 medicines and 14 diagnostic tests** are notified at the sub-centre-level AAM (172 and 63 at the PHC-level AAM); and **1,35,343 Jan Arogya Samitis** have been constituted, sitting beside the VHSNC as the community-oversight platform, with **team-based incentives** paid on the collective performance of the primary health team. Delayed payment and inadequate training were the key demotivators named by ASHAs themselves in Uttar Pradesh. [PMID 41816146] [PMID 41628200]

Current national data

- ▶ **NFHS-5 (2019–21), ASHA utilisation:** among 232,920 women, 63.3 per cent reported using ASHA services in their most recent pregnancy (70.6 per cent in the empowered action group states and Assam) and 88.6 per cent had an institutional delivery; ASHA use raised the odds of institutional delivery **1.6-fold (95% CI 1.5–1.7)**, 1.8-fold in EAG states, with a propensity-matched treatment effect of +5.1 percentage points nationally and +7.4 in EAG states. A

difference-in-difference analysis of conditional cash incentives combined with community-health-worker support found a **5-percentage-point** rise in women completing the whole antenatal-delivery-postnatal continuum, and the accompanying interviews found sustained worker engagement mattered more than the cash alone. [PMID 38227565] [PMID 41773900]

- **Anganwadi services — reach ahead of quality:** 1.4 million anganwadi workers serve about 158 million ICDS beneficiaries; NFHS-5 data on 59,115 children aged 6–23 months show ICDS use associated with 19 per cent higher minimum dietary diversity (aPR 1.19; 95% CI 1.13–1.25) and 27 per cent higher egg or flesh-food intake, but a 14 per cent **lower** chance of minimum meal frequency (aPR 0.86; 95% CI 0.83–0.90) and no gain in minimum acceptable diet; in Bihar and Madhya Pradesh only 48 per cent of beneficiaries got growth monitoring, 52 per cent take-home rations, 37 per cent an adequate number of home visits and 45 per cent infant-feeding counselling, with training, timely salary and supervision significantly associated with delivery; and a block assessment in eastern Odisha found VHND sessions held at 91.3 per cent of sites but iron and folic acid stocked at only 73.8 per cent and calcium at only 42.9 per cent of them. [PMID 41372851] [PMID 40679992] [PMID 40949520]

Elimination, surveillance and disease status

- **Malaria elimination target 2030:** in Nuh, Haryana — a district with over 12 times the state annual parasite incidence — ASHAs achieved 100 per cent same-day treatment initiation and completion and 87.5 per cent follow-up in the 8 confirmed malaria cases, but only 1.8 per cent of mothers went to an ASHA first for childhood fever, only 4 per cent of febrile children were tested within 24 hours, and 73.6 per cent said they usually approach informal providers such as hakims and traditional healers — fidelity is not the bottleneck, first-contact reach is. [PMID 41356870]
- **The non-communicable disease role is now evidence-backed:** a systematic review of 13 Indian community-health-worker interventions (9 pooled) found systolic blood pressure fell by 5.5 mmHg (95% CI –9.0 to –1.9; –7.5 mmHg among hypertensives) and BMI by 0.63 kg/m², with no significant change in HbA1c, fasting glucose or lipids and GRADE certainty low to very low; and in a cluster-randomised trial across 16 villages served by 8 PHCs in Guntur and Rohtak, ASHAs using a mobile clinical decision-support system achieved an HbA1c fall of at least 0.5 per cent in 21.8 versus 10.3 per cent on usual care, with better medication adherence (63.0 vs 43.1 per cent). [PMID 42555695] [PMID 41120559]

Guideline and legislative updates

- **The CBAC cut-off the ASHA applies is under challenge:** in 32,686 adults aged 30 years and above screened in urban Mysuru, CBAC discriminated poorly against point-of-care testing (AUC 0.62, 95% CI 0.61–0.63 for diabetes; AUC 0.61, 95% CI 0.60–0.61 for hypertension) and the score >4 threshold gave a Youden index of only 0.17 and 0.15 respectively; the authors propose that frontline workers add random capillary blood glucose and blood pressure measurement alongside the checklist. Quote score >4 as the current programme rule, but know this validation critique. [PMID 42555696]
- **PCPNDT enforcement rests on the same village workers:** among 213 ASHAs and ANMs in Jodhpur, 93.8 per cent knew of the PCPNDT Act but only 50.2 per cent knew the Rajasthan **Mukhbir Yojana** (informer scheme, 2015; Rs 3 lakh on a successful decoy operation, split 50 per cent to the pregnant woman, 33.3 per cent to the informer and 16.7 per cent to her supporter)

and only 23 per cent had been trained for it; NFHS-5 places the sex ratio at birth at 929 females per 1,000 males against 919 in 2015–16. [PMID 42257158]

Global and WHO developments

- ▶ **WHO recognition of the cadre:** India's one million ASHAs received a **Global Health Leaders Award** at the **75th World Health Assembly** — one of six Global Health Leaders Awards announced by the WHO Director-General on 22 May 2022, and one of only two given to a collective rather than an individual — for linking communities living in rural poverty to primary care, as shown through COVID-19; the accompanying commentary notes they still work 6–7 hours a day (10–12 at the pandemic peak) for roughly USD 20–60 a month as "volunteers", leaving fair remuneration, training and grievance redress as the unfinished agenda. [PMID 37384262]
- ▶ **Global evidence for investing in the cadre:** scoping reviews of 2015–2024 economic evaluations found community-health-worker programmes cost-effective in 35 of 44 non-communicable-disease scenarios and in the majority of 100 reproductive, maternal, newborn and child health scenarios in low- and middle-income countries, though few assessed affordability and worker salary data were often missing; India's UHC service coverage index rose from 57 in 2015 to 63 in 2021, just above the South-East Asia regional average of 62. [PMID 41371924] [PMID 41570010] [PMID 41816146]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community; Ch 11 Nutrition and Health); Park's Textbook of Preventive and Social Medicine, 27e (Ch 2, for the 1977 launch of the Village Health Guides' scheme); Sunder Lal — Textbook of Community Medicine (Ch 5 Nutrition; Ch 8; Ch 10 RMNCH+A; Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community); Jugal Kishore — National Health Programmes of India (Ch 1, for ASHA selection, training and village health committee functions); Integrated Child Development Services (ICDS) operational guidance. **Recent-advances sources:** 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam', J Family Med Prim Care 2026, DOI [10.4103/jfmprc.jfmprc.1966.24](https://doi.org/10.4103/jfmprc.jfmprc.1966.24); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, J Family Med Prim Care 2026, DOI [10.4103/jfmprc.jfmprc.478.25](https://doi.org/10.4103/jfmprc.jfmprc.478.25); 41628200 — Factors affecting the motivation of community health workers: perspectives from ASHAs in Uttar Pradesh, PLoS One 2026, DOI [10.1371/journal.pone.0341811](https://doi.org/10.1371/journal.pone.0341811); 38227565 — Association between the use of ASHA services and uptake of institutional deliveries in India, PLOS Glob Public Health 2024, DOI [10.1371/journal.pgph.0002651](https://doi.org/10.1371/journal.pgph.0002651); 41773900 — Conditional cash incentives, community health workers, and continuum of maternal and child healthcare: evidence from India, Health Policy Plan 2026, DOI [10.1093/heapol/czag019](https://doi.org/10.1093/heapol/czag019); 41372851 — Integrated Child Development Services and complementary feeding practices among children aged 6-23 months in India, BMC Public Health 2025, DOI [10.1186/s12889-025-25884-2](https://doi.org/10.1186/s12889-025-25884-2); 40679992 — Factors associated with Anganwadi Workers' service delivery of ICDS in rural India, PLoS One 2025, DOI [10.1371/journal.pone.0326971](https://doi.org/10.1371/journal.pone.0326971); 40949520 — Service gap assessment of Village Health and Nutrition Day services in a rural block of eastern Odisha, Indian J Community Med 2025, DOI [10.4103/ijcm.ijcm.293.23](https://doi.org/10.4103/ijcm.ijcm.293.23); 41356870 — Implementation fidelity of ASHA workers in malaria case management in a high-burden district of north India, Cureus 2025, DOI [10.7759/cureus.96103](https://doi.org/10.7759/cureus.96103); 42555695 — Community health worker-led interventions for cardio-metabolic diseases in India: systematic review and meta-analysis, Indian J Med Res 2026, DOI [10.25259/ijmr.3058.2025](https://doi.org/10.25259/ijmr.3058.2025); 41120559 — Preliminary effectiveness and feasibility of ASHA-led mobile health intervention for diabetes care in Indian primary health care settings, Sci Rep 2025, DOI [10.1038/s41598-025-20728-w](https://doi.org/10.1038/s41598-025-20728-w); 42555696 — Screening performance of the community-based assessment checklist compared with point-of-care screening tests in urban Mysuru, Indian J Med Res 2026, DOI [10.25259/ijmr.78.2026](https://doi.org/10.25259/ijmr.78.2026); 42257158 — Perspective of frontline health workers on implementation of the PCPNDT Act and Mukhbir Yojana in Jodhpur, Rajasthan, J Family Med Prim Care 2026, DOI [10.4103/jfmprc.jfmprc.1650.25](https://doi.org/10.4103/jfmprc.jfmprc.1650.25); 37384262 — India's one million ASHAs win the Global Health Leaders award at the 75th World Health Assembly, Lancet Reg Health Southeast Asia 2022, DOI [10.1016/j.lansea.2022.100029](https://doi.org/10.1016/j.lansea.2022.100029); 41371924 — Costs and cost-effectiveness of community health worker programmes focussed on non-communicable diseases in LMICs (2015-2024), BMJ Glob Health 2025, DOI [10.1136/bmjgh-2024-018035](https://doi.org/10.1136/bmjgh-2024-018035); 41570010 — Costs and cost-effectiveness of community health worker programs on reproductive, maternal, newborn

and child health in LMICs (2015-2024), PLOS Glob Public Health 2026, DOI [10.1371/journal.pgph.0004893](https://doi.org/10.1371/journal.pgph.0004893).



Community participation in NRHM

Asked in: Paper IV Nov 2017 — RGUHS MD Community Medicine

Definition — Community participation (community involvement) is defined by the World Health Organization (WHO) and the United Nations Children's Fund (UNICEF) as "the process by which individuals and families assume responsibility for their own health and welfare and for those of the community, and develop the capacity to contribute to their and the community's development" (Park 28e, Ch 1, pp 21-22). It further implies participation of the community in the planning, organization and management of their own health services — summed up as "**Health by the People**" (Park 28e, Ch 1, p 22).

Meaning and scope

- ▶ It is one of the **principles on which primary health care is built**: universal coverage cannot be achieved without the local community, so there must be continuing effort to secure meaningful involvement in the **planning, implementation and maintenance** of health services, with maximum reliance on local manpower, money and materials (Park 28e, Ch 10, p 720).
- ▶ It marks the shift from "**health care for the people**" to "**health care by the people**" — active involvement of families and communities in planning, implementation, utilization, operation and evaluation of services, i.e. a deliberate "demedicalization" of health (Park 28e, Ch 2, p 34).
- ▶ The *Declaration of Alma-Ata* holds that people have a **right and a duty** to participate individually and collectively in planning and implementing their health care; without it, health programmes are unlikely to succeed (Park 28e, Ch 22, p 1019). **No standard pattern** can be prescribed, since economic, social, political and cultural traits vary widely between and within communities — flexibility of approach is essential (Park 28e, Ch 2, p 34).
- ▶ Indian precedents: the **Village Health Guides scheme (1977)** and trained *dais*, now the **Accredited Social Health Activist (ASHA)** and *Anganwadi* worker under the National Health Mission; the concept drew heavily on China's **barefoot doctors** (Park 28e, Ch 2, p 34; Ch 10, p 720).

Types (degrees) of community participation

#	Type	What the community does	Indian example
i	Management	Takes responsibility to manage health services at village level	Management of sub-centres by village <i>Panchayat</i> / women members of Panchayati Raj Institutions (PRIs)

#	Type	What the community does	Indian example
ii	Resource generation	Generates its own resources to construct and maintain buildings	Construction and upkeep of sub-centres and schools
iii	Needs assessment and priority setting	Itself determines health needs and priorities and works to meet them	Village health action plan built on the household/village health register
iv	Implementation	Organises and implements the programme with its own resources plus government resources	Village Health, Nutrition and Sanitation Day (VHSND) at the <i>Anganwadi</i> centre
v	Monitoring	Monitors government-run programmes and gives feedback to the system	Community monitoring system, <i>Rogi Kalyan Samiti</i> monitoring of primary health centre (PHC) functions
vi	Evaluation	Evaluates programmes and issues a "report card" for corrective action	Community-based evaluation feeding into replanning of services

(Sunder Lal, Ch 2, pp 68-69)

Community-participation platforms already built into the NRHM

The NRHM was launched on 5 April 2005 for 2005-2012 (later extended to 2017), with focus on 18 high-focus states — 8 Empowered Action Group states (Bihar, Jharkhand, Madhya Pradesh, Chhattisgarh, Uttar Pradesh, Uttarakhand, Orissa, Rajasthan), 8 North-Eastern states (Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura), Himachal Pradesh and Jammu & Kashmir (Park 28e, Ch 7, pp 537-538). Increasing participation and ownership by the community (decentralization) is the first of its five pillars, and means transfer of funds, functions and functionaries to PRIs (Sunder Lal, Ch 17, p 737).

Level	Community institution	Composition / norm	Untied money	Participatory function
Village	Village Health, Sanitation and Nutrition Committee (VHSNC)	<i>Sarpanch</i> , ASHA (member secretary), <i>Anganwadi</i> worker, multipurpose health worker female / auxiliary nurse midwife (ANM), school teacher, self-help group member, non-governmental organisation (NGO)	Rs 10,000 per annum per village <i>panchayat</i>	Prepares and implements the village health action plan; monitors ASHA, <i>Anganwadi</i> worker and sub-centre; runs monthly VHSND
Village	ASHA	Resident woman, married/widow/divorced, 25-45 years, schooling up to 8th class; 1 per 1000 population (relaxed to 1 per habitation in tribal, hilly and desert areas)	Performance-based compensation (honorary volunteer)	Interface between community and public health system; accountable to the village <i>Panchayat</i>

Level	Community institution	Composition / norm	Untied money	Participatory function
Village	<i>Mahila Swasthya Sangh / Mahila Arogya Samiti (MAS)</i>	Women's group; MAS covers 50-100 households (urban)	—	Community-based women's platform for demand generation
PHC / CHC / district hospital	<i>Rogi Kalyan Samiti</i> (hospital management committee / users' group)	One at each PHC, CHC and district hospital; PHC to be transferred to elected PRIs for management and control	PHC: Rs 25,000 untied + Rs 50,000 maintenance per annum; community health centre (CHC): Rs 1 lakh maintenance per year	Community control of the facility; power to generate and use its own funds; monitors facility functioning
District / state	District Health Society and District Health Mission; State Health Mission	PRIs, officials, NGOs	Resource envelope against an approved plan	Decentralized district and village health planning with PRIs and people

(Sunder Lal, Ch 17, pp 737-738, 744, 751, 754-755; Park 28e, Ch 7, pp 535, 539)

Strategies to improve community participation in the NRHM

- **Complete the devolution to Panchayati Raj Institutions.** Transfer funds, functions and functionaries to PRIs; give PRIs an increased role in the ASHA programme (selection, supervision, grievance handling); progress the proposed transfer of PHCs to elected PRIs for management and control (Sunder Lal, Ch 17, pp 737, 751-752).
- **Make the VHSNC a working planning body rather than a paper committee.** Ensure every village has a constituted committee with the full statutory composition, that the Rs 10,000 untied grant actually reaches and is spent by it, and that it meets monthly, prepares the village health action plan and reviews the sub-centre, ASHA and *Anganwadi* worker. Move towards **one common village committee** covering health, drinking water, sanitation and ICDS instead of parallel committees (Sunder Lal, Ch 16, p 697; Ch 17, p 739).
- **Strengthen the ASHA as the community's own worker.** Select her from and by the village to the prescribed criteria with adequate representation of disadvantaged groups, train her, pay performance-based incentives promptly, and use her position as **member secretary of the VHSNC, accountable to the village Panchayat**, to keep the community's voice inside the plan (Park 28e, Ch 7, p 539; Sunder Lal, Ch 17, p 738).
- **Activate the *Rogi Kalyan Samiti* at every facility.** Constitute it at each PHC, CHC and district hospital, let it generate and spend its own funds for service improvement, and let it monitor facility functioning — this converts the community from a passive recipient into a part-owner of the hospital (Sunder Lal, Ch 17, pp 751, 754-755).
- **Use flexible financing and untied funds as the lever.** Untied funds at every level — VHSNC, sub-centre, PHC, CHC and district hospital — plus a resource envelope against a locally-developed plan, give the community real discretion; **demand-side financing** such as *Janani*

Suraksha Yojana puts purchasing power directly in the user's hands (Sunder Lal, Ch 17, pp 737-738).

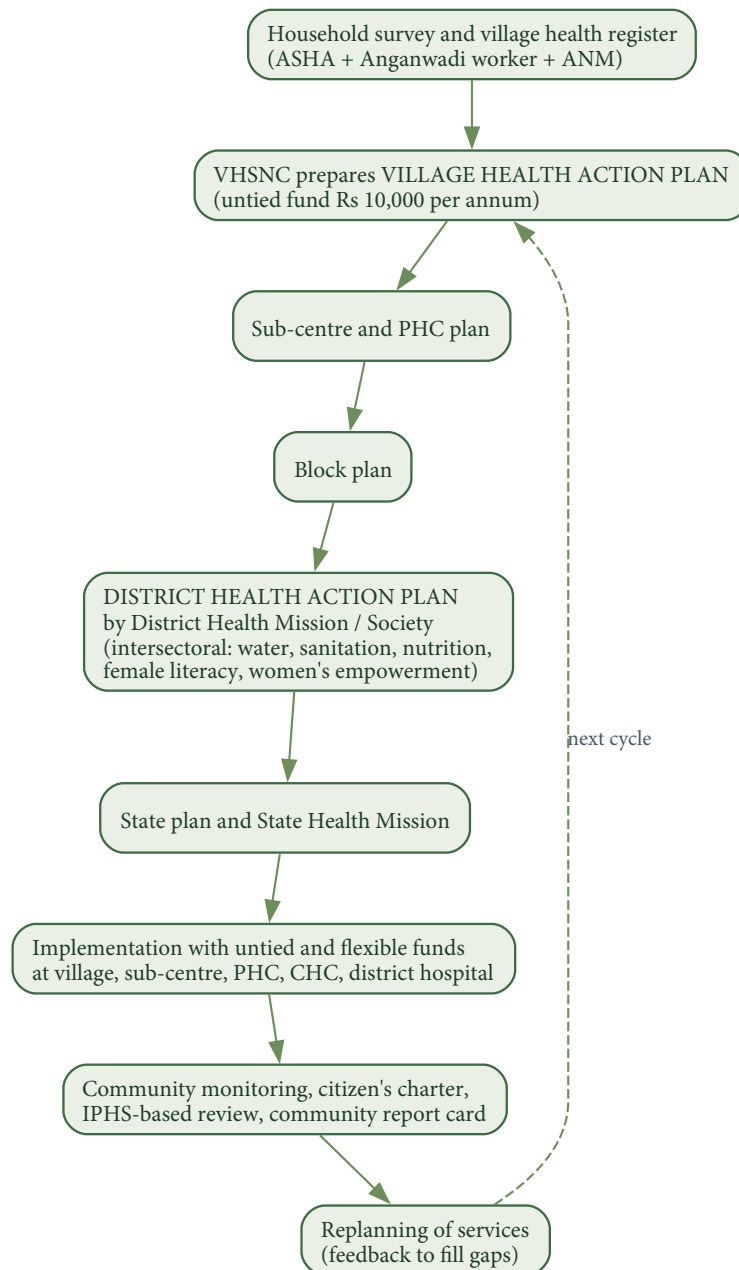
- ▶ **Build bottom-up participatory planning** (flowchart below). The village/census ward is the basic unit; village health action plans built on the household survey and village health register must aggregate upward, so that the **District Health Action Plan** is genuinely made from below and is intersectoral — drinking water, sanitation, hygiene, nutrition, early childhood development, female literacy and women's empowerment (Sunder Lal, Ch 16, p 697; Ch 17, p 739).
- ▶ **Institutionalise community-based monitoring and accountability**. Run the community monitoring system, display a **citizen's charter** of services, entitlements and grievance redressal at each PHC, and let the community measure the facility against the **Indian Public Health Standards (IPHS)**, feeding its "report card" back for corrective action and replanning of services (Sunder Lal, Ch 17, pp 737, 751-752; Ch 2, pp 68-69).
- ▶ **Converge, mobilise and educate**. Make the *Anganwadi* centre and the school the principal hubs of village health action; hold the monthly VHSND with ASHA, *Anganwadi* worker, ANM, self-help groups, community-based organisations and PRIs; use behaviour-change communication and universal health literacy alongside user-friendly services to raise health-seeking behaviour; and give **greater space to NGOs, faith-based and community-based organisations and public-private partnership** (Sunder Lal, Ch 17, pp 739, 744; Ch 5, pp 229-230; Ch 2, p 85; Ch 8, p 384).

Constraints the strategies must overcome

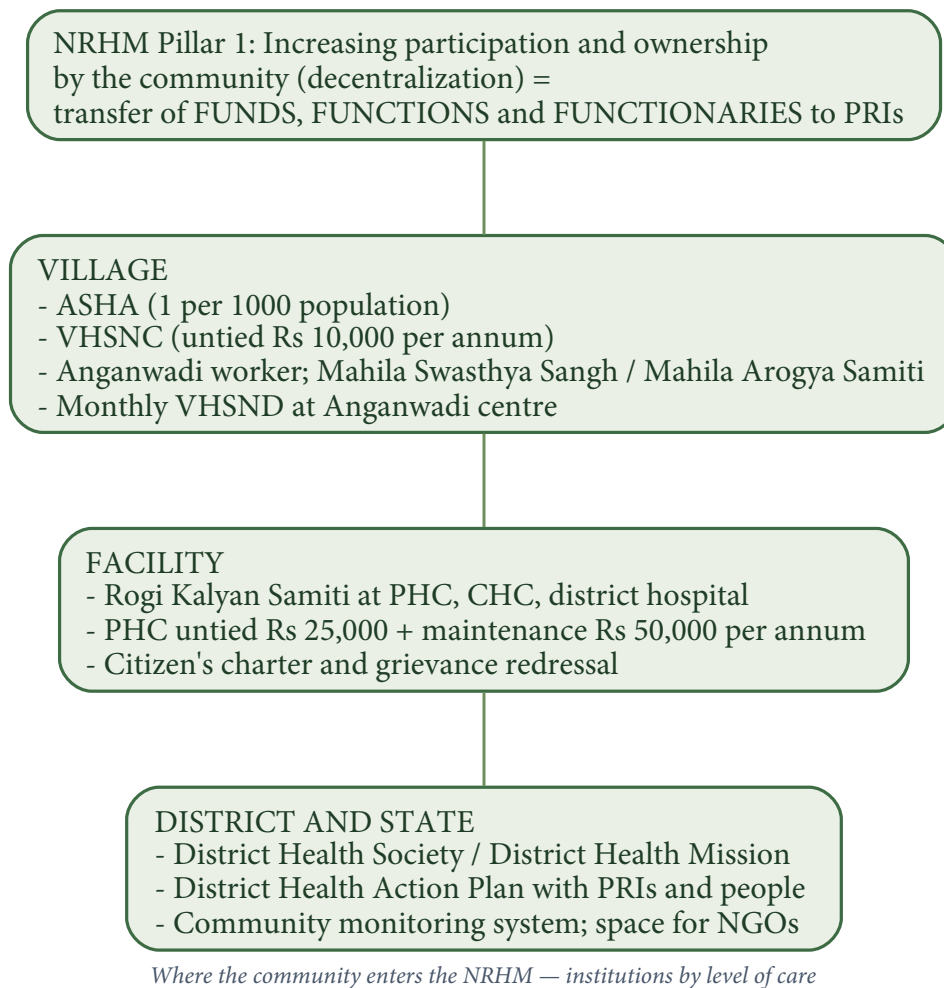
- ▶ Traditional Indian society is cut across by rigid **religion and caste lines**, and fixing an appropriate role for each caste group has been a serious obstacle to complete community participation (Park 28e, Ch 2, p 34).
- ▶ The **greatest resistance** to health-guide involvement in primary health care came from the **medical profession**, not from the lay public (Park 28e, Ch 2, p 34).
- ▶ Community participation "has become an aphorism that is still awaiting genuine realization" — committees exist on paper without funds, training or authority (Park 28e, Ch 2, p 34).
- ▶ Facility weakness undercuts it: PHCs function sub-optimally, essential drugs are unavailable in the majority, and only about one-third provide delivery services (Sunder Lal, Ch 17, p 751).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Village Health, Sanitation and Nutrition Committee (VHSNC) — the village unit of participation**. Draw a central box "VHSNC" with spokes to its members and arrows to its outputs. Exact labels: *centre* — "VHSNC"; *members* — "Sarpanch (Panchayat)", "ASHA — member secretary", "Anganwadi worker (ICDS)", "ANM / MPHW (Female)", "School teacher", "Self-help group member", "NGO representative"; *resource* — "Untied fund Rs 10,000 per annum"; *outputs* — "Village Health Action Plan", "Monthly Village Health, Nutrition and Sanitation Day at Anganwadi centre", "Monitoring of ASHA, AWW and sub-centre".



Bottom-up participatory planning under the NRHM — the village health action plan is the building block



High-yield exam pointers

- ▶ **Write the WHO/UNICEF definition verbatim:** "the process by which individuals and families assume responsibility for their own health and welfare and for those of the community, and develop the capacity to contribute to their and the community's development" — then the phrase "Health by the People".
- ▶ **Slogan:** *health care FOR the people* → *health care BY the people*; Indian examples Village Health Guide (1977) and ASHA; Chinese example barefoot doctors.
- ▶ **Numbers:** NRHM launched 5 April 2005; 18 high-focus states; ASHA 1 per 1000 population; VHSNC untied Rs 10,000 per annum; PHC untied Rs 25,000 + maintenance Rs 50,000 per annum; CHC maintenance Rs 1 lakh per year; 5.55 lakh VHSNCs (by 2021); over 10.72 lakh ASHAs.
- ▶ **Five pillars of NRHM, mnemonic "PICFH":** Participation/ownership by the community (decentralization), Indian Public Health Standards 2006, Capacity building and programme management, Flexible financing (untied funds), Human-resource innovations.
- ▶ **The five sub-strategies of Pillar 1:** PRIs in the ASHA programme → VHSNC → *Rogi Kalyan Samiti* → District Health Society → community monitoring system, plus greater space for NGOs.
- ▶ **Alma-Ata line worth a mark:** people have a *right and a duty* to participate individually and collectively in planning and implementing their health care.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: since May 2013 the NRHM is one of the two sub-missions of the **National Health Mission (NHM)**, the other being the National Urban Health Mission (NUHM), whose community platforms are the *Mahila Arogya Samiti* (50-100 households) and the urban ASHA/link worker (1 per 1000-2500 urban poor); the question uses the older name NRHM, so write NHM once in parentheses and continue with NRHM (Park 28e, Ch 7, pp 533-535). Coverage note: the retrieved corpus prints the ASHA count (over 10.72 lakh) and the *Mahila Swasthya Sangh* count (79,512) without a year, so quote them as "current textbook figures" rather than dating them.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW Ayushman Arogya Mandir portal, World Health Assembly 2024 resolution on social participation for health

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandirs, and each carries its own community committee — the Jan Arogya Samiti (JAS).** JAS is the primary-care counterpart of the *Rogi Kalyan Samiti* and is the newest storey of the participation architecture described above. A mixed-methods appraisal in a Punjab block found the JAS did lead health-promotion activity and support service quality, but stayed passive on backing village-level committees, monitoring services and enabling social accountability, largely because of awareness gaps; perceived decision space, capacity and accountability were low to moderate overall, with **decision space and accountability** significantly higher among elected representatives than among service recipients [PMID 41074751].
- ▶ **The community-monitoring stream now runs under the name Community Action for Health (CAH).** In Assam it began in 2007 in Dhemaji, Chirang and Kamrup districts (alongside eight other states) and later covered 27 districts; a 2021-22 survey of 219 respondents across an aspirational and a non-aspirational district found 94.1% of community members in the aspirational district and 53.6% in the other had never heard of CAH, with district type significantly associated with the need for support in training VHSNC members (chi-square 20.880, $P = 0.010$) and in funds for VHSNC functioning (chi-square 13.609, $P = 0.010$) [PMID 41045525].
- ▶ **Kerala demonstrates what completed devolution to PRIs actually buys.** A witness-seminar realist analysis of the People's Planning Campaign and the N(R)HM found untied plan funds to Local Self-Governments democratised health planning and let communities define local problems and solutions; there the ward-level committee — the **Ward Health, Sanitation and Nutrition Committee (WHSNC)**, not the village VHSNC — was designed as a sub-committee of the LSG with about fifteen members, at least half of them women, and with the **ASHA as chairperson** rather than only member secretary, and ASHAs subsequently emerged as elected LSG leaders. The authors caution that centralised health planning still contradicts the decentralisation ethos [PMID 38439025].

- ▶ **Participation has a documented downside that the strategy must anticipate.** Village-level fieldwork in Dhubri, Assam found community participation acting as a double-edged sword: it widens programme reach but simultaneously generates unrealistic demands on ASHAs and *Anganwadi* workers, draws their households into unpaid service delivery ("frontline families"), and is dominated by better-off households because labouring women cannot spare the time to attend meetings [PMID 42367022].
- ▶ **Participatory learning and action (PLA) through self-help groups is now trial-tested mobilisation, not just exhortation.** *Gram Varta* delivered a 20-meeting PLA cycle through 3,129 Jeevika self-help groups in Madhepura, Bihar; in a cluster-randomised trial across 68 *gram panchayats*, difference-in-differences estimates showed a **marginally significant** 5.8 percentage-point rise in contraceptive use (95% CI 0.00 to 0.12) and a significant 7.2 percentage-point fall in unmet need for limiting childbirth (95% CI -0.14 to 0.00), but **no significant effect on overall unmet need** and none on spacing — the authors conclude the effect on unmet need was inconsistent [PMID 40597148].

Current national data

- ▶ **1,86,891 Ayushman Arogya Mandirs were functional on 17 August 2026** — 1,34,685 sub-health centres, 24,510 PHCs, 5,471 urban PHCs, 9,968 urban health and wellness centres and 12,257 AYUSH centres — which makes the facility-level JAS the most widely replicated participation platform after the VHSNC [MoHFW 2026-Aug].
- ▶ **First state-wide evaluation of Mahila Arogya Samitis (MAS) under the National Urban Health Mission.** Chhattisgarh has 3,700 MAS covering about 3 million urban poor, each a collective of ten women nominated by roughly twenty households apiece, with an annual untied fund of Rs 5,000 and facilitation by the *Mitanin* community health worker cadre. In a representative survey of 401 MAS, 97.3% had held a meeting in the preceding month and 95.3% had acted on at least one of six domains over three years — food security and nutrition 74.1%, sanitation 70.8%, social environment 64.1%, gender-based violence 60.8%, health care 59.1%, drinking water 56.4% — but their action stopped at the immediate neighbourhood and did not shift accountability at higher levels [PMID 40349012].

Elimination, surveillance and disease status

- ▶ **Village-level workers now carry measurable NCD outcomes, which is the strongest current argument for investing in the participation platform.** An Indian systematic review and meta-analysis of 13 interventional studies (10 randomised) found community-health-worker-led interventions lowered systolic blood pressure by 5.5 mmHg (95% CI -9.0 to -1.9; $P = 0.041$; I-squared 79.3%), rising to 7.5 mmHg among those already hypertensive, and body mass index by 0.63 kg/m² ($P = 0.038$); diastolic pressure, fasting glucose, HbA1c and lipids were unchanged, and GRADE certainty was low to very low [PMID 42555695].
- ▶ **Peer-group community engagement is being proposed for the national NCD programme.** Across two primary-care clinics in Karnataka and Telangana, weekly peer-group self-care sessions over 12 weeks (142 of the 202 who underwent baseline investigations completed all sessions; 284 patients were enrolled) produced a median 3 mmHg fall in systolic blood pressure among hypertensive patients, and the authors recommend the model be taken up under the

National Programme for Prevention and Control of Non-Communicable Diseases [PMID 40964732].

Global and WHO developments

- The World Health Assembly passed a 2024 resolution calling for institutionalisation of "social participation for health." India's CAH under the NRHM is cited as a sustained national-scale precedent as countries operationalise it; the same witnessed history warns of *sarkarikaran* — governmentalisation of community processes, in which the community slides from claim-holder to agent and provider of the state — and concludes that institutionalising, standardising and centralising community action is neither lasting nor desirable [PMID 40629314].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1 Man and Medicine: Towards Health for All, pp 21-22; Ch 2 Concept of Health and Disease, p 34; Ch 7 Health Programmes in India, pp 533-540; Ch 10 Health Care of the Community, pp 718-720; Ch 22 Communication for Health Education, p 1019); Sunder Lal — Textbook of Community Medicine (Ch 2 Relationship of Social and Behavioural Sciences to Health and Disease, pp 68-69, 72, 85; Ch 5 Nutrition, pp 229-230; Ch 8 Epidemiology of Communicable and Non-Communicable Diseases, p 384; Ch 16 Health Planning and Management, p 697; Ch 17 Healthcare of the Community, pp 737-739, 744, 751-755); Declaration of Alma-Ata 1978; Indian Public Health Standards 2006.

Recent-advances sources: 41074751 — Exploring Community Participation through Local Health Facility Committees in Punjab: Extent of Decision Space, Capacity, and Accountability, Indian Journal of Public Health 2025, doi:[10.4103/ijph.ijph.421.24](https://doi.org/10.4103/ijph.ijph.421.24) (abstract-only); 41045525 — Status of Community Action for Health in Selected Districts of Assam, India: A Quantitative Study, Indian Journal of Public Health 2025, doi:[10.4103/ijph.ijph.1006.23](https://doi.org/10.4103/ijph.ijph.1006.23) (abstract-only); 38439025 — National Rural Health Mission reforms in light of decentralised planning in Kerala, India: a realist analysis of data from three witness seminars, BMC Public Health 2024, doi:[10.1186/s12889-024-18181-x](https://doi.org/10.1186/s12889-024-18181-x); 42367022 — A river, frontline families, and the double-edged sword of community participation: how nutrition interventions are shaped at the village level in Dhubri, India, Health Policy and Planning 2026, doi:[10.1093/heapol/czag011](https://doi.org/10.1093/heapol/czag011); 40597148 — The impact of a participatory learning and action intervention on unmet need for contraception: a cluster-randomized controlled trial in rural Bihar, India, Reproductive Health 2025, doi:[10.1186/s12978-025-02055-5](https://doi.org/10.1186/s12978-025-02055-5); 40349012 — Community participation through women's health collectives promoted by India's National Urban Health Mission: a realist evaluation in Chhattisgarh state, International Journal for Equity in Health 2025, doi:[10.1186/s12939-025-02498-z](https://doi.org/10.1186/s12939-025-02498-z); 42555695 — Community health worker-led interventions for cardio-metabolic diseases in India: Evidence from a systematic review and meta-analysis, Indian Journal of Medical Research 2026, doi:[10.25259/IJMR.3058.2025](https://doi.org/10.25259/IJMR.3058.2025); 40964732 — Impact of a Community Engagement Program on Treatment Outcomes among People with Noncommunicable Diseases Attending Primary Care Clinics in Two South Indian States, Indian Journal of Public Health 2025, doi:[10.4103/ijph.ijph.1280.24](https://doi.org/10.4103/ijph.ijph.1280.24) (abstract-only); 40629314 — How communitization begets and endures sarkarikaran: a witnessed history of community action for health in India's national rural health mission, BMC Health Services Research 2025, doi:[10.1186/s12913-025-13058-0](https://doi.org/10.1186/s12913-025-13058-0). Also MoHFW Ayushman Arogya Mandir portal, figures as on 17 August 2026; World Health Assembly resolution on social participation for health, 2024.

♦ ♦ ♦

Inter-sectoral coordination in health care

Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — Inter-sectoral coordination is the planned working together of the health sector with all other sectors of national and community development — agriculture, animal husbandry, food, industry, education, housing, public works, communication and others — so that their combined action produces a health gain the health sector alone cannot deliver (Park 28e, Ch 10, p 720). Named by the *Alma-Ata* Declaration (1978), it is the third of the four principles of primary health care, alongside equitable distribution, community participation and appropriate technology (Park 28e, Ch 10, pp 719-720). In Indian programme language it is called **convergence** — "working together or coming together or coordinating two or more sectors to produce a combined effect greater than the sum of their separate effects" (Sunder Lal, Ch 16, p 711).

Related terms distinguished

Term	Meaning
Intra-sectoral coordination	Coordination <i>within</i> the health sector — between its own programmes, levels and personnel (auxiliary nurse midwife, accredited social health activist, medical officer). Listed with inter-sectoral coordination among the World Health Organization's eight supportive activities for primary health care (Sunder Lal, Ch 17, p 741)
Inter-sectoral coordination	Coordination <i>between</i> the health sector and non-health sectors; a programme of community action must utilise all available resources, coordinate the efforts of all other agencies, and secure full community participation (Park 28e, Ch 2, pp 68-69)
Convergence	The Indian National Health Mission term for the same idea; the village is the first level of convergence of the programmes of all ministries (Sunder Lal, Ch 16, p 711)
Health in All Policies (HiAP)	The current, most-evolved form: "an approach to public policies across sectors that systematically takes into account the health implications of decisions, seeks synergies, and avoids harmful health impacts in order to improve population health and health equity" (World Health Organization 2014). National health governance evolved in three waves — inter-sectoral action → healthy public policy → Health in All Policies (Oxford Textbook of Public Health 7e, Ch 51, p 1900)
One Health	"An approach to designing and implementing programmes, policies, legislation and research in which multiple sectors communicate and work together to achieve better public health outcomes" — the human–animal–environment application of the same principle (Sunder Lal, Ch 1, p 36)

Sectors involved and what each contributes to health

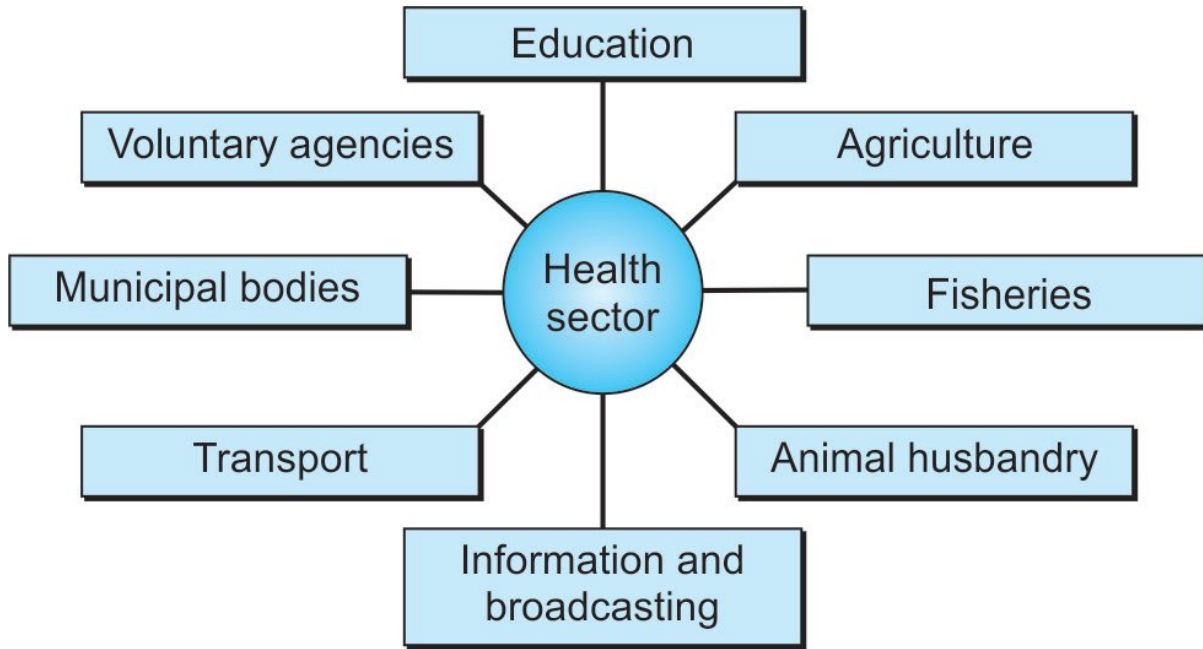


Fig 34.3 — Intersectoral co-ordination of primary health care (Suryakantha Environment Best, p.781)

Sector / ministry	Contribution to health care delivery
Agriculture and food production	Food security, dietary diversity, food fortification; nutrition-sensitive agriculture (Park 28e, Ch 10, p 720)
Women and child development	Integrated Child Development Services (ICDS), supplementary nutrition, <i>Saksham Anganwadi</i> , <i>Mission Shakti</i> , <i>Mission Vatsalya</i> , <i>Poshan 2.0</i> (Sunder Lal, Ch 5, p 230)
Drinking water and sanitation (<i>Jal Shakti</i>)	<i>Swachh Bharat Abhiyan</i> — safe water and improved sanitation for all; piped water supply and toilets for all households by 2024; reduces water- and vector-borne disease and worm infestation (Sunder Lal, Ch 8, pp 339-340)
Education	Female literacy, School Health Programme, <i>Rashtriya Bal Swasthya Karyakram</i> , health education through schools (Sunder Lal, Ch 16, p 711)
Animal husbandry, fisheries	Zoonoses control, animal-source protein, One Health surveillance (Park 28e, Ch 10, p 720)
Urban development and housing	Convergence partners of the National Urban Health Mission — Ministry of Urban Development, Ministry of Housing, basic services for the urban poor, water and sanitation, <i>Rajiv Awas Yojana</i> (Sunder Lal, Ch 17, pp 759-760)
Labour, industry, occupational safety	Industrial and occupational safety, minimum-wage administration, reduction of occupational injury (Jugal Kishore, Ch 2, pp 743, 798)
Information and broadcasting, communication	Mass behaviour-change communication for health programmes (Jugal Kishore, Ch 2, p 798)
Transport — <i>Yatri Suraksha</i>	Preventing deaths due to rail and road traffic accidents (Sunder Lal, Ch 16, p 698)

Sector / ministry	Contribution to health care delivery
Panchayati Raj institutions, urban local bodies, voluntary agencies	Local planning, ownership, monitoring, community mobilisation (Sunder Lal, Ch 17, p 739)

Importance of inter-sectoral coordination in health care delivery in India

- ▶ **The eight essential components of primary health care cannot be delivered by the health sector alone.** Safe water and basic sanitation, food supply and proper nutrition, and health education all belong to other sectors; the *Alma-Ata* goal of Health for All is therefore achievable only through inter- and intra-sectoral coordination (Park 28e, Ch 10, pp 719-720; Sunder Lal, Ch 17, p 741).
- ▶ **India's health indicators are driven by non-medical determinants.** Indicators of health depend as much on drinking water, sanitation, nutrition, female literacy, early childhood development and women's empowerment as they do on hospitals and a functional health system (Sunder Lal, Ch 16, p 711). Most health challenges originate beyond health care, which is why a whole-of-society governance approach is required (Oxford Textbook of Public Health 7e, Ch 51, pp 1882-1883).
- ▶ **Single-sector inputs fail without the complementary sector.** Supplementary nutrition has little effect if diarrhoeal disease in children remains unchecked; safe water and sanitation at the *Anganwadi* centre in convergence with *Swachh Bharat Abhiyan* makes the nutrition input work (Sunder Lal, Ch 16, p 711).
- ▶ **It removes duplication and saves scarce resources.** Joint planning with other sectors — the key element of the inter-sectoral approach — avoids unnecessary duplication of activities and permits reallocation of resources (Park 28e, Ch 10, p 720).
- ▶ **It extends reach to the underserved.** Equity, wider coverage, individual and community involvement and inter-sectoral coordination are the principles that make primary health care a radical departure from conventional health care systems (Park 28e, Ch 1, p 23); in urban areas, where multiple agencies operate, convergence is essential for better outcomes (Sunder Lal, Ch 17, pp 759-760).
- ▶ **It is a stated national policy obligation.** The National Health Policy 2017 mandates the Ministry of Health and Family Welfare to provide a roadmap of coordinated policy initiatives and practical actions implemented **across all sectors** — the "Health in All" approach as a complement to "Health for All" (Jugal Kishore, Ch 2, p 743). The policy identifies **seven priority areas** to address the social determinants of health through inter-sectoral coordination by a "whole of the government" and "whole of the society" approach (Sunder Lal, Ch 16, p 698), and separately sets a group of **five cross-sectoral goals related to health** (Park 28e, Ch 23, p 1030) — both lists are given below.
- ▶ **It is the basis of disease control targets.** Under the malaria programme of the National Vector Borne Disease Control Programme, inter-sectoral coordination happens through coordination committees constituted at various levels, by regular and periodical meetings at policy, planning and implementation level (Sunder Lal, Ch 8, p 384).
- ▶ **The *Alma-Ata* Conference required governments to legislate for it.** Countries may have to review their administrative system, reallocate resources and introduce suitable legislation to

ensure coordination takes place — this requires strong political will to translate values into action (Park 28e, Ch 10, p 720).

National Health Policy 2017 — the seven priority areas for inter-sectoral action (Sunder Lal, Ch 16, p 698)

- ▶ *Swachh Bharat Abhiyan* (Mission) — reduction of water-, excreta- and vector-borne diseases.
- ▶ Balanced healthy diets and regular exercise.
- ▶ Addressing tobacco, alcohol and substance abuse.
- ▶ *Yatri Suraksha* — preventing deaths due to rail and road traffic accidents.
- ▶ *Nirbhay Nari* — action against gender violence.
- ▶ Reduced stress and improved safety at the workplace.
- ▶ Reduction of indoor and outdoor air pollution.

National Health Policy 2017 — cross-sectoral goals related to health (a separate set of five)

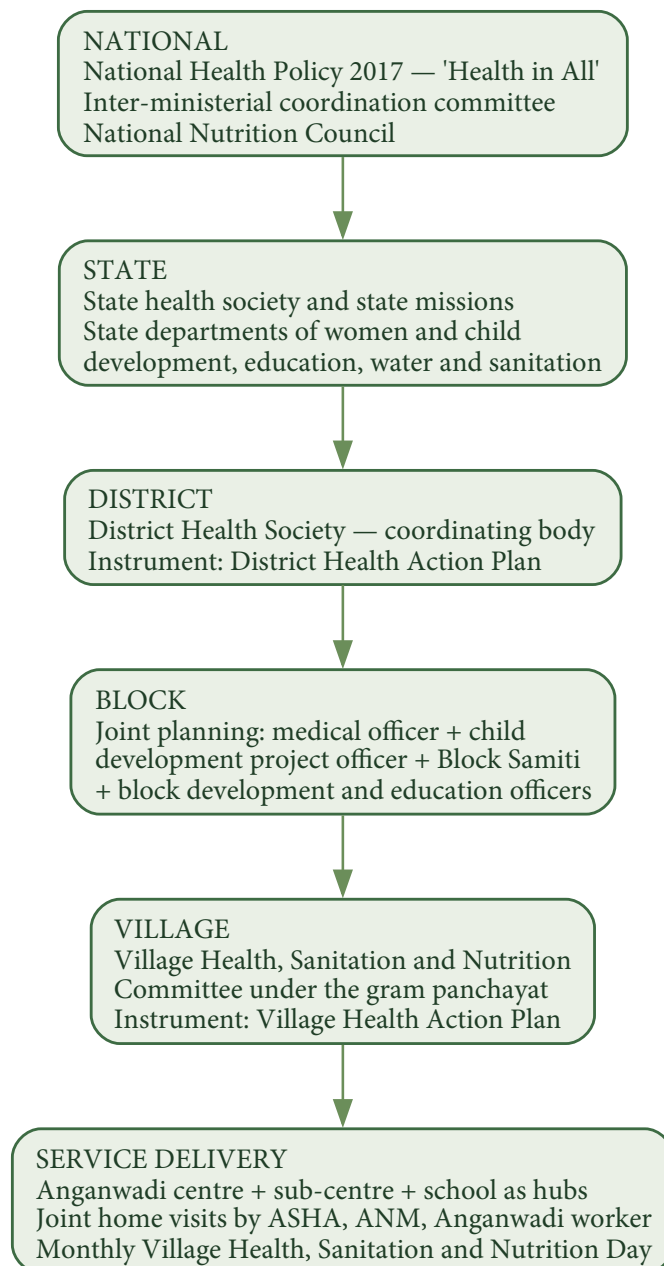
Cross-sectoral goal	Target
Relative reduction in prevalence of current tobacco use	15% by 2020 and 30% by 2025
Reduction in prevalence of stunting of under-five children	40% by 2025
Access to safe water and sanitation to all (<i>Swachh Bharat</i> Mission)	By 2020
Reduction of occupational injury by half from current level of 334 per lakh agricultural workers	By 2020
National/state level tracking of selected health behaviour	Ongoing

(All five: Park 28e, Ch 23, p 1030.)

Machinery of coordination at each administrative level in India

Level	Coordinating body	Instrument of convergence
National	Inter-ministerial coordination committee; National Nutrition Council for policy coordination, review and direction — the National Plan of Action on Nutrition 1995 carried sectoral commitments of 14 nutrition-related ministries and departments (Jugal Kishore, Ch 2, p 798)	National policy and mission documents
District	District Health Society — the coordinating body for decentralised district and village health planning with Panchayati Raj institutions and people; also the coordinating body for urban health (Sunder Lal, Ch 17, pp 737, 760)	District Health Action Plan (Sunder Lal, Ch 16, p 711)
Block	Joint planning by the medical officer, child development project officer (CDPO), <i>Block Samiti</i> , block development and <i>Panchayat</i> offices and block education officer (Sunder Lal, Ch 16, p 711)	Block action plan

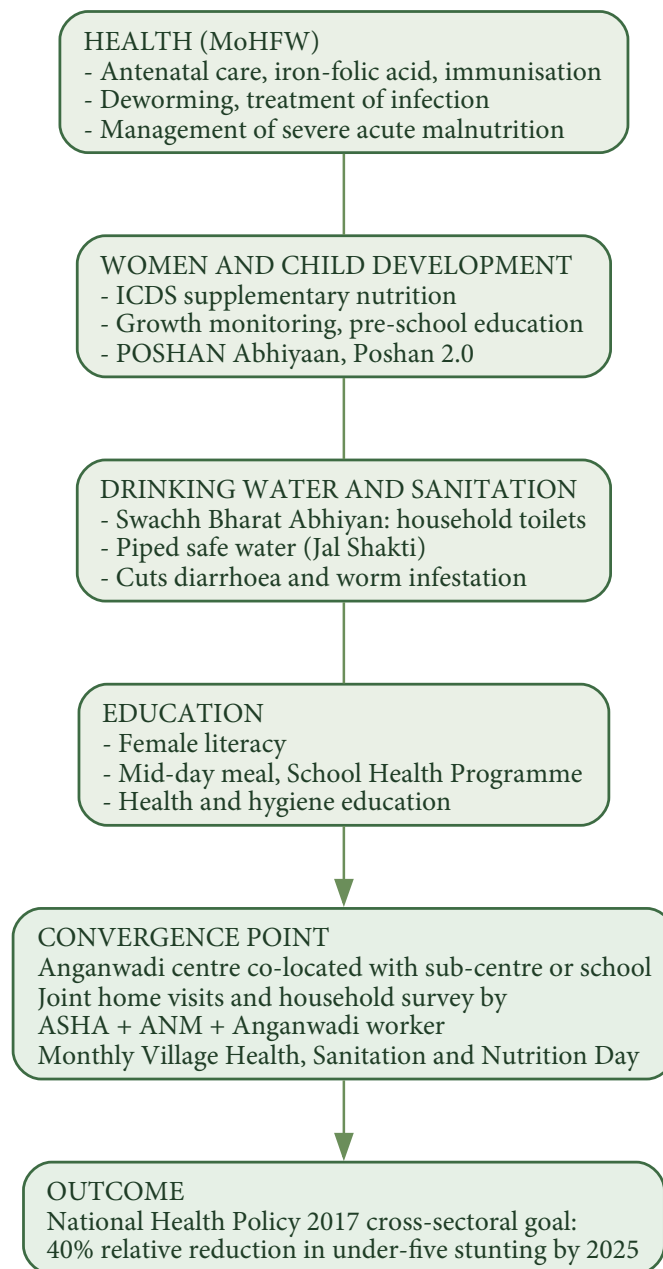
Level	Coordinating body	Instrument of convergence
Village	Village Health, Sanitation and Nutrition Committee (VHSNC) — one common committee replacing separate committees for drinking water, sanitation and ICDS; Panchayati Raj institutions fully involved (Sunder Lal, Ch 17, p 739)	Village Health Action Plan; village health register
Service delivery	<i>Anganwadi</i> centre and the school are the principal hubs for health action; co-location of <i>Anganwadi</i> centres with sub-centres or schools is a best management practice; joint home visits and joint household surveys by <i>Anganwadi</i> worker, auxiliary nurse midwife and accredited social health activist (Sunder Lal, Ch 16, pp 706, 711)	Monthly Village Health, Sanitation and Nutrition Day (VHSND) at the <i>Anganwadi</i> centre (Sunder Lal, Ch 16, p 706)



Machinery of inter-sectoral coordination for health in India, national down to service-delivery level

Worked Indian examples of inter-sectoral coordination

Programme / issue	Sectors brought together	Coordination mechanism and outcome sought
ICDS — "the finest example of intersectoral coordination in operation and achieving results" (Sunder Lal, Ch 5, p 230)	Women and child development (nodal), health and family welfare, education, Panchayati Raj institutions	<i>Anganwadi</i> centre as the single hub; two lakh <i>Anganwadis</i> being upgraded to <i>Saksham Anganwadis</i> with <i>Mission Shakti</i> , <i>Mission Vatsalya</i> and <i>Poshan 2.0</i> (Sunder Lal, Ch 5, p 230)
National Health Mission convergence	ICDS, education, agriculture, rural development, self-help groups, drinking water and sanitation	One common village health and sanitation committee; village and district action plans as the major instruments of convergence (Sunder Lal, Ch 17, p 739)
Nutrition — <i>Poshan Abhiyan</i> , <i>Anemia Mukh Bharat</i> , food fortification	Agriculture, civil supplies and public distribution, food, education, women and child development, health, rural development, information and broadcasting (Sunder Lal, Ch 5, p 217)	<i>Anemia Mukh Bharat</i> and the National Nutrition Mission (<i>Poshan Abhiyan</i>) with the Food Safety and Standards Act for fortification and food safety (Sunder Lal, Ch 8, pp 339-340); convergence action plan drawn up from top to village level (Sunder Lal, Ch 5, p 218)
Sanitation and water	<i>Jal Shakti</i> ministry, rural and urban development, health	<i>Swachh Bharat Abhiyan</i> , piped water and household toilets; success measured by reduction in water- and vector-borne disease and decline in improperly managed solid waste (Jugal Kishore, Ch 2, p 743)
Air pollution and clean fuel	Environment, petroleum, health	National Clean Air Programme; <i>Pradhan Mantri Ujjwala Yojana</i> subsidised liquefied petroleum gas to poor households (Sunder Lal, Ch 8, pp 339-340)
National Urban Health Mission	Urban local bodies, education, ICDS, HIV and AIDS control, urban development, housing, water and sanitation, women and child development	District health society as coordinating body; urban local bodies as units of planning; <i>Mahila Arogya Samiti</i> and <i>Rogi Kalyan Samiti</i> (Sunder Lal, Ch 17, pp 759-760)
Malaria control (National Vector Borne Disease Control Programme)	Sectors represented on programme coordination committees; Panchayati Raj institutions; public-private partnership	Coordination committees at policy, planning and implementation levels with regular periodical meetings; inter-sectoral convergence as a listed supportive intervention (Sunder Lal, Ch 8, pp 372, 384)
School-based programmes — School Health, Adolescent Health, National Tobacco Control, National Oral Health, National Anaemia Control, National Mental Health Programme	Education department, district early intervention centre	Delivered by convergence with the education department (Sunder Lal, Ch 16, p 711)



Convergence pathway — how four sectors act together to reduce child undernutrition

Constraints in the Indian setting and measures to strengthen coordination

- ▶ **Insufficient political commitment**, marginalisation of the health department within government, and limited capacity of public health to influence policy-making are the recognised barriers (Oxford Textbook of Public Health 7e, Ch 51, p 1900).
- ▶ **Vertical, silo-based programmes** with separate funds and reporting lines resist cross-cutting action; broader cross-cutting approaches aligned to country needs and priorities are needed (Oxford Textbook of Public Health 7e, Ch 100, p 3800).

- ▶ **Weak implementation of existing structures** — nutrition has the status of neither a ministry nor a department, all existing departments have their own mandates, and nutrition is not seen as an explicit goal by the concerned sectors (Jugal Kishore, Ch 2, pp 798-799).
- ▶ **Measures to strengthen:** review the administrative system, reallocate resources and enact suitable legislation as the *Alma-Ata* Conference called for; make joint planning with other sectors routine, to prevent duplication (Park 28e, Ch 10, p 720).
- ▶ **Institutionalise coordination fora** with a mandate and capacity for horizontal and vertical coordination, and develop technical capacity, since high turnover of the limited trained personnel is a major limitation in developing countries (Oxford Textbook of Public Health 7e, Ch 10, p 428).
- ▶ **Use decentralised participatory planning as the operating tool** — village, sub-centre, primary health centre, block and district action plans aggregating upward into the district and state plan (Sunder Lal, Ch 16, pp 697, 711).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Inter-sectoral coordination of primary health care** (the plate embedded above): draw the health sector as a central circle with eight boxes radiating from it, labelled **Education, Agriculture, Fisheries, Animal husbandry, Information and broadcasting, Transport, Municipal bodies, Voluntary agencies** (Suryakantha, Fig 34.3, p 781).
- ▶ **Four principles of primary health care:** a square with the corners labelled **Equitable distribution, Community participation, Inter-sectoral coordination, Appropriate technology**, with "Primary health care" at the centre (Park 28e, Ch 10, pp 719-720).

High-yield exam pointers

- ▶ **Four principles of primary health care** (mnemonic ECIA): **Equitable distribution, Community participation, Inter-sectoral coordination, Appropriate technology** (Park 28e, Ch 10, pp 719-720).
- ▶ *Alma-Ata* wording to reproduce: primary health care involves, in addition to the health sector, **agriculture, animal husbandry, food, industry, education, housing, public works, communication and other sectors**.
- ▶ **Convergence** = coordinating two or more sectors to produce a combined effect **greater than the sum of their separate effects**.
- ▶ **NHP 2017 = "Health in All"** complementing **"Health for All"**; **seven** priority areas; **whole-of-government and whole-of-society** approach; public health expenditure to rise to **2.5% of gross domestic product** by 2025.
- ▶ Dated **NHP 2017 cross-sectoral goals (five)**: tobacco use **–15% by 2020, –30% by 2025**; stunting **–40% by 2025**; safe water and sanitation for all by **2020**; occupational injury halved from **334 per lakh agricultural workers** by 2020; national/state-level tracking of selected health behaviour.
- ▶ **ICDS** is the textbook-named "finest example of inter-sectoral coordination"; **VHSNC** is the village-level coordinating body and the **Anganwadi centre** the hub; **District Health Society** is the district-level coordinating body.

- **WHO supportive activities for primary health care (eight):** (1) community involvement and participation, (2) intra- and inter-sectoral coordination, (3) provision of effective referral support, (4) development and mobilisation of resources, (5) managerial processes (policy, goals and objectives), (6) medical and health services research including innovative approaches, (7) development and application of appropriate technology, (8) health manpower development (Sunder Lal, Ch 17, p 741).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the National Health Policy 2017's seven priority areas and its five cross-sectoral goals are two separate lists in the source and are not paired one-to-one — write them as separate lists. The 2020 and 2025 target years quoted are the policy's stated goals with their years attached, not achieved figures. The National Nutrition Mission is written as *Poshan Abhiyan* (now *Poshan 2.0*) with National Nutrition Mission given once in parentheses.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, National Family Health Survey (NFHS-4, NFHS-5), Comprehensive Annual Modular Survey 2022-23, National Action Plan for dog-mediated Rabies Elimination 2021, National Action Plan on Antimicrobial Resistance, Global Fund

Programme and policy updates (India)

- A qualitative study of 52 in-depth interviews at the Ahmedabad Municipal Corporation mapped inter-sectoral collaboration onto the World Health Organization health-system building blocks during COVID-19: police, fire services, solid waste, estate and tax, transport, education and ICDS staff were deployed under the Municipal Commissioner's leadership, while the animal husbandry department converted its nitrogen plant into an oxygen plant and protected the milk supply chain during lockdown. The authors conclude that *structured and systematic* collaboration, not need-based improvisation, is what sustains service delivery in an emergency — a directly quotable modern illustration of the *Alma-Ata* principle [PMID 39552345].
- The **Pradhan Mantri Ayushman Bharat Health Infrastructure Mission (PM-ABHIM)**, announced 25 October 2021, is funding four zonal National Institutes of Virology through the Department of Health Research and ICMR. A March 2024 consultation across the four zones drew 186 participants from state health departments, the Integrated Disease Surveillance Programme, medical colleges and the Virus Research and Diagnostic Laboratory network, and used the modified **One Health Zoonotic Disease Prioritisation tool** applied through multicriteria decision analysis to fix zone-specific viral priorities — inter-sectoral coordination applied to surveillance and research planning [PMID 42522023].
- Current policy writing now calls the Ayushman Bharat Health and Wellness Centre an **Ayushman Arogya Mandir**. National Comprehensive Annual Modular Survey 2022-23 data on 302,086 households show publicly funded health insurance (mainly Ayushman Bharat-PM-JAY) was associated with higher outpatient (adjusted OR 1.40, 95% CI 1.33-1.48) and inpatient use (adjusted OR 1.17, 95% CI 1.13-1.21) and lower catastrophic inpatient expenditure (adjusted OR 0.79, 95% CI 0.74-0.85), but outpatient out-of-pocket spending stays largely unprotected; the

authors' remedy is to anchor outpatient and preventive care in the Ayushman Arogya Mandirs and link them to insurance reimbursement [PMID 42167946].

Current national data

- ▶ ICDS utilisation among children aged 6-59 months rose from **58% (NFHS-4, 2015-16) to 71% (NFHS-5, 2019-21)**, while underweight prevalence fell from 37% to 32%; a difference-in-differences decomposition attributes **9-12%** of that fall to ICDS strengthening, with health check-ups the fastest-growing service (+17 percentage points). The textbook's "finest example of inter-sectoral coordination" now carries a measured, if modest, effect size [PMID 38586943].
- ▶ *Swachh Bharat* Mission: across 640 districts over 2011-2020, districts where more than 30% of households received a toilet recorded **5.3 fewer infant deaths and 6.8 fewer under-five deaths per 1,000 live births** ($p < 0.05$), equating to roughly 60,000-70,000 infant deaths averted annually; mean district toilet coverage rose from 46.7% (2003) to 81.2% (2020). This is the strongest quantitative Indian evidence that a non-health ministry's programme buys a health gain the health sector could not deliver alone [PMID 39223201].

Elimination, surveillance and disease status

- ▶ Rabies: the **National Action Plan for dog-mediated Rabies Elimination (NAPRE, 2021)** aligns India with the global *Zero by 2030* target, and Sikkim and Goa have achieved rabies-controlled status through a One Health approach. Nationally, however, about **17.4 million animal bites and 20,565 human rabies deaths** occur annually; a 2022-23 survey found 71% of bite victims received at least three vaccine doses but only **43%** completed the full post-exposure prophylaxis schedule, one in five government facilities had no rabies immunoglobulin and about 30% had no vaccine stock; public health, veterinary, municipal and wildlife data remain poorly interoperable despite the Integrated Health Information Portal [PMID 42494549].
- ▶ Tuberculosis: under the Global Fund cycle 2021-2024, contact investigation across **208 districts in 23 states** symptom-screened 91% of 3,097,831 household contacts of 851,974 people with pulmonary TB; 54% of those screened began preventive treatment, with **84% overall completion** (92% with the shorter 3HP regimen against 83% with 6H, which accounted for most initiations), and household coverage rose from 28% to 98%. Health and Wellness Centres, initially under-used, became the key delivery point once District TB Officers and Chief Medical Officers coordinated capacity building; near-home models saw minimal pre-treatment loss against attrition of up to 40% with centralised facilities [PMID 42583442].

Guideline and legislative updates

- ▶ Animal-welfare law now cuts directly across rabies control: the Prevention of Cruelty to Animals Act 1960 and the **Animal Birth Control Rules (2001, revised 2023)** mandate sterilisation and vaccination over culling, but **August 2025 Supreme Court proceedings** permitting forcible removal of street dogs in defined circumstances were perceived by many stakeholders as conflicting with them. Conflicting state legislation and recurrent culling campaigns divert attention from the $\geq 70\%$ dog-vaccination coverage needed to interrupt transmission — a live example of the *Alma-Ata* demand that governments legislate coherently for coordination [PMID 42494549].
- ▶ Antimicrobial resistance: stakeholder interviews on pharmaceutical manufacturing effluent found India's action on environmental AMR fragmentary — lack of ownership of the National

Action Plan, and a government approach to setting antibiotic discharge limits marked by limited stakeholder engagement, were the major barriers, compounded by economic and capacity constraints that keep more "visible" issues ahead of AMR. The authors argue for strong leadership and **institutionalisation of One Health within governmental structures**, with collaboration across government, academia and industry and an explicit balancing of the interests and power of government, industry and civil society [PMID 40308153].

Global and WHO developments

- For the period beyond 2030, a six-function framework for **integrated governance** names the institutional capacities that inter-sectoral rhetoric usually lacks — coherence, arbitration between competing sectoral claims, shared-outcome financing, interlinkage-aware measurement, equity-sensitive accountability and adaptive learning. Its premise is examinable in its own right: SDG-3 progress depends on decisions taken across environmental, social, economic and fiscal policy, while institutions stay organised by sector, each with its own mandate, budget, indicators and lines of accountability [PMID 42539888].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1, Ch 2, Ch 10, Ch 23); Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 5, Ch 8, Ch 16, Ch 17); Jugal Kishore — National Health Programmes of India (Ch 2); Oxford Textbook of Public Health 7e (Ch 10, Ch 51, Ch 100); Suryakantha — Community Medicine with Recent Advances (Fig 34.3, p 781); Declaration of Alma-Ata 1978; National Health Policy 2017; National Plan of Action on Nutrition 1995.

Recent-advances sources: 39552345 — Intersectoral collaboration and health system resilience during COVID-19: learnings from Ahmedabad, India, *Health Policy and Planning* 2024, DOI [10.1093/heapol/czae045](https://doi.org/10.1093/heapol/czae045); 42522023 — Strengthening India's pandemic preparedness with the four zonal institutes of virology, *Infectious Diseases of Poverty* 2026, DOI [10.1186/s40249-026-01484-z](https://doi.org/10.1186/s40249-026-01484-z); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from CAMS 2022-2023, *BMJ Open* 2026, DOI [10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 38586943 — Utilization of Integrated Child Development Services (ICDS) and its linkages with undernutrition in India, *Maternal & Child Nutrition* 2024, DOI [10.1111/mcn.13644](https://doi.org/10.1111/mcn.13644); 39223201 — Toilet construction under the Swachh Bharat Mission and infant mortality in India, *Scientific Reports* 2024, DOI [10.1038/s41598-024-71268-8](https://doi.org/10.1038/s41598-024-71268-8); 42494549 — One Health implementation for rabies elimination in India by 2030: progress, governance, challenges and policy alignment, *One Health* 2026, DOI [10.1016/j.onehlt.2026.101520](https://doi.org/10.1016/j.onehlt.2026.101520); 42583442 — Lessons learned from implementing a large-scale contact investigation and TB preventive treatment project in India, *IJTLD Open* 2026, DOI [10.5588/ijtldopen.26.0168](https://doi.org/10.5588/ijtldopen.26.0168); 40308153 — 'Mind the gaps': stakeholder perspectives on addressing antimicrobial resistance in the environment in the Indian context, *Global Health Action* 2025, DOI [10.1080/16549716.2025.2491200](https://doi.org/10.1080/16549716.2025.2491200); 42539888 — Public health and integrated governance across the sustainable development goals beyond 2030: a six-function framework, *Frontiers in Public Health* 2026, DOI [10.3389/fpubh.2026.1892248](https://doi.org/10.3389/fpubh.2026.1892248).



Health tourism

Asked in: Paper IV 08-Jan-2025 — RGUHS MD Community Medicine

Definition — Health tourism (also called medical tourism, medical value travel, medical journeys, global or cross-border health care) is the practice of travelling to another country for the purpose of obtaining medical treatment in that country (Park Plus, Ch 24, pp 343-344). Operationally it is the provision of cost-effective medical care, with due consideration to quality, in collaboration with the tourism industry, for foreign patients needing specialised treatment and surgery (Jugal Kishore, Ch 2, p 736). In trade language it is the **consumption of health services abroad** — one recognised mode of international trade in health services (Health Economics, Ch 2, pp 45-46).

Streams of health tourism and India's position

Stream	What it covers	India's position
Curative / specialised surgical care	Cardiology and cardiothoracic surgery, joint replacement and orthopaedic surgery, organ and liver transplantation, ophthalmology, oncology, urology, neurology and neurosurgery, gastroenterology, dentistry, cosmetic surgery, reproduction, weight-loss surgery, scans, health screening and second opinions	India's dominant segment; the selling point is " <i>First World treatment at Third World prices</i> " (Jugal Kishore, Ch 2, pp 737-738; Park Plus, Ch 24, p 345)
Wellness and traditional-medicine tourism	Spas, stress relief and rejuvenation centres; Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy (AYUSH)	Kerala pioneered health tourism in India, capitalising on Ayurveda and its cultural heritage (Jugal Kishore, Ch 2, p 737)
Tele-outsourcing of diagnostics	X-rays and scans transmitted electronically and reported by lower-fee physicians in a different time zone	India is a major offshore reporting destination for Western emergency departments (Textbook of International Health, Ch 12, p 673)
Source of the patients	High-income countries (United States, United Kingdom, Canada) seeking cost saving and escape from waiting lists; neighbouring low-income countries (Bangladesh, Iran, Pakistan) where the treatment is simply not available	Both streams converge on Indian corporate tertiary hospitals (Jugal Kishore, Ch 2, pp 736-737)

Scale of health tourism — every figure with its year

Indicator	Figure (year)	Source
Global patients consuming health services abroad	about 4 million patients per year; global market US\$ 40–60 billion (2003 estimate)	Health Economics, Ch 2, pp 45-46
Saving to the travelling patient	75–90% below source-country cost	Textbook of International Health, Ch 12, p 673
Foreign patients treated in India	about 10,000 (2000) → about 1,80,000 (2004) → about 4,00,000 (2013)	Jugal Kishore, Ch 2, p 737; Park Plus, Ch 24, p 345

Indicator	Figure (year)	Source
Indian health-tourism market size	Rs 15.48 billion (2004), growing about 25% per year; Rs 7,500 crore (2013), projected Rs 12,000 crore by 2015	Jugal Kishore, Ch 2, p 737; Park Plus, Ch 24, p 345
Projected share of the health-care delivery market	Rs 5,000–10,000 crore of additional revenue for upmarket tertiary hospitals by 2012, equal to 3–5% of the total health-care delivery market (McKinsey)	Jugal Kishore, Ch 2, p 737
Illustrative price differential	Heart surgery US\$ 4,500 (India) vs US\$ 45,000 (USA); liver transplant US\$ 45,000 (India) vs US\$ 1.5 lakh (Europe); weight-loss surgery US\$ 500–800 vs US\$ 25,000–30,000	Jugal Kishore, Ch 2, p 737
Accredited Indian hospitals	19 Joint Commission International (JCI) and 63 National Accreditation Board for Hospitals (NABH) accredited hospitals (2011); JCI-accredited hospitals more than 19 and rising (2013)	Park Plus, Ch 24, pp 345–346
Single-institution volume	Apollo Hospitals alone treated 95,000 international patients (2013)	Park Plus, Ch 24, p 345

Why India is a destination — SWOT analysis

Quadrant	Points
Strengths	1. World-class range of tertiary services (cardiology, transplants, orthopaedics, oncology, neurosurgery). 2. Quality treatment at about one-fifth the cost abroad. 3. Large supply of qualified doctors, nurses and paramedics with an English-language advantage. 4. High success rates in advanced care and international reputation of hospitals and doctors. 5. Diversity of tourist destinations and luxury resorts. 6. Attraction of the Indian Systems of Medicine — Ayurveda and Yoga. 7. Absence of racial discrimination, which suits patients from Africa (Jugal Kishore, Ch 2, pp 737–738; Park Plus, Ch 24, pp 345–346)
Weaknesses	1. No strong government support or promotional initiative. 2. Poor coordination between airline operators, hotels and hospitals. 3. No uniform pricing policy across hospitals. 4. Too few internationally accredited hospitals, which shrinks the market from developed countries. 5. Non-medical costs such as accommodation inhibit patients from low-income economies. 6. Shortage of trained linguists for non-English-speaking patients (Jugal Kishore, Ch 2, p 738; Park Plus, Ch 24, p 346)
Opportunities	1. Rising demand from the ageing populations of the USA and UK. 2. Supply shortage and waiting lists in the National Health Systems of the UK and Canada. 3. Demand from countries with underdeveloped health facilities. 4. Western employers and insurers seeking lower-cost care and offering tie-up packages abroad. 5. Growing demand for wellness tourism and alternative cures. 6. Low medicine and labour costs (Jugal Kishore, Ch 2, p 738; Park Plus, Ch 24, p 346)
Threats	1. Strong competition from Thailand, Malaysia and Singapore. 2. Lack of international accreditation. 3. Overseas care not reimbursed by foreign insurers. 4. Under-investment in health infrastructure. 5. Difficulty in identifying a genuine practitioner and exploitation of tourists by illegal money changers. 6. Visa and infrastructure problems (Jugal Kishore, Ch 2, p 738; Park Plus, Ch 24, p 346)

Impact on the Indian health care system — favourable

Domain	Effect on the system
Foreign exchange and economy	Attracts foreign currency into a developing country; India earned about US\$ 30 crore a year at the turn of the century from roughly 1,50,000 foreign patients, projected to reach about US\$ 200 crore by 2010 — earned despite foreign insurers not reimbursing treatment taken in India (Sathe & Sathe, Ch 30, pp 335-336; Textbook of International Health, Ch 12, p 673)
Infrastructure and employment	The influx of foreigners <i>"generated increased demands for these services resulting in creation of additional infrastructure, employment generation, development of manpower, and building up of foreign exchange resources"</i> (Lok Sabha Question No. 5515 of 16-5-2007, quoted in Sathe & Sathe, Ch 30, pp 335-336)
Technology upgradation	Medical technology, equipment, facilities and infrastructure are brought up to international standard in Indian tertiary hospitals (Park Plus, Ch 24, p 345)
Quality and accreditation movement	Market forces — medical tourism, insurance and the corporate sector — accelerated the demand for quality in health care and drove hospital accreditation; NABH was constituted under the Quality Council of India to set a uniform national hospital standard in line with worldwide accreditation practice (Jugal Kishore, Ch 1; Sathe & Sathe, Ch 30, pp 336-337)
Revival and export of AYUSH	Foreigners are drawn to Ayurveda, Yoga and the other Indian Systems of Medicine, and the Task Force recommended integrating traditional-medicine facilities into tourism packages (Jugal Kishore, Ch 2, p 737; Sathe & Sathe, Ch 30, p 336)
Explicit policy endorsement	National Health Policy 2002, item 25 (<i>Providing Medical Facilities to Users from Overseas — Health Tourism</i>), strongly encourages providing such services on a payment basis to service seekers from overseas, to be used for development of the economy and of better health services (Jugal Kishore, Ch 2, p 731)

Impact on the Indian health care system — adverse

Domain	Effect on the system
Internal brain drain	Medical tourism draws health professionals away from the public health sector through higher remuneration, an effect already evident in India and Thailand; corporate hospitals "poach" doctors from government and teaching institutions with large sums, which paralyses government health infrastructure and leaves posts vacant in remote areas (Textbook of International Health, Ch 12, pp 673-674; Jugal Kishore, Ch 1, p 158). Health tourism drains trained manpower to the private sector and encourages privatisation, which is dangerous for public health while the private sector remains unregulated (Park Plus, Ch 21, p 278)
Resource diversion and system distortion	It draws resources such as expensive technologies away from the public health sector and distorts already fragile developing-country health systems, with particular damage to primary care and to services for the poor (Textbook of International Health, Ch 12, pp 673-674)
Two-tier (segmented) system	Even where the public sector receives some of the revenue, foreign patients are treated in separate, better wards with dedicated facilities, equipment and health professionals; as the middle classes opt out of public provision, the public system is left as a <i>"poor service for poor people"</i> (Textbook of International Health, Ch 12, pp 673-674; Jugal Kishore, Ch 1, p 24)

Domain	Effect on the system
Cost escalation for Indian patients	The effect on local patients is to make treatment more expensive, at a time when the cost of care is already out of reach of the average Indian patient, who often incurs debt for it (Sathe & Sathe, Ch 30, pp 336-337). Medical tourism and the entry of the corporate sector have raised the cost of health care in the country and generate pressure for larger budgetary allocations to government hospitals merely to stay competitive (Jugal Kishore, Ch 1, p 111). The backdrop is that out-of-pocket spending was 62.4% of total health expenditure (2014 estimate), which National Health Policy 2017 aims to reduce to 50% (Park 28e, Ch 23, p 1033)
Widening inequity	More than 70% of Indians live in rural areas where only 20% of the total hospital beds are located, and with the influx of medical tourists health-care inequality is bound to widen (Jugal Kishore, Ch 1, p 158); urban residents, who are 28% of the population, already access 66% of the country's hospital beds (Jugal Kishore, Ch 1, p 109). There is no evidence that a system catering to foreign medical tourists is more equitable or efficient; the greater likelihood is increased inequality (Jugal Kishore, Ch 1, p 24)
Strain on deficient public facilities	Encouraging public hospitals themselves to attract medical tourists is problematic because existing public health facilities are barely able to meet the requirement of the people within the country (Jugal Kishore, Ch 2, p 736)
Cross-border health security	Frequent cross-border movement, rapid transport and tourism let an outbreak in one country spread easily — in the multi-country SARS outbreak more than 8,000 cases occurred in 26 countries from a single infectious case in a hotel, which led to the International Health Regulations, legally effective in 2007 (Oxford Textbook of Public Health 7e, Ch 28, pp 1071-1072). Travellers import infection in both directions, the subject of <i>emporiatics</i> , the science of the health of travellers (Park 28e, Ch 3, p 152); the liberal medical-visa route has itself been flagged as a possible security concern (Jugal Kishore, Ch 2, p 737)
Consumer protection and ethics	Lack of uniform pricing across hospitals, cheating, difficulty in identifying a genuine practitioner and exploitation of tourists by illegal money changers; transplantation and reproductive services are among the commonest procedures sought, which is precisely why a uniform grading and accreditation system is the government's first stated role (Jugal Kishore, Ch 2, pp 736, 738)

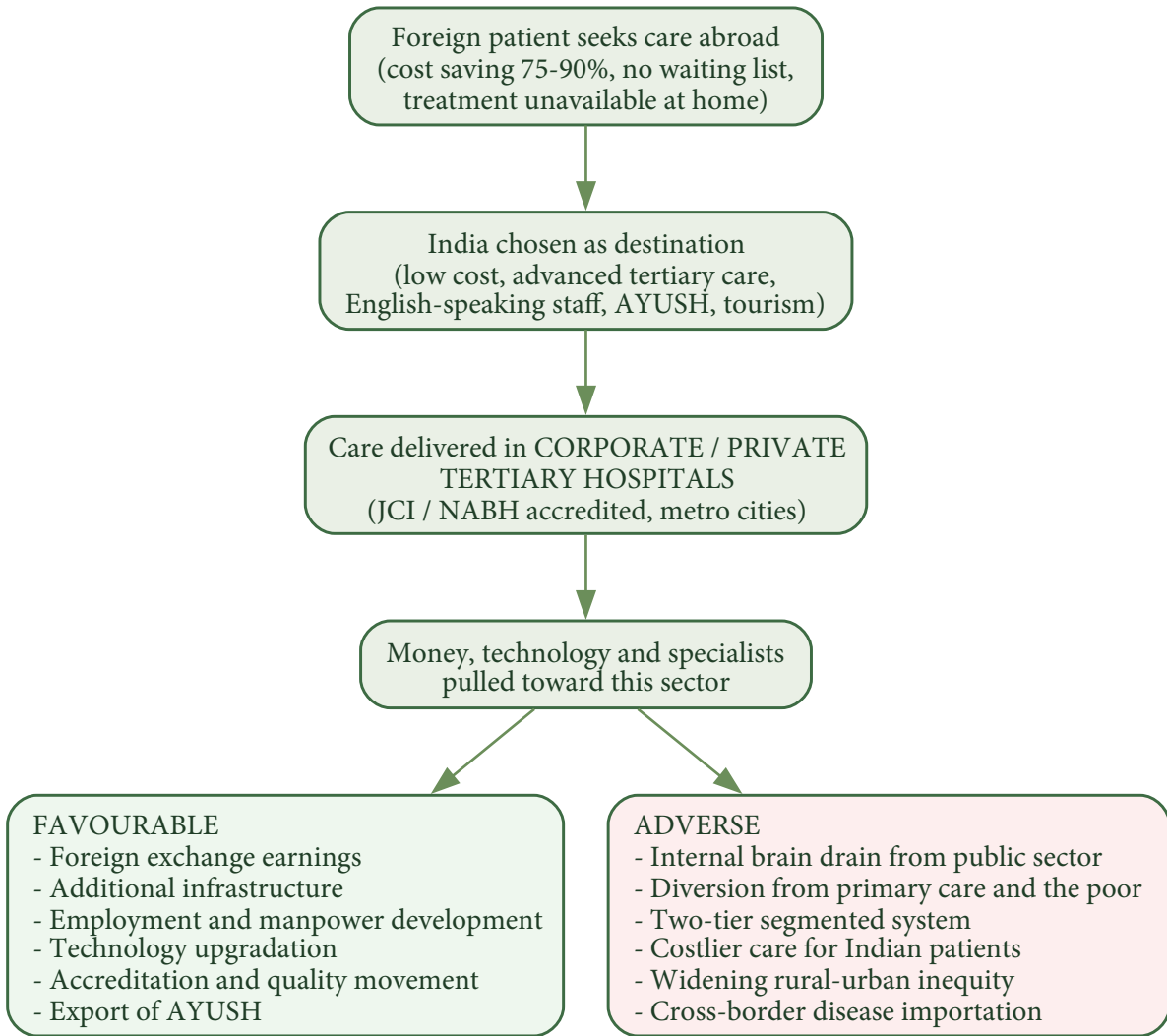
Regulation, stewardship and the public-health response

- **Regulator role** — institute a uniform grading and accreditation system for hospitals to build consumers' trust (Jugal Kishore, Ch 2, p 738); the Clinical Establishments Act 2010 was passed to regulate clinical standards in both the public and the private sector (Park Plus, Ch 21, p 278).
- **Facilitator role** — encourage private investment in medical infrastructure: recognise health care as an infrastructure sector with benefits under section 80-IA of the Income Tax Act, actively promote foreign direct investment, provide low-interest loans, reduce import and excise duty on medical equipment, and simplify clearances and certification (Jugal Kishore, Ch 2, p 738).
- **Travel facilitation** — the Ministry of Home Affairs introduced a separate **Medical ('M') visa** category for foreigners coming to India for treatment, with fast-track clearance for medical patients at airports; further recommendations were visa-on-arrival, an open-sky policy and medical attachés in Indian embassies (Jugal Kishore, Ch 2, pp 737-738; Sathe & Sathe, Ch 30, pp 335-336).

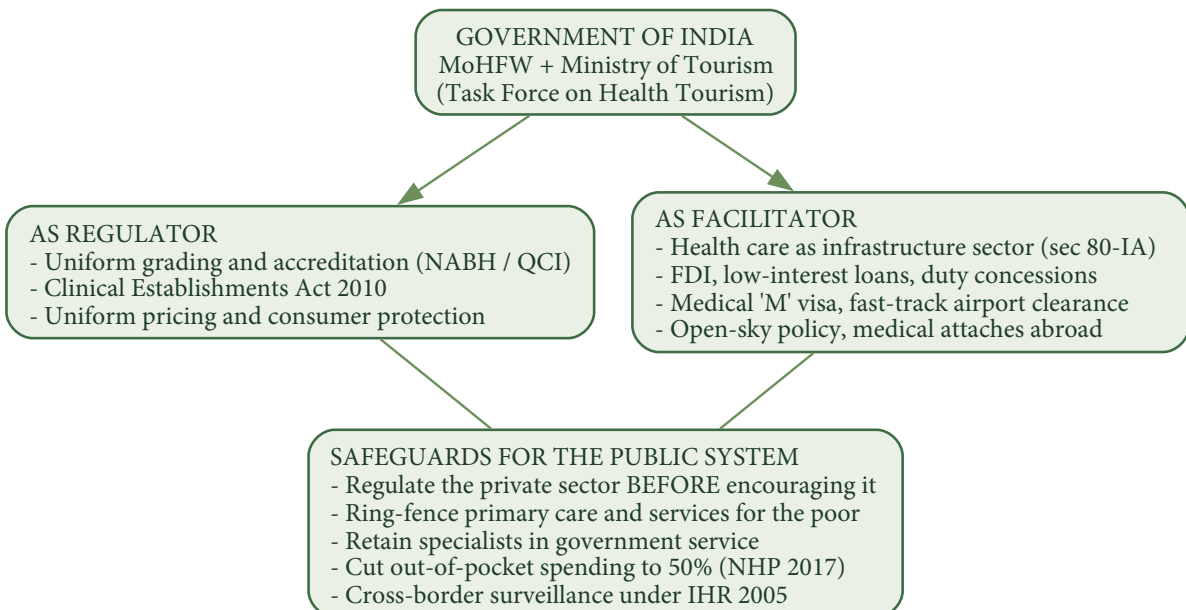
- ▶ **Inter-sectoral machinery** — a **Task Force on Health Tourism** was constituted to promote India as a Health Destination, with members from concerned ministries, State Governments, the Federation of Indian Chambers of Commerce and Industry, the Confederation of Indian Industry and reputed public and private hospitals; it recommended standardisation of hospitals, zone-specific promotional packages, expeditious issue of visas, integration of traditional systems of medicine into tourism packages and concessional land from State Governments. The Ministry of Health and Family Welfare and the Ministry of Tourism jointly evolve the strategy (Sathe & Sathe, Ch 30, p 336; Jugal Kishore, Ch 2, p 737).
- ▶ **Quality assurance** — the National Accreditation Board for Hospitals and Healthcare Providers (NABH) under the Quality Council of India for a uniform national standard, alongside JCI accreditation for the international market (Sathe & Sathe, Ch 30, pp 336-337).
- ▶ **Safeguards the public system needs** — regulate the private sector before encouraging it further, ring-fence primary care and services for the poor against resource diversion, retain specialists in government service, and maintain cross-border disease surveillance under the International Health Regulations.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **DRAWN figure** — "**Balance sheet of health tourism on the Indian health system**". Draw a horizontal beam resting on a central fulcrum labelled **Indian health care system**. Label the left pan **GAINS** and write inside it: *Foreign exchange; Additional infrastructure; Employment and manpower development; Technology upgradation; Accreditation and quality (NABH / JCI); AYUSH promotion*. Label the right pan **COSTS** and write inside it: *Internal brain drain to private sector; Resource diversion from primary care; Two-tier segmented system; Rising cost for Indian patients; Widening rural-urban inequity; Cross-border disease importation*. Under the fulcrum write the four stabiliser labels: *Regulation of the private sector; Uniform accreditation; Protection of primary care; International Health Regulations*.



Pathway by which health tourism acts on the Indian health care system



Twofold role of the Government of India in health tourism, and the safeguards the public system needs

High-yield exam pointers

- ▶ **Definition to write first:** travel to another country to obtain medical treatment there; in trade language, the *consumption of health services abroad*. Synonyms to list: medical tourism, medical value travel, medical journeys, cross-border health care.
- ▶ **Dated numbers worth quoting:** foreign patients in India about 10,000 (2000) → 1,80,000 (2004) → 4,00,000 (2013); global about 4 million patients and a US\$ 40–60 billion market (2003); saving to the patient 75–90%; out-of-pocket spending 62.4% of total health expenditure (2014), National Health Policy 2017 target 50%.
- ▶ **Slogan and counter-slogan:** "*First World treatment at Third World prices*" against the public system reduced to a "*poor service for poor people*".
- ▶ **Three mechanisms of harm** — the spine of the impact answer: internal brain drain → resource diversion from primary care → two-tier segmented system, ending in costlier care and widening inequity for Indians.
- ▶ **Structures and instruments to name:** Task Force on Health Tourism; National Health Policy 2002 item 25; Medical ('M') visa of the Ministry of Home Affairs; NABH under the Quality Council of India and JCI; Clinical Establishments Act 2010; International Health Regulations (legally effective 2007).
- ▶ **Equity anchor:** more than 70% of Indians are rural but hold only 20% of hospital beds; urban residents, 28% of the population, access 66% of beds.
- ▶ **Kerala** is the Indian pioneer of health (wellness) tourism, built on Ayurveda and AYUSH.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook of Preventive and Social Medicine 28e, the spine of this bank, carries no dedicated section on health tourism, so this answer is grounded on Jugal Kishore's National Health Programmes of India, Sathe & Sathe, Park Plus, the Textbook of International Health and Health Economics; the market and patient-volume figures those books quote date from 2000 to 2013 and are historical rather than current, so quote each with the year exactly as given and do not present them as today's position.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, National Health Accounts 2021-22, NOTTO, WHO International Health Regulations amendments 2024, WHO Pandemic Agreement 2025

Programme and policy updates (India)

- ▶ **Heal in India, Heal by India and Heal from India** is the three-part plan by which the Government of India and the hospital industry now promote medical value travel; name this branding, not only the older Task Force on Health Tourism, when the question asks what the government is doing today. *Heal in India* brings foreign patients to Indian hospitals — the inbound transplant flow is growing, with more patients arriving from Africa, the Middle East and South-East Asia, helped by the Ayushman Bharat Digital Mission and simplified medical-visa processing. *Heal by India* sends Indian doctors, surgeons and nurses to countries short of

that expertise. *Heal from India* exports medicines, devices, immunosuppressants and teleconsultation so the patient need not travel at all [PMID 41148243].

- ▶ India holds about **18% of the global medical-tourism market**, and roughly **6,97,453 foreign tourists came for medical treatment in 2019 — 6.4% of all foreign tourist arrivals**; quote this in place of the textbook's 4,00,000 (2013). Medical value travel is one of the 12 Champion Services sectors identified by the Government of India. The **Service Exports from India Scheme (SEIS)**, which gave duty credit on exported hospital, health and paramedical services, was discontinued in 2023 — in its final year (2022-23) it awarded Rs 2,455 crore of duty credits against service exports worth about Rs 49,100 crore — so promotion has shifted to the Market Access Initiative Scheme (2021), the AYUSH Export Promotion Council (2022) and the Ministry of Ayush (created 2014), with the absence of global recognition for Indian health certifications the main remaining barrier to wellness-service export [PMID 40409225].

Current national data

- ▶ Out-of-pocket spending fell from 60.6% of total health expenditure (2014-15) to **39.4%** (**National Health Accounts estimates for India, 2021-22**), and the UHC service-coverage index rose from 57 (2015) to 63 (2021), above the South-East Asia regional average of 62; Health and Wellness Centres have been renamed **Ayushman Arogya Mandir**. Quote these dated figures rather than the 62.4% (2014) estimate when arguing that medical tourism raises costs on a financing base that is improving but still fragile [PMID 41816146].
- ▶ Transplantation, the flagship medical-tourism service, is overwhelmingly private: NOTTO's register lists **682 kidney-transplant centres, 597 (87.5%) private and 85 (12.5%) government**. The **deceased-donor rate is 0.77 per million population** and deceased donation supplies only **15%** of transplants, so India ranks first in the world for living-donor transplants and third for total transplants but **68th for deceased-donor transplants** [PMID 41148243]. About **one in ten transplants in the reported year were done on foreigners** — high by world standards and almost entirely in the private sector, the sharpest single number available for the two-tier argument [PMID 39649309].

Elimination, surveillance and disease status

- ▶ The home system absorbs the complications, and imported resistance patterns defeat local protocols: a prospective four-centre Irish study of 37 patients presenting with surgical-site infection after cosmetic or bariatric surgery abroad (57% operated in Turkey) found Gram-negative organisms in 68% of the culture-positive cases and resistance to the locally recommended empirical antibiotics in 77%, with hospital stays of 1-107 days [PMID 42203044]. A UK rapid review of **655 patients treated by the NHS for complications after surgery abroad** — metabolic or bariatric (n = 385), cosmetic (n = 265) and ophthalmic (n = 5) — put the cost at **£1,058-£19,549 per patient at 2024 prices on very low certainty evidence**, and found no routine data collection on people travelling out for surgery [PMID 41529913]. The mirror image applies to India's inbound flow, which makes cross-border infection and antimicrobial-resistance surveillance part of the health-tourism answer, not an aside.
- ▶ A single facility abroad can seed a multi-jurisdiction outbreak: after *Fusarium solani* meningitis followed epidural anaesthesia at two clinics in Matamoros, Mexico in 2023, US public health officials contacted **233 potentially exposed US residents and reached 170 (73%)**; of those

reached, 104 (61%) reported epidural anaesthesia and were judged at risk, 24 developed fungal meningitis and 12 (50%) died, with genomically closely related isolates from both clinics [PMID 40696772]. This is the modern, citable illustration of why International Health Regulations tracing applies to elective cross-border surgery and not only to epidemics.

Guideline and legislative updates

- ▶ India's current surrogacy and assisted-reproduction law bans commercial surrogacy and bars non-national couples, allowing gestational surrogacy only for couples with a verified medical indication; the reproductive-tourism stream that older texts list among the commonest procedures sought by foreigners has effectively been closed, and the residual disputes are over citizenship, nationality, parentage and the rights of the child [PMID 42579864].
- ▶ Bariatric and metabolic surgery tourism remains largely unregulated worldwide, on a wider medical-tourism industry worth **more than US\$ 400 billion a year with anticipated growth of about 25% annually**; in a global survey of nearly 400 such providers **71% of patients had self-referred**, over half the providers had never heard of international health-care accreditation and fewer than 20% of their hospitals held it. A global-health commentary therefore calls for transnational, multi-actor regulation covering accreditation, marketing transparency, case selection and long-term follow-up [PMID 40664433]. Use this market figure in place of the 2003 US\$ 40-60 billion estimate.

Global and WHO developments

- ▶ The amended International Health Regulations (2005), completed at the 77th World Health Assembly on **1 June 2024** and entering into force in **September 2025**, insert a new "**pandemic emergency**" tier into the previously binary PHEIC alert mechanism and add provisions on equity, financing and implementation [PMID 40883832] [PMID 40589223]. The **WHO Pandemic Agreement** and its 35 articles were adopted by 124 governments at the 78th World Health Assembly on **20 May 2025** — only the second instrument negotiated under Article 19 of the WHO Constitution, after the Framework Convention on Tobacco Control [PMID 40589223]. This is the current legal frame for the cross-border health-security paragraph, in place of the bare "IHR legally effective 2007" line.
- ▶ India is the default destination for oncology referral out of sub-Saharan Africa: in a survey of **52 cancer clinicians from 17 African countries**, **94% named India as their commonest referral destination**, 75% had themselves referred patients abroad and 92% knew patients who self-referred, the commonest reason (93%) being a perception of better quality of care in foreign institutions; the authors frame the outflow as financial toxicity and as lost return on investment in home-grown workforce and cancer infrastructure — the source-country cost of India's gain [PMID 39447099].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, p 152; Ch 23, p 1033); Jugal Kishore — National Health Programmes of India (Ch 1, pp 24, 109, 111, 158; Ch 2, pp 731, 736-738); Sathe & Sathe — Epidemiology and Management for Health Care (Ch 30, The Future, pp 335-337); Park Plus, Amrutha & Vijayalakshmi (Ch 21, p 278; Ch 24, pp 343-346); Textbook of International Health (Ch 12, pp 673-674); Health Economics (Ch 2, pp 45-46); Oxford Textbook of Public Health 7e (Ch 28, pp 1071-1072); National Health Policy 2002 and National Health Policy 2017, Government of India; Lok Sabha Question No. 5515 of 16-5-2007.

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Universal health coverage in India

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — Universal health coverage (UHC) means that all people can obtain the health services they need — health promotion, disease prevention, treatment, rehabilitation, long-term and palliative care, and the technologies these require — of sufficient quality to be effective, without incurring financial hardship in paying for them (Oxford Textbook of Public Health 7e, Ch 107, pp 4044-4045; Ch 103, pp 3915-3916). It is Sustainable Development Goal (SDG) target 3.8: "achieve universal health coverage, including financial risk protection, access to quality essential health-care services and access to safe, effective, quality and affordable essential medicines and vaccines for all" (Park 28e, Ch 14, p 855). It entered global health policy through World Health Assembly resolution WHA58.33 (2005) on sustainable health financing, universal coverage and social health insurance (Oxford 7e, Ch 4, p 246).

The three dimensions of coverage (the WHO "coverage cube")

Axis of the cube	What it asks	Direction of expansion
Breadth — population cover	<i>Who</i> is covered?	Extend cover to the vulnerable and the uncovered
Depth — service cover	<i>Which</i> services are covered?	Scale up primary health care to a full comprehensive package
Height — cost cover (cost sharing)	<i>What proportion of the cost</i> is prepaid and pooled?	Raise public expenditure on health, reduce out-of-pocket (OOP) payment

(Sunder Lal, Ch 17, p 740, after WHO 2009)

Five conditions must be met simultaneously for coverage to be genuinely universal (Oxford 7e, Ch 103, p 3916):

Criterion	Requirement
Availability	Adequate essential services, with enough skilled health workforce on decent working conditions
Affordability	No financial hardship — out-of-pocket payment at the point of service minimised
Accessibility	Equitable geographical distribution; primary and community-based care near the home
Acceptability	Culturally appropriate, gender-sensitive, non-discriminatory, responsive to special needs
Quality and accountability	Quality of care, patient safety, responsiveness to demand, people's engagement

India's instruments for UHC

Instrument	Year	Contribution to UHC
National Health Policy (NHP) 2017	2017	Goal is to "progressively achieve universal health coverage" — free comprehensive primary care for all; affordable secondary and tertiary care through public and private providers; reduction of out-of-pocket and catastrophic health expenditure (Sunder Lal, Ch 16, p 697)
National Health Mission (NHM)	2013 (National Rural Health Mission 2005 + National Urban Health Mission)	"The primary vehicle to move towards universal health coverage"; promotes universal access to a cashless continuum of care from primary to tertiary level (Sunder Lal, Ch 17, pp 739-740)
Ayushman Bharat (AB)	Feb 2018	Paradigm shift from selective to comprehensive care; two inter-related pillars below (Sunder Lal, Ch 17, pp 739-741)
Pillar 1 — Ayushman Arogya Mandir (AAM), formerly AB Health and Wellness Centre	2018	All 1,50,000 sub-health centres (SHC) and all primary health centres (PHC) upgraded to deliver comprehensive primary health care (CPHC) — a package of 12 services, "time to care" within 30 minutes of travel; ~1.1 lakh SHCs transformed by March 2022 (Sunder Lal, Ch 17, pp 742, 746; Park 28e, Ch 7, p 535)
Pillar 2 — Pradhan Mantri Jan Arogya Yojana (PM-JAY)	23 Sep 2018	₹5 lakh per family per year for secondary and tertiary hospitalisation , 10.74 crore poor and vulnerable families (~50 crore people); cashless and paperless at empanelled public and private hospitals; subsumed the Rashtriya Swasthya Bima Yojana (Park 28e, Ch 10, p 745; Sunder Lal, Ch 17, p 763)
Indian Public Health Standards (IPHS) 2022	2022	Minimum "essential" and aspirational "desirable" facility standards, explicitly to accelerate progress towards UHC and SDG-3 in alignment with NHP 2017 (Park 28e, Ch 10, p 728)
Ayushman Bharat Digital Mission (ABDM)	Sep 2021	Health account number and longitudinal electronic health record for every citizen; portability and telemedicine; National Health Authority is the implementing agency (Sunder Lal, Ch 8, p 535; Ch 17, p 763)
Ayushman Bharat Health Infrastructure Mission (PM-ABHIM)	Oct 2021	Health-system infrastructure and pandemic-response capacity in all 733 districts (Sunder Lal, Ch 20, pp 805-806)

PM-JAY design features — no cap on family size or age; entitlement-based on the Socio-Economic and Caste Census (SECC) database; 3 days pre-hospitalisation and 15 days post-hospitalisation expenses covered; an *Ayushman Arogya Mitra* deployed at every empanelled hospital; centre:state funding 60:40 (90:10 for north-eastern states) (Sunder Lal, Ch 17, pp 763-764).

The 12-service CPHC package at the Ayushman Arogya Mandir (Sunder Lal, Ch 17, pp 749-750, 764; Park 28e, Ch 10, pp 735-737) — each with a promotive, preventive and curative component, delivered on a life-cycle continuum-of-care approach:

- ▶ Care in pregnancy and childbirth
- ▶ Neonatal and infant health care
- ▶ Childhood and adolescent health care
- ▶ Family planning, contraception and other reproductive health care
- ▶ Management of communicable diseases and national health programmes
- ▶ General out-patient care for acute simple illnesses and minor ailments
- ▶ Screening, prevention, control and management of non-communicable diseases
- ▶ Screening, basic management and referral for mental health ailments
- ▶ Care for common ophthalmic and ear, nose and throat problems
- ▶ Basic oral health care
- ▶ Elderly and palliative health care
- ▶ Emergency medical services

Supported by free essential medicines and diagnostics, teleconsultation, yoga and wellness activities, and electronic health records. At the SHC-AAM the team is ASHA and auxiliary nurse midwife (ANM) led by a **mid-level health provider (MLHP)** — a **community health officer (CHO)**: a BSc community health graduate, a GNM nurse with BSc, or an Ayurveda practitioner certified through a six-month community-health programme (Sunder Lal, Ch 17, p 742).

Critical appraisal of implementation

Dimension	What has been achieved	Where implementation falls short
Breadth (population)	PM-JAY is the largest government-funded health protection scheme in the world, covering ~40% of the population; ECHS, CGHS and ESIS cover organised sectors	Historically 86% of rural and 82% of urban people had no health expenditure support; the "missing middle" above the SECC cut-off and below organised employment remains outside prepayment (Sunder Lal, Ch 17, p 763)
Depth (services)	AAM extends the package from selective reproductive-and-child-health and communicable disease work to 12 comprehensive services including NCDs, mental health, elderly and palliative care	PM-JAY buys only secondary and tertiary hospitalisation ; out-patient consultation, drugs and diagnostics — where most household spending occurs — sit outside the benefit package
Height (financial protection)	Households facing health spending above 10% of total expenditure fell from 17.3% (2012–20) to 7% (2022)	India's OOP expenditure is the highest in the world — 53.23% of current health expenditure, 1.52% of GDP, ₹2,155 per capita per year; ~4.2% (~50 million people) are pushed into poverty annually by health costs (Sunder Lal, Ch 16, pp 727–729)
Public financing	15th Finance Commission recommends states raise health to 8% of their budget and spend 67% of health expenditure on primary care	Total health spending was 3.16% of GDP (2018–19) with government spending only 1.28% of GDP (40.6% of the total), against the NHP 2017 target of 2.5% of GDP by 2025; global evidence holds that basic needs are seldom met below 5–6% of GDP. "Failure to attain minimum levels of public health expenditure remains the single most important constraint" (Sunder Lal, Ch 16, p 727)

Dimension	What has been achieved	Where implementation falls short
Mixed provision and regulation	Public-private partnership is accepted as essential — no government can reach UHC alone (Oxford 7e, Ch 21, pp 834-836)	The private sector holds 80% of doctors, 25% of nurses and 49% of beds, is the dominant curative provider and is financed mainly by out-of-pocket payment; regulatory frameworks and accreditation for strategic purchasing are weak (Sunder Lal, Ch 16, p 729)
Equity of distribution	Government facilities provide preventive and promotive services to over 95% of the population	Up to 73.6% of doctors are concentrated in urban areas serving 36% of the population; ~66% of India is rural and half the urban population may live in slums, so building facilities and workforce to IPHS norms is an uphill task (Park 28e, Ch 10, p 725; Sunder Lal, Ch 1, p 52)
Efficiency of purchasing	—	Thailand spends about the same total on health as India but 77.7% of it is public, ~95% strategically purchased from public facilities; Brazil spends 9% of GDP with 4.1% public. Brazil, Thailand and Sri Lanka are close to UHC; India is not (Sunder Lal, Ch 16, p 727)

Verdict — India has built the *architecture* of UHC (a stated policy goal, a comprehensive primary care platform, and the world's largest publicly funded hospitalisation cover) but not yet its *substance*: coverage remains shallow in services, leaky in financing, and unequal in distribution, because public spending has not risen to the level the policy itself specifies.

Way forward

- ▶ Raise government health expenditure to 2.5% of GDP by 2025 and state health spending above 8% of state budgets, with two-thirds of it on primary health care.
- ▶ Extend the prepaid benefit package beyond hospitalisation to out-patient care, essential medicines and diagnostics — the main source of out-of-pocket spending.
- ▶ Move from passive reimbursement to **strategic purchasing** with accreditation and provider-payment reform, as Thailand has done.
- ▶ Close the human-resource gap — the WHO/World Bank SDG index threshold is **4.45 doctors, nurses and midwives per 1,000 population** (Oxford 7e, Ch 107, p 4047) — and correct rural-urban maldistribution.
- ▶ Regulate and contract the private sector into public health goals rather than leaving it to be financed by household cash.

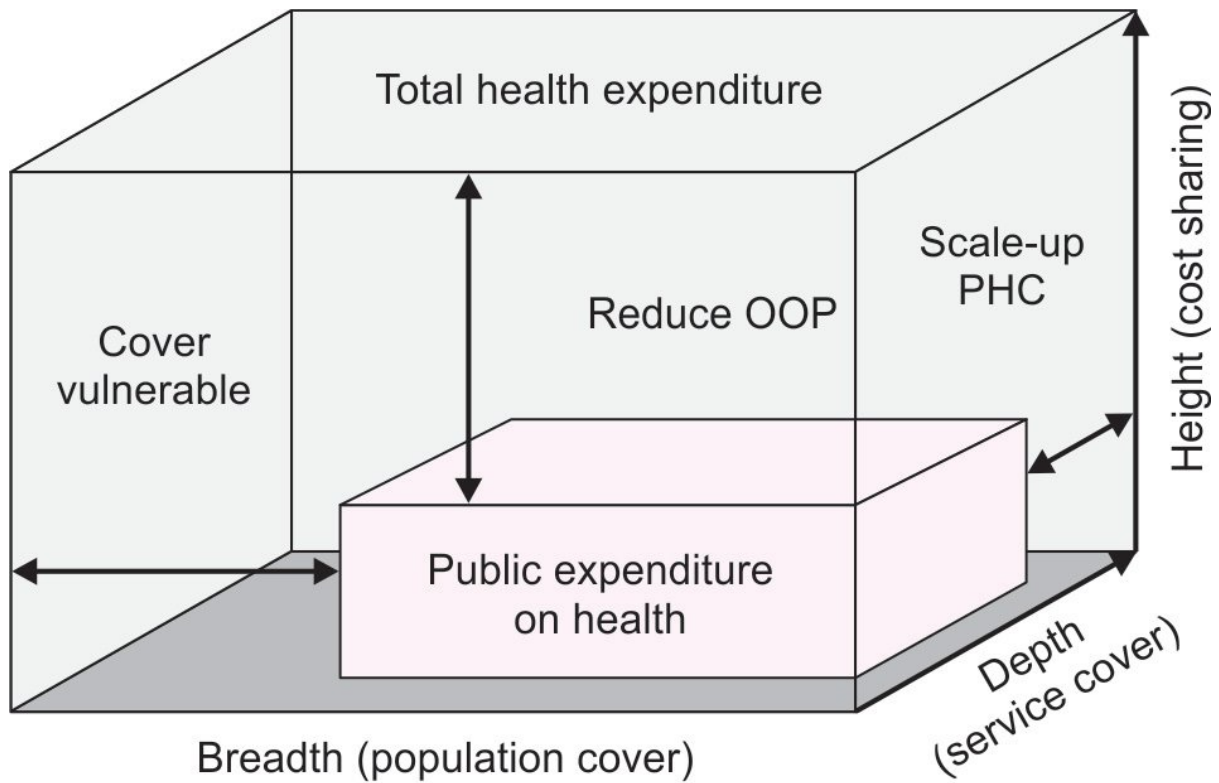
Monitoring UHC — indicators and their definitions

Indicator	Definition / denominator
SDG 3.8.1 — UHC service coverage index	Composite of tracer indicators on service access: hospital access, health workforce density by cadre, access to medicines and vaccines, International Health Regulations core capacities (Park 28e, Ch 14, p 858)

Indicator	Definition / denominator
Reproductive, maternal, newborn and child health tracers	Family planning, antenatal and delivery care, full immunisation coverage, health-seeking behaviour for suspected child pneumonia (Park 28e, Ch 14, p 855)
SDG 3.8.2 — financial protection	Proportion of the population with large household expenditure on health as a share of total household consumption or income (Park 28e, Ch 2, pp 42-43)
Catastrophic health expenditure	Household health cost exceeding 10% of total monthly consumption expenditure or 40% of non-food consumption expenditure (Sunder Lal, Ch 16, p 729)
Impoverishing health expenditure	Proportion of the population pushed below the poverty line by out-of-pocket health payments
Out-of-pocket expenditure share	Out-of-pocket payment as a percentage of current health expenditure; NHP 2017 target is to bring it to 50% of total health expenditure (2014 estimate 62.4%) (Park 28e, Ch 23, p 1033)

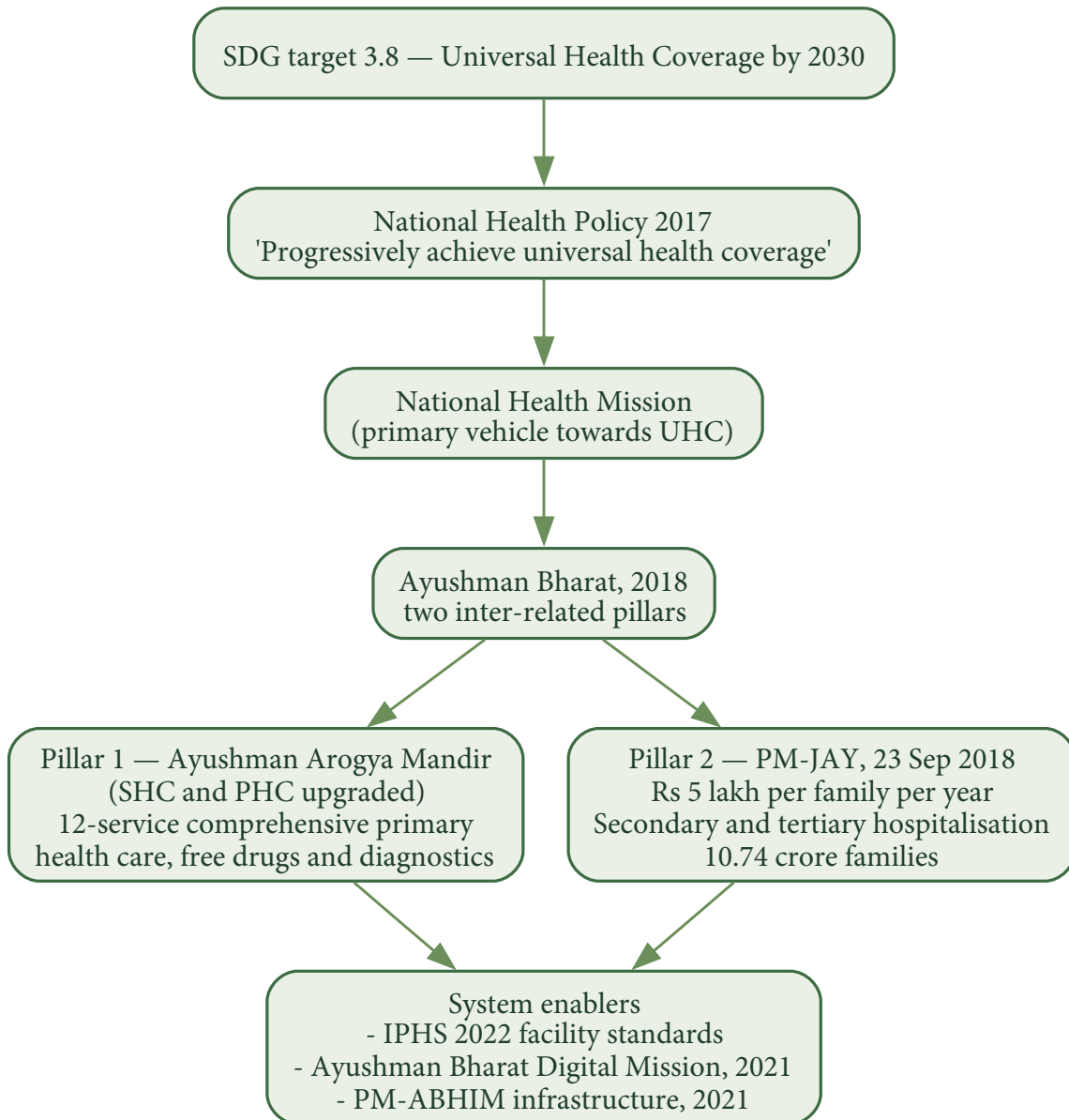
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce on the cube: outer box = *Total health expenditure*; inner solid box = *Public expenditure on health*; horizontal axis = *Breadth (population cover)* with arrow labelled "Cover vulnerable"; depth axis = *Depth (service cover)* with arrow labelled "Scale-up primary health care"; vertical axis = *Height (cost sharing)* with arrow labelled "Reduce out-of-pocket payment".

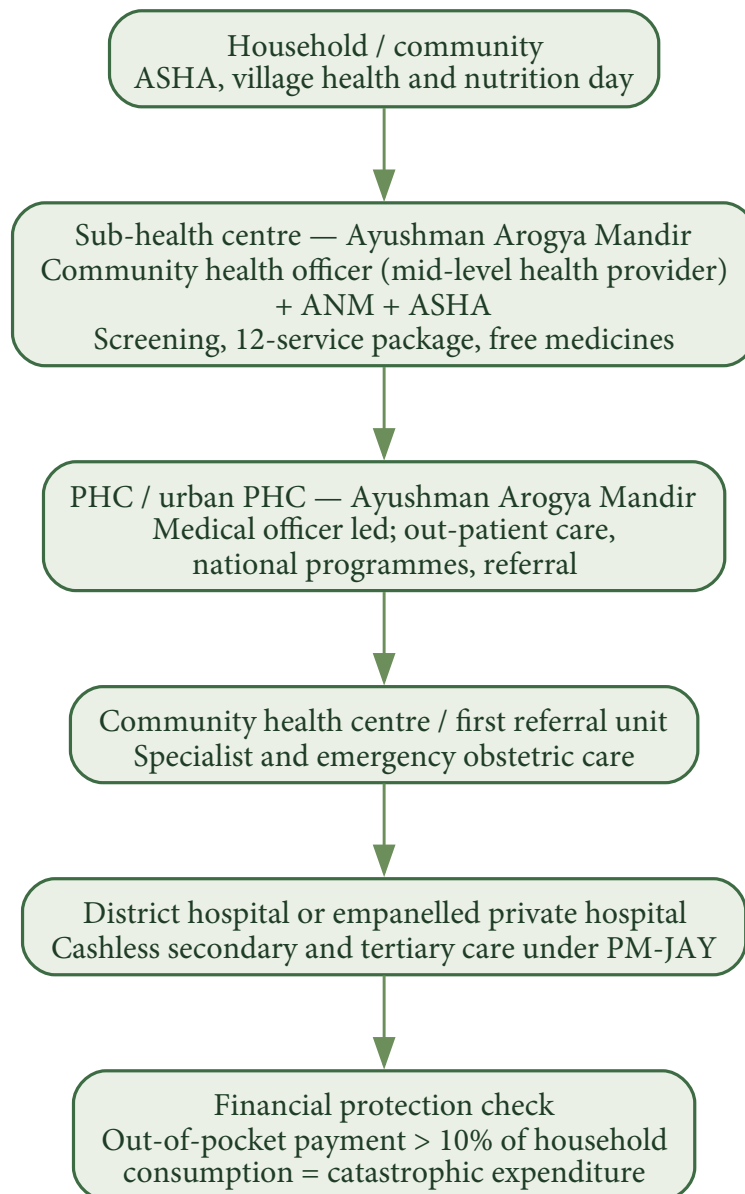


OOP: Out of pocket
PHC: Primary health care

Fig 17.2 — Attaining universal coverage (Source: WHO, 2009) (Sunderlal, p.740)



Policy and programme architecture for universal health coverage in India



Where a household meets universal health coverage — the Indian service and financing pathway

High-yield exam pointers

- ▶ UHC = all people get needed quality services without financial hardship = SDG target 3.8; first framed in WHA58.33 (2005).
- ▶ Cube mnemonic — B-D-H: Breadth (population), Depth (services), Height (cost sharing).
- ▶ Ayushman Bharat = two pillars: Ayushman Arogya Mandir (comprehensive primary care, 12 packages) + PM-JAY (₹5 lakh/family/year, secondary and tertiary hospitalisation, 10.74 crore families ≈ 50 crore people, launched 23 Sep 2018).
- ▶ Catastrophic health expenditure = >10% of total monthly consumption expenditure OR >40% of non-food consumption expenditure.
- ▶ Financing numbers to quote with years: total health spending 3.16% of GDP (2018–19), government 1.28% of GDP; out-of-pocket 53.23% of current health expenditure, 1.52% of GDP, ₹2,155 per capita per year; NHP 2017 target 2.5% of GDP by 2025; households with >10% health spending fell 17.3% (2012–20) → 7% (2022).

- **Workforce benchmark:** WHO/World Bank SDG index threshold 4.45 doctors, nurses and midwives per 1,000 population.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Ayushman Bharat Health and Wellness Centres have been renamed **Ayushman Arogya Mandir**, and the retrieved textbooks use both names — write the current name first with the former name in parentheses once. The financing percentages above are the National Health Accounts values reproduced in the retrieved editions; Sunder Lal quotes government health expenditure as 1.28% of gross domestic product (2018–19) while Park quotes the National Health Policy 2017 baseline as 1.15%, so always attach the year to whichever figure you write.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, National Health Accounts (India) 2021–22

Programme and policy updates (India)

- **Ayushman Arogya Mandir has crossed saturation but not staffing** — more than 1.75 lakh AAM are now operational, 82% of them delivering the full 12-package comprehensive primary care basket by November 2024 (up from 50% in March 2023); the free-drug and free-diagnostic entitlement has widened to 106 medicines and 14 tests at the sub-centre AAM and 172 medicines and 63 tests at the PHC/UPHC AAM. Against 1,16,978 community health officers in position (30 November 2024) there remains a shortfall of about 11,111 posts, with retention in remote postings the recurring constraint [PMID 41816146].
- **PM-JAY's benefit package has been restructured four times since launch** — HBP 2.0, 2.1, 2.2 and HBP 2022, expanding from 1,394 procedures at inception to 1,949 currently, and halving the proportion of significantly under-priced packages from 42% (HBP 1.0) to 20% (HBP 2.0). An interrupted-time-series analysis of 1.35 million claims from all 886 empanelled hospitals (222 government, 664 private) in Punjab found the revision raised utilisation, with price correction the strongest single lever and the effect no different in private than in public hospitals — evidence that strategic purchasing works through prices, not merely through empanelling more hospitals [PMID 39252993]. Scheme scale: 34.7 crore Ayushman cards issued nationally by August 2024, with the central allocation raised 28.84% from ₹7,300 crore (FY 2025) to ₹9,406 crore (FY 2026) [PMID 40976675].
- **Ayushman Bharat Digital Mission is the newest UHC instrument and is unevenly adopted** — about 40 crore beneficiary (ABHA) registrations and 27.3 crore health records linked by June 2023, with 2.08 lakh health facilities and 1.9 lakh health professionals verified; inter-state variation in facility and professional registration is the principal weakness, and registration is the gateway to using digital data for payment, organisation and regulation [PMID 39437234].

Current national data

- **Quote the National Health Accounts round with the figure** — out-of-pocket spending fell from 60.6% of current health expenditure in 2014–15 to **39.4% in 2021–22** (NHA estimates for India 2021–22), India's **UHC service coverage index rose from 57 (2015) to 63 (2021)**, marginally

above the WHO South-East Asia regional average of 62, and public-facility use rose from 91.6 crore outpatient contacts (2012–13) to 134.5 crore (2021–22) and 4.3 crore to 6.3 crore inpatient admissions [PMID 41816146]. Write the newer NHA value in the appraisal and label the older 62.4%/53.23% figures with their own years rather than presenting them as current.

- **The first district-level UHC index for India puts the national picture far lower than the state averages suggest** — applying the WHO–World Bank framework to 687 districts using 24 tracer indicators, the median district index was 43.9% (range 26.4% in Baksa, Assam to 69.4% in the Nilgiris, Tamil Nadu), and only 21.1% of districts exceeded 50%. Health insurance coverage was 40.7%, 8.0% of the district population incurred catastrophic and 11.2% impoverishing expenditure, the poorest wealth quintile scored 36.9% against the district figure of 43.9%, and 69.4% of all variation lay *between* states — so pathways to UHC in India are set at state and district level, not nationally [PMID 39219770].

Elimination, surveillance and disease status

- **Population-based NCD and cancer screening is now the largest surveillance activity in the primary care system, but throughput is not coverage** — cumulative screening at AAM to 30 November 2024 reached 84.91 crore for hypertension, 74.24 crore for diabetes, 48.98 crore for oral cancer, 21.8 crore women for breast cancer and 14.31 crore women for cervical cancer, alongside 30.76 crore eSanjeevani teleconsultations [PMID 41816146]. Set against this, the district index found the non-communicable disease tracer domain the weakest of all five (median 9.8%) and ever-screened coverage for cervical and breast cancer below 1% — the denominator, not the count of tests done, is what the UHC service coverage index measures [PMID 39219770].

Guideline and legislative updates

- **Health technology assessment has been institutionalised inside the purchaser** — the Health Financing and Technology Assessment (HeFTA) unit within the National Health Authority now supplies economic evidence for inclusion and exclusion decisions in the health benefit package, for setting reimbursement rates and for framing standard treatment guidelines, with a recommended progression to value-based pricing as the next financing reform [PMID 38715207]. This is the concrete mechanism behind the textbook prescription of "strategic purchasing" and is worth naming in the way-forward section.
- **The evidence on whether publicly funded insurance actually protects households is genuinely divided, and an examiner will reward the tension** — a case-control study of 772 hospitalised adults in six Haryana districts found direct medical out-of-pocket spending 65% lower in PM-JAY beneficiaries (₹11,131 vs ₹31,675) and catastrophic expenditure in 13.3% of beneficiaries versus 45.9% of non-beneficiaries (OR 7.15; 95% CI 4.74–10.80), yet beneficiaries incurred 19% *higher* non-medical costs because travel, food, lodging and wage loss sit outside the package [PMID 40976675]. A national instrumental-variable analysis of NSS rounds 60, 71 and 75 found no statistically significant association between enrolment in publicly funded insurance and out-of-pocket expenditure per hospitalisation, and no effect at all in the bottom two consumption quintiles, casual-labour households or rural areas [PMID 41913189]. The safe examination position: insurance demonstrably cuts the *covered* component of hospital cost while leaving outpatient care, drugs, diagnostics and indirect costs untouched.

Global and WHO developments

- ▶ **Global progress towards UHC has stalled since 2015** — the global UHC service coverage index rose from 54 (2000) to 71 (2023), but the annual rate of improvement fell from 1.5% in 2000–2015 to 0.5% in 2015–2023, roughly one-third of the pre-SDG pace. About 4.6 billion people still lacked full coverage of essential services in 2023, and 2.1 billion faced financial hardship from health spending in 2022, of whom 1.6 billion were pushed into or deeper into poverty. On present trends the index reaches only 74 by 2030, with about a quarter of the world's population still facing financial hardship — the world is not on track for SDG target 3.8 [WHO 2025].
- ▶ **South Asian projections show coverage and protection moving in opposite directions** — Bayesian projections for seven South Asian countries put regional primary-care service coverage at 53.5% (2000) rising to 81.5% by 2030, while the incidence of catastrophic health expenditure is projected to *rise* from 7.2% (2000) to 18.6% (2030), with India among the countries showing the largest increases; India's impoverishment rate improves only modestly, from 3.5% to 2.6%, against 0.5% for Sri Lanka and 0.3% for Bhutan. Expanding services without matching prepayment and risk pooling increases, rather than reduces, financial hardship [PMID 41192927].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 14 Sustainable Development Goals; Ch 23 Health Planning and Management); Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 2, Ch 8, Ch 16 Health Planning and Management, Ch 17 Healthcare of the Community, Ch 20); Oxford Textbook of Public Health 7e (Ch 4, Ch 8, Ch 21 Health Policy in Developing Countries, Ch 103, Ch 107 Global Health in the Era of Sustainable Development). Named documents: National Health Policy 2017; Ayushman Bharat/PM-JAY guidelines 2018; Indian Public Health Standards 2022; WHA58.33 (2005); WHO coverage cube (2009); 15th Finance Commission recommendations.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, J Family Med Prim Care 2026, DOI [10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 39252993 — Impact of health benefit package policy interventions on service utilisation under government-funded health insurance in Punjab, India: analysis of PM-JAY, Lancet Reg Health Southeast Asia 2024, DOI [10.1016/j.lansea.2024.100462](https://doi.org/10.1016/j.lansea.2024.100462); 40976675 — Impact of India's publicly funded health insurance scheme on financial risk protection: a case-control study from Haryana state in India, BMJ Open 2025, DOI [10.1136/bmjopen-2024-093304](https://doi.org/10.1136/bmjopen-2024-093304); 39437234 — The Ayushman Bharat Digital Mission of India: an assessment, Health Systems and Reform 2024, DOI [10.1080/23288604.2024.2392290](https://doi.org/10.1080/23288604.2024.2392290) (abstract-only); 39219770 — District-level monitoring of universal health coverage, India, Bulletin of the World Health Organization 2024, DOI [10.2471/BLT.23.290854](https://doi.org/10.2471/BLT.23.290854); 38715207 — Establishing a health technology assessment evidence ecosystem in India's Pradhan Mantri Jan Arogya Yojana, Health Systems and Reform 2024, DOI [10.1080/23288604.2024.2327097](https://doi.org/10.1080/23288604.2024.2327097) (abstract-only); 41913189 — Do publicly funded health insurance schemes reduce out-of-pocket expenditure? Evidence from India, BMC Health Services Research 2026, DOI [10.1186/s12913-026-14456-8](https://doi.org/10.1186/s12913-026-14456-8); 41192927 — Progress towards universal health coverage in South Asia, 2000–2030: an examination of the twin elements of primary healthcare provision and financial protection, BMJ Global Health 2025, DOI [10.1136/bmjgh-2025-020052](https://doi.org/10.1136/bmjgh-2025-020052). Plus WHO universal health coverage fact sheet, 2025.



Indian Public Health Standards (IPHS) for CHCs

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — Indian Public Health Standards (IPHS) are the benchmark standards of the Directorate General of Health Services, Ministry of Health and Family Welfare, defining the assured (**essential**) and achievable (**desirable**) services, and the infrastructure, human resources, drugs, diagnostics and equipment needed to deliver them, at each level of government health facility. The **community health centre (CHC)** is the secondary-level referral facility of the rural health system, sited at the community development block, serving **1,20,000 population in plains and 80,000 in tribal/hilly areas** and covering four primary health centres (Sunder Lal, Ch 17, p 752; Park 28e, Ch 13, p 850). IPHS were issued in 2006, revised in 2012 and **last revised in 2022**.

Objectives of IPHS and the domains standardised

#	Objective of IPHS (Sunder Lal, Ch 17, p 744)
1	Define a uniform standard to ensure high-quality services
2	Specify the minimum assured (essential) and achievable (desirable) services at sub-centre, PHC, CHC, sub-district and district level
3	Specify adequate workspace (buildings), human resources for health, drugs, diagnostics, equipment, administrative and logistic support for expected outcomes
4	Achieve and maintain an acceptable quality of care
5	Facilitate monitoring and supervision of health facilities
6	Serve as a tool for governance, leadership and evaluation — the standard converts into a facility-wise gap analysis

- IPHS 2022 covers district and sub-district hospitals, CHCs (rural and urban), Health and Wellness Centre–PHC (now *Ayushman Arogya Mandir*) and Health and Wellness Centre–sub health centre; each service package rests on **eight strengthening components** — infrastructure, human resources, drugs, diagnostics, equipment, quality, monitoring/supervision, governance — set at a **minimum acceptable functional standard** aligned to universal health coverage, Sustainable Development Goal 3 and the National Health Policy 2017 (Park 28e, Ch 10, p 728).

IPHS 2022 — categories, population norm and beds at CHC

Parameter	Non-FRU CHC (rural)	FRU CHC (rural)	FRU U-CHC (urban)
Population covered	1,20,000 plain / 80,000 tribal-hilly, one CD block	Same	2.5 lakh non-metro; 5 lakh metro
Essential beds	30	30 (50-bedded variant exists)	50; 100 in metros/million-plus cities
Oxygen-supported beds	18 of 30	20 of 30	—

Parameter	Non-FRU CHC (rural)	FRU CHC (rural)	FRU U-CHC (urban)
Specialist services	Physician/family medicine specialist essential (1); surgeon, obstetrician-gynaecologist, paediatrician, anaesthesiologist desirable only — no operative or blood-transfusion services	Maternity + surgical: all five essential (physician, surgeon, obstetrician-gynaecologist, paediatrician, anaesthesiologist) + functional operation theatre and blood storage unit	Maternity + surgical essential in all cities
Desirable additions	—	Ophthalmic, orthopaedic, ENT (50-bedded FRU CHC)	Ophthalmic, orthopaedic (100-bedded)
Scope of care	Preventive, promotive, curative, palliative, rehabilitative — normal delivery, infectious disease, nutritional disorders including iron-deficiency anaemia, mental health, non-communicable disease, eye/oral/ENT care, stabilisation of emergencies	The above plus elective and emergency secondary-level surgery, operative services and blood transfusion	As FRU

(Sunder Lal, Ch 17, pp 752–754, 759; Park 28e, Ch 10, p 728)

- **Secondary care specialist services are envisaged at BOTH types of CHC (FRU and non-FRU);** only operative and blood-transfusion services are confined to the FRU (Sunder Lal, Ch 17, p 754).
- A rural CHC may be **non-FRU or FRU**; every urban CHC is an **FRU U-CHC**. Block-headquarter CHCs become **Block Public Health Units (BPHU)** — clinical service delivery, public health functions, block public health laboratory, health management information system unit. **6,064 CHCs** (5,480 rural, 584 urban), 31 March 2022 (Sunder Lal, Ch 17, p 752).

Human resources for health at CHC (IPHS 2022)

Staff category	Non-FRU CHC, essential	Non-FRU, desirable	FRU CHC, essential	FRU, desirable
Physician / family medicine specialist	1	1	1	1
Surgeon	—	1	1	—
Obstetrician and gynaecologist	—	1	1	—
Paediatrician	—	1	1	—
Anaesthesiologist	—	1	1	—
Microbiologist	—	—	—	1
General duty medical officer (GDMO)	3	1	6	2
Medical officer, AYUSH	—	1	—	1
Medical officer, dental	1	—	1	—

Staff category	Non-FRU CHC, essential	Non-FRU, desirable	FRU CHC, essential	FRU, desirable
Staff nurses	29	—	46	—
Dietician, social worker/counsellor, physiotherapist, clinical psychologist	5	2	6	1
Technicians (laboratory, operation theatre, ECG, radiology) and assistants	9	4	13	4
Pharmacist and store keeper	2	1	2	2
Administrative staff (hospital manager, nursing in-charge, accountant, data entry operator, sanitary inspector, registration clerk)	7	3	10	1
Cleaning staff	5	4	8	1

(Sunder Lal, Ch 17, Table 17.8, pp 753–754. Staffing assumes an average bed occupancy of 60% and is raised if occupancy rises.)

Assured (essential) and desirable services at CHC

Domain	Assured (essential) services	Desirable
Clinical	Routine and emergency care in surgery, medicine, obstetrics and gynaecology and paediatrics; outpatient and indoor (admission) services; specialist and referral support to its four primary health centres	—
Maternal, newborn, child	24-hour delivery (normal and assisted); essential and emergency obstetric care including caesarean section; newborn care stabilisation unit; routine and emergency care of sick children	—
Reproductive health	Full range of family planning including laparoscopic sterilisation and no-scalpel vasectomy; reproductive tract and sexually transmitted infection services; integrated counselling and testing centre for HIV/AIDS	Medical termination of pregnancy — facility approved for the second trimester
Diagnostics and support	Cold chain maintenance for vaccines; designated microscopy centre for tuberculosis; essential laboratory and X-ray; blood storage facility; dental health services; round-the-clock referral transport	Biopsy services; intraocular pressure by tonometry; public-private partnership for laboratory services
Public health	School and adolescent health; integrated disease surveillance and outbreak control; training and continuing education of health teams; intersectoral convergence; community participation; public-private partnership; services for all national health programmes, with standard treatment protocols at every CHC	Link antiretroviral therapy centre

(Sunder Lal, Ch 17, pp 754–755)

Governance, quality and accountability standards

- ▶ Under the National Rural Health Mission — Rogi Kalyan Samiti at every CHC for ownership, accountability and monitoring; untied grant ₹50,000 per annum; maintenance grant ₹1 lakh per year; ₹20 lakh per CHC initially for IPHS upgradation; contractual anaesthetist and public health programme manager, multiskilling and public–private partnership to close the specialist gap; aim — all CHCs as 24-hour FRUs (Sunder Lal, Ch 17, pp 754–755).
- ▶ Citizen's charter at the entrance — services, timings, free drugs and diagnostics, referral vehicles, grievance redressal (Sunder Lal, Ch 17, p 754; Park 28e, Ch 10, pp 728–729).
- ▶ Support services (laboratory, transport, nursing, diet, central sterile supply, laundry, blood storage) strengthened; quality certified under **Kayakalp / National Quality Assurance Standards / LaQshya** against the *Operational Guidelines for Improving Quality in Public Healthcare Facilities under IPHS 2022* (Sunder Lal, Ch 17, pp 753–754, 759; Park 28e, Ch 10, p 729).
- ▶ Persisting constraint — an 80% shortfall of specialists, so most CHCs do not function as FRUs (Sunder Lal, Ch 17, p 753).

Criteria for deciding upgradation of a CHC to a First Referral Unit

An existing CHC (or sub-district/district hospital) is declared a fully operational FRU only if it provides round-the-clock comprehensive emergency obstetric and newborn care (CEmONC) (Sunder Lal, Ch 17, pp 754–755).

A. The three critical determinants (Park 28e, Ch 7, p 543) — available on a 24-hour basis: (1) surgical interventions (caesarean section); (2) newborn care; (3) a blood storage facility.

B. Physical and infrastructural prerequisites for the full range of FRU functions (Park 28e, Ch 7, p 543):

#	Requirement
a	Minimum bed strength of 20–30; relaxable initially to 10–12 beds in difficult areas — North-East states and underserved areas of Empowered Action Group states
b	A fully functional operation theatre
c	A fully functional labour room
d	An area earmarked and equipped for newborn care in the labour room and in the ward
e	A functional laboratory
f	Blood storage facility
g	24-hour water supply and electricity supply
h	Arrangements for waste disposal
i	Ambulance facility

C. The ten minimum services that must be assured before designation (Park 28e, Ch 7, p 543; Sunder Lal, Ch 17, p 755):

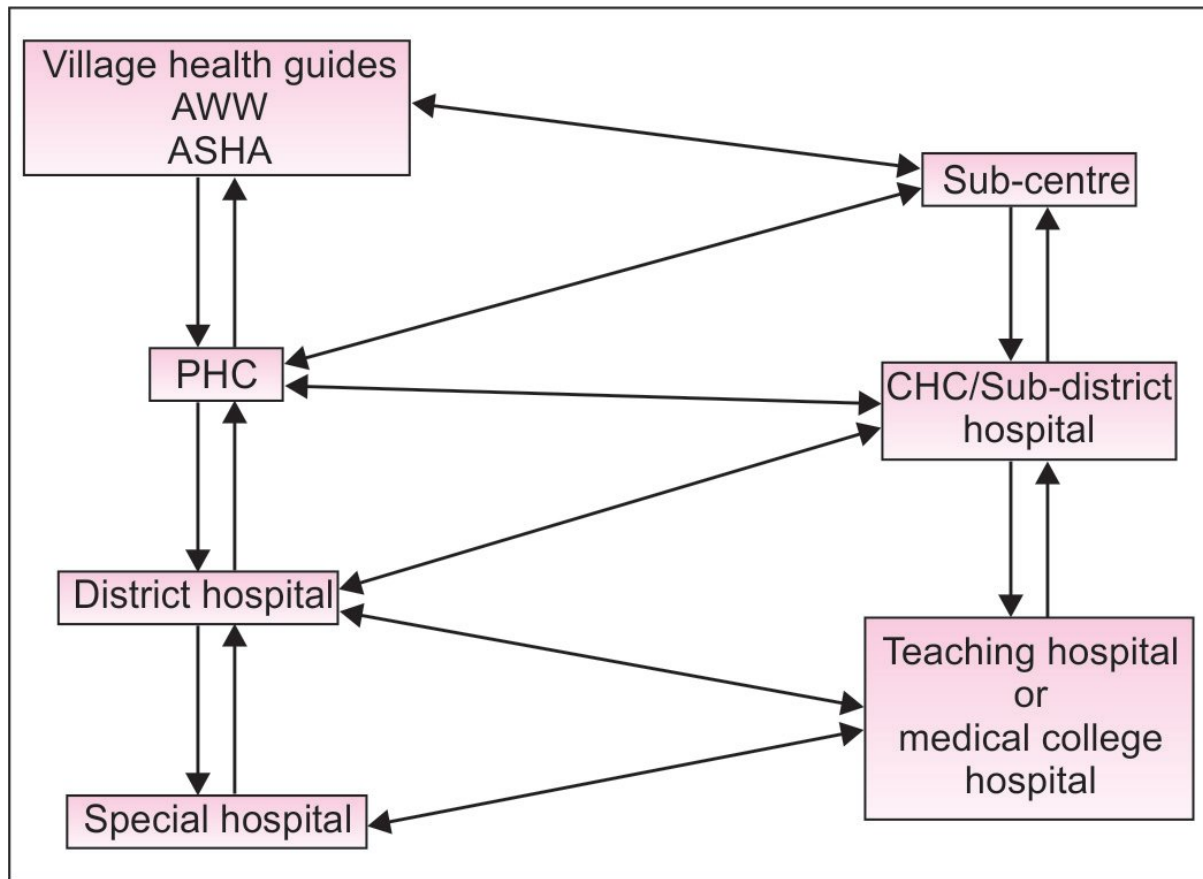
#	Minimum FRU service	#	Minimum FRU service
1	24-hour delivery — normal and assisted	6	Safe abortion services
2	Emergency obstetric care including caesarean section	7	Treatment of sexually transmitted / reproductive tract infections
3	Newborn care	8	Blood storage facility
4	Emergency care of sick children	9	Essential laboratory services
5	Full range of family planning including laparoscopic services	10	Referral (transport) services

D. Human-resource criterion — the specified specialists (obstetrician-gynaecologist, surgeon, paediatrician, anaesthetist, physician) in position; where the gap persists the CHC qualifies by **multiskilling of MBBS doctors** — 18 weeks in life-saving anaesthetic skills, 16 weeks in emergency obstetric skills including caesarean section — or by contractual hiring and public–private partnership (Sunder Lal, Ch 10, pp 580–581).

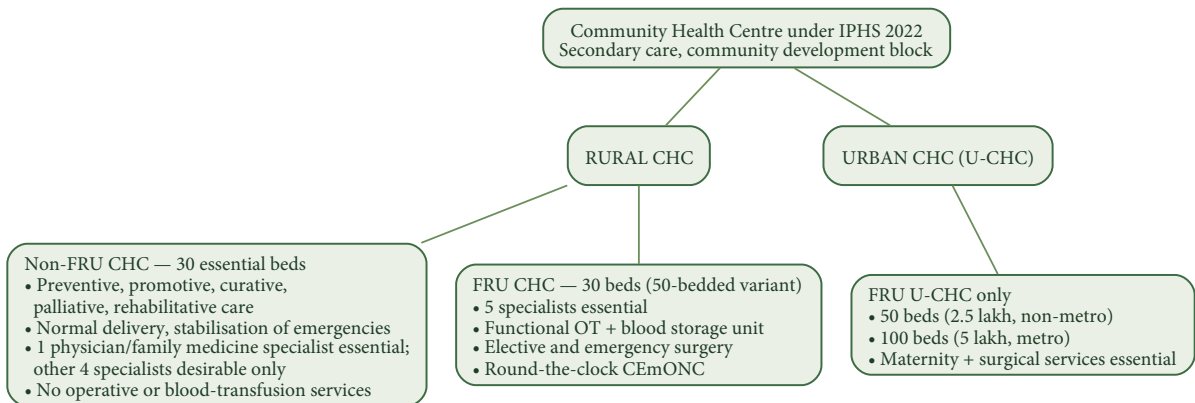
E. District-level siting criteria — two apply together:

- ▶ **Number needed** — the norm is **one FRU per 5,00,000 (500,000) population**, against which the National Health Policy 2017 indicator "proportion of facilities functioning as FRUs" is measured; the district upgrades CHCs until this norm is met (Park 28e, Ch 2, p 45).
- ▶ **Placement** — since **12–15% of all deliveries need emergency obstetric services** and such emergencies are unpredictable, FRUs must be so distributed that a pregnant woman reaches one **within two hours of travel time** (Sunder Lal, Ch 10, pp 580, 586).

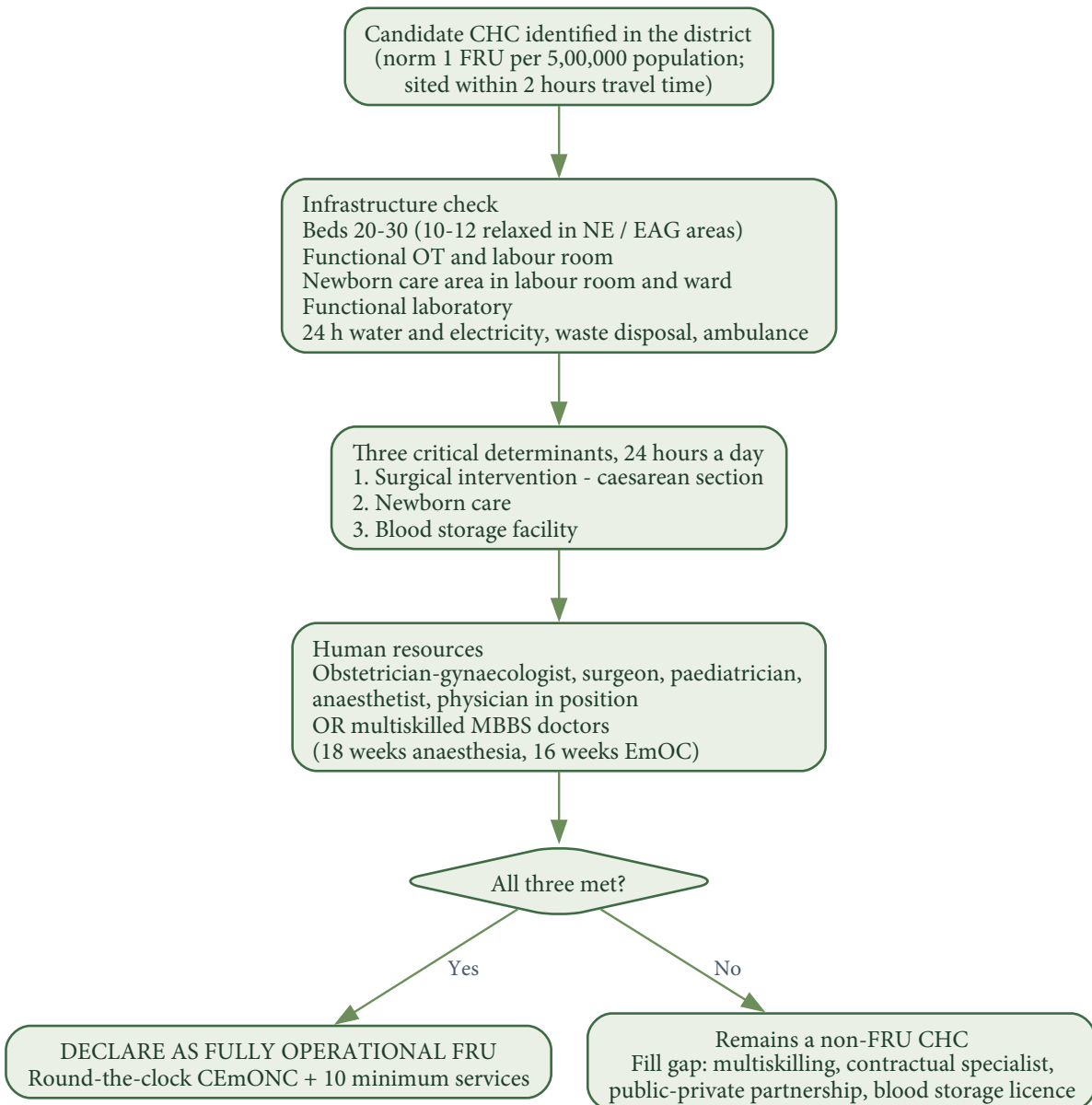
Diagrams



Referral system in the Indian health care delivery structure, showing the position of the CHC/sub-district hospital (Sunder Lal — Textbook of Community Medicine, Ch 17, p 762)



IPHS 2022 classification of community health centres and their assured service level



Decision pathway for upgrading a community health centre to a First Referral Unit

High-yield exam pointers

- ▶ CHC norm 1,20,000 plain / 80,000 tribal-hilly, one CHC per 4 PHCs; 6,064 CHCs (5,480 rural + 584 urban), 31 March 2022.
- ▶ IPHS 2006 → 2012 → 2022; every standard graded essential vs desirable; eight strengthening domains.
- ▶ Beds non-FRU 30 (18 oxygen-supported); FRU CHC 30 (20); FRU U-CHC 50, metro 100; staffing assumes 60% occupancy; non-FRU CHC still carries 1 essential physician/family medicine specialist.
- ▶ FRU mnemonic — "Surgery, Sick newborn, Storage of blood" = the three critical determinants, all 24-hourly.
- ▶ Planning norms (National Health Policy 2017, indicator 3.1.3): one FRU per 5,00,000 and one 24×7 PHC per 1,00,000; FRU beds 20–30 (10–12 in North-East / Empowered Action Group states); reachable within 2 hours; 12–15% of deliveries need emergency obstetric care.

- FRUs founded by the Child Survival and Safe Motherhood Programme, 1992; 1,685 of 3,559 designated FRUs functional in 2022.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Government of India publication *Indian Public Health Standards — Community Health Centre, 2022, Vol. II* carries the full facility-wise lists of drugs, diagnostics and equipment, which the textbooks summarise rather than reproduce; the retrieved corpus supports the service, bed, staffing and planning norms above but not that list.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, MoHFW, National Health Mission, NHSRC, Registrar General of India (SRS), WHO Global TB Report

Programme and policy updates (India)

- IPHS 2022 is now a scored, dashboard-monitored standard, not just a printed norm — MoHFW released an open-source (ODK) self-assessment toolkit and a public dashboard at iphs.mohfw.gov.in on 28 June 2024, converting the facility-wise gap analysis into a tracked compliance score for district hospitals, sub-district hospitals, CHCs, PHCs and sub-health centres; as on 30 November 2025 all public health facilities had completed IPHS self-assessment and 61% scored an IPHS compliance score above 50%. [MoHFW 2025]
- PM-ABHIM has inserted a new tier above and beside the CHC — of the 3,382 Block Public Health Units envisaged, administrative approval now stands for 2,151 BPHUs, 621 Critical Care Blocks and 744 district Integrated Public Health Laboratories, ₹32,928.82 crore approved to States/UTs; every district above 5 lakh population is to get a 50–100-bedded critical care block, the remaining districts only referral linkages. [MoHFW 2025]
- Health and Wellness Centres are formally *Ayushman Arogya Mandir*, 1,81,873 operational on 30 November 2025, with a 12-service package and teleconsultation. A state-representative assessment of 404 such centres in Chhattisgarh found they met only 31% of the population's outpatient need, 26% for hypertension and 21% for diabetes care — the CHC referral load falls only if the primary tier actually performs. [MoHFW 2025; PMID 39367295]

Current national data

- The national facility census has been renamed, and its numbers have moved on — *Rural Health Statistics* now appears as *Health Dynamics of India (Infrastructure and Human Resources)*; the 2022-23 edition (released 9 September 2024, data as on 31 March 2023) records 6,359 CHCs, 1,69,615 sub-centres, 31,882 PHCs, 714 district hospitals, with 26,280 specialists and medical officers in position at CHCs. On the mission returns, FRUs have roughly tripled — 3,108 functioning FRUs as on 30 June 2024 against 940 in 2005, alongside 12,453 PHCs functioning 24×7 — superseding the older "1,685 of 3,559 designated FRUs" figure. Always quote a figure with its edition year. [MoHFW 2024-Sep; NHM 2024]
- Tribal areas still sit below the IPHS facility norm — a decade-long analysis of the annual facility returns against IPHS requirement (one CHC per 80,000 in tribal areas) found the tribal CHC capacity index essentially static at 0.79 (2011) → 0.78 (2021), with sub-centres falling 0.96 → 0.79

and PHCs 0.89 → 0.79; any value below 1 means fewer facilities present than IPHS requires. [PMID 42192389]

- **Maternal mortality ratio 88 per lakh live births — SRS Special Bulletin 2021-23**, down from 130 in 2014-16; the National Health Policy target of <100 is met, the SDG 3.1 target of <70 by 2030 is not. The newborn capacity behind the FRU criterion now stands at **3,215 Newborn Stabilisation Units at FRU/CHC level** and 1,114 Special Newborn Care Units. [RGI 2025; MoHFW 2025]

Elimination, surveillance and disease status

- **Current declared status and the target years that replaced the textbook ones — kala-azar**: all **633 endemic blocks** below 1 case per 10,000 population since end-2023 and sustaining it to October 2025 (424 cases January–October 2025 against 444 in 2024), WHO validation requiring three sustained years; **malaria**: indigenous transmission to be interrupted by **2027**; **measles-rubella**: Zero Measles Rubella Campaign launched April 2025 for elimination by **2026**; **TB**: incidence **187 per 1,00,000 in 2024** (237 in 2015), treatment coverage 92%. [MoHFW 2025]
- **The outbreak stream the CHC feeds is increasingly zoonotic** — analysis of Integrated Disease Surveillance Programme portal reports for 2010–2023 counted **1,158 outbreaks** across ten zoonotic diseases, Japanese encephalitis/acute encephalitis syndrome commonest, with anthrax, JE/AES, Crimean-Congo haemorrhagic fever, leptospirosis and scrub typhus occurring year-round rather than seasonally — the case for One Health surveillance at block level. [PMID 41738949]

Guideline and legislative updates

- **NQAS certification is the operative quality benchmark, and the CHC is the weak link** — **36,967 public facilities certified by 30 June 2025** (13,535 national, 23,432 state), of which only 842 were CHCs against **30,069 Ayushman Arogya Mandir sub-centres**; the total rose to **45,577 by 30 November 2025**. Virtual NQAS assessment of AAM sub-centres began **28 June 2024** with 10% on-site physical verification, against an MoHFW target of **100% certification by December 2026**. [NHSRC 2025; MoHFW 2025]

Global and WHO developments

- **The signal-function framework behind the FRU definition is under revision** — basic and comprehensive emergency obstetric care signal functions, used as an international monitoring framework since **1997**, are the yardstick from which India's CEmONC-based FRU criterion derives; a meta-narrative review of 19 papers, done as part of revising that framework, grouped the critique into clinical, health-systems and human-rights traditions across five measurement themes. [PMID 38820693]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2, p 45; Ch 7, p 543; Ch 10, pp 728–729; Ch 13, p 850); Sunder Lal — Textbook of Community Medicine (Ch 17, pp 744, 752–755, 759, 762; Ch 10, pp 580–581, 586); Revised Indian Public Health Standards (IPHS) 2022, DGHS, Ministry of Health and Family Welfare; National Health Policy 2017 indicator framework.

Recent-advances sources: 39367295 — Performance of health and wellness centre in providing primary care services in Chhattisgarh, India, *BMC Primary Care* 2024, doi:[10.1186/s12875-024-02603-1](https://doi.org/10.1186/s12875-024-02603-1); 42192389 — Public health system capacity of tribal and rural areas in India: a retrospective state-level analysis from 2011 to 2021, *BMC Public Health* 2026, doi:[10.1186/s12889-026-27725-2](https://doi.org/10.1186/s12889-026-27725-2); 41738949 — Trend of zoonotic diseases outbreaks in India: a secondary data analysis of Integrated Disease Surveillance Programme data, 2010–2023, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2026, doi:[10.1093/trstmh/trag011](https://doi.org/10.1093/trstmh/trag011) (abstract-only); 38820693 — Evolving narratives on signal functions for monitoring maternal and newborn health services: a meta-narrative inspired review, *Social Science & Medicine* 2024, doi:[10.1016/j.socscimed.2024.116980](https://doi.org/10.1016/j.socscimed.2024.116980) (abstract-only). Ministry of Health and Family Welfare, *Initiatives and Achievements 2025* (data as on 30 November 2025); *Health Dynamics of India (Infrastructure and Human Resources) 2022-23*, MoHFW, released 9 September 2024; National Health Mission infrastructure returns, 30 June 2024; NHSRC Quality and Patient Safety Division, *Quality Darpan* Vol VI No 1, January–June 2025; Registrar General of India, SRS Special Bulletin on Maternal Mortality in India 2021-23.



Health care management, planning and economics



Hospital records – advantages and limitations

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — Hospital records are the permanent clinical, administrative and statistical documents generated by the in-patient and out-patient services of a hospital — case sheets, out-patient tickets, admission and discharge registers, discharge sheets, operation theatre and laboratory registers and death certificates — maintained by the medical record department (MRD). They form the fifth of the routine sources of health information in a health information system, defined as the mechanism for collection, processing, analysis and transmission of information required for organizing and operating health services and for research and training (Park 28e, Ch 21, pp 995-997). The eighth report of the WHO Expert Committee on Health Statistics recommended that hospital statistics be regarded in all countries as an integral and basic part of the national statistical programme (Park 28e, Ch 21, p 997).

Place of hospital records in the health information system

A comprehensive health information system needs information on seven subjects; hospital records feed five of them directly.

Component of a health information system	Fed by hospital records?	What the record supplies
Demography and vital events	Partly	Births and deaths occurring in the institution; medically certified cause of death
Environmental health statistics	No	Requires separate environmental data
Health status — mortality, morbidity, disability, quality of life	Yes	Diagnosis-wise morbidity, case fatality, complications
Health resources — facilities, beds, manpower	Yes	Bed strength, staffing, equipment in use
Utilization and non-utilization of services — attendance, admissions, waiting lists	Yes	Out-patient attendance, admissions, referrals
Indices of outcome of medical care	Yes	Discharge status, death, duration of stay
Financial statistics — cost and expenditure	Yes	Cost of hospital care

(Park 28e, Ch 21, pp 995-997)

Advantages

Advantage	Basis
Basic and primary source of morbidity data in India — because registration of vital events is defective and notification of infectious disease inadequate, hospital data constitute the principal available account of diseases prevalent in the community	Park 28e, Ch 21, p 997
Rich clinical content — discharge sheets carry diagnosis, medical and surgical procedures, complications, length of stay and laboratory data	Park 28e, Ch 21, p 997
Diagnostically the most secure routine source — the recorded diagnosis is accompanied by the laboratory data of the same episode, unlike lay-reported or notified data	Park 28e, Ch 21, p 997
Cheapest method of data collection — a health records survey uses data already generated in the course of care, needing no fresh field enumeration	Park 28e, Ch 21, p 999
Continuous and ongoing — available at any time without waiting for a decennial census or a periodic survey round	Park 28e, Ch 21, pp 996, 999
Yields hospital efficiency and utilization indices — bed-occupancy rate, duration of stay and cost-effectiveness of treatment policies are useful in monitoring the use of hospital facilities	Park 28e, Ch 21, p 997
Only feasible source for rare and chronic disease — the base for disease registers and for hospital-based cancer registries, which include all patients treated by a particular institution	Park 28e, Ch 21, pp 997-998; Ch 6, pp 459-460
Base for medical record linkage — hospital admission and discharge are among the events assembled when the important records of one individual are brought together, allowing study of associations between diseases that may have aetiological significance	Park 28e, Ch 21, p 998

Advantage	Basis
Source of cases for epidemiological research — hospital patients and hospital-maintained registries are a standard sampling frame for case-control studies	Gordis 7e, Ch 7, p 177
Basis of mortality statistics — the death certificate signed in hospital, coded by the International Classification of Diseases, is the foundation of cause-of-death data	Park 28e, Ch 3, p 76; Ch 2, p 69

Efficiency indices derived from hospital records (Park 28e, Ch 2, p 39):

Index	Formula
Bed-occupancy rate	Average daily in-patient census / average number of beds
Average length of stay	Days of care rendered / discharges
Bed turnover ratio	Discharges / average beds

What a study of hospital data yields

- ▶ Geographic sources of patients.
- ▶ Age and sex distribution of different diseases and duration of hospital stay.
- ▶ Distribution of diagnosis.
- ▶ Association between different diseases.
- ▶ The period between disease and hospital admission.
- ▶ The distribution of patients according to different social and biological characteristics.
- ▶ The cost of hospital care.

Such information is of great value in the planning of health care services (Park 28e, Ch 21, p 997).

Limitations

Limitation	Explanation
"Tip of the iceberg"	They inform only on those patients who seek medical care, not on a representative sample of the population; mild cases may not attend hospitals and subclinical cases are always missed (Park 28e, Ch 21, p 997)
Highly selective	Admission policy varies from hospital to hospital, so the case mix reflects the institution rather than the community (Park 28e, Ch 21, p 997)
Numerator without a denominator	The population served by a hospital — the population at risk — cannot be defined; there are no precise boundaries to the catchment area, so hospital statistics provide only the cases (Park 28e, Ch 21, p 997)
Poor guide to disease frequency	Extrapolation of hospital data to an entire community is highly conjectural in estimating frequency rates of disease (Park 28e, Ch 21, p 997)
Not population-based	Violates the first requirement laid down by the WHO Expert Committee for a health information system, that the system should be population-based (Park 28e, Ch 21, p 995)

Limitation	Explanation
Berksonian bias	Named after Joseph Berkson; arises because of the different rates of admission to hospital for people with different diseases, distorting the exposure-disease association in hospital-based case-control studies (Park 28e, Ch 3, pp 93-94)
Reliability and standardisation	Estimates obtained from records are not population based, reliability of the data is open to question, and there is a lack of uniform procedures and standardisation in the recording of data (Park 28e, Ch 21, p 999)
Kept for administration, not for monitoring	Facility records are mostly maintained for administrative purposes rather than for monitoring, so they are rarely analysed or fed back (Park 28e, Ch 21, p 998)
Inaccurate cause of death	Medical certification is often inaccurate or incomplete from lack of diagnostic evidence, inexperience of the certifying doctor and absence of postmortem; only a single underlying cause is usually tabulated (Park 28e, Ch 3, p 76)
Duplication without linkage	Repeat admissions and inter-institutional referrals inflate counts unless records are linked; record linkage is applied only on a limited scale because of the volume of data and is beyond the reach of many developing countries (Park 28e, Ch 21, p 998)
Restricted segment of the population	Institutional records relate only to a certain segment of the general population, so social, occupational and geographic groups that do not use that hospital are invisible (Park 28e, Ch 21, p 998)
No non-quantifiable information	Health policy, legislation, public attitudes, procedures and technology, which planners also require, are not captured (Park 28e, Ch 21, p 999)

Measures to strengthen hospital records as a data source

- ▶ Establish a **medical record department in every hospital** — its importance for the development of hospital statistics cannot be overemphasized (Park 28e, Ch 21, p 997).
- ▶ **Computerize medical records**, which enables medical care to be more effectively rendered, better planned and better evaluated (Park 28e, Ch 21, p 997).
- ▶ **Standardise diagnostic coding** using the International Classification of Diseases; ICD-11 was adopted by the World Health Assembly in 2019 and came into effect from 1st January 2022 (Park 28e, Ch 2, p 69).
- ▶ **Convert records into disease registers** — registration requires a permanent record, follow-up of cases and statistical tabulation of both frequency and survival, giving duration of illness, case fatality and survival (Park 28e, Ch 21, pp 997-998).
- ▶ **Institute medical record linkage** so that the important records of one individual, originating at different times and places, are assembled into one file (Park 28e, Ch 21, p 998).
- ▶ **Supplement with population surveys**, since routine statistics do not provide all the information about health and disease in the community and a health information system must be population-based (Park 28e, Ch 21, p 998).
- ▶ **Feed a health management information system (HMIS)** — the process whereby health data are recorded, stored, retrieved and processed for decision-making on planning, organizing and

control of health care facilities at national, state and institutional levels (Jugal Kishore, National Health Programmes of India, p 262).

- **Provide for feedback of data** to the reporting hospital, one of the six WHO requirements for a health information system (Park 28e, Ch 21, p 995).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Iceberg of disease — labels to reproduce: water line; above water — "Clinical cases, apparently sick" (the part hospital records capture); below water — "Apparently healthy: subclinical cases, undiagnosed cases, asymptomatic cases, carriers, latent cases" (the part hospital records always miss).

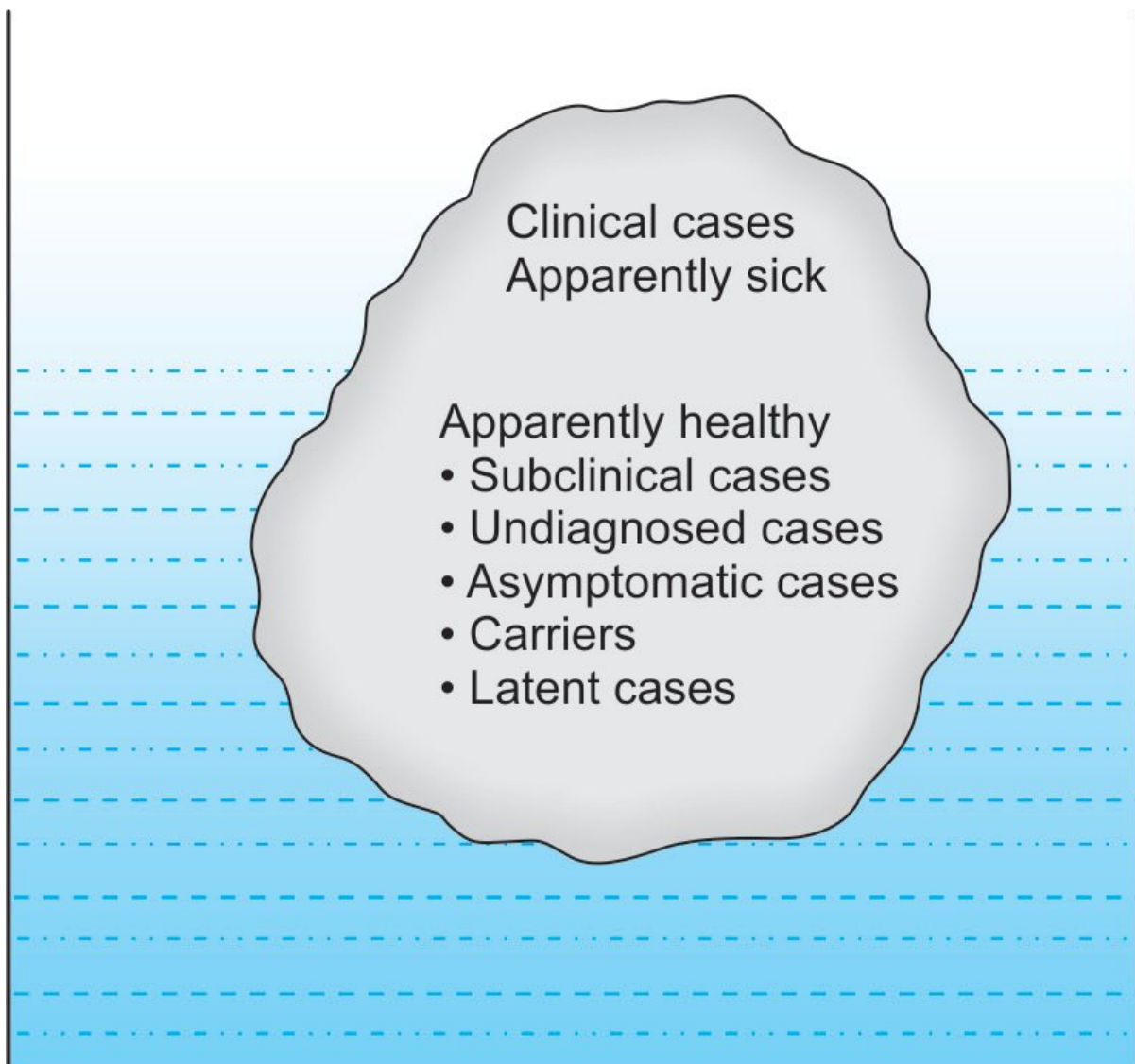
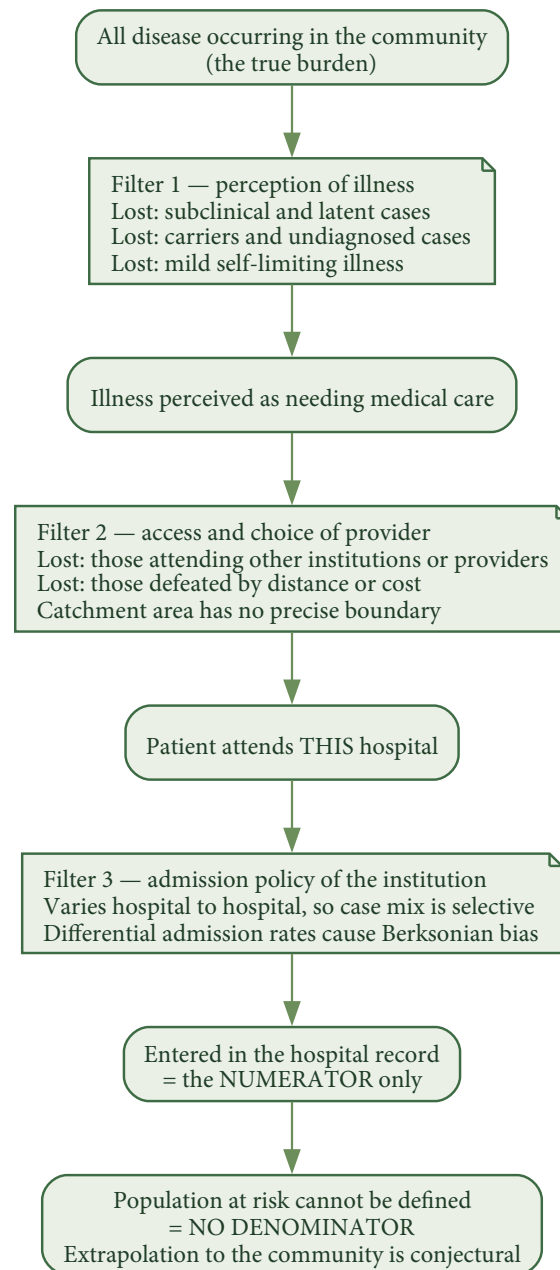
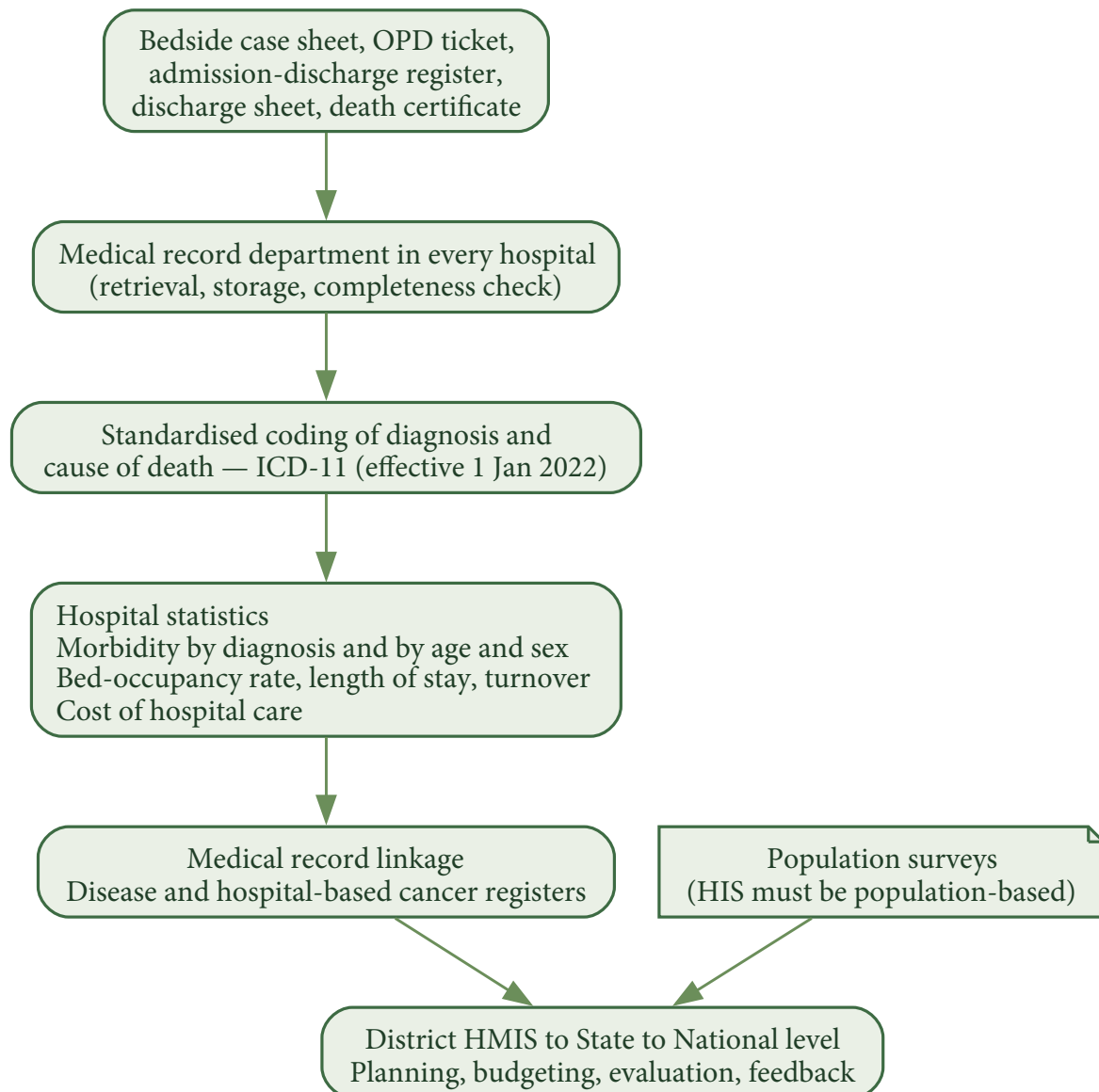


Fig 3.2 — Iceberg of diseases (Suryakantha Environment Best, p.35)



Successive filters between community disease and the hospital record — the numerator survives, the denominator is lost



Pathway from the bedside case sheet to usable health information, and where each strengthening measure acts

High-yield exam pointers

- ▶ Park lists hospital records as **source no. 5** of health information; **WHO Expert Committee on Health Statistics, 8th report** — hospital statistics to be an integral and basic part of the national statistical programme.
- ▶ Three classic drawbacks, in Park's own order: **tip of the iceberg** → **selective admission policy** → **population at risk undefined**, ending in the one-line verdict "hospital statistics provide only the numerator, not the denominator".
- ▶ Efficiency formulas: **bed-occupancy rate** = average daily in-patient census / average number of beds; **average length of stay** = days of care rendered / discharges; **bed turnover ratio** = discharges / average beds.
- ▶ **Berkson's bias** = different rates of hospital admission for people with different diseases — the reason hospital controls distort case-control studies.

- ▶ Seven yields of a hospital data study (mnemonic **G-A-D-A-P-S-C**): Geographic source, Age-sex, Diagnosis distribution, Association between diseases, Period from disease to admission, Social and biological characteristics, Cost of care.
- ▶ Remedies to quote: **medical record department + computerisation + ICD-11 coding + record linkage + disease registers + population surveys**.
- ▶ WHO requirement no. 1 for a health information system is that it be **population-based** — the single requirement hospital records inherently fail.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e states that ICD-11 was adopted by the World Health Assembly in 2019 and came into effect from 1st January 2022, and that hospital record computerisation is the recommended direction of travel; India's current operational vehicles for this — the Health Management Information System (HMIS) portal, the Integrated Health Information Platform and the Ayushman Bharat Digital Mission — are named in the corpus only in outline, so the answer stays with the textbook's generic recommendation rather than quoting programme detail the retrieved passages do not carry.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, PIB India, Registrar General of India (MCCD 2023), Ministry of Home Affairs (Registration of Births and Deaths (Amendment) Act 2023), MeitY (DPDP Rules 2025), WHO ICD-11.

Programme and policy updates (India)

- ▶ **Medical record linkage is no longer aspirational in India** — the Ayushman Bharat Digital Mission has crossed **100 crore health records linked to Ayushman Bharat Health Accounts (ABHA)**, doubling from 50 crore in about 15 months, with more than 450 public and private health-technology solutions integrated with the ecosystem; Uttar Pradesh (over 15 crore linked records) and Andhra Pradesh (nearly 12 crore) lead. This is the operational answer to Park's caution that record linkage is applied only on a limited scale and is beyond the reach of many developing countries. [PIB 2026-May]
- ▶ **Facility-level uptake still lags the national headline** — across three tertiary hospitals in Agra, only 129,007 of 537,278 OPD visits over eight months (23.96%) completed ABHA registration; mean digital health literacy was moderate (DHLE 2.94), the weakest domains being appraisal of reliability and privacy protection, only about 55% were comfortable with data-privacy safeguards and 58.6% lacked adequate knowledge of the mission despite attending a participating facility. Useful as the contemporary illustration that hospital records are still generated for administration rather than for monitoring. [PMID 41787376]
- ▶ **Standardised coding and registry interoperability are being built onto hospital records** — the Ministry of Health and Family Welfare has adopted **HL7 FHIR Release 4 as the base data-exchange standard under ABDM**, and the first comprehensive Indian oncology data model for cancer screening publishes 25 oncology-specific FHIR profiles and 50 standardised terminology value sets mapped to SNOMED CT, LOINC and ICD; a parallel proposal would run hospital-based cancer registries on ABHA-linked mobile capture instead of the present retrospective,

manual and semi-digital workflows that leave registry data delayed and non-standardised. Existing hospital electronic records and registries are not yet FHIR-enabled, so bridging mechanisms are still needed. [PMID 41329957]; [PMID 41771200]

Current national data

- **MCCD Report 2023 (Registrar General of India)** — about 19 lakh deaths were medically certified, roughly **22% of registered deaths**, with **diseases of the circulatory system the leading certified cause at 36.4%**; certification remains overwhelmingly institutional, urban and male. A 15-year cluster analysis of MCCD across states and union territories found the national rate stuck at 22.5% with only a 2.5 percentage-point gain over a decade: the largest cluster of 23 states averaged an 18% MCCD rate, 0.14 doctors per 1000 population and only 27.4% of hospitals actively reporting, against clusters averaging 60–63% where over 80% of hospitals reported. Quote these to update the textbook line that medical certification is inaccurate and incomplete. [RGI MCCD 2023]; [PMID 41484187]
- **Death registration itself is gendered (NFHS-5, 2019–21)** — national death registration was **73% for males versus 64% for females**, the gap widest in Bihar, Jharkhand, Uttar Pradesh, Arunachal Pradesh, Nagaland and Manipur, and near-universal (over 94% for both sexes) in Goa, Kerala and Lakshadweep. The gender gap narrows from 14 percentage points in the poorest quintile to 4 in the richest; female sex independently lowered the odds of registration (OR 0.65 in the 15–59 group, 0.58 in those 60+). A second denominator problem sitting upstream of the hospital record. [PMID 41296742]

Elimination, surveillance and disease status

- **Hospital case records can be turned into an early-warning system** — an event-based surveillance pilot in six private hospitals in Kasaragod, Kerala (May–December 2023) had nurses abstract admission records of acute febrile illness into an algorithm that flagged spatiotemporal clusters: 3294 of 4512 patients were analysed, 88 signals generated 10 verified events and **nine outbreaks, five of which the routine indicator-based IDSP reporting never detected**, with a median lead time of 5 days. The authors note the Integrated Health Information Platform can report events but cannot yet auto-detect signals from a patient line list. Directly counters the "records are rarely analysed or fed back" limitation. [PMID 40036102]

Guideline and legislative updates

- **Registration of Births and Deaths (Amendment) Act, 2023** — in force from 1 October 2023 — the Registrar General now maintains a **national database of registered births and deaths**, with chief registrars maintaining state databases and mandatorily sharing data upward; certificates may be issued electronically, and the birth certificate becomes the single document for admission to educational institutions, driving licence, voter list, Aadhaar, marriage registration and government appointment. Field work in Uttar Pradesh found only 55% of births registered within 21 days and 20% delayed beyond 30 days, driven by low awareness (81%) and **absent hospital certificates (30%)**; timeliness rose to about 80% where registration was anchored in the health facility with ANM/ASHA support — the argument for treating the hospital as the registration platform rather than a paper source. [MHA 2023-Oct]; [PMID 41783707]
- **Digital Personal Data Protection Act 2023, with Rules notified 13 November 2025** — hospitals now function as **data fiduciaries** over patient records, requiring specific, informed and revocable

consent, published privacy notices, rights of access, correction and erasure, breach safeguards, and board-registered consent managers who retain consent records for seven years without reading the data. This is the governance layer that computerisation of medical records must now satisfy, and the counterweight to unrestricted secondary use of hospital data for research. [MeitY 2025-Nov]

- **Electronic death certification measurably improves record quality** — at a tertiary institute in Puducherry, an in-house mortality module linked to the hospital information system (built on the ICMR-NCDIR e-Mortality project) raised fully complete MCCD forms from **4% to 19%**, cut forms with major errors from **88% to 42%**, reduced minor errors from 96% to 81%, and improved ICD-compatibility of the underlying cause of death from **73% to 96%** (all $P < 0.001$ except minor errors, $P < 0.002$). For the 60–70% of deaths occurring outside hospital, an ICMR-NCDIR physician-derived cause-of-death (PhyCoD) tool with separate neonatal, child, maternal and adult questionnaires is being validated in Kolar, Karnataka against hospital case records and autopsy. [PMID 40536381]; [PMID 40522917]

Global and WHO developments

- **ICD-11 has continued to expand since the textbook's 1 January 2022 effective date** — the Traditional Medicine Module 2 (TM2), mapping 1941 Indian Ayush morbidity codes developed through the NAMASTE portal, was released on the WHO ICD-11 browser in **February 2025**, with double coding piloted in the Ayush hospital management information system across more than 100 hospitals; ICMR-led work in Madhya Pradesh, Uttar Pradesh and Odisha is simultaneously training certifiers in MCCD and community health workers in the **WHO 2022 Verbal Autopsy instrument** to extend cause-of-death information to the population level. [WHO 2025-Feb]; [PMID 39705697]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 21 (Health Information and Basic Medical Statistics), pp 995-999; Ch 3 (Principles of Epidemiology and Epidemiologic Methods), pp 76, 93-94; Ch 2 (Concept of Health and Disease), pp 39, 69; Ch 6 (Epidemiology of Chronic Non-communicable Diseases), pp 459-460. Gordis Epidemiology 7e, Ch 7, p 177. Jugal Kishore — National Health Programmes of India, p 262. WHO Expert Committee on Health Statistics, 8th report (as cited in Park 28e). Figure: Suryakantha — Community Medicine with Recent Advances, Fig 3.2, p 35.

Recent-advances sources: 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission: an analytical cross-sectional study among outpatient department attendees, BMC Health Serv Res 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 41329957 — FHIR Standard-Based Oncology Data Model for Cancer Screening: Design and Implementation Study, JMIR Cancer 2025, doi:[10.2196/79011](https://doi.org/10.2196/79011); 41771200 — The need for mobile applications in hospital-based cancer registries in India: A framework for ABDM-integrated surveillance, Curr Probl Cancer 2026, doi:[10.1016/j.currproblcancer.2026.101284](https://doi.org/10.1016/j.currproblcancer.2026.101284) (abstract-only); 41484187 — A retrospective cluster analysis of regional disparities and healthcare factors influencing causes of death certification and mortality statistics in India, Sci Rep 2026, doi:[10.1038/s41598-025-27634-1](https://doi.org/10.1038/s41598-025-27634-1); 41296742 — Who gets counted? Understanding low female death registration in India, PLoS One 2025, doi:[10.1371/journal.pone.0337224](https://doi.org/10.1371/journal.pone.0337224); 40036102 — Piloting an event-based surveillance model in private hospitals for early detection of disease clusters, Kerala, India, Indian J Med Res 2025, doi:[10.25259/IJMR_1395_2024](https://doi.org/10.25259/IJMR_1395_2024); 41783707 — From delivery to identity: applying transaction cost economics to health-enabled civil registration in India, Front Public Health 2026, doi:[10.3389/fpubh.2026.1743341](https://doi.org/10.3389/fpubh.2026.1743341); 40536381 — Improving completeness & reducing errors in medical certification of cause of death: The impact of electronic mortality software in a tertiary care centre in South India, Indian J Med Res 2025, doi:[10.25259/IJMR_1394_2024](https://doi.org/10.25259/IJMR_1394_2024); 40522917 — Establishing a System for Medical Certification of Cause of Death for Noninstitutional Deaths in a Selected Area of Kolar District, Karnataka, India: Protocol for a Population-Based Feasibility and Validation Study, JMIR Res Protoc 2025, doi:[10.2196/72330](https://doi.org/10.2196/72330); 39705697 — Strengthening Cause of Death Statistics in Selected Districts of 3 States in India: Protocol for an Uncontrolled, Before-After, Mixed Method Study, JMIR Res Protoc 2024, doi:[10.2196/51493](https://doi.org/10.2196/51493). Press Information Bureau, Government of India — 100 crore health records linked with ABHA under ABDM, 22 May 2026. Office of the Registrar General of India — Report on Medical Certification of Cause of Death 2023. Ministry of Home Affairs — Registration of Births and Deaths (Amendment) Act 2023, in force 1 October 2023. Ministry of Electronics and Information Technology — Digital Personal Data Protection Act 2023 and Digital Personal Data Protection Rules notified 13 November 2025. WHO — ICD-11 Traditional Medicine Module 2 released on the ICD-11 browser, February 2025.

♦ ♦ ♦

Cost-benefit and cost-effectiveness analysis

Asked in: Paper IV 19-Nov-2021 — RGUHS MD Community Medicine

Definition — **Economic evaluation** is the comparative analysis of two or more alternative courses of action in terms of **both their costs and their consequences**, undertaken to judge value for money when resources are scarce (Health Economics, Ch 13, p 201). **Cost-benefit analysis (CBA)** expresses both the costs and the outcomes of a health programme in **monetary units**; **cost-effective(ness) analysis (CEA)** expresses costs in money but outcomes in **natural health units** such as lives saved, deaths averted or days free from disease (Park 28e, Ch 23, p 1028). Both are **quantitative managerial methods**, drawn from economics, operations research and budgeting, used by the health manager to set priorities among competing programmes and to obtain the maximum health gain from a fixed budget.

Place among the quantitative methods of health management

Park lists **ten** quantitative methods; cost-benefit and cost-effective analysis are the first two.

Quantitative method	What it does for the manager
1. Cost-benefit analysis	Compares the economic benefits of a programme with its cost, both in money, to decide whether the programme is economically sound and to select the best of several alternatives (Park 28e, Ch 23, p 1028)
2. Cost-effective analysis	Same comparison, but benefit expressed as results achieved (lives saved, disease-free days); "a more promising tool for application in the health field than cost-benefit analysis" (Park 28e, Ch 23, p 1028)
3. Cost-accounting	Provides basic data on the cost structure of a programme; three purposes — cost control, planning and allocation of people and financial resources, and pricing for cost reimbursement (Park 28e, Ch 23, p 1028)
4. Input-output analysis	Relates inputs (manpower, money, materials) to useful outputs (lives saved, inoculations performed) and shows how much of each input is needed per unit of output (Park 28e, Ch 23, pp 1028-1029)
5. Model	An abstraction of reality used to understand how factors in a situation affect one another; the decision process uses a model (Park 28e, Ch 23, p 1029)
6. Systems analysis	Investigates the problem, sets objectives, finds alternatives and evaluates the alternatives in terms of cost-effectiveness to select the most cost-effective one — i.e. CEA is the engine of systems analysis (Park 28e, Ch 23, p 1029)
7. Network analysis	A graphic plan of all events and activities needed to reach an objective; two types — PERT (Programme Evaluation and Review Technique) and CPM (Critical Path Method) (Park 28e, Ch 23, p 1029)
8. Planning-Programming-Budgeting System (PPBS)	A system helping decision-makers allocate resources so that they are used most effectively in achieving objectives; needs no change in the existing organisation, only the grouping of activities into programmes related to each objective . A related approach is the Zero Budget Approach — all budgets start at zero and no one gets any budget he cannot specifically justify on a year-to-year basis (Park 28e, Ch 23, p 1029)

Quantitative method	What it does for the manager
9. Work sampling	Systematic observation and recording of the activities of one or more individuals at predetermined or random intervals, giving quantitative measurement of the type of activity performed and the time needed per specified job ; done on doctors, nurses, pharmacists and laboratory technicians. Used to judge the appropriateness of current staff job descriptions and training, standardise methods of work and determine manpower needs (Park 28e, Ch 23, p 1029)
10. Decision making	Likened to differential diagnosis in clinical practice — decisions should be made at the level where the best decision can be made (not necessarily at the top of the organisation) and never on incomplete data ; in the health sector they concern resource development, optimum workload and care strategies (Park 28e, Ch 23, pp 1029-1030)

Cost-benefit analysis

- Use — programmes with **diverse objectives** are compared by CBA, because reducing every outcome to money creates a **common measure of benefit**; this allows comparison even across sectors of the economy (Reinke, Ch 20, p 274; Health Economics, Ch 13, pp 206-207).
- Summary measures and decision rules:

Measure	Formula	Decision rule
Benefit-cost ratio (BCR)	Present value of total benefits ÷ present value of total costs	$BCR > 1$ = worth considering; among competing programmes, the highest ratios get priority for limited funds (Reinke, Ch 20, p 274, pp 275-276)
Net present value (NPV)	Total monetised benefits – total costs	Positive NPV = the programme is a net gain to society (Health Economics, Ch 16, pp 247-248)

- Methods of putting a money value on health benefit:

Approach	Basis	Chief criticism
Human capital approach	Values a life by expected lifetime earnings (market wage rates for productivity lost through illness)	Assumes earnings equal what society values; assigns lower values to women and to members of minority and low-income groups , and to those outside the prime working ages (Reinke, Ch 20, pp 281, 286)
Market/replacement value	Values each duty a homemaker performs — market wage for each service × hours from a time-motion study	Ignores the psychic value of the homemaker to family and society (Reinke, Ch 20, p 281)
Willingness to pay (stated preference)	Maximum sum an individual would pay out of income for a stated reduction in the risk of death or illness	Surveys are costly and need respondents who understand probability; unreliable when benefits are complex (Reinke, Ch 20, pp 281-282)
Revealed preference	Observed market behaviour, e.g. the extra wages needed to attract workers into risky occupations	Applied so far only to the simple case of reduced mortality risk (Reinke, Ch 20, p 282)

Cost-effective analysis

- **Use** — CEA evaluates **alternative ways of reaching the same objective** and is the technique of choice in the health field (Reinke, Ch 20, p 274). It answers two managerial questions: (a) what is the **least costly means** of achieving a defined objective, e.g. protecting 10,000 infants against measles; and (b) what is the **maximum benefit obtainable from a fixed budget**, e.g. how many children can be immunised for a stated sum (Reinke, Ch 20, p 274).
- **Cost-utility analysis (CUA)** is a broader variant of CEA in which outcomes are expressed in units capturing **both quantity and quality of life** — the **quality-adjusted life year (QALY)**, scored 1.0 for full health to zero for death, or the **disability-adjusted life year (DALY)** (Health Economics, Ch 13, p 201; Ch 15, pp 230-231). **Cost-minimisation analysis (CMA)** is a narrow subset of CEA used only when outcomes of the alternatives are known to be identical, so only input costs are compared (Health Economics, Ch 13, p 208).
- **The three cost-effectiveness ratios:**

Ratio	Formula	When used
Average cost-effectiveness ratio (ACER)	Total cost of intervention A ÷ total effects of intervention A	A single intervention against its baseline (no programme or current practice) (Health Economics, Ch 16, p 250); independent interventions are ranked, lower ACER = higher priority (Health Economics, Ch 13, p 207)
Marginal cost-effectiveness ratio (MCER)	Change in cost ÷ change in effect on expanding or contracting an existing programme	Deciding how far to scale a programme up or down (Health Economics, Ch 16, pp 247-248)
Incremental cost-effectiveness ratio (ICER)	(Cost A – Cost B) ÷ (Effect A – Effect B)	Two mutually exclusive alternatives; gives the extra cost per additional unit of effect (DALY averted, QALY gained, case prevented) (Health Economics, Ch 16, p 253)

Worked example — costs £26,000 vs £5,000 and effects 10 vs 3 cases prevented give ICER = £21,000 ÷ 7 = **£3,000 per disease case prevented** (Health Economics, Ch 16, pp 251-252).

- **Decision rules** — first apply **strong dominance**: a strategy that is both more effective *and* less costly is adopted outright, and one both less effective and more costly is rejected (Health Economics, Ch 16, pp 251-252). In the trade-off quadrants the ICER is compared with a **willingness-to-pay threshold** — the maximum the decision-maker will pay per unit of health effect. WHO, following the Commission on Macroeconomics and Health (2001), classifies an intervention as "**highly cost-effective**" for a country if it averts a DALY for less than that country's **per capita gross domestic product**; the National Institute for Health and Care Excellence in the United Kingdom uses £20,000–30,000 per QALY gained (Health Economics, Ch 16, pp 253-254, pp 256-257).

Costing conventions common to both methods

Item	Content
Direct costs	Direct health care (drugs, staff, supplies, hospital stay) and direct non-health care (patient transport, child care) (Health Economics, Ch 14, pp 219-220)

Cost-benefit versus cost-effective analysis

Feature	Cost-benefit analysis	Cost-effective analysis
Question answered	Is this programme worth doing at all , and which of several differently-aimed programmes should be funded?	Which is the least costly way of reaching one agreed objective ?
Objectives compared	Diverse, even across sectors	Must share a single common unit of effect
Summary measure	Benefit-cost ratio; net present value	Cost per unit of effect — ACER, MCER, ICER
Type of efficiency assessed	Allocative efficiency (is the goal itself worth pursuing?) — and economic efficiency as well , since it assesses all relevant costs and benefits (Health Economics, Ch 13, p 209)	Economic efficiency only — "what is the least costly way to achieve a particular goal?" (Health Economics, Ch 13, p 209)
Main strength	Single comparable yardstick permits inter-sectoral priority setting	Avoids valuing life in money; results such as "cost per life-year gained" are readily understood by planners (Reinke, Ch 20, p 286; Health Economics, Ch 13, p 207)
Main limitation	Health benefits often cannot be expressed in money , so the scope of application is vague; valuing human life is contentious and the single money figure is opaque to decision-makers (Park 28e, Ch 23, p 1028; Health Economics, Ch 13, pp 206-207)	Restricted to interventions with one common outcome; cannot say whether the objective itself deserves funding; "even cost-effective analysis is not possible in many cases" (Park 28e, Ch 23, p 1028)
Data demand	Very high — earnings, employment rates, valuation of homemakers' services	Lower; the preferred technique in developing-country settings (Reinke, Ch 20, p 286)

Applications and pitfalls in health service management

- **Priority setting** — ranking competing programmes for a limited budget; the basis of cost-effectiveness **league tables** (cost per QALY, per life-year, per DALY) (Health Economics, Ch 16, pp 254-255).
- **Programme design** — choosing between service packages. In the *Narangwal* study in rural Punjab, India, CEA showed that infant and 1–3-year mortality fell at least cost with **health care alone**, perinatal mortality at least cost with **nutrition services**, and growth and psychomotor development at least cost with the **combined nutrition-plus-health-care package** (Reinke, Ch 20, pp 285-286).
- **Rationing and reimbursement** — thresholds used to decide which technologies a publicly funded system will pay for (Health Economics, Ch 16, pp 256-257).
- **Pitfalls** — league tables mix ICERs computed with different comparators, discount rates, time horizons and sub-groups, so the ranking may not reflect true relative value for money, and they usually omit uncertainty (Health Economics, Ch 16, pp 254-255). CBA is a tool of **microanalysis** and assumes the programme has only a marginal impact on the economic and demographic fabric — an assumption that fails for a large-scale eradication programme (Reinke, Ch 20, pp 275-276).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Reproduce the cost-effectiveness plane — the four quadrants with their dominance labels, and the slope of the line from the origin to a plotted point = the ICER.

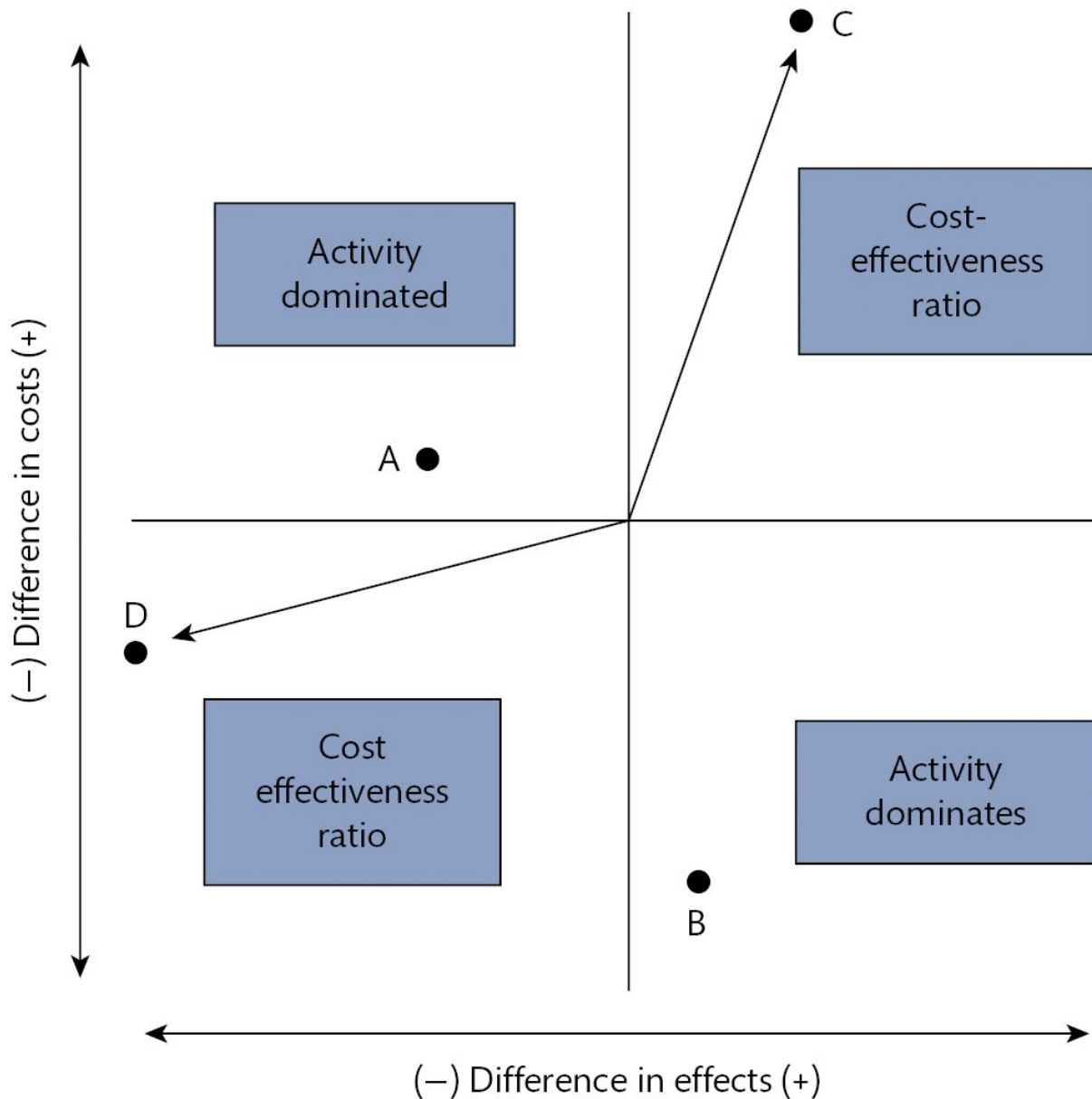
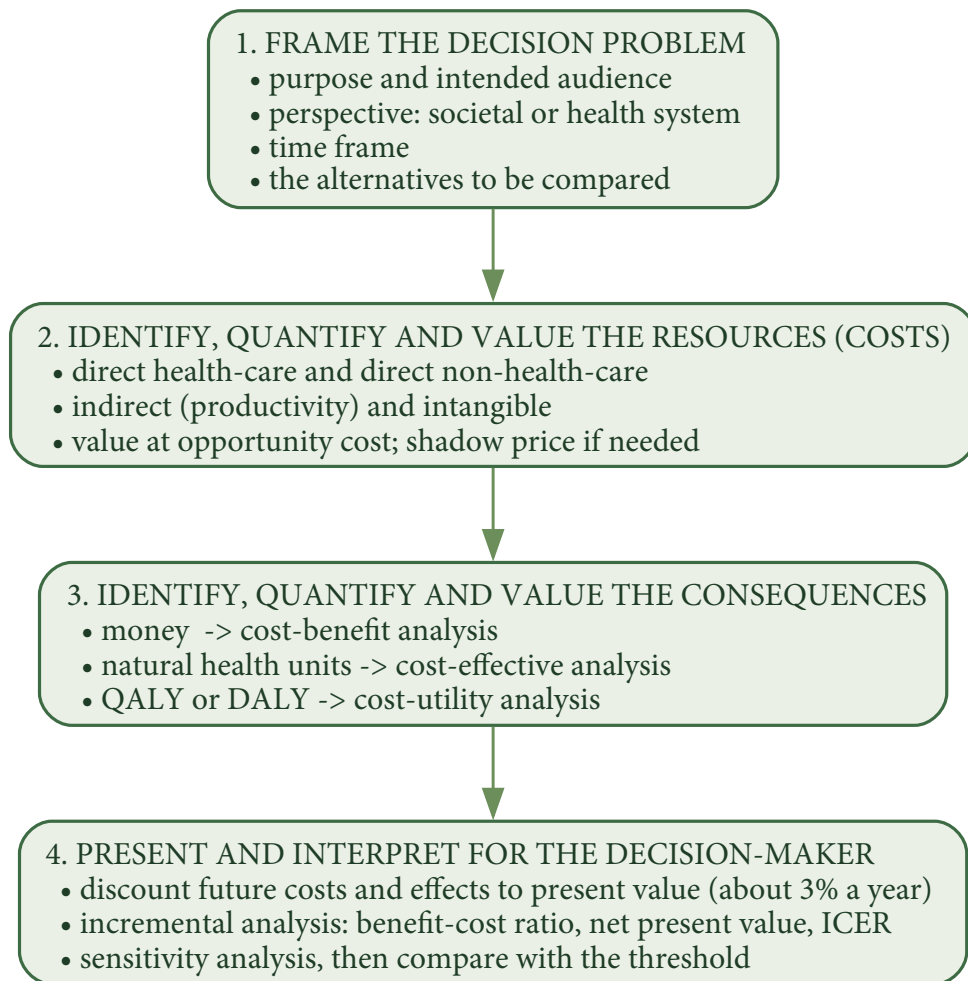
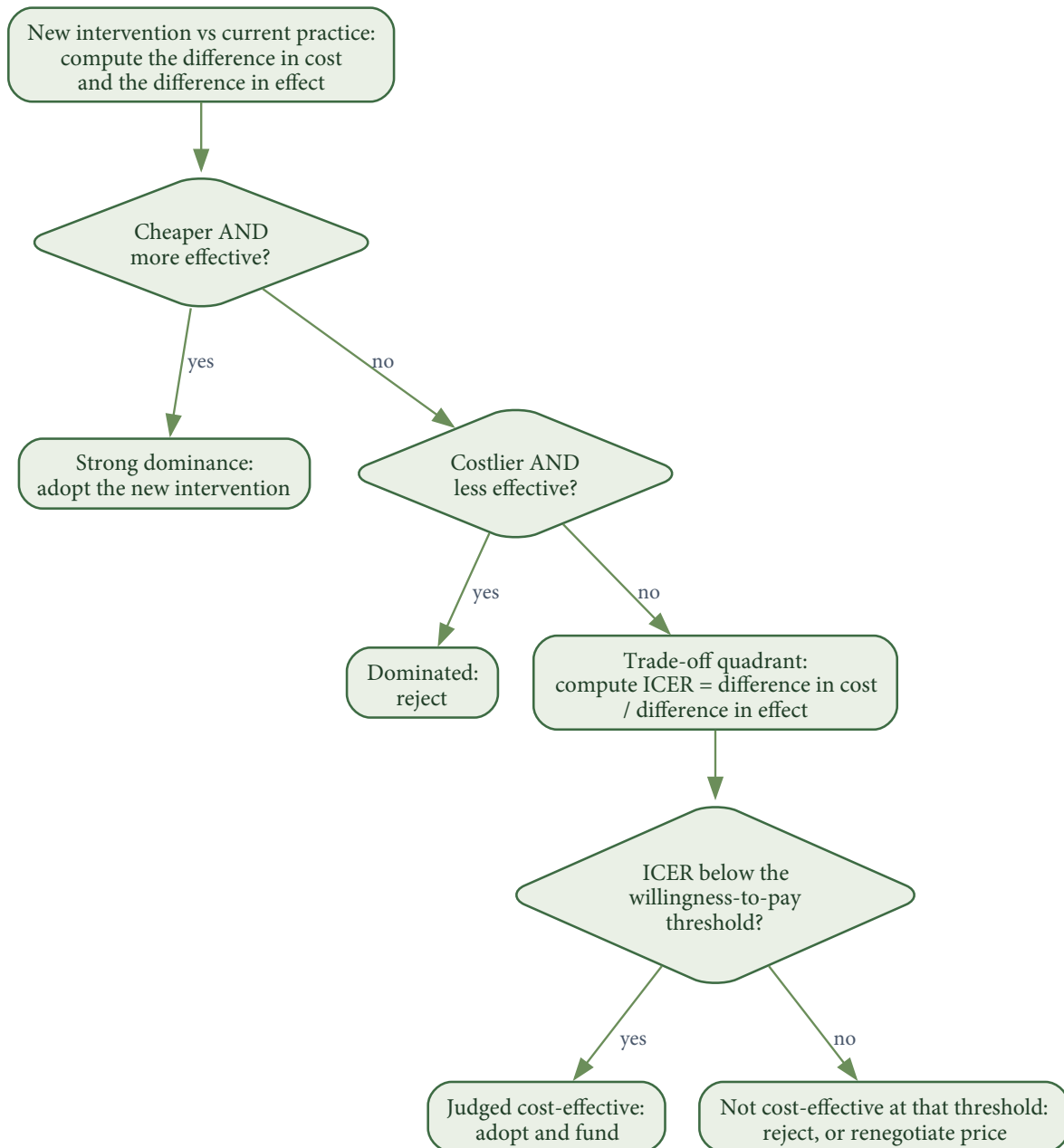


Fig 6.6 — 1 The cost-effectiveness plane. An intervention can be placed anywhere (Oxford public health 7th ed, p.1824)



The four steps of an economic evaluation of a health programme



Decision rule when a new intervention is compared with current practice

High-yield exam pointers

- ▶ **One-line discriminator:** CBA compares programmes with **different objectives** (benefit in rupees); CEA compares **different ways of reaching the same objective** (benefit in natural units).
- ▶ **Formulas to write:** $BCR = \text{present value of benefits} \div \text{present value of costs}$ (adopt if > 1); $NPV = \text{benefits} - \text{costs}$; $ACER = \text{total cost} \div \text{total effect}$; $ICER = \Delta \text{cost} \div \Delta \text{effect}$.
- ▶ **Threshold rule:** WHO/Commission on Macroeconomics and Health 2001 — **highly cost-effective** if cost per DALY averted is less than per capita GDP; NICE ceiling £20,000–30,000 per QALY.
- ▶ **Discount rate** = 3% per year after controlling for inflation; the higher the rate, the lower the present value of distant benefits.

- **Cost triad:** direct (health-care + non-health-care), indirect (productivity), intangible (pain, suffering, anxiety); **economic cost** = **opportunity cost**, financial cost = money actually spent.
- **Park's TEN quantitative managerial methods** (mnemonic C-C-C-I-M-S-N-P-W-D): Cost-benefit, Cost-effective, Cost-accounting, Input-output, Model, Systems analysis, Network analysis (PERT/CPM), PPBS (with the Zero Budget Approach), Work sampling, Decision making.
- **Indian example:** the *Narangwal* study, rural Punjab — CEA of health care versus nutrition versus combined services on infant and child mortality, morbidity and growth.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: Park's 28th edition uses the term "cost-effective analysis" (the wording of this question stem), while the standard economics literature calls the same technique cost-effectiveness analysis. Park's ten quantitative methods are a list separate from the chapter's parallel list of methods based on the behavioural sciences (organizational design, communication and the rest), which this question does not ask about. The retrieved corpus supports the World Health Organization rule that an intervention costing less than one per capita gross domestic product per disability-adjusted life year averted is "highly cost-effective", but does not carry the commonly quoted one-to-three times gross domestic product band, so that band is omitted; the corpus also yielded nothing on India's Health Technology Assessment agency, so no Indian threshold is quoted.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, HTAIn (Department of Health Research), National Health Authority, Indian Reference Case, WHO-CHOICE

Programme and policy updates (India)

- **India now has a standing agency that does this appraisal.** Health Technology Assessment in India (HTAIn) sits under the Department of Health Research, Ministry of Health and Family Welfare, set up after the National Health Policy 2017; it runs a Secretariat, a Board and a Technical Appraisal Committee that prioritises topics received from central and state departments, allocates each to a research centre for economic evaluation, and issues a policy brief to government. Regional resource centres grew from 1 in 2017 to 30 by 2024 with about 72 scientific and technical staff, and six regional resource hubs now handle quality assurance, capacity building and dissemination. [PMID 41316276] [PMID 40913689]
- **The appraisal machinery has itself been put through a benefit-cost test.** Three assessments commissioned by HTAIn between 2018 and 2020 were valued against a counterfactual of no appraisal: the return on investment was 9:1, rising to 71:1 if recommendations were fully implemented, and ranging 5:1 to 40:1 across interventions — a worked benefit-cost ratio applied to the priority-setting system rather than to a treatment. [PMID 40113231]
- **The insurance side has a separate unit.** A Health Financing and Technology Assessment (HeFTA) unit was created within the National Health Authority to feed HTA evidence into inclusion and exclusion decisions for the Ayushman Bharat PM-JAY health benefit package,

into standard treatment guidelines and into package pricing, with the stated direction of travel towards value-based pricing in strategic purchasing. [PMID 38715207]

Current national data

- **Comprehensive Annual Modular Survey (CAMS) 2022–23** covered 302,086 households across all states and union territories; 59.9% reported some insurance. Publicly funded insurance was associated with greater outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128–1.210) and with less catastrophic expenditure at the 20% threshold for inpatient care (adjusted OR 0.790, 95% CI 0.737–0.846). Mean inpatient out-of-pocket spending was ₹29,875 in PM-JAY or state-scheme households against ₹37,572 in the uninsured; outpatient care remains the uncovered gap, and the authors call for primary care strengthening through Ayushman Arogya Mandirs. [PMID 42167946]
- **The costing and valuation inputs an Indian analysis needs now exist.** Out-of-pocket spending stood at 49.8% of current health expenditure in 2021 — the standing case for explicit priority setting. The national health system cost database (Cost of Health Services in India, CHSI) and an Indian EQ-5D-5L value set now supply domestic unit costs and utility weights in place of imported ones, although disaggregated data for equity analysis remain scarce. [PMID 41316276] [PMID 40913689]

Elimination, surveillance and disease status

- **Shorter drug-resistant TB regimens evaluated for cost-effectiveness under NTEP.** A Markov model from the health system perspective, 3% discounting, willingness to pay set at one-time per capita GDP (₹1,15,746 for 2022–23), compared BPAL and BPALM with the current mixed standard of care for MDR/RR-TB: BPAL gave an ICER of –₹379 per QALY (cheaper *and* more effective, i.e. strong dominance) and BPALM +₹37 per QALY. Budget impact analysis: –₹106 million a year for BPAL, +₹13 million for BPALM. The authors conclude that countries should consider programmatic uptake and that the India TB programme should decide how to turn the finding into policy. [PMID 41520273]
- **Lymphatic filariasis — the stopping rule is now an economic decision.** Tightening the transmission assessment survey threshold from <2% to <1% antigenaemia raised the probability of local elimination (83.8% versus 78.8% at 5–10% baseline prevalence) and was cost-effective at 80% mass drug administration coverage but not at 65%. The threshold used for India was a country-specific opportunity-cost threshold based on per capita government expenditure, expressed as a proportion of per capita GDP and adjusted for purchasing power parity: US\$446.07 per DALY averted. [PMID 38662697]

Guideline and legislative updates

- **The Indian Reference Case is the national methods standard** for economic evaluation, covering decision problem, comparator, perspective, effectiveness, costs, health outcome, time horizon, discounting, heterogeneity, uncertainty and equity. It prescribes an **abridged societal perspective** — all direct and indirect costs to health system and patient, but excluding productivity loss — and **3% discounting**. An audit of 33 HTAIn studies scored adherence at 95.6% for the decision problem, 94.9% for the comparator, 86.8% for perspective and 82.4% for discounting, and lowest for **equity (25.7%) and heterogeneity (24.3%)**. The companion HTA Quality Appraisal

Checklist (HTA-QAC) is now the review instrument; of 26 studies from 2018–2023, 14% were excellent, 50% good, 32% average and 4% poor. [PMID 41316276] [PMID 40913689]

- **The threshold rule is in transition.** Every HTAIn study that used a threshold applied **one times per capita GDP**, and 84.8% of interventions were judged cost-effective on it (21.2% cost-saving outright). The WHO-CHOICE band of **one to three times per capita GDP is now cautioned against as the sole threshold**, by several experts including those at WHO, because a GDP multiple reflects neither opportunity cost nor the budget constraint. The recommendation for India is a threshold derived from the opportunity cost of current public health spending, used as a decision aid rather than a rule, with **health technology reassessment** identifying less cost-effective existing services for disinvestment — if more cannot be spent, it must be spent better. [PMID 41316276]

Global and WHO developments

- **The Expanded Programme on Immunization at 50 is the largest modern cost-benefit analysis in public health.** Across 194 WHO member states and 14 pathogens from 1974 to 2024, EPI cost **US\$937 billion** (95% CI 699–1187) against **\$15,050 billion** (12,609–17,605) in averted productivity loss from deaths prevented — an aggregate **benefit-cost ratio of 16.06** (95% CI 10.62–25.20), with the ICER cost-saving. Measles vaccination gave the highest ratio at **73.97** (45.78–124.06); the ratio exceeded 1 in all 194 member states and, across World Bank income groups, was highest in low-income countries at 23.63 (16.87–34.32). [PMID 42242275]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 23, Health Planning and Management, pp 1025-1030); Reinke — Health Planning for Effective Management (Ch 20, Cost-Benefit and Cost-Effectiveness Analysis of Health Programs, pp 274-288); Health Economics (Ch 13 What is economic evaluation, pp 200-213; Ch 14 Counting the costs, pp 215-228; Ch 15 Identifying, measuring and valuing consequences, pp 230-231; Ch 16 Economic evaluation and decision-making, pp 247-260; Ch 17 glossary, p 280); Oxford Textbook of Public Health 7e (Ch 49, Economic appraisal in public healthcare: assessing efficiency and equity, pp 1806-1852 — cost-effectiveness plane, p 1824). Named sources within these: WHO Commission on Macroeconomics and Health 2001 (per capita gross domestic product threshold); the Narangwal study of child and maternal health services in rural Punjab, India.

Recent-advances sources: 41316276 — Insights and implications: an analysis of the studies undertaken by Health Technology Assessment in India (HTAIn), Health Research Policy and Systems 2025, DOI [10.1186/s12961-025-01422-9](https://doi.org/10.1186/s12961-025-01422-9); 40913689 — A Mixed-Methods Assessment of India's Health Technology Assessment Ecosystem, Applied Health Economics and Health Policy 2025, DOI [10.1007/s40258-025-00995-4](https://doi.org/10.1007/s40258-025-00995-4); 40113231 — Is health technology assessment value for money? Estimating the return on investment of health technology assessment in India (HTAIn), BMJ Evidence-Based Medicine 2025, DOI [10.1136/bmjebm-2023-112487](https://doi.org/10.1136/bmjebm-2023-112487) (abstract-only); 38715207 — Establishing a Health Technology Assessment Evidence Ecosystem in India's Pradhan Mantri Jan Arogya Yojana, Health Systems & Reform 2024, DOI [10.1080/23288604.2024.2327097](https://doi.org/10.1080/23288604.2024.2327097) (abstract-only); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022–2023), BMJ Open 2026, DOI [10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 41520273 — Cost-effectiveness of BPAL/BPaLM as compared to mixed standard of care bedaquiline containing regimen for MDR/RR-TB, Indian Journal of Medical Research 2025, DOI [10.25259/IJMR_158_2025](https://doi.org/10.25259/IJMR_158_2025); 38662697 — Reducing the Antigen Prevalence Target Threshold for Stopping and Restarting Mass Drug Administration for Lymphatic Filariasis Elimination: A Model-Based Cost-effectiveness Simulation in Tanzania, India and Haiti, Clinical Infectious Diseases 2024, DOI [10.1093/cid/ciae108](https://doi.org/10.1093/cid/ciae108); 42242275 — Global, regional, and national impact of the Expanded Programme on Immunization against 14 pathogens from 1974 to 2024: an economic evaluation, The Lancet Global Health 2026, DOI [10.1016/j.langlo.2026.103965](https://doi.org/10.1016/j.langlo.2026.103965) (abstract-only).

♦ ♦ ♦

Cost-effectiveness analysis in health

Asked in: Paper IV Nov 2016 — RGUHS MD Community Medicine

Definition — Cost-effectiveness analysis (CEA) is a technique of **economic evaluation** in which the costs of alternative health interventions are compared with their consequences expressed in **natural health units** — lives saved, deaths averted, cases detected — instead of in money (Health Economics, Ch 13, pp 200-201). Park calls it "a more promising tool for application in the health field than cost-benefit analysis... benefit, instead of being expressed in monetary terms, is expressed in terms of results achieved" (Park 28e, Ch 23, p 1028). It is the commonest form of economic evaluation in the health sector, used when alternative ways of reaching the **same objective** compete for a limited budget (Reinke, Ch 20, p 274).

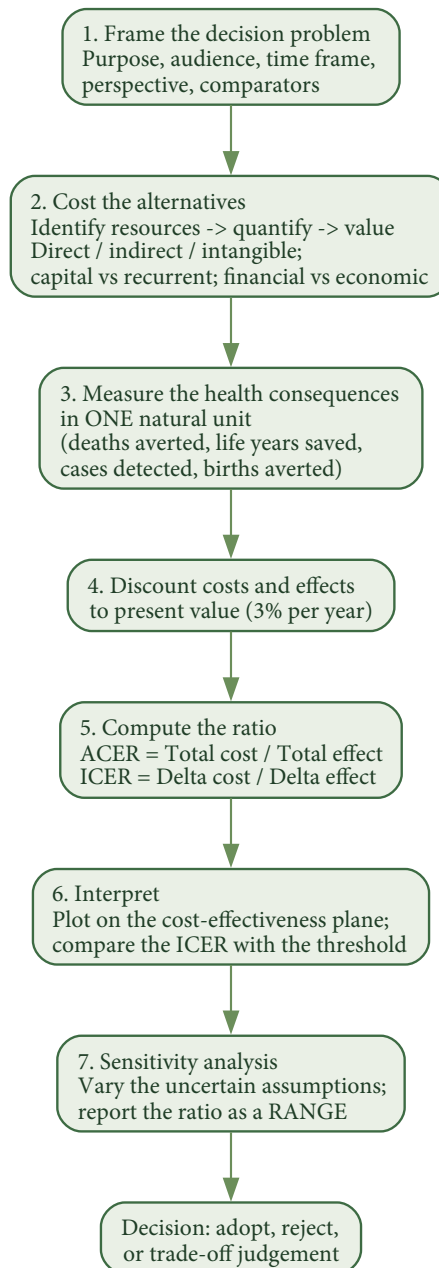
Place of CEA among the techniques of economic evaluation

Technique	How the consequence is valued	Question it answers	Summary measure
Cost-minimisation analysis (CMA)	Outcomes assumed identical, so only costs compared; a narrow subset of CEA	Which alternative is the cheapest way to reach an outcome already proved equal?	Total cost per patient
Cost-effectiveness analysis (CEA)	Natural health units (life years saved, deaths averted, cases detected)	What is the least costly way of achieving a given health goal?	Cost per unit of health effect
Cost-utility analysis (CUA)	Health units capturing both quantity and quality of life	Which option gives most health when morbidity and mortality both change?	Cost per quality-adjusted life year (QALY) gained or disability-adjusted life year (DALY) averted
Cost-benefit analysis (CBA)	Money	Is the goal itself worth pursuing, and how does it compare with non-health sectors?	Net present value (NPV); benefit-cost ratio (BCR)

(Health Economics, Ch 13, pp 200-201, 208-209; Ch 16, pp 247-248)

- CEA and CUA measure **economic efficiency** — the least costly way to a stated goal; CBA additionally addresses **allocative efficiency** — whether the goal is worth pursuing at all. Economic evaluation is *not* about choosing the cheapest option: if it were, doing nothing would always be most "efficient" (Health Economics, Ch 13, p 209).

Steps in conducting a cost-effectiveness analysis



Sequence of steps in a cost-effectiveness analysis (Health Economics, Ch 13, pp 209-210; Ch 14, pp 214-228; Ch 16, pp 253-254)

Key methodological choices within those steps

- **Perspective** decides whose costs and outcomes are counted: *societal* (all costs and outcomes, whoever bears or gains them) or *health system* (only those falling on the health sector) (Health Economics, Ch 13, p 211).
- **Discounting** converts future costs and effects to present value; national and international guidelines recommend **3 per cent per year**, after controlling for inflation (Health Economics, Ch 14, p 228).
- **Sensitivity analysis** tests robustness by varying the discount rate, unit costs and effect size, so the ratio is reported as a **range**, not a point value (Health Economics, Ch 14, p 228; Ch 16, p 254).

Costs to be identified and measured

Classification	Categories	Examples
By whom borne	Direct health care; direct non-health care; indirect; intangible	Drugs and staff; patient transport; value of resources spent by patients and carers to obtain care; pain, anxiety, discomfort
By life of the resource	Capital vs recurrent	Building, vehicle vs drugs, salaries (useful life under one year)
By behaviour with output	Fixed vs variable; marginal cost = change in total cost when one extra unit of output is produced	Rent vs consumables
By concept of value	Financial (budgetary) cost = money actually paid; economic (opportunity) cost = value in the next best alternative, using a shadow price where the market price is distorted (donated goods, taxes, subsidies)	Purchase price of vaccine vs true societal value of a donated vaccine
Support costs	Overhead cost — not incurred directly in patient care but needed to run the organisation	Personnel and administrative functions

(Health Economics, Ch 14, pp 215-220, 228)

Intangible costs matter to patients but are omitted from most evaluations because they are hard to measure (Health Economics, Ch 14, pp 219-220).

Measuring effectiveness and computing the ratio

Ratio	Formula	Use
Average cost-effectiveness ratio (ACER)	Total costs of intervention A ÷ total effects of intervention A	A single intervention against its baseline (no programme, or current practice)
Incremental cost-effectiveness ratio (ICER)	(Cost of A – Cost of B) ÷ (Effect of A – Effect of B)	Two competing alternatives; the cost of producing one additional unit of effect
Marginal cost-effectiveness ratio (MCER)	Change in cost ÷ change in effect on expanding or contracting a programme	Scaling an existing programme up or down

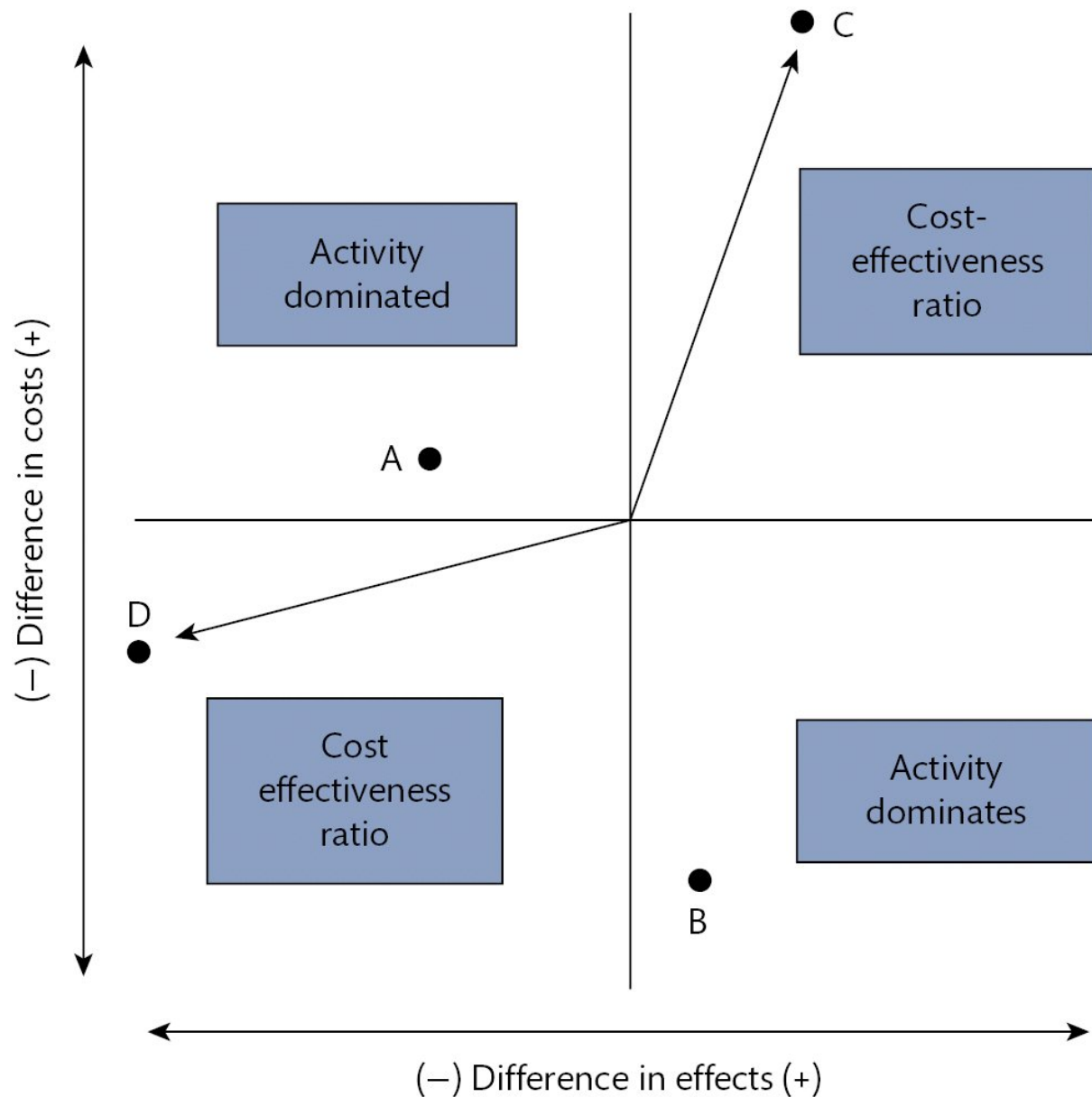
(Health Economics, Ch 13, p 207; Ch 16, pp 247-250)

Effect is stated in one natural unit — **years of life saved, cases of blindness prevented, births averted, deaths averted, cases detected, days of illness averted** (Reinke, Ch 20, pp 283-284). Where interventions are independent, a ratio is computed for each and ranked, priority going to the lower ACER (Health Economics, Ch 13, p 207). **Worked illustration** (Health Economics, Ch 16, p 253, Table 16.4):

Intervention	Cost per patient (£)	QALYs per patient	ICER (£/QALY)
Standard care	50	1.0	—

Intervention	Cost per patient (£)	QALYs per patient	ICER (£/QALY)
B	100	2.0	50
D	350	4.0	125
E	550	5.0	200

Interpreting the result — the cost-effectiveness plane and the threshold



The cost-effectiveness plane (Oxford public health 7th ed, p.1824)

Quadrant (incremental cost, incremental effect)	Interpretation	Decision
South-east: cost negative, effect positive	New intervention dominates	Always cost-effective — adopt

Quadrant (incremental cost, incremental effect)	Interpretation	Decision
North-west: cost positive, effect negative	New intervention is dominated	Never cost-effective — reject
North-east: cost positive, effect positive	Trade-off	Cost-effective only if ICER falls below the threshold
South-west: cost negative, effect negative	Trade-off	Judgement required

(Health Economics, Ch 16, pp 253-254)

- The **threshold (ceiling ratio)** is the most a decision-maker will pay for one extra unit of health. On the recommendation of the Commission on Macroeconomics and Health, the World Health Organization (WHO) classes an intervention as **highly cost-effective** if it averts a DALY for **less than that country's per-capita gross domestic product** (WHO 2001). Ceilings quoted elsewhere: United Kingdom £20,000–30,000 per QALY, United States US\$50,000 per QALY (1982 dollars) (Health Economics, Ch 16, pp 253-254).
- **League tables** rank interventions by cost per life year, QALY or DALY, but combine studies using different comparators, discount rates, time horizons and sub-groups and omit uncertainty, so the ranking may not reflect true value for money (Health Economics, Ch 16, pp 254-255).

Applications and limitations in public health

Applications

- **Priority setting between competing programmes** with the same objective — e.g. which disease control programme gives the lowest cost per death averted (Reinke, Ch 20, pp 283-284).
- **Programme evaluation and health planning** — evaluation "measures how much output or cost-effectiveness is achieved", permitting reallocation of priorities and resources (Park 28e, Ch 23, p 1027); Park's tool of **systems analysis** is "essentially finding the cost-effectiveness of the available alternatives" (Park 28e, Ch 23, p 1029).
- **Indian field example — the Narangwal experiment, rural Punjab**: villages received nutrition plus health care (NUTHC), nutrition only (NUT) or health care only (HC). Cost per perinatal death averted was lowest at US\$7.75 with NUT; health care alone was most cost-effective for infant (US\$25.35) and older-child (US\$30.65) deaths, against US\$72 for NUT and US\$101 for NUTHC (Reinke, Ch 20, pp 284-286).
- **Institutional resource allocation** — the National Institute for Health and Care Excellence (NICE), England and Wales, uses cost per QALY to decide what the health service funds (Health Economics, Ch 16, pp 256-257).

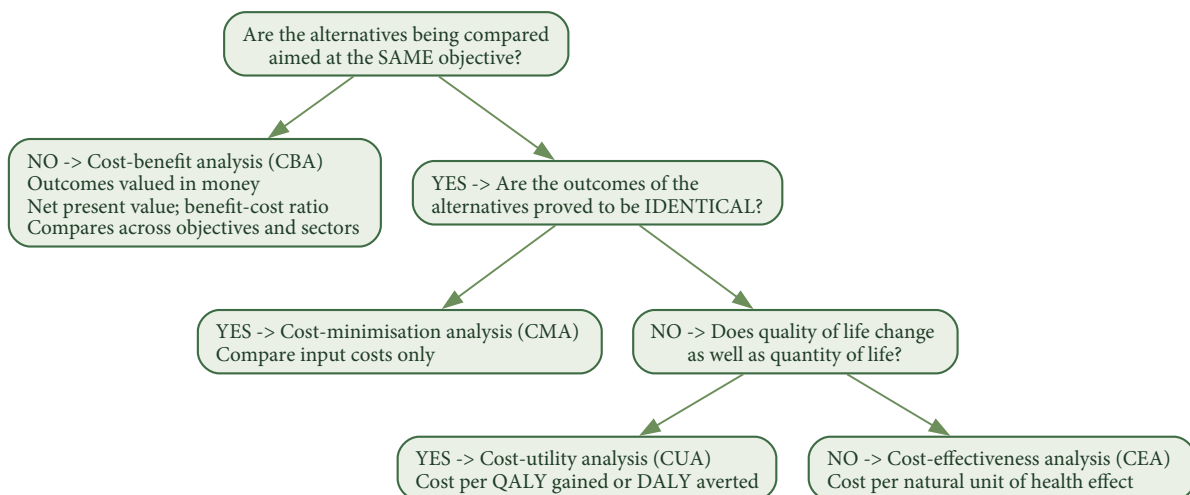
Limitations

- Considers **only one measure of effectiveness** at a time, omitting other valued consequences (Health Economics, Ch 13, p 209).
- Cannot compare programmes with **different objectives** or across sectors — that needs CBA; it answers "the least costly way to achieve a goal", not "whether the goal is worth pursuing".

- Sensitive to the **comparator, discount rate, time horizon and perspective** chosen, limiting comparability between studies.
- Intangible costs and process characteristics (respect, autonomy, information) are usually left out (Health Economics, Ch 13, p 208; Ch 14, pp 219-220).
- Silent on **equity** — extended cost-effectiveness analysis (ECEA) and distributional cost-effectiveness analysis (DCEA) were developed to quantify how costs and effects fall across population subgroups (Oxford Textbook of Public Health 7e, Ch 49, pp 1844-1845).
- Park notes plainly that "even cost-effective analysis is not possible in many cases" (Park 28e, Ch 23, p 1028).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The cost-effectiveness plane** — two axes crossing at the origin. Vertical axis "Difference in costs (– below, + above)"; horizontal axis "Difference in effects (– left, + right)". Quadrant labels: top-left "Activity dominated", top-right "Cost-effectiveness ratio — trade-off", bottom-left "Cost-effectiveness ratio — trade-off", bottom-right "Activity dominates". Add a dotted diagonal through the origin labelled "threshold / ceiling ratio".



Choosing the appropriate technique of economic evaluation

High-yield exam pointers

- **Four techniques, one difference** — how the outcome is valued: CMA (outcomes equal), CEA (natural units), CUA (QALY/DALY), CBA (money).
- **Formulas:** $ACER = \text{total cost} \div \text{total effect}$; $ICER = (\text{Cost A} - \text{Cost B}) \div (\text{Effect A} - \text{Effect B})$; $MCER = \text{change in cost} \div \text{change in effect}$. BCR and NPV belong to CBA, not CEA.
- **Discount rate 3 per cent** per year on both costs and effects; always follow with **sensitivity analysis** and report the ratio as a range.
- **WHO threshold rule:** costing less than one per-capita GDP per DALY averted = "highly cost-effective" (Commission on Macroeconomics and Health, 2001). NICE ceiling £20,000–30,000 per QALY.

- ▶ **Plane quadrants:** south-east = dominates (adopt); north-west = dominated (reject); north-east and south-west = trade-off.
- ▶ **DALY = years of life lost (YLL) + years lived with disability (YLD);** one DALY = one year of healthy life lost; one QALY = one year lived in perfect health (Park 28e, Ch 2, p 38).
- ▶ **Indian example:** Narangwal, rural Punjab — cost per older-child death averted US\$30.65 (health care) vs US\$72 (nutrition) vs US\$101 (both).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the retrieved corpus carries no India-specific cost-effectiveness threshold and no account of Health Technology Assessment in India (HTAIn), so the thresholds above are the World Health Organization per-capita gross domestic product rule and the country ceilings recorded in the textbooks; the Narangwal figures are historical United States dollars and illustrate the method, not current costs.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, HTAIn (Department of Health Research, MoHFW), National Health Authority, Indian Reference Case 2023, NICE

Programme and policy updates (India)

- ▶ **India now has a standing HTA agency, and it pays for itself** — Health Technology Assessment in India (HTAIn) was institutionalised in **2017 under the Department of Health Research, MoHFW**, to make public spending choices explicit [PMID 37694178]. An independent evaluation of three HTAIn assessments put the return on investment at **9:1**, rising to **71:1** with full implementation of the recommendations (5:1 to 40:1 across different interventions) [PMID 40113231].
- ▶ Within the **National Health Authority**, a **Health Financing and Technology Assessment (HeFTA) unit** applies cost-effectiveness evidence to **inclusions and exclusions in the PM-JAY health benefit package** and to **standard treatment guidelines**, with reported cost savings to the scheme; the authors recommend that future strategic-purchasing reforms integrate HTA with **value-based pricing** [PMID 38715207].
- ▶ **Screening only pays if treatment follows it.** Population-based screening for diabetes and hypertension in India was **not** cost-effective at a threshold of $1 \times$ per-capita GDP under existing patterns of care-seeking; 3- or 5-yearly screening of those aged 30–65 became cost-effective once 20% of newly diagnosed uncomplicated cases were treated at Health and Wellness Centres, and annual screening became **cost-saving** if that share rose to 70% (from the existing 4% at sub-centres and PHCs) [PMID 34774219]. Those centres have since been **renamed Ayushman Arogya Mandir** [PMID 40341602].

Current national data

- ▶ **First national willingness-to-pay estimate for India.** A six-state survey of 5,460 respondents (21,640 observations, time trade-off plus willingness-to-pay, discounted at 3%) put mean weighted WTP per QALY at ₹2,12,307 (≈US\$2,535), ranging ₹1,70,414–₹2,58,985 across health

states — equivalent to **1 to 1.52 times per-capita GDP**, and proposed as India's cost-effectiveness threshold [PMID 41176029].

- **Audit of what HTAIn has actually produced:** of 33 published HTAIn studies, 45.5% evaluated screening and 33.3% evaluated devices; every study addressed economic efficiency but only 45.5% addressed efficacy, effectiveness and efficiency together, and **equity (25.7%) and heterogeneity (24.3%) were the worst-performing domains** against the Indian Reference Case. The intervention was judged cost-effective in **28 of 33 (84.8%)** — 7 of them cost-saving — and every study that used a threshold used **1× per-capita GDP** [PMID 41316276]. A parallel appraisal of 26 studies from ten regional resource centres rated 14% excellent and 50% good quality, with the same weak spots in uncertainty analysis, equity and budget impact [PMID 40913689].

Elimination, surveillance and disease status

- **Single-dose HPV vaccination is cost-effective in every Indian state.** Vaccinating 10-year-old girls at 90% coverage gave an ICER of **US\$406 (₹30,000) per DALY averted** versus no vaccination (state range US\$67–593), below an opportunity-cost threshold of 30% of per-capita GDP in all states and far below the WHO 100%-of-GDP figure (US\$1,995). Two-dose versus no vaccination was US\$1,404, and two-dose versus single-dose at least US\$2,282 per DALY averted [PMID 37931940].
- **Intensified Mission Indradhanush was cost-effective as well as impactful.** Across 40 sampled districts the campaign cost an incremental **US\$327.63 per DALY averted**, US\$6.21 per additional dose delivered, **US\$82.99 per zero-dose child reached**, US\$360.72 per life-year saved and US\$9,701 per under-5 death averted, with a **90% probability of being cost-effective** at 1× per-capita GDP per DALY averted [PMID 38590052].

Guideline and legislative updates

- **The Indian Reference Case (2023) is now the national standard for economic evaluation — 12 recommendations across 11 principles** (decision problem, comparator, perspective, source of effectiveness evidence, measurement of costs, health outcomes, time horizon, discounting, heterogeneity, uncertainty analysis, equity) plus presentation of results. It requires an **abridged societal perspective**, with results also reported from the healthcare payer perspective because India is a multi-payer system with high out-of-pocket spending, and a **3% discount rate** on both costs and outcomes, varied 0–5% in sensitivity analysis [PMID 37694178]. Companion national **budget impact analysis** guidelines require a payer perspective under both multi-payer and single-payer scenarios, a **1–4 year horizon**, top-down estimation of the eligible population, and **no discounting** — inflation adjustment only, because BIA reports actual affordability rather than value for money [PMID 34184237].

Global and WHO developments

- **Empirically derived thresholds sit far below the GDP-per-capita rule the textbooks quote.** Opportunity-cost thresholds calculated for **174 countries** from growth in health expenditure and life expectancy fell below **1× GDP per capita in 168 of the 174 countries**, and below **0.5× GDP per capita** in 96% of low-income and 76% of lower-middle-income countries — lower than the ceilings many countries currently apply [PMID 37202020].
- **NICE has replaced its end-of-life premium with a disease-severity modifier** in its revised methods manual, weighting QALYs by severity rather than by proximity to death, and rejected

the other modifiers it consulted on [PMID 37268059]. Applied to 138 appraisal decisions published 2020–2022, 39% would qualify for a severity multiplier, overwhelmingly oncology indications (4.04 times more likely to qualify than non-oncology); but the modifier is harder to earn and less generous, only a fraction of appraisals reaching the top effective **£51,000 per QALY**, and indications previously accepted at around £50,000 per QALY would now need an ICER below roughly **£36,000** [PMID 38447743].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 23 Health Planning and Management, pp 1027-1029; Ch 2 Concept of Health and Disease, p 38); Health Economics (Ch 13, pp 200-211; Ch 14 Counting the costs, pp 214-228; Ch 15, pp 230-231; Ch 16 Economic evaluation and decision-making, pp 247-257); Reinke — Health Planning for Effective Management (Ch 20 Cost-Benefit and Cost-Effectiveness Analysis of Health Programs, pp 274-288); Oxford Textbook of Public Health 7e (Ch 49 Economic appraisal in public healthcare, pp 1813-1851). Threshold rule from the WHO Commission on Macroeconomics and Health, 2001.

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Digital health initiatives in India

Asked in: Paper IV 08-Jan-2025 — RGUHS MD Community Medicine

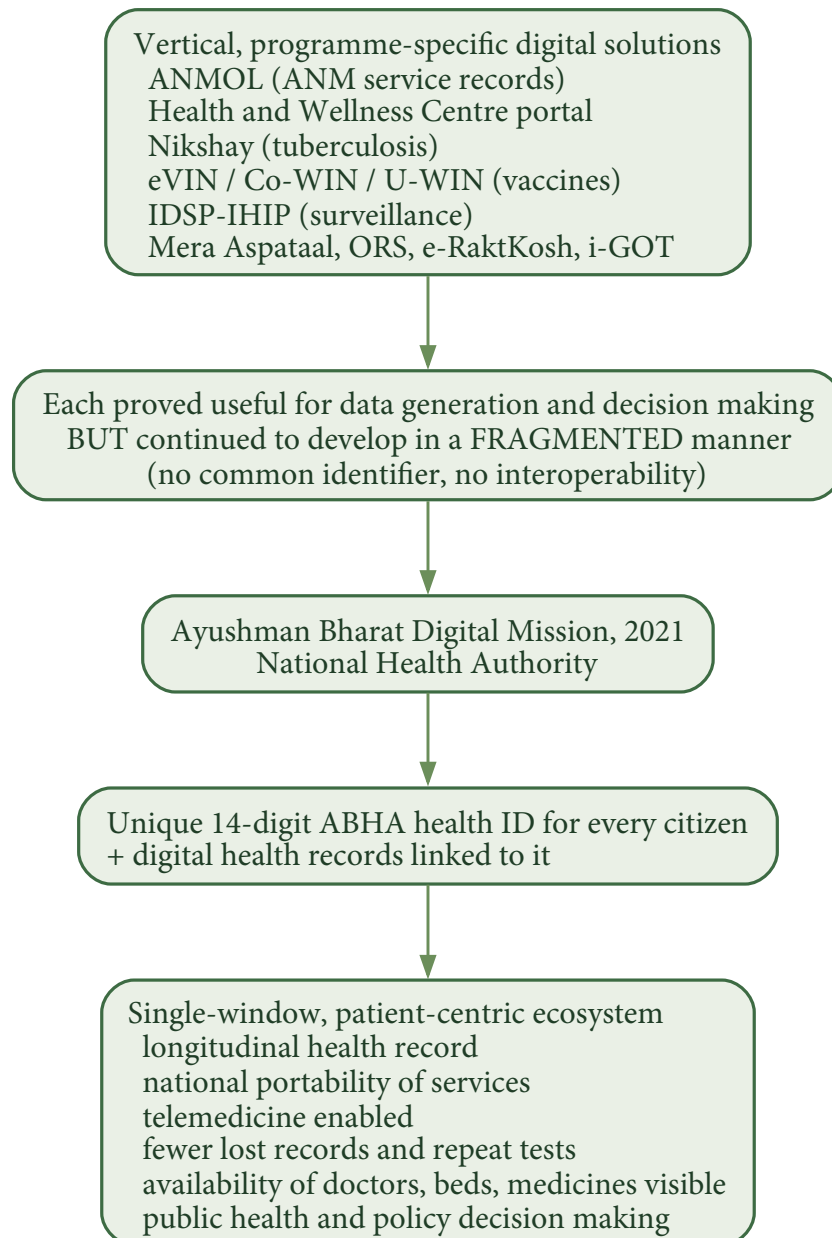
Definition — Digital health (eHealth) is "the cost-effective and secure use of information and communication technologies (ICT) in support of health and health-related fields, including healthcare services, health surveillance, health literature, and health education, knowledge and research" — World Health Assembly resolution WHA58.28 of 2005 (Oxford Textbook of Public Health 7e, Ch 25, p 961). Its clinical arm, **telemedicine**, is "the delivery of health care services, where distance is a critical factor, by all health care professionals using information and communication technologies for the exchange of valid information, for diagnosis, treatment and prevention of disease and injuries, research and evaluation" — WHO 2002 (Suryakantha, Ch 50, p 927). To *critically appraise* these initiatives is to weigh them against the standard elements used to evaluate any health service — **relevance, adequacy, accessibility, acceptability, effectiveness, efficiency and impact** (Park 28e, Ch 23, pp 1037–1038).

The major digital health initiatives in India

Initiative	Launched / owner	What it does
Ayushman Bharat Digital Mission (ABDM) (<i>formerly National Digital Health Mission</i>)	September 2021; National Health Authority is the implementing agency	Citizens create an Ayushman Bharat Health Account (ABHA) — a unique 14-digit randomly generated health ID — to which digital health records are linked, creating a longitudinal record, improving clinical decision-making, enabling telemedicine and giving national portability of health services (Sunder Lal, Ch 8, p 535; Ch 17, p 763)
eSanjeevani — National Telemedicine Service	Ministry of Health and Family Welfare	Two variants: eSanjeevani AB-HWC (doctor-to-doctor) connecting Health and Wellness Centres to specialists and superspecialists, and eSanjeevani OPD (patient-to-doctor), upgraded during the COVID-19 pandemic to give free consultation to citizens at home (Sunder Lal, Ch 17, p 748)
HMIS 2.0 (Health Management Information System)	December 2020, with WHO technical support	Real-time data entry and monitoring, works on mobile/tablet, integrates with other portals through an application programming interface, interactive dashboards, GIS layer down to village level, Local Government Directory compliance. Captures ~400 service-delivery elements monthly and ~700 infrastructure and human-resource elements annually ; daily data entry from April 2020 (Sunder Lal, Ch 8, p 532; Ch 20, p 786)

Initiative	Launched / owner	What it does
RCH portal / Mother and Child Tracking System (MCTS) with the Integrated RCH Register v2.0	Register developed 2021, launched June 2022; a step towards the Integrated Health Information Platform (IHIP)	Name-based identification and tracking of each beneficiary through the reproductive lifecycle; consolidates family planning, maternal health, child health, immunisation and mortality records into one register, ending duplicate entry by the ANM; feeds e-Janma, ANMOL, Kilkari and direct benefit transfer of Janani Suraksha Yojana and ASHA payments. Records validated by the Mother and Child Tracking Facilitation Centre at NIHFWS (Sunder Lal, Ch 8, pp 532, 535)
ANMOL (ANM Online)	7 April 2016	Tablet application letting the auxiliary nurse midwife enter and update beneficiary service records in real time ; operational in 24 States/UTs and used by more than 97,883 ANMs (Park 28e, Ch 7, p 541)
e-VIN (Electronic Vaccine Intelligence Network)	2015, MoHFW	Real-time visibility of vaccine stock, supply and storage temperature across the immunisation supply chain — where the vaccines are, whether quantities are adequate, whether storage conditions are correct. It was expanded into Co-WIN for COVID-19 beneficiary registration and tracking, and its learnings now run U-WIN for routine immunisation (Routine Immunization Manual for Medical Officers 2024, Ch 1, pp 29–30; Ch 5, p 149)
Nikshay	Central TB Division with the National Informatics Centre	Case-based, web-based TB surveillance covering notification to treatment outcome; registers private establishments, links patients with Aadhaar and the public financial management system for direct benefit transfer (Nikshay Poshan Yojana), and its dashboard drives real-time programme monitoring and district TB-score ranking (Park 28e, Ch 7, pp 510–511, 514)
IDSP with satellite–broadband network and the Media Scanning and Verification Cell	IDSP 2004; the cell from July 2008	Weekly S (syndromic) / P (probable) / L (laboratory-confirmed) reporting every Monday, a 24×7 call centre on toll-free 1075 for disease alerts, and electronic scanning of media for early warning signals — now migrating onto IHIP (Sunder Lal, Ch 7, pp 306–308)
Kilkari	MoHFW	One system-generated weekly voice call of time-sensitive audio messages on pregnancy and child health to the mobile phones of pregnant women, mothers of young children and their families, keyed to the month of pregnancy or the child's age, available in regional dialects (Park 28e, Ch 7, pp 540–541)

Initiative	Launched / owner	What it does
Other citizen- and provider-facing platforms	MoHFW	Mera Aspataal (patient-experience feedback), Online Registration System (ORS) for appointments, e-RaktKosh (blood banks), i-GOT (training), Anemia Mukh Bharat dashboard, and GIS-enabled facility mapping by the National Informatics Centre, disseminated through the Central Bureau of Health Intelligence website, launched March 2016 (Sunder Lal, Ch 8, pp 534–535)



From fragmented programme-specific applications to a single ABDM digital ecosystem

Critical appraisal against the elements of evaluation

Element (Park 28e, Ch 23, pp 1037–1038)	Credit — what has been achieved	Debit — where it falls short
Relevance (does the need genuinely exist?)	Squarely addresses the oldest weakness of the Indian health system — civil registration has been unreliable, the data grossly deficient in accuracy, timeliness and completeness (Park 28e, Ch 21, p 996) — and a real-time platform is the only way to serve a population of this size	A digital record cannot substitute for the missing manpower, drugs and infrastructure that actually limit rural care; the routine returns are themselves "incomplete and coverage of population may not be total, particularly in urban areas" (Sunder Lal, Ch 8, p 534), and digitising an incomplete denominator does not complete it
Adequacy (is the effort proportionate to the need?)	HMIS 2.0 captures ~400 monthly service elements; more than 97,883 ANMs use ANMOL; over one lakh Health and Wellness Centres are telemedicine-enabled (Sunder Lal, Ch 8, pp 534–535)	Coverage is still partial — ANMOL is operational in only 24 States/UTs (Park 28e, Ch 7, p 541); scaling is the documented failure point, few digital programmes having scaled successfully (Oxford 7e, Ch 4, p 248)
Accessibility (physical, economic, cultural barriers)	eSanjeevani OPD is free of cost and delivered into the home; the hub-and-spoke design brings a specialist opinion to the sub-centre (Sunder Lal, Ch 17, pp 748, 751)	The digital divide is the central charge — India and sub-Saharan Africa account for 42% of the world's unconnected , with more than 60% of their populations not yet on the internet (Oxford 7e, Ch 25, p 961); electricity and internet access are inconsistent and mobile data expensive (Oxford 7e, Ch 4, p 248)
Acceptability	Voice-based, dialect-friendly services such as Kilkari bypass literacy; patients gain access to their own medical history	Lack of confidence among patients about the outcome, technophobia among handlers, and cultural and generational barriers to adoption (Park Plus, Ch 24, p 376; Suryakantha, Ch 50, p 932)
Effectiveness (attainment of stated objectives)	Nikshay made TB notification and outcome auditable across the public and private sectors; e-VIN gave stock and temperature visibility; Co-WIN carried the COVID-19 vaccination drive	Evidence for information-system interventions including electronic health records is " less well established " in middle-income countries (Oxford 7e, Ch 4, p 248), and the Indian system remains data-driven rather than action-driven — reports are "seldom read, least understood and data collected locally are seldom used for planning and evaluation" (Sunder Lal, Ch 8, p 534)
Efficiency (resources used per unit of effect)	Removes duplicate registers, prevents repeat investigations and loss of records, and cuts travel and waiting time	Cost of telemedicine is high, infrastructure underdeveloped, technical expertise and posted staff lacking; frequent format changes force workers to prepare a separate report for each programme (Park Plus, Ch 24, p 376; Sunder Lal, Ch 8, p 534)

Element (Park 28e, Ch 23, pp 1037–1038)	Credit — what has been achieved	Debit — where it falls short
Impact (overall effect on health status)	Potential for equitable access, portability of care and evidence-based policy	No population-level impact evaluation of any Indian initiative is available in the retrieved corpus, and the digital transition itself risks worsening inequity — if patient information and notifications switch to digital systems, people without smartphones may lose access altogether (Oxford 7e, Ch 19, p 762)

Do they satisfy the WHO requirements of a health information system?

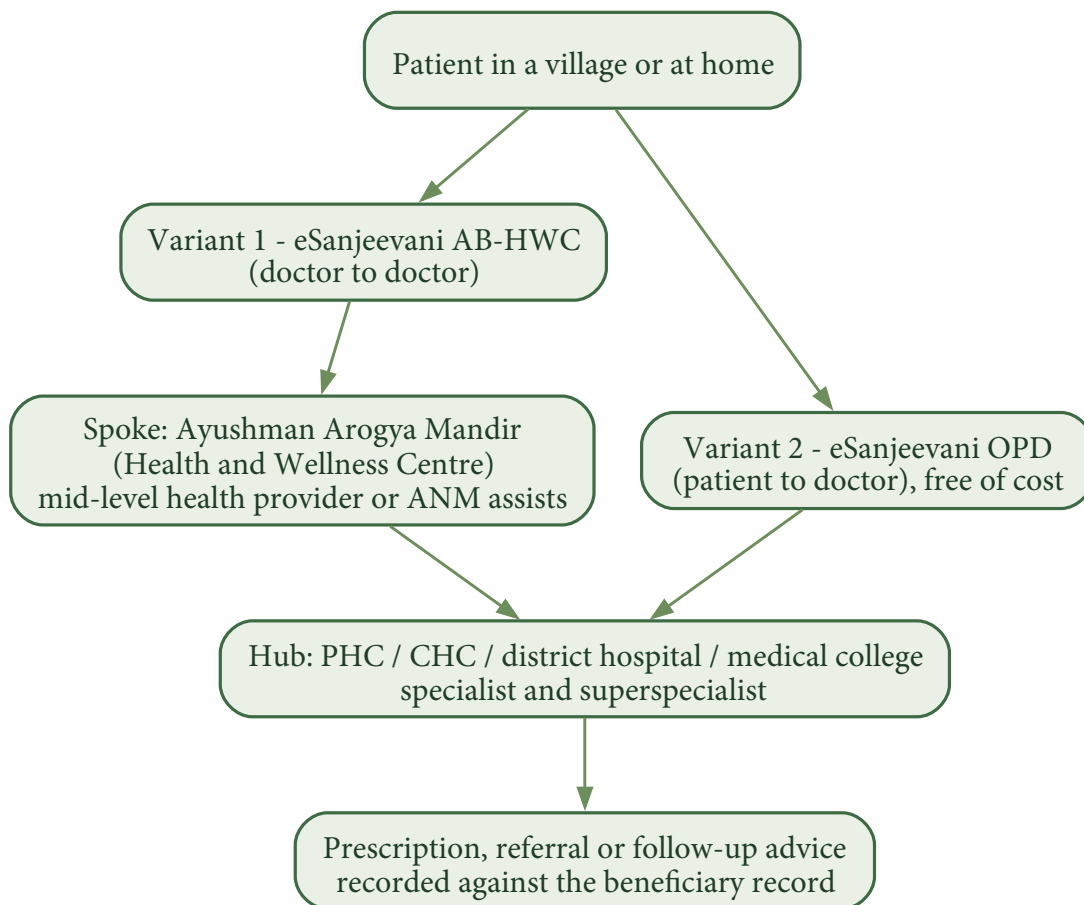
A WHO Expert Committee laid down six requirements (Park 28e, Ch 21, p 995); the Indian platforms meet them unevenly.

WHO requirement	Verdict
Should be population-based	Partly met — ABHA and the RCH portal are individual-name-based and universal in design, but enrolment is voluntary and incomplete
Should avoid unnecessary agglomeration of data	Not met — about 400 monthly plus 700 annual data elements, with a recognised tendency to over-report and inflate figures (Sunder Lal, Ch 8, p 534)
Should be problem-oriented	Met for disease-specific tools (Nikshay, e-VIN); weak for the generic monthly HMIS return
Should employ functional and operational terms	Met — standardised facility identifiers using the National Identification Number and Local Government Directory coding
Should express information briefly and imaginatively	Met — interactive dashboards, GIS dynamic maps, grading of CHCs and PHCs, automatic alerts for sensitive data items (Sunder Lal, Ch 8, pp 534–535)
Should make provision for feedback of data	Weakest link — "feedback to health workers who collect huge or voluminous data is not available and consequently there is no sustained motivation" (Sunder Lal, Ch 8, p 534)

Constraints and barriers documented for Indian conditions

- **Infrastructure** — underdeveloped connectivity, poor bandwidth, unreliable power supply and poor data-communication infrastructure (Park Plus, Ch 24, p 376; Suryakantha, Ch 50, p 932).
- **Cost** — the cost of telemedicine is too high, and scaling any digital intervention needs fresh investment (Park Plus, Ch 24, p 376; Oxford 7e, Ch 4, p 248).
- **Human resources** — no doctor posted at the telemedicine unit, lack of technical expertise, need for staff training, technophobia among handlers.
- **Literacy and language** — literacy and language remain barriers, only a small minority being versed in English (Suryakantha, Ch 50, p 932).
- **Governance and law** — medico-legal concerns, absence of a proper governing body to frame guidelines, and the scope for malpractice (Suryakantha, Ch 50, p 932).

- **Data integrity and security** — information may be lost through a software glitch or hardware failure; transmission must comply with recognised standards of security and encryption, secondary use of data must be limited, and geolocated results must be suppressed to prevent indirect identification of users (Oxford 7e, Ch 25, p 969).
- **Equity and rights** — data ownership and the inequitable future use of Big Data are right-to-health threats in their own right, and a human-rights impact assessment should precede large data-driven rollouts (Oxford 7e, Ch 19, pp 759–760).
- **Use of the information generated** — planning and management staff still rely on "gut feeling" for ad hoc decisions rather than the database (Sunder Lal, Ch 8, p 534).



eSanjeevani teleconsultation pathway and its two service variants

Way forward

- **Close the feedback loop** — the sub-centre should report performance to the panchayat, the hospital to the Rogi Kalyan Samiti and the district health mission to the Zila Parishad, and an **annual district report card on people's health** should be prepared and shared with the community (Sunder Lal, Ch 8, p 534).
- **Supervise for data quality** — supervisory visits and monthly meetings with sample re-checks of household surveys, verification of reports and spot correction, backed by community-based monitoring (Sunder Lal, Ch 8, p 534).
- **Train and re-train** — build health-information-system content into the induction and continuing education of health workers and medical officers, which is presently weak.

- ▶ **Stabilise the formats** — stop introducing a fresh reporting format with every new programme, and consolidate into the Integrated RCH Register and IHIP.
- ▶ **Protect equity** — retain a working non-digital route so that people without smartphones are not excluded from services or entitlements.
- ▶ **Regulate** — a statutory framework for teleconsultation, data privacy, informed consent and secondary use of health data.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce the same wheel by hand — a central hexagon labelled **Telemedicine** with eight spokes:

- ▶ Second opinion and complex interpretations
- ▶ Remote consultation and critical care monitoring
- ▶ Disease surveillance and programme tracking
- ▶ Continuing medical education and public awareness
- ▶ Disaster management
- ▶ Disease management
- ▶ Home care and ambulatory monitoring
- ▶ Teleremediated procedures / surgery — robotics

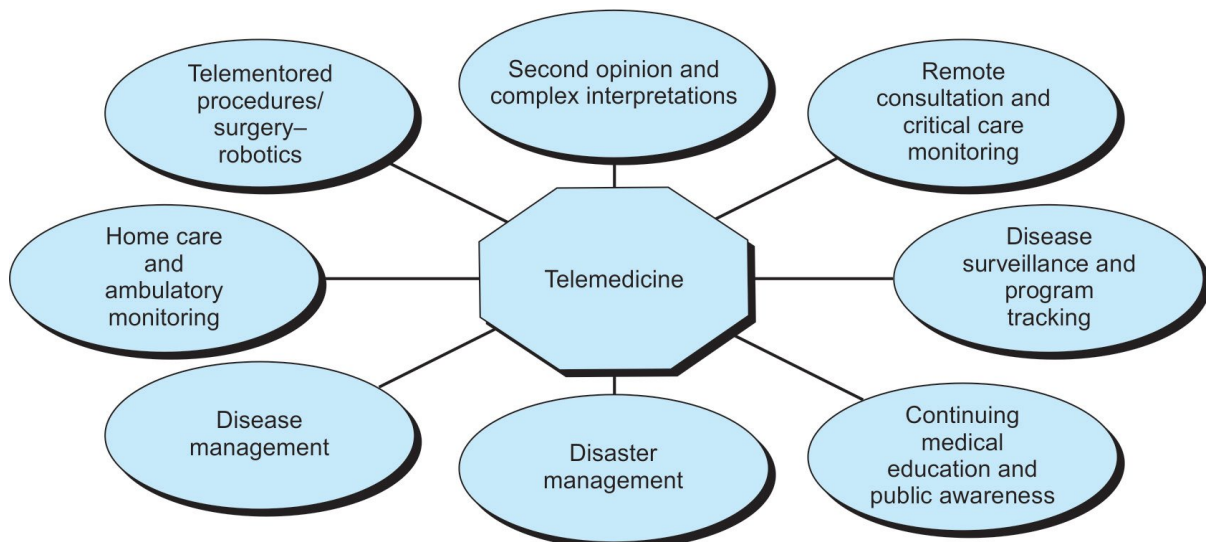


Fig 50.4 — Applications of telemedicine in public health (Suryakantha Environment Best, p.930)

High-yield exam pointers

- ▶ **Definitions to open with** — WHA58.28 (2005) for eHealth; WHO 2002 for telemedicine. *Tele* (Greek) = distance, *mederi* (Latin) = to heal.
- ▶ **The number that must be written** — ABHA is a unique 14-digit randomly generated health ID; ABDM was launched in September 2021 under the National Health Authority.
- ▶ **Appraisal skeleton (mnemonic RAAA-EEI)** — Relevance, Adequacy, Accessibility, Acceptability, Effectiveness, Efficiency, Impact.

- ▶ **The six WHO requirements of a health information system** — population-based; avoids unnecessary agglomeration of data; problem-oriented; functional and operational terms; brief and imaginative display; **provision for feedback of data**.
- ▶ **The single best critique sentence** — the system remains "**data driven rather than action driven**", and the solutions "continued to develop in a fragmented manner" before ABDM.
- ▶ **Dated facts** — e-VIN 2015 → Co-WIN (COVID-19) → U-WIN for routine immunisation; HMIS 2.0 December 2020; ANMOL 7 April 2016 (24 States/UTs, more than 97,883 ANMs); Integrated RCH Register v2.0 in 2022; IDSP 2004 with toll-free 1075.
- ▶ **Digital-divide figure** — India and sub-Saharan Africa together are **42% of the world's unconnected**, more than 60% of their populations not yet online.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e itself carries only a one-line entry on this topic — "2021 : Digital health ID project" in its history of public health (Ch 23, p 1040) — so the substantive description of the Ayushman Bharat Digital Mission and eSanjeevani above comes from Sunder Lal's Textbook of Community Medicine; two places in that book give slightly different counts of telemedicine-enabled Health and Wellness Centres (over 1.17 lakh at Ch 8, pp 534–535 and over 1.5 lakh at Ch 17, p 748), so write "over one lakh" unless you can date the figure. Ayushman Bharat Health and Wellness Centres have been renamed **Ayushman Arogya Mandirs**, so give the current name first and the former name once in parentheses.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, PIB India, MoHFW, National Health Authority, MeitY Digital Personal Data Protection Rules 2025, WHO Global Strategy on Digital Health 2020–2027

Programme and policy updates (India)

- ▶ **ABDM is a set of building blocks, not a single health ID** — and the published count differs, so say so — the Bulletin of the World Health Organization enumerates six: the 14-digit ABHA identifier, individual-controlled **personal health record** apps, the **Healthcare Professionals Registry**, the **Health Facility Registry**, a **drug registry**, and the **Unified Health Interface**, an open protocol for discovering facilities, booking appointments and teleconsulting, deliberately modelled on the Unified Payments Interface. The npj Digital Medicine account of the same architecture describes only **five key innovations**, leaving the drug registry out — so write "five to six building blocks, depending on whether the drug registry is counted", and name the components, which is what separates a Paper IV answer from a definition. Dated scale from the ABDM public dashboard, **19 October 2024: 3,41,973 facilities** in the facility registry (56.5% public) and **4,86,879 health workers** in the professionals registry (40.4% of them doctors, 71% government), with **2,074 active developers** in the sandbox, 1,545 of them private; npj puts private-sector integration at over 1,000 companies. Health-insurance claims run on the same rails through the **National Health Claims Exchange (NHCX)**, built by the National Health Authority with IRDAI [NHA 2024]. [PMID 39882498]; [PMID 39414924]

- **Scan and Share** — the one measured efficiency gain worth quoting — the National Health Authority built this QR-code, paperless OPD registration service in 2022; by 19 October 2024 it ran in 16,939 health facilities across 35 states and union territories, had generated 5,68,01,213 tokens for about 2.2 lakh users a day, and had cut registration waiting time from roughly one hour to 2–5 minutes. The critique sits in the same paper: only 1.3% of tokens (7,14,014) were issued in the private sector, state uptake is markedly uneven, and the residual barriers named are digital infrastructure, connectivity, device access and training — the same divide that limits every other platform, now being met with a **Digital Health Incentive Scheme**. [PMID 39882498]

Current national data

- **ABHA coverage** — never write the figure without its date — the peer-reviewed anchors are 56.8 crore ABHA accounts with over 35 crore records linked as of March 2024, and 41 crore records linked by 19 October 2024, when more than half of all accounts belonged to people aged 19–45. The National Health Authority subsequently announced that ABDM had crossed 90 crore ABHA accounts with over 100 crore linked health records on 30 May 2026 [PIB 2026-May]. An undated "crore" is a fault; the same number carrying its month is a strength. [PMID 39414924]; [PMID 39882498]
- **eSanjeevani** — the national programme-data analysis — over 163 million teleconsultations between the November 2019 national rollout and September 2023 across all 28 states and 8 union territories, the authors noting that few studies had until then examined the platform on nationally representative programme data. The provider-assisted **AB-HWC (doctor-to-doctor) model carried over 93%** of use; hubs grew from 397 to 14,036 and spokes from 6,868 to 108,610, with 1,08,840 of India's 1,39,389 registered AB-HWCs on the platform by 2022; women were 57–70% of users and adults aged 25–45 the largest single group, and diabetes and hypertension follow-up joined fever, cold and headache as leading reasons. Three critical caveats come from the paper itself: **five states — Tamil Nadu, Uttar Pradesh, Madhya Pradesh, Karnataka and Andhra Pradesh — supplied about 70.6%** of all consultations, so the national total masks sharp geographic concentration; the mean *active* doctor–patient interaction was only about **1 minute 15 seconds** in September 2023; and the aggregate data cannot show how gender intersects with caste, rurality or socioeconomic status. MoHFW's dashboard puts the cumulative Ayushman Arogya Mandir total at **41.93 crore by 30 November 2025**. [PMID 41158553]; [MoHFW 2025-Nov]
- **Adoption, not enrolment, is the real weak point** — three studies in a ladder — (i) in an ICMR task-force baseline survey of 5,709 **underprivileged residents** of Bhopal and Raigarh districts (November 2022 – May 2023) only 3.4% of **rural** and 2.3% of **urban** respondents held an ABHA ($\chi^2 = 5.8$, $p = 0.01$), Aadhaar was not linked to a mobile number in **40.6% rural** and **51.2% urban**, and the qualitative themes were no mobile phone, unlinked Aadhaar, unwillingness and plain unawareness; (ii) among 150 urban Bengaluru outpatients — the likeliest early adopters — only **8% had heard of ABDM** (against 88.7% aware of Aarogya Setu and 74% of eSanjeevani), just one had created an ABHA, only **24.7% knew of any health-data privacy law**, and the stated fears were data safety (60%), privacy (58.7%) and misuse by insurers (18%); (iii) at three tertiary hospitals in Agra, **1,29,007 of 5,37,278 OPD visits** ended in an ABHA registration over eight months to April 2025 — an adoption rate of **23.96%** — and among 425 attendees digital health literacy was only moderate (DHLI 2.94 ± 0.72 , males higher, education the strongest correlate),

over **80% found registration easier** but only **55% were comfortable about data privacy**, with barriers of preferring in-person care (85.6%), data-plan cost (81.9%), difficulty using the app (65.6%), language (64.0%) and reluctance to share an OTP (60.0%), and **88% asking for training**. This is the sharpest evidence for the *accessibility* and *acceptability* rows of the appraisal table. [PMID 41098267]; [PMID 39629395]; [PMID 41787376]

Elimination, surveillance and disease status

- **National Sickle Cell Anaemia Elimination Mission (2023) — a digital-first elimination programme** — it targets elimination as a public health problem by **2047**, with universal screening of **7 crore (70 million) people up to 40 years of age in states of high prevalence**, delivered through the existing RBSK and PMSMA platforms. Its architecture is deliberately digital: an **offline-capable screening application** letting frontline workers register beneficiaries and record solubility tests in low-connectivity tribal areas and sync when the network returns, a facility-level **Sickle Cell Treatment Module** recording SNOMED-CT-coded prescriptions, investigations, transfusions and vaccinations, a dashboard tracking case counts, hydroxyurea therapy and transfusion frequency, role-based access with device-integrity checks, and **mandatory ABHA generation** so every screened person gains a portable longitudinal record. Flagged risks are sustained workforce training, last-mile rural internet, the "lost to follow-up" flag at 3 months of inactivity being mistaken for a clinical discharge instead of a field trigger for the ASHA, and the socio-cultural sensitivity of pre-marital screening and genetic counselling. [PMID 41630077]
- **U-WIN nationwide rollout completed (November 2024)** — the Co-WIN successor now digitises **all** Universal Immunisation Programme vaccination for pregnant women and children from birth to 16 years against 12 vaccine-preventable diseases, against an annual target of about 2.9 crore pregnant women and 2.6 crore infants. It issues a **QR-based digital vaccination certificate** in place of the paper card, permits "anywhere" vaccination at any centre in the country, and auto-generates ABHA and child-ABHA — so the e-VIN → Co-WIN → U-WIN line in the body now ends in a fully national electronic immunisation registry, not a pilot. [MoHFW 2024-Nov]

Guideline and legislative updates

- **Digital Personal Data Protection Act 2023, with the DPDP Rules notified 13 November 2025** — the Act runs to **44 sections in 9 chapters** and, unlike the 2019 Bill, defines **no separate "sensitive personal data" category**, so health records sit under ordinary personal data. A hospital becomes a **data fiduciary** owing specific, informed and revocable consent, a notice in the patient's preferred language, and — under Rule 6 — encryption, access control and access logs kept for at least a year, with immediate breach intimation to the affected people and to the **Data Protection Board of India**; significant data fiduciaries additionally need a Data Protection Officer, an independent auditor and data protection impact assessments. Commencement is staged: the Rules were notified alongside the Act's own commencement notification on **13 November 2025**, registration of **consent managers** opens twelve months later on **13 November 2026**, and the substantive fiduciary duties phase in over eighteen months [MeitY 2025-Nov]. The documented practice problems are a **double consent burden** alongside the Mental Healthcare Act 2017, a right of erasure that collides with retaining drug-allergy and suicide-risk records,

notice fatigue among poorly educated, elderly or very ill patients, undefined thresholds for "reasonable" safeguards, and broad government exemption under s.17(2) while private practitioners carry the full penalty. This closes the legislative gap the body flags. [PMID 40949048]

- **Telemedicine Practice Guidelines (March 2020) and Telepsychiatry Operational Guidelines (May 2020) — get the two kinds of restriction right** — MoHFW issued the first; the Indian Psychiatric Society and the Telemedicine Society of India, with NIMHANS, issued the second. Drugs are sorted into **List O** (over-the-counter, e.g. nicotine replacement), **List A** (relatively safe; a video consultation is required for the first prescription), **List B** (add-on medicines for optimising treatment — allowed by teleconsultation but **not at the first visit**) and **List C** (**prohibited outright**, under the NDPS Act 1985 and Schedule X of the Drugs and Cosmetics Act 1940). Only List C is an absolute bar — diazepam, lorazepam, nitrazepam, chlordiazepoxide, methadone, buprenorphine, tramadol, methylphenidate and amphetamines — while **clonazepam and clobazam are List A**, added by an amendment on **4 April 2020**. The published critique therefore runs two ways: addiction care loses **opioid substitution altogether** (no buprenorphine, methadone or tramadol), and separately loses the **first visit** for anti-craving agents (naltrexone, acamprosate, disulfiram, baclofen, topiramate) and for **bupropion and varenicline** in tobacco cessation, none of which is addictive or legally controlled. Alcohol-withdrawal management is the sharpest inconsistency — **lorazepam, diazepam and chlordiazepoxide are barred** while two other benzodiazepines are freely prescribable, with no stated rationale — which leads the authors to propose abolishing List B and moving the non-abusable agents into List A. Field data support the governance worry: among **78 health workers** at rural Karnataka PHCs on a hub-and-spoke telemedicine service, 81% called it a promising way forward, 67% agreed it eased specialist access and 64% that it cut travel cost, but **59% feared a breach of patient confidentiality** (only 15% thought it safe), 46% feared loss of jobs and 20% were nervous using it. [PMID 41377753]; [PMID 41648971]

Global and WHO developments

- **WHA78(22), adopted at the seventh plenary meeting on 27 May 2025** — the Seventy-eighth World Health Assembly extended the **Global Strategy on Digital Health 2020–2025 by two years to 2027** and mandated a successor strategy for 2028–2033. India's own contribution to the global architecture is the **Global Initiative on Digital Health (GIDH)**, launched by WHO on **19 August 2023** at the G20 Health Ministers' Meeting in Gandhinagar under India's G20 presidency — a WHO-managed network matching country digital-health needs to donor resources and quality-assured products, and a named deliverable worth citing in any global-health or health-diplomacy answer. [WHO 2025-May]; [WHO 2023-Aug]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India; Ch 21 — Health Information and Basic Medical Statistics; Ch 23 — Health Planning and Management); Sunder Lal — Textbook of Community Medicine (Ch 7 — Epidemiology; Ch 8 — Epidemiology of Communicable and Non-Communicable Diseases and Related National Health Programmes; Ch 17 — Healthcare of the Community; Ch 20 — Recent Advances); Oxford Textbook of Public Health 7e (Ch 4, Ch 19, Ch 25); Suryakantha — Community Medicine with Recent Advances (Ch 50 — Telemedicine in Public Health); Park Plus (Ch 24 — Miscellaneous: telemedicine and cyber medicine); Routine Immunization I Manual for Medical Officers 2024 (Ch 1 and Ch 5 — e-VIN, Co-WIN, U-WIN). Coverage note: the retrieved corpus does not describe the National Digital Health Blueprint, the Health Facility and Healthcare Professionals Registries, the Telemedicine Practice Guidelines 2020 or the Digital Personal Data Protection Act, and carries no population-level impact evaluation of an Indian digital health initiative — these gaps are stated rather than filled.

Recent-advances sources: 39414924 — India's evolving digital health strategy, npj Digital Medicine 2024, doi:[10.1038/s41746-024-01279-2](https://doi.org/10.1038/s41746-024-01279-2); 39882498 — Scan and share to register outpatients, India, Bulletin of the World Health Organization 2025, doi:[10.2471/BLT.24.292828](https://doi.org/10.2471/BLT.24.292828); 41158553 — Adoption and utilization of India's eSanjeevani national telemedicine service, Oxford Open Digital Health 2025, doi:[10.1093/oodh/oqaf025](https://doi.org/10.1093/oodh/oqaf025); 41098267 — Bridging the healthcare gap: impact of the Ayushman Bharat Health Account (ABHA) on underprivileged populations of Bhopal and Raigarh districts, Cureus 2025, doi:[10.7759/cureus.92272](https://doi.org/10.7759/cureus.92272); 39629395 — Knowledge, attitude and practice about Ayushman Bharat Digital Mission and digital health among hospital patients, Journal of Family Medicine and Primary Care 2024, doi:[10.4103/jfmprc.jfmprc_255_24](https://doi.org/10.4103/jfmprc.jfmprc_255_24); 41787376 — Adoption, digital health literacy and patient satisfaction of the Ayushman Bharat Digital Mission among outpatient department attendees, BMC Health Services Research 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 41630077 — India's digital-first approach to sickle cell disease elimination, Archives of Public Health 2026, doi:[10.1186/s13690-025-01819-6](https://doi.org/10.1186/s13690-025-01819-6); 40949048 — The Digital Personal Data Protection Act 2023: implications for mental healthcare practice in India, Indian Journal of Psychological Medicine 2025, doi:[10.1177/02537176251370651](https://doi.org/10.1177/02537176251370651); 41377753 — Telepsychiatry in addiction treatment: evaluating India's telemedicine guidelines, Indian Journal of Psychiatry 2025, doi:[10.4103/indianjpsychiatry_634_25](https://doi.org/10.4103/indianjpsychiatry_634_25); 41648971 — Telemedicine in rural settings: a cross-sectional study on its promoters & barriers, Indian Journal of Medical Research 2025, doi:[10.25259/IJMR_2513_2025](https://doi.org/10.25259/IJMR_2513_2025). Non-journal sources: PIB / National Health Authority release on the 90-crore ABHA milestone, 30 May 2026; National Health Authority National Health Claims Exchange, built with IRDAI, 2024; MoHFW statement on completion of the nationwide U-WIN rollout, November 2024; MoHFW eSanjeevani / Ayushman Arogya Mandir dashboard, 30 November 2025; MeitY Digital Personal Data Protection Rules, notified 13 November 2025; WHO Global Initiative on Digital Health, 19 August 2023; World Health Assembly decision WHA78(22), seventh plenary meeting, 27 May 2025.



Data, information, intelligence — sources of health info

Asked in: Paper IV Nov 2017 — RGUHS MD Community Medicine

Definition — **Data** are observations of attributes or events that carry little meaning when considered alone; data as collected from health care systems or institutions are inadequate for planning. **Information** is data transformed by reducing, summarising and adjusting them for variations such as the age and sex composition of the population, so that comparisons over time and place become possible. **Intelligence** is the further transformation of information through integration and processing with experience and with perceptions based on social and political values (Park 28e, Ch 21, pp 995-996).

Data, information and intelligence — the distinguishing features

Feature	Data	Information	Intelligence
Nature	Raw observations of attributes or events	Data reduced, summarised and standardised	Information integrated with experience and social/political values
Meaning standing alone	Little or none	Meaningful for a defined population	Meaningful for a decision
Operation applied	Collection and recording	Reduction, summarisation, adjustment for age and sex composition	Integration and processing with judgement, values and context
Comparability	Not comparable across time or place	Comparable over time and place after standardisation and quality control	Comparison already interpreted against policy and feasibility
Typical example	Entries in a subcentre birth register; individual case sheets	Infant mortality rate (IMR) of a district for a stated year	Decision to shift the district's newborn care budget to facility-based care
User	Recording health worker	Epidemiologist, statistician	Decision-maker, policy-maker, planner, administrator
If not transformed further	Data not transformed into information are a waste	Information not transformed into intelligence is a waste to decision-makers, planners and health care personnel	Acted upon — the end-point of the system

Note the terminology in Indian practice: the *Central Bureau of Health Intelligence (CBHI)*, established in 1961, exists precisely to centralise the collection, compilation, analysis, evaluation and dissemination of national health statistics and to disseminate **epidemic intelligence** to States and to international bodies (Park 28e, Ch 23, p 1034).

SOURCE & COVERAGE

#	Source	What it yields	Chief limitation
1	Census	Total population and its breakdown by age and sex — the denominator for all vital rates, plus occupation, education, housing. First regular Indian census 1881, decennial thereafter; last census held March 2011 ; legal basis the Census Act, 1948 ; directed by the Census Commissioner for India (Park 28e, Ch 21, p 996)	Massive undertaking; full results not available quickly
2	Registration of vital events	Live births, deaths, foetal deaths, marriages, divorces, adoptions, legitimations and legal separations (United Nations definition). Governed by the Registration of Births and Deaths Act, 1969 (in force 1 April 1970); Aadhaar number mandatory for death registration since October 2018 (Park 28e, Ch 21, p 996)	Grossly deficient in accuracy, timeliness and completeness — illiteracy, lack of motivation, and multiple registering agencies (health, panchayat, police, revenue)
3	Sample Registration System (SRS)	Reliable birth and death rates, age-specific fertility and mortality rates, infant, under-five and adult mortality at national and State level; initiated mid-1960s , now covers the entire country (Park 28e, Ch 21, p 997)	Sample-based — designed to yield national and State-level estimates, not district-level ones
4	Notification of diseases	Incidence and distribution of specified notifiable diseases; historically the first health information sub-system established; primary purpose is prevention/control. Legally driven (Epidemic Diseases Act 1897; State Acts such as the Madras Public Health Act 1930). Internationally, cholera, plague and yellow fever are notifiable to WHO under the International Health Regulations (IHR); louse-borne typhus, relapsing fever, polio, influenza, malaria, rabies and salmonellosis are under international surveillance (Park 28e, Ch 21, p 997)	Covers only a small part of total sickness; heavy under-reporting; mild, atypical and subclinical cases escape; accuracy depends on laboratory facilities
5	Hospital records	In India, a basic and primary source of disease data — geographic source of patients, age and sex distribution of diseases, duration of stay, diagnosis distribution, association between diseases, cost of care; also bed-occupancy rate and length of stay. The 8th report of the WHO Expert Committee on Health Statistics recommended hospital statistics be an integral part of the national statistical programme (Park 28e, Ch 21, p 997)	Only the " tip of the iceberg " — covers care-seekers, not a representative sample; admission policy varies between hospitals; catchment population undefined

#	Source	What it yields	Chief limitation
6	Disease registers	A permanent record with follow-up and tabulation of frequency and survival — hence more than notification. Maintained for cancer, stroke, myocardial infarction, blindness, congenital defects, congenital rubella, tuberculosis and leprosy; yields duration of illness, case fatality and survival (Park 28e, Ch 21, pp 997-998)	Exist only for certain diseases and conditions; morbidity rates emerge only if reporting is effective and coverage national or representative
7	Record linkage	Assembly for each individual (or family) of a file of the more important health records arising at different times and places — birth, marriage, death, hospital admission and discharge, sickness absence, prophylaxis, use of social services; particularly suited to studying disease associations of aetiological significance (Park 28e, Ch 21, p 998)	Enormous data volume; applied only on a limited scale (twin studies, chronic disease epidemiology, family and genetic studies); beyond the reach of many developing countries
8	Epidemiological surveillance	Morbidity and mortality data generated by national disease control/eradication programmes — malaria, tuberculosis, leprosy, filariasis — reporting new cases and control efforts such as immunisations performed (Park 28e, Ch 21, p 998)	Disease-specific and programme-bound, not comprehensive
9	Other health service records	Records of outpatient departments, primary health centres, subcentres, polyclinics, private practitioners, mother and child health (MCH) centres, school health services, diabetic and hypertensive clinics. MCH records give birth weight, weight, height, arm circumference, immunisation and disease-specific mortality and morbidity (Park 28e, Ch 21, p 998)	Relate only to a segment of the population; kept for administrative purposes rather than for monitoring
10	Environmental health data	Statistics on air, water, harmful food additives, industrial toxicants, noise, radiation, waste disposal, and the production, combination and consumption of materials (Park 28e, Ch 21, p 998)	Collection of environmental data remains a major problem
11	Health manpower statistics	Numbers of physicians (by age, sex, speciality and place of work), dentists, pharmacists, veterinarians, hospital nurses and medical technicians — from State medical, dental and nursing councils and the Directorates of Medical Education; occupation data from the census; admissions and output of qualified persons published annually in <i>Health Information of India</i> by the Ministry of Health and Family Welfare (Park 28e, Ch 21, p 998)	Held by several separate bodies, so the number of <i>active</i> practitioners has to be estimated by reconciling sources — the Institute of Applied Manpower Research does this after allowing for mortality and fresh output

#	Source	What it yields	Chief limitation
12	Population surveys	The population-based supplement to all routine statistics — morbidity, mortality, nutritional status, disability, health expenditure, attitudes. The household is the commonest sampling unit; the National Family Health Survey (NFHS) and the District Level Household Survey (DLHS) are Indian examples (Park 28e, Ch 21, pp 998-1000)	Costly and periodic; recall and non-response errors
13	Other routine statistics related to health	<i>Demographic</i> — population density, movement, educational level; <i>economic</i> — consumption of tobacco, dietary fats and domestic coal, sales of drugs and remedies, per capita income, employment and unemployment; <i>social security schemes</i> — medical insurance data, sickness absence, sickness and disability benefit rates (Park 28e, Ch 21, p 999)	The list is expressly not comprehensive — these are examples of non-health data already put to good use by epidemiologists
14	Non-quantifiable information	Health policies, health legislation, public attitudes, programme costs, procedures and technology — the multi-disciplinary input planners and decision-makers actually need, requiring proper storage, processing and dissemination (Park 28e, Ch 21, p 999)	Neglected hitherto — health information systems have concentrated mainly on quantifiable (statistical) data

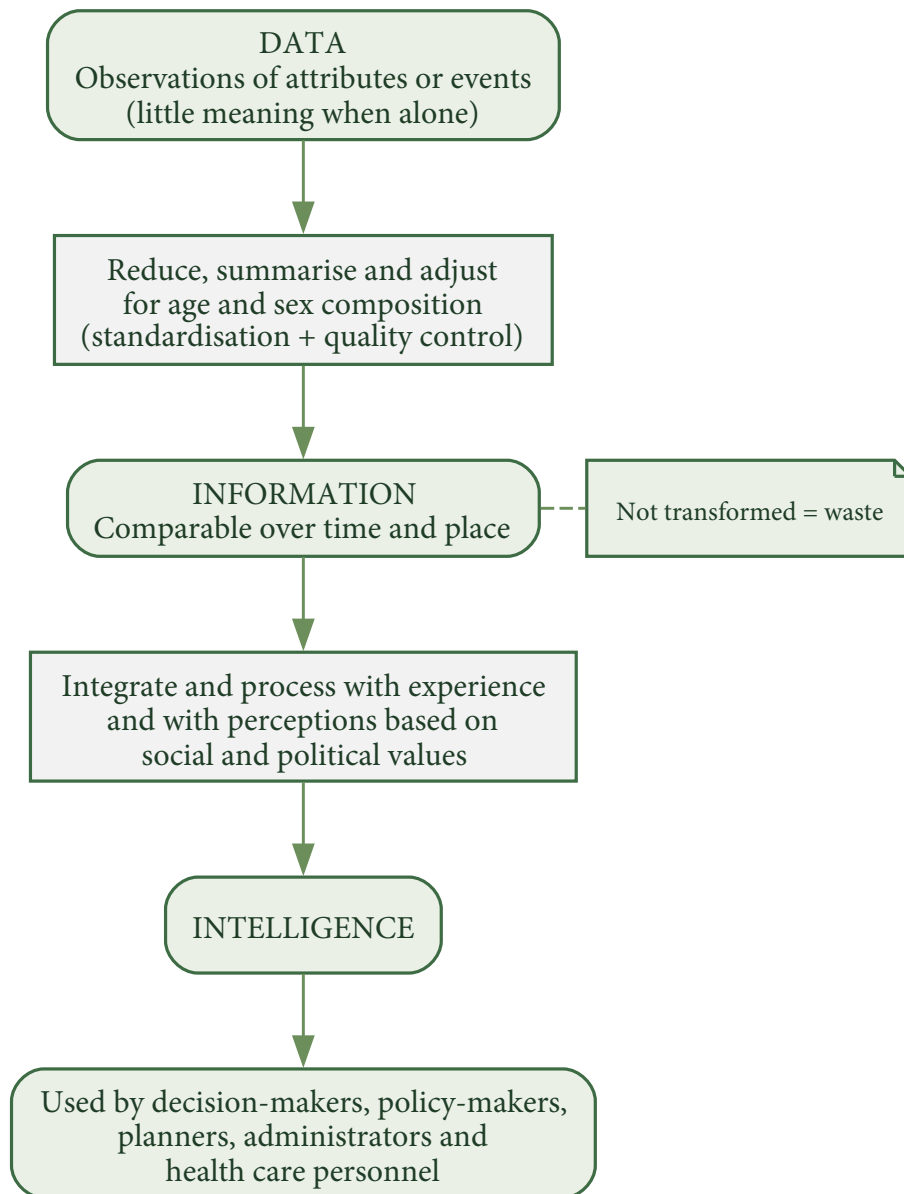
Population surveys — the four methods of data collection

Method	Best used for	Strength	Limitation
Health interview (face-to-face) survey	Subjective phenomena — perceived morbidity, disability and impairment, economic loss from illness, medical care expenditure, opinions, beliefs, attitudes, behaviour	Population-based data; the National Sample Survey Organization (NSSO) has used it for country-wide morbidity, family planning and vital events data	Morbidity data unreliable because of the interview method — hence often combined with examination or laboratory measurement
Health examination survey	Objective morbidity	Generally more valid morbidity information than an interview survey; conducted by a team of doctors, technicians and interviewers	Expensive; cannot be done on an extensive scale; obliges the team to treat those found diseased
Health records survey	Data already recorded in health services	Cheapest method	Estimates are not population-based; reliability open to question; no uniform procedure or standardisation of recording

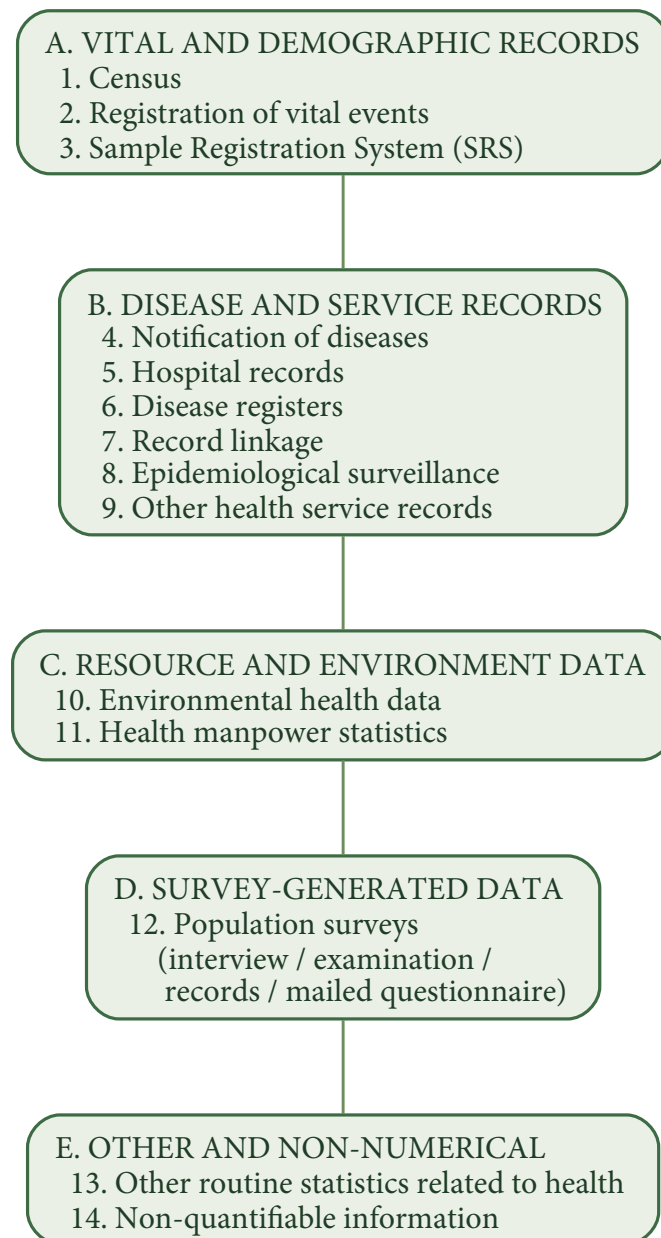
Method	Best used for	Strength	Limitation
Mailed (self-administered) questionnaire survey	Subjective phenomena at low cost	Simpler and cheaper; no interviewer needed	Requires literacy and skill of the respondent; usually a high non-response rate

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

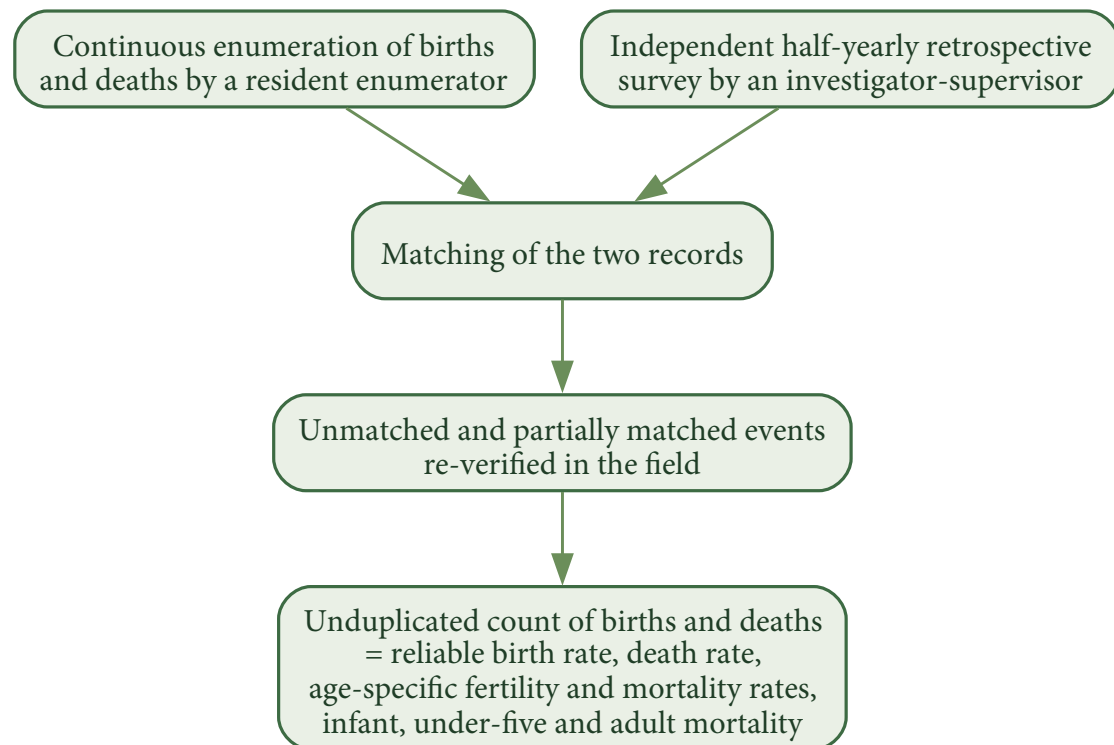
- **Labelled figure (draw and label):** the "data — information — intelligence" ladder as three ascending steps. Labels required: Step 1 "DATA" — raw observations, no meaning alone; Step 2 "INFORMATION" — reduced, summarised, age- and sex-adjusted, comparable over time and place; Step 3 "INTELLIGENCE" — integrated with experience and with social and political values, used for decision; a side arrow marked "untransformed data and information = waste", and the end arrow marked "decision-maker / policy-maker / planner / administrator".



Transformation of data into intelligence in a health information system



Classification of the sources of health information (Park 28e, Ch 21)



The Sample Registration System as a dual record system

High-yield exam pointers

- ▶ **One-line definitions to reproduce verbatim:** data = observations of attributes or events with little meaning alone; information = data reduced, summarised and adjusted for age and sex so comparisons over time and place are possible; intelligence = information integrated with experience and with perceptions based on social and political values.
- ▶ **The clinching sentence:** "Data not transformed into information, and information not transformed into intelligence, are a waste."
- ▶ **Mnemonic for the fourteen sources** — "C-R-S-N-H-D-R-E-O-E-M-P-O-N": Census, Registration of vital events, SRS, Notification, Hospital records, Disease registers, Record linkage, Epidemiological surveillance, Other health service records, Environmental health data, Manpower statistics, Population surveys, Other routine statistics, Non-quantifiable information.
- ▶ **Dated Indian anchors:** Census Act 1948, first census 1881, last census March 2011; Registration of Births and Deaths Act 1969 (in force 1 April 1970), Aadhaar mandatory for death registration since October 2018; SRS begun mid-1960s; CBHI established 1961.
- ▶ **Notifiable to WHO under the International Health Regulations:** cholera, plague, yellow fever (three only).
- ▶ **Hospital data = "tip of the iceberg"; registration is more than notification** (permanent record + follow-up + survival tabulation).
- ▶ **Survey methods:** examination survey gives the most valid morbidity data but is expensive; interview survey is best for subjective phenomena (NSSO); records survey is the cheapest but not population-based; mailed questionnaire has high non-response.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not part of the written answer: Park's enumeration is the examiner's expected list, but service-delivery data in India now flow largely through the **Health Management Information System (HMIS)** portal and the **Integrated Health Information Platform** of the Integrated Disease Surveillance Programme, with a citizen-level digital record being built under the Ayushman Bharat Digital Mission — these sit within Park's categories 8 and 9 rather than replacing them, and the census due after 2011 had not been conducted at the time of the retrieved edition.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, ICMR, Digital Personal Data Protection Act 2023, NFHS-5, Sample Registration System

Programme and policy updates (India)

- ▶ **A fifteenth stream has been added in practice — the Ayushman Bharat Digital Mission (ABDM, launched 2021)**, which builds a citizen-level health-record layer plus verified registries of facilities and professionals on top of Park's fourteen sources; assessed against the health-system control-knob framework, the ABDM dashboard showed 400 million beneficiary (ABHA) registrations as on 3 June 2023 and 273 million health records linked, with over 208,000 health facilities and 190,000 health-care professionals verified, but marked inter-State variation in facility and professional registration [PMID 39437234].
- ▶ **Real-world uptake, not the architecture, is the bottleneck** — an eight-month record audit at the main tertiary care hospital in Agra (September 2024 to April 2025) found 129,007 completed ABHA registrations against 537,278 outpatient visits, a below-average adoption rate of 23.96%; the accompanying survey of 425 attendees across three tertiary hospitals in Agra found digital health literacy only moderate (mean DHLI 2.94 on a 4-point scale), weakest in the domains of evaluating reliability (2.18) and protecting privacy (2.22), and 88% of participants said they needed training or support to use ABDM [PMID 41787376].
- ▶ **The routine service-record sub-system is unevenly digitised at the periphery** — a scored assessment of the digital platforms in Chandigarh health and wellness centres rated eVIN and the RCH portal effectively implemented (70–90%), IDSP–IHIP and eSanjeevani fairly implemented (51–70%), the HMIS portal, HWC portal, CPHC–NCD app and FP–LMIS poorly implemented (25–50%) and the ANMOL app not implemented at all, with 89.2% of health workers agreeing that dual data entry had increased their workload [PMID 40854229]; the corrective model demonstrated in Uttar Pradesh was a State HMIS portal replacing manual paper-based reporting across all districts, data-validation committee meetings at block, district and State level, and a State health dashboard ranking blocks and districts on priority health and data-quality indicators — the operational meaning of converting data into information [PMID 36041830].

Current national data

- ▶ **The two national sources disagree sharply on stillbirths, which is the standard illustration of why source choice matters** — the SRS reported an average stillbirth rate of 3.8 per 1000 births

over 2016–2020, against 9.7 per 1000 (95% CI 9.2–10.1) from NFHS-5 (2019–21), a 2.6-fold difference; neonatal mortality rates from the two sources were similar, so the discrepancy is definitional (gestation documentation, and classification of miscarriage and abortion) rather than sampling, and it matters because the India Newborn Action Plan target is a stillbirth rate below 10 per 1000 by 2030 [PMID 36865608].

- ▶ **Civil registration has improved but is not yet universal, and the governing Act has been amended** — the Registration of Births and Deaths Act 1969 was amended with effect from 1 October 2023; between NFHS-4 (2015–16) and NFHS-5 (2019–21) registration of institutional births rose from 85.9% to 91.8% and of non-institutional births from 57.7% to 68.8%, an institutional delivery carried adjusted odds of registration of 2.93 (95% CI 2.80–3.07), and the poorest-to-richest registration gap narrowed from 28.6 to 14.4 percentage points [PMID 39183961].
- ▶ **Peripheral health facilities are now registration points, not merely informants** — in many States these facilities function directly as Registrars of births and deaths under the amended Act, or as critical informants to the civil registration authority; the share of institutional deliveries occurring in PHCs and CHCs rose from 30.5% to 34.7% between NFHS-4 and NFHS-5, registration of births in those facilities rose from 80.8% to 90.2% (the largest gain of any delivery setting), and such births carried 38% higher odds of registration than births in private facilities — the mechanism by which India is approaching SDG target 16.9 (legal identity for all by 2030) [PMID 41752339].

Elimination, surveillance and disease status

- ▶ **Notification has moved from paper to the Integrated Health Information Platform (IHIP) under IDSP**, with syndromic (S), presumptive (P) and laboratory (L) forms reported digitally; sensitivity is the new limiting factor — IHIP captured only 44% of laboratory-tested malaria cases from primary health centres in Raigarh, Chhattisgarh in September 2023; a quality-improvement project traced this to laboratory technicians' inadequate knowledge of IHIP data entry and poor feedback from the district surveillance unit, and refresher training plus a weekly feedback mechanism raised sensitivity to 100% within five weeks in the four selected PHCs [PMID 41629241].
- ▶ **Parallel programme reporting remains the main structural weakness of Indian surveillance** — a qualitative study in two high-burden districts of a western Indian State found IHIP–IDSP and the vector-borne disease programme (NVBDGP) running as parallel information flows with different outbreak-detection protocols and separate digital and paper workflows (daily digital S-form syndromic reporting alongside weekly paper-based larval surveys), against a background in which only 13% of sanctioned IDSP district public health laboratories were functional [PMID 42245358]; the same platform nonetheless yields usable trend data, with 1158 outbreaks of ten zoonotic diseases reported to IDSP between 2010 and 2023, Japanese encephalitis/acute encephalitis syndrome the commonest and most of the diseases reported only during the rainy season (July to September), a core One Health input [PMID 41738949].

Guideline and legislative updates

- ▶ **Health data now sit inside a statutory privacy framework** — the Digital Personal Data Protection Act 2023 defines the individual as Data Principal and the collecting entity as Data

Fiduciary, requires consent that is free, specific, informed, unconditional and unambiguous, creates the Data Protection Board of India with mandatory breach notification to the Board and to each affected Data Principal, and imposes extra duties (Data Protection Officer, independent audits, impact assessments) on Significant Data Fiduciaries; its section 7 "legitimate uses" permit processing without consent in a medical emergency or during an epidemic or other public-health threat, and the ICMR Ethical Guidelines for Application of Artificial Intelligence in Biomedical Research and Healthcare (2023) supply the health-sector detail — anonymisation, access control, human-in-the-loop oversight and representative training data [PMID 41845374].

Global and WHO developments

- **The textbook's three-disease list was superseded not by the recent amendments but by IHR (2005) itself** — the 1969 list of cholera, plague and yellow fever gave way nearly two decades ago to the Annex 2 decision instrument, under which a State Party must notify WHO of any event that may constitute a public health emergency of international concern (PHEIC), together with four events notifiable in every case irrespective of the instrument: smallpox, poliomyelitis due to wild-type poliovirus, human influenza caused by a new subtype, and SARS [WHO IHR-2005 Annex 2].
- **What the 2024 amendments actually added is a new tier of alert above the PHEIC** — the Seventy-seventh World Health Assembly adopted the amendment package on 1 June 2024 (resolution WHA77.17), creating a "pandemic emergency" for a communicable-disease event with wide multi-country spread that exceeds health-system response capacity and requires coordinated international action; the amendments entered into force on 19 September 2025 [WHO 2024-Jun].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 21 (Health Information and Basic Medical Statistics), pp 995-1000, and Ch 23 (Health Planning and Management), pp 1028, 1034; Sunder Lal — Textbook of Community Medicine, Ch 8 (health management information system, HMIS portal and Ayushman Bharat Digital Mission). Dated Indian anchors as printed in the above: Census of India 2011; Registration of Births and Deaths Act 1969; Sample Registration System (Government of India, Statistical Report 2020, published 2022); National Family Health Survey-5, 2019-21.

Recent-advances sources: 39437234 — The Ayushman Bharat Digital Mission of India: An Assessment, Health Systems & Reform 2024, DOI [10.1080/23288604.2024.2392290](https://doi.org/10.1080/23288604.2024.2392290) (abstract-only); 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission: an analytical cross-sectional study among outpatient department attendees, BMC Health Services Research 2026, DOI [10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 40854229 — Implementation Status and Usability of Digital Health Interventions Among Health Care Workers and End Users at the Primary Health Care Level in Chandigarh, North India, JMIR Formative Research 2025, DOI [10.2196/69824](https://doi.org/10.2196/69824); 36041830 — Optimizing the Health Management Information System in Uttar Pradesh, India: Implementation Insights and Key Learnings, Global Health: Science and Practice 2022, DOI [10.9745/GHSP-D-21-00632](https://doi.org/10.9745/GHSP-D-21-00632); 36865608 — Stillbirth undercount in the sample registration system and national family health survey, India, Bulletin of the World Health Organization 2023, DOI [10.2471/BLT.22.288906](https://doi.org/10.2471/BLT.22.288906); 39183961 — Impact of Institutionalisation of Births on Health Policies and Birth Registration in India, Annals of Global Health 2024, DOI [10.5334/aogh.4474](https://doi.org/10.5334/aogh.4474); 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, International Journal of Environmental Research and Public Health 2026, DOI [10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 41629241 — Quality Improvement Initiative to Improve the Sensitivity of Malaria Reporting under Integrated Health Information Platform, Raigarh, Chhattisgarh, 2023, Indian Journal of Public Health 2026, DOI [10.4103/ijph.ijph.648.25](https://doi.org/10.4103/ijph.ijph.648.25) (abstract-only); 42245358 — Informational challenges of dengue surveillance and response: a case study from India, Frontiers in Public Health 2026, DOI [10.3389/fpubh.2026.1809584](https://doi.org/10.3389/fpubh.2026.1809584); 41738949 — Trend of zoonotic diseases outbreaks in India: a secondary data analysis of Integrated Disease Surveillance Programme data, 2010-2023, Transactions of the Royal Society of Tropical Medicine and Hygiene 2026, DOI [10.1093/trstmh/trag011](https://doi.org/10.1093/trstmh/trag011) (abstract-only); 41845374 — AI ethics in Indian healthcare: a scoping review of national and international guidelines on privacy, data protection, and security, BMC Medical Ethics 2026, DOI [10.1186/s12910-026-01435-1](https://doi.org/10.1186/s12910-026-01435-1). WHO: International Health Regulations (2005), Annex 2 decision instrument and the four events notifiable in every case (Article 6); and amendments to the International Health Regulations (2005) adopted by the Seventy-seventh World Health Assembly, resolution WHA77.17, 1 June 2024, entered into force 19 September 2025. All PubMed-sourced bullets re-grounded in PubMed Central full text except the three marked abstract-only (39437234, 41629241, 41738949), which are abstract-consistent.

♦ ♦ ♦

Health information system and census

Asked in: Paper IV 18-Nov-2019 — RGUHS MD Community Medicine

Definition — A **health information system (HIS)** is defined as *a mechanism for the collection, processing, analysis and transmission of information required for organizing and operating health services, and also for research and training* (Park 28e, Ch 21, p 995). It is an integral part of the national health system and a basic tool of management. Its **primary objective** is to provide reliable, relevant, up-to-date, adequate, timely and reasonably complete information to health managers at all three levels — central, intermediate and local — and, at periodic intervals, data showing the general performance of services and the trends in demand and workload (Park 28e, Ch 21, p 995).

Requirements and components of a health information system

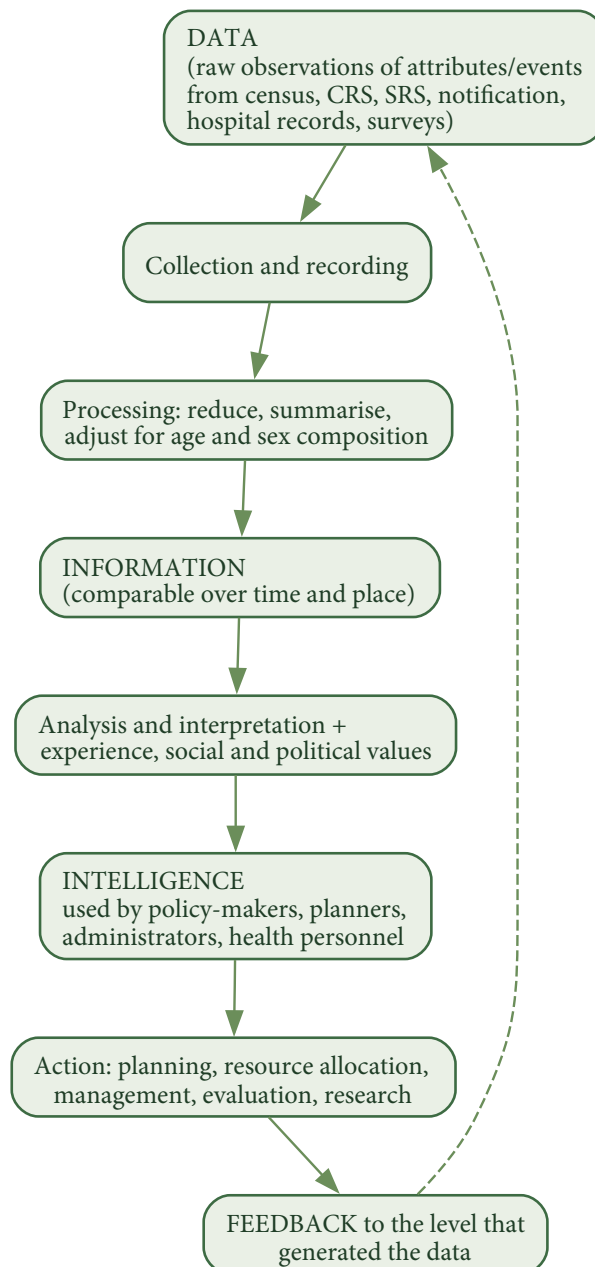
A. Requirements to be satisfied (WHO Expert Committee) (Park 28e, Ch 21, p 995):

- ▶ The system should be **population-based**.
- ▶ It should **avoid unnecessary agglomeration** of data.
- ▶ It should be **problem-oriented**.
- ▶ It should employ **functional and operational terms** (e.g. episodes of illness, treatment regimens, laboratory tests).
- ▶ It should express information **briefly and imaginatively** (e.g. tables, charts, percentages).
- ▶ It should make provision for the **feedback of data**.

B. Components (related subsystems) — a comprehensive HIS requires information and indicators on seven subjects (Park 28e, Ch 21, p 995):

#	Subsystem	Illustrative content
1	Demography and vital events	Population size and structure, births, deaths, marriages, migration
2	Environmental health statistics	Air, water, food additives, industrial toxicants, noise, waste disposal
3	Health status	Mortality, morbidity, disability, quality of life
4	Health resources	Facilities, beds, manpower
5	Utilization and non-utilization of health services	Attendance, admissions, waiting lists
6	Indices of outcome of medical care	— (Park lists this component without illustrative content)
7	Financial statistics	Cost and expenditure related to the particular objective

Planners also need **non-quantifiable information** — health policies, legislation, public attitudes, programme costs, procedures and technology — so an HIS has multidisciplinary inputs, with proper storage, processing and dissemination (Park 28e, Ch 21, p 999).



From data to intelligence — the working cycle of a health information system

(Park 28e, Ch 21, pp 995–996)

Uses of health information

The important uses to which health information may be applied are (Park 28e, Ch 21, pp 995–996):

- To **measure the health status** of the people and quantify their health problems and medical and health care needs.
- For **local, national and international comparisons** of health status — for which the data must be subjected to rigorous standardization and quality control.
- For **planning, administration and effective management** of health services and programmes.
- For **assessing whether health services are accomplishing their objectives** in terms of effectiveness and efficiency.

- For assessing the attitudes and degree of satisfaction of the beneficiaries with the health system.
- For research into particular problems of health and disease.

Data sources feeding the system

Park lists fourteen sources of health information; which one is used depends on the administrative level at which the requirement arises (Park 28e, Ch 21, pp 996–999).

#	Source	What it chiefly yields
1	Census	Denominators — population by age and sex, plus social and economic characteristics
2	Registration of vital events (civil registration, under the Registration of Births and Deaths Act 1969)	Births, deaths, fetal deaths; foundation of vital statistics
3	Sample registration system (SRS)	Dual-record system: continuous enumeration by an enumerator plus an independent half-yearly retrospective survey; now covers the whole country
4	Notification of diseases	Incidence and distribution of specified notifiable diseases; the oldest HIS subsystem
5	Hospital records	Diagnosis mix, duration of stay, bed-occupancy, cost of care
6	Disease registers	Permanent record with follow-up — duration of illness, case fatality, survival
7	Record linkage	Assembly of all key health records of one individual/family across time and place
8	Epidemiological surveillance	Morbidity and mortality generated by disease control/eradication programmes
9	Other health service records	Outpatient, primary health centre, subcentre, maternal and child health and school health records
10	Environmental health data	Air, water, food additives, industrial toxicants, noise, waste disposal
11	Health manpower statistics	Numbers of doctors, dentists, pharmacists, nurses, technicians (councils, Directorates of Medical Education)
12	Population surveys	Morbidity, nutritional status and other data that routine statistics cannot supply
13	Other routine statistics related to health	Demographic and economic data, sales of tobacco and dietary fats, social security and sickness-absence data
14	Non-quantifiable information	Policies, legislation, public attitudes, procedures, technology

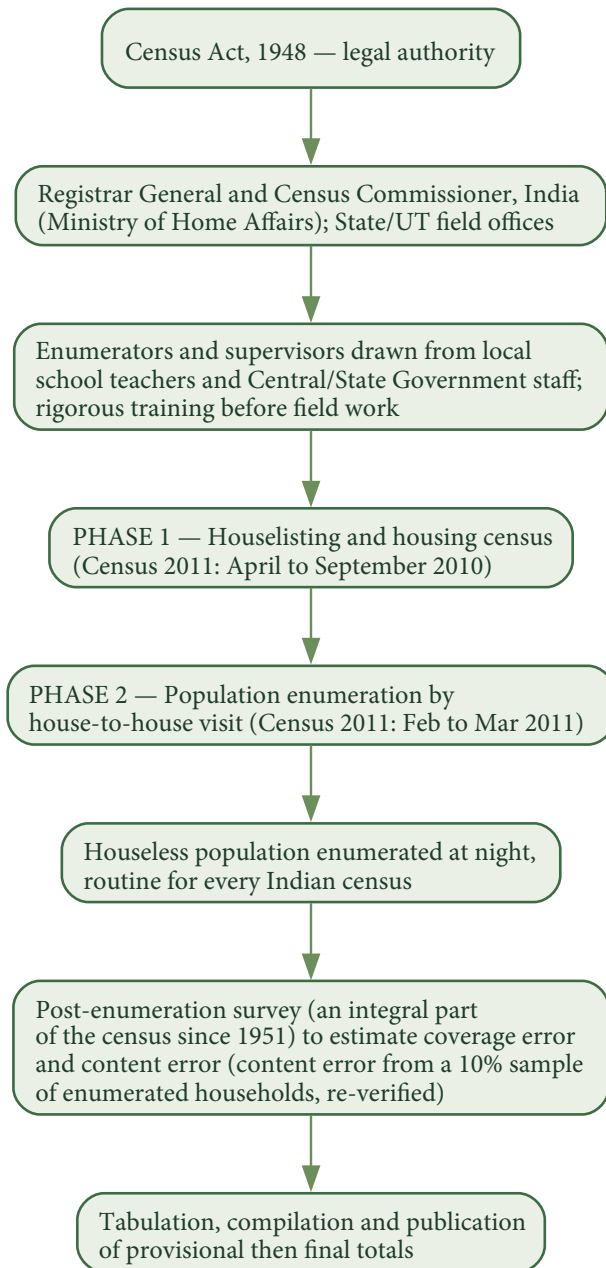
Census — definition, legal basis and conduct in India

A **census** is defined by the United Nations as *the total process of collecting, compiling and publishing demographic, economic and social data pertaining, at a specified time or times, to all persons in a country or delimited territory* (Park 28e, Ch 21, p 996). It is an **important source of health information**, and the principal source of the **denominator** — population by age and sex — without which vital rates and most health, demographic and socio-economic indicators cannot be computed (Park 28e, Ch 21, p 996).

- ▶ **Periodicity** — taken in most countries at a regular interval of 10 years; in India at the end of the first quarter of the first year of each decade, because most people are then resident in their own homes (Park 28e, Ch 21, p 996).
- ▶ **History** — earliest references are in Kautilya's *Arthashastra* (321–296 BC) and Abul Fazl's *Ain-e-Akbari* (1595–96); the non-synchronous exercise of 1865–1872 culminating in **1872** is called the first census of India, and the first **synchronous** census was **1881**. Census 2011 was the **15th in the continuous series since 1872 and the 7th since Independence** (Park Plus, Ch 19, p 228).
- ▶ **Legal basis** — the **Census Act, 1948**; the supreme officer who directs, guides and operates the census is the **Census Commissioner for India**, in the office of the Registrar General and Census Commissioner, India, under the Ministry of Home Affairs (Park 28e, Ch 21, p 996).
- ▶ **Two methods of enumeration** —

Feature	De facto	De jure
Person assigned to	Place where found at the time of enumeration	Usual place of residence , irrespective of where found
Used in	Developing countries, including India	Developed countries, e.g. USA
Merit	Simpler, quicker	Better indication of the permanent population and its socio-demographic profile
Demerit	Does not reflect the permanent population	More expensive; needs much better-trained enumerators

(Red Book — *Public Health and Preventive Medicine*, Ch 1, p 20; Park Plus, Ch 19, p 228)



Conduct of the decennial Census of India

(Census Act and Census Commissioner: Park 28e, Ch 21, p 996; two-phase field work and dates: Mahajan & Gupta 4e, Ch 25, p 479; enumerators, houseless enumeration and post-enumeration survey: Handbook of Epidemiology (Ahrens & Pigeot), Ch 38, pp 1547–1549)

Census 2011 at a glance, and its uses and limits in health

Item	Census 2011
Total population	1,210,193,422 (males 623,724,248; females 586,469,174)
Absolute decadal increase	+181 million over 2001–11
Decadal growth rate 2001–11	+17.64% (average annual exponential growth rate 1.64%) — the lowest since Independence

Item	Census 2011
Density	382 persons per sq km
Sex ratio (females per 1000 males)	943
Child sex ratio, 0–6 years	914 — the lowest since 1961, down 13 points from 927 in 2001
Literacy (age 7+, can read and write with understanding in any language)	74.04% overall; males 82.14%, females 65.46%
Highest / lowest literacy	Kerala 93.91% / Bihar 63.82%

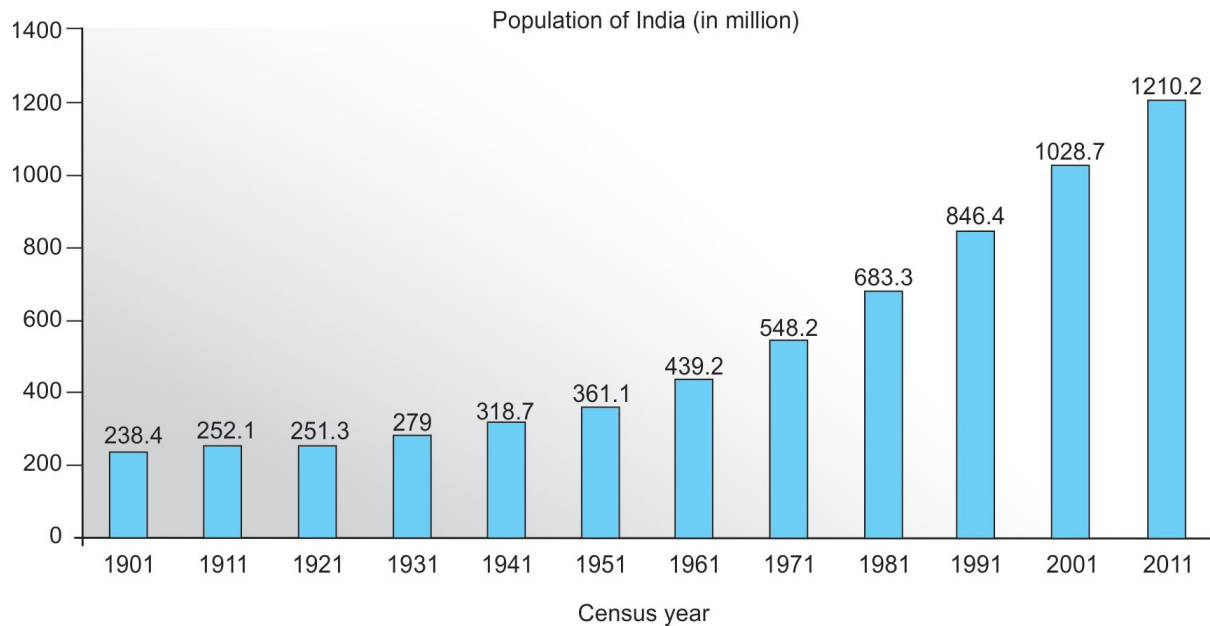
(Park Plus, Ch 19, pp 228–230; Sunder Rao & Richard, p 178; Park 28e, Ch 8, pp 582–585; Sunder Lal, Ch 9, pp 545–547)

Uses in health	Limitations
Provides the denominator for all vital rates and health indicators	Full results are not available quickly — the main drawback of the census as a data source
Planning of health services and of health manpower; delimitation of constituencies	Data are 10 years apart; intercensal years need projections
Evaluation of health interventions, e.g. family welfare programmes	Terms in the proforma may be interpreted differently by different enumerators
Baseline data on determinants of health — literacy, occupation, income, housing and amenities, disability	Shortage of trained manpower; tedious fieldwork; dishonest recording
The only source of primary data at village, town and ward level	Age heaping — age often recorded in multiples of 5 years

(Park 28e, Ch 21, p 996; Sunder Lal, Ch 9, p 560; Jugal Kishore, Ch 1, p 31; Mahajan & Gupta 4e, Ch 25, pp 478–479)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Mark 1921 as the "**big divide**" on the curve — the only decade with a fall in population, after which growth has been continuous (Park 28e, Ch 8, p 581; Suryakantha, Ch 23, p 635).



Population of India by census year, 1901–2011 (Suryakantha — Community Medicine with Recent Advances, Fig 23.3, p 636)

High-yield exam pointers

- **Definition to reproduce verbatim** — HIS = "a mechanism for the collection, processing, analysis and transmission of information required for organizing and operating health services, and also for research and training."
- **Data → Information → Intelligence** is the one-line chain that carries the "define" mark.
- **Six WHO requirements** (mnemonic P-A-P-F-B-F): Population-based, Avoid agglomeration, Problem-oriented, Functional terms, Brief and imaginative, Feedback.
- **Seven components**: demography and vital events; environmental health statistics; health status; health resources; utilization; outcome indices; financial statistics.
- **Census legal basis** = Census Act 1948; first regular (synchronous) census 1881; first census of India 1872; last census held March 2011; India uses the **de facto** method, USA the **de jure** method.
- **Census 2011 numbers**: population 1210.2 million, decadal growth 17.64%, density 382/sq km, sex ratio 943, child sex ratio 914, literacy 74.04% (M 82.14, F 65.46).
- **Census literacy** is defined for age 7 years and above, able to *both read and write with understanding* in any language.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e records the last census as March 2011 and covers none after it, so quote 2011 as the latest completed census. Park 28e Table 9 tabulates the 2011 sex ratio as the provisional 940 (as does Park Plus) whereas the final figure of 943 is the one used by Sunder Lal, so either value is acceptable provided the source is named. On the method of enumeration the corpus conflicts — Park Plus and the Red Book describe the Indian census as de facto, while Mahajan and Gupta 4e states that India has followed the de jure system since 1941; write the de facto answer, which is what Park and the majority of Indian texts teach.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, Registrar General of India gazette notifications, NFHS-5, Civil Registration System, WHO SCORE framework, SDG indicator 17.19.2

Programme and policy updates (India)

- ▶ **Ayushman Bharat Digital Mission (ABDM), launched in 2021, is now the national digital layer of the HIS** that Park's fourteen-source list predates — it develops and integrates health data records and registries covering beneficiaries, health facilities and healthcare professionals. Dashboard figures showed 400 million beneficiary registrations as of 3 June 2023 and 273 million linked health records, with over 208,000 health facilities and 190,000 healthcare professionals verified — but inter-state variation is significant, especially in facility and professional registration [PMID 39437234].
- ▶ **The Integrated Health Information Platform (IHIP) has replaced IDSP's weekly paper S, P and L form reporting** — soft-launched pan-India from 1 April 2021 after the 2015 Joint Monitoring Mission report urged a review and redesign of IDSP, it covers 33 epidemic-prone diseases and syndromes, is near-real-time and web-based, geo-tags symptomatic cases and health facilities through GIS, and captures disaggregate rather than aggregate data at each level of care [PMID 37082408]. In Odisha the IHIP "P form" registered 255,071 patients in Ganjam and 63,895 in Jagatsinghpur between April 2021 and October 2022, acute respiratory illness carrying the highest notification rate in both districts [PMID 41629233].
- ▶ **Real-world ABHA uptake still lags the registration headline** — across three government tertiary hospitals in Agra only 129,007 of 537,278 OPD visits (23.96%) completed an ABHA (Ayushman Bharat Health Account) registration between September 2024 and April 2025. Among 425 attendees the leading barriers were preference for in-person care (85.6%), cost of the data plan (81.9%), difficulty using health apps (65.6%), language difficulty (64.0%) and reluctance to share the OTP (60.0%); 88% wanted training or support and 41.4% had no knowledge of ABDM at all [PMID 41787376].

Current national data

- ▶ **Civil registration of deaths is still incomplete, which is the practical limit on India's vital statistics** — 70.8% of 84,390 deaths were registered (NFHS-5, 2019–21; 95% CI 70.4–71.2), falling to 34.7% for deaths at 0–4 years and 65.9% in rural versus 83.3% in urban areas [PMID 36733620]; 73% of male but only 64% of female deaths were registered [PMID 41296742]; and 152 districts form two contiguous low-registration "cold-spot" clusters — in Arunachal Pradesh,

Nagaland and Manipur, and in Jharkhand, Bihar and Uttar Pradesh (Moran's I 0.69) [PMID 40348997].

- **Birth registration has moved much closer to universal, driven by institutional delivery** — institutional births rose from 86% (NFHS-4) to 92% (NFHS-5) and under-five birth registration from 80% to 89%; registration of babies delivered at CHCs and PHCs rose from 80.8% to 90.2%, the largest rise of any delivery setting, and such births had 38% higher odds of being registered than private-hospital births [PMID 41752339].
- **Total fertility rate has fallen below replacement level** — from 3.4 in NFHS-1 to below replacement (TFR 2) in NFHS-5, a 41.17% national decline, though Bihar (3.0), Meghalaya (2.9) and Uttar Pradesh (2.4) remain above it; Sikkim (62.96%) and Arunachal Pradesh (57.14%) fell fastest, and Kerala least (10%), having already reached replacement level at NFHS-1 [PMID 41793333].

Elimination, surveillance and disease status

- **Surveillance completeness, not the platform, is the bottleneck** — IHIP held only 44% of laboratory-tested malaria cases from primary health centres in Raigarh, Chhattisgarh in September 2023, the root causes being weak laboratory-technician knowledge of data entry and poor district-surveillance-unit feedback; refresher training plus a weekly feedback mechanism raised sensitivity to 100% within five weeks [PMID 41629241].

Guideline and legislative updates

- **Census 2011 remains the last COMPLETED count, so every intercensal denominator in use is a projection** — the 2021 census was not held, being reported as indefinitely delayed in 2023 and as a three-year delay in 2024 [PMID 37716764] [PMID 39020196]. **The next count is now formally notified.** Gazette notification S.O. 2681(E) of 16 June 2025, issued under section 3 of the Census Act 1948 by the Registrar General and Census Commissioner, India, declares that a census of the population of India shall be taken during the year 2027, with the **reference date 00:00 hours of 1 March 2027**; for the Union territory of Ladakh and the snow-bound areas of Jammu and Kashmir, Himachal Pradesh and Uttarakhand the reference date is **00:00 hours of 1 October 2026** [ORGI 2025-Jun]. Quote 2011 as the last completed census, but name the 2027 reference date as the notified next one.
- **Census 2027 keeps Park's two-phase field design, and phase one is running now** — the first phase, **houcelisting and housing census (HLO)**, was notified on 8 January 2026 for the period **1 April 2026 to 30 September 2026** [ORGI 2026-Jan]. Within that window each State/UT has its own staggered slot, and each slot begins with a **self-enumeration period** in which households may fill their own schedule before the enumerator's visit [ORGI 2026-Apr]. Park's account of houcelisting followed by population enumeration therefore still holds; what has changed is the self-enumeration option that now precedes the enumerator.
- **The population-enumeration schedule now includes caste** — notified on 14 August 2026 as S.O. 4505(E) under section 8(1) of the Census Act 1948, the household schedule for the second phase carries **40 items**, of which item 10 is *Scheduled Caste (SC)/Scheduled Tribe (ST)/Caste*, so caste itself is recorded and not merely SC/ST status. Other notable items are digital literacy alongside literacy (item 15), place of COVID-19 vaccination (34), total number of bank accounts (35), and mobile, Aadhaar, voter-ID, passport and driving-licence particulars (36–40) [ORGI 2026-Aug].

- ▶ **The cost of stale denominators is measurable** — recomputing Anemia Mukht Bharat adolescent WIFS coverage in Faridabad on census-projected denominators instead of programme-dashboard denominators cut it from 85.1% to 14.1% and halved the district AMB index from 35.2 to 17.6 [PMID 39764332].
- ▶ **Medical certification of cause of death (MCCD) quality remains poor** — of 1,147 certificates audited at a North Indian tertiary hospital, only 10.38% were correctly filled, 74.80% had major errors and 9.77% differed from the autopsy diagnosis; after structured training the proportion of junior residents making errors fell from 72% to 28%, while untrained consultants stayed at 7% [PMID 40406755].

Global and WHO developments

- ▶ **SDG indicator 17.19.2 sets the benchmark India is measured against — 100% birth and 80% death registration by 2030 — and the WHO SCORE report anchors the wider push to strengthen health data systems** [PMID 36733620]; applying the WHO data-quality-review dimension of external consistency across 695 districts showed HMIS agreeing well with NFHS-5 for institutional births (r 0.758) and caesarean section (r 0.892) but poorly for sex ratio at birth (r 0.148, ICC 0.066) and anaemia in pregnancy (r 0.135), so HMIS-derived district comparisons need caution [PMID 40547776].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 21, pp 995–999; Ch 8, pp 581–585); Park Plus, Ch 19 (Health information and basic medical statistics), pp 222–252; Red Book — Public Health and Preventive Medicine, Ch 1 (pp 20–21); Sunder Lal — Textbook of Community Medicine (Ch 9, pp 545–560); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 25 (pp 478–479); Sunder Rao & Richard — Biostatistics and Research Methods (p 178); Jugal Kishore — National Health Programmes of India (Ch 1, p 31); Handbook of Epidemiology (Ahrens & Pigeot), Ch 38 (pp 1547–1549); Suryakantha — Community Medicine with Recent Advances (Ch 23, pp 635–636, Fig 23.3). All census figures are Census of India 2011 as reported in these texts.

Recent-advances sources: 39437234 — The Ayushman Bharat Digital Mission of India: An Assessment, Health Systems and Reform 2024, doi:[10.1080/23288604.2024.2392290](https://doi.org/10.1080/23288604.2024.2392290) (abstract-only); 37082408 — IHIP: A Leap into India's Dream of Digitalizing Healthcare, Indian Journal of Community Medicine 2023, doi:[10.4103/ijcm.ijcm_739_22](https://doi.org/10.4103/ijcm.ijcm_739_22); 41629233 — Epidemiological Profile of Diseases Reported in the P Form of Integrated Health Information Platform in Two Districts of Odisha, Indian Journal of Public Health 2026, doi:[10.4103/ijph.ijph_660_25](https://doi.org/10.4103/ijph.ijph_660_25) (abstract-only); 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission, BMC Health Services Research 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 36733620 — Death registration coverage 2019–2021, India, Bulletin of the World Health Organization 2023, doi:[10.2471/BLT.22.288889](https://doi.org/10.2471/BLT.22.288889); 41296742 — Who gets counted? Understanding low female death registration in India, PLoS One 2025, doi:[10.1371/journal.pone.0337224](https://doi.org/10.1371/journal.pone.0337224); 40348997 — Spatial disparities in death registration across states and districts of India, 2019–21, BMC Public Health 2025, doi:[10.1186/s12889-025-22707-2](https://doi.org/10.1186/s12889-025-22707-2); 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, International Journal of Environmental Research and Public Health 2026, doi:[10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 41793333 — Total Fertility Rate Over Two Decades in India: Trend Analysis from a National Survey, Indian Journal of Public Health 2026, doi:[10.4103/ijph.ijph_1155_24](https://doi.org/10.4103/ijph.ijph_1155_24) (abstract-only); 41629241 — Quality Improvement Initiative to Improve the Sensitivity of Malaria Reporting under Integrated Health Information Platform, Raigarh, Chhattisgarh, 2023, Indian Journal of Public Health 2026, doi:[10.4103/ijph.ijph_648_25](https://doi.org/10.4103/ijph.ijph_648_25) (abstract-only); 37716764 — India's indefinitely delayed census, The Lancet 2023, doi:[10.1016/S0140-6736\(23\)01477-0](https://doi.org/10.1016/S0140-6736(23)01477-0) (abstract-only); 39020196 — Three-year delay for Indian census frustrates researchers, Nature 2024, doi:[10.1038/d41586-024-02321-9](https://doi.org/10.1038/d41586-024-02321-9) (abstract-only); 39764332 — Digital Monitoring of Anemia Control Measures in a District Using Anemia Mukht Bharat Health Management Information System Indicators, Cureus 2024, doi:[10.7759/cureus.75248](https://doi.org/10.7759/cureus.75248); 40406755 — The Final Diagnosis: Evaluating Errors in Cause-of-Death Reporting, Cureus 2025, doi:[10.7759/cureus.82785](https://doi.org/10.7759/cureus.82785); 40547776 — Study on external consistency of health management information system data at district level across India, Journal of Family Medicine and Primary Care 2025, doi:[10.4103/jfmpc.jfmpc_1967_24](https://doi.org/10.4103/jfmpc.jfmpc_1967_24); ORGI 2025-Jun — Gazette of India Extraordinary, Ministry of Home Affairs (Office of the Registrar General, India), Notification S.O. 2681(E) dated 16 June 2025, declaring the Census of India 2027 and its reference dates; ORGI 2026-Jan — Office of the Registrar General, India, Gazette Notification: First Phase

Houselisting Operations (HLO) Period (1 April 2026 to 30 September 2026), published 8 January 2026; ORGI 2026-Apr — Office of the Registrar General, India, State/UT-wise Houselisting and Housing Census and Self-Enumeration periods, April 2026; ORGI 2026-Aug — Gazette of India Extraordinary, Ministry of Home Affairs (Office of the Registrar General, India), Notification S.O. 4505(E) dated 14 August 2026, questionnaire for the second phase, Population Enumeration.



Health information system and its sources in India

Asked in: Paper I 19-Nov-2020 — RGUHS MD Community Medicine

Definition — A **health information system (HIS)** is defined as *"a mechanism for the collection, processing, analysis and transmission of information required for organizing and operating health services, and also for research and training"* (Park 28e, Ch 21, p 995). It is an integral part of the national health system and a basic tool of management. Its **primary objective** is to provide reliable, relevant, up-to-date, adequate, timely and reasonably complete information to health managers at all levels — central, intermediate and local — and to give technical and scientific (including bibliographical) information to all health personnel in the services of a country.

Data, information and intelligence

Data are observations of attributes or events and carry little meaning alone; data as collected from existing health care systems or institutions are inadequate for planning. Data are transformed into **information** by reducing, summarising and adjusting them for variations such as the age and sex composition of the population, so that comparisons over time and place become possible. Information integrated and processed with experience and with perceptions based on social and political values becomes **intelligence** — the form usable by decision-makers, policy-makers, planners, administrators and health care personnel (Park 28e, Ch 21, p 995). Data not transformed into information, and information not transformed into intelligence, are of no use to the system.

Requirements and components of a health information system

WHO Expert Committee — requirements of a health information system	Components (subsystems) of a comprehensive HIS
1. The system should be population-based	1. Demography and vital events
2. It should avoid unnecessary agglomeration of data	2. Environmental health statistics
3. It should be problem-oriented	3. Health status — mortality, morbidity, disability, quality of life
4. It should employ functional and operational terms (episodes of illness, treatment regimens, laboratory tests)	4. Health resources — facilities, beds, manpower
5. It should express information briefly and imaginatively (tables, charts, percentages)	5. Utilization and non-utilization of health services — attendance, admissions, waiting lists

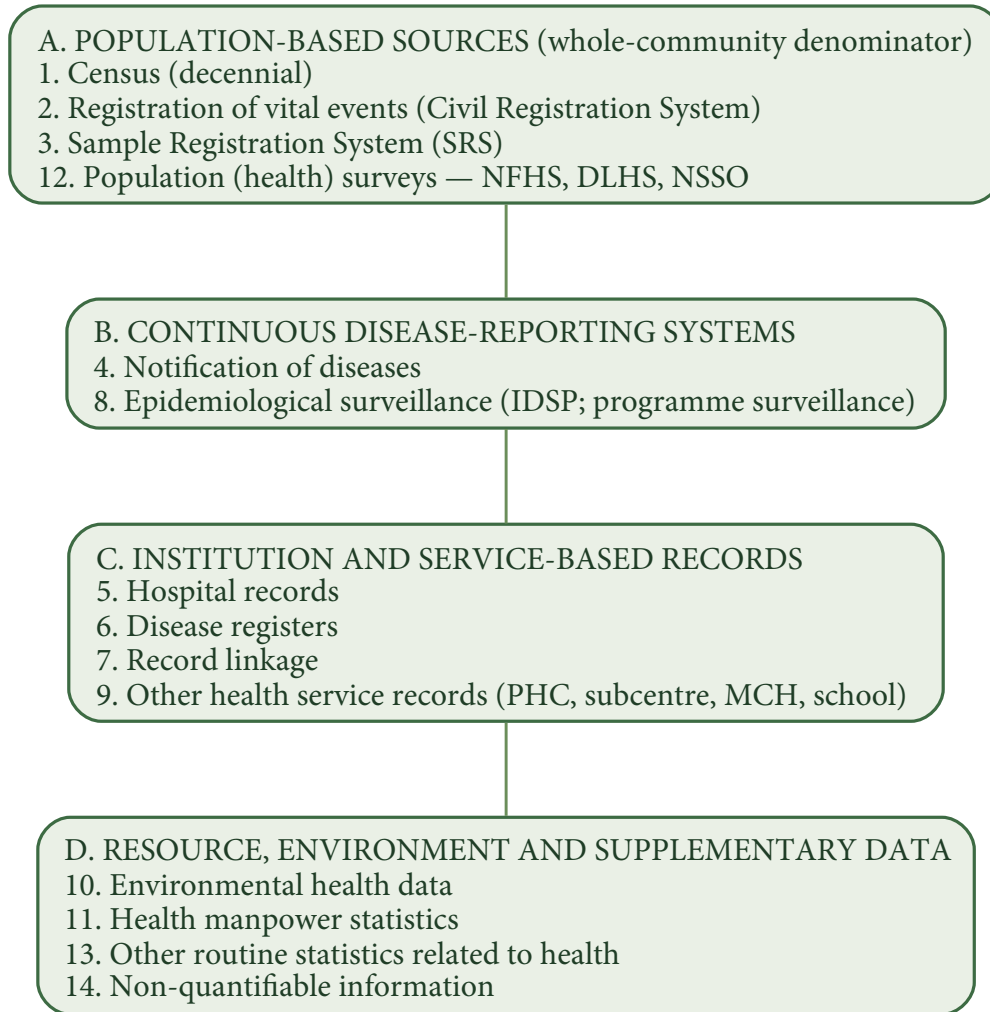
WHO Expert Committee — requirements of a health information system	Components (subsystems) of a comprehensive HIS
6. It should make provision for the feedback of data	6. Indices of outcome of medical care
	7. Financial statistics — cost and expenditure

(Park 28e, Ch 21, p 995)

Uses of health information

- ▶ To **measure the health status** of the people and to quantify their health problems and their medical and health care needs.
- ▶ For **local, national and international comparisons** of health status — such comparisons require rigorous standardization and quality control of the data.
- ▶ For **planning, administration and effective management** of health services and programmes.
- ▶ For **assessing whether health services are accomplishing their objectives** in terms of effectiveness and efficiency.
- ▶ For assessing the **attitudes and satisfaction of the beneficiaries** with the health system.
- ▶ For **research** into particular problems of health and disease (Park 28e, Ch 21, p 995).

Health information requirements differ by the administrative level at which the system is envisaged — what a public health administrator needs differs from what a hospital administrator needs — and these different contexts call for different sources (Park 28e, Ch 21, p 996).



The 14 sources of health information (list and order from Park 28e, Ch 21, pp 996-999), grouped here for recall

SOURCE & COVERAGE

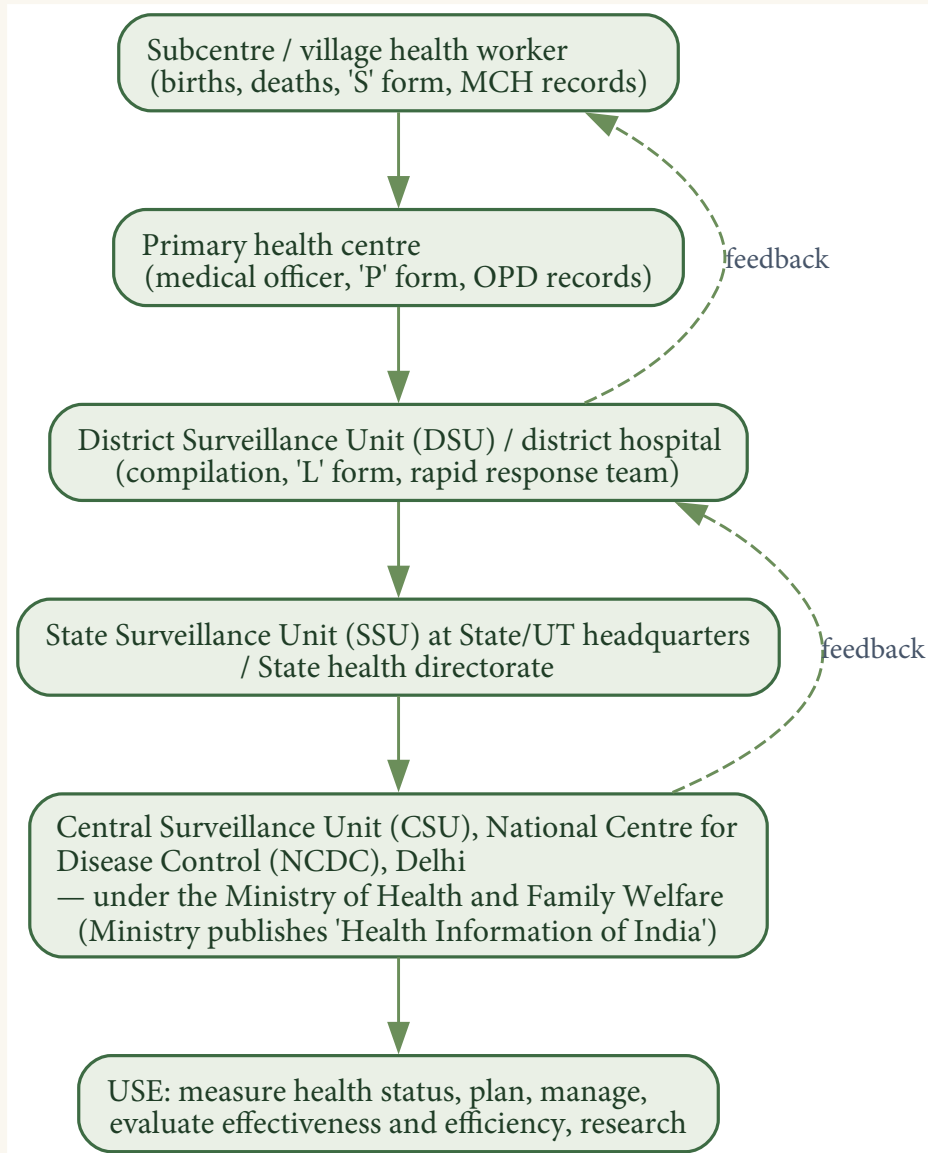
Source	What it yields in India	Key limitation
1. Census	Decennial total process of collecting, compiling and publishing demographic, economic and social data for all persons; first regular census 1881, last completed census March 2011; legal basis Census Act 1948 ; conducted by the Office of the Registrar General and Census Commissioner, India , Ministry of Home Affairs. Without census data no denominator exists, so no rate can be computed	Full results not available quickly; a vast undertaking taking years to compile and analyse
2. Registration of vital events (Civil Registration System)	Continuous, legally compulsory recording of live births, deaths, foetal deaths, marriages, divorces, adoptions, legitimations and legal separations under the Registration of Births and Deaths Act 1969 (in force 1 April 1970), administered by the Office of the Registrar General of India (RGI) ; reporting time limit 21 days ; Aadhaar number mandatory for death registration since October 2018	Grossly deficient in accuracy, timeliness and completeness — illiteracy, ignorance, lack of motivation, non-uniform procedures for rural and urban areas, and multiple registration agencies (health, panchayat, police, revenue)
3. Sample Registration System (SRS)	Initiated by the Office of the Registrar General of India in the mid-1960s (1964–65) because civil registration was deficient; a dual-record system — continuous enumeration of births and deaths by a resident part-time enumerator plus an independent half-yearly retrospective survey by an investigator-supervisor; covers the entire country and yields birth and death rates, infant mortality rate and other fertility and mortality indicators annually	Sample-based; gives national and State (sub-national) estimates only, never district or local ones; a supplement to, not a substitute for, civil registration
4. Notification of diseases	Historically the first HIS subsystem ; operates through legal Acts (e.g., Madras Public Health Act 1930; several States invoke the Epidemic Diseases Act 1897); reporting has shifted from the village chowkidar to the health worker; internationally, cholera, plague and yellow fever are notifiable to WHO under the International Health Regulations, with louseborne typhus, relapsing fever, polio, influenza, malaria, rabies and salmonellosis under international surveillance; notification has been extended to some non-communicable conditions — cancer, congenital malformations, mental illness, stroke	Covers only a small part of total sickness; heavy under-reporting; atypical and subclinical cases escape; a uniform national notification Act on the model of the Registration of Births and Deaths Act 1969 is still lacking

Source	What it yields in India	Key limitation
5. Hospital records	Where vital registration is defective and notification inadequate, hospital data are a basic and primary source on prevalent disease; the WHO Expert Committee on Health Statistics recommended hospital statistics be an integral part of the national statistical programme; discharge sheets give diagnosis, procedures, complications, length of stay and laboratory data; yield bed-occupancy rate, duration of stay and cost-effectiveness indices	Only the " tip of the iceberg " — numerator without denominator; the population served cannot be defined; mild and subclinical cases never attend; admission policy varies between hospitals, so data are highly selective
6. Disease registers	Registration implies more than notification — a permanent record, follow-up of cases, and tabulation of frequency and survival; morbidity registers exist only for certain diseases and conditions — stroke, myocardial infarction, cancer, blindness, congenital defects and congenital rubella, while tuberculosis and leprosy are registered in many countries where they are common; India's principal example is the National Cancer Registry Programme (NCRP) , started by the Indian Council of Medical Research in 1982 with three population-based and three hospital-based registries, since expanded into a national network (Jugal Kishore, pp 584–585)	Useful only if the reporting system is effective and coverage national or representative — India's population-based cancer registries cover about 7.45% of the country's population
7. Record linkage	Bringing together records of one individual or family originating at different times or places — birth, marriage, death, hospital admission and discharge, sickness absence, prophylactic procedures; especially suited to studying associations between diseases, which may have aetiological significance	Volume of accumulated data; applied only on a limited scale (twin studies, chronic disease and genetic studies); beyond the reach of many developing countries
8. Epidemiologic surveillance	Surveillance systems built into national disease control programmes (malaria, tuberculosis, leprosy, filariasis) reporting new cases and control efforts; the Integrated Disease Surveillance Programme (IDSP, 2004) collects weekly data on three formats — 'S' by the health worker, 'P' by the clinician and 'L' by the laboratory — with any rising trend investigated by the medical officer or rapid response team; three parallel streams of case definition are used — syndromic, presumptive and laboratory-confirmed	Programme-specific; depends on standard case definitions, regularity of reporting and on action actually being taken on the reports

Source	What it yields in India	Key limitation
9. Other health service records	Records of hospital outpatient departments, primary health centres and subcentres, polyclinics, private practitioners, maternal and child health (MCH) centres, school health records, diabetic and hypertensive clinics — e.g., MCH records give birth weight, weight, height, arm circumference, immunization and disease-specific mortality and morbidity	Relate only to a segment of the population; kept mainly for administrative purposes rather than for monitoring
10. Environmental health data	Data on air, water and noise pollution, harmful food additives, industrial toxicants, waste disposal, and production and consumption of materials	Collection of environmental data remains a major problem
11. Health manpower statistics	Numbers of physicians (by age, sex, speciality, place of work), dentists, pharmacists, veterinarians, nurses and technicians, maintained by State medical, dental and nursing councils and the Directorates of Medical Education; the census also records occupation; statistics of medical admissions and output are published every year in "Health Information of India", compiled by the Central Bureau of Health Intelligence (CBHI) under the Directorate General of Health Services, Ministry of Health and Family Welfare (Sunder Lal, Ch 7, p 274)	Multiple custodians; manpower figures are estimates rather than a single register
12. Population (health) surveys	Because an HIS must be population-based, routine statistics are supplemented by health surveys — for (a) evaluating health status, i.e., community diagnosis , (b) investigating factors affecting health and disease, and (c) administration, utilization, expenditure and unmet need. Indian examples: the National Family Health Survey (NFHS-5, 2019–21) , conducted by the International Institute for Population Sciences, Mumbai, for the Ministry of Health and Family Welfare, giving estimates for 28 States, 8 union territories and 707 districts ; the District Level Household and Facility Survey (DLHS) ; and the National Sample Survey Organization (NSSO) , long active in interview surveys yielding country-wide data on general morbidity, family planning and vital events	Costly and intermittent; NSSO morbidity data are not reliable because of the limitation of the interview method , which is why interviews are combined with examination or laboratory surveys; health-record surveys are cheap but not population-based, with reliability open to question and no standardization

Source	What it yields in India	Key limitation
13. Other routine statistics related to health	Demographic — population density, movement, educational level; economic — consumption of tobacco, dietary fats and domestic coal, drug sales, per capita income, employment and unemployment; social security schemes — medical insurance data, sickness absence, sickness and disability benefit rates	Collected for non-health purposes; indirect indicators
14. Non-quantifiable information	Health policies, health legislation, public attitudes, programme costs, procedures and technology — an HIS has multidisciplinary inputs and needs proper storage, processing and dissemination	Not amenable to statistical summary

Lay reporting is an additional approach used where civil registration is slow to develop: first-line, non-professional health workers (e.g., village health guides) collect health, demographic and socio-economic information in their own community and transmit it to higher levels of the health system (Park 28e, Ch 21, p 996). Whatever the source, the quality of a health information system is finally judged on the **accuracy, reliability, completeness, timeliness, precision and integrity** of the data it produces (Oxford Textbook of Public Health 7e, Ch 21, p 823).



Upward flow and downward feedback of health information in the Indian health system

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Figure: The health information cycle. Draw a closed loop of five boxes with arrows, and label exactly:

- ▶ **Collection** (census, civil registration, SRS, notification, hospital and service records, surveys)
- ▶ **Processing / compilation**
- ▶ **Analysis** (reduce, summarize, standardize for age and sex)
- ▶ **Transmission and dissemination** (tables, charts, percentages)
- ▶ **Use and FEEDBACK** (to the peripheral worker who generated the data)

Alongside, draw the three-step ladder and label: **DATA → INFORMATION → INTELLIGENCE**.

High-yield exam pointers

- ▶ **Definition to reproduce verbatim:** "a mechanism for the collection, processing, analysis and transmission of information required for organizing and operating health services, and also for research and training."
- ▶ **Six WHO requirements mnemonic — "P-A-P-F-B-F":** Population-based, Avoid agglomeration of data, Problem-oriented, Functional/operational terms, Brief and imaginative presentation, Feedback provision.
- ▶ **Seven components:** demography and vital events; environmental health statistics; health status; health resources; utilization; outcome indices; financial statistics.
- ▶ **Fourteen sources** in Park's order — census; registration of vital events; SRS; notification; hospital records; disease registers; record linkage; epidemiological surveillance; other health service records; environmental health data; health manpower statistics; population surveys; other routine statistics; non-quantifiable information.
- ▶ **Who owns what in India:** census and the Civil Registration System and SRS → **Registrar General of India**; "Health Information of India" → **Central Bureau of Health Intelligence (CBHI)** under the Directorate General of Health Services; NFHS → **IIPS Mumbai** for the Ministry of Health and Family Welfare; cancer registration → **ICMR (NCRP)**.
- ▶ **IDSP hierarchy to name correctly:** **Central Surveillance Unit (CSU) in the National Centre for Disease Control (NCDC), Delhi** → **State Surveillance Units (SSU) at State/UT headquarters** → **District Surveillance Units (DSU) in all districts**, with weekly S, P and L forms.
- ▶ **Dated Indian facts:** Census Act 1948, first regular census 1881, last completed census **March 2011**; Registration of Births and Deaths Act 1969 (effective 1 April 1970), registration within **21 days**, Aadhaar mandatory for death registration from **October 2018**; SRS began **mid-1960s** as a **dual-record system**; Epidemic Diseases Act 1897; IDSP launched **2004**; NCRP started **1982**; NFHS-5 2019–21 covering **707 districts**.
- ▶ **The one-line criticism examiners look for:** hospital data give the numerator but not the denominator — the "tip of the iceberg".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: Park 28e cites the census of March 2011 as the last completed Indian census, and the Integrated Disease Surveillance Programme has since been migrated to a digital reporting platform; the retrieved corpus does not describe that platform, so it is not stated in the answer body.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, Census of India, Civil Registration System, NFHS-5, WHO Verbal Autopsy Instrument 2022, International Health Regulations 2024 amendments, WHO Pandemic Agreement 2025

Programme and policy updates (India)

- ▶ **IDSP is now IHIP-IDSP.** Following the 2015 mid-term review, the programme moved in 2018 from aggregate paper-based returns to real-time case-based reporting on the **Integrated Health Information Platform (IHIP)**, and is now formally styled IHIP-IDSP; the S, P and L formats survive as digital forms [PMID 42245358]. Integration remains incomplete. In a two-district qualitative case study in one western Indian State, weekly paper reporting to the vector-borne disease programme ran in parallel with IHIP, facility staff maintained physical registers alongside digital entry because supervisors verified the registers rather than the portal, and the district public health laboratories created under IHIP-IDSP were functional in only three of that State's 36 districts, four of which also lacked sentinel-hospital cover [PMID 42245358]. At primary-care level in Chandigarh the IDSP-IHIP portal scored 69% against an implementation checklist ("fairly implemented", 51–70%), while the HMIS portal (42%), the Health and Wellness Centre portal (38%), the CPHC-NCD app (39%) and FP-LMIS (36%) were all only poorly implemented (25–50%), and 107 of 120 health workers (89.2%) agreed that dual entry into paper and digital systems had increased their workload [PMID 40854229].
- ▶ **The National Digital Health Mission (launched 2020) was renamed the Ayushman Bharat Digital Mission (ABDM) in 2021**, with a fourteen-digit **Ayushman Bharat Health Account (ABHA)** as the personal identifier that makes Park's "record linkage" operationally possible by tying a patient to providers, insurers and records across facilities; 8.79 crore ABHA numbers were created in financial year 2025–26 (to 28 January 2026). Real-world uptake at the point of care is still low — at a government tertiary hospital in Agra only 129,007 of 537,278 outpatient visits over eight months used ABHA registration (23.96%), and among 425 attendees interviewed only about 55% felt comfortable about data privacy on the platform [PMID 41787376].

Current national data

- ▶ **Census 2027 is notified and under way, but the 2011 count is still the last completed enumeration.** The notified reference dates are 1 October 2026 for Ladakh and the snow-bound non-synchronous areas of Jammu & Kashmir, Himachal Pradesh and Uttarakhand, and 1 March 2027 for the rest of India; the houselisting and housing census phase runs from 1 April to 30 September 2026 and population enumeration is scheduled for February 2027. It is planned as India's first digital census with a self-enumeration option, and the first to enumerate caste since 1931. Until enumeration completes, national denominators remain projections carried forward

from the 2011 count — the 2021 excess-mortality analysis below had to apply observed death rates to *projected* 2021 populations because no later enumerated base exists [PMID 41232123].

- **Civil Registration System report for 2021, released May 2025** — 10.2 million registered deaths against 8.1 million in 2020, a 26% rise, versus 335,004 officially reported COVID-19 deaths; excess mortality over the pre-pandemic trend was about 2.4 million, an excess-to-reported ratio of 7.2. The same report records that 47.3% of the deceased had received no medical attention at the time of death. The four-year publication lag is itself the standing criticism of the system [PMID 41232123].
- **Registration completeness from NFHS-5 (2019–21)** — only 71% of deaths were registered nationally, with strong district-level clustering (Moran's I 0.69) and two contiguous low-registration "cold-spot" clusters spanning 152 districts, in eastern India (Uttar Pradesh, Bihar, Jharkhand) and the north-east (Arunachal Pradesh, Nagaland, parts of Manipur) [PMID 40348997]; registration was 73% for male but 64% for female deaths, the gap widest in the poorest quintile (57% vs 44%) and narrowest in the richest (89% vs 85%) [PMID 41296742].

Surveillance data quality

- **Digital reporting does not by itself guarantee case capture.** In Raigarh, Chhattisgarh, IHIP held only 44% of the laboratory-tested malaria cases entered in primary health centre registers; refresher training of laboratory technicians plus weekly district surveillance unit feedback raised sensitivity to 100% within five weeks [PMID 41629241] (abstract-only).

Cause-of-death certification and civil registration

- **Medical certification of cause of death (MCCD) still covers only 22.5% of registered deaths,** because 60–70% of deaths occur outside a medical institution and so carry no certifying doctor; form 4 covers institutional deaths and form 4A non-institutional ones, but 4A is in practice neither insisted upon nor guided [PMID 40522917]. Electronic certification helps: an in-house mortality module in a Puducherry teaching hospital raised fully complete forms from 4% to 19%, cut forms carrying any major error from 88% to 42%, and improved ICD-10 codability of the underlying cause from 73% to 96% [PMID 40536381].
- **WHO's Verbal Autopsy Instrument 2022** supports electronic capture, and an ICMR mobile application built on it is projected to shorten the interval from verbal-autopsy interview to assigned cause of death from 6–8 months to 1–2 months; health and wellness centres are projected to handle 35–40 verbal autopsies a year and primary health centres 150–200 a year per medical officer [PMID 39846203]. The Registrar General has run physician-coded verbal autopsy under the SRS since 1999.
- **Designating primary care facilities as registrars works.** Birth registration for under-fives delivered in primary and community health centres rose from 80.8% (NFHS-4) to 90.2% (NFHS-5), the largest gain of any delivery setting, and such births had 38% higher odds of registration than births in private hospitals — the biggest State-level gains were in States that formally designated PHCs and CHCs as registrars of births and deaths [PMID 41752339].

Global and WHO developments

- **Amendments to the International Health Regulations (2005)** were agreed at the 77th World Health Assembly on 1 June 2024, closing a 21-month process, and entered into force in September 2025; they insert a "pandemic emergency" tier above the PHEIC within the same alert

framework — an event must first meet the PHEIC criteria, and the tier applies only to communicable-disease events — thereby normalising rather than replacing the Director-General's emergency powers [PMID 40883832]. The **WHO Pandemic Agreement**, with 35 articles, was adopted by 124 governments at the 78th World Health Assembly on 20 May 2025; its Pathogen Access and Benefit-Sharing annex is still under negotiation and goes to the 2026 World Health Assembly, so the Agreement takes effect only thereafter [PMID 40589223]. These are the current international obligations behind the "notification of diseases" source in the table above.

SOURCE & COVERAGE

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20 May 2025.



Monitoring and evaluation — Midday Meal Programme

Asked in: Paper IV 13-May-2022 — RGUHS MD Community Medicine

Definition — Monitoring is the periodic, continuous collection and analysis of selected indicators that lets managers judge whether key activities are being carried out as planned and are producing the expected effect in the target population; in management terms it is "the episodic oversight of the implementation of an activity, seeking to ensure that input deliveries, work schedules, targeted outputs and other required actions are proceeding according to plan" (Park 28e, Ch 2, pp 58-59; Sunder Lal, Ch 16, p 723). **Evaluation** is the process of assessing the results of a programme to determine whether its stated goals and objectives have been achieved, at what cost, and with what quality (Sunder Lal, Ch 16, p 724). The WHO Expert Committee on National Health Planning in Developing Countries states that evaluation "measures the degree to which objectives and targets are fulfilled and the quality of the results obtained... it measures the productivity of available resources... it makes possible the reallocation of priorities and of resources on the basis of changing health needs" (Park 28e, Ch 23, p 1027).

Monitoring compared with evaluation

Feature	Monitoring	Evaluation
Question answered	Are we doing the activities as planned?	Did we achieve what we planned, and at what cost?
Timing	Continuous / episodic, day-to-day operations (Park 28e, Ch 23, p 1027)	Terminal or periodic — progress and efficiency at least yearly, impact at least 5 years after start (Sunder Lal, Ch 16, p 725)
Level of the results chain	Inputs and processes (Sunder Lal, Ch 16, p 723)	Outputs and outcomes, i.e. effects and impacts (Sunder Lal, Ch 16, p 724)
Data source	Routine service records, monthly progress reports, supervisory field visits	Baseline household and facility surveys plus concurrent, mid-term and end-line surveys (Sunder Lal, Ch 8, p 340)
Purpose of the finding	Immediate mid-course correction of deviations	Decide whether to continue, modify or abandon the programme; reallocate resources
Chief actor	Programme manager and supervisor — an internal accounting of programme operations	Usually an external, post-facto audit needing independent outside appraisal, with periodic provision for external review (Reinke, Ch 4, pp 64-65)

The programme being evaluated — objectives and service norms

Evaluation can only be done against the programme's own stated objectives and prescribed standards, so these must be set out first.

Attribute	Detail
Names	Midday Meal Scheme, now PM POSHAN ; also called the National Programme of Nutritional Support to Primary Education ; earlier the School Lunch Programme, in operation since 1961 (Park 28e, Ch 11, pp 803-804)
Launched	Centrally sponsored scheme, 15 August 1995 ; revised 2004; 2,408 blocks in year 1, whole country by 1997-98 (Park 28e, Ch 11, p 804)
Ministry	Ministry of Education (Park 28e, Ch 11, Table 42)
Stated objectives	(1) Universalisation of primary education by increasing enrolment, retention and attendance ; (2) simultaneously improving the nutritional status of students in primary classes and their scholastic achievement (Park 28e, Ch 11, p 804; Sunder Lal, Ch 5, p 215)
Beneficiaries	Classes I-V in government, local body and government-aided schools; extended Oct 2002 to Education Guarantee Scheme and Alternative and Innovative Education Centres (Park 28e, Ch 11, p 804)
Nutritional norm	Cooked meal of minimum 300 kcal and 8-12 g protein per child per school day (Park 28e, Ch 11, p 804); Sunder Lal states 10-12 g protein (Ch 5, p 215)
Feeding days	Meal served for 200 days in a year (Sunder Lal, Ch 5, p 215); the National Institute of Nutrition holds that the minimum number of feeding days in a year should be 250 to have the desired impact on the children (Park 28e, Ch 11, pp 804-805)
Central assistance	Free foodgrain from the nearest Food Corporation of India godown at 100 g per student per day , plus transport subsidy; cooking cost Rs 2 per child per day, of which Rs 1 from the State (Park 28e, Ch 11, p 804; Sunder Lal, Ch 5, p 215)
Model menu (per child per day)	Cereals and millets 75 g, pulses 30 g, oils and fats 8 g, leafy vegetables 30 g, non-leafy vegetables 30 g; only iodised salt to be used (Park 28e, Ch 11, Table 43, p 804)
Design principles	Meal to be a supplement, not a substitute for the home diet; to give at least one-third of energy and half of protein needs; low cost; easily cooked in school; locally available foods; menu changed often to avoid monotony (Park 28e, Ch 11, p 804; Ch 9, p 684)

Steps in evaluation of the National Midday Meal Programme

Park's seven general steps of evaluation of health services (Park 28e, Ch 23, p 1037) applied to the Midday Meal Programme.

- **Determine what is to be evaluated.** Decide whether the object is the whole scheme, a State's implementation, or one district or school, and decide the type of evaluation — of **structure** (kitchen-cum-store, utensils, cooks, foodgrain stock, funds), of **process** (whether cooking, serving, tasting and recording actually follow the guidelines, appraised by an audit-type comparison against predetermined performance), or of **outcome** (whether the child is measurably better off — better enrolment, retention, attendance, weight-for-age and scholastic performance). Frame the evaluation questions from the scheme's two stated objectives, educational and nutritional.
- **Establish standards and criteria.** Standards drawn from the guidelines are the yardstick without which evaluation is impossible. *Structural criteria*: a functioning kitchen-cum-store of

recommended design, trained cook-cum-helper, weighing scale, safe drinking water, foodgrain at 100 g per child per day. *Process criteria*: meal served on every prescribed feeding day; the teacher and a school management committee member taste the meal before it is served; menu rotated; iodised salt used (Jugal Kishore, p 519). *Outcome criteria*: meal delivering not less than 300 kcal and 8-12 g protein, and a measurable change in enrolment, attendance and nutritional status.

- **Plan the methodology.** Prepare a written format for gathering the information wanted, with the standards and criteria built in at this planning stage. Fix the design — a baseline household and school survey, followed by concurrent, mid-term and end-line surveys (Sunder Lal, Ch 8, p 340) — the sampling frame of schools and children, the tools (record-review proforma, school observation checklist, child anthropometry, 24-hour dietary recall, exit interviews of children and parents), and the comparison group of schools not yet covered. Park notes that the **randomised controlled trial** is an important advance for the evaluation of the effectiveness and efficiency of health care programmes (Park 28e, Ch 11, p 805).
- **Gather information.** Collect what the plan calls for: school registers and monthly progress reports, foodgrain and expenditure records, the cook's menu register, direct observation of cooking and serving hygiene, anthropometry on a sample of children, and the political, cultural, economic, environmental and administrative factors influencing implementation. The amount collected is decided by the purpose and intended use of the evaluation (Park 28e, Ch 23, p 1037).
- **Analyse the results.** Compute the indicators, compare each against its standard, and quantify the gap between actual and expected performance; then find the root cause of each gap — irregular foodgrain lifting, delayed release of cooking cost, absent or untrained cook, or breaches of kitchen hygiene (Jugal Kishore, p 518). Analysis, interpretation and feedback must reach all concerned in the shortest feasible time (Park 28e, Ch 23, p 1037).
- **Take action.** Evaluation is productive only if it leads to action that supports, strengthens or modifies the service — correcting supply-chain failure, retraining cooks, constructing kitchen-cum-stores, tightening food-quality monitoring after any incident of food poisoning, or shifting priorities and revising objectives, such as extending the scheme to upper primary and secondary classes (Park 28e, Ch 23, p 1037; Jugal Kishore, p 519).
- **Re-evaluate.** Evaluation is a continuing cycle, not a single event; the corrected programme is monitored again and re-evaluated to make the service progressively more relevant, more efficient and more effective (Park 28e, Ch 23, p 1037).

Criteria of evaluation applied to the Midday Meal Programme

Criterion (Park 28e, Ch 23, p 1038)	Question asked of the Midday Meal Programme	Measure used
Relevance	Does the service still relate to a real need?	Persisting undernutrition and school drop-out in the covered population
Adequacy	Is sufficient attention paid to the previously determined need?	Foodgrain, cooking cost and cook-cum-helper allotted per child against the norm

Criterion (Park 28e, Ch 23, p 1038)	Question asked of the Midday Meal Programme	Measure used
Accessibility	What proportion of the target group can be expected to use the service, after geographic, economic, social and organisational barriers?	Proportion of eligible schools and enrolled children actually served
Acceptability	Is the service acceptable to the recipients and providers?	Proportion of enrolled children eating the meal; parent and teacher satisfaction; absence of discrimination in serving
Effectiveness	To what degree are the predetermined objectives and targets attained?	Rise in enrolment, attendance and retention; fall in undernutrition prevalence
Efficiency	How well are money, personnel, material and time used to achieve that effect?	Cost per child per meal served; foodgrain wastage; cost per unit rise in attendance
Impact	What is the overall effect on health status and socio-economic development?	Reduction in underweight and anaemia, improved scholastic performance, reduced child labour and gender gap in schooling

Indicators used at each level of the results chain

Level	Indicator for the Midday Meal Programme	Used in
Input	Foodgrain stock, funds released against sanctioned, cooks-cum-helpers in position, kitchens-cum-stores built, weighing scales, safe water	Monitoring
Process	Number of school days meal actually served against 200 planned; menu rotation; meal tasted before serving; hygiene checks	Monitoring
Output	Number and proportion of enrolled children served the meal per day (coverage); calories and protein actually delivered per child	Evaluation
Effect	Enrolment, attendance and retention rates; children's and parents' knowledge, attitude and practice on diet and hygiene	Evaluation
Impact	Prevalence of underweight, stunting and anaemia in school children; drop-out rate; scholastic achievement	Evaluation

Diagrams

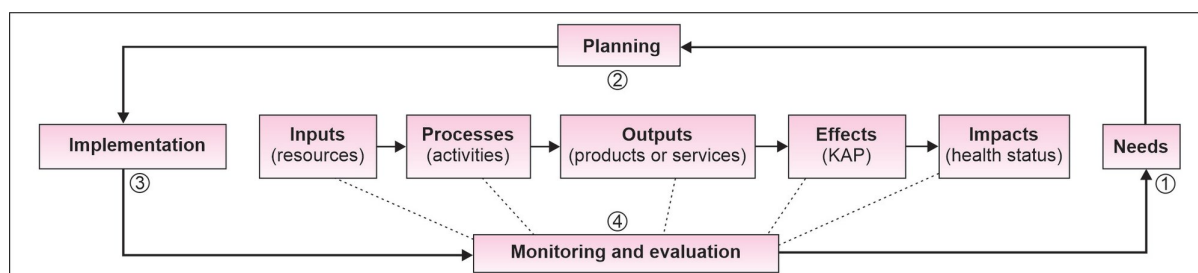
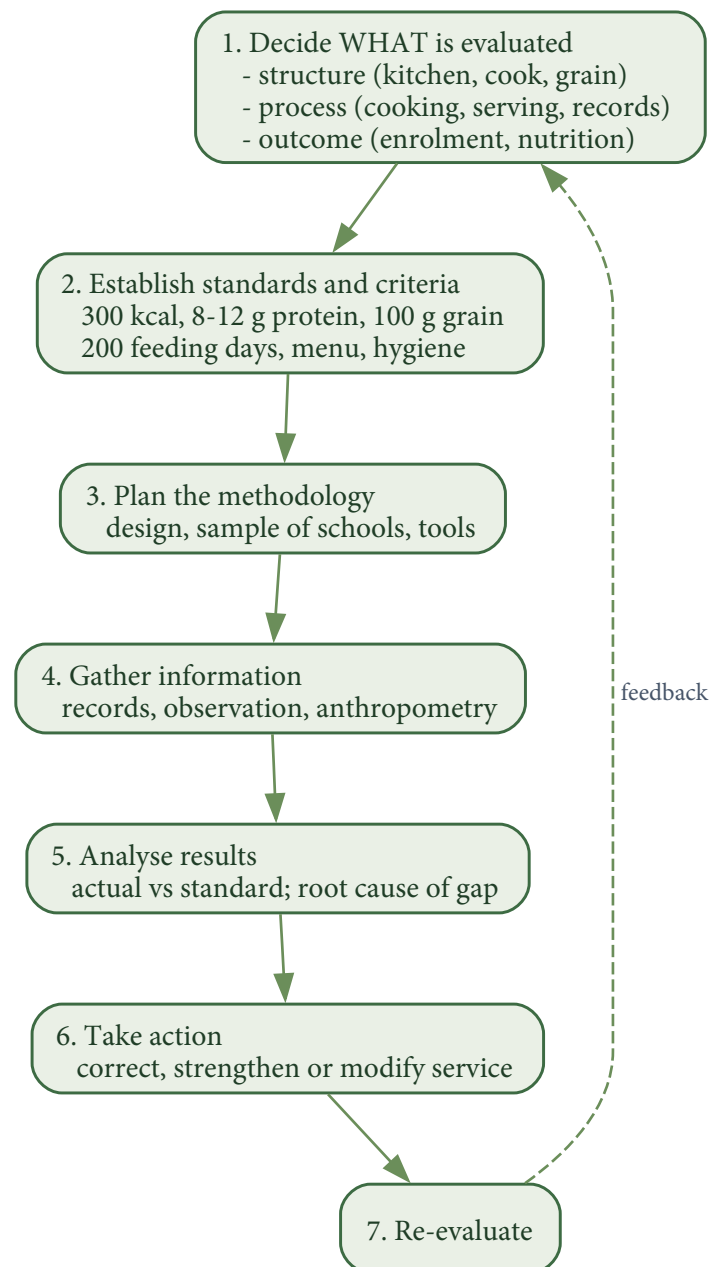
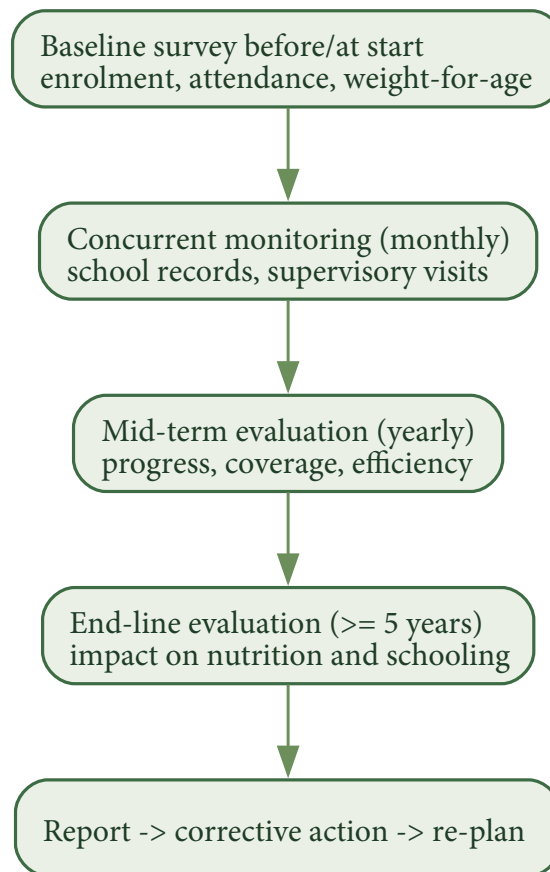


Fig 16.6 — Place of monitoring and evaluation in planning cycle (Sunderlal, p.722)



The seven steps in evaluating the National Midday Meal Programme (PM POSHAN)



Survey sequence for evaluating the Midday Meal Programme against its objectives

High-yield exam pointers

- ▶ **Monitoring** = inputs and processes, continuous, by the manager; **evaluation** = outputs and outcomes, periodic/terminal, often external. Monitoring asks "are we doing it right?", evaluation asks "did it work, and at what cost?"
- ▶ **Seven steps (Park)**: what to evaluate → standards and criteria → methodology → gather information → analyse → take action → re-evaluate.
- ▶ **Three types of evaluation**: structure, process (medical/nursing audit), outcome — outcome traditionally covering the **5 Ds of ill-health**: disease, discomfort, dissatisfaction, disability, death (Park 28e, Ch 23, p 1037).
- ▶ **Seven criteria (mnemonic RAAA-EEI)**: Relevance, Adequacy, Accessibility, Acceptability, Effectiveness, Efficiency, Impact.
- ▶ **Midday Meal numbers**: launched 15 August 1995; 100 g foodgrain per student per day; minimum 300 kcal and 8-12 g protein; 200 feeding days against the National Institute of Nutrition's recommended minimum of 250; meal must give one-third of energy and half of protein needs and be a supplement, not a substitute.
- ▶ **Effectiveness ≠ efficiency**: effectiveness is degree of attainment of objectives; efficiency is output achieved per unit of money, personnel, material and time.
- ▶ **Frequency**: evaluate progress and efficiency at least once a year; impact needs at least 5 years from programme start.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e records that the Midday Meal Scheme is "now known as PM POSHAN" but gives no year for the renaming, and lists the scheme in Table 42 under the Ministry of Human Resource Development, the former name of the Ministry of Education. Coverage note — the retrieved corpus carries nutritional norms only for primary classes I-V (minimum 300 kcal and 8-12 g protein) and no upper-primary norm, so quote only the primary-class figure; Park (8-12 g) and Sunder Lal (10-12 g) differ slightly on protein, so quote Park's figure with its source.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NFHS-5, CNNS 2016-18, ICMR, FSSAI, National Quality Assurance Standards 2024

Programme and policy updates (India)

- ▶ **The scheme has been renamed and its foodgrain re-specified.** The National Programme of Mid-Day Meal in Schools is now **Pradhan Mantri Poshan Shakti Nirman (PM POSHAN)**, and rice fortified with iron (28-42.5 mg/kg), folic acid and vitamin B12 entered PM POSHAN and ICDS in phase I during 2021-22, was extended through the public distribution system to 291 aspirational and high-anaemia-burden districts by March 2023, with phase III scheduled to reach all remaining districts by March 2024. Fortified rice is therefore a new *structural* standard at step 2 — the grain lifted must be fortified, not merely 100 g per child per day. [PMID 39110751]
- ▶ **The companion under-six meal norm was revised in 2023.** Supplementary Nutrition Programme standards for children aged 36-72 months moved from 300 kcal and 8-10 g protein (until 2009) to 500 kcal and 12-15 g protein (until 2023); the 2023 revision added fat and micronutrient recommendations (iron, calcium, zinc, vitamins A and B), protein-quality correction and dietary diversity, but left the per-child allocation of Rs 8 per day unchanged — the classic evaluation finding of a raised standard without a raised input. Linear-programming optimisation of meals actually served in 108 Anganwadis across 27 States and Union Territories (2022-23) showed the gap could be closed with small additions of eggs, milk and green leafy vegetables, but only at a further Rs 8 per child per day in Karnataka and Maharashtra and Rs 12 in Madhya Pradesh. [PMID 42199722]
- ▶ **Anaemia Mukht Bharat is being redesigned as AMB 2.0 with evaluation designed in from the start.** AMB (2018) set a target of a three-percentage-point annual fall in anaemia across six beneficiary groups. The strengthened package, drawn up by ICMR and NITI Aayog and submitted to the Health Ministry for consideration, has six pillars — test-treat-track, prophylactic IFA, anaemia-relevant health interventions, fortified rice, dietary diversification and Jan Andolan behaviour change — and the ICMR PRAKASH implementation-research study now operationalises them through iterative cycles of implementation and performance monitoring using implementation-science frameworks, aiming at anaemia of 20% or lower with coverage of 80% or more. This is prospective, concurrent evaluation replacing the post-facto audit model. [PMID 42313808]

Current national data

- **Quasi-experimental designs now carry step 3 (plan the methodology).** A birth-cohort fixed-effects analysis of seven national datasets spanning 1993-2016 (NSS consumer expenditure rounds, NFHS, IHDS), with a controlled interrupted time-series as robustness check, found height-for-age Z score 0.40 SD higher in children born to mothers living where Midday Meal coverage was 100% than in children of unexposed mothers, largest in the poorest households (0.5 SD), and attributed 13-32% of India's HAZ improvement between 2006 and 2016 to the scheme — an *intergenerational* impact level beyond the usual results chain; the time-series check gave a smaller 0.038-0.044 SD per year and a 28.6% share, and the plausibility pathway ran through maternal education, later first birth, fewer children and more antenatal care. Separately, coarsened exact matching on NFHS-5 data for 148,265 children showed ICDS supplementary nutrition in pregnancy reduced low birth weight by 1.35 percentage points (7.5% relative) and very low birth weight by 20.2%, with only 62.6% of mothers always receiving it. [PMID 34253719] [PMID 40830047]
- **Quote burden figures with their survey round and year.** NFHS-5 (2019-21): anaemia in 67% of children 6-59 months, 57% of women of reproductive age and 52% of pregnant women; CNNS (2016-18) gives the school-age figures the Midday Meal actually targets — 23.5% in children 5-9 years and 28.4% in adolescents. Over the same period under-five overweight rose from 1.9% (NFHS-3) through 2.3% (NFHS-4) to 4.0% (NFHS-5), a roughly 60% rise over NFHS-4, and was highest in stunted children (6.5% against 1.9% in the non-stunted) — so impact indicators for a school feeding programme must now capture overnutrition alongside underweight if SDG 2.2 is to be tracked honestly. [PMID 39110751] [PMID 39295437]

Elimination, surveillance and disease status

- **The COVID-19 school closure gave an unplanned natural experiment in withdrawal.** The lockdown from 24 March 2020 simultaneously stopped the Midday Meal, weekly iron-folic acid supplementation and biannual deworming; in 290 rural primary-school children in Koppal, Karnataka, anaemia almost doubled from 21% to 40% within a year (older girls 10% to 53%) while iron deficiency was unchanged (48.8% to 51.9%) — evidence that the meal's dietary diversity, not iron alone, was doing the work. [PMID 36619337]
- **Effectiveness under routine conditions is not efficacy.** Among 230 children aged 1-6 years at 23 Anganwadi centres in rural Andhra Pradesh only 22.2% were actually taking the fortified supplement Balamrutham, anaemia prevalence was 85.1%, and consumption showed no independent association with ferritin (adjusted OR 0.83, 95% CI 0.42-1.62). The qualitative arm located the failure in delivery rather than formulation — intra-household sharing, dose dilution through recipe changes, perceived side effects and supply irregularity — which is precisely the root-cause analysis demanded at step 5. [PMID 42483365]

Guideline and legislative updates

- **Length of stay has become an auditable quality standard.** Government operational guidelines for nutrition rehabilitation centres, following the WHO 10-step protocol, stipulate a minimum 14-day stay with weight gain of at least 8 g/kg/day; across five centres in Gwalior only 11.1% of 332 children completed 14 days, and mean weight gain rose from 5.32 to 11.47 g/kg/day and recovery from 58.3% to 86.5% across the length-of-stay strata — adjusted, each extra inpatient

day added 0.71 g/kg/day of weight gain and cut the odds of defaulting by 12%. The Comptroller and Auditor General's 2023 performance audit of POSHAN Abhiyan flagged premature discharge, and the National Quality Assurance Standards for these centres, introduced in 2024, now explicitly score length-of-stay compliance. [PMID 42434167]

► **The school food environment is now regulated, and mandatory fortification is contested.**

FSSAI's 2020 regulations restrict high-fat, salt and sugar foods in and around schools, working alongside the Ayushman Bharat School Health Programme and PM POSHAN — new process criteria against which a school can be evaluated. Against that, the mandatory iron-fortification-of-rice policy has been criticised on public-health-ethics grounds: it was adopted after an evidence review had concluded that rice fortification was not effective against anaemia or iron deficiency at population level, it carries a risk of iron excess in groups with high habitual intake already receiving weekly IFA, and no end-point has been fixed for the policy. [PMID 41974918] [PMID 38375654]

Global and WHO developments

- **Universal versus targeted provision is the live global question, and it bears on the acceptability criterion.** About 418 million children received a free or subsidised school meal in 2022. A 2025 scoping review of 44 studies across 13 countries found school feeding associated with healthier weight, better dietary intake and improved social engagement, but means-tested targeted programmes were more often linked to stigma and poorer mental-health outcomes than universal ones — an argument for evaluating a school meal on acceptability and equity, not calories alone. [PMID 41043853]

SOURCE & COVERAGE

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Planning cycle

Asked in: Paper IV 13-May-2022 — RGUHS MD Community Medicine

Definition — Planning is the process of analysing a system or defining a problem, assessing the extent to which the problem exists as a need, formulating goals and objectives to alleviate or ameliorate those identified needs, choosing from among alternative intervention strategies, initiating the necessary action for implementation, monitoring the system and evaluating the results of intervention in the light of the stated objectives (Park 28e, Ch 23, p 1026). Health planning is part of national development planning, and **national health planning** is "the orderly process of defining community health problems, identifying unmet needs and surveying the resources to meet them, establishing priority goals that are realistic and feasible and projecting administrative action to accomplish the purpose of the proposed programme" (Park 28e, Ch 23, p 1025).

Its purpose is threefold — to match limited resources with many problems, to eliminate wasteful or duplicated expenditure, and to develop the best course of action to accomplish a defined objective (Park 28e, Ch 23, p 1025).

Objective, target and goal — the end-results of planning

Term	Meaning (Park 28e, Ch 23, pp 1025–1026)
Objective	A precise, planned end-point of all activities — it is either achieved or not achieved. Objectives are concerned directly with the problem itself
Target	A discrete activity (number of blood films collected, number of vasectomies done); it permits the concept of a degree of achievement. Targets are concerned with the factors involved in the problem
Goal	The ultimate desired state towards which objectives and resources are directed. Unlike objectives and targets it is not constrained by time or existing resources and is not necessarily attainable. Stated as — what is to be attained, to what extent, in which population, in which geographic area, over what length of time

Pre-planning — the four preconditions

Pre-planning is the preparation *for* planning; without these the cycle cannot start (Park 28e, Ch 23, p 1026):

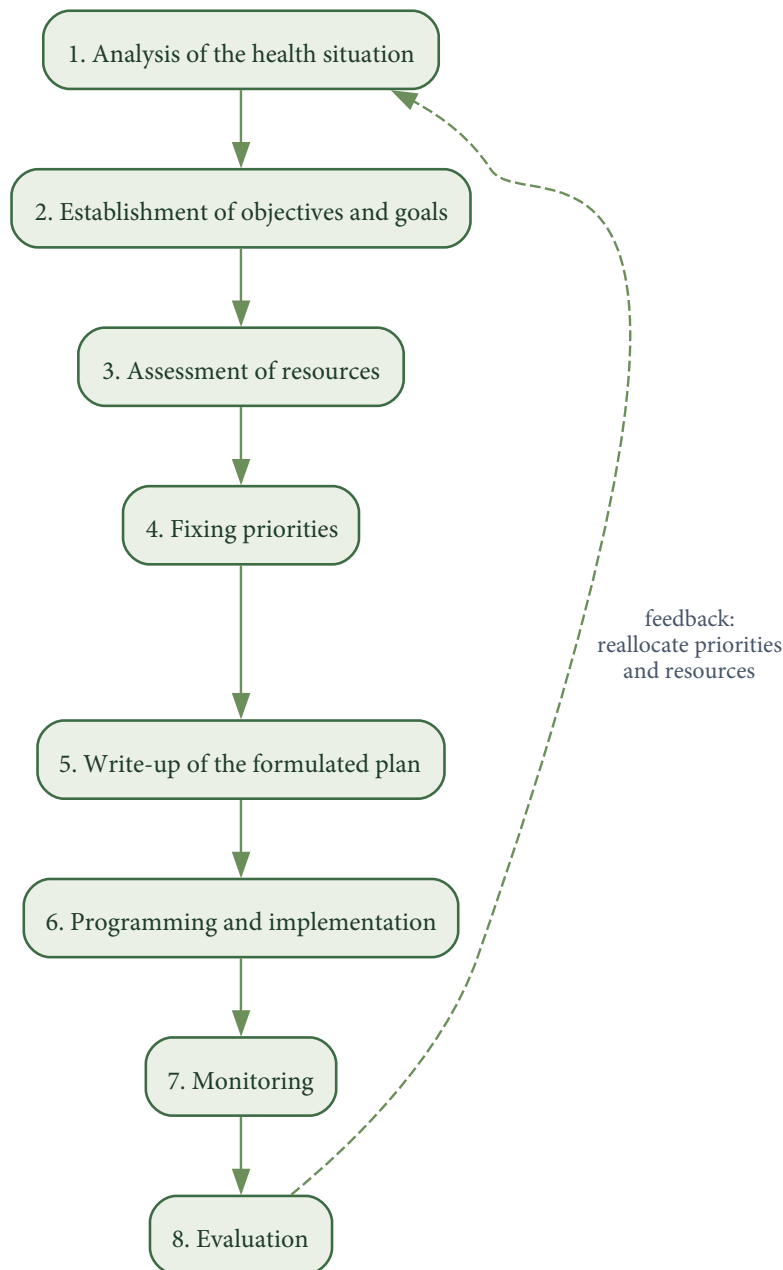
- ▶ **Government interest** — the health and welfare of the people must be accepted as a national responsibility, backed by strong political will manifested in clear directives.
- ▶ **Legislation** — political will translated into law; the *Medical Termination of Pregnancy Act, 1971*, enacted by the Indian Parliament to protect the health of mothers, is the classical example.
- ▶ **Organisation for planning** — a standing organisational structure to prepare the various parts of the plan, composed of full-time planners advised by technical experts in social and economic development. In India the **Planning Commission** served this function; it was replaced by NITI Aayog (National Institution for Transforming India) on 1 January 2015 (Park 28e, Ch 23, p 1031).
- ▶ **Administrative capacity** — capacity for proper coordination and implementation of the plan at all levels; for the health plan this is vested in the Central and State Ministries of Health.

Steps of the planning cycle

#	Step	What is done (Park 28e, Ch 23, pp 1026–1028)
1	Analysis of the health situation	Collection, assessment and interpretation of information to give a clear picture of the health situation. Minimum essential data — (a) population with age and sex structure; (b) morbidity and mortality statistics; (c) epidemiology and geographical distribution of diseases; (d) medical care facilities — hospitals, health centres and other health agencies, public and private; (e) technical manpower of various categories; (f) training facilities available; (g) attitudes and beliefs of the population towards disease, its cure and prevention. Analysis brings out the health problems, needs and demands of the population
2	Establishment of objectives and goals	Objectives guide effort and are also the yard-stick to measure work after it is done; without them there is haphazard activity, uneconomical use of funds and poor performance. Set at all levels down to the smallest organisational unit — general at upper levels, progressively more specified and detailed lower down; short-term or long-term, with time and resources as the limiting factors. Cost-benefit analysis and input-output study of health services help define them in definite terms
3	Assessment of resources	Resources = the manpower, money, materials, skills, knowledge and techniques needed or available for implementing the health programme. These are assessed and a balance is struck between what is required and what is available or likely to be available
4	Fixing priorities	Resources always fall short of the total requirement, so problems are ranked in order of importance or magnitude. Criteria — financial constraints; mortality and morbidity data; diseases preventable at low cost; saving the lives of younger people in whom there has been considerable social investment; political and community interests and pressures. Alternative plans are then formulated and assessed for practicability and feasibility, and the alternative with greater effectiveness is chosen
5	Write-up of the formulated plan	Preparation of the detailed plan — complete in all respects for execution, with resources related to results, the basis of the plan defined, guidance given to all those responsible for execution, and a "built-in" system of evaluation
6	Programming and implementation	Once the plan is approved, execution begins; it depends on an effective organisation with well-defined procedures, sufficient delegation of authority and fixation of responsibility. Main considerations — (a) definition of roles and tasks; (b) selection, training, motivation and supervision of manpower; (c) organisation and communication; (d) efficiency of individuals and institutions such as hospitals or health centres. Many well-considered plans break down at this stage from delays in critical supplies or inappropriate use of staff
7	Monitoring	Day-to-day follow-up of activities during implementation to ensure they are proceeding as planned and are on schedule. A continuous process of observing, recording and reporting — keeping track of the course of activities, identifying deviations and taking corrective action if deviations are excessive

#	Step	What is done (Park 28e, Ch 23, pp 1026–1028)
8	Evaluation	Assesses achievement of the stated objectives and the adequacy, efficiency and acceptability of the programme to all parties involved. Whereas monitoring is confined to day-to-day operations, evaluation is concerned with the end outcome and the factors associated with it. WHO Expert Committee on National Health Planning in Developing Countries — evaluation "measures the degree to which objectives and targets are fulfilled and the quality of the results obtained... the productivity of available resources in achieving clearly defined objectives... how much output or cost-effectiveness is achieved. It makes possible the reallocation of priorities and of resources on the basis of changing health needs"

Evaluation is not the end-point: it leads directly into a fresh planning cycle, which is why planning is not a single movement up a rigid stairway of steps but "an unending upward spiral of incremental efforts toward improvement" (Reinke, Health Planning for Effective Management, Ch 1, pp 24, 30).



The eight steps of the planning cycle, closed by the feedback arm from evaluation into a fresh situation analysis

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Reproduce the two features that make this plate a *cycle* rather than a list: the **two feedback arms** out of *Evaluate results* — *goals achieved? goals satisfactory?* (YES back to *Operate programme*; NO back to *Formulate health problems*), and the two side brackets labelled **Implementation** and **Training**.

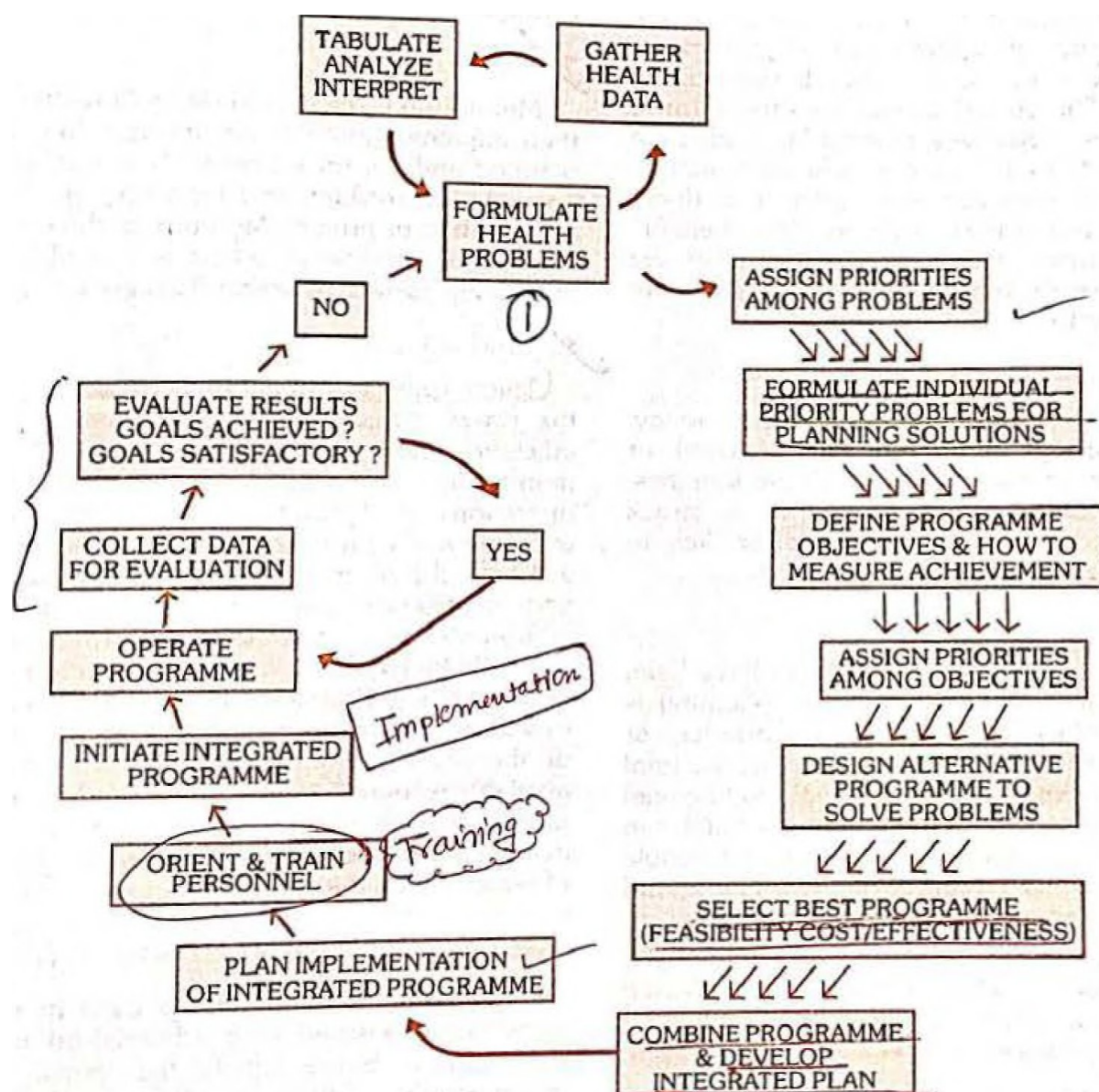
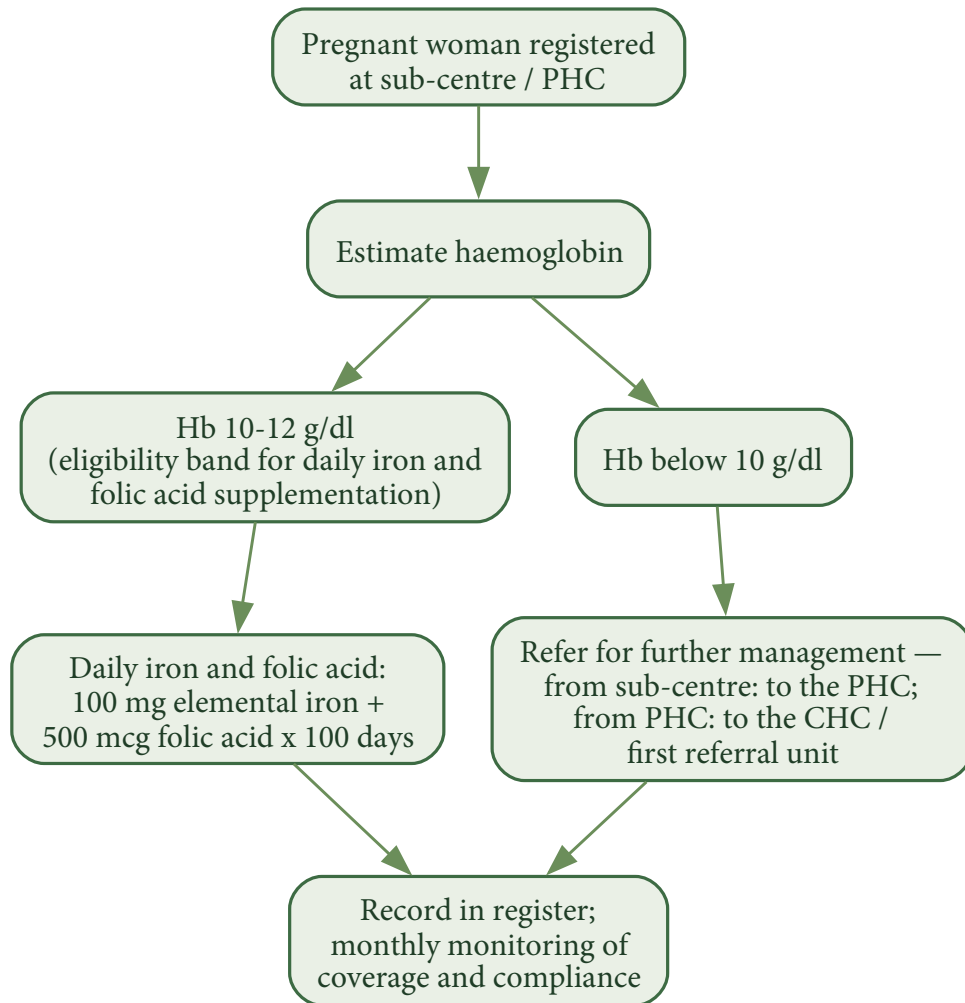


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, Ch 23 (Health Planning and Management), p 1026)

Worked example — planning for control of anaemia in pregnancy in a primary health centre (PHC) area

Step	Applied to anaemia in pregnancy, PHC covering 30,000 population
1. Situation analysis	Compile PHC and sub-centre records — pregnant women registered, haemoglobin estimations done, antenatal case load, staffing (medical officer, auxiliary nurse midwives, accredited social health activists), iron and folic acid (IFA) stock position, laboratory capacity, and local dietary and help-seeking beliefs. Benchmark against NFHS-5 (2019–21): anaemia in women aged 15–49 years about 57 per cent, in men about 25 per cent (Park 28e, Ch 11, p 785)

Step	Applied to anaemia in pregnancy, PHC covering 30,000 population
2. Objectives and goals	Goal — eliminate anaemia as a cause of maternal death in the PHC area (19 per cent of maternal deaths in India are attributed to anaemia; Park 28e, Ch 11, p 785). Objectives — screen 100 per cent of registered pregnant women for haemoglobin by 12 weeks of gestation, and reduce anaemia prevalence in registered antenatal women by a stated proportion in 2 years. Targets — number of haemoglobin estimations done and number of women completing 100 IFA tablets
3. Assessment of resources	Manpower (medical officer, auxiliary nurse midwives, accredited social health activists, laboratory technician), IFA tablet supply, haemoglobinometers and reagents, transport for referral, training days, and the untied PHC funds available — balanced against what each activity requires
4. Fixing priorities	Anaemia scores high on every Park criterion — large magnitude, high maternal and perinatal mortality and morbidity, preventable at very low cost, affects a young population with considerable social investment, and carries community and political support. Alternatives appraised — universal supplementation vs screen-and-treat vs food fortification vs deworming-plus-supplementation; the most effective feasible mix is chosen
5. Write-up of the plan	A written PHC plan naming each activity, the worker responsible, the schedule and the supply chain, with the built-in evaluation specified — who will compute coverage, from which register, at what interval
6. Programming and implementation	Deliver 100 mg elemental iron with 500 µg folic acid daily for 100 days to pregnant women through antenatal clinics, primary health centres and their sub-centres (Park 28e, Ch 9, p 627); extend weekly iron and folic acid supplementation (WIFS) to adolescents on a fixed-day approach — same 100 mg elemental iron and 500 µg folic acid weekly, plus albendazole 400 mg six-monthly — to break the intergenerational cycle (Park 28e, Ch 7, p 556); select, train, motivate and supervise the staff; run health education on diet and compliance
7. Monitoring	Monthly review of registers — women registered, haemoglobin done, IFA tablets issued and consumed, stock-outs, defaulters traced. Referral rule applied at every contact (National Nutritional Anaemia Prophylaxis Programme eligibility criteria, Park 28e, Ch 11, p 785): haemoglobin 10–12 g/dl → daily IFA supplementation; haemoglobin below 10 g/dl → refer — from the sub-centre to the PHC, and from the PHC onward to the community health centre / first referral unit, which serves as the referral centre for four PHCs (Park 28e, Ch 8, p 611). Park grades haemoglobin below 10 g/dl as severe , needing high doses of iron or blood transfusion (Park 28e, Ch 11, p 785)
8. Evaluation	At the end of the plan period compare pre- and post-intervention anaemia prevalence in registered antenatal women, the proportion completing 100 IFA tablets, and cost per case of anaemia averted. Findings feed back into the next situation analysis — a high prevalence despite good tablet coverage redirects the next cycle towards compliance, deworming or malaria control



Screen-and-treat and referral algorithm applied at steps 6 and 7 of the worked example. Anaemia in pregnancy is haemoglobin below 11 g/dl (non-pregnant adult females below 12, adult males below 13) — Park 28e, Ch 11, Table 12, p 764

High-yield exam pointers

- ▶ **Eight steps, mnemonic "A-O-R-P-W-I-M-E"** — Analysis of the health situation, Objectives and goals, Resources, Priorities, Write-up, Implementation (programming), Monitoring, Evaluation.
- ▶ **"Planning is for tomorrow, management is for today."** Planning in its broadest sense = **plan** formulation → execution → evaluation (Park 28e, Ch 23, p 1025).
- ▶ **Objective** is achieved or not achieved; **target** permits a degree of achievement; **goal** is the ultimate desired state, not bound by time or resources.
- ▶ **Monitoring** = day-to-day, is it on schedule; **evaluation** = end outcome, were the objectives met. This distinction is the most commonly asked one-liner.
- ▶ Four pre-planning preconditions — government interest, legislation (Medical Termination of Pregnancy Act 1971), organisation for planning (Planning Commission → NITI Aayog, 1 January 2015), administrative capacity.
- ▶ Priority-fixing criteria — financial constraints, mortality and morbidity data, low-cost preventability, saving younger lives with high social investment, political and community pressure.

- Datable Indian figures for the worked example — NFHS-5 (2019–21) anaemia in women 15–49 years ≈ 57 per cent, men ≈ 25 per cent; IFA in pregnancy = 100 mg elemental iron + 500 µg folic acid × 100 days.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the Planning Commission that Park describes as the organisational precondition for planning was replaced by NITI Aayog on 1 January 2015 and the Five Year Plan cycle ended with the Twelfth Plan (2012–2017), so a current answer should name NITI Aayog first and the Planning Commission only as the historical body; separately, the 10–12 g/dl and below-10 g/dl bands quoted above are the National Nutritional Anaemia Prophylaxis Programme's *eligibility and referral* thresholds, not the diagnostic cut-off, which for a pregnant woman is haemoglobin below 11 g/dl.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoHFW Ayushman Arogya Mandir portal, National Health Accounts estimates 2021–22, Indian Public Health Standards 2022, National Tuberculosis Elimination Programme, WHO Fourteenth General Programme of Work

Programme and policy updates (India)

- **Ayushman Bharat Health and Wellness Centres were renamed Ayushman Arogya Mandir in November 2023**, carrying the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth") and an AYUSH-informed holistic wellness remit; the exercise was budgeted at ₹3,000 per facility against a ₹48 crore outlay, with photographs of the renamed centre uploaded to the portal for verification. A current planning answer should name the facility **Ayushman Arogya Mandir** at the write-up and implementation steps, not "Health and Wellness Centre" [PMID 42257165].
- **The renamed platform now carries the peripheral planning machinery** — more than 1.75 lakh Ayushman Arogya Mandirs operational, 1,16,978 community health officers in position and 82 per cent of centres delivering all 12 expanded service packages by 30 November 2024 (up from 50 per cent in March 2023), with 1,35,343 Jan Arogya Samitis constituted as the facility-level community body that convenes an Arogya Sabha twice a year. Money for step 3 no longer comes from Five Year Plan outlays: PM-Ayushman Bharat Health Infrastructure Mission and Fifteenth Finance Commission health grants routed through Panchayati Raj Institutions and urban local bodies now sit alongside National Health Mission funds [PMID 41816146].
- **Priority-fixing has been institutionalised inside PM-JAY** — the Health Financing and Technology Assessment unit within the National Health Authority applies health technology assessment to decide inclusions and exclusions from the health benefit package and to frame standard treatment guidelines, with strategic purchasing and value-based pricing proposed as the next reform [PMID 38715207].

Current national data

- **Out-of-pocket expenditure fell from 60.6 per cent of total health expenditure in 2014–15 to 39.4 per cent in 2021–22** (National Health Accounts estimates); the UHC service coverage index for India rose from 57 in 2015 to 63 in 2021, above the South-East Asia regional average of 62,

and outpatient use of public facilities rose from 91.6 crore (2012–13) to 134.5 crore (2021–22), inpatient use from 4.3 crore to 6.3 crore [PMID 41816146].

- ▶ **A national evaluation of the National Health Mission is step 8 in real life:** time-series analysis found a significant post-NHM fall in infant mortality rate with narrowing regional inequity, a moderate effect on stillbirth rate, but no significant effect on neonatal mortality rate or maternal mortality ratio — exactly the kind of result that redirects the next cycle towards the newborn and the mother [PMID 39901042].
- ▶ **Resource assessment against Indian Public Health Standards 2022** in 53 Ayushman Arogya Mandirs of a Punjab border district graded staffing very good in 51 per cent, good in 15 per cent, average in 28 per cent and poor in 6 per cent; medical officers were 39 per cent short and staff nurses 45 per cent short in 24×7 primary health centres, while accountants and data-entry operators were 78 per cent short. The imbalance between cadres, not the aggregate headcount, is what step 3 has to surface [PMID 41017888].

Elimination, surveillance and disease status

- ▶ **India missed its pledged 2025 tuberculosis elimination target**, and the End TB milestone of a 50 per cent fall in incidence by 2025 was also missed globally; a two-phase DOST framework — intensive disease reduction, then focused progression reduction, calibrated to whether a state sits above or below 100 cases per 100,000 — has been proposed to sequence the National Strategic Plan. The 100-day TB Mukh Bharat Abhiyaan started in December 2024 across 347 high-burden districts and went nationwide from March 2025, detecting more than 2.6 million TB episodes including over 900,000 asymptomatic; the Ni-kshay Poshan Yojana transfer was doubled from ₹500 to ₹1,000 per month, domestic TB funding rose from about US\$350 million (2018) to about US\$710 million (2025–26), and AI-guided vulnerability mapping now covers 682 districts [PMID 41868818] [PMID 41868819].
- ▶ **Kala-azar and lymphatic filariasis show how an elimination target is defined and how it slips** — kala-azar elimination as a public health problem means fewer than one case per 10,000 population per year at block or sub-district level; Bangladesh reached it in 2016 and Nepal in 2023, while sustaining it in India turns on detecting post-kala-azar dermal leishmaniasis and on vector control, which has lagged. Lymphatic filariasis is still endemic in 345 districts against a 2030 elimination goal, with public–private partnership proposed to strengthen mass drug administration and post-MDA surveillance [PMID 40577231] [PMID 41951149].

Guideline and legislative updates

- ▶ **India now has a defensible cost-effectiveness threshold for the objective-setting and priority-fixing steps:** the first national willingness-to-pay estimate, from 21,640 observations, put mean willingness to pay per quality-adjusted life-year at ₹2,12,307 (about US\$2,535), ranging ₹1,70,414–₹2,58,985 across health states, or 1 to 1.52 times GDP per capita. Topic selection for Health Technology Assessment in India works at two levels — user departments such as the National Health Mission, the tuberculosis programme and the National Health Authority nominate topics on clinical effectiveness, policy relevance and budget impact, and HTAIn then re-prioritises; returns on investment in HTAIn have been estimated at 5:1 to 40:1 [PMID 41176029] [PMID 40348399].

Global and WHO developments

- ▶ WHO's Fourteenth General Programme of Work (GPW 14) is the current global planning frame, with 98 outcome indicators of which 43 are health-related SDG indicators mapped to the Triple Billion targets. Across 194 Member States for 2015–2024, 8 of 27 targets carried more than 90 per cent missing data points, and only target 3.2 (under-five and neonatal mortality) was near-complete at 10 per cent missing — a reminder that step 1 situation analysis is bounded by what the information system actually returns [PMID 42058604].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 23 (Health Planning and Management, pp 1025–1031), Ch 11 (Nutrition and Health, Table 12 p 764 and pp 785–786), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, p 627), Ch 8 (Demography and Family Planning, p 611), Ch 7 (Health Programmes in India, p 556); Reinke — Health Planning for Effective Management, Ch 1 (The Process, Structure, and Functions of Planning, pp 24–30); WHO Expert Committee on National Health Planning in Developing Countries (definition of evaluation, as quoted in Park 28e); National Family Health Survey-5 (NFHS-5), 2019–21.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, J Family Med Prim Care 2026, doi:[10.4103/jfmpc.jfmpc.1966.24](https://doi.org/10.4103/jfmpc.jfmpc.1966.24); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, J Family Med Prim Care 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 38715207 — Establishing a Health Technology Assessment Evidence Ecosystem in India's Pradhan Mantri Jan Arogya Yojana, Health Syst Reform 2023, doi:[10.1080/23288604.2024.2327097](https://doi.org/10.1080/23288604.2024.2327097) (abstract-only); 39901042 — A population-level impact assessment of the National Health Mission on maternal and child health outcomes in India, J Public Health Policy 2025, doi:[10.1057/s41271-025-00558-w](https://doi.org/10.1057/s41271-025-00558-w) (abstract-only); 41017888 — Human resource compliance with Indian Public Health Standards 2022: an assessment of Health and Wellness Centers in a district of northwestern India, Indian J Community Med 2025, doi:[10.4103/ijcm.ijcm.636.24](https://doi.org/10.4103/ijcm.ijcm.636.24); 41868818 — From DOTS to DOST: a new framework for TB elimination in India, Lancet Reg Health Southeast Asia 2026, doi:[10.1016/j.lansea.2026.100721](https://doi.org/10.1016/j.lansea.2026.100721); 41868819 — TB far from elimination: an enduring reality in the southeast Asia region, Lancet Reg Health Southeast Asia 2026, doi:[10.1016/j.lansea.2026.100750](https://doi.org/10.1016/j.lansea.2026.100750); 40577231 — Challenges in sustaining the elimination of visceral leishmaniasis in India, Int Health 2025, doi:[10.1093/inthealth/ihaf063](https://doi.org/10.1093/inthealth/ihaf063); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, Acta Trop 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 41176029 — Estimating the willingness-to-pay per quality-adjusted life-year to aid health technology assessment in India, Value Health 2026, doi:[10.1016/j.jval.2025.10.007](https://doi.org/10.1016/j.jval.2025.10.007) (abstract-only); 40348399 — Prioritisation of candidates for HTA: insights from the Indian healthcare landscape, BMJ Evid Based Med 2025, doi:[10.1136/bmjebm-2023-112566](https://doi.org/10.1136/bmjebm-2023-112566); 42058604 — Data gaps affecting health-related sustainable development goal indicators, Bull World Health Organ 2026, doi:[10.2471/BLT.25.294190](https://doi.org/10.2471/BLT.25.294190). Non-journal sources: National Health Accounts estimates for India 2021–22; Indian Public Health Standards 2022; MoHFW Ayushman Arogya Mandir portal (figures as of 30 November 2024); WHO Fourteenth General Programme of Work (GPW 14) results framework.

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Community needs assessment in health projects

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — Community needs assessment (also called **community health assessment**) is a procedure used to collect data describing the needs and strengths of a defined group, community or population, so that the health programme can be tailored to the needs and distinctive characteristics of that group (Health Program Planning and Evaluation, Ch 1, p 48; Ch 3, p 100). It is the earliest substantive phase of the programme planning and evaluation cycle, and its goal is to guide decisions on **problem prioritisation and programme development**. In Indian public health practice it is also the formal name of the **Community Needs Assessment Approach (CNAA)**, the bottom-up, target-free planning method adopted in the family welfare/Reproductive and Child Health programme (Sunder Lal, Ch 10, p 574).

1. Its place in the health planning cycle

Planning is "a process of analysing a system, or defining a problem, assessing the extent to which the problem exists as a need, formulating goals and objectives to alleviate those identified needs, examining and choosing from among alternative intervention strategies, initiating action for implementation, monitoring, and evaluating the results in the light of stated objectives" (Park 28e, Ch 23, pp 1026-1027). Needs assessment is the engine of steps 1 and 4.

Step of the planning cycle	Contribution of community needs assessment
1. Analysis of the health situation	The assessment itself — supplies the health problems, health needs and health demands of the population
2. Establishment of objectives and goals	Converts each prioritised need into a measurable objective and target
3. Assessment of resources	Manpower, money, materials, skills, knowledge and techniques are balanced against assessed need
4. Fixing priorities	Needs ranked, since resources always fall short of total requirement
5. Write-up of formulated plan	Need statement becomes the justification section of the project document
6. Programming and implementation	Assessed workload fixes staffing, supplies and schedules
7. Monitoring	Day-to-day follow-up against the planned coverage of the assessed population
8. Evaluation	Post-project measurement is made against the baseline established by the assessment

(Park 28e, Ch 23, pp 1026-1028)

Park's parallel concept is **community diagnosis** — identification of the basic health needs and health problems of the community, the needs *as felt* by the community being investigated and listed in order of priority; **community treatment** is then the sum of steps decided upon to meet those needs, taking into account the resources available and the wishes of the people (Park 28e, Ch 2, pp 67-68).

2. Types of need to be assessed (Bradshaw, 1972)

Type of need	What it is	How it is measured
Normative need	Deficit or excess defined by experts against a scientific norm	Infant mortality rate above the national average; body mass index above 29.9
Expressed need	Need revealed by health-care-seeking behaviour (demand)	Number requesting services, types of service sought, utilisation rates
Perceived (felt) need	Sense of lack as experienced by the target audience	What the community says it wants; stated deficits
Relative (comparative) need	Gap identified by contrasting an advantaged with a disadvantaged group	Most health disparities, e.g. one group's infant mortality rate twice another's

(*Health Program Planning and Evaluation, Ch 3, pp 91-92*)

A balanced assessment also records **community strengths** — community competence, community resilience and community assets (schools, self-help groups, local institutions) — not needs alone (ibid, Ch 3, p 92).

3. Steps in conducting the assessment, and the data used

- ▶ **Involve community members** in designing and executing the assessment — mandatory where the funder requires it, where insiders' perceptions are needed for completeness, or where the community is to own sustained implementation.
- ▶ **Define the community/population** to be assessed — the basic unit of planning (in India, the village or census ward).
- ▶ **Decide what data to collect** — magnitude of the problem, precursors/determinants, and demographic and behavioural characteristics.
- ▶ **Collect data from multiple sources** (table below).
- ▶ **Analyse** using epidemiological and statistical procedures to obtain rates and comparisons.
- ▶ **Write a health problem statement** summarising the problem, the at-risk group and its causal and mediating factors.
- ▶ **Prioritise** the problems by an explicit, agreed method (Section 4).

(*Health Program Planning and Evaluation, Ch 3, pp 106-111; Ch 4, p 120*)

Source	Content	Caveat
Public/secondary data	National surveys, vital statistics, census social indicators	May be several years old; may not reflect current local problem
Primary data	Interviews, household surveys, community forums, focus groups, key-informant and provider interviews	Data from programme participants are rarely used as the sole source — participants are already enrolled and so may be dissimilar to the audience the programme targets
Archival and proprietary data	Service registers, hospital and insurer databases, medical records	Limited to what was already collected

Source	Content	Caveat
Published literature	Comparison against professional standards and national goals	Best for normative and relative need
"Beyond the street lamp" data	Unobtrusive observation (e.g. walking a locality to judge safety), local media	Not readily available; needs deliberate effort

(*Health Program Planning and Evaluation, Ch 4, pp 120-122*)

Park's **minimum essential data for health planning** are: (a) population with age and sex structure; (b) morbidity and mortality statistics; (c) epidemiology and geographical distribution of diseases; (d) medical care facilities — hospitals, health centres and other health agencies, public and private; (e) technical manpower of various categories; (f) training facilities available; and (g) attitudes and beliefs of the population towards disease, its cure and prevention (Park 28e, Ch 23, pp 1026-1027). Need and resource data must both be translated into **service terms** if they are to be compared and reallocated rationally (Reinke, Ch 5, p 82).

4. Prioritising the assessed needs

Method	Basis	Key detail
Hanlon's Basic Priority Rating System (BPRS)	Objective scoring	$BPR = (A + 2B) \times C$; A = size/magnitude, B = seriousness (urgency, severity, economic consequences, willingness/involvement of others), C = intervention effectiveness; each factor rated on a 1-to-10 scale (size: 1 = small to 10 = endemic)
PEARL screen	Feasibility filter applied to the high-priority problems after BPRS	Propriety, Economics, Acceptability, Resources, Legality; each dimension scored yes (1) or no (0)
Nominal group technique	Structured group voting	Round-robin: three votes each to shortlist, then one vote each to select the problem
Delphi technique	Anonymous expert consensus	Median of ratings fed back over repeated rounds; seldom more than 3-4 rounds needed
Importance × changeability	Simple two-way grid	High importance + high changeability = first priority

(*Health Program Planning and Evaluation, Ch 4, pp 138-143; Reinke, Ch 15, p 219*)

5. Translating the assessment into a formulated health project

Element of the problem statement	Becomes in the project	Becomes in the evaluation
Risk of: health problem or condition	Programme goal	Outcome variables
Among: at-risk population or group, target audience	Recipients	Intervention group
As demonstrated in: health indicators	Programme objectives	Outcome and impact variables

Element of the problem statement	Becomes in the project	Becomes in the evaluation
Resulting from causal factors: specific processes, conditions and factors	Interventions or treatments for the target population	Outcome evaluation
But is mediated by: factors that must be present for the health problem to occur	Possible intervention	Possible control variables
Given moderation of the causes by: factors that increase or decrease the potency of the causative factors	Possible intervention	Possible control variables
And required existing factors of: sociodemographic characteristics and social-ecological factors	Programme eligibility criteria	Control variables or comparison groups

(*Health Program Planning and Evaluation, Ch 4, pp 138-139*)

6. The Community Needs Assessment Approach (CNA) in India

Why the change. Family welfare achievement was formerly judged against centrally fixed contraceptive targets, which became ends in themselves. The top-down target approach had three drawbacks: user preference was not reflected in the targets and there was no authentic feedback on which contraceptive to promote in which area or age group; quality of care became secondary (e.g. screening for reproductive tract and sexually transmitted infections before intrauterine device insertion was compromised); and staff were tempted into false reporting — together inflating performance statistics and deteriorating service quality (Park 28e, Ch 8, p 614).

How it evolved. In 1995-96 one district in each of 18 states (with pilot projects in Kerala and Tamil Nadu) was made target free; the fixing of method-specific contraceptive targets was given up from **April 1996**. Because the calculations proved too complex for health workers to fix their own performance norms, the target-free manual was modified and renamed the **Community Needs Assessment Approach Manual** (Park 28e, Ch 8, p 614; Table 12, Ch 9, pp 659-660 dates the target-free approach to 1996 and RCH-1 to 1997).

How it works. Health workers work out their own targets by consulting clients and the community in their area, and CNA with **decentralised participatory planning (DPP)** became a listed component of the RCH programme (Sunder Lal, Ch 10, p 574).

- ▶ **Annual household survey**, conducted once a year in **February or March** — the surveyor is the ASHA and the anganwadi worker, with the auxiliary nurse midwife (ANM) as supporting/supervisory staff — recording total population, eligible couples, pregnant women, infants and young children; the register is updated yearly and identifies high-risk households.
- ▶ **Continuous surveillance** thereafter — registration of all marriages, pregnancies, births, deaths and eligible couples must be complete, since one-time survey data go stale.
- ▶ **Village health action plan** prepared by the Village Health, Sanitation and Nutrition Committee (sarpanch, ASHA, multipurpose health worker — female, school teacher, anganwadi worker, self-help group member), which holds an untied fund of Rs 10,000 per annum per village panchayat; **over 5.55 lakh such committees** have been made functional and given the untied grant.

- **Upward aggregation:** village plans → sub-centre action plan → primary health centre plan → community health centre/block plan → **District Health Action Plan** → State Programme Implementation Plan; a parallel **facility survey** maps available inputs at each level.

(Sunder Lal, Ch 2, p 72; Ch 10, pp 574-575; Ch 16, pp 697, 701-702, 705-706)

Advantages — needs-based participatory planning; community investment in planning, monitoring and surveillance; multisectoral participation; self-estimated goals by health workers; an integrated package of services; better quality of care (Suryakantha, Fig 43.11). **Limitations** — computation is demanding for field staff (Park 28e, Ch 8, p 614); planning groups tend to "frontload" the cycle, over-investing in data collection and analysis at the cost of implementation (Health Program Planning and Evaluation, Ch 3, p 111); and felt needs may conflict with the normative needs stated by providers, and must be reconciled before the project is written (ibid, Ch 4, pp 121-122).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Bottom-up (needs-based) planning pyramid.** Draw the tiers from the base upward and write the role opposite each: *community/village (below the sub-centre) — participate in goal-setting; Sub-centre — set goals; Primary health centre — facilitate implementation; District level — support and monitor implementation; State level — report results; National level (apex) — strategies and computations*; left-hand principles: *needs-based participatory planning, community investment in planning, monitoring and surveillance, multisectoral participation, self-estimated goals by health workers, integrated package of services, good quality of care, education of community.*
- **Diagram 2 — Bradshaw's four needs as overlapping circles.** Labels: *Normative need (expert-defined), Expressed need (demand/utilisation), Felt/perceived need (community-stated), Comparative need (disparity between groups)*; shade the central overlap as *"real" need for programme action.*

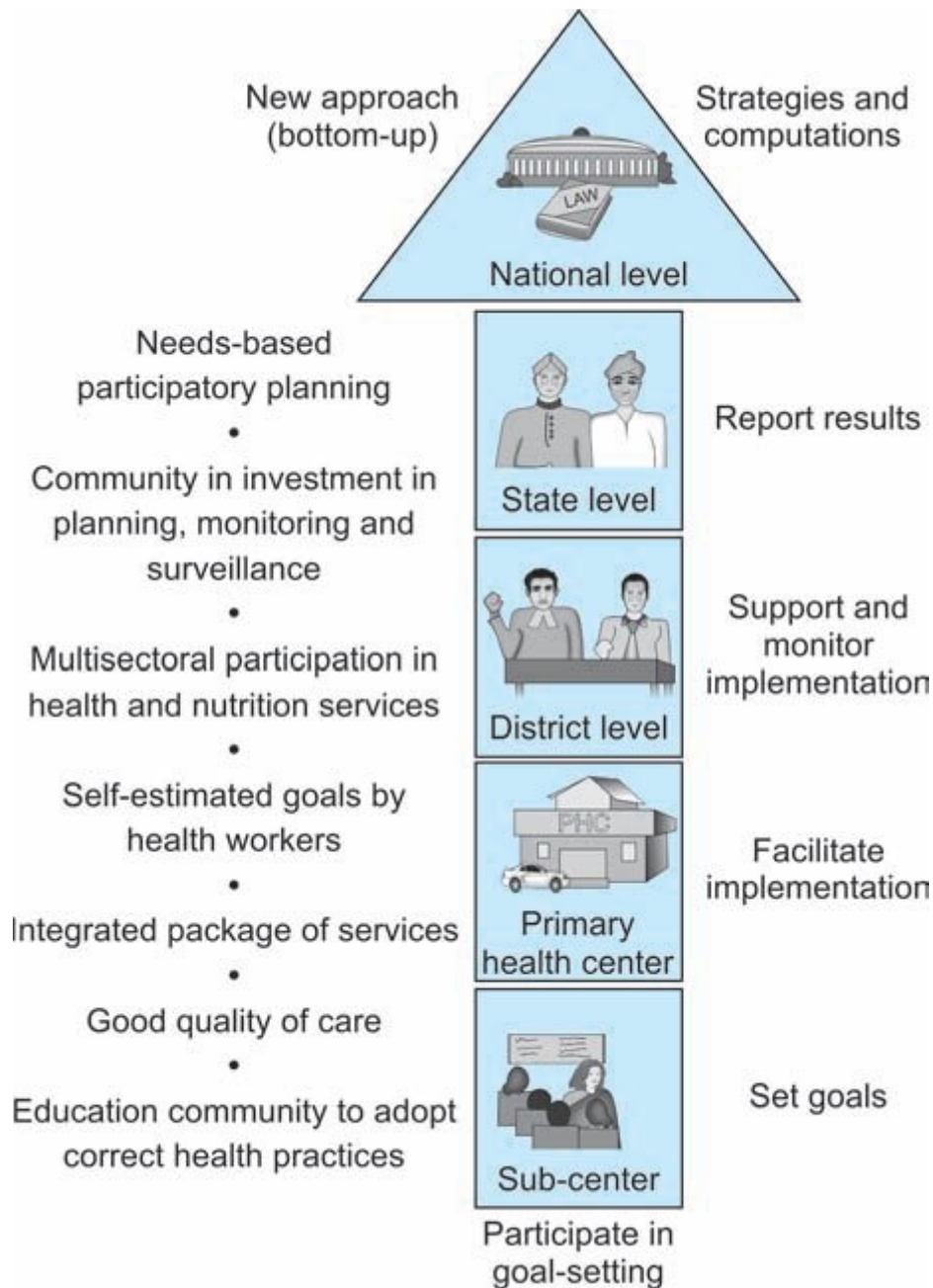


Fig 43.11 — 'Bottom-up' planning of RCH program (Suryakantha Environment Best, p.882)

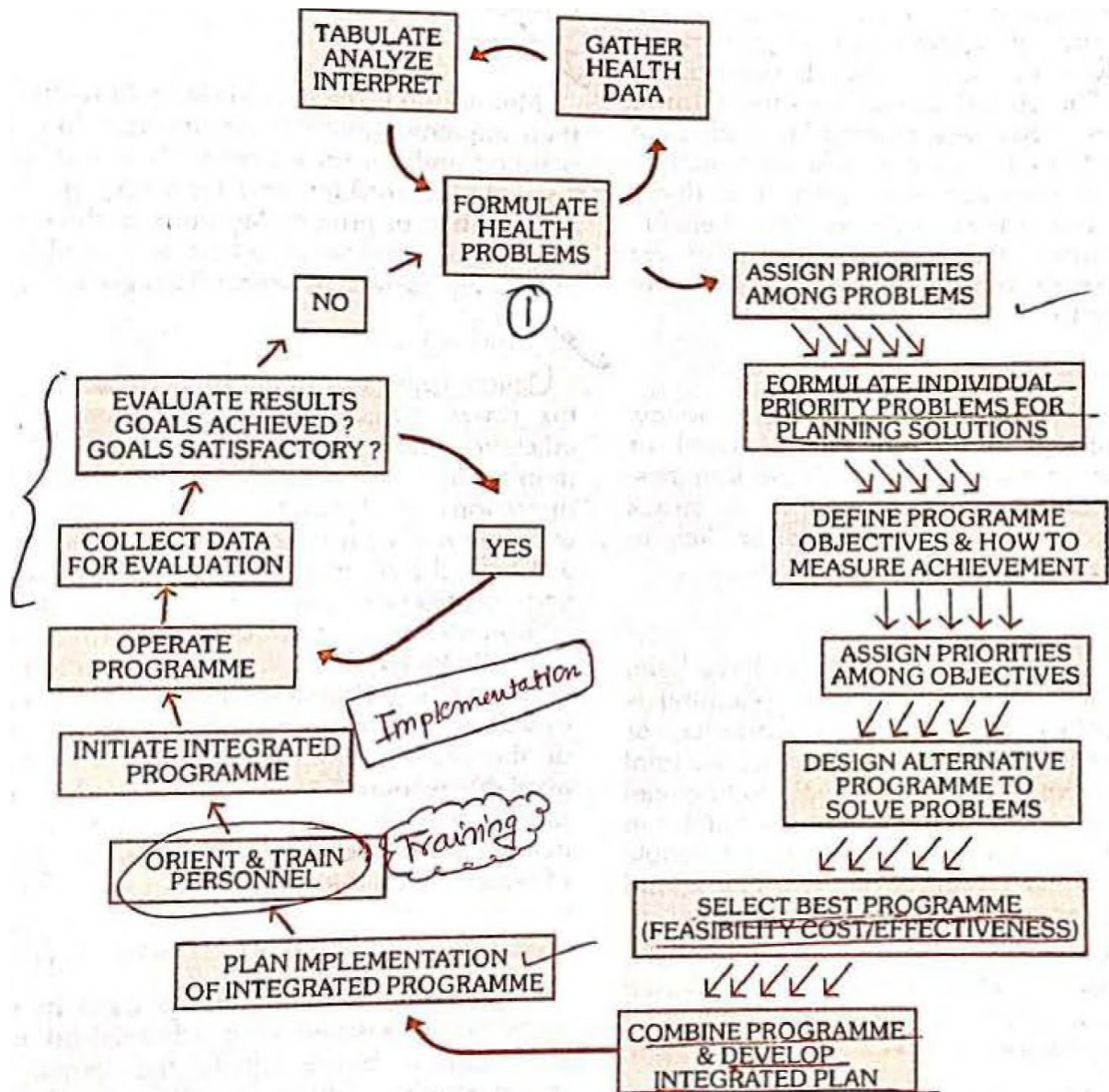
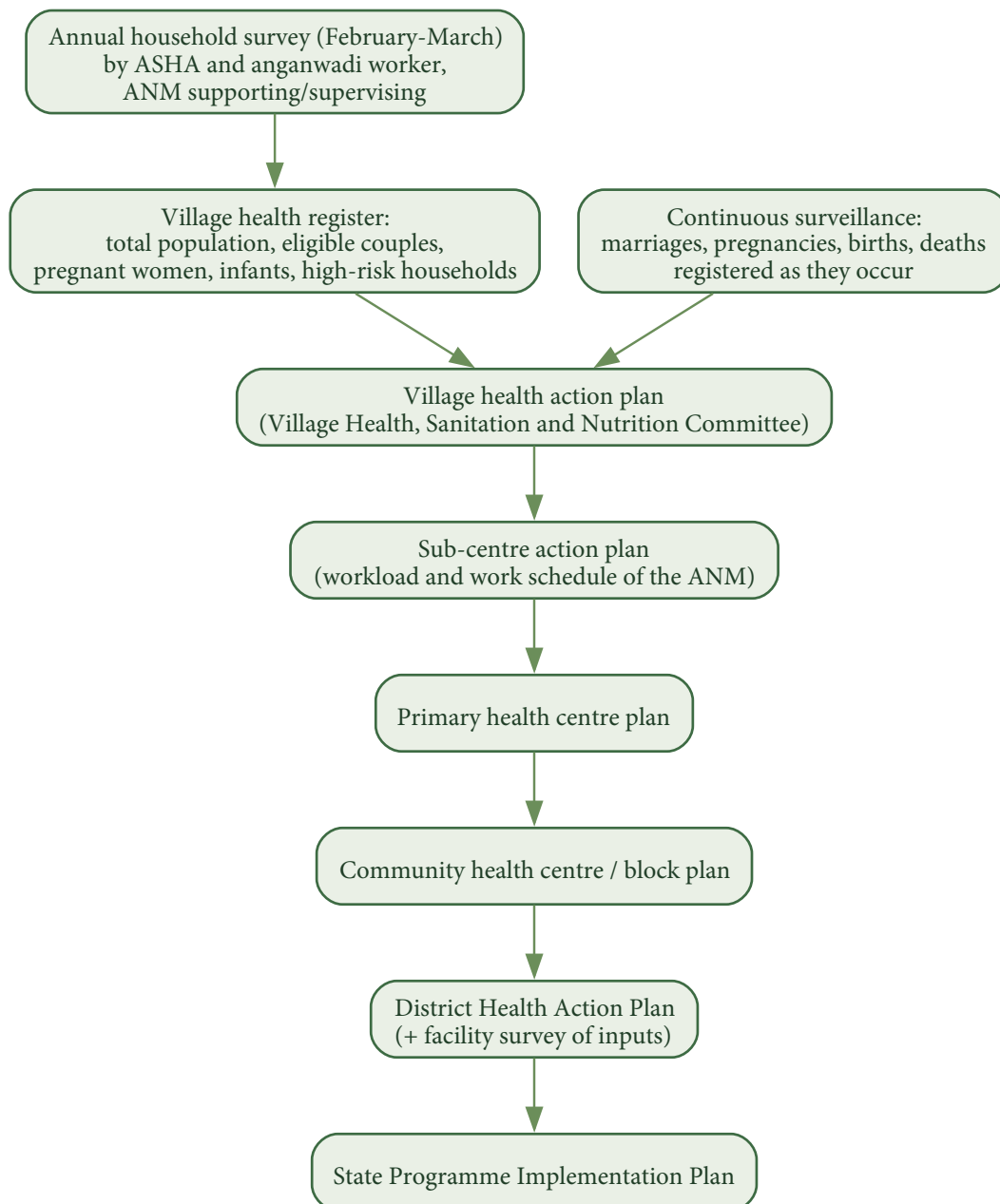
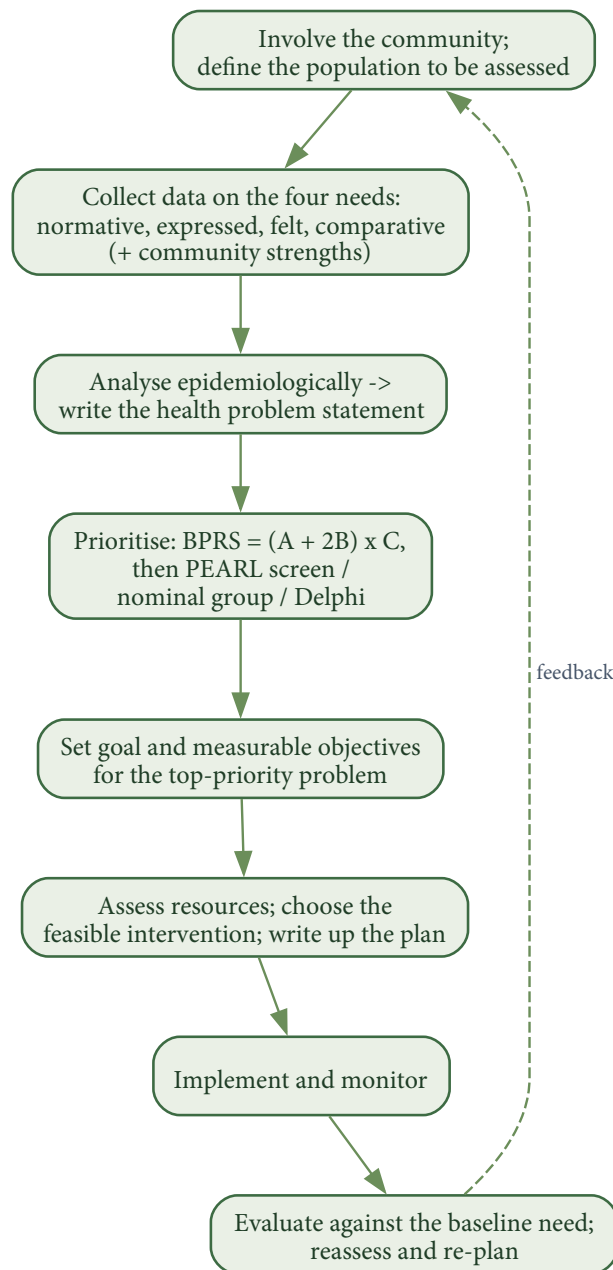


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, p.1026)



Community Needs Assessment Approach in India — bottom-up aggregation of village need into the district health action plan



From community needs assessment to a formulated health project

High-yield exam pointers

- ▶ **Bradshaw's four needs (1972)** — mnemonic NEFC: Normative, Expressed, Felt (perceived), Comparative (relative).
- ▶ **Hanlon's Basic Priority Rating System:** $BPR = (A + 2B) \times C$, each factor scored 1-10; A = size, B = seriousness, C = intervention effectiveness. Follow with PEARL (Propriety, Economics, Acceptability, Resources, Legality), scored 1/0.
- ▶ **Park's 8-step planning cycle:** analysis of health situation → objectives and goals → assessment of resources → fixing priorities → write-up of plan → programming and implementation → monitoring → evaluation.
- ▶ **Indian dates:** method-specific contraceptive targets abandoned **April 1996** (pilot 1995-96 in one district of each of 18 states); target-free manual renamed the **Community Needs Assessment**

Approach Manual; CNAA + decentralised participatory planning is a component of the RCH programme (RCH-1, 1997).

- ▶ **Annual household survey is done in February-March by ASHA and the anganwadi worker under ANM supervision;** village plan → sub-centre → PHC → block → **District Health Action Plan** → State Programme Implementation Plan.
- ▶ **VHSNC untied fund = Rs 10,000 per annum per village panchayat;** over 5.55 lakh committees functional; committee = sarpanch, ASHA, multipurpose health worker (female), school teacher, anganwadi worker, self-help group member.
- ▶ **Park's minimum essential data for planning:** population structure; morbidity and mortality; epidemiology and geographical distribution of disease; medical care facilities (public and private); technical manpower; training facilities; community attitudes and beliefs.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: the term "community needs assessment" carries two linked senses — the generic programme-planning phase described in the health-management literature, and the specific Indian **Community Needs Assessment Approach** that replaced centrally fixed contraceptive targets; a good answer names both, as the RGUHS stem asks about health *project* formulation generally. Sunder Lal's opening chapter records the committee as since reconstituted — a standing subcommittee of the panchayat with a minimum of 15 members, the village panch (mahila panch, scheduled caste preferred) as chairperson and the ASHA as member secretary, over 5.55 lakh constituted by March 2021 (Sunder Lal, Ch 1, p 42). The Rs 10,000 untied fund and the count of functional Village Health, Sanitation and Nutrition Committees are as stated in Sunder Lal's textbook and should be quoted as textbook figures rather than as a current-year value.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoHFW Ayushman Arogya Mandir programme data, National Health Accounts 2021-22, NSSO 2021-22, WHO UHC Service Coverage Index, IPHS 2022

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandir (AAM)** — the sub-centre and PHC tier that the Community Needs Assessment Approach plans for has been renamed and re-scoped: more than 1.75 lakh AAM operational, a new Community Health Officer / mid-level health provider cadre 1,16,978 strong in position as of 30 November 2024 leading the primary-care team with clinical, public-health and managerial roles, and 12 expanded service packages of which 82% of AAM were delivering all (up from 50% in March 2023) [PMID 41816146].
- ▶ **The bottom-up plan chain has gained a facility-level rung.** 1,35,343 Jan Arogya Samitis (JAS) stood constituted at AAM by 30 November 2024, created to strengthen people's participation in planning and monitoring primary-care delivery. The direction of flow is upward, as in Section 6: the **monthly village-level Samvad** (consultation) held in the villages under an AAM feeds the **Arogya Sabha** that the JAS facilitates twice every year at the AAM. JAS therefore sits *alongside* — not above — the Village Health, Sanitation and Nutrition Committee (rural) and Mahila Arogya

Samiti (urban) community platforms described above; the review's own recommendation is to strengthen the community-based MAS/VHSNC platforms together with the community-driven facility-based platform of JAS. Untied money is unchanged (Rs 10,000 per sub-centre, Rs 25,000 per PHC for local health action plus a Rs 50,000 annual maintenance grant per PHC, Rs 1 lakh to the Rogi Kalyan Samiti), but two new streams — the **PM-Ayushman Bharat Health Infrastructure Mission** and **XV Finance Commission** health grants routed through Panchayati Raj Institutions and urban local bodies — now finance the infrastructure gaps the assessment identifies, benchmarked against **IPHS 2022** [PMID 41816146].

- **NPCDCS was renamed the National Programme for Non-Communicable Diseases (NPNCD) in 2023.** Its household instrument, the **Community Based Assessment Checklist (CBAC)** introduced by the Government of India in 2016, is administered by the ASHA (or the ANM where no ASHA is available) to every resident aged 30 years and above; it scores six risk factors — age, family history of high blood pressure/diabetes/heart disease, tobacco, alcohol, physical inactivity and waist circumference — out of 10, and an individual scoring above 4 is treated as high risk and is then prioritised and actively mobilised for blood sugar and blood pressure assessment at the nearest primary health care centre. The same checklist also covers oral, breast and cervical cancer, chronic obstructive pulmonary disease, asthma, tuberculosis and leprosy [PMID 42555696].

Linking comment (mine, not a claim of the cited papers): for the NCD component, this annual CBAC household round is the practical counterpart of the family-welfare annual household survey described in Section 6 — the same enumerate-then-risk-list logic, applied to a different disease group.

Current national data

- **India's UHC Service Coverage Index rose from 57 (2015) to 63 (2021)**, above the South-East Asia regional average of 62; **out-of-pocket expenditure fell from 60.6% (2014-15) to 39.4% (2021-22)** per National Health Accounts; and public-facility utilisation rose from 91.6 to 134.5 crore outpatient and 4.3 to 6.3 crore inpatient episodes between 2012-13 and 2021-22 per NSSO — the current normative-need benchmarks against which a district plan is now justified [PMID 41816146].
- **Assessment output is measurable at national scale.** Cumulative screening at AAM to 30 November 2024: 84.91 crore persons for hypertension, 74.24 crore for diabetes, 48.98 crore for oral cancer, 21.8 crore women for breast cancer and 14.31 crore for cervical cancer; the facility denominator (Health Dynamics of India 2022-23) is 1,69,615 sub-centres and 31,882 PHCs, against 1,53,655 and 25,308 respectively in 2014-15 [PMID 41816146].

Elimination, surveillance and disease status

- **Episodic surveys are being supplemented by continuous surveillance.** NFHS, the Comprehensive National Nutrition Survey, ICMR-INDIAB and WHO STEPS run at five-yearly or longer intervals with delays between collection and dissemination frequently exceeding two to three years — too slow for routine institutional feedback, early detection of emerging risk patterns or rapid programme adjustment. An ICMR-National Institute of Nutrition life-course, institute-based surveillance system for overweight, obesity and metabolic risk is being built across kindergartens, schools, colleges and primary-care platforms in the Hyderabad and Medchal-Malkajgiri districts of Telangana, with the first surveillance round scheduled for **June**

2026 and analytical outputs expected in 2027. Within that protocol the Health and Wellness Centre (AAM) functions as a **data aggregation, validation and counselling hub**, with ABDM/ABHA identifier linkage proposed where feasible, subject to ethical approvals and data-governance safeguards [PMID 41973943].

Guideline and legislative updates

- ▶ **The CBAC's discriminative ability is poor and its cut-off is under challenge.** In what its authors describe as likely the first large-scale community-based validation of the tool (32,686 residents aged 30+ who did not already report diabetes or hypertension, five urban PHC catchment areas in Mysuru, screened 2019-20), CBAC yielded an AUC of 0.62 (95% CI 0.61-0.63) against random capillary blood glucose ≥ 140 mg/dL and 0.61 (95% CI 0.60-0.61) against BP $\geq 140/90$ mmHg; at the mandated cut-off above 4 sensitivity was only 63.0% for diabetes and 58.6% for hypertension, and the authors recommend that trained frontline health staff administer random blood sugar and blood pressure tests alongside the checklist rather than triage on the score alone [PMID 42555696]. An earlier facility-based study in Jodhpur, Rajasthan (942 attenders aged 30+ at three rural health training centres) found CBAC and the Indian Diabetes Risk Score statistically indistinguishable as metabolic-syndrome predictors (AUC 0.73 each; difference between AUCs $p = 0.833$), CBAC at ≥ 4 giving 84.7% sensitivity but only 48.8% specificity — the authors judged both inadequate to qualify as metabolic-syndrome screening tools [PMID 36972242].
- ▶ **Felt need is only as good as community awareness.** In a qualitative evaluation of the cancer component of NPCDCS in Meghalaya (56 health staff, 20 in-depth interviews and 4 focus group discussions, July-October 2021), frontline health workers in **one of the two study districts — East Jaintia Hills** — reported that public participation in filling the CBAC form was low because those screened were reluctant to share personal information such as substance use, and women were also hesitant to disclose abnormal vaginal discharge; the result was missing data and delayed identification of high-risk individuals. This was reported from East Jaintia Hills and not from the other study district, East Khasi Hills, and the study covered only **2 of Meghalaya's then 11 districts**, which the authors state limits generalisability. Hypertension and diabetes had also taken priority over the cancer component, and the village health council — community representatives already working on tribal community health — is proposed as the bridge between the health system and the community [PMID 41648973].
- ▶ **Population norms are a weak allocation rule where terrain matters.** In Chamoli and Rudrapur (Uttarakhand), enforcing the one-PHC-per-20,000 hill norm required 18 new PHCs at an incremental cost-effectiveness ratio of Rs 19,671 per additional person covered, whereas a coverage-optimised placement of only 6 PHCs reached 96.0% coverage at Rs 8,373 per additional person covered and dominated all alternatives — explicit coverage-cost optimisation, not headcount, should drive AAM siting in difficult geography [PMID 42482081].

Global and WHO developments

- ▶ **Priority setting is moving beyond cost-effectiveness to explicit equity and financial-protection criteria.** Modelling 30 priority infectious-disease interventions for Ethiopia's essential benefits package, an incremental annual coverage rise of 10 percentage points would gain 390,000 health-adjusted life-years and avert 65,000 cases of catastrophic health expenditure; 22% of the health

benefit would accrue to the poorest income quintile against 20% to the richest, but a third of the catastrophic-expenditure cases averted would be among the poorest. The authors conclude that examining these trade-offs and weighing each outcome enables tailored prioritisation of interventions within the package [PMID 41707995].

Linking comment (mine, not a claim of the cited paper): read this as the distributional counterpart to the Hanlon magnitude-and-seriousness scoring in Section 4 — a line worth adding if the stem asks how priorities are set today.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — community diagnosis and community treatment; Ch 8 — Community Needs Assessment Approach and the target-free approach; Ch 9 Table 12 — milestones; Ch 23 — planning cycle, minimum essential data and evaluation of health services); Health Program Planning and Evaluation: A Practical, Systematic Approach (Ch 1, Ch 3 — types of need, steps in community health assessment, frontloading of the planning cycle; Ch 4 — data sources, health problem statement, Table 4-6, BPRS and PEARL); Sunder Lal — Textbook of Community Medicine (Ch 1 — reconstituted VHSNC; Ch 2 — household survey register; Ch 10 — CNAA, annual household survey timing and decentralised participatory planning; Ch 16 — village to district health action plan, Table 16.6 implementation matrix); Reinke — Health Planning for Effective Management (Ch 5 — matching need to resources; Ch 15 — Delphi and nominal group techniques); Suryakantha — Community Medicine with Recent Advances (Fig 43.11 — bottom-up planning of the RCH programme).

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmprc.jfmprc.478.25](https://doi.org/10.4103/jfmprc.jfmprc.478.25) (PMC12975087, full text); 42555696 — Screening performance of community-based assessment checklist compared with point of care screening tests to identify high-risk individuals for diabetes and hypertension in urban Mysore, India, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR.78.2026](https://doi.org/10.25259/IJMR.78.2026) (PMC13466864, full text); 41973943 — Life-course framework for institute-based surveillance of overweight, obesity and metabolic risk factors in India: protocol for surveillance system development, *JMIR Research Protocols* 2026, doi:[10.2196/86084](https://doi.org/10.2196/86084) (PMC13075776, full text); 36972242 — Application of Indian Diabetic Risk Score (IDRS) and Community Based Assessment Checklist (CBAC) as metabolic syndrome prediction tools, *PLoS One* 2023, doi:[10.1371/journal.pone.0283263](https://doi.org/10.1371/journal.pone.0283263) (PMC10042346, full text); 41648973 — Barriers in implementing cancer prevention programme in North Eastern India: a case study from Meghalaya, *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR.742.2025](https://doi.org/10.25259/IJMR.742.2025) (PMC12883105, full text); 42482081 — Cost-effective primary health-centre allocation in Himalayan terrain: a spatial and resource-allocation analysis of Chamoli and Rudraprayag districts, India, *Cost Effectiveness and Resource Allocation* 2026, doi:[10.1186/s12962-026-00778-1](https://doi.org/10.1186/s12962-026-00778-1) (no PMC record — abstract-only, bullet kept because every figure quoted appears in the abstract); 41707995 — Incorporating equity and financial protection criteria into essential benefits package design: a case study of infectious diseases interventions in Ethiopia, *Value in Health* 2026, doi:[10.1016/j.jval.2026.01.027](https://doi.org/10.1016/j.jval.2026.01.027) (no PMC record — abstract-only, same basis). Programme figures cited from the MoHFW Ayushman Arogya Mandir portal (30 November 2024), National Health Accounts 2021-22, NSSO 2021-22, Health Dynamics of India 2022-23 and IPHS 2022 as reported in those papers. The two italic *Linking comment* lines in the Recent advances section are the answer author's bridge back to the textbook body and are deliberately outside every PMID tag — no cited paper makes those claims.

♦ ♦ ♦

Evaluation of health services

Asked in: Paper IV 19-Nov-2021 — RGUHS MD Community Medicine

Definition — Evaluation is the systematic assessment of the extent to which a health service has achieved its stated objectives, at what cost, and with what acceptance by those involved. The WHO Expert Committee on National Health Planning in Developing Countries defined it as measuring "the degree to which objectives and targets are fulfilled and the quality of the results obtained... the productivity of available resources in achieving clearly defined objectives... how much output or cost-effectiveness is achieved", thereby making possible reallocation of priorities and resources on the basis of changing health needs (Park 28e, Ch 23, p 1027). Evaluation is distinct from **monitoring**: monitoring is the day-to-day follow-up of activities at the input and process stages, whereas evaluation is concerned with the final outputs and outcomes and the factors associated with them (Park 28e, Ch 23, p 1027; Sunder Lal, Ch 16, p 723).

1. Monitoring versus evaluation — deciding the frame

Feature	Monitoring	Evaluation
Timing	Continuous, day-to-day, during implementation	Periodic/terminal — progress and efficiency at least yearly, impact at ≥ 5 years from start
Level of the results chain	Inputs and processes	Outputs, effects and impacts
Question asked	Are activities on schedule and as planned?	Were the objectives achieved, at what cost, should the service continue, be modified or be abandoned?
Data source	Routine monthly sub-centre / PHC / CHC reports, supervisory field visits, Gantt chart, community monitoring	Coverage evaluation surveys, special surveys, SRS, NFHS, sentinel surveillance, published reports
Purpose	Detect deviation early and take mid-course corrective action	Improve the system, strengthen infrastructure, guide allocation of resources for current and future programmes

(Sunder Lal, Ch 16, pp 722-725)

2. Step 1 — determine what is to be evaluated: the three types

The classical approach, introduced by *Donabedian* (1966), evaluates a health service as a system of **structure, process and outcome** (Health Program Planning and Evaluation, Ch 10, pp 276-277; Park 28e, Ch 23, pp 1037-1038).

Type	What is assessed	Yardstick / tool	Illustrative criterion
Structure	Whether facilities, equipment, manpower and organisation meet a standard accepted by experts as good	Indian Public Health Standards (IPHS) 2022 — "essential" (minimum acceptable functional standard) and "desirable" (aspirational) norms for Sub-Health Centre and Primary Health Centre level Ayushman Arogya Mandir, community health centre, sub-district and district hospital	Physical facilities and equipment; sanctioned versus in-position staff
Process	The way the activities of the service are carried out — recognition of the problem, diagnostic procedures, treatment and clinical management, care and prevention — compared with a predetermined standard	Medical (or nursing) audit — an objective and systematic way of evaluating physician or nurse performance	Every antenatal mother must receive 6 check-ups; every laboratory technician must examine 100 blood smears
Outcome	The end result — whether persons using the service experience measurable benefit such as improved survival or reduced disability	The traditional outcome components, the " 5 Ds of ill-health — disease, discomfort, dissatisfaction, disability, death	Alteration in patient health status (cured, dead, disabled); satisfaction/dissatisfaction; cessation of smoking, acceptance of small family norm

(Park 28e, Ch 23, pp 1037-1038; IPHS 2022 as summarised in Park 28e, Ch 10, p 728)

3. The seven general steps of an evaluation

- ▶ **Determine what is to be evaluated** — a programme, a service, a health centre or a hospital; and whether structure, process or outcome is the focus.
- ▶ **Establish standards and criteria** — a prerequisite for evaluation, developed to match the focus chosen: structural criteria, process criteria and outcome criteria as in the table above.
- ▶ **Plan the methodology** — prepare a format for gathering the information wanted; standards and criteria must be built in at this planning stage.
- ▶ **Gather information** — mortality and morbidity statistics plus the political, cultural, economic, environmental and administrative factors influencing the health situation, and the extent, scope and use of health systems, services and institutions. The amount of data collected depends on the purpose of the evaluation.
- ▶ **Analyse the results** — analysis, interpretation and feedback to all concerned within the shortest feasible time, with opportunity provided for discussing the results.
- ▶ **Take action** — actions designed to support, strengthen or modify the services, which may mean shifting priorities, revising objectives, or developing new services to meet previously unidentified needs.
- ▶ **Re-evaluate** — evaluation is an ongoing process aimed at rendering health activities more relevant, more efficient and more effective.

(Park 28e, Ch 23, pp 1037-1038)

4. Elements (components) of the evaluation process

These seven elements are the substance of the answer — each is a question I would ask of the service, with the measurement I would use.

Element	Question asked of the service	How I would measure it
Relevance (requisiteness)	Is the service needed at all, in relation to essential human needs, priorities and policies? If there is no need, the service can hardly be of value	Comparison of services offered against the current disease burden and national health goals — e.g. smallpox vaccination is now irrelevant because the disease no longer exists
Adequacy	Has sufficient attention been paid to previously determined courses of action?	Staff, supplies, equipment and budget against workload, targets and IPHS norms — e.g. staff allocated to a programme is inadequate if the quantum of workload was not considered
Accessibility	What proportion of the given population can be expected to use the facility?	Barriers analysed as physical (distance, travel time), economic (travel cost, fee charged) and social/cultural (caste or language barrier)
Acceptability	A service may be accessible yet not acceptable to all	Uptake of male sterilisation, uptake of screening for rectal cancer; client-satisfaction studies
Effectiveness	To what extent is the underlying problem prevented or alleviated — the degree of attainment of predetermined objectives and targets	Ultimate measure is reduction in morbidity and mortality rates
Efficiency	How well are resources — money, men, material and time — used to achieve a given effectiveness?	Immunisations given in a year against an accepted norm; percentage bed occupancy; cost per day in hospital; cost per patient treated
Impact	The overall effect of the service on health status and socio-economic development	Malaria control in India lowered malaria incidence <i>and</i> improved agricultural, industrial and social life; 100% immunisation must also reduce or eliminate vaccine-preventable disease; village water supply must reduce diarrhoeal disease

(Park 28e, Ch 23, pp 1037-1038)

5. Indicators used, by level of the results chain

Level	Indicator type	Examples I would use
Inputs	Resource availability	Doctor–population ratio, doctor–nurse ratio, population–bed ratio, population per health centre/sub-centre, population per trained birth attendant; percentage of sub-centres without essential drugs or weighing machine; money spent against sanctioned budget (Park 28e, Ch 2, p 39; Sunder Lal, Ch 16, p 723)

Level	Indicator type	Examples I would use
Processes	Activity indicators	Outreach/Village Health, Sanitation and Nutrition Day sessions held against planned; antenatal clinic days held against planned; home visits made against schedule
Outputs	Service-utilisation indicators — the proportion of people in need of a service who actually receive it in a given period, usually a year	Proportion of infants fully immunised; proportion of pregnant women receiving antenatal care or delivery by a trained attendant; bed-occupancy rate (average daily in-patient census ÷ average number of beds); average length of stay (days of care rendered ÷ discharges); bed turnover ratio (discharges ÷ average beds) (Park 28e, Ch 2, p 39)
Effects	Knowledge, attitude, practice and coverage	Percentage of eligible couples using contraception; proportion of mothers initiating breastfeeding within one hour of birth; proportion of institutional deliveries
Impacts	Health status change	Decline in incidence of diarrhoea and acute respiratory infection; decline in under-5 underweight; decline in birth rate, death rate and infant mortality rate — from the Sample Registration System, National Family Health Survey and special surveys

An indicator chosen for evaluation must be **valid, reliable, sensitive, specific, feasible and precise**, and the list must be kept short and specific (Sunder Lal, Ch 16, p 723). Where the whole system is being appraised rather than one service, the WHO framework of six health-system building blocks — service delivery, health workforce, health information and research, medicines and technologies, health financing, and governance/leadership — provides the headings (Oxford Textbook of Public Health 7e, Ch 51, pp 1877-1878).

6. Designs and management techniques used to carry out the evaluation

Design (E = experimental group, C = control, R = random allocation, O = observation, X = intervention)	Notation	Comment
One-time study	E X O	Commonest, fast, inexpensive; no baseline, no control — weakest
One group before-and-after	E O X O	Fast and inexpensive; no control group, no randomisation
Time-series study	E O O O X O O O	Shows impact over time; no control group, no randomisation
Non-equivalent control ("with and without")	E O X O / C O — O	Quasi-experimental; baseline and repeat in both groups, no randomisation
Post-programme with randomised control	E R X O / C R — O	True experimental design; no baseline data

Design (E = experimental group, C = control, R = random allocation, O = observation, X = intervention)	Notation	Comment
Baseline and repeat with randomised control	E R O X O / C R O — O	True experimental design with control, randomisation and baseline — strongest

(Sunder Lal, Ch 16, pp 724-725). Randomised controlled trials have been extended specifically to assess the effectiveness and efficiency of health services; the Indian controlled trials of tuberculosis chemotherapy, which showed domiciliary treatment to be as effective as costlier sanatorium treatment, are the classic example and ushered in the era of domiciliary treatment worldwide (Park 28e, Ch 3, p 105).

Management techniques that support evaluation (Park 28e, Ch 23, p 1028; Health Economics, Ch 13, p 201):

- ▶ **Cost-benefit analysis** — benefits expressed in monetary terms, to judge whether a programme is economically sound and to choose between alternatives.
- ▶ **Cost-effective analysis** — a more promising tool in the health field; benefit expressed as results achieved (lives saved, days free from disease) rather than money.
- ▶ **Cost-utility analysis** — outcome expressed in units capturing both quantity and quality of life (quality-adjusted life year, QALY; disability-adjusted life year, DALY).
- ▶ **Cost accounting** — provides basic data on the cost structure of the programme.
- ▶ **Input-output analysis, systems analysis** — relates resources consumed to results produced and finds the most cost-effective alternative.
- ▶ **Work sampling** — systematic observation and recording of the activities of staff at predetermined or random intervals, to judge appropriateness of current staffing, job description and training.
- ▶ **Medical audit** — objective, systematic review of provider performance against standards (process evaluation).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — Place of monitoring and evaluation in the planning cycle. Labels: Needs (1) → Planning (2) → Implementation (3) → Inputs (resources) → Processes (activities) → Outputs (products or services) → Effects (KAP) → Impacts (health status); a single box "Monitoring and evaluation" (4) sits below and receives dotted feed lines from inputs, processes, outputs, effects and impacts, and feeds back to Needs and Planning.

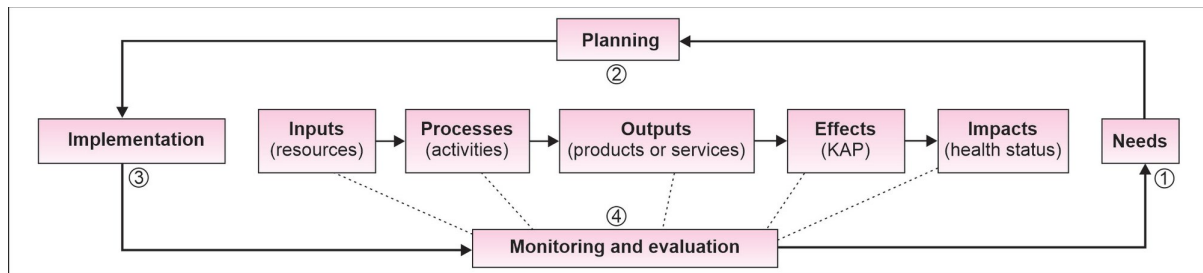


Fig 16.6 — Place of monitoring and evaluation in planning cycle (Sunderlal, p.722)

EVALUATION OF HEALTH SERVICES

Step 1: determine what is to be evaluated

1. EVALUATION OF STRUCTURE

facilities and buildings
equipment and supplies
manpower and skill mix
organisation and finance
standard: IPHS 2022 essential / desirable

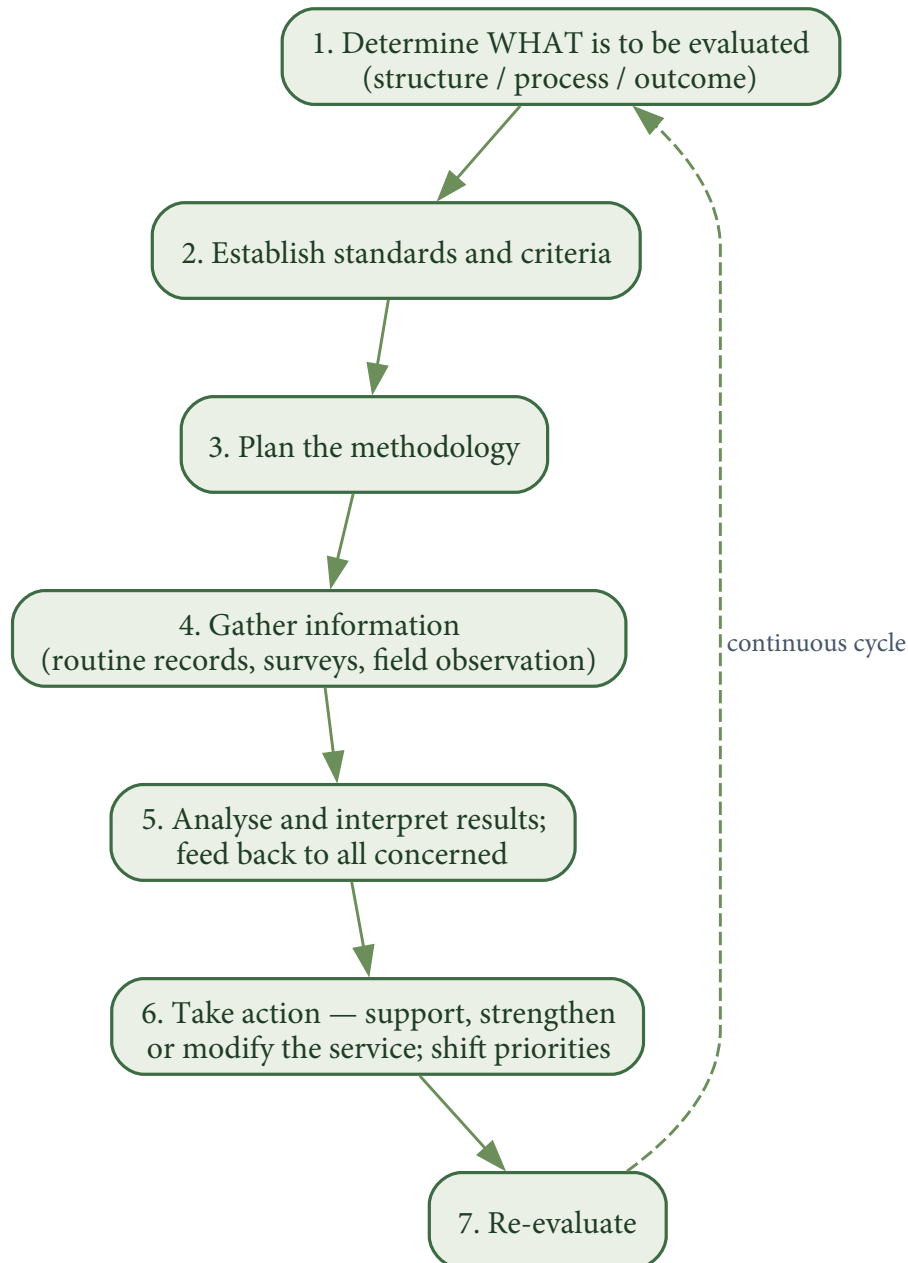
2. EVALUATION OF PROCESS

recognition of the problem
diagnostic procedures
treatment and clinical management
care and prevention
tool: medical (or nursing) audit

3. EVALUATION OF OUTCOME

measurable benefit to the user
improved survival, reduced disability
the 5 Ds of ill-health: disease,
discomfort, dissatisfaction,
disability, death

Determining what is to be evaluated — the three types of evaluation of a health service



The seven general steps in evaluating a health service, closing as a continuous cycle

High-yield exam pointers

- ▶ **Elements of evaluation, mnemonic "RAAA-EEI"** — Relevance, Adequacy, Accessibility, Acceptability, Effectiveness, Efficiency, Impact (Park 28e, Ch 23).
- ▶ **Three types of evaluation (Donabedian 1966)** — structure, process, outcome; process is judged by **medical audit**, outcome by the 5 Ds — disease, discomfort, dissatisfaction, disability, death.
- ▶ **Seven steps** — determine what → standards and criteria → methodology → gather → analyse → act → re-evaluate.
- ▶ **Formulas to write:** bed-occupancy rate = average daily in-patient census ÷ average number of beds; average length of stay = days of care rendered ÷ discharges; bed turnover ratio = discharges ÷ average beds; utilisation (actual coverage) = people receiving the service ÷ people in need of it, in a year.

- **Monitoring** = inputs and processes, continuous; **evaluation** = outputs, effects and impacts, progress yearly and impact at ≥ 5 years.
- **Current Indian structural yardstick** — Indian Public Health Standards 2022 (replacing IPHS 2012), with "essential" and "desirable" levels for each facility.
- **Qualities of an indicator** — valid, reliable, sensitive, specific, feasible, precise.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Indian Public Health Standards were revised in 2022 (replacing the 2012 standards) and Health and Wellness Centres have since been renamed Ayushman Arogya Mandir; public facilities in India are additionally certified against the National Quality Assurance Standards (NQAS), Kayakalp and LaQshya schemes, which Park 28e mentions on the model citizen's charter but does not describe in detail, so their criteria are not quoted here.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, Indian Public Health Standards 2022, WHO Service Availability and Readiness Assessment, WHO Standards for Maternal and Newborn Care, LaQshya guidelines, National Quality Assurance Standards

Programme and policy updates (India)

- **Single Nodal Agency reform changes how NHM money is evaluated** — assessment of Single Nodal Agency implementation in Bihar and Odisha (key-informant interviews, February 2023 to February 2024) found genuine gains in cash management, expenditure transparency and alignment of central releases with actual scheme spending, but weak budget credibility, patchy digital connectivity and thin financial staffing; the reform has shifted scrutiny onto fund-utilisation compliance rather than health outcomes — the classic trap of evaluating inputs and processes and calling it evaluation. [PMID 42480037]
- **A costed answer to "is the district hospital adequately resourced?"** — a blended-payment financing framework (risk-adjusted global budget for outpatients, case-based bundled payment for inpatients, per-bed global budget for indirect services), built on Costing of Health Services in India data from 27 district hospitals across nine states, put the mean annual requirement of an Indian district hospital at ₹326 million (US\$3.35 million), inpatient care taking 78 per cent; staffing the same hospital to Indian Public Health Standards norms raises it to ₹751 million, roughly a 131 per cent increase, and upgrading to a medical college raises it by 91.5 per cent. [PMID 39478753]
- **Ayushman Bharat Digital Mission — adoption is the weak link, not the technology** — across three government tertiary hospitals in Agra, only 129,007 of 537,278 outpatient visits over eight months (23.96%) generated an ABHA registration; among the 425 attendees interviewed, 58.6 per cent lacked adequate knowledge of the mission despite attending a participating facility, 88 per cent wanted training and support, and only 55 per cent felt comfortable about data privacy, with digital health literacy lower in women, older people and the less educated. [PMID 41787376]

Current national data

- **Comprehensive Annual Modular Survey 2022-23 (NSS 79th round), 302,086 households** — publicly funded health insurance, mainly AB-PM-JAY, was associated with higher outpatient (adjusted OR 1.404, 95% CI 1.328-1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128-1.210), and with less catastrophic health expenditure at the 20 per cent threshold for outpatient (adjusted OR 0.869, 95% CI 0.810-0.932) and inpatient care (adjusted OR 0.790, 95% CI 0.737-0.846); outpatient out-of-pocket spending remained the dominant residual burden even among the insured. [PMID 42167946]
- **Citizens Survey 2022-23 — a new national data source for acceptability and satisfaction** — the Lancet Commission on a Citizen-Centred Health System for India interviewed 50,000 households across 125 districts in 29 states and union territories (November 2022 to April 2023) with 141 questions on utilisation, perceived quality, cost, willingness to pay, provider preference and technology use; districts were sampled in tertiles of a district-level UHC index, participation was 98 per cent, and the anonymised dataset is publicly archived. It gives an acceptability and satisfaction denominator that SRS and NFHS do not carry. [PMID 41688486]
- **Facility readiness quantified against IPHS and the WHO readiness manual** — an ICMR-NCDIR survey of 415 public and private facilities in 19 districts of seven states scored diabetes and hypertension service readiness by the WHO Service Availability and Readiness Assessment method against Indian Public Health Standards, taking ≥ 70 per cent as "prepared": sub-centres 61 per cent, community health centres 59 per cent, primary health centres 73 per cent, private primary care 57 per cent; the medicines domain was weakest, with half the sub-centres reporting glimepiride stockouts and 44.8 per cent amlodipine stockouts lasting a median of up to seven months. This is structure evaluation reduced to a single reportable score. [PMID 40536370]

Elimination, surveillance and disease status

- **Routine HMIS returns repurposed as an evaluation and targeting tool** — district-level geospatial analysis of Karnataka Health Management Information System data for 2018-19 and 2019-20 mapped snakebite incidence (24.57 falling marginally to 24.49 per 100,000), localising the burden to the southern dry and southern transition agroclimatic zones with a monsoon peak, and so identifying where antivenom stocking and awareness effort should go. It illustrates the step from monitoring data to evaluative use. [PMID 41575973]

Guideline and legislative updates

- **LaQshya audited against the WHO maternal and newborn quality standards** — India's Labour Room Quality Improvement Initiative (2017) works through ten sequential steps to certification, external assessment at 18 months, and incentives released only on achieving at least 75 per cent of facility-level indicator targets plus an 80 per cent beneficiary satisfaction rate; its checklists cover 269 of the WHO's 352 quality measures (76 per cent) — strongest for input measures (80 per cent), then outcome (74 per cent) and process (72 per cent) — but five quality statements score 60 per cent or less: prevention of unnecessary or harmful practices in labour and the postnatal period (1.9), elimination of mistreatment of women and newborns (5.2), support for women during childbirth (WHO quality statement 6.2), availability of competent skilled birth attendants and support staff (7.2), and managerial and clinical leadership for continuous quality

improvement (7.3). A ready-made example of a process-evaluation tool and its limits. [PMID 38575944]

- **Accreditation training measurably raises NQAS compliance** — a checklist-based comparison in a north Indian teaching hospital found blocks whose staff had undergone NABH-recommended training scored higher on all seven National Quality Assurance Standards areas — patient rights, inputs, support services, clinical services, infection control, quality management and outcome — than untrained blocks, in both intensive care units and wards. [PMID 39670930]

Global and WHO developments

- **South Asia will meet the coverage target and miss the financial-protection one** — Bayesian projections to 2030 put the regional Composite Coverage Index at 81.5 per cent, up from 53.5 per cent in 2000 and just past the 80 per cent SDG mark, while catastrophic health expenditure incidence more than doubles from 7.2 to 18.6 per cent; India's per-capita primary-health-care spending (US\$28.93) trails Bhutan and Sri Lanka, and India is named among the countries facing higher catastrophic and impoverishing expenditure. Coverage and financial protection must therefore be evaluated as two separate outcomes, not one. [PMID 41192927]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 23 Health Planning and Management, pp 1027-1028, 1037-1038; Ch 2 Concept of Health and Disease, p 39; Ch 3 Principles of Epidemiology, p 105; Ch 10 Health Care of the Community, p 728); Sunder Lal — Textbook of Community Medicine (Ch 16 Health Planning and Management, pp 722-725); Health Program Planning and Evaluation — A Practical, Systematic Approach (Ch 1, pp 42-49; Ch 10, pp 276-288); Oxford Textbook of Public Health 7e (Ch 51, pp 1877-1878); Health Economics (Ch 13, p 201); Indian Public Health Standards 2022, Ministry of Health and Family Welfare; WHO Expert Committee on National Health Planning in Developing Countries (definition of evaluation, as quoted in Park 28e).

Recent-advances sources: 42480037 — Challenges to Achieving Universal Health Coverage Through Public Financial Management Reforms: Insights from the Implementation of Single Nodal Agency Reform in India, Health Systems and Reform 2026, doi:[10.1080/23288604.2026.2693357](https://doi.org/10.1080/23288604.2026.2693357) (abstract-only); 39478753 — Framework for responsive financing of district hospitals of India, Frontiers in Public Health 2024, doi:[10.3389/fpubh.2024.1398227](https://doi.org/10.3389/fpubh.2024.1398227); 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission: an analytical cross-sectional study among outpatient department attendees, BMC Health Services Research 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), BMJ Open 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 41688486 — The Citizens Survey 2022-23: a household-level dataset on Universal Health Coverage in India, Scientific Data 2026, doi:[10.1038/s41597-026-06775-6](https://doi.org/10.1038/s41597-026-06775-6); 40536370 — Preparedness of public & private health facilities for management of diabetes & hypertension in 19 districts in India, Indian Journal of Medical Research 2025, doi:[10.25259/IJMR.755.2024](https://doi.org/10.25259/IJMR.755.2024); 41575973 — Geospatial Analysis of Snakebite Incidence Rate in Karnataka: A District-Level Analysis Across Agroclimatic Zones Using Health Management Information System Data, Wilderness & Environmental Medicine 2026, doi:[10.1177/10806032251413140](https://doi.org/10.1177/10806032251413140) (abstract-only); 38575944 — Appraising LaQshya's potential in measuring quality of care for mothers and newborns: a comprehensive review of India's Labor Room Quality Improvement Initiative, BMC Pregnancy and Childbirth 2024, doi:[10.1186/s12884-024-06450-x](https://doi.org/10.1186/s12884-024-06450-x); 39670930 — Effect of National Accreditation Board for Hospital-Recommended Trainings on Patient Care Standards: A Comparative Study Based on the National Quality Assurance Standards Guidelines in a Tertiary Health-care Institute, Indian Journal of Public Health 2024, doi:[10.4103/ijph.ijph.1262.23](https://doi.org/10.4103/ijph.ijph.1262.23) (abstract-only); 41192927 — Progress towards universal health coverage in South Asia, 2000-2030: an examination of the twin elements of primary healthcare provision and financial protection, BMJ Global Health 2025, doi:[10.1136/bmjgh-2025-020052](https://doi.org/10.1136/bmjgh-2025-020052).



Health information sources in India — GIS

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — A health information system is "a mechanism for the collection, processing, analysis and transmission of information required for organizing and operating health services, and also for research and training" (Park 28e, Ch 21, p 995). Its lifeblood is the set of routine health statistics drawn from many separate sources; data become **information** only when reduced, summarised and adjusted for variations such as the age and sex composition of the population, and information becomes **intelligence** when integrated with experience and with social and political perceptions (Park 28e, Ch 21, pp 995-996).

SOURCE & COVERAGE

#	Source	What it yields in India	Key limitation
1	Census	Total population and its breakdown by age and sex; also social and economic characteristics, living and working conditions, income; the denominator for every rate	Decennial, so full results are not available quickly
2	Registration of vital events (civil registration system)	Live births, deaths, foetal deaths, marriages, divorces, adoptions, legitimations, legal separations	Grossly deficient in accuracy, timeliness and completeness; multiple registering agencies
3	Sample Registration System (SRS)	Reliable birth and death rates, age-specific fertility and mortality rates, infant, under-five and adult mortality at national and State level	Sample-based; no district-level estimates
4	Notification of diseases	Incidence and distribution of specified notifiable diseases; triggers prevention and control	Covers only a small part of total sickness; heavy under-reporting; no uniform national Act
5	Hospital records	Geographic source of patients, age-sex and diagnostic distribution, duration of stay, association between diseases, cost of care; bed-occupancy rate and length of stay	"Tip of the iceberg"; selective admission policy; catchment population undefined
6	Disease registers (e.g. cancer registries)	Duration of illness, case fatality, survival; permits follow-up and a continuous account of disease frequency	Coverage often not national or representative
7	Record linkage	Assembles birth, marriage, death, hospital admission and discharge for one individual or family; studies associations between diseases with possible aetiological significance	Volume of data; applied only on a limited scale and beyond the reach of many developing countries
8	Epidemiological surveillance	Morbidity and mortality from national disease control programmes — malaria, tuberculosis, leprosy, filariasis — plus immunizations performed	Disease-specific and programme-bound

#	Source	What it yields in India	Key limitation
9	Other health service records	Subcentre, primary health centre, out-patient, maternal and child health, school health, diabetic and hypertensive clinic records — birth weight, height, arm circumference, immunization, disease-specific morbidity	Relate to only a segment of the population; kept for administration, not monitoring
10	Environmental health data	Air, water and noise pollution; harmful food additives; industrial toxicants; waste disposal; production and consumption of materials	Collection of environmental data remains a major problem
11	Health manpower statistics	Numbers of physicians, dentists, pharmacists, nurses and technicians by age, sex, speciality and place of work; medical education admissions and out-turn	Held by many different councils and directorates
12	Population surveys	Morbidity, mortality, nutritional status, health care utilization, attitudes and behaviour — the only population-based source	Costly; interview-based morbidity data unreliable
13	Other routine statistics related to health	Demographic (density, migration, literacy); economic (tobacco, dietary fat and coal consumption, drug sales, per capita income, employment); social security (insurance, sickness absence, disability benefit rates)	Collected for non-health purposes
14	Non-quantifiable information	Health policies, health legislation, public attitudes, programme costs, procedures and technology — a health information system has multi-disciplinary inputs	Not statistical; hard to store and disseminate

(Park 28e, Ch 21, pp 996-999)

Census, civil registration, sample registration and notification — in detail

Source	Legal or operational basis	Method	Current position
Census	Census Act 1948; directed by the Census Commissioner for India	Complete enumeration of every person in the country, at the end of the first quarter of the first year of each decade, when most people are resident at home	First regular census 1881, thereafter at 10-year intervals; last census held March 2011 (Park 28e, Ch 21, p 996)

Source	Legal or operational basis	Method	Current position
Civil registration	<i>Central Births and Deaths Registration Act, 1969</i> , in force from 1 April 1970; some States such as Tamil Nadu, Karnataka and Assam had their own earlier Acts	Compulsory registration of births and deaths throughout the country; the public reports household events, heads of hospitals, nursing homes, hotels, jails and dharmashalas report institutional events; time limit 21 days uniformly all over India, late fee on default	From October 2018 the Aadhaar number is compulsory for registration of a death (Park 28e, Ch 21, p 996)
Sample Registration System	Initiated in the mid-1960s because civil registration was deficient	Dual record system — continuous enumeration of births and deaths by a resident enumerator, plus an independent half-yearly retrospective survey by an investigator-supervisor; unmatched events are field-verified	Now covers the entire country and is the main source of reliable Indian vital rates (Park 28e, Ch 21, p 997)
Notification	Historically the first health information sub-system to be established; lists vary between and within countries	Reporting duty in India has shifted from the village chowkidar to the health worker; a uniform national Act on the model of the 1969 Act is still needed	Internationally, cholera, plague and yellow fever are notifiable to WHO under the International Health Regulations; louse-borne typhus, relapsing fever, poliomyelitis, influenza, malaria, rabies and salmonellosis are under international surveillance (Park 28e, Ch 21, p 997)

Survey-based sources in India

- Population surveys supplement routine statistics because a health information system must be population-based. The types are: surveys to evaluate health status (community diagnosis); **interview surveys; health examination surveys; questionnaire surveys; and health records surveys** — the last the cheapest, but not population-based and of questionable reliability. National household samples typically cover **5,000 to 10,000 households** (Park 28e, Ch 21, p 999).
- The **National Sample Survey Organization** has given country-wide data on general morbidity, family planning and vital events; its morbidity data are unreliable because of the limitations of the interview method, which is why interviews are combined with examination and laboratory measurement (Park 28e, Ch 21, p 999).
- Health surveys on a permanent basis operate in only a few countries — **Japan since 1953, the USA since 1957, the United Kingdom since 1971**; India's first methodological general health survey was at the **Singur Health Centre** (Park 28e, Ch 21, p 999). The national datasets used for Indian indicator estimation are the Sample Registration System, Census, **National Family Health Survey** (NFHS-5, 2019-2021), District Level Household and Facility Survey, Annual Health Survey and Coverage Evaluation Survey (Park 28e, Ch 7, p 558; Ch 9, p 716).

Routine reporting and digital health information systems in India

System	Launched	Structure and content
Integrated Disease Surveillance Programme (IDSP)	Decentralised, State-based; networked with the National Informatics Centre and the Indian Space Research Organisation	Central Surveillance Unit at the National Centre for Disease Control, Delhi → State Surveillance Unit at every State and Union Territory headquarters → District Surveillance Unit in every district. Weekly data on epidemic-prone disease from subcentres, PHCs, CHCs, government and private hospitals and medical colleges in S (syndromic), P (presumptive) and L (laboratory) formats using standard case definitions; over 90 per cent of districts report weekly by e-mail or portal (Park 28e, Ch 7, pp 568-569)
Health Management Information System (HMIS) — health data recorded, stored, retrieved and processed for decision-making at national, State and institutional level (Jugal Kishore, p 262)	HMIS 2.0 in 2020-21, with WHO technical support	Real-time data entry and monitoring; interactive dashboards; application-programming-interface integration with other portals; GIS integration with a village layer; National Identification Number and Local Government Directory compliance identifying facilities to village and block level. About 400 service-delivery elements monthly and 700 infrastructure and human-resource elements annually ; daily data entry from April 2020 (Sunder Lal, Ch 8, p 532)
Mother and Child Tracking System (MCTS) , upgraded to the RCH portal	December 2009	Name-based tracking of each beneficiary through the reproductive lifecycle; feeds e-Janma (Karnataka), ANMOL, Kilkari and direct benefit transfer of Janani Suraksha Yojana and ASHA payments (Sunder Lal, Ch 8, p 535)
Integrated RCH Register version 2.0	Released 2022	Village- or ward-wise population, eligible couples, estimated pregnancies and infants, and village-level human resources, for workload assessment and workplan preparation; a step towards the Integrated Health Information Platform (IHIP) (Sunder Lal, Ch 8, p 532)

Weaknesses of the Indian health information system

- Civil registration is unreliable for accuracy, timeliness and completeness because of illiteracy, ignorance, lack of concern and motivation, non-uniform collection and transmission between rural and urban areas, and multiple registering agencies — health, panchayat, police and revenue (Park 28e, Ch 21, p 996).
- Health workers maintain a large number of registers whose data are rarely used for management decisions, so registers are poorly kept and workers complain of time spent on record keeping (Jugal Kishore, p 262).

- No feedback reaches the worker who collects the data, so motivation is not sustained; training on HMIS is weak; performance is over-reported and figures inflated; data are seldom analysed or acted on locally — the system remains **data-driven rather than action-driven** (Sunder Lal, Ch 8, p 534).

Role of Geographic Information Systems (GIS) in public health

Definition — GIS are systems of computer hardware and software capable of combining high-precision location information with health databases and analysis tools, so as to visualise the geographic distribution of health indicators such as morbidity, mortality, prevalence and incidence (Oxford Textbook of Public Health 7e, Ch 24, p 946). They are digital spatial-data management systems that integrate, store, adjust, analyse and arrange geographically referenced information (Park Plus, Ch 24, p 360). With remote sensing, cartography, photogrammetry and geodesy they constitute **geomatics** (Mahajan & Gupta 4e, Ch 3, p 36). The conceptual ancestor is *John Snow's* spot map of cholera deaths around the Broad Street pump in the Golden Square district of London, 1854 (Park 28e, Ch 3, pp 86-87).

Function of GIS	Public health application
Data storage and management	Electronic database of health facilities, educational institutions and training centres, mapped and disseminated through the Central Bureau of Health Intelligence website; the National Informatics Centre provides GIS mapping of government health facilities (Sunder Lal, Ch 8, pp 534-535)
Query and display	Health services, diseases and health information displayed on a map and correlated with environmental and social data; trends and correlations that conventional processing cannot reveal (Park Plus, Ch 24, p 361)
Multi-layer overlay	Considers multiple map layers and examines relationships among the variables on those layers at varying geographic scales (Lilienfeld's Foundations of Epidemiology, Ch 14, p 291)
Cluster detection	Evaluates whether patterns are random, clustered or dispersed; geospatial assessment of disease against locations of risk exposures and of preventive or therapeutic services (Lilienfeld's, Ch 14, p 291; Oxford 7e, Ch 42, p 1569)
Prediction with remote sensing	Climate, vegetation and satellite imagery combined with epidemiological data to predict vector occurrence and to identify where vectors pose the greatest risk, where toxic exposures are highest and where crop failure may cause famine (Handbook of Epidemiology, Ch 33, pp 1320-1321; Lilienfeld's, Ch 14, p 291)
Spatial optimisation	Identifies where and how emergency services are best located, and models movement of pollutants through air and water so that populations can be evacuated (Lilienfeld's, Ch 14, p 291)
Targeting and decision-making	Identifies areas at risk so that resources, screening and treatment are directed to the at-risk population; supports strategic planning, research and evaluation, emergency preparedness and siting of health care services (Nelson & Williams, Ch 7, p 201; Park Plus, Ch 24, pp 361-362)
Programme monitoring in India	GIS-enabled HMIS application launched in March 2016 to depict data on dynamic maps, with grading of CHCs and PHCs and automatic alerts for sensitive data items (Sunder Lal, Ch 8, pp 534-535)

Cautions — GIS should be used with care: it is useful to generate hypotheses and identify possible associations between disease risk and environmental exposure, but not to prove them (Handbook of Epidemiology, Ch 33, pp 1320-1321). The quality of modelling is critically dependent on the quality of the data collected and stored, and the disease model itself is built outside the GIS — the GIS only identifies where the stated conditions exist (Nelson & Williams, Ch 7, pp 188, 199). Recognition of clusters is common but hard to distinguish from random variation without formal statistical testing (Oxford 7e, Ch 42, p 1569).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce for the information flow diagram: Subcentre → PHC → District Surveillance Unit; feeding into the District Surveillance Unit from the left — CHC, District Hospital, Medical College, Public Health Laboratory; from the right — Private Practitioner, Private Health Facilities, Other Hospitals, Private Laboratory; District Surveillance Unit → State Surveillance Unit → Central Surveillance Unit.

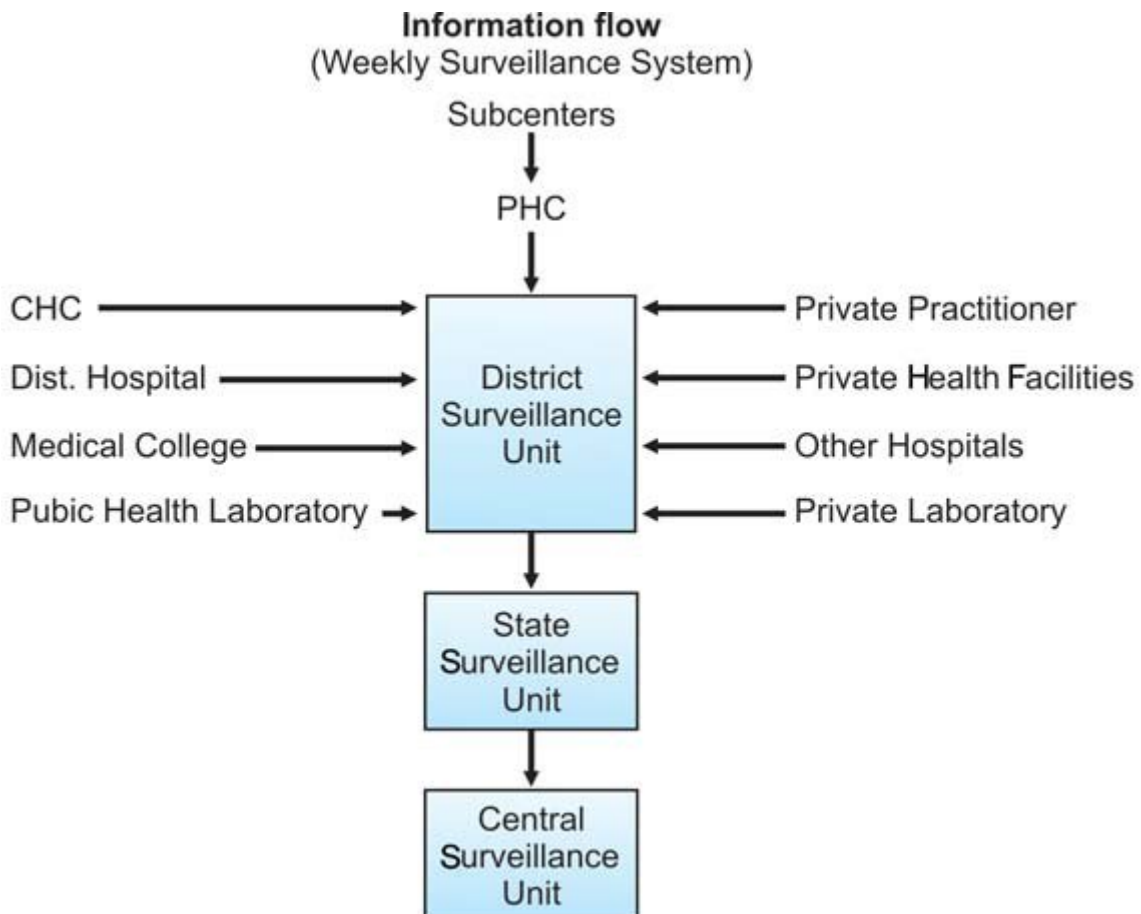


Fig 21.1 — Information flow (Weekly Surveillance System) (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.398)

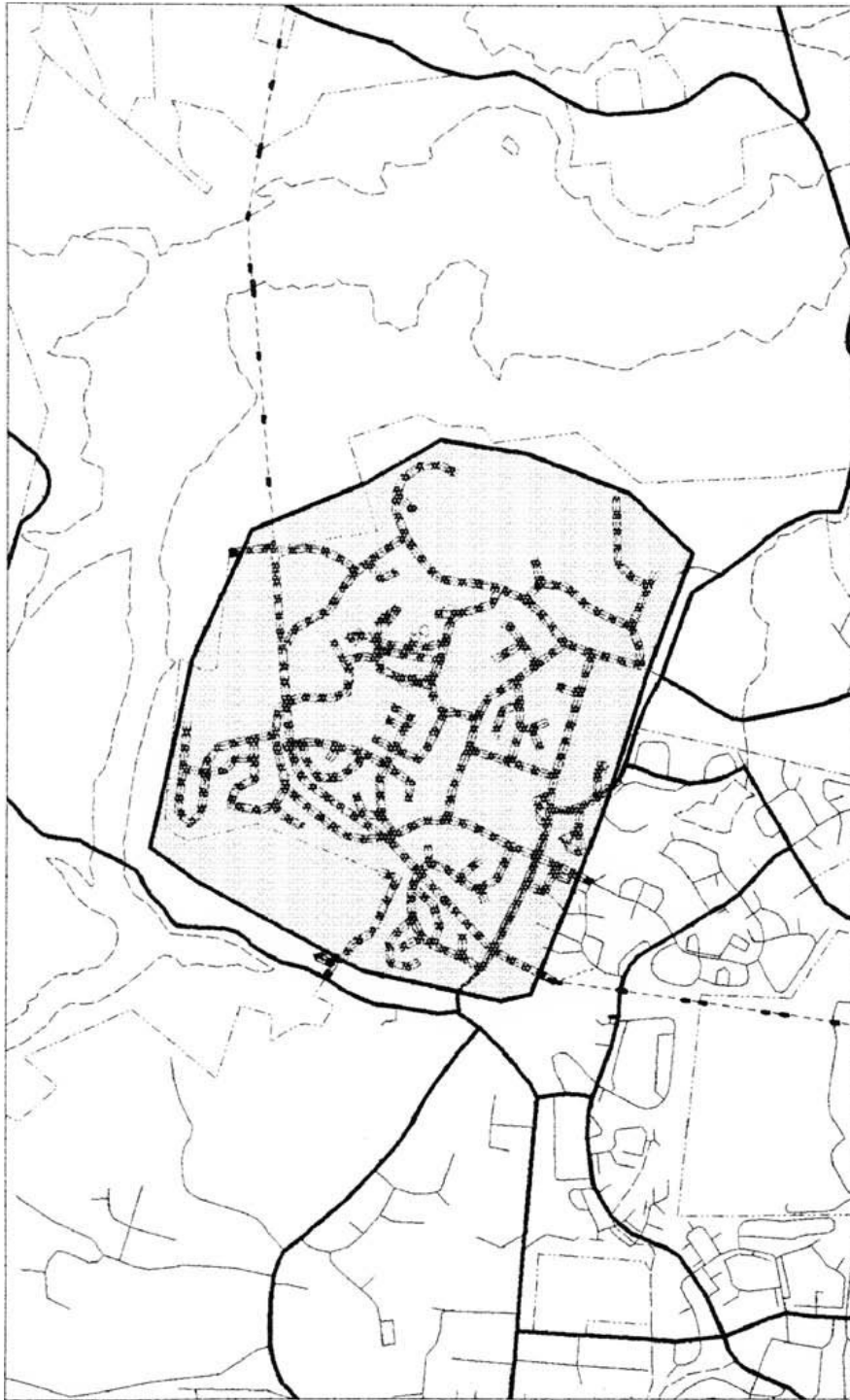
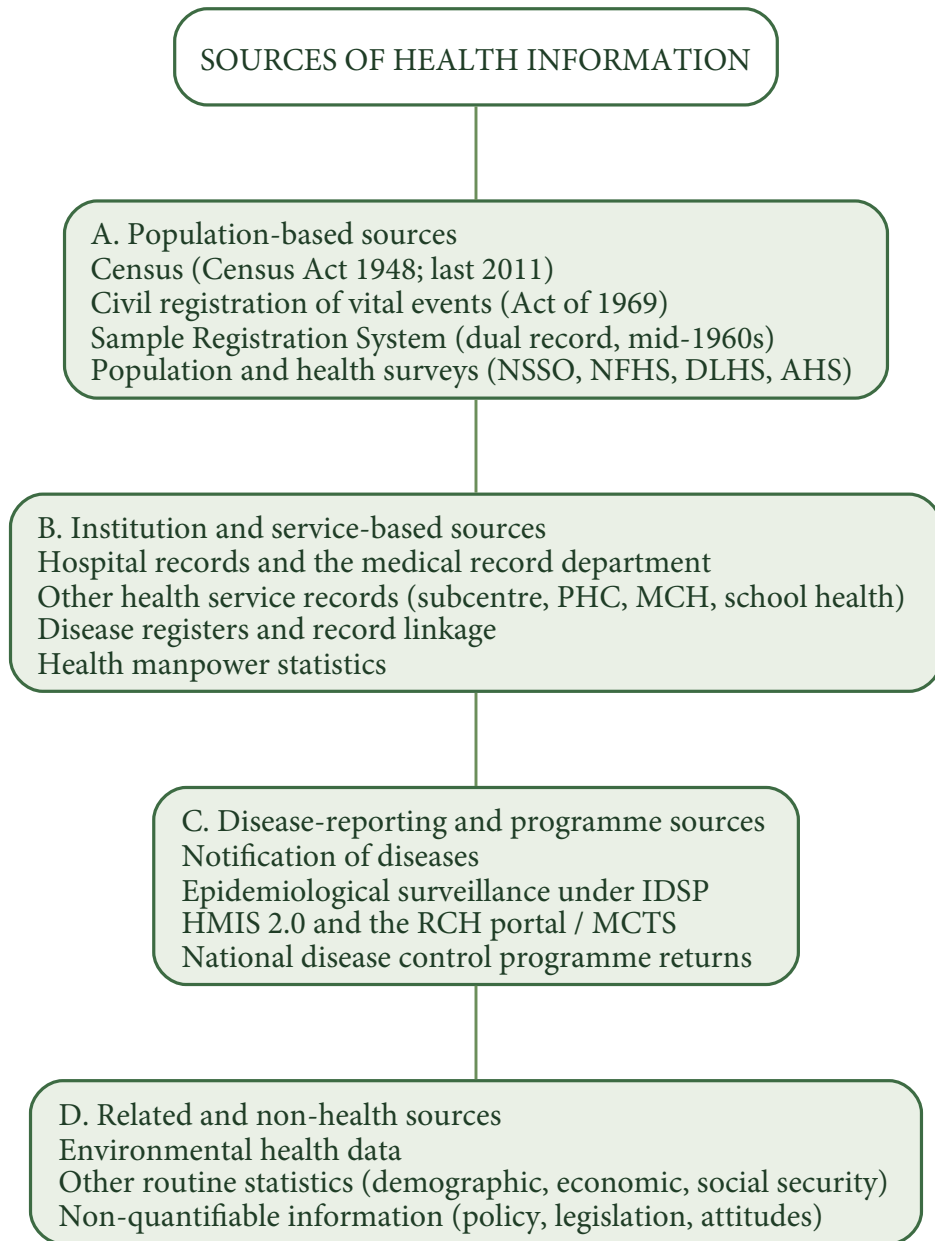
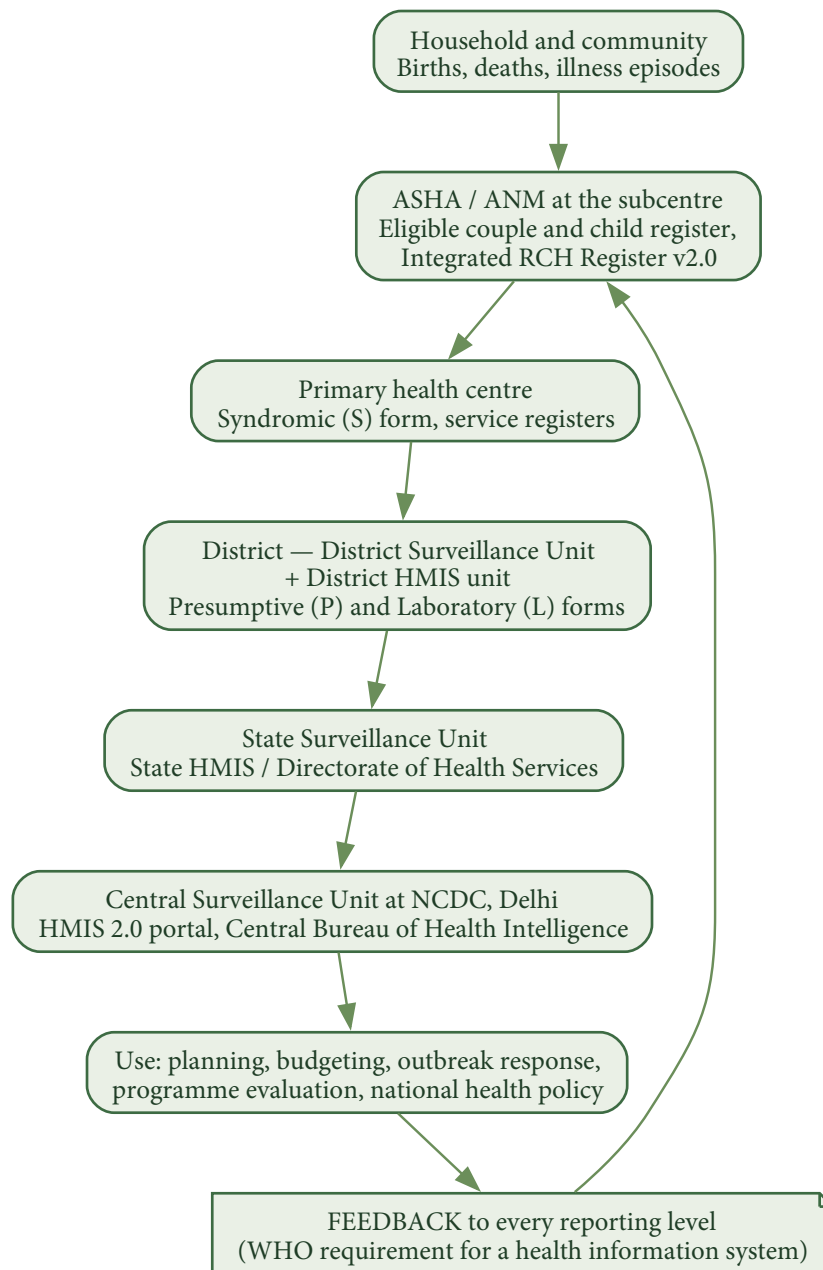


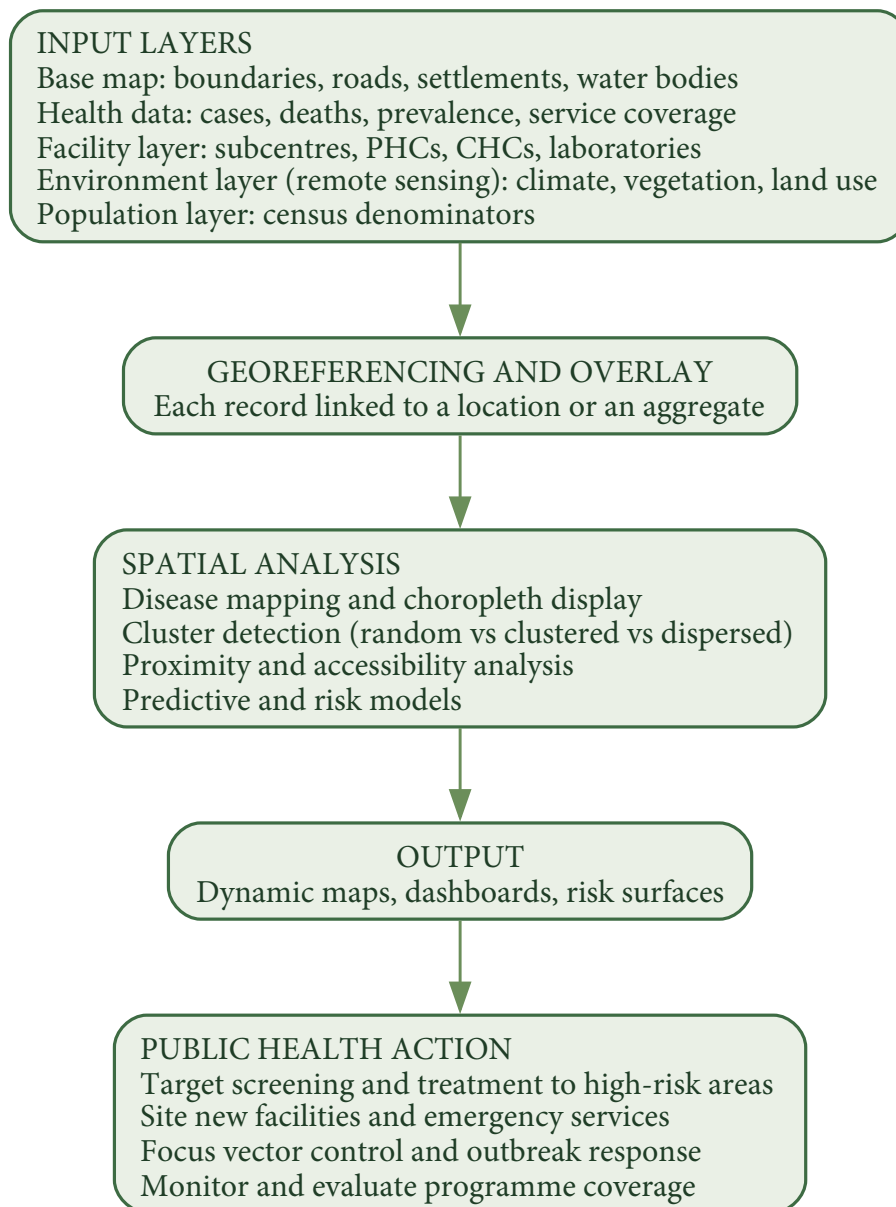
Fig 7.10 — Application of geographic information system to decision making showing the high-risk area (Nelson & Williams — *Infectious Disease Epidemiology*, p.202)



Classification of the sources of health information in India



Flow of routine health information in India, from the household to national policy



How a geographic information system converts health data into public health action

High-yield exam pointers

- ▶ Park's list is **14 sources**, in order: census, registration of vital events, sample registration system, notification, hospital records, disease registers, record linkage, epidemiological surveillance, other health service records, environmental health data, health manpower statistics, population surveys, other routine statistics, non-quantifiable information.
- ▶ Dates to write: **Census Act 1948**; **first regular census 1881**; **last census March 2011**; **Central Births and Deaths Registration Act 1969**, in force 1 April 1970; **registration within 21 days**; **Aadhaar compulsory for death registration from October 2018**; **SRS from the mid-1960s**.
- ▶ **SRS = dual record system** — continuous enumeration by the enumerator + independent half-yearly retrospective survey by the investigator-supervisor, with field verification of unmatched events.
- ▶ **IDSP reporting formats: S = syndromic, P = presumptive, L = laboratory**; Central Surveillance Unit at NCDC Delhi → State Surveillance Unit → District Surveillance Unit.

- ▶ **HMIS 2.0 (2020-21):** ~400 monthly service-delivery elements, ~700 annual infrastructure and manpower elements, GIS-integrated with a village layer; **GIS-enabled HMIS launched March 2016; MCTS launched December 2009**, now the RCH portal.
- ▶ Under the International Health Regulations only **cholera, plague and yellow fever** are notifiable to WHO.
- ▶ One-line GIS definition to reproduce: computer hardware and software combining **high-precision location information with health databases and analysis tools** to visualise the geographic distribution of health indicators; the ancestor is **John Snow's Broad Street spot map, 1854**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the retrieved corpus dates the last decennial census to March 2011 and does not carry a completed subsequent census round, so the answer quotes 2011 as Park 28e states it; likewise, the Ayushman Bharat Digital Mission and the Integrated Health Information Platform appear in the corpus only as the stated direction of travel beyond HMIS 2.0 and the RCH portal, so no operational detail for them has been asserted.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, PIB India, Registration of Births and Deaths (Amendment) Act 2023, WHO International Health Regulations (2005) as amended

Programme and policy updates (India)

- ▶ **Census 2027 supersedes the March 2011 round quoted above.** The Union Cabinet has approved the scheme for the next decennial census: house listing and housing census from April to September 2026, population enumeration in February 2027, with the snow-bound districts of Ladakh, Himachal Pradesh, Jammu & Kashmir and Uttarakhand enumerated from September 2026. It is the first digital census — an enumerator mobile application in 16 languages plus a self-enumeration web portal — and the first since 1931 to record caste (*jati*), following the Cabinet decision of 30 April 2025. [PIB 2025-Dec]
- ▶ **The Ayushman Bharat Digital Mission, launched September 2021, is now the national spine for health information**, built on four blocks — the ABHA (Ayushman Bharat Health Account) unique patient identifier; the Healthcare Professionals Registry, which pools the medical, nursing and paramedical councils into one database; the Health Facility Registry covering public and private hospitals, clinics, diagnostic centres and pharmacies across all systems of medicine; and a Health Information Exchange with a consent manager for personal health records. Cyber-resilience is therefore now a health-system risk, the November 2022 ransomware attack on AIIMS Delhi and the subsequent ICMR data breach being cited as evidence that health-sector cyberattacks are neither hypothetical nor rare. [PMID 42382219]
- ▶ **IDSP has migrated to the Integrated Health Information Platform (IDSP-IHIP)**, which the corpus carries only as a stated direction of travel; IHIP metadata now underpins operational work such as the genomic surveillance of 7,457 SARS-CoV-2 samples in central India over 2020-2022. The old weakness persists — a Karnataka malaria-elimination audit found IDSP/IHIP Form-P and Form-L returns from the private sector so poor that investigators had to survey

private practitioners and pharmacies directly to reconstruct the missing data. [PMID 38527669]
[PMID 39070449]

Current national data

- ▶ **Civil registration data for 2021 were released only in May 2025 — a four-year lag.** The CRS recorded 10.2 million deaths in 2021, a 26 per cent rise over the 8.1 million of 2020, against 335,004 officially reported COVID-19 deaths; excess deaths were about 2.4 million, a 7.2-fold undercount, with state-level ratios ranging from under 2 in Kerala and Goa to over 40 in Gujarat. Excess death rates were higher in men than women (2.2 vs 1.3 per 1000) and in urban than rural areas (2.3 vs 1.4 per 1000). The same report records that 47.3 per cent of the deceased received no medical attention at the time of death, which is why cause-of-death data from the CRS remain unusable. [PMID 41232123]
- ▶ **Completeness of death registration rose from 58 per cent in 2000 to 81 per cent in 2018 — an extra 3.2 million registered deaths a year,** helped by the online birth and death registration process introduced by the Registrar General in 2015. But male completeness (85 per cent in 2018) far exceeds female (74 per cent), and seven states holding 41 per cent of the population still reported no age at death, which is why the Indian death registration system is not used in Global Burden of Disease or United Nations mortality estimates. [PMID 34461914]
- ▶ **SRS and NFHS disagree sharply on stillbirths.** NFHS-5 gives 9.7 stillbirths per 1000 births (95% CI 9.2-10.1) against an SRS average of 3.8 for 2016-2020 — a 2.6-fold gap — while neonatal mortality rates from the two sources agree, pointing to failure in the stillbirth definition and in documentation of gestation rather than to sampling. The India Newborn Action Plan target is a stillbirth rate below 10 per 1000 births by 2030, with every state reaching it by 2035. [PMID 36865608]

Elimination, surveillance and disease status

- ▶ **GIS is now run directly on routine Indian health-service data, and travel-time isochrones have displaced simple facility counts as the measure of access.** A district-level analysis of Karnataka HMIS snakebite records mapped incidence by agroclimatic zone (24.57 per 100,000 in 2018-19, 24.49 in 2019-20), placing the heaviest burden in the southern dry and southern transition zones with a monsoon peak — source 9 above (other health service records) driving spatial targeting. A Kerala study geocoded every licensed inpatient psychiatric facility against a 100 m WorldPop population grid and an OpenStreetMap road network: 67.96 per cent of the population could reach a facility within 15 minutes and 96.85 per cent within 45 minutes, yet beds ranged from 0.75 per 100,000 in Kasargod to 68 in Idukki — showing that spatial accessibility and resource adequacy are separate questions GIS can pull apart. [PMID 41575973] [PMID 42303405]
- ▶ **One Health surveillance is the current direction of travel for notification systems.** Applying the One Health Surveillance Matrix to Gujarat, where IDSP-IHIP covers human disease and the National Animal Disease Reporting System covers a livestock population of 536.76 million nationally, showed the human system weak on data sharing, sampling and external communication and the animal system weak on dissemination to decision-makers and on data storage and interpretation — the integration gap that matters because about 75 per cent of infectious diseases are zoonotic. [PMID 39453563]

Guideline and legislative updates

- **The Registration of Births and Deaths (Amendment) Act, 2023** rewrites the 1969 Act described above. Passed in August 2023, it creates a national database of registered births and deaths under the Registrar General with matching State databases, provides for certificates issued electronically, requires the Aadhaar numbers of parents and informants, permits sharing of registration data with electoral rolls and other approved databases, and makes the birth certificate the single document proving date and place of birth for school admission, voter registration, government appointment and other notified purposes. Delivery-point registration is the practical lever: PHC and CHC deliveries rose from 30.5 to 34.7 per cent of institutional births between NFHS-4 and NFHS-5 while registration of those births rose from 80.8 to 90.2 per cent, giving 38 per cent higher odds of registration than private facilities, with Delhi, Rajasthan and Uttar Pradesh designating these centres as registrars — the operational route to SDG 16.9, legal identity for all by 2030. [RBD Amendment Act 2023] [PMID 41752339]

Global and WHO developments

- **The International Health Regulations were amended on 1 June 2024 by the Seventy-seventh World Health Assembly (resolution WHA77.17), the amendments entering into force on 19 September 2025.** They insert a "pandemic emergency" tier above the public health emergency of international concern — for a communicable disease of wide geographical spread across multiple States that exceeds health-system capacity, causes substantial disruption and demands rapid, equitable, coordinated international action — and require each State Party to designate a National IHR Authority. The textbook line that only cholera, plague and yellow fever are notifiable to WHO therefore now sits inside a substantially rewritten alert architecture. [WHO 2024-Jun] [WHO 2025-Sep]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 21 (Health Information and Basic Medical Statistics), pp 995-999; Ch 7 (Health Programmes in India), pp 558, 568-570; Ch 3 (Principles of Epidemiology and Epidemiologic Methods), pp 86-87. Sunder Lal — Textbook of Community Medicine, Ch 8, pp 532-535. Jugal Kishore — National Health Programmes of India, pp 262-264. Oxford Textbook of Public Health 7e — Ch 24 (Community diagnosis and health information systems in low- and middle-income countries), pp 946-947; Ch 42 (Public health surveillance), pp 1568-1569. Nelson & Williams — Infectious Disease Epidemiology, Ch 7 (Geographic Information Systems), pp 188, 199-201. Lilienfeld's Foundations of Epidemiology, Ch 14 (Field Epidemiology), p 291. Handbook of Epidemiology (Ahrens & Pigeot), Ch 33, pp 1320-1321. Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 3, p 36. Park Plus (Amrutha & Vijayalakshmi), Ch 24, pp 360-362. Figures: Mahajan & Gupta 4e, Fig 21.1, p 398; Nelson & Williams, Fig 7.10, p 202.

Recent-advances sources: 41232123 — India's death toll in 2021 during the COVID-19 pandemic: insights from delayed official civil registration data, Journal of Global Health 2025, DOI [10.7189/jogh.15.03045](https://doi.org/10.7189/jogh.15.03045); 34461914 — Have inequalities in completeness of death registration between states in India narrowed during two decades of civil registration system strengthening?, International Journal for Equity in Health 2021, DOI [10.1186/s12939-021-01534-y](https://doi.org/10.1186/s12939-021-01534-y); 36865608 — Stillbirth undercount in the sample registration system and national family health survey, India, Bulletin of the World Health Organization 2023, DOI [10.2471/BLT.22.288906](https://doi.org/10.2471/BLT.22.288906); 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, International Journal of Environmental Research and Public Health 2026, DOI [10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 42382219 — An assessment of cyber security and resilience of the National Digital Health Mission of India, Oxford Open Digital Health 2026, DOI [10.1093/oodh/oqag012](https://doi.org/10.1093/oodh/oqag012); 38527669 — COVID-19 genome surveillance: a geographical landscape and mutational mapping of SARS-CoV-2 variants in central India over two years, Virus Research 2024, DOI [10.1016/j.virusres.2024.199365](https://doi.org/10.1016/j.virusres.2024.199365); 39070449 — Evaluation of Outbreak Preparedness to Prevent the Reintroduction of Malaria in Ramanagara District: A Descriptive Epidemiological Study, Cureus 2024, DOI [10.7759/cureus.63162](https://doi.org/10.7759/cureus.63162); 41575973 — Geospatial Analysis of Snakebite Incidence Rate in Karnataka: A District-Level Analysis Across Agroclimatic Zones Using Health Management Information System Data, Wilderness & Environmental Medicine 2026, DOI [10.1177/10806032251413140](https://doi.org/10.1177/10806032251413140) (abstract-only); 42303405 — Spatial access to inpatient mental health services in Kerala, India: a Geospatial Information System-based travel-time isochrone analysis, BMJ Open 2026, DOI [10.1136/bmjopen-2025-113170](https://doi.org/10.1136/bmjopen-2025-113170); 39453563 — Application of the One Health Surveillance (OHS) Matrix to Evaluate the Disease Surveillance Systems in Gujarat, India: A Policy Content Analysis, Journal of Epidemiology and Global Health 2024, DOI [10.1007/s44197-024-00317-2](https://doi.org/10.1007/s44197-024-00317-2). Union Cabinet approval of the scheme for the Census of India 2027 (PIB, 12 December 2025); Phase-1 gazette notification (PIB, January 2026); and the Cabinet decision to include caste enumeration (30 April 2025). The Registration of Births and Deaths (Amendment) Act, 2023 (passed August 2023). International Health Regulations (2005) amendments adopted by the Seventy-seventh World Health Assembly, resolution WHA77.17, 1

June 2024, in force 19 September 2025.



Management and administrative techniques in public health

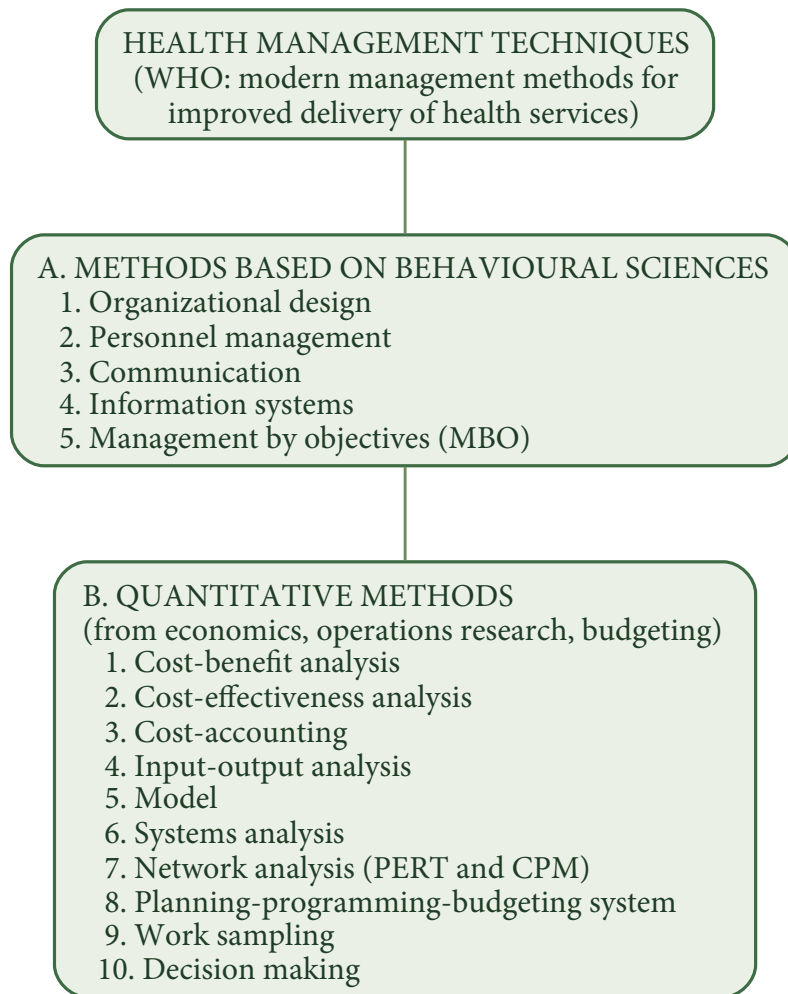
Asked in: Paper IV 08-Jan-2025 — RGUHS MD Community Medicine

Definition — *Administration* broadly means "getting things done", whereas *management* is "the purposeful and effective use of resources — manpower, materials and finances — for fulfilling a pre-determined objective" (Park 28e, Ch 23, pp 1027–1028). In health planning the term *resources* is wider: manpower, money, materials, skills, knowledge, techniques and time (Park 28e, Ch 23, p 1025); health management is accordingly defined as the purposeful and efficient use of health resources — manpower, money, material and time — to achieve predetermined goals (Sunder Lal, Ch 16, p 707). Management techniques are based on the principles of the behavioural sciences and on quantitative methods, and were developed by experts of management science to help any organisation achieve its stated goals more efficiently; the World Health Organization (WHO) is actively making them more popular for application in the health field (Park 28e, Ch 23, p 1028).

The four managerial functions within which every technique operates

Function	What the health administrator does	Example in Indian public health practice
1. Planning	Determines what is to be done, matching limited resources to many problems and eliminating wasteful or duplicated expenditure	Preparing the sub-centre or block action plan after assessing local needs and resources (Sunder Lal, Ch 16, pp 704–707)
2. Organizing	Sets up the framework and makes it possible for groups to do the work — defined procedures, delegation of authority, fixation of responsibility at each level	The integrated directorate recommended by the <i>Bhore Committee</i> (1946) — a single administrative officer for the curative and preventive departments of health (Park 28e, Ch 23, p 1035)
3. Communicating	Motivates people to do the work and removes roadblocks between doctor and patient, senior and junior, directorate and ministry	Establishing vertical and horizontal channels so that notification, compilation of statistics, release of supplies and salaries are not delayed (Park 28e, Ch 23, pp 1027–1028)
4. Monitoring (controlling)	Checks that work is progressing satisfactorily — continuous observing, recording and reporting, with corrective action when deviation is excessive	Monthly monitoring reports, supervisory field visits to the sub-centre, and the Gantt chart (Park 28e, Ch 23, p 1027; Sunder Lal, Ch 16, pp 706, 723)

Classification of health management techniques



Classification of management and administrative techniques used in health services

Techniques based on the behavioural sciences

Technique	What it involves	Public health example
Organizational design	Poor organization wastes resources; the structure must suit the current situation and the needs to be serviced, and must be reviewed every few years as purpose, problems and technology change	Redesign of peripheral health services from a unipurpose to a multipurpose worker strategy, and the current re-designation of sub-centres as health and wellness centres (Park 28e, Ch 23, p 1028; Sunder Lal, Ch 16, p 704)
Personnel management	Skilful use of human resources — selection, training and motivation; division of responsibility; distribution of roles; eliminating "square pegs in round holes"; incentives for better work; avenues for promotion; effective design of health teams	Not loading a clinically-inclined specialist with administrative charge; building the medical officer–health assistant–auxiliary nurse midwife (ANM)–accredited social health activist (ASHA) team (Park 28e, Ch 23, p 1028)

Technique	What it involves	Public health example
Communication	Establishing suitable vertical and horizontal channels; communication barriers cause delays in reporting and notification, in compiling statistics, in releasing supplies and salaries, and in instituting remedial measures	Fixed reporting formats and periodic review meetings linking the directorate, the district and peripheral health institutions (Park 28e, Ch 23, p 1028)
Information systems	Collection, classification, transmission, storage, retrieval, transformation and display of information, tailored to the management needs of the individual health service; provides data for monitoring and evaluation and gives feedback to administrators and planners at all levels; computers greatly improve it	The health management information system (HMIS) , organised around key indicators that measure a programme's progress towards its goals (Park 28e, Ch 23, p 1028; Sunder Lal, Ch 8, p 530)
Management by objectives (MBO)	Objectives are set for each unit and subunit, each preparing its own plan of action, usually short-term; objectives are general at upper levels and progressively specific lower down, and act both as a guide to action and as a yardstick to measure work	Immunisation and antenatal-care targets cascaded from state to district to primary health centre and into the sub-centre action plan (Park 28e, Ch 23, pp 1027–1028)

Quantitative and economic techniques

Technique	Principle	Public health example
Cost-benefit analysis (CBA)	Economic benefits of a programme are compared with its cost, both expressed in money , to judge whether it is economically sound and to select the best of several alternatives; an action with a benefit/cost ratio >1 is selected. Main drawback — health benefits often cannot be monetised. Allied terms: lead time = the interval from the start of the activity (use of resources) to the start of benefits; opportunity cost = the loss to the community from failure to use available resources	CV Jetanovic, Grab and Umera used cost-benefit analysis to compare the benefits of sanitary measures against vaccination in the control of typhoid (Park 28e, Ch 23, p 1028; Suryakantha, Ch 37, p 807)
Cost-effectiveness analysis (CEA)	As for CBA, except that benefit is expressed in results achieved rather than money; a more promising tool in the health field	Cost per life saved , per disease-free day, or per disability-adjusted life year (DALY) averted ; WHO calls an intervention "highly cost-effective" if it averts a DALY for less than the country's per-capita gross domestic product (Park 28e, Ch 23, p 1028; Health Economics, Ch 16, pp 253–254)
Cost-accounting	Financial records are kept so that costs attach to the purpose for which they were incurred; three purposes — (a) cost control, (b) planning and allocation of people and financial resources, (c) pricing for cost reimbursement	Costing a district hospital's laboratory or drug budget activity by activity rather than as a single line (Park 28e, Ch 23, p 1028)

Technique	Principle	Public health example
Input–output analysis	An economic technique: <i>input</i> = all health service activities which consume resources (manpower, money, materials and time); <i>output</i> = useful outcomes such as cases treated, lives saved or inoculations performed ; the input–output table shows how much of each input is needed to produce a unit amount of each output, and permits calculation of the effects of changing the inputs	How many ANM-hours and vaccine vials are consumed per 100 fully immunised children (Park 28e, Ch 23, pp 1028–1029)
Model	A basic concept of management science — an abstraction of reality, not reality itself — used to understand how the factors in a situation affect one another; the decision process includes the use of a model	The input–process–output model of community health care used to project manpower or bed requirements (Park 28e, Ch 23, p 1029)
Systems analysis	Helps the decision maker choose a course of action by investigating the problem, searching out objectives, finding alternative solutions, evaluating them for cost-effectiveness, re-examining objectives if necessary, and selecting the most cost-effective alternative	Analysing a hospital supply system, an information system, an outpatient clinic, or an entire community health service system (Park 28e, Ch 23, p 1029)
Network analysis — (a) PERT, (b) CPM	A network is a graphic plan of all events and activities to be completed in order to reach an end objective; it brings greater discipline into planning. PERT (programme evaluation and review technique) makes possible more detailed planning and more comprehensive supervision — its essence is an arrow diagram of the logical sequence in which events must take place, from which the time by which each activity must be completed and the critical activities are calculated, so that all concerned know what is expected of them and delays are minimised. CPM (critical path method) — the longest path of the network is the <i>critical path</i> ; if any activity along it is delayed the entire project is delayed	Scheduling the construction and commissioning of a new primary health centre, or the sequence of activities for a mass immunisation round (Park 28e, Ch 23, p 1029)
Gantt (bar) chart	The project's time frame drawn as a bar chart of each activity against a calendar time scale; network analysis grew out of Gantt charting and the two are used together — the network fixes the logical sequence, the Gantt chart fixes the actual dates and shows (by dotted extension) how far an activity may slip without delaying the project	The time frame of a health programme drawn as activities <i>a–h</i> against dated columns (Suryakantha, Ch 37, p 805, Fig 37.1; Reinke, Ch 15, pp 224–225); also listed as a routine monitoring tool alongside monthly reports and supervisory visits (Sunder Lal, Ch 16, p 723)

Technique	Principle	Public health example
Planning-programming-budgeting system (PPBS)	Helps decision makers allocate resources so that they are used most effectively for the organisation's objectives; needs no change in the existing organization, only the grouping of activities into programmes related to each objective . Park's allied approach is the " Zero Budget Approach " (commonly called zero-based budgeting) — all budgets start at zero and no one gets any budget he cannot specifically justify on a year-to-year basis	Budgeting a national health programme by objective — case detection, treatment, supervision — instead of by conventional line item (Park 28e, Ch 23, p 1029)
Work sampling	Systematic observation and recording of the activities of one or more individuals at predetermined or random intervals; measures the type of activity and the time needed for specified jobs; judges the appropriateness of the current job description and training, standardises methods and determines manpower needs	Work-sampling studies carried out on doctors, nurses, pharmacists and laboratory technicians (Park 28e, Ch 23, p 1029)
Decision making	Analogous to differential diagnosis in clinical practice; decisions should be made at the level where the best decision can be made — not necessarily at the top of the organization — and never on incomplete data	Decisions on development of resources, optimum workload for medical and paramedical workers, and strategies for providing health care (Park 28e, Ch 23, pp 1029–1030)

Constructing the PERT network — the procedure

- Identify the **key activities** in the project.
- Devise the **sequence** of activities and arrange them in a flow diagram.
- Assign a **duration of time** for performance of each phase of the work.
- **Develop the PERT network** — circles at the beginning and end of each activity are the **events**; the lines joining them are the **activities**.
- **Identify the critical path** — begin at the right where the large task is complete, take the last task that consumed the most time, connect it to the precedent activity that required the most time, and continue right to left across the page.
- **Perform network time calculations** — add the time allotments along the critical path; the sum is the total time required to complete the project.
- **Re-plan** — try other routes, always ensuring that no activity commences before its precedents are complete (e.g. *Anganwadi* workers cannot be trained before they are recruited and appointed).

(Sunder Lal, Ch 16, p 726)

Limitations and refinements. The biggest disadvantage of PERT is over-emphasis on time and almost no attention to cost (Sunder Lal, Ch 16, p 726). CPM is the earliest and simplest form of network analysis and does not allow for error in the time estimates; PERT adds that feature — the estimator supplies a most **optimistic**, a most **pessimistic** and an **expected** time for each activity, and the required time is a weighted average in which the best estimate carries four times the weight of either extreme (Reinke, Ch 15, pp 225–226).

Operational research — the umbrella over the quantitative techniques

Operational research (OR) is the application of scientific methods, techniques and tools to the operations of a system so as to obtain the optimum solution to the problem, giving the executive department a quantitative basis for decision making; its purpose is to improve the efficiency of the health system or of the health team. Its techniques include network analysis, PERT, critical path analysis, linear programming, computer simulation, queuing theory, cost-effectiveness analysis and work study (Sunder Lal, Ch 6, p 265). Indian examples: the directly observed treatment, short-course (DOTS) strategy of the tuberculosis programme was developed after operational research studies, as were joint ANM–Anganwadi worker home visits and home management of diarrhoea with available fluids (Sunder Lal, Ch 6, p 266).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **PERT / CPM arrow diagram** — labels: numbered event circles (0, 1, 2 ... 9); arrows as activities, each carrying its duration in weeks; "Start plan" at the first event; "Complete plan and start service" at the terminal event; the longest chain marked **critical path**.
- ▶ **The planning cycle** — labels: Gather health data → Tabulate, analyse, interpret → Formulate health problems → Assign priorities among problems → Define programme objectives and how to measure achievement → Assign priorities among objectives → Design alternative programmes → Select best programme (feasibility, cost/effectiveness) → Combine programmes and develop integrated plan → Plan implementation → Orient and train personnel → Initiate integrated programme → Operate programme → Collect data for evaluation → Evaluate results (goals achieved? satisfactory?) → Yes = continue operating, No = re-formulate health problems.

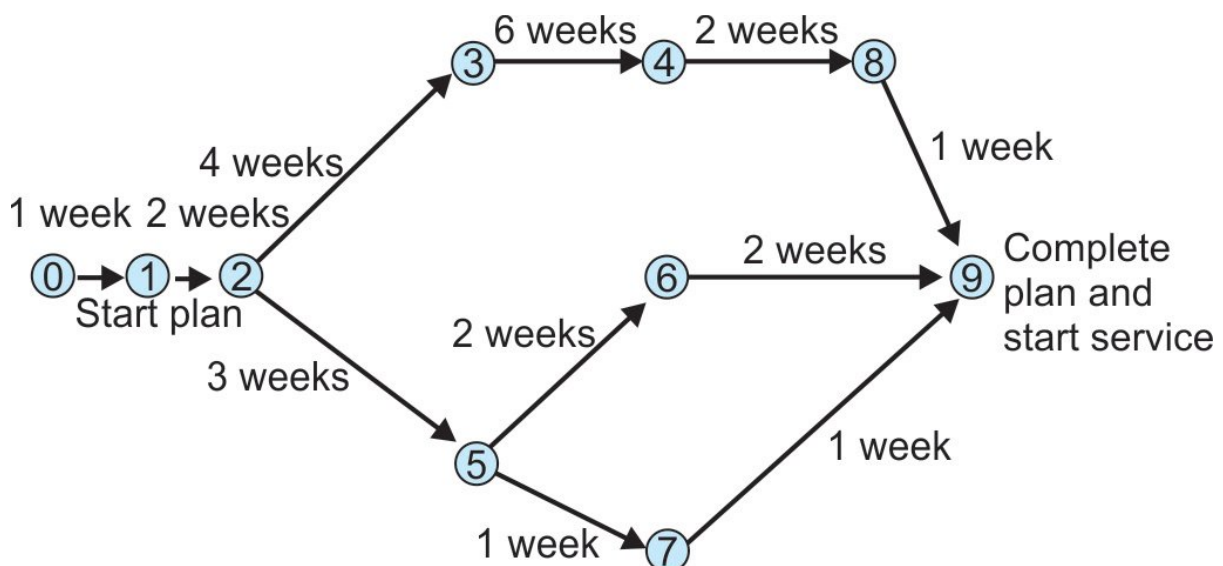


Fig 37.3 — Flow diagram of critical path (Suryakantha Environment Best, p.806)

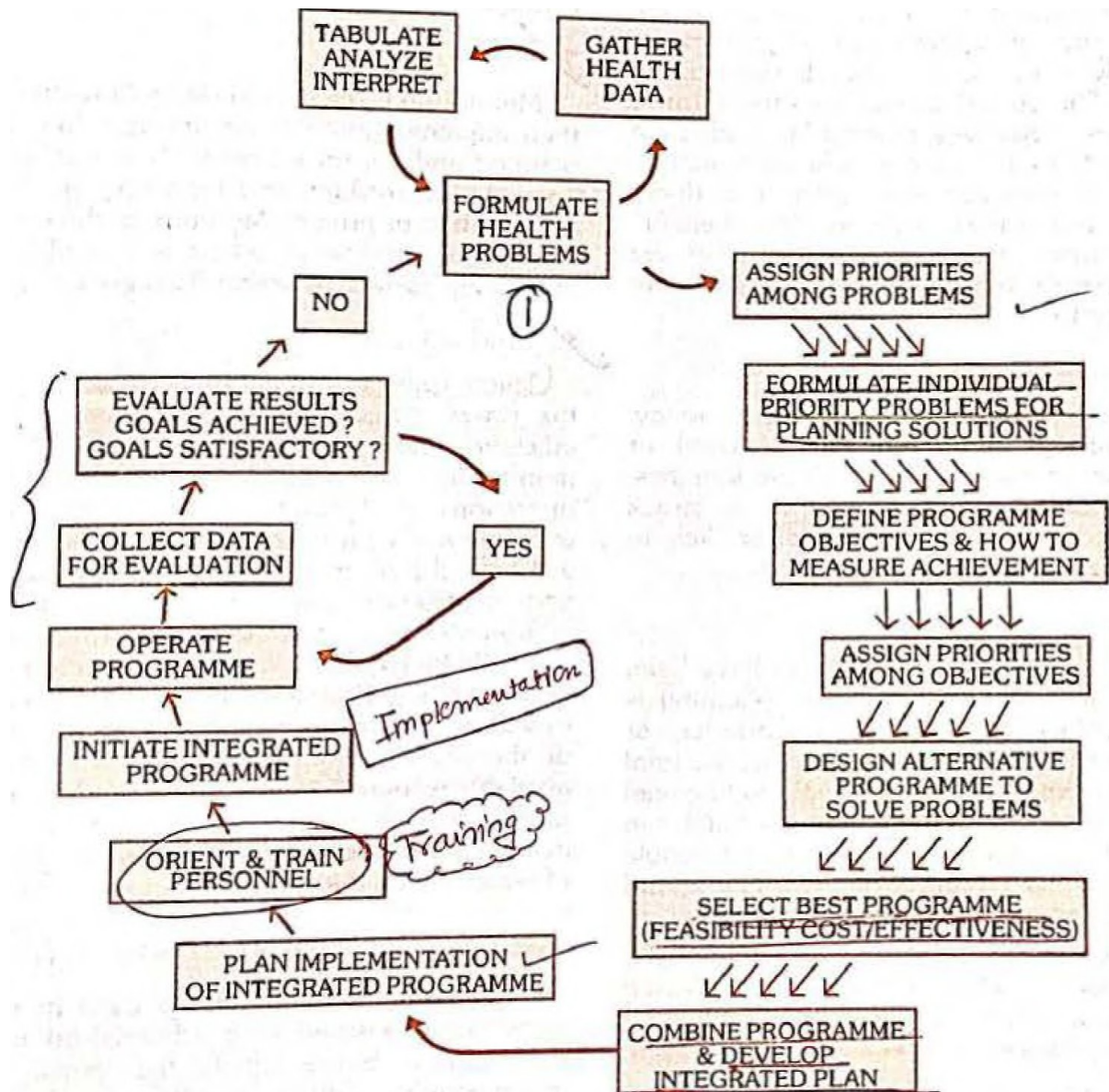
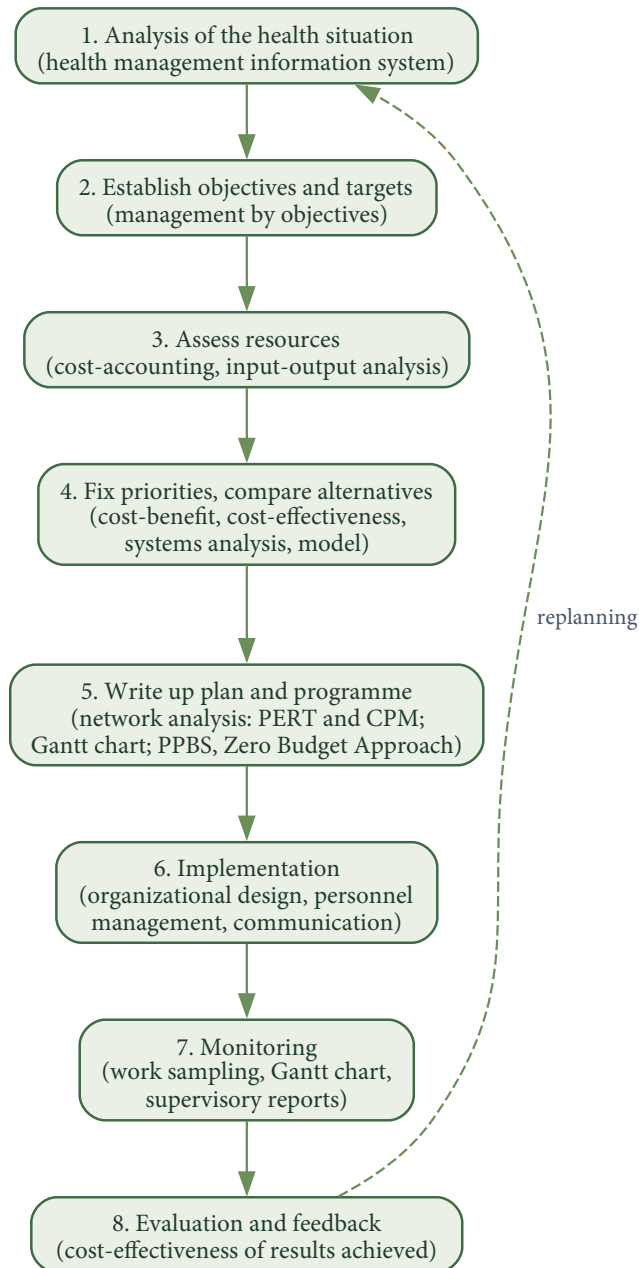


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, p.1026)



Where each management technique is applied along the health planning and management cycle

High-yield exam pointers

- ▶ Park's split: 5 behavioural + 10 quantitative techniques.
- ▶ Four managerial activities: **planning, organizing, communicating, monitoring (controlling)**.
- ▶ CBA = money; CEA = results achieved; select when benefit/cost ratio >1.
- ▶ Critical path = longest path; CPM = single time estimate, PERT = three time estimates.
- ▶ Park's term is the "Zero Budget Approach" (= zero-based budgeting).
- ▶ WHO threshold: highly cost-effective = averts one DALY for less than per-capita GDP.
- ▶ Decide at the level where the best decision can be made, never on incomplete data.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park 28e presents this set as the techniques promoted by the World Health Organization (Application of Modern Management Methods and Techniques for the Improved Delivery of Health Services, SEA/PHA/120, New Delhi 1973); a further managerial tool taught alongside them in Indian community medicine, **SWOT analysis** (strengths, weaknesses, opportunities, threats), appears in the corpus in Suryakantha's information, education and communication chapter rather than within this management-techniques list (Suryakantha, Ch 27, p 747).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e, Ch 23 (Health Planning and Management), pp 1025–1030 and p 1035; Sunder Lal — Textbook of Community Medicine, Ch 16 (Health Planning and Management), pp 704–707, 723 and 726, Ch 6 pp 265–266, Ch 8 p 530; Suryakantha — Community Medicine with Recent Advances, Ch 37 (Health Planning and Management), pp 805–808 and Ch 27 p 747; Reinke — Health Planning for Effective Management, Ch 15 (Tools for Improving the Process of Planning), pp 222–226; Health Economics, Ch 16 (Economic evaluation and decision-making), pp 253–255; WHO, Application of Modern Management Methods and Techniques for the Improved Delivery of Health Services, SEA/PHA/120, New Delhi 1973.

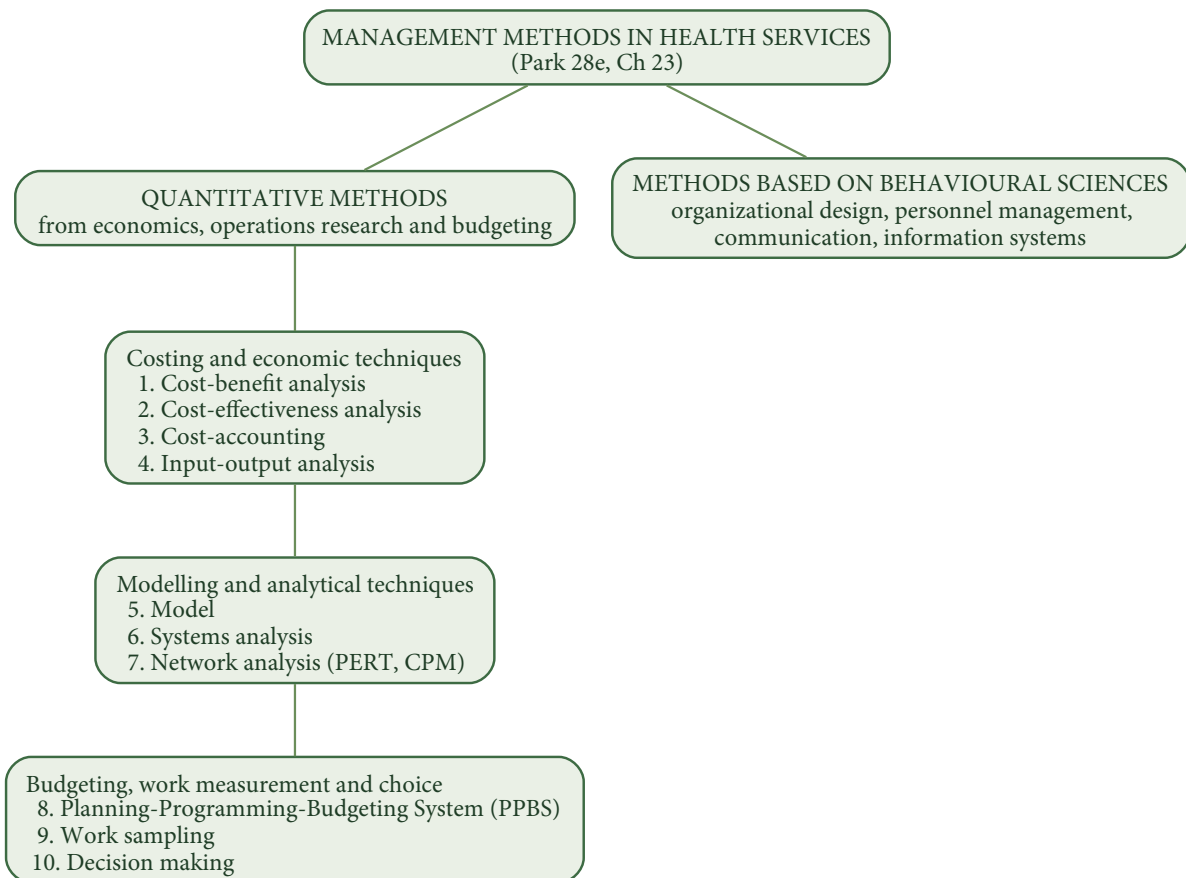


Quantitative management methods in health care

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — Quantitative management methods are the techniques of health service management derived from the fields of economics, operations research and budgeting, which express a management problem in measurable terms — money, time, manpower, materials and output — so that alternatives can be compared objectively and scarce resources allocated to best effect (Park 28e, Ch 23, p 1028). Park classifies management methods into **quantitative methods** and **methods based on the behavioural sciences**; the quantitative group is dealt with here.

Classification of quantitative management methods



Park's classification of management methods, with the quantitative group expanded

The ten quantitative methods and their health care applications

#	Method	Principle	Application in health care
1	Cost-benefit analysis (CBA)	Economic benefits of a programme are compared with its cost, both expressed in monetary terms, to decide whether the programme is economically sound and to select the best of several alternatives (Park 28e, Ch 23, p 1028)	Choosing between two alternative programmes for the same budget. Limitation: health benefits (births or deaths prevented, illness avoided) cannot always be put in money terms, so the scope of the method is vague (Park 28e, Ch 23, p 1028)

#	Method	Principle	Application in health care
2	Cost-effectiveness analysis (CEA)	Same as CBA except that benefit is expressed in results achieved — number of lives saved, number of days free from disease — instead of money; a more promising tool in the health field than CBA (Park 28e, Ch 23, p 1028)	Cost per death averted when comparing two vector-control options; cost per case detected when comparing two screening strategies
3	Cost-accounting	Provides basic data on the cost structure of a programme; financial records are kept so that costs are associated with the purpose for which they are incurred. Three purposes — (a) cost control, (b) planning and allocation of people and financial resources, (c) pricing of cost reimbursement (Park 28e, Ch 23, p 1028)	Deriving the unit cost of an outpatient visit or an inpatient bed-day for reimbursement under an insurance scheme; controlling drug expenditure at a district hospital
4	Input-output analysis	An economic technique in which <i>input</i> refers to resources consumed (manpower, money, materials) and <i>output</i> to useful outcomes such as lives saved or inoculations performed; the input-output table shows how much of each input is needed to produce one unit of each output (Park 28e, Ch 23, pp 1028-1029)	Estimating the auxiliary nurse midwife (ANM) time, vaccine and cold-chain input required per 1000 fully immunised children
5	Model	A basic concept of management science — an abstraction of reality, not the reality itself, used to understand how the factors in a situation affect one another; the decision process includes the use of a model (Park 28e, Ch 23, p 1029)	A transmission model used to size the response to an outbreak; a manpower projection model for a district
6	Systems analysis	Helps the decision maker choose an appropriate course of action by investigating the problem, searching out objectives, finding alternative solutions, evaluating the alternatives on cost-effectiveness, re-examining the objectives if necessary and finding the most cost-effective alternative. The "system" may be a hospital supply system, an information system, a total community health service system or an outpatient clinic, and may be made of independent subsystems (Park 28e, Ch 23, p 1029)	Redesigning the district drug supply chain; re-engineering outpatient department flow
7	Network analysis (PERT and CPM)	A network is a graphic plan of all events and activities to be completed in order to reach an end objective; it brings greater discipline into planning (Park 28e, Ch 23, p 1029)	Commissioning a new primary health centre; scheduling a pulse polio round or a mass drug administration campaign (detailed below)

#	Method	Principle	Application in health care
8	Planning-Programmi ng-Budgeting System (PPBS)	Primarily a system to help decision makers allocate resources so that the available resources of an organisation are used in the most effective way in achieving its objectives. It does not call for changes in the existing organization; it calls for grouping of activities into programmes related to each objective . A related approach is the zero budget approach — all budgets start at zero and no one gets any budget that he cannot specifically justify on a year-to-year basis (Park 28e, Ch 23, p 1029)	Presenting a district health budget by programme (maternal health, immunisation, tuberculosis) rather than by line item such as salaries and drugs
9	Work sampling	Systematic observation and recording of the activities of one or more individuals, carried out at predetermined or random intervals; it gives a quantitative measurement of the various activities. The parameters analysed are the type of activity performed and the time needed to do specified jobs (Park 28e, Ch 23, p 1029)	Studies have been done on doctors, nurses, pharmacists and laboratory technicians; it permits judgement on the appropriateness of the current staff's job description and training, helps standardise methods of performing jobs, and determines manpower needs — e.g. an activity study of an ANM to revise her job description
10	Decision making	Likened to the basic discipline of differential diagnosis in medical practice. Decisions should be made at the level where the best decisions can be made — it does not follow that the best decision is always made at the top of the organization — and decisions should not be made with incomplete data (Park 28e, Ch 23, pp 1029-1030)	In the health sector decisions concern development of resources, optimum work load for medical and paramedical workers, and strategies for providing health care (Park 28e, Ch 23, pp 1029-1030)

Economic evaluation techniques compared

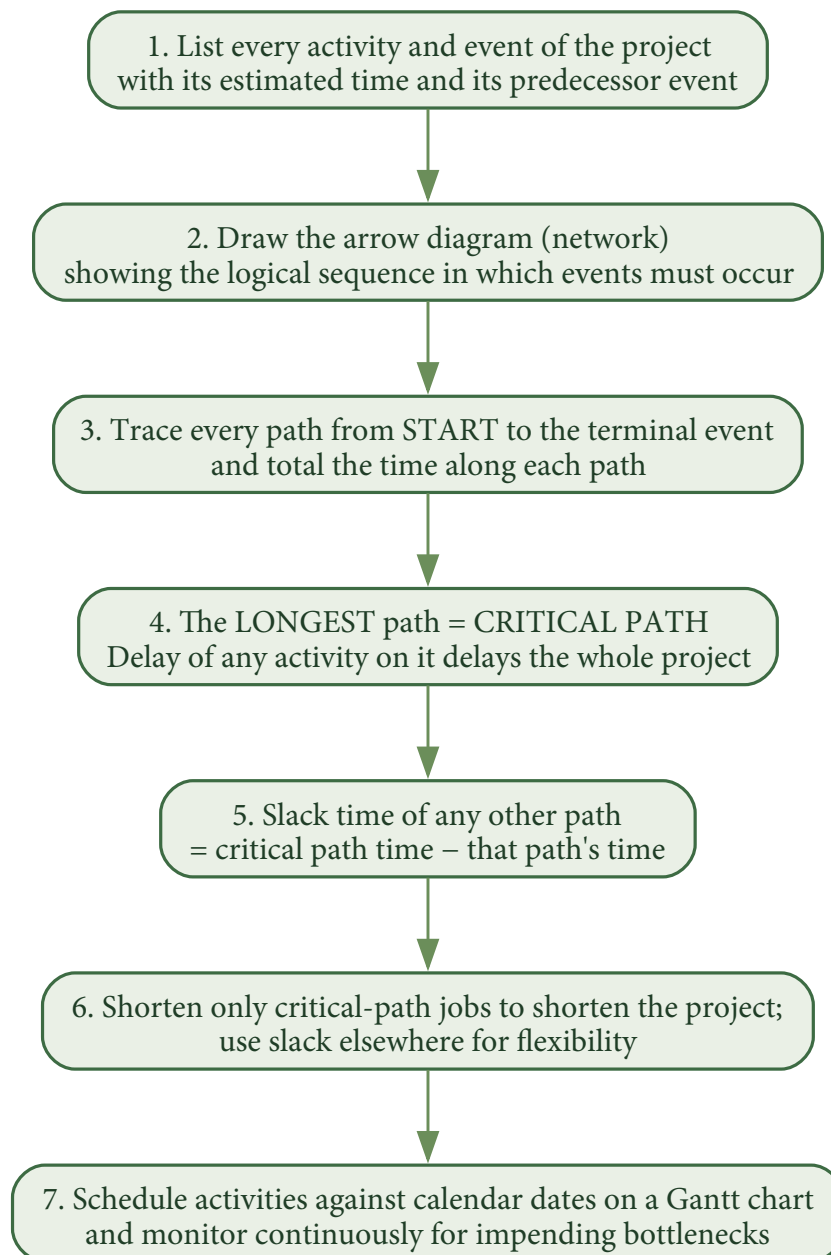
Feat ure	Cost-benefit analysis (CBA)	Cost-effectiveness analysis (CEA)	Cost-utility analysis (CUA)
Costs valued in	Money	Money	Money
Conse quenc es valued in	Money	Natural health units, e.g. life years saved (Health Economics, Ch 13, p 201)	Health units capturing quantity and quality of life — quality-adjusted life year (QALY), disability-adjusted life year (DALY) (Health Economics, Ch 13, p 201; Ch 15, pp 230-231)

Feature	Cost-benefit analysis (CBA)	Cost-effectiveness analysis (CEA)	Cost-utility analysis (CUA)
Summary measure	Benefit-cost ratio (BCR) = total monetised benefits ÷ total costs; net present value (NPV) = total monetised benefits – costs (Health Economics, Ch 16, pp 247-248)	Average cost-effectiveness ratio (ACER) = total cost of intervention A ÷ total effects of A; incremental cost-effectiveness ratio (ICER) = difference in cost between two programmes ÷ difference in effectiveness; marginal cost-effectiveness ratio (MCER) = change in cost and effect when a programme is expanded or contracted (Health Economics, Ch 16, pp 247-248, 250)	Cost per QALY gained or cost per DALY averted, judged against a ceiling ratio (Health Economics, Ch 16, p 254)
Answers the question	Is the programme worth doing at all in money terms?	Which programme buys the most health per rupee?	Which programme buys the most health per rupee once quality of life is counted?
Health example	Appraising a piped water supply scheme against its economic returns	Cost per tuberculosis case cured under two treatment delivery models	Cost per QALY of two antihypertensive strategies

Three cross-cutting rules apply to all three forms:

- **Perspective** must be stated at the outset — whose costs and whose outcomes are counted (Health Economics, Ch 13, p 211).
- **Discounting** converts future costs and outcomes to their present value; national and international guidelines recommend a **3 per cent discount rate** after controlling for inflation (Health Economics, Ch 14, p 228).
- **Sensitivity analysis** deliberately varies the uncertain assumptions to test whether the conclusion is robust; cost-effectiveness ratios are therefore reported as a range, not a point (Health Economics, Ch 13, p 212; Ch 16, p 254).

Network analysis — PERT and CPM



Steps in network analysis (PERT/CPM) applied to a health project

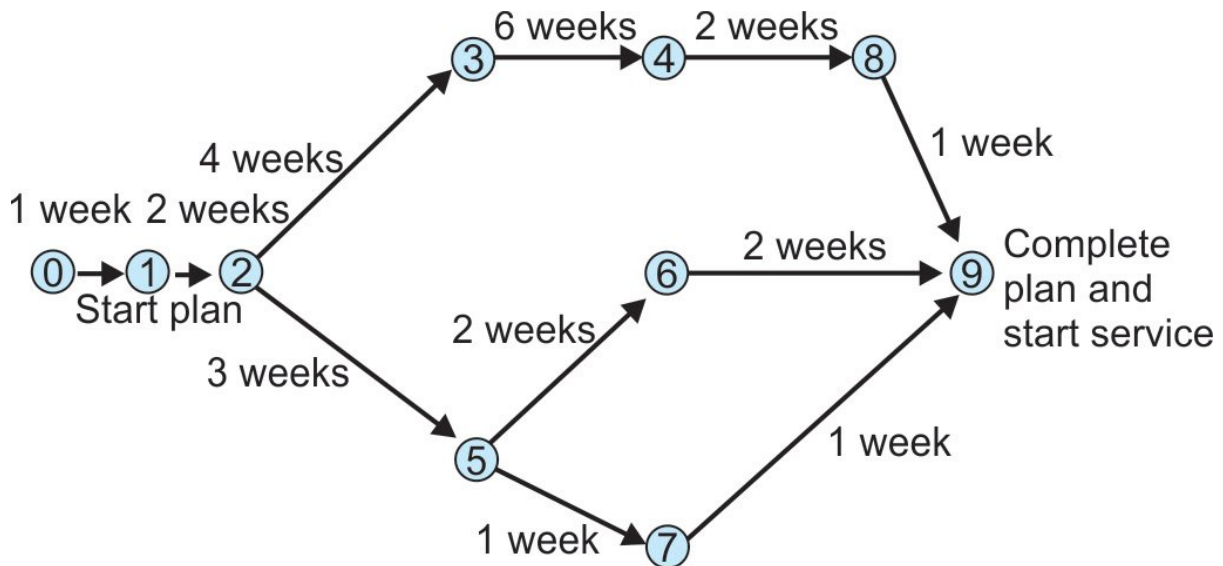


Fig 37.3 — Flow diagram of critical path (Suryakantha Environment Best, p.806)

Feature	Critical Path Method (CPM)	Programme Evaluation and Review Technique (PERT)
Time estimate per activity	A single estimate; the earliest and simplest form of network analysis (Reinke, Ch 15, pp 225-226)	Three estimates — most optimistic, expected (most likely) and most pessimistic (Reinke, Ch 15, pp 225-226)
Expected activity time	The estimate itself	A weighted average in which the best estimate is given four times as much weight as either of the other two, i.e. (optimistic + 4 × most likely + pessimistic) ÷ 6 (Reinke, Ch 15, pp 225-226)
Allows for error in time estimates	No — this is its main failing	Yes; it is the best-known approach that includes the error feature (Reinke, Ch 15, pp 225-226)
Critical path	The longest path through the network; if any activity on it is delayed, the entire project is delayed (Park 28e, Ch 23, p 1029)	Same
Best suited to	Projects with well-known, repetitive activity times — construction of a health centre, hospital wing or laboratory	Projects with uncertain activity times — a new programme launch, a research project, an epidemic response
Health application	Reinke's worked example schedules an emergency health centre activation: framework, wiring, plumbing, flooring, walls, roof, drug authorisation, ordering and receipt, and personnel authorisation, recruitment and training, with the critical path running through the plastering job (Reinke, Ch 15, pp 222-223)	Planning a mass immunisation round or a new district programme where microplanning, indent, training and cold-chain readiness must converge on one date

Related scheduling tools — the **Gantt chart** schedules the network's activities against calendar dates and shows by how much each may be delayed without harm; **PERT/COST** and **PERT/MANPOWER**

add cost and manpower dimensions; **decision CPM** analyses alternative sets of activities; **RAMPS** handles resource allocation across multiple projects (Reinke, Ch 15, pp 224-226).

Operations research models

Model	What it optimises	Application in health care
Linear programming	Has three features — an objective function to be optimised, constraints on the resources (inputs) available to produce the desired outputs, and the assumption that all relationships are linear and additive (Reinke, Ch 3, p 50)	Determining the optimal service mix or staffing mix . In Reinke's worked example the optimum was to train and employ 22 government workers and 64 community workers, yielding 41,200 service contacts at an average cost of \$1.69 each (Reinke, Ch 3, pp 51-52)
Distribution models	The least costly means of supplying several distributors from existing supply depots or production facilities — which depot should supply which distributor (Reinke, Ch 3, p 52)	Drug and vaccine supply logistics; deciding which peripheral clinic should refer to which referral point when referral capacity is insufficient for all patients from the nearest centres
Queuing models	Analysis of waiting lines — people waiting for service or hospital beds waiting for use; establishes the appropriate balance between patient waiting time and personnel idle time and the potential benefit of an appointment system (Reinke, Ch 3, p 52)	Outpatient department and health centre patient flow, where walk-in arrivals are random and cluster unevenly, so extra service capacity must be held in readiness

Honest limitation — health status is difficult to quantify and even more difficult to relate precisely to specified levels of service input, so the mathematical techniques of operations research have limited applicability in health matters; their value lies as much in forcing careful structuring of the problem as in producing a definitive solution (Reinke, Ch 3, pp 49-50).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Arrow (network) diagram for PERT/CPM** — draw circles numbered 0 to 9 as *events*, joined by arrows as *activities*, each arrow labelled with its time. Labels: START ("plan service"), "staff recruited", "staff trained", "equipment ordered", "equipment installed", TERMINAL EVENT ("start providing service"); mark the **critical path** with a double line and write the **slack time** beside each non-critical path.
- ▶ **Cost-effectiveness plane** — draw two axes crossing at the origin: horizontal = difference in effect (new – comparator), vertical = difference in cost. Label the four quadrants: north-east "more costly, more effective — judge against threshold", north-west "dominated (more costly, less effective)", south-east "dominant (less costly, more effective)", south-west "less costly, less effective". Draw the threshold (ceiling ratio) line through the origin.

High-yield exam pointers

- ▶ **Park's two families** — quantitative methods (economics, operations research, budgeting) versus methods based on behavioural sciences (organizational design, personnel management, communication, information systems).

- ▶ **The ten quantitative methods, in Park's order** — cost-benefit analysis, cost-effectiveness analysis, cost-accounting, input-output analysis, model, systems analysis, network analysis, PPBS, work sampling, decision making.
- ▶ **PERT expected time** = (optimistic + 4 × most likely + pessimistic) ÷ 6; **critical path** = the longest path; **slack time** = critical path time – that path's time.
- ▶ **CEA is more promising in health than CBA** because health benefit resists monetary valuation — the single most quotable line of this answer.
- ▶ **Cost-accounting's three purposes** — cost control; planning and allocation of people and financial resources; pricing of cost reimbursement.
- ▶ **PPBS** groups activities into programmes related to each objective and needs no change in the existing organization; the **zero budget approach** starts every budget at zero each year.
- ▶ **Economic-evaluation ratios** — ACER (single intervention vs baseline), MCER (expansion or contraction), ICER (one programme vs another), BCR and NPV (cost-benefit analysis only); discount rate **3 per cent**; always report a sensitivity analysis.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the cost-per-QALY ceiling ratios quoted in the source (about £20,000–30,000 per QALY in the United Kingdom, and roughly US\$50,000 per QALY in the United States) are the source text's illustrations of how a threshold is used — India has no single officially notified cost-effectiveness threshold, so quote them as examples of the method, not as an Indian standard. Comparing interventions in "league tables" of cost per QALY or DALY is common but unreliable, because the ratios come from studies using different comparators, discount rates, time horizons and population subgroups (Health Economics, Ch 16, pp 254-255).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Indian Reference Case (HTAIn), WHO South-East Asia Regional Office

Programme and policy updates (India)

- ▶ India has institutionalised health technology assessment inside its flagship insurance scheme — a Health Financing and Technology Assessment (HeFTA) unit now sits within the National Health Authority and uses HTA evidence to decide inclusions and exclusions in the Ayushman Bharat PM-JAY health benefit package and to frame standard treatment guidelines, with reported savings to the scheme; the stated next step is value-based pricing linked to strategic purchasing [PMID 38715207].
- ▶ Cost-accounting has moved from textbook principle to national price-setting. The Cost of Health Services in India (CHSI) study costed 63 hospitals across 14 states using combined top-down and bottom-up methods; at the health-benefit-package 1.0 → 2.0 revision 61 per cent of package rates rose and 18 per cent fell, and the share of procedures reimbursed at under half their measured cost (42 per cent under HBP 1.0) was halved. The 2022 revision raised 832 package rates, cut 122 and left 630 unchanged, and introduced differential pricing by city tier. CHSI data are now public as the National Health System Cost Database, and the argument is to replace one-off

costing surveys with routine cost surveillance built into hospital information systems [PMID 37963613].

- Cost-effectiveness analysis is now applied to national programme strategy and not only to technologies — an evaluation of Intensified Mission Indradhanush across 40 districts found an incremental cost of US\$6.21 per additional vaccine dose delivered and US\$82.99 per zero-dose child reached, about US\$328 per DALY averted, cost-effective against a threshold of one per-capita GDP per DALY with 90 per cent probability (costs and effects discounted at 3 per cent) [PMID 38590052].

Current national data

- District-hospital budgeting now has an evidence-based alternative to historical line-item allocation. Costing of 27 district hospitals in nine states supports a blended payment model — risk-adjusted global budget for outpatient care, case-based bundled payment for inpatient care, per-bed global budget for indirect services. Applied to India's 845 district hospitals the mean annual pay-out is ₹326 million (US\$3.35 million), range ₹66 million to ₹2.57 billion, with inpatient care taking 78 per cent; upgrading to Indian Public Health Standards would raise budgets by 131 per cent and upgrading to medical colleges by 91.5 per cent [PMID 39478753].
- Bottom-up unit costs for Indian primary care (51 subcentres and 33 primary health centres across four states): mean annual cost ₹0.69 million (US\$11,392) per subcentre and ₹5.1 million (US\$83,837) per PHC, salaries taking 74 and 63 per cent respectively; per antenatal visit ₹221 at subcentre level, per outpatient consultation ₹121 and per inpatient bed-day ₹1168 at PHC level. These are the input–output figures to quote for an Indian example rather than foreign illustrations [PMID 36588362].
- Work sampling has a national-scale successor in the WHO Workload Indicators of Staffing Need (WISN) method. A retrospective WISN analysis of 93 rural facilities in five states, modelled up to national estimates for 2019, found shortages of nurses at both PHCs and community health centres and of every specialist cadre at CHCs, with very high workload pressure, and poor concordance between posts sanctioned under existing norms and workload-based requirement — every cadre was under-sanctioned [PMID 35090494].

Elimination, surveillance and disease status

- Programme costing now underpins resource allocation in the National TB Elimination Programme. Bottom-up unit costs from 20 facilities (2018 US\$, Global Health Cost Consortium reference case): Ziehl–Neelsen smear microscopy \$2.45, digital chest radiograph \$2.85, Xpert MTB/RIF \$17.36, outpatient screening visit \$1.62, inpatient bed-day \$5.44, BCG vaccination \$0.76, and a full first-line regimen \$103 for new pulmonary and \$98 for extrapulmonary tuberculosis (drugs \$54 of each), rising to \$196 and \$183 respectively in previously treated disease [PMID 34886932].

Guideline and legislative updates

- India now has its own methodological standard for economic evaluation — the **Indian Reference Case**, developed with Health Technology Assessment in India (HTAIn) under the Department of Health Research. It issues twelve recommendations across eleven principles: decision problem, comparator, perspective, source of effectiveness evidence, measurement of costs, measurement of health outcomes, time horizon, discounting, heterogeneity, uncertainty analysis and equity

analysis, plus presentation of results. It was written because Indian evaluations had been using incompatible methods — societal perspective in 38 per cent against payer perspective in 48 per cent, QALYs in 29 per cent and DALYs in only 9 per cent — which made their ICERs non-comparable [PMID 37694178].

- The gap flagged in the note above — that India has no officially notified cost-effectiveness threshold — now has a first national estimate. A time-trade-off and willingness-to-pay survey of 5460 respondents across six states (21,640 observations, health gains discounted at 3 per cent) put mean weighted willingness to pay per QALY at ₹2,12,307 (US\$2535), ranging ₹1,70,414 to ₹2,58,985 across health states, or 1 to 1.52 times per-capita GDP, and proposes this as the cost-effectiveness threshold for health technology assessment in India [PMID 41176029].

Global and WHO developments

- WHO's WISN method has been reviewed across six South-East Asia Region countries including India, where the government had piloted it for the first time. The binding constraint proved to be the inputs rather than the arithmetic — none of the six had pre-defined service activity standards or reliable workload data — so the recommendation is to digitise workload standards, strengthen health workforce information systems, and involve the treasury and the employing ministry from the planning stage so that the staffing recommendations are affordable when they arrive [PMID 40512697].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e, Ch 23 (Health Planning and Management), pp 1027-1030 — the classification and the ten quantitative methods; Reinke — Health Planning for Effective Management, Ch 3 (Health Systems Research in Relation to Planning), pp 46-52 and Ch 15 (Tools for Improving the Process of Planning), pp 222-227 — operations research models, network analysis, PERT/CPM, slack time and Gantt charts; Health Economics, Ch 13 (What is economic evaluation), Ch 14 (Counting the costs), Ch 15 (Identifying, measuring and valuing consequences) and Ch 16 (Economic evaluation and decision-making), pp 201-260 — CBA/CEA/CUA, QALY and DALY, ACER/MCER/ICER/BCR/NPV, discounting and sensitivity analysis; Suryakantha — Community Medicine with Recent Advances, p 806 (Fig 37.3) — critical path flow diagram.

Recent-advances sources: 38715207 — Establishing a Health Technology Assessment Evidence Ecosystem in India's Pradhan Mantri Jan Arogya Yojana, *Health Systems and Reform* 2024, doi:[10.1080/23288604.2024.2327097](https://doi.org/10.1080/23288604.2024.2327097) (abstract-only); 37963613 — National hospital costing systems matter for universal healthcare: the India PM-JAY experience, *BMJ Global Health* 2023, doi:[10.1136/bmjgh-2023-012987](https://doi.org/10.1136/bmjgh-2023-012987); 38590052 — Health impact and cost-effectiveness of expanding routine immunization coverage in India through Intensified Mission Indradhanush, *Health Policy and Planning* 2024, doi:[10.1093/heapol/czae024](https://doi.org/10.1093/heapol/czae024); 39478753 — Framework for responsive financing of district hospitals of India, *Frontiers in Public Health* 2024, doi:[10.3389/fpubh.2024.1398227](https://doi.org/10.3389/fpubh.2024.1398227); 36588362 — Cost of delivering primary healthcare services through public sector in India, *Indian Journal of Medical Research* 2022, doi:[10.4103/ijmr.IJMR_67_19](https://doi.org/10.4103/ijmr.IJMR_67_19); 35090494 — Workforce problems at rural public health-centres in India: a WISN retrospective analysis and national-level modelling study, *Human Resources for Health* 2022, doi:[10.1186/s12960-021-00687-9](https://doi.org/10.1186/s12960-021-00687-9); 34886932 — Costs of TB services in India, *International Journal of Tuberculosis and Lung Disease* 2021, doi:[10.5588/ijtld.21.0105](https://doi.org/10.5588/ijtld.21.0105); 37694178 — Development of the Indian Reference Case for undertaking economic evaluation for health technology assessment, *Lancet Regional Health – Southeast Asia* 2023, doi:[10.1016/j.lansea.2023.100241](https://doi.org/10.1016/j.lansea.2023.100241); 41176029 — Estimating the Willingness-to-Pay per Quality-Adjusted Life-Year to Aid Health Technology Assessment in India, *Value in Health* 2025, doi:[10.1016/j.jval.2025.10.007](https://doi.org/10.1016/j.jval.2025.10.007) (abstract-only); 40512697 — The implementation of the Workforce Indicators of Staffing Needs (WISN) method to improve access to health workforce in selected South-East Asian countries, *PLOS Global Public Health* 2025, doi:[10.1371/journal.pgph.0004727](https://doi.org/10.1371/journal.pgph.0004727).



Cataract blindness control programme planning

Asked in: Paper IV Nov 2017 — RGUHS MD Community Medicine

Definition — Planning is the process of analysing a system, defining a problem, assessing the extent to which the problem exists as a need, formulating goals and objectives, choosing among alternative intervention strategies, initiating the system to ensure proper implementation, and evaluating the results against the stated objectives (Park 28e, Ch 23, p 1025). It proceeds as a **succession of eight steps that close into a cycle**, the evaluation of one round feeding the situation analysis of the next. The worked example below designs a cataract blindness reduction programme for **one rural block of 1,00,000 population** served by a community health centre (CHC), within the National Programme for Control of Blindness and Visual Impairment (NPCBVI).

Pre-planning (the preconditions that must exist before the cycle can start) — government interest and political will, enabling legislation, an organisation for planning, and administrative capacity for coordination and implementation (Park 28e, Ch 23, p 1025). For this programme these are already met: NPCBVI is an existing centrally sponsored scheme (60:40 centre-state share; 90:10 in North-Eastern states), and the District Health Society, into which the erstwhile District Blindness Control Society was merged, is the standing planning organisation (Park 28e, Ch 7, p 527; Ch 13, p 850).

Step 1 — Analysis of the health situation (situation analysis)

The minimum essential data for health planning are collected, assessed and interpreted to yield the health problems, health needs and health demands of the block (Park 28e, Ch 23, p 1027).

Minimum essential datum (Park 28e, Ch 23)	How it is obtained in the block	Illustrative finding for 1,00,000 population
Population, its age and sex structure	Sub-centre family registers, Census 2011 projections	Size of the 50 years and above cohort — the target denominator for cataract screening
Statistics of morbidity and mortality	Health management information system; block survey	Blindness prevalence 0.36% (National Blindness and Visual Impairment Survey, NBVIS, 2015–19) → ~360 blind persons
Epidemiology and geographical distribution of the disease	Village-wise line listing of the blind aged 50+ by trained health staff, non-governmental organisations (NGOs), <i>Panchayat</i> and volunteers given one day's training	Untreated cataract causes 66.2% of blindness in India (NBVIS 2015–19) → ~238 cataract-blind persons as the backlog; hamlets and tribal habitations clustered
Medical care facilities, public and private	Mapping of all fixed eye-care facilities and eye surgeons in government, voluntary and private sector in the district	One CHC without an operation theatre for eye surgery; nearest district hospital eye unit 40 km away
Technical manpower of various categories	District manpower register	One ophthalmic assistant at CHC; district eye surgeons shared
Training facilities available	District training centre, regional institute of ophthalmology	IOL (intraocular lens) surgery training available at the medical college

Minimum essential datum (Park 28e, Ch 23)	How it is obtained in the block	Illustrative finding for 1,00,000 population
Attitudes and beliefs of the population towards disease, its cure and prevention	Focus group discussions, key informant interviews	Fear of surgery, "wait until the cataract is ripe", loss of daily wage, no escort — the demand-side barriers

Determinants that shape the block's problem: **age** (senile cataract occurs about a decade earlier in India than in Europe and America), **sex** (higher prevalence in females), **social class** (blindness is twice as prevalent in the poorer classes as in the well-to-do), and **occupation** (exposure to X-rays, ultraviolet rays or heat waves causes premature cataract) (Park 28e, Ch 6, pp 477–478).

Step 2 — Establishment of goals, objectives and targets

A **goal** is the ultimate desired state towards which objectives and resources are directed, unconstrained by time or existing resources; an **objective** is a precise planned end-point, either achieved or not achieved; a **target** is a discrete activity that permits a degree of achievement (Park 28e, Ch 23, p 1025).

Level	Statement for this block
Goal	Elimination of avoidable blindness — <i>Vision 2020: The Right to Sight</i> , launched by WHO on 18 February 1999
National objective	Reduce prevalence of blindness to 0.25% (NPCBVI; original goal 1.4% → 0.3%)
Block objective (3 years)	Clear the existing cataract backlog of ~238 persons and attain "Cataract Backlog Free" status
Service target	Cataract surgical rate (CSR) of 400 operations per 1,00,000 population per year = 400 surgeries/year for this block
Technical target	Intraocular lens (IOL) implantation in at least the current national proportion of operations (94.2% in 2017–18)
Social/equity targets	>50% of operations in women; SC/ST and minority share of operations to match their census share; all below-poverty-line patients operated free

Step 3 — Assessment of resources, and Step 4 — Fixing priorities

Resources mean the **manpower, money, materials, skills, knowledge and techniques** needed or available, and a balance is struck between what is required and what is available (Park 28e, Ch 23, p 1027).

Resource	Requirement for 400 surgeries/year	Availability and gap
Manpower	1 eye surgeon (norm: 700 cataract operations per surgeon per year), 1 ophthalmic assistant, 1 counsellor	Surgeon shared from district hospital — gap met by fixed monthly operating days
Money	400 × ₹2,143 (government camp unit cost) ≈ ₹8.6 lakh; via the most cost-effective route — NGO screening camp with surgery at a base hospital, ₹1,128 per case — ≈ ₹4.5 lakh	District Health Society untied funds; cataract surgery packages under AB-PMJAY and state insurance schemes

Resource	Requirement for 400 surgeries/year	Availability and gap
Material	Operating microscope, IOLs, autoclave, dark room, transport	Mobile ophthalmic unit (80 central units attached to medical colleges and 341 district mobile units exist nationally)
Skills/ technique	IOL surgery training for the district surgeon; teacher training for school eye screening	Regional Institute of Ophthalmology

Priorities must be fixed because **resources always fall short of the total requirement**; attention is paid to financial constraints, mortality and morbidity data, diseases preventable at low cost, saving the lives of the young, and political and community pressures (Park 28e, Ch 23, p 1027). Two formal tools are used to keep this from being purely intuitive:

- **Hanlon's Basic Priority Rating System (BPRS)** = $(A + 2B) \times C$, where A = magnitude of the problem, B = seriousness, C = potential effectiveness of the intervention (Health Program Planning and Evaluation, Ch 4, p 140). Cataract scores highest of all eye conditions in the block: large magnitude (66.2% of blindness), high seriousness (bilateral, disabling, economically blinding), and very high intervention effectiveness (a single curative operation).
- **PEARL** screen of each high-scoring problem for Propriety, Economics, Acceptability, Resources and Legality, each scored yes (1) or no (0); a zero on any element removes the problem however high its rating (Health Program Planning and Evaluation, Ch 4, pp 142–143).

Where numerical data are thin, group consensus methods — the **nominal group technique** and the **Delphi technique** — are used to rank problems with the block-level stakeholders (Reinke, Ch 15, p 219).

Step 5 — Write-up of the formulated plan, and Step 6 — Programming and implementation

Alternative plans for achieving the priority are formulated and assessed for practicability and feasibility, and the alternative with greater effectiveness is chosen (Park 28e, Ch 23, p 1027). Here the choice lies between a **reach-in strategy** (surgery at fixed institutions) and a **reach-out strategy** (eye camps); the accepted mix is that **75–80% of the target is achieved through institutions and hospitals to ensure quality of service**, camps being reserved for remote, inaccessible or difficult areas and held only under conditions for safe surgery (Sunder Lal, Ch 8, p 519).

The written plan specifies **what will be done, when, and by whom**, and must contain a built-in system of evaluation (Park 28e, Ch 22, p 1024; Ch 23, p 1027). Its components are:

- **Primary level** — preventive and promotive eye care and information, education and communication (IEC) activities through Ayushman Arogya Mandirs (erstwhile Health and Wellness Centres); house-to-house vision screening of all persons aged 50+ by the accredited social health activist (ASHA) and multipurpose worker with a distance vision chart; village-wise blind register.
- **School eye screening** — teachers first screen children (6–7% of children aged 10–14 years have an eyesight problem affecting learning; teachers identify 80–90% of children with poor vision), suspected cases seen by the ophthalmic assistant, free spectacles to those below the poverty line.

- ▶ **Secondary level** — fixed operating days at the district hospital eye unit and accredited NGO eye hospital, with free transport and escort allowance; IOL implantation, now available up to taluka level.
- ▶ **Tertiary referral** — Regional Institutes of Ophthalmology and medical colleges for complications, glaucoma, diabetic retinopathy, vitreo-retinal disease and keratoplasty.
- ▶ **Partnership** — Lions International, Rotary International and NSPB India permitted to run camps under state guidelines; nationally the voluntary and private sector has covered over 61% of the backlog against 39% by government.
- ▶ **Plan execution** depends on defining responsibilities and tasks; selecting, training, motivating and supervising the manpower; and organisation and communication — many well-considered plans break down because of delays in critical supplies and inappropriate use of staff (Park 28e, Ch 23, p 1027).

Administratively the **district is the planning and implementation unit** and the District Collector leads the campaign, with district-wise targets and self-sufficiency in surgical capacity (Park 28e, Ch 7, p 527).

Step 7 — Monitoring, and Step 8 — Evaluation

Monitoring is the day-to-day follow-up of activities to ensure they are proceeding as planned and on schedule — observing, recording and reporting, identifying deviations and taking corrective action. **Evaluation** assesses achievement of the stated objectives, the programme's adequacy, efficiency and acceptance by all parties involved, and is mostly concerned with the end outcome (Park 28e, Ch 23, p 1027). In the words of the WHO Expert Committee on National Health Planning in Developing Countries, evaluation "measures the degree to which objectives and targets are fulfilled and the quality of the results obtained... It makes possible the reallocation of priorities and of resources on the basis of changing health needs."

Type	Indicator	Formula / definition
Input	Eye-care manpower and equipment availability	Eye surgeons and ophthalmic assistants per lakh population; functioning operating microscopes
Process	Screening coverage	Persons aged 50+ screened ÷ total 50+ population × 100
Output	Cataract surgical rate (CSR)	Cataract operations performed in a year ÷ population × 1,00,000 (target ≥400)
Output	Backlog clearance	Registered cataract-blind operated ÷ registered cataract-blind × 100
Quality	IOL implantation rate	Operations with IOL ÷ total cataract operations × 100 (national figure rose from 20% in 1997–98 to 94.2% in 2017–18)
Quality	Good visual outcome	Operated eyes with post-operative presenting vision 6/18 or better ÷ total operated eyes × 100
Equity	Sex and social-class equity	Operations in women ÷ total operations × 100 (target >50%)

Type	Indicator	Formula / definition
Outcome/ impact	Prevalence of blindness	Persons with presenting visual acuity worse than 3/60 in the better eye ÷ population surveyed × 100
Efficiency	Cost-effectiveness ratio	Cost of the programme ÷ health outcome achieved (e.g. cost per sight restored)

Evaluation results answer "goals achieved? goals satisfactory?" — if **no**, the problem is reformulated and the cycle re-enters at situation analysis; if **yes**, the programme continues to operate (Park 28e, Ch 23, Fig 1, p 1026).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce by hand (the planning cycle): Gather health data → Tabulate, analyze, interpret → Formulate health problems → Assign priorities among problems → Formulate individual priority problems for planning solutions → Define programme objectives and how to measure achievement → Assign priorities among objectives → Design alternative programmes to solve problems → Select best programme (feasibility, cost/effectiveness) → Combine programmes and develop integrated plan → Plan implementation of integrated programme → Orient and train personnel → Initiate integrated programme → Operate programme → Collect data for evaluation → Evaluate results (goals achieved? goals satisfactory?) → YES back to "Operate programme"; NO back to "Formulate health problems".

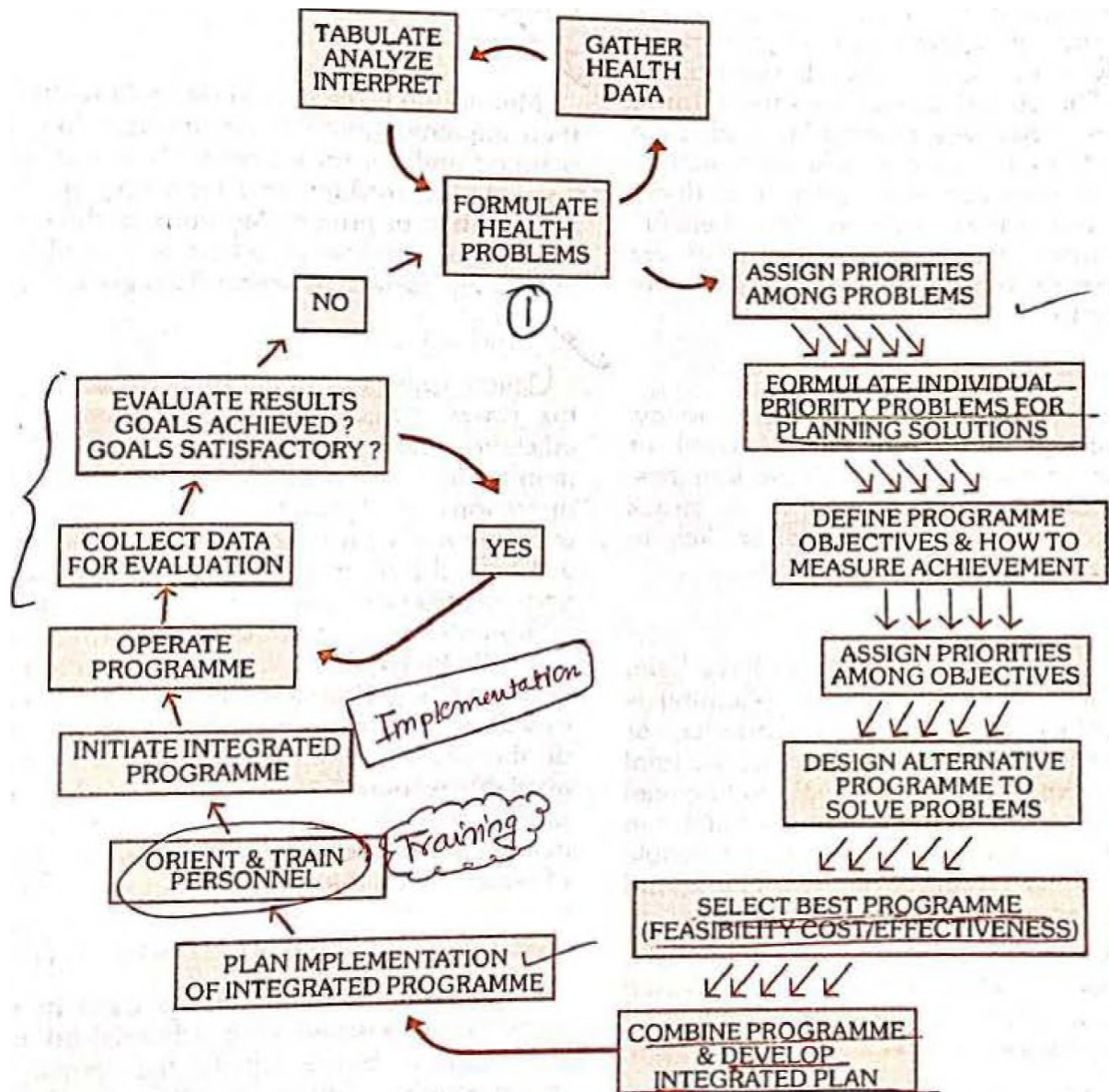
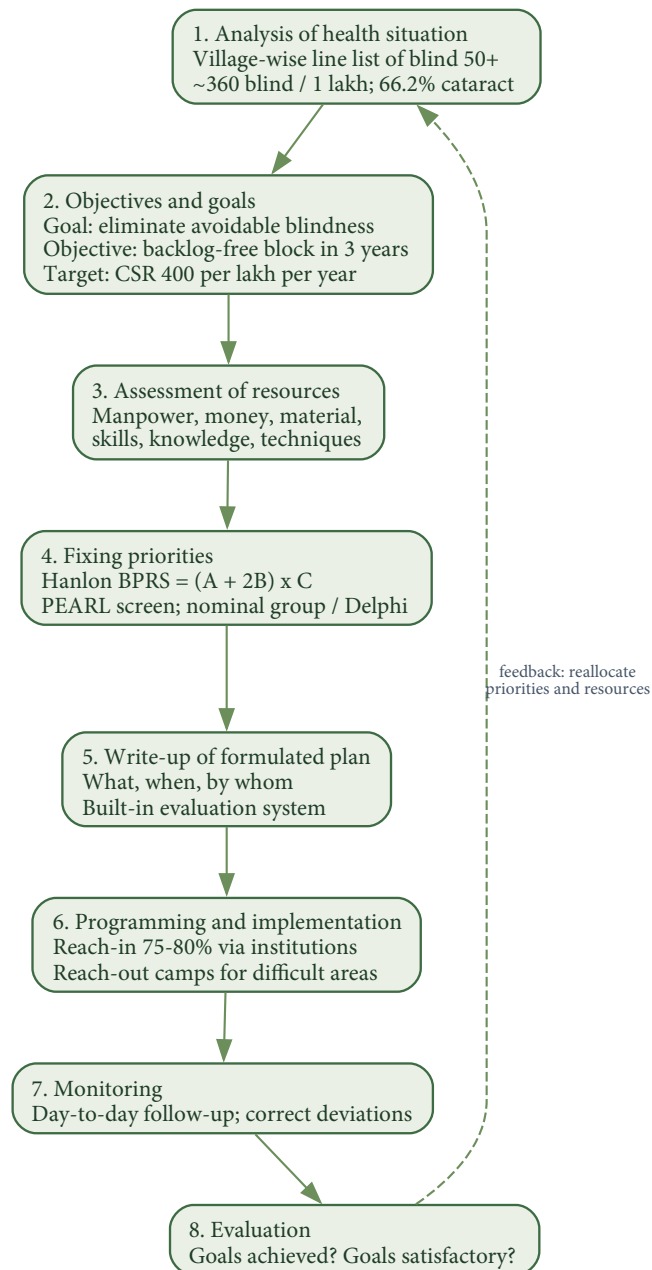
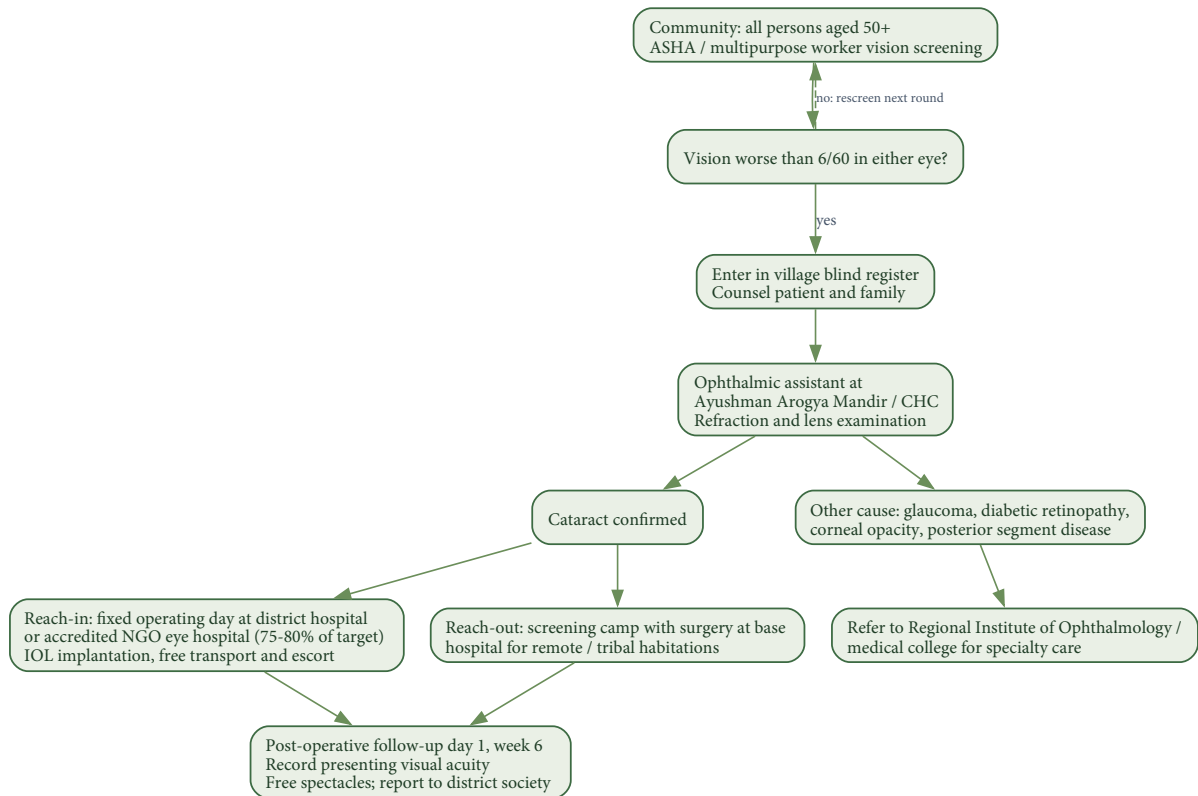


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, p.1026)



The eight steps of the planning cycle applied to cataract blindness in a rural block



Case-detection to follow-up pathway for cataract blindness in the rural block

High-yield exam pointers

- ▶ **Eight steps (Park):** situation analysis → objectives and goals → assessment of resources → fixing priorities → write-up of plan → programming and implementation → monitoring → evaluation. Pre-planning = political will, legislation, organisation for planning, administrative capacity.
- ▶ **Goal vs objective vs target:** goal is time- and resource-unconstrained; objective is precisely achieved or not; target permits a degree of achievement.
- ▶ **CSR = cataract operations in a year ÷ population × 1,00,000; minimum target 400 per lakh per year.** One eye surgeon = 700 operations per year.
- ▶ **Cataract = 66.2% of blindness in India (NBVIS 2015–19); blindness prevalence 0.36%.** WHO blindness = presenting visual acuity worse than 3/60 in the better eye.
- ▶ **NPCBVI:** launched 1976 as the National Programme for Control of Blindness, 100% centrally sponsored, absorbed the 1963 Trachoma Control Programme; goal now 0.25% prevalence; 60:40 funding (90:10 in North-Eastern states).
- ▶ **Hanlon BPRS = (A + 2B) × C** with the PEARL screen (propriety, economics, acceptability, resources, legality).
- ▶ **Reach-in 75–80% through institutions; reach-out camps only for inaccessible areas.** Social target: >50% of operations in women.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam — the programme was renamed from the National Programme for Control of Blindness to the National Programme for Control of Blindness and Visual Impairment, and the older technical norm of "minimum 30% of operations by intraocular lens surgery" is long superseded (94.2% of operations used an intraocular lens in 2017–18), so quote the current figure; the unit-cost figures (₹2,143 government camp, ₹1,128 non-governmental screening camp with base-hospital surgery) are from Jugal Kishore's National Health Programmes of India and are dated, so present them as illustrative costing rather than current rates, and the block-level case counts above are arithmetic worked from the National Blindness and Visual Impairment Survey 2015–19 percentages applied to a 1,00,000 population, not a surveyed figure.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, WHO SPECS 2030, WHA74 global eye-care targets

Programme and policy updates (India)

- ▶ **Mission-mode Netra Jyoti Abhiyan** is now the vehicle for clearing the cataract backlog, and both the campaign and the *cataract blindness backlog-free* certification operate at **block and district level** — so the block plan designed above is exactly the unit that certification assesses. Coverage varies so widely between regions that decentralised district-level planning, rather than a single uniform national target, is the recommended approach, with a dedicated regional task force proposed for the North-Eastern states [PMID 41460124].
- ▶ **The primary-level vision centre is the strongest single determinant of coverage.** In Theni district, Tamil Nadu (7,127 people aged ≥ 40), residents living within 5 km of a vision centre versus beyond it had visual impairment 18.1% vs 25.2%, eye-care utilisation 66.7% vs 52.3%, cataract surgical coverage 63.9% vs 55.0%, effective cataract surgical coverage 44.8% vs 32.6%, and effective refractive error coverage 47.9% vs 24.5%. A 5 km catchment is therefore a defensible siting norm for the eye-care tier of the Ayushman Arogya Mandir [PMID 40912902].
- ▶ **The leak is between screening and surgery, not in screening itself.** Across 50 villages in Uttar Pradesh, 45,286 people were screened, 16.8% were referred, and only 38.5% actually reached the vision centre; attendance was lowest (35.3% vs 42.5%) among those aged over 16 who had dependants and were referred for non-cataract conditions. Free transport, personalised counselling and subsidised spectacles were the measures the non-attenders themselves ranked highest — these belong inside the written plan, not in a remedial campaign afterwards [PMID 41224295].
- ▶ **Artificial-intelligence screening by frontline workers** is being used to pursue backlog-free status in Uttar Pradesh, where about 36 million people are aged 50 or older. The "Roshni" mobile application classifies segmented-pupil images with 96% sensitivity and 97% specificity, and achieved 86.6% sensitivity, 93.3% specificity and 98.5% negative predictive value on field testing — a direct upgrade to the ASHA distance-vision-chart step of the plan [PMID 39187013].

Current national data

- **Effective cataract surgical coverage (eCSC) has displaced the cataract surgical rate as the headline indicator**, because it counts only those operated people who actually end up seeing well; the residual is expressed as the relative quality gap. Pooled Indian estimates (four studies, 34 survey reports, searched to November 2024) are cataract surgical coverage 57% (95% CI 52–61), eCSC 36% (95% CI 32–40) and relative quality gap 37% (95% CI 33–40), with the North-East worst at CSC 36% and eCSC 23% (95% CI 16–31). A quality gap of that size means improving outcomes must run alongside — not after — any expansion of surgical volume [PMID 41460124].

Elimination, surveillance and disease status

- **Good outcomes are attainable in rural secondary centres, but presenting vision lags well behind corrected vision.** Across 123,685 surgeries in 20 rural secondary centres of a four-state network (2016–20; 82% manual small-incision surgery), best-corrected vision reached 6/12 or better in 91.1% at the spectacle visit, yet uncorrected vision reached 6/12 or better in only 67.1%; major intraoperative complications occurred in 3.14%. Since eCSC is scored on *presenting* vision, post-operative refraction and the actual supply of spectacles are the binding constraint rather than surgical volume; MSICS carried higher odds of poor vision than phacoemulsification (adjusted OR 1.93, 95% CI 1.79–2.09 for <6/12), largely through induced astigmatism [PMID 41288633].
- **Population-based outcome audit is where camp surgery under-performs.** Among 11,238 people examined across four Telangana districts, of 1,850 operated eyes 17.4% had borderline (<6/18 to 6/60) and 11.2% had poor (<6/60) presenting vision; surgery in eye camps and in government hospitals, absence of formal education and age ≥70 years were associated with worse outcomes, the causes being posterior-segment disease, uncorrected refractive error and surgical complications. This is the current evidence base for the 75–80% reach-in rule and for ruling out posterior-segment pathology before listing a patient [PMID 41705358].

Guideline and legislative updates

- **NPCBVI returns are not disaggregated by age or sex.** A review of national programme indicators found that NPCBVI — along with the national mental health and musculoskeletal disorder programmes and the Ayushman Arogya Mandir (erstwhile Health and Wellness Centre) reporting system — captures service-utilisation data with neither age nor sex breakdown, so the ">50% of operations in women" equity target cannot be verified from routine returns. Name this gap when writing the monitoring and evaluation section of the plan [PMID 39702198].

Global and WHO developments

- **The World Health Assembly's 2030 targets have superseded *Vision 2020* as the goal statement.** WHA74 (2021) adopted a 30 percentage-point increase in eCSC and a 40 percentage-point increase in effective refractive error coverage by 2030, as the operational expression of integrated people-centred eye care. Global eCSC was modelled at 48.2% (95% CI 39.7–57.2) for 2025, with only an 8.4 percentage-point rise projected across 2020–2030 (43.9% → 52.3%) — the world is not on track. Country estimates ranged from 2.1% (Burundi, 2024) to 77.7% (Qatar,

2023), and uncorrected refractive error accounted for a median 26.4% of non-good outcomes, correction of which alone would add roughly 3.7 percentage points [PMID 41687671].

- **WHO SPECS 2030**, launched in May 2024, is the delivery vehicle for the refractive-error half of those targets and is built on five pillars — services, personnel, education, cost and surveillance. WHO's current statement of magnitude is that at least **2.2 billion people** have near or distance vision impairment, in about **1 billion** of whom the vision loss is preventable or has not yet been addressed [WHO 2024-May; WHO 2026-Feb].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 23 Health Planning and Management, pp 1025–1031; Ch 6 Epidemiology of Chronic Non-communicable Diseases and Conditions, pp 477–480; Ch 7 Health Programmes in India, p 527; Ch 13 Tribal Health in India, p 850; Ch 22 Communication for Health Education, p 1024); Sunder Lal — Textbook of Community Medicine (Ch 8, pp 517–519); Jugal Kishore — National Health Programmes of India (pp 533–535); Health Program Planning and Evaluation: A Practical, Systematic Approach (Ch 1, pp 41–49; Ch 3, pp 111–112; Ch 4, pp 140–143; Ch 9, p 252); Reinke — Health Planning for Effective Management (Ch 1, p 27; Ch 15, p 219); Health Economics (Ch 13, pp 206–207). Named documents: National Blindness and Visual Impairment Survey 2015–19; National Programme for Control of Blindness and Visual Impairment; Vision 2020: The Right to Sight (WHO, 1999); WHO Global Eye Health Action Plan 2014–2019.

Recent-advances sources: 41460124 — Regional disparities in effective cataract surgical coverage and relative quality gap in India: a systematic review and meta-analysis, Indian Journal of Ophthalmology 2026, doi:[10.4103/IJO.IJO_130_25](https://doi.org/10.4103/IJO.IJO_130_25); 40912902 — Impact of Vision Centres on achieving universal eye health coverage: prevalence, coverage and utilisation of eye care services in Southern India, British Journal of Ophthalmology 2026, doi:[10.1136/bjo-2024-326637](https://doi.org/10.1136/bjo-2024-326637) (abstract-only); 41224295 — Rapid sequential mixed-method study to identify barriers and explore solutions for improving equitable access to community-based eye care services in Uttar Pradesh, India, BMJ Open 2025, doi:[10.1136/bmjopen-2025-108422](https://doi.org/10.1136/bmjopen-2025-108422); 39187013 — Development and testing of artificial intelligence-based mobile application to achieve cataract backlog-free status in Uttar Pradesh, India, Asia-Pacific Journal of Ophthalmology 2024, doi:[10.1016/j.apjo.2024.100094](https://doi.org/10.1016/j.apjo.2024.100094) (abstract-only); 41288633 — Visual outcomes of cataract surgeries performed at a network of rural eye centres in southern and eastern India: a model for scaling effective cataract surgical coverage, Indian Journal of Ophthalmology 2025, doi:[10.4103/IJO.IJO_1185_24](https://doi.org/10.4103/IJO.IJO_1185_24); 41705358 — Population-based assessment of visual outcomes after cataract surgery in four districts in Telangana, India, Ophthalmic Epidemiology 2026, doi:[10.1080/09286586.2026.2624449](https://doi.org/10.1080/09286586.2026.2624449) (abstract-only); 39702198 — Age- and sex-disaggregated disease burden among the older persons in India, BMC Geriatrics 2024, doi:[10.1186/s12877-024-05614-w](https://doi.org/10.1186/s12877-024-05614-w); 41687671 — Effective cataract surgical coverage in adults aged 50 years and older: empirical estimates from population-based surveys in 68 countries and modelled estimates for 2000–30, The Lancet Global Health 2026, doi:[10.1016/S2214-109X\(25\)00435-8](https://doi.org/10.1016/S2214-109X(25)00435-8) (abstract-only). WHO SPECS 2030, launched May 2024; WHO blindness and vision impairment fact sheet, 10 February 2026.



Health financing in India

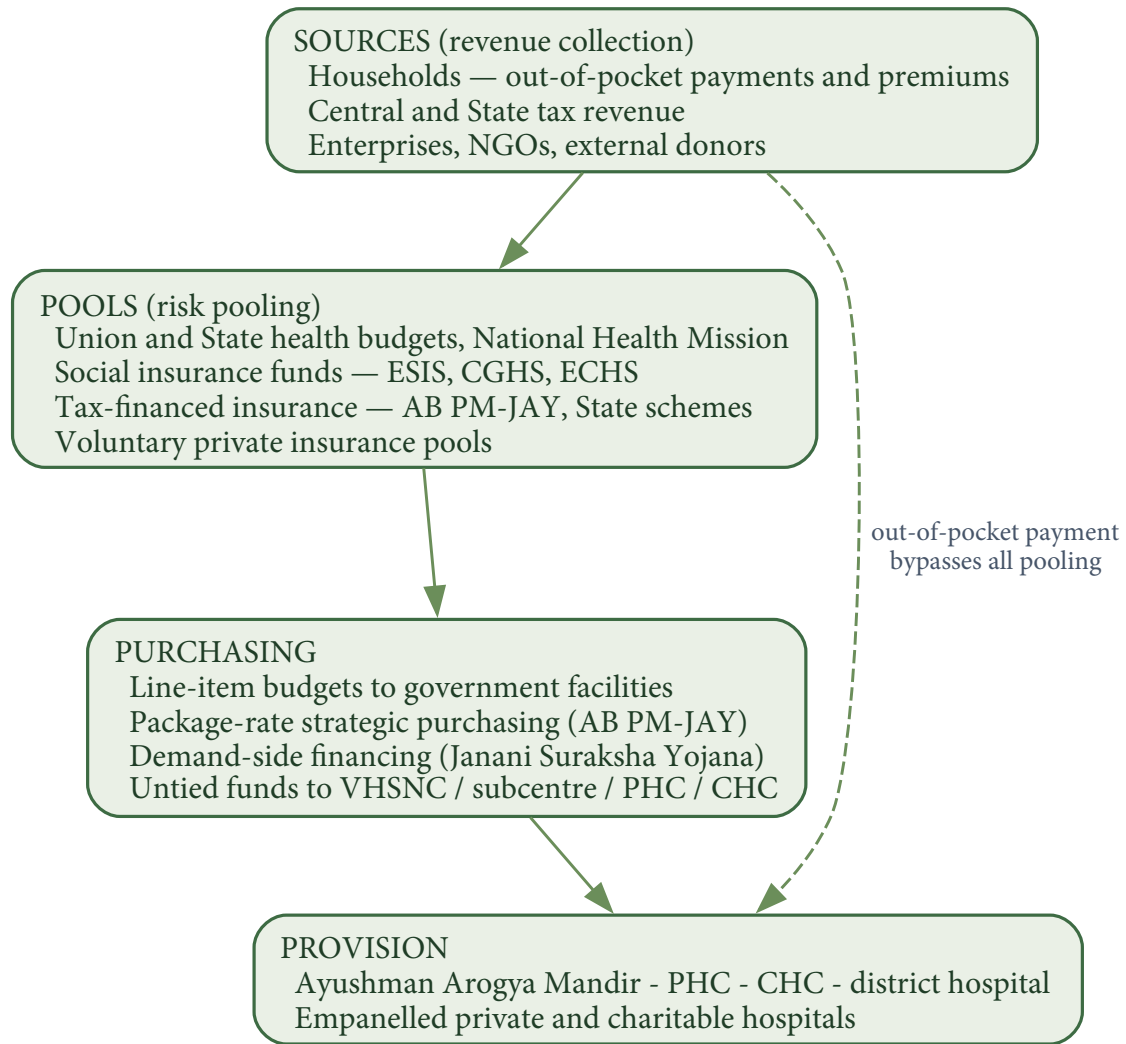
Asked in: Paper IV 17-Apr-2026 — RGUHS MD Community Medicine

Definition — Health system financing is the process by which revenues are collected from primary and secondary sources, accumulated in fund pools, and allocated to provider activities (Health Economics, Ch 9, pp 148-149). It therefore has three functions — **revenue collection**, **fund pooling** (so that contributors share risk) and **purchasing** (allocating funds to providers). India's financing mix is dominated by household out-of-pocket expenditure (OOPE), which makes financial protection the central policy problem rather than the total quantum of spending.

1. The three financing functions and the instruments India uses for each

Function	Meaning	Instruments operating in India
Revenue collection	Raising funds directly from individuals or indirectly through governments/donors	General taxation (Union + State), payroll contributions (Employees' State Insurance), voluntary premiums, user charges, external donor and non-governmental funds
Fund pooling	Collecting funds so that contributors share risk	Union and State health budgets, National Health Mission pools, Employees' State Insurance Scheme (ESIS)/Central Government Health Scheme (CGHS)/Ex-servicemen Contributory Health Scheme (ECHS) funds, Ayushman Bharat–Pradhan Mantri Jan Arogya Yojana (AB PM-JAY), private insurance pools
Purchasing	Allocating funds to providers	Line-item budgets to government facilities; package-rate strategic purchasing by AB PM-JAY from public and empanelled private hospitals; demand-side financing (<i>Janani Suraksha Yojana</i>); untied funds to village health committee, subcentre, primary health centre (PHC), community health centre (CHC) and district hospital; direct fee-for-service payment by the patient

Internationally, prepaid systems follow either the tax-based *Beveridge* model or the payroll-contribution *Bismarck* social-insurance model (Health Economics, Ch 12, pp 188-189). India runs a tax-based public system with small Bismarckian islands — the compulsory social insurance schemes CGHS, ESIS and ECHS — superimposed on a very large out-of-pocket market (Sunder Lal, Ch 16, p 729).



Flow of funds in the Indian health system — collection, pooling, purchasing and provision

2. Where India's health rupee comes from — National Health Accounts

Health accounts describe expenditures and the flow of funds through the health system in a financial year, answering what the **sources** are, **who** manages the funds and **who** provides the services. Current health expenditure (CHE), the recurrent spending that affects health outcomes in that year, is over **91%** of total health expenditure (THE); capital expenditure (buildings, equipment, vehicles) is about **9%** (Sunder Lal, Ch 16, p 727).

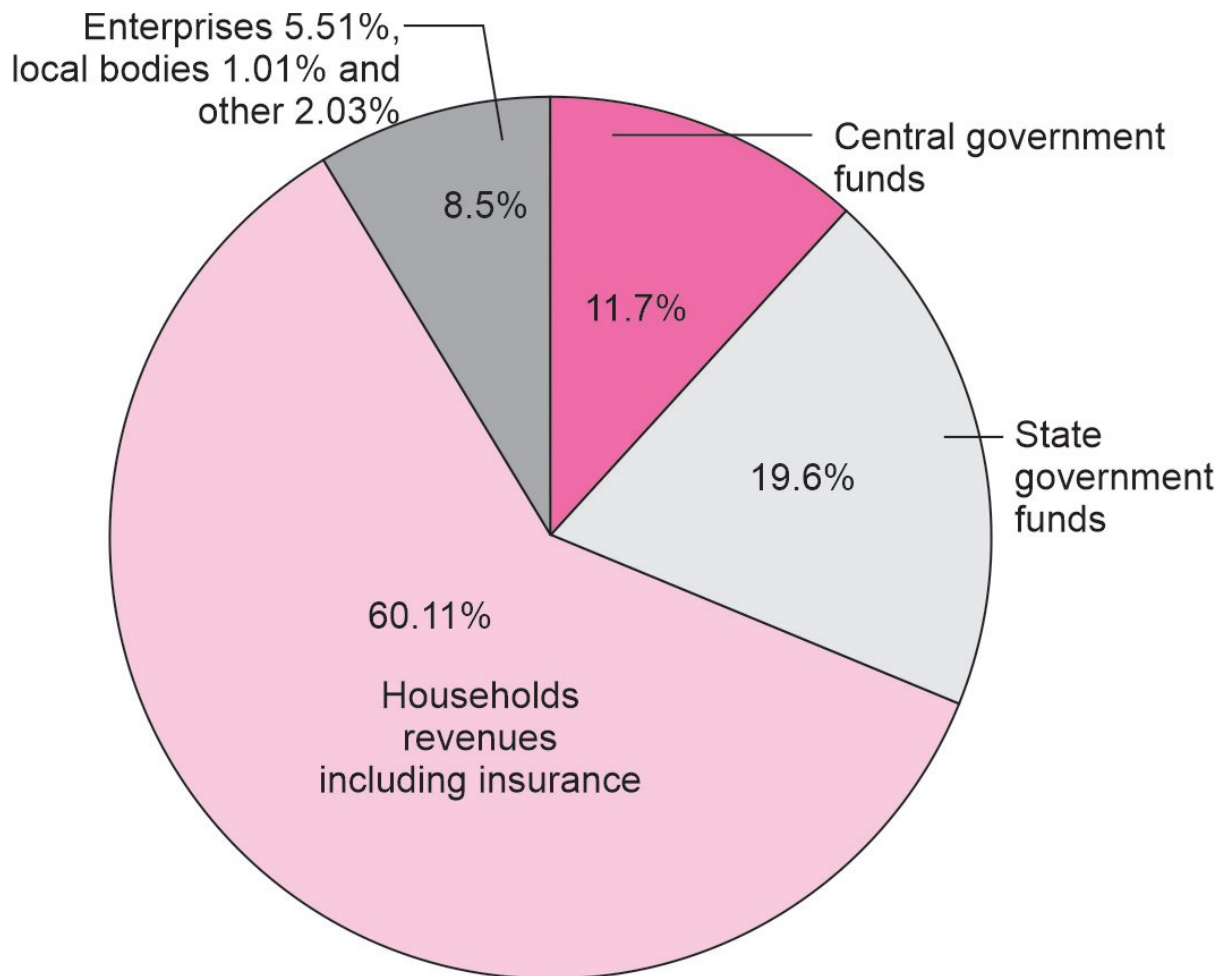


Fig 16.7 — Sources of health financing in India (NHA 2018-19) (Sunder Lal — Textbook of Community Medicine, Ch 16 (Health Planning and Management), p 728)

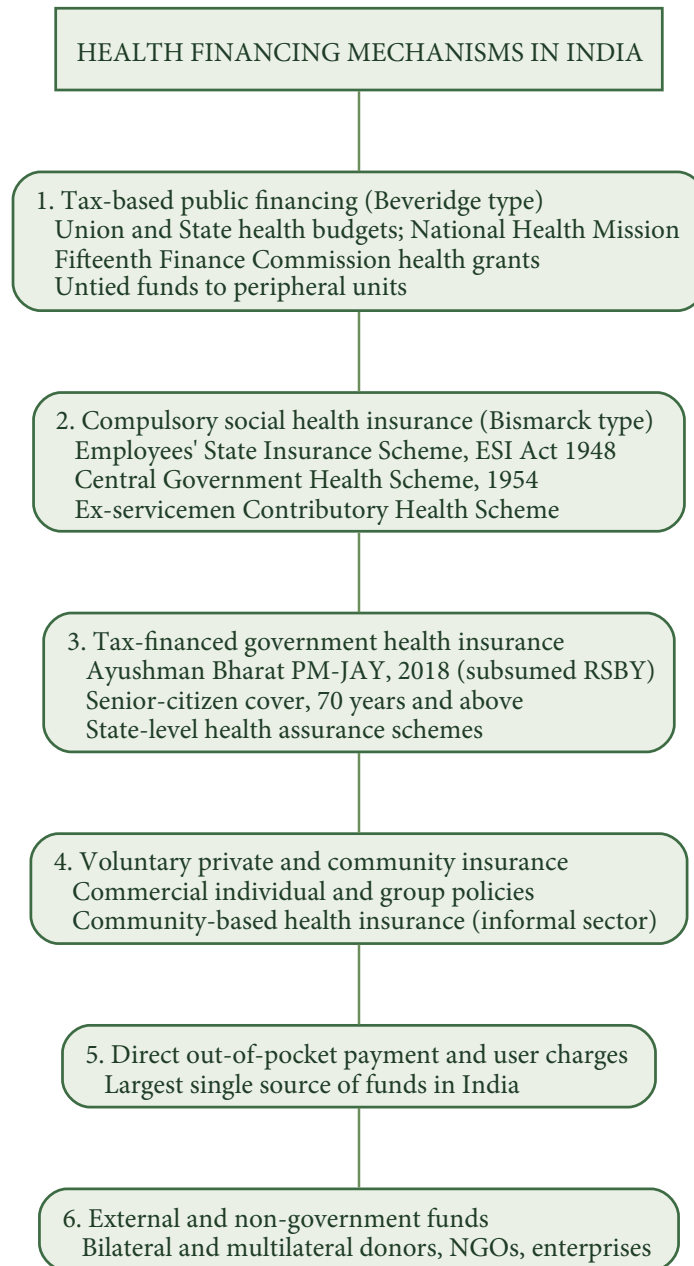
Source of current health expenditure	Share of CHE (National Health Accounts 2018-19)
Household revenues, including insurance contributions	60.11% — of which pure out-of-pocket expenditure alone is 53.23%
State government funds	19.63%
Central government funds	11.71%
Enterprises	5.51%
Local bodies	1.01%
Other (including external donors)	2.03%

Macro indicator	Value (year)
Total health expenditure	~3.16% of gross domestic product (GDP), 2018-19
Government health expenditure	1.28% of GDP = ~40.6% of total health spending

Macro indicator	Value (year)
Out-of-pocket expenditure	1.52% of GDP; ₹2,155 per capita per year
Social security expenditure on health (premiums paid by government for PM-JAY, RSBY, CGHS, ESIS, ECHS)	9.6% of THE
Private health insurance expenditure	6.6% of THE

Global evidence is that basic health needs are seldom met unless a country spends at least **5-6% of GDP** on health with the major part from government; failure to reach minimum public health expenditure is the **single most important constraint** on the Indian system (Sunder Lal, Ch 16, p 727). The comparison is instructive — Thailand's total spending resembles India's, but 77.7% of it is public and about 95% is strategically purchased from public facilities; Brazil spends 9% of GDP with public expenditure at 4.1% of GDP (Sunder Lal, Ch 16, p 727).

3. Prepayment and insurance mechanisms in India



Classification of health financing mechanisms operating in India

Scheme	Population covered	Financing and benefit
Employees' State Insurance Scheme (ESIS), ESI Act 1948	Employees of non-seasonal factories/establishments employing 10 or more persons for wages, and their families; wage ceiling ₹21,000 per month	Contributory and tripartite — employer 3.25% and employee 0.75% of wages (workers on daily wages below ₹176 are exempt from contributing), plus the State government's share of one-eighth of the cost of medical care; outpatient, inpatient and emergency care, maternity, employment-injury, dependants' and family-planning benefits (Park 28e, Ch 10, p 745; Ch 18, p 971; Sunder Lal, Ch 17, p 763)

Scheme	Population covered	Financing and benefit
Central Government Health Scheme (CGHS) , from 1954 — began with 16 allopathic dispensaries and 2.3 lakh beneficiaries, now in 72 cities	Central government employees and pensioners, Members and ex-Members of Parliament, sitting and retired judges, ex-Governors, freedom fighters (free of cost), accredited journalists	Contributory (formerly the <i>Contributory Health Service Scheme</i>) — a graded monthly contribution by the employee, ₹250 to ₹1,000 by pay level , plus the government's employer share, on the principle of cooperative effort by employee and employer; pensioners may buy a " whole-life " card by paying 10 years' subscription . Allopathic and AYUSH care through 417 wellness centres (329 allopathic + 88 AYUSH), 4 CGHS hospitals in Delhi and 1,403 empanelled hospitals (Park 28e, Ch 10, pp 745-746; Sunder Lal, Ch 17, pp 764-765)
Rashtriya Swasthya Bima Yojana (RSBY) , 2008 — <i>subsumed into PM-JAY on 23.09.2018</i>	Below-poverty-line families and 11 categories of unorganised workers; ~2.75 crore families	₹30,000 per family per year on a family-floater basis (maximum 5 members), 1,516 packages; launched by the Ministry of Labour and Employment, transferred to the Ministry of Health and Family Welfare on 01.04.2015 (Park 28e, Ch 10, p 745)
Ayushman Bharat–Pradhan Mantri Jan Arogya Yojana (AB PM-JAY) , launched 23 September 2018	10.74 crore poor and vulnerable families (~50 crore people); ~16 crore individual beneficiaries verified	₹5,00,000 per family per year for secondary and tertiary hospitalisation; cashless and portable; no cap on family size, age or gender; all pre-existing conditions covered; 1,685 procedures ; 3 days pre- and 15 days post-hospitalisation cover. Centre:State share 60:40 (90:10 for north-eastern States) . Over 21,800 hospitals empanelled; over 1.82 crore hospitalisations, saving over ₹22,500 crore of out-of-pocket expenditure (Park 28e, Ch 10, pp 744-746; Sunder Lal, Ch 17, p 763)
AB PM-JAY senior-citizen cover	All citizens aged 70 years and above , <i>regardless of socio-economic status</i> , on a distinct card	₹5 lakh per year on a family basis if not already in a PM-JAY family; a separate top-up of up to ₹5 lakh, not shared with members under 70, if already covered. Those on CGHS/ECHS/Central Armed Police Force schemes choose one; those with private insurance or ESI remain eligible (Park 28e, Ch 7, pp 571-572)

Despite these, coverage remains thin — about **86% of the rural and 82% of the urban population** were not covered by any health-expenditure support scheme, and although Ayushman Bharat has raised health protection to about **35% of the population**, with the aim of covering all by 2025, the private sector still holds **80% of doctors, 25% of nurses and 49% of beds**, dominates curative care and is financed almost entirely by out-of-pocket payment (Sunder Lal, Ch 16, p 729).

4. Consequences of the present financing mix — financial protection

Measure	Definition	Indian position
Catastrophic health expenditure	Household health spending exceeding 10% of total monthly consumption expenditure , or 40% of non-food consumption expenditure	17.3% of people spent more than 10% of household total expenditure on health during 2012-20, declining to 7% in 2022 (Sunder Lal, Ch 16, p 729)
Impoverishment	Households pushed below the poverty line by health spending	About 4.2% of the population, roughly 50 million people a year (Sunder Lal, Ch 16, pp 728-729)
Average expenditure per hospitalisation	Cost borne per admission, public versus private	₹4,450 in a public hospital against ₹31,845 in a private hospital — about 7 times higher; ₹16,676 in rural against ₹26,475 in urban areas (Sunder Lal, Ch 16, p 729)
Equity in financing	Contributions should follow ability to pay, not the event of falling ill	Out-of-pocket payment is regressive: it bypasses pooling entirely, and the sick and poor pay most

Universal health coverage is reached by expanding pooled prepayment along the three dimensions drawn in the diagram below, so that richer groups cross-subsidise poorer ones (Health Economics, Ch 12, pp 188-189).

5. Policy targets and the reform agenda

Commitment	Target
National Health Policy 2017	Raise government health expenditure from 1.15% to 2.5% of GDP by 2025 (Park 28e, Ch 23, p 1030)
National Health Policy 2017	State sector health spending to more than 8% of the State budget by 2020 (Park 28e, Ch 23, p 1030)
National Health Policy 2017	Reduce the proportion of households facing catastrophic health expenditure by 25% by 2025 (Park 28e, Ch 23, p 1030)
NITI Aayog Three Year Action Agenda (2017-18 to 2019-20), goal for 2020	Reduce out-of-pocket spending to 50% of total health expenditure (2014 estimate: 62.4%) (Park 28e, Ch 23, p 1033)
Fifteenth Finance Commission	State health spending to 8% of the State budget by 2022 , and 67% of total health expenditure on primary health care (Sunder Lal, Ch 16, pp 727-729)

Reform therefore runs along five lines:

- **Raise public spending** and protect it as recurrent, not capital-only, funding, since financial protection tracks the absolute and relative level of public health expenditure (Sunder Lal, Ch 16, p 727).
- **Shift the pooled money to primary health care** through Ayushman Arogya Mandirs and the Ayushman Bharat Health Infrastructure Mission (Sunder Lal, Ch 16, pp 728-729).
- **Replace passive line-item budgeting with strategic purchasing** — the government acting as a single payer buying defined packages of secondary and tertiary care from public and private hospitals (Sunder Lal, Ch 16, p 729).

- ▶ **Widen prepayment to the informal sector**, since compulsory social insurance reaches only formal employees while community-based insurance suffers from small risk pools, regressive flat-rate premiums and difficulty enrolling the poorest (Health Economics, Ch 12, p 194).
- ▶ **Regulate the private insurance market**, whose administration costs alone run at **10-14% of total costs** when unregulated, and whose small separate pools fragment risk (Health Economics, Ch 11, pp 177-178).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The universal health coverage cube** — draw a cube with three labelled axes: *breadth* (proportion of the population covered), *depth* (range of services covered), *height* (proportion of health care costs met by the financing system); shade the pooled-funds block and label the residual outside the block as "out-of-pocket payment".

High-yield exam pointers

- ▶ Three financing functions: **revenue collection** → **fund pooling** → **purchasing** (Murray and Frenk). Two prepayment models: **Beveridge (tax)** and **Bismarck (payroll social insurance)**.
- ▶ **Catastrophic health expenditure = health spending >10% of total household consumption expenditure, or >40% of non-food consumption expenditure**. Write both denominators.
- ▶ National Health Accounts 2018-19: total health expenditure **3.16% of GDP**, government **1.28% of GDP**, out-of-pocket **53.23% of current health expenditure** (1.52% of GDP, ₹2,155 per capita per year); current expenditure is 91% of total, capital 9%.
- ▶ National Health Policy 2017: government health spending **1.15% → 2.5% of GDP by 2025**; States **>8% of budget**; catastrophic expenditure down **25% by 2025**. NITI Aayog Three Year Action Agenda: out-of-pocket spending down to **50% of total health expenditure by 2020**.
- ▶ AB PM-JAY: **₹5 lakh per family per year**, launched **23.09.2018**, **10.74 crore families**, 1,685 procedures, 3 days pre- and 15 days post-hospitalisation, Centre:State **60:40 (90:10 north-east)**; **70 years and above** covered irrespective of income.
- ▶ ESI Act **1948** — establishments with **10 or more workers**, wage ceiling ₹21,000/month, employer 3.25% + employee 0.75%. CGHS from **1954** — **contributory**, ₹250-1,000 per month by pay level.
- ▶ Fifteenth Finance Commission: **8% of State budget on health**, **67% of health expenditure on primary care**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the National Health Accounts figures quoted are the 2018-19 estimates carried in the retrieved textbooks; later National Health Accounts rounds show government health expenditure rising and the out-of-pocket share falling further, so quote the year with every figure, and the retrieved corpus did not contain a costed account of the Pradhan Mantri Ayushman Bharat Health Infrastructure Mission or of provider payment rates under PM-JAY.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, PIB / MoHFW, National Health Accounts 2022-23, Comprehensive Annual Modular Survey 2022-23, Sixteenth Finance Commission, WHO/World Bank UHC Global Monitoring Report 2025

Programme and policy updates (India)

- ▶ **Ayushman Vay Vandana Card** — the 70-years-and-above expansion of AB PM-JAY was cleared by the Union Cabinet on **11 September 2024** and rolled out from **29 October 2024**, targeting about **6 crore senior citizens in 4.5 crore families** irrespective of income; seniors outside an existing PM-JAY family get ₹5 lakh on a family basis, while those already inside get a **separate ₹5 lakh top-up not shared with members under 70**. West Bengal signed on in June 2026 and the scheme went live there on **16 August 2026**, making it the 36th and last State/UT — PM-JAY now runs in all 36 States and Union Territories [PIB 2024-Oct; PIB 2026-Aug].
- ▶ **PM-JAY extended to gig and platform workers** in the Union Budget 2025-26 — identity cards plus **e-Shram registration** and PM-JAY cover for about **1 crore** workers, the first route by which informal-sector labour is folded into the tax-financed pool rather than left to out-of-pocket payment [PIB 2025-Feb].
- ▶ **Fund flow, provider payment and the fiscal frame are all being rebuilt, not just the pool**. The **Single Nodal Agency (SNA)** public financial management reform for centrally sponsored schemes including the National Health Mission, assessed in Bihar and Odisha on interviews run between February 2023 and February 2024, delivered real gains in cash management, expenditure transparency and the matching of central releases to actual sub-national spending; but weak budget credibility, patchy digital connectivity and thin finance staffing risk widening inequity in access to scheme funds, and the pressure to show utilisation has crowded out attention to health outcomes. On the purchasing side, costing of **27 district hospitals across nine States** puts the mean annual district-hospital budget at **₹326 million** (range ₹66 million in Himachal Pradesh to ₹2.57 billion in Andhra Pradesh), of which **inpatient care takes ₹221 million** — the largest single block — and proposes a **blended payment model** (a risk-adjusted global budget for outpatient work, case-based bundled payments for inpatient work, and a per-bed global budget for indirect services) to replace historical line-item allocation; staffing these hospitals to Indian Public Health Standards would lift the mean annual pay-out to **₹751 million** (range ₹218 million to ₹5.42 billion), and upgrading them to medical colleges would raise the mean pay-out by **91.5%**. Above both sits the **Sixteenth Finance Commission** award for **2026-31**, tabled on **1 February 2026**, which holds the States' share of the divisible pool at **41%** — so the Fifteenth Finance Commission health benchmarks quoted in the table above (8% of the State

budget on health, 67% of health spending on primary care) belong to the closed 2021-26 award and should be dated as such [PMID 42480037 (abstract-only); PMID 39478753; Finance Commission 2026-Feb].

Current national data

- ▶ **National Health Accounts 2022-23 is the current official round — use it, and update the textbook's 2018-19 table.** The Ministry of Health and Family Welfare released it in May 2026 (PIB press release PRID 2265816). Government health expenditure has risen nearly threefold, ₹1.30 lakh crore (1.15% of GDP) in 2013-14 → ₹3.85 lakh crore (1.43% of GDP) in 2022-23, or 1.48% of GDP measured on the new GDP series with base year 2022-23; per head it went ₹1,042 → ₹2,786. The government share of total health expenditure is now 43.7% against 28.6% in 2013-14, and the out-of-pocket share 43.4%, down nearly 21 percentage points from 64.2%. Social security expenditure on health rose 6% → 9.9% of total health expenditure, government health spending is 4.89% of general government expenditure (3.78% in 2013-14), and primary-care spending more than doubled, ₹0.5 → ₹1.4 lakh crore [PIB 2026-May].
- ▶ **Read the rounds as one series and attach the round-year to every figure.** The 2021-22 round is the one the peer-reviewed literature still works from: government share 48% of total health expenditure and out-of-pocket 39.4% (from 28.6% and 64.2% in 2013-14), government health expenditure 1.84% of GDP against 0.84% in 2004-05 — the figure usually rounded to "about 1.9%" — total health expenditure 3.83% of GDP and ₹6,602 per head, and health taking 6.12% of general government expenditure (3.94% in 2014-15). That 48% belongs to the peak pandemic-spending year; the 2022-23 round settles back to 43.7% government and 43.4% out-of-pocket, and the percentage-of-GDP denominator itself moved with the new 2022-23 base year. These are successive years of one revised series, not rival estimates of one year — 1.84-1.9% of GDP is 2021-22 and 1.43-1.48% is 2022-23. Beware the widely quoted 47.1% out-of-pocket share: it is a National Health Accounts figure, not an independent survey estimate, and the systematic review carrying it dates it to 2019-20 in its abstract but to 2021-22 in its text, where the official round says 39.4%; the same review prints government health expenditure as 1.84% of GDP in its text and 1.60% in its own table. Quote 47.1% as the 2019-20 round or leave it alone. The sharpest single-year fall in the out-of-pocket share came between 2016-17 and 2017-18, before the pandemic, which argues for a structural rather than a crisis effect [PMID 42548603; PMID 40552236; PIB 2026-May].
- ▶ **Comprehensive Annual Modular Survey 2022-23** (302,086 households; 173,096 rural, 128,990 urban; surveyed July 2022 to June 2023): 59.9% of households had some insurance and 49.7% were covered by a publicly funded scheme; 32.4% used outpatient and 13.1% inpatient care. Among households that actually had an inpatient episode, catastrophic expenditure at the 10% threshold was 46.3% — 41.3% where the household held PM-JAY or a State scheme against 54.2% in the uninsured — and mean inpatient out-of-pocket spending in those households was ₹29,875 with cover against ₹37,572 without (all-India ₹33,928). The denominator is the point: these are rates among the hospitalised, not among all Indian households. On adjusted analysis, cover was associated with more outpatient (adjusted OR 1.404, 95% CI 1.328-1.484) and inpatient care use (adjusted OR 1.168, 95% CI 1.128-1.210) and less catastrophic expenditure at the 20% threshold for outpatient (adjusted OR 0.869, 95% CI 0.810-0.932) and inpatient care (adjusted OR 0.790, 95% CI 0.737-0.846), with inpatient out-of-pocket spending falling

appreciably (coefficient -0.205 , 95% CI -0.258 to -0.153) but outpatient spending barely moving. **Enrolment is not utilisation** — among 320 enrolled PM-JAY beneficiaries surveyed in Jammu and Baramulla only **48.4%** had ever used the scheme, and in a Puducherry urban slum **49% of eligible households had registered and only 14% had drawn benefits** [PMID 42167946; PMID 41878109; PMID 41805171 (abstract-only)].

- **Financial risk protection is now tracked district by district, not only nationally** — a district-level universal health coverage index built on the WHO-World Bank framework covers **687 of 707 districts** using 24 tracer indicators across five domains, one of which is financial risk protection. Median index value was **43.9%** (26.4% in Baksa, Assam to 69.4% in the Nilgiris, Tamil Nadu) and only **21.1%** of districts exceeded 50%. Health insurance coverage was **40.7%**, **8.0%** of the district population faced catastrophic expenditure and **11.2%** impoverishment. Ranked by median district coverage, **infectious disease was the strongest domain (82.8%)**, financial risk protection next at **66.6%**, then reproductive, maternal, newborn and child health **64.8%** and service capacity and access **54.9%**, with **non-communicable disease the weakest at 9.8%** (cervical and breast cancer screening under 1%). **69.4% of all variation lay between States** rather than within them — the argument for district-level financing decisions rather than a single national prescription [PMID 39219770].
- **PM-JAY's benefit package has grown well past the 1,685 procedures in the textbook table** — the 2022 Health Benefit Package lists over **1,949 procedures**, of which **1,284 (65.8%) are surgical**, against 1,350 procedures (about 21.5% surgical) in the first list; **95%** of empanelled private hospitals and **75%** of public ones provide surgical care, and in the scheme's first eight months surgical packages were **61% of claims by volume and 75% by value**. The Centre:State split is 60:40, but the **90:10 concession covers Jammu and Kashmir, Himachal Pradesh and Uttarakhand as well as the north-eastern States**. For 2021-22 the scheme's allocation was **8.65% of the Union health budget** — about **US\$871 million**, i.e. **₹6,400 crore** at the source's own rate of ₹73.5 to the dollar (the source also prints ₹64,000 crore, ten times its own conversion and ten times 8.65% of that year's Union health budget, so quote the share and drop the rupee figure if in any doubt). Equity gaps persist: men raised **23.6%** more surgical claims than women once obstetric and gynaecological surgery is excluded, and only **7%** of hospitals in Aspirational Districts offer cardiothoracic and vascular surgery against 17% elsewhere [PMID 41393311].

Guideline and legislative updates

- **The National Health Policy 2017 target has slipped, the money is aimed at the wrong gap, and the national average hides the States**. Government health spending is **1.43% of GDP on the 2022-23 accounts (1.48% on the new base series)**, and was 1.84-1.9% on the 2021-22 round, against the policy's **2.5% by 2025**; catastrophic spending is commoner for **outpatient (47.8%) than inpatient (43.1%)** care, and **medicines alone absorb about two-thirds** of out-of-pocket spending — precisely what PM-JAY's hospitalisation benefit does not cover, and the private sector still delivers about two-thirds of all outpatient care. On 2011-12 survey data, **29 million of the 46 million households** crossing the 10%-of-consumption threshold did so on medicines alone. The proven domestic remedy is pooled public procurement of free essential medicines on the **Tamil Nadu Medical Services Corporation** model (running since 1995, with Rajasthan following), which a benefit-incidence analysis found raised public-facility use and was progressive towards the poorest; what makes it work is the machinery — autonomous tendering,

direct delivery to district warehouses, computerised stock tracking, quality clearance before payment — and not the budget line alone. Underneath the national figure the States diverge sharply: health insurance coverage runs from **13.8% (Jammu and Kashmir) to 87.8% (Rajasthan)** and per-head health spending from ₹1,588 (Bihar) to ₹10,607 (Kerala); across the **21 States with complete data** the *unweighted mean* out-of-pocket share is **51.2%** and the government share **39.4%** — a mean of States, not the national figure, built from National Family Health Survey-5 for coverage and National Health Accounts **2019-20** for the shares, so it is not comparable with the 2021-22 or 2022-23 national rounds. Insurance provision itself is fragmented (a State fragmentation index of **0.06 to 0.78** around a national average of 0.44, lowest in States such as Assam and Andhra Pradesh, highest in Maharashtra and Punjab), and public or prepaid State financing systems out-performed out-of-pocket-dominated ones on child mortality, immunisation and water-and-sanitation access. Out-of-pocket payment still impoverishes: on NSS 75th-round data (2017-18, inflation-adjusted Tendulkar poverty lines) it lifted the poverty headcount from **12.26% to 13.51%** (rural 17.71% → 19.05%, urban 6.80% → 7.97%), about 3.1 million households, with only **21.4%** of households holding any cover and an insured household 3.78 percentage points less likely to be impoverished [PMID 42548603; PMID 40552236; PMID 42218447; PMID 41938958].

Global and WHO developments

- **WHO/World Bank *Tracking Universal Health Coverage: 2025 Global Monitoring Report*** (December 2025) — the UHC service coverage index rose from **54 to 71** between 2000 and 2023 but is projected to reach only **74 by 2030**; **2.1 billion** people face financial hardship from out-of-pocket health payments, of whom **1.6 billion** are pushed into or deeper into poverty, and **4.6 billion** lack essential services. The hardship share fell from 34% to 26% (2000-2022), but in 2022 hardship struck **three in four of the poorest against fewer than one in twenty-five of the richest**. This is the **first round using the revised SDG 3.8.1 and 3.8.2 indicators**, the new financial-protection definition having been adopted at the **56th session of the UN Statistical Commission (March 2025)** — so cite the indicator version alongside the figure. A structured comparison of 12 low- and middle-income countries against the same World Bank and WHO indicators found stronger performance tied to one pooled risk fund rather than many, purchasing that is strategic rather than passive, integrated delivery built on primary care, a whole-system view of public and private provision, deliberate investment in infrastructure and workforce, and governance that can be held to account — which maps almost exactly onto the five reform lines set out above [WHO 2025-Dec; PMID 41401513 (abstract-only)].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 — Health Programmes in India, pp 571-572; Ch 10 — Health Care of the Community, pp 744-746; Ch 18 — Occupational Health, p 971; Ch 23 — Health Planning and Management, pp 1030, 1033); Sunder Lal — Textbook of Community Medicine (Ch 16 — Health Planning and Management, pp 727-729; Ch 17 — Healthcare of the Community, pp 763-765); Health Economics (Ch 9 — The changing world of health care finance, pp 148-150; Ch 11 — Private health insurance, pp 177-178; Ch 12 — Achieving universal coverage, pp 188-194); National Health Accounts estimates for India 2018-19; National Health Policy 2017; NITI Aayog Three Year Action Agenda 2017-18 to 2019-20; Fifteenth Finance Commission recommendations.

Recent-advances sources: 42548603 — Can a bigger budget shrink the bill at the bedside? Public health financing as a route to lower out-of-pocket spending in India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1903360](https://doi.org/10.3389/fpubh.2026.1903360); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 41938958 — Impact of out-of-pocket health expenditure on poverty in India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1762886](https://doi.org/10.3389/fpubh.2026.1762886); 42218447 — The role of health financing system and pluralism in achieving sustainable development goal health outcomes in India: a multilevel analysis, *International Journal for Equity in Health* 2026, doi:[10.1186/s12939-026-02888-x](https://doi.org/10.1186/s12939-026-02888-x); 41878109 — Beyond coverage: why Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana struggles to deliver healthcare in Jammu and Kashmir, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1783458](https://doi.org/10.3389/fpubh.2026.1783458); 42480037 — Challenges to Achieving Universal Health Coverage Through Public Financial Management Reforms: Insights from the Implementation of Single Nodal Agency Reform in India, *Health Systems and Reform* 2026, doi:[10.1080/23288604.2026.2693357](https://doi.org/10.1080/23288604.2026.2693357) (abstract-only); 41805171 — From card to care: gaps in implementation of publicly funded health insurance among a vulnerable population in South India, *Journal of Public Health (Oxford)* 2026, doi:[10.1093/pubmed/fdag021](https://doi.org/10.1093/pubmed/fdag021) (abstract-only); 41393311 — Assessing Surgical Care under the Government-Funded Health Insurance Scheme —Pradhan Mantri Jan Arogya Yojana in India, *Annals of Global Health* 2025, doi:[10.5334/aogh.4908](https://doi.org/10.5334/aogh.4908); 39478753 — Framework for responsive financing of district hospitals of India, *Frontiers in Public Health* 2024, doi:[10.3389/fpubh.2024.1398227](https://doi.org/10.3389/fpubh.2024.1398227); 39219770 — District-level monitoring of universal health coverage, India, *Bulletin of the World Health Organization* 2024, doi:[10.2471/BLT.23.290854](https://doi.org/10.2471/BLT.23.290854); 40552236 — Understanding out-of-pocket expenditure in India: a systematic review, *Frontiers in Public Health* 2025, doi:[10.3389/fpubh.2025.1594542](https://doi.org/10.3389/fpubh.2025.1594542); 41401513 — Global Progress Toward Universal Health Coverage: Learning from Successes and Failures, *Annual Review of Public Health* 2025, doi:[10.1146/annurev-publhealth-100824-103712](https://doi.org/10.1146/annurev-publhealth-100824-103712) (abstract-only). Press Information Bureau / Ministry of Health and Family Welfare — *Union Health Ministry Releases the National Health Accounts Estimates for India 2022-23*, May 2026, PRID 2265816 (the current official round); Ministry of Health and Family Welfare / National Health Systems Resource Centre — National Health Accounts Estimates for India, 2021-22 round, as reported in PMID 42548603 and PMID 40552236, and 2019-20 round, as reported in

PMID 42218447; PIB — Ayushman Vay Vandana Card / AB PM-JAY expansion to 70 years and above, Union Cabinet approval 11 September 2024, rollout 29 October 2024; PIB — *West Bengal Becomes 36th State/UT to Implement Ayushman Bharat PM-JAY*, August 2026, PRID 2270375; PIB — Union Budget 2025-26, PM-JAY and e-Shram cover for gig and platform workers, February 2025; Sixteenth Finance Commission report for 2026-31, tabled 1 February 2026; WHO and World Bank — *Tracking Universal Health Coverage: 2025 Global Monitoring Report*, December 2025.



Material management — ABC and VED analysis

Asked in: Paper IV 04-Aug-2021 — RGUHS MD Community Medicine

Definition — Material management is "the systematic and scientific process of planning, implementing and controlling the efficient and effective flow and storage of resources (goods and services) from the point of origin to the point of consumption in order to meet the customer's requirements" (Park Plus, Ch 21, p 282). **Inventory control**, its core activity, is a scientific system that decides *what* to order, *when* to order, *how much* to order and *how often* to order, so that purchase cost plus storage cost is kept as low as possible without interfering with supply (Jugal Kishore — National Health Programmes of India, p 261; Sathe, Ch 13, p 144). It matters because 25% to 40% of the total expenditure of hospitals, dispensaries and health centres is spent on the purchase of material, so even a small gain in efficiency releases a large sum for service delivery (Sathe, Ch 13, p 139).

Scope, objectives and components

Item	Content
Aim — the "five Rights"	Right quality, right quantity, at the right time, at the right place, for the right cost (Park Plus, Ch 21, p 282)
Objectives (nine)	Low prices; high inventory turnover; low cost of acquisition and possession; continuity of supply; consistency of quality; low payroll cost; favourable relations with suppliers; development of personnel; good records (Sathe, Ch 13, p 140). Park Plus words two of these as "consistency of quality and quantity " and "low procurement and storage cost" (Park Plus, Ch 21, p 282)
Components of the cycle	(1) Identification of items, (2) estimating demand, (3) purchase and procurement, (4) inspection and quality control, (5) storage, (6) inventory control , (7) issue procedure, (8) maintenance, repair and condemnation, (9) information system (Sathe, Ch 13, pp 140–141)
Inventory-related costs	Purchase cost — the amount actually paid for the material; inventory carrying (holding) cost — the hidden cost of money blocked in idle stock, made up of cost of borrowed money, space, additional manpower, obsolescence, deterioration and pilferage; ordering (replenishment) cost — cost of placing an order with a supplier; shortage cost — direct and indirect cost of not having the item in stock (Park Plus, Ch 21, pp 282–283; Red Book — Public Health and Preventive Medicine, Managerial Sciences, p 297). Total yearly carrying cost is generally taken as 20–25% of inventory value (Sathe, Ch 13, p 144)

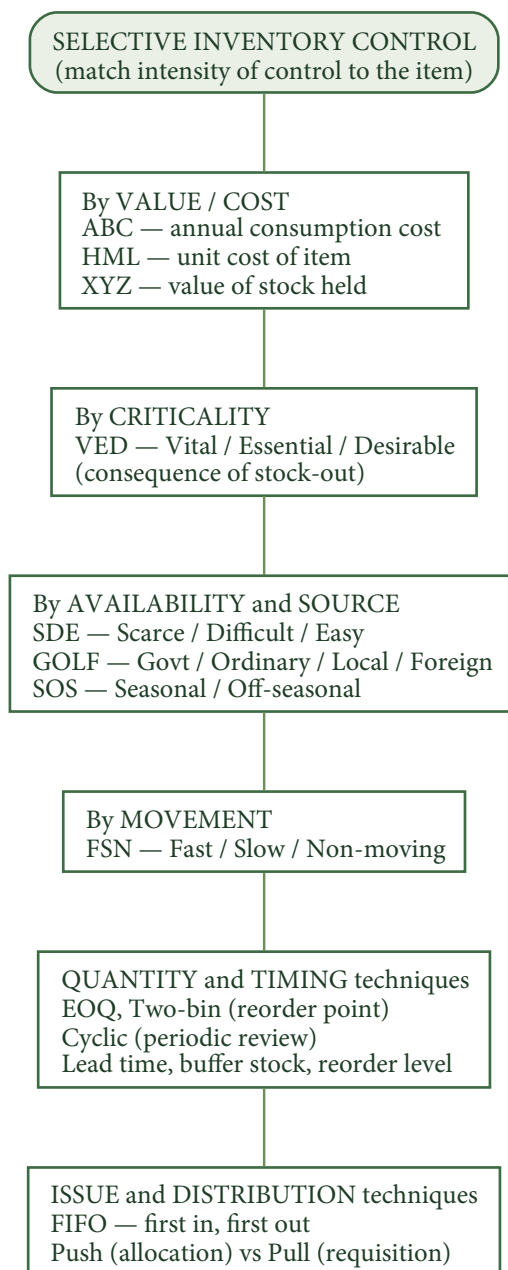
Because the number of stocked items is very large, applying strict control to every item is prohibitively expensive; hence **selective inventory control** — items are grouped by value, by criticality and by usage frequency, and the intensity of control is matched to the group (Red Book, Managerial Sciences, pp 297–298).

Material management techniques — enumeration

Technique	Basis of classification	Categories
ABC analysis	Total annual consumption cost of the item (not unit cost); rests on <i>Pareto's law</i> , that in any large quantity there are the "significant few and insignificant many" (Red Book, Managerial Sciences, p 298)	A = ~10% of items consuming ~70% of expenditure; B = ~20% items, ~20%; C = ~70% items, ~10% (Park Plus, Ch 21, p 283; Sathe, Ch 13, pp 144–145; Jugal Kishore, p 261)
VED analysis	Criticality — consequence of a stock-out for patient care	Vital, Essential, Desirable
HML analysis	Unit cost of the individual item (independent of consumption)	High, Medium, Low
SDE analysis	Ease of availability / procurement	Scarce, Difficult to obtain, Easy to obtain
GOLF analysis	Source of supply	Government, Ordinary (open market), Local, Foreign
FSN analysis	Rate of issue / movement from stores	Fast-moving, Slow-moving, Non-moving
SOS analysis	Seasonality of availability	Seasonal, Off-seasonal
XYZ analysis	Value of stock held in store at a point of time	X, Y, Z (Park Plus, Ch 21, p 283)
ABC–VED coupling (matrix)	Cost and criticality together	Category I, II, III
Economic order quantity (EOQ)	Balances ordering cost against carrying cost — the quantity at which the two are equal and their sum least	$EOQ = \sqrt{(2AP \div CU)}$; A = annual consumption in units, P = purchase cost per order, C = carrying cost as a proportion, U = unit price (Park Plus, Ch 21, p 285; Sathe, Ch 13, p 146)
Two-bin / lot-size reorder-point system	Order placed when stock falls to a fixed level, not on fixed dates	Lead time, buffer stock, reorder level (Sathe, Ch 13, pp 145–146)
Cyclic (periodic review) system	Stock reviewed at a fixed periodicity and topped up to a maximum level	Review period vs lead time; if lead time exceeds the review period, add the quantity "in the pipeline" (Sathe, Ch 13, p 146)
FIFO — "First In First Out" issue principle	Order of issue from the store	Newly arrived stock is racked at the back so that the earliest stock is issued first, preventing time-barred stock and condemnation (Sathe, Ch 13, pp 146–147, Fig 13.1)

Technique	Basis of classification	Categories
"Push" (allocation) vs "Pull" (requisition) distribution	Who decides the flow of commodities down the supply chain	Push — the higher level allocates stock down to the outlets (used for contraceptives; needs little field training but needs timely peripheral reporting); Pull — the periphery indents and draws stock; mixed systems are common (Sathe, Ch 13, p 147)

Three definitions underpin the ordering techniques: **lead time** is the average interval between placing an order and actual receipt of the material; **buffer (safety) stock** is the stock held as insurance against variation in supply and consumption; and the **reorder level** equals the consumption expected during the lead time *plus* the buffer stock (Sathe, Ch 13, pp 145–146).



Selective inventory control — the material management techniques grouped by the basis of classification

VED analysis — basis and categories

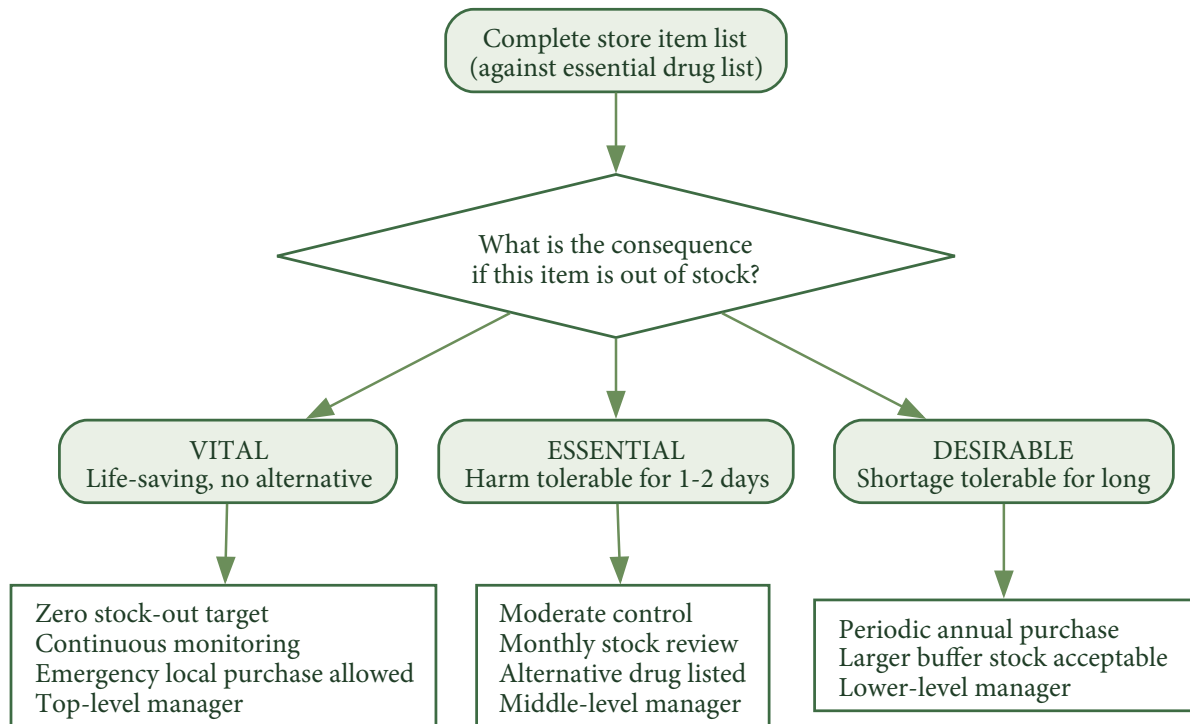
VED groups every stocked item by the **seriousness of the consequences of its running out**, not by its price (Red Book, Managerial Sciences, p 298). It is therefore the only technique that speaks the clinician's language, and it is indispensable where a cheap item can nevertheless kill a patient by its absence.

Category	Criterion	Approx. share of items	Tolerance of stock-out	Control implication
V — Vital	Life-saving, no acceptable alternative; shortage may be disastrous for the organisation	~10%	Nil — every effort, at whatever cost, to avoid an out-of-stock position	Continuous monitoring, top-level responsibility, emergency local purchase permitted
E — Essential	Quite critical for functioning; absence causes appreciable harm but alternatives exist	~40%	Short stretches only — must be restored within a day or two	Moderate control, middle-level manager, regular review of stock
D — Desirable	Non-availability has no serious consequence; largely for chronic or less serious illness	~50%	Can be tolerated for long periods	Periodic purchase, low priority, lower-level management, larger buffer stock acceptable

(Sathe, Ch 13, p 145; Red Book, Managerial Sciences, p 298)

Carrying out a VED analysis — steps and control action

- **Compile the complete item list** of the store (drugs, vaccines, consumables, sutures, reagents), preferably against the *essential drug list* after standardisation and codification, so that the exercise is done on a short rational formulary and not on the thousands of proprietary preparations in the market (Sathe, Ch 13, p 141).
- **Constitute a committee** — pharmacologist/clinician, hospital pharmacist, store officer and the medical superintendent — because criticality is a clinical judgement, not an accounting one.
- **Assign each item to V, E or D** by asking one question: *what happens to the patient and to the hospital if this item is not on the shelf today?*
- **Fix differential stock norms** — a high service level (near-zero stock-out) and frequent review for V items; moderate for E; relaxed for D.
- **Fix differential purchase authority** — V items may be bought by local or emergency purchase without waiting for the tender cycle, the situation Sathe illustrates with anti-rabies vaccine at a primary health centre and with pralidoxime for organophosphorus poisoning; D items are bought only in the annual cycle (Sathe, Ch 13, pp 142–143).
- **Monitor by exception** — daily or weekly stock-out reports for V items reach the top manager; E and D items are reviewed monthly and quarterly.
- **Revise periodically** — the first classification is laborious, but thereafter it is easily verified, revised or modified (Sathe, Ch 13, p 145); revision is needed whenever the case-mix, the standard treatment guidelines or the essential drug list changes.



VED analysis — from classification of an item to the control action taken in the store

Coupling VED with ABC — the two-axis matrix

Neither cost nor criticality alone is sufficient: an anti-snake venom is costly *and* vital, whereas a vitamin syrup may be costly but merely desirable. Cross-tabulating the two axes yields nine cells that collapse into three management categories, because Category I is defined as the items that are **either vital or expensive** (Sathe, Ch 13, p 145, Table 13.1; Park Plus, Ch 21, pp 284–285).

	V (Vital)	E (Essential)	D (Desirable)
A (high annual cost)	AV → Cat I	AE → Cat I	AD → Cat I
B (medium cost)	BV → Cat I	BE → Cat II	BD → Cat II
C (low cost)	CV → Cat I	CE → Cat II	CD → Cat III

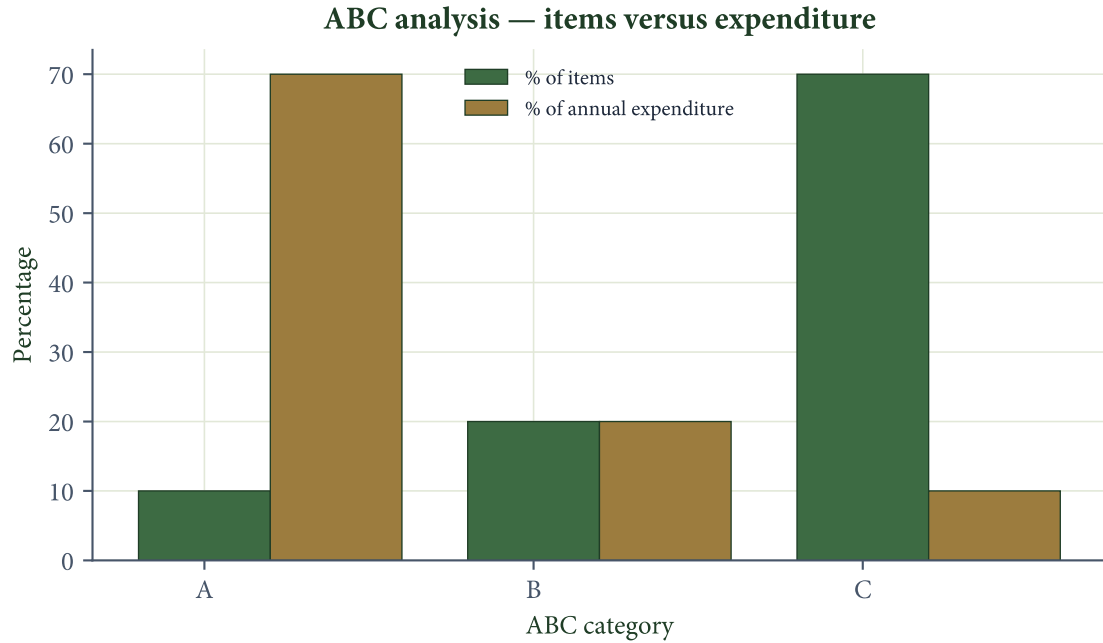
Category	What it contains	Share of items	Management action
Category I	Everything that is vital or expensive — the whole A row together with the vital items of B and C; shortage cripples the hospital, and overstocking or pilferage causes financial loss — e.g. anti-rabies vaccine, anti-snake venom, costly vaccines	~15%	"Management by exception" — continuous monitoring by the senior manager himself; safety stock kept low to reduce carrying cost
Category II	Items that are essential, or desirable but of average cost — essential drugs of medium and low cost and desirable drugs of medium cost	~40%	Moderate control by middle-level managers; consumption watched

Category	What it contains	Share of items	Management action
Category III	Desirable and inexpensive items only; their absence would not affect hospital functioning even for a long time	~45%	Periodic purchase, higher buffer stock permitted, managed at the lower level

(Sathe, Ch 13, p 145; Park Plus, Ch 21, pp 284–285)

Uses and limitations of VED analysis

Uses	Limitations
Guarantees availability of life-saving items in a resource-limited store, whatever their price	Classification is subjective and clinician-dependent; the same item may be vital in an intensive care unit and desirable in a sub-centre
Allows scarce managerial attention and budget to be concentrated on a small "vital" fraction	Ignores cost entirely, so a vital but cheap item may still be over-stocked; must be coupled with ABC
Justifies emergency and local purchase powers and delegation of authority by category	Needs periodic revision with every change in case-mix or standard treatment guidelines
Complements ABC (cost), FSN (movement) and SDE (availability) to produce a complete stock policy	Laborious the first time it is done for a large formulary
Directly usable to fix zero-stock-out targets in programme logistics — the Ministry of Health and Family Welfare aims at 0% stock-out at district level, 90% of facilities receiving supplies on time, and drug loss from expiry held to 5% of overall stock turnover (Jugal Kishore, p 261)	Fails if the store information system is weak — categorisation without a live stock ledger and bin cards has no effect



Distribution of stock items and annual expenditure by ABC category (Park Plus, Ch 21, p 283; Sathe & Sathe, Ch 13, pp 144-145)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **ABC (Pareto) curve** — x-axis "Cumulative number of inventory items (%)" marked at 10, 30 and 100; y-axis "Cumulative cost of inventory (%)" marked at 70, 90 and 100; a rising concave curve with the three zones labelled A (10% items, 70% cost), B (20% items, 20% cost), C (70% items, 10% cost); annotate "significant few" over A and "insignificant many" over C.

High-yield exam pointers

- ▶ Mnemonic for the techniques: **A-V-H-S-G-F-S-X** — ABC (annual cost), VED (criticality), HML (unit cost), SDE (availability), GOLF (source), FSN (movement), SOS (season), XYZ (stock value).
- ▶ ABC = 10% items : 70% cost / 20% : 20% / 70% : 10%; VED = V ~10%, E ~40%, D ~50% of items.
- ▶ ABC-VED matrix: Cat I = AV + AE + AD + BV + CV (~15% of items); Cat II = BE + BD + CE (~40%); Cat III = CD (~45%) — Category I is everything vital or expensive.
- ▶ Reorder level = consumption during lead time + buffer stock; $EOQ = \sqrt{(2AP \div CU)}$.
- ▶ Five Rights of material management: right quality, quantity, time, place, cost.
- ▶ One-line distinction for the examiner: ABC is based on cost, VED on criticality, HML on unit price, FSN on movement, SDE on availability, GOLF on source, SOS on season, XYZ on value of stock held.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park's Textbook of Preventive and Social Medicine 28e carries no separate section on material management or VED analysis; the content above is grounded in Sathe & Sathe (Chapter 13, Materials Management), the Red Book of Public Health and Preventive Medicine (Managerial Sciences), Park Plus (Health Planning and Management) and Jugal Kishore's National Health Programmes of India, and the percentage splits quoted for ABC, VED and the ABC-VED categories are conventional working figures that differ slightly between texts (the Red Book, for instance, illustrates group A as 10% of items costing 90% of inventory value).

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO Model List of Essential Medicines 2025, NLEM 2022, IPHS 2022, NTEP National Strategic Plan 2020–2025

Programme and policy updates (India)

- ▶ **The autonomous state medical supplies corporation is now the reference model for public drug logistics** — Uttar Pradesh replaced its decentralised rate-contract system with the Uttar Pradesh Medical Supplies Corporation (UPMSC), built on seven pillars adapted from the Tamil Nadu corporation: an essential drug list pruned from over a thousand items to about 300 (since expanded to 393 and tiered separately for DH, CHC, PHC and SC), warehouses covering all 75 districts, centralised tendering and supply orders, a **passbook system** giving every facility a virtual budget and the freedom to pull the drugs it actually wants, double-blinded batch testing in NABL-accredited laboratories, centralised payment to vendors, and real-time e-tracking through the **Drugs and Vaccine Distribution Management System (DVDMS)**. Availability of essential drugs in the warehouses rose from about 34% to 88% between December 2020 and July 2024 (mean 275 of 287 items per warehouse) and annual purchase orders from roughly US\$58 million to US\$112 million. [PMID 40786162]
- ▶ **ABC is now being coupled with XYZ (demand variability), not only with VED** — a six-month intervention in a public tertiary cancer hospital in northern India classified 286 medicines worth ₹12.71 crore of annual consumption on both axes and matched the ordering rule to the cell: scheduled orders plus buffer stock for stable high-value items (AX, BX) and consumption-based forecasting for volatile ones (AZ, BZ). Mean monthly stock-outs fell from 4.33 ± 1.51 to 0.50 ± 0.55 ($P < 0.001$, an 88.5% improvement) and monthly wastage from expiry from ₹28,333 to ₹6,667 (76.5% reduction, not statistically significant). [PMID 42583974]

Current national data

- ▶ **India Hypertension Control Initiative, 2018-19 versus 2023-24** — repeat cross-sectional surveys in nine districts of five states found antihypertensives "always available" in government facilities rising from 72% (425/588) to 81% (844/1041), the proportion of patients paying for their medicines falling from 47% to 9%, BP control improving from 37% to 48%, and care-seeking in government facilities rising from 20% to 32% (adjusted PR 1.5, 95% CI 1.4–1.7). Standardising to a drug- and dose-specific protocol shrank the procured basket to a few molecules (amlodipine 5 mg, telmisartan 40 mg), which is selective inventory control applied at programme scale.

NPCDCS was renamed the **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)** in 2023. [PMID 41146104]

- ▶ **Medicines are the weakest readiness domain in Indian public facilities** — ICMR-NCDIR assessed 415 public and private facilities across 19 districts of seven states (2021 and 2023) against **IPHS 2012 and 2022** using the WHO Service Availability and Readiness Assessment method: equipment scored highest and medicines lowest at every level. Of 105 sub-centres, 68.5% reported stock-outs of atenolol, 66.7% of enalapril, 50.5% of glimepiride, 44.8% of amlodipine and 35.2% of metformin, with median stock-out durations of one to seven months; overall readiness was 61% at sub-centres, 59% at CHCs and 73% at PHCs. Ayushman Bharat Health and Wellness Centres have since been renamed **Ayushman Arogya Mandirs**. [PMID 40536370; PMID 40320454]
- ▶ **NLEM 2022 is the current national reference list and carries 384 medicines** — a government tertiary care hospital procured 85.4% (328/384) of them, significantly above the WHO benchmark of 80% ($Z = 2.65$); 18 NLEM sections reached $\geq 80\%$ compliance and eight were fully compliant, but antidotes, diagnostic agents and dermatological medicines remained deficient — precisely the cheap-but-vital blind spot that VED, and not ABC, is designed to catch. [PMID 40449229]

Elimination, surveillance and disease status

- ▶ **The TB elimination deadline and the commodity gap behind it** — NTCP (1962) became RNTCP (1997) and then the **National TB Elimination Programme (NTEP)** in 2020, working to the National Strategic Plan 2020–2025 and a national elimination target of 2025, five years ahead of SDG 3.3. Incidence is still 199 per lakh with only a 3% annual decline against the roughly 15% required, and India carries about 27% of global cases. Frontline staff interviewed in Mumbai named drug shortages and shortages of TruNat cartridges among their chronic resource constraints, and India has 20.6 health workers per 10,000 population against the WHO threshold of 44.5 — a reminder that a stock-out is usually a systems failure, not a store-keeper's failure. [PMID 41275220]

Guideline and legislative updates

- ▶ **FEFO is displacing plain FIFO as the issue rule for medicines** — a national assessment covering all 9,831 primary health centres and 514 district health offices in Indonesia found 91.9% applying FIFO and FEFO (**First-Expired, First-Out**) together; average availability of 50 essential medicines was 66%, ranging from 83% in the best districts to 43% in the worst. Availability was independently higher where the centre had a pharmacist, adequate storage infrastructure, written guidelines, inventory-management principles in use and autonomy to procure, and it clustered spatially within 2 km (Moran's $I 0.67$). For a store holding short-dated items, expiry date rather than date of receipt is the correct basis of issue. [PMID 41651558]
- ▶ **ABC–VED is being extended into multi-criteria classification with EOQ and forecasting** — a 2025 hospital study integrated ABC and VED with lead time, storage limits, stock-out risk and supplier reliability, then applied demand forecasting and the EOQ model to the resulting categories, cutting total inventory cost by 17.63%. The direction of travel is away from the plain two-axis ABC–VED matrix towards multi-criteria scoring with a forecasting engine behind it. [PMID 41390789]

Global and WHO developments

- ▶ **WHO Model List of Essential Medicines, 24th list (September 2025)** — the biennial revision is the yardstick against which national lists such as NLEM are benchmarked, and it is the standard reference used when a bloated state EDL is pruned to a size that pooled procurement can actually service. [WHO 2025-Sep]
- ▶ **Digital logistics management information systems are the current answer to vaccine wastage** — a 2026 scoping review of vaccine management in low- and middle-income countries concluded that electronic systems reduced wastage, improved record-keeping and optimised supply-chain efficiency; India's contribution to that evidence base is the **Electronic Vaccine Intelligence Network (eVIN)**, alongside electronic immunisation registries in Vietnam, Tanzania and Zambia. Most vaccines must be held between +2 °C and +8 °C throughout, so cold-chain integrity is an inventory-control problem as much as an engineering one. [PMID 42302058]

SOURCE & COVERAGE

Sathe & Sathe — Epidemiology and Management for Health Care, Ch 13 (Materials Management), pp 139–147; Red Book of Public Health and Preventive Medicine — Managerial Sciences, pp 297–298; Park Plus (Amrutha & Vijayalakshmi), Ch 21 (Health Planning and Management), pp 282–285; Jugal Kishore — National Health Programmes of India, p 261; Park's Textbook of Preventive and Social Medicine, 28e (spine, no dedicated section on this topic).

Recent-advances sources: 40786162 — Lessons learned from redesigning public health medicines supply chain model in Uttar Pradesh, India, *Frontiers in Public Health* 2025, doi:[10.3389/fpubh.2025.1588227](https://doi.org/10.3389/fpubh.2025.1588227); 42583974 — Optimizing drug availability and reducing wastage through ABC-XYZ inventory management: insights from a public tertiary cancer hospital in Northern India, *Indian Journal of Pharmacology* 2026, doi:[10.4103/ijp.ijp_871_25](https://doi.org/10.4103/ijp.ijp_871_25); 41146104 — Changing patterns in health-seeking behaviour and prescription practices for treatment of hypertension in nine districts: insights from India Hypertension Control Initiative, 2018-19 and 2023-24, *BMC Public Health* 2025, doi:[10.1186/s12889-025-24898-0](https://doi.org/10.1186/s12889-025-24898-0); 40536370 — Preparedness of public & private health facilities for management of diabetes & hypertension in 19 districts in India, *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR_755_2024](https://doi.org/10.25259/IJMR_755_2024); 40320454 — Prevalence and determinants of oral potentially malignant disorders in western Rajasthan, *Scientific Reports* 2025, doi:[10.1038/s41598-025-98096-8](https://doi.org/10.1038/s41598-025-98096-8) (abstract-only); 40449229 — Conformance to the National List of Essential Medicines 2022 in a government tertiary care hospital, *Journal of Healthcare Quality Research* 2025, doi:[10.1016/j.jhqr.2025.101141](https://doi.org/10.1016/j.jhqr.2025.101141) (abstract-only); 41275220 — Frontline workers in India's tuberculosis (TB) elimination efforts: a street-level bureaucracy perspective, *BMC Public Health* 2025, doi:[10.1186/s12889-025-25431-z](https://doi.org/10.1186/s12889-025-25431-z); 41651558 — Local systems, local solutions: which factors drive essential medicine availability in public health facilities across Indonesia?, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-019616](https://doi.org/10.1136/bmjgh-2025-019616); 41390789 — The application of ABC-VED with multi-criteria analysis for drug inventory management, *Scientific Reports* 2025, doi:[10.1038/s41598-025-32068-w](https://doi.org/10.1038/s41598-025-32068-w); 42302058 — Vaccine management systems with a focus on cold chain management and delivery of immunization services in low- and middle-income countries: a scoping review, *PLOS ONE* 2026, doi:[10.1371/journal.pone.0350231](https://doi.org/10.1371/journal.pone.0350231). WHO Model List of Essential Medicines, 24th list, September 2025.



Leadership patterns and their use

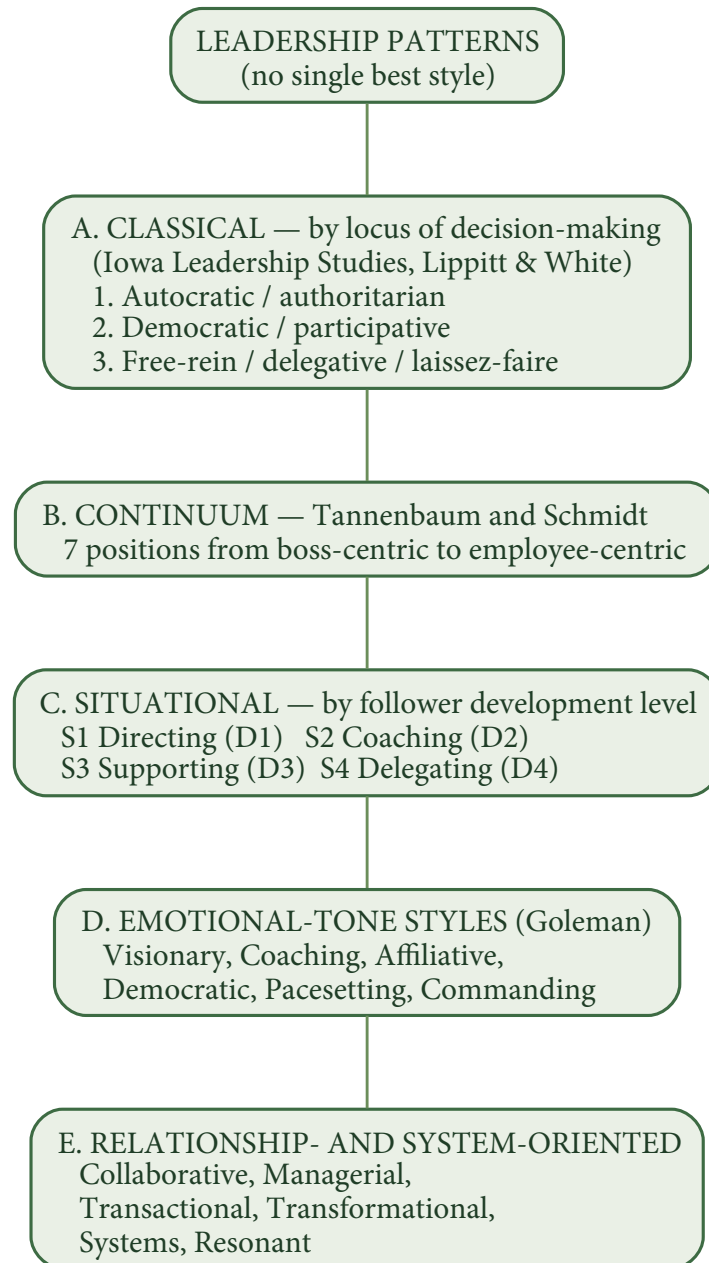
Asked in: Paper I Nov 2017 — RGUHS MD Community Medicine

Definition — Leadership is the art of motivating a group of people to act towards achieving a common goal, defined academically as "a process of social influence in which a person can enlist the aid and support of others in the accomplishment of a common task" (Oxford Public Health 7e, Ch 17, p 658). A **leadership pattern (style)** is *the manner in which the leader influences the behaviour of the health team* (Sunder Lal, Ch 16, p 712). Choosing the appropriate pattern is essentially a question of **deciding the extent to which a manager should be dictatorial and the extent to which he should be participatory** (Red Book, Ch 47, p 291).

Evolution of leadership thought and the basis of "appropriateness"

The **leader** deals with interpersonal influence and vision, the **manager** with acquiring and allocating resources for implementation — the distinction is widely perceived as the difference between **vision and implementation** (Oxford Public Health 7e, Ch 17, p 660).

- ▶ **Great Man theory** (over a century ago) and the **trait/attribute approach** (early twentieth century) — leadership capacity held to be **innate, fixed and cross-contextual**, the leader a solitary actor; research focused on the **individual, not the context** (Oxford Public Health 7e, Ch 17, pp 661-662).
- ▶ **Behavioural approach** — leaders are described by two prime concerns: **concern for people (humanistic)** and **concern for the task (mechanistic)**; *balanced leadership* has equal concern for both (Sunder Lal, Ch 16, p 712).
- ▶ **Contingency/situational approach** (mid-twentieth century, Van Wart) — takes into account the attributes of the leader, of the **followers** and of the **context**, and accepts that "**no single leader trait or style explains leadership in all circumstances**" (Oxford Public Health 7e, Ch 17, pp 662-663). Hence "appropriate use": the style is **fitted to the situation, the followers and the time available**, and every style becomes part of one leader's repertoire (Oxford Public Health 7e, Ch 17, p 670).



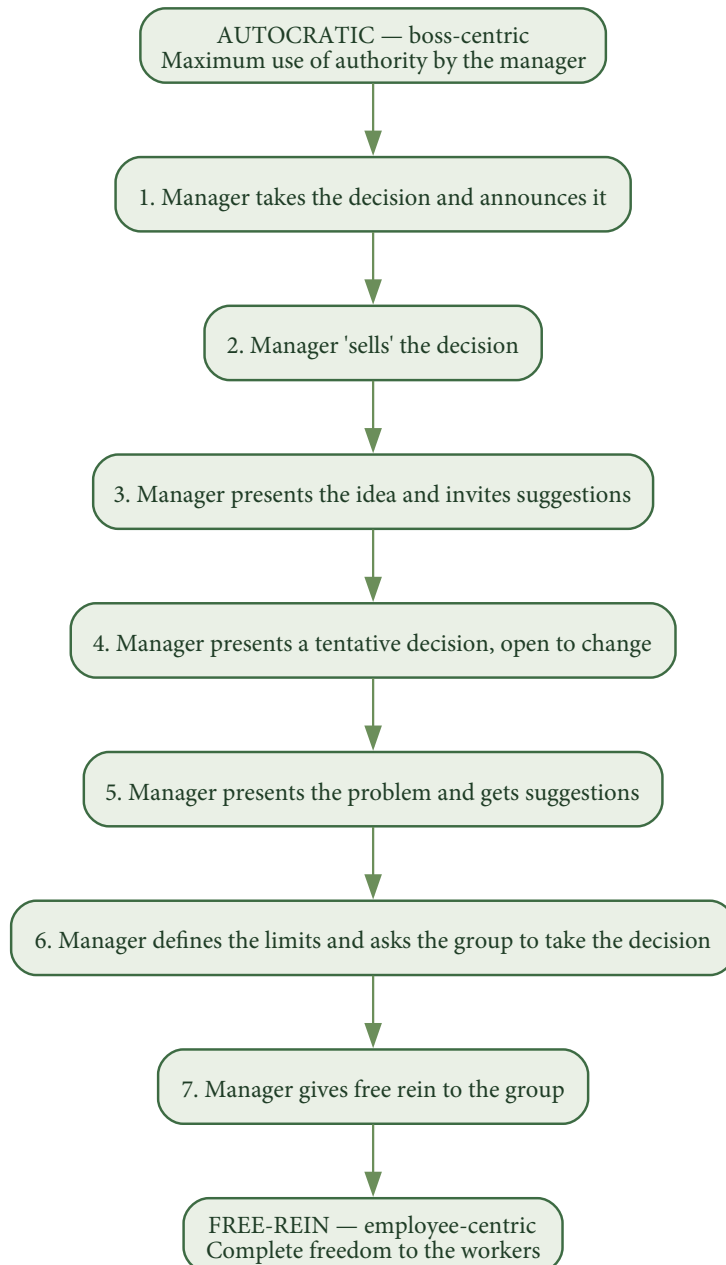
Classification of leadership patterns used in health services

The three classical patterns and where each fits

Patt ern	How the leader behaves	Advantage	Limitation	Appropriate use
Auto cratic (auth orita rian)	Gives orders and insists they be obeyed, often by instilling fear of punishment for non-compliance; formulates policies without consulting subordinates; tells the group only the immediate steps, not the future plan; awards praise or criticism on his own initiative and remains aloof from the group (Red Book, Ch 47, pp 291-292). Maintains authority by force, intimidation, threat, reward and punishment, or position (Oxford Public Health 7e, Ch 17, p 673)	Quick decision-making ; eliminates argument over how and why things are done; at its best a stable, secure and decisive work environment (Oxford Public Health 7e, Ch 17, p 674)	Narrows the range of ideas; leaves fear and mistrust ; sacrifices initiative, new ideas and staff development; problems of morale and does not promote initiative (Oxford Public Health 7e, Ch 17, p 674; Sathe & Sathe, Ch 10, p 115)	Workers low in the hierarchy , who may be better led by one who issues clear-cut, unambiguous and direct orders (Sathe & Sathe, Ch 10, p 115). For its use in an emergency see the <i>Commanding</i> row of Goleman's six styles below
Dem ocrat ic (parti cipati ve)	Gives orders only after consulting the group ; policies are discussed and decided by group discussion; explains the long-term plan before asking for work; the leader participates as an active member of the group (Red Book, Ch 47, pp 291-292). Participation extends beyond decisions to shaping the organisation's vision , but the final decision still rests with one person or a small management team/executive committee (Oxford Public Health 7e, Ch 17, pp 671-672)	Brings in everyone's ideas and so enriches the organisation's possibilities ; builds staff ownership of the goals	Requires time ; assumes followers have the requisite knowledge and skill to participate, which may not be true (Sathe & Sathe, Ch 10, p 115); if ideas are canvassed but never really discussed it carries the drawback of autocracy (no buy-in) without the advantage of quick, clear decisions (Oxford Public Health 7e, Ch 17, p 672)	Programme decisions where staff are competent and there is time to build consensus — monthly staff meeting, block-level review, community consultation. For its danger in a crisis see the <i>Democratic</i> row of Goleman's six styles below

Pattern	How the leader behaves	Advantage	Limitation	Appropriate use
Free-rein (delegative, laissez-faire)	The leader does not lead and leaves the group entirely to itself , is incapable of assuming control over subordinates and does not participate with the group (Red Book, Ch 47, p 292). In its constructive form the leader keeps a low profile , presents what is to be accomplished and leaves it to the followers to find ways of achieving the objectives (Sathe & Sathe, Ch 10, p 115)	Maximum autonomy, initiative and innovation from the team	May produce chaos if the followers are not capable (Sathe & Sathe, Ch 10, p 115)	Teams of professionals and intellectuals — senior specialists, faculty, a research team (Sathe & Sathe, Ch 10, p 115); the same ground as S4 Delegating for staff of high competence and high commitment (Sunder Lal, Ch 16, p 712)

Leadership as a continuum (Tannenbaum and Schmidt). The three classical patterns are not three boxes but the ends and middle of one scale: as the **use of authority by the manager falls, the freedom of the workers rises**. The manager picks the position the task, the team and the time allow (Red Book, Ch 47, pp 291-292).



Leadership as a continuum (Tannenbaum and Schmidt) — use of authority by the manager falls as freedom of the workers rises

The Indian anchor. The medical officer is the **captain of the health team at the primary health centre**, whose success depends largely on the team leadership he provides — he is the **planner, promoter, director, supervisor, coordinator and evaluator** (Park 28e, Ch 10, pp 738-739). He moves along the continuum within a single working day: direct orders for a new multipurpose worker, consultation at the monthly staff meeting, delegation of maternal and child health work to the health workers (Sunder Lal, Ch 16, p 712).

Situational leadership — matching the style to the follower

Competence means sound scientific knowledge and skills; **commitment** means a highly motivated and confident person. There is no one best style — the **most important criterion in deciding the style is the characteristics of the subordinates or health team**, and the appropriate style is always a combination of *directive* and *supportive* behaviour, with a shift from "ME to WE" (Sunder Lal, Ch 16, p 712).

Style	Leader behaviour	Follower development level	Appropriate use in the health team
S1 — Directing (directive behaviour)	Telling people what to do, how, where and when; specific instructions (written job description and tasks) with close supervision of task accomplishment	D1 — low competence (newly recruited), high commitment (eager, enthusiastic)	The newly posted health worker, ASHA or intern who is eager but untrained
S2 — Coaching	Directing plus supporting — continues to direct and supervise closely, but also explains decisions, solicits suggestions and supports progress	D2 — some competence, low commitment	The worker who has learnt the task but lost enthusiasm, or a supervisor demoralised by a poor programme review
S3 — Supporting	Listening , providing support and encouragement, facilitating involvement in problem-solving and decision-making , sharing responsibility for decisions	D3 — high competence, variable commitment	The experienced auxiliary nurse midwife or health assistant who is skilled but unsure of taking responsibility
S4 — Delegating	Turns over responsibility and authority for day-to-day decision-making and problem-solving to the subordinate	D4 — high competence, high commitment	Delegation of maternal and child health activities — antenatal care, deliveries, postnatal care, immunisation — to health workers at the sub-centre

(Sunder Lal, Ch 16, p 712.) The same progression appears as a **hierarchy of leadership components** — **Directing** → **Participating** → **Delegating** → **Transforming** — with directive leaders using authoritative command, participating leaders coaching and acting as role models, delegating leaders affording independence and self-direction, and transformational leaders at the top empowering the group (Harrison, Ch 3, pp 78-79).

Goleman's six styles — the moment each one suits

The most effective leaders **move among these styles**, adopting the one that meets the needs of the moment (*Primal Leadership*, cited in Oxford Public Health 7e, Ch 17, pp 669-670).

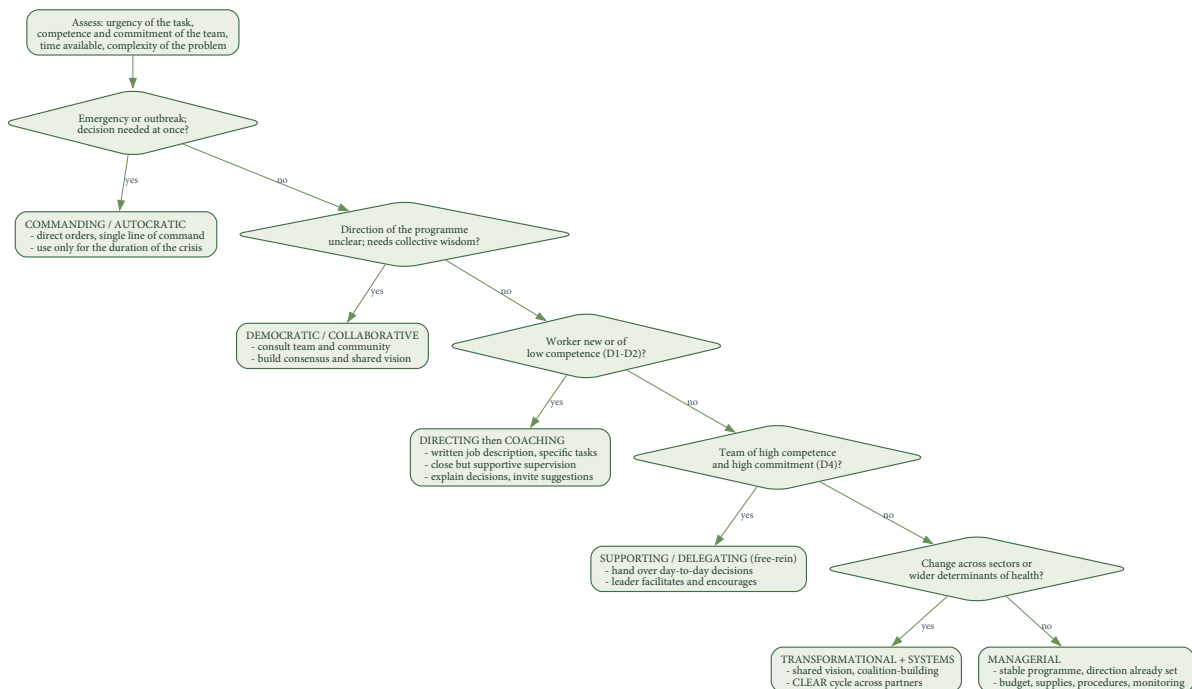
Style	Works best when	Caution
Visionary	The organisation needs a new direction ; articulates <i>where</i> the group is going but not <i>how</i> , setting people free to innovate, experiment and take calculated risks	Fails where the task, not the direction, is the problem
Coaching	One-to-one development of staff who show initiative and want professional development ; connects individual goals to organisational goals	Backfires if perceived as micromanaging ; undermines self-confidence

Style	Works best when	Caution
Affiliative	Heightening team harmony, raising morale, improving communication or repairing broken trust	Used alone, its group praise lets poor performance go uncorrected — "mediocrity is tolerated"
Democratic	The direction is unclear and the leader needs the group's collective wisdom	Disastrous in a crisis demanding quick decisions
Pacesetting	Short bursts with a highly competent team; the leader sets high standards, obsessive about doing things better and faster	Use sparingly — "pacesetting poisons the climate" and makes people feel they are failing
Commanding	Crisis, when an urgent turnaround is needed — the classic "military" style, the most often used but least often effective	Rarely praises, frequently criticises; undercuts morale and job satisfaction

Relationship- and system-oriented patterns

Pattern	Core idea	Appropriate use
Collaborative	The leader is "first among equals" — initiates discussion and pinpoints issues, but decisions are made by discussion and majority or consensus ; fosters trust and teamwork. Comes closest to servant leadership and to the equality and empowerment in the mission of grass-roots and community-based organisations, removing the distrust between line staff and administrators (Oxford Public Health 7e, Ch 17, p 671)	Community-based and voluntary organisations , village health sanitation and nutrition committees, intersectoral coordination committees, Rogi Kalyan Samitis
Managerial	Concerned primarily with running the organisation — funding, systems and infrastructure (policies, positions, equipment) and day-to-day operations; on top of the budget and the procedures manual — but does not steer the organisation (Oxford Public Health 7e, Ch 17, pp 672-673)	A stable programme already in the implementation phase , where direction is settled and efficiency is the need; good management without a clear vision leaves an organisation with no sense of purpose
Transactional	Emphasises rewarding good performance and corrective action for poor performance ; assumes employees are motivated by self-interest — the promise of reward or the threat of punishment (Harrison, Ch 3, pp 74-76)	Target-driven, measurable, routine work where incentives and sanctions are the available levers
Transformational	Introduced by James MacGregor Burns (1978) and extended by Bernard Bass ; the leader distils the values, hopes and needs of followers into a shared vision and then empowers them to pursue it, using charisma, intellectual stimulation, individual attention and emotional appeal ; followers trust, admire and respect the leader (Harrison, Ch 3, pp 76-79; Oxford Public Health 7e, Ch 17, p 673)	A political, social or economic environment in crisis and requiring radical reform — launching or relaunching a national programme, a health-sector reform, an eradication drive

Pattern	Core idea	Appropriate use
Systems	Leading across boundaries — departmental, organisational and sectoral; "how you lead when you are not in charge" and must influence rather than pull a management lever; combines collaborative leadership, coalition-building and systems insight (Oxford Public Health 7e, Ch 17, pp 674-676)	Wider determinants of health — transport, housing, food systems — and any large, complex, seemingly intractable problem no one organisation can solve alone
Resonant	The modern style for unprecedented change and uncertainty, built on emotional intelligence: self-awareness, self-management, social awareness and relationship management; the leader manages his own and others' emotions and builds trusting relationships (Oxford Public Health 7e, Ch 17, pp 668, 677)	Prolonged uncertainty, workforce fatigue and burnout — as in a protracted epidemic response



Choosing the leadership pattern for the situation, the follower and the time available

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Situational leadership grid.** Draw two axes — *x-axis: directive behaviour (low → high)*, *y-axis: supportive behaviour (low → high)* — and label the four quadrants: *S1 Directing — high directive, low supportive*; *S2 Coaching — high directive, high supportive*; *S3 Supporting — low directive, high supportive*; *S4 Delegating — low directive, low supportive*. Beneath the grid draw a bar reading, right to left, *D1 low competence + high commitment*, *D2 some competence + low commitment*, *D3 high competence + variable commitment*, *D4 high competence + high commitment*, with arrows from each D to its matching S.
- ▶ **Managerial Grid (Blake and Mouton, 1978).** Draw a square with *x-axis: task orientation (1 → 9)* and *y-axis: person orientation (1 → 9)*, and mark five points: *(9,1) Task Management leader*; *(1,9) Country Club Management leader*; *(1,1) Impoverished Management leader*; *(5,5) Middle of the Road Management leader*; *(9,9) Team Management leader — high and balanced concern for both task and people, the ideal but hardest to attain*.

High-yield exam pointers

- ▶ **Classical triad = A-D-L:** Autocratic (quick decisions, poor morale), Democratic (commitment, but needs time), Laissez-faire / free-rein (autonomy, but chaos if followers are not capable) — from the Iowa Leadership Studies (Lippitt and White).
- ▶ **Tannenbaum and Schmidt continuum:** seven positions from *manager takes the decision and announces it* to *manager gives free rein to the group*; **use of authority by the manager falls as freedom of the workers rises.**
- ▶ **Situational pairing:** S1 Directing–D1 (low competence, **high** commitment); S2 Coaching–D2 (some competence, low commitment); S3 Supporting–D3 (high competence, **variable** commitment); S4 Delegating–D4 (high competence, high commitment). **Competence = knowledge + skills; commitment = motivation + confidence.**
- ▶ **The crisis rule, both ways:** commanding/autocratic is effective **only** in a crisis; democratic is **disastrous** in a crisis. Pacesetting "poisons the climate".
- ▶ **Goleman's six (Primal Leadership):** Visionary, Coaching, Affiliative, Democratic, Pacesetting, Commanding. **Transformational = Burns 1978, extended by Bass; transactional = reward-and-punishment exchange.** Hierarchy: **Directing → Participating → Delegating → Transforming.**
- ▶ **Emotional intelligence** — four domains (self-awareness, self-management, social awareness, relationship management); accounts for **85–90% of the difference between outstanding leaders and their average peers** (Oxford Public Health 7e, Ch 17, pp 667-668). **CLEAR** framework for systems leadership (five elements, not necessarily sequential) = Convene and commit, Look and learn (system mapping), Engage and energize, Act with accountability, Review and revise (Ch 17, pp 675-676).
- ▶ **Closing line:** "*There is no one best leadership style*" — the deciding criterion is the **characteristics of the subordinates and the situation**; the primary health centre medical officer is the **captain of the health team** and carries all patterns in one repertoire.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the four situational styles S1–S4 are conventionally attributed to Hersey and Blanchard, but the corpus text describing them (Sunder Lal, Ch 16, p 712) does not itself name those authors. The Red Book also lists the Hawthorne Studies (Mayo and Roethlisberger), Theory X and Theory Y (McGregor), the Michigan Studies (Likert), the Ohio State Studies (Stogdill) and Taylor's scientific manager's style as the wider literature behind these patterns (Red Book, Ch 47, p 292).

SOURCE & COVERAGE

Red Book — Public Health and Preventive Medicine (Armed Forces Medical College), Ch 47 (Managerial Sciences), pp 291-292; Oxford Textbook of Public Health 7e — Ch 17 (1 Leadership in public health), pp 658-677; Sunder Lal — Textbook of Community Medicine, Ch 16 (Health Planning and Management), pp 711-713; Sathe & Sathe — Epidemiology and Management for Health Care, Ch 10 (Organizational behaviour), p 115; Harrison — Essentials of Health Care Organization, Ch 3 (Transformational leadership maximizes strategic planning), pp 74-79; Park's Textbook of Preventive and Social Medicine, 28e — Ch 10 (Health Care of the Community), pp 738-739. Coverage note: Park 28e carries no section on leadership patterns, so the classification and the appropriateness of each pattern are grounded on the Red Book and the health-management texts in the corpus, with Park supplying the Indian primary-health-centre application.

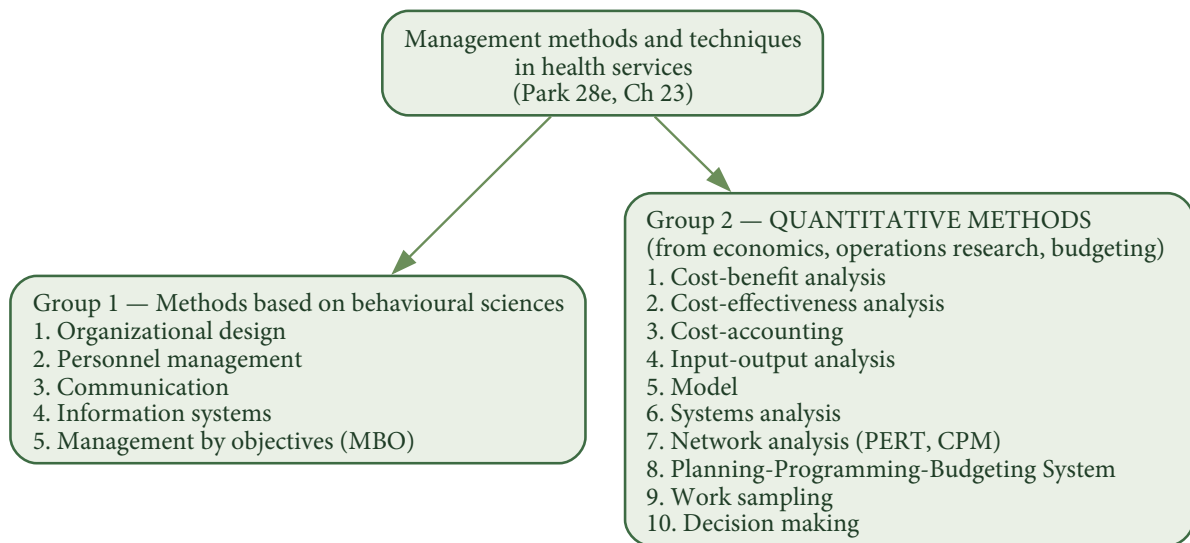


Quantitative methods of management

Asked in: Paper IV 20-Jul-2020 — RGUHS MD Community Medicine

Definition — Quantitative methods of management are those management techniques **derived from economics, operations research and budgeting** that give the health administrator a *numerical* basis for choosing between alternative courses of action (Park 28e, Ch 23, p 1028). They stand in contrast to the behavioural-science group of techniques, and their common purpose is to convert a management problem into measurable inputs, outputs, costs, times and probabilities so that resources are used in the most effective way in achieving stated objectives.

Classification of management techniques



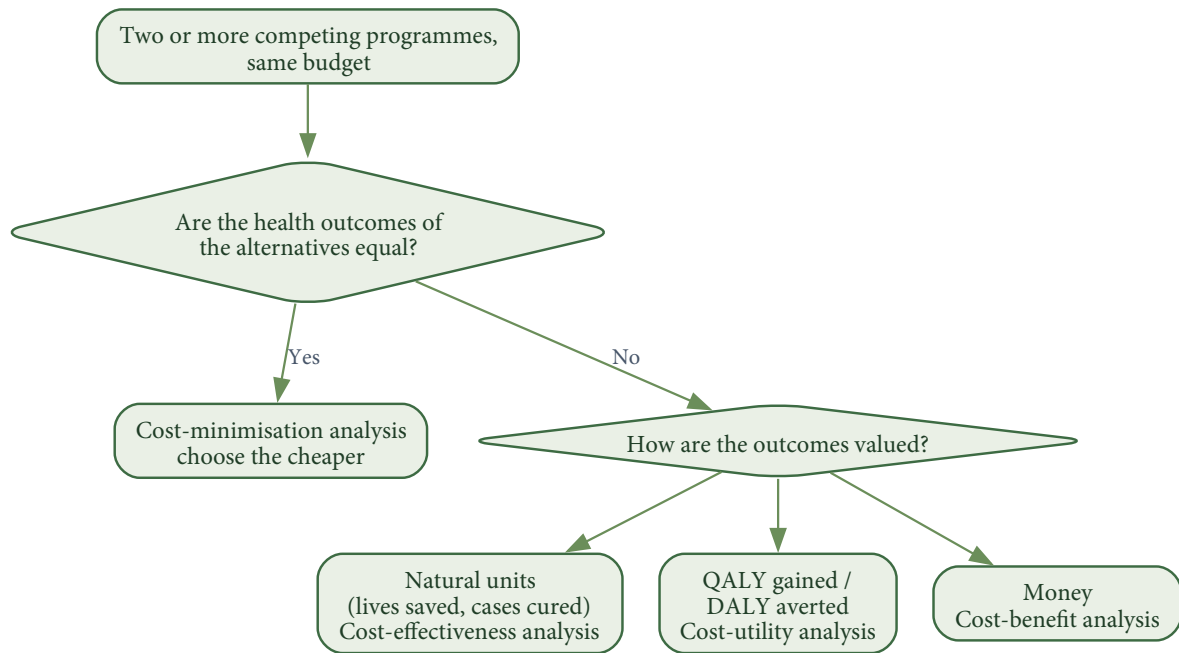
Park's two groups of health management techniques, with the ten quantitative methods expanded

Economic and budgeting methods, with examples

#	Method	Principle	Suitable example
1	Cost-benefit analysis (CBA)	Economic benefits of a programme are compared with its cost, both expressed in money ; benefit–cost ratio >1 means the action is worth doing (Park 28e, Ch 23, p 1028; Suryakantha, Ch 37, p 807)	First used in health by <i>Jetanovic, Grab and Umera</i> to compare sanitary measures with vaccination in typhoid control; historically conceived by <i>Jules Dupuit</i> , French engineer, 1848 (Suryakantha, Ch 37, p 807)
2	Cost-effectiveness analysis (CEA)	Same as CBA except benefit is expressed as results achieved, not money — lives saved, days free from disease, cases cured; more promising in health than CBA (Park 28e, Ch 23, p 1028)	Directly observed treatment, short-course (DOTS) versus hospital treatment of tuberculosis — DOTS is the cost-effective option; safe water supply versus treating water-borne disease — purification wins (Suryakantha, Ch 37, p 807)

#	Method	Principle	Suitable example
2a	Cost-utility analysis (CUA) <i>(refinement of CEA)</i>	Outcome measured as quality-adjusted life years (QALYs) gained or disability-adjusted life years (DALYs) averted ; the incremental cost-effectiveness ratio (ICER) = difference in cost ÷ difference in effect between two programmes (Health Economics, Ch 15–16, pp 230, 247–253)	Cost per DALY averted used to rank interventions in a "league table" for national priority-setting (Health Economics, Ch 16, pp 254–255)
2b	Cost-minimisation analysis (CMA) <i>(special case)</i>	Used only when two interventions are assumed to give equal benefit ; the cheaper one is chosen (Sunder Lal, Ch 2, p 88)	Generic drug versus branded equivalent of proven equal effectiveness in the district drug formulary
3	Cost-accounting	Financial records kept so that costs attach to the purpose for which they were incurred; three uses — (a) cost control, (b) planning and allocation of people and money, (c) pricing for cost reimbursement (Park 28e, Ch 23, p 1028)	Working out the true per-patient cost of an outpatient visit or a cataract surgery for reimbursement under a publicly funded insurance scheme
4	Input-output analysis	An economic technique: inputs are the resources consumed (manpower, money, materials); outputs are useful outcomes such as cases treated, lives saved or inoculations performed. An input-output table shows how much of each input yields a unit of output, and so forms the base of an evaluation and checking system (Park 28e, Ch 23, pp 1028–1029)	Auxiliary nurse midwife (ANM) hours, vaccine vials and cold-chain running cost set against the number of children fully immunised at a subcentre
8	Planning-Programming-Budgeting System (PPBS)	Helps decision-makers allocate resources so that available resources are used most effectively for the objectives; requires no change in existing organisation — only regrouping of activities into programmes tied to each objective (Park 28e, Ch 23, p 1029)	Budgeting a district health society by programme (maternal health, immunisation, vector control) instead of by line item
8a	Zero-based budgeting ("zero budget approach")	All budgets start at zero ; no unit gets any budget it cannot specifically justify afresh, year to year (Park 28e, Ch 23, p 1029)	Requiring a district malaria unit in a now low-transmission district to re-justify its spray-wing budget each year

Choosing the economic evaluation, and a worked cost-benefit example



Selecting the appropriate form of economic evaluation for two competing health programmes

Principles of a valid CBA (Suryakantha, Ch 37, p 807): 1. a common unit of measurement, usually money; 2. the planner's valuation must represent the beneficiary's valuation; 3. the comparison must be "with action versus without action"; 4. no double counting of costs or benefits; 5. long-term benefits must be counted.

Worked example — should a subcentre keep its own refrigerator for the cold chain? (Suryakantha, Ch 37, p 807)

Side	Item	Value
Cost	Refrigerator	– ₹50 per week
Cost	Electric power	– ₹10 per week
Benefit	Travel charge saved for the female health worker (subcentre to primary health centre)	+ ₹30 per week
Benefit	Work not lost at the subcentre during that travel	+ ₹40 per week
Net	Benefit	+ ₹10 per week = ₹520 per subcentre per year

The benefit–cost ratio exceeds 1, so keeping the refrigerator at the subcentre is the cost-beneficial choice. Two terms belong to this calculation: **lead time**, the interval between starting the activity (spending the resources) and the benefits beginning, and **opportunity cost**, the loss to the community from failing to use an available resource. CBA aids budgeting and long-term investment planning; its weakness is that health benefits often cannot honestly be priced, so the decision-maker's bias and missed indirect costs creep in.

Operations-research and scheduling methods, with examples

#	Method	Principle	Suitable example
5	Model	A basic concept of management science — an abstraction of reality, not the reality itself , used to understand how the factors in a situation affect one another; every decision process uses a model (Park 28e, Ch 23, p 1029)	A transmission model used to project how many malaria cases follow a given level of insecticide-treated-net coverage
6	Systems analysis	Helps the decision-maker choose a course of action by defining the problem, setting objectives, listing alternatives, evaluating them on cost-effectiveness, re-examining objectives, and picking the most cost-effective alternative; the "system" may be a hospital supply system, an information system, an outpatient clinic or a whole community health service (Park 28e, Ch 23, p 1029)	Redesigning a district hospital drug supply system after analysing its indent, storage and distribution subsystems
7	Network analysis	A graphic plan of all events and activities needed to reach an end objective; brings discipline into planning. Two forms — programme evaluation and review technique (PERT) and critical path method (CPM) (Park 28e, Ch 23, p 1029)	Scheduling the commissioning of a new primary health centre (see the network below)
9	Work sampling	Systematic observation and recording of the activities of one or more workers at predetermined or random intervals ; the parameters analysed are the type of activity performed and the time each job needs. It tests whether current job descriptions and training are appropriate, standardises how jobs are done, and determines manpower needs (Park 28e, Ch 23, p 1029)	Work-sampling studies on doctors, nurses, pharmacists and laboratory technicians to fix staffing norms (Park 28e, Ch 23, p 1029)
10	Decision making	Analogous to differential diagnosis: decisions should be made at the level where the best decision can be made — not necessarily at the top — and never on incomplete data (Park 28e, Ch 23, p 1029)	Deciding the optimum workload for medical and paramedical workers, or the strategy for delivering a service, at district rather than state level
+	Linear programming	Optimises an objective function subject to resource constraints , assuming all relationships are linear and additive (Reinke, Ch 3, p 50)	Fixing the service mix of a health centre when a particular skilled cadre is in short supply
+	Queuing (waiting-line) models	Analyse waiting lines produced by random arrivals, balancing patient waiting time against staff idle time (Reinke, Ch 3, p 52)	Sizing an outpatient department and testing whether an appointment system would cut waiting; scheduling elective admissions against available beds
+	Distribution models	Determine the least costly way of supplying many outlets from a few depots (Reinke, Ch 3, p 52)	Deciding which regional drug warehouse supplies which block, and which peripheral clinic refers to which first-referral unit

Network analysis in detail — CPM and PERT

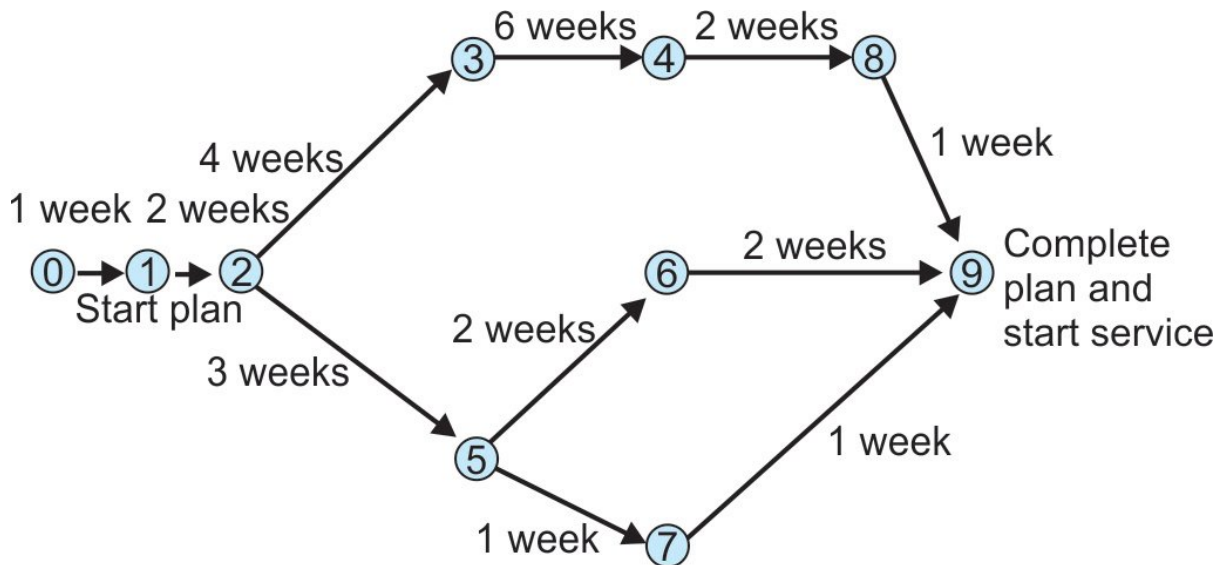


Fig 37.3 — Flow diagram of critical path (Suryakantha Environment Best, p.806)

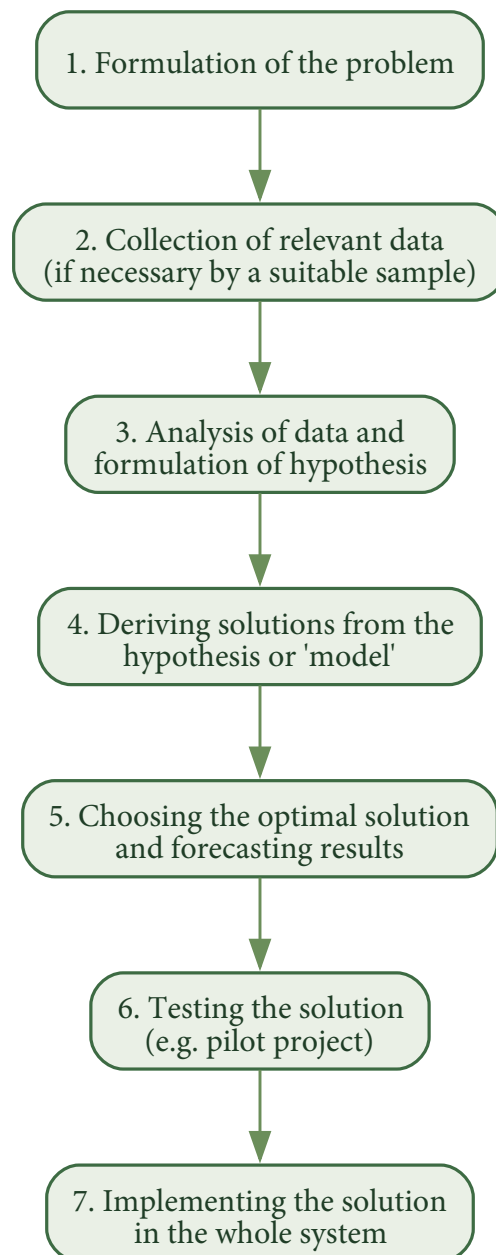
In the network above, the sequence 0→1→2→3→4→8→9 totals 16 weeks and is the **critical path** — the longest path through the network. The branch 0→1→2→5→6→9 takes 10 weeks and therefore carries 6 weeks of **slack**. Only shortening an activity *on the critical path* shortens the project; any delay on it delays the whole project (Park 28e, Ch 23, p 1029; Reinke, Ch 15, pp 222–224).

Feature	Critical path method (CPM)	Programme evaluation and review technique (PERT)
Time estimate	One deterministic estimate per activity	Three estimates — optimistic, most likely, pessimistic
Expected activity time	As estimated	Weighted average, the most likely estimate given four times the weight of either extreme, i.e. (optimistic + 4 × most likely + pessimistic) ÷ 6 (Reinke, Ch 15, pp 225–226)
Handles uncertainty	No	Yes — gives the probability of finishing by the target date
Chief use	Finding and compressing the longest (critical) path	Detailed planning and comprehensive supervision of a project whose activity times are uncertain
Extensions	Gantt chart, used with the network for actual calendar scheduling	PERT/COST — adds resource cost, finds the cheapest way to meet a deadline and ties progress to expenditure (Reinke, Ch 15, pp 224–226)

Operational research — the parent discipline

The term was coined during the Second World War over the best use of radar, and it means the **application of scientific methods of investigation to the study of complex human organisations and services**, with the deliberate object of inducing beneficial change (Park 28e, Ch 12, p 839). Its toolbox is exactly the quantitative set above — network analysis, PERT, critical path analysis, linear programming, computer simulation, queuing theory, cost-effectiveness analysis and work study (Sunder Lal, Ch 6, p

265). The minimum team in social-medicine applications is a public health administrator, an epidemiologist, a statistician and a social scientist, under a director (Park 28e, Ch 12, p 839).



The seven phases of an operational research study in health services

Health-service problems typically tackled (Park 28e, Ch 12, p 839): optimal size of area and population to be covered by a midwife or a basic health unit; the ideal vehicle for local health workers; architectural design of hospitals and health centres; queuing in outpatient departments and hospital waiting lists; bed load and nursing services in teaching versus non-teaching hospitals; length of stay in hospital and length of sickness absence; the extent to which a programme's stated objective is achieved (for example in malaria eradication or family planning); quality of medical care; and investigation of epidemic outbreaks.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Network (arrow) diagram with the critical path** — label: numbered event circles 0 to 9; activity arrows with duration in weeks written on each; "Start plan" at event 0; "Complete plan and start service" at the terminal event; the critical path marked with a **double line**; slack shown against one non-critical branch.
- ▶ **Cost-effectiveness plane** — label: horizontal axis "incremental effect (– to +)", vertical axis "incremental cost (– to +)", origin = comparator, the four quadrants marked, and the south-east quadrant labelled "new intervention dominates — always cost-effective" (Health Economics, Ch 16, p 253).

High-yield exam pointers

- ▶ **Park's ten quantitative methods, in order:** cost-benefit analysis, cost-effectiveness analysis, cost-accounting, input-output analysis, model, systems analysis, network analysis, planning-programming-budgeting system, work sampling, decision making.
- ▶ **Decision rules:** benefit–cost ratio $>1 \rightarrow$ accept; incremental cost-effectiveness ratio = $\Delta\text{cost} \div \Delta\text{effect}$; cost-utility outcome unit = QALY gained or DALY averted.
- ▶ **PERT expected time** = (optimistic + 4 × most likely + pessimistic) ÷ 6; **critical path** = longest path = shortest possible project duration; **slack** = critical path time – that path's time.
- ▶ **PPBS** groups activities into programmes tied to objectives and needs no reorganisation; **zero-based budgeting** starts every budget at zero each year.
- ▶ **Cost-accounting's three purposes:** cost control; planning and allocation of people and money; pricing for cost reimbursement.
- ▶ **Operational research** — coined in the Second World War (radar); seven phases from problem formulation to system-wide implementation; team = administrator + epidemiologist + statistician + social scientist.
- ▶ Ready examples: DOTS versus hospital care for tuberculosis (CEA); subcentre refrigerator saving ₹520 per year (CBA); work sampling of nurses to fix staffing norms; queuing analysis of an outpatient department.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's textbook lists cost-accounting, input-output analysis and the planning-programming-budgeting system as separate "quantitative methods", while health-economics texts fold cost-minimisation and cost-utility analysis into the same family as cost-benefit and cost-effectiveness analysis — both groupings are acceptable in the examination, and the cost-effectiveness thresholds quoted in economics texts (for example £20,000–30,000 per quality-adjusted life year in the United Kingdom) are not Indian figures and should not be quoted as such.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, Health Technology Assessment in India (HTAI), National Health Authority — PM-JAY, Indian Reference Case 2023

Programme and policy updates (India)

- ▶ **Health Technology Assessment in India (HTAI)** — the national agency for choosing between competing health technologies, set up under the Department of Health Research, Ministry of Health and Family Welfare. A historical and thematic study of how it came into being found that the decisive factors were a coalition of champions senior enough to carry political authority across several organisations, careful thought about where in government the body should sit, and the building of a network of institutions to generate technical evidence and train people, strengthened by international networks and funders [PMID 38715206]. For the examiner, this is the institutional home of cost-benefit, cost-effectiveness and cost-utility analysis in Indian public health.
- ▶ **Ayushman Bharat PM-JAY health benefit packages (HBP) have been revised four times** since the original HBP 1.0 of September 2018 (1,393 packages, 16 surgical and 8 medical specialties) — HBP 2.0 (2019), HBP 2.1 (November 2020), HBP 2.2 (October 2021) and HBP 2022 (April 2022, 1,949 packages). Successive rounds drew increasingly on empirical cost evidence, HBP 2.1 excepted, which was driven purely by expert consultation with no cost-effectiveness or HTA input at all. HBP 2022 introduced differential pricing to allow for structural variation in the cost of delivering a service: a higher reimbursement rate for the higher-cost city tiers (tier 1 and tier 2) with the national rate used everywhere else, and higher rates for tertiary than for secondary procedures. That revision also prompted the National Health Authority to set up a Health Financing and Technology Assessment (HeFTA) unit [PMID 38857943].
- ▶ **A national programme costed and judged by cost-effectiveness analysis:** Intensified Mission Indradhanush, the four-month campaign run from October 2017 to January 2018, cost US\$6.21 per incremental dose delivered, US\$82.99 per zero-dose child reached and US\$327.63 per DALY averted, and averted an estimated 1,413 child deaths across the 40 sampled districts in five states; at a threshold of one per-capita GDP per DALY averted it was cost-effective with 90% probability [PMID 38590052]. This is the modern worked example to quote instead of, or alongside, the subcentre-refrigerator calculation.

Current national data

- ▶ **Costing of Health Services in India (CHSI)** is the nationally representative facility-costing study that now anchors PM-JAY rate-setting — top-down and bottom-up micro-costing in 63 hospitals across 14 states. It showed that about 42% of HBP 1.0 packages were reimbursed at less than half their actual cost; applying the cost evidence halved that proportion, and the 2022 revision raised 832 package rates, reduced 122 and left 630 unchanged [PMID 37963613]. Cost-accounting and input-output analysis at national scale.
- ▶ **A hospital cost-surveillance pilot began in April 2022** in five states, capturing patient-level resource use through the PM-JAY transaction management system coded with ICD-11 for diagnoses, SNOMED CT for drugs and consumables and LOINC for diagnostics, as groundwork for a diagnosis-related-group style provider payment. Providers reported about two hours of data entry per inpatient admission — roughly 30 extra person-hours a day in a high-volume hospital — so the authors conclude that sustainability depends on building cost capture into existing billing and hospital information systems, and on the Ayushman Bharat Digital Mission, rather than running it as separate parallel paperwork [PMID 39260852].

Elimination, surveillance and disease status

- **Lymphatic filariasis — triple-drug (ivermectin + DEC + albendazole, IDA) versus two-drug (DEC + albendazole) mass drug administration:** modelled for a population of 100,000, IDA cost an extra US\$578 and averted 34 additional DALYs, an incremental cost-effectiveness ratio of US\$17 (₹1,413) per DALY averted, with 95% probability of being cost-effective against India's per-capita GDP threshold of US\$2,424; the budget-impact analysis found switching to IDA would save US\$66.92 million and cut overall programme cost by 46%, supporting IDA for the 2027 elimination goal [PMID 41817502] (*paywalled — grounded on the PubMed abstract, whose numbers are internally consistent*).
- **Cervical cancer elimination — single-dose HPV vaccination:** a cost-utility analysis of free single-dose immunisation of 11-year-old girls in Rajasthan put the cost at ₹6,641 (US\$75.68) per QALY gained and returned a **negative ICER (reported as −837,822)**, meaning the programme is cost-saving — cheaper as well as more effective than not vaccinating — at a subsidised Gavi/UNICEF price of ₹395 (US\$4.50) per dose. Immunising one cohort costs ₹1.28 crore, 0.64% of the state health budget, and halving or quadrupling the vaccine price changed the ICER by under 1%, because the result is driven by the cervical-cancer treatment costs avoided rather than by the vaccine price [PMID 41501796].

Guideline and legislative updates

- **The Indian Reference Case (2023)** is the national standard for conducting economic evaluation: twelve recommendations across eleven principles — current practice as comparator; an *abridged societal perspective* (direct medical plus direct non-medical costs borne by payer and patient, with indirect productivity costs only in sensitivity analysis); cost-utility analysis with QALYs as the preferred outcome, DALYs permitted where QALYs cannot be measured; a lifetime horizon for interventions affecting mortality; and 3% annual discounting of both costs and outcomes, varied 0–5% in sensitivity analysis [PMID 37694178].
- **Threshold discipline in the answer script:** India has no explicit or implicit cost-effectiveness threshold of its own, so Indian evaluations fall back on the WHO-CHOICE rule of thumb — an ICER under three times per-capita GDP counts as cost-effective, under one times as highly cost-effective [PMID 41501796]. Empirically derived opportunity-cost thresholds are far lower, and the confidence in a verdict shifts sharply with the threshold chosen: Intensified Mission Indradhanush was cost-effective with 90% probability at one times per-capita GDP per DALY averted (US\$1,927 in that study), but with only 54% probability against an empirically derived threshold of US\$349 per DALY averted [PMID 38590052]. State the threshold you are applying rather than calling an ICER "cost-effective" without one.

Global and WHO developments

- **WHO's global survey of 104 countries**, the largest of its kind, found health technology assessment used mainly for planning and budgeting (78%), clinical practice guidelines (75%) and design of health benefit packages (65%); only 24% of countries have guidelines or a reference case for economic evaluation and only 19% use an officially endorsed cost-effectiveness threshold, while 52% have a legislative requirement to consider the results in coverage decisions. Lack of awareness of HTA and weak institutionalisation were the top two barriers to using such

evidence. The mandate traces to World Health Assembly resolution WHA67.23 (2014) on health intervention and technology assessment [PMID 40236923].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 23, Health Planning and Management, pp 1026–1029; Ch 12, Medicine and Social Sciences, p 839); Suryakantha — Community Medicine with Recent Advances (Ch 37, Health Planning and Management, pp 805–807); Reinke — Health Planning for Effective Management (Ch 3, Health Systems Research in Relation to Planning, pp 46–52; Ch 15, Tools for Improving the Process of Planning, pp 222–227); Health Economics (Ch 13, 15, 16, pp 200–255); Sunder Lal — Textbook of Community Medicine (Ch 2, p 88; Ch 6, p 265).

Recent-advances sources (all re-checked against PubMed/PMC full text on 2026-08-18 unless marked otherwise): 38857943 — Understanding the extent of economic evidence usage for informing policy decisions in the context of India's national health insurance scheme: Ayushman Bharat Pradhan Mantri Jan Aarogya Yojana (PM-JAY), *BMJ Global Health* 2024, doi:[10.1136/bmjgh-2024-015079](https://doi.org/10.1136/bmjgh-2024-015079) (PMC11168173); 38590052 — Health impact and cost-effectiveness of expanding routine immunization coverage in India through Intensified Mission Indradhanush, *Health Policy and Planning* 2024, doi:[10.1093/heapol/czae024](https://doi.org/10.1093/heapol/czae024) (PMC11145919); 37963613 — National hospital costing systems matter for universal healthcare: the India PM-JAY experience, *BMJ Global Health* 2023, doi:[10.1136/bmjgh-2023-012987](https://doi.org/10.1136/bmjgh-2023-012987) (PMC10649618); 39260852 — Establishing national hospital costing systems: insights from the qualitative assessment of cost surveillance pilot in Indian hospitals, *BMJ Open* 2024, doi:[10.1136/bmjopen-2023-082965](https://doi.org/10.1136/bmjopen-2023-082965) (PMC11409349); 41501796 — Cost-utility analysis of free HPV immunization for girls in Rajasthan, India, *Cost Effectiveness and Resource Allocation* 2026, doi:[10.1186/s12962-025-00708-7](https://doi.org/10.1186/s12962-025-00708-7) (PMC12870112); 37694178 — Development of the Indian Reference Case for undertaking economic evaluation for health technology assessment, *The Lancet Regional Health — Southeast Asia* 2023, doi:[10.1016/j.lansea.2023.100241](https://doi.org/10.1016/j.lansea.2023.100241) (PMC10485782); 40236923 — The global landscape of country-level health technology assessment processes: A survey among 104 countries, *Health Policy OPEN* 2025, doi:[10.1016/j.hopen.2025.100138](https://doi.org/10.1016/j.hopen.2025.100138) (PMC11999493). Paywalled, grounded on the PubMed abstract only: 38715206 — Organizational Formation for Priority Setting: Historical Perspectives and Thematic Analysis of India's Health Technology Assessment Agency, *Health Systems and Reform* 2024, doi:[10.1080/23288604.2024.2327414](https://doi.org/10.1080/23288604.2024.2327414); 41817502 — Cost-Effectiveness of Triple Versus Dual-Drug Therapy for Elimination of Lymphatic Filariasis in India, *Value in Health Regional Issues* 2026, doi:[10.1016/j.vhri.2026.101604](https://doi.org/10.1016/j.vhri.2026.101604). Also cited: World Health Assembly resolution WHA67.23 (2014); Department of Health Research (HTAIn) for the agency's ministerial placement.

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Modern management techniques — critical path method

Asked in: Paper IV 22-Dec-2023 — RGUHS MD Community Medicine

Definition — Management techniques are methods developed by management scientists, based on the principles of the **behavioural sciences** and on **quantitative methods**, to help an organisation achieve its stated goals more efficiently (Park 28e, Ch 23, p 1028). They were evolved in business, industry and defence; the World Health Organization (WHO) and many governments now emphasise their application to improve the efficiency of health care delivery systems. Management itself consists of four basic activities — **planning** (determining what is to be done), **organizing**, **communicating** and **motivating** people to do the work, and **monitoring or controlling** (checking that work is progressing satisfactorily) (Park 28e, Ch 23, p 1028).

Enumeration of modern management techniques

Park classifies them into two groups (Park 28e, Ch 23, pp 1028–1030).

Group	#	Technique	Essence in one line
A. Methods based on behavioural sciences	1	Organizational design	Organisation must suit its current situation and the needs served; must be reviewed every few years as purpose, problems and technology change.
	2	Personnel management	Skilful use of human resources — selection, training, motivation, division of responsibility, incentives, promotion, effective design of health teams; avoiding "square pegs in round holes".
	3	Communication	Two-way exchange of information within the organisation; vertical (upward and downward) and horizontal channels.
	4	Information systems	Collection, classification, transmission, storage, retrieval, transformation and display of information; provides data for monitoring, evaluation and feedback at all levels.
	5	Management by objectives (MBO)	Objectives are set for each unit and subunit, each preparing its own short-term plan of action.
B. Quantitative methods (derived from economics, operations research and budgeting)	1	Cost-benefit analysis	Benefits expressed in monetary terms and compared with programme cost; limited because health benefits often cannot be monetised.
	2	Cost-effective analysis	Benefit expressed as results achieved (lives saved, disease-free days) instead of money; more applicable to health than cost-benefit analysis.
	3	Cost-accounting	Provides basic data on the cost structure of a programme from financial records.

Group	#	Technique	Essence in one line
	4	Input-output analysis	Relates inputs (manpower, money, materials) to outputs (lives saved, inoculations performed); shows how much input is needed per unit of output.
	5	Model	An abstraction of reality, not reality itself, used to understand how factors in a situation affect one another.
	6	Systems analysis	Finding the most cost-effective alternative among available courses of action for a system (hospital supply system, outpatient clinic, information system).
	7	Network analysis	Graphic plan of all events and activities needed to reach an end objective; two types — PERT and CPM.
	8	Planning–Programming–Budgeting System (PPBS)	Allocates resources to programmes grouped by objective; the related zero-base budget approach starts every budget at zero, justified yearly.
	9	Work sampling	Systematic observation and recording of activities of workers at predetermined or random intervals; used to fix job descriptions, standardise methods and determine manpower needs.
	10	Decision making	"Decisions should be made at the level where the best decisions can be made" — not necessarily at the top, and never on incomplete data.

Operational research groups several of these together — network analysis, PERT, critical path analysis, linear programming, computer simulation, queuing theory, cost-effective analysis and work study (Sunder Lal, Ch 6, p 265).

Critical path method — concept and terminology

A **network** is a graphic plan of all the events and activities to be completed in order to reach an end objective; it brings greater discipline into planning (Park 28e, Ch 23, p 1029). Any project of consequence involves many discrete activities, and failure of even a seemingly insignificant link may produce a bottleneck that jeopardises the whole project (Reinke, Ch 15, p 221). The **critical path method** (CPM) is the earliest and simplest form of network analysis (Reinke, Ch 15, pp 225–226).

Critical path = the longest path through the network. If any activity along the critical path is delayed, the entire project is delayed (Park 28e, Ch 23, p 1029).

Term	Meaning
Activity	A task that consumes time and resources; drawn as a line/arrow between two circles.

Term	Meaning
Event (node)	An instantaneous point marking the beginning or completion of an activity, e.g. "x-ray unit in place", "Pap kits on hand"; drawn as a circle (Reinke, Ch 15, p 222).
Predecessor event	The event that must be completed before an activity can begin (blueprints before installing the x-ray unit).
Path	Any continuous chain of activities from the start node to the terminal node.
Critical path	The path requiring the most time; it fixes the minimum total project time.
Slack (float) time	Difference between the critical-path time and the time of the path of interest (Reinke, Ch 15, p 223).
Late start / late finish	The maximum delay tolerable in starting or completing an activity without affecting total project time (Reinke, Ch 15, pp 224–225).
Gantt chart	Bar-chart representation of the same network against calendar dates; used operationally alongside the network for actual scheduling (Reinke, Ch 15, p 224).

Steps in constructing a critical path network

- Identify the key activities in the project and the events that begin and end each.
- Devise the sequence of activities and arrange them in a flow diagram, respecting all sequencing constraints.
- Assign a duration of time for the performance of each phase of the work.
- Develop the network — list activities, put circles at the beginning and end of each activity (events), and draw lines between the circles (activities); add circles for other events.
- Identify the critical path — begin at the right (project completion), choose the last task that took the most time, connect it to the preceding activity requiring the most time, and continue right to left; then review by reading left to right.
- Perform network time calculations — add the time allotments along the critical path; the sum is the total time required to complete the project.
- Re-plan — try other routes, always ensuring that no activity commences before its precedents are complete (Anganwadi workers cannot be trained unless first recruited and appointed).

(Sunder Lal, Ch 16, p 726)

Worked example — activating an emergency health centre

A crisis requires the construction, staffing and outfitting of an emergency health centre, ideally within 15 days (Reinke, Ch 15, pp 222–224).

Event	Description	Time (days)	Predecessor event
0	Start	—	—
1	Building framework in place	4	0
2	Electrical wiring installed	2	1

Event	Description	Time (days)	Predecessor event
3	Plumbing installed	3	0
4	Cement floor ready for use	2	3
5	Brick outer wall completed	6	1
6	Inside walls plastered and dry	10	1
7	Roof installed	2	1
8	Drug list authorized	4	0
9	Drugs ordered	1	8
10	Drugs received	7	9
11	Personnel authorized	2	0
12	Personnel recruited	3	11
13	Personnel trained	5	12
14	Health centre finished	2	2, 4, 5, 6, 7, 10, 13

Path	Total time (days)	Slack (days)	Interpretation
0–1–6–14	16	0	Critical path ; project needs 16 days against a 15-day target
0–8–9–10–14 (drugs)	14	2	Second-longest; least slack
0–11–12–13–14 (personnel)	12	4	Four days of flexibility

Two managerial lessons from the calculation (Reinke, Ch 15, pp 223–225):

- **Only shortening a critical activity shortens the project.** If plastering (job 6) is cut from 10 to 6 days by using costlier prefabricated material, the former critical path falls by 4 days (to 12), but total project time falls by only 2 days (16 → 14), because the drug path 0–8–9–10–14 then becomes critical. The gain is capped by the slack of the next-longest path.
- **Slack gives flexibility for administrative sequencing.** Because the personnel path has 2 days more slack than the drug path, the project officer may expedite drug authorisation at the cost of a slight delay in personnel authorisation without any project delay.

CPM compared with PERT

Feature	CPM (critical path method)	PERT (Programme Evaluation and Review Technique)
Time estimate per activity	Single, deterministic estimate	Three estimates — most optimistic, best (most likely) and most pessimistic
Computation of activity time	Estimate used as given	Weighted average , the best estimate given four times the weight of either other estimate

Feature	CPM (critical path method)	PERT (Programme Evaluation and Review Technique)
Handling of uncertainty	Fails to recognise possible error in time estimates	Incorporates error; yields probability of completion by the target date and the most probable completion date
Typical output	Critical path, slack, late start/late finish, Gantt schedule	The above plus probabilistic completion estimates
Refinements	Decision CPM	PERT/COST (adds resource cost, finds cheapest way to meet a deadline and links progress to expenditure), PERT/MANPOWER, RAMPS

(Reinke, Ch 15, pp 225–226 and index p 300)

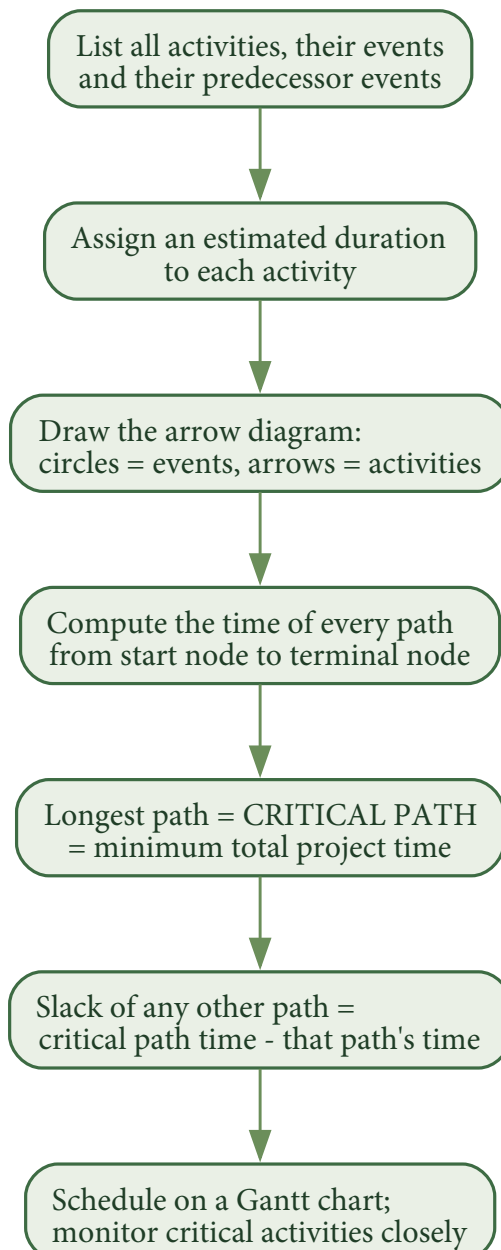
Uses and limitations in health management

Critical path analysis answers	Limitations
What tasks must be carried out	CPM's single time estimate ignores the possibility of error in the estimate
Where parallel activity can be performed	Over-emphasis on time , with almost no attention to cost (corrected by PERT/COST)
The shortest time in which the project can be completed	Requires realistic activity times; garbage in, garbage out
Resources needed to execute the project	Benefits in health projects are not always reducible to schedulable activities
The sequence of activities, scheduling and timings	Needs re-planning whenever assumptions change during implementation
Task priorities and the most efficient way of shortening time on urgent projects	Complex networks need computerised analysis

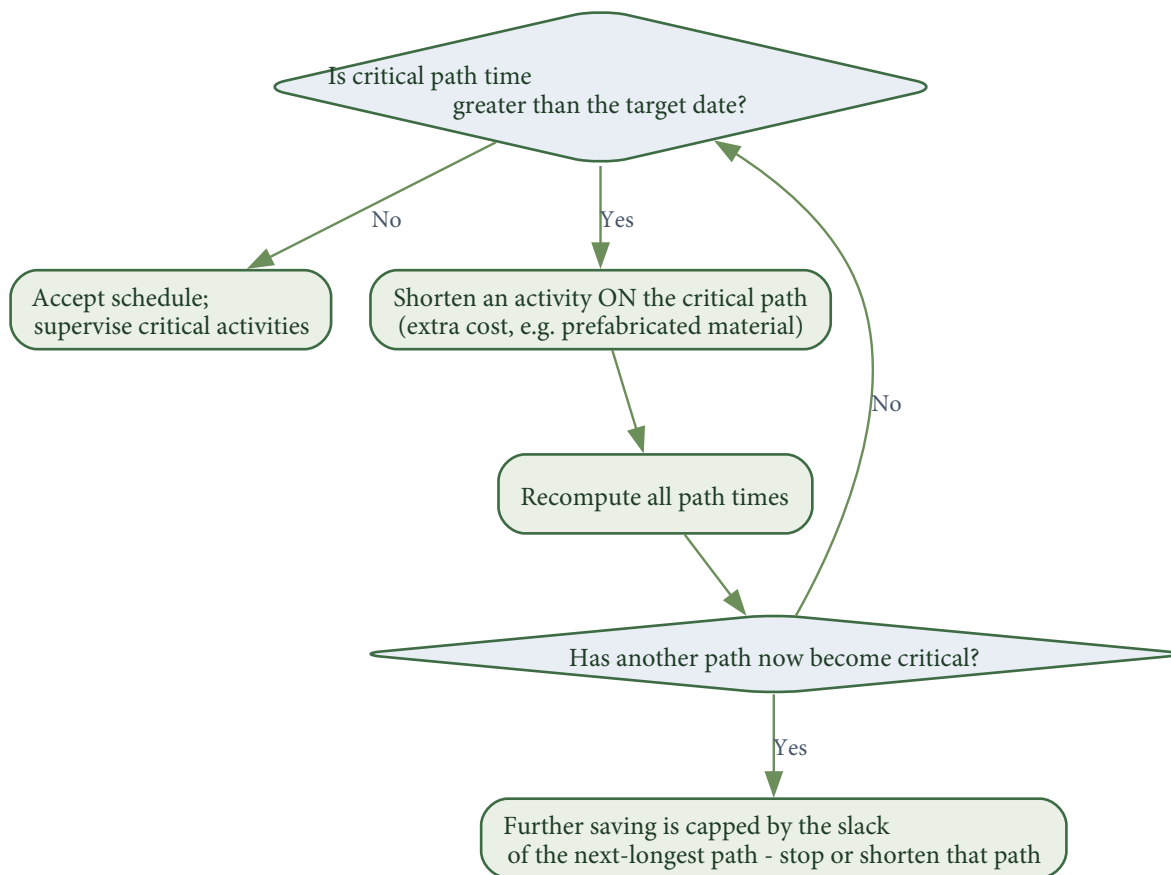
(Sunder Lal, Ch 16, p 726; Reinke, Ch 15, pp 221, 226)

Typical community-medicine applications: commissioning a new primary health centre or district hospital wing, launching a pulse polio or measles–rubella campaign, organising a mass immunisation or screening camp, and setting up a field research project.





Constructing and using a critical path network



Decision algorithm when the critical path exceeds the target date

High-yield exam pointers

- ▶ **Two groups of techniques** — behavioural (organizational design, personnel management, communication, information systems, MBO) and quantitative (cost-benefit, cost-effective, cost-accounting, input-output, model, systems analysis, network analysis, PPBS, work sampling, decision making).
- ▶ **One-line definition to write first:** the critical path is the **longest** path in the network; delay of any activity on it delays the whole project.
- ▶ **Slack time = critical path time – time of the path concerned.** Slack of the critical path itself = zero.
- ▶ **PERT weighting:** expected time is a weighted average with the best (most likely) estimate given **four times** the weight of the optimistic or the pessimistic estimate.
- ▶ **Trap answered correctly:** shortening a critical activity saves project time only up to the slack of the next-longest path — quote the 16 → 14 day example.
- ▶ **Network analysis = PERT + CPM**, item 7 among Park's quantitative methods; PPBS and the zero-base budget are items 8.
- ▶ **Draw the arrow diagram** — circles = events, arrows = activities with durations, bold line = critical path. A CPM answer without the network diagram loses marks.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: in the scanned copy of Reinke's Table 15.1 the duration for event 8 ("drug list authorized") is illegible; the value of 4 days used above is the only figure consistent with the book's own stated path arithmetic (drug path slack 2 days, personnel path slack 4 days, critical path 16 days). Some Indian textbooks (Sunder Lal) treat PERT and CPM as one technique under the single heading "PERT, sometimes called critical path method", whereas Park and Reinke keep them distinct as two types of network analysis — follow Park's separation in the RGUHS answer.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW National Health Mission, Health Technology Assessment in India, National Quality Assurance Standards, WHO

Programme and policy updates (India)

- ▶ **Single Nodal Agency (SNA) fund-flow reform** — the public financial management change now governing all centrally sponsored schemes including the National Health Mission. An assessment in Bihar and Odisha (key-informant work, February 2023 to February 2024) found real gains in cash management, expenditure transparency and alignment of central releases with actual scheme spending, but weak budget credibility, patchy digital connectivity and thin finance staffing threaten equitable access to funds, and the new pressure to demonstrate utilisation has drawn attention away from health outcomes. This is the live example to quote for "monitoring and controlling", the fourth basic management activity. [PMID 42480037]
- ▶ **Blended provider payment proposed for district hospitals** — costing of 27 district hospitals across nine states put the average annual district-hospital budget at ₹326 million (US\$3.35 million; range ₹66 million to ₹2.57 billion), of which inpatient care took ₹221 million, outpatient care ₹68 million and indirect services ₹36 million; inpatient care is the largest single share of the budget. The model pays outpatient care by a risk-adjusted global budget, inpatient care by case-based bundled payment and indirect services by per-bed global budget — the working successor to the textbook's cost-accounting and input-output analysis, replacing historical budgeting with need-based allocation. Upgrading every district hospital to Indian Public Health Standards would raise budgets by 131%, and to medical-college level by 91.5%. [PMID 39478753]
- ▶ **Ayushman Bharat Digital Mission (ABDM)** is now the national health information system that a management answer should name alongside HMIS. Among outpatients at three tertiary hospitals in Agra (September 2024 to April 2025), only 129,007 of 537,278 outpatient visits produced an ABHA ID registration (monthly adoption 23.96%); over 80% found registration easier, yet only 55% were comfortable about data privacy, and the leading barriers were preference for in-person care (85.6%), data-plan cost (81.9%) and difficulty using the apps (65.6%). [PMID 41787376]

Current national data

- ▶ **Comprehensive Annual Modular Survey (CAMS) 2022–23**, 302,086 households: publicly funded health insurance (mainly Ayushman Bharat–PM-JAY) was associated with greater outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (adjusted OR 1.168, 95%

CI 1.128–1.210), and with lower catastrophic health expenditure at the 20% threshold for both outpatient (adjusted OR 0.869, 95% CI 0.810–0.932) and inpatient care (adjusted OR 0.790, 95% CI 0.737–0.846); outpatient out-of-pocket spending remains the unprotected gap. [PMID 42167946]

- **NFHS-4 (2015–16) to NFHS-5 (2019–21)** — institutional births occurring in PHCs and CHCs rose from 30.5% to 34.7%, and birth registration among children delivered there from 80.8% to 90.2%, the largest gain of any delivery setting; such births had 38% higher odds of being registered than private-facility births. Always attach the survey round to any figure quoted. [PMID 41752339]

Elimination, surveillance and disease status

- **100-Day TB Elimination Campaign (7 December 2024 to 17 March 2025)** — a difference-in-differences analysis of Ni-kshay data comparing 53 campaign with 27 non-campaign districts in Maharashtra found notification 10% higher in campaign districts relative to control (IRR 1.10, 95% CI 1.03–1.17, $p = 0.003$); notification actually fell 1% in campaign districts against 10% in non-campaign districts, so the campaign stabilised case-finding rather than raised it. The benefit was confined to rural areas (IRR 1.21, 95% CI 1.11–1.32) and the public sector (IRR 1.17, 95% CI 1.06–1.29), with no effect on urban or private-sector notification. A time-bound national campaign is itself a networked project with a fixed deadline — the natural modern illustration for a critical-path answer. [PMID 42457294]
- **Lymphatic filariasis remains endemic in 345 districts** against the 2030 elimination target; ICMR's vector-control centre argues that public–private partnership, already proven in polio eradication and TB control, is the under-used lever for mass drug administration compliance, post-MDA surveillance, and morbidity management and disability prevention. [PMID 41951149]

Guideline and legislative updates

- **Health Technology Assessment in India (HTAIn) now drives national procurement policy** — a rapid HTA commissioned by the health ministry for the National Sickle Cell Anaemia Elimination Mission (announced in the 2023–24 Union Budget, elimination target 2047, universal screening of the 0–40-year tribal population in six endemic states) found point-of-care tests dearer per case screened — the cheapest of the four cost US\$2.57 and the dearest US\$3.85, against US\$0.67 for the solubility-plus-HPLC standard of care — but detecting 21,125 to 41,458 extra cases. The National Health Mission then directed states to procure by public tender; competitive bidding and domestic manufacture brought kits below US\$1.21 per test, worth about US\$37.86 million in savings at full coverage. Textbook cost-effectiveness analysis and budget-impact analysis in live policy use. [PMID 42594133]
- **National Quality Assurance Standards (NQAS) and NABH accreditation** are the current quality-management instruments layered on top of the classical techniques. In a comparative checklist study at a north-Indian tertiary institute, hospital blocks whose staff had undergone NABH-recommended training showed higher compliance across all seven assessed areas — patient rights, inputs, support services, clinical services, infection control, quality management and outcome — in both intensive care units and wards. [PMID 39670930]

Global and WHO developments

- ▶ **SDG monitoring is limited by missing data, not by method** — a WHO analysis of the United Nations SDG indicators database covering 194 Member States from 2015 to 2024 found 41 of 43 health-related indicators with over 90% missing data points for 2024, against 11 indicators in 2019, and 8 of 27 targets above 90% missing; 49% of indicators rely on modelled estimates rather than empirical data. An information system is only as good as the reporting behind it. [PMID 42058604]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 23, Health Planning and Management, pp 1025–1030 (management methods and techniques; network analysis; PERT and CPM; PPBS; work sampling; decision making). Reinke — Health Planning for Effective Management, Ch 15, Tools for Improving the Process of Planning, pp 221–227 (network construction, critical path, slack time, Gantt charts, PERT, PERT/COST) and index p 300. Sunder Lal — Textbook of Community Medicine, Ch 16, p 726 (steps in developing a network; uses of critical path analysis) and Ch 6, p 265 (operational research techniques). Suryakantha — Community Medicine with Recent Advances, Ch 37, p 806 (Figure 37.3, flow diagram of critical path).

Recent-advances sources: 42480037 — Challenges to Achieving Universal Health Coverage Through Public Financial Management Reforms: Insights from the Implementation of Single Nodal Agency Reform in India, Health Systems & Reform 2026, doi:[10.1080/23288604.2026.2693357](https://doi.org/10.1080/23288604.2026.2693357) (abstract-only); 39478753 — Framework for responsive financing of district hospitals of India, Frontiers in Public Health 2024, doi:[10.3389/fpubh.2024.1398227](https://doi.org/10.3389/fpubh.2024.1398227); 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission: an analytical cross-sectional study among outpatient department attendees, BMC Health Services Research 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), BMJ Open 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, International Journal of Environmental Research and Public Health 2026, doi:[10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 42457294 — Effect of the 100 Days TB Elimination Campaign on TB notification in Maharashtra, Indian Journal of Tuberculosis 2025, doi:[10.1016/j.ijtb.2025.08.014](https://doi.org/10.1016/j.ijtb.2025.08.014) (abstract-only); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, Acta Tropica 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, PLoS One 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 39670930 — Effect of National Accreditation Board for Hospital-Recommended Trainings on Patient Care Standards: A Comparative Study Based on the National Quality Assurance Standards Guidelines in a Tertiary Health-care Institute, Indian Journal of Public Health 2024, doi:[10.4103/ijph.ijph.1262.23](https://doi.org/10.4103/ijph.ijph.1262.23) (abstract-only); 42058604 — Data gaps affecting health-related sustainable development goal indicators, Bulletin of the World Health Organization 2026, doi:[10.2471/BLT.25.294190](https://doi.org/10.2471/BLT.25.294190).

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Programme evaluation in health education

Asked in: Paper II 15-Nov-2021 — RGUHS MD Community Medicine

Definition — Evaluation is the process of assessing the results of a programme to determine whether its stated goals and objectives have been achieved (Sunder Lal, Ch 16, p 724). The WHO Expert Committee on National Health Planning states that evaluation "measures the degree to which objectives and targets are fulfilled and the quality of the results obtained... it measures the productivity of available resources in achieving clearly defined objectives" (Park 28e, Ch 23, p 1027). Monitoring is continuous and is done at the **input** and **process** stages; evaluation is the terminal step of the planning cycle and is done at the **output** and **outcome** (effect and impact) levels (Sunder Lal, Ch 16, pp 723-724).

Types of programme evaluation

A. By what is evaluated — the structure/process/outcome triad (Park 28e, Ch 23, pp 1037-1038)

Type	What is examined	Standard/criterion used
Evaluation of structure	Physical facilities, equipment, manpower and organisation — whether these are accepted by experts as appropriate for the community served	Structural criteria, e.g. physical facilities and equipment against a norm
Evaluation of process	The actual delivery of care — recognition of the problem, diagnostic procedures, treatment and clinical management. Compared against a predetermined standard; the objective and systematic form of this is <i>medical audit</i> (and <i>nursing audit</i>)	Process criteria, e.g. every antenatal mother must receive the prescribed check-ups; every laboratory technician must examine 100 blood smears
Evaluation of outcome	End results — measurable benefits such as improved survival or reduced disability. Traditional outcome components are the 5 Ds of ill-health : disease, discomfort, dissatisfaction, disability and death	Outcome criteria, e.g. cured/death/disability; or behaviour resulting from health care, e.g. cessation of smoking, acceptance of the small-family norm

B. By stage in the programme cycle and purpose (Health Program Planning and Evaluation, Ch 1, pp 48-49; Ch 9, p 250; Ch 10, p 288; Reinke, Ch 4, p 55)

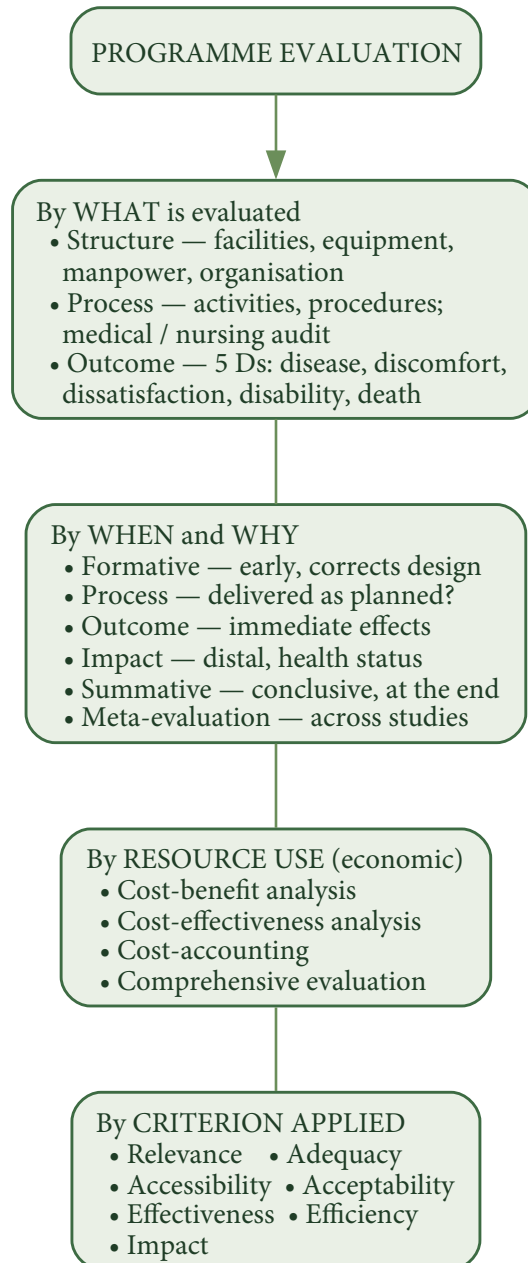
Type	When it is done	Question it answers
Formative evaluation	Early, while the programme is still "forming"	Are there early, correctable faults in the programme's design and delivery?
Process (implementation) evaluation	Throughout implementation	Was the programme delivered as designed — its coverage, intensity and fidelity?
Outcome evaluation	On completion of the intervention	Did the programme produce its immediate, closely linked effects?
Impact evaluation	Long after the intervention	Did it produce the temporally and causally distal effects on health status?
Summative evaluation	At the conclusion of the programme	What is the conclusive statement about the programme's overall effect?
Meta-evaluation	Across several published evaluations	Which programmatic interventions are most likely to succeed?

C. By resource use — economic evaluation (Park 28e, Ch 23, p 1028; Health Program Planning and Evaluation, Ch 1, pp 48-49)

Type	Basis
Cost-benefit analysis	Benefit expressed in monetary terms
Cost-effectiveness analysis	Benefit expressed as results achieved, e.g. lives saved or disease-free days — more applicable to the health field than cost-benefit analysis
Cost-accounting	Provides basic data on the cost structure of a programme
Comprehensive evaluation	Needs-assessment, process, effect and cost data analysed together as one set

D. By the criterion applied — the elements of evaluation (Park 28e, Ch 23, p 1038)

Element	Meaning	Example
Relevance (requisiteness)	Whether the service is needed at all	Vaccination against smallpox is now irrelevant because the disease no longer exists
Adequacy	Whether sufficient attention has been paid to a previously determined course of action	Staff allocated to an activity may not be sufficient
Accessibility	Proportion of the population that can be expected to use a specified facility, given physical (distance, travel, time), economic and social barriers	Distance of the subcentre from the served hamlets
Acceptability	Whether a service that is available and accessible is acceptable to those for whom it is intended	Refusal of a screening service on cultural grounds
Effectiveness	Extent to which the underlying problem is prevented or alleviated; degree of attainment of the predetermined objectives and targets	Ultimate measure is reduction in morbidity and mortality rates
Efficiency	How well resources (money, men, material and time) are used to achieve a given effectiveness	Immunisations given in a year against an accepted norm; bed occupancy percentage, cost per day in hospital, cost per patient treated
Impact	Overall effect of the programme on health status and socio-economic development	With malaria control in India not only did incidence fall, but agricultural, industrial and social life improved



Classification of programme evaluation

Plan to evaluate a proposed community education intervention programme

Take a proposed behaviour-change education intervention in a rural block (say, to raise exclusive breastfeeding, handwashing or contraceptive use). Because it is only *proposed*, evaluation can be designed in from the outset — "successful evaluation may also depend upon whether the means of evaluation were built into the design of the programme before it was implemented" (Park 28e, Ch 23, p 1038). The plan follows the seven general steps of evaluation (Park 28e, Ch 23, pp 1037-1038).

- **State the objectives in measurable terms.** Define what behaviour must change, in whom, by how much and within what period — e.g. "to increase acceptance of condoms from the current 9.5% to 12% within four years" — and set a single over-riding communication objective to guide the strategy (Sunder Lal, Ch 4, pp 152-153).

- **Determine what is to be evaluated.** Decide in advance whether structure, process or outcome is the focus, and at which indicator level; formative and process evaluation will run during delivery, summative outcome and impact evaluation after it.
- **Establish standards and criteria.** Fix structural, process and outcome criteria before the programme starts; without such standards evaluation is not possible (Park 28e, Ch 23, p 1037). Indicators must be **valid, reliable, sensitive, specific, feasible and precise**, and the list must be kept short and specific (Sunder Lal, Ch 16, p 725).
- **Choose the evaluation design** (Sunder Lal, Ch 16, pp 725-726), preferring the strongest that resources allow.

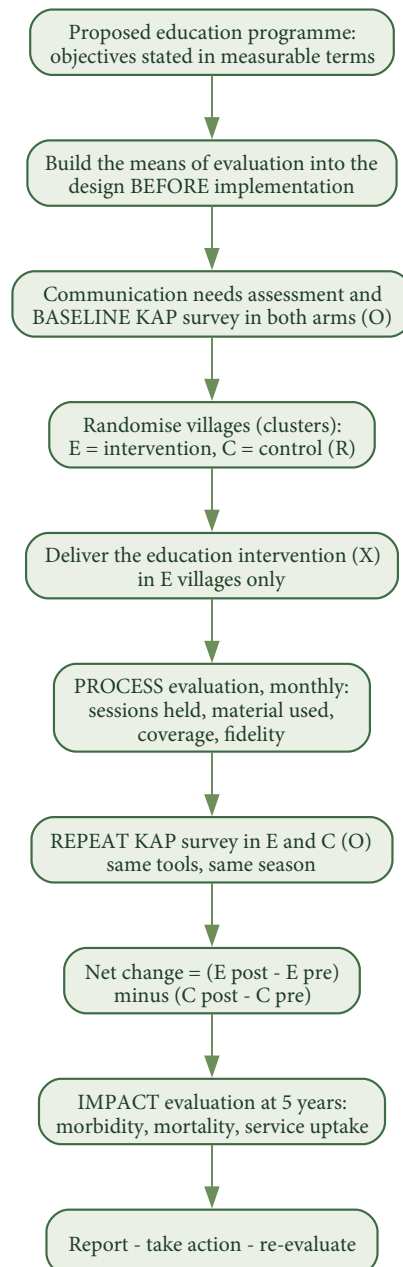
#	Design	Notation	Random allocation	Control group	Baseline data
1	One-time study	E X O	No	No	No — weakest design
2	One group, before-and-after	E O X O	No	No	Yes
3	Time-series study	E O O O X O O O	No	No	Yes, serial
4	Non-equivalent before-and-after with control ("with and without")	E O X O / C O ... O	No	Yes	Yes — quasi-experimental
5	Post-programme observation with control	E R X O / C R ... O	Yes	Yes	No — true experimental
6	Baseline/repeat observation, control group study	E R O X O / C R O ... O	Yes	Yes	Yes — true experimental; design of choice

- **Fix the unit of allocation and the study population.** Education reaches whole villages, so the **village (cluster) is the unit of randomisation** — a community trial, in which the intervention is allocated to communities rather than individuals (ICMR National Ethical Guidelines 2017, section 8.2.6). Match intervention and control villages for size and literacy, keep them geographically separate to prevent contamination, and calculate sample size on the expected change in the key behaviour, inflated for the design effect of clustering.
- **Select the indicators, one row per level of the results chain.**

Level	Indicator (example)	Numerator / denominator	Source
Input	Trained communicators and IEC material available	Workers trained in communication / sanctioned posts	Training and stock registers
Process	Education sessions actually held	Sessions held / sessions planned	Monthly subcentre and PHC reports
Output	Target group contacted; handouts distributed	Persons contacted / persons eligible	Clinic and outreach reports

Level	Indicator (example)	Numerator / denominator	Source
Effect (behaviour)	Change in knowledge, attitude and practice	Persons practising the behaviour / target group surveyed	Repeat KAP survey, direct observation
Coverage	Proportion of the target group receiving the service correctly and completely	Fully immunised children, mothers with full antenatal care / eligible population	Household survey
Impact	Decline in incidence of diarrhoea and acute respiratory infection; fall in birth rate, infant mortality rate, underweight prevalence	As defined for each rate	Sample Registration System, National Family Health Survey, District Level Household Survey

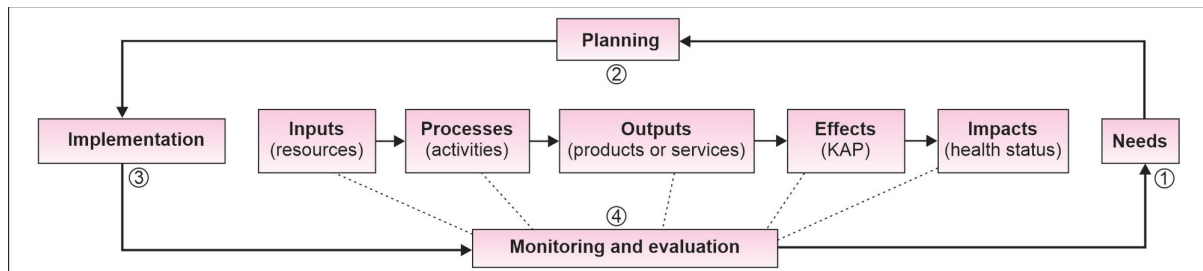
- **Plan the methodology and tools.** Use a pre-tested KAP schedule administered identically at baseline and repeat survey. Knowledge is measured by standard questions on the specific problem; attitude by a **Likert scale** (3-, 5- or 7-point) recording degree of agreement with a statement; practice by **direct observation** — handwashing, cord care, food handling — not by self-report (Sunder Lal, Ch 4, pp 153-154). Supplement with exit interviews, focus group discussions, workers' diaries and routine service records.
- **Fix the timetable.** Baseline survey immediately before the intervention in both arms; process monitoring monthly; effect (KAP) survey after a defined interval; progress and efficiency assessed at least annually, and impact not earlier than about five years from the start of the programme (Sunder Lal, Ch 16, p 725).
- **Gather and analyse the information.** Analyse and feed back the results in the shortest feasible time and provide an opportunity to discuss them (Park 28e, Ch 23, p 1037). Estimate the net effect attributable to the programme as: **net change = (mean post-intervention score in participants – mean pre-intervention score in participants) – (mean post score in non-participants – mean pre score in non-participants)** (Health Program Planning and Evaluation, Ch 14, p 368). Test the difference in behaviour-change proportions between arms, adjusting for clustering.
- **Satisfy the ethical requirements.** Obtain institutional ethics committee approval; take gatekeeper/community permission from the panchayat or community leaders followed by individual informed consent, since community traditions do not substitute for individual consent; and share the findings in aggregated form with the community (ICMR National Ethical Guidelines 2017, pp 110, 123, 126).
- **Take action and re-evaluate.** Use the findings to strengthen, modify or discontinue the intervention and to revise priorities and objectives; evaluation is an on-going process aimed at making health activities more relevant, more efficient and more effective (Park 28e, Ch 23, p 1037).



Plan for evaluating a proposed community education intervention programme

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce by hand — Needs (1) → Planning (2) → Implementation (3), then the results chain Inputs (resources) → Processes (activities) → Outputs (products or services) → Effects (knowledge, attitude, practice) → Impacts (health status), with Monitoring and evaluation (4) as a box below, receiving dotted feedback lines from every stage of the chain and looping back to Needs.



Place of monitoring and evaluation in the planning cycle (Sunder Lal — Textbook of Community Medicine, Ch 16, p 722)

High-yield exam pointers

- ▶ Three types by what is evaluated: **structure, process, outcome** — outcome components are the 5 Ds: disease, discomfort, dissatisfaction, disability, death.
- ▶ Seven **elements of evaluation** — Relevance, Adequacy, Accessibility, Acceptability, Effectiveness, Efficiency, Impact (mnemonic RA-AAEEI).
- ▶ Seven **steps of evaluation** — determine what is to be evaluated; establish standards and criteria; plan the methodology; gather information; analyse the results; take action; re-evaluate.
- ▶ **Monitoring = inputs and processes, continuous; evaluation = outputs and outcomes (effects and impacts), terminal.** Impact evaluation needs at least five years.
- ▶ Design of choice for an education intervention: **E R O X O / C R O ... O** — randomised, controlled, with baseline; village as the unit of randomisation.
- ▶ Net programme effect = (E post – E pre) – (C post – C pre).
- ▶ Indicator qualities: valid, reliable, sensitive, specific, feasible, precise (**KISS** — keep it short and specific). **Coverage** = proportion of the target group receiving the service correctly and completely.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: much of the published literature uses *outcome evaluation* and *impact evaluation* interchangeably; in the stricter usage followed above, outcome means the programme's immediate effects and impact its more distal effects on health status.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 23, Health Planning and Management); Sunder Lal — Textbook of Community Medicine (Ch 16, Health Planning and Management; Ch 4, Principles of Health Promotion and Education); Health Program Planning and Evaluation: A Practical, Systematic Approach (Ch 1, 9, 10, 12, 14); Reinke — Health Planning for Effective Management (Ch 4, Health Services Program Evaluation); ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (Public Health Research, sections 8.1 and 8.2.6).

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Health and implementation planning — planning cycle

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — Health planning is part of national development planning and has been defined as "the orderly process of defining community health problems, identifying unmet needs and surveying the resources to meet them, establishing priority goals that are realistic and feasible, and projecting administrative action to accomplish the purpose of the proposed programme" (Park 28e, Ch 23, pp 1025-1026). **Implementation planning** is the operational half of the same exercise: it converts the approved plan into a dated matrix showing *who* will do *what* activity, at *what* time and at *which* place, with what inputs — the framework against which monitoring is then done (Sunder Lal, Ch 16, p 675). Planning in its broadest sense therefore has three parts — plan formulation, execution and evaluation (Park 28e, Ch 23, p 1025).

Health planning — purpose, terminology and pre-requisites

Purpose of planning (Park 28e, Ch 23, p 1025): (1) to match limited resources with many problems; (2) to eliminate wasteful or duplicated expenditure; and (3) to develop the best course of action to accomplish a defined objective. Planning is a team exercise — the team includes not only health specialists but economists, statisticians, sociologists and managers.

Term	Meaning (Park 28e, Ch 23, pp 1025-1026)
Health need	A deficiency in health that calls for preventive, curative, control or eradication measures; needs as perceived by people differ from needs as seen by experts
Health demand	A need that people themselves present as a claim on the services
Resources	Manpower, money, materials, skills, knowledge, techniques and time available for, or needed by, a specified objective
Objective	A precise planned end-point — it is either achieved or not achieved
Target	A discrete activity (e.g. number of blood films collected, vasectomies done); it permits <i>degree</i> of achievement. Targets concern the factors involved in a problem, objectives concern the problem itself
Goal	The ultimate desired state towards which objectives and resources are directed; not constrained by time or existing resources and not necessarily attainable
Plan	A blue-print for action, with five elements — objectives, policies, programmes, schedules and budget
Programme	A sequence of activities designed to implement policies and accomplish objectives; a step-by-step approach to reach a predetermined goal
Schedule	The time sequence for the work to be done
Policies	Guiding principles stated as an expectation, not as a commandment
Procedures	A set of rules for carrying out work which, when observed by all, ensure maximum use of resources and effort

Pre-planning is the preparation for planning; four preconditions must exist before a health plan can be written (Park 28e, Ch 23, p 1026):

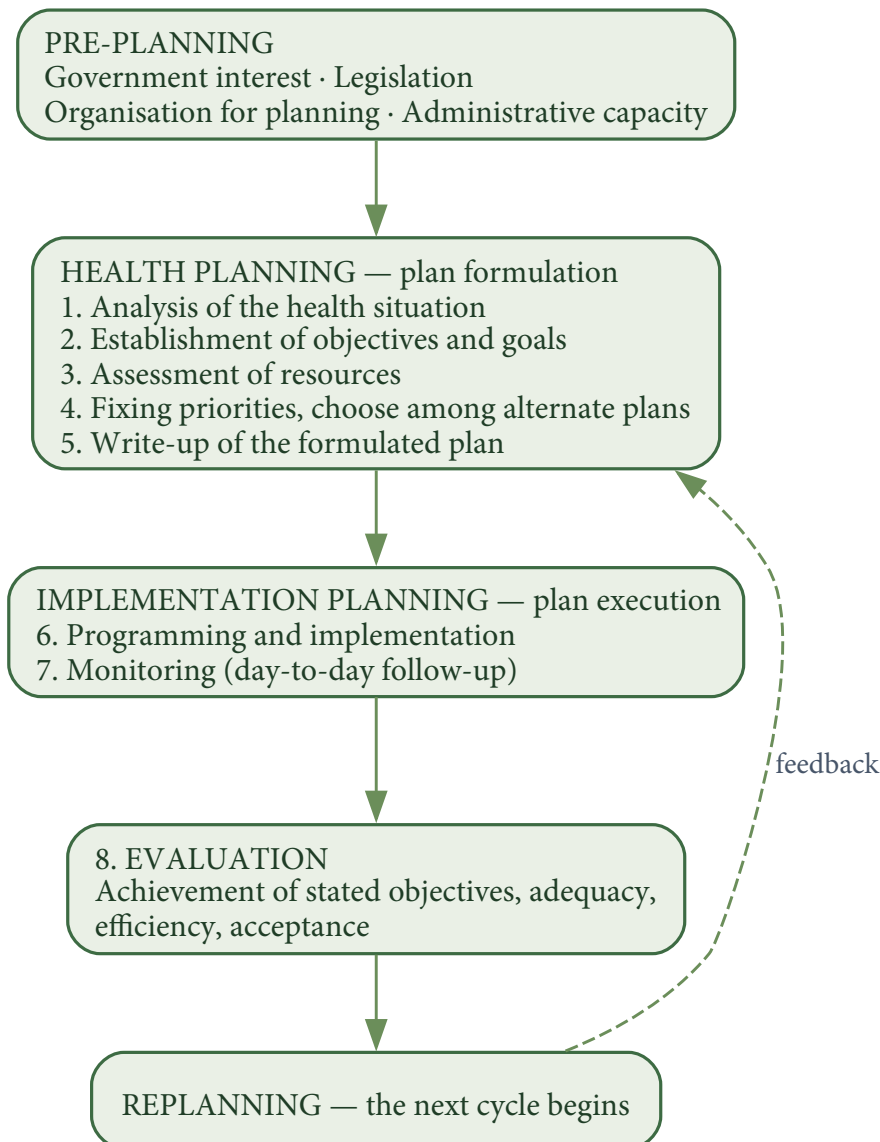
Precondition	What it means	Indian illustration
Government interest	A strong political will shown by clear directives or policies from the political authority	National Health Policy 2017 (Park 28e, Ch 23, pp 1029-1030)
Legislation	Social and health policies translated into law	Medical Termination of Pregnancy Act, passed by Parliament in 1971
Organisation for planning	A structure of full-time planners, advised by technical experts and political leaders, to prepare the parts of the plan	Planning Commission (set up 1950), replaced by NITI Aayog (National Institution for Transforming India) on 1 January 2015; Bureau of Planning in the Ministry of Health since 1965, which compiles the health Five Year Plans
Administrative capacity	Capacity to coordinate activities and implement the plan at all levels	Central and State Ministries of Health; State and District Health Societies

Health planning versus implementation planning

Feature	Health planning (plan formulation)	Implementation planning (plan execution)
Question answered	What is the problem, what shall we achieve, with what resources and at what priority?	Who does what activity, when, where, and with which inputs?
Steps of the cycle covered	Steps 1-5	Steps 6-7
Output document	A plan — objectives, policies, programmes, schedules, budget	An implementation/action matrix plus a fixed work schedule
Time horizon	Perspective and medium-term (5-15 years)	Annual, monthly, weekly, day-wise
Predominant level	Centre and State (macro)	District, block, primary health centre, sub-centre (micro)
Accountability	A planning body or committee	A named worker for each activity
Indian example	National Health Policy 2017 goals — life expectancy at birth 67.5 to 70 years, under-five mortality rate to 23, maternal mortality ratio to 100, infant mortality rate to 28, total fertility rate to 2.1, government health spending 1.15% to 2.5% of gross domestic product, all by the stated year (Park 28e, Ch 23, pp 1029-1030)	District Health Action Plan and State Programme Implementation Plan under the National Health Mission; the sub-centre annual action plan and the monthly work plan of the female health worker (Sunder Lal, Ch 16, pp 674-675)
Typical failure	Wrong priorities; goals left as vague platitudes	Delays in critical supplies, inappropriate use of staff — "it is at the implementation stage that shortcomings often appear in practice" (Park 28e, Ch 23, p 1027)

Reinke stresses that implementation must be treated as an integral part of planning, not handed over to service organisations afterwards: "no error in planning is more common or more serious" than that

narrow view, and the plan itself must incorporate detailed planning for implementation (Reinke, Ch 1, pp 16-17).



The two limbs of the planning cycle — plan formulation (health planning) and plan execution (implementation planning)

The planning cycle — eight steps (Park 28e, Ch 23, pp 1026-1028)

Planning is a process of analysing a system or defining a problem, assessing the extent to which the problem exists as a need, formulating goals and objectives, examining and choosing among alternative intervention strategies, initiating action for implementation, monitoring the system, and evaluating results in the light of stated objectives. It is a cycle, not a staircase — evaluation feeds straight back into replanning.

- **Analysis of the health situation** — collection, assessment and interpretation of information to give a clear picture of the health situation. The minimum essential data are: (a) the population with its age and sex structure; (b) statistics of morbidity and mortality; (c) the epidemiology and geographical distribution of different diseases; (d) medical care facilities such as hospitals and health centres, both public and private; (e) technical manpower of the various categories; (f) training facilities available; and (g) attitudes and beliefs of the population towards disease, its

cure and prevention. Their analysis yields the health problems, health needs and health demands of the population.

- ▶ **Establishment of objectives and goals** — without them there is haphazard activity, uneconomical use of funds and poor performance. Objectives are set at every level down to the smallest organisational unit; general at upper levels, progressively more specific and detailed at lower levels, and either short-term or long-term. They are both a guide to action and a yard-stick to measure work after it is done. Cost-benefit analysis and input-output study of health services are used to define them in definite terms.
- ▶ **Assessment of resources** — manpower, money, materials, skills, knowledge and techniques needed or available for implementing the programmes are assessed, and a balance is struck between what is required and what is (or is likely to be) available.
- ▶ **Fixing priorities** — because resources always fall short of the total requirement, problems are ranked by importance or magnitude. Priority setting weighs financial constraints, mortality and morbidity data, diseases preventable at low cost, saving the lives of younger people in whom there has been considerable social investment, and political and community interests and pressures. **Alternate plans** for achieving the priorities are then formulated and tested for practicability and feasibility, and the more effective alternative is chosen.
- ▶ **Write-up of the formulated plan** — preparation of the detailed plan document, complete in all respects for execution. For each proposed programme the resources (inputs) required are related to the results (outputs) expected; each stage is defined and costed, the time needed to implement is specified, working guidance is given to all those responsible for execution, and a "built-in" system of evaluation is included. Modification of the plan and reallocation of resources rest with the central planning authority and the government.
- ▶ **Programming and implementation** — begun once the plan is approved by the policy-making authorities. Execution depends on an effective organisation with well-defined procedures, sufficient delegation of authority and fixation of responsibility on named workers for the prescribed period. The main considerations are (a) definition of roles and tasks, (b) selection, training, motivation and supervision of the manpower involved, (c) organisation and communication, and (d) the efficiency of individual institutions such as hospitals or health centres.
- ▶ **Monitoring** — the day-to-day follow-up of activities during implementation to ensure they are proceeding as planned and on schedule. It is a continuous process of observing, recording and reporting on the activities of the organisation or project — keeping track of the course of activities, identifying deviations, and taking corrective action if deviations are excessive.
- ▶ **Evaluation** — assesses the achievement of the stated objectives, the adequacy, efficiency and acceptance of the programme by all parties involved. Monitoring is confined to day-to-day or ongoing operations; evaluation is mostly concerned with the final outcome and with the factors associated with it. The WHO Expert Committee on National Health Planning in Developing Countries states that evaluation "measures the degree to which objectives and targets are fulfilled and the quality of the results obtained... the productivity of available resources in achieving clearly defined objectives... how much output or cost-effectiveness is achieved. It makes possible the reallocation of priorities and of resources on the basis of changing health needs."

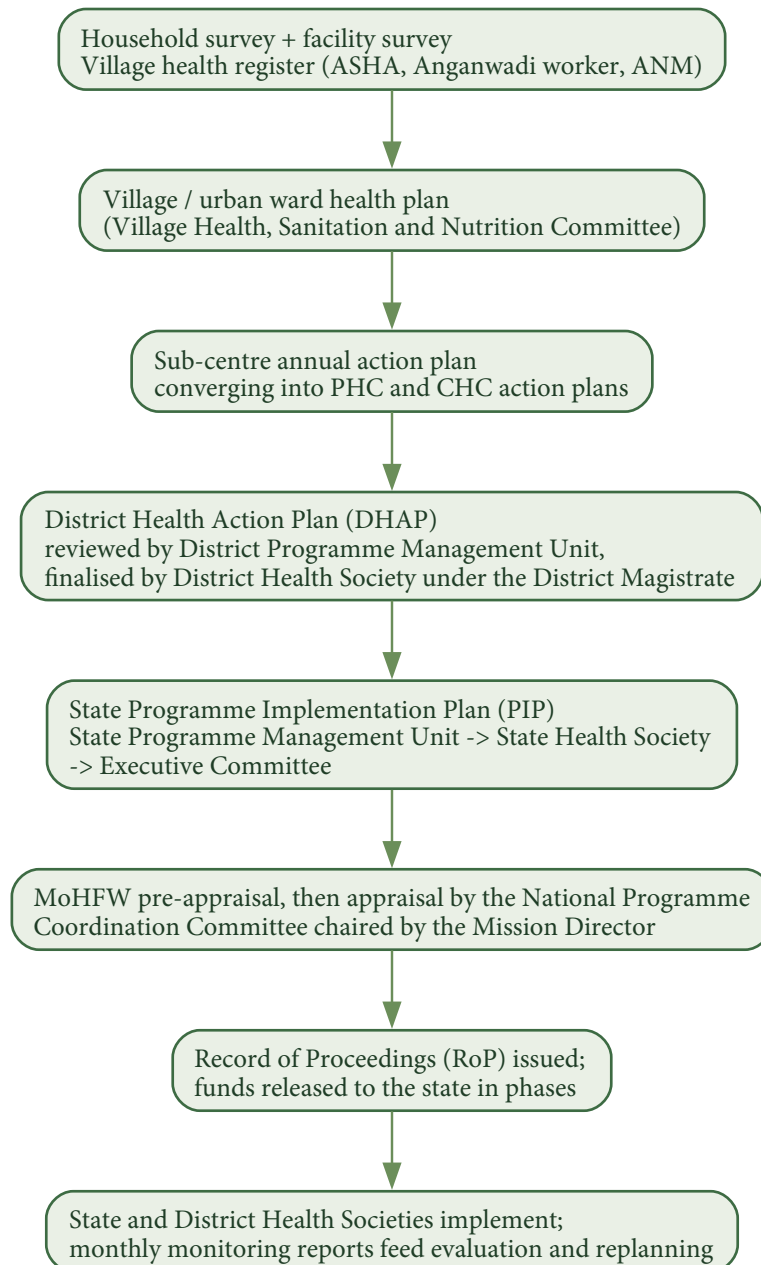
Worked Indian example — the cycle applied at district and sub-centre level

Step	Concrete action in an Indian district (National Health Mission)
1. Situation analysis	Annual household survey by accredited social health activists (ASHAs), <i>Anganwadi</i> workers and auxiliary nurse midwives (ANMs) enumerating households, eligible couples, antenatal and postnatal women, under-fives, those above 30 years and above 60 years, plus drinking-water source, latrine use and health-seeking behaviour; facility survey of sub-centre, primary health centre and community health centre mapping human resources, drugs, diagnostics and space (Sunder Lal, Ch 16, pp 670-671)
2. Objectives and goals	Adopt National Health Policy 2017 targets locally — antenatal care coverage above 90%, skilled attendance at birth above 90%, fully immunised infants above 90%, 80% of known hypertensives and diabetics maintaining controlled disease status (Park 28e, Ch 23, p 1030)
3. Assessment of resources	Compare available inputs against the public health standards to determine gaps in manpower, material and money; count untied funds; treat time and health information as resources too (Sunder Lal, Ch 16, pp 671-674)
4. Fixing priorities	Rank the felt needs identified from the village health register and community leaders' meetings, weighted by local mortality and morbidity
5. Write-up of plan	Village/urban ward health plan → sub-centre annual action plan (in matrix form, with annual targets per service package and an indent for drugs, contraceptives, equipment and manpower) → PHC and CHC action plans → District Health Action Plan (DHAP) (Sunder Lal, Ch 16, p 674; Ch 17, p 739)
6. Programming and implementation	DHAPs consolidate into the State Programme Implementation Plan (PIP) ; after approval a Record of Proceedings (RoP) releases funds and the State and District Health Societies execute the plan; at field level a fixed day-wise, village-wise work schedule is followed (Routine Immunization Manual for Medical Officers 2024, Ch 16, p 332; Sunder Lal, Ch 16, p 675)
7. Monitoring	Monthly monitoring reports on the revised format; deviations are fed back to workers for corrective action; supervisors do spot supervision and on-the-job training (Sunder Lal, Ch 16, p 675)
8. Evaluation and replanning	Indicators such as outreach sessions planned versus held, percentage of antenatal mothers given four check-ups, percentage of infants fully immunised, percentage of those above 30 years screened for non-communicable diseases; gaps found go back into the next cycle (Sunder Lal, Ch 16, pp 675-676)

Implementation matrix — the standard format of an implementation plan (adapted from Sunder Lal, Ch 16, p 675, Table 16.6):

Activity	Inputs / resources required	Person responsible	Supporting staff	Time (from-to)
Household survey	Training, household register	ASHA and <i>Anganwadi</i> worker	ANM	February to March every year

Activity	Inputs / resources required	Person responsible	Supporting staff	Time (from-to)
Antenatal care	Integrated RCH register, weighing scale, pregnancy test kit, blood pressure apparatus, iron-folic acid tablets, haemoglobinometer, tetanus toxoid, examination table	ANM	ASHA, <i>Anganwadi</i> worker	One outreach session per <i>Anganwadi</i> per month, Tuesday and Thursday at sub-centre
Screening for non-communicable diseases	Registers, glucometer, blood pressure apparatus	ANM	ASHA, <i>Anganwadi</i> worker	Every Monday at village level
Immunisation	Vaccines, vitamin A, vaccine carrier, syringes and needles, emergency kit, mother-child protection card, registers, reporting format	ANM	ASHA, <i>Anganwadi</i> worker	Outreach session every Wednesday at <i>Anganwadi</i>



Bottom-up implementation planning in India — from the village health register to the Record of Proceedings under the National Health Mission

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- The planning cycle (Park's Fig. 1) — draw as a closed loop with two limbs. **Planning limb (right side, top to bottom):** Gather health data → Tabulate, analyze, interpret → Formulate health problems → Assign priorities among problems → Formulate individual priority problems for planning solutions → Define programme objectives and how to measure achievement → Assign priorities among objectives → Design alternative programme to solve problems → Select best programme (feasibility, cost/effectiveness) → Combine programme and develop integrated plan. **Implementation limb (left side, bottom to top):** Plan implementation of integrated programme → Orient and train personnel → Initiate integrated programme → Operate programme → Collect data for evaluation → Evaluate results: goals achieved? goals satisfactory? → YES returns to Operate programme, NO returns to Gather health data.

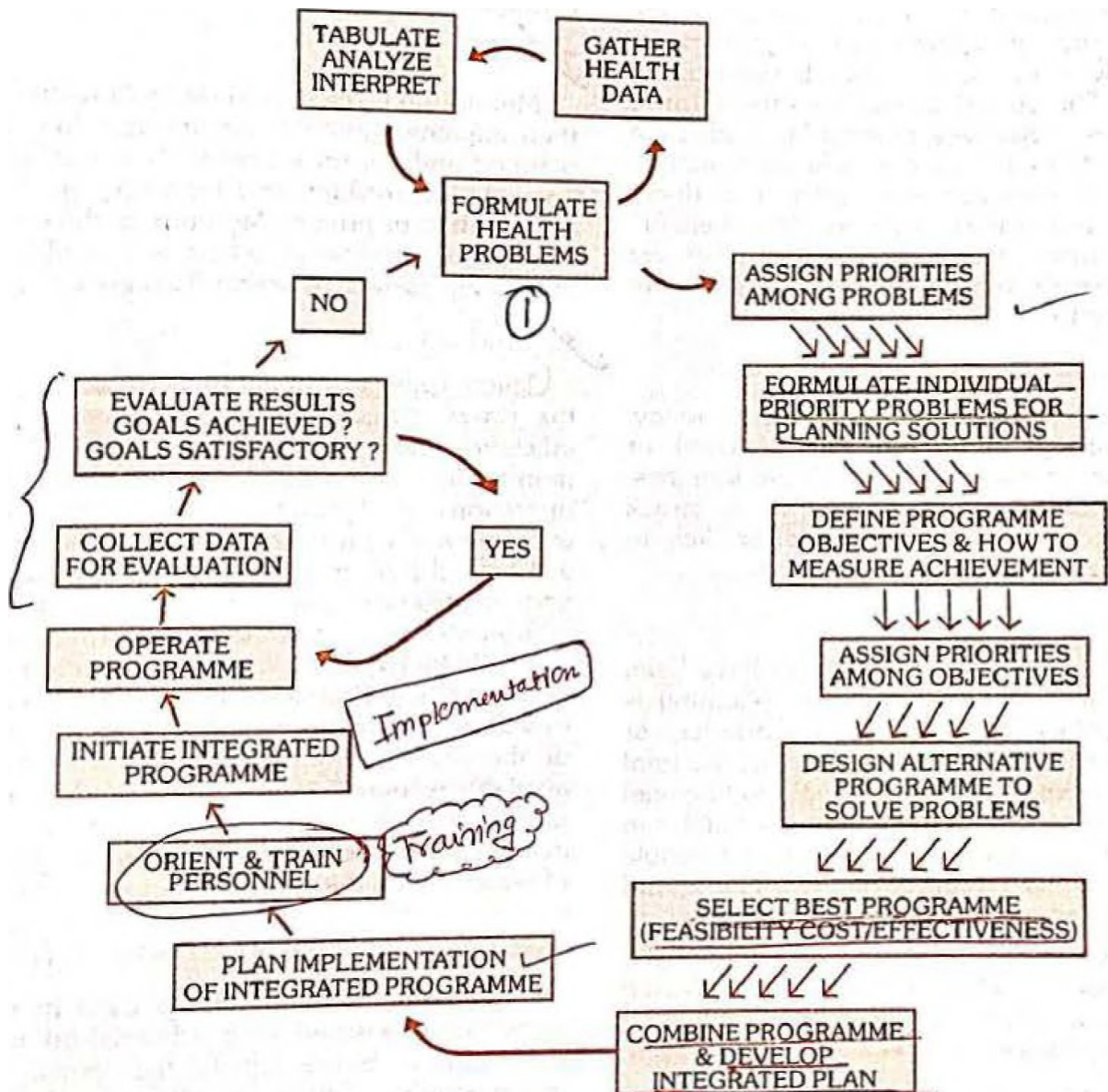


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, p.1026)

High-yield exam pointers

- ▶ Planning in its broadest sense = **plan formulation + execution + evaluation**; "planning is for tomorrow, management is for today" (Park 28e, Ch 23, p 1025).
- ▶ Mnemonic for the eight steps — **A-O-R-P-W-P-M-E**: Analysis of health situation, Objectives and goals, Resources, Priorities, Write-up, Programming and implementation, Monitoring, Evaluation.
- ▶ The five elements of a **Plan**: objectives, policies, programmes, schedules, budget.
- ▶ **Objective** = achieved or not achieved; **target** = permits degree of achievement; **goal** = ultimate desired state, not bound by time or resources.
- ▶ Four pre-planning preconditions: government interest, legislation, organisation for planning, administrative capacity. MTP Act 1971 is Park's own example of the legislation precondition.
- ▶ **Monitoring** = day-to-day follow-up during implementation; **evaluation** = final outcome and its determinants — the commonest one-mark distinction asked.
- ▶ Implementation plan answers **who, what, when, where, with what** — Planning Commission (1950) was replaced by NITI Aayog on **1 January 2015**; state PIPs are approved through the **Record of Proceedings**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Five Year Plan mechanism ended with the Twelfth Five Year Plan (2012-2017) and central planning now runs through NITI Aayog's vision and strategy documents rather than a Planning Commission Five Year Plan, so quote the Planning Commission in the historical past tense; Park 28e gives government health expenditure as 1.15% of gross domestic product rising to 2.5% by 2025 while Sunder Lal gives the baseline as 1.35%, and the Park figure is used above as this bank's spine.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Mission, National Health Accounts, Comprehensive Annual Modular Survey 2022-23, National Family Health Survey, National Action Plan on Antimicrobial Resistance, WHO

Programme and policy updates (India)

- ▶ Ayushman Bharat Health and Wellness Centres have been **renamed Ayushman Arogya Mandir** (tagline *Arogyam Parmam Dhanam*), with the mandate restated as preventive, promotive, curative, rehabilitative and palliative care aligned to Sustainable Development Goal 3 — so name the sub-centre-level and PHC-level facility by its current name, not "HWC". A Jammu and Kashmir survey of 320 *enrolled* PM-JAY beneficiaries found only 48.4% had actually used the scheme, which is Park's point about shortcomings appearing at the implementation stage, quantified. [PMID 41878109]
- ▶ **Fund flow under the National Health Mission has been re-engineered by the Single Nodal Agency reform**, which routes centrally sponsored scheme money through a single state account to improve budget execution. Key-informant work in Bihar and Odisha (February 2023 to February 2024) found gains in cash management, expenditure transparency and alignment of

central releases with actual spending, but flagged weak budget credibility, patchy digital connectivity and thin financial-management staffing, plus a drift of attention towards fund utilisation rather than health outcomes. This sits squarely on step 6, immediately after the Record of Proceedings. [PMID 42480037]

- **PM-ABHIM (Pradhan Mantri Ayushman Bharat Health Infrastructure Mission)** is establishing four zonal National Institutes of Virology. Zonal prioritisation workshops in March 2024 — 186 participants drawn from state health departments, the Integrated Disease Surveillance Programme, the Virus Research and Diagnostic Laboratory network and ICMR — ranked 40 viral diseases using a modified One Health Zoonotic Disease Prioritisation tool applied through multicriteria decision analysis. A citable modern example of step 4 done formally instead of by committee opinion. [PMID 42522023]

Current national data

- **The National Health Accounts trajectory supersedes the 1.15%-of-GDP figure carried in the textbook:** the government share of total health expenditure has risen from 28.6% to 48% and the out-of-pocket share has fallen from 64.2% to 39.4%, yet government health expenditure stands at roughly 1.9% of gross domestic product against the 2.5% target India set for itself. Outpatient care and medicines absorb about two-thirds of out-of-pocket spending and lie outside the flagship hospitalisation scheme. [PMID 42548603]
- **Comprehensive Annual Modular Survey 2022-23 (NSS 79th round), 302,086 households** — publicly funded health insurance was associated with greater outpatient (adjusted OR 1.404, 95% CI 1.328-1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128-1.210) and with less catastrophic expenditure at the 20% threshold (inpatient adjusted OR 0.790, 95% CI 0.737-0.846), while outpatient care remained poorly protected. Always attach the survey round to the figure. [PMID 42167946]
- **Five NFHS rounds (1993-2021)** show early initiation of breastfeeding rising from 9.6% to 41.2% and exclusive breastfeeding from 43.5% to 63.9%; against the 2030 Global Nutrition Target of 60% exclusive breastfeeding, 23 states and union territories have already achieved it, five are on-target and six are projected to miss it. The "Required Annualized Change" used to classify them is a clean contemporary illustration of step 2 — turning a goal into a measurable, time-bound target. [PMID 42134478]

Elimination, surveillance and disease status

- **National Sickle Cell Anaemia Elimination Mission** — a rapid health technology assessment for six endemic states costed point-of-care screening at US\$2.38-3.85 per case screened against US\$0.67 for the existing solubility-plus-HPLC pathway, but showed 21,125-41,458 additional cases detected. The National Health Mission then directed states to procure point-of-care kits publicly; competitive tendering and domestic manufacture brought the price below US\$1.21 per test, with an estimated US\$37.86 million saving at full coverage. Steps 3, 4 and 5 — resource assessment, priority setting and a costed write-up — in a single programme. [PMID 42594133]
- **Programme money tracks programme outcomes** — a decade of National Tuberculosis Programme Programme Implementation Plans and Records of Proceedings for Maharashtra (2011-2020) showed fund utilisation positively correlated with case notification rate (p 0.663, p 0.037) and treatment success rate (p 0.906, p 0.001), and negatively with case fatality rate (p

-0.855, p 0.002); median utilisation was highest for supervision and monitoring (85.53%) and public-private mix (85.25%). Good evidence that the PIP-to-RoP machinery described above is not merely administrative. [PMID 42457287]

Guideline and legislative updates

- **The National Action Plan on Antimicrobial Resistance has been cascaded into State Action Plans** in six states — the AMR analogue of the state Programme Implementation Plan. A 2025 review of the Indian response found policy and surveillance advances offset by enforcement gaps, over-the-counter antibiotic availability, pharmaceutical and hospital wastewater contamination and fragmented data, and argued for One Health surveillance embedded within universal health coverage rather than run as a vertical programme. [PMID 41443506]

Global and WHO developments

- **Global health law now runs on a treaty layer, not the International Health Regulations alone** — successive IHR revisions, the Framework Convention on Tobacco Control and the post-COVID Pandemic Agreement sit alongside one another, while WHO's authority is being tested by contested multilateralism, including the United States' withdrawal. Name the Pandemic Agreement as the newest instrument. [PMID 42312797]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e, Chapter 23 (Health Planning and Management), pp 1025-1031; Sunder Lal — Textbook of Community Medicine, Chapter 16 (Health Planning and Management), pp 666-676 and Chapter 17, p 739; Reinke — Health Planning for Effective Management, Chapter 1 (The Process, Structure, and Functions of Planning), pp 10-18; Routine Immunization I Manual for Medical Officers 2024, Chapter 16 (Financial provisions for Universal Immunization Programme), p 332; National Health Policy 2017 goals as reproduced in Park 28e, Ch 23, pp 1029-1030.

Recent-advances sources: 41878109 — Beyond coverage: why Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana struggles to deliver healthcare in Jammu and Kashmir, *Frontiers in Public Health* 2026, [DOI](#); 42480037 — Challenges to Achieving Universal Health Coverage Through Public Financial Management Reforms: Insights from the Implementation of Single Nodal Agency Reform in India, *Health Systems & Reform* 2026, [DOI](#) (abstract-only); 42522023 — Strengthening India's pandemic preparedness with the four zonal institutes of virology, *Infectious Diseases of Poverty* 2026, [DOI](#); 42548603 — Can a bigger budget shrink the bill at the bedside? Public health financing as a route to lower out-of-pocket spending in India, *Frontiers in Public Health* 2026, [DOI](#); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), *BMJ Open* 2026, [DOI](#); 42134478 — Trends in breastfeeding indicators across Indian states and Union Territories (1993-2021): Estimating progress towards the 2030 Global Nutrition Target, *Preventive Medicine* 2026, [DOI](#) (abstract-only); 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, *PLoS One* 2026, [DOI](#); 42457287 — Decadal trends (2011-2020) in financial allocation and utilization, and their relationship with outcome indicators under the National Tuberculosis Programme in Maharashtra, *Indian Journal of Tuberculosis* 2025, [DOI](#) (abstract-only); 41443506 — Antimicrobial resistance in India: Integrating the response into health systems for universal health coverage, *Journal of Global Antimicrobial Resistance* 2025, [DOI](#) (abstract-only); 42312797 — Eighty years of postwar multilateralism: Global health law's emergence and its contemporary challenges, *Global Public Health* 2026, [DOI](#) (abstract-only).



Information technology in community health care

Asked in: Paper IV 13-May-2022 — RGUHS MD Community Medicine

Definition — Information technology (IT) in health care is the application of computers, telecommunication networks and digital software to the collection, classification, transmission, storage, retrieval, transformation and display of health information (Park 28e, Ch 23, p 1028). Applied to community health care it converts the paper registers of the peripheral worker into a networked system that supports service delivery, disease surveillance, programme management and evaluation. Park notes that the world has moved from the post-industrial age into an information age, with two interconnected revolutions in information and in communication that give instant access to medical information once difficult to retrieve (Park 28e, Ch 2, p 32).

Why information technology is needed in community health care

- ▶ **Data are generated but not used.** Health workers maintain a large number of registers, yet the data generated are rarely the basis of management decisions; the flow of data is primarily unidirectional and much of what is collected is redundant or never adequately utilised (Jugal Kishore, National Health Programmes of India, pp 262, 264).
- ▶ **Manual systems duplicate work.** Efficient management of data is difficult in a manual system and often involves duplication of effort and wastage of time; a computerised management information system is one of the ways IT can improve the health system, aiding the worker in service provision, data collection, storage, analysis and dissemination (Jugal Kishore, p 264).
- ▶ **Information is the basic tool of management.** A good information system supplies data for monitoring and evaluation of health programmes and gives the requisite feedback to administrators and planners at all levels; computers can play a great role in improving it (Park 28e, Ch 23, p 1028).
- ▶ **Distance denies specialist care.** People in remote rural areas struggle to get timely, good-quality specialist care, and telemedicine has the potential to bridge this gap (Suryakantha, Ch 50, p 927).

Domains of use — with the Indian platform for each

Domain	What information technology does	Indian platform (year)
Routine health information	Facility-wise capture of service delivery, infrastructure and human-resource data for monitoring national programmes	Health Management Information System (HMIS) 2.0, launched 2020–21 with World Health Organization technical support (Sunder Lal, Ch 8, p 532)
Beneficiary tracking	Name-based tracking of each pregnant woman, infant and child through the reproductive life-cycle	Mother and Child Tracking System (MCTS), upgraded to the Reproductive and Child Health (RCH) portal; Integrated RCH Register version 2.0 launched June 2022 (Park 28e, Ch 7, p 540; Sunder Lal, Ch 8, pp 532, 535)
Disease surveillance	Weekly syndromic, presumptive and laboratory reporting; early warning signals of outbreaks	Integrated Disease Surveillance Programme (IDSP), begun 2004, migrating to the Integrated Health Information Platform (IHIP) (Jugal Kishore, p 487; Routine Immunization Manual 2024, Ch 12, p 270)

Domain	What information technology does	Indian platform (year)
Immunisation supply chain	Real-time monitoring of vaccine stock, supply and storage temperature; beneficiary records	Electronic Vaccine Intelligence Network (e-VIN) 2015, National Cold-chain Management Information System (NCCMIS), Co-WIN, now U-WIN (Routine Immunization Manual 2024, Ch 1, pp 29–30; Ch 5, pp 148–149)
Clinical care at a distance	Doctor-to-doctor and patient-to-doctor teleconsultation	eSanjeevani — eSanjeevani AB-HWC and eSanjeevani OPD (Sunder Lal, Ch 17, p 748)
Spatial analysis	Mapping of facilities, cases and risk factors on dynamic maps	Geographic information system (GIS)-enabled HMIS, launched March 2016; GIS mapping of government facilities by the National Informatics Centre (Sunder Lal, Ch 8, pp 534–535)
Longitudinal personal records	A unique digital health identity to which health records are linked	Ayushman Bharat Digital Mission (ABDM), launched September 2021; National Health Authority is the implementing agency (Sunder Lal, Ch 8, p 535; Ch 17, p 763)
Communication, training, administration	Behaviour-change messaging, worker training, grievance redress, blood-stock and appointment portals	Kilkari, Mobile Academy, ANM Online (ANMOL), i-GOT, Mera Aspatal, Online Registration System (ORS), e-Rakt Kosh (Sunder Lal, Ch 8, p 535)
Records and research	Computerised medical records, disease registers, record linkage	Hospital medical record departments; morbidity and cancer registers (Park 28e, Ch 21, pp 997–998)

Health Management Information System — the computerised backbone

- **Concept.** A health management information system is a process whereby health data (input) are recorded, stored, retrieved and processed for decision-making (output) covering planning, organising and control of health care facilities at national, State and institutional levels (Jugal Kishore, p 262).
- **Coverage.** About 2.17 lakh government and private health facilities across 747 districts are mapped as sub-centre / primary health centre / community health centre / sub-district hospital / district hospital; service data are entered each month by the 5th of the following month (Sunder Lal, Ch 8, p 532).
- **Data volume.** HMIS captures about **400 data elements of service delivery monthly** and about **700 elements of infrastructure and human resources annually**; daily data entry began in April 2020 (Sunder Lal, Ch 8, p 532).
- **HMIS 2.0 features.** Real-time data entry and monitoring, use on multiple devices, integration with other portals through application programming interfaces, interactive dashboards, GIS integration to village layer, and unique facility identification through the National Identification Number and Local Government Directory (Sunder Lal, Ch 8, p 532).

- ▶ **Value additions on the portal.** Grading of community health centres and primary health centres, automatic alerts for sensitive data items, and photographs of the health facility (Sunder Lal, Ch 8, pp 534–535).
- ▶ **Records.** Computerisation of medical records enables medical care to be more effectively rendered, better planned and better evaluated; **record linkage** assembles the important records of one individual or family originating at different times and places (Park 28e, Ch 21, pp 997–998).

Telemedicine and tele-health

- ▶ **Definition (World Health Organization, 2002).** The delivery of health care services, where distance is a critical factor, by all health care professionals using information and communication technologies for the exchange of valid information for diagnosis, treatment and prevention of disease and injuries, research and evaluation, and for the continuing education of health care providers — all in the interest of advancing the health of individuals and their communities (Suryakantha, Ch 50, pp 927–928).
- ▶ **Modes.** *Asynchronous* (store-and-forward, provider to provider) and *synchronous* (real-time, provider to patient) (Suryakantha, Ch 50, p 927).
- ▶ **Scope.** Telemedicine covers teleradiology, telepathology, teledermatology and telepsychiatry; tele-health focuses on disease management, monitoring and patient compliance (Suryakantha, Ch 50, p 928).
- ▶ **Set-up.** A Telemedicine Consultation Centre comprising the patient, the immediate health care provider, a telemedicine technician and the equipment that transmits the patient's medical information, linked to a specialist hub (Suryakantha, Ch 50, p 928).
- ▶ **India.** The Indian Space Research Organisation network links remote rural hospitals to superspeciality hospitals and supplies mobile telemedicine units for rural health camps (Suryakantha, Ch 50, p 932; Textbook of International Health, Ch 12, pp 663–664). Under eSanjeevani, **eSanjeevani AB-HWC** provides doctor-to-doctor teleconsultation across more than 1.5 lakh Health and Wellness Centres (Ayushman Arogya Mandirs), and **eSanjeevani OPD**, upgraded during the COVID-19 pandemic, provides free patient-to-doctor teleconsultation from home (Sunder Lal, Ch 17, p 748).

Geographic information system and remote sensing

- ▶ **What it is.** A geographic information system is computer hardware and software that combines high-precision location information with health databases and analytic tools to visualise the geographic distribution of morbidity, mortality, prevalence and incidence (Oxford Textbook of Public Health 7e, Ch 24, p 946). GIS with remote sensing, cartography, photogrammetry and geodesy together constitute **geomatics** (Mahajan & Gupta 4e, Ch 3, p 36).
- ▶ **Uses.** Judging whether disease patterns are random, clustered or dispersed; siting emergency services; modelling movement of pollutants; identifying where vectors pose greatest risk (Lilienfeld's Foundations of Epidemiology, Ch 14, p 291).
- ▶ **Indian application.** Remote sensing coupled with GIS allows macro-stratification of vast malarious areas so that control measures are prioritised cost-effectively (Jugal Kishore, p 420); the health-facility database mapped by the National Informatics Centre is disseminated through the Central Bureau of Health Intelligence website (Sunder Lal, Ch 8, pp 534–535).

- **Historical anchor.** John Snow's 1854 spot map of cholera deaths around the Broad Street pump is the forerunner of GIS in public health (Handbook of Epidemiology, Ch 33, pp 1320–1321).

Advantages against constraints

Advantage	Constraint / limitation
Real-time data access, rapid aggregation and display for decision-making at district, State and national level (Routine Immunization Manual 2024, Ch 12, p 270)	High cost of telemedicine, underdeveloped infrastructure, poor bandwidth and unreliable power supply (Park Plus, Ch 24, p 376; Suryakantha, Ch 50, p 932)
Ends duplication of registers and frees worker time (Jugal Kishore, p 264)	Poor uptake of electronic records by doctors and lack of inter-operability between systems and devices (Park Plus, Ch 7, p 83)
Role- and hierarchy-based feedback and alert mechanisms with GIS-enabled graphics (Routine Immunization Manual 2024, Ch 12, p 270)	Technophobia among handlers, need for continuous staff training, data lost to software or hardware failure (Suryakantha, Ch 50, p 932)
Named tracking ensures every mother and child receives the due service (Park 28e, Ch 7, p 540)	Breach of privacy and confidentiality; stigmatised conditions such as mental illness, infertility and HIV can damage reputation if disclosed (Park Plus, Ch 7, p 83)
Linked records are the most powerful method of studying rare and long-latency diseases (Maxcy-Rosenau-Last 15e, Ch 3, p 57)	Databases linked by internet or cloud computing pose additional risks to privacy beyond paper repositories; contractual liability and audit are required when data are outsourced (ICMR National Ethical Guidelines 2017, p 151)
Telemedicine reaches the unreached and serves continuing medical education (Suryakantha, Ch 50, p 932)	Medico-legal concerns, malpractice, absence of a governing body for guidelines, language barriers (Suryakantha, Ch 50, p 932; Park Plus, Ch 24, p 376)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Label box — components of a telemedicine consultation set-up (draw two boxes joined by a communication link):

- Nodal (peripheral) hospital: patient, immediate health care provider, telemedicine technician, data-entry console
- Peripheral equipment: microscope and medical instruments, scanner, video-conference camera, printer
- Link: modem — telephone line / Integrated Services Digital Network / satellite
- Referral (specialist) hospital: specialist, video-conference unit, patient record display
- Return arrow labelled "e-prescription and advice"

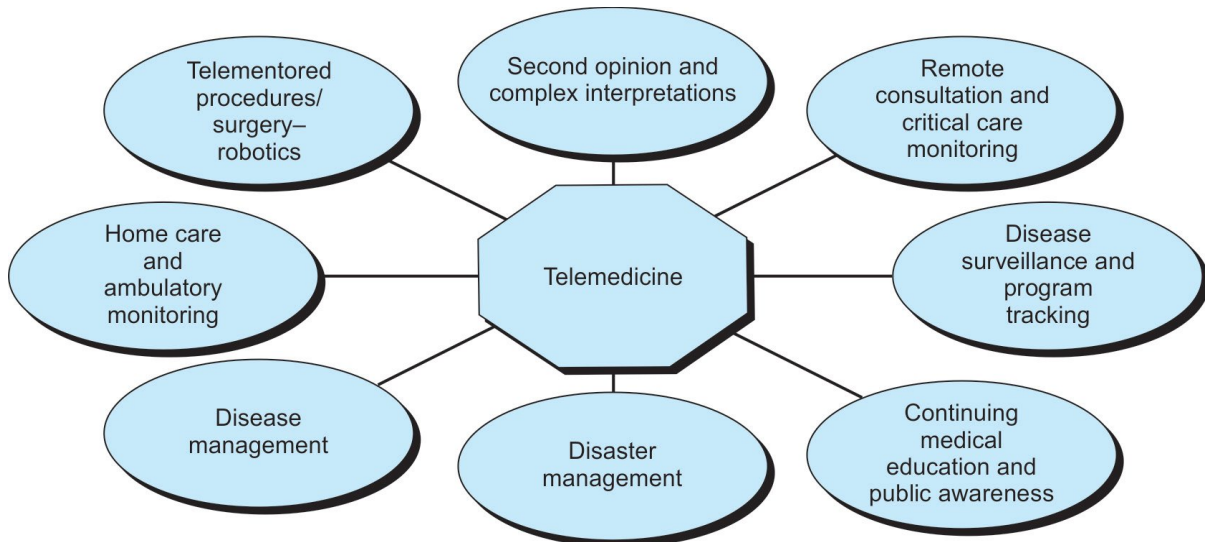
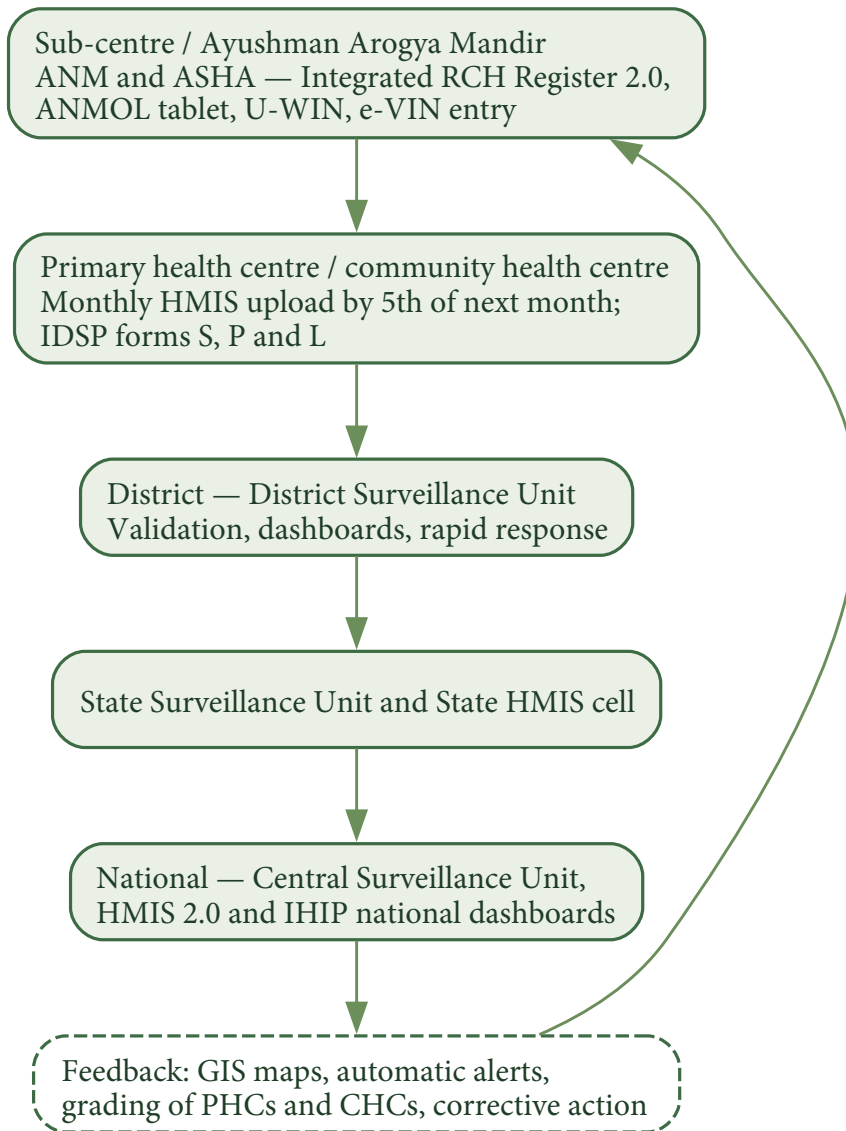
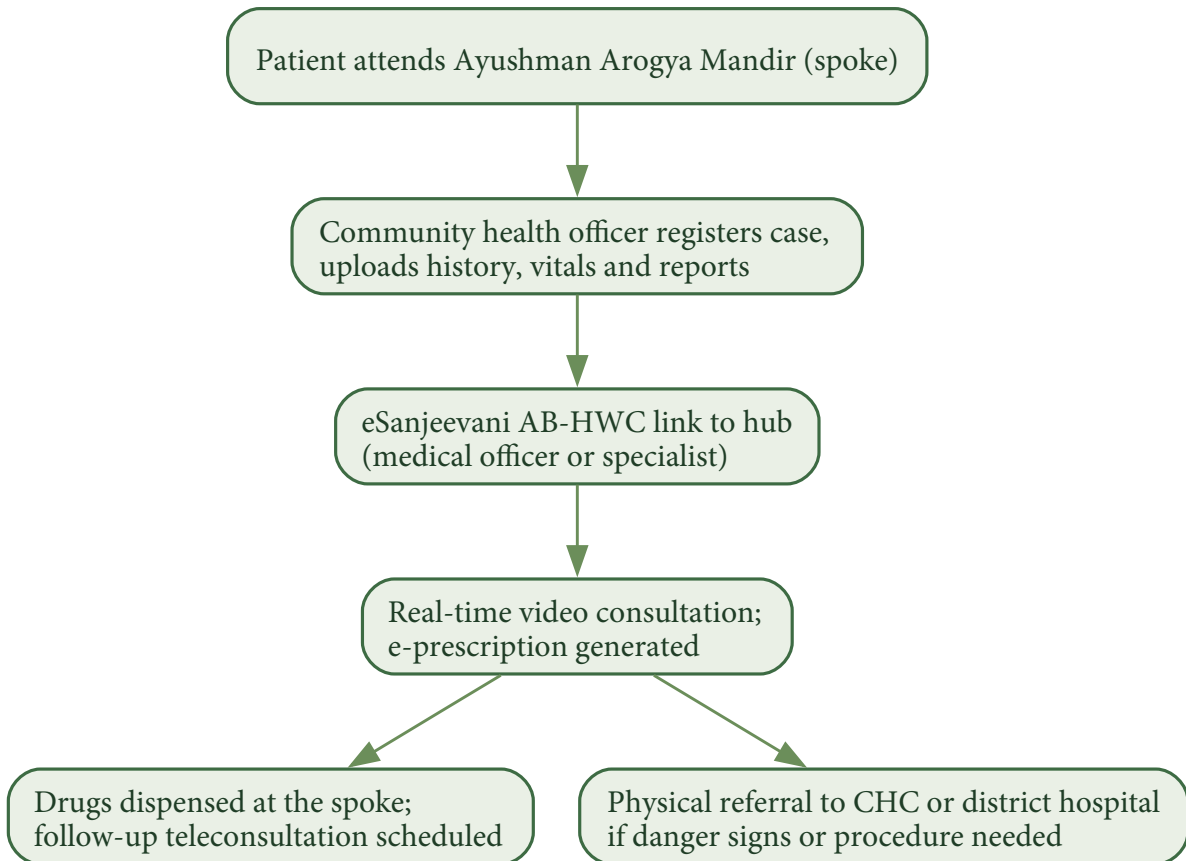


Fig 50.4 — Applications of telemedicine in public health (Suryakantha Environment Best, p.930)



Flow of digitised health data and feedback in the Indian public health system



eSanjeevani teleconsultation pathway from spoke to hub

High-yield exam pointers

- ▶ **Functions of an information system (Park):** collection, classification, transmission, storage, retrieval, transformation, display.
- ▶ **HMIS numbers:** HMIS 2.0 launched 2020–21; ~400 service-delivery elements monthly; ~700 infrastructure and human-resource elements annually; 2.17 lakh facilities across 747 districts; entry by the 5th of the following month.
- ▶ **Dates to quote:** IDSP 2004; e-VIN 2015; GIS-enabled HMIS March 2016; ABDM September 2021; Integrated RCH Register 2.0 June 2022.
- ▶ **Telemedicine:** World Health Organization 2002 definition; asynchronous = provider-to-provider, synchronous = provider-to-patient; eSanjeevani AB-HWC (doctor-to-doctor, >1.5 lakh Health and Wellness Centres) versus eSanjeevani OPD (patient-to-doctor, free).
- ▶ **Record linkage** = assembling for each individual a file of the important records originating at different times and places.
- ▶ **Geomatics** = GIS + remote sensing + cartography + photogrammetry + geodesy.
- ▶ **Ethics line to write:** internet- and cloud-linked health databases pose privacy and confidentiality risks beyond paper repositories, requiring contractual liability and audit (ICMR 2017).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Health and Wellness Centres have been renamed Ayushman Arogya Mandir, and the Ayushman Bharat Digital Mission was earlier called the National Digital Health Mission; the Integrated Disease Surveillance Programme is progressively migrating from the older weekly portal to the Integrated Health Information Platform, so numbers of connected centres and portals change year to year and should be quoted with the source year attached.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, ICMR Ethical Guidelines for AI in Biomedical Research and Healthcare 2023, Digital Personal Data Protection Act 2023

Programme and policy updates (India)

- ▶ **Ayushman Bharat Digital Mission — first formal assessment.** Applying a health-system control-knob framework to the mission launched in 2021, dashboard data showed about 400 million beneficiary registrations and 273 million health records linked as of 3 June 2023, with over 208,000 health facilities and 190,000 health-care professionals verified; progress varied sharply between States, particularly for facility and professional registration [PMID 39437234].
- ▶ **eSanjeevani needs a redesigned tele-referral pathway.** The national service now runs on a hub-and-spoke network of more than 154,000 Health and Wellness Centres at roughly 300,000 consultations a day, but usage has fallen since the pandemic surge and some authorities set daily referral targets for spoke workers; analysis of over 60 million consultations found absent screening tools, unstandardised symptom recording and free-text referrals, and the platform runs only in English — hence proposals for vernacular history entry and an electronic record visible to the hub clinician [PMID 39323563].
- ▶ **Ground-level maturity of the ministry's platforms differs widely.** Logical-framework scoring of nine government digital interventions at Health and Wellness Centres in Chandigarh graded e-VIN and the RCH portal as effectively implemented (71–90%), eSanjeevani and the IDSP–IHIP portal as fairly implemented (51–70%), HMIS, the HWC portal, the CPHC–NCD app and FP–LMIS as poorly implemented (25–50%), and ANMOL as not operational because of technical failure; HMIS and the HWC portal were dead at two centres for want of a data-entry operator [PMID 40854229].

Current national data

- ▶ **eSanjeevani volumes, with their dates attached.** Programmatic data from the Centre for Development of Advanced Computing record more than 163 million teleconsultations between November 2019 and September 2023 across all 28 States and 8 Union Territories, of which over 93% went through the provider-assisted eSanjeevani AB–HWC route rather than the direct-to-patient OPD route; hubs grew from 397 to 14,036 and spokes from 6,868 to 108,610 over the same period, 108,840 of 139,389 registered Health and Wellness Centres were on the platform by 2022, and five States — Tamil Nadu, Uttar Pradesh, Madhya Pradesh, Karnataka and Andhra Pradesh — contributed about 70.6% of all consultations. Use is highest in women and in the 25–45 age group, and the cumulative total crossed 276 million by November 2024, the largest documented government-led telemedicine service worldwide [PMID 41158553].

- ▶ **ABHA linkage fails mainly for want of a signal or a phone.** Over six months of outpatient work at a tertiary centre in eastern India (June–November 2024), 80,497 attendances yielded 30,786 ABHA tokens and 13,085 completed linkages; of 5,873 recorded failures, absent network signal accounted for 41.3%, no smartphone for 30.7% and refusal for 22.2%, the first two alone making up 72% — the case for assisted registration counters rather than smartphone-only design [PMID 41869122].

Elimination, surveillance and disease status

- ▶ **Digital surveillance still runs in parallel with paper.** Fifty-seven interviews plus observation in two high-burden districts of western India (2023–2025) found IHIP–IDSP digital syndromic (S), presumptive (P) and laboratory (L) reporting operating alongside the vector-borne programme's paper-based weekly epidemiological and entomological returns; the laboratory L form was filled separately and not linked to the matching S or P form, complete-blood-count platelet results from the state-contracted private laboratory went into that laboratory's own portal and into neither surveillance stream, district public health laboratories set up under IHIP–IDSP were functional in only three districts while four of 36 districts had no sentinel hospital, reporting was chased against fixed S, L and P completion targets of 92%, 90% and 80% rather than against data quality, and WhatsApp served as the real coordination layer between levels [PMID 42245358].
- ▶ **Vaccine-preventable disease resurgence is being read off IDSP.** A review drawing on IDSP and WHO Global Health Observatory data recorded spikes in reported measles across Indian States and a notable rise in mumps among older children even as the vaccine basket widened, with diphtheria re-emerging in Kerala and Assam and the burden shifting towards adolescents and adults; the authors call Co-WIN and U-WIN pivotal for real-time tracking and vaccine delivery — precisely the surveillance-plus-coverage read-out a digital system exists to deliver [PMID 40278999].

Guideline and legislative updates

- ▶ **The legal frame has moved on from the 2017 ethics code.** Governance of health data now rests on three pillars: the Digital Personal Data Protection Act 2023 (Data Principal and Data Fiduciary roles; consent that is free, specific, informed, unconditional and unambiguous; registered Consent Managers; a Data Protection Board of India; and for a Significant Data Fiduciary a resident Data Protection Officer, independent audit and periodic data-protection impact assessment), the ICMR Ethical Guidelines for Application of Artificial Intelligence in Biomedical Research and Healthcare 2023 (ten principles, human-in-the-loop oversight, shared developer-clinician liability, bias named the greatest threat), and NITI Aayog's #AIforAll strategy; section 7 of the Act permits processing without consent in a medical emergency or a public-health threat, an exemption that leaves data-hungry algorithms in a grey zone [PMID 41845374; ICMR 2023].
- ▶ **Digitisation can widen the gaps it claims to close.** A framework built on the Co-WIN and ABDM roll-outs sets out eight types of digital exclusion — access, skills, usage, outcome, motivation, physical, economic and cultural/social — noting that Co-WIN's smartphone-dependent registration systematically left out elderly, rural and non-smartphone users while ABDM met fragmented governance, uneven State adoption, limited provider capacity and privacy concerns. Digital health equity is framed as a socio-technical rather than a technical

problem, and the remedies proposed are participatory co-design with marginalised groups, privacy- and safety-by-design, multi-channel delivery that preserves a non-digital pathway, routine disaggregated equity monitoring and transparent algorithmic governance with redress [PMID 42215992].

Global and WHO developments

- **Global digital-health governance has been restructured.** The World Health Organization formally launched the Global Initiative on Digital Health on 20 February 2024, a WHO-managed network of governments, institutions and technical agencies supporting national digital-health transformation, backed by the G20 presidency; its four aims are assessing and prioritising country needs, aligning country-level resources against unfunded priorities, delivering the strategic objectives of the Global Strategy on Digital Health, and building local capacity to develop, maintain and adapt the technology [WHO GIDH page, accessed 2026-08-18].
- **The next global strategy is still only a draft — quote the current one.** WHO published the Global Strategy on Digital Health for 2020–2025 and now carries it on its Digital Health and Innovation page with the span extended to 2027; the successor **Global Strategy on Digital Health 2028–2033 has not been adopted by the World Health Assembly** and as of 2026 exists only as a draft strategy in open consultation, with a published consultation schedule and guidance. In an answer, write "Global Strategy on Digital Health 2020–2027, successor for 2028–2033 under consultation" — not that a 2028–2033 strategy has been approved [WHO Digital Health and Innovation page, accessed 2026-08-18].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2, Ch 7, Ch 21, Ch 23); Sunder Lal — Textbook of Community Medicine (Ch 8, Ch 17); Jugal Kishore — National Health Programmes of India; Suryakantha — Community Medicine with Recent Advances (Ch 50, Telemedicine in Public Health); Routine Immunization I Manual for Medical Officers 2024 (Units 1, 5, 12); Oxford Textbook of Public Health 7e; Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e; Lilienfeld's Foundations of Epidemiology; Handbook of Epidemiology (Ahrens & Pigeot); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e; ICMR National Ethical Guidelines 2017; Textbook of International Health; Park Plus (Amrutha & Vijayalakshmi).

Recent-advances sources: 39437234 — The Ayushman Bharat Digital Mission of India: An Assessment, *Health Systems & Reform* 2024, doi:[10.1080/23288604.2024.2392290](https://doi.org/10.1080/23288604.2024.2392290) (abstract-only); 39323563 — Reimagining India's National Telemedicine Service to improve access to care, *The Lancet Regional Health — Southeast Asia* 2024, doi:[10.1016/j.lansea.2024.100480](https://doi.org/10.1016/j.lansea.2024.100480); 40854229 — Implementation Status and Usability of Digital Health Interventions Among Health Care Workers and End Users at the Primary Health Care Level in Chandigarh, North India, *JMIR Formative Research* 2025, doi:[10.2196/69824](https://doi.org/10.2196/69824); 41158553 — Adoption and utilization of India's eSanjeevani national telemedicine service, *Oxford Open Digital Health* 2025, doi:[10.1093/oodh/oqaf025](https://doi.org/10.1093/oodh/oqaf025); 41869122 — Ayushman Bharat Health Account (ABHA) Integration in Action: Identifying Operational Barriers at a Tertiary Care Center in Eastern India, *Cureus* 2026, doi:[10.7759/cureus.103882](https://doi.org/10.7759/cureus.103882); 42245358 — Informational challenges of dengue surveillance and response: a case study from India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1809584](https://doi.org/10.3389/fpubh.2026.1809584); 40278999 — Vaccine-Preventable Diseases in Pediatric Age Group in India: Recent Resurgence, Implications and Solutions, *Indian Journal of Pediatrics* 2025, doi:[10.1007/s12098-025-05531-9](https://doi.org/10.1007/s12098-025-05531-9) (abstract-only); 41845374 — AI ethics in Indian healthcare: a scoping review of national and international guidelines on privacy, data protection, and security, *BMC Medical Ethics* 2026, doi:[10.1186/s12910-026-01435-1](https://doi.org/10.1186/s12910-026-01435-1); 42215992 — Toward equitable digital health: an integrated framework addressing exclusion, ethics, and implementation across healthcare systems, *International Journal for Equity in Health* 2026, doi:[10.1186/s12939-026-02903-1](https://doi.org/10.1186/s12939-026-02903-1). Also ICMR Ethical Guidelines for Application of Artificial Intelligence in Biomedical Research and Healthcare 2023; Digital Personal Data Protection Act 2023; WHO Global Initiative on Digital Health page, who.int/initiatives/global-initiative-on-digital-health, giving the 20 February 2024 launch and the four objectives; WHO Digital Health and Innovation page, who.int/teams/digital-health-and-innovation, which carries the Global Strategy on Digital Health with its span extended to 2027 and lists the 2028–2033 successor only as a draft strategy with a consultation schedule and guidance — both accessed 2026-08-18. No World Health Assembly adoption of a 2028–2033 strategy has occurred, and the earlier claim of one has been removed.

♦ ♦ ♦

Community health needs assessment

Asked in: Paper III 11-May-2022 — RGUHS MD Community Medicine

Definition — Community health needs assessment is the systematic procedure used to collect and interpret data describing the health problems, needs and strengths of a defined population, so that priorities can be set and a health programme planned for it (Health Program Planning and Evaluation, Ch 3, p 100). It is the operational form of **community diagnosis**, defined as the pattern of disease in a community described in terms of the important factors which influence that pattern (Park 28e, Ch 2, p 68). A **health need** is a deficiency in health that calls for preventive, curative, control or eradication measures (Park 28e, Ch 23, p 1025).

Purpose and place in health planning

- ▶ It is the **first step of the planning cycle** — analysis of the health situation, without which objectives, resources and priorities cannot be defined (Park 28e, Ch 23, p 1027).
- ▶ It **identifies and quantifies** health problems in terms of mortality and morbidity rates and ratios, and identifies their correlates, thereby defining the individuals or groups **at risk** or in need of health care (Park 28e, Ch 3, p 111).
- ▶ It allows **priorities** to be laid down in disease control and prevention when resources always fall short of total requirement (Park 28e, Ch 10, pp 721-722).
- ▶ It provides the **baseline or benchmark** against which health services are later evaluated (Park 28e, Ch 3, p 111).
- ▶ It answers, at community level, the question "What are the key health problems, and which resources are available to address them?" — so it examines community **assets and services** as well as deficits (Health Program Planning and Evaluation, Ch 3, p 106).
- ▶ It ends in **community treatment** — the sum of steps decided upon to meet the health needs of the community, taking into account the resources available and the wishes of the people (Park 28e, Ch 2, p 68).

Types of need assessed (Bradshaw, 1972)

Type of need	Whose view	How it is recognised / measured	Example
Normative need	Expert / health professional	Deviation of a measured value from an accepted norm or standard	Infant mortality rate above the national average; body mass index >29.9 as a need for weight reduction
Perceived (felt) need	The community itself	What people say they want; stated deficits and inadequacies, elicited by interview, focus group or community forum	Residents asking for a swimming pool and meeting place rather than the clinic planners proposed
Expressed need	Health-care-seeking behaviour	Demand for services — number requesting services, types of services sought, utilization rates	Out-patient attendance, antenatal registrations, bed occupancy

Type of need	Whose view	How it is recognised / measured	Example
Comparative (relative) need	Contrast between groups	Gap demonstrated between an advantaged and a disadvantaged group; most health disparities are stated this way	One racial or district group having twice the infant mortality rate of another

Source for the whole table: *Health Program Planning and Evaluation*, Ch 3, pp 91-92.

Park adds the practical caution that needs as seen by the people are **not the same** as needs seen by experts: some needs are not perceived at all, others only vaguely, and in a democratic society people's needs may be presented as **demands** (Park 28e, Ch 23, p 1025).

Data required and their sources

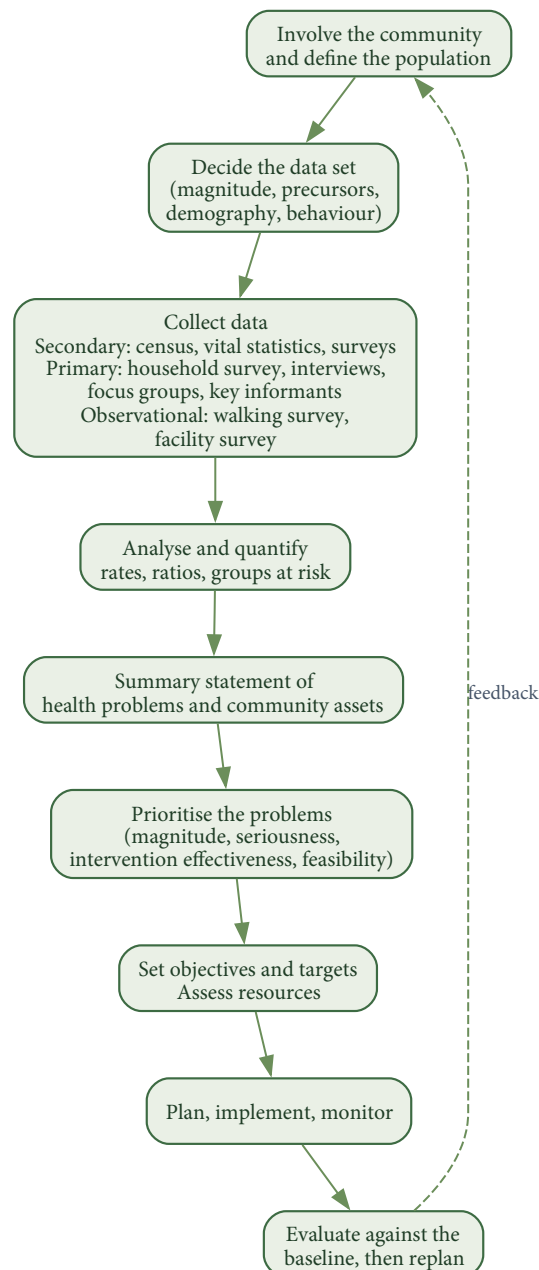
Data required (Park 28e, Ch 10, pp 721-722)	Usual source in Indian practice
Morbidity and mortality statistics	Civil registration of births and deaths with cause of death, verbal autopsy, hospital and programme returns (Sunder Lal, Ch 16, p 702)
Demographic conditions of the population	Census, annual household survey register maintained by accredited social health activist (ASHA), anganwadi worker (AWW) and auxiliary nurse midwife (ANM) (Sunder Lal, Ch 16, p 701)
Environmental conditions bearing on health	Household survey items — drinking-water source, sanitary latrine, cooking fuel; direct observation / walking survey
Socio-economic factors	Household survey — literacy, income, occupation
Cultural background, attitudes, beliefs, practices	Key-informant interviews, focus group discussions, health-seeking behaviour questions
Medical and health services available	Facility survey of sub-centre, primary health centre and community health centre against Indian Public Health Standards, to map gaps in human resources, drugs, diagnostics, equipment, space and finance (Sunder Lal, Ch 16, p 702)
Other (non-health) services available	Anganwadi under Integrated Child Development Services, school, panchayat, self-help groups

Data are of three kinds — **public/secondary data** (census, vital statistics, national surveys), **primary data** (surveys, interviews, community forums, focus groups, key informants, service providers) and **observational (unobtrusive) data** such as a walking or windshield survey. Provider-derived data reveal **normative need only** and, used alone, produce programmes the intended audience does not want (Health Program Planning and Evaluation, Ch 4, pp 120-122).

Steps in conducting the assessment

- ▶ **Involve community members** in the design and execution of the assessment from the outset.
- ▶ **Define the community or population** to be assessed — geographical, administrative or a defined subgroup.
- ▶ **Decide which data to collect** — magnitude of the problem, its precursors, and the demographic and behavioural characteristics of those affected.

- **Collect the data** using several sources and methods (secondary, primary and observational).
- **Analyse the data statistically** to arrive at quantified statements about the health problems of the population.
- **Develop a summary problem statement** and rank the problems in order of priority for action (Health Program Planning and Evaluation, Ch 3, pp 106-107).
- **Feed the ranked problems into the planning cycle** — objectives and goals, assessment of resources, alternative plans, implementation, monitoring and evaluation (Park 28e, Ch 23, p 1027).



Steps in a community health needs assessment and its place in the planning cycle

Setting priorities among the identified needs

Method	How it works
Park's criteria	Financial constraints; mortality and morbidity data; diseases preventable at low cost; saving lives of younger people in whom there has been considerable social investment; political and community interest and pressure (Park 28e, Ch 23, p 1027)
Hanlon's Basic Priority Rating System (BPRS)	Basic priority rating = $(A + 2B) \times C$, where A = size/magnitude of the problem, B = seriousness (urgency, severity, economic consequences, willingness of others), C = potential effectiveness of the intervention; each scored 1-10 (Health Program Planning and Evaluation, Ch 4, pp 140-141)
PEARL feasibility filter	Applied after BPRS: Propriety, Economics, Acceptability, Resources, Legality, each health problem is scored yes = 1 or no = 0 on all five dimensions to test whether acting on it is feasible at all (Health Program Planning and Evaluation, Ch 4, pp 142-143)
Importance × changeability matrix	Highly important + highly changeable = high priority; important but less changeable = high priority only if an innovative programme can be developed; unimportant and unchangeable = no programme warranted (Health Program Planning and Evaluation, Ch 4, pp 143-144)
Nominal group process	Structured group ranking by stakeholders, who must agree beforehand that the vote will be honoured (Health Program Planning and Evaluation, Ch 4, p 140)

Community health needs assessment in Indian practice

Tool / level	What it assesses
Annual household survey register (ASHA, AWW, ANM)	Households, families, total population, demographic profile, determinants of health, health-seeking behaviour, and the number of beneficiaries — eligible couples, antenatal and postnatal women, young children, adolescents, persons aged 30 years and above, and 60 years and above (Sunder Lal, Ch 16, p 701)
Community-Based Assessment Checklist (CBAC)	Risk score for non-communicable disease in all persons aged 30 years and above; six variables — age, family history, waist circumference, physical inactivity, tobacco use, alcohol use; score above 4 = high risk , prioritised for screening (Sunder Lal, Ch 8, p 503; Ch 16, p 703)
Facility survey	Inputs at sub-centre, primary health centre and community health centre compared against Indian Public Health Standards to locate gaps (Sunder Lal, Ch 16, p 702)
Vital events and routine service reports	Births, deaths and causes of death; past-year facility performance — to establish disease burden and unmet service needs (Sunder Lal, Ch 16, p 702)
Village Health, Sanitation and Nutrition Committee (VHSNC)	Sarpanch, ASHA, multipurpose health worker (female), school teacher, anganwadi worker and a self-help group member; untied fund of Rs 10,000 per annum per village panchayat; prepares the village health action plan (Sunder Lal, Ch 16, p 697)



Aggregation of community health needs assessment into the District Health Action Plan under the National Health Mission

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — The planning cycle (needs assessment is its opening arc). Labels: Gather health data → Tabulate, analyze, interpret → Formulate health problems → Assign priorities among problems → Formulate individual priority problems for planning solutions → Define programme objectives and how to measure achievement → Assign priorities among objectives → Design alternative programmes to solve problems → Select best programme (feasibility, cost/effectiveness) → Combine programmes and develop integrated plan → Plan implementation → Orient and train personnel → Initiate integrated programme → Operate programme → Collect data for evaluation → Evaluate results: goals achieved? goals satisfactory? → YES (continue) / NO (return to formulate health problems).

Diagram 2 — The four types of need as overlapping circles. Labels: Normative need (expert-defined); Perceived or felt need (community-defined); Expressed need (demand and utilization); Comparative or relative need (gap between groups); shade the central overlap and label it "real need on which programmes should be built".

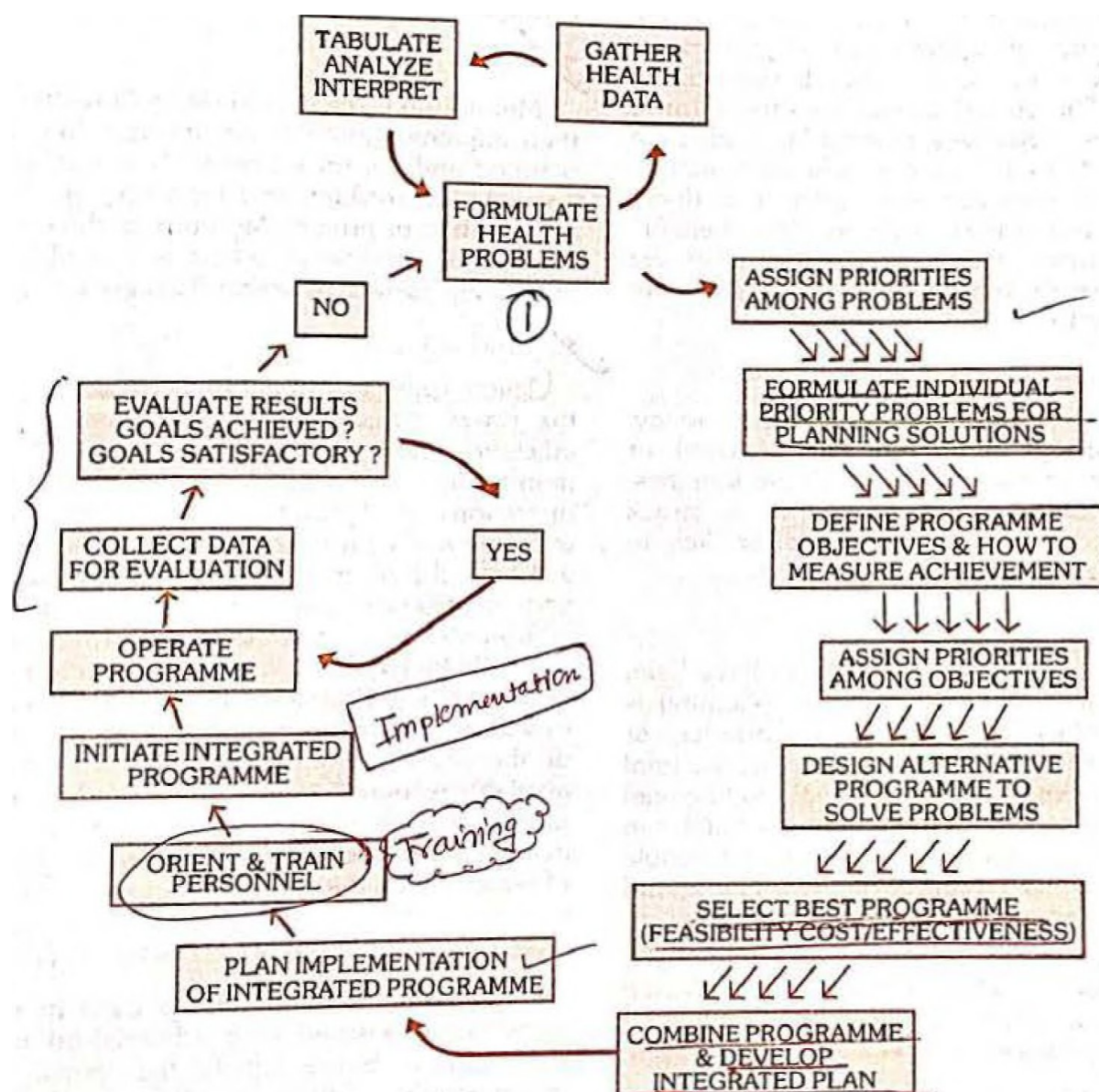


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, p.1026)

High-yield exam pointers

- **Definition to write first:** community diagnosis = "the pattern of disease in a community described in terms of the important factors which influence this pattern"; health need = "deficiencies in health that call for preventive, curative, control or eradication measures" (Park 28e).
- **Bradshaw's four needs** — mnemonic N-F-E-C: Normative (expert), Felt/perceived (people), Expressed (demand and utilization), Comparative (gap between groups).
- **BPRS formula:** Basic priority rating = $(A + 2B) \times C$ — A magnitude, B seriousness, C intervention effectiveness, each 1-10; follow with **PEARL** (Propriety, Economics, Acceptability, Resources, Legality), scored 0 or 1.
- **Park's seven data items** for community diagnosis: morbidity and mortality statistics, demography, environment, socio-economic factors, culture and beliefs, health services available, other services available.

- ▶ **Indian chain to quote:** household and facility survey → village health register → village health action plan (VHSNC, Rs 10,000 untied fund) → sub-centre plan → PHC/CHC plan → District Health Action Plan.
- ▶ **CBAC cut-off:** all persons ≥ 30 years enumerated; CBAC score >4 = **high risk** for non-communicable disease and prioritised for screening.
- ▶ **Park's priority criteria:** cost, mortality and morbidity data, low-cost preventability, social investment in the young, and political/community pressure.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the National Rural Health Mission is now the rural arm of the National Health Mission, and sub-centres and primary health centres delivering this assessment are designated Ayushman Arogya Mandir (formerly Health and Wellness Centres); the retrieved textbooks still use the older names, and the CBAC form and Indian Public Health Standards are periodically revised, so quote the version year you have read.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 3 Principles of Epidemiology; Ch 10 Health Care of the Community; Ch 23 Health Planning and Management); Health Program Planning and Evaluation — A Practical, Systematic Approach (Ch 3 Community Health Assessment for Program Planning; Ch 4 Characterizing and Defining the Health Problem); Sunder Lal — Textbook of Community Medicine (Ch 8 Epidemiology of Communicable and Non-Communicable Diseases; Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community); Reinke — Health Planning for Effective Management (Ch 5 An Overview of the Planning Process). Bradshaw's typology of need (1972) and Hanlon's Basic Priority Rating System (1973) are cited as reproduced in Health Program Planning and Evaluation.



Leadership and motivation

Asked in: Paper IV 04-Aug-2021 — RGUHS MD Community Medicine

Definition — Leadership is the ability to persuade others to seek defined objectives enthusiastically, by influencing the activities of individuals or groups for achieving the organisational goal in a given situation (*Keith Davis*); operationally, any person in an organization who **influences the behaviour of others** is a leader (Red Book — Public Health and Preventive Medicine, Ch 2, p 291; Sunder Lal, Ch 16, pp 711-712). **Motivation** is goal-directed behaviour — an inner force driving an individual to a certain action; the **motives** are the felt needs, wants, drives and impulses that arouse and maintain activity and so determine the direction of behaviour (Sunder Lal, Ch 16, p 713; Park 28e, Ch 12, p 818). **Motivating people to do the work** is one of the four basic activities of management, with *planning* and *monitoring/controlling* (Park 28e, Ch 23, p 1028).

Leadership — elements, styles and the managerial grid

Four elements make up the leadership process: the leader **influences an individual in a particular way**; leadership is rooted in feelings and so is **interpersonal influence**; it is **dynamic and ever-evolving**, so a leader must lead continuously; and it is exercised **in a particular situation**, so a non-leader may lead in another (Red Book, Ch 2, p 291). Choosing a style is essentially deciding **how far a manager should be dictatorial and how far participatory**.

Style	How the leader behaves	Use at the CHC
Autocratic (authoritarian, boss-centric)	Gives orders and insists they be obeyed , often by instilling fear of punishment for non-compliance; formulates policies without consulting subordinates; tells the group only the immediate steps, not the future plan; praise or criticism on the leader's initiative alone; remains aloof from the group for the greater part of the time	Reserve for an emergency — outbreak response, disaster triage, a mass-casualty event
Democratic (participative)	Gives orders only after consulting group members; policies discussed and decided by group discussion ; explains the long-term plan before asking staff to act; participates in the group as an active member	The working style for routine CHC functioning; the monthly sector meeting is its vehicle
Laissez-faire (free-rein, employee-centric)	Does not lead and leaves the group entirely to itself; is incapable of assuming control over subordinates and does not participate with the group	Legitimate only with a fully competent, committed team; otherwise it is abdication

(Red Book — Public Health and Preventive Medicine, Ch 2, pp 291-292; the continuum of use of authority from autocratic (boss-centric) to free-rein (employee-centric) is credited to Tannenbaum and Schmidt)

A leader has **two prime concerns** — the **people** (humanistic approach) and the **task** (mechanistic approach); balanced leadership weighs both equally (Sunder Lal, Ch 16, p 712). *Blake and Mouton's Managerial Grid* (1978) grades each concern 1–9; of 81 combinations, five are basic.

Grid position	Name	Character
(9, 1)	Task management leader	Concerned with the task at hand, little concern for the welfare of subordinates; completes the work at any cost
(1, 9)	Country club management leader	Concerned mainly with the welfare of subordinates, least concern for the task; seeks a harmonious, safe and pleasant atmosphere
(1, 1)	Impoverished management leader	Concerned with neither task nor people; avoids the conflict between goals and atmosphere; remains aloof from the organisation
(5, 5)	Middle of the road leader	Compromises between attainment of organisational goals and welfare of subordinates
(9, 9)	Team management leader	Highly concerned with both objectives and welfare — leaders are most effective at a high and balanced concern for both task and people ; an ideal state, difficult to attain

(Red Book, Ch 2, p 292)

Situational leadership and the qualities of a good leader

There is **no one best leadership style**: it is chosen to fit the follower's **development level (D1–D4)**, where *competence* means sound scientific knowledge and skills and *commitment* means a motivated, confident person (Sunder Lal, Ch 16, p 712).

Development level	Competence and commitment	Style	What the leader does
D1	Low competence (newly recruited), high commitment (enthusiastic)	Directing	Telling — written job description, specific instructions on what/how/where/when, close supervision of task accomplishment
D2	Some competence, low commitment	Coaching	Directing plus supporting — still directs and supervises closely, but explains decisions, solicits suggestions, supports progress
D3	High competence, variable commitment	Supporting	Listening — encourages, facilitates involvement in problem-solving, shares responsibility for decision-making
D4	High competence, high commitment	Delegating	Hands over responsibility and authority for day-to-day decision-making and problem-solving

(Sunder Lal, Ch 16, p 712)

- **LEAD model** — Lead with clear purpose; Empower to participate; Aim for consensus; Direct the process; i.e. set clear goals, involve people, reach consensus, and attend to both task and relationship, shifting from "ME to WE" (Sunder Lal, Ch 16, pp 712-713).
- **Qualities of a good leader** (Sunder Lal, Ch 16, p 713): sets an example of correct behaviour and practises what he teaches; facilitates work, listens actively and solves problems; takes responsibility for his own actions and those he supervises, and knows his own limits; decides

soundly and in time, on data and facts; gives credit and recognition for high performance and minimises punishment; is firm about goals but flexible about process.

- Park's community-facing attributes: takes the initiative, identifies himself with the community, is selfless, honest, impartial and accessible (Park 28e, Ch 22, p 1019).

Motivation — the process, motives and incentives

Motivation begins with a **felt need** (deficiency or deprivation), which generates a **drive** (behaviour directed at that deficiency), which is reduced by an **incentive** that fulfils the deficiency; a new need is then felt and the cycle repeats (Sunder Lal, Ch 16, pp 713-714). A **motivated person accomplishes the goal in a shorter time and produces quality work at lower cost**, though motivating raises only the *probability*, never the guarantee, of goal-directed behaviour.

Axis	Category	Content
Type of motive	Primary (inborn)	Sex, hunger, survival — driving forces initiating people into action
	Secondary (created by outside forces/incentives)	Praise, love, rivalry, rewards and punishment, recognition
Direction	Positive ("the carrot")	Reward-based; more successful than negative motivation
	Negative ("the stick")	Punishment-based
Nature of incentive	Intrinsic	Benefit that comes from solving one's own problem
	Extrinsic	Rewards not directly related to the goal; material or psychological, self-determined or selected by others

(Park 28e, Ch 12, p 818 and Ch 22, p 1019)

Two cautions: **motivation is not manipulation** — a motivated person acts willingly and knowingly; and **motivation is contagious** — one motivated person spreads motivation through a group (Park 28e, Ch 12, p 818).

Theories of motivation

Theory	Core proposition	What it directs the CHC in-charge to do
Maslow's hierarchy of needs (Abraham Maslow)	Needs are ordered — physiological → safety → social/belongingness → esteem → self-actualization; each tier is sought once the one below is met	Identify which tier a given worker is on; for a salaried government worker the live tiers are belongingness and esteem/recognition

Theory	Core proposition	What it directs the CHC in-charge to do
McGregor's Theory X and Theory Y (Douglas McGregor)	Superimposed on Maslow: Theory X assumes lower-order needs (physiological, safety) dominate the individual — he is seen as needing direction and control; Theory Y assumes higher-order needs dominate — he exercises self-direction and accepts responsibility	Do not manage a competent, committed D4 worker with Theory X assumptions; delegating authority and untied funds is Theory Y in practice
Herzberg's motivation-hygiene (two-factor) theory (Frederick Herzberg)	Factors producing job satisfaction are separate and distinct from those producing dissatisfaction — the opposite of job satisfaction is not job dissatisfaction but NO job satisfaction . Motivators/satisfiers (intrinsic): achievement, recognition for achievement, intrinsic interest in the work, responsibility, advancement and growth. Hygiene factors/dissatisfiers (extrinsic): organizational policy and administrative practices, supervision, interpersonal relationships, working conditions, salary, security, status	Absence of hygiene factors promotes dissatisfaction, but providing them does not motivate workers to do better work — removing them merely placates rather than motivates; real motivation must come from achievement, recognition, responsibility and growth
McClelland's achievement, affiliation and power motives (David McClelland); expectancy and psychological approaches	Named as the other recognised theories of motivation	Match the incentive to the dominant motive of the individual worker

(Red Book, Ch 2, p 291; Sathe & Sathe, Ch 10, p 109; Park Plus, Ch 20, pp 257-258)

The CHC in-charge — what he commands

Item	Fact
Level and size	Secondary-level 30-bedded referral hospital; specialist care in medicine, obstetrics and gynaecology, anaesthesia, surgery, paediatrics
Coverage	Referral centre for 4 primary health centres ; 1,20,000 population in plains, 80,000 in tribal areas
Number in India	6,064 CHCs — 5,480 rural + 584 urban , as on 31 March 2022
Human resource under the senior medical officer (SMO)	40–60 skilled technical staff in 20–25 sub-centres , plus health guides, trained birth attendants, ASHAs and over 100 Anganwadi workers with their supervisors
Five job roles of the SMO	Manager of manpower and material; trainer for continuing education; clinician; accountable to the community of 1,20,000; leader — health team building

Item	Fact
The recognised weak link	Doctors spend about 80% of their time in daily clinics and are unwilling or unable to take on management and leadership functions

(Sunder Lal, Ch 17, pp 752-755; Ch 16, p 720)

How I would motivate my subordinates as in-charge of the CHC

- ▶ **Know each member first** — personality, aptitude, attitude, family responsibility, training background, motivational level and job satisfaction; only this analysis makes the choice of style and of incentive correct for that individual (Sunder Lal, Ch 16, pp 711-712).
- ▶ **Give every worker a written job description** and match my style to their development level (D1–D4 above); the commonest local failure is a worker with no written job description and no guidance on her deficient skills.
- ▶ **Lead democratically, not autocratically** — decide policy by group discussion, explain the long-term plan before asking for the work, participate as an active member; reserve the autocratic style for outbreak or disaster response.
- ▶ **Convert supervision into supportive supervision** — a **fixed schedule of visits**, going prepared with that sub-centre's facts and figures, a **checklist per sub-centre** against Indian Public Health Standards, and **the deficient supplies carried on the visit** (Sunder Lal, Ch 16, p 720). The supportive supervisor acts as **consultant and teacher** and never takes over the task, which would undermine the auxiliary's status and confidence (Reinke, Ch 2, p 39).
- ▶ **Correct deficient skills by on-the-job training, not lectures** — need-based, by demonstration and return demonstration, with **worker-to-worker training** by a better-skilled colleague (Sunder Lal, Ch 16, p 721).
- ▶ **Use the monthly sector meeting as the team-building and recognition forum** — 25–30 Anganwadi workers, 6 female and as many male health workers, the health supervisor, the *Mukhya Sevika*, the child development project officer and the medical officer meet at sector headquarters on a fixed date; each narrates successes and failures, problems are solved, next month's work is planned jointly and continuing education is held (Sunder Lal, Ch 16, p 715).
- ▶ **Give deliberate feedback, positive and negative**, to improve performance, and **public credit and recognition for high performance** — Herzberg's first two motivators, and Maslow's esteem tier, are what a worker on a fixed government pay scale can actually be given.
- ▶ **Delegate with real authority and with money** — maternal and child health activities (antenatal care, deliveries, postnatal care, immunization) are delegated to health workers, and the **untied grant of Rs 10,000 per annum to the female health worker** is cited as a best practice of delegating financial powers; this supplies the motivators of **responsibility and achievement** (Sunder Lal, Ch 16, pp 712-713).
- ▶ **Work Herzberg's two factors separately** — the hygiene factors within my reach are the **quality of supervision, collegial relationships, working conditions and fair administrative practice**; correcting them only removes dissatisfaction, so each is paired with a satisfier, chiefly **opportunities for advancement** (Reinke, Ch 2, p 39; Sathe & Sathe, Ch 10, p 109).

- **Lead by example** — be punctual and available, practise what I teach, and behave as a manager and not only as a clinician.

The seven causes of dissatisfaction described for Indian health workers are, in Herzberg's terms, a **hygiene-factor list** — correcting them prevents dissatisfaction but does not itself motivate.

Common cause of dissatisfaction (hygiene factor)	Corrective action by the CHC in-charge
Inefficient administration (organizational policy and administrative practices)	Management by objectives, each subunit preparing its own short-term plan of action; timely supplies, drugs and transport
Incompetent supervision	Fixed-schedule supportive supervisory visits with a checklist; supervisor as teacher, not inspector
Poor personal relationship (interpersonal relationships)	Monthly sector/staff meeting as an open forum; intersectoral coordination with ICDS
Poor leadership qualities	Lead by example; LEAD model; situational style matched to D1–D4
Low pay (salary)	Not alterable locally — substitute the motivators (recognition, credit for achievement, delegated responsibility, advancement)
Bad working conditions	Functional equipment, residential and transport facilities, oxygen-supported beds, strengthened support services
Non-participation in plans of work	Joint planning of next month's work at the sector meeting; sub-centre action plan prepared by the worker herself

(Causes — Sunder Lal, Ch 16, pp 714-715; hygiene/motivator classification — Sathe & Sathe, Ch 10, p 109; management by objectives — Park 28e, Ch 23, p 1028)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Continuum of use of authority (Tannenbaum and Schmidt)** — one long horizontal bar, left end labelled *Autocratic (boss-centric)* — use of authority by the manager, right end labelled *Free rein (employee-centric)* — complete freedom to workers, with seven tick-marks left to right: *Manager takes decisions and announces it; Manager sells decisions; Manager presents idea and invites suggestions; Manager presents tentative decisions; Manager presents problems and gets suggestions; Manager defines limits and asks group to take decision; Manager gives free rein to group.*

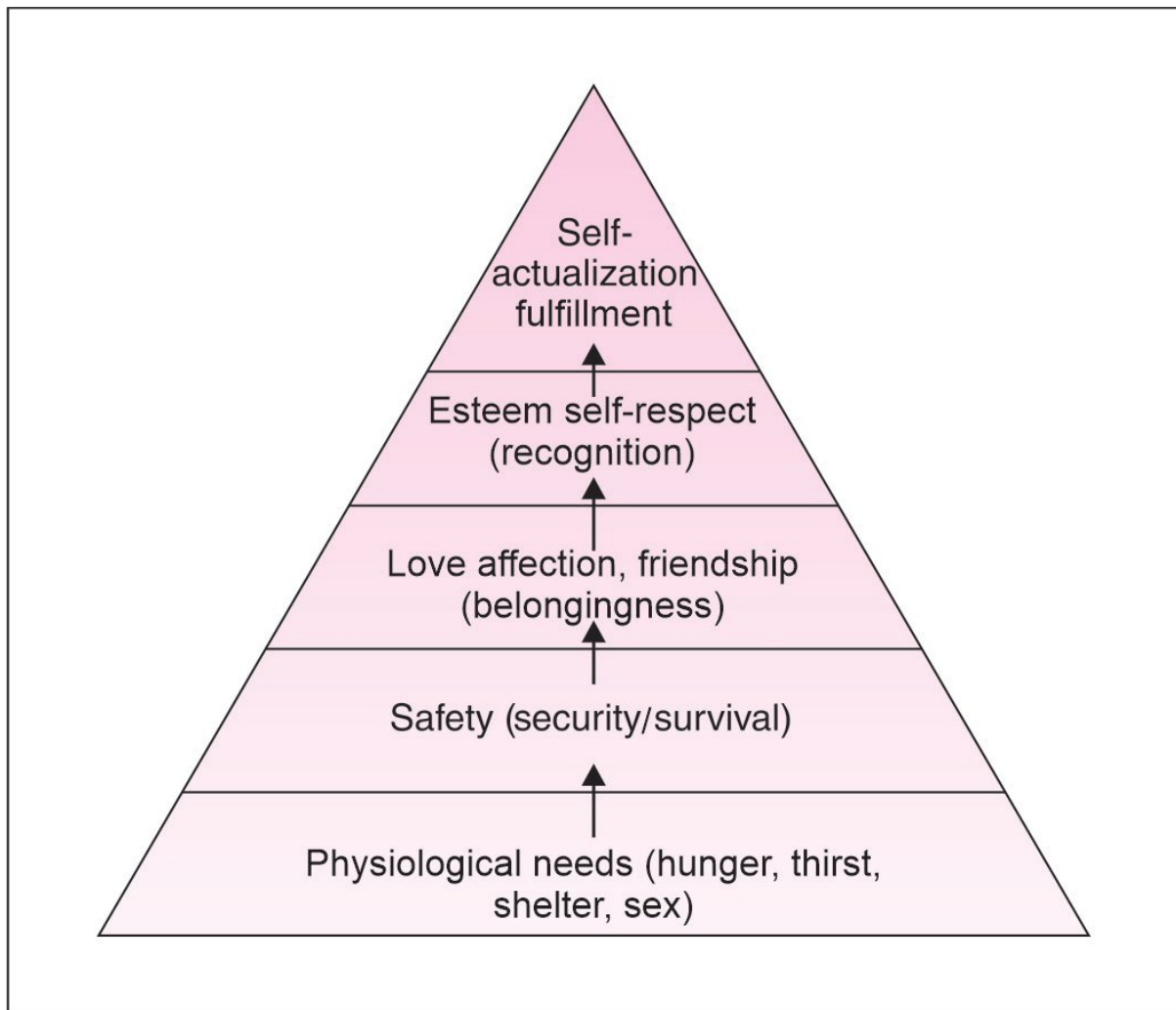


Fig 16.5 — Maslow's hierarchy of needs (Sunderlal, p.714)

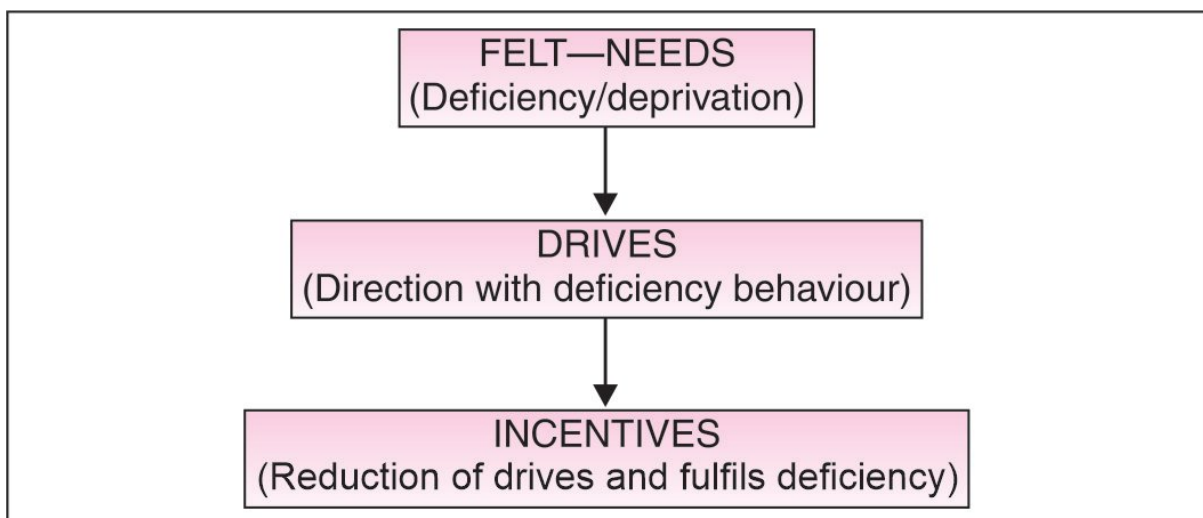
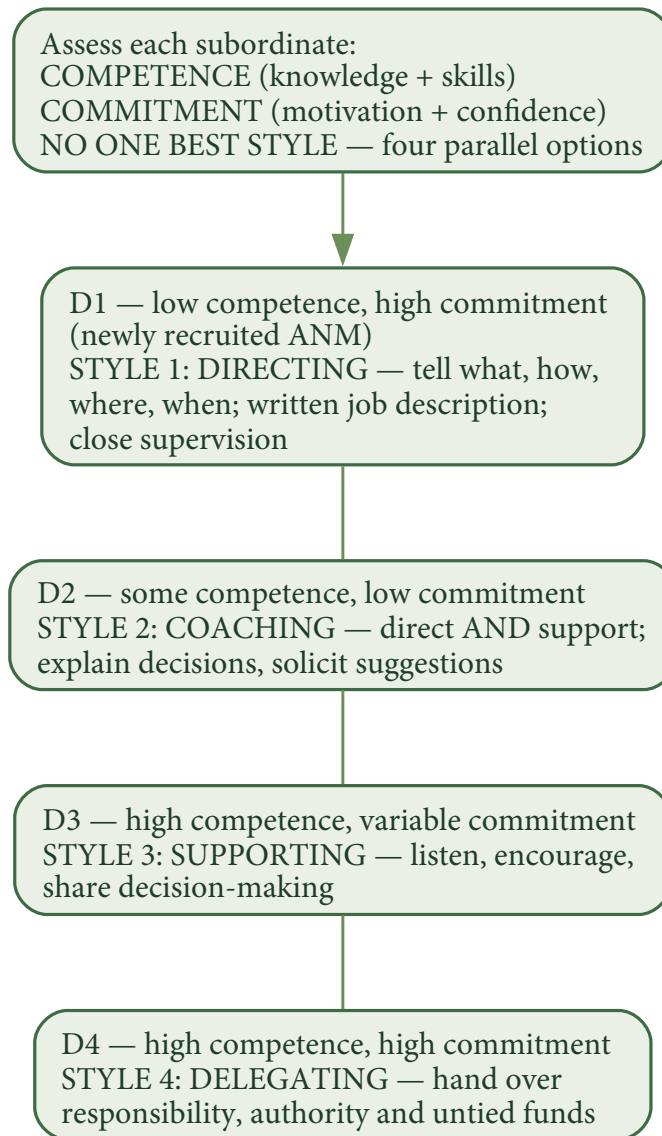
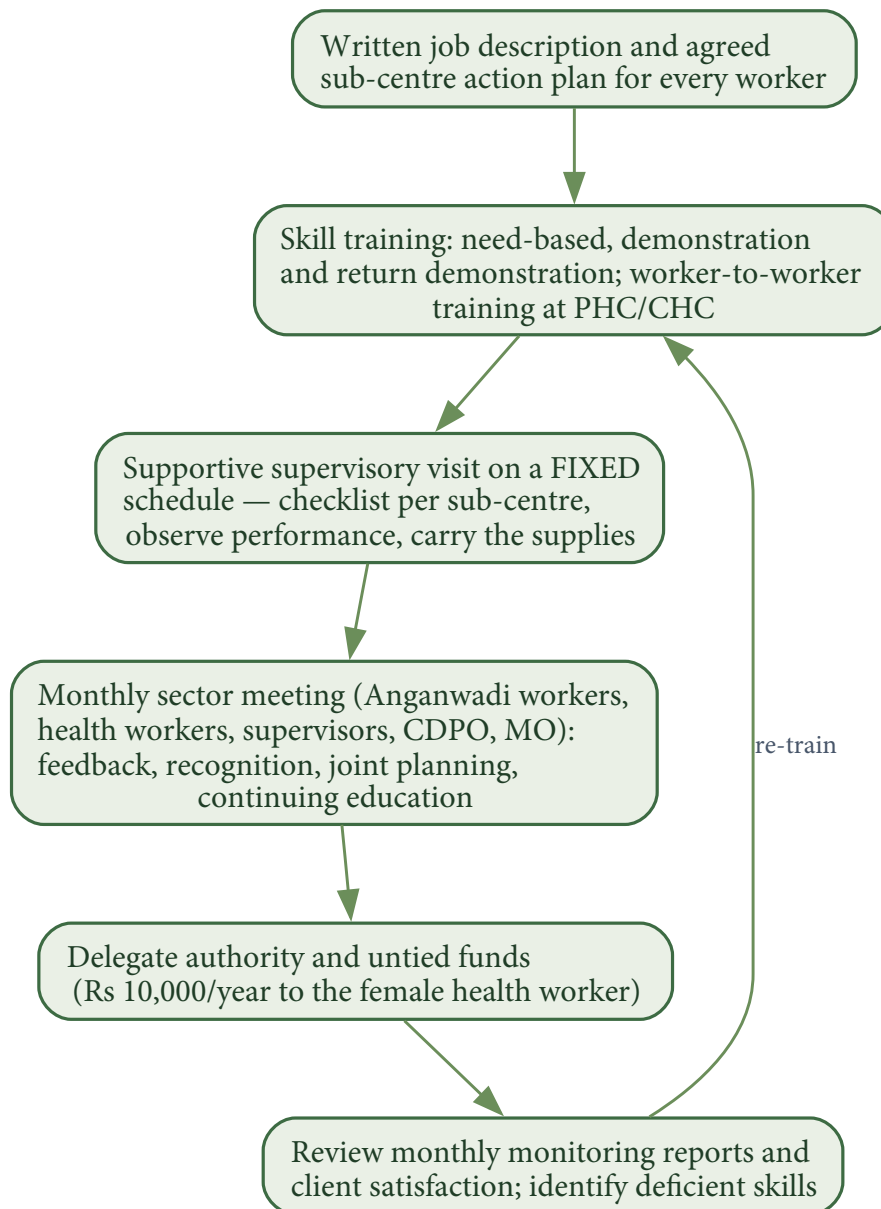


Fig 16.4 — Basic motivation process (Sunderlal, p.714)



Choosing the leadership style at the CHC by the follower's development level (Sunder Lal, Ch 16, p 712)



The CHC in-charge's monthly cycle of supportive supervision and motivation

High-yield exam pointers

- **Keith Davis** — leadership = **persuading others to seek defined objectives enthusiastically**; four elements — influence, interpersonal, dynamic/continuous, situational.
- **Three classical styles**: **autocratic** (orders + fear of punishment, no consultation, aloof); **democratic** (orders after consultation, policy by group discussion); **laissez-faire/free-rein** (does not lead). Continuum of authority **autocratic (boss-centric) → free rein (employee-centric)** — *Tannenbaum and Schmidt*.
- **Managerial Grid** (*Blake and Mouton, 1978*): **9,1** task management; **1,9** country club; **1,1** impoverished; **5,5** middle of the road; **9,9** team management (ideal). Two prime concerns — **people (humanistic)**, **task (mechanistic)**.
- **Four situational styles** — parallel options, not steps: **D1 Directing**, **D2 Coaching**, **D3 Supporting**, **D4 Delegating**. **LEAD** = Lead with purpose, Empower, Aim for consensus, Direct the process.

- **Motivation cycle:** felt need → drive → incentive → new need. **Maslow:** physiological → safety → belongingness → esteem → self-actualization. **McGregor:** Theory X = lower-order needs dominate; Theory Y = higher-order. "Carrot" beats "stick"; motivation is contagious, and is *not* manipulation.
- **Herzberg:** opposite of job satisfaction is **no job satisfaction**, not dissatisfaction. **Motivators** = achievement, recognition, responsibility, advancement, growth. **Hygiene** = policy/administration, supervision, interpersonal relations, working conditions, salary, security, status — the **seven causes of dissatisfaction** are this hygiene list.
- **CHC numbers:** 30 beds, referral for 4 PHCs, 1,20,000 (plains) / 80,000 (tribal); 6,064 CHCs — 5,480 rural + 584 urban, 31 March 2022; SMO commands 40–60 technical staff in 20–25 sub-centres and >100 Anganwadi workers.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Indian Public Health Standards were revised in 2022 and now classify community health centres as non-first-referral-unit (30 essential beds) and first referral unit (30 or 50 beds), with urban community health centres of 50 beds (100 in metropolitan and million-plus cities), and all block-headquarters centres are to become Block Public Health Units; a shortfall of about 80% of specialists at this level is documented, which is itself the reason the in-charge must rely on non-monetary motivation of the team he already has.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Indian Public Health Standards, Kayakalp, WHO WASH FIT, WHO operational framework for primary health care

Programme and policy updates (India)

- **Health and Wellness Centres have been renamed Ayushman Arogya Mandir**, and the team the CHC in-charge leads now includes the **Community Health Officer (CHO)** — a nurse-graduate mid-level provider created under Ayushman Bharat. In a 404-facility state-representative survey every centre had a CHO posted; 55.2% had completed the 6-month community health nursing bridge course after a nursing degree and the rest an integrated course, 90.6% were women, about 83% were under 30 years, and 72.3% were posted in their native district [PMID 39367295] [PMID 40341602].
- **Supportive supervision is the declared national model, but the supervisors themselves are untrained for it:** a realist evaluation of all six frontline supervisors of a block in Raisen, Madhya Pradesh (two lady health visitors, two male health assistants, two ASHA Sahyoginis; median 6.5 years in post) found that none had ever received formal management training and only one used a supervisory checklist. Authoritative, one-way communication and public reprimand produced fear, concealment of problems and reduced motivation, whereas collaborative, hands-on leadership produced open feedback loops and joint problem-solving; the authors argue that interpersonal and leadership competence is a core requirement for mid-level managers rather than a soft skill, and should be a selection criterion for supervisors [PMID 41254588].
- **A cash incentive works through the worker, not instead of her:** a difference-in-difference analysis of nationally representative survey data found that conditional cash transfer eligibility

raised completion of the full antenatal–institutional delivery–postnatal continuum by 8.9 percentage points unadjusted and by 5 percentage points after covariate adjustment ($p < 0.01$), and the qualitative arm found that sustained ASHA engagement mattered more for continuity of care than the cash alone [PMID 41773900].

Current national data

- ▶ **What an Ayushman Arogya Mandir actually delivers (Chhattisgarh, 404 centres, 2024):** a mean of 358 outpatients per month (95% CI 336–382), of whom 128 were NCD patients (81 hypertension, 40 diabetes); the centres met 31% of the catchment population's need for outpatient care, 26% of the need for hypertension care and 21% for diabetes care. A **Jan Arogya Samiti** had been constituted and was meeting monthly in 99.2% of them — the statutory community-participation forum the in-charge should work exactly as the monthly sector meeting is worked [PMID 39367295].
- ▶ **The classic weak link is now documented for the new cadre as well:** a time-motion study of CHOs in a rural block of central India found 64.2% of their working time went to curative work, 21.3% to managerial work and only 14.5% to promotive and preventive work, with focus on clinical care, absence of microplanning, insufficient resources and lack of job security named as the principal barriers — the same displacement of leadership by clinic work that the textbook describes for the medical officer [PMID 40214313].
- ▶ **Absenteeism, vacancy and burnout quantified:** unannounced visits to 43 public facilities in Jashpur, Chhattisgarh recorded absenteeism of 8% at PHC level against 1.14% at CHC level, with more than a third of sanctioned posts vacant at both, and attributed it to geographical isolation, missing equipment and amenities, doctor shortage with poor patient turnover acting as a demotivator, and unmet personal and professional expectations [PMID 39415651]. Its measured counterpart is burnout — 36.1% (95% CI 31.1–41.4) of 349 nursing professionals surveyed in Kerala scored high on the Burnout Assessment Tool, risk rising with direct clinical duty (adjusted prevalence ratio 1.75; 95% CI 1.18–2.59), urban posting (APR 1.41) and travel time of 30 minutes or more (APR 1.36) [PMID 41985970].

Elimination, surveillance and disease status

- ▶ **Delegation plus training, verified against a specialist standard:** 46 CHOs in Jodhpur given one day of training in oral visual examination raised mean knowledge scores from 5.93 to 9.46 out of 10 ($p < 0.001$), retaining 7.60 at two months; over eight months of field screening for oral potentially malignant disorders, with remote ENT-specialist validation, they achieved sensitivity 83.87%, specificity 98.31% and overall diagnostic accuracy 96.17% [PMID 40341602]. This is the current evidence for handing screening and surveillance work down to the sub-centre team instead of holding it at the CHC.

Guideline and legislative updates

- ▶ **Supportive supervision against a written standard changes facility practice:** across 6 PHCs and 5 CHCs of Luni block, Jodhpur, a package of structured training, formation of an Infection Control Committee under the block medical officer, and repeated unannounced supervisory visits carrying a checklist with on-site correction produced statistically significant improvement in all four WASH FIT domains at three and six months. The same assessment found that no

facility met the Indian Public Health Standards staffing norm — CHCs averaged 18.4 staff against the 46 required and PHCs 10 against 13 [PMID 40890711].

- **The CHC is now the least-prepared level for NCD care:** a 415-facility survey across 19 districts of seven states, scored by the WHO Service Availability and Readiness Assessment method against Indian Public Health Standards, put overall readiness for hypertension and diabetes services at 59% for CHCs and 61% for sub-centres — both below the 70% threshold for a facility to count as prepared — while PHCs reached 73%; the authors attribute the gap to human resources and uninterrupted essential-medicine supply, both of which are the in-charge's task-and-people responsibility [PMID 40536370].

Global and WHO developments

- **Non-financial incentives are half the lever:** a narrative review of large-scale primary health care workforce reform in five middle-income countries (Thailand, Brazil, Türkiye, Iran, China), organised on WHO's operational framework for primary health care, found that all of them paired short-term task sharing and in-service training of the existing workforce with a deliberate mix of financial *and* non-financial incentives to motivate and redistribute staff, and that performance-based incentives depend on careful selection of indicators, sustained political will and adequate public expenditure [PMID 41187459].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 12 (Medicine and Social Sciences), Ch 22 (Communication for Health Education), Ch 23 (Health Planning and Management); Sunder Lal — Textbook of Community Medicine, Ch 16 (Health Planning and Management) and Ch 17 (Healthcare of the Community); Red Book — Public Health and Preventive Medicine, Ch 2 (Managerial Sciences section, pp 291-292) for the leadership definition, elements, styles and Managerial Grid and for the list of theories of motivation; Sathe & Sathe — Epidemiology and Management for Health Care, Ch 10 (Organizational Behaviour) for Herzberg's motivation-hygiene concept; Park Plus (Amrutha & Vijayalakshmi), Ch 20, for McGregor's Theory X and Theory Y superimposed on Maslow's hierarchy; Reinke — Health Planning for Effective Management, Ch 2 (Management Analysis); Health Program Planning and Evaluation, Ch 7 (Process Theory for Program Implementation); Indian Public Health Standards 2022 as cited in Sunder Lal Ch 17.

Recent-advances sources: 39367295 — Performance of health and wellness centre in providing primary care services in Chhattisgarh, India, *BMC Primary Care* 2024, doi:[10.1186/s12875-024-02603-1](https://doi.org/10.1186/s12875-024-02603-1); 40341602 — Capacity building of community health officers for optimizing the screening and early diagnosis of oral cancer in Jodhpur, Rajasthan, India, *Scientific Reports* 2025, doi:[10.1038/s41598-025-00225-w](https://doi.org/10.1038/s41598-025-00225-w); 41254588 — Understanding the drivers of effective supportive supervision: a realist evaluation of community health worker programs in Madhya Pradesh, India, *BMC Health Services Research* 2025, doi:[10.1186/s12913-025-13567-y](https://doi.org/10.1186/s12913-025-13567-y); 41773900 — Conditional cash incentives, community health workers, and continuum of maternal and child healthcare: evidence from India, *Health Policy and Planning* 2026, doi:[10.1093/heapol/czag019](https://doi.org/10.1093/heapol/czag019); 40214313 — "Want a Job with a Safer Future": findings from an exploratory qualitative inquiry with community health officers using time-motion study, *Indian Journal of Public Health* 2025, doi:[10.4103/ijph.ijph.1175.23](https://doi.org/10.4103/ijph.ijph.1175.23) (abstract-only); 39415651 — Exploring health worker absenteeism at public healthcare facilities in Chhattisgarh, India, *Primary Health Care Research & Development* 2024, doi:[10.1017/S1463423624000343](https://doi.org/10.1017/S1463423624000343); 41985970 — Burnout, associated factors and coping strategies among nursing professionals in Kerala, India: a cross-sectional study, *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-113120](https://doi.org/10.1136/bmjopen-2025-113120); 40890711 — Water, sanitation and infection control practices in rural health facilities of Jodhpur District, India: an intervention through supportive supervision, *BMC Health Services Research* 2025, doi:[10.1186/s12913-025-12689-7](https://doi.org/10.1186/s12913-025-12689-7); 40536370 — Preparedness of public & private health facilities for management of diabetes & hypertension in 19 districts in India, *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR.755.2024](https://doi.org/10.25259/IJMR.755.2024); 41187459 — Implementing large-scale primary health care workforce reforms: a narrative literature review of middle-income country experiences, *Social Science & Medicine* 2025, doi:[10.1016/j.socscimed.2025.118717](https://doi.org/10.1016/j.socscimed.2025.118717) (abstract-only). Standards referenced: Indian Public Health Standards; Kayakalp and WHO WASH FIT assessment tools; WHO Service Availability and Readiness Assessment method; WHO operational framework for primary health care.

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Role of HMIS in RMNCH-A

Asked in: Paper III 11-May-2022 — RGUHS MD Community Medicine

Definition — The **Health Management Information System (HMIS)** is the routine, facility- and field-based reporting arm of India's health information system under the National Health Mission: a mechanism for the collection, processing, analysis and transmission of information needed for organising and operating health services (Park 28e, Ch 21, p 995). **RMNCH+A** (Reproductive, Maternal, Newborn, Child and Adolescent Health) is the continuum-of-care strategy adopted in 2013, built on five thematic areas and guided by equity, universal care, entitlement and accountability (Park 28e, Ch 7, p 553). HMIS is the measurement backbone of that strategy — "use of data for monitoring and review, including scorecards based on HMIS" is itself listed as a cross-cutting RMNCH+A intervention (Park 28e, Ch 7, pp 554-555).

1. What HMIS captures for RMNCH+A, and at which level

Level / responsibility centre	Record or report	RMNCH+A content captured	Period icity
Village, sub-centre — auxiliary nurse midwife (ANM), accredited social health activist (ASHA), Anganwadi worker	Integrated Reproductive and Child Health (RCH) Register version 2.0 (released 2022) — village/ward-wise, 5 sections, two financial years	Area profile: village population, eligible couples, estimated pregnant women, estimated infants and children 1–2 years; village-level human resources — used to compute annual workload and the work plan	Continu ous
Sub-centre / health and wellness centre	Monthly HMIS reporting format (April 2022) — Part A Reproductive and Child Health: M1 antenatal care, M2 deliveries, M3 pregnancy outcomes, M4 postnatal care, M5 family planning, M6 child immunisation, M7 childhood diseases, M8 vector-borne disease	Inputs (staff, equipment, stock of drugs and vaccines), processes and outputs (antenatal, natal, postnatal, newborn care, referral, immunisation, contraception), impact (births, deaths)	Monthl y
Same format, Parts B and C	Part B health facility services: M9 outpatients, M10 laboratory testing, M11 stock data. Part C mortality details with probable cause	Neonatal, infant, child and maternal deaths	Monthl y
Primary health centre (information assistant) / community health centre (data entry operator)	Compilation and online transmission to district programme officer	Facility-wise aggregated RMNCH+A service data	Monthl y

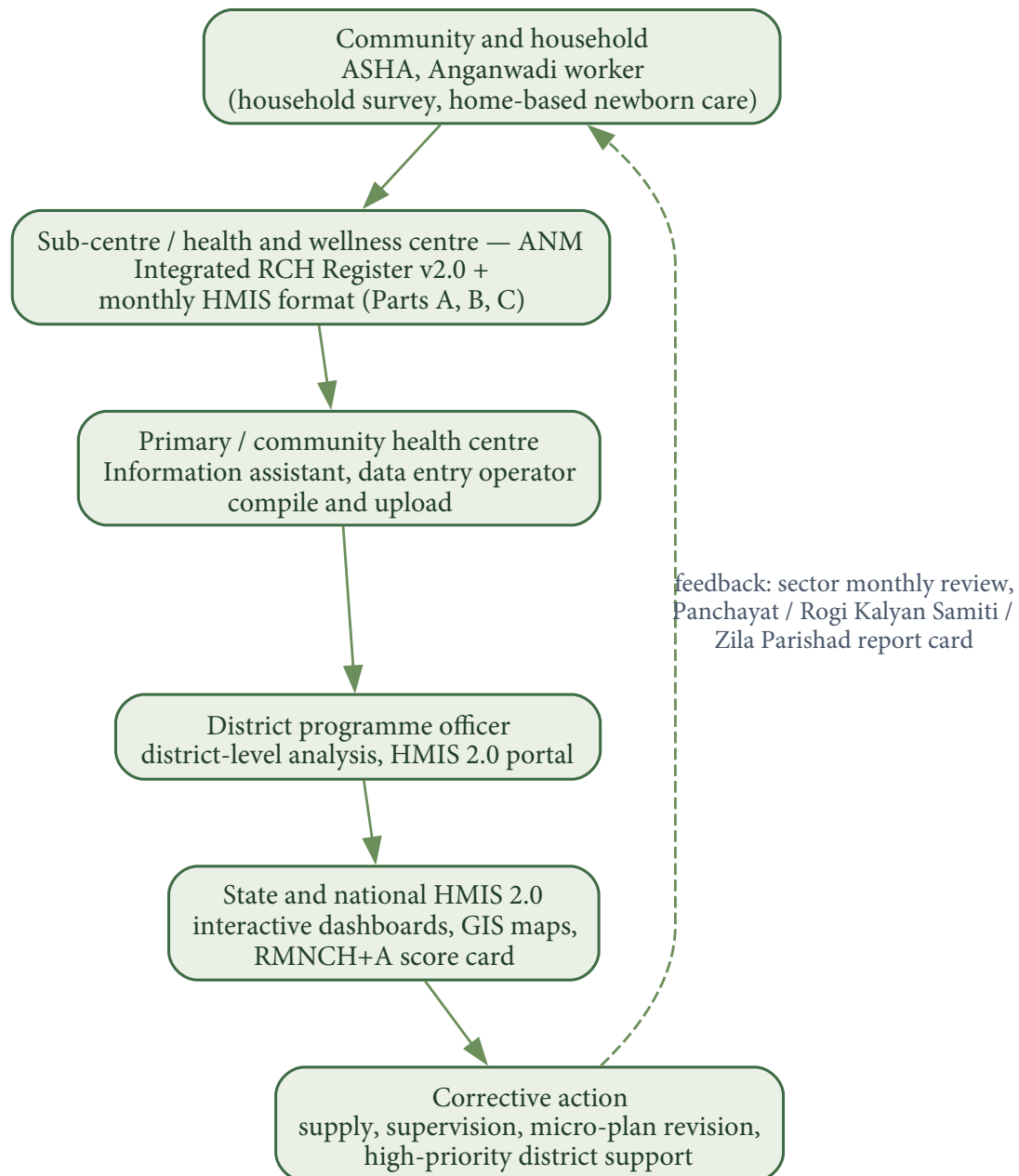
Level / responsibility centre	Record or report	RMNCH+A content captured	Period icity
National HMIS 2.0 portal (launched 2020 –21 with WHO technical support)	~400 data elements of service delivery monthly; ~700 elements of infrastructure and human resources annually, updated monthly; daily data entry since April 2020	Interactive dashboards, application-programming-interface integration with other portals, geographic information system (GIS) layer down to village, unique facility identification	Daily / monthly / annual

(Sub-centre formats, registers and HMIS 2.0 features: Sunder Lal, Ch 8, pp 532-533; reporting flow and feedback: Sunder Lal, Ch 8, pp 533-534)

2. The roles HMIS performs within RMNCH+A

Role	How HMIS delivers it in RMNCH+A
Measures health status and quantifies need	Village-wise estimates of pregnant women, infants and eligible couples in the Integrated RCH Register convert population denominators into a service workload and an annual work plan
Enables comparison between blocks, districts and states	All-India average of each HMIS indicator is the reference point against which states are scored (Park 28e, Ch 7, p 558)
Planning, administration and management	Facility-wise stock, human-resource and infrastructure elements drive commodity supply for the RMNCH+A "minimum essential commodities" arm of the 5×5 matrix
Assesses effectiveness and efficiency of services	Coverage and quality proportions (first-trimester registration, 3 antenatal visits, 100 iron-folic-acid tablets, institutional delivery, breastfeeding within 1 hour) show whether the high-impact interventions actually reached the beneficiary
Targets the underserved	Disaggregated HMIS output identifies the weak-performing districts — the strategy prioritised 184 high-priority districts across 29 states on RMNCH+A indicators (Park 28e, Ch 7, p 554)
Individual-level tracking and follow-up	Name-based tracking closes the loop for the individual mother and child, not merely the aggregate
Accountability and feedback	Sector (circle) monthly review meeting of ANMs, Anganwadi workers, ASHAs and supervisors; sub-centre reporting to <i>Panchayat</i> , hospital to <i>Rogi Kalyan Samiti</i> , district health mission to <i>Zila Parishad</i>
Research and evaluation	HMIS provides the routine denominator series against which survey rounds and special studies are interpreted

(Six generic uses of health information: Park 28e, Ch 21, pp 995-996; accountability channels: Sunder Lal, Ch 8, p 534)



3. HMIS-based score card — the formal monitoring instrument of RMNCH+A

The RMNCH+A score card has two arms (Park 28e, Ch 7, pp 557-558).

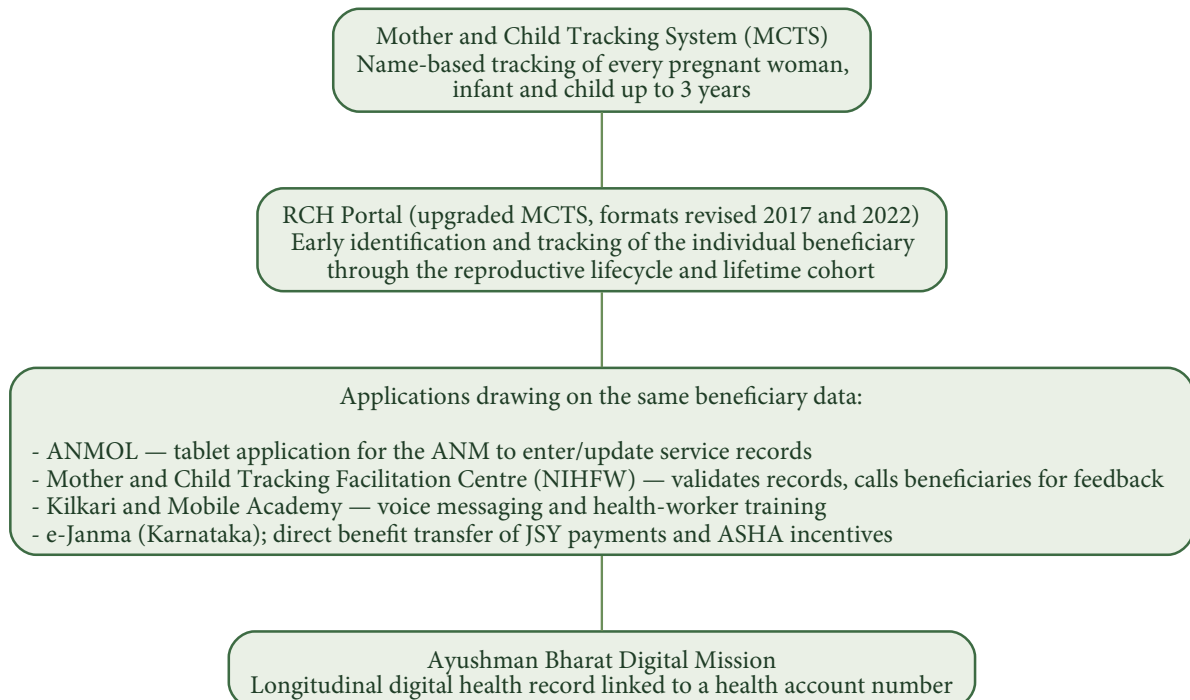
Arm	Data source	Indicator type	Examples
A. HMIS-based dashboard monitoring system — indicators chosen on the life-cycle approach	Routine HMIS returns	Process and coverage proportions, reported monthly	Proportion of first-trimester registration to total antenatal registration; women receiving 3 antenatal check-ups; women given 100 iron–folic-acid tablets; women receiving tetanus toxoid 2 or booster; obstetric complications attended to reported deliveries; institutional deliveries; postpartum sterilisation to total female sterilisation; intrauterine device insertions in public plus accredited private institutions to all family planning methods; newborns breastfed within 1 hour of birth to total live births; women discharged in <48 hours of delivery in public institutions; newborns <2.5 kg to newborns weighed at birth; newborns visited within 24 hours of home delivery; infants 0–11 months given measles vaccine to reported live births
B. Survey-based score card	Sample Registration System, Coverage Evaluation Survey, District Level Household and Facility Survey, National Family Health Survey, Census, Annual Health Survey	19 outcome and coverage indicators on health, nutrition and sanitation	Under-five, infant and neonatal mortality rates; maternal mortality ratio (per 100,000 live births); total fertility rate; births to women aged 15–19 of total births; children with birth weight <2.5 kg; children under 3 years underweight; child sex ratio 0–6; full immunisation of children 12–23 months (1 BCG, 3 pentavalent/oral polio vaccine, 1 inactivated polio vaccine each, 2 measles)

How the HMIS score card is scored (Park 28e, Ch 7, p 558):

- ▶ The all-India average of each indicator is taken as the reference point.
- ▶ Positive scores of +1 to +4 are given to states above the national average for positive indicators (and below it for negative indicators).
- ▶ Negative scores of –1 to –4 are given for the converse.
- ▶ All indicators carry the **same weightage**; indicator scores are consolidated into a single state score.
- ▶ States are classified into **four categories** on that state score.

Indicators being added include proportion of pregnant women <19 years to total antenatal registrations, home-based newborn care visits by ASHA to planned visits, children 9–11 months fully immunised to those due, children with diarrhoea treated with oral rehydration solution (and with oral rehydration solution plus zinc), children discharged alive from special newborn care units to admissions, and children with acute respiratory infection treated to those reported.

4. Name-based tracking — HMIS at the level of the individual mother and child



From aggregate HMIS reporting to name-based individual tracking under RMNCH+A

Its RMNCH+A value is that it converts a monthly count into a **named due-list**: timely antenatal care, institutional delivery and postnatal care for the mother, and immunisation for the child, with explicit tracking of **severely anaemic women, low-birth-weight babies and sick neonates** (Park 28e, Ch 7, p 540). The Mother and Child Tracking Facilitation Centre at the National Institute of Health and Family Welfare independently verifies records and telephones beneficiaries for feedback on Janani Shishu Suraksha Karyakram, Janani Suraksha Yojana, Rashtriya Bal Swasthya Karyakram and contraceptive distribution — an in-built data-validation loop (Sunder Lal, Ch 8, p 535).

5. Limitations that blunt the role, and the corrective

Limitation of routine HMIS	Consequence for RMNCH+A	Corrective
Incomplete reporting; coverage of population not total, especially urban; most care-seeking may be in the private sector	Denominator and coverage proportions are inflated or unstable	Private-sector and accredited-facility reporting; population-based surveys as the second score-card arm
Poor quality of cause-of-death data and of weighing of children	Newborn and nutrition indicators unreliable	Part C mortality reporting with probable cause; verbal-autopsy-based cause assignment

Limitation of routine HMIS	Consequence for RMNCH+A	Corrective
Tendency to over-report performance; figures inflated	Score card ranks the reporting, not the service	Data Integrity Measure — percentage deviation of HMIS-reported institutional deliveries and first-trimester antenatal registration from National Family Health Survey values, a scored indicator of the Health Index (Park 28e, Ch 2, pp 44-45)
Data seldom analysed or used locally; no feedback to the worker who collected it; used only for upward transmission	System stays "data driven rather than action driven"	Sector-level monthly review; annual district report card shared with the community
Weak training of workers and medical officers in HMIS; erratic supply of registers and formats	Errors at source	Continuing education on HMIS; integrated single register (Integrated RCH Register v2.0)

(Sunder Lal, Ch 8, p 534; private-sector care-seeking, Sunder Lal, Ch 7, p 303)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The RMNCH+A information pyramid** — label, bottom to top: *Community (ASHA, Anganwadi worker) → Sub-centre / health and wellness centre — ANM, Integrated RCH Register v2.0 → PHC / CHC — compilation and upload → District programme officer → State → National HMIS 2.0 portal*; with a downward arrow on the right labelled *Feedback and action* and a side box labelled *Score card: A. HMIS-based dashboard | B. Survey-based*.
- **The RMNCH+A 5×5 matrix** (Park 28e, Ch 7, Fig 16) — five columns labelled *Reproductive, Maternal, Newborn, Child, Adolescent*; five rows of high-impact interventions; plus a band beneath labelled *Cross-cutting and health-systems strengthening* — includes use of data for monitoring and review through HMIS-based score cards and a band labelled *Minimum essential commodities*.

High-yield exam pointers

- HMIS 2.0 launched 2020–21 with WHO technical support: ~400 service-delivery data elements monthly, ~700 infrastructure and human-resource elements annually, daily data entry since April 2020; GIS-enabled HMIS since March 2016.
- Score card = A (HMIS-based dashboard, life-cycle indicators) + B (19 survey-based outcome/coverage indicators); scoring +1 to +4 / –1 to –4 against the all-India average, equal weightage, states graded into **four categories**.
- RMNCH+A adopted 2013; **five thematic areas**; **5×5 matrix**; **184 high-priority districts** across 29 states; tenets — equity, universal care, entitlement, accountability.
- Sub-centre monthly HMIS format: **Part A** reproductive and child health (M1–M8), **Part B** facility services (M9–M11), **Part C** mortality with probable cause.

- ▶ **Data Integrity Measure** = percentage deviation of HMIS-reported data from survey (NFHS) data for institutional deliveries and first-trimester antenatal registration — quote this whenever asked about HMIS data quality.
- ▶ Name-based chain: **MCTS** → **RCH Portal** → **ANMOL** / **MCTFC** / **Kilkari** → **Ayushman Bharat Digital Mission**.
- ▶ WHO requirements of a health information system (Park 28e, Ch 21, p 995): population-based; avoids unnecessary agglomeration of data; problem-oriented; uses functional and operational terms; expresses information briefly and imaginatively; provides for **feedback**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e indexes the strategy in its current form as **RMNCAH+N** — reproductive, maternal, newborn, child and adolescent health *plus* nutrition — so the older "RMNCH+A" of the question stem and the current "RMNCAH+N" refer to the same strategy; the question's "RMNCH-A" is a typographical rendering of "RMNCH+A".

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Sample Registration System bulletins, NFHS-5, WHO ICD-11, Global Burden of Disease 2023

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandirs, and the RMNCH+A digital stack sitting on them is unevenly alive.** A 2025 audit of government digital platforms at primary-care level in Chandigarh graded the RCH portal and eVIN as effectively implemented (70–90%), eSanjeevani and the IDSP-IHIP portal as fair (51–70%), but the **HMIS portal**, the AB-HWC portal, CPHC-NCD and FP-LMIS as poorly implemented (25–50%), with **ANMOL not operational at all**; HMIS and the HWC portal were dead at two centres purely because no data entry operator had been posted — the reporting flow in the textbook diagram fails at exactly the post that fills it [PMID 40854229]
- ▶ **Anaemia Mukh Bharat scores districts entirely off HMIS returns:** 18 anaemia indicators and 6 key performance indicators reported district-wise each quarter on the AMB dashboard, consolidated into an AMB index. For 2018–19 Faridabad scored **35.2** (Haryana 28.6, India 34.0); the district HMIS held the numerators but its only denominator was the number of registered pregnant women, and substituting Census-derived denominators for AMB dashboard ones cut adolescent IFA coverage from 85.1% to 14.1% and halved the index to 17.6 — on an HMIS-driven score card the denominator, not the numerator, is usually where it breaks [PMID 39764332]
- ▶ **Maternal death surveillance and response (MDSR) is the correction now being layered onto HMIS Part C mortality reporting.** Rajasthan reported 8,784 maternal deaths over 2014–21 and the HMIS-derived MMR trend actually *rose* across the period; verbal autopsy of 248 deaths in Jodhpur and Udaipur attributed about 40% to obstetric haemorrhage. The authors read a low reported MMR as an artefact of weak reporting and absent surveillance rather than a real field value [PMID 42374271]

Current national data

- ▶ Always quote MMR with its SRS round, and pair the figure with the right round. SRS Special Bulletins give 130 (2014–16) → 122 (2015–17) → 113 (2016–18) → 103 (2017–19) → **97 per 100,000 live births (2018–20)**, and at that round a maternal mortality rate of 6.0 per 100,000 women aged 15–49 with a lifetime risk of 0.2%; the SDG 3.1 target is under 70 by 2030 [PMID 40547788]. The next round, SRS 2019–21, takes MMR to **93 per 100,000 live births** [PMID 41299336]
- ▶ **SRS 2023 puts IMR at 25 per 1,000 live births, down from 40 in 2013.** HMIS monthly mortality data for 2018–20 were used to derive a pneumonia-specific infant mortality rate — national monthly mean 36.4 (July) to 66.7 (January) per 100,000 live births, and 12.4 (August) to 107.7 (November) in Himachal Pradesh — showing HMIS supporting seasonal micro-planning, though the authors themselves caution that HMIS series need substantial correction before analysis [PMID 42056953]

Elimination, surveillance and disease status

- ▶ **India's stillbirth burden is far larger than the routine systems record.** GBD 2023 estimated a stillbirth rate of 25.9 (95% UI 21.3–32.0) at the WHO ICD-11 threshold of ≥22 weeks versus 16.1 (13.2–19.8) at ≥28 weeks — 565,900 against 347,300 stillbirths — with a four-fold spread between states (Mizoram 9.3, Uttar Pradesh 38.2). SRS 2023 late-gestation SBR was **2.3 times lower** than GBD while NMR matched almost exactly (ratio 1.0): it is stillbirths specifically, not mortality in general, that the vital systems miss [PMID 42565145]
- ▶ **HMIS itself can localise the same problem to the district, which is its distinctive value over SRS.** District HMIS data gave an all-India stillbirth rate of 13.4, 13.1 and 12.4 per 1,000 total births in 2017–18, 2018–19 and 2019–20; local indicator of spatial association (LISA) mapping found a persistent contiguous high-SBR belt across **Odisha–Madhya Pradesh–Rajasthan–Chhattisgarh**, and bivariate LISA against NFHS-4 linked it positively to low maternal BMI (Moran's I 0.24) and anaemia in pregnancy (0.20) and inversely to full ANC (–0.24), 100-day IFA (–0.23) and caesarean delivery (–0.29). Implausibly low figures from Uttar Pradesh and Bihar track their 70% share of home deliveries — a reminder that HMIS is facility-based [PMID 37383033]

Guideline and legislative updates

- ▶ The "external consistency" dimension of the WHO data quality report card has now been applied to Indian HMIS at scale — the evidence base behind the Data Integrity Measure. Across 695 districts, HMIS agreed closely with NFHS-5 for caesarean section ($r = 0.892$) and institutional births ($r = 0.758$), moderately for 4 antenatal visits (0.532), but weakly for early (first-trimester) antenatal registration (0.356), tetanus toxoid immunisation (0.225), sex ratio at birth (0.148) and anaemia in pregnancy (0.135, 564 districts); the intraclass correlation coefficient was below 0.5 and near zero for every indicator except institutional births and caesarean section [PMID 40547776]
- ▶ **Birthweight — the score-card element "newborns <2.5 kg to newborns weighed at birth" — remains the least trustworthy link in the chain.** HMIS reported low birth weight at 12.5% in both 2015–16 and 2018–19 while covering only 58% of live births (birthweight missing for roughly 40% throughout, 55% in Uttar Pradesh), against 18% from NFHS-4 and the

Comprehensive National Nutrition Survey; missing birthweight was 22% in NFHS-4 and 29% in CNNS, concentrated among the poorest (42% vs 6% in the richest quintile) and Scheduled Tribes and Castes, with 17–20% of newborns recorded at exactly 2,500 g — heaping on the low-birth-weight cut-off itself [PMID 36160303]

Global and WHO developments

- **Digital-health equity is the emerging critique of platforms of the RCH portal / ABDM type.** Co-WIN's smartphone-dependent registration systematically excluded elderly and rural users lacking devices or digital literacy, while the Ayushman Bharat Digital Mission (launched 2021) has met fragmented governance, uneven state-level adoption, limited provider capacity and adoption concentrated in urban tertiary hospitals with little penetration into rural primary health centres — the warning being that digitising RMNCH+A beneficiary tracking can widen the very inequities the score card exists to expose [PMID 42215992]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 7 (Health Programmes in India, pp 534, 540, 553-558, 563), Ch 21 (Health Information and Basic Medical Statistics, pp 995-998), Ch 2 (Concept of Health and Disease, pp 44-45), Ch 23 (Health Planning and Management, p 1028); Sunder Lal — Textbook of Community Medicine, Ch 7 (Epidemiology, pp 302-303) and Ch 8 (pp 532-535); Government of India RMNCH+A Strategic Approach 2013 and Integrated RCH Register version 2.0 (2022) as cited in those chapters.

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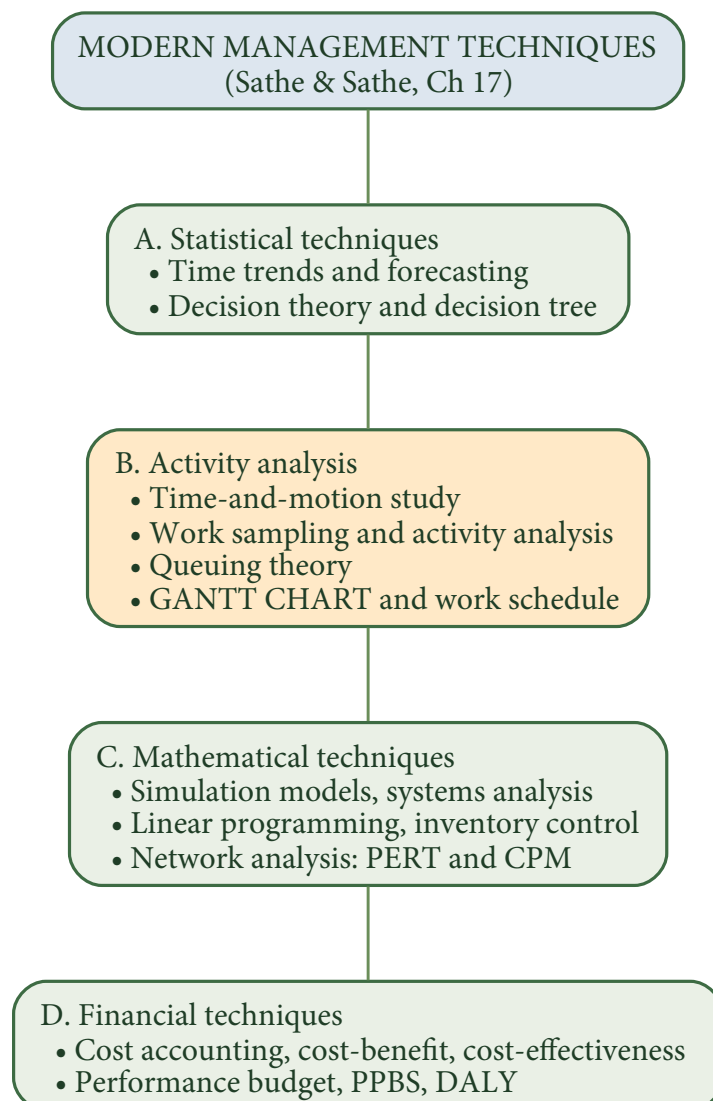
♦ ♦ ♦

Gantt chart

Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — A **Gantt chart** (bar chart or work-schedule chart) is a project-management tool in which every activity of a project is drawn as a horizontal bar against a calendar time-scale: the left end of the bar marks when the activity begins and the length of the bar is proportional to the time the activity takes. It is used in planning, coordinating and monitoring the progress of an activity that involves several parallel sub-activities, and a glance at the chart shows the progress achieved on any given date compared with what was to have been achieved according to plan (Sathe & Sathe, Ch 17, pp 184-185). It is named after **Henry L. Gantt**, one of the pioneers of scientific management, and belongs to the **activity-analysis** group of modern management techniques.

Place among management methods



Where the Gantt chart sits among modern management techniques

How a Gantt chart is constructed

- **Enumerate the activities** — break the project into discrete sub-activities, each with a clearly defined end point (an *event*, e.g. "surveyors trained", "forms printed").

- ▶ **Estimate the time** required for each activity, in the unit that suits the project — days, weeks or months.
- ▶ **Fix the sequence** — decide which activities must follow one another and which can be run in parallel.
- ▶ **Draw the time-flow scale along the top of the chart** (X-axis, the calendar).
- ▶ **List the activities down the left margin** (Y-axis), in the order in which they are to be taken up.
- ▶ **Draw one horizontal bar per activity** — the left end at the date of initiation, the length proportional to the period indicated on the time scale at the top.
- ▶ **Mark the current date with a heavy vertical line**, which is moved one step to the right day by day.
- ▶ **Post actual progress** against each bar (a second or shaded bar), so that the gap between work planned and work carried out is visible on any date (Sathe & Sathe, Ch 17, pp 184-185).
- ▶ **Show slack where the chart is used with network analysis** — a dotted extension of the bar indicates how far completion may be delayed without repercussions on total project time (Reinke, Ch 15, pp 224-226); dark bars are conventionally used for critical tasks and light bars for non-critical tasks (Handbook of Epidemiology, Ch 12).
- ▶ **For a campaign with a fixed deadline, plan backwards** from the deadline to the present so that each activity acquires its latest permissible start date (Intensified Mission Indradhanush guidelines).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Gantt chart** — label: X-axis "Time" (calendar divisions across the top: March 20–24, March 25–31, April 1–10, April 11–20, April 21–24, April 25); Y-axis "Nature of activity" (activities a to h listed downwards); one solid horizontal bar per activity, left end = start date, length = duration; overlapping bars = activities running in parallel; heavy vertical line = today's date; dotted extension of a bar = permissible delay (slack).

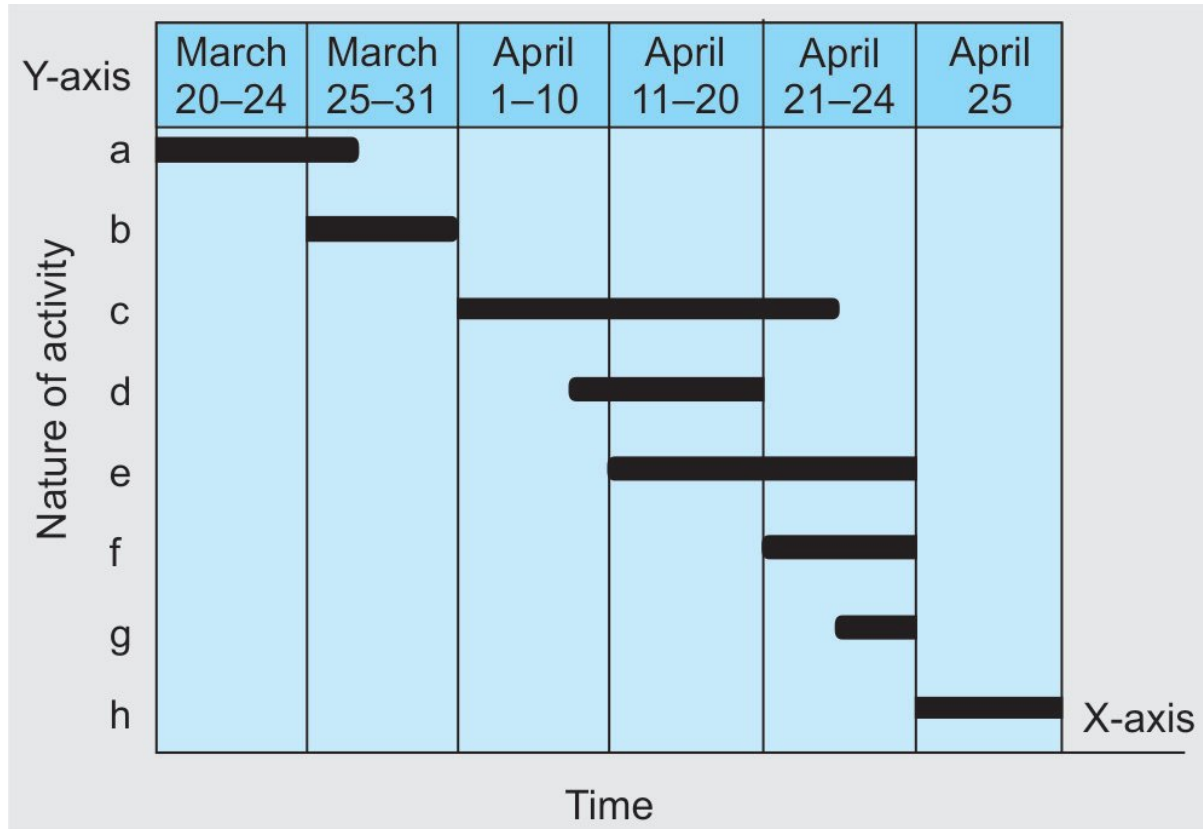
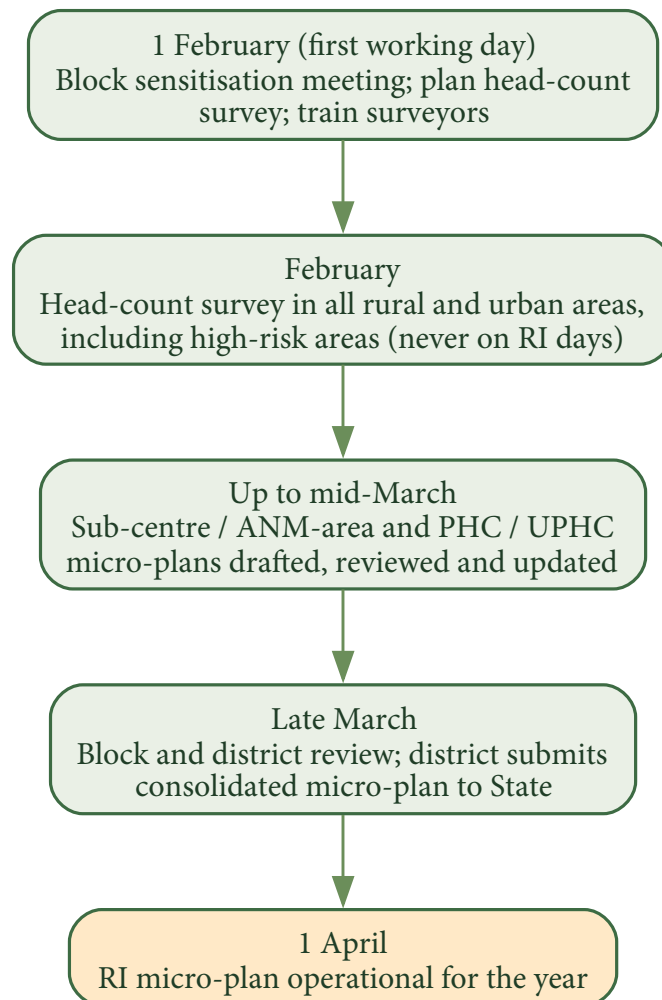


Fig 37.1 — Gantt chart (Suryakantha Environment Best, p.805)

Uses in health management, with programme examples

Managerial function	How the Gantt chart serves it	Example from practice
Planning	Converts a plan of action into dated sub-activities with named responsibility, and exposes activities for which time is insufficient	Modified Gantt chart used to time all activities preceding an eye camp so that surgery starts on 1 April (Sathe & Sathe, Ch 17)
Scheduling and coordination	Displays which sub-activities run in parallel and which must wait, so that manpower and supplies are matched to dates	Routine immunisation micro-plan timeline, Fig 4.3 (Routine Immunization Manual for Medical Officers 2024, Ch 4)
Monitoring	Is one of the four recognised sources of monitoring information, along with routine sub-centre/PHC/CHC reports, supervisory field visits and community monitoring	Sunder Lal, Ch 16, p 723
Control	The moving current-date line makes slippage visible the day it occurs, allowing corrective action before the deadline is lost	Sathe & Sathe, Ch 17
Evaluation	Listed among the monitoring tools for programme evaluation, with management by objectives, PERT and CPM	Jugal Kishore, National Health Programmes of India

Managerial function	How the Gantt chart serves it	Example from practice
Research management	Lays out the time-line of a study — protocol clearance, staff training, data collection, entry, analysis, report — and marks the critical tasks that would delay the whole study	Handbook of Epidemiology, Ch 12
Campaign management	Mandated for demand-generation and communication activities of intensified immunisation drives, prepared by working backwards from the deadline	Intensified Mission Indradhanush guidelines



A Gantt chart in use — the routine immunisation micro-plan calendar (Routine Immunization Manual for Medical Officers 2024, Ch 4)

Gantt chart compared with network techniques (PERT and CPM)

Feature	Gantt chart	Network analysis — PERT / CPM
Form	Horizontal bars of activities plotted on a calendar scale	Arrow diagram of events and activities (Park 28e, Ch 23, p 1029)
Primary information	<i>When</i> each activity starts, how long it lasts, and how much of it is done	<i>What depends on what</i> — the logical sequence in which events must take place

Feature	Gantt chart	Network analysis — PERT / CPM
Interdependence of activities	Not displayed; must be known to the manager beforehand	Displayed explicitly; the two common types are PERT and CPM
Critical activities	Not identified unless the chart is drawn alongside a network	Longest path through the network is the critical path ; delay of any activity on it delays the whole project
Time estimate	Single deterministic estimate per activity	CPM uses a single estimate; PERT uses a weighted average of the optimistic, most likely and pessimistic estimates, the most likely being given four times the weight of either other (Reinke, Ch 15, pp 225-226)
Best suited to	Projects whose activities are relatively limited and repetitive	Large, complex, one-off projects with many interdependent tasks
Relationship	Historically the parent technique — network analysis grew out of Gantt charting, and the two are used together, the network fixing the sequence and the Gantt chart fixing the calendar (Reinke, Ch 15, pp 224-225)	

Advantages and limitations

Advantages	Limitations
Simple, pictorial and immediately intelligible; needs no statistical training	Does not show precedence or interdependence between activities
Shows planned work and work actually carried out on the same scale, so progress is read off on any date	Does not by itself identify the critical path or the slack available on non-critical activities
Fixes accountability — who is to do what, by when	Suitable only when activities are relatively limited and repetitive; unwieldy for very large projects
Permits backward planning from a fixed deadline	Concentrates on time; gives almost no attention to cost — the same criticism made of PERT (Sunder Lal, Ch 16, p 726)
Can be modified to track sub-activities that are unrelated to one another but all essential to one project	Depicts duration but not the quantum of manpower, money or material consumed
Doubles as a monitoring and supervisory instrument at sub-centre, PHC and district level	Becomes misleading unless the progress bars and the current-date line are updated continuously

High-yield exam pointers

- Gantt chart = activity on the Y-axis, time on the X-axis, one horizontal bar per activity; named after Henry L. Gantt of the scientific-management school.
- Classified under **activity analysis** (with time-and-motion study, work sampling and queuing theory) — *not* under network analysis.

- ▶ Heavy vertical line = today's date; the gap between the planned bar and the progress bar = slippage; dotted extension = permissible delay.
- ▶ Gantt tells you **when**; PERT/CPM tells you **what depends on what**. Critical path = longest path through the network; a delay on it delays the entire project (Park 28e, Ch 23, p 1029).
- ▶ PERT expected time = (optimistic + 4 × most likely + pessimistic) ÷ 6 — the most likely estimate carries four times the weight of either other.
- ▶ Sunder Lal's four sources of monitoring information: routine reports, supervisory field visits, **Gantt chart**, community monitoring.
- ▶ Indian programme use to quote: Intensified Mission Indradhanush requires a Gantt chart for all activities worked backwards from the deadline; the Routine Immunization Manual for Medical Officers 2024 gives a Gantt chart for micro-planning — head-count survey through February, micro-plans updated by mid-March, plan operational from 1 April.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook 28e, Chapter 23 covers management methods and network analysis (PERT and CPM) but does not name the Gantt chart, so the description above is grounded in Sathe & Sathe, Suryakantha, Sunder Lal, Reinke and the Handbook of Epidemiology; indexed textbooks also differ on the originator's given name (Sathe & Sathe write Henry L. Gantt, the Handbook of Epidemiology writes "Charles Gantt, 1917"), the standard attribution being Henry Laurence Gantt.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed

Evidence on the chart itself

- ▶ **How well the bar chart is actually read** — two in-lab experiments with 93 participants compared Gantt charts, extended Gantt charts and stringline charts on event sequences of fixed order, scoring task error rate and completion time. On a single sequence, Gantt and extended Gantt charts gave comparable error rates in the duration-comparing task, Gantt charts were read in the same time or faster than extended Gantt charts, and both were faster than stringline charts; but where the number of event types was high, stringline charts made fewer errors in the comparing task, and across multiple point-event sequences stringline charts took less time than Gantt charts to reveal temporal patterns. On this evidence the plain bar-per-activity form is the quicker read for comparing how long one activity lasts, and the stringline form the quicker read for spotting patterns across many sequences [PMID 38277251].
- ▶ **The chart is now computed rather than drawn** — Staggr, an open-source R/Shiny application, treats a staggered laboratory protocol as a flow-shop scheduling problem: the user enters each operation's hands-on time and the waiting time that follows it, and the program raises the start-to-start interval in fixed increments (default 0.25 min) until it finds the shortest interval at which the single operator is never booked for two tasks at once. The output is a colour-coded Gantt chart of the work schedule, a downloadable step-by-step timetable, and a live timer that highlights the current step while the protocol runs; if the user forces a shorter interval, the clashes are drawn in red. In the worked eight-time-course example a 6-minute harvest step

needed a 28-minute stagger and a 10-minute harvest step an 84-minute stagger, which the author flags as a potential bottleneck [PMID 41822906].

Programme and policy updates (India)

- ▶ **A national framework whose stated first remedy is a Gantt chart** — India's **National Patient Safety Implementation Framework (2017)** set six strategic objectives, each with a timeline, responsible organisations, short-, mid- and long-term priority areas and expected outputs, to be delivered by 2025. A qualitative study of 80 key-informant and in-depth interviews across 18 secondary- and tertiary-level public facilities in Tamil Nadu (August–October 2021) found the facilities largely unprepared — staff shortages, deficient infrastructure and equipment, low awareness of patient safety, non-compliance with standard guidelines — and concluded that full-scale implementation by 2025 would be difficult. Its recommended first step is to form a state-level core patient safety committee and draw up a Gantt chart for implementation ordered by priority [PMID 38135519].
- ▶ **The Gantt chart is a required component of a research grant application** — a guidance article written for Indian biomedical investigators, who receive little formal training in grant writing, lists the Gantt chart among the contents a research grant application should carry, alongside aims and objectives, preliminary work, novelty and translational value, materials and methods with standard operating procedures, budget, investigator and referee details, mentor, approvals and permissions, and facilities available [PMID 37778964].

Global and WHO developments

- ▶ **Operational microplanning for high-quality vaccination activities** — the PAHO/WHO planning model adopted as national guidance by Brazil's health ministry in 2023 — was scored on a composite adherence indicator built from self-reported REDCap returns (a coordination committee, regular meetings between health surveillance and primary care, and use of a management tool for decision-making and strategic planning such as SWOT). Full adherence was reported by 45.75% of municipalities and 37.04% of states. In states with full adherence, most municipalities met the 2023 coverage targets for every vaccine examined, reaching statistical significance for pentavalent 3rd dose, polio 3rd dose and PCV10 2nd dose — an association from an ecological design, not proof of causation: the municipal-level analysis did not yield statistically significant results, and the authors' own primary concern is that financial incentives tied to microplanning and self-reported data collection may have led to over-reported compliance [PMID 39827108].
- ▶ **Micro-planning with a geospatial layer** — a cross-sectional household survey of 4,560 households across 31 regions of Tanzania found 85.5% of children aged 12–23 months fully immunised on the recommended schedule, 14.5% having failed to complete it in 2019; coverage was associated with a household of one to six members, higher household wealth, subsistence farming by the head of the family, caretaker's age, child's age, educational level, the child holding a clinic card, and receipt of DPT-3. The authors advise using the periodic micro-planning tool together with satellite image technology, particularly where immunisation coverage is below 80% [PMID 39378889].

Research management, surveillance and evidence synthesis

- **The chart repurposed to detect duplicate patients** — while preparing a systematic review of COVID-19 in pregnancy, investigators plotted each published study as a bar spanning its participant-enrolment dates, ordered by continent, country, city and hospital, with national registries as the darkest “parent” bars and the smaller hospital series treated as their sub-activities. Of 132 articles covering 12,877 pregnancies, 16 (12%) were excluded for missing dates, leaving 116 articles and 12,833 pregnancies (99.7%) charted. Bars overlapping within one hospital exposed probable double reporting: in a New York City worked example the caesarean-section incidence read 39% across all 11 studies, 40% across four studies with no evidence of overlap, and 35% when only the largest series from each hospital was retained [PMID 35908036].
- **Gantt charts in outbreak investigation** — a scoping review screened 1,957 Medline records and analysed 30 healthcare-outbreak papers carrying 37 transmission visualisations (median cluster size 20.5 people, range 7–1,963; median duration 214 days). Transmission networks accounted for 11 (30%), floor-plan transmission maps 10 (27%), timelines 6 (16%), cluster maps 4 (11%), Gantt charts 3 (8%) and other types 4 (11%). Gantt charts, timelines, cluster maps and transmission networks were used most often for viral clusters, while bacterial clusters favoured case-proximity maps, floor-plan maps and transmission networks. Across all 37 visualisations none carried the contagious period, only 2 (5%) carried the symptom-onset date, and only 2 named the software that had produced them [PMID 36483443].
- **Drafting the project timeline with a large language model** — a case study comparing a Cochrane-guided systematic review workflow with a ChatGPT-4-assisted one reported substantial time savings on the operational tasks, the Gantt chart among them, together with generating search terms and suggesting selection criteria, while the stages requiring interpretative judgement — analysing results, assessing risk of bias and final drafting — raised critical problems [PMID 41041739].

SOURCE & COVERAGE

Sathe & Sathe — Epidemiology and Management for Health Care, Ch 17 (Modern management techniques); Suryakantha — Community Medicine with Recent Advances, Ch 37 (Health Planning and Management), Fig 37.1; Sunder Lal — Textbook of Community Medicine, Ch 16 (Health Planning and Management); Reinke — Health Planning for Effective Management, Ch 15 (Tools for Improving the Process of Planning); Park's Textbook of Preventive and Social Medicine 28e, Ch 23 (Health Planning and Management); Handbook of Epidemiology (Ahrens & Pigeot), Ch 12 (Design and Planning of Epidemiological Studies); Jugal Kishore — National Health Programmes of India; Routine Immunization I Manual for Medical Officers 2024 (Ch 4, micro-planning); Intensified Mission Indradhanush operational guidelines.

Recent-advances sources: 38277251 — A Comparative Study on Fixed-Order Event Sequence Visualizations: Gantt, Extended Gantt, and Stringline Charts, IEEE Transactions on Visualization and Computer Graphics 2024, DOI [10.1109/TVCG.2024.3358919](https://doi.org/10.1109/TVCG.2024.3358919) (abstract-only); 41822906 — Staggr: an interactive R/Shiny application for planning and visualizing staggered experimental protocols, F1000Research 2026, DOI [10.12688/f1000research.168987.3](https://doi.org/10.12688/f1000research.168987.3); 38135519 — Barriers and Facilitators to Implementing the National Patient Safety Implementation Framework in Public Health Facilities in Tamil Nadu: A Qualitative Study, Global Health: Science and Practice 2023, DOI [10.9745/GHSP-D-22-00564](https://doi.org/10.9745/GHSP-D-22-00564); 37778964 — How to write a research grant proposal, Indian Journal of Medical Microbiology 2023, DOI [10.1016/j.ijmmb.2023.100482](https://doi.org/10.1016/j.ijmmb.2023.100482) (abstract-only); 39827108 — Evaluation of the adherence of municipalities and states to the Ministry of Health's microplanning for high-quality vaccination activities and the increase in vaccination coverage in Brazil, BMC Public Health 2025, DOI [10.1186/s12889-025-21326-1](https://doi.org/10.1186/s12889-025-21326-1); 39378889 — Routine Immunization Status and Factors Associated with Immunization Coverage among Children Aged 12-23 Months in Tanzania, American Journal of Tropical Medicine and Hygiene 2024, DOI [10.4269/ajtmh.23-0563](https://doi.org/10.4269/ajtmh.23-0563); 35908036 — Exploring the Gantt chart as a tool to highlight double report in case series published during the first wave of the COVID-19 pandemic, Systematic Reviews 2022, DOI [10.1186/s13643-022-02024-0](https://doi.org/10.1186/s13643-022-02024-0); 36483443 — Transmission visualizations of healthcare infection clusters: A scoping review, Antimicrobial Stewardship & Healthcare Epidemiology 2022, DOI [10.1017/ash.2022.237](https://doi.org/10.1017/ash.2022.237); 41041739 — Leveraging ChatGPT-4 for Evidence Synthesis: A Case Study on the Use of a Large Language Model in a Systematic Review, Studies in Health Technology and Informatics 2025, DOI [10.3233/SHTI251488](https://doi.org/10.3233/SHTI251488) (abstract-only).



Global positioning system (GPS) and health

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — The **global positioning system (GPS)** is a satellite-based navigation technology that returns the exact spatial co-ordinates — **latitude and longitude** — of any point on the earth's surface, in real time, from a hand-held receiver or a mobile phone. In public health it is the **data-capture arm** of spatial epidemiology: GPS supplies the geo-reference (the "where"), a **geographic information system (GIS)** stores, links and analyses that reference against health databases, and **remote sensing (RS)** supplies the environmental backdrop. GIS is defined as a system combining high-precision location information with health databases and analysis tools, so as to visualise the geographic distribution of health indicators such as morbidity, mortality, prevalence and incidence (Oxford Textbook of Public Health 7e, Ch 24, pp 946–947).

The three spatial technologies — what each contributes

Technology	What it actually delivers	Health use it makes possible
Global positioning system (GPS)	Point co-ordinates (latitude, longitude) of a household, session site, facility, vehicle or breeding site; also real-time tracking of a moving object	Geo-tagging of cases and facilities; household mapping for repeat follow-up; verifying that a primary health care provider is where he or she reports being (Oxford Textbook of Public Health 7e, Ch 102, p 3911)
Geographic information system (GIS)	A store of spatial data that also holds the spatial relationships between geographic features, organised as overlaid information layers for one area — base/satellite layers plus thematic layers such as cases and health services (Nelson & Williams, Ch 7, p 188; Park Plus, Ch 24, p 360)	Spot maps, risk maps, service-distribution maps, cluster detection, catchment and accessibility analysis
Remote sensing (RS)	Satellite imagery of land use, vegetation, water bodies and temperature. Landsat Thematic Mapper: each region every 16 days, 7 spectral bands, ~30 m resolution; AVHRR: twice daily, 5 bands, 1.1 km resolution (Nelson & Williams, Ch 7, pp 189–190)	Environmental stratification of vector-borne disease risk; identifying breeding habitats without ground visits

Two data formats inside a GIS

Format	Structure	Suits
Vector	Features stored as discrete points, lines and polygons with exact co-ordinates (Nelson & Williams, Ch 7, p 188)	Addresses, roads, administrative boundaries, GPS-captured points
Raster	A uniform grid of cells , each cell carrying one value; location is the row–column position, so feature edges are "fuzzy" and accuracy equals grid resolution (Nelson & Williams, Ch 7, pp 188–189)	Satellite imagery and continuous environmental surfaces

Historical line: Hippocrates' *On Airs, Waters and Places* first linked place to health; *John Snow's* 1854 **spot map** of the Golden Square district, London, localised cholera deaths around the **Broad Street pump**

and let him hypothesise water-borne spread long before the birth of bacteriology (Park 28e, Ch 3, pp 86–87); Howard T. Fisher's **SYMAP** (Synagraphic Mapping System, 1963) was the first computerised mapping system (Park Plus, Ch 24, pp 360–361). GPS is simply the step that made the "spot" on Snow's map an objective, reproducible co-ordinate instead of a hand-drawn dot.

Applications of GPS in health

Domain	How GPS is used	Grounding
Descriptive epidemiology and outbreak investigation	Geo-tagging every case at the point of interview replaces address-matching, so the analysis "by place" and the spot map are built from true co-ordinates; distance of cases from a suspected source and from one another (the "problem of adjacency") can then be queried statistically	Nelson & Williams, Ch 7, p 194; Park 28e, Ch 3, p 160
Field surveys and demographic surveillance (DSS)	Hand-drawn maps combined with hand-held GPS map and number every household in a fixed geographical area, so the same house is relocated at each follow-up visit; the longitudes and latitudes are then used within a GIS platform to explore environmental and health-service determinants of risk	Gilles & Warrell, <i>Essential Malariology</i> 4e, Ch 5, pp 109–110
Immunisation service delivery	" GPS coordinates of the facility " is a recorded header field on the Sub District Cold Chain Monitoring Checklist	Routine Immunization Manual 2024, Ch 17, pp 410–411
Campaign and outbreak-response vaccination	Electronic tablets for data collection combined with GPS tracking allowed remote review of data quality, recording of household visits, and spatial tracking with instant feedback to vaccination teams in the 2015 Iraq cholera response (two-dose oral cholera vaccine coverage 87%, 95% CI 85–89, in internally-displaced-person and refugee camps)	WHO — <i>Integrating Migrants and Refugees into National Immunization Programmes</i> , pp 55–56
Vaccine supply chain	With GPS maps available, supply-chain models can be run against the actual geography and demography of a country or district — vial presentation, storage decentralisation, distribution circuits and shipment frequency	Plotkin's <i>Vaccines</i> 7e, Ch 76, p 2154
Health workforce monitoring (mHealth)	Mobile phones used with GPS monitor primary health care provider performance by assuring they are where they report being , with check-in and check-out for each home visit — a low-cost accountability tool for developing countries pursuing universal health coverage	Oxford Textbook of Public Health 7e, Ch 102, p 3911
Environmental health and waste management	GPS synchronised with GIS gives real-time data on vehicle movement , bin pickup and delivery of waste to the treatment or disposal facility. <i>Indian example</i> : Nashik Municipal Corporation fitted GPS on all 124 ghantagadis (2011, under JnNURM) with a centralised monitoring unit, tracking stoppage time and generating daily/weekly reports; Pimpri-Chinchwad Municipal Corporation runs a GPS-based real-time vehicle-tracking system	Solid Waste Management, Ch 79, p 518; Ch 80, pp 519, 521–523

Domain	How GPS is used	Grounding
Occupational health	Lone-worker protection: a link to a central control room, with or without GPS, for mobile workers as a control measure against workplace violence	Oxford Handbook of Occupational Health 2e, Ch 5, p 168
Humanitarian and disaster settings	Surveys, interviews and GPS locators in post-coup Haiti generated estimates of 8,000 murders and 35,000 sexual assaults in the Port-au-Prince area over 22 months (Kolbe and Hutson 2006)	Textbook of International Health, Ch 8, p 436
Community participation	In public participatory GIS (PPGIS) community members themselves use cell phones or portable GPS units to mark informal gathering places, environmental contaminants and known disease vectors — "community asset mapping"; it guided siting of primary care for nomadic tribes in Chad and malaria vector suppression in Tanzania	Lilienfeld's <i>Foundations of Epidemiology</i> , Ch 14, pp 291–292

GPS written into Indian rules and policy

Instrument	The GPS requirement
Bio-Medical Waste Management Rules, 2016 — Rule 5(c), duties of the operator of a common bio-medical waste treatment and disposal facility	"establish bar coding and global positioning system for handling of bio-medical waste within one year " (BMW Rules 2016, pp 5–6)
Bio-Medical Waste Management Rules, 2016 — Rule 8(4), segregation, packaging, transportation and storage	"Bar code and global positioning system shall be added by the Occupier and common bio-medical waste treatment facility in one year time " (BMW Rules 2016, p 8)
Bio-Medical Waste Management Rules, 2016 — Form II (see rule 10), Application for authorisation or renewal of authorisation, item 4(ii)	"GPS coordinates of health care facility (HCF) or common bio-medical waste treatment facility (CBWTF) "; the applicant must also attach "a map with GPS locations of CBMWTF and area of coverage" (BMW Rules 2016, pp 30–31)
Bio-Medical Waste Management Rules, 2016 — Form IV (see rule 13), Annual report (due to the prescribed authority by 30 June each year), item 1(viii)	"GPS coordinates of HCF or CBMWTF" (BMW Rules 2016, pp 33–34)
Routine Immunization Manual for Medical Officers, 2024 — Sub District Cold Chain Monitoring Checklist	"GPS coordinates of the facility" recorded in the identification header, alongside cold chain point name, urban/rural location and facility type (Ch 17, pp 410–411)
Twelfth Five Year Plan (2012–17), health chapter — drug regulation	"E-governance systems that inter-connect all licensing and registration offices and laboratories, GPS based sample collection systems and online applications for licensing would be introduced" (Jugal Kishore, Ch 1, p 89)

Frequently quoted with GPS, but actually GIS or remote sensing

Examiners reward the candidate who does not blur the three technologies. These three commonly-cited Indian and international examples are **not** GPS results.

Example	What the source actually says
ICMR–ISRO Delhi study	A feasibility study using satellite (remote sensing) data with the Indian Space Research Organisation in and around Delhi found a significant correlation of changes in land-use features (water bodies, vegetation) with mosquito density; coupled with GIS , remote sensing is expected to drive macrostratification of malarious areas for cost-effective prioritisation of control (Jugal Kishore, Ch 1, p 420; ICMR 2000). GPS is not named
Integrated Disease Surveillance Programme (IDSP)	A recommendation in the vaccine-policy discussion: "Environmental surveillance using technology like Geographical Imaging System (GIS) and Remote Sensing (RS) should also be used to support sentinel surveillance" (Jugal Kishore, Ch 2, p 772) — a proposal, not a rule, and no GPS
IPEN project (International Physical Activity and the Environment Network, from 2004, 12 countries)	Measured physical activity by accelerometer and employed geographic information system databases to give objective indices of built- and natural-environment attributes (Oxford Textbook of Public Health 7e, Ch 63, pp 2369–2370) — GIS-derived, not GPS

Limitations, sources of error and ethical concerns

Problem	Explanation
Spatial error	Misplacement of a mapped feature relative to its true co-ordinate — from data-entry mistakes or from address-matching methods. Digitised street maps using an "arc-node" design place all house numbers of a block within its lower one-fifth (Nelson & Williams, Ch 7, pp 189–190)
Resolution error	Every database has a minimal mapping unit ; features finer than that unit simply do not exist in the map
Interpretive error	Land cover classified from satellite reflectance shows 15–20% error rates when "ground-truthed", especially in complex residential or rural environments (Nelson & Williams, Ch 7, p 191)
Temporal error	Spatial databases age; the map may not represent the environment at the time of exposure — worst where the incubation or latent period is long
Attribution of place	For diseases such as La Crosse encephalitis, the site of exposure and the place of residence differ, so a residence-based map misleads (Nelson & Williams, Ch 7, pp 190–191)
Confidentiality and privacy	A co-ordinate is effectively a direct identifier — mapping to household level can re-identify a person. People have suffered legal, social, physical and psychological harm after public disclosure of health data , and population-level health data have led to mandatory quarantine, isolation of groups and seizure of property (Oxford Textbook of Public Health 7e, Ch 42, pp 1576–1577)
Equity risk	Integrated spatial data can site emergency health services where they are most needed, but the same data can produce biased algorithms and discriminatory targeting, violating rights to privacy and justice (Oxford Textbook of Public Health 7e, Ch 19, p 762)
No hypothesis testing	Spatial data suit inferential statistics poorly — each geographic unit is a population, not a sample, and places are neither randomly located nor independent of their neighbours (Lilienfeld's <i>Foundations of Epidemiology</i> , Ch 14, p 291)

Problem	Explanation
Practical constraints	Cost of receivers and software, need for trained interpreters of remotely sensed imagery, poor data-communication infrastructure, and staff training

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The GIS layer-stack (GPS as the bottom input).** Draw four stacked transparent rectangles over one map outline, labelled bottom to top: (1) **Base map layer** — roads, rivers, boundaries; (2) **Remote-sensing layer** — land use, water bodies, vegetation; (3) **Thematic health layer** — GPS-tagged cases, sub-centre, PHC, cold chain point, session sites; (4) **Denominator layer** — census population by ward. Add a hand-held GPS receiver at the side with an arrow labelled "latitude, longitude" feeding layer 3, and an output arrow at the top labelled "overlay → risk map → targeted action".
- **Snow's spot map (the ancestor).** A street grid of the Golden Square district with **dots** = cholera deaths, **crosses** = water pumps, the **Broad Street pump** ringed and labelled as the source (Park 28e, Ch 3, Fig 6, p 87).

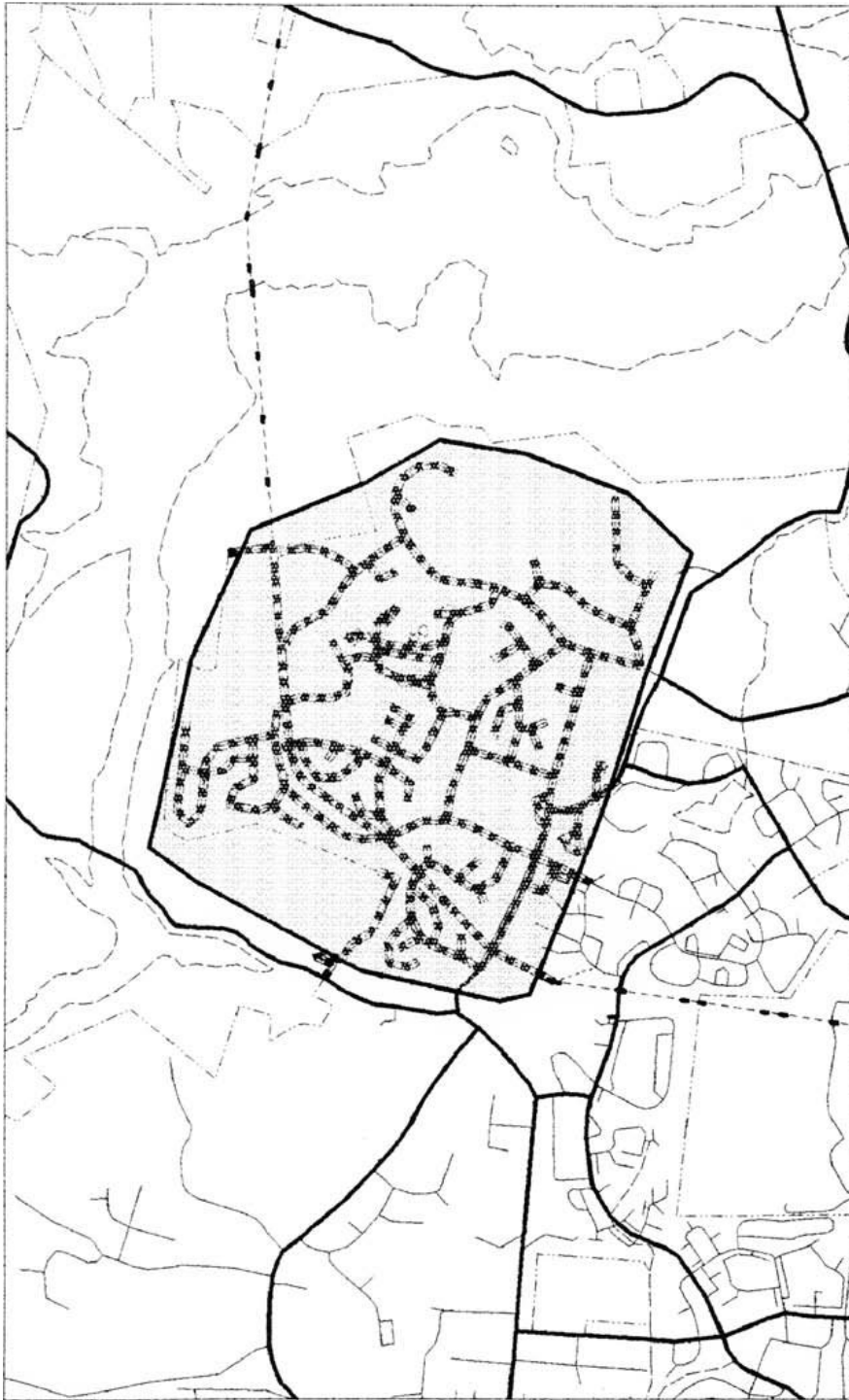
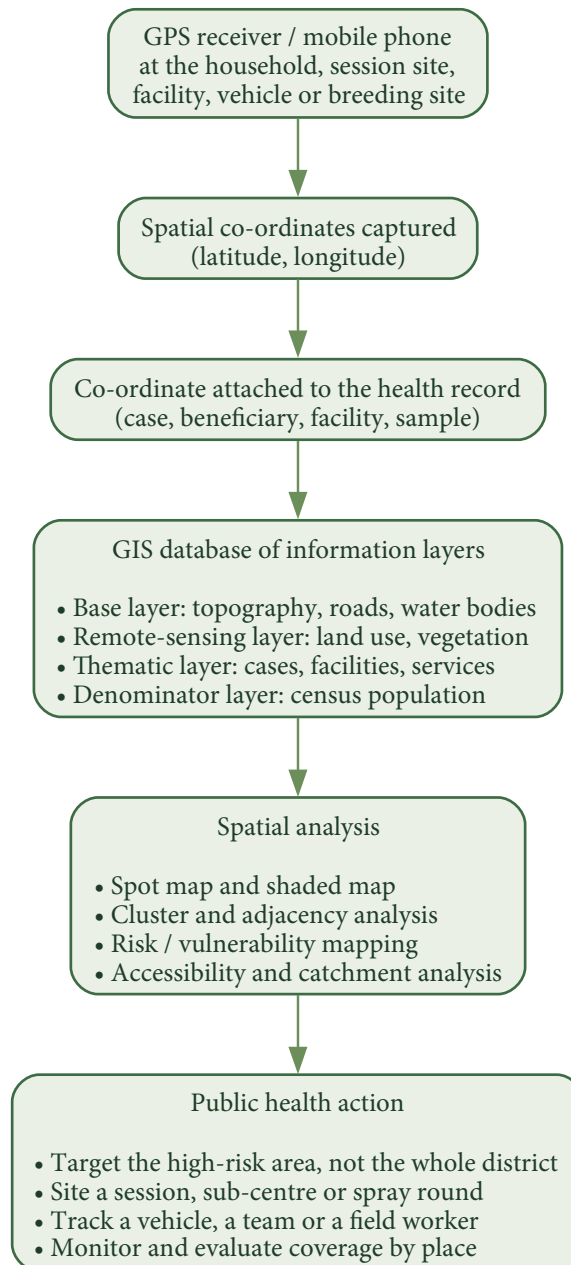
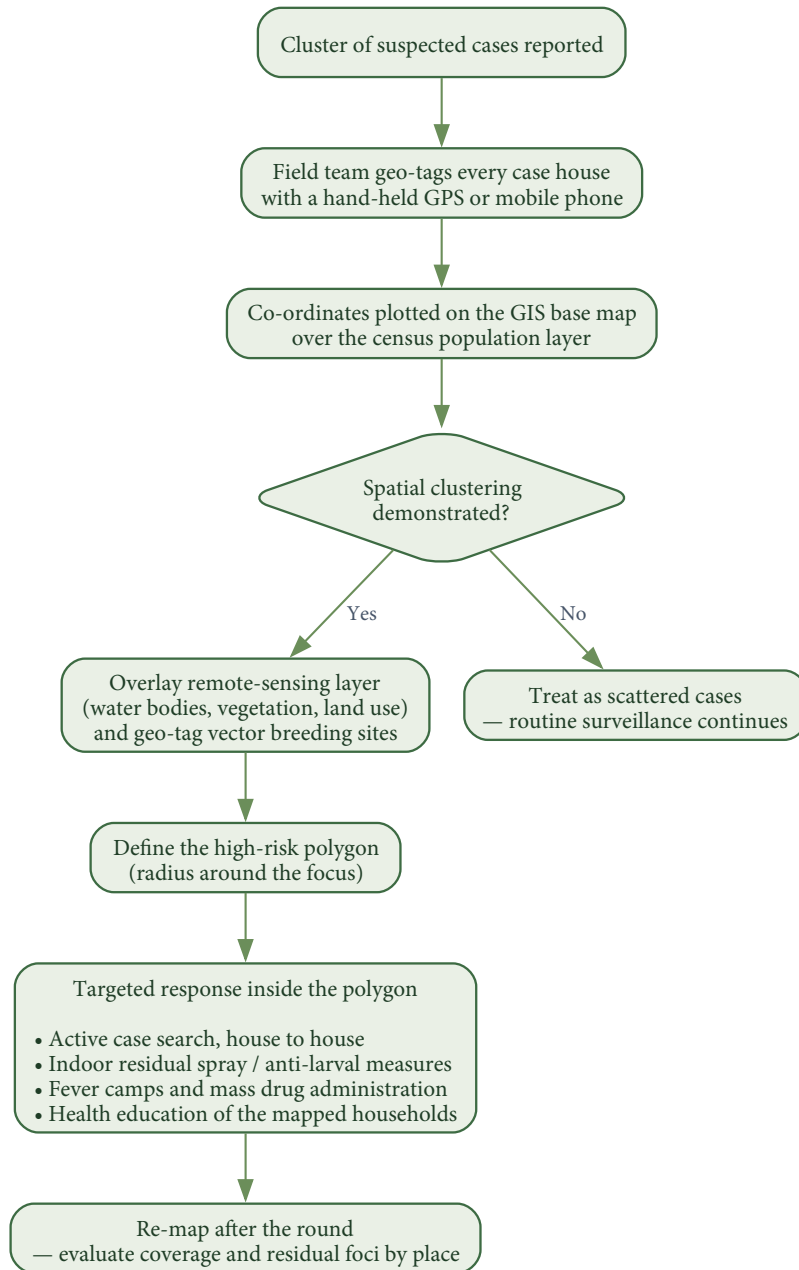


Fig 7.10 — Application of geographic information system to decision making shown (Nelson & Williams — *Infectious Disease Epidemiology*, p.202)



From a satellite fix to public health action — the GPS-GIS pathway



GPS-enabled response to a suspected vector-borne disease outbreak

High-yield exam pointers

- ▶ **One-line distinction to write first:** GPS = **where** (co-ordinates captured); GIS = **so what** (layers, overlay, analysis); remote sensing = **what surrounds it** (satellite imagery of the environment).
- ▶ **Snow, 1854, Golden Square, Broad Street pump** — the spot map is the ancestor of every GIS application; SYMAP (Howard T. Fisher, 1963) was the first computerised mapping system.
- ▶ **Bio-Medical Waste Management Rules 2016 — the four GPS clauses:** Rule 5(c) (operator) and Rule 8(4) (occupier + common facility) — bar code plus GPS within **one year**; **Form II (rule 10), application for authorisation**, item 4(ii) — GPS coordinates of the HCF or CBWTF; **Form IV (rule 13), annual report**, item 1(viii) — GPS coordinates of HCF or CBMWTF.
- ▶ **Routine Immunization Manual 2024** — "GPS coordinates of the facility" is a header field on the **Sub District Cold Chain Monitoring Checklist** (not on any district checklist).

- **Satellite resolutions to quote:** Landsat Thematic Mapper — every 16 days, 7 bands, ~30 m; AVHRR — twice daily, 5 bands, 1.1 km.
- **Error list (mnemonic S-R-I-T):** Spatial, Resolution, Interpretive (15–20% land-cover misclassification on ground-truthing), Temporal — plus confidentiality as the ethical fifth.
- **Five functions of GIS to list:** collect and store data, manage data, query data, model the processes generating the data, and make programmatic decisions (Nelson & Williams, Ch 7, p 188).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook of Preventive and Social Medicine 28e — the spine for this bank — covers spot maps and shaded maps but does not discuss the global positioning system by name; the GPS content above is therefore grounded on the Bio-Medical Waste Management Rules 2016, the Routine Immunization Manual for Medical Officers 2024, Nelson & Williams, the Oxford Textbook of Public Health 7e, Essential Malariology 4e and the Solid Waste Management manual, with Park 28e supplying the spot-map lineage. The GPS-based drug-sample collection proposal belongs to the Twelfth Five Year Plan (2012–17) health chapter, not to the National Health Policy.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, ICMR, NCVBDC, National Framework for Malaria Elimination 2016, National Strategic Plan for Malaria Elimination 2023–27, WHO Immunization Agenda 2030

Programme and policy updates (India)

- **The vector-borne programme is now cited as the National Center for Vector Borne Diseases Control (NCVBDC) — the renamed NVBDCP — and its National Framework for Malaria Elimination (2016) makes the district the strategic unit for planning and intervention, with elimination targeted by 2030.** A 635-district analysis of 2018 NCVBDC data combined **Getis-Ord Gi* clustering** with Sentinel-2 land-cover at 10 m resolution and MERRA-2 meteorological reanalysis: hotspots sat in the Chhattisgarh–Odisha–Telangana tribal belt and the north-east, and **not owning a mobile phone** carried the largest risk-increasing socio-economic risk ratio reported (RR 1.32, 95% CI 1.28–1.37) — the very digital gap that limits GPS-based field data capture. The authors recommend targeting **clusters of contiguous districts** rather than single districts, and note that the Mandla (Madhya Pradesh) Malaria Elimination Demonstration Project cut indigenous cases by about 91% using mobile-based surveillance. [PMID 41088173]
- **National Strategic Plan for Malaria Elimination 2023–2027 sets zero indigenous cases by 2027 and WHO certification of elimination by 2030, and grades states by annual parasite incidence — Goa sat in category 1 (annual parasite incidence <1 in all districts) and joined category 0, zero indigenous cases, in 2023** (national caseload 2.09 million in 2001 falling to 0.2 million in 2023). Yet imported cases more than doubled, from 585 in 2022 to 1,209 in 2023 — largely attributed by the authors to migrant construction workers, though the study could not quantify that proportion. PHC/UHC-level ArcGIS choropleths plus Landsat 5 and Landsat 9 land-use classification showed built-up area rising from about 70 km² (1991) to about 260 km² (2024), spatially overlapping the case clusters — a working demonstration of the GPS → GIS → remote-sensing stack in an elimination end-game. [PMID 39953530]

- **GPS-tagged mobile platforms are now embedded in Indian vector-borne programme delivery.** A 2025 scoping review of 25 mHealth malaria surveillance systems found India contributed the largest single-country share, naming **SOCH**, **MoSQuIT**, the GIS-enabled **Malaria Control System** (begun as a Mangaluru municipal project and since operational across several districts under the national programme) and **FeverTracker**, a community app piloted in north-east India that reports febrile illness in real time with **GPS tagging** to support active case detection. Only 2 of the 25 systems worldwide had been scaled into a national programme, and poor connectivity limited over half the studies — making **offline-first capture with later synchronisation** the operational requirement, not an optional extra. [PMID 41257723]

Current national data

- **Geospatial access mapping has entered national service planning.** Drive-time catchments built in ArcGIS around 806 reporting blood banks on the MoHFW **e-RaktKosh** platform, across the 661 million people of the eight **Empowered Action Group** states (data window January–February 2022), found **25.72% living within 30 minutes**, **61.44% within 60 minutes** and **92.46% within 90 minutes** of a blood bank; availability was **0.68 units per 1,000 population** at 60 minutes, against the WHO benchmark of 10 donations per 1,000 and the Lancet Commission on Global Surgery target of 15 units per 1,000. Regions failing both the time test and the stock test are termed "**blood deserts**". [PMID 39433405]
- **NFHS-5 (2019–21) linked to the Vulnerability Atlas of India 2019** across 202,194 children showed flood-zone residence carried **8% higher risk of low birth weight** and **27% higher risk of very low birth weight**; LISA cluster maps located the high-risk pockets and **geographically weighted regression** attributed **52% of the variability in LBW occurrence to flood hazard** — the climate-and-health application of the same spatial toolkit, and a reminder to quote the survey round with the figure. [PMID 38151131]

Elimination, surveillance and disease status

- **Kala-azar: India is transitioning into the post-elimination phase, and transmission has not stopped.** Quarterly sampling in one Bihar village and one Jharkhand village (December 2021 to May 2023) found **rK39 seroprevalence of 20–30%**, and *Leishmania*-infected sand-fly pools despite indoor residual spraying — **1.4% (4/285) in Bihar** and **1.2% (5/431) in Jharkhand**, collected predominantly from vegetation in one village and cattle enclosures in the other. Focal, microhabitat-specific residual transmission of exactly this kind is what **GPS-referenced micro-focus surveillance** exists to detect. [PMID 41324335]
- **Micro-level hotspot detection is now the operational unit for malaria elimination.** In Sonbhadra district, Uttar Pradesh, 18,217 cases over 2017–2024 fell by 95% and API dropped from 2.7 to 0.1, but annual falciparum incidence rose again to 0.09 in 2024 after near-total suppression in 2022–23; **Kulldorff spatial scan statistics with a discrete Poisson model** located the principal high-risk cluster in **Chopan block**, and mean and maximum annual rate of change were used to project the year of zero-case status. [PMID 41871706]
- **GIS hotspot statistics are being applied to Indian outbreak surveillance data.** An ICMR-NICED study funded by NCVBDC serotyped 3,249 DENV-RNA-positive samples from 20 West Bengal districts (2015–19) and ran **Getis-Ord Gi* hotspot cluster analysis at a 500 m distance threshold** on residential co-ordinates converted to UTM, with Moran's I confirming clustering

in every study year: **Kolkata was the hotspot in all five years**, and the predominant serotype shifted DENV-3 (2015) → DENV-1 (2016) → DENV-2 (2017–19), raising secondary-infection and severe-dengue risk precisely within the mapped clusters. [PMID 38577856]

Guideline and legislative updates

- A **geoprivacy ethics framework for researchers and ethics committees** (2026) breaks the problem across five research phases — collection, storage, sharing, analysis and dissemination — into **60 guiding questions graded high, moderate or low importance**, and adds an **ethical-review applicability matrix** keyed to data type, granularity, linkage potential and participant vulnerability. It argues that data-protection statutes set a regulatory baseline but not practical guidance; that even anonymised trajectory data can be re-identified from movement patterns; and that GeoAI and foundation models can now infer sensitive attributes from sparse location traces. This is the operational answer to the confidentiality objection against household-level GPS mapping. [PMID 41772605]

Global and WHO developments

- **GBD 2023 childhood immunisation estimates place India among the eight countries holding 51.1% (95% UI 47.7–53.6) of the world's zero-dose children in 2023.** Global coverage of the original EPI vaccines — DTP1, DTP3, MCV1, Pol3 and BCG — had still not recovered to pre-pandemic levels, and pandemic disruption alone added 12.8 million extra zero-dose children over 2020–23. Meeting the **Immunization Agenda 2030** zero-dose goal means halving the 2019 count to **7.35 million by 2030**, which for south Asia implies **95.9% DTP1 coverage**, 2.2 percentage points above anything the region has historically achieved. The authors argue that reaching missed children needs local, timely coverage estimates and a better understanding of **where** zero-dose children sit within each country, and that assessing coverage **by geography at subnational scales** exposes disparities the national averages mask — the case for geo-referenced microplanning. [PMID 40578370]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods); Nelson & Williams — Infectious Disease Epidemiology (Ch 7, Geographic Information Systems); Oxford Textbook of Public Health 7e (Ch 19 the right to health; Ch 24 community diagnosis and health information systems; Ch 42 public health surveillance; Ch 63 physical activity and public health; Ch 102 training of public health professionals in developing countries); Gilles & Warrell — Essential Malariology 4e (Ch 5, The epidemiology of malaria); Jugal Kishore — National Health Programmes of India (Ch 1, Twelfth Plan and remote sensing in malaria; Ch 2, vaccine policy and surveillance); Park Plus (Amrutha & Vijayalakshmi), Ch 24 — Miscellaneous (Geographic Information Systems in health); Oxford Handbook of Occupational Health 2e (Ch 5, Psychosocial hazards); Textbook of International Health (Ch 8, Health under Crisis); Lilienfeld's Foundations of Epidemiology (Ch 14, Field Epidemiology); Plotkin's Vaccines 7e (Ch 76, Immunization in Developing Countries); Solid Waste Management (Ch 79 and Ch 80). Named documents: **Bio-Medical Waste Management Rules, 2016** (Rules 5(c), 8(4), 10 and 13; Form II and Form IV); **Routine Immunization I Manual for Medical Officers, 2024** (Ch 4 microplanning; Ch 17 Sub District Cold Chain Monitoring Checklist); **Twelfth Five Year Plan (2012–17), health chapter**; WHO — **Integrating Migrants and Refugees into National Immunization Programmes**.

Recent-advances sources: 41088173 — Determinants of malaria transmission in Indian districts in 2018: insights from ensemble models, *Malaria Journal* 2025, DOI [10.1186/s12936-025-05546-9](https://doi.org/10.1186/s12936-025-05546-9); 39953530 — Spatiotemporal epidemiology of indigenous and imported malaria cases in Goa, western India, *Malaria Journal* 2025, DOI [10.1186/s12936-024-05232-2](https://doi.org/10.1186/s12936-024-05232-2); 41257723 — Mobile app-based malaria surveillance and vector control integration: a scoping review of evidence from endemic settings, *Malaria Journal* 2025, DOI [10.1186/s12936-025-05656-4](https://doi.org/10.1186/s12936-025-05656-4); 39433405 — Defining blood deserts and access to blood products for 660 million people: a geospatial analysis of eight states in Northern India, *BMJ Global Health* 2024, DOI [10.1136/bmjgh-2024-015637](https://doi.org/10.1136/bmjgh-2024-015637); 38151131 — Investigating the association between floods and low birth weight in India: using the geospatial approach, *Science of the Total Environment* 2024, DOI [10.1016/j.scitotenv.2023.169593](https://doi.org/10.1016/j.scitotenv.2023.169593) (abstract-only); 41324335 — High seropositivity to rK39 and capture of *Leishmania*-infected sand flies in Bihar and Jharkhand emphasize the need for post-elimination surveillance of visceral leishmaniasis in India, *Journal of Infectious Diseases* 2026, DOI [10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543) (abstract-only); 41871706 — Identification of micro-level malaria hotspots and its implications for achieving malaria elimination target in a remote district of India, *Acta Tropica* 2026, DOI [10.1016/j.actatropica.2026.108066](https://doi.org/10.1016/j.actatropica.2026.108066) (abstract-only); 38577856 — Geographic information system-aided evaluation of epidemiological trends of dengue serotypes in West Bengal, India, *Indian Journal of Medical Research* 2024, DOI [10.4103/ijmr.ijmr_1055_23](https://doi.org/10.4103/ijmr.ijmr_1055_23); 41772605 — Ethical guidelines for geoprivacy: a framework for researchers and ethics committees, *International Journal of Health Geographics* 2026, DOI [10.1186/s12942-026-00460-y](https://doi.org/10.1186/s12942-026-00460-y); 40578370 — Global, regional, and national trends in routine childhood vaccination coverage from 1980 to 2023 with forecasts to 2030: a systematic analysis for the Global Burden of Disease Study 2023, *Lancet* 2025, DOI [10.1016/S0140-6736\(25\)01037-2](https://doi.org/10.1016/S0140-6736(25)01037-2). Programme documents: **National Framework for Malaria**

Elimination (2016) and National Strategic Plan for Malaria Elimination 2023–27, NCVBDC; WHO Immunization Agenda 2030.



Health economics

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — Robbins, who propounded economics as the study of scarcity, defined it as "the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses" (Mahajan & Gupta 4e, Ch 27, p 539). **Health economics** is the sub-discipline of economics that applies economic theory and method to every aspect of health and health care; it is concerned with the choices over how best to spend the limited resources available for health care so that health gain is maximised (Sunder Lal, Ch 2, pp 86-87; Introduction to Health Economics 2e, Ch 1, p 33). Its two governing objectives are **efficiency** — maximising benefit from available resources, or minimising cost for a given benefit — and **equity**, fairness in the way resources and health are distributed (Introduction to Health Economics 2e, Ch 1, p 29).

Core concepts and scope

Concept	Meaning	Application in health
Scarcity	Resources are always fewer than wants, so choices are unavoidable	India spent about 3.16% of gross domestic product (GDP) on health in 2018-19 against unlimited need (Sunder Lal, Ch 16, p 727)
Opportunity cost (economic cost)	The value of the benefit of the next best alternative forgone	Providing service X necessarily displaces service Y; the health gain lost from Y is the true cost of X (Introduction to Health Economics 2e, Ch 1, p 32)
Margin / marginal analysis	The additional benefit or cost of one extra unit of consumption or production	Is one more auxiliary nurse midwife (ANM) worth more health gain than one more ambulance?
Technical (operational) efficiency	A given output produced with the least inputs — minimising wastage	Same number of institutional deliveries with fewer staff-hours
Economic (productive) efficiency	A given output produced at least cost	Choosing the cheapest mix of staff and drugs for a fixed output
Allocative efficiency	Producing the mix of outputs that maximises society's welfare	Deciding between an immunisation drive and a dialysis unit
Horizontal equity	Equal treatment of equals — equal access, use or expenditure for equal need	Same antenatal package for every pregnant woman

Concept	Meaning	Application in health
Vertical equity	Unequal but fair treatment of unequals — preferential treatment in proportion to disadvantage	Resource-allocation formulae weighted by a deprivation index; aspirational-district funding
Positive vs normative economics	Describes how things are vs recommends how things should be (contains a value judgement)	"Out-of-pocket spending is 53% of expenditure" vs "user fees should be abolished"
Equity–efficiency trade-off	The most equitable option is often not the most efficient, and vice versa	Reaching scattered tribal hamlets costs far more per case detected

Scope — demand and supply of health care; market failure and the role of government; financing and universal coverage; economic evaluation; equity and provider payment (Introduction to Health Economics 2e, Ch 1-17).

Health care as an imperfect market — market failures and the case for government

A **market failure** is a situation in which the market does not result in an efficient allocation of resources. Health care markets exist but are far from perfectly competitive, and are characterised by four classical failures (Introduction to Health Economics 2e, Ch 8, pp 130-132).

Market failure	How it appears in health care	Consequence	Corrective action
Monopoly	Barriers to entry through professional licensing and few close substitutes; specialists may hold a genuine monopoly on a service	Price raised by restricting output; allocative inefficiency and deadweight loss	Regulation of licensing and prices, public provision, limiting the restrictive powers of professional bodies
Externality — a cost or benefit of one person's consumption or production falling on others	Positive: immunisation and treatment of infectious disease protect the unvaccinated; negative: tobacco, alcohol, industrial pollution	Free market under-produces goods with positive externalities and over-produces those with negative externalities	Subsidy or free provision (Universal Immunization Programme); Pigouvian tax set to internalise the negative externality (tobacco taxation)
Public good — consumed simultaneously by everyone, no one can be excluded	Vector control, sanitation, health education, disease surveillance, clean air	Free-riding; the private market will not supply it at all	Tax-financed public provision
Asymmetry of information — participants do not have equal information	See sub-table below; the doctor–patient and doctor–payer relationships are agency relationships	Adverse selection, moral hazard, supplier-induced demand, cost escalation	Cost-sharing, community rating, compulsory insurance, reform of provider payment, standard treatment guidelines

Informed party	Uninformed party	Resulting market failure
Patient	Payer (insurer, social insurance fund or government)	Consumer moral hazard; adverse selection
Doctor	Patient	Imperfect agency
Doctor	Payer	Provider moral hazard = supplier-induced demand

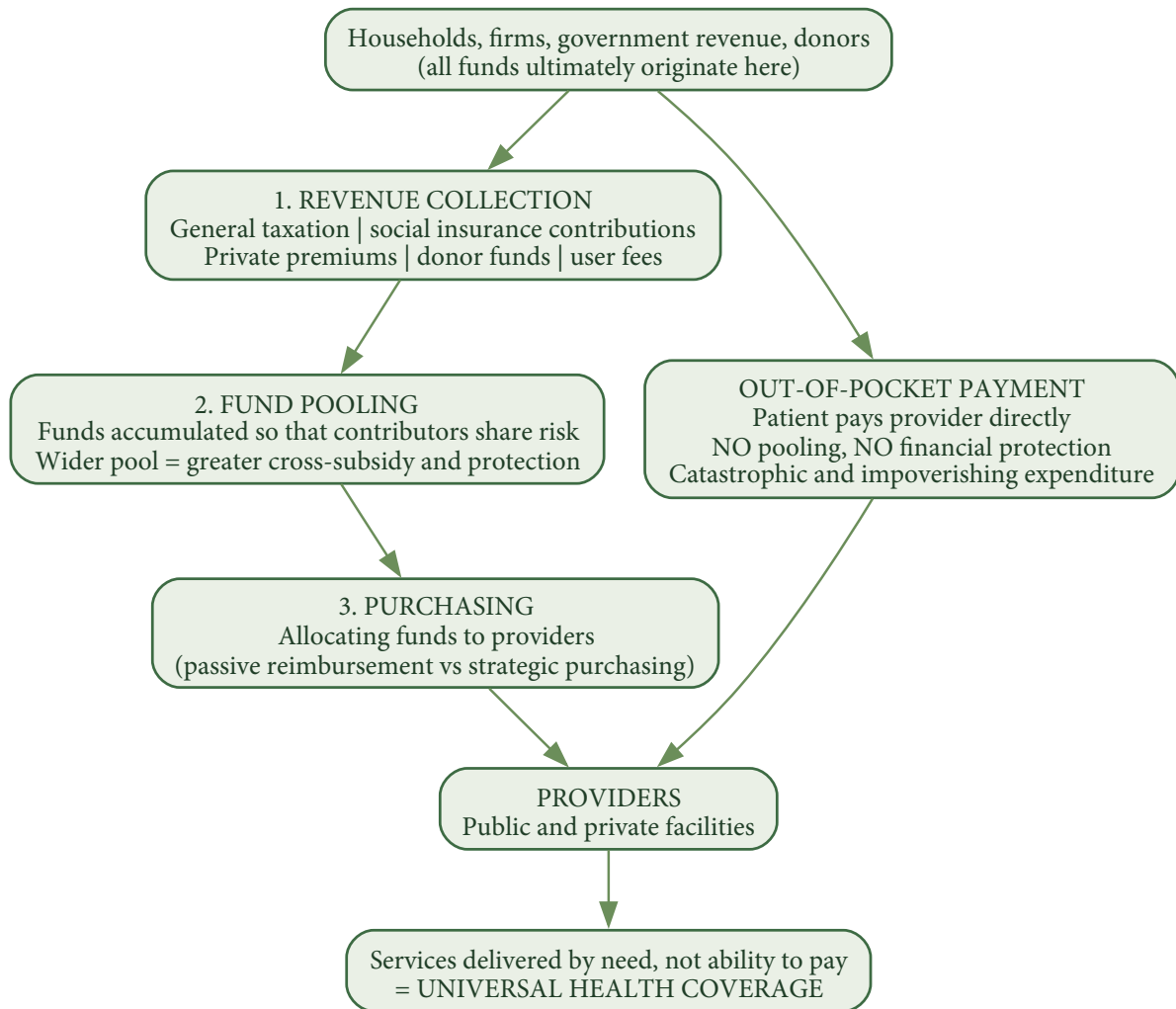
(Introduction to Health Economics 2e, Ch 8, Table 8.1, pp 143-144)

Term	Definition	Effect on the health system
Adverse selection	A party enters an agreement using their own private information to the disadvantage of the other	High-risk consumers buy the most generous cover, so voluntary insurance fails to cover the sick and the poor — inequity
Moral hazard	After the agreement is made, the informed party acts to gain at the other's expense	Insured take less care of their health and over-consume care that is free at the point of use — cost escalation
Supplier-induced demand	Demand beyond what consumers would have asked for had they been perfectly informed	Over-investigation and over-treatment, especially under fee-for-service payment
Diseconomies of scale	Small fragmented risk pools raise administrative cost per member	Third key failure of private health insurance (Introduction to Health Economics 2e, Ch 11, pp 177-178, 184)

Because of these failures — plus the objectives of equity, redistribution and macroeconomic stability — **government intervention in health care is economically justified**, through financing, regulation and direct provision.

Health care financing — functions, models and the Indian picture

Financing is analysed as three sequential functions, which out-of-pocket payment bypasses altogether (Introduction to Health Economics 2e, Ch 9, pp 149-151).



The three functions of health system financing and the unprotected out-of-pocket route

Universal coverage extends health services to the whole population according to need, regardless of income, social status or residence, and requires expansion along **three dimensions** (Introduction to Health Economics 2e, Ch 12, Fig 12.1, pp 188-189) — **breadth** of coverage (the proportion of the population with access to affordable quality care), **depth** of coverage (the range of services available to the population in need) and **height** of coverage (the proportion of health care costs met by the financing system).

Financi ng mecha nism	Revenue source	Equity and pooling	Main strength	Main weakness	Indian example
General taxation (<i>Beveridge</i> model)	Direct and indirect taxes from all within the tax net	Widest pool; generally progressive	No limitation on population coverage in principle; administrative ly simple; reaches hard- to-reach groups	Health must compete with defence, housing and social security for the budget	Tax-financed public system under the National Health Mission; free care at Ayushman Arogya Mandirs

Financing mechanism	Revenue source	Equity and pooling	Main strength	Main weakness	Indian example
Social health insurance (<i>Bismarck</i> model)	Compulsory employer + employee contributions in the formal sector	Funds earmarked for health; less progressive as it encourages under-reporting of salary	Income ring-fenced; employer and employee stake improves accountability and reduces moral hazard	A form of employment tax; coverage tied to size of the formal sector; high collection and administrative cost; misses the informal sector	Employees' State Insurance Scheme , created by Act of Parliament in 1948 — first Indian scheme on the contributory principle
Tax-funded purchasing / government-financed insurance	General taxation, purchasing from empanelled public and private hospitals	Large pool targeted at the poor and vulnerable	Financial protection against catastrophic hospitalisation, cashless and portable	Covers hospitalisation, not the outpatient and drug spending that drives most out-of-pocket expenditure	AB PM-JAY — cover of ₹5 lakh per family per year for 10.74 crore poor and vulnerable families (~50 crore people), launched 23 September 2018; RSBY subsumed into it
Voluntary private health insurance	Risk-rated premiums paid to an insurer	Small, fragmented pools; regressive	Choice; supplements the public system	Adverse selection, moral hazard, diseconomies of scale; in low-income countries attracts only the affluent	Private policies = 6.6% of total health expenditure (2018-19)
Community financing	Collective action of a local community pooling out-of-pocket payments — cash, in kind, partial or delayed	Small local pool	Reaches informal-sector and rural households	Pool too small to absorb catastrophic cost	Community-based prepayment schemes
Out-of-pocket (direct) payment	Patient pays the provider at the point of use	No pooling at all — most regressive	Simplest and earliest form of transaction	Deters use (Zambia, RAND experiments), causes catastrophic and impoverishing expenditure	53.23% of current health expenditure (2018-19)

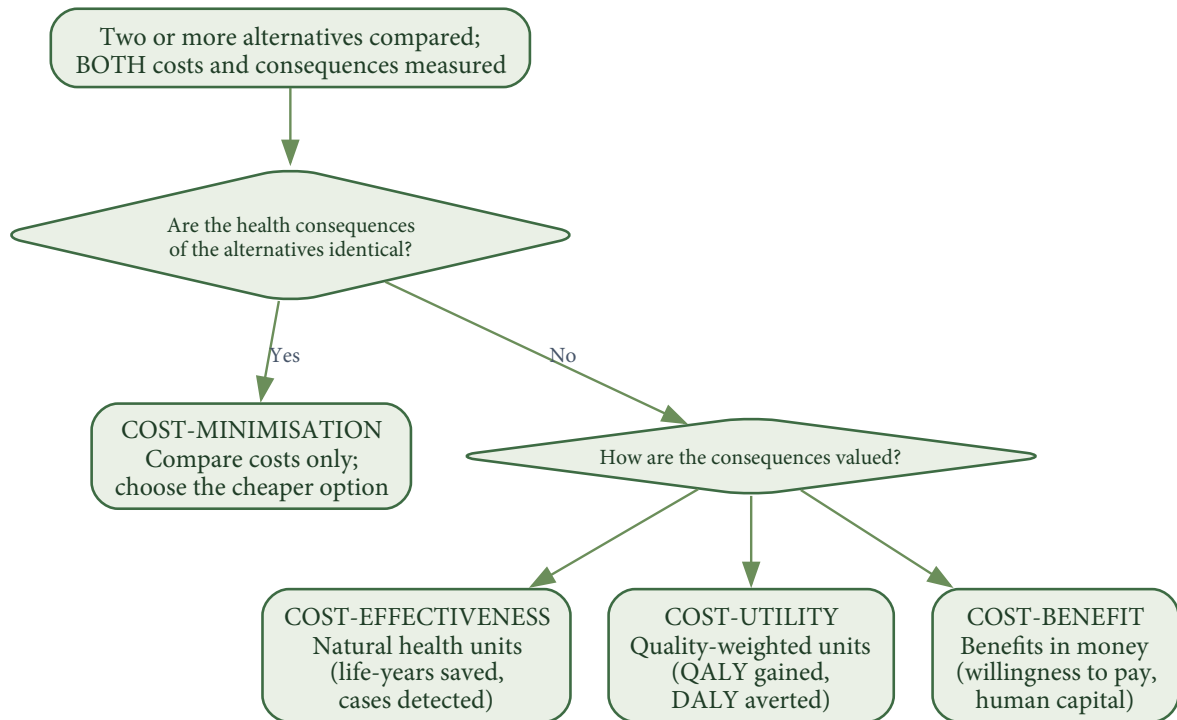
(Introduction to Health Economics 2e, Ch 9, pp 148-151; Ch 11, pp 177-178, 184; Ch 12, Table 12.1 (Mills 2007), pp 191-192; Park 28e, Ch 10, pp 745-746; Sunder Lal, Ch 16, p 729)

Indian health-financing indicator	Value and year	National Health Policy 2017 target
Total health expenditure (THE) as % of GDP	3.16% (2018-19)	—
Government health expenditure as % of GDP	1.28% (2018-19); 1.15% when the policy was framed	2.5% of GDP by 2025
Government share of total health spending	40.6% (2018-19)	—
Out-of-pocket expenditure	53.23% of current health expenditure (2018-19); 62.4% of THE (2014 estimate)	Reduce out-of-pocket spending to 50% of THE
Current health expenditure (recurrent) as % of THE	>91%; capital expenditure ~9% (2018-19)	—
State health spending as % of state budget	—	>8% of state budget by 2020 (15th Finance Commission: 8% by 2022, with 67% of health spending on primary health care)
Households facing catastrophic health expenditure	—	Reduce by 25% from current levels by 2025
Private sector share of provision	80% of doctors, 25% of nurses, 49% of beds (2018-19), financed chiefly by out-of-pocket expenditure	—

(Sunder Lal, Ch 16, pp 727-729; Park 28e, Ch 10, p 745; Ch 23, pp 1030, 1033)

Economic evaluation — the central tool of health economics

Economic evaluation compares the **costs and consequences** of alternative health care interventions to assess their value for money (Introduction to Health Economics 2e, Ch 13, p 201). Both limbs — *cost and consequence* — and at least two alternatives are required for a *full* evaluation.



Choosing the type of full economic evaluation according to how consequences are valued

Technique	Consequences measured as	Result reported as	Comment
Cost-minimisation analysis (CMA)	Assumed identical — only the inputs are compared	Least-cost option	Valid only where there is good evidence that outcomes are identical, which is rarely so (Mahajan & Gupta 4e, Ch 27, p 540; Introduction to Health Economics 2e, Ch 13, p 208)
Cost-effectiveness analysis (CEA)	Natural health units — life-years saved, deaths averted, cases detected, disease-free days	Cost per unit of effect	Park: "a more promising tool for application in the health field than cost-benefit analysis" (Park 28e, Ch 23, p 1028)
Cost-utility analysis (CUA)	Health units capturing both quantity and quality of life — quality-adjusted life years (QALY) gained, disability-adjusted life years (DALY) averted	Cost per QALY gained / DALY averted	Allows comparison across entirely different diseases and programmes
Cost-benefit analysis (CBA)	Money — willingness to pay (WTP), human capital approach using market wage rates	Net present value = monetised benefits – costs; benefit-cost ratio	Widest scope, but health benefits often cannot be expressed in money — Park's stated drawback (Park 28e, Ch 23, p 1028)

Outcome measures

- QALY — life-years weighted by quality on a scale of 1.0 = full health to 0 = death; a year at utility 0.5 = 0.5 QALY (Park 28e, Ch 2, p 38).
- DALY = years of life lost (YLL) + years lived with disability (YLD); developed by Harvard University for the World Bank in 1990, adopted by WHO in 2000 (Park 28e, Ch 2, p 38).

- Utility weights are elicited by the **standard gamble**, **time trade-off**, **person trade-off** and **visual analogue scale** — e.g. if 30 years in a compromised state equals 20 years in full health, that state is valued $20/30 = 0.67$ (Introduction to Health Economics 2e, Ch 15, pp 236-238).

Costing — the cost side must be classified before it is counted (Introduction to Health Economics 2e, Ch 14, pp 214-220).

Basis of classification	Categories
Resource life	Capital (useful life >1 year — buildings, vehicles, equipment; annualised over their life with depreciation) vs recurrent (life <1 year — salaries, drugs, fuel, electricity)
Who bears them	Direct (resources used in design, implementation, receipt and continuation of the intervention) · indirect (value of resources, chiefly time, expended by patients and carers) · intangible (pain, anxiety, discomfort, inconvenience — rarely quantified but often decisive for the patient)
Valuation basis	Financial (budgetary) cost = amount actually paid vs economic (opportunity) cost = value in the next best use; donated or subsidised inputs revalued at shadow prices
Behaviour with output	Fixed vs variable ; overhead costs support the organisation rather than direct patient care
Derived measures	Average cost = total cost ÷ quantity; marginal cost = change in total cost when one extra unit is produced

The **perspective must be declared** — patient, provider, health system, or **societal** (the broadest, alone capturing indirect and intangible costs).

Comparing costs with effects (Introduction to Health Economics 2e, Ch 14, p 228; Ch 16, pp 250, 253; Ch 17, p 280)

Term	Formula / meaning
Average cost-effectiveness ratio (ACER)	Difference in cost ÷ difference in effect for a single intervention measured against its baseline option (no programme or current practice)
Marginal cost-effectiveness ratio (MCER)	The change in cost and effect when a programme is expanded or contracted
Incremental cost-effectiveness ratio (ICER)	$(\text{Cost}_A - \text{Cost}_B) \div (\text{Effect}_A - \text{Effect}_B)$ = extra cost per additional unit of effect (one more DALY averted, one more QALY gained)
Dominance	An option that is less costly AND more effective dominates and is always cost-effective
Discounting	Converts costs and outcomes arising at different times to present value; guidelines recommend a 3% discount rate after controlling for inflation
Sensitivity analysis	Varies uncertain parameters — unit costs, discount rate, disability weights — within justified ranges to test robustness of the conclusion

Steps in conducting an economic evaluation (Introduction to Health Economics 2e, Ch 16, pp 259-260)

- ▶ Pose a well-defined question stating the alternatives compared and the **viewpoint** adopted.
- ▶ Describe the competing alternatives comprehensively and establish that effectiveness is already proven.
- ▶ **Identify** all relevant costs and consequences for each alternative.
- ▶ **Measure** them credibly in appropriate physical units.
- ▶ **Value** them, adjusting to market value where inputs were donated or subsidised.
- ▶ **Discount** future costs and consequences to present value.
- ▶ Perform an **incremental** analysis.
- ▶ Perform a **sensitivity analysis** over justified ranges, then present the results with their generalisability to other settings.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The cost-effectiveness plane** — draw two axes crossing at an origin marked "Comparator (current practice)". Label the horizontal axis "**Incremental effectiveness (ΔE)**" and the vertical axis "**Incremental cost (ΔC)**". Label the four quadrants: **north-east** — "More costly, more effective → ICER decides against the threshold"; **south-east** — "Less costly, more effective → NEW INTERVENTION DOMINATES, adopt"; **south-west** — "Less costly, less effective → trade-off"; **north-west** — "More costly, less effective → DOMINATED, reject". Draw a straight line through the origin into the north-east quadrant and label it "**Cost-effectiveness threshold**".

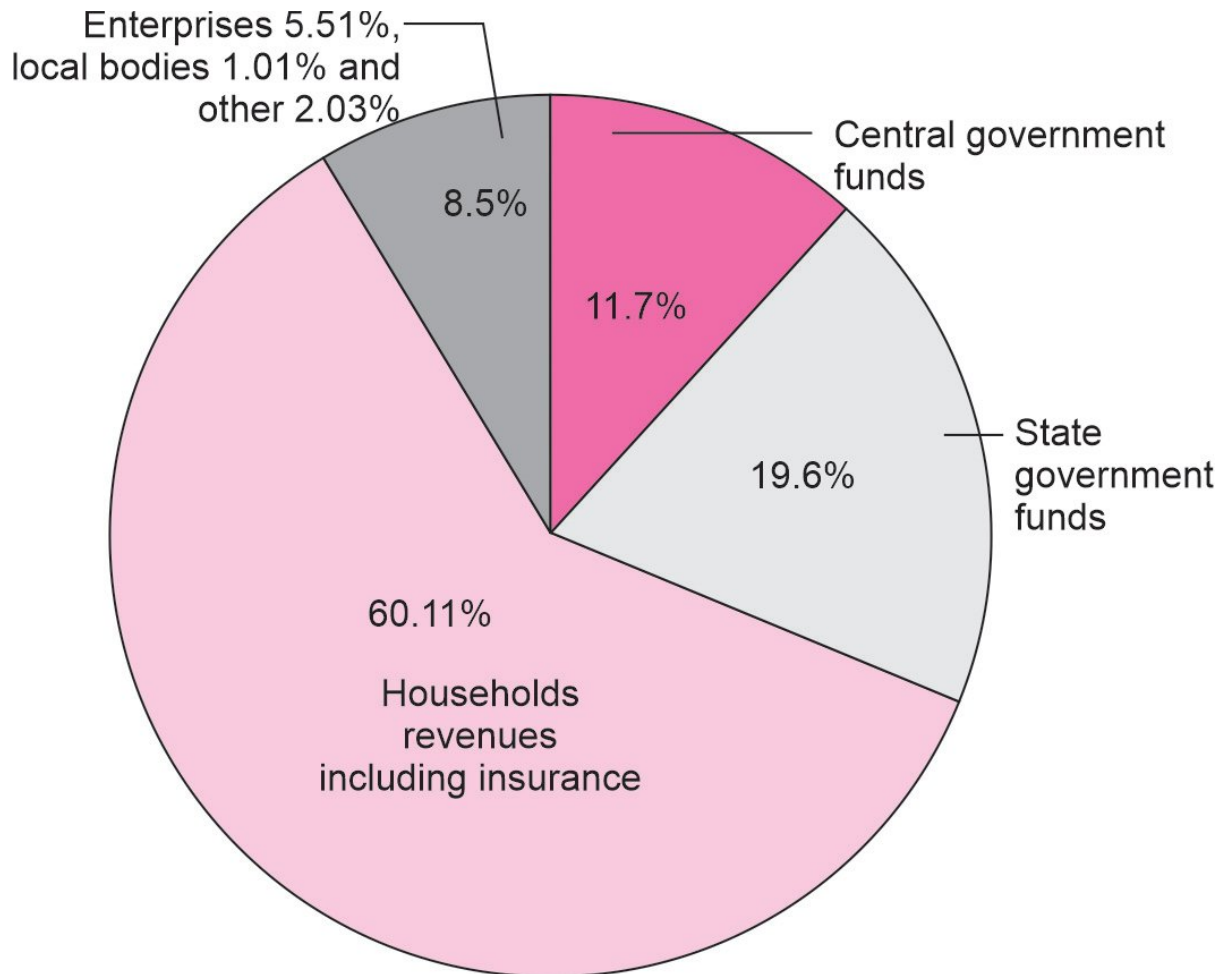


Fig 16.7 — Sources of health financing in India (NHA 2018-19) (Sunderlal, p.728)

High-yield exam pointers

- ▶ **Definition to open with:** health economics applies economic theory and method to health and health care — the allocation of **scarce** resources to competing health needs, judged on **efficiency** and **equity**.
- ▶ **Two words that earn marks anywhere:** *opportunity cost* (value of the next best alternative forgone) and *the margin* (the extra benefit of one more unit).
- ▶ **Four market failures (mnemonic MEPA):** Monopoly, Externality, Public good, Asymmetry of information (→ adverse selection, moral hazard, imperfect agency, supplier-induced demand).
- ▶ **Three financing functions:** revenue collection → fund pooling → purchasing; **Beveridge** = tax-funded, **Bismarck** = social health insurance.
- ▶ **Four evaluations and their outcome units:** CMA (identical) · CEA (natural units) · CUA (QALY/DALY) · CBA (money). $ICER = \Delta Cost \div \Delta Effect$; discount rate 3%.
- ▶ **DALY = YLL + YLD** (Harvard for the World Bank, 1990; WHO adopted 2000); **QALY scale 0 = death to 1 = full health**.
- ▶ **Dated Indian figures:** total health expenditure 3.16% of GDP, government 1.28% of GDP / 40.6% of total, out-of-pocket 53.23% of current health expenditure (all 2018-19); National Health Policy 2017 targets 2.5% of GDP by 2025; AB PM-JAY ₹5 lakh per family per year, 10.74 crore families, launched 23 September 2018.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: cost-minimisation analysis is counted as the fourth main type of economic evaluation by Mahajan & Gupta 4e Ch 27, while Introduction to Health Economics 2e Ch 13 treats it as a narrow subset of cost-effectiveness analysis used only when outcomes are proven identical; Park's older statement that India spends "about 3 per cent of gross national product on health and family welfare" is superseded by the National Health Accounts 2018-19 figures quoted above, and Ayushman Bharat Health and Wellness Centres have been renamed **Ayushman Arogya Mandirs**.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Accounts 2022-23 (MoHFW), Health Technology Assessment in India (DHR, MoHFW), WHO

Programme and policy updates (India)

- **Health Technology Assessment in India (HTAIn)** is now the institutional home of economic evaluation for public policy — its regional resource centres grew from 1 in 2017 to 30 by 2024, and 26 assessments from ten centres (2018-2023) were reviewed on the Indian HTA Quality Appraisal Checklist; of the 22 scored as full assessments, 14% were excellent, 50% good, 32% average and 4% poor. Supporting infrastructure now in place includes national methodological guidelines, a **reference case** for economic evaluation, a national **cost database** and an Indian **EQ-5D-5L value set**. The residual weaknesses are the examinable part: only 32% of studies addressed methodological uncertainty, 59% examined equity and 45% carried out a budget-impact analysis, while staff turnover keeps draining trained capacity. [PMID 40913689]
- **Provider-payment reform for district hospitals** — costing of 27 district hospitals across nine states supports replacing historical line-item budgets with a **blended payment** model: risk-adjusted global budget for outpatients, case-based bundled payment for inpatients, per-bed global budget for indirect services. The mean annual requirement is **₹326 million (US\$3.35 million) per district hospital** (range ₹66 million–₹2.57 billion), of which **inpatient care is the largest share at about 63%** (outpatient 21%, indirect 16%; the paper's abstract states 78%, a figure its own Results and Discussion contradict). Staffing to Indian Public Health Standards would raise the mean to ₹751 million, about 131% higher. This is **strategic purchasing applied inside the public sector**, not only through PM-JAY. [PMID 39478753]

Current national data

- **National Health Accounts 2022-23 supersede the 2018-19 figures quoted in the body above.** Total health expenditure **₹8,81,359 crore = 3.37% of GDP** (₹6,373 per capita); government health expenditure **₹3,85,332 crore = 43.72% of total health expenditure and 1.48% of GDP** (4.89% of general government expenditure); out-of-pocket expenditure **₹3,82,629 crore = 43.41% of total health expenditure** (49.90% of *current* health expenditure); private health insurance 9.19%; current expenditure 87% and capital 13% of total. Always quote the round, because the trend is **not** monotonic — out-of-pocket share of total health expenditure ran 48.2% (2018-19) → 47.1% (2019-20) → 44.4% (2020-21) → **39.4% (2021-22)** → 43.4% (2022-23). [MoHFW NHA 2022-23]

- **Catastrophic health expenditure in India is measured inconsistently**, which is itself the examinable point: a meta-analysis of 36 studies (2010-2023) found reported incidence ranging 5.1% to 69.9%, with a **pooled incidence of 30% (95% CI 25-35%) at the 10% threshold**. Disease-specific estimates (52%) far exceeded all-cause estimates (21%), and health surveys reported far more (33%) than consumption surveys (14%), while inpatient and outpatient estimates did not differ significantly. Standardised definitions of recall period, out-of-pocket components and "capacity to pay" remain the outstanding methodological need. [PMID 38727917]
- **Field evidence that AB PM-JAY delivers financial risk protection** — a case-control study of 772 respondents (386 beneficiaries and 386 neighbourhood-matched controls) across six Haryana districts found direct medical cost per hospitalisation 65% lower among beneficiaries (₹11,131 vs ₹31,675) and catastrophic expenditure ($\geq 30\%$ of household income) in **13.3% of beneficiaries vs 45.9% of non-beneficiaries** (OR 7.15; 95% CI 4.74–10.80). The residual gap is what a hospitalisation-only package cannot close: **total non-medical expenditure was 19% higher in beneficiaries** (₹2,872 vs ₹2,419, $p < 0.01$), **driven by transport and other out-of-hospital spending** — food and accommodation were in fact lower — and **wage loss was 1.9 times higher** (₹12,345 vs ₹6,637) but not statistically significant ($p = 0.27$). Neither non-medical cost nor productivity loss is in the benefit package. [PMID 40976675]

Elimination, surveillance and disease status

- **A rapid HTA changed national procurement policy under the National Sickle Cell Anaemia Elimination Mission** (announced in the 2023-24 Union Budget; universal screening of ages 0-40 years in endemic tribal populations, elimination target 2047). Point-of-care tests cost more per person screened than the standard solubility test (US\$2.57–3.85 vs US\$0.67, and US\$158–231 vs US\$42.53 per case detected) but detected 21,125–41,458 additional cases; on the strength of the assessment the **National Health Mission directed states to procure kits by public tender**, competitive bidding and in-house manufacture cut the price below **US\$1.21 per test**, and full coverage now yields about **US\$37.86 million** in savings. A clean worked example of a higher measured cost per case becoming the argument that wins a lower price. [PMID 42594133]
- **Nutritional support is a cost-effective tuberculosis intervention, not merely welfare** — a Markov model of household in-kind food baskets for **impoverished** persons with TB in India returned an **ICER of US\$141 per DALY averted** (95% UI 44–836) against India's willingness-to-pay threshold of **US\$550 per DALY**, and was cost-effective in 94% of simulations. Scaled to the 2,517,810 TB cases notified in 2023, it projects about **108,000 deaths averted at 100% coverage** (95% UI 54,700–147,500) — a **7–36% reduction in nationwide TB mortality** across 20-100% coverage. This is the economic case for **Nikshay Mitra in-kind food-basket support** within the National Tuberculosis Elimination Programme. Note what defines the modelled population: the effect size comes from a study among people mostly below the poverty line, so *impoverished*, not *undernourished*, is the eligibility criterion — undernutrition is the background rationale. [PMID 42082204]

Guideline and legislative updates

- **India now has its first empirical cost-effectiveness threshold**. A time-trade-off and willingness-to-pay survey of 5,460 respondents across six states (21,640 observations, health gains discounted at 3%) estimated mean **willingness to pay per QALY at ₹2,12,307 (US\$2,535)**, ranging ₹1,70,414

₹2,58,985 across health states — about **1 to 1.52 times GDP per capita**. It replaces the borrowed WHO-CHOICE "1–3 × GDP per capita" rule of thumb when judging whether an ICER is acceptable in India. [PMID 41176029]

Global and WHO developments

- **UHC monitoring has moved below the state level.** A district UHC index built from 24 tracer indicators in five domains — reproductive, maternal, newborn and child health; infectious disease; non-communicable disease; service capacity and access; and financial risk protection — across **687 districts** gave a median of **43.9%** (range 26.4–69.4), with only 21% of districts above 50% and the non-communicable disease domain lowest at a median of 9.8%. About **10% of the district population faced catastrophic and impoverishing health expenditure**, and the poorest wealth quintile sat at 36.9% against that district median. Wealth was the sharpest inequality dimension, and about 69% of the variation lay between states rather than within them. The index is a practical instrument for SDG 3.8 tracking and for identifying district hotspots for prioritisation. [PMID 39219770]
- **An interim WHO-authored protocol now supplies value-per-statistical-life estimates** for benefit–cost analysis of environment- and climate-health action, extending the 2019 benefit–cost reference case: global average **Int\$3.76 million (2024)**, South-East Asia Region Int\$1.76 million, covering 182 of 194 Member States and projected to reach Int\$9.77 million by 2100. The authors are explicit that these are interim and are **not** WHO's recommended method, which WHO is currently revising. The examinable distinction stands regardless: WHO's minimum standard for *health-sector* interventions remains **cost-effectiveness analysis**, with benefit–cost analysis reserved for action outside the health sector; and because the value rises with income, any such analysis must be supplemented by distributional analysis across income groups. [PMID 41924340]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2, 7, 10, 23); Guinness & Wiseman — Introduction to Health Economics, 2e (Ch 1, 8, 9, 11-17); Mahajan & Gupta — Textbook of Preventive and Social Medicine, 4e (Ch 27, Health Economics); Sunder Lal — Textbook of Community Medicine (Ch 2, 16); National Health Policy 2017 (Government of India); National Health Accounts Estimates for India 2018-19; Fifteenth Finance Commission recommendations on state health spending.

Recent-advances sources: PMID 40913689 — A Mixed-Methods Assessment of India's Health Technology Assessment Ecosystem, *Applied Health Economics and Health Policy* 2025, doi:[10.1007/s40258-025-00995-4](https://doi.org/10.1007/s40258-025-00995-4); PMID 39478753 — Framework for responsive financing of district hospitals of India, *Frontiers in Public Health* 2024, doi:[10.3389/fpubh.2024.1398227](https://doi.org/10.3389/fpubh.2024.1398227); PMID 38727917 — Measurement of Catastrophic Health Expenditure in India: A Systematic Review and Meta-Analysis, *Applied Health Economics and Health Policy* 2024, doi:[10.1007/s40258-024-00885-1](https://doi.org/10.1007/s40258-024-00885-1) (abstract-only); PMID 40976675 — Impact of India's publicly funded health insurance scheme on financial risk protection: a case-control study from Haryana state in India, *BMJ Open* 2025, doi:[10.1136/bmjopen-2024-093304](https://doi.org/10.1136/bmjopen-2024-093304); PMID 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, *PLoS One* 2026, doi:[10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); PMID 42082204 — Cost-effectiveness of in-kind nutritional support for impoverished persons with tuberculosis to reduce mortality and disengagement from care in India: a modelling study, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-020335](https://doi.org/10.1136/bmjgh-2025-020335); PMID 41176029 — Estimating the Willingness-to-Pay per Quality-Adjusted Life-Year to Aid Health Technology Assessment in India, *Value in Health* 2025, doi:[10.1016/j.jval.2025.10.007](https://doi.org/10.1016/j.jval.2025.10.007) (abstract-only); PMID 39219770 — District-level monitoring of universal health coverage, India, *Bulletin of the World Health Organization* 2024, doi:[10.2471/BLT.23.290854](https://doi.org/10.2471/BLT.23.290854); PMID 41924340 — Valuing reductions in the risk of death in benefit-cost analyses of environment- and climate-health actions, *Bulletin of the World Health Organization* 2026, doi:[10.2471/BLT.25.294080](https://doi.org/10.2471/BLT.25.294080). National Health Accounts Estimates for India 2022-23, Ministry of Health and Family Welfare / National Health Systems Resource Centre.



Hospital-acquired infections in tertiary care hospitals

Asked in: Paper II 09-May-2022 — RGUHS MD Community Medicine

Definition — A hospital-acquired infection (HAI, or nosocomial infection) is an infection acquired during hospital care that was neither present nor incubating at admission; it includes infection appearing after discharge and infection among hospital staff (Park 28e, Ch 5, p 430; Suryakantha, Ch 19, p 322). It is cross infection of one patient by another or by hospital staff, and a high frequency of it is evidence of poor quality of health service delivery (Park 28e, Ch 5, p 430). Tertiary care — the most specialised level, delivered by regional and apex institutions such as medical college hospitals and All India Institutes (Park 28e, Ch 10, p 719) — is where this risk concentrates.

1. Magnitude and causative agents

Item	Figure
Prevalence — developed countries	About 7–12%
Prevalence — India	About 25–40%
Staphylococci	~50% of all HAIs
Gram-negative bacilli — <i>Escherichia coli</i> , <i>Klebsiella</i> , <i>Proteus</i> , <i>Pseudomonas</i> , <i>Salmonella</i> , <i>Shigella</i>	~45%
Viruses, protozoa, rickettsiae, fungi	~5%; emergence of hepatitis B virus (HBV) and human immunodeficiency virus (HIV) has worsened the situation

(Suryakantha, Ch 19, pp 322–323)

2. Why the tertiary care hospital is at special risk

Feature of a tertiary hospital	Mechanism	Consequence
Referral of the sickest patients	Hospitalised patients are often immunocompromised; the severely ill and those on corticosteroid therapy are the most susceptible recipients	Highest attack rates in intensive care units, urological and geriatric wards (Park 28e, Ch 5, pp 430–431)
Intensive use of invasive devices and procedures	Catheterisation, intravenous lines, ventilation, dressings, sputum cups, bed pans and urinals all breach host defences	The four commonest HAIs — urinary tract infection, surgical wound infection, pneumonia and primary bloodstream infection — are each device- or procedure-associated (Park 28e, Ch 5, p 431)
Heavy, broad-spectrum antibiotic use	Selective pressure of intense antibiotic use	Emergence and spread of antibiotic-resistant organisms (Park 28e, Ch 5, p 430)

Feature of a tertiary hospital	Mechanism	Consequence
Large bed strength, teaching load, high bed turnover, overcrowding	Many hands per patient; dust, linen, furniture, sinks, door handles and air are laden with organisms	Sustained endemic transmission punctuated by outbreaks (Park 28e, Ch 5, pp 430–431)
Large volume of hazardous waste	Indian hospitals generate ½–4 kg of solid waste per bed per day (Bangalore survey, Rao 1995 — private hospitals ½–2 kg, nursing homes ½–1 kg) (Park 28e, Ch 16, p 944). Average Indian hospital-waste composition (National Environmental Engineering Research Institute 1997; 10 large hospitals in Mumbai, Kolkata, Delhi and Nagpur, 1993–96): infectious waste 1.5%, metal including sharps 1%, paper 15%, plastics 10%, rags 15%, glass 4.0%, general waste 53.5% (Park 28e, Ch 16, p 943). The World Health Organization (WHO) planning estimate for developing countries that have not done their own survey is 80% general, 15% pathological and infectious, 1% sharps, 3% chemical and pharmacological (Park 28e, Ch 16, p 943)	In Bangalore ~40 tonnes/day is generated, 45–50% of it infectious, yet segregation is practised in only ~30% of hospitals — risk of HIV and hepatitis B and C transmission by sharps injury to staff, waste handlers and scavengers (Park 28e, Ch 16, p 944)
Operating theatres, transplant, oncology, dialysis and neonatal units	Prolonged neutropenia, implants, extracorporeal circuits; renovation and construction work	Special risk of <i>Legionella</i> and <i>Aspergillus</i> infection during renovations (Park 28e, Ch 5, p 432)

3. Epidemiological chain: source, route, recipient

Park considers hospital-acquired infection from three angles — source, routes of spread, and recipients (Park 28e, Ch 5, pp 430–431).

Link	Components in a tertiary hospital
Source — patients	Viral infections (measles, german measles, influenza, viral hepatitis); skin infections (discharging wounds, infected skin lesions, eczema, psoriasis, boils, bed sores); respiratory infections (sore throat, pulmonary tuberculosis, chest infection); urinary tract infection (<i>Escherichia coli</i>)
Source — staff	<i>Staphylococcus aureus</i> carried in the nose or on the skin; haemolytic streptococci in the throat; salmonella in the gut
Source — environment	Hospital dust, linen, bed clothes, furniture, sinks, basins, door handles and even the air
Source — the patient's own flora	Self-infection (autoinfection): an organism normally non-pathogenic assumes pathogenic property because of a hospital procedure (Suryakantha, Ch 19, p 323)

Link	Components in a tertiary hospital
Routes of spread	(a) Direct contact — organism transferred from the hands of a nurse or doctor to a susceptible patient; (b) droplet infection from nose and throat on coughing or sneezing; (c) air-borne particles; (d) release of hospital dust into the air during sweeping, dusting and bed-making; (e) hospital procedures — catheterisation, intravenous procedures, infected catgut, dressings, sputum cups, bed pans, urinals
Recipients	All hospitalised patients are potential recipients; the severely ill and those on corticosteroid therapy are most susceptible; cross infection is greatest in intensive care units and in urological and geriatric wards

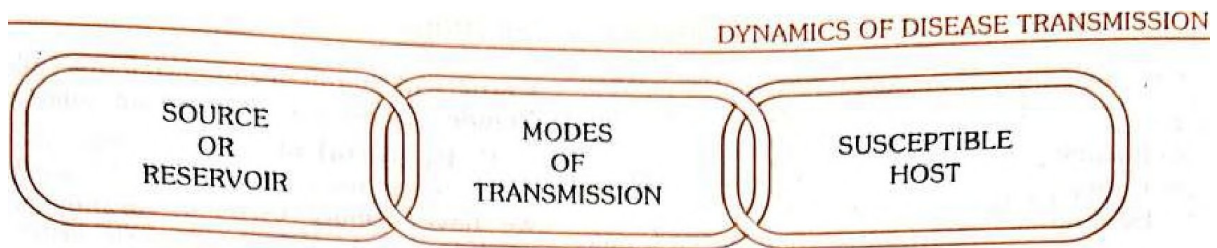
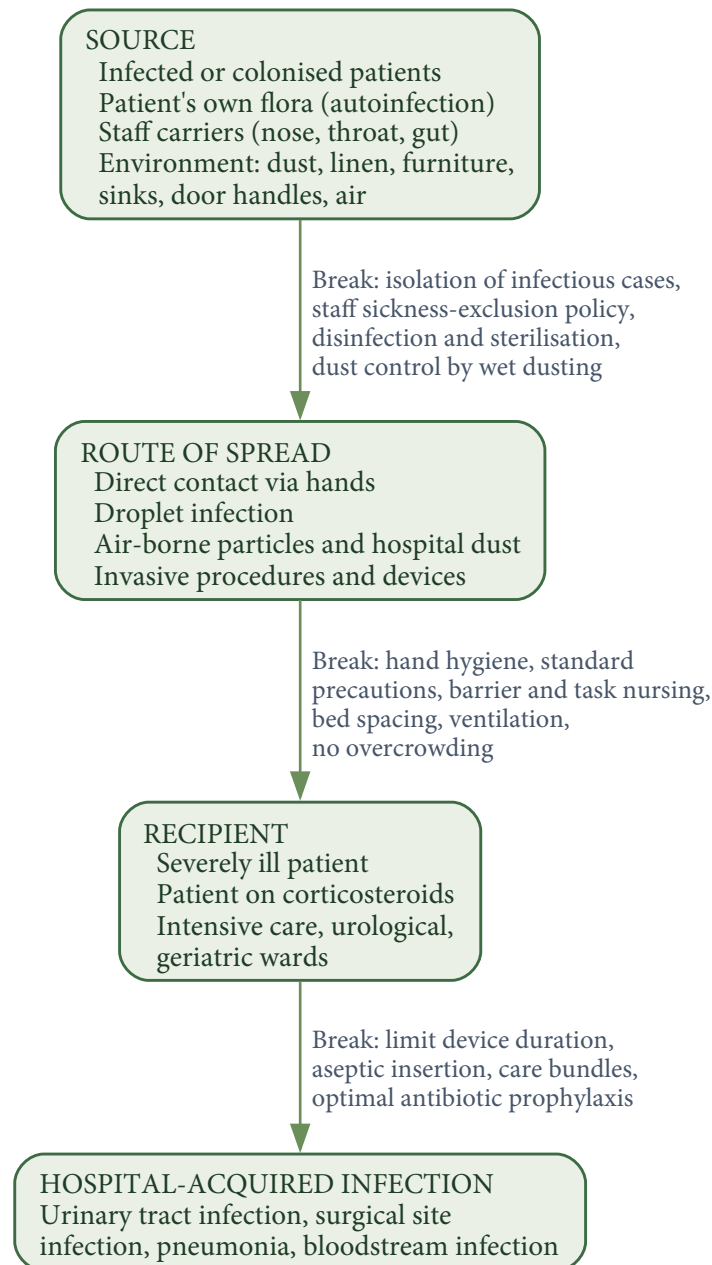


Fig 16 — Chain of Infection (Park's Textbook of Preventive and Social Medicine, 28e, p.115)



Chain of hospital-acquired infection in a tertiary care hospital, with the control measure that breaks each link

4. Case definitions used for surveillance

Simplified criteria for surveillance of nosocomial infections, usable in facilities with limited access to sophisticated diagnostics (Park 28e, Ch 5, p 430):

Type of nosocomial infection	Simplified criterion
Surgical site infection	Any purulent discharge, abscess, or spreading cellulitis at the surgical site during the month after the operation
Urinary infection	Positive urine culture (1 or 2 species) with at least 10^5 bacteria/ml, with or without clinical symptoms

Type of nosocomial infection	Simplified criterion
Respiratory infection	Respiratory symptoms with at least two of the following appearing during hospitalisation: cough, purulent sputum, new infiltrate on chest radiograph consistent with infection
Vascular catheter infection	Inflammation, lymphangitis or purulent discharge at the insertion site of the catheter
Septicaemia	Fever or rigors and at least one positive blood culture

5. Prevention and control

(a) General measures (Park 28e, Ch 5, p 431)

- ▶ **Isolation** — isolate infectious patients; never place a susceptible patient in a bed next to a source of infection.
- ▶ **Staff health** — staff with skin disease, sore throat, common cold, ear infection, diarrhoea or dysentery are kept off work until cured, with personal hygiene and regular change of aprons.
- ▶ **Hand-washing** — hands are the commonest route; washing must be thorough, and where soap and water is insufficient a suitable alcohol-based disinfectant is used.
- ▶ **Dust control** — wet dusting and vacuum cleaning, since dust is released during sweeping and bed-making.
- ▶ **Disinfection and sterilisation** — of the patient's articles, urine, faeces and sputum, and of instruments.
- ▶ **Control of droplet infection** — face masks, proper bed spacing, no overcrowding, adequate lighting and ventilation.
- ▶ **Nursing techniques** — barrier nursing and task nursing.
- ▶ **Protection of the susceptible** — admit highly susceptible patients in isolation wards and immunise where a vaccine exists, e.g. tetanus and hepatitis B (Suryakantha, Ch 19, p 324).

Standard (routine) precautions, applied to *all* patients (Park 28e, Ch 5, pp 431–432): wash hands promptly after contact with infective material; use a no-touch technique wherever possible; wear gloves for contact with blood, body fluids, secretions, excretions, mucous membranes and contaminated items, washing hands immediately after removal; add gown, mask and eye protection where contamination of clothes or face is anticipated; handle sharps with extreme care; clean spills promptly; discard, disinfect or sterilise equipment, supplies and linen between each patient use; ensure appropriate waste handling; boil soiled linen if no washing machine is available.

(b) Site- and device-specific measures for the four commonest infections

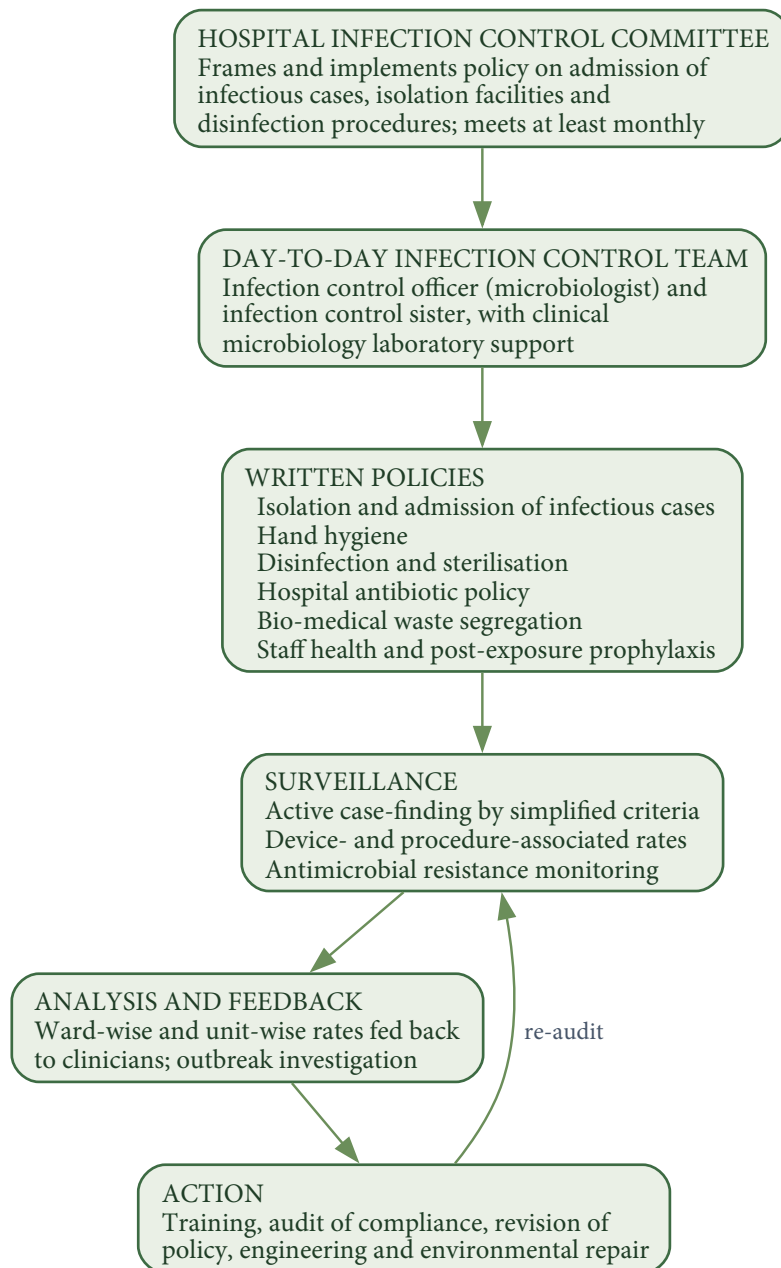
Infection	Proven effective	Proven not effective
Urinary tract infection	Limit duration of catheter; aseptic technique at insertion; maintain closed drainage	Antiseptic added to the drainage bag; antimicrobial-coated catheter; daily antiseptic perineal cleaning

Infection	Proven effective	Proven not effective
Surgical site infection	Surgical technique; clean operating environment; staff attire; limiting preoperative hospital stay; preoperative shower and local skin preparation; optimal antibiotic prophylaxis; aseptic practice in the operating room; surgical wound surveillance	Topical antibiotic prophylaxis; systemic irrigation or instillation of normal saline; fumigation; preoperative shaving; preoperative decontamination for all patients
Pneumonia	Ventilator-associated: aseptic intubation and suctioning, limit duration, non-invasive ventilation. Others: influenza vaccination for staff, isolation policy, sterile water for oxygen and aerosol therapy, prevention of <i>Legionella</i> and <i>Aspergillus</i> during renovations	Changing the ventilator circuit every 48 or 72 hours
Vascular device infection	All catheters: closed system, limit duration, local skin preparation, aseptic technique at insertion, removal if infection suspected. Central lines: surgical asepsis for insertion, limitation of frequency of dressing change, antibiotic-coated catheter for short-term use	Antimicrobial creams for skin preparation

(Park 28e, Ch 5, Table 2, pp 431–432)

Additional intensive-care bundle measures — the WHO prevention-of-complications bundle reproduced in Park's COVID-19 section (Park 28e, Ch 5, p 211): oral rather than nasal intubation in adolescents and adults; head of bed elevated 30–45°; closed suctioning with periodic draining of condensate; a new ventilator circuit per patient, changed only if soiled or damaged; heat-moisture exchanger changed when it malfunctions, when soiled, or every 5–7 days; and an insertion checklist verified by a real-time observer, prompting daily review for catheter removal.

6. Organising infection control in a tertiary care hospital



Hospital infection control programme in a tertiary care hospital — the surveillance-to-action loop

- **Committee and team** — the committee formulates policy on admission of infectious cases, isolation facilities, disinfection procedures and every other matter relating to control of hospital-acquired infection (Park 28e, Ch 5, p 431); it also conducts surveillance, frames guidelines for high-risk procedures, detects and controls outbreaks, and meets at least monthly. Its composition is set out in the organogram below (Suryakantha, Ch 19, p 324).
- **Surveillance** — continuous case-finding by the simplified criteria above, expressed as unit-wise device-associated rates and fed back to clinicians, with antimicrobial-resistance monitoring by the clinical microbiology laboratory.
- **Antibiotic policy** — a written, regularly revised hospital antibiotic policy with restricted-drug approval, because intense antibiotic use exerts selective pressure promoting resistance (Park 28e,

Ch 5, p 430), and resistance to commonly used drugs is itself a predisposing factor for HAI (Suryakantha, Ch 19, p 322).

- **Staff health programme** — review at recruitment covering immunisation history, previous exposure to communicable disease such as tuberculosis, and immune status, with varicella-zoster serology where indicated and a baseline Mantoux test. Immunisations: hepatitis A and B, yearly influenza, measles, mumps, rubella, tetanus and diphtheria, with varicella in specific cases. Written post-exposure policies are mandatory for HIV, hepatitis A, B, C and E viruses, *Neisseria meningitidis*, *Mycobacterium tuberculosis*, varicella-zoster virus, *Corynebacterium diphtheriae*, *Bordetella pertussis* and rabies (Park 28e, Ch 5, p 432).
- **Bio-medical waste management** — segregation at the point of generation into colour-coded, non-chlorinated bags in foot-operated bins under the Bio-Medical Waste Management Rules, 2016: **yellow** (anatomical, soiled, expired medicines, chemical, contaminated linen, laboratory waste) for incineration, plasma pyrolysis or deep burial; **red** (contaminated recyclable waste); **white translucent** (waste sharps including metals); **blue** (glassware and metallic implants). A waste register is maintained with monthly display, and all waste handlers get a health check-up at induction and at least yearly (Park 28e, Ch 16, pp 945–948; Bio-Medical Waste Management Rules, 2016).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Sites of the commonest nosocomial infections** (after Park 28e, Ch 5, Fig 1, p 431 — French National Prevalence Survey) — draw an outline human figure and label the sites: *surgical site, lower respiratory tract, skin and soft tissue, urinary tract, ear-nose-throat and eye, catheter site, other sites*.
- **Hospital Infection Control Committee organogram** (Suryakantha, Ch 19, p 324) — draw a top box *Chairperson: Medical Superintendent*; below it *Member Secretary: Infection Control Officer (microbiologist)*; a row of member boxes labelled *Chiefs of all clinical units / heads of department, Chief of blood bank service, Microbiologist, Medical record officer, Chief of nursing services, Infection control sister*; and a side box *Invited members: chiefs of all supportive services — operation theatre, kitchen, laundry*.

High-yield exam pointers

- Definition anchor: *not present or incubating at admission*; includes infection appearing after discharge and infection in staff.
- Magnitude: prevalence **25–40% in India** versus **7–12% in developed countries**; staphylococci ~50%, gram-negative bacilli ~45%, viruses/protozoa/rickettsiae/fungi ~5% (Suryakantha).
- Framework to write: **Source → Route → Recipient → Prevention** (Park's three angles), then the Hospital Infection Control Committee.
- The **four commonest HAIs**, all device- or procedure-associated: urinary tract infection, surgical wound infection, pneumonia, primary bloodstream infection.
- Cut-offs to quote exactly: urinary infection = positive urine culture (1 or 2 species), $\geq 10^5$ **bacteria/ml**, with or without symptoms; surgical site infection = purulent discharge, abscess or spreading cellulitis **within the month after operation**.

- **Proven not effective:** fumigation, preoperative shaving, routine ventilator-circuit change every 48–72 hours, antiseptic in the urinary drainage bag.
- **Waste:** Indian hospitals ½–4 kg per bed per day (Bangalore survey); Indian composition (NEERI 1997) infectious 1.5%, sharps 1%, general 53.5% — the often-misquoted "15% infectious, 1% sharps" is the WHO planning estimate for countries without their own survey, **not** Indian data.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park uses the older term "Control of Infection Committee", whereas Suryakantha and current Indian hospital and accreditation practice use "Hospital Infection Control Committee" with an infection control officer and infection control sister — write the current name and mention the textbook term once. No national antimicrobial-resistance containment programme text was retrieved, so none is quoted.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, WHO global action plan on infection prevention and control 2024–2030

Programme and policy updates (India)

- **India now has a national HAI surveillance network.** It was started in 2017 by AIIMS New Delhi with technical coordination by the Indian Council of Medical Research (ICMR) and the National Centre for Disease Control (NCDC) and support from the US Centers for Disease Control and Prevention, and runs active prospective bloodstream-infection surveillance in intensive care units of tertiary hospitals using case definitions modified for the Indian setting from the US CDC National Healthcare Safety Network (NHSN). Reporting began in May 2017 with 20 sites (66 ICUs) and reached 39 hospitals — 29 public and 10 private — with 131 surveillance ICUs across 22 of India's 36 states and union territories by December 2021, and about 200 adult, paediatric and neonatal ICUs in 54 hospitals by April 2024. The 2022 evaluation found the case definition correctly applied in 280 of 284 case report forms (98%), an automatic alert raised whenever an ICU-specific bloodstream-infection rate exceeds 20 per 1,000 patient days, and 12 of the 39 sites (31%) using the data to implement targeted infection-control measures; funding, limited human resources, absent digitisation, inconsistent information sharing and variable blood-culture practice — blood culture within 24 hours in only 44% of the febrile episodes reviewed — were the main constraints. [PMID 39623512] [PMID 40845882]
- **Ventilator-associated pneumonia now has an India-specific surveillance definition.** The CDC ventilator-associated event criteria need electronic medical records and daily review of ventilator settings, so the network adapted the CDC paediatric VAP criteria — keeping a positive culture as an essential criterion alongside respiratory, radiological and clinical findings — and pilot-tested the algorithm in 34 ICUs of 13 hospitals from February 2021 to April 2023. This is the first AIIMS New Delhi–ICMR collaboration to develop and pilot an India-specific VAP surveillance algorithm; sites reported interpretation of chest radiograph findings as its hardest component. [PMID 39130755]
- **Infection control is a scored domain in India's public-facility quality programmes.** A facility entering Kayakalp (2015) must first form a cleanliness and infection control committee and then

reach at least 70% of the criteria on peer assessment; infection control and waste management are two of its six thematic areas (facility upkeep, sanitation and hygiene, waste management, infection control, support services, hygiene promotion). Infection control is also a separate area of concern under the National Quality Assurance Standards (from 2013), LaQshya (2017) and MusQan (2021), and infection rates — surgical site, urinary tract, bloodstream and ventilator-associated pneumonia — are the clinical-care and safety outcome indicator under AB-PMJAY Bronze certification. Kayakalp enrolment rose from 100 facilities in 2015–16 to 19,488 in 2022–23, and NQAS-certified facilities from 10 in 2016–17 to 829 in 2022–23. [PMID 41280589]

Current national data

- ▶ **Central line-associated bloodstream infection (network data, May 2017 – April 2024):** 8,629 laboratory-confirmed events over 977,052 central-line days and 3,054,124 patient days — a pooled rate of 8.83 per 1,000 central-line days with a central-line utilisation ratio of 0.32, being 8.68 in adult, 6.71 in paediatric and 13.86 per 1,000 central-line days in neonatal ICUs. Of 10,042 pathogens reported 89.4% were bacterial and 10.6% fungal; *Klebsiella pneumoniae* (22.8%) and *Acinetobacter baumannii* (20.4%) were the commonest, and carbapenem resistance reached 87.1% in *A. baumannii* and 77.7% in *K. pneumoniae* among the isolates tested. [PMID 40845882]
- ▶ **Ventilator-associated pneumonia (network pilot, February 2021 – April 2023):** 261 events over 133,445 patient days and 40,533 ventilator days, giving 6.4 per 1,000 ventilator days with a device-utilisation ratio of 0.30; most events occurred between the third and seventh day in the unit (median 6 days) and 126 of the 261 events (48.3%) had a fatal outcome. Of 344 bacterial isolates spanning 20 species the two commonest supplied 29.6% and 26.7% of isolates; carbapenem resistance reached 98% in the most resistant pathogen group, 85.5% in Enterobacterales and 64.6% in a third group, and colistin resistance was already 6% in Enterobacterales. [PMID 39130755]
- ▶ **Candidaemia has become a measurable ICU HAI in India:** across 67 network ICUs, 377 events over 401,601 patient days gave an overall rate of 0.93 per 1,000 patient days, significantly higher in COVID-19 than in non-COVID-19 ICUs (2.52 versus 1.05 per 1,000 patient days), as was central line-associated candidaemia (4.53 versus 1.73 per 1,000 central-line days); mortality associated with candidaemia was 61% in COVID-19 versus 41% in non-COVID-19 ICUs. Of 435 isolates *Candida tropicalis* (26.7%) was commonest and *Candida auris*, which carried a high mortality, accounted for 17.5%. [PMID 39278818]

Elimination, surveillance and disease status

- ▶ **IPC structural readiness is now measured, not assumed.** A self-assessment using the WHO Infection Prevention and Control Assessment Framework (IPCAF) across 32 tertiary public and private facilities of the network returned a median score of 620, with 59% of hospitals rated advanced, 28% intermediate and 13% basic; multimodal improvement strategies showed the widest spread of scores between hospitals and IPC guidelines the narrowest. The network's own later report identifies surveillance for HAIs and human resources as the common lacunae that this baseline IPCAF assessment exposed in Indian hospitals. [PMID 34600081] [PMID 39130755]

Guideline and legislative updates

- ▶ **An integrated stewardship framework for tertiary hospitals in developing countries.** The Society of Antimicrobial Stewardship Practices in India has published 42 Integrated Antimicrobial Stewardship practice statements spanning administrative, diagnostic, infection-prevention and antimicrobial-use domains, summarised as a ten-point "PRESCRIBES" checkpoint — policy for AMR containment, resource integration, education of the different cadres, surveillance of practices, the right culture of sending and acting on diagnostics, rational antimicrobial use, information through AMR dashboards, behavioural interventions, engagement across sections and cadres, and sustainability plans. Unlike conventional siloed models it makes hospital leadership, the diagnostic laboratory and prescribers jointly accountable, and it is offered for regular self-audit against the National Action Plan on AMR and the WHO global action plan. [PMID 41525923]
- ▶ **Kerala's sub-national model shows what a state-level programme looks like.** KARSAP (2018) was India's first state AMR action plan, drafted jointly by the health, animal husbandry, environment, fisheries and agriculture departments on a One Health platform; its surveillance arm KARS-NET (2018, established with help from the WHO country office for India and the NCDC) has expanded to 59 tertiary satellite centres and 185 primary and secondary spokes with Government Medical College Thiruvananthapuram as focal point, Kerala published India's first state antibiogram in 2022 and the first district antibiogram (Ernakulam) in 2023, and AMR committees now exist in every district and in 190 health blocks. An Antibiotic Smart Hospital at tertiary level must have a functional hospital infection control committee and antimicrobial stewardship committee, HAI surveillance, quarterly prescription audit and antibiotic-utilisation metrics, 85% Access-group use in the outpatient department, NQAS/Kayakalp/NABH certification and a sewage or effluent treatment plant. Under Operation AMRITH the public can report on a toll-free number any antibiotic dispensed without a prescription, and such dispensing costs the pharmacist their licence — Kerala being the only Indian state to enforce the stringent Schedule H rule. [PMID 40657033]

Global and WHO developments

- ▶ **Infection prevention and control is now a formal global target, not merely good practice.** The World Health Assembly adopted a global strategy on IPC in 2023 and, in 2024, a global action plan and monitoring framework on IPC for 2024–2030 setting out actions, indicators and targets for national and facility-level IPC [WHO 2024]. The UN General Assembly political declaration on AMR of 26 September 2024 added complementary 2030 targets — a 10% reduction in the estimated 4.95 million annual deaths associated with bacterial AMR, at least 70% of antibiotics used for human health drawn from the WHO Access group, basic water, sanitation, hygiene and waste-management services in all health-care facilities in 100% of countries, and 90% of countries meeting WHO's minimum requirements for IPC programmes — together with sustainable national financing and US\$100 million in catalytic funding for national action plans. [PMID 40657033]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases: Hospital-Acquired Infection, Tables 1 and 2; Ch 10 — Health Care of the Community; Ch 16 — Hospital Waste Management, Table 1 and the Schedule I colour-code table); Suryakantha — Community Medicine with Recent Advances (Ch 19 — Epidemiology of Infectious Diseases: Hospital-Acquired Infections); Bio-Medical Waste Management Rules, 2016 (G.S.R. 343(E), Ministry of Environment, Forest and Climate Change), Schedule I; NEERI hospital-waste composition survey, 1997; Rao H.V.N. (1995), Bangalore hospital-waste survey; WHO (2002), *Prevention of Hospital-Acquired Infections: A Practical Guide*, 2e; WHO (2009), *Guidelines on Hand Hygiene in Health Care*.

Recent-advances sources: 39623512 — Implementing a healthcare-associated bloodstream infection surveillance network in India: a mixed-methods study on the best practices, challenges and opportunities, 2022, *Antimicrobial Resistance and Infection Control* 2024, doi:[10.1186/s13756-024-01501-6](https://doi.org/10.1186/s13756-024-01501-6); 40845882 — Profile of central line-associated bloodstream infections in adult, paediatric, and neonatal intensive care units of hospitals participating in a health-care-associated infection surveillance network in India: a 7-year multicentric study, *The Lancet Global Health* 2025, doi:[10.1016/S2214-109X\(25\)00221-9](https://doi.org/10.1016/S2214-109X(25)00221-9) (abstract-only); 39130755 — Surveillance of ventilator associated pneumonia in a network of Indian hospitals using modified definitions: a pilot study, *The Lancet Regional Health – Southeast Asia* 2024, doi:[10.1016/j.lansea.2024.100450](https://doi.org/10.1016/j.lansea.2024.100450); 41280589 — To gain insights and compare different quality certification programs for public health facilities in India, *Journal of Family Medicine and Primary Care* 2025, doi:[10.4103/jfmpc.jfmpc.1591.24](https://doi.org/10.4103/jfmpc.jfmpc.1591.24); 39278818 — Candidaemia and central line-associated candidaemia in a network of Indian ICUs: impact of COVID-19 pandemic, *Mycoses* 2024, doi:[10.1111/myc.13790](https://doi.org/10.1111/myc.13790) (abstract-only); 34600081 — Self-reported survey on infection prevention and control structures in healthcare facilities part of a national level healthcare associated infection surveillance network in India, 2019, *American Journal of Infection Control* 2022, doi:[10.1016/j.ajic.2021.09.019](https://doi.org/10.1016/j.ajic.2021.09.019) (abstract-only); 41525923 — An integrated antimicrobial stewardship approach in tertiary care hospitals in developing countries using a multidomain framework ("PRESCRIBES" checkpoint): a call for action, *International Journal of Infectious Diseases* 2026, doi:[10.1016/j.ijid.2026.108388](https://doi.org/10.1016/j.ijid.2026.108388) (abstract-only); 40657033 — When policy-makers have your back: the Kerala experience with state-wide antimicrobial resistance mitigation efforts, *Antimicrobial Stewardship & Healthcare Epidemiology* 2025, doi:[10.1017/ash.2025.10060](https://doi.org/10.1017/ash.2025.10060). WHO global action plan and monitoring framework on infection prevention and control 2024–2030 (World Health Assembly, 2024); WHO global strategy on infection prevention and control (2023).

♦ ♦ ♦

Management information system

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — A **management information system (MIS)** in health, usually called the **health management information system (HMIS)**, is a process whereby health data (input) are recorded, stored, retrieved and processed for decision-making (output) — the decisions being the managerial ones of planning, organizing and control of health care facilities at national, state and institutional levels (Jugal Kishore, National Health Programmes of India, p 262). The parent **health information system** is defined by WHO as *a mechanism for the collection, processing, analysis and transmission of information required for organizing and operating health services, and also for research and training* (Park 28e, Ch 21, p 995). Operationally it is a system that supplies **up-to-date, reliable, complete and timely** information to health managers at every level so they can take well-informed decisions about programme performance and operations (Sunder Lal, Ch 8, p 500).

Position of the MIS in health management

Park classifies management methods into those based on the **behavioural sciences** — organizational design, personnel management, communication, **information systems**, management by objectives — and **quantitative methods** such as cost-benefit and cost-effectiveness analysis, cost-accounting, input-output analysis, models, systems analysis, network analysis, planning-programming-budgeting system, work sampling and decision making (Park 28e, Ch 23, pp 1028-1029). The MIS is therefore not a statistical exercise but a **management method**: information is needed for the day-to-day running of the health system, comes from both formal and informal sources, and must be **tailored to the management needs of the individual health service** rather than collected for its own sake (Park 28e, Ch 23, p 1028).

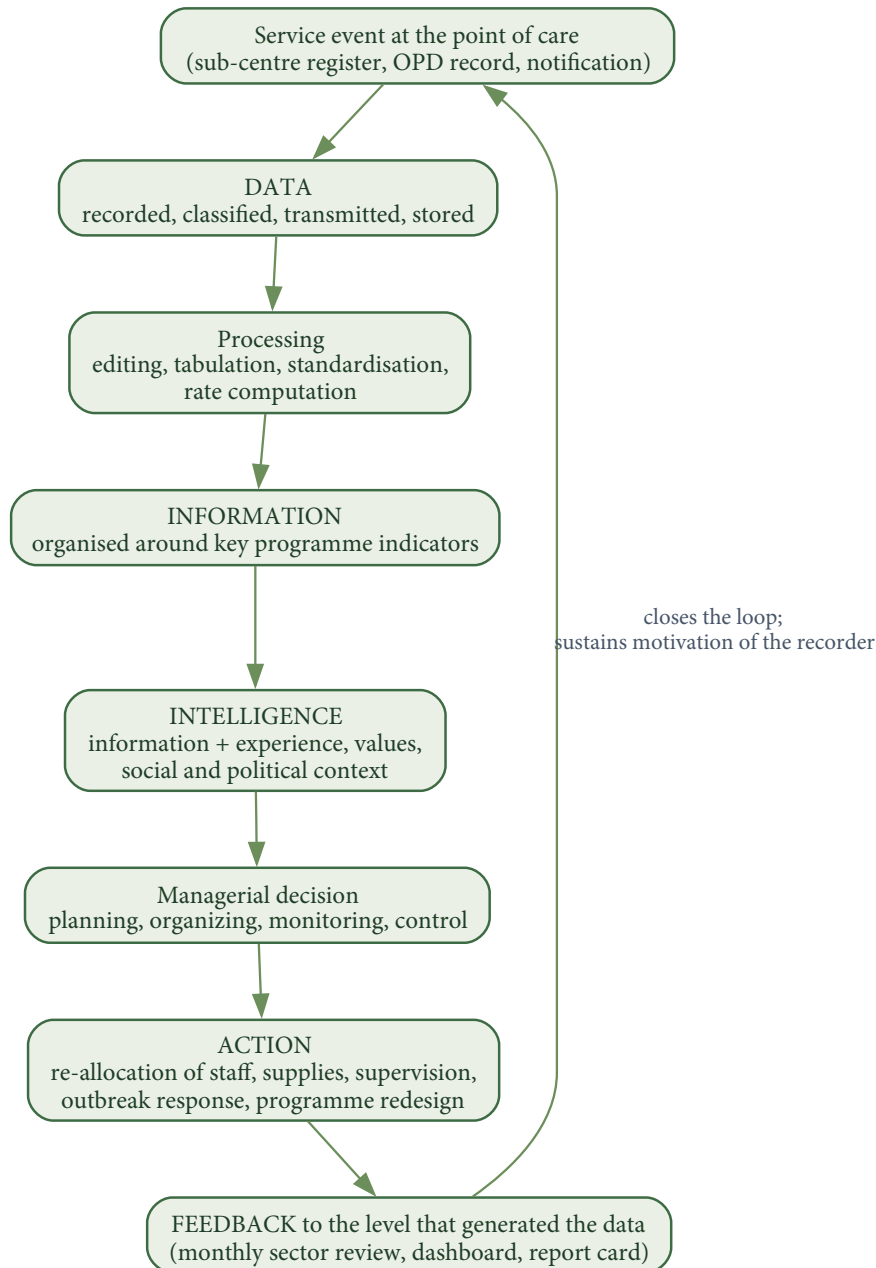
Element	Content
Functions of an information system	Collection → classification → transmission → storage → retrieval → transformation → display of information (Park 28e, Ch 23, p 1028)
Primary objective	Provide reliable, relevant, up-to-date, adequate, timely and reasonably complete information to health managers at central, intermediate and local levels, and technical/scientific information to all health personnel (Park 28e, Ch 21, p 995)
Design question	Who should gather what information, who receives it for decision purposes, and who needs to know the decision made (Reinke, Health Planning for Effective Management, Ch 2, p 41)
Governing principles	Management by objectives and management by exception — the routine system surveys all potential problem areas, and the "exceptions" it flags are then investigated selectively by more intensive second-stage data gathering (Reinke, Ch 2, p 41)
Cardinal failure	Information systems are typically characterised by "a superabundance of data and a dearth of useful information"; much effort goes into report preparation but the reports are seldom used for decision making (Reinke, Ch 2, pp 40-41)

Data, information and intelligence

The distinction is more than semantic and is the conceptual core of any MIS (Park 28e, Ch 21, p 995).

Stage	What it is	How it is produced
Data	Observations of attributes or events; carry little meaning when considered alone	Recorded in registers, formats, returns
Information	Data reduced, summarised and adjusted for variations such as age and sex composition, so that comparisons over time and place are possible	Editing, tabulation, standardisation, computation of rates
Intelligence	Information integrated and processed with experience, values and perceptions based on social and political factors	Interpretation by the manager for a specific decision

Data not transformed into information, and information not transformed into intelligence, are of no use to decision-makers, policy-makers, planners, administrators or health care personnel (Park 28e, Ch 21, p 995).



Requirements and components of a health information system

A WHO Expert Committee laid down **six requirements** that a health information system must satisfy, and specified the **subsystems** on which a comprehensive system must carry information and indicators (Park 28e, Ch 21, p 995).

WHO requirement	Components (subsystems) required
1. Should be population-based	1. Demography and vital events
2. Should avoid unnecessary agglomeration of data	2. Environmental health statistics
3. Should be problem-oriented	3. Health status — mortality, morbidity, disability, quality of life

WHO requirement	Components (subsystems) required
4. Should employ functional and operational terms (episodes of illness, treatment regimens, laboratory tests)	4. Health resources — facilities, beds, manpower
5. Should express information briefly and imaginatively (tables, charts, percentages)	5. Utilization and non-utilization of services — attendance, admissions, waiting lists
6. Should make provision for the feedback of data	6. Indices of outcome of medical care
	7. Financial statistics — cost and expenditure related to the particular objective

Alongside quantifiable data, planners need **non-quantifiable information** — health policies, health legislation, public attitudes, programme costs, procedures and technology — so the system has multidisciplinary inputs requiring proper storage, processing and dissemination (Park 28e, Ch 21, p 999).

Uses of health information

Use	What it answers
1. Measure health status	Quantify the people's health problems and their medical and health care needs
2. Comparison	Local, national and international comparison of health status — demands rigorous standardization and quality control
3. Planning and management	Planning, administration and effective management of health services and programmes
4. Performance	Whether health services are accomplishing their objectives, in terms of effectiveness and efficiency
5. Beneficiary view	Attitudes and degree of satisfaction of beneficiaries with the health system
6. Research	Research into particular problems of health and disease

(Park 28e, Ch 21, pp 995-996)

SOURCE & COVERAGE

#	Source	Chief value / chief limitation
1	Census (last held March 2011; Census Act 1948)	Denominators and socio-economic profile / results not available quickly
2	Registration of vital events (Registration of Births and Deaths Act 1969; reporting within 21 days)	Legal foundation of vital statistics / grossly deficient in accuracy, timeliness, completeness
3	Sample Registration System (SRS) — dual record system, now covers the whole country	Reliable birth, death, infant, under-five and adult mortality rates
4	Notification of diseases	Early warning of outbreaks / only a small part of total sickness; under-reporting; mild and subclinical cases escape
5	Hospital records	Basic primary source in India / only the "tip of the iceberg" — numerator without a definable denominator
6	Disease registers (cancer, stroke, blindness, congenital defects, tuberculosis, leprosy)	Permanent record with follow-up — duration, case fatality, survival
7	Record linkage	Studies associations between diseases / huge volume of data, limited scale
8	Epidemiological surveillance (disease control programmes)	Programme-specific morbidity and mortality data
9	Other health service records — OPD, PHC and sub-centre, MCH, school health, NCD clinics	Service delivery data / segment-specific, kept for administration not monitoring
10	Environmental health data	Air, water, food additives, industrial toxicants, noise, waste disposal
11	Health manpower statistics (councils, Directorates of Medical Education)	Doctors, dentists, nurses, technicians; published in <i>Health Information of India</i>
12	Population surveys (health interview, health examination, health records, mailed questionnaire)	The only population-based check on routine statistics
13	Other routine statistics — demographic, economic, social security schemes	Consumption of tobacco and fats, drug sales, sickness absence, disability benefit
14	Non-quantifiable information	Policies, legislation, attitudes, costs, technology

(Park 28e, Ch 21, pp 996-999)

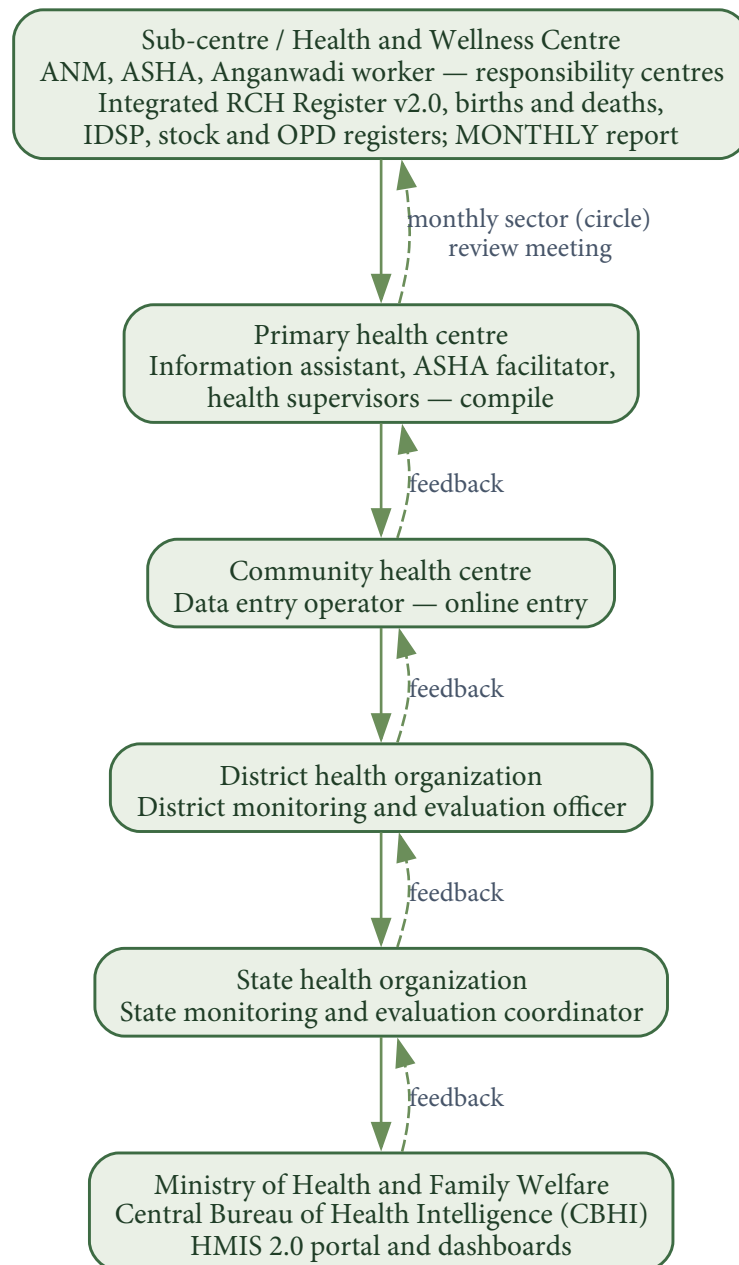
HMIS in India — structure, evolution and data flow

Milestone	Detail
Comprehensive HMIS model evolved	1992, after a test run; 13 separate registers designed for the sub-centre, all maintained by the auxiliary nurse midwife (ANM)

Milestone	Detail
Registers consolidated	Village-wise Integrated Reproductive and Child Health Register (IRCHR) , 2008-09, when HMIS version 1.0 began functioning with district-level reporting
Revamp	2019-20; HMIS version 2.0 launched December 2020 with WHO technical support; further features added 2022
HMIS 2.0 features	Real-time data entry and monitoring; multi-device compatibility; integration with other portals through application programming interface (API); interactive dashboards; geographic information system (GIS) layer with village-level and Local Government Directory (LGD) compliant unique facility identification
Data volume	~ 400 data elements of service delivery captured monthly; ~ 700 elements of infrastructure and human resources annually with monthly update; daily data entry from April 2020
Facility mapping	2.17 lakh health facilities mapped as sub-centre / PHC / CHC / sub-district hospital / district hospital across 747 districts ; service data entered by the 5th of the following month
Client tracking	Mother and Child Tracking System (MCTS), launched December 2009, name-based tracking; formats upgraded 2017 and 2022 into Integrated RCH Register version 2.0 (launched June 2022)
Digital layer	GIS-enabled HMIS application March 2016; Ayushman Bharat Digital Mission (ABDM) , formerly the National Digital Health Mission, launched September 2021 — a unique health account number linking longitudinal digital health records, implemented by the National Health Authority; a step towards the Integrated Health Information Platform (IHIP)

(Sunder Lal, Ch 8, pp 500-504 and Ch 17, p 763; Jugal Kishore, pp 262-264)

National Health Policy 2017 targets for health management information — a district-level electronic database of information on health system components by 2020, a strengthened health surveillance system, and a federated integrated health information architecture with health information exchanges and a national health information network by **2025** (Park 28e, Ch 23, pp 1030-1031).



Upward reporting and downward feedback in the Indian HMIS

Constraints of the Indian HMIS and measures to strengthen it

Constraint	Corrective measure
Incomplete reporting; coverage of the population not total, especially in urban areas	Facility mapping and mandatory monthly online entry; private-sector reporting
Poor quality of data on causes of death and on weighing of children	Supervisory visits with sample re-check of household survey and spot correction
Information seldom used at local level by health workers or medical officers; system remains "data driven rather than action driven"	Monthly sector review meetings that begin from the sub-centre's own indicators; sub-centre annual action plan
No feedback to the worker who collects the data → loss of motivation	Structured downward feedback at every level; display of sub-centre and Anganwadi data at the village centre

Constraint	Corrective measure
Information not shared with the community	Sub-centre reports to the <i>Panchayat</i> , hospital to the <i>Rogi Kalyan Samiti</i> , District Health Mission to the <i>Zila Parishad</i> ; annual district report card on people's health
Over-reporting and inflation of performance figures	Triangulation against population surveys (Sample Registration System, National Family Health Survey), community-based monitoring under the National Health Mission
Multiple parallel vertical-programme formats and duplication of registers	Integration — Integrated RCH Register v2.0, HMIS 2.0 with API integration, movement towards the Integrated Health Information Platform
Erratic supply of registers and formats; weak training	Assured logistics of formats; HMIS content in induction and continuing education of workers and medical officers
Managers rely on "gut feeling" rather than the database	Management by exception — act on the flagged indicator; scorecard and dashboard review at district level

(Sunder Lal, Ch 8, pp 503-504)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The management information cycle** — label: *Data → Processing → Information → Intelligence → Decision → Action → Feedback*, drawn as a closed loop, with "point of care" at the start of the loop and "feedback to the recorder" as the returning arm.
- **HMIS reporting pyramid** — a four-tier triangle with, from base to apex: *Sub-centre (ANM, ASHA, Anganwadi worker) → PHC (information assistant) → CHC (data entry operator) → District M&E officer → State M&E coordinator → MoHFW / Central Bureau of Health Intelligence*; label solid up-arrows "monthly reports" and dashed down-arrows "feedback".

High-yield exam pointers

- **Definition to write first:** HMIS = health data (input) recorded, stored, retrieved and processed for decision-making (output) — Jugal Kishore; WHO health information system = mechanism for collection, processing, analysis and transmission of information for organizing and operating health services and for research and training.
- **Functions (7):** collection, classification, transmission, storage, retrieval, transformation, display.
- **WHO's 6 requirements** (population-based, no unnecessary agglomeration, problem-oriented, functional/operational terms, brief and imaginative presentation, feedback) and the 7 components.
- **6 uses and the 14 sources** — the examiner expects the Indian sources named: Census 2011, Registration of Births and Deaths Act 1969 (21-day limit), Sample Registration System, notification, hospital records.

- **Dated Indian figures:** HMIS v1.0 2008-09; HMIS v2.0 December 2020; ~400 monthly service elements and ~700 annual infrastructure/human-resource elements; 2.17 lakh facilities in 747 districts; reporting by the 5th of the following month; Integrated RCH Register v2.0 June 2022; ABDM September 2021.
- **Data ≠ information ≠ intelligence** — write this distinction explicitly; it is the single point that separates a management answer from a statistics answer.
- **Weakest link:** non-use of data at the level that generates it; the cure is feedback, not more formats.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not written in the exam — the health management information system has been repeatedly renamed and re-versioned (HMIS v1.0 2008-09, HMIS v2.0 December 2020, further features 2022) and is progressively being folded into the Integrated Health Information Platform and the Ayushman Bharat Digital Mission (formerly the National Digital Health Mission, launched September 2021), so quote each figure with the year attached and give the current name first.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, National Health Authority (Ayushman Bharat Digital Mission), MoHFW HMIS, Registrar General of India (Civil Registration System), Registration of Births and Deaths (Amendment) Act 2023

Programme and policy updates (India)

- **Scan and Share**, the National Health Authority's QR-code paperless outpatient registration service built on ABDM in 2022, was operating in 16,939 facilities across 35 of 36 states and union territories by 19 October 2024, had issued 56,801,213 tokens (only 1.3% from the private sector), and cut registration waiting from about an hour to 2-5 minutes; 410 million health records had been linked to Ayushman Bharat Health Account numbers, and a Digital Health Incentive Scheme now funds facility uptake — the operational successor to the HMIS 2.0 "API integration" line in the textbook. [PMID 39882498]
- ABDM progress assessed against the health-system control-knob framework: 400 million ABHA registrations and 273 million linked health records by June 2023, with 208,000 facilities and 190,000 health professionals verified — but registration of facilities and professionals varies sharply between states, and that, not beneficiary enrolment, is the rate-limiting step for payment, organisation and regulation. [PMID 39437234]

Current national data

- India's Civil Registration System report for 2021 was released only in May 2025, a four-year lag: 10.2 million registered deaths (26% above 2020) against 335,004 officially reported COVID-19 deaths, giving about 2.4 million excess deaths and a 7.2-fold undercount; state ratios ran from under 2 (Kerala, Goa) to over 40 (Gujarat), and 47.3% of the deceased had received no medical attention before death — the sharpest available illustration of the textbook's "grossly deficient in accuracy, timeliness and completeness" verdict on vital registration. [PMID 41232123]

- First national external-consistency check of HMIS against a population survey: across 695 districts, HMIS and NFHS-5 agreed well only for institutional births ($r = 0.758$) and caesarean section ($r = 0.892$); agreement was weak for sex ratio at birth ($r = 0.148$), anaemia in pregnancy ($r = 0.135$) and tetanus protection ($r = 0.225$), intraclass correlation was below 0.5 for five of seven indicators, and HMIS over-reported four antenatal visits by a mean 15.1 percentage points — quantifying the "over-reporting and inflation of performance figures" constraint. [PMID 40547776]

Elimination, surveillance and disease status

- The Integrated Health Information Platform is only as sensitive as its data entry: in Raigarh district, Chhattisgarh (September 2023) just 44% of laboratory-confirmed malaria at primary health centres reached the platform; a Plan-Do-Study-Act cycle pairing refresher training for laboratory technicians with weekly feedback from the district surveillance unit raised sensitivity to 100% within five weeks — feedback, not new formats, was the corrective. [PMID 41629241]
- Anaemia Mukht Bharat monitors districts through 18 anaemia indicators and six key performance indicators drawn from routine HMIS and consolidated into an AMB index on an online dashboard; a Faridabad audit (April 2018-March 2019) returned an index of 35.2, IFA prophylaxis in 22.5% of children aged 6-59 months and 180-day IFA in 33.1% of pregnant women, and identified number-only reporting with mismatched population-projection denominators (deworming coverage computing above 100%) as the limiting defect. [PMID 39764332]

Guideline and legislative updates

- The **Registration of Births and Deaths (Amendment) Act 2023** underpins a shift towards *suo motu*, health-facility-based registration; CRS 2022 recorded only 74.4% of births and 73.8% of deaths registered inside the statutory 21 days, and a Uttar Pradesh field study found 55% registering within 21 days and 20% beyond 30 days, driven by low awareness (81%), hospitals not issuing certificates (30%) and access difficulty (20%) — with registration delayed beyond one year still needing an Executive Magistrate's order under section 13(3) of the 1969 Act. [PMID 41783707]
- Interoperability is moving from portals to standards: the first Indian oncology data model built on HL7 FHIR Release 4 — 25 resource profiles and 50 terminology value sets mapped to SNOMED-CT, LOINC and ICD-10, with FHIR Questionnaire and QuestionnaireResponse profiles for community screening of five cancers — is designed to plug into ABDM so that screening, hospital and registry data share one syntax rather than parallel vertical formats. [PMID 41329957]

Global and WHO developments

- WHO's 2022 consolidated guidelines on person-centred HIV strategic information define a minimum individual-level dataset; a survey of 18 countries including India found 83% using some form of unique identification and all collecting HIV testing elements, but harm-reduction data elements in only 4-6 countries and only two countries able to compute linkage to prevention — the current global template for what "person-centred" routine data actually requires. [PMID 41059651]

- A nine-country landscape analysis of routine health information systems, using India's HMIS data-element list among others, found only 14 of 29 (48%) globally proposed surgical-obstetric and family-planning indicators actually tracked, with just caesarean delivery and modern-method family-planning users common to all nine, alongside indicator profusion, duplication and vague definitions — the international restatement of Reinke's "superabundance of data and a dearth of useful information". [PMID 40795307]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 21 — Health Information and Basic Medical Statistics, pp 995-999; Ch 23 — Health Planning and Management, pp 1028-1031); Sunder Lal — Textbook of Community Medicine (Ch 8 — Health Management Information System, pp 500-504; Ch 17, p 763); Jugal Kishore — National Health Programmes of India (pp 262-264); Reinke — Health Planning for Effective Management (Ch 2 — Management Analysis, pp 40-41); National Health Policy 2017 (health management information targets, as cited in Park 28e).

Recent-advances sources: 39882498 — Scan and share to register outpatients, India, *Bull World Health Organ* 2025, doi:[10.2471/BLT.24.292828](https://doi.org/10.2471/BLT.24.292828); 39437234 — The Ayushman Bharat Digital Mission of India: An Assessment, *Health Syst Reform* 2024, doi:[10.1080/23288604.2024.2392290](https://doi.org/10.1080/23288604.2024.2392290) (abstract-only); 41232123 — India's death toll in 2021 during the COVID-19 pandemic: insights from delayed official civil registration data, *J Glob Health* 2025, doi:[10.7189/jogh.15.03045](https://doi.org/10.7189/jogh.15.03045); 40547776 — Study on external consistency of health management information system data at district level across India, *J Family Med Prim Care* 2025, doi:[10.4103/jfmprc.jfmprc_1967_24](https://doi.org/10.4103/jfmprc.jfmprc_1967_24); 41629241 — Quality Improvement Initiative to Improve the Sensitivity of Malaria Reporting under Integrated Health Information Platform, Raigarh, Chhattisgarh, 2023, *Indian J Public Health* 2026, doi:[10.4103/ijph.ijph_648_25](https://doi.org/10.4103/ijph.ijph_648_25) (abstract-only); 39764332 — Digital Monitoring of Anemia Control Measures in a District Using Anemia Mukta Bharat Health Management Information System Indicators, *Cureus* 2024, doi:[10.7759/cureus.75248](https://doi.org/10.7759/cureus.75248); 41783707 — From delivery to identity: applying transaction cost economics to health-enabled civil registration in India, *Front Public Health* 2026, doi:[10.3389/fpubh.2026.1743341](https://doi.org/10.3389/fpubh.2026.1743341); 41329957 — FHIR Standard-Based Oncology Data Model for Cancer Screening: Design and Implementation Study, *JMIR Cancer* 2025, doi:[10.2196/79011](https://doi.org/10.2196/79011); 41059651 — Harnessing digital health data for person-centred HIV prevention monitoring: a survey of national health information systems, *J Int AIDS Soc* 2025, doi:[10.1002/jia2.70039](https://doi.org/10.1002/jia2.70039); 40795307 — Surgical indicators for obstetrics and family planning in routine health information systems: a landscape analysis, *Health Policy Plan* 2025, doi:[10.1093/heapol/czaf052](https://doi.org/10.1093/heapol/czaf052). Also: Ayushman Bharat Digital Mission public dashboard, National Health Authority (October-November 2024); Civil Registration System report 2021, Registrar General of India (released May 2025) and CRS 2022; Registration of Births and Deaths (Amendment) Act 2023; WHO Consolidated guidelines on person-centred HIV strategic information (2022).



Material management in a primary health centre

Asked in: Paper IV 28-Jun-2024 — RGUHS MD Community Medicine

Definition — Material (materials) management is the system that ensures the supply of the **required quantity and quality of material, when and where it is needed**, at the lowest total cost; **logistics** is the narrower function of moving, storing and distributing those goods (Sathe & Sathe, Ch 13). In a primary health centre (PHC) — the first tier of medical care serving 30,000 population in the plains and 20,000 in tribal and hilly areas (Park 28e, Ch 7) — it covers everything from drugs, vaccines, contraceptives and reagents to instruments, furniture, linen and the ambulance.

Why it matters at a PHC

- ▶ **Money:** 25–40% of the total expenditure of hospitals, dispensaries and health centres goes on purchase of material, and the yearly cost of *carrying* inventory is a further 20–25% of its value (Sathe & Sathe, Ch 13).
- ▶ **Availability:** studies of Indian PHCs show quantities short of requirement, erratic supply, common drugs out of stock for long periods, and stock received close to expiry (Sathe & Sathe, Ch 13).
- ▶ **Cold chain:** vaccine potency depends entirely on storage and stock discipline at the PHC cold-chain point.
- ▶ **Variety:** thousands of pharmaceutical preparations, plus consumables and non-consumable equipment across many specialities, must be handled by one medical officer and one pharmacist.
- ▶ **Contraceptive and programme supply:** excess stock in one centre and stock-out in another, deterioration, and unreconciled physical stock are the recurring failures (Sathe & Sathe, Ch 13).
- ▶ **Accountability:** pilferage and near-expiry condemnation are the commonest audit objections against a PHC medical officer.

The nine primary objectives are low prices, high inventory turnover, low cost of acquisition and possession, continuity of supply, consistency of quality, low payroll cost, favourable relations with suppliers, development of personnel, and good records (Sathe & Sathe, Ch 13).

Components of the material management cycle

Component	What the PHC medical officer actually does
1. Identification / selection of items	Adopt a short PHC formulary from the National List of Essential Medicines (NLEM) — India's list was first issued 1996 and revised 2003, 2011, 2015 and 2022 (Park 28e, Ch 24, p 1041). Use generic (non-proprietary) names ; avoid irrational fixed-dose combinations.
2. Estimating demand	Work out the time trend of consumption over the previous 2–3 years, quarter by quarter to capture seasonality; discount unusual events such as an epidemic; allow for past stock-outs and for new activities (new sub-centre, new programme).
3. Purchase and procurement	Indent against rate contract; local purchase only for an out-of-stock vital drug.

Component	What the PHC medical officer actually does
4. Inspection and quality control	Check quantity, quality and expiry against the supply order before acceptance; prefer BIS (ISI) / ISO-marked items; obtain warranty or maintenance contract for equipment.
5. Storage	Separate medical and non-medical stores; ground floor / basement; light, ventilation, racks; ice-lined refrigerator (ILR) and deep freezer for vaccines.
6. Inventory control	ABC / VED and the min-max system (below).
7. Issue procedure	Written indent from a competent person; FIFO (first-in-first-out).
8. Maintenance, repair, condemnation	Preventive maintenance of cold-chain equipment and vehicles; condemnation board; depreciation fund for replacement.
9. Information system	Stock register, bin card, physical verification.

(Components list: Sathe & Sathe, Ch 13.)

Purchase and procurement — the three patterns

Pattern	How it works	Advantage	Disadvantage
Centralised	Needs processed and purchased centrally (e.g. Medical Store Depots under the Directorate General of Health Services, supplying ~16,000 institutions from the <i>Vocabulary of Medical Stores</i> , charged at cost + 10% processing)	Bulk order → better price; direct dealing with manufacturer; uniform inspection and quality control	Long lead time (may run into months); slow distribution; user not involved → dissatisfaction
Decentralised	Purchase authority delegated to the PHC	No red tape; flexible; user satisfaction; local budget control	Duplicated office work; inexperience; poor specifications → inferior goods; costlier small lots; vital irregular-demand items (e.g. anti-rabies vaccine) get condemned once and then never re-ordered
Rate contract (best of both)	Government fixes rates with suppliers, usually 2-yearly; the PHC places the order directly with the contracted firm and takes direct delivery	Bulk bargaining power with decentralised, delay-free ordering	Only covers listed items; others need direct/local purchase

(Sathe & Sathe, Ch 13.) Tendering should be by the **two-bid system** — technical specifications evaluated first, lowest financial bid chosen only among the technically qualified; the supply order must specify that goods carry a minimum of **5/6th of the shelf life** at supply, and a repeat order is limited to 50% of the original quantity, once, within one year, at the same price (Red Book, Managerial Sciences, p 382).

Storage at the PHC

- **Medical and non-medical stores separate;** store on ground floor or basement, in one place, with adequate light, ventilation, cupboards and racks.

- **Cool, away from autoclave, heating apparatus and laundry** so that thermal deterioration is minimised.
- **Cold chain:** ILR at +2°C to +8°C for all vaccines, deep freezer at –15°C to –25°C for ice-pack preparation (Routine Immunization Manual for Medical Officers 2024, Ch 5, p 117). Never store anti-snake venom, other drugs or eatables in the ILR.
- **Poisons and narcotics in a separate locked cupboard;** combustible and non-combustible material separated, with fire extinguishers at both.
- **Rodent- and vermin-proof,** with space for the associated paperwork. (Sathe & Sathe, Ch 13.)

Inventory control

Inventory control is maintenance of stock at that level at which purchasing and stocking costs are the lowest possible without interfering with supply (Sathe & Sathe, Ch 13). The three relevant costs are **carrying (holding)**, **shortage** and **ordering (replenishment)** costs (Red Book, Managerial Sciences, p 297).

Selective inventory control techniques

Technique	Basis of classification	Groups
ABC	Annual <i>consumption cost</i> (Pareto's law)	A = 10% of items, 70% of cost; B = 20% of items, 20% of cost; C = 70% of items, 10% of cost
VED	Criticality in patient care	V = vital, ~10%, no alternative, stock-out unacceptable; E = essential, ~40%, tolerable for a day or two; D = desirable, ~50%, tolerable longer
SDE	Ease of procurement	Scarce / Difficult / Easy
FSN	Rate of issue	Fast / Slow / Non-moving (10–15% of items non-moving in some large hospitals)
HML	Unit cost of the individual item	High / Medium / Low
GOLF	Source of supply	Government / Ordinary / Local / Foreign
SOS	Availability by season	Seasonal / Off-seasonal
XYZ	Value of stock held	X / Y / Z

(ABC, VED, SDE, FSN: Sathe & Sathe, Ch 13; HML, GOLF, SOS, XYZ: Park Plus, Ch 21, p 283.)

ABC–VED matrix — the coupled classification actually used for drugs

	V (vital)	E (essential)	D (desirable)
A (costly)	AV → I	AE → I	AD → I
B (moderate)	BV → I	BE → II	BD → II
C (cheap)	CV → I	CE → II	CD → III

Category	Composition	Share of items	Control
I — vital or expensive	AV, AE, AD, BV, CV	~15%	Maximum attention
II — essential, average cost	BE, BD, CE	~40%	Moderate
III — desirable, inexpensive	CD	~45%	Lowest

(Sathe & Sathe, Ch 13, Table 13.1.) Category I is run by "management by exception" — continuous monitoring by the medical officer personally, very accurate estimates, most frequent purchase, low safety stock to cut carrying cost. Category II gets moderate control at middle level; Category III is bought periodically with a high buffer stock and managed at the lowest level (Park Plus, Ch 21, p 284).

Key stock terms (Sathe & Sathe, Ch 13)

Term	Definition
Lead time	Average interval between placing the order and actual receipt of the material; differs from item to item
Buffer (safety) stock	Stock held as insurance against variation in supply and consumption — i.e. over and above the consumption expected during lead time
Reorder level	Consumption expected during lead time + buffer stock
Economic order quantity (EOQ)	The order size at which annual ordering cost equals annual carrying cost, so their sum is lowest; computed from annual consumption (A), purchase cost per order (P), carrying cost as a percentage (C) and unit price (U) — $EOQ = \sqrt{(2AP \div CU)}$ (Park Plus, Ch 21, p 285)

Methods of ordering: in the **two-bin system** an order is placed whenever stock falls to the reorder level, so orders are staggered evenly through the year — simple and suitable for field level. In the **cyclic (periodic) system** stock is reviewed at a fixed **review period** and enough is ordered to restore the maximum level; if lead time exceeds the review period, quantity already "in the pipeline" must be added.

Worked PHC example — vaccines (Routine Immunization Manual for Medical Officers 2024, Ch 5, Table 5.9, pp 145–146): at a PHC the working stock is **1 month**, buffer stock **0.25 month (7 days)** and lead time **0.25 month (7 days)**, giving a **maximum stock of 1.5 months and a minimum of 0.5 month**; the district holds a maximum of 2.75 months. For a PHC needing 280 doses of pentavalent monthly, buffer = 70 doses and maximum stock = 420 doses.

Issue, records, condemnation

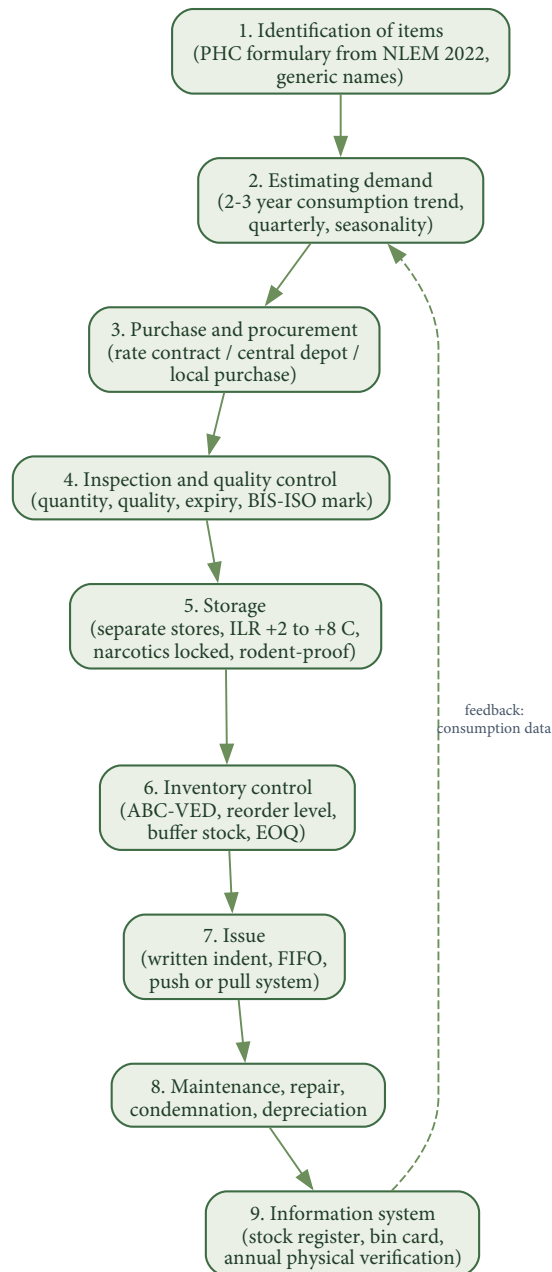
- **Issue** only against a written indent from a competent person; special rules for costly, poisonous and narcotic items.
- **FIFO principle** — new arrivals go to the back of the rack so older stock is issued first and does not become time-barred.
- **Push (allocation) vs pull (requisition) systems:** in the push system higher levels allocate stock downwards (needs little field training, but needs timely peripheral reporting); in the pull system the PHC's indent draws supplies down. Choice depends on management skill of staff, availability of usage information, and the number and quantity of items (Sathe & Sathe, Ch 13).

- **Records:** a stock register for receipt and issue, plus a **bin card per item** showing receipt, issue and balance on hand, kept by the officer in charge of stores; physical **stock verification at least annually** with random spot-checks of expensive items to detect pilferage; a list of items becoming time-barred ~2 months ahead so they can be used or loaned out.
- **Maintenance and condemnation:** preventive maintenance of cold-chain equipment and of vehicles (State Health Transport Organisation); log books for equipment; condemnation by a committee or a certifying officer — generally the officer competent to purchase an item is competent to condemn it — with depreciation calculated so that replacement can be funded (Sathe & Sathe, Ch 13).

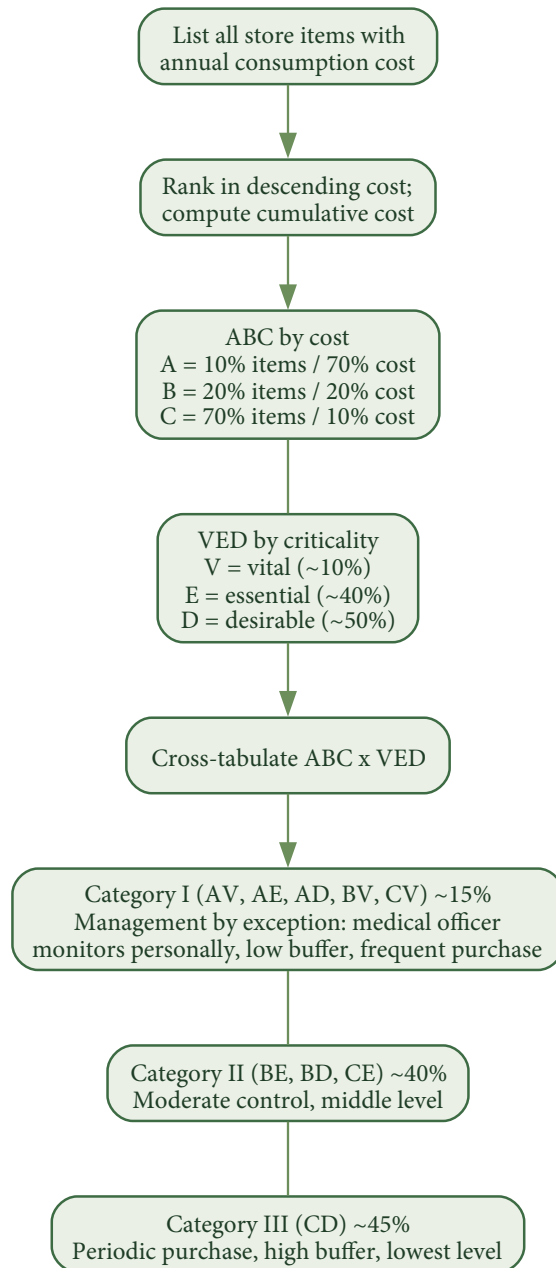
Information technology now in use: the **Electronic Vaccine Intelligence Network (e-VIN)** gives real-time vaccine stock and temperature visibility to cold-chain handlers (Routine Immunization Manual 2024, Ch 5, p 149), and the **Drug and Vaccine Distribution Management System (DVDMS)** does the same for drug supply at PHC/urban-PHC level (Operational Guidelines for Oral Health Care at Health and Wellness Centres, 2020).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Labelled line diagram to sketch** — "Stock level over time (min-max system)": label the *x*-axis **time**, *y*-axis **stock in hand**; mark the horizontal lines **Maximum stock (1.5 months at PHC)**, **Reorder level**, **Buffer stock**, **Minimum stock (0.5 month at PHC)**; mark the saw-tooth consumption slope, the arrow **Lead time** between *order placed* and *stock received*, and the vertical jump labelled **Economic order quantity**.
- **Labelled ABC curve to sketch:** *x*-axis **cumulative % of number of items (10 – 30 – 100)**, *y*-axis **cumulative % of annual cost (70 – 90 – 100)**; label the three segments **A, B, C** on the curve.



The nine-step material management cycle at a primary health centre



Coupling ABC and VED analysis to decide how tightly a PHC drug is controlled

High-yield exam pointers

- **Nine components** (mnemonic *I-E-P-I-S-I-M-I*): Identification, Estimating demand, Purchase, Inspection, Storage, Inventory control, Issue, Maintenance/condemnation, Information system.
- **ABC = cost** (10/70, 20/20, 70/10); **VED = criticality** (V 10%, E 40%, D 50%); **SDE = availability**; **FSN = movement**; **HML = unit cost**; **GOLF = source**; **SOS = season**; **XYZ = stock value**.
- **Reorder level = consumption during lead time + buffer stock**; **EOQ** = $\sqrt{(2AP \div CU)}$, the point where ordering cost = carrying cost.
- **ABC-VED matrix**: Category I ~15%, II ~40%, III ~45%; Category I is run by *management by exception*.
- **Vaccine stock at a PHC (2024 norms)**: working 1 month + buffer 7 days + lead time 7 days → **maximum 1.5 months, minimum 0.5 month**; district maximum 2.75 months.

- **Money figures:** 25–40% of health-facility expenditure is on material; carrying cost 20–25% per year.
- **NLEM India 2022** is the current national essential medicines list (1996 → 2003 → 2011 → 2015 → 2022).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the corpus's dedicated chapter (Sathe & Sathe, *Epidemiology and Management for Health Care*, Ch 13) predates the current digital supply chain, so the e-VIN and DVDMS lines above are drawn from the Routine Immunization Manual 2024 and the 2020 Health and Wellness Centre operational guidelines respectively; the National Health Mission's stated route for PHC drug supply is a state-level autonomous procurement corporation handling transparent procurement, quality assurance and distribution down to facility level (Jugal Kishore — National Health Programmes of India).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO Model List of Essential Medicines 2025, WHO–UNICEF Effective Vaccine Management tool, MoHFW programme guidance, NHSRC free-drugs guidelines

Programme and policy updates (India)

- **Health and Wellness Centres were renamed Ayushman Arogya Mandir (AAM) in November 2023**, tagline *Arogyam Paramam Dhanam*, at ₹3,000 per facility and a total outlay of ₹48 crore, with photographs uploaded to the AB-HWC portal for verification. Field visits to renamed centres found **mismatch between the e-Aushadhi electronic stock and the physical inventory** — the reason annual physical stock verification stays mandatory even after the store is digitised. [PMID 42257165] [MoHFW 2023-Nov]
- **The state medical-services-corporation model is now the reference design for PHC drug supply.** Uttar Pradesh rebuilt UPMSC on the Tamil Nadu (TNMSC) template around seven pillars — a rationalised essential drug list (pruned from ~1,300 items to ~300), district warehouses, centralised procurement, a **virtual passbook** giving each facility a notional budget to draw against, quality control, centralised payment, and DVDMS e-tracking for real-time inventory. Average availability of essential medicines across the **75 district warehouses rose from 27% to 97%** (June 2020 to July 2024), the average warehouse stocking 275 essential medicines by April 2024; last-mile delivery and prompt vendor payment remain the weak links. [PMID 40786162]
- **Procurement is shifting to the Government e-Marketplace (GeM, launched 2016)**, but one of the first Indian hospital audits of it — a 699-bed tertiary hospital in the north-east, financial year 2023–24 — found only **695 of 1,507 (46.1%) medicines and 945 of 1,219 (77.5%) consumables** actually materialised. Bottlenecks were vendor non-quotation or technical rejection (501 medicines, 33.2%), price-negotiation rejection (13.8%), and outright failure to supply — a modern restatement of the classic limitation that a rate contract covers only listed items. [PMID 40036111]

- **Morbidity-based forecasting has replaced incremental forecasting** in the India Hypertension Control Initiative: requirement is projected from expected patient load on a single drug-and-dose protocol instead of adding a percentage to last year's consumption. Across 22 districts in 4 states, antihypertensive procurement rose from 16 million tablets (2017–18) to 160 million (2019–20), amlodipine stock-out stayed **below 5%** of facilities, and amlodipine stock was held for ≥ 60 patient-days. [PMID 38096180]

Current national data

- **Essential-medicine availability across major Indian states is 17–51%** — against the WHO benchmark of 80% — with a **4–14 week lag** in receipt of drugs driving repeated stock-outs; a systematic review of 11 Indian studies attributes this to differing state procurement mechanisms, weak distribution and inventory systems, and non-adherence to NHSRC guidance, which sets a **3–5 day cycle time** for quality-check clearance of a consignment. [PMID 38882835]
- **ABC–VED applied at PHC level (Puducherry, 10 public facilities including 9 PHCs, financial year 2019–20)**: in State-government PHCs only **11 items (7.8%) were Class A** yet took 70% of the drug budget, while Vital items were 26.4% of items but 37.3% of expenditure and Essential items 47.8% of items and 49.8% of expenditure — **V + E together absorbed ~87% of a PHC's drug money**, and anti-rabies vaccine alone took 20.6% of the State PHC drug budget, confirming that the coupled matrix, not ABC alone, is what the medical officer should apply [PMID 40396070]. The newer **ABC–XYZ coupling** (XYZ = demand variability, used in place of VED) has so far been piloted only outside primary care — in a **public tertiary cancer hospital in northern India**, on 286 high-cost medicines worth ₹12.71 crore a year — where 6 months of category-matched procurement cut mean monthly stock-outs from 4.33 to 0.50, an **88.5% reduction** ($p < 0.001$), but cut monthly wastage only from ₹28,333 to ₹6,667, i.e. 76.5% at $p = 0.078$, not statistically significant [PMID 42583974].

Elimination, surveillance and disease status

- **Anti-rabies vaccine and rabies immunoglobulin — the textbook example of a vital, irregular-demand item — remain the worst-stocked.** In a nationwide facility survey of 467 public facilities across 60 districts in 15 states, ARV was available in 79.7% (95% CI 75.7–83.2) but in only **58.9% of urban PHCs**; RIG was available in just **20.3% (95% CI 16.8–24.3)** overall, 5.9% of primary-care facilities and **1.8% of urban PHCs** — against ~5,700 dog-mediated human rabies deaths a year in India and the global target of **zero dog-mediated rabies deaths by 2030**. [PMID 40584691]
- **Cold-chain stock and temperature are now audited electronically and against a WHO benchmark.** e-VIN has been scaled to all **731 districts** across 36 states and union territories; stakeholders in Nainital and Indore districts report real-time stock visibility that has virtually ended facility-level vaccine shortages, continuous logger-based temperature monitoring and standardised registers, with residual failures from unreliable PHC power supply, poor connectivity, cold-chain-equipment wear in hill blocks, logger and smartphone malfunction, and duplicated paper-plus-electronic records [PMID 40115572] [PMID 42569652]. Scored against the WHO–UNICEF **Effective Vaccine Management (EVM)** tool — 9 criteria (arrival, temperature, storage capacity, buildings/equipment/transport, maintenance, stock management, distribution, vaccine management, MIS), benchmark 80% — a six-district Gujarat assessment of

6 district vaccine stores and 90 service points reached temperature control 99%, storage capacity 98% and vaccine management 96%, but **MIS 79% and vehicle infrastructure 80%**: transport and information, not the ILR, are now the rate-limiting steps [PMID 41296592].

Guideline and legislative updates

- **Inspection and quality control on receipt are no longer a formality.** WHO has issued repeated product alerts since 2022 on contaminated oral liquid medicines, almost all manufactured in India, with **over 300 deaths** reported; the **October 2025 alert** covered three Indian-made syrups whose glycerine and propylene glycol excipients were contaminated with diethylene and ethylene glycol, and the 2022 Gambia tragedy had already killed **66 children**. The Drugs and Cosmetics Act and **Schedule M have been revised** to require mandatory inspections and manufacturer proof of excipient stability and safety in line with WHO GMP, with tighter imprisonment terms and penalties; compliance timelines for local manufacturers have been extended, so batch- and expiry-level verification at the PHC store remains the last line of defence. [PMID 41497199] [WHO 2025-Oct]

Global and WHO developments

- **WHO released the 24th Model List of Essential Medicines and the 10th list for children on 5 September 2025** — the reference against which India's NLEM 2022 and every state EDL is benchmarked. In parallel, a 2026 scoping review of 28 primary-care supply-chain studies concluded that what works is digital stock-tracking layered on strengthened governance, supervision and human-resource capacity, not technology alone: an LMIS cut stock-outs from 45% to 22%, a stock-tracker-plus-GIS package from 46% to 4%, and a prime-vendor system raised tracer-medicine availability from 69% to 94%. [WHO 2025-Sep] [PMID 42058231]

SOURCE & COVERAGE

Sathe & Sathe — Epidemiology and Management for Health Care for All, Ch 13 (Materials management); Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 24 Essential Medicines and Counterfeit Medicines); AFMC Red Book — Public Health and Preventive Medicine (Managerial Sciences: inventory systems, medical store purchase, equipment management); Park Plus (Amrutha & Vijayalakshmi), Ch 21 Health planning and management; Jugal Kishore — National Health Programmes of India; Routine Immunization I Manual for Medical Officers, Government of India, 2024 (Ch 5 Cold-chain and Logistics Management); National List of Essential Medicines, Government of India, 2022; Operational Guidelines for Oral Health Care at Health and Wellness Centres, 2020.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmpc.jfmpc.1966.24](https://doi.org/10.4103/jfmpc.jfmpc.1966.24); 40786162 — Lessons learned from redesigning public health medicines supply chain model in Uttar Pradesh, India, *Frontiers in Public Health* 2025, doi:[10.3389/fpubh.2025.1588227](https://doi.org/10.3389/fpubh.2025.1588227); 40036111 — Challenges in medicine procurement through GeM in a tertiary care hospital in northeast India, *Indian Journal of Medical Research* 2025, doi:[10.25259/ijmr.944.2024](https://doi.org/10.25259/ijmr.944.2024); 38096180 — Improving the availability of antihypertensive drugs in the India Hypertension Control Initiative, India, 2019–2020, *PLoS One* 2023, doi:[10.1371/journal.pone.0295338](https://doi.org/10.1371/journal.pone.0295338); 38882835 — Factors affecting the availability and utilization of essential medicines in India: a systematic review, *Journal of Pharmacy and Bioallied Sciences* 2024, doi:[10.4103/jpbs.jpbs.1198.23](https://doi.org/10.4103/jpbs.jpbs.1198.23); 40396070 — Assessment of drug inventory using ABC-VED matrix analysis in selected public health facilities of Puducherry, India, *Journal of Family Medicine and Primary Care* 2025, doi:[10.4103/jfmpc.jfmpc.1544.24](https://doi.org/10.4103/jfmpc.jfmpc.1544.24); 42583974 — Optimizing drug availability and reducing wastage through ABC-XYZ inventory management: insights from a public tertiary cancer hospital in northern India, *Indian Journal of Pharmacology* 2026, doi:[10.4103/ijp.ijp.871.25](https://doi.org/10.4103/ijp.ijp.871.25); 40584691 — Availability of anti-rabies vaccine and rabies immunoglobulin in Indian health facilities: a nationwide cross-sectional health facility survey, *The Lancet Regional Health – Southeast Asia* 2025, doi:[10.1016/j.lansea.2025.100608](https://doi.org/10.1016/j.lansea.2025.100608); 40115572 — Assessment of eVIN (Electronic Vaccine Intelligence Network) in utility and constraints of various stakeholders in Nainital district: a mixed method study, *Journal of Family Medicine and Primary Care* 2025, doi:[10.4103/jfmpc.jfmpc.1015.24](https://doi.org/10.4103/jfmpc.jfmpc.1015.24); 42569652 — Perceptions, challenges and operational bottlenecks in Electronic Vaccine Intelligence Network implementation: a qualitative study among immunization stakeholders in Indore district, India, *Cureus* 2026, doi:[10.7759/cureus.112254](https://doi.org/10.7759/cureus.112254); 41296592 — Evaluating supply chain management of cold storage for preventive child healthcare programmes: a comparative study from western India, *Indian Journal of Medical Research* 2025, doi:[10.25259/ijmr.1930.2024](https://doi.org/10.25259/ijmr.1930.2024); 41497199 — Tiny lives, big failures: India's drug regulation crisis, *The Lancet Regional Health – Southeast Asia* 2025, doi:[10.1016/j.lansea.2025.100705](https://doi.org/10.1016/j.lansea.2025.100705); 42058231 — Determinants, models, performances and strategies of supply chain to address medicine stockouts in primary health care: a scoping review,

Journal of Pharmaceutical Policy and Practice 2026, doi:[10.1080/20523211.2026.2659077](https://doi.org/10.1080/20523211.2026.2659077). Also used: WHO 24th Model List of Essential Medicines and 10th Model List of Essential Medicines for Children, 5 September 2025; WHO medical product alert on contaminated oral liquid medicines, October 2025; MoHFW renaming of Ayushman Bharat Health and Wellness Centres to Ayushman Arogya Mandir, November 2023.



Medical audit

Asked in: Paper IV 21-Nov-2022 · Paper IV Nov 2017 (2×) — RGUHS MD Community Medicine

Definition — Medical audit is the **evaluation of the quality of medical care through the analysis of medical records**; it is a regulatory mechanism that raises the quality of care given to hospitalised patients by having the work of the medical staff reviewed by a committee drawn from the medical staff itself (Mahajan & Gupta 4e, Ch 27, p 545). Stated as a programme, it is a **planned programme which objectively monitors and evaluates the clinical performance of all practitioners, identifies opportunities for improvement, and provides a mechanism through which action is taken to make and sustain those improvements** (Park Plus, Ch 21, p 304). It is therefore a **service review, not a trial** — it measures whether agreed procedures were actually carried out, using explicit criteria applied to clinical records (Abramson, Ch 21, p 208).

Purpose, need and objectives

Head	Content
Professional motive	Health care providers identify their own lacunae and deficiencies and make the necessary corrections (Park Plus, Ch 21, p 305)
Social motive	To ensure the safety of the public and protect them from care that is inappropriate, suboptimal or harmful (Park Plus, Ch 21, p 305)
Pragmatic motive	To reduce patient suffering, and to avoid both denial of an available service and injury from excessive or inappropriate service (Park Plus, Ch 21, p 305)
To plan future action	Provides baseline information, through evaluation of achievements, against which services can be compared and improved (Park Plus, Ch 21, p 305)
Regulatory function	Ensures full and effective utilisation of the staff and facilities available (Park Plus, Ch 21, p 305)
Effectiveness and efficiency	Assesses the effectiveness and efficiency of the health programmes and services actually put into practice (Park Plus, Ch 21, p 305)
Broad objectives (Mahajan & Gupta)	(a) improved quality of care; (b) a clear definition of responsibilities; (c) stimulation of research and development; (d) improved safety; (e) providing a basis for assessing care (Mahajan & Gupta 4e, Ch 27, p 545)
Educational value	Results translate directly into practice recommendations — if the audit shows X is <i>not</i> being done, the recommendation is that X <i>should</i> be done; a new doctor or nurse quickly learns that X is expected (Abramson, Ch 21, p 209)

Medical audit also **minimises the professional and administrative practices that hinder the delivery of effective medical care**, and streamlines hospital procedures by identifying bottlenecks in hospital services. Its proceedings **must be kept confidential**, and only the concerned medical staff are informed of the committee's observations (Mahajan & Gupta 4e, Ch 27, p 545). In evaluation terms audit is the **external, post-facto appraisal of performance**, complementary to the internal accounting of routine monitoring, and like financial accounting it needs independent outside appraisal to be credible (Reinke, Ch 4, p 64).

What is audited — structure, process and outcome

Audit follows Donabedian's categorisation of the measurement of quality — **structure (personnel characteristics and institutional features)**, **process (activities in providing care)** and **outcome (results of care)** (Maxcy-Rosenau-Last 15e, Ch 78, p 1303). Mahajan & Gupta name the same three as structure quality, process quality and product quality (Ch 27, p 545).

Dimension	What it measures	Examples of indicators audited
Structure	The resources of the health system — the setting in which care occurs and the capacity of that setting to produce quality	Bed complement; diagnostic and treatment facilities; staff available; qualification of personnel and staff-to-patient ratio; licensure, certification and accreditation status; availability of technology; patient volume (Park Plus, Ch 21, p 305; Handbook of Epidemiology, Ch 37, p 1495)
Process	The care patients actually receive — the net effect of clinicians' decision-making	Days of care, operations, deliveries, deaths, out-patient and laboratory investigations; generic measures such as average length of stay, number of prescriptions, day-case surgery rate ; condition-specific measures such as antibiotics within eight hours in community-acquired pneumonia, thromboembolism prophylaxis, beta-blockers after myocardial infarction. Because they are derived from evidence-based clinical guidelines, process problems are detected long before outcome differences become demonstrable (Park Plus, Ch 21, p 305; Handbook of Epidemiology, Ch 37, p 1496)
Outcome	The results of care, favourable or adverse, attributable to prior or concurrent care	The "five Ds" — death, disease, disability, discomfort, dissatisfaction ; also intermediate outcomes such as establishing the correct diagnosis or a change in the patient's health behaviour (Handbook of Epidemiology, Ch 37, pp 1496-1497; Abramson, Ch 21, p 209). Broader definitions add psychosocial functioning, quality of life, and resource utilisation and costs of care

Each dimension has a known weakness. Structural compliance guarantees quality only at the level of minimum standards — above that, the structure–outcome link weakens (specialist care does not always produce better outcomes). Process measures are not necessarily good predictors of outcome, collecting process data is elaborate, and consensus on the recommended process is often unattainable. Outcome differences may take years to appear and may reflect patient factors, admission practice or chance rather than the quality of care, so **case-mix adjustment is essential** before outcomes are compared between providers (Handbook of Epidemiology, Ch 37, pp 1495-1497).

The audit cycle — five stages, and the criteria used

Medical audit is cyclical, not a one-off inspection: the five stages are preparing for audit, selecting criteria, measuring the level of performance, making improvements and sustaining improvements (Park Plus, Ch 21, pp 305-308).

- ▶ **Preparing for audit.** Identify the problem — from routine service statistics, **direct observation of care** and direct conversations. **Select a topic** that is of concern to service users with potential to improve outcomes, of clinical concern (known variation in practice, a high-risk procedure, complex management), financially important (very common or very expensive), of local or national importance, practically viable (measurable, and change can actually be implemented), and supported by new research evidence — for example, the incidence of wound infection after hernia repair. **Define the purpose** with an action verb — to improve, to enhance, to increase, to change, to ensure. **Plan** it: involve all the people concerned, fix time and resources, access the evidence, design the data-collection instrument and method, **pilot** it, and decide in advance on the report, the action and the re-audit.
- ▶ **Selecting criteria.** The **criteria state what should be happening**; the **standards set the minimum acceptable performance** against those criteria. Both must be specific — clear, understandable, measurable, achievable, relevant to the aims of the audit — and theoretically sound, based on current research.
- ▶ **Measuring the level of performance.** Collect the data, analyse them, compare against the standards set, and disseminate the findings as feedback.
- ▶ **Making improvements.** Identify the barriers to change, then implement change.
- ▶ **Sustaining improvements.** Monitor and evaluate, **re-audit** to close the loop, and maintain and reinforce the improvement.

Prerequisites. The audit rests on **hospital operational statistics** of two kinds — hospital *resources* (bed complement, diagnostic and treatment facilities, staff available) and hospital *utilisation rates* — and on clinical records complete enough to answer the question asked (Park Plus, Ch 21, p 305).

Basis of the criteria	Meaning
Statistical (empirical)	Derived from what is actually done in clinical facilities that are of an acceptable level (Abramson, Ch 21, p 208)
Normative (consensus)	Experts' opinion on what procedures <i>should</i> be carried out in a specified type of case (Abramson, Ch 21, p 208)
Optimal care (general consensus)	The best achievable care, agreed by general consensus
Essential (critical)	The irreducible minimum that must be done in that situation
Scientific (validated)	Criteria validated against research evidence

(Five bases of criteria: Park Plus, Ch 21, pp 306-307.)

Standards may be drawn from — national guidance or standards; college or professional organisation guidelines; laws; observed and assessed current practice; standards used locally by colleagues or comparable institutions; research evidence; a literature review of clinical audits that have published their standards and results; and current knowledge from clinical experience. Whichever is used, the

evidence must first be appraised for validity, reliability and importance — its aim, methodology, results and applicability to your own patient group (Park Plus, Ch 21, pp 307-308).

Types of medical audit

Basis	Types
By subject matter	Morbidity audit; audit of operated cases; audit of obstetric cases; audit of deaths in hospital (mortality review); on-the-spot medical audit (Park Plus, Ch 21, pp 308-309)
By who performs it	Self-audit (by the practitioner); peer review (by others); internal audit (by colleagues in the same institution); external audit (by an outside body); nursing audit; medical care evaluation studies (Abramson, Ch 21, pp 217-218)
By coverage	All cases cared for; a representative sample; defined categories such as patients with selected conditions; or marker procedures chosen as indicators of the quality of care (Abramson, Ch 21, p 208)
By data source	Routine records; specially modified or designed records; or special investigations, including direct observation of practitioners at work (Abramson, Ch 21, pp 217-218)

A particularly productive variant is to review the past history of patients with poor outcomes — preventable disorders or complications — because this identifies deficiencies not only in the care given, but also in patient compliance and in the availability and use of services (Abramson, Ch 21, p 208).

Evolution of medical audit, and its place in Indian public health practice

Year	Landmark
1750 BC	Hammurabi, the sixth king of Babylon, instituted audits for clinicians
1853-1855	Florence Nightingale conducted the first clinical audit during the Crimean War; strict sanitary routine and hygiene standards reduced mortality from 40% to 2%
1912	Ernest Codman, the first true medical auditor, monitored surgical outcomes — his "end result idea" was to follow every patient's case history after surgery to identify errors made by individual surgeons
1987 and 1989	Formal audit accelerated by the Confidential Enquiry into Perioperative Deaths (CEPOD), 1987 and the United Kingdom government white paper "Working for Patients", 1989

(All four rows: Park Plus, Ch 21, p 305.)

Milestone or mechanism in India	Content
Mudaliar Committee, 1961	Report stressed the encouragement of medical audit in India (Park Plus, Ch 21, p 305)
1969	Then Union Health Minister Dr Sushila Nayyar introduced medical audit in India; it became operational only in 2007, after the establishment of the National Accreditation Board for Hospitals and Healthcare Providers (NABH) in 2005 (Park Plus, Ch 21, p 305)

Milestone or mechanism in India	Content
Indian Public Health Standards (IPHS)	Facility-based Maternal Death Review (MDR) is included in the service list for the community health centre / first referral unit, and is listed as a desirable service at the primary health centre (Jugal Kishore, Ch 1, pp 143-145)
PC & PNDT Act enforcement	Appropriate Authorities must take up the medical audit of Form F records kept by prenatal diagnostic clinics under the national scheme (Jugal Kishore, Ch 2, p 900)
Disease-control programmes	Medical audit to measure the effectiveness of the programme is a named secondary-level activity in the vector-borne disease control strategy, alongside training of medical officers and laboratory technicians and equipping district hospitals (Jugal Kishore, Ch 1, pp 382-383)
National health policy critique	Only the National Rural Health Mission carried a provision for audit; audit should be part of all national health programmes , and safety audit, clinical audit and death audit should be promoted (Jugal Kishore, Ch 2, p 746)
Present coverage	Medical audit has been introduced in some non-governmental and private hospitals, but not yet in most government hospitals in India (Mahajan & Gupta 4e, Ch 27, p 545)

An illustrative Indian audit: a medical audit of the management of diarrhoea in a teaching hospital showed that **53 per cent of infants with diarrhoea were treated with drugs alone, without oral rehydration solution or advice on home-available fluids** (Mahajan & Gupta 4e, Ch 27, p 545).

Limitations, and how medical audit differs from research

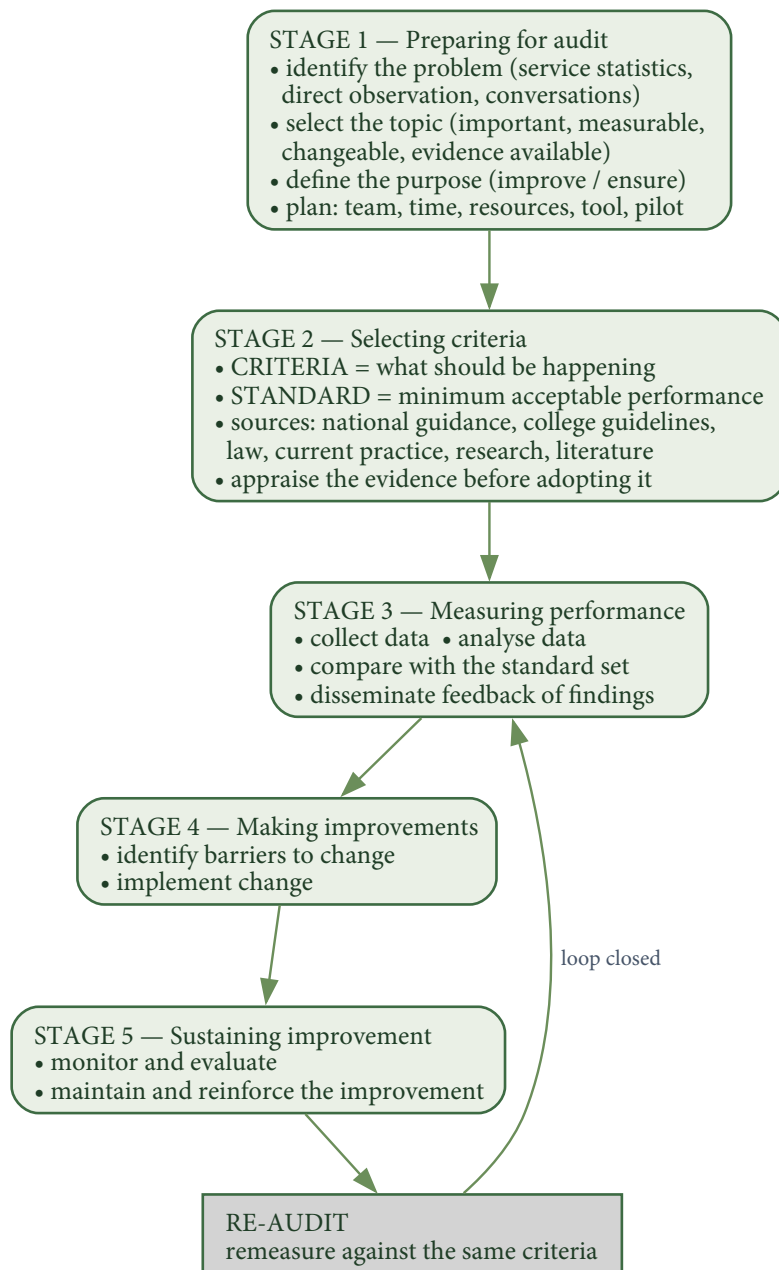
- **Commitment and participation.** In the Indian setting audits are still largely treated as an obligation, done to satisfy accreditation or other external agencies rather than to improve hospital processes and quality; the **number of trained auditors is small** (Park Plus, Ch 21, p 309).
- **Technique not standardised.** The techniques are imperfect and not standardised, despite the apparently clear-cut methods set out in official publications (Park Plus, Ch 21, p 309).
- **Limited by the record.** Being **retrospective and entirely dependent on the information in the record**, audit assesses only limited aspects of technical quality; if records were not planned for audit the required information is simply absent, and **fuller notes do not mean better care** — in myocardial infarction, documentation of history, examination and tests bore no relation to time lost from work, re-infarction or death (Park Plus, Ch 21, p 309; Abramson, Ch 21, pp 209-210).
- **Assumptions are untested.** Audit assumes that performing the chosen procedures benefits patients, and does not itself test that assumption; the evaluation is valid only in so far as the assumption is valid, and proof of a cause–effect link between the recommended procedure and the outcome is often lacking (Abramson, Ch 21, pp 208-209). This is the **core difference from research**: research asks *what should we be doing*, audit asks *are we doing what we have already agreed should be done*.
- **Gaming the indicator.** Procedures are sometimes chosen because data on them are easy to get rather than because they are good markers of quality; where audit results carry rewards or penalties, practitioners may increase performance of those procedures, **improving the audit score without improving care** (Abramson, Ch 21, p 209).

- ▶ **A minimum, not an optimum.** Quality assurance uses the minimum acceptable requirements as the criteria for corrective action, so it **maintains a minimum standard rather than fostering overall improvement**; continuous quality improvement, which treats problems as system failures and employees as the source of solutions, is the complementary approach (Health Program Planning and Evaluation, Ch 10, p 277; Maxcy-Rosenau-Last 15e, Ch 78, p 1303).

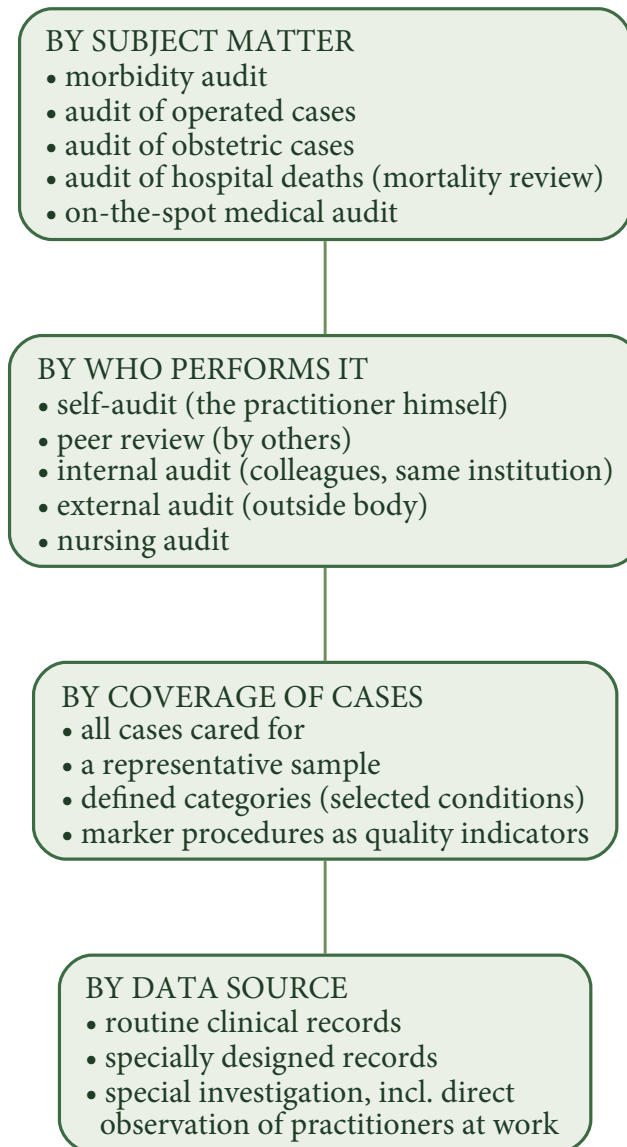
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Label box — the Donabedian triangle of quality assessment (draw a triangle, one dimension per corner, with an arrow running structure → process → outcome across the base):

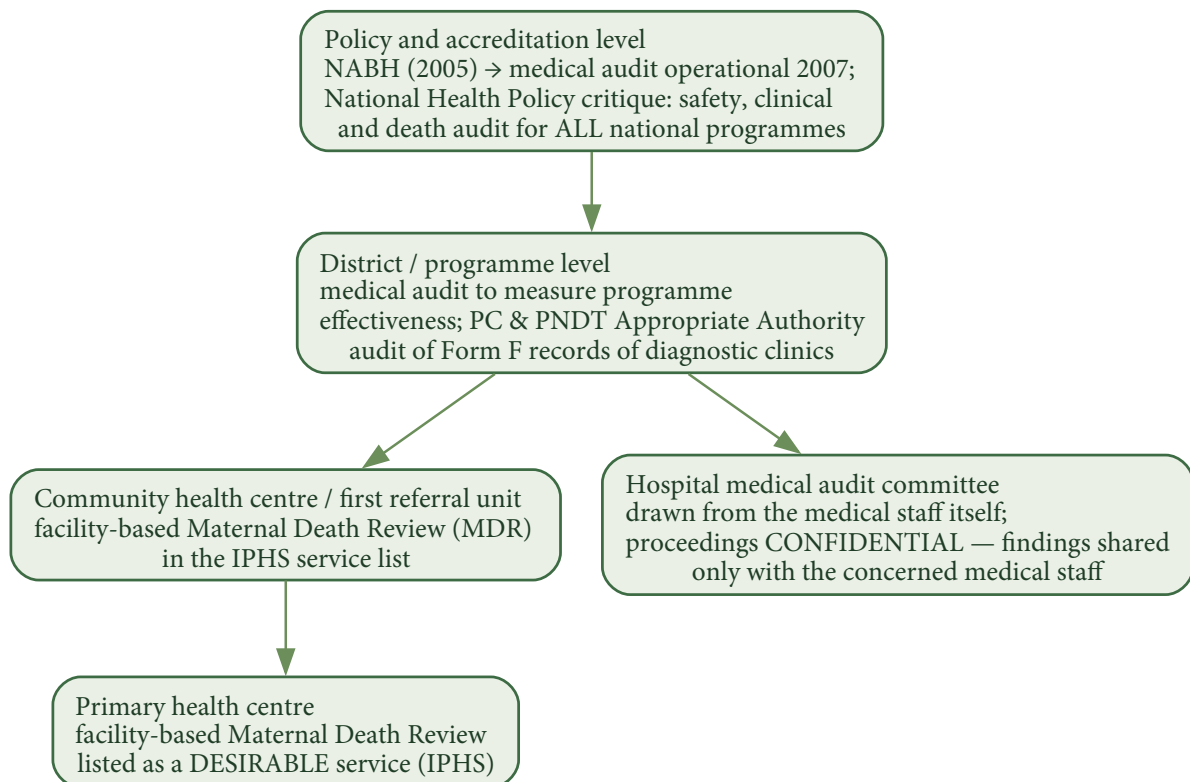
- ▶ Corner 1 — **STRUCTURE**: beds, equipment, drugs, staff number and qualification, accreditation status, patient volume
- ▶ Corner 2 — **PROCESS**: what was actually done — investigations ordered, drug given, operation performed, length of stay, adherence to the guideline
- ▶ Corner 3 — **OUTCOME**: the five Ds — death, disease, disability, discomfort, dissatisfaction; plus intermediate outcomes (correct diagnosis, changed health behaviour)
- ▶ Write across the base: *"the implicit assumption is that structure affects process and process affects outcome"*
- ▶ Write beside the triangle: **criteria = what should happen; standard = minimum acceptable level of performance**



The five-stage medical audit cycle — the loop is closed only by re-audit



Classification of medical audit — by subject, by auditor, by coverage and by data source



Where medical audit sits in the Indian health system

High-yield exam pointers

- ▶ **One-line definition to write first** — medical audit is the *evaluation of the quality of medical care through the analysis of medical records* (Mahajan & Gupta); the fuller programme definition adds *objectively monitors and evaluates the clinical performance of all practitioners, identifies opportunities for improvement, and provides a mechanism through which action is taken to make and sustain those improvements.*
- ▶ **Five stages (mnemonic "P-S-M-M-S")** — Preparing for audit → Selecting criteria → Measuring performance → Making improvements → Sustaining improvement, closed by **re-audit**.
- ▶ **Donabedian triad** — structure (personnel and institutional features), process (activities in providing care), outcome (results of care); outcome quality = the **five Ds** — **death, disease, disability, discomfort, dissatisfaction**.
- ▶ **Criteria versus standard** — the criterion states *what should be happening*; the standard sets the *minimum acceptable performance* for that criterion. Criteria may be statistical (empirical), normative (consensus), optimal care, essential (critical) or scientific (validated).
- ▶ **Names and dates to quote** — history: **Hammurabi 1750 BC**, **Florence Nightingale, Crimea 1853-1855** (mortality 40% → 2%), **Ernest Codman 1912** and the "**end result idea**", **CEPOD 1987**. India: **Mudaliar Committee 1961** recommended medical audit; **Dr Sushila Nayyar** introduced it in 1969; **NABH established 2005**, medical audit operational from 2007; facility-based **Maternal Death Review** is an IPHS service at community health centre / first referral unit level.
- ▶ **Indian audit finding to quote** — **53% of infants with diarrhoea in a teaching hospital were treated with drugs alone**, without oral rehydration solution or advice on home-available fluids.

- **Audit versus research in one line** — research asks *what should we be doing*; audit asks *are we doing what we have already agreed should be done*, and does not test that assumption.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the historical landmarks (Hammurabi, Florence Nightingale's Crimean audit, Codman's "end result idea", the 1987 Confidential Enquiry into Perioperative Deaths and the 1989 white paper "Working for Patients") and the Indian dates come from a single retrieved source, so write them as history rather than as verified public health data; the current Ministry of Health and Family Welfare quality initiatives that carry medical audit forward in India today — the National Quality Assurance Standards, Kayakalp and LaQshya, and Maternal Death Surveillance and Response as the successor to facility-based Maternal Death Review — are not described in the retrieved corpus and have therefore been left out of the answer.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Cochrane Library, MoHFW National Quality Assurance Standards, WHO Global Patient Safety Action Plan 2021–2030, Mandaluyong Declaration 2025

Programme and policy updates (India)

- **Health and Wellness Centres have been renamed Ayushman Arogya Mandir (AAM)** — the programme renaming that post-dates the textbook edition. More than 1.75 lakh AAM have been operationalised, with 1,16,978 community health officers in position and the expanded package of services rolled out at 82% of AAM as of 30 November 2024 (up from 50% in March 2023); the stated next step is to ensure AAM functionality as per IPHS norms with due certification under the National Quality Assurance System — making quality certification, not the hospital audit committee, the working audit instrument for primary care. [PMID 41816146]
- **The single medical audit committee has been overtaken by a layered national quality-certification architecture** — NQAS (district hospitals 2013; PHC and CHC 2014; health and wellness centres 2020–21), Kayakalp (2015), Swachh Swasth Sarvatra (December 2016), the Mera Aspataal patient-feedback platform (August 2016), LaQshya (November 2017), MusQan for child-friendly care (2021), the SaQushal patient-safety self-assessment tool (2022) and the SaQsham national certification portal of NHSRC (2023). Scale of the shift: NQAS-certified facilities rose from 10 in 2016–17 to 829 in 2022–23, and Kayakalp enrolment from 100 facilities in 2015–16 to 19,488 in 2022–23. PHC-level NQAS carries 8 areas of concern, 50 standards and 1,382 checkpoints; Kayakalp requires at least 70% on peer assessment; AB-PMJAY quality certification (National Health Authority with the Quality Council of India) is tiered **Bronze / Silver / Gold**, Bronze carrying 53 standards and 182 checkpoints and being open to non-empanelled hospitals as well. [PMID 41280589]
- **A current Indian audit cycle worth quoting alongside the older diarrhoea example** — at a 960-bed tertiary teaching hospital in eastern India the Kayakalp checklist was run through a **Plan-Do-Check-Act** loop (gap assessment → structure, process and outcome interventions → monitoring → review of the action plan), with final assessment by external government-nominated experts. The checklist score rose from 73.68% to 95.0%, hand-hygiene compliance

from 46% to 95% and overall patient satisfaction from 63% to 81%, while needle-stick injuries fell by 53%; the institution took the Kayakalp national first prize in its category. This is the audit cycle of the textbook, executed under a named national programme. [PMID 39534235]

Current national data

- **PM-JAY after four years has not yet delivered the quality gain it was designed for** — two representative household survey waves in Chhattisgarh (about 15,000 individuals each, 2021 and 2022) found enrolment reaching roughly **90% of the population** but no association with inpatient utilisation, patient satisfaction or length of stay. Among enrolled people using private hospitals, **78.1% (2021) and 70.9% (2022)** still incurred catastrophic health expenditure at the 10%-of-consumption threshold; mean out-of-pocket spend was ₹23,691 against mean annual non-health household consumption of ₹97,365. Purchasing was ineffective in regulating provider behaviour, and the authors' own remedy is that **renewal of provider contracts should turn on their track record of adhering to them** — audit as a condition of continued empanelment. [PMID 39135015]
- **Maternal death review coverage in India is high on paper and thin on completeness** — across 282 Indian facilities, 48 of 58 facility maternal deaths (82.8%) had evidence of a review, but only 41.1% of Indian maternal deaths (24 of 58) had a complete review meeting all seven criteria of the WHO maternal and perinatal death surveillance and response standard. Secondary-level facilities reviewed 61.1% of deaths against 92.1% at tertiary level; in Gonda district, applying the completeness definition dropped the indicator from 81.8% to 13.6%. A direct modern illustration of the textbook limitation that audit is only as good as the record. [PMID 38768186]

Elimination, surveillance and disease status

- **Maternal Death Review has been superseded by Maternal Death Surveillance and Response (MDSR)** — WHO's September 2013 technical guidance converted a retrospective review into a continuous cycle of identification and notification, review, analysis, dissemination and response. Content analysis of 56 MDSR reports from 32 low- and middle-income countries (2011-2020) shows how often the loop stays open: 59% described their data as incomplete, only 29% calculated notification rates and 30% review-completion rates, and although 86% gave recommendations for improving MDSR, 71% of subsequent reports carried no evidence of action on the previous ones. This is the current, quotable statement of the textbook's point that the audit loop is closed only by re-audit. [PMID 38442110]
- **Death audit is now a named tool inside the TB elimination programme** — a community-based verbal-autopsy death audit of 149 notified TB deaths across six Gujarat districts (2023-24) found a median delay of about five weeks from symptom onset to treatment initiation, with nearly 80% of deaths occurring within 16 weeks of confirmed diagnosis. Shorter survival was independently associated with comorbidity (HR 2.0, $p < 0.001$), drug-resistant TB (HR 1.7, $p = 0.003$) and key-population status (HR 1.5, $p = 0.01$). The authors' central recommendation is a **TB Death Surveillance and Response System (TBDSR)** joining community-based death reviews to facility-based medical audits with digital reporting through the Ni-kshay portal — the audit cycle written into a national programme. [PMID 41758813]

Guideline and legislative updates

- The 2025 Cochrane update is the current evidence base for what audit actually achieves — 292 randomised trials, 678 arms. Audit and feedback produced a mean absolute increase in desired practice of 6.2% (95% CI 4.1 to 8.2) and an odds ratio of 1.47 (95% CI 1.31 to 1.64), both moderate-certainty, with a median absolute improvement of 2.7% (IQR 0.0 to 8.6) across 558 outcomes. Effects were larger where baseline performance was low; where feedback measured **the individual rather than the team**; where performance was compared with **top peers or a benchmark**; where a **local champion with an existing relationship** delivered it; where delivery was **interactive and multi-modal** rather than didactic or written alone; and where an **explicit action plan** was attached. Prescribing (47%) and test-ordering (35%) were the commonest targets, and only 9% of trials came from low- and middle-income countries — the evidence gap for Indian practice. [PMID 40130784]
- Audit and feedback has become an antimicrobial-stewardship instrument — meta-analysis of 56 randomised trials in primary care found an 11% relative reduction in antibiotic prescribing volume (RR 0.89, 95% CI 0.84-0.95), 23% in unnecessary initiation (RR 0.77, 95% CI 0.68-0.87), 13% in prolonged course duration (RR 0.87, 95% CI 0.81-0.94) and 17% in broad-spectrum selection (RR 0.83, 95% CI 0.75-0.93). Effects were larger with **monthly feedback** (RR 0.78) than quarterly (0.96) or annual (0.93), and larger in lower-income countries and where baseline antibiotic use was high. The authors conclude that repeated audit-and-feedback interventions **should be prioritised in national action plans on antimicrobial resistance**. [PMID 39657007]

Global and WHO developments

- Patient safety is now the global frame within which medical audit sits — the 7th Global Ministerial Summit on Patient Safety (Manila, April 2025) issued the **Mandaluyong Declaration**, reaffirming World Health Assembly resolution 72.6 and endorsing the **WHO Global Patient Safety Action Plan 2021-2030** with four pledges: stronger global collaboration; leadership and governance in patient safety; integration of patient safety into disaster and emergency preparedness and climate resilience; and a people-centred approach. **World Patient Safety Day 2025 (17 September)** carried the theme "Safe Care for Every Newborn and Every Child", and one core action-plan indicator measures whether **a patient representative has been appointed to the governing board in at least 60% of hospitals** — audit widened from record review to system accountability. [PMID 40796350] [PMID 40934177]

SOURCE & COVERAGE

Park Plus — Health planning and management, Medical auditing section (Ch 21, pp 304-309) · Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 27, Health Economics — Medical Quality and Audit, p 545) · Abramson — Research Methods in Community Medicine (Ch 21, The Use of Documentary Sources — Medical Audit, pp 208-210 and notes pp 217-218; Ch 2, Types of Investigation; Ch 5, The Objectives of Evaluative Studies) · Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 78, Public Health Management Tools — Quality Assurance and Quality Improvement, pp 1303-1306) · Handbook of Epidemiology, Ahrens & Pigeot (Ch 37, Health Services Research — indicators of structural, process and outcome quality, pp 1495-1497) · Health Program Planning and Evaluation (Ch 10, Program Quality and Fidelity, p 277) · Reinke — Health Planning for Effective Management (Ch 4, Health Services Program Evaluation, p 64) · Jugal Kishore — National Health Programmes of India (Ch 1, Indian Public Health Standards pp 142-145, vector-borne disease control strategy pp 382-383; Ch 2, National Health Policy critique p 746, PC & PNDT medical audit of Form F p 900).

Coverage note: Park's Textbook of Preventive and Social Medicine 28e — the spine for this bank — carries no "medical audit" heading, so this answer is grounded on the other retrieved sources named above.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25) · 41280589 — To gain insights and compare different quality certification programs for public health facilities in India, *J Family Med Prim Care* 2025, doi:[10.4103/jfmpc.jfmpc.1591.24](https://doi.org/10.4103/jfmpc.jfmpc.1591.24) · 39534235 — Transforming Hospital Housekeeping: The Kayakalp Journey, *Glob J Qual Saf Healthc* 2024, doi:[10.36401/JQSH-23-54](https://doi.org/10.36401/JQSH-23-54) · 39135015 — The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) after four years of implementation — is it making an impact on quality of inpatient care and financial protection in India?, *BMC Health Serv Res* 2024, doi:[10.1186/s12913-024-11393-2](https://doi.org/10.1186/s12913-024-11393-2) · 38768186 — Validating the indicator "maternal death review coverage" to improve maternal mortality data: a retrospective analysis of district, facility, and individual medical record data, *PLoS One* 2024, doi:[10.1371/journal.pone.0303028](https://doi.org/10.1371/journal.pone.0303028) · 38442110 — Maternal death surveillance and response system reports from 32 low-middle income countries, 2011-2020: what can we learn from the reports?, *PLOS Glob Public Health* 2024, doi:[10.1371/journal.pgph.0002153](https://doi.org/10.1371/journal.pgph.0002153) · 41758813 — Factors accelerating time to death among persons with tuberculosis in Western India: evidence from a community-based retrospective death audit, *PLoS One* 2026, doi:[10.1371/journal.pone.0343271](https://doi.org/10.1371/journal.pone.0343271) · 40130784 — Audit and feedback: effects on professional practice, *Cochrane Database Syst Rev* 2025, doi:[10.1002/14651858.CD000259.pub4](https://doi.org/10.1002/14651858.CD000259.pub4) (abstract-only) · 39657007 — Audit and feedback interventions for antibiotic prescribing in primary care: a systematic review and meta-analysis, *Clin Infect Dis* 2025, doi:[10.1093/cid/ciae604](https://doi.org/10.1093/cid/ciae604) · 40796350 — Why patient safety is akin to weaving textiles: an International Society for Quality in Health Care perspective on the 7th Global Ministerial Summit on Patient Safety 2025, *Int J Qual Health Care* 2025, doi:[10.1093/intqhc/mzaf072](https://doi.org/10.1093/intqhc/mzaf072) · 40934177 — Ensuring the safety of newborns and children through

community and healthcare actions, *PLoS Med* 2025, doi:[10.1371/journal.pmed.1004730](https://doi.org/10.1371/journal.pmed.1004730). Also used: WHO Global Patient Safety Action Plan 2021-2030; Mandaluyong Declaration, Manila, April 2025; World Patient Safety Day 2025 (17 September).



NITI Aayog

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — NITI Aayog (**National Institution for Transforming India**) is the premier policy **think tank** of the Government of India, established on **1st January 2015** to replace the Planning Commission (instituted in 1950). It provides governments at the central and state levels with strategic and technical advice across the spectrum of key elements of policy, monitors and evaluates the implementation of programmes, and focuses on technology upgradation and capacity building (Park 28e, Ch 23, p 1031). It is an **enabler and a directional body, not a controlling or resource-allocating one** (Jugal Kishore, Ch 1, p 74).

Genesis and the need for the change

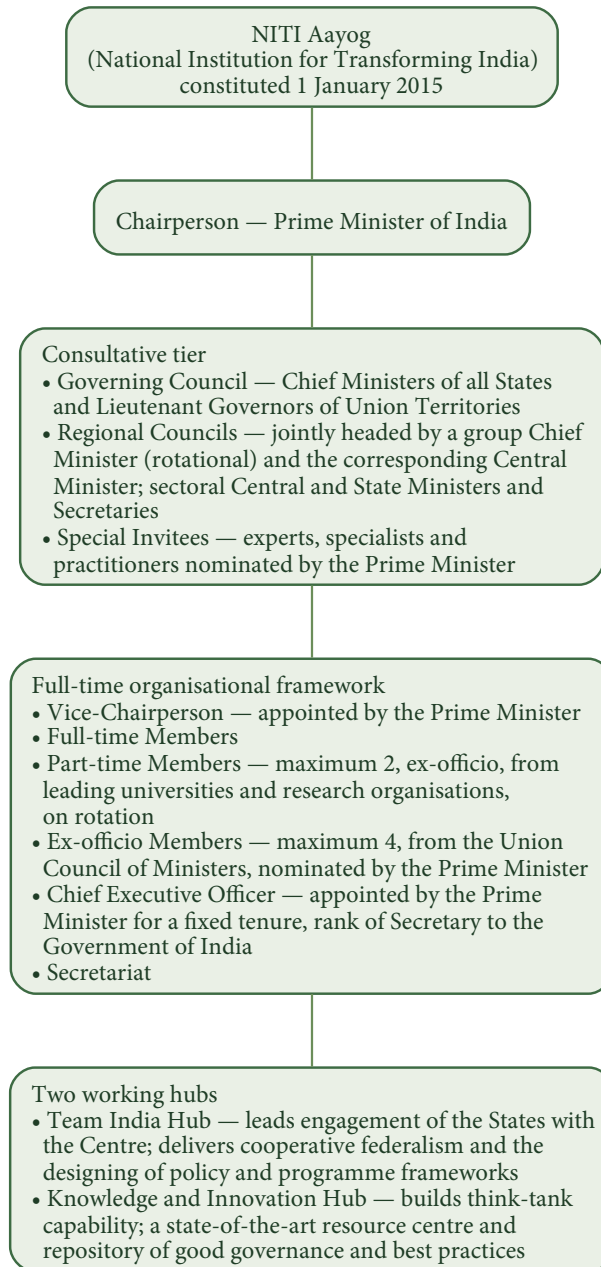
The Government of India constituted NITI Aayog as part of its reform agenda, as an evolutionary change from the past rather than a cosmetic one: the body was meant to act as the platform on which states come together to act in the national interest (Sunder Lal, Ch 16, pp 696–697).

Point	Content
Body replaced	Planning Commission, set up in 1950 to assess the material, capital and human resources of the country and draft developmental plans; it worked through five divisions — Programme Advisers, General Secretariat and Technical Divisions — and had a Perspective Planning Division making 20–25 year projections (Park 28e, Ch 23, p 1031; Sunder Lal, Ch 16, pp 696–697)
Announcement	Prime Minister announced the scrapping of the Planning Commission in his first Independence Day address, 2014; NITI Aayog announced and constituted in 2015 (Jugal Kishore, Ch 1, pp 73–74)
Stated reason	India had undergone a paradigm shift over six decades — politically, economically, socially, technologically and demographically — and the role of government in national development had evolved in parallel (Jugal Kishore, Ch 1, pp 73–74)
Core criticism of the old body	A "one size fits all" approach to planning was obsolete for a diversified country whose states are at very different stages of economic development (Park Plus, Ch 21, p 279)
Central intent	To bring states to act together in the national interest and thereby foster cooperative federalism (Sunder Lal, Ch 16, pp 696–697)

Organisational structure

Two hubs sit at the core of the institution's design, and NITI Aayog is additionally developing itself as a state-of-the-art resource centre with the resources, knowledge and skills to act with speed, promote

research and innovation, provide strategic policy vision to the government, and deal with contingent issues (Sunder Lal, Ch 16, pp 696–697).



Organisational structure of NITI Aayog

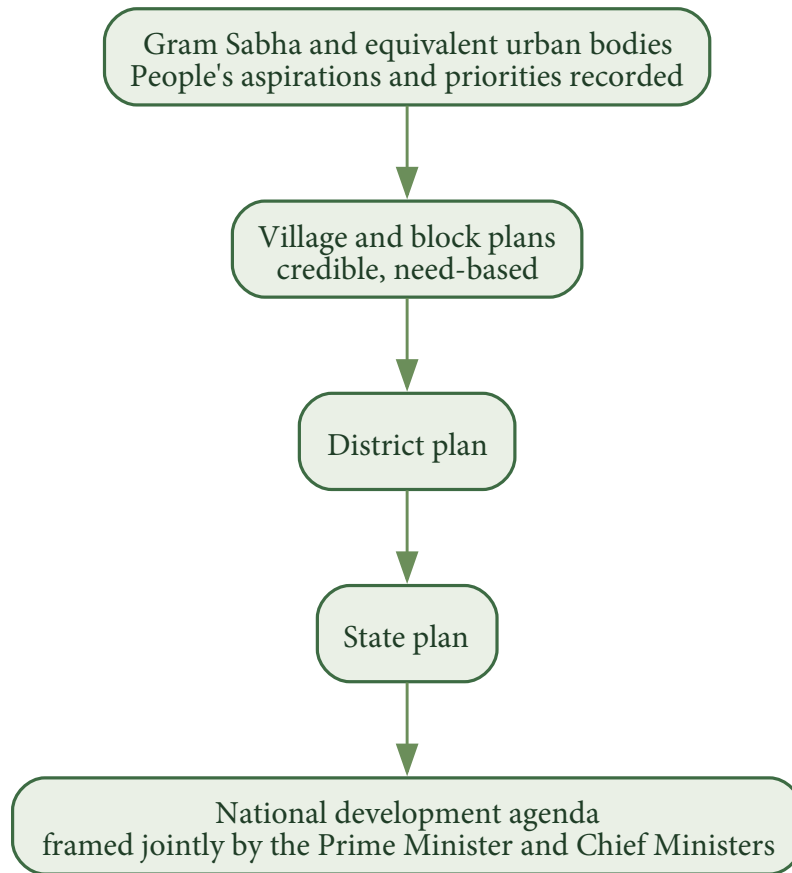
Objectives and functions

Objectives (nine)	Functions (fifteen)
1. Foster cooperative federalism with active involvement of states	1. Cooperative and competitive federalism — states participate in framing national policy and in time-bound implementation of quantitative and qualitative targets through the combined authority of the Prime Minister and Chief Ministers
2. Formulate plans at village level and aggregate them at higher levels	2. Shared national agenda — a shared vision of national development priorities with active state involvement

Objectives (nine)	Functions (fifteen)
3. Give special attention to sections at risk of not benefiting adequately from economic progress	3. State's best friend at the Centre — coordinating with ministries, championing state ideas, consultancy support, capacity building
4. Economic policy that incorporates national security interest	4. Decentralised planning — restructuring planning into a bottom-up model
5. Feedback for constant innovative improvement	5. Vision and scenario planning; 6. Domain strategies
6. Partnership with national and international think tanks	7. Sounding board — objective criticism and counter-views to refine government positions
7. Create a knowledge, innovation and entrepreneurial support system	8. Network of expertise; 9. Knowledge and innovation hub
8. Platform for resolution of inter-sectoral and inter-departmental issues	10. Harmonisation across layers of government; 11. Conflict resolution — inter-sectoral, inter-departmental, inter-state and Centre-State
9. Focus on technology upgradation and capacity building	12. Coordinating interface with the world; 13. Internal consultancy; 14. Capacity building; 15. Monitoring and evaluation

(Objectives — Jugal Kishore, Ch 1, p 74; functions — Park Plus, Ch 21, p 280 with detail from Jugal Kishore, Ch 1, p 74.)

The guiding **principles** are *Antyodaya* (service and uplift of the poorest first), **inclusion** of vulnerable and marginalised sections, the **village** as the bedrock of development, the **demographic dividend**, and **people's participation** (Park Plus, Ch 21, p 280).



Bottom-up planning under NITI Aayog — plans aggregate upward from the Gram Sabha

Planning Commission compared with NITI Aayog

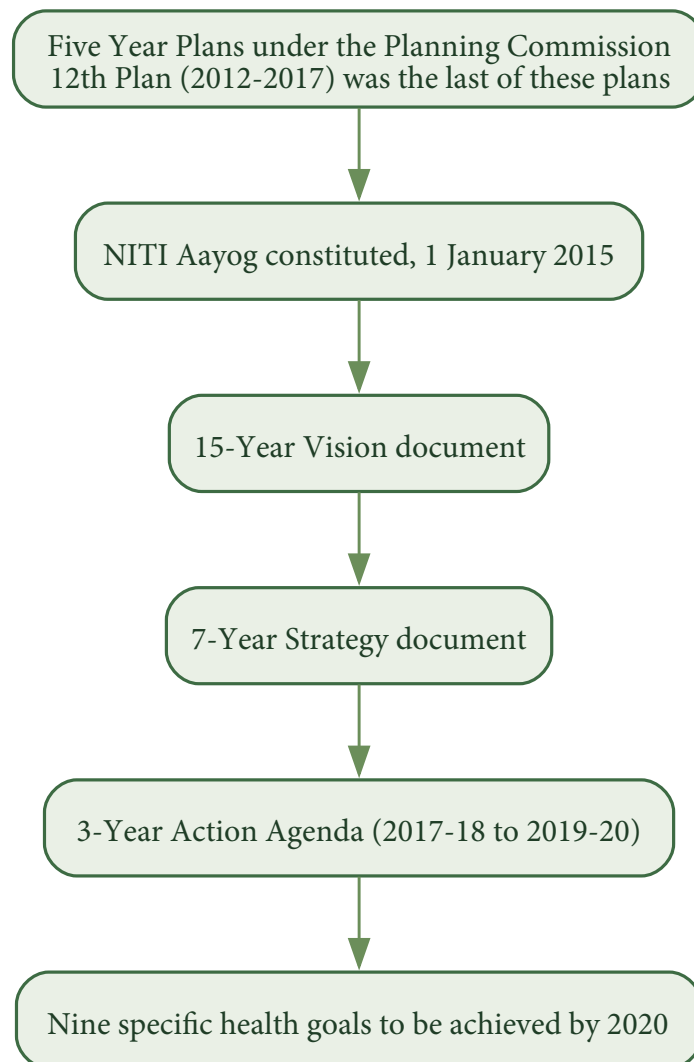
Feature	Planning Commission (1950–2014)	NITI Aayog (from 1 January 2015)
Character	Planning body that drafted developmental plans and reviewed progress	Think tank giving strategic and technical advice; an <i>enabler</i> , not a "provider of first and last resort"
Role in the system	Occupied the controlling seat of the planning process	Provides direction rather than control
Direction of planning	Largely top-down, Centre-driven	Bottom-up — village plans aggregated progressively upward
Leadership	Chairman, Deputy Chairman and Members; worked through Programme Advisers, General Secretariat and Technical Divisions	Chairperson (Prime Minister), Vice-Chairperson, full-time / part-time / ex-officio Members, and a Chief Executive Officer at the rank of Secretary
State interface	National Development Council — Prime Minister as chairman, Chief Ministers as members, Planning Commission members ex-officio	Governing Council of all Chief Ministers and Lieutenant Governors, plus Regional Councils for groups of states
Planning instrument	Successive Five Year Plans ; Perspective Planning Division projecting 20–25 years ahead	15-Year Vision, 7-Year Strategy and 3-Year Action Agenda — a deliberate departure from the Five Year Plan process

Feature	Planning Commission (1950–2014)	NITI Aayog (from 1 January 2015)
Federal philosophy	Uniform, "one size fits all" prescription	Cooperative and competitive federalism

(Park 28e, Ch 23, pp 1031, 1033; Jugal Kishore, Ch 1, pp 73–74; Park Plus, Ch 21, p 279.)

Planning documents and the health goals of the Three Year Action Agenda

NITI Aayog was tasked with preparing a 15-Year Vision, a Seven-Year Strategy and a Three-Year Action Agenda; this triad is an explicit departure from the Five Year Plan process, and the Action Agenda offers ambitious proposals for policy change over a relatively short period (Park 28e, Ch 23, p 1033). From the end of the 12th Plan in March 2017 the intent was that people's aspirations, as determined by every Gram Sabha and equivalent urban body, would form the basis of planning — "planning by the people, for the people and with the people", health included (Sunder Lal, Ch 16, p 697).



From the Five Year Plan process to the NITI Aayog planning documents

#	Health goal to be achieved by 2020	Baseline stated in the Agenda
1	Maternal mortality ratio (MMR) to 120 per 100,000 live births	167 per 100,000 (2013 estimate)
2	Infant mortality rate (IMR) to 30 per 1,000 live births	40 per 1,000 (2013 estimate)
3	Under-five mortality rate to 38 per 1,000 live births	48 per 1,000 (2015 estimate)
4	Total fertility rate (TFR) to 2.1	2.3 (2013 estimate)
5	Incidence of tuberculosis (TB) to 130 per 100,000	217 per 100,000 (2015 estimate)
6	Annual parasite incidence (API) of malaria < 1 per 1,000 in 90% of districts	74% of districts had API < 1 (2016 estimate)
7	Eliminate kala-azar and lymphatic filariasis	80% of endemic blocks (kala-azar) and 87% of endemic districts (filariasis) had eliminated (2015 estimate)
8	Reduce premature mortality from cardiovascular disease, cancer, diabetes or chronic respiratory disease by one-fourth	National Family Health Survey-4 (NFHS-4) levels
9	Reduce out-of-pocket spending to 50% of total health expenditure	62.4% (2014 estimate)

(Park 28e, Ch 23, pp 1032–1033, citing NITI Aayog, *Three Year Action Agenda 2017-18 to 2019-2020*.)

Work in the health sector, and limitations

Health is treated as a critical constituent of the development process and as vital to the nation's economic growth and internal stability, so NITI Aayog's health work runs along three lines — target-setting, measurement and ranking of state performance, and technical support to flagship schemes (Park 28e, Ch 23, p 1031).

Contribution	Content
Health Index — "Healthy States, Progressive India"	NITI Aayog's annual ranking of all states and Union Territories to measure health performance. States are grouped into 21 larger states, 8 smaller states and 7 Union Territories so that like is compared with like (Park 28e, Ch 2, p 44)
Composition of the Health Index	A weighted composite index of indicators in three domains — health outcomes 70%, governance and information 12%, key inputs and processes 18% . Within a domain or sub-domain the weight is distributed equally among the indicators (Park 28e, Ch 2, p 44; Park 27e, Ch 2, p 36)
Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana	In partnership with NITI Aayog, a robust, modular, scalable and interoperable information-technology platform was made operational for paperless, cashless transactions (Oxford Textbook of Public Health 7e, Ch 21, p 842)

Contribution	Content
Programme oversight	Monitors and evaluates the implementation of programmes and drives technology upgradation and capacity building across sectors including health (Park 28e, Ch 23, p 1031)
Limitation — structural	Restructuring of the planning process is welcome, but renaming without substantial structural reform or a holistic approach may not be fruitful (Jugal Kishore, Ch 1, pp 75–76)
Limitation — sectoral representation	Key ministries such as the Ministry of Health and Family Welfare and the Ministry of Science and Technology were omitted from the ex-officio composition, which weakens the transformation agenda (Jugal Kishore, Ch 1, pp 75–76)
Limitation — authority	It advises and directs but does not sit in the controlling seat, so translation of its advice into allocation and action depends wholly on the ministries and the states (Jugal Kishore, Ch 1, p 74)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **NITI Aayog organogram** — draw a vertical tree and label exactly: *Chairperson — Prime Minister; Governing Council — all Chief Ministers + Lieutenant Governors of Union Territories; Regional Councils; Special Invitees; Vice-Chairperson; Full-time Members; Part-time Members (max 2); Ex-officio Members (max 4 Union Ministers); Chief Executive Officer (rank of Secretary); Secretariat; and the two hubs — Team India Hub and Knowledge and Innovation Hub.*
- ▶ **Health Index pie** — draw one circle divided into three sectors and label exactly: *Health outcomes — 70%; Key inputs and processes — 18%; Governance and information — 12%;* title it *NITI Aayog Health Index: domain weights.*

High-yield exam pointers

- ▶ NITI = National Institution for Transforming India; born 1 January 2015; replaced the Planning Commission (1950); the 12th Five Year Plan (2012–2017) was the last Five Year Plan.
- ▶ Document triad to write first: 15-Year Vision → 7-Year Strategy → 3-Year Action Agenda (2017–18 to 2019–20).
- ▶ Composition mnemonic "PM chairs, VC runs, CEO executes" — plus *Governing Council* (all Chief Ministers + Lieutenant Governors), *Regional Councils*, *max 2 part-time*, *max 4 ex-officio*, CEO at Secretary rank.
- ▶ Two hubs: **Team India Hub** (states, cooperative federalism) and **Knowledge and Innovation Hub** (think tank).
- ▶ Health Index weights — 70 : 18 : 12 (health outcomes : key inputs and processes : governance and information); grouping 21 larger states, 8 smaller states, 7 Union Territories; report title *Healthy States, Progressive India*.
- ▶ Three Year Action Agenda 2020 health targets worth quoting: MMR 120, IMR 30, under-five mortality 38, TFR 2.1, TB incidence 130/100,000, out-of-pocket spending 50%.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the retrieved corpus does not cover NITI Aayog's later initiatives such as the Transformation of Aspirational Districts Programme or the SDG India Index, so these have been left out rather than stated from memory; the Health Index weights and state groupings above are those of the report edition cited in Park (Government of India, 2019, *Healthy States, Progressive India*), and later editions may regroup states as the number of Union Territories has since changed.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-5, Sample Registration System, National Health Accounts 2021-22, NITI Aayog National Multidimensional Poverty Index 2023, SDG India Index, WHO UHC Service Coverage Index, IPHS 2022

Programme and policy updates (India)

- ▶ **Health and Wellness Centres have been renamed Ayushman Arogya Mandir (AAM)** — this is the name to write now. More than 1.75 lakh were operational by late 2024, led by a new **Community Health Officer / mid-level health provider** cadre (1,16,978 in position on 30 November 2024). All twelve expanded service packages had been rolled out in 82% of centres, with 106 free medicines and 14 diagnostics at sub-centre level and 172 medicines and 63 diagnostics at primary health centre level [PMID 41816146]
- ▶ **NITI Aayog now also publishes the National Multidimensional Poverty Index** — *A Progress Review 2023* applies the Alkire–Foster method (NITI Aayog with MoSPI, and OPHI and UNDP as technical partners) over three dimensions — health and nutrition, education, standard of living — and 12 indicators, each dimension weighted one-third and a household counting as poor when deprived on at least one-third of the weighted indicators. The headcount ratio fell from **24.85% (2015–16) to 14.96% (2019–21)**, and because the index is computed down to district level it doubles as a targeting tool: districts such as Satna, Banda, Simdega, Bastar and Purnia cluster high poverty with poor child health and have been proposed as High-Priority Development Zones [PMID 41291549]
- ▶ **First causal evaluation of the Aspirational Districts Programme (launched 2018)** — propensity-score-matched difference-in-differences across NFHS-4 and NFHS-5 found a **2–4 percentage-point fall in under-five underweight** in aspirational districts on pre-pandemic data; no significant effect was demonstrable on most other health and nutrition indicators, so the programme's health dividend is real but narrow [PMID 39018957]

Current national data

- ▶ **The Three Year Action Agenda's maternal and fertility targets have already been overtaken** — the Sample Registration System now puts **MMR at 97 per 100,000 live births** (the Agenda sought 120 by 2020 from a 167 baseline), and NFHS-5 records **TFR 2.0** (target 2.1) with institutional births at 88.6%. The operative benchmark is now the SDG target of under 70 by 2030 [PMID 40547788]
- ▶ **Full immunisation coverage converged between district categories** — coverage at 12–23 months rose from 60.9% to 76.0% in non-aspirational districts and from 57.9% to 74.7% in

aspirational districts between NFHS-4 and NFHS-5. Wealth-related inequality fell in both but remained wider in aspirational districts, the main contributors being mother's education, use of health services, residence and region [PMID 41369871]

- **The SDG India Index is now used as an outcome benchmark in its own right** — Global Burden of Disease 2023 correlated NITI Aayog's state **SDG-3 index scores (2023–24)** inversely with the stillbirth rate (a modest association the authors themselves caution against over-reading), and found India's late-gestation stillbirth rate in SRS 2023 to be **2.3 times lower** than the modelled estimate — a direct caution about how complete the registration data behind such targets actually is [PMID 42565145]

Elimination, surveillance and disease status

- **Kala-azar: programme renamed, deadline already slipped once** — the Kala-azar Control Programme became the **National Kala-Azar Elimination Programme**, run by the **National Centre for Vector Borne Diseases Control (NCVBDC, the renamed NVBDCP)**. The elimination target is **under 1 case per 10,000 population at block level** in India, and the National Health Policy 2002 deadline of 2010 was revised to 2015. District programme data show visceral leishmaniasis still falling while **PKDL now dominates the caseload** (55 PKDL against 7 VL in 2022 in one endemic district), making PKDL detection the present bottleneck [PMID 42591813]
- **Filariasis and malaria targets remain open** — lymphatic filariasis is still endemic in **345 districts** against a **2030** elimination goal, with mass drug administration compliance and post-MDA surveillance the limiting steps [PMID 41951149]; and in Mizoram the species mix has shifted from *P. falciparum* to *P. vivax* (23% of cases in 2021 to 47% in 2024) with perennial rather than monsoon-linked transmission — directly relevant to the Agenda's district-level annual parasite incidence target [PMID 42310537]

Guideline and legislative updates

- **Standards and financing for primary care were restructured** — **Indian Public Health Standards 2022** now set the infrastructure benchmark for AAM, funded beyond National Health Mission grants by the **PM-Ayushman Bharat Health Infrastructure Mission** and **XV Finance Commission** health grants routed through Panchayati Raj Institutions and Urban Local Bodies. **Team-based performance incentives were added alongside the existing untied facility grants** — the untied fund of Rs 10,000 per sub-centre, the untied grant of Rs 25,000 and annual maintenance grant of Rs 50,000 per primary health centre, and Rs 1 lakh to the Rogi Kalyan Samiti all continue unchanged, so this is an addition to facility financing and not a replacement of it. Quality certification runs under the National Quality Assurance System [PMID 41816146]

Global and WHO developments

- **India's UHC Service Coverage Index rose from 57 (2015) to 63 (2021)**, above the South-East Asia regional average of 62, while **out-of-pocket expenditure fell from 60.6% (2014–15) to 39.4% (2021–22)** on National Health Accounts estimates — so the Action Agenda's 50% out-of-pocket goal is already met and should be quoted as surpassed [PMID 41816146]. Separately, WHO's ICD-11 definition of stillbirth at 22 weeks or more is not yet routinely reported in India, which still counts from 28 weeks, understating the burden [PMID 42565145]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 23, Health Planning and Management, pp 1031–1033; Ch 2, Concept of Health and Disease, p 44); Park's Textbook of Preventive and Social Medicine 27e (Ch 2, p 36 — clean text of the Health Index domain weights); Jugal Kishore — National Health Programmes of India (Ch 1, pp 73–76); Sunder Lal — Textbook of Community Medicine (Ch 16, Health Planning and Management, pp 696–697); Park Plus (Ch 21, Health planning and management, pp 278–280); Oxford Textbook of Public Health 7e (Ch 21, p 842). Named documents: NITI Aayog, *Three Year Action Agenda 2017-18 to 2019-2020* (Government of India, 2017); Government of India, *Healthy States, Progressive India — Report of the NITI Aayog on the Ranks of States and Union Territories* (2019).

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmmpc.jfmmpc.478.25](https://doi.org/10.4103/jfmmpc.jfmmpc.478.25); 41291549 — A study on spatial association of poor child health outcomes and multidimensional poverty in India, *BMC Public Health* 2025, doi:[10.1186/s12889-025-25222-6](https://doi.org/10.1186/s12889-025-25222-6); 39018957 — Health impact evaluation of Aspirational Districts Program in India: evidence from National Family Health Survey, *Economics and Human Biology* 2024, doi:[10.1016/j.ehb.2024.101411](https://doi.org/10.1016/j.ehb.2024.101411) (abstract-only); 40547788 — Decadal trends of maternal mortality and utilization of maternal health care services in India, *Journal of Family Medicine and Primary Care* 2025, doi:[10.4103/jfmmpc.jfmmpc.916.24](https://doi.org/10.4103/jfmmpc.jfmmpc.916.24); 41369871 — Decomposition of wealth-related inequality in child immunization among aspirational and non-aspirational districts in India, *Indian Journal of Pediatrics* 2025, doi:[10.1007/s12098-025-05877-0](https://doi.org/10.1007/s12098-025-05877-0) (abstract-only); 42565145 — Stillbirths at 22- and 28-weeks' gestation or longer for the states in India: findings from the Global Burden of Disease Study 2023, *The Lancet Regional Health — Southeast Asia* 2026, doi:[10.1016/j.lansea.2026.100819](https://doi.org/10.1016/j.lansea.2026.100819); 42591813 — Sustaining the elimination of visceral leishmaniasis: a secondary data analysis of the National Kala-Azar Elimination Programme in Malda district, West Bengal (2018–2022), *Cureus* 2026, doi:[10.7759/cureus.112553](https://doi.org/10.7759/cureus.112553); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42310537 — Rising burden of *Plasmodium vivax* in Mizoram: an emerging threat to malaria elimination in Northeast India, *BMC Infectious Diseases* 2026, doi:[10.1186/s12879-026-13756-y](https://doi.org/10.1186/s12879-026-13756-y) (abstract-only). Named documents: NITI Aayog, *National Multidimensional Poverty Index — A Progress Review* 2023; NITI Aayog, *SDG India Index* (SDG-3 state scores 2023–24); Ministry of Health and Family Welfare, *Indian Public Health Standards 2022*; *National Health Accounts Estimates for India 2021-22*; International Institute for Population Sciences, *National Family Health Survey-5 (2019–21)*; Office of the Registrar General of India, *SRS Special Bulletin on Maternal Mortality*.

♦ ♦ ♦

Principles of health manpower planning

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — "Health manpower" (human resources for health) includes both **professional and auxiliary health personnel** needed to provide health care; the World Health Organization (WHO) defines an **auxiliary** as a "technical worker in a certain field with less than full professional training" (Park 28e, Ch 10, p 724). **Health manpower planning** is the process of estimating the number, category-mix and geographic distribution of health workers a population will require at a defined future date, and of matching that requirement against the supply that training and retention will actually deliver. It is an integral part of community health planning and rests on a series of accepted ratios — doctor–population, nurse–population, bed–population (Park 28e, Ch 10, p 724).

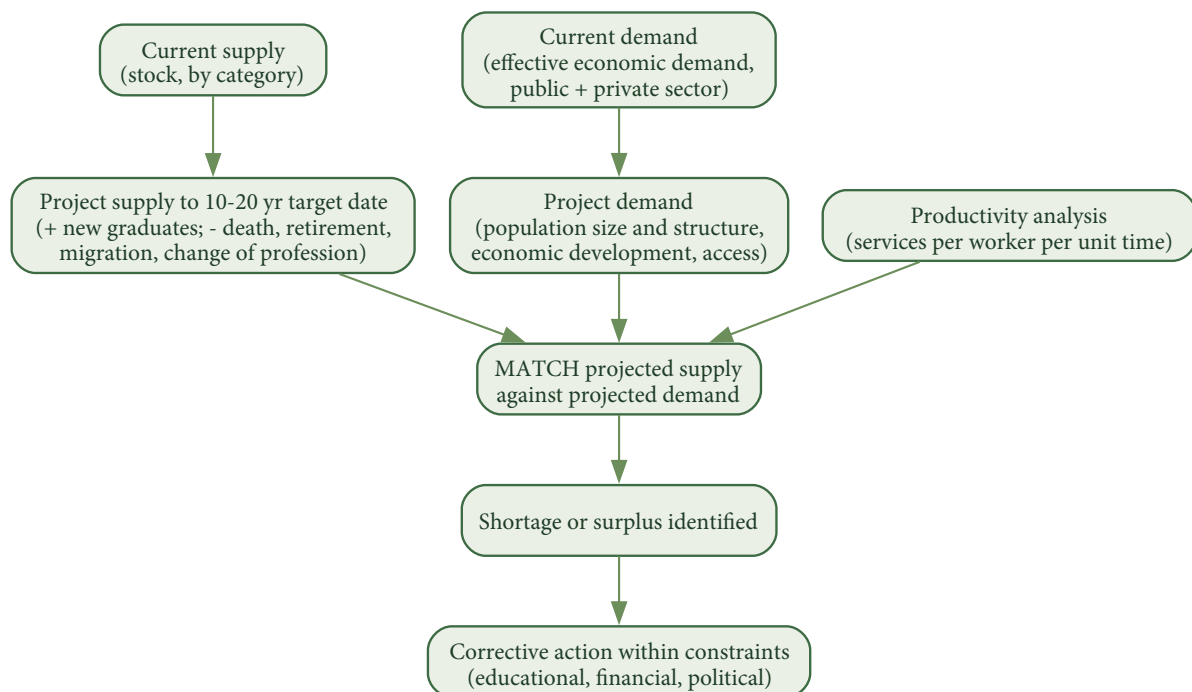
Governing principles

#	Principle	Basis
1	Plan for effective demand, not for need alone. Requirement means perceived need <i>backed by willingness and ability to commit resources</i> ; unused clinics standing in areas of continuing need are the proof of planning to need alone	Reinke, Ch 10, p 144, pp 151-152
2	Plan long, because lead times are long. "Hospitals can be built in months; it takes a decade to train a doctor" — hence projections are made to target dates 10–20 years ahead	Reinke, Ch 10, p 144, pp 144-145
3	Plan at two levels simultaneously — macro (how many workers) and micro (what functions and tasks each performs). "To ask how many nurses are needed is an idle question until service loads have been established"	Reinke, Ch 10, p 144; Ch 19, pp 272-273
4	Productivity is the fulcrum. Supply is balanced against demand either by <i>increasing supply</i> or by <i>increasing productivity</i> ; productivity gain is capped by acceptable quality	Reinke, Ch 10, p 145
5	Plan the team and the skill mix, not one category. Three levels — professional, assistant and aide — suffice for most planning; over-classification into many subgroups only confuses	Reinke, Ch 10, pp 145-146
6	Distribution matters as much as total number. A country may have an adequate overall number of workers so poorly distributed that corrective action is needed	Reinke, Ch 10, p 149
7	Requirements are dynamic, changing qualitatively and quantitatively as new programmes and philosophies enter the system (unipurpose → multipurpose worker; Health for All; disease-control programmes)	Park 28e, Ch 10, p 725
8	Personnel are the costliest input — in the labour-intensive health sector up to two-thirds of costs may be personnel, so errors are expensive and slow to correct	Reinke, Ch 10, p 144
9	The planner poses alternatives; government decides. Final choices (rural–urban equity, subsidy of education, retirement age, degree of state control) sit at higher levels of government	Reinke, Ch 10, pp 156-157

The four-part analytic framework

Step	What is done	Output
1. Supply analysis and projection	Measure current stock of every type of health worker; project forward with additions (new graduates) and subtractions (death, retirement, migration, change of profession)	Projected supply at target date
2. Demand analysis and projection	Assess effective economic demand for services from both private and public sectors; project it forward	Projected demand at target date
3. Productivity analysis and projection	Estimate average number of services per worker per unit time, and project change in it	Services obtainable per worker
4. Matching supply and demand	Compare the two projections and recommend adjustments to effect a balance within real constraints	Shortage/surplus and the corrective plan

(Reinke, Ch 10, pp 144-145)



The four-part analytic framework of health personnel planning (Reinke, Ch 10, pp 144-145)

Supply analysis in detail

- **Whom to count** — categories costly to society (numbers × average earnings): doctors, nurses, dentists, technicians, pharmacists; midwives and traditional practitioners where numerically important. Untrained institutional staff (drivers, maids, clerks) are part of the general labour pool and are not counted (Reinke, Ch 10, p 145).
- **Define each category** — one group of "doctors" comprising all persons licensed to practise modern scientific medicine; indigenous practitioners counted separately (Reinke, Ch 10, p 146).
- **Sources for the count** — graduate output of training institutions (corrected for death, retirement, migration); licensing bodies; annual/biennial professional registries; narcotic-use registration; professional society rosters; census; and, most accurate but costly, a **special survey** (Reinke, Ch 10, pp 146-147).

- **Characteristics to record** — age, sex, educational background, income, type and hours of practice, specialisation, productivity and **geographic location**; age composition is essential, since a gap in one age band manifests only decades later (Reinke, Ch 10, pp 147-149).
- **Losses** — retirement is usually the **greatest** loss (retirement age 55–70 across countries); then change of profession (for women, marriage may interrupt work temporarily or permanently); then migration; death is comparatively small (Reinke, Ch 10, pp 149-150).
- **Four determinants of increase** — qualified applicants ("raw material"), educational plant capacity (including teaching beds), capital, and **availability of teachers** — the last being the most important because of the long lag in preparing a teacher (Reinke, Ch 10, pp 150-151).

Approaches to estimating requirement

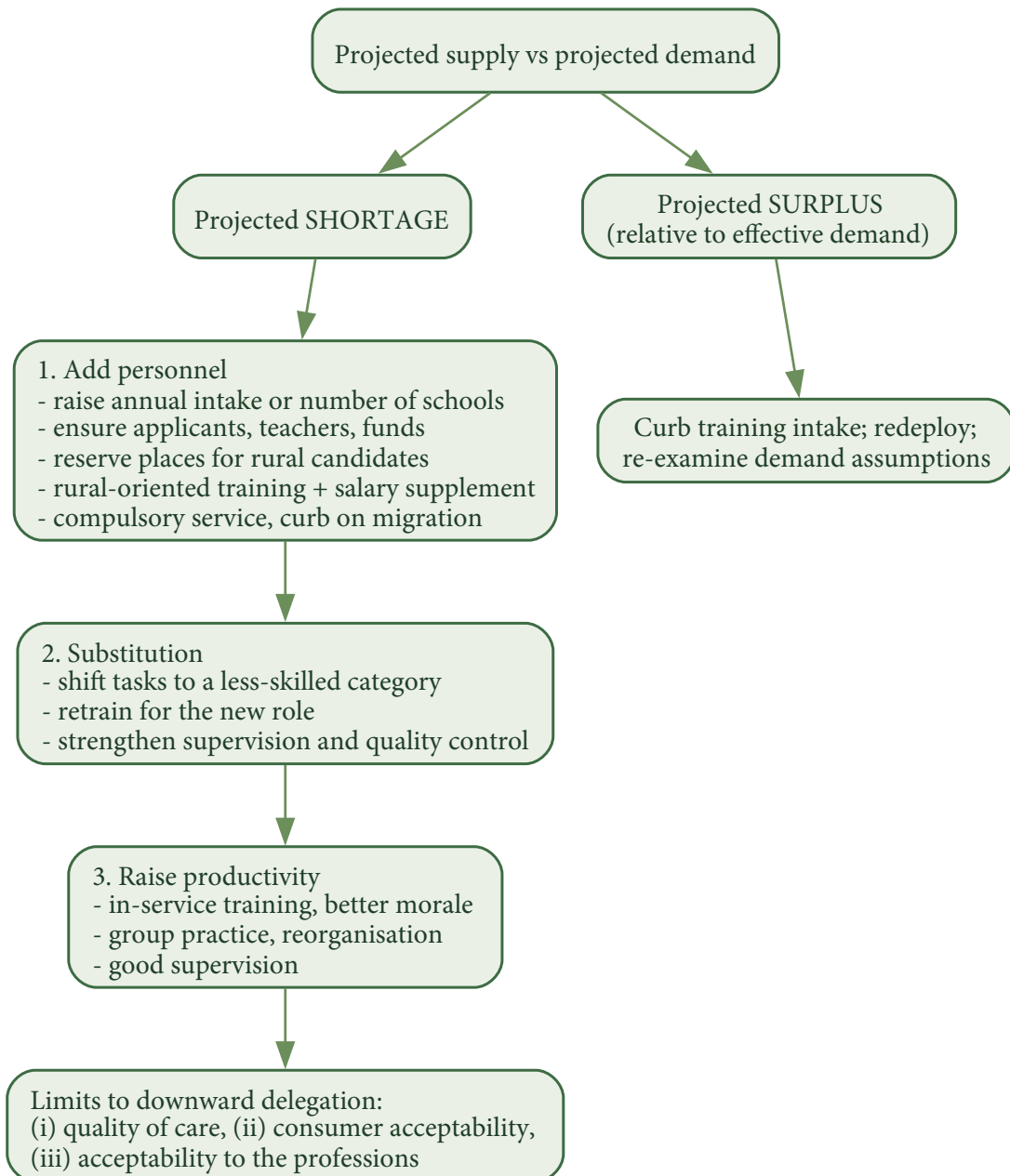
Approach	Basis of estimate	Main flaw
Basic biologic (health) need (<i>Lee and Jones</i>)	Morbidity/mortality level $\times$ professional time per case $\div$ hours worked per professional per year	Needs accurate morbidity data; one problem has alternative solutions using different workers; need $\neq$ demand
Status quo (normative) — existing worker-to-population ratios enshrined as norms	Present ratios projected onto future population	Never tests whether present ratios are suitable; ignores technological and demographic change
"Parkinsonian" modification	"Our doctors and nurses are busy now, therefore we need more" (public health nurse ratio thus revised from 1:10,000 to 1:5,000)	Workload perception substitutes for measurement
Comparative method	Adopts ratios of a reference country	Very few countries are truly comparable
Expert opinion	Consensus of authorities	Subjective; but often the best available for small specialised cadres (health education, sanitary engineering)
Established standards of practice (service targets)	Observed norms of practice — e.g. an outpatient doctor seeing 5 patients/hour, 5.5 hours/day, 240 days/year $\approx$ 6,600 visits/year; workers allocated on observed visits per 1,000 population	Complex and exacting to operate
Effective economic demand	Random sample survey of demand patterns, quantified by demographic and socio-economic attributes; in the public sector, budgeted vacancies index unmet demand	Demand reflects ability to pay, not need
Mixed / composite method	Elements of all relevant methods, results cross-checked	Recommended, since no single method is adequate

(Reinke, Ch 10, pp 151-153)

Projecting demand forward

Determinant	Effect on demand
Population size and composition	More people → more services; a higher proportion of elderly and young children raises demand disproportionately; urbanisation changes its size and character
Economic development	Higher income and education raise expectations and utilisation; more tax revenue expands public-sector demand; shift from indigenous to modern practitioners
Supply and accessibility	Effective demand does not exist where a doctor is unknown or unreachable; a new road markedly changes demand
Unpredictable factors	Change in disease pattern, technological advance, and organisational change — largely unforeseeable

(Reinke, Ch 10, pp 153-155)



Correcting a projected personnel imbalance (Reinke, Ch 10, pp 155-157)

Indian application — norms and current position

Category of personnel	Norm suggested
Nurses	1 per 5,000 population
Health worker (female and male)	1 per 5,000 population in plain areas; 1 per 3,000 in tribal and hilly areas
Trained <i>dai</i>	One for each village
Health assistant (male and female)	1 per 30,000 in plain areas; 1 per 20,000 in tribal and hilly areas; supervises 6 health workers
Pharmacist	1 per 10,000 population
Laboratory technician	1 per 10,000 population

Category of personnel	Norm suggested
Accredited Social Health Activist (ASHA)	1 per 1,000 population

(Park 28e, Ch 10, p 725)

Indicator, India	Value (2013–2021)
Doctors per 10,000 population	7.3
Beds per 10,000 population	7.0
Nurses and midwives per 10,000 population	17.3

(Park 28e, Ch 10, Table 3, p 725)

National averages conceal **maldistribution**: up to 73.6 per cent of doctors are concentrated in urban areas, where only 36 per cent of the population lives. Park attributes this to absence of amenities in rural areas, lack of job satisfaction, professional isolation, lack of rural experience, and inability to adjust to rural life (Park 28e, Ch 10, p 725).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Supply–demand balance on the fulcrum of productivity (after Reinke, Fig 10.1, Ch 10, p 145) — draw a see-saw/balance beam with these exact labels:

- ▶ Left pan: **SUPPLY of health workers** (stock + new graduates – death, retirement, migration, change of profession)
- ▶ Right pan: **DEMAND for health services** (population size and structure + economic development + accessibility)
- ▶ Fulcrum wedge beneath the beam: **PRODUCTIVITY** (services per worker per unit time)
- ▶ Arrow raising the left pan labelled **increase supply**; arrow shifting the fulcrum labelled **increase productivity**
- ▶ Ceiling line above the fulcrum labelled **limit = acceptable quality of care**

High-yield exam pointers

- ▶ **Four-part framework (mnemonic S-D-P-M)** — Supply analysis → Demand analysis → Productivity analysis → Matching; all projected to **10–20 year** target dates.
- ▶ **WHO definition of an auxiliary**: "technical worker in a certain field with less than full professional training" (Park 28e, Ch 10, p 724).
- ▶ **Seven approaches to estimating requirement**: biologic need (Lee and Jones) · status quo/normative ratio · Parkinsonian · comparative · expert opinion · established standards of practice · effective economic demand — plus the recommended **mixed method**.
- ▶ **Three ways to avert a shortage**: add personnel · substitute a lower category · raise productivity; downward delegation limited by quality, consumer acceptability and professional acceptability.
- ▶ **Supply losses in descending order**: retirement > change of profession > migration > death; **increase determinants**: applicants, plant capacity, capital, **teachers (rate-limiting)**.

- **Indian norms to write:** nurse 1:5,000 · health worker 1:5,000 plain (1:3,000 tribal/hilly) · health assistant 1:30,000 plain (1:20,000 tribal/hilly), supervising 6 workers · pharmacist and laboratory technician each 1:10,000 · ASHA 1:1,000 (Park 28e, 2021 data).
- **India 2013–2021:** 7.3 doctors, 7.0 beds and 17.3 nurses-and-midwives per 10,000 population; 73.6 per cent of doctors urban.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written: the term "manpower" has been largely replaced in current policy documents by "human resources for health", and India's cadre position is now reported through Rural Health Statistics and the Indian Public Health Standards rather than the older "rural health scheme" tables; the substance of the planning principles above is unchanged. Coverage note — the retrieved corpus contains no passage on the WHO Workload Indicators of Staffing Need (WISN) method or on the National Medical Commission's current seat-expansion figures, so neither is stated here.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, MoHFW, Indian Public Health Standards 2022, Public Health Management Cadre guidelines 2022, National Nursing and Midwifery Commission Act 2023

Programme and policy updates (India)

- **Health and Wellness Centres renamed Ayushman Arogya Mandir (November 2023)** by the Union Health Ministry, with the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth"). The rebrand was budgeted at ₹3,000 per facility, total outlay about ₹48 crore, and each centre still serves 3,000–5,000 population; facility and staffing changes were minor, so for manpower planning this is a renaming and a shift of service focus toward holistic/AYUSH-informed wellness rather than a re-sanctioning of posts [PMID 42257165].
- **Public Health Management Cadre (PHMC) guidelines, MoHFW 2022** operationalise the National Health Policy 2017 proposal for a dedicated public-health cadre; implementation from 2020 with roll-out to 2026 was estimated to create about **33,236 posts** spanning block, district and state levels, open to medical, paramedical and non-medical professionals (sociology, economics, anthropology, hospital management, communication) who have undergone public-health management training. A four-state situational analysis (Maharashtra, West Bengal, Gujarat, Odisha) found every state now has a dedicated cadre with its own hierarchy and directorate — though only Odisha and Gujarat have clearly demarcated roles and responsibilities — while penetration below district level remains limited; funding was not reported to be the problem, the binding constraint being a workforce skilled in public health management [PMID 37884946].
- **Electronic HR management systems are displacing the classical registry-and-survey count.** Uttar Pradesh's *Manav Sampada* eHRMS covers about **160,000 health personnel** across more than 30,000 facilities serving 235 million people, using a lifecycle-based minimum dataset and workflow design, and is positioned to become part of the Ayushman Bharat Digital Mission —

giving real-time supply data of the kind Reinke's supply analysis could only approximate from licensing bodies and special surveys [PMID 38821871].

Current national data

- **National Health Workforce Account (NHW) 2018 read against the Periodic Labour Force Survey 2017–18:** a *stock* of **5.76 million** health workers (allopathic doctors 1.16 million; nurses/midwives 2.34 million; pharmacists 1.20 million; dentists 0.27 million; AYUSH 0.79 million) but only **3.12 million active**. Stock density is 8.8 doctors and 17.7 nurses/midwives per 10,000, which falls to **5.0 and 6.0 per 10,000** once activity *and* adequate qualification are both applied — far below the WHO threshold of 44.5 doctors-nurses-midwives per 10,000. Among those holding a degree in medicine, **27 per cent are not active in the labour market** at all and a further 4 per cent are unemployed and job-seeking — a supply loss the older framework does not name [PMID 33752675].
- **District-level assessment of HR compliance with the revised Indian Public Health Standards 2022** (sampled health and wellness centres — 28 sub-health centres, 17 of 28 rural PHCs and 8 of 16 urban PHCs, drawn by simple random sampling, Amritsar, Punjab): sub-health centres showed only 3 per cent HR deficiency and 24×7 urban PHCs 7 per cent, but **rural PHCs ran 45–46 per cent deficient** and non-24×7 urban PHCs 48 per cent; 66 per cent of centres overall rated good or very good. IPHS 2022 supplies the current yardstick, and the urban–rural maldistribution is measurable facility by facility against a written standard [PMID 41017888].

Guideline and legislative updates

- **National Nursing and Midwifery Commission Act 2023 (India)** is expected to introduce uniform standards, entry and exit examinations, recognition of midwifery as a distinct discipline, a determined scope of practice for nurses and midwives, and a Nurse Practitioner programme with guidelines for limited prescribing authority — in planning terms, a legal route to the "substitution / downward delegation" option, with the classical limits (quality, consumer acceptability, professional acceptability) reappearing as stated concerns over commission composition, stakeholder representation, provisions for continued competence and clearly defined roles [PMID 39161310] (abstract-only).
- **WISN (Workload Indicators of Staffing Need) is the WHO workload method and has now been piloted by the Government of India.** The National Health Systems Resource Centre led district-hospital pilots in Kerala and Chandigarh (Meghalaya was dropped during COVID-19), finding gynaecologists adequate, paediatricians in surplus and nurses and laboratory technicians short; India has since approved the budget for the additional staff. The six-country regional review (Bangladesh, Bhutan, India, Indonesia, Nepal, Sri Lanka) notes IPHS numbers were formulated on expert opinion rather than an evidence base, and advises that facilities with known workforce shortages should first reach IPHS staffing, using WISN afterwards to fine-tune [PMID 40512697].
- **Nationally modelled WISN estimates for rural India (2019)** showed shortages in every cadre examined, very high workload pressure on PHC and CHC nurses and on all CHC specialist doctors, and poor concordance between current sanctioning norms and workload-based requirement, with every cadre under-sanctioned at national level — PHC and CHC nurses and CHC physicians and surgeons were under-sanctioned in *every* state, while PHC doctors were

under-sanctioned in 9 states but over-sanctioned in 13. The argument is that the sanctioning norms themselves are the error, and that deployment should be workload-based rather than ratio-based [PMID 35090494].

- **Five workforce-planning approaches identified across the rural and remote primary-care literature** — WISN, geospatial accessibility and catchment models, stock-flow forecasting, skill-mix and competency-based planning, and synthetic estimation modelling. A scoping review of 26 studies from 10 countries (January 2005 to July 2025) concludes that planning done in isolation from the model of care actually in use is likely to worsen workforce maldistribution, and calls for integrated, longitudinal, competency-linked workforce datasets rather than headcount alone [PMID 41803853].

Global and WHO developments

- **Fifth Global Forum on Human Resources for Health (WHO, Geneva, April 2023)** — "protect, invest, together": the global health and care workforce is projected to reach **84 million** by 2030 with a residual gap of **10 million**, concentrated in low- and middle-income countries of the African and Eastern Mediterranean regions and in small island developing states. **55 countries** appear on the 2023 WHO health workforce support and safeguards list — those whose health systems are most threatened by the growing migration of health workers to high-income countries; high-income countries hold 6.5 times the health-worker density of low-income countries, and Africa carries a quarter of the global disease burden with 4 per cent of the workforce; women make up 67 per cent of the workforce yet face a 24 per cent gender pay gap [PMID 37209702].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10, Health Care of the Community, pp 724-725; definitions, suggested personnel norms and Indian manpower position as on March 2021, with the same passage cross-checked in the 27e where the 28e scan is unclear); Reinke — Health Planning for Effective Management (Ch 10, Health Personnel Planning, pp 144-157, Timothy D. Baker; Ch 19, Personnel Analysis, pp 261-273; Ch 21, Synthesis of Health Needs and Resources, pp 289-296).

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of "Arogyam Paramam Dhanam" — a step towards holistic care, J Family Med Prim Care 2026, DOI [10.4103/jfmpc.jfmpc.1966.24](https://doi.org/10.4103/jfmpc.jfmpc.1966.24); 37884946 — Assessment of needs and gaps in public health cadre in India: a situational analysis, BMC Health Serv Res 2023, DOI [10.1186/s12913-023-10132-3](https://doi.org/10.1186/s12913-023-10132-3); 38821871 — Learnings from the implementation of an electronic human resource management system for the health workforce in Uttar Pradesh, India, Glob Health Sci Pract 2024, DOI [10.9745/GHSP-D-23-00312](https://doi.org/10.9745/GHSP-D-23-00312); 33752675 — Size, composition and distribution of health workforce in India: why, and where to invest?, Hum Resour Health 2021, DOI [10.1186/s12960-021-00575-2](https://doi.org/10.1186/s12960-021-00575-2); 41017888 — Human resource compliance with Indian Public Health Standards 2022: an assessment of health and wellness centers in a district of northwestern India, Indian J Community Med 2025, DOI [10.4103/ijcm.ijcm.636.24](https://doi.org/10.4103/ijcm.ijcm.636.24); 39161310 — National Nursing and Midwifery Commission Act, 2023 in India: issues and the way forward, Policy Polit Nurs Pract 2024, DOI [10.1177/15271544241271422](https://doi.org/10.1177/15271544241271422) (abstract-only); 40512697 — The implementation of the Workforce Indicators of Staffing Needs (WISN) method to improve access to health workforce in selected South-East Asian countries, PLOS Glob Public Health 2025, DOI [10.1371/journal.pgph.0004727](https://doi.org/10.1371/journal.pgph.0004727); 35090494 — Workforce problems at rural public health-centres in India: a WISN retrospective analysis and national-level modelling study, Hum Resour Health 2022, DOI [10.1186/s12960-021-00687-9](https://doi.org/10.1186/s12960-021-00687-9); 41803853 — Health workforce planning methods in rural and remote primary care: a scoping review, Hum Resour Health 2026, DOI [10.1186/s12960-026-01060-4](https://doi.org/10.1186/s12960-026-01060-4); 37209702 — Prioritising the health and care workforce shortage: protect, invest, together, Lancet Glob Health 2023, DOI [10.1016/S2214-109X\(23\)00224-3](https://doi.org/10.1016/S2214-109X(23)00224-3). Also: MoHFW Public Health Management Cadre guidelines 2022; Indian Public Health Standards 2022; National Nursing and Midwifery Commission Act 2023; WHO Fifth Global Forum on Human Resources for Health, Geneva, April 2023; WHO health workforce support and safeguards list 2023.



Quality-adjusted life years

Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

Definition — The **quality-adjusted life year (QALY)** is a composite measure of disease burden that combines **both the quality and the quantity of life lived**, and is used to assess the value for money of a health intervention (Park 28e, Ch 2, p 38). It is a health outcome measure based on **survival weighted by quality of life**, where quality of life is scored on a cardinal scale between **1.0 for full health and 0 for death** (Health Economics, Ch 15, pp 230-231). **One QALY is one year of life lived in perfect health**; a year lived at a utility value of 0.5 equals 0.5 QALY (Park 28e, Ch 2, p 38).

How a QALY is constructed

Step	What is done	Note
1. Fix the anchors	Death = 0, full health = 1.0 on a <i>cardinal</i> scale	On a cardinal scale the interval 0.2→0.3 means the same to the respondent as 0.7→0.8 (Health Economics, Ch 15, p 235)
2. Describe the health state	Define the compromised state the patient lives in, with and without the intervention	The description must be detailed enough for a respondent to picture living in that state (Health Economics, Ch 15, p 236)
3. Attach a utility weight (U)	Elicit U for that state by a direct or indirect method (next section)	Respondents may be the general population, patients with the disease, or health-sector personnel (Health Economics, Ch 15, p 236)
4. Multiply by time (T)	QALY = T (years) × U (utility of the state lived)	Sum across successive health states when utility changes over time
5. Discount future years	Convert future life years to present value, as is done for costs	National and international guidance recommends a 3 per cent discount rate , after controlling for inflation (Health Economics, Ch 14, p 228)
6. Take the difference	QALYs gained = QALYs with intervention – QALYs without intervention	The benefit of a health intervention is the <i>gain</i> in QALYs it produces, not the absolute total (Oxford Public Health 7e, Ch 49, pp 1831-1832)

Worked example. Two treatments each extend life by 10 years. Treatment A leaves the patient in full health (U = 1.0) and yields $10 \times 1.0 = 10$ QALYs. Treatment B leaves the patient in a state valued at U = 0.5 and yields $10 \times 0.5 = 5$ QALYs — half the health gain for the same survival (Health Economics, Ch 15, pp 234-235).

Measuring the quality weight (utility)

Approach	Method	How the weight is obtained	Point to remember
Direct	Standard gamble (SG)	Respondent living in a compromised state for X years is offered a treatment giving full health with probability P or immediate death with probability 1-P; P is varied until the respondent is indifferent. Utility = P	The method most consistent with conventional economic theory because it incorporates uncertainty (Health Economics, Ch 15, p 236)
Direct	Time trade-off (TTO)	Choice between X years in the compromised state and T years in full health; T is varied until indifference. Utility = T/X	30 years compromised judged equal to 20 years in full health → $20/30 = 0.67$ (Health Economics, Ch 15, pp 237-238)
Direct	Visual analogue scale (VAS)	A thermometer-type instrument on which the respondent places health states — mild problems near the top, severe states near the bottom; the endpoints are framed as "full health" and "death"	Simplest, but respondents tend merely to rank rather than scale, so it does not reflect strength of preference ; also subject to end-of-scale bias (respondents avoid 0 and 100) (Health Economics, Ch 15, p 238)
Direct	Person trade-off (PTO)	Respondent acts as a budget holder choosing between two mutually exclusive interventions helping two different groups	Used to elicit the disability weights for DALYs in the Global Burden of Disease study; extending 1,000 healthy lives judged equal to extending 1,700 blind lives → one year lived blind = $1,000/1,700 = 0.59$ (Health Economics, Ch 15, pp 238-239)
Indirect ("off-the-shelf")	Multi-attribute questionnaires	Patient completes a generic instrument; each health state already carries a value from general-population surveys	EuroQol-5D (EQ-5D) — five dimensions: mobility, self-care, usual activities, pain/discomfort, anxiety/depression , three levels each. Also SF-36 (Health Economics, Ch 15, p 239)

Use of the QALY in economic evaluation

Type of evaluation	Costs measured as	Consequences measured as	Use
Cost-effectiveness analysis (CEA)	Money	Natural units — lives saved, cases prevented, days free from disease	Compares alternative ways of reaching the same objective ; more applicable in health than cost-benefit analysis (Park 28e, Ch 23, p 1028; Reinke, Ch 20, p 274)
Cost-utility analysis (CUA)	Money	QALYs gained (or DALYs averted)	A special form of CEA; allows comparison across different health problems , and is especially useful where an intervention extends life at the cost of side-effects, e.g. cancer chemotherapy (Health Economics, Ch 13, pp 207-208)

Type of evaluation	Costs measured as	Consequences measured as	Use
Cost–benefit analysis (CBA)	Money	Money (benefit–cost ratio, net present value)	Compares programmes with diverse objectives ; a benefit–cost ratio above one is worth considering (Reinke, Ch 20, p 274)

The decision ratio. The distinctive output of a CUA is the **incremental cost-effectiveness ratio (ICER)** — the difference in cost between two alternatives divided by the difference in effect, expressed as the **cost per QALY gained** (Oxford Public Health 7e, Ch 49, pp 1826-1827; Health Economics, Ch 16, pp 247-248). The **average cost-effectiveness ratio (ACER)** relates to a single intervention against its baseline; the **incremental ratio** compares relative costs and effects of competing alternatives (Health Economics, Ch 13, p 207). Before any ratio is calculated, apply **dominance**: an option that is both more costly and less effective is *strongly dominated* and dropped; one that is bettered by a mix of two others is *weakly (extendedly) dominated* and also dropped (Health Economics, Ch 16, pp 252-253).

Intervention	Cost per patient (£)	QALYs per patient	ICER (£ per QALY)
Standard care	50	1.0	—
B	100	2.0	50
D	350	4.0	125
E	550	5.0	200

(Health Economics, Ch 16, Table 16.4, p 253 — after A is strongly dominated and C weakly dominated)

Thresholds (ceiling ratio). An intervention is judged cost-effective when the ICER falls **below** the threshold the decision-maker is willing to pay for a unit of health gain (Health Economics, Ch 16, pp 253-254).

Setting	Threshold quoted	Source and year
WHO / Commission on Macroeconomics and Health	"Highly cost-effective" if it averts one DALY for less than the per-capita national gross domestic product	WHO Commission on Macroeconomics and Health, 2001 (Health Economics, Ch 16, pp 253-254)
National Institute for Health and Care Excellence (NICE), UK	£20,000–30,000 per QALY gained as the ceiling ratio; in practice the threshold at which technologies are actually rejected has been found to be £35,000–48,000	Devlin and Parkin, 2004 (Health Economics, Ch 16, pp 256-257)
USA	US\$50,000 per QALY gained	Hirth <i>et al.</i> , 2000; expressed in 1982 US\$ (Health Economics, Ch 16, p 254)
Canada	CAN\$20,000–120,000 per QALY gained	Laupacis <i>et al.</i> , 1992; expressed in 1990 CAN\$ (Health Economics, Ch 16, p 254)

Applied example. In a cost–utility analysis of pre-adolescent cervical cancer (human papillomavirus) vaccination in Canada, vaccination cost Can\$739 versus Can\$476 unvaccinated (difference Can\$263)

and gave 30.5098 versus 30.5015 QALYs (gain 0.0083), so the cost per QALY gained was $\text{Can\$}263/0.0083 = \text{about Can\$}31,687$; allowing for cross-protection and herd immunity together it fell to Can\$18,672 (Oxford Public Health 7e, Ch 49, Table 6.6.2, pp 1827-1828, 1850-1851).

QALY compared with DALY

Feature	QALY	DALY
What it measures	Health gained — a "good" to be maximised	Health lost — a "health gap" or burden to be minimised
Weighting scale	Utility: 1 = full health, 0 = death	Disability weight: 0 = perfect health, 1 = dead (inverted)
Composition	Years lived $\times$ utility of the state	DALY = YLL + YLD (years of life lost + years lived with disability) (Park 28e, Ch 2, p 38)
Origin	Health economics, for cost-utility analysis of interventions	Harvard University for the World Bank, 1990; adopted by WHO in 2000 (Park 28e, Ch 2, p 38)
Reference for premature death	Not applicable (individual survival is modelled)	Global standard life expectancy — Japanese life expectancy statistics, the longest (Park 28e, Ch 2, p 38)
How weights were derived	Standard gamble, time trade-off, VAS, EQ-5D from patients or population	Person trade-off by international public health expert panels in the Global Burden of Disease study; updates now blend community and expert assessment (Health Economics, Ch 15, pp 235-236)
Chief use	Value-for-money of a specific intervention (cost per QALY)	Burden of disease, priority setting, Global Burden of Disease reporting

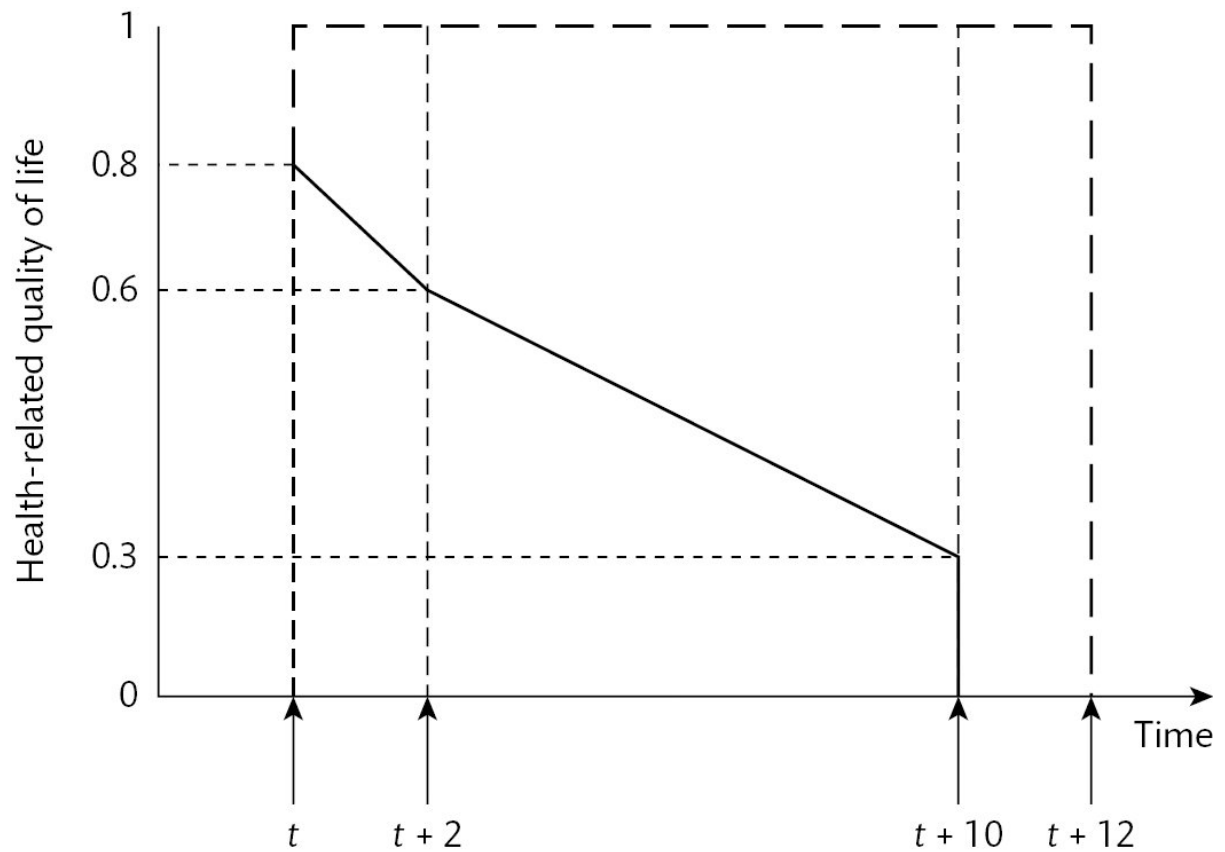
Strengths and limitations

Strengths	Limitations
Combines length and quality of life in one number , so interventions for entirely different diseases become comparable (Oxford Public Health 7e, Ch 49, pp 1831-1832)	Some argue length and quality of life cannot or should not be compared in the same metric (Oxford Public Health 7e, Ch 49, pp 1833-1834)
Deals with efficiency in the production of health itself , not merely of health care	The number of years over which quality can be adjusted is naturally shorter in the old, so QALYs and DALYs mathematically discriminate against the elderly (Health Program Planning and Evaluation, Ch 4, pp 126-127)
Being a population-level composite score , it is indifferent to individual preferences — the same weight applies whoever the patient is (Health Program Planning and Evaluation, Ch 4, pp 126-127)	For the same reason it is blind to who gains the QALY : the distribution of health gain between rich and poor, or advantaged and disadvantaged, is not captured, so equity must be brought in separately (Health Economics, Ch 17, p 265)
Essential where an intervention prolongs life at the cost of morbidity, e.g. cancer chemotherapy (Health Economics, Ch 13, pp 207-208)	QALYs do not capture the process characteristics of care — respect, autonomy, provision of information — which patients demonstrably value (Health Economics, Ch 13, p 208)

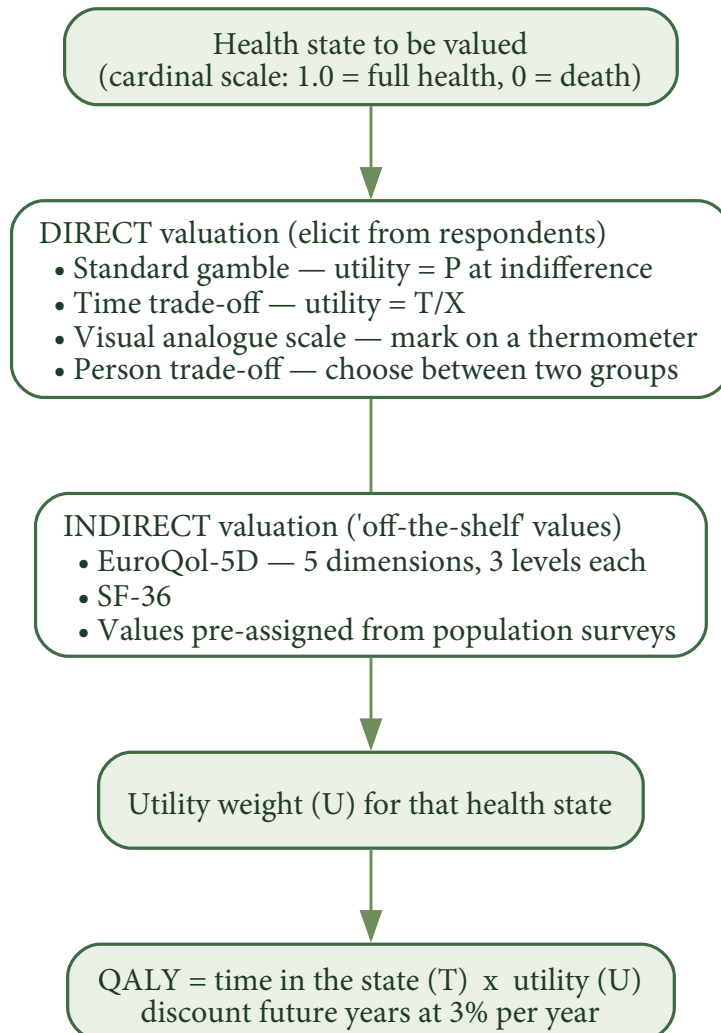
Strengths	Limitations
Permits ICER-based league tables and explicit, transparent priority setting (Health Economics, Ch 16, p 254)	Direct valuation is resource-intensive , needing described health states, experienced interviewers and cognitively intact respondents (Health Economics, Ch 15, p 236)
Uncertainty can be handled by sensitivity analysis on discount rate, utility weights and costs (Health Economics, Ch 15, p 236; Ch 14, p 228)	Choice of discount rate and of weighting assumptions materially changes the result (Health Economics, Ch 15, p 236)
Equity can be reintroduced by equity-weighted QALYs (Health Economics, Ch 17, p 265), the " <i>fair innings</i> " argument (Health Economics, Ch 17, p 274 — Williams 1997), NICE's end-of-life premium , or multi-criteria decision analysis (Oxford Public Health 7e, Ch 49, p 1844)	The " <i>fair innings</i> " route itself requires greater discrimination against the elderly than efficiency criteria alone would (Health Economics, Ch 17, p 274)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

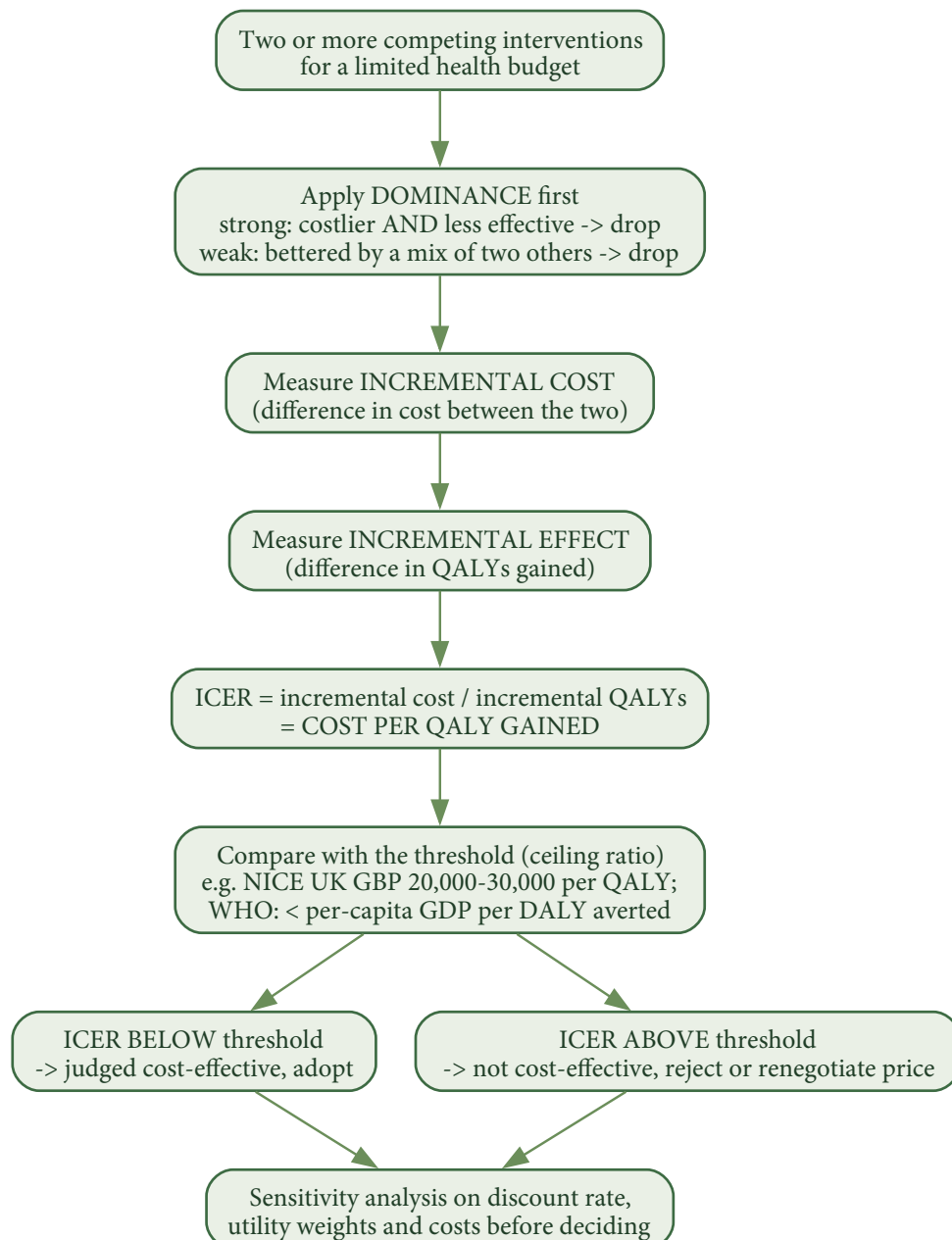
- **Diagram 1 — The QALY as the area between two prognosis profiles** (after Oxford Public Health 7e, Ch 49, Figure 6.6.3, pp 1832-1833). Draw a graph. Label the y-axis "**Health-related quality of life (utility): 1.0 = full health, 0 = death**" and the x-axis "**Time (years)**" with tick marks at **t, t+2, t+10 and t+12**.
 - **SOLID line = prognosis WITHOUT the intervention.** Start at 0.8 at time t, fall steeply and at a constant rate to 0.6 at (t+2), then fall more gently to 0.3 at (t+10), and drop vertically to 0 at (t+10) = death after a further 10 years.
 - **DASHED line = prognosis WITH the intervention.** A horizontal line at 1.0 (full health) from t to (t+12), then drop vertically to 0 = life expectancy raised to 12 years, each lived in full health.
 - **Shade the whole area between the two profiles and label it "QALYs gained = gain in quality of life + gain in length of life".**



Calculating gains in quality-adjusted life years (Oxford public health 7th ed, p.1833)



How the quality (utility) weight of a health state is obtained



Using QALYs in cost-utility analysis to reach a funding decision

High-yield exam pointers

- **Formula:** $\text{QALY} = \text{years of life} \times \text{utility weight of the health state}$. 1 QALY = 1 year in perfect health; 1 year at utility 0.5 = 0.5 QALY (Park 28e, Ch 2, p 38).
- **Anchors are opposite in the two measures** — QALY utility 1 = full health, 0 = death; DALY disability weight 0 = perfect health, 1 = death. $\text{DALY} = \text{YLL} + \text{YLD}$.
- **Four direct utility methods** — mnemonic "S-T-V-P": Standard gamble (utility = P), Time trade-off (utility = T/X), Visual analogue scale (thermometer; end-of-scale bias), Person trade-off ($1,000/1,700 = 0.59$ for blindness). Indirect = EQ-5D (mobility, self-care, usual activities, pain/discomfort, anxiety/depression) and SF-36.
- $\text{ICER} = (\text{Cost}_2 - \text{Cost}_1) \div (\text{QALY}_2 - \text{QALY}_1)$ = cost per QALY gained; ICER relates to a single intervention against baseline. Apply strong and weak dominance before ranking.

- **Thresholds to quote:** NICE UK £20,000–30,000/QALY (observed rejection range £35,000–48,000, Devlin and Parkin 2004); USA US\$50,000/QALY (1982 US\$); Canada CAN\$20,000–120,000 (1990 CAN\$); WHO/Commission on Macroeconomics and Health 2001 — highly cost-effective if it averts a DALY for less than per-capita GDP.
- **Discount future costs and health effects at 3 per cent per year** (after controlling for inflation), then run a sensitivity analysis.
- **Standard criticism to write:** QALYs mathematically **discriminate against the elderly** and are blind to *who* gains the health — answered by equity-weighted QALYs, the "*fair innings*" argument (Williams 1997) and NICE's end-of-life premium.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Coverage note, for reference only and not to be written in the exam: the retrieved corpus carries no Indian cost-per-QALY threshold, so no Indian willingness-to-pay figure has been quoted; the World Health Organization per-capita gross domestic product rule (Commission on Macroeconomics and Health, 2001) is the benchmark applicable to India in the retrieved text.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2, Concept of Health and Disease — Indicators of health; Ch 23, Health Planning and Management); Health Economics (Understanding Public Health series) — Ch 13 What is economic evaluation and what questions can it help to answer, Ch 14 Counting the costs, Ch 15 Identifying, measuring and valuing consequences, Ch 16 Economic evaluation and decision-making, Ch 17 Promoting equity and the role of government; Oxford Textbook of Public Health 7e — Ch 6.6, Economic appraisal in public healthcare: assessing efficiency and equity (Figure 6.6.3, Table 6.6.2); Health Program Planning and Evaluation (Ch 4, Characterizing and Defining the Health Problem); Reinke — Health Planning for Effective Management (Ch 20, Cost-Benefit and Cost-Effectiveness Analysis of Health Programs). Named benchmarks and sources cited within these texts: WHO Commission on Macroeconomics and Health, 2001; Devlin and Parkin, 2004 (NICE threshold); Hirth *et al.*, 2000 (USA); Laupacis *et al.*, 1992 (Canada); Williams, 1997 ("fair innings"); Dolan and Olsen, 2001 (equity-weighted QALYs); Anonychuk *et al.*, 2009 (Canadian human papillomavirus vaccination cost–utility analysis).



Quality assurance in health care

Asked in: Paper IV 28-Jun-2024 — RGUHS MD Community Medicine

Definition — Quality assurance is "the formal and systematic exercise of identifying problems in medical care delivery, designing activities to overcome these problems, and carrying out follow-up steps to ensure that no new problems have been introduced and that corrective actions have been effective" (Maxcy-Rosenau-Last 15e, Ch 78, p 1303). It works by fixing a **minimum acceptable requirement** for processes and a **standard** for outputs, measuring actual performance against that standard, and taking corrective action where the two diverge (Health Program Planning and Evaluation, Ch 10, p 277). Its measurement framework throughout is Donabedian's triad of structure, process and outcome (Donabedian 1966, *Evaluating the quality of medical care*).

Dimensions of quality that quality assurance protects

The whole field was pushed forward by the Institute of Medicine reports *To Err Is Human* (2000), which described the scale of the patient-safety problem, and *Crossing the Quality Chasm* (2001) (Harrison — Essentials of Health Care Organization, Ch 13, p 194); as many as 100,000 Americans a year were estimated to die of medical errors, most of them correctable (Maxcy-Rosenau-Last 15e, Ch 78, p 1303). *Crossing the Quality Chasm* named six dimensions in which health care must improve (Oxford Textbook of Public Health 7e, Ch 101, p 3864):

Dimension	What it means
Safe	Reducing harm to patients
Effective	Care based on scientific knowledge, avoiding underuse and overuse of services
Patient-centred	Care based on what matters to the patient — their preferences, needs and values
Timely	Reducing delays that compromise the outcome of care
Efficient	Avoiding waste, not only of material resources but of ideas and energy
Equitable	Consistent quality regardless of sex, ethnicity, geography or socio-economic status

Park approaches the same question through the **seven elements of evaluation** of a health service (Park 28e, Ch 23, pp 1025–1026):

Element	Question it answers	Example
Relevance (requisiteness)	Is the service needed at all?	Smallpox vaccination is now irrelevant — the disease no longer exists
Adequacy	Has sufficient attention been paid to the agreed course of action?	Staff allotted without reference to workload and targets
Accessibility	What proportion of the population can be expected to use the facility?	Barriers: physical (distance, travel time), economic (fee, travel cost), social/cultural (caste, language)

Element	Question it answers	Example
Acceptability	Is an accessible service actually acceptable?	Male sterilisation; screening for rectal cancer
Effectiveness	Is the underlying problem prevented or alleviated?	Ultimate measure — fall in morbidity and mortality rates
Efficiency	How well are money, men, material and time used for that effect?	Bed-occupancy rate, cost per patient treated, cost per bed-day
Impact	Overall effect on health status and socio-economic development	Malaria control improved agriculture, industry and social life, not only malaria incidence

What is measured — Donabedian's structure–process–outcome triad

Level	What it covers	Indicators used
Structure	Personnel characteristics and institutional features — the setting of care and its capacity to produce quality	Licensure and board certification, ownership, bed strength, teaching status, availability of technology, staff-to-patient ratio, patient volume, presence of a clinical information system (Handbook of Epidemiology, Ch IV.5, p 1513)
Process	Recognition of the problem, diagnostic procedures, treatment, clinical management, care and prevention — how the activity is actually carried out, compared against a predetermined standard	Antibiotic within 8 hours for community-acquired pneumonia, prophylaxis for venous thrombo-embolism, beta-blocker after myocardial infarction, average length of stay; assessed objectively as medical (or nursing) audit (Park 28e, Ch 23, pp 1025–1026; Handbook of Epidemiology, Ch IV.5, p 1514)
Outcome	End results — whether persons using the service experience measurable benefit such as improved survival or reduced disability	The "5 Ds" of ill-health: disease, discomfort, dissatisfaction, disability, death; also patient health status (SF-36), quality of life, patient satisfaction (Park 28e, Ch 23; Maxcy-Rosenau-Last 15e, Ch 78, p 1303)

The structure–outcome link is dependable only at the level of **minimum standards** (for example rules for operation-theatre hygiene); above that level structure predicts outcome poorly, which is why process measurement is preferred (Handbook of Epidemiology, Ch IV.5, p 1513). Outcome measures in turn suffer from long latency, low statistical power for rare events, and the need for **case-mix adjustment**, and they invite "upcoding" and "gaming" (Handbook of Epidemiology, Ch IV.5, pp 1514–1515).

Methods and mechanisms of quality assurance

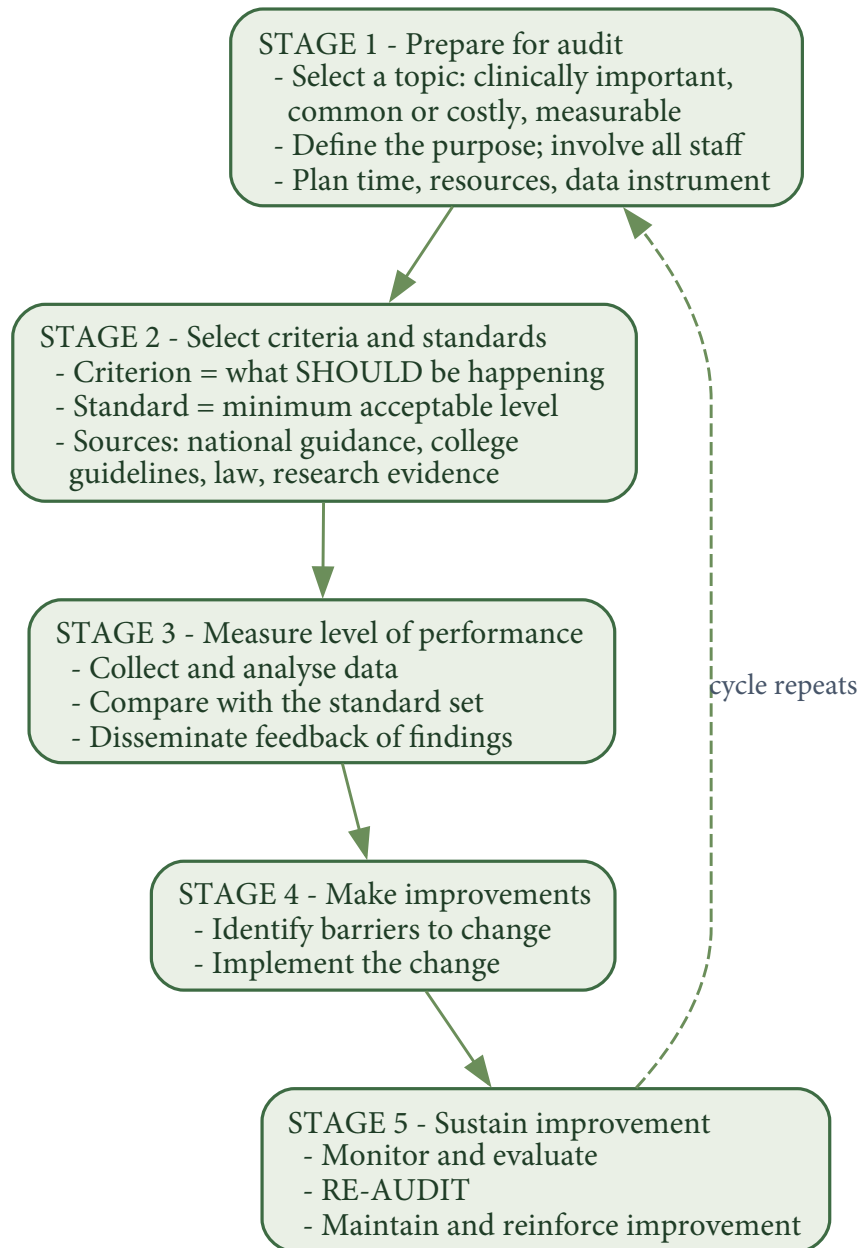
Mechanism	How it assures quality
Standards and practice guidelines	Statements of the diagnostic or therapeutic manoeuvres that should or should not be performed in defined clinical circumstances; developed from empirical evidence or, where evidence is thin, by consensus methods such as the <i>Delphi</i> technique (Maxcy-Rosenau-Last 15e, Ch 78, p 1304)

Mechanism	How it assures quality
Licensure and certification	Earliest quality-assessment procedures — professional and organisational structural measures applied to individuals, institutions and training programmes; necessary but not sufficient (Maxcy-Rosenau-Last 15e, Ch 78, p 1303)
Accreditation	"A self-assessment and external peer assessment process used by health care organizations to accurately assess their level of performance in relation to established standard and then to implement ways to continuously improve it" (Jugal Kishore, pp 121–122)
Medical / clinical audit	Objective, systematic evaluation of physician or nurse performance against a predetermined standard (Park 28e, Ch 23)
Peer review and utilisation review	Concurrent and post-hoc assessment by hospital committees; state boards, insurer audits, malpractice litigation (Maxcy-Rosenau-Last 15e, Ch 78, p 1304)
External quality assessment of laboratories	Tiered laboratory network with quality assurance at every tier — as in the National Tuberculosis Elimination Programme: peripheral microscopy centres → district/sub-district rapid molecular testing → culture and drug-susceptibility laboratories → Intermediate Reference Laboratory (state) → National Reference Laboratory (Park 28e, Ch 7, pp 508–511)
Patient satisfaction surveys and report cards	"Standardized, publicly released reports on the quality of care"; the Health Plan Employer Data and Information Set (HEDIS) of the National Committee for Quality Assurance is the widely cited example (Maxcy-Rosenau-Last 15e, Ch 78, p 1303)
Lot quality assurance sampling (LQAS)	An industrial quality-assurance technique adapted to public health — rapidly classifies areas as acceptable or unacceptable, e.g. identifying low-immunisation-coverage pockets (Sunder Lal, Ch 6, p 252)

Medical audit and the quality assurance cycle

Medical audit is "a planned programme which objectively monitors and evaluates the clinical performance of all practitioners, identifies opportunities for improvement, and provides mechanism through which action is taken to make and sustain those improvements" (Park Plus, Ch 21, p 305).

Landmarks (Park Plus, Ch 21, p 305): *Florence Nightingale* conducted the first clinical audit in the Crimean War (1853–1855), cutting mortality from 40% to 2%; *Ernest Codman's* "end result idea" (1912) followed every operated patient to identify surgical error; the **Mudaliar Committee (1961)** urged medical audit in India, Dr Sushila Nayyar introduced it in **1969**, and it became operational only in **2007**, after the National Accreditation Board for Hospitals and Healthcare Providers (NABH) was set up in **2005**.



The five-stage quality assurance (clinical audit) cycle — the loop is closed only by re-audit

Types of medical audit (Park Plus, Ch 21, p 308):

- ▶ Morbidity audit
- ▶ Audit of operated cases
- ▶ Audit of obstetric cases
- ▶ Audit of deaths in hospital (mortality review)
- ▶ On-spot medical audit

Quality assurance versus continuous quality improvement

	Quality assurance	Continuous quality improvement (CQI) / Total quality management (TQM)
Aim	Maintain a minimum acceptable standard	Raise the level of performance of key processes continuously
Focus	Identify errors and bring processes into compliance	Attention to the system as a whole
View of failure	An individual's failure to perform, correctable by training or discipline — "identifying bad apples" and cutting off one tail of the bell curve	Failures arise in the process or its suppliers, not in people; only a small minority are idiosyncratic
Timing	Largely retrospective review	Prospective, ongoing, iterative
Who does it	Inspectors, external reviewers, standing committees	Standing quality-improvement teams that include the employees doing the work — they are the best source of solutions
Net effect	Maintains a floor; does not by itself foster overall improvement	Reduces cost while improving quality
Source	Maxcy-Rosenau-Last 15e, Ch 78, pp 1304–1306	Health Program Planning and Evaluation, Ch 10, pp 277–278

CQI derives from *W. Edwards Deming*, *Joseph Juran* and *Philip Crosby* in the United States and *Genichi Taguchi* and *Kaoru Ishikawa* in Japan; *Donald Berwick* adapted it to health services. Its engine is the **Shewhart or Plan–Do–Study–Act (PDSA) cycle**, elaborated by the Hospital Corporation of America as **FOCUS–PDSA** (Maxcy-Rosenau-Last 15e, Ch 78, pp 1304–1305). **Six Sigma** reduces variation in clinical and business processes, and **Lean Six Sigma** adds a focus on minimising waste (Health Program Planning and Evaluation, Ch 10, p 280).

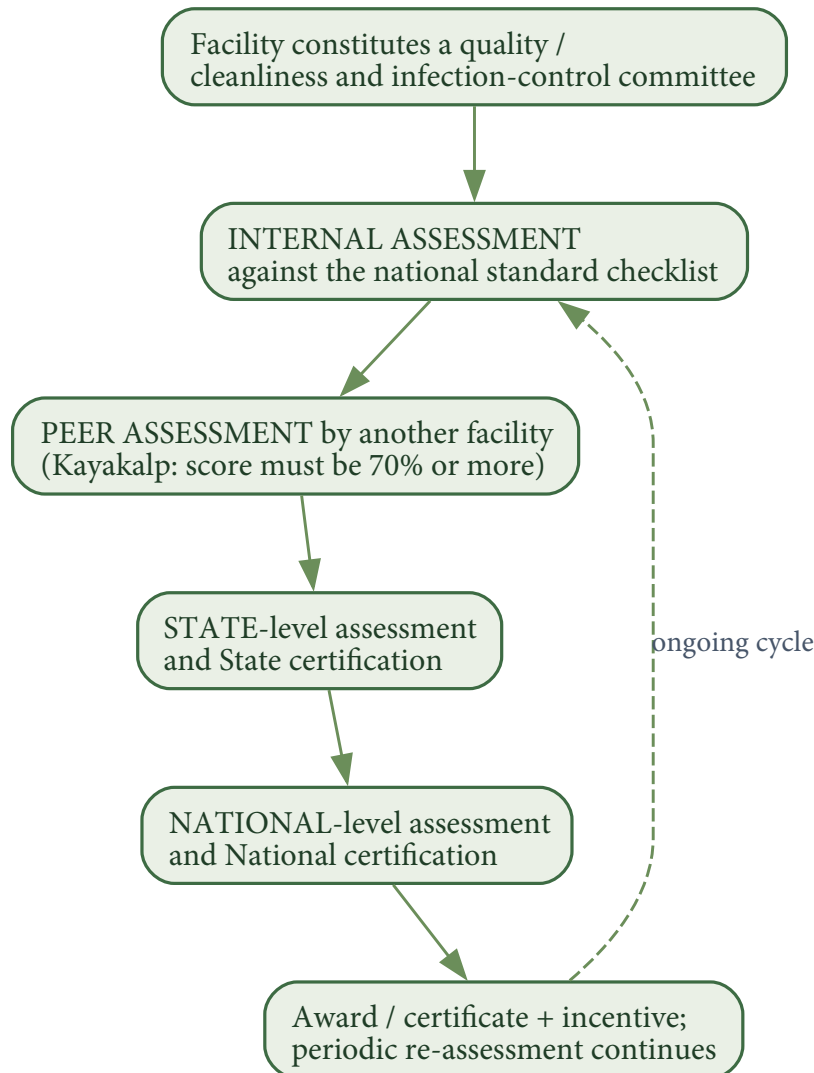
Seven basic quality-improvement tools — cause-and-effect (*Ishikawa* or fishbone) diagram, check sheet, control chart, histogram, Pareto chart, scatter diagram, flowchart (Health Program Planning and Evaluation, Ch 10, pp 277–280).

Limitations of a pure quality-assurance approach

- ▶ Being retrospective and dependent entirely on the record, audit can assess only limited aspects of the **technical** quality of care (Park Plus, Ch 21, p 308).
- ▶ Techniques are imperfect and not standardised, despite apparently clear-cut official methods.
- ▶ Auditors are few, and audits are often done to satisfy an accrediting agency rather than to improve the process.
- ▶ Quality assurance maintains a minimum standard but does not by itself produce system-wide improvement (Health Program Planning and Evaluation, Ch 10, p 277).
- ▶ There is limited evidence that standards of care by themselves change physician behaviour, unless there is consensus on clinical content, feedback framed against the physician's own performance, credible presenters such as opinion leaders, and reinforcement by academic detailing (Maxcy-Rosenau-Last 15e, Ch 78, p 1304).

Quality assurance in the Indian public health system

Initiative	What it assures
National Quality Assurance Standards (NQAS)	The national standard against which public health facilities are assessed and certified; the National Health Mission (NHM) tracks the "proportion of public health facilities with accreditation certificates by a standard quality assurance program (NQAS/NABH/ISO/AHPI)" as a formal programme indicator (Park 28e, Ch 2, p 46). Standards now extend down to the Health and Wellness Centre–Sub Centre, covering the 12 service packages (Sunder Lal, Ch 17, p 743). Certification status is displayed publicly — the model Citizen's Charter of a facility states that it is "IPHS & Kayakalp/NQAS/LaQshya certified" (Park 28e, Ch 10, pp 728–730)
Kayakalp (launched May 2015, with the Swachhta Guidelines for Public Health Facilities)	Awards for cleanliness, hygiene and infection control. Six themes — facility upkeep, sanitation and hygiene, waste management, infection control, support services, hygiene promotion. Prerequisites: a cleanliness and infection-control committee, a mechanism of periodic internal/peer assessment, and a score of at least 70% on peer assessment (Park Plus, Ch 7, pp 71–72)
LaQshya — Labour Room Quality Improvement Initiative (Ministry of Health and Family Welfare, 2017)	Quality of intrapartum and immediate postpartum care in labour rooms and maternity operation theatres of District Hospitals, Sub-District Hospitals, high-caseload Community Health Centres, First Referral Units and medical colleges; works through infrastructure upgradation, essential equipment, human resources, capacity building and adherence to clinical guidelines. 2,445 facilities including 193 medical colleges identified; 441 labour rooms and 392 operation theatres state-certified, and 447 labour rooms and 365 operation theatres nationally certified up to December 2021 (Park 28e, Ch 7, pp 533–534)
SUMAN — <i>Surakshit Matritva Aashwasan</i> (10 October 2019)	Assured, dignified, respectful and quality care at no cost with zero tolerance for denial of service to any woman and newborn attending a public facility (Park 28e, Ch 7, p 545)
Indian Public Health Standards (IPHS), revised 2022	Benchmarks of structure and assured services for Sub-Centres, Primary Health Centres, Community Health Centres (rural and urban), Sub-District and District Hospitals; the monitoring process and quality assurance mechanism are built into the standard (Jugal Kishore, p 145; Sunder Lal, Ch 17)
NABH under the Quality Council of India	Functionally autonomous board that sets and operates accreditation for hospitals and nursing homes. International counterparts: Joint Commission International and the Joint Commission on Accreditation of Health Care Organizations (USA), Canadian Council on Health Service Accreditation, Australian Council on Healthcare Standards, Council on Health Services Accreditation for South Africa, and the Malcolm Baldrige National Quality Award (USA) (Jugal Kishore, pp 121–122)



Certification pathway of an Indian public health facility under Kayakalp / LaQshya

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Donabedian's triad of quality assessment** — three stacked boxes with a downward arrow through them. Label exactly: box 1 **STRUCTURE** (personnel characteristics; institutional features — buildings, equipment, staffing, licensure); box 2 **PROCESS** (recognition, diagnostic procedures, treatment, clinical management, care and prevention — compared with a predetermined standard = medical/nursing audit); box 3 **OUTCOME** (end results — the 5 Ds: disease, discomfort, dissatisfaction, disability, death). Side-label the arrow "structure → outcome link holds firmly only at minimum-standard level".

High-yield exam pointers

- **Donabedian 1966** — structure, process, outcome. Outcome components = the 5 Ds: Disease, Discomfort, Dissatisfaction, Disability, Death.
- **Park's seven elements of evaluation**, in order: Relevance, Adequacy, Accessibility, Acceptability, Effectiveness, Efficiency, Impact — "R, three A's, two E's, I".

- ▶ Institute of Medicine six dimensions — mnemonic STEEEP: Safe, Timely, Effective, Efficient, Equitable, Patient-centred.
- ▶ Audit cycle in five stages: prepare → select criteria and standards → measure performance → make improvements → **re-audit**. Criterion = what should happen; standard = the minimum acceptable level.
- ▶ CQI engine = PDSA (Shewhart) cycle; founders Deming, Juran, Crosby, Taguchi, Ishikawa; Berwick adapted it to health care.
- ▶ Indian dates to quote: NABH established **2005**; medical audit operational in India **2007**; Kayakalp **May 2015** (six themes, ≥70% peer-assessment score); LaQshya **2017**; SUMAN **10 October 2019**; IPHS revised **2022**; LaQshya certification status **December 2021** (447 labour rooms, 365 operation theatres nationally certified).
- ▶ Quality assurance sets a **floor**; continuous quality improvement raises the **ceiling** — quality assurance blames the "bad apple", CQI blames the process.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Park section on evaluation of health services quoted here (structure/process/outcome, the 5 Ds, and the seven elements of evaluation) sits in Chapter 23 of the 28th edition, but the indexed 28e page scan is optically degraded, so the wording was cross-checked against the cleaner 27th-edition scan of the same section (27e, Ch 23, pp 1012–1013); the content is identical. Coverage note: the indexed corpus carries the National Quality Assurance Standards (NQAS) only as a named certification programme and an NHM indicator, not as a chapter with its eight areas of concern and its scoring rules, so those detailed NQAS internals are deliberately not stated here.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO Global Patient Safety Report 2024, WHO Global Patient Safety Action Plan 2021–2030, Global Ministerial Summit on Patient Safety 2025

Programme and policy updates (India)

- ▶ **Health and Wellness Centres have been renamed Ayushman Arogya Mandir (AAM)** — the same comprehensive-primary-care platform under a new name. More than 1.75 lakh are operationalised, 1,16,978 community health officers were in position and 82% of AAMs had rolled out all 12 expanded service packages by 30 November 2024; the stated next policy step is a quality step — saturate the remaining sub-centres and PHCs, then certify every AAM against IPHS norms under the National Quality Assurance Standards. [PMID 41816146]
- ▶ **The certification landscape has widened well beyond NQAS–Kayakalp–LaQshya.** MusQan (2021) certifies child-friendly services from birth to 12 years; SaQushal (2022) is a facility-level patient-safety self-assessment tool (4 areas of concern, 16 standards, 80 checkpoints); the SaQsham portal (2023, National Health Systems Resource Centre) runs the national certification workflow online; Mera Aspataal (29 August 2016) captures patient feedback; Swachh Swasth Sarvatra (December 2016) links Kayakalp to the Swachh Bharat Mission; and AB PM-JAY quality certification, run jointly by the National Health Authority and the Quality Council of India, awards Bronze, Silver and Gold, with Bronze open even to non-empanelled hospitals. [PMID 41280589]

- **Scale-up figures for the two flagship schemes:** Kayakalp enrolment rose from 100 facilities in 2015–16 to 19,488 in 2022–23, and NQAS-certified facilities from 10 in 2016–17 to 829 in 2022–23; the Kayakalp checklist for a PHC now carries a seventh theme, "beyond hospital boundary", giving 56 standards and 180 checkpoints. [PMID 41280589]

Current national data

- **NQAS certification remains the weak link.** Only 6.4% of India's rural primary health facilities (1,596 of 24,918) had been certified. Certification requires a score of at least 70% across the eight areas of concern — service provision, patient rights, inputs, support services, clinical services, infection control, quality management and outcomes — is valid for three years, and carries score-linked financial and non-financial incentives. Bhavnagar district in Gujarat is the positive-deviant case, having certified over 80% of its rural primary facilities — achieved by treating the checklist as the working document and aiming squarely at the 70% pass mark rather than at perfection. [PMID 39836671]
- **LaQshya has moved past the December-2021 figures:** by December 2022 about 47% of the 2,445 targeted facilities held national certification — 644 labour rooms and 504 maternity operation theatres — led by Madhya Pradesh, Maharashtra and Karnataka, while Meghalaya had not begun. Benchmarked against WHO's maternal and newborn quality standards, the LaQshya checklists capture 269 of 352 (76%) WHO quality measures, ahead of the Service Provision Assessment (62%) and the World Bank Service Delivery Indicator (12%). The WHO quality statements it covers least (60% or under) are prevention of unnecessary or harmful practices during labour and the postnatal period (46%), mistreatment of women and newborns (56%), support for women during childbirth (50%), competence of skilled birth attendants and support staff (39%) and managerial and clinical leadership (57%); offering a birth companion of the woman's choice, by contrast, is captured completely. [PMID 38575944]
- **At sub-centre level the Kayakalp headline score flatters the reality:** of 36 Health and Wellness Centres assessed in Ganjam, Odisha, 26 (72%) cleared the 70% mark, yet mean domain scores were 49% for biomedical waste management, 52% for infection control and 45% for support services; 72% had no designated cleaning staff and 61% no functional waste-disposal infrastructure. [PMID 40978827]

Guideline and legislative updates

- **NABH accreditation now covers more than 1,500 Indian hospitals,** and a record review of six Ahmedabad hospitals assessed in June–July 2024 (two seeking full accreditation, four entry-level) found documentation the commonest non-compliance — 27.7% of the 94 lapses at the full-accreditation hospitals (26 of 94), followed by process (25.5%) and policy (23.4%). The high-severity failures were absent fire plans, no method of patient identification in wards, undocumented verification of high-risk medication and no structured adverse-event monitoring — accreditation gaps cluster in records and systems rather than in clinical skill. [PMID 42260529]
- **Point-of-Care Quality Improvement (POCQI) — a four-step, team-based method disseminated in India through the Nationwide Quality of Care Network (NQOCN) — is being used for facility-level improvement.** Frontline staff define the problem, analyse causes (fishbone diagram, process-flow mapping), test change ideas through PDSA cycles, and fold what works

into routine practice. In the coronary care unit of KLE's Dr Prabhakar Kore Hospital, Belagavi, an NQOCN-mentored team raised compliance with WHO's five moments of hand hygiene from 63.8% to 80.5% over five PDSA cycles, holding about 80% for ten more weeks after mentoring ended, with no MRSA, *Pseudomonas* or *Klebsiella* bloodstream infections during the project. Note the modest claim the authors themselves make: structured QI capacity-building is offered as something that *could* be built into national infection-prevention programmes, not as established national practice. [PMID 41856546]

Global and WHO developments

- ▶ **WHO released its first Global Patient Safety Report on 30 May 2024**, benchmarking countries against the Global Patient Safety Action Plan 2021–2030. The standing global estimates it works from: about 1 in 10 patients is harmed in health care, over 3 million deaths a year follow unsafe care, more than half of that harm is preventable, and in primary and ambulatory care as many as 4 in 10 patients are harmed with up to 80% of it avoidable. [WHO 2024-May; WHO 2023-Sep]
- ▶ **The 7th Global Ministerial Summit on Patient Safety (Manila, 3–4 April 2025)** issued the Mandaluyong Declaration, with four pledges — stronger global collaboration, leadership and governance, patient safety embedded in disaster preparedness and climate resilience, and people-centred safety systems — reaffirming the 2021–2030 action plan; World Patient Safety Day is observed on 17 September. [PMID 40796350]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 7 — Health Programmes in India; Ch 10 — Health Care of the Community; Ch 23 — Health Planning and Management), cross-checked against Park 27e Ch 23; Park Plus — Amrutha & Vijayalakshmi (Ch 7 — Health Programmes in India; Ch 21 — Health Planning and Management); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 78 — Public Health Management Tools); Health Program Planning and Evaluation: A Practical, Systematic Approach (Ch 10 — Program Quality and Fidelity); Handbook of Epidemiology, Ahrens & Pigeot (Ch IV.5 — Health Services Research); Oxford Textbook of Public Health 7e (Ch 101 — Strategies for Health Services); Jugal Kishore — National Health Programmes of India (Accreditation of the Health System; Indian Public Health Standards); Sunder Lal — Textbook of Community Medicine (Ch 6 — Basic Statistics and Research Methodology; Ch 17 — Healthcare of the Community); Harrison — Essentials of Health Care Organization (Ch 13). Named documents: Institute of Medicine, *To Err Is Human* (2000) and *Crossing the Quality Chasm* (2001); Swachhta Guidelines for Public Health Facilities (2015); Indian Public Health Standards, revised 2022.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 41280589 — To gain insights and compare different quality certification programs for public health facilities in India, *J Family Med Prim Care* 2025, doi:[10.4103/jfmpc.jfmpc.1591.24](https://doi.org/10.4103/jfmpc.jfmpc.1591.24); 39836671 — What enabled the successful implementation of a quality certification initiative in Bhavnagar, Gujarat? A policy analysis case study, *PLOS Glob Public Health* 2025, doi:[10.1371/journal.pgph.0004180](https://doi.org/10.1371/journal.pgph.0004180); 38575944 — Appraising LaQshya's potential in measuring quality of care for mothers and newborns: a comprehensive review of India's Labor Room Quality Improvement Initiative, *BMC Pregnancy Childbirth* 2024, doi:[10.1186/s12884-024-06450-x](https://doi.org/10.1186/s12884-024-06450-x); 40978827 — Integrating facility metrics with grassroots insights: a mixed-methods study to improve Kayakalp outcomes at Health and Wellness Centres in Ganjam, Odisha, *Cureus* 2025, doi:[10.7759/cureus.90532](https://doi.org/10.7759/cureus.90532); 42260529 — A comprehensive mapping of NABH non-compliances in hospitals undergoing entry-level and full accreditation, *BMC Health Serv Res* 2026, doi:[10.1186/s12913-026-14809-3](https://doi.org/10.1186/s12913-026-14809-3); 41856546 — Enhancing hand hygiene compliance to reduce healthcare-associated infections in a coronary care unit, *BMJ Open Qual* 2026, doi:[10.1136/bmjopen-2025-003838](https://doi.org/10.1136/bmjopen-2025-003838); 40796350 — Why patient safety is akin to weaving textiles: perspective on the 7th Global Ministerial Summit on Patient Safety 2025, *Int J Qual Health Care* 2025, doi:[10.1093/intqhc/mzaf072](https://doi.org/10.1093/intqhc/mzaf072). Plus WHO Global Patient Safety Report, 30 May 2024; WHO patient safety fact sheet, 11 September 2023.



Scope of operational research in health services

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — Operational research is the **application of scientific methods of investigation to the study of complex human organizations and services** (Park 28e, Ch 12, p 839). It is also defined as "a scientific method for providing the executive department a quantitative basis for decision-making regarding the operation under control", or "the application of scientific methods, techniques and tools to operations of systems, with an optimum solution to the problem" (Sunder Lal, Ch 6, p 265). The term was coined during the Second World War in connection with the best use of a new invention, the radar; since then it has come to mean the study of whole systems of services rendered in industry, administration, education and health services (Park 28e, Ch 12, p 839).

What qualifies as operational research — the boundary of its scope

Feature	Pure research	Applied research	Operational research
Park's own illustration	A mathematician working on atomic structure	An engineer designing a new plant for an industry	A team studying the activities of a group of people with the object of inducing beneficial changes (Park 28e, Ch 12, p 839)
Unit studied	A phenomenon	A product or process	A system or a part of a system of services
Distinguishing content	—	—	An immense sociological content, which is what separates it from pure or applied research
Purpose	New knowledge	Useful application	To develop new knowledge about institutions, programmes, facilities, the individuals working in these activities and the communities served by them , so as to ensure optimal utilisation of resources — " men, material and money "
Output wanted by the user	A publication	A prototype	A prompt, "good enough", action-oriented solution the manager can implement (Reinke, Ch 3, p 46)

Park's governing statement fixes the scope: *practically everything within the broad field of public health and social medicine — either a whole system or a part of it — could be the subject of operational research*, and whenever social medicine passes from the stage of observation and classification to that of discovering and recommending appropriate action, it is engaged in operational research (Park 28e, Ch 12, p 839).

Scope — domains of application in health services

Domain	Operational research questions actually studied
Health manpower and deployment	Optimal size of area and population to be covered by a midwife or a basic health unit; leaving maximum time for technical personnel to utilise their skills; reallocation of worker roles as services extend to the periphery; innovative methods of training; alternative supervisory schemes (Park 28e, Ch 12, p 839; Reinke, Ch 3, pp 46–47)
Physical resources and logistics	Ideal vehicle for local health workers; architectural design of hospitals and health centres; criteria for siting facilities to promote access; replenishment cycles for drugs and supplies; vehicle maintenance policies and schedules
Hospital and outpatient operations	Queuing problems in out-patient departments and hospital waiting lists; study of bed load and nursing services in teaching and non-teaching hospitals; length of stay in hospital and length of sickness absence
Organisation, integration and information	Solutions to integration problems where specialised services have developed on an emergency basis outside the general health services; how to make decentralisation work; getting minimal information in the right form to the right place at the right time (management information systems)
Programme effectiveness	Effectiveness, i.e. the extent to which the stated objective of a programme is achieved — e.g. malaria eradication, family planning
Quality of care	Quality of medical care services
Epidemic control	Investigation of outbreaks of epidemic diseases
Community participation	Role of village health committees; selection and training of community workers; cost recovery through community financing schemes (Reinke, Ch 3, pp 46–47)
Financing	How services can be paid for without inhibiting utilisation by the needy — prepayment and insurance schemes
Causality and choice of intervention	Why certain providers are utilised or not utilised for certain conditions; whether an intervention (e.g. nutrition surveillance, village health workers) actually alleviates the problem identified

Techniques — what the discipline is able to tackle

Technique	Question it answers	Health service illustration
Linear programming	Which mix of services maximises benefit under fixed resource constraints?	Optimal service mix when a required skill is in short supply (Reinke, Ch 3, p 50)
Distribution models	Which supply depot should serve which outlet?	Drug supply systems; deciding which clinic refers to which referral point when the referral centre cannot take all patients (Reinke, Ch 3, p 52)
Queuing (waiting-line) theory	How to balance patient waiting time against staff idle time?	Patient flow through a health centre; value of an appointment system; scheduling elective hospital admissions

Technique	Question it answers	Health service illustration
Network analysis — PERT and CPM	In what sequence and by when must activities be completed?	An arrow diagram of a project; the longest path is the critical path , and delay of any activity on it delays the whole project (Park 28e, Ch 23, p 1029)
Systems analysis	Which of the available alternatives is most cost-effective?	A hospital supply system, an information system, an out-patient clinic, or a whole community health service system (Park 28e, Ch 23, p 1029)
Input–output analysis	How much of each input is needed to produce one unit of output?	Relating manpower, money and materials consumed to outputs such as lives saved or inoculations performed
Cost–benefit and cost-effectiveness analysis	Is the programme economically sound; which alternative gives most result per rupee?	Benefit expressed in money (cost–benefit) or in results achieved — lives saved, disease-free days (cost-effectiveness) (Park 28e, Ch 23, p 1028)
Work sampling / work study	How is staff time actually spent, and how many staff are needed?	Systematic observation of doctors, nurses, pharmacists and laboratory technicians at predetermined or random intervals; used to fix job descriptions and manpower needs (Park 28e, Ch 23, p 1029)
Computer simulation	What would happen under conditions not yet tried?	Simulating several years of "experience" from a sample of real data to infer optimal arrangements (Sunder Lal, Ch 6, p 265; Reinke, Ch 3, pp 48–49)
Planning–programming–budgeting system (PPBS)	How should resources be allocated across objectives?	Grouping activities into programmes related to each objective; the zero-based budget variant (Park 28e, Ch 23, p 1029)

Study approaches available within this scope

- ▶ **Natural experiments** — comparative analysis of variation that already exists (e.g. health centres with two, one or no nurse-midwives); quick and cheap, but the reasons for the variation may be uncontrolled (Reinke, Ch 3, p 48).
- ▶ **Synthetic analyses and simulations** — comparing a range of existing conditions to predict outcomes under more favourable, untried arrangements.
- ▶ **Systematic programme monitoring** — longitudinal follow-up of providers and clients, with measurement of the community factors that could confound the findings.
- ▶ **Focused testing of a single intervention** — the design closest to classical research, e.g. fixed bimonthly supervision versus supervision targeted by recent worker performance.
- ▶ **Testing of comprehensive programmes** — needed when piecemeal change is inadequate, e.g. selective versus fully integrated primary health care.
- ▶ **Mathematical modelling** — the operations research techniques tabulated above.

Indian applications

- ▶ **Tuberculosis** — the DOTS strategy is a product of Indian operational research. The National Tuberculosis Institute, Bengaluru showed that case detection should be by **sputum microscopy and not by X-ray**; the Tuberculosis Research Centre, Chennai showed the safety of **home treatment**, the effectiveness of **intermittent chemotherapy** (two or three times a week) and the feasibility of **direct observation** in the community. These three cardinal principles were adopted worldwide and, in India, through the Revised National Tuberculosis Control Programme, with a reported cure rate of 86–90% (Sunder Lal, Ch 6, p 266).
- ▶ **Maternal and child health** — integrating the health and Integrated Child Development Services (ICDS) systems to raise the coverage and quality of routine immunisation and antenatal, postnatal and neonatal care; **joint home visits by the anganwadi worker (AWW) and the auxiliary nurse midwife (ANM)**; delivery rooms ("delivery huts") at sub-centres and primary health centres to promote institutional delivery (Sunder Lal, Ch 6, p 266).
- ▶ **Newborn care** — operations research by the Indian Council of Medical Research (ICMR) for the provision of **home-based neonatal care through community-level providers** (Park 28e, Ch 9).
- ▶ **Diarrhoeal disease** — operational research showed that about **50% of community diarrhoea cases can be managed by mothers with home-available fluids**, most of the rest by health guides with oral rehydration salts (ORS), and only a few need referral; this underpins advance distribution of ORS packets during the Intensified Diarrhoea Control Fortnight (Sunder Lal, Ch 6, p 266).
- ▶ **Service delivery and governance** — public–private partnership in health care delivery; devolution of powers to village *panchayats* to manage sub-centres; alternative village-level delivery of ORS, vaccines, malaria diagnosis and treatment, and contraceptives; people's participation in primary health care.
- ▶ **Written into national programmes** — operational research is a named component of the National Vector Borne Disease Control Programme (studies on drug resistance and insecticide susceptibility), of joint TB–HIV collaborative activities, and of the mandate of medical colleges and tertiary cancer centres under the national non-communicable disease programme (Park 28e, Ch 7; Sunder Lal, Ch 8).

Composition of the operational research team

Member	Contribution
Public health administrator	Defines the operational problem and owns the decision
Epidemiologist	Study design, case definitions, measurement of outcome
Statistician	Sampling, analysis, modelling
Social scientist	Behaviour of providers and clients; acceptability
Ancillary staff (clerks, field workers)	Data collection and processing
Director	Heads the team and is responsible for the whole project

The minimum composition above is the one Park specifies for social-medicine applications; the actual composition varies with the type of research (Park 28e, Ch 12, p 839).

Limits of the scope

- ▶ Health status is **difficult to quantify** and even harder to relate precisely to specified levels of service input, so the mathematical techniques of operations research have **limited applicability** in health matters (Reinke, Ch 3, pp 49–50).
- ▶ Benefits in the health field cannot always be expressed in monetary terms, which makes the scope of **cost–benefit analysis rather vague**; cost-effectiveness analysis is the more promising tool (Park 28e, Ch 23, p 1028).
- ▶ Because the work is done in the community, **multiple variables cannot be controlled**; the research is necessarily multidisciplinary and there is **no fixed methodology** (Reinke, Ch 3, p 46).
- ▶ Researchers seek optimum conditions while administrators need prompt answers, so **timeliness must often be bought at the cost of sophistication**; findings are useless if they arrive after the decision.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **DRAWN figure** — network (arrow) diagram for network analysis. Label: numbered event circles 0 to 9; arrows as activities with duration in weeks written on each; "Start plan" at event 1; "Complete plan and start service" at the terminal event; mark the **critical path** as the longest path through the network; mark **slack time** on the non-critical branches.

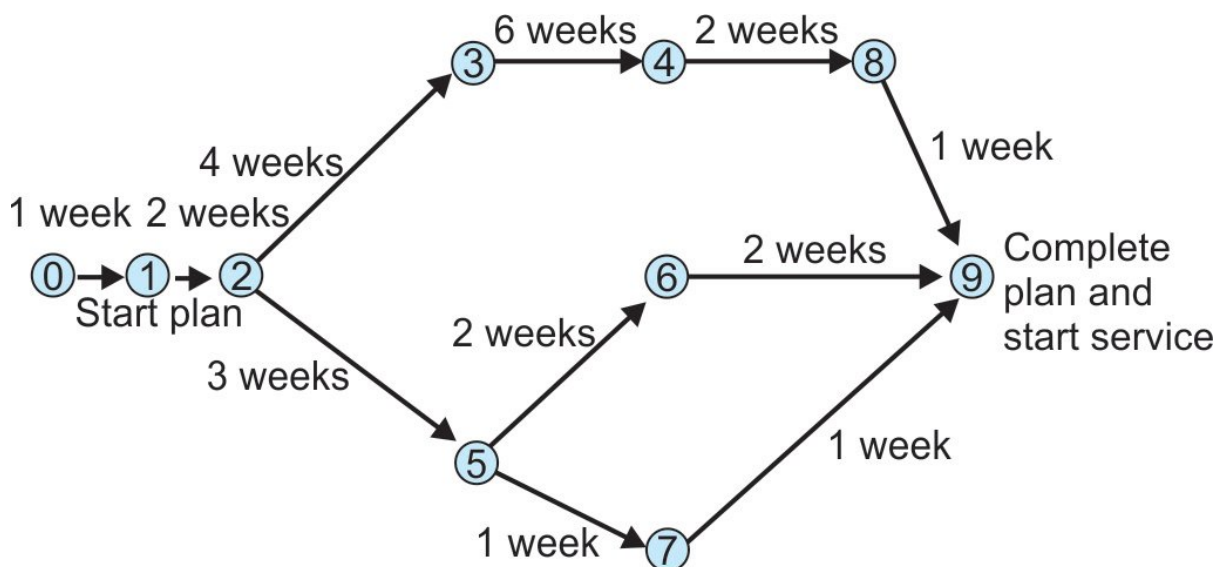
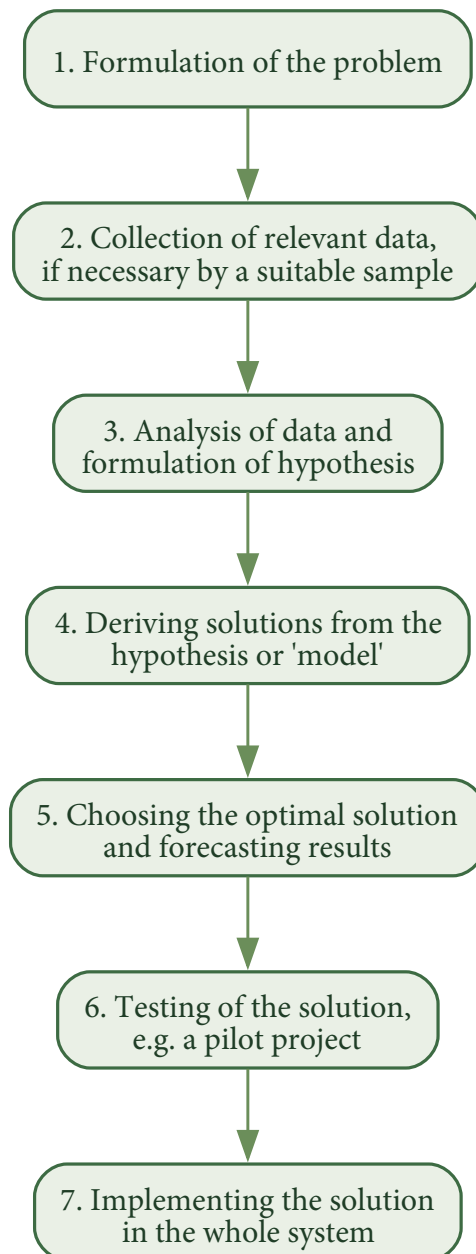
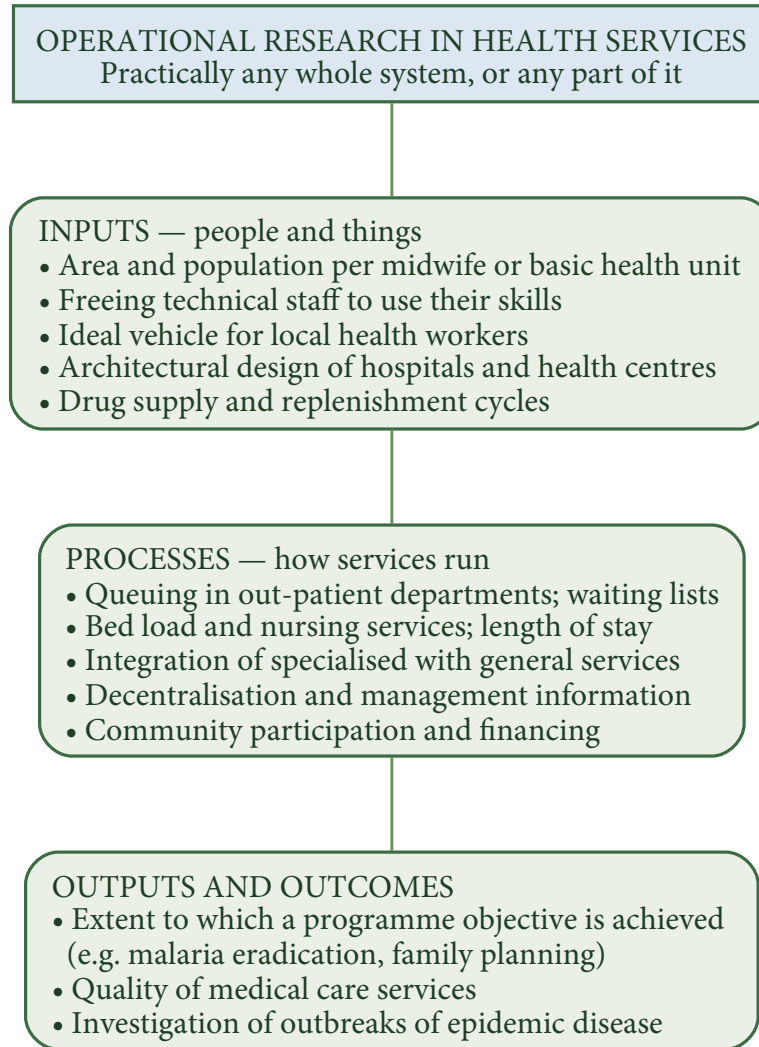


Fig 37.3 — Flow diagram of critical path (Suryakantha Environment Best, p.806)



The seven phases of an operational research study (Park 28e, Ch 12, p 839)



Scope of operational research in health services — a whole system, or any part of it

High-yield exam pointers

- ▶ **Definition to write verbatim:** "the application of scientific methods of investigation to the study of complex human organizations and services" — term coined in **World War II** over the best use of the radar.
- ▶ **Objective mnemonic — the three M's:** optimal utilisation of **men, material and money**.
- ▶ **Scope in one line:** practically everything in public health and social medicine, *either a whole system or a part of it*.
- ▶ **Seven phases:** Formulate → Collect → Analyse and hypothesise → Derive solutions → Choose optimum and forecast → Test by pilot → Implement system-wide.
- ▶ **Minimum team:** public health administrator + epidemiologist + statistician + social scientist, headed by a director.
- ▶ **Techniques:** linear programming, distribution and queuing models, network analysis (PERT/CPM — **critical path = longest path**), systems analysis, input–output analysis, cost–benefit and cost-effectiveness analysis, work sampling, simulation, PPBS.

- **Indian showpiece:** the three cardinal principles of DOTS — sputum microscopy, intermittent chemotherapy, direct observation — came from operational research at the National Tuberculosis Institute Bengaluru and the Tuberculosis Research Centre Chennai.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Revised National Tuberculosis Control Programme named in the operational research literature is now the **National Tuberculosis Elimination Programme (NTEP)**, and "operational research" is used interchangeably in the retrieved texts with "operations research" and "health systems research", the last term being preferred by some authors because it covers the whole system and not only the delivery of services.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 12 Medicine and Social Sciences, p 839; Ch 23 Health Planning and Management, pp 1028–1029; Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics); Reinke — Health Planning for Effective Management (Ch 3, Health Systems Research in Relation to Planning, pp 44–54); Sunder Lal — Textbook of Community Medicine (Ch 6, Basic Statistics and its Applications and Research Methodology, pp 265–266; Ch 8); Suryakantha — Community Medicine with Recent Advances (Fig 37.3, p 806).



System analysis

Asked in: Paper IV 28-Jun-2024 — RGUHS MD Community Medicine

Definition — Systems analysis is a quantitative management technique whose purpose is to help the decision maker choose an appropriate course of action by investigating the problem, searching out objectives, finding out alternative solutions, evaluating the alternatives in terms of cost-effectiveness, re-examining the objectives if necessary, and selecting the most cost-effective alternative (Park 28e, Ch 23, p 1029). It is essentially the finding of the **cost-effectiveness of the available alternatives** for a defined system. A *system* is an arrangement and set of relationships among multiple parts functioning as a whole, and a health system may itself be made of several subsystems (Sathe & Sathe, Ch 17, p 178; Park 28e, Ch 23, p 1029).

The systems approach versus the component approach

The technique rests on a prior conceptual shift: the performance of the whole depends on the harmonious, balanced and integrated functioning of its parts, so the output of the whole is more than the sum of its components (Sathe & Sathe, Ch 17, pp 177–178).

Feature	Component (reductionist) approach	Systems approach
Unit studied	Each component separately (registration counter, laboratory, ward)	The whole system plus the interactions between components
Assumption	Improving each part automatically improves the whole	Interaction, interdependence and interrelation determine the outcome
Result sought	Better performance of a part	Coordinated, synergistic performance of the total system
Weakness	A well-run part inside a badly-linked chain still fails	Needs more data, more time and specialist skills

Components of a health system for analysis

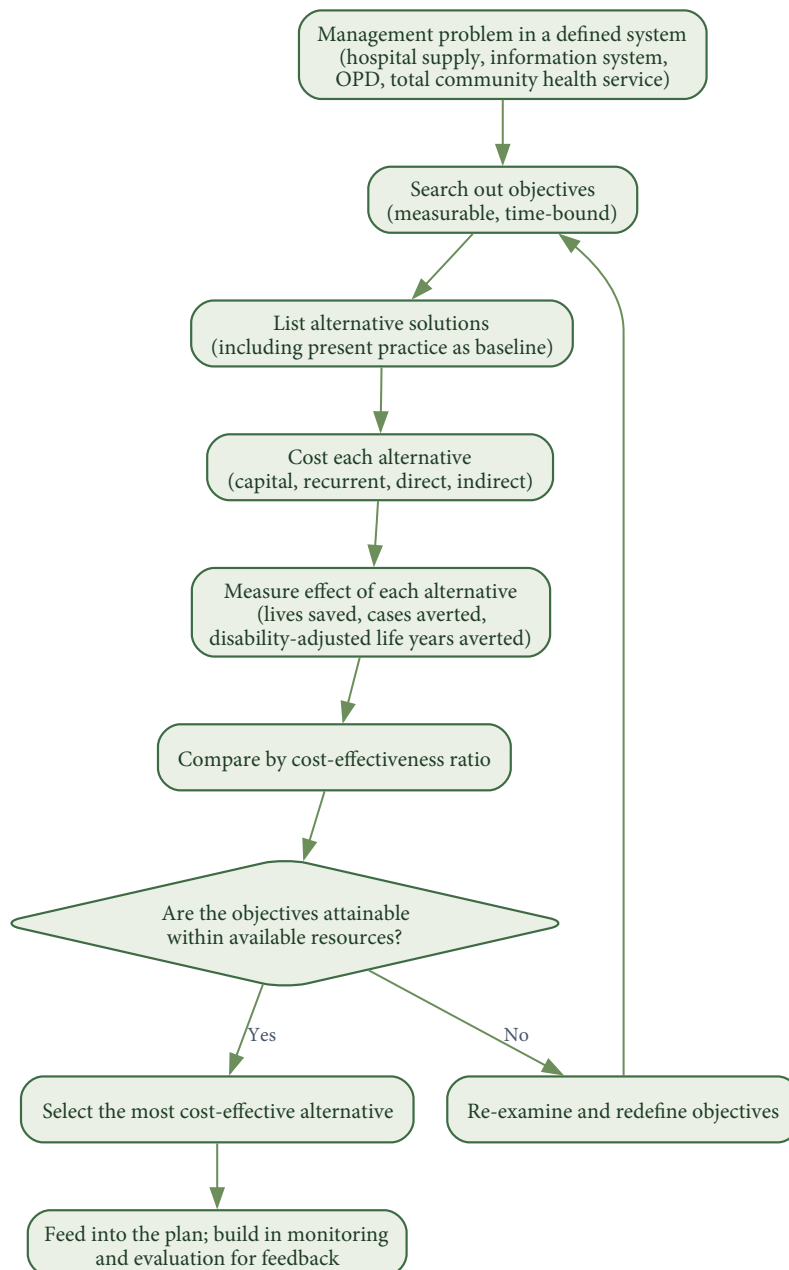
Component	Meaning	Example — district health services
Input	Resources committed	Money, manpower, facilities, drugs, vehicles (Jugal Kishore, National Health Programmes of India, pp 716-717)
Process	Programme organisation and activity	Outreach sessions held, supervisory visits, referral linkages
Output	Units of service produced	Immunisations given, antenatal check-ups, operations done
Outcome	Units of morbidity or mortality prevented	Measles cases averted, maternal deaths averted
Benefit / impact	Ultimate value to the individual or society	Improved health status and socio-economic development
Feedback	Information returned to the manager	Health management information system reports for monitoring and evaluation (Park 28e, Ch 23, p 1028)
Environment	Other sectors acting on health	Agriculture and nutrition, sanitation, income, education, cultural practices (Sathe & Sathe, Ch 17, p 178)

Efficiency is judged from input against output, effectiveness from input against outcome, and cost-benefit analysis from input against ultimate benefit (Jugal Kishore, pp 716-717). Reinke frames the same system as health problems in a defined population drawing on resources to produce services (output) that alter health status (outcome), and warns that **output must not be read as outcome** — output is easier to measure and is often used only as its proxy (Reinke, Ch 4, p 59; Ch 5, pp 70-71).

Steps in systems analysis

- **Investigate the problem** — define the system and its subsystems, and state the management problem precisely (a hospital supply system, a health information system, a total community health service system, an outpatient clinic).
- **Search out the objectives** — state what the system is meant to achieve, in measurable terms.

- **Find out alternative solutions** — list every practicable course of action, including continuing the present practice as the baseline comparator.
- **Evaluate the alternatives in terms of cost-effectiveness** — cost each alternative and express its benefit as results achieved (lives saved, days free of disease) rather than in money (Park 28e, Ch 23, pp 1028-1029).
- **Re-examine the objectives if necessary** — revise objectives that the analysis shows to be unattainable, and repeat the comparison.
- **Choose the most cost-effective alternative** and feed it into the plan (Park 28e, Ch 23, p 1029; Mahajan & Gupta 4e, Ch 26, pp 509-510).



Steps in systems analysis — from the management problem to the most cost-effective alternative

Tools used to carry out the analysis

Systems analysis is the framework; operations research supplies the tools that fill the costing and effect steps (Sathe & Sathe, Ch 17, pp 173, 177-184).

Tool	What it contributes	Health example
Cost-effectiveness analysis	Benefit expressed as results achieved, not money — the core comparison	Directly observed treatment for tuberculosis versus hospital treatment (Suryakantha, Ch 37, p 807)
Cost-benefit analysis	Benefit expressed in money; tests whether a programme is economically sound	Limited in health because benefits often cannot be monetised (Park 28e, Ch 23, p 1028)
Cost-accounting	Cost structure of a programme — cost control, resource allocation, pricing	Costing an outpatient visit or a hospital bed-day
Model / simulation	Abstraction of reality showing how factors affect one another	Typhoid fever model comparing mass vaccination against sanitation improvement (Sathe & Sathe, Ch 17, p 177)
Linear programming	Optimum mix of activities under fixed constraints	Choice of strategy for tuberculosis control (Sathe & Sathe, Ch 17, p 178)
Input-output analysis	How much of each input is needed per unit of output	Manpower and vaccine needed per 1,000 immunisations (Park 28e, Ch 23, p 1029)
Queuing theory	Balances cost of extra service points against cost of waiting	Registration clerk to doctor to dispensary flow in an outpatient department (Sathe & Sathe, Ch 17, p 184)
Network analysis (PERT, CPM)	Sequences activities and identifies the critical path	Commissioning a new district hospital block (Park 28e, Ch 23, p 1029)
Planning-Programming-Budgeting System	Groups activities into programmes tied to each objective for resource allocation	Programme-wise state health budget (Park 28e, Ch 23, p 1029)

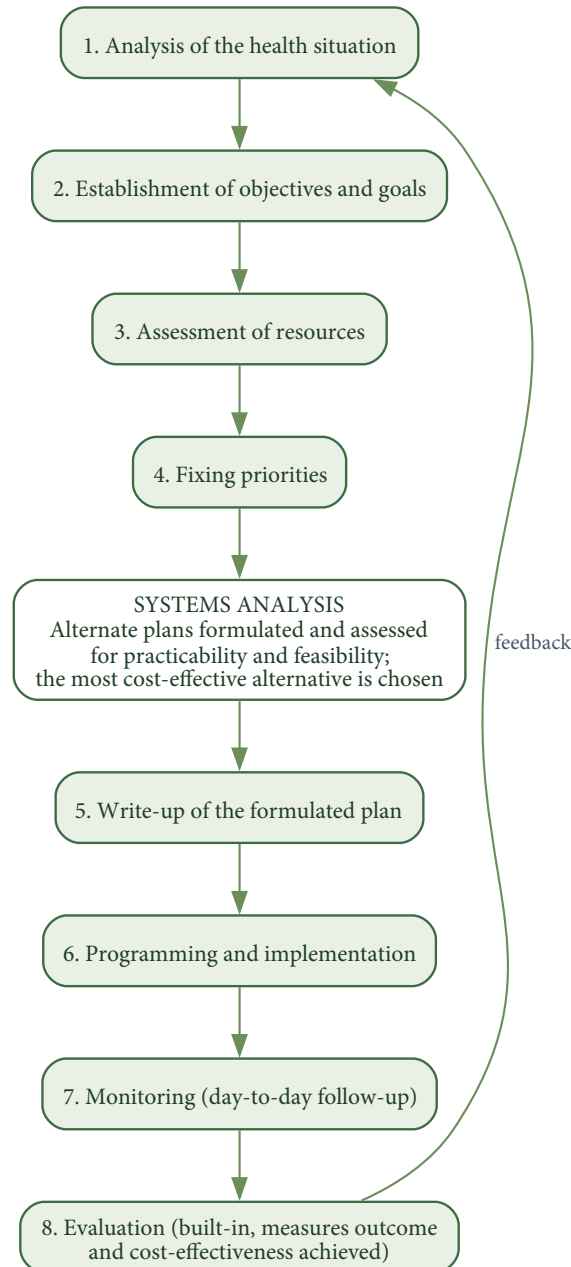
Expressing the result — the cost-effectiveness ratios

Ratio	What it compares	Use
Average cost-effectiveness ratio (ACER)	Total cost of one intervention ÷ total effects of that intervention, against a baseline of no programme or current practice	Screens whether an existing service is an efficient use of resources
Marginal cost-effectiveness ratio (MCER)	Change in cost ÷ change in effect when a programme is expanded or contracted	Should the existing programme be scaled up?
Incremental cost-effectiveness ratio (ICER)	Difference in cost between two alternatives ÷ difference in effectiveness between them	The decision rule when choosing between two competing alternatives

Source for all three: Health Economics, Ch 16, pp 247-250. Effect is measured as an additional unit of health — a quality-adjusted life year (QALY) gained or a disability-adjusted life year (DALY) averted; a

QALY weights each life year gained by a utility score from 0 (death) to 1 (perfect health) (Sunder Lal, Ch 2, p 89; Health Economics, Ch 16, p 253). A ratio is compared against a **threshold** or ceiling ratio set by the decision maker, and is reported as a range generated by sensitivity analysis rather than as a single figure (Health Economics, Ch 16, p 254).

Where systems analysis sits in the health planning cycle



Systems analysis within Park's health planning cycle — it is the tool that chooses between alternate plans

Applications, advantages and limitations

Application in the health sector	Source
Hospital supply system; health information system; outpatient clinic; total community health service system	Park 28e, Ch 23, p 1029

Application in the health sector	Source
Health systems research — policy and systems analysis, primary health care, manpower and training needs assessment	Sunder Lal, Ch 6, p 235
Health sector reform and area development projects; linking the policy-programme-project chain	Sathe & Sathe, Ch 17, p 178

Advantages	Limitations
Makes the choice between competing alternatives explicit and defensible instead of intuitive	Benefits in health often cannot be expressed in money, and even effectiveness cannot always be measured (Park 28e, Ch 23, p 1028)
Forces objectives to be stated in measurable terms before resources are committed	Identification and valuation of benefit can be subjective, and indirect costs and benefits are easily missed (Suryakantha, Ch 37, p 807)
Directs limited resources to the alternative giving most health per rupee	Models rest on assumptions of a stable community; conditions or a better technology change and the analysis must be redone (Sathe & Sathe, Ch 17, p 177)
Builds in feedback, so priorities and resources can be reallocated as needs change	Ratios computed with different comparators, discount rates and time horizons are not comparable across studies (Health Economics, Ch 16, pp 254-255)
Applicable at any level — a single clinic, a district, or a national programme	Requires a specialist, good baseline data and time; data are often the weakest link in Indian district settings

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The system evaluation model** (after Jugal Kishore, Fig 39.1) — draw a horizontal chain of five boxes and label exactly: **Input** (money, manpower, facilities), **Process** (programme organisation, activity), **Output** (units of service provided), **Outcome** (units of morbidity or mortality prevented), **Benefit** (ultimate value to individual or society). Below the chain draw three brackets and label them: *Efficiency* (input to output), *Effectiveness* (input to outcome), *Cost-benefit analysis* (input to benefit). Draw a feedback arrow from Output back to Input and label it *process evaluation (audit)*.
- **The cost-effectiveness plane** — draw two axes crossing at the origin: horizontal axis = incremental effect (positive to the right), vertical axis = incremental cost (positive upwards). Label the four quadrants; mark the south-east quadrant *dominant — always cost-effective* and the north-west quadrant *dominated — reject* (Health Economics, Ch 16, p 253).

High-yield exam pointers

- Park's one-line definition: *systems analysis is essentially finding the cost-effectiveness of the available alternatives* — write this first.
- Six steps in order: investigate problem → search out objectives → find alternatives → evaluate by cost-effectiveness → re-examine objectives → choose the most cost-effective alternative.

- ▶ System chain: **Input** → **Process** → **Output** → **Outcome** → **Benefit**; efficiency = output per unit input, effectiveness = outcome per unit input.
- ▶ ICER = (Cost A – Cost B) ÷ (Effect A – Effect B); ACER = total cost ÷ total effect of a single intervention against baseline; MCER applies to expansion or contraction of a running programme.
- ▶ Park's four named example systems: hospital supply system, information system, total community health service system, outpatient clinic.
- ▶ Placement: systems analysis is quantitative management method **number 6** in Park Ch 23, and operates at the "alternate plans" stage of the planning cycle, after fixing priorities and before the plan write-up.
- ▶ Output is not outcome — a high immunisation coverage (output) does not by itself prove reduced measles mortality (outcome).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park lists systems analysis as the sixth of the quantitative management methods in the Health Planning and Management chapter, and the underlying World Health Organization documents are Public Health Paper No. 55, 1974 (*Modern management methods and the organization of health services*) and Technical Report Series No. 596, 1976 (*Application of system analysis to health management*), both cited by Sathe & Sathe. Coverage note: the retrieved corpus gives no formal textbook list of "limitations of systems analysis", so the limitations column above is assembled from the stated limitations of its constituent tools — cost-benefit analysis, cost-effectiveness analysis and simulation modelling — each attributed to its own source, and no Indian cost-effectiveness threshold value was found in the corpus.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, HTAIn / Department of Health Research, National Health Authority, National Health Accounts, Indian Reference Case 2023, CHEERS 2022, HTAi 2025

Programme and policy updates (India)

- ▶ Systems analysis is no longer only a textbook management technique in India — it is an institutionalised government function. **Health Technology Assessment in India (HTAIn)** was set up in 2017 under the **Department of Health Research, Ministry of Health and Family Welfare**, acting on the National Health Policy 2017 commitment to build an institutional framework for health technology assessment. Its structure is a Secretariat, a Board, a **Technical Appraisal Committee (TAC)**, Regional Resource Centres and Technical Partners; the TAC receives topics from central and state departments (and private agents), prioritises them, allots each to a centre for economic evaluation, and the Board converts the finished study into a policy brief advising government — in effect, the six steps of systems analysis run as a standing national process [PMID 41316276].
- ▶ The network has scaled rapidly: Regional Resource Centres rose from 1 (2017) to 30 (2024) with roughly 72 scientific and technical staff, six zonal **HTA resource hubs** (north, west-and-central,

south, union territories, north-east) now oversee quality and dissemination, and a Master's-level course in Health Economics and Technology Assessment has been started. On the Indian HTA Quality Appraisal Checklist, 26 HTAIn studies done between 2018 and 2023 rated 14% excellent, 50% good, 32% average and 4% poor, with mean adherence to the Indian Reference Case only 7.13 out of 10 — capacity, staff turnover and data access, not the method itself, are the binding constraints [PMID 40913689].

- The best current Indian example of systems analysis applied to a whole programme is the **Ayushman Bharat PM-JAY health benefit package** (over 1,900 packages), revised as HBP 1.0 (September 2018) → 2.0 (2019) → 2.1 (November 2020) → 2.2 (October 2021) → HBP 2022 (April 2022). The national **Costing of Health Services in India (CHSI)** study supplied costs for 855 packages across eight specialties; a cost–price differential analysis found about 42% of packages were being reimbursed at under half their actual cost, and rates for 61% of packages were revised, with differential rates for tier-1 and tier-2 cities and higher rates for tertiary than secondary care. A dedicated **Health Financing and Technology Assessment (HeFTA) unit** was then created inside the National Health Authority to route assessment evidence into package inclusion and exclusion, reimbursement rates and standard treatment guidelines, with value-based pricing proposed as the next step in strategic purchasing [PMID 38857943; PMID 38715207].

Current national data

- **National Health Accounts, 2013–14 to 2021–22:** the government share of total health expenditure rose from 28.6% to 48.0% while the out-of-pocket share fell from 64.2% to 39.4%, the two lines crossing by the end of the period. Government health expenditure is still only about 1.9% of gross domestic product against the National Health Policy target of 2.5%, and total health expenditure about 3.8% of GDP. Medicines alone absorb roughly two-thirds of out-of-pocket spending, catastrophic expenditure is commoner for outpatient care (47.8% of households) than for hospitalisation (43.1%), and about 55 million people are pushed into poverty by health costs each year — which is why an input–output analysis confined to hospital admissions misses most of the burden [PMID 42548603].
- A national economic model of **population-based screening for diabetes and hypertension** put the incremental cost per QALY gained at roughly US\$0.02–0.03 million; at the existing pattern of care-seeking, no screening interval (annual, 3-yearly, 5-yearly) was cost-effective at a threshold of one times per-capita GDP. Screening the 30–65 year age group every 3 or 5 years became cost-effective once Health and Wellness Centres treated 20% of newly diagnosed uncomplicated cases, and annual screening became cost-saving at 70% (against the existing 4% at sub-centres and primary health centres) — a textbook demonstration that the cost-effectiveness of a component depends on how the rest of the system is linked to it [PMID 34774219].

Elimination, surveillance and disease status

- A transmission-model cost-effectiveness simulation tested tightening the **lymphatic filariasis** transmission assessment survey stopping rule from <2% to <1% antigenaemia prevalence in *Culex*-transmission settings including India. The stricter threshold raised the probability of true local elimination (83.8% versus 78.8% of simulations at 5–10% baseline prevalence) and, valued against India's willingness-to-pay of **US\$446.07 per DALY averted** — per-capita government expenditure adjusted for purchasing power parity, used as an opportunity-cost threshold — was

cost-effective at 80% mass drug administration coverage across all baseline prevalences, but not at 65% coverage because of the extra treatment rounds required [PMID 38662697].

Guideline and legislative updates

- The **Indian Reference Case (2023)** is the standard that now governs how an economic evaluation for India must be conducted: twelve recommendations across eleven principles — decision problem, comparator, perspective, source of effectiveness, costs, health outcome, time horizon, discounting, heterogeneity, uncertainty and equity, plus presentation of results. Key content for examinations: **current practice** is the comparator; an **abridged societal perspective** (direct medical and non-medical costs of payer plus patient) is the base case, reported alongside the healthcare payer perspective, with indirect productivity costs confined to sensitivity analysis; costs and outcomes beyond one year discounted at 3% (varied 0–5%); a **lifetime horizon** where the intervention changes mortality; and **QALYs** valued with the Indian EQ-5D-5L value set. Supporting infrastructure includes the National Health System Cost Database, the HTAIn process manual, the health system costing handbook and budget impact analysis guidelines [PMID 37694178].
- An audit of 33 published HTAIn studies found 84.8% of the interventions cost-effective and 21.2% outright cost-saving, but **every study that used a threshold used one times GDP per capita** — the WHO-CHOICE 1–3× GDP rule that WHO itself now cautions against because it ignores opportunity cost and budget limits. Adherence was weakest on **equity** (25.7%) and **heterogeneity/subgroup analysis** (24.3%). The authors call for an India-specific opportunity-cost threshold and for **health technology reassessment** — the systematic re-examination of technologies already in use, so that funding a new cost-effective service is matched by disinvestment from a less efficient one [PMID 41316276].

Global and WHO developments

- **CHEERS 2022** replaced the 2013 reporting standard with a **28-item checklist** and, unlike its predecessor, is written to cover any intervention intended to improve health — public health and social-care programmes as well as drugs — with added emphasis on stakeholder and patient involvement; it is the checklist against which an economic evaluation quoted in an answer should be judged [PMID 35031096].
- The **HTAi 2025 meeting (Buenos Aires, June 2025)** identified three next-generation methods entering health technology assessment: **environmental sustainability** (health care contributes about 5% of global greenhouse-gas emissions, with 143 countries committed at COP28 to low-carbon climate-resilient health systems), **adaptive or rapid assessment** for decisions that cannot wait for a full evaluation, and **artificial-intelligence-enabled real-world evidence** drawn from electronic health records — all three rated moderate-to-high on need, rigour and implementability by the audience [PMID 42411102].

SOURCE & COVERAGE

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Planning cycle for geriatric health care delivery

Asked in: Paper IV May 2019 — RGUHS MD Community Medicine

Definition — Planning is the process of analysing a system or defining a problem, assessing the extent to which the problem exists as a need, formulating goals and objectives to ameliorate that need, examining and choosing among alternative intervention strategies, initiating the action necessary for implementation, monitoring the system, and evaluating the results against the stated objectives (Park 28e, Ch 23, p 1026). Because the last step feeds back into the first, these steps form a closed loop called the **planning cycle**. Applied to my community, the cycle is worked through once for the health care of persons aged 60 years and above.

Pre-planning — the four pre-conditions

Pre-planning is the preparation for planning; Park lists four essential pre-conditions, all of which must be satisfied before the cycle is entered (Park 28e, Ch 23, p 1026).

Pre-condition (Park 28e, Ch 23, p 1026)	How it is satisfied for my geriatric plan
(a) Government interest — strong political will manifested as clear directives	<i>National Policy on Older Persons</i> (1999), which identifies financial security, health care, nutrition, shelter, education, welfare and protection of life and property as areas of intervention (Park 28e, Ch 9, p 704); the National Programme for Health Care of the Elderly (NPHCE) is its operational vehicle
(b) Legislation — the directive translated into an enactment	Section 20 of the <i>Maintenance and Welfare of Parents and Senior Citizens Act, 2007</i> , which places medical care of senior citizens on the State (Jugal Kishore, pp 909–910)
(c) Organization for planning — an organisational structure to prepare the various parts of the plan; the Planning Commission (now NITI Aayog) serves this function nationally	At my level, the District Health Society with the National Health Mission district planning cell, which drafts and consolidates the district Programme Implementation Plan; a block-level geriatric planning team of the medical officer, block health education officer and a panchayat representative
(d) Administrative capacity — for proper coordination and implementation at all levels; vested nationally in the Central and State Ministries of Health	The District Health Officer for inter-sectoral coordination and fund flow, and the primary health centre (PHC) medical officer as the accountable implementing authority for the field-practice area

Step 1 — Analysis of the health situation (community diagnosis of the elderly)

The first step is collection, assessment and interpretation of information so as to give a clear picture of the health situation (Park 28e, Ch 23, p 1027). I would conduct a house-to-house enumeration of all persons aged 60 years and above in the field-practice area through the accredited social health activist (ASHA) and auxiliary nurse midwife (ANM), supplemented by the sources below.

Data required	From which list	Source in my community	What it tells me about the elderly
Population, its age and sex structure	Ch 23 (a); Ch 10 (2) demographic conditions	Census, Civil Registration System, sub-centre family register	Size of the 60+ block, sex ratio, old-age dependency ratio (10.1 per cent for India, 2023) (Park 28e, Ch 8, p 583)
Statistics of morbidity and mortality	Ch 23 (b); Ch 10 (1)	PHC records, verbal autopsy, hospital returns	Main causes of illness in the aged in India — arthritis, cataract, bronchitis, avitaminosis, ear disease, hypertension, diabetes, rheumatism, helminthic infestation, accidents (Park 28e, Ch 9, p 704)
Epidemiology and geographical distribution of different diseases	Ch 23 (c)	Line list from Comprehensive Geriatric Assessment, spot map	Clustering of bedridden, blind and deaf elderly by hamlet
Medical care facilities — hospitals, health centres and other health agencies, both public and private	Ch 23 (d); Ch 10 (6)	Facility survey of sub-centre / health and wellness centre (HWC), PHC, community health centre (CHC) , district hospital, private clinics, Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy (AYUSH) units	Existing geriatric clinics, physiotherapy, drugs, referral transport
Technical manpower of various categories	Ch 23 (e)	Staffing position of the block; Indian Public Health Standards gap analysis	Whether a geriatric-trained medical officer, rehabilitation worker, physiotherapist, counsellor and staff nurse are in position or vacant
Training facilities available	Ch 23 (f)	District training centre, medical college department of community medicine, regional geriatric centre	Capacity to run geriatric-care modules for medical officers, staff nurses, ANM, ASHA and family caregivers
Attitudes and beliefs of the population towards disease, its cure and prevention	Ch 23 (g); Ch 10 (5) cultural background, attitudes, beliefs and practices	Focus group discussions with elderly and caregivers	Fatalism about ageing, neglect and elder abuse, health-seeking delay, preference for AYUSH or faith healing, family caregiver availability
Environmental conditions bearing on health	Ch 10 (3)	Household survey, home hazard checklist	Unsafe housing, poor lighting, slippery floors and steps — the falls risk; water and sanitation access for the frail

Data required	From which list	Source in my community	What it tells me about the elderly
Socio-economic factors directly affecting health	Ch 10 (4)	Household survey	Living alone, widowhood, below-poverty-line status, pension coverage, dependence for income
Other services available	Ch 10 (7)	Panchayat, Department of Social Justice and Empowerment	Old age homes, day-care centres, mobile medicare units, old age pension

(The table is a synthesis of the two lists Park gives: the seven minimum essential data requirements for health planning, items (a)–(g) — Park 28e, Ch 23, p 1027; and the seven data heads for community diagnosis, items (1)–(7) — Park 28e, Ch 10, pp 721–722. The "From which list" column names the source list of each row.)

Disabilities to be screened for specifically, being those incident to ageing: senile cataract, glaucoma, nerve deafness, osteoporosis affecting mobility, emphysema, failure of special senses and changes in mental outlook (Park 28e, Ch 9, p 703).

Step 2 — Establishment of goals and objectives

A goal states what is to be attained, the extent, the population involved, the geographic area, and the time required; objectives must be established at all levels and must be measurable, failing which there is haphazard activity, uneconomical use of funds and poor performance (Park 28e, Ch 23, p 1027).

Level	Statement for my community
Goal	Comprehensive, accessible and affordable promotive, preventive, curative and rehabilitative care to all persons aged 60 years and above in the PHC area over three years, promoting active and healthy ageing
Objective 1	Enumerate and issue a health card to 100 per cent of the 60+ population within 6 months
Objective 2	Subject 90 per cent of enrolled elderly to Comprehensive Geriatric Assessment twice a year for cognitive decline, limited mobility, malnutrition, visual impairment, hearing loss and depressive symptoms (Park 28e, Ch 10, p 737)
Objective 3	Run a weekly geriatric clinic at the PHC and a twice-weekly dedicated elderly clinic with a rehabilitation unit at the CHC, as under NPHCE (Park 28e, Ch 7, p 564)
Objective 4	Ensure a monthly domiciliary visit to every bedridden elderly person and train one family caregiver per such household
Objective 5	Link 100 per cent of eligible elderly to old age pension and social-security entitlements
Objective 6	Refer and operate 90 per cent of detected operable senile cataract within 3 months of detection

Step 3 — Assessment of resources, and Step 4 — Fixing priorities

Resources mean the manpower, money, materials, skills, knowledge and techniques needed or available; a balance is struck between what is required and what is or is likely to be available (Park 28e, Ch 23, p 1027).

Resource	Available	Additional requirement
Men (manpower)	Medical officer, ANM, male health worker, ASHA, community health officer at HWC	Trained rehabilitation worker or physiotherapist, counsellor, geriatric-trained staff nurse
Money	PHC untied funds, NPHCE grant-in-aid, National Health Mission programme implementation plan, panchayat funds	One-time grant to PHC for equipment; recurring cost of drugs and transport
Materials	Existing out-patient department (OPD) space, essential drug list	Walking aids, spectacles, hearing aids, weighing scale, glucometer, physiotherapy equipment, geriatric ward beds
Minutes (time)	Fixed weekly clinic day, monthly outreach day	Protected ANM time for domiciliary visits
Methods/skills	Standard treatment guidelines	Training modules in geriatric care for medical, paramedical and family caregivers

Once problems, resources and objectives are known, priorities are fixed in order of importance or magnitude, since resources always fall short of the total requirement. Park's criteria are financial constraints, mortality and morbidity data, diseases preventable at low cost, saving the lives of younger persons in whom there is considerable social investment, and political and community interests and pressures (Park 28e, Ch 23, p 1027). Priority setting is regarded by most health administrators as the heart of the planning process, and benefits more than any other step from being made an explicit, clearly defined exercise rather than an intuitive one (Reinke, Ch 1, p 27); the epidemiologically derived problem list should therefore be placed beside the community's own perception of its priorities and the two ranked jointly with the elderly and their caregivers (Reinke, Ch 13, p 195). Applying these, my priority order is: (1) cataract and refractive error — high magnitude, low-cost, immediately reversible disability; (2) hypertension and diabetes — high prevalence, opportunistic screening already funded; (3) falls, fractures and mobility loss; (4) hearing impairment; (5) depression, dementia and elder abuse; (6) palliative and end-of-life care. **Alternate plans** for each priority are then formulated and assessed for practicability and feasibility, and the alternative with greater effectiveness is chosen (Park 28e, Ch 23, p 1027).

Step 5 — Write-up of the formulated plan, and Step 6 — Programming and implementation

The written plan must be complete in all respects, define the results required, and be so clear as to give guidance to everyone responsible for execution; it must contain a **built-in system of evaluation** (Park 28e, Ch 23, p 1027). My written plan therefore specifies, for each priority, *what* will be done, *when*, *by whom* and *at what cost*, together with the service package at each level of care. Implementation depends on effective organisation and structure with well-defined procedures, sufficient delegation of authority and fixation of responsibility (Park 28e, Ch 23, p 1027).

Level of care	Service package for the elderly
Household / sub-centre–HWC	Health education on healthy ageing; domiciliary visits for homebound and bedridden elderly; training of family caregivers; supply of callipers and supportive devices obtained from the PHC to make the disabled elderly ambulatory; linkage with day-care centres and support groups (Jugal Kishore, p 621); Comprehensive Geriatric Assessment twice a year; <i>Sanjeevani</i> elderly self-help groups; yoga and activity sessions at the HWC; monthly visit to bedbound elderly; identification and reporting of elder abuse; promotion of inter-generational bonding (Park 28e, Ch 10, p 737)
PHC	Weekly geriatric clinic by a trained medical officer, with referral to the first referral unit for diseases needing further investigation and treatment (Park 28e, Ch 7, p 564); management of common geriatric ailments, counselling and supportive treatment; pain management and provision of palliative care with support of ASHA (Park 28e, Ch 10, p 737)
CHC	Dedicated clinic for the elderly twice a week; rehabilitation unit with a rehabilitation worker providing physiotherapy (Park 28e, Ch 7, p 564); advanced Comprehensive Geriatric Assessment; diagnostics, equipment, consumables and drugs leveraged from other national programmes; referral of cases needing secondary or tertiary care (Park 28e, Ch 10, p 737)
District hospital	Dedicated geriatric OPD and a 10-bedded geriatric ward in 80–100 district hospitals, with additional human resources, machinery, equipment, drugs, training and information–education–communication (IEC) (Jugal Kishore, pp 620–621)
Regional Geriatric Centre	Specialised geriatric OPD and a 30-bedded geriatric ward , with dedicated beds for the elderly in surgery, orthopaedics, psychiatry, urology, ophthalmology and neurology; a special sample-collection centre in the OPD block; tertiary care for cases referred from medical colleges, district hospitals and below; postgraduate training in Geriatric Medicine and research (Jugal Kishore, pp 621–622)
National Centre of Ageing	AIIMS New Delhi and Madras Medical College, Chennai — training of health professionals, research and geriatric health care delivery (Park 28e, Ch 7, p 563)

Convergence is built in: with the National Programme for Prevention and Control of Non-Communicable Diseases for hypertension, diabetes and opportunistic screening; the National Programme for Control of Blindness and Visual Impairment for cataract surgery; mainstreaming of AYUSH and revitalising of local health traditions; and the Ministry of Social Justice and Empowerment for pensions, old age homes and day-care centres (Jugal Kishore, pp 620–621).

Step 7 — Monitoring, and Step 8 — Evaluation and re-planning

Monitoring is the day-to-day follow-up of activities during implementation to ensure they are proceeding as planned and on schedule; it is a continuous process of observing, recording and reporting, and of identifying deviations and taking corrective action (Park 28e, Ch 23, p 1027). I would monitor through the monthly PHC review meeting, the ANM's diary, and the geriatric line list.

Evaluation measures the degree to which objectives and targets are fulfilled and the quality of the results obtained, measures the productivity of available resources in achieving clearly defined objectives, measures how much output or cost-effectiveness is achieved, and makes possible the reallocation of priorities and resources on the basis of changing health needs — WHO Expert Committee on National Health Planning in Developing Countries (quoted in Park 28e, Ch 23, p 1027). The input and process rows below are an **implementation (process) evaluation**, whose purpose is to verify that the programme

was in fact delivered and to quantify how faithfully it followed its design, so that a null result on the outcome rows can be read correctly; **efficiency** is computed from the same data as cost per unit of output (Health Program Planning and Evaluation, Ch 9, p 252).

Type	Indicator (numerator ÷ denominator)
Input	Number of trained geriatric care providers ÷ number of posts sanctioned
Process	Number of geriatric clinics held ÷ number of clinics scheduled; number of domiciliary visits made to bedridden elderly ÷ number planned
Output (coverage)	Number of 60+ persons given Comprehensive Geriatric Assessment ÷ total enumerated 60+ population
Efficiency	Recurring programme expenditure ÷ number of elderly given a complete assessment (cost per unit of output)
Outcome	Number of detected cataract cases operated ÷ number of operable cataract cases detected; proportion of hypertensive elderly with controlled blood pressure
Impact	Change in prevalence of avoidable blindness, of dependency in activities of daily living, and in elderly hospitalisation rate in the community

Evaluation results are compared with the stated objectives. If goals are achieved and satisfactory, the programme continues; if not, health data are gathered afresh, problems are reformulated and the **cycle is repeated** — this feedback is what makes planning a cycle rather than a line (Park 28e, Ch 23, p 1026, Fig. 1).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce Fig. 1 as a closed loop of its boxes, from "gather health data" round to "evaluate results", with the YES arm continuing the programme and the NO arm returning to "gather health data"; mark the segment from "plan implementation" to "operate programme" as **implementation**.

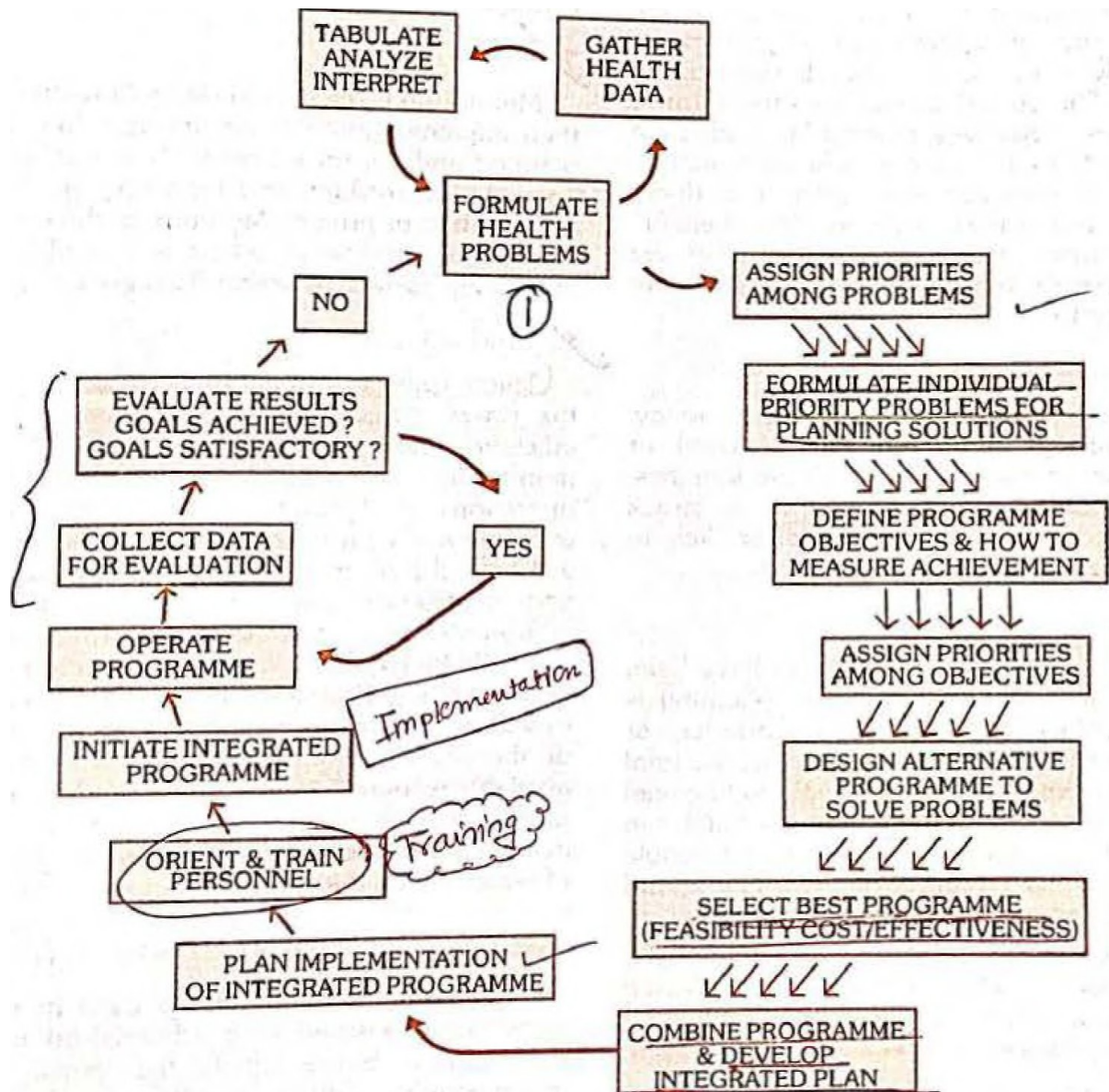
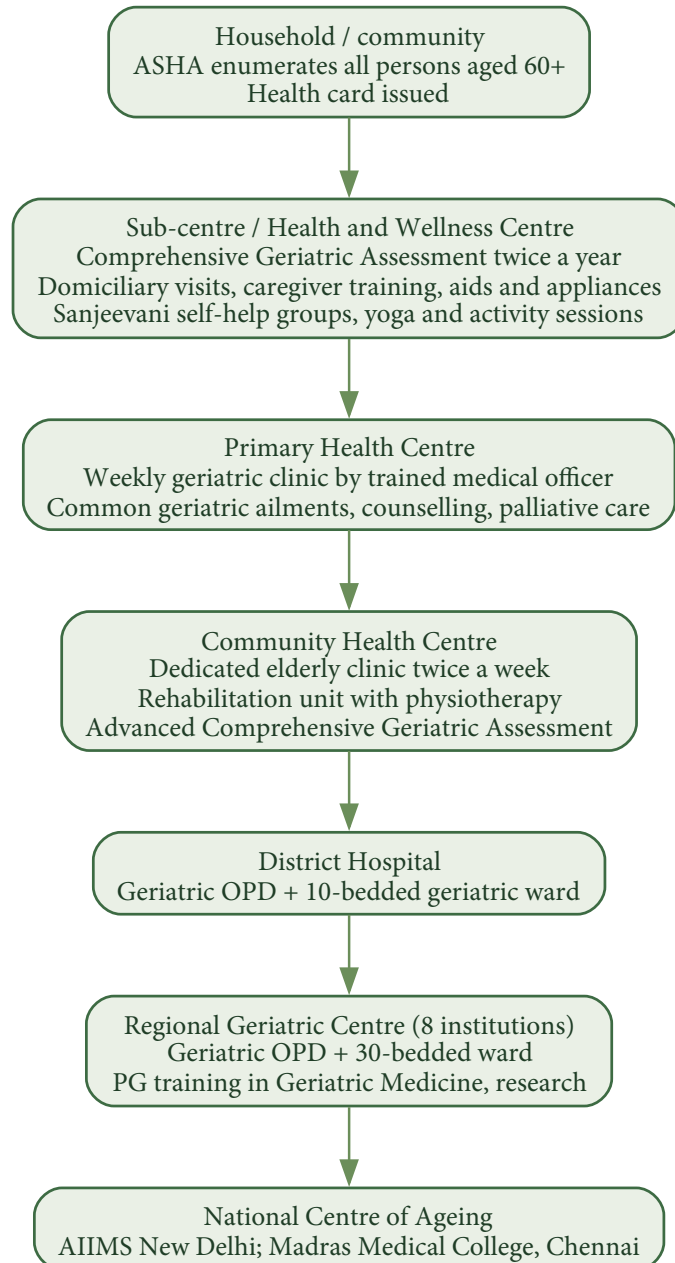
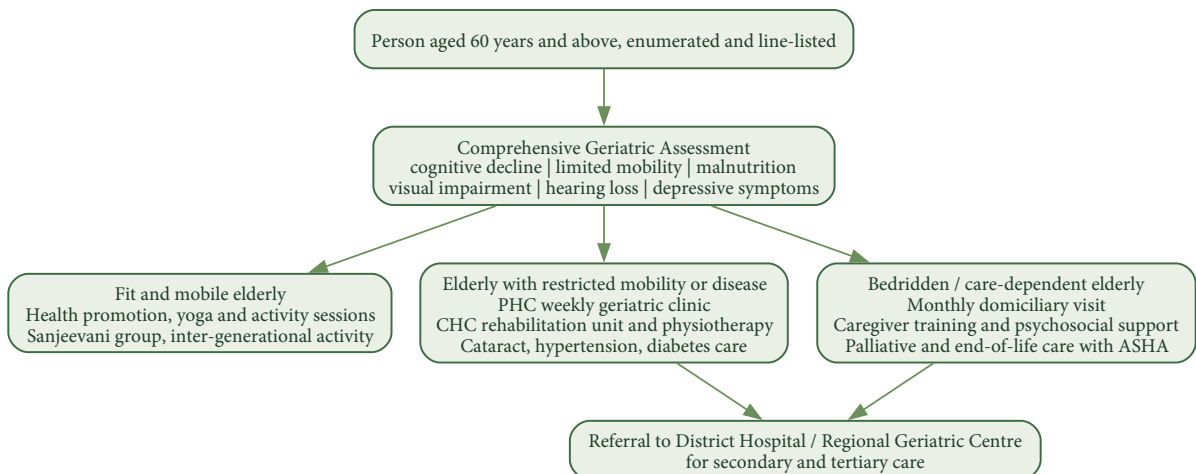


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, Ch 23 (Health Planning and Management), p 1026)



Delivery of geriatric health care services in the community — referral pathway by level of care



Case-finding to care pathway for an elderly person in the field-practice area

High-yield exam pointers

- ▶ **Eight steps of the planning cycle (Park):** analysis of the health situation → establishment of objectives and goals → assessment of resources → fixing priorities → write-up of the formulated plan → programming and implementation → monitoring → evaluation, with feedback to step 1; **pre-planning** precedes step 1 and needs all four pre-conditions — government interest, legislation, organization for planning, administrative capacity.
- ▶ **Minimum essential data for health planning — all seven (Park 28e, Ch 23, p 1027):** (a) population, its age and sex structure; (b) statistics of morbidity and mortality; (c) epidemiology and geographical distribution of different diseases; (d) medical care facilities such as hospitals, health centres and other health agencies, both public and private; (e) technical manpower of various categories; (f) training facilities available; (g) attitudes and beliefs of the population towards disease, its cure and prevention.
- ▶ **Resources = manpower, money, materials, skills, knowledge and techniques** (the "5 M's" of management: men, money, materials, minutes, methods).
- ▶ **Priority-setting criteria:** financial constraints, mortality and morbidity data, diseases preventable at low cost, social investment in the lives saved, political and community pressures.
- ▶ **NPHCE ladder to quote:** sub-centre domiciliary visits → PHC weekly geriatric clinic → CHC twice-weekly clinic with rehabilitation unit → district hospital 10-bedded geriatric ward → Regional Geriatric Centres with 30-bedded wards (8 strengthened; 12 envisaged) → 2 National Centres of Ageing (AIIMS New Delhi; Madras Medical College, Chennai). Started in 100 identified districts of 21 states, covering 20,000 sub-centres and 700 CHCs; the tertiary component is renamed *Rashtriya Varishth Jan Swasthya Yojana*.
- ▶ **Monitoring vs evaluation:** monitoring is day-to-day, continuous, "are we on schedule?"; evaluation is periodic, compares results with stated objectives and measures cost-effectiveness.
- ▶ **Ageing burden figures with their years:** old-age dependency ratio for India 10.1 per cent (2023) (Park 28e, Ch 8, p 583); the 60+ population is projected to rise from 8 per cent in 2010 to 19 per cent (about 323 million) by 2050 (Park 28e, Ch 12, p 826).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: on the number of Regional Geriatric Centres the corpus differs — 8 Regional Geriatric Centres in 8 regional medical institutions are recorded as strengthened under the NPHCE document (Jugal Kishore, pp 620–621), whereas Park 28e, Ch 7, p 563 states that 12 regional geriatric centres in selected medical colleges are *expected* to be developed, so quote 8 as strengthened and 12 as envisaged. Also, the NPHCE is now delivered largely through the Comprehensive Primary Health Care package of Ayushman Bharat Health and Wellness Centres (renamed *Ayushman Arogya Mandir*), so the elderly service package described at sub-centre level in Park 28e, Ch 10 is the Health and Wellness Centre package rather than a stand-alone vertical service; and the indicator set in the evaluation table is an illustrative plan-specific set, since the retrieved corpus does not prescribe a fixed NPHCE indicator list.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Longitudinal Ageing Study in India, Comprehensive Annual Modular Survey 2022–23, Global Burden of Disease 2019, WHO Integrated Care for Older People framework

Programme and policy updates (India)

- ▶ **Ayushman Bharat Health and Wellness Centres have been renamed *Ayushman Arogya Mandir*** (tagline *Arogyam Parmam Dhanam*), with an explicit mandate covering preventive, promotive, curative, rehabilitative and palliative care — so the sub-centre and PHC elderly package described above is now delivered under that name. Coverage is not access: of 320 enrolled beneficiaries surveyed in Jammu and Baramulla districts, only 48.4 per cent had ever used the scheme; poor road quality was the one physical barrier that independently reduced use (OR 0.761), while knowing the entitlement (OR 1.253) and knowing the empanelled-hospital list (OR 1.245) raised it, and the financial and paperwork barriers people complained of most were not independent predictors at all [PMID 41878109].
- ▶ **Publicly funded insurance protects the hospital bill, not the out-patient one, and does not by itself prevent impoverishment of elderly households.** In the Comprehensive Annual Modular Survey 2022–23 (302,086 households), Ayushman Bharat–PM-JAY or State scheme cover raised out-patient (adjusted OR 1.404, 95% CI 1.328–1.484) and in-patient (adjusted OR 1.168, 95% CI 1.128–1.210) utilisation and reduced catastrophic in-patient expenditure at the 20 per cent threshold (adjusted OR 0.790, 95% CI 0.737–0.846), while out-patient spending remained catastrophic even among the insured [PMID 42167946]; in LASI, among the 10,963 households (77.9 per cent of 14,067 older-adult households living with a non-communicable disease) that used any care in the past year, 41.2 per cent faced catastrophic expenditure and 28.4 per cent were impoverished, government insurance carrying *higher* odds of impoverishment than private cover (AOR 1.43, 95% CI 1.06–1.92) [PMID 42000994]; and at the last mile only 14 per cent of eligible households in a Puducherry urban slum had used the scheme despite 59 per cent awareness [PMID 41805171]. The resource step of any geriatric plan must therefore budget separately for drugs, diagnostics and transport.

- **Plan–Do–Check–Act has been applied as the audit frame for Health and Wellness Centre readiness** in 16 rural Lucknow centres: dedicated diagnostic space and disability-friendly facilities in 6.25 per cent each, biomedical waste facilities 18.75 per cent, mains-linked or backup electricity and cervical cancer screening in none, against diabetes and hypertension screening at 100 per cent and better hypertension screening in HWC villages (96.4 versus 84.8 per cent) — a ready-made checklist for the facility survey and the process-evaluation rows of the plan [PMID 42257150].

Current national data

- **Cataract, the plan's first priority, is quantified nationally by LASI (2017–18, 45 years and above):** prevalence 14.25 per cent (95% CI 13.75–14.77), higher in women (15.38 versus 12.92 per cent); 27.52 per cent of those with cataract had received no treatment, and among the treated 28.94 per cent did not achieve effective treatment coverage, the gap widening in women (30.80 versus 26.34 per cent) and the poorest (33.18 per cent) — the shortfall is in *effective* coverage, not merely surgical numbers, which justifies an outcome indicator rather than an output one [PMID 39161857].
- **One in five older Indians falls.** In 31,345 LASI participants aged 60 and above the prevalence of falls was 20.25 per cent, while only 25.0 per cent met all five "successful ageing" standards — no chronic disease, no disability, high cognition, no depressive symptoms and active social engagement. Successful ageing was negatively associated with having fallen (OR 0.70, 95% CI 0.65–0.76) and with repeated falls (OR 0.70, 95% CI 0.63–0.78), but not with fall-related injury (OR 0.86, 95% CI 0.72–1.04) [PMID 38914970].
- **Palliative need and its cost are now measurable at national level.** Applying the SPICT-LIS tool to LASI, 42.7 per cent of older adults had at least one indicator of poor or deteriorating health and 12.2 per cent had two or more, i.e. about one in eight needs supportive or palliative care, rising to 17 per cent in West Bengal [PMID 39616320]; for the 2,903 older adults so identified, mean annual expenditure was ₹5,002 for out-patient and ₹36,263 for hospitalisation, 94 and 82 per cent of it out of pocket, with 21.6 per cent of households facing catastrophic expenditure and about a quarter each liquidating savings (24.4 per cent) or taking loans (26.5 per cent) [PMID 41578256].

Elimination, surveillance and disease status

- **The burden of the 60-plus block is disaggregated by age and sex, but the programme data are not — and closing that gap is an evaluation-step task.** GBD 2019 puts older persons at 136.1 million DALYs in India — 29.1 per cent of the national total, 72.7 per cent of it from years of life lost — of which 77.9 per cent came from non-communicable diseases, 14.8 per cent from communicable, maternal, neonatal and nutritional causes and 7.3 per cent from injuries, with 6–8 of the ten leading causes being NCDs across all age groups in both sexes. Against this, every NPHCE service-delivery indicator — elderly enrolled, health cards issued, home visits, geriatric clinics held, geriatric-ward inpatients, referrals — is recorded with no age and no sex disaggregation, as are those of the blindness, mental health and musculoskeletal programmes and of the Health and Wellness Centres; the NCD programme carries sex but not age [PMID 39702198]. LASI places cardiometabolic multimorbidity at 27.3 per cent (95% CI 26.8–27.8) in

women versus 23.0 per cent (95% CI 22.5–23.5) in men among adults aged 45 and above, peaking at 60–74 years and higher in urban and wealthier groups [PMID 41120881].

Guideline and legislative updates

- **India now has a national intrinsic-capacity score, operationalising the WHO healthy-ageing framework for Comprehensive Geriatric Assessment.** Built by confirmatory factor analysis from 14 LASI items across cognitive, locomotor, psychological, sensory and vitality domains in 60,591 adults aged 45 and above, the 0–100 score averaged 69.7 (95% CI 69.6–69.8) and ranged from 63.4 in Sikkim to 77.8 in Chandigarh; a higher score was protective against poor self-rated health and functional limitation after adjustment — a single comparable metric usable as an outcome indicator in a district plan [PMID 41387631].

Global and WHO developments

- **Fall prevention delivered through primary health care works in a low-resource rural setting.** The FAMILY cluster randomised trial (128 villages in four provinces of rural China, 2,610 adults aged 60 and above at risk of falls) tested balance and functional exercises plus community-engaged health education integrated into the rural primary health care system: 29.7 per cent of the intervention group reported a fall over 12 months versus 38.3 per cent of controls (OR 0.67, 95% CI 0.48–0.91), with all secondary outcomes except the Timed Up and Go favouring the intervention — a directly transferable model for the sub-centre and HWC level [PMID 40853561].
- **The WHO Integrated Care for Older People (ICOPE) screening tool has been tested at national scale,** in 185,287 participants in Taiwan's 2023 programme linked to national insurance data: the six-domain instrument resolved into body functionality and physical–mental vitality factors, with impairment predicting frailty, emergency visits and unplanned hospitalisation, supporting its use as the screening front end of a population-level geriatric programme [PMID 42214510].

SOURCE & COVERAGE

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Direct and indirect costs in hospital management

Asked in: Paper IV 13-May-2022 · Paper IV 20-Jul-2020 (2×) — RGUHS MD Community Medicine

Definition — Cost is the value of the resources used up to produce a service (Sunder Lal, Ch 2, p 87). **Direct costs** are the resources actually consumed in designing, delivering, receiving or continuing a hospital service, and are traceable to a named patient, procedure or department (Health Economics, Ch 14, pp 217-218). **Indirect costs** are the resources lost or consumed *because of* that episode of care but not traceable to it — in economic terms the output lost through illness, disability or premature death, and in hospital accounting the shared overhead that supports the organisation as a whole and must be apportioned (Reinke, Ch 20, pp 277-278; Health Program Planning and Evaluation, Ch 8, p 231).

Direct costs in a hospital

Sub-type	Hospital examples	Key feature
Direct health care cost	Salaries of ward doctors, nurses and technicians; drugs, fluids, blood, sutures, implants; laboratory and imaging consumables; operation-theatre and intensive-care use; physicians' fees (Health Economics, Ch 14, p 218; Reinke, Ch 20, p 277)	Resources essential to the implementation, receipt and continuation of the health service
Direct non-health care cost	Transport of the patient to and from the facility; hospital diet and catering; bus fare of family members visiting the child in hospital (Health Economics, Ch 14, pp 218, 220-221)	Used in connection with care but not a health-sector resource
Borne by the provider	Salary of the nurse running the immunisation clinic; hospital cost of a child admitted for vaccine side-effects (Health Economics, Ch 14, pp 220-221)	Traceable to one cost centre without apportionment
Borne by the patient/family	Out-of-pocket expenditure on drugs, investigations, user charges (Health Economics, Ch 17, p 268)	Both provider and patient direct costs are counted unless the study perspective dictates otherwise (Health Economics, Ch 14, pp 217-218)

Indian magnitude: average medical expenditure per hospitalisation was ₹31,845 in a private hospital against ₹4,450 in a public hospital — about seven times higher — and medical plus related non-medical expenditure per hospitalisation was ₹16,676 in rural against ₹26,475 in urban areas (National Sample Survey 75th round, 2017-18; Sunder Lal, Ch 16, p 729). About 53.2% of current health expenditure in India is out-of-pocket, households contributing over 60% in all (Sunder Lal, Ch 16, pp 727-729).

Indirect costs in a hospital

Sense in which the term is used	What it covers	How it is valued
Economic (societal) sense — loss of output	Output lost through premature death or disability attributable to the disease (Reinke, Ch 20, pp 277-278)	Wages and earnings lost; where the person is not earning, alternative valuation of their time (Health Economics, Ch 14, p 219)
Patient and carer time	Patient's time spent travelling to and waiting in hospital instead of working; a family member's altered work schedule; cost of caring for siblings so the mother can bring the sick child; time lost taking a child for immunisation (Health Economics, Ch 14, pp 219, 220-221)	Wage loss / opportunity cost of unpaid time
Accounting (overhead) sense — hospital management usage	Costs not incurred directly in providing patient care but necessary to support the organisation: hospital administration, human-resource and legal staff, shared clerical support, rent, utilities, telephone, laundry, security, housekeeping, facilities management, equipment depreciation (Health Program Planning and Evaluation, Ch 8, p 231; Health Economics, Ch 14, p 215)	Pooled and apportioned to cost centres on an agreed base
Excluded to avoid double counting	Transfer payments (pension, relief payments) and income-tax losses (Reinke, Ch 20, p 278)	Adding these to lost earnings double- or triple-counts the same loss

Indian magnitude: an Indian worker with tuberculosis loses on average **83 working days**; with about 2 million new cases notified annually this is **166 million lost workdays** worth roughly **US\$200 million (₹694 crore)**, and once indirect costs are added tuberculosis is estimated to cost India about **₹12,000 crore a year** (Sunder Lal, Ch 2, p 87).

Direct versus indirect cost — points of difference

Basis	Direct cost	Indirect cost
Nature	Resources used up in providing or receiving care	Output/productivity lost, or shared support resources
Traceability to a cost centre	Traceable without apportionment	Not traceable; must be apportioned
Falls on	Hospital budget and the patient's pocket	Chiefly the patient, family and the economy; in accounts, the whole institution
Measurement	Expenditure records, purchase price, salary bills	Wages and earnings lost; for overheads, an allocation base
Typical relative size	44.8% of the total cost of illness (United States, 1975)	55.2% of the total cost of illness (United States, 1975) — indirect exceeded direct (Reinke, Ch 20, p 278)

Basis	Direct cost	Indirect cost
Effect on hospital pricing	Enters the unit cost of a service directly	Enters only after apportionment; drives the "overhead" component of a package rate

Other cost terms the examiner expects alongside

Term	Meaning
Intangible cost	Pain, discomfort, anxiety, suffering and inconvenience — the third component of the cost of illness, usually left out for want of data and method (Reinke, Ch 20, p 277; Health Economics, Ch 14, p 215)
Fixed vs variable cost	Fixed cost does not vary with the number of patients served (rent, administrative salaries, insurance); variable cost does (Health Program Planning and Evaluation, Ch 8, p 231)
Capital vs recurrent cost	Capital = resources with a useful life over one year (buildings, vehicles, equipment, one-off training); recurrent = used up within a year (salaries, drugs, fuel, electricity) (Health Economics, Ch 14, p 221; Sunder Lal, Ch 2, p 88)
Opportunity cost	Benefit forgone from the next best alternative use of the same rupee (Sunder Lal, Ch 2, p 88)
Average and marginal cost	Average = total cost ÷ units of output; marginal = extra cost of one more unit, e.g. of extending hospital stay by one day (Sunder Lal, Ch 2, p 88)

Apportioning indirect (overhead) cost to hospital cost centres

- **Define cost centres** — wards, operation theatre, laboratory, radiology, laundry, kitchen, administration.
- **Charge every directly traceable cost** to its own centre — ward staff salaries, drugs, consumables (Reinke, Ch 12, pp 179-180).
- **Pool the remaining overheads** that serve the whole institution.
- **Choose an allocation base** and apportion each pool (Health Economics, Ch 14, pp 222-223): buildings and utilities by **percentage of floor space**; staff by **percentage of time spent**; equipment by **percentage of time used**.
- **Annualise capital items** so they can be charged to a single year — the simplest method, straight-line depreciation, divides the initial cost by the number of years of useful life, so an X-ray machine costing £10,000 with a useful life of 10 years carries an annualised cost of £1,000 per year (Health Economics, Ch 14, p 221).
- **Divide total cost by output** to obtain the unit cost — cost per bed-day, cost per patient treated.

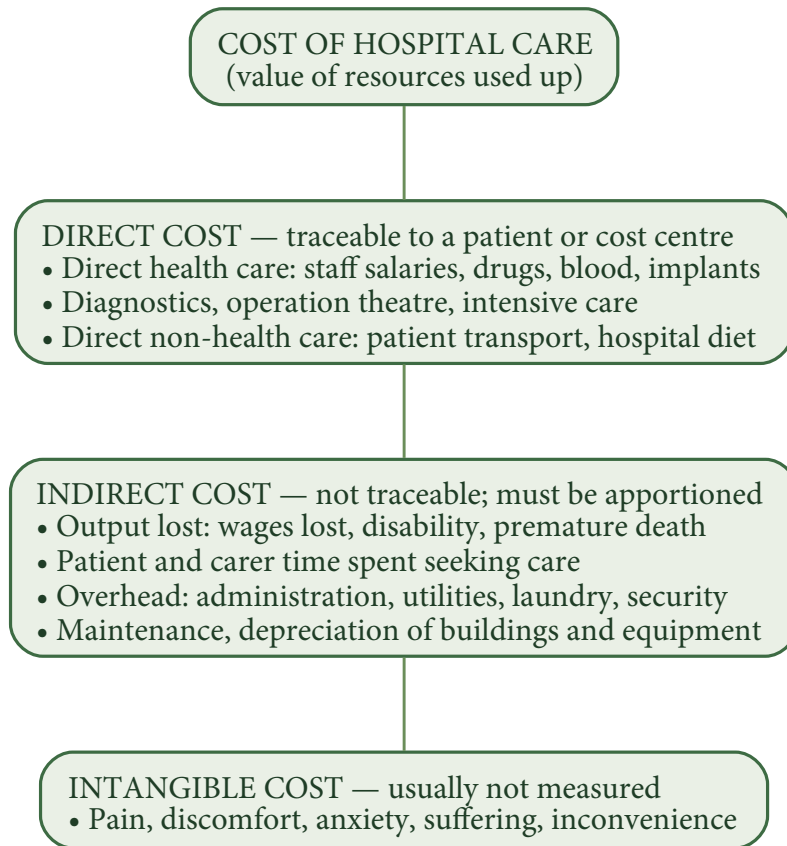
Two approaches exist: **macro- or top-down costing**, which splits the hospital's total cost among specialties by their share of cases, and **micro- or bottom-up costing**, which counts every item of service and consumable used by each patient; in practice a mixture is used (Oxford Textbook of Public Health 7e, Ch 49, pp 1829-1830).

Why the distinction matters in hospital management

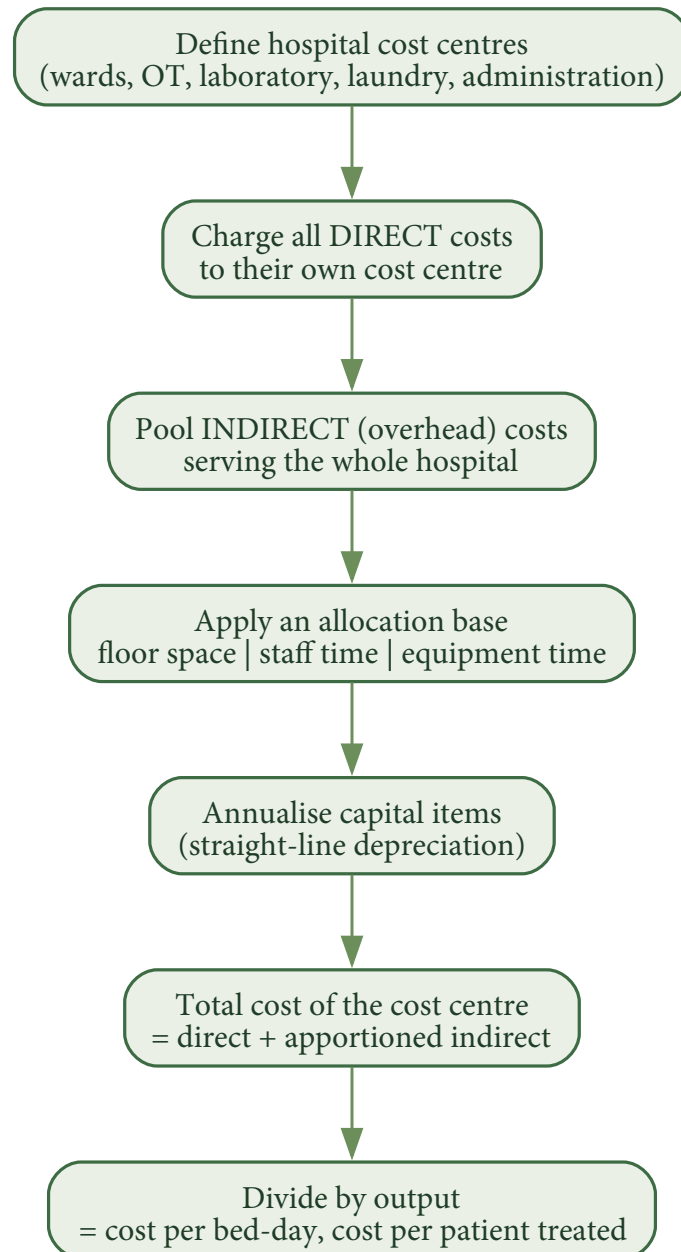
- ▶ **Cost accounting** supplies the basic data on cost structure and serves three purposes — cost control, planning and allocation of human and financial resources, and pricing for cost reimbursement (Park 28e, Ch 23, p 1028).
- ▶ **Efficiency indicators** of a hospital — bed occupancy, cost per day in hospital, cost per patient treated — can only be computed once direct and apportioned indirect costs are separated (Park 28e, Ch 23, p 1038).
- ▶ **Budgeting and fiscal monitoring** rest on a line-item budget of personnel, expendables and fixed capital (Reinke, Ch 12, pp 179-180); classifying by fixed versus variable makes the budget usable for management.
- ▶ **Economic evaluation** uses total (direct plus indirect) cost of a disease as the measure of benefit obtainable from a programme that controls it, in cost-benefit analysis and in cost-effectiveness analysis where benefit is expressed as results achieved rather than money (Reinke, Ch 20, p 278; Park 28e, Ch 23, p 1028).
- ▶ **Equity and financial protection** — free care at the point of delivery does not abolish cost to the patient: fee-exempted patients in the Cambodian study still incurred direct costs of US\$4.3 per hospitalisation episode plus indirect costs, and only 7% had sufficient cash (Health Economics, Ch 17, p 268). In India catastrophic health expenditure pushes about **63 million people below the poverty line every year** (National Sample Survey 75th round; Sunder Lal, Ch 17, p 742).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Label box** — "Components of the cost of hospital care": draw three stacked horizontal bars of one total, labelled **Direct cost (health care + non-health care)**, **Indirect cost (output lost / overhead)** and **Intangible cost (pain and suffering)**; mark the arrow "Total cost of illness = direct + indirect (+ intangible)" and annotate the direct/indirect split as 44.8% : 55.2% (United States, 1975).



Classification of costs in hospital management



Step-down apportionment of hospital costs to a unit cost

High-yield exam pointers

- ▶ **Total cost of illness = direct + indirect (+ intangible);** in the United States 1975 the total exceeded \$265 billion, about 17% of gross national product, split **44.8% direct : 55.2% indirect** — indirect exceeds direct.
- ▶ **Direct = resources used up; indirect = output lost (and, in hospital accounts, apportioned overhead); intangible = pain and suffering.**
- ▶ Direct costs split into **direct health care** and **direct non-health care** (transport, hospital catering).
- ▶ Indirect costs are valued by **wages and earnings lost**; never add **transfer payments or taxes** — that is double counting.

- Allocation bases: **floor space** for buildings and utilities, **staff time** for personnel, **equipment time** for machines; capital annualised by straight-line depreciation (cost ÷ years of useful life).
- Cost accounting has **three purposes**: cost control, resource planning and allocation, pricing for cost reimbursement (Park 28e, Ch 23).
- Indian figures to quote with their year: **NSS 75th round (2017-18)** — ₹31,845 private versus ₹4,450 public per hospitalisation; **out-of-pocket 53.2% of current health expenditure**; **63 million pushed below the poverty line per year** by catastrophic expenditure.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Sunder Lal (Ch 2, p 87) estimates that a hospital bed in a government teaching hospital costs ₹50,000–75,000 per annum, of which about 75% is staff salary and about 20% building maintenance (water and electricity), but the textbook does not state the year of that estimate, so quote it only as an illustration of how heavily direct salary cost dominates a hospital's recurrent budget.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Health Accounts India, Ayushman Bharat PM-JAY health benefit package, HTA in Indian Reference Case, WHO TB patient cost survey methodology

Programme and policy updates (India)

- **PM-JAY package rates are now set from measured hospital cost rather than from existing scheme charges.** The Costing of Health Services in India (CHSI) study collected mixed top-down and bottom-up cost data from 63 hospitals across 14 states; before it, about 42% of packages were reimbursed at less than half their actual cost, and applying the cost evidence halved that proportion. The 2022 revision raised 832 package rates, reduced 122 and left 630 unchanged, and brought in differential pricing by city tier after bed-day and procedure costs proved to differ between tier-1/tier-2 and tier-3 cities. CHSI unit costs are now public as the **National Health System Cost Database**, the first free repository of Indian facility costs by state, facility level, cost centre and service. [PMID 37963613]
- **National Health Authority cost-surveillance pilot, April 2022** — 61 empanelled hospitals in five states enter patient-level resource use (drugs, consumables, implants, diagnostics) with ICD-11 diagnoses through the existing claims and transaction system, so that price weights can reflect case mix and severity and, in the long run, support a diagnosis-related group payment system. The process evaluation found duplicated data entry, data-entry staff without clinical training and poor connectivity to be the main obstacles, and recommends fetching the same data automatically from Ayushman Bharat Digital Mission-compliant hospital information systems instead of running repeated large costing surveys. [PMID 37963613]
- **Health and Wellness Centres have been renamed Ayushman Arogya Mandirs (AAMs)**, and the first nationally representative costing of comprehensive primary health care at them is under way — top-down micro-costing of 72 facilities across six states, with cost centres separated into **care service centres** such as OPD, antenatal care, NCD screening and immunisation and **support service centres** such as medical records and administration; joint costs apportioned by staff time

allocation, floor area in square feet and patient footfall; capital annualised at a 3% discount rate as required by current Indian methodological guidelines; and data envelopment analysis used to score technical efficiency. [PMID 40044205]

Current national data

- **Quote National Health Accounts figures, not the NSS 75th-round shares, for out-of-pocket burden.** Between 2013-14 and 2021-22 the out-of-pocket share of total health expenditure fell from 64.2% to 39.4% while the government share rose from 28.6% to 48%; total health expenditure is about 3.8% of GDP and government health expenditure about 1.9%, still short of the National Health Policy target of 2.5%, and roughly 55 million people are pushed into poverty by health payments each year. Catastrophic expenditure is now commoner for outpatient care (47.8% of households) than for hospitalisation (43.1%), and **medicines alone account for about two-thirds of out-of-pocket spending** — precisely the costs that lie outside PM-JAY, which reimburses only secondary and tertiary admission; Tamil Nadu's pooled free-medicines procurement is cited as the domestic remedy with a three-decade track record. [PMID 42548603]
- **An ICMR institute-coordinated multicentre IVF study measures cost exactly along the direct/indirect axis used in this answer** — direct health cost, direct non-medical cost (travel, food and lodging of the woman and her attendant) and indirect cost (wage loss) counted separately at five tertiary hospitals. Mean out-of-pocket expenditure per cycle was ₹1,10,104 in public against ₹2,30,668 in private hospitals; about 30% of couples crossed the catastrophic threshold (private facility OR 7.7, 95% CI 2.5-23.2); **drug costs alone made up 55% of total expenditure**; and lost wages of the husband accounted for over 80% of the indirect cost. [PMID 42455781]

Elimination, surveillance and disease status

- **Programme costing under NTEP confirms how heavily direct staff salary dominates a public facility's cost** — mean annual cost of delivering NTEP services in northern India was US\$5,243 at a community health centre and US\$1,032 at a primary health centre, with human resources contributing 72.9% and 85.9% of the total respectively, and one-way sensitivity analysis showing human-resource cost as the largest influence on cost per treated case. [PMID 37100581]
- **On the patient side, catastrophic cost (>20% of annual household income) affected 4% of a Gujarat cohort for tuberculosis alone but rose to 12% with HIV co-infection and 5% with diabetes comorbidity**; against that burden Nikshay Poshan Yojana now pays ₹1,000 per month, doubled from the earlier ₹500, although a scoping review of 16 implementation studies found the first instalment still delayed by a median of 43 days to 5.2 months, with absence of a bank account the most connected barrier and the amount itself still perceived as inadequate. [PMID 39075416; PMID 41209996]

Guideline and legislative updates

- **The Indian Reference Case is the standard now expected for costing a health intervention in India.** It prescribes an *abridged societal perspective* — all direct and indirect costs falling on both the health system and the patient, but **productivity loss deliberately excluded** — with 3% discounting of costs and outcomes. An audit of 33 HTAIn studies found decision problem (95.6%) and comparator (94.9%) well reported but **equity (25.7%) and heterogeneity (24.3%) the weakest domains**; 84.8% of interventions were judged cost-effective at a 1× GDP-per-capita

threshold, a rule of thumb WHO now cautions against using on its own because it ignores opportunity cost and budget constraints. [PMID 41316276]

- **India now has its first empirical cost-effectiveness threshold.** Willingness to pay per quality-adjusted life-year, from 5,460 respondents across six states, was ₹2,12,307 (about US\$2,535), between 1 and 1.52 times per-capita GDP. Investment in the HTAIn machinery itself returns about 9:1, rising to 71:1 if its recommendations were fully implemented — the practical argument for why hospital cost accounting is worth its administrative overhead. [PMID 41176029; PMID 40113231]

Global and WHO developments

- **WHO's End TB target of zero catastrophic costs defines the indicator as total costs — direct medical plus direct non-medical plus indirect — exceeding 20% of annual household income or expenditure,** and Thailand's first national patient-cost survey shows the non-medical and indirect components dominating even where treatment is free at the point of use: mean cost per TB episode was US\$903, of which direct non-medical was 44.5% and productivity loss 43.6%, while direct medical was only 12%; 29.9% of households crossed the threshold, against a global pooled figure of 48% across 27 surveyed countries. [PMID 38755216]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 23, Health Planning and Management); Sunder Lal — Textbook of Community Medicine (Ch 2, Ch 16, Ch 17); Health Economics (Ch 14, Counting the costs; Ch 17); Reinke — Health Planning for Effective Management (Ch 12, Financial Analysis; Ch 20, Cost-Benefit and Cost-Effectiveness Analysis of Health Programs); Health Program Planning and Evaluation (Ch 8); Oxford Textbook of Public Health 7e (Ch 49, Economic appraisal in public healthcare); National Sample Survey 75th round on Social Consumption: Health, 2017-18. Coverage note: Park 28e covers hospital cost accounting only briefly, so the cost-centre and overhead-apportionment content comes from Health Economics Ch 14, Reinke Ch 12 and Health Program Planning and Evaluation Ch 8.

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Geographical information system (GIS)

Asked in: Paper IV 04-Aug-2021 — RGUHS MD Community Medicine

Definition — A geographical information system (GIS) is a set of computerised procedures to **input, store, retrieve, manipulate, analyse and output data carrying spatial (locational) attributes**, the result usually presented as a map (Nelson & Williams — Infectious Disease Epidemiology, Ch 7). In public health it is hardware and software that **combines high-precision location information with health databases and analysis tools**, so morbidity, mortality, prevalence, incidence and coverage can be visualised geographically (Oxford Textbook of Public Health 7e, Ch 24). What sets it apart from any other database is that it stores the **spatial relationships between features** and provides methods to study them.

Components and data structure of a GIS

Component	Content
Hardware and software	Computer hardware plus GIS software carrying a mapping module and spatial-analysis tools (Oxford Textbook of Public Health 7e, Ch 24; Handbook of Epidemiology, Ch 33)
Spatial (locational) data	The base map — physical layout of land, towns, buildings, roads, administrative boundaries, postal/PIN codes, census tracts (Handbook of Epidemiology, Ch 33)
Attribute (health) data	Cases, deaths, prevalence, coverage, denominators, facilities, manpower, land use, rainfall — linked to the map through a common key field in a relational database (Nelson & Williams, Ch 7)
Remotely sensed data	Satellite/aerial imagery giving updated environmental data; Landsat Thematic Mapper scans 7 spectral bands, ~30 m resolution, every 16 days; AVHRR scans 5 bands, 1.1 km resolution, twice daily (Nelson & Williams, Ch 7)
People and procedures	Trained personnel for geocoding, interpretation of imagery and quality control; interpretation of remotely sensed data requires substantial training outside GIS itself (Nelson & Williams, Ch 7)

Feature	Vector format	Raster format
Representation	Points, lines, polygons	Uniform grid of cells (rows × columns)
Locating a feature	Exact <i>x, y</i> co-ordinates	Row/column address of a cell
Health example	Point = subcentre or case; line = road/canal; polygon = village or PHC catchment	Satellite image; each cell records reflected light for one patch of earth
Assumption	Area inside a polygon is homogeneous	Feature edges are "fuzzy"; accuracy = grid resolution
Trade-off	Compact files; boundaries look sharper than reality	Finer resolution means far larger database and computing load

(Nelson & Williams — Infectious Disease Epidemiology, Ch 7)

Core functions of a GIS in public health

- ▶ **Collection and storage** — any variable tied to a place (address, village, PIN code, census tract, district) can be entered and held.
- ▶ **Management** — a relational structure links disease register, census, environment and facility files through a common key, and exposes errors such as one patient registered twice by a private and a public provider.
- ▶ **Querying** — the commonest use: how many cases lie in an area, how far from a suspected source, how close to one another ("problem of adjacency").
- ▶ **Modelling** — covariates around cases and controls are analysed statistically (e.g. logistic regression) and the fitted risk mapped across the area as a **predictive risk map**.
- ▶ **Decision making** — the risk map directs limited resources to high-risk pockets. (All five: Nelson & Williams, Ch 7.)

Use of GIS in various health programmes

Programme / area	What GIS contributes
Disease surveillance	Extends the traditional surveillance map into geospatial analysis of disease against locations of risk exposures and of preventive or therapeutic services; detects and characterises clusters and supports syndromic/early-warning systems (Oxford Textbook of Public Health 7e, Ch 42)
Outbreak investigation	Spot maps of individual cases and shaded (choropleth) maps of rates; sequences of maps show a changing distribution over time (Maxcy-Rosenau-Last 15e, Ch 2). John Snow's Broad Street map is the prototype — a manually constructed GIS (Nelson & Williams, Ch 7)
Vector-borne disease control	Combines rainfall, temperature, land use, settlement pattern and permanent water sources with case data to build malaria risk maps ; combined GIS and remote sensing are the emerging tools for defining epidemic-prone areas for targeted control and resource allocation (Gilles & Warrell — Essential Malariology 4e, Ch 5)
Neglected tropical disease / helminth control	Stores dense ecological survey data and queries it to locate the few transmission foci, so that snail or vector control can be concentrated where it matters (Nelson & Williams, Ch 7)
Sexually transmitted infection and HIV programmes	Geocoding of cases merged with census denominators yields area-specific incidence rates and high-risk area maps; mapping of vulnerable populations locates them, sizes them and tracks their movement for evidence-based prioritisation (Nelson & Williams, Ch 7; Jugal Kishore — National Health Programmes of India)
Immunisation programmes	Local working maps for house-to-house campaigns show booth sites, team movement and high-risk pockets — migrant clusters, unauthorised colonies, areas with poor routine immunisation (Sunder Lal, Ch 6; Jugal Kishore)
Health planning and management	Overlaying disease rates on the location of services shows that the highest rates of chronic disease often lie where service density is lowest, guiding facility siting and accessibility analysis (Health Program Planning and Evaluation, Ch 4)

Programme / area	What GIS contributes
Environmental health	Small-area, correlational and ecological GIS studies have demonstrated the concentration of air pollutants, hazardous waste sites, water pollution and lead exposure in deprived neighbourhoods (Maxcy-Rosenau-Last 15e, Ch 42)
Research and trials	Identifies a high-incidence population so that a field trial can be run with far fewer participants than random population-based recruitment (Nelson & Williams, Ch 7)

Worked programme examples

- **Schistosomiasis control, Egypt** — the Schistosomiasis Research Program (Ministry of Health with USAID) sampled six irrigation-canal sites monthly for over a year, recording **84 variables every 50 metres**. A GIS query for the snail host *Biomphalaria* found heavy infestation at only **9 of 1450 sites**, showing that focused snail control at a handful of locations could substantially cut the host population.
- **Sexually transmitted infection programme, Baltimore** — cases from public clinics and private providers were geocoded, validated and merged with census tract data to generate rate maps and identify high-risk areas. The GIS also exposed a reporting gap: the required field was reported for **99% of cases by public clinics but only 72% by private providers**.
- **Risk mapping and trial targeting, Lyme disease** — land use, soil, distance to forest, elevation and drainage around case and control residences were regression-weighted into a risk surface; cases proved **16 times more likely in high-risk than low-risk areas**. A vaccine trial needing a seasonal incidence of **500 per 100,000** (80% assumed efficacy, 4000 per arm) against a state crude incidence of about **9 per 100,000 (1995)** recruited only from high-risk streets; control-group incidence reached **2% per year**, allowing efficacy to be shown with a small sample.
- **Malaria stratification** — remote-sensed settlement, land-use, river and permanent-water data at about **1 × 1 km resolution** are combined with meteorological data into provisional risk maps, though rainfall-and-temperature maps ignore many other spatial determinants.

(1–3: Nelson & Williams — *Infectious Disease Epidemiology*, Ch 7; 4: Gilles & Warrell — *Essential Malariology 4e*, Ch 5)

GIS in India's health information system and national programmes

- **Facility mapping** — an electronic database of health facilities, training centres and other establishments is being mapped by GIS for dissemination through the Central Bureau of Health Intelligence (CBHI) website; the **National Informatics Centre provides GIS mapping of government health facilities**.
- **GIS-enabled Health Management Information System (HMIS)** — launched **March 2016** to display HMIS data on dynamic maps, with **grading of community health centres and primary health centres** and automatic alerts for sensitive data items.
- **Other national priorities** — the same approach is applied to endemic **fluorosis and safe water sources**. (1–3: Sunder Lal, Ch 8.)
- **Social mapping at field level** — health workers draw local-area working maps for mass campaigns such as pulse polio and malaria, marking facilities, water sources and **high-risk hot**

spots (Sunder Lal, Ch 6). Polio micro-planning used exactly this: booth siting, team movement, cold-chain points, and identification of high-risk areas — wild-virus or compatible areas, migrant clusters, unauthorised colonies, areas of poor routine immunisation (Jugal Kishore).

- ▶ **HIV programme** — vulnerable populations mapped for location, size and movement; **30 states** had completed detailed mapping to earmark priority areas (Jugal Kishore).

Limitations and cautions

- ▶ **Spatial error** — address-matching interpolates a house number within a block range, displacing mapped residences.
- ▶ **Resolution error** — every database has a minimal mapping unit; smaller features do not appear.
- ▶ **Interpretive error** — ground-truthing of remotely sensed land cover shows misclassification of 15–20%.
- ▶ **Temporal error** — land cover changes within years and rainfall within hours, so a database may not describe conditions at the time of study.
- ▶ **Wrong place plotted** — place of exposure is often unknown; for long-latency conditions (HIV, AIDS, tuberculosis) residence or workplace misleads, and surveillance records only where the patient was diagnosed (Giesecke).
- ▶ **Over-interpretation** — mapping is never more than an initial step; bright colour palettes "silence a statistical conscience" about chance differences in raw data (Handbook of Epidemiology, Ch 33).
- ▶ **Unstable small-area rates** — with a small denominator a tiny numerator change swings the rate (Health Program Planning and Evaluation, Ch 4).
- ▶ **Missing data** — updated spatial databases do not exist in much of the world, hence reliance on remote sensing. (1–4, 8: Nelson & Williams, Ch 7.)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — the GIS "layer cake"** (draw stacked transparent sheets over one base map, top to bottom). Labels: *Output thematic/risk map*; *Disease layer (geocoded cases — points)*; *Population layer (census polygons, denominators)*; *Health facility layer (subcentre, PHC, CHC — points)*; *Environment layer (land use, water bodies, rainfall — raster)*; *Infrastructure layer (roads, rivers, canals — lines)*; *Base map (administrative boundaries — polygons)*.

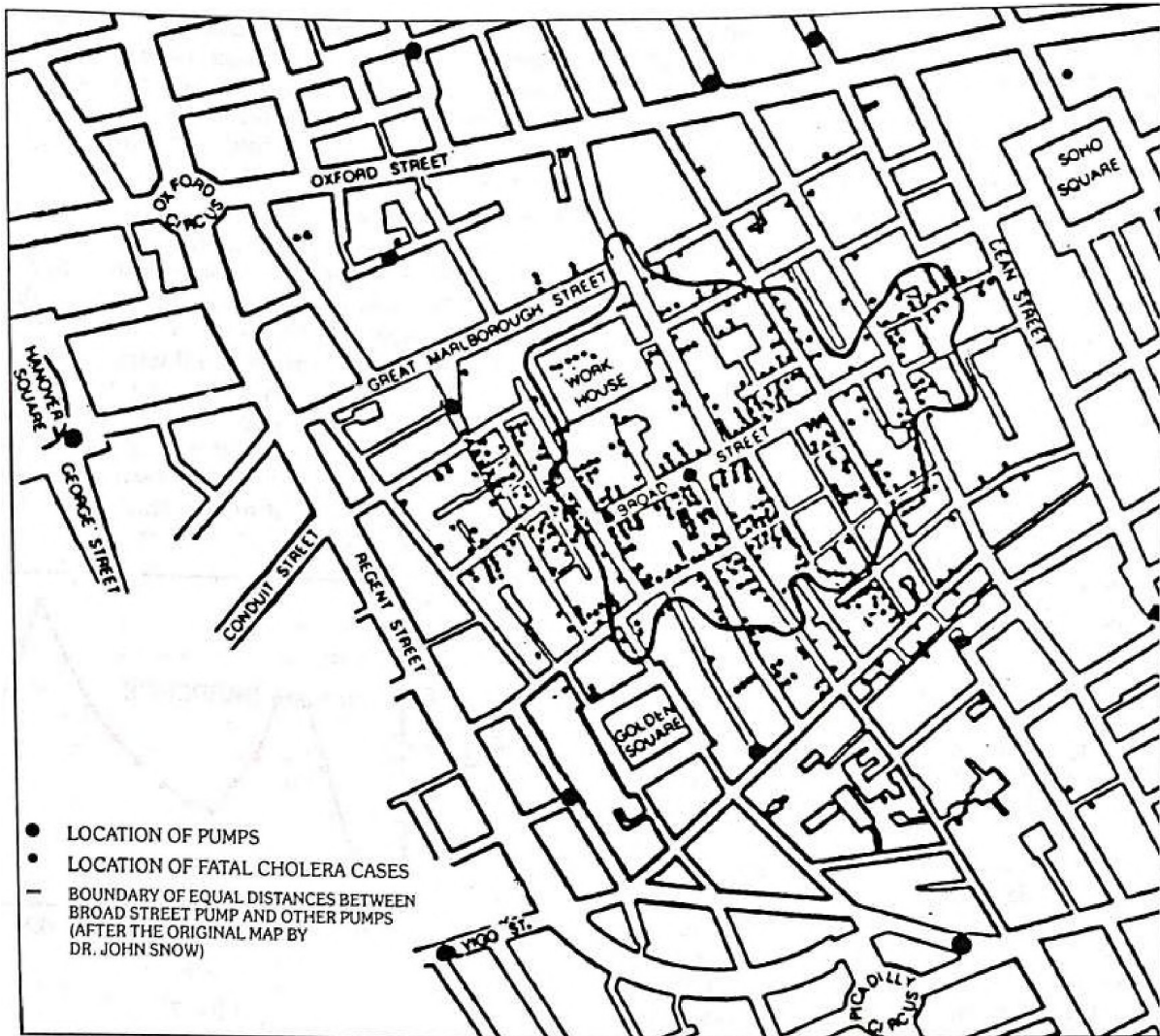
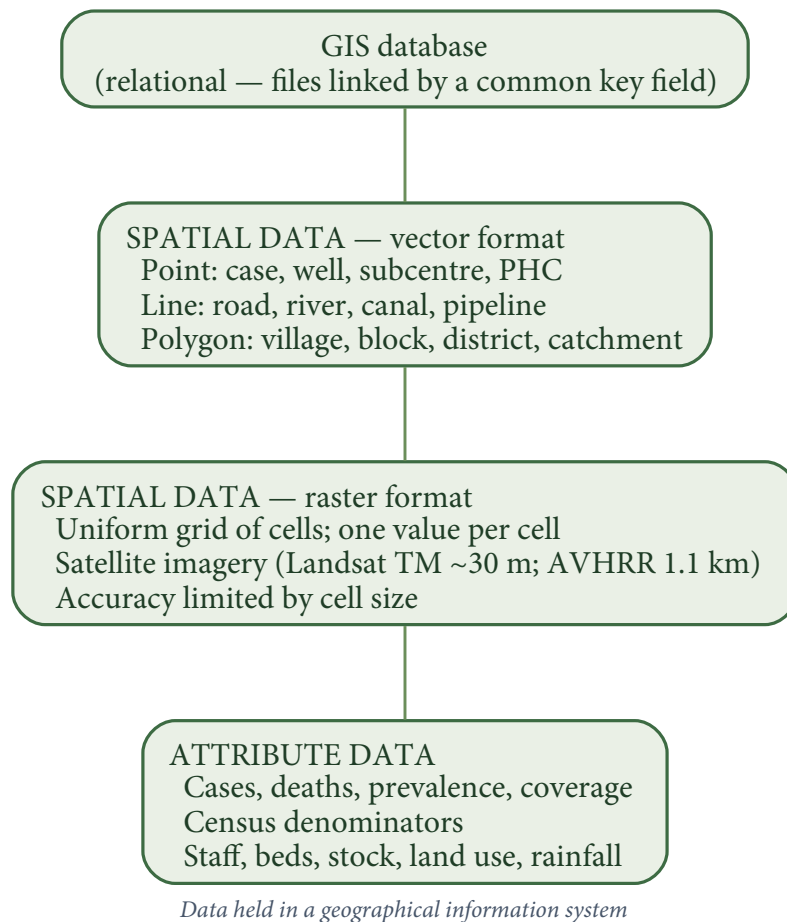
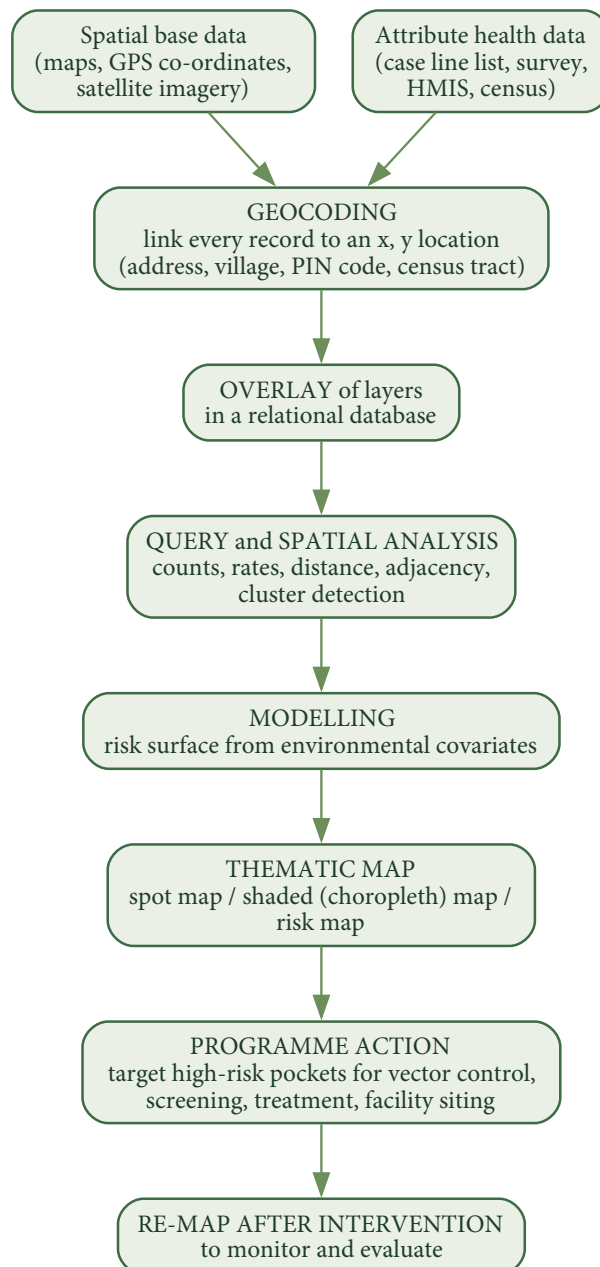


Fig 6 — Spot map of Asiatic cholera in London (Park's Textbook of Preventive and Social Medicine, 28e, p.87)





How a geographical information system converts health data into programme action

High-yield exam pointers

- ▶ **Definition to write first:** procedures to input, store, retrieve, manipulate, analyse and output data having **spatial attributes**, output as maps.
- ▶ **Five functions (S-M-Q-M-D):** Store → Manage → Query → Model → Decide.
- ▶ **Two data formats:** vector (point, line, polygon) versus raster (grid cells); attribute data linked by a **common key field**.
- ▶ **Indian specifics with dates:** GIS-enabled HMIS launched **March 2016**; NIC maps government health facilities; CBHI website hosts the database; grading of CHCs/PHCs.
- ▶ **Prototype:** John Snow's Broad Street spot map — a manually constructed GIS; it summarised and persuaded, it did not generate his waterborne theory.

- **Numbers:** Landsat TM ~30 m/16 days; AVHRR 1.1 km/twice daily; land-cover error 15–20%; Lyme high-risk areas carried **16 times** the risk.
- **Always end with limitations** — a map is a first step, not proof.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook of Preventive and Social Medicine 28e carries no dedicated section on geographical information systems, so this answer is grounded chiefly on the GIS chapter of Nelson & Williams, the Oxford Textbook of Public Health 7e and the Indian material in Sunder Lal and Jugal Kishore.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, MoHFW, IDSP-IHIP, NFHS-6, NCVBDC, WHO subnational tailoring reference manual 2025

Programme and policy updates (India)

- **Integrated Disease Surveillance Programme has moved from paper forms to a geospatial platform** — the Integrated Health Information Platform (IHIP) now carries case-based, near-real-time surveillance on **GIS-enabled dashboards with geo-tagged health facilities**, with built-in feedback and alert systems and integration with other national programmes; this is the operational descendant of the hand-drawn spot map, and is the single most quotable Indian example of GIS inside a national programme [MoHFW 2024-Nov].
- **Health and Wellness Centres have been renamed Ayushman Arogya Mandir** — more than **187,000 operational** with cumulative footfall above 5,000 million, and the ministry attributes the rise in first-trimester antenatal registration (70.0% to **76.2%**) explicitly to reduced *geographical barriers* in rural and tribal terrain; immunisation is now tracked on the geo-referenced **U-WIN** platform (173.9 million registered beneficiaries, 753.6 million doses as on 2 June 2026) [MoHFW 2026-May], although a 2026 review of Co-WIN and the Ayushman Bharat Digital Mission records that Co-WIN's smartphone-dependent registration systematically excluded elderly people, rural residents and those without devices or digital literacy [PMID 42215992] — the corollary for GIS being that a map drawn only from platform data under-counts exactly the pockets it should be targeting.

Current national data

- **NFHS-6 (2023–24) is the current round — quote it by name and year.** Fact sheets released **29 May 2026**; about **679,000 households across 715 districts**, with IIPS as nodal agency: full immunisation **87.1%** (NFHS-5 83.8%), institutional delivery **90.6%**, TFR **2.0**, stunting **29.3%** (35.5%), rotavirus vaccine **85.4%** (36.4%), health insurance coverage **60.2%** (41.0%). Because estimates are produced down to district level, NFHS is the standard attribute layer for district choropleth and hotspot maps [MoHFW 2026-May].
- **NFHS district data are already being mapped, not merely tabulated** — anaemia among rural men across 707 districts gave Moran's I **0.66** with **97 high-high hotspot districts** clustered in West Bengal, Jharkhand, Odisha, lower Assam and the Madhya Pradesh–Maharashtra border, against **96 cold-spot districts** mainly in the south [PMID 38243230]; in the north-eastern states neonatal mortality fell from **45.1 to 21.1 per 1,000 live births** between NFHS-1 and NFHS-5,

Getis-Ord Gi* placed the sharpest cluster in Hailakandi (Assam), and geographically weighted regression identified fewer than five antenatal visits as the strongest spatial correlate [PMID 40302668].

- **A systematic review of GIS in Indian health research (58 studies, 2000–2022)** grouped the work into disease surveillance and mapping, health risk assessment, and healthcare access; the commonest methods were inverse distance weighting (13 studies), Getis-Ord Gi* hotspot analysis (11), Moran's I (9) and overlay analysis (6), with ArcGIS used in 40 studies and SaTScan for cluster detection. It concluded that GIS is still not used to its full potential in Indian health systems and that ethical handling of geolocated health data remains an unaddressed gap — a ready-made "limitations" point [PMID 39574096].

Elimination, surveillance and disease status

- **Elimination targets are now written in spatial units, which makes mapping the monitoring tool rather than an illustration** — kala-azar elimination is defined as fewer than **one case per 10,000 population at block level**, with programme support concentrated in 33 districts of Bihar, 4 of Jharkhand and 11 of West Bengal [NCVBDC 2025]; GIS analysis of national malaria programme data showed reported cases falling from **1.17 million in 2015 to 0.18 million in 2022, 10.1- and 18.7-fold below the WHO estimates of 11.93 million and 3.38 million** for those two years — mapped state by state (Mizoram, Meghalaya, Tripura, Odisha, Chhattisgarh and Jharkhand carrying the persistent load), the authors themselves flagging the absence of district-level mapping as a limitation of their own work. That admission is the GIS lesson in miniature: the unit you map is the unit you can act on, and a state-level map cannot direct a primary health centre [PMID 38772359].
- **Karnataka worked example, directly usable in an RGUHS answer** — GIS analysis of **38,229 dengue and 8,094 chikungunya cases across 31 districts (2021–2024)** showed dengue rising from 5,619 cases in 2021 to **17,373 in 2024**, with the highest 2024 incidence in Chikkamagaluru (0.58 per 1,000), Mandya (0.47) and Udupi (0.45); spatial autocorrelation and cluster analysis identified persistent hotspots, and the authors argue for spatially targeted fogging and source reduction in place of uniform state-wide control [PMID 40836677].
- **Hotspot-guided active case finding measurably outperforms untargeted screening** — across 14 high-burden states of Nigeria, screening in 121 geospatially mapped tuberculosis hotspot wards versus 213 non-hotspot wards (3.99 million people screened) yielded **3.76 versus 2.14 confirmed cases per 10,000** ($p < 0.001$), precision 10.5% versus 7.5%, and a **number-needed-to-screen of 146 versus 193 per 10,000** [PMID 36753328]; within India, spatial analysis of NFHS-5 data detected **62 paediatric tuberculosis hotspots** nested within districts and states, supporting district-targeted rather than uniform NTEP effort [PMID 40023534].

Guideline and legislative updates

- **WHO's subnational tailoring reference manual (13 October 2025)** converts geospatial stratification from good practice into a formal planning step: national programmes are directed to use local data, stratification, mathematical modelling and cost-effectiveness analysis to select a *different* intervention mix for each subnational unit instead of one uniform national package. This is the doctrinal shift a modern GIS answer should cite — the map now determines the intervention, not merely displays the disease [WHO 2025-Oct].

Global and WHO developments

- Geospatial access metrics are entering health-system planning, and prediction is being augmented by machine learning — an analysis of the eight Empowered Action Group states (661 million people) found **25.7% living within 30 minutes' drive of a blood bank, 61.4% within 60 minutes and 92.5% within 90 minutes**, with only **0.68 units per 1,000 population** available inside the 60-minute catchment against the WHO benchmark of 10 donations per 1,000 and the Lancet Commission on Global Surgery target of 15 units per 1,000, leading the authors to argue that governments should report time-bound geographic access rather than national donation rates alone [PMID 39433405]; and a 2026 malaria risk-zonation model for flood-prone districts of **Pakistan**, combining eleven environmental, meteorological and socio-demographic layers over 250 union-council test-positivity records (2014–2024), reached **93.3% accuracy (kappa 0.86)** with a random forest and resolved localised hotspots better than the analytical hierarchy process weighted overlay [PMID 42302577].

SOURCE & COVERAGE

Nelson & Williams — Infectious Disease Epidemiology (Ch 7, Geographic Information Systems); Oxford Textbook of Public Health 7e (Ch 24; Ch 42, Public health surveillance); Sunder Lal — Textbook of Community Medicine (Ch 6, Ch 8); Jugal Kishore — National Health Programmes of India; Gilles & Warrell — Essential Malariology 4e (Ch 5); Handbook of Epidemiology, Ahrens & Pigeot (Ch 21, Ch 33); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 2, Ch 42); Health Program Planning and Evaluation (Ch 4); Giesecke — Modern Infectious Disease Epidemiology; Park's Textbook of Preventive and Social Medicine 28e (figure, p 87).

Recent-advances sources: 39574096 — Applications of geographical information system and spatial analysis in Indian health research: a systematic review, BMC Health Serv Res 2024, [DOI](#); 38243230 — A district-level geospatial analysis of anaemia prevalence among rural men in India, 2019-21, Int J Equity Health 2024, [DOI](#); 40302668 — Geospatial analysis of neonatal mortality in north-eastern India: a multilevel Bayesian approach, Epidemiol Health 2025, [DOI](#); 40836677 — Spatial epidemiology of dengue and chikungunya in Karnataka using GIS-based analysis, Glob Health Action 2025, [DOI](#); 38772359 — Spatiotemporal Epidemiology of Malaria in India from 2007 to 2022, Am J Trop Med Hyg 2024, [DOI](#) (PMC record returns abstract only; analysis and all maps are state/UT-level, district-level mapping named as a limitation by the authors); 40023534 — Spatial patterns and multilevel analysis of factors associated with paediatric tuberculosis in India, Indian J Tuberc 2024, [DOI](#) (abstract-only); 36753328 — Identifying Hot Spots of Tuberculosis in Nigeria Using an Early Warning Outbreak Recognition System, JMIR Public Health Surveill 2023, [DOI](#) (Nigeria); 39433405 — Defining blood deserts and access to blood products for 660 million people: a geospatial analysis of eight states in Northern India, BMJ Glob Health 2024, [DOI](#); 42215992 — Toward equitable digital health: an integrated framework addressing exclusion, ethics, and implementation across healthcare systems, Int J Equity Health 2026, [DOI](#); 42302577 — Geospatial and machine learning approaches for malaria risk mapping in flood-prone districts, Public Health 2026, [DOI](#) (abstract-only; Pakistan). MoHFW — Integrated Health Information Platform (IHIP) GIS-enabled surveillance dashboards and geo-tagged health facilities, November 2024. MoHFW/IIPS — National Family Health Survey 6 (2023–24) fact sheets, released 29 May 2026. NCVBDC — National Kala-Azar Elimination Programme, block-level elimination criterion, 2025. WHO — Subnational tailoring of malaria strategies and interventions: reference manual, 13 October 2025.



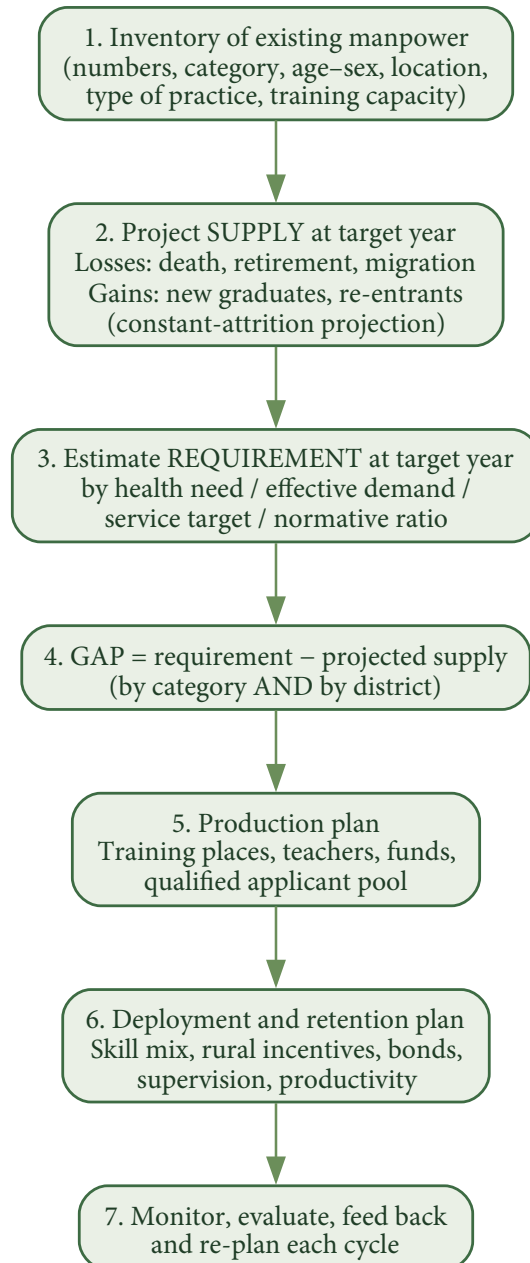
Health manpower planning

Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — Health manpower planning is the component of health planning that estimates the **number, categories, competencies and geographical distribution** of health personnel a country will need at a defined future date, and matches that requirement against the supply which existing training, deployment and retention arrangements will actually deliver. The term **health manpower** (now usually written *human resources for health*) includes both **professional and auxiliary** health personnel; the World Health Organization (WHO) defines an *auxiliary* as a worker in a certain field with less than full professional qualification (Park 28e, Ch 10, p 724). Manpower requirements of a country are based on (i) **the health needs and demands of the population** and (ii) **the desired outputs**, and are conventionally expressed as ratios such as the doctor–population ratio (Park 28e, Ch 10, p 724).

Why manpower planning dominates health planning

Attribute of the health workforce	Planning implication
Labour-intensive sector — up to two-thirds of health-sector cost may be personnel (Reinke, Ch 10, p 144); human resources consume 60–70% of public health expenditure in low- and middle-income countries (Oxford 7e, Ch 3, p 223)	The single largest recurring budget head; a manpower error is also the costliest financial error
Long production lead time — "Hospitals can be built in months; it takes a decade to train a doctor" (Reinke, Ch 10, p 144)	Decisions must be taken 10–15 years ahead; a wrong decision cannot be corrected within one plan period
Two planning levels — macro (enough workers to meet, but not exceed, future effective demand) and micro (functions and task assignment of those workers) (Reinke, Ch 10, p 144)	Asking "how many nurses do we need?" is idle until service loads and skill mix are defined (Reinke, Ch 19, pp 272–273)
Facilities are meaningless unless staffed (Reinke, Ch 21, p 289)	Unstaffed health centres and empty hospitals stand as monuments to inadequate personnel planning (Reinke, Ch 10, p 144)
Over-reaction risk — the more severe the shortage, the more drastic the remedy and the more serious the eventual surplus (Reinke, Ch 19, p 261)	Shortage and surplus alternate unless supply is projected, not guessed
It is not only a mathematical exercise — policy decisions and resistance to change are inherent, because careers and lives are at stake (Reinke, Ch 10, p 157)	Political feasibility must be planned for alongside the arithmetic



Steps in the health manpower planning cycle (requirement–supply–gap–production–deployment)

Approaches to estimating manpower requirements

Approach	Basis of the estimate	Critical limitation
Health-need approach	Professionally assessed disease burden converted into services, then into worker-hours	Morbidity cannot be neatly equated to a worker's time; need is not a measure of public demand — clinics stand unused in areas of continuing need (Reinke, Ch 10, pp 151–152)
Effective economic demand	Random sample survey of utilisation patterns by demographic and socio-economic group, where a private sector finances care	Reproduces existing inequity — the poor cannot express need as economic demand (Reinke, Ch 10, pp 152–153)

Approach	Basis of the estimate	Critical limitation
Status quo (normative) approach	Existing professional-to-population ratios enshrined as norms — the commonest method	Never tests whether the present ratio is <i>suitable</i> ; ignores technological and demographic change. Its "Parkinsonian" variant argues "our doctors are busy, so we need more" (Reinke, Ch 10, p 152)
Comparative method	Adopts the ratio of a reference country	Very few countries are truly comparable (Reinke, Ch 10, p 152)
Expert opinion	Panel judgement	Subjective; but often the best available for small specialised cadres — health education, sanitary engineering (Reinke, Ch 10, p 152)
Established standards of practice	Observed norms of practice — e.g. in the Soviet model one out-patient doctor saw 5 patients/hour, 5.5 hours/day, 240 days/year $\approx$ 6,600 visits/year, and doctors were allocated against observed visits per 1,000 population	Complex and exacting to operate (Reinke, Ch 10, pp 152–153)

Because population-to-resource ratios are commonly specified but their realism "is by no means assured", the sounder sequence is to begin with the community's principal health problems, translate these into service requirements, and only then derive personnel requirements (Reinke, Ch 21, p 296).

Health manpower situation in India — the record

(a) The national stock (Park 28e, Ch 10, p 746)

- Qualified allopathic physicians expanded from about 50,000 at Independence to about 13.08 lakhs in 2022.
- The doctor–population ratio for the country as a whole was 1 : 836 (2022).
- General practitioners constitute 70 per cent of the medical profession; most tend to congregate in urban areas and provide mainly curative services available to those who can pay.
- Ayurvedic physicians alone are estimated at about 7.73 lakhs, and nearly 90 per cent of them serve rural areas, most being local residents socially and culturally close to the people — so the largest rural-serving clinical cadre sits outside the allopathic system that manpower plans usually count.

(b) Suggested Indian norms for health personnel (Park 28e, Ch 10, pp 724–725)

Category of personnel	Norm suggested
Nurse	1 per 5,000 population
Health worker, female and male (multipurpose worker)	1 per 5,000 population in plain areas; 1 per 3,000 in tribal and hilly areas
Trained <i>dai</i>	One for each village
Health assistant, male and female	1 per 30,000 in plain areas; 1 per 20,000 in tribal and hilly areas — provides supportive supervision to 6 health workers

Category of personnel	Norm suggested
Pharmacist	1 per 10,000 population
Laboratory technician	1 per 10,000 population
Accredited social health activist (ASHA)	1 per 1,000 population

(c) India against her neighbours (Park 28e, Ch 10, p 725, Table 3, data 2013–2021)

Country	Doctors per 10,000	Beds per 10,000	Nurses and midwives per 10,000
India	7.3	7.0	17.3
Bangladesh	6.7	8.0	6.1
Sri Lanka	11.9	36.0	24.4
Thailand	9.3	21.0	17.7
Myanmar	7.5	9.0	11.0

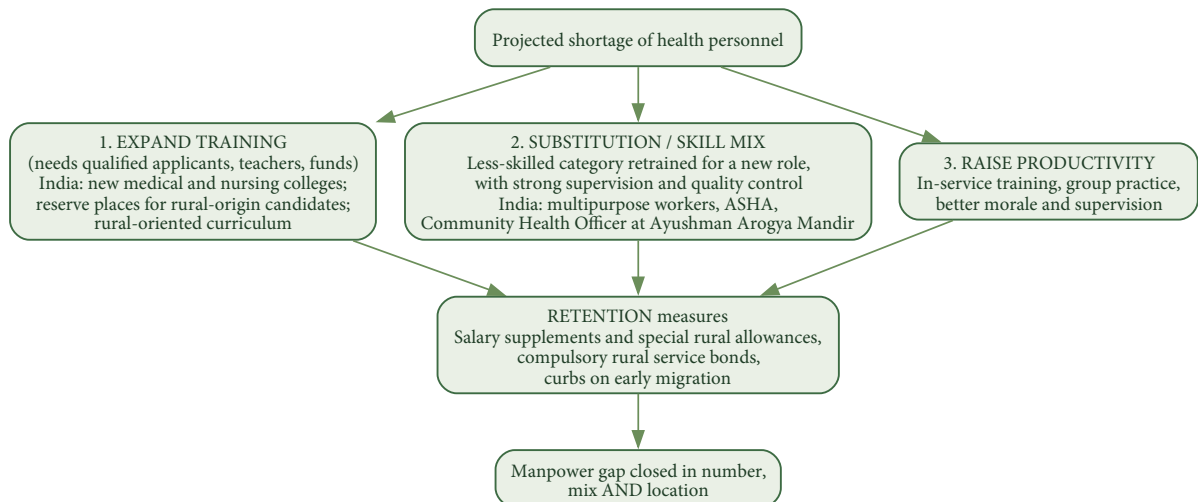
(d) Manpower in position, March 2021 (Park 28e, Ch 10, p 725, Table 4)

Cadre and level	Rural	Urban
Health workers (male + female) at sub-centres	2,39,911	—
Doctors / medical officers at primary health centres (PHCs)	32,901	7,682
Health worker female at PHCs	27,003	21,623
Staff nurses at PHCs	37,736	10,196
Pharmacists at PHCs	20,118	5,466
Laboratory technicians at PHCs	15,026	4,813
Surgeons at community health centres (CHCs)	913	—
Obstetricians and gynaecologists at CHCs	1,442	—
Physicians at CHCs	992	—
Paediatricians at CHCs	1,066	—
Doctors and specialists at sub-district hospitals	17,723	
Doctors and specialists at district hospitals	27,304	
Paramedical staff at district hospitals	92,884	

Critical comment on the Indian situation

Fault line	Evidence and comment
Averages look adequate; distribution does not	"Although the averages are satisfactory on a national basis, they vary widely within the country" (Park 28e, Ch 10, p 725). Adding India's 7.3 doctors and 17.3 nurses/midwives per 10,000 gives 24.6 (an addition made here, not printed in Park) — marginally above the threshold of 23 doctors, nurses and midwives per 10,000 population needed to deliver essential maternal and child health services (Oxford 7e, Ch 86, p 3263). The arithmetic passes; the service does not
Rural–urban maldistribution	Studies in India show a concentration of doctors of up to 73.6% in urban areas, where only 36% of the population lives (Park 28e, Ch 10, p 725)
Why doctors will not stay rural	Absence of amenities, lack of job satisfaction, professional isolation, lack of rural experience, and inability to adjust to rural life (Park 28e, Ch 10, p 725)
Specialist collapse at the community health centre	A CHC is established to be manned by four medical specialists — surgeon, physician, gynaecologist and paediatrician — supported by 21 paramedical and other staff , with 30 in-door beds, serving as referral centre for four PHCs; Indian Public Health Standards raise this to 6 specialists including an anaesthetist and an eye surgeon (Park 28e, Ch 8, p 611). Against 5,491 rural CHCs (Park 28e, Ch 10, p 725), only $913 + 1,442 + 992 + 1,066 \approx 4,400$ specialists were in position against roughly 22,000 required on the four-specialist establishment. First-referral care therefore exists on paper only in most blocks
Tribal and remote areas worst hit	In about half the states, health institutions in tribal areas were deficient in number by 27–40% against present norms, and the human-resource gap is attributed to limited professional growth, social and professional isolation, weak human-resource policies and poor working conditions (Park 28e, Ch 13, pp 850–851)
Skill mix flatters India by ratio, not by density	India's nurses-and-midwives-to-doctors ratio works out to about 2.4 : 1 ($17.3 \div 7.3$) — the highest of the five countries in Park's own table, so the ratio itself is not the defect. The defect is absolute density : 7.3 doctors, 17.3 nurses/midwives and only 7.0 beds per 10,000, against Sri Lanka's 11.9, 24.4 and 36.0 (Park 28e, Ch 10, p 725, Table 3, 2013–2021). Indian establishments are moreover fixed by the status-quo/normative method rather than by measured service loads — which the National Health Policy 2017 itself proposes abandoning, shifting human-resource development "from normative approach to targeted approach to reach under-served areas with a time-to-care approach" (Park 28e, Ch 10, p 728)
Curative and urban bias of production	The obstacles to implementing primary health care in India are named as shortage of health manpower, entrenchment of a curative culture, and high concentration of health services and personnel in urban areas (Park 28e, Ch 2, p 49)
Requirements keep shifting	Manpower requirements change qualitatively and quantitatively as new programmes and philosophies arrive — the shift from unipurpose to multipurpose workers, Health for All, and disease-control programmes each demanded new categories (Park 28e, Ch 10, p 725)

Fault line	Evidence and comment
Weak information base	Stock estimates depend on State medical, dental and nursing council registers, Directorates of Medical Education, the Census and the Institute of Applied Manpower Research (Park 28e, Ch 21, p 998); registers count <i>registration</i> , not <i>active practice</i> . Park's own two figures do not agree — a register-based 1 : 836 doctor–population ratio for 2022 (Ch 10, p 746) against a WHO-based 7.3 doctors per 10,000 , i.e. about 1 : 1,370, for 2013–2021 (Ch 10, p 725). A discrepancy of that size in the denominator makes every Indian manpower target soft



The three routes to averting a health manpower shortage, with their Indian applications

The Indian expression of route 2 is the strongest current corrective: the primary health-care team at the sub-health-centre-level **Ayushman Arogya Mandir** is headed by a **community health officer (CHO)** — a B.Sc. or general nursing and midwifery nurse, or an Ayurveda practitioner, trained in primary care and public health skills and certified through a six-month certificate programme in community health, supported by multipurpose workers and ASHAs; **Ayushman Arogya Mandirs of all types numbered 1,72,148 operational as on 31 March 2024** (Park 28e, Ch 7, p 570). For tribal districts the Expert Committee on Tribal Health additionally recommended creating mid-level care providers for sub-centres, substantially raising total salary to attract doctors, and dedicated medical colleges in tribal districts for tribal students (Park 28e, Ch 13, p 851).

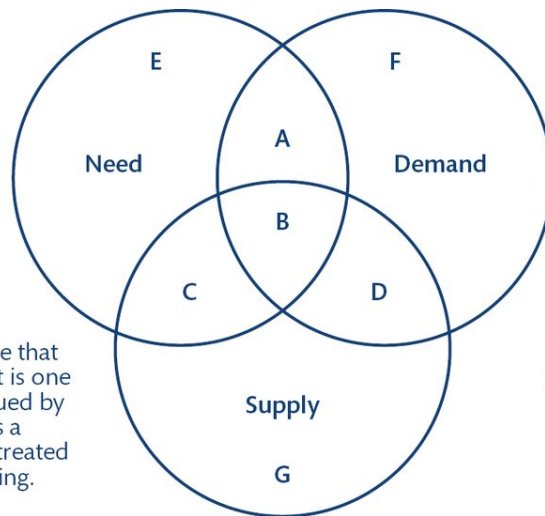
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the three-circle Venn of Need – Demand – Supply, labelling the zones exactly:

- ▶ **Need** circle: E = unmet need not expressed (not recognised, neglected, not demanded)
- ▶ **Demand** circle: F = demand without need
- ▶ **Supply** circle: G = supply meeting neither need nor demand
- ▶ A = need + demand, not supplied (unmet expressed need)
- ▶ B = need + demand + supply (need that is met)
- ▶ C = need + supply, not demanded (service not wanted or valued by patients)
- ▶ D = demand + supply without need (e.g. antibiotics prescribed for a cold)

Zone E represents unmet need that is not expressed for whatever reason (not recognized, neglected, or not demanded).

Zone C represents a service that meets a legitimate need but is one which is not wanted or valued by patients. An example is a terminally ill patient being treated in an acute hospital setting.



The overlap between need and demand represents healthcare needs that are expressed. Zone B represents need which is met while Zone A represents need that is not.

Zone D represents service that are provided to meet a demand but do not meet a healthcare need. A classic example is prescribing antibiotics for a cold.

Need, demand and supply of healthcare (Oxford Textbook of Public Health 7e, Ch 101, p 3879)

High-yield exam pointers

- ▶ **Six approaches, mnemonic "SEE-NEC"**: Status quo (normative), Effective economic demand, Established standards of practice, Need, Expert opinion, Comparative.
- ▶ **India, Park 28e Table 3 (2013–2021)**: doctors 7.3, nurses and midwives 17.3, beds 7.0 per 10,000 population — every figure quoted with this year range.
- ▶ **Registry figures (Park 28e, Ch 10, p 746)**: allopathic physicians ~50,000 at Independence → 13.08 lakhs in 2022; doctor–population ratio 1 : 836; general practitioners 70% of the medical profession; Ayurvedic physicians ~7.73 lakhs, ~90% serving rural areas.
- ▶ **The killer statistic**: up to 73.6% of doctors in urban areas serving 36% of the population (Park 28e, Ch 10, p 725).
- ▶ **Norms to reproduce**: health worker 1 per 5,000 (1 per 3,000 tribal/hilly); health assistant 1 per 30,000 (1 per 20,000 tribal/hilly) supervising 6 workers; ASHA 1 per 1,000; nurse 1 per 5,000; pharmacist and laboratory technician 1 per 10,000 each.
- ▶ **Reinke's three routes out of a shortage**: expand training, substitute skill mix, raise productivity — plus retention (Reinke, Ch 10, pp 155–156).

- **Quotable line:** "Hospitals can be built in months; it takes a decade to train a doctor" (Reinke, Ch 10, p 144).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the term *health manpower* has largely been replaced internationally by *human resources for health*, and the **Health and Wellness Centre** has been renamed **Ayushman Arogya Mandir**; the Medical Council of India was replaced by the National Medical Commission on 25 September 2020 (Park 28e, Ch 23, p 1040), so the regulator named in older answers is out of date. Park 28e's manpower stock table is dated March 2021 and its cross-country table 2013–2021, so quote both with those years. Three numbers above are arithmetic derived here from Park's printed counts rather than printed figures — the community health centre specialist shortfall, the combined density of 24.6 doctors-plus-nurses-and-midwives per 10,000 compared with the Oxford threshold of 23, and the 2.4 : 1 nurse-to-doctor ratio together with the 1 : 1,370 restatement of 7.3 doctors per 10,000.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Global Burden of Disease Study 2019, WHO National Health Workforce Accounts, WHO Health Workforce Support and Safeguards List, MoHFW Ayushman Arogya Mandir portal, Health Dynamics of India 2022–23, National Health Accounts 2021–22, National Ayush Mission

Programme and policy updates (India)

- **The community health officer is now a counted national cadre, and its own shortfall is the current bottleneck.** More than 1.75 lakh Ayushman Arogya Mandirs are operational and **1,16,978 CHOs were in position on 30 November 2024**, leading the sub-health-centre team on clinical, public-health and managerial roles; 12 expanded service packages have been rolled out, with 82 per cent of centres delivering all of them by November 2024, and teams now draw **team-based performance incentives**. Facilities are upgraded against **Indian Public Health Standards 2022**, financed through the **PM-Ayushman Bharat Health Infrastructure Mission** and XV Finance Commission grants routed via Panchayati Raj Institutions and urban local bodies. Against this, a **CHO deficit of about 11,111** persists, with retention failing in remote postings — substitution has produced the post, not yet the person in it [PMID 41816146].
- **Ayush has been converted from an uncounted parallel stock into a planned public cadre.** Under the Ministry of Ayush (created 2014) and the National Ayush Mission, **over 12,500 Ayush Health and Wellness Centres**, now branded Ayushman Arogya Mandirs, are led by qualified Ayurveda, Unani, Siddha, Sowa-Rigpa and homoeopathy doctors; India has **7,55,780 registered Ayush practitioners**, 932 Ayush colleges with an annual intake of roughly 70,000 undergraduates and 7,600 postgraduates, and Ayush services co-located in **7,011 PHCs, 3,051 CHCs and 460 district hospitals**. Park's Ayurvedic-physician figure should now be quoted as the wider Ayush register, and as a cadre inside the public system rather than outside it [PMID 41229518].

Current national data

- **Quote the infrastructure and manpower report by its current name.** *Rural Health Statistics* is now issued as **Health Dynamics of India (Infrastructure and Human Resources)**; the 2022–23

edition counts 1,69,615 sub-centres and 31,882 PHCs, against 1,53,655 and 25,308 in Rural Health Statistics 2014–15. Any sub-centre or PHC count written without that report name and year is dated [PMID 41816146].

- **India's coverage and financial-protection indicators have both moved.** The UHC service coverage index rose from 57 in 2015 to 63 in 2021, above the South-East Asia regional average of 62; out-of-pocket expenditure fell from 60.6 per cent (2014–15) to 39.4 per cent (2021–22) on National Health Accounts estimates; and public-facility outpatient attendance rose from 91.6 crore (2012–13) to 134.5 crore (2021–22) on NSSO data. Rising utilisation against a static establishment is itself a manpower-planning argument [PMID 41816146].
- **State-level audit shows the real Indian shortage is the unfilled sanctioned post, not the national register.** A survey of 377 public facilities and 1,749 providers across 21 districts of Haryana found only 3.6 core health workers (doctors, staff nurses, auxiliary nurse-midwives) per 10,000 population, ranging from 1.35 in Faridabad to 6.57 in Panchkula, with about 78 per cent of sanctioned posts occupied; inpatient load ran at 17 admissions per clinician per month at CHC level and 29 at district-hospital level. District spread, not the national average, is the planning unit [PMID 37843177].

Elimination, surveillance and disease status

- **Surveillance manpower now has an explicit district norm.** The India Epidemic Intelligence Service, begun in 2012 at the National Centre for Disease Control with US-CDC support, works to a target of **one trained field epidemiologist in each of roughly 770 districts**; ICMR's National Institute of Epidemiology added a non-communicable-disease track — a two-year advanced programme (2018) and a one-year intermediate programme (2021) — with state-owned versions in Chhattisgarh and Odisha aiming at one NCD-trained officer per district. The reported failures are of deployment rather than production: graduates are pulled back onto clinical duty and no career pathway follows the training, which is the classic manpower-planning error of producing a category without creating its post [PMID 36480803].

Guideline and legislative updates

- **The regulator set has been rebuilt beyond the National Medical Commission.** The Central Council of Indian Medicine and the Central Council for Homoeopathy were **replaced in 2020 by the National Commission for Indian System of Medicine (NCISM) and the National Commission for Homoeopathy (NCH)**; Indian Public Health Standards were revised in 2022 and now incorporate Ayush; and the National List of Essential Medicines has been extended with Ayush formulations alongside system-specific essential-medicine lists. An answer that updates only MCI to NMC leaves the other two councils out of date [PMID 41229518].
- **The norm India plans against is itself contested.** The "WHO requirement of one doctor per 1,000 population", cited throughout Indian policy and used to justify rapid private medical-college expansion without matching faculty or infrastructure, is argued to have **no actual WHO source** [PMID 41195915]; the same ratio is nonetheless still quoted as the recommended benchmark alongside India's figure of about **one doctor per 1,456 people** [PMID 39156627]. The examinable point is Reinke's — the status-quo/normative method never tests whether the ratio it enshrines is the right one — and the safe exam practice is to state the target and then name whose target it is.

Global and WHO developments

- ▶ The 23-per-10,000 threshold quoted in textbooks has been superseded. Global Burden of Disease 2019 modelling puts the minimum densities needed for a UHC effective-coverage score of 80 out of 100 at 20.7 physicians, 70.6 nurses and midwives, 8.2 dentistry personnel and 9.4 pharmaceutical personnel per 10,000 population — against the 2006 *World Health Report* benchmark of 22.8 skilled health workers per 10,000. Measured this way, South Asia alone was short 2.6 million physicians and 11.0 million nurses and midwives in 2019, and the global gap was 6.4 million physicians and 30.6 million nurses and midwives. India's 7.3 doctors and 17.3 nurses/midwives per 10,000 clear the old bar and fail the new one [PMID 35617980].
- ▶ WHO's own shortage estimate has been revised down, and its migration instrument renamed. National Health Workforce Accounts put the 2020 global stock at 65.1 million health workers and cut the projected 2030 shortage from 18 million to 15.4 million in 2020 and about 10.2 million by 2030, with the currently quoted projection near 11.1 million by 2030; about two-thirds of that residual shortage will sit in the countries on WHO's safeguards list [PMID 35760437; PMID 42602674]. The 2010 Global Code of Practice on the International Recruitment of Health Personnel was reworked in 2021 into the Health Workforce Support and Safeguards List — entry now set by a skilled-worker density below 44.6 per 10,000 combined with a UHC service-coverage cut-off — covering 47 countries in 2021 and 55 since 2023, 37 of them African [PMID 40498654]. India remains the principal source country for migrating doctors: 34.5 per cent of doctors on the UK register in 2019 were internationally trained, India first among the source countries [PMID 37924092].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 7 Health Programmes in India; Ch 8 Demography and Family Planning; Ch 10 Health Care of the Community; Ch 13 Tribal Health in India; Ch 21 Health Information and Basic Medical Statistics; Ch 23 Health Planning and Management); Reinke — Health Planning for Effective Management (Ch 10 Health Personnel Planning; Ch 19 Personnel Analysis; Ch 21 Synthesis of Health Needs and Resources); Oxford Textbook of Public Health 7e (Ch 3 Public health in low- and middle-income countries; Ch 86 Child health; Ch 101 Strategies for health services, Fig 11.8 Need, demand and supply of healthcare); Report of the Expert Committee on Tribal Health, Ministry of Health and Family Welfare, as cited in Park 28e; National Health Policy 2017, as cited in Park 28e.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 41229518 — India's journey in mainstreaming Ayush in primary health care: from tradition to integration, *Front Med* 2025, doi:[10.3389/fmed.2025.1629515](https://doi.org/10.3389/fmed.2025.1629515); 37843177 — Human Resources for Health in Haryana, India: What can be Done Better?, *WHO South-East Asia J Public Health* 2023, doi:[10.4103/WHO-SEAIPH.WHO-SEAIPH.11.22](https://doi.org/10.4103/WHO-SEAIPH.WHO-SEAIPH.11.22) (abstract-only); 36480803 — Building Noncommunicable Disease Workforce Capacity Through Field Epidemiology Training Programs: Experience From India, 2018-2021, *Prev Chronic Dis* 2022, doi:[10.5888/pcd19.220208](https://doi.org/10.5888/pcd19.220208); 41195915 — Medical education in India: disturbing trends, *Indian J Med Ethics* 2025, doi:[10.20529/IJME.2025.072](https://doi.org/10.20529/IJME.2025.072) (abstract-only); 39156627 — The Imperative for a Paradigm Shift in India's Healthcare Workforce, *Ann Neurosci* 2024, doi:[10.1177/09727531241271428](https://doi.org/10.1177/09727531241271428); 35617980 — Measuring the availability of human resources for health and its relationship to universal health coverage for 204 countries and territories from 1990 to 2019 (GBD 2019), *Lancet* 2022, doi:[10.1016/S0140-6736\(22\)00532-3](https://doi.org/10.1016/S0140-6736(22)00532-3); 35760437 — The global health workforce stock and distribution in 2020 and 2030, *BMJ Glob Health* 2022, doi:[10.1136/bmjgh-2022-009316](https://doi.org/10.1136/bmjgh-2022-009316); 42602674 — The global health workforce crisis toward 2030: a scoping review and systems-based policy framework, *Public Health Pract (Oxf)* 2026, doi:[10.1016/j.puhip.2026.100836](https://doi.org/10.1016/j.puhip.2026.100836); 40498654 — Health Workers in Sub-Saharan Africa: Concurrent Skilled Health Worker Shortages and Under-Employment, *Int J Health Plann Manage* 2025, doi:[10.1002/hpm.70001](https://doi.org/10.1002/hpm.70001); 37924092 — Exploring recent patterns of migration of doctors to the United Kingdom, *BMC Health Serv Res* 2023, doi:[10.1186/s12913-023-10199-y](https://doi.org/10.1186/s12913-023-10199-y). Plus WHO Health Workforce Support and Safeguards List (2021 revision; 2023 update); WHO National Health Workforce Accounts; Health Dynamics of India (Infrastructure and Human Resources) 2022-23, MoHFW; National Health Accounts estimates for India 2021-22; MoHFW Ayushman Arogya Mandir portal (data as on 30 November 2024); Indian Public Health Standards 2022.



Integrated health information platform (IHIP)

Asked in: Paper IV 17-Apr-2026 — RGUHS MD Community Medicine

Definition — The **Integrated Health Information Platform (IHIP)** is the online, web-based digital platform used for disease-surveillance data entry and management under the **Integrated Disease Surveillance Programme (IDSP)**. It provides real-time data access, rapid aggregation and display of data for decision-making, and advanced data modelling and analysis for users at district, state and national level; it carries role- and hierarchy-based feedback and alert mechanisms, geographic information system (GIS)-enabled graphical representation of data, and geo-tagging of reporting sites (Routine Immunization Manual for Medical Officers 2024, Ch 12, p 270). It commenced under IDSP in **2018** and is the electronic backbone that converts India's paper-and-e-mail weekly surveillance returns into a single national reporting-and-response platform (One Health Booklet, AIIMS).

What IHIP is — its placement, ownership and mandate

Attribute	Detail
Nature	Web-based digital platform for surveillance data entry, management, analysis and display; not a separate programme but the information technology (IT) layer of IDSP (Routine Immunization Manual 2024, Ch 12, p 270)
Parent programme	IDSP — launched November 2004 with World Bank assistance, continued under the National Health Mission (Sunder Lal, Ch 7, p 306)
Custodian	Ministry of Health and Family Welfare; Central Surveillance Unit (CSU) integrated with the National Centre for Disease Control (NCDC), Delhi , with a State Surveillance Unit (SSU) in every state/union territory and a District Surveillance Unit (DSU) in every district (Park 28e, Ch 7, p 568)
Year of commencement	2018, as "IDSP's Integrated Health Information Platform" (One Health Booklet, AIIMS)
What it digitises	The weekly IDSP returns — form "S" (syndromic, by health workers), form "P" (probable/presumptive, by clinicians) and form "L" (laboratory-confirmed, by laboratory staff), reported every Monday (Sunder Lal, Ch 7, p 307)
Core technical features	Real-time data access; rapid aggregation and display; advanced data modelling and analysis; role- and hierarchy-based feedback and alerts; GIS-enabled graphics; geo-tagging of reporting sites (Routine Immunization Manual 2024, Ch 12, p 270)
Policy mandate	National Health Policy 2017 — establish a federated integrated health information architecture, health information exchanges and a national health information network by 2025 , a district-level electronic database of health-system information, and registries for diseases of public health importance (Park 28e, Ch 23, pp 1030–1031)
Current expansion	Under the Pradhan Mantri–Ayushman Bharat Health Infrastructure Mission (PM-ABHIM, 2021), IHIP is being expanded to all states/union territories to connect all public health laboratories (Sunder Lal, Ch 20, p 805)

Attribute	Detail
Programme integration	National Leprosy Eradication Programme Pillar 5 — rollout of <i>Nikusth 2.0</i> with integration of reporting on IHIP (Park 28e, Ch 7, p 507); malaria web-based national reporting on IHIP with GIS mapping of high-malaria-prone areas and hotspots (Park 27e, Ch 5); IHIP as an impact-data source for immunisation monitoring (Routine Immunization Manual 2024, Ch 8, p 204); IDSP/IHIP field data analysed by NCDC for the National One Health Programme for Prevention and Control of Zoonoses (One Health Booklet, AIIMS)

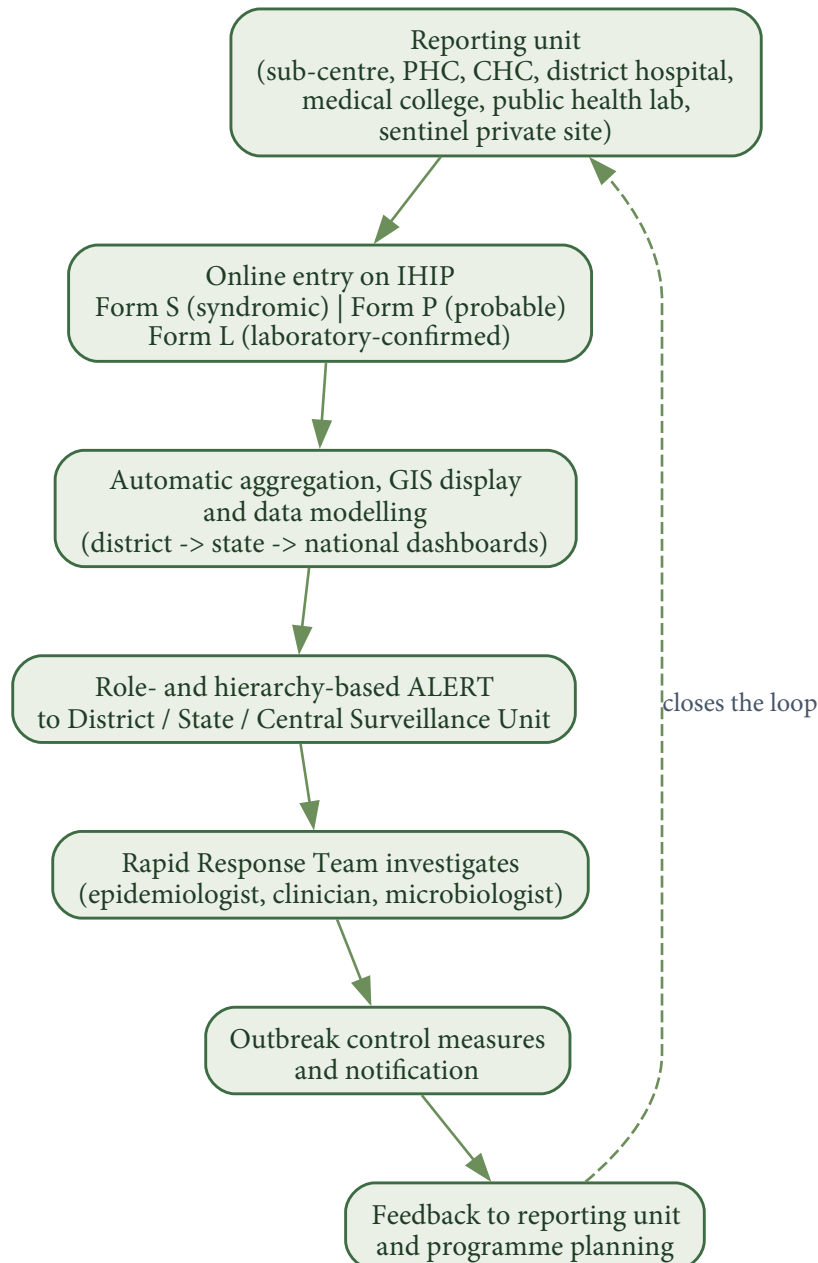
What changed: the pre-IHIP IDSP system versus IHIP

Feature	IDSP before IHIP	IDSP on IHIP
Reporting mode	Weekly S/P/L formats sent by e-mail/portal from reporting units; more than 90% of districts reporting (Park 28e, Ch 7, p 568)	Direct online entry on a single web platform at the reporting site (Routine Immunization Manual 2024, Ch 12, p 270)
Data granularity	Aggregate weekly counts, giving time trends only (Sunder Lal, Ch 7, p 307)	Real-time data access with rapid aggregation and display, plus geo-tagging of reporting sites
Analysis	Manual compilation and analysis at DSU/SSU	Advanced data modelling and analysis available at district, state and national level
Feedback	Sending regular feedback to reporting units as a DSU task (Jugal Kishore — National Health Programmes of India, p 490)	Automated, role- and hierarchy-based feedback and alert mechanism
Spatial display	Line lists and spot maps drawn locally	GIS-enabled graphical representation and hotspot mapping (Park 27e, Ch 5)
Programme silos	Separate vertical reporting for malaria, leprosy, tuberculosis, HIV	Vertical programme reporting progressively integrated onto one platform (Park 28e, Ch 7, p 507)

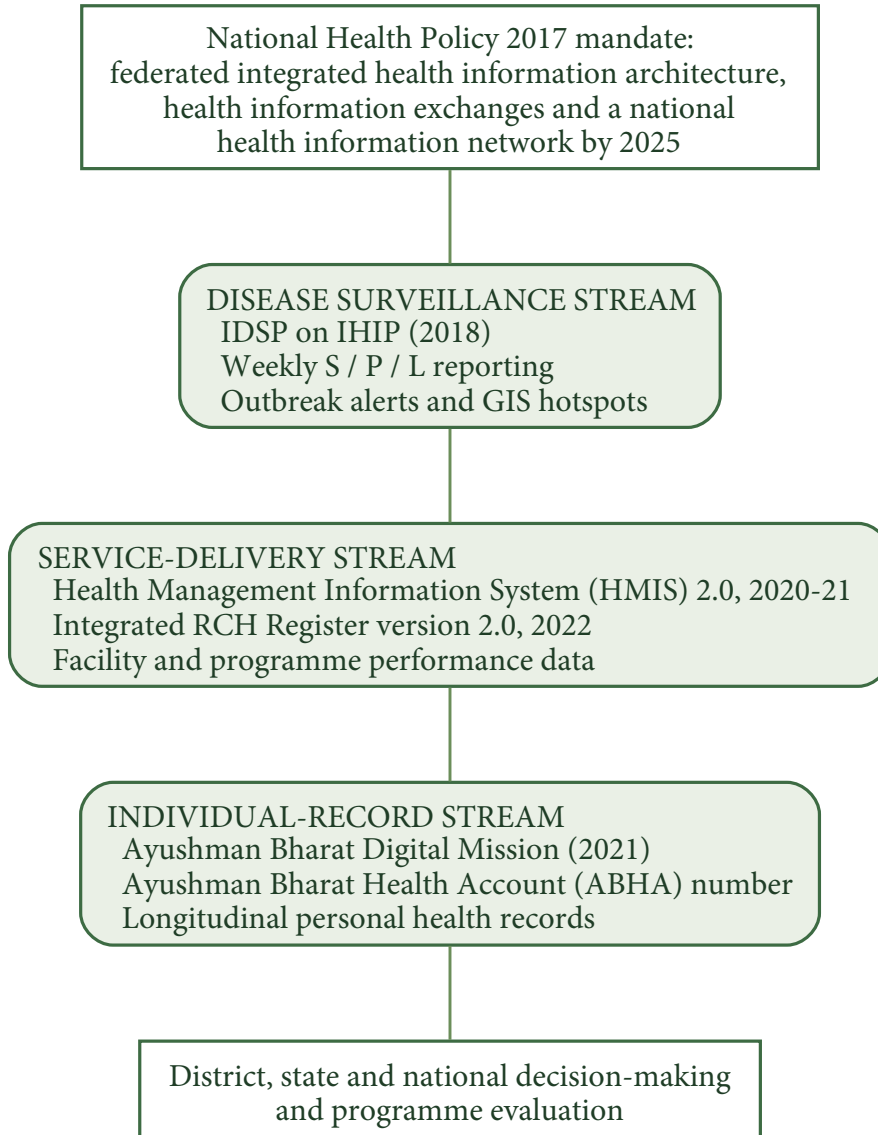
How IHIP works — from reporting unit to public health action

- **Reporting units enter data** — rural and urban public sector units (sub-centre, primary health centre, community health centre, district hospital, medical college, public health laboratory) and the private sector through **selected sentinel private sites (SSPS)**, since over 70% of first-contact providers are in the private sector (Sunder Lal, Ch 7, p 307).
- **Three parallel streams are captured** — *syndromic* (symptom-based, by paramedical staff and community volunteers, reported as "suspect" cases against lay case definitions), *presumptive/probable* (history and clinical examination, by medical officers) and *laboratory-confirmed* (Jugal Kishore, p 490; Sunder Lal, Ch 7, p 307).
- **Automatic aggregation and display** at DSU, SSU and CSU with GIS mapping and modelling.
- **Alert generation** — a rising trend triggers a role-based alert to the surveillance officer of that level; alerts also arrive from the **24 × 7 call centre (toll-free 1075)** and the **Media Scanning and Verification Cell (MSVC, established July 2008 at the National Institute of Communicable Diseases, now NCDC)**, which scans leading national dailies for outbreak reports and feeds them back to surveillance units (Sunder Lal, Ch 7, p 307).

- **Response** — the medical officer or the multidisciplinary **Rapid Response Team (RRT)** — epidemiologist, clinician, microbiologist and other specialists as required — investigates, diagnoses and controls the outbreak (Park 28e, Ch 7, p 568).
- **Feedback and evaluation** — analysis is fed back to the reporting units, closing the surveillance loop, since the prerequisites of effective surveillance are a **standard case definition, regularity of reports, and action on reports** (Park 28e, Ch 7, p 569).



Data-to-action pathway on the Integrated Health Information Platform (IHIP)



Where IHIP sits in India's national health information architecture

Advantages

Domain	Advantage
Timeliness	Real-time data access replaces weekly aggregate returns, so early-warning signals of impending outbreaks are detected earlier and effective response can be initiated in a timely manner — the founding purpose of IDSP (Park 28e, Ch 7, p 568)
Analysis at every level	Rapid aggregation, display and advanced data modelling put analysis in the hands of district, state and national users, not only the centre (Routine Immunization Manual 2024, Ch 12, p 270)
Automatic feedback	Role- and hierarchy-based feedback and alert mechanisms make feedback automatic instead of a manual DSU task, satisfying the surveillance prerequisite of "action on reports" (Park 28e, Ch 7, p 569)
Spatial epidemiology	GIS-enabled display and geo-tagging of reporting sites allow hotspot identification and micro-planning, already applied to malaria elimination (Park 27e, Ch 5)

Domain	Advantage
Integration and de-duplication	One platform for communicable and non-communicable disease surveillance across rural and urban systems, public and private sectors, and medical colleges — the central integration objective of IDSP (Sunder Lal, Ch 7, p 306); vertical programme reporting such as leprosy is being migrated onto it (Park 28e, Ch 7, p 507)
One Health and zoonoses	IDSP/IHIP field data are collected and analysed by NCDC to counter emerging and re-emerging zoonoses under the National One Health Programme (One Health Booklet, AIIMS)
Programme monitoring	Serves as an impact-data source for immunisation alongside vaccine-preventable-disease surveillance and HMIS (Routine Immunization Manual 2024, Ch 8, p 204)
Laboratory linkage	PM-ABHIM funds expansion of IHIP to all states/union territories to connect all public health laboratories, strengthening laboratory-confirmed surveillance (Sunder Lal, Ch 20, p 805)
Policy compliance	Directly delivers the National Health Policy 2017 targets of a district-level electronic health database and a national health information network (Park 28e, Ch 23, pp 1030–1031)

Challenges

Domain	Challenge
Private-sector capture	Gaps persist in capturing data from the private sector (Routine Immunization Manual 2024, Ch 12, p 270), yet over 70% of first-contact providers are private, so surveillance sensitivity depends on sentinel private sites that are voluntary (Sunder Lal, Ch 7, p 307)
Data quality	Health data in developing countries commonly fail on accuracy, reliability, completeness, timeliness, precision and integrity; weak linkage between monitoring systems and the point of data generation is a documented contributor (Oxford Textbook of Public Health 7e, Ch 21.5, p 823)
Duplication and under-use	Poorly coordinated monitoring systems with duplicated data collection and under-utilisation of nationally defined tools (Oxford Textbook of Public Health 7e, Ch 21.5, p 823) — in India, parallel entry into HMIS, programme portals and IHIP
Interoperability	Similar information is recorded and stored differently by different systems, so complex bridges are required to make them interoperable and to exchange data with public health agencies (Oxford Textbook of Public Health 7e, Ch 23.1, pp 911–912)
Human resources	Shortage of trained personnel is the most frequently documented barrier to quality health data; sensitivity and quality of surveillance depend entirely on district and sub-district staff, requiring continuous capacity building (Oxford Textbook of Public Health 7e, Ch 21.5, p 823; Routine Immunization Manual 2024, Ch 12, p 270)
Infrastructure and connectivity	Adequate laboratories, electronic record systems and connectivity are prerequisites; IHIP is still being extended to all states/union territories and all public health laboratories under PM-ABHIM (Oxford Textbook of Public Health 7e, Ch 21.5, pp 823–824; Sunder Lal, Ch 20, p 805)

Domain	Challenge
Privacy and data security	Health data are sensitive; anonymisation or pseudo-anonymisation, staff training, standard operating procedures and external audit of data security are needed to prevent breaches, and data privacy must be balanced against data sharing (Oxford Textbook of Public Health 7e, Ch 23.1, pp 912–914, Ch 21.5, p 823)
Equity and the digital divide	Where notification and services shift to digital systems, people without a smartphone or internet access can lose access; unequal electronic infrastructure widens health inequity (Oxford Textbook of Public Health 7e, Ch 19.3, pp 760–762)
Specificity of syndromic data	Syndromic entries are only "suspect" cases against lay case definitions — for example, fever with rash is entered as a syndrome, not as measles — so alerts have low specificity and require verification (Sunder Lal, Ch 7, p 307)
Data-to-action gap	The platform improves data flow but not automatically the response; effectiveness still requires standard case definitions, regularity of reporting and demonstrable action on reports (Park 28e, Ch 7, p 569)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the IDSP–IHIP reporting network with these exact labels:

- ▶ Central column (vertical, with two-way arrows): CSU (Central Surveillance Unit, NCDC Delhi) — SSU (State Surveillance Unit) — DSU (District Surveillance Unit)
- ▶ Left oval, **Rural surveillance**: PHC sub-centres and informers; rural medical colleges; Rural SSPS (selected sentinel private sites)
- ▶ Right oval, **Urban surveillance**: district hospital / infectious disease hospital / dispensaries / informers; district TB laboratory; district HIV/AIDS unit; district malaria unit; medical colleges; police; **Urban SSPS**; and a boxed list — ESI, Railway hospital, Water department, CGHS, Corporation hospital, Pollution control, ICMR laboratories
- ▶ Mark the **arrowheads** as two-way between the DSU and every reporting unit, to show that feedback (not only reporting) is part of the loop.

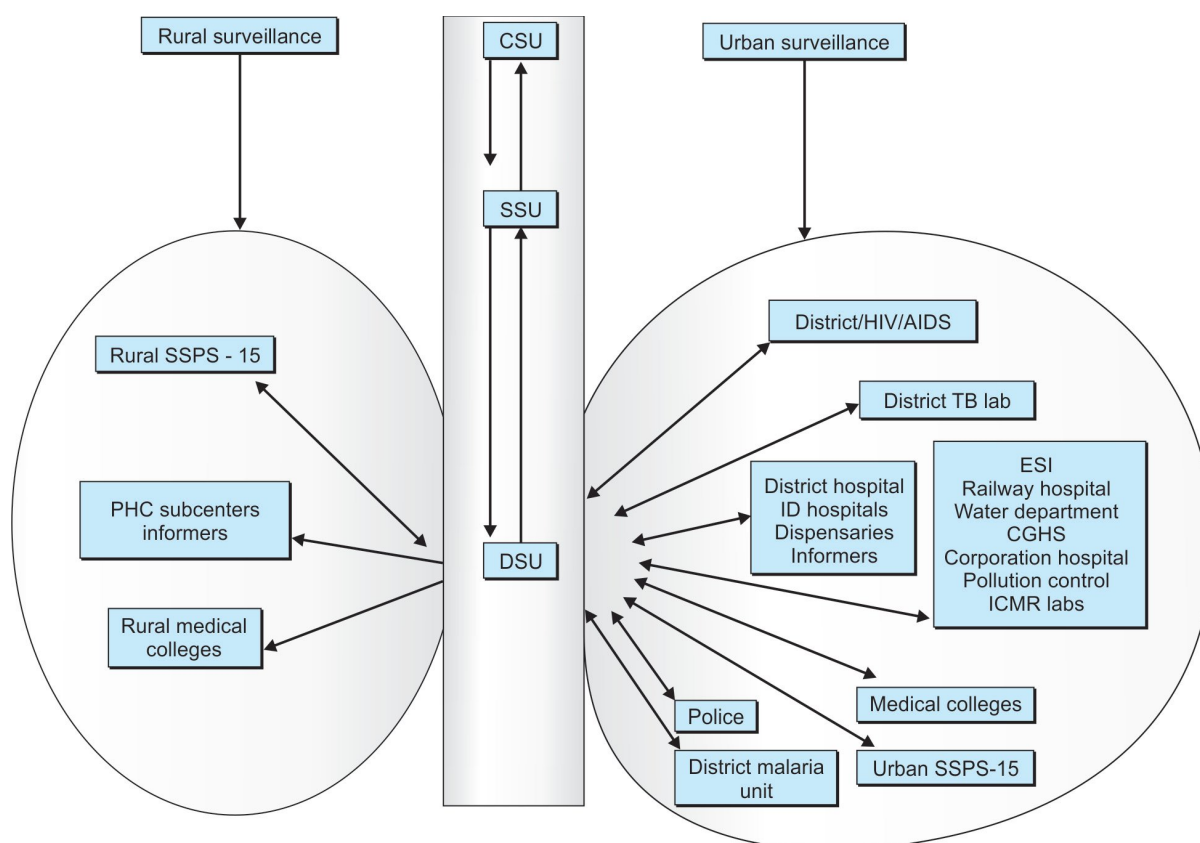


Fig 46.1 — Structural framework of integrated disease surveillance project Abbrev (Suryakantha Environment Best, p.905)

High-yield exam pointers

- ▶ IHIP = the web platform of IDSP; commenced 2018; CSU sits at NCDC, Delhi; IDSP itself launched November 2004 with World Bank assistance.
- ▶ Five IHIP features to write verbatim: real-time data access; rapid aggregation and display; advanced data modelling and analysis; role- and hierarchy-based feedback and alerts; GIS-enabled display with geo-tagging of reporting sites.
- ▶ Three weekly IDSP formats: S = syndromic (health worker), P = probable/presumptive (clinician), L = laboratory-confirmed (laboratory staff); reported every Monday.
- ▶ Two non-facility alert sources: 24 × 7 call centre, toll-free 1075 and the Media Scanning and Verification Cell (July 2008).
- ▶ Policy anchor to quote: National Health Policy 2017 — federated integrated health information architecture, health information exchanges and national health information network by 2025.
- ▶ Two strongest challenges to lead with: private-sector data capture gap (>70% of first-contact providers are private) and data quality (accuracy, reliability, completeness, timeliness, precision, integrity).
- ▶ Five components of surveillance activity (Park): collection — compilation — analysis and interpretation — follow-up action — feedback; prerequisites are standard case definition, regularity of reports, action on reports.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Coverage note: the indexed corpus describes IHIP only briefly — the substantive description is the Routine Immunization Manual for Medical Officers 2024 (Ch 12, p 270), supplemented by Sunder Lal, Park 28e/27e and the AIIMS–NCDC One Health booklet — so version numbers, the exact count of diseases under IHIP surveillance and the individual/case-based reporting claim are deliberately not stated here; also note that Sunder Lal writes the abbreviation as "Integrated Health Information Portal" while the Routine Immunization Manual and Park use "Platform", the latter being the accepted expansion.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed

Programme and policy updates (India)

- ▶ **A renaming worth memorising:** the **National Digital Health Mission**, launched **15 August 2020**, was renamed the **Ayushman Bharat Digital Mission (ABDM)** on **27 September 2021**; its citizen identifier is the 14-digit **ABHA** (Ayushman Bharat Health Account). About **50.39 crore ABHAs** had been created nationally by **December 2023** (Andhra Pradesh highest at 77.6%), and a **Digital Health Incentive Scheme** has paid facilities to create and link records digitally since **2023** [PMID 39382504]. ABDM runs on six building blocks — ABHA, personal health record apps, Health Professional Registry, Health Facility Registry, drug registry and the Unified Health Interface [PMID 39882498].
- ▶ **What the 2018 change actually was, and the platform's current name:** IHIP is now formally styled **IHIP–IDSP**, and the 2018 transition is best written as a move from an **aggregate, paper-based weekly system to real-time, case-based (person-disaggregated) reporting** — a restructuring triggered by the **2015 mid-term review of IDSP** [PMID 42245358]. IHIP–IDSP carries the electronically linked **S, P and L forms** and monitors **more than 33 health conditions**, with disaggregated individual data at every level [PMID 39453563]; each record is a line-list entry carrying age, sex, district and reporting date [PMID 40600283].
- ▶ **Rollout was staggered, and the published start year is not agreed:** two sources give **2018** — the IDSP–IHIP portal was launched in 2018 [PMID 39453563], and the 2018 aggregate-to-case-based transition [PMID 42245358] — while the Chhattisgarh surveillance analysis instead dates the platform's launch to **2021** [PMID 40600283]. Write **2018** (the textbook year) but flag that state go-live came later: Chhattisgarh began IHIP reporting in **April 2021** and Odisha in **2021**, so state surveillance series straddle two different information systems [PMID 40600283; PMID 41629233]. On the service-delivery side, ABDM's **Scan and Share** (National Health Authority, 2022) reached **16,939 facilities across 35 of 36 states and union territories** by 19 October 2024, cutting outpatient registration waits from about an hour to 2–5 minutes — but only **1.3%** of its 56.8 million tokens came from the private sector, mirroring IHIP's own private-sector gap [PMID 39882498].

Current national data

- ▶ **Chhattisgarh malaria under IDSP, 2015–2023 (IHIP from April 2021):** 391,387 laboratory-confirmed cases, of which only **34,768 (April 2021 – December 2023)** were IHIP-sourced —

January 2015 to March 2021 came from the weekly aggregate **L-form** of the pre-IHIP IDSP, so the series itself straddles two information systems. Annual parasite incidence 0.4–3.2 per 1000 with a **25% annual decline** (IRR 0.75; 95% CI 0.71–0.79); incidence far higher in Bastar (IRR 12.5; 95% CI 3.7–43.0) and Surguja (IRR 7.4; 95% CI 2.0–27.4) than in the central region. The annual blood examination rate stayed **below 10%** throughout, and the IHIP dashboard recorded **no malaria deaths at all in 2022 and 2023** — a completeness warning, not a success [PMID 40600283].

- ▶ **IHIP P-form field data (Odisha, April 2021 – October 2022):** the P form registered 255,071 patients in Ganjam (notification rate 6618 per lakh) and 63,895 in Jagatsinghpur (5146 per lakh), highest in males and in those over 60 years, with **acute respiratory illness the top-notified condition** in both districts [PMID 41629233]. Across the IDSP portal 2010–2023, **1158 zoonotic outbreaks** were reported for 10 diseases, **Japanese encephalitis / acute encephalitis syndrome commonest**; most diseases were reported only in the rainy season (July–September), while anthrax, JE/AES, Crimean-Congo haemorrhagic fever, leptospirosis and scrub typhus occurred year-round [PMID 41738949].

Elimination, surveillance and disease status

- ▶ **A malaria elimination target that may slip:** a separate modelling study, cited in the discussion of the Chhattisgarh surveillance analysis, projected Bastar, Dantewada, Bijapur and Narayanpur to **fall short of the zero-malaria-case target by 2030**; the state runs a *Malaria Mukh Bastar* test–treat–track campaign screening hard-to-reach populations in phases from **2020** [PMID 40600283]. For rabies, the **National Action Plan for Dog-Mediated Rabies Elimination (NAPRE)** targets elimination by **2030** on a One Health model, with mass dog vaccination aimed at 70% coverage in high-risk areas [PMID 39453563].
- ▶ **IHIP's real weakness is sensitivity, not software:** in Raigarh, Chhattisgarh (September 2023) IHIP captured only **44%** of laboratory-tested malaria cases from primary health centres. Root causes were inadequate laboratory-technician knowledge of IHIP data entry and poor district-surveillance-unit feedback; refresher training plus a weekly feedback mechanism from the district surveillance unit raised sensitivity to **100% within five weeks** across the four selected primary health centres — a quotable demonstration that the data-to-action gap closes with supervision, not with more software [PMID 41629241].
- ▶ **Vaccine-preventable disease resurgence is being picked up through IDSP:** measles spikes in several states, a marked rise in mumps among older children, and re-emergent diphtheria in Kerala and Assam shifting towards adolescents and adults, despite record overall coverage — with **Co-WIN and U-WIN** named as the real-time immunisation tracking platforms and life-course vaccination as the direction of travel for the Universal Immunisation Programme [PMID 40278999].

Guideline and legislative updates

- ▶ **Legislation is the named remedy for the trust gap:** digital health ID uptake was 78% in Chandigarh, yet only 28% had heard of electronic health records; uptake tracked confidence in government data security, and **enacting data-protection law** alongside digital health literacy is the stated lever for acceptance [PMID 39382504]. Real-world ABHA adoption remains low — **23.96%** of 537,278 outpatient visits over eight months in 2024–25 [PMID 41787376]. On the

surveillance side, IHIP–IDSP and the vector-borne programme (NVBDCP) still run **parallel, overlapping reporting flows** — NVBDCP applies static outbreak criteria while IHIP–IDSP triggers off the previous three years' data, entomological data stay on paper and Excel outside both digital systems, and P and L forms link automatically only when the diagnostic test is entered on the clinical diagnosis form, generating duplicate entries. Infrastructure lags too: district public health laboratories established under IHIP–IDSP were functional in only **three districts** of a 36-district state, and **four of those 36 districts** had no sentinel hospital [PMID 42245358].

Global and WHO developments

- **India's surveillance scored against a One Health yardstick:** applying the **One Health Surveillance Matrix** to IDSP–IHIP and the National Animal Disease Reporting System, IDSP scored management 57.4%, support 54.6% and operation 58.7%, with high stability and inclusiveness but **adaptability rated 0%** and persistent weakness in data sharing, sampling and external communication; 25 states have hired a veterinary consultant under the IDSP plan for intersectoral coordination [PMID 39453563]. Globally, an evidence map of 15 systematic reviews covering 82 primary studies confirms that **event-based surveillance** improves early outbreak detection and alert response in low- and middle-income countries, supporting its use alongside India's indicator-based IHIP stream [PMID 40388686].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India, pp 507, 568–570; Ch 23 Health Planning and Management, pp 1030–1031); Park's Textbook of Preventive and Social Medicine 27e (Ch 5, malaria elimination — IHIP web-based reporting); Sunder Lal — Textbook of Community Medicine (Ch 7 Epidemiology, pp 302–307; Ch 8, p 532; Ch 20 Recent Advances, p 805); Routine Immunization I Manual for Medical Officers 2024 (Ch 8 Sources and Use of Data, p 204; Ch 12 Vaccine Preventable Diseases and VPD Surveillance, p 270); Jugal Kishore — National Health Programmes of India (IDSP, pp 487–490); Oxford Textbook of Public Health 7e (Ch 19.3, pp 760–762; Ch 21.5, pp 822–824; Ch 23.1, pp 899–914); Suryakantha — Community Medicine with Recent Advances (Fig 46.1, p 905); One Health Booklet, AIIMS; National Health Policy 2017; Pradhan Mantri–Ayushman Bharat Health Infrastructure Mission 2021.

Recent-advances sources: 42245358 — Informational challenges of dengue surveillance and response. A case study from India, *Frontiers in Public Health* 2026, DOI [10.3389/fpubh.2026.1809584](https://doi.org/10.3389/fpubh.2026.1809584); 39453563 — Application of the One Health Surveillance (OHS) Matrix to Evaluate the Disease Surveillance Systems in Gujarat, India: A Policy Content Analysis, *Journal of Epidemiology and Global Health* 2024, DOI [10.1007/s44197-024-00317-2](https://doi.org/10.1007/s44197-024-00317-2); 40600283 — Epidemiology of malaria in Chhattisgarh, India: a surveillance data analysis, 2015–2023, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2025, DOI [10.1093/trstmh/traf072](https://doi.org/10.1093/trstmh/traf072); 39882498 — Scan and share to register outpatients, India, *Bulletin of the World Health Organization* 2024, DOI [10.2471/BLT.24.292828](https://doi.org/10.2471/BLT.24.292828); 39382504 — Perception towards electronic health records & uptake of digital health IDs among the urban residents in northern India: A mixed methods study, *Indian Journal of Medical Research* 2024, DOI [10.25259/ijmr_664_23](https://doi.org/10.25259/ijmr_664_23); 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission: an analytical cross-sectional study among outpatient department attendees, *BMC Health Services Research* 2026, DOI [10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 41629241 — Quality Improvement Initiative to Improve the Sensitivity of Malaria Reporting under Integrated Health Information Platform, Raigarh, Chhattisgarh, 2023, *Indian Journal of Public Health* 2026, DOI [10.4103/ijph.ijph_648_25](https://doi.org/10.4103/ijph.ijph_648_25) (abstract-only); 41629233 — Epidemiological Profile of Diseases Reported in the P Form of Integrated Health Information Platform in Two Districts of Odisha, *Indian Journal of Public Health* 2026, DOI [10.4103/ijph.ijph_660_25](https://doi.org/10.4103/ijph.ijph_660_25) (abstract-only); 41738949 — Trend of zoonotic diseases outbreaks in India: a secondary data analysis of Integrated Disease Surveillance Programme data, 2010–2023, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2026, DOI [10.1093/trstmh/traf011](https://doi.org/10.1093/trstmh/traf011) (abstract-only); 40278999 — Vaccine-Preventable Diseases in Pediatric Age Group in India: Recent Resurgence, Implications and Solutions, *Indian Journal of Pediatrics* 2025, DOI [10.1007/s12098-025-05531-9](https://doi.org/10.1007/s12098-025-05531-9) (abstract-only); 40388686 — Event-based surveillance in middle- and low-income countries: An evidence map, *Medwave* 2025, DOI [10.5867/medwave.2025.04.3031](https://doi.org/10.5867/medwave.2025.04.3031) (abstract-only).

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Network analysis in health management

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — A **network** is a graphic plan of all the **events** and **activities** that must be completed in order to reach an end objective, and it brings greater discipline into planning (Park 28e, Ch 23, p 1029). **Network analysis** is the group of operational-research techniques that draws this plan, attaches a time to every activity, and identifies the sequence of activities that controls the total project duration — so that a project is finished in minimum time at minimum cost, and corrective action can be taken the moment a delay that would upset the schedule is detected (Sathe & Sathe, Ch 17, p 188). The two common network techniques are the **programme evaluation and review technique (PERT)** and the **critical path method (CPM)** (Park 28e, Ch 23, p 1029); both belong to operational research alongside linear programming, queuing theory, simulation and work study (Sunder Lal, Ch 6, p 265).

Building blocks of a network

Term	What it means
Event (node)	A point marking the accomplishment of a task, written as a circle or square — e.g. "x-ray unit in place", "Pap kits on hand"; it consumes no time (Reinke, Ch 15, p 222; Sathe & Sathe, Ch 17, p 189)
Activity (arrow)	The time-consuming element of the project that joins two events; drawn as an arrow connecting them (Sathe & Sathe, Ch 17, p 189)
Predecessor event	For each task, the last job that must be completed before that task can start — e.g. the brick outer wall cannot be laid until the building framework is in place (Reinke, Ch 15, p 222)
Arrow diagramming method (ADM)	Network drawn with arrows as activities and nodes showing dependence; also called <i>activity-on-arrow</i> , and less commonly used than the precedence diagramming method (Sathe & Sathe, Ch 17, p 188)
Dummy activity	Technically not an activity at all, but a device for representing a dependency between two nodes (Sathe & Sathe, Ch 17, p 188)
Critical path	The longest path through the network; if any activity on it is delayed, the entire project is delayed (Park 28e, Ch 23, p 1029)
Slack (float)	Difference between the critical-path time and the time of the path of interest — the maximum an activity can slip without pushing out the completion date (Reinke, Ch 15, p 223; Sathe & Sathe, Ch 17, p 190)
Earliest and latest start time	Computed for every job; together they fix the maximum delay that can be tolerated in starting or finishing any activity without lengthening the project (Reinke, Ch 15, p 224; <i>Red Book</i> , p 294)

The two network techniques — PERT versus CPM

Point of comparison	PERT (programme evaluation and review technique)	CPM (critical path method)
Origin	Developed 1957–58 for the <i>Polaris</i> project; its use is estimated to have saved two years on that project (Sathe & Sathe, Ch 17, pp 188–189)	Developed at about the same time by an independent group, to establish chemical plants most economically (Sathe & Sathe, Ch 17, p 190)
Network is built on	Events	Activities (Sathe & Sathe, Ch 17, p 190)
Activity time	Not known — estimated from three figures, so completion time is uncertain and probabilistic (<i>Red Book</i> , p 294)	Known with certainty from past experience, and assumed controllable by adding inputs (Sathe & Sathe, Ch 17, p 190)
Primary objective	Shortest possible time, at any cost	Optimum mix of cost and time for economic return (Sathe & Sathe, Ch 17, p 190)
Best suited to	Non-repetitive, one-time projects of a complex nature without precedent (Sathe & Sathe, Ch 17, p 189)	Complex but previously experienced projects, where durations are known (Sathe & Sathe, Ch 17, p 190)
Focus of computation	Probability of completing by the target date, and the most probable completion date (Reinke, Ch 15, p 226)	Float , to identify the activities with least scheduling flexibility (Sathe & Sathe, Ch 17, p 190)

PERT time estimation. Three estimates are made for every activity — **optimistic time**, the time if all goes well; **most likely time**, the time taken most frequently; and **pessimistic time**, the time under the most adverse conditions. The expected time is their weighted average, the most likely estimate carrying four times the weight of either of the other two (Reinke, Ch 15, p 226):

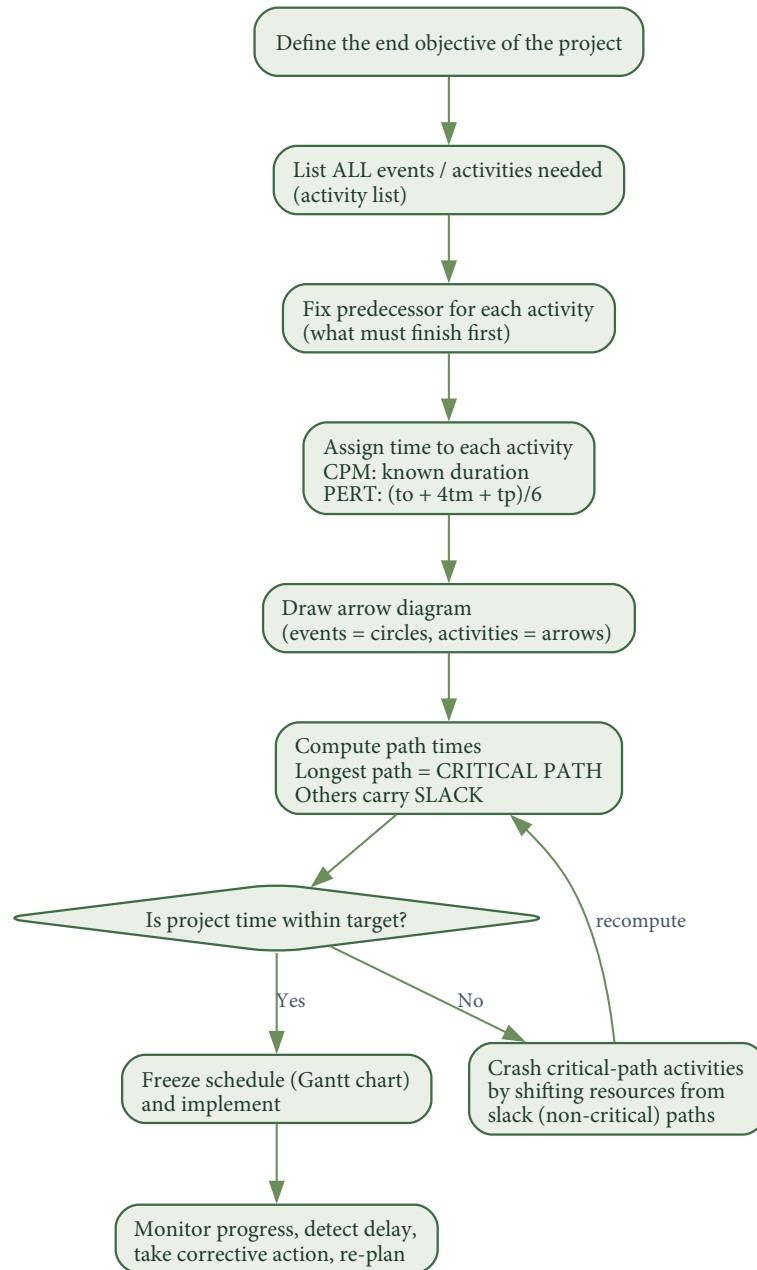
Expected time for an activity = [Optimistic time + (4 × Most likely time) + Pessimistic time] ÷ 6 (*Red Book*, p 294)

These three distinctions have blurred with time: variable time estimates are now used in CPM, cost reduction is now considered in PERT, and the concepts of critical path and slack have been introduced into PERT as well (Sathe & Sathe, Ch 17, p 190).

Steps in constructing and using a network

- **Identify the key activities** in the project and list them comprehensively; the order of listing is unimportant, since events are simply noted as they come to mind until all requisite tasks are enumerated (Sunder Lal, Ch 16, p 726; Reinke, Ch 15, p 222).
- **Devise the sequence** — document which activities can run in parallel and which can only run step by step, identifying for each task its predecessor (Sathe & Sathe, Ch 17, p 188).
- **Assign a duration** to each phase of the work, in days, weeks or any convenient unit (Sunder Lal, Ch 16, p 726).

- ▶ **Develop the network (arrow diagram)** — put circles at the beginning and end of each activity to represent events, and join them by lines representing the activities; add further circles for events not tied to an activity (Park 28e, Ch 23, p 1029; Sunder Lal, Ch 16, p 726).
- ▶ **Identify the critical path** — begin at the right, where the large task is complete, choose the last task that took the most time, connect back to the activity requiring the most time, and continue right to left; then read it left to right (Sunder Lal, Ch 16, p 726).
- ▶ **Perform network time calculations** — add the time allotments along the critical path to obtain total project time, and compute slack for every other path (Sunder Lal, Ch 16, p 726; Reinke, Ch 15, p 223).
- ▶ **Re-plan** — try other routes, always ensuring no activity starts before its precedents are complete; for example, training of *Anganwadi* workers cannot occur unless they have first been recruited and appointed (Sunder Lal, Ch 16, p 726). The schedule must then be **constantly monitored** to spot impending bottlenecks and adjust project targets (Reinke, Ch 15, p 226).



Sequence of steps in network analysis of a health project

Utilization in health management, with examples

Where it is used in health management	Suitable example	What the network delivers to the manager
National disease-control and eradication programmes	Application to the smallpox eradication programme was useful; application to a measles vaccination programme is a standard illustration (Sathe & Sathe, Ch 17, pp 188–189)	Sequences surveillance, vaccine supply, team deployment and search operations so no link becomes a bottleneck
Commissioning a new health facility	Construction, staffing and outfitting of an emergency health centre to a 15-day deadline (Reinke, Ch 15, p 222); starting a new medical college or teaching/non-teaching hospital (Sathe & Sathe, Ch 17, pp 189–190)	Tells which of construction, drug procurement or personnel recruitment actually governs the opening date

Where it is used in health management	Suitable example	What the network delivers to the manager
Programme planning exercises and surveys	PERT diagram for a safe motherhood needs assessment ; Gantt chart for holding an eye camp (Sathe & Sathe, Ch 17, Figs 17.1, 17.2, 17.6)	Converts a plan into a dated work schedule that every department can be held to
Manpower planning within a programme	<i>Anganwadi</i> worker recruitment → appointment → training in sequence (Sunder Lal, Ch 16, p 726); PERT/MANPOWER adds constraints on staff availability (Reinke, Ch 15, p 226)	Prevents two tasks being scheduled together when the same staff must do both
Financial monitoring of a project	PERT/COST relates project progress to budget expenditure — a two-year project that has spent one-third of its budget but completed only one-fourth of the work is in financial <i>and</i> scheduling difficulty (Reinke, Ch 15, p 226)	Determines the least expensive way to meet a stated deadline and flags cost overrun early
Managing several projects in one plan	RAMPS (resource allocation and multiproject scheduling); decision CPM where the result can be achieved in more than one way (Reinke, Ch 15, p 226)	Allocates scarce district resources across competing projects
Supervision and communication	CPM is also the standard method of communicating project plans, progress and costs (<i>Red Book</i> , p 294)	Lets all concerned know what is expected of them, minimising delays and crises in implementation (Park 28e, Ch 23, p 1029)

Worked health-service example (emergency health centre, target 15 days). Building framework (4 days) → inside walls plastered and dry (10 days) → health centre finished (2 days) forms the critical path of **16 days**, one day more than allotted; the personnel path (authorised 2 + recruited 3 + trained 5 + finish 2 = 12 days) therefore carries **4 days of slack**. If plastering is crashed from 10 to 6 days by using a costlier prefabricated substitute, the old critical path shortens by 4 days but total project time falls by only **2 days**, because the drug-procurement path then becomes critical — the classic demonstration that only the critical path, and only up to the next-least-slack path, is worth spending money on (Reinke, Ch 15, pp 223–224).

Advantages and limitations

Advantages	Limitations
Simple to understand, inexpensive, and does not need highly qualified specialists (Sathe & Sathe, Ch 17, p 188)	Overemphasis on time , with almost no attention to cost , is the biggest disadvantage of PERT (Sunder Lal, Ch 16, p 726)
Gives a clear diagrammatic view of all activities and their interrelationships (Sathe & Sathe, Ch 17, p 188)	CPM in its simple form ignores the possibility of error in the time estimates (Reinke, Ch 15, p 226)

Advantages	Limitations
Identifies the few critical activities, so costly supervision of non-critical activities is avoided (<i>Red Book</i> , p 294)	Greater sophistication needs more and better information and far greater analytical effort (Reinke, Ch 15, p 226)
Applies to any project whose activities can be described sequentially, and has been used in health services with advantage (Sathe & Sathe, Ch 17, p 188)	An initial schedule is not enough — it must be constantly monitored, or it becomes a paper exercise (Reinke, Ch 15, p 226)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce this **arrow diagram** and mark on it:

- ▶ Numbered event circles 0 → 1 → 2, then two branches; terminal event 9 = "Complete plan and start service"; event 1 labelled "Start plan"
- ▶ Activity times written on every arrow: 0-1 1 week, 1-2 2 weeks; upper branch 2-3 4 weeks, 3-4 6 weeks, 4-8 2 weeks, 8-9 1 week; lower branch 2-5 3 weeks, 5-6 2 weeks, 6-9 2 weeks, 5-7 1 week, 7-9 1 week
- ▶ Double-line the path 0-1-2-3-4-8-9 and label it "CRITICAL PATH = 16 weeks"
- ▶ Label the lower branches "0-1-2-5-6-9 = 10 weeks, slack 6 weeks" and "0-1-2-5-7-9 = 8 weeks, slack 8 weeks"
- ▶ Side box: "Delay on the critical path delays the whole project; delay within slack does not"

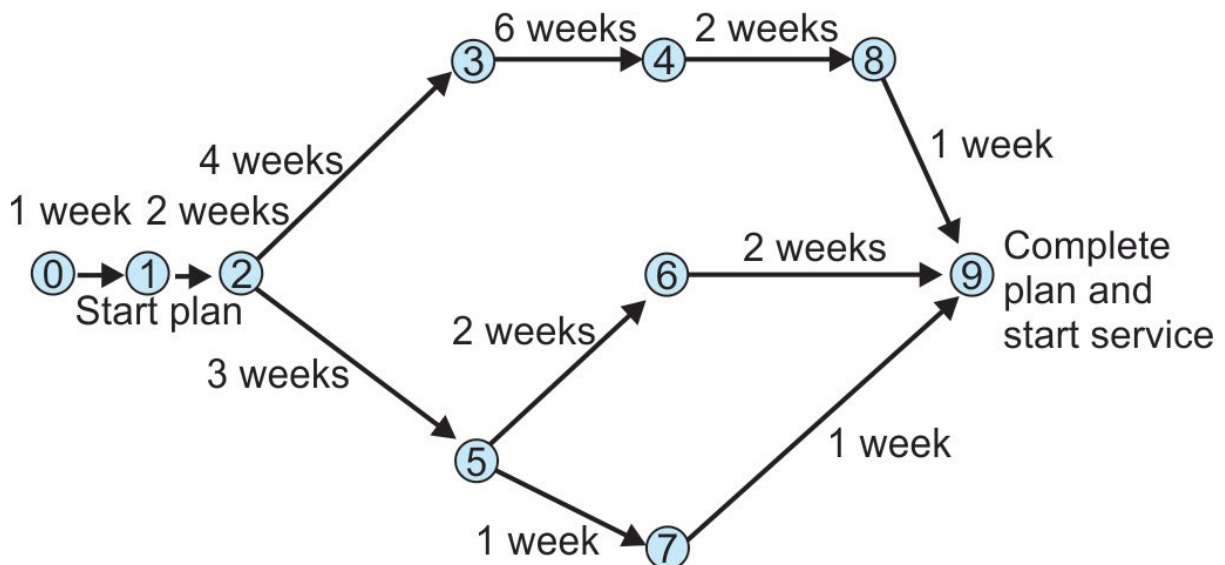


Fig 37.3 — Flow diagram of critical path (Suryakantha Environment Best, p.806)

High-yield exam pointers

- ▶ Park's one-line definition: "A network is a graphic plan of all events and activities to be completed in order to reach an end objective"; the two techniques are PERT and CPM.
- ▶ PERT expected time = (Optimistic + 4 × Most likely + Pessimistic) ÷ 6 — optimistic, most likely (×4), pessimistic, divided by 6.

- ▶ **Critical path** = longest path through the network; **slack** = critical-path time – time of that path.
- ▶ Three original PERT-versus-CPM differences: **events vs activities**, **estimated vs known time**, **time-at-any-cost vs cost–time optimum**.
- ▶ Health examples worth naming: **smallpox eradication**, **measles vaccination programme**, **safe motherhood needs assessment**, **commissioning a new hospital or medical college**, **eye camp scheduling**.
- ▶ Refinements to name: **PERT/COST**, **PERT/MANPOWER**, **RAMPS**, **decision CPM**; the companion display tool is the **Gantt chart**, from which network analysis grew.
- ▶ Network analysis is technique 7 among Park's quantitative management methods, sitting between systems analysis (6) and the planning-programming-budgeting system (8).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Textbooks in this corpus use "network analysis" strictly in its **operational-research/project-scheduling** sense (PERT and CPM); the newer social-science sense of "social network analysis" of contacts or organisations is not what the RGUHS health-management question asks for, and no retrieved passage supports that reading.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW National Patient Safety Implementation Framework 2017, IPHS 2022, National Framework for Malaria Elimination, National Kala-azar Elimination Programme, PM-ABHIM, WHO

Programme and policy updates (India)

- ▶ **Ayushman Bharat Health and Wellness Centres are now Ayushman Arogya Mandir (AAM)** — the February 2018 decision to convert 150,000 sub-centres and PHCs is monitored exactly as a network manager monitors a project, against a published year-wise target-versus-achievement curve; rollout of the full expanded service package rose from 50% of AAMs in March 2023 to 82% by November 2024, with infrastructure gaps closed against IPHS 2022 and financed through PM-ABHIM and XV Finance Commission grants. Manpower, not construction, is the activity that governs the opening date — a national community health officer shortfall of about 11,111 persists against 1,16,978 in position, and in one Chhattisgarh block 41.2% of sub-centre AAMs had no community health officer, the single input most strongly associated with poor facility performance ($p < 0.001$). [PMID 41816146] [PMID 40821268]
- ▶ **PM-ABHIM (announced 25 October 2021) is aimed at strengthening health systems and institutions at all levels of care** — under it the Department of Health Research and ICMR are establishing four zonal National Institutes of Virology at Bengaluru, Jammu, Dibrugarh and Jabalpur, with delivery pegged to World Bank Disbursement-Linked Indicator 8 and sustained by quarterly progress monitoring and annual priority updates through an independent verification agency; zonal viral-disease priorities were fixed by a multicriteria decision analysis consensus across 186 stakeholders rather than by opinion, the same operational-research family to which PERT and CPM belong. [PMID 42522023]

- **Vertical programmes are the parallel paths that never converge** — NTEP, the NP-NCD and the National Mental Health Programme still run in vertical silos although nearly 52% of persons with TB have multimorbidity (depression 33.1%, diabetes 14.5%, hypertension 11.4%); a SWOT analysis against the WHO six building blocks proposes the AAM as the single node at which these paths should merge, names isolated digital platforms and absent cross-programme training as the dependencies that must be cleared first, and attaches durations to the merge — digital interoperability in 12–18 months, bidirectional screening targets in two years, shared-financing pilots in 2–3 years, an institutionalised integrated care package in five. [PMID 42397826]

Current national data

- **Malaria** — the National Framework for Malaria Elimination (2016) grades States and UTs into categories 0–3 by annual parasite incidence and is aligned to the WHO Global Technical Strategy for Malaria 2016–2030; under the companion National Strategic Action Plan 678 districts were stratified on mean API, and districts in the two worst categories fell from 46 (category 2) and 109 (category 3) in 2017 to 32 and 10 in 2019 — a target-versus-achievement curve of exactly the kind a network schedule is monitored against, set against an economic burden put at about \$1.9 billion. Cases nonetheless rose again in 2023 and 2024 (pandemic-era disruption, drug and insecticide resistance, under-detection of asymptomatic carriers and *pfhrp2*-deleted parasites), making malaria-free-by-2030 the national milestone most likely to slip. [PMID 39295839] [PMID 40168983]
- **Ni-kshay Poshan Yojana, 2022 NTEP cohort** — 92.7% (95% CI 89.8–94.8) of 3,201 notified persons with TB were credited at least one instalment, but the median time to the first instalment was 105 days (IQR 60–174), only 42% received it within two months of diagnosis, and 24.1% received it only after their treatment outcome had been declared; non-receipt carried a 4.93-fold risk of an unfavourable outcome (95% CI 3.61–6.75). An activity that is completed, yet completed late enough to be worthless to everything downstream of it. [PMID 39034827]

Elimination, surveillance and disease status

- **Kala-azar — elimination threshold reached, validation pending** — endemic blocks still above the target of <1 case per 10,000 population fell from 633 (2005) to 136 (2015), 8 (2021) and 0 by December 2023, the last being in Pakur district, Jharkhand; annual cases fell from 34,803 (2005) to 8,243 (2015) and 524 (2023). India has entered a consolidation phase and must hold the threshold for three consecutive years before submitting its dossier for WHO validation; Bangladesh was validated first, in 2023. Post-kala-azar dermal leishmaniasis and HIV–VL co-infection are the human reservoirs that could restart transmission. [PMID 40828749]
- **Mass drug administration — albendazole and the limits of community-wide MDA** — India carries close to 40% of the global lymphatic filariasis burden, losing about 2 million DALYs and US\$811 million in wages a year, which is what keeps the MDA activity on every district schedule [PMID 41451596]. **DeWorm3** (2025), a cluster-randomised trial across India, Benin and Malawi in units that had already completed at least five rounds of community-wide MDA for lymphatic filariasis, gave biannual community-wide albendazole for three years against school-based deworming — but the co-primary outcomes measured were **soil-transmitted helminth** endpoints, not LF. *Necator americanus* prevalence was much lower with community-wide MDA in India (adjusted prevalence ratio 0.41; 95% CI 0.32–0.52), yet transmission interruption was

reached in only 1 of 20 intervention clusters there: interruption may be attainable in isolated pockets but is not programmatically achievable on the timescale tested, though community-wide MDA is still put forward as an alternative to school-based deworming to improve equity and outcomes. [PMID 40752908]

Guideline and legislative updates

- **National Patient Safety Implementation Framework (MoHFW 2017; target year 2025)** — the framework is itself written in network language: six strategic objectives, each carrying a timeline, a responsible organisation, short-, mid- and long-term priority areas, and expected outputs. A Tamil Nadu evaluation across 18 secondary and tertiary facilities (80 interviews) found the ground situation not conducive to full implementation by 2025, and its stated way forward was a State-level core multidisciplinary patient-safety committee plus a **Gantt chart** sequencing the framework by priority — the direct remedy for the standing criticism that an unmonitored schedule becomes a paper exercise. [PMID 38135519]

Global and WHO developments

- **CPM is still being applied, and published, for health-service projects** — a 2025 case study scheduled a hospital outpatient-registration subsystem by forward and backward pass, identified demand analysis → basic data preparation → subsystem test → simulation test as the critical path, then crashed the test activity from 7 weeks to 4, finishing ahead of schedule and saving 1,000 yuan [PMID 39618106]; an arrow diagram of the 2015 Taiwan burn mass-casualty incident (208 minutes from the emergency call to the last casualty reaching hospital) showed that fast-tracking the National Army's incident command post into parallel with the fire department's, inside the first hour, would have delivered the last casualty about an hour earlier, 23:58 → 22:53 [PMID 34235987]. Both are crashing decisions taken only on the critical path.
- **In current public-health writing "network analysis" usually means social network analysis of organisations** — stakeholder mapping now reports density, connectedness, centralisation, reciprocity and broker roles to show who is missing from a health system: 66 respondents from 27 organisations were mapped to design implementation-science infrastructure in Alberta, where the support network was both denser (17% vs 11%) and more reciprocal (27% vs 18%) than the research network [PMID 41280436]; and in Kenya's assistive-technology ecosystem 37 of 76 organisations responded, with metrics reported by organisation *type* rather than by individual organisation for confidentiality — government ministries and agencies, then organisations of persons with disabilities, ranked highest on betweenness centrality, while innovation-training organisations were not yet integrated into the network at all [PMID 38224051]. Keep the two senses apart in the exam — this is the sociometric sense, not PERT/CPM.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 23, Health Planning and Management); Sathe & Sathe — Epidemiology and Management for Health Care (Ch 17, Modern Management Techniques); Reinke — Health Planning for Effective Management (Ch 15, Tools for Improving the Process of Planning); Sunder Lal — Textbook of Community Medicine (Ch 16, Health Planning and Management; Ch 6, Operational Research); Suryakantha — Community Medicine with Recent Advances (Ch 37, Fig 37.3); Red Book — Public Health and Preventive Medicine (p 294).

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Operational planning

Asked in: Paper IV 17-Apr-2026 — RGUHS MD Community Medicine

Definition — Operational planning is the short-term, action-level stage of the planning cycle in which agreed goals are converted into a dated, costed and individually-assigned schedule of activities for a defined population or facility. Operating plans "set strategic plans in motion and carry out the tasks they prescribe", each plan being assigned to a named leader and linked to specific activities with deadlines (Harrison — Essentials of Health Care Organization, Ch 4, pp 102-103). In health services it is usually written as an **annual work plan with its budget**, and it answers four questions in one document — who will do what activity, at what time, and at which place (Sunder Lal, Ch 16, p 705).

Strategic versus operational planning

Feature	Strategic planning	Operational planning
Time horizon	Perspective plans ~20 years; intermediate-term plans ~5 years	Annual plan, tied to the yearly budgeting cycle (Reinke, Ch 1, p 18)
Orientation	Longer term, conceptual, "helicopter" perspective	Immediate term, concrete, "on the ground" (Zuckerman, Ch 11, pp 168-169)
Central concern	Effectiveness — identifying key issues and opportunities	Efficiency — resolving existing performance problems
Mode	Reflective, breaking new ground, "hands off"	Action and doing, routine and ongoing, "hands on"
Output document	Policy directions, vision, priority statement	Targeted plan of work for direct administrative action (Reinke, Ch 1, p 18)
Budget type	Development budget supporting the 5-year plan	Routine budget prepared with the annual work plan (Reinke, Ch 12, p 178)
Indian counterpart	National Health Policy 2017; State perspective plan	Sub-centre and village annual action plans, District Health Action Plan, State Programme Implementation Plan

An operational plan is therefore not a separate exercise but the implementation-facing arc of the **planning cycle**, and it feeds the next cycle through evaluation and replanning.

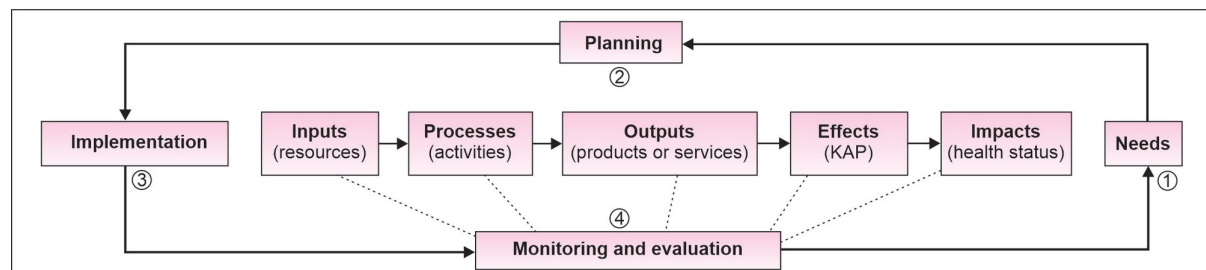


Fig 16.6 — Place of monitoring and evaluation in planning cycle (Sunderlal, p.722)

Steps in operational planning in health care

- **Situation analysis** — "where are we now?" Collect, assess and interpret information on the population: demography, mortality and morbidity, health services and their utilisation, technical manpower of the various categories, training facilities available, and the attitudes and beliefs of

the population towards disease, its cure and prevention (Park 28e, Ch 23, pp 1026-1027). Sources are the household and facility survey, service statistics and routine reports, supplemented by listening to community leaders, women's groups, *Anganwadi* workers and accredited social health activists (ASHAs) (Sunder Lal, Ch 16, p 701). The analysis yields the health **problems, needs and demands**.

- ▶ **Assessment of health service needs and prioritisation.** Resources always fall short of total requirement, so problems are ranked. Priority is guided by financial constraints, mortality and morbidity data, diseases preventable at low cost, saving the lives of younger people in whom there has been considerable social investment, and political and community interest (Park 28e, Ch 23, p 1027). Priority setting is regarded as the heart of the planning process (Reinke, Ch 1, p 27).
- ▶ **Setting objectives and targets — "what do we want to achieve?"** Objectives are the specific results expected from the programme and must be **SMART** — Specific, Measurable, Appropriate/relevant, Realistic and Time-bound (Sunder Lal, Ch 16, p 704). They are set at every level down to the smallest organisational unit, becoming more detailed at lower levels, and serve both as a guide to action and as the yard-stick to measure work after it is done (Park 28e, Ch 23, p 1027).
- ▶ **Assessment of resources and choice among alternatives.** Resources mean the manpower, money, materials, skills, knowledge and techniques needed or available; a balance is struck between what is required and what is likely to be available (Park 28e, Ch 23, p 1027). Time and information are equally resources at a health and wellness centre (Sunder Lal, Ch 16, pp 704-705). Alternate plans are then formulated and assessed for practicability and feasibility, and the one with greater effectiveness is chosen (Park 28e, Ch 23, p 1027).
- ▶ **Programming — "how will we get there?"** A programme is a grouping of continuous and related **activities** designed to achieve the set objective — home visits, immunisation sessions, maternal and child health and family planning clinics, growth monitoring, counselling, *Anganwadi* visits, screening for non-communicable diseases, and recording and reporting (Sunder Lal, Ch 16, pp 704-705). Activities are arranged in order and against a time frame, and a fixed day-wise, village-wise work schedule is drawn up covering the whole sub-centre population, flexible enough to be readjusted for outbreaks or campaigns (Sunder Lal, Ch 16, pp 701, 705).
- ▶ **Budgeting.** The needs are translated into a schedule of financial outlays. A **line-item budget** organises planned costs by input category — personnel, supplies, equipment (Reinke, Ch 12, pp 178-179). The link between the medium-term plan and the annual work plan and budget is crucial; a medium-term plan cannot be implemented without first phasing it into annual plans (Reinke, Ch 12, p 170).
- ▶ **Write-up and approval of the plan.** The detailed plan document is prepared, complete in all respects for execution, with targets and standards, and with a built-in system of evaluation (Park 28e, Ch 23, pp 1026-1027). Increasing detail and specificity is introduced for the most immediate period, such as the next year (Reinke, Ch 1, pp 28-29). In India the written plan is approved and funded as the Programme Implementation Plan.
- ▶ **Implementation.** A plan is only as good as its implementation (Sunder Lal, Ch 16, p 705). The main considerations at this stage are definition of responsibilities and tasks; the selection,

training, motivation and supervision of the manpower involved; organisation and communication; and the efficiency of individual institutions such as hospitals or health centres (Park 28e, Ch 23, p 1027). Many well-considered plans break down through delays in critical supplies and inappropriate use of staff.

- **Monitoring, supervision and coordination.** Monitoring is the day-to-day follow-up of activities during implementation to ensure they are proceeding as planned and are on schedule — a continuous process of observing, recording and reporting, keeping track of the course of activities, identifying deviations and taking corrective action when deviations are excessive (Park 28e, Ch 23, p 1027). Operationally it is the periodic collection and analysis of selected **input and process indicators** so that managers can improve operational plans and make mid-course corrections (Sunder Lal, Ch 16, p 723).
- **Evaluation and replanning.** Evaluation measures the degree to which objectives and targets are fulfilled and the quality of the results obtained, measures the productivity of available resources, and makes possible the reallocation of priorities and resources on the basis of changing health needs — WHO Expert Committee on National Health Planning in Developing Countries (Park 28e, Ch 23, p 1027). It uses **output, effect and impact indicators** (Sunder Lal, Ch 8, p 340). Its findings expose gaps in the planning process, which **replanning** then corrects, closing the cycle (Sunder Lal, Ch 16, pp 706-707).

Scheduling and costing tools used at the operational stage

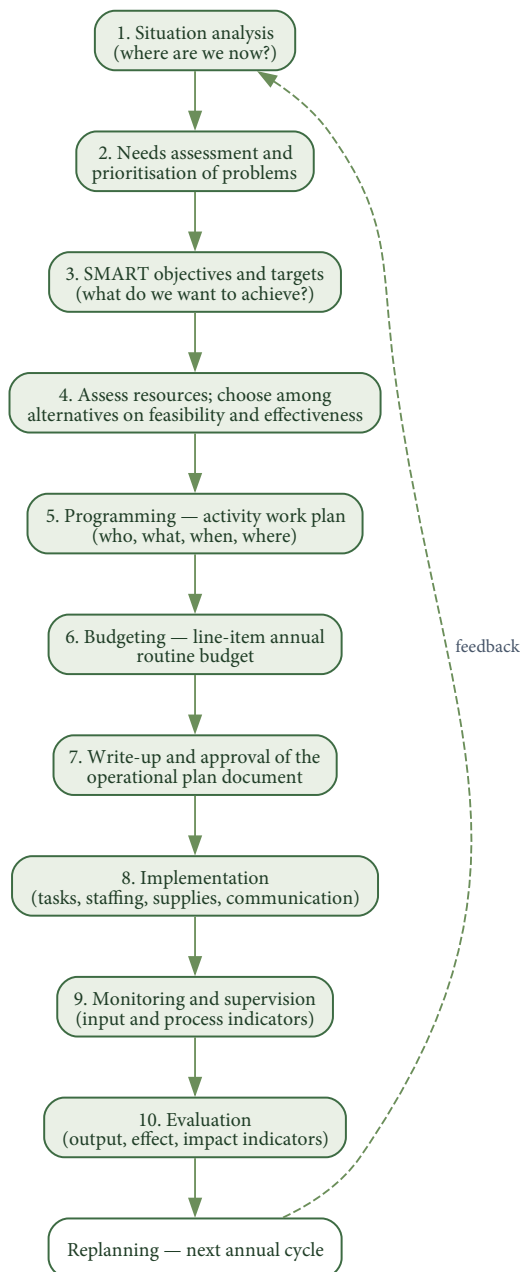
Tool	What it does	Key point
Gantt chart	Bars each activity against calendar dates; dotted lines show permissible delay	Used operationally alongside network analysis in actual scheduling (Reinke, Ch 15, pp 224-225)
Critical path method (CPM)	Network of events and activities; identifies the longest (critical) path and the slack time of non-critical jobs	Earliest and simplest form of network analysis; ignores error in time estimates (Reinke, Ch 15, pp 225-226)
Programme evaluation and review technique (PERT)	Uses three time estimates — most optimistic, expected, most pessimistic; required time is a weighted average giving the best estimate four times the weight of either other	Permits more detailed planning and more comprehensive supervision (Park 28e, Ch 23, p 1029; Reinke, Ch 15, pp 225-226)
PERT/ MANPOWER, RAMPS, decision CPM	Add personnel-availability constraints, multi-project resource allocation, and alternative activity sets	Refinements for plans containing several projects (Reinke, Ch 15, p 226)
Line-item and standard costing	Budgets by input head; adjusts budget to achieved output to separate performance from efficiency	Financial control at the implementation stage (Reinke, Ch 12, pp 178-181)

Operational planning in the Indian health system

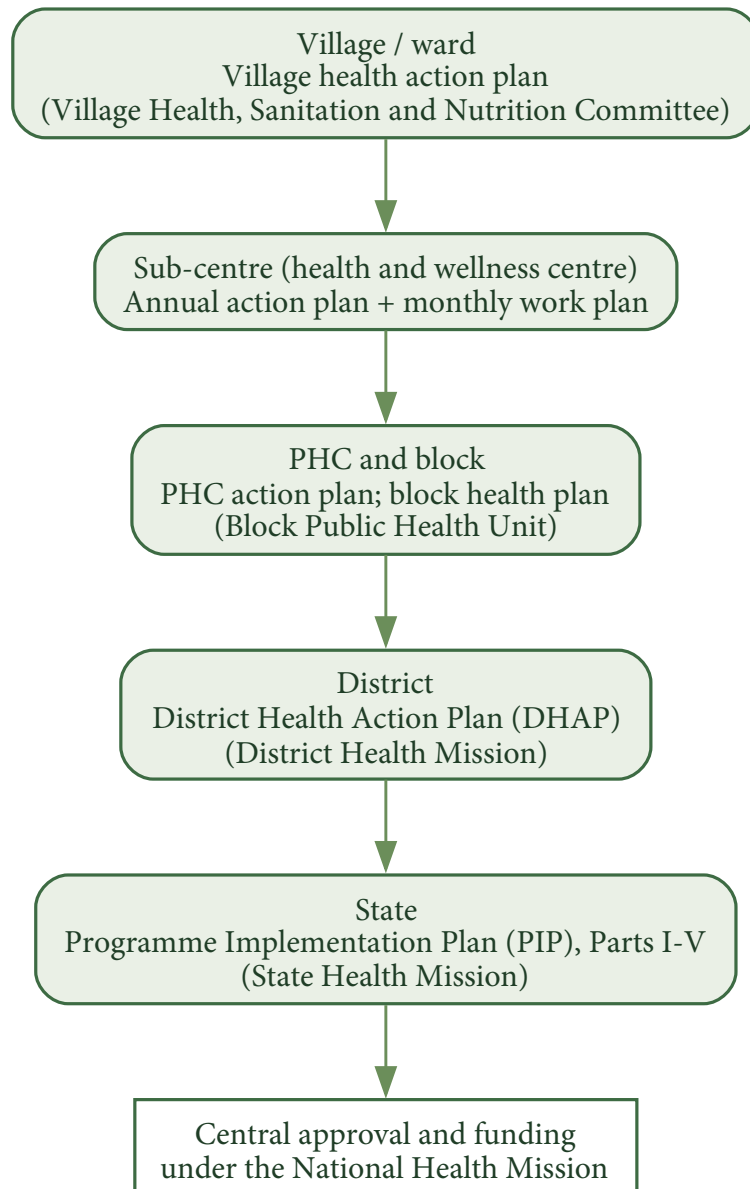
Level	Operational plan produced	Prepared by
Village / census ward	Village health action plan	Village Health, Sanitation and Nutrition Committee — <i>Sarpanch</i> , ASHA, male/female multipurpose health worker, school teacher, <i>Anganwadi</i> worker, self-help group member; untied fund of ₹10,000 per year per village <i>panchayat</i> (Sunder Lal, Ch 16, p 697)
Sub-centre (health and wellness centre)	Sub-centre annual action plan plus the monthly work plan of the female health worker	Sub-centre team; the first step of decentralised planning (Sunder Lal, Ch 16, p 705)
Primary health centre / block	PHC action plan; block health plan through the Block Public Health Unit, supported by XV Finance Commission funds	Medical officer with Child Development Project Officer, <i>Block Samiti</i> , block development and education officers (Sunder Lal, Ch 16, p 711; Ch 17, p 752)
District	District Health Action Plan (DHAP) — the main instrument for planning, intersectoral convergence, implementation and monitoring under the National Health Mission	District Health Mission (Sunder Lal, Ch 17, p 739)
State	Programme Implementation Plan (PIP) — Part I NRHM–RCH Flexipool, Part II NUHM Flexipool, Part III communicable diseases, Part IV non-communicable diseases/injury and trauma, Part V infrastructure maintenance	State Health Mission; central financing to the state is based on the approved PIP (Jugal Kishore, Ch 1, pp 126, 130)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Also draw **Park's planning cycle** (Park 28e, Ch 23, Fig 1, p 1026) with the labels: gather health data → tabulate, analyse, interpret → formulate health problems → assign priorities among problems → formulate individual priority problems for planning solutions → define programme objectives and how to measure achievement → assign priorities among objectives → design alternative programmes to solve problems → select best programme (feasibility, cost/effectiveness) → combine programmes and develop integrated plan → plan implementation of integrated programme → orient and train personnel → initiate integrated programme → operate programme → collect data for evaluation → evaluate results (goals achieved? goals satisfactory?) → YES back to operate programme, NO back to formulate health problems.



The ten steps of operational planning in health care, closing as a cycle



Bottom-up tiers of operational plans in the Indian health system under the National Health Mission

High-yield exam pointers

- ▶ **Mnemonic for the ten steps** — "SPO-RaP-BWIME": Situation analysis, Prioritisation, Objectives (SMART), Resources and alternatives, Programming, Budgeting, Write-up, Implementation, Monitoring, Evaluation → replanning.
- ▶ **SMART objectives** = Specific, Measurable, Appropriate/relevant, Realistic, Time-bound (Sunder Lal, Ch 16, p 704).
- ▶ **Planning horizons**: perspective ~20 years, intermediate-term ~5 years, operational/annual = 1 year tied to the budget cycle (Reinke, Ch 1, p 18).
- ▶ **Monitoring vs evaluation**: monitoring is continuous and uses **input and process** indicators; evaluation is terminal and uses **output, effect and impact** indicators (Sunder Lal, Ch 8, p 340; Ch 16, p 723).

- ▶ PERT weights the expected time estimate four times as heavily as the optimistic or the pessimistic estimate; CPM is the simpler network method that ignores estimation error (Reinke, Ch 15, pp 225-226).
- ▶ **Indian ladder:** village health action plan → sub-centre annual action plan → PHC/block plan → District Health Action Plan → State Programme Implementation Plan (Parts I-V) → National Health Mission funding.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — *operational planning* (converting goals into a dated, costed work plan) must not be confused with *operational research*, which is the application of scientific methods and tools such as network analysis, linear programming, queuing theory and work study to obtain an optimum solution for an operation under control (Sunder Lal, Ch 6, p 265); and the Programme Implementation Plan retains the older National Rural Health Mission and National Urban Health Mission labels because the National Health Mission covers both the rural and the urban missions.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW, WHO, IPHS 2022, NFHS-5, NSS 79th round (CAMS)

Programme and policy updates (India)

- ▶ The Health and Wellness Centre has been renamed the **Ayushman Arogya Mandir**, with the tag-line *Arogyam Parmam Dhanam*, retaining the same brief of preventive, promotive, curative, rehabilitative and palliative care; the national dashboard showed **1,86,917 functional Ayushman Arogya Mandirs** — 1,34,707 sub-health centres, 24,510 PHCs, 5,473 urban PHCs, 9,969 urban health and wellness centres and 12,258 AYUSH facilities. The sub-centre annual action plan in the table above should now be written for an *Ayushman Arogya Mandir*; the older label is the superseded one [PMID 41878109] [MoHFW 2026-Aug].
- ▶ **Single Nodal Agency reform has changed how an approved Programme Implementation Plan actually reaches the field.** Funds for centrally sponsored schemes including the National Health Mission are now routed through one state nodal account instead of multiple society accounts. Key-informant work across administrative levels in Bihar and Odisha (February 2023 to February 2024) found real gains in cash management, expenditure transparency and alignment of central releases with actual sub-national spending, but weak budget credibility, patchy digital connectivity and thin financial-management staffing; the reform also raised compliance pressure to spend while attention to health-system outcomes stayed limited [PMID 42480037].

Current national data

- ▶ The **Comprehensive Annual Modular Survey 2022-23 (NSS 79th round)** is the **current national read on financial protection** — over 302,000 households, four years after PM-JAY rollout. Sixty per cent of households had some health insurance and 49.7% publicly funded cover; PM-JAY or state-scheme cover raised the odds of outpatient use (adjusted OR 1.404, 95% CI 1.328-1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128-1.210). Inpatient catastrophic expenditure at the 10% threshold was 41.3% in publicly insured households against 54.2% in the uninsured, so

cover shifts utilisation more convincingly than it removes financial hardship. Always quote the survey round with the figure [PMID 42167946].

- **Vital-event data feeding situation analysis have improved but remain incomplete.** NFHS-5 (2019-21) recorded 92% institutional births against 86% in NFHS-4 (2015-16), and birth registration of children under five rose from 80% to 89% over the same interval. Under the Registration of Births and Deaths Act 1969 as amended in 2023, many PHCs and CHCs now function as registrars or statutory informants, which converts every institutional delivery into a registration opportunity but adds documentation load to ANMs and ASHAs already carrying maternal, immunisation, nutrition and family-planning targets [PMID 41752339].

Elimination, surveillance and disease status

- **Lymphatic filariasis remains endemic in 345 districts, against a national elimination goal of 2030.** Mass drug administration, morbidity management and disability prevention, and post-MDA surveillance are straining programme capacity, and public-private partnership is being proposed for advocacy, social mobilisation, MDA non-compliance, case identification, digital tools and vector control — the same partnership route already used in polio and TB [PMID 41951149].
- **Malaria planning has moved from district-wide to sub-district micro-stratification.** In Sonbhadra district 18,217 cases were reported between 2017 and 2024 with a 95% fall; annual parasite incidence dropped from 2.7 to 0.1, but annual falciparum incidence rebounded to 0.09 in 2024 after near-total suppression, and scan statistics localised the residual high-risk cluster to a single block. This is the operational argument for allocating resources to identified hotspots rather than uniformly across a district [PMID 41871706].

Guideline and legislative updates

- **District-hospital budgeting is being reworked from historical allocation to need-based payment.** Costing of 27 district hospitals across nine states proposed a blended model — risk-adjusted global budget for outpatient care, case-based bundled payment for inpatient care, per-bed global budget for indirect services. Mean annual pay-out worked out to Rs 326 million (US\$ 3.35 million) per district hospital across 845 hospitals, with inpatient care taking 78% of the budget; upgrading them to IPHS-compliant secondary hospitals would raise budgets by 131%, and to medical colleges by 91.5%. This is the current successor to the classical line-item budget of step 6 [PMID 39478753].
- **Indian Public Health Standards 2022 is now the benchmark against which resources are assessed.** An ICMR-led survey of 415 facilities across 19 districts in seven states scored readiness for diabetes and hypertension care against IPHS 2012 and 2022 and the WHO Service Availability and Readiness Assessment manual, taking 70% as the threshold for a prepared facility: sub-centres 61%, community health centres 59%, PHCs 73% and private primary facilities 57%, while district hospitals, medical colleges and private tertiary facilities cleared 70%. Stockouts of essential medicines were reported from sub-centre up to sub-district hospital — the classic resource-assessment finding of step 4 [PMID 40536370].

Global and WHO developments

- **WHO's Fourteenth General Programme of Work (GPW 14) runs 2025-2028 and was adopted at the Seventy-seventh World Health Assembly in May 2024, superseding GPW 13.** Its six

strategic objectives are climate change as a health threat, determinants and root causes of illness through cross-sector policy, primary health care and essential system capacities for universal coverage, service coverage with financial protection, all-hazards risk preparedness, and rapid detection and response to emergencies. The triple-billion targets by end-2028 are 6 billion people with better health and well-being, 5 billion with universal health coverage without financial hardship, and 7 billion better protected from health emergencies [WHO 2024-May].

► **The amended International Health Regulations entered into force on 19 September 2025.**

Adopted by consensus at the Seventy-seventh World Health Assembly on 1 June 2024, they insert a "pandemic emergency" tier above the public health emergency of international concern, triggered by wide geographical spread across multiple states, health-system capacity exceeded or at risk of being exceeded, substantial social and economic disruption, and a need for coordinated international action; temporary recommendations remain non-binding and lapse after three months unless reissued. National and state operational plans must now build in this higher alert level [WHO 2024-Jun].

SOURCE & COVERAGE

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Recent-advances sources: 41878109 — Beyond coverage: why Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana struggles to deliver healthcare in Jammu and Kashmir, Front Public Health 2026, doi:[10.3389/fpubh.2026.1783458](https://doi.org/10.3389/fpubh.2026.1783458); 42480037 — Challenges to Achieving Universal Health Coverage Through Public Financial Management Reforms: Insights from the Implementation of Single Nodal Agency Reform in India, Health Syst Reform 2026, doi:[10.1080/23288604.2026.2693357](https://doi.org/10.1080/23288604.2026.2693357) (abstract-only); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022-2023), BMJ Open 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, Int J Environ Res Public Health 2026, doi:[10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, Acta Trop 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 41871706 — Identification of micro-level malaria hotspots and its implications for achieving malaria elimination target in a remote district of India, Acta Trop 2026, doi:[10.1016/j.actatropica.2026.108066](https://doi.org/10.1016/j.actatropica.2026.108066) (abstract-only); 39478753 — Framework for responsive financing of district hospitals of India, Front Public Health 2024, doi:[10.3389/fpubh.2024.1398227](https://doi.org/10.3389/fpubh.2024.1398227); 40536370 — Preparedness of public & private health facilities for management of diabetes & hypertension in 19 districts in India, Indian J Med Res 2025, doi:[10.25259/IJMR_755_2024](https://doi.org/10.25259/IJMR_755_2024). Programme and global sources: MoHFW Ayushman Arogya Mandir national dashboard, August 2026; WHO Fourteenth General Programme of Work 2025-2028, adopted May 2024; WHO International Health Regulations amendments, adopted 1 June 2024, in force 19 September 2025.

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Social audit

Asked in: Paper IV 22-Dec-2023 — RGUHS MD Community Medicine

Definition — Social audit is a **local public review of the quality of government decision making** (Narayanasamy — *Participatory Rural Appraisal*, Glossary). Operationally, it is the process in which the intended beneficiaries of a public programme themselves examine the programme's records, verify them against what was actually received on the ground, and place the findings before an open public forum such as the *gram sabha*, so that the implementing agency answers to the people and not only to its own hierarchy. It is one expression of **accountability** — the duty of decision-makers in government, the private sector and civil society organisations to be answerable to the public as well as to institutional stakeholders (Oxford Textbook of Public Health 7e, Ch 51, pp 1894–1895).

How social audit differs from the other audits in a health system

Feature	Financial audit	Performance audit	Medical / nursing audit	Social audit
Who audits	Accounts officer / chartered accountant, internal to government	Independent, outside appraisal body (Reinke, Ch 4, p 64)	Peers — doctors and nurses	The beneficiary community itself, facilitated by an independent team
Question asked	Were funds managed as per the budget and General Financial Rules? (Sunder Lal, Ch 16, pp 716–717)	Did the programme achieve its stated objectives?	Was the clinician's performance up to a predetermined standard? (Park 28e, Ch 23, p 1037)	Did the promised entitlement actually reach the person it was promised to, in the promised quality?
Evidence used	Vouchers, receipts, bills paid, percentage variance from budget (Health Program Planning and Evaluation, Ch 10, p 276)	Programme output and outcome data	Case records	Records + door-to-door verification + beneficiary testimony
Forum	Closed; report goes upward to authority	Report to management/legislature	Departmental meeting	Open public hearing (<i>gram sabha</i> / <i>jan sunwai</i>)
Domain of accountability	Fiscal	Efficiency and legal	Service delivery quality	Coverage, equity and service delivery — did the target population get it? (Health Program Planning and Evaluation, Ch 10, p 276)

Essential features are therefore: (i) the auditor is the citizen, not the accountant; (ii) the yardstick is the declared entitlement, not the ledger; (iii) verification is by triangulating written records against people and physical assets; (iv) findings are read out publicly, in the local language, in the presence of officials; and (v) a time-bound action-taken report closes the loop.

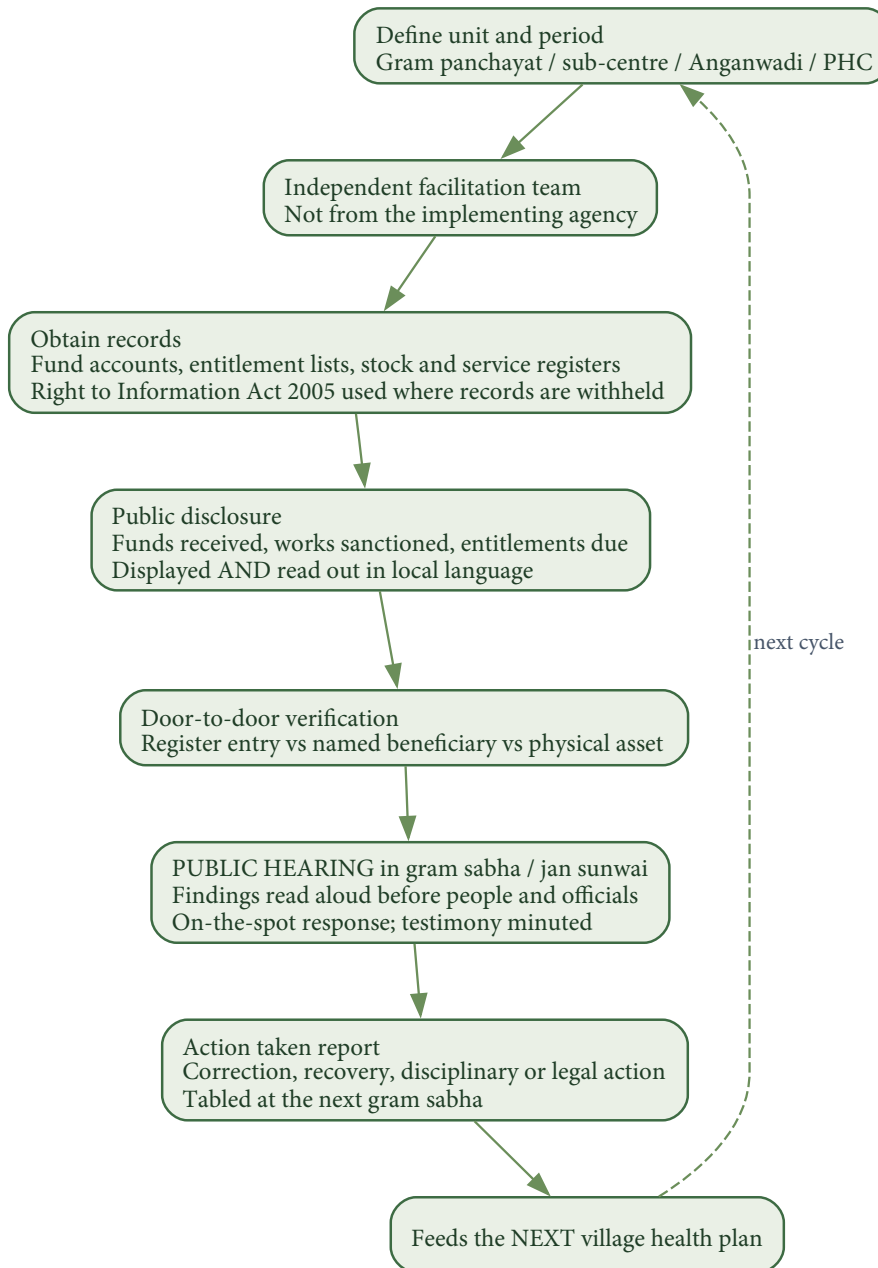
Statutory and policy basis of social audit in India

Instrument	Provision that mandates or enables social audit
73rd and 74th Constitutional Amendment Acts, 1992	Made health, family welfare and education a responsibility of village <i>panchayats</i> and <i>nagarpalikas</i> ; Panchayati Raj Institutions became the vehicle of decentralised planning and programme implementation (Sunder Lal, Ch 9, p 553)
<i>Gram sabha</i>	All adult residents above 18 years, with voting rights, who "keep a watch over the functions of the village <i>panchayat</i> " (Sunder Lal, Ch 2, p 67); it must meet at least twice a year (Park 28e, Ch 23, p 1036) — this is the constitutional forum in which a social audit is held
Right to Information Act, 2005	Provides the practical regime for citizens to secure access to information held by public authorities "in order to promote transparency and accountability in the working of every public authority"; Public Information Officers must supply information in time (Red Book — Public Health and Preventive Medicine, Ch 2, p 377) — this is how audit teams obtain the records
MGNREGA, 2005	All implementing agencies are accountable to the public for their work; social audit and right to information apply to each aspect of implementation ; the <i>gram sabha</i> identifies works and the <i>panchayats</i> hold principal responsibility for planning, implementation and monitoring (Jugal Kishore, Ch 3, p 918) — the template later copied by health and nutrition schemes
National Food Security Act, 2013	Transparency provisions: all Public Distribution System records open to public inspection; periodic social audits of the PDS and other welfare schemes ; end-to-end computerisation; vigilance committees at state, district, block and fair-price-shop level (Park Plus, Ch 11, p 140)
National Rural Health Mission / Indian Public Health Standards	" Social audit by involvement of the community through Rogi Kalyan Samiti is being recommended "; quality maintained by external monitoring through Panchayati Raj Institutions plus internal monitoring at appropriate intervals; a Rogi Kalyan Samiti is mandatory in every community health centre, with the Charter of Patients' Rights displayed at the entrance and grievance redressal under its supervision (Jugal Kishore, Ch 1, p 142)
Saksham Anganwadi and Poshan 2.0	Active participation of Village Health Sanitation and Nutrition Committees and mothers' groups for "regular monitoring and social audit and local community ownership of the programme" (AFMC Red Book of Community Medicine 2024, Ch 38, p 1207)
Draft National Health Bill	Community-based monitoring framework under the proposed National Public Health Board, and dispute resolution through public dialogues and public hearings on health — <i>Swasthya Jan Sunwais</i> — as a non-adversarial community-level redressal mechanism, backed by in-house complaint forums and designated district courts (Jugal Kishore, Ch 2, p 842)

How a social audit is conducted

- **Define the unit and period** — one *gram panchayat*, sub-centre, Anganwadi centre or primary health centre, for a stated financial year or programme cycle.
- **Constitute an independent facilitation team** — resource persons and community volunteers who are *not* drawn from the implementing agency, since an audit of performance requires independent, outside appraisal (Reinke, Ch 4, p 64).

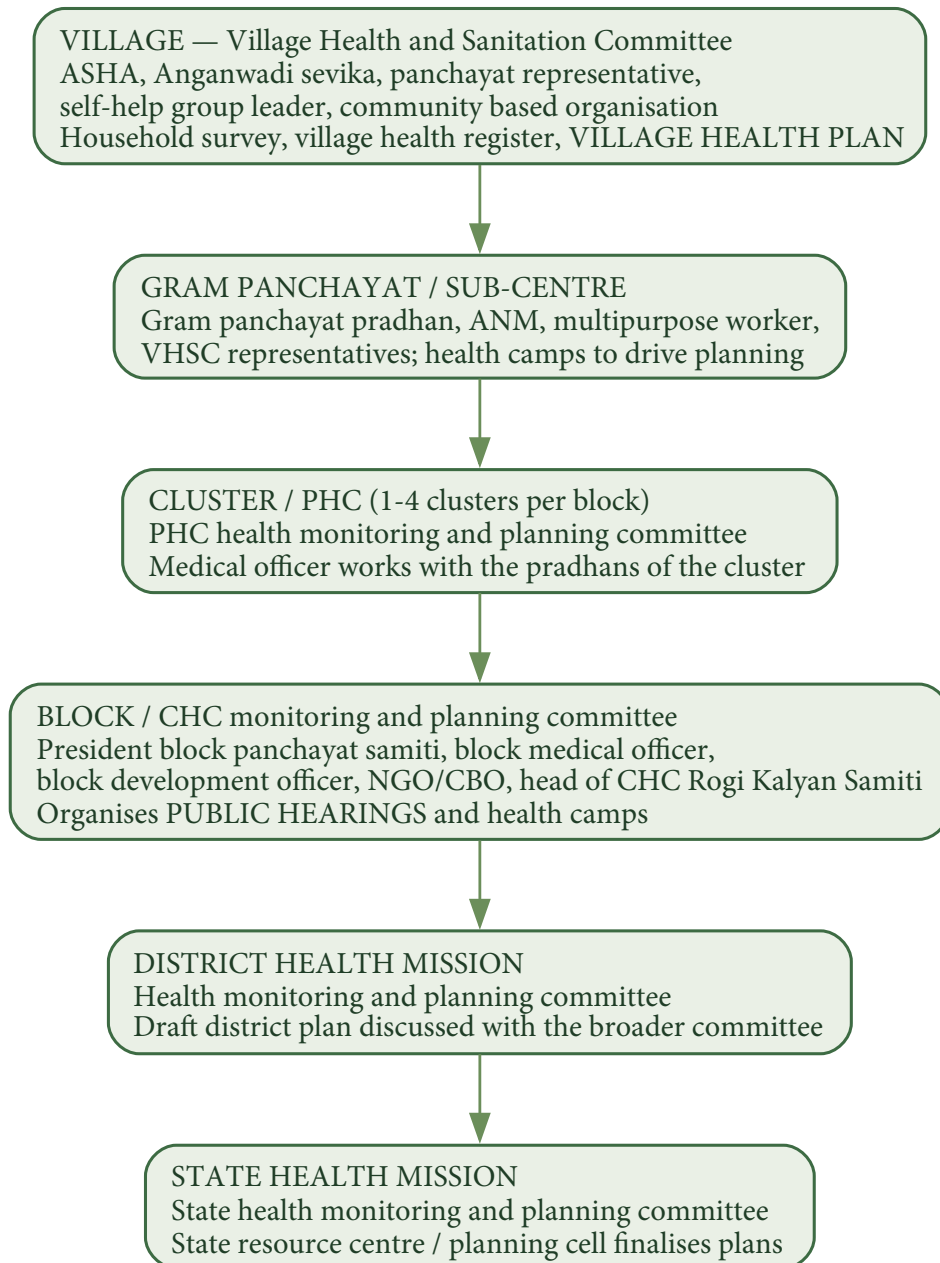
- ▶ **Obtain the records** — untied-fund and village-health-fund accounts, beneficiary and entitlement lists, drug and stock registers, service registers; secured under the Right to Information Act, 2005 where they are withheld.
- ▶ **Public disclosure** — funds received, works sanctioned and entitlements due are displayed at the *panchayat* office or facility and read out, so that the illiterate are not excluded.
- ▶ **Door-to-door verification** — each register entry is cross-checked against the named beneficiary and against the physical asset or commodity, and objections are recorded.
- ▶ **Public hearing in the *gram sabha* / *jan sunwai*** — findings are read aloud before beneficiaries, elected representatives and officials; officials respond on the spot; testimony is minuted.
- ▶ **Action-taken report** — corrective action, recovery of misused funds or disciplinary/legal action with a stated deadline, tabled at the next *gram sabha*.
- ▶ **Feed back into planning** — the corrected picture becomes an input to the next village health plan, closing the planning–implementation–evaluation loop.



The social audit cycle — from disclosure of records to a time-bound action taken report

Social audit in health sector planning

Indian health planning under the National Health Mission is **bottom-up**: the village health plan is the first brick and every higher plan is an aggregation of the one below, which is precisely what makes a community audit of the plan possible. The community's role is threefold — to contribute facilities, manpower and funds; to be actively involved in planning, management and evaluation; and to use the services (Park 28e, Ch 2, p 34). Health planning has accordingly shifted from **health care for the people** to **health care by the people** (Park 28e, Ch 2, p 34).



Bottom-up NHM planning ladder and the level at which social audit enters

At each rung the plan is a public document, so the audit questions are the same at every level: *was the household survey actually done, does the plan reflect what the survey found, were the untied funds spent on what the plan said, and did the deprived hamlet get anything?* Block and district committees are explicitly required to **organise public hearings and health camps "in order to make the planning process activity intensive"** (Jugal Kishore, Ch 1, pp 135–136).

Social audit in programme implementation

Platform	Level	What it audits and how
Village Health, Sanitation and Nutrition Committee (VHSNC)	Village	Creates public awareness of programmes; develops the village health plan; ensures the ANM and multipurpose worker visit on the fixed day and oversees the ANM, multipurpose worker and Anganwadi worker; takes a bimonthly health-delivery report from providers; discusses, analyses and gets registered in the <i>panchayat</i> every maternal and neonatal death in the village; manages the village health fund (Jugal Kishore, Ch 1, p 138)
VHSNC as the community monitoring centre under ICDS	Village	Functions as a sub-committee of the Panchayati Raj Institution; membership fixed as village <i>sarpanch</i> , ASHA, Anganwadi worker, two pregnant and lactating mothers on rotation , community members, two village organisations and the village secretary (ICDS District Manual, Ch 9, p 17)
Anganwadi feedback loop	Anganwadi centre	Feedback collected through ICDS accreditation systems and community public hearings (ICDS District Manual, Ch 9, pp 24–25); community-based monitoring is encouraged alongside administrative monitoring "to bring in transparency and accountability in delivery of services" (ICDS District Manual, Ch 6, p 14)
Rogi Kalyan Samiti / patient welfare society	PHC, CHC, district hospital	Community management of public hospitals; the named vehicle for social audit of facility services , for supervising grievance redressal and for displaying the Charter of Patients' Rights (Jugal Kishore, Ch 1, pp 138, 142)
<i>Jan sunwai</i> / public hearing	Block and district	Public hearings organised by block and district planning committees (Jugal Kishore, Ch 1, p 136); <i>Swasthya Jan Sunwais</i> proposed as the standing health grievance forum (Jugal Kishore, Ch 2, p 842)
Civil society and non-governmental organisations	All levels	Engaged under the Mission specifically "for monitoring, evaluation, and social audit", alongside the standing monitoring group for ASHA and task-force membership (Jugal Kishore, Ch 1, p 140); India's <i>Centre for Health and Social Justice</i> carries out social auditing and presses for public health policy to be formulated and implemented on social-justice principles, as ActionAid International does internationally (Textbook of International Health, Ch 14)

Social audit thus supplies the community's answer to Park's three evaluation questions — **structure** (is the sub-centre staffed and stocked?), **process** (did the ANM come on the fixed day?) and **outcome** (did the entitlement change anything?) (Park 28e, Ch 23, pp 1037–1038).

Strengths, limitations and requirements for success

Strengths	Limitations
Detects leakage, ghost beneficiaries and non-delivery that record-based audit cannot see	Powerful local interests can capture the <i>gram sabha</i> ; the poorest may not speak
Empowers the community and converts passive recipients into active participants (Park 28e, Ch 2, p 34)	Findings are non-binding unless the scheme's own law makes action mandatory

Strengths	Limitations
Cheap, uses an existing constitutional forum, needs no new bureaucracy	Needs literacy, facilitation skill and independent facilitators, who are scarce
Improves equity by asking who was left out , not only what was spent	Risk of becoming a ritual meeting with a pre-written report
Complements the governance toolkit of grievance redress, internal/external audit and performance-based contracts (Oxford 7e, Ch 51, pp 1894–1895)	Poor follow-through: without a time-bound action-taken report the exercise loses credibility

Requirements for success: proactive public disclosure of records; an independent facilitation team; the *gram sabha* or *jan sunwai* as the forum; guaranteed protection for those who testify; and a mandatory, time-bound action-taken report placed before the next public meeting.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — "The accountability triangle of a health programme" (draw a triangle with a label at each corner and arrows along each side). Corner labels, exactly: **Citizens / beneficiaries** (*gram sabha*, VHSNC, Rogi Kalyan Samiti); **Elected representatives** (*panchayat*, *block samiti*, *zila parishad*); **Health service providers** (ASHA, ANM, medical officer, district health administration). Side-arrow labels, exactly: from citizens to representatives — "*vote and voice*"; from representatives to providers — "*compact and oversight*"; from providers back to citizens — "*service delivery*"; and draw a **short direct arrow from citizens to providers** labelled "**SOCIAL AUDIT — the short route of accountability**". Add the base caption: *Right to Information Act 2005 supplies the information; the gram sabha supplies the forum.*

High-yield exam pointers

- **One-line definition to write first:** social audit = "a local public review of the quality of government decision making" — the beneficiaries audit the entitlement, not the accountant auditing the ledger.
- **Four legal anchors, in order:** 73rd/74th Constitutional Amendment Acts **1992** → Right to Information Act **2005** → MGNREGA **2005** (social audit + RTI apply to *each aspect* of implementation) → National Food Security Act **2013** (periodic social audits of the PDS; vigilance committees at state, district, block and fair-price-shop level).
- **The health-specific sentence examiners look for:** under the Indian Public Health Standards, **social audit through the Rogi Kalyan Samiti** is recommended, with external monitoring by Panchayati Raj Institutions and internal monitoring at intervals.
- **Village-level mnemonic** — VHSNC does "**PLAN-VISIT-DEATH-FUND**": village health **PLAN**; ensure the fixed-day **VISIT** of ANM and multipurpose worker; discuss and register every maternal and neonatal **DEATH**; manage the village health **FUND**.
- **Forum names to use:** *gram sabha* (meets at least twice a year, all residents above 18 years), *jan sunwai* / public hearing, and *Swasthya Jan Sunwai* in the draft National Health Bill.

- ▶ **Link to evaluation theory:** social audit is community-conducted evaluation of **structure, process and outcome** (Park's three types of evaluation) and delivers **coverage and service-delivery accountability**, which financial audit cannot.
- ▶ **Steps to reproduce (8):** define unit → independent team → obtain records under RTI → public disclosure → door-to-door verification → public hearing → action-taken report → feed the next plan.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam — the retrieved Indian corpus documents social audit mainly through the statutes and programme documents that mandate it (MGNREGA 2005, National Food Security Act 2013, Indian Public Health Standards/Rogi Kalyan Samiti, ICDS and Poshan 2.0) rather than as a stand-alone textbook chapter, so the eight-step conduct sequence and the accountability-triangle diagram above are the standard, universally-taught operational framing rather than a verbatim textbook list; the Village Health and Sanitation Committee was renamed the Village Health, Sanitation and Nutrition Committee when nutrition was added to its remit, and Integrated Child Development Services now runs as Saksham Anganwadi and Poshan 2.0, so give the current name first and the older one in parentheses once.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHA77 2024 resolution on social participation, National Health Accounts 2021–22, National Quality Assurance Standards and Kayakalp, Indian Public Health Standards 2022

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandir** — give the current name first and the old one once in brackets. More than 1.75 lakh have been operationalised and **1,35,343 Jan Arogya Samitis** stood constituted at them as on 30 November 2024, with the VHSNC in rural and the **Mahila Arogya Samiti** in urban areas as the paired community platform. The JAS itself convenes an **Arogya Sabha twice every year** at the facility, following the monthly village-level *Samvad* consultation in the villages it covers — the facility-level counterpart of the *gram sabha* hearing. Money for these facilities now also arrives through XV Finance Commission health grants routed via Panchayati Raj Institutions and Urban Local Bodies and through PM-ABHIM, alongside National Health Mission grants — which is what puts an auditable health account inside the panchayat itself. [PMID 41816146]
- ▶ **"Community Action for Health" (CAH) is the National Health Mission's strategy for placing people at the centre of the process so that their health needs and rights are fulfilled** — first implemented in 2007 in three districts of Assam (Dhemaji, Chirang and Kamrup) and in eight other Indian states, later expanded to 27 districts. A comparative survey of one aspirational district (Goalpara) and one non-aspirational district (Nagaon), run from August 2021 to March 2022 across 219 respondents, found 94.1% of community respondents in the aspirational district and 53.6% in the non-aspirational district unaware of CAH: the audit platform exists on paper well ahead of the community's awareness of it, and the authors' remedy is effective use of the

funds already allocated for CAH plus periodic monitoring of CAH activities at every level of the health system. [PMID 41045525]

Current national data

- ▶ **Jan Arogya Samitis are doing health promotion but not accountability.** A 2025 sequential mixed-methods appraisal in a Punjab block found the JAS performing the functions of leading health promotion and enabling quality service delivery, but playing a passive role in supporting village-level committees, monitoring services and facilitating social accountability, largely from awareness gaps. Perceived decision space, capacity and accountability were low to moderate, with decision space and accountability significantly higher among elected representatives than among the service recipients they are supposed to speak for. [PMID 41074751]
- ▶ **Figures worth quoting only with their year attached:** UHC service coverage index 57 in 2015 rising to 63 in 2021 (WHO); out-of-pocket expenditure 60.6% in 2014–15 falling to 39.4% in 2021–22 (National Health Accounts estimates for India 2021–22); public-facility outpatient utilisation 91.6 crore in 2012–13 rising to 134.5 crore in 2021–22. [PMID 41816146]
- ▶ **The strongest current argument for auditing empanelled private hospitals.** State-representative household surveys in Chhattisgarh in 2021 and 2022 found PM-JAY enrolment was not associated with greater inpatient utilisation, better quality or reduced out-of-pocket spending. Among enrollees using private hospitals, 78.1% (2021) and 70.9% (2022) crossed the catastrophic-expenditure threshold of 10% of annual consumption; mean out-of-pocket expenditure was ₹23,691 against mean annual non-health household consumption of ₹97,365; and *double billing* — cash taken from the patient while the same service is claimed from the scheme — persisted despite empanelment contracts prohibiting copayment. [PMID 39135015]

Elimination, surveillance and disease status

- ▶ **Community-based monitoring and planning (CBMP) is the working form of the social audit under the Mission.** In Maharashtra and Chhattisgarh, villagers, health workers and local officials meet quarterly to review service data and issue public *report cards*; sites running CBMP recorded higher outpatient attendance, better drug availability and greater user satisfaction, and Maharashtra evaluations linked training villagers to independently audit facilities with improvement in maternal health indicators, complaint resolution and service utilisation. Citizen-sourced surveillance is also entering the formal stream — the Integrated Disease Surveillance Programme has added social-media listening for early outbreak warning. [PMID 41907566]

Guideline and legislative updates

- ▶ **Know the quality-certification stack the community now audits *against*:** Kayakalp (2015; a facility must clear 70% on peer assessment; enrolment grew from 100 facilities in 2015–16 to 19,488 in 2022–23), NQAS (district hospitals 2013, PHC and CHC 2014, health and wellness centres 2020–21; 10 certified facilities in 2016–17 to 829 in 2022–23), LaQshya (2017), MusQan (2021), SaQushal patient-safety self-assessment (2022), the SaQsham certification portal (2023) and AB-PMJAY Bronze/Silver/Gold certification with the National Health Authority and Quality Council of India; revised IPHS came in 2022, and *Mera Aspataal* (My Hospital) has collected patient feedback on the MyGov platform since 29 August 2016. Almost all of these carry patient-safety or patients'-rights standards and a satisfaction survey — "**Patients' Rights**" is a named area of concern specifically in the NQAS family (NQAS for PHC, NQAS for SHC-HWC, LaQshya

and MusQan) — which is the documentary hook a social audit pulls on. [PMID 41280589] But the certification score is not itself the audit: a facility-versus-community comparison of Kayakalp in 36 health and wellness centres of Ganjam, Odisha found 72% cleared the 70% mark while 72% had no designated cleaning staff, 61% lacked functional biomedical waste disposal, 38% had irregular or no piped water and 25% sat outside the village settlement (some on stigmatised sites), with domain means of 49% for waste management, 52% for infection control and 45% for support services — only the frontline and community accounts explained *why*, and the recommendation is explicitly community-led monitoring. [PMID 40978827]

- **Grievance redressal is the audit's legal counterpart.** A critical synthesis anchored in Supreme Court of India patient-rights litigation lists the implementation tools as patient-rights charters, ombudsmen, tribunals, health-professional councils, blame-free reporting, **direct community monitoring** and the courts, and sets the normative test for any redressal body — administrative, financial and statutory autonomy, so that it is genuinely impartial and independent of the provider it judges. It also documents precisely the group a social audit exists to protect: the poor, rural, elderly, migrants and those with low social capital are the least likely to lodge a complaint at all. [PMID 34133734]

Global and WHO developments

- **The 77th World Health Assembly (2024) adopted a resolution on social participation for universal health coverage, health and well-being,** calling for inclusive health-system decision-making with marginalised communities — the global mandate now sitting behind India's *gram sabha* / VHSNC / JAS route. A realist review of 150 studies found the strategies that make participation genuinely inclusive are securing access to the platform, tailored recruitment and culturally safe methods; the recurring failure mode is tokenism — communities invited to speak but never empowered to decide. [PMID 42216032]
- **Human-rights-based social accountability yields most where exclusion is worst.** A Dalit women's collective in Raichur, Karnataka — about 4,600 members drawn from 50 villages across 25 *gram panchayats* and 10 PHC areas — combined community enquiry with repeated public campaign cycles, using the ICESCR standard and General Comment 14's availability, accessibility, affordability and quality tests as the audit yardstick, and changed both primary care and specific maternal-health practices. A parallel monitoring framework built on the WHO–UNICEF primary-health-care measurement model makes the complementary point that community engagement must itself be measured with its own structure, input, process and output indicators, not inferred from programme outcomes. [PMID 36414960; PMID 41476785]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 Concept of Health and Disease, p 34; Ch 23 Health Planning and Management, pp 1036–1038); Jugal Kishore — National Health Programmes of India (Ch 1, pp 135–136, 138, 142; Ch 2, p 842; Ch 3, p 918); Sunder Lal — Textbook of Community Medicine (Ch 2, p 67; Ch 9, p 553; Ch 16, pp 716–717); Oxford Textbook of Public Health 7e (Ch 51 Governance and management of public health programmes, pp 1893–1895); Health Program Planning and Evaluation — A Practical Systematic Approach (Ch 10, p 276); Reinke — Health Planning for Effective Management (Ch 4, p 64); N. Narayanasamy — Participatory Rural Appraisal: Principles and Methods (Glossary); Red Book — Public Health and Preventive Medicine (Ch 2, p 377, Right to Information Act 2005); AFMC Red Book of Community Medicine 2024 (Ch 38, p 1207, Poshan 2.0); Integrated Child Development Services District Manual (Ch 6, p 14; Ch 9, pp 17, 24–25); Park Plus (Ch 11, p 140, National Food Security Act 2013); Textbook of International Health (Ch 14). **Coverage note:** no indexed textbook in this corpus carries a dedicated "social audit" chapter, so the definition is taken from the participatory-appraisal glossary and the health application is assembled from the programme documents cited above.

Recent-advances sources: 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *J Family Med Prim Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 41045525 — Status of Community Action for Health in selected districts of Assam, India: a quantitative study, *Indian J Public Health* 2025, doi:[10.4103/ijph.ijph.1006.23](https://doi.org/10.4103/ijph.ijph.1006.23) (abstract-only); 41074751 — Exploring community participation through local health facility committees in Punjab: extent of decision space, capacity, and accountability, *Indian J Public Health* 2025, doi:[10.4103/ijph.ijph.421.24](https://doi.org/10.4103/ijph.ijph.421.24) (abstract-only); 39135015 — The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana after four years of implementation — is it making an impact on quality of inpatient care and financial protection in India?, *BMC Health Serv Res* 2024, doi:[10.1186/s12913-024-11393-2](https://doi.org/10.1186/s12913-024-11393-2); 41907566 — Strengthening the structure of public health: suggesting the incorporation of 'People' component as the 'seventh block', *Ann Glob Health* 2026, doi:[10.5334/aogh.5012](https://doi.org/10.5334/aogh.5012); 41280589 — To gain insights and compare different quality certification programs for public health facilities in India, *J Family Med Prim Care* 2025, doi:[10.4103/jfmpc.jfmpc.1591.24](https://doi.org/10.4103/jfmpc.jfmpc.1591.24); 40978827 — Integrating facility metrics with grassroots insights: a mixed-methods study to improve Kayakalp outcomes at health and wellness centres in Ganjam, Odisha, *Cureus* 2025, doi:[10.7759/cureus.90532](https://doi.org/10.7759/cureus.90532); 34133734 — Multilevel governance framework on grievance redressal for patient rights violations in India, *Health Policy Plan* 2021, doi:[10.1093/heapol/czab066](https://doi.org/10.1093/heapol/czab066); 42216032 — Inclusiveness of social participation for health reform: findings from a realist review, *Int J Equity Health* 2026, doi:[10.1186/s12939-026-02847-6](https://doi.org/10.1186/s12939-026-02847-6); 36414960 — Employing innovative evidence-backed community processes for maternal health services by Dalit women, *Int J Equity Health* 2022, doi:[10.1186/s12939-022-01776-4](https://doi.org/10.1186/s12939-022-01776-4); 41476785 — Accelerating community engagement: measuring results, *Int J Public Health* 2025, doi:[10.3389/ijph.2025.1608499](https://doi.org/10.3389/ijph.2025.1608499). Also used: World Health Assembly resolution on social participation for universal health coverage, health and well-being (WHA77, 2024); National Health Accounts estimates for India 2021–22; National Quality Assurance Standards, Kayakalp,

LaQshya, MusQan, SaQushal and SaQsham; Indian Public Health Standards 2022.



Telemedicine in India

Asked in: Paper IV 03-Oct-2025 — RGUHS MD Community Medicine

Definition — The World Health Organization (WHO, 2002) defines telemedicine as "the delivery of health care services, where distance is a critical factor, by all health care professionals using information and communication technologies for the exchange of valid information, for diagnosis, treatment and prevention of disease and injuries, research and evaluation and for the continuing education of health care providers, all in the interest of advancing the health of individuals and their communities" (Suryakantha, Ch 50, p 927). The word combines *tele* (Greek, "distance") and *mederi* (Latin, "to heal"). It began with the National Aeronautics and Space Administration (NASA) efforts of 1960, and its primary public-health aim is **to reach the unreached** by moving information rather than moving the patient.

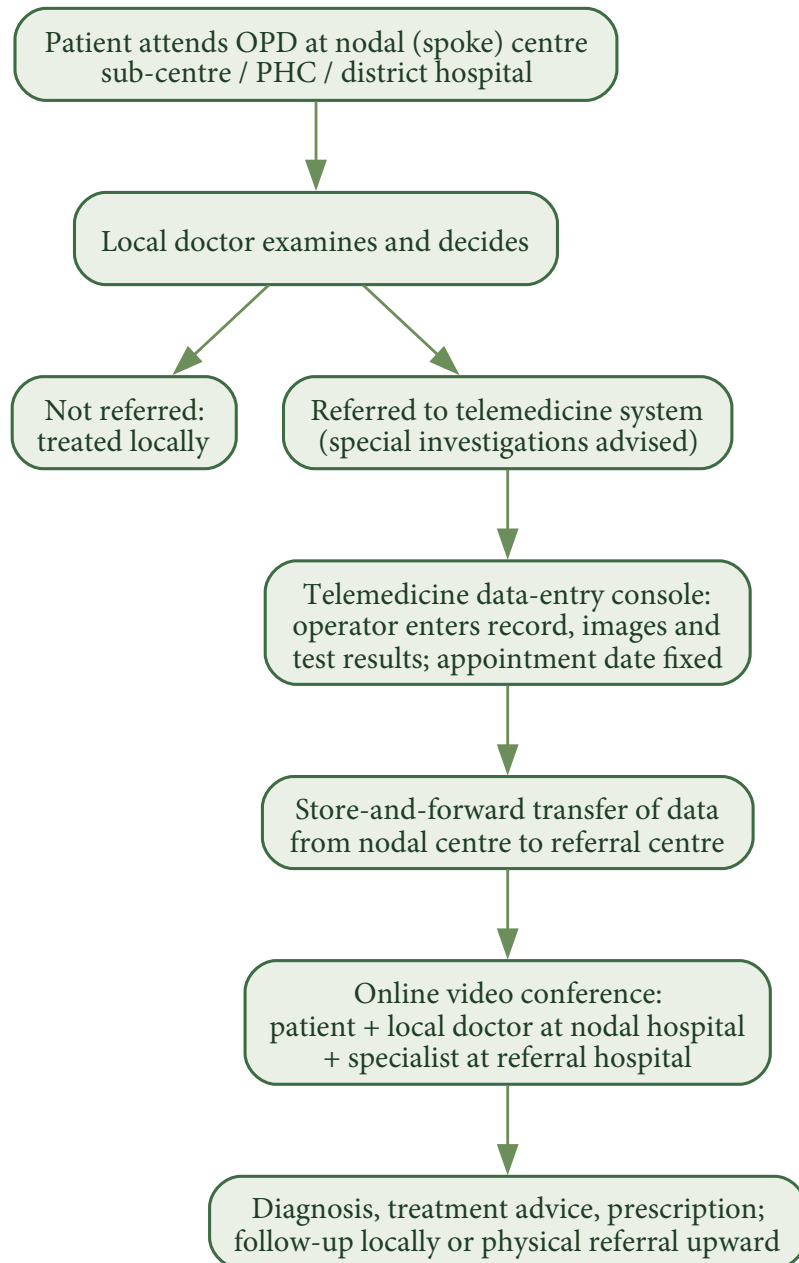
Types and modes of telemedicine

Axis of classification	Categories	Content
By timing of exchange	Store-and-forward (asynchronous)	Patient data and digital images e-mailed to the specialist for later diagnosis and management advice; typically provider-to-provider (Park Plus, Ch 24, pp 374-375; Suryakantha, Ch 50, p 927)
	Real-time (synchronous)	Videoconferencing for a live consultation involving the patient, the primary-care provider and the specialist; typically provider-to-patient
By purpose	Clinical	Teleconsultation, second opinion, complex image interpretation, remote critical-care monitoring
	Educational	Tutorial notes, audio/video resources by e-mail; lectures transmitted by videoconference to multiple sites simultaneously — continuing medical education (CME)
	Administrative	Meeting notices and requirements circulated electronically; audio/video conferencing for interactive discussion between participants
By discipline	Teleradiology, telepathology, teledermatology, telepsychiatry, teleophthalmology	Telemedicine is organised around such clinical disciplines, whereas telehealth is the broader term covering disease management, monitoring and patient compliance (Suryakantha, Ch 50, pp 927-928)

Axis of classification	Categories	Content
By link	Doctor-to-doctor (hub-and-spoke) vs patient-to-doctor (direct-to-consumer)	The two service streams of India's eSanjeevani platform (Sunder Lal, Ch 17, p 748)

Essential components and the teleconsultation pathway

- **Telemedicine Consultation Centre (TCC)** at the peripheral or *spoke* end — the patient, the immediate health-care provider, a trained telemedicine technician, and the equipment that captures and transmits the patient's medical information (Suryakantha, Ch 50, p 928).
- **Nodal (spoke) hospital equipment** — computer with customised medical software, scanner, digital camera, microscope and other medical instruments, videoconferencing unit, printer, modem.
- **Connectivity** — ordinary telephone line, ISDN, broadband, or, in the Indian Space Research Organisation (ISRO) model, a commercial Very Small Aperture Terminal (VSAT) satellite link at each location (Jugal Kishore, Ch 2, p 739).
- **Referral (hub) hospital** — the specialist or super-specialist who receives the record, studies it, and advises during the videoconference.
- **Information management back-end** — patient information, clinical and investigation data, appointment scheduling, and archival and retrieval of records (Suryakantha, Ch 50, p 927).



Patient pathway through a telemedicine consultation, nodal centre to referral hospital

Telemedicine in India — the existing architecture

Initiative	Year / agency	Key features
ISRO telemedicine pilot	2001, Indian Space Research Organisation	Proof-of-concept using customised medical software + hardware + diagnostic instruments on a commercial VSAT link (Jugal Kishore, Ch 2, p 739)
ISRO national network	ISRO with State governments and private hospitals	Grew to 100 hospitals — 78 remote/rural/district hospitals linked to 22 super-speciality hospitals (Jugal Kishore, Ch 2, p 739); a later count records 245 hospitals — 205 district/rural and 40 super-speciality (Suryakantha, Ch 50, p 931). Covers Jammu & Kashmir (including Leh and Kargil), Lakshadweep, Andaman & Nicobar, North-Eastern States, Army field hospitals; also mobile telemedicine units for rural camps, especially ophthalmology (Park Plus, Ch 24, p 375)

Initiative	Year / agency	Key features
Karnataka network	State + ISRO	11 district/taluk hospitals linked to five super-speciality hospitals in Bengaluru and Mysuru; the pilot delivered over 10,000 teleconsultations, and Rabindranath Tagore International Institute of Cardiac Sciences (Kolkata) and Narayana Hrudayalaya (Bengaluru) act as free-of-cost hubs for seven States (Suryakantha, Ch 50, pp 931-932; Jugal Kishore, Ch 2, p 740)
Onconet-India	National Cancer Control Programme	Telemedicine project linking 27 Regional Cancer Centres, each to 4-5 peripheral centres (Jugal Kishore, Ch 1, p 584)
eSanjeevani AB-HWC	MoHFW, National Health Mission	Doctor-to-doctor teleconsultation; all 1,50,000 Ayushman Bharat Health and Wellness Centres (AB-HWCs) connected to specialists and super-specialists (Sunder Lal, Ch 17, p 748)
eSanjeevani OPD	MoHFW, upgraded during COVID-19	Patient-to-doctor teleconsultation from the citizen's home, free of cost, to ensure continuum of care (Sunder Lal, Ch 17, p 748)
Teleconsultation in the service package	Ayushman Bharat HWCs	Teleconsultation is an explicit component at sub-centre-HWC, PHC/UPHC-HWC, CHC/sub-district hospital, district hospital and medical college levels under the National Programme for Prevention and Control of Non-Communicable Diseases (AFMC Red Book 2024, Ch 38, Table 38.13); HWCs are designated platforms for teleconsultation and expanded diagnostics (Park 28e, Ch 10, p 731)
Ayushman Bharat Digital Mission (ABDM)	Launched September 2021; National Health Authority	Digital health identity (Ayushman Bharat Health Account) for every citizen, longitudinal electronic health records, national portability — explicitly intended to encourage the use of technologies such as telemedicine (Sunder Lal, Ch 17, p 763 and Ch 8, p 535)

Advantages in the Indian context

Domain	Advantage
Equity and access	Makes national health services more accessible, available and equitable ; potentially the cheapest and fastest way to bridge the rural-urban health divide (Suryakantha, Ch 50, p 932). India — 32,87,268 sq km, 1.4 billion people, hilly and island terrain, large tribal populations — is described as an ideal setting for telemedicine-assisted delivery (Jugal Kishore, Ch 2, p 739)
Reaching the unreached	Rural doctors reach specialists over secure links carrying voice and video, allowing more accurate assessment for patient management; improves specialist access for peripheral and non-ambulant patients (Jugal Kishore, Ch 2, p 739; Textbook of International Health, Ch 12, p 664)
Cost and time to the patient	Patient travel time, effort and expense are significantly reduced — directly reducing out-of-pocket and indirect costs
Emergency and disaster care	Swift consultation lets emergency doctors decide on trauma-team mobilisation or air evacuation; by reducing helicopter transport it greatly reduces the cost of emergency care; also used in disaster management (Jugal Kishore, Ch 2, p 739; Suryakantha, Fig 50.4)

Domain	Advantage
Efficient use of scarce specialists	Optimises the value of highly trained sub-specialists by making their expertise available to a wider population through physician-to-physician telemedicine; improves efficiency in use of scarce equipment (Oxford Textbook of Public Health 7e, Ch 101; Textbook of International Health, Ch 12, p 664)
Public-health surveillance	Gives new insight into geographical distribution and gradients of disease, identifies populations at risk and risk factors, aids interventional planning, helps anticipate epidemics, and enables real-time disease monitoring locally and globally (Park Plus, Ch 24, p 376)
Health communication and self-care	Relays information, enables informed decision-making, supports peer exchange and emotional support, and promotes self-care and domiciliary care (Park Plus, Ch 24, p 376)
Education and capacity building	Continuing medical education, continuing professional development and consultation with senior colleagues; supports new competencies among health workers (Suryakantha, Ch 50, p 932; Oxford Textbook of Public Health 7e, Ch 101)
Continuity of care	Home care and ambulatory monitoring, chronic disease management and programme tracking; national portability of records once linked to ABDM (Suryakantha, Fig 50.4; Sunder Lal, Ch 8, p 535)

Challenges in the Indian context

Domain	Challenge
Financial	Initial investment is too high; the recurring cost of telemedicine is a barrier to implementation (Suryakantha, Ch 50, p 932; Park Plus, Ch 24, p 376)
Infrastructure	Poor data communication infrastructure, inadequate bandwidth, unreliable power supply and connectivity; correct diagnosis and data pacing need advanced biological sensors and greater bandwidth (Suryakantha, Ch 50, p 932; Park Plus, Ch 24, p 376)
Socio-economic	Nearly 40 per cent of the population lives below the poverty line, and basic amenities — transport, electricity, safe drinking water, primary care — are themselves lacking, so the technology risks widening rather than narrowing inequity (Suryakantha, Ch 50, p 932)
Literacy and language	Literacy of 65.38 per cent with only about 2 per cent well-versed in English; interfaces and content in English exclude most users (Suryakantha, Ch 50, p 932); language is a listed barrier (Park Plus, Ch 24, p 376)
Human resource	Doctors are not fully convinced of or familiar with e-medicine; technophobia among handlers; no doctor posted to the telemedicine unit; staff need dedicated training; lack of technical expertise
Demand side	Lack of patient confidence in the outcome of a remote consultation; a perceived lack of demand
Regulatory and medico-legal	Medico-legal concerns and legal issues around liability and consent; malpractice can occur; standard-setting and governance were historically weak (Suryakantha, Ch 50, p 932; Park Plus, Ch 24, p 376)

Domain	Challenge
Data integrity and privacy	Information may be lost through a software glitch or hardware failure; electronic linkage of person-identifiable health data poses added risks to privacy and confidentiality and demands physical security, password protection, role-based access and encryption (Suryakantha, Ch 50, p 932; ICMR National Ethical Guidelines 2017, Sec 11.7 and Box 11.6)
Clinical limitation	Separating assessment from service provision removes physical examination and the doctor-patient relationship; not all conditions are triageable remotely (Oxford Textbook of Public Health 7e, Ch 101)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Applications of telemedicine (spokes radiating from a central "Telemedicine" box).** Labels: *Telementored procedures / surgery — robotics; Second opinion and complex interpretations; Remote consultation and critical care monitoring; Disease surveillance and programme tracking; Continuing medical education and public awareness; Disaster management; Disease management; Home care and ambulatory monitoring.*
- **Diagram 2 — A modern telemedicine link (two boxes joined by the network).** Labels: left box *Nodal (spoke) hospital — microscope and medical instruments, scanner, digital camera, videoconference unit, computer, modem, telephone, printer;* the connecting line labelled *POTS / ISDN / VSAT satellite link;* right box *Referral (hub) hospital — specialist workstation, videoconference unit, modem, printer.*

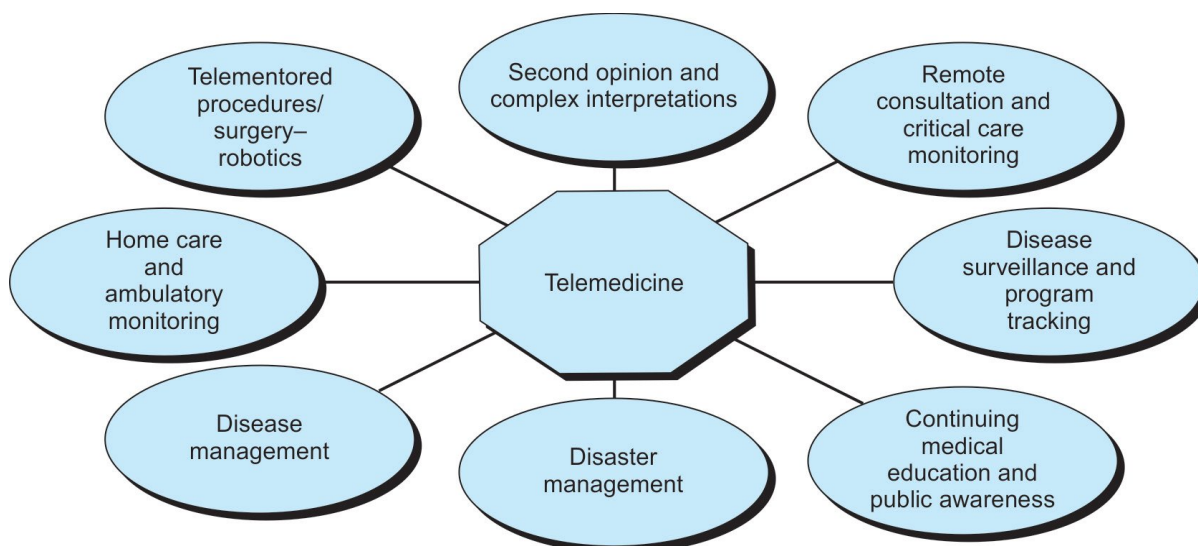
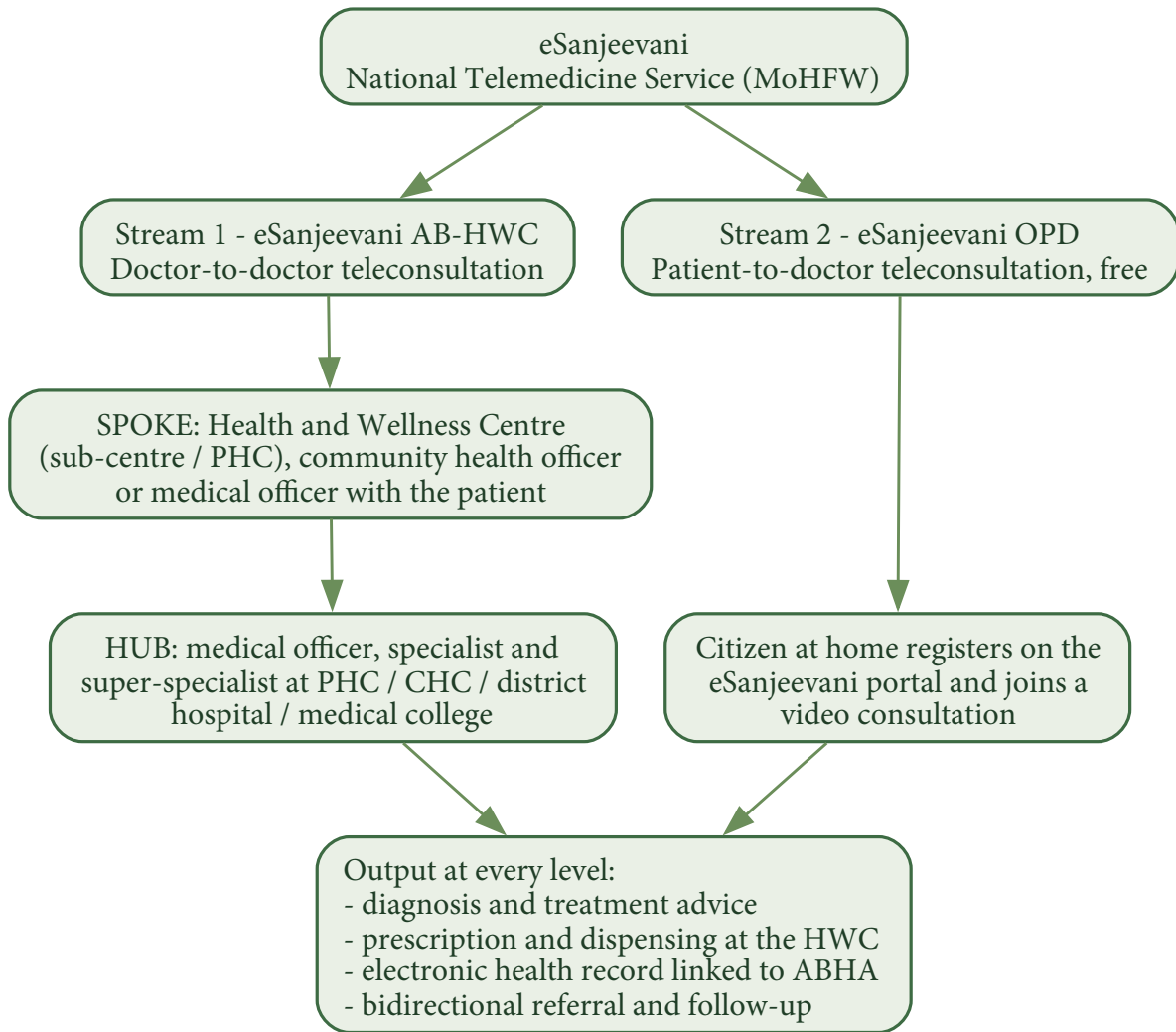


Fig 50.4 — Applications of telemedicine in public health (Suryakantha Environment Best, p.930)



eSanjeevani National Telemedicine Service - the two streams and the levels of care they link

High-yield exam pointers

- ▶ **Definition to write verbatim:** WHO 2002 — "delivery of health care services, where distance is a critical factor... for the exchange of valid information, for diagnosis, treatment and prevention of disease and injuries, research and evaluation and for the continuing education of health care providers." Etymology: *tele* (Greek) = distance, *mederi* (Latin) = to heal; began with NASA, 1960.
- ▶ **Two modes mnemonic** — "SAFE-R": Store-And-Forward = asynchronous (provider-to-provider); REal-time = synchronous (provider-to-patient).
- ▶ **Telemedicine vs telehealth:** telemedicine = clinical disciplines (teleradiology, telepathology, teledermatology, telepsychiatry); telehealth = disease management, monitoring, patient compliance.
- ▶ **Indian numbers to quote with their source:** ISRO pilot 2001 on VSAT; 78 rural/district hospitals linked to 22 super-speciality hospitals (Jugal Kishore); eSanjeevani has **two variants** — AB-HWC (doctor-to-doctor) and OPD (patient-to-doctor); **1,50,000 AB-HWCs connected** for specialist teleconsultation (Sunder Lal, 2023); ABDM launched **September 2021** under the National Health Authority.
- ▶ **Challenge figures from Suryakantha:** literacy 65.38 per cent with only 2 per cent well-versed in English; nearly 40 per cent below the poverty line.

- **Structural must-draw:** the nodal (spoke) → referral (hub) pathway, and the applications wheel with its eight spokes.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written as fact in the exam: the retrieved textbook chapter (Suryakantha, Ch 50) predates the Board of Governors, Medical Council of India *Telemedicine Practice Guidelines* of 2020, so its statement that "there is no proper governing body to form guidelines" and "no initiation taken by Government" is now dated — this regulatory correction is standard current knowledge but is not supported by any passage in the retrieved corpus, and the same is true of any current eSanjeevani consultation count. Also note an internal inconsistency in one source: Sunder Lal states 1,50,000 Health and Wellness Centres are connected for teleconsultation in Ch 17 (p 748) but "over 1.17 lakh telemedicine-enabled" in Ch 8 (p 504); quote the Ch 17 figure and attach the source. The socio-economic figures used in the challenges table (literacy 65.38 per cent, about 40 per cent below the poverty line) are the textbook's own and are considerably older than the current census and survey estimates, so quote them with the book's name attached.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, NMC Telemedicine Practice Guidelines 2020, IRDAI, ICMR

Programme and policy updates (India)

- **Ayushman Bharat Health and Wellness Centres have been renamed Ayushman Arogya Mandir** — write the new name in the exam. A facility audit of 20 centres in Kurud block, Dhamtari (Chhattisgarh) found teleconsultation initiated in 70% of centres but internet connectivity in only 20%, and a 41.2% vacancy of community health officers. Scoring the sub-centre HWCs on the programme's own A-to-E performance grading, the high-performing half were significantly those with a community health officer ($p < 0.001$), a working teleconsultation facility ($p = 0.002$) and a laptop and a web camera (each $p = 0.050$) — so a live teleconsultation link tracks closely with overall centre performance. [PMID 40821268]
- **eSanjeevani 2.0 was rolled out in March 2023**, adding audio-video stability, prescription synchronisation and point-of-care/electronic-record integration; the hub-and-spoke architecture now formally links Ayushman Arogya Mandirs and PHCs (spokes) to district and sub-district hospitals (hubs). Platform infrastructure grew from 397 hubs and 6,868 spokes in 2020 to 14,036 hubs and 108,610 spokes by September 2023, with 29,114 registered medical practitioners consulting on it. [PMID 41158553]
- **Tele-MANAS — Tele Mental Health Assistance and Networking Across States — was launched in October 2022** as the National Tele Mental Health Programme: a 24×7 multilingual helpline delivering stepped care, escalating from counsellor to psychiatrist or to local services. It has answered more than 3.43 million calls from every State and Union Territory, mostly for anxiety, depression and psychosocial stress with a critical minority reporting suicidal distress, and WHO has assessed it as a scalable, equity-focused model — the largest government-led tele-mental-health service anywhere. [PMID 42467064]

Current national data

- ▶ **The eSanjeevani figure to quote, with its date, and who actually uses it:** over 163 million teleconsultations between November 2019 and September 2023 across all 28 States and 8 Union Territories, rising to over 276 million by November 2024 — the largest documented government-led telemedicine service in the world. More than 93% of that volume came through the provider-assisted doctor-to-doctor AB-HWC stream, not the direct-to-citizen OPD stream. Women were the majority on both streams (about 57–58% of AB-HWC consultations, rising to 70.1% of OPD consultations by 2023) and adults aged 25–45 formed the largest age band. Use is shifting from acute illness towards chronic care — 3,27,491 diabetes and 2,71,253 hypertension follow-up consultations on AB-HWC 2.0 between March and September 2023. Mean active doctor–patient time was about 1 minute 15 seconds, and only 38% of Indian households are digitally literate. [PMID 41158553]
- ▶ **Correct the textbook's "1,50,000 HWCs connected":** 108,840 of 139,389 registered AB-HWCs (78.1%) were actually operating the platform in 2022, and just five States accounted for about 70.6% of all consultations, so national coverage figures hide a steep State gradient. Ranked by total volume in the paper's Table 3, those five are Andhra Pradesh, West Bengal, Tamil Nadu, Karnataka and Uttar Pradesh (the paper's text names Madhya Pradesh in place of West Bengal; its own Table 3 totals support the list given here, so quote the States only if you are asked for them). [PMID 41158553]
- ▶ **The Ayushman Bharat Health Account is the weak link in the digital chain.** In an ICMR task-force baseline survey of 5,709 people in Bhopal and Raipur districts (November 2022 – May 2023), only 3.4% of rural and 2.3% of urban participants held an ABHA. The commonest barrier was an Aadhaar not linked to a mobile number (40.6% rural vs 51.2% urban), followed by no mobile phone and simple unawareness; ABHA generation was significantly associated with gender, residence, education and employment. [PMID 41098267]

Elimination, surveillance and disease status

- ▶ **Digital surveillance and programme portals are unevenly implemented at the periphery.** A facility audit of nine government digital health interventions at Chandigarh HWCs scored eVIN (vaccine cold chain) and the Reproductive and Child Health portal 70–90% (effectively implemented), the IDSP–IHIP disease-surveillance portal and eSanjeevani 51–70% (fairly implemented), and HMIS, the HWC portal, CPHC–NCD and FP–LMIS only 25–50%. Community awareness of these platforms was 1–18.3% (against Aarogya Setu 78.3% and CoWIN 35.8%), and 89.2% of health workers said duplicate paper-plus-digital entry had increased their workload — the practical reason routine surveillance data quality suffers. [PMID 40854229]
- ▶ **Teleophthalmology is now doing real disease-control work at the periphery.** Across 41 rural vision centres in 9 States, each serving 20,000–50,000 people and staffed by an optometrist trained for a year, 1,91,194 patients were examined over 18 months (April 2022 – September 2023): 16.13% were refracted and dispensed spectacles, 30.03% were prescribed medicines and followed up locally, 25.44% were referred to the base hospital, 6.14% were detected with diabetic retinopathy, and 23,623 (12.35%) below-poverty-line patients underwent free cataract surgery. The teleconsultation charge was ₹75, subsidised to ₹30 in tribal areas. [PMID 40880155]

Guideline and legislative updates

- **The regulatory vacuum the older textbooks describe has closed — do not write "there is no governing body" in the exam.** The Government of India released the Telemedicine Practice Guidelines on 25 March 2020; the National Medical Commission carries them as Appendix-5 to the erstwhile Medical Council of India's Professional Conduct (Etiquette and Ethics) Regulations 2002, and on 11 June 2020 the Insurance Regulatory and Development Authority of India recognised teleconsultation for insurance claims, which is what drove the proliferation of platforms and apps. The residual gap now under scrutiny is the issue of medical and fitness certificates on the strength of a teleconsultation alone. [PMID 39817299]

Global and WHO developments

- **WHO-hosted consensus on cross-border telehealth (2026).** A three-round modified Delphi with 18 experts in health policy, regulation, telehealth implementation and trade, followed by public consultation on the WHO website, produced a matrix of 20 key challenges grouped into seven domains — policy and governance (which the panel said underpins the rest); knowledge and awareness; technology and operations; data privacy; quality and ethics; financing; and resilience in emergencies. The panel's conclusion is directly quotable in a "challenges" answer: the binding constraint is fragmented policy, governance and financing rather than technology. India appears in WTO 2022 data as both a leading exporter and a leading importer of cross-border telehealth services. [PMID 42232833]

SOURCE & COVERAGE

Suryakantha — Community Medicine with Recent Advances (Ch 50, Telemedicine in Public Health, pp 927-932, including Fig 50.4); Sunder Lal — Textbook of Community Medicine (Ch 17, pp 748, 763; Ch 8, pp 504, 535); Jugal Kishore — National Health Programmes of India (Ch 2, pp 739-740; Ch 1, p 584); Park Plus — Amrutha & Vijayalakshmi (Ch 24, pp 374-377); Textbook of International Health (Ch 12, p 664); Oxford Textbook of Public Health 7e (Ch 101, Strategies for Health Services); AFMC Red Book of Community Medicine 2024 (Ch 38, Table 38.13); ICMR National Ethical Guidelines 2017 (Sec 11.7-11.8); Park's Textbook of Preventive and Social Medicine 28e (Ch 10, p 731). **Coverage note:** Park 28e carries no telemedicine chapter — its only relevant line is that Health and Wellness Centres serve as a platform for teleconsultation (Ch 10, p 731); the substantive grounding for this answer therefore rests on Suryakantha Ch 50 and the Indian programme sources listed above.

Recent-advances sources: 41158553 — Adoption and utilization of India's eSanjeevani national telemedicine service, *Oxford Open Digital Health* 2025, DOI [10.1093/oodh/oqaf025](https://doi.org/10.1093/oodh/oqaf025); 40821268 — Assessment of the Functioning of Health and Wellness Centers in Kurud Block, Dhamtari District, Chhattisgarh: A Facility-Based Cross-Sectional Study, *Cureus* 2025, DOI [10.7759/cureus.88071](https://doi.org/10.7759/cureus.88071); 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, *British Journal of Psychiatry* 2026, DOI [10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 41098267 — Bridging the Healthcare Gap: A Mixed-Method Study on the Impact of Ayushman Bharat Health Account (ABHA) on Underprivileged Populations of Bhopal and Raisen Districts, *Cureus* 2025, DOI [10.7759/cureus.92272](https://doi.org/10.7759/cureus.92272); 40854229 — Implementation Status and Usability of Digital Health Interventions Among Health Care Workers and End Users at the Primary Health Care Level in Chandigarh, North India, *JMIR Formative Research* 2025, DOI [10.2196/69824](https://doi.org/10.2196/69824); 40880155 — Teleconsultation: Ophthalmology reaching every nook and corner, *Indian Journal of Ophthalmology* 2025, DOI [10.4103/IJO.IJO_3004_24](https://doi.org/10.4103/IJO.IJO_3004_24); 39817299 — Unverified medical certifications surge amid telemedicine guideline ambiguities, *Indian Journal of Medical Ethics* 2024, DOI [10.20529/IJME.2024.054](https://doi.org/10.20529/IJME.2024.054) (abstract-only); 42232833 — Policy analysis and expert consultation on cross-border telehealth services, *Bulletin of the World Health Organization* 2026, DOI [10.2471/BLT.25.294599](https://doi.org/10.2471/BLT.25.294599). Also: Telemedicine Practice Guidelines, Government of India, 25 March 2020, carried by the National Medical Commission as Appendix-5 to the erstwhile MCI's Professional Conduct (Etiquette and Ethics) Regulations 2002; IRDAI recognition of teleconsultation for insurance claims, 11 June 2020.



International and global health



Bill and Melinda Gates Foundation

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — The Bill and Melinda Gates Foundation is a private, grant-making philanthropic foundation established in 2000 by Bill Gates, the founder of Microsoft, together with his wife; it is the largest and one of the newest of the philanthropies working in international health (Textbook of International Health, Ch 3, p 137). It delivers no services itself: it makes large grants to governments, United Nations agencies, non-governmental organisations, research institutes, vaccine manufacturers and public-private partnerships, with the stated aim of reducing inequality through the development and application of technologies (Textbook of International Health, Ch 3, p 138). In the taxonomy of global health actors it is the archetypal **philanthropic organisation**, sitting alongside the World Health Organization (United Nations agency), the World Bank (multilateral development bank) and the GAVI Alliance (global health initiative) (Oxford Textbook of Public Health 7e, Ch 51, p 1897).

Origin, endowment and scale

Item	Detail
Year established	2000, by Bill Gates and Melinda Gates (Textbook of International Health, Ch 3, p 137)
Endowment	US\$38.7 billion, supplemented in 2006 by a US\$31 billion donation from the investor <i>Warren Buffett</i> , spread out over time (Textbook of International Health, Ch 3, pp 137–138); listed later at US\$42.3 billion, the second wealthiest charitable foundation in the world (Oxford Textbook of Public Health 7e, Ch 106, pp 4023–4024)
Annual spending	Over US\$3 billion, of which approximately 60% goes to global health ; the remainder to educational and local efforts in the Pacific Northwest of the United States (Textbook of International Health, Ch 3, pp 137–138)
Comparative scale	Its global health budget has surpassed that of the World Health Organization in some years (Textbook of International Health, Ch 3, p 138)
Administrative style	Makes large grants with a relatively small bureaucratic infrastructure — a nimble, risk-taking, entrepreneurial business model, deliberately unlike risk-averse government bureaucracy (Textbook of International Health, Ch 3, p 138; Oxford Textbook of Public Health 7e, Ch 106, pp 4023–4024)
Presence in India	A country office has been established in India (Textbook of International Health, Ch 3, p 138)
Wider influence	Has singlehandedly redefined the concept of individual social responsibility and inspired similar giving worldwide, including among wealthy individuals in developing countries (Oxford Textbook of Public Health 7e, Ch 106, p 4025)

Programme divisions and priority areas

Division	Priority areas
Global Health	Increasing access to childhood immunisation; prevention of and research into tuberculosis (TB), malaria and human immunodeficiency virus (HIV) infection; poliomyelitis eradication; development of an acquired immunodeficiency syndrome (AIDS) vaccine, topical microbicides and pre-exposure prophylaxis (Oxford Textbook of Public Health 7e, Ch 106, pp 4024–4025, 4036)

Division	Priority areas
Global Development	Alleviation of hunger and poverty; microfinance to empower women (support to the Grameen Foundation in Bangladesh); novel agricultural practices for food security, in partnership with The Rockefeller Foundation (Oxford Textbook of Public Health 7e, Ch 106, pp 4024–4025, 4036)
United States programme	Enhancement of secondary and post-secondary education within the United States (Oxford Textbook of Public Health 7e, Ch 106, pp 4024–4025)
Stated disease priorities (grant listing)	Infectious diseases, vaccines, HIV/AIDS and TB; reproductive and child health (Textbook of International Health, Ch 3, Table 3–10, pp 136–137)

Modes of working

- ▶ **Direct large grants** to governments, non-governmental organisations and research institutions, administered with a deliberately lean staff (Textbook of International Health, Ch 3, p 138).
- ▶ **Funding of public–private partnerships:** US\$1.5 billion to the GAVI Alliance for vaccine development and delivery, and US\$258 million to the Malaria Vaccine Initiative (Textbook of International Health, Ch 3, p 138). Its money into the Global Fund to Fight AIDS, Tuberculosis and Malaria stimulated further large donations from Warren Buffett and the William J. Clinton Foundation (Oxford Textbook of Public Health 7e, Ch 106, p 4028).
- ▶ **Product development partnerships** — not-for-profit bodies that carry a product from laboratory to field: the International AIDS Vaccine Initiative, the Malaria Vaccine Initiative, Aeras for TB vaccines, and the Program for Appropriate Technology in Health (PATH) (Plotkin's Vaccines 7e, Ch 4, pp 81–82).
- ▶ **Challenge grants with delimited, measurable goals** — exemplified by the *Grand Challenges in Global Health* initiative created in 2003 to stimulate researchers to solve critical scientific and technological problems of diseases of the developing world; fourteen Grand Challenges were framed (Textbook of International Health, Ch 3, p 138 and Ch 13, p 703, Box 13–3).
- ▶ **Advocacy and price negotiation** — its funding, with lobbying by Médecins Sans Frontières, secured free supply of eflornithine from Aventis for African trypanosomiasis, after which estimated cases fell from about 450,000 in 1997 to about 50,000 (Nelson & Williams, Infectious Disease Epidemiology, Ch 25, pp 872–873).
- ▶ **Agenda setting** — it has challenged the public health community by calling for renewed efforts to achieve **malaria eradication** and committing large resources to that goal (Nelson & Williams, Infectious Disease Epidemiology, Ch 27, p 900).

Contributions to global health, with Indian examples

Area	Contribution
Immunisation	Principal financial and advocacy backer of <i>Gavi, the Vaccine Alliance</i> , which has been critical to introducing new vaccines in developing countries (Plotkin's Vaccines 7e, Preface); GAVI is a network of governments, bilateral and technical agencies, WHO, UNICEF, the World Bank, pharmaceutical companies and non-governmental organisations including the Gates and Rockefeller Foundations (Jugal Kishore, National Health Programmes of India, Ch 1, p 207)

Area	Contribution
Meningococcal A vaccine	Meningococcal Vaccine Project (2001) — Gates Foundation funded PATH in collaboration with WHO; technology transfer enabled the Serum Institute of India to develop the group A conjugate vaccine, first tested for safety and immunogenicity in India (Nelson & Williams, Infectious Disease Epidemiology, Ch 17, p 535)
Japanese encephalitis	PATH, funded by the Gates Foundation, enabled routine use of SA 14-14-2 JE vaccine; large quantities were imported to combat epidemics in India (Plotkin's Vaccines 7e, Ch 33, p 768)
Oral cholera vaccine	Contributor to the public-private partnership that developed and disseminated <i>Sanchol</i> , alongside the National Institute for Cholera and Enteric Diseases (India) and Shantha Biotechnics (India) (Oxford Textbook of Public Health 7e, Ch 106)
Measles and rubella	Partner in the Measles & Rubella Partnership under Immunization Agenda 2030, with the American Red Cross, United Nations Foundation, Centers for Disease Control and Prevention, GAVI, UNICEF and WHO; launched in 2001 as the Measles and Rubella Initiative (AFMC Red Book of Community Medicine 2024, Ch 31, p 927)
Health information in India	Co-funded the National Family Health Survey (NFHS)-3, 2005–06 and NFHS-4 , with the United States Agency for International Development, the Department for International Development, UNICEF, the United Nations Population Fund and the Ministry of Health and Family Welfare (Jugal Kishore, National Health Programmes of India, Ch 1, pp 57–63)
HIV/AIDS	Gates Foundation–Merck Botswana Comprehensive HIV/AIDS Partnership, cited as a public-private partnership in which health services were strengthened (Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e, Ch 76, p 1280); with the Clinton Foundation and PEPFAR, credited with reducing HIV deaths in developing countries (Oxford Textbook of Public Health 7e, Ch 106)
Poliomyelitis	Sustained large-scale funding for global polio eradication (Oxford Textbook of Public Health 7e, Ch 106, p 4036)

Criticisms and limitations

Criticism	Substance
Technological reductionism	Views the determinants of health in narrow terms, separating the scientific and technical from the political, social and economic; the Grand Challenges deliberately excluded integrated social-science approaches and remain "technological quick fixes" that cannot, alone, meet the Foundation's own goal of reducing health inequalities (Textbook of International Health, Ch 3, p 138 and Ch 13, p 703)
Undemocratic agenda setting	Because other bilateral, multilateral, public and private donors are keen to match its funding to be associated with a high-profile success, the Foundation exercises enormous public policy-making power as a private actor that is not democratically accountable (Textbook of International Health, Ch 3, pp 138–139)
Distortion of national priorities	Foundations set goals from the personal perspectives of their founders, often without consulting public health leaders; a donation can dwarf a recipient country's own health budget and divert scarce staff and infrastructure to donor priorities (Oxford Textbook of Public Health 7e, Ch 106, pp 4034–4035)

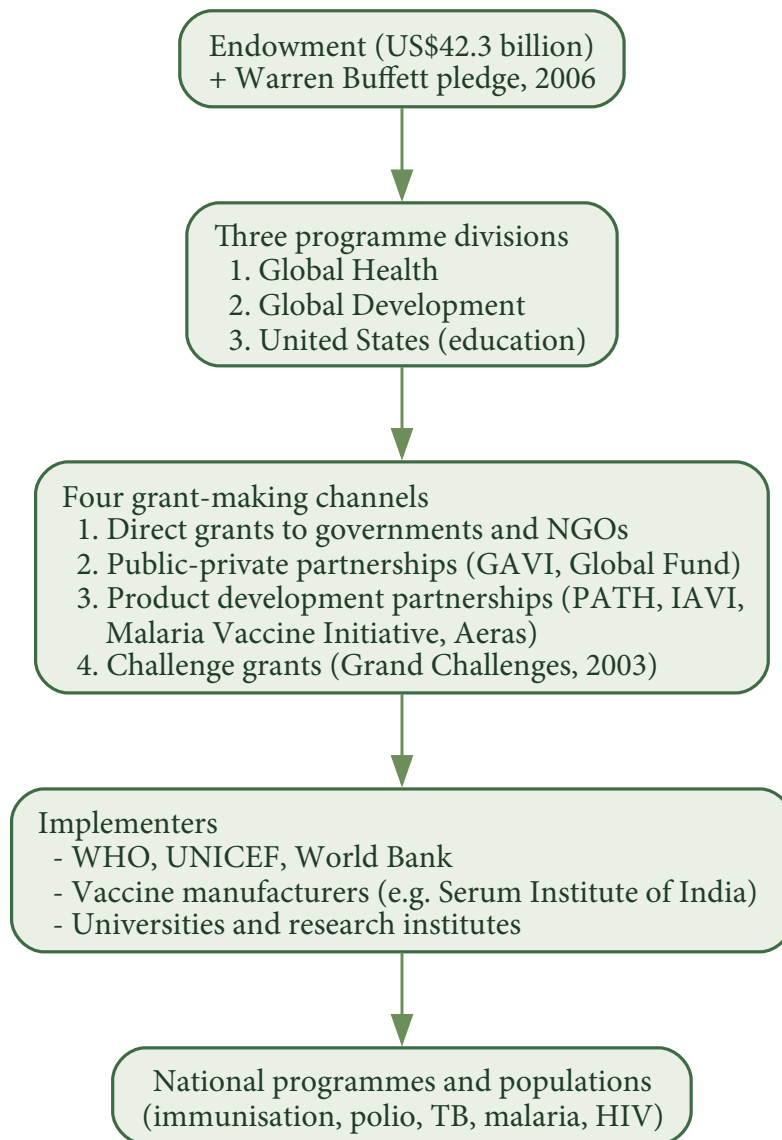
Criticism	Substance
Weakening of primary health care	With the Global Fund, it has been accused of diverting local health workers from primary health care roles and of overlooking poverty, nutrition, transport and housing (Textbook of International Health, Ch 3, p 161). Donor programmes offering higher salaries draw staff out of the public sector — in Haiti, improvement in HIV/AIDS consumed most of the health workforce and other measures, including infant mortality, deteriorated (Oxford Textbook of Public Health 7e, Ch 106, pp 4034–4035)
Conflict of interest	Accused of investing its endowment in polluting industries and profiteering pharmaceutical companies, and of defending intellectual property rights over affordable life-saving medicines; the head of its global health programme was a former executive and serving board member of GlaxoSmithKline, a conflict rarely articulated because critics fear offending a powerful funder (Textbook of International Health, Ch 3, p 138)
Vertical, not horizontal	Funds disease-specific vertical programmes; "no health system in the world is actually built on vertical programs" (Textbook of International Health, Ch 13, p 703)

Four inter-related concerns about sustainability of such private funding (Oxford Textbook of Public Health 7e, Ch 106, pp 4035–4037):

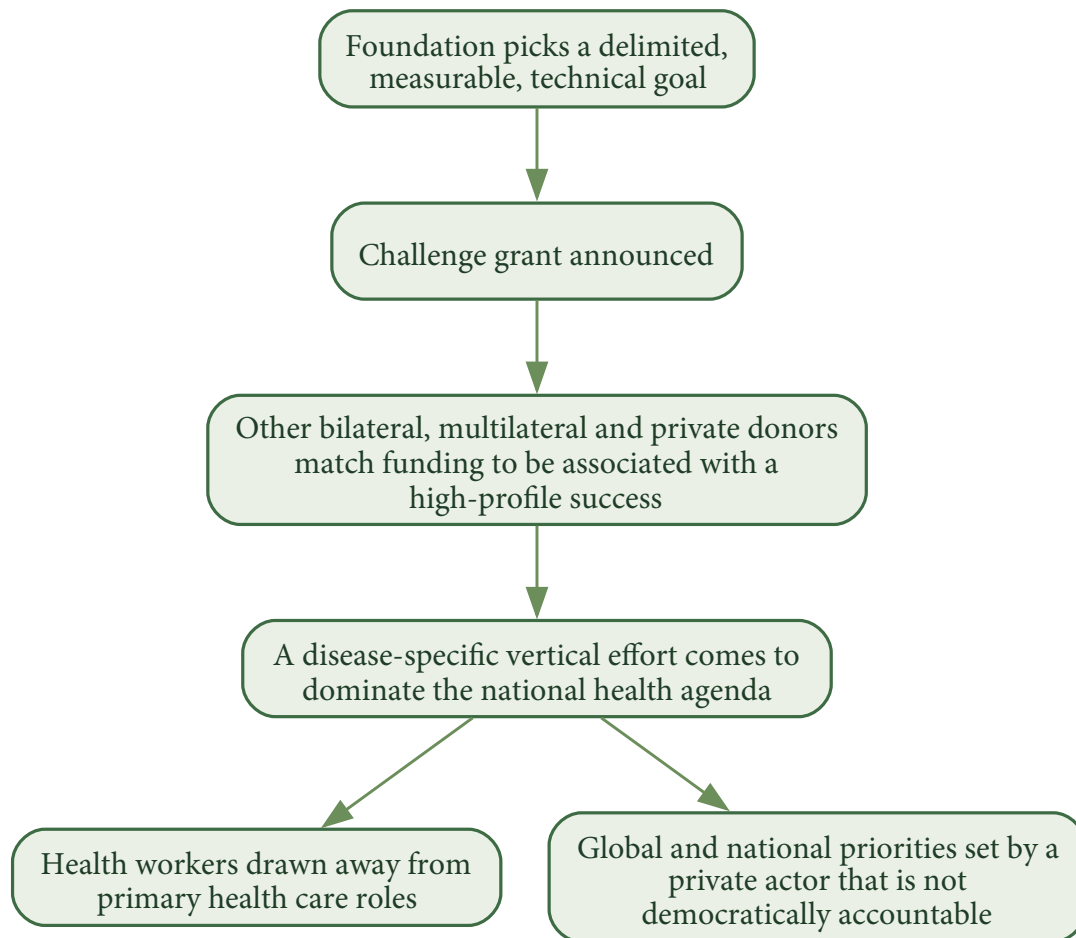
- ▶ Withdrawal of funds for a chronic condition needing long-term investment — an interruption in antiretroviral supply promotes drug-resistant HIV; goals such as polio eradication must be funded until achieved.
- ▶ A diverse portfolio invites shifting money from an incomplete initiative to a newer one, which is normal corporate re-prioritisation but counterproductive in public health.
- ▶ Priority setting itself — heavy AIDS funding may be at the expense of conditions with a greater mortality burden, such as diarrhoea and respiratory disease.
- ▶ Lack of harmonisation among donors at country level, producing duplication, inefficiency and diversion of scarce human resources away from government-identified priorities.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ Drawn label box — "Actors in global health governance, and where the Gates Foundation sits" (six labelled rows, one example each, after Oxford Textbook of Public Health 7e, Ch 51, p 1897): United Nations agencies → *World Health Organization, UNICEF, United Nations Population Fund*; National governments → *Ministries of health*; Multilateral development banks → *World Bank*; Global health initiatives → *GAVI Alliance*; Philanthropic organisations → *Bill and Melinda Gates Foundation*; Civil society organisations → *International Diabetes Federation*.



How Gates Foundation money reaches a population — endowment to beneficiary



The catalytic effect of a Gates Foundation grant and its accountability problem



The Bill and Melinda Gates Foundation ranks second among the wealthiest charitable foundations. Source: Oxford Textbook of Public Health 7e, Ch 106, Table 11.13.1, pp 4023-4024.

High-yield exam pointers

- ▶ **Founded 2000**; endowment US\$38.7 billion plus the **2006 Warren Buffett pledge of US\$31 billion**; later listed at **US\$42.3 billion**, second only to the Chan-Zuckerberg Foundation.
- ▶ **Over US\$3 billion spent annually, ~60% on global health**; its global health budget has exceeded WHO's in some years — the single most quotable fact.
- ▶ **Three divisions**: Global Health, Global Development, United States (education).
- ▶ **Landmark grants**: US\$1.5 billion to the GAVI Alliance; US\$258 million to the Malaria Vaccine Initiative; *Grand Challenges in Global Health*, 2003 (14 challenges).
- ▶ **Indian links to name**: country office in India; Serum Institute of India meningococcal A conjugate vaccine through PATH; SA 14-14-2 Japanese encephalitis vaccine imported for Indian epidemics; co-funder of NFHS-3 (2005–06) and NFHS-4.
- ▶ **Criticism triad**: technological reductionism → vertical programmes weakening primary health care → agenda setting by an actor that is not democratically accountable.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the corpus's most recent Indian text (AFMC Red Book of Community Medicine 2024) names the organisation the "Bill and Melinda French Gates Foundation", so an examiner may encounter either form of the name. Park's Textbook of Preventive and Social Medicine 28e carries no separate section on this Foundation, so the answer above is grounded on the international-health and public-health texts in the corpus, and the endowment and spending figures quoted are those printed in those editions rather than a current-year balance sheet.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed

Programme and policy updates (India)

- ▶ India's position changed sharply after the 2025 contraction in development assistance for health, which fell by about 21% in a single year — most of the drop being a 67% cut in United States contributions worth more than US\$9 billion — while the Global Fund's 2024–2026 grant cycle was trimmed by US\$1.4 billion, about 11%. USAID's contribution of over US\$140 million to TB Mukht Bharat, and a tuberculosis "buddy" scheme that reached some 5.1 million patients in four high-burden states, both stopped in 2025: treatment services carried on, but the case-finding, contact tracing and adherence support that the elimination drive rests on did not [PMID 42529073].
- ▶ The Indian presence that the older texts call simply "a country office" is now one node in an embassy-like network: alongside the Seattle headquarters the Foundation runs offices in Washington DC, London, Berlin, Beijing, Abuja, Addis Ababa, Johannesburg, Dakar, Nairobi and **Delhi**, employs more than 2,000 staff working with partners in over 130 countries, and spent US\$328 million on advocacy and policy grants in 2023 alone [PMID 40275294].

Current national data

- ▶ Indian health-financing figures to quote with their years: the government share of total health expenditure rose from 29% in 2014–15 to 48% in 2021–22, out-of-pocket spending is down to about 39%, and public health spending sits at roughly 1.9% of gross domestic product — still below the National Health Policy 2017 target of 2.5% by 2025. India supplies close to a fifth of the world's generic medicines by volume and about 60% of its vaccine doses, with pharmaceutical exports past US\$30 billion in financial year 2025, yet about 70% of the active pharmaceutical ingredients and key starting materials it uses are imported from China [PMID 42529073].
- ▶ The Foundation's own scale figures now supersede those printed in the textbooks. Its endowment stands at almost US\$78 billion [PMID 41151830]; grant disbursements approach US\$80 billion over its first 25 years, and in 2024 it ranked **third among all contributors of development assistance for health**, behind only the United States and Germany [PMID 40275294]. Its partnership commitments likewise dwarf the textbook's US\$1.5 billion GAVI figure — across 2000 to 2024 GAVI received 16 grants totalling US\$6.1 billion and the Global Fund 39 grants totalling US\$3.9 billion [PMID 41151830].

Elimination, surveillance and disease status

- ▶ Poliomyelitis, the Foundation's single largest commitment, is still not eradicated and the target year has repeatedly slipped. Type 1 is the only wild serotype left and survives only in Pakistan and Afghanistan, while circulating vaccine-derived poliovirus type 2 has become the bigger problem, at 4,819 of 5,585 circulating vaccine-derived cases (86.3%) spread over 57 countries [PMID 42214662]. Wild type 1 cases in the two endemic countries went from 12 in 2023 to 98 in 2024, years in which the Foundation gave WHO over a quarter of a billion dollars and then US\$225.8 million for polio [PMID 41151830]. Note that the two sources disagree on the 2024 wild type 1 count — the 10-year global analysis puts it at 871 — so name the source whenever the figure is quoted [PMID 42214662; PMID 41151830].
- ▶ Malaria vaccination has moved from research to programme, and the Foundation sits inside both products. RTS,S/AS01 came out of a collaboration with GlaxoSmithKline and PATH and was recommended by WHO in October 2021; R21/Matrix-M, developed at Oxford and manufactured by the **Serum Institute of India**, followed in October 2023, showing 75% efficacy where transmission is highly seasonal and 68% where it is perennial, over the 12 months after the three-dose primary series. Worldwide in 2024 there were an estimated 282 million cases and 610,000 deaths, the WHO African Region carrying over 94% of cases and 95% of deaths, with roughly 75% of those deaths among children under five [PMID 42482135].

Guideline and legislative updates

- ▶ Novel oral poliovirus vaccine type 2 (nOPV2), introduced for outbreak response in 2021, has passed two billion doses administered [PMID 42217520], and the Strategic Advisory Group of Experts on Immunization has endorsed a bivalent OPV cessation policy framework: repeated high-quality supplementary campaigns before cessation, a routine schedule of at least two immunologically effective inactivated poliovirus vaccine doses, then sensitive surveillance and adequate vaccine stockpiles afterwards [PMID 42208276].
- ▶ Vaccine financing rules have shifted. A 2025 Gavi/UNICEF procurement deal brings R21/Matrix-M down to roughly US\$2.99 a dose — about US\$12 for the four-dose course —

which the Gavi subsidy cuts to around US\$0.20 a dose for the poorest countries; and Gavi's African Vaccine Manufacturing Accelerator, launched in 2024, makes up to US\$1.2 billion available over ten years towards the African Union and Africa CDC target of producing more than 60% of the continent's vaccine needs locally by 2040, up from about 1% now, with first disbursements due in 2026 [PMID 42482135]. Under Immunization Agenda 2030 the co-financing and transition policy is pushing recipient countries towards self-financing, with lower-income countries committing a record US\$1.1 billion of their own resources to immunisation in 2026 against a projected US\$38.4 billion funding gap for 16 vaccines across 94 low- and middle-income countries over 2011–2030 [PMID 42519657].

Global and WHO developments

- ▶ The organisation now trades simply as the **Gates Foundation**, having dropped "Bill & Melinda" from its name [PMID 40275294], and the governing board long criticised as too small and too narrow for an actor of its reach — for years just Bill and Melinda Gates, Bill Gates Senior and Warren Buffett — has been rebuilt: Gates Senior died in 2020, Buffett stepped down as a trustee in 2021, and six new trustees have been added since 2022 to widen it by gender, geography and expertise [PMID 41151830]. In 2025 the Foundation announced that it will close early, spending about **US\$200 billion over the next two decades** instead of existing in perpetuity — so the textbook's four "concerns about sustainability of private funding" now have a stated end date attached to them [PMID 40345811].
- ▶ A grants audit puts numbers on the textbook's charge of technological reductionism and vertical programming. Between 2000 and 2024 the Foundation made 640 grants to WHO worth US\$5.5 billion, 6.4% of everything it disbursed in those years; its actual payments to WHO, US\$4.0 billion over 2010–2023, were 9.5% of WHO's total revenue, making it WHO's second largest funder after the United States, which gave notice of withdrawal in January 2025. Infectious disease took 82.6% of that money and polio alone 58.9%, against 0.7% for health-systems strengthening, under 1% for non-communicable diseases, 0.2% for water and sanitation, and nothing at all for mental health. All ten of its largest WHO grants funded polio immunisation, and in 2022–23 it paid 56.5% of what WHO spent on polio [PMID 41151830].

SOURCE & COVERAGE

Textbook of International Health, Ch 3 "International Health Agencies, Activities, and Other Actors" and Ch 13 "Toward Healthy Societies"; Oxford Textbook of Public Health 7e, Ch 106 "Private support of public health" (incl. Table 11.13.1), Ch 51 "Governance and management of public health programmes" and Ch 5 "Globalization"; Plotkin's Vaccines 7e (Preface, Ch 4, Ch 33); Nelson & Williams, Infectious Disease Epidemiology (Ch 17, Ch 25, Ch 27); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 76); Jugal Kishore, National Health Programmes of India (NFHS-3 and NFHS-4 funding; GAVI); AFMC Red Book of Community Medicine 2024 (Ch 31, Measles & Rubella Partnership under Immunization Agenda 2030). Coverage note: Park's Textbook of Preventive and Social Medicine 28e carries no section on this Foundation, and no captioned figure in the indexed corpus matched the topic, so structural flowcharts, a label box and a data chart are given in place of a textbook plate.

Recent-advances sources: 41151830 — Who's leading WHO? A quantitative analysis of the Bill and Melinda Gates Foundation's grants to WHO, 2000-2024, BMJ Global Health 2025, doi:[10.1136/bmjgh-2024-015343](https://doi.org/10.1136/bmjgh-2024-015343); 40275294 — The Gates Foundation's network diplomacy in European donor countries, Globalization and Health 2025, doi:[10.1186/s12992-025-01112-9](https://doi.org/10.1186/s12992-025-01112-9); 40345811 — Gates Foundation to shut early, spending \$200bn in next two decades, BMJ 2025, doi:[10.1136/bmj.r957](https://doi.org/10.1136/bmj.r957) (abstract-only); 42529073 — Insulated at home, indispensable abroad: positioning India's global health strategy after the 2025 contraction in development assistance, Frontiers in Health Services 2026, doi:[10.3389/frhs.2026.1903108](https://doi.org/10.3389/frhs.2026.1903108); 42482135 — Who will pay for the malaria vaccines in Africa?, Archives of Public Health 2026, doi:[10.1186/s13690-026-02020-z](https://doi.org/10.1186/s13690-026-02020-z); 42519657 — From aid dependence to domestic ownership: risks, challenges, and policy pathways for sustaining immunisation gains in lower-income countries, Preventive Medicine Reports 2026, doi:[10.1016/j.pmedr.2026.103560](https://doi.org/10.1016/j.pmedr.2026.103560); 42214662 — WPV1 resurgence and the dominant threat of cVDPV2: a 10-year global epidemiological analysis (2017-2026), International Journal of Infectious Diseases 2026, doi:[10.1016/j.ijid.2026.108839](https://doi.org/10.1016/j.ijid.2026.108839) (abstract-only); 42217520 — Quantifying the benefits of improved operations for polio outbreak response: a model-based analysis, Vaccine 2026, doi:[10.1016/j.vaccine.2026.128689](https://doi.org/10.1016/j.vaccine.2026.128689) (abstract-only); 42208276 — Global cessation of bivalent oral polio vaccine (bOPV): a comprehensive review of modeling studies from 2000 to 2025, Vaccine 2026, doi:[10.1016/j.vaccine.2026.128707](https://doi.org/10.1016/j.vaccine.2026.128707) (abstract-only). Where 41151830 and 42214662 report different 2024 wild poliovirus type 1 counts (98 versus 871), both figures are given with their source rather than reconciled.

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International Labour Organization (ILO)

Asked in: Paper IV 20-Jul-2020 — RGUHS MD Community Medicine

Definition — The International Labour Organization (ILO) is the autonomous specialised agency of the United Nations (UN) system that sets and monitors international employment standards, created on the recognition that "problems of industry, like disease, know no frontiers" (Park 27e, Ch 25, p 1034) and mandated to improve the working and living conditions of the working population all over the world (Park 28e, Ch 25, p 1059; Textbook of International Health, Ch 9, p 491). Its defining feature is **tripartism** — government, employer and worker representatives share its decision-making.

Identity and mandate at a glance

Item	Detail
Year established	1919, immediately after the First World War (Park 28e, Ch 25, p 1059)
Founding instrument	Treaty of Versailles, 1919; began as an affiliate of the League of Nations (Textbook of International Health, Ch 9, p 491)
Present status	Autonomous UN specialised agency — "sets and monitors employment standards" (Textbook of International Health, Ch 3, pp 107–108)
Headquarters	Geneva, Switzerland (Park 28e, Ch 25, p 1059)
Membership	A membership organisation of States, as with the UN; 175 member States (Park Plus, Ch 23, p 322)
Principal instrument	The International Labour Code — a collection of international minimum standards on the health, welfare, living and working conditions of workers worldwide (Park 28e, Ch 25, p 1059)
India	Founder member; Area Office at New Delhi (Park Plus, Ch 23, p 323)

Purposes and strategic objectives

Constitutional purposes (1919)	Four strategic objectives (current)
1. Contribute to the establishment of lasting peace by promoting social justice	1. Promote and realise standards and fundamental principles and rights at work
2. Improve, through international action, labour conditions and living standards	2. Create greater opportunities for women and men to secure decent employment and income
3. Promote economic and social stability	3. Enhance the coverage and effectiveness of social protection for all
(Park 28e, Ch 25, p 1059)	4. Strengthen tripartism and social dialogue (Park Plus, Ch 23, p 322)

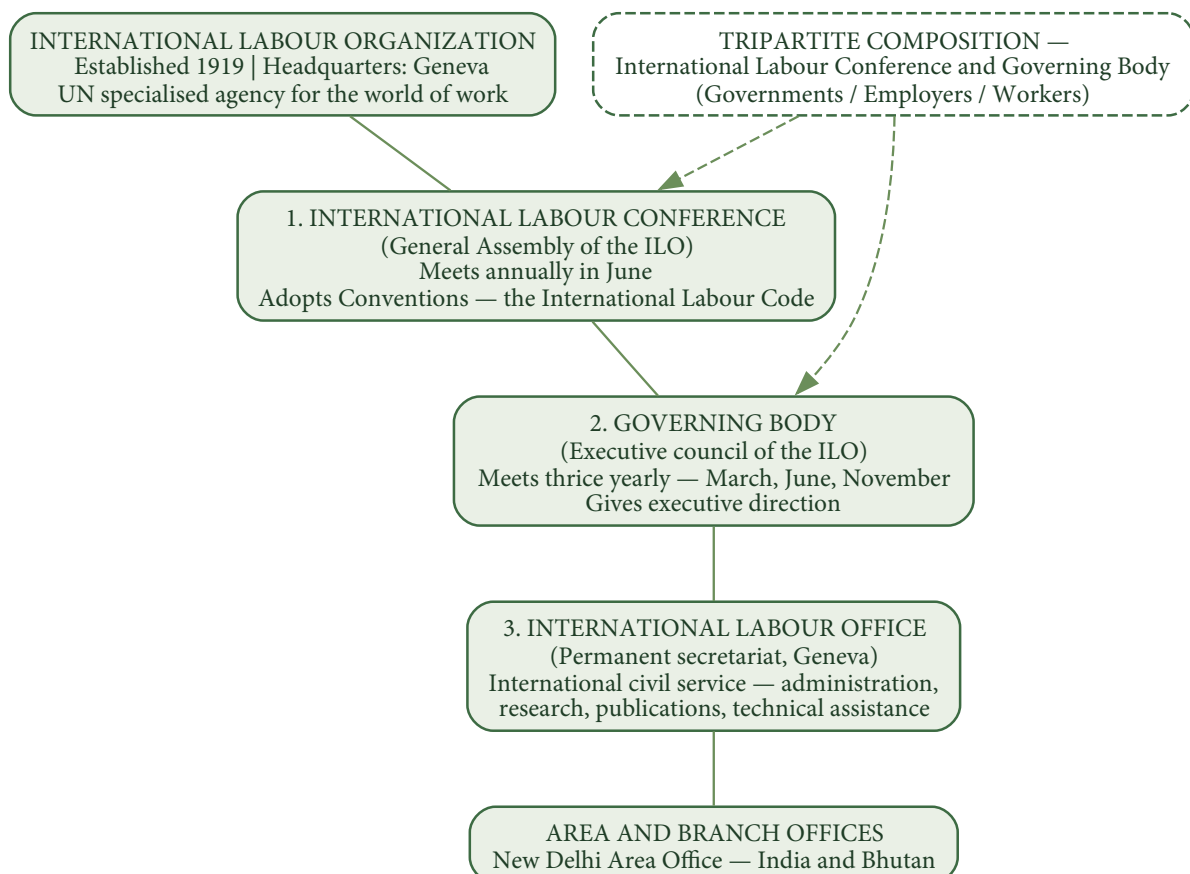
Structure — the three organs and the field network

The administration and functioning of the ILO is carried out by three organs (Park Plus, Ch 23, pp 322–323); tripartism — government, employer and worker representation — operates in the two deliberative and executive organs, the International Labour Conference and the Governing Body, while the International Labour Office is the permanent secretariat staffed by an international civil service. Each

member State sends four delegates to the Conference in a 2 : 1 : 1 ratio — two government, one employer and one worker delegate — each voting independently.

Organ	Nature	Frequency of meeting	Principal role
International Labour Conference	The General Assembly of the ILO	Annually, in June	Deliberative body; adopts the Conventions that constitute the International Labour Code and sets general policy
Governing Body	The executive council	Three times a year — March, June and November	Executive direction of the Organization between Conferences
International Labour Office	The permanent secretariat, at Geneva	Continuous	Day-to-day administration; technical work, research, publications and technical assistance to member States

Beneath the three organs sits a **field network of area and branch offices** delivering country-level technical assistance — for India, the New Delhi Area Office serving India and Bhutan (Park Plus, Ch 23, p 323).



Structure of the International Labour Organization — the three organs, the field network and where tripartism operates

Functions

- **Standard setting — the International Labour Code.** The Conference adopts **Conventions**, treaty instruments that lay down minimum desirable standards for the health, welfare, safety, and living

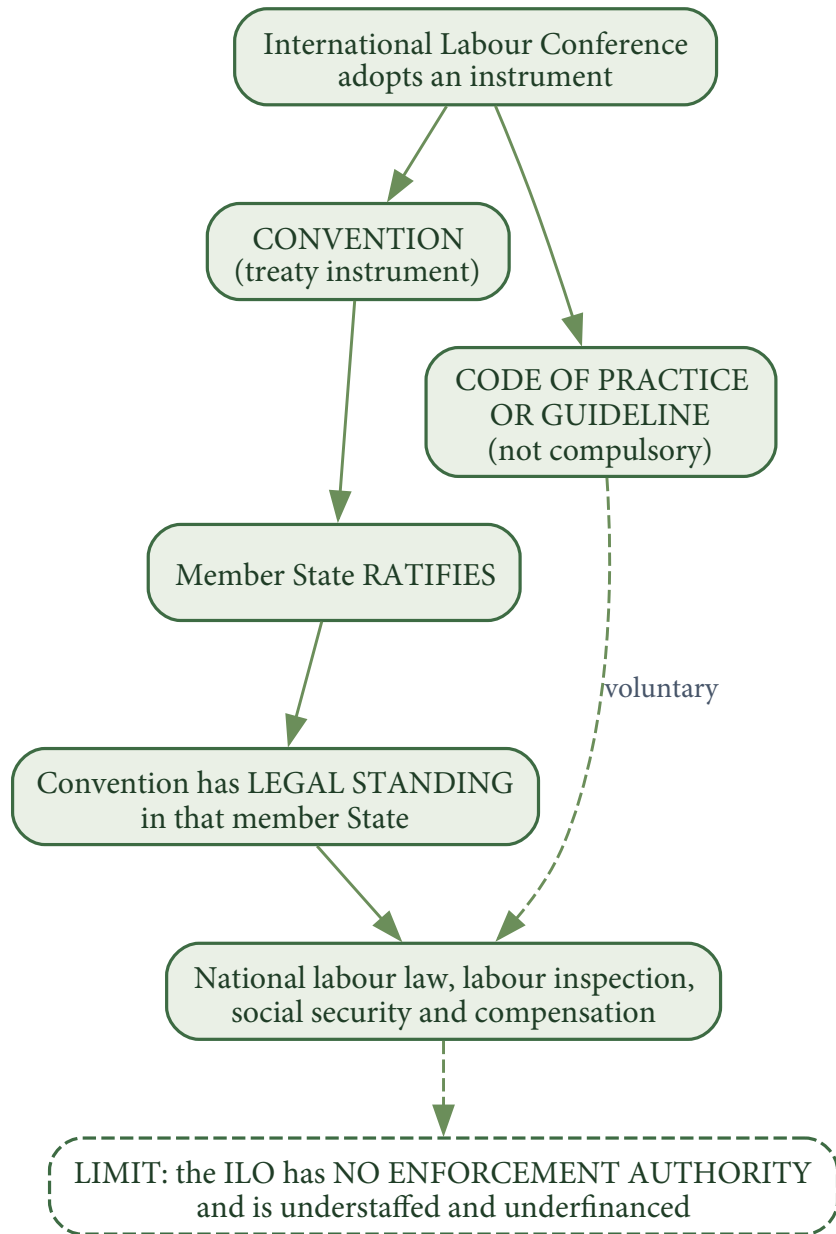
and working conditions of workers throughout the world, and that have legal standing in those member countries which have ratified them (Park 28e, Ch 25, p 1059; Suryakantha, Ch 42, p 832; Textbook of International Health, Ch 9, p 491). Of the **184 Conventions** adopted, **more than half deal with occupational safety and health** (Textbook of International Health, Ch 9, pp 491–492).

- ▶ **Promotion of workers' health and safety** — to promote the health and safety of the working population, to improve the living standards of workers, and to promote their economic and social stability (Suryakantha, Ch 42, p 832).
- ▶ **Technical assistance and technical cooperation** — assistance to governments and to organisations interested in the betterment of living and employment standards (Park 28e, Ch 25, p 1059), delivered through the country and area offices.
- ▶ **Guidance instruments that are not compulsory** — besides Conventions, the ILO issues **codes of practice and guidelines**, which carry no obligation to ratify (Textbook of International Health, Ch 9, pp 491–492).
- ▶ **Standardised technical tools for occupational health practice** — most importantly the **ILO International Classification of Radiographs of the Pneumoconioses**, sponsored by the International Labour Office at Geneva since 1930 and modified several times, most recently in 2002. It is used to describe the **pattern and severity of the change in groups of workers**, and is a tool for epidemiological, research and statutory-compliance purposes, **not** a method of diagnosis or of estimating severity in an individual case (Hunter's Diseases of Occupations, Ch 75, p 1012 and Ch 68, p 911). The ILO has also prepared **standard time losses** for computing the accident severity rate — death or permanent disability is taken as equivalent to **48,000 hours lost** (Sunder Lal, Ch 11, p 646).
- ▶ **Research, statistics and global reporting** — the ILO generates the global estimates on which occupational health policy rests. It reported an estimated **2.78 million deaths from occupational causes in 2015**, of which 2.40 million were from fatal work-related illnesses and the remainder from work-related accidents — on average **over 7,500 work-related deaths every day** (Oxford Textbook of Public Health 7e, Ch 55, p 2051).
- ▶ **Protection of vulnerable working groups** — child labour, young persons, women workers and migrant workers. In 1973 the ILO passed a Convention establishing **15 years as the minimum working age** for most sectors, while permitting light work from **age 13** provided such work was unlikely to harm the child's health, morals and safety or prejudice school attendance (Park 28e, Ch 9, p 695).
- ▶ **Collaboration with the World Health Organization (WHO)** in the field of health and labour (Park 28e, Ch 25, p 1059). The working definition of **occupational health** — "the promotion and maintenance of the highest degree of physical, mental and social wellbeing of workers in all occupations" — is that of a **joint WHO/ILO committee** (Oxford Textbook of Public Health 7e, Ch 55, pp 2048–2049). The division of labour is that the ILO sets and monitors labour standards, while the WHO is responsible for the technical aspects of occupational safety and health — medical standards, medical examinations and hygienic standards (Textbook of International Health, Ch 9, p 492).

Selected ILO Conventions relevant to public health

Convention No.	Subject
No. 81	Labour inspection
No. 87	Freedom of association
No. 98	Freedom to organise and bargain collectively
No. 105	Freedom from forced labour
No. 155	Occupational safety and health
No. 161	Occupational health services
No. 174	Prevention of major industrial accidents
Employment Injury Benefits Convention (1964)	Compensation for occupational injury — ratified by only 23 countries

(Textbook of International Health, Ch 9, pp 491–492)



How an international labour standard becomes binding — and where the ILO's authority stops

ILO in India

- ▶ India is a **founder member** of the ILO.
- ▶ A **Branch Office** was opened at New Delhi in 1929; it became an **Area Office** of the ILO in 1970.
- ▶ The Area Office coordinates and provides technical assistance to **India and Bhutan** in rural labour, women workers, employment generation and occupational safety and health.
- ▶ On **child labour**, the ILO works within the framework of the **Child Labour (Prohibition and Regulation) Act, 1986** and the **National Policy on Child Labour**, funding local and industry-specific projects. Two major projects are the **Child Labour Action and Support Programmes (CLASP)** and the **International Programme on the Elimination of Child Labour (IPEC)**. The contribution of IPEC in India has been to generate a critical consciousness among all three **social partners** towards eliminating child labour (Park Plus, Ch 23, pp 323–324).

- ▶ India is a signatory to many international treaties and ILO Conventions on occupational health (Jugal Kishore, National Health Programmes of India, p 496).

Constraints on effectiveness

- ▶ **Ratification is uneven**, so a Convention binds only the States that have taken it on — the Employment Injury Benefits Convention is the extreme example, and inadequate detection and compensation make a mockery of the standard even where it is in force (Textbook of International Health, Ch 9, p 491).
- ▶ The ILO remains **under-recognised by public health professionals**, even though it is an appropriate forum through which occupational health could be improved (Textbook of International Health, Ch 9, pp 491–492).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The ILO tripartite triangle.** Draw an equilateral triangle with a central circle. Labels: apex — **GOVERNMENT**; left base angle — **EMPLOYERS**; right base angle — **WORKERS**; central circle — **TRIPARTISM AND SOCIAL DIALOGUE**; below the triangle — "represented in the International Labour Conference and the Governing Body"; caption — "The three social partners of the ILO".

High-yield exam pointers

- ▶ **1919, Treaty of Versailles, Geneva** — begun as an affiliate of the League of Nations, now an autonomous UN specialised agency; **three constitutional purposes** (social justice and lasting peace; labour conditions and living standards; economic and social stability) and **four strategic objectives** (rights at work, decent employment, social protection, tripartism and social dialogue).
- ▶ **Three organs (mnemonic C-G-O):** Conference (General Assembly, June, annual) → Governing Body (executive, thrice yearly — March/June/November) → Office (permanent secretariat). Tripartism operates in the Conference and the Governing Body; the Office is an international civil service.
- ▶ **International Labour Code** = the collection of minimum standards on health, welfare, living and working conditions; **184 Conventions**, more than half on occupational safety and health; codes of practice and guidelines are additional but not compulsory.
- ▶ **Number-tagged Conventions worth quoting:** 81 labour inspection, 155 occupational safety and health, 161 occupational health services, 174 major industrial accidents.
- ▶ **ILO classification of radiographs of pneumoconioses** — since 1930, latest revision 2002; describes pattern and severity in *groups* of workers for epidemiological and statutory use, not individual diagnosis. **Accident severity rate:** death or permanent disability = 48,000 hours lost.
- ▶ **ILO 2015 estimate:** 2.78 million deaths from occupational causes (2.40 million from work-related illness); over 7,500 work-related deaths per day.
- ▶ **India:** founder member; New Delhi Branch Office 1929 → Area Office 1970 (India and Bhutan); IPEC and CLASP on child labour; minimum working age 15, light work from 13, under the 1973 Convention.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the membership figure of 175 member States and the count of 184 Conventions are as printed in the retrieved textbooks and both rise over time, and the 2 : 1 : 1 delegate composition of the International Labour Conference is standard ILO Constitution content that the retrieved books do not themselves state.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO/ILO Joint Estimates, Bulletin of the World Health Organization, ILO (International Labour Conference 2022; global occupational safety and health estimates 2023; Decent Work Country Programme for India 2023–2027), ILO/UNICEF global child labour estimates 2025

Programme and policy updates (India)

- ▶ **The New Delhi office has outgrown the "Area Office, 1970" description.** It now functions as the ILO Decent Work Team for South Asia and Country Office for India, and its work is framed by the Decent Work Country Programme for India 2023–2027 rather than by ad-hoc technical assistance; India has additionally held a seat on the **Governing Body since 1922** [ILO 2023].
- ▶ **The ILO now reaches India's informal economy through participatory training packages, not the labour inspectorate.** WARM (Work Adjustment for Recycling and Managing Waste) trained waste-collection union trainers in Ahmedabad, who within one year had trained about **1,000 workers, mostly women**; WIND (Work Improvement in Neighbourhood Development) trained cotton-farmer trainers in Telangana, who reached **647 farmers across 18 villages**. Sister packages are **WISE** (small enterprises), **WISH** (home-based work) and **WISCON** (small construction sites), all delivered through trade unions, employer organisations and farmer associations [PMID 39697308].

Current national data

- ▶ **India, 2016 (WHO/ILO Joint Estimates): 341,389 deaths** (95% UR 310,810–371,968) from diseases attributable to selected occupational risk factors — a mortality rate of **35.8 per 100,000 working-age population** (95% UR 32.6–39.0) against a global rate of **27.7**. The **South-East Asia Region is the worst of the six WHO regions** at 36.5 per 100,000; the African Region is the best at 11.4 [PMID 37265682].
- ▶ **Informality is the defining Indian occupational-health statistic.** Informal employment is **88.2% of total employment in India** (Nepal 94.3%, Bangladesh 89.0%, Pakistan 82.4%, Sri Lanka 70.4%), and about **2 billion workers — 61.2% of global employment — work informally**, largely outside the reach of occupational safety and health legislation and inspection [PMID 39697308].

Elimination, surveillance and disease status

- ▶ **Occupational-cancer surveillance in India has effectively failed.** Over 2004–2024 the Directorate General of Mines Safety recorded **zero notified mesothelioma cases**, and across 27 responding States and Union Territories only **one occupational cancer** was notified to a Chief Inspector of Factories — against substantial published Indian hospital case series. The National Cancer Registry Programme covers only **16% of the population**, and India has ratified neither

the Occupational Cancer Convention, 1974 nor the Asbestos Convention, 1986, while asbestos import and processing continue [PMID 41652913].

- **Child labour — the elimination deadline has been missed.** ILO/UNICEF global estimates put **138 million children in child labour in 2024**, over 100 million fewer than in 2000 and no worse than the 2020 estimate despite fears of COVID-19 reversal; but **SDG target 8.7's 2025 deadline for ending child labour has now passed unmet**, so IPEC-type programmes continue past their target year [ILO/UNICEF 2025].

Guideline and legislative updates

- **The single most examinable ILO update — the four fundamental principles became five.** At the **110th Session of the International Labour Conference, June 2022**, "a safe and healthy working environment" was added as the **fifth category** of fundamental principles and rights at work (the earlier four being freedom of association and collective bargaining, abolition of forced labour, abolition of child labour, and elimination of discrimination). **Convention No. 155 (Occupational Safety and Health, 1981)** and **Convention No. 187 (Promotional Framework for OSH, 2006)** thereby became **fundamental Conventions**, which obliges **every member State to respect, promote and realise them whether or not it has ratified them** [PMID 39697308; PMID 40029770].
- **India's Occupational Safety, Health and Working Conditions Code, 2020** subsumes **13 central labour laws**, including the **Factories Act, 1948** and the **Mines Act, 1952**, and merges their separate lists of notifiable occupational diseases into a single **Third Schedule**. Enforcement remains split — factories with the State governments (advised centrally by DGFASLI), mines with the Centre (DGMS) — and **mesothelioma lost its separate listing**, surviving only inside the broad "occupational cancer" entry [PMID 41652913].

Global and WHO developments

- **The 2015 mortality figure quoted in the textbooks has been superseded.** The ILO's 2023 global assessment estimates **nearly 3 million work-related deaths a year — about 2.6 million from disease and 330,000 from accidents — a 5% rise on 2015**, with circulatory disease, malignant neoplasms and respiratory disease together accounting for over 75% of the total. Membership now stands at **187 member States** (not the 175 printed in the older editions) [ILO 2023-Nov].
- **Long working hours are the largest single occupational risk factor yet quantified.** In 2016, **488 million people — 8.9% of the global population — worked ≥55 hours/week**, causing **745,194 deaths** and **23.3 million DALYs** from ischaemic heart disease and stroke (population-attributable fraction 3.7% for ischaemic heart disease, 6.9% for stroke). Of all **1,879,890 work-related deaths** that year only **19% were injuries**, which is why WHO and the ILO have proposed a new indicator on work-related **disease mortality** to sit beside the existing **SDG indicator 8.8.1** on occupational injuries [PMID 34011457; PMID 37265682].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 25 International Health; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics); Park's Textbook of Preventive and Social Medicine, 27e (Ch 25, for the founding-rationale wording); Textbook of International Health (Ch 3 International Health Agencies, Activities, and Other Actors; Ch 9 Globalization, Trade, Work, and Health); Oxford Textbook of Public Health 7e (Ch 55 Occupational health); Park Plus — Amrutha & Vijayalakshmi (Ch 23 International Health); Suryakantha — Community Medicine with Recent Advances (Ch 42 International Health Organizations); Hunter's Diseases of Occupations (Ch 68 Imaging in occupational lung disease; Ch 75 Inorganic dusts); Sunder Lal — Textbook of Community Medicine (Ch 11 Occupational Health); Jugal Kishore — National Health Programmes of India. Named instruments: ILO Constitution purposes; International Labour Code; ILO International Classification of Radiographs of the Pneumoconioses (2002 revision); ILO minimum-age Convention 1973; Child Labour (Prohibition and Regulation) Act 1986 (India).

Coverage note: Park 28e gives the ILO only a single paragraph and does not describe its organs, so the structural detail comes from Park Plus and the standard-setting detail from the Textbook of International Health.

Recent-advances sources: 39697308 — Participatory Training Activities in South Asia for Promoting Safety and Health in Informal Economy Workplaces — Roles of Trade Unions and Employer Organizations, *Safety and Health at Work* 2024, doi:[10.1016/j.shaw.2024.09.002](https://doi.org/10.1016/j.shaw.2024.09.002); 37265682 — New global indicator for workers' health: mortality rate from diseases attributable to selected occupational risk factors, *Bulletin of the World Health Organization* 2023, doi:[10.2471/BLT.23.289703](https://doi.org/10.2471/BLT.23.289703); 34011457 — Global, regional, and national burdens of ischemic heart disease and stroke attributable to exposure to long working hours for 194 countries, 2000–2016: a systematic analysis from the WHO/ILO Joint Estimates of the Work-related Burden of Disease and Injury, *Environment International* 2021;154:106595, doi:[10.1016/j.envint.2021.106595](https://doi.org/10.1016/j.envint.2021.106595); 41652913 — Commentary on Notification and Recordkeeping of Occupational Mesothelioma in India, *American Journal of Industrial Medicine* 2026, doi:[10.1002/ajim.70060](https://doi.org/10.1002/ajim.70060); 40029770 — Psychosocial risks and mental health of health and social care workers in the context of the COVID-19 pandemic, *Archivos de Prevención de Riesgos Laborales* 2024;27(3):229–232, doi:[10.12961/aprl.2024.27.03.01](https://doi.org/10.12961/aprl.2024.27.03.01) (abstract-only). Institutional sources: ILO — resolution of the 110th International Labour Conference including a safe and healthy working environment in the framework of fundamental principles and rights at work (June 2022); ILO — *A Call for Safer and Healthier Working Environments* (November 2023); ILO Decent Work Team for South Asia and Country Office for India — Decent Work Country Programme for India 2023–2027; ILO/UNICEF — global child labour estimates for 2024 (published 2025).

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World Health Organization (WHO)

Asked in: Paper IV 04-Aug-2021 — RGUHS MD Community Medicine

Definition — The World Health Organization (WHO) is the specialised agency of the United Nations (UN) whose first Constitutional function is to act as the **directing and coordinating authority on all international health work**. Its Constitution was drafted in 1946 by a Technical Preparatory Committee chaired by *Rene Sand*, approved the same year by an International Health Conference of **51 nations** at New York, and came into force on **7 April 1948** — observed every year as **World Health Day** (Park 28e, Ch 25, p 1055). Its objective, set out in Article 1 of the Constitution, is "**the attainment by all peoples of the highest possible level of health**", now read as a level of health that permits every person to lead a socially and economically productive life (Park 28e, Ch 25, p 1055; Textbook of International Health, Ch 3, p 110).

Constitutional basis, membership and headquarters

Item	Detail
Headquarters	Geneva, Switzerland (Park 28e, Ch 25, p 1055)
Constitution in force	7 April 1948 — World Health Day; a theme is chosen each year to focus attention on one aspect of public health (Park 28e, Ch 25, p 1055)
Membership	Open to all countries. In 1948 there were 56 members ; WHO now has 194 Member States and 2 Associate Members (Park 28e, Ch 25, p 1056)
Associate Members	Territories not responsible for the conduct of their own international relations; they participate in deliberations without a vote (Park 28e, Ch 25, p 1056)
Obligation of a member	Each Member State contributes yearly to the budget and is entitled to the services and aid the Organization can provide (Park 28e, Ch 25, p 1056)
Finance	A fixed (regular) budget from assessed contributions of Member States on a sliding scale based on national population and gross national product (GNP) , plus extrabudgetary funds voluntarily donated by members and other partners (Textbook of International Health, Ch 3, pp 111-112)

Structure I — the three principal organs

The WHO consists of three principal organs — the **World Health Assembly**, the **Executive Board** and the **Secretariat** (Park 28e, Ch 25, p 1057).

Organ	Composition	Meetings	Main functions
(a) World Health Assembly (WHA) — the "Health Parliament of Nations" and the supreme governing body	Delegates representing all Member States; each Member State has one vote	Annually, usually in May, generally at the Geneva headquarters and from time to time in other countries	(i) Determine international health policy and programmes; (ii) review the work of the past year; (iii) approve the budget for the following year; (iv) elect Member States to designate a person to serve for three years on the Executive Board and replace retiring members. The Assembly also appoints the Director-General on the nomination of the Executive Board, and it is the practice to hold "technical discussions" on a subject of world interest at each Assembly (Park 28e, Ch 25, p 1057)
(b) Executive Board (EB)	34 members, each designated by a Member State (originally 18, later raised to 24 and 30; the 1976 Assembly raised it from 30 to 31 with no fewer than three elected from each WHO region). Members are "technically qualified in the field of health"; they are designated by, but do not represent, their governments, and serve 3-year terms, one-third of the membership being renewed every year	At least twice a year — generally in January and shortly after the Assembly in May	Give effect to the decisions and policies of the Assembly; prepare the Assembly's agenda and work on the budget; facilitate the daily work of WHO; and take action itself in an emergency such as epidemics, earthquakes and floods where immediate action is needed. The Board may authorise the Director-General to combat epidemics, provide health relief to victims of a calamity and undertake urgent studies (Park 28e, Ch 25, p 1057; Textbook of International Health, Ch 3, p 111)
(c) Secretariat	Headed by the Director-General (DG), the chief technical and administrative officer of the Organization, elected by the World Health Assembly every 5 years and subject to the authority of the Executive Board; assisted by Assistant Directors-General and technical divisions. Staff grew from 250 persons in 1948 to 4,475 international public servants by 1985, and the Organization now has an international staff of about 8,000 across headquarters, the regional offices and 147 country offices	Continuous	Provide Member States with technical and managerial support for their health development programmes; execute the decisions of the Assembly and the Board (Park 28e, Ch 25, pp 1057-1058; Textbook of International Health, Ch 3, pp 110-111)

Structure II — regional and country machinery

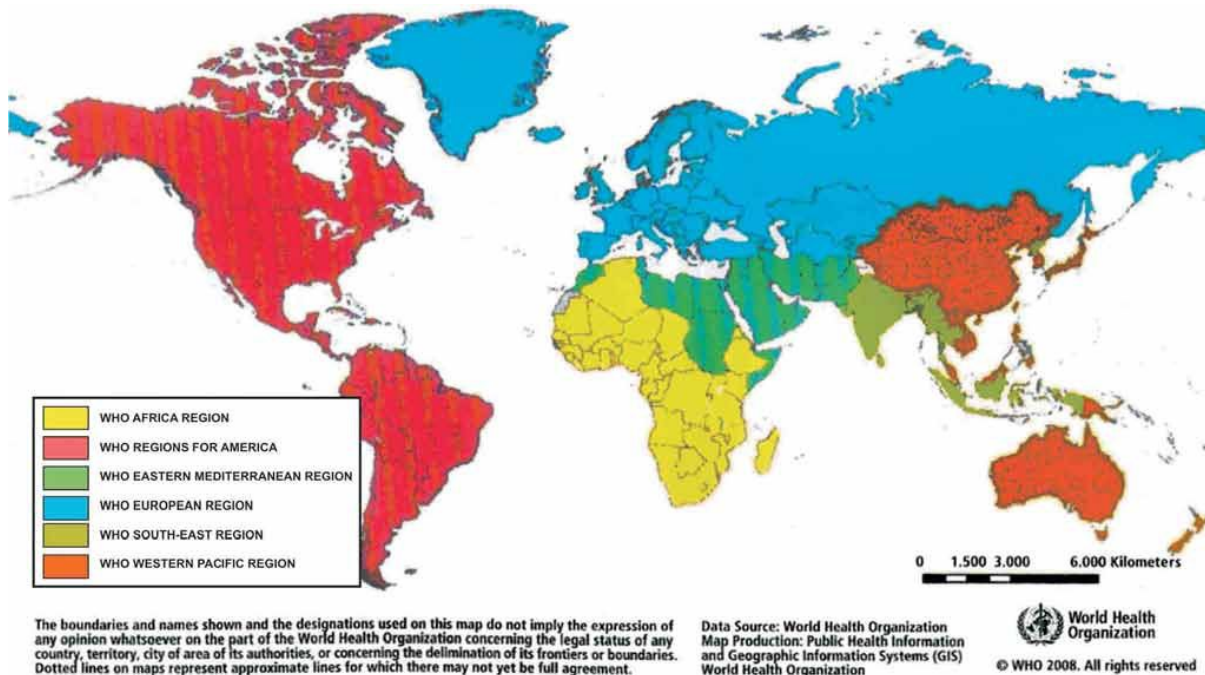
To meet the special health needs of different areas, WHO has established **six regional organisations**; these are provided for **under the Constitution** and are an integral part of the Organization (Park 28e, Ch 25, p 1057).

WHO Region	Regional office headquarters
1. South-East Asia (SEARO)	New Delhi, India — World Health House, Indraprastha Estate
2. Africa (AFRO)	Brazzaville, Congo
3. The Americas (AMRO / PAHO)	Washington D.C., USA
4. Europe (EURO)	Copenhagen, Denmark
5. Eastern Mediterranean (EMRO)	Cairo, Egypt
6. Western Pacific (WPRO)	Manila, Philippines

(Park 28e, Ch 25, pp 1057-1058; Textbook of International Health, Ch 3, pp 110-111)

Machinery at each level below headquarters:

- ▶ **Regional Committee** — composed of **representatives of the Member States of that region**; meets **once a year** to review the work in the region and to plan its continuation and development. Regional plans are then amalgamated into the overall plan of WHO at headquarters (Park 28e, Ch 25, p 1057).
- ▶ **Regional Office** — headed by the **Regional Director**, assisted by technical and administrative officers and members of the secretariat; regional offices are important in implementing the policies and programmes of WHO (Park 28e, Ch 25, p 1057).
- ▶ **Country office** — headed by the **WHO Country Representative (WR)**, assigned to one country or to several small adjacent countries. The WR works with national health authorities, other agencies and donors, and assists the government in **reviewing health needs and resources and in planning, coordinating, implementing and evaluating national health programmes and policies** (Textbook of International Health, Ch 3, p 111).
- ▶ **The South-East Asia Region has 11 Member States** — Bangladesh, Bhutan, Democratic People's Republic of Korea, India, Indonesia, Maldives, Myanmar, Nepal, Sri Lanka, Thailand and Timor-Leste (Park 28e, Ch 25, pp 1057-1058). WHO activity in this region covers malaria eradication, tuberculosis control, control of other communicable diseases, health laboratory services and vaccine production, health statistics, public health administration and rural health services, maternal and child health, nursing, environmental health and water supply, health education, nutrition, mental health, dental health, medical rehabilitation, quality control of drugs and medical education (Park 28e, Ch 25, p 1058).
- ▶ **PAHO is a special case** — founded in **1902 as the International Sanitary Bureau**, later the Pan American Sanitary Bureau, it retains **more independence than the other regional offices** (Textbook of International Health, Ch 3, p 110).



WHO regions (Suryakantha — *Community Medicine with Recent Advances*, p.356)

Functions I — the Constitutional functions (Article 2)

WHO's functions are laid down in Article 2 of its Constitution (Textbook of International Health, Ch 3, p 110):

- ▶ To act as the **directing and coordinating authority on international health work.**
- ▶ To assist governments in strengthening health services and in emergency aid.
- ▶ To **promote maternal and child health and welfare.**
- ▶ To foster activities in the mental health field.
- ▶ To promote the **improvement of nutrition, housing, sanitation and recreation**, and of economic, working and environmental conditions.
- ▶ To **study and report on public health and medical care.**
- ▶ To promote research and health training.
- ▶ To advance work to eradicate epidemic, endemic and other diseases, and to prevent injuries.
- ▶ To **propose conventions, agreements and regulations** and make recommendations on international health matters.
- ▶ To **standardise diagnostic procedures** and revise the international nomenclatures of diseases, causes of death and public health practices.
- ▶ To **develop, establish and promote international standards** for food, biologicals, pharmaceuticals and similar products.

The first function is the pivotal one: it allows Member States **collectively** to identify priority health problems, define health policies and targets, and devise the strategies, principles and programmes to attain them (Park 28e, Ch 25, p 1056).

Functions II — the broad areas of WHO's work in practice

Area of work	What WHO actually does
1. Prevention and control of specific diseases	Communicable and non-communicable disease control; immunisation against the common diseases of childhood (the Expanded Programme on Immunization) is a priority programme (Park 28e, Ch 25, p 1056)
2. Development of comprehensive health services	Described as WHO's most important single function — supporting national health policy and comprehensive national health programmes; organising health systems based on primary health care ; health manpower development and utilisation; building long-term national capability in health infrastructure, managerial capacity (including monitoring and evaluation) and health services research. Appropriate Technology for Health (ATH) was launched to encourage self-sufficiency in solving health problems as part of building up primary health care (Park 28e, Ch 25, p 1056)
3. Family health	A major programme activity since 1970, subdivided into maternal and child health care, human reproduction, nutrition and health education ; the chief concern is improving the quality of life of the family as a unit (Park 28e, Ch 25, p 1056)
4. Environmental health	Advises governments on national programmes for basic sanitary services; protection of the quality of air, water and food , health conditions of work, radiation protection and early identification of new hazards from new technology, through the WHO Environmental Health Criteria Programme and the WHO Environmental Health Monitoring Programme (Park 28e, Ch 25, p 1056)
5. Health statistics	Dissemination of morbidity and mortality statistics since 1947, published in the Weekly Epidemiological Record, World Health Statistics Quarterly and World Health Statistics Annual . To make statistics comparable between countries WHO publishes the International Classification of Diseases (ICD) , revised about every 10 years — ICD-10 came into effect on 1 January 1993 , and ICD-11 was adopted by the World Health Assembly in 2019 and came into effect from 1 January 2022 . WHO also helps countries improve medical records and plan and operate national health information systems (Park 28e, Ch 25, p 1056; Ch 2, p 69)
6. Biomedical research	WHO does not itself do research — it stimulates and coordinates it, through a worldwide network of WHO Collaborating Centres , grants to research workers and institutions, Regional Advisory Committees on health research that set regional priorities and a Global Advisory Committee for policy issues of global import. Six tropical diseases — malaria, schistosomiasis, trypanosomiasis, filariasis, leishmaniasis and leprosy — are the target of the WHO Special Programme for Research and Training in Tropical Diseases (Park 28e, Ch 25, pp 1056-1057)
7. Health literature and information	Acts as a clearing house for information on health problems ; publications run to hundreds of titles; the WHO library is a satellite centre of MEDLARS (Medical Literature Analysis and Retrieval System) of the U.S. National Library of Medicine; a public information service operates at headquarters and at each of the six regional offices (Park 28e, Ch 25, p 1057)
8. Cooperation with other organisations	Collaborates with the UN and the other specialised agencies with varying degrees of working relationship, and has established relations with a number of international governmental organisations (Park 28e, Ch 25, p 1057)

Functions III — how the functions are delivered, and their limits

WHO's work is delivered through **two channels** — *central technical services* and *services to governments* — drawn as a flowchart below (Textbook of International Health, Ch 3, p 110).

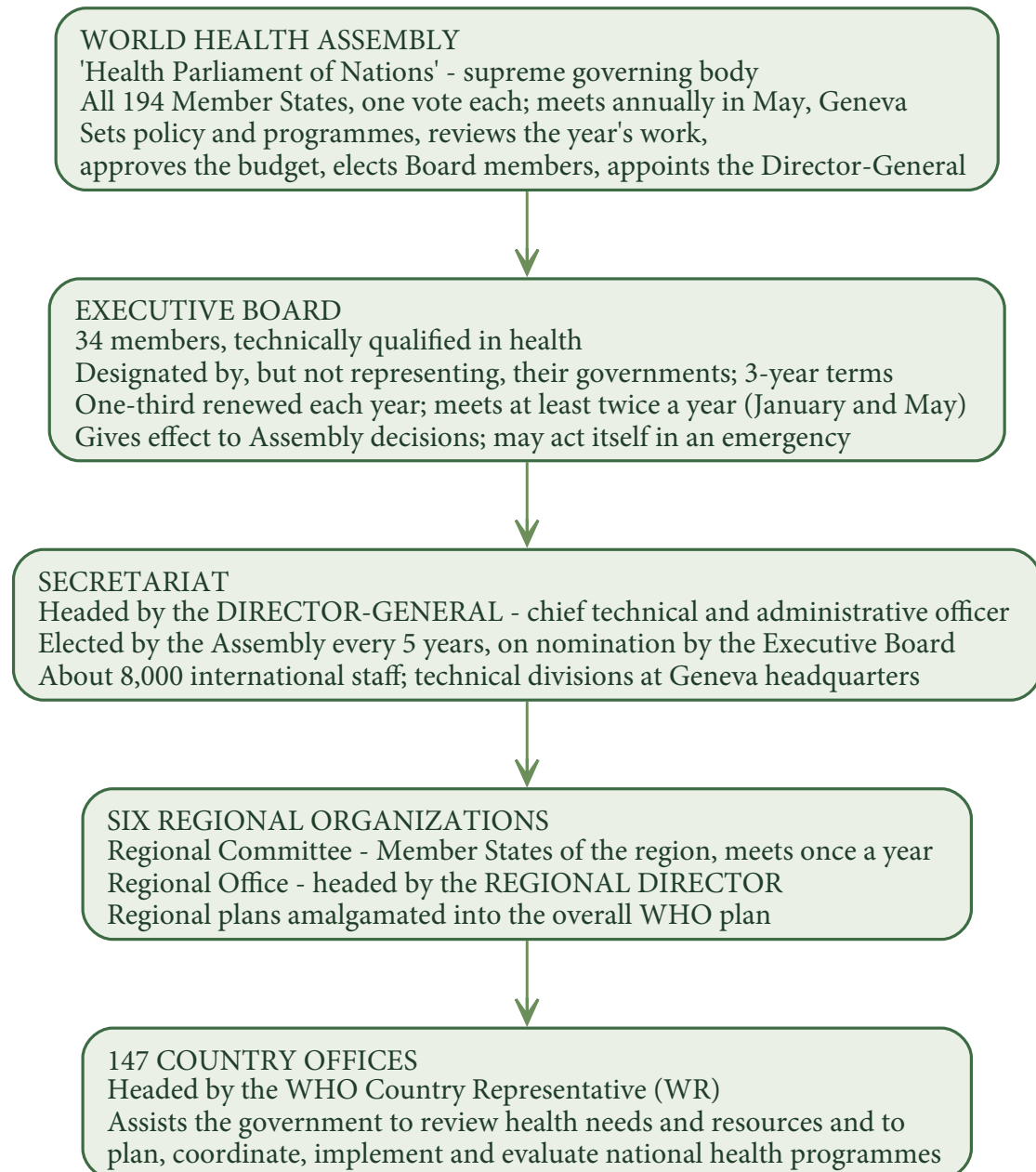
Its **regulatory and emergency function** rests on the **International Health Regulations (IHR) 2005**, the legally binding instrument made under the Constitution's power to propose regulations. The sequence from notification to a **public health emergency of international concern (PHEIC)** is drawn as a flowchart below. Points to add in writing:

- ▶ Member States must **notify WHO headquarters of important epidemic outbreaks**, must respond to public health emergencies, and must maintain a mechanism for **seeking technical assistance from WHO** (Textbook of International Health, Ch 5, pp 234-235).
- ▶ PHEICs so far declared include **pandemic influenza A(H1N1) in 2009, polio, Ebola and Zika virus infection** (Oxford Textbook of Public Health 7e, Ch 75, pp 2862-2863).
- ▶ Several hundred **WHO Collaborating Centres**, laboratories and institutes act as international reference centres and are mobilised in this response, together with the **Global Outbreak Alert and Response Network (GOARN)**, formed in 2000 (Textbook of International Health, Ch 3, p 110; Oxford Textbook of Public Health 7e, Ch 75, pp 2862-2863).

Constitutional limits of these functions: WHO can intervene in a country only when requested to do so, and its resolutions **urge but never oblige** Member States to act (Textbook of International Health, Ch 3, p 111).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Labelled diagram to sketch by hand:** the WHO governance pyramid — a three-tier triangle labelled from apex downwards "World Health Assembly (194 Member States, one vote each, meets May, Geneva)" → "Executive Board (34 technically qualified members, meets January and May)" → "Secretariat (Director-General, Geneva headquarters)", with two boxes hanging from the base labelled "6 Regional Offices — Regional Committee + Regional Director" and "147 Country Offices — WHO Representative", and side arrows marked *appoints Director-General, nominates Director-General, gives effect to Assembly decisions*.
- ▶ **World map of the six WHO regions** — outline map shaded and labelled AFRO (Brazzaville), AMRO/PAHO (Washington D.C.), EMRO (Cairo), EURO (Copenhagen), SEARO (New Delhi), WPRO (Manila), with India marked in SEARO.



Structure of the World Health Organization — from the World Health Assembly down to the country office

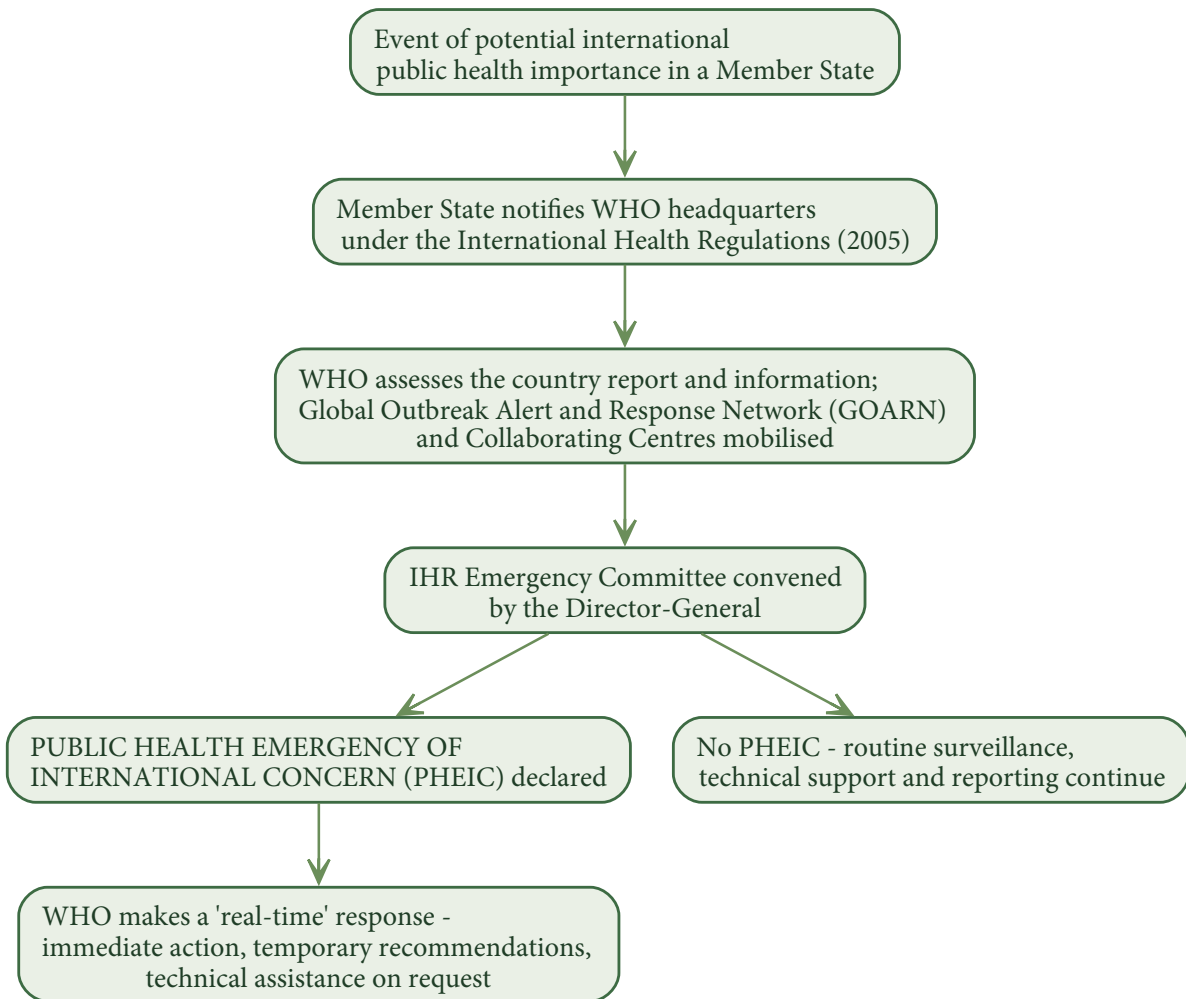
FIRST CONSTITUTIONAL FUNCTION (Article 2)
To act as the DIRECTING AND COORDINATING AUTHORITY
on all international health work

CENTRAL TECHNICAL SERVICES
(global, done by WHO itself)
Epidemiological intelligence
International health agreements and regulations
Standardisation of vaccines and pharmaceuticals
Knowledge dissemination through expert
committees and technical reports

SERVICES TO GOVERNMENTS
(national, only at the request of the country)
Delivered through the six regional offices
Inter-regional and intra-regional projects
Country office and WHO Representative support

Several hundred WHO COLLABORATING CENTRES,
laboratories and institutes act as
international reference centres for both channels

The two channels through which WHO delivers its constitutional functions



WHO's emergency function under the International Health Regulations 2005

High-yield exam pointers

- ▶ **Dates and numbers:** Constitution drafted 1946 (*Rene Sand* committee), approved by **51 nations**, in force **7 April 1948** = World Health Day; headquarters **Geneva**; **194 Member States** + **2 Associate Members**; **6 regions**; **147 country offices**.
- ▶ **Three organs mnemonic "A-B-S"** — Assembly (policy, budget, elects), Board (executes, 34 members, emergency powers), Secretariat (Director-General, technical support).
- ▶ **Executive Board arithmetic:** 18 → 24 → 30 → **31 in 1976** (≥ 3 per region) → **34 now**; **3-year term**, one-third renewed yearly, meets **January and May**.
- ▶ **Director-General** — appointed by the **World Health Assembly on the nomination of the Executive Board**, for **5 years**.
- ▶ **SEARO** — New Delhi, **11 Member States**; regional committee meets **once a year**.
- ▶ **ICD** — revised about every 10 years; **ICD-10 from 1 January 1993**; **ICD-11 adopted by the WHA in 2019**, in effect from **1 January 2022**.
- ▶ **Six tropical diseases** of the WHO Special Programme for Research and Training in Tropical Diseases: **malaria, schistosomiasis, trypanosomiasis, filariasis, leishmaniasis, leprosy**.
- ▶ **Key limit to quote:** WHO intervenes **only on request**; its resolutions **urge, never oblige**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park 28e's Table 1 still prints **Alexandria** as the Eastern Mediterranean regional headquarters, whereas the Textbook of International Health lists **Cairo**, which is the current location — write Cairo. Similarly, the staff figures Park quotes (250 in 1948, 4,475 in 1985) are historical illustrations of growth, not current numbers, and any current programme-budget figure should be quoted from the latest World Health Assembly resolution rather than from a textbook. WHO's current five-year strategy is expressed as the "triple billion" targets — one billion more people benefiting from universal health coverage, one billion more better protected from health emergencies and one billion more enjoying better health and well-being (Oxford Textbook of Public Health 7e, Ch 107, pp 4044-4045).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, International Health Regulations (2005) as amended 2024, WHO Pandemic Agreement 2025, World Health Assembly decision WHA79(7) 2026, WHO Traditional Medicine Strategy 2025-2034, UN General Assembly political declaration on antimicrobial resistance 2024

Programme and policy updates (India)

- ▶ **TB preventive treatment delivered through Ayushman Bharat Health and Wellness Centres** — a Global Fund-supported National TB Elimination Programme project (2021-2024) across **208 districts in 23 states** (about 400 million people) traced **3,097,831 household contacts** of 851,974 pulmonary TB patients; **91% were symptom-screened, 54% of those screened started preventive treatment and 84% completed** (92% on the 3-month weekly isoniazid-rifapentine **3HP** regimen). Health and Wellness Centres, community health officers and ASHAs/Mitanins/Sahiyas were the decisive delivery platform; fewer than 1% later progressed to active TB, and the project drove national scale-up of **Cy-Tb** skin testing and 3HP. This is the "services to governments" channel of WHO's work in visible operation, delivered jointly with the WHO Country Office for India [PMID 42583442].
- ▶ **WHO Traditional Medicine Strategy 2025-2034** supersedes the 2014-2023 strategy: it embeds a people-centred model, risk-based regulation for safety and informed choice, integration of traditional, complementary and integrative medicine into primary health care, and acceptance of patient-reported outcome measures as evidence. India's Ministry of Ayush is directly engaged with its implementation, making this the most India-relevant of WHO's current normative products [PMID 41762002].

Current national data

- ▶ **India's TB diagnostic network, appraised against the latest WHO Global TB Report** — a **61.56% national deficit** in culture and drug-susceptibility-testing laboratory capacity relative to projected need, worst in rural and remote areas; an audit of **327 health facilities across seven states and NCR-Delhi** found the **WHO Essential Diagnostics List** poorly implemented at primary level, with nucleic acid amplification testing insufficient for early case detection. The authors tie this to India carrying the world's largest multidrug-resistant TB burden and question the "TB-Free India by 2025" target [PMID 42578696].

- **Zero TB in Kids, Indian congregate settings, 2017-2024** — 20,068 people in 63 schools, monasteries and nunneries over 67,637 person-years: **TB incidence fell 83%, from 576 (95% CI 455-718) to 97 (47-179) per 100,000**, and TB infection prevalence fell from 22% to 15.5%. Preventive treatment cut TB risk in schoolchildren by **82% (aHR 0.18; 95% CI 0.06-0.48)**, and disease prevalence fell 84% even among those who did *not* receive it — a herd effect. Institutes never screened had incidence of 640/100,000 [PMID 41705172].

Elimination, surveillance and disease status

- **Mpox** — WHO declared a **public health emergency of international concern again in August 2024** for clade I virus. India's first **clade Ib** cluster was in Kerala: ten laboratory-confirmed cases between September 2024 and March 2025, 90% with recent international travel (chiefly the UAE) and **one documented local transmission**; APOBEC3 mutational signatures indicated sustained human-to-human spread. The textbook PHEIC list (H1N1 2009, polio, Ebola, Zika) should now be written as including **COVID-19 (2020) and mpox (2022 and 2024)** [PMID 41558442].
- **Antimicrobial resistance is now written into the Pandemic Agreement** — AMR was directly responsible for about **1.27 million deaths** and associated with **4.95 million** in 2021. The **79th UN General Assembly political declaration (2024)** commits to cutting AMR-associated deaths by **10% by 2030**, with 62 recommendations, 45 commitments and **US\$100 million** to help at least 60% of countries fund National Action Plans; oversight sits with the **Quadripartite — WHO, FAO, WOA and UNEP**. A review of 56 papers concluded the forces impeding implementation currently outweigh the drivers [PMID 41815751].

Guideline and legislative updates

- **The International Health Regulations (2005) have been amended** — the **77th World Health Assembly agreed the package on 1 June 2024** and it **entered into force on 19 September 2025**. Amended Articles 1 and 12 empower the Director-General to declare a "**pandemic emergency**", a *higher* tier sitting **inside** the PHEIC framework and restricted to communicable disease; the long-debated intermediate or "traffic-light" tier was considered and **rejected**. Revised **Article 13** obliges the Secretariat to remove barriers to timely, equitable access to health products during an emergency, and revised **Article 44** creates a **Coordinating Financial Mechanism** under the Health Assembly for developing countries. Legally this is a real expansion of WHO's emergency powers, not a cosmetic change [PMID 40589223; PMID 40883832].
- **A new National IHR Authority** — amended **Article 4(1)bis** requires every State Party to *establish or designate* a national IHR authority to **coordinate implementation of the Regulations across sectors and subnational levels**, alongside (and possibly combined with) the older **National IHR Focal Point**; contact details must be given to WHO and confirmed annually, and states must adapt domestic legislation. For India this reaches the state and district tier of surveillance, not just the central focal point [PMID 40953844; PMID 39402755].
- **The WHO Pandemic Agreement** was adopted by the **78th World Health Assembly on 20 May 2025** — **124 governments** adopted its **35 Articles** after nearly three years of negotiation, at a World Health Assembly plenary chaired by Director-General Dr Tedros Adhanom Ghebreyesus. The high-yield point: it is only the **second instrument ever negotiated under Article 19 of the WHO Constitution**, after the **Framework Convention on Tobacco Control** (adopted 2003, in force 2005). It provides for a Coordinating Financial Mechanism and a **Global Supply Chain**

and Logistics Network, but its equity core — the Pathogen Access and Benefit-Sharing (PABS) system under Article 12 — sits in an annex that missed its April 2026 deadline: after six formal meetings the Intergovernmental Working Group sent the 79th Assembly only a draft it flagged as not final agreed text, and decision WHA79(7) of 22 May 2026 told the Group to keep negotiating and report to the 80th Assembly, or to an earlier special session in 2026. Adoption of that annex is the trigger for opening the Agreement to signature and ratification — so it is still not in force [PMID 40589223; PMID 41837256; PMID 41832575; WHO 2026-May].

Global and WHO developments

- **United States withdrawal and the financing of WHO** — announced 20 January 2025, repeating the July 2020 move that was reversed in 2021. In 2022-2023 the US provided about **US\$218 million in assessed contributions plus roughly US\$1.065 billion in voluntary funds**, close to **20% of the budget**; the Bill & Melinda Gates Foundation is now the second-largest voluntary funder, raising the concern that a small group of donors could steer priorities away from country needs. Commentators call for a broader assessed-contribution base and reform of governance and representation, and WHO's **2026 humanitarian appeal seeks close to US\$1 billion** for fragile and conflict-affected states — the practical answer to the textbook line that WHO's finance is "assessed contributions plus extrabudgetary funds" [PMID 40767200; PMID 41580716; PMID 41998654].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 25 (International Health, pp 1055-1058) and Ch 2 (Concept of Health and Disease, p 69). Textbook of International Health — Ch 3 (International Health Agencies, Activities, and Other Actors, pp 107-112) and Ch 5 (Data on Health, pp 234-235). Oxford Textbook of Public Health 7e — Ch 75 (Emerging and Re-emerging Infections, pp 2862-2863) and Ch 107 (Global Health in the Era of Sustainable Development, pp 4044-4045). WHO Constitution, Articles 1 and 2; International Health Regulations (2005). Figure from Suryakantha — Community Medicine with Recent Advances, p 356.

Recent-advances sources: 40589223 — Editorial: The 2025 World Health Assembly Pandemic Agreement and the 2024 Amendments to the International Health Regulations Combine for Pandemic Preparedness and Response, *Med Sci Monit* 2025, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411); 40883832 — To tier or not to tier: the institutionalization of the World Health Organization's power to determine pandemic emergency in the amended International Health Regulations (2005), *Glob Health Res Policy* 2025, doi:[10.1186/s41256-025-00438-6](https://doi.org/10.1186/s41256-025-00438-6); 40953844 — National IHR authorities in the 2024 amendments to WHO's International Health Regulations: transforming the opportunity into reality, *BMJ Glob Health* 2025, doi:[10.1136/bmjgh-2025-019284](https://doi.org/10.1136/bmjgh-2025-019284); 39402755 — The amended international health regulations: Implications and challenges for domestic legal frameworks, *Int J Health Plann Manage* 2024, doi:[10.1002/hpm.3853](https://doi.org/10.1002/hpm.3853) (abstract-only); 41837256 — Pathogen Access and Benefit-Sharing: Can the WHO Pandemic Agreement Bridge the Equity Divide?, *China CDC Wkly* 2026, doi:[10.46234/ccdcw2026.009](https://doi.org/10.46234/ccdcw2026.009) (abstract-only); 41832575 — Africa's Critical Role in Shaping and Implementing the Pandemic Agreement's PABS Annex in an Era of Fragmentation, *Global Health* 2026, doi:[10.1186/s12992-026-01190-3](https://doi.org/10.1186/s12992-026-01190-3); 41815751 — Inclusion of antimicrobial resistance in a pandemic agreement: why it matters and what comes next?, *Health Aff Sch* 2026, doi:[10.1093/haschl/qxag044](https://doi.org/10.1093/haschl/qxag044); 40767200 — The United States Withdrawal From the World Health Organization: Implications and Challenges, *Int J Health Policy Manag* 2025, doi:[10.34172/ijhpm.9086](https://doi.org/10.34172/ijhpm.9086); 41580716 — China's global health diplomacy through the World Health Organization: a qualitative study, *Global Health* 2026, doi:[10.1186/s12992-025-01165-w](https://doi.org/10.1186/s12992-025-01165-w); 41998654 — WHO's 2026 emergency appeal and global health security, *Global Health* 2026, doi:[10.1186/s12992-026-01211-1](https://doi.org/10.1186/s12992-026-01211-1); 42583442 — Lessons learned from implementing a large-scale contact investigation and TB preventive treatment project in India, *IJTL D Open* 2026, doi:[10.5588/ijtdopen.26.0168](https://doi.org/10.5588/ijtdopen.26.0168); 41705172 — Comprehensive tuberculosis screening and preventive treatment in schools and congregate settings of India (2017-2024): a prospective study, *Lancet Reg Health Southeast Asia* 2026, doi:[10.1016/j.lansea.2026.100725](https://doi.org/10.1016/j.lansea.2026.100725); 42578696 — Evaluating the nationwide availability of essential diagnostics for tuberculosis toward achieving a TB-free India, *Microbiol Spectr* 2026, doi:[10.1128/spectrum.00703-26](https://doi.org/10.1128/spectrum.00703-26) (abstract-only); 41558442 — The evolving landscape of Mpox: Clinical and genomic insights into Clade Ib emergence in India, *Virology* 2026, doi:[10.1016/j.virol.2026.110800](https://doi.org/10.1016/j.virol.2026.110800) (abstract-only); 41762002 — WHO Traditional Medicine Strategy 2025-2034: A Policy Analysis of Its People-Centered Framework, *J Integr Complement Med* 2026, doi:[10.1177/27683605261426230](https://doi.org/10.1177/27683605261426230) (abstract-only). WHO 2026-May — Seventy-ninth World Health Assembly, decision WHA79(7), "Outcome of the open-ended Intergovernmental

Working Group on the WHO Pandemic Agreement in relation to the drafting and negotiation of the Annex described in Article 12 of the WHO Pandemic Agreement", adopted at the sixth plenary meeting, 22 May 2026 (A79/VR/6), with document A79/8 (13 May 2026).



Organisation and functions of WHO

Asked in: Paper IV 22-Dec-2023 — RGUHS MD Community Medicine

Definition — The **World Health Organization (WHO)** is the specialized, non-political health agency of the United Nations, with headquarters at Geneva, whose constitutional objective is "*the attainment by all peoples of the highest possible level of health*" (Park 28e, Ch 25, p 1055). Its constitution came into force on **7 April 1948**, observed every year as **World Health Day**. It works through three principal organs — the World Health Assembly, the Executive Board and the Secretariat — decentralised into six regional organizations.

Origin and constitutional basis

Milestone	Detail
Proposal	April 1945, San Francisco conference to set up the United Nations; delegates of Brazil and China proposed an international health organization (Park 28e, Ch 25, p 1055)
Constitution drafted	1946, by the Technical Preparatory Committee under the chairmanship of Rene Sand
Constitution approved	1946, International Health Conference of 51 nations , New York; an <i>Interim Commission</i> carried out urgent tasks and took over the health activities and assets of UNRRA (United Nations Relief and Rehabilitation Administration) at the end of 1946
Came into force	7 April 1948 — formal existence of WHO; celebrated as World Health Day , with a theme chosen each year
Predecessor bodies	Office International d'Hygiène Publique, Pan American Sanitary Bureau, and the Health Organization of the League of Nations — WHO owes much to the work and methods of the last, whose <i>Weekly Epidemiological Record</i> was never suspended even after the League was dissolved in 1939
Preamble (key clauses)	"Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity"; "the enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, economic or social condition"

Membership — open to all countries. Territories not responsible for the conduct of their international relations may be admitted as **associate members**, who participate **without a vote**. Each member state contributes yearly to the budget and is entitled to the services and aid of the organization. WHO had **56 members in 1948** and now has **194 member states and 2 associate members** (Park 28e, Ch 25, p 1056).

Organization — the three principal organs

Organ	Composition	Meets	Main functions
World Health Assembly (WHA) — the "Health Parliament of Nations", supreme governing body	Delegates representing member states; each state has one vote	Annually, usually in May , generally at Geneva headquarters, occasionally in other countries	(i) determine international health policy and programmes; (ii) review the work of the past year; (iii) approve the budget for the following year; (iv) elect member states to designate a person to the Executive Board for three years; also appoints the Director-General on the nomination of the Executive Board ; holds "technical discussions" on a subject of world interest at each Assembly (Park 28e, Ch 25, p 1057)
Executive Board (EB)	34 members , technically qualified in the field of health, designated by but not representing their governments ; membership raised progressively from the original 18 → 24 → 30, and in 1976 to 31 with not fewer than three elected from each WHO region ; one-third renewed every year	At least twice a year — generally January, and shortly after the WHA in May	Gives effect to the decisions and policies of the Assembly; prepares the WHA agenda and works on budgets; has power to act on its own in an emergency — epidemics, earthquakes, floods — where immediate action is needed (Park 28e, Ch 25, p 1057; Textbook of International Health, Ch 3, p 111)
Secretariat	Headed by the Director-General (DG) , the chief technical and administrative officer, elected by the WHA (5-year term); an international civil service of about 8000 staff across headquarters, the regional offices and country offices — from only 250 persons in 1948	Continuous	Provides member states with technical and managerial support for their health development programmes; runs the technical divisions, the library and public information service (Park 28e, Ch 25, pp 1056-1057; Textbook of International Health, Ch 3, p 111)

Regional organization and country level

To meet the special health needs of different areas, WHO has established **six regional organizations** (Park 28e, Ch 25, p 1057, Table 1).

WHO Region	Regional office headquarters
South-East Asia (SEAR)	New Delhi (India) — <i>World Health House</i> , Indraprastha Estate; the region has 11 member states
Africa	Brazzaville (Congo)
The Americas	Washington D.C. (U.S.A.)
Europe	Copenhagen (Denmark)
Eastern Mediterranean	Alexandria (Egypt)

WHO Region	Regional office headquarters
Western Pacific	Manila (Philippines)

- ▶ Each regional office is headed by a **Regional Director**, assisted by technical and administrative officers and members of the secretariat.
- ▶ Each region has a **Regional Committee** of representatives of the member states of that region; it **meets once a year** to review the work in the region and to plan its continuation and development.
- ▶ **Regional plans are amalgamated into the overall plans** of headquarters — the regional organizations are an integral and important part of implementing WHO's policies and programmes.
- ▶ At country level, a **WHO Representative (WR)** is assigned to a country (or to several small adjacent countries) and works with national health authorities, other agencies and donors in reviewing health needs and resources, and in planning, coordinating, implementing and evaluating national health programmes and policies (Textbook of International Health, Ch 3, pp 111-112).

Financing — a regular budget of **assessed contributions** from member states, levied on a sliding scale based on national population and gross national product, plus **extrabudgetary (voluntary) contributions** from member states and other donors. Many projects are paid for jointly by the regular budget, the country concerned, and United Nations, bilateral or private partners. Because extrabudgetary funds are not subject to regulation and priority-setting by the WHA, rising dependence on them shifts priority-setting towards donors and creates funding instability (Textbook of International Health, Ch 3, p 112).

Functions — Articles 1 and 2 of the Constitution

Article 1 states the **objective**; Article 2 lists the **specific functions** (Textbook of International Health, Ch 3, p 110).

#	Constitutional function
1	To act as the directing and coordinating authority on international health work — WHO's <i>first</i> constitutional function
2	To assist governments in strengthening health services and in emergency aid
3	To promote maternal and child health and welfare
4	To foster activities in the field of mental health
5	To promote improvement of nutrition, housing, sanitation, recreation , and of economic, working and environmental conditions
6	To study and report on public health and medical care
7	To promote research and health training
8	To advance work to eradicate epidemic, endemic and other diseases , and to prevent injuries

#	Constitutional function
9	To propose conventions, agreements and regulations and make recommendations on international health matters
10	To standardize diagnostic procedures and revise the international nomenclatures of diseases, causes of death and public health practices
11	To develop, establish and promote international standards for food, biologicals, pharmaceuticals and similar products

The first function is the pivot: it permits member states **collectively** to identify priority health problems, define health policies and targets, and devise the strategies, principles and programmes to attain them (Park 28e, Ch 25, p 1056). Operationally the work divides into **central technical services** (epidemiological intelligence, international health agreements, standardization of vaccines and pharmaceuticals, dissemination of knowledge through expert meetings and technical report series) and **services to governments**, delivered on request through the six regional offices.

Functions in practice — the eight broad areas of work

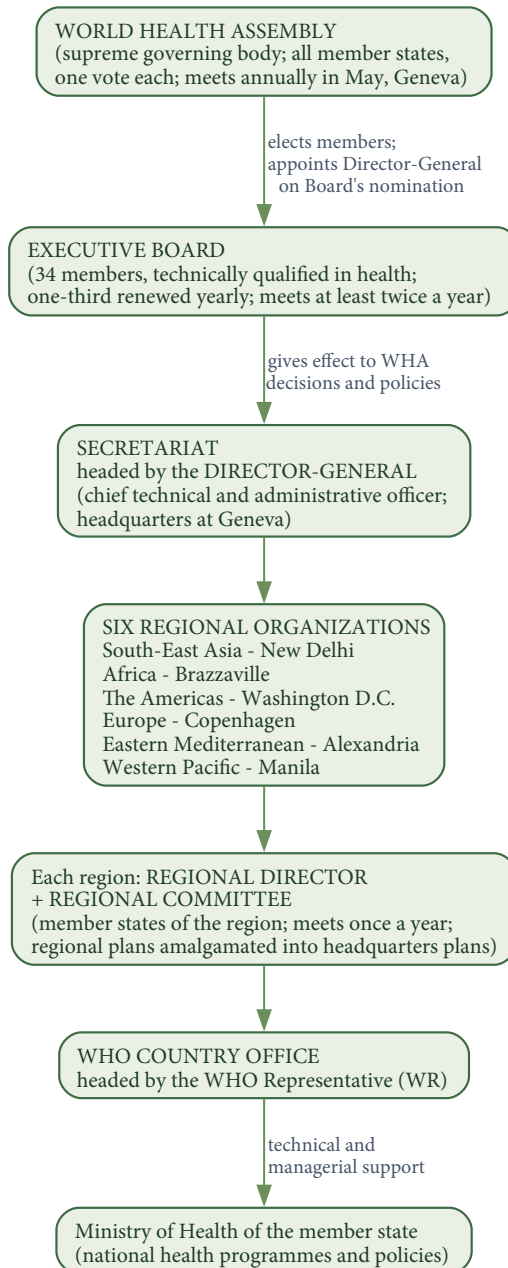
Area of work	What WHO actually does
1. Prevention and control of specific diseases	Almost all communicable diseases have been the subject of WHO activity; global eradication of smallpox is the outstanding example, and the same effort now directs the global battle against poliomyelitis . Runs epidemiological surveillance , collecting and disseminating information on diseases subject to the International Health Regulations through the <i>Weekly Epidemiological Record</i> ; the Expanded Programme on Immunization against common childhood diseases is a priority programme (Park 28e, Ch 25, p 1056)
2. Development of comprehensive health services	WHO's " most important single function " — supporting national health policy and comprehensive national health programmes: organizing health systems based on primary health care , health manpower development and utilization, health infrastructure, managerial capability including monitoring and evaluation, and health services research; Appropriate Technology for Health (ATH) encourages self-sufficiency in solving health problems
3. Family health	A major programme activity since 1970 , subdivided into maternal and child health care, human reproduction, nutrition and health education; the chief concern is the quality of life of the family as a unit
4. Environmental health	Advises governments on national programmes for basic sanitary services; protection of the quality of air, water and food , health conditions of work, radiation protection, and early identification of hazards from new technology; the <i>WHO Environmental Health Criteria Programme</i> and <i>WHO Environmental Health Monitoring Programme</i>
5. Health statistics	Disseminates morbidity and mortality statistics through the <i>Weekly Epidemiological Record</i> , <i>World Health Statistics Quarterly</i> and <i>World Health Statistics Annual</i> ; publishes the International Classification of Diseases (ICD) , updated every 10th year (ICD-10 in force from 1 January 1993), so that statistics from different countries are comparable; assists countries with medical records and national health information systems

Area of work	What WHO actually does
6. Biomedical research	WHO does not itself do research — it stimulates and coordinates it, through a worldwide network of WHO collaborating centres, grants to workers and institutions, Regional Advisory Committees on health research and a Global Advisory Committee. The Special Programme for Research and Training in Tropical Diseases targets six diseases — malaria, schistosomiasis, trypanosomiasis, filariasis, leishmaniasis and leprosy
7. Health literature and information	Acts as a clearing house for health information; publications run to hundreds of titles; the WHO library is a satellite centre of MEDLARS (Medical Literature Analysis and Retrieval System) of the U.S. National Library of Medicine; a public information service operates at headquarters and at each of the six regional offices
8. Cooperation with other organizations	Collaborates with the United Nations and its other specialized agencies, and maintains working relations with international governmental and non-governmental organizations

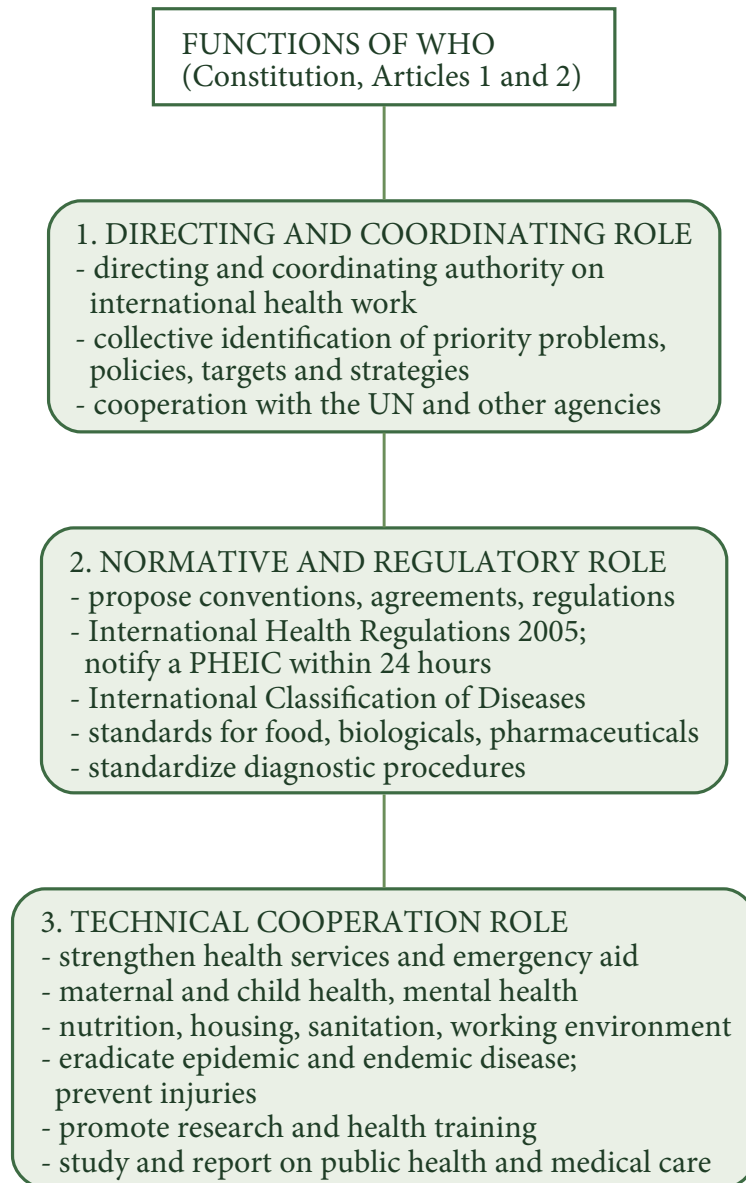
Normative and regulatory role today — under the **International Health Regulations (IHR) 2005**, member states must build and maintain core capacity to detect, assess and respond to public health events, and must notify WHO through their designated **National IHR Focal Point within 24 hours** of an event that may constitute a **public health emergency of international concern (PHEIC)** (Oxford Textbook of Public Health 7e, Ch 105, pp 3996-3997). WHO resolutions **urge but never oblige** member states, and WHO can intervene in a country only when requested (Textbook of International Health, Ch 3, p 111). WHO's current five-year strategy sets the "**triple billion**" targets — one billion more people benefiting from **universal health coverage**, one billion more better protected from **health emergencies**, and one billion more enjoying **better health and well-being** (Oxford Textbook of Public Health 7e, Ch 107, pp 4044-4045).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Drawn figure — label these exactly: *Organogram of WHO* — World Health Assembly (supreme body, one vote per member state) → Executive Board (34 members) → Secretariat headed by Director-General, Geneva → six Regional Offices (SEARO New Delhi, AFRO Brazzaville, AMRO Washington D.C., EURO Copenhagen, EMRO Alexandria, WPRO Manila), each with a Regional Director and a Regional Committee → WHO Country Office headed by the WHO Representative → Ministry of Health of the member state.



Organizational structure of the World Health Organization — from the World Health Assembly down to the country office



The functions of WHO grouped into its three working roles

High-yield exam pointers

- **Dates and numbers:** Constitution in force 7 April 1948 (World Health Day); 56 members in 1948 → 194 member states + 2 associate members now; 34 Executive Board members, one-third renewed yearly; 6 regions; 11 member states in the South-East Asia Region.
- **Three organs mnemonic — "A-B-S":** Assembly (policy and budget) → Board (executive, acts in emergencies) → Secretariat (Director-General, technical and managerial support).
- **Who appoints whom:** the WHA appoints the Director-General on the nomination of the Executive Board; the WHA elects member states which then designate Board members for three years.
- **Regional headquarters in order:** New Delhi, Brazzaville, Washington D.C., Copenhagen, Alexandria, Manila.

- **Eight areas of work:** specific disease control · comprehensive health services (the "most important single function") · family health · environmental health · health statistics · biomedical research · health literature and information · cooperation with other organizations.
- **Research caveat:** WHO does **not** conduct research itself — it stimulates and coordinates it through WHO **collaborating centres**; the tropical-disease programme targets **malaria, schistosomiasis, trypanosomiasis, filariasis, leishmaniasis, leprosy**.
- **Current strategy:** the "**triple billion**" targets — universal health coverage, protection from health emergencies, and better health and well-being, one billion more people each.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park's 28th edition table prints the Eastern Mediterranean Regional Office at Alexandria, whereas that office now functions from Cairo; the same chapter's health-statistics section still describes ICD-10 (in force from 1 January 1993), while the Eleventh Revision has since come into use — the retrieved corpus supports only the older statements, which are the ones reproduced above.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, International Health Regulations (2024 amendments), WHO Pandemic Agreement 2025, Bulletin of the World Health Organization, WHO South-East Asia Regional Office

Programme and policy updates (India)

- **A new organ at country level — the "national IHR authority".** Amended Article 4 obliges every State Party, India included, to *establish or designate* an entity — a new or an existing body — through domestic **legislative and/or administrative arrangements when appropriate**, to coordinate IHR implementation across the whole jurisdiction (Article 4(1)(bis)). This is a coordinating function **additional to** that of the older **National IHR Focal Point** (the communication channel that must notify WHO within 24 hours), but States may use **one or two entities**, so the same body may discharge both roles. The focal point had persistently lacked seniority, intersectoral reach and domestic authority to report events to WHO; the Assembly deliberately dropped the earlier draft wording that would have made the new authority formally accountable for overall IHR compliance. Contact details must be given to WHO and confirmed annually [PMID 40953844].
- **WHO's Global Traditional Medicine Centre is based in India**, working from an **interim office on the Institute of Teaching and Research in Ayurveda (ITRA) campus at Jamnagar, Gujarat**. WHO's new traditional medicine strategy aims to advance evidence-based traditional, complementary and integrative medicine towards the highest attainable standard of health, and its implementation is the theme of the **second WHO Global Summit on Traditional Medicine in December 2025**. Two exam-relevant spin-offs: **Chapter 26 of ICD-11** lets traditional-medicine disorders and patterns be coded alongside biomedical diagnoses, filling long-standing data gaps in health information systems; and **less than 1% of global health research funding** goes to traditional medicine, an inequity that undermines building the evidence base [PMID 41180266].

Current national data

- ▶ **WHO South-East Asia Region (11 member states, regional office New Delhi) — measles and rubella, 2013 to 2023.** Five of the eleven countries (Bhutan, DPR Korea, Maldives, Sri Lanka, Timor-Leste) had been verified as having eliminated both diseases by 2023; regional coverage in 2022 was MCV1 92%, MCV2 85%, RCV1 92%; reported measles incidence *rose* from 16.2 to 40.9 per million between 2013 and 2023, driven largely by India and Indonesia, while estimated measles deaths fell by 96% and the region's share of global measles deaths fell from **26% (2013) to 7% (2022)**. WHO-proficient laboratories in the region grew from 23 to 58 over the decade [PMID 39460260].

Elimination, surveillance and disease status

- ▶ **The regional measles-rubella deadline has slipped twice** — the 2013 Regional Committee resolution set "measles elimination and rubella control by 2020", this was revised in 2019 to "measles and rubella elimination by 2023", and a 2023 regional consultation named **2026** as the next feasible target date. Quote the current target, not the textbook one [PMID 39460260].
- ▶ **Kala-azar (visceral leishmaniasis) — India met the elimination threshold in December 2023.** All 633 endemic blocks fell below **1 case per 10,000 population at block (sub-district) level**; reported cases fell from 34,803 in 2005 to 524 in 2023, and the WHO South-East Asia Region now carries only about 6% of the global burden. India is in the mandatory **three-year consolidation phase** before it can submit a dossier to WHO for validation; **Bangladesh was validated first, in 2023**. Single-dose liposomal amphotericin B (introduced into the programme in 2014) was the decisive tool; PKDL and HIV-VL co-infection remain the human reservoirs that could restart transmission [PMID 40828749].
- ▶ **PHEIC — seven declarations since the IHR (2005) came into force**, the most recent being **mpox**; a document analysis of 101 IHR Emergency Committee statements found the three declaration criteria applied inconsistently, though justification and structure of the statements have improved since COVID-19 [PMID 40022025].

Guideline and legislative updates

- ▶ **IHR (2005) amendments — adopted 1 June 2024 at the 77th World Health Assembly, in force 19 September 2025.** Key changes: a new "**pandemic emergency**" tier *within* the PHEIC framework (Articles 1 and 12), applying only to communicable-disease events, which the Director-General may now declare — an intermediate/lower alert tier was debated and rejected; the **national IHR authority** under Article 4; a revised **Article 13** requiring the WHO Secretariat to facilitate access to health products during a PHEIC or pandemic emergency; and a revised **Article 44** creating a **Coordinating Financial Mechanism** under the Health Assembly's authority, with an advisory subcommittee established under Article 54 [PMID 40589223; PMID 40883832; PMID 40953844].
- ▶ **WHO Pandemic Agreement — adopted 20 May 2025 at the 78th World Health Assembly by 124 governments**, running to **35 Articles** after three years of negotiation. It is only the **second instrument ever negotiated under Article 19 of the WHO Constitution**, after the Framework Convention on Tobacco Control (adopted 2003, in force 2005) — a high-yield pairing for any question on WHO's normative and regulatory powers. It also provides for a Coordinating Financial Mechanism and a Global Supply Chain and Logistics Network — member states

directed the intergovernmental working group to begin setting both up — and **antimicrobial resistance** — the "constant pandemic" — is written into it, linking the treaty to the 2024 UN General Assembly AMR declaration that targets a 10% cut in bacterial-AMR deaths by 2030 [PMID 40589223; PMID 41815751].

- **The Agreement is not yet complete — the PABS annex is the missing piece.** The **Pathogen Access and Benefit-Sharing** system under Article 12 (sharing of pathogen materials and sequence data in return for equitable access to vaccines, diagnostics and therapeutics) was left to an Intergovernmental Working Group, which first met on **9 July 2025** with an **April 2026** target for the annex before it goes to the Health Assembly. Until the annex is agreed, the Agreement's equity provisions remain largely aspirational [PMID 41832575; PMID 40589223].

Global and WHO developments

- **United States withdrawal from WHO announced 20 January 2025**, repeating the July 2020 move that had been reversed in 2021. Scale of the loss: in 2022–2023 the USA provided about **US\$218 million in assessed contributions plus US\$1.065 billion in voluntary (largely earmarked) funds**, roughly a fifth of the WHO budget; the Bill & Melinda Gates Foundation is the second-largest voluntary funder. This is the live illustration of the textbook point that dependence on extrabudgetary money shifts priority-setting from the Health Assembly towards donors and destabilises the budget [PMID 40767200].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 25 (International Health), pp 1055-1058 and Chapter 2, p 35; Textbook of International Health — Chapter 3 (International Health Agencies, Activities, and Other Actors), pp 110-112; Oxford Textbook of Public Health 7e — Chapter 105, pp 3996-3997 and Chapter 107, pp 4044-4049; WHO Constitution (1946, in force 1948), Articles 1 and 2; International Health Regulations (2005).

Recent-advances sources: 40589223 — Editorial: The 2025 World Health Assembly Pandemic Agreement and the 2024 Amendments to the International Health Regulations Combine for Pandemic Preparedness and Response, *Med Sci Monit* 2025, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411); 40883832 — To tier or not to tier: the institutionalization of the World Health Organization's power to determine pandemic emergency in the amended International Health Regulations (2005), *Glob Health Res Policy* 2025, doi:[10.1186/s41256-025-00438-6](https://doi.org/10.1186/s41256-025-00438-6); 40953844 — National IHR authorities in the 2024 amendments to WHO's International Health Regulations: transforming the opportunity into reality, *BMJ Glob Health* 2025, doi:[10.1136/bmjgh-2025-019284](https://doi.org/10.1136/bmjgh-2025-019284); 41832575 — Africa's Critical Role in Shaping and Implementing the Pandemic Agreement's PABS Annex in an Era of Fragmentation, *Global Health* 2026, doi:[10.1186/s12992-026-01190-3](https://doi.org/10.1186/s12992-026-01190-3); 41815751 — Inclusion of antimicrobial resistance in a pandemic agreement: why it matters and what comes next?, *Health Aff Sch* 2026, doi:[10.1093/haschl/qxag044](https://doi.org/10.1093/haschl/qxag044); 40767200 — The United States Withdrawal From the World Health Organization: Implications and Challenges, *Int J Health Policy Manag* 2025, doi:[10.34172/ijhpm.9086](https://doi.org/10.34172/ijhpm.9086); 40022025 — How have PHEIC determinations changed since the COVID-19 pandemic? A document analysis of 101 WHO International Health Regulations Emergency Committees' statements, *BMC Public Health* 2025, doi:[10.1186/s12889-025-22056-0](https://doi.org/10.1186/s12889-025-22056-0); 39460260 — Progress towards Measles and Rubella Elimination in the South-East Asia Region—2013-2023, *Vaccines (Basel)* 2024, doi:[10.3390/vaccines12101094](https://doi.org/10.3390/vaccines12101094); 40828749 — The story of elimination of visceral leishmaniasis (kala-azar) in India—Challenges towards sustainment, *PLoS Negl Trop Dis* 2025, doi:[10.1371/journal.pntd.0013321](https://doi.org/10.1371/journal.pntd.0013321); 41180266 — Traditional medicine and its contributions to science, health equity and sustainability, *Bull World Health Organ* 2025, doi:[10.2471/BLT.25.295061](https://doi.org/10.2471/BLT.25.295061).



Emporiiatrics

Asked in: Paper I 29-Jul-2021 — RGUHS MD Community Medicine

Definition — In Greek, *emporos* means traveller and *iatrike* means medicine, so **emporiatics** is the branch of medical science dealing with the health of international travellers (Suryakantha — Community Medicine with Recent Advances, Ch 19, p 324); Park defines it as the term coined to describe **the science of the health of travellers** (Park 28e, Ch 3, p 152). It assesses, prevents and manages the health risks arising from international travel, and it operates within the legal framework of the **International Health Regulations (IHR) 2005**, whose stated aim is to ensure maximum **security** against the international spread of diseases with the minimum interference with world traffic (Park 28e, Ch 25, p 1056).

Why the international traveller is at special risk

Category of risk	Content (Park 28e, Ch 3, p 152)
1. Stress of jet travel	Crowding, long hours of waiting, disruption of eating habits, change of climate and of time zone — lowering resistance and provoking nausea, indigestion, extreme fatigue and insomnia
2. Exposure to diseases NOT covered by the IHR	Malaria, giardiasis, dengue, influenza, filariasis, sexually transmitted disease and acquired immunodeficiency syndrome (AIDS), intestinal parasites, typhoid and paratyphoid fever, viral hepatitis
3. Delayed manifestation	Many of these do not manifest immediately but appear over a varying period after the traveller has returned to his normal way of life — hence the need for post-travel vigilance
4. Environmental/hygiene factors at destination	Poor hygiene by the food handler, poor water quality, improper disposal of wastes
5. Separation from care	The traveller is cut off from familiar and accessible sources of medical care
6. Host and destination factors	Younger travellers are at greater risk than older ones, and the greater the climatic and cultural contrast between the native and the destination country, the higher the risk; diarrhoea and vomiting are the commonest problems (Suryakantha, Ch 19, p 324)
7. Legal/administrative risk	Denial of entry, or detention, where a legally required yellow fever certificate is absent (Park 28e, Ch 5, p 348)

The traveller carries a **personal responsibility** to recognise these risks, which can be minimised by immunisation and by chemoprophylaxis or chemotherapy (Park 28e, Ch 3, p 152).

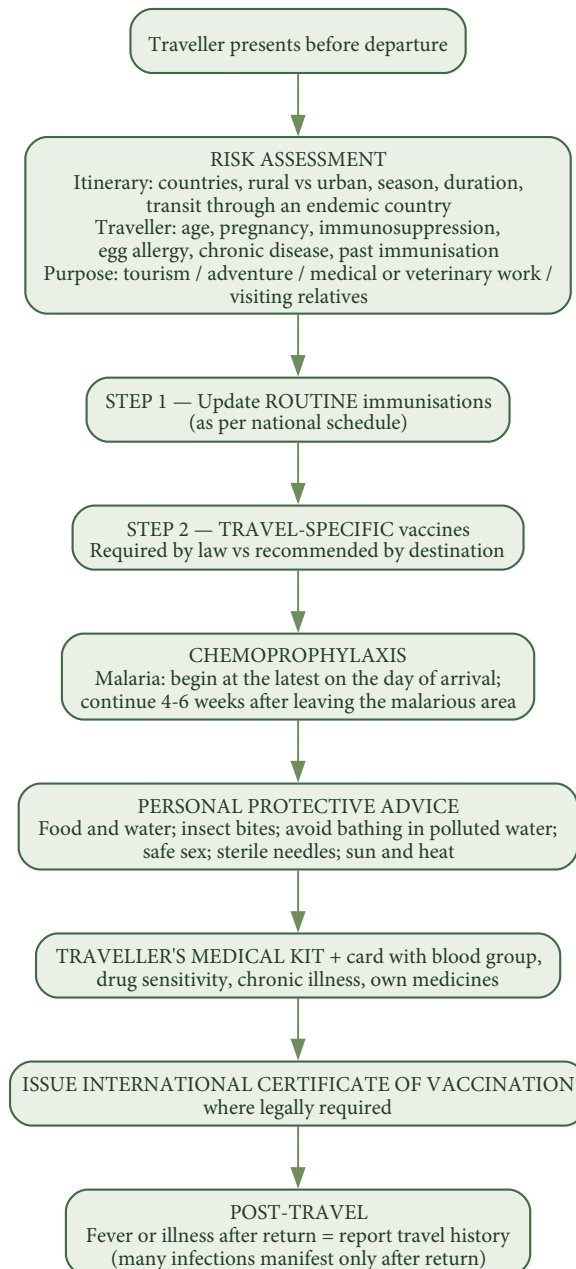
Jet lag (Suryakantha, Ch 19, pp 324-325)

- **What it is** — the symptoms caused by disruption of the body's internal clock and the approximately 24-hour (**circadian**) rhythms it controls, occurring when **multiple time zones** are crossed travelling east to west or west to east.
- **Symptoms** — indigestion, bowel disturbance, general malaise, daytime sleepiness, difficulty sleeping at night, and reduced physical and mental performance; they wear off gradually as the body adapts to the new time zone.

- **Cannot be prevented**, but the effects are reduced by: resting well before and during the flight; light meals with limited alcohol; short-acting sleeping pills where needed; and exposure to daylight at the destination, which aids adaptation.

The pre-travel consultation

Immunising and advising a traveller is not a rote selection of vaccines by destination country; it requires the itinerary, the living conditions and mode of travel, the purpose of travel and the medical history to be taken together (Plotkin's Vaccines 7e, Ch 71, p 2003).



The pre-travel consultation — sequence from risk assessment to certification

Immunisation of the traveller

Tier	Vaccines	Key points
Required by law (certificate)	Yellow fever	The only certificate required for international travel (Park 28e, Ch 3, p 152); yellow fever was the first disease for which the IHR required a certificate for travellers entering an <i>Aedes aegypti</i> -infested receptive area from an endemic country (Plotkin's Vaccines 7e, Ch 63, p 1728)
Required by the receiving country	Quadrivalent meningococcal (A, C, W, Y) for pilgrims to Mecca	Risk in pilgrims reached up to 2000 per million before quadrivalent immunisation; Saudi health authorities grant the conjugate certificate 8 years validity (Plotkin's Vaccines 7e, Ch 71, pp 2013-2015)
Recommended by destination	Hepatitis A (immunoglobulin reserved for infants under 1 year or vaccine allergy), hepatitis B, typhoid, rabies pre-exposure prophylaxis, Japanese encephalitis; cholera only in the rare high-risk traveller	Japanese encephalitis vaccine for travel of ≥1 month in enzootic rural Asia in the transmission season, or shorter stays with extensive outdoor exposure (Plotkin's Vaccines 7e, Ch 33, p 801). Rabies pre-exposure does not remove the need for post-exposure doses (Ch 71, p 2015). Cholera vaccine is not recommended routinely for travellers and no country requires it for entry ; it is offered only to the rare traveller to areas of high risk and poor sanitation, such as relief workers in a cholera outbreak (Ch 71, p 2008)
Routine, to be updated	Diphtheria, tetanus booster if ≥10 years since the last complete course or booster, poliomyelitis, measles, annual influenza; pneumococcal and zoster in the older traveller	Park 28e, Ch 3, p 152; Plotkin's Vaccines 7e, Ch 71, p 2004

Yellow fever certificate — the rules to reproduce

Item	Rule
Vaccine	17D live attenuated, 0.5 mL subcutaneously at the insertion of the deltoid, irrespective of age (Park 28e, Ch 5, p 348)
Onset and duration of immunity	Immunity appears on the 7th day and lasts possibly for life
Certificate becomes valid	10 days after the date of vaccination
Validity	Lifetime . The World Health Assembly amendment to IHR (2005) entered into force on 11 July 2016 ; since then no booster or revaccination is required as a condition of entry, and lifetime validity applies automatically in India (Park 28e, Ch 5, p 349)
Where given	Only at a designated centre , the certificate bearing the official stamp of the Ministry of Health, Government of India
No certificate	Traveller quarantined for 6 days in a mosquito-proof ward from the date of leaving the infected area

Item	Rule
Contraindications	Age under 9 months for routine use (under 6 months in an epidemic); pregnancy, except when outbreak risk is high; severe egg allergy; severe immunodeficiency (symptomatic HIV/AIDS, thymus disorder)
Spacing	Cholera and yellow fever vaccines interfere if given together or within 3 weeks — give ≥3 weeks apart

Chemoprophylaxis and personal protective advice

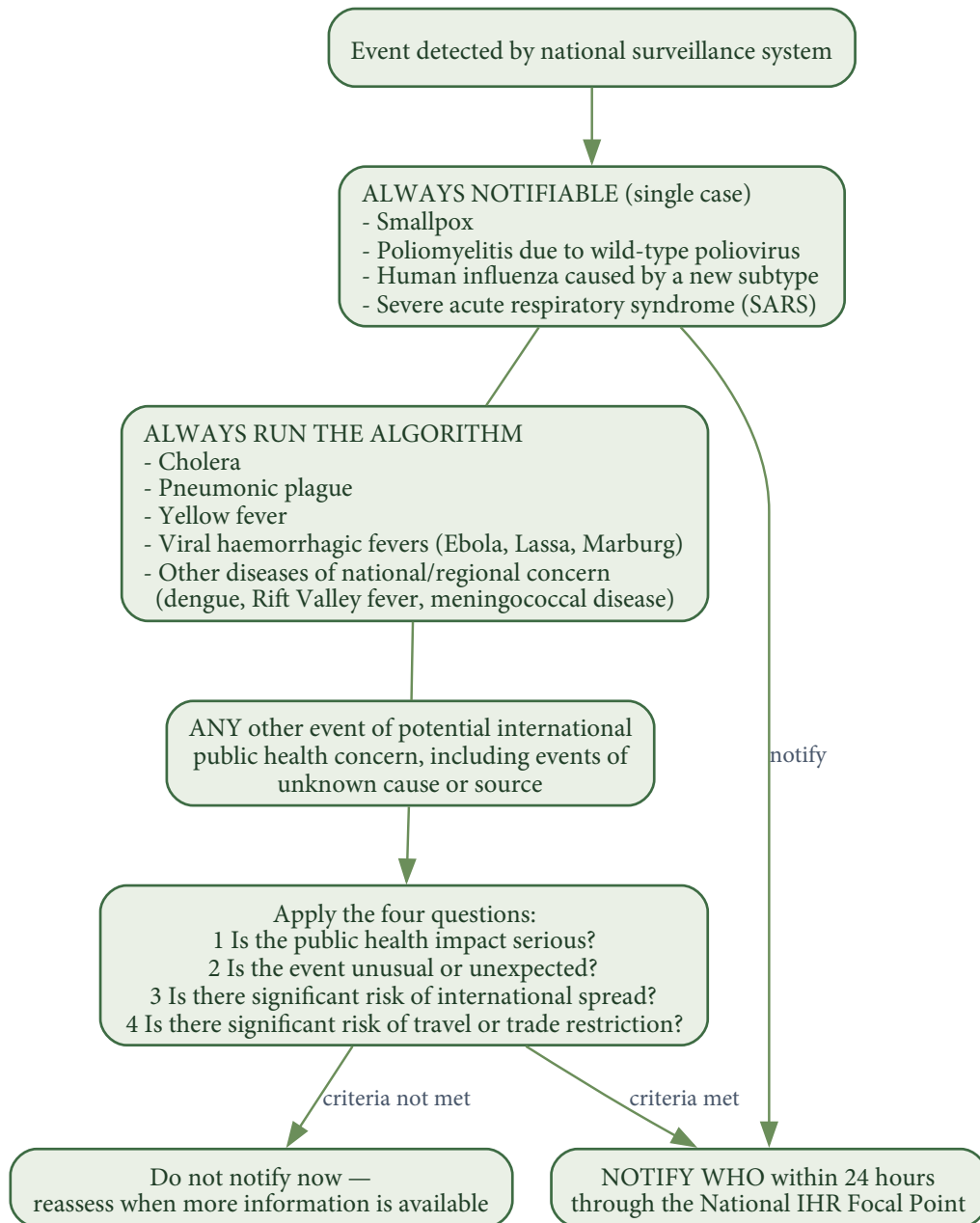
Hazard	Advice to the traveller (Park 28e, Ch 3, p 152)
Travellers' diarrhoea	Affects an estimated 20-50 per cent of all travellers ; contaminated food and drink are the commonest source. Appearance of food is no guide to its safety. Regard unpasteurised milk, non-bottled drinks and uncooked food (except fruit and vegetables that can be peeled or shelled) as unsafe; eat food thoroughly and freshly cooked; use boiled or bottled water; know the value of oral rehydration fluid containing salt and glucose
Malaria	High risk in endemic areas. Chemoprophylaxis must begin at the latest on the day of arrival in the malarious area and be continued for 4-6 weeks after leaving it
Insect bites	Repellents, mosquito coils, fumigation mats, bed nets, covering clothing
Water contact	Avoid bathing in polluted water — causes ear, eye and skin infection. Excessive heat, humidity or over-exertion causes exhaustion from loss of water and salt
Hepatitis A	Where vaccine cannot be used, normal human immunoglobulin 0.02-0.05 mL/kg body weight (3.2-8.0 mg/kg) repeated every 4 months for travellers exposed to unhygienic conditions in tropical or developing countries (Park 28e, Ch 3, Table 3, pp 128-129). Immunoglobulin should not be given within 3 weeks before, or until 2 weeks after, a live vaccine. The inactivated vaccine is highly safe and is preferred for pre-exposure protection
Hepatitis E	No hepatitis E vaccine is available to international travellers. A recombinant HEV 239 vaccine has been licensed in China since 2012 but is licensed nowhere else and is not recommended for travellers (Plotkin's Vaccines 7e, Ch 27, pp 577-578). Immunoglobulin prepared in Europe and the USA gives little protection, so avoidance of contaminated food and water is the only effective measure (Park 28e, Ch 3, p 152)
Hepatitis B	Vaccine is safe and effective — three doses, the first two one month apart, the third about 6 months later
Sexually transmitted infection and AIDS	Same measures as at home — abstinence or a single faithful uninfected partner, condom use; avoid injecting drugs; ensure needle and syringe are sterile
Medical kit	Disinfectant, easily applied dressing, sun cream, mosquito repellent, oral rehydration salts, first-aid articles; a card noting blood group, drug sensitivity and chronic illness; and enough of the traveller's own medicines (diabetes, cardiac disease) to avoid a break in medication

The legal framework — IHR (2005) and the traveller

- Notification within 24 hours to the World Health Organization (WHO) through the designated National IHR Focal Point of any event that may constitute a **public health emergency of**

international concern (PHEIC) — defined as an extraordinary event constituting a public health risk to other States through international spread and potentially requiring a coordinated international response (Oxford Textbook of Public Health 7e, Ch 105, pp 3996-3997; Textbook of International Health, Ch 5, p 234).

- ▶ **Core capacities** for surveillance and response must be developed, strengthened and maintained by every State Party so that events are rapidly detected, assessed, notified, reported and contained; capacity is now assessed through the voluntary **joint external evaluation (JEE)** (Oxford Textbook of Public Health 7e, Ch 75, pp 2859-2866).
- ▶ **Older classification (still examinable)** — diseases subject to the IHR: **cholera, plague and yellow fever**; diseases under surveillance by WHO: louse-borne typhus fever, relapsing fever, paralytic poliomyelitis, malaria, viral influenza-A, severe acute respiratory syndrome (SARS) and smallpox (Park 28e, Ch 3, p 144).
- ▶ **Measures at Indian ports and airports** — aircraft and ships arriving from endemic areas are subjected to **aerosol disinsection**; airports and seaports are kept free of vector breeding over an area extending at least **400 metres** around their perimeters. The ***Aedes aegypti* index** is the percentage of houses in a limited well-defined area on whose premises actual breeding of *Aedes aegypti* is found, out of the total houses examined in that area; it is kept **at zero at all ports** (Park 28e, Ch 15, p 925).
- ▶ **WHO's traveller publication** — WHO publishes annually "**International Travel and Health: Vaccination Requirements and Health Advice**", giving the main health risks by region and the precautions against them (Park 28e, Ch 3, p 152).



IHR (2005) Annex 2 decision instrument — is the event a public health emergency of international concern?

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **International certificate of vaccination against yellow fever** — draw a rectangular certificate and label: *name and nationality of the traveller; vaccine used and manufacturer's batch number; date of vaccination; signature and professional status of the supervising clinician; period of validity.*

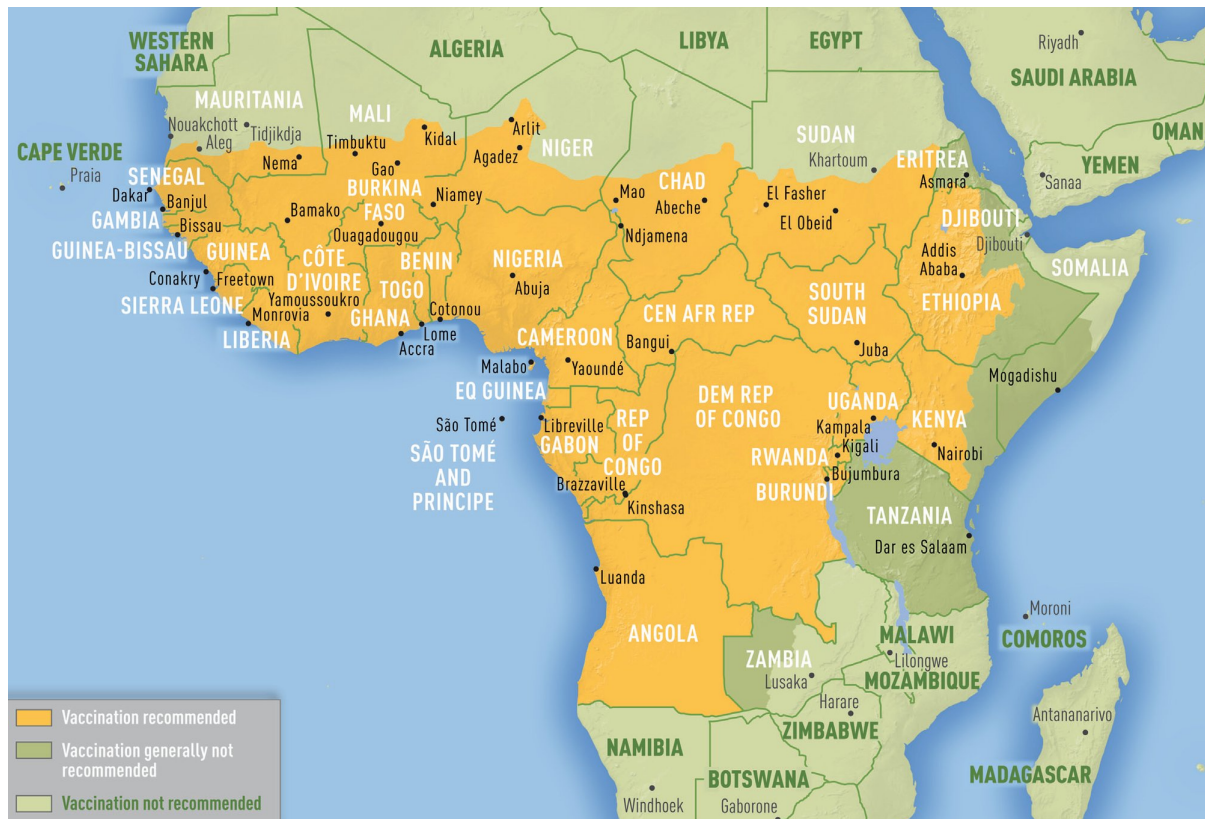


Fig 71.5 — Yellow fever vaccine recommendations in Africa. (From Centers for Disease Control and Prevention) (Plotkin's Vaccines, 7ed, p.2018)

High-yield exam pointers

- ▶ **Emporiatrics** — Greek *emporos* (traveller) + *iatrike* (medicine) = **the science of the health of travellers** (Park 28e, Ch 3, p 152) — open the answer with this one line.
- ▶ **Three reasons travellers fall ill:** (1) stress of jet travel, (2) exposure to diseases not covered by the IHR, (3) separation from accessible medical care.
- ▶ **Travellers' diarrhoea** 20-50 per cent of all travellers; **malaria chemoprophylaxis** from the day of arrival, continued 4-6 weeks after leaving; **tetanus booster** if ≥ 10 years; **hepatitis A immunoglobulin** 0.02-0.05 mL/kg every 4 months.
- ▶ **Yellow fever** is the only vaccination certificate required for international travel: 17D, 0.5 mL subcutaneous; valid from the 10th day; lifetime validity since 11 July 2016; no certificate $\rightarrow$ 6 days quarantine in a mosquito-proof ward.
- ▶ **IHR 2005:** notify WHO within **24 hours** through the National IHR Focal Point; **four always-notifiable diseases** = smallpox, wild poliovirus poliomyelitis, human influenza of a new subtype, SARS.
- ▶ **Airport/seaport sanitation:** vector-free zone of at least **400 metres** around the perimeter; *Aedes aegypti* index kept at zero at ports.
- ▶ **Jet lag** — circadian disruption on crossing multiple time zones; not preventable, but reduced by rest, light meals with limited alcohol, short-acting hypnotics and daylight at the destination.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written as exam content: Park 28e still reproduces the pre-2005 list of "diseases subject to the International Health Regulations" (cholera, plague, yellow fever), whereas the IHR (2005) replaced disease-specific notification with the Annex 2 all-hazards decision instrument shown above — write both, giving IHR (2005) as the current position. Park is also internally inconsistent on the *Aedes aegypti* index target: Ch 15 (p 925) says it is kept at zero at all ports, while Ch 5 (p 348) sets a target of not more than 1 per cent in large towns and seaports.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO International Health Regulations (2005) as amended 2024, WHO Pandemic Agreement 2025, ICMR, GeoSentinel, Swiss Expert Committee for Travel Medicine

Programme and policy updates (India)

- ▶ **Kala-azar elimination reached, and the programme renamed.** India crossed the elimination-as-a-public-health-problem threshold — fewer than **1 case per 10,000 population per year at block level** — in **2023**; the 2002 National Health Policy had originally set 2010 and the deadline was deferred repeatedly. By 2021 only **7 of 633 endemic blocks** were still above threshold. Field operations (house-to-house active search, indoor residual spraying, the Kala-azar Management Information System) now run under the **National Centre for Vector Borne Diseases Control** — the renamed NVBDCP — within the Accelerated Kala-azar Elimination Plan; liposomal amphotericin B was introduced in 2014 [PMID 40988627].
- ▶ **PM-ABHIM is building four zonal virology institutes with an explicit importation remit.** Under the **Pradhan Mantri Ayushman Bharat Health Infrastructure Mission** (announced 25 October 2021), the Department of Health Research through ICMR is establishing four zonal National Institutes of Virology at **Bengaluru, Jammu, Dibrugarh and Jabalpur**. A March-2024 multicriteria prioritisation across 186 stakeholders ranked 40 viruses in three groups, of which **Group C is defined by risk of emergence, evolution or importation** — the traveller-facing arm of India's preparedness architecture. Dengue, influenza, measles, Japanese encephalitis and hepatitis A were the top five priorities across zones [PMID 42522023].

Current national data

- ▶ **Measles, India, August 2023 – January 2024: 13,220 cases, fourth highest in the world.** Coverage alone is not protection — in a tribal cohort in Kandhamal and Boudh (Odisha), second-dose measles-containing-vaccine coverage was **97.5%**, yet only **55.1%** of 1,185 children reached a protective IgG titre (≥ 250 IU/L) three months after the second dose; male sex predicted a suboptimal response [PMID 42555691].
- ▶ **About one in four Indian under-fives is still incompletely immunised.** Pooled prevalence **26%** (**95% CI 21–32%**) across 45 cross-sectional studies and more than 428,000 children, 2015–2025. The dominant determinant was absence of an immunisation card (OR 9.46, 95% CI 3.28–27.28), followed by non-institutional delivery, maternal illiteracy and low socioeconomic status — a

direct argument for the "update routine immunisations first" step of the pre-travel consultation [PMID 42320383].

Elimination, surveillance and disease status

- **Measles–rubella elimination target is December 2026, and transmission is still running.** India, with the rest of the WHO South-East Asia Region, is committed to rubella elimination by **December 2026**; but in central India (Jabalpur, 1 May 2023 – 30 April 2024) rubella **IgM seropositivity was 7.2%**, confirming ongoing transmission, against IgG of 81.0% overall and 89.6% in pregnant women [PMID 42159053]. Travellers keep re-seeding measles internationally: GeoSentinel logged **53 imported measles cases into 15 countries, May 2019 – June 2025**, of whom 74% were aged 21 years or older, 61% were hospitalised and 79% had no or unknown vaccination history [PMID 40738382].
- **Mpox clade Ib has entered India, and the route was air travel.** Ten laboratory-confirmed cases were profiled in Kerala between September 2024 and March 2025; median age 34 years, 80% male, and **90% had recent international travel, chiefly to the United Arab Emirates**, with one documented instance of local transmission. Viral DNA cleared at 20–28 days, and the Indian sequences clustered with those from Oman, Pakistan and Thailand [PMID 41558442].

Guideline and legislative updates

- **The IHR (2005) were amended on 1 June 2024 (WHA77) and the amendments entered into force in September 2025** — the first substantive revision since 2005 and the single most examinable change to this topic. Three things to reproduce: (1) a new "**pandemic emergency**" tier was inserted *within* the PHEIC framework as the **highest level of alert**, applying only to communicable-disease PHEICs — no lower or intermediate tier was adopted, so the framework is now **PHEIC → pandemic emergency** [PMID 40883832]; (2) the Article 4 **National IHR Focal Point is unchanged**, but every State Party must additionally designate a **National IHR Authority** to coordinate domestic implementation, which may or may not be the same body as the focal point [PMID 40497081]; (3) a revised **Article 13** obliges the WHO Secretariat to facilitate access to health products during a PHEIC or pandemic emergency, and a revised **Article 44** creates a coordinating financial mechanism for developing countries [PMID 40589223].
- **A dengue vaccine now exists for travellers, but with a serostatus gate.** Qdenga (TAK-003) has been licensed for travellers in several European countries since December 2022. Expert travel-medicine guidance is explicit: it is **not** recommended for the dengue-naïve traveller, and may be offered only to those **aged 6 years and above with laboratory or serological evidence of previous dengue infection** who are going to an area of significant transmission. Dengvaxia carries no recommendation for travellers at all [PMID 39463237].

Global and WHO developments

- **The WHO Pandemic Agreement was adopted on 20 May 2025 at the 78th World Health Assembly** — 124 governments, 35 Articles, after three years of negotiation. It is only the **second treaty ever negotiated under Article 19 of the WHO Constitution**, following the Framework Convention on Tobacco Control (adopted 2003, in force 2005), and it sits alongside — not instead of — the amended IHR, adding a Coordinating Financial Mechanism, a Global Supply Chain and Logistics Network, and a Pathogen Access and Benefit-Sharing system [PMID 40589223].

- **Current PHEIC status — write this, not the textbook list.** Mpox was declared a PHEIC a second time on **14 August 2024** for clade Ib and that PHEIC was **lifted in September 2025**; in the WHO European Region 82 clade I cases were reported up to 23 November 2025, of which **45 were imported and 37 acquired locally** [PMID 42318655]. **Ebola disease due to Bundibugyo virus was declared a PHEIC on 17 May 2026** (outbreak in Ituri, DRC, with exported cases in Uganda) — the ninth PHEIC since 2007. By 17 June 2026 there were **896 confirmed cases and 232 deaths (case-fatality about 26%)**; there is **no licensed vaccine against Bundibugyo virus**, and WHO advises that Ervebo not be used outside research settings because cross-protection evidence is insufficient, while a ChAdOx1 Bundibugyo candidate is being scaled with the Serum Institute of India [PMID 42383297; PMID 42173889].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3 — Principles of Epidemiology and Epidemiologic Methods, Table 3 pp 128-129, p 144, p 152; Ch 5 — Epidemiology of Communicable Diseases, pp 347-349; Ch 15 — Environment and Health, p 925; Ch 25 — International Health, p 1056); Suryakantha — Community Medicine with Recent Advances (Ch 19 — Epidemiology of Infectious Diseases, pp 324-325); Plotkin's Vaccines, 7ed (Ch 71 — Vaccines for International Travel, pp 2003-2018; Ch 63 — Yellow Fever Vaccines, p 1728; Ch 33 — Japanese Encephalitis Vaccines, p 801; Ch 27 — Hepatitis E Vaccines, pp 577-578); Oxford Textbook of Public Health 7e (Ch 75 — Emerging and re-emerging infections, pp 2859-2866; Ch 105, pp 3996-3997); Textbook of International Health (Ch 5, p 234); WHO International Health Regulations (2005), Second Annotated Edition, with the yellow fever amendment in force from 11 July 2016; WHO *International Travel and Health* (annual).

Coverage note: the indexed corpus carries emporiatrics as a short Park section plus the Suryakantha travel section and the travel-vaccine and IHR chapters cited above; it contains no Indian traveller-clinic service statistics, so no Indian coverage figures are quoted.

Recent-advances sources: PMID 40988627 — Integrating community efforts, innovative government interventions and data in the fight against kala-azar: lessons from Dumka, India, *International Health* 2026, doi:[10.1093/inthealth/ihaf083](https://doi.org/10.1093/inthealth/ihaf083); PMID 42522023 — Strengthening India's pandemic preparedness with the four zonal institutes of virology, *Infectious Diseases of Poverty* 2026, doi:[10.1186/s40249-026-01484-z](https://doi.org/10.1186/s40249-026-01484-z); PMID 42555691 — Protective antibody response following two doses of measles vaccination and factors for non-response among children from tribal Odisha, India, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_3672_2025](https://doi.org/10.25259/IJMR_3672_2025); PMID 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: a systematic review and meta-analysis (2015–2025), *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only); PMID 42159053 — Monitoring Rubella Immunity and Transmission in Central India Amidst Elimination Efforts, *Microbiology and Immunology* 2026, doi:[10.1111/1348-0421.70065](https://doi.org/10.1111/1348-0421.70065) (abstract-only); PMID 40738382 — Measles importations by international travelers, GeoSentinel 2019–2025, *Travel Medicine and Infectious Disease* 2025, doi:[10.1016/j.tmaid.2025.102885](https://doi.org/10.1016/j.tmaid.2025.102885) (abstract-only); PMID 41558442 — The evolving landscape of Mpox: clinical and genomic insights into Clade Ib emergence in India, *Virology* 2026, doi:[10.1016/j.virol.2026.110800](https://doi.org/10.1016/j.virol.2026.110800) (abstract-only); PMID 40589223 — Editorial: The 2025 World Health Assembly Pandemic Agreement and the 2024 Amendments to the International Health Regulations Combine for Pandemic Preparedness and Response, *Medical Science Monitor* 2025, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411); PMID 40883832 — To tier or not to tier: the institutionalization of the World Health Organization's power to determine pandemic emergency in the amended International Health Regulations (2005), *Global Health Research and Policy* 2025, doi:[10.1186/s41256-025-00438-6](https://doi.org/10.1186/s41256-025-00438-6); PMID 40497081 — Exercise Crystal: simulations that drive National IHR Focal Point capacity-strengthening, *Western Pacific Surveillance and Response Journal* 2025, doi:[10.5365/wpsar.2025.16.2.1240](https://doi.org/10.5365/wpsar.2025.16.2.1240); PMID 39463237 — Vaccination against dengue fever for travellers, *Swiss Medical Weekly* 2024, doi:[10.57187/s.3858](https://doi.org/10.57187/s.3858) (abstract-only); PMID

42318655 — Epidemiology and characteristics of cases with monkeypox virus clade I in the WHO European Region, 2024 to 2025, *Eurosurveillance* 2026, doi:[10.2807/1560-7917.ES.2026.31.24.2500955](https://doi.org/10.2807/1560-7917.ES.2026.31.24.2500955); PMID 42383297 — Editorial: The WHO Identifies Ebola Disease Due to Bundibugyo Virus as a Public Health Emergency of International Concern (PHEIC) as Vaccine Development Accelerates, *Medical Science Monitor* 2026, doi:[10.12659/MSM.954627](https://doi.org/10.12659/MSM.954627); PMID 42173889 — The current Ebola outbreak is a public health emergency of international concern, *Nature Communications* 2026, doi:[10.1038/s41467-026-73746-1](https://doi.org/10.1038/s41467-026-73746-1). Plus WHO International Health Regulations (2005) as amended by WHA77 on 1 June 2024 (in force September 2025) and the WHO Pandemic Agreement adopted by WHA78 on 20 May 2025.



Sustainable Development Goals (SDGs) — health goals

Asked in: Paper IV 17-Apr-2026 — RGUHS MD Community Medicine

Definition — Sustainable development is "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (Park 28e, Ch 14, p 852). On 25 September 2015 the United Nations General Assembly adopted *Transforming our world: the 2030 Agenda for Sustainable Development*, comprising **17 Sustainable Development Goals (SDGs)** and **169 targets**, which came into force on 1 January 2016 and run to 2030 (Park 28e, Ch 14, p 852; Oxford Textbook of Public Health 7e, Ch 10, pp 419-420). The Agenda integrates the three dimensions of sustainable development — **economic, social and environmental** — around the **five Ps: people, planet, prosperity, peace and partnership**, with "leaving no one behind" as its overarching theme.

The 17 Sustainable Development Goals

Goal	Short title	Goal statement
1	No poverty	End poverty in all its forms everywhere
2	Zero hunger	End hunger, achieve food security and improved nutrition, and promote sustainable agriculture
3	Good health and well-being	Ensure healthy lives and promote well-being for all at all ages
4	Quality education	Ensure inclusive and equitable quality education and promote life-long learning opportunities for all
5	Gender equality	Achieve gender equality and empower all women and girls
6	Clean water and sanitation	Ensure availability and sustainable management of water and sanitation for all
7	Affordable and clean energy	Ensure access to affordable, reliable, sustainable and modern energy for all
8	Decent work and economic growth	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

Goal	Short title	Goal statement
9	Industry, innovation and infrastructure	Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
10	Reduced inequalities	Reduce inequality within and among countries
11	Sustainable cities and communities	Make cities and human settlements inclusive, safe, resilient and sustainable
12	Responsible consumption and production	Ensure sustainable consumption and production patterns
13	Climate action	Take urgent action to combat climate change and its impacts
14	Life below water	Conserve and sustainably use the oceans, seas and marine resources for sustainable development
15	Life on land	Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt and reverse land degradation and halt biodiversity loss
16	Peace, justice and strong institutions	Promote peaceful and inclusive societies, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
17	Partnerships for the goals	Strengthen the means of implementation and revitalize the global partnership for sustainable development

(Park Plus, Ch 8, p 96; Sunder Lal, Ch 17, Table 17.10)

Which goals are "direct" health goals

- **SDG 3 is the one comprehensive, directly health goal** — "Ensure healthy lives and promote well-being for all at all ages", carrying **13 targets** that cover every major health priority (Park 28e, Ch 14, p 852).
- This is a deliberate structural change from the Millennium Development Goals (MDGs), which had **3 of 8 goals directly health related** (child mortality, maternal mortality, communicable disease); the SDGs replace these with **one comprehensive goal** that also embraces well-being and non-communicable disease (Park Plus, Ch 8, p 96).
- **Two further goals set targets that are themselves health outcomes** and are therefore counted among the health-related SDGs — **SDG 2** (ending stunting, wasting and all forms of malnutrition) and **SDG 6** (safe drinking water, sanitation and hygiene, end of open defecation) (Sunder Lal, Ch 17, Table 17.11).
- All the **remaining goals act indirectly**, through the social determinants of health — income, education, gender, work, housing, energy, climate and governance (Park 28e, Ch 1, p 23).

SDG 3 and its 13 targets

Target	Statement (by 2030 unless stated)	Core indicator
3.1	Reduce the global maternal mortality ratio to less than 70 per 100,000 live births	Maternal mortality ratio; proportion of births attended by skilled health personnel
3.2	End preventable deaths of newborns and under-fives — neonatal mortality ≤ 12 per 1000 and under-five mortality ≤ 25 per 1000 live births	Neonatal mortality rate; under-five mortality rate
3.3	End the epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases (NTDs); combat hepatitis, water-borne and other communicable diseases	New human immunodeficiency virus (HIV) infections; tuberculosis incidence; malaria incidence; new hepatitis B infections; persons needing NTD interventions
3.4	Reduce by one third premature mortality from non-communicable diseases (NCDs) and promote mental health and well-being	Mortality from cardiovascular disease, cancer, diabetes or chronic respiratory disease; suicide mortality rate
3.5	Strengthen prevention and treatment of substance abuse, including narcotic drugs and harmful use of alcohol	Coverage of treatment interventions; alcohol per capita consumption (≥ 15 years, litres of pure alcohol)
3.6	By 2020 , halve global deaths and injuries from road traffic accidents	Road traffic fatal injury deaths per 100,000 population
3.7	Ensure universal access to sexual and reproductive health-care services, including family planning, information and education, integrated into national programmes	Need for family planning satisfied with modern methods; adolescent birth rate
3.8	Achieve universal health coverage (UHC) — financial risk protection, quality essential services, and safe, effective, affordable essential medicines and vaccines for all	Coverage of tracer interventions; population protected against catastrophic/impoverishing out-of-pocket health expenditure
3.9	Substantially reduce deaths and illness from hazardous chemicals and air, water and soil pollution and contamination	Mortality attributed to household and ambient air pollution; mortality attributed to unsafe water, sanitation and hygiene
3.a	Strengthen implementation of the WHO Framework Convention on Tobacco Control (FCTC) in all countries	Age-standardised prevalence of current tobacco use, ≥ 15 years
3.b	Support research and development of vaccines and medicines for diseases that primarily affect developing countries; provide access to affordable essential medicines and vaccines under the Doha Declaration on the TRIPS Agreement and Public Health	Population with access to affordable medicines and vaccines; net official development assistance to medical research and basic health
3.c	Substantially increase health financing and the recruitment, development, training and retention of the health workforce in developing countries	Health worker density and distribution per 10,000 population

Target	Statement (by 2030 unless stated)	Core indicator
3.d	Strengthen capacity of all countries for early warning, risk reduction and management of national and global health risks	Percentage of the 13 International Health Regulations (IHR) core capacities attained

(Park 28e, Ch 14, Table 1, p 853; Park Plus, Ch 8, p 97)

Discussion of the health goal

- **Four targets carry forward and expand the unfinished MDG agenda** — 3.1 maternal mortality, 3.2 newborn and child mortality, 3.3 infectious disease, and 3.7 sexual and reproductive health care. The expansion is substantive: 3.2 separates out **neonatal** mortality for the first time, and 3.3 adds **hepatitis, water-borne disease and NTDs** to the MDG trio of HIV, tuberculosis and malaria (Park 28e, Ch 14, p 852).
- **Four targets are entirely new** to the global development agenda and reflect the epidemiological transition — 3.4 NCDs and mental health, 3.5 substance abuse, 3.6 road traffic injuries, and 3.9 environmental and chemical exposures.
- **Target 3.8 (universal health coverage) is the pivot of the whole goal.** UHC is described as the key to achieving all other targets and to the development of a strong, resilient health system; it cuts across every target and is measured both by service coverage and by financial protection (Park 28e, Ch 14, pp 852, 858).
- **Four "means of implementation" targets** (3.a, 3.b, 3.c, 3.d) are levers rather than outcomes — tobacco control, research and access to medicines, health financing and workforce, and health-security capacity.
- **Universality.** Unlike the MDGs, which were framed for developing countries and financed by rich countries, SDG monitoring covers **all** countries; some targets are more relevant to developing countries but all should in principle be reported (Park 28e, Ch 14, p 852; Park Plus, Ch 8, p 96).
- **Acknowledged gaps in SDG 3.** The 2030 Agenda repeatedly invokes human rights but nowhere states that **health is a human right**; **ageing** is not addressed except indirectly through NCDs and mental health; and **sexual and reproductive rights, violence and discrimination against women and girls** are dealt with under Goal 5 rather than Goal 3 (Park 28e, Ch 14, pp 852, 858).

How the health goal is operationalised

- **A single global indicator framework.** SDG 3 is monitored globally with indicators developed by the Inter-Agency and Expert Group on SDG Indicators, while countries additionally set their own national indicators; the World Health Assembly has adopted strategies and plans of action that give the goal its operational content (Park 28e, Ch 14, p 858).
- **The Global Strategy for Women's, Children's and Adolescents' Health 2016-2030** is the worked example of implementation: it is built around the themes "**Survive — Thrive — Transform**" and deliberately draws on health targets in several goals at once, not SDG 3 alone (Park 28e, Ch 14, p 855).

- **Adolescents are a population newly emphasised by the SDGs.** Although generally healthy, they are exposed to risks whose behavioural consequences run through the rest of the life course; globally, **road injuries, HIV/AIDS and self-harm** are the leading causes of adolescent death (Park 28e, Ch 14, p 855).
- **Reproductive, maternal, newborn and child health is one of the four categories of universal health coverage tracer indicators,** measured by family planning coverage, four or more antenatal visits with skilled attendance at birth, full child immunisation coverage, and care-seeking for suspected childhood pneumonia (Park 28e, Ch 14, p 855).

India's position against the quantified SDG 3 targets

Target	Level to be reached by 2030	India's latest value (year)
3.1 Maternal mortality ratio	< 70 per 100,000 (lakh) live births	97 per lakh live births (Sample Registration System, SRS 2018-20)
3.2 Neonatal mortality rate	≤ 12 per 1000 live births	20 per 1000 live births (SRS 2020)
3.2 Under-five mortality rate	≤ 25 per 1000 live births	32 per 1000 live births (SRS 2020)
3.4 Premature NCD mortality	Reduce by one third	21.9% vs global 17.8% (2019)
3.4 Suicide mortality rate	Promote mental health	12.7 vs global 9.2 per 100,000 (2019)
3.5 Alcohol consumption (≥15 y)	Reduce harmful use	4.9 vs global 5.8 litres pure alcohol per capita (2019)
3.6 Road traffic fatal injury deaths	Halve	15.6 vs global 16.7 per 100,000 (2019)
3.9 Air-pollution mortality	Substantially reduce	184.3 vs global 114.1 per lakh population (2016)
3.a Current tobacco use (≥15 y)	Strengthen FCTC implementation	27.2% vs global 22.3% (2020)
3.d IHR core capacities attained	Strengthen health security	95% vs global 71% (2010-17)

(Park 28e, Ch 14, Table 1, p 853; Ch 9, pp 663-676, Tables 20 and 27)

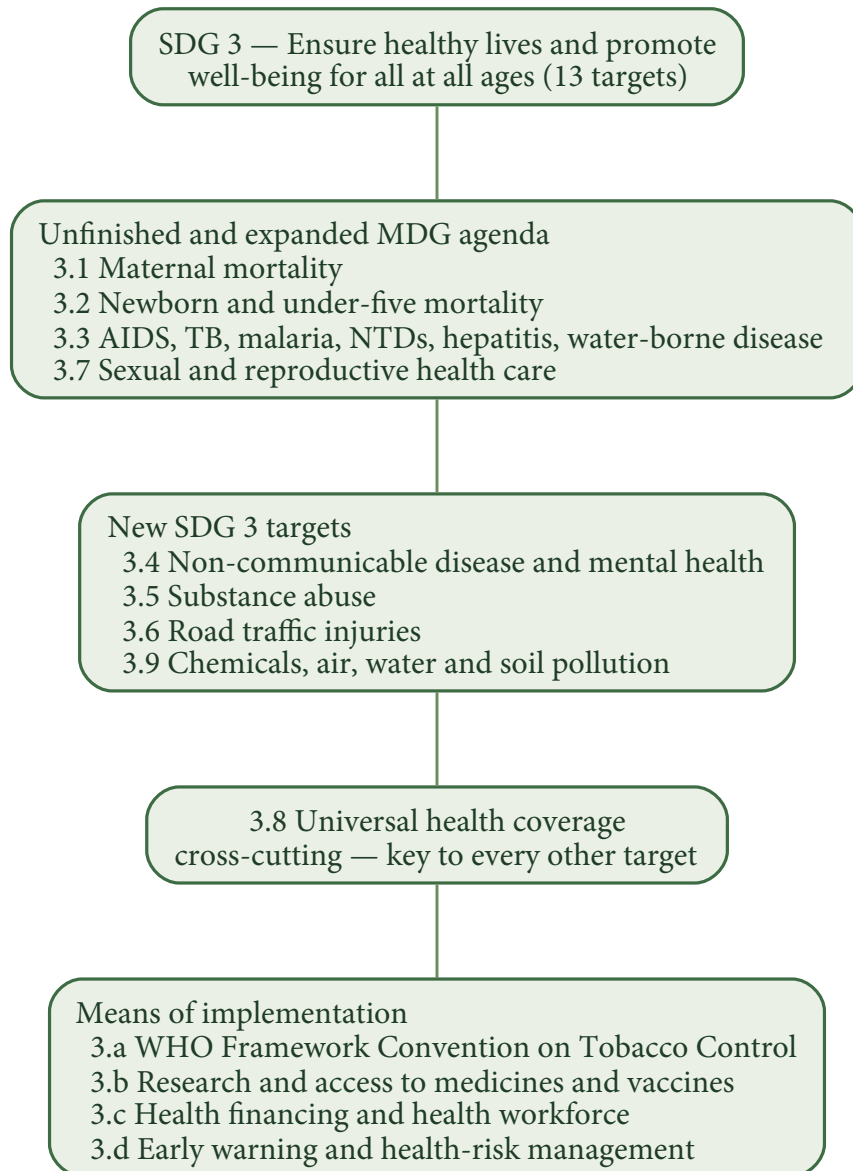
India's stated route to these targets is high political commitment behind *Sabka Saath Sabka Vikas*, investment in public health to finish the maternal and child health agenda, confronting NTDs, eliminating malaria, AIDS and hepatitis while intensifying tuberculosis control, implementing universal health care through a strengthened rural and urban primary health care system, and building a robust system for monitoring, evaluation and accountability (Park 28e, Ch 14, p 859).

Health-related targets located in other goals

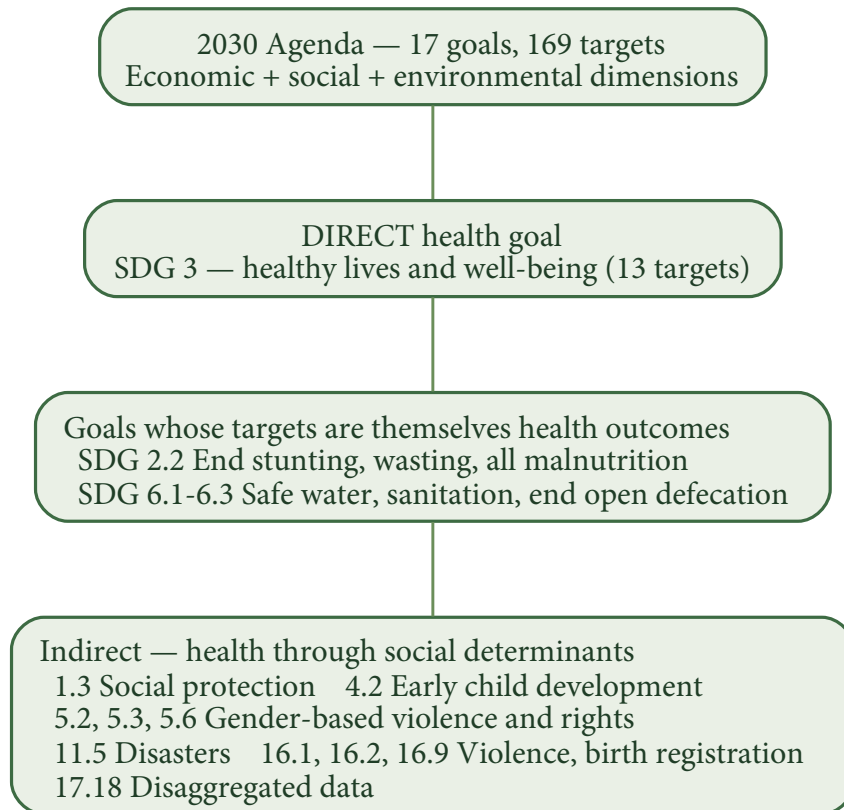
Goal	Target	Health relevance
1	1.3	Nationally appropriate social protection systems and floors covering the poor and vulnerable

Goal	Target	Health relevance
2	2.2	End all forms of malnutrition; internationally agreed stunting and wasting targets in under-fives by 2025; nutrition of adolescent girls, pregnant and lactating women and older persons
4	4.2, 4.a	Early childhood development, care and pre-primary education; child-, disability- and gender-sensitive, safe, non-violent learning environments
5	5.2, 5.3, 5.6	Eliminate violence against women and girls; eliminate child, early and forced marriage and female genital mutilation; universal access to sexual and reproductive health and reproductive rights
6	6.1, 6.2, 6.3	Universal safe and affordable drinking water; adequate sanitation and hygiene, ending open defecation; improved water quality and halved untreated wastewater
10	10.4	Fiscal, wage and social protection policies to achieve greater equality
11	11.5	Reduce deaths and people affected by disasters, including water-related disasters
16	16.1, 16.2, 16.6, 16.9	Reduce violence and related death rates; end abuse and trafficking of children; accountable institutions; legal identity for all including birth registration
17	17.18	Capacity building for high-quality, timely, reliable data disaggregated by income, sex, age, disability and geography

(Park 28e, Ch 14, Table 2, pp 858-859)



Architecture of the 13 targets of Sustainable Development Goal 3



Where health sits in the 2030 Agenda — one direct goal, two goals with health outcomes, and the rest acting through determinants

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — The SDG grid.** Draw a 3 x 6 grid of numbered boxes. Labels: 1 No poverty; 2 Zero hunger; 3 Good health and well-being; 4 Quality education; 5 Gender equality; 6 Clean water and sanitation; 7 Affordable and clean energy; 8 Decent work and economic growth; 9 Industry, innovation and infrastructure; 10 Reduced inequalities; 11 Sustainable cities and communities; 12 Responsible consumption and production; 13 Climate action; 14 Life below water; 15 Life on land; 16 Peace, justice and strong institutions; 17 Partnerships for the goals. Shade box 3 to mark the direct health goal.
- **Diagram 2 — The five Ps of the 2030 Agenda.** A central circle labelled 2030 Agenda — leave no one behind, with five spokes labelled People, Planet, Prosperity, Peace, Partnership, and three outer arcs labelled Economic, Social, Environmental.



The 17 Sustainable Development Goals of the 2030 Agenda (Oxford public health 7th ed, p.934)

High-yield exam pointers

- ▶ 17 goals, 169 targets, 232 indicators; adopted 25 September 2015, in force 1 January 2016, to be met by 2030. Five Ps = people, planet, prosperity, peace, partnership.
- ▶ SDG 3 = 13 targets = 4 (unfinished MDG agenda: 3.1, 3.2, 3.3, 3.7) + 4 (new: 3.4, 3.5, 3.6, 3.9) + 1 (UHC: 3.8) + 4 (means of implementation: 3.a-3.d).
- ▶ Numerical targets to reproduce exactly: **maternal mortality ratio < 70 per 100,000 live births**; **neonatal mortality ≤ 12 per 1000 live births**; **under-five mortality ≤ 25 per 1000 live births**; **premature NCD mortality reduced by one third**; **road traffic deaths halved by 2020**.
- ▶ India's gap (dated): **maternal mortality ratio 97 per lakh (SRS 2018-20)**, **neonatal mortality 20** and **under-five mortality 32 per 1000 live births (SRS 2020)** — all three still short of the 2030 targets.
- ▶ MDG vs SDG in one line: **8 goals / 18 targets / 48 indicators, 3 directly health related, for developing countries** versus **17 goals / 169 targets, 1 comprehensive health goal, universal**.
- ▶ Two goals beyond SDG 3 with true health outcomes: **SDG 2 (malnutrition)** and **SDG 6 (water and sanitation)**. **Target 3.8, universal health coverage, is the key to all the rest.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the global and Indian indicator values in Park's Table 1 carry different reference years (2016 to 2021) and are the latest printed in the 28th edition — always write the year alongside the figure, and expect the National Institution for Transforming India (NITI Aayog) SDG India Index to report more recent state-level values than the textbook.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, ICMR, NFHS-5, National Health Accounts, UN IGME, WHO South-East Asia Region

Programme and policy updates (India)

- **PM-ABHIM is now the operational vehicle for SDG target 3.d (health security and early warning).** The Pradhan Mantri Ayushman Bharat Health Infrastructure Mission, announced 25 October 2021 and World Bank supported, is funding four zonal National Institutes of Virology — Bengaluru (South), Jammu (North), Dibrugarh (East) and Jabalpur (Central) — through the Department of Health Research and ICMR, decentralising the workload previously carried by ICMR-NIV Pune. A four-zone prioritisation exercise in March 2024 with 186 stakeholders from state health departments, the Virus Research and Diagnostic Laboratory network and the Integrated Disease Surveillance Programme ranked dengue, influenza, measles, Japanese encephalitis and hepatitis A as the shared top-five priority viral diseases. [PMID 42522023]
- **The distance between PM-JAY eligibility and actual use is where target 3.8 is lost in practice.** A 2023 community survey in an urban coastal slum in Puducherry (273 adults) found 59% aware of the scheme and 65% of households eligible; among eligible households only 49% had registered and just 14% had ever drawn a benefit, with a difficult registration process and limited empanelment and package cover reported as the barriers. Financial risk protection therefore has to be examined as a cascade — awareness, enrolment, utilisation — not as a card-issue figure. [PMID 41805171]

Current national data

- **India's financing mix still fails the financial-protection half of universal health coverage.** Linking NFHS-5 (2019–21) with the National Health Accounts 2019–20, out-of-pocket spending averaged 51.2% of total health expenditure against 39.44% government and only 9.35% prepaid; health-insurance coverage ranged from 13.84% (Jammu and Kashmir) to 87.84% (Rajasthan) and per-capita health expenditure from Rs 1,588 (Bihar) to Rs 10,607 (Kerala). States whose dominant financing was government or prepaid showed better child mortality, immunisation and water-and-sanitation outcomes. Fragmentation of the insurance pool raised expenditure and administrative inefficiency even though it was itself associated with better vaccination (beta 1.23), modern-contraceptive (beta 0.51), under-five mortality (beta -0.57) and neonatal mortality (beta -0.56) outcomes — spreading people across many schemes buys access but not efficiency, which is why the authors argue for one nationally pooled scheme rather than more schemes. Always cite the survey round alongside the figure. [PMID 42218447]

Elimination, surveillance and disease status

- **Kala-azar (visceral leishmaniasis) has entered a post-elimination phase, which is not the same as a finished one.** Longitudinal epidemiological, serological and entomological surveys in endemic villages of Bihar and Jharkhand (December 2021 to May 2023) found rK39 seroprevalence of 20–30% and *Leishmania*-infected sand-fly pools of 1.4% and 1.2% despite indoor residual spraying. Low-level transmission persists, so the programme task has shifted from case reduction to sustained post-validation surveillance. [PMID 41324335]
- **Lymphatic filariasis remains endemic in 345 districts** against the national elimination goal of 2030; mass drug administration, morbidity management and disability prevention, and post-MDA surveillance are the operationally strained components, and public-private partnership is being proposed to close the gap as it did for polio and tuberculosis. [PMID 41951149]

- ▶ **India, with the other countries of the WHO South-East Asia Region, has set December 2026 as the deadline for rubella elimination** — a date most textbook editions predate. ICMR laboratory serology from central India (1 May 2023 to 30 April 2024) reported rubella IgM positivity of 7.2%, indicating ongoing transmission, against IgG of 81.0% overall and 89.6% among pregnant women; the residual IgM positives were concentrated in young children, marking the immunity gap that could push the deadline again. [PMID 42159053]

Guideline and legislative updates

- ▶ **Antimicrobial resistance containment is being written into the universal health coverage agenda.** India now runs a National Action Plan on AMR alongside six state action plans covering stewardship programmes, laboratory network strengthening, awareness and intersectoral coordination; enforcement gaps, limited awareness and fragmented data are the acknowledged weaknesses, with One Health surveillance expansion and embedding AMR strategy within UHC named as the priorities. [PMID 41443506]
- ▶ **Food regulation is India's principal statutory lever on SDG targets 2.2 and 3.4.** Roughly 9–12% of Indian children and adolescents are now overweight or obese, coexisting with undernutrition; the operative instruments are the FSSAI 2020 restrictions on high fat, salt and sugar foods in and around schools, the Ayushman Bharat School Health Programme, PM POSHAN, and higher effective taxation of sugar-sweetened beverages, with a front-of-pack nutrition labelling system still under development. [PMID 41974918]

Global and WHO developments

- ▶ **SDG-era child survival progress has slowed sharply, and this is the headline global update against target 3.2.** In 2024 an estimated 4.9 million children died before age five (90% uncertainty interval 4.7–5.2 million), including 3.6 million infant deaths (75%) and 2.3 million neonatal deaths (47%). The annual rate of reduction in under-five mortality fell from 3.9% during 2000–15 to 1.5% during 2015–24 — 61% slower — and the neonatal rate slowed from 3.0% to 1.3%. Sixty countries are projected to miss the under-five target and 66 the neonatal target, with 27.3 million under-five deaths expected during 2025–30 on current trends. South Asia carries 25% of under-five and 33% of neonatal deaths. [PMID 42242779]
- ▶ **Service coverage is rising in South Asia while financial protection deteriorates.** Projections to 2030 put India's Composite Coverage Index at 72% in 2016 and 76% in 2020, and the South Asian regional index at 53.5% in 2000 rising to 81.5% by 2030 — past the 80% UHC service-coverage benchmark; but catastrophic health expenditure in India is projected to reach 26.3% by 2030 against a regional rise from 7.2% (2000) to 18.6% (2030), with rural rates about 6 percentage points above urban. Impoverishing health expenditure in India is projected to fall only from 3.5% to 2.6%. Target 3.8 must therefore be answered in both halves — coverage and financial protection. [PMID 41192927]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 14 — Sustainable Development Goals, pp 852-859; Ch 1, p 23; Ch 2, pp 40-41; Ch 9, pp 663-676); Park Plus (Ch 8 — From Millennium Development Goals to Sustainable Development Goals, pp 95-97); Sunder Lal — Textbook of Community Medicine (Ch 17, Tables 17.10 and 17.11); Oxford Textbook of Public Health 7e (Ch 10, pp 419-420; Ch 24, p 937; Ch 100, p 3799). United Nations, *Transforming our world: the 2030 Agenda for Sustainable Development*, 2015. Indian mortality figures from the Sample Registration System (SRS) as reproduced in Park 28e, 2020 and 2018-20.

Recent-advances sources: 42522023 — Strengthening India's pandemic preparedness with the four zonal institutes of virology: stakeholder consultations for outbreak response and collaborative research priorities, *Infectious Diseases of Poverty* 2026, doi:[10.1186/s40249-026-01484-z](https://doi.org/10.1186/s40249-026-01484-z); 41805171 — From card to care: gaps in implementation of publicly funded health insurance among a vulnerable population in South India, *Journal of Public Health (Oxford)* 2026, doi:[10.1093/pubmed/fdag021](https://doi.org/10.1093/pubmed/fdag021) (abstract-only); 42218447 — The role of health financing system and pluralism in achieving sustainable development goal health outcomes in India: a multilevel analysis, *International Journal for Equity in Health* 2026, doi:[10.1186/s12939-026-02888-x](https://doi.org/10.1186/s12939-026-02888-x); 41324335 — High seropositivity to rK39 and capture of *Leishmania*-infected sand flies in Bihar and Jharkhand emphasize the need for post-elimination surveillance of visceral leishmaniasis in India, *Journal of Infectious Diseases* 2026, doi:[10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543) (abstract-only); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42159053 — Monitoring rubella immunity and transmission in central India amidst elimination efforts, *Microbiology and Immunology* 2026, doi:[10.1111/1348-0421.70065](https://doi.org/10.1111/1348-0421.70065) (abstract-only); 41443506 — Antimicrobial resistance in India: integrating the response into health systems for universal health coverage, *Journal of Global Antimicrobial Resistance* 2025, doi:[10.1016/j.jgar.2025.12.001](https://doi.org/10.1016/j.jgar.2025.12.001) (abstract-only); 41974918 — Public health approaches to combating childhood obesity in India: a mini-review, *European Journal of Clinical Nutrition* 2026, doi:[10.1038/s41430-026-01742-1](https://doi.org/10.1038/s41430-026-01742-1) (abstract-only); 42242779 — Global, regional, and national levels and trends in under 5, infant, and neonatal mortality during 1990-2024 with scenario based projections to 2030: modelling study, *BMJ* 2026, doi:[10.1136/bmj-2025-088684](https://doi.org/10.1136/bmj-2025-088684); 41192927 — Progress towards universal health coverage in South Asia, 2000-2030: an examination of the twin elements of primary healthcare provision and financial protection, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-020052](https://doi.org/10.1136/bmjgh-2025-020052).

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Bilateral health agencies

Asked in: Paper IV 13-May-2022 — RGUHS MD Community Medicine

Definition — A **bilateral health agency** is the official development-assistance arm of a single donor government that funds health and development activity in a single recipient country, i.e. a project involving **one donor and one recipient government** (Textbook of International Health, Ch 3, pp 126-127). Such agencies are usually subsidiary to, or dependent on, the donor's foreign affairs and trade ministry, and they work through counterpart government channels in the recipient country. Park deals with them under the heading "**Health work of bilateral agencies**", separately from the multilateral United Nations agencies (Park 28e, Ch 25, p 1059).

Bilateral versus multilateral agencies

Feature	Bilateral agency	Multilateral agency
Source of funds	One donor government's own budget	Assessed and voluntary contributions pooled from many member governments
Channel	Donor government → recipient government (government-to-government)	Member states → the agency → member countries
Parent body	Donor's ministry of foreign affairs / trade	The United Nations system or an international financial institution
Examples	United States Agency for International Development (USAID), Swedish International Development Agency (SIDA), Danish International Development Agency (DANIDA), Norwegian Agency for Development Cooperation (NORAD), UK Department for International Development (DFID), Japanese International Cooperation Agency (JICA), Canadian International Development Agency (CIDA), European Community Humanitarian Aid (ECHO)	World Health Organization (WHO) and other United Nations agencies; World Bank, International Monetary Fund, World Trade Organization
Priority setting	Set largely by the donor; may be restricted to selected sectors or activities	Set by the governing body of member states
Accountability	To the donor country's own parliament and taxpayers	To the collective membership

(Typology and examples: Textbook of International Health, Ch 3, p 105; Park 28e, Ch 25, p 1059.)

How bilateral agencies work

- **Official development assistance (ODA)** is the vehicle. Most of it comes from members of the **Organisation for Economic Co-operation and Development (OECD)**; annual flows have run in the range of **US\$100 billion** in recent years (Textbook of International Health, Ch 3, pp 126-127).
- At the OECD's **Development Assistance Committee (DAC)** — **22 donor countries plus the European Commission** — donors review and compare their national contributions to both

bilateral ("foreign aid") and multilateral aid programmes (Textbook of International Health, Ch 3, pp 126-127).

- ▶ Most bilateral agencies were **established after the Second World War**, and most **sponsor, oversee and assess rather than implement**, operating through contracts with intermediary universities, research organisations, for-profit companies and non-governmental organisations (Textbook of International Health, Ch 3, p 127).
- ▶ Countries are **targeted for strategic reasons** — conflict situations, a particularly heavy burden of disease, and historical and political ties. ODA forms a substantial share of smaller recipient governments' health expenditure (Textbook of International Health, Ch 3, p 127).
- ▶ Procurement of goods, equipment and technical personnel is typically **"tied aid"** (also called phantom aid) — spending restricted to the donor country (Textbook of International Health, Ch 3, p 127).

Example — USAID, and its health work in India

Item	Detail (Park 28e, Ch 25, pp 1059-1060; Textbook of International Health, Ch 3, pp 127-128)
Full name	United States Agency for International Development
Created	1961 , taking charge of activities previously administered by the Technical Cooperation Mission (TCM)
Parent	Government of the United States — the largest bilateral donor in the world
Three channels of US aid to India	(1) USAID; (2) the Public Law 480 (Food for Peace) Programme ; (3) the US Export-Import Bank
Instruments	Grants and loans extended by the USAID mission in the recipient country
Stated priorities (agency budget table, 2008: US\$36 billion)	Economic growth, agriculture and trade; global health ; democracy, conflict prevention and humanitarian assistance
Global health portfolio	Child nutrition and child survival, health care financing, environmental health (sanitation), infectious diseases, HIV/AIDS control
Projects assisted in India	Malaria eradication; medical education; nursing education; health education; water supply and sanitation; control of communicable diseases; nutrition; family planning — with the recent trend of US assistance shifting towards agriculture and family planning and away from the general public health field
Flagship bilateral health programme	PEPFAR — the President's Emergency Plan for AIDS Relief, launched 2003 , based in the US Department of State, US\$15 billion for the initial 5-year initiative and a further US\$37 billion committed for 2009-2013

Bilateral assistance to Indian national health programmes

Bilateral agency (country)	Indian programme assisted	What is given
SIDA (Sweden)	National Tuberculosis Control Programme, since 1979	Procurement of supplies — X-ray units, microscopes, anti-tuberculosis drugs (Park 28e, Ch 25, p 1060)
DANIDA (Denmark)	National Programme for Control of Blindness, since 1978	Development of eye-care services (Park 28e, Ch 25, p 1060)
USAID and DANIDA, alongside DFID	Revised National Tuberculosis Control Programme	External financial and technical assistance, together with the World Bank, WHO, the Global Drug Facility and the Global Fund to Fight AIDS, Tuberculosis and Malaria (Park 28e, Ch 7, p 516)
USAID through its Maternal Child Health Integrated Programme (MCHIP)	Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A) strategy	Technical support for introduction and implementation, with UNICEF and UNFPA (Park 28e, Ch 7, p 553)
NORAD (Norway)	Named by Park among the foreign agencies giving technical and material assistance to India's national health programmes (Park 28e, Ch 7, p 493)	Technical and material assistance
The Colombo Plan (a regional cooperative arrangement Park places in this section)	Health promotion, mainly through fellowships	The All India Institute of Medical Sciences, New Delhi was established with financial assistance from New Zealand ; Canada supplied Cobalt Therapy Units to Indian medical institutions (Park 28e, Ch 25, pp 1059-1060)

Merits and criticisms

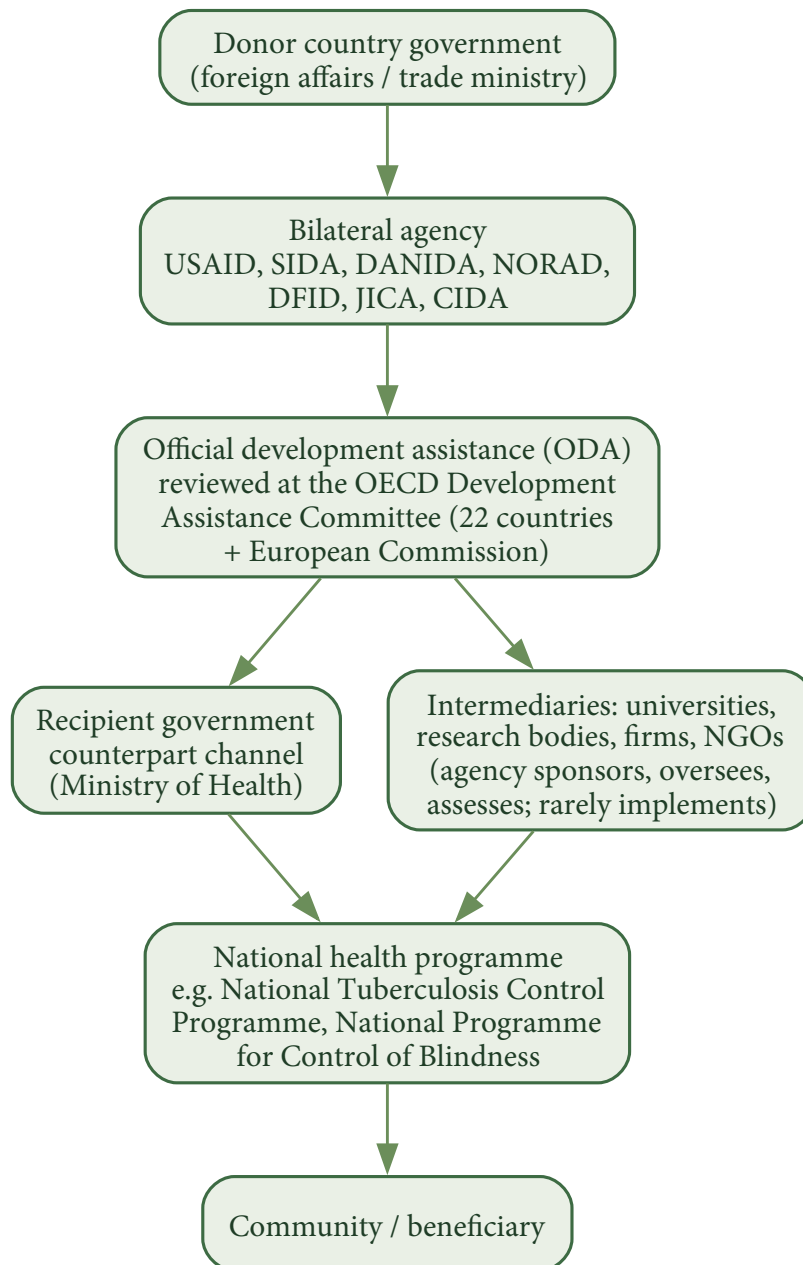
Merits	Criticisms
Collectively the largest single source of development funding , and often the channel through which national contributions to multilateral agencies are routed (Textbook of International Health, Ch 3, p 127)	Bilateral ODA may be limited to sectors or activities that are not the recipient country's own first priority (Textbook of International Health, Ch 3, p 127)
Direct government-to-government relationship allows targeted, sustained support to a named national programme	Tied aid : about 90% of aid flows back to the donor for the United States and France; the UNDP records that such arrangements cost developing countries up to 20% more than open-market purchase, the "tied aid tax" costing Africa alone US\$1.6 billion a year
Brings technical expertise, commodities and equipment that a recipient budget cannot absorb	Poor harmonisation : Vietnam received about 400 separate donor missions in 2003 , only 2% of them joint ; donor use of national monitoring and evaluation systems was 13%, procurement 18%, auditing 9%

Merits	Criticisms
Can respond quickly to an emergency or an epidemic (e.g. PEPFAR's scale-up of antiretroviral therapy)	Assistance is concentrated on HIV, malaria and tuberculosis , overlooking neglected tropical diseases, chronic disease, maternal and child health, nutrition and basic sanitary infrastructure; PEPFAR has been faulted for creating almost no public health infrastructure and for hiring workers away from other health activities

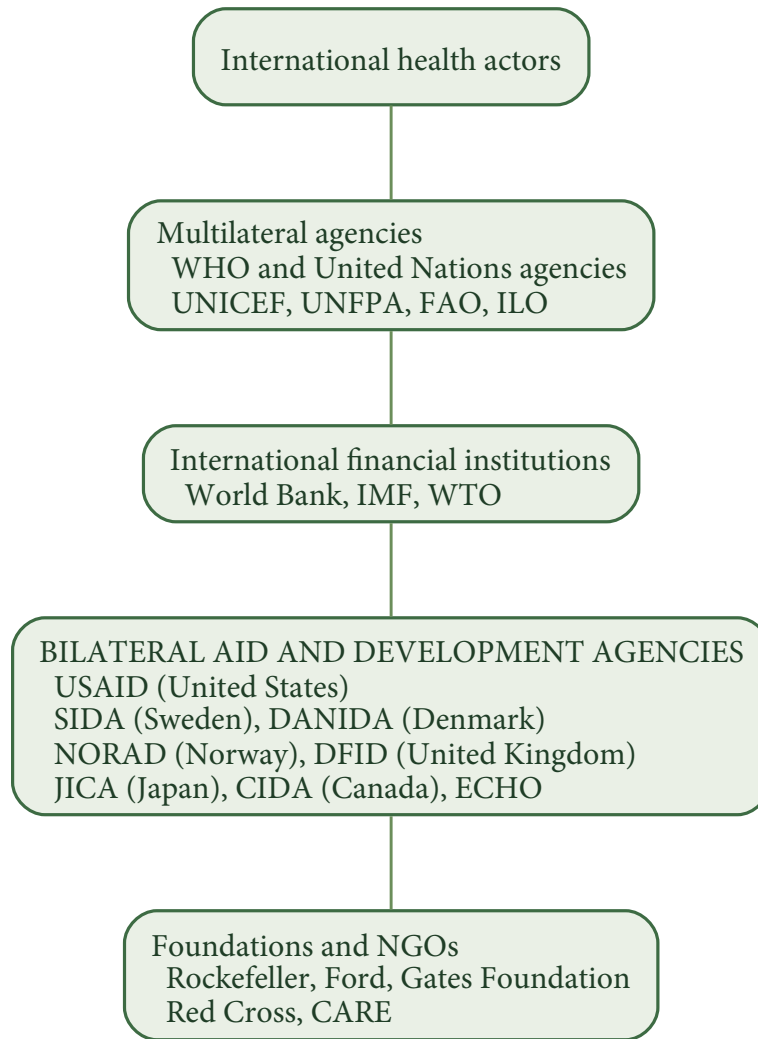
Corrective measures agreed by donors — the Rome Declaration on Harmonization (2003) and the Paris Declaration on aid effectiveness (2005), the latter setting targets that 75% of aid be disbursed according to agreed schedules and at least 25% be committed to long-term programmes rather than projects (Textbook of International Health, Ch 11, pp 601-602).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Bilateral versus multilateral aid — a two-arm sketch.** Labels: *Donor government A* → *bilateral agency of A* → *recipient government X* (one arrow, one donor, one recipient); alongside it *Donor governments A, B, C, D* → *pooled multilateral agency (WHO / UNICEF / World Bank)* → *recipient governments X, Y, Z* (many arrows in, many arrows out). Mark the bilateral arm "government-to-government, donor sets priorities, aid often tied" and the multilateral arm "pooled funds, priorities set by member states".



Route taken by bilateral health aid, from donor treasury to the recipient country's national health programme



Where bilateral agencies sit among international health actors

High-yield exam pointers

- ▶ **One donor, one recipient = bilateral; many donors pooled = multilateral.** That single line earns the definition mark.
- ▶ Mnemonic for the bilateral agencies India deals with — "USA, Sweden, Denmark, Norway" = USAID, SIDA, DANIDA, NORAD (Park 28e, Ch 7, p 493).
- ▶ The two dated Indian examples Park gives: SIDA → **National Tuberculosis Control Programme since 1979** (X-ray units, microscopes, anti-tuberculosis drugs); DANIDA → **National Programme for Control of Blindness since 1978**.
- ▶ **USAID created 1961**; US aid reaches India through **three channels** — USAID, PL-480 (Food for Peace) and the US Export-Import Bank.
- ▶ **PEPFAR 2003**, US\$15 billion for the first five years — quote it as the highest-profile bilateral *health* programme.
- ▶ **Tied aid** and the **Paris Declaration 2005** (75% of aid on agreed schedules; ≥25% to long-term programmes) are the criticism-and-remedy pair.
- ▶ **AIIMS New Delhi** was built with **New Zealand** money under the **Colombo Plan** — a favourite one-line factual question.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park 28e and the Textbook of International Health still use the older agency names and pre-2010 budget years, so always attach the year when quoting a budget figure; the United Kingdom's Department for International Development (DFID) was merged into the Foreign, Commonwealth and Development Office (FCDO) in 2020, and USAID was restructured in 2025 with its surviving programmes absorbed into the United States Department of State — neither reorganisation appears in the indexed corpus, so state the textbook position first and add the change only as a remark.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Lancet, Lancet Global Health, BMJ Global Health, PEPFAR programme data, Global Fund programme reports, Gavi

Programme and policy updates (India)

- ▶ **The dateable modern *bilateral* example for India is USAID and TB Mukht Bharat — and it has been withdrawn.** After USAID was dissolved in 2025, more than **US\$140 million** committed to the **TB Mukht Bharat (TB-Free India)** initiative and a tuberculosis "buddy" scheme reaching about **5.1 million patients across four high-burden states** were both halted. One donor government, one recipient government, one named national programme — quote this when the examiner asks for a modern bilateral example alongside Park's SIDA-1979 and DANIDA-1978 entries. Treatment itself continued because India pays for most of its own National TB Elimination Programme — the government share of total health expenditure rose from **29% to 48% between 2014-15 and 2021-22**, out-of-pocket spending fell to about **39%**, and public health spending is roughly **1.9% of GDP** against the National Health Policy 2017 target of 2.5% — but community case-finding, contact tracing and adherence support were the casualties, leaving India shielded on treatment and exposed on elimination. [PMID 42529073]
- ▶ **The Global Fund's TB-prevention project in India is the multilateral channel, not a bilateral one — label it correctly.** Under the Global Fund's sixth grant cycle (2021-2024) India ran a **large-scale contact-investigation and TB preventive treatment project across 208 districts in 23 states**, covering about 400 million people. From 851,974 people with pulmonary TB, **3,097,831 household contacts** were identified, **91% were symptom-screened**, **54% of those screened started preventive treatment and 84% completed it** — completion reached 92% with the 3-month weekly isoniazid-rifapentine regimen (3HP) against 83% on 6H, the regimen behind most initiations. Delivery ran through **Health and Wellness Centres** with ASHAs, Mitans and Sahiyas, and the project directly informed the national scale-up of Cy-Tb testing, 3HP expansion and community-health-worker incentives. [PMID 42583442] The Global Fund and Gavi are **multilateral "vertical funds"**, not bilateral agencies: the Global Fund drew about **45% of its 2019-2022 funding from the United States** and Gavi more than a third, which is why donor retrenchment struck them as well. [PMID 41248938]
- ▶ **India is moving from recipient to supplier and emerging donor.** It supplies close to a fifth of the world's generic-medicine volume and around **60% of global vaccine doses**; pharmaceutical exports crossed **US\$30 billion in FY2025**, and Indian-made medicines are supplied to over 200

countries. Indian generic antiretrovirals cut the annual cost of HIV treatment from about US\$10,000 per patient in 2001 to under US\$300. The strategy now argued for is to build on **Vaccine Maitri** and anchor a Global South pooled-procurement mechanism on the **PAHO Revolving Fund** model, which secures savings of roughly 50%. [PMID 42529073]

Current national data

- **Development assistance for health fell about 21% in a single year in 2025**, most of it a **67% fall in United States contributions worth more than US\$9 billion**; the United Kingdom, France and Germany cut official development assistance for the first time in decades, and by mid-2025 the Global Fund had lost **US\$1.4 billion, roughly 11%**, of what it had allocated for 2024-2026. [PMID 42529073]
- **The United States supplied about 45 of every 100 dollars spent on global health in 2019-2022** — US\$13.4 billion a year against US\$29.3 billion from all 32 OECD Development Assistance Committee members, so the textbook's "22 donor countries plus the European Commission" is superseded and should be quoted with its year. US bilateral health aid averaged **US\$9.4 billion a year**, two-thirds disbursed through USAID and 20% through Health and Human Services, with about 57% going to **HIV and other sexually transmitted disease programmes**; 14 countries received US health assistance worth 10% or more of their total health expenditure. [PMID 41248938]

Elimination, surveillance and disease status

- **The modelled cost of withdrawal is now quantified.** Complete cessation of US funding without replacement is projected to cause **4.1 million additional AIDS deaths (range 1.6-6.6 million)** across 55 countries and **606,900 additional tuberculosis deaths (95% UI 466,000-768,800)** across 79 countries between 2025 and 2030 [PMID 40975076]; a tuberculosis-specific model puts USAID termination alone at **1.4 million excess TB episodes (95% UI 1.1-1.7 million)** and **537,700 excess deaths (451,900-662,300)** by 2035 [PMID 40659023]. Looking backwards, USAID funding across 133 countries was associated with **15% lower age-standardised all-cause mortality (RR 0.85, 95% CI 0.78-0.93)** and **32% lower under-five mortality (RR 0.68, 0.57-0.80)**, about 91.8 million deaths averted over 2001-2021, with more than 14 million additional deaths forecast by 2030 if the cuts stand [PMID 40609560].
- **PEPFAR's own facility data show which part of a bilateral programme breaks first.** Across 31,628 facilities and community sites, between the last quarters of 2024 and 2025 the number of people on HIV treatment fell only 1%, but HIV testing fell 18%, pre-exposure prophylaxis initiations 33%, and the directly supported health workforce by **62,541 posts (24%)**; infant HIV testing fell 60% at intermittently reporting facilities. Treatment coverage alone is therefore a poor measure of programme health. [PMID 42507823]

Guideline and legislative updates

- **The "America First Global Health Strategy" rewrites the bilateral compact.** US engagement is shifting from open-ended programme support to **time-bound bilateral health agreements** negotiated country by country — the government-to-government bargain of the textbook definition being rewritten in real time. Where the memorandum of understanding broke down, as in Zimbabwe, modelling projects about **75,440 additional HIV infections within one year** of full PEPFAR withdrawal, roughly 61% of them from treatment disruption. [PMID 42456026]

Global and WHO developments

- ▶ **The multilateral side is being reset at the same time.** The United States notified withdrawal from WHO and cut roughly 83% of the development assistance it channelled through USAID [PMID 41761883]; it had been WHO's largest single donor at about 13% of revenue (around US\$500 million a year) and had supplied nearly 40% of UNAIDS income, and UNAIDS has since cut staff by 54% [PMID 41248938].
- ▶ **Gavi's co-financing and transition model is the hand-over being tested at scale.** Lower-income countries committed a record US\$1.1 billion of their own resources to immunisation in 2026 (announced 15 April 2026), in line with Immunization Agenda 2030; but a projected US\$38.4 billion funding gap for 16 vaccines across 94 low- and middle-income countries over 2011-2030, and evidence that bilateral aid tends to fall at the same moment as Gavi graduation, make the transition fragile. [PMID 42519657]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 25 (International Health), Ch 7 (Health Programmes in India). Textbook of International Health — Ch 3 (International Health Agencies, Activities, and Other Actors), Ch 11 (Health Economics and the Economics of Health). Oxford Textbook of Public Health, 7e — Ch 3 and Ch 51 (actors in supranational health governance). Rome Declaration on Harmonization 2003; Paris Declaration on aid effectiveness 2005. Coverage note: the corpus carries no post-2010 bilateral-agency budget series and no entry for the DFID-to-FCDO or the 2025 USAID reorganisation, so those two points appear only in the reference-only Note.

Recent-advances sources: 42529073 — Insulated at home, indispensable abroad: positioning India's global health strategy after the 2025 contraction in development assistance, Front Health Serv 2026, doi:[10.3389/frhs.2026.1903108](https://doi.org/10.3389/frhs.2026.1903108); 42583442 — Lessons learned from implementing a large-scale contact investigation and TB preventive treatment project in India, IJTLD Open 2026, doi:[10.5588/ijtldopen.26.0168](https://doi.org/10.5588/ijtldopen.26.0168); 41248938 — US divestment in global health: disruption, uncertainty and response, BMJ Glob Health 2025, doi:[10.1136/bmjgh-2025-019990](https://doi.org/10.1136/bmjgh-2025-019990); 40975076 — Effects of reductions in US foreign assistance on HIV, tuberculosis, family planning, and maternal and child health: a modelling study, Lancet Glob Health 2025, doi:[10.1016/S2214-109X\(25\)00281-5](https://doi.org/10.1016/S2214-109X(25)00281-5); 40659023 — The potential impact of reductions in international donor funding on tuberculosis in low-income and middle-income countries: a modelling study, Lancet Glob Health 2025, doi:[10.1016/S2214-109X\(25\)00232-3](https://doi.org/10.1016/S2214-109X(25)00232-3) (abstract-only); 40609560 — Evaluating the impact of two decades of USAID interventions and projecting the effects of defunding on mortality up to 2030, Lancet 2025, doi:[10.1016/S0140-6736\(25\)01186-9](https://doi.org/10.1016/S0140-6736(25)01186-9); 42507823 — The impact of the United States foreign aid freeze on HIV service delivery in PEPFAR-supported countries: a facility-level analysis of 2024-2025 programme data, J Int AIDS Soc 2026, doi:[10.1002/jia2.70182](https://doi.org/10.1002/jia2.70182); 42456026 — Projected impact of U.S. PEPFAR withdrawal on Zimbabwe and implications for the global HIV response, J Acquir Immune Defic Syndr 2026, doi:[10.1097/QAI.0000000000003925](https://doi.org/10.1097/QAI.0000000000003925) (abstract-only); 41761883 — Islamic finance as a strategic instrument for strengthening health systems, East Mediterr Health J 2026, doi:[10.26719/2026.32.1.3](https://doi.org/10.26719/2026.32.1.3) (abstract-only); 42519657 — From aid dependence to domestic ownership: risks, challenges, and policy pathways for sustaining immunisation gains in lower-income countries, Prev Med Rep 2026, doi:[10.1016/j.pmedr.2026.103560](https://doi.org/10.1016/j.pmedr.2026.103560).

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Functions of World Health Organization (WHO)

Asked in: Paper IV May 2019 — RGUHS MD Community Medicine

Definition — The World Health Organization (WHO) is the specialized, non-political health agency of the United Nations (UN), with headquarters at Geneva. Its Constitution was drafted in 1946 by a Technical Preparatory Committee chaired by *Rene Sand*, approved the same year by an International Health Conference of 51 nations at New York, and came into force on 7 April 1948 — observed annually as World Health Day. Its objective, set out in Article 1, is "the attainment by all peoples of the highest possible level of health"; WHO is unique among the UN specialized agencies in having its own constitution, governing bodies, membership and budget, and is part of, but not subordinate to, the United Nations (Park 28e, Ch 25, pp 1055-1056; Textbook of International Health, Ch 3, p 110).

Constitutional functions (Article 2 of the WHO Constitution)

#	Function as written in Article 2	How it is discharged
1	Act as the directing and coordinating authority on international health work	Member States collectively identify priority health problems, define health policies and targets, and devise strategies and programmes to attain them (Park 28e, Ch 25, p 1056)
2	Assist governments in strengthening health services and in emergency aid	Technical and managerial support to national health development programmes, on request
3	Promote maternal and child health and welfare	The family health programme, a major WHO activity since 1970
4	Foster activities in the mental health field	Dedicated technical division and normative guidance
5	Promote improvement of nutrition, housing, sanitation, recreation , and of economic, working and environmental conditions	Environmental health criteria and monitoring programmes; advice on basic sanitary services
6	Study and report on public health and medical care	World Health Statistics series; expert committees and technical reports
7	Promote research and health training	Collaborating centres, research grants, health manpower development
8	Advance work to eradicate epidemic, endemic and other diseases , and to prevent injuries	Smallpox eradication; the current global drive against poliomyelitis
9	Propose conventions, agreements and regulations and make recommendations on international health matters	International Health Regulations (2005); WHO Framework Convention on Tobacco Control
10	Standardize diagnostic procedures and revise international nomenclatures of diseases, causes of death and public health practices	International Classification of Diseases (ICD)

#	Function as written in Article 2	How it is discharged
11	Develop and promote international standards for food, biologicals, pharmaceuticals and similar products	Standardization of vaccines and pharmaceuticals; Model List of Essential Medicines

(Article 2 list: *Textbook of International Health, Ch 3, p 110, Box 3-2*)

These eleven functions are delivered in two operational categories — **central technical services** (epidemiological intelligence, international health agreements, standardization of vaccines and pharmaceuticals, and dissemination of knowledge through expert meetings and technical reports) and **services to governments**, provided on request through the regional and country offices (*Textbook of International Health, Ch 3, p 110*).

The eight broad areas of work

Broad area	What WHO actually does in it
1. Prevention and control of specific diseases	Almost all communicable diseases have been the subject of WHO activity; global eradication of smallpox is the outstanding example of international health cooperation, and the same effort is now directed at poliomyelitis. Epidemiological surveillance is a core activity — WHO collects and disseminates information on diseases subject to the International Health Regulations through an Automatic Telex Reply Service and the Weekly Epidemiological Record , and Member States may call on the WHO emergency scheme. The Expanded Programme on Immunization (EPI) against common childhood diseases is a priority programme
2. Development of comprehensive health services	WHO's most important single function — support to national health policy development and comprehensive national health programmes: organizing health systems based on primary health care, health manpower development and utilization, and building long-term national capability in health infrastructure, managerial capacity (including monitoring and evaluation) and health services research. Appropriate Technology for Health (ATH) encourages self-sufficiency in solving health problems as part of building up primary health care
3. Family health	A major programme activity since 1970, subdivided into maternal and child health care, human reproduction, nutrition and health education; the chief concern is improving the quality of life of the family as a unit
4. Environmental health	Advises governments on national programmes for basic sanitary services; protection of the quality of air, water and food; health conditions of work; radiation protection; and early identification of new hazards from new technology. Delivered through the WHO Environmental Health Criteria Programme and the WHO Environmental Health Monitoring Programme
5. Health statistics	Concerned since 1947 with disseminating morbidity and mortality statistics, published in the <i>Weekly Epidemiological Record</i> , <i>World Health Statistics Quarterly</i> and <i>World Health Statistics Annual</i> . So that statistics from different countries are comparable, WHO publishes the International Classification of Diseases (ICD) , updated every tenth year (the Tenth Revision came into effect on 1 January 1993). It also assists countries to improve medical records and to plan and operate national health information systems

Broad area	What WHO actually does in it
6. Biomedical research	WHO does not itself do research — it stimulates and coordinates it, through a worldwide network of WHO collaborating centres , grants to research workers and institutions, Regional Advisory Committees on health research and a Global Advisory Committee. The Special Programme for Research and Training in Tropical Diseases targets six diseases: malaria, schistosomiasis, trypanosomiasis, filariasis, leishmaniasis and leprosy
7. Health literature and information	Acts as a clearing house for information on health problems; publications run to hundreds of titles, and the WHO library is one of the satellite centres of the Medical Literature Analysis and Retrieval System (MEDLARS) of the US National Library of Medicine. A public information service operates at headquarters and at each of the six regional offices
8. Cooperation with other organizations	Collaborates with the UN and the other specialized agencies at various degrees of working relationship, and maintains relations with a number of international governmental organizations

(All eight areas: Park 28e, Ch 25, pp 1056-1057)

Machinery through which the functions are discharged

Membership is open to all countries; territories not responsible for the conduct of their international relations may be admitted as **associate members**, who participate without a vote. From 56 Members in 1948, WHO now has **194 Member States and two associate members**, each contributing yearly to the budget and each entitled to the Organization's services (Park 28e, Ch 25, p 1056).

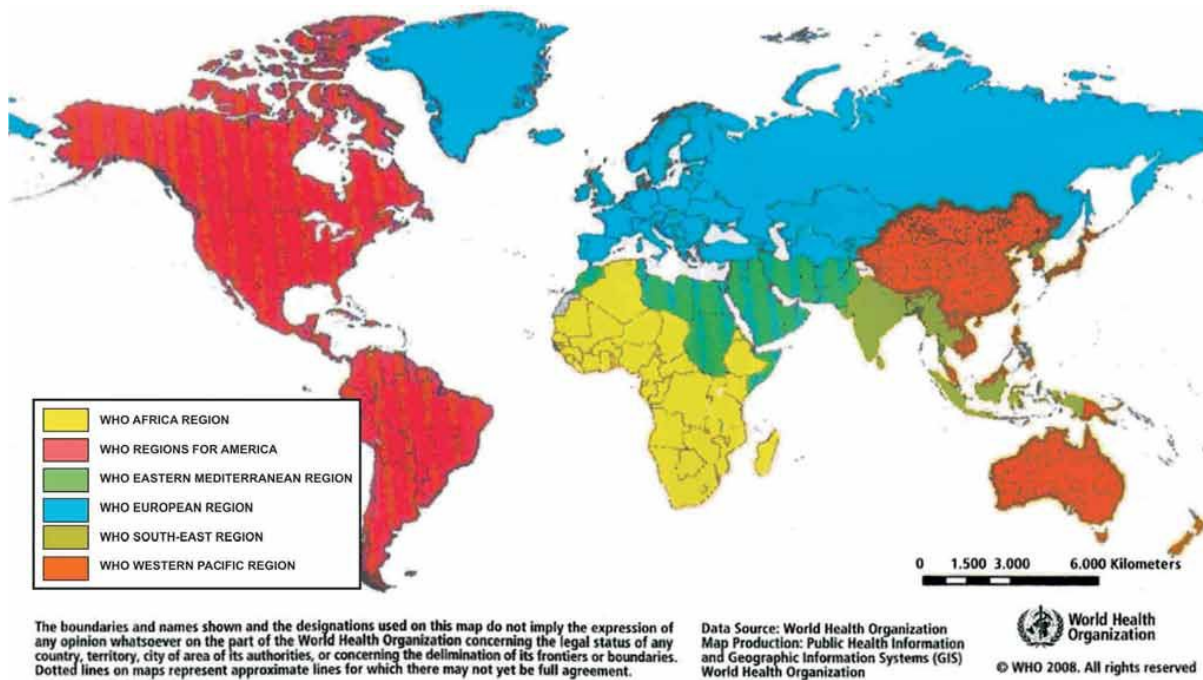
Organ	Composition	Its functions
World Health Assembly (WHA) — the "Health Parliament of Nations" and supreme governing body	Delegates of all Member States, one vote each ; meets annually, usually in May, generally at Geneva	(i) determine international health policy and programmes; (ii) review the work of the past year; (iii) approve the budget for the following year; (iv) elect Member States to designate persons to the Executive Board and replace retiring members; appoints the Director-General on the nomination of the Executive Board; holds "technical discussions" on subjects of world interest
Executive Board	Now 34 members , technically qualified in the field of health, designated by but not representing their governments; three-year terms with one-third renewed each year; no fewer than three elected from each region; meets at least twice a year (January, and after the WHA in May)	Gives effect to the decisions and policies of the Assembly; prepares the WHA agenda and works on budgets; may itself act in an emergency such as epidemics, earthquakes and floods where immediate action is needed
Secretariat	Headed by the Director-General, the chief technical and administrative officer, elected by the WHA every five years; about 8,000 staff across headquarters, the regional offices and 147 country offices	Provides Member States with technical and managerial support for their health development programmes

(Park 28e, Ch 25, pp 1057-1058; Textbook of International Health, Ch 3, pp 110-111)

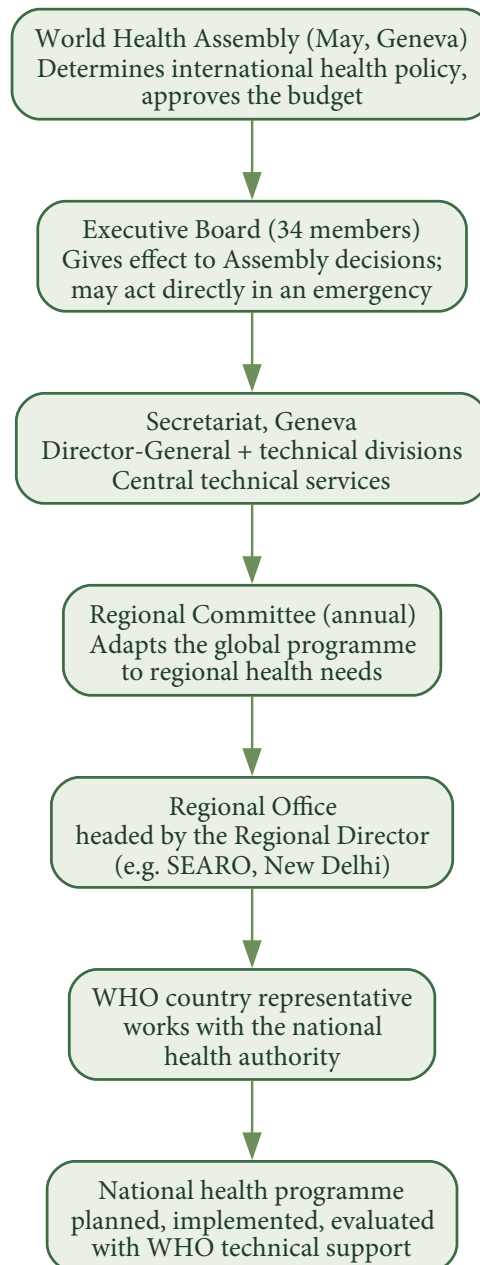
To meet the special health needs of different areas, WHO has established **six regional organizations**, each with a Regional Office headed by a **Regional Director** and a **Regional Committee** of that region's Member States, meeting once a year to review and plan the regional work; regional plans are amalgamated into the overall programme. A **WHO country representative** works with national health authorities to plan, coordinate, implement and evaluate national health programmes.

Region	Headquarters
South-East Asia (SEARO)	New Delhi, India
Africa (AFRO)	Brazzaville, Congo
The Americas (PAHO/AMRO)	Washington DC, USA
Europe (EURO)	Copenhagen, Denmark
Eastern Mediterranean (EMRO)	Cairo, Egypt
Western Pacific (WPRO)	Manila, Philippines

(Park 28e, Ch 25, p 1057, Table 1; Textbook of International Health, Ch 3, pp 110-111, Table 3-5)



The six WHO regions (Suryakantha Environment Best, p.356)

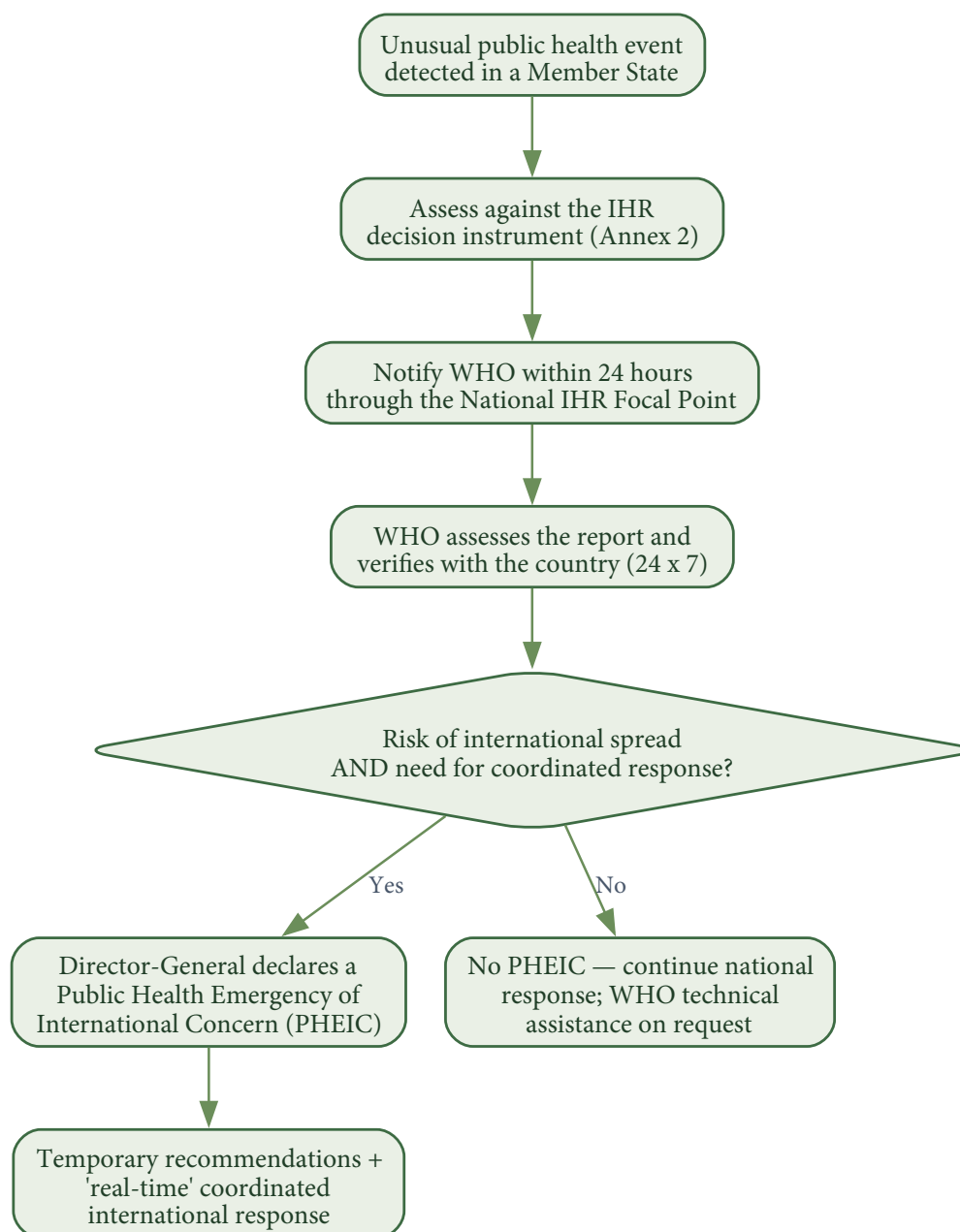


How WHO's directing and coordinating function reaches a country

Regulatory function under the International Health Regulations (2005)

- ▶ The International Health Regulations (IHR), in force since 2007, are the world's **first legally binding agreement on disease notification** — the concrete expression of Article 2's power to propose regulations.
- ▶ Member States must notify WHO of **all events that may constitute a public health emergency of international concern (PHEIC)**, a scope far wider than the fixed disease list of the 1969 Regulations.
- ▶ A PHEIC is an event that (a) constitutes a public health risk to other States through the **international spread of disease**, and (b) potentially requires a **coordinated international response**.

- Notification is due to WHO **within 24 hours** of the first official case of a listed disease, through National IHR Focal Points operating 24 hours a day, seven days a week.
- WHO's reciprocal duty is to assess country reports, recommend concrete action, provide technical assistance on request, and — once a PHEIC is determined — mount a "real-time" response.
- Under the earlier IHR (1969) only **cholera, plague and yellow fever** were internationally notifiable; louse-borne typhus, relapsing fever, paralytic poliomyelitis, malaria, influenza A, severe acute respiratory syndrome and smallpox were kept under WHO surveillance (Park 28e, Ch 3, p 144; Ch 21, p 997).
- **Limitation to state in the answer:** WHO resolutions *urge* but never *oblige* Member States, and WHO can intervene in a country only when requested (Textbook of International Health, Ch 3, p 111).



IHR (2005) notification pathway — WHO's emergency and regulatory function

These functions as experienced in India

- ▶ India is a **founder member**, joining in 1948; SEARO, at World Health House, Indraprastha Estate, New Delhi, now has **11 member countries** (Park 28e, Ch 25, pp 1057-1058).
- ▶ WHO provides **technical and material assistance** for the implementation of India's national health programmes, alongside UNICEF, UNFPA and the World Bank (Park 28e, Ch 7, p 493).
- ▶ The **National Guinea Worm Eradication Programme** was launched in 1984 with WHO technical assistance, integrated into the national health system at village level with intersectoral coordination and close collaboration with WHO and UNICEF (Park 28e, Ch 7, p 572).
- ▶ **Normative and statistical support** — the ICD makes Indian mortality and morbidity data internationally comparable, and WHO assists in improving medical records and in planning and operating the national health information system.
- ▶ **Essential medicines** — WHO's *Selection and Use of Essential Medicines* (Technical Report Series No. 933, 2005) provides the model; India's current list is the National List of Essential Medicines, 2022 (Park 28e, Ch 24, p 1053).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **WHO organogram (label exactly):** World Health Assembly (supreme governing body, meets annually in May, one vote per Member State) → Executive Board (34 technically qualified members) → Secretariat, Geneva (Director-General) → six Regional Offices, each with a Regional Director and a Regional Committee → WHO country representative → Ministry of Health of the Member State. Mark **South-East Asia Regional Office, New Delhi** and label India's year of joining, **1948**.

High-yield exam pointers

- ▶ **Dates and names:** Constitution drafted 1946 (Technical Preparatory Committee, chaired by *Rene Sand*), approved by 51 nations at New York, **in force 7 April 1948 = World Health Day**. Objective (Article 1): "the attainment by all peoples of the highest possible level of health".
- ▶ **First constitutional function** is the one to write first: *to act as the directing and coordinating authority on international health work*.
- ▶ **Membership:** 56 Members in 1948 → **194 Member States + 2 associate members** today.
- ▶ **Three organs:** World Health Assembly (May, Geneva, one vote each) — Executive Board (34 members, 3-year terms, one-third renewed yearly, meets ≥ twice a year) — Secretariat (Director-General, elected by the WHA every 5 years).
- ▶ **Six regional headquarters, in order:** New Delhi, Brazzaville, Washington DC, Copenhagen, Cairo, Manila.
- ▶ **Park's eight broad areas — mnemonic "D-S-F-E-S-R-L-C":** Diseases, Services, Family health, Environment, Statistics, Research, Literature, Cooperation.
- ▶ **TDR's six tropical diseases:** malaria, schistosomiasis, trypanosomiasis, filariasis, leishmaniasis, leprosy. **IHR 2005:** notify within 24 hours; **PHEIC** = international spread + need for coordinated response.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e prints Alexandria as the Eastern Mediterranean regional office headquarters whereas the Textbook of International Health gives Cairo (the current seat, used above), and Park's health-statistics section still describes ICD-10 as the current revision (in force 1 January 1993) although WHO has since issued ICD-11; staffing and country-office counts quoted here are those of the retrieved texts and change from year to year.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, International Health Regulations (2024 amendments), WHO Pandemic Agreement 2025, MoHFW national programmes

Programme and policy updates (India)

- ▶ **National Action Plan for Rabies Elimination (NAPRE), launched 2021** — commits India to zero dog-mediated human rabies deaths by 2030, matched to the global *Zero by 30* target, and moves dog-bite and rabies reporting from the Integrated Disease Surveillance Programme onto the real-time Integrated Health Information Portal (IHIP). About 17.4 million animal bites and 20,565 human rabies deaths are still estimated each year; **Sikkim and Goa have reached rabies-controlled status** through One Health working. A 2022-23 national survey found 71% of bite victims received at least three vaccine doses but only 43% completed the full post-exposure schedule, one in five government facilities held no rabies immunoglobulin and roughly 30% had no vaccine stock [PMID 42494549].
- ▶ Under the **National Tuberculosis Elimination Programme (NTEP)**, the ten-year TB-Free Haryana public-private partnership (state government with a corporate tertiary hospital; piloted in Rewari in 2014, statewide by 2019) paired mobile digital chest radiography and AI-assisted reading with point-of-care molecular testing to give same-day diagnosis and treatment start. Over 62,000 chest X-rays gave a 5.3% diagnostic yield, improved smear-negative case detection and kept running through the COVID-19 pandemic [PMID 42529833].

Current national data

- ▶ India's **Universal Immunisation Programme** now targets about 26 million infants and 29 million pregnant women a year against 12 diseases; full-immunisation coverage rose from **62% in 2015-16 to 76.4% in 2019-21** (the NFHS-4 and NFHS-5 rounds) — still short of the 90% Immunization Agenda 2030 target. Polio elimination was achieved in 2014 and maternal and neonatal tetanus elimination in 2015. Since 2023 fractional IPV is given at 6 weeks, 14 weeks and 9 months, and a **hexavalent (pentavalent + IPV) switch** is under economic appraisal: at current prices it would raise programme cost, but a 50% cut in unit price offsets it [PMID 42295252].
- ▶ A 2026 meta-analysis of 45 cross-sectional studies (>428,000 children, 2015-2025) put **incomplete immunisation among under-fives at 26% (95% CI 21-32%)**, higher in northern and multi-state samples than in southern and north-eastern ones. The strongest determinants were absence of an immunisation card (OR 9.46, 95% CI 3.28-27.28), non-institutional delivery (OR 3.59), maternal illiteracy (OR 3.25), low socioeconomic status (OR 2.03), higher birth order (OR 1.88) and female sex (OR 1.34) [PMID 42320383].

- **Measles** — India reported 13,220 cases between August 2023 and January 2024, the fourth highest count globally. A community-based, cluster-randomised study in tribal Kandhamal-Boudh, Odisha, recorded first-dose coverage 99.6% and two-dose coverage 97.5%, yet only 55.1% seropositivity (IgG $\geq$ 250 IU/L) three months after the second dose, and a high-titre response ($\geq$ 500 IU/L) in just 27.3%; male sex was associated with poorer response — a reminder that coverage figures alone overstate population protection [PMID 42555691].

Elimination, surveillance and disease status

- **Kala-azar** — the tri-country Kala-azar Elimination Programme (India, Nepal, Bangladesh, launched 2005) set the threshold at fewer than 1 case per 10,000 population at block (sub-district) level. Blocks above that threshold fell from 633 in 2005 to 136 in 2015, 8 in 2021 and zero in 2023 — the last being in Pakur district, Jharkhand — with reported cases down from 34,803 to 524. The rK39 rapid test, oral miltefosine and later single-dose liposomal amphotericin B made the difference. **Post-kala-azar dermal leishmaniasis and HIV-VL co-infection remain human reservoirs** able to restart transmission in the post-elimination phase [PMID 40828749] [PMID 41502015].
- **Trachoma** — WHO validated India in 2024 as having eliminated trachoma as a public health problem, the twentieth country to be so validated. The criteria are trachomatous trichiasis unknown to the health system below 0.2% in adults aged $\geq$ 15 years, trachomatous inflammation-follicular below 5% in children aged 1-9 years sustained for two years without mass antibiotic treatment in every formerly endemic district, and a health system able to detect and treat incident trichiasis; post-validation impact and surveillance surveys remain obligatory. By contrast **lymphatic filariasis is still endemic in 345 districts** against a 2030 elimination goal, with public-private partnership proposed for MDA compliance, post-MDA surveillance and morbidity management [PMID 39462916] [PMID 41951149].

Guideline and legislative updates

- **The International Health Regulations (2005) were amended on 1 June 2024 at the 77th World Health Assembly, the amendments entering into force in September 2025** — the single most examinable change to the regulatory section above. Amended Articles 1 and 12 allow the Director-General to declare a "**pandemic emergency**", a new tier above a PHEIC; amended Article 13 obliges the WHO Secretariat to facilitate access to health products during a PHEIC or pandemic emergency; amended Article 44 establishes a **Coordinating Financial Mechanism** for developing countries under the authority of the Health Assembly; and a new advisory subcommittee sits under Article 54. Equity, financing and implementation are the organising themes of the revision [PMID 40589223].

Global and WHO developments

- **The WHO Pandemic Agreement was adopted by 124 governments at the 78th World Health Assembly on 20 May 2025, after three years of negotiation** — 35 Articles, and only the **second instrument ever negotiated under Article 19 of the WHO Constitution**, after the Framework Convention on Tobacco Control (adopted 2003, in force 2005). It covers surveillance, data sharing, protection of health workers, sustained research and development (Article 9) and equitable access to vaccines, diagnostics and therapeutics, and provides for a Coordinating Financial Mechanism and a Global Supply Chain and Logistics Network. The Pathogen Access

and Benefit-Sharing annex (Article 31.2) goes to the 2026 Health Assembly, after which the Agreement takes effect. Commentators flag vague drafting, WHO's limited enforcement powers and notable abstentions as real limits on its bite [PMID 40589223] [PMID 40443067].

- **WHO financing** — on 20 January 2025 the United States announced withdrawal from WHO, repeating the July 2020 move that was reversed in 2021. In 2022-23 the US provided roughly US\$218 million in assessed and US\$1.065 billion in voluntary contributions, close to a fifth of the budget, so the exit threatens pandemic preparedness and programme delivery in low- and middle-income countries; the Gates Foundation is now the second-largest voluntary funder, raising the concern that priorities follow a narrow donor base rather than country need. This is the practical face of the textbook limitation that WHO resolutions urge but never oblige [PMID 40767200].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 25 (International Health), pp 1055-1058; Ch 3, p 144; Ch 7, pp 493, 572; Ch 21, p 997; Ch 24, p 1053. Textbook of International Health — Ch 3 (International Health Agencies, Activities, and Other Actors), pp 110-112, 115-116; Ch 5, pp 234-235. Oxford Textbook of Public Health 7e — Ch 24, Box 4.2.1 (Constitution of the WHO). Suryakantha — Community Medicine with Recent Advances, p 356 (WHO regions map, © WHO 2008). Named instruments: WHO Constitution (1946, in force 1948); International Health Regulations (2005, in force 2007); WHO Technical Report Series No. 933 (2005).

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♦ ♦ ♦

Globalisation and public health

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — **Globalization** is the closer integration or interconnectedness of human societies across geographical or territorially-defined boundaries; what distinguishes its contemporary form since the late twentieth century is the vast increase in the **types, scale and speed** of transboundary flows — of people, other life forms, goods, capital, microbes and information — and the reach of their consequences (Oxford Textbook of Public Health 7e, Ch 1.5, pp 254-255). **Economic (neoliberal) globalization** is the narrower term for the increasing connectedness of the world economy through removal of barriers to trade and capital flows (Textbook of International Health, Ch 1, p 44). Globalization is not intrinsically good or bad for health: it **redistributes the determinants of health**, creating winners and losers within and between countries, and the public health task is to maximise its benefits and minimise its costs equitably (Oxford 7e, Ch 1.5, pp 254-255).

Dimensions of globalization and its drivers

Dimension of change	What it means	Why it matters to public health
Spatial change	How people organise and interact across physical or territorial space — the "global village", the "death of distance", virtual communities	Pathogens, products and people cross borders within an incubation period; a local outbreak becomes a global risk (Oxford 7e, Ch 1.5, p 255)
Temporal change	Acceleration of the timeframe in which phenomena occur — currency transactions in seconds, faster travel, mechanised and digitalised production	Spread outpaces classical notification and quarantine; response must be real-time (Oxford 7e, Ch 1.5, p 256)
Cognitive change	Global production and diffusion of knowledge, ideas, values and beliefs through mass and social media, advertising, research and educational institutions; ascendancy of English	Global marketing of tobacco, alcohol and ultra-processed food; cultural homogenisation of lifestyle (Oxford 7e, Ch 1.5, pp 256-257)
Drivers (three competing explanations)	(i) Technology — information, communication and transport technologies as enablers; (ii) Economics — producers seeking cheap inputs and new markets, the "invisible hand"; (iii) Ideology — <i>neoliberalism</i> , i.e. trade liberalisation, market competition, foreign direct investment and a minimal state	Determines whether health harms are treated as unavoidable side-effects or as products of a policy choice that can be reversed (Oxford 7e, Ch 1.5, pp 257-258)

Pathways by which globalization affects health

Globalization influences health and generates health inequity through **five major pathways** (Labonté and Schrecker, in Textbook of International Health, Ch 9, p 463).

Pathway	Mechanism	Health consequence
1. Trade liberalisation and the world trade regime (World Trade Organization, WTO)	Farmers displaced by agri-business; workers de-industrialised into unregulated informal work; professionals (including health workers) emigrate	Loss of livelihood and safety net, especially for agricultural workers, small entrepreneurs, women and children; workforce depletion in low- and middle-income countries (Textbook of International Health, Ch 9, p 463)
2. Global reorganisation of production and labour markets	Export of jobs and whole industries to areas of lax regulation — a "race to the bottom" in wages, labour, environmental and safety protection	Unemployment and stress in industrialised countries; hazardous, unorganised, low-wage work in poorer countries; rising occupational injury and pesticide exposure (Textbook of International Health, Ch 9, pp 454-455, 463)
3. Debt crisis and structural adjustment	Debt-servicing and loan conditionalities force cuts in food subsidies, public social programmes and public-sector incomes; user fees introduced	Reduced government investment in water, sanitation, nutrition and health; in Zimbabwe user fees under structural adjustment reduced service use, raised the infant mortality rate and increased malnutrition in rural children (Textbook of International Health, Ch 9, pp 464, 475)
4. Environmental damage	Commodification of public goods such as water; unenforced environmental regulation; transboundary dumping of hazardous waste and relocation of risky occupations (ship-breaking, asbestos disposal)	Cross-border pollution; a cholera outbreak in South Africa in 2000-2001 with about 120,000 cases and 265 deaths followed water-utility privatisation and user fees under structural adjustment (Oxford 7e, Ch 1.5, p 265)
5. Financial liberalisation	Volatile capital flows and currency speculation; sudden devaluations	Purchasing power evaporates, deepening the effect of inequality on the social determinants of health (Textbook of International Health, Ch 9, pp 464-465)

WTO trade agreements and their public health effects

The General Agreement on Tariffs and Trade (GATT) was established in 1948; membership grew from 23 countries in 1947 to 125 by 1994, and the **World Trade Organization was created in 1995** (164 member states in 2018), its remit now extending beyond goods to services, agriculture and intellectual property (Oxford 7e, Ch 1.5, p 260).

Agreement	Health effect through loss of domestic regulatory space
TRIPS (Trade-Related Aspects of Intellectual Property Rights)	Extended patent protection limits access to essential medicines; higher drug cost drains money from primary health care and other determinants of health
GATS (General Agreement on Trade in Services)	Locks in existing privatisation of health care, education, water and environmental services; once committed, no cost-free way to restore public provision
SPS (Sanitary and Phytosanitary Measures)	Requires costly and imperfect scientific risk assessment, while foreign goods may not be treated differently from domestic goods

Agreement	Health effect through loss of domestic regulatory space
TBT (Technical Barriers to Trade)	Any regulatory barrier must be "least trade restrictive"; many disputes over domestic health and safety rules have invoked it (France's asbestos ban was challenged and upheld — the only WTO ruling to date favouring health over trade)
Agreement on Agriculture	Permits continued export and producer subsidies by rich countries, depressing world prices and undermining growers in developing countries
TRIMs (Trade-Related Investment Measures) and Government Procurement	Bar governments from using local-purchase requirements and public contracts for domestic employment, regional or employment equity

(Source: *Textbook of International Health*, Ch 9, Table 9-2, p 467; GATS and TBT commentary p 468.)

Balance sheet — gains and harms for public health

Domain	Benefits	Harms
Communicable disease	Faster global sharing of epidemiological data (ProMED-mail reaches over 70,000 subscribers in at least 185 countries); airline-travel data used to model Ebola transmission	Rapid international spread — SARS reached about 25 countries within weeks in 2002-2003; 60% of emerging infectious diseases are zoonotic and over two-thirds of these originate in wildlife; global spread of antimicrobial-resistant organisms (Oxford 7e, Ch 1.5, pp 271-273)
Non-communicable disease	Global norms — the WHO Framework Convention on Tobacco Control (FCTC)	Nutrition transition and globalisation of sedentary lifestyle; NCDs killed 40 million people in 2017 (70% of all deaths) — cardiovascular disease 17.7 million, cancers 8.8 million, respiratory disease 3.9 million, diabetes 1.6 million; about 1.9 billion adults overweight and 650 million obese , and at least 340 million school-age children overweight or obese (Oxford 7e, Ch 1.5, pp 274-275)
Medicines and technology	Economies of scale, higher research spending and improved distribution lower prices of essential products	Research skewed to profitable markets; a new prescription drug now costs about US\$2.6 billion to bring to market; 1 in 10 medical products in low- and middle-income countries is substandard or falsified (WHO 2017) (Oxford 7e, Ch 1.5, pp 261, 265)
People and workforce	New training and employment opportunities; remittances; 258 million international migrants in 2017 , up from 173 million in 2000	Overcrowded refugee settlements without water and sanitation; depletion of health workers from low- and middle-income countries against a global shortage of over 4 million health workers (Oxford 7e, Ch 1.5, p 264; Textbook of International Health, Ch 12, p 671)
Equity	Rising average incomes and living standards in several economies	The richest 1% held 50.1% of world wealth by 2017, up from 42.5% in 2008 ; illicit trade flourishes — illicit tobacco is about 9% of retail sales (Oxford 7e, Ch 1.5, pp 258, 265)

What globalization means for India

- ▶ **Access to medicines.** India's patent law demands a high standard of *novelty* — a new drug must bring significant efficacious benefit beyond existing drugs — one of the most restrictive standards in the world; this kept generic production going and enabled access to medicines for millions of the country's poor. A patent-denial challenge by Novartis alleging violation of TRIPS was ruled outside the court's jurisdiction in 2007, leaving India's policy intact (Textbook of International Health, Ch 9, pp 493-494).
- ▶ **Medical tourism.** India is the **largest medical tourism market**; patients save 75-90% of home costs. The trade expands the private sector, draws expensive technology and health professionals away from the public sector, and creates two-tier care — damage already evident in India and Thailand (Textbook of International Health, Ch 12, p 673).
- ▶ **Brain drain.** In absolute terms far more doctors emigrate from the Indian subcontinent than anywhere else — almost **80,000** (Textbook of International Health, Ch 12, p 672). Diagnostic outsourcing (radiology and pathology read across time zones) is a related flow.
- ▶ **Antimicrobial resistance.** Because of the **globalization of food supply and international travel**, antimicrobial resistance among animal bacteria can affect consumers anywhere in the world (Park 28e, Ch 5, p 430).
- ▶ **Non-communicable disease.** Globalization, along with urbanisation and population ageing, is listed among the major forces driving the underlying determinants of cardiovascular disease — "the causes of the causes" (Park 28e, Ch 6, p 438). Trade liberalisation is considered to be driving the NCD epidemic in Asia through market penetration by transnational tobacco, alcohol and ultra-processed food corporations (Oxford 7e, Ch 2.8, p 526).
- ▶ **System strengthening.** India, like Brazil and China, has invested in building public health system capacity over the past decade in response to these globalised risks (Oxford 7e, Ch 1.5, p 266).

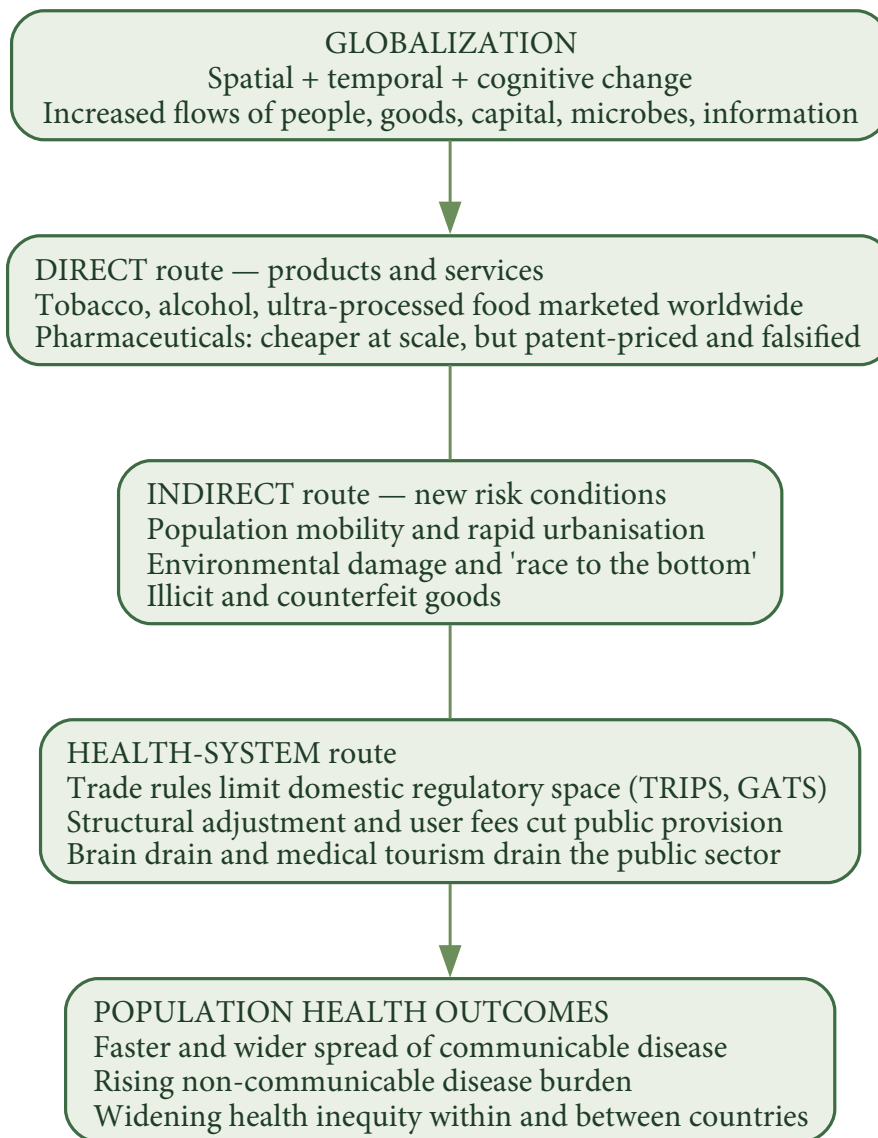
Public health response — governing globalization for health

- ▶ **International Health Regulations (2005)** — revised after SARS and in force from 2007; scope widened from a handful of named diseases to any **public health emergency of international concern**, with binding **core capacity requirements** for every country to detect, assess, notify, report and contain events, and to share information (Oxford 7e, Ch 8.17, pp 2859-2860).
- ▶ **WHO Framework Convention on Tobacco Control** — negotiated expressly because global marketing had limited the effectiveness of national action; adopted 2003, in force 2005, **181 parties by 2018**; the Protocol to Eliminate Illicit Trade in Tobacco Products supplements it (Oxford 7e, Ch 1.5, p 279).
- ▶ **Doha Declaration on the TRIPS Agreement and Public Health (2001)** and the WTO Decision on Implementation of Paragraph 6 (2003) — affirm that TRIPS shall not prevent members from protecting public health, and clarify **compulsory licensing** and **parallel importation**; TRIPS compliance was deferred for 30 least-developed countries to 2016. This is written into **Sustainable Development Goal target 3.b** (Textbook of International Health, Ch 9, p 466; Park 28e, Ch 14, p 853).

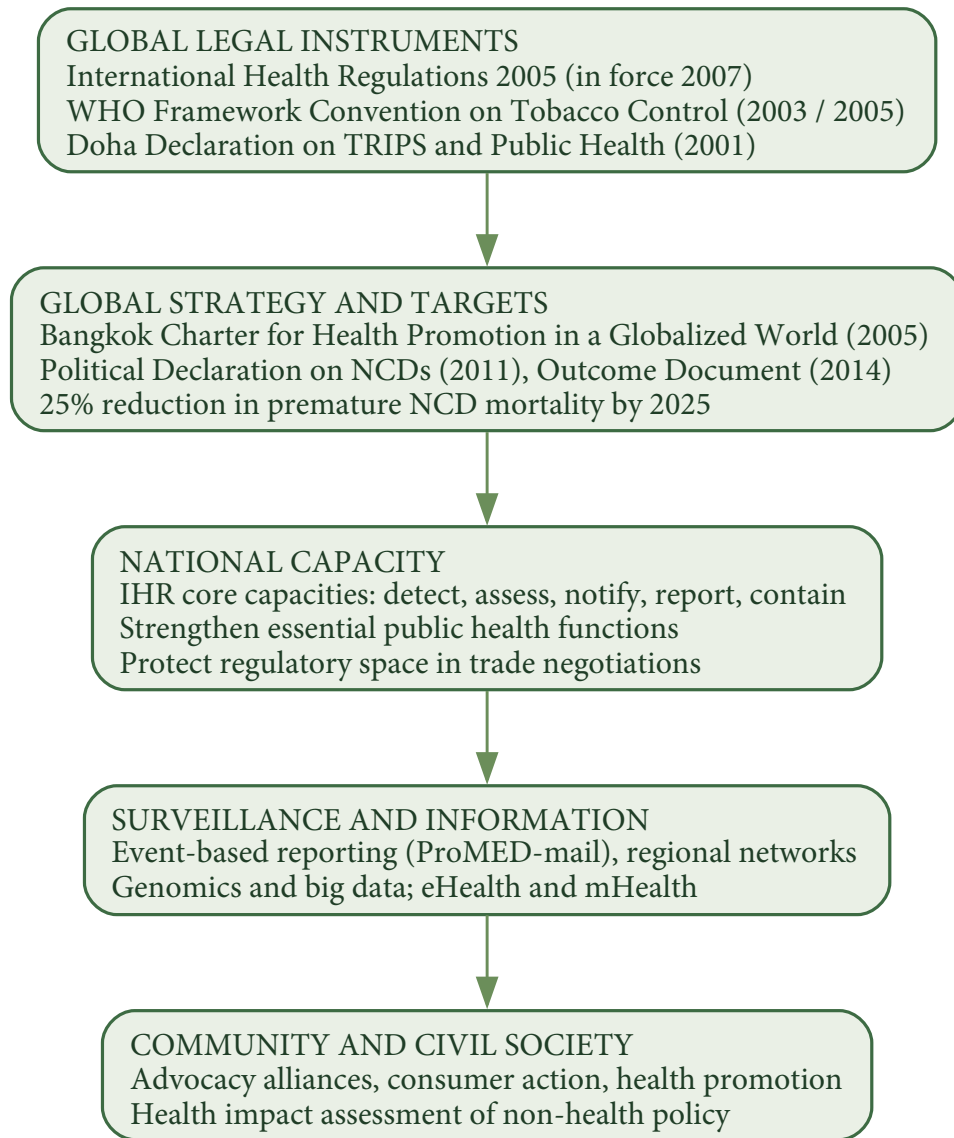
- ▶ **Bangkok Charter for Health Promotion in a Globalized World (2005)** — the Sixth Global Conference on Health Promotion; sets out the actions and commitments needed to address the determinants of health in a globalized world (Oxford 7e, Ch 1.5, pp 274-275).
- ▶ **Collective action on NCDs** — UN General Assembly High-Level Meetings gave the Political Declaration (2011) and Outcome Document (2014), with the target of a **25% reduction in premature NCD mortality by 2025**, and the WHO Global Action Plan for the Prevention and Control of NCDs 2013-2020 (Oxford 7e, Ch 1.5, pp 275, 280).
- ▶ **Adapt essential public health functions** — surveillance beyond national aggregates, event-based and internet-based reporting (ProMED-mail), regional surveillance networks, genomics and "big data"; national agencies created for this purpose include the Public Health Agency of Canada and the European Centre for Disease Prevention and Control, both in 2004 (Oxford 7e, Ch 1.5, pp 266, 273).
- ▶ **Treat key public health functions as global public goods** — outbreak information, disease surveillance and immunisation will be under-produced if left to the market, and require collective action to correct that market failure (Oxford 7e, Ch 1.5, p 287).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram — pathways of neoliberal globalization and their effect on health** (after Labonté and Schrecker; Textbook of International Health, Ch 9, Fig 9-3, p 464). Draw a vertical funnel with these exact labels, top to bottom: box 1 "*Neoliberal global and national political, economic and ideological processes*" → box 2 "*Relations of power (class, gender, race) and unequal distribution of resources and wealth*" → a row of five boxes: "*Environmental degradation*", "*Macroeconomic processes: commodity pricing, unfair trade liberalization*", "*Reorganization and deregulation of labour, waste production*", "*Debt crises, international financial instruments, loan conditionalities, divestment of public social programmes*", "*Financial liberalization and currency speculation*" → box 3 "*Deterioration of social policies and neighbourhood / workplace / household living conditions*" → final box in capitals "**UNEQUAL HEALTH OUTCOMES**".



Three routes by which globalization reaches population health



Ladder of the public health response to globalization, from global instruments to community action

High-yield exam pointers

- ▶ **Definition to write first:** globalization = closer integration/interconnectedness of societies across territorial boundaries, distinguished by the **types, scale and speed** of transboundary flows.
- ▶ **Three changes (mnemonic S-T-C):** Spatial, Temporal, Cognitive.
- ▶ **Five pathways (Labonté and Schrecker):** trade liberalisation and the world trade regime; reorganisation of production and labour; debt crisis and structural adjustment; environmental damage; financial liberalisation.
- ▶ **Name the agreements:** TRIPS (medicines), GATS (health services and water), SPS, TBT, Agreement on Agriculture, TRIMs, Government Procurement. GATT 1948 → WTO 1995.
- ▶ **Dated global figures:** NCDs 40 million deaths, 70% of all deaths (2017); 1.9 billion overweight and 650 million obese; 258 million international migrants (2017); FCTC 181 parties (2018); richest 1% held 50.1% of world wealth (2017).

- **Indian anchors:** largest medical tourism market; ~80,000 doctors emigrated from the subcontinent; strict *novelty* standard in patent law protects generic supply; Novartis case ruling 2007.
- **Response instruments to list:** IHR 2005, FCTC, Doha Declaration 2001, Bangkok Charter 2005, UN High-Level Meetings on NCDs, global public goods for health.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook of Preventive and Social Medicine 28e has no dedicated chapter on globalization and mentions it only in passing (as a driver of antimicrobial resistance through the global food supply, and as an underlying determinant of cardiovascular disease); this answer is therefore grounded mainly in the Oxford Textbook of Public Health 7e chapter on globalization and the Textbook of International Health chapter on globalization, trade, work and health, with Park supplying the Indian and Sustainable Development Goal anchors.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, amended International Health Regulations (2024), WHO Pandemic Agreement WHA78.1 (2025), MoHFW / National Health Mission, National Family Health Survey, AOGIN India 2026 policy statement

Programme and policy updates (India)

- **100-Day TB Elimination Campaign (7 December 2024 – 17 March 2025)** — a nationwide National Tuberculosis Elimination Programme drive of intensified case detection, early diagnosis, targeted screening and nutritional/financial support; a difference-in-differences comparison of 53 campaign against 27 non-campaign districts in Maharashtra found the relative change in TB notification rate 10% higher in campaign districts (DID IRR 1.10; 95% CI 1.03–1.17; $p = 0.003$), significant in the rural setting (IRR 1.21) and the public sector (IRR 1.17) but not in urban areas (IRR 0.95) or private health care (IRR 0.97); notification fell 10% in non-campaign districts against 1% in campaign districts, so the campaign chiefly stabilised rural, public-sector surveillance [PMID 42457294].
- **National Sickle Cell Anaemia Elimination Mission** — announced in the Union Budget 2023–24 with an elimination target of 2047 and universal screening of the 0–40-year population of endemic, largely tribal, areas; a Health Technology Assessment in India appraisal of four point-of-care tests led the National Health Mission to instruct states to procure kits by public tendering, which encouraged in-house Indian manufacturing and competitive bidding, brought the price below US\$1.21 per test and projects US\$37.86 million of savings at full coverage of the 41.1 million people aged 0–40 in the six endemic states (Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha, Chhattisgarh) — a worked example of domestic manufacturing and public procurement power lowering a diagnostic price [PMID 42594133].

Current national data

- **Smokeless tobacco use in women aged 15–49 is falling across survey rounds** — 8.4% in NFHS-3 (2005–06), 5.6% in NFHS-4 (2015–16) and 3.8% in NFHS-5 (2019–21), with gutkha/paan masala the commonest product in NFHS-5 (1.3%) and Mizoram highest at 45%, followed by Manipur at 41.4%; more than 248 million people use smokeless tobacco worldwide and about 90% of them

are from the Indian subcontinent, while the gutka and paan masala bans under the Food Safety and Standards Act 2006 and its 2011 Regulations have not been effectively enforced in many states, the products staying widely available informally — the gap is enforcement, not absence of law [PMID 42545929].

- **NFHS-5 quantifies the two-tier effect of a commercialised delivery market** — the all-India caesarean section rate was 19.21% (public facilities 13.95%, private facilities 47.18%), and the public/private nature of the institution was the strongest determinant (adjusted OR 4.11; 95% CI 3.98–4.24 nationally, rising to 16.19 in Bihar and 8.62 in Uttar Pradesh), while socio-economic factors mattered only where medical infrastructure was lacking; the authors attribute the excess to economic incentives on private providers. The usual comparator is the WHO 1985 population benchmark of 10–15%, which WHO has since clarified is not an ideal rate — these authors argue the ideal band is 10–19% [PMID 42406715].

Elimination, surveillance and disease status

- **Lymphatic filariasis remains endemic in 345 districts** of India against a 2030 elimination goal, with mass drug administration, morbidity management and disability prevention, and post-MDA surveillance placing considerable strain on public health infrastructure; public-private partnership — already used effectively for polio eradication and TB control in India — is argued to be a strategic but underused lever for advocacy, social mobilisation, treatment non-compliance during MDA, post-MDA surveillance, patient identification and care, digital innovation, operational research and vector control [PMID 41951149].
- **Rubella elimination is targeted for December 2026 in India and the rest of the WHO South-East Asia Region**, but state-level laboratory surveillance in central India (1 May 2023 – 30 April 2024) found rubella IgM seropositivity of 7.2%, indicating ongoing transmission, alongside IgG seropositivity of 81.0% overall and 89.6% in pregnant women; the continuing IgM-positive cases, particularly in young children, mark the immunity gaps that could delay the declared deadline [PMID 42159053].
- **Malaria caused an estimated 610,000 deaths globally in 2024**, predominantly in sub-Saharan Africa; using India as the case study, the authors document gaps in district-level availability of intervention data (number of rapid diagnostics used, antimalarials used, insecticide-treated net and other vector-control distribution) and argue for a global policy mandating open access to such data as a public health good essential to innovation, transparency, resource allocation and health equity [PMID 42167215].

Guideline and legislative updates

- **AOGIN India policy statement on HPV vaccination (2026)** — holds that integrating HPV vaccine into the Universal Immunization Programme is necessary if India is to meet the WHO requirement that 90% of adolescent girls receive the vaccine by age 15, and offers evidence-based recommendations to health care professionals, policymakers and programme managers on HPV vaccination dosage, schedules and delivery methods, developed from the most recent Indian and worldwide evidence including data from single-dose efficacy trials; the stated aim is to reduce disparities in access and accelerate India's progress towards the WHO 90-70-90 elimination targets. Cervical cancer accounts for nearly one-fifth of the cancer-death burden among Indian women [PMID 42169564].

Global and WHO developments

- ▶ **Amended International Health Regulations (2005), adopted 1 June 2024 at the 77th World Health Assembly** — no intermediate alert tier was added, and the regional-PHEIC idea was rejected as fragmenting the response; instead **pandemic emergency** was introduced as a higher tier within the PHEIC alert framework, applying only to communicable-disease PHEICs, which formalises and legally expands WHO's emergency powers; Article 13(8) obliges WHO to remove barriers to timely and equitable access to relevant health products once a PHEIC, including a pandemic emergency, is determined, and Article 13(9) obliges States Parties to support that work [PMID 40883832].
- ▶ **WHO Pandemic Agreement (resolution WHA78.1) adopted 20 May 2025** — it sets up a **Pathogen Access and Benefit-Sharing (PABS)** system that ties fast sharing of pathogen samples and genetic sequence data to guaranteed access to the resulting vaccines, tests and medicines, a dedicated global supply and allocation mechanism, a **Coordinating Financial Mechanism** for sustainable financing so that low- and middle-income countries can scale manufacturing and distribution, and a **Conference of the Parties** to oversee implementation and compliance; a scoping review with document analysis finds it addresses coordination and financing but advances little on enforcement, technology transfer and independent monitoring — intellectual-property provisions rest on existing TRIPS flexibilities and voluntary licensing with no compulsory-licensing trigger, and the PABS modalities are deferred to future Conference of the Parties decisions, so the Doha Declaration argument on access to medicines is reopened rather than settled [PMID 41464441]. Determining a pandemic emergency under the amended IHR is the normative trigger that activates the Agreement's pillars — research and development (**Article 9**), technology transfer and know-how for making pandemic-related health products (**Article 11**), PABS (**Article 12**) and procurement and distribution (**Article 14**) [PMID 40883832].

SOURCE & COVERAGE

Oxford Textbook of Public Health 7e (Ch 1.5 Globalization, pp 254-289; Ch 2.1 Determinants of health: overview, pp 306-308; Ch 2.8 Food and nutrition, p 526; Ch 8.17 Emerging and re-emerging infections, pp 2859-2860); Textbook of International Health (Ch 1 Introduction, p 44; Ch 9 Globalization, Trade, Work and Health, pp 454-501, incl. Table 9-1, Table 9-2 and Fig 9-3; Ch 12 Understanding and Organizing Health Care Systems, pp 671-674); Park's Textbook of Preventive and Social Medicine 28e (Ch 5 Epidemiology of Communicable Diseases, p 430; Ch 6 Epidemiology of Chronic Non-communicable Diseases, p 438; Ch 14 Sustainable Development Goals, p 853). Named instruments cited with their years: International Health Regulations (2005), WHO Framework Convention on Tobacco Control (2003/2005), Doha Declaration on the TRIPS Agreement and Public Health (2001), Bangkok Charter for Health Promotion in a Globalized World (2005). **Recent-advances sources:** 42457294 — Effect of the '100 days TB elimination campaign' on TB notification in Maharashtra, Indian Journal of Tuberculosis 2025, DOI [10.1016/j.ijtb.2025.08.014](https://doi.org/10.1016/j.ijtb.2025.08.014) (abstract-only); 42594133 — Strengthening the National Sickle Cell Elimination Mission through universal access to point-of-care screening test, PLoS One 2026, DOI [10.1371/journal.pone.0355845](https://doi.org/10.1371/journal.pone.0355845); 42545929 — Trends and correlates of smokeless tobacco use among Indian women: evidence from three rounds of the National Family Health Survey 2005–2021, PLoS One 2026, DOI [10.1371/journal.pone.0353704](https://doi.org/10.1371/journal.pone.0353704); 42406715 — Caesarean section epidemic in India: is private sector to blame? A multivariate logistic regression analysis of the National Family Health Survey, PLoS One 2026, DOI [10.1371/journal.pone.0352156](https://doi.org/10.1371/journal.pone.0352156); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, Acta Tropica 2026, DOI [10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42159053 — Monitoring rubella immunity and transmission in central India amidst elimination efforts, Microbiology and Immunology 2026, DOI [10.1111/1348-0421.70065](https://doi.org/10.1111/1348-0421.70065) (abstract-only); 42167215 — Public availability of malaria intervention data will be vital for the acceleration of its control and elimination, American Journal of Tropical Medicine and Hygiene 2026, DOI [10.4269/ajtmh.25-0412](https://doi.org/10.4269/ajtmh.25-0412) (abstract-only); 42169564 — AOGIN India policy statement on the use of HPV vaccination for cervical cancer elimination, Asian Pacific Journal of Cancer Prevention 2026, DOI [10.31557/APJCP.2026.27.5.1603](https://doi.org/10.31557/APJCP.2026.27.5.1603) (abstract-only); 40883832 — To tier or not to tier: the institutionalization of the World Health Organization's power to determine pandemic emergency in the amended International Health Regulations (2005), Global Health Research and Policy 2025, DOI [10.1186/s41256-025-00438-6](https://doi.org/10.1186/s41256-025-00438-6); 41464441 — Global health governance and the WHO Pandemic Agreement: a scoping review of challenges and analysis of reforms, International Journal of Environmental Research and Public Health 2025, DOI [10.3390/ijerph22121807](https://doi.org/10.3390/ijerph22121807). Instruments cited with their dates: amended International Health Regulations (2005) adopted 1 June 2024 at the 77th World Health Assembly; WHO Pandemic Agreement, resolution WHA78.1, adopted 20 May 2025.



Health problems of nomads

Asked in: Paper III May 2017 — RGUHS MD Community Medicine

Definition — Nomads are populations without a fixed residence who move cyclically over a defined territory, following pasture for domesticated herds, game and wild vegetation, or seasonal trade and labour. Sociologically they are the **foraging (hunter–gatherer)** and **pastoral** societies — the Pygmies of Central Africa and the Bedouins of Africa and the Middle East respectively (Sociology: The Essentials 6e, Ch 5, pp 131-132). In Indian public health practice a nomadic settlement is not a residual curiosity but a formally coded service-delivery unit: **high-risk area (HRA) code 2** — "Nomads" in the routine immunisation microplan (Routine Immunization Manual for Medical Officers 2024, Unit 4, p 72).

Determinants that generate the health problems

Determinant	Mechanism	Health consequence
Mobility itself	Periodic migration of nomads and wandering tribes is an outstanding example of internal migration (Park 28e, Ch 5, p 321)	Break in continuity of care; re-introduction of parasites into cleared areas
No fixed residence	Nomads without a fixed residence and without identity cards have strong disincentives to civil registration (Textbook of International Health, Ch 5, p 261)	Invisible in census and vital statistics; no entitlement documents
Livestock dependence	Constant close contact with cattle, sheep and goats and their birth products	Zoonoses — taeniasis, brucellosis, anthrax
Shelter and habits	Sleeping out of doors, simple dwellings, refusal to accept residual spraying, replastering of sprayed walls, non-use of bed nets (Park 28e, Ch 5, p 321)	Sustained man–vector contact; failure of vector control
Harsh thermal environment	Hot arid or very cold terrain, with behavioural adaptation and clothing microclimate as the only available control (Hunter's Diseases of Occupations, Ch 49, p 550)	Heat and cold strain
Political and economic marginalisation	Nationalisation of land has forced nomadic hunters and pastoralists onto unproductive land (Textbook of International Health, Ch 7, p 373)	Loss of livelihood, distress migration to towns
Fear of the State	Families that fear contact with government — those lacking documents, nomadic groups, homeless families (Routine Immunization Manual 2024, Unit 11, p 259)	Active avoidance of health services

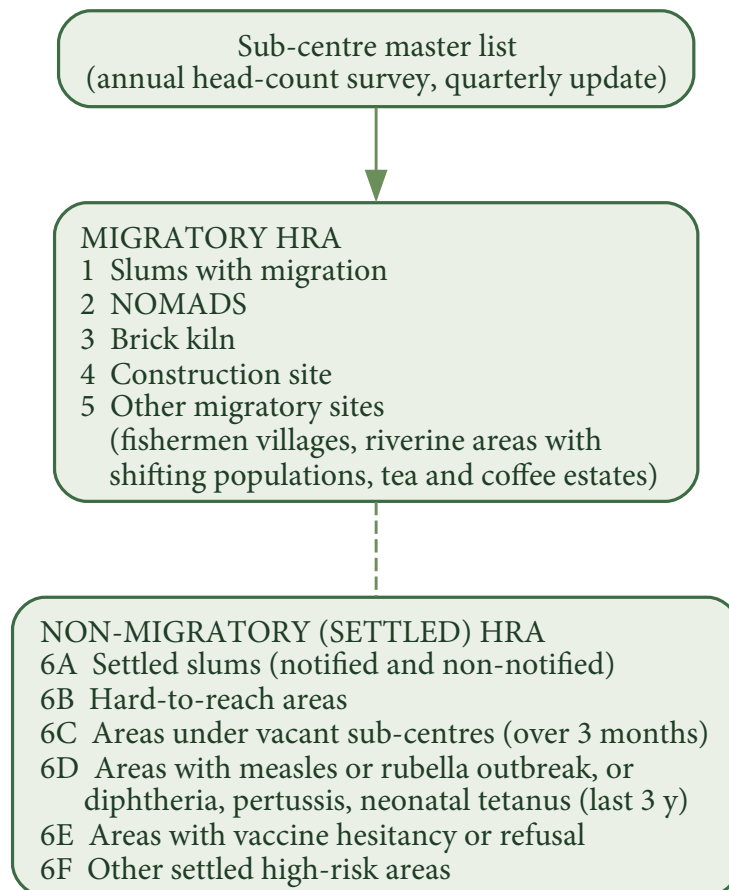
Health problems

Group	Specific problems and evidence
Vector-borne	Nomads may import malaria parasites in their blood and re-introduce malaria into areas where it has been controlled or eliminated (Park 28e, Ch 5, p 321). Nomadism is itself listed among the socio-economic obstacles to malaria control, alongside refusal to spray dwellings and replastering of sprayed walls (Gilles & Warrell, Essential Malariology 4e, Ch 6, pp 195-196)

Group	Specific problems and evidence
Vaccine-preventable	A named cause of failure to vaccinate susceptible persons against measles is "populations who are difficult to reach (e.g. street children, nomadic persons)" (Nelson & Williams, Ch 16, p 519). For polio , nomads are cited as populations that are difficult to survey, so virus circulation can be missed (Plotkin's Vaccines 7e, Ch 49, pp 1280-1281). Historically, smallpox seasonality was amplified by festivals, marriage parties and the seasonal movement of nomads (Plotkin's Vaccines 7e, Ch 54, p 1470)
Zoonoses (One Health)	Taeniasis — the highest rates of human <i>Taenia saginata</i> infection have been reported in Africa among nomadic cattle herders (Maxcy-Rosenau-Last 15e, Ch 16, p 473). Brucellosis — largely occupational, by contact with infected tissues, blood, urine, vaginal discharge, aborted fetuses and especially placenta, entering through abraded skin, mucosa or conjunctiva (Park 28e, Ch 5, p 356). Anthrax — soil is the natural reservoir, and epizootics persist wherever livestock vaccination is not comprehensive (Maxcy-Rosenau-Last 15e, Ch 16, p 455)
Tuberculosis	Migrants, displaced and undocumented people and refugees are named as people with limited access to quality tuberculosis services (Park 28e, Ch 5, p 255); transit camps, migrants and the homeless are explicit targets of campaign-mode active case finding under the national programme (Sunder Lal, Ch 8, p 359)
Nutrition and food insecurity	Conflict-driven food insecurity drove rural and nomadic tribes into towns in southern Sudan in the 1990s, with an estimated 78,000 deaths in 1998 alone (Textbook of International Health, Ch 8, p 416)
Occupational and environmental	Thermal extremes of hot arid and very cold terrain, managed only by activity patterning and layered clothing rather than by any engineering control (Hunter's, Ch 49, p 550)

Information and health-system problems — the distinctive difficulty

- **Denominator failure.** The existence of nomadic populations is specifically named as a factor that makes a census count a formidable task (Textbook of International Health, Ch 5, p 245), and under-reporting is more severe among nomads than in the general population (Reinke, Ch 5, p 85).
- **Distorted rates.** Births among nomads are consistently under-reported; where a birth is followed by early death, neither event is registered, so numerator and denominator are both corrupted (Reinke, Ch 9, p 139). Any coverage or mortality indicator quoted for these groups is therefore uninterpretable at face value.
- **Disincentives to registration** — no identity documents, suspicion that records will be used for taxation or enforced military service, fees or bribes demanded for certificates, and the need of a death certificate before burial in remote areas (Textbook of International Health, Ch 5, p 261).
- **Discontinuity of care across the migration cycle.** Migration health has three distinct but interdependent phases — **pre-departure, the journey itself, and arrival at destination** — each shaping outcomes through social networks, language, food security, shelter, sanitation and access to services (Gushulak and MacPherson, in Oxford Textbook of Public Health 7e, Ch 92, p 3455).



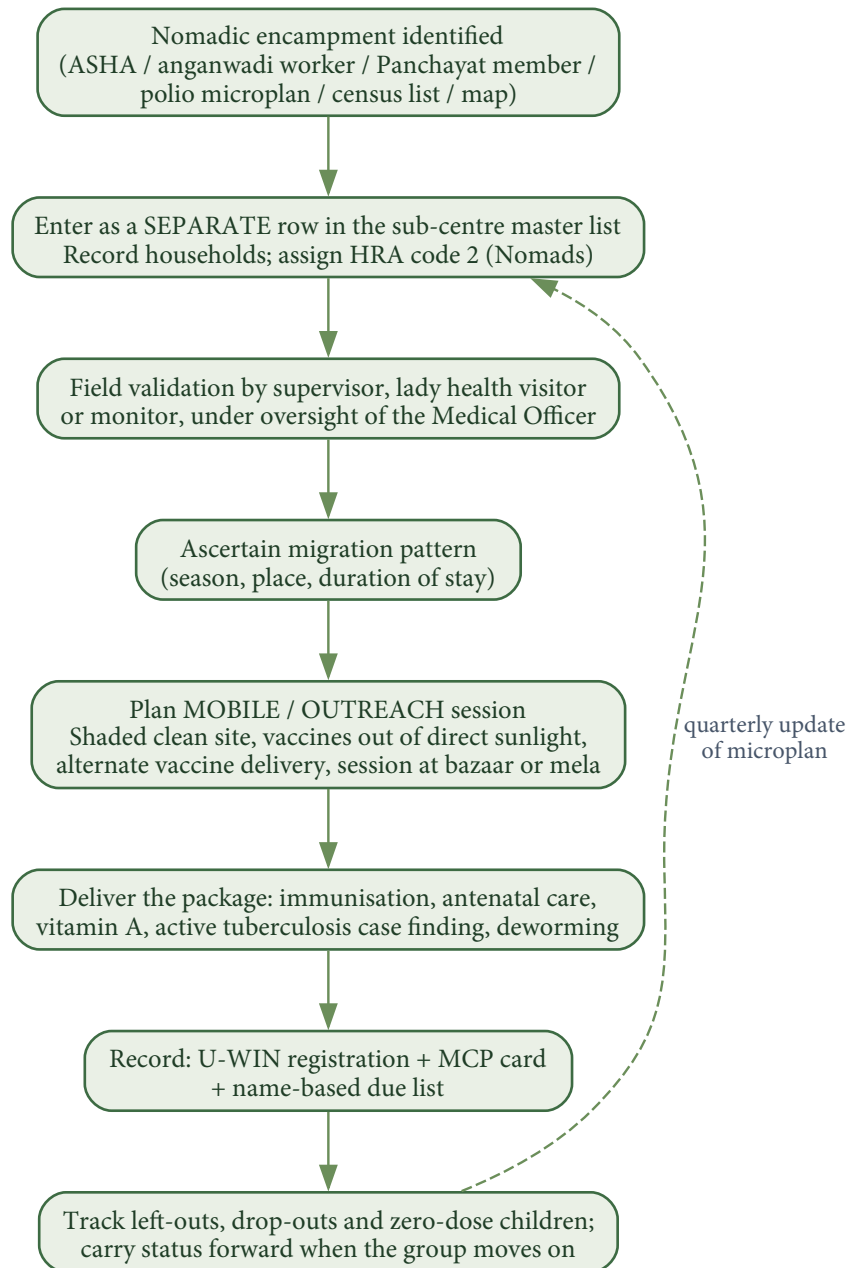
Where nomads sit in the high-risk area (HRA) classification of the Indian routine immunisation microplan

Prevention and control — the programme response in India

- **Enumerate before you plan.** Every nomadic site must be entered as a **separate row** in the sub-centre master list with its household count recorded or approximated; the sources are existing polio microplans, census lists, maps, high-risk-area lists and interaction with the accredited social health activist (ASHA), the anganwadi worker and Panchayati Raj Institution members (Routine Immunization Manual 2024, Unit 4, p 72).
- **Validate in the field.** Migratory sites missing from the microplan are added immediately with revision of the plan, validation being done by supervisors under the oversight of the medical officer (Unit 14, p 303). Nomad high-risk areas are a stated priority for supervisory visits (Unit 10, pp 239-240).
- **Fit the service to the migration pattern.** Ascertain the season, place and duration of movement and include those beneficiaries in the microplan and due list; use alternate vaccine delivery and hold outreach sessions at local bazaars, melas and festivals (Unit 14, p 315). At nomadic sites and brick kilns choose a clean, convenient, shaded place with the vaccines protected from direct sunlight, and plan mobile sessions (Unit 4, pp 110-111).
- **Make the record portable.** Register the beneficiary on U-WIN, the digital immunisation platform, which permits registration of a child by linkage to a parent or guardian and issues a QR-based vaccination certificate, and is interoperable with e-VIN, the Ayushman Bharat Digital Mission, the reproductive and child health portal and POSHAN tracker (Unit 15, p 324). With

the Mother and Child Protection (MCP) card and the name-based due list, this lets a child's vaccination status follow the family when the group moves (Unit 8, p 197).

- ▶ **Overcome the fear of the State.** Determine where these populations reside, visit them, and work through local mobilisers, educators, community groups and leaders; offer "on-demand vaccination" with day, time and venue agreed with the community (Unit 11, pp 259-260).
- ▶ **Campaign platform — Intensified Mission Indradhanush.** An equity-based drive built on the strategies, tools, techniques and human resources of the Pulse Polio Programme, explicitly focused on **nomadic sites**, brick kilns, construction sites, other migrant settlements (fishermen villages, riverine areas with shifting populations) and hard-to-reach forested, tribal and hilly populations; it aims for **more than 90 per cent full immunisation coverage**, paired with routine-immunisation system strengthening to sustain the gains (Mission Indradhanush Guidelines, pp 19-20, 23-24).
- ▶ **Case-finding platform.** Vulnerability mapping followed by periodic screening, community screening at a mobile or fixed facility, and door-to-door household screening are the programme's tools for reaching vulnerable groups for tuberculosis (Park 28e, Ch 5, p 226).
- ▶ **Relaxed infrastructure norms** apply to the terrain such groups traverse — one health sub-centre per 3,000, one primary health centre per 20,000 and one community health centre per 80,000 population in tribal and hilly areas (Park 28e, Ch 13, p 850); the Health and Wellness Centre–Sub-Health Centre norm is 3,000 in hilly and tribal areas against 5,000 in the plains (Sunder Lal, Ch 17, p 746). Indian Public Health Standards 2022 and the National Urban Health Mission both name **temporary migrants, the homeless, street children, rag-pickers, and construction and lime-kiln workers** as the target vulnerable population (Park 28e, Ch 10, p 728 and Ch 7, pp 535-536).
- ▶ **One Health for the zoonoses.** Annual immunisation of livestock is recommended in areas of endemic anthrax, using the avirulent **Sterne strain live spore vaccine** given **2–4 weeks before the season in which outbreaks are expected** (Maxcy-Rosenau-Last 15e, Ch 16, p 456).
- ▶ **Global practice worth citing.** Cross-border vaccination, permanent vaccination posts at **transit points and on market days**, and increased surveillance of migrant and nomadic populations (WHO, Integrating Migrants and Refugees into National Immunization Programmes, p 42). Participatory geographic information systems using global positioning system units have been used to site primary health care services for nomadic tribes in Chad (Lilienfeld's Foundations of Epidemiology, Ch 14, pp 291-292).



Operational algorithm for delivering primary health care to a nomadic encampment

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The three-phase migration health model** (Gushulak and MacPherson) — three linked boxes in a cycle, labelled exactly: **1. Pre-departure phase** (baseline disease burden, immunisation status, nutrition); **2. The migratory journey** (exposure en route, interruption of treatment, injury, no health record); **3. Arrival at destination** (shelter, sanitation, safe water, food security, language, entitlement, access to services); with a curved **return / onward movement** arrow from box 3 back to box 1 to show the annual cycle of the nomad.

High-yield exam pointers

- ▶ **HRA codes to reproduce verbatim:** 1 slums with migration, 2 nomads, 3 brick kiln, 4 construction site, 5 other migratory sites; 6A–6F are the settled high-risk categories (Routine Immunization Manual 2024, Unit 4, p 72).
- ▶ **Intensified Mission Indradhanush mnemonic:** *Identify, enlist, mobilize, vaccinate, monitor and report* — target more than **90 per cent full immunisation coverage**, focused on nomadic sites, brick kilns, construction sites and other migrant settlements.
- ▶ **Park's malaria determinants that name nomads:** population mobility (re-introduction of parasites) and human habits (sleeping out of doors, nomadism, refusal of spraying, replastering of sprayed walls, non-use of bed nets) — Park 28e, Ch 5, p 321.
- ▶ **Zoonosis one-liner:** highest human *Taenia saginata* rates reported in Africa among **nomadic cattle herders**; anthrax livestock vaccine = **Sterne strain, 2–4 weeks before the outbreak season**.
- ▶ **Denominator caution:** nomads are under-enumerated in both census and civil registration, so any rate quoted for them is distorted in numerator *and* denominator.
- ▶ **Relaxed Indian norms:** sub-centre 1:3,000, primary health centre 1:20,000, community health centre 1:80,000 in tribal and hilly areas; Health and Wellness Centre–Sub-Health Centre 3,000 (hilly/tribal) versus 5,000 (plain).
- ▶ **Portable record** = U-WIN + MCP card + **name-based due list** — the single most examinable modern answer to discontinuity of care in a mobile population.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Textbook 28e has no dedicated chapter on nomadic health and mentions nomads only within malaria epidemiology, so this answer rests on the migratory high-risk-area guidance of the Routine Immunization Manual 2024 and the Mission Indradhanush guidelines as the operational Indian sources; a national estimate of the size of India's nomadic population, and any scheme named specifically for de-notified and nomadic communities, could not be verified from the corpus and is therefore not quoted.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5, Ch 7, Ch 10, Ch 13); Routine Immunization I Manual for Medical Officers, Ministry of Health and Family Welfare, 2024 (Units 4, 8, 10, 11, 14, 15); Mission Indradhanush / Intensified Mission Indradhanush Guidelines, Government of India; Sunder Lal — Textbook of Community Medicine (Ch 8, Ch 17); Textbook of International Health (Ch 5, Ch 7, Ch 8); Oxford Textbook of Public Health 7e (Ch 92); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 16); Nelson & Williams — Infectious Disease Epidemiology (Ch 16); Plotkin's Vaccines 7e (Ch 49, Ch 54); Gilles & Warrell — Essential Malariology 4e (Ch 6); Hunter's Diseases of Occupations (Ch 49); Reinke — Health Planning for Effective Management (Ch 5, Ch 9); Lilienfeld's Foundations of Epidemiology (Ch 14); Sociology: The Essentials 6e (Ch 5); WHO — Integrating Migrants and Refugees into National Immunization Programmes.



International agencies promoting health in India

Asked in: Paper IV Nov 2017 — RGUHS MD Community Medicine

Definition — International health agencies are intergovernmental, bilateral, philanthropic or non-governmental bodies that operate across national borders to improve health, and that support a host country with technical advice, trained manpower, supplies, equipment and funds. In India most of them work through the Ministry of Health and Family Welfare and the State health departments, providing technical and material assistance in the implementation of the national health programmes (Park 28e, Ch 7, p 493), while a few voluntary bodies such as the Red Cross and CARE also deliver services directly (Park 28e, Ch 10, p 747; Ch 25, pp 1060–1061).

Classification of international agencies working in India

Category	Agencies active in India	Nature of assistance
Multilateral — United Nations system	World Health Organization (WHO), United Nations Children's Fund (UNICEF), United Nations Development Programme (UNDP), United Nations Fund for Population Activities (UNFPA), Food and Agriculture Organization (FAO), International Labour Organization (ILO), United Nations Educational, Scientific and Cultural Organization (UNESCO)	Technical norms, standards, surveillance, supplies, grants
International financial and economic institutions	World Bank, International Monetary Fund (IMF), World Trade Organization (WTO)	Setting macroeconomic policy and establishing and overseeing trade rules, together with sector-specific grants and loans carrying incentives and conditionalities (Textbook of International Health, Ch 3, pp 117–118)

Category	Agencies active in India	Nature of assistance
Bilateral aid and development agencies	United States Agency for International Development (USAID), Swedish International Development Agency (SIDA), Danish International Development Agency (DANIDA), Norwegian Agency for Development Cooperation (NORAD); the Colombo Plan as a regional cooperative arrangement	Government-to-government grants, loans, equipment, fellowships
Foundations and non-governmental organizations	Rockefeller Foundation, Ford Foundation, Cooperative for Assistance and Relief Everywhere (CARE), International Red Cross, Bill and Melinda Gates Foundation	Philanthropic grants, institution-building, research, relief; some direct service delivery
Global public-private partnerships	GAVI Alliance (formerly the Global Alliance for Vaccines and Immunization), The Global Fund to Fight AIDS, Tuberculosis and Malaria	Pooled disease-specific financing

(Indian agency list from Park 28e, Ch 7, p 493 and Ch 10, p 748.)

World Health Organization — the principal agency

WHO is the specialized, non-political health agency of the United Nations, with headquarters at Geneva. Its Constitution was drafted in 1946 by a Technical Preparatory Committee chaired by *Rene Sand* and approved by an International Health Conference of 51 nations at New York; it came into force on **7 April 1948**, observed every year as World Health Day. Its objective, from the preamble, is "the attainment by all peoples of the highest possible level of health". Membership is open to all countries; from 56 Members in 1948 WHO now has **194 Member States and two associate members**, associate members participating without a vote. India has been a Member State since 1948 (Park 28e, Ch 25, pp 1055–1058).

Organ	Composition	Main functions
World Health Assembly — the "health parliament of nations", the supreme governing body	Delegates of all Member States, one vote each; meets annually, usually in May, at Geneva	Determine international health policy and programmes; review the past year's work; approve the next year's budget; elect Member States to designate persons to the Executive Board; appoint the Director-General on the Board's nomination; hold "technical discussions"
Executive Board	34 members, technically qualified in health, designated by but not representing their governments; not fewer than three from each region; one-third renewed yearly; meets at least twice a year (January, and after the Assembly in May)	Give effect to the decisions and policies of the Assembly; act on its own in emergencies such as epidemics, earthquakes and floods
Secretariat	Headed by the Director-General, the chief technical and administrative officer; international public servants at Geneva and the regional offices	Provide Member States with technical and managerial support for their health development programmes

Six regional organizations (Park 28e, Ch 25, p 1057) — each headed by a Regional Director, with a Regional Committee of the Member States of the region that meets once a year:

Region	Regional office
South-East Asia (India's region)	New Delhi — World Health House, Indraprastha Estate
Africa	Brazzaville, Congo
The Americas	Washington D.C., USA
Europe	Copenhagen, Denmark
Eastern Mediterranean	Alexandria, Egypt
Western Pacific	Manila, Philippines

Eight broad areas of WHO work (Park 28e, Ch 25, pp 1056–1057):

- ▶ **Prevention and control of specific diseases** — epidemiological surveillance of communicable diseases, disseminated through the *Weekly Epidemiological Record*; global eradication of smallpox and the ongoing battle against poliomyelitis; the Expanded Programme on Immunization.
- ▶ **Development of comprehensive health services** — WHO's single most important function, built around primary health care and Appropriate Technology for Health.
- ▶ **Family health** — a major programme since 1970; maternal and child health care, human reproduction, nutrition and health education.
- ▶ **Environmental health** — advice on basic sanitary services; protection of air, water and food quality; the WHO Environmental Health Criteria and Environmental Health Monitoring Programmes.
- ▶ **Health statistics** — morbidity and mortality data; the *International Classification of Diseases*.
- ▶ **Biomedical research** — including the Special Programme for Research and Training in Tropical Diseases.
- ▶ **Health literature and information** — WHO acts as a clearing house; its library is a satellite centre of the Medical Literature Analysis and Retrieval System (MEDLARS).
- ▶ **Cooperation with other organizations** — with the United Nations, its specialized agencies and international governmental organizations.

Other United Nations agencies and their work in India

Agency	Founding / office	Work in India
UNICEF (United Nations Children's Fund; established 1946 by the General Assembly as the International Children's Emergency Fund for children of war-ravaged countries, renamed 1953 with the initials retained)	Headquarters at the United Nations, New York; regional office at Kathmandu, Nepal (South Central Asian Region — Afghanistan, Sri Lanka, India, Maldives, Pakistan, Bhutan, Bangladesh, Nepal); 36-nation Executive Board	Supported India's BCG vaccination programme from its inception ; aid for production of vaccines and sera; a penicillin plant near Pune, a donated DDT plant, plants for triple vaccine and iodized salt; dairy plant equipment in several States; vitamin A where xerophthalmia is prevalent, iodized salt in endemic goitre, iron and folate for anaemia; rural water supply and environmental sanitation. Promotes the GOBI campaign for a "child health revolution" — G growth charts to monitor child development, O oral rehydration for all mild and moderate dehydration, B breast feeding, I immunization against measles, diphtheria, polio, pertussis, tetanus and tuberculosis; and since 1976 participates in Urban Basic Services (UBS) to upgrade health, nutrition, water supply, sanitation and education for children and women in selected cities and towns (Park 28e, Ch 25, pp 1057–1059)
UNDP (established 1966)	Funded by annual pledges from rich and poor member countries; the main source of funds for technical assistance	Helps poorer nations develop human and natural resources; projects across agriculture, industry, education and science, health and social welfare (Park 28e, Ch 25, p 1059)
UNFPA	Assisting India since 1974	Funds national-level schemes and Area Projects for intensive development of health and family welfare infrastructure in 11 districts of Bihar and 4 districts of Rajasthan ; national capability for contraceptive manufacture; population education; strengthening programme management (Park 28e, Ch 25, p 1059)
FAO — formed 1945, the first UN specialized agency created for world cooperation; headquarters Rome	Aims to raise living standards, improve nutrition, increase the efficiency of farming, forestry and fisheries and better the condition of rural people	Runs the Freedom from Hunger Campaign (FFHC) to combat malnutrition and disseminate information and education; collaborates in the Applied Nutrition Programmes; joint WHO/FAO expert committees underpin nutritional surveys and training on brucellosis and other zoonoses; with UNICEF and WHO promotes national food and nutrition policies that provide for child nutrition (Park 28e, Ch 25, pp 1058–1059)

Agency	Founding / office	Work in India
ILO — established 1919 as an affiliate of the League of Nations; headquarters Geneva	Purposes: lasting peace through social justice, better labour conditions and living standards, economic and social stability	Maintains the International Labour Code , a collection of international minimum standards on the health, welfare, living and working conditions of workers — the basis of occupational health practice; assists organizations working to better living and employment standards, in close collaboration with WHO (Park 28e, Ch 25, p 1059)
UNESCO	UN specialized agency	With UNICEF, assists India in expanding and improving science teaching — laboratory equipment, workshop tools, library books, audiovisual aids (Park 28e, Ch 25, p 1058)

World Bank and bilateral agencies

Agency	Assistance to India
World Bank — a UN specialized agency whose powers are vested in a Board of Governors	Loans for projects that lead to economic growth, e.g. India's Population Projects ; projects on electric power, roads, railways, agriculture, water supply, education and family planning, with health and environmental components added; cooperative programmes with WHO on water supply, the World Food Programme and population control (Park 28e, Ch 25, p 1059)
USAID — created 1961, taking over activities of the Technical Cooperation Mission; the United States also aids India through the Public Law 480 (Food for Peace) Programme and the US Export-Import Bank	Grants and loans for malaria eradication, medical education, nursing education, health education, water supply and sanitation, control of communicable diseases, nutrition and family planning (Park 28e, Ch 25, pp 1059–1060)
The Colombo Plan — drawn up at Colombo in January 1950 ; 20 developing member countries within the region plus 6 non-regional members (Australia, Canada, Japan, New Zealand, UK, USA)	Bulk of assistance goes to industrial and agricultural development, health promotion mostly through fellowships and visits by experts; the All India Institute of Medical Sciences, New Delhi was established with financial assistance from New Zealand ; Canada supplied cobalt therapy units to Indian medical institutions (Park 28e, Ch 25, p 1060)
SIDA (Sweden)	Assisting the National Tuberculosis Elimination Programme (formerly the National Tuberculosis Control Programme) since 1979 — X-ray units, microscopes and anti-tuberculosis drugs (Park 28e, Ch 25, p 1060; renaming, Ch 7, p 508)
DANIDA (Denmark)	Assisting development of services under the National Programme for Control of Blindness and Visual Impairment (formerly the National Blindness Control Programme) since 1978 (Park 28e, Ch 25, p 1060; current name, Ch 6, p 480)

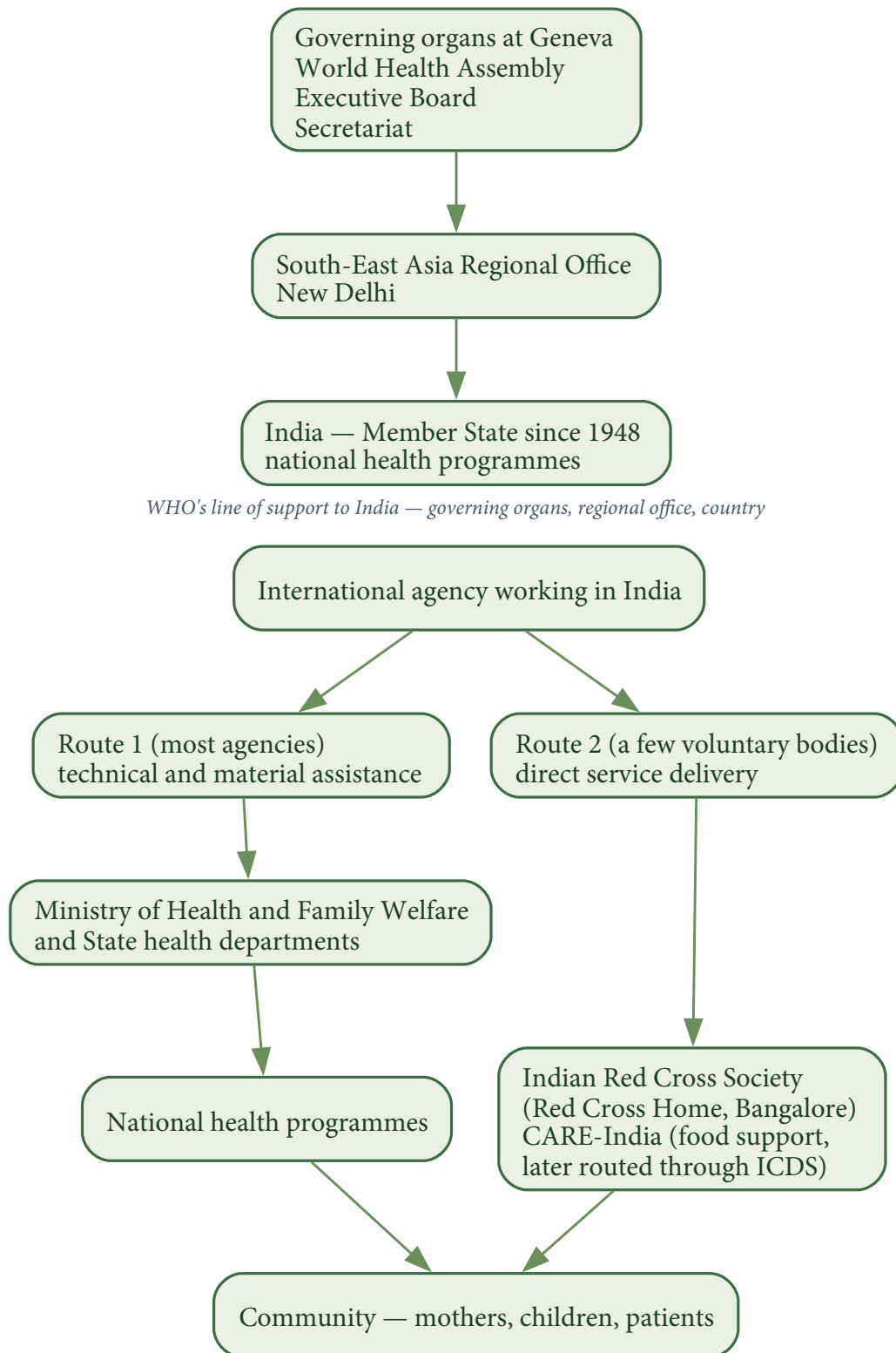
Agency	Assistance to India
NORAD (Norway)	Technical and material assistance to the national health programmes (Park 28e, Ch 7, p 493)

Foundations, non-governmental agencies and partnerships

Agency	Contribution in India
Rockefeller Foundation — chartered 1913, endowed by John D. Rockefeller	Work in India began 1920 with a scheme for hookworm control in the then Madras Presidency; the All India Institute of Hygiene and Public Health, Kolkata owes its establishment in large measure to the Foundation; training of teachers and research workers, overseas fellowships and travel grants, visits of medical specialists from the USA, grants-in-aid, medical college libraries, population studies, support to the National Institute of Virology, Pune and to the field practice area at Ballabgarh attached to AIIMS; now directs support to agriculture, family planning, rural training centres and medical education (Park 28e, Ch 25, p 1060)
Ford Foundation — active in the development of rural health services and family planning rather than in universities	(1) Orientation training centres at Singur, Poonamalle and Najafgarh , giving public health training courses to medical and paramedical personnel from all over India; (2) research-cum-action projects in environmental sanitation, e.g. the hand-flushed sanitary latrine for rural areas; (3) pilot project in rural health services, Gandhigram (Tamil Nadu) — a coordinated health service for a rural population of 100,000 as a model for health administrators; (4) establishment of the National Institute of Health Administration and Education, Delhi for staff-college type training of health administrators; (5) master plan for water supply, sewerage and drainage for Kolkata; (6) research in reproductive biology and family planning fellowships (Park 28e, Ch 25, p 1060)
CARE (Cooperative for Assistance and Relief Everywhere) — founded in North America in 1945	Began in India in 1950; till the end of the 1980s its primary object was to supply food directly to children aged 6–11 years ; from the mid-1980s it focused its food support within the Integrated Child Development Services (ICDS) ; runs the Integrated Nutrition and Health, Better Health and Nutrition, Anaemia Control, Child Survival, Improving Women's Health, Adolescent Girls' and Konkan Integrated Development projects in partnership with the Government of India, State governments and NGOs (Park 28e, Ch 25, pp 1060–1061)

Agency	Contribution in India
International Red Cross — founded by <i>Henry Dunant</i>, a young Swiss businessman, after the battle of Solferino in North Italy in 1859; the First Geneva Convention of 1864 created the International Committee of the Red Cross; the League of Red Cross Societies followed in 1919 at Geneva	Service to armed forces and war veterans, disaster service, first aid and nursing, health education, and maternity and child welfare. The Indian Red Cross Society, created by an Act of the Indian Legislature in 1920 and with a network of branches all over India, works for improvement of health, prevention of disease and mitigation of suffering; it gives relief in earthquakes, floods, epidemics and famine, and itself runs a well-equipped hospital, the "Red Cross Home" at Bangalore for permanently disabled ex-servicemen — the only one of its kind in India and the Far East; it administers or affiliates maternity and child welfare centres, runs family planning clinics and blood banks, and its Junior Red Cross does village uplift, first aid and anti-epidemic work (Park 28e, Ch 25, p 1061; Ch 10, pp 746–747)
Other NGOs	Oxfam, Save the Children Fund, Family Planning Association of India, International Planned Parenthood Federation, the Population Council, All India Women's Conference, Indian Medical Association, Trained Nurses Association of India, the International Agency for the Prevention of Blindness (built in 1974 at WHO's invitation, its primary task the prevention of blindness), World Federation of the Deaf, International Leprosy Association, International Union against Cancer. <i>Vision 2020: The Right to Sight</i>, the global initiative to eliminate avoidable blindness, was launched by WHO on 18 February 1999 (Park 28e, Ch 25, p 1061; Ch 6, p 480)
Global public–private partnerships	The GAVI Alliance, Geneva — a partnership of philanthropic foundations, UNICEF, WHO, the World Bank, private business and NGOs that builds on WHO's Expanded Programme on Immunization to widen children's access to vaccines. The Global Fund to Fight AIDS, Tuberculosis and Malaria, the largest such partnership, was established as an independent Swiss foundation in 2002; it raises money, reviews proposals and disburses grants rather than implementing programmes itself, and with the World Bank, the Bill and Melinda Gates Foundation and the United States government accounts for a large share of global health financing (Textbook of International Health, Ch 3, pp 143–144; Ch 11, p 599)

Diagrams



Two routes by which international assistance reaches people in India

High-yield exam pointers

- WHO: Constitution in force 7 April 1948 (World Health Day); drafted 1946 under *Rene Sand*; approved by 51 nations at New York; headquarters Geneva; 194 Member States + 2 associate

members; three organs — Assembly, Executive Board (34 members), Secretariat; six regions; SEARO at World Health House, New Delhi; India a member since 1948.

- ▶ Bilateral aid mnemonic — "Sweden for TB, Denmark for the eyes": SIDA → National Tuberculosis Elimination Programme since 1979; DANIDA → National Programme for Control of Blindness and Visual Impairment since 1978.
- ▶ UNICEF — GOBI = Growth charts, Oral rehydration, Breast feeding, Immunization; Urban Basic Services since 1976; regional office Kathmandu, headquarters New York; supported India's BCG programme from its inception.
- ▶ UNFPA in India since 1974 — Area Projects in 11 districts of Bihar and 4 of Rajasthan.
- ▶ Institutions built with foreign help — AIIMS New Delhi (Colombo Plan, New Zealand funds), All India Institute of Hygiene and Public Health, Kolkata (Rockefeller), NIHAIE Delhi and the Singur–Poonamalle–Najafgarh orientation training centres (Ford Foundation), Gandhigram rural health pilot project (Ford Foundation).
- ▶ CARE: founded 1945, India since 1950, food for children aged 6–11 years, later routed through ICDS.
- ▶ Red Cross: Henry Dunant, Solferino 1859 → First Geneva Convention 1864 → ICRC → League of Red Cross Societies 1919; Indian Red Cross Society by Act of 1920.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's chapter on international health is written largely in the historical present, so quote its figures as the textbook prints them. Park still prints Alexandria as the Eastern Mediterranean regional office; the office has been at Cairo since 1998. Current programme partnerships — GAVI support for pentavalent and rotavirus vaccine introduction, and Global Fund support to the National AIDS Control Programme and the National Tuberculosis Elimination Programme — are described in the retrieved corpus only in general terms, without Indian programme detail.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, International Health Regulations (2005) 2024 amendments, WHO Pandemic Agreement (WHA78.1), Gavi, The Global Fund

Programme and policy updates (India)

- ▶ The USAID row of the bilateral table is now historical rather than current. Executive Order 14169 of 20 January 2025 suspended United States foreign assistance, and on 10 March 2025 the cancellation of 83% of USAID's programmes was announced; the forecasting scenario models outright termination of the agency's funding from 2026. A *Lancet* impact evaluation across 133 low- and middle-income countries estimated that USAID funding averted 91,839,663 deaths (95% CI 85,690,135–98,291,626) between 2001 and 2021, including 30.4 million in children under five, and projected 14,051,750 (uncertainty interval 8,475,990–19,662,191) excess deaths by 2030 — 4.54 million of them under five — if the cuts stand, putting SDG target 3.2 (25 or fewer under-five deaths per 1000 live births by 2030) out of reach for the poorest countries. [PMID 40609560]

- **Gavi's organising idea is now the "zero-dose" child** — operationally, a child who has not received a first dose of DTP-containing (pentavalent) vaccine — rather than the introduction of new antigens. Reaching these children sits at the core of Gavi 5.0/5.1 (the 2021–2025 strategy) and Immunization Agenda 2030, both aligned to the IRMMA framework — Identify, Reach, Monitor, Measure, Advocate. Gavi's Zero-Dose Learning Hubs run prospectively from 2022 to 2025 in Mali, Nigeria, Uganda and Bangladesh, with the International Institute of Health Management Research, New Delhi sitting in the global learning consortium — India supplying technical expertise rather than receiving it. Gavi separately announced a US\$250 million partnership with India in February 2023 to reach zero-dose children. [PMID 41805772] [Gavi 2023-Feb]
- **India now hosts a WHO global centre instead of only receiving assistance.** The WHO Global Traditional Medicine Centre was established at Jamnagar, Gujarat in 2022 — WHO's only centre of its kind — with foundational funding from the Government of India; a useful counter-example when the question implies that assistance flows only one way. [WHO 2022]

Current national data

- **The current international datasets on India's burden are the Global Burden of Disease Study 2023 and the kala-azar programme returns.** GBD 2023, published in 2025, covers 375 diseases and injuries and 88 risk factors and produces subnational (State-level) estimates for 660 locations in 20 countries, India among them; globally DALYs rose 6.1% (95% UI 4.0–8.1) from 2.64 billion in 2010 to 2.80 billion in 2023 while the age-standardised DALY rate fell 12.6% (11.0–14.1), non-communicable disease DALYs rising from 1.45 to 1.80 billion as communicable, maternal, neonatal and nutritional DALYs fell from 874 to 681 million. For kala-azar, as of January 2025 India had 165.4 million people at risk in 633 endemic blocks across 54 districts, 458 of those blocks in Bihar, with deaths down from 1419 in 1992 to four in 2023. [PMID 41092926] [PMID 40037366]

Elimination, surveillance and disease status

- **Trachoma validated, kala-azar not yet.** WHO announced in October 2024 that India had eliminated trachoma as a public health problem; 32 countries have now been validated worldwide. Kala-azar reached the WHO threshold of fewer than one case per 10,000 population at block (sub-district) level in December 2023 and has entered a consolidation phase — the threshold must hold for three consecutive years before India can submit a dossier to WHO for validation. The programme has run since a 2005 memorandum of understanding with Bangladesh and Nepal, financed mainly by the Government of India with implementation support from the World Bank, the Bill and Melinda Gates Foundation and the UK Department for International Development (now the Foreign, Commonwealth and Development Office) between 2014 and 2019. [PMID 39389595] [PMID 40037366] [WHO 2026]
- **The South-East Asia Region's measles–rubella target has slipped twice.** The 2013 regional goal of measles elimination and rubella control by 2020 was revised in 2019 to measles and rubella elimination by 2023; that date was missed, and a regional consultation in 2023 set **2026** as the next feasible target. Only five of the eleven Member States — Bhutan, DPR Korea, Maldives, Sri Lanka and Timor-Leste — had been verified as of 2023, and regional measles incidence rose from 16.2 to 40.9 cases per million population between 2013 and 2023. [PMID 39460260]

Guideline and legislative updates

- **The International Health Regulations (2005) were amended** on 1 June 2024 by the Seventy-seventh World Health Assembly (resolution WHA77.17) and came into force on 19 September 2025. The amendments add a **pandemic emergency** tier above the public health emergency of international concern, triggered when a communicable disease event spreads or risks spreading widely across countries, exceeds or threatens to exceed the capacity of health systems, causes substantial social or economic disruption, and calls for rapid, equitable and coordinated international action. [WHO 2024-Jun]

Global and WHO developments

- **The WHO Pandemic Agreement was adopted on 20 May 2025** at the Seventy-eighth World Health Assembly (resolution WHA78.1) with 124 votes in favour, none against and 11 abstentions. It creates binding obligations on outbreak reporting, national public health capacities and compliance, and establishes three mechanisms worth naming, numbered here from the treaty text annexed to WHA78.1: a **Pathogen Access and Benefit-Sharing (PABS) System (Article 12)**, a **supply chain and logistics network (Article 13)** and a **Coordinating Financial Mechanism for sustainable financing (Article 18)**. The PABS annex that will make Article 12 operational was left under negotiation, for finalization at the 2026 World Health Assembly; 60 ratifications are required for entry into force. [PMID 41464441] [WHA78.1 2025-May]
- **The United States announced its withdrawal from WHO on 20 January 2025**. In 2022–23 it had provided about US\$218 million in assessed and US\$1.065 billion in voluntary contributions, close to a fifth of the WHO budget, and the Gates Foundation is now WHO's second-largest voluntary contributor — a shift that concentrates agenda-setting in a small group of private donors and raises questions about transparency and about priorities being set away from the countries served. Quote the membership as 194 Member States, as Park does, but flag the withdrawal whenever the question turns to WHO finance or governance. [PMID 40767200]
- **The Global Fund's Eighth Replenishment raised US\$12.64 billion** at the pledging summit held in Johannesburg on 21 November 2025, covering 2026–2028 — the cycle that funds the pooled AIDS, tuberculosis and malaria grants described in the partnerships table above. [Global Fund 2025-Nov]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 25 (International Health), pp 1055–1063; Ch 10 (Health Care of the Community), pp 746–748; Ch 7 (Health Programmes in India), pp 493, 508; Ch 6 (Epidemiology of Chronic Non-communicable Diseases and Conditions), p 480. Textbook of International Health — Ch 3 (International Health Agencies, Activities, and Other Actors), pp 117–126, 143–144; Ch 11 (Health Economics and the Economics of Health), p 599.

Recent-advances sources: 40609560 — Evaluating the impact of two decades of USAID interventions and projecting the effects of defunding on mortality up to 2030, *Lancet* 2025, doi:[10.1016/S0140-6736\(25\)01186-9](https://doi.org/10.1016/S0140-6736(25)01186-9); 41805772 — Gavi's Zero-Dose Learning Hubs: enhanced approach for evidence generation and use from local to global, *Health Policy and Planning* 2026, doi:[10.1093/heapol/czag028](https://doi.org/10.1093/heapol/czag028); 41092926 — Burden of 375 diseases and injuries, risk-attributable burden of 88 risk factors, and healthy life expectancy, 1990–2023 (GBD 2023), *Lancet* 2025, doi:[10.1016/S0140-6736\(25\)01637-X](https://doi.org/10.1016/S0140-6736(25)01637-X); 40037366 — Kala-azar elimination in India: reflections on success and sustainability, *International Health* 2025, doi:[10.1093/inthealth/ihaf013](https://doi.org/10.1093/inthealth/ihaf013); 39389595 — India has eliminated trachoma as a public health problem, reports WHO, *BMJ* 2024, doi:[10.1136/bmj.q2232](https://doi.org/10.1136/bmj.q2232) (news item; the PubMed record carries no abstract, so only the fact in its title is cited); 39460260 — Progress towards Measles and Rubella Elimination in the South-East Asia Region 2013–2023, *Vaccines* 2024, doi:[10.3390/vaccines12101094](https://doi.org/10.3390/vaccines12101094); 41464441 — Global Health Governance and the WHO Pandemic Agreement: a scoping review of challenges and analysis of reforms, *International Journal of Environmental Research and Public Health* 2025, doi:[10.3390/ijerph22121807](https://doi.org/10.3390/ijerph22121807); 40767200 — The United States Withdrawal From the World Health Organization: Implications and Challenges, *International Journal of Health Policy and Management* 2025, doi:[10.34172/ijhpm.9086](https://doi.org/10.34172/ijhpm.9086). Also: WHO Global Traditional Medicine Centre, Jamnagar (2022); WHO trachoma fact sheet (2026); International Health Regulations (2005) amendments, WHA77.17, adopted 1 June 2024, in force 19 September 2025; WHO Pandemic Agreement, resolution WHA78.1, 20 May 2025, with the treaty text annexed to that resolution (used in preference to the scoping review for article numbering, which is internally inconsistent in PMID 41464441 — that paper gives Article 12 to the supply system and Articles 13 and 18 to the financial mechanism, whereas the treaty text places PABS at Article 12, supply chain and logistics at Article 13, and sustainable financing at Article 18, the last confirmed by operative paragraph 9(3) of WHA78.1); Gavi–India partnership, February 2023; Global Fund Eighth Replenishment Summit, Johannesburg, 21 November 2025.



International Health Regulations, 2015

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — The **International Health Regulations (IHR)** are an international legal instrument, adopted by the World Health Assembly and binding on its States Parties, whose stated purpose and scope are "to prevent, protect against, control and provide a public health response to the international spread of disease in ways that are commensurate with and restricted to public health risks and which avoid unnecessary interference with international traffic and trade" (Suryakantha, Ch 19, pp 307-309). The currently operative version is the **IHR (2005)**, adopted by the 58th World Health Assembly in 2005 and in force from **15 June 2007** — for India from **8 August 2007** (Suryakantha, Ch 19, pp 307-309). They are the world's first legally binding agreement on disease notification (Textbook of International Health, Ch 5, p 234).

1. Evolution of the Regulations

Year	Milestone
1851	First International Sanitary Conference , Paris — 12 states; produced a 137-article international sanitary code covering cholera, plague and yellow fever, which never came into force as too few countries ratified it (Park 28e, Ch 25, p 1054)
1892	First International Sanitary Convention , dealing with cholera; a plague convention followed at the 10th conference in 1897 (Oxford Public Health 7e, Ch 75, p 2856)
1951	WHO adopts the revised International Sanitary Regulations (ISR) — still confined to cholera, plague and yellow fever (Oxford Public Health 7e, Ch 75, p 2856)
1969	22nd World Health Assembly renames them the International Health Regulations (IHR) ; four notifiable diseases — cholera, plague, yellow fever and smallpox — with pre-set control measures at international borders (Oxford Public Health 7e, Ch 75, p 2857; Suryakantha, Ch 19, pp 307-309)
1973	Amended so that cholera vaccination shall no longer be required of any traveller (Sunder Lal, Ch 18, p 771)
1981	34th World Health Assembly deletes smallpox after eradication (Suryakantha, Ch 19, pp 307-309; Park Plus, Ch 23, p 336)
1995	48th World Health Assembly calls for revision — the three-disease list was too narrow, countries under-reported for fear of stigma and trade loss, and emerging/re-emerging infections were not covered (Suryakantha, Ch 19, pp 307-309; Oxford Public Health 7e, Ch 75, p 2857)
2005	Fully revised IHR (2005) adopted, the process accelerated by severe acute respiratory syndrome (SARS) in 2003 , which showed how fast a new disease could spread (Oxford Public Health 7e, Ch 105, pp 3996-3997)
15 June 2007	IHR (2005) enter into force ; India from 8 August 2007 (Suryakantha, Ch 19, pp 307-309)

2. Purpose, objectives and guiding principles

- **Purpose and scope** — as in the definition above: maximum security against the international spread of disease with the **minimum interference with world traffic and trade** (Park 28e, Ch 25; Jugal Kishore, Ch 78, p 961).
- **Objectives** (Park Plus, Ch 23, p 336):

a. Appropriate application of **routine preventive measures** at ports and airports, and use by all countries of internationally approved documents such as vaccination certificates. b. **Notification to WHO** of all events that may constitute a public health emergency of international concern (PHEIC). c. Implementation of any **temporary recommendations** issued once the WHO Director-General has determined that such an emergency is occurring.

- **Guiding principles** (Suryakantha, Ch 19, pp 307-309): implementation shall be with **full respect for the dignity, human rights and fundamental freedoms of persons**, and shall be guided by the **Charter of the United Nations and the Constitution of WHO**.
- **Legal reach** — binding on **196 States Parties** (Oxford Public Health 7e, Ch 105, p 3996); the Regulations are structured in **10 parts**, of which Part 5 (four chapters) deals with public health measures (Suryakantha, Ch 19, pp 307-309).

3. What the 2005 revision changed

Feature	IHR (1969)	IHR (2005)
Trigger for notification	Occurrence of 3 named diseases — cholera, plague, yellow fever (smallpox until 1981)	Any "event that may constitute a PHEIC", from any origin or source
Basis of the obligation	Disease-specific list	Non-disease-specific; Annex 2 decision instrument applied to the event
Subject matter	Communicable disease only	Also chemical spills, leaks and dumping, radionuclear events and food/environmental contamination (Suryakantha, Ch 19, pp 307-309; Oxford Public Health 7e, Ch 105, p 3996)
Core strategy	Pre-established control measures at borders	National core capacities to detect, assess, notify, report and contain events
Information accepted	Official State reports only	Also non-state sources , which often precede official notification (Oxford Public Health 7e, Ch 75, pp 2861-2862)
WHO's power	Essentially passive dissemination	Director-General may declare a PHEIC and issue temporary recommendations on the advice of an IHR Emergency Committee (Oxford Public Health 7e, Ch 75, pp 2862-2863)
Consequence for the State	None specified	Duty of collective action, free sharing of information and collaboration between all States (Oxford Public Health 7e, Ch 75, p 2860)

4. Notification and reporting obligations

A **public health emergency of international concern (PHEIC)** is defined as an extraordinary event which (a) constitutes a public health risk to other States through the **international spread of disease**, and (b) potentially requires a **coordinated international response** (Textbook of International Health, Ch 5, p 234; Park Plus, Ch 23, p 336).

Obligation	What must be done	Time limit
Assessment	Assess all reports of urgent events received from the intermediate level	Within 48 hours (Sunder Lal, Ch 18, p 771)
Notification	Notify WHO of any event that may constitute a PHEIC, through the National IHR Focal Point	Within 24 hours (Oxford Public Health 7e, Ch 105, p 3996)
Mandatory (Group 1)	A single case of smallpox, poliomyelitis due to wild-type poliovirus, human influenza caused by a new subtype, or SARS — notified irrespective of context	Immediately (Park Plus, Ch 23, pp 336-338)
Algorithm-dependent (Group 2)	Cholera, pneumonic plague, yellow fever, viral haemorrhagic fevers (Ebola, Lassa, Marburg), West Nile fever and other diseases of special national or regional concern (dengue, Rift Valley fever, meningococcal disease) — always run the Annex 2 algorithm, notify if criteria met	Per algorithm (Sunder Lal, Ch 18, p 742)
Risks outside own territory	Inform WHO of evidence of a public health risk outside its territory that may cause international spread — imported or exported human cases, infected/contaminated vectors, or contaminated goods	Within 24 hours of receipt of evidence (Park Plus, Ch 23, pp 336-338)
Consultation	Confidential dialogue with WHO on further event assessment and appropriate investigative or response measures, where formal notification is not (yet) required	Ongoing (Park Plus, Ch 23, pp 336-338)
National IHR Focal Point	Must give WHO timely, accurate information flow on a 24 hours a day / 7 days a week basis	Continuous (Textbook of International Health, Ch 5, p 234)

5. Health measures at points of entry and for travellers

Provision	Requirement under IHR (2005)
Point of entry	"A passage for international entry or exit of travellers, baggage, cargo, containers, conveyances, goods and postal parcels, as well as agencies and areas providing services to them on entry or exit". Three types: international airports, ports and ground crossings (Park Plus, Ch 23, pp 336-338)
Competent authority	States Parties designate authorities to develop core capacities at designated points of entry , maintain hygiene and sanitation with effective vector, rodent and environmental control, and apply health measures in affected areas (Park Plus, Ch 23, pp 336-338)
On arrival and departure (Art 23)	Traveller's destination and itinerary; review of required health documents; a non-invasive medical examination — the least intrusive that achieves the objective ; inspection of baggage, cargo, containers, conveyances, goods, postal parcels and human remains (Suryakantha, Ch 19, pp 307-309)
Informed consent (Art 23.3)	No medical examination, vaccination, prophylaxis or health measure without the prior express informed consent of the traveller or parent/guardian, except as provided in Art 31.2 (Suryakantha, Ch 19, pp 307-309)
Traveller who refuses (Art 31)	Entry may be denied ; on evidence of imminent public health risk the State Party may compel the least invasive examination, vaccination or prophylaxis, isolation, quarantine or public health observation (Suryakantha, Ch 19, pp 307-309)

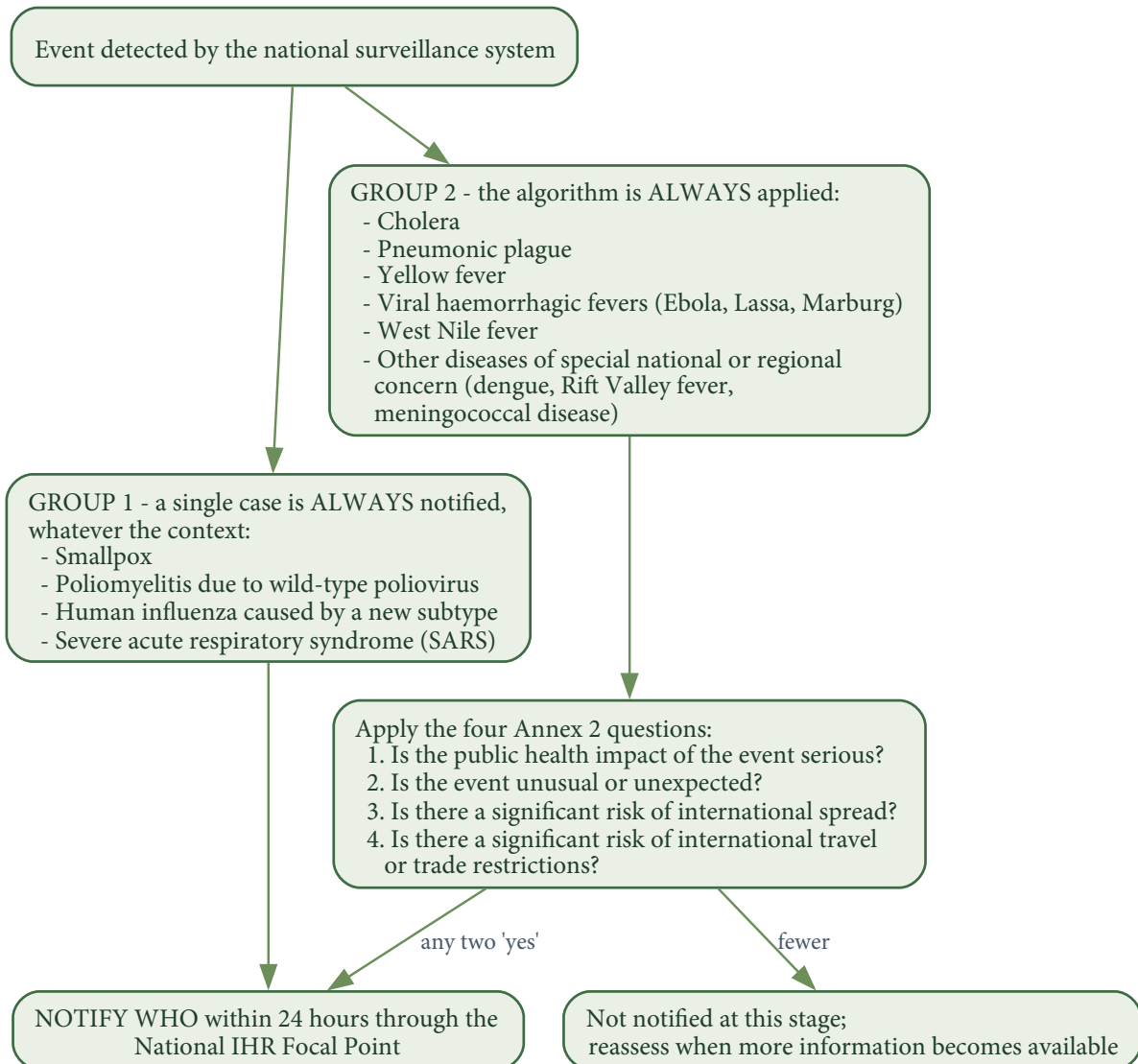
Provision	Requirement under IHR (2005)
Treatment of travellers (Art 32)	Courtesy and respect; regard to gender, socio-cultural, ethnic and religious concerns; accommodation, food, water, clothing, protection of luggage, medical treatment and communication in an understandable language (Suryakantha, Ch 19, pp 307-309)
Conveyances	Disinsecting, derating and disinfection of ships and aircraft specified for vector-borne disease; the old Deratting and Deratting Exemption certificates are replaced by the Ship Sanitation Control Certificate and Ship Sanitation Control Exemption Certificate , covering a broader range of public health risks (Sunder Lal, Ch 18, p 771; Park Plus, Ch 23, pp 336-338)
Vaccination certificate	Yellow fever is the only certificate that may be required in international travel. On 11 July 2016 an amendment entered into force giving the certificate lifetime validity — revaccination or a booster is no longer required as a condition of entry, and this applies in India (Park 28e, Ch 5, p 349; Sunder Lal, Ch 18, p 771)
Vaccination not to be required	Plague vaccination shall not be required for admission; cholera vaccination requirement removed in 1973; smallpox vaccination no longer indicated (Sunder Lal, Ch 18, pp 740-771)
Quarantine incubation periods	Plague 6 days ; cholera 5 days ; yellow fever 6 days (Sunder Lal, Ch 18, pp 740-771)

6. Public health emergencies of international concern declared to date

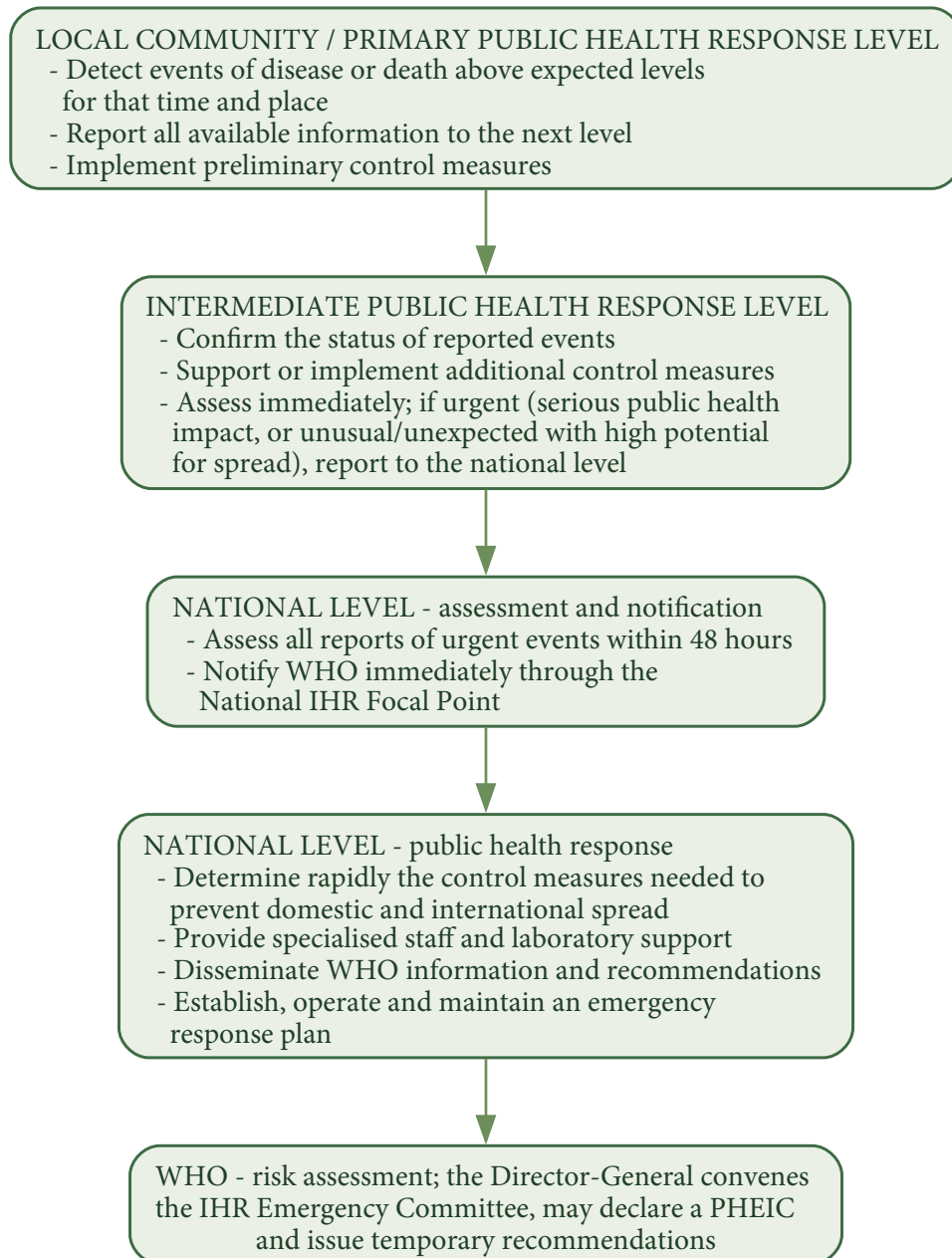
PHEIC	Declared	Status
Influenza A (H1N1) pandemic	2009 — the first use of the IHR (2005), declared within weeks of onset on the advice of the IHR Emergency Committee	Ended (Oxford Public Health 7e, Ch 75, pp 2862-2863; Ch 107, p 4048)
Poliomyelitis (international spread of wild poliovirus)	2014	Emergency Committee continued to review it into 2018 (Oxford Public Health 7e, Ch 107, pp 4048, 4070)
Ebola virus disease, West Africa	2014 (2013–16 outbreak) — widely criticised as declared too late	Ended (Oxford Public Health 7e, Ch 107, pp 4048-4049)
Zika virus infection	2016	Ended (Oxford Public Health 7e, Ch 107, p 4048)
COVID-19 (SARS-CoV-2)	30 January 2020 ; declared a pandemic 11 March 2020	Global emergency declared over on 5 May 2023 (Sunder Lal, Ch 20, p 788; Harrison 22e, Ch 205, pp 1582-1583)
Mpox (monkeypox)	23 July 2022 (Clade IIb); lifted 10 May 2023; re-declared 14 August 2024 for Clade Ib and the African resurgence	Park 28e Updates, p 58

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Annex 2 decision instrument** (label the boxes exactly): *Event detected by national surveillance system* → *Group 1: single case always notified — smallpox / poliomyelitis due to wild-type poliovirus / human influenza caused by a new subtype / SARS* ; *Group 2: always apply the algorithm — cholera, pneumonic plague, yellow fever, viral haemorrhagic fevers (Ebola, Lassa, Marburg), West Nile fever, other diseases of special national or regional concern* ; the **four questions** — *serious public health impact?* / *unusual or unexpected?* / *significant risk of international spread?* / *significant risk of international travel or trade restrictions?* → *Event shall be notified to WHO under the International Health Regulations.*
- **Diagram 2 — the core-capacity chain:** three stacked tiers labelled *local community / primary public health response level* → *intermediate level* → *national level (assessment, notification and response)* → *WHO*.



Annex 2 decision instrument for assessing and notifying an event that may constitute a PHEIC



IHR (2005) core capacity requirements - the escalation chain from local detection to WHO response

High-yield exam pointers

- **Purpose (verbatim)** — "to prevent, protect against, control and provide a public health response to the international spread of disease in ways that are commensurate with and restricted to public health risks and which avoid unnecessary interference with international traffic and trade".
- **Dates** — ISR 1951 → IHR 1969 → amended 1973 and 1981 → revised 2005 (58th World Health Assembly) → in force 15 June 2007; India 8 August 2007.
- **Two clocks** — assess an urgent event **within 48 hours**; notify WHO **within 24 hours** through the National IHR Focal Point (which works 24 × 7).
- **Group 1 mnemonic** — "SPHS": Smallpox, Polio (wild-type), Human influenza of a new subtype, SARS — a single case is always notifiable.

- **Annex 2 rule** — four questions (serious impact / unusual or unexpected / risk of international spread / risk of travel or trade restriction); any two "yes" ⇒ notify.
- **Points of entry** — three types: international airports, ports, ground crossings; deratting certificates replaced by the **Ship Sanitation Control (Exemption) Certificate**; yellow fever is the only certificate that may be demanded, with **lifetime validity since 11 July 2016**.
- **PHEIC roll-call** — H1N1 2009, polio 2014, Ebola 2014, Zika 2016, COVID-19 30 Jan 2020 (ended 5 May 2023), mpox 23 Jul 2022, lifted 10 May 2023, re-declared 14 Aug 2024.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The year in the question stem is a slip — there is no "International Health Regulations 2015"; the instrument in force is the **IHR (2005)**, and that is what this answer describes. Coverage note: the corpus states the number of States Parties as 196 (Oxford Textbook of Public Health 7e) and as 194 (Suryakantha), so quote either with its source; and the "any two of the four questions" rule is read off the Annex 2 decision instrument itself (WHO, International Health Regulations 2005, second edition), which the textbooks reproduce as a figure rather than in running text.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, WHA78.1, IHR (2005) 2024 amendments, WHO Pandemic Agreement 2025, ICMR, Africa CDC

Programme and policy updates (India)

- **National One Health Mission** — an inter-departmental national BSL-3 laboratory network. India has pooled high-containment laboratories from ICMR, DBT, CSIR, ICAR and the Department of Animal Husbandry into a single network of **22 BSL-3 laboratories, two of them mobile**, trained by the apex facilities **ICMR-NIV Pune** and **ICAR-NIHSAD Bhopal**, so that any laboratory can process outbreak samples of human, animal, wildlife or environmental origin irrespective of its parent department. A new scoring checklist covering staff training, sample transport, testing, data management, biomedical waste, emergency preparedness and reporting has been built on the IHR, the Global Health Security Agenda and the DBT 2024 BSL-3 certification guidelines. This is India's operational answer to the IHR core-capacity requirements for laboratory capacity and multisectoral coordination [PMID 41797819].
- **India's mpox response was run through the IHR (2005) machinery** — strengthened surveillance, scale-up of diagnostics and alignment with WHO guidance — with the residual gaps reported as vaccine availability and risk communication; the global response exposed wide between-country disparities in surveillance capacity and regulatory readiness even though the IHR provided the coordinating framework [PMID 40483795].

Current national data

- **How core capacity is actually scored now** — SPAR and JEE. Article 54 reporting rests on the **mandatory State Party Self-Assessment Annual Report (SPAR)** — 2nd edition 2021, **15 capacities and 35 indicators**, submitted on the e-SPAR platform — and the **voluntary Joint External Evaluation (JEE)**, 3rd edition 2022, **19 technical areas**. Across 108 paired assessments by 93 States Parties (2016–2023; 1,445 matched indicator pairs), self-assessed SPAR scores ran

consistently higher than JEE, but the mean gap narrowed from about **20 points in the early editions to about 6 points in 2022–23**, with 25 of 28 matched indicators no longer differing significantly. Persisting disagreement lies in infection prevention and control, healthcare-associated infection surveillance and **National IHR Focal Point function** [PMID 41640577].

- **Current States-Party count and a regional capacity benchmark.** As of 1 May 2025 the IHR bind **196 States Parties, which include all 194 WHO Member States** — quote this figure when a textbook gives 194. Average SPAR scores in the WHO Western Pacific Region were 68% (2021), 72% (2022) and 66% (2023); the weakest core capacities were **zoonotic disease, food safety, chemical events and radiation emergencies**, with financing and infection prevention weakest in low- and middle-income States Parties [PMID 40778190].

Elimination, surveillance and disease status

- **Mpox clade Ib is the live test of the notification system.** 89 travel-related clade Ib cases were detected in 33 countries between August 2024 and July 2025, about one third of them producing secondary transmission; the secondary attack rate was **73% among sexual contacts and 17% among household contacts**, with none recorded among community or healthcare contacts [PMID 41714599]. In the WHO European Region **82 clade I cases were reported through IHR channels between 14 August 2024 and 23 November 2025** — 45 imported and 37 acquired locally, no deaths, with autochthonous cases in men rising from October 2025, suggesting undetected spread. **The second mpox PHEIC, declared 14 August 2024, was lifted in September 2025** — update the PHEIC roll-call in section 6 accordingly [PMID 42318655; PMID 41639867].
- **Nipah virus in India — the standing test case for the Annex 2 algorithm in section 4.** After Siliguri (2001) and Nadia (2007) in West Bengal, NiV re-emerged in **Kerala in 2018** and has recurred in 2019, 2021, 2023, 2024 and 2025, with two further West Bengal cases in **January 2026**. Case-fatality in Indian outbreaks is **45–100% (68–94% across the major spillovers)**, against the **global WHO priority-pathogen figure of 40–75%** — quote the Indian range when the question is about India. Spillover in India is direct bat-to-human with no intermediate amplifying host, and no licensed vaccine or antiviral exists. Deployment of a **mobile BSL-3 laboratory in the 2023 Kerala outbreak**, Truenat point-of-care micro-PCR giving confirmation within an hour, and national guidelines mandating **21-day contact tracing** mark the shift from reactive containment to standing One Health surveillance [PMID 42160364].

Guideline and legislative updates

- **The IHR (2005) have been amended — the single most examinable update on this topic.** The Working Group on Amendments considered **over 300 proposals**; the package was adopted by consensus at the **77th World Health Assembly on 1 June 2024** and **entered into force on 19 September 2025**. Amended Article 4 now requires every State Party to establish or designate a **National IHR Authority** to coordinate overall implementation of the Regulations — a function distinct from the **National IHR Focal Point**, which remains the 24 × 7 communication channel with WHO; a State may use one entity or two, and must give WHO the contact details and confirm them annually. **Article 43 was left largely unchanged**, so a State Party may still apply additional health measures beyond WHO's temporary recommendations provided they are justified and reported [PMID 40953844; PMID 40778190; PMID 41361765].

- A new "pandemic emergency" tier sits above PHEIC, and equity is now a core principle. The proposed intermediate or regional alert was rejected; instead a **higher tier was inserted inside the PHEIC framework**. A pandemic emergency must first satisfy the PHEIC criteria, applies only to **communicable disease**, and lapses with the underlying PHEIC (Articles 1, 12 and 49). New **Article 13(8)** obliges the Director-General to work to remove barriers to timely and equitable access to relevant health products once such an emergency is determined [PMID 40883832]. Alongside this, equity was written in as a guiding principle with operational provisions on access to health products, a **Coordinating Financial Mechanism** was created to fund core-capacity development in developing countries, and a new **Article 54 bis** established a **States Parties Committee** to support implementation, with collaboration and financing as its focus — but no independent external monitoring or sanctions [PMID 41361765].

Global and WHO developments

- **The WHO Pandemic Agreement, adopted on 20 May 2025 at the 78th World Health Assembly**, is the second limb of pandemic prevention, preparedness and response and now sits alongside the amended IHR. **124 governments** adopted its **35 Articles**; it is only the **second instrument ever negotiated under Article 19 of the WHO Constitution**, after the Framework Convention on Tobacco Control (adopted 2003, in force 2005). It commits members to sustained **research and development** (Article 9), a **Coordinating Financial Mechanism**, a **Global Supply Chain and Logistics Network**, and equitable, timely and affordable access to vaccines, diagnostics and therapeutics [PMID 40589223]. **Antimicrobial resistance** enters only obliquely — through prevention and surveillance under **Article 4** — and AMR-specific rules remain absent from the IHR altogether, a noted missed opportunity [PMID 41815751].
- **What is still unsettled about the Agreement**. Its **Pathogen Access and Benefit-Sharing (PABS)** system under **Article 12** was deferred to an annex due for finalisation in **April 2026**; Article 12 as adopted requires legal contracts with manufacturers guaranteeing **at least 20% of relevant pandemic products — half donated, the rest at affordable prices** — distributed by WHO by public health risk, with safeguards for low- and middle-income countries [PMID 41832575]. **The treaty opens for signature only once the Health Assembly adopts that annex, and enters into force after 60 ratifications** [WHA78.1 2025-May]. It was concluded **without the United States**, which is withdrawing from WHO and will therefore neither take part in it nor remain bound by the IHR; **Argentina** has signalled the same intention [PMID 40273267; PMID 41361765; PMID 41832575]. **Travel measures were left out of the treaty altogether** and Article 43 was not substantively changed, so the border free-for-all of COVID-19 — **over 9,000 measures interfering with international traffic or trade reported to WHO by January 2022**, most without the scientific rationale Article 43 demands — remains unresolved [PMID 39880415; PMID 41361765].
- **Regional emergency declarations now run in parallel with the PHEIC**. The African Union elevated **Africa CDC** to an autonomous entity in **2022**, with power to declare a **Public Health Emergency of Continental Security (PHECS)**. It was first used in **August 2024 for mpox** — 17,541 cases and 517 deaths across 13 African Union member states, a 160% rise in cases on the same period of 2023 — and WHO reinstated the mpox PHEIC later that same month; Africa CDC then kept its own declaration in force after WHO lifted the PHEIC in September 2025. This is the first working example of a continental alert tier operating beside the IHR's single global

one, with tiered activation, cross-sectoral oversight and its own financing — the African Epidemics Fund, activated in **February 2025** [PMID 41639867].

SOURCE & COVERAGE

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Internationally notifiable diseases

Asked in: Paper IV 18-Nov-2019 — RGUHS MD Community Medicine

Definition — Internationally notifiable diseases are those diseases and events which the national health authority of a member state is legally obliged to report to the World Health Organization (WHO) at Geneva under the **International Health Regulations (IHR)**, the world's first legally binding agreement on disease notification (Textbook of International Health, Ch 5, p 234). They are **cholera, plague and yellow fever, besides notification of any event that may constitute a public health emergency of international concern (WHO, 2005)** (Sunder Lal, Ch 7, p 286). The purpose and scope of the IHR is to prevent, protect against, control and provide a public health response to the international spread of disease in ways restricted to the public health risk and avoiding unnecessary interference with international traffic and trade (Suryakantha, Ch 20, p 325) — Park's phrasing is maximum security against the international spread of disease with minimum interference with world traffic (Park 28e, Ch 25, p 1056).

Evolution of the international notification list

Instrument (year)	Diseases notifiable to WHO	Diseases kept "under surveillance" by WHO
International Sanitary Regulations, 1951	Cholera, plague, yellow fever, smallpox, relapsing fever, louse-borne typhus (Textbook of International Health, Ch 5, p 234)	—
International Health Regulations, 1969 (Third Annotated Edition, 1983)	Four quarantinable diseases — cholera, plague, yellow fever and smallpox ; relapsing fever and louse-borne typhus dropped in 1969 (Oxford 7e, Ch 75, p 2857; Textbook of International Health, Ch 5, p 234)	Louse-borne typhus, relapsing fever, paralytic poliomyelitis, malaria, influenza A, rabies, salmonellosis, later severe acute respiratory syndrome (SARS) (Park 28e, Ch 3, p 144; Ch 21, p 997)

Instrument (year)	Diseases notifiable to WHO	Diseases kept "under surveillance" by WHO
Amendment of 1981 (34th World Health Assembly)	Cholera, plague, yellow fever only — smallpox removed from the list after it was certified eradicated in 1980 (Oxford 7e, Ch 75, pp 2857-2858; Textbook of International Health, Ch 5, p 234; Suryakantha, Ch 20, pp 325-326)	As above
International Health Regulations, 2005 — in force 15 June 2007, binding international law for 194 States Parties (India: entry into force 8 August 2007) (Suryakantha, Ch 20, pp 325-326); Oxford 7e, Ch 105, pp 3996-3997 gives 196 countries	Any event that may constitute a public health emergency of international concern (PHEIC) , assessed by the Annex 2 decision instrument; four diseases are notifiable on a single case. Smallpox was re-added 25 years after its deletion (Textbook of International Health, Ch 5, p 234)	Superseded — surveillance is now event-based and syndrome-triggered, and does not supersede or interfere with existing national or international surveillance (Jugal Kishore, pp 970-971)

Diseases notifiable to WHO today — IHR (2005), Annex 2

Group	Diseases	Rule
A. Always notified	Smallpox; poliomyelitis due to wild-type poliovirus; human influenza caused by a new subtype; severe acute respiratory syndrome (SARS)	A single case is by definition unusual or unexpected and may have serious public health impact — notify immediately, irrespective of context, without applying the algorithm (Oxford 7e, Ch 75, pp 2857-2859; Park Plus, Ch 23, p 338)
B. Algorithm always applied	Cholera; pneumonic plague; yellow fever; viral haemorrhagic fevers (Ebola, Lassa, Marburg); West Nile fever; other diseases of special national or regional concern, e.g. dengue fever, Rift Valley fever, meningococcal disease	These have demonstrated the ability to cause serious public health impact and to spread rapidly internationally — the decision instrument must be used every time, and the event is notified only if the algorithm is satisfied (Oxford 7e, Ch 75, pp 2858-2859; Park Plus, Ch 23, p 338)
C. Any other event	Any event of potential international public health concern, including events of unknown cause or source, and events beyond communicable disease (chemical spills, radiation incidents)	The decision instrument is applied to decide whether notification is required (Oxford 7e, Ch 75, p 2859; Suryakantha, Ch 20, p 325)

The Annex 2 operating rule — the event is assessed against four questions (serious public health impact; unusual or unexpected; significant risk of international spread; significant risk of international travel or trade restrictions). A "yes" to any TWO of the four obliges notification to WHO.

The notification obligation — who notifies, to whom, and how fast

- **Detection** — the event is picked up by the national surveillance system, whose minimum core capacities are laid down in IHR Annex 1 (Oxford 7e, Ch 75, pp 2859-2860).
- **Assessment** — the state assesses the event against the Annex 2 decision instrument using the WHO case definitions; the assessment is made by the country in which the event is occurring,

even if the report of it came from elsewhere (Oxford 7e, Ch 75, pp 2857-2858). At intermediate level reported events are assessed immediately; at national level all reports of urgent events are assessed **within 48 hours** (Sunder Lal, Ch 18, p 771).

- ▶ **Notification within 24 hours** — WHO must be notified within 24 hours of assessment, through the country's **National IHR Focal Point**, which is accessible 24 hours a day, 7 days a week (Textbook of International Health, Ch 5, p 234; Oxford 7e, Ch 105, pp 3996-3997; Red Book, Ch 2, p 1024).
- ▶ **Continued reporting** — the state keeps sending case definitions, laboratory results, source and type of risk, numbers of cases and deaths, conditions affecting spread, the health measures employed, and the difficulties faced and support needed (Red Book, Ch 2, p 1024).
- ▶ **Consultation** — short of formal notification a state may hold a confidential dialogue with WHO, and must inform WHO within 24 hours of evidence of a public health risk *outside* its territory (Park Plus, Ch 23, p 338).
- ▶ **WHO response** — WHO verifies and assesses the report, the Director-General (advised by an IHR Emergency Committee) decides whether it is a PHEIC, and WHO recommends concrete action and provides assistance (Textbook of International Health, Ch 5, pp 234-235; Oxford 7e, Ch 75, pp 2862-2863).
- ▶ **Dissemination** — epidemiological information on diseases subject to the IHR is circulated to all member states through the **Weekly Epidemiological Record** and the Automatic Telex Reply Service (Park 28e, Ch 25, p 1056).

Public health emergencies of international concern declared under IHR (2005)

A PHEIC is an extraordinary event determined to constitute a public health risk to other states through the international spread of disease and to potentially require a coordinated international response (Oxford 7e, Ch 107, pp 4048-4049). The table is the **WHO record of declarations to date**: Oxford 7e lists only the four events up to 2016 — it predates the rest, stating that as of June 2019 no emergency had yet been declared for the Ebola epidemic in the Democratic Republic of the Congo — and the later declarations carry their own sources.

Event	Declared	Subsequent action
Pandemic influenza A (H1N1) — the first use of the IHR (2005)	2009	Declared a PHEIC within weeks of onset on the recommendation of the IHR Emergency Committee (Oxford 7e, Ch 20, pp 790-791; Ch 75, pp 2862-2863)
Poliomyelitis	2014	Standing PHEIC (Oxford 7e, Ch 107, pp 4048-4049)
Ebola virus disease (West Africa)	8 August 2014	WHO issued detailed advisories to countries to protect their citizens (Sunder Lal, Ch 8, p 474; Oxford 7e, Ch 107, pp 4048-4049)
Zika virus disease	2016	— (Oxford 7e, Ch 107, pp 4048-4049)
Ebola virus disease (eastern Democratic Republic of the Congo, outbreak detected August 2018)	17 July 2019	The world's second-largest Ebola outbreak; declared over in June 2020 with ~3,500 cases and two-thirds of them fatal (Sunder Lal, Ch 8, p 474; Harrison 22e, Ch 485, pp 3892-3893)

Event	Declared	Subsequent action
COVID-19 (SARS-CoV-2)	30 January 2020	Declared a pandemic on 11 March 2020; the PHEIC was declared over on 5 May 2023 (Sunder Lal, Ch 20, p 788; Harrison 22e, Ch 205, pp 1582-1583)
Mpox (formerly monkeypox) — global Clade IIb outbreak, largely by sexual contact	23 July 2022	Outbreak subsided and the PHEIC was lifted on 10 May 2023 (Park 28e, Ch 5, p 172)
Mpox — emergence of the new Clade Ib with resurgence of cases in Africa	14 August 2024	PHEIC declared once again. India's first case of this emergency period, reported 9 September 2024 in a returned traveller, was Clade II and therefore outside the Clade Ib emergency (Park 28e, Ch 5, p 172)

Disease-specific international requirements — the three classical quarantinable diseases

Disease	Incubation period recognised for quarantine	Current notification rule	Principal international measure
Cholera	5 days (Indian Aircraft (Public Health) Rules, 1954 — Jugal Kishore, pp 867-868; Sunder Lal, Ch 18, p 771)	Notifiable locally and nationally; since 2005 international notification of every case is no longer mandatory , but every cholera event must be assessed against the Annex 2 decision instrument (Park 28e, Ch 5, pp 292-293)	Cases and deaths are reported daily and weekly until the area is declared free of cholera — an area is declared free when twice the incubation period (10 days) has elapsed since the death, recovery or isolation of the last case (Park 28e, Ch 5, pp 290-291)
Plague	6 days (Jugal Kishore, pp 867-868)	WHO must be informed promptly of any epidemic or isolated case; case notification is required by the IHR (Park 28e, Ch 5, pp 362-363)	Surveillance of natural foci replaces quarantine, which has been shown to be ineffective; special attention to seaports and airports near natural foci and to cargo from enzootic areas (Park 28e, Ch 5, p 363)

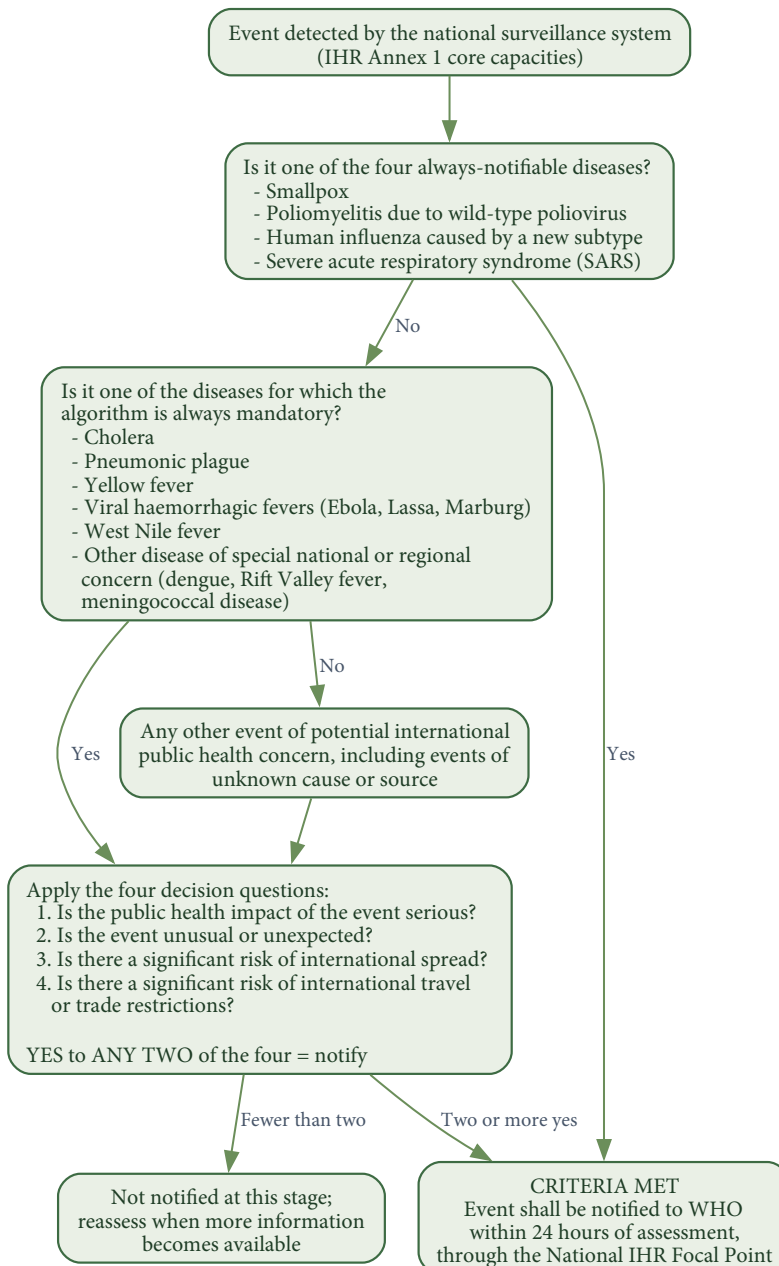
Dis eas e	Incubation period recognised for quarantine	Current notification rule	Principal international measure
Yell ow fever	6 days recognised under the IHR (clinical incubation 3-6 days) (Park 28e, Ch 5, pp 347-348)	Notifiable; India is a yellow fever receptive area — the disease does not exist but conditions would permit its development if introduced (Park 28e, Ch 5, p 348)	Valid international certificate of vaccination (17D live attenuated vaccine) demanded of travellers from endemic zones — by India even if only in transit; valid for the lifetime of the person vaccinated since the amendment of 11 July 2016 (previously 10 years), beginning 10 days after vaccination, with no booster demandable as a condition of entry (Park 28e, Ch 5, pp 348-349). Without a certificate the traveller is quarantined in a mosquito-proof ward for 6 days from the date of leaving an infected area; the area within 400 metres of the perimeter of international airports and seaports is kept free of mosquitoes and the <i>Aedes aegypti</i> index kept at zero at all ports (Park 28e, Ch 15, p 925)

Implementation in India

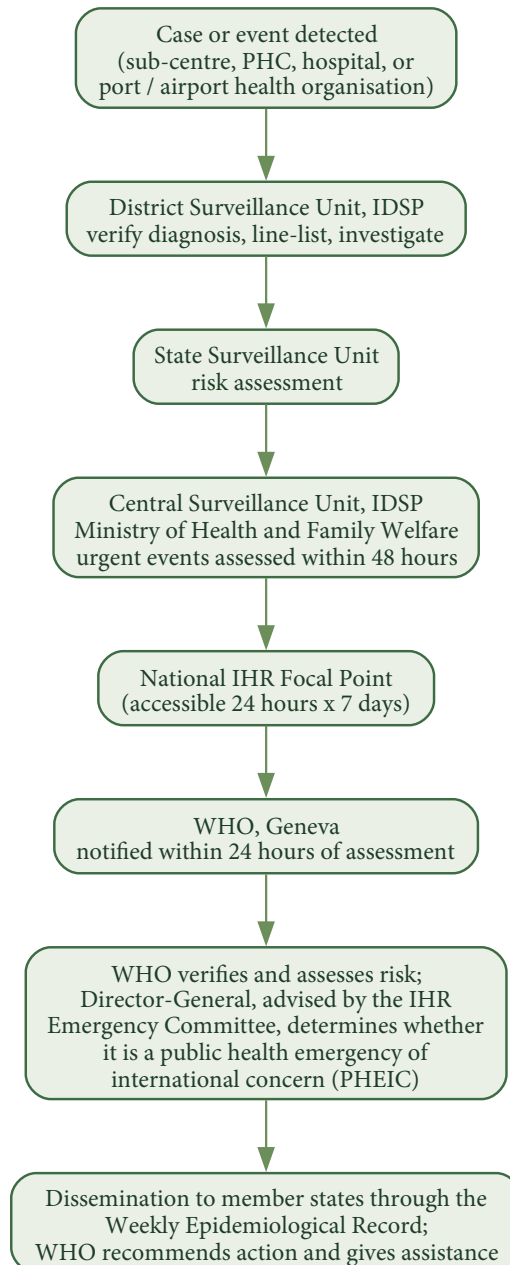
- **International health relations and quarantine** are a specific function of the Directorate General of Health Services (DGHS), Union Ministry of Health and Family Welfare; the major seaports (Kolkata, Visakhapatnam, Chennai, Cochin, Mumbai, Kandla) and international airports (Mumbai–Santa Cruz, Kolkata–Dum Dum, Chennai–Meenambakkam, Tiruchirapalli, Delhi–Palam) are directly controlled by the DGHS (Park 28e, Ch 23, p 1033).
- **Indian Aircraft (Public Health) Rules, 1954 (amended 1969)**, framed under the Aircraft Act 1934, define infected aircraft, infected area and isolation, and fix the quarantine incubation periods — yellow fever 6 days, plague 6 days, cholera 5 days, smallpox 14 days, typhus 14 days, relapsing fever 8 days. Contravention is punishable with imprisonment up to 3 months or fine up to Rs 1,000 or both (Jugal Kishore, pp 867-868).
- **Import of human remains** of persons who died of yellow fever, plague, anthrax or glanders is prohibited unless properly cremated (Jugal Kishore, p 867).
- **Epidemic Diseases Act, 1897** is the domestic legal instrument for containment — it was invoked for COVID-19 in 2020 (Park 28e, Ch 5, p 205); it is being replaced by a **Public Health Act** for the prevention, control and management of epidemics, bioterrorism and disaster response (Sunder Lal, Ch 7, p 286).
- **Integrated Disease Surveillance Programme (IDSP)** carries plague and yellow fever under the category "other international commitments", and its syndromic surveillance (haemorrhagic fevers, respiratory distress, diarrhoea, jaundice, meningoencephalitis) is the detection arm feeding the IHR obligation (Jugal Kishore, p 488; pp 970-971).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Drawn figure — "Timeline of the international notification list", a single horizontal line with six labelled marks: 1851 — *first International Sanitary Conference, Paris*; 1951 — *International Sanitary Regulations: 6 diseases*; 1969 — *IHR: 4 quarantinable diseases (cholera, plague, yellow fever, smallpox)*; 1981 — *smallpox deleted after eradication, 3 quarantinable diseases remain (cholera, plague, yellow fever)*; 2005 — *IHR revised: event-based notification of any possible PHEIC*; 15 June 2007 — *IHR (2005) enters into force (India, 8 August 2007)*.



IHR (2005) Annex 2 decision instrument for the assessment and notification of events that may constitute a public health emergency of international concern



Route of an internationally notifiable event from the Indian periphery to WHO

High-yield exam pointers

- ▶ **Aim of the IHR** (one line worth writing): maximum security against the international spread of disease with minimum interference with world traffic. In force **15 June 2007**, binding on **194 States Parties** (India, 8 August 2007).
- ▶ **Classical three** (IHR 1969 as amended 1981): **cholera, plague, yellow fever**. Mnemonic **C-P-Y**. The 1969 list had **four** — smallpox was the fourth, deleted in 1981 after eradication; the 1951 ISR list had two more still, relapsing fever and louse-borne typhus.
- ▶ **IHR (2005) always-notifiable four**: **smallpox, wild poliovirus polio, human influenza due to a new subtype, SARS** — a single case is notified without applying the algorithm.
- ▶ **Annex 2 rule**: four decision questions — serious impact, unusual/unexpected, risk of international spread, risk of travel or trade restrictions; **"yes" to any TWO obliges notification**.

- **Timing:** notify WHO **within 24 hours** of assessment, through the **National IHR Focal Point** (24 h × 7 days); urgent events assessed at national level within **48 hours**.
- **Quarantine incubation periods** (Indian Aircraft (Public Health) Rules 1954): yellow fever 6, plague 6, cholera 5, smallpox 14, typhus 14, relapsing fever 8 days. Cholera area declared free after **10 days (twice the incubation period) from the death, recovery or isolation of the last case**.
- **Yellow fever:** India is a *receptive area*; no certificate → 6 days' quarantine in a mosquito-proof ward; certificate valid for **life** since 11 July 2016; 400 m mosquito-free zone around ports and airports.
- **PHEICs declared to date:** H1N1 (2009), polio (2014), Ebola West Africa (8 Aug 2014), Zika (2016), Ebola Democratic Republic of the Congo (17 July 2019), **COVID-19 (30 Jan 2020; pandemic 11 Mar 2020; PHEIC ended 5 May 2023)**, mpox (23 July 2022, lifted 10 May 2023; re-declared 14 Aug 2024 for Clade Ib).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the number of States Parties bound by the IHR (2005) is given as 194 by Suryakantha (Ch 20, p 325) and as 196 countries by the Oxford Textbook of Public Health 7e (Ch 105, pp 3996-3997), the difference being the non-member states that also acceded — write either with its source; and West Nile fever belongs on the official IHR (2005) Annex 2 second list but is lost from the scanned rendering of that figure in the retrieved Oxford 7e text.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO International Health Regulations (2005, as amended 2024), WHO Pandemic Agreement 2025, WHO State Party Self-Assessment Annual Reporting, Africa CDC, IDSP-IHIP India

Programme and policy updates (India)

- **IDSP now runs on the Integrated Health Information Platform (IHIP-IDSP)** — the 2018 move from aggregate paper returns to real-time case-based reporting — but the detection arm that feeds India's IHR notification duty remains thin on the ground: fieldwork through 2023–2025 in two high-burden districts of a western Indian state found the district public health laboratories set up under IHIP-IDSP functional in only three districts, four of the state's 36 districts without sentinel hospital coverage, and parallel reporting into IHIP-IDSP and the National Vector Borne Disease Control Programme fragmenting outbreak detection, with harmonisation of outbreak criteria across the two programmes identified as the fix [PMID 42245358].

Current national data

- **Zoonotic outbreaks reported through IDSP, 2010–2023:** 1,158 outbreaks across ten zoonoses, with Japanese encephalitis / acute encephalitis syndrome the most frequently reported. Anthrax, JE/AES, Crimean-Congo haemorrhagic fever, leptospirosis and scrub typhus were reported throughout the year, whereas most of the remaining diseases clustered in the monsoon months of July to September [PMID 41738949].

Elimination, surveillance and disease status

- ▶ **The ninth PHEIC since 2007 was declared on 17 May 2026** — Ebola disease due to **Bundibugyo virus**, following an outbreak in Ituri province of the Democratic Republic of the Congo that extended to North and South Kivu, with imported cases and limited local transmission in Uganda. By 17 June 2026 there were 896 confirmed cases and 232 deaths (case fatality 26%), with more than 90% of confirmed cases still within Ituri and children accounting for nearly a fifth of them. No vaccine is licensed for Bundibugyo virus; WHO's May 2026 emergency guidance judged the evidence for cross-protection by Ervebo — the one available Ebola vaccine, a live recombinant vesicular stomatitis virus product approved by the US FDA in November 2019 — still insufficient, and recommended against its use in this outbreak or outside research settings [PMID 42383297].
- ▶ **IHR core capacities are now graded annually by self-assessment** through the State Party Self-Assessment Annual Reporting (SPAR) tool — 15 capacities scored on a 1–5 scale, a legal obligation under Article 54. In the Western Pacific Region the regional average rose from 68% (2021) to 72% (2022) and then fell to 66% (2023); the weakest domains were zoonotic disease, food safety, chemical events and radiation emergencies, with financing and infection prevention and control lagging in low- and middle-income States Parties. Joint External Evaluation supplies the external, qualitative counterpart to this self-report [PMID 40778190].

Guideline and legislative updates

- ▶ **The IHR (2005) amendments adopted on 1 June 2024 at the 77th World Health Assembly add a "pandemic emergency" tier** at the top of the alert framework. Defined in Article 1 as communicable in nature, it must first satisfy the PHEIC criteria and runs for the same duration as the underlying PHEIC (Articles 12 and 49), so the Director-General assesses both together without reconvening the Emergency Committee; the lower "intermediate" or regional alert tier widely proposed after COVID-19 was ultimately not adopted, and the alert ladder now reads PHEIC → pandemic emergency [PMID 40883832]. The two-of-four rule and the four always-notifiable diseases of Annex 2 are unchanged — the Annex 2 middle box gained a clause on clusters of severe acute respiratory disease of unknown or novel cause [WHA77.17 2024-Jun].
- ▶ **Amended IHR (2005) Article 13(8) now obliges WHO itself to remove barriers to timely and equitable access to relevant health products** once a PHEIC — including a pandemic emergency — has been determined and for as long as it lasts, with Article 13(9) obliging States Parties to support WHO in doing so. Note that this is an *IHR* article: the equity duties sit on WHO, not only on States Parties [PMID 40883832]. The amended Regulations entered into force in September 2025, and a revised Article 44 adds a commitment to sustainable financing for developing countries through a coordinating financial mechanism under the authority of the Health Assembly [PMID 40589223].
- ▶ **A National IHR Authority (NIA) is now a required national entity, distinct from the National IHR Focal Point.** The NIA is mandated to drive policy, resource allocation and multisectoral coordination for IHR implementation inside the country, while the National IHR Focal Point continues as the round-the-clock communication channel with WHO for notification, verification and reporting. Writing both, and distinguishing them, is the safest way to answer the "who notifies" step now [PMID 40778190].

- **Article 43 (additional health measures)** was left substantively unchanged by the 2024 revisions. Of the four changes member states put forward, only the one protecting essential travel and trade — health and care workers, humanitarian and life-threatening situations, and supply chains for health products and food — was agreed; proposals to formalise travel measures based on regular risk assessment, to create a WHO review of measures that are disproportionate or excessive, and to set up a WHO-facilitated consultation process were rejected over sovereignty concerns, and travel measures were likewise excluded from the Pandemic Agreement negotiations. Uncoordinated unilateral travel restrictions therefore remain the likely default in the next emergency [PMID 39880415].

Global and WHO developments

- **The WHO Pandemic Agreement was adopted on 20 May 2025 by the 78th World Health Assembly** — 35 Articles, adopted by 124 governments after three years of negotiation. It is only the second instrument negotiated under Article 19 of the WHO Constitution, after the Framework Convention on Tobacco Control, and it complements rather than replaces the amended IHR; member states directed the intergovernmental working group to take the steps needed to establish a Coordinating Financial Mechanism and a Global Supply Chain and Logistics Network [PMID 40589223]. That Network is established under the Agreement's **own Article 13** — a wholly different provision from IHR Article 13(8) above, and the two are easily confused — while the Agreement's **Pathogen Access and Benefit-Sharing (PABS) annex under Article 12, set for finalisation by April 2026**, is the equity test of the treaty: participating manufacturers are to make at least 20% of pandemic-related products available to WHO, half as donation and half at affordable prices, allocated by public health risk with explicit safeguards for low- and middle-income countries, with traceability of pathogen samples and of digital sequence information, and standardised contracts, the unresolved sticking points [PMID 41832575].
- **A parallel regional alert mechanism now exists alongside the PHEIC.** The African Union made Africa CDC an autonomous institution in 2022, Article 3 of its Statutes empowering it to declare a Public Health Emergency of Continental Security (PHEC) in consultation with the affected member states. It was exercised for the first time in August 2024 for sustained mpox transmission — by then 17,541 cases and 517 deaths across 13 African Union member states, a 160% rise in cases on the same period of 2023 — and WHO reinstated the mpox PHEIC later that same month. WHO then lifted that PHEIC in September 2025, the month Africa CDC's emergency consultative group met again on whether to maintain the continental emergency and pressed for continued vigilance: regional and global determinations can now diverge for the same event [PMID 41639867].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3 Principles of Epidemiology, p 144; Ch 5 Epidemiology of Communicable Diseases, pp 172, 205, 290-293, 347-349, 362-363; Ch 15 Environment and Health, p 925; Ch 21 Health Information and Basic Medical Statistics, p 997; Ch 23 Health Planning and Management, p 1033; Ch 25 International Health, p 1056); Oxford Textbook of Public Health 7e (Ch 20 Law and the public's health, pp 790-791; Ch 75 Emerging and re-emerging infections, pp 2856-2863 — the IHR and the Annex 2 decision instrument; Ch 105, pp 3996-3997; Ch 107, pp 4048-4049); Textbook of International Health (Ch 5, pp 233-235 — evolution of the notification list and IHR 2005 obligations); Sunder Lal — Textbook of Community Medicine (Ch 7 Epidemiology, pp 285-286 — internationally notifiable diseases; Ch 8, p 474 — Ebola PHEICs 2014 and 2019; Ch 18 International Health and International Health Regulations, pp 771-772; Ch 20 Recent Advances, p 788 — COVID-19 PHEIC and pandemic dates); Suryakantha — Community Medicine with Recent Advances (Ch 20, pp 325-326 — purpose, scope, legal status and evolution of the IHR 2005); Harrison's Principles of Internal Medicine 22e (Ch 205 SARS-CoV-2 and COVID-19, pp 1582-1583; Ch 485 Global Issues in Medicine, pp 3892-3893); Park Plus (Ch 23 International Health, pp 336-338 — Group 1 and Group 2 notification); Red Book — Public Health and Preventive Medicine (Ch 2, p 1024 — notification and National IHR Focal Point functions); Jugal Kishore — National Health Programmes of India (International Health Regulations, pp 970-971; Indian Aircraft (Public Health) Rules 1954, pp 867-868; Integrated Disease Surveillance Programme, p 488). International Health Regulations (2005), WHO — Annex 1 and Annex 2, as reproduced in the above texts.

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secondary data analysis of Integrated Disease Surveillance Programme data, 2010-2023, Transactions of the Royal Society of Tropical Medicine and Hygiene 2026, doi:[10.1093/trstmh/trag011](https://doi.org/10.1093/trstmh/trag011) (abstract-only). Primary instrument: International Health Regulations (2005) as amended by resolution WHA77.17 of 1 June 2024 — Annex 2 decision instrument.



Global Health Sector Strategy on Viral Hepatitis

Asked in: Paper II 10-Jul-2023 — RGUHS MD Community Medicine

Definition — The **Global Health Sector Strategy (GHSS) on viral hepatitis** is the World Health Organization's time-bound framework for **eliminating hepatitis B virus (HBV) and hepatitis C virus (HCV) as public health threats by 2030**. The first strategy (2016–2021) was endorsed by the World Health Assembly in May 2016 and set the elimination goal of a **90 per cent reduction in new infections and a 65 per cent reduction in mortality by 2030** (Oxford Textbook of Public Health 7e, Ch 70, pp 2631–2632). It has been succeeded by the **GHSS on viral hepatitis 2022–2030**, which prescribes ten key strategic and operational shifts together with dated impact and coverage targets (Park 28e, Ch 5, pp 274–275). Hepatitis elimination is also written into **Sustainable Development Goal target 3.3**, with hepatitis B incidence as its impact indicator (Park 28e, Ch 14, p 856).

1. The baseline the strategy has to act on

Element of the baseline	Figure (year)
People living with chronic hepatitis B and C, combined	About 304 million (2022) — 254 million hepatitis B and 50 million hepatitis C (Park 28e, Ch 5, p 271)
New infections	2.2 million in 2022, down from 2.5 million in 2019 (Park 28e, Ch 5, p 271)
Deaths from viral hepatitis	About 1.3 million in 2022 — similar to tuberculosis, and the second leading infectious cause of death after COVID-19 (Park 28e, Ch 5, p 271)
Age and sex distribution of the chronic burden	Half the burden is in people aged 30–54 years; men account for 58 per cent of all cases; an estimated 12 per cent of the burden is in children , in particular for hepatitis B (Park 28e, Ch 5, p 271)
The diagnostic gap the strategy targets	More than 80 per cent of people with chronic hepatitis B and more than 90 per cent of those with hepatitis C are undiagnosed (Park 28e, Ch 5, p 275)

2. The ten interventions — key strategic and operational shifts, 2022–2030

#	Strategic / operational shift	What it means on the ground
a	Promote greater public and political awareness of the importance of hepatitis B and C prevention, testing and treatment	Advocacy and information–education–communication; political commitment that puts hepatitis on the national health agenda
b	Allocate increased financial resources to hepatitis B and C	External catalytic funding plus domestic funding; hepatitis prevention, testing and treatment written into the essential national health benefit package
c	Scale up universal access to the hepatitis B birth-dose vaccine and to improved services for testing of pregnant women, to prevent vertical (mother-to-child) transmission	Birth dose for every newborn within 24 hours; antenatal hepatitis B surface antigen (HBsAg) testing
d	Ensure continued investment in primary prevention	Improved safety of medical injections and procedures; comprehensive prevention including harm reduction and other evidence-based measures for people who inject drugs; hepatitis B vaccination for infants and at-risk populations
e	Substantially increase access to hepatitis B and C testing	Reach the people living with chronic infection who remain undiagnosed — more than 80 per cent of hepatitis B and more than 90 per cent of hepatitis C
f	Substantially increase treatment access	Expand treatment by building on existing community-based and health-facility-based services
g	Promote simplified service-delivery models	Decentralise hepatitis B and C testing and treatment to lower-level facilities including primary care; integrate with other services such as harm-reduction and HIV services; task-share , with care and treatment delivered by non-specialists and nurses
h	Address the barriers faced by populations most affected and at-risk	The populations named are people who inject drugs, people in prisons and other closed settings, sex workers, gay men and other men who have sex with men, indigenous populations, and mobile and migrant populations from countries of higher prevalence (Park 28e, Ch 5, p 271)
i	Strengthen community and civil-society engagement and innovative partnerships	Peer outreach, community-led testing and demand generation, partnership with non-governmental organisations
j	Advance the research agenda	Curative treatment strategies for hepatitis B and a preventive vaccine for hepatitis C , for which no vaccine yet exists

(Park 28e, Ch 5, pp 274–275)

3. The technical package that delivers these shifts

Pillar	Content of the intervention
Hepatitis B immunisation	All infants receive the first dose as soon as possible after birth, ideally within 24 hours , followed by 2 or 3 additional doses to complete the primary series (Park 28e, Ch 5, p 274). India uses the four-dose schedule — birth dose, then doses with pentavalent vaccine at 6, 10 and 14 weeks (Park 28e, Ch 5, p 270). The complete series induces protective antibody in more than 95 per cent of infants, children and young adults (Park 28e, Ch 5, pp 270–271)
Prevention of vertical transmission	All pregnant women tested for HIV, syphilis and HBsAg at least once and as early as possible in pregnancy ; timely birth dose, with hepatitis B immunoglobulin (HBIG) for the newborn of a carrier mother — simultaneous HBIG and vaccine being more efficacious than HBIG alone (Park 28e, Ch 5, pp 271, 274)
Blood and injection safety	Screening of every blood unit for blood-borne infection and rejection of HBsAg-positive donors; voluntary blood donation , since purchased blood carries a higher risk of post-transfusion hepatitis; adequate sterilisation of instruments, safe injection practice and safe sharps disposal (Park 28e, Ch 5, pp 271, 276)
Harm reduction	Sterile injecting equipment for people who inject drugs, with needles and syringes distributed per person counted as a coverage indicator (Park 28e, Ch 5, pp 275–276)
Testing services — the WHO "5Cs"	Consent, Confidentiality, Correct test results, Counselling and Connection (linkage to prevention, treatment and care). Testing must always be voluntary , with consent informed by pre-test information (Park 28e, Ch 5, p 271)
Treatment	Chronic hepatitis B — oral tenofovir or entecavir , one tablet a day, the most potent suppressors of the virus; they slow progression of cirrhosis, reduce liver-cancer incidence and improve long-term survival, rarely cause resistance, but mostly suppress rather than cure , so treatment is long-term. Chronic hepatitis C — direct-acting antivirals (DAAs) , more effective and better tolerated than interferon-based therapy, curing most infections in a shorter (usually 12-week) and safer course (Park 28e, Ch 5, pp 271, 276)
Secondary and tertiary prevention	Education and counselling on care and treatment options; immunisation against hepatitis A and B to prevent co-infection; early and appropriate antiviral therapy; regular monitoring for early diagnosis of chronic liver disease (Park 28e, Ch 5, p 276)

4. Impact and coverage targets and milestones for viral hepatitis by 2030

Type	Indicator	Baseline 2020	Target 2025	Target 2030
Impact	HBsAg prevalence in children under 5 years	0.94%	0.5%	0.1%
Impact	New hepatitis B infections per year	1.5 million (20 per 100,000)	850,000 (11 per 100,000)	170,000 (2 per 100,000)
Impact	New hepatitis C infections per year	1.575 million (20 per 100,000)	1 million (13 per 100,000)	350,000 (5 per 100,000)
Impact	New hepatitis C infections among people who inject drugs, per year	8 per 100	3 per 100	2 per 100

Type	Indicator	Baseline 2020	Target 2025	Target 2030
Impact	Deaths from hepatitis B per year	820,000 (10 per 100,000)	530,000 (7 per 100,000)	310,000 (4 per 100,000)
Impact	Deaths from hepatitis C per year	290,000 (5 per 100,000)	240,000 (3 per 100,000)	140,000 (2 per 100,000)
Coverage	Hepatitis B — % diagnosed / % treated	30% / 30%	60% / 50%	90% / 80%
Coverage	Hepatitis C — % diagnosed / % cured	30% / 30%	60% / 50%	90% / 80%
Coverage	Newborns benefiting from a timely birth dose and other interventions against vertical transmission	50%	70%	90%
Coverage	Hepatitis B vaccine third-dose coverage among children	90%	90%	90%
Coverage	Needles and syringes distributed per person who injects drugs	200	200	300
Coverage	Blood safety — proportion of blood units screened for blood-borne disease	95%	100%	100%
Coverage	Safe injections — proportion of safe health-care injections	95%	100%	100%

(Park 28e, Ch 5, p 275)

5. Surveillance that tracks the interventions

Infection with HBV or HCV evolves in three phases — acute infection, chronic infection, and sequelae (cirrhosis and hepatocellular carcinoma). Because 20–30 years may separate infection from death, all three phases must be captured before the epidemiological situation can be fully described, and the hepatitis programme must consolidate and triangulate data from these different systems (Park 28e, Ch 5, p 274).

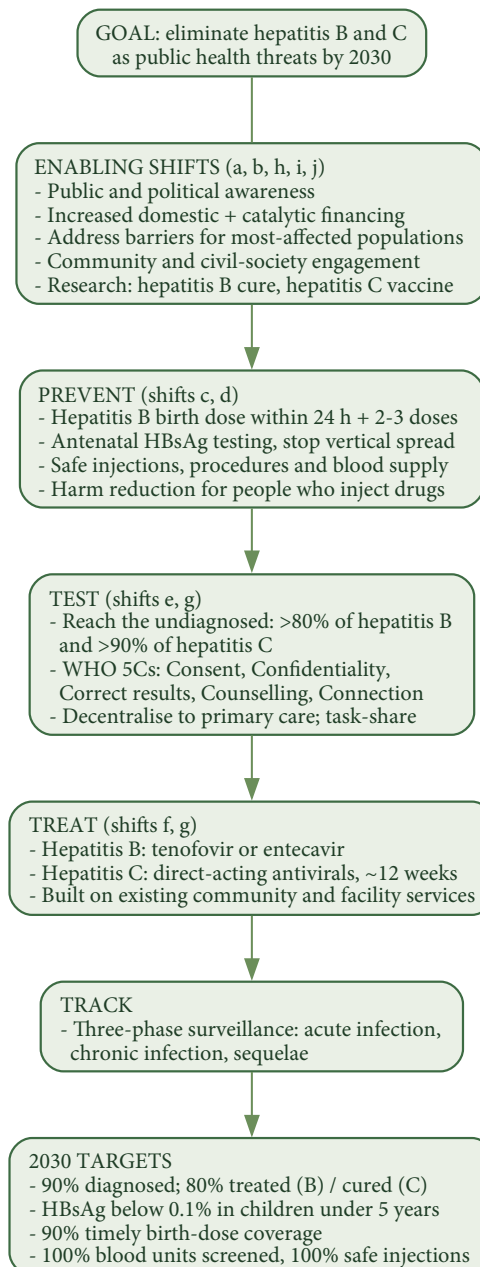
Component	1. Acute hepatitis (new infections)	2. Chronic, prevalent hepatitis	3. Sequelae
Activity	Syndromic surveillance in the general population; event-based surveillance; enhanced case reporting with in-vitro diagnosis and risk-factor data	Regular biomarker surveys	Death certificates combined with testing of patients with cirrhosis and hepatocellular carcinoma for HBV and HCV
Population under surveillance	Persons presenting to health facilities with discrete-onset acute hepatitis	Persons without acute symptoms, tested during population surveys	Persons diagnosed with cirrhosis or hepatocellular carcinoma

Component	1. Acute hepatitis (new infections)	2. Chronic, prevalent hepatitis	3. Sequelae
Usual implementer	Communicable disease surveillance; hepatitis programme where sentinel sites are used	Hepatitis programme, with the agencies running biomarker surveys	Vital registration; sentinel sites caring for liver disease; cancer registries
Case definition used	Presumptive and confirmed acute hepatitis, by type	Chronic HBV and HCV infection; serological evidence of past or present HCV infection	Cirrhosis or hepatocellular carcinoma with chronic HBV or HCV infection
Objective	Detect outbreaks; describe type-specific trends and identify risk factors	Estimate prevalence of infection; model incidence trends	Estimate mortality from HBV- and HCV-associated cirrhosis and liver cancer

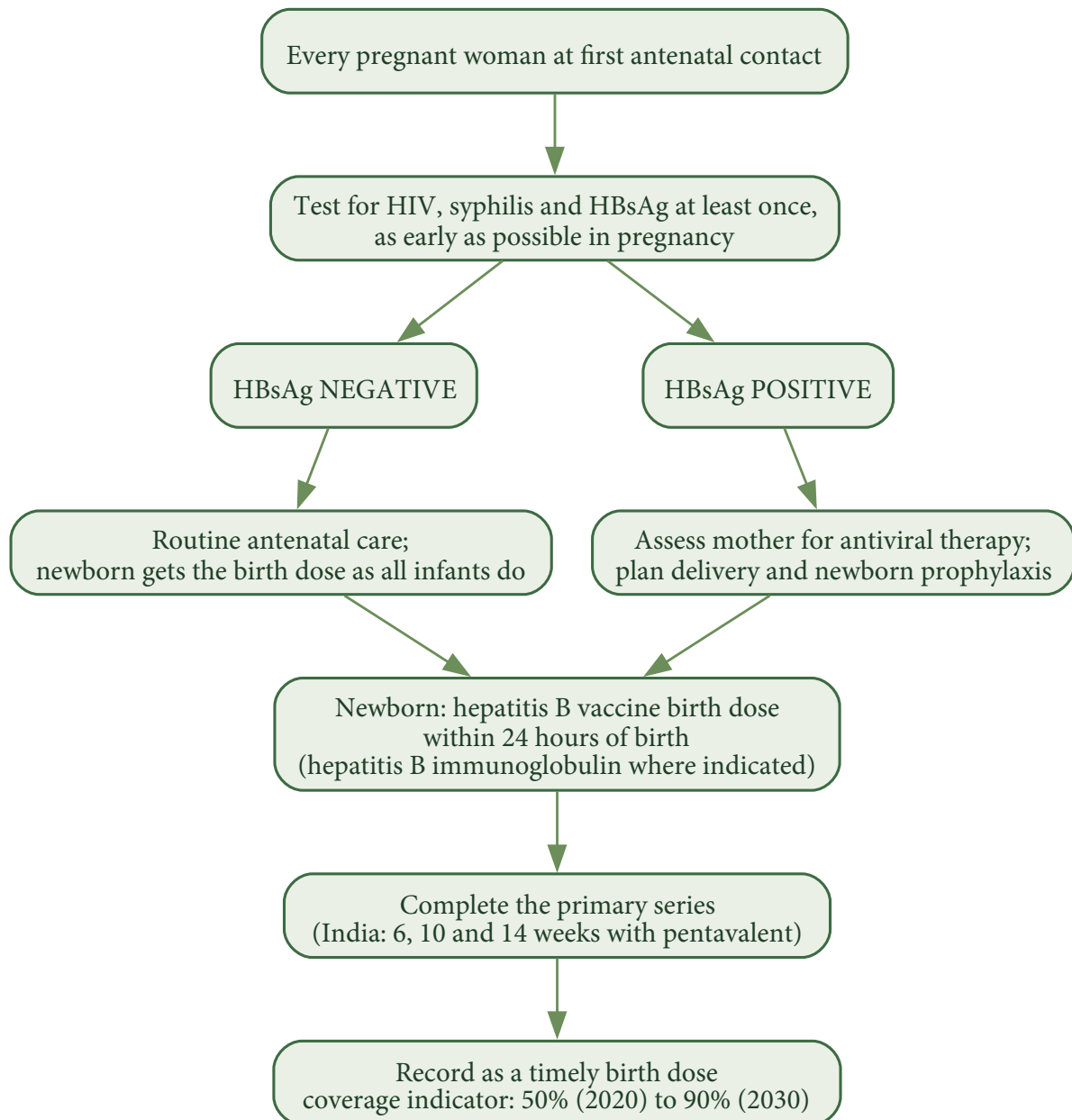
(Park 28e, Ch 5, p 274)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Elimination pathway of the Global Health Sector Strategy on viral hepatitis.** Labels, top to bottom: *Goal — eliminate hepatitis B and C as public health threats by 2030*; four stacked pillars *Prevent (birth-dose vaccine, prevention of vertical transmission, blood and injection safety, harm reduction)*, *Test (WHO 5Cs, decentralised testing)*, *Treat (tenofovir or entecavir for hepatitis B; direct-acting antivirals for hepatitis C)*, *Track (three-phase surveillance)*; a cross-cutting side bar marked *Awareness — Financing — Simplified service delivery — Equity — Community engagement — Research*; base labelled *Targets: 90 per cent diagnosed, 80 per cent treated or cured, HBsAg below 0.1 per cent in under-fives by 2030*.



How the ten strategic and operational shifts translate into a service pathway for hepatitis elimination by 2030



Interruption of mother-to-child transmission of hepatitis B, the coverage target rising from 50 per cent (2020) to 90 per cent (2030)

High-yield exam pointers

- ▶ **Two strategies, two dates** — GHSS on viral hepatitis 2016–2021 (World Health Assembly, May 2016; –90 per cent infections, –65 per cent deaths by 2030), then GHSS 2022–2030 carrying **ten** shifts, lettered a to j.
- ▶ **Three headline 2030 targets** — **90 per cent diagnosed**; **80 per cent treated (B) or cured (C)**; **HBsAg under 0.1 per cent in under-fives**, from 0.94 per cent in 2020.
- ▶ **The two 100 per cent targets by 2030** — all blood units screened, all health-care injections safe.
- ▶ **Birth dose within 24 hours**; timely birth-dose coverage **50 → 90 per cent**; third-dose coverage held at **90 per cent**; needles and syringes per person who injects drugs **200 → 300**.
- ▶ **5Cs of testing** — Consent, Confidentiality, Correct results, Counselling, Connection.

- **Burden, 2022** — 304 million (254 million B, 50 million C); 2.2 million new infections; 1.3 million deaths; men 58 per cent of cases.
- **India's delivery arm** — National Viral Hepatitis Control Programme, 2018.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written as textbook fact: the 2022–2030 hepatitis strategy is one arm of the World Health Organization's integrated **Global Health Sector Strategies on HIV, viral hepatitis and sexually transmitted infections, 2022–2030**, and India delivers it through the **National Viral Hepatitis Control Programme (NVHCP)**, launched on World Hepatitis Day, 28 July 2018 under the National Health Mission, with the National Centre for Disease Control as nodal centre (Sunder Lal — Textbook of Community Medicine, Ch 8, p 417).

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO 2024 chronic hepatitis B guidelines, WHO Global Health Sector Strategy 2022–2030, National Viral Hepatitis Control Programme (India), Lancet Gastroenterology & Hepatology Commission

Programme and policy updates (India)

- **India's first triple-elimination initiative for vertical transmission** — West Bengal has run the country's first Elimination of Vertical Transmission of HIV, Syphilis and Hepatitis B (EVTHSH) initiative, piloted in four districts with integrated antenatal screening, standardised treatment pathways, strengthened referral linkage, unified data recording and a single governance mechanism; after the pilot the State government decided to scale it up statewide. This is strategic shift (c) — antenatal HBsAg testing plus a timely birth dose — turned into an Indian programme structure. [PMID 42084996]
- **What the National Viral Hepatitis Control Programme actually delivers on the ground** — a mixed-method evaluation across six districts of Uttarakhand found the programme integrated well with existing health infrastructure, but COVID-19 cut hepatitis B screening by 73 per cent and hepatitis C screening by 76 per cent between 2019–20 and 2020–21; the biggest missing link in the hepatitis C cascade was the absence of PCR-based viral-load testing at treatment centres, compounded by no dedicated data-management portal, an inefficient supply chain, and no nomination of pharmacists or peer supporters. [PMID 42257146]

Current national data

- **India's share of the global burden, 2022 data** — of 254 million people living with chronic hepatitis B and 50 million with hepatitis C worldwide, India accounts for 35.3 million, 11.6 per cent of the combined global burden; five countries — China (83.7 million), India, Indonesia, Nigeria and Pakistan — together carry 55 per cent of it. Verified reports were received from 187 of 194 countries and territories. [PMID 41895313]
- **Hepatitis and tuberculosis co-infection in India** — pooled prevalence among patients with tuberculosis is 10 per cent (95% CI 5–16) for hepatitis B and 6 per cent (95% CI 5–7) for hepatitis C, rising to 17 per cent for hepatitis B in HIV-TB co-infection and 11 per cent in drug-resistant

tuberculosis, with the highest rates in southern and western India — a direct argument for screening integrated between the tuberculosis and hepatitis programmes. [PMID 42295712]

Elimination, surveillance and disease status

- ▶ **The 2030 target is off-track, but Indian policy has moved** — the 20 highest-burden countries carry more than 75 per cent of the global disability-adjusted life years from viral hepatitis; on the hepatitis elimination policy index, 14 of these 20 improved after 2018, with India among those showing the largest gains, mostly through a formal national action plan, publicly funded screening and government-subsidised antiviral treatment. [PMID 38367629]
- ▶ **The real care cascade against the strategy's coverage targets** — in 2022 only 13 per cent of the 254 million people with chronic hepatitis B were diagnosed and 3 per cent treated; pooling 110 cohorts from 50 countries, assessment for treatment eligibility (50.4 per cent) and initiation of antiviral therapy (49.3 per cent) were both lower in primary care than in hospital specialist care (73.9 and 78.1 per cent), so decentralisation and task-sharing must be paired with deliberate linkage and retention support. [PMID 40819651]
- ▶ **STOP-C (2026), India** — 3,000 people who inject drugs across seven cities received sofosbuvir-velpatasvir for 12 weeks with adherence support allocated by a prognostic risk score; high-intensity support (patient navigation plus flexible directly observed therapy) gave only a marginal gain in the lower-risk stratum (adjusted RR 1.09; 95% CI 1.00–1.19; $p = 0.04$) and no benefit in the higher-risk stratum. Sustained virological response peaked at 68.3 per cent (minimal-risk, high-intensity arm), was 45–51 per cent across the elevated-risk arms, and was 58 per cent overall on intention-to-treat — well below the levels reported in registration trials, so adherence support alone will not deliver elimination in this key population. [PMID 42379314]

Guideline and legislative updates

- ▶ **WHO guidelines on chronic hepatitis B, revised March 2024** — the core of the revision is simplification of care and expansion of treatment criteria, so a far larger share of people with chronic infection now qualify for antiviral therapy; it is written for high-endemicity, resource-limited settings where the older society algorithms were largely unusable, and the next EASL, AASLD and APASL revisions are expected to move partly towards these simplified algorithms, with some points of disagreement. [PMID 39647534]

Global and WHO developments

- ▶ **Hepatitis mortality has overtaken tuberculosis** — 1.3 million deaths from hepatitis B and C in 2022, 1.1 million of them from hepatitis B, a point estimate above that for tuberculosis deaths in 2023 (1.25 million); of 1.2 million new chronic hepatitis B infections worldwide, 62.7 per cent (771,000) were in the WHO African Region, so scaling hepatitis B vaccination there is the highest-yield single action. Only 34.1 million people with hepatitis B had been diagnosed and 6.6 million were on antiviral treatment in 2022. [PMID 41895313]
- ▶ **Self-testing as a WHO-recommended entry point for hepatitis C** — in a Republic of Korea dynamic Markov model for 2024–2030, universal hepatitis C self-testing was the only strategy able to reach the 2030 goal, cutting severe liver disease by 71 per cent and liver-related deaths by 69 per cent and averting 267,942 disability-adjusted life years at an incremental cost-effectiveness ratio of about US\$ 8,078 per DALY; risk-based screening was cheapest but fell short of the target. [PMID 39363405]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 5 (Epidemiology of Communicable Diseases), pp 270–271 and 274–276, for the ten strategic and operational shifts, the populations most affected and at-risk, the age and sex distribution of the chronic burden, Table 3 (activities that contribute to surveillance for viral hepatitis), Table 4 (impact and coverage indicators, targets and milestones for viral hepatitis by 2030), hepatitis B vaccination and immunoglobulin, the WHO 5Cs of testing, and hepatitis B and C treatment; and Chapter 14 (Sustainable Development Goals), p 856, for Sustainable Development Goal target 3.3 with hepatitis B incidence as an impact indicator. Oxford Textbook of Public Health 7e — Chapter 70 (Sexually transmitted infections), pp 2631–2632, for the World Health Assembly endorsement of the Global Health Sector Strategy on viral hepatitis 2016–2021 and its 90 per cent and 65 per cent reduction goals. Sunder Lal — Textbook of Community Medicine, Chapter 8, pp 417–418, for the National Viral Hepatitis Control Programme as India's implementing arm of the global strategy.

Recent-advances sources: 42084996 — First Initiative of Elimination of Vertical Transmission of Human Immunodeficiency Virus, Syphilis, and Hepatitis B in India: Learnings from West Bengal, Indian Journal of Public Health 2026, DOI [10.4103/ijph.ijph.1368.24](https://doi.org/10.4103/ijph.ijph.1368.24) (abstract-only); 42257146 — Evaluation of the National Viral Hepatitis Control Program in Uttarakhand: A mixed-method approach, Journal of Family Medicine and Primary Care 2026, DOI [10.4103/jfmpc.jfmpc.1733.25](https://doi.org/10.4103/jfmpc.jfmpc.1733.25); 41895313 — The burden of chronic hepatitis B and C in 2022 and progress towards elimination: a global report, Lancet Gastroenterology & Hepatology 2026, DOI [10.1016/S2468-1253\(25\)00375-9](https://doi.org/10.1016/S2468-1253(25)00375-9) (abstract-only); 42295712 — Viral hepatitis co-infections with tuberculosis in India: a systematic review and meta-analysis, Indian Journal of Medical Research 2026, DOI [10.25259/ijmr.3570.2025](https://doi.org/10.25259/ijmr.3570.2025); 38367629 — Progress towards elimination of viral hepatitis: a Lancet Gastroenterology & Hepatology Commission update, Lancet Gastroenterology & Hepatology 2024, DOI [10.1016/S2468-1253\(23\)00321-7](https://doi.org/10.1016/S2468-1253(23)00321-7) (abstract-only); 40819651 — Service delivery models and care cascade outcomes for people living with chronic hepatitis B: a global systematic review and meta-analysis, Lancet Gastroenterology & Hepatology 2025, DOI [10.1016/S2468-1253\(25\)00163-3](https://doi.org/10.1016/S2468-1253(25)00163-3) (abstract-only); 42379314 — A precision randomized trial of hepatitis C treatment support among people who inject drugs in India: the STOP-C Trial, Journal of Hepatology 2026, DOI [10.1016/j.jhep.2026.06.034](https://doi.org/10.1016/j.jhep.2026.06.034) (full text verified, PMC13420006); 39647534 — The 2024 updated WHO guidelines for the prevention and management of chronic hepatitis B: main changes and potential implications for the next major liver society clinical practice guidelines, Journal of Hepatology 2025, DOI [10.1016/j.jhep.2024.12.004](https://doi.org/10.1016/j.jhep.2024.12.004) (abstract-only); 39363405 — Self-testing strategy to eliminate hepatitis C as per World Health Organization's goal: analysis of disease burden and cost-effectiveness, Clinical and Molecular Hepatology 2025, DOI [10.3350/cmh.2024.0484](https://doi.org/10.3350/cmh.2024.0484). Plus WHO guidelines for the prevention, diagnosis, care and treatment of people with chronic hepatitis B infection, March 2024.

♦ ♦ ♦

Millennium Development Goals

Asked in: Paper I Nov 2016 — RGUHS MD Community Medicine

Definition — The Millennium Development Goals (MDGs) are the eight time-bound, quantified goals for development and poverty eradication that flowed from the United Nations Millennium Declaration of September 2000, each to be reached by the year 2015 against a 1990 baseline (Park 28e, Ch 1, p 23). They place health at the heart of development and represent commitments by governments throughout the world to reduce poverty and hunger and to tackle ill-health, gender inequality, lack of education, poor access to clean water and environmental degradation.

Genesis and structure of the framework

- ▶ **Millennium Summit, New York, September 2000** — representatives of 189 countries met and adopted the United Nations Millennium Declaration (Park 28e, Ch 1, p 23).
- ▶ The Declaration made specific commitments in **seven areas**: peace, security and disarmament; development and poverty eradication; protecting our common environment; human rights, democracy and good governance; protecting the vulnerable; meeting the special needs of Africa; and strengthening the United Nations.
- ▶ A "Road Map" then established goals and targets to be reached by 2015 in each of these seven areas. Those falling under development and poverty eradication are what is now universally called the Millennium Development Goals.
- ▶ In 2001, **189 countries** (nearly all United Nations member states) plus all major United Nations and other international agencies — including the World Bank and the International Monetary Fund — agreed to the eight interdependent goals with accompanying measurable targets (Textbook of International Health, Ch 4, p 210).
- ▶ **Monitoring architecture**: 8 goals, 18 targets, 48 indicators of progress, all measured against a 1990 baseline (Park 28e, Ch 1, p 23).
- ▶ **Revised framework**: agreed by the United Nations General Assembly at the 2005 World Summit and effective from **15 January 2008**, it expanded the framework to 21 targets monitored by 60 indicators, and added target 5B — "achieve, by 2015, universal access to reproductive health" under the goal of improving maternal health (Sunder Lal, Ch 17, p 767; Oxford Textbook of Public Health 7e, Ch 10, pp 419-420).
- ▶ **Share belonging to health**: three of the eight goals, eight of the 18 targets and 18 of the 48 indicators are *directly* health-related; every remaining goal acts on health *indirectly* through its determinants (Park 28e, Ch 1, p 23).

The eight goals and their principal targets

MD G	Goal	Principal target(s) to be reached by 2015 (baseline 1990)
1	Eradicate extreme poverty and hunger	Halve the proportion of people living on less than \$1 per day and of those who suffer from hunger
2	Achieve universal primary education	Ensure that all boys and girls complete a full course of primary schooling

MD G	Goal	Principal target(s) to be reached by 2015 (baseline 1990)
3	Promote gender equality and empower women	Eliminate gender disparity in primary and secondary education
4	Reduce child mortality	Reduce by two-thirds the mortality rate among children under five years
5	Improve maternal health	Reduce by three-quarters the maternal mortality ratio; (5B, added 2008) universal access to reproductive health
6	Combat HIV/AIDS, malaria and other diseases	Halt and begin to reverse the spread and incidence of these diseases
7	Ensure environmental sustainability	Integrate sustainable development into country policies and reverse loss of environmental resources; halve the proportion of people without sustainable access to safe drinking water and basic sanitation; improve the lives of at least 100 million slum-dwellers
8	Develop a global partnership for development	A non-discriminatory trade and financial system serving the least-developed countries; deal comprehensively with the debt problems of developing countries; cooperate with the pharmaceutical industry to provide access to affordable essential drugs and with the private sector to make new information and communication technologies available

(Textbook of International Health, Ch 4, pp 210-211)

The 18 health-related indicators of progress

Goal	Indicator number and definition
1. Eradicate extreme poverty and hunger	4. Prevalence of underweight children under five years of age; 5. Proportion of population below the minimum level of dietary energy consumption
4. Reduce child mortality	13. Under-five mortality rate; 14. Infant mortality rate; 15. Proportion of 1-year-old children immunized against measles
5. Improve maternal health	16. Maternal mortality ratio; 17. Proportion of births attended by skilled health personnel
6. Combat HIV/AIDS, malaria and other diseases	18. HIV prevalence among young people aged 15-24 years; 19. Condom use rate of the contraceptive prevalence rate; 20. Number of children orphaned by HIV/AIDS; 21. Prevalence and death rates associated with malaria; 22. Proportion of population in malaria-risk areas using effective malaria prevention and treatment measures; 23. Prevalence and death rates associated with tuberculosis; 24. Proportion of tuberculosis cases detected and cured under Directly Observed Treatment, Short-course (DOTS)
7. Ensure environmental sustainability	29. Proportion of population using solid fuel; 30. Proportion of population with sustainable access to an improved water source, urban and rural; 31. Proportion of urban population with access to improved sanitation
8. Develop a global partnership for development	46. Proportion of population with access to affordable essential drugs on a sustainable basis

(Park 28e, Ch 2, p 40)

Global scorecard at the close of the MDG era

Domain	Achievement over the MDG period	Verdict against the target
Maternal mortality (MDG 5)	Maternal mortality ratio fell about 44-45%, from 385 to 216 per 100,000 live births, 1990-2015; average annual reduction 2.6%	Missed — 5.5% annual reduction was needed for a three-quarters cut
Skilled birth attendance (MDG 5)	Births assisted by skilled health personnel rose from 59% to 71%	Substantial but incomplete
Child mortality (MDG 4)	Under-five mortality rate fell from 91 to 43 per 1000 live births (1990-2015), and to 39 by 2017; under-five deaths fell from 12.7 million (1990) to 5.9 million (2015)	Missed — well short of a two-thirds reduction
Newborn survival (MDG 4)	Newborn deaths declined <i>slowest</i> of all under-five deaths and now form 45% of them, up from 40% in 1990	Weakest component
Measles immunization (MDG 4)	Coverage with at least one dose rose from 73% (2000) to about 85% (2017)	Good progress
HIV/AIDS (MDG 6)	New HIV infections fell from 3.5 million (2000) to 1.8 million (2017); people on antiretroviral therapy rose from 800,000 (2003) to 21.7 million (2017)	Spread reversed
Malaria (MDG 6)	Incidence fell about 37% and mortality about 58%, 1990-2015	Substantial progress
Drinking water (MDG 7)	Target declared met by the United Nations, but the indicators failed to capture sustainability, accessibility and safety; over 1.3 billion still lacked safe water at end-2015	Contested
Sanitation (MDG 7)	The least progress of any MDG target — about 2.3 billion people remained without improved sanitation at the end of the MDG period	Missed

(Oxford Textbook of Public Health 7e, Ch 107 pp 4042-4043, Ch 21 pp 812-814, Ch 12 pp 477-478)

India's position at 2015 (base year 1990)

Indicator	India's level at the close of the MDGs (2015)
Under-five mortality rate	Fell from 123 to 48 per 1000 live births — good progress
Infant mortality rate	40 per 1000 live births
Maternal mortality ratio	167 per 100,000 live births — MDG 5 showed the least progress
Safe (institutional/attended) deliveries	About 81%
Adolescent birth rate	39 — remained high
Under-three children underweight	About 33%

Indicator	India's level at the close of the MDGs (2015)
Improved drinking water source	About 90% of the population
Improved sanitation facility	Only 40% of the population
Adult HIV prevalence	Stable at a low 0.26%
Tuberculosis	Prevalence 195 and incidence 167 per lakh population; mortality 17 per lakh; DOTS cure rate above 85%
Vector-borne disease including malaria	Continued high morbidity with recurring outbreak and epidemic threat

(Sunder Lal, Ch 17, p 767)

Delivery route in India — the goals were pursued not as a separate vertical programme but through existing machinery: the Reproductive and Child Health framework, the National Population Policy 2000, the National Health Policy 2002 and the National Rural Health Mission, since most MDG objectives and indicators map onto reproductive and child health. *Countdown to 2015*, formed in 2005 by scientists, policymakers, activists and institutions, tracked progress specifically towards MDG 4 (Sunder Lal, Ch 17, p 767).

Limitations of the MDG framework

- ▶ **Narrow definition of poverty**, with a focus on market-based approaches and debt servicing rather than on structural change (Textbook of International Health, Ch 4, p 211).
- ▶ **Failure to address the causes of, and solutions to, poverty** — the framework did not recognise the role of exploitation in creating and perpetuating poverty, and proposed no real redistributive strategy beyond renewed emphasis on economic growth (Textbook of International Health, Ch 4, p 213).
- ▶ **Priorities potentially in conflict with domestic needs**, since targets were set globally and uniformly while national health agendas differed.
- ▶ **Inequity blind** — the goals were framed as national averages and did not adequately address equity, so gains could be recorded while the poorest, and vulnerable groups such as forced migrants, were left behind (Oxford Textbook of Public Health 7e, Ch 92, p 3478).
- ▶ **Ownership deficit** — the goals were written primarily by the United States, Europe and Japan under heavy influence from large international aid organisations, rather than by the countries expected to achieve them (Oxford Textbook of Public Health 7e, Ch 24, pp 919-920).
- ▶ **Measurement failure** — deficiencies in availability, quality, timeliness and disaggregation of data, in statistical capacity and in national information systems were serious obstacles throughout 2000-2015; low- and middle-income countries with weak civil registration faced effectively unreachable standards of proof (Oxford Textbook of Public Health 7e, Ch 24, p 920).
- ▶ **Selective and aid-dependent** — the MDGs were closely associated with aid-spending and with vertical, donor-driven campaigns, so progress was hostage to donor priority and to absorptive capacity (Park 28e, Ch 14, pp 858-859; Textbook of International Health, Ch 13, p 699).
- ▶ Sen judged the goals **insufficiently reflective of the Millennium Declaration** from which they were drawn (Textbook of International Health, Ch 4, p 211).

Despite these, the MDGs are credited with accelerating development, saving lives, lifting people out of poverty and — by treating humanity as one community — with establishing "global health" as a working concept and unlocking large earmarked funds through PEPFAR, the President's Malaria Initiative, the GAVI Vaccine Alliance, the Bill & Melinda Gates Foundation and the Global Fund (Oxford Textbook of Public Health 7e, Ch 24, pp 919-920).

Transition from the MDGs to the Sustainable Development Goals

Feature	Millennium Development Goals (2000-2015)	Sustainable Development Goals (2016-2030)
Architecture	8 goals, 18 targets (21 from 2008), 48 indicators (60 from 2008)	17 goals, 169 targets, 232 indicators
Scope	A limited set of human development targets — poverty, education and health	Integrates the economic, environmental and social pillars of sustainable development
Applicability	A more or less singular purpose: improved human development outcomes in developing countries; closely tied to aid-spending	"Integrated and indivisible, global in nature and universally applicable" — relevant to all countries
Equity	Framed as national averages	Strong focus on equity — " no one will be left behind "
Health	3 separate health goals (MDG 4, 5, 6)	One comprehensive goal, SDG 3 — "ensure healthy lives and promote well-being for all at all ages" — with 13 targets
Continuity	—	MDG targets on maternal mortality (3.1), newborn and child mortality (3.2) and infectious disease (3.3) are retained and expanded to include neonatal mortality, hepatitis and waterborne disease; new targets added for non-communicable disease and mental health (3.4), substance abuse (3.5) and injuries (3.6)

(Park 28e, Ch 14, pp 852-859; Oxford Textbook of Public Health 7e, Ch 10, pp 419-420)

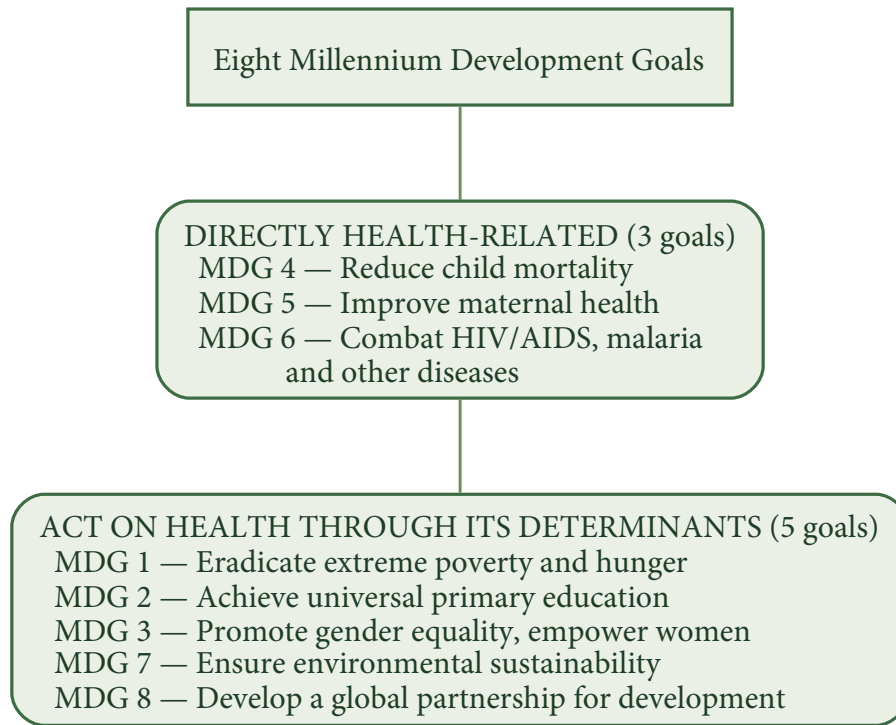
The MDGs came to the end of their term in December 2015 and the SDGs of the 2030 Agenda officially came into force on 1 January 2016 (Park 28e, Ch 14, p 852).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

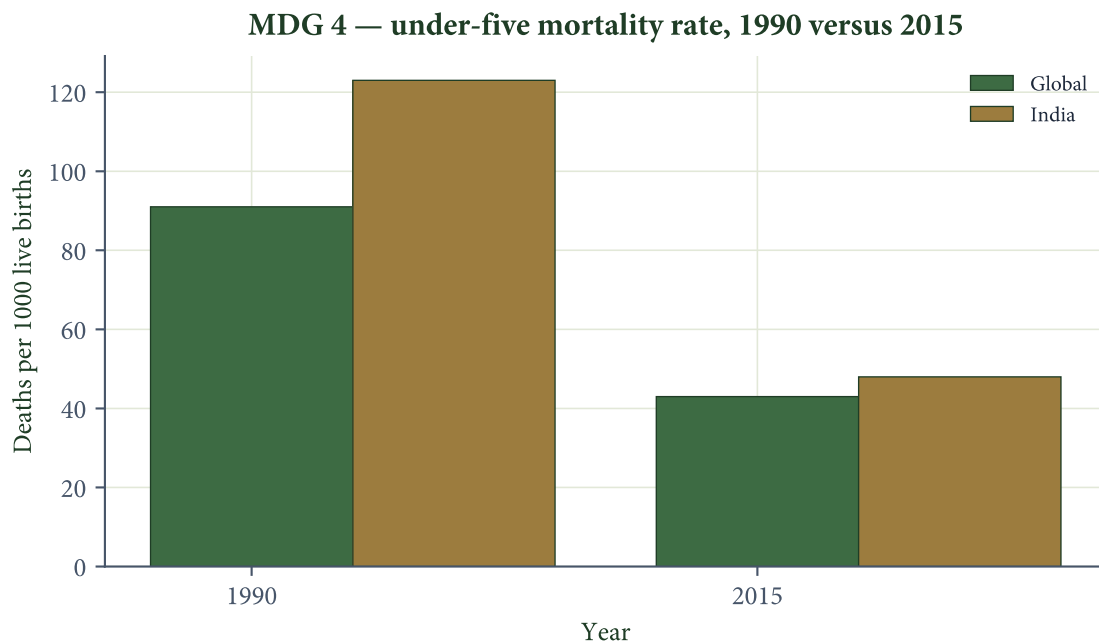
- **DRAWN figure** — the MDG monitoring pyramid (three stacked tiers, label exactly):
 - Apex: **8 Goals**
 - Middle tier: **18 Targets** (21 after the revision effective 15 January 2008)
 - Base tier: **48 Indicators of progress** (60 after the 2008 revision)
 - Right-side annotation: "**Directly health-related share = 3 goals, 8 targets, 18 indicators**"
 - Footer annotation: "**Baseline year 1990 → deadline 2015**"



Evolution of the Millennium Development Goals, from the Millennium Summit to the Sustainable Development Goals



The eight Millennium Development Goals grouped by their relation to health



Global under-five mortality fell from 91 to 43 per 1000 live births between 1990 and 2015 (Oxford Textbook of Public Health 7e, Ch 21); India's fell from 123 to 48 per 1000 live births over the same period (Sunder Lal, Ch 17, figures as at 2015). Neither reached the two-thirds reduction required by MDG 4.

High-yield exam pointers

- Order of the eight goals, mnemonic "P-E-G, C-M-D, E-P": Poverty and hunger, Education (primary), Gender; Child mortality, Maternal health, Disease (HIV/AIDS, malaria, others); Environment, Partnership.

- **Numbers to write:** 8 goals - 18 targets - 48 indicators (revised to 21 - 60 from 15 January 2008); health-related share 3 - 8 - 18; baseline **1990**, deadline **2015**; adopted at the Millennium Summit, September 2000, by **189 countries**.
- **Target arithmetic:** MDG 4 = reduce under-five mortality by **two-thirds**; MDG 5 = reduce maternal mortality ratio by **three-quarters**; MDG 1 and MDG 7 (water, sanitation) = **halve**; MDG 7 also = improve the lives of at least **100 million slum-dwellers**.
- **Target 5B** (added effective 15 January 2008) = "achieve, by 2015, universal access to reproductive health".
- **India at 2015:** under-five mortality 123 → 48 per 1000 live births; infant mortality rate 40; maternal mortality ratio 167 per 100,000 live births (least progress); safe deliveries 81%; adult HIV prevalence 0.26%; improved water 90% but improved sanitation only 40%.
- **Global verdict:** sanitation showed the least progress of any target; maternal mortality fell 44-45% against a required 75%; under-five mortality fell from 91 to 43 per 1000 live births against a required two-thirds cut.
- **Successor:** SDGs from 1 January 2016 — 17 goals, 169 targets, 232 indicators, deadline 2030; SDG 3 carries all 13 health targets; motto "no one will be left behind".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e states the original framework as 18 targets and 48 indicators while the Textbook of International Health states 16 targets, so write Park's figures in an RGUHS answer; the revised counts of 21 targets and 60 indicators date from 15 January 2008. India's 2015 values quoted above are the MDG-era closing figures and have since improved further under the Sustainable Development Goal framework — the sample registration system maternal mortality ratio for India was 97 per 100,000 live births for 2018-2020 (Park 28e, Ch 14, p 853).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, National Family Health Survey (NFHS-3 to NFHS-5), UN Inter-agency Group for Child Mortality Estimation, MoHFW / National Health Mission, WHO Framework Convention on Tobacco Control

Programme and policy updates (India)

- **Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana (PM-JAY)**, launched 2018, is now the vehicle through which India pursues the universal-health-coverage ambition the MDGs never carried: the world's largest publicly funded health insurance, covering the socio-economically bottom 40% (about 535 million people in 107.4 million households) up to ₹5,00,000 per household per year, financed 60:40 between the centre and the states (90:10 for the north-eastern and Himalayan states). The MDG-era equity criticism persists inside it — after excluding obstetric and gynaecological procedures, men raise 23.6% more surgical claims than women, and only 7% of hospitals in Aspirational Districts are empanelled for cardiothoracic and vascular surgery against 17% elsewhere [PMID 41393311].
- **National Sickle Cell Anaemia Elimination Mission**, announced in the Union Budget 2023-24, targets elimination of sickle cell disease by 2047 through universal screening of tribal populations

aged 0-40 years in six high-burden states (Madhya Pradesh, Maharashtra, Gujarat, Tamil Nadu, Odisha, Chhattisgarh). A Health Technology Assessment in India appraisal led the National Health Mission to direct states to procure point-of-care tests by public tender; competitive bidding and in-house manufacture cut the price to under US\$1.21 per test, projecting US\$37.86 million of savings at full coverage [PMID 42594133].

Current national data

- ▶ **Where India's child deaths now sit** — across five NFHS rounds (1993-2021) roughly 58% of under-five deaths now occur in the neonatal period, up from 42.5% in 1993. Relative inequality in early neonatal mortality between the poorest and richest wealth quintiles *widened* from 1.9 to 2.9 over that period, and most child deaths are now concentrated in the two lowest quintiles; the poorest and least-educated children are not on track for the SDG early-neonatal or post-neonatal targets [PMID 40316411].
- ▶ **Nutrition indicators diverge** — exclusive breastfeeding rose from 43.5% (1993) to 63.9% (2021) and early initiation from 9.6% to 41.2%, with 23 of 36 states and union territories already at the 60% Global Nutrition Target for 2030 and six projected off-target, though mixed milk feeding worsened by nine percentage points [PMID 42134478]. Against this, anaemia in children aged 6-59 months *rose* from 64% in NFHS-4 (2015-16) to 68% in NFHS-5 (2019-21), with maternal anaemia the strongest determinant and Scheduled Tribe children at about 1.5 times the risk [PMID 40579634].
- ▶ **Immunisation shortfall** — pooling 45 Indian studies (2015-2025, over 428,000 children), 26% (95% CI 21-32%) of under-fives remain incompletely immunised despite the Universal Immunization Programme and Mission Indradhanush. Strongest predictors were absence of an immunisation card (OR 9.46, 95% CI 3.28-27.28), non-institutional delivery (OR 3.59), maternal illiteracy (OR 3.25) and female sex (OR 1.34) [PMID 42320383].

Elimination, surveillance and disease status

- ▶ **Kala-azar (visceral leishmaniasis)** — the tri-country Kala-azar Elimination Programme launched by India, Bangladesh and Nepal with WHO in 2005 set the threshold of fewer than 1 new case per 10,000 population at block/upazila level. By 2023 India had cut cases 98.4% below pre-programme levels, Bangladesh 99.5% and Nepal 88.5%, leaving the region at about 2% of its earlier burden. Bangladesh was the first country validated by WHO (2023); India has held the threshold for two consecutive years but validation is still pending, and East Africa now carries most of the global caseload [PMID 41856659].
- ▶ **Rotavirus vaccine in the Universal Immunization Programme** — India was the first country in the WHO South-East Asia Region to add rotavirus vaccine to the UIP (Rotavac, March 2016; Rotasiil added 2018, nationwide by 2019). Eight-year sentinel surveillance of under-five diarrhoea admissions shows rotavirus positivity falling from 37% before rollout to 17.5% by 2023, the steepest fall in infants (29.4% to 15.1%). G3P[8] is now the dominant genotype (44.7%), but the same shift is seen in unvaccinated countries, so it reflects natural strain dynamics rather than vaccine pressure [PMID 41832515].

Guideline and legislative updates

- ▶ **Tobacco carried into the SDG framework as target 3.a** — SDG 3.a commits countries to strengthen implementation of the WHO Framework Convention on Tobacco Control, with the

UN target of a 30% relative reduction in tobacco use by 2030 against a 2016 baseline; global progress is projected to fall short by at least seven percentage points, and MPOWER's "O" (offer help to quit) reaches full cost coverage in only 26 countries (about 32% of the world's population). India's Chandigarh Declaration (Fifth National Conference on Tobacco or Health, 2021) commits to a tobacco endgame by 2030 [PMID 37955214]. Smokeless tobacco use among women aged 15-49 fell across NFHS rounds from 8.4% (2005-06) to 5.6% (2015-16) to 3.8% (2019-21), but reaches 45% in Mizoram and 41.4% in Manipur, Scheduled Tribe women carry more than twice the odds (AOR 2.12), and the gutka and pan-masala bans under the Food Safety and Standards Act 2006 remain weakly enforced in many states [PMID 42545929].

Global and WHO developments

- **Child and newborn survival against SDG 3.2** — UN Inter-agency Group estimates show global under-five mortality falling from 93.58 (1990) to 36.72 per 1000 live births (2023), yet the world is not projected to reach the target of 25 or fewer until 2035; India and Nigeria together account for about one-third of the 4.78 million under-five deaths, although India is among only nine countries expected to reach the under-five target by 2030 [PMID 41920787]. Neonatal mortality fell from 36.70 to 17.33 per 1000 over the same period, with the target of 12 or fewer not projected globally until 2036 — and India, which recorded the largest absolute number of neonatal deaths (401,499 in 2023), is projected to reach it only between 2031 and 2040 [PMID 40853132]. Newborn survival therefore remains the weak link it was under MDG 4.
- **Maternal mortality against SDG 3.1** — an analysis of 163 countries (1990-2023) finds no single convergence path but three distinct clusters; only 65 countries (40%) are forecast to reach the SDG 3.1 target of fewer than 70 maternal deaths per 100,000 live births by 2030, while 98 (60%) stay above it. The global maternal-mortality Gini coefficient has held at about 0.66 since 1990, with between-region inequality the dominant component — the same national-average blindness that undid MDG 5 [PMID 42106672].

SOURCE & COVERAGE

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Objectives and works of World Health Organization (WHO)

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — The World Health Organization (WHO) is the specialized, non-political health agency of the United Nations (UN), with headquarters at Geneva. Its Constitution was drafted in 1946 by a Technical Preparatory Committee under *Rene Sand*, approved the same year by an International Health Conference of 51 nations at New York, and came into force on **7th April 1948**, celebrated every year as World Health Day (Park 28e, Ch 25, p 1055). It is unique among UN specialized agencies in having its own constitution, governing bodies, membership and budget — part of, but not subordinate to, the UN (Park 28e, Ch 25, pp 1055–1056).

Objective and the constitutional principles behind it

Item	Content
Constitutional objective — set out in the preamble of the Constitution (Park 28e)	"The attainment by all peoples of the highest possible level of health " (Park 28e, Ch 25, p 1055)
Current/operational objective (30th World Health Assembly, May 1977)	Attainment by all people of the world of a level of health that will permit them to lead a socially and economically productive life — the Health for All (HFA) formulation (Park 28e, Ch 25, p 1055; Ch 10, p 720)
Current five-year strategic vision — the Thirteenth General Programme of Work (GPW 13), 2019–2023	One billion more people benefiting from universal health coverage (UHC); one billion more better protected from health emergencies; one billion more enjoying better health and well-being (WHO 2018) (Oxford Textbook of Public Health 7e, Ch 107, p 4043 and pp 4044–4045)

Nine principles of the preamble carry the objective (Park 28e, Ch 25, p 1055):

- ▶ **Health** is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.
- ▶ Enjoyment of the highest attainable standard of health is one of the **fundamental rights** of every human being, without distinction of race, religion, political belief, economic or social condition.
- ▶ The health of all peoples is fundamental to the attainment of **peace and security**, and is dependent upon the fullest cooperation of individuals and States.
- ▶ The **achievement of any State** in the promotion and protection of health is of **value to all**.
- ▶ **Unequal development** in different countries in the promotion of health and control of disease, especially communicable disease, is a common danger.
- ▶ **Healthy development of the child** is of basic importance; the ability to **live harmoniously in a changing total environment** is essential to such development.
- ▶ The **extension to all people of the benefits** of medical, psychological and related knowledge is essential to the fullest attainment of health.
- ▶ **Informed opinion and active cooperation** on the part of the public are of the utmost importance in the improvement of the health of the people.
- ▶ **Governments have a responsibility** for the health of their peoples, which can be fulfilled only by the provision of adequate health and social measures.

Constitutional functions (Article 2) — the mandate for the "works"

WHO's first constitutional function is to act as the directing and coordinating authority on all international health work; this lets Member States collectively identify priority health problems, define policies and targets, and devise strategies and programmes to attain them (Park 28e, Ch 25, p 1056).

Group	Functions listed in Article 2
Directing / coordinating	Act as directing and coordinating authority on international health work; collaborate with UN bodies, governments and professional groups
Technical cooperation	Assist governments in strengthening health services and in emergency aid; promote maternal and child health and welfare; foster mental health activities
Health promotion and determinants	Promote improvement of nutrition, housing, sanitation, recreation, and of economic, working and environmental conditions
Knowledge generation	Study and report on public health and medical care; promote research and health training
Disease control	Advance work to eradicate epidemic, endemic and other diseases, and to prevent injuries
Normative / standard-setting	Propose conventions, agreements and regulations and make recommendations on international health matters; standardize diagnostic procedures; revise international nomenclatures of diseases and causes of death; develop and promote international standards for food, biologicals, pharmaceuticals and similar products

(Textbook of International Health, Ch 3, p 110, Box 3–2)

Structure through which WHO works

Organ	Composition	Main functions
World Health Assembly — the "Health Parliament of Nations", supreme governing body	Delegates of all Member States, one vote each ; meets annually, usually in May, generally at Geneva	Determine international health policy and programmes; review the past year's work; approve the budget for the following year; elect Member States to designate persons to the Executive Board; appoint the Director-General on nomination of the Board; hold "technical discussions"
Executive Board	34 members , technically qualified in the field of health, designated by but not representing their governments; not fewer than three elected from each WHO region; one-third renewed yearly; meets at least twice a year (January, and after the Assembly in May)	Give effect to the decisions and policies of the Assembly; may itself act in an emergency such as epidemics, earthquakes and floods
Secretariat	Headed by the Director-General , chief technical and administrative officer, assisted by Assistant Directors-General and divisions at Geneva	Provide Member States with technical and managerial support for their national health development programmes

Organ	Composition	Main functions
Regional organizations (6)	Regional Office headed by a Regional Director ; a Regional Committee of Member States meets once a year	Review the region's work, plan its continuation and development; regional plans are amalgamated into the global programme
Country offices	WHO Representative (WR)	Work with national health authorities on reviewing health needs and resources, and planning, coordinating, implementing and evaluating national health programmes (Textbook of International Health, Ch 3, p 111)

(Park 28e, Ch 25, pp 1056–1057)

Membership and finance — Membership is open to all countries; territories not responsible for the conduct of their international relations may be **associate members**, who participate without a vote. From **56 members in 1948**, WHO now has **194 Member States and two associate members** (Park 28e, Ch 25, p 1056). It is financed by assessed contributions from Member States on a sliding scale based on population and gross national product, plus voluntary **extrabudgetary funds** (Textbook of International Health, Ch 3, pp 111–112).

WHO region	Regional Office headquarters
South-East Asia (SEARO)	New Delhi, India
Africa (AFRO)	Brazzaville, Congo
The Americas — Pan American Health Organization (PAHO)	Washington D.C., U.S.A.
Europe (EURO)	Copenhagen, Denmark
Eastern Mediterranean (EMRO)	Cairo, Egypt (Park 28e's table still prints Alexandria)
Western Pacific (WPRO)	Manila, Philippines

(Textbook of International Health, Ch 3, p 110, Table 3–5; Park 28e, Ch 25, p 1057, Table 1)

Works of WHO — the eight broad areas

#	Area	What WHO actually does — with named examples
1	Prevention and control of specific diseases	Global eradication of smallpox — the outstanding example of international health cooperation; now directing the global battle against poliomyelitis . Epidemiological surveillance and dissemination of information on diseases subject to the International Health Regulations (IHR) through the <i>Weekly Epidemiological Record</i> ; the "WHO Emergency Scheme for Epidemics". Non-communicable disease work on cancer, cardiovascular disease, genetic disorders, diabetes, blindness, mental disorders, drug addiction and dental disease. Vector biology and control, immunology, quality control of drugs and biological products, drug abuse monitoring, health laboratory technology; the Expanded Programme on Immunization (EPI) is a priority programme

#	Area	What WHO actually does — with named examples
2	Development of comprehensive health services	Supports national health policy development and comprehensive national health programmes — organizing health systems based on primary health care , health manpower development and use, building long-term national capability in health infrastructure, management (including monitoring and evaluation) and health services research; Appropriate Technology for Health to encourage self-sufficiency
3	Family health (a major programme since 1970)	Maternal and child health care, human reproduction, nutrition and health education — aimed at improving the quality of life of the family as a unit
4	Environmental health	Advises governments on basic sanitary services; protection of the quality of air, water and food; health conditions of work; radiation protection; early identification of new technological hazards — through the WHO Environmental Health Criteria Programme and the WHO Environmental Health Monitoring Programme
5	Health statistics	Dissemination of morbidity and mortality statistics through the <i>Weekly Epidemiological Record</i> , <i>World Health Statistics Quarterly</i> and <i>World Health Statistics Annual</i> ; publishes the International Classification of Diseases (ICD) , revised about every 10 years, to make national statistics comparable — the Eleventh Revision (ICD-11) was adopted by the World Health Assembly in 2019 and came into effect on 1 January 2022 , superseding ICD-10 which had been in force from 1 January 1993 (Park 28e, Ch 2, p 69); assists countries with medical records and national health information systems
6	Biomedical research	WHO does not itself do research — it stimulates and coordinates it, through a worldwide network of WHO Collaborating Centres , grants, Regional Advisory Committees and a Global Advisory Committee on health research; the Special Programme for Research and Training in Tropical Diseases targets malaria, schistosomiasis, trypanosomiasis, filariasis, leishmaniasis and leprosy
7	Health literature and information	Acts as a clearing house for health information; publishes hundreds of titles; the WHO library is a satellite centre of the Medical Literature Analysis and Retrieval System (MEDLARS) of the U.S. National Library of Medicine; public information services at headquarters and each regional office
8	Cooperation with other organizations	Collaborates with the UN and other specialized agencies, and maintains relations with international governmental and non-governmental organizations

(Park 28e, Ch 25, pp 1056–1057)

Landmark works — a chronology

Period	Work
1946–1948 (Interim Commission)	Control of post-Second World War epidemics
1947–1970s	Fellowships programme — over 50,000 doctors, nurses and sanitary engineers trained

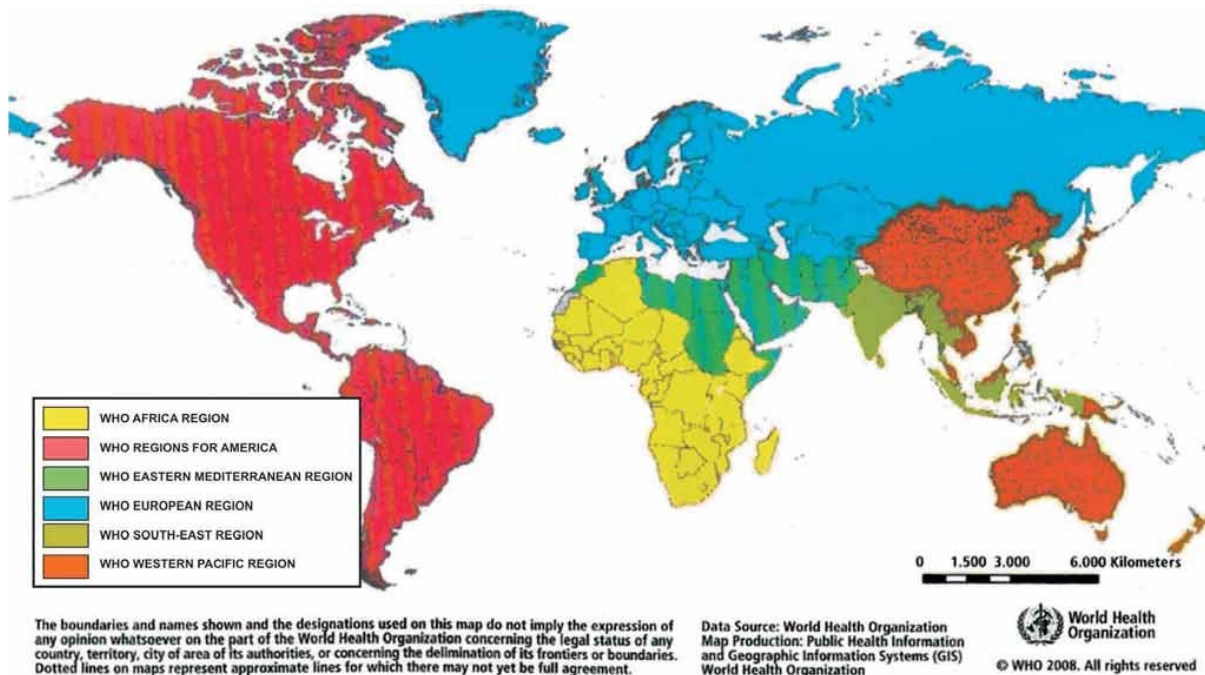
Period	Work
1950s	Mass bacille Calmette-Guerin (BCG) immunization of children against tuberculosis
1952–1970	Yaws campaign with the United Nations Children's Fund (UNICEF) — 300 million people treated in 50 countries, 95% fall in global prevalence
1955–1969	Global Malaria Eradication Campaign
1963	Codex Alimentarius created (food standards)
1967–1980	Smallpox Eradication Campaign
1974–	Expanded Programme on Immunization — diphtheria, pertussis, tetanus, measles, poliomyelitis, tuberculosis
1975–	Special Programme for Research and Training in Tropical Diseases
1977–	Essential Drugs Programme — first <i>Model List of Essential Medicines</i> ; Health for All declaration
1978	Alma-Ata International Conference on Primary Health Care (with UNICEF)
1981	International Code of Marketing of Breast-milk Substitutes
1982–	Selective Primary Health Care / Child Survival Programmes
1986	Ottawa Charter for Health Promotion; Global Programme on AIDS established
1988–	Global Polio Eradication Initiative with Rotary International, the Centers for Disease Control and Prevention (CDC) and UNICEF
1990s–	Rise of public–private partnerships — Roll Back Malaria (1998–), Stop TB Partnership (2000–), Global Alliance for Vaccines and Immunization (GAVI) Alliance (2000–)
1995–	Directly observed treatment, short-course (DOTS) programme for tuberculosis control, reaching 30 million people
2000–2002	Commission on Macroeconomics and Health
2001–	Measles Initiative (with the American Red Cross, UNICEF, CDC)
2003	Framework Convention on Tobacco Control adopted by the World Health Assembly
2003–2005	3 by 5 Initiative — 3 million people on antiretroviral treatment by 2005 (not achieved)
2005–2008	Commission on Social Determinants of Health
2007 (previous revisions 1951, 1969)	Implementation of the new International Health Regulations

(*Textbook of International Health*, Ch 3, pp 112–113, Table 3–6; *IHR implementation* — *Oxford Textbook of Public Health 7e*, Ch 105, pp 3996–3997)

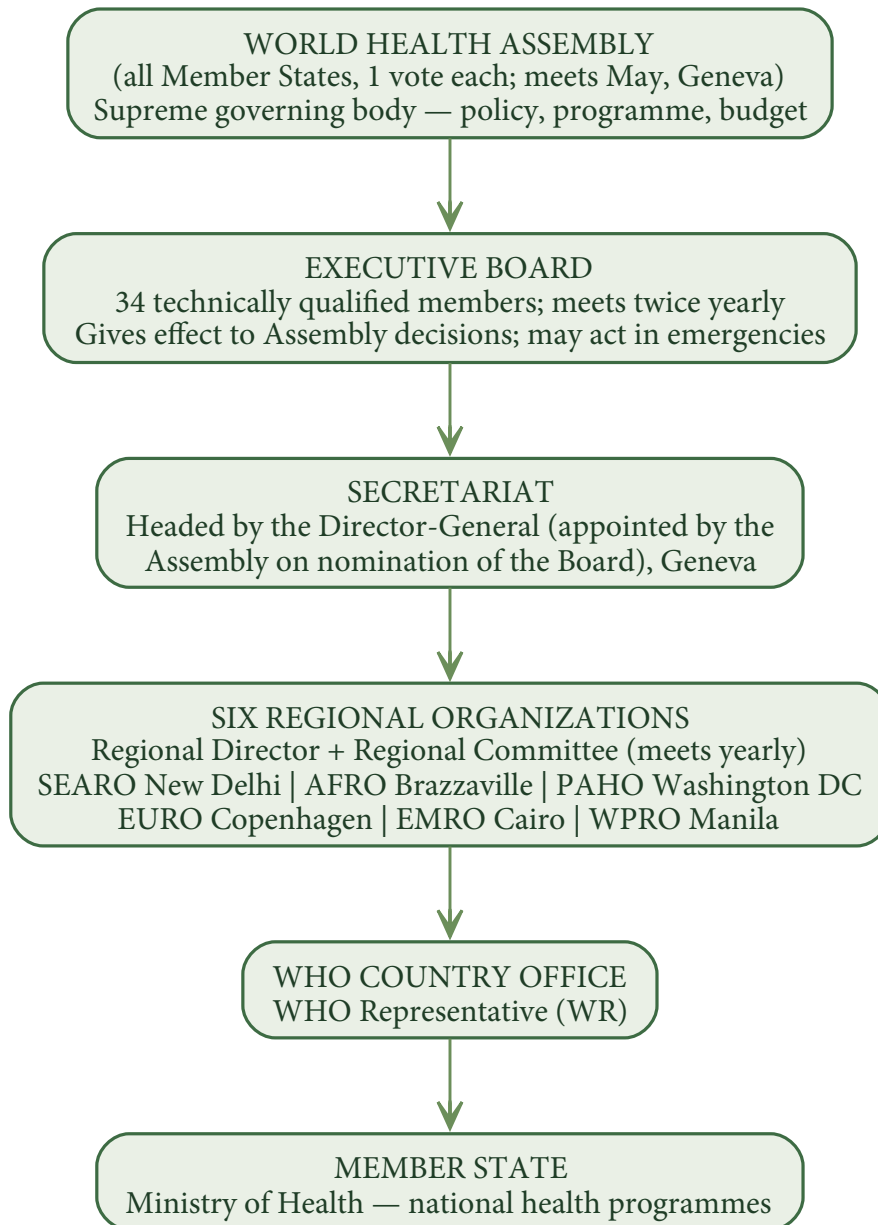
Work of WHO in the South-East Asia Region and India

- ▶ The South-East Asia Regional Office (SEARO) is at *World Health House, Indraprastha Estate, New Delhi*; the region has 11 Member States, and India joined in 1948 (Park 28e, Ch 25, pp 1057–1058).
- ▶ Regional activities cover malaria eradication, tuberculosis control, control of other communicable diseases, health laboratory services and vaccine production, health statistics, public health administration and rural health services, maternal and child health, nursing, environmental health and water supply, health education, nutrition, mental health, dental health, medical rehabilitation, quality control of drugs, and medical education (Park 28e, Ch 25, p 1058).
- ▶ Under the International Health Regulations (2005) each country must notify WHO of a potential public health emergency of international concern through its **National IHR Focal Point within 24 hours**, and must build core surveillance and response capacities (Oxford Textbook of Public Health 7e, Ch 105, pp 3996–3997).

Diagrams



WHO regions (Suryakantha — *Community Medicine with Recent Advances*, Fig 20.17, p 356)



Structure of WHO — the three principal organs and the regional and country tiers



How a WHO decision reaches the field — from World Health Assembly resolution to the sub-centre

High-yield exam pointers

- ▶ **Dates:** Constitution drafted 1946 (Rene Sand, Technical Preparatory Committee; 51 nations, New York) → in force 7 April 1948 = World Health Day. Members: 56 in 1948 → 194 Member States + 2 associate members now.
- ▶ **Objective, two versions:** constitutional (preamble) — "attainment by all peoples of the highest possible level of health"; current (30th World Health Assembly, 1977) — a level of health permitting a socially and economically productive life.
- ▶ **Nine preamble principles** — mnemonic *"Definition – Right – Peace – Value to all – Unequal danger – Child – Extension of knowledge – Informed public – Government responsibility"*.
- ▶ **Three principal organs:** World Health Assembly (annual, May, one vote per Member State) → Executive Board (34 members, twice yearly) → Secretariat (Director-General). Six regions: SEARO New Delhi, AFRO Brazzaville, PAHO Washington DC, EURO Copenhagen, EMRO Cairo, WPRO Manila.
- ▶ **Eight broad areas of work** (mnemonic *"Disease – Services – Family – Environment – Statistics – Research – Literature – Cooperation"*).

- **Named WHO products to quote:** ICD-11 (in effect 1 January 2022), International Health Regulations, *Weekly Epidemiological Record*, Model List of Essential Medicines, Codex Alimentarius, Framework Convention on Tobacco Control, WHO Collaborating Centres, Special Programme for Research and Training in Tropical Diseases.
- **Triple-billion targets of GPW 13 (2019–2023):** one billion more each for universal health coverage, protection from health emergencies, and better health and well-being — and remember that **WHO does not itself conduct research**, and is *part of, but not subordinate to*, the United Nations.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's international-health chapter still describes the Tenth Revision of the International Classification of Diseases as the current one while Park's own chapter on concepts of health and disease records the Eleventh Revision as superseding it from 1 January 2022, Park places the constitutional objective in the preamble whereas the Textbook of International Health places the same wording in Article 1, and the retrieved corpus carries no current Director-General's name and no dated post-2020 programme budget figure.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO Pandemic Agreement 2025 (Article 33), International Health Regulations 2024 amendments, WHO Fourteenth General Programme of Work 2025–2028

Programme and policy updates (India)

- **Immunisation programme lineage and the live HPV question** — WHO launched the Expanded Programme on Immunization in 1974 with bacille Calmette-Guerin, oral polio, diphtheria-pertussis-tetanus and measles vaccines for children plus tetanus toxoid for pregnant women, all free of cost; India adopted it in 1978 *without* measles vaccine, and only after measles was rolled out nationally during 1985–1990 was the programme renamed the Universal Immunisation Programme. The National Technical Advisory Group on Immunisation (NTAGI), established in 2000, is the body whose recommendations have governed every subsequent vaccine addition to the UIP [PMID 40252840] (abstract-only). The current addition under debate is human papillomavirus vaccine: WHO's cervical-cancer elimination framework requires 90% of adolescent girls vaccinated by age 15 (the "90-70-90" targets), and the AOGIN India policy statement argues that folding HPV vaccine into the UIP is necessary, drawing on single-dose efficacy trial data for dose and schedule [PMID 42169564] (abstract-only).
- **Tuberculosis programme renamed, and a target that does not match WHO's** — National Tuberculosis Control Programme (1962) → Revised National Tuberculosis Control Programme (1993) → **National Tuberculosis Elimination Programme (NTEP) since 2020**. The National Strategic Plan 2017–2025 runs on four pillars — Detect, Treat, Prevent, Build — and targets TB incidence, TB deaths and out-of-pocket expenditure. India's declared elimination year is **2025**, a decade ahead of WHO's global End TB target of **2035**; the WHO South-East Asia Region carries 45% of global TB incidence and 50% of global TB deaths, with India the single largest contributor [PMID 40707075] (abstract-only).

- **Ayushman Bharat's digital arm was renamed, and uptake lags the rollout** — the National Digital Health Mission, launched in 2020, was renamed the **Ayushman Bharat Digital Mission (ABDM)** in 2021, guided by NITI Aayog and the National Health Authority; its Ayushman Bharat Health Account (ABHA) is a 14-digit unique health identifier. Real-world adoption remains modest — at a government tertiary care hospital in Agra between September 2024 and April 2025, only 129,007 of 537,278 outpatient visits produced a completed ABHA registration (23.96%). Among 425 interviewed patients, 88% wanted training or support to use ABDM, 60% were reluctant to share the one-time password and about half reported privacy or security concerns [PMID 41787376].

Current national data

- **PM-JAY scale versus residual out-of-pocket spending** — Pradhan Mantri Jan Arogya Yojana, implemented in 2018, provides inpatient cover of ₹5,00,000 per household per year to more than 500 million socially and economically disadvantaged Indians, including 3 days of pre-hospitalisation and 15 days of post-hospitalisation costs, with *Ayushman Mitras* as in-hospital navigators. A mixed-methods evaluation across seven states (Bihar, Chhattisgarh, Gujarat, Kerala, Meghalaya, Tamil Nadu, Uttar Pradesh) and two districts each, with fieldwork in the early rollout period of November 2019 to March 2020, found 29% of exiting patients had still incurred pre-hospitalisation and 22% in-hospital out-of-pocket expenditure; beneficiaries in Chhattisgarh, Gujarat, Kerala, Meghalaya and Tamil Nadu were more appreciative of the scheme, while those in Bihar and Uttar Pradesh reported significant dissatisfaction over waiting times, informal payments and missing medicines [PMID 41316149].
- **Kala-azar — the numbers behind India's elimination claim** — cases fell from 77,102 (1992) to 599 (2023) and deaths from 1,419 (1992) to 58 (2018), 28 (2021) and 4 (2023); the South-East Asia Region's share of the global burden fell from 67% (2006) to 6% (2023). India reported 324 post-kala-azar dermal leishmaniasis cases and 75 kala-azar–HIV co-infections in 2023, with an at-risk population of 165.4 million across 54 districts of Bihar, Jharkhand, West Bengal and Uttar Pradesh, over 90% of cases coming from Bihar and Jharkhand [PMID 40037366].

Elimination, surveillance and disease status

- **Kala-azar elimination status and the validation pathway** — endemic blocks still above the elimination target fell from 633 (2005) to 136 (2015), 8 (2021) and **0 by December 2023**, the last being in Pakur district, Jharkhand, so every Indian implementation unit now meets the WHO threshold of **<1 case per 10,000 population at sub-district (block) level**. India has entered a *consolidation phase* and must hold the threshold for **three consecutive years** before submitting a dossier for WHO validation — Bangladesh was certified first, in 2023. The single-dose liposomal amphotericin B that underpinned the programme comes through a WHO-negotiated Gilead donation running only to the end of 2025, making validation of cheaper generic formulations and continued surveillance the stated risks to sustainment [PMID 40828749; PMID 40037366].

Guideline and legislative updates

- **International Health Regulations amended (2024)** — **supersedes the 2005 text taught in the body above** — adopted by consensus at the 77th World Health Assembly on 1 June 2024 and entering into force from **September 2025**. Key changes: a new "**pandemic emergency**" tier sitting above the public health emergency of international concern, which the Director-General may

declare (Articles 1 and 12); **National IHR Authorities** created alongside the existing National IHR Focal Point to coordinate implementation (Article 4); a duty on the WHO Secretariat to facilitate access to health products during a PHEIC or pandemic emergency (Article 13); and a **Coordinating Financial Mechanism** for core-capacity financing in developing countries (Article 44), with an implementation subcommittee under Article 54. Under Articles 59 and 61 a State Party may reject an amendment or enter a reservation, in which case it does not bind that state [PMID 40589223; PMID 40083009].

- **WHO Pandemic Agreement adopted, 20 May 2025** — the 78th World Health Assembly adopted a 35-article instrument negotiated under **Article 19 of the WHO Constitution**, only the second such treaty in WHO's history after the Framework Convention on Tobacco Control (adopted 2003, in force 2005); 124 governments voted to adopt it after three years of negotiation [PMID 40589223]. It commits parties to a **Pathogen Access and Benefit-Sharing (PABS)** system — under which at least **20% of pandemic health products go to WHO (10% as donation, 10% at affordable price)** — plus a **Global Supply Chain and Logistics Network** and a **Coordinating Financial Mechanism**; unlike the IHR, which binds all Member States on an opt-out basis, the Agreement is opt-in and requires each state to commit explicitly [PMID 40083009]. The PABS annex (Article 31.2) passed to an intergovernmental working group reporting to the 2026 Health Assembly [PMID 40589223]. **Under Article 33 the Agreement enters into force 30 days after the 60th instrument of ratification, acceptance, approval, formal confirmation or accession** [WHO Pandemic Agreement, Article 33; WHO news release, 20 May 2025]. Commentators caution that its practical effect may be blunted by vague wording, unresolved provisions and the absence or abstention of influential states [PMID 40443067] (abstract-only).

Global and WHO developments

- **GPW 14 has replaced GPW 13 and the triple-billion targets** — the **Fourteenth General Programme of Work**, approved by the 77th World Health Assembly in May 2024, covers 2025–2028. By end-2028 it targets **6 billion** people enjoying better health and well-being, **5 billion** covered by universal health coverage without financial hardship and **7 billion** better protected from health emergencies, delivered through six strategic objectives — climate change as a health threat, health determinants across sectors, primary health care for UHC, service coverage and financial protection, all-hazards risk preparedness, and rapid emergency detection and response [WHO, Fourteenth General Programme of Work 2025–2028].
- **WHO's financing is under acute strain** — the United States announced withdrawal on 20 January 2025, having contributed roughly **US\$218 million in assessed** and **US\$1.065 billion in voluntary funds in 2022–23**, and directed that negotiations on the Pandemic Agreement and IHR amendments cease [PMID 40767200; PMID 40083009]. Assessed contributions from Member States now supply **less than one-fifth** of WHO's budget; the 2026 health emergency appeal, launched on 3 February 2026, seeks **nearly US\$1 billion** across 36 emergencies including 14 Grade 3 crises, after 2025 funding limited coverage to a third of the 81 million people targeted [PMID 41998654].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 25 International Health, pp 1055–1058, Table 1; Ch 2 Concept of Health and Disease, p 69 on the International Classification of Diseases; Ch 10, p 720); Textbook of International Health (Ch 3, International Health Agencies, Activities and Other Actors, pp 110–113, Box 3–2, Table 3–5, Table 3–6); Oxford Textbook of Public Health 7e (Ch 107, Global health in the era of sustainable development, p 4043 and pp 4044–4045 on the Thirteenth General Programme of Work 2019–2023; Ch 105, pp 3996–3997 on the International Health Regulations 2005; Ch 24, Box 4.2.1, pp 918–919 on the WHO founding principles); Suryakantha — Community Medicine with Recent Advances (Fig 20.17, WHO regions, p 356). Coverage note: the corpus supports the constitutional objective, the nine preamble principles, the Article 2 functions, the structure and the eight areas of work in full, but carries no dated post-2020 WHO programme budget or leadership data.

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Public health emergency of international concern (PHEIC)

Asked in: Paper III 12-Jul-2023 — RGUHS MD Community Medicine

Definition — A public health emergency of international concern (PHEIC) is "an extraordinary event which is determined to constitute a public health risk to other States through the international spread of disease and to potentially require a coordinated international response" (Oxford Textbook of Public Health 7e, Ch 107, pp 4048-4049). It is a formal determination made by the **Director-General of the World Health Organization (WHO)** under the **International Health Regulations (IHR), 2005**, and is the strongest rating available under the Regulations as written. The duty to notify WHO arises from the Annex 2 assessment itself; declaring a PHEIC then triggers WHO's temporary recommendations and mobilises a coordinated international response.

1. Legal basis — the International Health Regulations (2005)

Item	Content
Instrument	IHR (2005), adopted by the World Health Assembly in 2005 and in force from 2007; a legally binding treaty accepted by 196 countries (Oxford 7e, Ch 42, pp 1577-1578; Ch 105, pp 3996-3997)
Predecessor	International Sanitary Regulations 1951 → IHR 1969, which covered only cholera, plague, yellow fever and smallpox and worked by pre-set control measures at international borders (Textbook of International Health, Ch 5, p 234; Oxford 7e, Ch 75, p 2857)
Why it was revised	Border posts could not stop diseases that crossed while still in the incubation period; countries under-reported cholera, plague and yellow fever for fear of stigma and trade loss; the 2003 severe acute respiratory syndrome (SARS) epidemic showed how fast a new disease spreads (Oxford 7e, Ch 75, pp 2857-2858; Ch 105, pp 3996-3997)
Purpose	"To ensure maximum security against international spread of diseases with the minimum interference with world traffic" (Park 28e, Ch 25, p 1056) — formally, to <i>prevent, protect against, control and provide a public health response to the international spread of disease</i> , commensurate with and restricted to public health risks, avoiding unnecessary interference with international traffic and trade (Suryakantha, Ch 20, p 325; Oxford 7e, Ch 28, pp 1071-1072)
Scope	ALL public health risks — infectious, food-borne, chemical, radiological and environmental contamination — not merely a fixed disease list (Oxford 7e, Ch 105, pp 3996-3997; AFMC Red Book 2024, Ch 26, p 769)
National machinery	Every State Party designates a National IHR Focal Point , accessible at all times, which sends urgent communications to the WHO IHR Contact Point and consolidates input from surveillance, points of entry, clinics, hospitals and other government departments (Red Book — Public Health and Preventive Medicine, Ch 2, p 1024)
Who declares a PHEIC	The WHO Director-General, on the advice of an IHR Emergency Committee convened for that event (Oxford 7e, Ch 75, pp 2862-2863)

2. What must be notified — the Annex 2 decision instrument

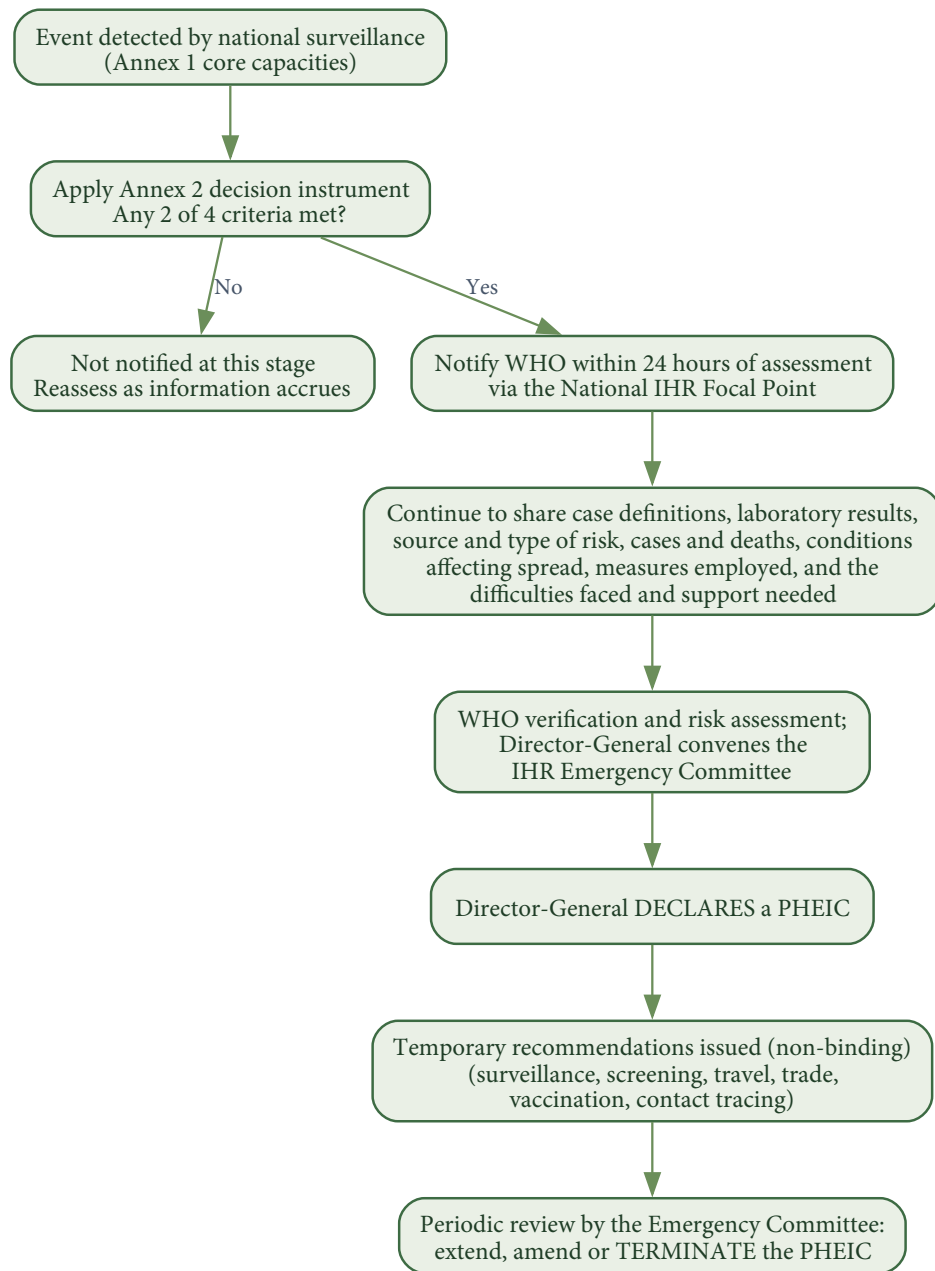
Category	Diseases / events	Action required
Always notifiable — a single case suffices	Smallpox; poliomyelitis due to wild-type poliovirus; human influenza caused by a new subtype; severe acute respiratory syndrome (SARS)	Notify WHO immediately; no algorithm needed (Oxford 7e, Ch 75, pp 2857-2858; AFMC Red Book 2024, Ch 26, p 769)
Always run the algorithm	Cholera; pneumonic plague; yellow fever; viral haemorrhagic fevers (Ebola, Lassa, Marburg); West Nile fever; other diseases of special national or regional concern e.g. dengue fever, Rift Valley fever, meningococcal disease	Apply the Annex 2 decision instrument to decide whether the event is a PHEIC (Sunder Lal, Ch 18, p 772; Red Book, Ch 2, pp 1025-1026)
Any other event	Any event of potential international public health concern, including those of unknown cause or source , and biological, chemical or radiological events meeting IHR criteria	Apply the Annex 2 decision instrument (AFMC Red Book 2024, Ch 26, p 769)

The four Annex 2 questions — the event is notified if the answer to any TWO of the four is "yes" (AFMC Red Book 2024, Ch 26, p 769):

- ▶ Is the public health impact of the event **serious**?
- ▶ Is the event **unusual or unexpected**?
- ▶ Is there a significant risk of **international spread**?
- ▶ Is there a significant risk of **international travel or trade restrictions**?

If the criteria are not met the event is *not notified at this stage* and is **reassessed when more information becomes available** (Sunder Lal, Ch 18, pp 772-773).

3. How a PHEIC is declared and what follows



Pathway from detection of an event to declaration and termination of a PHEIC under IHR (2005) (Red Book, Ch 2, p 1024; Oxford 7e, Ch 75, pp 2862-2863)

Three points the flowchart cannot carry:

- ▶ **Verification draws on unofficial sources** — WHO uses the Global Outbreak Alert and Response Network (GOARN) and its Collaborating Centres, and the IHR explicitly recognise **non-State sources of information**, which often precede official notification when a country is reluctant to reveal an event (Oxford 7e, Ch 75, pp 2861-2863).
- ▶ **The Regulations bind, the recommendations do not** — the IHR (2005) are a legally binding treaty on States Parties, whereas the temporary recommendations issued under a PHEIC are time-limited advice, and a State that ignores them faces no sanction (WHO International Health Regulations (2005), second edition; Suryakantha, Ch 20, p 325).

- **Termination** — when the Committee advises that the criteria are no longer met; mpox is the clearest example of a PHEIC formally lifted and later re-declared (Park 28e, Ch 5, p 172).

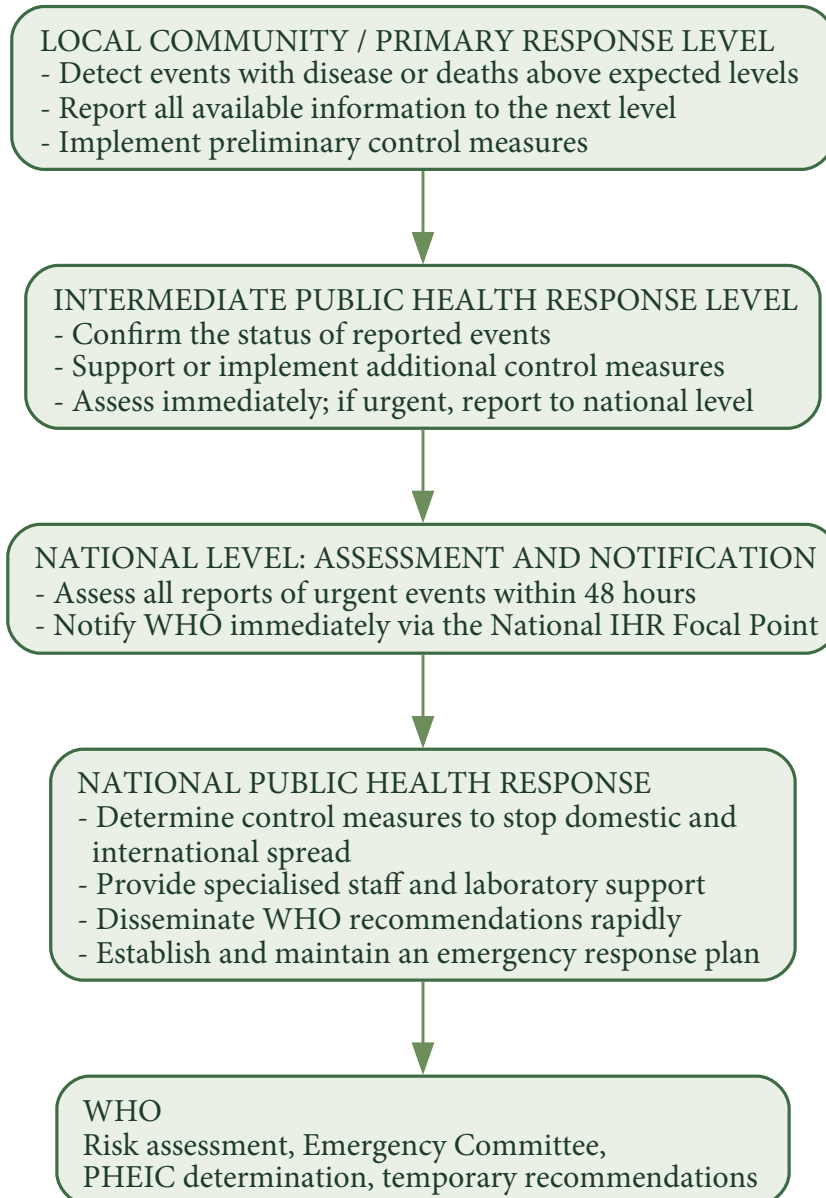
4. PHEICs declared so far — examples

Event	Date declared	Why it qualified and what happened
Pandemic influenza A (H1N1)	Late April 2009	First use of the IHR (2005); declared within weeks of onset on the Emergency Committee's advice. WHO declared Pandemic Phase 6 on 11 June 2009 and the pandemic over in August 2010 ; more than 4,00,000 laboratory-confirmed cases and 80,000 deaths reported to WHO (Sunder Lal, Ch 8, pp 470-472; Oxford 7e, Ch 75, pp 2862-2863)
Wild poliovirus — international spread	2014	Threat to global polio eradication; a single case of wild-type poliovirus is always notifiable under Annex 2 (Oxford 7e, Ch 20, pp 790-791; Ch 107, pp 4048-4049)
Ebola virus disease, West Africa	8 August 2014	Guinea, Liberia, Nigeria and Sierra Leone; close to 2000 cases with more than 1000 deaths at declaration — the largest number and widest geographical spread ever known for an Ebola outbreak (Sunder Lal, Ch 8, p 474; Jugal Kishore — National Health Programmes of India, Ch 1, pp 667-668)
Zika virus and congenital malformation	1 February 2016 (declared over November 2016)	Alarming link with severe birth defect (microcephaly) and Guillain-Barré syndrome; about 3 million cases estimated in Latin American countries , hardest hit Brazil with 1.5 million cases ; WHO imposed no travel or trade restriction as transmission was at a low level (Sunder Lal, Ch 8, pp 475-476; Oxford 7e, Ch 107, pp 4048-4049)
Ebola virus disease, Democratic Republic of the Congo	17 July 2019	Second Ebola PHEIC; detailed WHO advisories issued to other countries to protect their citizens (Sunder Lal, Ch 8, p 474)
COVID-19 (SARS-CoV-2)	30 January 2020	Emergenced in Wuhan, China in December 2019 and spread to over 220 countries; subsequently declared a pandemic on 11 March 2020 (Sunder Lal, Ch 20, p 788; Park 28e, Ch 5, p 205)
Mpox, Clade IIb	23 July 2022 (lifted 10 May 2023)	Global outbreak spread mainly through sexual contact, chiefly in adult gay and bisexual men and other men who have sex with men; over 97,000 laboratory-confirmed cases and 186 deaths in all 6 WHO regions between January 2022 and end-April 2024 (Park 28e, Ch 5, p 172)
Mpox, Clade Ib	14 August 2024	Re-declared on the emergence of the new Clade Ib and a resurgence of cases (Park 28e, Ch 5, p 172)

What the examples teach — the 2013-2016 West African Ebola outbreak is the standard critique: neither the affected countries nor WHO sounded the alarm early enough, the PHEIC rating was **overdue**, and coordination between WHO, ministries of health, United Nations agencies and non-governmental partners was weak. Although the IHR require countries to avoid disrupting travel and trade, **more than 40 countries imposed measures outside the scope of WHO's temporary recommendations** — possible precisely because those recommendations are advisory — causing severe

political, economic and social consequences and barriers to receiving personnel and supplies (Oxford 7e, Ch 107, p 4049). COVID-19 similarly exposed global inequality in emergency preparedness (Park 28e, Ch 5, p 205).

5. Obligations of States Parties — IHR core capacities



IHR (2005) core capacity requirements for surveillance and response, from the community to WHO

Obligation	Content
Definition of an "urgent" event	Serious public health impact and/or an unusual or unexpected nature with high potential for spread (Sunder Lal, Ch 18, p 771)
Points of entry	Core capacity requirements are laid down for airports, ports and ground crossings ; competent authorities monitor baggage, cargo, containers, conveyances, goods, postal parcels and human remains, keep traveller facilities sanitary, and hold contingency arrangements for unexpected public health events (Sunder Lal, Ch 18, p 771; Red Book, Ch 2, pp 1024-1026)

Obligation	Content
Prescribed measures	Quarantine; disinsection, deratting, disinfection and decontamination of aircraft and ships, carried out so as to avoid injury and, as far as possible, discomfort to persons (Sunder Lal, Ch 18, p 771; Red Book, Ch 2, pp 1025-1026)
Monitoring of capacity	Countries must assess their surveillance and response capacity and report whether it is sufficient; the voluntary Joint External Evaluation (JEE) gives them a platform to assess core capacities and then cost an action plan to mobilise resources (Oxford 7e, Ch 105, pp 3996-3997; Ch 75, pp 2866-2867)
Global indicator	The IHR core capacity index is one of WHO's 100 Core Health Indicators and the health-security indicator for Sustainable Development Goal 3.d.1 (Park 28e, Ch 2, pp 42-43); it is the percentage of the 13 core capacity attributes attained at a given point in time — Global 71, India 95 (2010-17) (Park 28e, Ch 14, pp 853-854)

India's application — for COVID-19 the outbreak was declared an epidemic and the **Epidemic Diseases Act, 1897** invoked, closing educational, religious, entertainment and commercial establishments, followed by unlocking in six phases up to 30 November 2020 (Park 28e, Ch 5, p 205). The 125-year-old Act is being replaced by the proposed **National Public Health Act, 2022**, which covers prevention, control and management of epidemics, bioterrorism, lockdown, quarantine and disaster management in the light of the IHR (2005), and proposes a **4-tier system of Public Health Authorities at national, State, district and block levels** (Sunder Lal, Ch 8, p 410; Ch 13, p 671). Under the **Ayushman Bharat Health Infrastructure Mission (October 2021)** India is strengthening the National Centre for Disease Control with 5 regional centres and 20 metropolitan surveillance units, operationalising **17 new and strengthening 33 existing public health units at points of entry** — **32 airports, 11 seaports, 7 land crossings**, setting up 15 health emergency operation centres, and expanding the Integrated Health Information Portal to all States and Union Territories (Sunder Lal, Ch 20, p 805).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Annex 2 decision instrument (IHR 2005)** — draw as a labelled tree, labels only. Header box: *Events detected by national surveillance system (Annex 1)*. Three parallel boxes below it, each carrying only its heading: *Always notifiable — a single case suffices* | *Always leads to use of the algorithm* | *Any event of potential international public health concern*. The left-hand box **bypasses the algorithm** — draw its arrow straight down to the notification exit box. Only the middle and right-hand boxes feed the question chain: below them draw the four question labels in a vertical chain — *Serious?* → *Unusual or unexpected?* → *Risk of international spread?* → *Risk of travel or trade restriction?* Two exit boxes at the foot: *Not notified at this stage — reassess later* and **EVENT SHALL BE NOTIFIED TO WHO UNDER THE INTERNATIONAL HEALTH REGULATIONS**.

High-yield exam pointers

- **Any 2 of 4 Annex 2 criteria** = notify — mnemonic **S-U-S-T**: Serious public health impact, Unusual or unexpected, Spread internationally, Travel or trade restriction risk.
- **Four always-notifiable diseases** (a single case suffices): smallpox, poliomyelitis due to wild-type poliovirus, human influenza caused by a new subtype, SARS.

- ▶ **Two clocks:** assess all reports of urgent events within **48 hours** at national level; notify WHO within **24 hours** of assessment through the National IHR Focal Point.
- ▶ **Who declares, and what binds** — the WHO Director-General on the advice of the IHR Emergency Committee, not the World Health Assembly and not the affected country; the Regulations are binding, the temporary recommendations issued under a PHEIC are NOT, and more than 40 countries breached them during West African Ebola without sanction.
- ▶ **Dates:** H1N1 influenza late Apr 2009 (Pandemic Phase 6 on 11 Jun 2009); wild poliovirus 2014; Ebola West Africa 8 Aug 2014; Zika 1 Feb 2016 and declared over Nov 2016; Ebola DR Congo 17 Jul 2019; COVID-19 PHEIC 30 Jan 2020 and pandemic 11 Mar 2020; mpox 23 Jul 2022, lifted 10 May 2023, re-declared 14 Aug 2024.
- ▶ **Zika magnitude:** about 3 million cases estimated in Latin America, Brazil worst hit with 1.5 million; the link with microcephaly is what made the event "extraordinary".
- ▶ **IHR core capacity index = Sustainable Development Goal indicator 3.d.1** (Global 71, India 95, 2010-17); India's points-of-entry network under the Ayushman Bharat Health Infrastructure Mission (2021) = 32 airports, 11 seaports, 7 land crossings.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the corpus dates the 2014 wild-poliovirus declaration by year alone and does not say whether that emergency has since been terminated, so write it as 2014. Sunder Lal prints the 2019 Ebola declaration as an outbreak in the "Republic of Congo" whereas the outbreak was in fact in the Democratic Republic of the Congo. Oxford Textbook of Public Health 7e states that 196 countries accepted the International Health Regulations while Suryakantha states they are binding on 194 countries — either figure is safe if written with its source. The 2024 amendments to the Regulations, outside this textbook corpus, added a higher tier called a "pandemic emergency" above the public health emergency of international concern. Park 28e dates the lifting of the mpox emergency to 10 May 2023, the day the Emergency Committee met, whereas WHO's own announcement is dated 11 May 2023 — write the textbook date.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO International Health Regulations (2005) 2024 amendments, WHO Pandemic Agreement 2025, MMWR, Africa CDC, NCDC, PM-ABHIM

Programme and policy updates (India)

- ▶ Under the **Pradhan Mantri Ayushman Bharat Health Infrastructure Mission (PM-ABHIM)**, announced 25 October 2021, India is establishing **four zonal National Institutes of Virology** with World Bank support — Jammu (North), Bengaluru (South), Dibrugarh (East) and Jabalpur (Central). Four stakeholder workshops held in March 2024, with 186 participants drawn from state health departments, medical colleges, research institutes and the Virus Research and Diagnostic Laboratory network, ranked 40 viral diseases by multicriteria decision analysis to give each zone its own priority list; **dengue, influenza, measles, Japanese encephalitis and hepatitis A** headed the combined top five. This is the research and outbreak-response arm behind the points-of-entry and NCDC expansion described above. [PMID 42522023]

- The **National Centre for Disease Control**, with the Rajasthan health department, ran India's first national One Health zoonotic disease prioritisation (Jaipur workshop, February 2020) — a modified One Health Zoonotic Disease Prioritization process with **50 experts spanning seven sectors** and analytic hierarchy weighting of five criteria — narrowing 40 zoonoses to a ranked top ten headed by **zoonotic influenza, anthrax, Japanese encephalitis, leptospirosis and brucellosis**, followed by dengue, rabies, scrub typhus, plague and Crimean-Congo haemorrhagic fever; this is now the working list against which IHR-linked surveillance and One Health capacity are planned. [PMID 41429420]

Current national data

- **Nipah virus** is India's recurring IHR-notifiable event: **six Kerala outbreaks between 2018 and 2025 — nine Indian NiV events in all since 2001** (Siliguri 2001 and Nadia 2007 in West Bengal; Kozhikode 2018, Ernakulam 2019, Kozhikode 2021, Kozhikode 2023, Malappuram 2024 and Malappuram–Palakkad 2025 in Kerala; and a **Barasat, West Bengal re-emergence in January 2026** after nearly two decades). Reported case-fatality across Indian outbreaks is 45–100% and there is still no licensed human vaccine or treatment; **four of the six Kerala outbreaks fell in El Niño years**, and control therefore rests on Integrated Disease Surveillance Programme syndromic reporting, contact tracing, mobile BSL-3 laboratories and Truenat point-of-care micro-PCR. [PMID 42160364]
- Self-reported core-capacity scores run systematically high: across **93 States Parties with 108 paired assessments (2016–2023)** the SPAR self-assessment averaged **13 points above** the externally verified Joint External Evaluation, the gap narrowing from 20 points in 2016–17 to 6 points in 2022–23, with early-warning surveillance, health-emergency response and infection prevention and control the worst-agreeing indicators — the caveat to attach to any quoted national core-capacity index. [PMID 41640577]

Elimination, surveillance and disease status

- **Ebola disease due to Bundibugyo virus is the current PHEIC**. The health ministries of the Democratic Republic of the Congo and Uganda declared outbreaks on **15 May 2026**, centred on Ituri Province, and MMWR records that WHO determined the event a PHEIC on **17 May 2026**, with 378 confirmed cases and 63 confirmed deaths reported by 2 June 2026 [PMID 42275261]; a commentary on the same determination dates it **16 May 2026**, so write the day with its source. There is **no licensed vaccine or therapeutic for this Ebola species** — the approved products cover Zaire ebolavirus only — leaving early detection, isolation, supportive care, infection prevention, contact tracing and safe burial as the whole of the response [PMID 42306416]. It is already the largest recorded Bundibugyo outbreak: CDC branching-process projections gave a 65% chance of exceeding 20,000 cases within three months if only 20% of patients are isolated, against roughly one chance in twenty of reaching 10,000 cases if 70% are isolated. [PMID 42275271]
- The **mpox PHEIC was lifted in September 2025**, WHO retaining its standing recommendations — so the clade Ib entry in the table above now reads declared 14 August 2024 and terminated September 2025. Across the WHO European Region 82 clade I cases were reported from 13 of the 53 countries between 14 August 2024 and 23 November 2025, 45% of them acquired locally, with no deaths and only four admissions on clinical grounds. [PMID 42318655]

- **Poliomyelitis has been a PHEIC continuously since 5 May 2014** and was reaffirmed at the Emergency Committee's thirty-seventh meeting in December 2023 — more than a decade unbroken, and the correct status to write rather than leaving the 2014 declaration undated as to outcome. India operationalises the temporary recommendations by requiring travellers coming to India from polio-endemic countries, from countries with importation-linked poliovirus circulation, and from countries reporting circulating vaccine-derived poliovirus in the previous three years to hold polio vaccination certification dated at least four weeks before departure. [PMID 39544851]

Guideline and legislative updates

- **The IHR (2005) amendments were adopted by consensus at the Seventy-seventh World Health Assembly on 1 June 2024 and came into force on 19 September 2025.** They add a **pandemic emergency** tier above a PHEIC as the highest level of global alert; amended Article 4 requires every State Party to establish or designate a **National IHR Authority** to coordinate national implementation of the Regulations — it may be a separate entity from, or the same entity as, the National IHR Focal Point, which carries only the communication function; and new provisions create a **Coordinating Financial Mechanism** for core capacities and a States Parties Committee that works by cooperation and consensus rather than sanction. The amended text says little about how the new authority should be structured, so the seniority and cross-sectoral reach it will need are left for States to settle. [PMID 40953844] [PMID 40083009] [PMID 42497312]

Global and WHO developments

- **The WHO Pandemic Agreement was adopted in May 2025 (resolution WHA78.1)** at the Seventy-eighth World Health Assembly, the United States having withdrawn from the Organization and from the negotiations months before they closed. Its Article 12 **Pathogen Access and Benefit-Sharing (PABS)** system would oblige each participating manufacturer to give WHO **20% of real-time production of pandemic vaccines, therapeutics and diagnostics — 10% as donation and 10% reserved at affordable prices**; the treaty enters into force only after **60 ratifications**, and the PABS annex that makes the system operable was still unagreed and due for review at the World Health Assembly of April 2026, leaving the equity commitments deferred. [PMID 41748273] [PMID 42395645]
- **Africa CDC declared mpox a Public Health Emergency of Continental Security (PHECS) in August 2024**, the first use of that continental mechanism; its legal foundation is **Article 3(e) and (f) of the Africa CDC Statute approved by African Union Heads of State in July 2022**, which empowers Africa CDC to declare a PHECS in consultation with affected Member States. At declaration 17,541 cases and 517 deaths had been reported across 13 AU member states, a 160% rise in cases on the same period of 2023, and WHO reinstated its own PHEIC later the same month — so a regional emergency tier now sits alongside the IHR determination, and examiners may ask how the two interact. [PMID 41639867]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2, Ch 5, Ch 14, Ch 25); Oxford Textbook of Public Health 7e (Ch 20, Ch 28, Ch 42, Ch 75, Ch 105, Ch 107); Sunder Lal — Textbook of Community Medicine (Ch 8, Ch 13, Ch 18, Ch 20); Suryakantha — Community Medicine with Recent Advances (Ch 20); AFMC Red Book of Community Medicine 2024 (Ch 26); Red Book — Public Health and Preventive Medicine (Ch 2); Textbook of International Health (Ch 5); Jugal Kishore — National Health Programmes of India (Ch 1). Named documents: WHO International Health Regulations (2005), second edition; Ayushman Bharat Health Infrastructure Mission, October 2021.

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025-01177-6. Named documents: International Health Regulations (2005) amendments, adopted 1 June 2024, in force 19 September 2025; WHO Pandemic Agreement, adopted May 2025; Africa CDC Public Health Emergency of Continental Security declaration, August 2024.



Roll Back Malaria initiative – rationale

Asked in: Paper II 09-May-2022 — RGUHS MD Community Medicine

Definition — Roll Back Malaria (RBM) is the global partnership established on **23 July 1998** as a priority project of the World Health Organization (WHO), founded jointly by WHO, the United Nations Children's Fund (UNICEF), the United Nations Development Programme (UNDP) and the World Bank (Park 28e, Ch 5, p 333). The name is taken from the *Oxford English Dictionary* sense of "roll back" — "**to cause to retreat or decrease**" — which is what the project set out to do to malaria globally, though concentrated on Africa; its declared aim was to **halve the world's malaria burden by 2010** (Gilles & Warrell 4e, Ch 6, p 185). The question asks *why* such an initiative became necessary in 1998; the reasons are epidemiological, biological, systemic, economic and political.

Reasons for launching Roll Back Malaria

#	Reason	Evidence that made it a reason
1	Collapse of the Global Malaria Eradication Programme (GMEP), 1955–1969	GMEP eliminated malaria from southern Europe and parts of the Americas but achieved very little in sub-Saharan Africa , where efforts were deliberately limited (Oxford Textbook of Public Health 7e, Ch 8.15, p 2718). By the mid-1960s failure was evident; it came to be regarded as a major public health failure (Nelson & Williams, Ch 27)
2	Epidemic resurgence after eradication was abandoned	Most countries sharply cut malaria expenditure after 1969, and the result in many areas was epidemic resurgence (Nelson & Williams, Ch 27). Resurgence was greatest in southern and south-eastern Asia, with deterioration in tropical Africa and a shift of morbidity to older age groups (Gilles & Warrell 4e, Ch 1, p 2). In India, cases fell to 0.1 million in 1965, then rose to 6.46 million by 1976 (Park 28e, Ch 7, p 493)
3	Parasite resistance to antimalarial drugs	Chloroquine-resistant <i>Plasmodium falciparum</i> spread throughout the African continent , and multidrug resistance became common in South-East Asia (Gilles & Warrell 4e, Ch 1, p 2), removing the cheap first-line cure on which control depended
4	Vector resistance to insecticides	Anopheline resistance to DDT and later compounds was one of the two biological developments that broke GMEP (Nelson & Williams, Ch 27). Resistance forced a shift to more expensive pyrethroids , thereby limiting coverage (Park 28e, Ch 5, p 333). No single method could by itself interrupt transmission, forcing integrated vector management (Park 28e, Ch 5, p 333)

#	Reason	Evidence that made it a reason
5	Weak health systems and the failure of integration	GMEP's complex logistical and operational demands exceeded the weak infrastructure of most tropical countries (Nelson & Williams, Ch 27). WHO dismantled the vertical approach and promoted integration, but effective systems to integrate into never materialised, so the gains could not be sustained (Gilles & Warrell 4e, Ch 6, pp 185–186)
6	Unmet access of the poorest to drugs and prevention	The single most important system needed for RBM to succeed was that all families, especially the marginalised and poorest, have ready access to adequate doses of antimalarials the moment they are needed , plus appropriate preventive measures (Gilles & Warrell 4e, Ch 6, p 186)
7	Unacceptable burden, concentrated in African children	Around the launch period, more than 1 million children a year, mainly under 5 years, died of malaria in sub-Saharan Africa alone — one child every 30 seconds (Textbook of International Health, Ch 6, p 310). Even in 2017 there were 219 million cases and 435,000 deaths, 61% in children under 5 , with the WHO African Region carrying 92% of cases and 93% of deaths (Oxford Textbook of Public Health 7e, Ch 8.15, p 2717)
8	Malaria as a brake on development and a cause of poverty	Malaria is an important cause of underdevelopment and poverty , reducing gross domestic product in sub-Saharan Africa by about 1.3% per year (Oxford Textbook of Public Health 7e, Ch 8.15, p 2717); direct treatment and prevention costs plus indirect costs of lost productivity exceeded US\$1 billion per year (Textbook of International Health, Ch 6, p 310)
9	Urbanisation and complex emergencies	Uncontrolled, rapid urbanisation created pockets of transmission in cities, enlarging vulnerable groups ; severe outbreaks with high mortality occurred in several African countries (Gilles & Warrell 4e, Ch 1, p 2)
10	A window of unprecedented political commitment	Political concern was mobilised at the Amsterdam Ministerial Conference on Malaria, 1992 ; the Harare Declaration of African heads of state (Organization of African Unity, June 1997) pledged full support; and at the G8 Birmingham summit (May 1998) the industrialised nations agreed to back the WHO-led antimalaria campaign. By 1998 there had been no comparable political support for global action against malaria since the eradication era (Gilles & Warrell 4e, Ch 6, p 185)
11	New science and new tools ready for deployment	By the end of the 1990s there were major advances in parasite and vector genomics, vector control methods, immunology, and rational development of novel antimalarials and vaccines (Nelson & Williams, Ch 27). Enough trial evidence had accumulated for RBM to recommend long-lasting insecticide-treated nets as a key method in high-transmission Africa (Nelson & Williams, Ch 27)

#	Reason	Evidence that made it a reason
12	No single agency could deliver it — the partnership rationale	Malaria is not only a health problem but a public health issue with social, economic and development components cutting across ministries and the private sector (Gilles & Warrell 4e, Ch 6, p 186). RBM was therefore constituted as a public-private partnership — one of the wave of such partnerships from the mid-1990s — and grew to more than 90 partners coordinating stakeholders, planning, and raising funds and political support (Nelson & Williams, Ch 27; Textbook of International Health, Ch 3, p 106)

Guiding concepts that followed from these reasons

RBM's design answers the failures listed above point for point. Its emphasis is on **partnership, evidence-based action, political mobilisation and participation of civil society**; its core concepts are to focus on results, prioritise effective action, stimulate attention, use innovative approaches, act on evidence, place stronger emphasis on the political context, provide a common platform for multi-disease action, and foster a social movement that puts concepts into practice (Gilles & Warrell 4e, Ch 6, p 185). **Political commitment** is a stated key priority of RBM (Suryakantha, Ch 43).

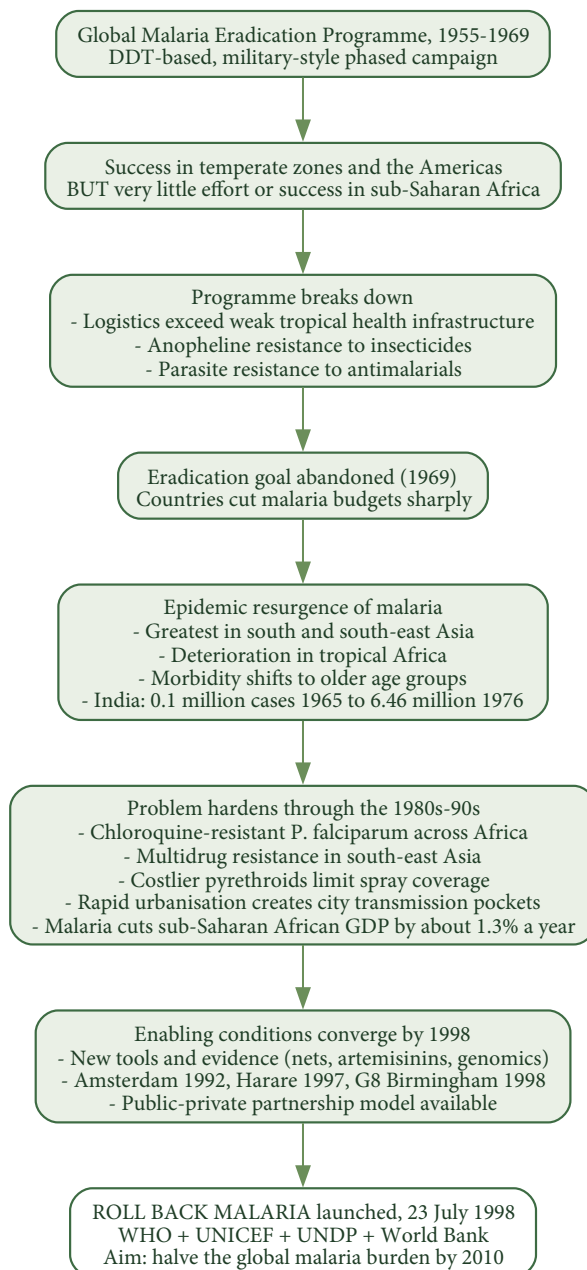
Targets set and the milestones that followed

Year	Milestone	Significance
1998	Roll Back Malaria launched (23 July) by WHO Director-General Dr Gro Harlem Brundtland	Set in motion increased investment and support to endemic countries (Oxford Textbook of Public Health 7e, Ch 8.15, p 2718)
1998	Target: halve the global malaria burden by 2010, especially deaths	The founding quantitative goal (Gilles & Warrell 4e, Ch 6, p 185; Suryakantha, Ch 43)
2000	UN Millennium Declaration; Abuja Declaration by African heads of state at the African Summit on Roll Back Malaria	Together with RBM, established indicators and targets for malaria control (Park 28e, Ch 5, p 333)
2005	World Health Assembly 2005 and the RBM global strategic plan 2005	Consolidated the indicator and target framework (Park 28e, Ch 5, p 333)
2016–2030	Action and Investment to Defeat Malaria (AIM), the RBM Partnership advocacy plan, complementing WHO's Global Technical Strategy (GTS) for Malaria 2016–2030	GTS requires global investment to triple to an estimated US\$8.7 billion annually by 2030 (Jugal Kishore, National Health Programmes of India)
2016–2030	GTS pillars: (1) universal access to prevention, diagnosis and treatment; (2) accelerate elimination; (3) transform surveillance into a core intervention	Targets vs 2015: reduce mortality rate and case incidence by ≥40% (2020), ≥75% (2025), ≥90% (2030); eliminate malaria from ≥35 countries by 2030 (Park 28e, Ch 5)

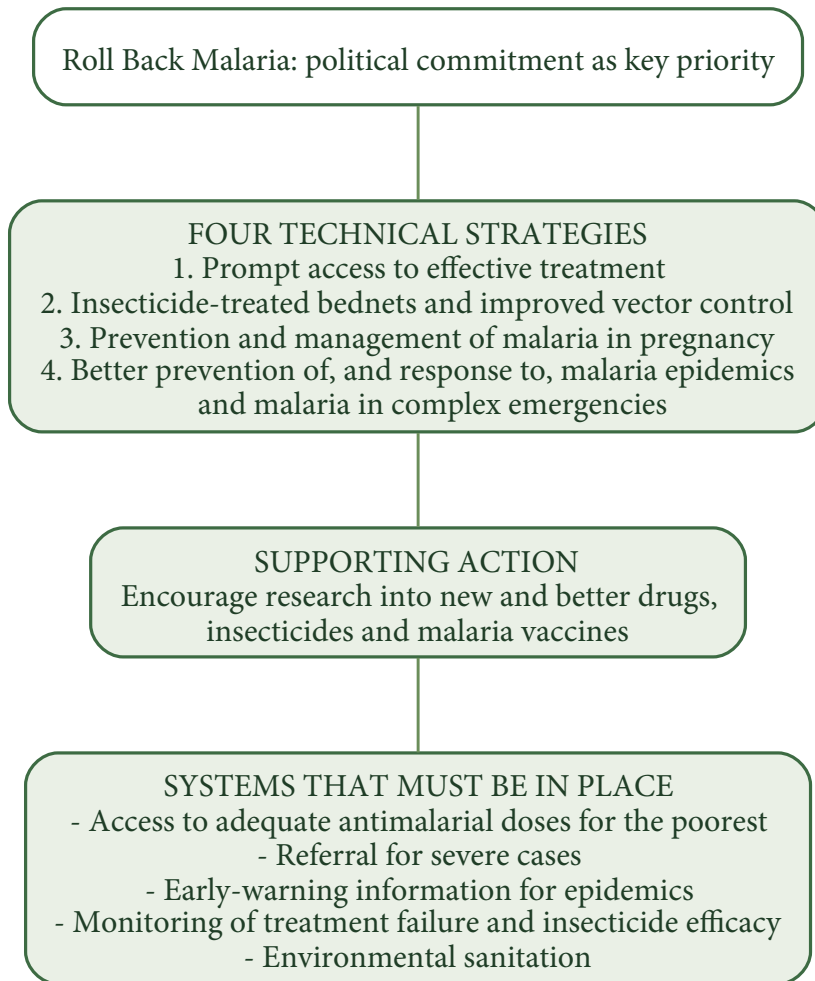
Result to date: global malaria deaths fell from 864,000 in 2000 to 608,000 in 2022, and case incidence from 81 to 58 per 1000 population at risk, though there were still 249 million cases in 2022 across 85 endemic countries, with about 95% of cases and deaths in sub-Saharan Africa (Park 28e, Ch 5, p 317).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — "Timeline of global malaria control". A single horizontal time line with these exact labels marked in order: 1955–1969 Global Malaria Eradication Programme (GMEP) → 1969 eradication goal abandoned; funding withdrawn → 1970s resurgence; chloroquine resistance spreads in Africa → 1992 Amsterdam Ministerial Conference — Global Malaria Control Strategy → 1997 Harare Declaration (OAU) → May 1998 G8 Birmingham summit → 23 July 1998 Roll Back Malaria launched (WHO + UNICEF + UNDP + World Bank) → 2000 Abuja Declaration and UN Millennium Declaration → 2005 RBM global strategic plan → 2016–2030 Global Technical Strategy for Malaria + AIM.

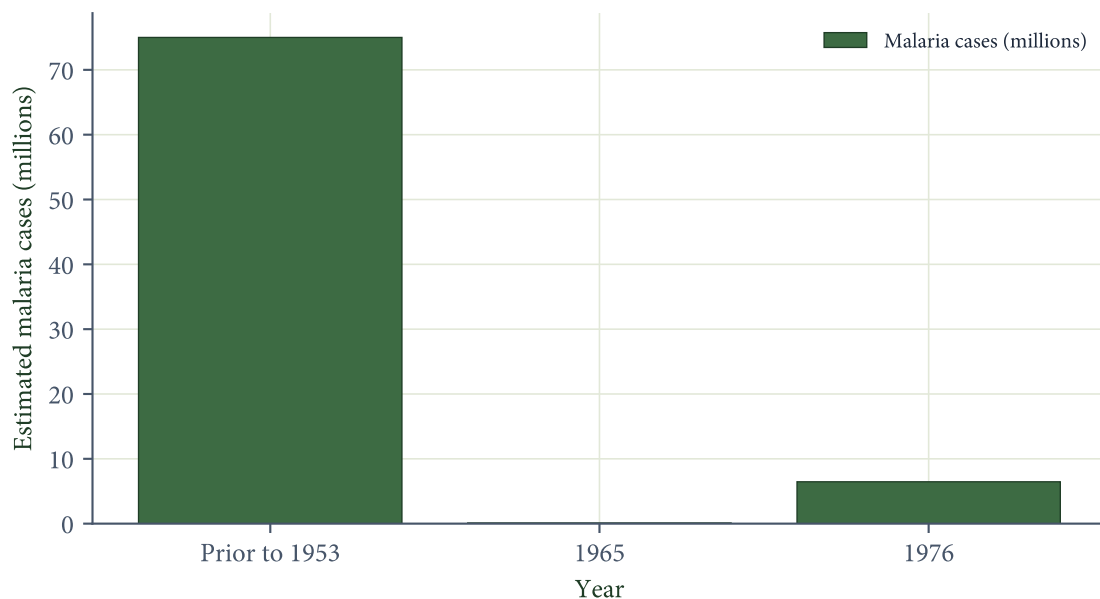


Why the eradication era ended and Roll Back Malaria became necessary



The four technical strategies of Roll Back Malaria and its supporting actions

Malaria in India: control, near-elimination, then resurgence



Milestones of malaria control activities in India - the post-eradication resurgence that mirrored the global collapse of the eradication strategy (Park's Textbook of Preventive and Social Medicine 28e, Ch 7, p 493).

High-yield exam pointers

- ▶ **Definition anchor:** RBM = WHO priority project, 23 July 1998; four founders — WHO, UNICEF, UNDP, World Bank; name from the *Oxford English Dictionary* meaning of "roll back" = to cause to retreat or decrease; goal halve the global malaria burden by 2010.
- ▶ **Two-word summary of why GMEP failed:** *resistance* (vector to insecticides, parasite to drugs) plus *weak infrastructure* — then budget withdrawal and resurgence.
- ▶ **Four RBM technical strategies (mnemonic T-N-P-E):** Treatment prompt and effective, Nets and vector control, Pregnancy — prevention and management of malaria in pregnancy, Epidemics and complex emergencies.
- ▶ **Numbers to quote with their year:** more than 1 million child deaths a year in sub-Saharan Africa, one child every 30 seconds (WHO 2007); 219 million cases and 435,000 deaths, 61% in under-5s, Africa 92% of cases (World Malaria Report 2018); malaria costs sub-Saharan Africa about 1.3% of GDP per year; 249 million cases and 608,000 deaths in 2022 (Park 28e).
- ▶ **Political chain to name:** Amsterdam 1992 → Harare Declaration 1997 → G8 Birmingham May 1998 → RBM July 1998 → Abuja Declaration 2000 → RBM global strategic plan 2005 → AIM with GTS 2016–2030.
- ▶ **India link:** the same resurgence logic — 75 million cases before 1953, 0.1 million in 1965, **6.46 million in 1976**, Modified Plan of Operations 1977 (Park 28e, Ch 7, p 493).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: the Roll Back Malaria Partnership's own advocacy plan is now **Action and Investment to Defeat Malaria (AIM) 2016–2030**, which runs alongside — and does not replace — WHO's Global Technical Strategy for Malaria 2016–2030, so a current answer should name the Global Technical Strategy as the operative technical framework and RBM as the partnership and advocacy platform behind it.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO Global Malaria Programme, World Malaria Report 2025, WHO Global Technical Strategy for Malaria 2016–2030, NCVBDC (MoHFW India), National Strategic Plan for Malaria Elimination 2023–2027

Programme and policy updates (India)

- ▶ **The Indian programme has been renamed and re-documented since most textbook editions.** The National Vector Borne Disease Control Programme (NVBDCP) now operates as the **National Centre for Vector Borne Diseases Control (NCVBDC)**, and the operative document under the National Framework for Malaria Elimination 2016–2030 is the **National Strategic Plan for Malaria Elimination 2023–2027**, which classifies states and districts by annual parasite incidence — **category 1 = elimination phase, API <1 per 1000**. Madhya Pradesh moved from category 3 in 2015 to category 1 in 2022; the national elimination target remains 2030 [PMID 41566118] [PMID 42096166]
- ▶ **RBM's "no single agency can deliver it" logic is now being applied inside India.** ICMR and the national programme are being pushed into a "virtual integration" — joint monitoring of drug

and insecticide resistance, digital near-real-time case reporting in place of paper aggregate returns, and site-specific last-mile strategies for residual foci [PMID 38889711]. Its first product is an ICMR-VCRC/NCVBDC multicentric longitudinal study in **12 districts across 10 states**, about **25,000 people**, spanning urban, peri-urban, rural, forest-foothill and coastal settings with linked entomology, insecticide-resistance and health-equity assessment [PMID 41663159]

Current national data

- **Quote the World Malaria Report 2025 figures, not the 2022 ones.** An estimated **282 million cases and 610,000 deaths in 2024** across **80 endemic countries**, with the WHO African Region carrying **95% of cases (265 million) and 95% of deaths (579,000)**; cases rose about 3% over 2023 and deaths from 598,000, leaving the Global Technical Strategy 2025 milestone ($\geq 75\%$ reduction against 2015) unmet [WHO 2025-Dec] [PMID 42166501]. Nigeria alone accounted for **24.3% of global cases and 30.3% of global deaths** in 2024 [PMID 42563162]
- **India's own current numbers.** NCVBDC reported **255,500 cases and 86 deaths (provisional) in 2024**, with **API 0.18 per 1000** against 0.117 in 2021 [NCVBDC 2025]. A national household survey by ICMR-NIMR with NCVBDC across ten endemic states (**37,849 households, 177,644 individuals**) found LLIN ownership **85.2%** but regular use only **73.6%**, and only **57.4%** of febrile people sought government care within 24–48 h; lack of formal education and tribal residence independently raised the odds of malaria [PMID 42096166]

Elimination, surveillance and disease status

- **The drug-resistance reason that created RBM has returned in African form.** The World Malaria Report 2025 confirms **partial artemisinin resistance in Eritrea, Rwanda, Uganda and Tanzania**, with suspected resistance in Ethiopia, Namibia, Sudan and Zambia [PMID 42563162]. A 2014–2024 East African meta-analysis (24 studies, 15,951 patients) pooled non-synonymous **Pfkelch13** mutations at **5.0% (95% CI 3.0–7.0%)**, rising to **10%** in both Rwanda (95% CI 4–16%) and Uganda (95% CI 6–14%); **R561H** and **A675V** were commonest, and carriers were significantly more likely to show day-3 parasitaemia after ACT [PMID 42520053]
- **India's own resistance position is currently reassuring but not static.** Nanopore-based targeted sequencing of 15 resistance genes in **238 *P. falciparum* isolates from six states** detected **no WHO-validated Pfk13 artemisinin-resistance mutation**, supporting continued ACT efficacy — but chloroquine resistance (Pfcr K76T, CVIET haplotype) and quintuple/sextuple Pfdhfr–Pfdhps haplotypes conferring full sulfadoxine-pyrimethamine resistance persist in the Northeast, while Pfmdr1 Y184F signals emerging lumefantrine tolerance in central India [PMID 41846600]
- **WHO now tracks four biological threats to malaria control**, all of which extend the original RBM rationale: antimalarial drug resistance, insecticide resistance, ***pfhrp2/3* gene deletions** producing false-negative rapid diagnostic tests, and **invasion by the urban vector *Anopheles stephensi*** (reported from Nigeria, Ghana, Sudan, Djibouti, Ethiopia, Eritrea, Somalia and Kenya) — compounded by climate change, armed conflict and displacement. Cases rose significantly in **five of the eleven High Burden to High Impact countries** over 2015–2024 (Nigeria, DR Congo, Cameroon, Uganda, Mali) [PMID 42166501]. Detection remains the weak link: **27% of 1,030 febrile patients in central India were PCR-positive with none detected by microscopy or RDT** [PMID 41566118], and only **58.7%** of *P. vivax* patients in tribal Odisha completed the 14-day primaquine radical cure [PMID 41505814]

Guideline and legislative updates

- ▶ WHO's 2024 implementation guide on multiple first-line therapies (MFT) asks endemic countries to deploy more than one ACT simultaneously so that drug pressure is spread rather than a still-working drug withdrawn; Rwandan modelling projects **49% fewer treatment failures** under MFT than under continued reliance on artemether-lumefantrine alone, alongside expanded sentinel therapeutic-efficacy sites and routine molecular genotyping [WHO 2024] [PMID 42563162]

Global and WHO developments

- ▶ **Malaria vaccines — the tool RBM did not have in 1998 — are now programme reality.** RTS,S/AS01 was recommended in October 2021 and R21/Matrix-M in October 2023, and malaria vaccine now sits in the routine schedule of **25 sub-Saharan African countries** [WHO 2025-Dec] [PMID 42107373]. The 46-month Malaria Vaccine Implementation Programme evaluation (158 clusters across Ghana, Kenya and Malawi; 1.29 million children given dose 1) found all-cause non-injury mortality **13% lower** in implementation areas — mortality rate ratio **0.87 (95% CI 0.77–0.97; p = 0.016)**, roughly **one death in eight averted** — achieved at third-dose coverage of only 71.1% and fourth-dose coverage of 39.9% [PMID 42107373]
- ▶ **Elimination is demonstrably still achievable, which is the strongest counter-argument to RBM-era pessimism.** WHO certified Cabo Verde malaria-free on 12 January 2024, the 44th country certified [PMID 42166501], followed by Egypt (2024) and Georgia, Suriname and Timor-Leste (2025); certification requires at least three consecutive years of zero indigenous cases. The Global Technical Strategy targets of **≥90% reduction in incidence and mortality and elimination in ≥35 countries by 2030** remain the operative framework, with RBM Partnership to End Malaria as the advocacy platform behind it [WHO 2025]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 5, Epidemiology of Communicable Diseases; Ch 7, Health Programmes in India); Gilles & Warrell — Essential Malariology 4e (Ch 1, Historical outline; Ch 6, Rationale and technique of malaria control); Nelson & Williams — Infectious Disease Epidemiology (Ch 27, The epidemiology and control of malaria); Oxford Textbook of Public Health 7e (Ch 8.15, Malaria); Textbook of International Health (Ch 3, International Health Agencies; Ch 6, Epidemiologic Profiles of Global Health and Disease); Suryakantha — Community Medicine with Recent Advances (Ch 43, National Health Programs); Jugal Kishore — National Health Programmes of India. Named documents: WHO World Malaria Reports (2018, 2019, and the 2022 data reproduced in Park 28e); Harare Declaration 1997; Abuja Declaration 2000; WHO Global Technical Strategy for Malaria 2016–2030.

Coverage note: the indexed corpus documents RBM's founding, rationale, core concepts, four technical strategies and successor framework, but contains no passage giving the numerical Abuja Declaration coverage targets, so those are not quoted here.

Recent-advances sources: 41566118 — High burden of febrile sub-microscopic *Plasmodium* mixed infections in central India: a cross-sectional study, *Infectious Diseases and Therapy* 2026, doi:[10.1007/s40121-025-01297-x](https://doi.org/10.1007/s40121-025-01297-x); 42096166 — Comprehensive analysis of health service utilization for febrile illness and malaria in malaria-endemic states in India: LLINs as a pillar of control strategies, *Infectious Diseases and Therapy* 2026, doi:[10.1007/s40121-026-01359-8](https://doi.org/10.1007/s40121-026-01359-8); 38889711 — For malaria elimination, India needs to integrate its research and control arms, *American Journal of Tropical Medicine and Hygiene* 2024, doi:[10.4269/ajtmh.24-0011](https://doi.org/10.4269/ajtmh.24-0011); 41663159 — Multicentric longitudinal study on malaria burden, vector bionomics and health system assessment in diverse eco-epidemiological settings in the context of malaria elimination in India: study protocol, *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-111645](https://doi.org/10.1136/bmjopen-2025-111645); 42166501 — Turning the tide against malaria in high-burden African countries: trends, threats, and solutions, *PLOS Global Public Health* 2026, doi:[10.1371/journal.pgph.0006494](https://doi.org/10.1371/journal.pgph.0006494); 42563162 — Artemisinin resistance and Nigeria's malaria burden: urgent policy imperatives from the WHO World Malaria Report 2025, *International Journal for Equity in Health* 2026, doi:[10.1186/s12939-026-02970-4](https://doi.org/10.1186/s12939-026-02970-4); 42520053 — *Plasmodium falciparum* non-synonymous Kelch13 mutations mediating artemisinin resistance in East Africa: a systematic review and meta-analysis, 2014–2024, *PLoS One* 2026, doi:[10.1371/journal.pone.0354429](https://doi.org/10.1371/journal.pone.0354429); 41846600 — Targeted genomic surveillance unveils genetic variations linked to regional malaria drug resistance dynamics in India, *Open Forum Infectious Diseases* 2026, doi:[10.1093/ofid/ofag106](https://doi.org/10.1093/ofid/ofag106); 41505814 — Suboptimal primaquine adherence in *Plasmodium vivax* malaria: evidence from high-burden tribal districts in Odisha, *Journal of Infection and Public Health* 2025, doi:[10.1016/j.jiph.2025.103123](https://doi.org/10.1016/j.jiph.2025.103123) (abstract-only); 42107373 — Impact of introducing RTS,S/AS01 malaria vaccine on mortality in young children in Ghana, Kenya, and Malawi: an observational evaluation of a cluster-randomised implementation programme, *The Lancet* 2026, doi:[10.1016/S0140-6736\(26\)00248-5](https://doi.org/10.1016/S0140-6736(26)00248-5) (abstract-only). Non-journal sources: WHO malaria fact sheet, updated 4 December 2025; WHO World Malaria Report 2025; WHO list of countries and territories certified malaria-free, 2025; WHO implementation guide

on multiple first-line therapies, 2024; NCVBDC (MoHFW) malaria situation data, 2025.



Role of World Bank in implementation of health programmes in India

Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — The World Bank is a specialized agency of the United Nations, created at the Bretton Woods Conference in 1944 with headquarters at Washington, established to help less developed countries raise their living standards; its powers are vested in a Board of Governors (Park 28e, Ch 25, p 1059; Jugal Kishore, Ch 3, p 971). It lends for projects that lead to economic growth — power, roads, railways, agriculture, water supply, education and family planning — to which health and environmental components have since been added (Park 28e, Ch 25, p 1059). In India it works not by running services but by lending money, buying equipment and drugs, and building programme management systems for national health programmes run by the Ministry of Health and Family Welfare.

Structure of the World Bank Group and its lending windows

Arm	Full name	What it does	Terms
IBRD	International Bank for Reconstruction and Development	Loans to middle-income and "credit-worthy poor" countries	15–20 year loans; per-capita income ceiling US\$6,055 (2007); interest below commercial bank rates
IDA	International Development Association	Low-interest loans ("credits") and grants to the poorest countries with little access to credit markets	Eligibility per-capita income < US\$1,065 (2007); maturity 35–40 years with a 10-year grace period before repayment of principal
IFC	International Finance Corporation	Promotes private investment in developing countries — the route by which private hospitals and insurers are financed	Equity and commercial-type investment
MIGA	Multilateral Investment Guarantee Agency	Guarantees IFC-type investments against political risk	Guarantee, not loan
ICSID	International Centre for Settlement of Investment Disputes	Settles investor–state disputes	Arbitration

(Textbook of International Health, Ch 3, p 118 and index p 81; Jugal Kishore, Ch 3, p 971 — Jugal Kishore restricts the term "World Bank" to IBRD + IDA, the other three being the wider Bank Group.)

How the Bank became a health actor — the four instruments of influence

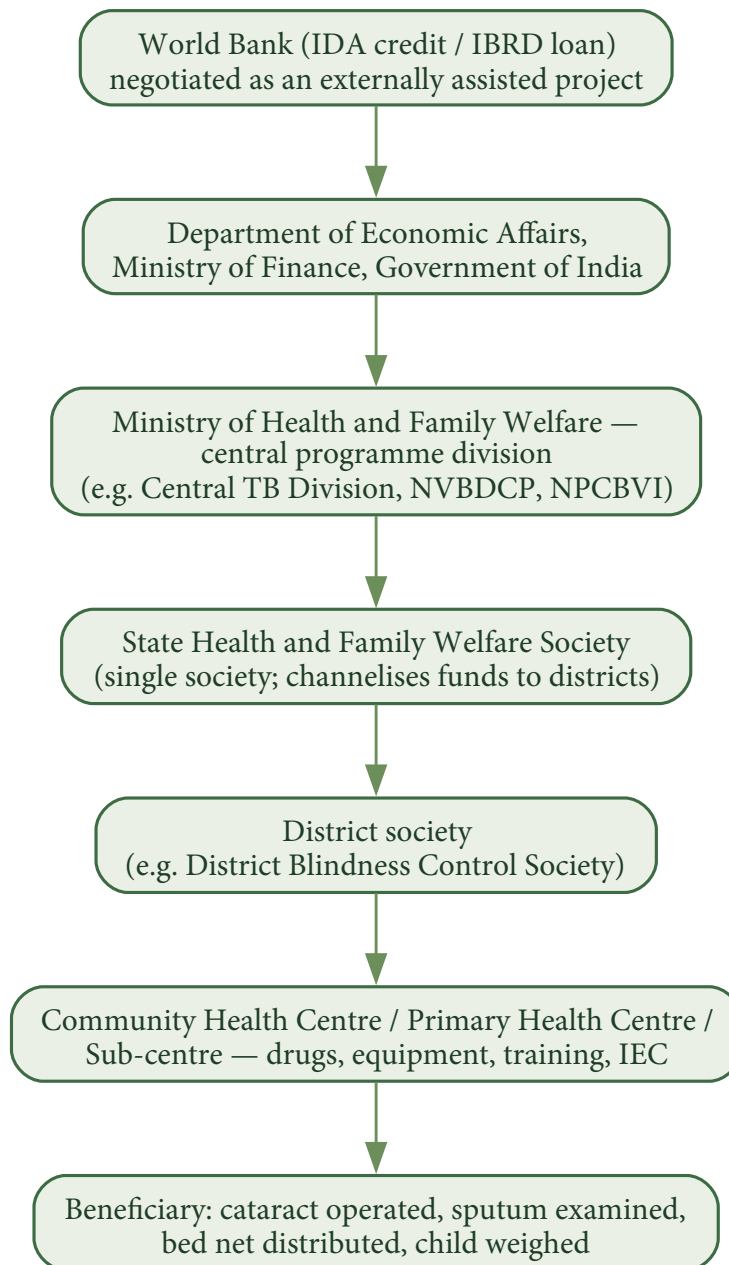
- **Direct project lending.** In the 1980s the Bank agreed to lend directly to health projects and created a **Health, Nutrition and Population (HNP)** branch from its earlier nutrition and population-control programmes; the first such loan went to **Tunisia in 1981** for health systems development (Textbook of International Health, Ch 3, p 118).
- **Scale.** By the mid-1990s the Bank was the **world's largest external funder of health**, securing over **US\$1 billion in health-related loans annually**; from 1970 to 2007 the HNP sector lent **US\$27 billion to over 100 countries for some 850 projects** (Textbook of International Health, Ch 3, p 119). **India is the largest single recipient of IBRD/IDA funds** — US\$3.7 billion for 67 projects in 2006–07, of which over US\$2 billion was for health projects (ibid).
- **Policy prescription.** Its **World Development Report 1993, *Investing in Health*** — the Bank's first health-oriented WDR — introduced the **disability-adjusted life year (DALY)** as the yardstick for priority setting and recommended that poor countries buy a "basket" of cost-effective interventions costing **US\$12 per capita per year (low-income) and US\$21.50 (middle-income)**, with user fees, private-sector competition and cost-effectiveness as governing principles (Textbook of International Health, Ch 11, pp 595–596; Ch 5, pp 266–267). The DALY itself was developed at Harvard University for the World Bank in 1990 and adopted by WHO in 2000 (Park 28e, Ch 2, p 38).
- **Conditionality and technical assistance.** Loans carry conditions on fiscal ceilings, cost recovery and outsourcing; the Bank also supplies research, consultancy, capacity building and independent procurement review (Jugal Kishore, Ch 1, p 23 and Ch 3, pp 971–972).

World Bank support to specific national health programmes in India

National programme	World Bank project and period	What the Bank actually financed
National Vector Borne Disease Control Programme (malaria)	Enhanced Malaria Control Project (EMCP), launched 30 September 1997; a further World Bank-supported National Malaria Control Project in 2008	Rs 891 crore over 5 years covering 100 selected districts and 19 urban areas (Park 28e, Ch 7, p 482 milestones table; Suryakantha)
Kala-azar elimination	Ongoing externally assisted support	World Bank support for Bihar, Jharkhand and West Bengal ; Bank support also for malaria control in non-North-Eastern (largely tribal) states (Park 28e, Ch 13, pp 849–850)
National Programme for Control of Blindness and Visual Impairment	Cataract Blindness Control Project, Rs 554 crore, seven years from 1994–95 (to 2002)	Upgrading ophthalmic services, rural and tribal outreach, creation of District Blindness Control Societies , ophthalmic manpower training, intraocular lens (IOL) implants, and major equipment — indirect ophthalmoscopes, slit lamps, anterior vitrectomy units, keratometers, operating microscopes, A-scan biometers, YAG lasers. IOL implantation rose from 3% (1993) to 75% (2002) (Jugal Kishore; Suryakantha)

National programme	World Bank project and period	What the Bank actually financed
National TB Elimination Programme (formerly RNTCP)	Joint Government of India–WHO–World Bank review of the National Tuberculosis Programme, 1992 → RNTCP pilot 1993, nationwide by 1997	The review itself produced the DOTS-based revision; the Bank funds drug procurement and appoints an independent consultant for quarterly post-procurement review of contracts at central and state level (Suryakantha; Jugal Kishore, Ch 9)
National AIDS Control Programme	Bank funding across successive phases since the early 1990s	Multisectoral collaboration and sustained programme financing; the Bank was among the first to force attention onto HIV/AIDS in India when it was still denied (Suryakantha; Jugal Kishore)
National Leprosy Eradication Programme	Financial assistance to the programme	Cash support alongside WHO's free multidrug-therapy blister packs and DANIDA (Suryakantha)
Family welfare — India Population Projects (IPP)	IPP-VIII (1993, extended to 2001); IPP-IX (ended 2001)	IPP-VIII: urban slum and <i>jhuggi-jhonpadi</i> maternal and child health in Delhi through the Municipal Corporation — 12.5 lakh population, 929 clusters, health posts of 1 auxiliary nurse midwife (ANM) + 5 Basti Sevikas + 5 trained dais per 10,000. IPP-IX: whole of Assam, 10 backward districts of Rajasthan, 13 of Karnataka — 4,844 new buildings, 2,280 renovated, 1.86 lakh personnel trained . These area projects were later absorbed into Reproductive and Child Health (RCH)-II (Jugal Kishore, p 677)
Reproductive and Child Health Programme	RCH-I onwards	Co-funded with the European Community, UNICEF, DANIDA, ODA and UNFPA (Mahajan & Gupta 4e)
Integrated Child Development Services (ICDS)	UDISHA, 1999–2004	World Bank-assisted countrywide training programme for all ICDS functionaries — regular job training, orientation and refresher components (Jugal Kishore)
Integrated Disease Surveillance Programme (formerly Project)	Planning from 1998	Designed with the World Bank together with WHO, CDC, NIH, USAID, DFID and NICD (Jugal Kishore)

The Bank's own summary of its Indian portfolio is that ongoing projects support national disease-control programmes for **malaria and other vector-borne diseases, polio, HIV/AIDS and tuberculosis**, plus **child nutrition and reproductive and child health**, and that projects also **strengthen state-level rural health systems and national food and drug regulation and disease surveillance** (Jugal Kishore, Ch 3, p 972). Currently the Bank is described as supporting the blindness, AIDS control and tuberculosis programmes in particular (Suryakantha, p 814).



Route by which a World Bank credit reaches a beneficiary under an Indian national health programme

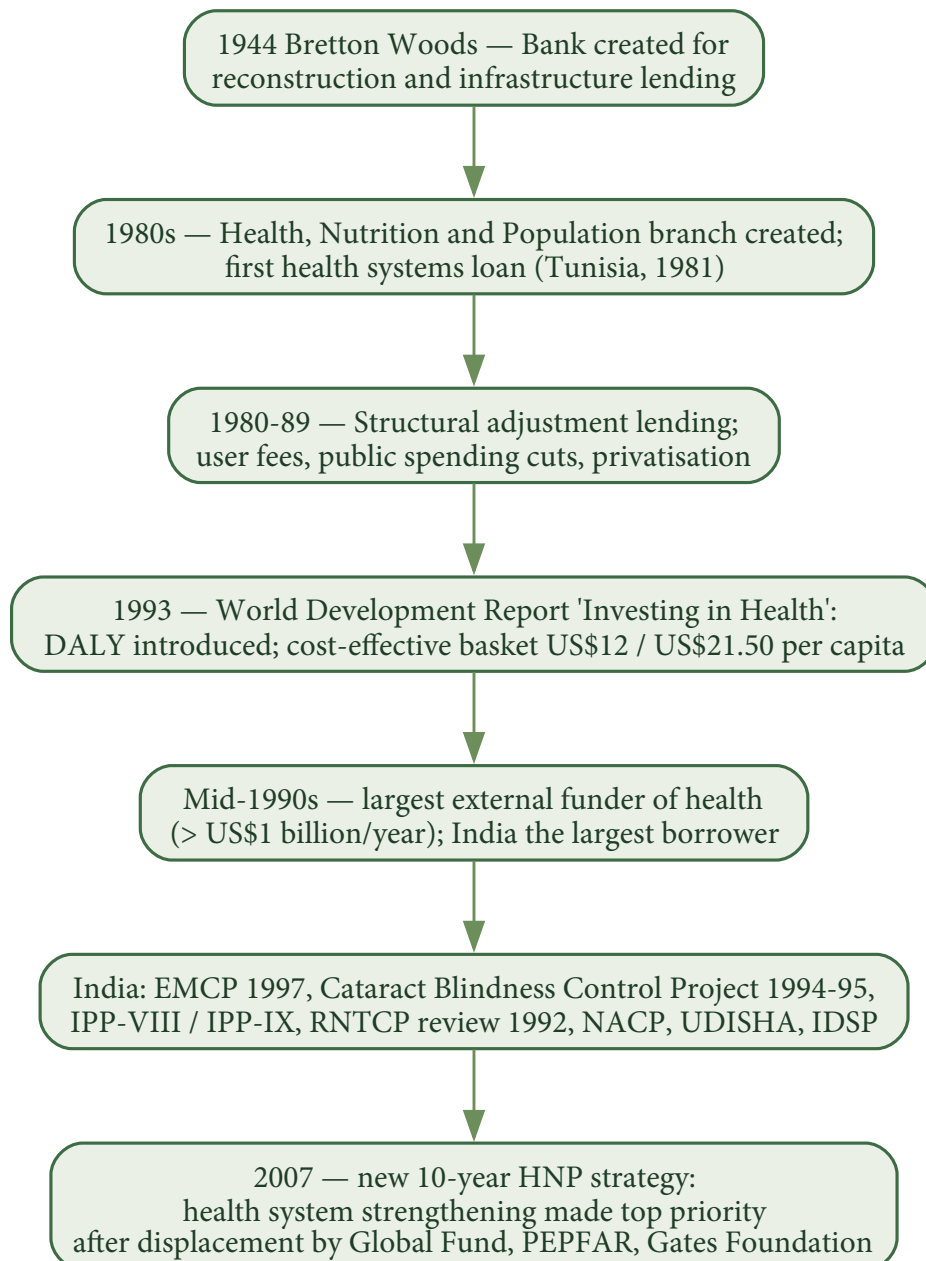
Modes of assistance — what the Bank contributes, in order of importance

- ▶ **Money as a credit, not a grant** — IDA credits and IBRD loans repayable by the Government of India, so every programme gain carries a debt-servicing tail.
- ▶ **Capital goods and consumables** — operating microscopes, slit lamps, IOLs, laboratory reagents, insecticide-treated nets, first-line anti-tuberculosis drugs.
- ▶ **Infrastructure** — sub-centre, primary health centre and staff-quarter construction, as under IPP-IX.
- ▶ **Human resource development** — UDISHA for ICDS functionaries, ophthalmic manpower training, programme-management training.

- **Management systems** — district and state societies, management information systems, monitoring and evaluation, independent procurement review and beneficiary assessment surveys.
- **Policy and analysis** — WDR-driven priority setting, disease-burden estimation, cost-effectiveness analysis, and joint programme reviews (as with the 1992 tuberculosis review).

Criticisms and limitations

Criticism	Evidence from the corpus
Structural adjustment damaged health	1980–89 lending to service third-world debt carried structural adjustment policies that streamlined economies "at the expense of health and social services"; UNICEF held them responsible for "reduced health, nutritional and educational levels for tens of millions of children in Asia, Latin America and Africa" (Jugal Kishore, Ch 3, p 981)
"Health sector reform" pushed user fees	Bank/IMF-promoted reform from the mid-1980s imposed tight fiscal limits on public health spending, direct cost recovery (user fees) and outsourcing to the private sector (Jugal Kishore, Ch 1, p 23)
User fees cut utilisation	A demonstrated 40–50% fall in health-service utilisation where user fees were introduced without working exemption policies, with worse adherence in chronic disease and HIV (Textbook of International Health, Ch 11, pp 586–587)
Eclipses WHO in policy	The Bank now exerts the single largest influence over health policy through loan conditions and advice, operating in direct competition with WHO (Jugal Kishore)
Loans, not entitlements	<i>Investing in Health</i> defined health as a private responsibility and health care as a private good, and its reforms dismantled already fragile public coverage in several countries (Textbook of International Health, Ch 11, p 596)
No longer pre-eminent	The Bank's position as chief international health financier has been partly displaced by the Global Fund, PEPFAR and the Gates Foundation; its 2007 HNP ten-year strategy therefore made health system strengthening its top priority (Textbook of International Health, Ch 3, p 119)



Evolution of the World Bank's engagement with health and its consequences for India

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — World Bank Group organogram.** Draw one top box "World Bank Group (Bretton Woods, 1944; HQ Washington)" with five boxes beneath, labelled exactly: IBRD (loans, middle-income); IDA (credits/grants, poorest, 35–40 yr maturity, 10 yr grace); IFC (private investment); MIGA (investment guarantee); ICSID (dispute settlement). Add a side label "Board of Governors — supreme body".
- ▶ **Diagram 2 — the two flowcharts above** (fund flow from Bank to beneficiary; and the historical evolution) are the diagrams to reproduce in the answer script.

High-yield exam pointers

- ▶ Bretton Woods 1944; HQ Washington; Board of Governors; five arms — IBRD, IDA, IFC, MIGA, ICSID (Bank proper = IBRD + IDA).
- ▶ WDR 1993 *Investing in Health* = the single most quotable date — gave the DALY and the US\$12 / US\$21.50 per capita cost-effective package.
- ▶ India is the largest recipient of IBRD/IDA funds — US\$3.7 billion, 67 projects, 2006–07, over US\$2 billion for health.
- ▶ Four dated Indian projects to write: EMCP 30 Sep 1997 (Rs 891 crore, 100 districts, 19 urban areas); Cataract Blindness Control Project 1994–95, Rs 554 crore, 7 years (IOL implantation 3% → 75%); IPP-VIII 1993–2001 and IPP-IX (ended 2001); UDISHA 1999–2004 for ICDS.
- ▶ 1992 GoI–WHO–World Bank review of the National Tuberculosis Programme is what produced RNTCP (pilot 1993, nationwide 1997).
- ▶ The balancing criticism sentence: **structural adjustment and user fees cut health-service utilisation by 40–50%**; assistance is a **loan, not a grant**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: several of these programmes have since been renamed — the Revised National Tuberculosis Control Programme is now the National Tuberculosis Elimination Programme, the National Programme for Control of Blindness is now the National Programme for Control of Blindness and Visual Impairment, and the Integrated Disease Surveillance Project is now the Integrated Disease Surveillance Programme; the retrieved corpus does not carry World Bank commitment figures more recent than the 2006–07 India portfolio, so quote the dated figures above rather than any current total.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, World Bank, MoHFW, National Health Accounts (India), NSSO

Programme and policy updates (India)

- ▶ A post-2010 World Bank lending figure for India at last. The textbook corpus stops at the 2006–07 portfolio, but an analysis of multilateral health financing for 2014–2022 gives dateable Bank numbers: IBRD/IDA disbursed US\$374.81 million for surgical care delivery and system

strengthening, with **India one of only three named country recipients** (alongside Ukraine and El Salvador, plus multi-country work in Africa) — paying for management of non-communicable diseases and injuries, outpatient and day surgery, cervical cancer prevention, early detection and treatment, and cancer registries. Separately the IFC — the private-investment arm in the structure table above — placed **US\$81.3 million in India for service delivery and health information systems**, part of 26 IFC surgical-care investments worth US\$1.64 billion [PMID 42049379]

- **Health and Wellness Centres have been renamed Ayushman Arogya Mandir** — over 1.75 lakh operationalised, 1,16,978 community health officers and 1,35,343 Jan Arogya Samitis in place by 30 November 2024, and 82% of centres delivering all 12 expanded service packages. **In addition to National Health Mission grants**, the Government of India has introduced new infrastructure financing through the **PM-Ayushman Bharat Health Infrastructure Mission** and **XV Finance Commission grants**, routed through urban local bodies and panchayati raj institutions [PMID 41816146]
- **eSanjeevani**, the national telemedicine service launched November 2019, delivered over 163 million consultations across all 28 states and 8 union territories between January 2020 and September 2023, more than 93% of them through the provider-assisted health-and-wellness-centre model; hubs grew from 397 (2020) to 14,036 and spokes from 6,868 to 108,610 [PMID 41158553]

Current national data

- **The WHO–World Bank universal health coverage monitoring framework** — the same framework used to produce the global monitoring reports that track UHC progress in 183 countries — has been adapted to Indian districts, giving a median district UHC index of **43.9%** across 687 districts, from 26.4% (Baksa, Assam) to 69.4% (Nilgiris, Tamil Nadu); only 21.1% of districts exceeded 50%. Non-communicable disease coverage was the weakest domain (median 9.8%), health insurance coverage 40.7%, and about one person in ten faced catastrophic or impoverishing health spending [PMID 39219770]
- **Out-of-pocket expenditure** has fallen from 60.6% of total health expenditure in 2014–15 to 39.4% in 2021–22 on National Health Accounts estimates, with the UHC service coverage index rising from 57 (2015) to 63 (2021) [PMID 41816146]; NSSO 75th-round analysis nonetheless finds out-of-pocket payments pushing the poverty headcount from 12.26% to 13.51% — about 3.1 million additional households impoverished — with use of a private facility raising the probability of impoverishment by 5.2 percentage points and household insurance cover lowering it by 3.8, though only 21% of households had any insured member [PMID 41938958]

Elimination, surveillance and disease status

- **Kala-azar** — recorded in the body above, on Park's authority, as World Bank-supported in Bihar, Jharkhand and West Bengal — **reached the elimination-as-a-public-health-problem threshold across India in 2023**, every block reporting under 1 case per 10,000 population; the 2002 National Health Policy had first set 2010 as the target and the deadline was revised repeatedly before 2023. Jharkhand cases fell from 2,515 (2013) to 160 (2023), a 93.6% decline, and in 2021 only 7 of 633 endemic blocks nationally remained above threshold [PMID 40988627]

- ▶ India has now entered the **post-elimination surveillance phase**, and low-level transmission persists: rK39 seroprevalence of 20–30% in study villages in Bihar and Jharkhand, with *Leishmania*-infected sand fly pools of 1.4% (4/285) and 1.2% (5/431) despite indoor residual spraying — making sustained surveillance, not case-finding alone, the current programme priority [PMID 41324335]

Guideline and legislative updates

- ▶ The **DALY**, which the Bank's 1993 *World Development Report* introduced and which is central to the account of its policy influence above, is under sustained methodological attack: the original 1990 Global Burden of Disease disability weights were set by a panel of health professionals drawn predominantly from high-income countries; modelled estimates built on sparse inputs carry wide uncertainty intervals that are routinely stripped away before reaching policymakers; and reporting incentives reward measurable target performance over the structural investments — workforce, supply chain, referral linkage — that decide whether a coverage rate means anything. India's own vertical, cost-effectiveness-justified disease programmes are cited as a case where programme-specific coverage rose while generalist primary care, the generalist workforce and surge capacity stayed underfunded — deficits COVID-19 exposed when oxygen, beds and rural diagnostics ran short. The proposed correction is to shift primary investment from international modelling to national data systems, civil registration and vital statistics [PMID 42243815]

Global and WHO developments

- ▶ The Bank now holds a **minority share of multilateral health money**: of US\$229 billion disbursed by multilateral organisations in 2014–2022, multilateral development banks including the World Bank Group accounted for **US\$50 billion (21.4%)**, against US\$87.3 billion (38%) from the UN system and US\$85.5 billion (37%) from vertical funds. Surgical care took only **US\$901.5 million, 0.4%** of that total, roughly 84% of it from development banks and led by the World Bank Group and the Inter-American Development Bank; the Bank Group put just 1% of its own health disbursements into surgical care, and — precisely the credit-not-grant model set out above — **disbursed entirely as loans, implemented 100% through recipient governments** [PMID 42049379]
- ▶ The World Bank-hosted **Pandemic Fund** scores country funding proposals through a **Technical Advisory Panel** on technical quality, sustainability plans and adherence to One Health principles, and the **2024 amendments to the International Health Regulations** acknowledge One Health and sustainability explicitly for the first time. Expert assessment from 29 global One Health specialists is that the WHO Joint External Evaluation and the WOAHP Performance of Veterinary Services tools still capture short-term technical capacity and legal compliance rather than durability, and that donor dependency — sharpened by post-2020 funding contractions from USAID and the UK FCDO — remains the central threat to capacities built with external money [PMID 41779720]

Linking the RA layer to the body — the answer's own framing, not a finding of any paper cited above: the Ayushman Arogya Mandir and eSanjeevani bullets describe domestically run national programmes and are not World Bank projects, so quote them as the current state of Indian primary care rather than as

Bank achievements. The 2014–2022 multilateral figures are what to set against the textbook's mid-1990s "largest external funder of health" line.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 25 International Health; Ch 7 Health Programmes in India; Ch 13 Tribal Health in India); Textbook of International Health (Ch 3 International Health Agencies, Ch 5 Data on Health, Ch 9 Globalization, Trade, Work and Health, Ch 11 Health Economics); Jugal Kishore — National Health Programmes of India (Ch 1, Ch 3 National and International Health Organizations, Ch 9, India Population Projects, National Programme for Control of Blindness); Suryakantha — Community Medicine with Recent Advances (International agencies; malaria, tuberculosis, AIDS, leprosy and blindness programmes); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (RCH funding partners); World Bank *World Development Report 1993: Investing in Health*.

Coverage note: the corpus documents the Bank's project-wise support up to the 2006–07 India portfolio in detail but carries no post-2010 India lending figures, so no current rupee or dollar commitment is quoted.

Recent-advances sources: 42049379 — Quantifying multilateral contributions to surgical system development in low- and middle-income countries: an analysis of trends and opportunities, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-021112](https://doi.org/10.1136/bmjgh-2025-021112); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmpc.jfmpc.478.25](https://doi.org/10.4103/jfmpc.jfmpc.478.25); 41158553 — Adoption and utilization of India's eSanjeevani national telemedicine service, *Oxford Open Digital Health* 2025, doi:[10.1093/oodh/oqaf025](https://doi.org/10.1093/oodh/oqaf025); 39219770 — District-level monitoring of universal health coverage, India, *Bulletin of the World Health Organization* 2024, doi:[10.2471/BLT.23.290854](https://doi.org/10.2471/BLT.23.290854); 41938958 — Impact of out-of-pocket health expenditure on poverty in India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1762886](https://doi.org/10.3389/fpubh.2026.1762886); 40988627 — Integrating community efforts, innovative government interventions and data in the fight against kala-azar: lessons from Dumka, India, *International Health* 2026, doi:[10.1093/inthealth/ihaf083](https://doi.org/10.1093/inthealth/ihaf083); 41324335 — High seropositivity to rK39 and capture of Leishmania-infected sand flies in Bihar and Jharkhand emphasize the need for post-elimination surveillance of visceral leishmaniasis in India, *Journal of Infectious Diseases* 2026, doi:[10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543) (abstract-only — no PMC full text; bullet checked against the published abstract); 42243815 — Escaping the metric-driven governance trap in global health: how DALYs and coverage indicators mislead policymakers in LMICs, *International Journal for Equity in Health* 2026, doi:[10.1186/s12939-026-02909-9](https://doi.org/10.1186/s12939-026-02909-9); 41779720 — Exploring measures of sustainability in the WHO Joint External Evaluation and the WOA performance of veterinary services tools, *PLoS One* 2026, doi:[10.1371/journal.pone.0343801](https://doi.org/10.1371/journal.pone.0343801). Also used: WHO–World Bank universal health coverage monitoring framework; National Health Accounts estimates for India 2021–22; MoHFW Ayushman Arogya Mandir portal (30 November 2024); NSSO 75th round (2017–18); International Health Regulations 2024 amendments.

Recent-advances verification note: every retained bullet was re-checked against the cited article's PMC full text (PMID 41324335 against its abstract, no PMC full text available). One bullet was dropped — a Meghalaya drone-delivery cost-effectiveness study (PMID 41403554), whose data were sound but whose full text never mentions the World Bank, so its World Bank-financing

framing was unsupported.



Sustainable Development Goals

Asked in: Paper IV 25-Nov-2020 — RGUHS MD Community Medicine

Definition — Sustainable development is "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (Park 28e, Ch 14, p 852; World Commission on Environment and Development 1987, cited in Oxford 7e, Ch 107). The **Sustainable Development Goals (SDGs)** are the 17 goals and 169 targets of the United Nations resolution *Transforming our world: the 2030 Agenda for Sustainable Development*, adopted by the UN General Assembly in September 2015 as the successor to the Millennium Development Goals (MDGs) (Park 28e, Ch 1, p 23). They integrate the three dimensions of sustainable development — **economic, social and environmental** — around the **five Ps: people, planet, prosperity, peace and partnership**, with "leaving no one behind" as the overarching theme (Park 28e, Ch 14, p 852).

Genesis and architecture of the 2030 Agenda

Element	Detail
Political foundation	Rio+20 Conference, Rio de Janeiro, 2012 (Oxford 7e, Ch 10, pp 419–420)
Predecessor	8 MDGs (2000–2015); 3 of 8 goals, 8 of 18 targets and 18 of 48 indicators were health-related (Park 28e, Ch 1, p 23)
Adoption	UN General Assembly, September 2015 ; MDGs ended December 2015
Came into force	1 January 2016 (Park 28e, Ch 14, p 852)
Deadline	2030
Structure	17 goals, 169 targets , measured by a global indicator framework of about 232 indicators (Oxford 7e, Ch 10, pp 419–420)
Monitoring body	Inter-Agency and Expert Group on SDG Indicators (IAEG-SDG); countries also frame their own national indicator sets (Park 28e, Ch 14, p 854)
Political oversight	High-Level Political Forum (HLPF) — "from silos to integrated policy making" (Oxford 7e, Ch 10, pp 421–422)
Financing	Addis Ababa Action Agenda , Third International Conference on Financing for Development, July 2015 (Oxford 7e, Ch 10, p 420)
Scope	Universal — applies to all countries, not only developing countries (Park 28e, Ch 14, p 859)

The 17 Sustainable Development Goals



The 17 Sustainable Development Goals (Oxford public health 7th ed, p.934)

No.	Goal	Direct link to health
1	No poverty	Social protection floors (target 1.3); catastrophic health spending
2	Zero hunger	Target 2.2 — stunting, wasting, overweight in under-fives; anaemia in women
3	Good health and well-being	The health goal — 13 targets
4	Quality education	Target 4.2 — early childhood development and pre-primary education
5	Gender equality	Targets 5.2, 5.3, 5.6 — violence, child marriage, female genital mutilation, reproductive rights
6	Clean water and sanitation	Targets 6.1, 6.2, 6.3 — safe water, sanitation, end open defecation
7	Affordable and clean energy	Target 7.1 — clean household energy (indoor air pollution)
8	Decent work and economic growth	Target 8.8 — occupational safety, migrant workers
9	Industry, innovation and infrastructure	Health technology and infrastructure
10	Reduced inequalities	Target 10.4 — fiscal, wage and social protection policy
11	Sustainable cities and communities	Targets 11.5, 11.6 — disaster deaths, ambient air pollution (PM 2.5, PM10)
12	Responsible consumption and production	Target 12.4 — sound management of chemicals and wastes
13	Climate action	Climate resilience and adaptation
14	Life below water	Marine pollution and biodiversity

No.	Goal	Direct link to health
15	Life on land	Terrestrial ecosystems, zoonotic spillover
16	Peace, justice and strong institutions	Targets 16.1, 16.2, 16.9 — violence, child abuse, birth registration
17	Partnerships for the goals	Targets 17.18, 17.19 — disaggregated data, civil registration and census

(Goal titles from the United Nations SDG plate reproduced in Oxford 7e, Ch 24, p 934; health linkages from Park 28e, Ch 14, pp 856–857 and WHO World Report on the Health of Refugees and Migrants, Table 5.1.)

SDG 3 and its 13 targets

Goal 3 — "Ensure healthy lives and promote well-being for all at all ages." Its 13 targets comprise four carried forward from the unfinished and expanded MDG agenda, four new outcome targets, four "means-of-implementation" targets, and the overarching target 3.8 on universal health coverage (UHC), which is the key to achieving all the others and to building a strong, resilient health system (Park 28e, Ch 14, p 852).

Targ et	Content	Numerical target / indicator
3.1	Reduce global maternal mortality	MMR < 70 per 100,000 live births by 2030; 3.1.2 births attended by skilled personnel
3.2	End preventable newborn and under-five deaths	Neonatal mortality ≤ 12 per 1,000 live births; under-five mortality ≤ 25 per 1,000 live births
3.3	End epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases (NTDs); combat hepatitis, water-borne and other communicable diseases	3.3.1 new HIV infections/1,000 uninfected; 3.3.2 TB incidence/100,000; 3.3.3 malaria cases/1,000 at risk; 3.3.4 hepatitis B; 3.3.5 persons needing NTD interventions
3.4	Reduce premature non-communicable disease (NCD) mortality by one third and promote mental health	3.4.1 probability of death from cardiovascular disease, cancer, diabetes or chronic respiratory disease between ages 30 and 70; 3.4.2 suicide rate
3.5	Strengthen prevention and treatment of substance abuse and harmful alcohol use	3.5.2 alcohol per capita consumption (≥15 y, litres of pure alcohol)
3.6	Halve global deaths and injuries from road traffic accidents by 2020	3.6.1 road traffic deaths per 100,000 (age-standardised)
3.7	Universal access to sexual and reproductive health care, family planning, information and education	3.7.1 demand for family planning satisfied by modern methods; 3.7.2 adolescent birth rate
3.8	Universal health coverage — financial risk protection, quality essential services, safe and affordable essential medicines and vaccines	3.8.1 coverage of tracer interventions; 3.8.2 population with catastrophic/impoverishing out-of-pocket health expenditure

Target	Content	Numerical target / indicator
3.9	Reduce deaths and illness from hazardous chemicals and air, water and soil pollution	3.9.1 mortality from household and ambient air pollution; 3.9.2 from unsafe water, sanitation and hygiene; 3.9.3 unintentional poisoning
3.a	Strengthen implementation of the WHO Framework Convention on Tobacco Control	3.a.1 age-standardised current tobacco use ≥ 15 y
3.b	Research and development of vaccines and medicines for diseases of developing countries; access to affordable essential medicines under the Doha Declaration on TRIPS and Public Health	3.b.1 access to affordable medicines and vaccines; 3.b.2 official development assistance to medical research
3.c	Increase health financing and the recruitment, development, training and retention of the health workforce	3.c.1 health worker density and distribution per 10,000 population
3.d	Strengthen capacity for early warning, risk reduction and management of health risks	3.d.1 International Health Regulations (IHR) — % of the 13 core capacities attained

(Target and indicator wording: Park 28e, Ch 14, Table 1, pp 852–853.)

Health-related targets in the other goals

Health cannot be delivered by SDG 3 alone; more than a dozen targets in other goals are health-related and are tracked alongside it (Park 28e, Ch 14, Tables 2–8, pp 854–858).

Type of indicator	SDG target	Proposed indicator
Impact	3.1, 3.2	Maternal, neonatal and under-five mortality
Impact	3.9, 11.5, 13.1	Mortality from unsafe water/sanitation, air pollution, disasters
Impact	16.1	Homicide and conflict-related deaths
Coverage	3.8	UHC reproductive, maternal, newborn and child health tracers — family planning, antenatal care, full immunisation, care-seeking for suspected child pneumonia
Risk factor	2.2	Child stunting, wasting, overweight
Risk factor	6.1, 6.2	Safely managed drinking water; safely managed sanitation
Risk factor	7.1, 11.6	Clean household energy; ambient air pollution (PM 2.5, PM10)
Risk factor	5.2	Women and girls subjected to physical, sexual or psychological violence
System	17.18, 17.19	Disaggregated data; birth and death registration, regular census

MDGs versus SDGs

Feature	Millennium Development Goals (2000–2015)	Sustainable Development Goals (2016–2030)
Number	8 goals, 18 targets, 48 indicators	17 goals, 169 targets, ~232 indicators
Health goals	3 of 8 directly health-related	1 comprehensive goal (SDG 3) with 13 targets
Origin	Millennium Declaration, drafted by UN experts	Negotiated by member states after Rio+20
Scope	Mainly developing countries; tied to aid spending	Universal — "integrated and indivisible, global in nature and universally applicable"
Pillars	Limited set of human development outcomes	Economic + social + environmental pillars
Equity	Emphasis on national averages	Explicit equity focus — "no one will be left behind"
Financing	Official development assistance dominant	Domestic resource mobilisation + Addis Ababa Action Agenda
Health system	Largely vertical, disease-specific	UHC (3.8) as the integrating framework

(Park 28e, Ch 14, pp 858–859; Oxford 7e, Ch 10, pp 419–420.)

India and the SDGs — status on selected SDG 3 indicators

- ▶ India adopted the SDGs alongside the national principle of *Sabka Saath, Sabka Vikas* ("together with all, development for all"), with the first claim on development belonging to the poor (Park 28e, Ch 14, p 859).
- ▶ Priorities identified for India: sustain the unfinished maternal and child health agenda; confront NTDs; eliminate malaria, AIDS and hepatitis; intensify the fight against tuberculosis; implement universal health care; strengthen both rural and urban health systems with primary health care at the centre; involve the private sector; and build a strong monitoring, evaluation and accountability system (Park 28e, Ch 14, p 859).
- ▶ Progress was already off-track before COVID-19; the pandemic pushed tens of millions back into extreme poverty, disrupted childhood immunisation, and interrupted care for NCDs, tuberculosis, malaria and HIV, hitting informal workers, older people, persons with disabilities, indigenous people, migrants and refugees hardest (Park 28e, Ch 14, p 859).

SDG 3 indicator	Global	India	Year
3.1.1 Maternal mortality ratio (per 100,000 live births)	223	97	2018–20
3.3.1 New HIV infections (per 1,000 uninfected)	0.19	0.05	2021
3.3.2 Tuberculosis incidence (per 100,000 population)	124	210	2021
3.3.4 Infants receiving 3 doses of hepatitis B vaccine (%)	80	85	2021
3.4.1 Premature NCD mortality — probability of dying age 30–70 (%)	17.8	21.9	2019

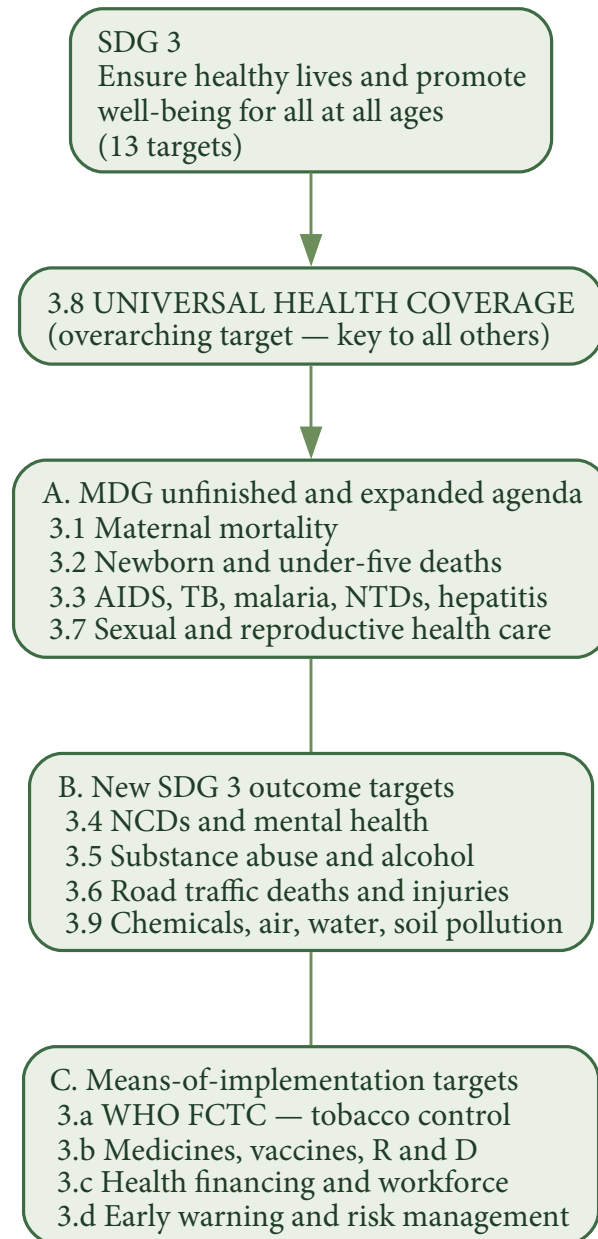
SDG 3 indicator	Global	India	Year
3.4.2 Suicide mortality rate (per 100,000)	9.2	12.7	2019
3.5.2 Alcohol consumption (litres pure alcohol, ≥ 15 y)	5.8	4.9	2019
3.6.1 Road traffic deaths (per 100,000, age-standardised)	16.7	15.6	2019
3.9.1 Mortality from household + ambient air pollution (per lakh)	114.1	184.3	2016
3.a.1 Current tobacco use, age ≥ 15 y (%)	22.3	27.2	2020
3.d.1 IHR core capacities attained (%)	71	95	2010–17

(Park 28e, Ch 14, Table 1, pp 852–853.)

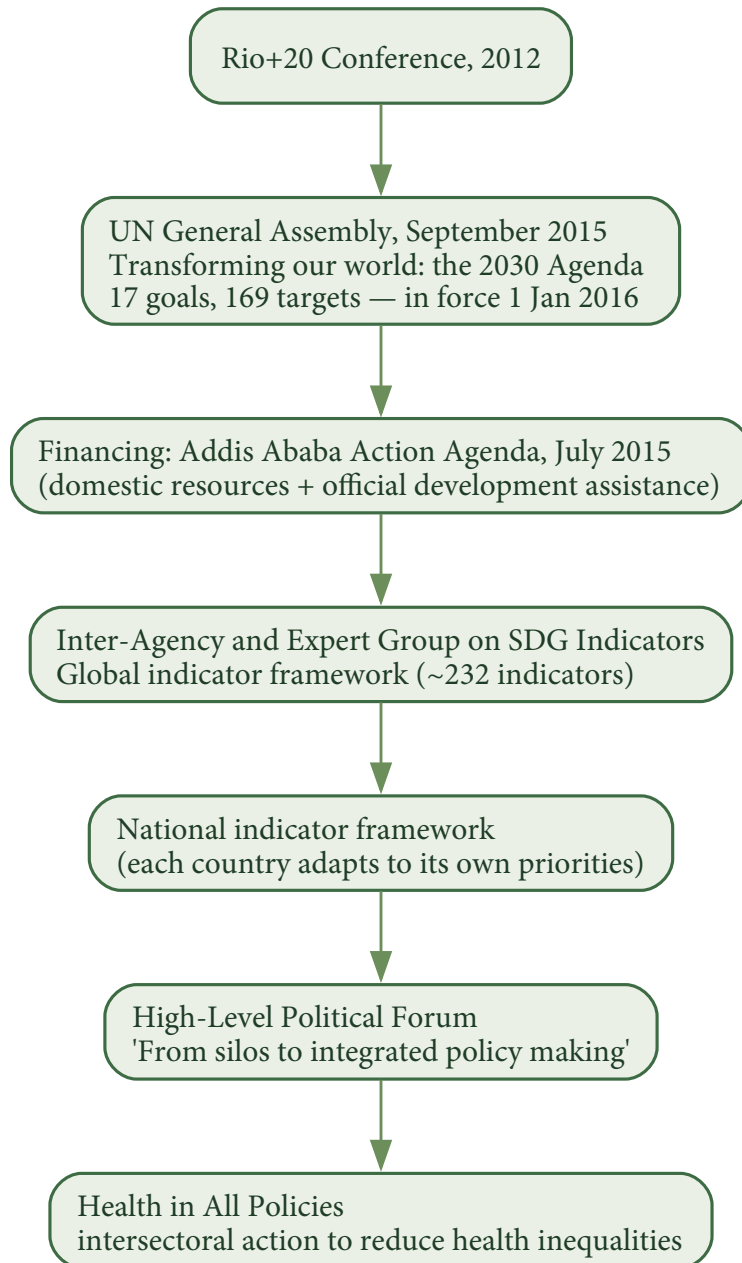
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Figure 1 — Framework of SDG 3 health targets (Park 28e, Ch 14, Fig 1, p 855). Draw three vertical columns under the heading "SDG 3: Ensure healthy lives and promote well-being for all at all ages", with UHC (3.8) as a band running beneath all three. Label exactly:

- ▶ Column 1 — **MDG unfinished and expanded agenda**: 3.1 maternal mortality; 3.2 newborn and child deaths; 3.3 AIDS, TB, malaria, NTDs, hepatitis, water-borne diseases; 3.7 sexual and reproductive health care.
- ▶ Column 2 — **New SDG 3 targets**: 3.4 NCDs and mental health; 3.5 substance abuse; 3.6 road traffic injuries; 3.9 hazardous chemicals, air, water and soil pollution.
- ▶ Column 3 — **Means of implementation**: 3.a tobacco control (WHO FCTC); 3.b medicines, vaccines and research and development; 3.c health financing and workforce; 3.d early warning and risk management.
- ▶ Base band — **3.8 Universal Health Coverage**; outer frame — **other social and environmental SDGs, and SDG 17 on means of implementation**.



Architecture of the 13 targets of Sustainable Development Goal 3



How SDG progress is financed, monitored and reviewed

High-yield exam pointers

- ▶ **Numbers to write:** 17 goals, 169 targets, ~232 indicators; SDG 3 has 13 targets; adopted September 2015, in force 1 January 2016, deadline 2030.
- ▶ **Five Ps** — People, Planet, Prosperity, Peace, Partnership; three dimensions — economic, social, environmental; motto — "leave no one behind".
- ▶ **Numeric SDG 3 targets:** MMR < 70/100,000 live births; neonatal mortality ≤ 12/1,000; under-five mortality ≤ 25/1,000; premature NCD mortality reduced by one third; road traffic deaths halved by 2020.
- ▶ **3.4.1 denominator trap:** premature NCD mortality is the probability of dying between exact ages 30 and 70 from cardiovascular disease, cancer, diabetes or chronic respiratory disease — not a crude death rate.

- ▶ **Target 3.8 (UHC) is the integrating target** — quote it verbatim; the WHO Thirteenth General Programme of Work sets the "**triple billion**" targets: one billion more people with UHC, one billion better protected from health emergencies, one billion enjoying better health and well-being (Oxford 7e, Ch 107).
- ▶ **MDG vs SDG one-liner:** MDGs were aid-financed, developing-country, outcome goals; SDGs are universal, equity-focused, and integrate economic, social and environmental pillars.
- ▶ **India, 2018–20:** MMR 97 per 100,000 live births (SDG target < 70); TB incidence 210 per 100,000 (2021); tobacco use 27.2% (2020).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not written in the exam: the indicator count is quoted as 232 in Oxford Textbook of Public Health 7e, Ch 10 and as 250 in the same book's Ch 107, because the global indicator framework has been revised several times since 2017 — write "about 230 indicators" if pressed. The NITI Aayog **SDG India Index**, which ranks Indian states and union territories on SDG performance, is not covered in the indexed corpus and is therefore not quoted here; the state ranking described in Park 28e, Ch 2, p 44 is the separate NITI Aayog **Health Index**, not the SDG India Index.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-5, Global Burden of Disease 2021, National Health Accounts 2019–20, WHO

Programme and policy updates (India)

- ▶ **Health and Wellness Centres are now Ayushman Arogya Mandirs (AAMs)** — the single most examinable renaming since the textbook edition. About 1.6 lakh AAMs plus roughly 6,400 block-level and 770 district NCD clinics screen more than eight crore adults a year for hypertension and diabetes, yet NTEP, NP-NCD and the National Mental Health Programme still run as vertical silos; a SWOT analysis argues that linking Ni-kshay, the NCD app and ABHA-linked records on the AAM platform is the practical route to targets 3.4 and 3.8. [PMID 42397826]
- ▶ **Financing pluralism is India's binding constraint on SDG 3** — a multilevel analysis combining NFHS-5 with National Health Accounts 2019–20 found out-of-pocket spending still at 51.2% of total health expenditure (government 39.4%, prepaid 9.4%) and insurance coverage ranging from 13.8% in Jammu and Kashmir to 87.8% in Rajasthan. Public and prepaid financing improved child survival, immunisation and WASH access; fragmented state schemes raised expenditure. The authors call for a nationally pooled insurance framework. [PMID 42218447]

Current national data

- ▶ **NFHS-5 (2019–21) — non-communicable disease clusters within households.** Across 636,699 households, two or more members had hypertension in 14.9% and diabetes in 7.7%, and these clustered households carried 49.8% of all hypertension and 39.3% of all diabetes cases in India. Spouses were the commonest dyad (about 54% for both). Every affected member was unaware of the diagnosis in 42.5% of hypertension-clustered and 55.5% of diabetes-clustered households —

an argument for household-level rather than individual screening under target 3.4. [PMID 41698756]

- **SDG 6.1 — Water Access Index across NFHS-1 to NFHS-5.** Improved drinking-water coverage rose from 75.8% (1992–93) to 95.9% (2019–21), and 98% of the 707 districts now rank "achiever" or "front runner" for improved sources on the NITI Aayog localisation scale. Progress is uneven within the target: only 1% of districts reach universal on-premises supply, 52% remain "aspirant" for household water treatment, and 221 districts (31%) are aspirant on the composite index, clustered in West Bengal, Bihar, Odisha, Jharkhand and the north-east. [PMID 40499970]

Elimination, surveillance and disease status

- **Kala-azar — elimination as a public health problem achieved in 2023.** India reported fewer than 1 case per 10,000 population at block level nationwide in 2023. The deadline had slipped repeatedly from the 2002 National Health Policy target of 2010; the decisive inputs were liposomal amphotericin B (2014), synthetic pyrethroid indoor residual spraying with hand-compression pumps (2015–16) and wage-loss compensation for patients. Jharkhand's caseload fell from 2,515 in 2013 to 160 in 2023 (a 93.6% decline). [PMID 40988627]
- **Lymphatic filariasis — still endemic in 345 districts against a 2030 elimination goal.** The economic and operational demands of mass drug administration, morbidity management and post-MDA surveillance strain the programme; public–private partnership, already used for polio and TB, is proposed to close gaps in social mobilisation, MDA compliance and post-MDA surveillance. Quote the 2030 deadline, not an earlier one. [PMID 41951149]
- **SDG 3.2 — South Asia is off track on neonatal mortality.** A Global Burden of Disease 2021 analysis found neonatal-disorder mortality in South Asia fell 40% and DALYs 35% between 1990 and 2021 despite a 15% rise in prevalence. ARIMA projections to 2031 place only Bangladesh on track for the ≤ 12 per 1,000 target; India, Nepal and Bhutan need accelerated effort, and Pakistan remains far behind. Preterm birth and neonatal sepsis dominate Indian neonatal deaths. [PMID 40080234]

Guideline and legislative updates

- **Registration of Births and Deaths (Amendment) Act 2023 and target 16.9.** Birth certificates are now mandatory for a widening range of services. Between NFHS-4 and NFHS-5 institutional deliveries rose from 86% to 92% and under-five birth registration from 80% to 89%; births at PHCs and CHCs were registered in 90.2% of cases (up from 80.8%, the largest gain of any setting) and carried 38% higher odds of registration than private facilities. Gains were greatest where states designate PHCs and CHCs as Registrars — Delhi, Rajasthan and Uttar Pradesh. [PMID 41752339]

Global and WHO developments

- **UHC in South Asia — service coverage rising, financial protection falling.** Modelled projections to 2030 put the regional composite coverage index at 81.5% (from 53.5% in 2000), with India projected to cross the 80% benchmark. Catastrophic health expenditure, however, is projected to more than double regionally from 7.2% to 18.6%, reaching 26.3% in India by 2030, while impoverishment from out-of-pocket spending in India falls only from 3.5% to 2.6%. Target 3.8 has two limbs and India is moving in opposite directions on them. [PMID 41192927]

- **Framing the post-2030 successor agenda.** With the Global Sustainable Development Report 2023 judging fragmented action inadequate and the Pact for the Future renewing the call, a six-function framework for integrated governance beyond 2030 has been proposed — coherence, arbitration, shared-outcome financing, interlinkage-aware measurement, equity-sensitive accountability and adaptive learning — extending Health in All Policies and One Health into whatever replaces the 2030 Agenda. [PMID 42539888]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 14 (Sustainable Development Goals, pp 852–859), Ch 1 (p 23), Ch 2 (pp 40–44); Oxford Textbook of Public Health 7e — Ch 10 (Reducing health inequalities in developing countries, pp 419–422), Ch 24 (p 934, United Nations SDG plate), Ch 100 (p 3799), Ch 107 (Global health in the era of sustainable development); WHO World Report on the Health of Refugees and Migrants (Table 5.1, SDGs most directly related to global health); United Nations, *Transforming our world: the 2030 Agenda for Sustainable Development* (A/RES/70/1, 2015) and the Addis Ababa Action Agenda (2015) as cited in the above texts. Coverage note: Textbook of International Health predates the 2030 Agenda and contributed nothing to this answer.

Recent-advances sources: PMID 42397826 — Integrated tuberculosis-multimorbidity management in India: a SWOT analysis, *Indian J Med Res* 2026, doi:[10.25259/IJMR.2790.2025](https://doi.org/10.25259/IJMR.2790.2025); PMID 42218447 — The role of health financing system and pluralism in achieving sustainable development goal health outcomes in India: a multilevel analysis, *Int J Equity Health* 2026, doi:[10.1186/s12939-026-02888-x](https://doi.org/10.1186/s12939-026-02888-x); PMID 41698756 — Clustering of hypertension and clustering of diabetes at the household level and variations in disease awareness within households in India, *BMJ Glob Health* 2026, doi:[10.1136/bmjgh-2024-018809](https://doi.org/10.1136/bmjgh-2024-018809); PMID 40499970 — Progress and determinants of household access to improved drinking water in India using a Water Access Index: insights from the National Family Health Survey towards achieving SDG 6.1, *BMJ Open* 2025, doi:[10.1136/bmjopen-2024-091269](https://doi.org/10.1136/bmjopen-2024-091269); PMID 40988627 — Integrating community efforts, innovative government interventions and data in the fight against kala-azar: lessons from Dumka, India, *Int Health* 2026, doi:[10.1093/inthealth/ihaf083](https://doi.org/10.1093/inthealth/ihaf083); PMID 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Trop* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); PMID 40080234 — Mortality projections, regional disparities in the burden of neonatal disorders, and the status of achieving SDG targets by 2030 in South Asia: insights from the Global Burden of Disease Study 2021, *J Epidemiol Glob Health* 2025, doi:[10.1007/s44197-025-00359-0](https://doi.org/10.1007/s44197-025-00359-0); PMID 41752339 — Strengthening civil registration through grassroots health institutions in India, *Int J Environ Res Public Health* 2026, doi:[10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); PMID 41192927 — Progress towards universal health coverage in South Asia, 2000–2030: an examination of the twin elements of primary healthcare provision and financial protection, *BMJ Glob Health* 2025, doi:[10.1136/bmjgh-2025-020052](https://doi.org/10.1136/bmjgh-2025-020052); PMID 42539888 — Public health and integrated governance across the sustainable development goals beyond 2030: a six-function framework, *Front Public Health* 2026, doi:[10.3389/fpubh.2026.1892248](https://doi.org/10.3389/fpubh.2026.1892248).

♦ ♦ ♦

UNICEF's contribution to child health

Asked in: Paper IV 28-Jun-2024 — RGUHS MD Community Medicine

Definition — The United Nations Children's Fund (UNICEF) is described by Park as "one of the specialized agencies of the United Nations", established in 1946 by the UN General Assembly to deal with the **rehabilitation of children in war-ravaged countries**; when the emergency functions were over in 1953 the General Assembly gave it the new name "U.N. Children's Fund" but **retained the original initials, UNICEF** (Park 28e, Ch 25, p 1058). Its activities today cover **child survival, protection and development** — immunisation, improved infant feeding, growth monitoring, home-based diarrhoea management, drinking water and environmental sanitation, birth spacing, education of girls and income generation for women (Park 28e, Ch 25, p 1059).

Origin, mandate and organisation

Item	Detail
Year of establishment	1946, by resolution of the UN General Assembly (Park 28e, Ch 25, p 1058)
Original purpose	Emergency relief and rehabilitation of children in countries devastated by the Second World War (hence the original expansion, <i>United Nations International Children's Emergency Fund</i>) (Park 28e, Ch 25, p 1058)
Renamed	1953 — "U.N. Children's Fund"; initials UNICEF retained (Park 28e, Ch 25, p 1058)
Headquarters	United Nations, New York (Park 28e, Ch 25, p 1058)
Regional office serving India	Kathmandu, Nepal — the South Central Asian Region, covering Afghanistan, Sri Lanka, India, the Maldives, Pakistan, Bhutan, Bangladesh and Nepal (Park 28e, Ch 25, p 1058)
Governing body	A thirty-six-nation Executive Board (as in 2010) (Park 28e, Ch 25, p 1058)
Works in close collaboration with	WHO, and the other specialised agencies — UNDP, FAO, UNESCO; in the early years jointly with WHO on malaria, tuberculosis and venereal diseases (Park 28e, Ch 25, p 1058)
Current mandate	Survival, protection and development of the child; full partner with WHO in primary health care , developing joint strategies to support its implementation at country level (Park 28e, Ch 25, p 1059)
Serial publication	<i>The State of the World's Children</i> — a UNICEF report series repeatedly cited as a source for child survival statistics (e.g. UNICEF, <i>State of the World's Children</i> , 2006, 2009, 2019 in Park 28e, Ch 9 reference list; 2016 in Oxford Textbook of Public Health 7e, Ch 86 reference list)
Universal Children's Day	November 14 ; started by the International Union for Child Welfare and UNICEF, and in 1954 the UN General Assembly passed a formal resolution establishing the day and assigning UNICEF the responsibility for promoting it (Park 28e, Ch 9, p 653)

Four areas of special emphasis — the operational mandate

Area	What UNICEF actually does
(a) Child health	Substantial aid for the production of vaccines and sera ; support to national immunisation programmes; assistance to environmental sanitation programmes emphasising safe and sufficient water for drinking and household use in rural areas; currently focused on delivering primary health care to mothers and children — immunisation, infant and young child care, family planning aspects of family health, safe water and adequate sanitation (Park 28e, Ch 25, pp 1057-1058)
(b) Child nutrition	Began as supplementary child feeding, expanded from the mid-1950s with low-cost protein-rich food mixtures ; with FAO aided " applied nutrition " programmes through community development, agricultural extension, schools and health services; specific aid against deficiency diseases — large doses of vitamin A where xerophthalmia is prevalent, enrichment of salt with iodine in endemic goitre areas, iron and folate supplements for the anaemias, and food enrichment; with FAO and WHO now encourages national food and nutrition policies (Park 28e, Ch 25, p 1058)
(c) Family and child welfare	Improving the care of children within and outside the home through parent education, day-care centres, child welfare and youth agencies and women's clubs — carried out not as separate projects but as part of health, nutrition and education programmes (Park 28e, Ch 25, p 1058)
(d) Education — formal and non-formal	With UNESCO, assistance for basic primary and science education : science laboratory equipment, workshop tools, library books and audiovisual aids, with emphasis on schooling relevant to the child's environment and future life (Park 28e, Ch 25, p 1058)
Cross-cutting — Urban Basic Services (UBS), since 1976	Upgrading basic services (health, nutrition, water supply, sanitation, education) for children and women in selected cities and towns, to improve the survival and development of children of urban low-income families (Park 28e, Ch 25, pp 1058-1059)
Policy shift — the "whole child"	UNICEF has turned away from single-disease eradication campaigns unless of direct benefit to mothers and children, towards the concept of the " whole child " — assistance geared to long-term personal development and to the development of the country; this approach is known as " country health programming " (Park 28e, Ch 25, p 1058)

Technical strategies UNICEF contributed to child health

Strategy	Year	Content
GOBI, and its augmented form GOBI-FFF	GOBI campaign, Park 28e; "child survival revolution" launched 1982 ; Task Force for Child Survival 1984 with WHO, World Bank, UNDP and the Rockefeller Foundation	Four simple technologies — Growth charts to monitor child development, Oral rehydration for all mild and moderate dehydration, Breast-feeding, Immunisation against measles, diphtheria, polio, pertussis, tetanus and tuberculosis, promoted by Park as a " child health revolution " (Park 28e, Ch 25, pp 1058-1059); the augmented package adds Food supplementation, Female literacy and Family planning (Textbook of International Health, Ch 3, p 116; Ch 13, pp 696-697)

Strategy	Year	Content
Oral rehydration therapy (ORT)	ORT approved for use by WHO in 1969; WHO/UNICEF Joint Statement 2004	Oral rehydration treatment can be used safely for acute diarrhoea of all aetiologies, in all age groups and in all countries; since January 2004 the reduced-osmolality ORS is the only formulation procured by UNICEF , and WHO and UNICEF recommend that countries manufacture and use it in place of the previous formulation (Park 28e, Ch 5, p 282; WHO, UNICEF (2004), <i>Clinical Management of Acute Diarrhoea</i> — WHO/UNICEF Joint Statement, Park 28e Ch 5 reference list; Textbook of International Health, Ch 6, p 306). India's Diarrhoeal Diseases Control Programme began in 1978 and the National Oral Rehydration Therapy Programme in 1985-86 (Park 28e, Ch 5, p 286)
Zinc in acute diarrhoea	Joint WHO/UNICEF recommendation	10 mg elemental zinc daily for infants under 6 months and 20 mg daily for children over 6 months, for 10–14 days ; reduces duration and severity, and lowers diarrhoea incidence for the next 2–3 months (Park 28e, Ch 5, p 285)
Baby Friendly Hospital Initiative (BFHI)	Launched jointly by UNICEF and WHO in 1991 ; implementation guidance revised 2018	The Ten Steps to Successful Breast-feeding — a package of policies and procedures for every facility providing maternity and newborn services, beginning with full compliance with the International Code of Marketing of Breast-milk Substitutes; Indian "baby friendly" hospitals must also practise clean delivery, essential newborn care, immunisation and ORT (Jugal Kishore — National Health Programmes of India; Park 28e, Ch 9, pp 640-641)
Integrated Management of Childhood Illness (IMCI)	WHO–UNICEF	Combined case management of the principal causes of child death — diarrhoeal disease, acute respiratory infection, measles, malaria and underlying malnutrition; the Indian adaptation is IMNCI , the central pillar of child health under RCH-II, which adds the 0–7 day age group and neonatal care (Park 28e, Ch 7, pp 551-552; Ch 9, p 678)
Accelerated measles mortality reduction strategy	WHO–UNICEF; implementation began 2001	Two doses of measles-containing vaccine for all children through routine services and supplementary immunisation activities (SIAs), plus improved disease surveillance (Park 28e, Ch 5, p 176)
Universal Child Immunization — 1990	Declaration sponsored by UNICEF in October 1985 , at the UN's 40th anniversary	Renamed and added impetus to the WHO Expanded Programme on Immunization (EPI, launched May 1974 globally, January 1978 in India); India's Universal Immunization Programme was launched on 19 November 1985 (Park 28e, Ch 3, p 147)

Strategy	Year	Content
Under-five mortality rate as the master indicator	—	UNICEF defines it as the "annual number of deaths of children aged under 5 years, expressed as a rate per 1000 live births", i.e. the probability of dying between birth and exactly 5 years, and regards it as the best single indicator of social development and well-being — better than gross national product per capita , because it reflects income, nutrition, health care and basic education (Park 28e, Ch 9, p 676)
Convention on the Rights of the Child (CRC)	Adopted by the UN General Assembly, 1989	The principal legal instrument protecting children's well-being; Article 24 addresses the child's right to health and health care ; ratified by all but two UN member states (Oxford Textbook of Public Health 7e, Ch 86, pp 3233-3234)
Conceptual framework of child undernutrition	UNICEF, 1990	Separates immediate, underlying and basic causes of maternal and child undernutrition — the standard framework used worldwide to plan multisectoral nutrition action (Oxford Textbook of Public Health 7e, Ch 21, pp 814-815)

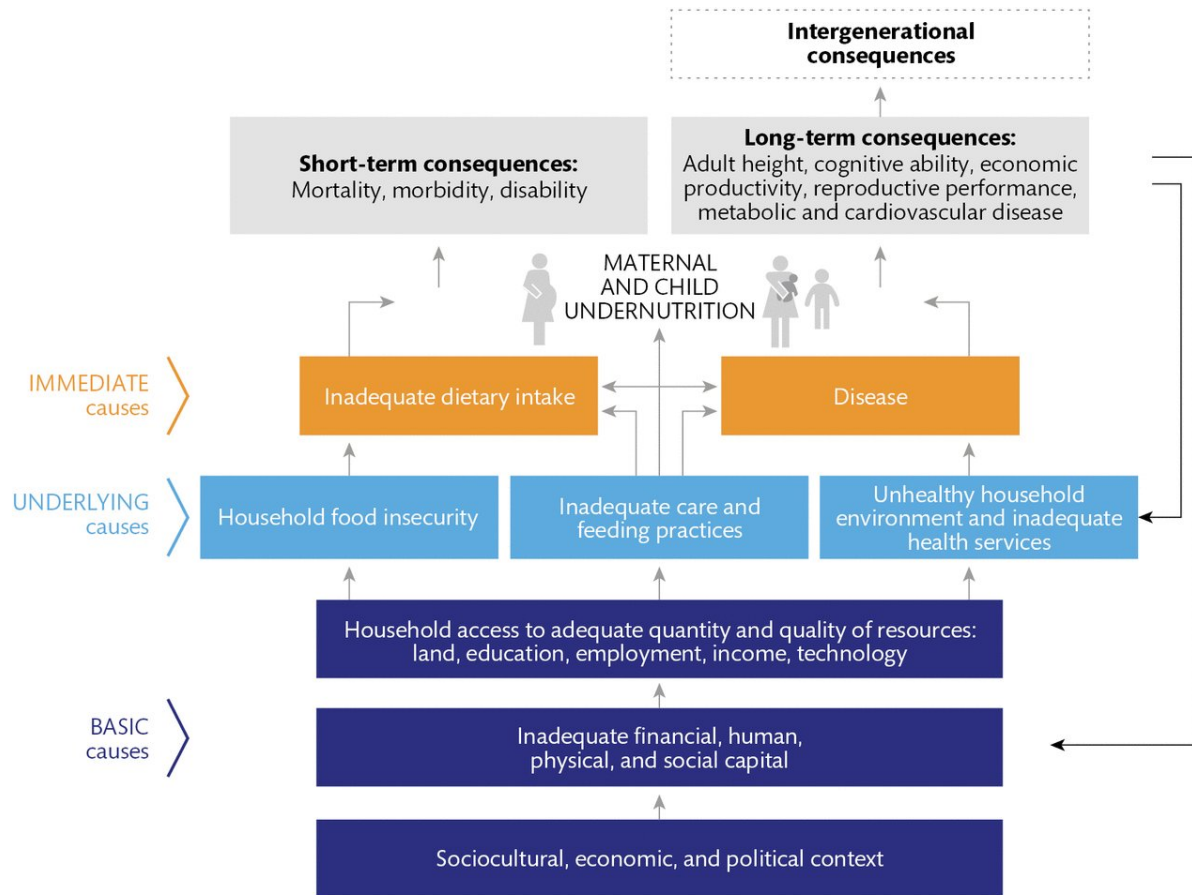
Contribution to child health in India

- **Immunisation** — UNICEF has supported **India's BCG vaccination programme** from its inception, has provided substantial aid for the **production of vaccines and sera**, and supports the Expanded Programme on Immunization, including protection of pregnant women with tetanus toxoid (Park 28e, Ch 25, pp 1057-1058).
- **Manufacturing capacity** — assisted the erection of a **penicillin plant near Pune**, donated a **DDT plant**, and provided **two plants for the manufacture of triple vaccine and of iodised salt** (Park 28e, Ch 25, p 1058).
- **Nutrition** — aided "**applied nutrition**" programmes with FAO through community development, agricultural extension, schools and health services, and supplied **equipment for modern dairy plants** in Maharashtra, Gujarat, Karnataka, Uttar Pradesh, West Bengal and Andhra Pradesh; supplies vitamin A, iodised salt and iron-folate against the deficiency diseases (Park 28e, Ch 25, p 1058).
- **Water, sanitation and hygiene** — assisted rural environmental sanitation and drinking-water programmes; played a **pivotal role in advancing and popularising the India Mark II and Mark III hand pumps**, a classical example of appropriate technology for drawing groundwater in rocky areas, and in training local men and women to maintain them (Park 28e, Ch 25, p 1058; Sunder Lal, Ch 3, p 93).
- **Disease elimination and eradication** — the **National Guinea Worm Eradication Programme** (launched 1984; close collaboration with WHO and UNICEF; zero cases since August 1996; India certified free of dracunculiasis transmission in February 2000) (Park 28e, Ch 7, p 572); founding partner of the **Global Polio Eradication Initiative, launched 1988** with WHO, Rotary International and the Centers for Disease Control (Textbook of International Health, Ch 3, p 113; Jugal Kishore — National Health Programmes of India); and **urban measles vaccination drives** initiated by UNICEF in 13 cities in 1998 and 50 more cities in 1999–2000 (Jugal Kishore).

- ▶ **Programme strategy** — with the Governments of India, Ethiopia and the USA, UNICEF convened the "Global Child Survival Call to Action: A Promise to Keep" summit (Washington DC, June 2012), where more than 80 countries pledged to reduce child mortality to **fewer than 20 child deaths per 1000 live births in every country by 2035**; India's own Call to Action Summit in February 2013 launched the **RMNCH+A strategy**, since then the heart of the National Health Mission (Park 28e, Ch 7, p 553).
- ▶ **Service delivery support** — carried out the **National Cold Chain Assessment (2008)**, which counted 66,765 operational ice-lined refrigerators and deep freezers and 199 walk-in cold and freezer rooms and quantified the shortfall for the Universal Immunization Programme; developed a **perinatal referral transport toolkit (2010)**; and, with the Ministry of Health and Family Welfare, drafted the *Mother and Baby Friendly Health Services* standards (Jugal Kishore — National Health Programmes of India).
- ▶ **Data for child health** — provided **supplemental funding for NFHS-1 and NFHS-2** (the principal funder being the United States Agency for International Development) (Jugal Kishore), and with the Ministry of Health and Family Welfare and the Population Council conducted the **Comprehensive National Nutrition Survey (CNNS), 2019** (Sunder Lal, Ch 5, pp 179-180).
- ▶ **Health education** — the *Facts for Life* messages, published by UNICEF with UNFPA, UNAIDS, WFP and the World Bank (editions 1989, 1993 and 2002), which put life-saving child-health information into every family's basic stock of knowledge (Sunder Lal, Ch 4, p 145).
- ▶ **Where the impact must be read** — India's infant mortality rate is 28, early neonatal mortality rate 15 and post-neonatal mortality rate 8 per 1000 live births (Sample Registration System, 2020), and about 62 per cent of under-five deaths occur in the neonatal period, so UNICEF-supported action has shifted decisively towards the newborn (Park 28e, Ch 9, pp 669-678).

Diagrams

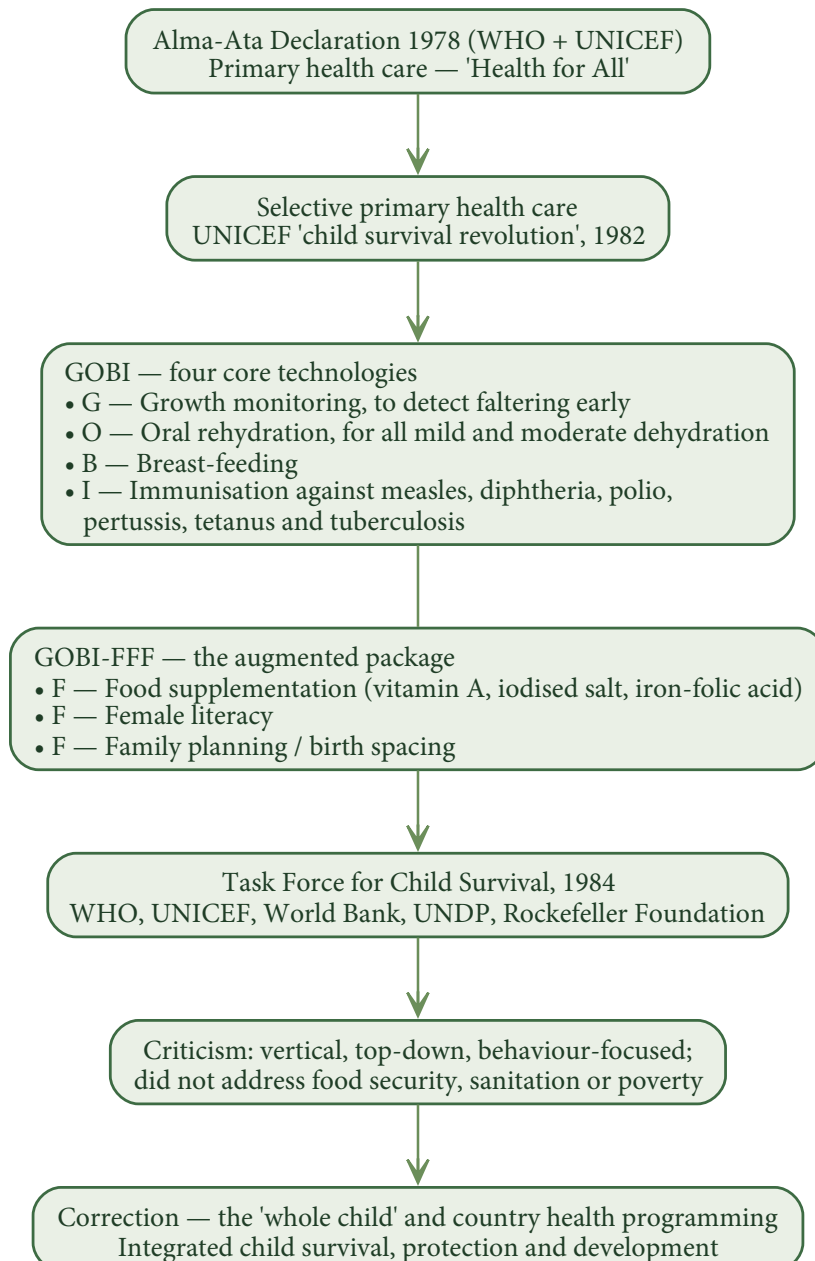
UNICEF CONCEPTUAL FRAMEWORK OF THE DETERMINANTS OF CHILD UNDERNUTRITION



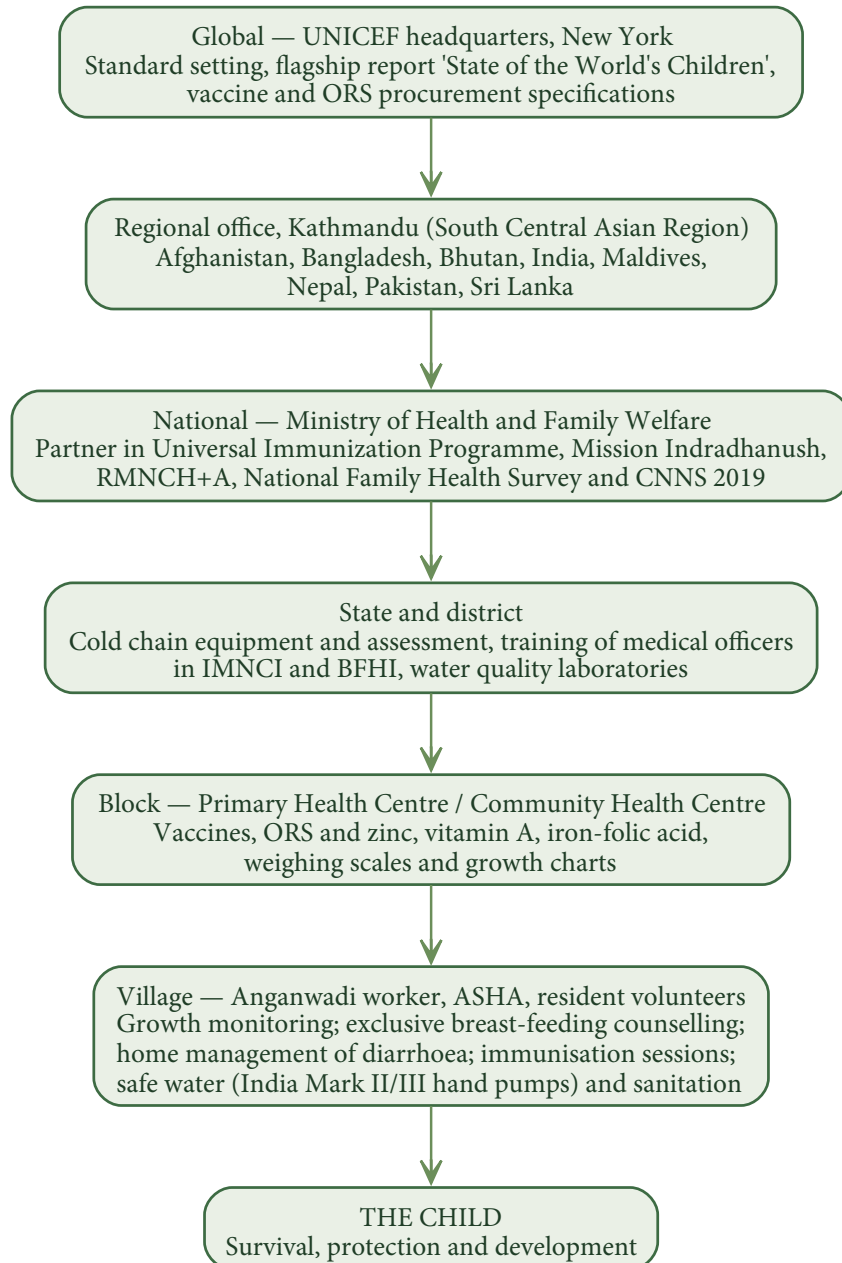
The black arrows show that the consequences of undernutrition can feed back to the underlying and basic causes of undernutrition, perpetuating the cycle of undernutrition, poverty and inequities.

Source: Adapted from UNICEF, 1990.

Fig 3.5 — 2 Conceptual framework for the causes of malnutrition in society. Repr (Oxford public health 7th ed, p.815)



UNICEF's child survival strategy, its criticism, and the separate policy shift



How UNICEF support reaches the Indian child — levels of action

High-yield exam pointers

- **Dates and places:** founded **1946**; renamed **1953** (initials kept); Universal Children's Day resolution **1954**; Urban Basic Services since **1976**; child survival revolution **1982**; Task Force for Child Survival **1984**; Universal Child Immunization declaration **October 1985**; Convention on the Rights of the Child **1989**; BFHI **1991**; reduced-osmolarity ORS the only UNICEF-procured formulation since **January 2004**; Call to Action **June 2012** — headquarters **New York**, regional office for India **Kathmandu** (South Central Asian Region), **36-nation Executive Board** (2010).
- **Mnemonic for the mandate — the four areas:** child Health, child Nutrition, family and child Welfare, child Education, plus Urban Basic Services.
- **GOBI-FFF** in full: Growth monitoring, Oral rehydration, Breast-feeding, Immunisation + Food supplementation, Female literacy, Family planning; and **India's four flagship gifts** — BCG

programme from its inception, penicillin plant near Pune, DDT plant, plants for triple vaccine and iodised salt.

- ▶ **U5MR formula (UNICEF)** = (number of deaths of children under 5 years in a given year ÷ number of live births in the same year) × 1000; UNICEF calls it the **best single indicator of social development**, ahead of gross national product per capita.
- ▶ **Zinc in diarrhoea (WHO/UNICEF)**: 10 mg/day if under 6 months, **20 mg/day if over 6 months**, for 10–14 days.
- ▶ **Current Indian denominators to quote**: infant mortality rate **28**, early neonatal **15**, post-neonatal **8** per 1000 live births (SRS 2020); **62 per cent** of under-five deaths are neonatal.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park 28e (Ch 25, p 1058) gives only the **four-component GOBI** and calls it a "child health revolution", whereas the three added Fs — food supplementation, female literacy and family planning — and the phrase "child survival revolution" (1982) come from the Textbook of International Health (Ch 3, p 116; Ch 13, pp 696–697), and some Indian textbooks expand the third F as "ferrum", i.e. iron and folic acid. Park 28e calls UNICEF "one of the specialized agencies of the United Nations", but UNICEF is strictly a subsidiary organ (a programme) created by a General Assembly resolution, unlike WHO which is a true specialised agency with its own constitution and budget (Park 28e, Ch 25, p 1055). Park 28e (Ch 9, p 653) dates Universal Children's Day to November 14, which is India's Children's Day, while the United Nations observes it on 20 November — the date on which the Declaration of the Rights of the Child was adopted in 1959. UNICEF was awarded the Nobel Peace Prize in 1965; this is standard general knowledge but was not verifiable in the retrieved corpus, so it is placed here rather than in the answer body. Park 28e still quotes the Sample Registration System 2020 round and a 2010 figure for the size of the Executive Board, so quote each with its year.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO/UNICEF estimates of national immunization coverage, Immunization Agenda 2030, WHO 2023 wasting recommendations, NFHS-5

Programme and policy updates (India)

- ▶ **Pneumococcal conjugate vaccine is now a full UIP antigen** — introduced in five states in 2017 and extended nationwide in **2021**, using PCV13 and **Pneumosil** (Serum Institute of India, whose serotype composition was built from Indian prevalence data). Early post-introduction surveillance already shows the classical serotype-replacement signal UNICEF-supported programmes must watch for: the non-vaccine serotype **15B rose from 0.5 to 11.3 per cent** (adjusted $p = 0.001$), while 19F and 19A persisted and were multidrug-resistant in **89.7 and 80 per cent** of isolates respectively [PMID 39632638].
- ▶ **Typhoid conjugate vaccine can be given with measles-rubella at the 9-month visit** — a phase IV randomised three-arm trial in **900 Indian infants aged 9–10 months** (CTRI/2022/02/040274) found anti-Vi seroconversion of **90.2 per cent** with concomitant TCV + MR versus 75.3 per cent when MR was given first, with anti-measles 80.4 per cent and anti-rubella 87.7 per cent and no

immunological interference; adverse events were fewer with co-administration (23.9 versus 32.0 and 32.7 per cent) and all but one were mild. This is the evidence base for adding TCV to the existing 9-month contact rather than creating a new one [PMID 41012665].

- **Intensified Mission Indradhanush (October 2017 – January 2018) is now formally costing** — across 40 sampled districts in Assam, Bihar, Maharashtra, Rajasthan and Uttar Pradesh it delivered an extra **2.2 million doses**, reached **165,000 zero-dose children** and averted an estimated **1,413 under-five deaths**, at **US\$82.99 per zero-dose child** and **US\$327.63 per DALY averted** — cost-effective at a threshold of $1 \times$ GDP per capita with 90 per cent probability. About **37 per cent** of children vaccinated in IMI sessions would have been reached anyway, and delivery volumes fell back to baseline once the campaign ended [PMID 38590052].

Current national data

- **Oral rehydration is still the weak link, and zinc far weaker** — ORS treatment for childhood diarrhoea rose from **50.6 per cent** (NFHS-4, 2015-16) to **60.7 per cent** (NFHS-5, 2019-21) while two-week diarrhoea prevalence fell from 9.2 to 7.3 per cent, but the gain was uneven: district-level ORS coverage ranged from **49.0 per cent in Haryana** to **83.4 per cent in Jammu and Kashmir**, and ORS use actually *fell* in **185 of 720 districts** [PMID 40810941]. In the same round only **30.5 per cent** of diarrhoeal under-fives received zinc and just **23.6 per cent** received ORS and zinc together, and first contact with a private provider carried **52 per cent lower odds** of receiving ORS than a government facility [PMID 39361593].
- **Childhood anaemia is the one nutrition indicator moving the wrong way** — prevalence in children aged 6–59 months rose from **64 per cent** (NFHS-4) to **68 per cent** (NFHS-5, 2019-21) even as stunting, wasting and underweight improved; maternal anaemia was the strongest determinant, followed by maternal BMI, and Scheduled Tribe children carried about **1.5 times** the risk irrespective of socioeconomic status [PMID 40579634].

Elimination, surveillance and disease status

- **Measles elimination has slipped globally** — first-dose measles-containing vaccine coverage was **83 per cent** in 2023, still below the pre-pandemic 86 per cent; estimated cases rose **20 per cent**, from **8,645,000 (2022)** to **10,341,000 (2023)**, estimated deaths were **107,500**, and countries with large or disruptive outbreaks increased from 36 to 57. **No WHO region has achieved and sustained elimination**; in the South-East Asia Region (India's region) MCV1 was 91 per cent and MCV2 85 per cent in 2023. Set against this, measles vaccination is credited with averting **60.3 million deaths during 2000–2023** [PMID 39541251].
- **India still carries the world's second-largest zero-dose burden** — of **14.5 million children worldwide who received no DTP1 dose in 2023**, **1,592,000 were in India**, second only to Nigeria (2,134,000), and ten countries accounted for 59 per cent of the global total; **22.2 million** children missed MCV1. Halving the number of zero-dose children is a core Immunization Agenda 2030 goal, pursued through the **Big Catch-Up** strategy for children missed during 2019–2022 [PMID 39480752].

Guideline and legislative updates

- **WHO's 2023 wasting recommendations introduced prioritisation criteria for moderate wasting** — specially formulated foods rather than counselling alone for children **under 2 years**, or **weight-for-age Z-score below -3 SD**, or **MUAC 115–119 mm**. Operationally the triage may

not discriminate: in 35,685 moderately wasted children **95 per cent met at least one priority criterion**, and recovery and anthropometric trajectories were similar in higher- and lower-priority groups once both received the food [PMID 41353631].

Global and WHO developments

- ▶ **Routine childhood vaccination has averted an estimated 154 million child deaths since EPI began in 1974**, but the zero-dose count fell to 14.7 million in 2019, peaked at **18.6 million in 2021** and stood at **15.7 million (95% UI 14.6–17.0) in 2023**, more than half of them in eight countries including India. Forecasts to 2030 suggest **only DTP3 would reach the Immunization Agenda 2030 target of 90 per cent global coverage**, and only under an optimistic scenario [PMID 40578370].
- ▶ **Breast-feeding is being re-costed as a non-communicable disease intervention** — modelling across **132 low- and middle-income countries**, using WHO-UNICEF exclusive breastfeeding coverage as the baseline, estimated that raising exclusive breastfeeding to **90 per cent** would delay **72,300 NCD deaths a year** (1.04 million years of life gained) and avert about **10 million type 2 diabetes** and 3.8 million hypertension cases across two generations, with ischaemic heart disease and stroke accounting for most delayed deaths — an argument for BFHI beyond infant survival [PMID 41109255].

SOURCE & COVERAGE

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Vaccination requirements for international travel

Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — *Emporiatics* is the science of the health of travellers (Park 28e, Ch 3, p 152). A traveller leaving India for Africa faces three separable obligations: a **legally enforceable vaccination requirement** under the **International Health Regulations (IHR) 2005** — for which the **yellow fever certificate is the only certificate required for international travel** (Park 28e, Ch 3, p 152); a set of **risk-based recommended vaccines and chemoprophylaxis**; and **non-vaccine personal health advice**. The World Health Organization (WHO) publishes these annually as the booklet *International Travel and Health* — *Vaccination Requirements and Health Advice* (Park 28e, Ch 3, p 152).

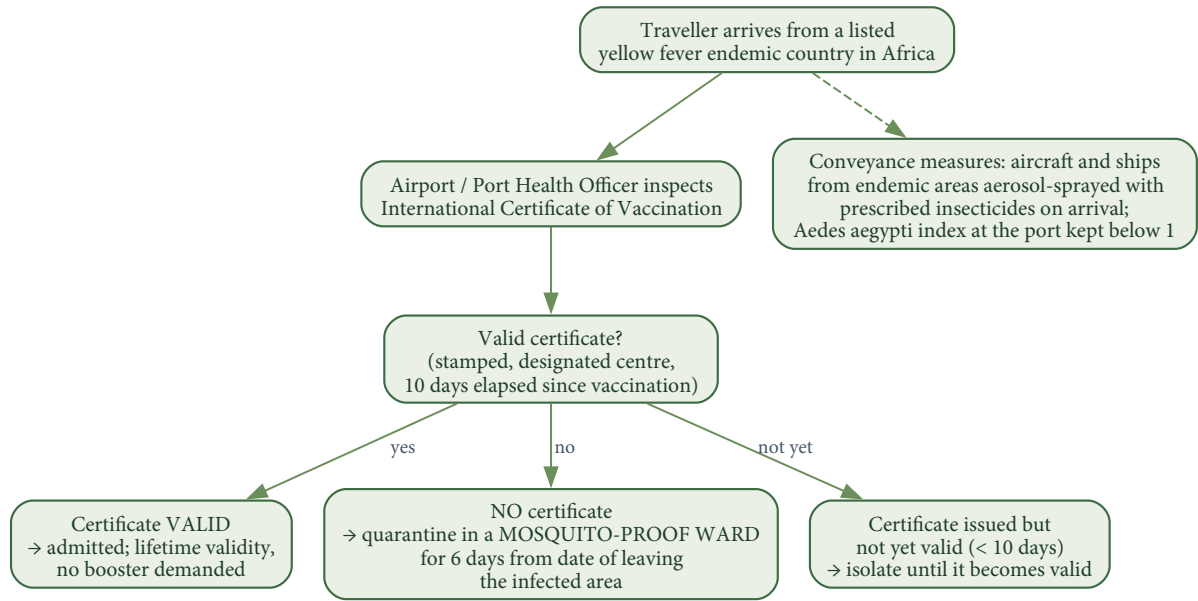
1. Why the India-to-Africa traveller needs a pre-travel consultation

- ▶ **Stress of jet travel** — crowding, long waiting, disrupted eating habit, change of climate and time zone, producing nausea, indigestion, extreme fatigue and insomnia (Park 28e, Ch 3, p 152).
- ▶ **Diseases NOT covered by the IHR** — malaria, giardiasis, dengue, influenza, filariasis, sexually transmitted disease (STD) and acquired immunodeficiency syndrome (AIDS), intestinal parasites, typhoid and paratyphoid fever, viral hepatitis; many declare themselves only after return. Poor food-handler hygiene, poor water quality and improper waste disposal add to the risk, and the traveller is **separated from familiar and accessible medical care** (Park 28e, Ch 3, p 152).
- ▶ **The African burden of yellow fever** — of 47 countries at risk in Africa and Latin America (combined population over 900 million), an estimated **508 million people in 32 African countries** are at risk (Park 28e, Ch 5, p 347).
- ▶ The traveller carries a **personal responsibility** to recognise these risks, which are minimised by immunisation, chemoprophylaxis or chemotherapy (Park 28e, Ch 3, p 152).

2. Mandatory (certificate-required) vaccination — yellow fever

Item	Requirement
Who must be vaccinated	All travellers, including infants, exposed to the risk of yellow fever or passing through endemic zones, before entry into a yellow fever "receptive" area (Park 28e, Ch 5, p 348)
Vaccine and dose	Live attenuated 17D vaccine, 0.5 ml subcutaneously, single dose irrespective of age; reconstituted with the physiological saline diluent supplied, kept on ice away from sunlight, discarded if unused within half an hour (Park 28e, Ch 5, p 348)
Onset and duration of immunity	Immunity begins to appear on the 7th day and lasts possibly for life (Park 28e, Ch 5, p 348)
Where given	At a designated centre, the certificate stamped by the Ministry of Health, Government of India (Park 28e, Ch 5, pp 348-349)
Certificate validity	Valid from 10 days after the date of vaccination. Under the World Health Assembly amendment to the IHR (2005) adopted in May 2014 and in force from 11 July 2016, validity changed from 10 years to the lifetime of the person vaccinated; no revaccination or booster may be demanded as a condition of entry, regardless of when the certificate was issued. India applies this lifetime validity automatically (Park 28e, Ch 5, p 349)
Contraindications	(a) Children under 9 months for routine immunisation (under 6 months during an epidemic); (b) pregnancy — except during an outbreak when risk is high; (c) severe allergy to egg protein; (d) severe immunodeficiency from symptomatic human immunodeficiency virus (HIV)/AIDS, other causes, or thymus disorder (Park 28e, Ch 5, p 348)
Adverse events	Mild post-vaccinal reactions (myalgia, headache, low-grade fever) in 2-5 per cent, appearing 5-10 days after vaccination; anaphylaxis very rare, mainly in the egg-allergic. Risk of death from yellow fever far exceeds the risk from the vaccine (Park 28e, Ch 5, p 348)
Spacing	Cholera and yellow fever vaccines interfere with each other if given together or within 3 weeks — give them 3 weeks or more apart (Park 28e, Ch 5, p 348)
Penalty for no certificate	Traveller quarantined in a mosquito-proof ward for 6 days from the date of leaving the infected area; if he arrives before his certificate becomes valid, he is isolated until it becomes valid (Park 28e, Ch 5, p 348)

African countries listed by the Government of India as yellow fever endemic (Park 28e, Ch 5, p 349): Angola, Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Chad, Congo, Côte d'Ivoire, Democratic Republic of the Congo, Equatorial Guinea, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Liberia, Mali, Mauritania, Niger, Nigeria, Rwanda, Senegal, Sierra Leone, Sudan, South Sudan, Togo and Uganda. When a case of yellow fever is reported from any country, India regards that country as a country with risk of transmission and adds it to this list.

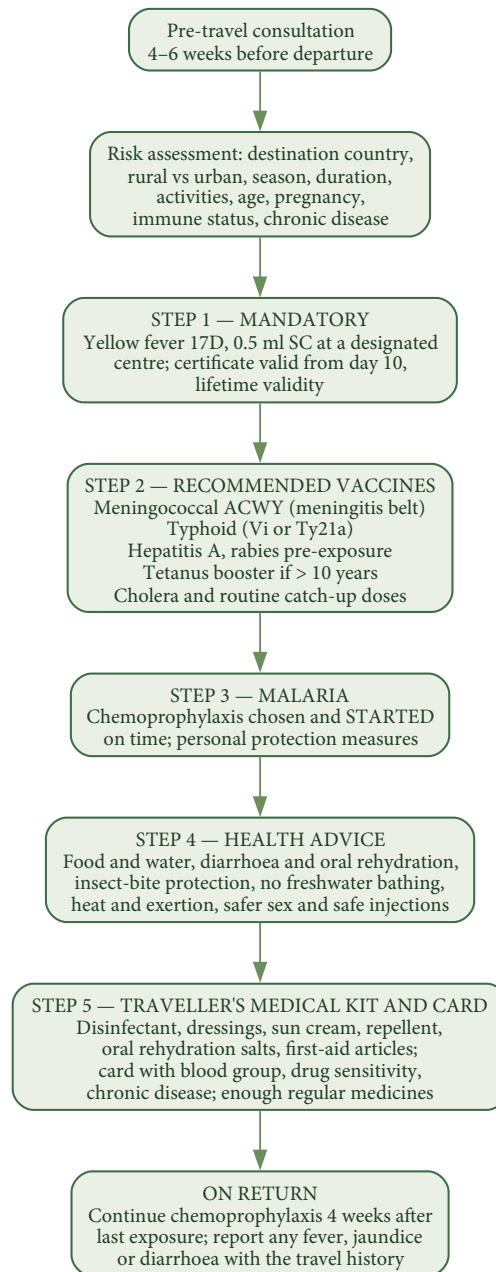


Yellow fever certificate check on arrival — the statutory pathway at an Indian point of entry

3. Recommended (risk-based) vaccinations before travel to Africa

Vaccine	Schedule, route and timing	Remarks for the Africa-bound traveller
Meningococcal (A, C, W135, Y)	Conjugate quadrivalent: 1 dose at ≥ 2 years; 9-23 months, 2 doses 12 weeks apart. MenA conjugate 9-18 months (5 μg), 1 dose. Polysaccharide (bivalent A,C / trivalent A,C,W135 / quadrivalent), 50 μg of each polysaccharide, single dose subcutaneously; store at 2-8°C (Park 28e, Ch 3, p 149; Ch 5, p 197)	Essential for the African meningitis belt — sub-Saharan Africa from Senegal in the west to Ethiopia in the east, ~400 million people, where serogroup A predominates. Conjugate vaccines are preferred over polysaccharide for herd protection and better immunogenicity, particularly under 2 years (Park 28e, Ch 5, pp 196-197)
Typhoid — Vi polysaccharide	Single dose, subcutaneous or intramuscular, ~25 μg antigen; protection after 7 days; licensed from age 2 years (no adequate response below 2 years) (Park 28e, Ch 3, p 146; Ch 5, pp 296-297)	May be co-administered with yellow fever and hepatitis A vaccines and with routine programme vaccines (Park 28e, Ch 5, p 297)
Typhoid — Ty21a	Oral live attenuated, enteric-coated capsules on days 1, 3 and 5; protective immunity 7 days after the last dose; store at 2-8°C (Park 28e, Ch 3, p 146; Ch 5, p 297)	Efficacy not documented below 3 years of age
Cholera (oral)	2 doses orally, 7 days to 6 weeks apart (Dukoral WC-rBS; Shanchol, mORCVAX, Euvichol from 1 year, 2 doses 14 days apart, revaccination after 2 years) (Park 28e, Ch 3, pp 146, 149; Ch 5, p 292)	No cholera certificate is required for international travel; give ≥ 3 weeks apart from yellow fever vaccine
Hepatitis A	Vaccine, at least 1 dose (Park 28e, Ch 3, p 149)	Food- and water-borne risk throughout Africa

Vaccine	Schedule, route and timing	Remarks for the Africa-bound traveller
Rabies — pre-exposure prophylaxis	Days 0, 7 and 21 or 28; 1.0 or 0.5 ml intramuscularly (vaccine-dependent) or 0.1 ml intradermally, one site per day (Park 28e, Ch 5, p 345)	No routine boosters for those travelling to or living in high-risk areas who have completed a primary series; if later exposed, only 1 dose on days 0 and 3 and no rabies immunoglobulin (Park 28e, Ch 5, pp 344-345)
Tetanus toxoid	Booster dose if 10 years or more have elapsed since the last injection of a complete course or booster (Park 28e, Ch 3, p 152)	A wise precaution for every traveller
Influenza (inactivated)	Two adequately spaced doses of 1.0 ml for primary immunisation (one dose if an epidemic threatens); immunity 3-6 months (Park 28e, Ch 3, p 146)	Seasonal vaccine, revaccinate annually
Plague	2 doses subcutaneously or intramuscularly, 7-14 days apart ; immunity starts 5-7 days after inoculation and lasts about 6 months (Park 28e, Ch 3, p 146)	Only for travel to a recognised plague focus
Routine / catch-up	Complete the national immunisation schedule appropriate to age before departure	Measles-rubella, polio, hepatitis B and diphtheria-tetanus status must be reviewed and completed at the pre-travel visit



Pre-travel consultation for a traveller leaving India for Africa — the sequence to follow

4. Malaria chemoprophylaxis

Chemoprophylaxis is recommended for **travellers from non-endemic areas** and as a short-term measure for soldiers, police and labour forces serving in highly endemic areas; it must be **complemented by personal protection and vector control**. With drug resistance it has become unreliable, but experts hold that it reduces the risk of fatal disease. "Short-term" means **less than 6 weeks** (Park 28e, Ch 5, p 331).

Drug	Adult prophylactic dose	When to start	Cautions
Chloroquine	300 mg base = 2 tablets of 150 mg once a week, on the same day each week	1 week before arrival	Nausea, vomiting, blurring of vision, headache; retinal damage only with large cumulative doses; never on an empty stomach (Park 28e, Ch 5, pp 331-332)
Proguanil	200 mg = 2 tablets of 100 mg once a day	Day before arrival	Recommended only in association with chloroquine (Park 28e, Ch 5, p 332)
Mefloquine	250 mg = 1 tablet once a week, same day each week	2-3 weeks before departure, to raise pre-travel blood levels and to unmask side-effects in time to switch	Contraindicated with a history of convulsions, neuropsychiatric problems or cardiac conditions; does not accumulate on long-term intake (Park 28e, Ch 5, pp 331-332)
Doxycycline	100 mg = 1 capsule once a day	The day before arrival in the risk area	Contraindicated in pregnancy and in children under 8 years ; long-term data limited (Park 28e, Ch 5, pp 331-332)

- Children's doses are calculated on body weight, never by age (Park 28e, Ch 5, p 331).
- All prophylactic drugs must be taken **with unfailing regularity** throughout the stay and **continued for 4 weeks after the last possible exposure**, since parasites may still emerge from the liver (Park 28e, Ch 5, p 331). Park's travel chapter states the same rule as **4-6 weeks after leaving the malarious area**, prophylaxis beginning **at the latest on the day of arrival** (Park 28e, Ch 3, p 152).
- **Personal protection against the vector** — insecticide-treated bed-nets, repellents, protective clothing, screening of houses, mosquito coils and fumigation mats (Park 28e, Ch 5, pp 332-333, 348).

5. Non-vaccine health advice to the traveller

Risk	Advice to be given
Traveller's diarrhoea	Affects an estimated 20-50 per cent of all travellers; contaminated food and drink are the commonest source. Appearance of food is no guide to its safety . Treat unpasteurised milk, non-bottled drinks and uncooked food (except fruit and vegetables that can be peeled or shelled) as unsafe; eat food thoroughly and freshly cooked ; drink boiled or bottled water ; know the value of oral rehydration fluid containing salt and glucose (Park 28e, Ch 3, p 152)
Bathing and recreational water	Avoid bathing in polluted water — it causes ear, eye and skin infections (Park 28e, Ch 3, p 152)
Heat and exertion	Excessive heat and humidity or over-exertion cause exhaustion from loss of water and salt (Park 28e, Ch 3, p 152)
Insect bites	Take the full package of measures for prevention of insect bites (Park 28e, Ch 3, p 152; Ch 5, pp 332-333)

Risk	Advice to be given
STD, HIV and hepatitis B	Preventive measures are the same whether abroad or at home — avoid casual sex or limit it to a single faithful, uninfected partner; use a condom ; avoid injecting drugs , and if an injection is essential, ensure the needle and syringe are sterile (Park 28e, Ch 3, p 152)
Traveller's medical kit	A disinfectant and a dressing that can be applied easily; sun cream, mosquito repellent, oral rehydration salts and first-aid articles are basic necessities (Park 28e, Ch 3, p 152)
The traveller with chronic disease	Patients with diabetes, cardiac disease and similar conditions should carry enough medicine to avoid a break in medication , and carry a card recording blood group, drug sensitivities and the chronic illness (Park 28e, Ch 3, p 152)
Reference document	Direct the traveller to the annual WHO booklet <i>International Travel and Health — Vaccination Requirements and Health Advice</i> , which gives the main health risks by region and the precautions against them (Park 28e, Ch 3, p 152)

6. The regulatory framework behind the advice

- ▶ Measures restricting the spread of yellow fever are specified in the WHO **International Health Regulations** and implemented by the Government of India through **stringent aerial and maritime traffic regulations** (Park 28e, Ch 5, p 348).
- ▶ **Conveyances**: aircraft and ships from endemic areas are **aerosol-sprayed with prescribed insecticides on arrival**; airports and seaports are kept free of vector breeding around their perimeters, the *Aedes aegypti* index kept below 1 (Park 28e, Ch 5, p 348).
- ▶ Under the IHR (2005) — the first legally binding agreement on disease notification — states must notify WHO of a **public health emergency of international concern (PHEIC)** through their **National IHR Focal Point within 24 hours**, and must build **core capacities** to detect, assess, notify, report and contain such events (Textbook of International Health, Ch 5, p 234; Oxford Textbook of Public Health 7e, Ch 75, pp 2859-2866; Ch 105, pp 3996-3997).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Outline map of Africa showing the two risk zones.** Labels to write: *yellow fever endemic zone — West, Central and East Africa (vaccination recommended); African meningitis belt — Senegal in the west to Ethiopia in the east, about 400 million people; southern Africa — yellow fever vaccination generally not recommended; malaria risk area;* and mark the **transit airports** at which a certificate may be demanded.
- ▶ **Diagram 2 — The International Certificate of Vaccination or Prophylaxis (yellow fever).** Labels: *name and passport number of traveller; vaccine — yellow fever 17D; date of vaccination; signature of the supervising clinician; official stamp of the designated vaccination centre / Ministry of Health, Government of India; "valid from 10 days after the date of vaccination"; "valid for the life of the person vaccinated (IHR amendment, in force 11 July 2016)".*

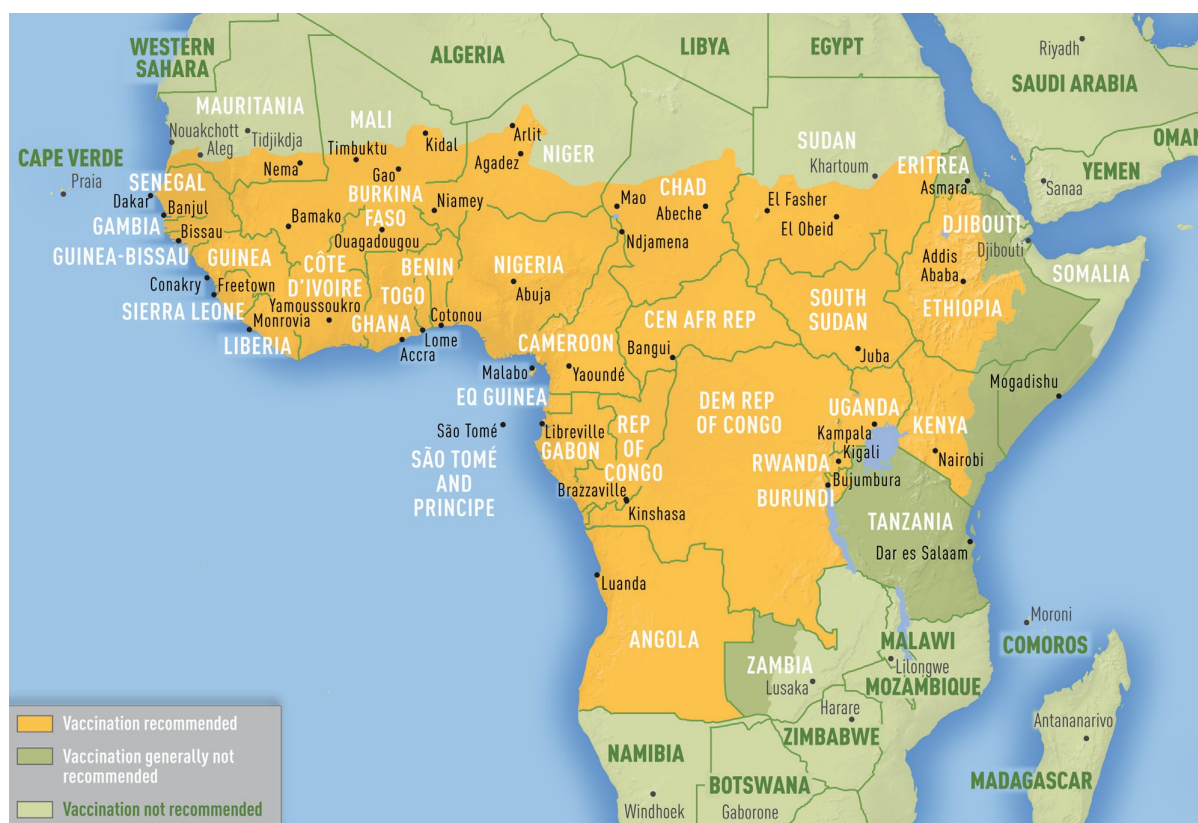


Fig 71.5 — Yellow fever vaccine recommendations in Africa. (From Centers for Disease Control and Prevention.) (Plotkin's Vaccines, 7ed, p.2018)

High-yield exam pointers

- ▶ Empiricists = the science of the health of travellers; the yellow fever certificate is the **ONLY** certificate required for international travel.
- ▶ Yellow fever numbers: 17D vaccine, 0.5 ml subcutaneous, single dose at any age; immunity from day 7; certificate valid from day 10; lifetime validity since 11 July 2016 (World Health Assembly amendment of May 2014); no certificate → 6 days quarantine in a mosquito-proof ward; not below 9 months (6 months in an epidemic); cholera and yellow fever vaccines 3 weeks apart.
- ▶ Malaria prophylaxis start times (the commonest question): doxycycline the day before, chloroquine 1 week before, mefloquine 2-3 weeks before; all continued 4 weeks after the last exposure; short-term prophylaxis = under 6 weeks.
- ▶ Meningitis belt = Senegal to Ethiopia, serogroup A; WHO epidemic threshold >100 cases per 100,000 per year; endemicity bands >10 (high), 2-10 (moderate), <2 (low) per 100,000 per year; conjugate preferred over polysaccharide.
- ▶ Typhoid: Vi polysaccharide — one dose, protection after 7 days, age ≥2 years; Ty21a — oral, days 1, 3 and 5, immunity 7 days after the last dose.
- ▶ Traveller's diarrhoea affects 20-50 per cent of travellers; "Be careful what you eat" — peel it, boil it, cook it or forget it; oral rehydration salts in every medical kit.
- ▶ IHR (2005): notify WHO of a public health emergency of international concern within 24 hours through the National IHR Focal Point.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: this answer states requirements as recorded in Park's 28th edition and the retrieved corpus; actual entry requirements are country-specific and change, so a candidate advising a real traveller must check the current WHO *International Travel and Health* list and the destination embassy's rules before departure, and note that Park's travel chapter and its malaria chapter word the post-travel continuation of chemoprophylaxis slightly differently (4-6 weeks after leaving the area, versus 4 weeks after the last possible exposure).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, WHO SAGE, World Health Assembly, CDSO, World Malaria Report 2024

Programme and policy updates (India)

- ▶ **The vaccine for the African meningitis belt is now an Indian product that also covers serogroup X.** MenFive (also called NmCV-5 or Men5CV), a pentavalent A, C, W, Y, X conjugate developed by PATH with the Serum Institute of India, received full Indian marketing authorisation in September 2022 for ages 18-85 years and WHO prequalification in July 2023 for ages 1-85 years, amended in March 2024 to include children from 9 months. It is the first and only licensed vaccine protecting against serogroup X. The Indian phase 2/3 trial in 1,640 adults aged 18-85 years across nine hospitals in seven cities met immunological non-inferiority to quadrivalent MenACWY-D for all four shared serogroups and showed lot-to-lot consistency. WHO SAGE recommended in September 2023 that every belt country give a single dose at 9-18 months with catch-up at 1-19 or 2-19 years, and the updated WHO position paper on multivalent meningococcal conjugate vaccines followed in January 2024. [PMID 40968077] [PMID 39521012] [CDSO 2022-Sep] [WHO 2023-Jul] [WHO SAGE 2023-Sep] [WHO 2024-Jan]
- ▶ **Typhoid conjugate vaccine has displaced plain Vi polysaccharide as the Indian standard.** Typbar-TCV, TYPHIBEV, PedaTyph and Zyvax TCV are all licensed Indian conjugates; unlike Vi polysaccharide they work below 2 years of age and give a T-cell dependent response with immune memory. The Serum Institute has carried a **bivalent** conjugate through phase 1 — Vi linked to tetanus toxoid plus *Salmonella* Paratyphi A O-specific polysaccharide linked to diphtheria toxoid — which would be the first vaccine to cover paratyphoid A, the serotype now causing an increasing share of Asian enteric fever. [PMID 41847975]

Current national data

- ▶ **World Malaria Report 2024** places India at more than half the malaria burden of the WHO South-East Asia Region, and the national count has risen rather than fallen — 161,753 cases in 2021, 176,522 in 2022 and 255,500 in 2024, about 58 per cent above the 2021 figure. This cuts two ways for the traveller: fever after return from Africa must be investigated together with the travel history, and India itself remains receptive enough for an imported falciparum infection to seed local transmission, which is the public-health reason the post-travel continuation of chemoprophylaxis is insisted on. [PMID 41176154]

Elimination, surveillance and disease status

- **The meningitis belt has changed serogroup, and the new vaccine has now been used at scale.** MenAfriVac reached over 350 million people in 24 belt countries and cut serogroup A disease by more than 99 per cent, the last confirmed belt case being in 2017, so current epidemics are serogroups C, W and X rather than A. MenFive was deployed for the first time in March 2024 in Nigeria against a serogroup C outbreak, after which Niger procured more than 800,000 doses for C and W outbreaks. Enhanced surveillance across the two 2024 campaigns covered 4,881,027 vaccinees, with 1,109 adverse events following immunisation reported (22.7 per 100,000) of which 11 (0.99 per cent) were serious including one death, and no new safety signal. A candidate should therefore qualify the classical teaching that serogroup A predominates in the belt. [PMID 40968077] [PMID 41791242]
- **Measles is the vaccine-preventable disease travellers most reliably carry home.** GeoSentinel recorded 53 imported measles cases in travellers across 15 countries between May 2019 and June 2025; 74 per cent were aged 21 years or more, 61 per cent were hospitalised, and 79 per cent had no or unknown vaccination history. This is the evidence for auditing measles-rubella status at every pre-travel visit in adults, not only in children. [PMID 40738382]

Guideline and legislative updates

- **"Immunity possibly for life" from a single yellow fever dose is being re-examined at the margins.** In a Brazilian cohort given 17DD vaccine, 78.2 per cent of 4,014 vaccinees remained seropositive at 4 years, but the figure was only 67.3 per cent among those vaccinated at 9-23 months against 94.8 per cent in adults aged 36-50 years, with mean neutralising titres falling steadily after the 30-45 day peak. WHO and PAHO separately accept a fractional 0.1 ml dose as an exceptional measure during vaccine shortage, and the booster debate continues for adults over 60, people living with HIV and other immunocompromised groups. The **legal position is unchanged** — the certificate is valid for the life of the person vaccinated and no booster may be demanded for entry — but the clinical advice for an infant or an immunocompromised traveller is not the same question as the legal one. [PMID 41830133] [PMID 42447166]
- **Oral cholera vaccine has been on a single-dose schedule since 2022**, one dose having replaced the standard two because global supply could not meet demand. Modelled single-dose effectiveness was 54.9 per cent (95% CrI 18.9-84.9) in Cameroon and 75.3 per cent (54.0-86.4) in Haiti, exceeding each country's herd-immunity threshold for much of the outbreak but dipping below it at times, which is why the single-dose strategy is framed as outbreak control rather than durable protection. [PMID 42191191]

Global and WHO developments

- **The IHR (2005) were amended by the World Health Assembly on 1 June 2024**, the first substantive revision of the framework described above. More than 300 proposals yielded a new **"pandemic emergency"** tier sitting above the ordinary PHEIC on the same continuum, equity written in as a core principle, a new **Article 54 bis** States Parties Implementation Committee, and a Coordinating Financial Mechanism to fund core-capacity building in developing countries. The Article 43 flexibility that allows a state to apply additional health measures beyond WHO's temporary recommendations — provided the measures and their rationale are reported to WHO — was deliberately retained. [PMID 41361765] [WHA 2024-Jun]

- ▶ **The WHO Pandemic Agreement was adopted in May 2025** after 13 formal negotiating rounds. It is only the **second treaty adopted under Article 19 of the WHO Constitution**, after the **Framework Convention on Tobacco Control** (adopted 2003, in force 2005), so it joins the FCTC and the IHR as a third legally binding instrument in global health law — the IHR being adopted as regulations under Article 21, not as a treaty. Its centrepiece is the Article 12 Pathogen Access and Benefit-Sharing system, under which manufacturers set aside 20 per cent of pandemic-related products for WHO-led distribution — 10 per cent donated and 10 per cent at affordable prices. The operational annex goes to the 2026 Assembly, and the treaty enters into force once 60 states ratify it. [PMID 41748273] [WHA 2025-May] [WHO FCTC 2003]
- ▶ **The list of PHEIC declarations a candidate should be able to recite has grown.** Nine have been declared since 2007; poliovirus remains under a standing declaration, mpox was declared in 2024, and on 17 May 2026 WHO declared Ebola disease due to **Bundibugyo virus** in Ituri province, Democratic Republic of the Congo, a PHEIC. By 17 June 2026 there were 896 confirmed cases and 232 deaths, a case fatality of about 26 per cent, with imported cases and secondary transmission reported in Uganda. This is the live illustration of why notification through the National IHR Focal Point within 24 hours, and the pre-travel risk assessment for an African itinerary, are not academic. [PMID 42383297]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3, pp 146, 149, 152 (health advice to travellers, emporiatrics, travel vaccines and schedules, tetanus booster, medical kit, WHO *International Travel and Health*); Ch 5, pp 196-197 (African meningitis belt and meningococcal vaccines), 292, 296-297 (oral cholera, Vi and Ty21a typhoid vaccines), 331-333 (malaria chemoprophylaxis and personal protection), 344-345 (rabies pre-exposure prophylaxis), 347-349 (yellow fever in Africa, 17D vaccine, international certificate and lifetime validity, 6-day quarantine, conveyance disinsection, Government of India list of endemic African countries). Plotkin's Vaccines, 7ed, p 2018 (map: yellow fever vaccine recommendations in Africa). Textbook of International Health, Ch 5, p 234; Oxford Textbook of Public Health 7e, Ch 75, pp 2859-2866, Ch 105, pp 3996-3997 (International Health Regulations 2005, PHEIC, National IHR Focal Point, core capacities). **Coverage note:** the retrieved corpus did not carry the Government of India oral polio vaccine requirement for travel to polio-affected countries, nor country-by-country meningococcal certificate requirements, so neither is stated here.

Recent-advances sources: 40968077 — Development of MenFive®, an affordable pentavalent meningococcal conjugate vaccine (ACYWX) for Africa and beyond, *Human Vaccines & Immunotherapeutics* 2025, doi:[10.1080/21645515.2025.2559506](https://doi.org/10.1080/21645515.2025.2559506); 39521012 — Safety and immunogenicity of a pentavalent meningococcal conjugate vaccine versus a quadrivalent meningococcal conjugate vaccine in adults in India: an observer-blind, randomised, active-controlled, phase 2/3 study, *Lancet Infectious Diseases* 2024, doi:[10.1016/S1473-3099\(24\)00576-0](https://doi.org/10.1016/S1473-3099(24)00576-0) (abstract-only); 41847975 — Development and validation of immunological assays for preclinical evaluation of a novel bivalent typhoid conjugate vaccine, *Human Vaccines & Immunotherapeutics* 2026, doi:[10.1080/21645515.2026.2644731](https://doi.org/10.1080/21645515.2026.2644731); 41176154 — India should consider sero-epidemiology as an additional tool to bolster malaria elimination, *International Journal of Infectious Diseases* 2025, doi:[10.1016/j.ijid.2025.108169](https://doi.org/10.1016/j.ijid.2025.108169) (abstract-only); 41791242 — Enhanced passive safety surveillance during the first use of the meningococcal vaccine A/C/W/Y/X conjugate for outbreak response in Niger and Nigeria, *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128390](https://doi.org/10.1016/j.vaccine.2026.128390) (abstract-only); 40738382 — Measles importations by international travelers, GeoSentinel 2019-2025, *Travel Medicine and Infectious Disease* 2025, doi:[10.1016/j.tmaid.2025.102885](https://doi.org/10.1016/j.tmaid.2025.102885) (abstract-only); 41830133 — A cohort study of the duration of immunity to yellow fever after 4 years of primary vaccination in children and adults, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2026, doi:[10.1093/trstmh/trag026](https://doi.org/10.1093/trstmh/trag026); 42447166 — Immunoprevention and immunomodulation of yellow fever: a scoping review of global and Latin American evidence, *PLoS One* 2026, doi:[10.1371/journal.pone.0352755](https://doi.org/10.1371/journal.pone.0352755); 42191191 — Estimating time-varying cholera transmission and oral cholera vaccine effectiveness in Haiti and Cameroon, 2021-2023, *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-113520](https://doi.org/10.1136/bmjopen-2025-113520); 41361765 — When one size does not fit all: New Zealand perspectives on the International Health Regulations during the COVID-19 pandemic, *Globalization and Health* 2025, doi:[10.1186/s12992-025-01160-1](https://doi.org/10.1186/s12992-025-01160-1); 41748273 — The WHO pandemic agreement: securing Africa's leadership in a fragmenting global order, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-020634](https://doi.org/10.1136/bmjgh-2025-020634); 42383297 — Editorial: the WHO identifies Ebola disease due to Bundibugyo virus as a public health emergency of

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Millennium Development Goals for health

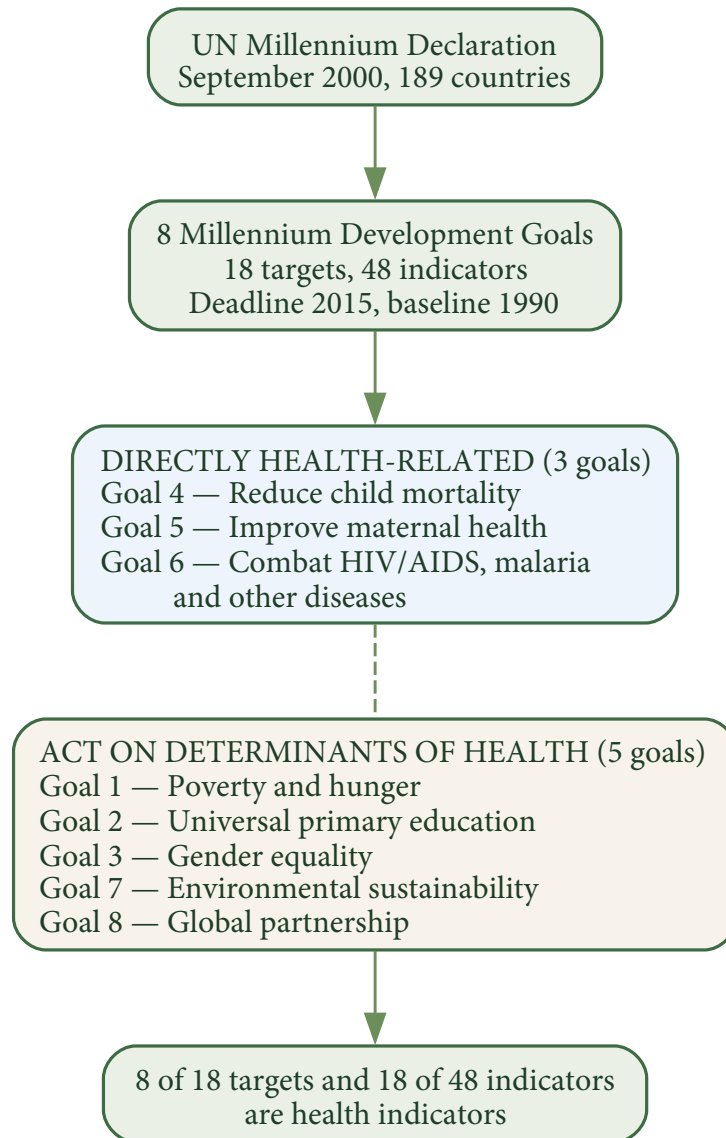
Asked in: Paper I 29-Jul-2021 — RGUHS MD Community Medicine

Definition — The Millennium Development Goals (MDGs) are the eight time-bound, measurable development goals derived from the United Nations Millennium Declaration of September 2000, to be achieved by the year 2015 against a 1990 baseline. They placed health at the heart of development: three of the eight goals are **directly health-related** (goals 4, 5 and 6), while all the remaining goals act on the underlying determinants of health (Park 28e, Ch 1, p 23). Of the framework's 18 targets, 8 are health targets, and of its 48 indicators of progress, 18 are health indicators (Park 28e, Ch 10, p 721).

Origin and structure of the framework

- ▶ In **September 2000**, representatives of **189 countries** met at the Millennium Summit in New York and adopted the **United Nations Millennium Declaration** (Park 28e, Ch 1, p 23).
- ▶ The leaders made commitments in **seven areas** — peace, security and disarmament; development and poverty eradication; protecting our common environment; human rights, democracy and good governance; protecting the vulnerable; meeting the special needs of Africa; and strengthening the United Nations.
- ▶ A "**Road Map**" set goals and targets to be reached by **2015** in each of these areas; the goals falling under *development and poverty eradication* became known as the Millennium Development Goals.
- ▶ In the following year all major United Nations agencies plus the World Bank and the International Monetary Fund endorsed the eight interdependent goals with measurable targets (Textbook of International Health, Ch 4, p 210).
- ▶ **Baseline year is 1990** for most indicators; the deadline was 31 December 2015 (Textbook of International Health, Ch 4, p 213).
- ▶ The framework was **revised at the 2005 World Summit, effective 15 January 2008**, adding MDG 5, **Target 5B — "Achieve, by 2015, universal access to reproductive health"** (Sunder Lal, Ch 17, p 767).

- **Countdown to 2015** was formed in 2005 by scientists, policymakers and institutions to track progress towards MDG 4 (Sunder Lal, Ch 17, p 767).



Architecture of the Millennium Development Goals — from Declaration to indicators

The eight goals and their principal targets

Goal	Principal target	Health relevance
1. Eradicate extreme poverty and hunger	Halve the proportion of people living on less than US\$ 1 per day and who suffer from hunger	Indirect — undernutrition, food security
2. Achieve universal primary education	Ensure that all boys and girls complete a full course of primary schooling	Indirect — literacy, health-seeking behaviour
3. Promote gender equality and empower women	Eliminate gender disparity in primary and secondary education	Indirect — women's status, reproductive health

Goal	Principal target	Health relevance
4. Reduce child mortality	Reduce by two-thirds, between 1990 and 2015, the under-five mortality rate	Direct
5. Improve maternal health	Reduce by three-quarters, between 1990 and 2015, the maternal mortality ratio; Target 5B — universal access to reproductive health by 2015	Direct
6. Combat HIV/AIDS, malaria and other diseases	Halt and begin to reverse the spread and incidence of these diseases	Direct
7. Ensure environmental sustainability	Integrate sustainable development into country policies; reverse loss of environmental resources; halve the proportion of people without sustainable access to safe drinking water and sanitation; improve the lives of at least 100 million slum-dwellers	Indirect — water, sanitation, housing
8. Develop a global partnership for development	Non-discriminatory trade and financial system for least-developed countries; debt sustainability; cooperate with the pharmaceutical industry to provide access to affordable essential drugs ; make new information and communication technologies available	Indirect, with one health target

Sources: *Textbook of International Health*, Ch 4, pp 210–211 (Box 4–3); Park 28e, Ch 2, p 50 and Ch 9, p 657.

The 18 health-related indicators

Goal	Indicator no.	Indicator
1. Poverty and hunger	4	Prevalence of underweight children under five years of age
	5	Proportion of population below minimum level of dietary energy consumption
4. Child mortality	13	Under-five mortality rate
	14	Infant mortality rate
	15	Proportion of 1-year-old children immunized against measles
5. Maternal health	16	Maternal mortality ratio
	17	Proportion of births attended by skilled health personnel
6. HIV/AIDS, malaria, other diseases	18	HIV prevalence among young people aged 15–24 years
	19	Condom use rate of the contraceptive prevalence rate
	20	Number of children orphaned by HIV/AIDS
	21	Prevalence and death rates associated with malaria

Goal	Indicator no.	Indicator
	22	Proportion of population in malaria-risk areas using effective malaria prevention and treatment measures
	23	Prevalence and death rates associated with tuberculosis
	24	Proportion of tuberculosis cases detected and cured under Directly Observed Treatment, Short-course (DOTS)
7. Environmental sustainability	29	Proportion of population using solid fuel
	30	Proportion of population with sustainable access to an improved water source, urban and rural
	31	Proportion of urban population with access to improved sanitation
8. Global partnership	46	Proportion of population with access to affordable essential drugs on a sustainable basis

Source: Park 28e, Ch 2, p 40, Table 3 — "Health-related Millennium Development Goals and Indicators".

India: delivery mechanism and progress by 2015

- **Delivery vehicles:** most MDG health objectives and indicators map on to the **reproductive and child health (RCH)** framework; they were pursued through RCH and other national health programmes, the **National Population Policy 2000**, the **National Health Policy 2002** and the **National Rural Health Mission (NRHM)** (Sunder Lal, Ch 17, p 767).

Indicator	India's position at the close of the MDG period (2015)
Under-five mortality rate	Fell from 123 (1990 baseline) to 48 per 1000 live births
Infant mortality rate	Reduced to 40 per 1000 live births
Safe (skilled-attended) deliveries	About 81 per cent
Maternal mortality ratio	167 per 100,000 live births — MDG 5 showed the least progress
Adolescent birth rate	Remained high at 39
Underweight children under three years	About 33 per cent
Improved drinking water source	About 90 per cent of population
Improved sanitation	Only 40 per cent of population
Adult HIV prevalence	Stable and low at 0.26 per cent
Tuberculosis	Prevalence 195, incidence 167 and mortality 17 per lakh population; DOTS cure rate over 85 per cent
Malaria and other vector-borne diseases	Continued high morbidity with recurrent outbreaks

Source: Sunder Lal, Ch 17, p 767 — "Progress of MDGs in India so far (2015)".

Strengths and criticisms

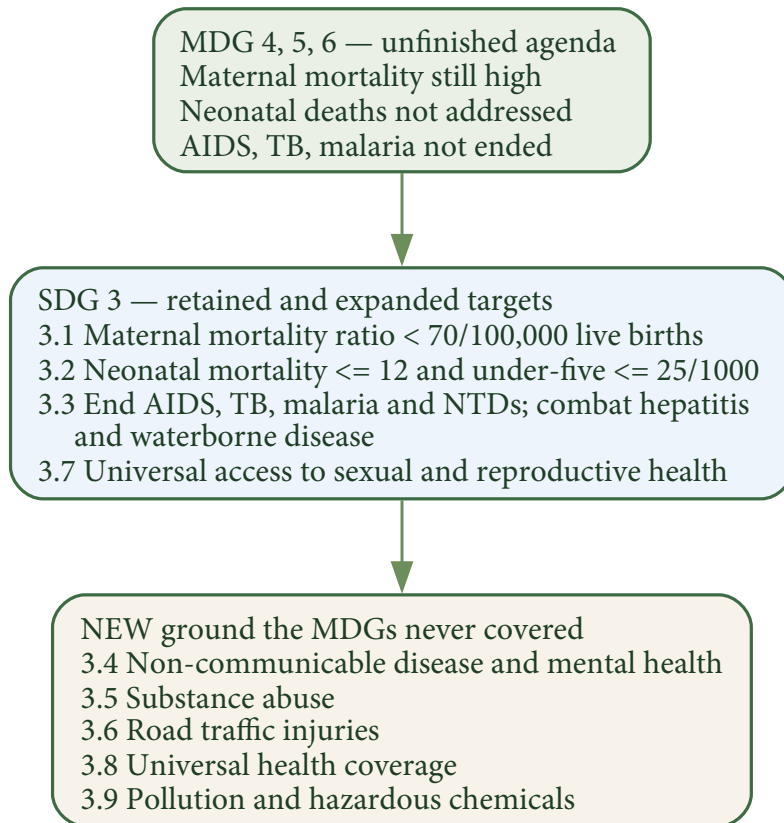
Strengths	Criticisms
Placed health at the centre of the development agenda and gave it a single, shared global target set	Written mainly by the United States, Europe and Japan with heavy influence of large aid agencies; neither the citizens nor the governments of the developing nations they were meant to help were involved
Galvanised political commitment and unlocked major financing — PEPFAR, President's Malaria Initiative, GAVI, the Gates Foundation and the Global Fund	Global goals but country-level performance assessment, so many countries were unfairly labelled "off-track"
Concrete, measurable, time-bound targets made accountability possible and gave birth to the concept of "global health"	Narrow definition of poverty (the US\$ 1/day line); market- and debt-focused, with no strategy addressing the causes of poverty
Drove unprecedented falls in maternal and child mortality and in infectious disease burden	Many health targets difficult or impossible to measure — weak civil registration and vital statistics in most low- and middle-income countries
Exposed massive inequities between low- and high-income regions	Benchmark (proportional-reduction) approach penalised countries starting from the worst baselines; MDG 5 in particular was widely missed

Sources: *Oxford Textbook of Public Health 7e*, Ch 4.2, pp 919–920 and pp 935–936; *Textbook of International Health*, Ch 4, pp 211–213.

Succession by the Sustainable Development Goals

In December 2015 the MDGs came to the end of their term and, from 1 January 2016, the 17 Sustainable Development Goals (SDGs) of the 2030 Agenda took their place; several SDG 3 targets are the direct continuation of the unfinished MDG agenda (Park 28e, Ch 14, pp 852, 855).

Feature	MDGs (2000–2015)	SDGs (2016–2030)
Number	8 goals, 18 targets, 48 indicators	17 goals, 169 targets
Health goals	3 separate health goals (4, 5, 6)	One comprehensive goal, SDG 3, with 13 targets
Scope	A limited set of human-development targets, primarily for developing countries; closely tied to aid spending	Integrated and indivisible, global in nature and universally applicable to all countries
Content	Maternal and child mortality and infectious disease	Adds non-communicable disease, mental health, substance abuse, injuries, universal health coverage
Guiding principle	Aggregate national averages	Strong focus on equity — "no one will be left behind"



Carry-over of the unfinished MDG health agenda into Sustainable Development Goal 3

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

"Three plus five" MDG wheel — a central hub with eight spokes; label the hub *Millennium Declaration 2000 — 8 goals by 2015*, and the spokes:

- ▶ Inner (shaded) spokes, labelled **Direct health goals**: Goal 4 *Reduce child mortality*; Goal 5 *Improve maternal health*; Goal 6 *Combat HIV/AIDS, malaria and other diseases*.
- ▶ Outer spokes, labelled **Determinants of health**: Goal 1 *Poverty and hunger*; Goal 2 *Primary education*; Goal 3 *Gender equality*; Goal 7 *Environmental sustainability*; Goal 8 *Global partnership*.
- ▶ Rim label: 18 targets · 48 indicators · 8 health targets · 18 health indicators.

High-yield exam pointers

- ▶ "3 – 8 – 18" of health in the MDGs: 3 of 8 goals, 8 of 18 targets, 18 of 48 indicators are health-related.
- ▶ **Dates and numbers**: Millennium Summit **September 2000**, 189 countries, seven areas of commitment; baseline **1990**, deadline **2015**; term ended **December 2015**, SDGs in force **1 January 2016**.
- ▶ **The three quantified health targets**: child mortality **down by two-thirds**; maternal mortality ratio **down by three-quarters**; HIV/AIDS, malaria and tuberculosis **halted and reversed** — all between 1990 and 2015.

- ▶ **Target 5B** (universal access to reproductive health by 2015) was a **later addition** — agreed at the 2005 World Summit, effective 15 January 2008.
- ▶ **India 2015 in one line:** under-five mortality 123 → 48, infant mortality 40, skilled attendance 81 per cent, but **maternal mortality ratio 167** — **MDG 5 the weakest goal**; sanitation coverage only 40 per cent.
- ▶ **Commonest paired question:** MDG *versus* SDG — 8 goals/3 health goals *versus* 17 goals/1 health goal (SDG 3) with 13 targets.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the corpus carries a minor discrepancy in the target count — Park 28e states 18 targets and 48 indicators for the original framework, while the Textbook of International Health cites 16 original targets and 48 indicators; quote Park's "8 goals, 18 targets, 48 indicators" as the spine figure. The Indian progress figures above are those reported at the close of the MDG period in 2015 and are therefore historical benchmarks, not current status — India's maternal mortality ratio has since fallen to 97 per 100,000 live births (Sample Registration System 2018–2020) and under-five mortality to 31 per 1000 live births (2021).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Sample Registration System 2023, NFHS-5 (2019–21), National Health Accounts 2019–20, National Health Policy 2017, UN Inter-agency Group for Child Mortality Estimation, WHO

Programme and policy updates (India)

- ▶ The MDG-era delivery vehicles named above (National Population Policy 2000, National Health Policy 2002, NRHM) have been superseded by the **National Health Policy 2017**, whose target of 2.5 per cent of GDP for public health spending remains unmet: spending moved only from 0.9 per cent of GDP in 2015–16 to 1.6 per cent in 2021–22, and India ranked 179th of 189 countries in the budget priority given to health [PMID 39668924].
- ▶ A multilevel analysis linking NFHS-5 to National Health Accounts 2019–20 showed that India's SDG-3 outcomes now track the financing route a state takes — government-financed and prepaid states performed better on child mortality, full immunisation and WASH access — while out-of-pocket spending still formed **51.2 per cent** of total health expenditure (government 39.4 per cent, prepaid 9.4 per cent) and insurance fragmentation raised cost without matching gains [PMID 42218447].

Current national data

- ▶ Sample Registration System 2023 places India's **infant mortality rate at 25 per 1000 live births**, down from 40 in 2013 — so the "IMR 40" figure in the 2015 progress table is now a decade out of date. Pneumonia remains the second commonest cause of infant death at about 16 per cent, with a September-to-December seasonal peak [PMID 42056953].
- ▶ **NFHS-5 (2019–21)**, 176,843 live births: national neonatal mortality 16.2 per 1000, rural 18.3 against urban 11.5 (a 1.6-fold gap), highest in Uttar Pradesh (25.9) and Bihar (23.2). Deaths were

concentrated among poorer households (Erreygers concentration index -0.0123 , $p < 0.001$), driven by unclean cooking fuel, high parity and inadequate antenatal care [PMID 42354325].

- ▶ District-level NFHS-5 estimates for **Uttar Pradesh** show under-five mortality of about 60 per 1000 — more than twice the SDG 3.2 target — ranging from 111 in Amethi to below 25 in Jaunpur and Deoria; the eight aspirational districts fared marginally better than the state as a whole (U5MR 56.5 versus 59.9) [PMID 41869150].
- ▶ **Karnataka** taluk-level mapping of maternal deaths (Moran's I 0.114, $p = 0.017$) identified nine high-high cluster taluks concentrated in the northern districts of Kalaburagi, Raichur and Ballari; **39.8 per cent of maternal deaths occurred within 48 hours postpartum**, with primary postpartum haemorrhage, sepsis and hypertensive disorders of pregnancy the leading direct causes [PMID 42386290].

Elimination, surveillance and disease status

- ▶ UN Inter-agency Group for Child Mortality Estimation data to 2023: global under-five deaths fell from 12.98 million (1990) to 4.78 million, and the under-five mortality rate from 93.58 to 36.72 per 1000 live births. At the 1990–2023 rate of decline the world reaches **SDG 3.2 only in 2035 — five years late** — and India alone still accounted for 0.64 million under-five deaths in 2023 [PMID 41920787].

Guideline and legislative updates

- ▶ **BLOW2 (2025)**, a 5420-newborn multicentre randomised trial in three south Indian neonatal intensive care units, found that BCG Danish plus oral polio vaccine given within 48 hours to babies weighing under 2000 g reduced all-cause neonatal mortality (adjusted HR 0.83, 95 per cent CI 0.69–0.98; NNT 21) and deaths from infection (adjusted HR 0.53, 95 per cent CI 0.40–0.70). The authors ask WHO to extend the birth dose to unstable and very preterm newborns and to shift the BCG timeliness indicator from 12 months to 1 month of age [PMID 40983468].

Global and WHO developments

- ▶ A convergence analysis of 163 countries (1990–2023) found maternal-mortality inequality essentially static — Gini coefficient about 0.66 — with countries splitting into three "clubs" rather than converging on one path; only **65 of 163 countries (40 per cent) are projected to meet SDG 3.1 (MMR below 70 per 100,000 live births) by 2030** [PMID 42106672].
- ▶ An updated systematic review of health and health-related SDG implementation (192 publications, June 2019 to January 2025) reports high political commitment and a shift towards decentralised, multisectoral delivery, but names COVID-19 service disruption, poor-quality data and unsustainable financing as the binding constraints — the same measurement weakness that undermined MDG accountability [PMID 41629068].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 1 — Man and Medicine: Towards Health for All; Ch 2 — Concept of Health and Disease, Table 3 "Health-related Millennium Development Goals and Indicators"; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 — Health Care of the Community; Ch 14 — Sustainable Development Goals); Textbook of International Health (Ch 4 — The Political Economy of Health and Development, Box 4–3 "MDGs and Selected Targets"); Sunder Lal — Textbook of Community Medicine (Ch 17 — Healthcare of the Community, "Progress of MDGs in India so far, 2015"); Oxford Textbook of Public Health 7e (Ch 4.2 — Community diagnosis and health information systems in low- and middle-income countries). Named documents: United Nations Millennium Declaration 2000; 2005 World Summit revised MDG framework (effective 15 January 2008); Transforming our World — the 2030 Agenda for Sustainable Development, 2015.

Recent-advances sources: 39668924 — Five Years of National Health Policy in India: Critical Analysis of the Public Health Expenditure from 2017 to 2022 and Way Forward, *Indian Journal of Community Medicine* 2024, DOI [10.4103/ijcm.ijcm.823.22](https://doi.org/10.4103/ijcm.ijcm.823.22); 42218447 — The role of health financing system and pluralism in achieving sustainable development goal health outcomes in India: a multilevel analysis, *International Journal for Equity in Health* 2026, DOI [10.1186/s12939-026-02888-x](https://doi.org/10.1186/s12939-026-02888-x); 42056953 — Seasonality of pneumonia mortality among infants in India from 2018–2020: a joinpoint analysis, *BMC Infectious Diseases* 2026, DOI [10.1186/s12879-026-13229-2](https://doi.org/10.1186/s12879-026-13229-2); 42354325 — Decomposing Wealth-Based Inequalities in Neonatal Mortality in India: Evidence from National Family Health Survey (2019–2021), *International Journal of Environmental Research and Public Health* 2026, DOI [10.3390/ijerph23060795](https://doi.org/10.3390/ijerph23060795); 41869150 — District-Level Estimates of Child Mortality in Uttar Pradesh, India, *Cureus* 2026, DOI [10.7759/cureus.103889](https://doi.org/10.7759/cureus.103889); 42386290 — Spatial autocorrelation and regression approach for delineating maternal mortality and its associated factors in Karnataka, India, *BMJ Open* 2026, DOI [10.1136/bmjopen-2024-094191](https://doi.org/10.1136/bmjopen-2024-094191); 41920787 — Tracking under-five mortality from 1990 to 2023: global, regional, and national trends, inequities, and projections toward achieving SDG Target 3.2 by 2030, *PLoS One* 2026, DOI [10.1371/journal.pone.0343745](https://doi.org/10.1371/journal.pone.0343745); 40983468 — Effect of BCG Danish and oral polio vaccine on neonatal mortality in newborn babies weighing less than 2000 g in India: multicentre open label randomised controlled trial (BLOW2), *BMJ* 2025, DOI [10.1136/bmj-2025-084745](https://doi.org/10.1136/bmj-2025-084745); 42106672 — Convergence in maternal mortality ratios: evidence and implications for SDG 3.1, *BMC Public Health* 2026, DOI [10.1186/s12889-026-27485-z](https://doi.org/10.1186/s12889-026-27485-z); 41629068 — Implementation of health and health-related sustainable development goals: progress, challenges and opportunities — a systematic literature review update, *BMJ Global Health* 2026, DOI [10.1136/bmjgh-2025-021623](https://doi.org/10.1136/bmjgh-2025-021623). Named data sources cited within these bullets: Sample Registration System 2023; National Family Health Survey-5 (2019–21); National Health Accounts 2019–20; National Health Policy 2017; UN Inter-agency Group for Child Mortality Estimation.



WHO theme for 2024 — national strategies

Asked in: Paper IV 03-Oct-2025 — RGUHS MD Community Medicine

Definition — The Constitution of the World Health Organization (WHO) came into force on **7 April 1948**, and that date is observed every year as **World Health Day**, for which a theme is chosen to focus world attention on one specific aspect of public health (Park 28e, Ch 25, p 1055). The **World Health Day theme for 2024 (WHO's 76th anniversary)** was "**My health, my right**" — a campaign asserting that the enjoyment of the highest attainable standard of health is a fundamental right of every human being, and that States are duty-bound to make health services and the underlying determinants of health available to all.

The 2024 theme at a glance

Element	Detail
Day and date	World Health Day, 7 April 2024 (anniversary of the WHO Constitution coming into force, 1948) (Park 28e, Ch 25, p 1055)
Theme / slogan	"My health, my right"
Central assertion	"The enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, economic or social condition" — WHO Constitution preamble (Park 28e, Ch 25, p 1055)
Scope claimed	Right of access not only to quality health services, but to the underlying determinants — safe and potable water, sanitation, safe food and nutrition, housing, healthy occupational and environmental conditions, and health-related education and information (General Comment on the Right to Health, 2000) (Park 28e, Ch 2, p 33)
Why chosen	Globally at least half the population still lacks access to essential health services; about 800 million people spend $\geq 10\%$ of the household budget on health and nearly 100 million are pushed into extreme poverty by health costs each year (Oxford Textbook of Public Health 7e, Ch 107, pp 4044-4045)
Global anchor	Sustainable Development Goal target 3.8 — universal health coverage (UHC) with financial risk protection; WHO "triple billion" targets — one billion more people with UHC, one billion more protected from health emergencies, one billion more enjoying better health and well-being (Oxford PH 7e, Ch 107, pp 4044-4045)

What the right to health legally means (the content a government must deliver)

Instrument / element	Content
Universal Declaration of Human Rights, 1948 (Art. 25)	Links health and well-being to an adequate standard of living (Park 28e, Ch 2, p 33)
International Covenant on Economic, Social and Cultural Rights, 1966 (Art. 12)	Central instrument; recognises the right of everyone to the highest attainable standard of physical and mental health, given equal weight (Park 28e, Ch 2, p 33)
Convention on Elimination of all forms of Racial Discrimination, 1965 (Art. 5)	Non-discrimination in health entitlements (Park 28e, Ch 2, p 33)

Instrument / element	Content
General Comment on Right to Health, 2000 (UN Committee on Economic, Social and Cultural Rights)	Steps to full realisation — reduce stillbirth rate and infant mortality and secure healthy child development; environmental and industrial hygiene; prevention, treatment and control of epidemic, endemic and occupational disease; and creation of conditions assuring all medical service in sickness. Defines a " core content " = minimum essential level, including supply of essential drugs (Park 28e, Ch 2, p 33)
AAAQ criteria	Hospitals, clinics, medicines and doctors' services must be Available, Accessible (non-discriminatory, physically accessible, affordable, information accessible), Acceptable (culturally appropriate) and of good Quality, for everyone on an equitable basis (Park 28e, Ch 2, p 33; Oxford PH 7e, Ch 19, p 754)
Five key principles	Universality; Equity; Accountability; Transparency; Participation (Park 28e, Ch 2, p 33)

India's own legal obligation — the basis for national strategy

Provision	Obligation created
Article 21 (Fundamental Right to life)	Requires the State to ensure the health and nutritional well-being of all people (Sunder Lal, Ch 17, p 743)
Directive Principles of State Policy, Part IV	State shall regard raising the level of nutrition and standard of living and the improvement of public health as among its primary duties (Park 28e, Ch 2, p 35)
Directive Principles (contd.)	Health and strength of workers, men, women and children of tender age not to be abused; provision for just and humane conditions of work and maternity relief; public assistance in sickness, old age and disablement within the limits of economic capacity (Park 28e, Ch 2, p 35; Ch 12, p 824)
Article 24 (Fundamental Right)	No child below 14 years to be employed in any factory, mine or hazardous employment (Park 28e, Ch 18, p 967)
Mental Healthcare Act, 2017	Enacted to align Indian law with the UN Convention on the Rights of Persons with Disabilities, which India ratified — makes it obligatory on Government to protect the rights of persons with mental illness (Park 28e, Ch 7, pp 568-569)
Rights of Persons with Disabilities Act, 2016	Replaced the PWD Act 1995; disability defined on an evolving basis, with expanded entitlements (Park 28e, Ch 9, p 689)

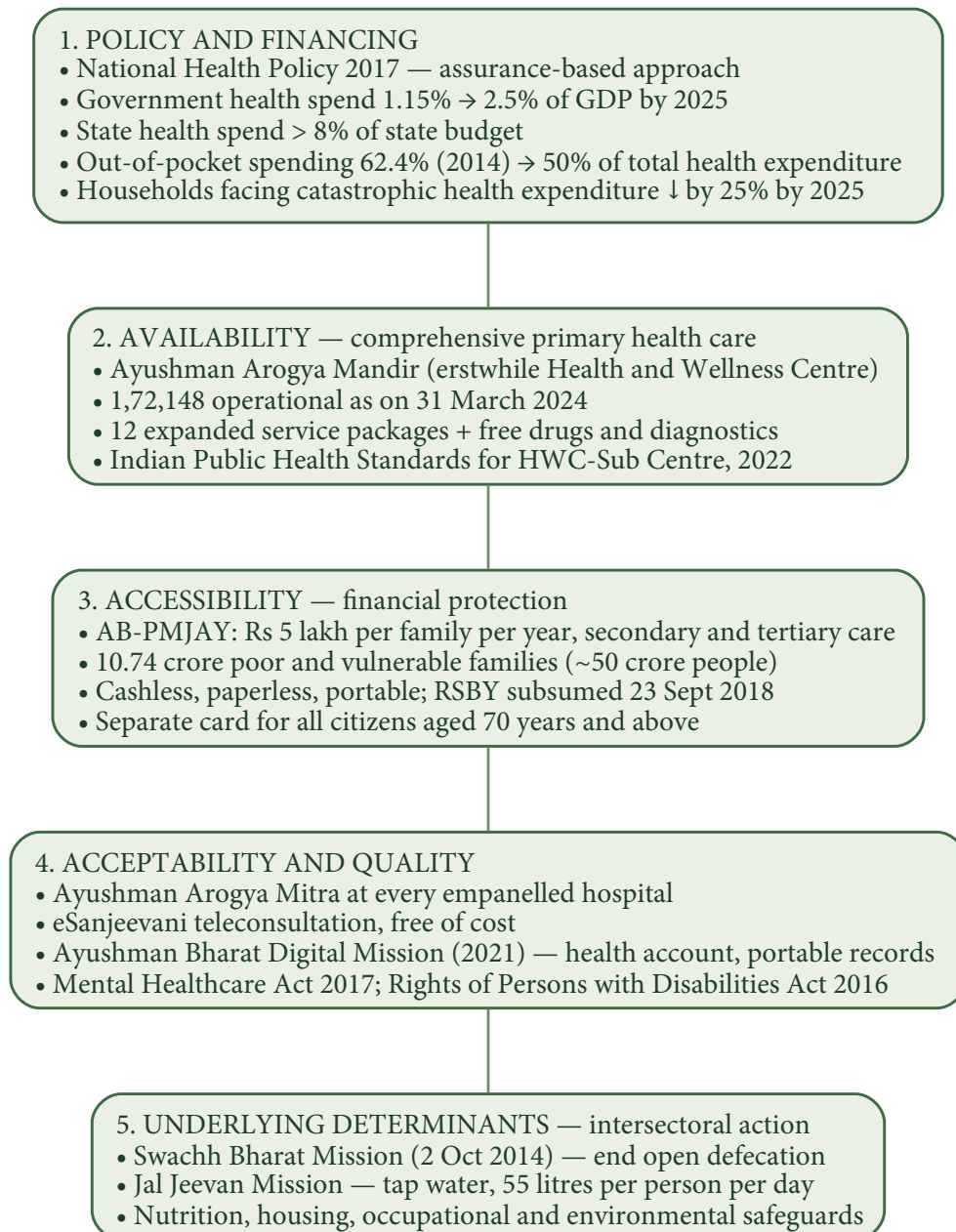
The minimum core obligations a national government must meet at once

The right to health is realised progressively, but a **minimum core** is immediate and non-derogable; national strategy is first judged against these six obligations (Oxford PH 7e, Ch 19, p 754), each of which India discharges through a named instrument.

- **A national plan of action for health, including sexual and reproductive health** — met in India by the National Health Policy 2017 and the Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A) strategy.

- ▶ **Provision of essential drugs as identified by WHO** — met through free drugs supplied against the Essential Drug List at every Ayushman Arogya Mandir.
- ▶ **Provision of reproductive, maternal and child health care** — met through Janani Suraksha Yojana, Janani Shishu Suraksha Karyakram and Pradhan Mantri Surakshit Matritva Abhiyan, delivered free at the Ayushman Arogya Mandir (Park 28e, Ch 10, pp 729-730).
- ▶ **Immunisation against the major infectious diseases** — met through the Universal Immunization Programme and Mission Indradhanush.
- ▶ **Prevention, treatment and control of epidemic and endemic disease** — met through the Integrated Disease Surveillance Programme and the disease-specific national programmes, including the National Tuberculosis Elimination Programme.
- ▶ **Training of health personnel** — met through the mid-level health provider cadre, ASHA and auxiliary nurse midwife training, and the human-resource targets of the Indian Public Health Standards.

Strategies of the Government of India for achieving the theme

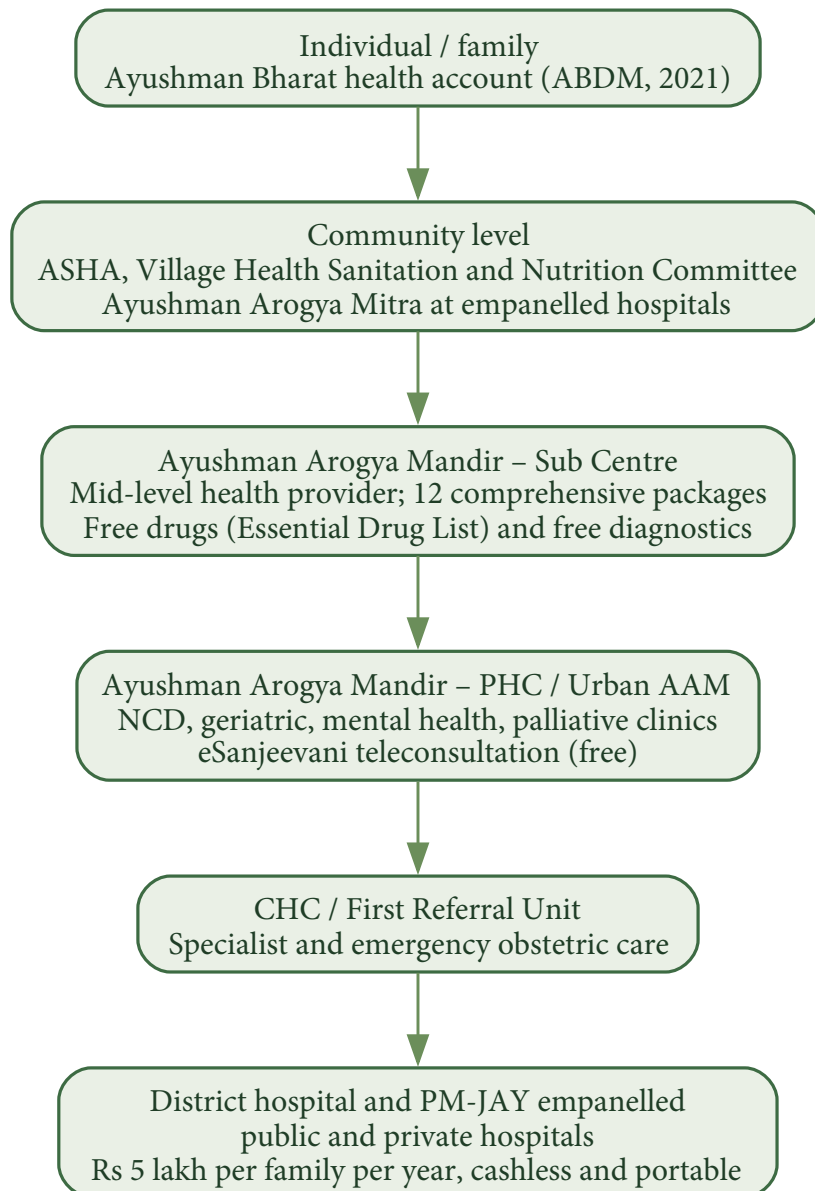


Five pillars of the Government of India's strategy to realise "My health, my right"

Pillar	Key instruments and what each delivers
Legal and policy guarantee	National Health Policy 2017 — aims to inform, clarify, strengthen and prioritise the role of Government in health systems, financing, service organisation, disease prevention, cross-sectoral action, access to technology and financial protection (Park 28e, Ch 23, p 1030)
Assured service availability	Ayushman Arogya Mandir (AAM), launched 2018 as Health and Wellness Centres — a paradigm shift from a selective to a comprehensive continuum-of-care approach across primary, secondary and tertiary levels; first centre inaugurated 14 April 2018 at Bijapur, Chhattisgarh; 1,72,148 operational as on 31 March 2024 (Park 28e, Ch 7, pp 541, 570)

Pillar	Key instruments and what each delivers
Financial risk protection	AB-PMJAY, launched 23 September 2018 — cover up to Rs 5 lakh per family per year for secondary and tertiary hospitalisation for 10.74 crore families (~50 crore people) , cashless and paperless, including 3 days pre- and 15 days post-hospitalisation; centre-state funding 60:40 (90:10 for North-Eastern states); the largest government-funded health protection scheme in the world (Park 28e, Ch 10, p 745; Sunder Lal, Ch 17, p 763)
Removing point-of-care cost	Free drugs as per the Essential Drug List and free diagnostics as per the Free Diagnostic Initiative guidelines, 24×7 ambulance with drop-back, generic drug store at the facility (Park 28e, Ch 10, p 729)
Digital access and portability	Ayushman Bharat Digital Mission (launched September 2021) — health account number, linked longitudinal digital health records, national portability, telemedicine; eSanjeevani AB-HWC (doctor-to-doctor) and eSanjeevani OPD (patient-to-doctor) free of cost (Sunder Lal, Ch 17, pp 748, 763)
Reaching the last mile	Ayushman Bhav campaign, launched 13 September 2023 — saturation of selected health services in every village and town, target-free approach, IEC, specially designed packages for urban slums and tribal areas (Park 28e, Ch 7, p 541)
Non-discrimination for vulnerable groups	Mental Healthcare Act 2017 and Rights of Persons with Disabilities Act 2016; urban outreach targeted at slum dwellers and other vulnerable groups through female health workers based at urban primary health centres (Park 28e, Ch 7, p 535)
Underlying determinants	Swachh Bharat Mission (2 October 2014) — elimination of open defecation, eradication of manual scavenging, scientific municipal solid waste management; Jal Jeevan Mission — a Functional Household Tap Connection delivering ≥55 litres per person per day of water meeting prescribed quality standards (Park 28e, Ch 7, pp 573-574)

How the entitlement reaches the citizen



Level-wise delivery of the health entitlement in India, from household to tertiary care

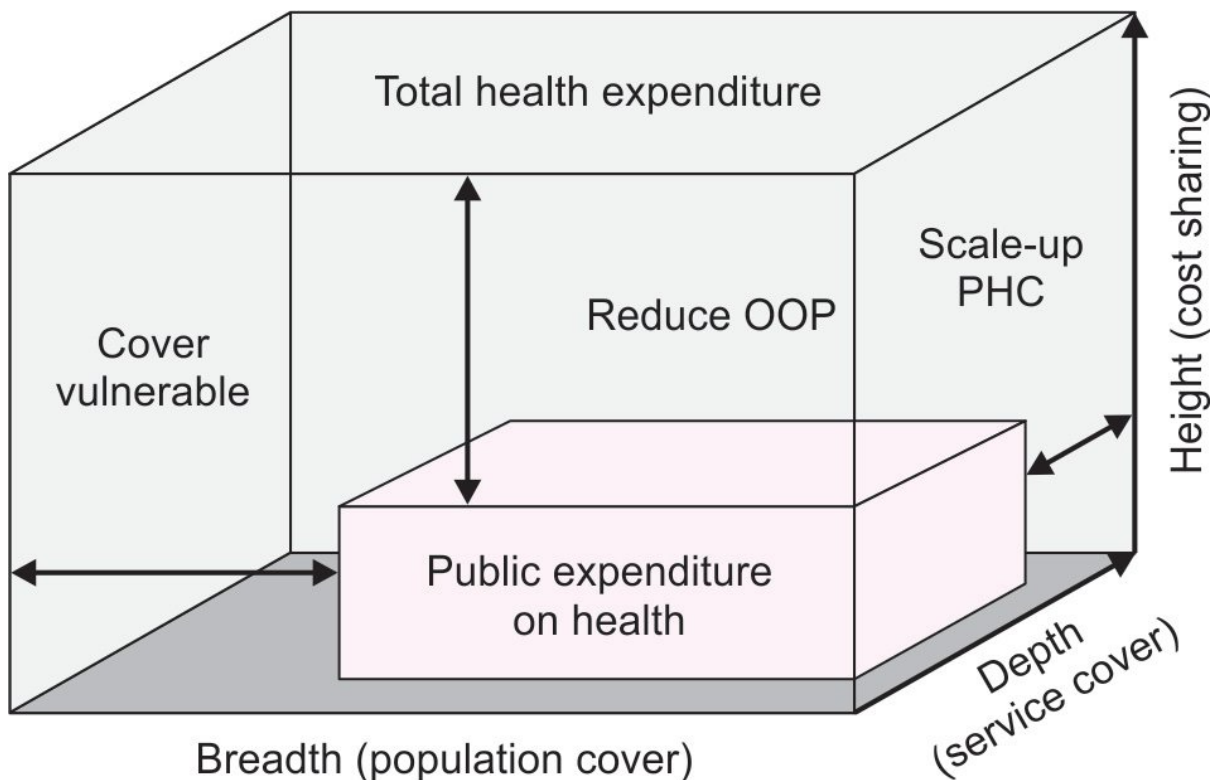
Indicators by which achievement is judged

Indicator (National Health Policy 2017 target)	Target
Government health expenditure as % of gross domestic product	1.15% → 2.5% by 2025 (Park 28e, Ch 23, p 1030)
State sector health spending as % of state budget	>8% by 2020 (Park 28e, Ch 23, p 1030)
Out-of-pocket spending as % of total health expenditure	62.4% (2014 estimate) → 50% (Park 28e, Ch 23, p 1033)
Households facing catastrophic health expenditure	Reduce by 25% by 2025 (Park 28e, Ch 23, p 1030)
Life expectancy at birth	67.5 → 70 years by 2025 (Park 28e, Ch 23, p 1030)

Indicator (National Health Policy 2017 target)	Target
Under-five mortality rate / maternal mortality ratio	23 by 2025 / 100 by 2020 (Park 28e, Ch 23, p 1030)
Access to improved drinking water / adequate sanitation, India	90% / 71% (2020) (Park 28e, Ch 15, p 861)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The universal health coverage box (three dimensions)** — label: outer box "Total health expenditure"; inner box "Public expenditure on health"; **Breadth (population cover)** → "Cover vulnerable"; **Depth (service cover)** → "Scale-up primary health care"; **Height (cost sharing)** → "Reduce out-of-pocket expenditure".
- **AAAQ wheel** — a central circle "Right to health" with four spokes labelled *Available*, *Accessible* (non-discriminatory / physical / affordable / information), *Acceptable*, *Good Quality*, ringed by the five principles — Universality, Equity, Accountability, Transparency, Participation.



OOP: Out of pocket
PHC: Primary health care

Attaining universal coverage — the three dimensions (Sunderlal, p.740)

High-yield exam pointers

- ▶ **World Health Day = 7 April**, the day the WHO Constitution came into force in 1948; **2024 theme = "My health, my right"** (WHO's 76th anniversary).
- ▶ Quote the WHO Constitution preamble verbatim — "the enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, economic or social condition."
- ▶ **AAAQ + UEATP** — Available, Accessible, Acceptable, Quality; Universality, Equity, Accountability, Transparency, Participation.
- ▶ Indian legal hooks: **Article 21**, Directive Principles Part IV ("raising the level of nutrition... and improvement of public health as among its primary duties"), Article 24, Mental Healthcare Act 2017, Rights of Persons with Disabilities Act 2016.
- ▶ Dated Indian numbers to write: **AAM 1,72,148 as on 31 March 2024; PM-JAY Rs 5 lakh/family/year for 10.74 crore families; out-of-pocket 62.4% (2014) with a 50% target; government health spend 1.15% → 2.5% of GDP by 2025.**
- ▶ **SDG 3.8** is the global anchor — UHC with financial risk protection; WHO "triple billion" targets.
- ▶ Two components of Ayushman Bharat = Ayushman Arogya Mandir (comprehensive primary health care) + AB-PMJAY (financial protection); ABDM (2021) is the digital spine.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e records that a World Health Day theme is chosen each year but does not print the 2024 slogan; "My health, my right" and the WHO campaign's stated scope are taken from the WHO World Health Day 2024 campaign and are not sourced from the indexed corpus — every other fact, figure and target above is corpus-grounded and carries its year. Health and Wellness Centres were formally renamed **Ayushman Arogya Mandir**, so write the current name first and the former name once in parentheses.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO World Health Day campaign, WHO Pandemic Agreement, International Health Regulations, NHA Estimates (India), National Health Authority, NCVBDC

Programme and policy updates (India)

- ▶ **PM-JAY health benefit package (HBP) governance has been rebuilt around cost evidence** — the package list and reimbursement rates have been revised four times since the September 2018 launch (HBP 2.0 November 2019, HBP 2.1 November 2020, HBP 2.2 October 2021, then HBP 2022 in April 2022). The national Costing of Health Services in India (CHSI) study showed about 42% of packages were priced below half the actual cost of delivering the service; after re-pricing only 20% remained so, and rates for 400 procedures were revised in HBP 2.2. A **Health Financing and Technology Assessment (HeFTA) unit** now sits within the National Health Authority to bring health technology assessment into package inclusion and pricing decisions [PMID 38857943].

- **Ayushman Bharat Digital Mission — adoption is the weak link, not the architecture** — across three tertiary care hospitals in Agra (September 2024 to April 2025) only **129,007 ABHA registrations** were completed against **537,278 outpatient visits**, an adoption rate of **23.96%**. Among the 425 attendees interviewed the leading barriers were preference for in-person care (85.6%) and the cost of a mobile data plan (81.9%), and **88% asked for training and support** before they could use ABDM [PMID 41787376].

Current national data

- **Out-of-pocket spending — quote the national estimate with its round** — OOP expenditure was **48.2% of total health expenditure in 2018-19, down from 64.2% in 2013-14** (NHA Estimates). Write this beside the National Health Policy 2017 baseline of 62.4% (2014) and its 50% target rather than in place of it; duration of hospitalisation, use of private hospitals, household size and an older adult in the household remain significant predictors of health-expenditure impoverishment [PMID 41938958].
- **Service coverage is improving while financial protection is deteriorating** — Bayesian projections for seven South Asian countries put the composite service coverage index at 53.5% in 2000 rising to a projected 81.5% by 2030, but catastrophic health expenditure rising from 7.2% to a projected 18.6% over the same period, with **India projected to reach 26.3% by 2030**. This is the single most quotable challenge to the "accessibility" pillar of the 2024 theme [PMID 41192927].

Elimination, surveillance and disease status

- **Malaria — India works to the National Framework for Malaria Elimination 2016-2030 and the National Strategic Plan 2023-2027** — as of 2024 the National Centre for Vector Borne Diseases Control classified 103 districts as Category 0 (annual parasite incidence zero), 634 as Category 1 (API <1), 9 as Category 2 (API 1-2) and 28 high-burden districts as Category 3 (API >2) [PMID 41663159].
- **Kala-azar and lymphatic filariasis — declared status is ahead of transmission reality** — India is moving into the visceral leishmaniasis post-elimination phase, yet endemic villages in Bihar and Jharkhand showed rK39 seropositivity of 20-30% and *Leishmania*-positive sand fly pools (1.4% and 1.2%) despite indoor residual spraying, so surveillance cannot be stood down [PMID 41324335]. Lymphatic filariasis is still endemic in **345 districts** against a **2030** elimination goal, with mass drug administration, morbidity management and post-MDA surveillance the operational bottlenecks [PMID 41951149].

Guideline and legislative updates

- **Rajasthan Right to Health Care Act, 2022 — the clearest worked example of the distance between a declared right and an enforceable entitlement** — private hospital, insurer and physician coalitions united rapidly behind a public campaign (#RollBackRTH), pairing direct and indirect tactics with deliberate framing choices to win the public narrative and secure a dominant place in the policy process, leaving the Act's supporters on the defensive. Cite it whenever an answer asks how "my health, my right" is to be made legally operational in India [PMID 41362954].

Global and WHO developments

- **The World Health Day theme has moved on twice since 2024 — quote the year the question asks for** — 2025: "Healthy beginnings, hopeful futures" (7 April 2025), a year-long campaign on

maternal and newborn health, citing roughly 260,000 maternal deaths a year, over 2 million newborn deaths plus about 2 million stillbirths, and 4 of every 5 countries off track for the 2030 maternal survival target [WHO 2025-Apr]. **2026: "Together for health. Stand with science"** — the day itself stays fixed at **7 April 2026**, commemorated in Lyon alongside the One Health Summit (5-7 April 2026) and the inaugural Global Forum of WHO Collaborating Centres (7-9 April 2026), with WHO's call-to-action release dated 6 April 2026; the campaign is about choosing evidence over misinformation and embedding science in health, climate, food and environment policy [WHO 2026-Apr]. **"My health, my right" belongs to 2024 alone.**

- ▶ **WHO Pandemic Agreement adopted by the World Health Assembly in May 2025** after nearly three years of negotiation, and negotiated under **Article 19 of the WHO Constitution**. Article 12 requires legal contracts with manufacturers guaranteeing access to **at least 20% of pandemic-related products — half donated, half at affordable prices**, with distribution coordinated by WHO; the Pathogen Access and Benefit-Sharing (PABS) annex that operationalises it was set for finalisation by **April 2026** [PMID 41832575].
- ▶ **International Health Regulations — the June 2024 amendment package came into force on 19 September 2025** — amended Article 4 now requires every State Party to establish or designate a **national IHR authority** to coordinate implementation of the Regulations nationally: a new coordinating function separate from the national IHR focal point's communication role with the WHO IHR Contact Point and from the Article 22 point-of-entry competent authorities, though a State may designate a single entity to discharge both the authority and focal-point functions [PMID 40953844].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 12 Medicine and Social Sciences; Ch 15 Environment and Health; Ch 18 Occupational Health; Ch 23 Health Planning and Management; Ch 25 International Health); Oxford Textbook of Public Health 7e (Ch 19 The right to health supports global public health; Ch 107 Global health in the era of sustainable development); Sunder Lal — Textbook of Community Medicine (Ch 17 Healthcare of the Community); National Health Policy 2017 (Government of India); Indian Public Health Standards for Health and Wellness Centres, 2022. Coverage note: the 2024 World Health Day slogan itself is not present in the indexed corpus (see Note).

Recent-advances sources: 38857943 — Understanding the extent of economic evidence usage for informing policy decisions in the context of India's national health insurance scheme: Ayushman Bharat Pradhan Mantri Jan Aarogya Yojana (PM-JAY), *BMJ Global Health* 2024, doi:[10.1136/bmjgh-2024-015079](https://doi.org/10.1136/bmjgh-2024-015079); 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission: an analytical cross-sectional study among outpatient department attendees, *BMC Health Services Research* 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 41938958 — Impact of out-of-pocket health expenditure on poverty in India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1762886](https://doi.org/10.3389/fpubh.2026.1762886); 41192927 — Progress towards universal health coverage in South Asia, 2000-2030: an examination of the twin elements of primary healthcare provision and financial protection, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-020052](https://doi.org/10.1136/bmjgh-2025-020052); 41663159 — Multicentric longitudinal study on malaria burden, vector bionomics and health system assessment in diverse eco-epidemiological settings in the context of malaria elimination in India: study protocol, *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-111645](https://doi.org/10.1136/bmjopen-2025-111645); 41324335 — High Seropositivity to rK39 and Capture of Leishmania-Infected Sand Flies in Bihar and Jharkhand Emphasize the Need for Post-Elimination Surveillance of Visceral Leishmaniasis in India, *Journal of Infectious Diseases* 2026, doi:[10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543) (abstract-only); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 41362954 — #RollBackRTH: Tactics, strategies and framing in the Right to Health Care Act 2022 debate in Rajasthan, India, *Global Public Health* 2025, doi:[10.1080/17441692.2025.2597619](https://doi.org/10.1080/17441692.2025.2597619) (abstract-only); 41832575 — Africa's Critical Role in Shaping and Implementing the Pandemic Agreement's PABS Annex in an Era of Fragmentation, Globalization and Health 2026, doi:[10.1186/s12992-026-01190-3](https://doi.org/10.1186/s12992-026-01190-3); 40953844 — National IHR authorities in the 2024 amendments to WHO's International Health Regulations: transforming the opportunity into reality, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-019284](https://doi.org/10.1136/bmjgh-2025-019284). Plus WHO World Health Day 2025 campaign (April 2025) and WHO World Health Day 2026 campaign (April 2026).

♦ ♦ ♦

World Health Day theme 2023

Asked in: Paper IV 14-Jul-2023 — RGUHS MD Community Medicine

Definition — World Health Day is the global health awareness day observed on 7 April every year under the sponsorship of the World Health Organization (WHO), marking the date in 1948 on which the WHO Constitution came into force; a theme is chosen each year to focus world attention on one specific aspect of public health (Park 28e, Ch 25, p 1055). The theme for 7 April 2023 was "Health For All". It was chosen because 2023 was the 75th anniversary year of the WHO (1948–2023), an occasion the organisation used to call for a renewed drive for **health equity** (Sunder Lal, Ch 18, p 775).

World Health Day — origin and purpose

Item	Fact
Date observed	7 April, every year
Why 7 April	Date the WHO Constitution came into force, 1948 — the formal birth of the WHO (Park 28e, Ch 25, p 1055)
Who decided	The First World Health Assembly (1948) resolved to observe 7 April annually, with effect from 1950 (Park Plus, Ch 24 (Miscellaneous), p 381)
Sponsor	WHO — a specialised, non-political health agency of the United Nations, headquarters Geneva ; 194 member states and two associate members (Park 28e, Ch 25, p 1056)
Purpose	To draw worldwide attention each year to a subject of major importance to global health, and to mobilise governments, non-governmental organisations and communities around it (Park Plus, Ch 24 (Miscellaneous), p 381)
Status	One of the eight official global health campaign days of the WHO , alongside World Tuberculosis Day, World Immunization Week, World Malaria Day, World No Tobacco Day, World AIDS Day, World Blood Donor Day and World Hepatitis Day (Park Plus, Ch 24 (Miscellaneous), p 381)
Observed by	All member states of the WHO

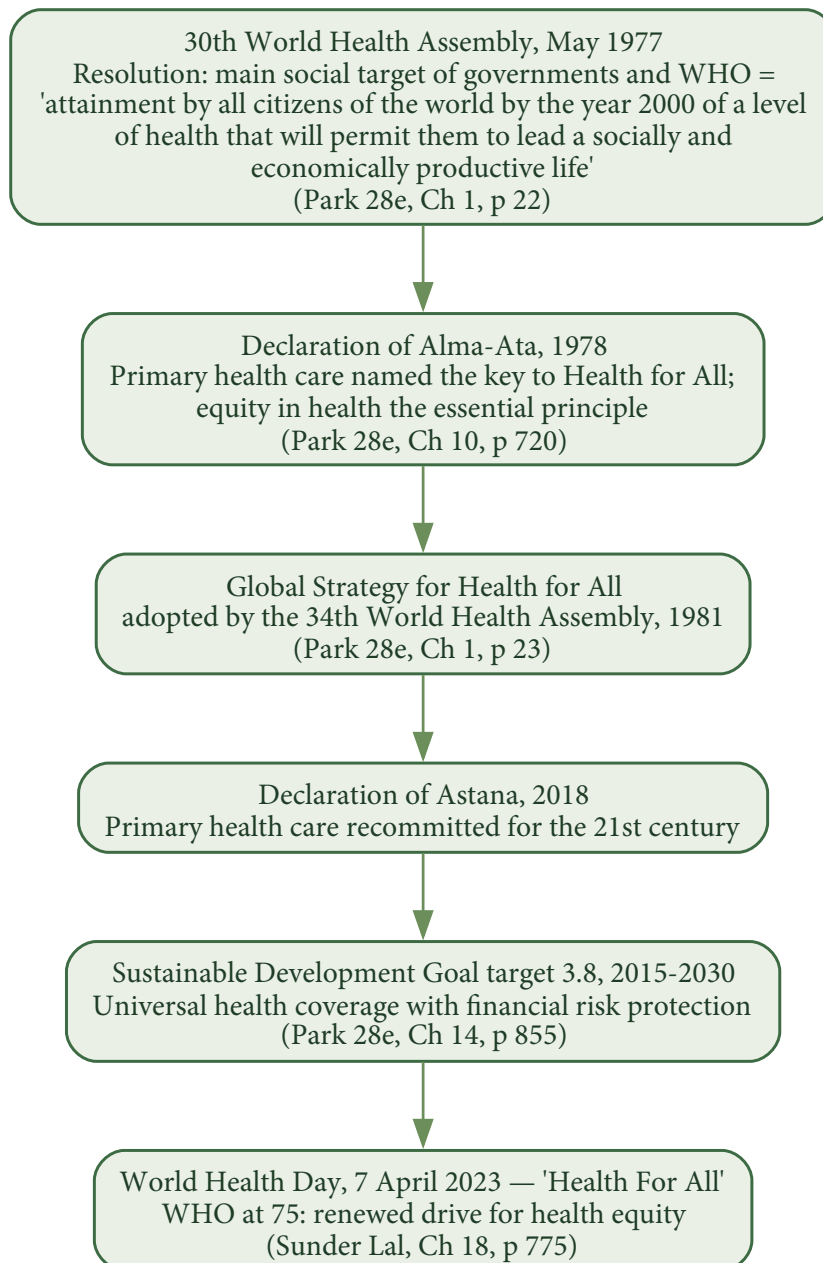
Theme 2023 — "Health For All": the message

- **Anniversary framing.** 2023 completed 75 years of the WHO, and the day looked back on the gains since 1948 — smallpox, the only disease eradicated globally, confirmed eradicated by the World Health Assembly in **May 1980** (Park 28e, Ch 5, p 171), and poliomyelitis now endemic in only Afghanistan and Pakistan (Park 28e, Ch 2, p 58).
- **Health as a right, not a privilege.** The theme restates the founding principle in the preamble of the WHO Constitution — *"the enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, economic or social condition"* (Park 28e, Ch 25, p 1055).
- **Equity is the operative word.** On its 75th anniversary the WHO explicitly called for a renewed drive for **health equity** (Sunder Lal, Ch 18, p 775) — the fundamental principle of the Health for All strategy, meaning an equal health status for people and countries ensured by an equitable distribution of health resources (Park 28e, Ch 10, p 720).
- **Route to Health for All = primary health care.** The theme deliberately re-invoked the **Declaration of Alma-Ata (1978)**, which reaffirmed Health for All as the major social goal of

governments and named primary health care as the best approach to reach it, especially for the vast majority of underserved rural people and the urban poor (Park 28e, Ch 10, p 720).

- **Measurable end-point = universal health coverage (UHC).** "Health For All" in 2023 was operationalised as **Sustainable Development Goal target 3.8** — achieve universal health coverage, including financial risk protection, access to quality essential health-care services, medicines and vaccines for all, by 2030 (Park 28e, Ch 14, p 855).
- **The gap the theme addresses.** At least half the world's population still lacks access to essential health services; as a step towards target 3.8 the WHO set out to extend coverage to an additional **one billion people** (Oxford Textbook of Public Health 7e, Ch 107, pp 4044–4045).

Health for All — the concept the theme revived



From Health for All (1977) to World Health Day 2023 — "Health For All"

Universal health coverage — the operational content of "Health For All"

UHC means that all people have access to needed health services of sufficient quality without incurring financial hardship (Oxford Textbook of Public Health 7e, Ch 107, pp 4044–4045); it spans health promotion, disease prevention, treatment, rehabilitation, long-term and palliative care (Oxford Textbook of Public Health 7e, Ch 103, pp 3915–3916).

Dimension of coverage	What it asks	Corresponding action
Breadth — population covered	Who is covered?	Extend cover to the uncovered and the vulnerable
Depth — services covered	Which services are covered?	Scale up primary health care and the essential services package
Height — proportion of cost covered	How much of the cost is met by pooled public funds?	Raise public expenditure on health, reduce out-of-pocket (OOP) spending

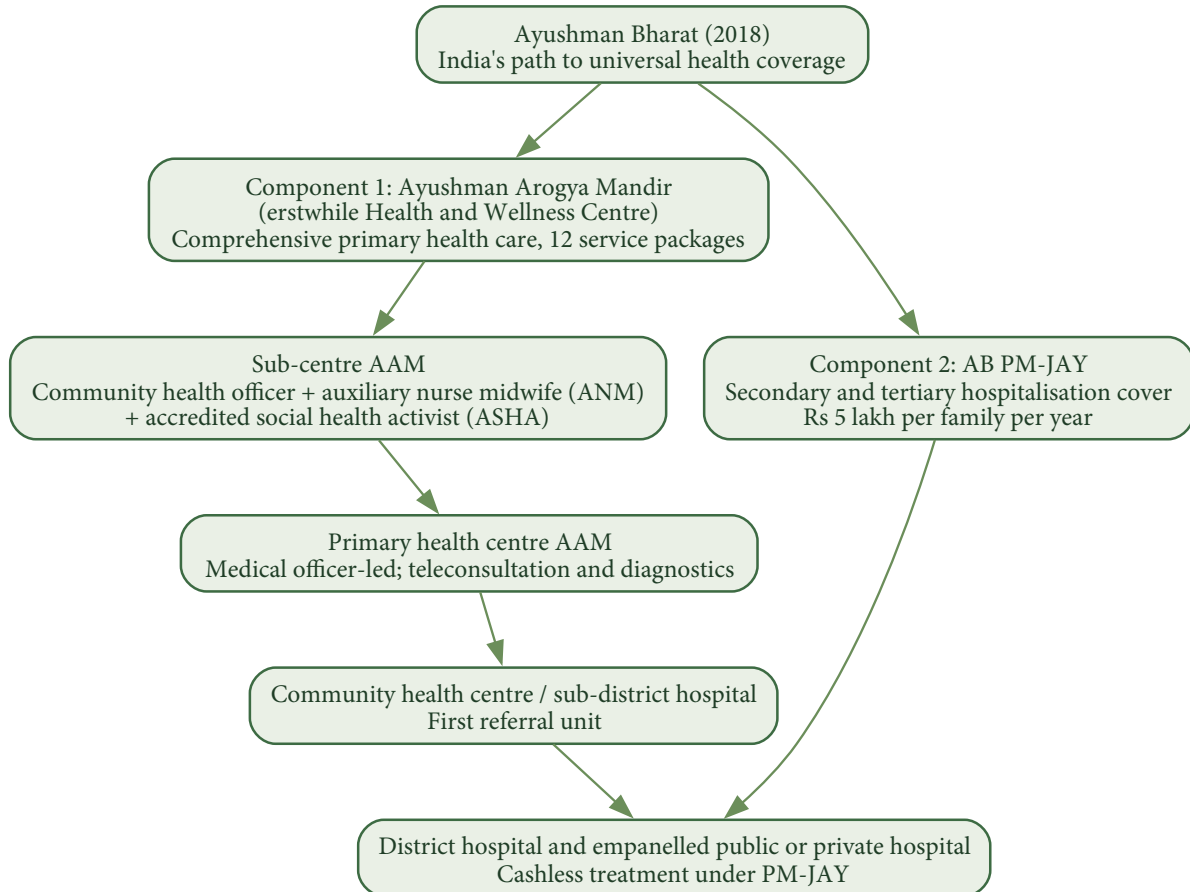
(Sunder Lal, Ch 17, pp 739–741)

WHO's own targets for the period (13th General Programme of Work, WHO 2018): one billion more people benefiting from universal health coverage; one billion more people better protected from health emergencies; one billion more people enjoying better health and well-being — the "triple billion" targets (Oxford Textbook of Public Health 7e, Ch 107, pp 4044–4045).

India's response — how "Health For All" is being delivered

Level	Instrument	Content
Policy	National Health Policy 2017	Goal: progressive achievement of universal health coverage ; raise government health expenditure from 1.15% to 2.5% of gross domestic product (GDP) by 2025; increase state sector health spending to more than 8% of the state budget by 2020; reduce the proportion of households facing catastrophic health expenditure by 25% by 2025 (Park 28e, Ch 23, p 1030)
Service delivery	Ayushman Arogya Mandir (AAM) — erstwhile Ayushman Bharat Health and Wellness Centre	Launched 2018; sub-centres and primary health centres upgraded to deliver comprehensive primary health care through an expanded package of 12 services — headed by care in pregnancy and childbirth, neonatal and infant care, family planning and reproductive health, management of communicable diseases under the national health programmes, and screening, prevention and management of non-communicable diseases (Park 28e, Ch 10, pp 730–731); first centre inaugurated 14 April 2018, Bijapur district, Chhattisgarh ; 1,72,148 centres operational as on 31 March 2024 (Park 28e, Ch 7, p 570)

Level	Instrument	Content
Financial protection	Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)	Health cover of ₹5 lakh per family per year for secondary and tertiary hospitalisation to about 10.74 crore poor and vulnerable families identified from Socio-Economic Caste Census data (Park 28e, Ch 7, p 570; Ch 10, p 745); extended to senior citizens aged 70 years and above irrespective of socio-economic status , who receive a distinct card and, if in an already-covered family, an additional top-up of up to ₹5 lakh per year (Park 28e, Ch 7, pp 570–572)



Ayushman Bharat — India's two-component route to Health for All

Recent World Health Day themes (high-yield recall)

Year	Theme
2015	Food safety
2016	Halt the rise: beat diabetes
2017	Depression: let's talk
2018	Universal health coverage: everyone, everywhere (WHO at 70)
2019	Universal health coverage: everyone, everywhere
2020	Support nurses and midwives — Year of the Nurse and Midwife

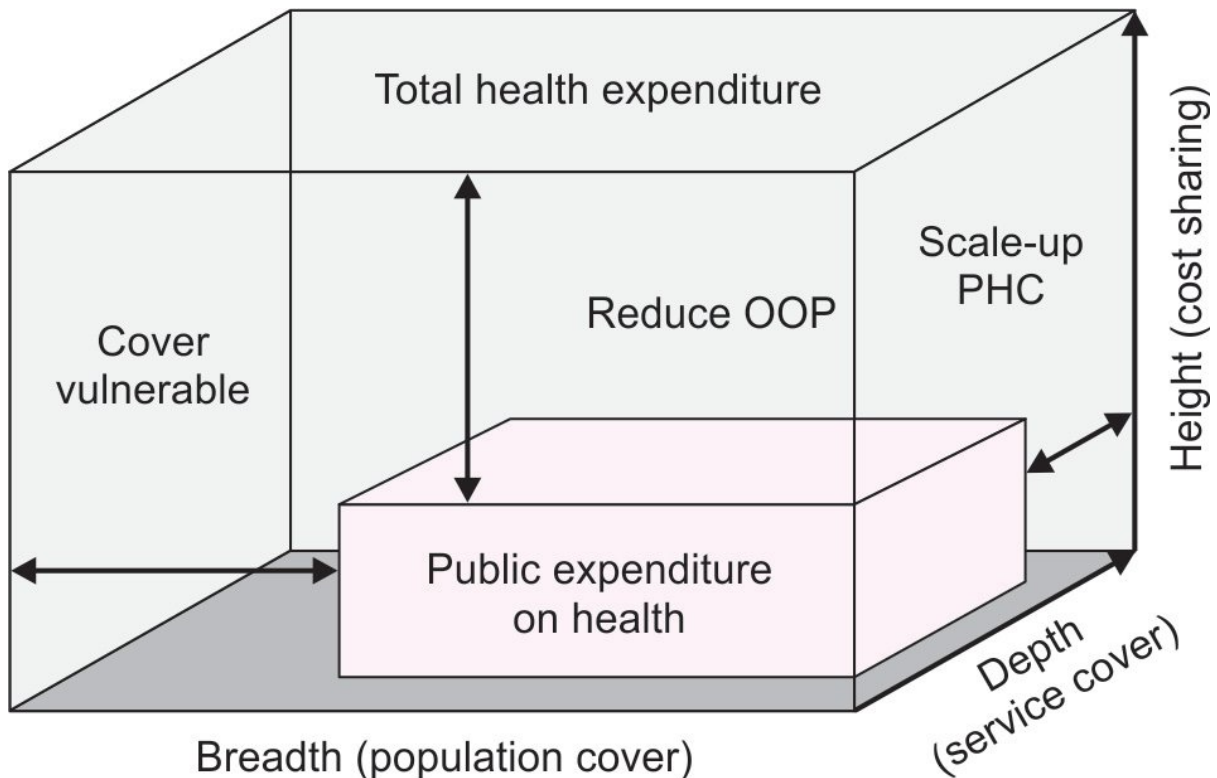
Year	Theme
2021	Building a fairer, healthier world
2022	Our planet, our health
2023	Health For All (WHO 75th anniversary)

(Park Plus, Ch 24 (Miscellaneous), p 381; Sunder Lal, Ch 1, p 35; Sunder Lal, Ch 18, p 775)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **universal health coverage cube** and label:

- ▶ Outer box — **Total health expenditure**
- ▶ Inner solid block — **Public expenditure on health**
- ▶ Horizontal axis — **Breadth (population cover)** → arrow labelled *Cover vulnerable*
- ▶ Depth axis — **Depth (service cover)** → arrow labelled *Scale up primary health care*
- ▶ Vertical axis — **Height (cost sharing)** → arrow labelled *Reduce out-of-pocket expenditure*
- ▶ Footnote key — OOP = out of pocket; PHC = primary health care



OOP: Out of pocket
PHC: Primary health care

Attaining universal coverage — the three dimensions of the coverage cube (Sunderlal, p.740)

High-yield exam pointers

- ▶ **7 April 1948** — WHO Constitution in force; observed as World Health Day **from 1950** (First World Health Assembly resolution); one of **eight** official WHO global health campaign days.
- ▶ **2023 theme** = "Health For All", marking **WHO at 75**; the campaign call was a renewed drive for **health equity**.
- ▶ Health for All definition (30th World Health Assembly, 1977): "*attainment of a level of health that will enable every individual to lead a socially and economically productive life*"; essential principle = **equity in health**.
- ▶ **SDG target 3.8** = universal health coverage with financial risk protection by 2030; WHO **triple billion** targets — 1 billion more under UHC, 1 billion more protected from emergencies, 1 billion more with better health and well-being.
- ▶ Scale of the financial-hardship problem the theme targets: about **800 million people** spend at least **10% of the household budget** on health and nearly **100 million** are pushed into extreme poverty by health costs each year (Oxford Textbook of Public Health 7e, Ch 107).
- ▶ Indian figures with their year: government health expenditure **1.15% → 2.5% of GDP by 2025** (National Health Policy 2017); **1,72,148 Ayushman Arogya Mandirs operational as on 31 March 2024**; PM-JAY ₹5 lakh per family per year for **10.74 crore families**.
- ▶ WHO has **194 member states**, headquarters Geneva, six regional offices; the South-East Asia Regional Office is at **New Delhi**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the indexed corpus prints themes only up to 2018 (Park Plus, p 381) plus 2019 (Sunder Lal, Ch 1) and 2020 (Sunder Lal, Ch 18), so the 2021 ("Building a fairer, healthier world"), 2022 ("Our planet, our health") and 2023 ("Health For All") theme strings are stated from standard public health knowledge, the corpus recording only the WHO 75th-anniversary health-equity call for 2023; note also the renaming of Ayushman Bharat Health and Wellness Centres to **Ayushman Arogya Mandirs**, so write the current name first and the former name once in parentheses.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, PIB India, MoHFW, WHO, National Health Accounts 2022-23, NFHS-6

Programme and policy updates (India)

- ▶ **Renaming is the examinable change:** the Union Health Ministry renamed Ayushman Bharat Health and Wellness Centres as **Ayushman Arogya Mandir (AAM)** in **November 2023**, adding the tagline **Arogyam Paramam Dhanam** ("health is the supreme wealth"). The rebranding carried a stated shift from illness-focused care to holistic wellness on AYUSH lines — yoga, lifestyle counselling, mental health support, nutrition and elderly care — was funded at ₹3,000 per facility (about ₹48 crore in all), and required a photograph of each renamed centre to be uploaded to the portal; the substantive gain is limited where staffing and facilities do not change

with the name. Write the current name first and the former name once in parentheses. [PMID 42257165] [MoHFW 2023-Nov]

- **Scale of the primary-care arm of Health for All:** more than 1.75 lakh AAMs operational; 1,16,978 community health officers in position and 1,35,343 Jan Arogya Samitis constituted as on 30 November 2024; centres delivering the *full* expanded service package rose from 50% (March 2023) to 82% (November 2024); free care is notified as 106 medicines + 14 diagnostics at sub-centre AAM and 172 medicines + 63 diagnostics at PHC-level AAM. India's UHC service coverage index rose from 57 (2015) to 63 (2021). Persisting constraints are workforce shortage, infrastructure gaps and supply-chain failures. [PMID 41816146]

Current national data

- **National Health Accounts 2022-23 (released 29 May 2026)** — the tenth NHA report, and the figures to quote with their year: government health expenditure **1.43% of GDP** (1.15% in 2013-14) and ₹3.85 lakh crore (₹1.30 lakh crore in 2013-14); government share of total health expenditure 43.7% (28.6%); social security expenditure on health 9.9% (6.0%); per-capita government health spending ₹2,786 (₹1,042). **Out-of-pocket expenditure = 43.4% of total health expenditure**, down from 64.2% in 2013-14. Examiner trap: the widely quoted **39.4% belongs to 2021-22** and reflected one-off COVID-era government spending, so 2022-23 reads higher without reversing the long-term fall. Government health expenditure remains well short of the National Health Policy 2017 target of **2.5% of GDP by 2025**. [PIB 2026-May]
- **NFHS-6 (2023-24) was released on 29 May 2026** — 6.79 lakh households across 715 districts, IIPS Mumbai the nodal agency. Against NFHS-5 (2019-21): households with any member covered by a **health insurance or financing scheme 41.0% → 60.2%** (the most directly UHC-relevant indicator in the survey); institutional births **88.6% → 90.6%**; four or more antenatal visits **58.5% → 65.2%**; children 12-23 months fully vaccinated **83.8% → 87.1%**; rotavirus vaccine coverage **36.4% → 85.4%**; stunting under-5 **35.5% → 29.3%**; severe wasting **7.7% → 5.2%**; current use of any contraceptive method **66.7% → 69.1%**; total fertility rate **2.0**. Always name the round and its years alongside the figure. [PIB 2026-May]
- **Coverage has outrun financial protection — the live criticism of India's route to Health for All.** Four years after launch, in representative Chhattisgarh household surveys, AB PM-JAY enrolment was associated with neither greater inpatient utilisation nor lower out-of-pocket or catastrophic expenditure; among enrollees admitted to *private* hospitals, **78.1% (2021) and 70.9% (2022)** still incurred catastrophic expenditure at the 10%-of-consumption threshold, with hospital type the dominant predictor [PMID 39135015]. A pan-India instrumental-variable analysis of National Sample Survey rounds 60, 71 and 75 likewise found **no statistically significant effect of publicly funded health insurance on out-of-pocket expenditure per hospitalisation** [PMID 41913189]. Bayesian projections for South Asia put India's composite coverage index past the SDG 80% mark by 2030 (72% in 2016, 76% in 2020) while catastrophic health expenditure climbs to about **26.3% by 2030**, rural exceeding urban by roughly 6 percentage points [PMID 41192927].

Elimination, surveillance and disease status

- **The elimination deadlines the textbook edition predates.** India reset its measles-rubella elimination target to **2026** after missing 2015, 2020 and 2023, and launched the National Zero

Measles-Rubella Elimination Campaign 2025-26 during World Immunization Week in April 2025 — MR first-dose coverage 93.7% and second-dose 92.2% (2024-25), with 332 districts reporting zero measles and 487 zero rubella cases in January-March 2025 [PIB 2025-Apr]. For tuberculosis, the WHO Global Tuberculosis Report 2025 records India's incidence falling from **237 to 187 per lakh between 2015 and 2024** (about 21%, roughly twice the global pace) — real progress, but the National TB Elimination Programme's **2025 elimination deadline has not been met**, so state the target year *and* the current status [WHO 2025].

Guideline and legislative updates

- ▶ **The WHO Pandemic Agreement was adopted by the 78th World Health Assembly on 20 May 2025** — only the second treaty negotiated under Article 19 of the WHO Constitution, after the Framework Convention on Tobacco Control (2003). It embeds a **One Health** approach (Article 5) and creates a **Pathogen Access and Benefit-Sharing (PABS)** system under which participating manufacturers make **20% of real-time production** of pandemic health products available to WHO, at least half of that as donations and the remainder at affordable prices. The PABS annex was deferred to the 79th World Health Assembly; the treaty opens for signature only after that annex is adopted and needs **60 ratifications** to enter into force. [PMID 41035553] [PMID 40589223]
- ▶ **The 2024 amendments to the International Health Regulations (2005)**, agreed at the 77th World Health Assembly on 1 June 2024, **entered into force on 19 September 2025**. They add a **"pandemic emergency"** tier above the previously binary PHEIC alert, oblige every State Party to designate a **National IHR Authority**, and add equity and solidarity provisions on access to medical products and financing; 11 of the 196 States Parties rejected the amendments and remain bound by the earlier text. [WHO 2025-Sep] [PMID 40883832] [PMID 40589223]

Global and WHO developments

- ▶ **World Health Day themes after 2023, to extend the recall table:** 2024 — *"My health, my right"*; 2025 — *"Healthy beginnings, hopeful futures"* (ending preventable maternal and newborn deaths and prioritising women's longer-term health); 2026 — *"Together for health. Stand with science"*, a One Health-framed campaign anchored to the One Health Summit and the first Global Forum of WHO Collaborating Centres. Update the denominator too: as of **2021 about 4.5 billion people** — over half the world's population — lacked full coverage of essential health services, **over 1 billion** faced catastrophic health spending and **344 million** were pushed into extreme poverty by health costs. [PMID 40951390] [PMID 40376441] [PMID 41192927] [WHO 2026-Apr]
- ▶ **The financing base of "Health For All" has weakened sharply.** The United States — historically WHO's largest donor and about **22% of assessed contributions** — began withdrawal by executive order on **20 January 2025** [PMID 40796848]. WHO then cut its **2026-27 programme budget from US\$ 5.3 billion to US\$ 4.2 billion** (about 21%), shrank headquarters from **11 divisions to 6** and **67 departments to 36**, and the 78th World Health Assembly approved a further **20% rise in assessed contributions** to part-offset the gap. This is the current "threat to Health for All" line for a 2026 answer. [WHO 2025-May]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1, Ch 2, Ch 5, Ch 7, Ch 10, Ch 14, Ch 23, Ch 25); Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 17, Ch 18); Oxford Textbook of Public Health, 7e (Ch 103 — Transformative learning for the future health workforce; Ch 107 — Global health in the era of sustainable development); Park Plus review companion (Ch 24, Miscellaneous — World Health Day and its themes); Declaration of Alma-Ata 1978 and Declaration of Astana 2018; National Health Policy 2017; Sustainable Development Goal target 3.8. Coverage note: the corpus carries the World Health Day mechanism and the WHO 75th-anniversary health-equity call but does not print the 2021, 2022 or 2023 theme strings, which are stated as standard, universally accepted public health fact.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmmpc.jfmmpc.1966.24](https://doi.org/10.4103/jfmmpc.jfmmpc.1966.24); 41816146 — Ayushman Arogya Mandir and India's progress toward Universal Health Coverage: a narrative review, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmmpc.jfmmpc.478.25](https://doi.org/10.4103/jfmmpc.jfmmpc.478.25); 39135015 — The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana after four years of implementation — is it making an impact on quality of inpatient care and financial protection in India?, *BMC Health Services Research* 2024, doi:[10.1186/s12913-024-11393-2](https://doi.org/10.1186/s12913-024-11393-2); 41913189 — Do publicly funded health insurance schemes reduce out-of-pocket expenditure? Evidence from India, *BMC Health Services Research* 2026, doi:[10.1186/s12913-026-14456-8](https://doi.org/10.1186/s12913-026-14456-8); 41192927 — Progress towards universal health coverage in South Asia, 2000–2030: an examination of the twin elements of primary healthcare provision and financial protection, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-020052](https://doi.org/10.1136/bmjgh-2025-020052); 41035553 — Implementation of the WHO Pandemic Agreement, *Bulletin of the World Health Organization* 2025, doi:[10.2471/BLT.25.294146](https://doi.org/10.2471/BLT.25.294146); 40589223 — Editorial: the 2025 World Health Assembly Pandemic Agreement and the 2024 amendments to the International Health Regulations combine for pandemic preparedness, *Medical Science Monitor* 2025, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411); 40883832 — To tier or not to tier: the institutionalization of the World Health Organization's power to determine pandemic emergency in the amended International Health Regulations (2005), *Global Health Research and Policy* 2025, doi:[10.1186/s41256-025-00438-6](https://doi.org/10.1186/s41256-025-00438-6); 40951390 — Editorial: World Health Day 2024 — "My health, my right", *Frontiers in Public Health* 2025, doi:[10.3389/fpubh.2025.1679401](https://doi.org/10.3389/fpubh.2025.1679401); 40376441 — World Health Day 2025: time to change mindset beyond global commitment to maternal health and women's well-being, *China CDC Weekly* 2025, doi:[10.46234/ccdcw2025.074](https://doi.org/10.46234/ccdcw2025.074); 40796848 — The United States withdrawal from the World Health Organization: its implications for global health governance, *Globalization and Health* 2025, doi:[10.1186/s12992-025-01137-0](https://doi.org/10.1186/s12992-025-01137-0). Non-journal sources used: MoHFW directive renaming Ayushman Bharat Health and Wellness Centres as Ayushman Arogya Mandir, November 2023; PIB India — National Health Accounts estimates for India 2022-23, 29 May 2026; PIB India — National Family Health Survey-6 (2023-24), 29 May 2026; PIB India — National Zero Measles-Rubella Elimination Campaign 2025-26, April 2025; WHO — amended International Health Regulations (2005) in force 19 September 2025; WHO Global Tuberculosis Report 2025; WHO

World Health Day campaigns 2024, 2025 and 2026; Seventy-eighth World Health Assembly decisions on the 2026-27 programme budget and assessed contributions, May 2025.



World Health Day theme 2019

Asked in: Paper IV 19-Nov-2021 · Paper IV 18-Nov-2019 · Paper IV May 2019 (3×) — RGUHS MD Community Medicine

Definition — World Health Day is the global health awareness day observed on **7 April** every year, the date on which the Constitution of the World Health Organization (WHO) came into force in 1948; a theme is chosen each year to focus world attention on one specific aspect of public health (Park 28e, Ch 25, p 1055). The First World Health Assembly (1948) resolved that the day be observed annually with effect from **1950** (Park Plus, Ch 24, p 381). The theme for **World Health Day 2019** was "**Universal health coverage: everyone, everywhere**" (Sunder Lal, Ch 1, p 35).

1. The observance at a glance

Item	Detail
Date	7 April, every year
Commemorates	Entry into force of the WHO Constitution on 7 April 1948; the Constitution was drafted in 1946 by the Technical Preparatory Committee under <i>Rene Sand</i> and approved by an International Health Conference of 51 nations at New York (Park 28e, Ch 25, p 1055)
First observed	1950, on a resolution of the First World Health Assembly, 1948 (Park Plus, Ch 24, p 381)
Observed by	WHO and all its Member States — 194 member states and 2 associate members (Park 28e, Ch 25, p 1055; Park Plus, Ch 24, p 381)
Status	One of eight official WHO global health campaigns , with World Tuberculosis Day, World Immunization Week, World Malaria Day, World No Tobacco Day, World AIDS Day, World Blood Donor Day and World Hepatitis Day (Park Plus, Ch 24, p 381)
Activities	International, regional and local events built round the year's theme; governments, non-governmental organisations and the media join in advocacy on the same theme (Park Plus, Ch 24, p 381)
Theme, 2019	" Universal health coverage: everyone, everywhere " (Sunder Lal, Ch 1, p 35)

2. The 2019 theme — universal health coverage (UHC)

Universal health coverage means that all people can access the health services they need — health promotion, disease prevention, treatment, rehabilitation, long-term and palliative care — of sufficient quality, without incurring financial hardship (Oxford Textbook of Public Health 7e, Ch 103, pp 3915–16; Ch 107, pp 4044–45). It carries **twin imperatives**: people must *receive the services they need* and must *not be impoverished by paying for them* (Oxford 7e, Ch 107, pp 4044–45). The 2019 slogan

"everyone, everywhere" restates these two imperatives as a demand for equity — coverage that reaches the vulnerable and the geographically remote, not only national averages.

UHC is conventionally shown as a cube with three dimensions that must each be expanded:

Dimension of the UHC cube	Question it answers	What must expand	Indian instrument
Breadth — population cover	<i>Who</i> is covered?	Cover the vulnerable and the still-uncovered	Ayushman Arogya Mandir for the whole population of the area; PM-JAY for 10.74 crore poor and vulnerable families
Depth — service cover	<i>Which services</i> are covered?	Scale up primary health care from a selective to a comprehensive package	The 12 comprehensive primary health care packages
Height — cost sharing	<i>What proportion of the cost</i> is met from pooled public funds?	Raise public expenditure on health, reduce out-of-pocket (OOP) payment	Free drugs and diagnostics at the Ayushman Arogya Mandir; cashless hospitalisation up to Rs 5 lakh per family per year

(Sunder Lal, Ch 17, pp 739–41, after WHO 2009)

The system elements that have to be strengthened to deliver on all three axes are shown below.

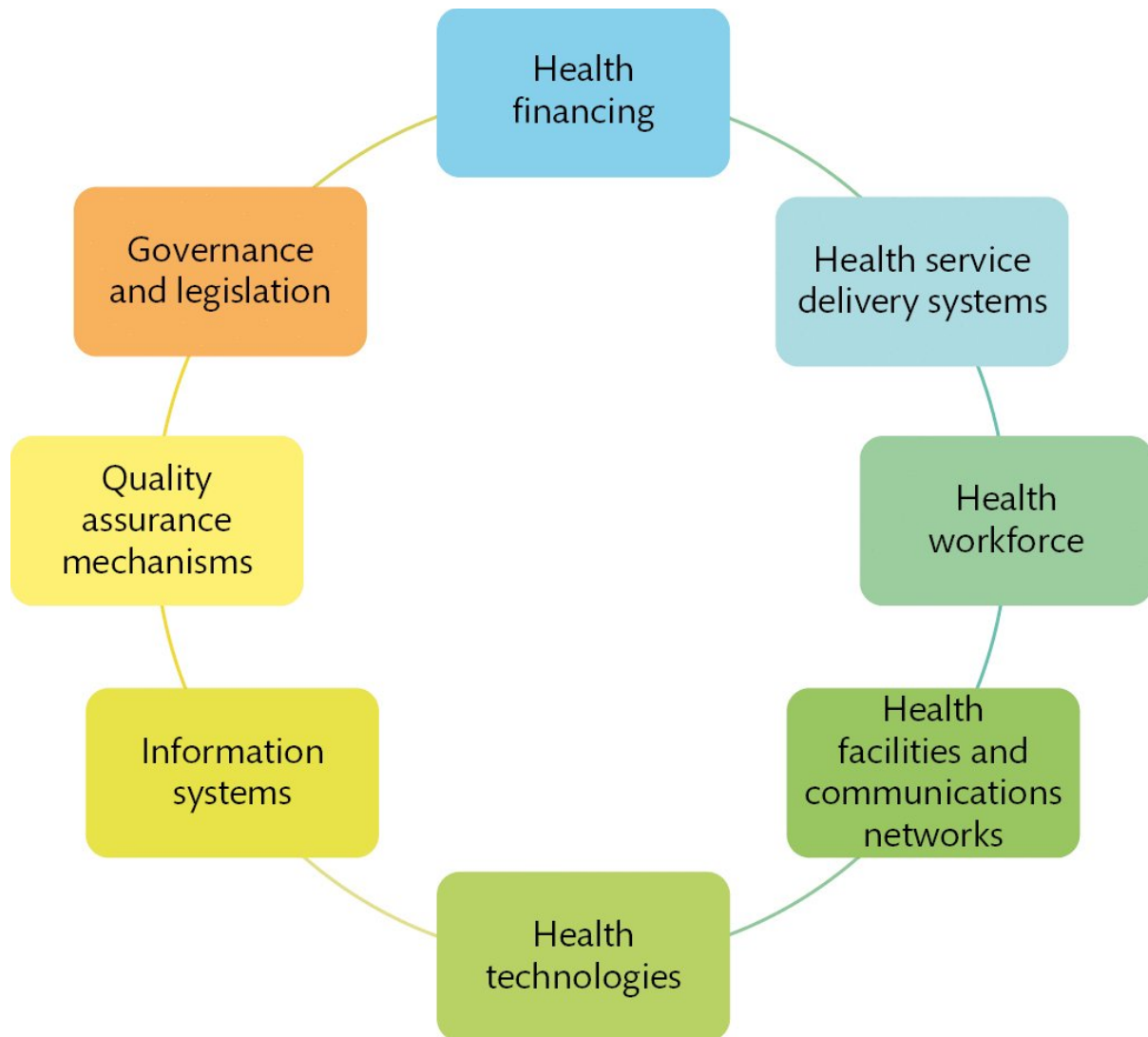


Fig 3.5 — 6 Components of Universal Health Coverage. (Oxford public health 7th ed, p.844)

3. Why universal health coverage was chosen as the 2019 theme

- ▶ **It is the central health target of the Sustainable Development Goals.** SDG target 3.8 is to "achieve universal health coverage, including financial risk protection, access to quality essential health-care services and access to safe, effective, quality and affordable essential medicines and vaccines for all"; the UHC target is the one on which achievement of all the other SDG-3 targets rests (Park 28e, Ch 1, p 23; Ch 14, p 853).
- ▶ **The service-access gap is enormous.** At least half the world's population still lacks access to essential health services (Oxford 7e, Ch 107, pp 4044–45).
- ▶ **The financial-hardship gap is equally large.** About 800 million people spend at least 10 per cent of the household budget on health, and for almost 100 million people each year these payments are large enough to push them into extreme poverty (WHO/World Bank 2017, quoted in Oxford 7e, Ch 107, pp 4044–45). Each year roughly 100 million people slide into poverty because of medical payments and another 150 million are forced to spend nearly half their income on medical expenses (Oxford 7e, Ch 8, p 347).
- ▶ **UHC is now measurable, and therefore actionable.** Progress is tracked on two indicators — SDG 3.8.1, coverage of tracer interventions (full child immunisation, antiretroviral therapy,

tuberculosis treatment, hypertension treatment, skilled attendance at birth), and **SDG 3.8.2**, the fraction of the population facing catastrophic or impoverishing out-of-pocket health spending (Park 28e, Ch 14, pp 853, 858).

- **It continues an unbroken policy line.** The Alma-Ata Declaration of 1978 on primary health care was renewed by the **Astana Declaration on Primary Health Care, 2018**, which restated comprehensive primary health care as the route to UHC (Park 28e, Ch 1, p 23; Oxford 7e, Ch 47, p 1774).
- **It anchors WHO's own five-year strategy.** WHO's General Programme of Work sets the "triple billion" vision — one billion more people benefiting from universal health coverage, one billion more better protected from health emergencies, and one billion more enjoying better health and well-being; UHC coverage is to be expanded to an additional one billion people by 2023 (Oxford 7e, Ch 107, pp 4044–45).
- **The political moment was right.** The WHO Director-General's message of the period was that "universal health coverage is a political choice", and later in the same year the United Nations General Assembly adopted a landmark political declaration on universal health coverage on **23 September 2019** (Oxford 7e, Ch 101, pp 3857–58, 3899).

4. India and the 2019 theme — the national response

India's own health-financing profile explains why the theme was directly relevant to it:

Indicator	Value (year)
Total health spending	3.16 per cent of GDP (2018–19)
Government health spending	1.28 per cent of GDP, about 40.6 per cent of total health spending
People pushed below the poverty line each year by catastrophic health expenditure	63 million per year (National Sample Survey 75th round)
Population seeking care in the public sector	32 per cent rural, 26 per cent urban
Share of care borne by the private sector	70 per cent of outpatient and 60 per cent of inpatient care

(Sunder Lal, Ch 16, p 727; Ch 17, p 742)

Ayushman Bharat (2018) is the vehicle by which India moves towards UHC; its two inter-related components together expand access, universality and equity (Sunder Lal, Ch 17, pp 739–41).

Component	Launch	Content	Coverage attained
Ayushman Bharat Health and Wellness Centres , now renamed Ayushman Arogya Mandir (AAM)	First centre inaugurated 14 April 2018 , Bijapur district, Chhattisgarh	All 1.5 lakh sub-centres and primary health centres upgraded to deliver 12 comprehensive primary health care packages with free essential drugs and diagnostics; sub-centre team headed by a Community Health Officer (B.Sc./GNM nurse or trained Ayurveda practitioner); "time to care" not more than 30 minutes of travel	1,72,148 centres operational as on 31 March 2024

Component	Launch	Content	Coverage attained
Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)	23 September 2018; Rashtriya Swasthya Bima Yojana was subsumed into it	Cashless secondary and tertiary care hospitalisation cover of Rs 5 lakh per family per year; the largest health protection scheme in the world	About 10.74 crore poor and vulnerable families (≈50 crore people), identified from Socio-Economic Caste Census data

(Park 28e, Ch 7, pp 541, 570; Ch 10, p 745; Sunder Lal, Ch 17, pp 742, 746)

The 12 comprehensive primary health care packages delivered at an Ayushman Arogya Mandir (Sunder Lal, Ch 17, p 749; Park 28e, Ch 7, p 571):

- ▶ Care in pregnancy and childbirth
- ▶ Neonatal and infant health care services
- ▶ Childhood and adolescent health care services
- ▶ Family planning, contraceptive services and other reproductive health care services
- ▶ Management of communicable diseases — the national health programmes
- ▶ General outpatient care for acute simple illnesses and minor ailments
- ▶ Screening, prevention, control and management of non-communicable diseases
- ▶ Care for common ophthalmic and ear, nose and throat problems
- ▶ Basic oral health care
- ▶ Elderly and palliative health care services
- ▶ Emergency medical services including burns and trauma
- ▶ Screening and basic management of mental health ailments

The **National Health Mission** remains the primary vehicle for moving towards universal health coverage, promoting universal access to a cashless continuum of care from primary to tertiary level with seamless referral linkage, and the **revised Indian Public Health Standards (IPHS) 2022** set the facility benchmarks against which that goal is measured (Sunder Lal, Ch 17, pp 739–42; Park 28e, Ch 10, p 728). One caution is stressed in the Indian literature: UHC must be built on universal access to the determinants of health — safe drinking water, sanitation, wholesome nutrition, basic education and safe housing — since coverage without these is unworkable (Sunder Lal, Ch 17, p 736).

5. Selected World Health Day themes (recall table)

Year	Theme
2014	Vector-borne diseases: small bite, big threat
2015	Food safety
2016	Halt the rise: beat diabetes
2017	Depression: let's talk
2018	Universal health coverage (WHO's 70th year)

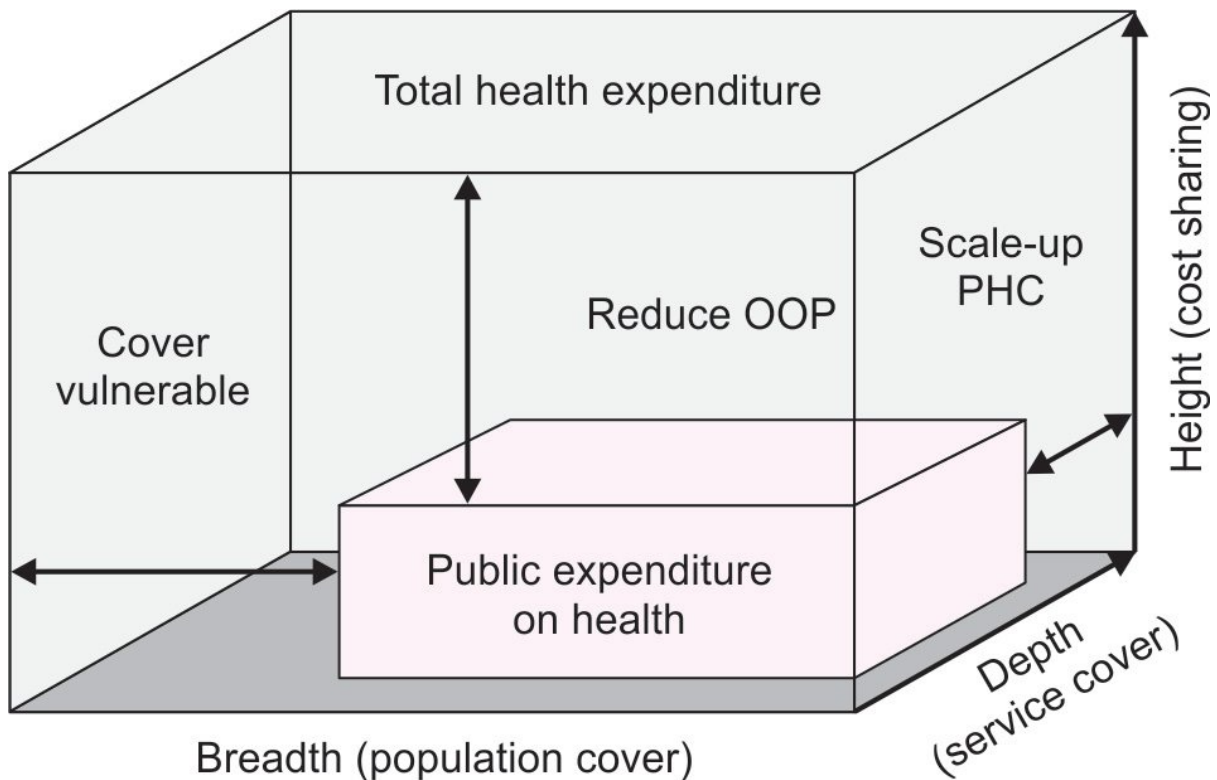
Year	Theme
2019	Universal health coverage: everyone, everywhere
2020	Year of the nurse and the midwife

(Park Plus, Ch 24, p 381; Sunder Lal, Ch 1, p 35; Ch 18, p 775)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

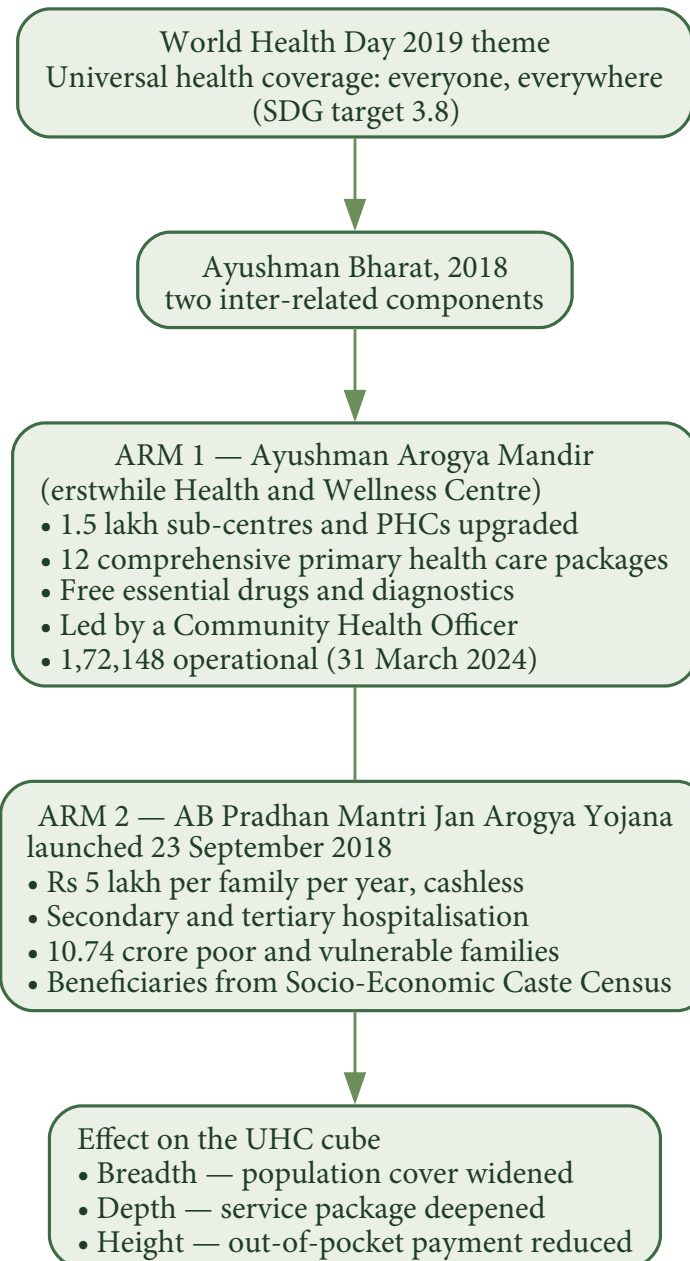
Real plate — the universal health coverage cube (the visual core of the 2019 theme):

Exact labels to reproduce by hand: outer box = **Total health expenditure**; inner solid box = **Public expenditure on health**; horizontal axis = **Breadth (population cover)** with arrow marked **Cover vulnerable**; depth axis = **Depth (service cover)** with arrow marked **Scale-up PHC**; vertical axis = **Height (cost sharing)** with arrow marked **Reduce OOP**; key = **OOP: out of pocket**; **PHC: primary health care**.

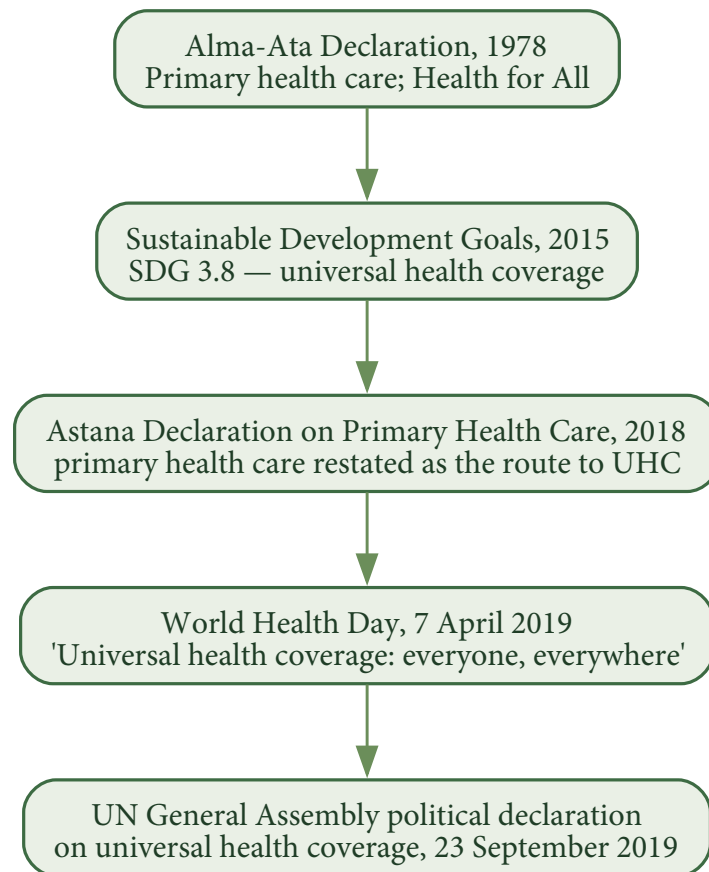


OOP: Out of pocket
PHC: Primary health care

Fig 17.2 — Attaining universal coverage (Source: WHO, 2009) (Sunderlal, p.740)



India's operational route to the 2019 theme — the two arms of Ayushman Bharat



Where World Health Day 2019 sits in the policy chain from Alma-Ata to universal health coverage

High-yield exam pointers

- ▶ **7 April 1948** — WHO Constitution in force; day observed from 1950; one of WHO's **eight** official global health campaigns; **2019 theme** = "**Universal health coverage: everyone, everywhere**".
- ▶ **UHC cube mnemonic** — **B-D-H**: Breadth (population cover) → cover the vulnerable; Depth (service cover) → scale up primary health care; Height (cost sharing) → reduce out-of-pocket spending.
- ▶ **SDG 3.8** with its two indicators: **3.8.1** service-coverage index of tracer interventions; **3.8.2** population with catastrophic/impoverishing out-of-pocket health spending.
- ▶ Global figures to quote: **at least half** the world's population lacks essential health services; **800 million** spend ≥10 per cent of household budget on health; **~100 million** pushed into extreme poverty every year (WHO/World Bank 2017).
- ▶ Indian figures to quote: total health spending **3.16 per cent of GDP (2018–19)**, government share **1.28 per cent of GDP**; catastrophic expenditure pushes **63 million** below the poverty line per year (NSS 75th round).
- ▶ **Ayushman Bharat** = **2 arms**: Ayushman Arogya Mandir (12 primary care packages; 1,72,148 operational, 31 March 2024) + **PM-JAY** (Rs 5 lakh per family per year; **10.74 crore** families; launched **23 September 2018**).
- ▶ WHO's "**triple billion**" vision: one billion more with UHC, one billion more protected from health emergencies, one billion more enjoying better health and well-being.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved corpus carries the universal health coverage theme against both 2018 and 2019 — WHO ran the same campaign across its seventieth-anniversary year and the following year — so if the examiner's year is 2018 the same theme text applies. Ayushman Bharat Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir**, so write the current name first and the former name in parentheses once. Coverage note: the indexed corpus supplies World Health Day themes only up to 2020, so no theme after that year is quoted here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, World Health Assembly, National Health Accounts (India), NFHS-5, Comprehensive Annual Modular Survey 2022–23

Programme and policy updates (India)

- ▶ PM-JAY has in practice become India's surgical-care financing instrument rather than a general hospitalisation cover: about 65 per cent of listed procedures and 82 per cent of empanelled hospitals are surgical, and on scheme transaction data to May 2019 surgical packages took 61 per cent of claim volume and 75 per cent of claim value — 77 per cent of claim value at private hospitals against 66 per cent at public. The disparities sit on the supply and utilisation side: only 7 per cent of hospitals in Aspirational Districts offer cardiothoracic and vascular surgery against 17 per cent elsewhere, and men raised 23.6 per cent more surgical claims than women once obstetric and gynaecological procedures were excluded. The remedies asked for are disaggregated district-level monitoring, better-targeted enrolment of the poorest quintiles within the bottom 40 per cent, and a larger allocation to surgical services [PMID 41393311].
- ▶ The **Ayushman Bharat Digital Mission (ABDM)** with its ABHA health account is the third arm added since the textbook account of the two-arm scheme, and its uptake is the new access barrier: in Agra's government tertiary outpatient departments only 129,007 of 537,278 visits (23.96 per cent) generated an ABHA registration between September 2024 and April 2025; among the 425 attendees interviewed, 88 per cent wanted training and support, and digital health literacy was significantly lower in women (Mann–Whitney U, $p < 0.001$), in older attendees (Spearman $\rho = -0.230$) and in the less educated ($\rho = 0.480$ for education) [PMID 41787376].

Current national data

- ▶ **National Health Accounts, 2013–14 to 2021–22 — the financing crossover.** The government share of total health expenditure rose from 28.6 to 48 per cent while the out-of-pocket share fell from 64.2 to 39.4 per cent, the two lines crossing by the end of the period; total health spending is now about 3.8 per cent of GDP and government health spending about 1.9 per cent against the National Health Policy target of 2.5 per cent, out-of-pocket payments still push roughly 55 million people into poverty a year, and medicines alone take about two-thirds of out-of-pocket spending. Quote these in place of the older 2018–19 figures, with the year attached [PMID 42548603].
- ▶ **First district-level UHC index for India** (687 of 707 districts; 24 tracer indicators drawn from NFHS-5 2019–21, NSS 75th round 2017–18 and programme data): median 44.0 per cent, ranging from 26.4 per cent in Baksa, Assam to 69.4 per cent in the Nilgiris, Tamil Nadu; only 21.1 per

cent of districts exceeded 50 per cent, 69.4 per cent of the variation lay between states, and the median for the poorest wealth quintile (36.9 per cent) sat below the district median (43.96 per cent) — a ready-made answer to "how is UHC measured below the national level?" [PMID 39219770].

- **Comprehensive Annual Modular Survey 2022–23** (302,086 households, surveyed July 2022 to June 2023): 59.9 per cent reported some health insurance and 49.7 per cent publicly funded cover. Publicly funded insurance was associated with greater outpatient (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (1.168, 1.128–1.210) and less catastrophic expenditure at the 20 per cent threshold (outpatient 0.869, 0.810–0.932; inpatient 0.790, 0.737–0.846) — but at the 10 per cent threshold the outpatient benefit disappeared (adjusted OR 1.048, not significant), so outpatient care remains the dominant residual burden even among the insured: the case for folding outpatient benefits into PM-JAY and linking Ayushman Arogya Mandirs to it [PMID 42167946].

Elimination, surveillance and disease status

- The tracer domains of the district index separate sharply and explain where the service-coverage deficit now sits: infectious disease coverage had the highest median (82.8 per cent — tuberculosis treatment success 88.0 per cent, water/sanitation/hygiene 79.4 per cent), whereas non-communicable disease coverage had the lowest (9.8 per cent), with cancer screening (cervical and breast) below 1 per cent and round-the-clock facility access only 48.7 per cent. India's communicable-disease and elimination-programme machinery is not matched on the NCD side, which is the binding constraint on SDG 3.8.1 [PMID 39219770].

Guideline and legislative updates

- Monitoring guidance for **SDG indicator 3.8.2** is being widened beyond direct medical costs: current commentary argues that national household surveys must also capture direct non-medical costs (transport) and indirect time costs, which fall disproportionately on rural and poorer households and cause foregone care; five priority actions are proposed, including survey redesign, methods for valuing time, and extending social protection to non-medical costs. Catastrophic expenditure thresholds themselves remain plural — 10 per cent of total household consumption, or 40 per cent of non-food expenditure — so always state which threshold a quoted figure uses [PMID 41251686].

Global and WHO developments

- **World Health Day themes since the recall table ends:** 2021 "Building a fairer, healthier world"; 2022 "Our planet, our health"; 2023 "Health for All" (WHO's 75th year); 2024 "My health, my right"; 2025 "Healthy beginnings, hopeful futures"; 2026 "Together for health. Stand with science". Note the line of descent — the 2019 UHC theme is restated as a rights claim in 2024 [WHO 2026-Apr].
- **The global figures in the body are superseded.** On WHO's current accounting the UHC service coverage index reached 71 in 2023 (54 in 2000), yet about 4.6 billion people were still not fully covered by essential health services; in 2022, 2.1 billion people (26 per cent of the world) faced financial hardship from health payments and 1.6 billion of them were living in poverty or pushed deeper into it. Critically, the annual rate of coverage improvement slowed from 1.5 per cent

(2000–2015) to 0.5 per cent after 2015 — the "half the world" and "800 million/100 million" figures of the 2019 campaign should be replaced by these, dated [WHO 2025-Dec].

- **The 2019 political declaration was not the end of the line.** The UN General Assembly held a second High-Level Meeting on universal health coverage in September 2023, deliberately framed as the follow-up to the September 2019 declaration quoted in this answer, and negotiated in parallel with the pandemic instrument and High-Level Meetings on pandemic preparedness and tuberculosis. Analysis of the successive negotiating texts shows UHC and global health security norms converging — primary health care, equity and access to health products entering the pandemic text, and outbreak preparedness, emergency response and One Health entering the UHC declaration; the **WHO Pandemic Agreement** was subsequently adopted at the Seventy-eighth World Health Assembly on 20 May 2025 [PMID 39994683; WHA78 2025-May].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 1 Man and Medicine; Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 14 Sustainable Development Goals; Ch 25 International Health); Sunder Lal — Textbook of Community Medicine (Ch 1 Concept of Health and Disease; Ch 16 Health Planning and Management; Ch 17 Healthcare of the Community; Ch 18 International Health and International Health Regulations); Oxford Textbook of Public Health 7e (Ch 8 Poverty, justice and health; Ch 21 Health policy in developing countries; Ch 47 Health promotion; Ch 101 Strategies for health services; Ch 103 Transformative learning; Ch 107 Global health in the era of sustainable development); Park Plus — Amrutha & Vijayalakshmi (Ch 24 Miscellaneous). Named documents: WHO Constitution 1948; Alma-Ata Declaration 1978; Astana Declaration on Primary Health Care 2018; UN Sustainable Development Goals 2015 (target 3.8); WHO/World Bank Global Monitoring Report on UHC 2017; National Sample Survey 75th round; Ayushman Bharat operational guidelines 2018; Indian Public Health Standards 2022.

Recent-advances sources: 41393311 — Assessing Surgical Care under the Government-Funded Health Insurance Scheme–Pradhan Mantri Jan Arogya Yojana in India, *Annals of Global Health* 2025, doi:[10.5334/aogh.4908](https://doi.org/10.5334/aogh.4908); 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission, *BMC Health Services Research* 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 42548603 — Can a bigger budget shrink the bill at the bedside? Public health financing as a route to lower out-of-pocket spending in India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1903360](https://doi.org/10.3389/fpubh.2026.1903360); 39219770 — District-level monitoring of universal health coverage, India, *Bulletin of the World Health Organization* 2024, doi:[10.2471/BLT.23.290854](https://doi.org/10.2471/BLT.23.290854); 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022–2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 41251686 — Time to fully account for cost in monitoring financial protection and universal health coverage in low- and middle-income settings, *Health Policy and Planning* 2026, doi:[10.1093/heapol/czaf085](https://doi.org/10.1093/heapol/czaf085); 39994683 — Normative convergence between global health security and universal health coverage, *Globalization and Health* 2025, doi:[10.1186/s12992-025-01099-3](https://doi.org/10.1186/s12992-025-01099-3). Also: WHO World Health Day campaign themes, 2021–2026 (April 2026); WHO universal health coverage fact sheet, updated 5 December 2025; WHO Pandemic Agreement adopted at the Seventy-eighth World Health Assembly, 20 May 2025.



World Health Day theme 2017

Asked in: Paper IV May 2018 · Paper IV Nov 2017 (2×) — RGUHS MD Community Medicine

Definition — World Health Day is the global health awareness day observed every year on **7 April** under the sponsorship of the World Health Organization (WHO), marking the day in 1948 on which the WHO Constitution came into force; a World Health Day theme is chosen each year to focus attention on a specific aspect of public health (Park 28e, Ch 25, p 1055). The First World Health Assembly (1948) resolved that 7 April be observed each year with effect from **1950** (Park Plus, Ch 24, pp 380–381). The theme for **2017** was "**Depression — Let's talk**" (Park Plus, Ch 18, pp 214–215).

World Health Day — origin, status and purpose

Item	Detail
Date observed	7 April, every year
Historical basis	WHO Constitution drafted 1946 by the Technical Preparatory Committee under <i>Rene Sand</i> , approved the same year by an International Health Conference of 51 nations at New York, and came into force 7 April 1948 (Park 28e, Ch 25, p 1055)
Decision to observe	Taken at the First World Health Assembly, 1948 ; observance effective from 1950 (Park Plus, Ch 24, pp 380–381)
Observed by	All Member States of WHO, with governments and non-governmental organisations
Purpose	To mark WHO's founding and to draw worldwide attention to a subject of major importance to global health each year through a chosen theme
Activities	International, regional and local events organised by WHO on the theme; media advocacy by partners such as the Global Health Council
Status	One of eight official WHO global health campaigns — with World Tuberculosis Day, World Immunization Week, World Malaria Day, World No Tobacco Day, World AIDS Day, World Blood Donor Day and World Hepatitis Day (Park Plus, Ch 24, pp 380–381)

Recent World Health Day themes (to place 2017 in sequence)

Year	Theme
2012	Good health adds life to years
2013	Healthy heart beat, healthy blood pressure
2014	Vector-borne diseases: small bite, big threat
2015	Food safety
2016	Halt the rise: beat diabetes
2017	Depression: Let's talk
2018	Universal health — full campaign slogan " Universal health coverage: everyone, everywhere "

(Theme list: Park Plus, Ch 24, pp 380–381. Fuller 2018 slogan wording: Sunder Lal — Textbook of Community Medicine, Ch 1. Earlier mental-health theme: 2001 — "Mental health: stop exclusion, dare to care".)

The 2017 theme — "Depression: Let's talk"

- ▶ **Slogan and focus** — "Depression – Let's talk" was the slogan for World Health Day 2017, making depression the single global health subject for the year; depression contributes significant disease burden at national and global levels, and at the individual and family level leads to poor quality of life with huge social and economic impact (Park Plus, Ch 18, pp 214–215).
- ▶ **Why talking matters** — National Mental Health Survey data show the interval between onset of symptoms and seeking help ranged from **1 month to 10 years**, with traditional, spiritual and household remedies often the first step (Park Plus, Ch 18, p 217); public education in mental health to increase awareness and reduce stigma is itself a stated component of the District Mental Health Programme (Park 28e, Ch 7, p 567).
- ▶ **Rationale of choice** — globally, depression is the top cause of illness and disability among young and middle-aged populations, while suicide ranks second among causes of death in the same age groups (Park Plus, Ch 18, pp 214–215).
- ▶ **Link with the rest of the disease burden** — depression is both a **cause and a consequence** of several non-communicable diseases (cancer, ischaemic heart disease, diabetes), of substance use disorders (alcohol, drugs) and of nutritional disorders (undernutrition, over-nutrition and obesity); it is also adversely associated with chronic communicable diseases such as tuberculosis and human immunodeficiency virus (HIV) infection (Park Plus, Ch 18, pp 214–215).
- ▶ **Relevance to India** — with globalization, urbanization, migration and modernization alongside a rapid socio-demographic transition, depression is likely to increase in India in the coming years (Park Plus, Ch 18, p 215).
- ▶ **Operational goal — recognition in general and primary care** — asking two questions routinely of all persons attending primary care settings can achieve early detection (Park Plus, Ch 18, p 217); training general practitioners and other health-care staff to identify depression and provide adequate treatment has been successful in reducing suicidality and suicides, and even minimal psychotherapy delivered in primary care can prevent full-blown depression (Oxford Textbook of Public Health 7e, Ch 65, pp 2430, 2433).

Why depression was chosen — magnitude of the problem

Level	Finding (with year/source)
Global — numbers	Tentative global estimates of major mental disorders: depression 300 million , bipolar affective disorder 60 million, schizophrenia and other psychoses 23 million, dementia 50 million (Park 28e, Ch 20, p 985)
Global — prevalence	4.4% of the world population; higher in females (5.1% vs 3.6% in males); peak in the 55–74 year age group in both sexes (Park Plus, Ch 18, p 215)
Global — disability	Single largest contributor to non-fatal health loss — 7.5% of global years lived with disability (YLD) and 2.0% of global disability-adjusted life years (DALY), 2015 (Park Plus, Ch 18, p 215)

Level	Finding (with year/source)
Global — share	Depressive disorders = nearly one-third (33.84%) of all DALYs from mental and substance use disorders, Global Health Estimates 2015 (Park Plus, Ch 18, p 216)
Global — projection	Projected 2nd leading cause of disease burden globally and 3rd in low- and middle-income countries by 2030 (Park Plus, Ch 18, p 215)
Global — mortality link	Depression and anxiety affect nearly 1 in 10 people; among young adults aged 15–29 years suicide accounts for 8.5% of all deaths , the 2nd leading cause after road traffic injuries (Park 28e, Ch 14, p 857)
Global — suicide and mental illness	About 90% of suicides are estimated to be associated with mental disorders, mostly mood disorders such as depression, which is associated with 60% of suicides (Oxford Textbook of Public Health 7e, Ch 65, p 2404)
India — prevalence	National Mental Health Survey (NMHS 2015–16, 12 states): current mental morbidity in adults ≥18 years 10.6% , lifetime 13.7% ; young adolescents 7.3% ; high suicidal risk 0.9% (Park 28e, Ch 20, p 985)
India — service need and severity	Nearly 150 million Indians aged ≥18 years are likely to have one or more mental health problems and are in need of services; less than 2% had a severe mental illness (psychosis or bipolar affective disorder), and among those with severe mental illness nearly 50% had moderate-to-severe disability (NMHS 2015–16; Park 28e, Ch 20, p 985)
India — disability from depression	Nearly two-thirds reported disability of varying severity — work life 67.3% , social life 68.6% , family life 70.2% ; over 50% reported that the condition interfered with daily activities; one-fifth faced severe restrictions in performing daily activities, present on average for 20 days in the month preceding the study (Park Plus, Ch 18, p 217)
India — cost	Median monthly expenditure per affected household ₹1000–2500 (NMHS 2015–16); families of persons with depression spend on average about ₹1500 per month on consultation, drugs and transport (Park 28e, Ch 20, p 985; Park Plus, Ch 18, p 217)
India — care-seeking delay	Interval between symptom onset and first help-seeking 1 month to 10 years , traditional, spiritual and household remedies commonly the first step (Park Plus, Ch 18, p 217)

Recognising depression — what "Let's talk" asks the health worker to do

Element	Content
Duration criterion	Symptoms present for a minimum of 2 weeks (simplified ICD-10 criteria, Box 1)
Major symptoms	Feeling sad or depressed most of the day, nearly every day; markedly reduced interest or pleasure in almost all activities; loss of energy or fatigue nearly every day
Minor symptoms	Loss of self-confidence or self-esteem; unreasonable guilt; recurrent thoughts of death or suicide, or suicidal behaviour; diminished ability to think or concentrate; agitation or slowness; disturbed sleep; gloomy view of the future; increased or decreased appetite and/or weight
Severity grading	Mild, moderate or severe, based on symptomatology

Element	Content
Presentation in children	Sadness, irritability, tantrums, physical complaints (headache, stomach ache), tearfulness, boredom, separation anxiety, poor academic performance, disturbed family relations
Presentation in adults	Marked change in level of functioning, with suicidal thoughts and behaviour
Presentation in late life	Psychomotor agitation or irritability rather than depressed mood; somatic complaints, poor appetite, sometimes psychotic features
Two-question primary-care screener (Box 2)	(1) During the past 4 weeks , have you often been bothered by feeling down, depressed or hopeless? (2) During the past 4 weeks , have you often been bothered by having little interest or pleasure in doing things? — "YES" to both warrants further assessment for depression
Validated Indian instruments	General Health Questionnaire (GHQ), Kessler Psychological Distress Scale (K6), Self-Reporting Questionnaire (SRQ); a higher cut-off is recommended in resource-poor primary settings
First-line management of mild depression	Psychotherapy and counselling , not pharmacological treatment

(All rows: *Park Plus*, Ch 18, pp 215–219.)

Determinants of depression

Group	Factors
Biological	Genetic factors; altered hypothalamo-pituitary-adrenal axis; altered circadian rhythm
Psychological	Negative parental influences; separation or early loss of maternal attachment/bond; personality and lifestyle factors
Socioeconomic	Poverty and low literacy; modernization, urbanization, migration and globalization; distressing social relationships; domestic and intimate partner violence; natural disasters and conflicts
Cultural	Religion, caste, beliefs, attitudes and interpretations, which influence onset of depression and symptom thresholds

(*Park Plus*, Ch 18, p 216.)

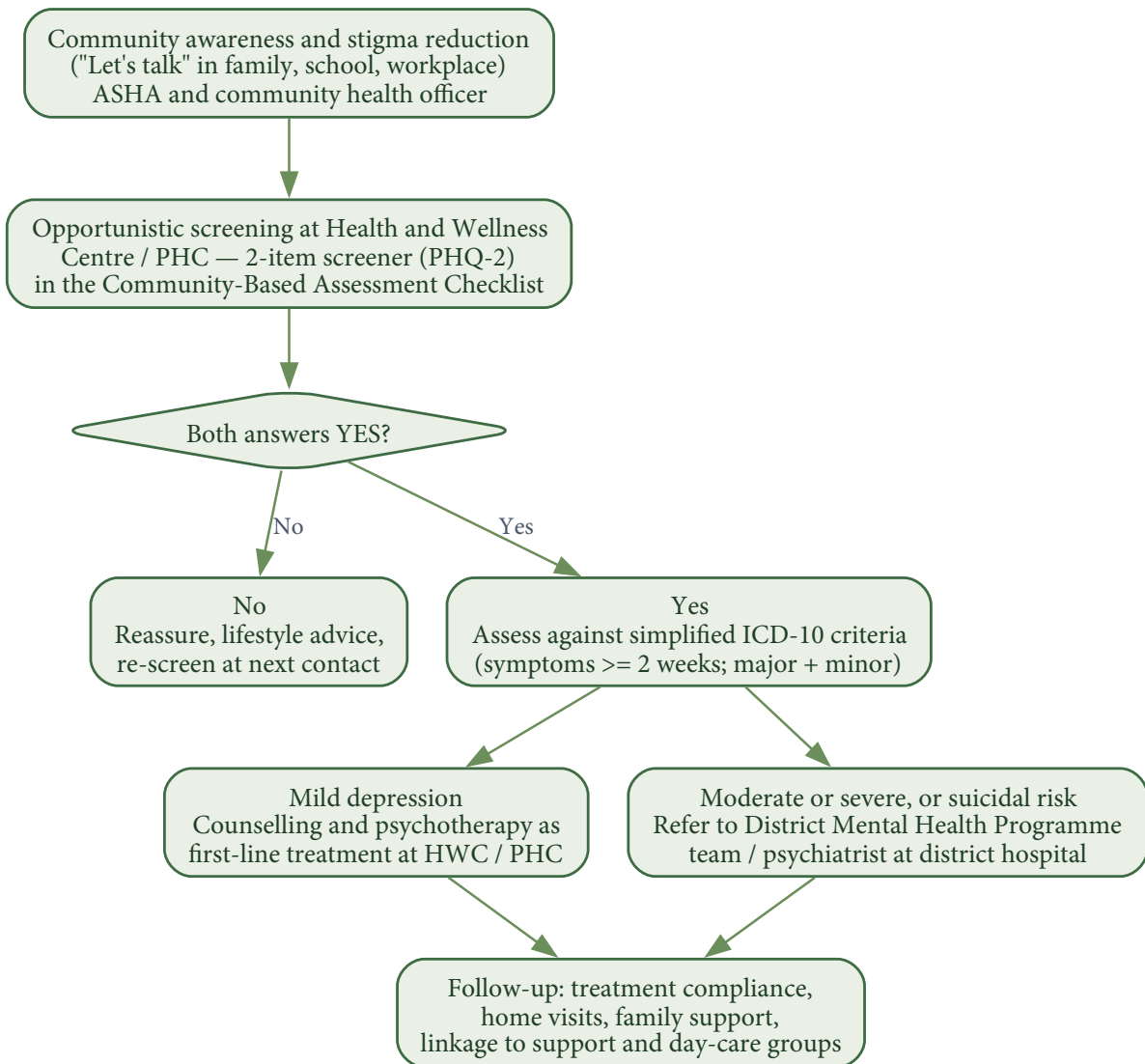
India's response, taken forward on the same day

Instrument	Content
National Mental Health Programme (NMHP), 1982	Aims: (a) prevention and treatment of mental and neurological disorders and associated disabilities; (b) use of mental health technology to improve general health services; (c) application of mental health principles in total national development to improve quality of life (Park 28e, Ch 7, pp 567–568)
NMHP objectives	(1) availability and accessibility of minimum mental health care for all, particularly the vulnerable and underprivileged; (2) application of mental health knowledge in general health care and social development; (3) promotion of community participation and self-help

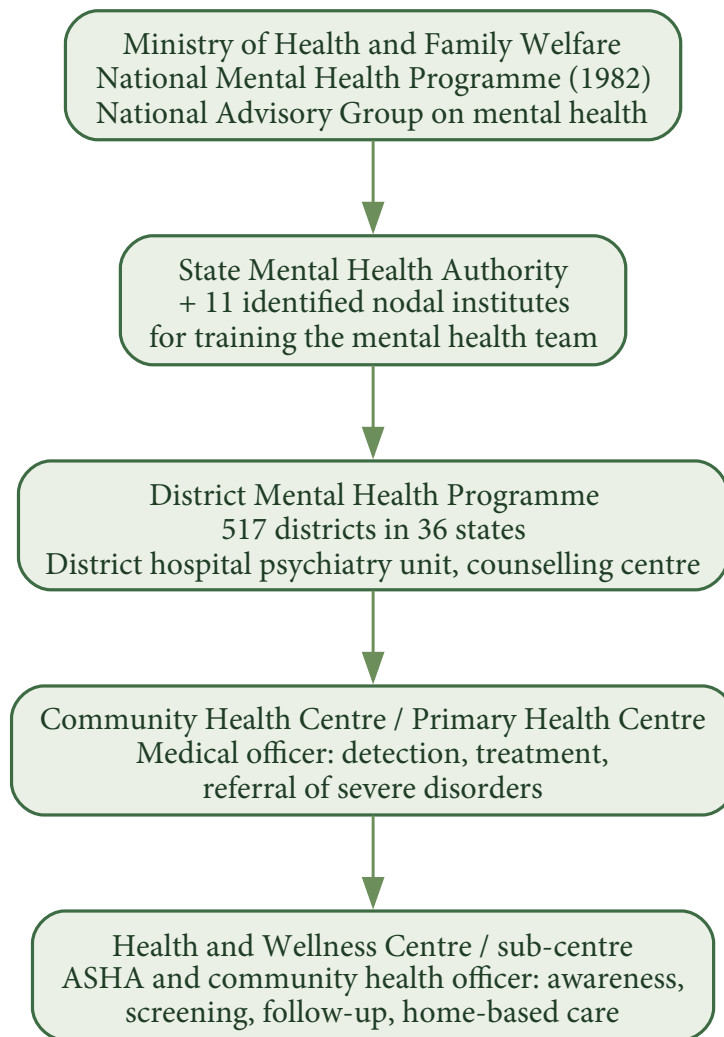
Instrument	Content
NMHP strategies	Integration of mental health with primary health care through the District Mental Health Programme (DMHP); tertiary care institutions for treatment of severe disorders; eradication of stigma and protection of rights of the mentally ill through the Central and State Mental Health Authorities
Coverage and training	DMHP covers 517 districts in 36 states ; eleven institutions identified nationally for imparting training in basic mental health knowledge and skills to primary health care physicians and paramedical personnel (Park 28e, Ch 7, p 567)
DMHP components	Training of the mental health team at the identified nodal institute in the state ; public education to increase awareness and reduce stigma ; outdoor and indoor services for early detection and treatment; community-level data for planning, service improvement and research
Thrust areas	Enlarged DMHP covering the whole country; modernization of mental hospitals away from a custodial role; upgrading psychiatry departments in medical colleges; workplace stress management; suicide-prevention services — district counselling centres, sensitisation workshops, information-education-communication and help lines (Park 28e, Ch 7, p 568)
Mental Healthcare Act, 2017	Received Presidential assent on 07.04.2017 — World Health Day 2017 itself ; supersedes the Mental Health Act 1987; decriminalises attempted suicide ; establishes the Central and State Mental Health Authorities and a Mental Health Review Board; mandates insurance for mental illness on par with physical illness; restricts electroconvulsive therapy to emergencies with anaesthesia and muscle relaxants and bars it in minors; aligns India with the United Nations Convention on the Rights of Persons with Disabilities (Park 28e, Ch 7, p 568; Park Plus, Ch 18, pp 219–221)
Service delivery today	Health and Wellness Centre package: community awareness on depression and other conditions, screening with the CIDT tool , Patient Health Questionnaire-2 (PHQ-2) included in the Community-Based Assessment Checklist (CBAC) , identification and referral to Health and Wellness Centre/primary health centre, treatment compliance, home visits and linkage to support groups and day-care centres (Park 28e, Ch 10, p 735)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

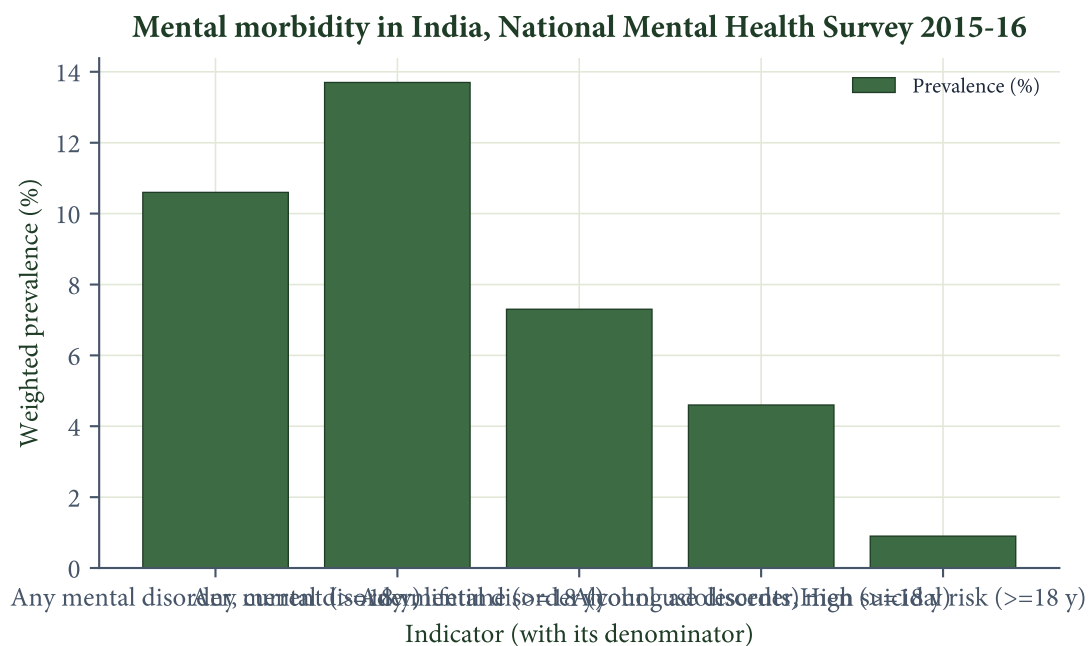
- Pie chart — DALYs due to mental and substance use disorders (Global Health Estimates 2015). Labels: *Depressive disorders — 33.84% (nearly one-third)*; *Anxiety disorders*; *Drug use disorders*; *Schizophrenia*; *Others*. Title the figure "Depressive disorders are the largest single share of the mental-disorder disease burden".



Community pathway for depression under the 2017 theme "Depression: Let's talk"



Levels of delivery of mental health care in India under the National Mental Health Programme



National Mental Health Survey 2015-16, 12 states (Park 28e, Ch 20, p 985). Tobacco use disorders excluded.

High-yield exam pointers

- ▶ 7 April = WHO Constitution in force, 1948; observance decided by the First World Health Assembly 1948, effective 1950; one of **eight** official WHO global health campaigns.
- ▶ 2017 theme = "Depression: Let's talk". Earlier mental-health theme: 2001 — "Mental health: stop exclusion, dare to care".
- ▶ Depression: **300 million** globally; **4.4%** of world population (F 5.1% vs M 3.6%, peak 55–74 y); **7.5% of global YLDs, 2.0% of global DALYs (2015)**; **one-third** of mental-disorder DALYs; projected **2nd** leading cause of global disease burden by 2030.
- ▶ NMHS 2015–16 denominators — **do not mix them up**: current mental morbidity **10.6%** and lifetime **13.7%** of adults ≥18 y, adolescents **7.3%**, high suicidal risk **0.9%**, **~150 million** adults need services; **<2%** have a **severe** mental illness, and it is **within that severe group** that nearly **50%** have moderate-to-severe disability.
- ▶ ICD-10 shortcut: symptoms ≥ **2 weeks**; three **major** symptoms = low mood, loss of interest/pleasure, fatigue.
- ▶ **Two separate two-item tools**: the Box 2 primary-care screener asks about the **past 4 weeks** (YES to both → assess further); the **Patient Health Questionnaire-2 (PHQ-2)** is the item built into the **Community-Based Assessment Checklist** at the Health and Wellness Centre.
- ▶ **Mental Healthcare Act 2017 received Presidential assent on 7 April 2017 — World Health Day 2017 itself** — and decriminalised attempted suicide. Quote this link; it is the mark-winning sentence.
- ▶ **Mild depression → counselling and psychotherapy first, not drugs.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

This is a theme-year question, so the figures above are those current at the time of the 2017 campaign and quoted in Park and Park Plus; later WHO estimates put all mental disorders at about 970 million (2019) and depressive disorders at 246 million, a rise of about 28% after the COVID-19 pandemic (AFMC Red Book 2024, Ch 36, pp 1161–1162). Sunder Lal's Textbook of Community Medicine gives the fuller universal health coverage slogan but dates it to 7 April 2019, while Park Plus lists it against 2018 and Sunder Lal's own source is the WHO release "WHO at 70 — working for better health for everyone, everywhere" of 5 April 2018. Coverage note: the indexed corpus carries the World Health Day theme list, the depression burden and the Indian programme response fully, but does not carry the WHO campaign's own operational documents, so campaign-specific targets and materials are not stated here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO World Health Day campaign listing, WHO Mental Health Atlas 2024, MoHFW National Suicide Prevention Strategy 2022, National Mental Health Survey-2, Mental Healthcare Act 2017

Programme and policy updates (India)

- ▶ **Tele-MANAS (Tele Mental Health Assistance and Networking Across States)**, launched **October 2022** is now the largest government-run tele-mental-health service in the world — a free

24×7 multilingual helpline operating in every state and union territory, working on a stepped-care model in which first contact gets immediate counselling and complex cases are escalated to psychiatrists or local services. It had answered more than **3.43 million calls** across all states and union territories in its first few years, most callers reporting anxiety, depression or psychosocial stressors and a critical minority reporting suicidal distress; uptake has been driven by linguistic inclusivity, anonymity and integration with India's digital health ecosystem, and is highest among young adults. A WHO assessment has recognised it as a scalable, equity-focused solution for low- and middle-income countries. This is the single most examinable operational answer to the 2017 "Let's talk" call [PMID 42467064].

- ▶ **Peripheral delivery now runs through the Ayushman Arogya Mandir** (the Health and Wellness Centre under its current name), where **mental-health screening is one of the twelve services** in the comprehensive primary health care package. The method is WHO-recommended **task-shifting**: Community Health Officers and ASHAs use community-based assessment checklists to pick up people at risk of communicable, non-communicable and mental disorders — the **Mental Health Evidence & Research Team (MERIT) tool** takes about five minutes per household and covers common and severe mental disorders and substance use — then refer to District Early Intervention Centres or the district-hospital mental health clinic and follow up for treatment adherence. The driver is workforce scarcity: India needs about **3 psychiatrists per 100,000** (roughly 13,000 clinicians) but has only about **3,500, a ratio near 0.7 per 100,000** [PMID 40534365].
- ▶ **Universal two-step depression screening has now been costed for the Indian primary-care system** — PHQ-2 by the ASHA at household level followed by PHQ-9 by the Community Health Officer at the sub-centre-Ayushman Arogya Mandir. In a government-funded economic model the incremental cost-utility ratio was **₹48,746 per QALY for screening at ≥30 years and ₹43,995 per QALY at ≥20 years**, both below the per-capita-GDP threshold of ₹1,71,498; net savings were **₹291–482 billion (0.19–0.32% of GDP)** and depression-related suicides were projected to fall by about **15% annually**. The programme stops yielding positive net health benefit if **PHQ-2 sensitivity falls to 26%** (or combined PHQ-2 + PHQ-9 sensitivity below 35%), and it saves money only if at least **60% of those treated after screening use public facilities** — so quality of training and supportive supervision, not the tool itself, is the rate-limiting step [PMID 41438244].

Current national data

- ▶ **Quote the survey round with the figure.** The 10.6% current mental morbidity used above is NMHS 2015–16. The **second National Mental Health Survey (NMHS-2) is being conducted in 2024–2026** across all **28 states and 8 union territories**, with over **2,25,000 adult assessments** by more than 240 lay interviewers — the survey is a once-a-decade national exercise and this is the largest mental-health epidemiological survey India has run. Its primary instrument is the **WHO Flexible Interview for ICD-11 (FLII-11)**, translated into **22 Indian languages**, administered by lay interviewers put through a certified **3-week structured training** (an 80% score against psychiatrist ratings is required to pass), so the next generation of Indian prevalence figures will be **ICD-11-based** [PMID 40371258].
- ▶ **Treatment gap remains the dominant service statistic** — **70–92%** across disorders nationally [PMID 40534365]. A re-analysis of NMHS 2015–16 data for adults aged ≥60 years found an

overall gap of 84% (76.9% for neurotic and stress-related disorders), rising to 90.9% for severe mental disorders; median duration of illness 24 months, with a delay of 1.5–8 months between symptom onset and first consultation. Read the two numbers separately — 24 months is how long the illness had run, not the wait before care. The gap was higher in men (85.5%) than women (83.6%), and urban-metro residence, male sex and lower household income were independently associated with psychiatric morbidity [PMID 41092637].

Elimination, surveillance and disease status

- ▶ India's first National Suicide Prevention Strategy was released by the MoHFW on 21 November 2022, with the target of reducing suicide mortality by 10% by 2030. Its four goals are institutional capacity and leadership, capacity of health services to deliver suicide-prevention care, community resilience with stigma reduction, and strengthened surveillance and evidence generation; actions are time-bound at 1–3, 4–7 and 8–10 years and reach outside the health sector into pesticide access restriction, safer farming practice and responsible media reporting. Note the mismatch worth quoting: WHO's Comprehensive Mental Health Action Plan 2013–2030 target is a one-third reduction in the suicide rate by 2030 [PMID 36874511].
- ▶ The national suicide rate has been rising, reversing the earlier decline — 9.9 per lakh in 2017, 10.2 in 2018, 10.4 in 2019, 11.3 in 2020 and 12.0 in 2021; suicide is the leading cause of death in the 15–29 year age group in India, exceeding road traffic deaths in men and maternal mortality in women [PMID 36874511].

Guideline and legislative updates

- ▶ The Mental Healthcare Act 2017's electroconvulsive therapy provisions are now formally contested. The Act prohibits unmodified ECT, bars ECT in under-18s without Mental Health Review Board permission, and bars ECT as an emergency treatment. NIMHANS authors argue that scarcity of anaesthetists and equipment confines modified ECT to higher-level centres and thereby limits access and affordability, and have made a published case for amending the Act to permit unmodified ECT where an informed patient or nominated representative chooses it [PMID 42495032].

Global and WHO developments

- ▶ World Health Day themes after 2017 (a standing follow-up question): 2018 "Universal health coverage: everyone, everywhere"; 2019 "Health for all: everyone, everywhere"; 2020 support nurses and midwives; 2021 "Building a fairer, healthier world"; 2022 "Our planet, our health"; 2023 "Health For All" (WHO's 75th year); 2024 "My health, my right"; 2025 "Healthy beginnings, hopeful futures" (maternal and newborn survival); 2026 "Together for health. Stand with science". WHO's own campaign listing dates the universal-health-coverage slogan to 2018 and "Health for all: everyone, everywhere" to 2019, which settles the textbook discrepancy flagged in the note below [WHO 2026-Apr].
- ▶ The evidence base behind the 2017 campaign has been superseded. WHO estimated a 25% rise in global anxiety and depression prevalence in the first year of the COVID-19 pandemic [PMID 40534365]; GBD 2021 counts 332.4 million people with depressive disorders and 359.2 million with anxiety disorders, together 9.1% of all disease and 63.1% of all mental-disorder cases, with the age-standardised DALY rate for depressive disorders up 13.4% since 1990 and cases projected to exceed 466 million by 2040, the growth concentrated in middle- and low-SDI

countries [PMID 40935255]. The WHO Mental Health Atlas 2024 (seventh edition, released 2 September 2025) reports from 144 countries and tracks progress against the Comprehensive Mental Health Action Plan 2013–2030 [WHO 2025-Sep].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 25 International Health; Ch 20 Mental Health; Ch 14 Sustainable Development Goals; Ch 10 Health Care of the Community; Ch 7 Health Programmes in India); Park Plus (Amrutha & Vijayalakshmi) — Ch 24 Miscellaneous (World Health Day and theme list) and Ch 18 Mental Health (Depression; Mental Health Care Act); Oxford Textbook of Public Health 7e, Ch 65 (Public mental health and suicide); Sunder Lal — Textbook of Community Medicine, Ch 1; AFMC Red Book of Community Medicine 2024, Ch 36. Named sources: WHO Global Health Estimates 2015; National Mental Health Survey of India 2015–16; Mental Healthcare Act 2017.

Recent-advances sources: 42467064 — India's Tele-MANAS: evolution, early outcomes and a scalable blueprint for digital public mental health, Br J Psychiatry 2026, doi:[10.1192/bjp.2026.10733](https://doi.org/10.1192/bjp.2026.10733) (abstract-only); 40534365 — Towards Mental Health Equity: Task Shifting Strategy to Overcome Barriers in Primary Healthcare Services in India, J Prev Med Public Health 2025, doi:[10.3961/jpmp.25.144](https://doi.org/10.3961/jpmp.25.144); 41438244 — Comprehensive primary health care for cost-effective scale-up of depression screening in India: an economic modelling study, Lancet Reg Health Southeast Asia 2025, doi:[10.1016/j.lansea.2025.100701](https://doi.org/10.1016/j.lansea.2025.100701); 40371258 — Lay-interviewer training protocol for the WHO Flexible Interview for ICD-11 for the National Mental Health Survey — 2 (India), Indian J Psychiatry 2025, doi:[10.4103/indianjpsychiatry.indianjpsychiatry_153_25](https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry_153_25); 41092637 — Treatment gap and care characteristics for mental disorders among older adults in India: Evidence from National Mental Health Survey 2015-16, Asian J Psychiatr 2025, doi:[10.1016/j.ajp.2025.104723](https://doi.org/10.1016/j.ajp.2025.104723) (abstract-only); 36874511 — National suicide prevention strategy of India: Great leap forward in field of mental health, Indian J Psychiatry 2023, doi:[10.4103/indianjpsychiatry.indianjpsychiatry_835_22](https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry_835_22); 42495032 — Mental healthcare act and electroconvulsive therapy: Making a case for review, Indian J Psychiatry 2026, doi:[10.4103/indianjpsychiatry_695_26](https://doi.org/10.4103/indianjpsychiatry_695_26); 40935255 — Global, regional and national burden of anxiety and depression disorders from 1990 to 2021, and forecasts up to 2040, J Affect Disord 2025, doi:[10.1016/j.jad.2025.120299](https://doi.org/10.1016/j.jad.2025.120299) (abstract-only). WHO World Health Day campaign listing (themes to 2026, April 2026); WHO Mental Health Atlas 2024, seventh edition, 2 September 2025.

♦ ♦ ♦

World Health Day theme 2015

Asked in: Paper I Nov 2016 — RGUHS MD Community Medicine

Definition — World Health Day is observed globally on **7 April** every year, the date on which the Constitution of the World Health Organization (WHO) came into force in 1948; a single theme is chosen each year to focus world attention on one aspect of public health (Park 28e, Ch 25, p 1055). The theme for **World Health Day 2015** was **FOOD SAFETY**, campaigned under the slogan "**From farm to plate, make food safe**" (Sunder Lal, Ch 5, p 219). It called on every link of the food chain — government sectors, farmers, manufacturers, retailers, health practitioners and consumers — to treat safe food as a shared responsibility rather than a purely regulatory one (Sunder Lal, Ch 5, p 224).

The observance and the 2015 theme at a glance

Item	Detail
Day and date	World Health Day, 7 April, annually (Park 28e, Ch 25, p 1055)
Origin	WHO Constitution, drafted 1946 by the Technical Preparatory Committee under <i>Rene Sand</i> and approved by an International Health Conference of 51 nations, came into force 7 April 1948 (Park 28e, Ch 25, p 1055)
Purpose of a theme	To focus attention on one specific aspect of public health for that year (Park 28e, Ch 25, p 1055)
Theme 2015	Food safety (Sunder Lal, Ch 5, p 219)
Slogan 2015	"From farm to plate, make food safe" (Sunder Lal, Ch 5, p 219)
Target audiences	Government sectors, farmers, manufacturers, retailers, health practitioners and consumers (Sunder Lal, Ch 5, p 224)
Central message	Food safety is a shared, cross-sectoral responsibility across the whole food chain from primary production to consumption

Why food safety was chosen — the burden of unsafe food

Level	Burden (with source)
Diseases carried by food	More than 200 diseases spread through contaminated food, ranging from diarrhoea to cancer and hepatitis (WHO, quoted in Sunder Lal, Ch 5, pp 222-223)
Global burden	600 million cases of food-borne disease and 4,20,000 deaths every year (AFMC Red Book 2024, Ch 30, p 871). The often-quoted 2 million deaths, mostly children , is the wider campaign-era WHO figure for deaths <i>linked to</i> unsafe food and is not the same estimate (Sunder Lal, Ch 5, pp 222-223) — quote 4,20,000 when asked for deaths <i>from</i> food-borne disease
WHO South-East Asia Region	150 million fall ill and 1,75,000 die each year from food poisoning (AFMC Red Book 2024, Ch 30, p 871)
India — projected caseload	Over 100 million food-borne disease cases annually at current standards, rising to 150-177 million by 2030 (AFMC Red Book 2024, Ch 30, p 871)

Level	Burden (with source)
India — relative rank	The Ministry of Health recognises food-borne illness as a greater health burden in India than malaria, HIV/AIDS and tuberculosis ; the Integrated Disease Surveillance Programme found food poisoning one of the commonest outbreaks reported in 2017 (Sunder Lal, Ch 5, p 223)
India — sanitation gap	Basic hygiene facilities available to only 68% of people (82% urban, 60% rural) (Sunder Lal, Ch 5, p 224)
India — food quality	23-28% of food samples non-conforming (unsafe, sub-standard or mislabelled) in 2016-17 and 2018-19, against a WHO norm of 5% (Sunder Lal, Ch 5, pp 219-220)
Parasitic contribution	The WHO Foodborne Disease Burden Epidemiology Reference Group (FERG), 2015 identified <i>Taenia solium</i> as a leading cause of food-borne deaths — 2.8 million disability-adjusted life years (DALYs) lost and 2.56-8.30 million neurocysticercosis cases (Park 28e, Ch 5, p 368); the same group put echinococcosis at 19,300 deaths and about 871,000 DALYs a year (Park 28e, Ch 5, p 369)

New and emerging threats to food safety (the campaign's rationale)

These four shifts were named as the reasons why national food safety systems are now under strain (Sunder Lal, Ch 5, p 219):

Threat	Public health implication
Changes in food production, distribution and consumption	Contamination is commonest during mass cooking, feasts and festivals , so one batch exposes a large population (Sunder Lal, Ch 5, p 223); cost-cutting along the chain — contaminated water, inadequate refrigeration, poor hygienic facilities for workers — is a common point of entry (Textbook of International Health, Ch 10, p 536)
Changes in the environment	Ecological change — agriculture, dams, altered water ecosystems, deforestation, flood and drought, famine and climate change — is a recognised driver of infectious disease emergence (Textbook of International Health, Ch 6, pp 333-334)
New and emerging pathogens	Livestock in overcrowded pens spread disease to one another and to humans, the prime route of emerging disease (Nipah virus, H5N1 influenza); several preventable zoonoses reach humans through food (brucellosis, tuberculosis) or food contamination (hydatidosis) (Textbook of International Health, Ch 10, p 536)
Antimicrobial resistance (AMR)	Animal feeds containing antimicrobials in subtherapeutic concentrations favour drug resistance, making <i>E. coli</i> and salmonellae of animal origin highly resistant; with a globalised food supply this reaches consumers anywhere (Park 28e, Ch 5, pp 429-430), so AMR surveillance must be integrated with food-borne disease surveillance (Sunder Lal, Ch 5, p 224)

Food-borne diseases — the entity the theme addresses

Food-borne disease is "a disease, usually either infectious or toxic in nature, caused by agents that enter the body through the ingestion of food" (Park 28e, Ch 11, p 799). A **food-borne disease outbreak** is the occurrence of **two or more cases of a similar illness from ingestion of a common food**, or when the observed number of cases exceeds the expected number; it is *confirmed* when at least one causal agent

is identified and *suspected* when based on clinical and epidemiological information alone (Sunder Lal, Ch 5, p 223).

Class	Subgroup	Examples
A. Food-borne intoxications	Naturally occurring toxins in food	Lathyrism (beta-oxalyl-amino-alanine), endemic ascites (pyrrolizidine alkaloids)
	Toxins produced by bacteria	Botulism, staphylococcal poisons
	Toxins produced by fungi	Aflatoxin, ergot, fusarium toxins
	Food-borne chemical poisoning	Heavy metals (mercury in fish, cadmium in shellfish, lead in canned food), oils, petroleum derivatives and solvents, migrant chemicals from packaging, asbestos, pesticide residues
B. Food-borne infections	Bacterial	Typhoid and paratyphoid fever, salmonellosis, staphylococcal intoxication , <i>Clostridium perfringens</i> illness, botulism , <i>Bacillus cereus</i> food poisoning, <i>E. coli</i> diarrhoea, non-cholera vibrio illness, <i>V. parahaemolyticus</i> infection, streptococcal infection , shigellosis, brucellosis
	Viral	Hepatitis A and E, norovirus gastroenteritis, rotavirus
	Parasitic	Taeniasis, hydatidosis, trichinosis, ascariasis, amoebiasis, oxyuriasis

(Classification per Park 28e, Ch 11, p 799. Park lists *staphylococcal intoxication* and *botulism* in the bacterial infection group as well as under intoxications — write them in both places.)

What the theme asked for — action along the chain

1. WHO's Five Keys to Safer Food (the consumer- and food-handler-level message; Sunder Lal, Ch 5, pp 223-224):

- ▶ Wash your hands before handling food and during food preparation.
- ▶ Separate raw and cooked food.
- ▶ Cook thoroughly.
- ▶ Keep food at a safe temperature.
- ▶ Use safe water and safe raw material.

2. Making safe and healthy food accessible to all (the system-level message; Sunder Lal, Ch 5, p 224):

- ▶ Strengthen food safety management and regulation.
- ▶ Adopt a risk-based approach to monitoring food safety.
- ▶ Strengthen surveillance of food-borne disease under the Integrated Disease Surveillance Programme (IDSP) to establish when, why and how food-borne hazards and illness occur.
- ▶ Include integrated surveillance of antimicrobial resistance in food-borne bacteria.

3. Preventive systems — Hazard Analysis and Critical Control Point (HACCP) is the systematic preventive approach that addresses physical, chemical and biological hazards at defined control points, as prevention rather than finished-product inspection (Sunder Lal, Ch 5, pp 223-224). Food hygiene in

its widest sense means hygiene in the **production, handling, distribution and serving** of all types of food, its primary aim being prevention of food-borne disease (Park 28e, Ch 11, p 797). Food handlers require pre-employment and day-to-day health appraisal, exclusion when ill, and education in personal hygiene, since they transmit diarrhoeas, dysenteries, typhoid and paratyphoid fevers, enteroviruses, viral hepatitis, protozoal cysts, helminth eggs, and streptococcal, staphylococcal and salmonella infections (Park 28e, Ch 11, p 799).

India's food safety architecture (the national translation of the theme)

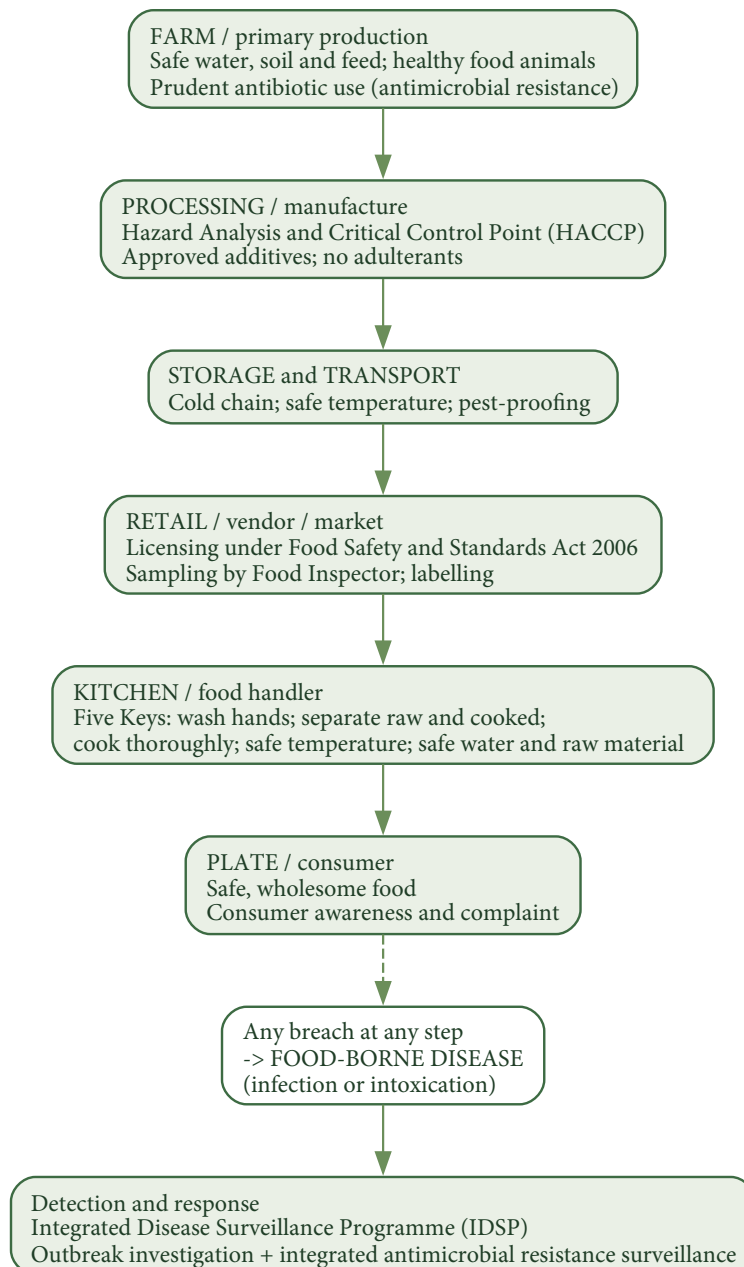
Instrument / body	Substance
Prevention of Food Adulteration (PFA) Act, 1954	Ministry of Health and Family Welfare responsible for safe, wholesome food; subject lies in the concurrent list ; enforced by the States (Sunder Lal, Ch 5, pp 218-219)
Food Safety and Standards Act (FSSA), 2006	Consolidates all food laws; in force from 5 August 2011 , repealing the PFA Act, Fruit Products Order 1955 and Solvent Extracted Oil Order 1967 (Sunder Lal, Ch 5, p 219)
Food Safety and Standards Authority of India (FSSAI)	Established 2008 ; lays down science-based standards and regulates manufacture, storage, distribution, sale and import of food; implemented by all States/Union Territories (Park 28e, Ch 11, p 802)
Enforcement machinery under the FSS Act 2006	The Food Authority and State Food Safety Authorities enforce the Act (Sec 29); the Food Safety Officer enforces and executes it within his area; the Commissioner of Food Safety (State) and the Designated Officer (district) hold the same powers, supervise the Designated Officer and food analysts, and sanction prosecution (Jugal Kishore, Ch 2, p 882)
Food Analyst	Appointed by the Commissioner of Food Safety for a local area; receives, analyses and reports on the article of food in a laboratory notified or accredited by the Food Authority (Jugal Kishore, Ch 2, p 882)
Codex Alimentarius Commission	Principal organ of the joint FAO/WHO Food Standards Programme , which formulates food standards for the international market; Indian food standards are based on Codex standards (Park 28e, Ch 11, p 802)
Agmark	Set by the Directorate of Marketing and Inspection; voluntary quality assurance (Park 28e, Ch 11, p 802)
Bureau of Indian Standards (ISI mark)	Voluntary ; expresses quality above PFA standards (Park 28e, Ch 11, p 802)
Laboratory network	137 food laboratories under States/Union Territories and local bodies; 45 State public laboratories being strengthened and 13 National Reference Laboratories approved (Sunder Lal, Ch 5, p 219). Above them stand the four regional APPELLATE Central Food Laboratories — Kolkata, Mysore, Ghaziabad and Pune — whose report is considered final (Park 28e, Ch 11, pp 801-802); the routine sample goes to the designated State/Union Territory laboratory, and the Central Food Laboratory report is final on appeal or reference
Surveillance	Food-borne disease and antimicrobial resistance surveillance under IDSP (Sunder Lal, Ch 5, p 224)

Recent World Health Day themes (context for the 2015 answer)

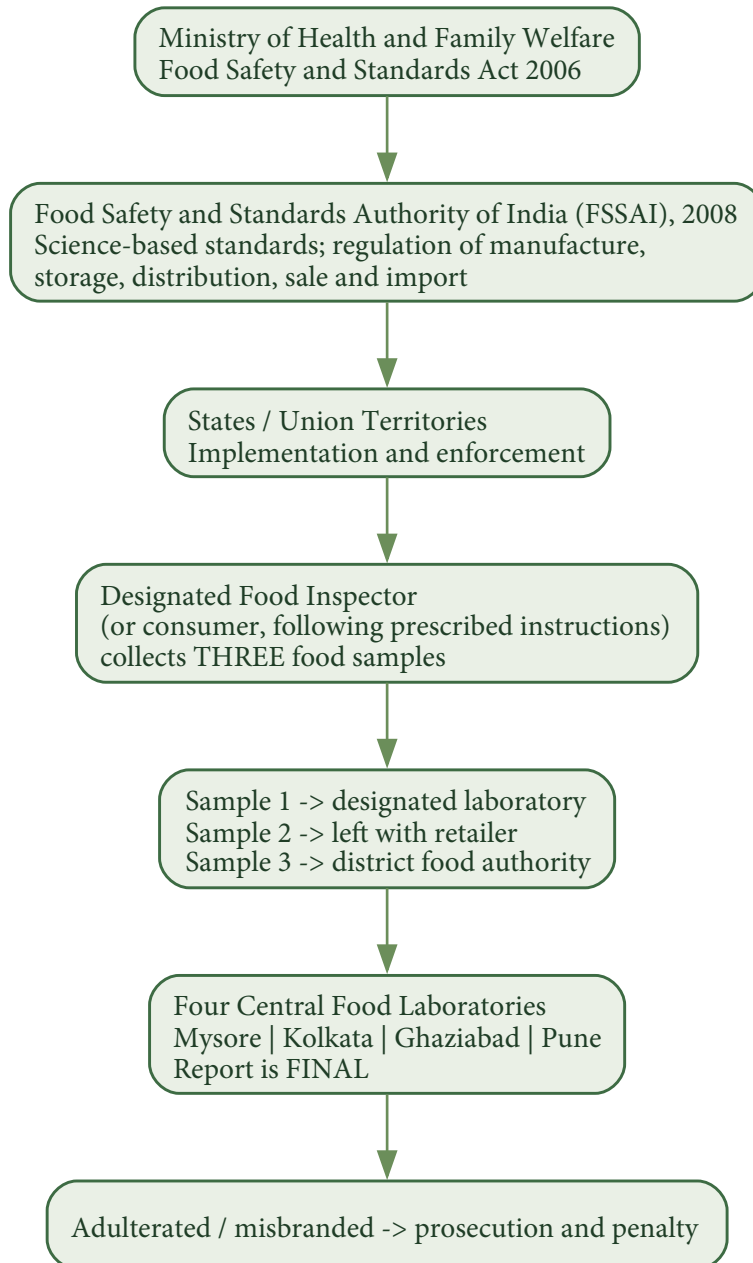
Year	Theme / slogan
2008	Protecting health from climate change (Mahajan & Gupta 4e, Ch 36, p 670)
2009	Make hospitals safe in emergencies (Mahajan & Gupta 4e, Ch 36, p 670)
2010	Urbanization and health — 1000 Cities, 1000 Lives (Mahajan & Gupta 4e, Ch 36, p 670)
2011	Antimicrobial resistance: no action today, no cure tomorrow (Mahajan & Gupta 4e, Ch 36, p 670)
2012	Ageing and health: good health adds life to years (Mahajan & Gupta 4e, Ch 36, p 670)
2013	High blood pressure — "a silent killer" (Sunder Lal, Ch 8, pp 490-491)
2014	Small bite, big threat — vector-borne diseases (Sunder Lal, Ch 8, p 371)
2015	Food safety — "From farm to plate, make food safe" (Sunder Lal, Ch 5, p 219)
2016	Halt the rise, beat diabetes (Sunder Lal, Ch 8, pp 486-487)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

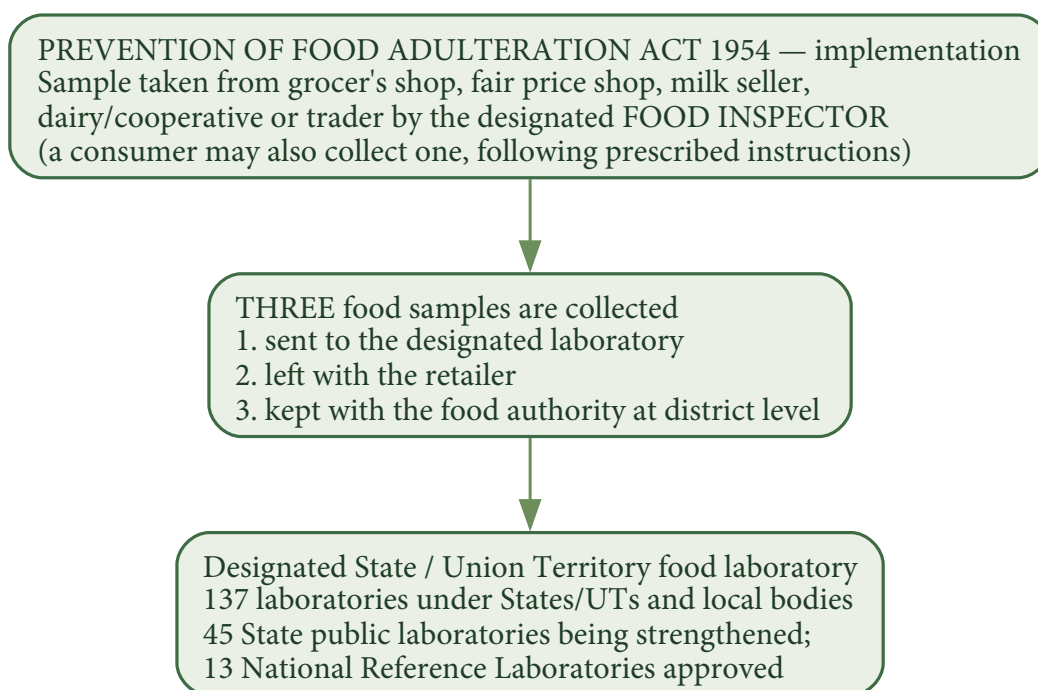
- **The Five Keys to Safer Food wheel** — a circle with a central hub labelled *Five Keys to Safer Food (WHO)* and five spokes labelled: *1 Wash your hands; 2 Separate raw and cooked; 3 Cook thoroughly; 4 Keep food at safe temperature; 5 Use safe water and raw materials.*



"From farm to plate, make food safe" — control points along the food chain, World Health Day 2015



Enforcement of the Food Safety and Standards Act 2006 — officers at each level, and where the appellate laboratory sits



The three-sample procedure described for the Prevention of Food Adulteration Act 1954 — PFA-era practice still carried in the textbooks

High-yield exam pointers

- ▶ **World Health Day** = 7 April, the day the WHO Constitution came into force in 1948; one theme every year.
- ▶ **2015 theme** = Food safety; slogan "From farm to plate, make food safe." Write the slogan verbatim — it is the mark-earning line.
- ▶ **Five Keys mnemonic** — "CLEAN, SEPARATE, COOK, CHILL, SAFE SOURCE" (wash hands; separate raw and cooked; cook thoroughly; safe temperature; safe water and raw material).
- ▶ **Global burden to quote**: 600 million illnesses and 4,20,000 deaths a year; >200 diseases carried by food; WHO South-East Asia Region 150 million ill, 1,75,000 deaths.
- ▶ **Indian anchors and law**: food-borne illness a bigger burden than malaria, HIV/AIDS and tuberculosis (Ministry of Health), >100 million cases a year rising to 150-177 million by 2030, hygiene facilities for only 68% (82% urban, 60% rural); PFA Act 1954 (concurrent list, State-enforced) → FSSA 2006, in force 5 August 2011 → FSSAI (2008); standards based on the **Codex Alimentarius Commission**; enforcement by the **Food Safety Officer** under a district Designated Officer and a State Commissioner of Food Safety; four regional **appellate** Central Food Laboratories = **Kolkata, Mysore, Ghaziabad, Pune** (report final on appeal).
- ▶ **Four new threats (name all four)**: changes in food production/distribution/consumption; environmental change; new and emerging pathogens; antimicrobial resistance.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the Prevention of Food Adulteration Act 1954 stands repealed by the Food Safety and Standards Act 2006 with effect from 5 August 2011, so the current answer must lead with the Food Safety and Standards Authority of India and the Food Safety Officer, while the Food Inspector and the three-sample procedure shown in the last flowchart are PFA-era practice that the textbooks still describe.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 25 International Health, p 1055; Ch 11 Nutrition and Health, pp 796-802; Ch 5 Epidemiology of Communicable Diseases, pp 368-369 and pp 429-430); Sunder Lal — Textbook of Community Medicine (Ch 5 Nutrition, pp 218-224; Ch 8, p 371, pp 486-487 and pp 490-491 for the neighbouring World Health Day themes); Jugal Kishore — National Health Programmes of India (Ch 2, pp 881-882 — enforcement of the Food Safety and Standards Act 2006); Textbook of International Health (Ch 10 Health and the Environment, p 536; Ch 6, pp 333-334 — factors in the emergence of infectious diseases); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 36 International Health, p 670, Table 36.1 — World Health Day slogans); AFMC Red Book of Community Medicine 2024 (Ch 30, p 871 — global and Indian food-borne disease burden); WHO Foodborne Disease Burden Epidemiology Reference Group, 2015.

Coverage note: The corpus carries the 2015 theme, its slogan, its rationale and the Five Keys, but not the full text of the WHO World Health Day 2015 campaign document; the campaign's stakeholder messages above are drawn from the textbook summaries of it, and the corpus does not state which WHO estimate the 600 million / 4,20,000 figures derive from.



World Health Day theme 2024

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — World Health Day is the global health awareness day observed on **7 April** every year under the sponsorship of the World Health Organization (WHO), marking the date in 1948 on which the WHO Constitution came into force (Park 28e, Ch 25, p 1055). A theme is chosen each year to focus world attention on one specific aspect of public health. The theme for **World Health Day 2024**, WHO's 76th anniversary, was "My health, my right".

1. World Health Day — origin and standing

Item	Detail
Basis of the date	WHO Constitution drafted 1946 by the Technical Preparatory Committee under <i>Rene Sand</i> ; approved by an International Health Conference of 51 nations at New York; came into force 7 April 1948 (Park 28e, Ch 25, p 1055)

Item	Detail
Formal decision	The First World Health Assembly (1948) resolved to observe 7 April each year, with effect from 1950, as World Health Day
Purpose	To draw worldwide attention each year to a subject of major importance to global health; WHO organises international, regional and local events on that theme
Observed by	All Member States of WHO, together with governments and non-governmental bodies
Company it keeps	One of WHO's official global public health campaigns, along with World TB Day, World Immunization Week, World Malaria Day, World No Tobacco Day, World AIDS Day, World Blood Donor Day and World Hepatitis Day
Its parent idea	The theme always operationalises the WHO objective — "the attainment by all people of the highest possible level of health" (Preamble, WHO Constitution)

2. The 2024 theme — "My health, my right"

Element	Content
Theme (slogan)	"My health, my right"
Date and occasion	7 April 2024 — the 76th anniversary of WHO
Root text invoked	The Preamble to the WHO Constitution: <i>"The enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, economic or social condition"</i> (Park 28e, Ch 25, p 1055)
Scope championed	The right of everyone, everywhere to quality health services, health education and information , and to the underlying determinants — safe drinking water, clean air, good nutrition, quality housing, decent working and environmental conditions, and freedom from discrimination
Problem it addressed	Health rights are constitutionally recognised in most countries, yet enabling legislation and financing are not translated into actual entitlement to services; more than half the world's population is still not fully covered by essential health services
Call to action	Move from constitutional recognition → enforceable law → financed, accountable service delivery, i.e. universal health coverage (UHC) as the delivery vehicle of the right

- **Why this theme in 2024** — the right to health is already the oldest normative commitment in international health (1946), but WHO's own review found the gap lies in *implementation*: rights recognised on paper, not claimable at the point of care.
- **Whose right** — framed in the first person ("my") deliberately, to shift the language from state benevolence and charity to an **individual entitlement** that the citizen may demand and the state is legally bound to deliver.
- **What it is not** — the right to health is **not a right to be healthy**; it is the right to the facilities, goods, services and conditions conducive to the highest attainable standard of physical and mental health (Oxford Public Health 7e, Ch 19, p 745).

3. Content of the "right to health"

Freedoms and entitlements (Oxford Public Health 7e, Ch 19, pp 745–746):

Freedoms (protections from)	Entitlements (claims to)
Freedom from discrimination in access to care	Maternal, child and reproductive health services
Freedom from non-consensual medical treatment	Prevention, treatment and control of diseases
Freedom from non-consensual experimentation	Access to essential medicines
Privacy of health information and access to information	Healthy workplaces and environments
Freedom to participate in health decisions affecting oneself	Safe and potable water and adequate sanitation

Legal architecture — the milestones an examiner expects named and dated:

Instrument	Year	Health provision
WHO Constitution	drafted 1946; in force 7 April 1948	Highest attainable standard of health is a fundamental right of every human being
Universal Declaration of Human Rights, Article 25(1)	1948	Right to a standard of living adequate for health and well-being — food, clothing, housing, medical care and necessary social services
International Covenant on Economic, Social and Cultural Rights (ICESCR), Article 12	adopted 1966, in force 1976	The cornerstone binding protection — states recognise the right of everyone to the highest attainable standard of physical and mental health; 164 states had ratified by 2018 (Oxford Public Health 7e, Ch 19, p 746)
CESCR General Comment 14	2000	Authoritative interpretation of Article 12; source of the AAAQ criteria and of the respect–protect–fulfil typology
Convention on the Rights of the Child, Article 24	1989	Child's right to the highest attainable standard of health; ratified by every country except the United States
Declaration of Alma-Ata	1978	Health for All through primary health care; India is a signatory (Park 28e, Ch 2, p 35)
Sustainable Development Goal target 3.8	2015	Universal health coverage including financial risk protection — the measurable expression of the right

4. AAAQ and the duties it places on the State

AAAQ — the four-part operational test of any health service, facility or good (Oxford Public Health 7e, Ch 19, pp 764–765):

Criterion	What must be asked
Availability	Are functioning facilities, trained and employed staff, medicines and supplies present in sufficient quantity, sustainably?

Criterion	What must be asked
Accessibility	Can everyone reach and use them — irrespective of gender, disability, caste, residence, income (physical, economic and information accessibility)?
Acceptability	Is care culturally appropriate, ethical, confidential, and delivered in the language of disadvantaged groups?
Quality	Is care scientifically and medically appropriate and of good quality?

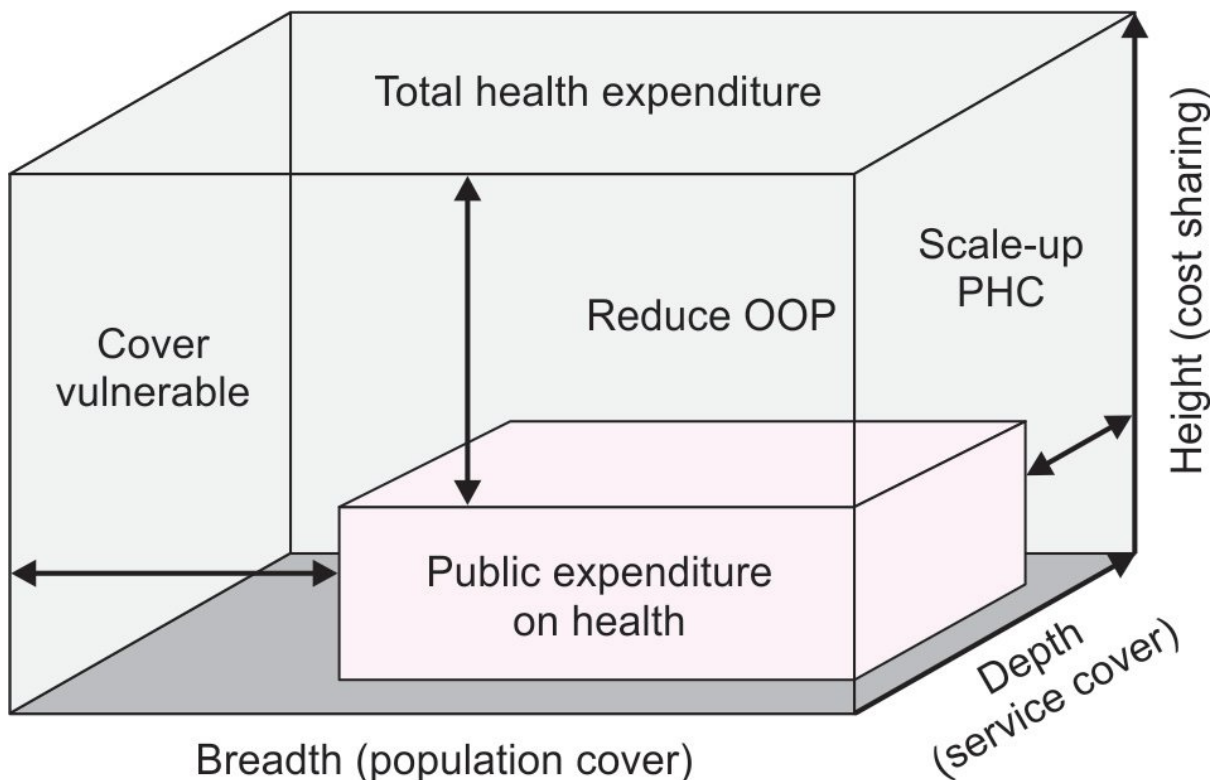
Obligations of the State (Oxford Public Health 7e, Ch 19, pp 752–754):

- **Respect** — refrain from directly or indirectly interfering with the enjoyment of the right to health.
- **Protect** — prevent third parties (employers, industry, private providers, insurers) from interfering with it.
- **Fulfil** — adopt the legislative, administrative and **budgetary** measures needed for its full realisation.
- **Progressive realisation** — the standard demanded rises with a state's resources; a state must be doing better in two years than today, measured by explicit indicators and benchmarks.
- **Non-retrogression** — existing levels of service provision and access may not be reduced; if retrogression occurs, the state must justify it to United Nations monitoring committees.
- **Obligations of immediate effect** — not subject to resource constraints: freedom from **discrimination**, a national plan of action for health, and provision of WHO-listed essential medicines and reproductive, maternal and child health services.
- **Cross-cutting features** — dignity and equity, **non-discrimination**, active and informed **participation** (a bottom-up approach), and **accountability** with a remedy when the right is violated (Oxford Public Health 7e, Ch 19, p 755).

5. Applying "My health, my right" to India

Instrument / scheme	Provision relevant to the right to health
Constitution, Article 21	Right to life; requires the State to ensure the health and nutritional well-being of all people — the judicially expanded basis of the right to health in India (Sunder Lal, Ch 17, p 743)
Constitution, Article 47 (Directive Principles, Part IV)	The State shall regard <i>"the raising of the level of nutrition and the standard of living of its people and the improvement of public health as among its primary duties"</i> (Park 28e, Ch 2, p 35)
Constitution, Article 24	No child below 14 years shall be employed in any factory, mine or other hazardous employment (Park 28e, Ch 18, p 967)
Health as a State subject	States deliver health services; the Centre frames broad policy and funds the national health programmes (Sunder Lal, Ch 17, p 743)
National Health Policy 2017	Government health expenditure to rise from 1.15% to 2.5% of GDP by 2025; state health spending >8% of budget by 2020; households facing catastrophic health expenditure to fall by 25% by 2025 ; life expectancy at birth 67.5 → 70 by 2025 (Park 28e, Ch 23, p 1030)

Instrument / scheme	Provision relevant to the right to health
Ayushman Bharat — Ayushman Arogya Mandir (AAM) (erstwhile Health and Wellness Centre)	Comprehensive primary health care — preventive, promotive, curative, rehabilitative and palliative — envisaged by the National Health Policy 2017 as the foundation of India's health system; first centre inaugurated 14 April 2018, Bijapur district, Chhattisgarh; 1,72,148 centres operational as on 31 March 2024 ; the sub-centre-level AAM is headed by a Community Health Officer (Park 28e, Ch 7; Oxford Public Health 7e, Ch 21, pp 841–842)
Ayushman Bharat — Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)	Financial risk protection: ₹5 lakh per family per year for secondary and tertiary hospitalisation to about 10.74 crore poor and vulnerable families identified from Socio-Economic Caste Census data; launched 23.09.2018 , subsuming the Rashtriya Swasthya Bima Yojana; a distinct card now extends cover to all persons aged 70 years and above (Park 28e, Ch 10, p 745; Ch 7; Ch 1, p 6)
Health for All (30th World Health Assembly)	"Attainment of a level of health that will enable every individual to lead a socially and economically productive life"; equity is its fundamental principle (Park 28e, Ch 10, p 720)



OOP: Out of pocket
PHC: Primary health care

Attaining universal coverage — the three dimensions along which the right to health is progressively realised (Sunderlal, p.740)

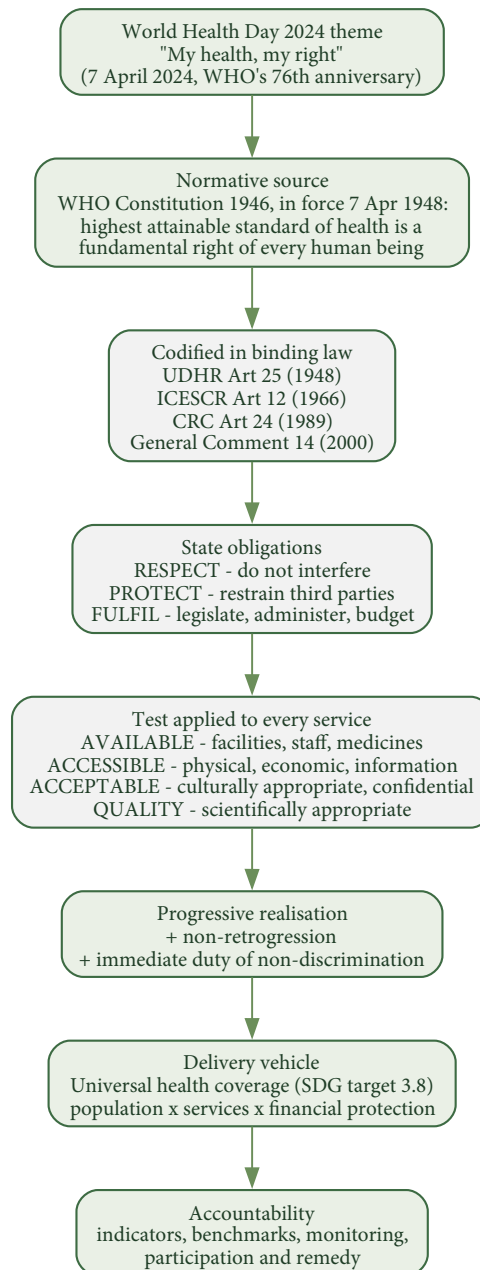
6. Recent World Health Day themes (rapid recall)

Year	Theme
2015	Food safety

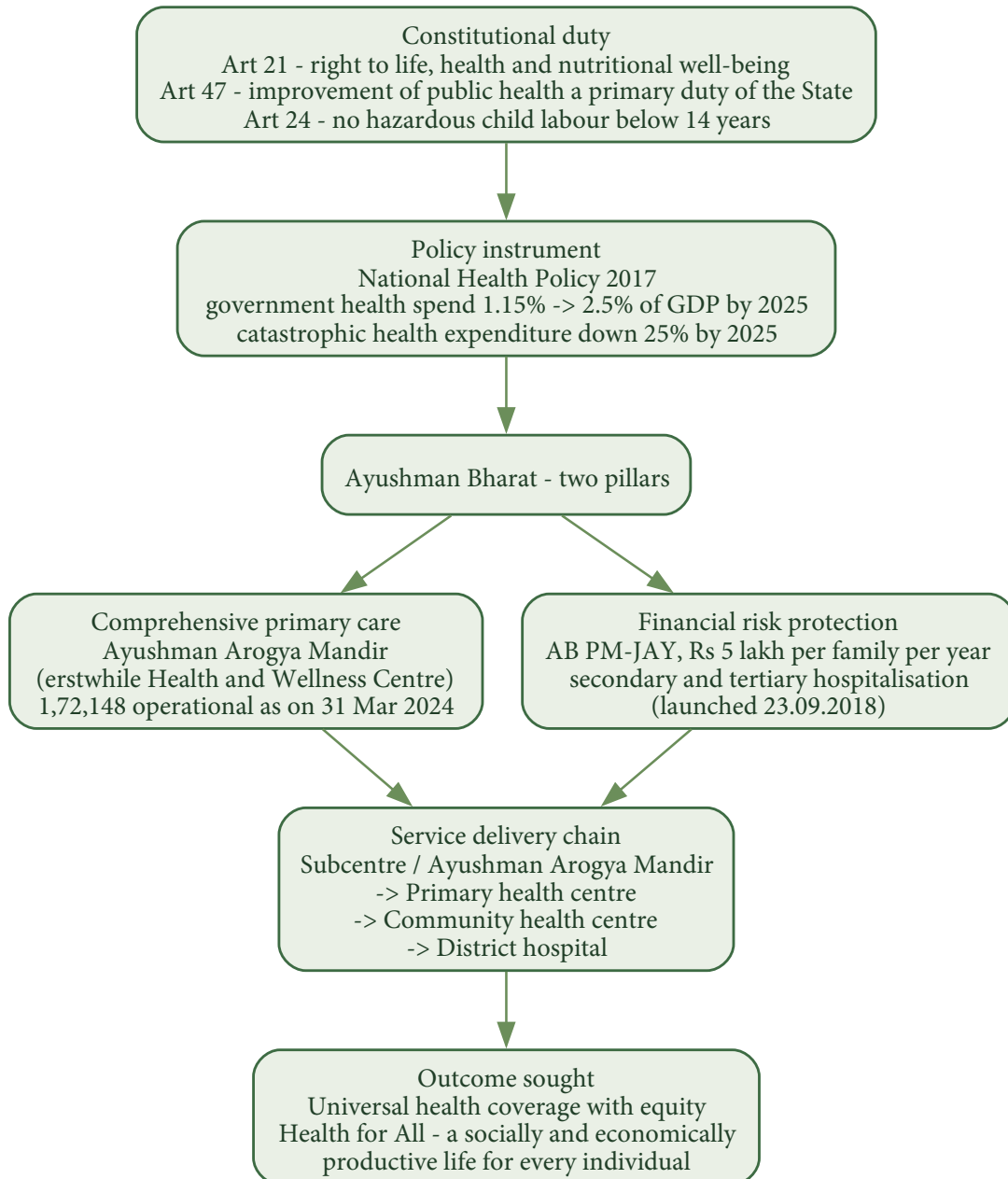
Year	Theme
2016	Halt the rise: beat diabetes
2017	Depression: let's talk
2018 and 2019	Universal health coverage: everyone, everywhere (carried over two years)
2020	Support nurses and midwives
2021	Building a fairer, healthier world
2022	Our planet, our health
2023	Health for All (75 years of WHO)
2024	My health, my right
2025	Healthy beginnings, hopeful futures

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — "The right to health: from norm to entitlement".** Labels to write: *WHO Constitution 1946 (in force 7 April 1948) → Universal Declaration of Human Rights, Article 25 (1948) → ICESCR Article 12 (1966) → CESCR General Comment 14 (2000) → State duties: respect / protect / fulfil → AAAQ: available, accessible, acceptable, quality → Accountability and remedy.*
- **Diagram 2 — "Universal health coverage cube"** (embedded above). Labels: *Breadth = population covered, Depth = services covered, Height = proportion of costs met (cost sharing), inner block = public expenditure on health, arrows = cover the vulnerable, scale up primary health care, reduce out-of-pocket expenditure.*



From "My health, my right" to a claimable entitlement — the normative chain and the State's duties



Realising the right to health in India — constitutional duty translated into financed service delivery

High-yield exam pointers

- **Theme 2024** = "My health, my right", observed 7 April 2024, WHO's 76th anniversary; theme 2023 = *Health for All* (75th anniversary), theme 2025 = *Healthy beginnings, hopeful futures*.
- **Why 7 April** — WHO Constitution came into force 7 April 1948 (drafted 1946 under *Rene Sand*, approved by 51 nations at New York); World Health Assembly 1948 fixed the day, observed from 1950.
- **Quote the Preamble verbatim**: "The enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, economic or social condition."
- **AAAQ** = Available, Accessible, Acceptable, Quality (General Comment 14, 2000). **Respect–Protect–Fulfil** = the three state obligations.

- ▶ The right to health is **not** the right to be healthy — it is the right to facilities, goods, services and conditions conducive to the highest attainable standard of health.
- ▶ Indian anchors: **Article 21** and **Article 47**; National Health Policy 2017 target of **2.5% of GDP by 2025**; AB PM-JAY cover of **₹5 lakh per family per year**; right to health/health care is constitutionally recognised in **at least 115 countries** (Park 28e, Ch 2, p 34).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written as textbook fact: the theme wording, the 76th-anniversary framing and the campaign's own statistics (at least 140 countries recognising health as a constitutional right, and 4.5 billion people not fully covered by essential health services in 2021) come from the WHO World Health Day 2024 campaign material and the WHO–World Bank universal health coverage global monitoring report 2023 — the indexed textbook corpus carries World Health Day themes only up to 2018, so themes for 2019 onward in the table above are taken from WHO campaign announcements; Health and Wellness Centres were renamed **Ayushman Arogya Mandir** in 2023, and both names may appear in older texts.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, National Health Accounts (India), IHR (2005) amendments 2024, WHO Pandemic Agreement 2025

Programme and policy updates (India)

- ▶ Health and Wellness Centres now carry the name **Ayushman Arogya Mandir** (tagline *Arogyam Parmam Dhanam*), but the rename has not closed the entitlement gap — among 320 enrolled PM-JAY beneficiaries surveyed in Jammu and Baramulla districts (Feb–Apr 2025) only **48.4%** had ever used the scheme; awareness of entitlements (OR 1.253) and knowing which hospitals are empanelled (OR 1.245) raised utilisation and poor road quality lowered it (OR 0.761), while financial and administrative barriers were not independent predictors. This is the textbook demonstration of the 2024 theme's central claim: coverage on paper is not a claimable right. [PMID 41878109]
- ▶ **Ayushman Bharat Digital Mission** supplies the informational limb of the right to health through **ABHA**, a unique **14-digit** health account ID, with 8.79 crore IDs generated in financial year 2025–26 (as on 28 January 2026); uptake at the counter still lags — only 1,29,007 of 5,37,278 outpatient visits (**23.96%**) were registered through ABHA across three Agra tertiary hospitals over eight months, and digital health literacy was strongly education-dependent, so the digital route can itself become a new accessibility barrier. [PMID 41787376]

Current national data

- ▶ **National Health Accounts, 2013–14 to 2021–22** — the government share of total health expenditure rose from 28.6% to **48.0%** while the out-of-pocket share fell from 64.2% to **39.4%**; the two curves crossed within the decade. Government health expenditure nonetheless remains at about **1.9% of GDP** against the National Health Policy 2017 target of 2.5%, medicines absorb roughly two-thirds of out-of-pocket spending, and catastrophic expenditure is commoner for outpatient care (47.8%) than for hospitalisation (43.1%) — exactly the spending PM-JAY does not cover. [PMID 42548603]

- ▶ CAMS 2022–23 (Comprehensive Annual Modular Survey, 3,02,086 households) — publicly funded insurance (AB PM-JAY and state schemes) raised outpatient care use (adjusted OR 1.404, 95% CI 1.328–1.484) and cut **inpatient** catastrophic expenditure (adjusted OR 0.721, 95% CI 0.676–0.769 at the 10% threshold). Against **outpatient** catastrophic expenditure it gave **no** protection at the 10% threshold (adjusted OR 0.999, 95% CI 0.936–1.066), though a significant reduction did appear at the stricter 20% threshold (adjusted OR 0.869, 95% CI 0.810–0.932) — outpatient care remains the scheme's weakly covered flank. [PMID 42167946]
- ▶ South Asia projections to 2030 — the regional composite coverage index rises from 53.5% (2000) to a predicted 81.5% (2030), with India projected to cross the 80% UHC benchmark, yet regional catastrophic health expenditure incidence more than doubles (7.2% to 18.6%) and India reaches **26.3% by 2030**; service coverage and financial protection are moving in opposite directions. [PMID 41192927]

Elimination, surveillance and disease status

- ▶ Antimicrobial resistance is being reframed as a universal health coverage obligation rather than a bedside problem — India now runs the **National Action Plan on AMR** together with six state action plans, covering stewardship programmes, laboratory network strengthening, awareness and One Health intersectoral surveillance; enforcement gaps, over-the-counter sales, pharmaceutical and hospital wastewater contamination and fragmented data remain the limiting factors. [PMID 41443506]

Guideline and legislative updates

- ▶ **Rajasthan's Right to Health Care Act (Bill 2022)** is the live Indian test case of "my health, my right" — a state statute converting the right into an enforceable claim on providers. Private-sector coalitions united rapidly, pairing direct and indirect tactics to dominate the public narrative and the policy process, while a published critique judged the drafting seriously deficient and pointed to Tamil Nadu's *Innuyir Kappom Thittam — Nammai Kaakum 48* scheme as a working alternative. Examiners use this to test the gap between constitutional recognition and enforceable entitlement. [PMID 41362954]; [PMID 38909300]

Global and WHO developments

- ▶ **IHR (2005) amendments adopted 1 June 2024** at the 77th World Health Assembly, completing a 21-month process — no intermediate alert tier was created; instead a higher "**pandemic emergency**" tier was inserted *within* the PHEIC framework, applying only to communicable-disease events and running for the duration of the underlying PHEIC, and new Article 13(8) duties oblige WHO to remove barriers to timely, equitable access to health products once such an emergency is determined. [PMID 40883832]
- ▶ **WHO Pandemic Agreement adopted May 2025** at the 78th World Health Assembly under Article 19 of the WHO Constitution — its **Pathogen Access and Benefit-Sharing (PABS)** annex under Article 12 was mandated for finalisation by April 2026 and would bind manufacturers by contract to make at least **20%** of pandemic-related products available to WHO (half donated, half at affordable prices), with a Global Supply Chain and Logistics Network under Article 13. This is the right to health translated into treaty obligation. [PMID 41832575]
- ▶ World Health Day themes after 2024 — **2025: "Healthy beginnings, hopeful futures"** (maternal and newborn survival) and **2026: "Together for health. Stand with science"**, observed 7 April

2026; quote the current year's theme alongside 2024's when the question asks for context. [WHO 2026]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 Concept of Health and Disease; Ch 7 Health Programmes in India; Ch 10 Health Care of the Community; Ch 18 Occupational Health; Ch 23 Health Planning and Management; Ch 25 International Health); Oxford Textbook of Public Health 7e (Ch 19.3 The right to health supports global public health; Ch 16.11 How access to healthcare affects population health; Ch 21.5 Health policy in developing countries); Textbook of International Health (Ch 3 International Health Agencies; Ch 4 The Political Economy of Health and Development); Sunder Lal — Textbook of Community Medicine (Ch 17 Healthcare of the Community); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 36 International Health); Park Plus (Miscellaneous — list of World Health Day themes); WHO Constitution 1946; CESC General Comment 14 (2000); National Health Policy 2017; WHO World Health Day 2024 campaign.

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♦ ♦ ♦

World Health Day theme 2025

Not previously asked — added for currency — RGUHS MD Community Medicine

Definition — World Health Day is observed on 7 April every year, the date on which the Constitution of the World Health Organization (WHO) came into force in 1948, and a theme is chosen each year to focus world attention on one specific aspect of public health (Park 28e, Ch 25, p 1055). The theme for 7 April 2025 is "**Healthy beginnings, hopeful futures**", which opens a year-long WHO campaign on **maternal and newborn health**. Its twin demands are to end preventable maternal and newborn deaths and stillbirths, and to look beyond survival to the longer-term health and well-being of women.

The observance and how the theme is set

Item	Detail
Date and origin	7 April annually; the WHO Constitution was drafted in 1946 by the Technical Preparatory Committee under <i>Rene Sand</i> , approved the same year by an International Health Conference of 51 nations at New York, and came into force on 7 April 1948 (Park 28e, Ch 25, p 1055)
Ratification	Ratified by 26 United Nations member states — a majority — on 7 April 1948; this date is commemorated annually (Textbook of International Health, Ch 3, p 102)
Purpose of an annual theme	A new theme is suggested every year to focus world attention on a current, specific health issue of international significance and to generate a favourable climate for its management and control (Suryakantha, Ch 41; Park 28e, Ch 25, p 1055)
Theme 2025	" Healthy beginnings, hopeful futures " — maternal and newborn health, run as a year-long campaign, not a single-day event
Related health days in the calendar	World TB Day 24 March (the one that precedes it), World Malaria Day 25 April, World Hypertension Day 17 May, World No Tobacco Day 31 May, Safe Motherhood Day 10 July, National Newborn Week 15–21 November (Suryakantha, Annexures; Red Book — Public Health and Preventive Medicine, Important Health Related Days, p 1083)

Recent World Health Day themes	Theme
1998	Pregnancy is special, let us make it safe (Red Book, p 1083)
2001	Mental health — stop exclusion, dare to care (Red Book, p 1083; Suryakantha, Ch 43)
2004	Road safety is no accident (Red Book, p 1083; Sunder Lal, Ch 8)
2005	Make every mother and child count (Red Book, p 1083)
2008	Protecting health from climate change (Red Book, p 1083; Suryakantha, Ch 48)
2010	Urban health matters (Sunder Lal, Ch 17, p 756)
2013	High blood pressure (Sunder Lal, Ch 8)

Recent World Health Day themes	Theme
2014	Small bite, big threat — vector-borne disease (Sunder Lal, Ch 8)
2015	From farm to plate, make food safe (Sunder Lal, Ch 5)
2016	Halt the rise — beat diabetes (Sunder Lal, Ch 8)
2019	Universal health coverage: everyone, everywhere (Sunder Lal, Ch 1)
2020	Year of the nurse and the midwife (Sunder Lal, Ch 18, p 775)
2025	Healthy beginnings, hopeful futures — maternal and newborn health

Read down that list and 2025 is the third time in under thirty years that WHO has given the day to mothers and babies — after *"Pregnancy is special, let us make it safe"* in 1998 and *"Make every mother and child count"* in 2005. That repetition is itself the argument: the subject keeps returning because the mortality gap has never closed.

Why maternal and newborn health was chosen — the burden addressed

Indicator	Global	India
Maternal deaths per year	287,000 (2020)	24,000 (2020) — with Nigeria (82,000) accounts for ~35% of all global maternal deaths
Maternal mortality ratio (MMR), per 100,000 live births	223 (2020); least developed countries 372; sub-Saharan Africa 377	97 (SRS 2018–20) , down from 167 (SRS 2011–13)
Lifetime risk of maternal death	1 in 210 (2020); 1 in 66 in least developed countries	0.3% overall; 0.5% in EAG states and Assam vs 0.1% in southern states (SRS 2018–20)
Maternal morbidity	For every maternal death, ~20 women suffer pregnancy-related illness — ~10 million women a year	—
Timing of maternal death	50–70% in the postpartum period, of which 45% within the first 24 hours after delivery; 11–17% during childbirth itself	—
Newborn deaths per year	2.4 million died before 4 weeks of age (2022); half within the first 24 hours; 98% in developing countries	Neonatal mortality rate 20 per 1000 live births (SRS 2020) — rural 23, urban 12
Neonatal share of child death	Global neonatal mortality rate 17 per 1000 live births (2022)	71.9% of all infant deaths are neonatal (rural 74.1%, urban 62.4%); 62% of under-5 deaths occur in the neonatal period, 20% on the day of birth, 49% within the first 7 days

Indicator	Global	India
Infant and under-5 mortality	India contributes ~25% of the world's 5.3 million under-5 deaths each year	IMR 28 (SRS 2020); under-5 mortality ranges from 8 in Kerala to 51 in Madhya Pradesh , rural 36 vs urban 21 (SRS 2020)
State spread of MMR (India)	—	Kerala 19 (lowest) to Assam 195 (highest); EAG states + Assam 137 vs "other" states 76 (SRS 2018–20)

(India and global values: Park 28e, Ch 9, pp 651–671; IMR for 2020: Sample Registration System Bulletin 2022, Vol 55 No 1.)

Causes of maternal death, India (SRS 2001–03)	Per cent
Haemorrhage	38
Other conditions (anaemia the largest component, 19%)	34
Sepsis	11
Abortion	8
Hypertensive disorders (toxaemia)	5
Obstructed labour	5

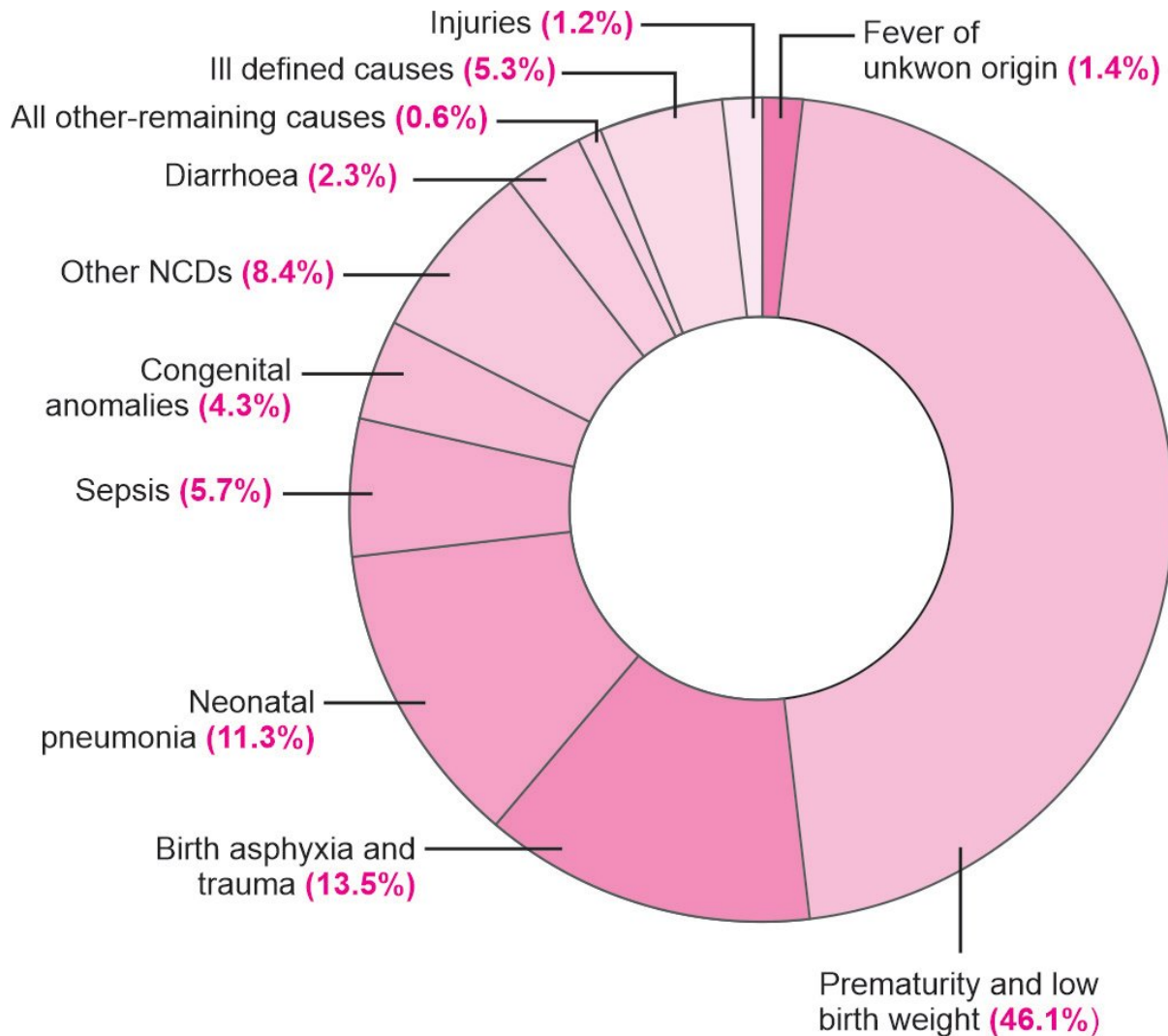


Fig 10.4 — Causes of neonatal deaths in India (%) SRS—2015–17 (Sunderlal, p.592)

Anaemia is both a cause of maternal death in its own right and an aggravating factor for haemorrhage, sepsis and toxæmia; unsafe abortion persists despite Medical Termination of Pregnancy facilities, pointing to unmet need for contraception (Park 28e, Ch 9, p 664). On the newborn side, prematurity with low birth weight alone accounts for nearly half of all neonatal deaths, and asphyxia and birth injury reflect poorly managed labour and lack of access to obstetric services; nearly three-quarters of neonatal deaths are preventable if the woman is adequately nourished and receives appropriate care during pregnancy, childbirth and the postnatal period (Park 28e, Ch 9, p 669).

The campaign therefore makes two explicit calls. WHO asks **governments** to invest in, staff and legislate for maternal and newborn services so that no woman dies giving life, and asks the **health community** to deliver quality care across the whole continuum rather than only at the moment of delivery — and then to look beyond survival, addressing postpartum haemorrhage, anaemia, hypertensive disease, mental health, family planning and the chronic conditions that follow a woman through the rest of her life course. A whole year, rather than a single day, was chosen precisely because a service redesign of that size cannot be carried by one observance.

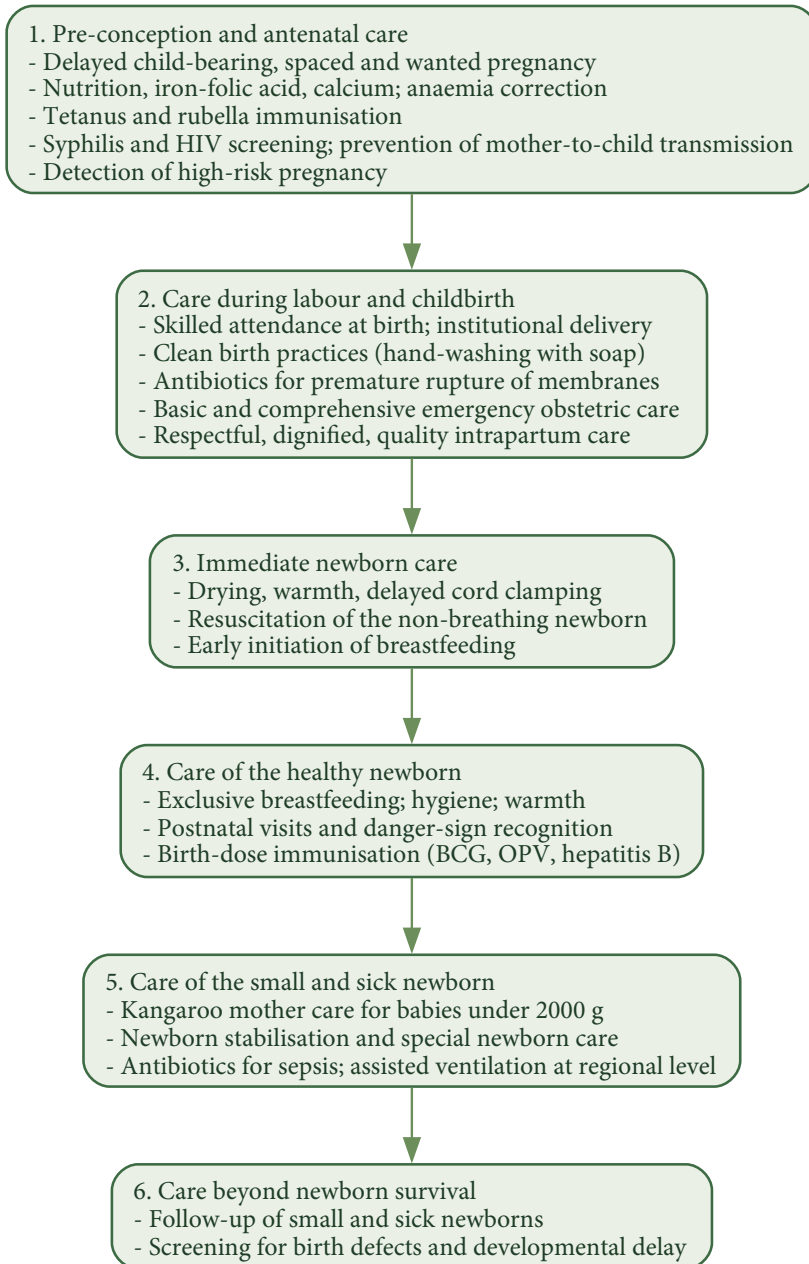
India is the pivotal country for this theme. With Nigeria it accounts for about **35% of all maternal deaths worldwide**, it contributes roughly a quarter of the world's under-5 deaths, and although it has

already met the National Health Policy maternal mortality goal, its **neonatal mortality of 20 per 1000 live births (SRS 2020)** remains above both the national target of 16 and the Sustainable Development Goal value of 12. Neonatal deaths make up 71.9% of India's infant deaths, and the widest inequity is not between countries but between states — Kerala against Madhya Pradesh, urban against rural. For an Indian candidate the theme therefore translates into an argument about **equity of newborn care within the country**, not merely about the national average.

Targets against which the 2025 theme is measured

Framework	Target
Sustainable Development Goal 3.1	Global MMR below 70 per 100,000 live births by 2030 (Park 28e, Ch 14, p 853)
Sustainable Development Goal 3.2	Neonatal mortality at least as low as 12 and under-5 mortality at least as low as 25 per 1000 live births by 2030 (Park 28e, Ch 9, p 653)
Global Strategy for Women's, Children's and Adolescents' Health (2016–2030)	SURVIVE — end preventable deaths; THRIVE — ensure health and well-being; TRANSFORM — expand enabling environments (Park 28e, Ch 9, p 663)
Every Newborn Action Plan (ENAP), WHO and UNICEF	Five strategic objectives for two measurable targets — by 2035 , every country to reach ≤10 newborn deaths per 1000 live births and ≤10 stillbirths per 1000 total births (Park 28e, Ch 9, p 661)
India Newborn Action Plan (INAP), June 2014	Single-digit neonatal mortality rate by 2030 and single-digit stillbirth rate by 2030 ; milestones reviewed every five years in sync with ENAP (Park 28e, Ch 7, p 560)
National Health Policy 2017 (India)	MMR 100 by 2020; IMR 28 by 2019; neonatal mortality 16 and stillbirth rate single digit by 2025; under-5 mortality 23 by 2025 (Park 28e, Ch 23, p 1030)

Translating the theme into service delivery — the continuum of care

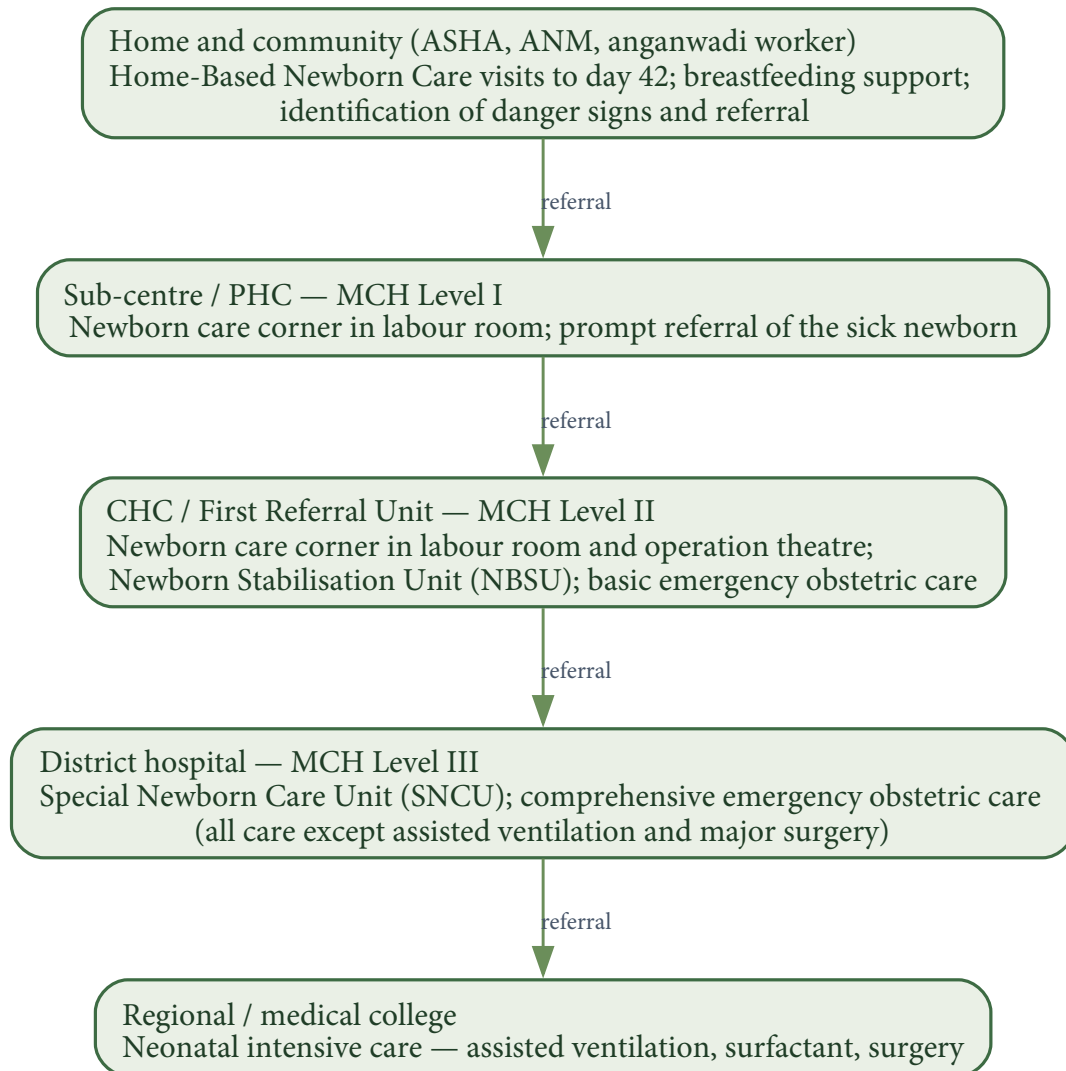


Continuum-of-care intervention packages that operationalise "Healthy beginnings, hopeful futures" (India Newborn Action Plan pillars)

India's programme response that the theme mobilises

Programme (year)	What it provides
Janani Suraksha Yojana (JSY), 2005	Conditional cash transfer to promote institutional delivery, with special focus on low-performing states (Park 28e, Ch 7, p 539)
Janani Shishu Suraksha Karyakram (JSSK), 1 June 2011	Entitlement-based free delivery including caesarean section — free drugs and consumables, diagnostics, blood, diet and transport home-to-facility, facility-to-facility and drop back; the same entitlements cover sick newborns treated in public institutions up to 30 days after birth, and the scheme now also covers complications during antenatal and postnatal care and sick infants (Park 28e, Ch 7, p 545)

Programme (year)	What it provides
Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA), June 2016	Fixed-day, free, assured, quality antenatal care with a minimum package of investigations and drugs on the 9th day of every month , using volunteer private specialists; about 2.88 crore antenatal check-ups conducted and over 23.60 lakh high-risk pregnancies identified (Park 28e, Ch 7, p 545)
LaQshya, 2017	Quality of care in labour rooms and maternity operation theatres, with respectful care in the intrapartum and postpartum periods (Park 28e, Ch 7, p 545)
SUMAN (Surakshit Matritva Aashwasan), 10 Oct 2019	Assured, dignified, respectful, free care with zero tolerance for denial of service , to end all preventable maternal and newborn deaths and give a positive birthing experience (Park 28e, Ch 7, p 545)
Navjaat Shishu Suraksha Karyakram (NSSK), 2009	A trained health person in basic newborn care and resuscitation at every delivery point — prevention of hypothermia and infection, early initiation of breastfeeding (Park 28e, Ch 7, p 547)
Facility-Based Newborn Care	22,147 newborn care corners at delivery points, 3,215 newborn stabilisation units at CHCs and first referral units, 1,054 special newborn care units at district/sub-district hospitals and medical colleges (Park 28e, Ch 7, pp 548, 560)
Home-Based Newborn Care (HBNC), 2011	ASHA home visits up to day 42 — six visits after institutional delivery (days 3, 7, 14, 21, 28, 42) and seven after home delivery (adding day 1) ; over 1.47 crore newborns visited in 2023–24 (Park 28e, Ch 7, p 550)
India Newborn Action Plan (INAP), 2014	Six intervention pillars, delivered within the RMNCH+A framework and graded Essential [E], Situational [S] and Advanced [A] (Park 28e, Ch 7, p 560)
Village Health and Nutrition Day; maternal death review	Monthly outreach antenatal and postpartum care at the anganwadi; facility- and community-based maternal death audit to improve obstetric care (Park 28e, Ch 7, p 545)



Where "Healthy beginnings, hopeful futures" is delivered — newborn care by level of the Indian health system

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Continuum of care for maternal and newborn health** — a horizontal life-course band crossed by a vertical service band. Label the life-course stages: *Adolescence and before pregnancy* → *Pregnancy* → *Labour, birth and first week* → *Postnatal* → *Child*. Label the service levels: *Community* → *First and secondary level facility* → *Referral and tertiary level facility*. Add the base bar *Intersectoral action: living and working conditions, water and sanitation, nutrition, education and empowerment of girls* (after Park 28e, Ch 9, Fig 17).

High-yield exam pointers

- **7 April 1948** — WHO Constitution in force; hence World Health Day, with a new theme every year. **2025 theme: "Healthy beginnings, hopeful futures" — maternal and newborn health, year-long campaign.** Earlier maternal themes: 1998 and 2005.
- **MMR** = maternal deaths ÷ live births × 100,000 (not total births). India **97** (SRS 2018–20); global **223** (2020); SDG 3.1 target **<70** by 2030.

- ▶ $NMR = \text{deaths under 28 days} \div \text{live births} \times 1000$. India **20** (SRS 2020); SDG 3.2 target **≤12 by 2030**; National Health Policy 2017 target **16 by 2025**; INAP single digit by 2030; ENAP **≤10 newborn deaths and ≤10 stillbirths by 2035**.
- ▶ Global: **287,000** maternal deaths (2020) and **2.4 million** newborn deaths (2022), half in the first 24 hours.
- ▶ India's top killers — mother: **haemorrhage 38%** (SRS 2001–03); newborn: **prematurity and low birth weight 46.1%** (SRS 2015–17).
- ▶ Programme mnemonic for the theme — "J-J-P-L-S": JSY (cash) → JSSK (free entitlement) → PMSMA (9th of every month) → LaQshya (labour-room quality) → SUMAN (zero denial of service).
- ▶ HBNC visit schedule: **3, 7, 14, 21, 28, 42** (institutional); add **day 1** for home delivery.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The 2025 theme title, its framing as a year-long maternal and newborn health campaign and the campaign's calls to action come from the World Health Organization's own campaign announcement for 7 April 2025 and not from the indexed textbook corpus, which predates it; every burden figure, target, programme detail and schedule quoted above is from the retrieved textbook passages, and the Indian mortality figures carry their Sample Registration System round because the survey year is what makes them examinable.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, Cochrane, MoHFW–National Health Mission operational guidelines, NFHS-6 (IIPS), WHO, WHO/UNICEF/UNFPA/World Bank Group/UNDESA maternal mortality estimates, UN IGME

Programme and policy updates (India)

- ▶ **Home-Based Newborn Care and Home-Based Care for Young Child have been merged into a single programme — Samagra Shishu Bal Swasthya Karyakram (SSBSK), operational guidelines 2026** ("पहले तीन साल, सम्पूर्ण देखभाल" — first three years, complete care): one continuum of home-based care from birth to 36 months in place of two separate schemes, with risk stratification of "at-risk" newborns and young children, team-based care by ASHA–ANM–AWW–CHO–Medical Officer, a monthly Shishu Shivar at the Ayushman Arogya Mandir, a Well-Baby Session on VHSND/UHND day up to age 3, and digital line-listing and tele-consultation through the JANANI portal and Shaishav app. This is the single most examinable renaming in this topic — the textbook still describes HBNC and HBYC as separate programmes. [MoHFW 2026]
- ▶ **The HBNC home-visit schedule has been rewritten** — the classic six visits on days 3, 7, 14, 21, 28 and 42 (seven with day 1 after home delivery) now apply only to at-risk babies in expanded form: a healthy newborn after institutional delivery gets **4 visits (day of discharge/day 3, 7, 14, 42)**, a healthy newborn after home delivery **5** (one within 24 hours plus days 3, 7, 14, 42), and an "at-risk" newborn **9 visits (days 1, 3, 5, 7, 10, 14, 21, 28, 42)** with joint ASHA + ANM/CHO visits on days 3 and 7; HBYC's 3–15 month visits become **8 quarterly visits at 3, 6, 9, 12, 18, 24, 30 and 36 months**, plus immunisation contacts at 1.5 and 2.5 months. [MoHFW 2026]

- **Antenatal contacts have been raised above the textbook four** — the 2026 continuum-of-care package specifies 5 ANC visits for a normal pregnancy (including the PMSMA contact) and 8 for a high-risk pregnancy, the three extra visits delivered under Extended PMSMA (E-PMSMA), alongside first-trimester registration with a digital MCH card, calcium and vitamin D3 with IFA, and syphilis/HIV/TB/hepatitis B/GDM screening; Health and Wellness Centres are now Ayushman Arogya Mandirs, and RBSK 2.0 and revised Anaemia Mukh Bharat Abhiyaan guidelines were both reissued in 2026. [MoHFW 2026]

Current national data

- **NFHS-6 (2023–24) has replaced NFHS-5 as the current round** — India fact sheets released May 2026 (IIPS): institutional births 90.6% (NFHS-5 88.6%), births attended by skilled personnel 91.3%, at least 4 ANC visits 65.2% (58.5%), first-trimester ANC 76.2%, IFA for ≥180 days 37.8% (26.0%), postnatal check of the mother within 2 days 82.8%, caesarean section 27.2% (21.5%) and 54.1% of private-facility births, total fertility rate 2.0. A Karnataka taluk-level study found 39.8% of maternal deaths occur within 48 hours postpartum, which is exactly the window these postnatal indicators cover. [NFHS-6 2023-24] [PMID 42386290]
- **NFHS-6 child health, and a change in what the survey reports** — full immunisation at 12–23 months 82.6% (76.6%), rotavirus 3 doses 85.4% (up from 36.4%), MCV2 71.8% (58.6%), hepatitis B birth dose 77.6%, early initiation of breastfeeding 50.1% (41.8%), stunting 29.3% (35.5%) and wasting 19.0%; exclusive breastfeeding under 6 months fell to 55.8% from 63.7%. Note for answer-writing: the NFHS-6 India fact sheet carries 101 indicators and **no longer reports anaemia prevalence or infant/neonatal/under-five mortality**, so those figures must still be quoted from SRS. [NFHS-6 2023-24]

Elimination, surveillance and disease status

- **The elimination endgames have shifted from case-finding to residual-reservoir and species surveillance** — in kala-azar, district data show visceral leishmaniasis falling (Malda, West Bengal: 19 cases in 2018 to 7 in 2022, all blocks under the elimination threshold) while **post-kala-azar dermal leishmaniasis rose from 40 to 55 and became the majority of reported cases**, making PKDL detection the main threat to sustained elimination; in malaria, Mizoram records a shift from *P. falciparum* to *P. vivax* (23% of cases in 2021 to 47% in 2024, 68% in Mamit district in 2025), which matters because vivax needs radical cure and perennial transmission; and for measles–rubella, NFHS-6 second-dose coverage of only 71.8% is the gap between programme intent and elimination-grade immunity. [PMID 42591813] [PMID 42310537] [NFHS-6 2023-24]

Guideline and legislative updates

- **WHO issued a new postpartum haemorrhage guideline in 2025**, built on the E-MOTIVE cluster-randomised trial — 80 secondary-level hospitals, 210,132 vaginal births, a calibrated blood-collection drape for objective diagnosis plus a first-response bundle (uterine massage, oxytocic, tranexamic acid, IV fluids, examination, escalation): the composite of severe PPH, laparotomy for bleeding or death fell from 4.3% to 1.6% (RR 0.40, 95% CI 0.32–0.50), with detection rising from 51.1% to 93.1% and bundle use from 19.4% to 91.2%. Objective measurement plus a bundle, not a new drug, is the practice change. [PMID 41161050] [PMID 37158447]

- **The uterotonic and tranexamic acid evidence has been re-ranked** — a Cochrane network meta-analysis of 122 trials and 121,931 women found **ergometrine plus oxytocin (RR 0.76, 95% CI 0.64–0.90)** and **misoprostol plus oxytocin (RR 0.70, 0.57–0.87)** better than oxytocin alone for PPH ≥ 500 mL, with carbetocin equivalent to oxytocin but the only agent without extra side effects; an individual-patient meta-analysis of 54,404 women showed tranexamic acid reduces life-threatening postpartum bleeding (**OR 0.77, 0.63–0.93**) with no excess thromboembolism (OR 0.96, 0.65–1.41). A 2025 Lancet meta-analysis of 327 studies confirms **uterine atony causes 70.6% of PPH** and puts **anaemia, prior PPH and caesarean birth** among the strongest risk factors — which is why anaemia control is a maternal-survival intervention, not only a nutrition one. [PMID 40237648] [PMID 39461793] [PMID 40188841]
- **Kangaroo mother care now starts before stabilisation** — the iKMC trial in 3,211 infants weighing 1.0–1.799 kg (sites including India) was stopped early when immediate KMC cut 28-day mortality from 15.7% to **12.0% (RR 0.75, 95% CI 0.64–0.89)**; this is the evidence behind starting skin-to-skin contact immediately after birth rather than after the baby is stable, and it is carried into the 2026 essential-newborn-care package. [PMID 34038632]

Global and WHO developments

- **The 2025 theme has been succeeded** — **World Health Day 2026 is "Together for health. Stand with science."**, a year-long campaign on scientific collaboration and evidence-based action across human, animal, plant and planetary health. The maternal–newborn year it follows produced the headline data now used in answers: WHO and partners released *Trends in maternal mortality 2000–2023* on 7 April 2025, giving **about 260,000 maternal deaths in 2023 and a global MMR of 197 per 100,000 live births** (over 700 deaths a day, ~87% in sub-Saharan Africa and southern Asia) against the SDG 3.1 target of <70 ; and UN IGME modelling puts **4.9 million under-five deaths in 2024, including 2.3 million neonatal deaths**, with the decline slowing to 1.5% a year since 2015 (3.9% in 2000–15), **66 countries off-track for the SDG neonatal target** and 27.3 million under-five deaths projected for 2025–30. [WHO 2026-04] [WHO 2025-Apr] [PMID 42242779]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India, Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, Ch 14 Sustainable Development Goals, Ch 23 Health Planning and Management, Ch 25 International Health); Park's Textbook of Preventive and Social Medicine 27e (Ch 9, consulted for the Every Newborn Action Plan target sentence, which is damaged in the 28e scan); Red Book — Public Health and Preventive Medicine (Important Health Related Days — year-wise World Health Day themes since 1950, and the calendar of health days, p 1083); Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 5 Nutrition, Ch 8 Epidemiology of Communicable and Non-Communicable Diseases, Ch 10 Reproductive, Maternal, Newborn, Child and Adolescent Health, Ch 17 Healthcare of the Community, Ch 18 International Health); Suryakantha — Community Medicine with Recent Advances (Ch 41 International Health Organizations, Ch 43, Ch 48, Annexures); Textbook of International Health (Ch 3 International Health Agencies, Activities, and Other Actors). Named documents: WHO Constitution 1948; Sustainable Development Goals 2030 Agenda; Global Strategy for Women's, Children's and Adolescents' Health 2016–2030; Every Newborn Action Plan; India Newborn Action Plan 2014; National Health Policy 2017; Sample Registration System 2018–20 special survey of deaths, SRS Bulletin 2022 (Vol 55 No 1) and SRS 2015–17. Coverage note: the corpus's year-wise theme table ends at 2008 and is extended only by scattered chapter mentions up to 2020, so the 2025 theme itself is taken from the WHO campaign announcement and is flagged as extra-corpus in the Note above.

Recent-advances sources: 42242779 — Global, regional, and national levels and trends in under 5, infant, and neonatal mortality during 1990–2024 with scenario based projections to 2030, *BMJ* 2026, [DOI](#); 37158447 — Randomized Trial of Early Detection and Treatment of Postpartum Hemorrhage (E-MOTIVE), *N Engl J Med* 2023, [DOI](#) (abstract-only); 40237648 — Uterotonic agents for preventing postpartum haemorrhage: a network meta-analysis, *Cochrane Database Syst Rev* 2025, [DOI](#); 39461793 — Tranexamic acid for postpartum bleeding: a systematic review and individual patient data meta-analysis, *Lancet* 2024, [DOI](#); 40188841 — Causes of and risk factors for postpartum haemorrhage: a systematic review and meta-analysis, *Lancet* 2025, [DOI](#) (abstract-only); 34038632 — Immediate "Kangaroo Mother Care" and Survival of Infants with Low Birth Weight (iKMC), *N Engl J Med* 2021, [DOI](#); 41161050 — Active labour management and multiprofessional teamwork are cornerstones in preventing postpartum haemorrhage: EBCOG commentary on the new WHO postpartum haemorrhage guideline 2025, *Eur J Obstet Gynecol Reprod Biol* 2025, [DOI](#) (abstract-only); 42591813 — Sustaining the Elimination of Visceral Leishmaniasis: National Kala-Azar Elimination Programme in Malda District, West Bengal (2018–2022), *Cureus* 2026, [DOI](#); 42310537 — Rising burden of *Plasmodium vivax* in Mizoram: an emerging threat to malaria elimination in Northeast India, *BMC Infect Dis* 2026, [DOI](#) (abstract-only); 42386290 — Spatial autocorrelation and regression approach for delineating maternal mortality and its associated factors in Karnataka, India, *BMJ Open* 2026, [DOI](#). Non-journal sources: MoHFW–National Health Mission, *Samagra Shishu Bal Swasthya Karyakram (SSBSK) Operational Guidelines* 2026 (Aug 2026), *RBSK 2.0 Operational Guidelines* 2026 and *Anaemia Mukta Bharat Abhiyaan Operational Guidelines* 2026; International Institute for Population

Sciences, *National Family Health Survey (NFHS-6), 2023-24: India and State/UT Fact Sheets*, May 2026; WHO World Health Day 2026 campaign, "Together for health. Stand with science." (Apr 2026); WHO/UNICEF/UNFPA/World Bank Group/UNDESA, *Trends in maternal mortality 2000 to 2023*, 7 April 2025.



World Health Day theme 2026

Not previously asked — added for currency — RGUHS MD Community Medicine

Definition — World Health Day is the annual global health observance held on **7 April**, the date on which the Constitution of the World Health Organization (WHO) came into force in 1948 (Park 28e, Ch 25, p 1055). A single theme is chosen each year to focus world attention on one specific aspect of public health. For 7 April 2026 the WHO theme is "**Together for health. Stand with science**" — a two-part advocacy message pairing *solidarity* in health action with *evidence* as the basis of that action.

World Health Day at a glance

Item	Detail
Date observed	7 April, every year
Why this date	The WHO Constitution came into force on 7 April 1948 (Park 28e, Ch 25, p 1055)
Constitution drafted	1946, by the Technical Preparatory Committee under the chairmanship of <i>René Sand</i> ; approved the same year by an International Health Conference of 51 nations at New York (Park 28e, Ch 25, p 1055)
Brought into force by	Ratification by 26 United Nations member states (Textbook of International Health, Ch 3, p 102)
Who sets the theme	WHO headquarters, Geneva — one theme per year, "to focus attention on a specific aspect of public health" (Park 28e, Ch 25, p 1055)
Theme for 2026	" Together for health. Stand with science "
Constitutional basis for the campaign	WHO's first constitutional function — "to act as the directing and coordinating authority on all international health work" (Park 28e, Ch 25, p 1056)
Other WHO campaign days (for contrast)	World TB Day 24 March; World Malaria Day 25 April; World No Tobacco Day 31 May; World AIDS Day 1 December

Unpacking the 2026 theme

Element	Public-health meaning	What it demands of a health system
"Toget her"	Solidarity and collective action — no country, sector or community can secure health alone; the WHO Constitution states that "the health of all peoples is fundamental to the attainment of peace and security and is dependent on the fullest co-operation of individuals and states" (Oxford 7e, Ch 24, pp 918–919)	Multilateral cooperation under the International Health Regulations (2005); whole-of-government and whole-of-society action; community participation
"for health"	Health as a fundamental human right, not a commodity — "the enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being" (Park 28e, Ch 25, p 1055)	Progress towards universal health coverage, the key target of Sustainable Development Goal 3 (Park 28e, Ch 1, p 23)
"Stand with"	Active defence, not passive agreement — public advocacy for science when it is politically attacked; "the mainstream media as a purveyor of trusted information to the public needs to be protected from politically motivated attacks" (Oxford 7e, Ch 7, pp 337–338)	Health professionals acting as advocates for health at all levels of society (Oxford 7e, Ch 47, pp 1742–1743)
"science"	Evidence-based public health — "decision-making based on the best available evidence, availability of resources, and population needs, values and preferences" (Oxford 7e, Ch 35, p 1264)	Sustained research funding, surveillance and data systems, and implementation science to close the know-do gap

Public-health rationale — why this theme now

Driver	Public-health consequence	Response embedded in the theme
Infodemic — health misinformation spreading faster than verified information over social media and messaging platforms	Delayed care-seeking, uptake of unproven remedies, erosion of programme credibility	Social listening, pre-bunking and rapid evidence-based rebuttal
Vaccine hesitancy driven by "social, cultural and religious issues and misinformation" and by organised anti-vaccine lobby groups "often based on unsubstantiated concerns" (Oxford 7e, Ch 96, p 3583)	Falling immunisation coverage, resurgence of measles and other vaccine-preventable disease, outbreaks in pockets of low coverage	Restoring vaccine confidence through trusted local messengers
Politicisation of science and weakening of multilateral institutions and health financing	Evidence-free policy; interrupted supply of vaccines, drugs and diagnostics to low-income settings	Renewed political and financial commitment to WHO and to global solidarity
The know-do gap — on average 17 years for less than 15% of original research to become practice or policy, and longer in low- and middle-income countries (Oxford 7e, Ch 52, p 1925)	Proven interventions never reach scale; avoidable mortality persists	Implementation science and knowledge-translation partnerships

Driver	Public-health consequence	Response embedded in the theme
Recurrent public-health emergencies — Public Health Emergencies of International Concern declared for pandemic influenza A (H1N1) 2009, polio, Ebola and Zika (Oxford 7e, Ch 75, pp 2862–2863)	Collective vulnerability where any one country's core capacities are weak	Full implementation of International Health Regulations (2005) core capacities

Objectives and key messages of the 2026 campaign

- **Advocacy** — place health at the top of the political agenda on one calendar day when media attention is guaranteed, and mobilise political will in favour of health.
- **Defend the evidence base** — assert that health policy, guidelines and programme decisions must rest on the best available scientific evidence, and that uncertainty should be communicated honestly rather than concealed.
- **Counter the infodemic** — equip health workers, teachers and community leaders to identify and correct health misinformation.
- **Rebuild trust** — trust in health institutions, in the health worker and in vaccines is itself a public-health resource that must be actively maintained.
- **Sustain investment in research and in the health workforce** — including local research capacity in low- and middle-income countries, since "the quality of public health is dependent on the competence and vision of the public health workforce" (Oxford 7e, Ch 1, p 137).
- **Reaffirm solidarity** — protect equitable access to vaccines, medicines and diagnostics, and honour obligations under the International Health Regulations (2005).

Recent World Health Day themes

Year	Theme
2008	Protecting health from climate change (Oxford 7e, Ch 107, p 4069)
2022	Our planet, our health
2023	Health for All (WHO's 75th anniversary)
2024	My health, my right
2025	Healthy beginnings, hopeful futures (maternal and newborn health)
2026	Together for health. Stand with science

Observing World Health Day 2026 in an Indian district

- **Plan in advance** — the district health society issues the calendar, budget and information–education–communication (IEC) plan under the National Health Mission at least four weeks ahead.

- ▶ **Translate and adapt** the WHO key messages into the local language and idiom; mass media alone are generally inadequate in changing human behaviour and must be combined with interpersonal methods (Park 28e, Ch 22, p 1023).
- ▶ **Use every channel** — newspapers, radio, television, direct mailing to panchayat leaders, plus social media, which now reaches and benefits more people than traditional channels (Oxford 7e, Ch 25, pp 1230–1231).
- ▶ **Deliver services, not only slogans** — screening camps for hypertension, diabetes and oral, breast and cervical cancer, and immunisation catch-up sessions at **Ayushman Arogya Mandirs** (erstwhile Ayushman Bharat Health and Wellness Centres, launched 2018) (Park 28e, Ch 7, p 541).
- ▶ **Engage the frontline** — accredited social health activists (ASHAs), auxiliary nurse midwives (ANMs) and anganwadi workers act as the trusted messengers who actually close the last mile.
- ▶ **Intersectoral involvement** — schools, workplaces, panchayati raj institutions, self-help groups, medical colleges and non-governmental organisations, reflecting the "together" pillar.
- ▶ **Evaluate** — record reach (persons contacted, camps held, services delivered) and, where feasible, change in knowledge or in service uptake, rather than counting rallies alone.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Structure of the World Health Organization** (draw as a vertical tree; the day commemorates its Constitution). Labels: *Constitution in force 7 April 1948 → World Health Day*; **World Health Assembly** (supreme governing body, "health parliament of nations", meets annually in May at Geneva, one vote per Member State — determines international health policy, reviews the past year's work, approves the budget, elects Executive Board members and appoints the Director-General); **Executive Board** (34 members, technically qualified in the field of health, three-year terms, one-third renewed yearly, meets at least twice a year); **Secretariat** (headed by the Director-General, the chief technical and administrative officer); **Six regional offices** — South-East Asia (New Delhi), Africa (Brazzaville), The Americas (Washington DC), Europe (Copenhagen), Eastern Mediterranean (Alexandria), Western Pacific (Manila) (Park 28e, Ch 25, p 1057).

"Together for health. Stand with science"
World Health Day, 7 April 2026

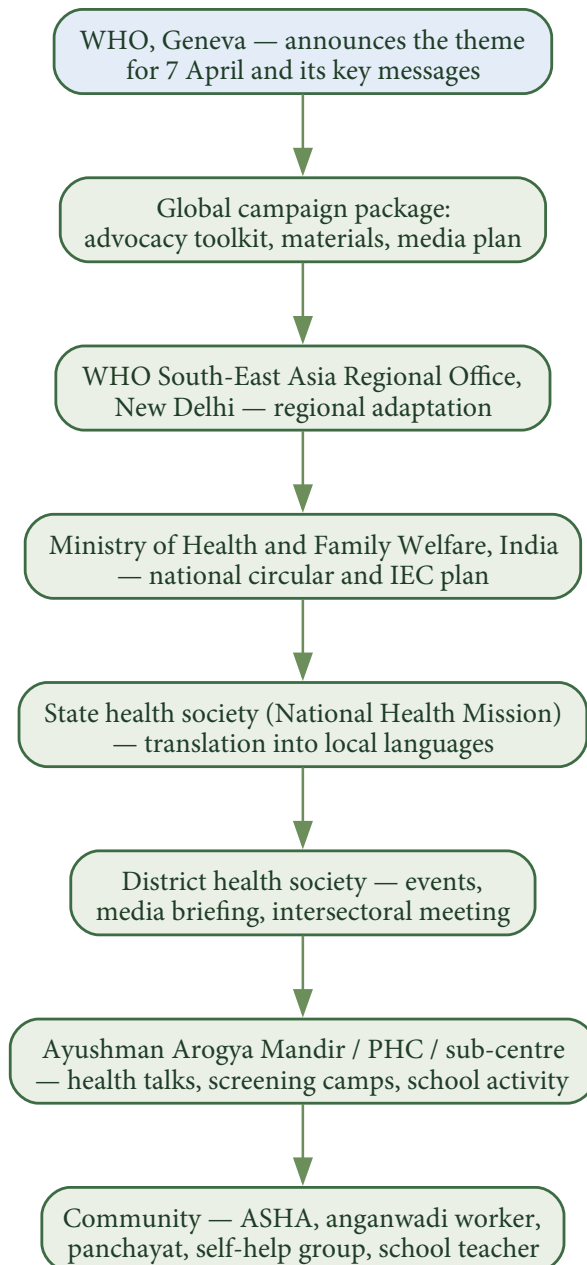
PILLAR 1 — TOGETHER FOR HEALTH (solidarity)

- Multilateral cooperation under International Health Regulations (2005)
- Whole-of-government, whole-of-society, multisectoral action
- Community participation within primary health care
- Equitable financing towards universal health coverage (SDG 3.8)
- Protecting and investing in the health workforce

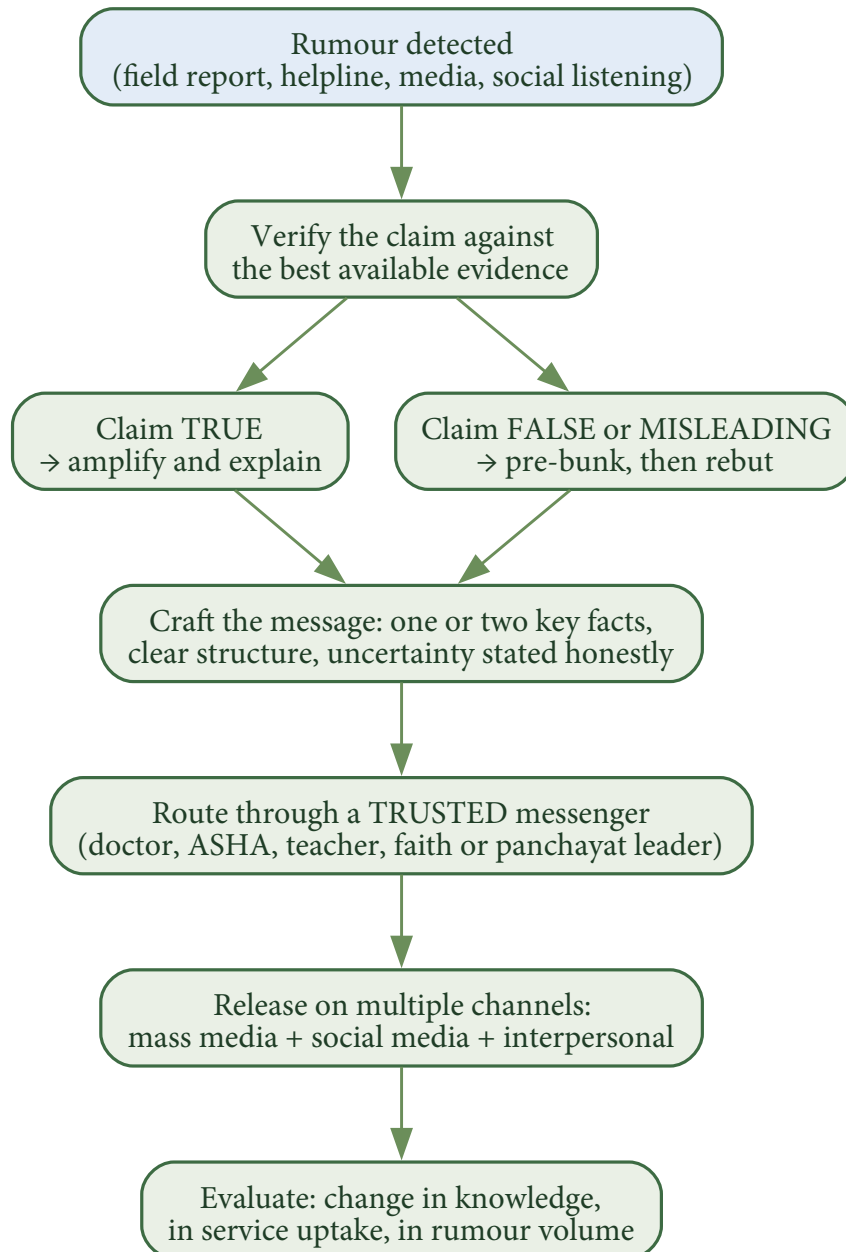
PILLAR 2 — STAND WITH SCIENCE (evidence)

- Evidence-based public health decision-making
- Countering the infodemic and health misinformation
- Restoring vaccine confidence and trust in institutions
- Sustained funding for research and local research capacity
- Closing the know-do gap through implementation science

The two pillars of the World Health Day 2026 theme and what each demands



How a World Health Day theme is translated into action, from Geneva to the village



Operational pathway for the "Stand with science" pillar — responding to health misinformation

High-yield exam pointers

- ▶ **7 April 1948** = WHO Constitution in force = World Health Day. Draft 1946 by the Technical Preparatory Committee under *René Sand*; approved by the International Health Conference of **51 nations**, New York, 1946; ratified by **26 United Nations member states**.
- ▶ **Theme 2026: "Together for health. Stand with science."** Write it verbatim, then split it into the two pillars — solidarity and evidence.
- ▶ WHO's three principal organs: **World Health Assembly** (May, Geneva) → **Executive Board** (34 members) → **Secretariat** (Director-General). Six regions; **South-East Asia Regional Office** is at **New Delhi** — India joined in 1948.
- ▶ WHO's first constitutional function: "**directing and coordinating authority on all international health work.**" WHO objective: "**the attainment by all peoples of the highest possible level of health.**"

- Evidence-based public health = best available **evidence + resources + population needs, values and preferences**. Know-do gap $\approx$ 17 years for <15% of research to reach practice.
- Link the theme to **SDG 3** ("Ensure healthy lives and promote well-being for all at all ages"; 13 targets), with **universal health coverage** as the key target underpinning all the others.
- **Mass media alone do not change behaviour** — combine with interpersonal communication through a trusted local messenger.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The 2026 theme wording is taken from the WHO World Health Day campaign announcement and postdates every indexed textbook; the retrieved corpus confirms the date, origin and purpose of World Health Day and the 2008 theme, but a year-by-year list of themes is not carried in any indexed book, so the themes given for 2022–2026 are from WHO campaign announcements rather than from the retrieved passages. Ayushman Bharat Health and Wellness Centres have since been renamed **Ayushman Arogya Mandir**; give the current name first in the examination and the former name once in parentheses.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, World Health Assembly resolutions WHA78 and WHA79, WHO Pandemic Agreement, Universal Immunisation Programme (MoHFW), IARC

Programme and policy updates (India)

- **HPV vaccine enters the Universal Immunisation Programme.** India launched a free nationwide HPV vaccination drive on 28 February 2026 aiming to vaccinate about 1.15 crore girls aged 14, joining the 160-plus countries that carry HPV vaccine in their national immunisation programme; the indigenous quadrivalent vaccine **Cervavac** is the vaccine proposed for Universal Immunisation Programme use in girls aged 9–14. India accounts for nearly 19% of the world's cervical cancer cases and 23% of cervical-cancer deaths (IARC 2024), so the drive is India's principal contribution to the WHO 90–70–90 cervical cancer elimination targets for 2030. This is the newest UIP change and the most likely 2026 viva question. Misinformation flooded social media soon after the launch — unverified safety claims falsely tied to a misattributed adolescent death case from the 2009 PATH demonstration project, spread through forwarded messages and edited videos. The proposed counter is the "stand with science" pillar in operational form: *pre-bunking* rooted in inoculation theory before rumours take hold, framing the vaccine as cervical-cancer prevention rather than a sexually transmitted infection vaccine, capacity-building for providers, and credible regional-language short videos, with culturally tailored material carried on the Ministry of Health and Family Welfare's **U-WIN** digital platform. [PMID 42544634] [PMID 42138645]
- **Rotavirus vaccine (Rotavac, Rotasiil) impact under the UIP is now measurable.** Eight-year multicentre hospital-based surveillance (2016–2023, 27,862 stool samples from under-fives admitted with acute gastroenteritis) found rotavirus positivity of 24.2%, against 37% in the pre-rollout multicentre study; positivity among infants fell from 29.4% (2016) to 15.1% (2023), a 48.6% reduction, with an overall fall of 43.9%. Genotype shifts (rise of G3P[8], decline of G1P[8])

have also occurred in countries without rotavirus vaccination, so the authors read them as natural strain dynamics rather than vaccine pressure. [PMID 41832515]

Immunisation coverage and vaccine confidence (India)

- **Incomplete immunisation, 2015–2025 pooled estimate:** 26% (95% CI 21–32) of Indian under-fives, from 45 cross-sectional studies and more than 428,000 participants — roughly one child in four despite Mission Indradhanush. Strongest determinants were absence of an immunisation card (OR 9.46, 95% CI 3.28–27.28), non-institutional delivery (OR 3.59, 95% CI 1.96–6.56), maternal illiteracy (OR 3.25, 95% CI 2.22–4.78) and low socioeconomic status (OR 2.03, 95% CI 1.51–2.71); northern and multistate studies showed higher prevalence than southern and north-eastern ones. [PMID 42320383]
- **Parental confidence in COVID-19 vaccination of children, Mysore district, Karnataka** (Behavioural and Social Drivers survey, n = 506 parents, fielded November 2021 – May 2022): 90.8% scored high on the Vaccine Confidence Index and 78.3% were likely to have their child vaccinated against COVID-19; higher confidence independently predicted that intent (adjusted OR 1.2, 95% CI 1.02–1.4). Among the 21.7% who were hesitant or unsure, fear of side-effects (56.9%) was the commonest reason. The **ASHA** was the only named information source whose association with vaccine trust reached statistical significance (95.7% trust, chi-square p = 0.039; every other named source p > 0.2) — note that 95.7% was not the highest trust figure in the table, only the significant one. A single-district, COVID-19-specific result from four-year-old fieldwork, so quote it as local support for the "trusted local messenger" principle, not as national childhood-immunisation data. [PMID 41990035]

Elimination, surveillance and disease status

- **Diphtheria is not a historical disease in India:** the country contributed nearly a quarter of the roughly 25,000 cases reported globally in 2024. A 15-year hospital series from a Delhi infectious diseases hospital (2007–2021, 5,493 suspected cases) found 38.8% culture positivity, a case-fatality ratio of 19.1%, and only 1.1% of confirmed cases fully vaccinated for age; partially vaccinated patients had an adjusted OR of 4.5 (95% CI 2.8–7.4) for confirmed diphtheria against the fully vaccinated. Median age rose from 5 years (2007–2011) to 7 years (2017–2021), pointing to **booster-dose** failure rather than primary-dose failure — and booster coverage is not routinely captured in national immunisation surveys. [PMID 42555698]
- **Malaria elimination is being complicated by a species shift in the north-east.** In Mizoram the *Plasmodium vivax* share rose from 23% (2021) to 47% (2024) and 42% (2025); in Mamit district *P. vivax* caused 68% of 2025 cases and 86% of infections in under-fives. Unlike *P. falciparum*, whose transmission followed the monsoon, *vivax* transmission was perennial, and probable cross-border spread from Myanmar raises the question of chloroquine-resistant *vivax*. [PMID 42310537]

Guideline and legislative updates

- **First WHO resolution devoted to stroke (EB158/CONF./9)** was advanced by the Executive Board at its 158th session on 3 February 2026 and adopted by Member States at the **Seventy-ninth World Health Assembly**. It sets 16 operative actions for Member States across three blocks — prevention, acute care and rehabilitation, and surveillance and data — plus eight requests to

the Director-General covering normative guidance, technical support, medicine access, research and progress reporting. [PMID 42095627]

- **Kidney health resolution of the Seventy-eighth World Health Assembly (May 2025)** asks Member States to fold kidney care into national health strategies, strengthen primary-care detection and expand access to kidney replacement therapy; chronic kidney disease affects about 850 million people worldwide. It was followed by the Political Declaration of the UN High-Level Meeting on NCDs and Mental Health. [PMID 42155504]

Global and WHO developments

- **The WHO Pandemic Agreement has been adopted by the World Health Assembly**, closing the legally binding negotiation WHO opened in December 2021 after the inequitable distribution of COVID-19 vaccines; the annex on **pathogen access and benefit sharing (PABS)** — the clause that decides whether low-income countries actually get vaccines next time — is still to be agreed. Quote it as the concrete expression of the "together" pillar and of International Health Regulations solidarity. [PMID 42312458]
- **The evidence base for countering the infodemic has firmed up.** A systematic review of reviews (31 systematic reviews, 17 of them published) put health-related misinformation at 0.2–28.8% of the social-media content studied, and summarised a correction meta-analysis in which corrections worked about twice as well when delivered by experts (effect 0.42, 95% CI 0.28–0.55) as by non-experts (0.24, 95% CI 0.13–0.34), were harder when the false claim came from a peer (0.24, 95% CI 0.11–0.36) than from a news agency (0.48, 95% CI 0.15–0.81), and were hardest of all for infectious-disease claims (0.28, 95% CI 0.17–0.39) versus other health topics (0.55, 95% CI 0.31–0.79). A 2026 systematic review of 15 empirical studies (January 2023 – August 2025) adds the newer threat: generative-AI health misinformation can be as persuasive as or more persuasive than the human-written kind, users struggle to tell the two apart, and detectors trained on human-authored misinformation lose accuracy against it — so labelling, provenance-aware detection and governance now belong in every infodemic-management plan, with the caveat that label effects are context-dependent. Over 60% of the studies reviewed concerned vaccine or pandemic misinformation, which the authors flag as limiting how far the synthesis generalises to other health topics. [PMID 36062247] [PMID 41606555]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 25 International Health, pp 1055–1058; Ch 1 Man and Medicine, pp 22–23; Ch 7 Health Programmes in India, pp 535, 541; Ch 22 Communication for Health Education, pp 1022–1023); Textbook of International Health (Ch 3 International Health Agencies, pp 102, 110–111); Oxford Textbook of Public Health 7e (Ch 1 Scope and concerns of public health, p 137; Ch 7 Politics of public health, pp 337–338; Ch 24 Community diagnosis, pp 918–919; Ch 25 New communication technologies and social media, pp 960, 1230–1231; Ch 35 Clinical epidemiology, p 1264; Ch 47 Health promotion and health education, pp 1742–1743; Ch 52 Implementation science, p 1925; Ch 75 Emerging and re-emerging infections, pp 2862–2866; Ch 96 Immunization and vaccination, p 3583; Ch 107 Global health in the era of sustainable development, p 4069); WHO Constitution 1948; International Health Regulations (2005). **Coverage note:** the annual campaign theme itself is a WHO announcement, not textbook content, and is dated here to 7 April 2026.

Recent-advances sources: 42544634 — Addressing HPV vaccine hesitancy and misinformation in India through public health communication strategies, *Human Vaccines & Immunotherapeutics* 2026, DOI [10.1080/21645515.2026.2711551](https://doi.org/10.1080/21645515.2026.2711551) (full text verified); 42138645 — Human Papillomavirus Vaccination in India: Paving the Way to Cervical Cancer Prevention, *Annals of African Medicine* 2026, DOI [10.4103/aam.aam.745.25](https://doi.org/10.4103/aam.aam.745.25) (abstract-only); 41832515 — Molecular epidemiology of rotavirus causing diarrhea among under-five children after the introduction of rotavirus vaccines, Rotavac and Rotasiil, into the national immunization program of India, *Virology Journal* 2026, DOI [10.1186/s12985-026-03126-0](https://doi.org/10.1186/s12985-026-03126-0) (full text verified); 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: a systematic review and meta-analysis (2015-2025), *Vaccine* 2026, DOI [10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only); 41990035 — Behavioural and Social Drivers (BeSD) of COVID-19 vaccination for children: a cross-sectional study of parental vaccine confidence and intent in Mysore, India, *PLoS One* 2026, DOI [10.1371/journal.pone.0316877](https://doi.org/10.1371/journal.pone.0316877) (full text verified; COVID-19 vaccine only, single district, fieldwork November 2021 – May 2022); 42555698 — Persistent high diphtheria positivity in North India: evidence from a tertiary hospital-based study, 2007-2021, *Indian Journal of Medical Research* 2026, DOI [10.25259/IJMR.249.2026](https://doi.org/10.25259/IJMR.249.2026) (full text verified); 42310537 — Rising burden of Plasmodium vivax in Mizoram: an emerging threat to malaria elimination in Northeast India, *BMC Infectious Diseases* 2026, DOI [10.1186/s12879-026-13756-y](https://doi.org/10.1186/s12879-026-13756-y) (abstract-only); 42095627 — First World Health Organization resolution on stroke at the Seventy-ninth World Health Assembly, *International Journal of Stroke* 2026, DOI [10.1177/17474930261452532](https://doi.org/10.1177/17474930261452532) (abstract-only); 42155504 — Implementing the commitments of the World Health Assembly kidney health resolution, *The Lancet* 2026, DOI [10.1016/S0140-6736\(26\)00755-5](https://doi.org/10.1016/S0140-6736(26)00755-5) (abstract-only); 42312458 — The World Expected More of Canada: Canada and the Pandemic Agreement, *International Journal of Social Determinants of Health and Health Services* 2026, DOI [10.1177/27551938261458695](https://doi.org/10.1177/27551938261458695) (abstract-only); 36062247 — Infodemics and health misinformation: a systematic review of reviews, *Bulletin of the World Health Organization* 2022, DOI [10.2471/BLT.21.287654](https://doi.org/10.2471/BLT.21.287654) (full text verified; the correction effect sizes are from the Walter et al. meta-analysis summarised within it);

41606555 — Generative AI and health misinformation: production, propagation, and mitigation — a systematic review, *BMC Public Health* 2026, DOI [10.1186/s12889-025-26148-9](https://doi.org/10.1186/s12889-025-26148-9) (full text verified). Also: World Health Assembly resolutions WHA78 (May 2025, kidney health) and WHA79 (2026, stroke; EB158/CONF./9, 3 February 2026); WHO Pandemic Agreement; India Universal Immunisation Programme HPV rollout, 28 February 2026; IARC 2024 cervical cancer estimates.



International Health Regulations (IHR)

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — The health regulations governing international travel are the **International Health Regulations (IHR) (2005)** of the World Health Organization (WHO), an instrument of international law binding on **196 States Parties** and in force since **15 June 2007** (Oxford 7e, Ch 105, pp 3996-3997; Suryakantha, Ch 20, pp 325-326). Their purpose and scope is "to prevent, protect against, control and provide a public health response to the international spread of disease in ways that are commensurate with and restricted to public health risks, and which avoid unnecessary interference with international traffic and trade" (Suryakantha, Ch 20, p 325) — in Park's compact phrasing, maximum security against international spread with minimum interference with world traffic (Park 28e, Ch 25).

Evolution of the Regulations

Year	Event	Significance
1851	First International Sanitary Conference, Paris	Ten conferences held 1851-1897 among 12 European countries, all concerned with containment of epidemics (Oxford 7e, Ch 75, p 2856)
1892	First International Sanitary Convention	Dealt with cholera; a plague convention followed at the 10th conference in 1897 (Oxford 7e, Ch 75, p 2856)
1902 / 1907	Pan American Sanitary Bureau / Office International d'Hygiène Publique	First permanent international health bodies (Oxford 7e, Ch 75, pp 2856-2857)
1951	WHO adopts a revised version of the International Sanitary Regulations (ISR)	Oxford records that it "remained focused on the control of cholera, plague, and yellow fever" (Oxford 7e, Ch 75, pp 2856-2857)
1969	22nd World Health Assembly renames ISR as International Health Regulations	Four notifiable diseases — cholera, plague, yellow fever, smallpox ; control by pre-set measures at borders (Suryakantha, Ch 20, pp 325-326; Oxford 7e, Ch 75, p 2857)
1980-81	Smallpox eradicated; 34th World Health Assembly removes it from the list	Left three "quarantinable diseases" — cholera, plague, yellow fever (Suryakantha, Ch 20, pp 325-326; AFMC Red Book 2024, Ch 9, p 226)

Year	Event	Significance
1995	48th World Health Assembly calls for revision	Three-disease scope too narrow; borders failed because disease crossed during incubation, or silently in insects, animals, food and goods (Oxford 7e, Ch 75, p 2857)
2005	IHR (2005) adopted; in force 15 June 2007	Any public health emergency of international concern (PHEIC) is notifiable; India bound from 8 August 2007 (Suryakantha, Ch 20, pp 325-326)

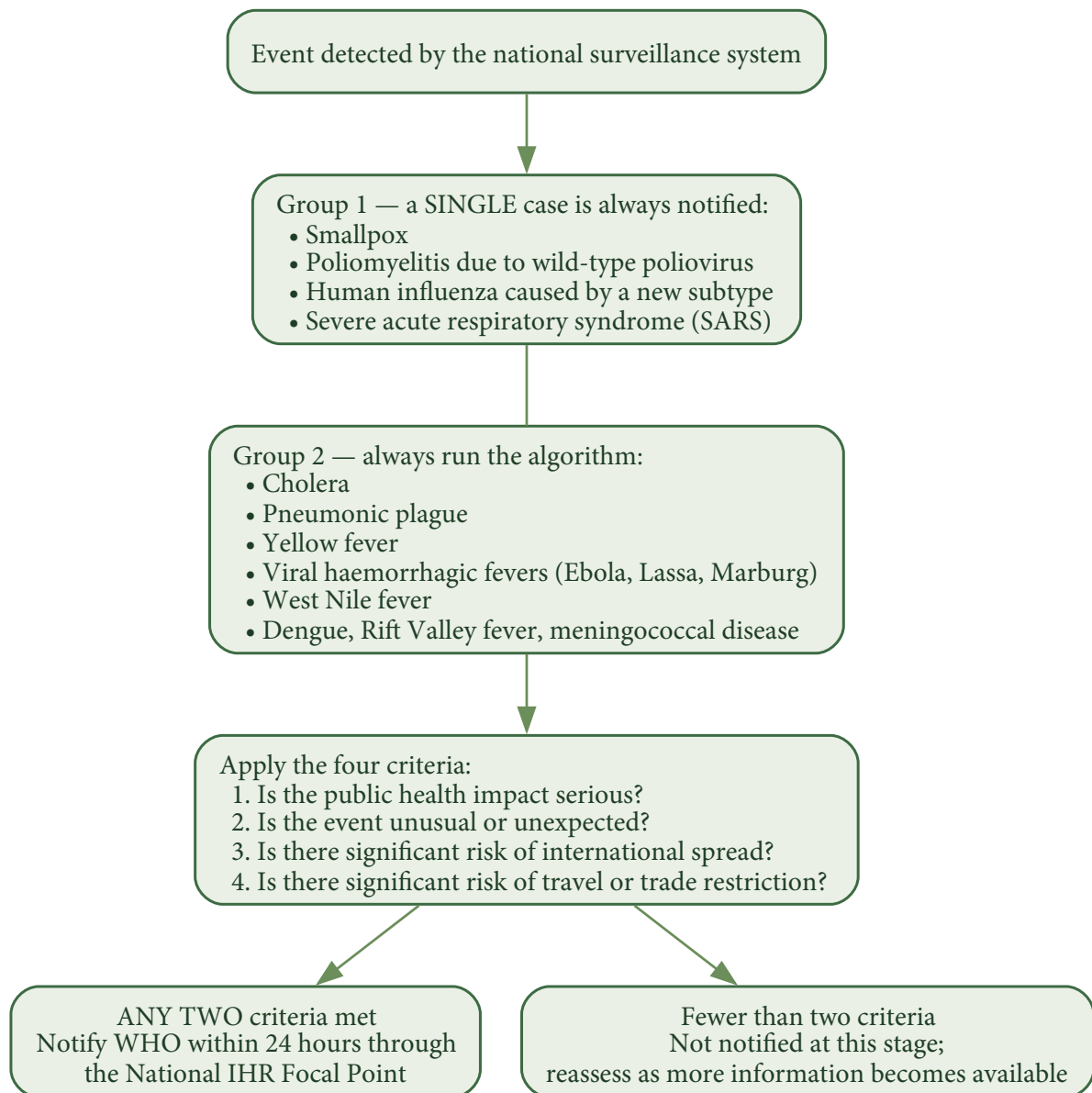
Purpose, scope and the obligations they create

- **Scope widened beyond infection** — the Regulations apply to any acute public health risk that may cross borders, including chemical spills, leaks, dumping and nuclear meltdowns, not only to a listed disease (Suryakantha, Ch 20, p 325).
- **Notification** — every State Party must notify WHO through its **National IHR Focal Point** **within 24 hours** of assessing an event that may constitute a PHEIC; the focal point must function 24 hours a day, 7 days a week (Oxford 7e, Ch 105, pp 3996-3997; Textbook of International Health, Ch 5, p 234).
- **Designated points of entry** — core capacities are separately prescribed for airports, ports and ground crossings, along with disinsection, deratting and disinfection of aircraft and ships (Sunder Lal, Ch 18, p 771).
- **Human-rights safeguard** — implementation must respect the dignity, human rights and fundamental freedoms of persons, and is guided by the Charter of the United Nations and the WHO Constitution (Suryakantha, Ch 20, pp 325-326).

Core capacity requirements for surveillance and response (Sunder Lal, Ch 18, pp 770-771)

Level	Capacities required
Local community / primary public health response level	Detect events involving disease or deaths above expected levels for that time and place; report all available information to the next level; implement preliminary control measures
Intermediate public health response level	Confirm the status of reported events and support or implement additional control measures; assess reported events immediately and report urgent events (serious public health impact, or unusual/unexpected with high potential for spread) to national level
National level — assessment and notification	Assess all reports of urgent events within 48 hours ; notify WHO immediately through the National IHR Focal Point; determine control measures rapidly; provide specialist staff and laboratory support; disseminate WHO recommendations; establish, operate and maintain an emergency response plan

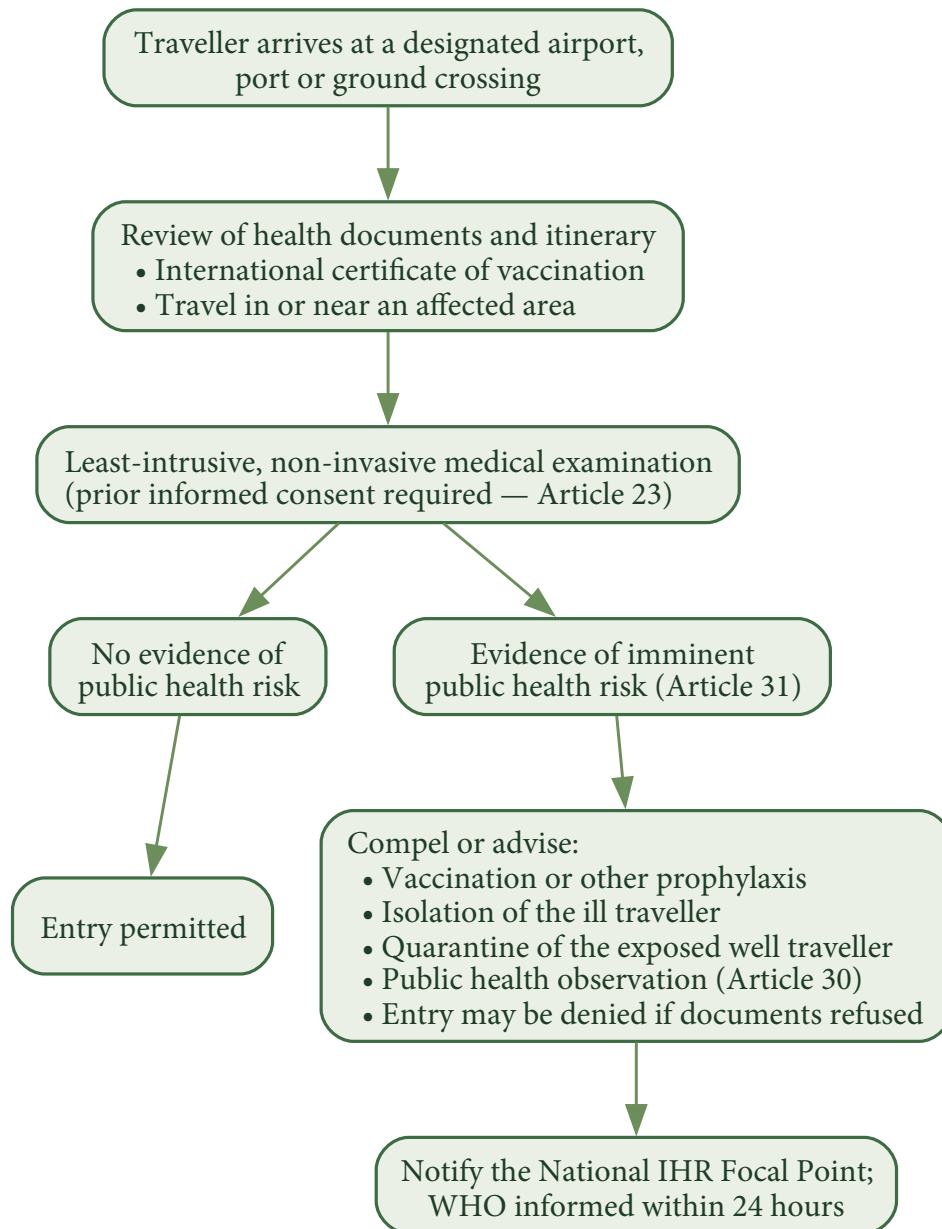
Notifiable events — the Annex 2 decision instrument



Annex 2 decision instrument — deciding whether an event must be notified to WHO

Category	Content
A PHEIC	"An extraordinary event which is determined to constitute a public health risk to other States through the international spread of disease and to potentially require a coordinated international response"; declared on any two of the four criteria in the algorithm above (Oxford 7e, Ch 107, pp 4048-4049)
Under WHO international surveillance (older list)	Louse-borne typhus fever, relapsing fever, paralytic polio, malaria, viral influenza-A, SARS (Park 28e, Ch 3, p 144)
Quarantinable diseases — incubation period fixed for the purposes of the Regulations	Plague 6 days ; cholera 5 days ; yellow fever 6 days (Sunder Lal, Ch 18, pp 770-771; Park 28e, Ch 5, pp 347-348)

Health measures applicable to the traveller



Health measures applied to an international traveller at a designated point of entry

Article	Provision
Article 23	Health measures on arrival and departure — destination and itinerary details, review of health documents, least-intrusive non-invasive examination; inspection of baggage, cargo, containers and conveyances; no examination, vaccination or prophylaxis without prior expressed informed consent (Suryakantha, Ch 20, p 326)
Article 30	A suspect traveller placed under public health observation may continue the voyage if not an imminent risk, with the destination point of entry informed (Suryakantha, Ch 20, p 326)
Article 31	Where imminent risk exists, the State Party may compel the least invasive examination, vaccination or prophylaxis, isolation, quarantine or observation, and may deny entry to a traveller refusing information or documents (Suryakantha, Ch 20, p 326)

Article	Provision
Article 32	Travellers to be treated with courtesy and respect, with regard to gender, socio-cultural, ethnic and religious concerns, and provided accommodation, food, water, clothing, baggage protection and medical treatment (Suryakantha, Ch 20, p 326)

International certificate of vaccination

Disease	Requirement under the Regulations	Validity / remarks
Yellow fever	The only certificate that may be required for international travel; required of all travellers exposed to the risk of yellow fever or passing through endemic zones (Park 28e, Ch 3, p 152; Ch 5, p 348)	Since 11 July 2016 the certificate is valid for the life of the person vaccinated , from 10 days after vaccination; no booster is required as a condition of entry, and lifetime validity applies automatically in India (Park 28e, Ch 5, p 349)
Yellow fever — no valid certificate	Traveller quarantined in a mosquito-proof ward for 6 days from the date of leaving the infected area; if the certificate is not yet valid, isolated until it becomes valid (Park 28e, Ch 5, p 348; Suryakantha, Ch 21, pp 462-463)	Contraindications: infants <9 months (<6 months in an epidemic), pregnancy (except in an outbreak), severe egg allergy, severe immunodeficiency (Park 28e, Ch 5, p 348)
Cholera	Vaccination shall not be required of any traveller (World Health Assembly amendment, 1973)	Parenteral vaccine gives incomplete, unreliable, short protection (Sunder Lal, Ch 18, p 771)
Plague	Vaccination shall not be required as a condition of admission to a territory (Sunder Lal, Ch 18, p 771)	Incubation period fixed at 6 days for quarantine
Smallpox	Not indicated since eradication (confirmed 1978) and may be dangerous to vaccinee and contacts (Sunder Lal, Ch 18, p 771)	Removed from the notifiable list in 1981

Measures on conveyances and at points of entry. Aircraft and ships from endemic areas undergo **disinsection**; around airports and seaports the *Aedes aegypti* index is held **below 1** over a zone of about **400 metres**, and ships moor about 400 metres from shore (Park 28e, Ch 5, pp 347-348; Suryakantha, Ch 21, pp 462-463). Port health authorities issue the **Bill of Health** at departure and grant **Pratique** on arrival; a ship from an infected port flies the yellow quarantine flag and anchors at quarantine moorings until cleared (AFMC Red Book 2024, Ch 9, pp 226-227). In India the **Directorate General of Health Services** directly controls the Port and Airport Health Organisations at the six major ports and five international airports (Park 28e, Ch 23, p 1033).

Role in preventing pandemics

- **Converts border defence into source containment.** Border posts alone failed, because infection crossed while still incubating, or silently in insects, animals, food and goods; the 2005 revision shifted the obligation to containing the event where it arises (Oxford 7e, Ch 75, p 2857).
- **Compels early, honest reporting.** The 24-hour notification duty removes the discretion that once let countries delay reporting for fear of stigma and economic loss; the Regulations also explicitly

recognise **non-state sources of information**, which often precede official notification (Oxford 7e, Ch 75, pp 2861-2862).

- **Builds the capacities that stop an outbreak becoming a pandemic** — surveillance, public health laboratories, rapid response teams and risk communication; the same systems that contain a natural outbreak also detect a deliberate release (Oxford 7e, Ch 75, pp 2861-2862).
- **Provides the trigger and the machinery for a coordinated response** — the Director-General convenes an **Emergency Committee**, declares a PHEIC and issues temporary recommendations; the **Global Outbreak Alert and Response Network (GOARN)** and the WHO Collaborating Centres supply the field response (Oxford 7e, Ch 75, pp 2862-2863).

Every PHEIC declared since the 2005 revision

Declared	Event	Source
2009	H1N1 (pandemic) influenza — first-ever use of the IHR (2005)	Oxford 7e, Ch 20, pp 790-791; Ch 75, pp 2862-2863
2014	Poliomyelitis	Oxford 7e, Ch 20, pp 790-791
8 August 2014	Ebola virus disease, West Africa	Sunder Lal, Ch 8, p 474
2016	Zika virus disease	Oxford 7e, Ch 20, pp 790-791
17 July 2019	Ebola virus disease, Democratic Republic of the Congo	Sunder Lal, Ch 8, p 474
30 January 2020	COVID-19 — pandemic declared 11 March 2020	Oxford 7e, Ch 75, p 2864; Park 28e, Ch 5, p 205
23 July 2022	Mpox — PHEIC lifted 10 May 2023	Park 28e, Ch 5, p 172
14 August 2024	Mpox again, on the emergence of Clade Ib and resurgence in the African region	Park 28e, Ch 5, p 172

- **Protects travel and trade deliberately** — measures must be commensurate with the risk, keeping supply lines and personnel moving, the condition on which containment itself depends.
- **Is externally verified** by the voluntary WHO **Joint External Evaluation (JEE)**, which lets a country assess its core capacities and then cost an action plan to mobilise resources (Oxford 7e, Ch 75, pp 2866-2867).

Where the Regulations have worked	Where they have failed
H1N1 2009 declared a PHEIC within weeks of onset, triggering collaborative obligations and activating national pandemic plans (Oxford 7e, Ch 75, pp 2862-2863)	In the 2013-2016 West African Ebola outbreak neither the affected countries nor WHO sounded the alarm early enough, and PHEIC declaration was overdue (Oxford 7e, Ch 107, pp 4048-4049)
Pandemic Influenza Preparedness Framework (2011) sets obligations for sharing influenza viruses, vaccines and medicines once a pandemic is under way (Oxford 7e, Ch 107, pp 4051-4052)	More than 40 countries applied measures significantly interfering with international traffic, outside WHO's temporary recommendations, with severe economic consequences for Guinea, Liberia and Sierra Leone (Oxford 7e, Ch 107, p 4049)

Where the Regulations have worked	Where they have failed
Core capacity requirement equips countries to detect and respond where outbreaks occur, preventing spread across borders (Oxford 7e, Ch 75, pp 2861-2862)	COVID-19 caused 14.9 million excess deaths and 336.8 million years of life lost in 2020-2021, and exposed inequality in health emergency preparedness worldwide (Park 28e, Ch 5, p 205)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **IHR core capacity pyramid (three tiers).** Labels: *Local community / primary public health response level* — detect events above expected levels, report, preliminary control; *Intermediate level* — confirm and assess events, report urgent events upward, support control measures; *National level* — assess urgent events within 48 hours, notify WHO through the National IHR Focal Point, deploy specialist staff and laboratories, maintain emergency response plan. Draw an upward arrow labelled "notification" on the left and a downward arrow labelled "WHO recommendations and technical support" on the right.

High-yield exam pointers

- Purpose clause, learn verbatim: "prevent, protect against, control and provide a public health response to the international spread of disease... avoiding unnecessary interference with international traffic and trade."
- Dates: IHR (2005) adopted 2005, **in force 15 June 2007**, India **8 August 2007**, **196 States Parties**.
- Numbers: notify WHO in **24 hours**; assess urgent events in **48 hours**; PHEIC needs **any 2 of 4** criteria — but a single Group 1 case (smallpox, wild-type poliovirus polio, human influenza of a new subtype, SARS) is notified outright without the algorithm.
- Quarantine periods under the Regulations: **plague 6 days, cholera 5 days, yellow fever 6 days**.
- Yellow fever certificate is the **only** certificate required; **lifetime validity from 11 July 2016**, effective 10 days after vaccination; unvaccinated traveller quarantined 6 days in a mosquito-proof ward; *Aedes aegypti* index **below 1** within about 400 metres of airports and seaports.
- PHEICs in order: H1N1 2009 → polio 2014 → Ebola West Africa Aug 2014 → Zika 2016 → Ebola DR Congo Jul 2019 → COVID-19 30 Jan 2020 (pandemic 11 Mar 2020) → mpox 23 Jul 2022 (lifted 10 May 2023) → mpox again 14 Aug 2024.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Oxford 7e (Ch 105) puts the Regulations at 196 States Parties while Suryakantha still prints 194, so quote 196; Oxford Ch 107 was written before 2020, so its list of "four such events" is a pre-COVID snapshot and not the current total; the retrieved textbooks predate the 2024 World Health Assembly amendments to the Regulations, so those are not described here; Sunder Lal still prints the pre-2016 ten-year validity of the yellow fever certificate, which Park 28e supersedes with lifetime validity in force since 11 July 2016, and records that India's Epidemic Diseases Act 1897 is proposed for replacement by a National Public Health Act.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO IHR (2005) 2024 amendments, WHO Pandemic Agreement 2025

Programme and policy updates (India)

- **India's first national One Health zoonotic-disease prioritisation.** The National Centre for Disease Control, with the Rajasthan health department and 50 experts (33 voters, 10 advisors and 7 facilitators, participation spanning seven major sectors), ranked 40 zoonoses on severity in humans (weight 0.45), capacity for prevention and control (0.22), potential for introduction or increased transmission into India (0.195), economic burden (0.188) and pandemic potential (0.187). The top ten are zoonotic influenza A, anthrax, Japanese encephalitis, leptospirosis, brucellosis, dengue, rabies, scrub typhus, plague and Crimean-Congo haemorrhagic fever. The authors offer the ranked list as the base for strengthening surveillance, laboratory capacity and intersectoral coordination, and for the state action plans developed under the National One Health Programme for Prevention and Control of Zoonoses. [PMID 41429420] Exam use: quote the top ten as India's own priority list beside the Annex 2 diseases in the body above.
- **How an imported case is worked up while a PHEIC is running.** The Kerala IDSP state surveillance unit with ICMR-NIV Pune investigated a 48-year-old man returning from Sharjah; specimens were screened for mpox, MERS-CoV, Crimean-Congo haemorrhagic fever and hepatitis viruses, and real-time PCR with next-generation sequencing confirmed fatal HSV-2 with multi-organ failure after antiviral therapy was delayed by anchoring on mpox. It is the standard cautionary example that a declared PHEIC can narrow rather than widen the differential in a returning traveller. [PMID 41056625]

Current national data

- **SPAR is the annual figure the Regulations themselves generate — always quote its year.** All 196 States Parties self-report their IHR core capacities every year through State Party Annual Reporting, each capacity scored from 20 (not established) to 100 (sustainably implemented). An 80-country analysis of 2021 SPAR scores against COVID-19 resilience found that capacity scores did not predict control outcomes; the authors attribute this to SPAR measuring only technical and institutional capacity and propose an added social-resilience indicator under Capacity 10 (risk communication and community engagement). The same paper dates the declared end of the COVID-19 pandemic to 5 May 2023 — the closing date the textbook tables omit. [PMID 40234990]

- **India's tuberculosis deadline is the live health-security illustration.** India carries about a quarter of the world's TB burden and in 2018 set a 2025 elimination target, five years ahead of the global goal; a 2025 appraisal judges that deadline unrealistic given multidrug-resistant TB, weak biosafety infrastructure, relapse without post-treatment monitoring, air pollution, and the absence of targeted strategies for tribal populations and undocumented migrants, and argues for realignment to 2030 alongside Pradhan Mantri TB Mukh Bharat Abhiyan and Ni-kshay Poshan Yojana. [PMID 40781735]

Elimination, surveillance and disease status

- **A ninth PHEIC — Ebola disease due to Bundibugyo virus, declared 17 May 2026.** WHO was alerted in early May 2026 to an outbreak in Ituri province, Democratic Republic of the Congo, with imported cases and limited local transmission in Uganda. By 17 June 2026 the cumulative count was 896 confirmed cases and 232 deaths (case fatality 26%), more than 90% of cases still confined to Ituri, and children about a fifth of confirmed cases. No vaccine is licensed for Bundibugyo virus; WHO emergency guidance of 28 May 2026 found the evidence for Ervebo cross-protection insufficient and advised against its use outside research settings. Append this event to the PHEIC table in the body. [PMID 42383297]
- **Mpox clade Ib shows what the 24-hour notification duty actually delivers.** Case data shared with WHO under IHR (2005) covered 127 mpox cases, of which 89 were travel-related; by 19 June 2025 clade Ib had been reported from 14 African countries and travel-related cases from 19 countries outside Africa, seeding secondary cases in 8 destination countries. Twenty-five imported cases produced 34 secondary cases; the secondary attack rate was 73% (95% CI 39-94) for sexual contacts and 17% (95% CI 8-31) for household contacts, with a mean serial interval of 12 days (95% CI 4-27) and no secondary cases among community or healthcare contacts. [PMID 41714599]

Guideline and legislative updates

- **The Regulations were amended on 1 June 2024 at the 77th World Health Assembly and entered into force in September 2025.** The amendments add a pandemic emergency tier at the top of the alert framework: it remains a PHEIC, applies only to communicable disease, must first satisfy the Article 12 PHEIC criteria and runs for the same duration, so the Emergency Committee is not reconvened for the same event. The lower "intermediate" or "early action" alert that was proposed was rejected. Revised Article 13 obliges the WHO Secretariat to facilitate access to health products during a PHEIC or pandemic emergency; revised Article 44 commits States Parties to sustainable financing for developing countries through a new Coordinating Financial Mechanism under the Health Assembly, with an advisory subcommittee under Article 54. [PMID 40589223; PMID 40883832]
- **Article 43, the travel-measures article, was left substantively unchanged.** Member-state proposals to require regular risk assessment, to let WHO review measures that are disproportionate or excessive, and to create a WHO-facilitated consultation on the scientific rationale for a measure were all dropped over sovereignty concerns; only protection of essential travel and trade was agreed. The existing rule therefore stands unaltered — additional health measures may be used only when other measures would be ineffective, and must be evidence-based, temporary, promptly notified to WHO and respectful of human rights. The remedy

proposed is a WHO-agreed standardised risk-based methodology plus a decision instrument added as an annex, on the model of the Annex 2 notification algorithm. [PMID 39880415]

Global and WHO developments

- ▶ **WHO Pandemic Agreement adopted 20 May 2025 by the 78th World Health Assembly.** After three years of negotiation, 124 governments adopted the 35-article instrument — only the second treaty negotiated under Article 19 of the WHO Constitution, after the Framework Convention on Tobacco Control (2003). It complements rather than replaces the amended Regulations, and a determination of pandemic emergency under the amended IHR is what invokes its key pillars: research and development (Article 9), transfer of technology and know-how for producing pandemic health products (Article 11), the pathogen access and benefit-sharing system (Article 12) and procurement and distribution (Article 14). Separately, a WHO-convened Global Supply Chain and Logistics Network is established under Article 13 to keep pandemic-related health products reaching countries in need. The Pathogen Access and Benefit-Sharing annex was left to an Intergovernmental Working Group first convened on 9 July 2025, for finalisation by April 2026 and submission to the 2026 Health Assembly; participating manufacturers would contract to make at least 20% of pandemic-related products available to WHO, half as donation and half at affordable prices, allocated by public health risk and need — with traceability of samples and sequence information, intellectual property and dispute resolution still unresolved. [PMID 40589223; PMID 40883832; PMID 41832575]
- ▶ **Two chikungunya vaccines have entered travel practice, and they are not equivalent.** VLA1553 (IXCHIQ), a single-dose live-attenuated vaccine, gave 98.9% seroprotection at day 28 but produced chikungunya-like illness in 11.7% of recipients against 0.6% on placebo, and in May 2025 the FDA and CDC advised pausing its use in people aged 60 and above after serious post-marketing cardiac and neurological events. PXVX0317 (Vimkunya), a non-live virus-like-particle vaccine licensed from age 12, gave a 97.8% immune response with no vaccine-associated viraemia and no chikungunya-like illness. Neither has efficacy data — phase 4 readouts are expected in 2029-2030 — and both are recommended for travellers to areas with a declared outbreak, which makes the virus-like-particle vaccine the safer travel-clinic choice for pregnant, immunocompromised and older travellers. [PMID 41733825]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, Ch 5, Ch 23, Ch 25); Oxford Textbook of Public Health 7e (Ch 20 Law and the Public's Health; Ch 75 Emerging and Re-emerging Infections; Ch 105 Public Health Emergency Response; Ch 107 Global Health in the Era of Sustainable Development); Sunder Lal — Textbook of Community Medicine (Ch 8; Ch 18 International Health and International Health Regulations); Suryakantha — Community Medicine with Recent Advances (Ch 20; Ch 21); AFMC Red Book of Community Medicine 2024 (Ch 9); Textbook of International Health (Ch 5); WHO International Health Regulations (2005), Second Annotated Edition; WHO Pandemic Influenza Preparedness Framework 2011.

Recent-advances sources: 41429420 — Transforming One Health in India: National Multisectoral Mixed Method Study on Prioritization of Zoonotic Diseases, JMIR Public Health Surveill 2025, 10.2196/77850; 41056625 — A fatal case of Herpes simplex virus-2 infection imported from Sharjah to India: Need for differential diagnosis amid Mpox surveillance, Diagn Microbiol Infect Dis 2025, 10.1016/j.diagmicrobio.2025.117128 (abstract-only); 40234990 — Why do International Health Regulations self-assessment capacities (SPAR) scores not predict COVID-19 control outcomes?, Global Health 2025, 10.1186/s12992-025-01111-w; 40781735 — Enhancing India's Health Security Efforts Against Mycobacterium tuberculosis: Gaps and Opportunities, Health Secur 2025, 10.1089/hs.2024.0109 (abstract-only); 42383297 — The WHO Identifies Ebola Disease Due to Bundibugyo Virus as a Public Health Emergency of International Concern (PHEIC) as Vaccine Development Accelerates, Med Sci Monit 2026, 10.12659/MSM.954627; 41714599 — Characteristics and Transmission Dynamics of Global Travel-Related Mpox Cases Caused by Clade Ib Monkeypox Virus, Emerg Infect Dis 2026, 10.3201/eid3202.251530; 40589223 — The 2025 World Health Assembly Pandemic Agreement and the 2024 Amendments to the International Health Regulations Combine for Pandemic Preparedness and Response, Med Sci Monit 2025, 10.12659/MSM.950411; 40883832 — To tier or not to tier: the institutionalization of the World Health Organization's power to determine pandemic emergency in the amended International Health Regulations (2005), Glob Health Res Policy 2025, 10.1186/s41256-025-00438-6; 39880415 — Latest revisions to the International Health Regulations will fail to prevent future travel chaos, BMJ Glob Health 2025, 10.1136/bmjgh-2024-017077; 41832575 — Africa's Critical Role in Shaping and Implementing the Pandemic Agreement's PABS Annex in an Era of Fragmentation, Global Health 2026, 10.1186/s12992-026-01190-3; 41733825 — Travel-Related Challenges of Chikungunya Virus and Vaccination Options, J Epidemiol Glob Health 2026, 10.1007/s44197-026-00518-x. Also used: WHO International Health Regulations (2005) as amended 1 June 2024 (in force September 2025); WHO Pandemic Agreement adopted 20 May 2025; WHO emergency guidance on Ebola vaccines, 28 May 2026; FDA/CDC IXCHIQ use pause, May 2025.

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UNICEF and its role in health

Asked in: Paper IV 02-Jun-2025 — RGUHS MD Community Medicine

Definition — The United Nations International Children's Emergency Fund (UNICEF) is one of the specialised agencies of the United Nations (UN), established in 1946 by the UN General Assembly to deal with the rehabilitation of children in war-ravaged countries (Park 28e, Ch 25, p 1058). In 1953, when its emergency functions were over, the General Assembly gave it the new name "U.N. Children's Fund" but retained the original initials, UNICEF (Park 28e, Ch 25, p 1058). It is the lead UN organisation working for the **survival, protection and development** of children, its programmes focusing on immunisation, primary health care, nutrition, maternal health and basic education (Textbook of International Health, Ch 3, pp 106-107).

Origin, mandate and organisation

Attribute	Detail
Year established	1946, by resolution of the UN General Assembly (Park 28e, Ch 25, p 1058)
Original purpose	Rehabilitation of children in countries ravaged by the Second World War (Park 28e, Ch 25, p 1058)
Renamed	1953 — "U.N. Children's Fund"; initials UNICEF retained (Park 28e, Ch 25, p 1058)
Headquarters	United Nations, New York (Park 28e, Ch 25, p 1058)
Regional office serving India	Kathmandu, Nepal — the South Central Asian Region , covering Afghanistan, Sri Lanka, India, the Maldives, Pakistan, Bhutan, Bangladesh and Nepal (Park 28e, Ch 25, p 1058)
Governing body	A thirty-six-nation Executive Board (as in 2010) (Park 28e, Ch 25, p 1058)
Principal partners	WHO, and the other specialised UN agencies — UNDP, FAO, UNESCO (Park 28e, Ch 25, p 1058)
Formal WHO link	The UNICEF/WHO Joint Committee on Health Policy (1975) (Park 28e, Ch 10, p 748)
Current mandate	Child survival, protection and development (Park 28e, Ch 25, p 1059)

Evolution of UNICEF's approach

- ▶ **Emergency phase (1946–53)** — relief and rehabilitation of children in war-ravaged countries (Park 28e, Ch 25, p 1058).
- ▶ **Disease-campaign phase** — in the early years UNICEF and WHO worked together on urgent problems such as **malaria, tuberculosis and venereal diseases** (Park 28e, Ch 25, p 1058).
- ▶ **Basic-services phase** — assistance widened to maternal and child health, nutrition, environmental sanitation (especially rural water supply), health centres and health education (Park 28e, Ch 25, p 1058).
- ▶ **The "whole child" phase** — UNICEF turned away from campaigns for the eradication of specific diseases unless of direct benefit to mothers and children, and geared aid to the child's long-term

personal development and to the development of the country; this approach is called '**country health programming**' (Park 28e, Ch 25, p 1058).

- **Child-survival phase** — the GOBI campaign for a "child health revolution", later expanded to GOBI-FFF (Park 28e, Ch 25, pp 1058-1059; Textbook of International Health, Ch 3, p 116).
- **Primary health care partnership** — as full partners in primary health care, UNICEF and WHO develop joint strategies supporting its implementation at country level (Park 28e, Ch 25, p 1059).

Contribution in the health sector — global, with examples

Area	UNICEF's contribution	Example
Primary health care	Co-convened, with WHO, the Joint WHO–UNICEF International Conference at Alma-Ata (USSR), 1978 , where the governments of 134 countries and many voluntary agencies called for a revolutionary approach to health care	The Declaration of Alma-Ata and the goal "Health for All" (Park 28e, Ch 1, pp 22-23)
Child survival	The GOBI strategy — Growth charts, Oral rehydration, Breast feeding , Immunisation against measles, diphtheria, polio, pertussis, tetanus and tuberculosis; augmented to GOBI-FFF by adding Food supplementation, Female literacy and Family planning	Task Force for Child Survival, 1984 — WHO, UNICEF, World Bank, UNDP and the Rockefeller Foundation (Park 28e, Ch 25, pp 1058-1059; Textbook of International Health, Ch 3, p 116)
Immunisation	Sponsored the declaration Universal Child Immunization, 1990 at the UN's 40th anniversary (October 1985), to add impetus to WHO's Expanded Programme on Immunization	UNICEF is a founding partner of the GAVI Alliance , with philanthropic foundations, WHO, the World Bank, private business and NGOs (Park 28e, Ch 3, p 147; Textbook of International Health, Ch 3, pp 143-144)
Measles mortality reduction	With WHO, developed the accelerated measles mortality reduction strategy — two doses of measles-containing vaccine through routine services and supplementary immunisation activities (SIAs), plus improved surveillance	Developed after the 1980 burden of an estimated 2.6 million measles deaths worldwide (Park 28e, Ch 5)
Diarrhoeal disease	With WHO, recommended that countries manufacture and use reduced-osmolarity oral rehydration salts (ORS) ; since January 2004 this is the only formulation procured by UNICEF . Jointly recommends zinc 10 mg/day (infants <6 months) and 20 mg/day (>6 months) for 10–14 days	WHO/UNICEF Global Action Plan for Pneumonia and Diarrhoea (GAPD), 2013 (Park 28e, Ch 5, pp 282, 285; Ch 5 references)
Breastfeeding	The Baby Friendly Hospital Initiative (BFHI) was created and promoted by WHO and UNICEF; implementation guidance revised in 2018 (the Ten Steps)	BFHI is supported by the major professional medical and nursing bodies in India (Park 28e, Ch 9, pp 640-641)
Sick-child care	With WHO, developed Integrated Management of Childhood Illness (IMCI) — combining improved case management with nutrition, immunisation and health promotion	India's adaptation, IMNCI , is the central pillar of child health interventions (Park 28e, Ch 9, p 680; Ch 7, pp 551-552)

Area	UNICEF's contribution	Example
Nutrition	The UNICEF conceptual framework of the determinants of child undernutrition (1990) — immediate, underlying and basic causes — is the standard analytic tool; with FAO and WHO, promotes national food and nutrition policies	Framework reproduced below (Oxford Textbook of Public Health 7e, Ch 3, p 815; Park 28e, Ch 25, p 1058)
Water and sanitation	Runs the WHO/UNICEF Joint Monitoring Programme (JMP) for Water Supply and Sanitation — the global source of drinking-water and sanitation coverage data	<i>Progress on Drinking Water and Sanitation</i> reports and the JMP "service ladders" (Textbook of International Health, Ch 10, pp 572-573; Oxford Textbook of Public Health 7e, Ch 2, p 481)
Child rights and advocacy	The UN General Assembly resolution of 1954 established Universal Children's Day and assigned UNICEF the responsibility for promoting it; UNICEF advocacy underpins the Convention on the Rights of the Child (CRC) , 1989, whose Article 24 addresses the child's right to health	Annual flagship report <i>The State of the World's Children</i> ; the World Summit for Children, 1990 goals (Park 28e, Ch 9, pp 653-654; Oxford Textbook of Public Health 7e, Ch 86, pp 3233-3234)
Measurement and definitions	Supplies internationally comparable child indicators — UNICEF defines the under-5 mortality rate as the "annual number of deaths of children aged under 5 years, expressed as a rate per 1000 live births", measuring the probability of dying between birth and exactly 5 years of age	Also defines <i>street children</i> as those "on the street" and those "of the street" (Park 28e, Ch 9, p 676; Ch 9, street children)
Global governance	Member of the H-8 — WHO, UNICEF, UNFPA, UNAIDS, the World Bank, the GAVI Alliance, the Global Fund and the Gates Foundation	Sets the global health agenda alongside private actors (Textbook of International Health, Ch 3, p 146)

Contribution in the health sector — India, with examples

Field (Park's "content of services")	UNICEF's assistance to India	Example
(a) Child health	Substantial aid for the production of vaccines and sera ; supported India's BCG vaccination programme from its inception ; assisted the erection of a penicillin plant near Pune ; donated a DDT plant ; provided two plants for the manufacture of triple vaccine and iodised salt	Vaccine and cold-chain self-sufficiency underpinning the Universal Immunization Programme (Park 28e, Ch 25, pp 1057-1058)
Environmental sanitation	Assisted sanitation programmes emphasising safe and sufficient water for drinking and household use in rural areas — to reduce child illness and death <i>and</i> improve village quality of life	Village-level delivery through resident volunteers or part-time primary health workers chosen with the agreement of the local community (Park 28e, Ch 25, pp 1057-1058)

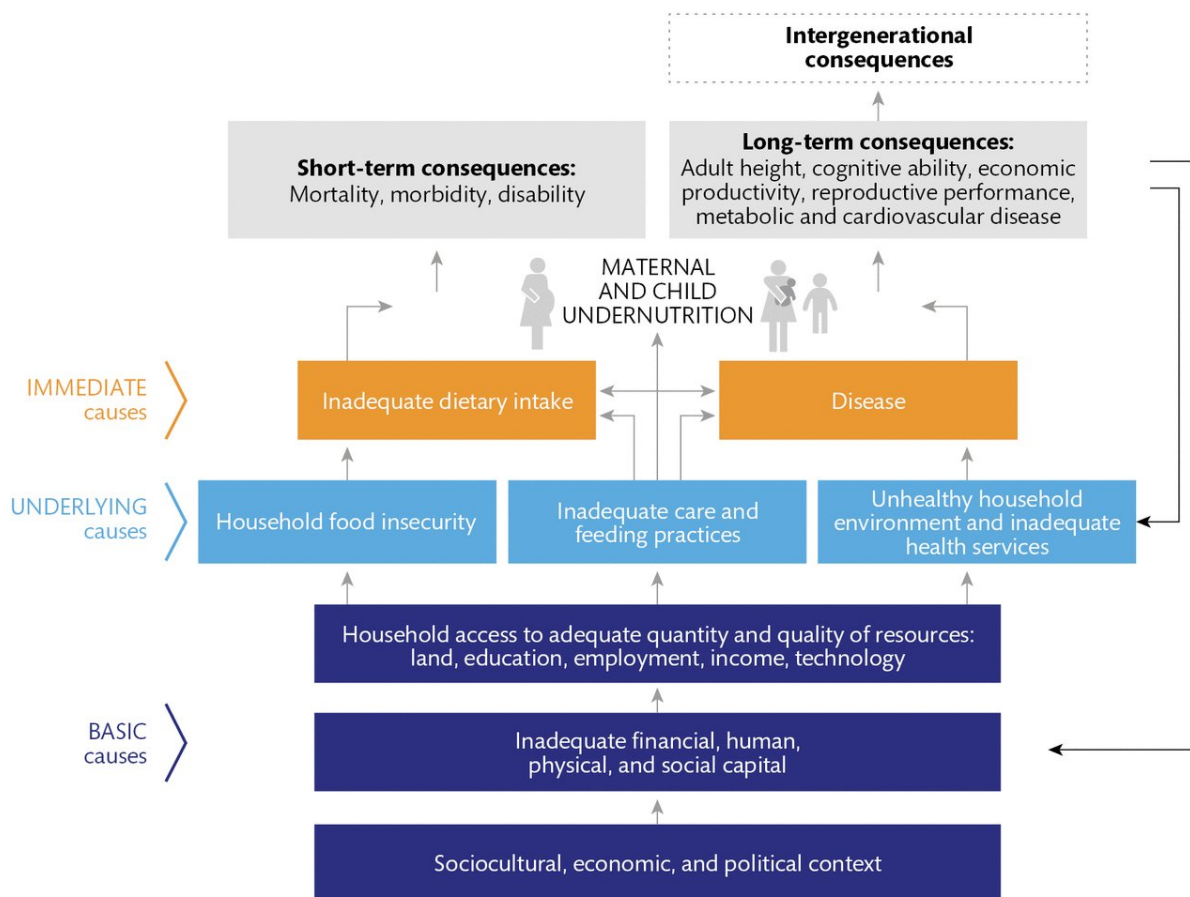
Field (Park's "content of services")	UNICEF's assistance to India	Example
(b) Child nutrition	Supplementary child feeding, expanding from the mid-1950s with low-cost protein-rich food mixtures ; with FAO, aided " applied nutrition " programmes through community development, agricultural extension, schools and health services; supplied equipment for modern dairy plants in Maharashtra, Gujarat, Karnataka, Uttar Pradesh, West Bengal and Andhra Pradesh	Specific aid against deficiency diseases — large doses of vitamin A where xerophthalmia is prevalent, iodisation of salt in endemic goitre areas, iron and folate for anaemia, and food enrichment (Park 28e, Ch 25, p 1058)
School health	Assists the Applied Nutrition Programme in schools with implements, seeds, manure and water-supply equipment	School gardens whose produce feeds the school feeding programme and nutrition education (Park 28e, Ch 9, p 684)
(c) Family and child welfare	Improving care of children within and outside the home through parent education, day-care centres, child welfare and youth agencies and women's clubs — delivered <i>within</i> health, nutrition and education programmes, not as separate projects	Support to the Integrated Child Development Services (ICDS) (Park 28e, Ch 25, p 1058; Ch 9 references)
(d) Education (formal and non-formal)	With UNESCO, assisting the expansion and improvement of science teaching — science laboratory equipment, workshop tools, library books and audiovisual aids	Schooling made relevant to the child's environment and future life (Park 28e, Ch 25, p 1058)
Urban health	Participation in Urban Basic Services (UBS) since 1976 — upgrading health, nutrition, water supply, sanitation and education for children and women in selected cities and towns	Objective: improve the degree and quality of survival and development of children of urban low-income families (Park 28e, Ch 25, pp 1058-1059)
Disease elimination	Partner with WHO India and other partners in designing and implementing the accelerated strategy to eliminate neonatal tetanus	UNICEF is likewise a partner, with WHO and GAVI, in the global Eliminate Yellow Fever Epidemics (EYE) strategy (Park 28e, Ch 5)

In short, UNICEF's activities cover programmes assisting **child survival, protection and development** — immunisation, improved infant feeding practices, child growth monitoring, home-based diarrhoea management, drinking water, environmental sanitation, birth spacing, education of girls and income-generating activities for women (Park 28e, Ch 25, p 1059).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — UNICEF conceptual framework of the determinants of child undernutrition** (reproduce the plate above). Labels, bottom upward: *Basic causes* — sociocultural, economic and political context; inadequate financial, human, physical and social capital; household access to adequate quantity and quality of resources (land, education, employment, income, technology). *Underlying causes* — household food insecurity; inadequate care and feeding practices; unhealthy household environment and inadequate health services. *Immediate causes* — inadequate dietary intake; disease. *Outcome* — maternal and child undernutrition. *Short-term consequences* — mortality, morbidity, disability. *Long-term consequences* — adult height, cognitive ability, economic productivity, reproductive performance, metabolic and cardiovascular disease → intergenerational consequences.

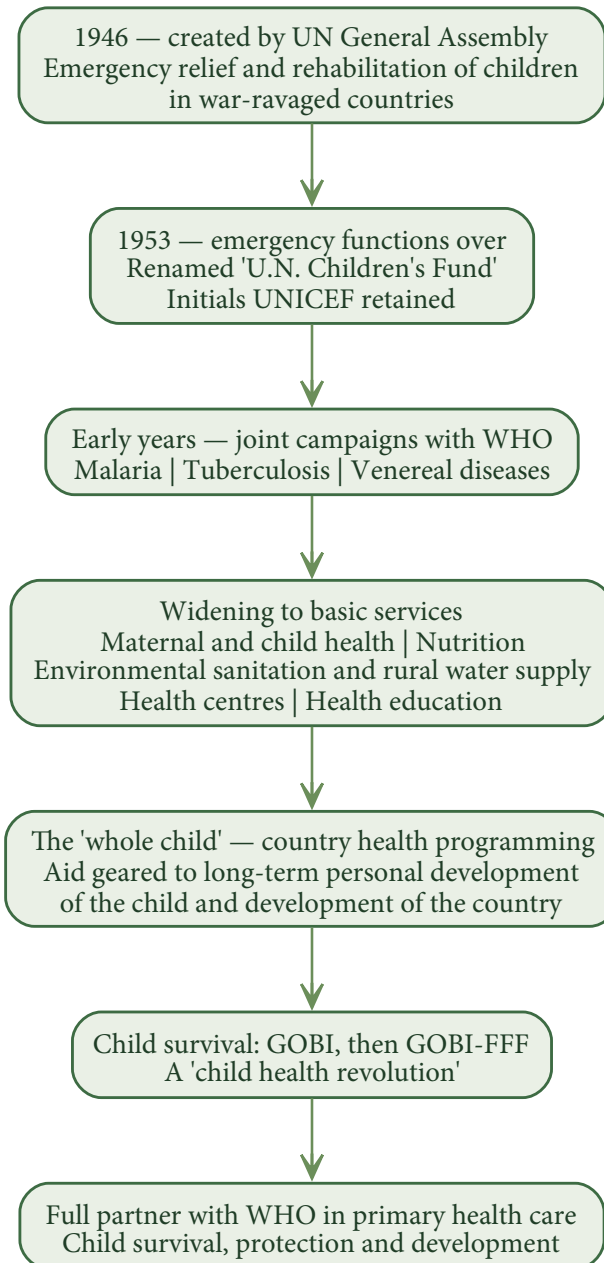
UNICEF CONCEPTUAL FRAMEWORK OF THE DETERMINANTS OF CHILD UNDERNUTRITION



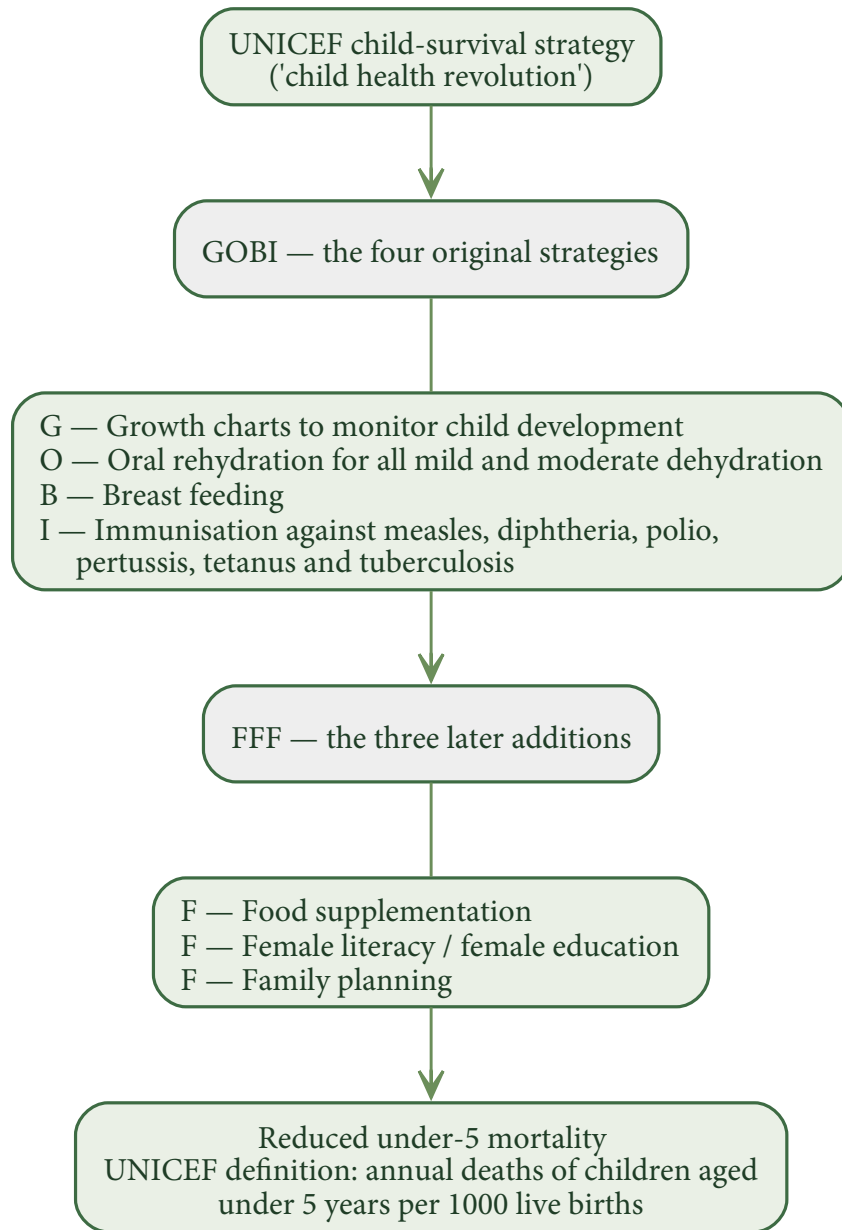
The black arrows show that the consequences of undernutrition can feed back to the underlying and basic causes of undernutrition, perpetuating the cycle of undernutrition, poverty and inequities.

Source: Adapted from UNICEF, 1990.

UNICEF conceptual framework of the determinants of child undernutrition (Oxford public health 7th ed, p.815)



Evolution of UNICEF's mandate, 1946 to the present



The GOBI-FFF child-survival strategy of UNICEF

High-yield exam pointers

- ▶ **Dates to write:** established 1946; renamed "U.N. Children's Fund" in 1953; headquarters New York; regional office for India at Kathmandu, Nepal (South Central Asian Region); **thirty-six-nation Executive Board.**
- ▶ **Mnemonic GOBI-FFF** — Growth charts, Oral rehydration, Breast feeding, Immunisation + Food supplementation, Female literacy, Family planning.
- ▶ **Under-5 mortality rate (UNICEF definition)** = annual number of deaths of children aged under 5 years ÷ 1000 live births — the probability of dying between birth and exactly 5 years of age.
- ▶ **Zinc in diarrhoea (WHO/UNICEF)** — 10 mg/day for infants <6 months, 20 mg/day for children >6 months, for **10–14 days**; reduced-osmolarity ORS is the **only** formulation procured by UNICEF since **January 2004.**

- **Joint WHO–UNICEF products to name:** Alma-Ata Conference 1978 (134 countries); Baby Friendly Hospital Initiative (revised 2018); IMCI (Indian adaptation IMNCI); accelerated measles mortality reduction strategy; JMP for water and sanitation; UNICEF/WHO Joint Committee on Health Policy.
- **Four "content of services" heads for India** — (a) child health, (b) child nutrition, (c) family and child welfare, (d) education, formal and non-formal; plus **Urban Basic Services since 1976**.
- **Concrete Indian examples** — BCG programme supported from its inception; penicillin plant near Pune; DDT plant; triple-vaccine and iodised-salt plants; dairy-plant equipment in six states; applied nutrition programmes with FAO.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved corpus (Park 28e, Ch 25) states UNICEF's Executive Board membership as of 2010 and does not give its financing structure, its current country-programme cycle for India, or post-2015 organisational changes, so those are not stated here; the current name in official use is the **United Nations Children's Fund**, the acronym UNICEF being retained from the original 1946 title.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed (PMC full text checked where available), WHO fact sheets, WHO South-East Asia Regional Office, WHO/UNICEF (WUENIC) estimates, UN IGME, NFHS-6, WHO 2023 wasting and nutritional oedema guideline

Programme and policy updates (India)

- **India launched a national HPV vaccination programme in 2026 — the largest addition to the UIP since the textbook edition.** The rollout offers a **free, voluntary, single-dose quadrivalent** vaccine to about **1.15–1.2 crore adolescent girls aged 14 years**, delivered through schools and government health facilities, with girls turning 15 within three months of launch also eligible. Until then HPV vaccination in India was sub-national — school-based programmes in **Punjab and Sikkim from 2016, covering roughly 97 per cent of target girls** — and India now also has an indigenous quadrivalent vaccine, **Cervavac**. New introductions of this kind are tracked through the **WHO/UNICEF Joint Reporting Form**, and **10 of the 11 South-East Asia Region countries have now incorporated HPV vaccine into their national immunisation programmes** — the DPRK is the only one that has not. [PMID 42518306]
- **UNICEF's classic cold-chain contribution has turned into a digital-logistics question.** A stakeholder study of the **Electronic Vaccine Intelligence Network (eVIN)** across 40 cold-chain points in Indore district, Madhya Pradesh, recorded real-time stock visibility, continuous temperature monitoring, reduced wastage and standardised records, alongside persistent bottlenecks — **no beneficiary-level immunisation data**, unreliable connectivity, device malfunction, workforce shortage and duplication of paper with electronic registers. This is the modern successor to the vaccine, triple-vaccine and iodised-salt plants UNICEF financed in the 1950s–70s. [PMID 42569652]

Current national data

- ▶ NFHS-6 (2023-24) has superseded NFHS-5, and the two breastfeeding indicators have moved in opposite directions. The MoHFW fact sheets give exclusive breastfeeding under 6 months at 55.8 per cent (63.7 per cent in NFHS-5) and early initiation within one hour at 50.1 per cent (41.8 per cent in NFHS-5), with persistent undernutrition, rising urban overweight and widening rural–urban and tribal inequity, and mortality indicators not yet released [PMID 42528306; PMID 42554934]. Across the five rounds from 1993 to 2021, early initiation rose from 9.6 to 41.2 per cent and exclusive breastfeeding from 43.5 to 63.9 per cent while mixed milk feeding worsened by about **nine percentage points** (these are the authors' own re-estimates from the survey series, so they sit marginally off the published fact-sheet values); on that trajectory **23 of 36 states and UTs already meet the 2030 Global Nutrition Target of 60 per cent exclusive breastfeeding** and six are projected to miss it [PMID 42134478]. Always write the round and its year beside the figure.
- ▶ "Zero-dose" is what the "I" of GOBI has become operationally, and India still carries a measurable share. NFHS-5 (2019-21) data on 41,132 children aged 12-23 months put zero-dose prevalence at 6.67 per cent, rising from 5.72 per cent in the richest wealth quintile to 8.85 per cent in the poorest (adjusted OR 2.13, 95% CI 1.72-2.65); decomposing the concentration index (−0.12) attributed 24.4 per cent of the wealth-related inequality to the wealth index itself, 12.1 per cent to maternal education, 10.9 per cent to media exposure, 10.6 per cent to antenatal-care visits and 9.8 per cent to place of delivery, with 53.7 per cent left unexplained [PMID 41372226]. Pooling 45 Indian studies and more than 428,000 children (2015-2025), 26 per cent (95% CI 21-32) of under-fives were incompletely immunised, the strongest determinants being absence of an immunisation card (OR 9.46, 95% CI 3.28-27.28), non-institutional delivery (OR 3.59) and maternal illiteracy (OR 3.25) [PMID 42320383].
- ▶ The UNICEF undernutrition framework still fits the Indian data, with maternal nutrition as the dominant basic cause. Three NFHS rounds put low birth weight at 20 per cent unadjusted and 40 per cent after correction for digit heaping (2000-2020); a low-birth-weight child had 1.5 times the odds of being underweight and 1.3 times the odds of stunting, and once low birth weight had occurred there was little scope to prevent later anthropometric failure — the case for nourishing girls from early childhood through adolescence and preconception [PMID 42442394]. Multilevel analysis of the same three rounds (39,355, 38,654 and 31,725 children) found **women's empowerment strengthening into an independent protective determinant of child undernutrition**: in the fully adjusted model it was non-significant in NFHS-3 (β −0.08, 95% CI −0.26 to 0.11), became significant in NFHS-4 (β −0.74, 95% CI −0.98 to −0.50) and persisted in NFHS-5 (β −0.29, 95% CI −0.51 to −0.07). Clustering was greatest at the **household level** — household ICC 0.62 in the NFHS-4 null model and 0.64 in NFHS-5, rising to about 0.78–0.79 after full adjustment, against negligible state-level ICC — and **wealth quintile was the dominant covariate in every round**, an empirical fit to the framework's "basic causes" tier [PMID 42473846].

Elimination, surveillance and disease status

- ▶ The measles-rubella elimination deadline has moved again — write 2026, not 2023. The WHO South-East Asia Region has reaffirmed elimination of measles and rubella by 2026, the regional target having already slipped from 2015 to 2020 to December 2023, and Nepal was verified as

having eliminated rubella in August 2025 [WHO SEARO 2025-Aug]. The strategic arithmetic is unchanged: measles needs about 95-96 per cent herd immunity (R_0 near 25 at a pre-vaccination median age of ~30 months), and one dose at 9-11 months seroconverts only about 94 per cent of infants, so the second MR dose in the second year of life at ≥ 95 per cent coverage in every district and block is the operative UIP target [PMID 37282389].

- India's measles figures suggest the residual gap is immunity, not only coverage. India reported 13,220 measles cases during 2023-2024, fourth highest in the world; in a cluster study of 1,375 children in tribal undivided Phulbani (Kandhamal and Boudh districts), Odisha, first-dose coverage 99.6 per cent and second-dose 97.5 per cent, yet only 55.1 per cent were seropositive ($\text{IgG} \geq 250 \text{ IU/L}$) three months after the second dose (of 1,185 children actually bled), with just 27.3 per cent reaching the high-titre threshold of $\geq 500 \text{ IU/L}$ and male sex predicting sub-optimal response [PMID 42555691]. Note that this paper's own abstract miscalls the 99.6 per cent figure "two-dose" coverage; its Results section is the one to quote. In the Pune city outbreak that began 14 November 2022 (266 cases), a 1:1 case-control study found that receiving no MR dose carried OR 3.09 (95% CI 2.37-4.04) and a single dose OR 2.03 (1.33-3.10), with the COVID-19 lockdown the commonest stated reason for non-vaccination [PMID 40964759].

Guideline and legislative updates

- The 2023 WHO wasting guideline changed who is treated first, and field data now question the extra stratification. WHO's 2023 guideline on wasting and nutritional oedema prioritises moderately wasted children with MUAC 115-119 mm, weight-for-age z-score below -3 , age under 24 months, or clinical and social vulnerability. Across 87,961 treatment episodes in Chad, Mali and Niger managed on a simplified MUAC-based protocol, 97.4 per cent met at least one higher-priority criterion, recovery was 85-90 per cent and mortality below 0.4 per cent in both priority groups, and WAZ below -3 was the strongest independent predictor of treatment failure — so where a simplified combined protocol is already running, further risk-stratification adds little. [PMID 42431696]

Global and WHO developments

- The current WHO/UNICEF (WUENIC) coverage estimates are for 2025, and the zero-dose count is the headline figure. Globally in 2025 there were 13.5 million zero-dose children plus roughly 6 million partially vaccinated, DTP3 coverage 85 per cent, first-dose measles 84 per cent, second-dose measles 77 per cent, and first-dose HPV in girls 33 per cent — up from 17 per cent in 2019 but far short of the 90 per cent 2030 target; measles vaccination is credited with averting about 59 million deaths between 2000 and 2024, against an estimated 95,000 measles deaths in 2024 [WHO 2026-Jul]. An analysis of the 2024 WUENIC revision across 194 reporting entities put DTP1 at 88.9 per cent, DTP3 at 84.6 per cent and MCV1 at 84.0 per cent, with a no-dose burden of 14.26 million children, 74.6 per cent of it concentrated in 20 countries [PMID 42520643].
- Under-five mortality is still falling, but the SDG 3.2 trajectory has flattened. UN IGME estimates 4.9 million under-five deaths in 2024, an under-five mortality rate of 37.4 per 1000 live births, with 2.3 million neonatal deaths making up 47 per cent of the total; the annual rate of reduction since 2015 is less than half that of 2000-2015, and sub-Saharan Africa with southern Asia carry over 80 per cent of the deaths [WHO 2026-May]. The same estimation stream now

reaches past the under-fives: **2.1 million deaths at ages 5-24 years in 2024, 31 per cent of all deaths under 25 years**, against 21 per cent in 1990 — the age band that classical child-survival framing has under-served [PMID 42242781].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 25 (International Health), Ch 1 (Man and Medicine: Towards Health for All), Ch 3 (Principles of Epidemiology), Ch 5 (Epidemiology of Communicable Diseases), Ch 7 (Health Programmes in India), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics), Ch 10 (Health Care of the Community); Textbook of International Health — Ch 3 (International Health Agencies, Activities and Other Actors), Ch 10 (Health and the Environment), Ch 13; Oxford Textbook of Public Health 7e — Ch 2 (water and sanitation, JMP), Ch 3 (UNICEF conceptual framework of child undernutrition), Ch 86 (Child health, Convention on the Rights of the Child 1989). Named documents: Declaration of Alma-Ata (WHO/UNICEF, 1978); Universal Child Immunization 1990 declaration (UNICEF, 1985); Baby Friendly Hospital Initiative implementation guidance (WHO/UNICEF, 2018); WHO/UNICEF Global Action Plan for Pneumonia and Diarrhoea (2013); UNICEF *The State of the World's Children* (annual).

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children from tribal Odisha, India, *Indian Journal of Medical Research* 2026, doi:[10.25259/ijmr_3672_2025](https://doi.org/10.25259/ijmr_3672_2025) (full text checked; its abstract conflicts with its Results — Results used here). 40964759 — Risk factors for measles outbreak in Pune city: a case-control study, *Indian Journal of Public Health* 2025, doi:[10.4103/ijph.ijph_240_24](https://doi.org/10.4103/ijph.ijph_240_24) (abstract-only). 42431696 — Assessing WHO prioritisation criteria for children 6-59 months treated for moderate wasting in a MUAC-based protocol: a multicountry analysis in West and Central Africa, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-023264](https://doi.org/10.1136/bmjgh-2025-023264). 42520643 — Country-level immunization recovery and DTP1 no-dose proxy burden after COVID-19: a descriptive ecological analysis of WUENIC, 2010-2024, *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128973](https://doi.org/10.1016/j.vaccine.2026.128973) (abstract-only). 42242781 — Global, regional, and national levels and trends in older child, adolescent, and youth (5-24 years) all-cause mortality from 1990 to 2024: modelling study, *BMJ* 2026, doi:[10.1136/bmj-2025-088685](https://doi.org/10.1136/bmj-2025-088685). Plus: WHO — Immunization coverage fact sheet, updated 15 July 2026 (WHO/UNICEF WUENIC 2025 estimates); WHO — Measles fact sheet, updated 15 July 2026; WHO — Child mortality (under 5 years) fact sheet, updated 1 May 2026 (UN IGME 2024 estimates); WHO South-East Asia Regional Office — regional measles and rubella elimination by 2026, and verification of rubella elimination in Nepal, 19 August 2025; WHO — Guideline on the prevention and management of wasting and nutritional oedema in infants and children under 5 years, 2023.

♦ ♦ ♦

Health problems in refugee camps

Asked in: Paper II May 2019 — RGUHS MD Community Medicine

Definition — A refugee camp is a planned temporary settlement providing shelter, protection and basic services to a forcibly displaced population; under the 1951 United Nations Convention a **refugee** is a person outside the country of his nationality owing to a well-founded fear of persecution for reasons of race, religion, nationality, membership of a particular social group or political opinion (Oxford Textbook of Public Health 7e, Ch 92, pp 3441–3443). Because a camp concentrates an uprooted, under-immunised and often malnourished population in overcrowded conditions with disrupted water, sanitation and health services, its health problems are highly predictable. Management is therefore a standard sequenced public health package, not an improvised medical relief effort.

Health problems in refugee camps — why they arise and their magnitude

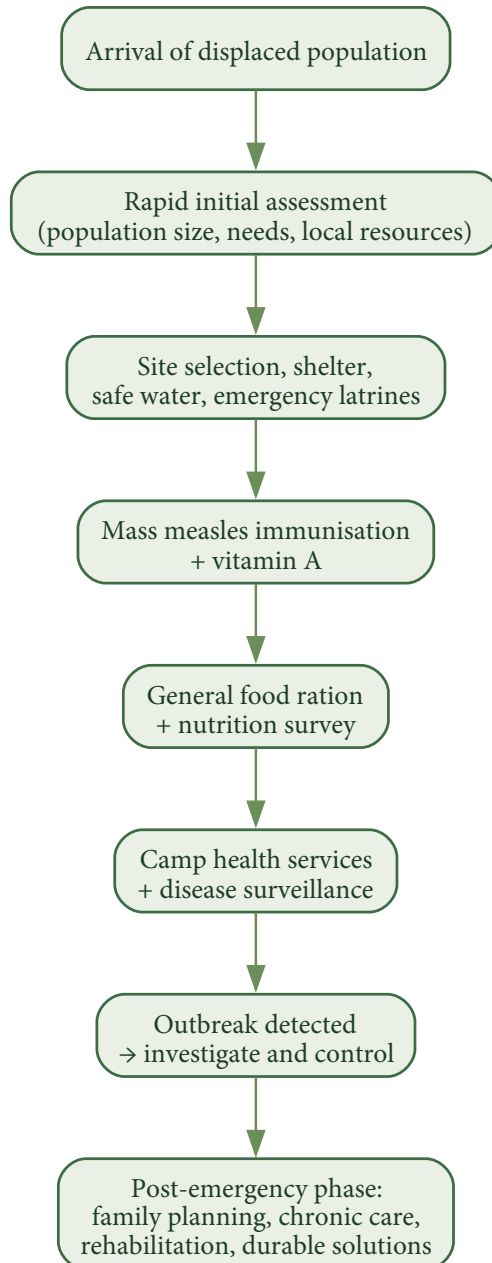
Health problem	Why it arises in the camp	Magnitude in the evidence
Diarrhoeal disease — cholera, bacillary and amoebic dysentery	Faecal contamination of scarce water and food; damaged sewerage; scarcity of soap	The greatest single threat to life: more than 40% of deaths in the acute phase may be diarrhoeal, over 80% of them in children under 2 years. Of the 50,000 deaths among the 800,000 Rwandan arrivals in North Kivu in the acute phase following the early-July 1994 influx, 85% were diarrhoeal — 60% cholera and 40% shigella dysentery; 70% of deaths among Kurdish refugees were diarrhoeal; dysentery case-fatality up to 10% in the elderly and under-fives (Oxford 7e, Ch 92, pp 3458–3459)
Measles	Crowding + low prior coverage + malnutrition	Risk of death is exacerbated by malnutrition; measles caused 53% and 42% of refugee deaths in eastern Sudan and Somalia respectively in 1985, and vaccination is now accorded priority (Oxford 7e, Ch 92, p 3459)
Acute respiratory infection (ARI)	Lack of shelter, overcrowding with poor ventilation, exposure and cold stress	Ranks among the leading causes of death in displaced populations ; in Kabul, 1993, ARI caused 30% of under-5 deaths and 23% of all deaths in displaced persons (Oxford 7e, Ch 92, pp 3459–3460)
Malaria and other vector-borne disease	Ecological change creating breeding sites; non-immune migrants entering endemic areas; overcrowding increases biting frequency	Does not appear immediately — takes several weeks to reach epidemic levels (Park 28e, Ch 17, p 952)
Acute malnutrition and micronutrient deficiency	Broken food chain, inadequate and monotonous ration	11–81% of under-fives acutely malnourished in surveys of conflict-affected populations, 1988–1995; iron and vitamin A deficiency commonest, plus scurvy, pellagra and beriberi (Oxford 7e, Ch 92, pp 3462–3463)
Zoonoses	Displaced domestic and wild animals sheltering with owners	Leptospirosis after the Odisha super-cyclone 1999; rickettsiosis, anthrax, rabies (Park 28e, Ch 17, p 952)

Health problem	Why it arises in the camp	Magnitude in the evidence
Reproductive health problems	Loss of services, insecurity, gender-based violence	Unsafe delivery and abortion, no family planning, sexually transmitted infections including HIV (Oxford 7e, Ch 92, p 3463)
Mental health problems	Violence, bereavement, loss of livelihood and social support	Documented rates range widely — post-traumatic stress disorder (PTSD) 4–70%, depression 3–88%, anxiety 2–80%. The pooled meta-analysis of 7000 refugees resettled in Western countries suggests 1 in 10 has PTSD, 1 in 20 major depression, 1 in 25 generalised anxiety disorder — still ten times the general-population PTSD rate (Fazel <i>et al.</i> 2005; Oxford 7e, Ch 92, pp 3467–3468)
Non-communicable and chronic disease	Interrupted treatment; emerges as the population ages and communicable disease comes under control	Chronic disease now requires explicit attention in displaced populations (Oxford 7e, Ch 92, pp 3457–3458). A World Health Organization Eastern Mediterranean Regional Office analysis found 79% of Syrian national mortality was NCD-related , and delivering that care to the displaced has challenged the humanitarian system (Oxford 7e, Ch 104, p 3966)

Priorities of management in the acute (emergency) phase

The emergency phase is addressed through the following standard priorities (Textbook of International Health, Ch 8, p 415):

- ▶ **Initial rapid assessment** — a rapid damage and needs assessment to size the population and identify needs and available resources before supplies are ordered (Park 28e, Ch 17, p 952).
- ▶ **Measles immunisation** — critical and highly effective, given with vitamin A (Maxcy-Rosenau-Last 15e, Ch 79, p 1318).
- ▶ **Water and sanitation.**
- ▶ **Food and nutrition.**
- ▶ **Shelter and site planning.**
- ▶ **Health care** — curative services for casualties and common illness.
- ▶ **Control of communicable diseases** — acute respiratory infection, measles, diarrhoea and malaria.
- ▶ **Surveillance** — a reliable disease reporting system with prompt investigation of every reported outbreak.
- ▶ **Training** of relief workers and community health workers.
- ▶ **Coordination.**
- ▶ **Camp management.**



Sequence of the public health response in a refugee camp, from arrival to the post-emergency phase

Environmental health management

Component	Action and standard
Site and shelter	After water and food, shelter is the most immediate need, and sudden severe cold stress kills the young and the elderly; mass care and shelter is typically provided by the Red Cross / Red Crescent, with physician, nursing and ancillary health support from medical planners (Maxcy-Rosenau-Last 15e, Ch 79, p 1318)
Water quantity	At least 15–20 litres of potable water per person per day as a baseline, with extra allotments for clinics and feeding centres (Maxcy-Rosenau-Last 15e, Ch 79, p 1317). The World Health Organization minimum for basic health protection is 20 L/person/day, of which 7.5 L is for drinking and food preparation (Oxford 7e, Ch 12, pp 493–494)

Component	Action and standard
Water access	Water availability is assessed as distance: normally 1 km or a 30-minute round-trip collection time, but 0.5 km in disaster response (Sphere Project 2004; Oxford 7e, Ch 12, pp 493–494)
Water quality	Chlorination; all water tankers inspected, cleaned and disinfected before transporting water (Park 28e, Ch 17, p 953)
Excreta disposal	Emergency latrines as pits, trenches or chemical toilets; the whole aim is to reduce faecal contamination of food and water supplies (Maxcy-Rosenau-Last 15e, Ch 79, p 1317)
Personal hygiene	Washing, cleaning and bathing facilities for the displaced; community hygiene education (Park 28e, Ch 17, p 953)
Food hygiene	Where mass feeding programmes run, kitchen sanitation is critical and food handlers must be monitored — poor food hygiene is the major cause of food-borne disease in camps (Park 28e, Ch 17, p 953)
Vector control	Intensify control in the emergency and rehabilitation phases: dengue (mosquitoes), leptospirosis and rat-bite fever (rats), typhus (lice, fleas), plague (fleas); flood water provides breeding opportunities for mosquitoes (Park 28e, Ch 17, p 953)
Animals and carcasses	Veterinary evaluation of zoonotic risk; carcasses foul water supplies and unsecured surviving animals act as reservoirs (Maxcy-Rosenau-Last 15e, Ch 79, p 1318)
Dead bodies	Health hazard is minimal when death results from trauma — corpses are unlikely to cause outbreaks of typhoid, cholera or plague, though bodies contaminating wells or streams may cause gastroenteritis or food poisoning. Identification and respectful disposal matter more than disinfection (Park 28e, Ch 17, pp 951–952)

Nutrition, communicable disease control and immunisation

The immediate cause of death in most malnourished persons is not starvation itself but infection, and malnutrition and measles have a **synergistic relationship** — each worsens the other — so feeding and communicable disease control must run as one integrated response (Maxcy-Rosenau-Last 15e, Ch 79, p 1318; Oxford 7e, Ch 92, pp 3462–3463).

Four steps make an emergency feeding programme effective (Park 28e, Ch 17, pp 952–953):

- ▶ **Assess** the food status and stocks of the affected area.
- ▶ **Gauge** the nutritional needs of the affected population.
- ▶ **Calculate** the daily food ration and the needs of large population groups.
- ▶ **Monitor** the nutritional status of the affected population.

Level of feeding	Target group	Content
General ration	Whole camp population	Adequate energy and protein; a fortified blended cereal added to prevent micronutrient deficiency (Oxford 7e, Ch 92, pp 3462–3463)

Level of feeding	Target group	Content
Supplementary (selective) feeding	Moderate acute malnutrition — weight-for-height/length Z-score between -3 and -2 , or mid-upper arm circumference (MUAC) 115 up to 125 mm; plus infants, young children, pregnant and nursing women and the sick, the groups Park identifies as most prone (Park 28e, Ch 9, p 712)	Extra ration on top of the general ration
Therapeutic feeding	Severe acute malnutrition — weight-for-height/length Z-score below -3 SD of the median child growth standards, or MUAC <115 mm , or the presence of nutritional oedema (Park 28e, Ch 11, p 783)	Intensive nutritional rehabilitation with ready-to-use therapeutic food and treatment of concurrent infection
Micronutrient measures	Whole population	Vitamin A with measles vaccination; correction of iron and vitamin A deficiency, scurvy, pellagra and beriberi; exchange of rations for locally produced fresh fruit and vegetables where feasible (Oxford 7e, Ch 92, p 3463)

Park lists **seven mechanisms** by which displacement increases transmission, and each mechanism dictates its own control measure (Park 28e, Ch 17, p 952):

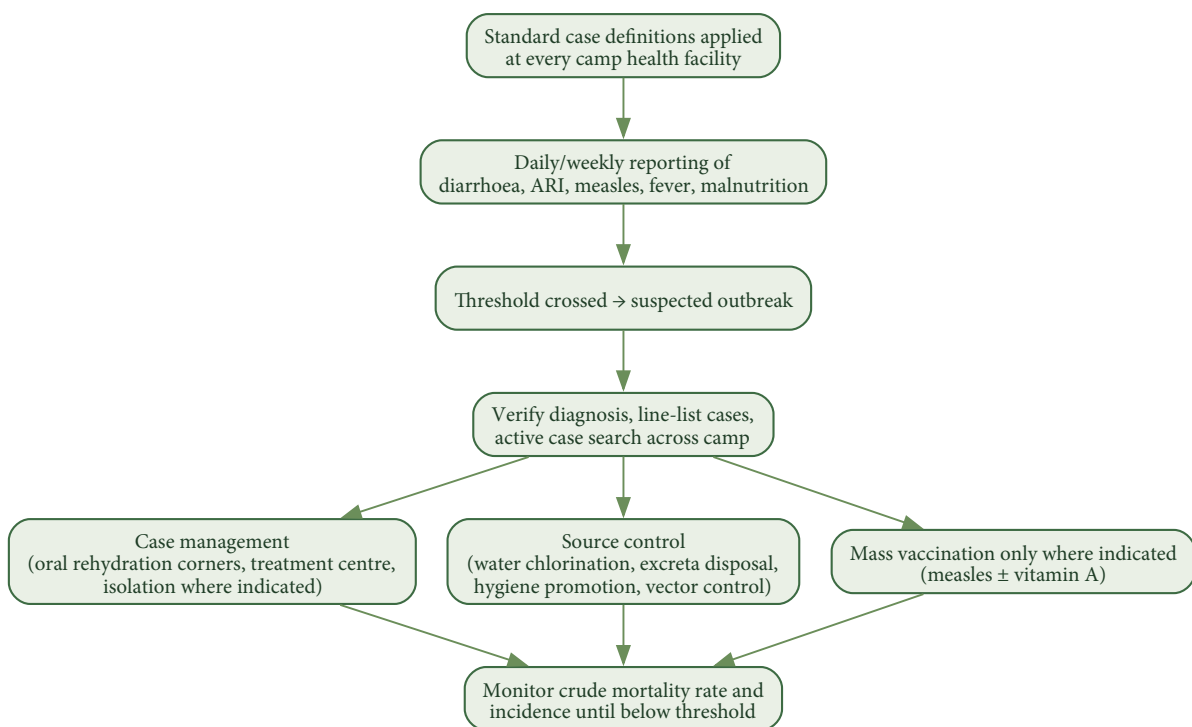
Mechanism	Control measure
1. Overcrowding and poor sanitation in temporary resettlements	Site planning, adequate shelter space, sanitation
2. Population displacement introducing disease to which migrant or host population is susceptible	Screening, immunisation, surveillance covering both camp and host community
3. Disruption and contamination of water supply, sewerage and power	Safe water and excreta disposal — "the most practical and effective strategy"
4. Disruption of routine control programmes as funds and staff divert to relief	Protect and monitor routine polio and measles programmes
5. Ecological change favouring vector breeding and higher vector density	Intensified vector control
6. Displacement of domestic and wild animals carrying zoonoses	Veterinary services and animal control
7. Emergency food, water and shelter from a new source	Quality control of all relief commodities

Immunisation policy in camps:

- **Measles vaccine with vitamin A** — emergency measles vaccination together with vitamin A is a critical and highly effective measure to prevent measles and to reduce the morbidity and mortality it causes (Maxcy-Rosenau-Last 15e, Ch 79, p 1318).
- **Mass typhoid vaccination is not recommended** — the World Health Organization does not advise it in disaster settings: a mass campaign consumes workers who could be deployed

elsewhere, supervision of sterilisation may be impossible, and it creates a false sense of security about the risk of disease that leads to neglect of effective control measures (Park 28e, Ch 17, p 952).

- **Oral cholera vaccine (OCV) — now recommended** — the World Health Organization's revised 2010 recommendation is that OCVs be used where cholera is endemic, and considered where outbreaks are a risk (**preemptive vaccination**) and where outbreaks are occurring (**reactive vaccination**); a WHO OCV stockpile now exists. Two-dose killed OCV has been feasibly delivered **in refugee camps and during complex emergencies** (Uganda, Indonesia) and in campaigns in Bangladesh, India, Haiti, South Sudan and Guinea, where reactive *Shanchol* gave **86% protection over 6 months** (Plotkin's Vaccines 7e, Ch 14, pp 284–285). OCV is an **adjunct to, never a replacement for**, safe water, sanitation and hygiene.
- **Tetanus** — mass vaccination is usually unnecessary; protection rests on high routine immunisation coverage plus adequate wound cleaning and treatment, with a toxoid booster for a wounded patient whose last dose was more than 5 years ago. It is prudent to include tetanus vaccination in camps holding large numbers of children (Park 28e, Ch 17, p 952).
- **Cold chain and authority** — inadequate cold-chain facilities must be requested at the same time as the vaccines, and the vaccination policy to be adopted is decided at **senior level only** (Park 28e, Ch 17, p 952).



Outbreak detection and response pathway within a refugee camp

Reproductive health, mental health and vulnerable groups

Need	Management
Reproductive health	Deliver the Minimum Initial Service Package (MISP) , the agreed emergency reproductive health package produced to address the reproductive health needs of women affected by complex emergencies and forced migration (Oxford 7e, Ch 92, pp 3463–3464). The documented needs it must meet are sanitary napkins and condoms, safe birthing and referral for obstetric emergencies, access to family planning, and management of sexually transmitted infections including HIV (Oxford 7e, Ch 92, p 3463). Where the response has been strong, reproductive health outcomes in camps have exceeded those of host and origin populations
Gender-based violence	Prevention through camp lighting, layout and safe access to water and latrines; clinical and psychosocial care for survivors; violence rises where societal norms cannot be enforced and armed men control resources (Oxford 7e, Ch 92, pp 3463–3464)
Mental health	Respond to three distinct groups : those who are mentally ill and need treatment; those with particularly horrific experiences who need care and support to recover; and the majority, who need a stable social and political environment in which to resume normal daily activity (Oxford 7e, Ch 92, p 3464). Use culturally valid assessment tools
Children under five	Highest-risk group — most camp deaths occur in this age band (Oxford 7e, Ch 92, p 3457); priority for measles vaccine, vitamin A, feeding programmes and growth monitoring
Internally displaced persons and the host population	The response must extend beyond the camp to the surrounding host community and to internally displaced persons, who have not crossed an international border and whose needs may be greater while their access to assistance is poorer (Oxford 7e, Ch 92, pp 3456–3457, 3465)

Surveillance, evaluation and coordination

Indicator	Definition	Interpretation
Crude mortality rate (CMR)	Deaths per 10,000 population per day during emergencies (Textbook of International Health, Ch 8, p 403)	>1 death per 10,000 per day defines the acute phase of a complex emergency, based on a doubling of the sub-Saharan African baseline (<i>Toole and Waldman</i> 1990; Oxford 7e, Ch 92, p 3457); the most-used measure of whether the camp is being managed adequately
Under-five mortality rate	Deaths in children <5 years per 10,000 under-fives per day	More sensitive than CMR, because the highest-risk group is young children and most camp deaths occur in the under-5 age group (Oxford 7e, Ch 92, p 3457)
Prevalence of wasting (acute malnutrition, moderate + severe)	% of under-fives with weight-for-height below –2 Z score (Textbook of International Health, Ch 8, p 403)	Repeat surveys track the effect of the feeding programme; wasting ≥20% with CMR ≥1/10,000/day marks famine conditions (Textbook of International Health, Ch 8, p 412)

Indicator	Definition	Interpretation
Measles vaccination coverage	% of the target camp child population vaccinated	Measles vaccination is accorded priority in camps, so coverage should approach completeness within days of camp establishment (Oxford 7e, Ch 92, p 3459)
Litres of water per person per day; persons per latrine	Direct environmental measures	Audited against Sphere minimum standards (Textbook of International Health, Ch 8, pp 413–414)
Incidence of priority diseases	Cases per week by standard case definition	Diarrhoea, acute respiratory infection, measles and malaria — the commonest causes of high mortality in complex emergencies (Textbook of International Health, Ch 8, p 415)

Coordinating framework	What it contributes
Sphere Project Handbook (2004 edition)	Minimum standards in water supply and sanitation, nutrition, food aid, shelter and site planning, and health services, prefaced by the Humanitarian Charter — all people in all circumstances have the right to live with dignity and to protection and assistance (Textbook of International Health, Ch 8, pp 413–414)
Cluster approach	A lead agency per sector, with the World Health Organization leading the Health Cluster (operating in 27 countries in 2018) (Oxford 7e, Ch 104, pp 3963–3964)
United Nations High Commissioner for Refugees (UNHCR)	Holds the mandate for international protection of refugees under its Statute adopted by the United Nations General Assembly in December 1950 (Oxford 7e, Ch 92, p 3451)
Operational partners	International Organization for Migration, UNICEF, the United Nations Population Fund, the International Committee of the Red Cross, the International Federation of Red Cross and Red Crescent Societies, UNAIDS and non-governmental organisations such as Médecins Sans Frontières (Oxford 7e, Ch 92, pp 3471–3472)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Crude mortality rate curve of a refugee emergency.** Labels: y-axis "Crude mortality rate (deaths per 10,000 population per day)"; x-axis "Weeks since displacement"; horizontal dashed line at 1/10,000/day marked "**emergency threshold**"; lower horizontal line marked "**baseline mortality of country of origin**"; a steep peak in the first weeks marked "**acute phase — highest risk, deaths mainly in under-fives**"; a falling limb marked "**effect of water, sanitation, measles vaccination and feeding programmes**"; the tail crossing below the threshold marked "**post-emergency phase**".
- ▶ **The malnutrition–infection synergy triangle.** Three corners labelled "**Acute malnutrition**", "**Measles / diarrhoea / acute respiratory infection**" and "**Excess mortality**", with double-headed arrows between malnutrition and infection labelled "*each worsens the other*", and both feeding into mortality.

High-yield exam pointers

- ▶ **Emergency-phase priorities in two halves** — the five life-saving inputs, "*A Man Wants Food and Shelter*" (Assessment, Measles, Water and sanitation, Food and nutrition, Shelter and site planning); then the system functions (Health care, Communicable disease control, Surveillance, Training, Coordination, Camp management).
- ▶ **CMR > 1 death per 10,000 population per day** defines the acute phase of a complex emergency (Toole and Waldman, 1990) — quote the denominator exactly.
- ▶ **Severe acute malnutrition** = weight-for-height/length Z-score **below –3 SD**, or MUAC <115 mm, or nutritional oedema → therapeutic feeding; the –2 Z cut-off is **wasting** (acute malnutrition overall), used as the camp surveillance indicator.
- ▶ **Water:** at least 15–20 L per person per day; WHO basic-health-protection minimum 20 L/person/day, 7.5 L of it for drinking and cooking; source within 0.5 km in disaster response.
- ▶ **Goma, 1994:** 50,000 deaths among 800,000 arrivals — **85% diarrhoeal, of which 60% cholera and 40% shigella**. **Kabul, 1993:** acute respiratory infection caused **30% of under-5 deaths**.
- ▶ **Mass typhoid vaccination is not recommended** in camps and mass campaigns give a false sense of security (Park); but **oral cholera vaccine IS recommended** — preemptively and reactively — since the WHO 2010 revision and the WHO stockpile, always as an **adjunct** to safe water and excreta disposal, which remain the most practical and effective strategy.
- ▶ **Park's seven mechanisms** of increased communicable disease transmission after displacement — quote them as a list; each one supplies its own control measure.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e Ch 17 p 952 states that mass typhoid and cholera vaccination is not a proven large-scale public health measure in disasters, but this is now out of date for cholera — the World Health Organization revised its recommendation in 2010 and established an oral cholera vaccine stockpile, so oral cholera vaccine is currently recommended for preemptive and reactive use, including in refugee camps and complex emergencies (Plotkin's Vaccines 7e, Ch 14, p 285). The retrieved corpus covers refugee camp health chiefly through global sources and Park's generic Disaster Management chapter, and contained no India-specific refugee camp figures or Indian policy on refugee health, so none is quoted here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO 2023 wasting guideline, Disaster Management (Amendment) Act 2025

Programme and policy updates (India)

- ▶ **Hillchol (BBV131), an Indian single-strain oral cholera vaccine** — phase III randomised trial across 10 Indian sites, 1,800 participants stratified as adults, children 5 to <18 years and infants 1 to <5 years, allocated 3:1 against Shanchol with two oral doses on days 0 and 14. A ≥ 4 -fold rise in vibriocidal antibody occurred in 68.25% (Ogawa) and 69.52% (Inaba), meeting non-inferiority, with adverse events in 13.1% and no difference between vaccines. The single stable O1 Hikojima strain is designed to raise production efficiency and affordability, and adding it to the stockpile is argued to improve supply and simplify logistics against the vaccine shortage that has accompanied the global surge in cholera outbreaks [PMID 40174256].
- ▶ **Pre-emptive two-dose OCV delivered entirely through the routine Indian public health system** — Bishnupur-II block, South 24 Parganas, West Bengal, October–November 2024: household census microplanning and door-to-door delivery by ASHAs and anganwadi workers enumerated 45,073 people; first-dose coverage 66.4%, two-dose coverage 63.1%, second-dose retention 95.1%, coverage above 75% in children 1–5 years, only mild adverse events following immunisation (0.3% of 58,390 doses). Total cost per fully vaccinated person was INR 416 (about USD 5), of which delivery was INR 13 and vaccine procurement 97% of all expenditure. The authors present it as a replicable framework for high-risk, flood-prone settings across South and Southeast Asia under the WHO Ending Cholera roadmap to 2030 — which is the same district machinery a relief camp would have to vaccinate through [PMID 42092212].

Current national data

- ▶ **MUAC screening thresholds re-derived for Indian children** — community cross-sectional surveys of 2,335 children aged 6–59 months in five districts of Jharkhand, Odisha, Uttar Pradesh and Madhya Pradesh found wasting (weight-for-height $Z < -2$) in 30.9% and severe wasting in 5.5%. MUAC correlated strongly with the Z-score (0.72; 95% CI 0.69–0.74) but agreed poorly with the current cut-offs (kappa 0.28 for wasting, 0.36 for severe wasting); ROC-optimal values were 12.8 cm for severe wasting (sensitivity 0.80, specificity 0.87, AUC 0.91) and 13.8 cm for wasting. The authors propose field screening at MUAC <14.0 cm for wasting and <13.0 cm for

severe wasting, with Z-score confirmation afterwards — directly relevant to camp nutrition screening in India [PMID 41369874].

- **Chronic disease dictates quality of life in a protracted camp** — 1,058 Rohingya adults surveyed in Kutupalong, the largest camp in Cox's Bazar, about five years after displacement. The sample was deliberately stratified — 558 people with chronic disease recruited from camp medical centres and 499 healthy household residents — so the resulting 52.8%/47.2% split is a sampling design, **not** a prevalence, and must not be quoted as one. Among those with chronic disease, 46.6% rated quality of life poor or very poor against 11.6% of the healthy group, and Tobit regression put their physical-domain score 10.57 points lower (95% CI –12.97 to –8.17) and psychological-domain score 7.20 points lower (95% CI –9.71 to –5.93); people in work scored higher in both domains. The same paper puts the global forcibly displaced total above 110 million — about 1 person in 74 [PMID 39287945].

Elimination, surveillance and disease status

- **The camp epidemic profile has not improved in three decades** — scoping review of 92 epidemics in 32 countries in complex emergencies, 1990–2022. Shigellosis had the highest incidence at 241.0 per 1,000 at risk (mortality 11.7 per 1,000), then measles 25.0 per 1,000 (mortality 0.76) and cholera 13.0 per 1,000 (mortality 0.22); yellow fever, though uncommon at 0.8 per 1,000, carried a 20.1% case-fatality. There was **no statistically significant change in incidence, mortality or fatality trend across the whole period**, and the authors classify every epidemic identified as fundamentally preventable — water, sanitation, shelter, food and health services being the determining factors [PMID 38481292].
- **Measles still breaks out inside refugee settlements** — outbreak investigation in Palabek refugee settlement, Lamwo District, Uganda, July–September 2022: 56 cases (48 probable, 8 confirmed), 89% South Sudanese, overall attack rate 10 per 10,000, rising to 72 per 10,000 in the worst-affected parish and 51 per 10,000 in infants under 1 year. Vaccination coverage was only 63% and vaccine effectiveness 72%; the risk factors were malnutrition (OR 3.73, 95% CI 1.22–11.43) and having had one or no doses of measles vaccine (OR 7.0, 95% CI 2.8–17.6). The response recommended was mass vitamin A plus a measles dose for every child under five irrespective of vaccination status [PMID 41211032].
- **Single-dose oral cholera vaccine protects only modestly** — four years of epidemiological and genomic surveillance in the Democratic Republic of the Congo put single-dose killed whole-cell OCV effectiveness over 24 months at 59.8% (95% CI 19.7–79.9). A large clonal outbreak occurred 14 months after a campaign of more than a million doses because coverage in informal settlements reached only 9%, and household clinical and water isolates were closely related, confirming both person-to-person and water-to-person spread. With global shortage forcing single-dose campaigns, this reinforces that OCV remains an adjunct to water, sanitation and hygiene, never a substitute [PMID 39983698].

Guideline and legislative updates

- **Simplified, combined protocol for acute malnutrition** — against WHO's 2023 definitions (severe acute malnutrition = Z-score <–3, MUAC <115 mm or bilateral oedema; moderate = Z <–2 or MUAC <125 mm), a single combined protocol admitting on MUAC <125 mm or oedema and giving two ready-to-use therapeutic food sachets daily for severe and one for moderate cases

was run at scale in two Central African Republic health districts. Of 7,909 children, 81.2% recovered overall and 67.9% of those admitted with severe acute malnutrition, mean stay 38.7 days and mean 43.4 sachets per child; defaulting varied sharply between districts (8.2% versus 31.1%). This collapses the textbook's separate supplementary and therapeutic feeding streams into one product and one admission criterion — operationally far easier in a camp [PMID 38956431].

- **Disaster Management (Amendment) Act 2025 (India)** — passed by Parliament in March 2025, it transfers disaster-plan preparation from executive committees to the National and State Disaster Management Authorities themselves, obliges them to maintain national and state disaster databases covering risk severity, fund allocation and preparedness, empowers states to constitute Urban Disaster Management Authorities for state capitals and municipal corporation cities under the Municipal Commissioner and their own State Disaster Response Forces, and gives statutory status to the National Crisis Management Committee under the Cabinet Secretary. This is the legal frame within which Indian relief camps are now planned, resourced and audited [Disaster Management (Amendment) Act 2025].

Global and WHO developments

- **Refugee mental illness burden restated, and the scalable response to it** — a meta-analysis restricted to studies diagnosing by clinical interview with a validated instrument (26 studies, 5,143 adults, 15 countries) found PTSD in 31.46% (95% CI 24.43–38.50), depression in 31.51% (95% CI 22.64–40.38), anxiety disorders in 11.09% (95% CI 6.75–15.43) and psychosis in 1.51% (95% CI 0.63–2.40), with PTSD higher in women (34.02%) — roughly three times the "1 in 10 with PTSD, 1 in 20 with depression" estimates of the 2005 resettled-refugee analysis quoted in most textbooks, and an argument for care extending well beyond initial resettlement [PMID 32956381]. WHO's Self-Help Plus, five group sessions designed for delivery by briefly trained non-specialists, was then tested in an individual participant data meta-analysis of three randomised trials (1,795 refugees and asylum seekers). **Immediately after the course** it improved self-identified problems (β -1.56, 95% CI -2.54 to -0.59) and wellbeing (β 6.22, 95% CI 1.60 to 10.90) but **not** depressive symptoms (β -1.47, 95% CI -3.19 to 0.26, non-significant). **At 5–6 months** it outperformed enhanced care as usual for depressive symptoms (β -1.13, 95% CI -1.99 to -0.26), with wellbeing (β 4.15, 95% CI 1.44 to 6.87) and self-identified problems (β -0.78, 95% CI -1.53 to 0.04) still favouring it — the authors attribute the delayed effect to the "incubation" period of acceptance and commitment therapy. Separately, on the **postintervention** depression outcome, being unemployed moderated the effect (β 1.60, 95% CI 0.20 to 3.01), meaning the unemployed benefited more than the employed, as did those with lower baseline wellbeing. This is the delivery model that fits camps, where mental health specialists are almost never available [PMID 37524517].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 17 — Disaster Management; Ch 11 — Nutrition and Health; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics); Oxford Textbook of Public Health 7e (Ch 92 — Forced migrants and other displaced populations; Ch 104 — Humanitarian emergencies; Ch 12 — Water and sanitation); Textbook of International Health (Ch 8 — Health under Crisis); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 79 — Disaster Preparedness and Response); Plotkin's Vaccines 7e (Ch 14 — Cholera Vaccines); Sphere Project Handbook 2004 (minimum standards and Humanitarian Charter); 1951 UN Convention Relating to the Status of Refugees and the 1967 Protocol; UNHCR Statute 1950.

Recent-advances sources: 40174256 — A double-blind, randomised phase III clinical trial to evaluate safety, immunogenicity, non-inferiority and lot-to-lot consistency of single component oral cholera vaccine BBV131 (Hillchol) in comparison to Shanchol, *Vaccine* 2025, doi:[10.1016/j.vaccine.2025.126998](https://doi.org/10.1016/j.vaccine.2025.126998) (abstract-only); 42092212 — Mass vaccination with two-dose oral cholera vaccine in vulnerable populations, using existing public health infrastructure from planning to delivery through multi-sectoral collaboration in West Bengal, India, *Journal of Infection and Public Health* 2026, doi:[10.1016/j.jiph.2026.103237](https://doi.org/10.1016/j.jiph.2026.103237) (abstract-only); 41369874 — Optimizing Mid-Upper Arm Circumference (MUAC) Thresholds for Screening Wasting and Severe Wasting in Children Aged 6–59 Months in Community Settings in India, *Indian Journal of Pediatrics* 2025, doi:[10.1007/s12098-025-05869-0](https://doi.org/10.1007/s12098-025-05869-0) (abstract-only); 39287945 — Chronic Illness and Quality of Life 5 Years After Displacement Among Rohingya Refugees in Bangladesh, *JAMA Network Open* 2024, doi:[10.1001/jamanetworkopen.2024.33809](https://doi.org/10.1001/jamanetworkopen.2024.33809); 38481292 — The epidemiological profile of incidence and mortality from epidemics in complex humanitarian emergencies from 1990 to 2022 — a scoping review, *Tropical Medicine and International Health* 2024, doi:[10.1111/tmi.13982](https://doi.org/10.1111/tmi.13982) (abstract-only); 41211032 — Epidemiological investigation of measles outbreak in a refugee settlement in Lamwo District, Uganda, *Pan African Medical Journal* 2025, doi:[10.11604/pamj.supp.2025.51.1.47672](https://doi.org/10.11604/pamj.supp.2025.51.1.47672); 39983698 — Epidemiologic and Genomic Surveillance of *Vibrio cholerae* and Effectiveness of Single-Dose Oral Cholera Vaccine, Democratic Republic of the Congo, *Emerging Infectious Diseases* 2025, doi:[10.3201/eid3102.241777](https://doi.org/10.3201/eid3102.241777); 38956431 — Effectiveness of acute malnutrition treatment with a simplified, combined protocol in Central African Republic: an observational cohort study, *Maternal and Child Nutrition* 2024, doi:[10.1111/mcn.13691](https://doi.org/10.1111/mcn.13691); 32956381 — The prevalence of mental illness in refugees and asylum seekers: a systematic review and meta-analysis, *PLoS Medicine* 2020, doi:[10.1371/journal.pmed.1003337](https://doi.org/10.1371/journal.pmed.1003337); 37524517 — Self-Help Plus for refugees and asylum seekers: an individual participant data meta-analysis, *BMJ Mental Health* 2023, doi:[10.1136/bmjment-2023-300672](https://doi.org/10.1136/bmjment-2023-300672). Also the Disaster Management (Amendment) Act 2025, India (passed by Parliament March 2025), and the WHO 2023 wasting guideline definitions of severe and moderate acute malnutrition.

♦ ♦ ♦

Yellow fever vaccination

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — Yellow fever vaccination is active immunisation with the live attenuated 17D vaccine, the only product currently approved for immunisation against yellow fever. It is the single most effective control measure against the disease and the **only vaccination for which a certificate is required for international travel** under the International Health Regulations (IHR) 2005 (Park 28e, Ch 3, p 152; Park 28e, Ch 5, p 348).

The 17D vaccine — composition, cold chain and administration

Attribute	Specification
Type	Live attenuated viral vaccine, derived by serial passage of the virulent <i>Asibi</i> strain in mouse-embryo and then chick-embryo tissue culture (<i>Theiler and Smith</i>) (Plotkin 7e, Ch 63, pp 1756, 1759)
Substrains in manufacture	17DD and 17D-204, from passage levels 195 and 204 of the original 17D virus (Plotkin 7e, Ch 63, p 1759)
Substrate and form	Grown in chick embryo , subsequently freeze-dried (lyophilised) (Park 28e, Ch 5, p 348)
Cold chain	Heat sensitivity of the lyophilised vaccine is its major drawback in mass campaigns in tropical countries ; store between +5 °C and –30 °C, preferably below 0 °C, until reconstituted with the sterile physiological saline diluent supplied (Park 28e, Ch 5, p 348)
After reconstitution	Keep on ice, away from sunlight; discard if not used within half an hour (Park 28e, Ch 5, p 348)
Dose and route	0.5 ml subcutaneously at the insertion of the deltoid, single dose irrespective of age (Park 28e, Ch 5, p 348)
Onset and duration of immunity	Immunity appears from the 7th day and lasts possibly for life ; neutralising antibodies develop in more than 90 per cent of vaccinees (Park 28e, Ch 5, p 348; Plotkin 7e, Ch 63, p 1772)
Spacing from other vaccines	Cholera and yellow fever vaccines interfere if given together or within 3 weeks — give 3 weeks or more apart (Park 28e, Ch 5, p 348)
Minimum potency	All 17D vaccines contain a minimum of 3 log₁₀ international units per dose (Plotkin 7e, Ch 63, p 1776)

Indications — who is vaccinated

Vaccination is targeted because **47 countries in Africa and Latin America, with a combined population of more than 900 million, are at risk** — 508 million in 32 African countries and the remainder in 13 Latin American countries, with an estimated 84,000–170,000 cases and 29,000–60,000 deaths worldwide (Park 28e, Ch 5, p 347).

- **All inhabitants of endemic countries or endemic areas within countries** — routinely, preferably at **9 months of age**, given in the Expanded Programme on Immunization (EPI) along with measles vaccine (Plotkin 7e, Ch 63, pp 1804, 1808).

- **Persons migrating from non-endemic to endemic areas** — high risk, should be immunised (Plotkin 7e, Ch 63, p 1804).
- **Occupationally exposed groups** — wood cutters, hunters and forest workers whose work brings them into contact with jungle (sylvatic) transmission (Park 28e, Ch 5, p 347).
- **All international travellers, including infants, exposed to the risk of yellow fever or passing through endemic zones**, who must hold a valid international certificate before entering a yellow fever receptive area (Park 28e, Ch 5, p 348).
- **Populations at risk during an urban outbreak** — rapid mass immunisation is the **most effective control strategy** for urban yellow fever (Park 28e, Ch 5, p 348).

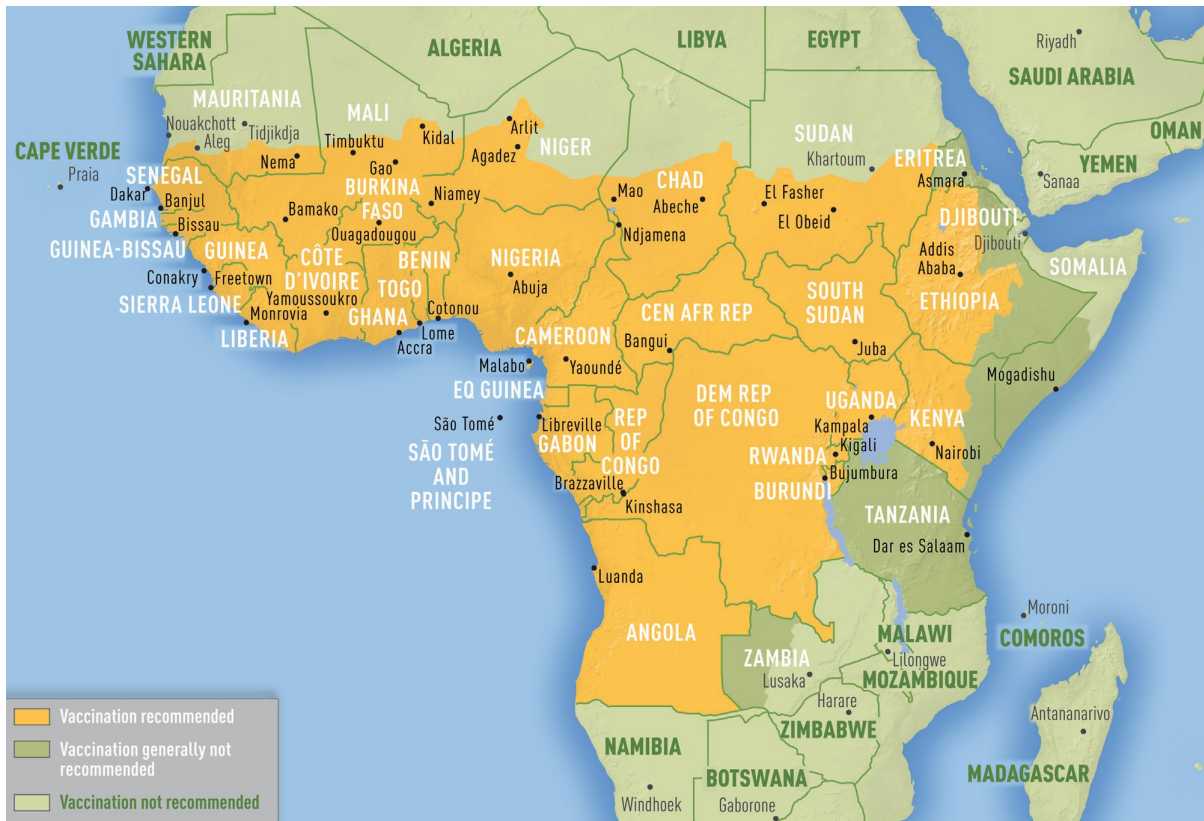


Fig 71.5 — Yellow fever vaccine recommendations in Africa. (From Centers for Disease Control and Prevention (Plotkin's Vaccines, 7ed, p.2018))



Fig 71.6 — Yellow fever vaccine recommendations in the Americas. (From Centers for Disease Control and Prevention (Plotkin's Vaccines, 7ed, p.2019)

Contraindications, precautions and adverse events

The risk of death from yellow fever is much higher than the risk related to the vaccine, so contraindications must be applied strictly but not liberally (Park 28e, Ch 5, p 348).

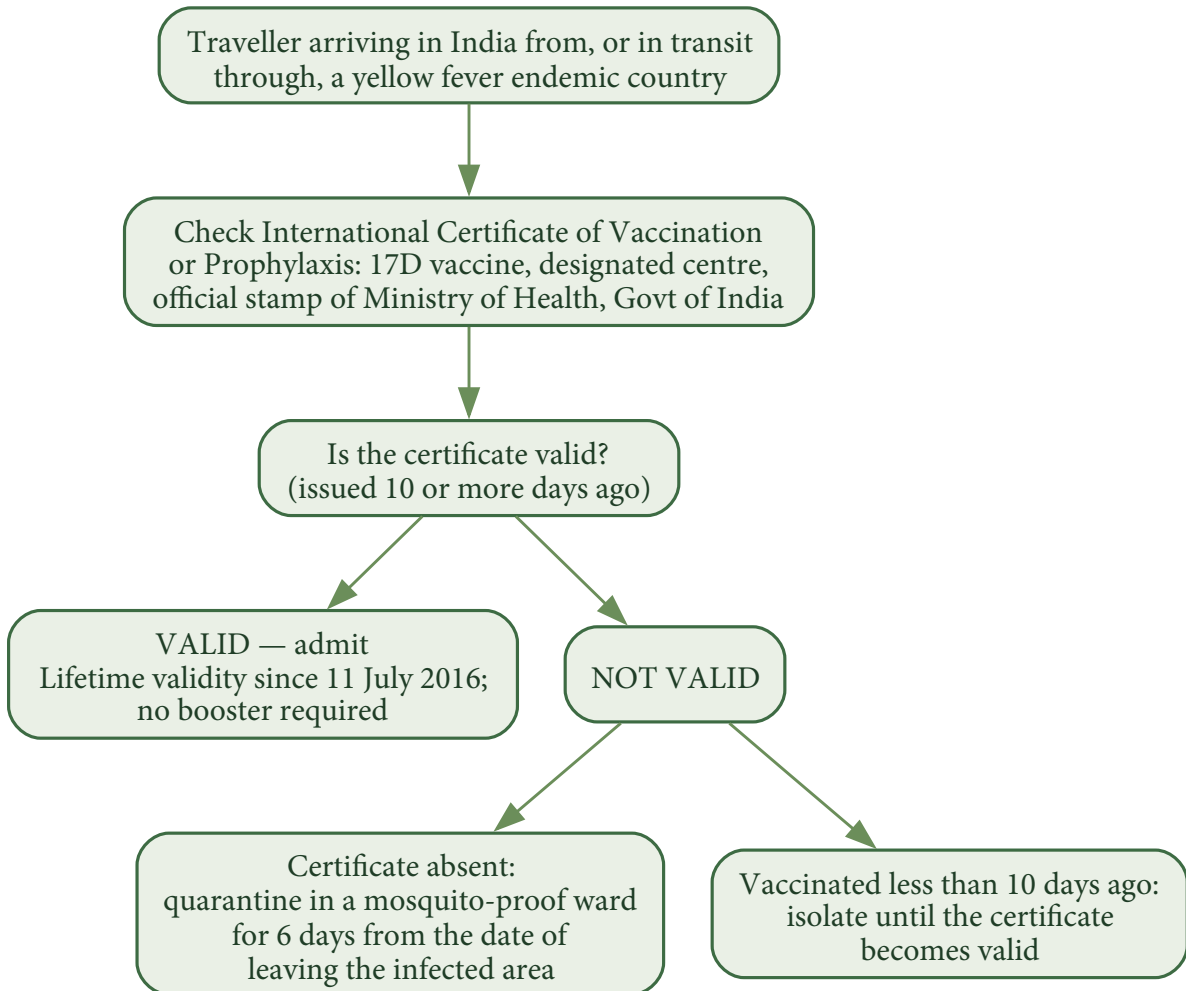
Category	Groups
Contraindications (Park 28e, Ch 5, p 348)	Children under 9 months for routine immunisation (under 6 months during an epidemic); pregnancy , except during an outbreak when the risk of infection is high; severe allergy to egg protein ; severe immunodeficiency from symptomatic HIV/AIDS or other causes, or a thymus disorder
Contraindications (Plotkin 7e, Ch 63, p 1805)	Allergy to vaccine components (including egg); age < 6 months ; symptomatic HIV or CD4 ⁺ T-lymphocytes < 200/mm³ (< 15 per cent of total in children under 6 years); thymus disorder with abnormal immune function or history of thymectomy ; primary immunodeficiencies; malignant neoplasms; transplantation; immunosuppressive or immunomodulatory therapy
Precautions (Plotkin 7e, Ch 63, p 1805)	Age 6–8 months ; age ≥ 60 years ; asymptomatic HIV with CD4 ⁺ 200–499/mm³ (15–24 per cent in children under 6 years); pregnancy ; breastfeeding

Adverse event following immunisation	Frequency
Mild post-vaccinal reactions — myalgia, headache, low-grade fever, 5–10 days after vaccination	2–5 per cent of vaccinees (Park 28e, Ch 5, p 348)
Anaphylaxis / life-threatening allergic reaction, mainly in those allergic to eggs	Very rare; 1 per 55,000 doses (Park 28e, Ch 5, p 348; Plotkin 7e, Ch 71, p 2020)
Vaccine-associated neurotropic disease (YEL-AND)	1 per 125,000 doses ; risk is highest in young infants and inversely proportional to age (Plotkin 7e, Ch 71, p 2020; Ch 63, p 1805)
Vaccine-associated viscerotropic disease (YEL-AVD) — resembles wild-type yellow fever, similar lethality; first described 2001	1 per 250,000 doses (Plotkin 7e, Ch 71, p 2020; Ch 63, p 1728)
Serious adverse events by age (reporting rate ratio vs 19–29 years)	5.9 (95% CI 1.6–22.2) in first-time vaccinees ≥ 60 years; 10.4 (95% CI 2.7–40.2) in those ≥ 70 years (Plotkin 7e, Ch 71, p 2020)

The international certificate and the International Health Regulations (2005)

- **Certificate required** — India and most other countries require a valid certificate of vaccination against yellow fever from travellers coming from infected areas; **a few countries, including India, require it even if the traveller has only been in transit** (Park 28e, Ch 5, p 348).
- **Where it is issued** — vaccination for international travel must be given at an **approved / designated yellow fever vaccination centre**, and the certificate must be signed by the authorised official and bear the **official stamp of the Ministry of Health, Government of India** (Park 28e, Ch 5, pp 348–349).
- **When it becomes valid** — the certificate is valid from **10 days after the date of vaccination** (Park 28e, Ch 5, pp 348–349).
- **How long it is valid** — in May 2014 the World Health Assembly adopted an amendment to the IHR (2005) changing the term of validity from **10 years to the life of the person vaccinated**; it entered into force on **11 July 2016** and is legally binding on all IHR States Parties (Park 28e, Ch 5, p 349).

- **No booster** — since 11 July 2016, **revaccination or a booster dose is not required** of international travellers as a condition of entry into a State Party, regardless of when the certificate was first issued; **in India this lifetime validity applies automatically** (Park 28e, Ch 5, p 349).
- **Infants** — it rests with each country to decide whether a certificate is required for infants under one year of age (Park 28e, Ch 5, p 348).



Yellow fever certificate check for a traveller arriving in India, a receptive area

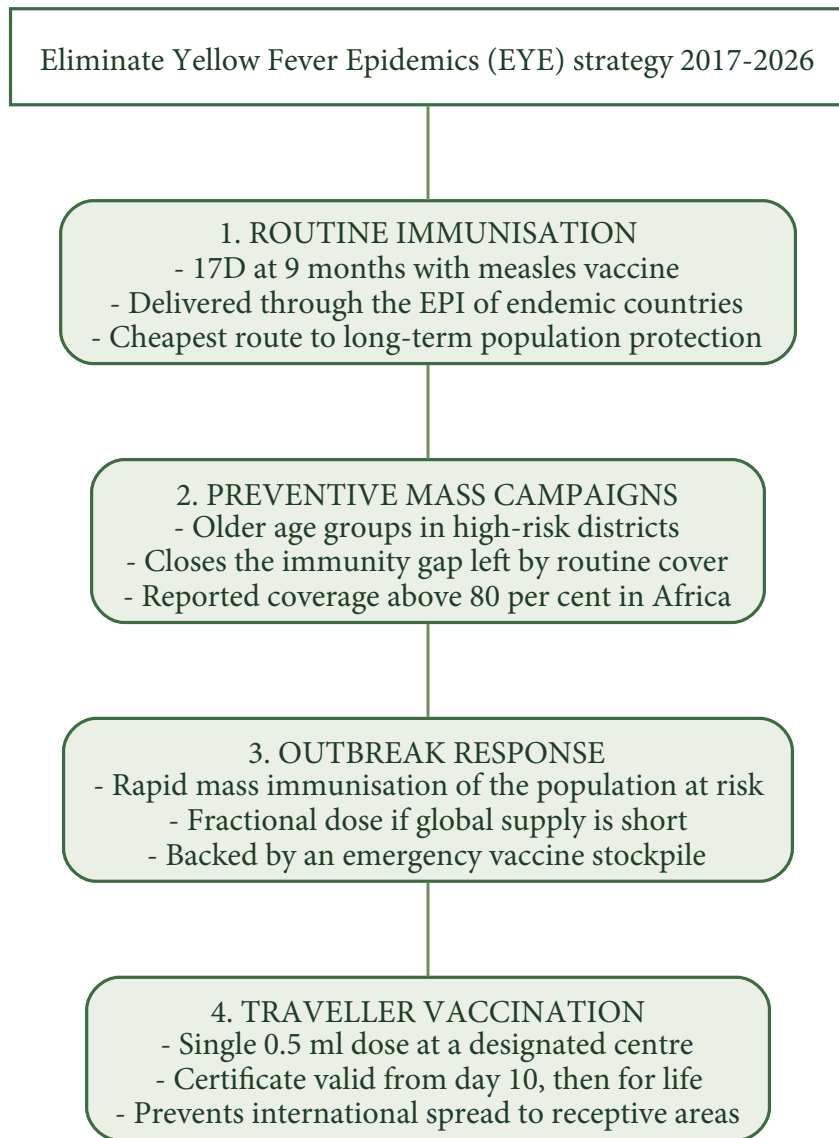
Accompanying international measures at ports (implemented by the Government of India through aerial and maritime traffic regulations, because India is a **receptive area** — "an area in which yellow fever does not exist but where conditions would permit its development if introduced", with an unvaccinated susceptible population, abundant *Aedes aegypti* and favourable climate):

Measure	Standard
Aircraft and ships arriving from endemic areas	Aerosol spraying with prescribed insecticides on arrival to destroy insect vectors (Park 28e, Ch 5, p 348)
Vector-free zone around ports	Airports and seaports kept free from breeding of vectors over an area extending at least 400 metres around their perimeters (Park 28e, Ch 15, p 925)

Measure	Standard
<i>Aedes aegypti</i> index — percentage of houses in a limited, well-defined area on whose premises actual breeding of <i>Aedes aegypti</i> is found, out of houses examined	Kept below 1 (Park 28e, Ch 5, p 348); under the IHR it is kept at zero at all ports (Park 28e, Ch 15, p 925)
Basis of the 6-day quarantine	Incubation period is 3–6 days , with 6 days recognised under the IHR (Park 28e, Ch 5, pp 347–348)

Vaccination as a control strategy — global initiatives

- **WHO Yellow Fever Initiative (2006)** — three aims: include yellow fever vaccine in the **EPI at 9 months and older**, run **preventive mass immunisation campaigns** for older age groups in high-risk areas, and **ensure adequate vaccine supplies**; it protected more than 95 million people in Africa and raised production capacity from 30 million doses (2000) to over 75 million doses (2009) (Plotkin 7e, Ch 63, p 1809).
- **Eliminate Yellow Fever Epidemics (EYE)** — a **global strategy 2017–2026** (WHO, *Weekly Epidemiological Record* 2017;92(16):193–204), a response to the increased threat of urban outbreaks with international spread. Its **three strategic objectives** are to (i) **protect at-risk populations**, (ii) **prevent international spread**, and (iii) **contain outbreaks rapidly** (Park 28e, Ch 5, pp 349–350). It carries forward the plan WHO launched in 2016 to **immunise 1.4 billion people against yellow fever in 40 countries by 2026** (Oxford Textbook of Public Health 7e, Ch 107, p 4050).
- **Five competencies underpinning EYE success** — affordable vaccines and a sustained vaccine market; strong political commitment at global, regional and country levels; high-level governance with long-term partnerships; synergies with other health programmes and sectors; and research and development for better tools and practices (Park 28e, Ch 5, pp 348–349).
- **Fractional dosing** — after the 2016 Angola and Democratic Republic of the Congo outbreaks, WHO with the approval of the Strategic Advisory Group of Experts (SAGE) **endorsed fractional doses for outbreak response**; a 10-fold reduction in dose produced seroconversion rates and viraemia similar to the full dose. It was used successfully in Angola, the Democratic Republic of the Congo (2016) and Brazil (2017) during the global vaccine shortage (Plotkin 7e, Ch 63, pp 1765, 1777; Oxford Textbook of Public Health 7e, Ch 75, p 2837).



The four delivery routes of yellow fever vaccination under the EYE strategy 2017-2026

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Labelled diagram: the yellow fever vaccination certificate pathway.** Labels to write: Traveller from endemic country → Designated yellow fever vaccination centre → 17D vaccine, 0.5 ml subcutaneous, deltoid insertion → Certificate signed and stamped by Ministry of Health, Government of India → Valid from day 10 → Valid for life (IHR amendment, 11 July 2016).
- ▶ **Labelled diagram: the three cycles of yellow fever transmission that vaccination interrupts.** Labels to write: Sylvatic (jungle) cycle — monkey ↔ forest mosquito, spillover to forest workers; Intermediate (savannah) cycle — semi-domestic mosquitoes, monkey and man, commonest African outbreak type; Urban cycle — man ↔ *Aedes aegypti* ↔ man, the cycle that causes large epidemics and that mass vaccination is aimed at.

High-yield exam pointers

- ▶ "17D at 9 months, 0.5 ml subcutaneous, deltoid, single dose, immunity from day 7, certificate from day 10, valid for life."
- ▶ **The four dates:** vaccine strain from *Asibi* by Theiler; May 2014 World Health Assembly amendment; 11 July 2016 entry into force of lifetime validity; 2017–2026 EYE strategy.
- ▶ **Storage:** +5 °C to –30 °C, preferably below 0 °C; reconstituted vaccine **discarded within half an hour.**
- ▶ **Three adverse-event denominators:** anaphylaxis 1 in 55,000; YEL-AND 1 in 125,000; YEL-AVD 1 in 250,000 doses.
- ▶ **Contraindication mnemonic — "9-P-E-I":** age under 9 months, Pregnancy (unless outbreak), Egg allergy (severe), Immunodeficiency (symptomatic HIV/AIDS, thymus disorder).
- ▶ **India is a receptive area, not an endemic area** — no indigenous yellow fever has ever been reported in Asia, but *Aedes aegypti* and the climate would permit establishment; hence the 6-day quarantine, the 400-metre vector-free port perimeter and the *Aedes aegypti* index.
- ▶ **Cholera and yellow fever vaccines must be given 3 weeks or more apart.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's summary immunisation table in Chapter 3 (pp 146–147) still states that immunity after 17D vaccine begins 10–12 days after vaccination and extends up to 10 years; this is superseded by the same book's Chapter 5 text (immunity from day 7, lasting possibly for life) and by the 2016 International Health Regulations amendment giving the certificate lifetime validity, so write the Chapter 5 position in the exam.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, WHO EYE strategy 2017–2026, Global Burden of Disease 2021

Programme and policy updates (India)

- **An Indian laboratory route to measuring yellow fever immunity** — Serum Institute of India, Pune, developed and validated an in-house PRNT₅₀ for yellow fever neutralising antibodies, testing it for specificity, linearity, precision, accuracy and robustness. It is aimed squarely at the flavivirus cross-reactivity that makes ELISA-based tests unreliable, and gives India domestic capacity to evaluate 17D immunogenicity after vaccination and after natural infection [PMID 41442866].
- **What one kind of prior flavivirus exposure does to the 17D response** — in 250 healthy young adults in Munich, all naive to natural flavivirus infection and unvaccinated against Japanese encephalitis and yellow fever, **prior tick-borne encephalitis (TBEV) vaccination** left the early IgM-driven neutralising response to 17D intact — IgM carried about **75% of the neutralising titre at day 28** — but skewed the later IgG response towards the cross-reactive, poorly neutralising **fusion-loop epitope**, and those antibodies produced antibody-dependent enhancement of dengue and Zika virus infection in cell culture. Note what was **not** studied: dengue- and Japanese-encephalitis-experienced people were excluded by design, so applying this to an Indian population is an **extrapolation, not a finding**. What the paper does support directly is the laboratory point — after 17D, protection should be judged from neutralisation titres, not from total anti-flavivirus IgG [PMID 38402207].

Current national data

- **WHO has revised the risk denominator the textbooks quote** — the yellow fever fact sheet of 20 October 2025 classes 27 African and 13 Latin American countries as **high-risk** and estimates **67,000–173,000 severe infections and 31,000–82,000 deaths each year**, most of them in Africa. Write these figures with their date in place of the older "47 countries, 84,000–170,000 cases, 29,000–60,000 deaths" set [WHO 2025-Oct].
- **Global Burden of Disease 2021** — incident cases fell from 341,942 in 1990 to **86,509 in 2021**, a **74.7% decrease**, with the age-standardised incidence rate down from 6.08 to **1.15 per 100,000** (estimated annual percentage change –4.76; 95% CI –5.10 to –4.42). Western, Central and Eastern sub-Saharan Africa carry the highest rates, and children aged 5–10 years the highest age-specific incidence [PMID 40559602].

Elimination, surveillance and disease status

- **The EYE strategy reaches its 2026 endpoint with coverage as the binding constraint** — Africa-wide yellow fever vaccination coverage was still **44% in 2020**, and the two most recent field surveys both come in around 80%. In the Democratic Republic of the Congo, coverage was **80.8%** among 1,247 children aged 9–23 months across six provinces in 2024 — short of the country's coverage targets, which the authors discuss without putting a number on them — with the rainy season (adjusted OR 5.92), absence of community awareness activities (4.42) and reliance on a fixed-site strategy (5.73) as the dominant barriers. In Ghana, coverage was **80.3%** among 2,914 nomadic people in 414 households of the Savannah Region after an outbreak response campaign, significantly below the national figure of **88%** ($p < 0.001$) but, on the authors' own reading, still within the **WHO herd-immunity threshold of about 80%**; the commonest reason for non-vaccination was simply being away from the district during the campaign [PMID 42542050; PMID 42111508].

- **Vector surveillance shows how broad the competent-vector base is** — Nigeria's national survey of 16 outbreak states (2017–2020) found yellow fever virus in **33 of 127 Aedes pools (about 26%)** across six species, including *Ae. aegypti* and *Ae. albopictus*, and house and Breteau indices above the WHO risk thresholds of 4% and 5% in every state surveyed but one. This is the same index logic India applies at its ports, and both species are established in India [PMID 41174791].

Guideline and legislative updates

- **NIFTY (2025) put fractional dosing on a firm footing in adults** — a double-blind randomised non-inferiority trial in 480 adults in Kenya and Uganda using Institut Pasteur de Dakar 17D-204 found **500 IU and 1,000 IU non-inferior to the standard 13,803 IU dose** for seroconversion at day 28 (difference 0.01 percentage points, 95% CI –5.0 to 5.1 for both, intention to treat), while 250 IU failed per protocol (–6.7 points, 95% CI –11.7 to 1.6). The catch for travel medicine is day 10: seroconversion was **61% with 500 IU against 85% with the standard dose**, and day 10 is exactly when the international certificate becomes valid [PMID 39970397].
- **The adult minimum dose does not transfer to infants** — in 420 infants aged 9–12 months in Kenya and Uganda given 17D-204 alongside measles-rubella vaccine, 500 IU produced **93% seroconversion against 99%** for the standard dose (difference –6.15 percentage points, 95% CI –10.27 to –2.02), **failing non-inferiority**; the standard dose must be retained for the 9-month dose in the Expanded Programme on Immunization. Safety is not the obstacle — a Ugandan trial of one-fifth and one-half doses of 17DD in 1,053 children aged 9–23 months found no difference in the frequency or severity of adverse events between dose arms and no serious event causally related to vaccination [PMID 41544642; PMID 39153293].

Global and WHO developments

- **Fifty years of the Expanded Programme on Immunization (1974–2024)** — an economic evaluation across all 194 WHO member states put the cost of the programme at **US\$937 billion** (95% CI 699–1,187) against **US\$15,050 billion** (12,609–17,605) in averted productivity losses, an aggregate **benefit–cost ratio of 16.06** (95% CI 10.62–25.20). Yellow fever is one of the 14 pathogens covered, and the programme was cost-saving in every member state [PMID 42242275].
- **Climate has entered the yellow fever risk model** — a dynamical model of Brazil's first urban yellow fever epidemic in nearly a century showed that **drought** drove it, by pushing forest mosquitoes and non-human primates towards the city in search of water and making mosquitoes bite more often to avoid desiccation; of vector control, habitat conservation and vaccination, **vaccination had the strongest effect** in ending the outbreak. Droughts are projected to become more frequent in the region, so this is a One Health and climate-adaptation argument for sustained routine coverage rather than reactive campaigns [PMID 42600025].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods; Ch 5, Epidemiology of Communicable Diseases; Ch 15, Environment and Health); Plotkin's Vaccines 7e (Ch 63, Yellow Fever Vaccines; Ch 71, Vaccines for International Travel; Ch 9, General Immunization Practices); Oxford Textbook of Public Health 7e (Ch 75, Emerging and Re-emerging Infections; Ch 107, Global Health in the Era of Sustainable Development); International Health Regulations (2005), Annex 7 amendment in force 11 July 2016; WHO, *Eliminate Yellow Fever Epidemics: a global strategy 2017–2026*, Weekly Epidemiological Record 2017;92(16):193–204.

Recent-advances sources: 41442866 — Plaque reduction neutralization test (PRNT50) for the detection of anti-yellow fever antibodies from clinical samples, Vaccine 2026, doi:[10.1016/j.vaccine.2025.128151](https://doi.org/10.1016/j.vaccine.2025.128151) (abstract-only); 38402207 — Prior flavivirus immunity skews the yellow fever vaccine response to cross-reactive antibodies with potential to enhance dengue virus infection, Nature Communications 2024, doi:[10.1038/s41467-024-45806-x](https://doi.org/10.1038/s41467-024-45806-x); 40559602 — The Incidence and Trends of Yellow Fever from 1990 to 2021 in Major Endemic Regions: A Systematic Analysis Based on the 2021 Global Burden of Disease Study, Pathogens 2025, doi:[10.3390/pathogens14060594](https://doi.org/10.3390/pathogens14060594); 42542050 — Systemic barriers to achieving yellow fever vaccination coverage targets among children aged 9–23 months in the Democratic Republic of the Congo, Vaccine 2026, doi:[10.1016/j.vaccine.2026.128995](https://doi.org/10.1016/j.vaccine.2026.128995) (abstract-only); 42111508 — Yellow fever vaccination coverage among nomadic populations in the Savannah Region, Ghana: a cross-sectional study following an outbreak, Pan African Medical Journal 2026, doi:[10.11604/pamj.2026.53.24.49809](https://doi.org/10.11604/pamj.2026.53.24.49809); 41174791 — Nationwide surveillance detects yellow fever and chikungunya viruses in multiple Aedes mosquito species in Nigeria, Parasites & Vectors 2025, doi:[10.1186/s13071-025-07051-z](https://doi.org/10.1186/s13071-025-07051-z); 39970397 — Low-Dose Yellow Fever Vaccine in Adults in Africa (NIFTY), New England Journal of Medicine 2025, doi:[10.1056/NEJMoa2407293](https://doi.org/10.1056/NEJMoa2407293); 41544642 — Low-dose yellow fever vaccination in infants: a randomised, double-blind, non-inferiority trial, Lancet 2026, doi:[10.1016/S0140-6736\(25\)02069-0](https://doi.org/10.1016/S0140-6736(25)02069-0) (abstract-only); 39153293 — Adverse events following immunization (AEFI) with fractional one-fifth and one-half doses of yellow fever vaccine compared to full dose in children 9–23 months old in Uganda, Vaccine 2024, doi:[10.1016/j.vaccine.2024.126197](https://doi.org/10.1016/j.vaccine.2024.126197); 42242275 — Global, regional, and national impact of the Expanded Programme on Immunization against 14 pathogens from 1974 to 2024: an economic evaluation, Lancet Global Health 2026, doi:[10.1016/j.langlo.2026.103965](https://doi.org/10.1016/j.langlo.2026.103965) (abstract-only); 42600025 — Drought dynamics explain once in a century yellow fever virus outbreak in Brazil with implications for climate change, Science Advances 2026, doi:[10.1126/sciadv.adz6832](https://doi.org/10.1126/sciadv.adz6832); WHO yellow fever fact sheet, 20 October 2025; WHO Eliminate Yellow Fever Epidemics (EYE) strategy 2017–2026.

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